INORGANIC CHEMISTRY
for the JEE
(Mains and Advanced)

K. Rama Rao

ALWAYS LEARNING

PEARSON

Inorganic Chemistry
for the JEE
(Mains and Advanced)

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Preface

Inorganic Chemistry is an outcome of several years of teaching chemistry to the students preparing for different competitive examinations. When the thought of bringing out this book came to my mind, three pertinent questions required consideration.

- First, is there really a need to bring out yet another book on the subject 'inorganic chemistry' when there are already so many standard books available in the market?
- Second, how this book would tackle the limitations in the presentation of the subject in case of preparation of different competitive examinations after the completion of Class XII?
- Third, what must be the right order or sequence of topics so that the preparation will be easier for the students?

I believe that the answer to the first question lies in the rich content offered by this book. While dealing with various topics of inorganic chemistry, one needs to refer to different books for clarifications on various topics. Naturally, an absence of a one-stop solution to all the problems is felt by the students. This justifies the need to bring out a book which can serve as a single source of reference on inorganic chemistry and make learning simpler and facilitate problem-solving.

An analysis of the questions asked in various competitive examinations conducted over the last few years was done. It was seen that the knowledge required to answer the questions is much more than what is offered at the Class XII level. In fact, a deeper study of the concepts is needed to attempt these questions. This book goes a little deeper into the concepts/topics included at the Class XII level. This answers the second question mentioned above. Moreover, this analysis also fulfills the need to study the difficulty level of questions, types of questions and important topics etc., as the sole purpose of this book is to equip the student with sufficient knowledge to solve multiple-choice questions pertaining to inorganic chemistry. As far as the third question is concerned, my own personal experience in teaching the subject in a student-friendly manner has been helpful in planning the sequence in which the topics should be dealt with.

Earlier, the study of inorganic chemistry was thought to be a very complex and elongated process; one requiring the need to study and remember all the properties and uses of various elements and their compounds thoroughly. But with the passage of time, the requirements of teachers and students necessitated the need to deal with each topic in a logical manner. For instance, there has been a steady infiltration of physical chemistry into inorganic chemistry and it has resulted in the subject being made rigorous and more comprehensive for studying as compared to the past scenario. As such, it is a futile attempt if one writes a book on inorganic chemistry without laying stress on structural and energy considerations which are the two kingpins upon which a satisfactory development of the subject rests. The Class XII syllabus tends to reflect this trend. For this purpose, concepts such as enthalpy, entropy, free energy changes, equilibrium and equilibrium constant, acid base etc., are explicitly touched upon wherever necessary. In explaining the stability and solubility of inorganic compounds such as carbonates, sulphates, halides, oxides, hydroxides, etc., the concept of thermodynamics is aptly used. Similarly, an attempt has been made to explain the trend in solubility of inorganic compounds based on the acid base theory and thermodynamic data. Mainly, the focus is on explaining the different aspects with a logical approach so that students may retain a steady interest in the subject.

The book goes beyond the immediate needs of the existing Class XII syllabus and fulfills all the requirements of a preparatory tool required to attempt questions asked in various competitive examinations. I hope that not only students would benefit from this book but teachers would also find it as a valuable resource for referring to the explanations of various concepts in a simple and detailed manner.

I am grateful to all those who directly or indirectly encouraged me to author this book. I am also very grateful to the staff of Pearson Education, especially Rajesh Shetty, Bhupesh Sharma and Vamanan Namboodiri, for their continuous encouragement and hard work in bringing out this book in this fascinating manner.

Good Luck!

K. Rama Rao

1 CHAPTER

Structure of Atom

The universe is a concourse of atoms.
Marcus Aurelius

1.1 INTRODUCTION

The introduction of atomic theory by John Dalton early in the nineteenth century marks the beginning of modern era in the research initiatives in the field of Chemistry. The virtue of Dalton's theory was not that it was new or original, for theories of atoms are older than the science of chemistry, but that it represented **the first attempt to place the corpuscular concept of matter upon a quantitative basis**. The theory of the atomic constitution of matter dates back at least 2,500 years to the scholars of ancient Greece and early Indian philosophers who were of the view that atoms are fundamental building blocks of matter. According to them, the continued subdivision of matter would ultimately yield atoms **which would not be further divisible**. The word 'atom' has been derived from the Greek word **'a-tomio'** which means **'uncuttable'** or **non-divisible**. Thus, we might say that as far as atomic theory is concerned, Dalton added nothing new. He simply displayed a unique ability to crystallize and correlate the nebulous notions of the atomic constitution prevalent during the early nineteenth century into a few simple quantitative concepts.

John Dalton was born in England in 1766. His family was poor, and his formal education stopped when he was eleven years old. He became a school teacher. He was colour blind. His appearance and manners were awkward, he spoke with difficulty in public. As an experiment he was clumsy and slow. He had few, if any outward marks of genius.

In 1808, Dalton published his celebrated *New Systems of Chemical Philosophy* in a series of publications, in which he developed his conception of atoms as the fundamental building blocks of all matter. It ranks among two greatest of all monuments to human intelligence. No scientific discovery in history has had a more profund affect on the development of knowledge.

Dalton died in 1844. His stature as one of the greatest scientists of all time continues to grow.

1.2 ATOMIC THEORY

The essentials of **Dalton's atomic theory** may be summarized in the following postulates:

1. All matter is composed of very small particles called atoms.
2. Atoms are indestructible. They cannot be subdivided, created or destroyed.
3. Atoms of the same element are similar to one another and equal in weight.
4. Atoms of different elements have different properties and different weights.
5. Chemical combination results from the union of atoms in simple numerical proportions.

Thus, to Dalton, the atoms were solid, hard, impenetrable particles as well as separate, unalterable individuals. Dalton's ideas of the structure of matter were born out of considerable amount of subsequent experimental evidence as to the relative masses of substances entering into chemical combination. Among the experimental results and relationship supporting this atomic theory were Gay-Lussac's law of combination of gases by volume, Dalton's law of multiple proportions, Avagadro's hypothesis that equal volumes of gases under the same conditions contain the same number of molecules, Faraday's laws relating to electrolysis and Berzelius painstaking determination of atomic weights.

Modern Atomic Theory

Dalton's atomic theory assumed that the atoms of elements were indivisible and that no particles smaller than atoms exist. As a result of brilliant era in *experimental physics which began towards the end of the nineteenth century extended into the 1930s paved the way for the present modern atomic theory*. These refinements established that atoms can be divisible into sub-atomic particles, i.e., electrons, protons and neutrons—a concept very different from that of Dalton. The major problems before the scientists at that time were

(i) How the sub-atomic particles are arranged within the atom and why the atoms are stable?
(ii) Why the atoms of one element differ from the atoms of another elements in their physical and chemical properties?
(iii) How and why the different atoms combine to form molecules?
(iv) What is the origin and nature of the characteristics of electromagnetic radiation absorbed or emitted by atoms?

1.3 SUB-ATOMIC PARTICLES

We know that the atom is composed of three basic sub-atomic particles namely the electron, the proton and the neutron. The characteristics of these particles are given in Table 1.1.

It is now known that many more sub-atomic particles exist, e.g., the positron, the neutrino, the meson, the hyperon etc, but in chemistry only those listed in Table 1.1 generally need to be considered. The discovery of these particles and the way in which the structure of atom was worked out are discussed in this chapter.

1.3.1 Discovery of Electron

The term 'electron' was given to the smallest particle that could carry a negative charge equal in magnitude to

Table 1.1 The three main sub-atomic particles

Particle	Symbol	Mass	Charge
Electron	*e*	1/1837 of H-atom or 9.109×10^{-28} g or 9.1×10^{-31} kg	-4.8×10^{-10} esu or -1.602×10^{-19} coulombs (–1 unit)
Proton	*p*	1.008 amu or 1.672×10^{-24} g or 1.672×10^{-27} kg (1 unit)	$+4.8 \times 10^{-10}$ esu or $+1.602 \times 10^{-19}$ coulombs (+ 1 unit)
Neutron	*n*	1.0086 amu or 1.675×10^{-24} g or 1.675×10^{-27} kg (1 unit)	No charge

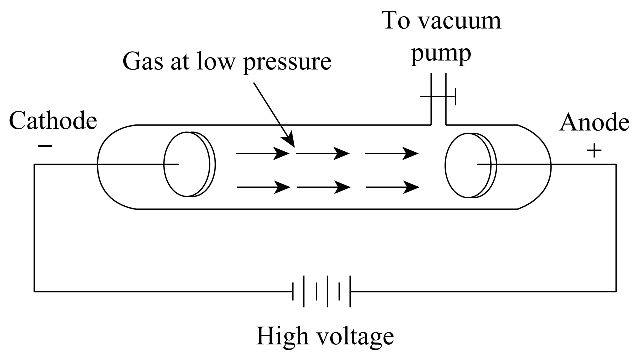


Fig 1.1 Cathode ray discharge tube

the charge necessary to deposit one atom of a 1-valent element by **Stoney** in 1891. In 1879, *Crookes discovered that when a high voltage is applied to a gas at low pressure streams of particles, which could communicate momentum, moved from the cathode to the anode*. It did not seem to matter what gas was used and there was strong evidence to suppose that the particles were common to all elements in a very high vacuum they could not be detected. The **cathode ray discharge tube** is shown in Fig 1.1.

The properties of the cathode rays are given below:

(i) When a solid metal object is placed in a discharge tube in their path, a sharp shadow is cast on the end of the discharge tube, showing that they travel in straight lines.
(ii) They can be deflected by magnetic and electric fields, the direction of deflection showing that they are negatively charged.
(iii) A freely moving paddle wheel, placed in their path, is set in motion showing that they possess momentum, i.e., particle nature.
(iv) They cause many substances to fluoresce, e.g., the familiar zinc sulphide coated television tube.
(v) They can penetrate thin sheets of metal.

J.J. Thomson (1897) extended these experiments and determined the velocity of these particles and their charge/mass ratio as follows.

The particles from the cathode were made to pass through a slit in the anode and then through a second slit. They then passed between two aluminium plates spaced about 5 cm apart and eventually fell onto the end of the tube, producing a well-defined spot. The position of the spot was noted and the magnetic field was then switched on, causing the electron beam to move in a circular arc while under the influence of this field (Fig 1.2).

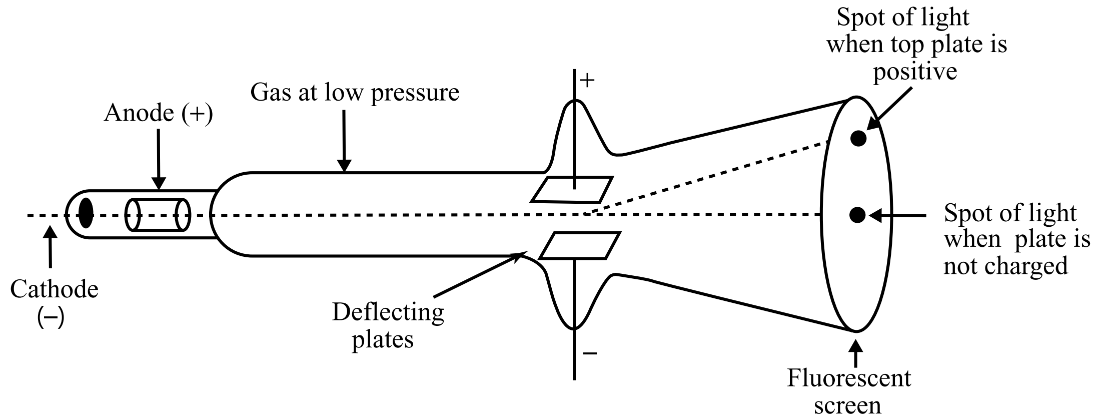


Fig 1.2 Thomson's apparatus for determining e/m for the electron

Thomson proposed that the amount of deviation of the particles from their path in the presence of electrical and magnetic field depends on

(i) Greater the magnitude of the charge on the particle, greater is the interaction with the electric and magnetic fields, and thus greater is the deflection.
(ii) Lighter the particle, greater the deflection
(iii) The deflection of electrons from its original path increase **in the voltage across the electrodes,** or the strength of the magnetic field.

By careful and quantitative determination of the magnetic and electric fields on the motion of the cathode rays, Thomson was able to determine the value of charge to mass ratio as

$$\frac{e}{m_e} = 1.758820 \times 10^{11}\ C\ kg^{-1} \tag{1.1}$$

m_e is the mass of the electron in kg and e is the magnitude of the charge on the electron in Coulomb.

1.3.2 Charge on the Electron

Thomson's experiments show electrons to be negatively charged particles. Evidence that electrons were discrete particles was obtained by Millikan by his well known oil drop experiment during the years 1910–14. By a series of very careful experiments Millikan was able to determine the value – electronic charge, and the mass. Millikan found the charge on the electron to be -1.6×10^{-19} C. The present-day accepted value for the charge on the electron is 1.602×10^{-19} C. When this value for 'e' is compared with the most modern value of e/m, the mass of the electron can be calculated.

Millikan Oil Drop Method

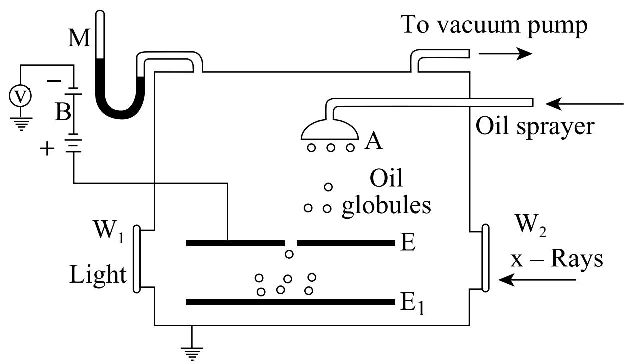


Fig 1.3 Millikan's apparatus for determining the value of the electronic charge

$$m_e = \frac{e}{e/m_e} = \frac{1.6022 \times 10^{-19}\ C}{1.758820 \times 10^{11}\ C\ kg^{-1}} \tag{1.2}$$

$$= 9.1094 \times 10^{-31}\ \text{kg} \tag{1.3}$$

Small droplets of oil from an atomiser are blown into a still thermostated airspace between parallel plates, and the rate of fall of one of these droplets under gravity is observed, from which its weight can be calculated. The airspace is now ionized with an X-ray beam, enabling the droplets to pick up charge by collision with the ionized air molecules. By applying a potential of several thousand volts across the parallel metal plates, the oil droplet can either be speeded up or made to rise, depending upon the direction of the electric field. Since, the speed of the droplet can be related to its weight, the magnitude of the electric field, and the charge it picks up, the value of the charge can be determined.

1.3.3 Discovery of Proton

If the conduction of electricity, through gases is due to particles, which are similar to those involved during electrolysis, it was to be expected that positive as well as negative ones should be involved, and that they would be drawn to the cathode. By using a discharge tube containing a perforated cathode, Goldstein (1886) had observed the formation of rays (shown to the right of the cathode in Fig 1.4).

J.J. Thomson (1910) measured their charge/mass ratio from which he was able to deduce that the particles were positive ions, formed by the loss of electrons from the residual gas in the discharge tube. The proton is the smallest positively charged particle equal in magnitude to that on the electron and is formed from the hydrogen atom by the loss of an electron.

$$H \rightarrow H^+ + e^-$$

Unlike cathode rays, the characteristics of positively charged particles depend upon the nature of the gas present in the cathode ray tube. These are positively charged ions. The charge to mass ratio of these particles is found to be dependent upon the gas from which these originate. Some of the positively charged particles carry a multiple of the fundamental unit of electrical charge. The behaviour of these particles in the magnetic or electric field is opposite to that observed for electron or cathode rays.

1.3.4 Discovery of Neutron

The neutron proved to be a very elusive particles to track down and its existence, predicted by Rutherford in 1920, was first noticed by **Chadwick** in 1932. Chadwick was bombarding the element beryllium with α-particles and noticed a particle of great penetrating power which was unaffected by magnetic and electric fields. It was found to have approximately the same mass as the proton (hydrogen ion). The reaction is represented as

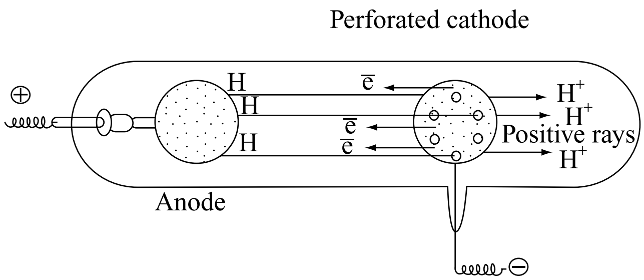


Fig 1.4

$$^{9}_{4}Be + ^{2}_{4}He \rightarrow ^{12}_{6}C + ^{1}_{0}n$$

Where the superscript refers to the atomic mass and the subscript refers to the atomic number (the number of protons in the nucleus). Notice that a new element, carbon, emerges from this reaction.

1.4 ATOMIC MODELS

The discovery that atoms contained electrons caused some consternation. Left to themselves, atoms were known to be electrically neutral. So, the negative charge of the electrons had to be balanced by an equal amount of positive charge. The puzzle was to work out how the two types of charges were arranged. To explain this, different atomic models were proposed. Two models proposed by J.J. Thomson and Earnest Rutherford are discussed here though they cannot explain about the stability of atoms.

1.4.1 Thomson Model of Atom

"A theory is a tool not a creed." J.J. Thomson

Sir Joseph John Thomson 1856–1940

Thomson's researches on discharge of electricity through gases led to the discovery of the electron and isotopes.

In 1898, Sir J.J. Thomson proposed that the electrons are embedded in a ball of positive charge (Fig 1.5). This model of the atom was given the name **plum pudding** or **raisin pudding** or **watermelon.** According to this model we can assume that just like the seeds of a watermelon are embedded within the reddish juicy material, the electrons are embedded in a ball of positive charge. It is important to note that in Thomson's model, the mass of the atom is

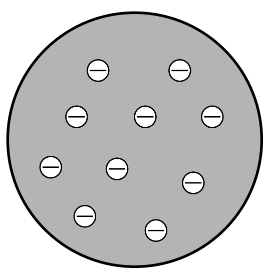


Fig 1.5 The Thomson model of atom. The positive charge was imagined as being spread over the entire atom and the electrons were put in this background

uniformly distributed over the atom. Though this model could explain the overall neutrality of the atom, it could not explain the results of later experiments.

The Discovery of X-rays and Radioactivity

In 1895, **Rontgen** noticed *when electrons strike a material in the cathode ray tubes, produces a penetrating radiation emitted from the discharge tubes, and it appeared to originate from the anode*. The radiation had the following properties:

(i) It blackened the wrapped photographic film.
(ii) It ionized gases, so allowing them to conduct electricity.
(iii) It made certain substances fluoresce, e.g., zinc sulphide.

Furthermore, the radiation was shown to carry no charge since it could not be deflected by magnetic or electric fields. Since Rontgen did not know the nature of the radiation, he named them as X-rays and the name is still carried on. The true nature of X-rays was not discovered until 1912, when it became apparent that its properties could be explained by assuming to be wave like in character, i.e., similar to light but is much smaller wavelength. It is now known that X-rays are produced whenever fast-moving electrons are stopped in their tracks by impinging on a target, the excess energy appearing mainly in the form of X-radiation.

The year after Rontgen discovered X-rays, **Henry Becquerel** observed that uranium salts emitted a radiation with properties similar to those possessed by X-rays. The **Curies** followed up this work and discovered that the ore pitchblend was more radioactive than purified uranium oxide; this suggested that something more intensely radioactive than uranium was responsible for this increased activity and eventually the Curies succeeded in isolating two new elements called polonium and radium, which were responsible for this increased activity.

In 1889, Becquerel reported that the radiation from the element radium could be deflected by a magnetic field and in the same year, Rutherford noticed that the radiation from uranium was composed of at least two distinct types. Subsequently, it was shown that the radiation from both sources contained three distinct components and are named as $\alpha - \beta$ and γ-rays. Rutherford found that α-rays consist of high energy particles carrying two units of positive charge and four units of atomic mass. He concluded that α-particles are helium nuclei **as when α-particles combined with two electrons yielded helium gas**. β-rays are negatively charged particles similar to electrons. The γ-rays are high energy radiations like X-rays, are neutral in nature and do not consist of particles. As regards penetrating power, α-particles have the least followed by β-rays (100 times that of α-particles) and γ-rays (1000 times that of α-particles).

1.4.2 Rutherford's Nuclear Model of Atom

Ernest Rutherford 1871–1937

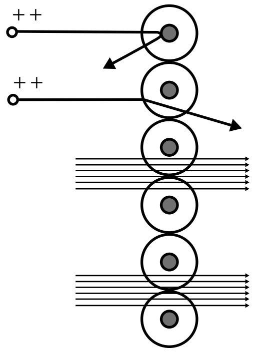
Rutherford was born in New Zealand in 1871. He was educated at the University of New Zealand and at Cambridge University. He taught at McGill University, and at the University of Manchester. In 1919, he became director of the celebrated Cavendish laboratory at Cambridge. In 1908, he received Nobel prize for chemistry. Rutherford made many of the basic discoveries in the field of radioactivity. With Bohr and others, he elaborated a theory of atomic structure. In 1919, he produced the first artificial transmutation of an element (that of nitrogen into oxygen). For many years. he was a vigorous leader in laying the foundation of the greatest developments in atomic science, which he did not live to see. He died in 1937.

In 1911, Ernest (later Lord) Rutherford demonstrated a classic experiment for testing the Thomson's model. Rutherford, Geiger and Harsden studied in detail the effect of bombarding a thin gold foil by high speed α-particle (positively charged helium particles). A thin parallel beam of α-particles was directed onto a thin strip of gold and the subsequent path of the particles was determined. Since the α-particles were very energetic, it was thought they would go right through the metal foils. To their surprise they observed some unexpected results which are summarized as follows:

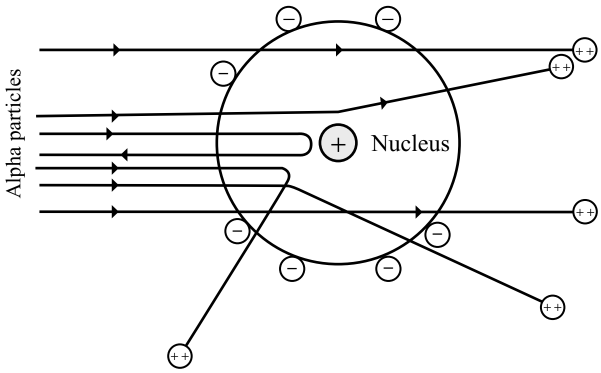
(i) It was observed that 99 per cent of the alpha particles passed straight through the foil and struck the screen at the centre.
(ii) A few of the alpha particles deflected from their original path through varying angles.
(iii) Hardly one out of the 20, 000 α-particles was bounced back on its path.

The results of Rutherford's experiment is represented more explicitly in Fig 1.6.

Rutherford's Explanation Rutherford pointed out that his results are not in agreement with Thomson's model.



(a)



(b)

Fig 1.6 Results of Rutherford's experiment (a) One layer of metal atoms each with nucleus (b) One atom of the metal with nucleus

He explained the above results by drawing the following conclusions:

(i) As most of the alpha particles passed very nearly straight through the foil, it means that the atom is extra ordinarily hollow with a lot of empty space inside.

(ii) The only way to account for the large deflection is to say that the positive charge and mass in the gold foil are concentrated in very small regions. Although most of the alpha particles can go through without any deflection, occasionally some of them which come closer to the region of positive charge, they repel each other, and the repulsion may be big enough to cause the α-particles to undergo large deviations from its original path.

(iii) Due to the rigid nature of the nucleus, some α-particles on colliding with the positive charge, turn back on their original paths.

On the basis of above observations and conclusions, Rutherford proposed the nuclear model of atom as follows:

(i) An atom has a centre of the nucleus, in which positive charge and mass are concentrated and he called this centre as the **nucleus**.

The quantitative results of scattering experiments such as Rutherford's indicate that the nucleus of an atom has a diameter of approximately one to six Fermis (1 Fermi = 10^{-10} m) and atoms have diameters about 100000 times as great as the size of the nucleus, i.e., of the order 10^{-5} m.

(ii) The atom as a whole is largely an empty space and the nucleus is located at the centre of the atom.

(iii) The nucleus is surrounded by the electrons which *are revolving round the nucleus in closed paths like the planets*.

(iv) Electrons and the nucleus are held together by electrostatic forces of attraction.

Rutherford, therefore, contemplated the dynamical stability and imagined the electrons to be whirling about the nucleus, **similar to the planetary motion.**

Drawbacks of the Rutherford Model

Following objections were raised against the Rutherford's model:

(i) It did not explain how the protons could be close-packed to give a stable nucleus.

(ii) When an electron revolves round the nucleus, it will radiate out energy, resulting in the loss of energy. This loss of energy will make the electron to move slowly and consequently, it will be moving in a spiral path and ultimately falling onto the nucleus. So, the atom should be unstable, but the atom is stable.

(iii) If an electron starts losing energy continuously, the observed spectrum would be continuous and have broad bands merging into one another. But most of the atoms give line spectra. Thus, Rutherford's model failed to explain the origin of line spectra.

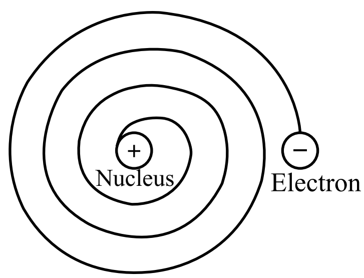


Fig 1.7 An electron that is accelerating, radiates energy. As it loses energy, it spirals onto the nucleus

1.5 ATOMIC NUMBER

In 1913, **H.G.J. Moseley**, a young English Physicist and one of Rutherford's brilliant students, examined the X-ray spectra of 38 elements. When Moseley bombarded different elements with cathode rays, the X-rays were generated which had different frequencies. He suggested that *the frequency* of X-*rays produced in this manner was related to the charge presented on the nucleus of an atom of the element used as anode*. When he took the frequencies of a particular line in all elements such as K_α or K_β lines, then the frequencies were related to each other by the equation.

$$\sqrt{v} = a\,(Z - b) \tag{1.3}$$

Where, v is the frequency of any particular K_α or K_β etc, line. Z is the atomic number of the element and 'a' and 'b' are constants for any particular type of line. A plot of $\sqrt{v}$ vs Z gives a straight line showing the validity of the above equation. This is shown in Fig 1.8.

However, no such relationship was obtained when the frequency was plotted against the atomic mass. Moseley further found that the nuclear charge increases by one unit in passing from one element to the next element arranged by Mendeleef in the order of increasing atomic weight. *The number of unit positive charges carried by the nucleus of an atom is termed as the atomic number of the element. Since, the atom as a whole is neutral, the atomic number is equal to the number of positive charges present in the nucleus.*

Atomic number = Number of protons present in the nucleus

= Number of electrons present outside the nucleus of the same atom.

While the positive charge of the nucleus is due to protons, the mass of the nucleus is due to protons and neutrons. The total number of protons and neutrons is called the **mass number**. The particles present in the nucleus are called **nucleons**.

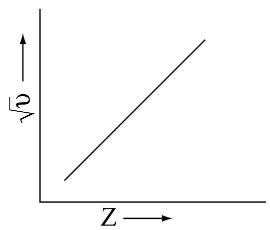


Fig 1.8 Variation of x–rays frequency with atomic number

Mass number (A) = Number of protons (Z) + Number of neutrons (n)

1.5.1 Isobars, Isotopes and Isotones

In modern methods, the symbols of elements are written as ${}^{A}_{Z}[X]^{n}_{x}$, where the left hand superscript A is the mass number, left hand subscript Z is the atomic number, right hand superscript n is the number of charges and right hand subscript is the number of atoms.

Different atoms having same mass number but with different atomic numbers are called **isobars,** e.g., ${}^{14}_{6}C$ and ${}^{14}_{7}N$. The atoms having same atomic number but with different mass numbers are called **isotopes.** The difference between isotopes is due to the difference in the number of neutrons present in the nucleus, e.g., ${}^{1}_{1}H$, ${}^{2}_{1}D$ and ${}^{3}_{1}T$ are isotopes of hydrogen namely protium, deuterium and tritium respectively.

Isotopes exhibit similar chemical properties because they depend on the number of protons in the nucleus. Neutrons present in the nucleus show very little effect on the chemical properties of an element. So, all the isotopes of a given element show same chemical behavior.

Sometimes atoms of different elements contain same number of neutrons. Such atoms are called **isotones.** For example, ${}^{13}_{6}C$ and ${}^{14}_{7}N$. Isotones differ in both atomic number and mass numbers but the difference in atomic number and mass number is the same.

1.6 DEVELOPMENTS LEADING TO THE BOHR MODEL OF ATOM

During the period of development of new models to improve Rutherford's model of atom, two new concepts played a major role. They are

(i) **Dual behaviour of electromagnetic radiation:** This means that light has both *particle* like and *wave* like properties.

(ii) **Atomic spectra:** The experimental results regarding atomic spectra of atoms can only be explained by assuming quantized (fixed) electronic energy levels in atoms.

Let us briefly discuss about these concepts before studying a new model proposed by **Niels Bohr** known as **Bohr Model of Atom.**

1.6.1 Nature of Light and Electromagnetic Radiation

A radiation is a mode of transference of energy of different forms. Light, X-rays and γ-radiations are examples of

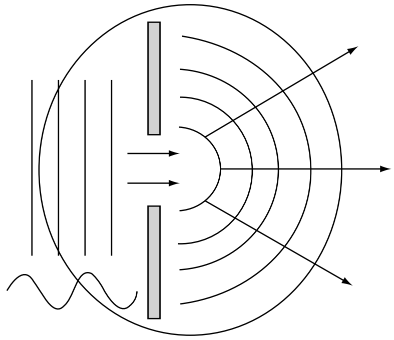


Fig 1.9 Diffraction of the waves while passing through a slit

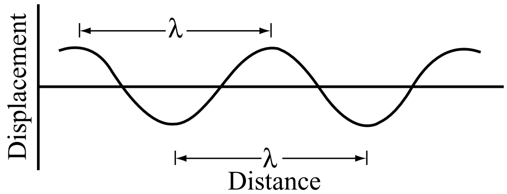


Fig 1.10 Propagation of wave

the radiant energy. The earliest view of light, due to Newton, **regarded light** as **made up of particles** (commonly termed as corpuscles of light). The particle nature of light explained some of the experimental facts such as reflection and refraction of light. However, it failed to explain the phenomenon of interference and diffraction. The corpuscular theory was therefore, discarded and Huygens proposed a wave-like character of light. With the help of wave theory of light, Huygens explained the phenomena of interference and diffraction.

We know that when a stone is thrown **into water of a quiet pond,** on the surface of water waves are produced. The waves originate from the centre of the disturbance and propagate in the form of up and down movements. *The point of maximum upward displacement is called* the **crest** and *the point of maximum downward displacement is called* **trough.** Thus, waves may be considered as disturbance which originate from some vibrating source and travel outwards as a continuous sequence of alternating crests and troughs as shown in Fig 1.10.

Characteristics of the Wave Motion

1. **Wavelength (λ):** It is the distance between two nearest crests or troughs. It is denoted by the Greek letter Lambda λ and is expressed in Angstrom units (Å). It is also expressed as micron meter (μ), milli micron meter (mμ), nanometer (nm), picometer (pm) etc. These units are related to SI unit (m) as

$$1\ \text{Å} = 10^{-10}\ \text{m};\ 1\ \mu = 10^{-6}\ \text{m},\ 1\ \text{m}\mu = 10^{-9}\ \text{m},$$
$$1\ \text{nm} = 10^{-9}\ \text{m};\ 1\ \text{pm} = 10^{-12}\ \text{m}$$

2. **Frequency (ν):** The number of waves passing through a given point in a unit time is known as its frequency. It is denoted by the Greek letter v (nu). The frequency is inversely proportional to the wavelength. Its unit in cycles per second (cps) of Hertz (Hz); 1 cps = 1 Hz. A cycle is said to be completed when a wave consisting crest and trough passes through a point.
3. **Velocity (c):** The distance travelled by the wave in one second is called the velocity of wave. It is equal to product of wavelength and frequency of the wave. Thus,

$$c = v\lambda$$
$$\text{or} \qquad v = \frac{c}{\lambda} \tag{1.4}$$

4. **Amplitude:** It is the height of crest or depth of wave's trough and is generally expressed by the letter '*a*'. The amplitude of the wave determines the intensity or brightness of radiation.
5. **Wave number ($\bar{v}$):** It is equal to the reciprocal of wavelength. In other words, it is defined as the number of wavelengths per centimeter. It is denoted by $\bar{v}$ and is expressed in cm^{-1}. Thus,

$$\bar{v} = \frac{1}{\lambda} \tag{1.5}$$
$$\text{But} \qquad v = \frac{c}{\lambda} \text{ or } \lambda = \frac{c}{v}$$
$$\bar{v} = \frac{\lambda}{c} \tag{1.6}$$

Electromagnetic Radiation

Maxwell in 1873 proposed that light and other forms of radiant energy propagate through space in the form of waves. These waves have electric and magnetic fields associated with them and are, therefore, called **electromagnetic radiations** and **electromagnetic waves.**

The important characteristics of electromagnetic radiations are

(i) These consist of electric and magnetic fields that oscillate in the directions perpendicular to the direction in which the wave is travelling as shown in Fig 1.11. The two field components have the same wavelength and frequency.

(ii) All electromagnetic waves travel with the same speed. In vacuum, the speed of all types of electromagnetic radiation is 3. 00 × 10^{8} ms^{-1}. This speed is called the velocity of light.

(iii) These electromagnetic radiations do not require any medium for propagation. For example, light reaches us from the sun through empty space.

Electromagnetic Spectrum

The different electromagnetic radiations have different wavelengths. The visible light in the presence of which our eyes can see, contains radiations having wavelength between 3800 Å to 7600 Å. The different colours in the visible light corresponds to radiations of different wavelengths. In addition to visible light there are many other electromagnetic radiations, such as X-rays, ultraviolet rays, infra-red rays, microwaves and radiowaves.

The arrangement of different types of electromagnetic radiations in the order of increasing wavelengths (or decreasing frequencies) is known as **electromagnetic spectrum.**

Different regions of electromagnetic spectrum are identified by different names. The complete electromagnetic spectrum is shown in Fig 1.12.

The various types of electromagnetic radiations have different energies and are being used for different purposes as listed in Table 1.2.

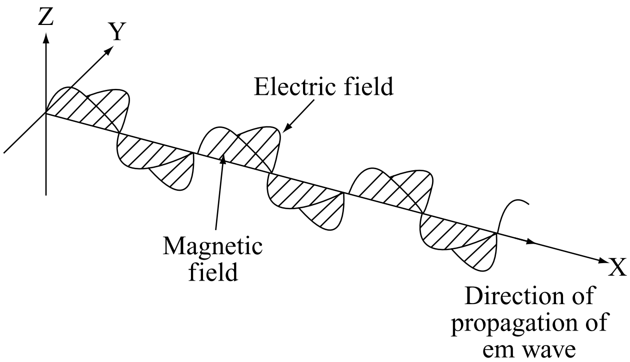


Fig 1.11 Electric and magnetic fields associated with an electromagnetic wave

Table 1.2 Some applications of electromagnetic waves

Name	Frequency	Wavelength	Uses
γ-rays	$10^{20} - 10^{21}$	10^{-12}	Cancer treatment
X-rays	$10^{17} - 10^{19}$	10^{-10}	Medical "pictures", material testing
Ultraviolet	$10^{15} - 10^{16}$	10^{-7}	Germicidal lamps
Visible	$10^{13} - 10^{14}$	10^{-6}	Illumination
Infrared	$10^{12} - 10^{13}$	10^{-4}	Heating
Microwave	$10^{9} - 10^{11}$	10^{-2}	Cooking, radar
Radio frequency	$10^{5} - 10^{8}$	10^{2}	Signal transmission

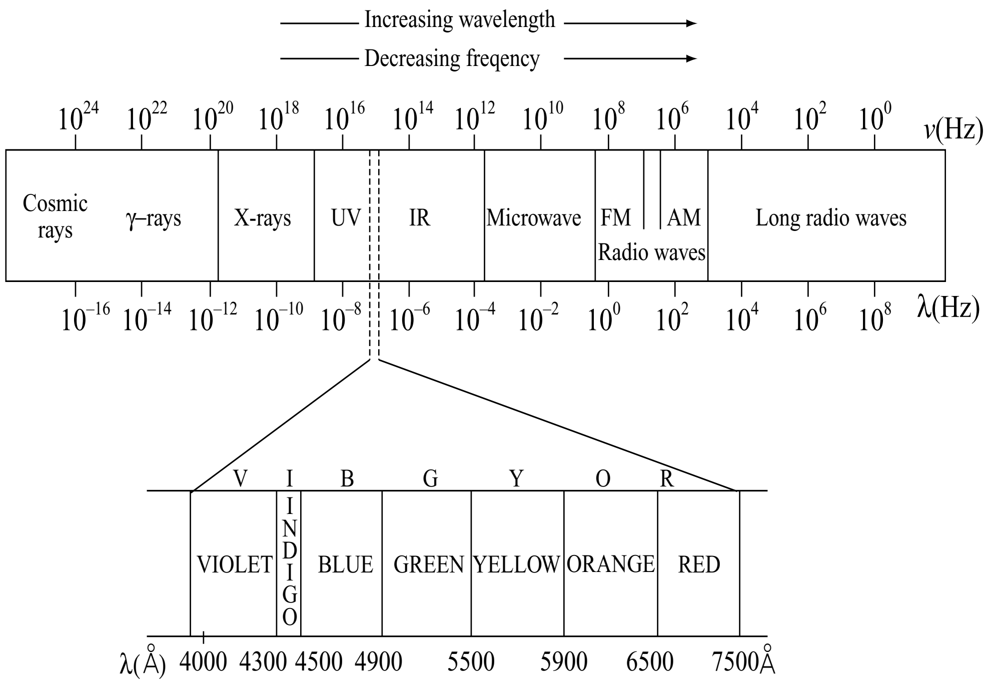


Fig 1.12 Complete electromagnetic spectrum

Solved Problem 1

Calculate and compare the energies of two radiations, one with a wavelength of 8000 Å and the other with 4000 Å.

Solution:

According to Planck's equation,

$$E = hv = h\frac{c}{\lambda}$$

$h = 6.60 \times 10^{-34}$ J s; $c = 3 \times 10^{10}$ ms^{-1}

Also $\lambda_1 = 8000$ Å $= 8000 \times 10^{-10}$ m

and $\lambda_2 = 4000$ Å $= 4000 \times 10^{-10}$ m

$$E_1 = \frac{6.62 \times 10^{-34}\ \text{J s} \times 3 \times 10^{10}\ \text{m s}^{-1}}{8000 \times 10^{-10}\,\text{m}}$$

$$= 2.475 \times 10^{-19}\ \text{J}$$

$$\text{and } E_2 = \frac{6.62 \times 10^{-34}\ \text{J s} \times 3 \times 10^{10}\ \text{m s}^{-1}}{4000 \times 10^{-10}\,\text{m}}$$

$$= 4.95 \times 10^{-19}\ \text{J}$$

$$\frac{E_1}{E_2} = \frac{2.475 \times 10^{-19}\ J}{4.95 \times 10^{-19}\ J}$$

$$\text{or } \frac{E_1}{E_2} = \frac{1}{2}$$

$$\text{or } 2E_1 = E_2$$

Solved Problem 2

A radio station is broadcasting a programme at 100 MHz frequency. If the distance between the radio station and the receiver set is 300 Km, how long would it take the signal to reach the receiver set from the radio station? Also calculate wavelength and wave number of these radio waves.

Solution:

All electromagnetic waves travel in vacuum or in air with the same speed of 3×10^{-8} m s^{-1}

$$\text{Time taken} = \frac{\text{Distance}}{\text{Velocity}} = \frac{300 \times 1000\,\text{m}}{3 \times 10^{8}\ \text{m s}^{-1}} = 1 \times 10^{-3}\,\text{s}$$

Calculate of wavelength (λ),

$c = v\lambda$

$$\lambda = \frac{c}{v} = \frac{3 \times 10^{8}\ \text{m s}^{-1}}{100 \times 10^{6}\ \text{s}^{-1}} = 3\,\text{m}$$

Calculation of wave number ($\overline{v}$)

$$\overline{v} = \frac{1}{\lambda} = \frac{1}{3\,\text{m}} = 0.33\ \text{m}^{-1}$$

1.6.2 Quantum Theory of Radiation

When an object is heated, its colour gradually changes. For example, a black coal becomes red, orange, blue and finally 'white hot' with increasing temperature. This shows that red radiation is most intense at a particular temperature, blue radiation is most intense at another temperature and so on. It indicates that the intensities of radiations of different wavelengths emitted by a hot body depend upon the temperature. The curves representing the distribution of radiation from a black body at different temperatures are shown in Fig 1.13.

A black body is one which will absorb completely the incident radiation of wavelength. The ideal black body does not reflect any energy, but it does radiate energy. Although no actual body is perfectly black, it is possible to postulate a condition where radiation is completely enclosed in a space surrounded by thick walls at a constant temperature. Thus, we may imagine a cavity inside a solid body containing a small hole in one wall whereby the investigator may study the enclosed radiation without altering its character. When a beam of light is passed through the hole, it would be absorbed by the inner walls of the enclosure and their temperature increases. As soon as the beam was removed, since the enclosed kept at a constant temperature the excess energy emerges through the hole in the form light. This gives a distinctive spectrum, usually referred to as the **black body spectrum.** If the energy emitted is plotted against its frequency or wavelength where ν = c / l, the curve shows a maximum, as indicated in Fig 1.13. The graph shows that the energy emitted is

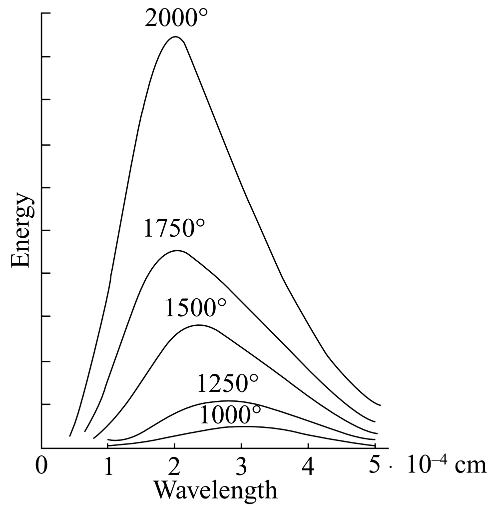


Fig 1.13 Energy distribution for black body radiation at different absolute temperatures

greatest at the middle wavelengths in the spectrum and least at the highest and lowest frequencies. If curves are plotted for a series of temperatures, as given in Fig 1.13. It is found that the maximum moves towards shorter wavelengths as the temperature rises.

Max Planck 1858–1947

A German physicist. He received his Ph.D. in theoretical physics from the University of Munich in 1879. Max Planck was well known for the Quantum Theory which won the Nobel Prize for physics in 1918. He also made significant contributions to thermodynamics and other areas of physics.

The shapes of the curves could not be explained on the basis of wave theory of radiation. Max Planck in 1900 resolved this discrepancy by postulating the assumption that the *black body radiates energy — not continuously but discontinuously in the form of energy packets called* **quanta**. The general quantum theory of electromagnetic radiations can be stated as

(i) When atoms or molecules absorb or emit the radiant energy, they do so in separate units of waves. These waves are called quanta or photons.

(ii) The energy of a quantum or photon is given by

$$E = hv \tag{1.7}$$

where, v is the frequency of the emitted radiation and h is the Planck's constant.

(iii) An atom or molecule emits or absorbs either one quantum of energy (hv) or any whole number multiples of this unit.

This theory provided the basis for explaining the photoelectric effect, atomic spectra and also helped in understanding the modern concepts of atomic and molecular structure.

Quantization of Energy

The restriction of any property to discrete values is called quantization. A quantity cannot vary continuously to have any arbitrary values but can change only discontinuously to have some specific values. For example, a ball moves down a staircase (Fig 1.14a) then the energy of the ball changes discontinuously and it can have only certain definite values of energy corresponding to the energies of various steps. Energy of the ball in this case is **quantized.** On the other hand, if the ball moves down a ramp (Fig 1.14b), then the energy of the ball changes continuously

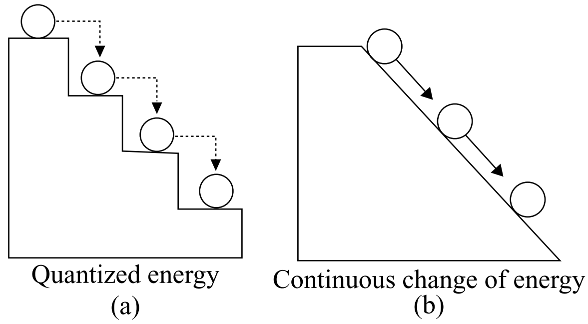


Fig 1.14

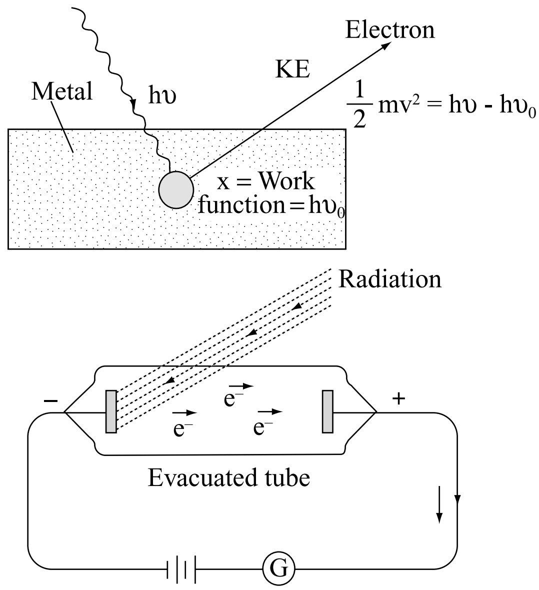


Fig 1.15 (a) Einstein's explanation of photoelectric effect, and (b) Experimental device for photoelectric effect

and the ball can have any value of energy corresponding to any point on the ramp. Energy in this case in not quantized.

1.6.3 Photoelectric Effect

When a beam of light falls on a clean metal plate in vacuum, the plate emits electrons. This effect was discovered by Hertz in 1887, and is known as the **photoelectric** effect. The metal surface emits electrons by *the action of light, that can be demonstrated using negatively charged gold leaf electroscope (Fig 1.15).* As the light from carbon arc falls on the metal plate, the diversion of leaves is reduced slowly. This shows that electrons come out of all metal plates.

The studies have revealed the following facts:

(i) Photoelectric effect is instantaneous, i.e., as soon as the light rays fall on the metal surface, electrons are ejected.

(ii) Only light of a certain characteristic frequency is required for expulsion of electrons from a particular metal surface.

(iii) The number of electrons emitted are directly proportional to the intensity of the light.

(iv) The gas surrounding the metal plate has no effect on photoelectric transmission.

(v) Kinetic energy of photoelectron depends upon the nature of metal.

(vi) For the same metal kinetic energy of photoelectrons varies directly with the frequency of light.

If the frequency is decreased below a certain value (Threshold frequency), no electrons are ejected at all.

The classical wave theory of light was completely inadequate to interpret all these facts. Einstein applied the quantum theory of radiation to explain this phenomenon.

An electron in a metal is found by a certain amount of energy. The light of any frequency is not able to cause the emission of electrons from the metal surface. There is certain minimum frequency called the **threshold frequency** which can just cause the ejection of electrons. Suppose the threshold frequency of the light required to eject electrons from the metal is v_0. When photon of light of this frequency strikes the metal surface, it imparts its entire energy (hv_0) to the electron. This enables the electrons to break away from the atom by overcoming the attractive influence of the nucleus. Thus, each photon can eject one electron. If the frequency of the light is less than ν_0, the electrons are not ejected. If the frequency of the light (say v) is more than v_0, more energy is supplied to the electron. The remaining energy, which will be the energy hv imparted by incident photon and the energy used up, i.e., hv_0, would be given as kinetic energy to the emitted electron. Hence,

$$hv = hv_0 + \frac{1}{2}mv^2 \tag{1.8}$$

where $\frac{1}{2}mv^2$ is the kinetic energy of the emitted electron and hv_0 is called the **work function**.

or
$$\frac{1}{2}mv^2 = h(v - v_0) \tag{1.9}$$

This equation is known as Einstein's photoelectric equation. If the frequency of light is less than the threshold frequency, there will be no ejection of electrons. Values of photoelectric work functions of some metals are given below.

Metal	**Li**	**Na**	**K**	**Mg**	**Cu**	**Ni**
W (eV)	2.42	2.3	2.25	3.7	4.8	4.3

Solved Problem 3

Calculate the energy in one photon of yellow light of wavelength 589 nm.

Solution:

The energy E, of one photon

$$E = h\nu = \frac{hc}{\lambda} w$$

Substituting the values

$$E = \frac{(6.626\times10^{-34}\ \text{Js})\ (3.00\times10^{8}\ \text{m s}^{-1})}{5890\times10^{-10}\,\text{m}} = 3.375\times10^{-19}\ \text{J}$$

Solved Problem 4

When a radiation of wavelength 200 nm falls on a metal surface, the work function of the metal being 5 eV, the electron is ejected. What is the velocity of the electron?

Solution:

Work function 5 $eV = 5 \times 1.602 \times 10^{-19}$ J $= 8.01 \times 10^{-19}$ J (\1eV = 1.602×10^{-19} J)

Work function = hv_0

$$v_0 = \frac{\text{Work function}}{h} = \frac{8.01\times10^{-19}\ \text{J}}{6.626\times10^{-34}\ \text{J}} = 1.21\times10^{-15}\ \text{S}^{-1}$$

$$v = \frac{c}{\lambda} = \frac{3.0\times10^{8}\,\text{m s}^{-1}}{200\times10^{-9}\,\text{m}}$$

$$= 1.5\times10^{15}\ s^{-1}\ \frac{1}{2}mv^2 = h(v - v_0)$$

$$\text{V}^2 = \frac{2h}{m}(v - v_0) = \frac{2\times6.626\times10^{-34}}{9.11\times10^{-31}\,\text{kg}}(1.5-1.21)\times10^{15}$$

$$\text{V} = 0.649\times10^{6}\ \text{m s}^{-1}$$

Solved Problem 5

A photon of wavelength 253.7 nm strikes a copper plate, the work function of the metal being 4.65 eV. Calculate the energy of photon and kinetic energy of the emitted photoelectron.

Solution:

(i) Energy of the photon

$$v = \frac{c}{\lambda} = \frac{3\times10^{8}\,\text{m s}^{-1}}{253.7\times10^{-9}\,\text{m}} = 11.824\times10^{14}\ \text{s}^{-1}$$

$$E = hv = (6.626\times10^{-34}\ \text{Js})(11.824\times10^{14}\ \text{s}^{-1})$$

$$= 7.83 \times 10^{-19}\ \text{J}$$

Since, 1 eV = 1.602×10^{-19} *J*

$$\therefore E = \frac{7.83\times10^{-19}\ \text{J}}{1.602\times10^{-19}\ \text{J}} = 4.887\ \text{eV}$$

(ii) Kinetic energy of the photon

Work function = 4.65 eV

$KE = h\nu$ – Work function

= 4.887 eV – 4.65 eV = 0.237 eV

1.6.4 Compton Effect

According to classical electromagnetic wave theory, monochromatic light falling upon matter should be scattered without change in frequency. But when X-rays are impinged on matter of low atomic weight, X-rays of slightly longer wavelength than those of impinging beam were produced. In other words when a photon collides with an electron, it may leave with a lower frequency, and the electron thereby acquires a greater velocity. In the collision the energy of the photon is reduced and the energy of the electron is increased. This is called **Compton effect** and can be explained as a result of the impact between two bodies, the photon and the electron and it is of course additional proof of the corpuscular nature of light. This impact is elastic, and both kinetic energy and momentum are conserved. After the encounter each body has a different energy and a different momentum from the values it had before the contact, but the sum of the energies and the sum of the momenta are unchanged.

1.6.5 Dual Nature of Electromagnetic Radiations

As described, photoelectric effect and Compton effect could be explained considering that electromagnetic radiations consisting of small packets of energy called quanta or quantum (or single photon). These packets of energy can be treated as particles, on the other hand, radiations exhibit phenomena of interference and diffraction which indicate that they possess wave nature. So, it may be concluded that electromagnetic radiation possesses the dual nature, i.e., particle nature as well as *wave* nature. Einstein (1905) even calculated the mass of the photon associated with a radiation of frequency ν as given below:

The energy E of the photon is given as $E = h\nu$

Albert Einstein 1879–1955

Born in Germany but later shifted to America. He was regarded as one of the two great physicists the world has known, the other being Isaac Newton. He is well known for his three research papers on special relativity, Brownian motion and photoelectric effect. These research papers were published in 1905 while he was working as a technical assistant in a Swiss patent office in Berne. He received the Nobel Prize for Physics in 1921 for his explanation of the photoelectric effect. His work has influenced the development of physics in significant manner.

"*To him who is discoverer in the field (or science), the products of his imagination appear so necessary and natural that he regards them, and would like to have them regarded by others, not as creations of thought but as given realities*"— Albert Einstein

Also according to Einstein's mass energy equation $E = mc^2$ where m is the mass of photon. From the above two equations, we get

$$h\nu = mc^2 \tag{1.10}$$

$$m = \frac{h\nu}{c^2} \tag{1.11}$$

$$m = \frac{h}{c^2}\cdot\frac{c}{\lambda} \tag{1.12}$$

$$m = \frac{h}{c\lambda} \tag{1.13}$$

Thus, the mass of the photon can be calculated.

1.6.6 Atomic Spectra

The velocity of light depends upon the nature of the medium through which it passes. Because of this reason while a beam of light passing from one medium to another, deviates or refracts from its original path, the radiations with shorter wavelength deviate more than the one with a longer wavelength. If a ray of sunlight or any white light is allowed to pass through a prism, it splits up into a continuous band of seven colours from red to violet as observed in a rainbow. This phenomenon is known as **dispersion** and the pattern of colours is called **spectrum.** The red colour light radiation having longest wavelength will be deviated least while the radiation of violet colour having shortest wavelength will be deviated more. The spectrum of white light that we can see ranges from violet at 7.50×10^{14} Hz to red at 4×10^{14} Hz.

In this spectrum one colour merges into another colour adjacent to it *viz* violet merges into blue, blue into green and so on. When electromagnetic radiation interacts with matter, atoms and molecules may absorb energy and reach to a higher energy state. While coming to the ground state (more stable lower energy state) the atoms and molecules emit radiations in various regions of the electromagnetic spectrum.

1.6.7 Types of Spectra

Spectra are of two types: Emission spectra and Absorption spectra

(i) Emission spectra: When energy is supplied to a sample by heating or irradiating it, the atoms or molecules or ions present in the sample absorb energy and the atoms, molecules or ions having higher energy are said to be excited. While coming to the ground (normal) state the excited species emit the absorbed energy. The spectrum of such emitted radiation is called the **emission spectrum**. Emission spectra are further classified according to its appearance as continuous, line and band spectra.

(a) Continuous spectrum: *When the source emitting light is an incandescent solid, liquid or gas at a high pressure, the spectrum so obtained is continuous. In other words, this type of spectrum is obtained whenever matter in the bulk is heated. For example, hot iron and hot charcoal gives the continuous spectra.*

(b) Line spectrum: *Line spectrum is obtained when the light emitting substance is in the atomic state. Hence, it is also called as the* **atomic spectrum**. *Line spectrum consists of discrete wavelengths extended throughout the spectrum and are generally obtained from the light sources like mercury, sodium, neon discharge tube, etc.*

(c) Band spectrum: *This type of spectrum arises when the emitter in the molecular state is excited. Each molecule emits bands which are characteristics of the molecule concerned and that is why we call this as* **molecular spectrum** *also. The sources of band spectrum are (i) carbon with a metallic salt in its core, (ii) vacuum tube, etc.*

In the emission spectra, bright lines on black background will appear.

It may be noted that the dark lines in absorption spectra appear exactly at the same place where the coloured lines appear in the emission spectra.

(ii) Absorption spectra: It is produced when the light from a source emitting a continuous spectrum is first passed through an absorbing substance and then recorded after passing through a prism in a spectroscope. In that spectrum, it will be found that certain colours are missing which leave dark lines or bands at their places. This type of spectrum is called the absorption spectrum. Similar to emission spectra, absorption spectra are also of three types:

(a) Continuous absorption spectrum: *This type of spectrum arises when the absorbing material absorbs a continuous range of wavelengths. An interesting example is the one in which red glass absorbs all colours except red and hence, a continuous absorption spectrum will be obtained.*

(b) Line absorption spectrum: *In this type, sharp dark lines will be observed when the absorbing substance is a vapour or a gas. The spectrum obtained from sun gives Fraunhofer absorption lines corresponding to vapours of different elements which are supposed to be present on the surface of the sun.*

(c) Band absorption spectrum: *When the absorption spectrum is in the form of dark bands, this is known as* **band absorption spectrum** *An interesting example is that of an aqueous solution of $KMnO_4$ giving five absorption bands in the green region.*

The pattern of lines in the spectrum of an element is characteristics of that element and is different from those of all other elements. In other words, *each element gives a unique spectrum irrespective of even the form in which it is present*. For example, we always get two important lines 589 nm and 589.6 nm in the spectrum of sodium whatever may be its source. It is for this reason that the line spectra are also regarded as the **fingerprints of atoms.**

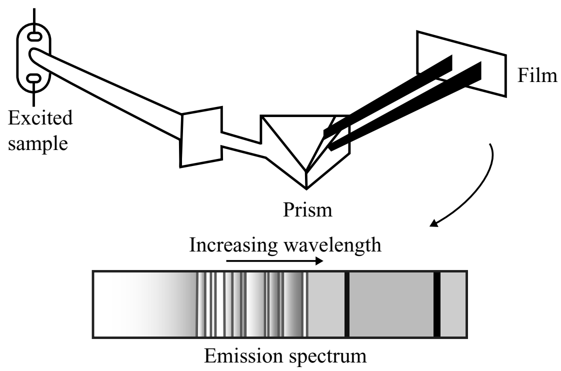


Fig 1.16

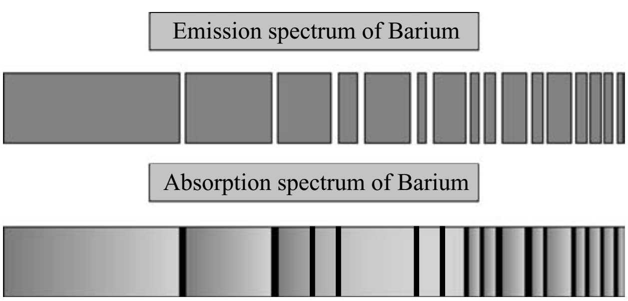


Fig 1.17 Illustrating emission and absorption spectra

Since atoms of different elements give characteristic sets of lines of definite frequencies, emission spectra can be used in chemical analysis to identify and estimate the elements present in a sample. The elements rubidium, caesium, thallium, gallium and scandium were discovered when their minerals were analyzed by spectroscopic methods. The element helium was discovered in the Sun by spectroscopic method.

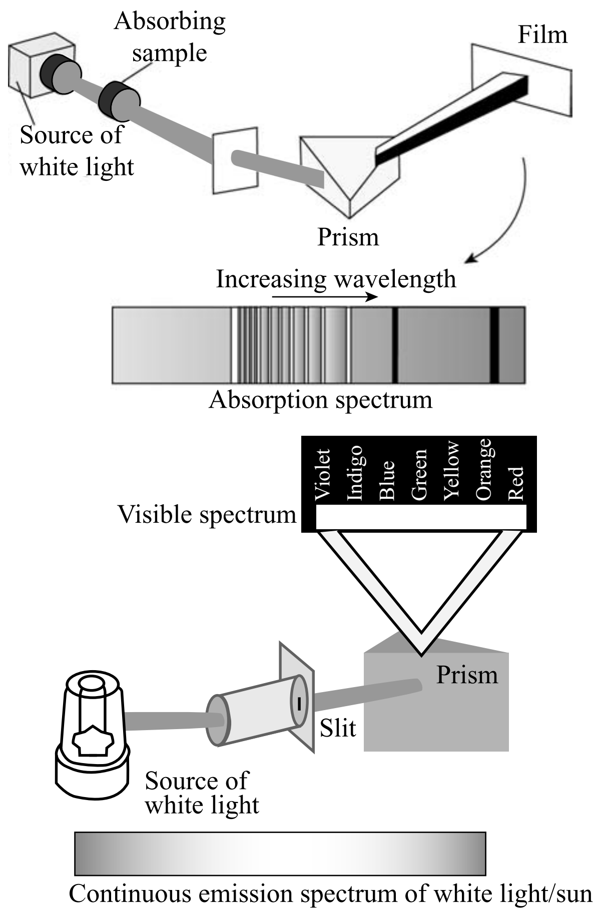


Fig 1.18 Emission and absorption spectra

Hydrogen Spectrum

As explained above, each element emits its own characteristic line spectrum which is different from that of any other element. Since, hydrogen contains only one electron, its spectrum is the simplest to analyze.

To get the spectrum of hydrogen, the gas is enclosed in a discharge tube under low pressure and electric discharge is passed through it. The hydrogen molecules dissociate into atoms and get excited by absorbing the energy: On the principle that what goes up must come down, sooner or later the atoms must loose their energy and fall back to the ground state, by loosing the energy as light. The light that is given out can be measured in a spectrometer and the pattern recorded on a photographic paper. The pattern is called hydrogen spectrum.

The hydrogen spectrum obtained consists of a series of lines in the visible, ultraviolet and infrared regions. These have been grouped into five series in Fig 1.19 which are named after their discoverers. These are given in Table 1.3 with the year when discovered. The Balmer series happened to be the first series of lines in the hydrogen spectrum. This was because the lines were in the visible part of the spectrum and therefore the easiest to observe.

Balmer (1885) discovered a relationship between the wave number and the position of the line in the series.

The relation is

$$\frac{1}{\lambda} = \bar{v} = R_H\left(\frac{1}{2^2} - \frac{1}{m^2}\right) \tag{1.14}$$

where, R_H is the Rydberg constant and is equal to 109677 cm^{-1}.

$\bar{v}$ is the wave number and m is an integer having values 3, 4, 5, 6, etc. **Ritz** (1908) gave a generalization known as **Ritz Combination principle** which was nicely applicable to hydrogen spectrum. According to this principle, the wave number in any line in a series can be represented as a difference of two square terms, one of which is consistent and the other varies throughout the series. Thus,

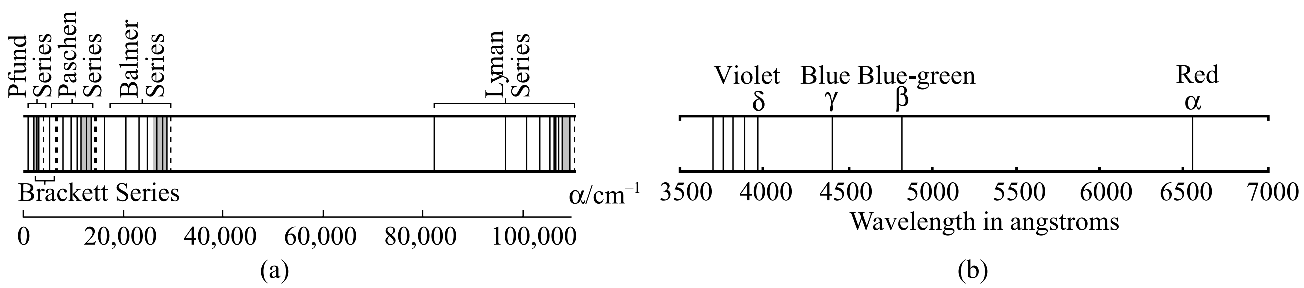


Fig 1.19 Hydrogen spectrum (a) Relative location of the Lyman, Balmer, Paschen, Brackett and Pfund series of hydrogen spectrum (b) The Balmer-series spectrum of hydrogen

Table 1.3

Series	Region of spectrum	Value of n	Values of m
Lyman (1914)	Ultraviolet	1	2, 3, 4, ...
Balmer (1885)	Visible	2	3, 4, 5, ...
Paschen (1908)	Infrared	3	4, 5, 6, ...
Brackett (1922)	Infrared	4	5, 6, 7, ...
Pfund (1924)	Infrared	5	6, 7, 8, ...

$$\bar{v} = R_H\left[\frac{1}{n^2} - \frac{1}{m^2}\right] \qquad (1.15)$$

where, R_H is Rydberg constant.

All the lines appearing in the hydrogen spectrum are governed by the above equation. For,

Lyman series $\bar{v} = R_H\left[\frac{1}{1^2} - \frac{1}{m^2}\right]; m = 2,3,4$ (1.16)

Paschen series $\bar{v} = R_H\left[\frac{1}{3^2} - \frac{1}{m^2}\right]; m = 4,5,6$ (1.17)

Brackett series $\bar{v} = R_H\left[\frac{1}{4^2} - \frac{1}{m^2}\right]; m = 5,6,7$ (1.18)

Pfund series $\bar{v} = R_H\left[\frac{1}{5^2} - \frac{1}{m^2}\right]; m = 6,7,8$ (1.19)

Solved Problem 6

An electronic transition from M shell (m = 3) to K shell (n = 1) takes place in a hydrogen atom. Find the wave number and the wavelength of radiation emitted (R = 1,09,677 cm^{-1}).

Solution:

$$\bar{v} = \frac{1}{\lambda} = R_H\left[\frac{1}{n^2} - \frac{1}{m^2}\right], n = 1 \text{ and } m = 3$$

$$= 1,09,677\left[\frac{1}{1^2} - \frac{1}{3^2}\right] = 109677 \times \frac{8}{9}$$

$$\bar{v} = 97,491 \text{ cm}^{-1}$$

Wavelength,

$$\lambda = \frac{1}{v} = \frac{1}{97491} = 1.026 \times 10^{-5} \text{ cm} = 1026 \text{ Å}$$

Solved Problem 7

In hydrogen atom, an electron jumps from fourth orbit to first orbit. Find the wave number, wavelength and the energy associated with the emitted radiation.

Solution:

$$\bar{v} = \frac{1}{\lambda} = R_H\left[\frac{1}{1^2} - \frac{1}{4^2}\right] \text{cm}^{-1} = 109677 \times \frac{15}{16} \text{ cm}^{-1}$$

$$= 102822 \text{ cm}^{-1}$$

1.7 BOHR'S MODEL OF THE ATOM

Neils Bohr 1885–1962

Born in Denmark in 1885. He received Ph.D. from the University of Copenhagen in 1911. He spent a year with J.J. Thomson and Ernest Rutherford in England. In 1913, he returned to Copenhagen and in 1920 he was named as Director of the Institute of Theoretical Physics.

Bohr's theory of atomic structure, which was presented in 1913, laid a broad foundation for the great atomic progress of recent years.

Second only in importance to his celebrated theory are the facts that it was in Bohr's laboratory that the implications of nuclear fission were first predicted and that he obtained an understanding of nuclear stability that contributed greatly to the spectacular development of atomic energy.

After the First World War, Bohr worked for peaceful uses of atomic energy. He received the first Atoms for Peace award in 1957. Bohr was awarded the Nobel Prize for Physics in 1922.

"...... The very word 'experiment' refers to a situation where we can tell others what we have done and what we have learned" **— Neils Bohr.**

"Our experiments are questions that we put to Nature." **— Neils Bohr.**

In order to explain the line spectra of hydrogen and the overcome to objections leveled against Rutherford's model of the atom, Neils Bohr (1913) proposed his quantum mechanical structure of the atom. He made use of quantum theory, according to which energy is lost or gained not gradually, but in bundles or quanta. The concept of nucleus explained by Rutherford was retained in Bohr's model. Through this model does not meet the modern quantum mechanics, it is still in use to rationalize many points in the atomic structure and spectra. The important postulates of the Bohr Theory are:

1. *Electrons move in certain fixed orbits associated with a definite amount of energy. The energy of an electron moving in an orbit remains constant as long as it stays in the same orbit called* ***stationary state*** *or* ***orbit****.*

Hence, a certain fixed amount of energy is associated with each electron in a particular orbit. Thus, stationary orbits are also known as **energy levels** or **energy shells**. Bohr gave number 1, 2, 3, 4 etc, (starting from the nucleus to these energy levels). The various energy levels are designated as K, L, M, N etc, and are termed as **principal quantum numbers**.

2. Energy is emitted or absorbed only when an electron jumps from one orbit to another, i.e., form one energy level (or principal quantum number) to another.

   Since each level is associated with a definite amount of energy, the farther the energy level from the nucleus, the greater is the energy associated with it.

3. *Permissible orbits are those for which the angular momentum is an integral multiple of (h/2π).*

   The electrons move in such orbits without any loss of energy. Thus, angular momentum

$$m_e\ vr = \frac{nh}{2\pi} \qquad (1.20)$$

where, n is an integer 1, 2, 3 .... m_e is mass of electron and r is the radius of orbit. The n values corresponds to the principal quantum number.

4. *When an electron gets sufficient energy from outside, an electron from an inner orbit of lower energy state E_1 moves to an outer orbit of higher energy state E_2.*

   The excited state lasts for about 10^{-8} s. The difference of energy $E_2 - E_1$ is thus radiated out. During the emission or absorption of radiant energy to Planck's, Einstein equation $E = hv$ is obeyed. Thus the frequency of emitted radiation is given by

$$hv = \Delta E = E_2 - E_1 \qquad (1.21)$$

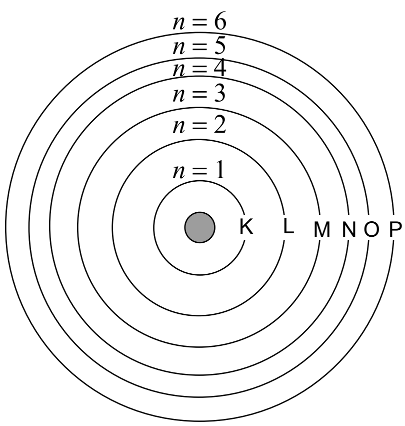
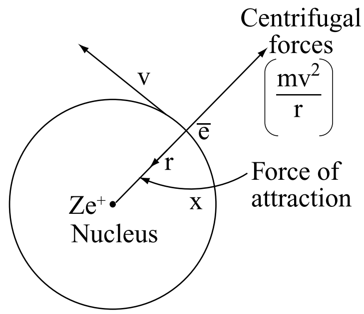


Fig 1.20 Bohr's orbit-like representation of various energy levels

Based on the above postulates, Bohr calculated the radii of the various orbits and the energies associated with the electrons present in those orbits. The frequencies of the spectral lines determined experimentally by Lyman, Balmer and others are in excellent agreement with those calculated by Bohr's theoretical equations.

1.7.1 Bohr's Theory of the Hydrogen Atom

Bohr pictured the hydrogen atom as a system consisting of a single electron with a charge designed as e, rotating in a circular orbit of radius r about the nucleus of charge Ze with a velocity v.



Tangential velocity of the revolving electron

Calculation of velocity V: The angular momentum of the electron is defined as the product of the velocity of the electron in its orbit, its mass, and the radius of its orbit. The product is symbolized as mvr. According to Bohr's postulates, the angular momentum are whole multiples of $h / 2\pi$.

Therefore, according to this quantum restriction, angular momentum may be restricted as

$$mvr = \frac{nh}{2\pi}$$

∴ The velocity of the electron in an orbit,

$$v = \frac{nh}{2\pi mr} \qquad (1.22)$$

Radius of an Orbit

According to Coulomb's law, the electrostatic force of attraction F_e between the charges may be evaluated mathematically as

$$F_e = \frac{Ze^2}{r^2} \qquad (1.23)$$

By definition, the magnitude of the centrifugal force F_c for an electron of mass m, with a velocity in its orbit of V and with an orbit radius r is given as

$$F_c = \frac{mv^2}{r} \qquad (1.24)$$

In order that the electron's orbit may remain stable, it is necessary to assume that the two forces, electrostatic and centrifugal, are equal and opposed to each other. From this assumption may be deduced the following mathematical relationship

$$\frac{mv^2}{r} = \frac{Ze^2}{r^2} \tag{1.25}$$

and upon solving for

$$v = \sqrt{\frac{Ze^2}{mr}} \tag{1.26}$$

Equations (1.26) and (1.22) may be equated since they are equal to each other

$$\sqrt{\frac{Ze^2}{mr}} = \frac{nh}{2\pi mr} \tag{1.27}$$

and upon solving for r, we obtain

$$r = \frac{n^2h^2}{4\pi^2 mZe^2} \tag{1.28}$$

Consequently, the radius r_0 of the smallest orbit (first orbit) for the hydrogen atom is

$$r_0 = \frac{h^2}{4\pi^2 me^2} \tag{1.29}$$

Energy of the Electron

By definition the kinetic energy of body is equal to $\frac{1}{2}mv^2$. The total energy E of the electron is the sum of its kinetic and potential energies. If the potential energy of an electron is taken as zero when it is at an infinite distance from the nucleus, the value at a distance r is given by $-Ze^2 / r$. This value may be obtained by integrating Eq. (1.24) between the limits of r and infinity. The negative sign indicates that work must be performed on the electron to transfer it to infinity. Therefore, the total energy of the electron is

$$E = \frac{mv^2}{2} - \frac{Ze^2}{r} \tag{1.30}$$

From Eq (1.25) we may obtain

$$\frac{mv^2}{2} = \frac{Ze^2}{2r} \tag{1.31}$$

and upon combining Eqs (1.30) and (1.31) the following expression may be derived

$$E = -\frac{Ze^2}{2r} \tag{1.32}$$

Now if we substitute the value r from Eq. (1.28) into (1.32) the total energy of the electron may be stated as

$$E_n = -\frac{2\pi^2\ me^4}{h^2} \times \frac{Z^2}{n^2} \tag{1.33}$$

where, E_n is the total energy of the electron in the orbit designated by the quantum number n.

Solved Problem 8

Calculate the (i) radius of first orbit (ii) velocity of electron in the first orbit, and (iii) energy of first orbit of hydrogen atom.

Solution:

(i) Radius of an orbit is given by $r = \frac{n^2h^2}{4\pi^2 mZe^2}$

Since, $n = 1$ and Z for hydrogen = 1

$$r = \frac{h^2}{4\pi^2 me^2}$$

$$r = \frac{\left(6.626\times10^{-27}\ \text{ergs}\right)^2}{4\times\pi^2\times\left(9.109\times10^{-28}\ \text{g}\right)\left(4.803\times10^{-10}\ \text{esu}\right)^2}$$

$= 0.529 \times 10^{-8}$ cm = 0.529 Å

(ii) Velocity of electron in the first orbit of hydrogen

$$V = \frac{nh}{2\pi mr}$$

$$V = \frac{1\times6.626\times10^{-27}\ \text{ergs}}{2\times\pi\times\left(9.109\times10^{-28}\ \text{g}\right)\left(0.529\times10^{-8}\ \text{cm}\right)}$$

$= 2.188 \times 10^8\ \text{cm}^{-1}$

(iii) Energy of electron

$$E_n = \frac{2\pi^2 me^4}{h^2} \times \frac{Z^2}{n^2}$$

Since, $n = 1$, and $Z = 1$

Energy of electron in the first orbit

$$E = \frac{2\pi^2 me^4}{h^2}$$

$$E = \frac{2\pi^2\left(9.109\times10^{-28}\ \text{g}\right)\left(4.803\times10^{-10}\ \text{esu}\right)^4}{\left(6.626\times10^{-27}\ \text{ergs}\right)^2}$$

$= 2.179 \times 10^{-11}$ erg / atom

The energy of an electron is usually expressed in Kcal or KJ mol^{-1} or electron volt (eV)

One erg/molecule = 1.44×10^{13} Kcal / mol
= 6.22×10^{13} KJ / mol

$\therefore\ E_1 = (-2.179 \times 10^{-11}) \times (1.44 \times 10^{13})$
$= -313.77$ Kcal / mol $= -1312.19$ KJ / mol

Now 1eV = 1.602×10^{-12} ergs

$$\therefore\ E_1 = \frac{2.179\times10^{-11}\ \text{erg}}{1.602\times10^{-12}} = -13.6\ eV/\text{atom}$$

1.7.2 Origin of Spectral Lines and the Hydrogen Spectrum

As described by Bohr, energy is radiated when an electron moves from one orbit of a quantum number of n_2 to an orbit of quantum number n_1, where n_1 is the inner orbit. Energy is absorbed if the electron moves in opposite direction, i.e., from inner orbit to outer orbit. The energy ΔE which is emitted may be represented as a difference in energy of the two electronic states and can be indicated by the equation

$$\Delta E = E_2 - E_1 = h\nu \qquad (1.34)$$

where, E_2 is the energy of orbit n_2 and E_1 is the energy of inner orbit n_1.

Spectral lines are produced by the radiation of photons, and the position of the lines on the spectral scale is determined by the frequency, or frequencies, of the photons emitted. Transitions to the innermost level n_1 from orbits n_2, n_3, n_4 etc, gives rise to the first, second, third etc, lines of the Lyman series in the UV region. Transitions from the outer most energy levels gives rise to spectral lines of higher frequencies. Transition of electron to n_2 level from outer level gives Balmer series of lines in the visible region. Similar transitions from higher orbits to the third orbit ($n = 3$) produces Paschen series in the infrared region. Other series of lines have been discovered for similar shifts in the far infrared region. A sketch indicating the transitions which produces these spectral lines is given in Fig 1.21.

From the Eq. (1.34) the difference between the energies of an electron in the two orbits n_2 and n_1 may be indicated as

$$\Delta E = E_{n_2} - E_{n_1} = h\nu = \frac{2\pi^2 Z^2 e^4 m}{h}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \qquad (1.35)$$

If the term wave number is used for frequency

$$\bar{\nu} = \frac{\nu}{c}$$

$$\bar{\nu} = \frac{2\pi^2 Z^2 e^4 m}{h^3 c}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \qquad (1.36)$$

All the terms in the fraction $2\pi^2 Z^2 e^4 m/h^3c$ are constant. For hydrogen atom Z is 1. If we evaluate the fraction from the constant terms, the following value is obtained:

$$\frac{2\pi^2 1^2 e^4 m}{h^3 c} = 109{,}700 \text{ cm}^{-1} = R \qquad (1.37)$$

where, R is the **Rydberg constant.** Moreover, when we substitute R in Eq. 1.36

$$\bar{\nu} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \qquad (1.38)$$

Equation 1.38 is identical with the general equation for the wave number of any spectral line within any of the spectral series of the hydrogen atom as given in Eq. (1.15).

1.7.3 Limitations of the Bohr's Model

Bohr's atomic model was a very good improvement over Rutherford's nuclear model and enabled in calculation of radii and energies of the permissible orbits in the hydrogen atom. The calculated values were in good agreement with the experimental values. Bohr's theory could also explain the hydrogen spectra successfully and also the spectra of hydrogen like atoms (He^+, Li^{2+}, Be^{3+}, etc). The theory was, therefore, largely accepted and Bohr was awarded Nobel prize in recognition of

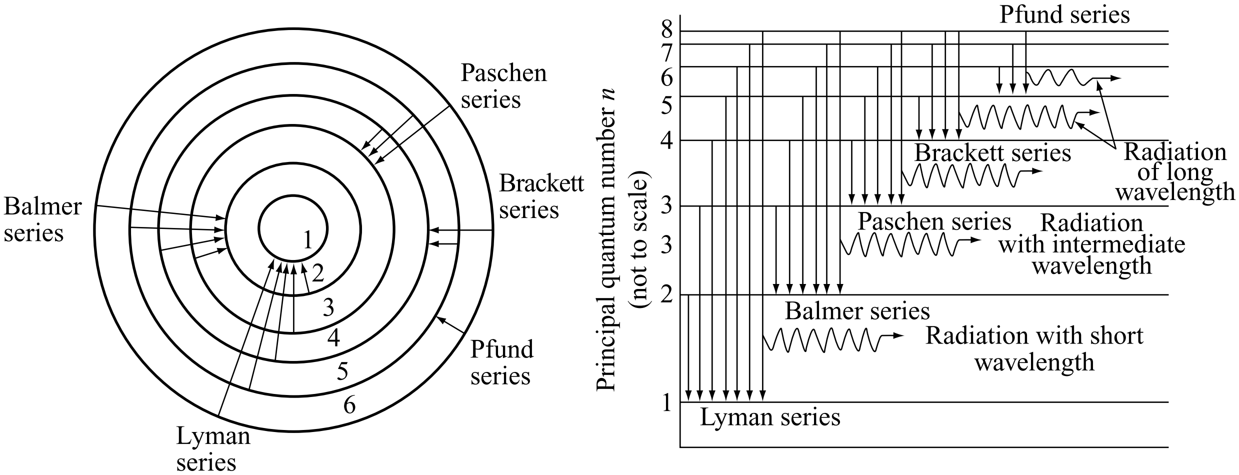


Fig 1.21 Origin of emission spectrum of hydrogen atom

this work. However, Bohr's model could not explain the following points:

(i) Bohr's model could not explain the spectra of atoms containing more than one electron. Also, it could not explain hydrogen spectrum obtained using high resolution spectroscopes. Each spectral line, on high resolution was found to consist of two closely spaced lines.

(ii) It was observed that in the presence of a magnetic field each spectral line gets splitted up into closely spaced lines. This phenomenon, known as **Zeeman effect**, could not be explained by the Bohr's model. Similarly, the splitting of spectral line under the effect of applied electric field, known as **Stark effect**, could not be explained by the Bohr's model.

(iii) It could not explain the spectra of atom of elements other than hydrogen.

(iv) Bohr's model could not explain the ability of atoms to form molecules and the geometry and shapes of molecule.

1.8 WAVES AND PARTICLES

Because of the limitations of Bohr's atomic model several scientists tried to develop a more subtle and general model for atoms. During that period two important proposals were contributed significantly for the modern quantum mechanical model of the atom. They are

(i) Dual nature, i.e., wave as well as particle nature of matter.

(ii) Heisenberg's uncertainty principle.

1.8.1 Dual Nature of Matter

According to **Maxwell's** concept, *light, radiation consists of waves while Planck's quantum theory considers photons as particles*. Thus, light, which consists of electromagnetic radiation, is both a wave and particle. Based on this analogy in 1924, the Frenchman, **Prince Louis de Broglie** published an exceedingly complicated account of the wave-particle duality. de Broglie stated that *any form of matter such as electron, proton, atom or molecules, etc, has a dual character.* The waves predicted by de Broglie are known as **matter waves**. These waves are quite different from electromagnetic waves in following two respects:

(i) Matter waves cannot radiate through empty space like the electromagnetic waves.

(ii) Speed of matter waves is different from that of electromagnetic waves.

Louis de Broglie 1892–1987

He was a French physicist. Studied history but while working on radio communications during the First World War as an assignment, he developed interest in science. He received his Dr Sc. from the University of Paris in 1924. He was professor of theoretical physics in the University of Paris from 1932 to 1962. He was awarded Nobel Prize for physics in 1929.

By making use of the Einstein's ($E = mc^2$) and Planck's quantum theory, ($E = hv$) de Broglie deduced a fundamental relation called the **de Broglie equation:**

$$\lambda = \frac{h}{mv} \tag{1.39}$$

This equation gives the relationship between the wavelength of the moving particle and its mass. In the Eq. (1.39), λ is the wavelength of the wave associated with an electron of mass 'm' moving with velocity v.

Eq. (1.39) can be written as

$$mv = \frac{h}{\lambda} \tag{1.40}$$

$$mv \propto \frac{1}{\lambda} \tag{1.41}$$

$$p \propto \frac{1}{\lambda} \tag{1.42}$$

In the E.q. (1.42) 'p' represents the momentum mv of particle; and λ corresponds to the wave character of matter and p its particle character. Thus, *the momentum (p) of a moving particle is inversely proportional to the wavelength of the waves, associated with it.*

The revolutionary postulate of de Broglie received direct experimental verification in 1927 by Davisson and Germer, G.P. Thomson and later by Stern. They showed that heavier particles (H_2, He etc.) showed diffraction patterns when reflected from the surface of crystals. Particularly Davisson and Germer found that electron beam was diffracted when striken on a single crystal of nickel, which proves the wave like character of electrons.

de Broglie's Relationship Concept and Bohr's Theory

Application of de Broglie's relationship to a moving electron around a nucleus puts some restrictions on the size of the orbits. It means that electron is not a mass particle moving in a circular path but instead a standing wave train (non-energy, radiating motion) extending around the nucleus in the circular path as shown in Fig 1.22 (a) and 1.22 (b).

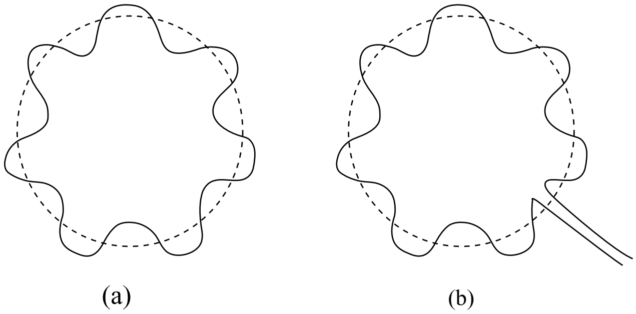


Fig 1.22 Diagrammatic representation of electron orbits one of which is (a) in phase; and the other (b) out of phase

For the wave to remain continually in phase, the circumference of the orbit should be an integral multiple of wavelength λ,

$$2\pi r = n\lambda \tag{1.43}$$

where, r is the radius of the orbit, and n is a whole number.

From Eq. (1.40) we know

$$\lambda = \frac{h}{mv} \tag{1.44}$$

Substituting the value of λ in Eq. (1.43), we get

$$2\pi r = \frac{nh}{mv}$$

$$\text{or} \quad mvr = \frac{nh}{2\pi} \tag{1.45}$$

which is the same as Bohr's postulate for angular moment of electron. From the E.q. (1.45) it can be known that *electrons can move only is such orbits for which the angular momentum must be an integral multiple of $h/2\pi$*. If the circumference is bigger or smaller than the value as given by in Fig 1.22 b, the electron wave will be out of phase. Thus, de Broglie relation provides a theoretical basis for the Bohr's postulate for angular momentum. In Fig 1.22 (a) the wave is in phase continually.

In the summer of 1927 physicists from all over the world arrived in Brussels at the Solvay Congress. At this congress de Broglie's concept on the relationship between waves and particles was totally rejected. For many years to come, a complete different representation of this relationship led the way. It was Heisenberg and Schrodinger who supported and strongly represented the concept of de Broglie at another congress and got it accepted.

de Broglie's equation is true for material particles of all sizes and dimensions. However, in the case of small micro objects like electrons, the wave character is of significance only. In the case of large macro-objects the wave character is negligible and cannot be measured properly. Thus, de Broglie equation is more useful for small particles.

Meaning of ψ and its Significance

ψ is the wave function or the amplitude of the wave. The value of amplitude increases and reaches the maximum which is indicated by peak in the curve. This is shown by the upward arrow in the figure. The value of the amplitude decreases after reaching the maximum value. This is shown by the downward arrow.

Above the X-axis the amplitude is shown as +ve, along the X-axis it is zero and below the X-axis it is negative. The intensity of light is proportional to the square of amplitude (or A^2). Therefore, A^2 can be taken as a measure of the intensity of light since light is considered to consist of photons (corpuscular theory), the density of photons is considered to be proportional to A^2. Thus, as far as light waves are concerned A^2 indicates the density of photons in space or the intensity of light.

This concept can be extended even to the ψ functions moving in the atom since electron moving with high speed is associated with a wave characteristics. Therefore, ψ^2 in the case of electron wave denotes the probability of finding an electron in the space around the nucleus or the electron density around the nucleus. If ψ^2 is maximum, the probability of finding an electron is also maximum.

Solved Problem 9

An electron with mass of 9.1×10^{-31} kg is moving with a velocity of 10^3 m/s. Calculate its kinetic energy and wavelength (Planck's constant $h = 6.6 \times 10^{-34}$ kg m^2 s^{-1}).

Solution:

Kinetic energy KE

$$= \frac{1}{2}mv^2 = \frac{1}{2} \times \left(9.1 \times 10^{-31} \text{ kg}\right)\left(10^3 \text{ m s}^{-1}\right)^2$$

$$= 4.55 \times 10^{-7} \text{ J}$$

$$\text{Wavelength } \lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}}{9.1 \times 10^{-31} \text{ kg} \times 10^3 \text{ ms}^{-1}}$$

$$= 7.25 \times 10^{-7} \text{ m}$$

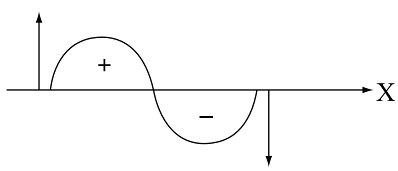


Fig 1.23

Solved Problem 10

A moving electron has 2.8×10^{-25} J of kinetic energy. Calculate its wavelength (mass of electron = 9.1×10^{-31} kg).

Solution:

Kinetic energy = $\frac{1}{2}mv^2$

$$V^2 = \frac{2 \cdot \text{ Kinetic energy}}{m}$$

$$= \frac{2\times 2.8\times 10^{-25}}{9.1\times 10^{-31}}\text{ m}^2\text{ s}^{-2}$$

$$= \frac{2\times 2.8\times 10^{-25}}{9.1\times 10^{-31}}\text{ m}^2\text{ s}^{-2}$$

$$= 61538.61\text{ m}^2\text{ s}^2$$

$\therefore\ v = 784.46\text{ m s}^{-1}$

By de Broglie's equation, we have

$$\lambda = \frac{h}{mv} = \frac{6.6\times 10^{-34}}{9.1\times 10^{-31}\times 784.46}\times\frac{\text{kgm}^2\text{ s}^{-2}\text{s}}{\text{kgm s}^{-1}}$$

$$= 9.2455 \times 10^{-7}\text{ m}$$

1.8.2 Heisenberg's Uncertainty Principle

Werner Heisenberg (1901–1976)

He received his Ph.D. in physics from the University of Munich in 1923. Later he worked with Max Born at Gottingen and with Niels Bohr at Copenhagen. He spent about 14 years (1927–1941) as professor of physics at the University of Leipzig. He led the German team working for atomic bomb during the Second World War. Once the war ended he became the director of Max Planck Institute for physics in Gottingen. He was awarded Nobel Prize for physics 1932.

The exact sciences also start from the assumption that in the end, it will always be possible to understand nature, even in every new field of experience but we make no priori assumptions about the meaning of the word understand.

W. Heisenberg

According to classical mechanics, a moving electron is considered to be a particle. Therefore, its position and momentum could be determined with a desired accuracy. On the other hand de Broglie also considered a moving electron to be wave-like. Therefore, it becomes impossible to locate the exact position of the electron on the wave because it is extending throughout a region of space. So, the following fundamental question arises:

"If an electron is exhibiting the dual nature, i.e.*, wave and particle, is it possible to know the exact location of the electron in space at some given instant?"*

The answer to the above question was given by **Heisenberg** in 1927 who stated that

"For a subatomic object like electron, it is impossible to simultaneously determine its position and velocity at any given instant to an arbitary degree of precision."

Heisenberg gave mathematical relationship for the uncertainty principle by relating the uncertainty in position (Δx) with uncertainty in momentum (Δp) as

$$\Delta p\times\Delta x \geq \frac{h}{4\pi} \tag{1.46}$$

The ≥ in the Eq. (1.46) means that the product of Δx and Δp can be either greater than or equal to but would be never smaller than $h / 4\pi$. But $h / 4\pi$ is constant. Therefore, it follows from Eq. (1.46) that smaller the uncertainty in locating the exact position (Δx), greater will be the uncertainty in locating the exact momentum (Δp) of the particle and *vice versa.*

As Δp is equal to $m\Delta v$ the Eq. (1.46) is equivalent to saying that position and velocity cannot be simultaneously determined to an arbitrary precision.

However, in our daily life, these principles have no significance. This is because we come across only large objects. The position and velocity of these objects can be determined accurately because in these cases the changes that occur due to the impact of light are negligible. The microscopic objects suffer a change in position or velocity as a result of the impact of light. For example, to observe an electron, we have to illuminate it with light or electromagnetic radiation. The light must have a wavelength smaller than the wavelength of electron. When the photon of such light strikes the electron, the energy of the electron changes. In this process, no doubt, we shall be able to calculate the position of the electron, but we would know very little about the velocity of the electron after the collision.

1.8.3 Significance of Uncertainty Principle

The most important consequence of the Heisenberg uncertainty principle is that *it rules out existence of definite paths or trajectories of electrons and other similar particles.* The trajectory of an object is determined by its location and velocity at various moments. At any particular instant if we know the position of a body and also its velocity and the forces acting on it at that instant we can predict at what position it will present at a particular time. This indicates that the position of an object, and its velocity completely determine its trajectory. Because, it is not possible simultaneously to determine the position and velocity of an electron at any given instant precisely it is not possible to talk of the trajectory of an electron.

The effect of Heisenberg's uncertainty principle is significant only for motion of microscopic objects and is negligible for that of macroscopic objects. This can be seen from these illustrated examples.

Solved Problem 11

Calculate the uncertainty (D*v*) *in the velocity of a cricket ball of mass 1 kg, if certainty* (D*x*) *in its position is of the order of 1 Å.*

Solution:

In the above example.

$$\Delta V = \frac{h}{4\pi\Delta x\, m}$$

But $h = 6.6 \times 10^{-34}$ kg m^2 s^{-2} s;

$\Delta x = 1$Å $= 10^{-10}$ m and

$m = 1$ kg

$$\Delta V = \frac{6.6\times10^{-34}}{4\times\pi\times10^{-10}\times1}\times\frac{\text{kg m}^2\text{s}^{-1}\text{ s}}{\text{m kg}}$$

$$= \frac{7\times6.6\times10^{-34}}{4\times22\times10^{-10}}\text{ m s}^{-1}$$

$$= 5.25 \times 10^{-25}\text{ m s}^{-1}$$

Solved Problem 12

A microscope using suitable photons is employed to locate an electron in an atom within a distance of 0.1 Å. What is the uncertainty involved in the measurement of its velocity?

Solution:

$$\Delta v = \frac{h}{4\pi\Delta x\, m}$$

But $h = 6.626\times10^{-34}$ kg m^2 s^{-2}s;

$\Delta x = 0.1$Å $= 1 \times 10^{-11}$ m and

$m = 9.11 \times 10^{-31}$ kg

$$\therefore \Delta v = \frac{6.626\times10^{-34}}{4\times\pi\times10^{-11}\times9.11\times10^{-31}}\times\frac{\text{kg } m^2\text{ s}^{-2}\text{s}}{\text{m kg}}$$

$$= 0.579 \times 10^{7}\text{ m s}^{-1}$$

$$= 5.79 \times 10^{6}\text{ m s}^{-1}$$

Why Bohr's Model was a Failure?

Bohr considered the electron as a charged particle moving in well defined circular orbits around the nucleus. So, the position and the velocity of the electron can be known exactly at the same time. Uncertainty principle shows that it is impossible to measure these variables simultaneously because the wave character of electron was not considered in the Bohr model. Calculation of the trajectory of an electron in an atom or molecule is, therefore, a futile exercise. We can now appreciate the major fault of the Bohr model. In calculating electron orbits precisely, Bohr was violating this fundamental requirement and hence his theory was only partially successful. So, a model which can account the wave-particle duality of matter and be consistent with Heisenberg uncertainty principle was in quest. This came with the advent of quantum machanics.

Concept of Probability

The consequences of the uncertainty principle are far reaching and probability takes the place of exactness in velocity (which is related to kinetic energy) of an electron. The Bohr concept of the atom, which regards the electrons as rotating in definite orbits around the nucleus, must be abandoned and should be replaced by a theory which considers probability of finding the electrons in a particular region of space. This means that, it is possible to state the probabilities of the electron to be various distances with respect to the nucleus. In the same way, probable values of velocity can also be given. It should be clearly understood, however, that a knowledge of probability for an electron 'moves' from one location to another.

1.8.4 Quantum Mechanical Model of Atom

Classical mechanics based on Newton's laws of motion successfully describes the motion of all macroscopic objects since the uncertainties in position and velocity are small enough to be neglected. However, it fails in the case of microscopic objects like electrons, atoms, molecules etc. So, while considering the motion of microscopic objects, the concept of dual behaviour of matter and the uncertainty principle are taken into account. The branch of science that takes into account this dual behaviour of matter is called **quantum mechanics**.

As far as the other dynamical variables of an electron are concerned, we can show that the uncertainty principle leads to the following result.

The total energy of an electron in an atom or molecule has a well-defined (sharp) value. The probability distribution as well as the sharp values can be calculated from a function designated as ψ (x) and called the **wave function** or the **psi function**. ψ (x) is obtained by solving the **Schrodinger equation** which is the fundamental equation in quantum mechanics in the same manner that Newton's equation is fundamental in classical mechanics.

Important Features of the Quantum Mechanical Model of Atom

Quantum mechanical model of the atom is the outcome of the application of Schrodinger wave equation to atoms. Its important features are

- Quantization of the energy of electron in atoms, i.e., it can have certain discrete values.

- The allowed solutions of **Schrödinger** wave equation tell about the existence of quantized electronic energy levels for electrons in atoms having wave like properties.
- As per Heisenberg's uncertainty principle, both the position and velocity of an electron cannot be determined simultaneously and exactly. So, the path of the electron can never be determined accurately. Because of this reason the concept of probability was introduced for finding the electron at different points in an atom.
- The wave function for an electron in an atom is the atomic orbital. Whenever we say about electron by a wave function it occupies that orbital. Because several wave functions are possible for an electron, there also as many atomic orbitals. These one electron orbital wave functions or orbital form the basis for electronic structure of atoms. Every orbital possess certain energy and can accommodate only two electrons. In atoms having several electrons, the electrons are filled in the order of increasing energy. In the multielectron atoms each electron has an orbital wave function characteristic of the orbital it occupies. All the information about the electron in an atom is stored in its orbital wave function ψ and quantum mechanics makes it possible to extract this information out of ψ.

  The probability of finding the electron at a point within an atom is proportional to ψ^2 at that point. Though ψ is sometimes negative, ψ^2 is always positive and is known as the **probability density**. The values of ψ^2 predicts the different points in a region within an atom at which the electron will be most probably found.

1.8.5 Schrödinger Wave Equation

Erwin Schrödinger 1887–1966

Erwin Schrödinger was born in Austria. He received his Ph.D. in theoretical physics from the University of Vienna in 1910. At the request of Max Planck, Schrödinger became his successor at the University of Berlin in 1927. Because of his opposition to Hitler and Nazi policies, he left Berlin and returned to Austria and in 1936, when Austria was occupied by Germany, he was forcibly removed from his professorship. Then he moved to Dublin, Ireland. He shared the 1933 Nobel Prize for physics with P.A.M. Dirac.

During 1920 there was a great deal of interest in wave-particle duality and de Broglie's matter waves. It was the Austrian physicist Erwin Schrödinger who invented a method of showing how the properties of waves could be used to explain the behaviour of electrons in atoms. Schrodinger published his ideas in January 1926. This date represents one of the milestones in the history of Chemistry. His work formed the basis of all our present ideas on how atoms bond together. The heart of his method was his prediction of the equation that governed the behaviour of electrons, and the method of solving it. Schrodinger's equation is

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi^2 m}{h^2}(E-U)=0 \qquad (1.43)$$

where, m = Mass of electron

E = Total energy of the electron (kinetic energy + potential energy)

U = Potential energy

h = Planck's constant

ψ = Wave function

This equation applies to stationary waves as it does not have the time dependence part of the wave function.

Schrödinger won the Nobel prize for physics in 1933. The solution of this equation are very complex and you will learn them for different systems in higher classes.

For a system (such as an atomic or a molecule whose energy does not change with time), the Schrödinger equation is written as $\hat{H}\psi = E\psi$, where $\hat{H}$ is a mathematical operator called **Hamiltonian.** Schrödinger gave a recipe of constructing this operator from the expression for the total energy of the system. The total energy of the system takes into account the kinetic energies of all the sub-atomic particles (electrons, nuclei) attractive potential between the electrons and nuclei and repulsive potential among the electrons and nuclei individually. Solutions of this equation gives E and ψ.

1.8.6 The Meaning of Wave Function

The wave function could only be used to provide information about the **probability** of finding the electron in a given region of space around the nucleus. **Max Born**, a German physicist proposed that we must give up ideas of the electron orbiting the nucleus at a precise distance. In this respect Bohr was wrong in thinking that the electron in the ground state of the hydrogen atom was always to be found at a distance a_0 form the nucleus. Rather, it was only ***most probably*** to be found at this distance. The electron had a smaller probability of being found at a variety of other distances as well.

It is important to realise that we should not try to talk about finding the electron at a given point. The reason for this is that there is an infinite number of points around the nucleus. So, the probability of finding the electron at any

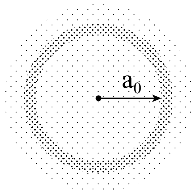


Fig 1.24 Electron density

one of these points is infinitely small, i.e., zero. Then to predict the position of electron, we can imagine taking out a series of photographs of an atom to give us an instantaneous picture of the whereabouts of the electron. If we combine all pictures we would end up with a picture as shown in Fig 1.24. The separate dots have overlapped to give regions in which the density of dots is very high and regions where the density of dots is much lower. In the high density regions, we say that there is a high probability density. The maximum in the density comes at exactly the same distance a_0, as Bohr predicted in his work. The circulated symmetry of the probability density is clear. However, we should remember that atoms exist in three dimensions, so really the diagram should be shown as a sphere.

It is easier to draw circles rather than spheres, so usually we draw the density diagram as a circle. Also, it is common practice not to include the shading and to agree that when a circle is drawn, it provides a boundary surface within which, say, there is a 95 per cent probability of finding the electron.

When Schrödinger equation is solved for hydrogen atom, the solution gives the possible energy levels the electron can occupy and the corresponding wave function(s) ψ of the electron associated with each energy level.

Accepted solutions to the wave functions are called **Eigen wave functions**. The probability of finding an electron at a point in space whose coordinates are x, y and z is given by $|\psi\ (x, y, z)|^2$dv. This three dimensional region obtained where the probability to find the electron is about 95 per cent, is called an orbital.

> Because atomic behavor is so unlike ordinary experience, it is very difficult to get used to and it appears peculiar and mysterious to everyone, both to the novice and to the experienced scientist. Even the experts do not understand it the way they would like to, and it is perfectly reasonable that they should not, because all of direct human experience and of human intution applies to large objects. We know how large objects will act, but things on a small scale just do not act that way. So, we have to learn about them in a sort of abstract or imaginative fashion and not by connection with our direct experience. *Feynman Lectures on Physics Vol. l Chapter 37.*

1.9 QUANTUM NUMBERS

The mathematical solution of three dimensional Schrödinger wave equation gives three values of E for acceptable values of ψ. These values are related to one another through whole numbers. These values are termed as **Quantum numbers** and represented with n, l, m — called **principal**, **azimuthal** and **magnetic quantum numbers,** respectively. These three quantum numbers together with the fourth called *spin quantum number* describe fully the location and energy of an electron. Thus, quantum numbers are *the numbers which determine the energy of electron, the angular momentum, shape of the electron orbital, the orientation of the orbital and spin of the electron.* Thus, each quantum number is associated with a particular characteristic of the electron. These quantum numbers are discussed below briefly.

1. **Principal Quantum Number** (n): The number allotted to Bohr's original stationary states, visualized as circular orbits is called the principal quantum number. The innermost orbit, i.e., that nearest to the nucleus has a principal quantum number 1, the second orbit has a quantum number of 2, and so on. So, the principal quantum number denoted by n have value 1, 2, 3, 4, ... Alternatively, letters are used to characterize the orbits, K, L, M, N, ... for 1, 2, 3, 4, ... The choice of letters originates from Mosely's work on the X-ray spectra of the elements. He called group of lines in the spectra the K, L, M, N, ... groups.

The number of electrons in an atom which can have the same principal quantum number is limited and is given by $2n^2$ where n is the principal quantum number concerned. Thus,

Principal quantum number (n)	1	2	3	4
Letter designation	K	L	M	N
Maximum number of electrons	2	8	18	32

This is the most important quantum number as it determines to a large extent the energy of an electron. It also determines the average distance of an electron from the nucleus. As the value of n increases, the electron gets farther away from the nucleus and its energy increases. The higher the value of n, the higher is the electronic energy. For hydrogen and hydrogen like species, the energy and size of the orbital are determined by the principal quantum number alone.

The principle quantum number also identifies the number of allowed orbitals within a shell. With the increase in the value of 'n' the number of allowed orbitals increases and are given by n^2. All the orbitals of a given value of 'n' constitute a single shell of atom.

2. **Azimuthal Quantum Number** (l): It is also known as *orbital angular momentum or subsidiary quantum number.* It defines the three dimensional shape of the

sub-shell. For each value of the principal quantum number there are several closely associated orbitals, so that the principal quantum number represents a group or shell of orbits. In any one shell, having the same principal quantum number the various subsidiary orbits are denoted as *s*, *p*, *d*, *f*, ... sub-shells. The letters originates from the *sharp*, *principal, diffuse and fundamental* series of the lines in spectra.

The number of sub-shells or sub-levels in a principal shell is equal to the value of *n*. The values of subshells is represented with *l*. *l* can have values ranging from 0 to $n-1$, i.e., for a given value of *n* the possible value of *l* can be 0, 1, 2......(n–1). For example when n = 1 value of *l* is only 0. For n = 2 the possible value of *l* can be 0 and 1. For $n = 3$ the possible values are 0, 1 and 2. Sub-shells corresponding to different values are as follows:

value for *l*	0	1	2	3	4	5........
Notation for subshell	*s*	*p*	*d*	*f*	*g*	*h* ...

The permissible values for '*l*' for a given principal quantum number and the corresponding sub-shell notation are as follows.

***l* values**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
0	1*s*	2*s*	3*s*	4*s*	5*s*	6*s*	7*s*
1		2*p*	3*p*	4*p*	5*p*	6*p*	7*p*
2			3*d*	4*d*	5*d*		
3				4*f*	5*f*		

3. **Magnetic Quantum Number** (m_l): This quantum number which is denoted by m_l refers to the different orientations of the electron cloud in a particular subshell. These different orientations are called **orbitals**. The number of orbitals in a particular sub-shell within a principal energy level is given by the number of values allowed to m_l which in turn depends upon on the value of *l*. The possible values of m_l range from $+l$ through 0 to $-l$, thus making a total of $(2l + 1)$ values. Thus, in a subshell, the number of orbitals is equal to $(2l + 1)$.

For $l = 0$ (i.e., **s-subshell**) m_l can have only one value $m_l = 0$. It means that s-**subshell** has only **one orbital**.

For $l = 1$ (i.e., **p-subshell**) m_l can have three values +1, 0 and −1. This implies that *p*-**subshell** has **three orbitals**.

For $l = 2$ (i.e., **d-subshell**) m_l can have five values +2, +1, 0, and −2. It means that *d*-**subshell** has **five orbitals.**

For $l = 3$ (i.e., **f-subshell**) m_l can have seven values +3, +2, +1, 0, −1, −2 and −3. It means that *f*-**subshell** has **seven orbitals.**

The number of orbitals in various types of subshells are as given below.

Sub-shell	*s*	*p*	*d*	*f*	*g*
Value of *l*	0	1	2	3	4
No. of orbitals $(2l + 1)$	1	3	5	7	9

The relationship between the principal quantum number (*n*), angular momentum quantum number (*l*) and magnetic quantum number (m_l) is summed up in Table 1.4.

4. **Spin Quantum Number** (m_s): This quantum number which is denoted by m_s does not follow from the wave mechanical treatment. Electron spin was first postulated in 1925 by Uhlenbeck and Goudsmit to account for the splitting of many single spectral lines into double lines when examined under a spectroscope of

Table 1.4 Relationship amongst the value of *n*, *l* and m_l

Energy level	**Principal quantum number (n)**	**Possible values of (l)**	**Designation of sub-shell**	**Possible values of m_l**	**Number of orbitals: In a given sub-shell**	**Number of orbitals: In a given energy level**
K	1	0	1*s*	0	1	1
L	2	0	2*s*	0	1	
		1	2*p*	+1, 0, –1	3	4
M	3	0	3*s*	0	1	
		1	3*p*	+1, 0, –1	3	9
		2	3*d*	+2, +1, 0, –1, –2	5	
N	4	0	4*s*	0	1	
		1	4*p*	+1, 0, –1	3	16
		2	4*d*	+2, +1, 0, –1, –2	5	
		3	4*f*	+3, +2, +1, 0, –1, –2, –3	7	

high resolving power. The electron in its motion about the nucleus also rotates or spins about its own axis. In other words, an electron has, besides charge and mass, an intrinsic spin angular quantum number.

Spin angular momentum of the electron: A vector quantity, can have two orientations relative to a chosen axis. The spin quantum number can have only two values which are $+\frac{1}{2}$ and $-\frac{1}{2}$. The $+\frac{1}{2}$ value indicates the clockwise spin, generally represented by an arrow upwards, i.e., $\uparrow$ and the other indicates anti-clockwise spin, generally represented by an arrow downwards, i.e., $\downarrow$. The electrons that have different m_s values (one $+\frac{1}{2}$ and the other $-\frac{1}{2}$) are said to have opposite spins. An orbital cannot hold more than two electrons and these two electrons should have opposite spins.

Orbit and Orbital

Orbit and orbitals are different terms. Bohr proposed an orbit as a circular path around the nucleus in which the electrons moves. As per Heisenberg's uncertainty principle the precise description is not possible. So, its physical existence cannot be demonstrated experimentally.

An orbital is a three-dimensional region calculated from the allowed solutions of the Schrödinger's equation. It is quantum mechanical concept and refers to one electron wave function ψ in an atom. It is characterised by three quantum numbers (n, l and m_l) and its value depends upon the coordinates of the electron. ψ has no physical meaning but its square $|\psi|^2$ at any point in an atom gives the value of probability density at that point. Probability density ($|\psi|^2$) is the probability per unit volume and the electron in that volume. $|\psi|^2$ varies from one region to another region in the space but its value can be assumed to be constant within a small volume. The total probability of finding the electron in a given volume can be calculated by the sum of all products of $|\psi|^2$ and the corresponding volume elements. It is thus possible to get the probable distribution of an electron in an orbital.

1.10 SHAPES OF ORBITALS

The nature of the Schrödinger equation is such that the wave function ψ may be regarded as an amplitude function. This corresponds, on a three-dimensional scale, to the function which expresses the amplitude of vibration of a plucked string.

As it is the square of amplitude of the vibrating string which measures the intensity of the wave involved, so $|\psi|^2$ measures the probability of an electron existing at a point, meaning therefore, that the chance of finding an electron at that point is zero. A high value of $|\psi|^2$ at a point means that there is a high chance of finding an electron at the point.

The value of the wave function ψ, for an electron in an atom is dependent, in general, both on the radial distance, r, of the electron from the nucleus of the atom and on its angular direction away from the nucleus.

The nucleus, itself, however, cannot provide any sense of direction until a set of arbitrary chosen axes are superimposed. If cartesian axes are chosen as in Fig 1.25 (a) then a point in space, P, around the nucleus, N, can be defined in terms of x–, y– and z-coordinates. Such axes do not, however, have any absolute directional significance until an external magnetic field is applied. The axes then have a definite direction in relation to the direction of the magnetic field. Alternatively, polar coordinates in Fig 1.25 (b) can be used.

The mathematical relationship between ψ and r and the angular direction can be established accurately for a single electron in a hydrogen like atom. For more complicated atoms containing more than one electron, the corresponding relationships can only be established approximately because of the difficulty involved in solving the mathematical equations.

The differences between s–, p– and d-orbitals depend on the ways in which ψ and / or ψ^2 vary with r and with the angular direction, as explained in the following sections.

The probability of finding an electron in a given volume of space is represented by radial probability distribution curves. These curves indicate how the probability of finding an electron varies with the radial distance from the nucleus without any reference to its

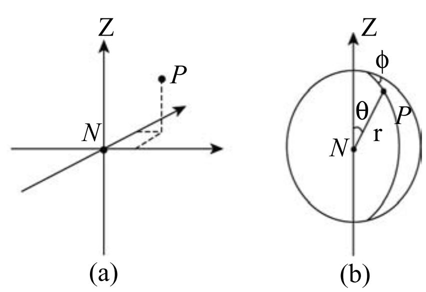


Fig 1.25 The position of a point P relative to point N can be expressed in terms of (a) Cartesian coordinates x, y, z or (b) Polar coordinates r, θ and ϕ.

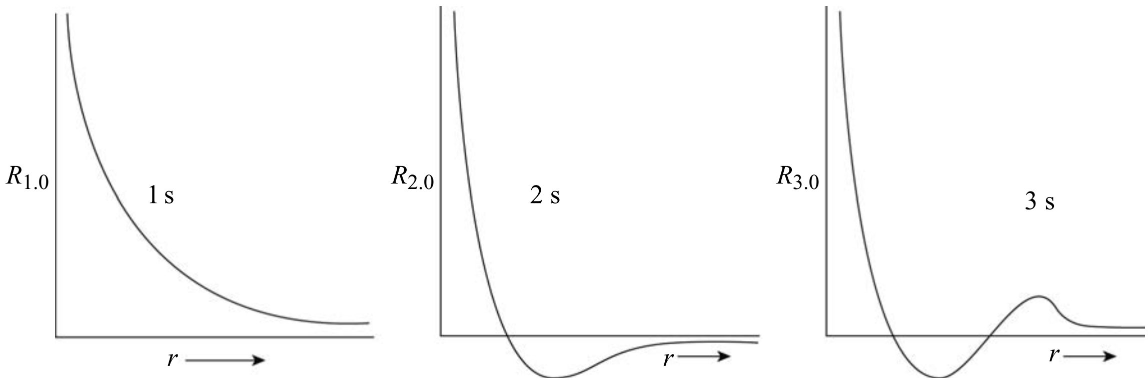


Fig 1.26 Plots of $|\psi_{n,l}|$ versus r for electron belonging to different orbitals

direction from the nucleus. The radial wave function is generally written as $\psi_{n,l}$. Let us now plot the function $\psi_{n,l}$ against r for electrons belonging to different orbitals and try to correlate them with the probability density around a point at a distance r from the nucleus. The plots are as shown in Fig 1.26.

From these plots, it is clear that $\psi_{n,1}$ cannot be related with probability density around any point at a distance r from the nucleus because

(i) $\psi_{n,l}$ is maximum at r = 0. If $\psi_{n,l}$ represents probability density *the electron will have maximum probability of occurring at the nucleus*. This cannot be true as the actual probability of finding an electron at the nucleus is zero.

(ii) $\psi_{n,l}$ has both positive as well as negative values (if probability curves for 2*s* and 3*s* electrons). However, *probability density cannot have a negative value*.

Objection No. (ii) can be removed by considering $|\psi_{n,l}|^2$ to be related to probability density instead of $\psi_{n,l}$. Now even if $\psi_{n,l}$ is negative at some space its square has to be positive. The plots of $\psi_{n,l}$ versus r for electrons belonging to different orbitals are as shown in Fig 1.28.

The curves do not exhibit negative value for the radial wave functions at any distance. But even $|\psi_{n,l}|^2$ cannot be related with probability density because $|\psi_{n,l}|^2$ is maximum at $r = 0$. This means that the probability of finding electron is maximum on the nucleus which again cannot be true (as already explained). Thus, neither $\psi_{n,l}$ nor $\psi^2_{n,l}$ can be directly related with probability of finding electron at a point which is at a distance r from the nucleus.

Let us consider the space around the nucleus (taken at the centre) to be divided into a large number of thin concentric spherical shells of thickness dr. Consider one of these shells with the inner radius r (Fig 1.27).

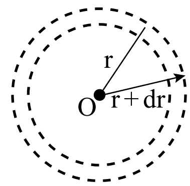


Fig 1.27 Division of space around the nucleus into small spherical shells of very small thickness

The volume of such a shell viz, dV, will be given by

$$dV = (4/3)\,\pi\,(r + dr)^3 - (4/3)\,\pi r^3 \quad (1.44)$$

$$= \left(\frac{4}{3}\right)\pi\,[r^3 + dr^3 + 3r^2dr + 3rd^2 - r^3] \quad (1.45)$$

Neglecting very small terms dr^2 and dr^3, we get

$$dV = 4\pi r^2\,dr \quad (1.46)$$

The probability of finding electron within the small radial shell of thickness dr around the nucleus, called **radial probability** will therefore be given by

$$\rho\,dV = \psi_{n,1} \times dV \quad (1.47)$$

$$= 4\pi r^2\,dr\,|\psi^2_{n,l}| \quad (1.48)$$

The radial probability distribution between $r = 0$ to $r = r$ will be given by the summation of all the probability distributions for concentric radial shells from $r = 0$ to $r = r$, i.e., by

$$\int_{r=0}^{r=r} 4\pi r^2\,dr\,\psi^2_{n,l} \quad (1.49)$$

In other words, the **radial probability distribution** of electron may be obtained by plotting the function $4\pi r^2\,|\psi^2_{n,l}|$ against r, its distance from the nucleus. Such graphs are radial probability distribution curves.

The radial probability distribution curves for 2*s*, 3*s*, 3*p* and 3*d* electrons are shown in Fig 1.29.

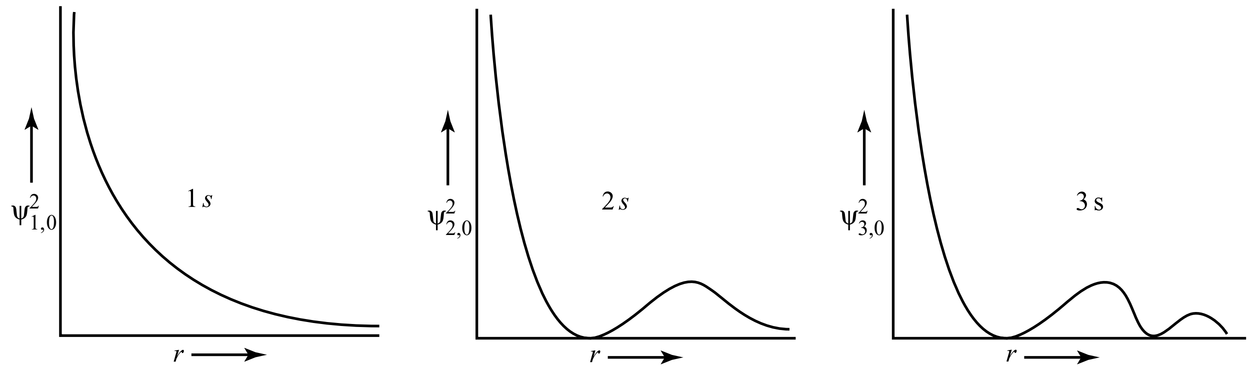


Fig 1.28 Plots of $\psi^2_{n,l}$ versus r for electrons belonging to different orbitals

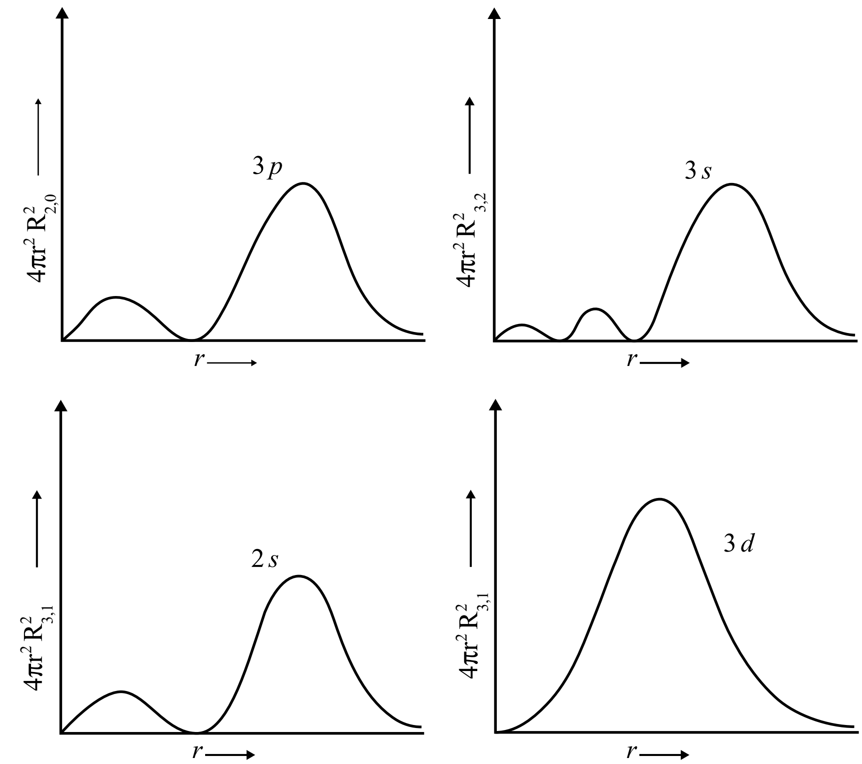


Fig 1.29 Radial probability distribution curves for 2*s*, 3*s*, 3*p* and 3*d* electrons.

It may be noted that for 1*s* orbital, the probability density is maximum at the nucleus and it decreases sharply as we move away from it. On the other hand, for 2*s* orbital the probability density first decreases sharply to zero and again starts increasing. After reaching a small maxima it decreases again and approaches zero as the value of *r* increases further. The region where this probability density function reduces to zero is called **nodal surfaces** or **nodes**. In general it has been found that *ns*-orbital has $(n-1)$ nodes, that is, number of nodes increases with increase of principal quantum number *n*. In other words, number of nodes for 2s orbital is one, two for 3s and so on.

These probability density variations can be visualized in terms of charge cloud diagrams (Fig 1.30). In these diagrams, the density of the dots in the region represents electron probability density in that region.

1.10.1 Boundary Surface Diagrams

***s*-orbitals:** The shapes of the orbitals in which constant probability density for different orbitals can be represented with **boundary surface diagram.** The value of probability density $|\psi|^2$ is constant in this boundary surface drawn in space. In principle, many such boundary surfaces

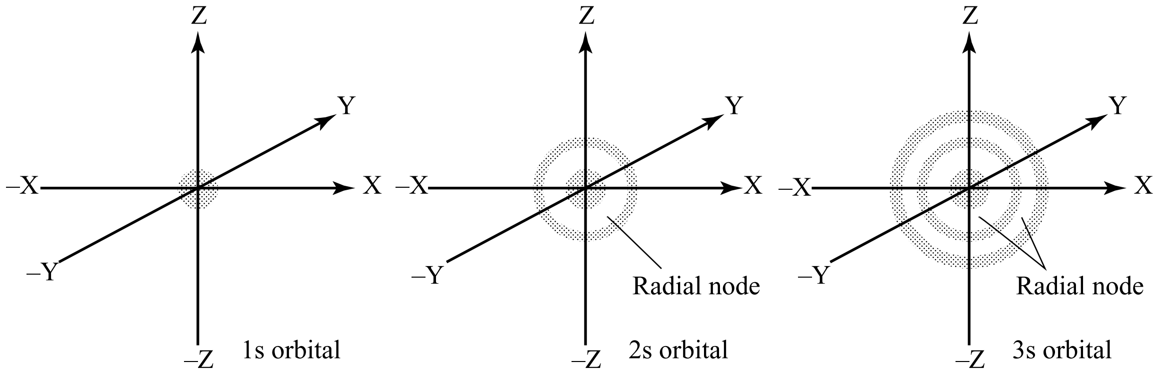


Fig 1.30 The shapes of various *s*-orbitals

may be possible. However, for a given orbital, only that boundary surface diagram of constant probability density is taken to be good representation of the shape of the orbital which encloses a region or volume in which the probability of finding the electron is very high, say 90 per cent, but cannot be 100 per cent because always there will be some value, however small it may be, at any finite distance from the nucleus. Boundary surface diagram for an '*s*' orbital is actually a sphere centred on the nucleus. In two dimensions, this sphere looks like a circle. The probability of finding the electron at any given distance is equal in all directions. It is also observed that the size of the *s*-orbital increases in *n*, that is, $4s > 3s > 2s > 1s$ and the electron is located further away from the nucleus as the principal quantum number increases.

***p*-orbitals:** The probability density diagrams for *p* orbitals are very different to those of *s*-orbitals. If you look back at quantum numbers you will find that we said that the magnetic quantum number can tell us the number of orbitals of a given type. The result is that for *s*-orbitals the magnetic quantum number, m_l, has only one value. This means that there is only one variety of *s*-orbitals that is the variety we have already met; the spherically symmetric ones. For *p*-orbitals there are three possible values of *m* (+1, 0, −1) — the three types of *p*-orbitals are called p_x, p_y and p_z. The boundary surface diagrams are known as *radial probability distribution curves* or *radial charge density curves* or simply as *radial distribution curves.* Such curves truly depict the variation of probability density of electronic charge with respect to *r* (distance of charge from the nucleus.)

Radial probability at a distance r is the probability of finding electron at all points in space which are at a distance r from the nucleus and the radial probability distribution is the graph of these probabilities as a function of r.

The radial probability distribution curves for 1*s* and 2*p* electrons are as shown in Fig 1.31.

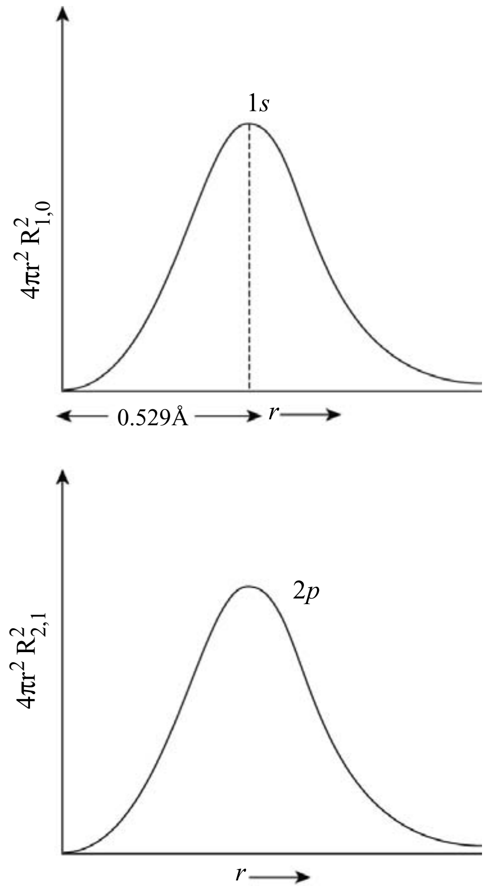


Fig 1.31 Radial probability distribution curves for 1*s* and 2*p* electrons

The radial probability function $4\pi r^2|\Psi_{n,l}|^2$ written for the sake of simplicity as $4\pi r^2|\psi_{n,l}|^2$ is evidently the product of two factors. While the probability factor ψ^2 decreases as *r* increases. This gives rise to curves of the type shown in Fig 1.31. At r = 0, though the factor $|\psi|^2$ is maximum,

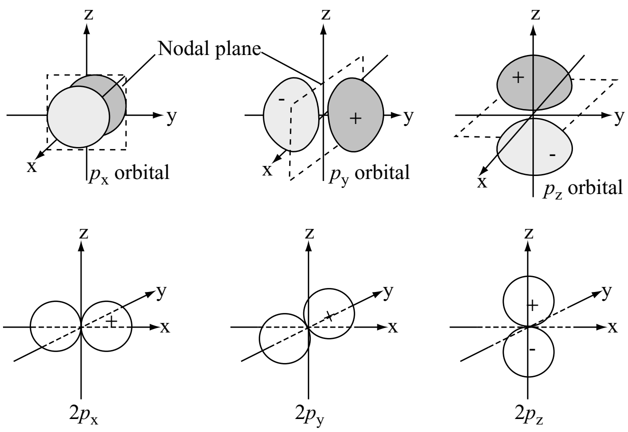


Fig 1.32 The shapes and boundary surface diagrams for $2p_x$, $2p_y$ and $2p_z$ orbitals

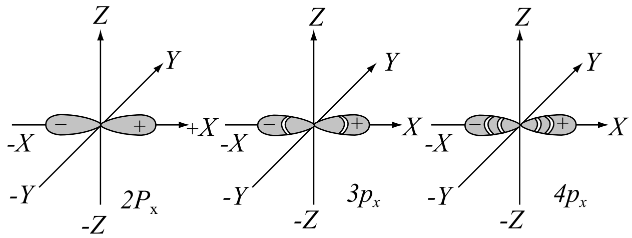


Fig 1.33 The exact shapes of $2p_x$, $3p_x$ and $4p_x$ orbitals showing the radial nodes

(Fig 1.28) the factor $4\pi r^2$ is zero. Similarly, when r is very large, the factor $4\pi r^2$ is no doubt very large but the probability factor $|\psi|^2$ is negligible so that the radial probability is exceedingly small as shown in Fig 1.32.

Here, unlike *s*-orbitals, the boundary surface diagrams are not spherical. Instead each *p*-orbital consists of two sections called **lobes** that are one either side of the plane that passes through the nucleus. The probability density function is zero on the plane where the two lobes touch each other. This plane is called **nodal plane.** The nodal plane for p_x orbital is *YZ*-plane, for p_y orbital, the nodal plane is *XZ* plane, and for p_z orbital nodal plane is *XY* plane. It should be understood, however, that there is no simple relation between the values of m_l (−1, 0 and +1) and the *x*, *y* and *z* directions. The size, shape and energy of the three orbitals are identical.

Like *s*-orbitals, *p*-orbitals increase in size and energy with increase in the principal quantum number and hence the order of the energy and size of various *p*-orbitals is $4p > 3p > 2p$. Further, like *s*-orbitals, the probability density functions for *p*-orbital also pass through value zero, besides at zero and infinite distance, as the distance increases. The number of nodes are given by $n - 2$, that is, number of radial nodes is 1 for 3*p*-orbital, two for 4*p*-orbital three and so on. The shapes of $2p_x$, $3p_x$ and $4p_x$ orbits with radial nodes are shown in Fig 1.33.

***d*-orbitals:** The orbitals having *l* value as 2 are known as *d*-orbitals. The minimum value of principal quantum number must be 3 to have *l* value 2 as the value of *l* cannot be greater than n − 1. There are five *d*-orbitals corresponding to m_l values (−2, −1, 0, +1 and +2) for $l = 2$. Their boundary surface diagrams are shown in Fig 1.34.

The five *d*-orbitals are designed as $d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}$ and d_{z^2} The orbital d_{z^2} has a dumbell shaped curve symmetric structure about the z-axis having a ring-like collar in *XY* plane. The dumbell shaped part of the curve has a positive geometric sign (because whatever be the sign of z positive or negative, its square is always positive), ring the collar in the *XY* plane has a negative geometric sign.

The orbital d_{xy} has a double dumbell shape. The quantity *XY* will be positive when both *x* and *y* are positive or when both are negative. However, *XY* will be negative when either of *x* or *y* is negative. Thus, the sign of the curve is positive in first and third quadrants while it is negative in second and fourth quadrants. The shape of d_{xz}, d_{yz} orbirals can be explained in a similar manner.

The orbital $d_{x^2-y^2}$ is also double dumbell shaped but its lobes lie on *X* and *Y* axes. The signs of the lobes on X-axis will always be positive (∵ whatever be the sign of *X*, X^2 will always be positive) whereas the sign of the lobes on *Y* axis will always be negative (∵ whatever be the sign of *Y*, $-Y^2$ will always be negative). The **exact shapes** of *d* orbitals are obtained by taking into consideration the total wave function. Accordingly $3d_{x^2-y^2}$ orbital would be similar in shape to $4d_{x^2-y^2}$ orbital or $5d_{x^2-y^2}$ orbital except for the fact that

1. A 5*d* orbital would have two, a 4*d* orbital have one and a 3*d* orbital have no radial node.
2. A 5*d* orbital would occupy more space than a 4*d* orbital and a 4*d* orbital would occupy more space than a 3*d* orbital as shown in Fig 1.35.

Every *d*-orbital also has two nodal planes passing through the origin and bisecting the *XY* plane containing *Z* axis. These are called angular nodes and are given by *l*.

For any orbital

(i) The number of **angular nodes** (nodal planes) is equal to *l* value.
(ii) The number of radial nodes (nodal surfaces or nodes) is equal to $(n–l–1)$.
(iii) The total number of radial nodes and angular nodes for any orbital is equal to $(n - l - 1) + l = n - 1$.

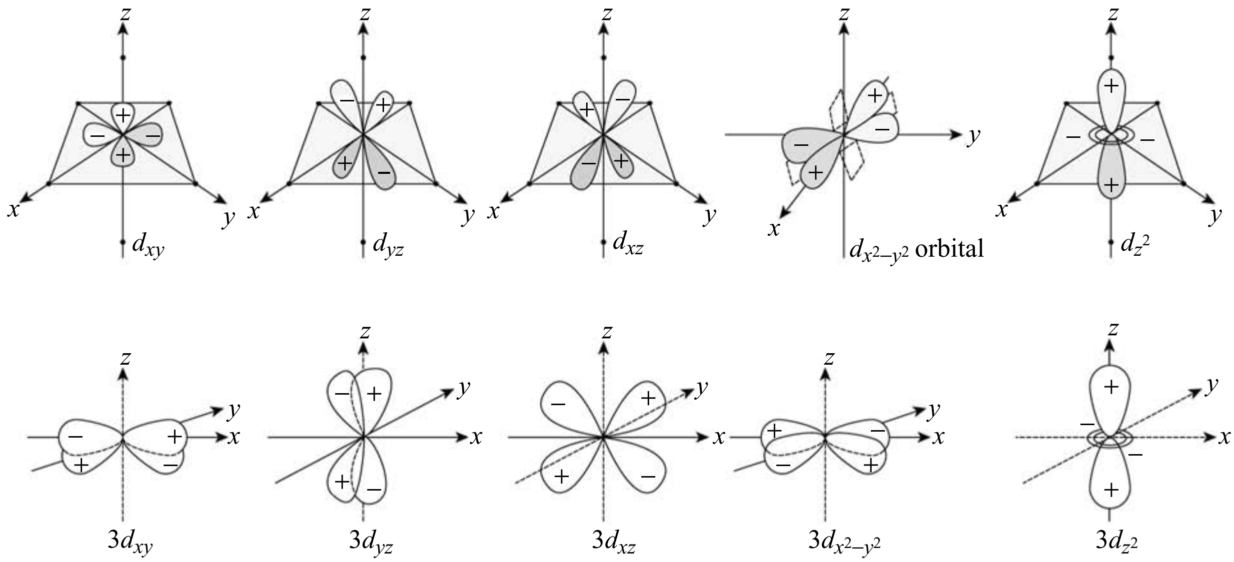


Fig 1.34 Shapes and boundary surface diagrams for $3d_{xy}, 3d_{yz}, 3d_{xz}, 3d_{x^2-y^2}$ and $3d_{z^2}$ orbitals. The nodal planes for $d_{x^2-y^2}$ are shown. Similarly there will be two nodal planes for each d_{xy}, d_{yz} and d_{xz} orbitals

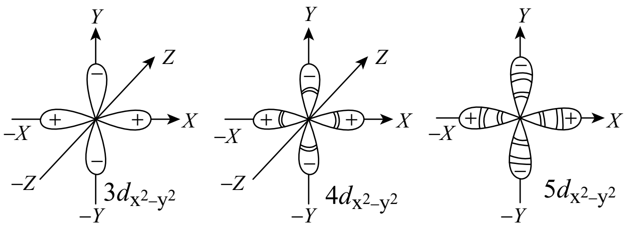


Fig 1.35 The $3d_{x^2-y^2}$, $4d_{x^2-y^2}$ and $5d_{x^2-y^2}$ orbitals in three dimensional space showing radial nodes

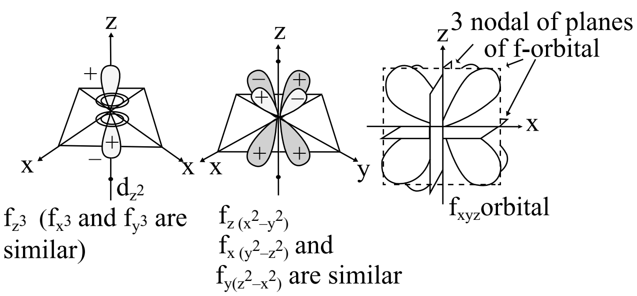


Fig 1.36 Shapes of *f*-orbitals

***f*-orbitals:** The orbitals having l value as 3 are known as *f*-orbitals. The minimum value of principal quantum number must be 4 to have l value 3 as the value of l cannot be greater than $n - 1$. There are seven *f*-orbitals corresponding to m_l values (−3, −2, −1, 0, +1, +2, +3) for $l = 3$. Their boundary surface diagrams are shown in Fig 1.36.

1.10.2 Energies of Orbitals

In the hydrogen atom the energy of an electron depends mainly on the principal quantum number. So, the energies of all the sub shells in a principal quantum number are equal. For example,

$$1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f$$

Through the shapes of different sub shells are different, the energy of the electron is the same when present in any one of the sub shells belonging to the same principal quantum number. The orbitals having same energy are called **degenerate orbitals**. In hydrogen atom when electron is present in 1s orbital, the hydrogen atom is considered in most stable condition and it is called the **ground state.** This is because the electron is nearer to the nucleus. But when electron is present in 2s, 2*p or* higher orbitals in the hydrogen atom, it is said to be in the excited state.

In multielectron atoms, unlike the hydrogen atom, the energy of an electron depends both on its principal quantum number (shell), and the azimuthal quantum number (sub shell). This means that different sub shells *viz.*, *s*, *p*, *d*, *f*, ... in a principal quantum number will have different energies, because of the mutual repulsion between the electrons. In hydrogen atom there is only attractive force between nucleus and electron but no repulsive forces. The stability of an electron in a multi-electron atom is because the total attractive interactions are more than the repulsive interactions experienced by every electron with other electrons. The attractive force

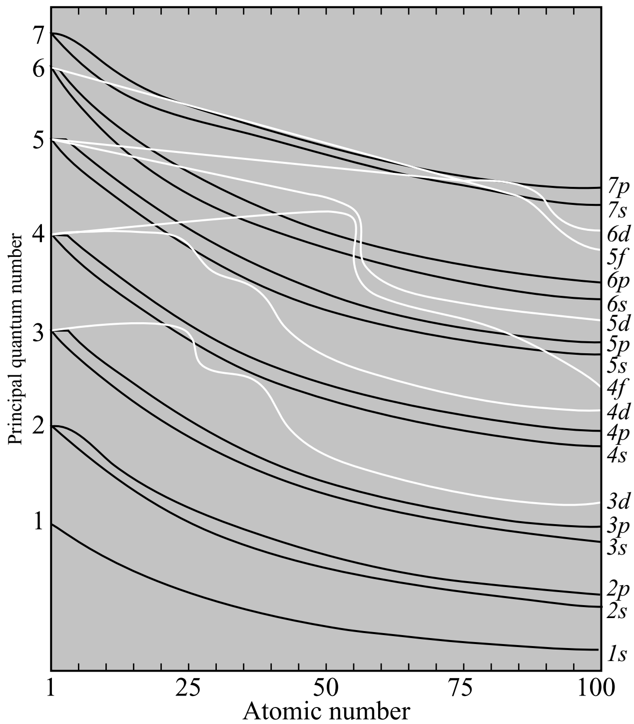


Fig 1.37 The relative energies of the atomic orbitals as a function of atomic number

of nucleus on the electrons increases with increase in the positive charge (Ze) of nucleus. The attractive power of nucleus on the electrons will be less due to screening effect of inner electrons. The net positive charge experienced by the electron from the nucleus is known as **effective nuclear charge** (Z^*).

The attractive and repulsive interactions depend upon the shell and shape of the orbital in which the electron is present. The spherical *s* orbital can shield the electrons from the nuclear attraction more effectively than the *p*-orbital with dumbell shape. This is because of the spherical nature of the *s*-orbital can shield nucleus from all directions equally but the electron in a *p*-orbital say p_x can shield the nucleus in only one direction, i.e., x-direction. Similarly, because of difference in their shape and more diffused character the *d*-orbitals have less shielding power than *p*-orbitals. Further because of their different shapes the electron in spherical orbital spends more time close to the nucleus when compared to *p*-orbital and the *p*-electron spends more time in the vicinity of nucleus when compared to *d*-electron. For this reason the effective nuclear charge experienced by different sub shell decreases with increase in the azimuthal quantum number (l). From this we can easily understand that *s*-electron is strongly attracted by the nucleus than the *p*-electron which in turn will be strongly attracted than the *d*-electron and so on. Thus, the different sub-shells belonging to same principal quantum number have different energies. However, in hydrogen atom, these have the same energy. In multielectron atoms, the dependence of energies of the sub shells on '*n*' and '*l*' are quite complicated but in simple way is that of combined value of *n* and *l*. *Lower the value n + l for a sub shell, lower is its energy. If the two sub-shells have the same n + l value, the orbital having lower n value have the lower energy.* The energies of sub-shells in the same shell decrease with increase in the atomic number (Z^*). For example, the energy of 2*s* orbital of hydrogen atom is greater than that of 2*s* orbital of lithium and that of lithium is greater than that of sodium and so on, i.e., E_{2s} (H) > E_{2s} (Li) > E_{2s} (Na) > E_{2s} (K).

1.11 FILLING OF ORBITALS

The distribution of electrons in various orbitals is known as **electronic configuration.** Having derived the energy level sequence, it is now a simple matter to write the electronic configurations of atoms by making use of Aufbau principle, Pauli's exclusion principle and the Hund's rule of maximum multiplicity.

Aufbau Principle

Aufbau is a German term meaning "building up". This principle is utilized to deduce the electronic structure of poly-electron atoms by building them up, by filling up of orbitals with electrons. The Aufbau principle states that *"in the ground state of the atoms, the orbitals are filled in order of their increasing energies."* In other words, in the ground state of atom, the orbital with a lower energy is filled up first before the filling of the orbital with a higher energy commences.

The increasing order of energy of various orbitals is

1*s*, 2*s*, 2*p*, 3*s*, 3*p*, 4*s*, 3d, 4*p*, 5*s*, 4*d*, 5*p*, 6s, 4*f*, 5*d*, 6*p*, 7*s*, ...

The above sequence of energy levels can be easily remembered with the help of the graphical representation shown in Fig 1.38.

Pauli's Exclusion Principle

The four quantum numbers define completely the position of electron in an atom. It is thus possible to identify an electron in an atom completely by stating the values of its four quantum numbers. **Wolfgang Pauli**, an Austrian scientist, put forward an ingenious principle which controls the number of electrons to be filled in various orbitals, and hence, it is named as the *exclusion principle*. It states that **no. two electrons in an atom can have the same set of four quantum numbers**.

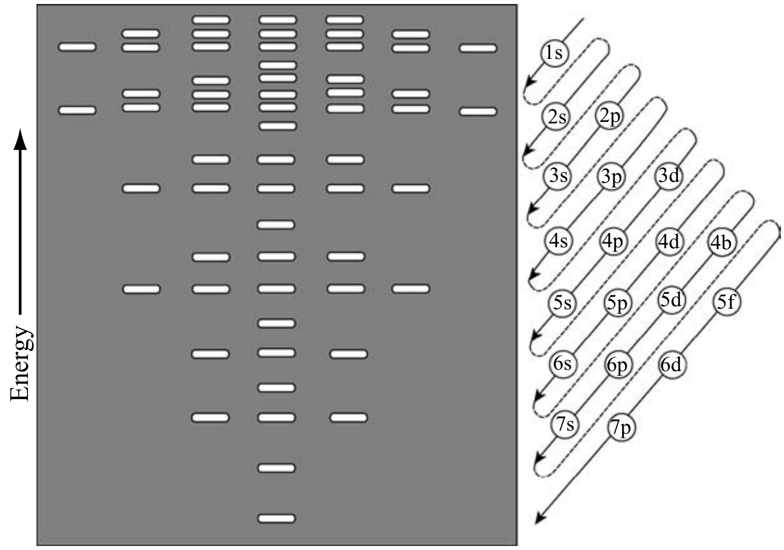


Fig 1.38 The order in which the atomic orbitals are used in building up the electron configuration of many electron atoms. The orbitals are used in sequence from the bottom in accordance with the Aufbau principle, Hund's rule and Pauli's exclusion principle

Thus, in the same atom any two electrons may have three quantum numbers identical but not the fourth which must be different. This means that the two electrons can have the same value of three quantum numbers, n, l and m_l but must have the opposite spin quantum numbers. This restricts that only two electrons may exist in the same orbital and these electrons must have opposite spins. Pauli's exclusion principle helps in calculating the number of electrons to be present in any sub shell. For example, the $1s$ sub-shell comprises one orbital and thus the maximum number of electrons present in $1s$ sub-shell can be two because of the following two possibilities.

$$n = 1, l = 0, m = 0, s = +\frac{1}{2}$$

$$n = 1, l = 0, m = 0, s = -\frac{1}{2}$$

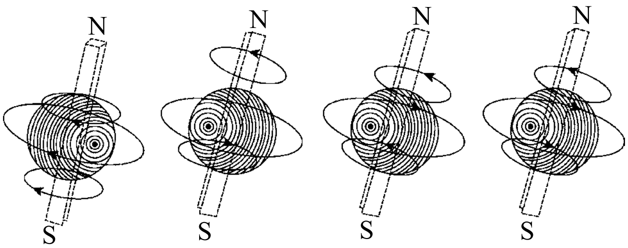
The p-sub-shell which is having 3 orbitals (p_x, p_y and p_z) can accommodate 6 electrons, the d-sub shell which having 5 orbitals can accommodate 10 electrons and the f-sub-shell having 7 orbitals can accommodate 14 electrons. This can be summed up as *the maximum number of electrons in the shell with principal quantum number n is equal to* $2n^2$.

Here a question arises? Why only two electrons can present in one orbital? It can be answered as follows. We may recall from our knowledge of elementary physics that the motion of an electric charge creates a magnetic field. In an orbital the electron spin may be clockwise or anti-clockwise as shown in Fig 1.39. This results in the formation of two magnets. Placing of two electrons together in the same orbital results in considerable repulsion due to same negative charge. By overcoming the repulsive forces and to keep the two electrons together in the same orbital some energy is required, called **pairing energy**. If the two electrons also have the same spin there will be magnetic repulsion also, causing the requirement of more pairing energy. But, if the two electrons are spinning in opposite directions and forming a pair in an orbital, mutually cancel their magnetic moments. Now, we can easily understand that for an electron if pairing energy is less than the energy of next higher energy level the electron will be paired in the orbital. But if the pairing energy is greater than the energy of next higher energy level the electron goes to the next higher energy level.

Hund's Rule of Maximum Multiplicity

The orbitals belonging to same sub-shell have same energy and are called degenerate orbitals. Hund's rule states that *the electron pairing in degenerate orbitals of a given sub shell will not take place unless all the available orbitals of a given shell contains one electron each.*

Since there are three p, five d and seven f orbitals, pairing of electrons will start in p, d and f orbitals with the entry



(a) Both electrons with parallel spins
(b) Both electrons with opposite spins

Fig 1.39 No two electrons in an atom can have the same set of four quantum numbers

of 4th, 6th and 8th electrons, respectively. It is also found that half-filled and fully filled degenerate set of orbitals acquire extra stability.

1.11.1 Electronic Configuration of Atoms

The filling of electrons in different orbitals in an atom are governed by the above three rules *viz*.,

(i) Aufbau principle,
(ii) Pauli's exclusion principle, and
(iii) Hund's rule of maximum multiplicity. The electronic configuration of different atoms can be represented in two ways:

(i) nl^x or $s^a p^b d^c$ ...... notation
(ii) Orbital diagram method

In the nl^x method, n represents the principle quantum number, l represents the azimuthal quantum number (sub-shell) and x represents the number of electron in that sub shell. In $s^a p^b d^c$ ... notation the sub shell is depicted with a superscript like a, b, c, ... etc, along with the principal quantum number before the respective sub shell. In the orbital diagram notation each orbital of the sub shell is represented by a box and the electron is represented by an arrow (↑) for positive spin and an arrow (↓) for negative spin. The advantage of the second notation over the first notation is that it represents all the four quantum numbers. Based on these notations, the electronic configurations of atoms of various elements of the Periodic Table in their ground state are as given in Table 1.5.

Hydrogen has only one electron (Z = 1). This electron will enter the lowest energy orbital which is 1s. Thus, the solitary electron of hydrogen atom, will occupy the 1s orbital. The electronic configuration of hydrogen atoms is, therefore, represented as $1s^1$.

The next element, **helium**, has two electrons ($Z = 2$). One of these electrons occupies the 1s orbital as in the case of the hydrogen atom. The second electron can also enter this orbital so as to fill it completely. The electronic configuration of helium is therefore, represented as $1s^2$. The two electrons occupying this orbital will have opposite spins as can be seen from the following orbital diagram:

	1s		1s
H	↑	He	↑↓

The next element is **lithium**. It has three electrons (Z = 3). The third electron of lithium (Li) is not allowed into the 1s orbital because of the Pauli exclusion principle. If third electron enters into 1s orbital its spin will

Table 1.5 Electronic configuration of first 10 elements

			1s	2s	2p		
1	Hydrogen	$1s^1$	↑				
2	Helium	$1s^2$	↑↓				
3	Lithium	$1s^2\,2s^1$	↑↓	↑			
4	Beryllium	$1s^2\,2s^2$	↑↓	↑↓			
5	Boron	$1s^2\,2s^2\,2p_x^1$	↑↓	↑↓	↑		
6	Carbon	$1s^2\,2s^2\,2p_x^1\,2p_y^1$	↑↓	↑↓	↑	↑	
7	Nitrogen	$1s^2\,2s^2\,2p_x^1\,2p_y^1\,2p_z^1$	↑↓	↑↓	↑	↑	↑
8	Oxygen	$1s^2\,2s^2\,2p_x^2\,2p_y^1\,2p_z^1$	↑↓	↑↓	↑↓	↑	↑
9	Fluorine	$1s^2\,2s^2\,2p_x^2\,2p_y^2\,2p_z^1$	↑↓	↑↓	↑↓	↑↓	↑
10	Neon	$1s^2\,2s^2\,2p_x^2\,2p_y^2\,2p_z^2$	↑↓	↑↓	↑↓	↑↓	↑↓

become similar with one of the two electrons due to which the charge repulsion and magnetic repulsion will become more. So more pairing energy will be required. This pairing energy will be more than the energy of the next higher energy level $2s$. So, the electron goes into $2s$. Hence, the electronic configuration of Li is $1s^2\ 2s^1$. The $2s$ orbital can accommodate one more electron. The configuration of beryllium (Be) atom is therefore $1s^2\ 2s^2$.

The next element **boron** ($Z = 5$) has five electrons. The $2s$ orbital has already been filled up. But the principal quantum number is now 2, there are three p-orbitals still available. The fifth electron, therefore, enters one of these p-orbitals. The electronic configuration of boron is $1s^2\ 2s^2\ 2p^1$.

The next five elements, namely carbon ($Z = 6$), nitrogen ($Z = 7$), Oxygen ($Z = 8$), fluorine ($Z = 9$) and neon ($Z = 10$) of this period get their p-orbitals successively filled up. Their electronic configurations are given in Table 1.5.

It may be noted that in carbon, the arrangement of electrons in p-orbitals is $2p_x^1, 2p_y^1$, and not the $2p_x^2$.

This is because to pair the electron in $2p_x$ pairing energy is required but to enter into the $2p_y$ orbital having energy equal (degenerate) to $2p_x$ pairing energy is not required. So, the electron goes into $2p_y$.

Similarly in nitrogen, the arrangement is $2p_x^1\ 2p_y^1\ 2p_z^1$ and not $2p_x^2\ 2p_y^1$. This is in accordance with Hund's rule of maximum multiplicity. In oxygen the last electron has two options. One to pair in one of three p-orbitals or to go the next higher energy level 3s. Since the pairing energy is less than the energy required to go to $3s$ orbital, pairing of electrons starts in oxygen and continues till all the three $2p$ orbitals are filled up.

The principal quantum 3 (M-shell) begins with sodium ($Z = 11$). the eleventh electron will evidently, enter, into the $3s$ orbitals, the electronic configuration being $1s^2\ 2s^2\ 2p^6\ 3s^1$.

The next seven elements namely magnesium ($Z = 12$) aluminum ($Z = 13$), silicon ($Z = 14$), phosphorus ($Z = 15$), sulphur ($Z = 16$), chlorine ($Z = 17$), and argon ($Z = 18$) follow the sequence described above.

Consider now the electronic configuration of the next element which is potassium ($Z = 19$). The additional electron (i.e., nineteenth electron) instead of entering $3d$ orbitals, all of which are lying unoccupied so far, goes into the $4s$ orbital. This is because $4s$ orbital is at lower energy level than $3d$ orbitals. The electronic configuration of potassium may be represented as $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$. The additional electron in calcium ($Z = 20$) follows the same course and goes into the same, i.e., $4s$ orbitals, to fill it up completely. So, the electronic configuration of calcium is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$.

The electrons in the completely filled shells are known as **core electrons** and the electrons that are added to the electronic shell with the highest principal quantum number are called the **valence electrons**. For example, the electrons in Ne are the core electrons and the electrons in the highest principal quantum number from Na to Ar are the valence electrons.

The next ten elements ($Z = 21$ to 30) which follow calcium are scandium (Sc), titanium (Ti), vanadium (V), chromium (Cr), manganese (Mn), iron (Fe), cobalt (Co), nickel (Ni), copper (Cu) and zinc (Zn). In these elements, addition of electrons takes place in the inner $3d$ orbitals, while the outer $4s$ orbitals remains fully occupied. The electronic configuration of scandium ($Z = 21$), the first element in this series, is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1\ 4s^2$ while that ($Z = 30$) of the last element of the series is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2$. Thus, while filling of $3d$ orbitals begins with scandium, it ends up with zinc.

Anomalous configurations of chromium and copper: It may be noted that the configurations of chromium ($Z = 24$) and copper ($Z = 29$) do not follow the general trend. The electronic configurations of Cr and Cu are expected to be as follows:

$$\text{Cr } 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^4\ 4s^2,$$
$$\text{and} \quad \text{Cu } 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9\ 4s^2$$

actually their configuration are

$$Cr\ \ 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$$
$$\text{and} \quad Cu\ 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$$

These anamolies are attributed to the fact that exactly half-filled and completely filled orbitals (i.e., p^3, p^6, d^5, d^{10}, f^7 and f^{14}) are more stable because they possess the lower energy. The cause of this extra stability has been discussed in detail in the subsequent sections. Thus, to acquire an increased stability, one of the electrons from $4s$ orbital goes into the nearby $3d$ orbital so that $3d$ orbitals become exactly half-filled in the case of chromium and completely filled up in the case of copper.

After completely filling the $3d$ orbitals, the filling up of $4p$ orbitals starts at gallium (Ga) and is complete at krypton (Kr). In the next 18 elements from rubidium (Rb) to xenon (Xe) the pattern of filling the $5s$, $4d$ and $5p$ orbitals are similar to that $4s$, $3d$ and $4p$ orbitals as discussed above. Then comes the turn of the $6s$ orbital. In caesium (Cs) and barium (Ba), this orbital contains one and two electrons respectively. After the filling up of $6s$ sub shell with barium ($Z = 56$) the next electron in lanthanum ($Z = 57$) goes to $5d$ sub shell and not the $4f$ sub shell. The subsequent electrons in the succeeding elements of the lanthanide series (from Ce to Lu $Z = 58$ to 71) however enters the $4f$ sub shell by following the normal sequence of energies. The filling up of $5d$ sub shell recommences only after the $4f$ sub shell is completely filled up with Lu ($Z = 71$).

For the sake of simplicity, we do not always write the complete electronic configuration. Instead, the configuration of the noble gas preceding the valence shell is not given; it is represented by its symbol in square bracket. For example, for Na ($Z = 11$), we may write [Ne] $3s^1$, and for Ca ($Z = 20$), we may write [Ar] $4s^2$. Thus, the configurations of La, Ce and Pr would be as follow:

La ($Z = 57$): [Xe] $4f^0\ 5d^1\ 6s^2$
Ce ($Z = 58$): [Xe] $4f^1\ 5d^1\ 6s^2$
Pr ($Z = 59$): [Xe] $4f^2\ 5d^1\ 6s^2$

Similarly, after filling up of $7s$ sub-shell with radium ($Z = 88$) the next electron in actinium ($Z = 89$) goes to $6d$ sub shell and not to $5f$ sub-shell. The subsequent electrons in the succeeding elements of the actinides (from Th to Lr $Z = 90$ to 103), however, enter the $5f$ sub-shell. Thus, the configurations of Ac, Th and Pa would be as follows:

Ac ($Z = 89$): [Rn] $6f^1\ 7s^2$
Th ($Z = 90$): [Rn] $5f^1\ 6d^1\ 7s^2$
Pa ($Z = 91$): [Rn] $5f^2\ 6d^1\ 7s^2$

The electronic configurations of the known elements (as determined by spectroscopic methods are given in Table 1.6).

Table 1.6 Electronic configuration of the elements

Element	Z	1s	2s	2p	3s	3p	3d	4s	4p	4d	4f	5s	5p	5d	5f	6s	6p	6d	7s
H	1	1																	
He	2	2																	
Li	3	2	1																
Be	4	2	2																
B	5	2	2	1															
C	6	2	2	2															
N	7	2	2	3															
O	8	2	2	4															
F	9	2	2	5															
Ne	10	2	2	6															
Na	11	2	2	6	1														
Mg	12	2	2	6	2														
Al	13	2	2	6	2	1													
Si	14	2	2	6	2	2													
P	15	2	2	6	2	3													
S	16	2	2	6	2	4													
Cl	17	2	2	6	2	5													
Ar	18	2	2	6	2	6													
K	19	2	2	6	2	6		1											
Ca	20	2	2	6	2	6		2											
Sc	21	2	2	6	2	6	1	2											
Ti	22	2	2	6	2	6	2	2											
V	23	2	2	6	2	6	3	2											
Cr*	24	2	2	6	2	6	5	1											
Mn	25	2	2	6	2	6	5	2											
Fe	26	2	2	6	2	6	6	2											
Co	27	2	2	6	2	6	7	2											
Ni	28	2	2	6	2	6	8	2											
Cu*	29	2	2	6	2	6	10	1											
Zn	30	2	2	6	2	6	10	2											
Ga	31	2	2	6	2	6	10	2	1										
Ge	32	2	2	6	2	6	10	2	1										
As	33	2	2	6	2	6	10	2	3										

Table 1.6 (Cont.)

Element	Z	1s	2s	2p	3s	3p	3d	4s	4p	4d	4f	5s	5p	5d	5f	6s	6p	6d	7s
Se	34	2	2	6	2	6	10	2	4										
Br	35	2	2	6	2	6	10	2	5										
Kr	36	2	2	6	2	6	10	2	6										
Rb	37	2	2	6	2	6	10	2	6			1							
Sr	38	2	2	6	2	6	10	2	6			2							
Y	39	2	2	6	2	6	10	2	6	1		2							
Zr	40	2	2	6	2	6	10	2	6	2		2							
Nb*	41	2	2	6	2	6	10	2	6	4		1							
Mo*	42	2	2	6	2	6	10	2	6	5		1							
Tc	43	2	2	6	2	6	10	2	6	5		2							
Ru*	44	2	2	6	2	6	10	2	6	7		1							
Rh*	45	2	2	6	2	6	10	2	6	8		1							
Pd*	46	2	2	6	2	6	10	2	6	10									
Ag*	47	2	2	6	2	6	10	2	6	10		1							
Cd*	48	2	2	6	2	6	10	2	6	10		2							
In	49	2	2	6	2	6	10	2	6	10		2	1						
Sn	50	2	2	6	2	6	10	2	6	10		2	2						
Sb	51	2	2	6	2	6	10	2	6	10		2	3						
Te	52	2	2	6	2	6	10	2	6	10		2	4						
I	53	2	2	6	2	6	10	2	6	10		2	5						
Xe	54	2	2	6	2	6	10	2	6	10		2	6						
Cs	55	2	2	6	2	6	10	2	6	10		2	6		1				
Ba	56	2	2	6	2	6	10	2	6	10		2	6		2				
La*	57	2	2	6	2	6	10	2	6	10		2	6	1	2				
Ce*	58	2	2	6	2	6	10	2	6	10	2	2	6		2				
Pr	59	2	2	6	2	6	10	2	6	10	3	2	6		2				
Nd	60	2	2	6	2	6	10	2	6	10	4	2	6		2				
Pm	61	2	2	6	2	6	10	2	6	10	5	2	6		2				
Sm	62	2	2	6	2	6	10	2	6	10	6	2	6		2				
Eu	63	2	2	6	2	6	10	2	6	10	7	2	6		2				
Gd*	64	2	2	6	2	6	10	2	6	10	7	2	6	1	2				
Tb	65	2	2	6	2	6	10	2	6	10	9	2	6		2				
Dy	66	2	2	6	2	6	10	2	6	10	10	2	6		2				
Ho	67	2	2	6	2	6	10	2	6	10	11	2	6		2				
Er	68	2	2	6	2	6	10	2	6	10	12	2	6		2				
Tm	69	2	2	6	2	6	10	2	6	10	13	2	6		2				
Yb	70	2	2	6	2	6	10	2	6	10	14	2	6		2				
Lu	71	2	2	6	2	6	10	2	6	10	14	2	6	1	2				
Hf	72	2	2	6	2	6	10	2	6	10	14	2	6	2	2				
Ta	73	2	2	6	2	6	10	2	6	10	14	2	6	3	2				
W	74	2	2	6	2	6	10	2	6	10	14	2	6	4	2				
Re	75	2	2	6	2	6	10	2	6	10	14	2	6	5	2				
Os	76	2	2	6	2	6	10	2	6	10	14	2	6	6	2				
Ir	77	2	2	6	2	6	10	2	6	10	14	2	6	7	2				
Pt*	78	2	2	6	2	6	10	2	6	10	14	2	6	9	2				
Au	79	2	2	6	2	6	10	2	6	10	14	2	6	10	2				
Hg	80	2	2	6	2	6	10	2	6	10	14	2	6	10	2				
Tl	81	2	2	6	2	6	10	2	6	10	14	2	6	10	2	1			

Element	Z	1s	2s	2p	3s	3p	3d	4s	4p	4d	4f	5s	5p	5d	5f	6s	6p	6d	7s
Pb	82	2	2	6	2	6	10	2	6	10	14	2	6	10	2	2			
Bl	83	2	2	6	2	6	10	2	6	10	14	2	6	10		2	3		
Po	84	2	2	6	2	6	10	2	6	10	14	2	6	10		2	4		
At	85	2	2	6	2	6	10	2	6	10	14	2	6	10		2	5		
Rn	86	2	2	6	2	6	10	2	6	10	14	2	6	10		2	6		
Fr	87	2	2	6	2	6	10	2	6	10	14	2	6	10		2	6		1
Ra	88	2	2	6	2	6	10	2	6	10	14	2	6	10		2	6		2
Ac	89	2	2	6	2	6	10	2	6	10	14	2	6	10		2	6		2
Th	90	2	2	6	2	6	10	2	6	10	14	2	6	10		2	6		2
Pa	91	2	2	6	2	6	10	2	6	10	14	2	6	10	2	2	6		2
U	92	2	2	6	2	6	10	2	6	10	14	2	6	10	3	2	6		2
Np	93	2	2	6	2	6	10	2	6	10	14	2	6	10	4	2	6		2
Pu	94	2	2	6	2	6	10	2	6	10	14	2	6	10	6	2	6		2
Am	95	2	2	6	2	6	10	2	6	10	14	2	6	10	7	2	6		2
Cm	96	2	2	6	2	6	10	2	6	10	14	2	6	10	7	2	6		2
Bk	97	2	2	6	2	6	10	2	6	10	14	2	6	10	8	2	6		2
Cf	98	2	2	6	2	6	10	2	6	10	14	2	6	10	10	2	6		2
Es	99	2	2	6	2	6	10	2	6	10	14	2	6	10	11	2	6		2
Fm	100	2	2	6	2	6	10	2	6	10	14	2	6	10	12	2	6		2
Md	101	2	2	6	2	6	10	2	6	10	14	2	6	10	13	2	6		2
No	102	2	2	6	2	6	10	2	6	10	14	2	6	10	14	2	6	1	2
Lr	103	2	2	6	2	6	10	2	6	10	14	2	6	10	14	2	6	2	2
Rf	104	2	2	6	2	6	10	2	6	10	14	2	6	10	10	2	6	3	2
Db	105	2	2	6	2	6	10	2	6	10	14	2	6	10	11	2	6	4	2
Sg	106	2	2	6	2	6	10	2	6	10	14	2	6	10	12	2	6	5	2
Bh	107	2	2	6	2	6	10	2	6	10	14	2	6	10	13	2	6	6	2
Hs	108	2	2	6	2	6	10	2	6	10	14	2	6	10	14	2	6	7	2
Mt	109	2	2	6	2	6	10	2	6	10	14	2	6	10	14	2	6	8	2
Ds	110	2	2	6	2	6	10	2	6	10	14	2	6	10	14	2	6	9	2
Rg**	111	2	2	6	2	6	10	2	6	10	14	2	6	10	14	2	6	10	1

* Elements with exceptional electronic configurations.

* * Elements with atomic number 112 and above have been reported but not yet fully authenticated and named.

1.11.2 Relative Stabilities of Electronic Configurations

Before we discuss the relative stabilities of various electronic configurations, we should clearly understand the difference between electronic configuration and electronic arrangement. For this, let us consider the nitrogen atom. Its outer electronic configuration is $2s^2\ 2p^3$. From this we get information that three electrons are present in a set of three $2p$ orbitals. This configuration gives us no information about the manner in which these three electrons are arranged in three $2p$ orbitals.

It can be easily shown through permutations and combinations that there are as many as 20 different ways of placing these three electrons in the three $2p$ orbitals. In other words, the electronic configuration $2p^3$ can have 20 different electronic arrangements as shown in Table 1.7 These different electronic arrangements would have different energies inspite of the fact that they belong to the same electronic configuration. (Some of the electronic arrangements may have the same energy due to similar inter electronic repulsions). Therefore, when we talk of the stability of the electronic configuration, we are *ipso* facto concerned with the stability of the electronic arrangement of that configuration. Of all the electronic arrangements permitted by a given configuration the most stable configuration can be predicted based on the following two factors:

1. Symmetrical distribution of charge
2. Hund's rule of maximum multiplicity

To explain these two factors we shall consider here the different electronic arrangements of p^3 configuration as shown in Table 1.7.

Table 1.7 Arrangements of three electrons in three *p*-orbitals

S. no.	p_x	p_y	p_z
1	↑	↑	↑
2	↑	↑	↓
3	↑	↓	↑
4	↓	↑	↑
5	↓	↓	↓
6	↓	↓	↑
7	↓	↑	↓
8	↑	↓	↓
9	↑↓	↑	
10	↑↓		↑
11		↑↓	↑
12		↑	↑↓
13	↑		↑↓
14	↑	↑↓	
15	↑↓	↓	
16	↑↓		↓
17		↑↓	↓
18		↓	↑↓
19	↓		↑↓
20	↓	↑↓	

Symmetrical Distribution of Charge

In order to understand the effect of the symmetrical distribution of charge in orbitals on the stability of electronic arrangements, let us recall the shapes of various sets of orbitals one by one. The *s*-orbital is spherical in shape which implies that the electronic charge would be distributed uniformly in all directions whether there is one or two electrons present in the orbital.

Of the three *p*-orbitals, the p_x is symmetrical along the X-axis while the orbitals p_y and p_z are symmetrical along the Y- and Z-axes respectively. If one electron is present in p_x -orbital, i.e., for p_x^1 configuration, the electronic charge would be mainly concentrated in the X-direction only. Likewise the electronic charge would be mainly concentrated in the Y-direction of p_y^1 configuration. Similarly, in $p_x^1\,p_y^1$ configuration, the electronic charge would be mainly concentrated in the *xy* plane. In a similar manner, the electronic charge would be concentrated more along the X-direction in the configuration $p_x^2\,p_y^1\,p_z^1$ and more in the plane xy in the configuration $p_x^2\,p_y^2\,p_z^1$. It is evident that in all the *p* configurations mentioned above, the charge is not evenly spread along all the directions. In other words, the distribution of charge is **non-uniform** or **unsymmetrical.** On the other hand, the distribution of charge in the configurations $p_x^1\,p_y^1\,p_z^1$ and $p_x^2\,p_y^2\,p_z^2$ would be **uniform** or **symmetrical** in all directions.

Table 1.8 Spin multiplicity

No. of unpaired electrons	S	Multiplicity	Name of state
0	0	1	Singlet
1	$\frac{1}{2}$	2	Doublet
2	1	3	Triplet
3	$\frac{3}{2}$	4	Quartet
4	2	5	Quintet

Configurations with even or uniform or symmetrical distribution of charge in all directions would evidently be associated with lower energy and hence higher stability than the configurations with unsymmetrical distribution of electronic charge.

To sum up: *A symmetrical distribution of electronic charge leads to a decrease in energy and hence an increase in the stability of the system.*

It can be easily seen that among 20 electronic arrangements of p^3 configurations of Table 1.7 1 to 8 arrangements would have a uniform or symmetrical distribution of charge in space. Therefore, these configurations would be more stable than all other p^3 configurations.

Now a question arises that among the 1 to 8 arrangements of p^3 configurations which will be more stable ? This can be explained based on Hund's rule of maximum multiplicity. Then what is spin multiplicity? The spin multiplicity is the value of 2S + 1 where S is the resultant spin quantum number. The relation between the number of unpaired electrons, the resultant spin quantum number S and multiplicity is given in Table 1.8.

Maximum spin multiplicity is possible only when all the degenerate orbitals occupy with electrons of parallel spin. In the Table 1.7 the 1 and 5 electronic arrangements have maximum number of unpaired electrons with parallel spin and thus have maximum multiplicity. So, these two (1 and 5) electronic arrangements are more stable than the other electronic arrangements. This is because of lowering of energy in those arrangements. Then again another question arises as to why such arrangements have less energy? The answer for this question can be explained involving the contribution of exchange energy as discussed below.

By advanced mathematical treatment, it can be shown that if on exchanging the position in space of two electrons with parallel spins, there is no change in the electronic arrangement, it would lead to decrease in energy. The pair of electrons is called the **exchange pair**. The larger the number of exchange pairs of electrons, the greater would be the decrease of energy. The energy decrease per exchange pair of electrons is termed as **Columbic exchange energy** and is represented with E. it would evidently, carry a negative sign.

In the electronic arrangements 1 and 5 of p^3 configurations in Table 1.7, three exchange pairs are possible (1, 2; 1, 3; and 2, 3). So the lowering of energy due to columbic exchange energy will be $-3E$. From 1-8 electronic arrangements of p^3 configurations (Table 1.7) except 1and 5 there is only one exchange pair since one electron is with anti-clockwise spin. So, the decrease in energy due to columbic exchange energy is only $-E$. In the remaining electronic arrangements 9 to 20 (Table 1.7) two electrons are paired in one of the p orbitals which requires pairing energy represented by P. Obviously, the pairing energy is positive.

Since the pairing energy has positive sign, while the Columbic exchange energy have negative sign, pairing energy will destabilize the atom while exchange energy will give stability to the atom. Thus, the overall stability of a system would be decided by the aggregate of exchange energy (E) and pairing energy (P). In the case of electronic arrangements 9 to 20 (Table 1.7) the net energy change will be $-E + P$, and is the least stable. The electronic arrangements 1 and 5 (Table 1.7) are most stable because of symmetrical distribution of charge and more number of columbic exchange energies.

Similarly, the stability of anomalous electronic configuration of chromium can be explained.

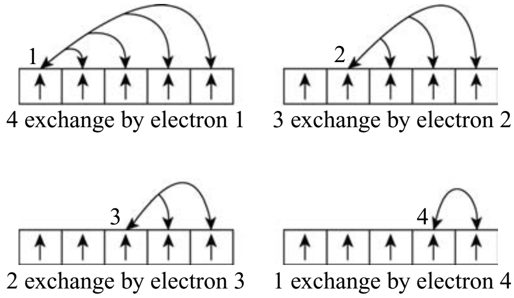


Fig 1.40 Possible exchange pairs for a d^5 configuration

	1s		2s			4s
Normal outer electronic configuration of chromium	↑	↑	↑	↑		↑↓
Anomalous electronic configuration of chromium	↑	↑	↑	↑	↑	↑

The $3d^4\,4s^2$ configuration has only 10 exchange pairs ($^5C_2$ pairs) whereas the $3d^5\,4s^1$ has 15 exchange pairs ($^6C_2$ pairs) because the $3d^4\,4s^2$ has 5 electrons with parallel spins while $3d^5\,4s^1$ arrangement has 6 electrons with parallel spins. Hence, in $3d^5\,4s^1$ arrangement the decrease in energy is $-15E$ while in $3d^4\,4s^2$, is $-10E$. Further, in $3d^4\,4s^2$ configuration, there is one pair of electrons in $4s$ orbital which requires the pairing energy. So, the net decrease in energy of $3d^4\,4s^2$ configuration is $-10E + P$. Also, the $3d^4\,4s^2$ configuration has no symmetrical distribution of charge. So, the $3d^5\,4s^1$ configuration for chromium is more stable than $3d^4\,4s^2$ configuration.

It can thus be concluded that electronic arrangements with exactly half-filled or completely filled degenerate orbitals would be more stable than any other electronic arrangement.

KEY POINTS

Sub-Atomic Particles

- The fundamental particles of every atom are
  (i) electron (ii) proton, and (iii) neutron
- The only element whose atoms do not contain a fundamental particle (neutron) is hydrogen.

Electron

- Discovered by **J.J. Thomson** in 1897.
- Experiment in a discharge tube: cathode rays are formed in a discharge tube voltage (10, 000 V) and low pressure.
- Electron is the lightest fundamental particle.
- Charge on electron is equal to -4.8×10^{-10} esu or -1.602×10^{-19} coulombs.
- **Mass of one electron** is equal to 1/1837 of hydrogen atom or 9.109×10^{-28} g or 9.1×10^{-31} kg.
- Mass of one mole of electrons is equal to 0.55 mg.
- Energy of electron is not influenced by any magnetic field.
- **Specific charge** decreases with increase in velocity of electron due to increase in mass.
- e/m ratio is constant for the cathode ray.

Proton

- Discovered by **Goldstein** in canal rays experiment.
- Properties of the canal rays are characteristics of the gas taken in the discharge tube.
- The canal rays contain **positively charged** ions of gas present in the discharge tube.
- The smallest +ve charged particle is proton.

- The **mass of proton** 1.008 u or 1.672×10^{-24} g or 1.672×10^{-27} kg.
- The **charge of proton** is equal to 4.8×10^{-10} esu or 1.602×10^{-19} coulombs.
- Proton is H+ ion.

Neutron

- Neutron was discovered by **Chadwick.**
- Neutron is emitted when lighter elements like beryllium and boron are bombarded with α-particles.
- Neutron has no charge.
- Mass of neutron is 1.00866 u or 1.675×10^{-24} g or 1.675×10^{-27} kg.
- Neutron is the heaviest fundamental particle of the atom.

Rutherford's Atomic Model

- Rutherford's experiment proved the presence of **nucleus**.
- The total mass of the atom is concentrated at the centre of the spherical atom known as **nucleus**.
- The positive charge of the nucleus is counter-balanced by the negatively charged electrons revolving round the nucleus.
- Rutherford model is also known as the **Planetary model**.
- As per the laws of electrodynamics a moving electron (charged particle) should continuously loose energy by emission of radiation. Consequently it should fall into the nucleus.
- If the electron loses energy continuously, the atomic spectrum should be band spectrum but the atomic spectra are line spectra.

Atomic Number

- The number of electrons or protons in an atom is equal to its **atomic number.**
- **Moseley** discovered the relation between the frequencies of the characteristic X-rays (v) of an element and its atomic number (z).

$$\sqrt{v} = a(z-b) \quad a \text{ and } b \text{ are constants.}$$

- Sum of the number of protons and neutrons in an atom is called its **mass number.**
- Mass number is the corrected atomic weight to a whole number.
- Mass number $A = Z + N$ where Z is atomic number and N is the number of neutrons.
- Atoms of an element which differ in their mass but have the same atomic number are called **isotopes**.
- Isotopes have same number of protons but differ in the number of neutrons present in them.
- Different atoms having same mass number but with different atomic numbers are called **isobars**.
- Atoms of different elements containing the same number of neutrons are called **isotones.**
- Isotones differ in both atomic numbers and mass number but difference in the atomic number and mass number is the same.

Developments Leading to the Bohr's Model of Atom, Nature of Light

- All radiant energy propagates in the form of waves.
- The radiant energy is in the form of electromagnetic waves.
- The radiations are associated with electric and magnetic field perpendicular to one another.
- On the propagation of an electromagnetic radiation there is only propagation of wave but not that of the medium.
- **Wavelength** is the distance between two successive crests or between two successive troughs of waves.
  (i) It is denoted by λ.
  (ii) It is measured in Å (Angstroms) or nm (nanometers) $1\text{Å} = 10^{-8}$ cm or $1\ nm = 10^{-9}$ m.
- **Frequency** is the number of waves per second passing at a given point. It is denoted by v. Units of frequency are Hertz or cycles s^{-1}.
- **Velocity of light** is the distance travelled by one wave in one second.
- Light or all electromagnetic radiations travel in vacuum or air with the same velocity.
- Velocity of light $c = 3 \times 10^8$ m s^{-1}.
- Wavelength is inversely proportional to the frequency of the wave.
- Velocity of light = Frequency × Wavelength or $c = \nu\lambda$.
- Wave number is the number of waves spread in one centimeter, denoted by $\overline{v}$.
- Wave number $\overline{v}$ is the reciprocal of wavelength.

$$\overline{v} = \frac{1}{\lambda}, \text{ but } v = \frac{c}{\lambda}, \ \therefore\ v = c\overline{v}$$

- Units of wave number are cm^{-1} or m^{-1}.

Quantum Theory of Radiation

- **Corpuscular theory** of light was proposed by **Newton**.
- According to corpuscular theory of Newton, light is propagated in the form of small invisible particles.
- **Quantum Theory** was proposed by Max Planck and was extended by Einstein.
- A hot vibrating body does not emit or absorb energy continuously but emits or absorbs energy discontinuously in the form of small energy packets called **quanta.**
- The energy of radiation (E) is proportional to its frequency (v).

$$E \propto v \qquad \text{(or)} \qquad E = hv$$

h is a constant known as **Planck's constant**. Its value is 6.6256×10^{-34} Js or kg $m^2\ s^{-1}$ or 3.6253×10^{-27} erg second or g cm^2.

- A hollow sphere coated inside with platinum black and having a small hole in its wall acts as a near **black body**.
- Black body is a perfect absorber and perfect emitter of radiant energy.
- At a given temperature the intensity of radiation increases with wavelength, reaches a maximum and then decreases.
- As the temperature increases the intensity of the radiation will be more towards the lower wavelengths.
- Planck's quantum theory explains only the black body radiations.
- Einstein extended the quantum theory to all types of electromagnetic radiations.
- Einstein called the energy packets of electromagnetic radiations as **photons**.
- The emission of electrons from metal surface when exposed to radiation of suitable wavelength is known as **photoelectric effect**.
- The photoelectric effect is readily exhibited by alkali metals like K and Cs.
- When photon strikes the metal, its energy is absorbed by the electron and emission of electrons takes place.
- A part of the energy of the photon is used up to escape the electron from the attractive forces and the remaining energy is used in increasing the kinetic energy of the electron.

$$hv = W + \text{KE}$$

hv = Energy of photon; W = Energy required to overcome the attractive forces on an electron in the metal; KE = Kinetic Energy of the emitter electron.

- A body can absorb light energy in the form of photons and goes to the excited state E_2 from the ground state E_1.
- Emission of energy will be accompanied during the transition from excited state to ground state.

$$E_2 - E_1 = \Delta E = hv$$

Types of Spectra

- The arrangement obtained by splitting of electromagnetic radiation into its component wave lengths when passed through a prism is called a **spectrum**.
- When white light is passed through the prism, it gives a **continuous spectrum** of seven colours (VIBGYOR).
- In a **continuous spectrum** each colour fades into the next colour as in a rainbow.
- The spectrum of incandescent white light obtained by heating a solid to very high temperature is a continuous spectrum.
- The spectrum formed by the emission of energy in the form of light radiation is called **an emission spectrum**.
- Emission spectrum consists of **bright lines** or **bands** on a **dark background.**
- Absorption spectrum is just opposite to emission spectrum.
- The dark lines in the absorption spectrum and bright lines in the emission spectrum of a particular substance appear at the same places (same wave length).
- The apparatus used to record the spectrum is called the **spectrometer** or **spectrograph**.
- Emission spectrum is due to the emission of light by the **excited atoms** or **molecules.**
- Absorption spectrum is due to the **excitation** of **atoms** or molecules by absorbing the energy.
- Each element has its own characteristic line spectrum by which it can be identified.
- Line spectra are given by atoms so known as **atomic spectra** and band spectra are given by molecules so known as **molecular spectra**.

Hydrogen Spectrum

- Of all the spectra, hydrogen spectrum is the simplest spectrum.
- Hydrogen spectrum consists of a groups of lines classified into various series.
- Only Balmer series is the **visible** series in hydrogen spectrum.
- Wavelength of a spectral line in hydrogen spectrum can be calculated by using **Rydberg's equation**.

$$\bar{v} = \frac{1}{\lambda} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

- R is Rydberg's constant and its value is 109677 cm^{-1} for hydrogen atom.
- In Balmer series first line is called as H_α, second line is H_β, third line as H_γ, fourth line as H_δ and so on.
- The wavelengths of H_α, H_β, H_γ and H_δ are obtained by substitution series in hydrogen spectrum are

Name of series	n_1	n_2	Spectral Region
Lyman series	1	2, 3, 4, 5, 6,........	Ultraviolet
Balmer series	2	3, 4, 5, 6,........	Visible
Paschen series	3	4, 5, 6,........	Near infrared
Brackett series	4	5, 6, 7,........	Infrared
Pfund series	5	6, 7	Far infrared

- The spectral lines get closer and closer as we move from $n_2 = 3$ to 4 to 5 to 6 etc.
- If n_2 is taken as $\propto$ the limiting wavelength of the lines in any series can be obtained.

- From Lyman series to Pfund series, wavelength increases but frequency and energy decrease.
- Maximum difference in energy for hydrogen atom is found when $n_1 = 1$ and $n_2 = 2$ or higher value.
- R_H value 109677 cm–^{1} is valid only for lines in the hydrogen spectrum.
- For a spectral line of any one electron species like He^+ or Li^{2+}, the value of R is equal to 109677 z^2 cm^{-1} where z is the atomic number.
- Number of spectral lines when an electron is coming to ground state or to the orbit n_1 from excited state in orbit n_2 can be calculated as.

$$\frac{n(n+1)}{2}, \text{ where } n = n_2 - n_1.$$

Bohr's Model of the Atom

- The basis for the Bohr's model of atom is the Planck's quantum theory and hydrogen spectrum.
- Electrons revolve round the nucleus in concentric circular orbits which are represented by K, L, M, N, O, P..... or 1, 2, 3, 4, 5, 6..... etc.
- Each orbit is associated with some energy. So they are known as **energy levels** or **stationary orbits**.
- The energy of the orbit increases with increase in the distance form the nucleus.
- The energy of an electron moving in an orbit remains constant.
- The electron can move from one orbit to another orbit only when they lose or gain the energy difference of the orbitals.
- $\Delta E = E_2 - E_1 = hv$
- An electron while moving from higher energy level to the lower energy level emits energy in the form of light.
- The **angular momentum** (mvr) is quantized. It is an integral multiple of $\frac{h}{2\pi}$ where 'h' is the Planck's constant.

$$mvr = \frac{nh}{2\pi}$$

- The **force of attraction** between the nucleus and the electron $= \frac{e^2}{r^2}$.
- The centrifugal force gained by the electron due the revolving round the nucleus $= -\frac{mv^2}{r}$.
- Bohr's equation to calculate the **radius of an orbit is**

$$r = \frac{n^2h^2}{4\pi^2 2e^2 m}$$

- For hydrogen, since $z = 1$, the radius of orbit is equal to $0.529 \times n^2$ Å. Radius of 1st orbit = 0.529 Å, radius of 2nd orbit = 2.116 Å etc.
- Kinetic energy of the electron due to motion $= \frac{1}{2}mv^2$.
- Potential energy of the electron due to position $= -\frac{e^2}{r}$.
- Total energy of the electron E = Kinetic energy + potential energy

$$= \frac{1}{2}mv^2 - \frac{e^2}{r} = \frac{e^2}{2r} - \frac{e^2}{r} \qquad \left[\because \frac{e^2}{r} = mv^2\right]$$

$$= -\frac{e^2}{2r}$$

- Bohr's equation for calculating the energy of an orbit

$$E_n = -\frac{2\pi^2 me^4}{h^2} \times \frac{z^2}{n^2}$$

- The energy of an electron in an atom is negative and inversely proportional to the square of the n value (n is the number of orbit).
- As the electron moves away from the nucleus the kinetic energy decreases and potential energy increases.
- The potential energy of an electron is zero when the electron is at infinite distance from the nucleus.
- As the n value of the orbit decreases, the size and energy of the orbit decreases.
- The kinetic energy of an electron (positive value) is greater than the potential energy (negative value).
- The energy of an electron in the hydrogen atom.

$$E_n = -\frac{313.6}{n^2} \text{ Kcal mol}^{-1}$$

$$= -\frac{1312}{n^2} \text{ KJ mol}^{-1}$$

$$= -\frac{2.18\times 10^{-11}}{n^2} \text{ ergs atom}^{-1}$$

$$= -\frac{13.6}{n^2} \text{ eV atom}^{-1}$$

- Bohr's equation to calculate the velocity of an electron in hydrogen atom,

$$V_n = \frac{nh}{2\pi mr} = \frac{2\pi e^2}{h} \times \frac{z}{n}$$

- The velocity of electron in first orbit of hydrogen is 2.188 ms^{-1}.
- The energy emitted or absorbed when an electron moves from one orbit to another orbit can be calculated from $\Delta E = hv = hc\overline{v}$.

- $\bar{v}$ can be calculated by using Rydberg's-Ritz equation.

$$\bar{v} = R_H\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

- The R_H value $= \dfrac{2\pi^2 me^4}{ch^3} = 109699\ \text{cm}^{-1}$.
- Bohr's theory could explain the spectra of atoms and ions like He^+, Li^{2+}. Be^{3+} having one electron only.
- Bohr's theory correlates the velocity of light, electronic mass, Planck's constant and electronic charge.
- The energy of the orbit, radius of the orbit and the value of Rydberg's constant R_H calculated from Bohr's theory are in good agreement with the experimental value.
- Bohr's theory could not explain the spectra of atoms having electrons more than one.
- Bohr's theory could not explain the wave nature of electron established by de Broglie.
- It could not explain Zeeman effect and Stark effect.
- Splitting of spectral lines in a strong magnetic field is called as the **Zeeman effect**.
- Splitting of spectral lines in strong electric field is called the **Stark effect**.
- The further splitting of ordinary spectrum when taken by super-spectrometer is called the **fine spectrum**.

Quantum Numbers

- To explain fine spectrum each electron in an atom is assigned with a set of four **quantum numbers**.
- Quantum numbers explain the **orbital concept** of atomic model i.e., size, shape, orientation and spin of the electron.

Principal Quantum Number

- It was proposed by **Bohr**, and is denoted by n.
- It will have any **integer value** except zero.
- It gives the *size* of the orbit and hence **energy** of orbit.
- As the value of n increases the size and energy of the orbit increases.
- It also represents the distance of the electron from the nucleus.
- The number of electrons that can be present in an orbit is equal to $2n^2$.

Azimuthal Quantum Number

- It was proposed by **Sommerfeld** denoted by l.
- It gives the **shape** of an orbital.
- 'l' values depend upon the 'n' values and are equal to '$n - l$' i.e., 0, 1, 2, 3,.....
- Depending upon the 'l' values the subshell in which the electron present can be known.
  If $l = 0$, the electron belongs to s-sub-shell.
  If $l = 1$, the electron belongs to p-sub-shell.
  If $l = 2$, the electron belongs to d-sub-shell.
  If $l = 3$, the electron belongs to f-sub-shell.
- The number of electrons that can be present in any sub-shell is equal to $2(2l + 1)$ or $4l + 2$.
- The number of sub-shells present in an orbit is equal to the value of principal quantum number 'n' value.
- Number of subs-shells in K shell ($n = 1$) = 1 (i.e., 1s)
- Number of sub-shells in L shell ($n = 2$) = 2 (i.e., 2s and 2p)
- Number of sub-shells in M shell ($n = 3$) = 3(i.e., 3s, 3p and 3d)
- Number of sub shells in N shell ($n = 4$) = 4(i.e., 4s, 4p, 4d and 4f)
- s-orbital is spherically symmetrical in shape.
- p-orbital is dumb-bell in shape.
- d-orbital is double dumb-bell in shape.
- f-orbital is four fold dumb-bell in shape.

Magnetic Quantum Number

- It was proposed by **Lande** to account for the Zeeman effect.
- It is denoted by m_l.
- It gives the **orientation** of the orbital.
- The values of 'm_l' depends upon 'l' and are equal to $(2l + 1)$ ranging from $-l$ passing through '0' to $+ l$.
- s-sub-shell will have only one value '0'. So contain only one orbital i.e., s-orbital.
- p-sub-shell will have three orbitals having 'm_l' value $-1, 0, +1$. These are p_x, p_y and p_z.
- d-sub-shell will have five orbitals having 'm_l' values $-2, -1, 0, +1, +2$.
- f-sub-shell will have seven orbitals having 'm_l' values $-3, -2, -1, 0, +1, +2, +3$.
- The + and – signs indicate only change in direction but the magnetic quantum number 'm_l' as such is neither positive nor negative.

Spin Quantum Number

- It was proposed by **Uhlenbeck** and **Goudshmit** and is denoted by 's'.
- Electron moving in an orbital can spin on its own axis.
- The spin of the electron may be clockwise or anti-clockwise.
- The **clockwise** spin is denoted by $+\frac{1}{2}$ or $\uparrow$ and **anti-clockwise** spin is denoted by $-\frac{1}{2}$ or $\downarrow$.

- The difference between the two spin quantum numbers is 1 which is equal to the difference between successive quantum numbers.
- Modern theory of atomic structure was proposed on the basis of quantum mechanics or wave mechanics proposed independently by de Broglie, E. Schrodinger and W. Heisenberg.

Waves and Particles

- Wave theory of moving particles was proposed by de Broglie.
- Wavelength of a moving particle can be calculated by the relation $\lambda = \frac{h}{mv} = \frac{h}{p}$

  where λ is wavelength of a moving particle h is Planck's constant, m is mass of the particle, v is velocity of particle, and p is the momentum of particle.
- The dimensions of the Planck's constant depend upon the dimensions of momentum. de Broglie derived the wave nature of a moving particle from Einstein's mass energy equation ($E = mc^2$) and Planck's quantum theory ($E = hv$).
- While the electron wave is moving in an orbit if the two ends meet to give regular series of crests and troughs, the electron wave is said to be in phase.
- For the electron to be in phase it is necessary that the circumference of the Bohr's orbit (= $2\pi r$) must be equal to the whole number of the wavelength (λ) of electron wave $n\lambda = 2\pi r$.

Heisenberg's Uncertainty Principle

- It is impossible to know exactly both position and momentum of an electron or any other smaller particle simultaneously and accurately.
- Mathematically the uncertainty principle can be expressed as

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

- For an electron revolving round the nucleus in an atom the value of n is 4.

$$\text{Hence,} \quad \Delta x \cdot \Delta p_x \geq \frac{h}{4\pi}$$

Schrödinger Wave Equation

- The consequence of Heisenberg uncertainty principle is that since the exact position of an electron in an atom cannot be predicated, only probability of finding an electron in space around the nucleus can be predicted.
- Schrodinger wave equation gives the probability of finding the electron in space around the nucleus.
- Schrodinger wave equation is

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E - U)\psi = 0$$

  Where ψ is the wave function, m is the mass of electron, E is the total energy of electron, U is the potential energy of the electron.

  Schrodinger wave equation indicates the variation of the ψ value along x, y and z axes.
- ψ is the **amplitude** of the wave function.
- $|\psi|^2$ denotes the **particle density** when applied to the particles.

Probability Concept

- p-orbital has both nodal regions equal to $n - 2$ and nodal planes equal to the value of l, i.e., = 1.
- The nodal plane for p_x orbital is yz; for p_y is xz and for p_z is xy.
- Each d-orbital has nodal regions equal to $n - 3$ and nodal planes equal to the value of l i.e., 2.
- For any orbital the **number of nodal regions** is equal to $n - l - 1$ and **nodal planes** equal to l.
- The **nodal regions** are known as **radial nodes** and the **nodal planes** are known as **angular nodes**.

Filling of Orbitals Pauli's Exclusion Principle

- No two electrons in the same atom can have the same values for all the four quantum numbers.
- The two electrons present in a given orbital may have the same values of n, l and m_l but they must have opposite spin, i.e., either $+\frac{1}{2}$ or $-\frac{1}{2}$
- Pauli's principle helps the determination of maximum number of electrons that can be placed in an orbital of sub-shell and a main shell.
- Pauli's principle is followed only while writing the electronic configuration of atoms in their ground state.

Aufbau Principle

- The electron configurations of successive elements differ only in the last electron which is known as **differentiating electron**.
- According to aufbau principle "the electrons tend to occupy orbitals of minimum energy" in the ground state of atom.

- The energies of different orbitals can be calculated from $n + l$ values.
- In case two orbitals have same $n + l$ values, the electron goes into the orbital whose n values is less.
- The sequence of filling different orbitals by electrons be known form **Moeller diagram.** (Fig 1.38).
- The energy of 4*s* orbital is less than that of 3*d* orbital in elements having atomic number up to 20, but in elements having atomic number above 20, the energy of 3*d* orbital is less than 4*s*.
- The energy of 4*f* orbital has more energy than 6*s* in the elements having atomic number up to 57, but in the elements having atomic number around 90 the energy of 4*f* is less than 5*s* orbital.
- Electronic configurations are written in nl^x method where n is the principal quantum number. l is the azimuthal quantum number and x is the number of electrons present in it.

Hund's Rule

- The orbitals in a sub-shell having equal energy are called **degenerate orbitals.**
- According to Hund's rule all the available orbitals are singly filled first with electrons of parallel spin and then only pairing of electrons starts.
- The half-filled and completely filled degenerate orbitals in atoms provide the stability.
- When two electrons are paired with opposite spins columbic repulsion energy is lowered.
- When two electrons are paired with parallel spins they remain farther apart due to columbic repulsion.

Stability of Atoms

- On exchanging the position in space of two electrons with parallel spins, if there is no change in the electronic arrangement it leads to decrease in energy. This pair is called the **exchange pair**.
- The decrease in energy per exchange pair of electrons is referred as **Columbic exchange energy**.
- As the number of exchange pairs increases the stability of an atom increases by the lowering of exchange energy E for each pair.
- For chromium with electronic configuration $3d^5\ 4s^1$ has five unpaired electron in *d*-orbitals and one electron in 4*s* orbital having parallel spin which gives 15 exchange pairs ($^6C_2$), so average lowering energy is $15E$.
- For chromium with electronic configuration $3d^4\ 4s^2$ has only four unpaired electrons in *d*-orbitals and which form 10 exchange pairs ($^5C_2$), so average lowering of energies is $10E$. Hence $3d^5\ 4s^1$ configuration rather than $3d^4\ 4s^2$ configuration is stable for chromium.

Magnetic Properties of Atoms

- Atoms which contain unpaired electrons exhibit **paramagnetism.**
- Paramagnetic substances are attracted by the external magnetic field.
- Atoms in which all the electrons are paired **exhibit diamagnetism.**
- Diamagnetic substances are repelled by the external magnetic field.
- The paramagnetic moments of atoms depend upon the number of unpaired electrons.
- The paramagnetic moment can be calculated by using the formula $\propto = \sqrt{n(n+2)}$ BM where n is the number of unpaired electrons.

PRACTICE EXERCISE

Multiple Choice Questions with Only One Answer

Level I

1. Which statement does not form part of Bohr's model of the hydrogen atom.
(a) energy of the electrons in the orbit is quantized.
(b) the electron in the orbit nearest the nucleus is in the lowest energy.
(c) electrons revolve in different orbits around the nucleus.
(d) the position and velocity of the electrons in the orbit cannot be determined simultaneously.

2. If S_1 be the specific charge (e/m) of cathode rays and S_2 be that of positive rays then which is true.
(a) $S_1 = S_2$ (b) $S_1 < S_2$
(c) $S_1 > S_2$ (d) either of these

3. The mass of an electron is m, its charge e and it is accelerated from rest through a potential difference V. The kinetic energy of the electron in joules will be
(a) V (b) eV
(c) MeV (d) none

4. In the above question, the velocity acquired by the electron will be
(a) $\sqrt{(V/m)}$ (b) $\sqrt{(eV/m)}$
(c) $\sqrt{(2eV/m)}$ (d) none

5. For an electron in a hydrogen atom, the wave function, ψ is proportional to exp (r/a_o), where a_o is the Bohr's radius. What is the ratio of the probability density the electron at the nucleus to the probability density at a_o

   (a) e (b) e^2 (c) $\frac{1}{e^2}$ (d) 0

6. Photoelectric effect is the phenomenon in which
   (a) photons come out of a metal when it is hit by a beam of electrons.
   (b) photons come out of the nucleus of an atom under the action of an electric field.
   (c) electrons come out of a metal with a constant velocity which depends on the frequency and intensity of incident light wave.
   (d) electrons come out of a metal with different velocities not greater than a certain value which depends only on the frequency of the incident light wave not on its intensity.

7. In photoelectric effect, the photo-current
   (a) increases with increase of frequency of incident photon.
   (b) decreases with increases of frequency of incident photon.
   (c) does not depend on the frequency of photon but depends only on the intensity of incident light.
   (d) depends both on the frequency of the incident photon.

8. Increase in the frequency of the incident radiations increases the
   (a) rate of emission of photo electrons
   (b) work function
   (c) kinetic energy of photo electrons
   (d) threshold frequency

9. A photo sensitive metal is not emitting photo electrons when irradiated. It will do so when threshold is crossed. To cross the threshold we need to increase.
   (a) intensity (b) frequency
   (c) wavelength (d) none

10. If E_1, E_2 and E_3 represent respectively the kinetic energies of an electron, an alpha particle and a proton each having same de Broglie wavelength then.
    (a) $E_1 > E_3 > E_2$ (b) $E_2 > E_3 > E_1$
    (c) $E_1 > E_2 > E_3$ (d) $E_1=E_2=E_3$

11. A surface ejects electrons when hitted by green light but not when hitted by yellow light. Will electrons be ejected if the surface is hitted by red light.
    (a) yes
    (b) no
    (c) yes, if the red beam is quite intense
    (d) yes, if the red beam continues to fall upon the surface for a long time.

12. The angular speed of the electron in the nth orbit of Bohr hydrogen is.
    (a) Directly proportional to n
    (b) inversely proportional to $\sqrt{n}$
    (c) inversely proportional to n^2
    (d) inversely proportional to n^3

13. The minimum energy required to excite a hydrogen atom from its ground state is.
    (a) 3.4 eV (b) 13.6 eV
    (c) −13.6 eV (d) 10.2 eV

14. The binding energy of the electron in the lowest orbit of the hydrogen atomis 13.6 eV. The energies required in eV to remove an electron from three lowest orbits of the hydrogen atom are
    (a) 13.6, 6.8, 8.4 eV
    (b) 13.6,10.2,3,4 eV
    (c) 13.6,27.2, 40.8 eV
    (d) 13.6,3.4,1.5 eV

15. The frequency of first line of Balmer series in hydrogen atom is ν_0. The frequency of corresponding line emitted by singly ionized helium atom is.
    (a) $2\nu_0$ (b) $4\nu_0$
    (c) $\frac{\nu_0}{2}$ (d) $\frac{\nu_0}{4}$

16. The orbital angular momentum of an electron in 2s-orbital is.
    (a) $\frac{h}{4\pi}$ (b) zero
    (c) $\frac{h}{2\pi}$ (d) $\sqrt{2}\frac{h}{2\pi}$

17. Which one represents an impossible arrangement.

	n	l	m	s
(a)	3	2	−2	½
(b)	4	0	0	½
(c)	3	2	−3	½
(d)	5	3	0	½

18. Two electrons in the same orbital may be identified with
    (a) n (b) l
    (c) m (d) s

19. Photoelectric emission is observed from a surface for frequencies ν_1 and ν_2 of the incident radiation $(\nu_1 > \nu_2)$. If the maximum kinetic energies of the photoelectrons in the two cases are in the ratio 1:k then the threshold frequency ν_0 is given by.
    (a) $\frac{\nu_2 - \nu_1}{k-1}$ (b) $\frac{k\nu_1 - \nu_2}{k-1}$
    (c) $\frac{k\nu_2 - \nu_1}{k-1}$ (d) $\frac{\nu_2 - \nu_1}{k}$

20. The ratio of the different in energy of electron between the first and second Bohr's orbits to that between second and third Bohr's orbit is.

(a) $\frac{1}{3}$ (b) $\frac{27}{5}$

(c) $\frac{9}{4}$ (d) $\frac{4}{9}$

21. The wavelength of a moving body of mass 0.1 mg is 3.31×10^{-29} m. The kinetic energy of the body in J would be

(a) 2.0×10^{-6} (b) 1.0×10^{-3}

(c) 4.0×10^{-3} (d) 2.0×10^{-4}

22. If the speed of electron in the Bohr's first orbit of hydrogen atom is x, the speed of the electron in the third orbit is

(a) $\frac{x}{9}$ (b) $\frac{x}{3}$

(c) 3x (d) 9x

23. If each hydrogen atom is excited by giving 8.4 eV of energy, then the number of spectral lines emitted is equal to.

(a) none (b) two

(c) three (d) four

24. When 4f- level of an atom is completely filled with electrons, the next electron will enter

(a) 5 s (b) 6 s

(c) 5d (d) 5p

25. The electron identified by quantum numbers n and l,

(i) $n = 4, l = 1$ (ii) $n = 4, l = 0$

(iii) $n = 3, l = 2$ (iv) $n = 3, l = 1$

can be placed in order of increasing energy from the lowest to highest

(a) iv < ii < iii < i (b) ii < iv < i < iii

(c) i < iii < ii < iv (d) iii < i < iv < ii

26. The energy of an electron in the Bohr's orbit of H atom is −13.6 eV. The possible energy value (s) of the excited state (s) for electrons in Bohr's orbits of hydrogen is (are)

(a) −3.4 eV (b) −4.2 eV

(c) −6.8 eV (d) +6.8 eV

27. The number of d-electrons in Fe^{+2} (at no of Fe = 26) is not equal to that of the

(a) p-electrons in Ne (at. no = 10)

(b) s-electrons in Mg (at. no = 12)

(c) d-electrons in Fe

(d) p-electrons in Cl^- (at. no = 17)

28. Which have the same number of s-electrons as the d-electrons in Fe^{2+}

(a) Li (b) Na

(c) N (d) P

29. If 'R_H' is the Rydberg constant, then the energy of an electron in the ground state of hydrogen atom is

(a) $\frac{R_H C}{h}$ (b) $\frac{1}{R_H ch}$

(c) $\frac{hc}{R_H}$ (d) $-R_H hc$

30. When atoms are bombared with α-particles, only a few in a million of the α-particles suffer deflections while others pass through undeflected. This is because

(a) the force of attraction on the α-particles by the oppositely charged electrons is not sufficient

(b) the nucleus occupies much smaller volume compared to the volume of the atom

(c) the force of repulsion on the fast moving α-particles small

(d) the effect in the nucleus do not have any effect on the α-particles

31. The nucleus and an atom can be assumed to be spherical. The radius of the nucleus of mass number A is given by $1.25 \times 10^{-13} \times A^{1/3} cm.$ The atomic radius of atom is one A^0. If the mass number is 64, the fraction of the atomic volume that is occupied by the nucleus is.

(a) 1.0×10^{-3} (b) 5.0×10^{-5}

(c) 2.5×10^{-2} (d) 1.25×10^{-13}

32. The nucleus of an atom is located at x = y = z = 0. If the probability of finding an s-orbital electron in a tiny volume around x = a, y = z = 0 is 1×10^{-5}. What is the probability of finding the electron in the same sized volume around x = z = 0, y = a?

(a) 1×10^{-5} (b) 1×10^{-5} x a

(c) 1×10^{-5} x a^2 (d) 1×10^{-5} x a^{-1}

33. What is the probability at the second site if the electron were in a p_z orbital for data in (Q. No. 32) above?

(a) 1×10^{-5} (b) 2×10^{-5}

(c) 4×10^{-5} (d) 0

34. If there are three possible values $\left(-\frac{1}{2}, 0, +\frac{1}{2}\right)$ for the spin quantum number m_s, then the electronic config of K(19) will be

(a) $1s^2 2s^3 2p^9 3s^3 3p^1$

(b) $1s^2 2s^2 2p^6 3s^2 4s^1$

(c) $1s^2 2s^2 2p^9 3s^2 4p^4$

(d) None is correct

35. If the Aufbau rule is not followed in filling of suborbitals then block of the element will change in

(a) K (19) (b) Sc (21)

(c) V(23) (d) Ni (28)

36. If Hund's rule is not followed, magnetic moment of Fe^{2+}, Mn^+ and Cr all having 24 electrons will be in order

(a) $Fe^{2+} < Mn^+ < Cr$ (b) $Fe^{2+} = Cr < Mn^+$

(c) $Fe^{2+} = Mn^+ < Cr$ (d) $Mn^+ = Cr < Fe^{2+}$

37. Following ions will be be coloured if Aufbau rule is not followed.
(a) Cu^{2+} (b) Fe^{2+}
(c) Sc^{3+} (d) 1,2 true

38. The wavelength is equal to the distance travelled by the electron in one second, then
(a) $\lambda = h/\nu$ (b) $\lambda = h/m$
(c) $\lambda = \sqrt{h/p}$ (d) $\lambda = \sqrt{h/m}$

39. Magnitude of the charge on the helium ion is.
(a) 4.8×10^{-10} esu (b) 2.4×10^{10} esu
(c) 9.6×10^{-10} esu (d) 1.6×10^{-10} esu

40. A photon was absorbed by a hydrogen atom in its ground state and the electron was promoted to the fifth orbit, then the excited atom returned to its ground state, visible and other quanta were emitted, other quanta are
(a) $2 \longrightarrow 1$ (b) $5 \longrightarrow 2$
(c) $3 \longrightarrow 1$ (d) $4 \longrightarrow 1$

41. Consider the following ions
(i) Ni^{2+} (ii) Co^{2+}
(iii) Cr^{2+} (iv) Fe^{3+}
(atomic numbers: Cr = 24, Fe = 26, Co = 27, Ni = 28)
The correct sequence of the increasing order of the number of unpaired electrons in these ions is
(a) a, b, c, d (b) d, b, c, a
(c) a, c, b, d (d) c, d, b, a

42. The ratio of the energy of the electron in ground state of hydrogen to the electron in first excited state of Be^{3+} is
(a) 1:4 (b) 1:8 (c) 1:16 (d) 16:1

43. Bohr model can explain spectrum of
(a) the hydrogen atom only
(b) the hydrogen molecule only
(c) an atom or iron having one electron only
(d) the sodium atom only

44. Of the following radiation with maximum wavelength is.
(a) UV (b) radio wave
(c) X-ray (d) IR

45. Zeeman effect explains splitting of lines in
(a) magnetic field (b) electric field
(c) both (d) none

46. In presence of magnetic field d-suborbit is.
(a) 5-fold degenerate
(b) 3-fold degenerate
(c) 7-fold degenerate
(d) 2-fold degenerate

47. Size of the nucleus is
(a) 10^{-13} cm (b) 10^{-10} cm
(c) 10^{-1} mm (d) all correct

48. The orbit and orbital angular momentum of an electron are $\frac{3h}{2\pi}$ and $\sqrt{\frac{3}{2}}.\frac{h}{\pi}$ respectively. The number of radial and angular nodes for the orbital in which the electron is present are respectively
(a) 0,2 (b) 2,0
(c) 1,2 (d) 2,2

49. Each orbital has a nodal plane. Which of the following statements about nodal planes are not true?
(a) a plane on which there is zero probability that the electron will be found
(b) a plane on which there is maximum probability that the electron will be found
(c) both
(d) none

50. The radial distribution curve of 2s sublevel consists of x nodes, x is
(a) 1 (b) 3
(c) 2 (d) 0

51. Magnetic moments of V (z = 23), Cr (z = 24), Mn (z = 25) are x, y, z Hence
(a) x = y = z (b) x < y < z
(c) x < z < y (d) z < y < x

52. Uncertainty in position and momentum are equal. Uncertainty in velocity is.
(a) $\sqrt{h/\pi}$ (b) $\sqrt{h/2\pi}$
(c) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$ (d) none

53. If Aufbau principle is not used, 19 th electron in Sc (z = 21) will have
(a) n = 3, l = 0 (b) n = 3, l = 1
(c) n = 3, l = 2 (d) n = 4, l = 0

54. Number of electrons that F(z=9) has in p-orbitals, is equal to
(a) number of electrons in s-orbitals in Na (11e)
(b) number of electrons in d-orbitals in Fe^{3+} (23e)
(c) number of electrons in d-orbitals in Mn (25e)
(d) 1, 2, 3 are true

55. If each orbital can hold a maximum of 3 electrons, the number of elements in 4^{th} period of periodic table (long form) is.
(a) 48 (b) 54
(c) 27 (d) 36

56. If the radius of first Bohr orbit is x, then de Broglie wavelength of electron in 3^{rd} orbit is nearly
(a) $2\pi x$ (b) $6\pi x$
(c) $9x$ (d) $\frac{x}{3}$

57. The series of lines present in the visible region of the hydrogen spectrum is
(a) Balmer (b) Lyman
(c) Paschen (d) Brackett

58. Which orbital gives an electron a greater probability of being found close to the nucleus
(a) 3p (b) 3d
(c) 3s (d) equal

59. Hund's rule deals with the distribution of electrons in
(a) quantum shell (b) an orbit
(c) an orbital (d) degenerate orbitals

60. s-orbital is spherically symmetrical hence
(a) it is directional independent
(b) angular dependent
(c) both correct
(d) both incorrect

61. In centro-symmetrical system, the orbital angular momentum, a measure of the momentum of a particle travelling around the nucleus, is quantized. Its magnitude is.

(a) $\sqrt{l(l+1)}\dfrac{h}{2\pi}$ (b) $\sqrt{l(l-1)}\dfrac{h}{2\pi}$

(c) $\sqrt{s(s+1)}\dfrac{h}{2\pi}$ (d) $\sqrt{s(s-1)}\dfrac{h}{2\pi}$

62. If travelling at equal speeds, the longest wavelength of the following matter is that of
(a) electron (b) proton
(c) neutron (d) alpha particle.

63. The line spectra are the characteristics of
(a) atoms in the excited state
(b) molecules in the excited state
(c) atoms in ground state
(d) molecules in ground state

64. In the Bohr model of the hydrogen atom, the ratio of the kinetic energy to the total energy of the electron in quantum state is.
(a) 1 (b) 2
(c) −1 (d) −2

65. The number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number +2
(a) 3 (b) 4
(c) 2 (d) 1

66. The distance of closest approach of an α-particle fired towards a nucleus with momentum p is r. What will be the distance of closest approach when the momentum of the α-particle is 2p?
(a) 2r (b) 4r
(c) r/2 (d) r/4

67. The radius of Bohr's first orbit in H atom is 0.53 mm. the radius of second orbit in He^+ would be
(a) 0.0265 mm
(b) 0.0530 mm
(c) 0.1060 mm
(d) 0.2120 mm

68. The ionization potential of hydrogen atom is 13.6 eV. The energy required to remove an electron from the n = 2 state of hydrogen atom is.
(a) 27.2 eV (b) 13.6 eV
(c) 6.8 eV (d) 3.4 eV

69. The value of charge on the oil droplets experimentally observed were -1.6×10^{-19} and -4×10^{-19} coulomb. The value of the electronic charge, indicated by these results is.
(a) 1.6×10^{-19} (b) -2.4×10^{-19}
(c) -0.4×10^{-19} (d) -0.8×10^{-19}

70. The ratio of energy of a photon of 2000 A^0 wave length radiation to that of 4000 A^0 radiation is

(a) $\dfrac{1}{4}$ (b) $\dfrac{1}{2}$

(c) 2 (d) 4

71. Atomic weight of Ne is 20.2. Ne is a mixture of ^{20}Ne and ^{22}Ne. Relative abundance of heavier isotope is.
(a) 90 (b) 20
(c) 40 (d) 10

72. The velocity of electron in the hydrogen atom is $2.2 \cdot 10^6 m/s$. The de Broglie wavelength for this electron is.
(a) 33 nm (b) 45.6 nm
(c) 23.3 nm (d) 0.33 nm

73. What is the energy in joule of a photon of light with wavelength 4.0×10^3 nm
(a) 7.5×10^{-20} (b) 5.0×10^{-20}
(c) 2.0×10^{-10} (d) 2.5×10^{-10}

74. The maximum wavelength of light that can excite an electron from first to third orbit of hydrogen atom is.
(a) 487 nm (b) 170 nm
(c) 103 nm (d) 17 nm

75. The specific charge of a proton is 9.6×10^7 C kg^{-1}, then for an α-particles it will be

(a) 2.4×10^7 C kg^{-1} (b) 4.8×10^7 C kg^{-1}

(c) 19.2×10^7 C kg^{-1} (d) 38.4×10^7 C kg^{-1}

76. The work function for a metal is 4 eV. To emit a photo electron of zero velocity from the surface of the metal, the wavelength of incident light should be
(a) 2700 A^0 (b) 1700 A^0
(c) 5900 A^0 (d) 3100 A^0

77. Ultraviolet light of 6.2 eV falls on aluminum surface (work function = 4.2 eV). The kinetic energy (in joule) of the fastest electron emitted is approximately
(a) 3×10^{-21} (b) 3×10^{-19}
(c) 3×10^{-17} (d) 3×10^{-15}

78. The threshold wavelength for photoelectric effect on sodium is 5000 A^0. Its work function is.
(a) 4×10^{-19} J (b) 1 J
(c) 2×10^{-19} J (d) 3×10^{-10} J

79. Photons of energy 6 eV are incident on a potassium surface of work function 2.1 eV. what is the stopping potential?
(a) -6V (b) -2.1 V
(c) -3.9 V (d) -8.1 V

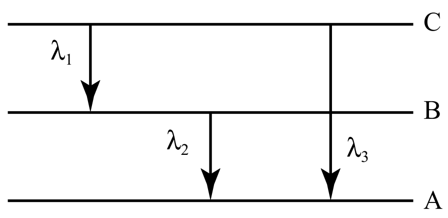
80. In an atom two electrons move around the nucleus in circular orbits of radii R and 4R. The ratio of the time taken by them to complete one revolution is.
(a) 1:4 (b) 4:1
(c) 1:8 (d) 8:7

81. In the series limit of wavelength of the Lyman series for the hydrogen atom is 912 A^0, then the series limit of wavelength for the Balmer series of the hydrogen atom is.
(a) 912 A^0 (b) $912 \times 2A^0$
(c) $912 \times 4A^0$ (d) 912/2A^0

82. The difference in angular momentum associated with the electron in two successive orbits of hydrogen atom is
(a) $\dfrac{h}{\pi}$ (b) $\dfrac{h}{2\pi}$
(c) $\dfrac{h}{2}$ (d) $(n-1)\dfrac{h}{2\pi}$

83. Ionization potential of hydrogen atom is 13.6 eV. Hydrogen atom in the ground state are excited by monochromatic light of energy 12.1 eV. The spectral lines emitted by hydrogen according to Bohr's theory will be
(a) one (b) two
(c) three (d) four

84. Energy levels A, B, C of a certain atom corresponding to increasing values of energy, i.e; $E_A < E_B < E_C$. If λ_1, λ_2 and λ_3 are the wavelengths of radiations corresponding to the transitions C to B, B to A and C to A respectively, which of the following statement is correct



(a) $\lambda_3 = \lambda_1 + \lambda_2$ (b) $\lambda_3 = \dfrac{\lambda_1\lambda_2}{\lambda_1 + \lambda_2}$
(c) $\lambda_1 + \lambda_2 + \lambda_3 = 0$ (d) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$

85. The ratio of the radius of the orbit for the electron orbiting the hydrogen nucleus to that of an electron orbiting a deuterium nucleus is.
(a) 1:1 (b) 1:2
(c) 2:1 (d) 1:3

86. Suppose 10^{-17} J of light energy is needed by the interior of human eye to see an object. The photons of green light (λ = 550 nm) needed to see that object are
(a) 27 (b) 28
(c) 29 (d) 30

87. A photon of 300 nm is absorbed by a gas and then reemits two photons. One reemitted photon has wavelength 496 nm, the wavelength of second reemitted photon is.
(a) 757 (b) 857
(c) 957 (d) 657

88. The shortest λ for the Lyman series is ……. (given (R_H = 109678 cm^{-1})
(a) 912 A^0 (b) 700 A^0
(c) 600 A^0 (d) 811 A^0

89. The longest λ for the Lyman series is…….. (given (R_H = 109678 cm^{-1})
(a) 1215 A^0 (b) 1315 A^0
(c) 1415 A^0 (d) 1515 A^0

90. The λ for H_α line of Balmer series is 6500 A^0. Thus λ for H_β line of Balmer series is.
(a) 4864 A^0 (b) 4914 A^0
(c) 5014 A^0 (d) 4714 A^0

91. The momentum of particle of wavelength 0.33 nm is…..kg m sec^{-1}
(a) 2×10^{-24} (b) 2×10^{-12}
(c) 2×10^{-6} (d) 2×10^{-48}

92. An oil drop has charge 6.39×10^{-19} C. The total number of electrons on oil drop are
(a) 1 (b) 2
(c) 3 (d) 4

Multiple Choice Questions with Only One Answer

Level II

1. Select the **correct** statement.
(a) Orbital angular momentum is $\dfrac{h}{\pi}$ for the electron having n = 5 and lower value of the azimuthal quantum number.
(b) If $n = 3, l = 0, m = 0$ for the last valence shell electron, then the possible atomic number may be 12 or 13.
(c) Total spin of electrons for the atom ($Z = 25$) Mn is $\pm\dfrac{7}{5}$.
(d) Magnetic moment due to spin for noble gas is zero.

2. Identify the **correct** statement (s)
I. The radial probability distribution curves for 1s, 2s, 2p and 3d are identical.

II. The number of nodal planes for $2p$, $3d$ and $4p$ orbitals is the same.

III. Theoretically calculated spin only magnetic moment is same for Ti^{2+} and Ni^{2+}.

IV. A 'p' orbital can accommodate a maximum of six electrons.

(a) I and II are correct.
(b) II and III are correct.
(c) III and IV are correct.
(d) Only IV is correct.

3. The magnetic moment of two ions M^{x+} and M^{y+} of the element $M(Z = 26)$ is found to be 5.916 BM. If $x > y$, then which of the following statement is correct?
(a) M^{y+} is more stable than M^{x+}.
(b) M^{y+} is less stable than M^{x+}.
(c) Both are equal stable.
(d) Cannot be predicted exactly.

4. If λ_0 is threshold wavelength for photo electronic emission, λ is the wavelength of light falling on the surface of metal, and m, mass of electron, then de Broglie wavelength of the emitted electron is

(a) $\left[\dfrac{h(\lambda\lambda_0)}{2mc(\lambda_0-\lambda)}\right]^{\frac{1}{2}}$ (b) $\left[\dfrac{h(\lambda_0-\lambda)}{2mc\lambda\lambda_0}\right]^{\frac{1}{2}}$

(c) $\left[\dfrac{h(\lambda-\lambda_0)}{2mc\lambda\lambda_0}\right]^{\frac{1}{2}}$ (d) $\left[\dfrac{h\lambda\lambda_0)}{2mc}\right]^{\frac{1}{2}}$

5. From the following observation of Mr. Gupta, Mr. Agarwal and Mr. Maheshwari, predict the type of orbital.
Mr. Gupta: The angular function of the orbital intersect the three axes at the origin only.
Mr. Agarwal: XY plane acts as a nodal plane.
Mr. Maheshwari: The graph of radial probability *vs* 'r' intersects the radial axis at three separate regions, points including origin.
(a) $5d_{yz}$ (b) $4p_z$
(c) $6d_{yz}$ (d) $6d_{xy}$

6. The probability of finding the electron is maximum for $d_{x^2-y^2}$ orbital
(a) along x-axis only
(b) along y-axis only
(c) along x, y-axis only
(d) between x, y axes

7. Which one of the following statement is **not** correct?
(a) Rydberg's constant and wave number have same units.
(b) Lyman series of hydrogen spectrum occurs in the UV region.
(c) The angular momentum of the electron into the ground state hydrogen atom is equal to $\dfrac{h}{2\pi}$.
(d) The radius of first Bohr orbit of hydrogen atom is 2.116×10^{-8} cm.

8. Which one of the following statement is correct?
(a) $2s$ orbital is spherical with two nodal planes.
(b) The de Broglie wavelength (λ) of a particle of mass m and velocity v is equal to $\dfrac{mv}{h}$.
(c) The principal quantum number n indicates the shape of an orbital.
(d) The electronic configuration of phosphorus is given by Ne $3s^2\ 3p_x^1 p_y^1 p_z^1$.

9. What is **true** regarding the length of photographic film occupied by various series in hydrogen emission spectrum?
(L_1: Length of Lyman series; L_2: Length of Balmer series; L_3 Length of Paschen series)
(a) $L_1 = L_2 = L_3$ (b) $L_1 < L_2 < L_3$
(c) $L_1 < L_2 > L_3$ (d) $L_1 > L_2 > L_3$

10. Photons having energy equivalent to binding energy of 4th state of He^+ ion are used the metal surface of work function 1.4 eV. If electrons are further accelerated through the potential difference of 4V, then the minimum value of de Broglie's wavelength associated with the electron is
(a) 1.1 Å (b) 9.15 Å
(c) 5 Å (d) 11 Å

11. If the angular momentum of an electron changes from $\dfrac{2h}{\pi}$ to $\dfrac{h}{\pi}$ during a transition in a hydrogen atom, it results in the formation of spectral lines in
(a) UV region (b) visible region
(c) IR region (d) Far IR region

12. Proton, alpha particles are accelerated with same potential. The de Broglie wavelength ratio is
(a) $2\sqrt{2}:1$ (b) $1:2\sqrt{2}$
(c) 4: 1 (d) 1: 4

13. A stationary He^+ ion emitted a photon corresponding to the first line of the Lyman series. The photon liberates a photo electron from a stationary hydrogen atom in ground state. The kinetic energy of the electron is
(a) 10.2 eV (b) 13.6 eV
(c) 20.4 eV (d) 27.2 eV

14. The uncertainty in position is of the order of 1 Å. The uncertainty in velocity of a cricket ball
(a) $1.7\ K \times 10^{-23}$ m/s (b) 1.7×10^{-24} m/s
(c) 3.4×10^{-23} m/s (d) 3.4×10^{-24} m/s

15. A standing wave in a string 35 cm long has a total of six nodes including ends. Hence, the wavelength is
(a) 14 cm (b) 5.82 cm
(c) 7 cm (d) 17.5 cm

16. The m value for an electron in an atom is equal to the number of m values for $l = 1$. The electron may be present in
(a) $3d_{x^2-y^2}$ (b) $5f_{xyz}$
(c) $2p_x$ (d) $3\,S$

17. An electron travels with velocity of v ms^{-1}. For a proton to have same de Broglie wavelength, the velocity will be approximately
(a) v ms^{-1} (b) 1840 v ms^{-1}
(c) v/1840 ms^{-1} (d) 1840 v ms^{-1}

18. The dye Acriflavine, when dissolved in water, has its maximum light absorption at 4530 Å and the maximum fluorescence emission at 5080 Å. The number of fluorescence quanta is, on the average 53 per cent of the number of quanta absorbed. Using the wavelengths of maximum absorption and emission, what percentage of absorbed energy is emitted as fluorescence?
(a) 100% (b) 47%
(c) 36% (d) 12%

19. A gaseous particle '*X*' has a proton (atomic) number '*n*' and a charge of +1. An another gaseous particle '*Y*' has a proton (atomic number) of $(n + 1)$ and is isoelectronic with *X*.
Which of the following statements **correctly** describes *X* and *Y*?
(a) *X* has larger radius than *Y*.
(b) *X* requires more energy than *Y* when a further electron is removed from each particle.
(c) *X* releases more energy than *Y* when an electron is added to each particle.
(d) *X* and *Y* both have the same electro negativity.

20. When an electron is transited from $2E$ to E energy level, the wavelength of photon produced is λ_1. If electronic transition involves 4/3 E to E level, the wavelength of resultant photon is λ_2. Which of the following is a correct relation?
(a) $\lambda_2 = \lambda_1$ (b) $\lambda_2 = 3\lambda_1$
(c) $\lambda_2 = \frac{3}{4}\lambda_1$ (d) $\lambda_2 = \frac{4}{2}\lambda_1$

21. If a certain metal was irradiated by using two different light radiations of frequency 'x' and '$2x$' the kinetic energies of the ejected electrons are 'y' and '$3y$' respectively. The threshold frequency of the metal will be
(a) $\frac{x}{3}$ (b) $\frac{x}{2}$
(c) $\frac{3x}{2}$ (d) $\frac{2x}{3}$

22. During the excitation of an electron, it travels at a distance of nearly 0.7935 nm in the hydrogen atom. The maximum number of spectral lines formed during the deexcitation is
(a) 1 (b) 3
(c) 6 (d) 10

23. The ratio between the wave numbers of spectral lines of Balmer series of hydrogen having the least and the highest wavelength values is
(a) 5: 27 (b) 27: 5
(c) 5: 9 (d) 9: 5

24. The number of nodal planes, radial nodes and peaks in the radial probability curve of $5d$ orbital are respectively
(a) 5, 2, 3 (b) 4, 3, 2
(c) 2, 3, 2 (d) 2, 2, 3

25. The number of revolutions made by electron in Bohr's 2nd orbit of hydrogen atom in one second is
(a) 6.55×10^{15} (b) 8.2×10^{14}
(c) 1.64×10^{15} (d) 2.62×10^{16}

26. When an excited state of H-atom emits a photon of wavelength λ and returns to the ground state, the principal quantum number of the excited state is given by
(a) $\sqrt{\frac{(\lambda R-1)}{\lambda R}}$ (b) $\sqrt{\lambda R(\lambda R-1)}$
(c) $\sqrt{\frac{\lambda R}{(\lambda R-1)}}$ (d) $\sqrt{\lambda R(\lambda R+1)}$

27. What is the maximum number of electrons in an atom that can have the quantum numbers $n = 4$, $m_l = +1$?
(a) 4 (b) 15
(c) 3 (d) 6

28. The wave function (ψ) of $2s$ is given by $(\psi)_{2s} = \frac{1}{2\sqrt{2\lambda}}\left(\frac{1}{a_0}\right)^{1/2}\left(2-\frac{r}{a_0}\right)e^{-r/2a_0}$. At $r = r_0$, radial node is formed. Thus, for $2s$, r_0 in terms of a_0 is
(a) $r_0 = a_0$ (b) $r_0 = 2a_0$
(c) $r_0 = a_0/2$ (d) $r_0 = 4a_0$

29. Which of the following statements in relation to the hydrogen atom is **correct**?
(a) $3s$, $3p$ and $3d$ orbitals all have the same energy.
(b) $3s$ and $3p$ orbitals are of lower energy than $3d$ orbital.
(c) $3p$ orbital is lower in energy than $3d$ orbital.
(d) $3s$ orbital is lower in energy than $3p$ orbital.

30. Choose the **wrong** statement.
(a) The shape of an atomic orbital depends upon the azimuthal quantum number.
(b) The orientation of an atomic orbital depends upon the magnetic quantum number.
(c) The energy of an electron in an atomic orbital of multi-electron atom depends upon the principal number only.

(d) The number of degenerate atomic orbital of one type depends upon the value of azimuthal and magnetic quantum number only.

31. The ground state electronic configurations of the elements, U, V, W, X and Y (the symbols do not have any chemical significance) are as follows:

U $1s^2\,2s^2\,2p^3$
V $1s^2\,2s^2\,2p^6\,3s^1$
W $1s^2\,2s^2\,2p^6\,3s^2\,3p^2$
X $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^5\,4s^2$
Y $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}\,4s^2\,4p^6$

Determine which sequence of elements that satisfy the following statements?

I. Element forms a carbonate which is not decomposed by heating.
II. Element is most likely to form coloured ionic compounds.
III. Element has largest atomic radius.
IV. Element forms only acidic oxide.

(a) *VWYU* (b) *VXYW*
(c) *VWYX* (d) *VXWU*

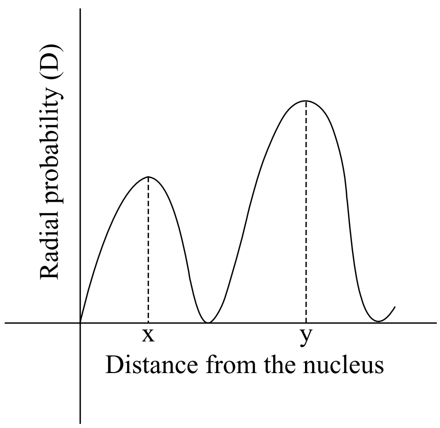
32. If the quantum numbers are defined as follows, $n = 1, 2, 3....$, $l = 0, 1, 2,...(n - 1)$, $m = -l$ in integral steps to $+l$, how many electrons could be fitted in Ist shell?

(a) 2 (b) 8
(c) 10 (d) 18

33. If electrons were to fill up progressively with orbit saturation (neglecting Aufbau rule) and each orbital were to accommodate the three electrons, instead of two; which of the following would NOT hold correct for the new electronic arrangement in zirconium atom ($Z = 40$)?

(a) The number of p-electrons would be double the number of s-electrons.
(b) Spin quantum number would become super flows.
(c) Zirconium would continue to belong to the 4th period (in new Periodic Table)
(d) Zirconium would show retention of block.

34. If the above radial probability curve indicates 2s orbital, the distance between the peak points x, y is



(a) 2.07 Å (b) 1.59 Å
(c) 0.53 Å (d) 2.12 Å

35. An electron is continuously accelerated in a vacuum tube by applying a potential difference. If the de Broglie's wavelength is decrease by 10 per cent the change in the kinetic energy of the electron is nearly

(a) decreased by 11%
(b) increased by 23.4%
(c) increased by 10%
(d) increased by 11.1%

36. The angular momentum of electron in Li^{2+} was found to be (7h/11). The distance of the electron from nucleus is

(a) 1.69 Å (b) 0.68 Å
(c) 2.82 Å (d) 8.19 Å

37. The frequency 'v' of certain line of the Lyman series of the atomic spectrum of hydrogen satisfies the following conditions

I. It is the sum of the frequencies of another Lyman and a Balmer line
II. It is the sum of the frequencies of a certain line, a Balmer line and a Paschen line
III. It is the sum of the frequencies of a Lyman and a Paschen line but no Brackett line

To which transition does v correspond?

(a) $n_2 = 3$ to $n_1 = 1$ (b) $n_2 = 3$ to $n_1 = 2$
(c) $n_2 = 2$ to $n_1 = 1$ (d) $n_2 = 4$ to $n_1 = 1$

38. A hydrogen atom sample in the ground state is excited by monochromatic light radiation of wavelength λ Å. The resulting spectrum consists of maximum 15 different lines. What is λ?
($R_H = 109677\ cm^{-1}$)

(a) 937.3 Å (b) 1025 Å
(c) 1236 Å (d) None of these

39. The ratio between time periods taken by electron in Bohr's 2nd and 3rd orbits for each revolution is

(a) 9: 4 (b) 4: 9
(c) 8: 27 (d) 27: 8

40. The velocity of an electron in n^{th} orbit of hydrogen atom bears the ratio 1: 411 to the velocity of light. The number of coloured lines formed when electron jumps from $(n + 3)$ state to ground state, is

(a) 4 (b) 3
(c) 5 (d) 6

41. The de Broglie wavelength (λ) of electron is related to its accelerating potential (V) by the equation

(a) $\lambda_{(\AA)} = \dfrac{12.27}{\sqrt{V}}$

(b) $\lambda_{(nm)} = \dfrac{12.27}{\sqrt{V}}$

(c) $\lambda_{(\mathring{A})} = \frac{150}{\sqrt{V}}$

(d) $\lambda_{(nm)} = \frac{150}{\sqrt{V}}$

42. Hydrogen-like ion has the wavelength difference between the first lines of Balmer and Lyman series equal to 33.4 nm. The atomic number of that ion is equal to

(a) 4 (b) 3
(c) 2 (d) 5

43. When a certain metal was irradiated with light of frequency 3.2×10^{16} Hz, the photo electrons emitted has twice the kinetic energy as did photoelectrons emitted when the same metal was irradiated with light of frequency 2.5×10^{16} Hz. The threshold frequency of the metal is ....10^{15} Hz.

(a) 18 (b) 20
(c) 22 (d) 16

44. What is the wavelength of the photon emitted by a hydrogen atom when an electron makes transition from $n = 2$ to $n = 1$? Given that ionization potential is 13.786 V. (Answer expressed in Angstrom units)

(a) 1200 (b) 900
(c) 800 (d) 1000

45. The ratio of the energy of the electron in ground state of hydrogen to the electron in first excited state of Be^{3+} is

(a) 1:4 (b) 1:8
(c) 1:16 (d) 16:1

46. The configuration of Cr atom is $3d^5\ 4s^1$ but not $3d^4\ 4s^2$ due to reason R_1 and the configuration of Cu atom $3d^{10}\ 4s^1$ but not $3d^9\ 4s^2$ due to Reason R_2. R_1 and R_2 are

(a) R_1: The exchange energy of $3d^5\ 4s^1$ is greater than that of $3d^4\ 4s^2$.
R_2: The exchange energy of $3d^{10}\ 4s^1$ is greater than that of $3d^9\ 4s^2$.
(b) R_1: $3d^5\ 4s^1$ and $3d^4\ 4s^2$ have same exchange energy but $3d^5\ 4s^1$ is spherically symmetrical.
R_2: $3d^{10}\ 4s^1$ is also spherically symmetrical.
(c) R_1: $3d^5\ 4s^1$ has a greater exchange energy than $3d^4\ 4s^2$.
R_2: $3d^{10}\ 4s^1$ has a spherical symmetry.
(d) R_1: $3d^5\ 4s^1$ has greater energy than $3d^4\ 4s^2$.
R_1: $3d^{10}\ 4s^1$ has a greater energy than $3d^9\ 4s^2$.

47. Maximum value $(n + 1 + m)$ for unpaired electrons in second excited state of chlorine $_{17}Cl$ is

(a) 28 (b) 25
(c) 20 (d) None of these

48. If r be the radius of first Bohr's orbit for hydrogen atom, the de Broglie wavelength in the n^{th} orbit of hydrogen atom is given by

(a) $2\pi / n$ (b) $2\pi r \cdot n$
(c) $2\pi r / n^2$ (d) $2\pi r n^2$

49. When two electrons are present in two degenerate orbitals of an atom, the energy is lower if their spin is parallel. The statement is based upon

(a) Pauli's exclusion principle
(b) Aufbau principle
(c) Hund's rule
(d) Bohr's rule

50. If the nitrogen atom had electronic configuration $1s^7$, it would have energy lower than that of the normal ground state configuration $1s^2 2s^2 2p^3$, because the electrons would be closer to the nucleus. yet $1s^7$ is not observed because it violates.

(a) Heisenberg uncertainly principle
(b) Hund's rule
(c) Pauli exclusion principle
(d) Bohr's postulate of stationary orbits

51. If R_H be the Rydberg constant, then the energy of an electron in the ground state of hydrogen atom is

(a) $\frac{R_H C}{h}$ (b) $\frac{hC}{R_H}$
(c) $\frac{1}{R_H Ch}$ (d) $-R_H hC$

52. When electronic transition occurs from higher energy state to a lower energy state with energy difference equal to ΔE eV, the wavelength of the line emitted is approximately equal to

(a) $\frac{12400\times10^{-10}}{\Delta E} m$ (b) $\frac{13397\times10^{-10}}{\Delta E} m$
(c) $\frac{14322\times10^{-10}}{\Delta E} m$ (d) $\frac{12387\times10^{-10}}{\Delta E} m$

53. In Mosley's equation given constants $a = b = 1$ then the frequency is $100 s^{-1}$. The element will be

(a) K (b) Na
(c) Rb (d) Cs

54. Which of the pair of orbitals have electronic density along the axis?

(a) $d_{xy} d_{yz}$ (b) $d_{x^2-y^2} d_{z^2}$
(c) $d_{xy} d_{xz}$ (d) $d_{xy} d_{z^2}$

55. If azimuthal quantum number could have value of n also in addition to normal value, then E. C. of Cr $(z = 24)$ would have been

(a) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^4$
(b) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$
(c) $1s^2\ 1p^6\ 2s^2\ 2p^6\ 3s^2\ 2d^6$
(d) $1s^2\ 1p^6\ 2s^2\ 2p^6\ 2d^6$

56. In radial probability distribution curve of 2*s* orbital, the distance between the two peaks is
(a) 0.53 (b) 2.1
(c) 1.1 (d) 1.57

57. If *n* and *l* are respectively principal and azimuthal quantum numbers, then the expression for calculating the total number of electrons in any energy level is
(a) $\sum_{l=0}^{l=n-1} 2(2l+1)$ (b) $\sum_{l=1}^{l=n} 2(2l+1)$
(c) $\sum_{l=1}^{l=n-1} 2(2l+1)$ (d) $\sum_{l=0}^{l=n} 2(2l+1)$

58. Consider the following statements concerning the sub shells:
I. The maximum value of *m* for *g* sub shell is 4
II. The maximum value number of unpaired electrons in *g* sub shell is 9
III. The lowest value for the shell having *g* sub shell is 5
Select correct statements
(a) II, I only (b) II, III only
(c) I, III only (d) I, II, III

59. If a 1 *g* body is travelling along the *X*-axis at 100 cm s^{-1} within 1 cm s^{-1} than 4 uncertainty in its position is
(a) 2.64×10^{-30} m (b) 5.28×10^{-30} m
(c) 1.32×10^{-30} m (d) 0.66×10^{-30} m

60. The photoelectric emission requires a threshold frequency V_0 for a certain metal λ_1 = 2000 Å and λ_2 = 1660 Å produce electrons with kinetic energy KE_1 and KE_2 of $KE_2 = 2KE_1$. The threshold frequency V_0 is
(a) $1.15 \times 10^{15}\ s^{-1}$ (b) $1.15 \times 10^{14}\ s^{-1}$
(c) $1.19 \times 10^{15}\ s^{-1}$ (d) $1.19 \times 10^{14}\ s^{-1}$

61. Calculate the wavelength emitted during the transition of electrons in between two levels of Li^{2+} ion whose sum is 4 and the difference is 2.
(a) 1.14×10^{-6} cm (b) 1.14×10^{-5} cm
(c) 2.12×10^{-6} cm (d) 2.14×10^{-5} cm

62. A certain dye absorbs light of λ = 4700 Å and then fluorescence a light *X* Å. Assuming that 47 per cent of absorbed light is reemitted as fluorescence, the ratio of quanta out to the number of quanta absorbed is 0.5. The value of *X* is
(a) 4000 Å (b) 5000 Å
(c) 5500 Å (d) 6000 Å

63. The bond dissociation energy of H—H bond is 400 KJ/mol. The energy required to excite 0.04 moles of H_2 gas to the first excited state is
(a) 94.6 KJ (b) 47.3 KJ
(c) 70.9 KJ (d) 80.4 KJ

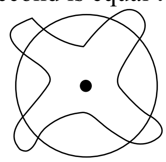
64. The angular momentum of an electron in a Bohr's orbit of hydrogen atom is 4.217×10^{-34} kg-m^2/s. The number of specified lines formed in visible region when electron falls from this level is
(a) 4 (b) 3
(c) 2 (d) 6

65. What transition in the hydrogen spectrum would have the same wavelength as the transition *n* = 4, *n* = 2 in He^+ ion?
(a) $4 \rightarrow 2$ (b) $3 \rightarrow 1$
(c) $3 \rightarrow 2$ (d) $2 \rightarrow 1$

66. If the radius of first Bohr orbit is X_1, then the de Broglie wavelength of electron in 3rd orbit is nearly
(a) $2\pi x$ (b) $6\pi x$
(c) $9x$ (d) $x/3$

67. The angular speed of the electron in n^{th} orbit of Bohr hydrogen atom is
(a) directly proportional to *n*
(b) inversely proportional to $\sqrt{n}$
(c) inversely proportional to n^2
(d) inversely proportional to n^3

68. In hydrogen atom, the electronic motion is represented as follows. The number of revolutions made by that electron in one second is equal to



(a) 6.5×10^{15} (b) 2×10^{14}
(c) 2×10^{13} (d) 1.01×10^{14}

69. The length of the minor axis corresponding to the elliptical path for which *n* = 4 and *k* = 2.
(a) 8.464 Å (b) 16.92 Å
(c) 2.116 Å (d) 4.23 Å

70. In which quantum level does the electron jumps in He^+ ion to ground state if it is given an energy corresponding to 99 per cent of the ionisation potential of He^+ ion.
(a) 4 (b) 6
(c) 8 (d) 10

71. Which among the following is **correct** of $_5B$ in normal state?
(a) $\boxed{\uparrow\uparrow}$ $\boxed{}\boxed{\uparrow}\boxed{\uparrow}$: Against Hund's rule
(b) $\boxed{\uparrow}$ $\boxed{\uparrow}\boxed{\uparrow}\boxed{}$: Against Aufbau principle as well as Hund's rule
(c) $\boxed{\uparrow\uparrow}$ $\boxed{\uparrow}\boxed{}\boxed{}$: Violation of Pauli's exclusion principle and not Hund's rule
(d) $\boxed{\uparrow\downarrow}$ $\boxed{}\boxed{}\boxed{}$: Against Aufbau principle

72. How many times does light travel faster in vacuum than an electron in Bohr's first orbit of hydrogen atom?
(a) 13.7 times (b) 67 times
(c) 137 times (d) 97 times

73. When a hydrogen atom emits a photon of energy 12.1 eV. The orbit angular momentum changes by
(a) 1.05×10^{-34} J-s (b) 2.11×10^{-34} J-s
(c) 3.16×10^{-34} J-s (d) 4.22×10^{-34} J-s

74. If kinetic energy of a particle is doubled; de Broglie wavelength becomes
(a) 2 times (b) 4 times
(c) $\sqrt{2}$ times (d) $1/\sqrt{2}$ times

75. Imagine an atom made up of a proton and a hypothetical particle of double the mass of the electron but having the same charge as the electron. Apply the Bohr atomic model and consider possible transitions of this hypothetical particle to the first excited level. The largest wavelength of the photon that will be emitted has wavelength λ equal to (*R* is the Rydberg constant.)
(a) $\frac{9}{5R}$ (b) $\frac{36}{5R}$
(c) $\frac{18}{5R}$ (d) $\frac{27}{5R}$

76. When a certain metal was irradiated with light having a frequency of 3.0×10^{16} s^{-1}, the photo electrons emitted had twice the kinetic energy as did photoelectrons emitted when the same metal was irradiated with light having a frequency of 2.0×10^{16} s^{-1}. The threshold frequency of the metal (in 10^{16} s^{-1}) is
(a) 1 (b) 2
(c) 3 (d) 4

77. Given that ionisation potential of hydrogen atom is 2.0×10^{-18} J, Planck's constant is 6.0×10^{-34} J-s. The frequency H_β line in Balmer series is in the values of (10^{12} Hz) is
(a) 625 (b) 840
(c) 925 (d) 520

78. The radius of the orbit in hydrogen atom is 0.8464 nm. The velocity of electron in this orbit (in the order of 10^3 m/s), is
(a) 547 (b) 637
(c) 342 (d) 832

79. The calculated magnetic moment of Cr^{2+} is
(a) 1.73 BM
(b) 2.8 BM
(c) 4.8 BM
(d) 5.9 BM

80. The number of *d* electrons in Fe^{2+} is not equal to that of
(a) *p*-electrons in Ne
(b) *s*-elelctrons in Mg
(c) *p*-electrons in Cl^-
(d) *p*-electrons in Mg

81. de Broglie wavelength of the Bohr's first orbit is λ. The circumference of the Bohr fourth orbit is
(a) 4 λ (b) 8 λ
(c) 16 λ (d) 2 λ

82. What hydrogen like ion has the wavelength difference between the first line of Balmer and Lyman series equal to 133 nm:
(a) He^+ (b) Li^{2+}
(c) Be^{3+} (d) B^{4+}

83. The shortest wavelength of hydrogen atom in Lyman series is *X*. The longest wavelength in Balmer series of He^+ is?
(a) $\frac{9x}{5}$ (b) $\frac{36x}{5}$
(c) $\frac{x}{4}$ (d) $\frac{5x}{9}$

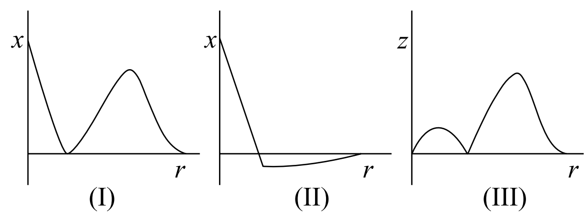
84. Pick out the correct option. Where T stands for True and F stands for False.
I. The energy of the 3*d* orbital is high compared to the 4*s* orbital in hydrogen atom.
II. The electron density in *xy* plane in $d_{x^2-y^2}$ orbital is zero.
III. 24th electron in Cr goes to 3*d* orbital.
IV. The three quantum numbers were clearly explained in terms of Schrodinger wave equation.
(a) TTFF (b) TFTT
(c) TFTF (d) FFTT

85. Who modified Bohr's theory by introducing elliptical orbits for electron path?
(a) Hund
(b) Thomson
(c) Rutherford
(d) Sommerfeld

86. The velocity of electron in a certain Bohr orbit of hydrogen atom bears the ratio 1: 275 to the velocity of light. So the number of orbits is
(a) 1 (b) 2
(c) 3 (d) 4

87. If there were three possibilities of electron spin, *K* (19) would be placed in
(a) *s*-block (b) *p*-block
(c) *d*-block (d) *f*-block

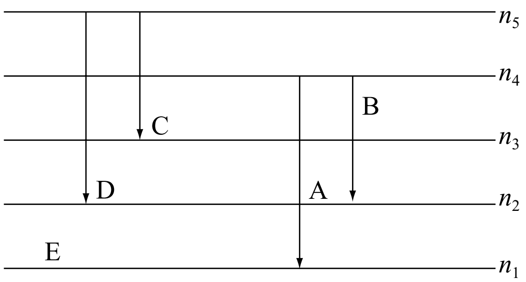
88. Consider the following plots for 2*s*-orbitals



x, *y* and *z* are respectively,
(a) ψ, ψ^2 and 4pr^2 ψ^2
(b) ψ^2, ψ and 4pr^2 ψ^2

(c) $4pr^2\ \psi^2$ and ψ^2 and ψ
(d) ψ^2, $4pr^2\ \psi^2$ and ψ

89. For a hypothetical H-like atom which follows Bohr's model, some spectral lines were observed as shown. If it is known that line '*E*' belongs to the visible region, then the lines possibly belonging to ultraviolet region will be (n_1 is not necessarily ground state) [*Assume for this atom, no spectral series shows overlaps with other series in the emission spectrum*]



(a) B and D (b) D only
(c) C only (d) A only

90. In any sub shell, the maximum number of electrons having same value of spin quantum number is
(a) $\sqrt{l(l+1)}$
(b) $l+2$
(c) $2l+1$
(d) $4l+2$

91. Given the wave function of an orbital of hydrogen atom; $\psi_{1.0} = \left(\frac{1}{\sqrt{\pi a_0^{3/2}}}\right) e^{-r/a_0}$, the most probable distance of the electron present in the given orbital from the nucleus is
(a) $a_0 \times e^{-2r}$ (b) a_0
(c) $(3/2)a_0$ (d) $2a_0$

92. Given the correct order of initials T (True) or F (False) for following statements:
(i) Maximum kinetic energy a photoelectron emitted from a metal [work function = 2 eV] by a photon of wavelength 310 nm is 3.2×10^{-19} J.
(ii) The ratio of magnitude of total energy; kinetic energy: Potential energy for any orbit is 1: 1: 2.
(iii) Number of maxima for the curve $4\pi r^2 R^2(r)$ *vs* are three for the $5p_x$ orbital.
(iv) The ratio of de Broglie wavelength of a 'H' atom, 'He' atom and CH_4 molecule moving with equal kinetic energies 4: 2: 1.
(a) FFTT (b) TTFT
(c) TFTF (d) FTTT

93. The difference of n^{th} and $(n + 1)^{th}$ Bohr's radius of H-atoms is equal to $(n - 1)^{th}$ Bohr's radius. Hence, the value of 'n' is
(a) 1 (b) 2
(c) 3 (d) 4

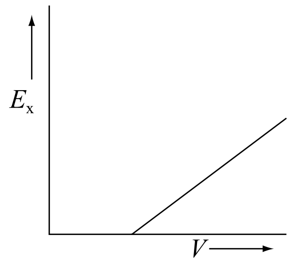
94. Which orbital is represented by complete wave function y_{420}?
(a) $4d_{z^2}$ (b) $3d_{x^2-y^2}$
(c) $4p_x$ (d) $4d_{yz}$

95. The wave function of the atomic orbital of an H-like species is given as

$$\psi_{2s} = \frac{1}{4\sqrt{2\pi}} Z^{1/2}(2 - Zr)e^{-Zr/2}$$

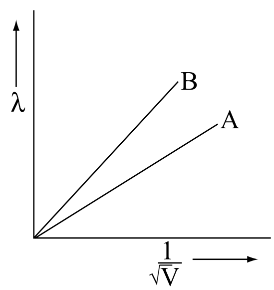
The radius for nodal surface for He^+ ion in Å is
(a) 1.5 Å (b) 1 Å
(c) 2 Å (d) 4 Å

96. For the photoelectric effect, the maximum kinetic energy E_K of the emitted photo electrons is plotted against the frequency v of the incident photons as shown in figure. The slope of the curve gives



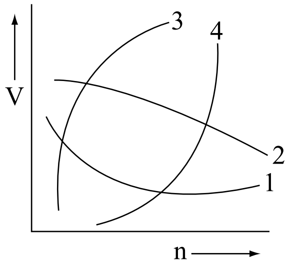
(a) charge of the electron
(b) work function of the metal
(c) Planck's constant
(d) ratio of the Planck's constant to electronic charge

97. de Broglie wavelength of two particles *A* and *B* are plotted against $\left(\frac{1}{\sqrt{V}}\right)$; where *V* is the potential on the particles. Which of the following relation is correct about the mass or the particles?



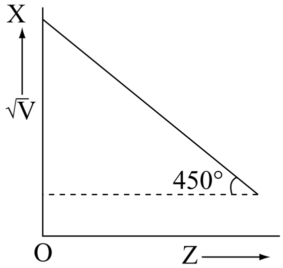
(a) $m_A = m_B$ (b) $m_A > m_B$
(c) $m_A < m_B$ (d) $m_A \leq m_B$

98. Which of the following curves represent the speed of the electron of hydrogen atom as a function of principal quantum number 'n'?



(a) 1 (b) 2
(c) 3 (d) 4

99. In the graph between $\sqrt{v}$ and Z for the Mosley's equation, $\sqrt{v} = a\,(Z - b)$, the intercept OX is 1 on $\sqrt{v}$ axis.



What is the frequency v when the atomic number Z is 52?

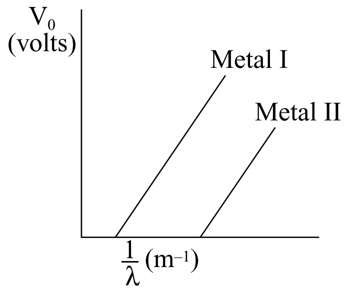
(a) 7.14 s^{-1} (b) 7 s^{-1}
(c) 2401 s^{-1} (d) 2601 s^{-1}

Multiple Choice Questions with One or More Than One Answer

1. Which of the following can be concluded correctly from the solution of Schrodinger equation?
(a) For all orbitals, the radial wave-function (ψ) approaches to value zero as r (distance from nucleus)
(b) The radial probability density $(\psi)^2$ for 1s orbital is maximum at nucleus.
(c) The radial distribution function ($4\pi r^2$, ψ^2) for 1s orbital is minimum at nucleus.
(d) In plot of radial wave function (ψ), ψ changes its sign at the nodes.

2. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photo electrons have maximum kinetic energy T_A (expressed in eV) and de Broglie wavelength λ_A. The maximum kinetic energy of the photo electrons liberated from another metal 'B' by photons of energy 4.20 eV is $T_B = T_A - 1.5$ (expressed in eV). If the de Broglie wavelength of these photo electrons is $\lambda_B = 2\lambda_A$, then
(a) work function of A is 2.25 eV.
(b) work function of B is 4.20 eV.
(c) $T_A = 2$ eV.
(d) $T_B = 2.50$ eV.

3. In a photoelectric experiment, the stopping potential is plotted against $\frac{1}{\lambda}$ of incident radiation for two different metals, the curve is like as shown in figure. Predict which of the following statements are correct?



(a) Slope of the curves for both the metals is 1.212×10^{-6} Jm C^{-1}
(b) When an electromagnetic radiation of wavelength 100 nm strikes the two metals separately, the stopping potential for metal I is 12.18 V.
(c) The stopping potential for metal I is more than that of metal II, if both the metals are exposed to electromagnetic radiations of wavelength 200 nm, separately.
(d) The stopping potential for metal II is more than that of metal I, if both the metals are exposed to electromagnetic radiation of wavelength 200 nm, separately.

4. Which of the following are the **correct** statements?
(a) Light waves were considered electromagnetic in nature.
(b) In vacuum, all types of electromagnetic radiation regardless of their wavelength, travel at the same speed, i.e., 3×10^8 ms^{-1}.
(c) The ideal body which emits and absorbs all frequencies, is called a black body.
(d) Quantum is also called photon.

5. Which of the following pairs have identical values of magnetic moment?
(a) Zn^{2+}, Cu^{+} (b) Co^{2+}, Ni^{2+}
(c) Mn^{4+}, Co^{2+} (d) Mg^{2+}, Sc^{+}

6. Which of the following statements are (is) **correct**?
(a) Electrons in motion behaves as if they were waves.
(b) s-orbital is non directional.
(c) An orbital can accommodate a maximum of two electrons with parallel spins.
(d) The energies of the various sub shells in the same shell are in the order $s > p > d > f$.

7. The nucleus of an atom is located at $x = y = z = 0$. If the probability of finding an 's' orbital electron in a tiny volume around $x = a, y = z = 0$ is 1.0×10^{-5}, choose the **correct** statement (s) regarding the probability of e^-

(a) The probability of finding the electron in the same-sized volume around $x = z = 0, y = a$ is the same i.e., 1×10^{-5}.
(b) The probability of finding the electron in the same-sized volume around $x = y = 0, z = a$ is zero.
(c) The probability at the second site if the electron were in a p_z orbital is zero.
(d) The probability at the second site if the electron were in a p_z orbital is the same i.e., 1.0×10^{-5}.

8. Choose the **correct** statement (s) regarding Bohr-Sommerfeld's model.
(a) All paths around the nucleus are elliptical.
(b) When an electron revolves around the nucleus following circular path, only the angle of rotation is changed.
(c) When an electron revolves around the nucleus following elliptical path, both the angle of rotation and the distance from the nucleus are changed.
(d) For an elliptical path, $k < n$.

9. If there were three possible values $\left(-\frac{1}{2}, 0, +\frac{1}{2}\right)$ for the spin magnetic quantum number (m_s), which of the following statement (s) is/are **correct** regarding a hypothetical Periodic Table based on this condition?
(a) The first period would have only 2 vertical columns.
(b) The second period would have 12 elements.
(c) The Periodic Table would contain 27 groups.
(d) The third period would have 12 elements.

10. Select the **wrong** statements.
(a) If the shortest wavelength of H-atom in Lyman series is 'x', then the longest wavelength in Balmer series of He^+ ion is $\frac{9x}{5}$
(b) When an electron jumps from 4^{th} excited state to ground state, maximum number of phrotons that may be emitted are 4
(c) Number of nodal planes and radial nodes in $5d_{x^2-y^2}$ orbital are 2 and 1 respectively
(d) All four quantum numbers are obtained from the Schrodinger wave equation when applied on the atom

11. Choose the **correct** statement among the following.
(a) Radial distribution function ($\psi^2 4\pi r^2$) gives the probability at a particular distance along one chosen direction.
(b) 'xz' plane acts as a nodal plane for $3p_x$ orbital.
(c) For 's' orbitals wave function is independent of θ and ϕ.
(d) '$2p$' orbital with quantum numbers $n = 2, l = 1, m = 0$, also shows the angular dependence.

12. According to Bohr's theory,
(a) When the atom gets the required energy from the outside, electrons jump from lower orbits to higher orbits and remain there.
(b) When the atom gets the required energy from outside, electrons jump from lower orbits to higher orbits and remain there for very short intervals of time and return back to the lower orbit, radiating the energy.
(c) Angular momentum of the electron is proportional to its quantum number.
(d) Angular momentum of the electron is independent of its quantum number.

13. Choose the **correct** statement (s) regarding the photoelectric effect:
(a) No electrons are ejected, regardless of the intensity of the radiation, unless the frequency equals or exceeds a threshold value characteristic of the metal.
(b) The kinetic energy of the ejected electrons varies linearly with the frequency of the incident radiation and its intensity.
(c) Even at low intensities, electrons are ejected immediately if the frequency is above the threshold value.
(d) An intense and a weak beam of monochromatic radiations differ in having number of photons and not in the energy of photons.

14. Mark out the **correct** statement (s)
(a) Electrons within a sub shell for which $l > 0$ mean that the relatively greater electrostatic repulsion between two electrons in the same orbital within a sub shell as compared with occupancy of two orbitals having different m_l values.
(b) Electrons in singly occupied orbitals tend to have their spins in the same direction so as to maximize the net magnetic moment.
(c) In the ground state, an atom adopts a configuration with the greatest number of unpaired electrons.
(d) Electrons in different orbitals with parallel spins show mutual attraction, so the electron-nucleus interaction is improved.

15. Which of the following statement is/are **correct**?
(a) For all values of 'n' the p-orbitals have the same shape, but the overall size increases as 'n' increases, for a given atom.
(b) The fact that there is a particular direction, along which each p-orbital has maximum electron-density, plays an important role in determining molecular geometries.

(c) The charge cloud of a single electron in $2p_x$ atomic orbital consists of two lobes of electron-density.
(d) The very fact that an e^- is present in both the lobes at equal times, and the fact that in order to move from one lobe to another, it must cross the node, implies, there is at least some probability of finding an e^- in the node.

16. In a hydrogen-like sample electron is in 2^{nd} excited state, the binding energy of 4^{th} state of this sample is 13.6 eV, then
(a) A 24.17 eV photon can set free the electron from the second excited state of this sample.
(b) 3 different types of photons will be observed if electrons make transition up to ground state from the second excited state.
(c) If 23 eV photon is used, then KE of the ejected electron is 1 eV.
(d) 2nd line of Balmer series of this sample has same energy value as 1st excitation energy of H-atoms.

17. Identify the **correct** statement (s) among the following.
(a) If Aufbau rule is not followed, then chromium will still remain in *d*-block.
(b) If there are three possible values for spin quantum numbers, then chromium would have been lying in *s*-block
(c) If Hund's rule of maximum multiplicity is not followed, then the magnetic moment of chromium would be zero.
(d) If Hund's rule of maximum multiplicity is not followed, then Cr^{3+} (*aq*) ion would be colourless.

18. The energy of an electron in the first Bohr orbit of H-atom −13.6 eV; then which of the following statement (s)/are **correct** for He^+?
(a) The energy of electron in second Bohr orbit is −13.6 eV.
(b) The KE of electron in the first orbit is 54.46 eV
(c) The KE of electron in second orbit is 13.6 eV.
(d) The speed of electron in the second orbit is 2.19×10^6 m/s.

19. If the electron of the hydrogen atom is replaced by another particle of same charge but of the double mass, then
(a) radii of different shells will increase.
(b) energy gap between two levels will become double.
(c) ionization energy of the atom will be double.
(d) speed of new particle in a shell will be lesser than the speed of electron in the same shell.

20. Which of the following is/are **correct** about the radial probability curves?
(a) $3d_z^2$ has three angular nodes.
(b) The number of angular nodes are *l*.
(c) The number of radial nodes is equal to $n - l - 1$.
(d) The number of maximum peaks in 2*s*-orbitals are two.

21. Select the **correct** statements about the wave function ψ.
(a) ψ must be real.
(b) ψ must be single values, continuous.
(c) ψ has no physical significance.
(d) ψ^2 gives the probability of finding the electrons.

Comprehensive Type Questions

Passage-I

Imagine a universe in which the four quantum numbers can have the same possible values as in our universe except the angular momentum quantum number '*l*' can have integral values of 0, 1, 2,..., *n* instead of 0, 1, 2,.....$(n - 1)$ and magnetic quantum number '*m*' can have integral values of $-(l + 1)$ to 0 to $(l + 1)$ instead of $-l$ to 0 to $+l$.

1. Electronic configuration of the element with atomic number 25, based on the above assumption is
(a) $1s^2\ 1p^6\ 2s^2\ 1d^{10}\ 2p^5$
(b) $1s^2\ 1p^{10}\ 2s^2\ 1d^{11}$
(c) $1s^3\ 1p^9\ 2s^3\ 2p^9\ 3s^1$
(d) $1s^6\ 1p^{10}\ 2s^6\ 2p^3$

2. Based on above assumption magnetic moment of the element with atomic number 18, is
(a) $\sqrt{24}$ BM (b) $\sqrt{8}$ BM
(c) Zero BM (d) $\sqrt{15}$ BM

3. Based on assumption after completion of 3*s* orbital electron enters into .... orbital
(a) 3*p* (b) 2*d*
(c) 4*s* (d) 1*d*

4. If spin quantum numbers are $-\frac{1}{2}$, 0, $+\frac{1}{2}$ based on the above assumption electronic configuration of the element with atomic number 30 is
(a) $1s^3\ 2s^3\ 2p^9\ 3s^3\ 3p^9\ 4s^3$
(b) $1s^3\ 1p^{15}\ 2s^3\ 1d^9$
(c) $1s^9\ 1p^{15}\ 2s^6$
(d) $1s^3\ 1p^9\ 2s^3\ 2p^9\ 1d^6$

Passage-II

Imagine an atom made up of a proton and a hypothetical particle of half the mass of the electron but having the same charge as the electron. Bohr's model is applicable and all transition of this hypothetical particle from higher level to ground state is possible.

1. The largest wavelength of photon that will be emitted when hypothetical particle jumps to first excited level is

(a) $\frac{18}{5R}$ (b) $\frac{72}{5R}$

(c) $\frac{36}{5R}$ (d) $\frac{8}{R}$

2. The velocity of this particle in second Bohr orbit is

(a) 2.188×10^6 m/s

(b) 1.094×10^6 m/s

(c) 0.547×10^6 m/s

(d) 4.376×10^6 m/s

3. The radius of third Bohr orbit is

(a) 9.522 Å (b) 4.761 Å

(c) 2.380 Å (d) 1.587 Å

4. The kinetic energy of the hypothetical particle in first orbit is

(a) −6.8 eV (b) 6.8 eV

(c) 27.2 eV (d) −27.2 eV

Passage-III

A single electron atom has nuclear charge +Ze. Here, Z is the atomic number and *e* is charge. It requires 47.2 eV to excite the electron from second Bohr orbit to third Bohr orbit.

1. Atomic number of the element is

(a) 2 (b) 3

(c) 4 (d) 5

2. The wavelength required to remove electron from the first Bohr orbit in above data, is

(a) 0.22×10^{-7} cm

(b) 1.0×10^{-7} cm

(c) 5.6×10^{-7} cm

(d) 3.7×10^{-7} cm

3. The kinetic energy of the electron in first Bohr orbit in above data, is

(a) 1.962×10^{-10} ergs

(b) 3.48×10^{-10} ergs

(c) 0.872×10^{-10} ergs

(d) 5.45×10^{-10} ergs

Passage-IV

A gas of identical H-like atom has some atoms in the lowest energy level *A* and some atoms in a particular upper (excited) energy level *B* and there are no atoms in any other energy level. The atoms of the gas make transition to a higher energy level by absorbing monochromatic light of photon of energy 2.55 eV. Subsequently, the atoms emit radiation of only six different photon energies. Some of the emitted photons, have energy 2.55 eV. Some have more and some have less than 2.55 eV.

1. The initially excited level *B* is

(a) 2nd orbit (b) 3rd orbit

(c) 4th orbit (d) 5th orbit

2. The ionization energy of the gaseous atom is

(a) 13.6 eV (b) 14.4 eV

(c) 27.2 eV (d) 54.4 eV

3. The minimum energy of the emitted photon is

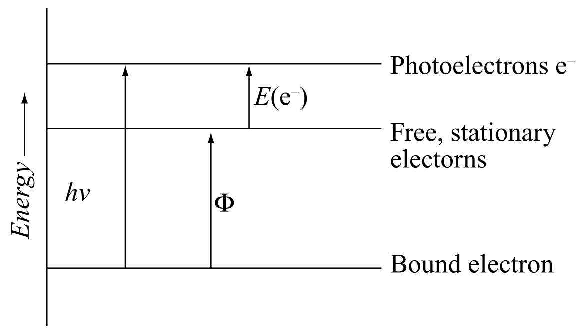
(a) 2.7 eV (b) 2.0 eV

(c) 0.66 eV (d) 0.54 eV

Passage-V

In the photoelectric effect, an incoming photon brings a definite quantity of energy, *hv*. It collides with an electron close to the surface of the metal target, and transfers its energy to it. The difference between the work function, ϕ, and the energy *hv* appears as the kinetic energy of the ejected electron.

The minimum potential at which the plate photoelectric potential. If V_0 is the stopping potential, then $eV_0 = hv.$ $- hv_0$ current becomes zero is called stoping.



1. Light of wavelength 4000 Å is incident on a metal whose work function is 2 eV. Then the maximum possible kinetic energy of the photoelectron is

(a) 3.09 eV (b) 1.9 eV

(c) 1.09 eV (d) None of these

2. The threshold frequency for emitting photoelectrons from a metal surface is 5×10^{14} s^{-1}. Which of the following could be the frequency of radiation to produce photoelectrons having twice the kinetic energy of those produced by the radiation of frequency 10^{15} s^{-1}?

(a) 2.15×10^{14} s^{-1} (b) 3.70×10^{13} s^{-1}

(c) 4.11×10^{14} s^{-1} (d) 15×10^{14} s^{-1}

3. If one photon has 25 eV energy and the work function of the material is 7 eV, then value of the stopping potential will be

(a) 32 V (b) 18 V

(c) 3.3 V (d) Zero

Passage-VI

According to Planck's quantum theory of radiation, the radiant energy is emitted or absorbed discontinuously in the form of quanta. The energy of a quantum is given by $E = h\nu$, where 'ν' is the frequency of radiation. Moreover, a body can emit or absorb energy only in an integral multiples of quanta.

1. Planck's quantum theory of radiation favors the following nature of electromagnetic radiation.
   (a) Particle (b) Wave nature
   (c) Dual nature (d) None of these
2. The energy associated with one photon of wavelength 3000 Å is
   (a) 5.42×10^{-19} J (b) 6.63×10^{-19} J
   (c) 3.73×10^{-19} J (d) 3.73×10^{-19} J
3. Which of the following is not the correct representation of Einstein's photoelectric equation?
   (a) $\frac{1}{2}mv^2 = h(\nu + \nu_0)$
   (b) $h\nu = h\nu_0 + \frac{1}{2}mv^2$
   (c) $h\nu = W + \frac{1}{2}mv^2$
   (d) $\frac{1}{2}mv^2 = h(\nu - \nu_0)$

Passage-VII

The properties of electrons indicate that they have a dual nature, i.e., they behave both as particle and as wave. de Broglie suggested that every object which possesses mass and velocity behaves both as a particle and as a wave. According to de Broglie, the wavelength λ of a particle of mass 'm', moving with a velocity 'v' is given by $\lambda = h/mv$. The dual nature of electron further gets support by Heisenberg's uncertainty principle which states that it is not possible to determine simultaneously and accurately the position and momentum of a moving electron or some other microscopic particle.

According to Heisenberg $\Delta x \Delta p \geq \frac{h}{4\pi}$, where Δx and Δp respectively represent the uncertainties involved in the simultaneous determination of position and momentum of a moving particle.

1. The de Broglie wavelength of a cricket ball of mass 0.2 kg moving with a velocity of 30 ms^{-1} would be
   (a) 1.8×10^{-34} m
   (b) 3.6×10^{-34} m
   (c) 2.2×10^{-34} m
   (d) 1.1×10^{-34} m
2. An electron can be located with an error equal to 10^{-10} m. What is the uncertainty in its momentum?
   (a) 5.28×10^{-25} kg ms^{-1} (b) 5.5×10^{-26} kg ms^{-1}
   (c) 3.34×10^{-14} kg ms^{-1} (d) 6.63×10^{-27} kg ms^{-1}
3. If the uncertainty in the position and momentum for an electron are equal, then the uncertainty in its velocity is
   (a) $\sqrt{\frac{h}{\pi}}$ (b) $\sqrt{\frac{h}{2\pi}}$
   (c) $\frac{1}{m}\sqrt{\frac{h}{2\pi}}$ (d) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$

Passage-VIII

In 1908, W. Ritz introduced his combination principle, which is virtually a generalization of Balmer formula. In its simplest form the principle states that the wave number of any spectral line may be represented as the combination of two terms, one of which is constant and the other variable throughout each spectral series.

$\bar{\nu} = \frac{R}{x^2} - \frac{R}{y^2}$; x and y integers for H-atom. The later on theories have modified the above principle as for H-atom.

$$\bar{\nu} = R_H \left\{ \frac{1}{n_2^1} - \frac{1}{n_2^2} \right\}$$

R_H is Rydberg number = 109,677.76 cm^{-1} and n_1 and n_2 are lower and higher energy levels of the H-atom.

1. What transition H-spectrum would have the same λ as the Balmer transition n = 4 to n = 2 of He$^+$ spectrum?
   (a) $2 \to 1$ (b) $3 \to 1$
   (c) $4 \to 1$ (d) $5 \to 1$
2. The wavelength of radiation of the longest transition in Lyman series of He$^+$ ion is
   (a) 22.8 nm (b) 32.2 nm
   (c) 40 nm (d) 48 nm
3. If the shortest λ of H-atom in Balmer series is λ_1, the longest λ in Lyman series is
   (a) 0.14 λ_1 (b) 0.33 λ_1
   (c) 0.44 λ_1 (d) 0.66 λ_1

Passage-IX

A hydrogen-like atom (atomic number Z) is in a higher excited state of quantum number 'H'. This excited atom can make a transition to the first excited state by successively emitting two photons of energies 10.2 eV and 17.00 eV respectively. Alternatively, the atom from the same excited state can make a transition to the second excited state by emitting two photons of energy 4.25 eV and 5.95 eV respectively.

1. The value of n is
   (a) 8 (b) 6
   (c) 5 (d) 4
2. The atomic number Z is
   (a) 1 (b) 2
   (c) 3 (d) 4

Passage-X

A formula analogous to the Rydberg formula applies to the series of spectral lines which arise from transitions from higher energy levels to the lower energy level of the hydrogen atom.

A muonic hydrogen ion is like a hydrogen atom in which the electron is replaced by a heavier particle the muon. The mass of the muon is 207 times the mass of an electron while charge is the double that the electron (assume charge of proton is same).

1. Radius of first Bohr orbit of muonic hydrogen ion is
   (a) $\frac{0.529}{207}$ Å (b) $\frac{0.529}{414}$ Å
   (c) $\frac{0.529}{828}$ Å (d) 0.529×828 Å
2. Ionization energy of muonic hydrogen ion is
   (a) 13.6×207 eV (b) 13.6×414 eV
   (c) 13.6×828 eV (d) 13.6×3312 eV
3. Distance between 1st and 3rd Bohr orbit of muonic hydrogen ion will be
   (a) $\frac{0.529}{207} \times 8$ Å (b) $\frac{0.529}{414} \times 8$ Å
   (c) $\frac{0.529}{207} \times 5$ Å (d) $\frac{0.529}{828} \times 8$ Å

Matching Type Questions

1. Match the Column I with Column II

Column I (Chemical Prop)	Column II (Metals)
(a) Distance of maximum probability of 1s electron	**(p)** 1.1 Å
(b) Radial node of 2s electron is	(q) 1.06 Å
(c) Radius of 2nd orbit in He^+ ion is	(r) 1.59 Å
(d) Radius of 3rd orbit in Li^{2+} ion is	(s) 0.53 Å

2. Match the following

Column I	Column II
(a) Emission spectrum	(p) Cold nitrogen gas
(b) Absorption spectrum	(q) Excited atomic hydrogen
(c) Band spectrum	(r) Very hot carbon dioxide
(d) Line spectrum	(s) Neon gas at room temperature

3. Match the following

Column I	Column II
(a) Magnetic moment of an atom or ion	(p) Principle QN
(b) Radial node	(q) Azimuthal QN
(c) Hund's rule	(r) Magnetic QN
(d) Orbital angular momentum of electron	(s) Spin QN

4. Match the following

Column I	Column II
(a) Hydrogen atom	(p) Principal quantum number
(b) Nitrogen atom	(q) Azimuthal quantum number
(c) Li^{2+} ion	(r) Exchange energy
(d) Helium atom	(s) Symmetry

5. Match the following

Column I	Column II
(a) Magnetic moment	(p) Principal quantum number
(b) Splitting of spectral lines in magnetic field	(q) Azimuthal quantum number
(c) Number of spherical nodes	(r) Magnetic quantum number
(d) Fine lines spectra	(s) Spin quantum number

6. $n \rightarrow$ orbit no; $Z \rightarrow$ at no; $r_{n,z} \rightarrow$ radius $V_{n,z} \rightarrow$ velocity; $T_{n,z} \rightarrow$ Time period of revolution $K_{n,z} \rightarrow$ kinetic energy of the electron.

Column I (of single electron species)	Column II (Ratio)
(a) $r_{2,1}: r_{1,2}$	(p) 9: 1
(b) $V_{1,3}: V_{3,1}$	(q) 8: 1
(c) $T_{1,2}: T_{2,1}$	(r) 16: 1
(d) $K_{1,2}: K_{2,1}$	(s) 1: 32

7. Match the following

Column I	Column II
(a) Violation of Aufbau rule	(p) [↑↑] [↑↑ | ↑ |]
(b) Violation of Pauli's Exclusion Principle	(q) [↑↑] [↑ | ↑ | ↓]
(c) Violation of Hund's rule	(r) [↑] [↑↑ | ↑↑ | ↑]
(d) Violation of above three principles	(s) [↑] [↑↑ | ↑ | ↓]

8. Match the following

Column I	Column II
(a) n	(p) Stark effect
(b) l	(q) Schrodinger's equation
(c) m	(r) Bohr's atomic model
(d) s	(s) Pauli's exclusivity
	(t) Aufbau sequencing

9. Match the following

Column I	Column II
(a) $2s$	(p) Sum of $(n + l)$ is 3
(b) $2p$	(q) Total number of radial nodes = 1
(c) $3s$	(r) no. of peaks in radial probability graph = 2
(d) $5f$	(s) Spherical symmetry of orbital around nucleus

10. Match the following Column I with Column II

Column I	Column II
(a) For transition: $n = 5$ to $n = 2$	(p) Only two Balmer lines are obtained
(b) For transition: $n = 5$ to $n = 3$	(q) Total three spectral lines are obtained
(c) For transition: $n = 4$ to $n = 2$	(r) At least one Lyman line is obtained
(d) For transition: $n = 4$ to $n = 1$	(s) H_b for Balmer series and H_a for Paschen series both are obtained

Assertion (A) and Reason (R) Type Questions

"In each of the following questions a statement of **Assertion (A)** is given followed by a corresponding statement of **Reason (R)** just below it. Mark the correct answer from the following statements.

a. If both assertion and reason are true and reason is the correct explanation of assertion
b. If both assertion and reason are true but reason is not the correct explanation of assertion
c. If assertion is true but reason is false
d. If assertion is false but reason is true

1. **Assertion (A):** The wave functions of p-type ($l = 1$) with the same value of 'n' but with different values of 'm' will have the same radial distribution function.
Reason (R): Radial distribution function depends only on the values 'n' and 'l' and not on the value of 'm'.

2. **Assertion (A):** According to Bohr IE_2 of He is 54.4 eV/atom.
Reason (R): IE of atom is given by $13.6\dfrac{Z_{eff}^2}{n^2}$ eV/atom

3. **Assertion (A):** An electron can never be found in the nucleus.
Reason (R): Velocity of the electron wave is less than the velocity of light.

4. **Assertion (A):** No two electrons of an atom can have the same set of all four quantum numbers.
Reason (R): An orbital can accommodate a maximum of two electrons.

5. **Assertion (A):** $3d_{xy}$, $4d_{xy}$, $5d_{xy}$.......... all are having the same shape i.e., the double dumb bell.
Reason (R): The size of orbitals are independent of the principal quantum number.

6. **Assertion (A):** $\dfrac{e^-}{m}$ ratio of anode rays is different for gasses.
Reason (R): Proton is the fundamental particle present in gases.

7. **Assertion (A):** ψ^2 measures the electron probability density at a point in an atom.
Reason (R): ψ and ψ^2 vary as a function of the three coordinates 'r' (radial part), θ and ϕ (angular part).

8. **Assertion (A):** Two orbitals with same value of l but different values of m_l will have same number of angular nodes.
Reason (R): Angular wave function of an orbital depends upon its l and m_l values.

9. **Assertion (A):** Aufbau principle is violated in writing electronic configuration of Pd.
Reason (R): Pd is diamagnetic in nature.

10. **Assertion (A):** Spectral lines would not be seen for a $2p_x - 2p_y$ transition.
Reason (R): p-orbitals are degenerate orbitals.

11. **Assertion (A):** Energy of orbitals in hydrogen atom is
$1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f$
Reason (R): The energy of orbitals in hydrogen atom is determined by using principal quantum number.

12. **Assertion (A):** Angular momentum of an electron in an orbit is quantized.
Reason (R): The electron wave propagating around the nucleus remains in the same phase, $2\pi r = n\lambda$.
13. **Assertion (A):** Fine lines are observed in spectra if an atom is placed in a strong magnetic field.
Reason (R): Degenerate orbitals split in the presence of a magnetic field.
14. **Assertion (A):** Spin quantum number can have the value +1/2 or –1/2.
Reason (R): Sign here signifies the wave function.
15. **Assertion (A):** The transition of electrons $n_3 \rightarrow n_2$ in the atom will emit greater energy than $n_4 \rightarrow n_3$.
Reason (R): n_3 and n_2 are closer to nucleus than n_4.

Integer Type Questions

1. In which quantum level does the electron jumps in He^+ ion to ground state if it is given an energy corresponding to 99 per cent of the ionization potential of He^+ ion?
2. When a certain metal was irradiated with light having a frequency of 3.0×10^{16} s^{-1}, the photo electron emitted had twice the kinetic energy as did photoelectrons emitted when the same metal was irradiated with light having a frequency of 2.0×10^{16} s^{-1}. The threshold frequency of the metal (in 10^{16} s^{-1}) is ______________.
3. Given ionization potential of hydrogen atom is 2.0×10^{-18} J, Planck's constant is 6.0×10^{-34} J-s. The frequency of H_β line in Balmer series has the values of (10^{12} Hz) is ______________.
4. The radius of the orbit in hydrogen atom is 0.8464 nm. The velocity of electron in this orbit (in the order of 10^3 m/s) is ______________.
5. The velocity of electron in a certain Bohr's orbit of hydrogen atom bears in the ratio 1: 275 to the velocity of light. The qunatum number (n) of the orbit is ___________.
6. The wavelength of a line in Balmer series of hydrogen atom is 4814 Å. In which orbit deexitation take place?
7. The de Broglie wave length of an electron in a certain orbit of hydrogen atom is 13.3 Å. So the number of waves present in an orbit is ______________.
8. Give the number of angular nodes of orbital belong to the subshell for which minimum value of magnetic quantum number is –2.
9. What will be the number of lines corresponding to Lyman series when an electron is in 3rd excited state of hydrogen atom?
10. In a single hydrogen atom, the electron is excited to its 6th orbit. The maximum no. of distinct lines possible, when it comes to the ground state is ______________.
11. For the hydrogen atom, $E_n = \frac{1}{n^2} R_H$.
Where, $R_H = 2.178 \times 10^{-18}$ J.
Assuming that the electrons in other shells exert no effect, find the minimum no. of Z (atomic no.) at which a transition from the second energy level to the first results in the emission of an X-ray. Given that the lowest energy rays have $\lambda = 4 \times 10^{-8}$ m.
12. Find the quantum number 'n' corresponding to the excited state of He^+ ion if on transition to the ground state that ion emits two photons in succession with wavelengths 1078.5 and 30.4 nm.
13. Energy required to stop the ejection of electrons from Cu-plate 0.27 eV. Calculate the work function (in eV) when radiation of λ = 235 nm strikes down the plate.
14. What is the degeneracy of the level of the hydrogen atom that has the energy $\frac{-R_H}{9}$?
15. The number of visible lines when an electron returns from the 5th orbit to ground state in the hydrogen spectrum is ______________.
16. A photon of wavelength 5000 Å strikes a metal surface, the work function of the metal being 0.475 eV. The kinetic energy of the emitted photo electron is ______________ eV.
17. The de Broglie wave length of on electron in a certain Bohr's orbit H-atom is 6.64 Å.The quantum number of orbit is.
18. In a nonconventional basis, there are three allowed value of spin quantum numbers, then how many more elements can be accommodated in the second period as compared to convertional periodic table.
19. The number of revolutions made by electron in 1 sec in H –atom in its orbit is twice of the number of revolutions made by electron in 1 sec in the 2nd orbit of He^+ ion, then on is.
20. An electron in a hydrogen atom in its ground state has 1-5 times as much energy as the minimum required for its escape from the atom. The velocity of electron in the scientific notation is $x \cdot 10^y\ m/\mathrm{sec}$. Then the value of Y is ______________.
21. In a collection of H-atoms, all the electrons tend to flow to n = 5 to ground level finally (directly or indirectly) without emitting any line in Balmer series the possible different radiations are.

Previous Years' IIT Questions

1. The electrons, identified by quantum numbers n and l, **(1999)**

I. $n = 4, l = 1$,
II. $n = 4, l = 0$,
III. $n = 3, l = 2$ and
IV. $n = 3, l = 1$ can be placed in order of increasing energy, from the lowest to highest as:

(a) IV < II < III < I
(b) II < IV < I < III
(c) I < III < II < IV
(d) III < I < IV < II

2. The number of nodal planes in a p_x orbital is: **(2000S)**

(a) one (b) two
(c) three (d) zero

3. The electronic configuration of an element is $1s^2$, $2s^22p^6$, $3s^23p^63d^5$, $4s^1$. This represents its: **(2000S)**

(a) excited state (b) ground state
(c) cationic form (d) anionic form

4. The wavelength associated with a golf ball weighing 200 g and moving at a speed of 5 m/h is of the order: **(2001S)**

(a) 10^{-10} m (b) 10^{-20} m
(c) 10^{-30} m (d) 10^{-40} m

5. The quantum numbers +1/2 and −1/2 for the electron spin represent: **(2001S)**

(a) rotation of the electron in clockwise and anticlockwise direction respectively
(b) rotation of the electron in anticlockwise and clockwise direction respectively
(c) magnetic moment of the electron pointing up and down respectively
(d) two quantum mechanical spin states which have no classical analogue

6. Rutherford's experiment, which established the nuclear model of the atom, used a beam of: **(2002S)**

(a) β-particles, which impinged on a metal foil and got absorbed.
(b) γ-rays, which impinged on a metal foil and ejected electrons.
(c) helium atoms, which impinged on a metal foil and got scattered.
(d) helium nuclei, which impinged on a metal foil and got scattered.

7. If the nitrogen atom has electronic configuration $1s^7$, it would have energy lower than that of the normal ground state configuration $1s^22s^22p^3$, because the electrons would be closer to the nucleus. Yet $1s^7$ is not observed because it violates: **(2002S)**

(a) Heisenberg uncertainty principle
(b) Hund's rule
(c) Pauli exclusion principle
(d) Bohr postulate of stationary orbits

8. The radius of which of the following orbit is same as that of first Bohr's orbit of hydrogen atom? **(2004S)**

(a) He^+ (n = 2) (b) Li^{2+} (n = 3)
(c) Li^{2+} (n = 2) (d) Be^{3+} (n = 2)

9. The number of radial nodes of 3s and 2p – orbitals are respectively: **(2005S)**

(a) 2,0 (b) 0.2
(c) 1,2 (d) 2,1

10. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius] **(2012)**

(a) $\dfrac{h^2}{4\pi^2 ma_0^2}$ (b) $\dfrac{h^2}{16\pi^2 ma_0^2}$
(c) $\dfrac{h^2}{32\pi^2 ma_0^2}$ (d) $\dfrac{h^2}{64\pi^2 ma_0^2}$

11. Match the following: **(2006)**

Column I	Column II
(a) V_n/K_n = ?	(p) 0
(b) If radius of nth orbital $\propto E_n$ x; x = ?	(q) −1
(c) Angular momentum in lowest orbital	(r) −2
(d) $\frac{1}{r_n} \propto z^y; y = ?$	(s) 1

12. Match the entries in Column-I with the correctly related quantum number(s) in column-II **(2008)**

Column I	Column II
(a) Orbital angular momentum of the electron in a hydrogen like atomic orbital.	(p) Principle quantum number
(b) A hydrogen-like one electron wave function obeying Pauli principle.	(q) Azimuthal quantum number
(c) Shape, size and orientation of hydrogen like atomic orbitals.	(r) Magnetic quantum number
(d) Probability density of electron at the nucleus in hydrogen-like atom.	(s) Electron spin quantum number

Comprehensive Type Questions

The hydrogen-like species Li^{2+} is in spherically symmetrical state S_1, with one radial node. Upon absorbing light the ion undergoes transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom.

(2010)

Answer the following questions:

13. The state S_1 is:
(a) 1s (b) 2s
(c) 2p (d) 3s

14. Energy of the state S_1 in units of the hydrogen atom ground state energy is:
(a) 0.75 (b) 1.50
(c) 2.25 (d) 4.50

15. The orbital angular momentum quantum number of the state S_2 is:
(a) 0 (b) 1
(c) 2 (d) 3

Integer Type Questions

Answer to the following question in a single digit integer, ranging from 0 to 9.

16. The maximum number of electrons that can have principle quantum number, n = 3 and spin quantum numbers $m_s = -\frac{1}{2}$ is ______.

(2011)

17. The work function of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is ______.

(2011)

Metal	**Li**	**Na**	**K**	**Mg**	**Cu**	**Ag**	**Fe**	**Pt**	**W**
eV	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

ANSWER KEYS

Multiple Choice Questions with Only One Answer

Level-I

1. d	**2.** c	**3.** b	**4.** c	**5.** c	**6.** d
7. c	**8.** c	**9.** b	**10.** a	**11.** b	**12.** d
13. d	**14.** d	**15.** b	**16.** b	**17.** c	**18.** d
19. b	**20.** b	**21.** d	**22.** b	**23.** a	**24.** c
25. a	**26.** a	**27.** d	**28.** d	**29.** d	**30.** b
31. d	**32.** a	**33.** d	**34.** d	**35.** a	**36.** b
37. d	**38.** d	**39.** c	**40.** a	**41.** a	**42.** a
43. c	**44.** b	**45.** a	**46.** d	**47.** a	**48.** a
49. b	**50.** a	**51.** b	**52.** c	**53.** c	**54.** d
55. c	**56.** b	**57.** a	**58.** c	**59.** d	**60.** a
61. a	**62.** a	**63.** a	**64.** c	**65.** a	**66.** d
67. c	**68.** d	**69.** d	**70.** c	**71.** d	**72.** d
73. b	**74.** c	**75.** b	**76.** d	**77.** b	**78.** a
79. c	**80.** c	**81.** c	**82.** b	**83.** c	**84.** b
85. a	**86.** b	**87.** a	**88.** a	**89.** a	**90.** c
91. a	**92.** d				

Multiple Choice Questions with Only One Answer

Level-II

1. d	**2.** b	**3.** b	**4.** a	**5.** a	**6.** c
7. d	**8.** d	**9.** b	**10.** c	**11.** b	**12.** a
13. d	**14.** d	**15.** a	**16.** b	**17.** c	**18.** b
19. a	**20.** b	**21.** b	**22.** c	**23.** d	**24.** d
25. b	**26.** c	**27.** d	**28.** b	**29.** a	**30.** c
31. b	**32.** d	**33.** c	**34.** a	**35.** b	**36.** c
37. d	**38.** a	**39.** c	**40.** a	**41.** a	**42.** a
43. a	**44.** a	**45.** a	**46.** c	**47.** b	**48.** b
49. c	**50.** c	**51.** d	**52.** a	**53.** b	**54.** b
55. c	**56.** b	**57.** a	**58.** d	**59.** a	**60.** c
61. a	**62.** b	**63.** a	**64.** c	**65.** d	**66.** b
67. d	**68.** d	**69.** a	**70.** d	**71.** c	**72.** c
73. b	**74.** d	**75.** c	**76.** a	**77.** a	**78.** a
79. c	**80.** c	**81.** c	**82.** b	**83.** a	**84.** d
85. d	**86.** b	**87.** b	**88.** b	**89.** d	**90.** c
91. b	**92.** b	**93.** d	**94.** a	**95.** b	**96.** c
97. b	**98.** a	**99.** d			

Multiple Choice Questions with One or More than One Answer

1. a, b, c, d	**2.** a, c	**3.** a, b, c
4. a, b, c, d	**5.** a, c	**6.** a, b
7. a, c	**8.** b, c, d	**9.** b, c, d
10. b, c, d	**11.** c, d	**12.** b, c
13. a, c, d	**14.** a, b, c	**15.** a, b, c
16. a, b	**17.** a, c	**18.** a, b, c, d
19. b, c	**20.** b, c, d	**21.** a, b, c, d

Comprehensive Type Questions

Passage – I	**1.** d	**2.** b	**3.** b	**4.** c
Passage – II	**1.** b	**2.** b	**3.** a	**4.** b
Passage – III	**1.** d	**2.** d	**3.** d	
Passage – IV	**1.** a	**2.** a	**3.** c	
Passage – V	**1.** c	**2.** d	**3.** b	
Passage – VI	**1.** a	**2.** b	**3.** a	
Passage – VII	**1.** d	**2.** a	**3.** d	

Passage – VIII	**1.** a	**2.** a	**3.** b
Passage – IX	**1.** b	**2.** c	
Passage – X	**1.** b	**2.** c	**3.** b

Matching Type Questions

1.	a-s	b-p	c-q	d-r
2.	a-r, s	b-p, s	c-p, s	d-q, s
3.	a-q, s	b-p, q	c-p, q, r, s	d-q, r
4.	a-p	b-p, q, r, s	c-p	d-p, q
5.	a-q, s	b-r	c-p, q	d-q
6.	a-q	b-p	c-s	d-r
7.	a-r, s	b-p, q, r	c-p, q, r	d-r
8.	a-q, s, r, t	b-q, s, t	c-p, q, s	d-s
9.	a-q, r, s	b-p	c-p, s	d-q, r
10.	a-s	b-q, c	c-pq, s-d	d-p, r, s

Assertion (A) and Reason (R) Type Questions

1. a	**2.** a	**3.** a	**4.** b
5. c	**6.** b	**7.** b	**8.** b
9. b	**10.** a	**11.** a	**12.** a
13. c	**14.** c	**15.** b	

Integer Type Questions

1. 10	**2.** 1	**3.** 625	**4.** 547	**5.** 2
6. 4	**7.** 4	**8.** 2	**9.** 3	**10.** 5
11. 2	**12.** 5	**13.** 5	**14.** 9	**15.** 3
16. 2	**17.** 2	**18.** 4	**19.** 1	**20.** 6
21. 6				

Previous Years' IIT Questions

1. a	**2.** a	**3.** b	**4.** c	**5.** d
6. d	**7.** c	**8.** d	**9.** a	**10.** c
11. (a) r	(b) q	(c) p	(d) s	
12. (a) q, r	(b) p, q, r, s	(c) p, q, r	(d) p, q	
13. 2	**14.** 3	**15.** 2	**16.** 9	**17.** 4

HINTS AND SOLUTIONS

Multiple Choice Questions with Only One Answer

Level-I

2. Mass of electron is less when compared to mass of positive rays

5. $\psi = e^{r/ao}$
Probability density means ψ^2

10. $\lambda = \dfrac{h}{\sqrt{2KEm}}$

12. Angular speed $= \dfrac{V}{r}$

16. Orbital angular momentum $= \sqrt{l(l+1)}\dfrac{h}{2\pi}$

19. $h\,v_1 = h v_o + x$

$h\,v_2 = h v_o + kx$

$$\frac{v_1 - v_o}{v_2 - v_o} = \frac{1}{k}$$

$$v_o = \frac{kv_1 - v_2}{k-1}$$

20. $$\frac{E_2 - E_1}{E_3 - E_2} = \frac{\frac{-13.6}{4} - \frac{-13.6}{1}}{\frac{-13.6}{9} - \frac{-13.6}{4}} = \frac{27}{5}$$

21. $\lambda = \dfrac{h}{\sqrt{2KEm}}$

$$3.31\times10^{-29} = \frac{6.625\times10^{-34}}{\sqrt{2(KE)\times0.1\times10^{-5}}}$$

$\therefore KE = 2\times10^{-4}\,J$

31. Radius of nucleus $= 1.25\times10^{-13}\times(64)^{1/3} = 5\times10^{-13}$cm

$$\text{Fraction of volume} = \frac{\frac{4}{3}\pi(5\times10^{-13})^3}{\frac{4}{3}\pi(10^{-8})^3} = 1.25\times10^{-13}$$

32. The probability of finding election in all directions are same

38. $V = \lambda$

$$\therefore \lambda = \frac{h}{m\lambda}$$

$$\therefore \lambda = \sqrt{\frac{h}{m}}$$

48. n = 3, $l = 2$
So orbital is 3d

52. $\Delta\chi = \Delta p$

$$\Delta p = \sqrt{\frac{h}{4\pi}}$$

$$\therefore \Delta V = \frac{1}{m}\sqrt{\frac{h}{4\pi}}$$

56. $2\pi r = n\lambda$
$2\pi x = \lambda$
In third orbit
$2\pi(9x) = 3\lambda$
$\therefore \lambda = 6\pi x$

66. When reaches nucleus kinetic energy converted to potential energy

$mv = p$

$v = \dfrac{p}{m}$

$\therefore \dfrac{1}{2}mV^2 = \dfrac{P^2}{2m}$

$\therefore \dfrac{p^2}{2m} = \dfrac{Ze^2}{r}$

$\therefore$ *If momentum is* $2P$

$\dfrac{4P^2}{2m} = \dfrac{Ze^2}{r_1}$

$\therefore 4\dfrac{Ze^2}{r} = \dfrac{Ze^2}{r_1}$

$\therefore r_1 = \dfrac{r}{4}$

80. The time taken by one revolution $= \dfrac{2\pi r}{V}$

$\therefore$ Ratio $= \dfrac{2\pi r}{2.188\times10^{-8}} : \dfrac{2\pi(4r)}{2.188\times158/2}$

$\therefore 1:8$

Multiple Choice Questions with Only One Answer

Level-II

1. Orbital angular momentum $= \sqrt{l(l+1)}\dfrac{h}{2\pi}$. For $n = 5$ l values are 0, 1, 2, 3, 4 and these cannot satisfy the given condition.

2. (i) The no. of peaks in a radial probability curve is equal to $n - l$.
So, no. of peaks in 1s and 2s are not equal.
(ii) No. of nodal planes equals to l, so it is same for 2p, 3p and 4p orbitals.
(iii) Ti^{2+} and Ni^{2+} both have same no. of unpaired e^- and have same magnetic moments.
(iv) Any orbital of any sub shell can accommodate only two e^-.

3. M^{+x} and M^{+y} both have same magnetic moment 5.916 which means that both have same no. (5) of unpaired e^-. It is possible only if x and y are +3 and +1. Electron configuration corresponding to these two oxidation states are
$M^{+x} \rightarrow 3s^2\, 3p^6\, 3d^5$
$M^{+y} \rightarrow 4s^1\, 3d^6$
So, M^{+x} is more stable due to the half-filled configuration.

4. According to photoelectric effect eq.

$$\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + E \text{ where } E = KE$$

$$E = hc\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right)$$

de Broglie's wavelength

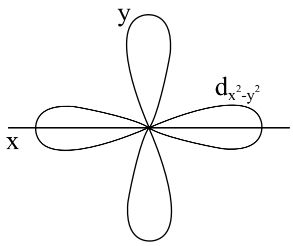
$$\frac{h}{\sqrt{2mE}}, \text{ where } M = \text{mass of } e^- \text{ and } E = KE$$

or
$$\frac{h}{\sqrt{2mhc\left(\dfrac{\lambda_0-\lambda}{\lambda\lambda_0}\right)}} = \sqrt{\frac{h^2(\lambda\lambda_0)}{2mhc(\lambda_0-\lambda)}}$$

$$= \left(\frac{h\lambda\lambda_0}{2mhc(\lambda_0-\lambda)}\right)^{1/2}$$

5. From the observations no of nodal regions for that orbital $n - l - 1 = 2$. xy is a nodal plane. So orbital is d_{yz}

6. Along x, y axis the shape of orbital is as shown below:



8. (i) No. of nodal planes = l = 0
(ii) de Broglie's wavelength $\lambda = \dfrac{h}{mv} = \dfrac{h}{p}$
(iii) The principal quantum no. n indicates the size and energy of orbit.
(iv) p has electronic configuration $3s^2\, 3p^3$.

9. The length of a series depends on the difference between wavelength of successive lines. It is more in Lyman series.

10. Energy given to the $e^- = \dfrac{13.6}{4^2} \cdot 2^2 = 3.4$ eV
Work function = 1.4 eV
The additional energy supplied due to acceleration is 4 eV.

So, net resultant KE = (3.4 – 1.4) + 4 = 6 eV

de Broglie's wavelength $= \dfrac{12.27}{\sqrt{V}}$ Å $= \dfrac{12.27}{\sqrt{6}}$ 5 Å

11. Angular momentum of $e^- = \dfrac{nh}{2\pi}$

So the transition occurs from 4 → 2; that means the transition lies in Balmer series and will be observed in the visible region.

12. de Broglie's wavelength $= \dfrac{h}{\sqrt{2meV}}$

where m is mass of particle, e is charge and V is stopping potential.

So, $\lambda_p : \lambda_a = \dfrac{h}{\sqrt{2m_p \times e \times 0}} : \dfrac{k}{\sqrt{2 \times 4m_p \times 2e \times v}}$

or $\lambda_p : \lambda_a = \dfrac{1}{\sqrt{2}} : \dfrac{1}{4} = 2\sqrt{2} : 1$

13. Energy of emitted photon

$= \left[\dfrac{13.6}{1^2} \times 4 - \dfrac{13.6}{2^2} \times 4\right]$ eV

= 3 × 13.6 eV

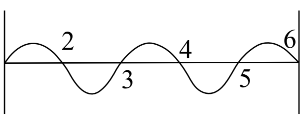
∴ KE of e^- = [3 × 13.6 – 13.6] = 27.2 eV

14. $\Delta X \times \Delta V = \dfrac{h}{4\pi m}$

$\therefore \Delta V = \dfrac{6.625 \times 10^{-34}}{4 \times 3.142 \times 0.1505} \times \dfrac{1}{10^{-10}}$

$\cong 3.4 \times 10^{-24}$ m/s

15. Six nodes are as shown:



So, it has 2.5 waves, so wavelength $= \dfrac{35}{2.5} = 14$ cm.

16. No. of m values = $2l + 1$
So, no. of m values for $l = 1$ is 3
This value is possible for m in case of f and higher sub shells only.

17. To have the same de Broglie's wavelength they must have the same momentum.
$\Rightarrow P_p = P_e \Rightarrow m_p \times v_p = m_e v_e$
$\Rightarrow 1837\, m_e \times v_p = m_e v_e$

So, $v_p = \dfrac{v_e}{1837}$

18. Let us suppose that 'n_a' is the no. of quanta absorbed and 'n_e' is the no. of quanta emitted. n_a and n_e were related as
$n_e = 0.53\, n_a$
So, the absorbed energy $E_a = n_a \times \dfrac{hc}{4530}$

Emitted energy $E_e = n_e \times \dfrac{hc}{5080}$

$\Rightarrow \dfrac{n_e}{n_a} 0.53 n_a \cdot \dfrac{hc}{5080} \times \dfrac{4530}{n_a hc}$

= 0.4726

19. From the data no. of e^- in both cases is $(n - 1)$. So, x has +1 charge and y has +2 charge.

20. Energy involved in transition $E_1 = 2E - E = E = \dfrac{hc}{\lambda}$.

Energy involved in transition $E_1 = \dfrac{4E}{3} - E = \dfrac{E}{3} = 1.$

$\therefore \dfrac{E_1}{E_2} = \dfrac{E}{E/3} = \dfrac{hc/\lambda_1}{hc/\lambda_2} \Rightarrow \dfrac{\lambda_2}{\lambda_1} = 3$

21. From the photoelectric effect

$$h(x) = h\nu_0 + y \qquad (1)$$
$$h(2x) = h\nu_0 + 3y \qquad (2)$$

from Eq. (1) $\rightarrow y = hx - h\nu_0$

$\Rightarrow h(2x) = h\nu_0 + 3[h(x) - h(\nu_0)]$

$\Rightarrow \nu_0 = \dfrac{x}{2}$

22. From the data the difference between radii of ground state and excited state orbit radius
$\Rightarrow r_n - r_1 = 0.7935 \times 10^{-9}$ m
$\Rightarrow r_n - (0.0529 \times 10^{-9}$ m$) = 0.7935 \times 10^{-9}$ m
$\Rightarrow r_n = 0.8464 \times 10^{-9}$ m $= 0.0529 \times 10^{-9} \times n^2$
$\Rightarrow n = 4$
So, the maximum spectral lines formed
$= \dfrac{(n_2 - n_1)(n_2 - n_1 + 1)}{2} = 6$

23. For Balmer series $n_1 = 2$ and for the least wavelength $n_2 = \infty$ and for the highest wavelength $n_2 = 3$.
So, wave no. for spectral line with least wavelength,
$\bar{\nu}_1 = R_H\left[\dfrac{1}{2^2} - \dfrac{1}{\infty^2}\right] = \dfrac{R_H}{4}$
wave no. for spectral line highest wavelength

$\bar{\nu}_2 = R_H\left[\dfrac{1}{2^2} - \dfrac{1}{3^2}\right] = \dfrac{5R_H}{36}$

So, $\bar{\nu}_1 : \bar{\nu}_2 = \dfrac{R_H}{4} : \dfrac{5R_H}{36} = 9 : 5$

24. No. of nodal planes = $l = 2$
No. of Radial nodes = $n - l - 1 = 5 - 2 - 1 = 2$
No. of peaks = $n - l = 5 - 2 = 3$

25. No. of revolutions made by one e^- per sec

$$= \frac{\text{velocity of } e^-}{2\pi r_n} \cong \frac{z^2}{n^3} \times 6.66 \times 10^{15}$$

26. From Rydberg's equation

$$\frac{1}{\lambda} = R\left[\frac{1}{1^2} - \frac{1}{n^2}\right]$$

$$\Rightarrow \frac{1}{n^2} = 1 - \frac{1}{\lambda R} \Rightarrow n^2 = \frac{\lambda R}{\lambda R - 1}$$

28. At radial node i.e., at $r = r_0$, $\psi^2 = 0$.

$$\Rightarrow \left[\frac{1}{2\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{1/2}\left(2 - \frac{r_0}{a_0}\right)e^{-r_0/2a_0}\right]^2 = 0$$

$$\Rightarrow 2 - \frac{r_0}{a_0} = 0 \Rightarrow r_0\, 2a_0$$

29. In case of unielectron systems energy can be determined only from n values i.e., orbital of same shell have same energy.

30. The energy of an e^- in a multielectronic atom can be determined from $(n + l)$ value.

31. (i) Carbonates with thermal stability can be produced by alkali metals. So, — V.
(ii) Coloured compounds will be formed by transition elements. So, — X.
(iii) Largest atomic radius is possible with maximum no. of shells. So, — Y.
(iv) Acidic oxides will be formed by non-metals. So, — W.

32. No. of l values possible for first shell
= 3 (0, 1, 2)
So, the total possible no. of m values are
$n_m = 9(1 + 3 + 5)$
[No. of m values for given $l = 2l + 1$]
So, the no. of e^- that can be fitted = 2 × n.18e^-.

33. According to the given rules
Zr (40) has electronic configuration as follows;
$1s^3\, 2s^3\, 2p^9\, 3s^3\, 3p^9\, 3d^{13}$

35. λ_1 is the wavelength corresponding to initial kinetic energy E_1.
λ_2 is the wavelength corresponding to final kinetic energy E_2 and $\lambda_2 = 0.9\lambda_1$.

$$\lambda_1 = \frac{hc}{\sqrt{2mE_1}} \quad : \quad \lambda_2 = \frac{hc}{\sqrt{2mE_2}}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{1}{0.9} = \sqrt{\frac{E_2}{E_1}}$$

$\Rightarrow E_2 = 1.23456\, E_1$
So, E_2 is 23.456 per cent is more than that of E_1.

36. Angular momentum = $\dfrac{nh}{2\pi} = \dfrac{7h}{11}$
$\Rightarrow \quad = n = 4$

Radius of n^{th} orbit $r_n = \dfrac{0.529 \cdot n^2}{2}$

$\Rightarrow \quad r_n = 2.82$ Å

37. The data indicates that the transition is only possible with Lyman, Balmer, Paschen series that means transition to 4th orbit is not possible which means that transition from 4th orbit occurs.

38. Maximum no. of spectral lines possible
$= \Sigma\,(n - 1) = 15$
$\Rightarrow \quad n = 6$
So, from Rydberg's equation

$$\bar{v} = \frac{1}{\lambda} = R_H\left[\frac{1}{1^2} - \frac{1}{6^2}\right]$$

$$\lambda = \frac{36}{35} \times 912 \text{ Å} \qquad \left[\frac{1}{R_H} = 912 \text{ Å}\right]$$

$= 938.05$

39. Time period $\propto \dfrac{n^3}{Z^2}$

$$\Rightarrow \frac{T_1}{T_2} = \frac{Z_2^2}{Z_1^2} \cdot \frac{n_1^3}{n_2^3}$$

[z is the same]

$$\Rightarrow \frac{2^3}{3^3} = 8 : 27$$

40. Velocity of an e^- in n^{th} orbit

$$= \frac{2.17 \cdot 10^6}{n} \text{ m/s}$$

From the data

$$\frac{2.17 \cdot 10^6}{n} = \frac{3 \cdot 10^8}{4}$$

$\Rightarrow \quad n = 3$
No. of lines in Balmer series is $6 - 2 = 4$

41. $\lambda = \dfrac{h}{\sqrt{2m_e V_0}}$

42. Wavelength of first line in Lyman series

$$\lambda_1 = \frac{1}{R_H} \cdot \frac{3}{4} \times \frac{1}{Z^2}$$

$$\Rightarrow \quad \frac{1}{\lambda} = 109677 \times 9\left[\frac{1}{1^2} - \frac{1}{3^2}\right]$$

$\Rightarrow \quad \lambda = 114$ Å $= 1.14 \times 10^{-6}$ cm

62. If n_a is the number of absorbed quanta and n_e is the no. of emitted quanta, then

$$\frac{n_e}{n_a} = 0.5$$

Emitted energy E_e is 47 per cent of the absorbed energy E_a

$$\text{So,} \qquad \frac{n_e \cdot \dfrac{hc}{X}}{n_a \dfrac{hc}{4700}} = 0.47$$

$$\Rightarrow \qquad \frac{4700}{X} \cdot \frac{n_e}{n_a} = 0.47$$

$$\Rightarrow \qquad X = \frac{4700 \cdot 0.5}{0.47} = 5000 \text{ Å}$$

63. Before the excitation the molecule should undergo the homolytic cleavage. So, energy required for dissociation $E_d = 0.04 \times 400 = 16$ KJ. On dissociation the sample consists of 0.08 moles of H gas atoms.
Energy required for excitation

$$= 0.08 \times 6.023 \times 10^{23}\left[\frac{-13.6}{4} - \left(\frac{-13.6}{1}\right)\right]$$

$= 4.914768 \times 10^{23}$ eV

$$= \frac{4.914768 \times 10^{23} \times 1.602 \times 10^{-19}}{10^3} \text{ KJ}$$

= 78.734 KJ
So, total energy required = 16 + 78.734
= 94.7345 KJ

64. Angular momentum of e^-

$$mvr = \frac{nh}{2\pi}$$

$$\text{or} \quad \frac{nh}{2\pi} = 4.217 \times 10^{-34} \text{ kg.m}^2/\text{s}$$

$\Rightarrow \quad n = 3.999 \cong 4$
The no. of spectral lines in visible region $= n - 2 = 4 - 2 = 2$

65. From the data

$$R_H \times 4 \times \left[\frac{1}{2^2} - \frac{1}{4^2}\right] = R_H\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

By comparison

$$\left(\frac{2}{2}\right)^2 = \frac{1}{n_1^2} \Rightarrow n_1 = 1$$

$$\left(\frac{2}{4}\right)^2 = \left(\frac{1}{n_2}\right)^2$$

$\Rightarrow \quad n_2 = 2$

66. de Broglie's wavelength in n^{th} orbit is
$n\lambda = 2\pi rn$
$\Rightarrow \quad n\lambda = 2\pi r_1 n$ [for hydrogen]
$\Rightarrow \quad \lambda = 2\pi r_1 n = 2\pi r(x)3$
$\Rightarrow \quad \lambda = 6\pi x$

67. If ω is angular velocity and r is radius of orbit, v is velocity of e^-.
$v = r\omega$

$$\Rightarrow \quad \omega = \frac{v}{r} \infty \frac{1}{n^3}$$

68. We know that the no. of waves made by an e^- in an orbit equals to its principle quantum number 'n'.
So, from the figure $n = 4$
No. of revolutions made by an e^- per sec (or)

$$\text{Frequency} = \frac{Z^2}{n^3} \times 6.66 \times 10^{15} \text{ s}^{-1}$$

$$= \frac{1}{(4)^3} \times 6.66 \times 10^{15} = 1 \times 10^{14} s^{-1}$$

69. In case of an elliptical orbit, we know that

$$\frac{\text{length of major axis}}{\text{length of minor axis}} = \frac{n}{k}$$

And length of major axis is double the radius of that circular orbit

$$\Rightarrow \quad \frac{2 \cdot 16 \cdot 0.529}{l} = \frac{4}{2}$$

$\Rightarrow \qquad l = 8.464$ Å

70. Ionization potential in H-like system means, it is the energy of e^- in ground state.

$$\text{So,} \quad E_n = (0.01)\, E_1 = \frac{E_1}{n^2}$$

$\Rightarrow \quad n = 10.$

73. $\Delta E = 12.1 = E_{n_2} - E_{n_1}$

$$\text{So,} \quad 12.1 = \frac{-13.6}{n^2} - (-13.6)$$

$\Rightarrow \quad n = 3$
Therefore, change in angular momentum

$$= \Delta n \frac{h}{2\pi}$$

$$= \frac{(3-1)\times 6.625\times 10^{-34}}{2\pi}$$

$\cong 2.11 \times 10^{-34}$ J s.

74. If E is KE of a particle, de Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mE}}$$

So, if KE doubles, λ becomes $\frac{1}{\sqrt{2}}$ times

75. $\frac{1}{\lambda} = R_H\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

As e^- has double the mass in hypothetical system. Rydberg's constant in the equality will be equal to $2R_H$.

So, $\frac{1}{\lambda} = 2R\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$ [transition is to the first excited level i.e., to 2^n orbit]

$$\Rightarrow \quad \lambda = \frac{18}{5R}$$

77. For, H_β line of Balmer series $n_1 = 2$ and $n_2 = 4$.

So, $hv = \frac{E_1}{n_2^2} - \frac{E_1}{n_1^2}$

[Here, E_1 is –ve of ionization Energy]

$$\Rightarrow 6.000\times 10^{-34}\times v = \left[\frac{-2\times 10^{-18}}{16} - \left(\frac{-2\times 10^{-18}}{4}\right)\right]$$

$\Rightarrow$ v = $6.25 \times 10^{14} = 625 \times 10^{12}$ Hz

78. Radius of n^{th} orbit $r_n = 0.529 \times n^2$ Å

$\Rightarrow$ 8.464 Å = $0.529 \times n^2$

$\Rightarrow$ $n = 4$

Velocity of e^- in n^{th} orbit = $\frac{2\pi Ze^2}{nh}$

$\cong \frac{2.18\cdot 10^6}{n}$ m/s

$\Rightarrow$ Velocity of e^{-1} in 4^{th} orbit = 545×10^3 m/s

79. Electronic configuration of Cr and Cr^{2+} are as given below:

Cr $\rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$

$Cr^{+2} \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^4$

So, no. of unpaired e^- = 4.

Magnetic moment μ = $\sqrt{n(n+2)}$

$= \sqrt{4(4+2)} = 4.89$ BM

80. Electronic configuration of

$Fe^{2+} = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$

So, no. of d e^- in Fe^{2+} = 6

Configuration of Ne — $1s^2\ 2s^2\ 2p^6$

” ” Mg — $1s^2\ 2s^2\ 2p^6\ 3s^2$

” ” Cl^- — $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

81. de Broglie e's wavelength

$\lambda_n = 2\pi r = n\lambda$

So, $\frac{\lambda_1}{\lambda_4} = \frac{2\pi r_1}{2\pi r_1 \times 4}$

$$\lambda_4 = 4\lambda_1 = \frac{2\pi r_4}{4}$$

$\Rightarrow$ Circumference of the fourth orbit = $4\lambda_4$

= $16\lambda_1$

83. Shortest wavelength is possible for any series if $n_2 = \infty$.

Longest wavelength is possible for any series if $n_2 = n_1 + 1$.

So, for hydrogen

$$\frac{1}{x} = R_H\left[\frac{1}{1^2} - \frac{1}{\infty^2}\right]$$

$$\Rightarrow \quad R_H = \frac{1}{x} \qquad \text{(i)}$$

For helium

$$\frac{1}{\lambda} = R_H\cdot Z^2\left[\frac{1}{2^2} - \frac{1}{3^2}\right] \qquad \text{(ii)}$$

From Eqs (i) and (ii),

$$\lambda = \frac{9x}{5}$$

84.

(a) In H-like systems, energy of electron in an orbital increases with increase in 'n'. So, energy of $3d$ is less than energy of $4s$.

(b) In $d_{x^2-y^2}$ orbitals cloud is along x and y-axes. So, it lies in xy plane.

(c) Electronic configuration of Cr — $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$

(d) Except the spin quantum no. the remaining Quantum number are obtained from Schrodinger equation.

86. From the data = $\frac{\text{Velocity of } e^{-1}}{\text{Velocity of light}} = \frac{1}{275}$

$\Rightarrow$ V = 1090909 m/s

Velocity of e^- in n^{th} orbit

$V_n = \frac{2.18\cdot 10^6}{n}$ m/s $\Rightarrow$ $n = 2$

87. If three spin quantum no. were possible, then each orbital can accommodate $3e^-$. So, electronic configuration of K is $1s^3\ 2s^3\ 2p^9\ 3s^3\ 3p^1$.

89. For the line in ultraviolet region, the energy must be more than that corresponding to E.
Energy of A > Energy of E

90. In the given sub shell, half of the max. no. of e^- in sub shell will have the same spin quantum number.

91. To find the distance at which finding probability of e^- is maximum, we have to count $\frac{d^2\psi}{dr^2} = 0$.

$\Rightarrow e^{-r/a_0} = 0 \Rightarrow -\text{In}\,(r/a_0) = -\text{In}\,1$

$\Rightarrow \quad r = a_0$

92. (i) $KE = \frac{hc}{\lambda}$ – Work function

$$= \frac{6.625\times10^{-34}\times3\times10^{8}}{310\times10^{-9}} - (2\times1.602\times10^{-19})$$

$= (2 \times 10^{-19})$

(ii) Total energy: Kinetic energy: Potential energy

$$\frac{-mv^2}{2r} : + \frac{mv^2}{2r} : -\frac{mv^2}{r}$$

1: –1: 2

If magnitude is only considered, ratio, 1: 1: 2.

(iii) No. of maximum (or) no. of peaks
$= n - l = 5 - 4 = 1$

(iv) In terms of KE (E) de Broglie's wavelength

$$\lambda = \frac{h}{\sqrt{2mE}}$$

So, $\lambda_{H}: \lambda_{He}: \lambda_{CH_4} = \frac{1}{\sqrt{1}} : \frac{1}{\sqrt{4}} : \frac{1}{\sqrt{16}}$

$1: \frac{1}{2} : \frac{1}{4} = 4: 2: 1$

93. From the data $r_{(n+1)} - r_n = r_{(n-1)}$

$\Rightarrow \quad [(0.529)(n+1)^2] - [0.529 \times n^2] = (0.529)(n-1)^2$

$\Rightarrow \quad n^2 + 2n + 1 - n^2 = n^2 - 2n + 1$

$\Rightarrow \quad n^2 = 4n \Rightarrow n = 4$

94. Orbital can be represented in terms of wave function as 'ψ_{nlm}' and we know that in general, 0 value for m can be assigned for orbital along with z axis.

95. At nodal surface, ψ^2 value must be equal to zero. By equating $\psi^2 = 0$ we will get the expresion as
$(2 - Zr) = 0$

$$\Rightarrow \quad r = \frac{2}{Z} = \frac{2}{2} = 1\ \text{Å}$$

96. Equation for photoelectric effect
$h\nu = h\nu_0 + \text{KE}$

$\Rightarrow \quad \nu = \nu_0 \times \frac{1}{h} \cdot \text{KE}$

or $\quad \text{KE} = h\nu - h\nu_0$

$y = mx - c$

So, slope of the curve $m = h$.

97. de Broglie' wavelength $\lambda = \frac{h}{\sqrt{2meV}}$

from the curve, we can say that at any point $\lambda_B > \lambda_A$, means that $m_B < m_A$.

98. Velocity of $e^- = \frac{2.18 \cdot 10^6}{n}$ m/s

So, graphic representation must be a hyperbola curve.

99. $\sqrt{\nu} = a(z - b)$

$\sqrt{\nu} = az - ab$

a = slope of the curve = tan 45° = 1

ab = Intercept on the axis = 1

$\Rightarrow \quad \sqrt{\nu} = 52 - 1 \Rightarrow \nu = 2601\ \text{s}^{-1}$

Multiple Choice Questions with One or More than One Answer

2. (a, c) From the data we can write the expression as follows:

$$4.25 = W_A + T_A \quad \text{(i)}$$
$$4.20 = W_B + T_B \quad \text{(ii)}$$
$$T_B = T_A - 1.5 \quad \text{(iii)}$$
$$\frac{h}{\sqrt{2mT_B}} = \frac{2h}{\sqrt{2mT_A}}$$
$$\Rightarrow \quad T_A = 4T_B \quad \text{(iv)}$$

So, from above equations

$T_B = 0.5$ eV

$T_A = 2$ eV

$W_A = 2.25$ eV

$W_B = 4.15$ eV

3. $h\nu = h\nu_0 + \text{KE}$ and stopping potential is equal to K.E.

$$\Rightarrow \quad \frac{hc}{\lambda} = \frac{hc}{\lambda_0} + \nu_0$$

$$\nu_0 = \frac{hc}{\lambda} - hc \cdot \frac{1}{\lambda_0}$$

$$\nu_{0_{(eV)}} = \frac{hc}{e} \cdot \frac{1}{\lambda} - \frac{hc}{e} \cdot \frac{1}{\lambda_0}$$

$y = mx - \text{c}$

So, slope $= m = \frac{6.625\times10^{-34}\times3\times10^{8}}{1.602\times10^{-19}}$

Work function for metal = 0.24 eV

$$\frac{hc}{\lambda} = \omega + v_0$$

$$\Rightarrow \quad v_0 = \frac{6.625\times10^{-34}\times3\times10^{8}}{100\times10^{-9}}$$

$-0.24 \times 1.602 \times 10^{-19}$
$= 1.949 \times 10^{-18}$ J = 12.1 eV

The work function for metal II is more than metal I. So, if both metals were exposed to same light, KE i.e., stopping potential for metal II is less than metal I.

5. (a, c) For the same magnetic moment ions (*a*) atoms should contain same no. of unpaired e^-.

16. BE = Ionization energy of H-like system

$$= \frac{13.6}{n^2}\cdot Z^2 = \frac{13.6}{4^2}\cdot Z^2 = 13.6$$

$$\Rightarrow \quad Z = 4$$

Comprehensive Type Questions

Passage-I

Based on the given conditions,
No. of sub shells = $n + 1$, and
No. of orbitals in a sub shell = $2l + 3$
That means 1st orbit consists of *s* and *p* sub shells with 3 and 5 orbitals and 2nd orbit consists of *s*, *p* and *d* sub shells with 3, 5 and 9 orbitals, respectively.

1. EC with atomic no. 25
$1s^2\ 1p^{10}\ 2s^6\ 2p^3$

2. EC corresponding to atomic No. 18 is
$1s^6\ 1p^{10}\ 2s^2$, so it has two unpaired e^- which causes a magnetic moment of $\sqrt{8}$ BM.

3. After 3*s*, 3*p* and 2*d* have the same ($n + l$) values. So, because of lower *n* value e^- enters into 2*d*.

4. If three spin quantum no. were possible each orbital can accommodate 3 electrons, So, configuration of atom with atomic no. is
$1s^9\ 1p^{15}\ 2s^6$

Passage-II

Because of half of the mass of –ve particle in the hypothetical system. Energy, velocity, radius, expression were as follows:

$$E = \frac{-2\pi^2 Z^2 me^4}{n^2h^2}, \text{ So } E_{\text{hypothetical}} = \frac{E}{2}$$

$$V = \frac{2\pi Ze^2}{nh}, \text{ So } V_{\text{hypothetical}} = V$$

$$r = \frac{n^2h^2}{4\pi^2 me^2}, \text{ So } r_h = 2r$$

Rydberg's constant, $R = \dfrac{2\pi^2 mZ^2e^4}{ch^3}$; So, $R_h = \dfrac{R}{2}$

1. Longest wavelength is possible if transition occurs from the immediate orbit. So, the given transition is $3 \rightarrow 2$.

$$\text{So, } \frac{1}{\lambda} = R_n\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$$

$$= \frac{R}{2}\left[\frac{9-4}{4\times9}\right] \Rightarrow \lambda = \frac{72R}{5}$$

2. $V_n = V = \dfrac{2.188\cdot 10^6}{n}$ m/s

$$= \frac{2.188\cdot 10^6}{2} = 1.094\cdot 10^6 \text{ m/s}$$

3. $r = r_n \times n^2 = (2r)\, n^2 = 0.529 \times 2 \times 9$
$= 9.522$ Å

4. We know that KE and total energy are with opposite sign and are of equal magnitude.

So, $KE_n = -E_n = \dfrac{E}{2} = \dfrac{13.6}{2} = 6.8$ eV/atom

Passage-III

1. We know that

$$E_3 - E_2 = \frac{-13.6\times2^2}{9} - \left(\frac{-13.6}{4}\times Z^2\right)$$

$$\text{So, } 47.2 = 13.6\times2\left[\frac{1}{4} - \frac{1}{9}\right] \Rightarrow Z = 5$$

2. $\dfrac{hc}{\lambda} = (-13.6 \times 25 \times 1.602 \times 10^{-19})$

$\Rightarrow \quad \lambda = 3.6489 \times 10^{-9}$ m

3. $KE = -E = -(13.6 \times 1.602 \times 10^{-12})$ ergs
$= 5.4468 \times 10^{-9}$ ergs

Passage-IV

From the data *A* is the 1st orbit and *B* is nth orbit. After absorption of 2.55 eV, e^{-1} excites to n^1 orbit. The possible no. of spectral lines for de-excitation is $\Sigma\,(n^1 - 1) = 6$.

Which mass that $n^1 = 4$ and *n* may be 2 (or) 3.

From the given data $E_{n^1} - E_n = 2.55$

$$\text{i.e., } \frac{-13.6}{16}\times Z^2 - \left(\frac{-13.6}{n^2}\right) n^2 = 2.55$$

The above expression is possible only if $n = 2$ and $Z = 1$.

3. Minimum energy is possible due to transition of e^- from 4th orbit to 3rd orbit.

$$E = -\frac{13.6}{4} - \left(\frac{-13.6}{9}\right) = 0.6611 \text{ V}$$

Passage-V

1. $\dfrac{hc}{\lambda} = W + KE$

$$\Rightarrow \left(\frac{6.625\times10^{-34}\times3\times10^{8}}{4000\times10^{-10}}\times\frac{1}{1.602\times10^{-19}}\right)\text{eV}$$

$\Rightarrow \quad 2 + KE \Rightarrow KE = 1.10$ eV

2. $KE_1 = hv_1 - hv_0 = h\,(10\times10^{14} - 5\times10^{14})$
$= h\,(5\times10^{14})$
In second case $KE_2 = 2KE_1$
$\Rightarrow \quad h\,(10\times10^{14}) = hv_1 - 5\times10^{14}\times h$
$\Rightarrow \quad v_1 = 15\times10^{14}\,s^{-1}$

3. Stopping potential is the –ve potential that must be applied on collection plate to stop 1 photoelectrons. It will be equal to KE of e^- in (eV.)
So, $V_0 = 25 - 7 = 18$ eV.

Passage-VI

2. $E = \dfrac{hc}{\lambda} = \dfrac{6.625\times10^{-34}\times3\times10^{2}}{3000\times10^{-80}} = 6.625\times10^{-19}$ J

Passage-VII

1. de Broglie wavelength of a cricket ball

$$= \frac{h}{mV} = \frac{6.625\times10^{-34}}{0.2\times30} = 1.1\times10^{-34}\text{ m}$$

2. $\Delta P = \dfrac{h}{4\pi\cdot\Delta x} = \dfrac{6.625\times10^{-34}}{4\pi\times10^{-10}} = 5.272\times10^{-2}$

3. $\Delta x = \Delta P = m\times\Delta V$
So, according to Heisenberg's principle

$$\Delta x\times\Delta P = m^2\,(\Delta V)^2 = \frac{h}{4\pi}$$

$$\Rightarrow \Delta V = \frac{1}{2m}\sqrt{\frac{h}{2\pi}}$$

Passage-VIII

2. Longest wavelength corresponding to the transition $(n+1)\rightarrow n$, so in case of Lyman series the transition is $2\rightarrow1$.

$$\text{So, } \frac{1}{\lambda} = R_H\cdot Z^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$$

$$\frac{1}{\lambda} = R_H\cdot\left[\frac{1}{1}-\frac{1}{\alpha}\right]$$

$$\Rightarrow \lambda = \frac{1}{4R_H} = \frac{912}{4}\text{ Å} = 22.8\text{ nm}$$

3. Shortest wavelength corresponds to the transition of $\infty\rightarrow n$.

$$\text{So, } \frac{1}{\lambda_1} = R_{\text{H}}\left[\frac{1}{2^2}-\frac{1}{\infty^2}\right] = \frac{R_{\text{H}}}{4}$$

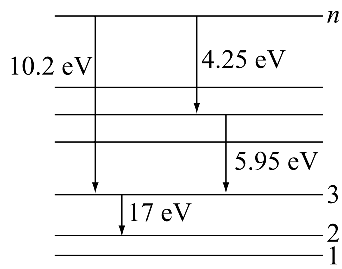
For the longest wavelength

$$\frac{1}{\lambda} = R_{\text{H}}\left[\frac{1}{1^2}-\frac{1}{2^2}\right] = \frac{3R_{\text{H}}}{4}$$

$$\Rightarrow \quad \lambda = \frac{\lambda_1}{3} = 0.33\,\lambda_1$$

Passage-IX

The data can be represented in the form of a diagram as follows:



So, $\quad E_n - E_2 = (10.2 + 17)$

$$\Rightarrow \quad \frac{-13.6}{n^2}\times Z^2 - \left(\frac{-13.6}{2^2}\right)Z^2 = 27.2$$

Similarly,

$$\frac{-13.6}{n^2}\times Z^2 - \left(\frac{-13.6}{3^2}\right)Z^2 = 10.2$$

By solving Eqs. (i) and (ii),
$n = 6$, $Z = 3$

Passage-X

1. Radius $r = \dfrac{n^2h^2}{4\pi^2 mZe^2}$, So for hypothetical, it can be modified as

$$r_m = \frac{n^2h^2}{4\pi^2(207m)(2e)} = \frac{0.529}{414}\text{ Å}$$

2. Ionization energy is the energy of e^- in first orbit with +ve sign.

$$E = \frac{-2\pi^2Z^2me^4}{n^2h^2}\text{ for hypothetical}$$

$$E_m = \frac{-2\pi^2Z^2(207m)(2e)}{n^2h^2}$$
$= 828\times13.6$ eV

3. $r_3 - r_1 = \dfrac{0.529}{414}\left[3^2-1^2\right] = \dfrac{0.529}{414}\times8$ Å

Matching Type Questions

2. **Note:** Molecules at very high temperature are mono-atomic gases that gives a line-spectrum.
6. We know that

$$\frac{r_1}{r_2} = \frac{n_1^2}{n_2^2} \cdot \frac{Z_2}{Z_1}$$

$$\frac{V_1}{V_2} = \frac{Z_1}{Z_2} \cdot \frac{n_2}{n_1}$$

$$\frac{T_1}{T_2} = \frac{n_1^3}{n_2^3} \cdot \frac{Z_2^2}{Z_1^2}$$

$$\Rightarrow \quad \frac{K_1}{K_2} = \frac{n_2^2}{n_1^2}$$

Assertion (A) and Reason (R) Type Questions

1. Peaks in radial distribution curve = $n - l$
   radial nodes = $n - l - 1$
   Which means that the radial probability curves depend only on 'n' and 'l' and not on 'm'.
3. An electron can never be found in the nucleus and this can be explained by uncertainty principle.
   Any object can not move with velocity of light.
4. Reason is the consequence of assertion and it can not explain assertion.
5. The shape of orbital is independent of principal quantum number.
6. Anode rays are +ve ions of different gases and they have different charges and masses. So, e/m ratio for anode rays depends on nature of gas.
7. ψ^2 is the square of amplitude of e^- wave and we know that $I \propto a^2$ where I is intensity and a is amplitude.
8. The no. of angular nodes = l and angular wave functions depend on 'l' and m
9. Diamagnetic nature is due to pairing of e^-
10. In the absence of electric and magnetic fields p_x, p_y and p_z orbitals will have same energy and transition in between p_x and p_y can not give spectral line.
11. In case of H like systems, energy of e^- can be determined only from 'n' values and all the sub shells with same 'n' value will have same energy.
12. The electron can revolve in a particular orbit if it's wave is in phase only. For the He, condition is $n\lambda = 2\pi r$ which can be modified as $mvr = \dfrac{nh}{2\pi}$.
14. + (or) − (ve) sign indicates that electrons are with opposite spins only.
15. As the orbit moves towards the nucleus, the energy difference between successive orbits increases.

Integer Type Questions

1. After the absorption of energy, energy of $e^- = -0.01\ E$
   So $\quad -0.01\ E = \dfrac{-E}{n^2}$
   $\Rightarrow \quad n = 10$
2. 1
3. H_β line is due to the transition of e^- from 4 to 2

   So, $\quad h\nu_z = \dfrac{-2\times10^{-18}}{16} - \left(\dfrac{-2\times10^{-18}}{4}\right)$

   $\nu = 6.25 \times 10^{14}$ or 625
4. Radius of orbit $r_n = 0.529 \times n^2 = 0.8464$
   $\Rightarrow \quad n = 4$

   Velocity of $e^- = \dfrac{2.188 \cdot 10^6}{n}$ m/s

   = 547.00 m/s
5. Velocity of e^-

   $$\frac{2.188 \cdot 10^6}{n} = \frac{3 \cdot 10^8}{275}$$

   $\Rightarrow \quad n = 2$
6. $$\frac{1}{\lambda} = R_H\left[\frac{1}{2^2} - \frac{1}{n^2}\right]$$

   $$\Rightarrow \quad \frac{1}{4814} = \frac{1}{912}\left[\frac{1}{4} - \frac{1}{n^2}\right]$$

   $\Rightarrow \quad n = 4$
7. If λ is de Broglie's wavelength and $\overline{n}$ is principal quantum no,
   $n\lambda = 2\pi r_n = 2\pi \times 0.529 \times n^2$
   $\Rightarrow \quad \lambda = 2\pi \times 0.529 \times \ddot{n}$
   $\Rightarrow \quad n = 4$
8. Minimum value of magnetic quantum no. = $-l$
   So, 'l' for the given orbital is 2.
   No. of angular nodes = $l = 2$
9. 3rd excited state means 4th orbit. So, spectral lines in Lyman series = $n - 1 = 4 - 1 = 3$
10. Max no. of spectral lines = $\Sigma\,(n_2 - n_1)$
    $(6 - 1) = 5$
11. $$\frac{hc}{\lambda} = E_{n_2} - E_{n_1}$$

    $$\frac{hc}{\lambda} = Z^2\left[\frac{1}{1} - \frac{1}{4}\right] \times 2.178 \times 10^{-8}$$

    $\Rightarrow \quad$ Minimum possible number for Z = 2
12. $$E_{n_2} - E_{n_1} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$$

$$\Rightarrow \left[-\frac{13.6}{n^2}-\left(\frac{-13.6}{1}\right)2^2\times 1.602\times 10^{-19}\right]$$

$$= 6.625 \times 10^{-34} \times 3 \times 10^8\left[\frac{1}{108.5\times 10^{-9}}+\frac{1}{30.4\times 10^{-9}}\right]$$

$$\Rightarrow \quad n = 5$$

13. $\frac{hc}{\lambda} = w + \mathbf{eV_0}$

$\Rightarrow$ w = 5 eV

14. Degeneracy means the total no. of degenerate orbitals. In H-like system all the orbitals present in a particular orbit will have the same energy. So, no. of degenerate orbitals (d) degeneracy
$n^2 = 9$

15. No. of visible lines means, lines in Balmer series,
$= n - 2 = 5 - 2 = 3$

16. 16.2

17. $\lambda = 2\pi r_1 n$

$$\Rightarrow \quad n = \frac{6.64}{2\pi\times 0.529} = 2$$

18. In second period 2*s*, 2*p* orbital are to be filled with three e^- in each orbital. So, total no. of elements in second will be 12.

19. No. of revolutions (a) frequency

$$f \propto \frac{z^2}{n^3}$$

$$\text{So, } \frac{f_1}{f_2} = \frac{Z_1^2}{Z_2^2}\cdot\frac{n_2^3}{n_1^3}$$

$$\Rightarrow \quad \frac{2}{1} = \frac{1}{4}\cdot\frac{2^3}{n^2} \Rightarrow n = 1$$

20. From the data KE of e^-
$= 13.6 \times 0.5$ eV

$\Rightarrow \frac{1}{2}mv^2 = 13.6 \times 0.5 \times 1.602 \times 10^{-19}$

$\Rightarrow v = 1.5473 \times 10^6$

So, the correct answer is 6.

21. (6)

Previous Years' IIT Questions

1. Smaller is the value of (n+l), lesser is the energy of sub shell. When (n+l) values are same then energy of sub shell depends on the principal quantum number 'n'; smaller will be the value of 'n' lesser is the energy of sub shell.

(iv) n = 3, $l = 1$ <
(ii) n = 4, $l = 0$ <
(iii) n = 3, $l = 2$ <
(i) n = 4, $l = 1$

2. p_x orbital has only one nodal plane i.e, yz-plane

3. $1s^2$, $2s^22p^6$; $3s^23p^63d^5$, $4s^1$ is ground state electronic configuration of Cr_{24}

4. $\lambda = \frac{h}{mv} = \frac{6.626\times 10^{-34}}{0.2\times 5/3600}$

6. Beam of $\propto\alpha$-particles (nuclei of helium) are used in the Rutherford's gold leaf experiment

7. An orbital can accommodate maximum two electrons with opposite spin and no two electrons in the atom can have same values of all four quantum numbers.

8. $r = \frac{n^2}{z}\cdot\ 0.529\,\mathring{A}$

For Be^{3+}, n = 2, z = 4
$\therefore$ r = 0.529 Å

9. Number of radial nodes = (n–l–1)
Number of radial nodes for 3s = (3–0–1) = 2
Number of radial nodes for 2p = (2–1–1) = 0

10. $$\text{KE} = \frac{2\pi^2 e^{4m}}{4h^2} = \frac{h^2 e^{4m}}{2h^2} = \frac{\pi^2 m}{2h^2}e^4$$

$$a_o = \frac{h^2}{4h^2e^2m}$$

$$\therefore e^4 = \frac{h^4}{16\pi^4 m^2 a^2}$$

$$\therefore \text{KE} = \frac{h^2}{32\pi\ ma^2}$$

Matching Type Questions

11. (a–r) (b–q) (c–p) (d–s)

$$E_n\,(\text{Total energy}) = \frac{-KZe^2}{2r};$$

$$K_n\,(Kinetic\,energy) = -\frac{Z^2}{n^2}\times 2.18\times 10^{-18}\,J = \frac{KZe^2}{2r}$$

$$V_n = potential\ energy = -\frac{KZe^2}{r}$$

$$r_n = Radius\ of\ Orbit = \frac{n^2}{Z}\times 0.529\,\mathring{A}$$

$$\frac{V_n}{K_n} = -2$$

$$r_n \propto E_n^{-1}$$

In the lowest orbit (1s), the angular momentum

$$=\sqrt{l(l+1)}\frac{h}{2\pi}$$

$$=0\,(for\, l=0)$$

$$r_n=\frac{n^2}{Z}\times 0.529\,\mathring{A}$$

$$\frac{1}{r_n}\propto Z^y\ i.e, \propto Z^1\ i.e., y=1$$

12. Orbital angular momentum = $\sqrt{l(l+1)}\frac{h}{2\pi}\cos\theta$

i.e., it depends on azimuthal quantum number, magnetic quantum number

Shape, size and orientation of hydrogen like atomic orbitals is determined by principal, azimuthal and magnetic quantum numbers

Probability density of electron at the nucleus is determined by the principal and azimuthal quantum number

Comprehensive Type Questions

13. 2s orbital is spherically symmetrical and it has one radial node.

Number of radial nodes of 2s orbital = n–l–1=2–0–1=1

14. $E_{S_1}=-\frac{Z^2}{n^2}\times 13.6\,eV=-\frac{3^2}{2^2}\times 13.6\,eV=-\frac{9}{4}\times 13.6\,eV$

$$E_H=-\frac{1^2}{1^2}\times 13.6\,eV$$

$$\therefore \frac{E_{S_1}}{E_H}=\frac{9}{4}=2.25$$

15. $E_{S_2}=E_H$

$$-\frac{Z^2}{n^2}\times 13.6\,eV=-13.6\,eV$$

$$Z^2=n^2$$

$$Z=n=3 \text{ for } Li^{2+}$$

Thus, S_2 will be 3p because it has one radial node

For $3p, n=3, l=1$

Number of radial node $=n-l-1=3-1-1=1$

Integer Type Questions

16. Total number of electrons with (n = 3) = $2n^2$ = 18

Out of these 18 electrons, 9 electrons will have anticlockwise spin $\left(m_s=-\frac{1}{2}\right)$

17. Photoelectrons are ejected only when the energy of absorbed quantum is greater than the threshold energy or work function (∅) of metal.

Absorbed Energy, $E=\frac{hc}{\lambda}=\frac{6.626\times 10^{-34}\times 3\times 10^{8}}{300\times 10^{-9}}$

$$=6.626\times 10^{-19}\,J=\frac{6.626\times 10^{-19}}{1.6\times 10^{19}}eV=4.17eV$$

Thus, four metals, i.e., Li, Na, K, Mg will show photoelectron emission.

Wavelength corresponding to Balmer series.

$$\lambda_2 = \frac{1}{R_H} \times \frac{36}{5} \times \frac{1}{Z^2}$$

$$\Rightarrow \quad \lambda_2 - \lambda_1 = \frac{1}{R_H} \cdot \frac{1}{Z^2}\left[\frac{36}{5} - \frac{3}{4}\right]$$

$$\Rightarrow \quad 91.2 \times \frac{1}{Z^2}\left[\frac{129}{20}\right] = 33.4$$

$$\Rightarrow \quad Z \cong 4$$

43. From the data
$h\,(3.2 \times 10^{16}) = h\nu_0 + 2E$ (i)
$h\,(2.5 \times 10^{16}) = h\nu_0 + E$ (ii)
By solving Eqs. (i) and (ii)
$\nu_0 = 18 \times 10^{15}$ Hz

44. $\dfrac{hc}{\lambda} = E_2 - E_1$

$$\Rightarrow \quad \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$

$$= 1.602 \times 10^{-19}\left[\frac{-13.786}{4} - \left(\frac{-13.786}{1}\right)\right]$$

$= \lambda = 1.199 \times 10^{-7}$ m = 1200 Å

45. Energy of e^- in n^{th} orbit of an H like system,

$$E_n = \frac{-13.6}{n^2} \times Z^2 \text{ eV/atom}$$

So, $\dfrac{E_1}{E_2} = \dfrac{Z_1^2}{Z_2^2} \cdot \dfrac{n_2^2}{n_1^2}$

$$= \frac{1}{16} \cdot \frac{4}{1} = 1:4$$

47. Electronic configuration in second excited state
$= 1s^2\, 2s^2\, 2p^6\, 3s^2\, 3p^3\, 3d^2$
So, max $(n + l + \text{m})$ value for this configuration
$= [[(3 + 1 + (-1)] + (3 + 1 + 0) + (3 + 1 + 1)] + (3 + 2 + 2) + (3 + 2 + 1)] = 25$

48. In any orbit the no. of waves by an e^- equals to its principal quantum no. and total length of all the waves must equal n circumference of the orbit.
$\Rightarrow \quad n\lambda = 2\pi rn$
$\Rightarrow \quad n\lambda = 2\pi r\,(n^2)$

51. Energy of e^- in ground state

$$E_n = \frac{-2\pi^2 Z^2 me^4}{n^2 n^2}$$

R_H = Rydberg constant $= \dfrac{2\pi^2 nZ^2 e^4}{Ch^3}$

So, energy of e^- in ground state configuration
$-R_H \times ch$

52. $\Delta E = \dfrac{hc}{\lambda}$

$$\Rightarrow \quad \lambda = \frac{hc}{\Delta E} = \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{(\Delta E_{e.V.}) \times 1.602 \times 10^{-19}}$$

$= 1.2406 \times 10^{-10}$ m

53. Moseley's equation is $\sqrt{\nu} = a\,(Z - \text{b})$.
So, from the data
$\sqrt{100} = 1\,(Z - 1) \Rightarrow Z = 11$

55. By the given condition, each primary shell should consist of $(n + 1)$ subshells. So, configuration will be $1s^2\, 1p^6\, 2s^2\, 2p^6\, 3s^2\, 2d^6$

57. No. of e^- corresponding for a given l value equals to $2\,(2l + 1)$. For a given n value the possible 'l' values are 0 to $n - 1$.
So, total no. of e^- in a given orbit

$$I = \sum_{i=0}^{n-1} 2(2l+1)$$

58. l value for g sub shell is 4.
(i) Maximum no. of m values $= 2l + 1$. So, for $g = 9$
(ii) Maximum no. of unpaired $e^{-1} = 2l + 1$. So, for $g = 9$.
(iii) The minimum principal quantum no. of $g_{\text{shell}} = 5$.

59. $\Delta x \cdot \Delta v = \dfrac{h}{4\pi m}$

$$\Rightarrow \quad \Delta x = \frac{6.625 \times 10^{-27}}{4 \times 3.142 \times 1} \times \frac{1}{2} = \frac{5.272 \times 10^{-28}}{2} \text{ cm}$$

$\cong 2.64 \times 10^{-30}$ m

60. From the data

$$\frac{hc}{\lambda_1} = h\nu_0 + E_1$$

$$\frac{hc}{\lambda_2} = h\nu_0 + 2E_1$$

By substitution and on solving 1/4 and 3/4
$\Rightarrow \quad \nu_0 = 1.19 \times 10^{15}\ s^{-1}$

61. From the data $n_1 + n_2 = 4$
and $n_1 - n_2 = 2$
$\Rightarrow \quad n_1 = 1$ and $n_2 = 3$
So, from Rydberg's constant

$$\frac{1}{\lambda} = R_H \cdot Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

CHAPTER 2

Periodic Classification

So I begin to look about and write down the elements with their atomic weights and typical properties.
Dmitri Mendeleev

2.1 INTRODUCTION

Chemists have long tried to find patterns in the properties of the elements. Once the atomic weights became available, attempts were made to discover if there was any pattern between these figures and the properties of the elements.

In 1829, **Dobereiner** noticed that particular groups of three elements with similar chemical properties (triads as he called them) had atomic weights, such that the atomic weight of the middle member of the triad was approximately the arithmetic mean of the other two. Some examples are shown in Table 2.1.

Table 2.1 Some Dobereiner triads

Lithium	7	Sulphur	32
Sodium	23	Selenium	79
Potassium	39	Tellurium	127.5
Calcium	40	Chlorine	35.5
Strontium	87.5	Bromine	80
Barium	137.5	Iodine	127

No further constructive work was possible until the appearance of Cannizzaro's unambiguous atomic weight table in 1858, with the consequent allotment of vacancies to atoms as a measure of combining power.

Newlands (1864) was the first to notice that if the known elements were written down in ascending order of atomic weight similar properties recurred in every eight elements, like notes in a musical scale. This system worked well for the lightest elements; for example, it brought the lithium-sodium-potassium triad together but by failing to allow for a bigger octave when dealing with the heavier elements, the theory broke down and was the object of a good deal of ridicule.

Table 2.2 Examples of Newlands octaves

Ist	Element	Li	Be	B	C	N	O	F
	Atomic weight	7	9	11	12	14	16	19
IInd	Element	Na	Mg	Al	Si	P	S	Cl
	Atomic weight	23	24	27	29	31	32	35.5
IIIrd	Element	K	Ca					
	Atomic weight	39	40					

However, Newlands, comparison of the recurrence of chemical properties among the elements with the repitition of musical notes in an octave was unfortunate. Newlands' proposal was received with ridicule and derison by members of the **Chemical Society** and they refused to permit his paper to be published in the journal of this scientific organization. It is said that one member, G.C. Foster by name, asked Newlands, sarcastically "When he had ever examined the elements according to the order of their initial letters". As a lesson in the history of science it might be added that professor Foster's foolish question is the only reason that his name is still remembered. Despite the many serious defects in Newlands, proposal, it did indicate more clearly than any other previous work, the recurrence of similar properties among the elements. In 1887, the Royal Society awarded Newlands, the Davy medal for his discovery of the periodicity existing among the elements.

2.2 LOTHER MEYER'S ARRANGEMENT

In 1870, a German chemist, Lother Meyer used the physical properties such as molar volume, melting point, boiling point, etc. to arrive at his arrangement of elements. He showed that **when the properties of the elements are plotted as a function of their atomic weights, the elements with similar properties occupied almost similar positions.** On this basis, Lother Meyer proposed that when the elements are arrranged in the increasing order of their atomic weights similarities in physical and chemical properties appear at regular intervals.

Figure 2.1 shows a modern representation of this curve using atomic number rather the atomic weight, since this gives more regular shape. Several generalizations can be made from the shape of the curve as follows.

(i) Gaseous, volatile and readily fusible elements are located on the ascending portions of the curve or at the peaks.

(ii) Elements with high melting points are found on the descending portion of the curve and at the broad maxima.

(iii) Chemically similar elements occupy similar positions on the curve; thus, the alkali metals (sodium, potassium, rubidium and caesium) occur at the peaks, the halogens (fluorine, chlorine, bromine and iodine) occur on the ascending portions, immediately followed by the noble gases (neon, argon, krypton and xenon).

Other properties of elements show a similar periodicity, e.g., first ionization energies of atoms.

2.3 MENDELEEV'S CLASSIFICATION

The most important step in developing a periodic classification of the elements was taken in 1869, when the Russian chemist, Mendeleev studied the relationship between the atomic weights and their properties with special emphasis on their valencies. He was led to the conclusion that **the properties of the elements are in periodic dependence on their atomic weights** a conclusion that had previously been cited at by Newlands but never developed. Mendeleev's periodic law can be stated as **the physical and chemical properties of elements are periodic function of their atomic weights.**

On the basis of his periodic law, Mendeleev arranged all the known elements in the form of a table known as **periodic table**.

Mendeleev wrote down the names of the elements in order of increasing atomic weight and noticed that the properties of the elements, with slight modifications, repeated themselves at intervals. Elements with similar properties were arranged vertically beneath each other and by this means a table was eventually constructed. A version that he published is shown in Fig. 2.2.

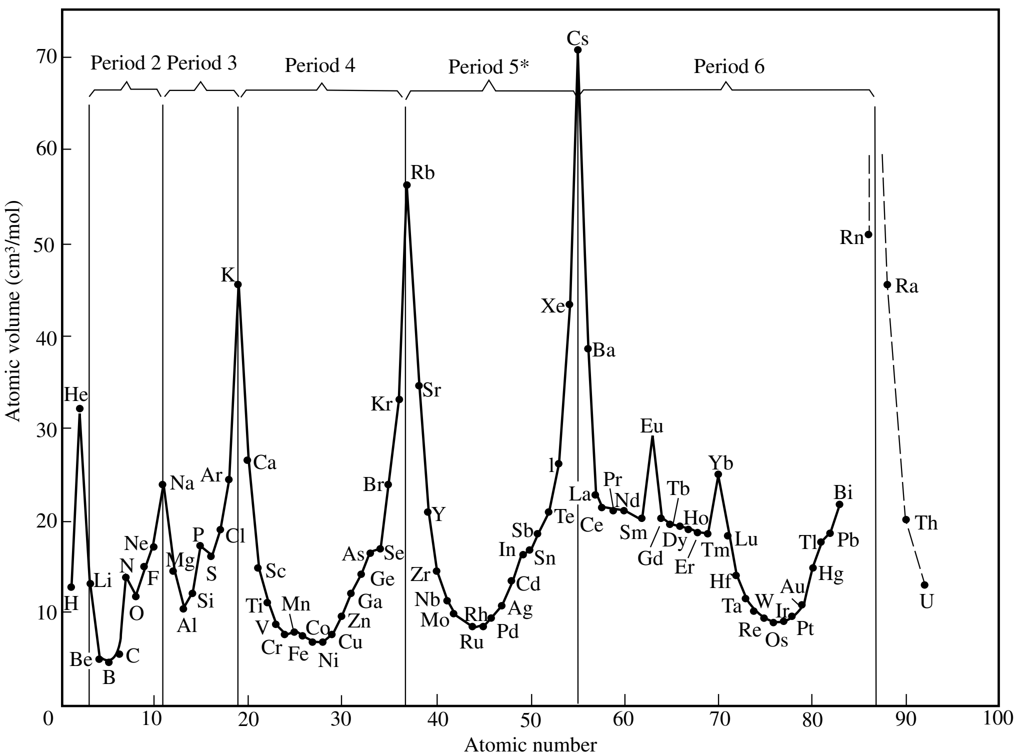


Fig 2.1 Atomic volume (cm^3/mol) plotted against atomic number

Periodic Systems of the Elements in Groups and Series											
Series	**Groups of Elements**										
	0	i	ii	iii	iv	v	vi	vii	viii		ix
1	—	Hydrogen H 1.008	—	—	—	—	—	—			
2	Helium He 4.0	Lithium Li 7.03	Beryllium Be 9.1	Boron B 11.0	Carbon C 12.0	Nitrogen N 14.04	Oxygen O 16.00	Fluorine F 19.0			
3	Neon Ne 19.9	Sodium Na 23.06	Magnesium Mg 24.3	Aluminium Al 27.0	Silicon Si 28.4	Phosphorus P 31.0	Sulphur S 32.06	Chlorine Cl 35.45			
4	Argon Ar 38	Potassium K 39.1	Calcium Ca 40.1	Scandium Sc 44.1	Titanium Ti 48.1	Vanadium V 51.4	Chromium Cr 52.1	Manganese Mn 55.0	Iron Fe 55.9	Cobalt Co 59	Nickel Ni 59
5		Copper Cu 63.6	Zinc Zn 65.4	Gallium Ga 70.0	Germanium Ge 72.3	Arsenic As 75	Selenium Se 79	Bromine Br 79.95			
6	Krypton Kr 81.8	Rubidium Rb 85.4	Strontium Sr 87.6	Yttrium Y 89.0	Zirconium Zr 90.6	Niobium Nb 94.0	Molybdenum MO 96.0	—	Ruthenium Ru 101.7	Rhodium Rh 103	Palladium Pd 106.5
7		Silver Ag 107.9	Cadmium Cd 112.4	Indium In 114.0	Tin Sn 119.0	Antimony Sb 120.0	Tellurium Te 127	Iodine I 127			
8	Xenon Xe 128	Caesium CS 132.9	Barium Ba 137.4			—	—	—	—	—	—
9		—		—	—	—	—	—			
10	—	—	—	Ytterbium Yb 173	—	Tantalum Ta 183	Tungsten W 184	—	Osmium OS 191	Iridium Ir 193	Platinum Pt 194.9
11		Gold Au 192.2	Mercury Hg 200.0	Thallium Tl 204.1	Lead Pb 206.9	Bismuth Bi 208	—	—			
12	—	—	Radium Ra 224	—	Thorium Th 232	—	Uranium U 239				
					HIGHER SALINE OXIDES						
	R	R_2O	RO	R_2O_3	RO_2	R_2O_5	RO_3	R_2O_7	RO_4		
					HIGHER GASEOUS HYDROGEN COMPOUNDS						
					RH_4	RH_3	RH_3	RH			

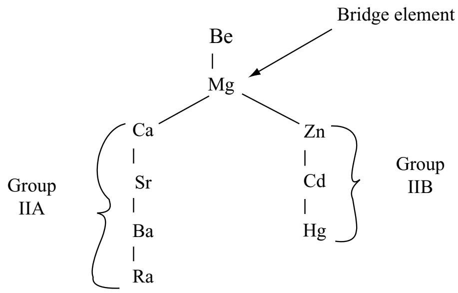
Fig 2.2 Mendeleev's periodic table

2.3.1 Salient Features of Mendeleev's Table

Mendeleev's table consists of nine vertical columns called **'groups'** and seven horizontal rows called **'periods'**. The groups are numbered 0 to VIII. The groups I to VII are subdivided into **A and B subgroups.** VIII group consists of nine elements arranged in three rows containing three elements in each row. The Ist period contains only two elements and is called **very short period.** IInd and IIIrd periods contain eight elements each and are called **short periods**. IVth and Vth periods contain eighteen elements each and are called **long periods**. In the VI period 14 lanthanides are placed in III group along with lanthanium and in VII period 14 actinides are placed in IIIrd group along with actinium. Thus the VI period contains 32 elements and is called the **longest period.** The VII period is the **incomplete period.** The 14 lanthanides and 14 actinides are shown separately at the bottom of the periodic table.

The elements of third period are called **typical elements** because the properties of the elements in a group are studied by taking the third period element as an example. Second period elements cannot be considered as typical elements because they show similarities in some properties with the elements of the same group and in some properties they show similarity with the third period element in the adjacent group (diagonal relationship).

Third period elements are also called as **bridge elements** as the division between two subgroups A and B



starts from these elements. In the second group magnesium acts as a bridge element. The properties of bridge element are some what mixed properties of the elements of two subgroups. For example, magnesium shows similarities with alkaline earth metals (IIA) on one hand and with zinc metals (IIB) on the other.

Mendeleev's periodic table was one of the greatest achievements in the development of chemistry. Some of its advantages and defects are discussed below.

2.3.2 Advantages of Mendeleev's Periodic Table

(i) **Systematic study of the elements:** Mendeleev's table simplified the study of the chemistry of elements. If the properties of one element in a group are known, the properties of other elements can be predicted, because the elements in a group behave similarly but with a gradual change due to increase in atomic number. Similarly, the properties of elements in a period also show a gradual change from left to right with increase in atomic number.

(ii) **Correction of atomic weights:** Mendeleev's table helped in correcting the atomic weights of several elements based on their position in the periodic table. For example, atomic mass of beryllium was corrected from 13.5 to 9.

(iii) **Prediction of new elements:** At the time of Mendeleev only 56 elements were known. While arranging these elements, he left some gaps. These gaps represented the undiscovered elements. From the study of adjacent elements and their compounds, he was able to predict the characteristics of certain elements. These predictions were found to be very accurate. Mendeleev named the elements to be filled in the gaps as Eka aluminium; Eka silicon and Eka boron which are named later as gallium, germanium and scandium, respectively (Eka = similar). Some of the properties predicted by Mendeleev for these elements and those found experimentally are listed in Table 2.3.

2.3.3 Defects of Mendeleev's Periodic Table

In spite of many advantages there are certain defects in Mendeleev's table, some of which are summarized as follows.

(i) **Position of hydrogen:** Hydrogen is placed in group I. However, it resembles the elements of group I (alkali metals) as well as the elements of group VIIA (halogens). So, the position of hydrogen in the periodic table is not certain.

(ii) **Anomalous pairs:** In certain cases the increasing order of atomic weight rule was violated. In these cases, Mendeleev arranged the elements according to similarities in the properties and not in increasing order of their atomic masses. For example, argon (at. wt. 39.9) and potassium (at. wt. 39.1); cobalt (at. wt. 59.9) and nickel (at. wt. 58.6) and tellurium (at. wt. 127.6) and iodine (at. wt. 126.9).

(iii) **Position of isotopes:** According to increasing order of atomic weight rule different isotopes of same element must be given different places in the periodic table because they have different atomic weights. However, isotopes have not been given separate places in the periodic table.

Table 2.3 Comparison of the properties predicted for Eka aluminium (gallium) and Eka silicon (germanium) by Mendeleev with those observed for gallium and germanium

Property	Eka aluminium (predicted)	Gallium (found)	Eka silicon (predicted)	Germanium (found)
Atomic weight	68	70	72	72.6
Density (gm cm^{-3})	5.9	5.94	5.5	5.36
Melting point	Low	30.2	High	1231
Formula of oxide	E_2O_3	Ga_2O_3	EO_2	GeO_2
Formula of chloride	ECl_3	$GaCl_3$	ECl_4	$GeCl_4$

(iv) Similar elements in different groups and dissimilar elements in the same group: Similar elements like copper and mercury which resembled in their properties are kept in different groups. At the same time alkali metals and coinage metals (Cu, Ag, Au) which have different properties are kept in the same group.

(v) Position of lanthanides and actinides: The 14 lanthanides and 14 actinides which have different atomic weights have not been given separate places in Mendeleev's table. They were shown at the bottom of the periodic table.

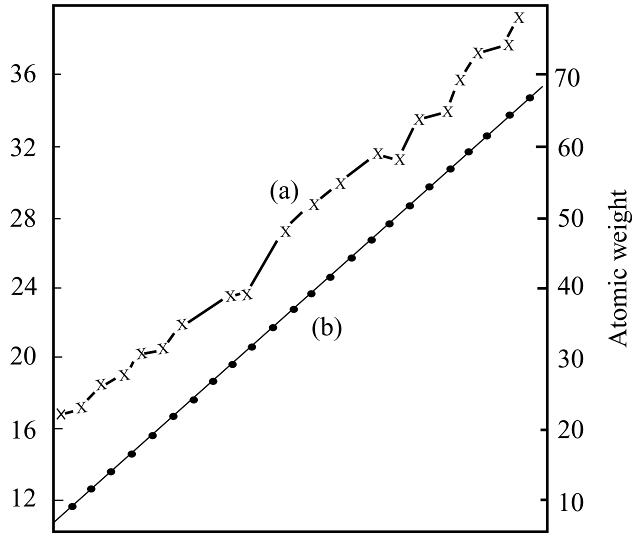
(vi) Elements in VIII group: Arrangement of nine elements in three rows each containing three elements in the VIII group is not justified.

(vii) Cause of periodicity: Mendeleev did not explain the cause of periodicity among the elements.

2.4 ATOMIC NUMBERS: MOSELEY'S PERIODIC LAW

In 1942, Moseley a young English physicist discovered the relationship between X-ray spectra and the atomic number of the elements. When high energy electrons were focussed on a target made of the elements under study, the excess energy appearing mainly in the form of X-radiation. Moseley studied the frequencies of X-rays emitted from the elements and observed that the frequency (ν) of the prominent X-rays emitted by an element was proportional to the atomic number and not the atomic weight. Then, frequencies are related to each other by the equation $\sqrt{\nu} = a(Z - b)$ where ν is the frequency of the X-ray, Z = atomic number of the element in the periodic table, i.e., nuclear charge a,b are the constants for any selected type of the line. A plot of ν against 'Z' gives a straight line as shown in Fig. 2.3. However, no such relationship was obtained when the plot was drawn between frequency and the atomic mass. The atomic number of the elements, according to Moseley stands for serial number (or order number) of the element in the periodic table. As the atomic numbers of the elements increase, the wave lengths of the characteristic X-rays decrease. This led Moseley to conclude that **atomic number and not atomic weight is the fundamental property of the atoms.** He therefore suggested that atomic number instead of atomic weight should be the basis of classification of elements.

> This led to the **modern periodic law** as the physical and chemical properties of the elements are periodic functions of their atomic numbers.



(a) Plot of atomic weight against $\sqrt{\text{Frequency}}$
(b) Plot of atomic number against $\sqrt{\text{Frequency}}$

Fig 2.3 Linear relationship between Z and $\sqrt{\nu}$

By changing the periodic law using atomic weight to atomic number two defects in Mendeleev's table were corrected.

(i) Anomalous pairs were eliminated. Though the atomic weights of anomalous pairs as discussed in section 2.3.3(ii) are in decreasing order but their atomic numbers are in increasing order.

(ii) Since all the isotopes have same atomic number (though they have different atomic weights) there is no need to give different places for them in the periodic table.

Though these two defects are eliminated the other defects in Mendeleev's periodic table remain. So, several scientists tried to develop better classification methods from time to time.

2.4.1 Nomenclature of Elements with Atomic Number Above 100

The elements after uranium do not occur naturally. They are synthesized and characterized in the laboratories containing highly sophisticated costly equipment. Generally in earlier days the elements are named traditionally by its discoverer or discoverers. However, in recent years disputes arise regarding the original discoveries of certain elements. For example, both American and Soviet scientists claimed credit for discovering the element 104. The Americans named it Rutherfordium, whereas Soviets named it Kurchatovium. To avoid such problems, the IUPAC recommended that until a new element's discovery is proved, and its name is officially recognized, a systematic nomenclature be derived directly from the atomic number of the element using the numerical roots for 0 and numbers 1 to 9 as shown in Table 2.4. The roots are put together in order of digits which make up the atomic number and "ium" is added at the end. The IUPAC names for elements with Z above 100 are shown in Table 2.5.

2.5 THE LONG FORM PERIODIC TABLE

The periodic table appearing most frequently in recent textbooks in general chemistry is the extended, or long form of the table. This extended form is shown in Fig. 2.4. It is sometimes referred to as the Bohr table, since it apparently follows Bohr theory of electronic arrangements for the various types of atoms. However, credit cannot be assigned to any single person for this widely used table. It was first used in a simple form by Rang in 1893, a more modern arrangement was advocated by Burry in 1921. The long form periodic table is constructed on the basis of repeating ***electronic configuration***.

The long form periodic table was constructed by unfolding the A,B subgroups of the long periods of Mendeleev's table. In this table, the vertical columns are termed as "groups" and the horizontal rows as "periods". There are 18 groups and 7 periods in this table. All the elements are arranged in the increasing order of their atomic numbers. The atomic number increases by one unit from one element to the immediate next element as we move from left to right in the periodic table.

This periodic arrangement is obviously an outgrowth of Mendeleev's table and is a noteworthy attempt to overcome some of the latter table's defects. In the extended form of the periodic table the first two periods are broken, inorder to place the elements of these short periods in the families to which they clearly belong. In this table Mendeleev's subgroups are eliminated. Transitional elements fall together in the middle portion of the table. It also affords satisfactory position for Fe, Co and Ni and the

Table 2.4 Notation for IUPAC nomenclature of elements

Digit	Name	Abbreviation
0	nil	n
1	un	u
2	bi	b
3	tri	t
4	quad	q
5	pent	p
6	hex	h
7	sept	s
8	oct	o
9	enn	e

Table 2.5 Nomenclature of elements with atomic number above 100

Atomic number	Name	Symbol	IUPAC official name	IUPAC symbol
101	Unnilunium	Unu	Mendelevium	Md
102	Unnilbium	Unb	Nobelium	No
103	Unniltrium	Unt	Lawrencium	Lr
104	Unnilquadium	Unq	Rutherfordium	Rf
105	Unnilpentium	Unp	Dubnium	Db
106	Unnilhexium	Unh	Seaborgium	Sg
107	Unnilseptium	Uns	Bohrium	Bh
108	Unniloctium	Uno	Hassnium	Hs
109	Unnilennium	Une	Metnerium	Mt
110	Ununnilliium	Uun	Darmstadium	Ds
111	Unununnium	Uuu	Rontgenium*	Rg*
112	Ununbium	Uub	*	*
113	Ununtrium	Uut	+	
114	Ununquadium	Uuq	*	*
115	Ununpentium	Uup	+	
116	Ununhexium	Uuh	*	*
117	Ununseptium	Uus	+	
118	Ununoctium	Uuo	+	

* Official IUPAC name yet to be announced; + Elements yet to be discovered.

d-Transition elements

1 H $1s^1$

Period Number	Representative elements Groups Number 1 IA	2 IIA	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8	9 VIII	10	11 IB	12 IIB	Representative elements Group Number 13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	Noble gases 18 0
1																		2 He $1s^2$
2	3 Li $2s^1$	4 Be $2s^2$											5 B $2s^2 2p^1$	6 C $2s^2 2p^2$	7 N $2s^2 2p^3$	8 O $2s^2 2p^4$	9 F $2s^2 2p^5$	10 Ne $2s^2 2p^6$
3	11 Na $3s^1$	12 Mg $3s^2$											13 Al $3s^2 3p^1$	14 Si $3s^2 3p^2$	15 P $3s^2 3p^3$	16 S $3s^2 3p^4$	17 Cl $3s^2 3p^5$	18 Ar $3s^2 3p^6$
4	19 K $4s^1$	20 Ca $4s^2$	21 Sc $3d^1 4s^2$	22 Ti $3d^2 4s^2$	23 V $3d^3 4s^2$	24 Cr $3d^5 4s^1$	25 Mn $3d^5 4s^2$	26 Fe $3d^5 4s^2$	27 Co $3d^7 4s^2$	28 Ni $3d^8 4s^2$	29 Cu $3d^{10} 4s^1$	30 Zn $3d^{10} 4s^2$	31 Ga $4s^2 4p^1$	32 Ge $4s^2 4p^2$	33 As $4s^2 4p^3$	34 Se $4s^2 4p^4$	35 Br $4s^2 4p^5$	36 Kr $4s^2 4p^6$
5	37 Rb $5s^1$	38 Sr $5s^2$	39 Y $4d^1 5s^2$	40 Zr $4d^2 5s^2$	41 Nb $4d^4 5s^1$	42 Mo $4d^5 5s^1$	43 Tc $4d^5 5s^2$	44 Ru $4d^7 5s^1$	45 Rh $4d^8 5s^1$	46 Pd $4d^{10}$	47 Ag $4d^{10} 5s^1$	48 Cd $4d^{10} 5s^2$	49 In $5s^2 5p^1$	50 Sn $5s^2 5p^2$	51 Sb $5s^2 5p^3$	52 Te $5s^2 5p^4$	53 I $5s^2 5p^5$	54 Xe $5s^2 5p^6$
6	55 Cs $6s^1$	56 Ba $6s^2$	57 *La $5d^1 6s^2$	72 Hf $4f^{14} 5d^2 6s^2$	73 Ta $5d^3 6s^2$	74 W $5d^4 6s^2$	75 Re $5d^5 6s^2$	76 Os $5d^6 6s^2$	77 Ir $5d^7 6s^2$	78 Pt $5d^9 6s^1$	79 Au $5d^{10} 6s^1$	80 Hg $5d^{10} 6s^2$	81 Tl $6s^2 6p^1$	82 Pb $6s^2 6p^2$	83 Bi $6s^2 6p^3$	84 Po $6s^2 6p^4$	85 At $6s^2 6p^5$	86 Rn $6s^2 6p^6$
7	87 Fr $7s^1$	88 Ra $7s^2$	89 Ac** $6d^1 7s^2$	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Uuu	112 Uub	—	114 Uuq	—	116 Uuh	—	—

f-Inner transition elements

6	* Lanthanide series $4f^n 5d^{0-1} 6s^2$	58 Ce $4f^1 5d^1 6s^2$	59 Pr $4f^3 5d^0 6s^2$	60 Nd $4f^4 5d^0 6s^2$	61 Pm $4f^5 5d^0 6s^2$	62 Sm $4f^6 5d^0 6s^2$	63 Eu $4f^7 5d^0 6s^2$	64 Gd $4f^7 5d^1 6s^2$	65 Tb $4f^9 5d^0 6s^2$	66 Dy $4f^{10} 5d^0 6s^2$	67 Ho $4f^{11} 5d^0 6s^2$	68 Er $4f^{12} 5d^0 6s^2$	69 Tm $4f^{13} 5d^0 6s^2$	70 Yb $4f^{14} 5d^0 6s^2$	71 Lu $4f^{14} 5d^1 6s^2$
7	** Actinide series $5f^n 6d^{0-1} 7s^2$	90 Th $5f^0 6d^2 7s^2$	91 Pa $5f^2 6d^1 7s^2$	92 U $5f^3 6d^1 7s^2$	93 Np $5f^4 6d^1 7s^2$	94 Pu $5f^6 6d^0 7s^2$	95 Am $5f^7 6d^0 7s^2$	96 Cm $5f^7 6d^1 7s^2$	97 Bk $5f^9 6d^1 7s^2$	98 Cf $5f^{10} 6d^0 7s^2$	99 Es $5f^{11} 6d^0 7s^2$	100 Fm $5f^{12} 6d^0 7s^2$	101 Md $5f^{13} 6d^0 7s^2$	102 No $5f^{14} 6d^0 7s^2$	103 Lr $5f^{14} 6d^0 7s^2$

Fig 2.4 Long form of the Periodic Table of the elements with their atomic numbers and atomic weights. The groups are numbered 1-18 in accordance with the 1984 IUPAC recommendations. This notation replaces the old numbering scheme of IA-VIIA, VIII, IB-VIIB and 0 for the elements

platinum metals, which in Mendeleev's table were artificially placed in group VIII. The inert gases are placed at the end of the periods where they fit more satisfactorily. There are many other apparent advantages of the long form over Mendleev's table.

2.5.1 Salient Features of Long form Periodic Table

(i) Periods: In terms of electronic structure of the atom a period constitutes a series of elements whose atoms have the same number of electron shells, i.e., principal quantum number (n). There are seven periods and each period starts with the filling of a new orbit in an alkali metal and ends with complete filling of the p-subshell of the same orbit in an inert gas.

The **first period** correspond, to n = 1 is unique because it contains only two elements. This is not surprising because first energy shell has only one orbital (i.e., 1s) which can accommodate only two electrons. This means that there can be only two elements in which one and two electrons are present in the first energy level. The first period contains hydrogen ($1s^1$) and helium ($1s^2$).

The **second period** contains 8 elements because for n = 2 there are four orbitals (one 2s and three 2p). In all, these four orbitals have a capacity of eight electrons and therefore second period has eight elements. It starts with lithium (Z = 3) in which one electron enters the 2s-orbital. The period ends with neon (Z = 10) in which the second shell is complete ($2s^22p^6$) with neon (Z = 10).

In the **third period** elements the outermost orbit is 3. It contains nine orbitals (one 3s, three 3p and five 3d). Since the energy of 3d-orbital is more than 4s-orbital, in the elements of this period only 4 orbitals (3s and 3p) are filled with electrons. So, this period contains eight elements from sodium (Z = 11) to argon (Z = 18).

The **fourth period** corresponding to n = 4 involves the filling of one 4s- and three 4p-orbitals. Further, 4d and 4f are higher in energy than 5s-orbital and are filled later. In between 4s- and 4p-orbitals five 3d-orbitals are filled which have energies between these orbitals. Thus in all 9 orbitals are to be filled and therefore, there are 18 **elements** in the fourth period from potassium (Z = 19) to krypton (Z = 36).

The **fifth period** similar to the fourth period also consists of 18 elements. It begins with rubidium (Z = 37) with filling of 5s-orbital and ends with xenon (Z = 54) with the completely filling of the 5p-orbitals.

The **sixth period** contains 32 elements (Z = 55 to 86) and successive electrons enter into 6s-, 4f-, 5d- and 6p-orbitals, in that order. It starts with caesium (Z = 55) and ends at radon (Z = 86).

The **seventh period** is an incomplete period and contains only 19 elements at present.

This arrangement of elements provides a theoritical justification for periodicity occurring at regular intervals of 2, 8, 8, 18, 18, 32 and 32. These numbers, i.e., 2, 8, 18 and 32 are called **magic numbers.**

(ii) Groups: In terms of electronic structure of the atom, the elements in a group have the same outer electronic configurations. For example, the electronic configuration of IA group elements is given in Table 2.6.

There are 18 groups in the long form of the periodic table. According to the recommendations of the IUPAC, these groups are numbered from 1 to 18. Previously, they were numbered only from I to VIII as A and B groups. Both the systems of numbering the groups is given in the periodic table. However, the old convention is still used in many books.

The first two groups on the extreme left and last six groups on the extreme right involve the filling of s- and p-orbitals, respectively. These groups are called **main groups** and are numbered as 1, 2, 13, 14, 15, 16, 17 and 18 corresponding to general configurations of ns^1, ns^2, ns^2np^1, ns^2np^2, ns^2np^3, ns^2np^4, ns^2np^5 and ns^2np^6, respectively. The elements present in these groups are called as **normal** or **representative** elements.

In the middle 10 groups of the long form periodic table the (n–1)d orbitals are filled with electrons. Since a d-orbital can accommodate 10 electrons, there are 10 groups and are numbered 3 to 12. The elements present in these groups are called **transition elements.**

In the 14 lanthanides and in 14 actinides, which are shown at the bottom of the periodic table separately as two rows, the f-orbitals are filled with electrons. Since an f-orbital can accommodate 14 electrons, each series also contain 14 elements.

The number of elements and the corresponding orbitals being filled are given in Table 2.7.

2.5.2 Classification of Elements into s-, p-, d- and f-block Elements

The electronic configurations of two successive elements differ only in the last electron. For example, electronic configurations of beryllium and boron are

Be	$1s^22s^2$
B	$1s^22s^2\,2p^1$

The last electron which distinguishes the electronic configurations of two successive elements is called a **differentiating electron**. Depending upon the orbital into which the differentiating electron enters, the elements are classified into s-, p-, d- and f-block elements.

Table 2.6 Electronic configuration of IA group elements (alkali metals)

Elements	At. No. (Z)	Electronic configuration
Li	3	$1s^2, 2s^1$
Na	11	$1s^2, 2s^2 2p^6, 3s^1$
K	19	$1s^2, 2s^2 2p^6, 3s^2 3p^6, 3d^{10}, 4s^1$
Rb	37	$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6, 5s^1$
Cs	55	$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6 4d^{10}, 5s^2, 5p^6, 6s^1$
Fr	87	$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6 4d^{10} 4f^{14}, 5s^2 5p^6 5d^{10}, 6s^2 6p^6, 7s^1$

Table 2.7 Number of elements (corresponding to electrons) in each period

Period (n)	Orbitals being filled up	Number of electrons or elements in the period	
First 1	1s	2	= 2
Second 2	2s 2p	2 + 6	= 8
Third 3	3s 3p	2 + 6	= 8
Fourth 4	4s 3d 4p	2 + 10 + 6	= 18
Fifth 5	5s 4d 5p	2 + 10 + 6	= 18
Sixth 6	6s 4f 5d 6p	2 + 14 + 10 + 6	= 32
Seventh 7	7s 5f 6d 7p	2 + 14 + 10 + 6	= 32

* At present, the 7th period is incomplete and contains about 23 elements.

(i) s-Block elements: The elements in which the differentiating electron (last electron) enters into s-orbital of their outermost orbit are called **s-block elements.**

Their general outer electronic configuration is either ns^1 or ns^2. The elements corresponding to ns^1 configuration are called **alkali metals**, while those corresponding to ns^2 configuration are called **alkaline earth metals**. The elements in the first two groups of the periodic table belong to s-block elements. The general characteristics are as follows.

(i) except hydrogen all these elements are metals

(ii) they are soft metals with low melting and boiling points

(iii) their ionization energies are low.

(iv) alkali metals exhibit +I and alkaline earth metals exhibit +II oxidation states

(v) they form ionic compounds

(vi) they are strong reducing agents

(ii) p-Block elements: The elements in which the differentiating electron enters into p-orbital of outermost orbit are called **p-block elements.** The last six groups of periodic table belong to p-block.

Their general outer electronic configuration is $ns^2\, np^{1-6}$. The only p-block element in which the differentiating electron does not enter into p-orbital is helium. Their general characteristics are as follows.

(i) p-block consists of metals, non-metals and metalloids. There is a gradual change from metallic to non-metallic character as we move from left to right in a period in p-block elements.

(ii) Their ionization energies are relatively high as compared to s-block elements.

(iii) Generally they form covalent compounds (mostly).

(iv) Most of these elements exhibit variable valence (show more than one oxidation state in their compounds).

(v) Gradually the reduction power decreases and oxidation power increases as we move from left to right in a period in the p-block elements.

p-block consists of six groups: (i) Group IIIA/13 is called **boron family.** The outer electronic configurtion of these elements is ns^2np^1. These elements exhibit +I and +III oxidation states.

(ii) Group IVA/14 is called **carbon family.** The outer electronic configuration of these elements is ns^2p^2. These elements exhibit + II and +IV oxidation states.

s-block

	I	II
1s	H	
2s	Li	Be
3s	Na	Mg
4s	K	Ca
5s	Rb	Sr
6s	Cs	Ba
7s	Fr	Ra

d-block

	⟵			Transition Elements				⟶	IB	IIB
3d	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
4d	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
5d	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
5d	Ac									

p-block

	III	IV	V	VI	VII	0
						He
2p	B	C	N	O	F	Ne
3p	Al	Si	P	S	Cl	Ar
4p	Ga	Ge	As	Se	Br	Kr
5p	In	Sn	Sb	Te	I	Xe
6p	TI	Pb	Bi	Po	At	Rn

f-block

Lanthanides 4f	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
Actinides 5f	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

Fig 2.5 Classification of elements into s-, p-, d- and f-block elements

(iii) Group VA/15 is called **nitrogen family.** The outer electronic configuration of these elements is ns^2np^3. These elements exhibit –III, +III and +V oxidation states.

(iv) Group VIA/16 is called **oxygen family.** The outer electronic configuration of these elements is ns^2np^4. These elements can gain two electrons to get stable inert gas configuration. As a result they can form –II ions. Also, they can share two electrons and form covalent molecules. They also exhibit +II, +IV and +VI oxidation states.

(v) Group VIIA/17 is called **halogen family.** The outer electronic configuration of these elements is ns^2np^5. These are more electronagative elements and their electron gain enthalpies are more. So, they can gain an electron easily to get stable inert gas configuration and can convert into –I ion. Also, they can share their outer electrons. So, they can exhibit –I, +I, +III, +V, +VII oxidation states in their covalent compounds.

(vi) Group 0/18 is called **inert gases.** The outer electronic configuration of these elements is ns^2np^6. These elements are very stable and so do not exhibit chemical reactivity.

If a diagonal line is drawn starting from boron to astatine, in the p-block, non-metals are found above the line, metals are found below the line while metalloids appear along the line. Thus we find the most non-metallic elements at extreme right top corner of the periodic table and the most metallic elements at the extreme left bottom side of the periodic table.

(iii) d-block elements: In these elements the differentiating electron enters into (n–1)d orbital. The middle 10 groups (3–12) of the periodic table belongs to d-block, since the five d-orbitals can accommodate totally a maximum of 10 electrons. The general outer electronic configuration of d-block elements is $(n–1)d^{1–10}\ ns^{1 \text{ or } 2}$ where $n \geq 4$. As 3d-, 4d- and 5d-orbitals are filled with electrons three series of elements each with 10 members are resulting. The fourth due to the filling of the 6d-orbitals is incomplete.

(a) 3d - series starting from scandium (z = 21) to zinc (z = 30) in the fourth period.
(b) 4d - series starting from yttrium (z = 39) to cadmium (z = 48) in the fifth period.
(c) 5d - series starting from lanthanum (z – 57) to mercury (z = 80) in the sixth period.
(d) 6d - series starting from actinium (z = 89) and is incomplete. These elements appear in the seventh period of the table.

The general characteristics of d-block elements are

(i) they are all metals having high melting and boiling points
(ii) they are good conductors of electricity
(iii) they form coloured compounds
(iv) they form complex compounds
(v) their compounds are generally paramagnetic
(vi) the metals and their compounds have catalytic activity
(vii) they exhibit variable valency

(iv) f-block elements: In these elements the differentiating electron enters into (n–2) f orbital. The two series of elements, viz., lanthanides and actinides which are placed at the bottom of the periodic table belong to f-block. Their general outer electronic configuration is $(n–2)\ f^{1–14}\ (n–1)\ d^{0\ or\ 1}\ ns^2$.

In the first series of 14 elements after lanthanum starting from cerium (at. no. 58) to lutetium (at. no. 71) the differentiating electron enters into 4f-orbital. So, they are known as 4f series. Since the properties of 4f series elements are similar to lanthanum they are known as **lanthanides** or **lanthanons** or **lanthanoids**.

In the second series of 14 elements after actinium starting from thorium (Z - 90) to lawrencium (Z = 103) the differentiating electron enters into 5f-orbital. So, they are known as 5f series. Since the properties of these elements are similar to actinium they are known as **actinides or actinons or actinoids**.

The general characteristics of f - block elements are

(i) they are heavy metals
(ii) they generally have high melting and boiling points
(iii) they exhibit variable valence
(iv) they form coloured compounds and their compounds are paramagnetic
(v) they have a tendency to form complex compounds
(vi) Actinides are radioactive in nature

2.5.3 Classification of the Elements Based on Properties

The elements in the periodic table can also be classified into four types on the basis of their properties and electronic configuration.

(i) Noble gases: The elements belonging '0' group or group 18 of the periodic table are called noble gases or inert gases or rare gases. They are He, Ne, Ar, Kr, Xe and Rn. Except He ($1s^2$), all the inert gases have the general outer electronic configuration ns^2np^6. All these elements are monoatomic gases. They are chemically inactive. Their chemical inactivity is attributed to the presence of an octet in their outer most orbit of their atoms. In case of He, the noble character is due to its completely filled K shell. In these elements the outermost orbit is considered to be completely filled.

(ii) Representative elements: All the elements of s- and p-block with the exception of noble gases are called representative elements. Their general outer electronic configuration is ns^1 to ns^2np^5. In these elements the outermost orbit is incompletely filled. So, they are highly reactive elements. These elements acquire the nearest inert gas configuration by losing or by gaining or by sharing electrons. Some elements may get pseudo inert gas (presence of 18 electrons in the outermost orbit) configuration. Representative elements consists of metals, non-metals and few metalloids.

(iii) Transition elements: The d-block elements are called transition elements because they show transition from metallic character of the s-block elements to non-metallic character of p-block elements and also from the ionic character of the compounds of s-block elements to the covalent character of the compounds of p-block elements. Their general outer electronic configuration is $(n–1)d^{1–10}\ ns^{1 or 2}$. In these elements the outer most two, ultimate and penultimate, shells are incompletely filled. Corresponding to the incompletely filled 3d-, 4d-, 5d- and 6d-orbitals, there are four transition series.

The small size of their atoms, high nuclear charge and presence of unpaired electrons in their d-orbitals impart some characteristic properties to the transition elements. They are as follows.

(i) They are hard and heavy metals with high densities.
(ii) They have high melting and boiling points.
(iii) They are good conductors of heat and electricity.
(iv) They exhibit variable valency.
(v) The elements and their compounds are paramagnetic in nature.
(vi) They form coloured compounds.
(vii) The elements and their compounds have catalytic activity.
(viii) They form alloys. **e.g.,** brass, bronze, german silver, etc.

Zinc, cadmium and mercury are not considered as transition elements because these elements do not exhibit the characteristic properties of transition elements due to the presence of completely filled d-orbitals. But they are studied along with the transition elements to rationalize their chemistry.

(iv) Inner transition elements: The f-block elements are called inner transition elements because in these elements the differentiating electron enters into one more orbit inner to the orbit in which the differentiating electron enters into the orbit in transition elements and form another transition series inside a transition series. Their general outer electronic configuration is $(n–2)f^{1–14}\ (n–1)\ d^{0 or 1}ns^2$. In these elements the outermost three ultimate, penultimate and antipenultimate shells are incompletely filled. As the outer two shell electronic configurations are same, their physical and chemical properties are closely similar. **E.g.,** oxidation state of +3 is shown by most of these elements.

There are two series of inner transition elements. The first series is known as lanthanides (4f-orbitals are filled) This series has 14 elements starting from cerium (Z = 58) to lutetium (Z = 71). The second series is called as actinides (5f-orbitals are filled). These actinides also include 14 elements starting from thorium (Z = 90) to lawrencium (Z = 103). The elements after uranium are called **transuranium elements.** They are all man made, synthetic and artificial. All the transuranium elements are radioactive and disintegrate into other elements.

2.5.4 Classification of Elements into Metals, Non-metals and Metalloids

In addition to the above two types of classifications the elements in the periodic table can also be classified into **metals**, **non-metals** and **metalloids** basing on their properties. Metals comprise more than 78 per cent of all known elements and appear on the left side of the periodic table. Except mercury all the metals are solids (Gallium and caesium also have very low melting points 303 and 302K, respectively). Metals usually have high melting and boiling points. They are good conductors of heat and electricity. They are malleable (can be made into thin sheets) and ductile (can be made into wires). They have bright lustre.

On the other hand, non-metals are present on the right-hand side of the periodic table. They are usually solids or gases. Bromine is the only liquid non-metal. Generally non-metals have low melting and boiling points. They are poor conductors of heat and electricity. Most non-metallic solids are brittle and are neither malleable nor ductile.

In general, the elements become more metallic as we go down a group and non-metallic character increases as we move from left to right along a period in the periodic table. It may be noted that the change from metallic to non-metallic character is not abrupt as shown by zig-zag line in Fig. 2.4. The elements bordering the line (silicon, germanium, arsenic, antimony, tellurium, etc) show properties characteristic of both metals and non-metals. These are called **metalloids or** semi-metals.

Prediction of period, group and block of an element:

The period, group or block of an element can be predicted from the electronic configuration of the elements.

1. The period number of an element is equal to the principal quantum number value of the valence shell.
2. Block of the element corresponds to the subshell into which the differentiating electron enters.
3. Group number of an element is equal to the number of electrons present in the valence shell or penultimate shell as follows.
   For s-block elements group number is equal to number of valence electrons (ns electrons)
   For p-block elements group number is equal to 10 + number of valence electrons (ns + np)
   For d-block elements group number is equal to number of electrons in (n–1)d and ns subshells.

2.6 PERIODIC TRENDS IN PROPERTIES OF ELEMENTS

There are many physical and chemical properties of the elements which show periodic variation with atomic number. For example within a period, chemical reactivity tends to be high in group I metals, lesser in the elements present in the middle of the table and increases to a maximum in the group 17 non-metals. Similarly with-in a group of representative metals (e.g., alkali metals) reactivity increases on moving down the group, whereas within a group of non-metals (halogens) reactivity decreases down the group.

Cause of periodicity: The recurrence of similar properties of the elements after certain regular intervals when they are arranged in the order of increasing atomic numbers is called periodicity. But why do the properties of elements follow these trends? and how can we explain the periodicity? To answer these questions, we must look into the theories of atomic structure and properties of the atom.

The physical and chemical properties of the elements depend mainly on the distribution of the electrons in the various shells of their atoms. Generally the properties of the elements do not depend much on the arrangement of electrons in the inner shells but mainly depends on the arrangement of electrons in the outermost shell (called the valence shell). For example, the electronic configurations of the atoms of alkali metals in group I (Table 2.8) shows that they all have same outer electronic configuration ns^1 preceded by the noble gas configuration (ns^2np^6 but for He, ns^2).

Thus the cause of periodicity of the properties of elements is the repetition of similar electronic configuration of their atoms in the outermost shell (or valence shell) after certain regular intervals.

The alkali metals have a great tendency to lose the single electron in order to acquire a stable noble gas configuration. Consequently, all of them are very reactive metals. Similarly all the members of the halogen family have outer electronic configuration ns^2np^5, one electron short of inert gas configuration. Naturally the halogens have a tendency to gain one electron to acquire stable inert gas configuration. Thus all the halogens form halide ion, X^-, readily in their compounds.

There are numerous physical properties of elements such as melting and boiling points, heats of fusion and vapourization, energy of atomization, etc., which show

Table 2.8 Electronic configuration of alkali metals and halogens

Element	At. No.	Electronic configurations	Element	At. No	Electronic configurations
Li	3	[He] $2s^1$	F	9	[He] $2s^22p^5$
Na	11	[Ne] $3s^1$	C*l*	17	[Ne] $3s^23p^5$
K	19	[Ar] $4s^1$	Br	35	[Ar]$3d^{10}$ $4s^24p^5$
Rb	37	[Kr] $5s^1$	I	53	[Kr]$4d^{10}$ $5s^25p^5$
Cs	55	[Xe] $6s^1$			
Fr	87	[Rn] $7s^1$			

periodic variations. However, the periodic trends with respect to some physical properties are discussed here.

Before going to discuss about the periodic trends of some properties we must have knowledge of effective nuclear charge and screening or shielding effect.

2.6.1 Shielding or Screening Effect: (Effective Nuclear Charge)

The energy of an electron in an atom is a function of Z^2/n^2. Since the nuclear charge (= atomic number) increases more rapidly than the principal quantum number, one might expect that the energy necessary to remove an electron from an atom would continually increase with increasing atomic number. This is not so, as can be shown by comparing hydrogen (Z = 1) with lithium (Z = 3). The ionization energy of hydrogen is 1312 KJ mol^{-1} while that of lithium is 520 KJ mol^{-1}. The ionization energy of lithium is less than hydrogen because of two reasons (i) The average radius of a 2s electron is greater than that of 1s electron and (ii) The $2s^1$ electron in lithium is repelled by the inner core $1s^2$ electrons. Another way of treating this inner core repulsion is to view it as shielding or screening of the nucleus by the inner electrons, so that the valence electron actually experience only a part of the total charge. **This decrease in the force of attraction exerted by the nucleus on the valence electron due to the presence of electrons in the inner shells is called screening effect or shielding effect**. The magnitude of the screening effect depends upon the number of inner electrons, i.e., higher the number of inner electrons, greater shall be the value of screening effect. The actual attractive power of the nucleus leaving the valence electrons after shielded by the inner electrons is called **effective nuclear charge** and represented by Z*. Z* is always less than Z (actual nuclear charge) and can be calculated by Z* = Z – S

S is the screening constant.

The shielding constant S can be calculated by following **Slater** rules which are empirical.

1. The electronic configuration of the elements should be written in the following order and groupings (1s) (2s,2p) (3s,3p) (3d) (4s,4p) (4d) (4f) (5s,5p), etc.
2. Electrons in any group which are right to the (ns,np) group do not contribute anything to the shielding constant.
3. All the electrons in the (ns,np) group (valence shell or outermost orbit) shield the valence electron to an extent of 0.35 each.
4. All the electrons in the n–1 shell (penultimate shell) shield to an extent of 0.85 each.
5. All the electrons in n–2 shell or lower, shield completely that is their contribution is 1.00 each.
6. All the electrons in groups lying to the left of 'nd' or 'nf' group contribute 1.00.

E.g. 1 Nitrogen ($_7N$): Electronic configuration $1s^22s^22p^3$ grouping the orbitals $(1s)^2(2s,2p)^5$

$S = (2 \times 0.85) + (4 \times 0.35) = 3.10$

$Z^* = Z - S = 7 - 3.10 = 3.9$

E.g. 2 Zinc ($_{30}$ Zn). Electronic configuration $1s^22s^22p^6 3s^23p^63d^{10}4s^2$

Grouping the orbitals $(1s)^2$ $(2s2p)^8$ $(3s3p)^8$ $(3d)^{10}$ $(4s)^2$

$S\ (10 \times 1.0) + (18 \times 0.85) + (1 \times 0.35) = 25.65$

$Z^* = Z - S = 30 - 25.65 = 4.35$

E.g. 3 Screening constant for 3d electron in zinc.

Grouping of orbitals is as in **E.g. 2.**

$S = (18 \times 1.00) + (9 \times 0.35) = 21.15$

$Z^* = 30 - 21.15 = 8.85$

It can be seen that the rules are an attempt to generalize and to quantify those aspects of the radial distributions discussed in structure of atom. For example, d and f electrons are screened more effectively (S = 1.00) than s and p electrons (S = 0.85) by the electrons lying immediately below them. Slater's rules assume that all electrons of s, p, d, or f shield electrons lying above them equally well (in computing shielding, the nature of shielded electron is ignored). This is not quite true and will lead to some error. For example, in the Ga atom

(....... $3s^2$ $3p^6$ $3d^{10}$ $4s^2$ $4p^1$) the rules imply that the 4p electron is shielded as effectively by the 3d electrons as by the 3s and 3p electrons contrary to radial probability functions of 3s, 3p and 3d orbitals.

Slater formulated these rules in proposing a set of orbitals for use in quantum mechanical calculations. Slater orbitals are basically hydrogen-like but differ in two important respects.

1. They do not contain nodes. This simplifies them considerbly but makes them less accurate.
2. They make use of Z* in place of Z and for heavier atoms n is replaced by n* where for n = 4, n* = 3.7; n = 5, n* = 4.0; n = 6, n* = 4.2. The difference between n and n* is referred to as quantum defect.

To remove the difficulties and inaccurancies in the simplified Slater treatment of shielding, Clementi and Raimondi have obtained effective nuclear charges from self-consistent field wave functions for atoms from hydrogen to krypton and have generalized these into a set of rules for calculating the shielding of any electron. The discussion of these rules is beyond the scope of this book, but such values are listed in Table 2.9.

2.6.2 Atomic Radii

The term radius for an atom or ion is usually considered as **the distance from the centre of the nucleus to the outer shell of the atomic particle**. It is obviously impossible to isolate an individual atom or ion and determine its radius, especially since its size or radius is affected by the association of other particles. But the internuclear distance of the bonded atoms can be measured using X-ray diffraction or electron diffraction techniques and it is taken as standard.

The atomic radius depends on many factors like

(i) the number of bonds formed by the atom
(ii) nature of bonding
(iii) the oxidation number of the atoms
(iv) the coordination number of the atoms
(v) the repulsive forces between atoms not directly bonded
(vi) the type of hybridization

Three types of atomic radii are considered based on the nature of bonding. They are (i) covalent radius, (ii) van der Waal's radius and (iii) crystal radius or atomic radius.

(i) Covalent radius: Covalent radius is half the internuclear distance of two atoms, held together by a covalent bond in the homoatomic molecule.

For example, the single-bond distance between carbon and carbon is 154 pm. The atomic radius of the carbon atom is one half of the carbon-carbon bond or 77 pm. In the case of heteroatomic molecules the bond length is also equal to the sum of the covalent radii of two atoms. So if we know the covalent radius of one atom and the bond length of the heteroatomic diatomic molecule, the covalent radius of other atom can be calculated. For example, the C–Si bond length in silicon carbide is 194 pm and the covalent radius of carbon as calculated above is 77 pm. So, the covalent radius of silicon is 194 – 77 = 114 pm which is equal to half the internuclear distance of Si – Si covalent bond of 234 pm. The covalent radii of those elements which normally form covalent bonds are summarized in Table 2.10.

Effect of number of bonds on covalent radii: If the number of bonds between two atoms increases the bond length decreases. Thus the covalent radius of same elements may change with change in the number bonds formed by the atom. Some numerical values are given in Table 2.11.

By comparing the values given in Table 2.11 with those for single-bond covalent radii given in Table 2.10, it will be seen that the double-bond radius of an atom is about 13 per cent and triple-bond radius about 22 per cent less than the corresponding single-bond radius.

Effect of ionic character on normal covalent radii: The sum of the covalent radii of two atoms in a hetero atomic diatomic molecule only gives the bond length, when the bond is purely covalent or nearly so. In other cases the measured bond length is less than the sum of the covalent radii for the bonded atoms, an effect which is attributed to the partial ionic character. In such cases the bond length of AB molecule is calculated as follows.

$$\text{Bond length of A} - \text{B} = r_A + r_B - 0.09\,(\chi_A - \chi_B)$$

where r_A and r_B are the covalent radii of A and B and χ_A and χ_B their respective electronegativities. This expression, which is purely empirical, gives good agreement between calculated and measured bond lengths in many cases, but it fails to account for all the measured bond lengths. The value of constant 0.09 depends on the type of atoms involved as follows. For bonds involving atoms of 2nd period it is 0.08 and for those Si, P and S bonded to more electronegative atoms not belonging to 1st period, it is 0.06.

Effect of hybridization on normal covalent radii: The single bond distance between two similar atoms is also effected by the hybridization. For example, the C – C single bond lengths in sp^3– sp^3, sp^3–sp^2, sp^2 – sp^2, sp^2 – sp and sp – sp hybrid orbital overlaps are given in Table 2.12.

It is obvious that the covalent radius calculated using C – C single bond lengths shown in Table 2.12 are different and depend on the type of hybridization.

Table 2.9 Effective nuclear charges for elements 1 to 36

Element	1s	2s	2p	3s	3p	4s	3d	4p
H	1.000							
He	1.688							
Li	2.691	1.279						
Be	3.685	1.912						
B	4.680	2.576	2.421					
C	5.673	3.217	3.136					
N	6.665	3.847	3.834					
O	7.658	4.492	4.453					
F	8.650	5.128	5.100					
Ne	9.642	5.758	5.758					
Na	10.624	6.571	6.802	2.50				
Mg	11.619	7.392	7.826	3.308				
A*l*	12.591	8.214	8.963	4.117	4.066			
Si	13.575	9.020	9.945	4.903	4.285			
P	14.558	9.825	10.961	5.642	4.886			
C*l*	16.524	11.430	12.993	7.068	6.116			
Ar	17.508	12.230	14.008	7.757	6.764			
K	18.490	13.006	15.027	8.680	7.726			
Ca	19.473	13.776	16.041	9.602	8.658	4.398		
Sc	20.457	14.574	17.055	10.340	9.406	4.632	7.120	
Ti	21.441	15.377	18.065	11.033	10.104	4.817	8.141	
V	22.426	16.181	19.073	11.709	10.785	4.981	8.983	
Cr	23.414	16.984	20.075	12.368	11.466	5.133	9.757	
Mn	24.396	17.794	21.084	13.018	12.109	5.283	10.528	
Fe	25.381	18.599	22.089	13.676	12.778	5.434	11.180	
Co	26.367	19.405	23.092	14.322	13.435	5.576	11.855	
Ni	27.353	20.213	24.095	14.961	14.085	5.711	12.530	
Cu	28.339	21.020	25.097	15.594	14.731	5.858	13.201	
Zn	29.325	21.828	26.098	16.219	15.369	5.965	13.878	
Ga	30.309	22.599	27.091	16.996	16.204	7.067	15.093	6.222
Ge	31.294	23.365	28.082	17.760	17.014	8.044	16.251	6.780
As	32.278	24.127	29.074	18.596	17.850	8.944	17.378	7.449
Sc	33.262	24.888	30.065	19.403	18.705	9.758	18.477	8.287
Br	34.247	25.643	31.056	20.218	19.571	10.553	19.559	9.028
Kr	35.232	26.398	32.047	21.033	20.434	11.316	20.626	9.769

Table 2.10 Single bond covalent radii of some elements (pm) values given in parentheses are crystal radius (metallic radius)

H	Be	B	C	N	O	F
30	107 (111)	80	77	74	74	72
	Mg 140 (160)	Al 126 (143)	Si 117	P 110	S 104	Cl 99
		Ga 126 (122)	Ge 122	As 121	Se 117	Br 114
		In 144 (162)	Sn 140	Sb 141	Te 137	I 133

Table 2.11 Multiple bond covalent radii (pm)

B =	C =	N =	O =
76	67	60	55
B ≡	C ≡	N ≡	O ≡
68	60	55	50
	Si =		S =
	107		94

(ii) van der Waal's radius: This type of radius is used only for molecular substances present in the solid state. The forces holding the molecules together are weak van der Waal's forces. Half the internuclear distance of two non-bonded neighbouring atoms of two adjacent molecules in the solid state which are close to each other due to van der Waal's attractive forces is called **van der Waal radius.**

For example, the distance between two adjacent chlorine atoms of different molecules is 360 pm. Then the van der Waal's radius of chlorine atom is 360/2 = 180 pm. Since the van der Waal's forces of attractions are weak, the atoms do not come into contact with one another. It is found that the van der Waal's radius is approximately 40 per cent more than the covalent radius.

(iii) Crystal radius: This type of radius is applicable for metal atoms. The metal atoms treated as spheres in close contact with each other in the metallic crystal.

Half the internuclear distance of two adjacent atoms in a metal crystal is called crystal radius or atomic radius. For example, the internuclear distance of two sodium atoms in a sodium metal crystal is 372 pm. So, the atomic radius of sodium is 372/2 = 186 pm. The atomic radii of some elements are given in the Table 2.14.

In metals, the atomic radius is slightly bigger than the corresponding covalent radius as shown in Table 2.15.

The metallic radii are generally more than the covalent radii of elements because in a crystal the component atoms are not as much close to one another as in the bonded atoms. So the order of different radii of same atom will be van der Waal's radius > crystal radius > covalent radius.

Variation of atomic radius in a group: The atomic radii of elements increases from top to bottom in a group. In a group the effective nuclear charge Z* increases very slowly from one element to another element. For example, using Slater's rules the effective nuclear charge Z* of IA group elements are as follows.

Element	H	Li	Na	K	Rb	Cs
Z*	1.0	1.3	2.2	2.2	`2.2	2.2

However, while going from one atom to another there is increase in the principal quantum level. The effect of increase in the size of the electron cloud is more pronounced; consequently, the distance of the outermost electron from the nucleus increases gradually down a group. In other words, the size of the atom goes on increasing as we move down a group. The variation in atomic radius of alkali metals (IA group) and halogens (VIIA or 17th group) are given in Table 2.16.

The van der Waal's radius of neon is more than the van der Waal's radius of flourine eventhough there is increase in the effective nuclear charge. The reason can be traced to greater electronic repulsions. In noble gases all the electrons in their atoms are paired. The repulsive forces become stronger than the attractive forces between the nucleus and the electrons. Hence, the radii increase.

Every period starts with an electron entering s-sub level of new orbit. After the first electron has entered the s-subshell, when the second electron enters the same subshell, the resulting decrease in the atomic radius is

Table 2.12 Effect of atom hybridization on the C – C bond length

C – C Hybridization	Valence – Bond structure	C – C single bond length (pm)	Covalent radius of C
sp^3– sp^3	$\geq C-C \leq$	154	77
sp^3–sp^2	$\geq C-C \lessdot$	150	75
sp^3–sp	$\geq C-C \equiv$	146	73
sp^2 – sp^2	$\gtrdot C-C \lessdot$	147	73.5
sp^2 – sp	$\gtrdot C-C \equiv$	142 – 147	71 – 73.5
sp – sp	$\equiv C-C \equiv$	138	69

Table 2.13 van der Waal's radii of some elements

				H	120		
N	150	O	140	F	135	Ne	160
P	190	S	185	C*l*	180	Ar	190
As	200	Se	200	Br	195	Kr	200
Sb	220	Te	220	I	215	Xe	220

Table 2.14 Atomic radii of some metals (pm)

Li	152	Be	112	Sc	160
Na	186	Mg	160	Fe	126
K	231	Ca	197	Zn	133
Rb	244	Sr	215		
Cs	262	Ba	217		

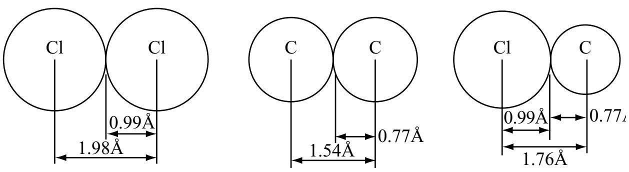


Fig 2.6 Covalent radius

Table 2.15 Comparison of atomic and covalent radii of element

	Radius (pm)			Radius (pm)	
Element	**Atomic**	**Covalent**	**Element**	**Atomic**	**Covalent**
K	231	203	C	77	77
Ba	217	198	P	110	110
La	183	169	S	104	104
Zr	157	145	Br	114	114

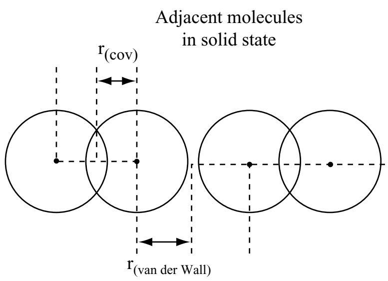


Fig 2.7 van der Waal's radius (comparison between covalent and van der Waal's radii)

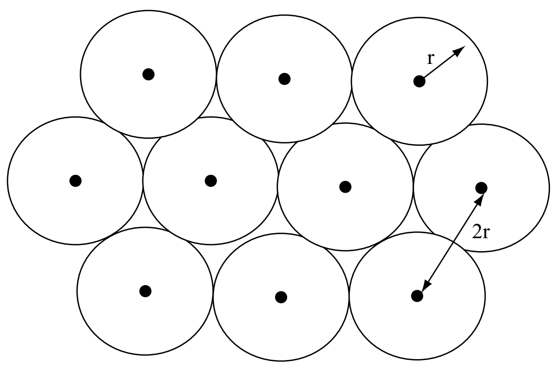


Fig 2.8 Crystal radius in a metal crystal

Table 2.16 Atomic radii (pm) in the groups

Group IA Alkali metals		Group VII A or 17 Halogens	
Element	**Atomic radius**	**Element**	**Atomic radius**
Li	152	F	72
Na	186	C*l*	*99*
K	231	Br	114
Rb	244	I	133
Cs	262	At	140

significant. But the decrease in the atomic radius, while the p-, d-, f-subshells are being filled is normal.

Variation of atomic radius in transition elements: The atomic radii of the transition elements are intermediate between those of s- and p-block elements. The atomic radii of the d-block elements are given in Table 2.17.

The atomic radii of elements of a particular transition series decreases with increase in atomic number but the decrease is very small. With increase in atomic number, nuclear charge goes on increasing progressively. However, the electrons enter the penultimate shell and the added d-electrons screen the outermost s-electrons. Since the d-orbitals have poor shielding effect, the net electrostatic attraction between the nuclear charge and the outermost electrons increases. Consequently, the atomic radius decreases but the decrease is little when compared to s- and p-block elements. After manganese, pairing of electrons starts. Between the paired electrons there will be repulsion which neutralizes the increased attractive force with increase in the nuclear charge. So the atomic radii remain constant in the middle of each series. At the end of each series there is a slight increase in the atomic radii. As the number of electron pairs increases, the electron-electron repulsions become predominant than the increased nuclear charge. This causes the increase in atomic radii at the end in every series.

Variation of atomic radius in inner-transition series: In the lanthanoid series, with increase in atomic number, the nuclear charge increases by one unit and one electron is added. The new electrons are added to the same inner 4f-sub shells. The shielding effect of 4f electrons on one another is quite poor due to the more diffused shapes of the f-orbitals. Hence with increase in nuclear charge, the effective nuclear charge experienced by the valence electrons increases. As a result there will be contraction in the sizes of the atoms of the successive elements. The contraction in the size from one element to the next element in the lanthanide series is fairly small. But the cumulative effect over 14 lanthanides from Ce to Lu is about 20 pm. This sum of the successive decrease in size of lanthanides is called **lanthanide contraction.**

> The cumulative decrease in the atomic sizes of 14 lanthanide elements due to the poor shielding effect of 4f orbitals is called lanthanide contraction.

Due to lanthanide contraction, the crystal structure and other properties of the elements and their compounds become very close. As a consequence, it becomes difficult to separate them from the mixture. Due to lanthanide contraction the atomic sizes of zirconium and hafnium become equal due to which the properties of these two elements and their compounds are almost similar. Because of this reason these two elements occur together and their separation from one another is very difficult. Because of lanthanide contraction the atomic sizes of post-lanthanide elements in the third transition (5d) series are almost become equal to the corresponding second transition (4d) series elements which are just above them. Therefore the 4d and 5d transition elements are more closer in properties than those exhibited by the elements of 3d and 4d transition elements. The atomic radii of lanthanides are given in Table 2.18.

The atomic sizes of the elements in actinides series also decrease slowly and gradually as in lanthanide elements due to poor shielding effect of 5f-orbitals. This results in **actinide contraction.** The atomic radii of some actinide elements are given in Table 2.19.

Table 2.17 Atomic radii of d-block elements (pm)

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
144	132	122	119	119	117	116	115	117	125
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
162	145	134	129	125	124	125	128	134	141
La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
169	144	134	130	128	126	126	129	134	144

2.6.3 Ionic Radii

The ions are formed as a result of addition or removal of electrons to or from the outermost shell of atoms. The ions formed by the loss of electrons acquire positive charge and are called **cations**, while the ions formed by gaining of electrons are called **anions**.

> Ionic radius may be defined as the distance from the nucleus of the ion to the point where the attractive force of nucleus on electrons ceases.

If the ions are regarded as spheres, the internuclear distance may be taken equal to the sum of the radius of the cation and the radius of the anion. If the radius of one ion is known the radius of the other ion can be calculated. Several methods are proposed to find the absolute value of at least one ion from which the ionic radii of the other elements can be calculated. The details of these methods are beyond the scope of this book.

Radius of cation: The radius of a cation is always smaller than the atom from which it is formed. This is because of the following reasons.

(i) When a cation is formed from a neutral atom by the loss of one or more electrons, the number of electronic shells decreases. For example in the conversion of Li → Li^+; Na → Na^+; Mg → Mg^{2+}, A*l* → Al^{3+}, all the electrons in the outermost shell are removed due to which the ions will have one shell less than in their corresponding neutral atoms.

(ii) Moreover, with the removal of electrons from an atom the magnitude of the nuclear charge remains same while the number of electrons decreases. As a result, the same nuclear charge now acts on less number of electrons. So the electrons are more strongly attracted and are pulled towards the nucleus.

Because of the above two reasons the cationic radius is always less than that of its neutral atom. Atomic and cationic radii of certain elements are given in Table 2.20.

Radius of anion: The negative ion is always larger than that of the corresponding atom. The negative ion is formed by the gain of one or more electrons by the neutral atom and the number of electrons increases while the magnitude of nuclear charge remains same. As a result the same nuclear charge acts on large number of electrons than were present in the neutral atom. Thus the effective nuclear charge decreases per electron and the electrons are held less tightly by the nucleus. This causes increase in the size of the ion. Atomic and anionic radii of certain elements are given in Table 2.21.

Trends in values of ionic radii: As a study of the numerical values given leads to the following pertinent conclusions, most of which simply serve to illustrate, numerically, general effects which have already been described.

(i) The ions of elements in any one group of the periodic table increase in size as the relative atomic mass of the element increases.

(ii) A series of ions with the same arrangement of electrons is called **isoelectronic series**. In isoelectronic series of ions the size of the ion decreases as the nuclear charge

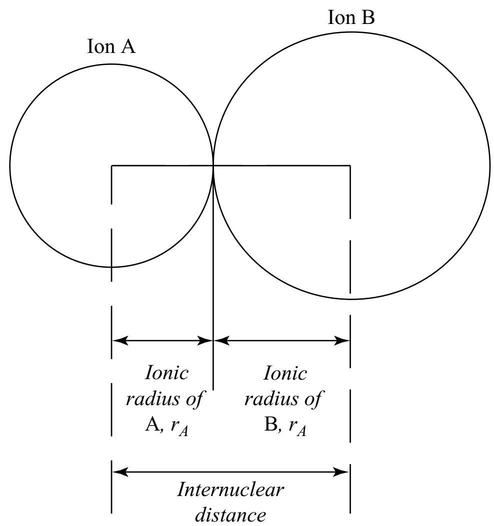


Fig 2.9 The meaning of ionic radius

Table 2.18 Atomic radii (pm) of lanthanides

Element	La	Ca	Pr	Nd	Pm*	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
Atomic radius	187	183	182	181	–	179	204	180	178	177	176	175	174	194	174

* Promethium is the only radioactive lanthanide.

Table 2.19 Atomic radii (pm) of actinides

Element	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf
Atomic radius	112	108	104	102.5	101	100	97.5	97	96	95

i.e., the atomic number increases, for, as the nuclear charge increases, the electrons are attracted more strongly and drawn inwards. This effect is observable in the series O^{2-}, F^-, Na^+, Mg^{2+}, Al^{3+} (all with 2, 8 structure) S^{2-}, Cl^-, K^+, Ca^{2+}, Sc^{3+} (all with a 2, 8, 8 structure) or Cu^+, Zn^{2+} Ga^{3+} (all with 2, 8, 18 structure).

(iii) With increase in the number of positive charges on an ion, the ionic radii decreases. This effect is to be expected for the ion with higher charge has fewer extra nuclear electrons and these are held more tightly, than the ion of lower charge.

Tl^+	Pb^{2+}	Mn^{2+}	Fe^{2+}
144	121	80	75
Tl^{3+}	Pb^{4+}	Mn^{3+}	Fe^{3+}
95	84	62	60

(iv) There is decrease in ionic radius as the atomic number of lanthanides increases as in the atomic radii of lanthanides called lanthanide contraction.

A similar decrease in the ionic radii with increase in atomic number is found in the actinides and in the transitional elements.

(v) Effect of coordination number on ionic radii: The ionic radii given so far are based on the assumption that an ion may be regarded as a sphere. But the ions are deformed by the surrounding ions in anionic crystal. The deformation of an ion is not the same in different types of crystal arrangements. For example, ammonium chloride crystallises with NaC*l* structure (coordination number 6) above 184.3°C and with the CsC*l* structure (coordination number 8) below that temperature and there is found to be an interionic distance about 3 per cent greater in the latter arrangement. Thus with increase in coordination number size of an ion increases. On these lines conversion factors can be worked out for converting the ionic radii for 6 coordinate to ionic radii for any other coordination. Thus for coordination number 6 to 8 the factor is 1.03 and from 6 to 4 it is 0.94. It will be seen that the changes are not very great.

La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
115	111	109	108	106	104	103	102	100	99	97	96	95	94	93

Table 2.20 Atomic and cationic radii (pm) of atoms

Li	Li^+	Be	Be^{2+}	B	B^{3+}
152	60	111	39		
Na	Na^+	Mg	Mg^{2+}	A*l*	Al^{3+}
186	95	160	65	143	50
K	K^+	Ca	Ca^{2+}	Ga	Ga^{3+}
231	133	197	100		62
Rb	Rb^+	Sr	Sr^{2+}	In	In^{3+}
244	148		113		81
Cs	Cs^+	Ba	Ba^{2+}	T*l*	Tl^{3+}
262	169		135		95

Table 2.21 Covalent and anionic radII (pm) of some elements

N	N^{-3}	O	O^{2-}	F	F^-
75	171	74	140	72	136
		S	S^{2-}	C*l*	Cl^-
P	P^{-3}	104	170	99	181
110	212	Se	Se^{2-}	Br	Br^-
		117	184	144	195
		Te	Te^{2-}	I	I^-
		137	221	133	216

2.6.4 Ionization Enthalpy

It is a well-known fact that the electrons in an atom are attracted by the positively charged nucleus. If energy is supplied to an atom, electrons in it are promoted to different higher energy levels depending on the amount of supplied energy. When sufficient energy is supplied, an electron in the outermost shell can be removed from the atom resulting in the formation of positive ion. This energy is referred to as ionization enthalpy.

> This can be defined as the amount of energy required to remove the most loosely bound electron from an isolated, neutral gaseous atom in its ground state.

$$M_{(g)} + \text{Energy (I.E.)} \rightarrow M^+_{(g)} + e^-$$

The ionization enthalpy is expressed either in terms of electron-volts (ev / atom) or kilo calories per mole (K. cal mol^{-1}) or kilo Joules per mole (kJ mol^{-1})

1 ev per atom = 23.06 K.cal mol^{-1} = 96.45 kJ mol^{-1}

The energy required to remove the first electron from the neutral gaseous isolated atom is called Ist ionization enthalpy. Once the first electron has been removed from the

Radii (in pm), listed in the order: r_{cov}, r_{vdw}, r_{ion}

(100 pm = 1Å)

1 H 37 130 208(1–)																	2 He 54 140 …
3 140 Li 180 90(1+)	4 Be 120 59(2+)											5 B 83 … …	6 C 77 170 …	7 N 73 155 …	8 O 70 140 126(2–)	9 F 54 135 119(1–)	10 Ne … 154 …
Na 154 230 116(1+)	12 Mg 148 170											13 Al 130 68(3+)	14 Si 117 210 54(4+)	15 P 110 185 212(3–)	16 S 103 185 170(2–)	17 Cl 97 180 167(1–)	18 Ar … 192 …
19 K 198 280 152(1+)	20 Ca 178 … 114(2+)	21 Sc 160 … 89(3+)	22 Ti … … 100(2+)	23 V 131 … …	24 Cr	25 Mn	26 Fe	27 Co 125 … 83(2+)	28 Ni 125 … 83(2+)	29 Cu 128 140 91(1+)	30 Zn 120 140 88(2+)	31 Ga 130 190 76(3+)	32 Ge 122 … 67(4+)	33 As 121 … …	34 Se 117 200 184(2–)	35 Br 114 190 187(1–)	36 Kr 110 200 …
37 Rb 220 … 166(1+)	38 Sr 192 … 132(2+)	39 Y 178 104(3+)	40 Zr 134 … 80(4+)	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd … 160 100(2+)	47 Ag 144 170 129(1+)	48 Cd 149 160 99(2+)	49 In 148 190 94(3+)	50 Sn 140 220 83(4+)	51 Sb 141 … 90(3+)	52 Te 137 220 207(2–)	53 I 133 200 206(1–)	54 Xe 130 220 …
55 Cs 265 … 181(1+)	56 Ba 218 … 149(2+)	57 La … … 115(3+)	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt … 175 94(2+)	79 Au 144 170 151(1+)	80 Hg 150 150 116(2+)	81 Ti … 200 103(3+)	82 Pb 144 … 92(4+)	83 Bi 152 … 117(3+)	84 Po	85 At	86 Rn
87 Fr	88 Ra … …	89 Ac 187 … 111(3+)	104	105	106	107	108	109									

Fig 2.10 Periodic trends in atomic and ionic radii for each element, the top value is the average single bond covalent radius, the middle value is the average van der Waal's radius; the bottom value is the ionic radius for the oxidation state that is specified in parentheses. Radii are given in pm

gaseous atom, it is possible to remove second and successive electrons from positive ions one after the other. For example

$$M_{(g)} + IE_1 \rightarrow M^{+}_{(g)} + e^{-}$$

$$M^{+}_{(g)} + IE_2 \rightarrow M^{2+}_{(g)} + e^{-}$$

$$M^{2+}_{(g)} + IE_3 \rightarrow M^{3+}_{(g)} + e^{-}$$

The amounts of energies required to remove most loosely bounded electron from unipositive, dipositive, tripositive ions of the element in gaseous state are called second, third, fourth .... ionization enthalpies, respectively.

The second ionization enthalpy (IE_2) is greater than the first ionization enthalpy (IE_1). After removing an electron from an atom, the unipositive ion formed will have more effective nuclear charge than the neutral atom. This decreases the repulsions between the electrons and at the same time increases the nuclear attractions on the electrons in the outer shells. As a result, more energy is required to remove an electron from the unipositive ion. Hence the second ionization enthalpy (IE_2) is greater than the first ionization enthalpy (IE_1). Due to similar reasons there will be progressive increase in the values of successive ionization enthalpies. An atom has as many ionization enthalpies as the number of electrons present in it. The order of successive ionization enthalpies is $IE_1 < IE_2 < IE_3 <$ .... $< IE_n$ where 'n' is the number of electrons in the atom.

The ionization enthalpies are determined from the spectral studies as well as from the discharge tube experiments. The discharge tube is filled with gas whose ionization enthalpy is to be determined. At low voltages, there is no flow of electricity. But on increasing the voltage between anode and the cathode, the gas ionizes at a particular voltage, which is indicated by a sudden increase in the flow of electricity. At a particular voltage, atom receives sufficient energy to lose an electron. This voltage is called ionization potential or ionization enthalpy. The ionization potentials of different elements are given in Table 2.22.

Factors influencing the ionization enthalpies: The magnitude of ionization enthalpy of an atom depends on the following factors.

(i) Atomic size: The attractive force of nucleus on the electrons decreases with increase in the distance. As the size of the atom increases, the distance from the nucleus to the outer electrons also increases causing the decrease in attractive force of nucleus on these electrons. Therefore, ionization energy decreases with increase in atomic size.

(ii) Nuclear charge: The attractive force between the nucleus and the electrons increases with the increase in nuclear charge provided their main energy shell remains the same. This is because the force of attraction is directly proportional to the product of charges on the nucleus and that on the electron. Hence ionization enthalpy increases with the increase in the nuclear charge.

(iii) Screening effect of the inner electrons: In an atom having more than one electron, the valence electrons are

Table 2.22 Ionization potential of some elements

Element	Atomic number (Z)	Atomic radius (A)	Ionization potentials (in ev)							
			I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8
Hydrogen	1		13,595							
Helium	2		24,580	54.40						
Lithium	3	1.225	5.390	75.6193	122.420					
Beryllium	4	0.869	9.320	18.206	153.850	217.657				
Boron	5	0.80	8.296	25.149	37.920	259.298	340.127			
Carbon	6	0.771	11.264	24.576	47.864	64.476	391.986	489.84		
Nitrogen	7	0.74	14.54	29.605	47.426	74.450	97.863	551.925	666.83	
Oxygen	8	0.74	13.614	35.146	54.934	77.394	113.813	138.08	739.114	871.12
Fluorine	9	0.72	17.42	34.98	62.646	87.23	114.214	157.117	185.139	953.60
Neon	10		21.559	41.07	64.0	97.16	126.4	157.91		
Sodium	11	1.572	5.138	47.29	71.65	98.86	138.60	172.36	208.44	264.155
Magnesium	12	1.364	7.644	15.03	80.12	109.29	141.23	186.86	225.31	265.97
Aluminium	13	1.248	5.984	18.823	28.44	119.96	153.77	190.42	241.93	285.13
Silicon	14	1.173	8.149	16.34	33.46	45.13	166.75	205.11	246.41	303.87
Phosphorus	15	1.10	11.0	19.65	30.156	51.354	65.007	220.414	263.31	309.26
Sulphur	16	1.04	10.357	23.4	35.0	47.29	72.5	88.029	280.99	328.80
Chlorine	17	0.934	13.01	23.80	39.90	53.5	67.80	96.7	114.27	348.3
Argon	18		15.755	27.62	40.90	59.79	75.0	91.3	124.0	143.46
Potassium	19	2.025	4.339	31.81	46.0	60.90		99.7	118.0	195.0
Calcium	20	1.736	6.111	11.67	51.21	67.0	84.39		128.0	147.0

attracted by the nucleus and at the same time repelled by the inner core of electrons. Thus, the outermost electrons are shielded or screened from the nucleus by the inner electrons. This is known as **screening effect** or **shielding effect.** Due to this screening effect, the outermost electrons are less attracted by the nucleus causing the decrease in their ionization enthalpy with increase in the number of electrons in the inner shells, the screening effect also increases. Consequently, ionization enthalpy decreases more.

(iv) Penetration effect of electrons: In the case of multi electron atoms, the electrons in the s-orbital have maximum probability to find them near the nucleus and this probability decreases from s to p to d to f-orbitals. In other words, s-electrons are more penetrating towards the nucleus. The penetration power decreases in a given shell in the order s > p > d > f. If penetration power of an electron is more, it will be nearer to the nucleus and experience less shielding effect by the inner electrons. Consequently, ionization enthalpy increases. Thus for the same shell the ionization enthalpy follows the sequence of the extent of penetration of orbitals, i.e., the order s > p > d > f.

(v) Electronic configuration: Completely filled or half-filled sub shells give more stability to the atoms. Such atoms have less tendency to lose the electron and consequently have high ionization enthalpies.

Variation of ionization enthalpy in a group: In a group of elements ionization enthalpy decreases from top to bottom. This is because of the increase in the atomic size down the group. Though the nuclear charge increases in a group, the size of an atom and the screening effect of the inner electrons also increase with an increase in the number of shells. As a result, the outermost electron is held less tightly by nuclear forces and hence, can be easily removed. Ionization enthalpy, therefore decreases down the group.

Table 2.23 Ionization enthalpies of alkali metals

Element	Li	Na	K	Rb	Cs
Ionization Enthalpy KJ mol^{-1}	520	496	419	403	375

Variation of ionization enthalpy in a period: Generally in a period the atomic size decreases. So the ionization enthalpy increases in a period from left to right. As the atomic number increases in a period electrons enter into either the same outermost shell or the inner shells. Nuclear attractions on the outer electrons increase. Hence the energy required to remove the outermost electron is greater. So, ionization enthalpy generally increases as we move from left to right in any period. For example, the variation of IE among the second period elements is given in Table 2.24.

However, some irregularities in the general trend have been noticed. These are due to half-filled and completely filled configurations which have extra stability. Certain examples are discussed here.

The ionization enthalpy of boron is less than beryllium though the nuclear charge of boron is more than beryllium. This is because that in boron the electron to be removed is from 2p orbital which is at slightly higher in energy than the 2s electron in beryllium. The 2p electron in boron is well shielded by two orbitals 1s and 2s where the 2s electron in beryllium is shielded by only one orbital 1s and is nearer to the nucleus. So, the ionization enthalpy of boron is less than that of beryllium. Due to similar reasons, the IE_1 of aluminium is less than the IE_1 of magnesium.

Another example of anomalous pair of elements is 'N' and 'O'. In this pair 'N' has more IE_1 (1403 KJ mol^{-1}) than that of 'O' (1314 KJ mol^{-1}). This is because that 'N' is having stable exactly half-filled $2p^3$ configuration, whereas 'O' has $2p^4$ configuration with paired electrons. The repulsion between the paired electrons results in lowering the IE_1 in oxygen while due to stable half-filled configuration, causes the increase in IE_1 of nitrogen. Due to similar reasons the IE_1 of sulphur is less than the IE_1 of phosphorus.

In a transition series, the differentiating electron enters the (n–1)d sublevel. The electrons that are lost on ionization are those that lie at highest energies and therefore require the least energy to remove. One might expect, therefore that electrons would be lost on ionization in the reverse order in which orbitals were filled. There is a tendency for this to be true. However, there are some very important exceptions notably in the transition elements. In the case of transition elements and for the heavier metals the ns^2 electrons are lost before (n–1)d or (n–2)f electrons which are filled last. If the 4s level is lower and fills first, then its electrons must be stable and be ionized last. This can be explained as follows.

As the atomic number increases, the effective nuclear charge (Z*) increases, and the energy levels approach more closely to those in a hydrogen atom, namely, all the sublevels in the same principle quantum number are degenerate and lie below those of the next quantum number. The effective nuclear charge in the ion increases markedly because of the net ionic charge and the reduced shielding effect. The formation of a dipositive ion accomplishes the lowering the energy of 3d level so far below the 4s that repulsion energies are overcome and the total energy is minimized if the 3d level rather than 4s is occupied. This tendency towards hydrogen-like orbitals is dramatic with increasing effective nuclear charge. For example, core electrons are scarcely differentiated energetically according to type of orbital - they closely approach the hydrogenic degeneracy. Similar effects also occur in inner transition elements.

In each period the last element, i.e., inert gas element has the highest ionization enthalpy and the alkali metal has the lowest ionization enthalpy. Thus of all the elements helium has the highest and caesium has the lowest ionization enthalpies.

2.6.5 Electron Gain Enthalpy

Electron gain enthalpy is another important property of elements. Just as energy is required to remove an electron from an atom, energy is released when an electron is added to the neutral atom. This energy is called electron gain enthalpy. It is defined as **the enthalpy change that occurs when an electron is added to an isolated neutral gaseous atom is called the electron gain enthalpy**

This can be represented as

$$X_{(g)} + e^- \rightarrow \Delta H = \Delta_{eg}H$$

Electron gain enthalpy may be either endothermic or exothermic. For many elements electron gain enthalpy is exothermic. So its value is negative. For example, the halogens have very high negative electron gain enthalpies. But the inert gases have positive electron gain enthalpy because the electron has to be added to the next higher energy level because the inert gases have stable octet in their outermost valence shell. The addition of electron to higher energy level leads to highly unstable configuration.

Electron gain enthalpies are expressed either in electron volts per atom or kilo joules per mole of atoms. For example, the electron affinity of chlorine is –348 kJ mol^{-1}, i.e.,

$$Cl_{(g)} + e^- \rightarrow Cl^-_{(g)} \cdot \Delta_{eg}H = -348 \text{ kJ}$$

Table 2.24 Ionization enthalpies of second period elements

Elements	Li	Be	B	C	N	O	F	Ne
Ionization enthalpy KJ mol^{-1}	520	899	801	1086	1403	1314	1681	2080

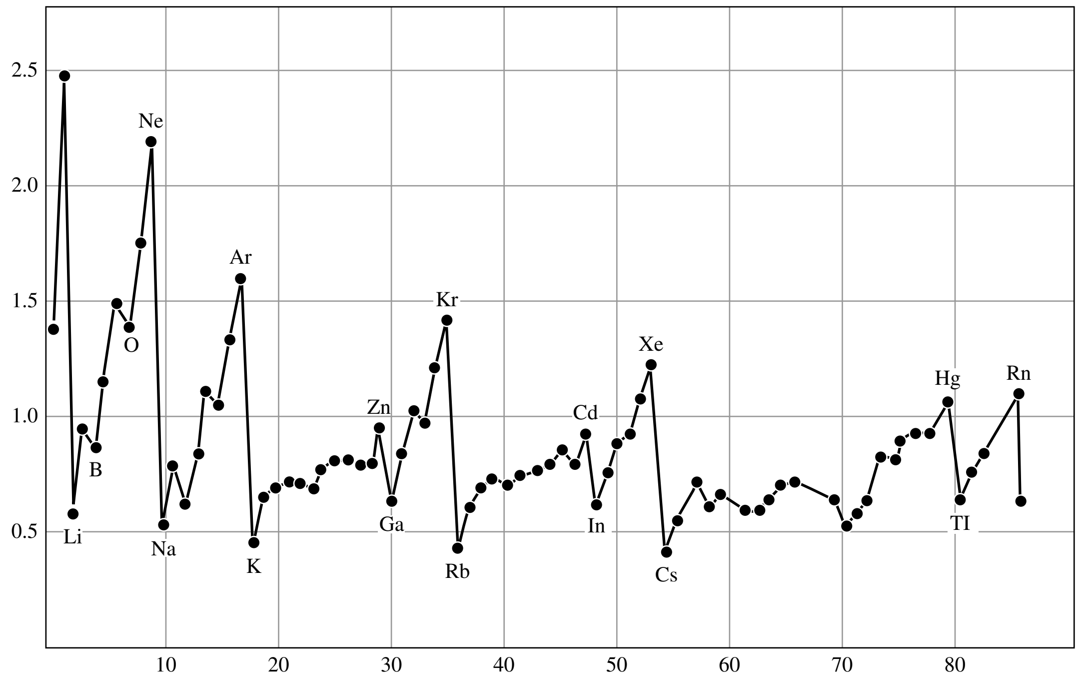


Fig 2.11 Periodic trends in the first ionization enthalpies

Electron gain enthalpy is also referred to as electron affinity (Ae). When an electron is added to a neutral isolated gaseous atom the energy released is represented with negative sign as electron gain enthalpy and with positive sign as electron affinity or vice versa if energy is absorbed contrary to the thermodynamic convention. However, electron affinity is defined at absolute zero. So at any other temperature (T) heat capacities of the reactants and the products have to be taken into account

$$\Delta_{eg}H = -A_e - 5/2\ RT$$

Factors influencing the electron gain enthalpy: (i) Effective nuclear charge: If the effective nuclear charge (Z*) is more, attraction towards the incoming electron will be more. As a result the electron gain enthalpy will be more negative.

(ii) Atomic size: If the size of atom is more, the distance between the outermost shell and nucleus increases. This decreases the attraction on the electron added to the outermost shell. So, the electron gain enthalpy will be less negative.

(iii) Electronic configuration: If an atom has stable electronic configuration, it will have less tendency to accept the electron. In such cases the electron gain enthalpy will be more positive. For example, the elements having completely filled sublevels of the valence shell have relatively stable configurations and consequently their electron gain enthalpies are more positive values.

The electron gain enthalpy values cannot be determined directly but are obtained indirectly using Born - Haber cycle. The electron gain enthalpy values for some elements are given in Table 2.25.

Variation of electron gain enthalpy in a group: In a group of elements in the periodic table the electron gain enthalpy values become less negative from top to bottom as the atomic size increases on moving down a group the atomic size as well as nuclear charge increases. But the effect of increase in atomic size is much more pronounced than that of nuclear charge and thus the additional electron feels less attraction. Consequently, electron gain enthalpy becomes less negative on going down the group.

But in general the second element in a group, i.e., 3rd period element (II to VII groups) has more negative electron gain enthalpy than the first element, i.e., 2nd period element. For example, the electron gain enthalpy of fluorine (–333 kJ mol^{-1}) is less negative than that of chlorine (–348 kJ mol^{-1}). This is because when an electron is added to fluorine atom, it enters into the relatively small second energy level. As a result it experiences significant repulsion from the other electrons present in the shell.

Table 2.25 Electron gain enthalpy values for some elements (kJ mol^{-1})

H							He
– 72							+48
Li	Be	B	C	N	O	F	Ne
–57	+66	–15	–121	+31	–142	–333	+116
Na	Mg	A*l*	Si	P	S	C*l*	Ar
–53	+67	–26	–135	–60	–200	–348	+96
K				As	Se	Br	Kr
–48				–77	–195	–325	+96
Rb				Sb	Te	I	Xe
–47				–101	–190	–295	+77
Cs				Bi	Po	At	Rn
–46				–97	–174	–270	+68

These repulsions are overcome at the expense of a part of the energy released. On the other hand, in chlorine atom, the added electron goes into the bigger third energy level where the electron - electron repulsions are less. Hence, the overall energy liberated is less than that of chlorine atom. Because of similar reasons the electron gain enthalpy values of second period elements are less negative.

The oxygen in VIth group elements has the least negative electron gain enthalpy. In Vth group electron gain enthalpy of nitrogen is even positive because of small size and stable exactly half-filled $2p^3$ configuration.

Variation of electron gain enthalpy in a period: In a period from left to right the atomic size decreases and effective nuclear charge increases. Both these factors results in greater attraction for the incoming electron. Therefore the electron gain enthalpies tend to become more negative as we move from left to right in a period. In any period halogen has the more negative electron gain enthalpy.

However, there are certain irregularities in the general trend. These are mainly due to the stable electronic configurations of certain elements.

(i) The electron gain enthalpies of noble gases are positive because they have stable ns^2np^6 configuration and electron has to enter the next higher priniciple quantum level as explained earlier.

(ii) The electron gain enthalpies of alkaline earth metals (Be and Mg) are also positive because they have completely filled stable ns^2 configuration.

(iii) The electron gain enthalpies of nitrogen family are less negative because they have stable exactly half-filled np^3 configuration.

When the atoms have stable electronic configuration in their valence shell, they have least tendency to gain an electron.

Successive electron gain enthalpies: Similar to ionization enthalpies successive electron gain enthalpies are possible, i.e., $\Delta_{eg}H_1$, $\Delta_{eg}H_2$, etc. But the second electron gain enthalpy is always positive due to the repulsion between the negatively charged ion and the negatively charged electron

Ex: $O_{(g)} + e^- \rightarrow O^-_{(g)} \quad \Delta_{eg}H_1 = -142 \text{ KJ mol}^{-1}$

$O^-_{(g)} + e^- \rightarrow O^{2-}_{(g)} \quad \Delta_{eg}H_2 = +780 \text{ KJ mol}^{-1}$

To overcome the repulsive forces between the uninegative ion (X^-) and the electron added (e^-), some energy is required. Hence, the second electron gain enthalpy is always positive.

The ionization enthalpy of a neutral atom is always equal to the electron gain enthalpy of its unipositive ion.

Ex: $M_{(g)} + e^- \rightarrow M^+_{(g)} + e^-,$

$M^+_{(g)} + e^- \rightarrow M_{(g)}, \quad \Delta_{eg}H$

Similarly the electron gain enthalpy of a neutral atom is equal to the ionization enthalpy of its uninegative ion.

$X_{(g)} + e^- \rightarrow X^-_{(g)} \quad \Delta_{eg}H$

$X^-_{(g)} \rightarrow X_{(g)} \quad IE$

Generally, metals have a tendency to lose electrons. So, the electron gain enthalpies of metals are less negative. Among metals gold has more negative electron gain enthalpy. Among all the elements chlorine has the maximum negative electron gain enthalpy.

2.6.6 Electronegativity

In a covalent bond between unlike atoms, the pair of shared electrons will not necessarily be shared equally by both atoms. For example in bond A – B, the atom B has a stronger attraction for electrons than A. Then the shared pair will be attracted towards B and away from A. Any permanent displacement of electrons of this sort in a covalent bond will give the bond some ionic character, and the actual bond will have to be represented as a resonance hybrid between covalent and ionic forms.

$$A - B \longleftrightarrow A^+ B^-$$

The tendency of the atom of an element to attract the shared electron pairs, more towards itself in a heteronuclear diatomic molecule or in a polar covalent bond is called electronegativity.

Electronegativity values are very difficult to measure. A particular atom may present in different type of environments in different molecules. So, the electronegativity of an atom may not remain constant irrespective of its environment. For example in C – H and in C – C*l* bonds the electronegativities of the same carbon appear to be different. But invariably electronegativity is assumed as constant. To overcome this difficulty several quantitative scales of electronegativities are presented.

Pauling scale: Pauling suggested that molecules formed from atoms of different electronegativity would be stabilized by **ionic resonance energy** resulting from resonance as shown above. Pauling suggested that electronegativity could be estimated from calculations involving this resonance energy. The ionic resonance energy can be calculated from the bond energies.

Actual bond energy measured experimentally	= H
Bond energy if the bond was truly covalent	= Q
Ionic resonance energy	= H – Q

Values of H can be measured experimentally. A truly covalent bond between unlike atoms is hypothetical and therefore the value of Q can only be obtained as either the arithmetic or the geometric mean of bond energies of the bonds A – A (E_{AA}) and B – B (E_{BB}), i.e.,

$$E_{AB} = \frac{E_{AA} + E_{BB}}{2} \text{ or } E_{AB} = \sqrt{(E_{AA} \cdot E_{BB})}$$

The difference between the values of H and Q is taken as ionic resonance energy of the bond A – B.

$$\Delta_{AB} = H - Q$$

According to pauling the electronegativity difference between two atoms is equal to square root of the ionic resonance energy of the bond A – B

$$\chi_A - \chi_B = 0.208 \text{ if } \Delta_{AB} \text{ is expressed in k.cal mole}^{-1}$$

or $$\chi_A - \chi_B = 0.1017 \text{ if } \Delta_{AB} \text{ is expressed in k.J mole}^{-1}$$

Pauling assigned the electronegativity values of 2.1 to hydrogen and 4.0 for the most electronegative element fluorine and calculated the electronegativity values of other elements.

Mulliken Scale: In 1934, R.S. Mulliken and Jaffe suggested that the electronegativity of an element is the average of its ionization energy and electron gain enthalpy.

$$\chi_M = \frac{IE + Ae}{2} = \frac{\Delta_{IE}H - \Delta_{eg}H}{2}$$

where IE = ionization potential and Ae = electron affinity. Mulliken electronegativity values are about 2.8 times larger than the Pauling's values.

$$\text{Thus, } \chi = \frac{IE + Ae}{2 \times 2.8} = \frac{IE + Ae}{5.6}$$

If IE and Ae are measured in KJ mol^{-1} then the above equation becomes as

$$\text{Thus, } \chi_M = \frac{IE + Ae}{5.6 \times 96.48}$$

($\because$ 1 ev / atom = 96.48 KJ mol^{-1} of atoms)

$$\chi_M = \frac{IE + Ae}{540}$$

Allred and Rochow scale: According to Allred and Rochow, electronegativity of an atom can be calculated as

$$\chi_M = 0.744 + \frac{0.359 Z^*}{r^2}$$

where χ_M = electronegativity of atom M, Z^* effective nuclear charge of M and r = radius of the atom M.

Factors influencing the electronegativity

(i) **Effective nuclear charge:** The higher is the effective nuclear charge of an atom the more is its power to attract the shared electron pair and thus have more electronegativity.

(ii) **Screening effect:** The more the screening effect of inner electrons in an atom, the lesser the attractive power of nucleus on valence electrons. So, electronegativity of such atoms will be less.

(iii) **Size of the atom:** With increase in the size of atoms, attractive power of the nucleus on valence electrons decreases due to increase in distance between the nucleus and valence shell. So, electronegativity decreases with increase in the size of atom.

(iv) **Oxidation state:** Atoms in higher oxidation states have more electronegativity values as they have more effective nuclear charge.

(v) **Type of hybridization:** With increase in the s-character of hybrid orbital, the size of hybrid orbital decreases and come nearer to the nucleus. So, the attractive power of nucleus on the electrons in such orbital is more. e.g.,

Hybrid orbital	s-character	Electronegativity
sp^3	25%	2.48
sp^2	33%	2.75
sp	50%	3.29

Pauling and Mulliken electronegativity values of different elements are given in Table 2.26.

Periodic Trends: Electronegativity is also a periodic function of atomic number of elements. Table 2.26 shows that electronegativity decreases down a particular group because the size of atoms becomes progressively larger and the attraction for electron decreases. On the other hand, electronegativity increases accross a particular period till the elements of group 17 (halogens).

Importance of electronegativity: Using electronegativity values, the following predictions can be made.

(i) Nature of bond: The nature of bond between two atoms can be predicted from the electronegativity difference of the two atoms.

(a) If the electronegativity difference of the two atoms in a bond is zero, the bond is purely covalent **Ex:** H_2, Cl_2, O_2, N_2, etc.

(b) If the electronegativity difference of the two atoms in a bond is small, the bond is polar covalent bond. **Ex:** HF, HC*l*, H_2O, etc.

(c) If the electronegativity difference of the two atoms in a bond is 1.7, the bond is 50 per cent covalent and 50 per cent ionic.

Table 2.26 Electronegativities of the elements, values at the top are calculated by Allred and Rochow method. Values in middle are estimated by Pauling's method. Values at the bottom are obtained by using Mulliken's method. Roman numeral at the top give the oxidation states used for the Pauling values

I	II	III	IV	II	II	II	II	II	II	II	II	III	IV	III	II	I	
H																	**He**
2.20																	
Li	**Be**											B	C	N	O	F	Ne
0.97	**1.47**											**2.01**	**2.50**	**3.07**	**3.50**	**4.10**	
0.98	1.57											2.04	2.55	3.04	3.44	3.98	
0.94	1.46											2.01	2.63	2.33	3.17	3.91	
Na	Mg											A*l*	Si	P	S	C*l*	Ar
1.01	**1.23**											**1.47**	**1.74**	**2.06**	**2.44**	**2.83**	
0.93	1.31											1.61	1.90	2.19	2.58	3.16	
0.93	1.32											1.81	2.44	1.81	2.41	3.00	
K	Ca	**Sc**	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
0.91	1.04	1.20	1.32	1.45	1.56	1.60	1.64	1.70	1.75	1.75	1.66	1.82	2.02	2.20	2.48	2.74	
0.82	1.00	1.36	1.54	1.63	1.66	1.55	1.88	1.88	1.91	1.90	1.65	1.65	2.01	2.18	2.55	2.96	
0.80										1.36	1.49	1.49		1.75	2.23	2.70	
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
0.89	**0.99**	**1.11**	**1.22**	**1.23**	**1.30**	**1.36**	**1.42**	**1.45**	**1.35**	**1.42**	**1.46**	**1.49**	**1.72**	**1.82**	**2.01**	**2.21**	
0.82	0.95	1.22	1.33		2.16					1.93	1.69	1.78	1.96	2.05		2.66	
										1.36	1.4	1.80		1.65	2.10	2.56	
Cs	Ba	*	**Hf**	**Ta**	**W**	**Re**	**Os**	**Ir**	**Pt**	**Au**	**Hg**	**T*l***	**Pb**	**Bi**	**Po**	**At**	**Rn**
0.86	**0.97**		**1.23**	**1.33**	**1.40**	**1.46**	**1.52**	**1.55**	**1.44**	**1.42**	**1.44**	**1.44**	**1.55**	**1.67**	**1.76**	**1.96**	
0.79	**0.89**	* *			2.36			2.20	2.28	2.54	2.00	2.33	2.02				
Fr	**Ra**																
0.86	0.97																
		*La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	
		103	**1.06**	**1.07**	**1.07**	**1.07**	**1.07**	**1.01**	**1.11**	**1.10**	**1.10**	**1.10**	**1.11**	**1.11**	**1.06**	**1.14**	
		1.10	1.12	1.13	1.14		1.17		1.20		1.22	1.23	1.24	1.25			
		**Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md			
		1.00	**1.11**	**1.14**	**1.22**	**1.22**	**1.22**										
					1.38	1.36	1.28										

(d) If the electronegativity difference of the two atoms in a bond is more than 1.7, the bond is more ionic and less covalent.

The percentage of ionic character of a bond between two atoms can be calculated from the difference in the electronegativities.

% ionic character = $16 [\chi_A - \chi_B] + 3.5 [\chi_A - \chi_B]^2$

where χ_A is the electronegativity of more electronegative element and χ_B is the electronegativity of less electronegative element.

(ii) Writing the formula of inorganic compounds: Electronegativity values are useful in writing the formulae of inorganic compounds. While writing the formulae of inorganic compounds less electronegative element should be written first and more electronegative element last, e.g.

OF_2 (correct) F_2O (wrong)
OCl_2 (wrong) Cl_2O (correct)

(iii) Sign of oxidation states: Oxidation states of more electronegative elements are assigned with negative sign and less electronegative elements are assigned with positive sign for different elements in a compound.

For example, oxidation state of oxygen in Cl_2O is – II while in OF_2 is + II.

(iv) Strength of bond: Increase in the difference in electronegativities makes the bond more stronger. For example, in the nitrogen trihalides, the order of the strength of N – X bond is

$$N – F > N – Cl > N – Br > N — I$$

(v) Nature of elements: Elements with more electronegativity values are non-metals and with less less electronegativity values are metals.

ELECTRONEGATIVITY AND ELECTRON AFFINITY (ELECTRON GAIN ENTHALPY)

Both electronegativity and electron affinity are a measure for the attractive power of an atom on the electrons but have different meaning.

Electron affinity refers to attraction of an isolated atom for an additional electron, whereas electronegativity is the attraction of a bonded atom on the shared pair of electrons.

Electron affinity can be measured experimentally while electronegativity is a relative value.

2.7 PERIODICITY OF VALENCE

The number of hydrogen atoms or halogen atoms or double the number of oxygen atoms that can combine with one atom of an element is known as **valence**.

In the main group elements the valence of an element is equal to the group number or 8-group number. When an element shows variable valence, the group number need not necessarily correspond to the most important valence. Then the group number is equal to the minimum valence (**e.g.:** coinage metals Cu, Ag, Au) or the maximum valence for some groups (**e.g.:** Group 15 N, P, As, Sb Bi the valence is 5).

The valence of elements shows periodicity. The variation of valence is illustrated by the fluorides, hydrides and the oxides of elements of 2nd, 3rd, and 4th periods are shown in Table 2.27.

The valency is more often replaced by oxidation state of the element. The valence is useful in writing the formulae of compounds.

Table 2.27 The periodicity of valence with respect to hydrogen, fluorine and oxygen of certain elements

		Formula of the compound						
Period	**Compound**	**IA**	**IIA**	**IIIA**	**IVA**	**VA**	**VIA**	**VIIA**
II	Oxide	Li_2O	BeO	B_2O_3	CO_2	N_2O_5	–	OF_2
	Hydride	LiH	BeH_2	B_2H_6	CH_4	H_3N	H_2O	HF
	Fluoride	LiF	BeF_2	BF_3	CF_4	NF_3	OF_2	–
III	Oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	SO_3	Cl_2O_7
	Hydride	NaH	MgH_2	$(AlH_3)_n$	SiH_4	PH_3	H_2S	HCl
	Fluoride	NaF	MgF_2	AlF_3	SiF_4	PF_5	SF_6	ClF_3
IV	Oxide	K_2O	CaO	Ga_2O_3	GeO_2	As_4O_{10}	SeO_3	oxides of Br
	Hydride	KH	CaH_2	Ga_2H_6	GeH_4	AsH_3	SeH_2	HBr
	Fluoride	KF	CaF_2	GaF_3	GeF_4	AsF_5	SeF_6	BrF_5

The periodicity of valence can be seen from successive valencies shown in hydrides or oxides or fluorides in each period.

Hydrides	1	2	3	4	3	2	1
Oxides	1	2	3	4	5	6	7
Fluorides	1	2	3	4	5	6	–

RELATIVISTIC EFFECTS

The knowledge of the structure of an atom based on Schrodinger equation are non-relativistic. The solutions of Schrodinger wave equation that do not incorporate relativity theory show that both the radial and the angular parts of the wave functions for the heavier atoms in the periodic table are appreciably altered by relativistic effects.

According to Einsten's theory of relativity, the mass 'm' of a particle increases over its rest mass, m_o, when its velocity V approaches the speed of light C.

$$m = \frac{m_o}{\sqrt{1-(v/c)^2}}$$

We can think of the electron as a particle that is accelerated to a certain radial velocity by the attraction of the nucleus (to escape from centripetal force). With increase in nuclear charge, i.e., atomic number, the attraction of the nucleus on the electron also increases. To escape from the increased attractive power of nucleus the electron should get more centrifugal force by increasing its velocity. Thus as the nuclear charge builds up, the radial velocity builds up too. The radial velocity approaches the speed of light as the nuclear charge approaches Z = 137.036. The average radial velocity of the 1s electrons of an atom is proportional to the ratio of the nuclear charge of that atom to 137. The 1s electron of a mercury atom move at 80/137 = 0.58 times the speed of light. Hence, the mass of 1s electrons in mercury is about 1.2 times its rest mass. But the radius of the Bohr orbit of an electron is inversely proportional to the mass of the electron. Hence, the radius of the 1s orbital to be about 20 per cent less than calculated by using Bohr's equation. This radial effect is known as **relativistic contraction**. It also effects the 2s and higher s orbitals roughly as much because of their inner lobes, which are close to the highly charged nucleus. The effect is present to a lesser extent among p-orbitals and is nearly absent among d- and f-orbitals which have fewer lobes near the nucleus. In the simplest case we can say that s- and p-orbitals will contract and that d- and f-orbitals will expand some what. The expansion of d- and f-orbitals is known as **relativistic expansion.**

It is surprising that the d- and f-orbitals expand instead of contract is an indirect effect. Since the relativistic effects on d- and f-orbitals are small because these orbitals do not have electron density near the nucleus. However, the **increased shieding** of d- and f-orbitals by relativistically contracted s- and p-orbitals tends to cancel the effect of increased Z. So the s- and p-electrons are moved close, to the nucleus, their energy is lowered (made more negative) and they are stabilized. The d- and f-orbitals are raised in energy (destabilized) and expand. Since the outer most orbitals are ns and np while the (n–1)d or (n–2)f are inner to these ns and np orbitals, each atom as whole contracts.

The relativistic effect goes approximately as Z^2 and this is the reason for its importance in the heavier elements. In terms of energy and size, it starts to become important in the vicinity of Z = 60 – 70, contributing perhaps an additional 10 per cent to the non-relativistic lanthanide contraction. As we have seen, this results in an almost exact cancellation of the expected increase in size with increase in n (principal quantum number) from zirconium to hafnium.

While the contraction resulting from the poor shielding of 4f electrons ceases at halfnium, the relativistic effect continues across the sixth row of the periodic table. It is largely responsible for the stabilisation of the 6s orbital and the inert 's' pair effect shown by the elements Hg to Bi.

Consequences of Relativistic Effects: While the 5d-orbitals are filling and expanding, they are doing their usual poor job of shielding the 6s-orbital from the nucleus. Once the 5d-orbitals are filled, they are no longer valence orbitals and the Pauling electronegativity must be determined by the 6s-orbitals of group II (copper group) and beyond. These orbitals are contracted strongly in gold and mercury. Beyond mercury, the less strongly affected and better shielding p-orbitals are valence orbitals, so the effect diminishes somewhat eventually, however, the increase in nuclear charge again causes atomic contraction.

Since relativistic contraction has its maximum effect in the sixth period at the 6s-orbital of gold and mercury, anomalous effects appear for these elements. One effect is the enhanced electronegativity of gold and mercury as compared with the elements above them, silver and cadmium. Although the lanthanide contraction is partially responsible, it cannot account for the fact that enhancement is at a maximum in this part of the period. Since the 6s-orbital is contracted and bound more tightly, its separation in energy from the 6p-orbitals is anomously large, so much so that the 6p-orbitals are almost not-valance orbitals, but rather are nearly post-valence orbitals. Mercury with $5d^{10}\ 6s^2$ valence electron configuration, has essentially filled all of its valence orbitals; thus makes it very near to being a noble gas (or rather a noble liquid). Gold may then be thought of as one electron short of a noble liquid electron configuration and thus might be expected to show some resemblence to a halogen such as iodine. Indeed, gold has stable Au^- ion similar in size to the Br^- ion, in the compounds RbAu, CsAu and $(Cr^+)_3(Au^-)(O^{2-})$ and in solution in liquid ammonia.

Beyond gold and mercury, the 6s-electrons are still difficult to ionize, which accounts for the low stability of the group oxidation state in the subsequent elements. These elements prefer the oxidation state equal to the group number minus two; this is referred as the **inert pair** (of 6s electrons) **effect**. Thus T*l* normally occurs as Tl^+, Pb as Pb^{2+} and Bi as Bi^{3+}. It also stabilizes one of the 6p-orbital of bismuth allowing the unusual +1 oxidation state in addition to +3 and +5.

The increased ionization energies of the heavier transition metals are also due to these relativistic effects. Although none of the coinage metals is very reactive, gold is well less reactive than copper or silver; iron, cobalt and nickel rust and corrode but osmium, iridium and platinum are noble and unreactive.

The heavier members of each family, thallium, lead, perhaps bismuth has a greater electronegativity than its lighter congener, indium, tin and antimony. This is mainly due to relativistic effect along with lanthanide contraction. Indeed, gallium, germanium, arsenic and perhaps selenium seem to have higher electronegativities than their lighter congeners.

Only enough, for some ions near the centre of the fifth and sixth periods, **e.g.,** Ag^+, Sn^{2+}, Hg^{2+} and Au^+ the preferred coordination number is often quite small - as low as 2. This anomaly is again due to relativistic effects. The acidity of the hydrated Hg^{2+} and Sn^{2+} ions are much higher than expected. The electrostatic force pulling the electrons of each water molecule is stronger than in normal hydrated ions.

Variation of oxidation states: The oxidation state or oxidation number of an element is defined as the possible charge which the atom of an element appears to have acquired in a given species.

Alkali metals (Group I) exhibit only one oxidation state +I by losing their only electron in the valence shell to get the stable octet.

Alkaline earth metals (Group II) exhibit a fixed valence +II by loosing two electrons present in the valence shell to get the octet.

Oxidation state of an element may be positive or negative depending on the electronegativity of the other element with which an element combines. More important is that the oxidation state is variable. Generally the p-block elements exhibit more than one oxidation states i.e., variable valence.

Elements of **boron family** (Group III) exhibit +I oxidation state in ground state and +III oxidation state, in excited state.

Elements of **carbon family** (Group IV) exhibit +II oxidation state in ground state and +IV oxidation state in excited state.

Elements of **nitrogen family** (Group V) exhibit –III or +III oxidation state in ground state and +V oxidation state in excited state.

Elements of **oxygen family** (Group VI) exhibit –II or +II oxidation state in ground state and +IV and +VI oxidation states in excited states (except oxygen).

Halogens (Group VII) exhibit –I or +I oxidation state in ground state and +III, +V and +VII oxidation states in excited states (except fluorine).

Elements of groups I to IV show positive oxidation states equal to their group number. But the elements of groups V to VIII show both positive and negative oxidation states in ground state depending on the electronegativity of the other element with which it combines but always exhibit positive oxidation states in excited states.

The oxidation number of an elements in groups V to VII is equal to 8 - Gp number if the electronegativity of the other element is more or Gp number - 8 if the electronegativity of the other element is less. For example, oxidation number of oxygen in OF_2 is $8 - 6 = +2$ and in Cl_2O $6 - 8 = -2$.

The d-block elements exhibit variable valence, i.e., more than one oxidation states because in these elements the electrons in both ns and (n-1)d orbitals can participate in bonding. All the d-block elements exhibits a common oxidation state +II. The minimum oxidation state that can be exhibited by a d-block element is equal to the number of electrons present in ns-orbital. The maximum oxidation state that can be exhibited by a d-block element is equal to the sum of the electrons present in ns and (n-1)d orbitals. In 3d series (first transition series), manganese exhibits a maximum oxidation state +VII, but ruthenium in 4d series (second transition series) and osmium in 5d series (third transition series) exhibits a maximum oxidation state +VIII in their compounds RuO_4 and OsO_4. The oxidation states of first transition series elements are given in Table 2.28.

Oxidation states underlined are more stable oxidation states while the oxidation states shown in parenthesis are unstable.

The f-block elements (inner transition elements) have the general outer electronic configuration $(n–2)\ f^{1–14}\ (n-1)d^{\,0\ or\ 1}\ ns^2$. All the inner transition elements exhibit a common oxidation state +III. Generally the inner transition elements do not exhibit variable valence because the (n–2) f electrons are deep-seated, i.e., present inside to (n–1)s and

Table 2.28 Oxidation states of first transition series elements

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Electrons in ns	2	2	2	1	2	2	2	2	1	2
Electrons in (n-1)d	1	2	3	5	5	6	7	8	10	10
Oxidation states	+II +III	+II +III +IV	+II +III +IV +V	(+I) +II +III (+IV) +V +VI	+II (+III) +IV +V (+VI) +VII	+II +III (+IV) +V (+VI)	+II +III (+IV) +V	+II (+III) +IV	+1 +II +III	+II

p-orbitals except whenever there is a possibility for anomalous electronic configurations occurs, i.e., whenever they get half-filled and completely filled f-orbitals.

2.8 PERIODICITY OF CHEMICAL PROPERTIES

There are three ways in which the periodicity of chemical properties are usually discussed.

(i) The first compares the properties of elements going down a group.
(ii) The second compares the properties of elements going across a period.
(iii) The third is a comparison of elements that are on a diagonal line traversing two groups and two periods.

Periodicity in a group: Owing to increase in shielding effect down a group, atoms towards the bottom of a group lose one or more of their outer electrons more easily than the atoms nearer the top. **The tendency to lose an electron is known as metallic character or electropositive character.** So, the following periodicity occurs in the group from top to bottom.

Periodicity in a period: Across a period the number of protons in the nucleus increases, as do the number of electrons. This also happens down a group; but the key difference is that across a period a new shell of electrons is not completed until the noble gas at the end of the period is reached. Complete shells of electrons screen the outer electrons from the full attraction of the nucleus. As we go across a period the efficiency of screening is not so great. Indeed the pattern is for the increasing nuclear charge to hold the electrons more tightly.

As a consequence the ionization of the elements become difficult and they also tend to attract electrons towards them; thus, we see a change from the metals (which give positive ions) on the left of a period to the non-metals (which give negative ions) on the right. With this change goes, a change in the nature of the hydrides, oxides and chlorides also occurs.

A comparison of the behaviour of oxides,hydroxides and chlorides of the 2nd and 3rd period elements are given in Tables 2.29, 2.30, respectively.

In Tables 2.29 and 2.30 the words ionic and covalent should not be taken too literally. Majority of the compounds show a mix of both extreme types of bonding. From the information given in Tables 2.29 and 2.30 the following conclusions can be made to the left of the periods, the oxides give hydroxides and to the right they give acids. For example with the IInd period elements we can see the change from strong alkaline character of LiOH to amphoteric nature of $Be(OH)_2$ and finally to strong acidic nature of nitric acid. However, if the bonding in hydroxides and acids are observed we find that they are more similar. In each case the hydrogen atoms are bonded to the oxygen atoms. Molecules which we regard as acids tend to give up hydrogen atoms (as hydrogen ions) to water molecules and the molecules which we regard as bases tend to give up hydroxide ions to water. The molecules which are regarded as amphoteric can give either hydrogen ion or hydroxide. To explain this behaviour we have to consider several factors like the polarity of H – O and O – E bonds, energetic (resonance) stabilization, entropy changes and kinetic factors.

Shielding of outer electrons increases → Metallic nature increases → Non-metallic nature decreases → Ionic character of compounds increases → Basic nature of oxides increases

Table 2.29

Li_2O	Be_2O	B_2O_3	CO_2	NO_2	O_2	OF_2
Ionic	Covalent/ionic	Covalent	Covalent	Covalent	Covalent	Covalent
Basic	Amphoteric	Acidic	Acidic	Acidic	Neutral	Acidic
Hydrolyses	Insoluble	Slightly soluble	Hydrolyses	Hydrolyses	Slightly soluble	Hydrolyses
$LiOH$	$Be(OH)_2$	$B(OH)_3$	H_2CO_3	HNO_3	H_2O	HF
Strong alkali	Amphoteric	Weak acid	Weak acid	Strong acid	Amphoteric	Weak acid
$LiCl$	$BeCl_2$	BCl_3	CCl_4	NCl_3	Cl_2O_7	ClF
Ionic	Covalent	Covalent	Covalent	Covalent	Covalent	Covalent
Dissolves	Dissolves	Hydrolyzes	No reaction	Hydrolyses	Hydrolyses	Hydrolyses
Ionizes	Ionizes					

* Sometimes, water is considered as amphiprotic rather than amphoteric.

Table 2. 30

Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_3	Cl_2O_7
Ionic	Ionic	Ionic	Covalent	Covalent	Covalent	Covalent
Basic	Basic	Amphoteric	Acidic	Acidic	Acidic	Acidic
Hydrolyses	Slightly soluble	Insoluble	Hydrolyses	Hydrolyses	Hydrolyses	Hydrolyses
$NaOH$	**$Mg(OH)_2$**	**$Al(OH)_3$**	**H_3SiO_3**	**H_3PO_4**	**H_2SO_4**	**$HClO_4$**
Strong alkali	Weak alkali	Amphoteric	Weak acid	Weak acid	Strong acid	Strong acid
$NaCl$	**$MgCl_2$**	**$AlCl_3$**	**$SiCl_4$**	**PCl_3**	**S_2Cl_2**	**Cl_2**
Ionic	Partly covalent	Covalent	Covalent	Covalent	Covalent	
		⟵ React with water	⟶	Acidic solution	⟶	
Dissolves	Dissolves	$Al(OH)_3$	$SiO_{2(S)}$	H_3PO_3	H_2SO_3	$HOCl$, HCl
Ionizes	ionizes					

2.9 DIAGONAL RELATIONSHIP

In the periodic table the elements in second period show similarities in properties with the elements in third period in the adjacent group which are diagonal to them. This relationship is known as the diagonal relationship.

These diagonal similarities are usually weaker than the similarities within a group, but quite pronounced in the pairs shown above. This phenomenon disappears after Group IV. The diagonal relationship is due to similar sizes of atoms (or of ions) and similar electronegativities of the respective elements. The diagonally similar elements possesses the same polarizing power. Polarizing power of a species is the ratio of its ionic charge to the square of the ionic radius.

$$\text{Polarizing power} \propto \frac{\text{Ionic charge}}{(\text{Ionic radius})^2}$$

We know that atomic or ionic size increases down the group and decreases across a period. If we take a diagonal route across the periodic table these two trends might be expected to cancel one another out partially. Further, ion charges increases in the diagonal route. So the polarizing power of the elements present on a diagonal line will become similar. So, they exhibit similarities in properties. The pairs that we shall consider are (i) lithium and magnesium, (ii) beryllium and aluminium and (iii) boron and silicon. Their similarities are summarized in sections 2.31 to 2.33.

Group	I	II	III	IV
2nd period	Li ↘	Be ↘	B ↘	C
3rd period	Na	Mg	Al	Si

Table 2.31 Comparison of the elements lithium and magnesium

Carbonates	:	Like $MgCO_3$, Li_2CO_3 is decomposed by heat (the other group I carbonates do not decompose). Both carbonates are insoluble (the other group I carbonates are soluble)
Chlorides	:	Both chlorides are hydrated $LiCl.\ 3H_2O$, $MgCl_2\ 6H_2O$ (the other group I chlorides are not hydrated)
Oxides	:	Both give their normal oxides Li_2O, MgO when they burn (the other group I metals give peroxides, e.g., Na_2O_2)
Hydration	:	Both Li^+ and Mg^{2+} ions are heavily hydrated in solution (they are both dense centres of charge).

Table 2.32 Comparison of beryllium and aluminium

With conc. HNO_3	:	Both metals are rendered passive and will not dissolve.
With alkali	:	Both dissolve liberating hydrogen
Amphoteric nature	:	Both metals are amphoteric (react with both acid and alkali)
Oxides	:	Both BeO and Al_2O_3 are amphoteric
Chlorides	:	Both $BeCl_2$ and $AlCl_3$ are electron deficient; both act as Lewis acids.

Table 2.33 Comparison of the elements boron and silicon

Hydrides	:	Both give a wide variety of unstable hydrides the boranes and the silanes **e.g.,** BH_4^-, B_2H_6, SiH_4, Si_2H_6
Oxides	:	Both are weakly acidic, e.g., $B_2O_3 \longrightarrow H_3BO_3$ $SiO_2 \longrightarrow H_2SiO_3$
Chlorides	:	Both BCl_3 and $SiCl_4$ are volatile and will undergo hydrolysis to give an acidic solution.

KEY POINTS

- **Dobereiner** proposed that particular groups of three elements with similar chemical properties (triads as he called them) had atomic weights, such that the atomic weight of the middle member of the triad was approximately the arithmetic mean of the other two.
- Some Dobereiner's **triads**

Li	7	Ca	40	S	32	Cl	35.5
Na	23	Sr	87.5	Se	79	Br	80
K	39	Ba	137.5	Te	127.5	I	127

- **Newlands law of octaves** states that if known elements are arranged in ascending order of atomic weights similar properties recurred in every eighth element, like notes in a musical scale.
- **Lother Meyer** proposed that when the properties of the elements are plotted as a function of their atomic weights, the elements with similar properties occupied almost similar positions.
- **Mendeleevs' periodic law** is that the physical and chemical properties of the elements are periodic function of their atomic weights.
- Mendeleev first prepared the short form of periodic table by arranging the elements in the increasing order of their atomic weights.
- Mendeleev's periodic table helped in
  (i) the correction of atomic weights of elements like beryllium and indium and
  (ii) the discovery of elements like gallium, germanium.
- Defects in the Mendeleev's table are
  (i) elements like lanthanides (rare earths) having different atomic weights are kept in the same group.
  (ii) **anomalous pairs** in which the arrangement of elements inverted with respect to their increasing order of atomic weights. They are potassium and argon; tellurium and iodine, cobalt and nickel.

- **Moseley** discovered the atomic numbers from X-ray spectra of elements.
- **Moseley's periodic law** is that the physical and chemical properties of the elements are periodic functions of their atomic numbers.

LONG FORM PERIODIC TABLE

- Long form periodic table is known as **Bohr's table**.
- Long form periodic table is based on the electronic configuration of element.
- The vertical columns of the table are called **groups** while the horizontal columns are called as **periods**.
- Long form periodic table contains 7 periods and 18 groups.
- Long form periodic table is the graphical representation of **Aufbau principle**.
- Every period starts with filling of the s-orbital of a new orbit and ends with completely filling of the p-orbital of the same orbit.
- The first period contains only two elements in which the first orbit (K-shell) is filled with the electrons. It is known as **shortest period**.
- The second period starts with the filling of differentiating electron into 2s-orbital of second orbit (L-shell) and ends with the completely filling of 2p-orbital of the same orbit in neon.
- The second and third periods contain eight elements each and are known as **short periods**.
- The fourth and fifth periods contain eighteen elements each and are known as **long periods.**
- The sixth period contains 32 elements and is known as **longest period**.
- The seventh period is the **incomplete period**.
- Every period starts with an alkali metal and ends with an inert gas.
- The number of period to which an element belongs is equal to the number of orbits filled in the atom of that element or to the value of principal quantum number.
- The number of group to which an element belongs is equal to the number of electrons present in its outer most orbit.
- Elements of III period are known as **typical elements** because they represent the properties of the elements in that group.
- III period elements are known as **bridge elements** because the division between two subgroups A and B starts from these elements. The properties of these elements are some what mixed properties of the elements of two sub groups.
- The elements in a group will have similarities in the chemical properties due to the similar outer electron configuration but shows gradation in their physical properties.

Classification of Elements into s-, p-, d- and f-Block Elements

- Depending on the type of orbital in which the differentiating electron enters, the elements are classified into s- p- d- and f-block elements.

s-Block elements

(i) Differentiating electron enters into s-orbital.
(ii) Their general electronic configuration is ns^1 or ns^2.
(iii) Since the maximum capacity of s-orbital is 2, the s-block contain only two groups.
(iv) The first two groups of the periodic table are s-block elements.
(v) I group contain hydrogen and alkali metals with outer electronic configuration ns^1.
(vi) II group contain alkaline earth metals with outer electronic configuration ns^2.
(vii) Except the hydrogen all the s-block elements are metals and form ionic compounds.
(viii) All the s-block elements exhibit a fixed valency.

p-Block elements

(i) The last six groups of the periodic table are p-block elements.
(ii) Differentiating electron enters into p-orbital.
(iii) Their general outer electronic configuration is ns^2np^1 to ns^2np^6.
(iv) p–block contain metals, non-metals and metalloids.
(v) The only p-block element in which the differentiating electron does not enter into p-orbital is He.
(vi) p-block elements exhibit variable valency.

d-Block elements

(i) The middle 10 groups of the periodic table are d-block elements.
(ii) The differentiating electron enters into (n–1)d orbital.
(iii) Their general outer electronic configuration is $(n-1)d^{1-10}$, $ns^{1 \text{ or } 2}$.
(iv) The d-block elements are placed in between the s and p-block elements.
(v) All the d-block elements are metals.

f-Block elements

(i) The lanthanides and actinides which are placed below the periodic table are known as f-block elements.

(ii) The differentiating electron enters into the f-orbital of antipenultimate shell.
(iii) Their general outer electronic configuration is $(n-2)f^{1-14}\ (n-1)d^{0 \text{ or } 1}\ ns^2$.
(iv) In the elements after lanthanum starting from cerium (Z = 58) to lutetium (Z = 71) the differentiating electron enters into 4f orbital so they are known as **4f-series**.
(v) Since the properties of elements of 4f-series are similar to lanthanum they are known as **lanthanides** or **lanthanoids** or **lanthanons**.
(vi) In the elements after actinium starting from thorium (Z = 90) to lawrencium (Z = 103) the differentiating electron enters into 5f-orbital. So, they are known as **5f-series.**
(vii) Since the properties of 5f-series elements are similar to actinium they are known as **actinides** or **actinoids** or **actinons.**
(viii) Each series of f-block contain 14 elements.

Classification of the Elements Based on Properties

- Depending on the chemical properties and electronic configuration, the elements are classified into four types: (a) inert gases, (b) representative elements, (c) transition elements and (d) inner transition elements.

Inert gases

(i) These are '0' group elements.
(ii) There are six elements He, Ne, Ar, Kr, Xe and Rn.
(iii) Except in He ($1s^2$) all the other elements have ns^2 np^6 outer electronic configuration.
(iv) All are monoatomic gases.
(v) All are chemically inert due to the presence of stable electronic configuration ns^2np^6 in their outermost shell.

Normal (or) Representative elements

(i) Except zero group all the remaining s- and p-block elements of the periodic table are called representative elements.
(ii) Their general outer electronic configuration is $ns^{1-2}\ np^{0\text{-}5}$.
(iii) Their outermost orbit is incomplete.
(iv) Metals, non-metals and metalloids are present in representative elements.
(v) All are chemically reactive and to get stable inert gas configuration (ns^2np^6) they lose or gain or share electrons.

Transition elements

(i) These are d-block elements.
(ii) Their general outer electronic configuration is $(n-1)d^{1-10}\ ns^{1 \text{ or } 2}$.
(iii) In these elements the outermost and penultimate shells are incompletely filled.
(iv) Differentiating electron enters into the d-orbital of the penultimate shell.
(v) There are four transition series and each series contain 10 elements.
(vi) The first transition series is from scandium (Z = 21) to zinc (Z = 30) in which differentiating electron enters into 3d–orbital and hence are known as 3d-series.
(vii) The 2nd transition series is from yttrium (Z = 39) to cadmium (Z = 48) in which the differentiating electron enters into 4d -orbital so known as 4d-series.
(viii) The 3rd transition series is from hafnium (Z = 72) to mercury (Z = 80) including lanthanum (Z = 71) in which the differentiating electron enters into 5d-orbital so known as 5d series.
(ix) The fourth transition series is incomplete.
(x) Due to small size, high effective nuclear charge, presence of unpaired - electrons, transition elements exhibit characteristic properties.
(xi) All the transition elements are hard and heavy metals.
(xii) The transition elements have high m.pts, b.pts, densities.
(xiii) They exhibit variable valence.
(xiv) Due to the d-d transition their compounds are coloured.
(xv) The transition elements and their compounds act as catalysts eg: Nickel in the hydrogenation of oils, iron powder mixed with molybdenum powder as promoter in the manufacture of ammonia are used as catalysts.
(xvi) Due to the presence of unpaired electrons in d-orbitals the transition elements are paramagnetic.
(xvii) Transition elements form alloys like brass, bronze, german silver etc.

Inner transition elements

(i) These are f-block elements.
(ii) Their outer most three orbits are incompletely filled.
(iii) Their general outer electronic configuration is $(n-2)f^{1-14}$, $(n-1)d^{0 \text{ or } 1}\ ns^2$.
(iv) Differentiating electron enters into the antipenultimate shell.
(v) Since the electronic configuration in the outermost two orbits is similar in these elements, they have very much similarities in their physical and chemical properties.
(vi) These are two series 4f- and 5f-series or lanthanides and actinides.
(vii) Lanthanides are also known as rare earths because of their rare occurence.

(viii) The common stable oxidation state of lanthanides is +3.
(ix) The elements after uranium in actinides or 5f-series are called **transuranium elements**.
(x) All the transuranium elements are man made, synthetic and radioactive.
(xi) The lanthanides and actinides are present in the IIIB group and 6 and 7 periods of long form periodic table.

Classification of Elements into Metals, Non-metals and Metalloids

- Elements are classified into **metals, non-metals and metalloids** basing on their properties.
- 78 per cent of the known elements are metals and appear on the left side of the periodic table.
- Metals usually have high m.pts. and b.pts.
- Mercury is liquid, gallium and caesium have very low m.pts.
- All metals are good conductors of heat and electricity.
- All metals are malleable and ductile.
- **Non-metals** are on the right-hand side of the periodic table.
- Non-metals are usually solids or gases. Only bromine is a liquid non-metal.
- Generally non-metals have low m.pts and b.pts.
- Non-metals are poor conductors of heat and electricity.
- Most non-metallic solids are brittle and are neither malleable nor ductile.
- In a group as we move down a group metallic character increases and non-metallic character decreases.
- In a period from left to right along a period metallic character decreases and non-metallic character increases.
- If a diagonal line is drawn starting from boron to astatine, the elements along this line are metalloids showing the properties of both metals and non-metals. The elements above the line are non-metals and elements below the line are metals.

PERIODIC TRENDS IN PROPERTIES OF ELEMENTS

- Periodicity means that when elements are arranged according to their electronic configuration, the properties of elements in the periodic table reoccur at regular intervals. Such properties which reoccur are called as **periodic properties**.
- Only the **outer electrons** are responsible for the chemical properties but not the electrons in the inner orbits.
- As the electronic configuration in the outer most orbit changes the chemical properties of the elements also changes.
- The elements having similar outer electronic configuration have same chemical properties.
- The periodicity occurs at regular intervals after every 2, 8, 8, 18, 18 and 32 elements.

Shielding Effect: Effective Nuclear Charge

- The decrease in the force of attraction exerted by the nucleus on the valence electrons due to the presence of electrons in the inner shells is called **screening effect** or **shielding effect.**
- The actual attractive power of the nucleus reaching the valence electrons after shielded by the inner electrons is called **effective nuclear charge** and represented by Z^*.
- Z^* is always less than Z (actual nuclear charge) and can be calculated by
  $Z^* = Z - S$ where S is the screening constant
- The screening constant S can be calculated by following **slater** rules.
  1. The electronic configuration of the elements should be written in the following order and groupings.
     (1s); (2s, 2p); (3s, 3p); (3d); (4s 4p); (4d); (4f); (5s 5p) etc.
  2. Electrons in any group which are right to the (ns, np) group do not contribute anything to the shielding effect.
  3. All the electrons in the (ns, np) group (valence shell or outermost orbit) shield the valence electron to an extent of 0.35 each.
  4. All the electrons in the n–1 shell (penultimate shell) shield to an extent of 0.85 each.
  5. All the electrons in n–2 shell or lower shield completely, i.e., their contribution is 1.00 each.
  6. All the electrons in groups lying to the left of 'nd' or 'nf' group contribute 1.00.

Atomic Radii and Ionic Radii

- The distance from the nucleus to the outermost shell is called as **atomic radius**.
- Atomic radius cannot be determined directly because according to wave mechanical picture, the electron cloud is more diffused.
- Atomic radius of atom is effected by (i) the type of bonding, (ii) its oxidation state, (iii) its coordination number and (iv) number of bonds between the atoms.

- **Metallic or crystal radius** is half the internuclear distance between the adjacent atoms in a metal crystal.
- The internuclear distance in sodium metal crystal is 372 pm. So, the atomic radius is 186 pm.
- **Van der Waal's radius** is half the internuclear distance of the two atoms of different molecules when they come close due to Van der Waal's attractive forces.
- Van der Waal's radius is 40 per cent greater than the radius measured when the atoms are in chemical bond.
- **Covalent radius** is half the internuclear distance of the two atoms of same element in a covalent bond.
- In heteroatomic molecules the covalent radius of one atom can be calculated if the covalent radius of another atom and the bond distance in that molecule is known

  E.g.: The C – H bond distance is 110 pm and the covalent radius of carbon is 77pm then the covalent radius of hydrogen is 110–77 = 33 pm
- The bond distance in a covalent molecule is equal to the sum of the covalent radii of the two atoms in that molecule.
- Covalent radius is affected by (i) the number of bonds between two atoms, (ii) the oxidation number and (iii) coordination number.
- As the number of covalent bonds between two atoms increases the covalent radius decreases. **E.g.:** The covalent radius of carbon decreases with increase in the number of bonds between carbon atoms.

Types of bond	Bond length	Covalent radius
C – C	154 pm	77 pm
C = C	134 pm	67 pm
C º C	120 pm	60 pm

- As the oxidation number of atom increases the covalent radius decreases. **E.g.:** Fe^{3+} is smaller than Fe^{2+}.
- In a group from top to bottom the atomic radius increases gradually due to the increase in the number of orbits.
- In a period from left to right the atomic radius decreases due to the increase in the effective nuclear charge.
- In every period the alkali metal have the maximum atomic size while the halogen have the minimum size.
- In a period atomic radius decreases upto halogen but suddenly increases in inert gases, because the atomic radii of inert gases is Van der Waal's radii.
- The decrease in the atomic size in a period is more when the differentiating electron enters into s-orbital.
- In transition elements the atomic radius decreases slowly in the beginning, remains constant in the middle and then increase a little at the end.
- The initial decrease in atomic radii of transition elements is due to the increase in effective nuclear charge.
- The constant atomic radii in the middle and a little increase at the end in a transition series is due to repulsion between paired electrons.
- In lanthanides the atomic radii decreases with increase in the atomic number in a regular manner due to the poor shielding capacity of f-electrons.This is known as **lanthanide contraction**.
- The lanthanide contraction is more pronounced in their trivalent ions than in lanthanides.
- Because of lanthanide contraction the crystal structures and their properties are very similar which makes their separation difficult.
- Due to lanthanide contraction the atomic sizes of 4d and 5d transition elements become almost equal due to which their properties are very close.
- **Ionic radius** is the effective distance from the nucleus of an ion upto which it has its influence on its electron cloud.
- Positive ion is always smaller than the parent atom because of the contraction of orbits by the increased effective nuclear charge.
- The size of the negative ion is always greater than the parent atom because of the decrease in the effective nuclear charge.
- Ions having the same number of electrons but with different nuclear charge are known as **isoelectronic series**.
- In isoelectronic series the size of the ion decreases as the number of positive charges increases and the number of negative charges decreases.

Ionization Enthalpy

- The amount of energy required to remove the most loosely bound electron from an isolated, gaseous atom in a ground state is known as ionization enthalpy.
- The ionization enthalpy is expressed either in terms of ev / atom or k.ca*l* mol^{-1} or K.J $mole^{-1}$ 1 ev $atom^{-1}$ = 23.06 k.Ca*l* mol^{-2} = 96.45 KJ $mole^{-1}$.
- With increase in the atomic size ionization enthalpy decreases due to the decrease in attractive power of nucleus on electrons of the outermost orbit.
- With increase in the effective nuclear charge ionization enthalpy increases.
- If the number of electrons in the inner shells are more, shielding capacity of the inner electrons on the nuclear charge will be more. Hence, ionization enthalpy increases.
- With increases in the no. of positive charges on an ion ionization enthalpy increases.
- With increase in the no. of negative charges on an ion ionization enthalpy decreases.
- If the valence electrons are more penetrated into inner shells ionization enthalpy increases.

- Penetration power of different orbitals is in the order s > p > d > f.
- The energy required to remove an electron from different types of orbitals of same orbit is in the order s > p > d > f.
- If the outermost electronic configuration is stable, i.e., either exactly half-filled or completely filled, ionization enthalpy is more.
- The element with highest ionization enthalpy is He.
- In a graph showing relation between ionization enthalpy and atomic number the inert gases appear at the peaks and alkali metals appear at the bottoms.
- The number of ionization enthalpies that an element possess is equal to its atomic number or the number of electrons possessed by that species.
- Ionization enthalpies are measured in discharge tube.
- If the ionization enthalpy of an element is less it is more reactive and acts as strong reducing agent.
- In a group from top to bottom ionization enthalpy decreases due to the increase in atomic size and increases in the screening effect of inner electrons.
- In a period from left to right ionization enthalpy increases due to the increase in the effective nuclear charge.
- Ionization enthalpy of Be is greater than B because the electron to be removed in B is from 2p orbital which is at slightly higher energy than the 2s electron in beryllium. The 2p electron in boron is well shielded by two orbitals 1s and 2s where as the 2s electron in beryllium is shielded by only one orbital 1s and is nearer to the nucleus.
- Due to similar reasons as explained above the first ionization enthalpy of aluminium is less than magnesium.
- The first ionization enthalpy of nitrogen is greater than oxygen because in nitrogen the electron to be removed is from stable exactly half-filled 2p-orbital. Due to similar reasons first ionization enthalpy of phosphorus is greater than sulphur.
- In any period the alkali metal have lowest ionization enthalpy and inert gas will have the highest ionization enthalpy.

Electron Gain Enthalpy

- The energy released when an electron is added to a neutral gaseous isolated atom to convert into anion is called **electron gain enthalpy** or **electron affinity**.
- Electron gain enthalpy is represented as negative value while electron affinity is represented as positive value, and are related as Δ_{eg} H = –Ae – 5/2 RT because at different temperatures heat capacities of the reactants and the products have to be taken into account.
- Electron gain enthalpies are expressed in electron volts per atom or kilo joules per mole of atoms.
- Electron gain enthalpies can be calculated from **Born – Haber cycle**.
- Electron affinity is the measure of **oxidation power** of an element.
- Electron affinity depends on size, effective nuclear charge and electronic configuration of an element.
- The electron affinities of inert gases are negative because they have no tendency to gain electron due to stable ns^2np^6 configuration.
- Electron affinities of Be and Mg (2nd group) are negative.
- The electron affinity in a period increases from left to right due to the increase in the effective nuclear charge.
- The electron affinity in a group from top to bottom decreases due to the increase in size.
- The electron affinity of chlorine is more positive than fluorine because of the repulsion between the extra electron added and the electrons already present in small second orbit.
- Due to similar reasons the electron affinity of oxygen is less positive than sulphur and that of nitrogen is less positive than phosphorus.
- The electron affinities of group – VA elements are very low because of stable exactly half-filled np^3 configuration.
- In metals gold has maximum electron affinity.
- The electron affinity of a neutral atom (X) is equal to the ionization - enthalpy of its negative ion (X^-).
- The ionization enthalpy of a neutral atom (M) is equal to the electron gain enthalpy of its cation (M^+).
- The second electron gain enthalpy of any element is endothermic (energy will be absorbed in the addition of second electron to a uninegative ion).
- In a period the halogens will have maximum negative electron gain enthalpy.
- Of all the elements chlorine has the maximum negative electron gain enthalpy.

Electronegativity

- The tendency of an atom in a compound to attract the shared pair of electrons in a covalent bond towards itself is known as **electronegativity**.
- Though both electron gain enthalpy and electronegativity refers to the measure of electron attracting power, the electron affinity refers to an isolated gaseous atom while electronegativity refers to bonded atom in a molecule.
- **Pauling scale** is the most widely used scale.
- Pauling scale is based on **bond energies**.

- Pauling assigned arbitrary values of electro negativities 4.0 for most electronegative element fluorine and 2.1 for hydrogen.
- Pauling calculated the electronegativities of other elements by using the formula $\chi_A - \chi_B = 0.208$

  where χ_A and χ_B are electronegativities of A and B and Δ is the difference in the actual bond energy and calculated bond energy.

  Δ = [Actual bond energy of $E_{A-B} - \frac{1}{2}$ $[E_{A-A} + E_{B-B}]$
- Δ indicates the bond polarity.
- According to **Mulliken's Scale** the electronegativity is the average of ionization enthalpy and electron gain enthalpy.

  Electronegativity = $\dfrac{IE_1 + EA_1}{2}$
- Mulliken electronegativity values are approximately **2.8 times** greater than Pauling electronegativity values.
- Mulliken scale is applicable only to **univalent elements.**
- The electronegativities of inert gases are almost zero.
- The nearest actual values of electronegativities in Mulliken scale can be obtained by

$$\chi_M = \frac{IE + Ae}{2 \quad 2.8} = \frac{IE + Ae}{5.6}$$

- If IE and Ae are measured in kJ mol^{-1} then the above equation becomes

$$\chi_M = \frac{IE + Ae}{5.6 \quad 96.48} = \frac{IE + Ae}{540} \quad (\because 1 \text{ ev/atom} =$$

  96.48 kJ mol^{-1} of atoms)

- According to **Allred** and **Rochow** electronegativity of an atom can be calculated as

$$\chi_M = 0.744 + \frac{0.359\,Z^*}{r^2}$$

  where χ_M is the electronegativity of atom M, Z* is the effective nuclear charge of M and r is the radius of the atom of M.
- In a group from top to bottom, electronegativities decreases due to the increase in the atomic size.
- In a period from left to right, the electronegativities increases due to the decrease in the atomic size and increase in effective nuclear charge.
- Smaller atoms will have more electronegativities than bigger atoms.
- The elements whose atoms have electronic configuration nearest to inert gas configuration will have more electronegativity values.
- Metals which acquire inert gas configuration by losing electron will have less electronegativity values.
- Next to fluorine the most electronegative element is oxygen.
- The least electronegative element is caesium.
- Electronegative values are useful in predicting the nature of bond between two atoms of different elements.
- If the electronegativity difference between two elements is more than 1.7, the bond formed between their atoms will be ionic.
- Electronegativity values are useful in assigning +ve or –ve sign to the oxidation numbers of elements in a compound.
- The oxidation number for more electronegative element is given –ve sign and for less electronegative element is given +ve sign.
- While writing the formulae of inorganic compounds the less electronegative element is written first and the most electronegative element is written last.
- Greater the electronegativity of an element greater is the tendency to gain electron and hence it acts as a strong oxidizing agent and will be a non-metal.
- The per cent ionic character of a bond between two atoms can be calculated from difference in electronegativities.

  % ionic character = $16\,[\chi_A - \chi_B] + 3.5\,[\chi_A - \chi_B]^2$

  where χ_A and χ_B are the electronegativities of more electronegative and less electronegative elements A and B, respectively.
- Increase in the difference in electronegativities makes the bond more stronger. **E.g.:** In the nitrogen trihalides the order of the strength of N – X bonds.

  N – F > N – C*l* > N – Br > N – I

Periodicity of Valence

- **Valence** of an element is the number of hydrogen atoms or double the number of oxygen atoms that can combine with one atom of that element.
- The valence of an element is not always constant.
- If an element exhibit more than one valence it is known as **variable valence**.
- The **maximum valence** of an element is equal to the number of electrons present in the outermost orbit of an atom.
- The number of possible charges that can present on an atom of an element is known as **oxidation number** or **oxidation state**.
- The oxidation number of s-block elements is equal to their group number, i.e., group IA elements exhibit +1 and group IIA elements exhibit +2 oxidation states.
- The oxidation states of elements may be positive (or) negative (or) zero (or) fractional.
- p-block elements exhibit variable valence to some extent.

- The different oxidation states of an element of p-block changes by two units.
- Group IIIA elements exhibit two oxidation states +1 and +3.
- Though there is a pair of ns^2 electrons in the valence shell but are reluctant to participate in the bond, the pair of electrons is called **inert pair** and the effect is called **inert pair effect**.
- **Group IVA** elements exhibit two types of oxidation states +**2** and +**4**.
- **Group VA** elements exhibit two types of oxidation states +3 and +5.
- Since the group VIA elements are more electronegative; their general oxidation state is –2, but they also exhibit +2, +4 and +6 oxidation states in excited states.
- The group VII A elements are most electronegative elements and they generally exhibit-1 oxidation state. Except fluorine the other group VII A elements can exhibit +1, +3, +5 and +7 oxidation states also.
- The elements of group IA to IVA always exhibit oxidation number equal to group number.
- The elements in groups IVA to VIIIA exhibit an oxidation number either equal to group number or 8-group number. The general oxidation number is always equal to 8-group number.
- In a period valence with respect to oxygen increase from 1 to 7 but with respect to hydrogen increases from 1 to 4 upto fourth group and then decreases to 1 up to VII A group.
- d-block elements exhibit common oxidation state +2 due to the presence of ns^2 electrons.
- d-block elements exhibit variable valence due to the participation of ns and (n–1)d electrons.
- In the first transition series manganese exhibit maximum oxidation state +7.
- Of all the elements Os and Ru exhibit maximum oxidation state +8 in OsO_4, RuO_4.

PERIODICITY OF CHEMICAL PROPERTIES

- The tendency to loose electrons by an element is called **electropositive character**.
- Electropositive character is opposite to electronegativity.
- Generally metals exhibit electropositive character.
- If the tendency to lose the electrons is more, the electropositive character is also more.
- Electropositive character can be measured in terms of ionization enthalpy.
- Electropositive character is inversely proportional to ionization enthalpy. With increase in the electropositive character they tend to form ionic compounds, reactivity with water and acids, the tendency to form basic oxides and hydroxides increases.
- More electropositive metals dissolve in water giving hydroxides.
- More electropositive metal ions do not hydrolyse in water.
- Electropositive character can be explained in terms of electrode potentials.
- In a group from top to bottom the electropositive character increases due to the increase in atomic size.
- In a period from left to right electropositive character decreases due to the decrease in atomic size, the electrons are more attracted by the nucleus.
- In every period the more electropositive element will be alkali metal.
- Generally the metal oxides are basic and form alkaline solutions when dissolved in water.
- The first group metal oxides are strongly basic oxides.
- Oxides of the non-metals are acidic and form acids when dissolved in water.
- Oxides of the group VIIB elements are strongly acidic.
- The oxides of metalloids are amphoteric. They have both acidic and basic properties.
- In a group from top to bottom basic nature of the oxides gradually increase while the acidic nature gradually decrease.
- In a period from left to right basic nature of the oxides gradually decrease while acidic nature gradually increase.

DIAGONAL RELATIONSHIP

- The second period elements show some similarities with the third period elements which are diagonal to them in the adjacent group and this known as **diagonal relationship:**

$$\begin{array}{ccccccc} \mathrm{Li} & & \mathrm{Be} & & \mathrm{B} & & \mathrm{C} \\ & \searrow & & \searrow & & \searrow & \\ \mathrm{Na} & & \mathrm{Mg} & & \mathrm{A}l & & \mathrm{Si} \end{array}$$

- Diagonal relationship do not occur after IVA group.
- The reason for the diagonal relationship is due to the similar ionic sizes and electronegativities.
- BeO and Al_2O_3 are amphoteric oxides.
- Carbides of beryllium and aluminium on hydrolysis gives same methane gas

$$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$$
$$Be_2C + 4H_2O \longrightarrow 2Be(OH)_2 + CH_4$$

- In modern concept diagonal relationship is due to similar polarizing power of the ions.

$$\text{Polarizing power} \propto \frac{\text{ionic charge}}{(\text{Ionic radius})^2}$$

EXERCISE–I

Single Answer Type Questions

1. What conclusions can be drawn from the following reactions?

$$H_2 + 4.5eV \longrightarrow H + H$$
$$H + 13.6eV \longrightarrow p^+ + e^-$$

(a) It is more difficult to break up an H_2 molecule than it is to break up a hydrogen atom.
(b) It is easier to break up an H_2 molecule than it is to breakup a hydrogen atom.
(c) The average energy of formation of H and p+ are the same.
(d) Electron and proton attraction in an H_2 molecule as well as an H atom are the same.

2. The outer electronic structure of lawrencium (atomic number 103) is
(a) Rn: $5f^{13}7s^27p^2$ (b) Rn: $5f^{13}6d^17s^17p^2$
(c) Rn: $5f^{14}\ 7s^1\ 7p^2$ (d) Rn: $5f^{14}6d^1\ 7s^2$

3. In the periodic table, metallic character of the elements shows one of the following trend.
(a) Decreases down the group and increases across the period.
(b) Increases down the group and decreases across the period.
(c) Increases across the period and also down the group.
(d) Decreases across the period and also down the group.

4. The ionic radii of Li^+, Be^{2+} and B^{3+} follow the order
(a) $Be^{2+} > B^{3+} > Li^+$ (b) $Li^+ > B^{3+} > Be^{2+}$
(c) $B^{3+} > Be^{2+} > Li^+$ (d) $Li^+ > Be^{2+} > B^{3+}$

5. Which of the following is arranged in order of increasing radius?
(a) K^+ (aq) < Na^+ (aq) < Li^+ (aq)
(b) Na^+ (aq) < K^+ (aq) < Li^+ (aq)
(c) K^+ (aq) < Li^+ (aq) < Na^+ (aq)
(d) Li^+ (aq) < Na^+ (aq) < K^+ (aq)

6. A, B and C are hydroxy compounds of the elements X, Y and Z, respectively. X, Y and Z are in the same period of the periodic table. A gives an aqueous solution of pH less than seven. B reacts with both strong acids and strong alkalis. C gives an aqueous solution which is strongly alkaline.
Which of the following statements is/are true?
I. The three elements are metals
II. The electronegativities decreases from X to Y to Z.
III. The atomic radius decreases in the order X, Y and Z.
IV. X, Y and Z could be phosphorus, aluminium and sodium, respectively.
(a) I, II, III only correct
(b) I and III only correct
(c) II, IV only correct
(d) II, III and IV only correct

7. Which of the following statements is incorrect?
(a) The second ionization energy of sulphur is greater than that of chlorine.
(b) The third ionization energy of phosphorus is greater than that of aluminium.
(c) The first ionization energy of aluminium is approximately the same as that of gallium.
(d) The second ionization energy of boron is greater than that of carbon.

8. The incorrect statement among the following is
(a) The first ionization potential of A*l* is less than the first ionization potential of Mg.
(b) The second ionization potential of Mg is greater than the second ionization potential of Na.
(c) The first ionization potential of Na is less than the first ionization potential of Mg.
(d) The third ionization potential of Mg is greater than the third ionization potential of A*l*.

9. Which of the following process refers to the ionization potential?
(a) $Y_{(s)} \longrightarrow Y^+_{(g)} + e^-$
(b) $Y_{(g)} + aq \longrightarrow Y^+_{(g)} + e^-$
(c) $Y_{(g)} \longrightarrow Y^+_{(g)} + e^-$
(d) $Y_{(g)}\ e^- \longrightarrow Y^-_{(g)}$

10. The first ionization potentials of Na, Mg, A*l* and Si are in the order
(a) Na < Mg > A*l* < Si
(b) Na > Mg > A*l* > Si
(c) Na < Mg < A*l* > Si
(d) Na > Mg > A*l* < Si

11. A metal from period 4 is added to water and a vigorous reaction takes place with the evolution of a gas. Which statements are correct?
I. Oxygen is evolved.
II. Hydrogen is evolved.
III. The resulting solution is acidic.
IV. The resulting solution is basic.
(a) I and II only (b) II and III only
(c) II and IV only (d) I and IV only

12. Lanthanides are called
(a) rare earths
(b) alkali metals
(c) alkaline earth metals
(d) pnicogens

13. Ionic radii are
(a) inversely proportional to effective nuclear charge
(b) inversely proportional to the square of the effective nuclear charge

(c) directly proportional to the effective nuclear charge
(d) directly proportional to the square of effective nuclear charge

14. If the ionic radii of K^+ and F^- are about 1.34 Å each, then the expected values of atomic radii of K and F should be, respectively
(a) 1.34 and 1.34 Å (b) 2.31 and 0.64 Å
(c) 0.64 and 2.31 Å (d) 2.31 and 1.34 Å

15. The species F^-, Ne and Na^+ all have the same number of electrons. Which is the correct order when they are arranged in the order of decreasing size (largest first)?
(a) $F^- > Ne > Na^+$ (b) $Ne > Na^+ > F^-$
(c) $Na^+ > F^- > Ne$ (d) $F^- > Na^+ > Ne$

16. Which of the following is correct order of size?
(a) $Na > N^{3-} > M_g^{2+}$ (b) $Mg^{2+} > Na > N^{3-}$
(c) $N^{3-} > Na > Mg^{2+}$ (d) $Mg^{2+} > N^{3-} > Na$

17. The radius of cation is r_+, anion is r_-, Z_a is nuclear charge of anion, Zc is nuclear charge of cation, 'S' is screening constant. Which of the following is correct?
(a) $\frac{r_c}{r_a} = \frac{Z_a - S}{Z_c - S}$ (b) $\frac{r_a}{r_c} = \frac{Z_a - S}{Z_c - S}$
(c) $\frac{r_a}{r_c} = \frac{Z_a}{Z_c}$ (d) $\frac{r_a}{r_c} = \frac{Z_c}{Z_a}$

18. A certain compound when burnt gave three oxides, the first one turned lime water milky, the second turned anhydrous cobalt chloride paper pink and the third formed an aqueous solution of pH nearly three. Which are the elements present in that compound?
(a) C, H, S (b) C, H, Ca
(c) C, H, Ca (d) C, S, O

19. The normal boiling points of four elements with consecutive atomic numbers are tabulated below.

	W	X	Y	Z
Atomic number	N	N + 1	N + 2	N + 3
Normal boiling point/°c	58	–152	688	1380

(Note that W, X, Y and Z are codes which are unrelated to element's real symbols)
Which of the following compounds would be the most stable?
(a) W_2X (b) W_2Y
(c) W_2Z (d) XY_2

20. Elements Q and R form both oxides and chlorides under laboratory conditions, QCl_2 is a cherry-red liquid while RCl_2 is a white solid. Which one of the following statements is most likely to describe the oxides of these elements?
(a) Q forms a basic oxide QO while R forms an acidic oxide RO.
(b) Q forms a basic oxide QO_2 while R forms two acidic oxides RO and RO_2.
(c) Q forms two acidic oxides QO_2 and QO_3 while R forms a basic oxide.
(d) Q forms an acidic oxide QO while R forms a basic oxide RO_2.

21. The correct order of 1st ionization potentials is
(a) $Ne > Cl > P > S > Al > Mg$
(b) $Ne > Cl > P > S > Mg > Al$
(c) $Ne > Cl > S = P > Mg > Al$
(d) $Ne > Cl < S < PS < Al > Mg$

22. Which of the following statements is incorrect?
(a) Second ionization potential of Li is more than second ionization potential of oxygen
(b) Electron gain enthalpy of sulphur is most exothermic in its group.
(c) Ionization potential of Cu > Ag > Au
(d) Electronegativity of In < Tl

23. The first ionization energy of Ar is less than that of Ne. An explanation of this fact is that
I. The effective nuclear charge experienced by a valence electron in Ar is much larger than that in Ne.
II. The effective nuclear charge experienced by a valence electron in Ar is much smaller than that in Ne.
III. The atomic radius of Ar is larger than that of Ne.
IV. The atomic radius of Ar is smaller than that of Ne.
(a) I and II (b) I and II
(c) II and III (d) III and IV

24. Which one of the following is an example of a positive ion and negative ion that is isoelectronic with F^-?
(a) S^{2-} and Mg^{2+} or Na^+
(b) O^{2-} and Ca^{2+} or K^+
(c) O^{2-} and Mg^{2+} or Li^+
(d) O^{2-} and Mg^{2+} and Na^+

25. Which of the following is an example of positive ion and negative ion that is isoelectronic with argon?
(a) K^+ and Cl^- or Ca^{2+} and S^{2-}
(b) Na^+ and Cl^- or Mg^{2+} and O^{2-}
(c) K^+ and F^- or Mg^{2+} and S^{2-}
(d) K^+ and Br^- or Mg^{2+} and O^{2-}

26. The screening effect of inner electrons of the nucleus causes
(a) a decrease in the ionization potential
(b) an increase in the ionization potential
(c) no effect on the ionization potential
(d) an increase in the attraction of the nucleus to the electrons

27. An element X which occurs in the first short period has outer electronic structure s^2p^1. What is the formula and acid base character of its oxide?

(a) XO_3 basic (b) XO_3 acidic
(c) X_2O_3 acidic (d) X_2O_3 basic

28. X, Y and Z are the elements in the same short period. The oxide of Z is ionic, the oxide of X is a giant molecule. Y forms acidic oxide. The arrangement of the elements in order of increasing atomic number would be
(a) X, Y, Z (b) X, Z, Y
(c) Z, X, Y (d) Y, Z, X

29. Which of the following conversions show minimum energy release?
(a) $C \longrightarrow C^-$ (b) $N \longrightarrow N^-$
(d) $O \longrightarrow O^-$ (d) $F \longrightarrow F^-$

30. Choose the incorrect statement.
(a) The third IE of P is greater than that of A*l*.
(b) The first IE of A*l* is same as that of Ga.
(c) The second IE of S is greater than that of C*l*.
(d) The second IE of B is greater than that of C.

31. The correct statement regarding BOH is (χ is electronegativity)
(a) If $\chi_o - \chi_B > \chi_o - \chi_H$, BOH will be basic.
(b) If $\chi_o - \chi_B > \chi_o - \chi_H$, BOH will be acidic.
(c) If $\chi_o - \chi_B = \chi_o - \chi_H$, BOH will be acidic.
(d) If $\chi_o - \chi_B < \chi_o - \chi_H$, BOH will be basic.

32. Periodicity in the properties of elements is due to
(a) a regular increase in atomic weight of elements
(b) successive increase in the atomic number of elements
(c) periodicity in the electronic configuration of atoms of elements
(d) existence of families of elements

33. The ions O^{2-}, F^-, Na^+, Mg^{2+} and Al^{3+} are isoelectronic. Their radii show
(a) an increase from O^{2-} to F^- and then decrease from Na^+ to Al^{3+}
(b) a decrease from O^{2-} to F^- and then increase from Na^+ to Al^{3+}
(c) a significant increase from O^{2-} to Al^{3+}
(d) a significant decrease from O^{2-} to Al^{3+}

34. Five ionization energy values in kJ/mol are listed below $E_1 = 870$, $E_2 = 830$, $E_3 = 1010$, $E_4 = 1290$, $E_5 = 376$. These are
(a) successive ionization energies for the element with atomic number 5
(b) the first IE of successive elements in group 15, 16, 17, 18 and 1, respectively
(c) the first IE of elements with atomic numbers 1-5
(d) successive IE for transition elements with four electrons in d-subshell

35. Consider the following changes
I. $M_{(s)} \longrightarrow M_{(g)}$
II. $M_{(s)} \longrightarrow M^{2+}_{(g)} + 2e^-$
III. $M_{(g)} \longrightarrow M^+_{(g)} + e^-$
IV. $M^+_{(g)} \longrightarrow M^{2+}_{(g)} + e^-$
V. $M_{(g)} \longrightarrow M^{2+}_{(g)} + 2e^-$
The second ionization energy of M could be calculated from the energy values associated with
(a) 1 + 5 (b) 2 + 4
(e) 5 – 3 (d) 2 – 1+ 3

36. Aqueous solutions of two compounds M – O – H and M′ – O – H have been prepared in two different beakers. If the electronegativity of M = 3.0, M′ = 1.72, O = 3.5 and H = 2.1, then the solutions respectively are
(a) acidic, acidic
(b) acidic, basic
(c) basic, basic
(d) basic, acidic

37. If an electron is transferred from A to B forming A^+, and B^-. The above reaction is possible when
(a) $(E_B + I_B) > (I_A - E_A)$
(b) $(I_A - E_B) > (I_B - E_A)$
(c) $(I_B + E_B) > (I_A + E_A)$
(d) $(I_A + E_A) > (I_B - E_B)$

38. The ionization energy of boron is less than that of beryllium because
(a) beryllium has a higher nuclear charge than that of boron
(b) beryllium has lower nuclear charge than boron
(c) the outermost electron in boron occupies a 2p-orbital
(d) the 2s and 2p – orbitals of boron are degenerate

39. Which of the following statements are correct?
I. Electron affinity is defined as the ionization energy of the uninegative gaseous ion.
II. First electron affinity of F is less than that of C*l* due to its small size.
III. Second electron affinity of O is endothermic.
IV. In a given period, noble gases have the highest electron affinity.
The correct statements are
(a) All are correct
(b) Only (II) is correct
(c) (I) and (IV) are correct
(d) (II) and (III) are correct

40. Choose the correct statements among the following.
I. The formation of ion from neutral atom is always endothermic.
II. More the charge on an ion, more stable it is.
III. Electron affinity values may be +ve or –ve
The correct statements are
(a) All are correct. (b) All are wrong.
(c) I, II are correct. (d) Only III is correct.

41. Choose the correct statement from the following.
I. In the periodic table the increase in electronegativity usually follow those of the first ionization energy.

II. In any period the highest ionization energy is possessed by a halogen.
III. The ability of an atom to form a stable anion is related to its electronegativity.
The correct statements are
(a) All are correct (b) All are wrong
(c) I and III are correct (d) I and II are correct

42. Exactly half-filled and completely filled subshell give stability to the atoms. Then which of the following is the correct order of stability?
(a) $p^3 < d^5 < d^{10} < p^6$ (b) $p^3 < d^{10} < d^5 < p^6$
(c) $d^5 < p^3 < d^{10} < p^6$ (d) $d^5 < d^{10} > p^3 < p^6$

43. Which of the isoelectronic ions K^+, Ca^{2+}, Cl^- and S^{2-} has the lowest ionization energy?
(a) K^+ (b) Ca^{2+}
(c) Cl^- (d) S^{2-}

44. The experimental bond energy of HY differs from its calculated value 2.0 kcal/mol. The electronegativity of 'Y' is equal to
(a) 1.5 (b) 1.8
(c) 1.6 (d) 1.9

45. Which of the following statements is false?
(a) Reducing power of elements decreases along a period.
(b) Oxidizing power of elements increases along a period.
(c) Basic nature of oxides increases along a period.
(d) Electronegativity of elements increases and along a period.

46. The true statement about the electronegativity and electron affinity is as
(a) Both represent the same quantity.
(b) EA is a characteristic of single atom while electronegativity is the characteristic of bonded atom.
(c) EA is always lesser than electronegativity.
(d) Both of them release or absorb energies during the process.

47. Choose the incorrect statement
(a) Van der Waal's radius of Iodine is more than its covalent radius.
(b) All isoelectronic ions belong to same period of Periodic Table.
(c) First IE of N is higher than that of O while IE_2 of O is higher than that of N.
(d) The electron affinity of N is 0 while that of P is the 3 kJ/mol.

48. Due to internal screening effect of electrons in atoms.
(a) ionization potential decreases.
(b) ionization potential increases.
(c) no change in ionisation potential.
(d) attraction of nucleus on electron increases.

49. The radius of isoelectronic species
(a) increases with increase in nuclear charge.
(b) decreases with increase in nuclear charge.
(c) same for all.
(d) first increases and then decreases.

50. Which of the following statements are correct?
(a) On moving across a period polarizing power generally increases.
(b) On moving down a group polarizing power generally increases.
(c) On moving across a period polarizing power generally decreases.
(d) On moving down a group polarizing power generally remain constant.

More Than One Correct Answers

1. The properties which are common to both groups 1 and 17 elements in the periodic table are
(a) Electropositive character increases down the groups.
(b) Reactivity decreases from top to bottom in these groups.
(c) Atomic radii increases as the atomic number increases.
(d) Electronegativity decreases on moving down a group.

2. Which of the following pairs have approximately the same atomic radii?
(a) Zr and Hf (b) Al and Mg
(c) Al and Ga (d) Na and Ne

3. Ionization energy of an element is
(a) equal in magnitude but opposite in sign to the electron gain enthalpy of the cation of the element.
(b) same as electron affinity of the element.
(c) energy required to remove one valence electron from an isolated gaseous atom in its ground.
(d) equal in magnitude but opposite in sign to the electron gain enthalpy of the anion of the element.

4. Consider the following ionization steps:
$$M_{(g)} \longrightarrow M^+_{(g)} + e^-; \Delta H = 100\text{ev}$$
$$M_{(g)} \longrightarrow M^{2+}_{(g)} + 2e^-; \Delta H = 250\text{ev}$$
Select the correct statement(s)
(a) IE_1 of $M_{(g)}$ is 100 eV
(b) IE_1 of $M^+_{(g)}$ is 150 eV
(c) IE_2 of $M_{(g)}$ is 250 eV
(d) IE_2 of $M_{(g)}$ is 150 eV

5. Sulphur is
(a) s-block element
(b) non-metal
(c) valency shell contains 6 electrons
(d) representative element

6. Amongst the following statements which is/are correct?
(a) The second ionization potential of boron is greater than that of carbon.
(b) First ionization potential of S is greater than that of 'P'.
(c) The second IP of Mg is greater than that of 'P'.
(d) First IP of A*l* and Ga are almost the same.

7. Choose the correct statement (s).
(a) The maximum positive oxidation state shown by any representative element is equal to the total number of electrons (s and p) in valence shell.
(b) The maximum oxidation state shown by elements in a group is also known as group oxidation number.
(c) Group oxidation number is the most common or most stable oxidation state for a particular element.
(d) All the elements in a group form some compounds in which they exhibit their group oxidation number.

8. In which of the following sets of elements Ist element is more metallic than second?
(a) Ba, Ca (b) Sb, Sn (c) Ge, S (d) Na, F

9. Which of the following affects the elctronegativity of an atom?
(a) s-character in hybridization
(b) Multiplicity of bond between atoms
(c) Oxidation number
(d) The number of neutrons in the nucleus

10. Which of the following is / are correct relation?
(a) Basic character of element
$$\propto \frac{1}{\text{effective nuclear charge}}$$
(b) Acidic character of element ∝ size of atoms which form anion
(c) Acidic character of element
$$\propto \frac{1}{\text{atomic size}}$$
(d) Acidic character of element α metallic character.

11. Which of the following is / are correct statement (s)?
(a) Tl^{3+} salts are better oxidizing agents.
(b) Ga^{+} salts are better reducing agents.
(e) Pb^{4+} salts are better oxidizing agents.
(d) As^{5+} salts are better oxidizing agents.

12. Which of the following is / are correct statement(s)?
(a) The members of the 2nd period have lower electron affinity than the next members in their respective group.
(b) Chlorine has the highest electron affinity among all elements.
(c) Fluorine has the highest electron affinity among halogens
(d) Noble gas has positive electron affinity

13. Which of the following is / are correct statement(s)?
(a) For $A^{-}_{(g)} + e^{-} \longrightarrow A^{2-}_{(g)}$; ΔH should be positive
(b) For $A^{-}_{(g)} + e^{-} \longrightarrow A^{2-}_{(g)}$; ΔH should be negative
(c) For $A^{-}_{(g)} + e^{-} \longrightarrow A^{2-}_{(g)}$; ΔH should be positive
(d) For $Ne_{(g)} + e^{-} \longrightarrow Ne^{-}_{(g)}$; ΔH should be positive

14. Which of the following is / are correct statement(s)?
(a) The electron affinity of Si is greater than that of P.
(b) Penetrating power of p-orbital is more than s-orbital.
(c) Second ionization energy of sodium is more than Mg.
(d) The numerical value of electronegativity of an atom depends on its ionization potential and electron affinity.

15. Which of the following increases the basic character of elements?
(a) Increasing metallic character of element.
(b) Increasing electronegativity of metal.
(c) Decreasing ionization potential of metal.
(d) Decreasing effective nuclear charge.

16. Which of the following statement(s) are not correct?
(a) It is possible to determine absolute values of electronegativity of an element.
(b) The second ionization energy of sodium is greater than that of magnesium.
(c) The first and second ionization energies of nitrogen are greater than those of oxygen
(d) The decreasing order of electron affinity of F, Cl, Br, is F > Cl > Br.

17. In which of the following are the orders of electron affinity of the elements or ions show correct?
(a) $S > O^{-}$ (b) $O > S^{-}$ (c) $O^{-} > S^{-}$ (d) $N^{-} > P$

18. An element of atomic weight 40 has an electronic configuration of [Ar] $4s^2$. Which of the following statements is correct regarding this element.
(a) the element forms only dipositive cation
(b) on reaction with hydrogen the element forms hydride which liberates H_2 gas at cathode on electrolysis
(c) the formula of its oxide will be MO_2
(d) the element will displace hydrogen from dilute acids.

19. The statement(s) in correct for f-block elements
(a) They belong to group 3rd (IIIA) of periodic table
(b) Their outermost three shells are incomplete
(c) They are collectively called as transuranic elements
(d) 4f elements are called rare earths or lanthanons

20. Which of the following statement(s) is / are correct for electrons gain enthalpy?

(a) Most electron gain enthalpies are negative but some are positive.
(b) A negative electron gain enthalpy implies that the univalent negative ion is lower in energy than the gaseous atom.
(c) A positive electron gain enthalpy mean that energy must be absorbed when an electron is added because the uninegative ion is higher in energy than the gaseous atom.
(d) Electron affinities are more difficult to measure than ionization energies.

Comprehension Questions

Passage-I

Study the following passage and answer the questions at the end of it.

Nuclear charge actually experienced by an electron is termed as effective nuclear charge. The effective nuclear charge Z* actually depends on the type of shell and orbital in which electron is actually present. The relative extent to which various orbitals penetrate the electron clouds of other orbitals is s > p > d > f (for the same value of 'n').

The phenomenon in which penultimate shell electrons act as screen or shield in between nucleus and valence shell electrons and thereby reducing nuclear charge is known as shielding effect. The penultimate shell electrons repel the valence shell electrons to keep them loosely held with nucleus. It is thus evident that more is the shielding effect, lesser is the effective nuclear charge and lesser is the ionization energy.

1. Which of the following valence electron experiences maximum effective nuclear charge?
(a) $4s^1$ (b) $4p^1$
(e) $3d^1$ (d) $2p^3$

2. Which of the following is not concerned to effective nuclear charge?
(a) Higher ionization potential of carbon than boron.
(b) Higher ionization potential of magnesium than aluminium.
(c) Higher values of successive ionization energy.
(d) Higher electronegativity of higher oxidation state.

3. Ionization energy is not influenced by
(a) size of atom
(b) effective nuclear charge
(c) electrons present in inner shell
(d) change in entropy

Passage-2

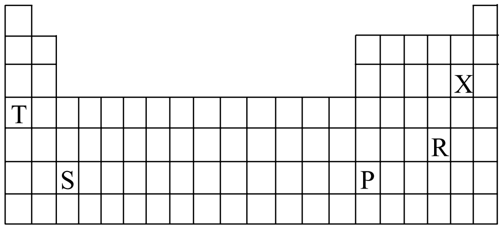
Properties that are directly or indirectly related to the electronic configuration of the elements and show a regular gradation when we move from left to right in a period or from top to bottom in a group are called periodic properties. Some of these properties are ionization energy, electron affinity and electronegativity.

1. Increasing order of second ionization energy is
(a) Ne > O > F > N > B > C > Be
(b) Be < C < B < N < F < O < Ne
(c) Be < B < C < N < F < O < Ne
(d) B < C < N < O < F < Be < Ne

2. Decreasing order of electron affinity is
(a) F > C*l* > Br > I > S > Si
(b) Si > S > F > C*l* > Br > I
(c) C*l* > F > Br > I > S > Si
(d) C*l* > F > Br > S > Si > I

3. Which of the following is arranged in the order of decreasing electropositive character?
(a) Fe, Mg, Cu (b) Mg, Cu, Fe
(c) Mg, Fe, Cu (d) Cu, Fe, Mg

Passage-3



Analyze the table and answer the following questions.

1. X is
(a) Halogen (b) Chalcogen
(c) Lanthanum (d) Actinium

2. Which one is called chalcogen?
(a) T (b) S
(c) R (d) M

3. 'S' represents an element named
(a) Yttrium (b) Cerium
(c) Lanthanum (d) Actinium

Passage-4

If a small amount of energy is supplied to an atom, then an electron may be promoted to higher energy level but if the amount of energy supplied is sufficiently large, the electron may be completely removed. The energy required to remove the most loosely bound electron from an isolated gaseous atom is called ionization energy. The ionization energy for the group 2 elements show that the first ionization energies almost double the value for the corresponding group - I elements. This is because the increase in nuclear charge results in a small size. The ionization energy also depends on the type of electron which is removed from s- p-, d- and f-sub orbitals.

1. The ionization energy of the orbital is
   (a) s > p > d > f (b) s > d > p > f
   (c) f > d > p > s (d) s = p = d = f
2. The second ionization energy is higher for
   (a) Mg (b) A*l*
   (c) Na (d) Si
3. Which of the following is correct?
   (a) Ionization energy of noble gas is highest in respective period.
   (b) Ionization energy of group - I metals is lowest in respective period.
   (c) Ionization energy of Al and Ga is almost same.
   (d) All the above are correct.

Passage-5

Ionic radius is defined as the distance between the nucleus and the outermost shell of an ion. The size of the cation of the same element decreases with the increase in positive charge. If Z/e ratio increases, the size decreases. According to Pauling, ionic radius depends on effective nuclear charge.

$$r_{ion} \; a \; \frac{1}{Z_{eff}} \qquad \text{or} \qquad r_{ion} = \frac{C}{Z_{eff}} \; A$$

For 10 electronic systems C = 6.14.

1. Which of the following is correct?
   (a) Sn^{2+} is smaller in the size than Sn^{4+}.
   (b) Fe^{3+} is bigger in size than Fe^{2+}.
   (c) F^- is smaller than F.
   (d) Al^{3+} is smaller than Al^{2+}.
2. Which one is largest in size?
   (a) Ne (b) F^-
   (c) Na^+ (d) Al^{3+}
3. The ionic radius of sodium is
   (a) 0.82A (b) 0.905
   (c) 0.35 (d) 0.865

Matching Types

1. Match the following

Columns I	Column II
(a) $3s^23p^6$	(p) Metal
(b) $6s^14f^{14}5d^{10}$	(q) Non-metal
(c) $4f^15s^25p^65d^16s^2$	(r) Noble gas
(d) $5s^24d^{10}5p^5$	(s) Lanthanides

2. The values of IE_1 and IE_2 (kJ mol^{-1}) of few elements designated by A, B, C and D are shown below in Column I. Match their characteristics listed in Column II.

Column I	Column II
(a) IE_1 2372, IE_2 5251	(p) A reactive metal
(b) IE_1 520, IE_2 7300	(q) A reactive non-metal
(c) IE_1 900, IE_2 1760	(r) A noble gas
(d) IE_1 1680, IE_2 3380	(s) A metal that forms an halide of formula Ax_2

3. In Column I the comparision of IE values of A and B are given and in Column II. The elements A and B are listed. Match them.

Column I	Column II	A	B
(a) IE_1 of A is more than that of B	(p)	Be	B
(b) IE_1 of A is less than that of B	(q)	B	C
(c) IE_3 of A is greater than that of B	(r)	C	N
(d) IE_3 of A is lessers than that of B	(s)	N	O

Integer Answer Type Questions

1. The total number of oxidation states that can be exhibited by manganese.
2. Number of ionization potential for uninegative oxide ion.

KEY

Single Answer Type Questions

1. b	**2.** a	**3.** b	**4.** d	**5.** a
6. c	**7.** a	**8.** b	**9.** c	**10.** a
11. c	**12.** a	**13.** a	**14.** b	**15.** a
16. c	**17.** a	**18.** a	**19.** c	**20.** c
21. b	**22.** c	**23.** c	**24.** d	**25.** a
26. a	**27.** c	**28.** c	**29.** b	**30.** c
31. a	**32.** c	**33.** d	**34.** b	**35.** c
36. b	**37.** c	**38.** c	**39.** d	**40.** d
41. b	**42.** c	**43.** d	**44.** b	**45.** c
46. b	**47.** b	**48.** a	**49.** b	**50.** a

Single Answer Type Questions

Explanations

4. With increase in the number of charges the size decreases.
5. The degree of hydration decreases down the group among alkali metals.
6. X is non-metal since its hydroxy compound is acidic. Y is metalloid as its hydroxy compound react with both acid and base. Z is metal.
7.

11. Metal when react with water gives H_2 and the solution is alkaline.
12.
13. (1)
14. K is bigger than K^+ and F is smaller than F^-
15.
16. (3)
18. First gas is CO_2, second one is H_2O and third one must be SO_2.
19. Z must be divalent alkaline earth metal and W must be halogen.
20. Q is SCl_2, R in RCl_2 is divalent metal.
21. 22. (3)
23. The effective nuclear charge in Ar is less due to shielding by more number of inner electrons and the atomic size of Ar is greater than Ne.
24. O^{2-}, Mg^{2+} and Na^+ are isoelectronic.
29. N is having stable half-filled outer electronic configuration.
31. If $c_0 - c_B$ is greater than $c_0 - c_H$, B – O bond ionizes more easily than O – H bond.
33. In isoelectronic species ionic radii decrease with increase in nuclear charge.
34. IE values are increasing gradually and suddenly decreased in the E_5 value indicating a change from noble gas to alkali metal.
35. $IE_2 = (IE_1 + IE_2) - IE_1$
37. In the formation of ionic bond the cation forming element should have lesser IE and EA values than the anion forming element.
40. Formation of ion means cation or anion. So, it may be endothermic or exothermic. More the charge on ion lesser is the stability. Electron affinities may be positive or negative.
44. $X_H - X_y = 0.208$
47. (2) Isoelectronic ions can be formed from different periods.
50. Across a period effective nuclear charge increases. So, polarising power increases.

More Than One Correct Answer

1. a, c, d	**2.** a, c	**3.** a, c
4. a, b, d	**5.** b, c, d	**6.** a, d
7. a, b	**8.** a, c, d	**9.** a, b, c
10. a, c	**11.** a, b, c	**12.** a, b, d
13. a, c	**14.** a, c, d	**15.** a, c, d
16. a, c, d	**17.** a, b	**18.** a, d
19. a, b, d	**20.** a, b, c, d	

More Than One Correct Answer

Explanations

2. (1, 3) Due to lanthanide contraction Zr and Hf have equal atomic radii. Due to poor shielding effect of 3d electrons in Ga, both A*l* and Ga have equal atomic radii.
9. (1, 2, 3) Increases in s character increases electronegativity. More the number of bonds also increases the attractive power of atom M bonded electrons increases due to decreases in bond length. Higher the oxidation state, more is the electronegativity.
10. (1, 2, 3) Increase in effective charge increase the acidic character and decreases the basic character of the elements.
11. (1, 2, 3) Due to inert pair effect Tl^{3+} and Pb^{4+} are unstable and act as oxidizing agents. Ga^+ is less stable it acts so, reducing agent.

Passage Comprehension Questions

Passage-1	**1.** d	**2.** b	**3.** d
Passage-2	**1.** b	**2.** c	**3.** c
Passage-3	**1.** a	**2.** c	**3.** c
Passage-4	**1.** a	**2.** c	**3.** d
Passage-5	**1.** d	**2.** b	**3.** b

Matching Types

1. a-q, r	b-p	c-p, s	d-q
2. a-r	b-p	c-s	d-q
3. a-p, s	b-q, r	c-p, r	d-q, s

Integer Answer Type Questions

1. 6 Manganese exhibit +2, +3, +4, +5, +6 and +7 oxidation states.
2. 9O^- ion contains 9 electrons, so, 9 IE values.

CHAPTER 3

Chemical Bonding

One of the most striking evidences of the reliability of two chemists, methods of determining molecular structure is the fact that he has never been able to derive satisfactory structures for supposed molecules which are infact non-existent.

Plutarch

3.1 INTRODUCTION

Except noble gases, no other elements or compounds exist as atoms under ordinary conditions. In all other cases, atoms are present as aggregates which have properties quite different from those of the individual atoms comprising them. Atomic aggregates occur as molecules, as macromolecules or as giant networks. The existence of matter in one of the three states is determined by the conditions in which the matter is present and the attractive force between the atomic aggregates. **An atom or a group of atoms having independent existence in nature is defined as a molecule.** The number of atoms present per molecule depends on the combining power of the constituent elements. The combining power is known as **valence** or **valence of the specific atoms** (i.e., the element). **The attractive force that holds two constituent atoms or oppositely charged ions together in different chemical species (i.e., molecules, or ions or atoms is called as chemical bond.**

The theory of the chemical bond.... is still far from perfect. Most of the principles that have been developed are crude, and only rarely can they be used in making an accurate quantitative prediction. However, they are the best that we have, as yet, and I agree with poincare' that "It is far better to foresee even without certainty than not to foresee at all ***Linus Pauling.***

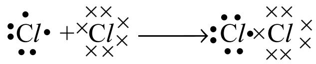
Chemical bonding is based on a single principle, **the principle of minimum energy**. The number of atoms in a molecule, their relative spacing and the overall molecular shape are all fixed because the energy of a molecule is minimum (relative to isolated atoms) only for a particular number of atoms and arrangement of the constituent atoms. The classical concept of a chemical bond is just a symbolic way to denote this energy lowering. The great merit of the "electronic" atom is that by providing an insight into the different mechanism of energy lowering in terms of electron behaviour, it provides an understanding of the nature and types of chemical bonding and of molecular structure.

3.2 KOSSEL–LEWIS THEORY

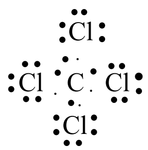
A number of attempts were made to explain the formation of chemical bond in terms of electrons, but it was only in 1916 when Kossel and Lewis explained independently in a successful manner.

Lewis discussed atomic structure in terms of positively charged **Kernel** (i.e., the nucleus plus the 'inner' electrons) and an outer shell that could contain upto eight electrons. He assumed that these outer electrons were arranged at the corners of a cube surrounding the kernel. This way the single electron in the outer shell of sodium would occupy one corner of the cube, whereas all eight corners would be occupied by the electrons in the outer shell of an inert gas. **This octet of electrons represented a particularly stable electronic arrangement, and Lewis suggested that when atoms were linked by chemical bonds, they achieved this stable octet of electrons in their outer shells.** Atoms such as sodium and chlorine could achieve an octet by the transfer of an electron from sodium to chlorine, forming Na^+ and Cl^- ions, respectively. This was essentially the mechanism proposed by Kossel.

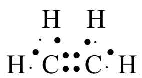
Lewis proposed a second mechanism to account for the formation of nonpolar molecules. Here, there was no transfer of electrons from one atom to another (and thus no ion formation) but the bond resulted from the sharing of a pair of electrons, each atom contributing one electron to the pair. This theory was considerably extended by **Langmuir** (1919). Although he abandoned the idea of the stationary cubical arrangement of the outer electrons, he introduced the term **covalent bond** to describe the Lewis **electron pair** bond of link. Lewis–Langmuir theory can be explained by considering the Cl_2 molecule. Chlorine, with the electronic configuration [Ne] $3s^2$ $3p^5$ is one electron short of the inert gas configuration of argon [Ne] $3s^2 3p^6$ and the formation of the stable diatomic chlorine molecule Cl_2, results from the sharing of electrons by the two chlorine atoms.



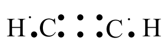
The atoms linked in this way need not be identical; thus, in carbon tetrachloride each of the four outer electrons of the carbon pairs with an electron from a chlorine atom to form four covalent bonds. The Lewis–Langmuir theory would represent the structure as follows.



Formation of double and triple bonds involve the sharing of four and six electrons, respectively. Based on this, the structures of ethene and ethyne would be written as



and



In these molecules the electron pair for each single bond is provided by the two combining atoms, each contributing one electron.

3.2.1 Lewis Dot Formulae

According to the definition of Lewis, one covalent bond between two atoms results from the sharing of a pair of electrons between the atoms. Such a pair of electrons are called **bonding electrons** and are considered to be localized or fixed between the two atoms and the bond is represented by a **line** connecting the atoms. Electrons that are not shared between atoms are localized as **lone pair** electrons on one or another atom(s) within a molecule. The electronic structure of the entire molecule is represented by the sum of all of the bonding pairs and the lone pairs of electrons. Based on these concepts, it is possible to represent the electronic structure of a molecule in diagrammatic form. Such representations are called **Lewis diagrams. The Lewis diagrams provide a picture of bonding in molecules and ions in terms of the shared pairs of electrons and the octet rule.** Though the Lewis diagrams may not explain the bonding and behaviour of a molecule completely, it does help in understanding the formation and properties of a molecule to a large extent. Lewis approach can also be extended with the use of hybridization theory, and with the VSEPR theory, to account for subtle aspects of geometry. These three concepts (i.e., the Lewis diagram, hybridization and VSEPR theory) in unison become extraordinarily powerful as an approach to structure and bonding. Eventually, however, the concepts fail because of the limitations of viewing the electrons in a strictly localized way. Resonance can be added to the paradigm, but this represents only a temporary "fix". The localized approach to bonding is useful because of its simplicity.

3.2.2 Writing Lewis Structures

For writing Lewis structures for a molecule or complex ion, only the valence electrons of the atoms or ions are considered. Lewis structure is complete when the atoms have been connected properly and the valence electrons have been distributed within the structure as either bonding or lone-electron pairs. The procedure can be summarized as follows.

First we have to draw the molecule. In drawing the molecule, several general principles can be helpful. There are exceptions to all of them, so they should be taken only as guidelines.

1. If one atom is different from others, it is positioned as the central atom with others arranged as surrounding atoms. Examples: NH_3, SO_3, CH_4, SO_4^{2-}. Hydrogen and oxygen atoms are usually found on the outside of the molecule.
2. If there are single atoms of two elements, the one with more atomic number is the central atom of the molecule with the other atoms arranged as surrounding atoms. Examples. $POCl_3$, $SOCl_2$.
3. The carbon family usually has four bonds, the nitrogen family three bonds, the oxygen family two bonds, and the halogens usually have one bond in neutral molecules.
4. When oxygen and hydrogen atoms are in the same molecule, they usually form the combination H – O – X where X is whatever other atom is in the molecule.
5. Three membered rings are unlikely for most molecules. Larger rings are possible, but still not as common as other structures.
6. The total number of electrons required for writing the structures are obtained by adding the valence electrons of the combining atoms. **Ex:** In the CH_4 molecule there are eight valence electrons available for bonding (4 from carbon and 4 from the four hydrogen atoms).
7. For anions, each negative charge would mean addition of one electron. For cations, each positive charge would result in subtraction of one electron from the total

number of valence electrons. For example for the CO_3^{2-} ion, two negative charges indicate that there are two additional electrons than those provided by the neutral atoms. For NH_4^+ ion, one positive charge indicates the loss of one electron from the group of neutral atoms.

8. By using the chemical symbols of the contributing atoms one can write the skeleton structure of the compound by knowledge or by guessing and distribute the total number of electrons as bonding share pairs between the atoms in proportion to the total bonds.

9. After accounting for the shared pairs of electrons for single bonds, the remaining electron pairs are either utilized for multiple bonding or remain as the lone pairs. The basic requirement being that each bonded atom gets an octet of electrons.

Illustrated Examples

3.1 Drawing Lewis dot formulae of NH_3 and NO_3^-

1. Calculate the number of valence electrons available in the atoms counting only those outside any noble gas core. If the molecule has a charge, add an electron for each negative charge and subtract an electron for each positive charge.

NH_3

3 hydrogens have 1 electron each	$3 \times 1 = 3$
1 nitrogen has 5 electrons	$1 \times 5 = 5$
No charge,	
	8

NO_3^-

3 oxygens have 6 electrons each	$3 \times 6 = 18$
1 nitrogen has 5 electrons	$1 \times 5 = 5$
charge of 1 –	1
	24

2. Calculate the number of electrons to satisfy the normal valence structure of the atoms if each were totally independent of the other.

NH_3

3 hydrogens require 2 electrons each	$3 \times 2 = 6$
1 nitrogen require 8 electrons	$1 \times 8 = 8$
Total electrons required	14

NO_3^-

3 oxygen require 8 electrons each	$3 \times 8 = 24$
1 nitrogen require 8 electrons	$1 \times 8 = 8$
Total electrons required	32

3. Find the difference between the number of electrons required and the number available. This is the number of bonding electrons, which must be shared by two atoms and counted twice, once for each atom. Because each bond uses 2 electrons, the number of bonds is half the number of bonding electrons.

NH_3	**NO_3^-**
14 electrons required	32 electrons required
– 8 electrons available	– 24 electrons available
6 electrons shared, for 3 bonds	8 electrons shared for 4 bonds.

4. Sketch the molecule with the number of bonds calculated. Several additional rules useful in determining how to draw the molecule are given later. In this case, ammonia has three bonds, just enough for one connecting each hydrogen to the nitrogen. If the number of bonding pairs exceeds the minimum needed to form single bonds between the atoms, double or triple bonds are used. For example, the nitrate ion requires a double bond.

5. Fill in electrons pairs around the atoms upto the total number of electrons available and the maximum of 8 around each atom (2 on hydrogen) to complete the structure as in figure given below. In the ammonia example, one lone pair is added to the nitrogen. In the nitrate example, the singly bonded oxygens each need 3 more lone pairs, and the doubly bonded oxygen needs two lone pairs.

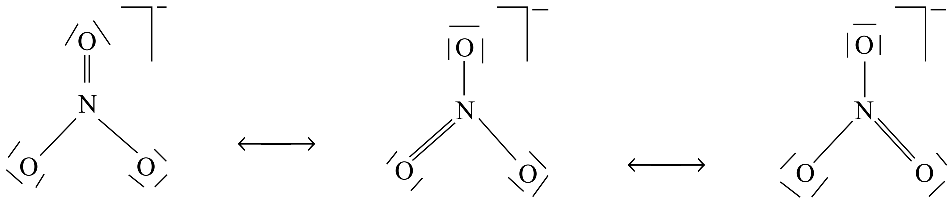
For ease in drawing, a line frequently designate a pair of electrons, but care must be taken to distinguish it from a minus sign. For this reason, we indicate on an ion as $\urcorner^-$

$$\begin{array}{c} \text{H}-\ddot{\text{N}}-\text{H} \\ | \\ \text{H} \end{array}$$

6. If there is more than one way to draw the structure, as with nitrate, all possible structures are drawn. The actual

NH_3	**N**	**H**	
(a) Electrons needed	8 +	3×2	= 14
(b) Electrons available	5 +	3×1	= 8
(c) Difference (a – b)			= 6
(d) Bonding electrons			= 6
(e) Bond pairs (bp) 6/2			= 3
(f) No. of *l*p = (b – d)/2 = (8–6)/2			= 1
Net 3bp and 1 *l*p, a total of 8 electrons			

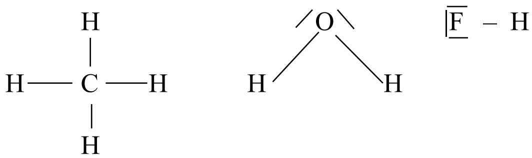
	N	**O**	
(a) Electrons needed	8 +	3×6	= 32
(b) Electrons available	5 +	$(3 \times 6) + 1$	= 24
(c) Difference (a–b)			8
(d) Bonding electrons			8
(e) Bond pairs (bp) 8/2			4
(f) No. of *l*p = (b–d)/2 = (8–6)/2			= 8
Net 4bp and 8 *l*p a total of 24 electrons.			



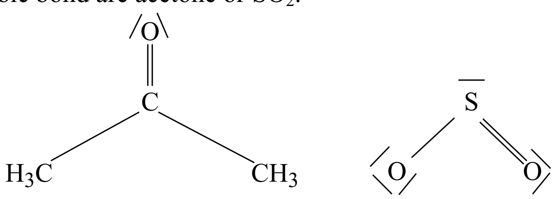
structure is a mixture of all possible structures (resonance) although formal charge (described later) may help in deciding which are more or less important.

If step 3 gives fewer electron pairs than the number of atoms surrounding the central atom, expand the shell on the central atom to give enough bonds to connect all the atoms together.

For many substances the number of valence electrons is just sufficient to provide an octet for each non-hydrogen atoms. These are **saturated** systems, and Lewis diagrams can be written using single bonds exclusively as explained for NH_3 in the example. The other examples are CH_4, H_2O and HF.



Unsaturated substances are those where the number of valence electrons that are available within a molecule or complex ion is not sufficient to allow Lewis diagrams to be written using single bonds only. Then, use of **multiple bonds** between selected atoms is required to complete the octet for each atom in the structure as explained for NO_3^- in the illustrated example. The other examples containing double bond are acetone or SO_2.



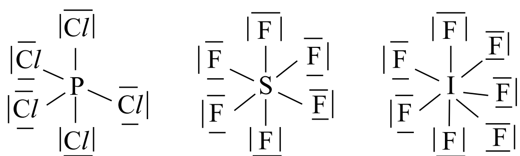
A triple bond (or two double bonds) is necessary when there is an extensive unsaturation as in CO_2 or thiocyanate ion.

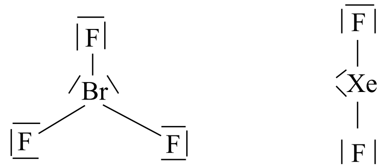
In each of these last examples, unsaturation requires the use of multiple bonds inorder to maintain an octet of electrons for each atom, without using more than the number of valence electrons that are actually available.

For certain molecules, all the available valence electrons are used before an octet is achieved for each non-hydrogen atom. Such molecules are called **electron deficient molecules.** The molecules that feature this electron deficiency usually involve the elements boron, beryllium and sometimes aluminium. The examples are shown here under.

Unsaturated systems are different from an electron deficient one. In the former, an octet is achieved through multiple bonding. In the latter Lewis diagram is properly written with less than an octet of electrons for certain atoms. The molecules in which the central atom is having an octet and all the electron pairs are bond pairs are called **electron precise** molecules. The molecules in which the central atom is having an octet but some are bond pairs and some are lone pairs are called **electron rich** molecules.

For molecules or ions involving atoms beyond the second row of the periodic table, the octet rule does not necessarily apply. These larger atoms may acquire more than an octet of electrons, Such molecules whose central atom is having more than an octet of electrons are called **hypervalent** molecules. This is also called **valence shell expansion or expanded octet** and it is made possible by the availability of valence d level orbitals on these atoms. For now, it is interesting to note that the octet rule finds only limited application, being replaced by the 18 electron rule while considering the coordination compounds formed by metals. Examples include PCl_5, SF_6, IF_7, BrF_3, XeF_2, etc.





Illustrated Examples

3.2 Write Lewis diagrams of CO molecule

Solution

1. Calculate the number of valence electrons available in the atoms by counting only those outside any noble gas core.

1 carbon has 4 electrons	$1 \times 4 =$	4
1 oxygen has 6 electrons	$1 \times 6 =$	6
Total electrons available		10

2. Write the skeletal structure of CO as C O
3. Calculate the number of electrons required to satisfy the normal valence structure of the atoms if each were totally independent of the others.

1 carbon requires 8 electrons	$1 \times 8 =$	8
1 oxygen requires 8 electrons	$1 \times 8 =$	8
Total electrons required		16

4. Find the difference between the number of electrons required and the number available. This is the number of bonding electrons which must be shared by two atoms and counted twice, once for each atom. Because each bond uses 2 electrons, the number of bonds is half the number of bonding electrons.

Electrons required	16
Electrons available	– 10
Electrons shared	6 for 3 bonds.

5. Fill in electron pairs around the atoms up to the total number of electrons available and the maximum of 8 of around each atom. Since the number of bonding pairs exceeds the minimum needed to form single bonds between the atoms double or triple bonds are used.

$$C \equiv O$$

6. Fill in electron pairs around the atoms up to the total number of electrons available and the maximum of 8 around each atom. In CO, one lone pair is added to the carbon and oxygen each to get an octet.

$$|C \equiv O|$$

3.3 Write Lewis diagram of CH_3F

Solution

1. Calculate the number of valence electrons available in the atoms by counting only those outside any noble gas core.

3 hydrogens have 1 electron each	$3 \times 1 =$	3
1 carbon has 4 electrons	$1 \times 4 =$	4
1 fluorine has seven electrons	$1 \times 7 =$	7
Total electrons available		14

2. Calculate the number of electrons required to satisfy the normal valence structure of the atoms if each were totally independent of the others. Hydrogen needs two electrons, all other atoms require 8 electrons.

3 hydrogens require 2 electrons each	$3 \times 2 =$	6
1 carbon require 8 electrons	$1 \times 8 =$	8
1 fluorine require 8 electrons	$1 \times 8 =$	8
Total electrons required		22

3. Find the difference between the number of electrons required and the number available. This is the number of bonding electrons which must be shared by two atoms and counted twice once for each atom. Because each bond uses 2 electrons, the number of bonds is half the number of bonding electrons.

Electrons required	22
Electrons available	14
Electrons shared	8 for 4 bonds.

4. Write the skeletal structure of CH_3F as

```
    H
H   C   F
    H
```

5. Fill in electron pairs around the atoms upto the total number of electrons available and the maximum of 8 around each atom (2 on hydrogen) to complete the structure of the total electrons available 8 electrons are filled as bond pairs. The remaining six electrons are to be added to fluorine to get an octet.

```
    H
    |    __
H – C – F|
    |    ‾‾
    H
```

3.4 Write Lewis diagrams of ClF_3 and SF_6

Solution

ClF_3	Normal	Expanded shell
	Cl 3F	Cl F
Electrons required	$8 + (3 \times 8) = 32$	$10 + (3 \times 8) = 34$
Electrons available	$7 + (3 \times 7) = 28$	$7 + (3 \times 7) = 28$
Shared electrons	4	Shared electrons 6

In normal case the 4 shared electrons are not enough for the 3 bonds required but in the expanded shell the 6 electrons are sufficient for forming 3 bonds.

Because at least 3 bonds, or 6 shared electrons, are required, the number of electrons that can be around the chlorine must be increased from the usual octet to 10 After the 6 bonding electrons are drawn in, 6 electrons (3 pairs) are added to each fluorine and 4 (2 pairs) are added to the chlorine as lone pairs. The chlorine thus be viewed as having an expanded shell of 10 electrons.

SF_6	Normal	Expanded shell
	S 6F	S 6F
Electrons required	$8 + (6 \times 8) = 56$	$12 + (6 \times 8) = 60$
Electrons available	$6 + (6 \times 7) = 48$	$6 + (6 \times 7) = 48$
Shared electrons	$= 8$	Shared electrons = 12
	Only 4 bonds possible	6 bonds are possible

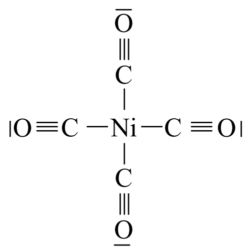
Because atleast 6 bonds are required, the number of electrons around sulphur must be increased from 8 to 12. Three lone pairs on each fluorine complete the picture. Octets may be exceeded only for elements of atomic number 14 (Si) or higher. This can be explained by invoking the use of d orbitals in the bonding.

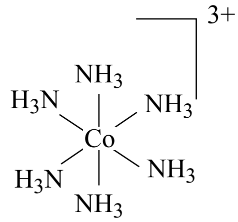
In expanding the octet, the number around the central atom is expanded 2 electrons at a time until the number of bonds is sufficient for all the atoms. There are exceptions, but this procedure is valid for most molecules.

3.2.3 Drawbacks of the Octet Theory

- Octet rule is based upon the chemical inertness of noble gases. But some noble gases like xenon and krypton react with oxygen and fluorine to form compounds like XeF_2, XeF_4, XeF_6, KrF_2, XeO_3, XeO_4, $XeOF_2$, etc.
- The theory could not explain the shape of molecules.
- The theory could not explain the relative stability of molecules being totally silent about the energy of molecule.
- The theory could not explain the formation of hyper valent or expanded octet molecules.
- The theory could not explain about the formation of electron deficient molecules.

In a preliminary discussion let us briefly learn about coordination compounds, which feature a central metal atom bonded to other groups. The groups that are bonded to central metal atoms are called ligands. Examples of coordination compounds are $Ni(CO)_4$, $[Co(NH_3)_6]^{3+}$ and $[Pt(NH_3)_2Cl_2]$. Lewis diagrams of simple coordination compounds of the transition metals may be written without taking into account the presence of the (n - 1)d electrons of the metal. The bonds are considered to be **coordinate covalent bonds** in which both electrons of the metal–ligand bond are supplied by the ligand. The ligands are considered as **Lewis bases** (electron pair donors) and the metal centres are considered to be **Lewis acids** (electron pair acceptors.) The octet rule does not apply, instead, the ligands add enough electrons to those of the metal to bring the total for the metal to that of the next noble gas, that is, 18 valence electrons in all. Hence, the octet rule is replaced by the **18 electron rule** or **pseudoinert gas configuration** because of the additional 10 electrons of the d–orbitals in any transition series. **Examples:**





In each case the metal electrons are not listed in Lewis diagram, but they are counted towards the 18–electron total. Note also many transition metal compounds have other than the closed shell 18–electron total and they are still perfectly stable. We shall learn more about this later in this chapter.

3.3 FORMAL CHARGE

Lewis diagrams, in general, do not represent the actual shapes of the molecules. In case of polyatomic ions, the net charge is possessed by the ion as a whole and not by a particular atom. It is, however, feasible to assign a formal charge on each atom. Formal charges can be used to help in the assessment of resonance structures and molecular topology. Formal charges can help in assigning bonding when there are several possibilities. This can eliminate the least likely forms when we are considering resonance structures and in some cases, suggest multiple bonds beyond those required by the octet rule. It is essential, however, to remember that formal charge is only a tool for assessing Lewis structures, not a measure of any actual charge on the atoms.

Formal charge is the apparent electronic charge of each atom in a molecule, based on the electron–dot structure. Formal charge on an atom is the number of valence electrons available in a free atom of an element minus the total for that atom in the molecule (determined by counting lone pairs as two electrons and bonding pairs as one assigned to each atom)

$$\text{Formal charge} = \begin{bmatrix}\text{No. of valence}\\ \text{electrons in a}\\ \text{free atom of the}\\ \text{element}\end{bmatrix} - \begin{bmatrix}\text{No. of unshared}\\ \text{electrons on the}\\ \text{atom}\end{bmatrix} - \begin{bmatrix}\text{No of bonds}\\ \text{to the atom}\end{bmatrix}$$

charge on the molecule or ion = sum of all the charges.

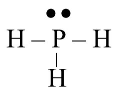
Illustated Examples

3.5 Calculate the formal charge on all atoms in (a) PH_3 (b) NH_4^+ and (c) BF_4^-

Solution

(a) PH_3

Lewis diagram of PH_3 (is)



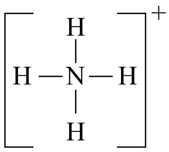
Using the formula $F = N_A - N_{LP} - 1/2\ N_{BP}$

For hydrogen $F = 1 - 0 - 1 = 0$

For phosphorous $F = 5 - 2 - 3 = 0$

(b) NH_4^+

Lewis diagram of NH_4^+ is

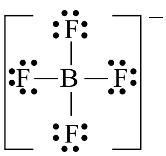


For hydrogen $F = 1 - 0 - 1 = 0$

For nitrogen $F = 5 - 0 - 4 = +1$

(c) BF_4^-

Lewis diagram of BF_4^- is



For fluorine $F = 7 - 6 - 1 = 0$

For boron $F = 3 - 0 - 4 = -1$

Some additional rules about formal charge can make it more useful in deciding between different possible structures.

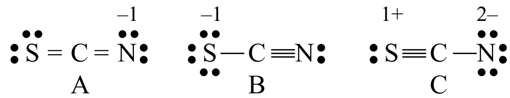
1. Structures with small formal charges (+z, –z or less) are more likely than those with larger formal charges.
2. Non-zero formal charge on adjacent atoms is usually of opposite sign.
3. More electronegative atoms (those in the upper right corner of the periodic table) should have negative rather than positive formal charges.
4. Formal charges of opposite signs separated by large distances are unlikely.
5. The most stable structures have the largest sum of the electronegativity differences for adjacent atom. For example, HOC*l* is more stable than HC*l*O. In other words, a bond with large polarity is more stable than a bond with smaller.

3.6 Write all the possible Lewis diagrams for SCN^- OCN^- and CNO^- assign the formal charges to all atoms in these ions and predict the most stable Lewis diagram for each ion

Solution

SCN^-

Three Lewis diagrams can be written for SCN^- ion as follows.



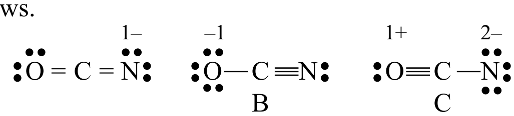
Structure A has only one negative formal charge on the nitrogen atom, the most electronegative atom in the ion, and fits the rules well. Structure B has a single negative charge on S, which is less electronegative than N. Structure C has charge of 2– on N and 1+ on S consistent with the relative electronegativities of the atoms but with a larger charge and greater charge separation than the first. Therefore, these structures lead to prediction that structure A is most important, structure B is next in importance and any contribution from C is minor. This prediction is in consistence with the bond length values for structures A and B. Protonation of the ion forms HNCS consistent with negative charge on N in SCN^-. The bond lengths in HNCS are those of double bonds, consistent with the structure H – N = C = S.

Table 3.1 Bond lengths of S – C and C – N (pm)

	S – C Bond length (pm)	C – N Bond length (pm)
SCN^-	165	117
HNCS	156	122
Single bond	181	147
Double bond	155	128 (approximate)
Triple bond	—	116

OCN^-

Three Lewis diagrams can be written for OCN^- ion as follows.



An alternative method for calculating formal charge is to use the equation.

$$F = [N_A - N_{LP} - 1/2N_{BP}]$$

Where F is the formal change, N_A is the number of electrons in the valence shell in the free atom (group number), N_{LP} is the number of electrons in lone pairs (unshared) and N_{BP} is the number of electrons in bond pairs (shared electrons).

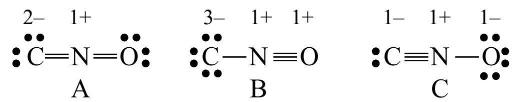
OCN^- ion has the same possibilities as SCN^- but the larger electronegativity of O makes structure B more important than in SCN^-. The protonated form contains 97 per cent HNCO and 3 per cent HOCN consistent with structure A and a small contribution from B. The bond lengths in OCN^- and HNCO are consistent with this but not perfectly.

Table 3.2 Bond lengths of O – C and C – N (pm)

	O – C Bond length (pm)	C – N Bond length (pm)
OCN^-	113	121
HNCO	118	120
Single bond	143	147
Double bond	119	128 (approximate)
Triple bond	113	116

CNO^-

The isomeric fulminate ion CNO^- can have the following three Lewis diagrams, similar to OCN^- but the resulting formal charges are unlikely.



Because the order of electronegativities is C < N < O, none of these are plausible structures and the ion is predicted to be unstable.

The only common fulminates are of mercury and silver which are explosive. Fulminic acid is linear HCNO in the vapour phase consistent with structure C, and coordination complexes of CNO^- with many transition metal ions are known with HCNO structures.

3.4 IONIC BOND

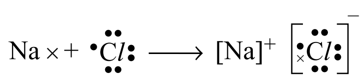
Kossel, Lewis and Langmuir theories explained the electrovalency, and the formation of electrovalent (i.e., ionic) bonds in terms of the electrostatic attraction between ions of opposite charges. These ions are formed by a complete transfer of electrons between atoms. The formation of ionic bond is thus related to the ease with which ions can be formed from neutral atoms, and to the way in which the ions are packed together in the crystal structure.

When two atoms of different elements approach each other, there will be attractive and repulsive forces between the two atoms. The attractive forces are due to the nucleus of one atom on the electrons of another atom or vice - versa. The repulsive forces are due to the positive charge of the nuclei of both atoms and negative charges of the electrons of both atoms. The atom having more electronegativity can remove the electron from the outer orbit of another atom converting it into positive ion while itself converting into negative ion by gaining that electron. This is possible if the electronegativity difference of the atoms of the two elements is greater than 1.7.

In 1916, Kossel suggested that those elements placed just before a noble gas in the periodic table could attain a noble gas structure by gaining electrons and forming negatively charged ions. Thus chlorine with a structure 2, 8, 7 could become a chloride ion Cl^- with a structure 2, 8, 8 by gaining an electron.

Similarly, an element placed just after a noble gas could achieve noble gas structure by losing electrons and forming positive charged ions, for example, sodium 2, 8, 1 could form a sodium ion, Na^+ with the inert gas structure 2, 8 by loosing an electron.

Both the chloride and sodium ions would have stable structures, and by combining would form sodium chloride. For two ions, sometimes known as an **ion–pair,** the state of affairs could be represented, showing only the outermost or valence electrons as:



The two ions being held together by electrostatic attraction. **The electrostatic attractive force existing between the oppositely charged ions is called electrovalent bond or ionic bond:** In the formation of ionic compounds. "the number of electrons, an atom of the element loses or gains" is known as its **electrovalence** of that element.

Many simple compounds can be formulated as ionic compounds containing ionic bonds. Calcium bromide and potassium sulphide provide typical examples.

Calcium bromide $CaBr_2$ Potassium sulphide K_2S

Generally, metals have a tendency to lose electrons due to low ionization potentials and convert easily into positive ions. Non–metals have a tendency to gain electrons due to high electronegativity values and convert easily into negative ions. Therefore, ionic bonds are formed between more electropositive metals and more electronegative elements. Compound having more ionic character will be formed between highly electropositive caesium and highly electronegative fluorine atoms. When an ionic bond is formed, the ions involved neither can come very close to one another nor can they be far away from each other. They just approach each other to come to an equilibrium position. In this state, the attractive forces (operating between the nuclei or the electrons) balance each other. Thus, the molecules (i.e., ion pairs) of the ionic compounds are formed.

3.4.1 Writing the Formulae of Ionic Compounds

To write the formulae of ionic compounds made up by two ions (simple or polyatomic) having electrovalencies X and Y, respectively, the following points may be followed.

1. Write the symbols of cation and anion side by side so that the cation is on the left while anion is on the right side.
2. Write their electrovalencies in figures on the top of each symbol.
3. By cross multiplying the ions with charges we can get the formulae of ionic compound.

Examples:

Magnesium fluoride	$Mg^{2+}\ Cl^{-}$	$= MgCl_2$
Sodium oxide	$Na^{+}\ O^{2-}$	$= Na_2O$
Calcium oxide	$Ca^{2+}\ O^{2-}$	$= Ca_2O_2 = CaO$ (simple ratio)
Ammonium sulphate	$NH_4^{+}\ SO_4^{2-}$	$= (NH_4)_2\,SO_4$

3.4.2 Factors Favourable for the Formation of Ionic Bond

The factors favourable for the formation of ionic bond are (a) Favourable conditions for the formation of cation and (b) Favourable conditions for the formation of anion.

(a) Favourable conditions for the formation of cation:

(i) Atomic size: With increase in the size of an atom the attractive force of the nucleus on outer electrons decreases. Therefore, they can lose their outer electrons easily since in a group of periodic table the atomic size increases down the group, the tendency to convert into positive ion increases.

Ex: The ease of formation of cations of alkali metals will be in the order $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$.

(ii) Ionization potential: Atoms having lesser ionization potentials can lose electrons easily. Therefore, the lesser the ionization energy, the easier the formation of cation. Again in a group of periodic table, the ionization potentials decrease down the group, so the tendency to convert into cation increases.

(iii) Number of charges on the cation: With the removal of each successive electron from an atom, the positive charge on it increases gradually. Further, the size of the ion decreases simultaneously. As a result of this the effective nuclear charge of the atom or ion increases, causing less tendency to convert into cation. This means that with increase in the number of charges on cation the tendency to convert into cation decreases. **Ex:** The relative ease of formation of ions Na^+, Mg^{2+} and Al^{+3} is in the order $Na^+ > Mg^{+2} > Al^{+3}$.

(iv) Electronic configuration: A cation with inert gas configuration can be formed easily than a cation with pseudoinert gas configuration (18 electrons) in their outermost orbit. **Ex:** Among Ca^{+2} and Zn^{+2} ions, Ca^{+2} ion can be formed easily than Zn^{+2} ion because Ca^{+2} ion has 8 electrons in its outermost orbit while Zn^{+2} ion has 18 electrons in its outermost orbit. When 18 electrons are present in the outermost orbit of an ion, 10 electrons are present in the d-orbitals which have poor shielding effect. Therefore, the attraction of the nucleus on electrons to be removed increases causing increase in their ionization potentials. Hence, their formation becomes difficult.

(v) Favourable conditions for the formation of anion:

(i) Atomic size: Small atoms hold the electrons gained by them strongly and hence form anions easily. In a group of periodic table, since the atomic size increases down the group the tendency of holding the electrons decreases. Therefore, the tendency to convert into anions decreases. For example, the formation of halide ions will be in the order $F^- > Cl^- > Br^- > I^-$.

(ii) Electronegativity and electron affinity: Atoms having more electronegativities and electron affinities can accept the electrons readily and convert into anions easily. For example, in the elements of the Vth group of the periodic table as the electronegativity decreases down the group the tendency to convert into anion decreases. Thus, N^{-3} can be formed easily than P^{-3}.

(iii) Number of charges on the ion: With increase in the number of charges on an anion, the holding capacity of electrons by that atom decreases due to decrease in effective nuclear charge. Thus with increase in number of negative charges on anion, the tendency to convert into anion decreases.

Ex: The relative ease of formation of F^-, O^{-2} and N^{-3} is in the order $F^- > O^{-2} > N^{-3}$

Table 3.3 Effect of ionic size and ionic charge on the formation of common cations

$\longleftarrow$ Decrease in ionic charge
Ions form more easily

Increase in ionic size $\downarrow$ / Ions form more easily $\downarrow$

Li^+	Be^{2+}		
Na^+	Mg^{2+}	Al^{3+}	
K^+	Ca^{2+}	Sc^{3+}	
Rb^+	Sr^{2+}	Y^{3+}	Zr^{4+}
Cs^+	Ba^{2+}	La^{3+}	Ce^{4+}

Decrease in ionic charge $\longrightarrow$
Ions form more easily

Ions form more easily $\uparrow$ / Decrease in ionic size $\uparrow$

N^{3-}	O^{2-}	F^-
P^{3-}	S^{2-}	Cl^-
		Br^-
		I^-

3.4.3 Covalent Character of Ionic Bonds

Every ionic bond contains a mixture of covalent and ionic character. Generally a compound is ionic or covalent as long as the compound in question is predominantly one or the other. In many cases, however, it is convenient to be able to say something about intermediate situations. In general, there are two ways of treating ionic–covalent bonding. The method that has proved most successful is to consider the bond to be covalent and then consider the effect of increasing charge displacement from one atom toward another. Another method is to consider the bond to be ionic and then allow for a certain amount of covalence to occur. The second method was due to **Fajans.** This theory has found no place in the repertoire of the theoretical chemist largely because it has not proved amenable to the quantitative calculations which other theories have developed. Even then the qualitative ideas embodied in "Fajans rules" offer simple if not exact approaches to the problem of partial covalent character in ionic compounds.

Fajans considered the effect which a small, highly charged cation would have on an anion. If the anion were large and soft enough, the cation should be capable of polarizing it, and the extreme of this situation would be the cation actually penetrating the anionic electron cloud giving a covalent (shared electrons) bond as shown in Fig 3.1.

Fajans suggested the following rules to estimate the extent to which a cation could polarize an anion and thus induce covalent character.

1. **Small cation with more charge:** Small, highly charged cations exert a greater effect in polarizing anions than large and / or singly charged cations. Thus in a group of periodic table down the group ionic character increases with a particular anion. **Eg:** The order of covalent character of alkali metal chlorides is $LiCl > NaCl > KCl > RbCl > CsCl$. Similarly in a period from left to right covalent character increases for the metal ions with a particular anion **Eg:** $LiCl < BeCl_2 < BCl_3 < CCl_4$.
2. **Larger the anion with more charge:** The polarizability of the anion will be related to its "softness" that is, to the deformability of the electron cloud. Both increasing charge and increasing size will cause this cloud to be less under the influence of the nuclear charge of the anion and more easily influenced by the charge on the cation. Thus the order of covalent character of halides with a particular metal ion is in the order $NaF < NaCl < NaBr < NaI$ and in a period the order of covalent character of compound from left to right will be in the order $Mg_3N_2 > MgO > MgF_2$.
3. **Electron configuration of the cation:** Cations with inert gas configurations form ionic compounds while those cations with pseudoinert gas configuration favour the covalent bond formation. The cations with pseudoinert gas configuration due to poor shielding effect of d–electrons, have more polarizing power on anion and thus considerably more covalent than those of cations with inert gas configuration.
Ex: $ZnCl_2$ is more covalent than $CaCl_2$, similarly the compounds like $CdCl_2$, $HgCl_2$, Cu_2Cl_2, $AgCl$ have covalent character.

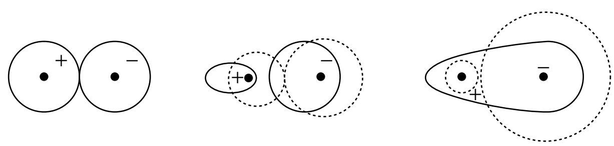


Fig 3.1 Polarization effects. (a) Idealized ion pair with no polarization, (b) Mutually polarization pair and (c) polarization sufficient to form covalent bond. Dashed lines represent hypothetical unpolarized ions

3.4.4 Crystal Lattice Energy

When cations formed by losing electrons and anions formed by gaining electrons approach each other in large numbers due to electrostatic attractive force they are arranged themselves in a regular pattern so that a close packed arrangement is acquired. This arrangement of ions to form a solid crystal is called crystal lattice. While attaining such a structure, energy is liberated and the system get stabilized.

The amount of energy liberated when one mole of gaseous cations and one mole of gaseous anions are brought together from infinite distance apart to their equilibrium position in the crystal lattice at 0 k i.e., the change in the internal energy ΔU is called **lattice energy**.

The lattice energy can also be defined as the amount of energy required to break one mole of ionic crystal and to separate the constituent ions from their position in the crystal to infinite distance but with positive sign to ΔU.

Lattice energy can be calculated for structures consisting of spherical ions. The interaction energy between two ions of charge $-Z^+Z^-e^2/R$ (in cgs units) or $-Z^+Z^-/4\pi\varepsilon R$ (in SI units).

Figure 3.2 shows part of sodium chloride structure, consider the interaction energy between a particular cation C and all other ions in one mole of crystalline NaCl. In this structure C has.

6 nearest neighbours (Cl^-) at a distance R.

12 second neighbours (Na^+) at a distance $\sqrt{2}$R

8 third neighbours (Cl^-) at a distance $\sqrt{3}$R

6 fourth neighbours (Na^+) at a distance $\sqrt{4}$R

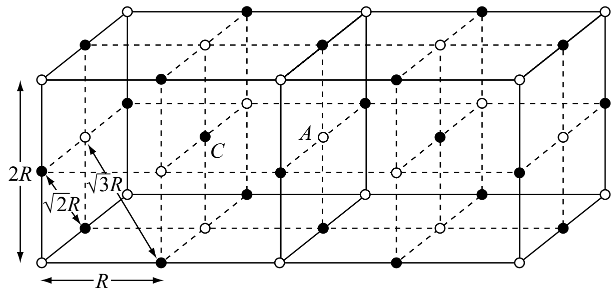


Fig 3.2 The Na*Cl* structure: Na^+ black circle, Cl^- open circles

So that the coulombic interaction energy at C becomes $-(6Z^+Z^-e^2/R) + [12(Z^+)^2 e^2/\sqrt{2}R] - (8Z^+Z^-e^2/\sqrt{3}R) + [6(Z^+)^2 e^2/\sqrt{4}R]$ (3.2)

OR

$-Z^+Z^-e^2/R \times [6 - (12Z^+/(\sqrt{2})Z^-) + (8/\sqrt{3}) - [6Z^+/2Z^-] + \ldots\ldots]$ (3.3)

The ratio of the ionic charges is constant for a given structure (e.g., $Z^+/Z^- = 1$ for Na*Cl*, 2 for TiO_2), so that the coulombic energy term can be written $U_c = -Z^+Z^-e^2A/R$ where A is called the Madelung constant is written for the series enclosed within the square bracket of equation 3.2.

We can repeat the analysis to get the coulombic interaction at a particular anion A in the sodium chloride structure. [This will also $-Z^+Z^-e^2A/R$ since cations and anions have the same arrangement of neighbours in the sodium chloride structure]. The total coulomb energy for a lattice of N cations and N anions, where N is the Avogadro number is then given by

$$U = \frac{-Z^+Z^-e^2AN}{R} \tag{3.4}$$

The coulombic interaction energy, which represents a net attraction between the ions, must be opposed by a repulsion energy since the ions maintain an equilibrium separation in the crystal structure. A number of expressions has been suggested for this repulsion. According to Born-Lande the expression for repulsion energy is BN/R^n so that the lattice energy becomes

$$U = -Z^+Z^-e^2AN/R + BN/R^n \tag{3.5}$$

The constant B can be eliminated if we make use of the fact that at $R = R_e$, the equilibrium separation of the ions in the crystal $(\delta U/\delta R) = 0$. Then

$$B = \frac{Z^+Z^-e^2A}{n} R_e^{n-1} \tag{3.6}$$

Then

$$U = \frac{-Z^+Z^-e^2AN}{R_e}\left[1 - \frac{1}{n}\right] \text{(cgs units)} \tag{3.7}$$

$$= \frac{-Z^+Z^-e^2AN}{4\pi\varepsilon R_e}\left[1 - \frac{1}{n}\right] \text{(SI units)} \tag{3.8}$$

The integer n, which depends upon the nature of the ions, can be estimated from compressibility measurements.

The development of quantum mechanics showed that electron wave functions decreased exponentially with increasing distance from the nucleus. Born and Mayer used the expression B′ N exponent $(-R/\rho)$ for the repulsion term. The constant B′ can again be eliminated by putting $(\delta U/\delta R)_{R = Re = 0}$ and the expression for the lattice energy now becomes.

$$U = -\frac{Z^+Z^-e^2AN}{r_+ + r_-}\left[1 - \frac{\rho}{r_+ + r_-}\right] \tag{3.8}$$

The constant ρ, for ions with inert–gas electron configuration is about 0.35×10^{-10} m.

Unfortunately, Madelung constants become increasingly difficult to calculate as the complexity of the structures increases. **A.F. Kapustinskii** observed that if the Madelung constants for a number of structures are divided by the number of ions per formula unit ‘n’, then approximately the same value is obtained for all of them. He also noted that the value so obtained increases with the coordination number, the variation in A/n $(r_+ + r_-)$ from one structure to another can be expected to be fairly small. From this observation Kapustinskii proposed that there exists a hypothetical rock–salt (Na*Cl*) structure that is energetically equivalent to the true structure of any ionic solid and therefore that the lattice enthalpy can be calculated by using the rock–salt Madelung constant and the appropriate ionic radii for (6, 6 - coordination). The resulting expression is called the **Kapustinskii equation.**

$$U = \frac{n(Z^+Z^-)}{r_+ + r_-}\left(1 - \frac{34.5}{r_+ + r_-}\right)\kappa \tag{3.9}$$

Where κ is a constant and its value is 1.21×10^5 KJ pm mole $^{-1}$, n is the number of ions in the formula considered (e.g., n = 2 for Na*Cl*, 3 for CaF_2, etc), r_+ and r_- are the ionic radii.

3.4.5 Born–Haber Cycle

In 1919, Born and Haber devised a cycle for the calculation of lattice energies. To calculate the heat of formation of sodium chloride, i.e., the heat of reaction

$$Na_{(s)} + \frac{1}{2}Cl_{2(g)} \longrightarrow NaCl_{(s)},\ \Delta H_f NaCl$$

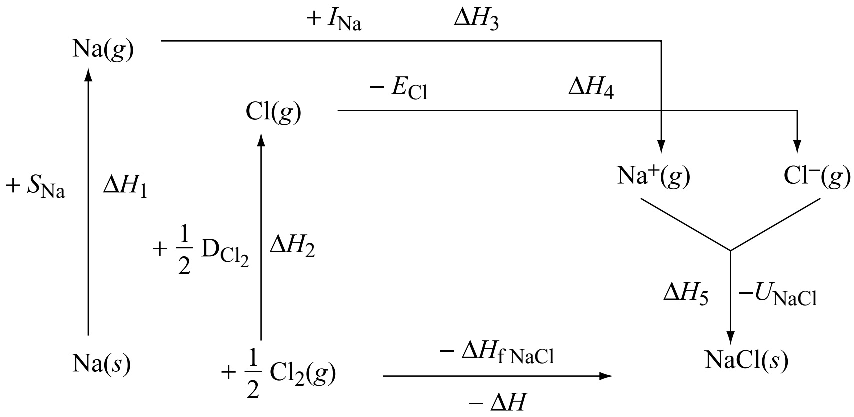


Fig 3.3 Born–Haber cycle for calculation of lattice energy of NaCl

it is necessary to take all the following energy factors into account.

(a) The ionization energy of sodium I_{Na}

(b) The electron affinity of chlorine $E_{Cl.}$

(c) The energy required to dissociate solid sodium into free atoms which is equal to the heat of sublimation S_{Na}

(d) The dissociation energy of gaseous chlorine D_{Cl}

(e) The crystal lattice energy of sodium chloride U_{NaCl}

All the above energy factors are interrelated in the Born - Haber cycle which can be summarized in Fig 3.3.

The thermochemical equation for the formation of solid sodium chloride from its constituent elements can be written as follows.

$$Na_{(s)} + \frac{1}{2} Cl_{2(g)} \xrightarrow[\Delta H]{\Delta H_f NaCl} NaCl_{(s),}\ \Delta H = -410.5\ \text{KJ mol}^{-1}$$

The heat liberated in this reaction is called heat of formation of NaC*l* since NaCl is formed from the constituent elements in their standard state. The heat involved in this reaction is -410.5 KJ mol^{-1}.

This reaction can be assumed to take place in several steps, which are as follows.

(i) The heat absorbed in the conversion of solid sodium to gaseous sodium is equal to its **sublimation energy** (+S). The sublimation energy of sodium is 108.7 KJ mol^{-1}.

$$Na_{(s)} \xrightarrow[\Delta H_1]{\Delta S_{Na}} Na_{(g),} \qquad \Delta H_1 = 108.7\ \text{KJ mol}^{-1}$$

(ii) Gaseous sodium atoms ionize by absorbing an energy equal to its ionization energy of 492.82 kJ/mol^{-1}

$$Na_{(g)} \xrightarrow[\Delta H_2]{+I_{1\,Na}} Na^+(g); \qquad \Delta H_2 = 492.82\ \text{KJ mol}^{-1}$$

(iii) Chlorine molecules should breakup into chlorine atoms for which the energy absorbed is equal to its bond energy. To break half the mole of Cl_2 molecules into atoms the energy absorbed is equal to half of its bond energy, i.e., +239.1/2 = +119.55 KJ.

$$\frac{1}{2} Cl_{2(g)} \xrightarrow[\Delta H_3]{+1/2\, D_{Cl_2}} Cl_{(g)}$$

$$\Delta H_3 = \frac{1}{2} D_{Cl_2} = 119.55\ \text{KJ mol}^{-1}$$

(iv) Chlorine atoms convert into chloride ions by gaining an electron due to which energy liberated is equal to its electron affinity.

$$Cl_{2(g)} + e^- \xrightarrow[\Delta H_4]{-E_{Cl}} Cl^-_{(g)}$$

$$\Delta H_4 = -E_{Cl} = 361.60\ \text{kJ mol}^{-1}$$

(v) The gaseous sodium and chloride ions come close together to form solid crystalline NaC*l*. The energy released in this process is called crystal lattice energy (-U). Crystal lattice energy cannot be measured directly. However, it can be calculated by using Born–Haber cycle.

Applying Hess's law of constant heat summation to the above sequence of reactions the heat of formation of NaC*l*(s) is given by

$$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4 + \Delta H_5 \tag{3.10}$$

$$\Delta H_{fNaCl} = + S_{Na} + I_{Na} + D_{Cl_2} - E_{Cl} - U_{NaCl} \tag{3.11}$$

Substituting experimental values of energy changes in various steps and solving for U we get

$U = -769.97$ kJ mol^{-1}

3.4.6 Structure of Ionic Crystals

Coordination Number

The number of oppositely charged ions immediately surrounding a particular ion is called its coordination number. In a binary compound AB the coordination number of A^+ and B^- must be equal, but in AB_2 type compound the coordination numbers of A^+ will be twice that of B^-.

The coordination number which will give the most stable structure for any pair of ions A^+ and B^- is dependent on the ratio of the radius of A^+ (r^+) to that of B^- (r^-), i.e., or r^+/r^-.

Consider, for example, a structure in which a central A^+ ion is surrounded octahedrally, by six B^- ions, the coordination number will be 6. Figure 3.4 shows the general arrangement in such a structure but indicates only the nucleii of the ions concerned and not their actual relative sizes. Figure 3.5 shows the possible relative sizes of A^+ and B^- in the limiting case. There will be B^- ions both directly above and below the central A^+ ion but these have been omitted for the sake of clarity. In this limiting case, the B^- ions are in contact with each other as well as with A^+. Such a state of affairs can only exist when $r^- / (r^+ + r^-)$ is equal to cos 45°, i.e., when r^+ / r^- is equal to 0.414.

If different ions with large r^- and / or smaller r^+ values are concerned, a stable octahedral arrangement much less likely for, as shown in Fig. 3.5b, is observed. The B^- ions would be in contact with each other, and would be repelling each other, without being in contact with, and being attracted by A^+. Such ions would rearrange into a tetrahedral structure with coordination number of 4. In simple language, the cation has become too small, or the anion too large, for the central cation to be surrounded by six anions. Instead, it is surrounded by four.

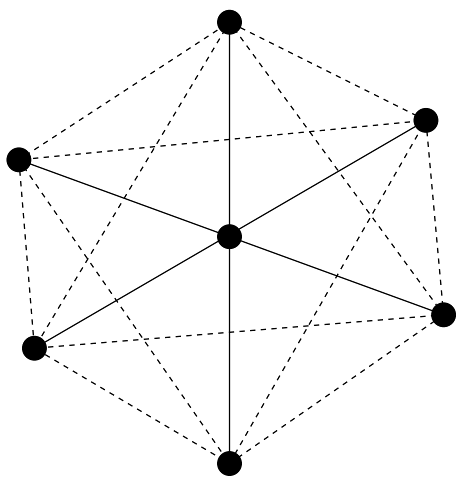


Fig 3.4 Octahedral arrangement of six ions around a central ion

The ratio r^+ / r^-,become smaller still, the central cation may only be able to be surrounded by three anions in a coplanar, triangular arrangement with a coordination number of 3. Such an arrangement is shown in Fig. 3.6 from which it can be seen that the limiting radius ratio for such a structure occurs when $r^- / (r^+ + r^-)$ is equal to cos 30°, i.e., when r^+ / r^- is equal to 0.155.

By similar arguments and calculations the limiting radius ratio for various coordination numbers can be summarized as follows.

Table 3.4 Relation between structure, coordination number and limiting radius ratio

Structure	Coordination number	Limiting radius ratio
Linear	2	0 – 0.155
Triangular	3	0.155 – 0.225
Tetrahedral	4	0.225 – 0.414
Octahedral	6	0.414 – 0.732
Cubic	8	0.732 – 1

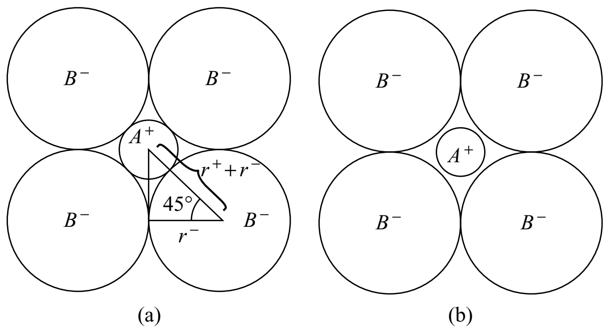


Fig 3.5 (a) Showing the limiting r^+/r^- value for an octahedral arrangement (b) the unstable octahedral arrangement r^+/r^- becomes less than 0.414

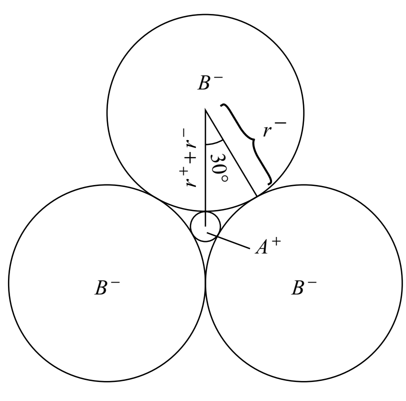


Fig 3.6 The limiting r^+/r^- value for a triangular arrangement

Coordination numbers of 5, 7, 9, 10 and 11 do not occur because of the impossibility of balancing the electrical charges of the ions concerned. When the radius ratio becomes equal to 1, ions of the same size are making up the crystal. Such a state of affairs is to be found in crystals of metals.

Radius ratio values are useful, but they do not apply absolutely accurately. This is only to be expected for the whole conception of ionic radius, based as it is on the idea that anion is always spherical, is only an approximation.

Sodium Chloride Crystal Structure: The radius ratio for Na^+ and Cl^- ions is 0.095/0.181 or 0.525 so that an octahedral structure with coordination numbers of 6 would be expected, and this, in fact is what is found.

The way in which the ions build up into a sodium chloride crystal can be followed in stages. An ion pair shown in Fig. 3.7 (a) has a strong residual field and will attract a second ion pair, just as two magnets would attract each other, as shown in Fig. 3.7, (b) four ion pairs will arrange themselves as in Fig. 3.7 (c) and a larger number will take up the arrangement as shown in Fig. 3.8.

Other compounds, made up of simple ions which have the same crystal structure as sodium chloride include the halides of lithium, sodium, potassium, rubidium and caesium (except the chloride, bromide an iodide of caesium), the oxides and sulphides of magnesium calcium, strontium and barium, the chloride, bromide and iodide of silver, the monoxide of cadmium, iron (II) and manganese (II) and the monosulphide of manganese (II) and lead (II).

One or both, of the simple ions in a sodium chloride crystal may, also, be replaced by a charged group of atoms. Ammonium chloride, bromide and iodide, for instance, have the sodium chloride structure above their transition temperatures while Ca^{2+} and CO_3^{2-} ions in calcite or iceland spar or Ca^{2+} and $(C \equiv C)^{2-}$ ions in calcium carbide, are also arranged with sodium chloride structure.

Caesium chloride crystal structure: The radius ratio Cs^+/Cl^- for caesium chloride is 0.169/0.181 or 0.93 and the crystal structure is cubic with a coordination number of 8.

Caesium bromide and iodide have the same structure, but caesium fluoride with the smaller F^- ion has a sodium

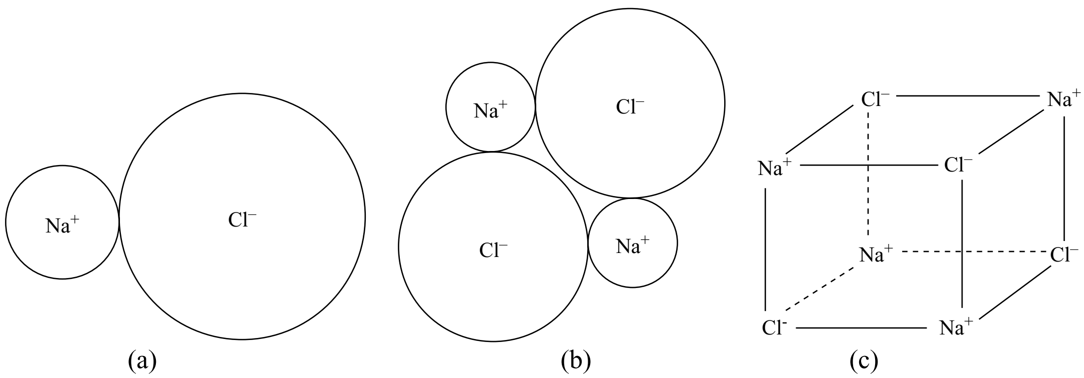


Fig 3.7 (a) Ion pair of sodium chloride, (b) two ion pairs of sodium chloride (c) four ion pairs sodium chloride

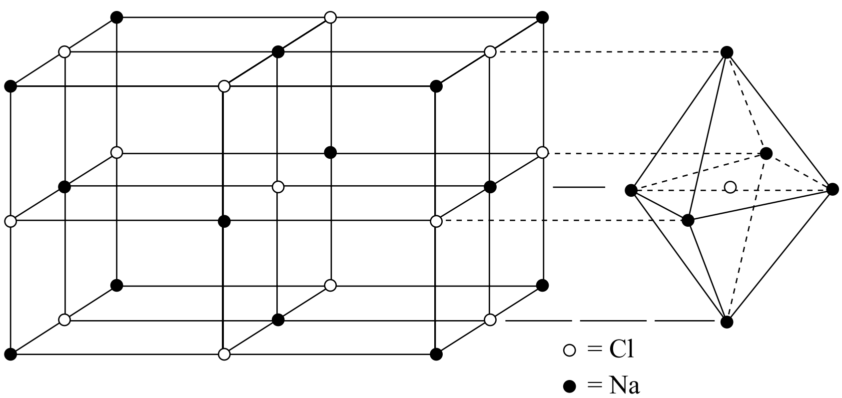


Fig 3.8 Crystal structure of sodium chloride, showing the octahedral arrangement of six sodium ions around one chloride ion

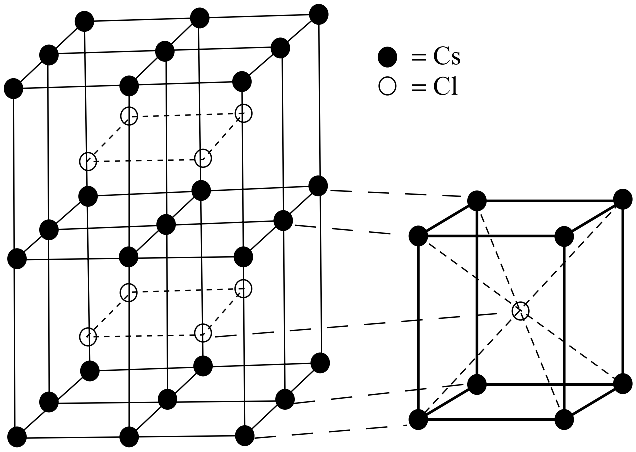


Fig 3.9 Crystal structure of caesium chloride showing (right) the cubical arrangement of eight caesium ions around one chloride ion

chloride structure. Other substances which crystallize with caesium chloride structure include caesium cyanide, thallium (I) cyanide, chloride, bromide and iodide and the chloride, bromide and iodide of ammonium below their transition points.

Contribution of Lattice Points to the Unit Cell: The orderly three dimensional pattern in which the ions in an ionic compound arranged in a crystal is called **crystal lattice**. The crystal lattice may be supposed to be build up of simple three dimensional units known as unit cell. The unit cell is repeated throughout in the crystal in three dimensional manner. Unit cells contain usually a small number of ions (1 to 12). The ions packed in a crystalline substance are shown by points in a crystalline lattice. These points are called lattice points. In different cubic unit cells there are mainly four kinds of lattice points. The four types of lattice points and the contribution of each ion at the lattice point to the unit cell are as follows:

1. An ion in the body of the unit cell belongs to that unit cell and counts 1.
2. An ion on a face is shared by two unit cells and contributes 1/2 to the unit cell.
3. An ion on an edge is shared by four unit cells and contributes 1/4.
4. An ion at a corner is shared by eight unit cells that share the corner, and so contributes 1/8.

Utilizing the contributions of ions for each unit cell as given above, the number of ions of each kind present per unit cell of NaCl can be calculated. As shown in the figure, NaCl has a face-centred cubic lattice.

Number of body-centred sodium ions in the lattice = 1.

Number of sodium ions on 12 edges of cube contributing to unit cell is $12 \cdot \frac{1}{4} = 3$.

Thus, total number of sodium ions belonging to unit cell = 4.

Number of chloride ions present at the 8 corners of the cube contributing to unit cell is $8 \cdot \frac{1}{8} = 1$.

Number of chloride ions present on six faces of the unit cell contributing to each unit cell is $6 \cdot \frac{1}{2} = 3$.

Thus, total number of chloride ions belonging to the unit cell is 4.

There are four Na^+ and four Cl^- ions per unit cell of sodium chloride. In other words, 4 ion pairs are present per unit cell. Similarly, the number of ions pairs per unit cell of CsCl is one, i.e., one Cs^+ and one Cl^- ions per unit cell.

Ionic compounds contain only ions. Vapourization of such ionic crystals produces only ion pairs. Hence, the formula of an ionic compound simply shows the relative numbers of each ion present. In such situations it is proper to express their quantities in terms of **formula weights** rather than in terms of molecular weights.

3.4.7 Characteristics of Ionic Compounds

General Properties

(i) Individual molecules of ionic compounds do not exist because the compounds are made up of an interlocking structure of ions. The ions are held together by strong electrostatic forces within an ionic crystal.

(ii) Ionic compounds are often hard solids having high melting points. This is because large amounts of energy are required to overcome the electrostatic attractive forces of attraction in the crystal lattice.

(iii) Ionic compounds are invariably electrolytes, for, in the presence of an ionizing solvent, such as water, the forces between the ions are so greatly reduced that the ions 'fall apart' and the free ions, in the resulting solution or in the fused state are able to move under the influence of an electrical field and undergo electrolysis.

(iv) The reactions of ionic compounds are reactions of their ions, i.e., they undergo ionic reactions. Therefore, they can participate in reactions fastly and have high reaction rates or instantaneous in solutions.

(v) Due to non-directional nature of the ionic bonds in an ionic compound, ionic compounds do not exhibit stereoisomerism but they exhibit polymorphism and isomorphism. **Polymorphism** is the existence of same ionic compound in different crystal structures and **isomorphism** is the existence of different ionic crystals in the same crystal structure.

(vi) Ionic compounds are soluble in polar solvents like water but their solubility in non-polar solvents like benzene or carbon tetrachloride is negligible. When an ionic compound is dissolved in a polar solvent, the constituent ions are solvated. This reduces the binding forces between the ions resulting in the disruption of the crystal lattice, thus the substance dissolves. The non-polar solvents which do not contain charged centres cannot get the ions solvated. Therefore, the substances do not dissolve. The solvent character is decided by the dielectric constant of the solvent. The higher the dielectric constant, the better it acts as a solvent for the ionic compounds.

3.4.8 Consequences of Lattice Energies

(a) Melting points and Boiling Points: Ionic compounds tend to have high melting points. Ionic bonds usually are quite strong and they are omnidirectional. **Ignoring the omnidirectional nature of ionic bond leads us to conclude that ionic bonding was much stronger than covalent bonding which is not the case.** It can be seen that substances containing strong, multidirectional covalent bonds, such as diamond also have very high melting points rather than only the ionic compounds.

Increasing the ionic charges will certainly increase the lattice energy of a crystal. For compounds which are predominantly ionic, increased ionic charges will result in increased melting and boiling points. **Examples:** NaF, mp = 997° C and MgO, mp = 2800° C.

According to Fajan's rules, increasing charge results, in increasing covalent character, especially when small cations combine with large anions. Increased covalent character decreases the melting and boiling points. For example, the alkali halides (expect Cs*Cl*, CsBr, CsI) and silver halides (except AgI) crystallize in the NaCl structure. The sizes of cations are comparable: Na^+ = 116 pm, Ag^+ = 129 pm and K^+ = 152 pm, yet the melting points differ considerably due to covalent character of the silver halides (resulting from the d^{10} electron configuration.)

A similar comparison can be made between the predominantly ionic species CsF and BaF_2 and the more covalent KBr and $CaBr_2$.

Table 3.5 Melting Points of Sodium, Potassium and Silver Halides

NaF	993° C	KF	858° C	AgF	435° C
NaCl	801° C	KCl	770° C	AgCl	455° C
NaBr	747° C	KBr	734° C	AgBr	432° C

The change from 1:1 to 1:2 composition in the highly ionic fluorides produces the expected increase in lattice energy and corresponding increase in the m.pts and b.pts.

Table 3.6 Melting Points of Some Alkali and Alkaline Earth Halides

Melting Points				Boiling Points			
KBr	734	$CaBr_2$	730	KBr	1435	$CaBr_2$	812
CsF	682	BaF_2	1355	CsF	1251	BaF_2	2137

For the more covalent bromides, the gaseous and to some extent in the liquid phases, the molecular $CaBr_2$ has sufficient stability due to covalent character. Therefore, the melting point is about the same as that of KBr and the boiling point is actually lower.

The extreme cases of Fajan's rules, as in BeI_2 and transition metal bromides and iodides, the stabilization due to covalent character is very large. For metal halides the boiling points of these compounds are comparatively low as expected (BeI_2 = 590° C, ZnI_2 = 624° C and $FeCl_3$ = 315° C). The extreme trend of covalent character is in Al_2Br_6, in which discrete Al_2Br_6 molecules exist even in the solid state. Its m.p = 97° C and b.p = 263° C.

For a clear understanding of covalent character in ionic compounds as per Fajan's rules the melting points, boiling points and equivalent conductivities in fused state of chlorides of typical A subgroup elements are given in Table 3.7.

The chlorides given in Table 3.7 clearly fall into two groups. Those beneath the diagonal line are electrolytes and probably containing ionic bonds, whilst those above the line are non-electrolytes and probably contain covalent bonds.

(b) Solubility: Lattice energies play an important role in solubilities, as the dissolution involves breaking up the lattice. One rule that is reasonably well obeyed is that **compounds that contain ions with widely different radii are soluble in water.** Conversely, the least water soluble salts are those of ions with similar radii. That is, in general, difference in size favours solubility in water. It is found empirically that an ionic compound M tends to be very soluble when the radius of M^+ is smaller than that of X^- by about 80 pm. The solubilities of the group II sulphates decrease from $MgSO_4$ to $BaSO_4$. By contrast, the solubility of the group II hydroxides increases down the group. In the case of sulphates a large anion requires a large cation for precipitation but in the case of hydroxides small anion requires a small cation for precipitation.

The solubility of an ionic compound depends on the Gibbs energy for the process.

$$MX(s) \rightarrow M^+(aq) + X^-(aq)$$

Table 3.7 The melting points °C (a), Boiling points °C (b) and equivalent conductivities in the fused state (c) of chlorides of typical A subgroup elements

Increasing Ionic Charge →

Increasing ionic size ↓

	LiCl	**BeCl$_2$**	**BCl$_3$**	**CCl$_4$**	**NCl$_5$**	**OCl$_6$**
(a)	606°	404°	–107°	–23°	Not	Not
(b)	1382°	488°	12.5°	76°	formed	formed
(c)	166	0.086	0	0	-	-
	NaCl	**MgCl$_2$**	**AlCl$_3$**	**SiCl$_4$**	**PCl$_5$**	**SCl$_6$**
(a)	800°	715°	-	–70°	148°	Not
(b)	1440°	(1410°)	183°	57°	Decomposes	formed
(c)	133	29	15×10^{-6}	0	0	
	KCl	**CaCl$_2$**				
(a)	768°	774°				
(b)	1415°	(1660°)				
(c)	103	52				
	RbCl	**SrCl$_2$**				
(a)	717°	870°				
(b)	1383	(1250°)				
(c)	78	56				
	CsCl	**BaCl$_2$**				
(a)	640°	955°				
(b)	1303°	(1800°)				
(c)	67	65				

In this process, the lattice energy of MX is replaced by hydration energy (solvation energy in general) of ions. If the cation (smaller) has a larger hydration energy than its anion partner or vice versa, then the dissolution of the salt is exothermic.

The variation in lattice energy can be explained using the ionic model. The lattice energy is inversely proportional to the distance between the centres of the ions.

$$\Delta H_L \propto \frac{1}{r_+ + r_-} \tag{3.12}$$

However, the hydration energy, with each ion being hydrated individually, is the sum of individual ion contributions.

$$\Delta_{hyd} H \propto \frac{1}{r_+} + \frac{1}{r_-} \tag{3.13}$$

The hydration energy of a small ion is very large. However, in the expression for lattice energy one small ion cannot make the denominator of the expression small by itself. Thus, one small ion can result in a large hydration energy but not necessarily lead to a high lattice energy. Therefore, ion size asymmetry can result in exothermic dissolution. If both ions are small, then both the lattice energy and hydration energy may be large and dissolution might not be very exothermic.

In many cases, the enthalpy of solution for ionic compounds in water is positive. In such cases cooling of solution takes places during the dissolution of solutes. The mixing tendency of entropy is forcing the solution to do work to pull the ions apart, and since in adiabatic process such work can be done only at the expense of internal energy, the solution cools. If the enthalpy of the solution is sufficiently positive, favourable entropy may not be able to overcome, and the compound will be insoluble.

We can summarize the energetics of a solution as follows. There will usually be an entropy driving force favouring solution. In cases where the enthalpy is negative, zero or slightly positive, dissolution of compound takes place. If the enthalpy changes and the accompanying solution is too positive, then the compound is insoluble. In qualitatively estimating the enthalpy effect, solute–solute, solvent–solvent and solute–solvent interactions must be considered.

$$\Delta H_{Solution} = \Delta H_{Solute–Solvent} - (\Delta H_{Solute–Solute} + \Delta H_{Solvent–Solvent})$$

(c) Thermal stability of ionic solids: The particular aspect we consider here is the temperature required for the thermal decomposition of ionic compounds containing anions like carbonate, sulphate, nitrate, superoxide, etc. The stabilizing influence of a large cation on an unstable anion can be explained in terms of trends in lattice energies. The standard Gibbs energy for the decomposition of a solid $\Delta G° = \Delta H° - T\Delta S°$ becomes more negative when $T\Delta S°$ exceeds the $\Delta H°$. This is possible when T is more if $\Delta H°$ is large.

$$T = \frac{\Delta H^\circ}{\Delta S^\circ} \tag{3.14}$$

Here, the energy changes are discussed by taking the carbonate as an example.

$$MCO_3(s) \longrightarrow MO\,(s) + CO_2(g)$$

In many cases it is sufficient to consider only trends in the reaction enthalpy, as the reaction entropy is essentially independent of M because it is dominated by the formation of gaseous CO_2. The standard enthalpy of decomposition of the solid is

$$\Delta H° \approx \Delta_{decomp} H + \Delta H_L\,[MCO_3\,(s)] - \Delta H_L\,[MO\,(s)] \tag{3.15}$$

where Δ_{decomp} H is the enthalpy change for the gas phase decomposition of CO_3^{2-}

$$CO_3^{2-}\,(g) \longrightarrow O^{2-}\,(g) + CO_2\,(g)$$

Because Δ_{decomp} H is large and positive the overall reaction enthalpy is positive (decomposition is endothermic) but it is less positive if the lattice energy of the oxide is markedly greater than that of the the carbonate because lattice energy of MCO_3(s) is less than that of MO(s). It follows that the decomposition temperature of carbonates will be low that have relatively high lattice energies for oxides. Thus, the compounds having small, highly charged cations, such as Mg^{2+}, are unstable because small cation increases the lattice energy of its oxide more than that of a carbonate. These arguments can easily extended to many inorganic solids like hydroxides, nitrates, sulphates, peroxides, superoxides, etc.

(d) Stabilities of oxidation states: A similar argument as used in the thermal stability of ionic solids can also be used to account for the general observation that high oxidation states are stabilized by small anions. In particular, fluorine has a greater ability than the other halogens to stabilize the high oxidation states of metals. Thus, the only known halides of Ag (II), Co (III) and Mn (IV) are the fluorides. The stability of the heavier halides of the metals in high oxidation states like iodides of Cu (II) and Fe (III) is less and decompose on standing at room temperature (to CuI and FeI_2). O^{2-} ion having small size and high charge can also stabilize the highest oxidation states of elements. Consider the reaction.

$$MX\,(s) + \frac{1}{2}X_2\,(g) \longrightarrow MX_2\,(g)$$

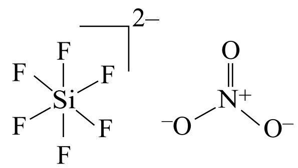
where X is halogen. Though electron affinity of fluorine is less than chlorine, conversion of $\frac{1}{2}X_2$ to X^- is more exothermic because bond enthalpy of F_2 is less than that of Cl_2. Therefore, lattice energies play a major role. In the conversion of MX to MX_2, the number of charges on cation increases from +1 to +2 and hence the lattice energy increases. If the size of anion increases the difference in the lattice energies of MX and MX_2 also decreases and the exothermic contribution to the overall reaction also decreases. Hence, both the lattice energy and X^- formation enthalpy lead to a less exothermic reaction with increase in the size of halogen from F to I. Provided the entropy factors are similar, we expect an increase in thermodynamic stability of MX relative to MX_2 on going from X = F to X = I. Thus, many iodides do not exist for metals in their higher oxidation states and compounds such as CuI_2, TII_3 and VI_5 are unknown, whereas CuF_2, TIF_3 and VF_5 are easily formed. In effect, the high oxidation state metal oxidizes I^- ions to I_2 leading to the formation of a lower metal oxidation state such as Cu (I), Tl(I) and V (III) in the iodides of these metals.

3.5 COVALENT BOND

Lewis concept of covalent bond was already discussed. The covalent bond energy arises due to the forces of attraction between nuclei and the shared electron pair. Covalent bond is rigid and directional in nature. Covalent bonds are formed between the atoms having same or almost nearly equal electronegativities. **Ex:** F_2, Cl_2, Br_2, HCl, ICl, etc.

The number of electrons contributed by an atom in the formation of covalent bonds is called **valence or oxidation number.** The sign + or –, of an oxidation number depends on the electronegativity of the other atom with which it is bonded. Certain elements exhibit more than one oxidation state. The exhibition of more than one valence by the same element is called the **variable valence.**

Example: Phosphorous exhibits +3, +5, sulphur exhibits, +2, +4 and +6. The number of electron pairs shared by an atom is called the **covalence**. For example in SiF_6^{2-}, the valence of silicon is 4 but its covalence is 6 because silicon contributed 4 electrons for bonding and shares 6 electron pairs.



Similarly, in NO_3^- ion, oxidation state of nitrogen is 5 since it contributes 5 electrons for bonding but its covalence is only 4 because it shares only 4 electron pairs.

The species having similar Lewis diagrams, i.e., having the same electronic structure except the identity of atoms are called **isoelectronic** species. Ex: CO_3^{2-}, NO_3^-, etc.

The molecules/ions having the same number of atoms and electrons are known as **isosters**. Their Lewis diagrams are also the same. Examples;

(i) N_2, CO, CN^-	(ii) NH_4^+, CH_4
(iii) NH_3 and H_3O^+	(iv) NH_2^+ and O_3^-
(v) NO_2^- and H_2O	(vi) NO_3^- and CO_3^{2-}

3.5.1 Characteristics of Covalent Compounds

(i) Physical state: In covalent compounds, molecules are held together by weak van der Waal's forces of attraction and that is why covalent compounds are found in all the three solid, liquid or gaseous states, and tendency to become solid increases with increase in molecular weight.

(ii) Melting and boiling points: Generally covalent compounds have low melting points and boiling points due to weak attractive forces between the covalent molecules. The low melting and boiling points of covalent compounds are due to the breaking of weak dispersion forces between molecules but not covalent bonds between the atoms within the molecule. Giant covalent molecules like silica, diamond etc, are very hard and have very high melting points and boiling points because in such cases the covalent bonds have to be broken and extended three dimensionally.

(iii) Isomerism: Covalent bonds are rigid and directional and hence the covalent compounds exhibit stereoisomerism.

(iv) Solubility: Generally covalent compounds (mainly non-polar) are insoluble in water but soluble in organic solvents. Some polar covalent compounds are water soluble, due to hydrogen bonding but insoluble in non-polar organic solvents like benzene, e.g., sugar in water.

(v) Electrical conductivity: Covalent compounds are non-conductors because they do not contain mobile ions/free electrons.

(v) Chemical reactions: In their chemical reactions, covalent molecules undergo molecular reactions. During the reactions, bond breaking and bond making takes place. Therefore, these reactions are generally slow and rarely proceed to completion.

Limitations of Lewis concept

(i) It could not explain the bond energies and bond lengths.

(ii) It could not explain the geometry and shapes of molecules.

(iii) It could not explain how the molecules having expanded octets and the molecules with incomplete octet are formed and stable.

3.5.2 Valence Shell Electron Pair Repulsion (VSEPR) Theory

In 1940, **Sidgwick** and **Powell** laid the basis for a new theory when they concluded that the arrangements of bonds around multi covalent atoms are related simply to the total number of valence shell electrons, the unshared as well as shared electron pairs. From 1950 onward Lennard Jones, Pople, Linnet, Mellish, Walsh and particularly Gillespie and Nyholm, to name only the most active contributors, have been stressing the importance of the lone pairs of electrons and the Pauli principle in stereochemistry. In reality, they have been building a new theory. The new theory, we call, following Gillespie's suggestion, as the Valence Shell Electron Pair Repulsion (VSEPR) Theory. This theory proposes that the stereochemistry of an atom in a molecule is determined primarily by the repulsive interactions amongst all the electron pairs in its valence shell. The theory assumes that the valence shell electrons occupy essentially localized orbitals spatially oriented so as to maximize their average distance part.

The theory may be summarized as follows:

1. The shape of a molecule is determined by the arrangement and the repulsion between all of the electron pairs present in the valence shell of the central atom.
2. A lone pair (LP) of electrons takes up more space around the central atom than a bond pair, since the lone pair is under the attraction of only one nucleus while the bond pair (BP) is shared by two nuclei and drawn out between two positive centres.

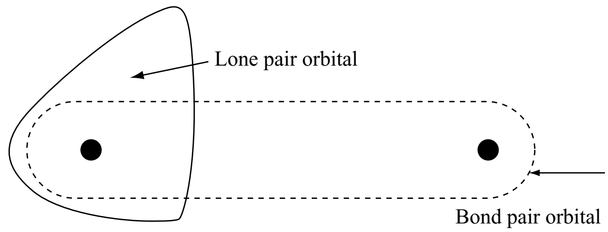


Fig 3.10 Diagramatic representation of the spatial differences between lone pair and bond pair orbitals

It results that the repulsion between two lone pairs is greater than the repulsion between two bond pairs. Thus the presence of lone pairs on the central atom causes slight distortion of the bond angles from the expected values. The repulsions are expected to increase in the order:

Bond pair - Bond pair < Bond pair - Lone pair < Lone pair - Lone pair

3. The magnitude of the repulsions between the bonding pairs of electrons depends on the electronegativity difference between the central atom and the other atoms bonde with it.
4. Multiple bonds do not affect the gross stereochemistry of a molecule. The geomerty of a molecule is determined primarily by the number of sigma bond pairs and the number of lone pairs. But the electron pairs in double bond cause more repulsion than one electron pair in single bond and three electron pairs in triple bond cause more repulsion than two electron pairs in the double bond. Since the geometry of a molecule does not depend on the multiple bonds the two or three electron pairs of a multiple bond are treated as a **single super pair**.
5. Repulsions between electron pairs in filled shells are greater than those between electron pairs in incomplete shells.
6. When an atom, with filled valence shell and one or more lone pairs, is bonded to an atom with an incomplete valence shell, or a valence shell that can become incomplete by electron shifts, there is a tendency for the lone pairs to be partially transferred from the filled to the unfilled shell.
7. In a valence shell containing five or seven electron pairs in which all electron pairs cannot have the same number of nearest neighbours, those pairs with largest number of nearest neighbours will be located at a greater average distance from the nucleus than the other electron pairs.

A molecule can be described by the generic formula AB_mE_n, where A is the central atom, B stands for any atom or group of atoms surrounding the central atom and E represents a lone pair of electrons. The steric number

[SN = m + n] is the number of positions occupied by atoms or lone pairs around a central atom. When all electron pairs are bond pairs, the structures for 2, 3, 4 and 6 electron pairs are completely regular with all bond angles and distances the same. Neither 5- nor 7- coordinate structures can have uniform angles and distances, because there are no regular polyhedral with these number of vertices. The 5- coordinate molecules have a trigonal bipyramidal structure, with a central triangular plane of three positions plus the other positions above and below the centre of the plane. The 7-coordinate molecules have a pentagonal bipyramidal structure, with a pentagonal plane of five positions and square antiprism structure (SN = 8) is like a cube with the top and bottom faces twisted 45° into the antiprism arrangement as shown in Fig. 3.11. It has three different bond angles for adjacent fluorines. $[TaF_8]^{3-}$ has a square antiprism symmetry but is distorted from this ideal solid. (A simple cube has only the 109.5° and 70.5° bond angles measured between two corners and the centre of the cube because all edges are equal and any square face can be taken as the bottom or top.)

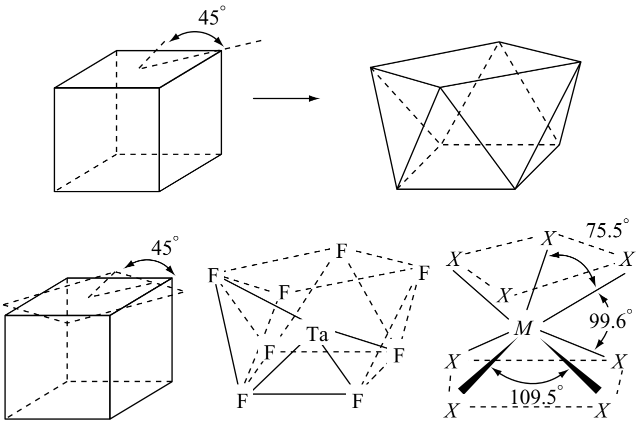


Fig 3.11 Conversion of a cube into a square antiprism

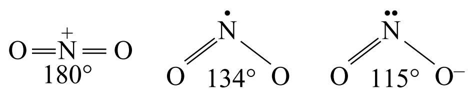
Table 3.8 Equilibrium arrangement of like particles (bond pairs) on the surface of an atom

Steric number	Arrangement	Calculated bond angles
2	Linear	180°
3	Planar triangular (trigonal)	120°
4	Tetrahedral	109°28'
5	Trigonal bipyramidal	120°, 90°
6	Octahedral	90°
7	Pentagonal bipyramidal	72°, 90°
8	Square antiprism	70.5°, 99.6°, 109.5°

We must keep in mind that we are always attempting to match our explanations to experimental data. The explanation that fits the data best should be the current favorites, but new theories are continuously being suggested and tested. Because we are working with such a wide variety of atoms and their molecular structures, it is unlikely that a single, simple approach will work for all of them. Although the fundamental ideas of atomic and molecular structures are relatively simple, their application to complex molecules is not. It is also helpful to keep in mind that for many purposes, prediction of exact bond angles is not usually required to a first approximation, lone pairs, single bonds, double bonds and triple bonds can all be treated similarly while predicting molecular shapes. However, better predictions of overall shapes can be made by considering some important differences between lone pairs and bond pairs. These methods are sufficient to show the trends and explain the bonding.

Lone Pair Repulsions

Steric number-3: The bond angles in NO_2^+, NO_2 and NO_2^- are as follows.



The lone pair in NO_2^- exert greater effect than a lone electron in NO_2 and thus the bond angle in NO_2^- is less than in NO_2, NO_2^+ ion is linear since it does not have lone pair or lone electron. In BF_3, the calculated bond length is 150 pm but the experimental value is 130 pm. According to rule 6 if the lone pair on fluorine is donated to boron atom through pπ-pπ bond the electron density at fluorine decreases and get stabilized, due to which B-F bond gets a double bond character lowering the bond length.

Steric number-4: The isoelectronic molecules CH_4, NH_3, H_2O and HF (Fig. 3.12). illustrate the lone pairs on molecular shape. Methane has four identical bonds between carbon and each of the hydrogens. When the four pairs of electrons are arranged as far from each other as possible according to rule No.1, the result is the familiar tetrahedral shape, with six regular tetrahedral angles 109.5°. Similar tetrahedral arrangement probably occur in covalent compounds of trivalent nitrogen, divalent oxygen and monovalent fluorine.

The nitrogen atom in NH_3 molecule is surrounded by three bond pairs and one lone pair, oxygen in water is surrounded by two bond pairs and two lone pairs and fluorine in hydrogen fluoride is surrounded by one bond pair and three lone pairs. The bond angles in ammonia and water are less than the tetrahedral bond angle due to the greater repulsion effect of the lone pairs as compared with the bond pairs. Also, it can be seen that as the number of lone pairs increases, decrease in bond angle is more as can be observed from bond angles of NH_3 and H_2O.

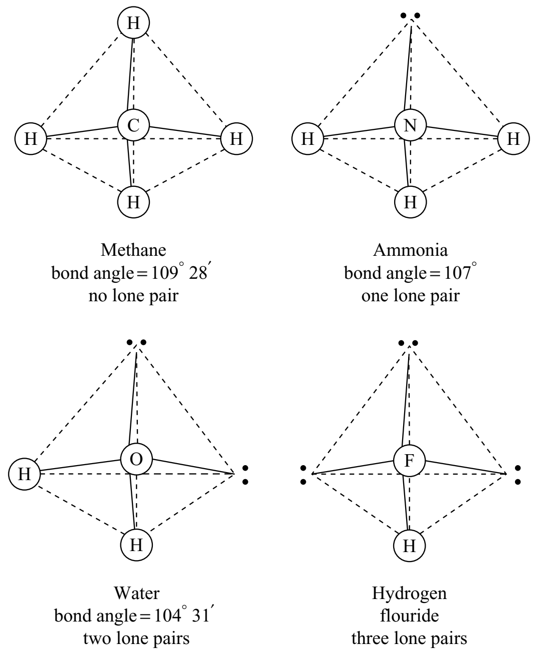
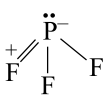


Fig 3.12 The correlation between molecules of methane, ammonia, water and hydrogen fluoride. The tetrahedral distribution of four electron pairs around the central atom.

As per rule 3 the bond angle in molecules having steric number 4, decreases with increase in the electronegativity of the bonded atom or decrease in the electronegativity of the central atom. In other words, the bond angle decreases with increase in the size of bonded atom or decrease in the size of central atom. This is because when electronegativity of bonded atom is more, the bond pair of electrons move away from the central atom, thereby contracting and thinning out the orbital, thus causing decrease in the repulsion between bond pairs and coming close to each other. The examples in this regard include

H_2O 104.5° > OF_2 103.2°
NH_3 107.3° > NF_3 102°
PI_3 102° > PBr_3 101.5° > PCl_3 100°
AsI_3 101° > $AsBr_3$ 100.5° > $AsCl_3$ 98.4°

If in a molecule the central atom is having vacant orbitals and the bonded atom has a lone pair, delocalization of electrons from the bonded atom towards the central atom not only strengthen the bond but also reduce the interatomic repulsions between the bonded atoms. Then, partial multiple bonding occurs and serves to increase the bond angles. For example



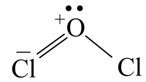
PH_3 93.3° < PF_3 104°

AsH_3 91.8° < AsF_3 102°

So far, our examples have been ones in which the central atom has the incomplete shell. We find examples also of this effect when the central atom has the filled shell and the bonded atoms have incomplete shells. Consider the angles.

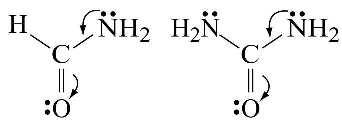
Cl_2O 110.8° > F_2O 103.2° or H_2O, 104.5°

The structure of Cl_2O can be postulated as



This would cause the opening of a bond angle. We know that the bond angle in SCl_2 is only 102°. Here, the sulphur atom is bigger in size than oxygen atom and will not have the same need to delocalize its non-bonding electrons. Another best example is that the molecule $N(SiH_3)_3$ is planar while $N(CH_3)_3$ is pyramidal. In $N(SiH_3)_3$, the lone pair on nitrogen can delocalize into the vacant d-orbital of silicon. This is not possible in the case of $N(CH_3)_3$ since carbon do not contain the d-orbital in its valence shell. Some other similar examples are

(i) The molecules $[Cl_5RuORuCl_5]$ and $[TiCl_2(C_2H_5)]_2O$ are linear suggesting the species such as $\bar{M} = \overset{++}{O} = \bar{M}$ contributing to the structure.
(ii) The ion $[O(HgCl)_3]^+$ is triangular rather than pyramidal which would have been expected if the oxygen lone pair remains on the oxygen.
(iii) SiOSi bond angles fall in the range of 130–150°.
(iv) SOS angles are invariably larger than SSS angles. For example, the angle in $(SO_3)_3$ is 114°, in $S_2O_7^{2-}$ is 124° but in S_8 it is 107°, in S_4O_6, SSS angle is 130° and in S_4^{2-} it is 104.5°.
(v) POP angles are larger than PSP angles. The angle in P_4O_{10} is 123.5°, but in P_4S_{10} it is 109.5° and all other known PSP angles fall below the tetrahedral angle, whereas all other known POP angles fall into 120–140° range.
(vi) The planar arrangement with 120° bond angles around the nitrogen atoms in urea and formamide may be explained similarly



Steric number-5: For trigonal bipyramidal geometry, there are two possible locations of lone pairs, i.e., axial and equatorial. If lone pairs are present, they occupy only the equatorial positions. The equatorial position provides the lone pair with more space and minimizes the interactions between the lone pairs and lone pairs and also between lone pairs and bond pairs. If the lone pairs were axial, it would have three 90° interactions with bonding pairs, although in an equatorial position it has only two such interactions as shown in Fig. 3.13. The actual structure is distorted by the lone pair as it spreads out in space and effectively squeezes the rest of the molecule together.

ClF_3 provides the second example of the influence of lone pairs in molecules having a steric number 5. There are three possible structures for ClF_3, as shown in Fig. 3.14. Lone pairs in Fig. 3.14 are designated as LP and bonding pairs as BP.

In determining the structure of molecules, the lone pair–lone pair interactions are most important, with the lone pair–bonding pair interactions next in importance. In addition, interactions at angles of 90° or less are most important; larger angles generally have less influence. In ClF_3, structure B can be eliminated quickly because of the 90° LP-LP angle. The LP-LP angles are large for A and C, so the choice must come from LP-LP and BP-BP angles.

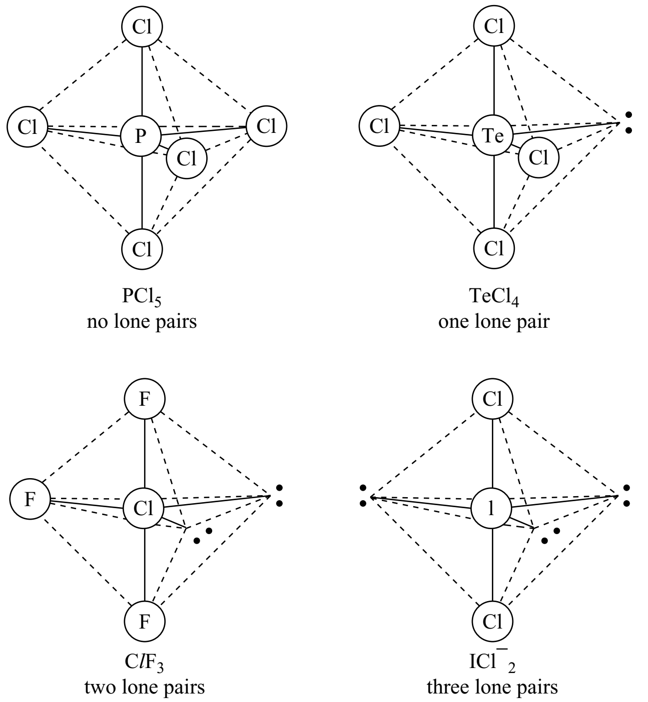


Fig 3.13 Distribution of five electron pairs around a central atom in a trigonal bipyramidal arrangement

Because the LP-LP angles are more important, C which has only four 90° LP-LP interactions, is favoured over A, which has six such interactions. Experiments have confirmed that the structure is based on C with slight distortions due to the lone pairs. The lone pair-bonding pair repulsions causes the LP-BP angles to be larger than 90° and the BP-BP angles less than 90° (actually 87.5°). The Cl-F bond distances show the repulsive effects as well, with the axial fluorines (approximately 90° LP-BP angles) at 169.8 pm and the equatorial fluorine (in the plane with two lone pairs) at 159.8 pm. Angles involving lone pairs cannot be determined experimentally. The angles in Fig. 3.14 are calculated assuming maximum symmetry consistent with the experimental shape.

XeF_2 is the third example having three lone pairs having a linear structure as shown in Fig. 3.15.

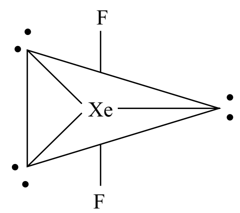
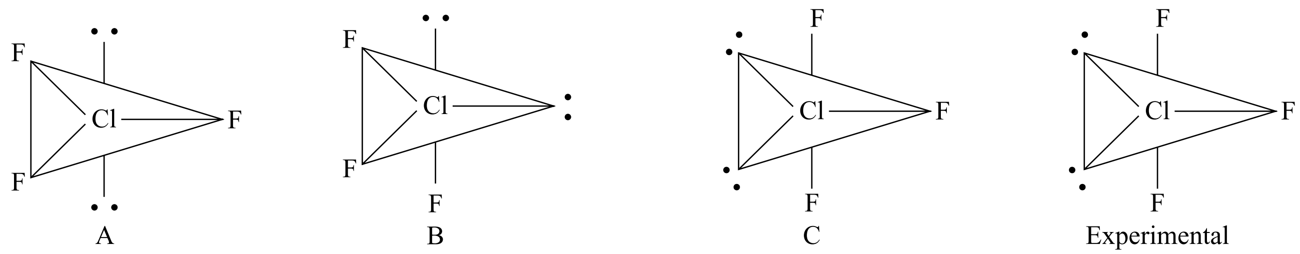


Fig 3.15 Structure of XeF_2



	Calculated			Experimental
	A	**B**	**C**	
Lp – Lp	180°	90°	120°	Cannot be determined
Lp – Bp	6 at 90°	3 at 90°	4 at 90	Cannot be determined
	2 at 120°	2 at 120		
Bp – Bp	3 at 120°	2 at 90°	2 at 120°	2 at 87.5°
		1 at 120°		Axial Cl – F 169.8 pm
				Equatorial Cl – F 139.8 pm

Fig 3.14 Distribution of five electron pairs around a central atom in a trigonal bipyramidal arrangement

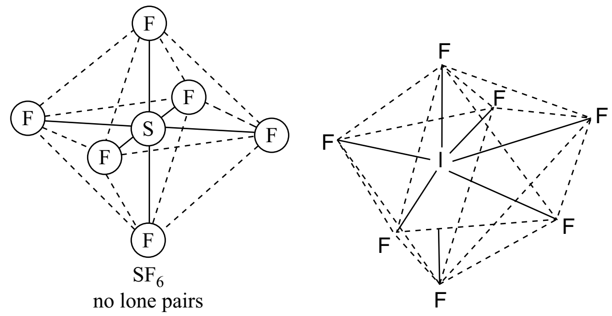


Fig 3.16 Structures of SF_6 and IF_7

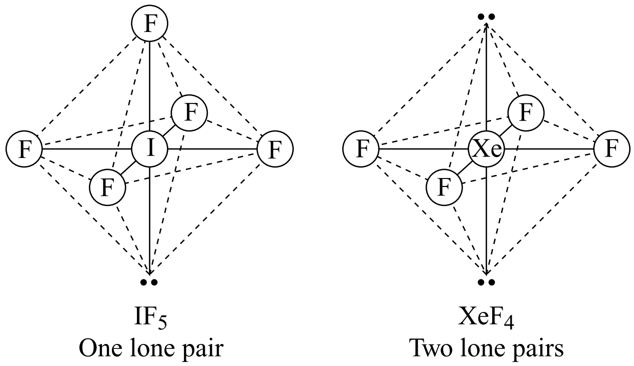


Fig 3.17 Structures of IF_5 and XeF_4

Steric numbers 6 and **7** The molecules having the steric numbers 6 and 7 will be in regular geometries with octahedral and pentagonal bipyramid shapes, respectively as shown in Fig. 3.16.

If a lone pair is present in the octahedral structure, it will occupy any one corner of the octahedron since all the LP-BP interactions are similar, but if there are two lone pairs they have to occupy the opposite corners of the octahedron as in IF_5 and XeF_4, respectively. Therefore, the structures of IF_5 will be square pyramid and XeF_4 will be square planar.

Due to the repulsion by lone pair, the 4 fluorine atoms raises above the plane. Therefore, all bond angles become less than 90° either in the square plane of the square pyramidal structure or with the fluorine at the apex of the square pyramid structure. Since two lone pairs on opposite sides of the octahedron exhibit equal repulsions on all the four fluorine atoms in XeF_4, FXeF bond angles remain at 90°.

No discussion of the VSEPR model of molecular structure would be complete without a brief discussion of

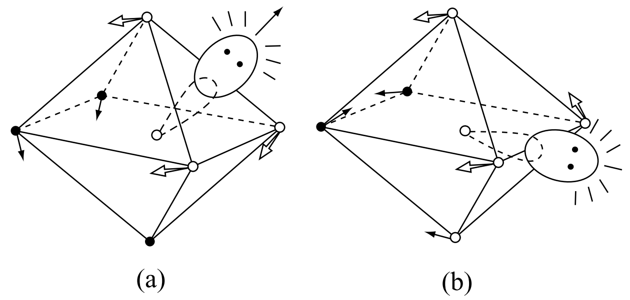


Fig 3.18 Possible molecular structures of xenon hexafluoride (a) Lone pair emerging through face of octahedron (b) Lone pair emerging through edge of octahedron

some remaining problems. One interesting problem is the molecular structure of XeF_6. The simplest MO treatment of this molecule predicted that the molecule would be perfectly octahedral. In contrast, the VSEPR model considers the fact that there will be seven pairs of electrons in the valence shell (six bonding pairs and one lone pair) and predicts a structure based on seven-coordination. Unfortunately, we have little to guide us in choosing the preferred arrangement. Gillespie suggested three possibilities for XeF_6, a distorted pentagonal bipyramid, a distorted octahedron or a distorted trigonal prism. The lone pair should occupy a definite geometric position and volume as great as or greater than the bonding pair.

Determining the exact structure of the gaseous XeF_6 molecule proved to be unexpectedly difficult. It is known to be a slightly distorted octahedron. The best model for the molecule as shown in Figure 3.18 appears to be a distorted octahedron in which the lone pair extends either through a face or through an edge.

The structure of XeF_6 was confirmed as shown in Figure 3.18 a by the X-ray diffraction studies of its analogous isoelectronic compound $Xe(OTeF_5)_6$ (for the oxygen coordination shell about the xenon) indicating a stereochemically active lone pair.

Multiple Bonds: We now look briefly at the effects that multiple bonding will have upon bond angles. In Table 3.9, some bond angles for molecules possessing multiple bonds are given.

The VSEPR model considers double and triple bonds to have slightly greater repulsive effects than single bonds because of the repulsive effect of π-electrons. It can be seen from Table 3.9 that for all of the phosphorus compounds POX_3 and PSX_3, the bond angles XPX fall not far from but definitely below the tetrahedral angle. Since there are four σ-bond pairs and no lone pairs, a tetrahedral angle would be predicted for the gross stereochemistry.

Table 3.9 Effect of multiple bonding on bond angles

Molecule	XAX angle	Molecule	XSX angle	XSO angle
$O = PF_3$	101.3°	$O = SF_2$	93°	107°
$O = PCl_3$	103.3°	$O = SCl_2$	114°	106°
$O = PBr_3$	105.5°	$O = SBr_2$	96°	108°
$S = PF_3$	100.3°	$O = S(CH_3)_2$	97°	107°
$S = PCl_3$	101.8°	$O = S(C_6H_5)_2$	97°	106°
$S = PBr_3$	106°	O_2SF_2	96°	124°
$O = CH_2$	118°	O_2SCl_2	111°	110.8°
$O = CF_2$	108°	$O_2S(CH_3)_2$	115°	110.4°
$O = CCl_2$	111.3°	$O_2S(NH_2)_2$	112.1°	125°
$O = C(CH_3)_2$	119.6°			
$O = C(NH_2)_2$	118°			

However, if one of the bonds is multiple, that is, it consists four electrons rather than just two, and the added electron density of the multiple bond exerts a greater effect than that exerted by a single bond and hence the reduction in bond angle. We see the same effect in the COX_2 compounds, where instead of the 120° angle predicted for three sigma and no lone pairs, we obtain smaller XCX angles due to the additional repulsion of the C=O bond electrons. The same effect must be responsible for the observed bond angles in the sulphur compounds SOX_2 and SO_2X_2 given in Table 3.9. Thus it would appear that a safe generalization would be that multiple bond orbitals repel other orbitals more strongly than single bond orbitals.

Additional examples of the effect of multiple bonds on molecular geometry are shown in Fig. 3.19.

From the structures given in Fig 3.19 it can be seen that multiple bonds tend to occupy the same positions as lone pairs. For example, the double bonds to oxygen in SOF_4, ClO_2F_3 and XeO_3F_2 are all equatorial as are the lone pairs in the matching compounds of steric number 5, SF_4, ClF_3 and XeF_2. Also multiple bonds, like lone pairs, tend to occupy more space than single bonds and to cause distortions that in effect squeeze the rest of the molecules together. In molecules that have both lone pairs and multiple bonds, these features may compete for space. Examples are shown in Fig. 3.19.

3.5.3 Writing the Structures of Molecules

Now, we will learn how to write the structures of molecules. To write the structure of a molecule we should know the total number of electron pairs (P), the number of bond pairs (*bp*) and lone pairs (*lp*) around the central atom in a molecule. These can be calculated by using the formula.

$$P = \frac{1}{2}(V + M - c + a)$$

'V' is the number of electrons in the valence shell of the central atom, 'M' is the number of monovalent atoms and 'c' is the number of positive charges, if the given species is cation and 'a' is the number of negative charges, if the given species is anion.

Then, the number of bond pairs (bp) and lone pairs (lp) can be known from the formula.

$$lp = P - bp$$

The bp in a molecule is equal to the number of atoms around the central atom. Knowing the number of bp and lp we can write the structure of molecules.

Illustrative Examples

A. For molecules containing only monovalent atoms and do not contain multiple bonds, calculation for SF_6 is given below.

Solution: $P = \frac{1}{2}(6 + 6 - 0 + 0)$; $BP = 6$

$lp = 6 - 6 = 0$

Since there are 6 *bp* and without lone pair the molecules SF_6 have regular octahedral structure. Similarly we can write the structures of BeF_2, BCl_3, CCl_4, PCl_3, PCl_5, NH_3, H_2O, OF_2, SF_4, ClF_3, IF_7, XeF_2, etc.

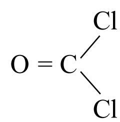
B. For molecules containing multiple bonds (π - bonds). The calculation taking example for $COCl_2$.

Solution: $P = \frac{1}{2}(4 + 2 - 0 + 0) = 3$

$bp = 3$

$lp = 3 - 3 = 0$

Since oxygen is divalent, there will be a double bond between carbon and oxygen.



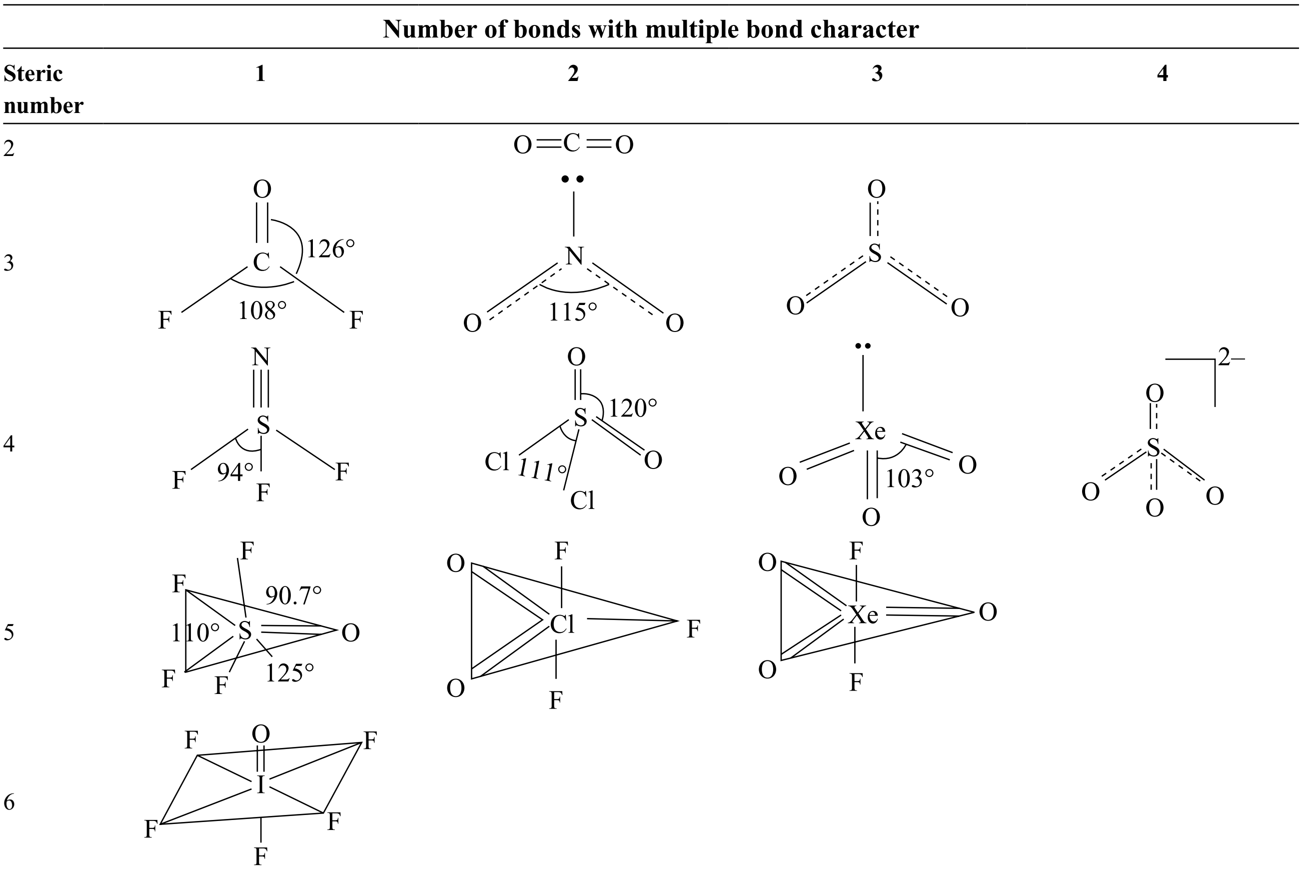


Fig 3.19 Structures of molecules containing multiple bonds

The ClCCl angle is less than 120° while OCCl bond angle is greater than 120°.

C. For cations like NH_4^+, CH_3^+, H_3O^+, etc. The calculation for H_3O^+ is as follows

Solution: $P = \frac{1}{2}(6+3-1+0) = 4$

$bp = 3$

$lp = 4 - 3 = 1$

H_3O^+ will have a pyramidal shape with one lone pair and three bond pairs.

D. For anions like $NO_3^-, NO_2^-, PO_4^{3-}, CO_3^{2-}, SO_4^{2-}, SO_3^{2-}$, etc. The calculation for NO_3^- is as follows.

Solution: $P = \frac{1}{2}(5+0-0+1) = 3$

$bp = 3$

$lp = 3 - 3 = 0$

Only *bp* are present, no *lp*. Therefore, the NO_3^- will have the planar triangular structure.

E. For ions like ICl_4^-, I_3^-, ClF_2^-, etc. The calculation for ICl_4^- is as follows.

Solution: $P = \frac{1}{2}(7+4-0+1) = 6$

$bp = 4$

$lp = 6 - 4 = 2$

Iodine atom in ICl_4^- contains 2 lp and 4 bp. Therefore, it will have a square planar structure since two lp occupies the opposite corners of the octahedron.

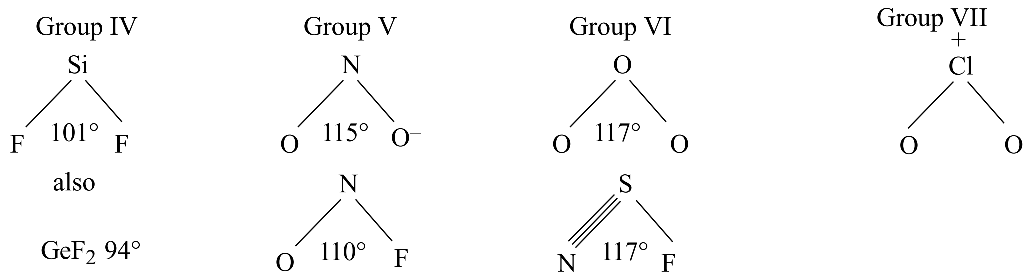
3.5.4 Isoelectronic and Isostructural Species

If Lewis diagrams of different species are identical, except for the identity of the central atom, and having the same electronic structure then they are called as isoelectronic species. These can be divided into different groups depending on the total number of valence electrons in a given species and steric number.

(i) Linear 16 - electron molecules and ions: This group includes linear molecules and ions of Ag and Au. $\left[Ag(NH_3)_2^+, Au(NH_3)_2^+, H_3N\cdot AuCl_2^-\right]$, mercuric halides and a group of molecules and ions containing C, N and O.

(ii) Steric number 3: There are two groups: 18-electron group and 24-electron group.

(a) The 18 electron group



Group IV	Group V	Group VI
	also	also
TiF_2 130°	NOCl 116°	SSO 118°
SnX_2	NOBr 117°	OSO 120°
PbX_2	NO(OH) 111°	

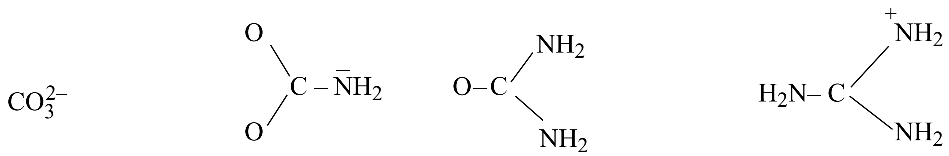
Note the remarkable similarity in interbond angles in all the compounds of N, O and S despite great difference in multiplicity of bonds.

(b) The 24-electron group

BO_3^{3-}	CO_3^{2-}	NO_3^-	SO_3
–	–	O_2NX	–
–	OCX_2	$(NOF_2)^+$	–
BX_3	–	–	–

All have equivalent triangles.

No complete vertical family in the chart is known with O or halogen ligands, but with the isoelectronic NH_2 instead of halogen we have

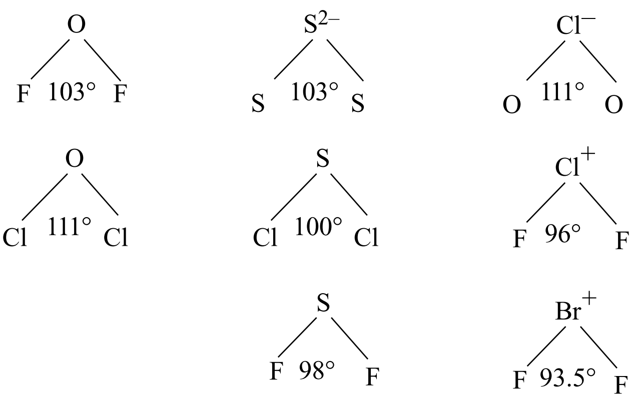


Carbonate ion | Carbamate ion | Carbamide | Guanidinium ion

All the above are shown to be planar.

(iii) Steric number - 4

(a) The 20-electron group



(b) **The 26-electron group:** Pyramidal AO_3 complexes include the SO_3^{2-}, SeO_3^{2-}, ClO_3^-, BrO_3^- and IO_3^- ions and XeO_3 molecule. Less symmetrical pyramidal molecules include the thionyl and chloryl halides; iodyl fluoride IO_2F.

SO_3^{2-}	ClO_3^-	XeO_3
SO_2F^-	ClO_2F	
$SOCl_2$		

(c) **The 32-electron group:** This group includes tetrahedral oxy-ions of Si, P, S and Cl, halogeno - ions MX_4 formed by several non-transition elements, many intermediate oxy - halogen ions and molecules.

Steric number 5:

(a) **The 22-electron group:** Examples here are confined to linear ions like, $ClBr_2^-$, $BrCl_2^-$, IX_2^-, $IX'X''$, Br_3^-, I_3^-, KrF_2 and XeF_2 molecules.

(b) **The 28-electron group:** The T - shaped configuration has been established for ClF_3 and BrF_3.
(c) **The 34-electron group:** This group includes SF_4, SCl_4, SeF_4, $TeCl_4$, IOF_3 and XeO_2F_2
(d) **The 40-electron group:** This group includes the pentahalides of the group - V and $SnCl_5^-$ ion.

Steric number 6:
(a) **The 36-electron group:** This group includes the species like BrF_4^-, ICl_4^- and XeF_4 having square planar structure.
(b) **The 42-electron group:** The species belonging to this group have square pyramidal structure.

SbF_5^{2-} BrF_5 $XeOF_4$
$SbCl_5^{2-}$ TeF_5^-
XeF_5^+

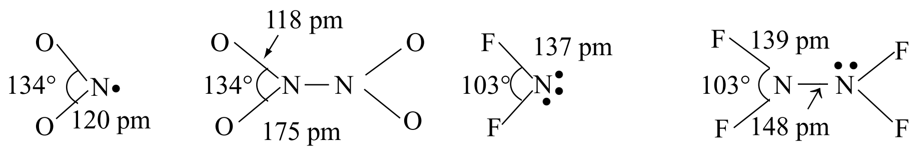
(c) **The 48-electron group:** Octahedral molecules and ions of this group are large in number. Examples.

$Te(OH)_6$ $IO(OH)_5$
TeO_6^{6-} IO_6^{5-} XeO_6^{4-}
IOF_5
AlF_6^{3-} SiF_6^{2-} PF_6^- SF_6 IF_6^+

ODD ELECTRON SYSTEMS AX_2 AND THEIR DIMERS $X_2A - AX_2$:

The 17-electron and 19-electron systems are of special interest because they might be expected to dimerise to the 34- and 38-electron molecules and so get rid of the unpaired electron.

17 electron   34 electrons   19 electrons   38 electrons.



3.5.6 Ligand Close Packing

Gillespie developed another approach **Ligand Close Packing (LCP)** to explain about the bond angles. The LCP model uses the distances between the outer atoms in molecules as a guide. For a series of molecules with the same central atom, the non-bonded distances between the outer atoms are consistent, but the bond angles and bond lengths change. For example, a series of BF_2X and BF_3X compounds where, X = F, OH, NH_2, Cl, H, CH_3, CF_3 and PH_3 have B-F bond distances of 130.7 to 142.4 pm and F-B-F bond angles of 105.4°, but the non-bonded F.....F distances remain nearly constant at 225 to 220 pm. Examples are shown in Table 3.10.

From the values in Table 3.10 it can be observed that in 3 coordinate boron compounds, the B-F bond lengths and FBF angles does not alter much because of the back bonding to boron. However, the B-F bond length values

Table 3.10 Ligand close packing data

Molecule	Coordination number of B	B-F distance (pm)	FBF angle (°)	F.........F distance (pm)
BF_3	3	130.7	120.0	226
BF_2OH	3	132.3	118.0	227
BF_2NH_2	3	132.5	117.9	227
BF_2Cl	3	131.5	118.1	226
BF_2H	3	131.1	118.3	225
BF_2BF_2	3	131.7	117.2	225
BF_4^-	4	138.2	109.5	226
$BF_3CH_3^-$	4	142.4	105.4	227
$BF_3CF_3^-$	4	139.1	109.9	228
BF_3PH_3	4	137.2	112.1	228
BF_3NMe_3	4	137.2	111.5	229

are higher in the case of 4-coordinate compounds when compared to 3-coordinate compounds due to absence of back bonding. The FBF bond angles decreases more in 4-coordinate complexes due to shifting of bond pair electron density towards more electronegative fluorine. Though the bond angles decrease, due to increase in bond lengths in 4-coordinate compounds, the non-bonded F..........F distances remain constant.

3.6 VALENCE BOND THEORY

Lewis theory could explain the writing, structures of molecules but failed to explain the formation of chemical bond. It also did not explain the reason for the difference in the bond dissociation enthalpies and bond lengths in molecules like H_2 (435.8 KJ mol^{-1}, 74 pm) and F_2 (150.6 KJ mol^{-1}, 42 pm) although in both cases a single covalent bond is formed by the sharing of an electron pair between the respective atoms. It also did not give an idea about the shapes of polyatomic molecules.

Similarly, the VSEPR theory gave geometry of simple molecules but could not explain about the bond energies, the reason for bond formation, bond lengths, etc. In modern chemistry there are two "contenders for the throne" of explaining the bonding theory. They are **valence bond theory (VBT)** and **molecular orbital theory (MOT)**. There is a big contention between the proposers of the two theories as to which was best. Sometimes, overzealous supporters of one theory have given that the other is "wrong". Each theory has its own merits and demerits. Given a specific question, one theory may prove distinctly superior in insight, ease of calculation, or simplicity and clarity of results, but a different question may reverse the picture completely. Surely the inorganic chemist must be familiar with both the theories. The valence bond (VB) theory grew directly out of the ideas of electron pairing by Lewis and others. In 1927, W. Heitler and F. London proposed a quantum-mechanical treatment of the hydrogen molecule. Their method has come to be known as the valence bond approach and was developed extensively by men such as Linus Pauling and J.C. Slater. Valence bond theory is mainly based on the knowledge of atomic orbitals, electronic configurations of elements, the overlap criteria of atomic orbitals, the hybridization of atomic orbitals and the principles of variation and super position. These include a lot of mathematical calculations. Therefore, for the sake of convenience, valence bond theory has been discussed in terms of qualitative and non-mathematical treatment only.

We will start with the H_2 molecule. Consider two hydrogen atoms *A* and *B* approaching each other having nuclei N_a and N_b and electrons present in them are represented by e_a and e_b. As the two atoms are gradually brought together, there will be attractive and repulsive forces between the two atoms as follows.

Attractive forces arise between:

(i) nucleus of one atom and its own electron, that is, N_a - e_a and N_b - e_b.

(ii) nucleus of one atom and electron of other atom, i.e., N_a - e_b and N_b - e_a.

Repulsive forces arise between

(i) electrons of two atoms e_a - e_b.

(ii) nuclei of two atoms N_a - N_b.

The two hydrogen atoms come close together due to attractive forces while they tend to move away due to repulsive forces as shown in Fig. 3.20.

Experimentally, it has been found that the magnitude of new attractive force is more than the new repulsive forces. As a result, two atoms approach each other and potential energy decreases. Ultimately a stage is reached where the net force of attraction balances the force of repulsion and the system acquires minimum energy. At this stage, two hydrogen atoms are said to be bonded together to form a stable molecule having the bond length of 74 pm.

Since the energy is liberated when the bond is formed between the hydrogen atoms, the hydrogen molecule is more stable than that of isolated hydrogen atoms. The energy liberated is known as bond formation enthalpy. This

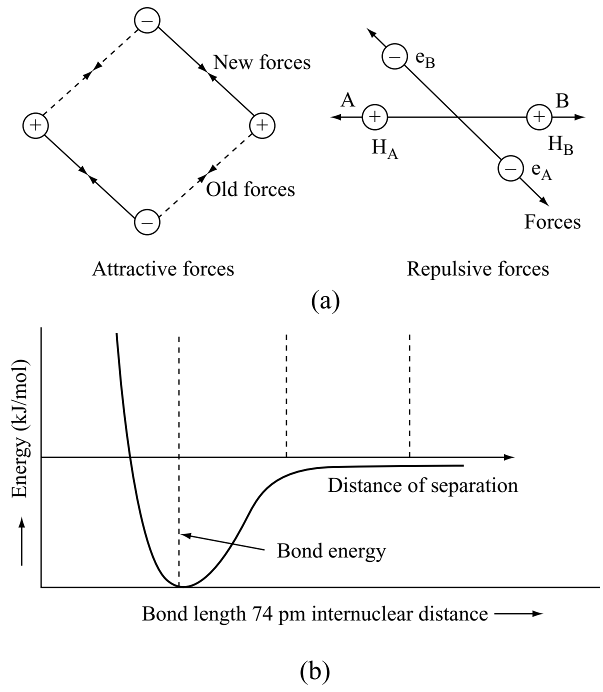


Fig 3.20 (a) Force of attraction and repulsion during the formation of H_2 molecule. (b) Potential energy curve for the formation of H_2 molecule as a function of internuclear distance of the H atoms. The minimum in the curve corresponds to the most stable state of H_2

is shown in Fig. 3.16b. To break the bond in hydrogen molecule, the same amount of energy (435.8 KJ) is required and is known as the **bond dissociation enthalpy**.

The important postulates of valence bond theory are as follows:

(i) A covalent bond is formed by the overlapping of atomic orbitals of valence shell of the two atoms.

(ii) The orbitals containing unpaired electrons can only participate in bonding and the spin of the two electrons must be opposite.

(iii) The greater the extent of overlap the stronger is the bond.

(iv) The bonds are formed in the direction in which the atomic orbitals are directed. The atoms with half-filled atomic orbitals must come close to one another with their axes in proper directions for overlapping. The bonds are formed in the direction in which the atomic orbitals are directed. The imaginary line joining the two nuclei in a molecule is known as **inter nuclear axis** or **molecular axis**.

(v) The maximum electron density in the overlapping orbitals is in between the nuclei of two atoms. The attractive force between this electron density and the nuclei is the c**ovalent bond**.

(vi) Electrons which are already paired in valence shell can also participate in bonding, if they are separated and excited to higher energy level of the same orbit.

(vii) The covalent bond formed with the linear overlap of atomic orbitals along the inter-nuclear axis is known as σ **(sigma) bond**. The covalent bond formed by the lateral overlap or sidewise overlap of two atomic orbitals perpendicular to the inter-nuclear axis is known as **π (pi) -bond.**

(viii) A sigma bond can exist independently but a π-bond cannot. π-bond will be formed only after the formation of a sigma bond. π-bond is weaker than σ -bond.

(ix) The shapes of molecules depend only on σ-bonds but not on π-bonds.

(x) The overlapping atomic orbitals can only form a bond, if they also have similar energy and similar symmetry. The strength of a covalent bond is closely related to the extent of overlap of the atomic orbitals concerned. This is known as the principle of maximum overlap and it is extremely useful in a general consideration of covalent bonding in molecules.

Two s-orbitals cannot overlap very strongly because of the spherical distribution of charge. What might be called an s-s bond, is therefore relatively weak.

Because *p*-orbitals are concentrated in a particular direction and because their lobes are longer than the radius of the corresponding *s*-orbital, they can overlap with other *s*- or *p*-orbitals more effectively than two s-orbitals can overlap, *s-p* σ-bonds are, therefore stronger than *s-s* bonds and *p-p* σ-bonds are stronger still. It can be shown that the relative bond strengths are of the order.

s – s	s – p	p – p
1	1.732	3

The σ s - s, σ s - p, σ p - p and π p - p overlaps will be as shown in Fig 3.22.

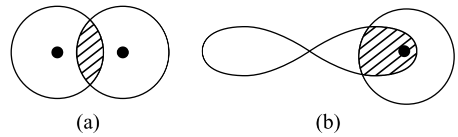


Fig 3.21 The difference in overlap between (a) Two s-orbitals and (b) one s- and one p-orbital

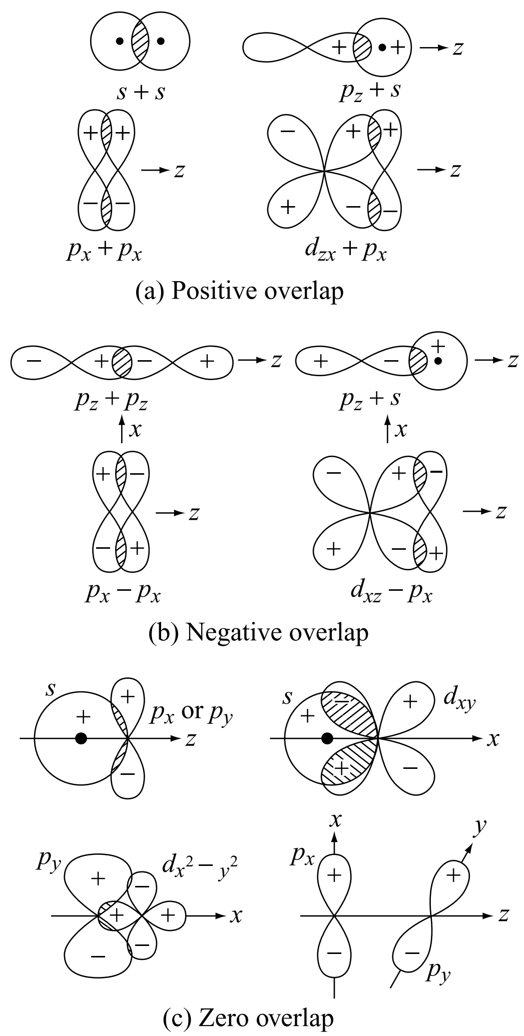


Fig 3.22 Depending on the atomic orbital characters, the overlap may be of one of the three kinds namely (a) Positive, (b) Negative and (c) Zero overlap

3.6.1 Directional Nature of Covalent Bonds

Because of the spherical symmetry of s-orbital it is not concentrated in any particular direction. The three *p*-orbitals in a shell are, however, concentrated along *x*-, *y*- and z-axes, mutually at right angles to each other. *d* and *f*-orbitals also have directional nature. It is the directional nature of *p*-, *d*- and *f*-orbitals, which accounts for the directional nature of the covalent bond. Let us now examine some examples.

Formation of hydrogen molecule (s-s overlap): Each hydrogen atom has one electron in 1s-orbital which is spherical. As the two hydrogen atoms that are combining approach each other, their s-orbitals overlap to form a bonding orbital which is a σ-bond.

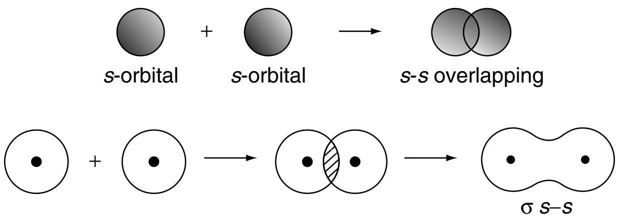


Fig 3.23 Formation of H_2 molecule

Formation of fluorine molecule (p-p overlap) The outer electronic configuration of fluorine atom is $2s^2$ $2p_x^2$ $2p_y^2$ $2p_z^1$. As the two fluorine atoms that are combining approach each other, their p-orbitals overlap along the

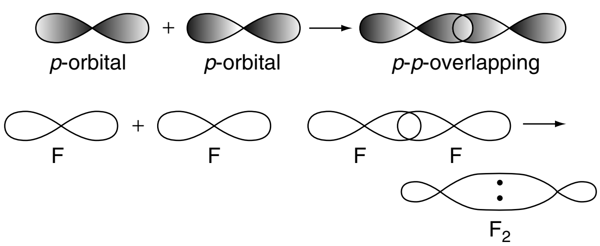


Fig 3.24 Formation of F_2 molecule

z-axis, forming a bonding orbital which is a σ-bond. All the remaining atomic orbitals which are not involved in bonding remain unaltered.

(a) Formation of HF, H_2O, NH_3 molecules (*s-p* overlap) (a) **Formation of HF molecule** The1s orbital of hydrogen overlaps with the p-orbital of fluorine having unpaired electron forming a sigma bond due to *s-p* overlap.

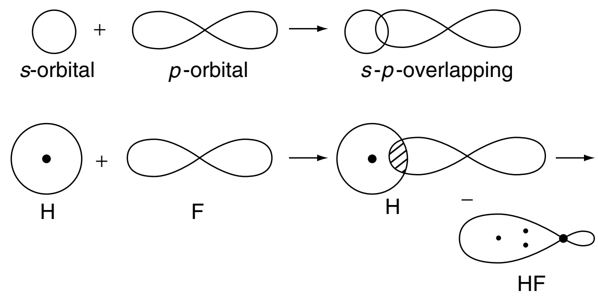


Fig 3.25 Formation of HF molecule

(b) Formation of water molecule: The two O-H bonds in a water molecule are essentially, *s-p* bonds like the bond in the hydrogen halides (as in HF given above) and the formation of the molecule from one oxygen and two hydrogen atoms can be represented as in Fig 3.26.

If one of the O-H bond is along the *x*-axis and the other along the *y*-axis, the $2p_x$ atomic orbital of the oxygen atom will be able to combine with a 1s-atomic orbital of one of hydrogen atom to form a σ bond, whilst the $2p_y$ orbital of the oxygen atom will combine, similarly, with a 1s-orbital of the other hydrogen atom.

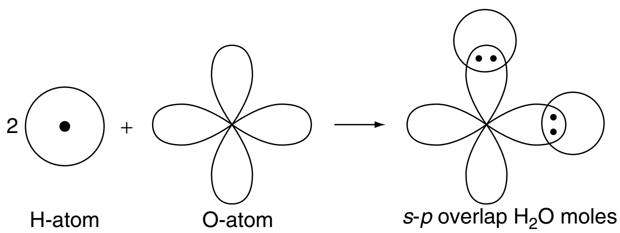


Fig 3.26 Formation of H_2O molecule

The bond angle in such a molecule would be expected to be 90° but the actual bond angles found in water and other similar molecules are

H_2O	H_2S	H_2Se	H_2Te
104°28′	93.3°	91°	90°

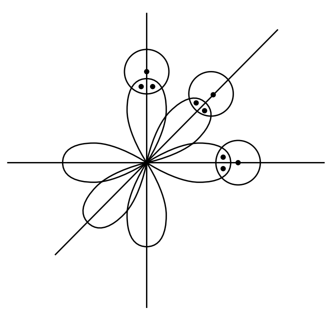


Fig 3.27 Formation of ammonia molecule

(c) Formation of ammonia molecule: A nitrogen atom with an outer electronic configuration $2s^2\ 2p^3$ has three *p*-electrons available for bond formation. As the three *p*-orbitals are mutually at right angles, they should form s-p bonds by overlap with 1s-atomic orbitals of three hydrogen atoms so that the resulting molecule of ammonia, NH_3, will be pyramidal with N-H bonds at 90° to each other.

The actual bond angles in ammonia and similar molecules are as follows.

NH_3	PH_3	AsH_3	SbH_3
107.3	93.3	91.8	91.3

Formation of oxygen molecule: The outer electronic configuration of oxygen atom is $2s^2 2p_x^2\ 2p_y^1\ 2p_z^1$. As the two oxygen atoms approach along the z-coordinate axis, then their p_z orbitals overlap along the internuclear axis to form a σ-bond. The p_y orbitals of these atoms are in a perpendicular direction to that of the internuclear axis. Therefore, they overlap laterally (i.e., sidewise overlap) forming a second bond. Lateral overlap of *p*-orbitals give a weak bond known as a π-bond.

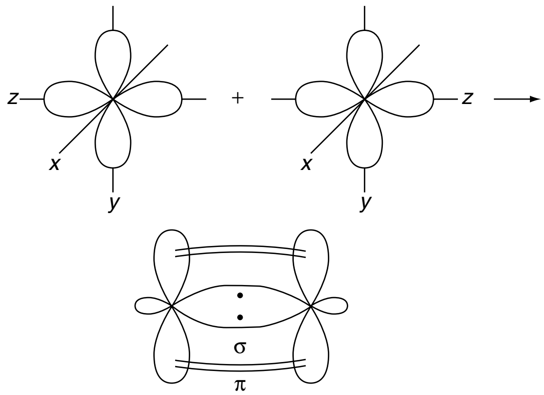


Fig 3.28 Formation of oxygen molecule

Formation of nitrogen molecule: In a nitrogen atom, three unpaired electrons are present one in each of the x, y and z orbitals. Giving the same reasons that were given in the formation of an oxygen molecule, we can show that nitrogen molecule has one σ bond and two π-bonds.

Strength of sigma and pi Bonds: The strength of a bond depends upon the extent of overlapping. In case of a sigma bond, the overlapping of orbitals takes place to a larger extent. Hence it is stronger as compared to the π-bond where the extent of overlapping occurs to a smaller extent. The order of the strength of bonds formed by different orbital overlaps will be in the order.

$\sigma\ p\text{-}p > \sigma\ s\text{-}p > \sigma\ s\text{-}s > \pi\ p\text{-}p$

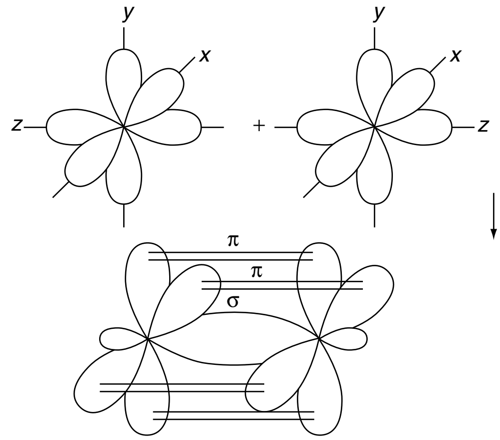


Fig 3.29 Formation of nitrogen molecule

3.6.2 Hybridization

The valence bond theory explains satisfactorily about the formation of various molecules but it failed to account the geometry and shapes of various molecules and bond angles. It could not explain why $BeCl_2$ is linear, BF_3 is planar, CH_4 is tetrahedral, NH_3 is pyramidal and water is angular. To explain these characteristic geometrical shapes of polyatomic molecules, Pauling introduced the concept of hybridization. According to this concept, the intermixing of atomic orbitals of nearly equivalent energies of an atom to form new set of equivalent orbitals is known as **hybridization** and the new orbitals formed are called **hybrid orbitals.** The main **postulates** of hybridization are as follows.

(i) The intermixing of atomic orbitals of similar or nearly similar energies and their redistribution into an equal number of identical orbitals is known as hybridization.

(ii) The atomic orbitals of only one atom can participate in hybridization but the atomic orbitals belonging to different atoms cannot participate in hybridization.

(iii) Number of the hybrid orbitals formed is always equal to number of atomic orbitals participated in the hybridization.

(iv) The hybrid orbitals are symmetrically arranged around the nucleus so that the repulsion between them will be minimum. The angle between any two hybrid orbitals in an atom will be equal except in sp^3d and sp^3d^3 hybrid orbitals.

(v) The distribution of electrons in different hybrid orbitals follow Hund's rules and Pauli's exclusion principle. Only atomic orbitals undergo hybridization but not the electrons present in them.

(vi) Hybrid orbitals always form σ-bonds. Each of the hybrid orbitals has a large lobe concentrated in a particular direction and is therefore, able to overlap very strongly with an orbital on another atom located at an appropriate distance and direction and hence the bond formed by a hybrid orbital is stronger than the bond formed by a pure orbital.

(vii) The atomic orbitals belonging to the valence shells of the atom only partcipate in hybridization. It is not necessary that only half-filled orbitals should participate in hybridization. In some cases even filled orbitals of valence shell take part in hybridization.

Promotion of Electrons: The formation of bonds by the overlap of two atomic orbitals, require that each of the two orbitals must contain one unpaired electron and that the two electrons must have different spins. This is because the molecular orbital formed by the overlap can only hold two electrons and they must have opposite spins by Pauli's principle.

In this way, then the number of bonds which an atom might be able to form depends, to some extent on the number of unpaired electrons in the atom. A study of the electronic arrangements in the atoms of atomic number 1 to 10 shows that the number of unpaired electrons is, in fact, equal to the commonest numerical covalency of the element concerned in all cases except beryllium, boron and carbon.

The significant data are repeated below.

In terms of unpaired electrons, beryllium would be expected to behave as an inert gas, boron might be expected to be monovalent like fluorine, and carbon would be divalent like oxygen.

	1s	*2s*	*$2p_x$*	*$2p_y$*	*$2p_z$*
Be	↿⇂	↿⇂			
B	↿⇂	↿⇂	↿		
C	↿⇂	↿⇂	↿	↿	

To meet the known chemical facts, it must be assumed that some of the paired electrons are uncoupled or unpaired before the atom participating in a chemical bond. This will require an input of energy but such energy will be available from the heat of reaction when covalent bonds are formed.

The unpairing electrons require the removal of 2s electron into a 2p-orbital of higher energy level. This process is referred to as promotion and the arrangement of electrons, after promotion is sometimes referred to as an excited valence state.

Element	**H**	**He**	**Li**	**Be**	**B**	**C**	**N**	**O**	**F**	**Ne**
No. of unpaired electrons	1	0	1	0	1	2	3	2	1	0
Numerical valency	1	0	1	2	3	4	3	2	1	0

The simplest excited valency states of beryllium, boron and carbon are as follows.

	1s	*2s*	*$2p_x$*	*$2p_y$*	*$2p_z$*
Be*	↿⇂	↿	↿		
B*	↿⇂	↿	↿	↿	
C*	↿⇂	↿	↿	↿	↿

Such excited valence states, generally denoted by an asterisk, provide enough unpaired electrons for beryllium to form two, boron three and carbon four, bonds, but they are not the bonds actually found in practice.

3.6.3 Types of Hybridization

(i) ***sp*-Hybridization:** The combination of *s*- and a *p*-orbital leads to two hybrid orbitals known as sp-hybrid orbitals. The suitable orbitals for *sp* hybridization are s and p_z, if the hybrid orbitals are to lie along the *z*-axis. Each *sp* hybrid orbital has 50 per cent s-character and 50 per cent *p*-character. The two *sp* hybrid orbitals point in the opposite direction along the *z*-axis with projecting large positive lobes and small negative lobes.

A hybrid *sp*-orbital is, therefore, able to form a stronger bond by overlap of its large positive lobe with another hybrid orbital of another atom, than *s*- or *p*-orbitals alone. The molecule will have linear geometry. This type of hybridization is also known as **diagonal hybridization**.

(a) Beryllium chloride: The ground state electronic configuration of Be is $1s^2\,2s^2$. In the excited state one of

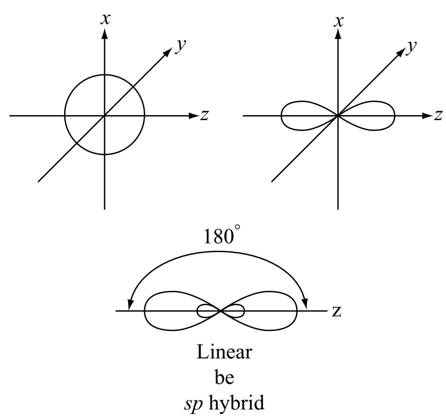


Fig 3.30 Two collinear *sp* hybrid orbitals

the 2*s*-electrons is promoted to vacant 2*p* orbital. These one 2*s*-and one 2*p*-orbitals participate in hybridization to form *sp* hybridized orbitals. These two *sp* hybrid orbitals are oriented in opposite direction at an angle of 180°. Each of these *sp* hybrid orbital overlaps with a 2*p*-orbital of chlorine and form two Be-Cl sigma bonds as shown in Fig. 3.31.

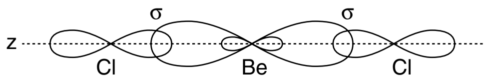


Fig 3.31 Formation of linear $BeCl_2$ molecule

(b) Carbon dioxide: The ground state electronic configuration of carbon is $1s^2\ 2s^2, 2p^2$.

In the excited state, one of the 2*s*-electron is promoted to vacant 2*p* orbital. Then, one 2*s* and one 2*p*-orbital participate in hybridization. The outer electronic configuration of oxygen is $2s^2\ 2p_x^2\ \ 2p_y^1\ \ 2p_z^1$.

	Outer electronic configuration 2s	2p		
Ground state of carbon	⇅	↑	↑	
Excited state of carbon	↑	↑	↑	↑
	sp - hybridization			
Ground state of oxygen	⇅	⇅	↑	↑

The $2p_z$ orbital of oxygen atom having unpaired electron overlap with the hybrid orbital to form a sigma bond. Similarly, second oxygen atom form another sigma bond with second hybrid orbital of carbon. In the carbon atom still there are two pure *p*-orbitals having unpaired electron that have not participated in hybridization. These orbitals overlap laterally with the *p*-orbitals of two oxygen atoms having unpaired electron on either side of carbon, forming two pi bonds.

The two pi bonds are perpendicular to each other. The CO_2 molecule is linear with a bond angle of 180°.

(c) Formation of ethyne (C_2H_2) molecule: In the formation of ethyne molecule, hybridization of one 2*s*- one 2*p*- carbon orbitals leads to each of the two carbon atoms having colinear *sp*-hybrid orbitals. These overlap to form a σ-bond between the two carbon atoms and each of the two *sp*-orbitals also overlaps with a 1*s*-atomic orbital of a hydrogen atom to form σ-bonds between the carbon and hydrogen atoms. It is because

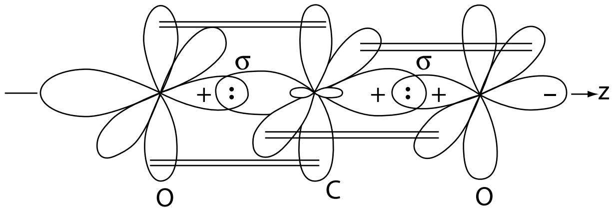


Fig 3.32 Formation CO_2 molecule

sp-hybrid orbitals are directed along the molecular axis and hence the ethyne molecule is colinear.

Each of the carbon atoms has two remaining *2p*-atomic orbitals which interact to form two π-bonds between the carbon atoms, these two bonds being in planes at right angles to each other. The C≡C bond consists, therefore, a σ-bond and two π-bonds.

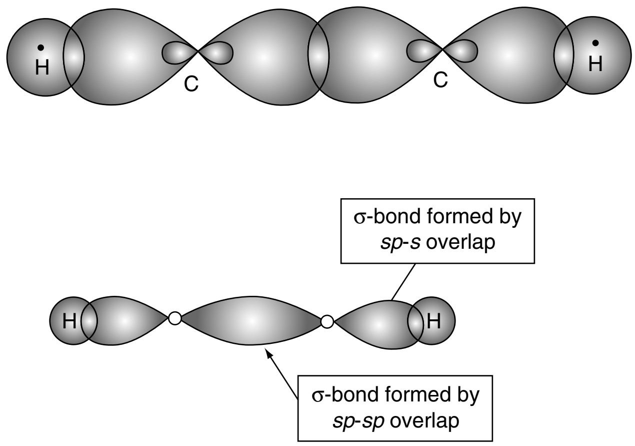
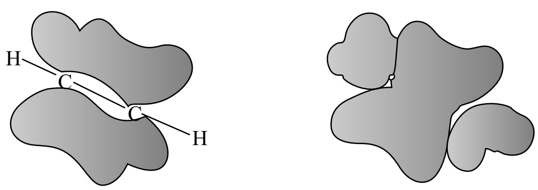


Fig 3.33 Bonds of ethyne formed by overlap of two sp hybrid orbitals



(a) The two carbon-carbon π-bonds formed by lateral overlap of *p*-orbitals.

(b) The triple bond has an electron density above and below and in front and in back of the inter nuclear axis of the molecule.

Fig 3.34 Formation pi - bonds in ethyne

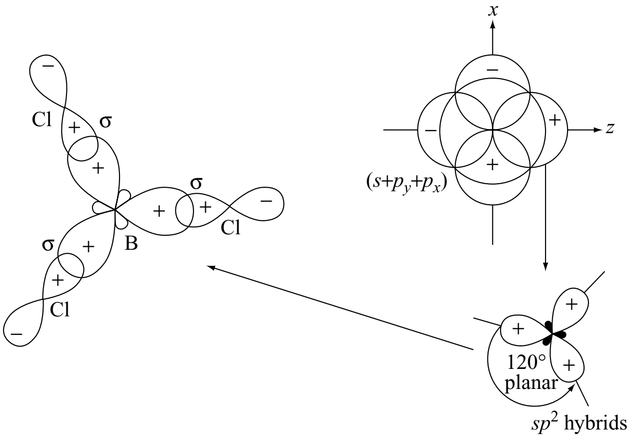


Fig 3.35 Formation of BCl_3 molecule

(ii) sp^2-Hybridization: Combination of one *s*- and two *p*-orbitals gives rise to three sp^2-hybrid orbitals which are coplanar and directed at angles of 120° to each other. These hybrid orbitals have 33 per cent *s*-character and 67 per cent *p*-character.

(a) Formation of BCl_3 molecule: The ground state electronic configuration of central boron atom is $1s^2\ 2s^2\ 2p^1$. In the excited state, one of the 2*s*-electrons is promoted to vacant 2*p* orbital and as a result boron has three unpaired electrons. These three orbitals (one 2*s* and two 2*p*) hybridize to form three sp^2 hybrid orbitals. The three hybrid orbitals, so formed are oriented in a trigonal planar arrangement and overlap with 2*p*-orbitals of chlorine to form three B-Cl bonds. Therefore, in BCl_3 (Fig. 3.35) the geometry is trigonal planar with ClBCl bond angle of 120°.

(b) Formation of ethene (C_2H_4) molecule: In ethene, each of the two carbon atoms form bonds through sp^2-orbitals. Two of the orbitals of each atom form σ-bonds with the 1s-orbitals of hydrogen atoms. The remaining sp^2 orbital of each carbon atom forms a σ-bond between the carbon atoms. The two carbon and four hydrogen atoms are all in the same plane as the bond angles are 120°. At right angles to this plane, there remains the unchanged *2p*-orbital of each carbon atom and these two 2p-orbitals interact to form a π-bond between the two carbon atoms. The double bond between the carbon atoms consists, therefore, of a σ-bond and a π-bond.

(a) Formation of carbon-carbon σ-bond in ethene by sp^2-sp^2 overlap and the carbon-hydrogen bonds by sp^2-*s* overlap.

(b) Formation of carbon-carbon π-bond by the lateral overlap of *p*-orbitals.

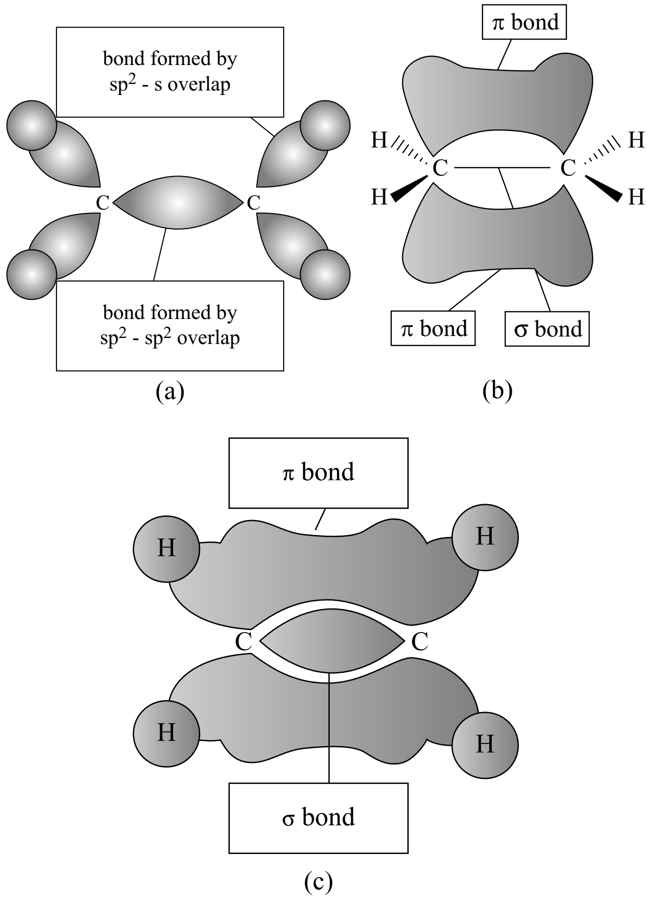
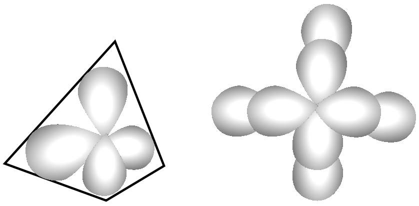


Fig 3.36 Formation of ethene molecule

(c) The electron density of the π-bond is above and below the plane containing the two carbons and four hydrogens.

(iii) **sp^3-Hybridization:** The orbitals formed by the hybridization of one *s*-and three *p*-orbitals are known as sp^3-orbitals, they are directed towards the corners of the tetrahedron. The bond angles in the resulting tetrahedral molecules are always close to the expected theoretical angle 109°28′ known as the tetrahdral angle. Each sp^3 hybrid orbital have 25 per cent s-character and 75 per cent p-character.

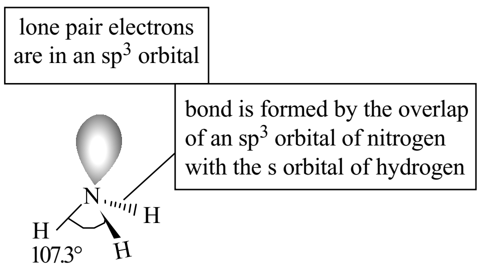
(a) Formation of methane molecule: The ground state electronic configuration of central carbon atom is $1s^2\ 2s^2\ 2p_x^1\ \ 2p_y^1\ \ 2p_z^0$. In the excited state, one of the 2*s* electrons is promoted to vacant 2*p*-orbital, and as a result carbon has four unpaired electrons. These four orbitals (one 2*s* and three 2*p*) hybridize to form four sp^3 hybrid orbitals. These four hybrid orbitals are directed towards the corners of a tetrahedron and overlap with the 1s orbital of four hydrogen atoms to form four C-H bonds. Therefore CH_4 (Fig. 3.37) geometry is tetrahedral with HCH bond angle 109°28′.



(a) (b)

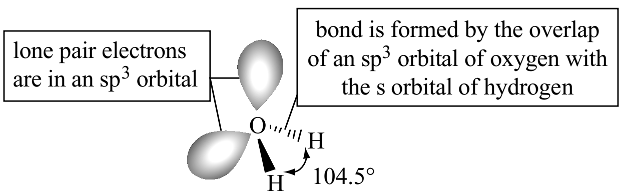
Fig 3.37 Formation of CH_4 molecule

(b) Formation of NH_3 molecule: The structure of NH_3 molecule can be explained by sp^3 hybridization as explained in the case of methane. The valence shell (outer) electronic configuration of nitrogen in the ground state is $2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1$. The four atomic orbitals (one 2*s* and three 2*p*) orbitals participate in hybridization forming four *sp*^{3} hybrid orbitals. One of these four orbitals contain a lone pair and the other three hybrid orbitals contain one electron each. These three hybrid orbitals overlap the *s*-orbitals of three hydrogen atoms and form three σ sp^3-s bonds. We know that the force of repulsion between a lone pair and a bond pair is more than the force or repulsion between two bond pairs of electrons. The molecule thus gets distorted and the bond angle is reduced to 107° from 109°5’. The geometry of such molecule will be pyramidal as shown in Fig. 3.38a.



NH_3

(a) ammonia



(b) H_2O Water

Fig 3.38 Formation of ammonia molecule and water molecule

(c) Formation of H_2O molecule The central atom in water molecules is oxygen. The outer electronic configuration of oxygen atom is $2s^2\ 2p_x^2\ 2p_y^1\ 2p_z^1$. These four oxygen orbitals (one *2s* and three *2p*) undergo sp^3 hybridization forming four sp^3 hybrid orbitals out of which two contain one electron each and the other two contain a pair of electrons. These four sp^3 hybrid orbitals acquire a tetrahedral geometry, with two corners occupied by hydrogen atoms while the other two by the lone pairs. The bond angle in this case is reduced to 104.5 and the molecule thus acquires a *V*-shape or angular geometry (Fig. 3.38b).

(d) Formation of ethane (C_2H_6) molecule: In ethane, both carbon atoms are tetravalent and are in an excited state.

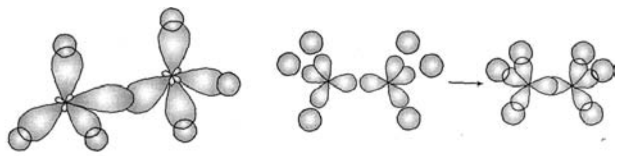


Fig 3.39 Formation of ethane molecule

Outer electronic configuration of C

	2s	*2p*		
Ground state	⥮	↿	↿	
Excited state	↿	↿	↿	↿

Now one *2s*-orbital and three *2p* orbitals participate in hybridization forming four hybrid orbitals arranged tetrahedrally, and each of these hybrid orbitals contain unpaired electron. One of the four sp^3 hybrid orbitals of carbon atom overlaps axially with similar orbital of other atom to form sp^3 - sp^3 sigma bond with the other three hybrid orbitals of each carbon atom. The structure may be as shown in Fig. 3.35.

3.6.4 Hybridization Involving *d*-orbitals

The elements in the third period contain *d*-orbitals in addition to *s*- and *p*-orbitals. Under certain conditions, these orbitals also participate in hybridization. The 3*d*-orbitals can participate in hybridization with *3s*- and *3p*-orbitals or with *4s*- and *4p*-orbitals, because the difference in energies of *3p, 3d* and *4s* orbitals is little. If the 3d-orbitals involve in hybridization with *3s*-*3p* orbitals, it is known as sp^3d hybridization, but if *3d*-orbitals involve in hybridization with *4s*- and *4p*-orbitals, it is known as dsp^3 hybridization.

(i) sp^3d-Hybridization: Formation of PCl_5 molecule: The ground state and excited state outer electronic configurations of phosphorus (z = 15) are as follows.

Outer electronic configuration of P

	3s	3p	3d
Ground state	↑↓	↑ ↑ ↑	_ _ _ _ _
Excited state	↑	↑ ↑ ↑	↑ _ _ _ _

sp^3d Hybridization

Now the five orbitals (one *3s*, three *3p* and one *3d*-orbitals) participate in hybridization to yield a set of five sp^3d hybrid orbitals which are directed to the corners of the trigonal bipyramid as shown in Fig. 3.40.

It should be noted that the bond angles in trigonal bipyramidal geometry are not equivalent. Three sp^3d hybrid orbitals form coplanar bonds with angles 120° and the remaining two sp^3d hybrid orbitals form two bonds at right angles to this plane one above and one below. The three coplanar bonds at 120° angle are called **equatorial bonds** while the other two bonds above and below the equatorial plane making an angle 90° are called **axial bonds.** The axial bond pairs suffer more repulsive interaction from the equatorial bond pairs (due to 90° angle) the axial bonds are slightly longer and weaker than the equatorial bonds. Therefore, PCl_5 molecule is more reactive.

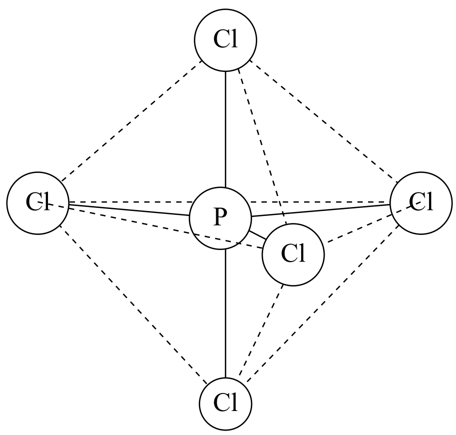


Fig 3.40 Trigonal bipyramidal geometry of PCl_5 molecule

(ii) sp^3d^2 hybridization Formation of SF_6 molecule: a combination of one *s*, three *p* and two *d* atomic orbitals leads to six hybrid orbitals which are directed to the corners of an octahedron. Sulphur hexafluoride provides an example. The central sulphur atom in SF_6 molecule has the ground state and excited state outer electronic configurations as follows.

Outer electronic configuration of S

	3s	*3p*	*3d*
Ground state	↑↓	↑↓ ↑ ↑	_ _ _ _ _
Excited state	↑	↑ ↑ ↑	↑ ↑ _ _ _

sp^3d^2 Hybridization

The six sp^3d^2 hybrid orbitals of sulphur overlap with the orbitals of fluorine containing unpaired electron forming six S-F bonds. Thus, SF_6 molecule has a regular octahedral geometry.

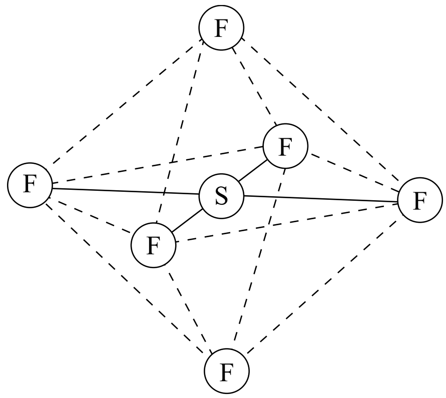


Fig 3.41 Octahedral geometry of SF_6 molecule

(iii) sp^3d^3 Hybridization: Formation of IF_7 molecule: A combination of one *s*, three *p* and three d orbitals leads to seven hybrid orbitals. Five sp^3d^3 hybrid orbitals are directed to the corners of a pentagon with an angle 72° and the remaining two sp^3d^3 hybrid orbitals are directed at right angle to the plane of pentagon one above and one below. These seven sp^3d^3 hybrid orbitals will overlap with seven orbitals of seven fluorine atoms having unpaired electrons forming seven sigma I-F bonds. Thus, IF_7 has a pentagonal bipyramidal structure as shown in Fig 3.42.

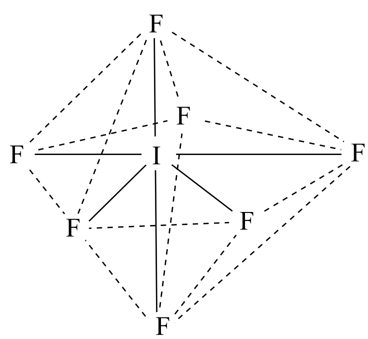


Fig 3.42 Pentagonal bipyramidal geometry of IF_7 molecule

3.6.5 Role of *d*-orbitals in Bonding

Spectroscopic data for free atoms such as Si, P, S and Cl as well as their heavier congeners, show that the valence shell *d*-orbitals are far higher in energy and much more diffuse than the *s*- and *p*-orbitals, and thus imply that they cannot contribute much to the bonding.

However, it can be shown that the size and energy of the *d*-orbitals are sensitive functions of both the oxidation state and the electron configuration of the atom. Some detailed conclusions, which will now be summarized are based on calculations for the sulphur atom. In a similar way, the formation of other compounds like PCl_5, IF_7, etc, can be explained.

(i) When two electrons promoted to give an sp^3d^2 configuration, the 3*d* orbitals contract remarkably. The average radius of the *d*-orbitals in the excited states of s^2p^4, sp^4d and s^2p^3d is 300–350 pm, but for an sp^3d^2 hybrid valence state the radius of d-orbital is decreased to –160 pm. This value is comparable with the radii of ~90 pm for the *3s* and *3p* orbitals. It can also be understood that as the number of electrons entering into *d*-orbitals increases, the size of *d*-orbitals decreases.

(ii) The most important factor is the charge on the atom. If the atom carries a formal positive charge then all the electrons are pulled towards the nucleus. This effect is greater for the outer electrons. If the central 'S' atom is bonded to highly electronegative element such as F, O or Cl, then the electronegative element attracts more than its share of the bonding electrons and the F, O or Cl atoms gets a δ negative charge while the *S* atom gets a δ+charge. This positive charge on S atom makes the orbitals contract. Since the *3d* orbital contracts in size very much more than the *3s* and *3p* orbitals so that the energies of the *3s*, *3p* and *3d* orbitals may become closer enough to allow hybridization to occur in SF_6.

(iii) However, the promotion of energy to states derived from the sp^3d^2 configuration is very large namely in the range of 30–35 eV. Such energies can only be compensated, if at all, by very strong bonds.

In SF_6 and similar compounds, the use of outer *d*-orbitals is plausible on the above considerations.

3.6.6 Bent's Rule

Unlike for the compounds with coordination numbers, 2, 3, 4 and 6 there is no unique, highly symmetrical set of equivalent orbitals that can be constructed for five coordination of the two sp^3d hybridizations with trigonal bipyramidal geometry and square pyramidal geometry as most compounds of non-metals favour the trigonal bipyramidal structure. Many coordination compounds are known with square pyramidal structures. There are many compounds that cannot be classified really into trigonal bipyramidal or square pyramidal geometries because in these compounds there is a continuous interchange of these geometries.

An sp^3d hybrid orbital set may be considered to be a combination of $p_zd_z^2$ hybrids and sp_xp_y hybrids. The former make two linear hybrid orbitals bonding axially and the latter form the trigonal equatorial bonds. The sp^2 hybrid orbitals are capable of forming stronger bonds and they are shorter than the weaker axial bonds. When the electronegativities of the substituents on the phosphorus atom differ as in the mixed chlorofluorides PCl_xF_{5-x} and the alkyl phosphorous fluorides R_xPF_{5-x} the arrangement of atoms or groups around phosphorus atom follows Bent's rule. Bent's rule states that more electronegative substituents "prefer" hybrid orbitals having less s-character, and more electropositive substituents "prefer" hybrid orbitals having more s-character. Thus the fluorine atoms occupy the axial positions and Cl atoms or alkyl groups occupy the equatorial position of the trigonal bipyramidal geometry of the phosphorus atom.

A second example of Bent's rule is provided by the fluoromethanes. In CH_2F_2, the F-C-F bond angle is less than 109.5° indicating less than 25 per cent s-character but the H-C-H bond angle is larger and C-H bond has more *s*-character. The bond angles in the other fluoromethanes yield similar results.

3.6.7 Calculation of Per cent Character of Hybrid Orbitals

Only the elements to the left in the periodic table have unambiguous hybridization assignable by structure. Thus few would argue with an assumption of sp^2 for boron in its tricovalent compounds, and organic chemistry is based on the successful assumption of diagonal (*sp*), trigonal (sp^2), and tetrahedral (sp^3) hybridizations for carbon. However, the hybridizations of nitrogen, oxygen, phosphorus and sulphur do not fit well into such simple schemes. This is because the hybrids are often some non-integral mix of *s* and *p* characters

The relation between hybridization and bond angles is simple for *s-p* hybrids. For two or more equivalent orbitals, the percent *s*-character or percent *p*-character is given by the relationship

$$\cos\theta = \frac{S}{S-1} = \frac{P-1}{P} \tag{3.17}$$

where θ is the angle between the equivalent orbitals (°) and the *s* and *p* characters are expressed as decimal fractions. In methane for example

$$\cos\theta = \frac{0.25}{-0.75} = -0.333 \quad \theta = 109.5° \tag{3.18}$$

In hybridizations involving non-equivalent hybrid orbitals, such as sp^3d, it is usually possible to resolve the set of hybrid orbitals into subsets of orbitals that are equivalent within the subset, as the sp^2 subset and the dp subset. The non-equivalent hybrid orbitals (having lone pairs and bond pairs) may have fractional bonding orbitals midway between pure p and sp^3 hybrids. For molecules such as water, we can divide the four orbitals into the bonding subset (the bond angle 104.5°) and the non-bonding subset (angle unknown). Now, we can calculate s and p character of hybrid orbitals.

$$\cos\theta = -0.25 = \frac{P-1}{P} \qquad (3.19)$$

$P = 0.80 = 80\%$ *p* character and 20 per cent s character.

Now, of course, the total p character summed over all four orbitals on oxygen must be 3.00 (= $p_x + p_y + p_z$) and the total s character must be 1.00. If the bonding orbitals contain proportionately more p-character, than the non-bonding orbitals (the two lone pairs) must contain proportionately less p character, 70 per cent, [0.8 + 0.8 + 0.7 + 0.7 = 3.0 (p); 0.2 + 0.2 + 0.3 + 0.3 = 1.0 (s)]. The opening of some bond angles and closing of others in nominally "tetrahedral" molecules is a common phenomenon. Usually the distortion is only a few degrees but it should remind us that the terms "trigonal", "tetrahedral", etc., usually are only approximation.

When a set of hybrid orbitals are formed, the energy of the resulting hybrids is a weighted average of the energies of the participating atomic orbitals. For example in phosphorus in ground state, that any hybridization will cost energy as a filled 3s orbital is raised in energy and half-filled 3p orbitals are lowered in energy. This energy of hybridization is of the order of magnitude of bond energies and can thus be important in determining the structure of molecules. The energy required for hybridization in phosphorus is about 600 KJ mol^{-1}. To decrease the hybridization energy the lone pairs occupy the pure s-orbital and only three p orbitals participate in bonding. Opposing this tendency is the repulsion of electrons, both bonding and nonbonding. This favours an approximate tetrahedral arrangement. In the case of small atoms like N and O, the steric effects are more prominent, but in larger atoms such as those of P, As, Sb, S, Se and Te these effects are somewhat related, allowing the reduced hybridization energy of more p character in the bonding orbitals to come into play.

Bond angles in the hydrides of groups VA and VIA

NH_3	107.2°	PH_3	93.8	AsH_3	91.8°	SbH_3	91.3°
H_2O	104.5°	H_2S	92°	H_2Se	91°	H_2Te	89.5°

3.6.8 Orbitals Participating in Different Types of Hybridization

If the molecular axis is selected as *z*-axis, the orbitals participating in different types of hybridization are shown in Table 3.11.

Table 3.11 Orbitals partcipating in different types of hybridization

Types of hybridization	Atomic orbitals
sp	$s + p_z$
sp^2	$s + p_z + p_x$ or p_y
sp^3	$s + p_x + p_y + p_z$
dsp^2	$d_{x^2-y^2} + s + p_x + p_y$
dsp^3 [Trigonal bipyramid]	$d_{z^2} + s + p_x + p_y + p_z$
dsp^3 [Square pyramid]	$d_{x^2-y^2} + s + p_x + p_y + p_z$
d^2sp^3	$d_{x^2-y^2} + d_z{}^2 + s + p_x + p_y + p_z$
sp^3d^3	$s + p_x + p_y + p_z + d_{xy} + d_{yz} + d_{xz}$

Generally, the orbitals directed towards the bond formation in a particular geometry will participate in hybridization.

3.7 MOLECULAR ORBITAL THEORY

Though valence bond theory could explain to some extent, yet it failed to stand the test of reasoning. According to valence bond theory, all the other orbitals except the bonding orbitals of an atom remain undisturbed. However, this seems to be an exaggeration because the nucleus of an approaching atom is found to affect the electron waves of nearly all orbitals of the other atom. Apart from this glaring contradiction, the valence bond theory also failed to explain a few anomalies in behaviour of the molecules like oxygen.

To explain these anomalies, molecular orbital theory was developed by Hund and Mulliken and later by Lennard Jones and Coulson which is modern and rational. According to molecular orbital (MO) theory, all the atomic orbitals of the atoms participating in molecule formation get mixed up to an equivalent number of new orbitals that belongs to the molecule, and now these are called **molecular orbitals.**

3.7.1 Linear Combination of Atomic Orbitals (LCAO) Method

The preferred combination way of working out the wave functions for molecular orbitals is to adopt a method known as the **linear combination of atomic orbitals** (LCAO) approximation. Like many other methods of wave mechanics, it is only an approximate method simply because of greater accuracy.

The linear combination of two atomic orbital wave functions can be brought about either by adding or by subtracting the two wave functions. This can be expressed in a simplified form by the equation

$$\psi = \psi_A \pm \lambda\psi_B$$

where ψ is the molecular orbital wave function obtained by the combination of atomic wave functions ψ_A and ψ_B. λ gives a measure of the ionic character of the bond between *A* and *B* and its value dependent on the nature of *A* and *B*. For diatomic molecule of an element A_2, ψ_A and ψ_B are equal and the value of λ is 1. In other cases, the value of λ cannot unfortunately be calculated from first principles, but it is possible to give quantitative value to the electronegativities of various atoms either by Pauling's scale or by Mulliken's scale.

If the lobes of the combining atomic orbitals with the same sign overlap the bonding (MO) ψ_b is formed.

$$\psi_b = \psi_A + \psi_B \tag{3.20}$$

When the lobes of the combining orbitals of opposite sign overlap the new MO is obtained. It is denoted by ψ_a the anti-bonding MO.

$$\psi_a = \psi_A - \psi_B \tag{3.21}$$

The electron distribution in an MO can be obtained by squaring the wave functions ψ_b and ψ_a

$$\psi_b^2 = \psi_A^2 + \psi_B^2 + 2\psi_A\,\psi_B \tag{3.22}$$

$$\psi_a^2 = \psi_A^2 + \psi_B - 2\psi_A\,\psi_B \tag{3.23}$$

These represent the probability functions for bonding and anti-bonding MOs. It may be seen from equation for ψ_b^2 that ψ_b^2 is greater than $\left(\psi_A^2 + \psi_B^2\right)$ by a term $2\psi_A\psi_B$. This indicates the energy of MO represented by ψ_b is lesser than the energy of two atomic orbitals ψ_A and ψ_B which leads to the formation of a stable bond termed as bonding MO. But in the equation for ψ_a^2, it may be seen that ψ_a^2 is less than $\left(\psi_A^2 + \psi_B^2\right)$ by a term $2\psi_A\psi_B$. This means that the MO wave function ψ_a has less charge density between the interacting atoms than the isolated atoms with wave functions ψ_A (or) ψ_B. Thus, the energy of the MO represented by ψ_a is greater than the energy of either of the atomic orbitals ψ_A and ψ_B. Therefore, such orbitals cannot form a stable chemical bond and hence termed as anti-bonding MO.

Qualitatively, the formation of molecular orbitals can be understood in terms of the constructive or destructive interferences of the electron waves of the combining atoms. In the formation of bonding MO, the electron waves of the bonding atoms reinforce each other due to constructive interference while in the formation of anti-bonding MO the electron waves cancel each other due to destructive interference. As a result, the electron density in a bonding MO is located between the nuclei of the bonded atoms because of which the repulsion between the nuclei is very less while in case of an anti-bonding MO most of the electron density is located away from the space between the nuclei. In fact, there is a nodal plane (the plane on which the electron density is zero) between the nuclei and hence the repulsion between the nuclei is high. In the bonding

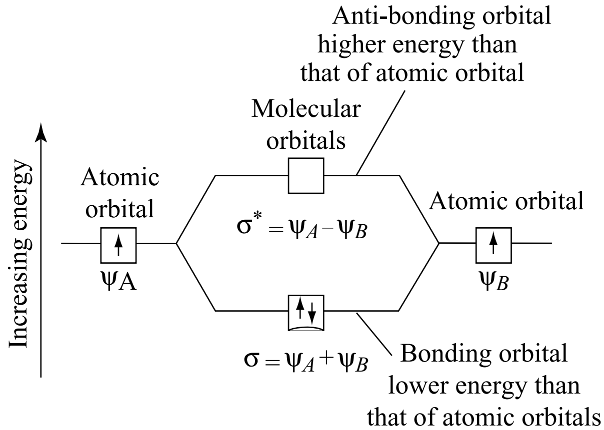


Fig 3.43 Formation of bonding (σ) and anti-bonding (σ*) MOs by the LCAO method

MO the electrons are placed between the nuclei hold the nuclei together and stabilize the molecule. Therefore, the bonding MO always possesses lower energy than either of the atomic orbitals while the energy of the anti-bonding MO always possesses higher energy. This is because of the attraction between nuclei and electrons in bonding MO, and repulsions of the nuclei in anti-bonding MO. The total energy of two MOs however remains the same as that of two original atomic orbitals.

An anti-bonding orbital is often more strongly anti-bonding than the corresponding bonding orbital is bonding after although the "binding" effect of a bonding electron and the "anti-bonding" effect of an anti-bonding electron are similar, the nuclei repel each other in both cases and this repulsion pushes both levels up in energy.

If two similar *s*-atomic orbitals are combined, the MOs formed will be as shown in Fig 3.44.

When *p*-orbitals on two atoms approach along the same axis (p_z-orbitals), they give rise to σ p-bonding and σ* *p*-anti-bonding orbital (Fig 3.45).

When *p*-orbitals of the two atoms are approaching in a manner that their axes are mutually parallel (e.g., $2p_x$ or

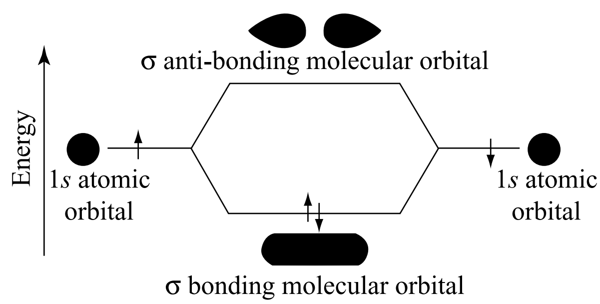


Fig 3.44 Fomation of molecular orbitals by combining of s - orbitals

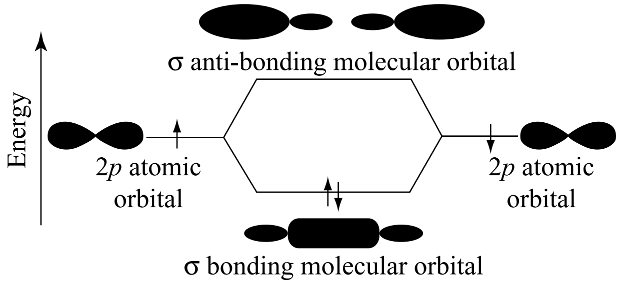


Fig 3.45 Formation of σ molecular orbitals by the approach of p orbitals along the same axis

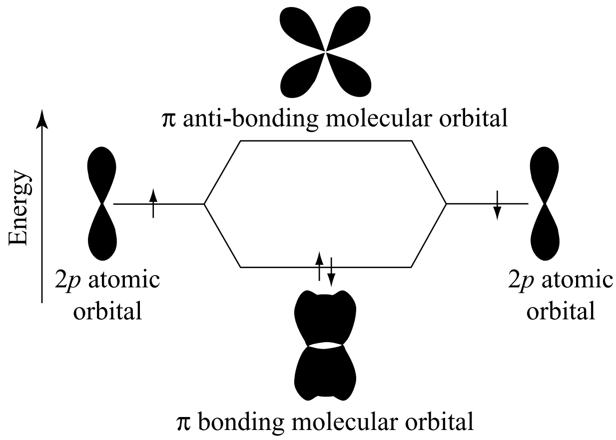


Fig 3.46 Formation of molecular orbitals by the approach of p - orbitals laterally

$2p_y$ orbitals) they interact to give rise to the formation of the MOs that are not symmetrical about nuclear axis and are called π MOs (Fig 3.46).

The σs and σp_z MOs does not have nodal planes. The σ^*s and $\sigma^* p_z$ MOs have one nodal plane perpendicular to molecular axis. The $\pi\, p$ MOs have one nodal plane along the molecular axis while the π^* MOs have two nodal planes perpendicular to each other one along the molecular axis.

3.7.2 Limitations to Combination of Atomic Orbitals

(i) The energies of the atomic orbitals must be similar in magnitude. The application of this principle means that 1*s* and 2*s* or *s*- and *p*-orbitals do not combine to form MOs in homonuclear diatomic molecules, i.e., A_2. The energy differences between 1*s*- and 2*s* or 2*s*- and 2*p*-atomic orbitals are too great. However, in the heteronuclear molecules such as *AB*, this might not be so.

(ii) The atomic orbitals must overlap to a considerable extent if they are going to combine to form an MO. The greater the overlap of the atomic orbitals, the greater the charge between the nuclei. This general idea is developed into the principle of **maximum overlap.**

(iii) The atomic orbitals must have the same symmetry about the molecular axis. This means that some atomic orbitals of comparable energy which do overlap can still not be combined to give MOs. A $2P_y$ atomic orbital, for example, will not combine with an s-atomic orbital. The *p*-orbital has positive and negative lobes, whereas *s*-orbital is everywhere positive.

The atomic orbitals which can be combined together from a symmetry point of view are given in Table 3.12.

Table 3.12 Permitted combination of atomic orbitals

1st orbital	IInd orbital allowed	Forbidden
s	s, p_z, d_{z^2}, $d_{x^2-y^2}$	px, py, d_{xy}, d_{yz}, d_{xz}
p_z	s, p_z, d_{z^2}, $d_{x^2-y^2}$	p_x, p_y, d_{xy}, d_{yz}, d_{xz}
p_x	p_x, d_{xy}	s, p_y, p_z, $d_{x^2-y^2}$, d_{yz}, d_{xz}
p_y	p_y, d_{xy}	s, p_x, p_z, $d_{x^2-y^2}$, d_{z^2}, d_{xz}, d_{yz}
d_{xz}	p_x	s, p_y, p_z, $d_{x^2-y^2}$, d_{z^2}, d_{xy}, d_{yz}
d_{yz}	p_y	s, p_x, p_z, $d_{x^2-y^2}$, d_{z^2}, d_{xy}, d_{yz}
d_{z^2}	s, p_z, $d_{x^2-y^2}$, d_{z^2}	p_x, p_y, d_{xy}, d_{yz}, d_x
$d_{x^2-y^2}$	s, p_z, d_{z^2}, $d_{x^2-y^2}$	p_x, p_y, d_{xy}, d_{yz}, d_{xz}

3.7.3 Energy Level Diagram For Molecular Orbitals

The energy levels of these molecular orbitals have been determined experimentally from spectroscopic data for homonuclear diatomic molecules of second row elements of the periodic table. The increasing order of energies of various MOs are as shown in Fig. 3.47.

The order of energies of different MOs will be as follows.

σ 1s < σ^*1s < σ2s < σ^*2s < $\sigma 2p_z$ < $\pi 2p_x \approx \pi 2p_y$ < $\pi^* 2p_x \approx \pi^* 2p_y < \sigma^* 2p_z$.

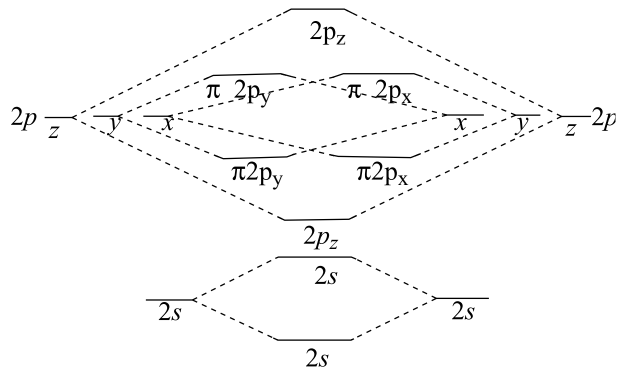


Fig 3.47 Schematic energy level representation of MOs for simple molecules of elements

3.7.4 Mixing of Orbitals

Spectroscopic studies have revealed that sequence of energy levels of MOs described in section 3.7.3 is not valid for all molecules. It has been observed that in the case of homonuclear diatomic molecules of second row elements of the periodic table upto nitrogen the $\sigma 2p_z$ MO is higher in energy than the $\pi 2P_x$ and $\pi 2P_y$ MOs. The reversal in sequence of energy levels of MOs is possible due to mixing of 2s and $2P_z$ atomic orbitals.

The $\sigma 2s$ and $\sigma^* 2s$ MOs are formed by the combination of 2s atomic orbitals of the two atoms while $\sigma\ 2P_z$ and $\sigma^* 2P_z$ MOs are formed by the combination of $2P_z$ atomic orbitals of two atoms as described earlier. However, the energy differences between 2s and 2p atomic orbitals being small there is a possibility of the mixing of these orbitals (hybridization of atomic orbitals) as a result of which σ 2s and σ^*2s MOs do not retain pure s-character and similarly the $\sigma 2P_z$ and $\sigma^* 2P_z$ MOs do not retain pure p-character. On the other hand, all the four orbitals acquire sp-character due to this s-p mixing, the energies of all four orbitals change in such a way that the MOs σ 2s and σ^*2s become more stable and are thus lowered in energy, whereas the MOs $\sigma 2P_z$ and $\sigma^* 2P_Z$ become less stable and are thus raised in energy. The MO energy level diagram of diatomic molecules involving mixing of 2s and sp orbitals is shown in Fig 3.48.

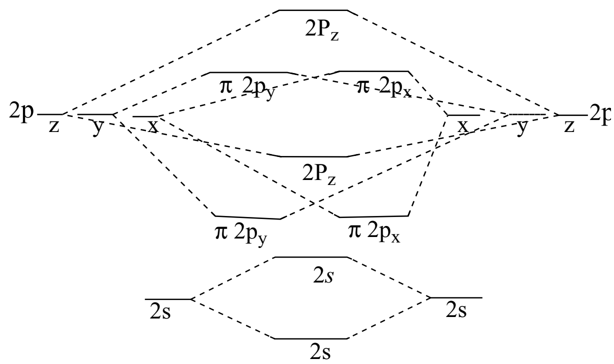


Fig 3.48 The energy levels of different MOs resulting from the mixing of 2s and 2p atomic orbitals

From the above discussion, a question arises as to why the 2s and 2p orbitals mix in the case of second row elements of periodic table upto nitrogen only and not in the case of oxygen and fluorine. This is due to the difference in the energies of 2s and 2p atomic orbitals of atoms of second row elements.

The new sequence of energy of MOs therefore becomes $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2P_x \simeq \pi 2P_y < \sigma 2P_z < \pi^*\ 2P_x \approx \pi^* 2p_y < \sigma^*\ 2p_x$.

When two molecular orbitals of the same symmetry have similar energies, they interact to lower the energy of the lower orbital and rise the energy of the higher. For example, in the homonuclear diatomics the $\sigma 2s$ and $\sigma 2p_z$ orbitals both have same symmetry and these orbitals interact to lower the energy of the $\sigma 2s$ and to rise the energy of the $\sigma 2p_z$. Similarly σ^*2s and σ^*2p_z orbitals interact to lower the energy of the σ^*2s and to rise the energy of σ^*2p_z. When two molecular orbitals of the same symmetry mix, the one with higher energy moves still higher and the one with lower energy moves lower in energy, As a practical matter, atomic orbitals with energy difference greater than 12 or 13eV usually do not interact significantly. For this purpose, the orbital potential energies are given in Table 3.13. These potential energies are negative, because they represent attraction between valence electrons and atomic nuclei. The values are the average energies for all electrons in the same level and are weighted average of all the energy states. They do not show the variation of the ionization energies, but steadily become more negative from left to right within a period as the increasing nuclear charge attracts all the electrons more strongly.

3.7.5 Bond Order

Like an atomic orbital, an MO cannot accomodate more than a maximum of two electrons and these two should have different spins. Similarly, in the MOs (like atomic orbitals) having equal energy the electrons will pair up only after they are filled with one electron each. In other words, the MOs also obey Pauli's exclusion principle and Hund's rule of maximum multiplicity. The electrons in the molecule enter the available MOs in the order of increasing energy i.e., Aufbau principle.

How many are the bonds in a molecule or **bond order** is correctly predicted by the MO theory. The number of bonds in a molecule is one-half of the difference of the number of electrons in the bonding MOs and the number of electrons in the anti-bonding MOs. Mathematically

The no. of bonds in a molecule =

[No. of electrons in the bonding MOs – No. of electrons in the anti-bonding MOs]

In common practice, only MOs formed from valence orbitals are considered for determining bond orders. The MO theory helps in establishing whether the existence of a bond is feasible or not. For example, the formation of He_2 molecule discussed later in this chapter, is completely ruled out as the bond order is predicted zero.

3.7.6 Significance of Bond Order

From the values of bond order, sufficient conclusions can be drawn. These are as follows.

(i) From the value of bond order of a molecule, its stability can be deduced. A negative or zero bond order

Table 3.13 Orbital potential energies

Atomic number	Element	Orbital potential energy eV 1s	2s	2p	Atomic number	Element	Orbital potential energy ev 3s	3p	Atomic number	Element	Orbital potential energy eV 4s	4p
1	H	–13.61			11	Na	–5.14		19	K	–4.34	
2	He	–24.59			12	Mg	–7.65		20	Ca	–6.11	
3	Li		–5.39		13	Al	–11.32	–5.98	30	Zn	–9.39	
4	Be		–9.32		14	Si	–15.89	–7.78	31	Ga	–12.61	–5.93
5	B		–14.05	–8.30	15	P	–18.84	–9.65	32	Ge	–16.05	–7.54
6	C		–19.43	–10.66		S	–22.71	11.62	33	As	–18.94	–9.17
7	N		–25.56	–13.18		Cl	–25.23	–13.67	34	Se	–21.37	–10.82
8	O		–32.38	–15.85		Ar	–29.24	–15.82	35	Br	–24.37	12.49
9	F		–40.17	–18.65					36	Kr	–27.51	–14.22
10	Ne		–48.47	–21.59								

means that a molecule is unstable. However, a postive bond order reveals that it is a stable molecule.

(ii) higher the bond order, the higher is the dissociation energy. For example, a molecule having a bond order of two has a greater dissociation energy than a molecule with bond order 1 or 1.5.

(iii) Bond orders may have integral as well as non-integral values. The integral bond order of 1, 2, or 3 correspond to single, double or triple bonds, respectively. In this respect bond order concept in the MO method corresponds to that of a chemical bond. However, this correspondence is not exact because non-integral values of bond order do not have corresponding fractional chemical bonds.

(iv) Bond order is inversely proportional to bond length. higher the bond order the shorter the bond length or vice versa.

3.7.7 Homonuclear Diatomic Molecules

H_2 Molecule: In the formation of H_2 molecule, two 1s orbitals each having one electron overlap to form σ1s and σ^*1s MOs. Both the electrons in the hydrogen molecule will move to the σ1s MO of lower energy but with opposite spin the MO electronic configuration of H_2 molecule is $\sigma 1s^2$. The bond order for H_2 molecule is $(2 - 0)/2 = 1$. It is diamagnetic since both electrons are paired.

H_2 molecule may be converted into H_2^+ or H_2^- ions either by removing or by adding electron from H_2 molecule, respectively. In the formation of H_2^+ ion, electron is removed from bonding σ Is^2 results in the decrease of bond order to 0.5. If electron is added to convert H_2 into H_2^- ion it should be added to σ^*1s MO. Then also the bond order decreases to 0.5. Among H_2^+ and H_2^- ions, H_2^- ion is less stable due to the presence of unpaired electron in antibonding MO which destabilizes the H_2^- ion. Both H_2^+ and H_2^- ions are paramagnetic. The order of stability is $H_2 > H_2^+ > H_2^-$.

He_2 Molecule: The electronic configuration of helium atom is $1s^2$. Each helium atom contains 2 electrons. Of the four electrons from two helium atoms, two electrons occupy σ1s and remaining two electrons occupy the σ^*1s MO. The MO electronic configuration of He_2 molecule is $\sigma 1s^2$, $\sigma^* 1s^2$ resulting in bond order $(2 - 2)/2 = 0$. The zero value of bond order reveals that such a molecule has no existence.

He_2^+ molecule ion has three electrons, two are occupying a σ1s orbital, whereas the third electron occupies σ^*1s MO. The MO electronic configuration of He_2^+ is $\sigma 1s^2$ $\sigma^* 1s^1$. The bond order will be 0.5. As the bond order of He_2^+ being the same as H_2^- the dissociation energies and bond lengths of two species H_2^- and He_2^+ are comparable which was confirmed experimentally. The existence of He_2^+ has been detected in a discharge tube.

Li_2 Molecule: Each lithium atom has electronic configuration of $1s^2$ $2s^1$. There are totally six electrons from the two lithium atoms which are to be filled. The MO electronic configuration of Li_2 molecule is $\sigma 1s^2$ $\sigma^* 1s^2$ $\sigma 2s^2$. It can also be written as KK*[$\sigma 2s^2$] where KK* represents the closed K shell structure $\sigma 1s^2$ $\sigma^* 1s^2$. The bond order is 1. Li_2 species are present in sufficient concentration at the boiling temperature of lithium. The bond energy (105 KJ mole^{-1}) is quite low. The bond length is 267 pm. The Li-Li bond is weaker than H – H bond because the 2s orbital of lithium is larger in size than 1s orbital of hydrogen and the outer electrons in σ2s MO are shielded by the inner shell.

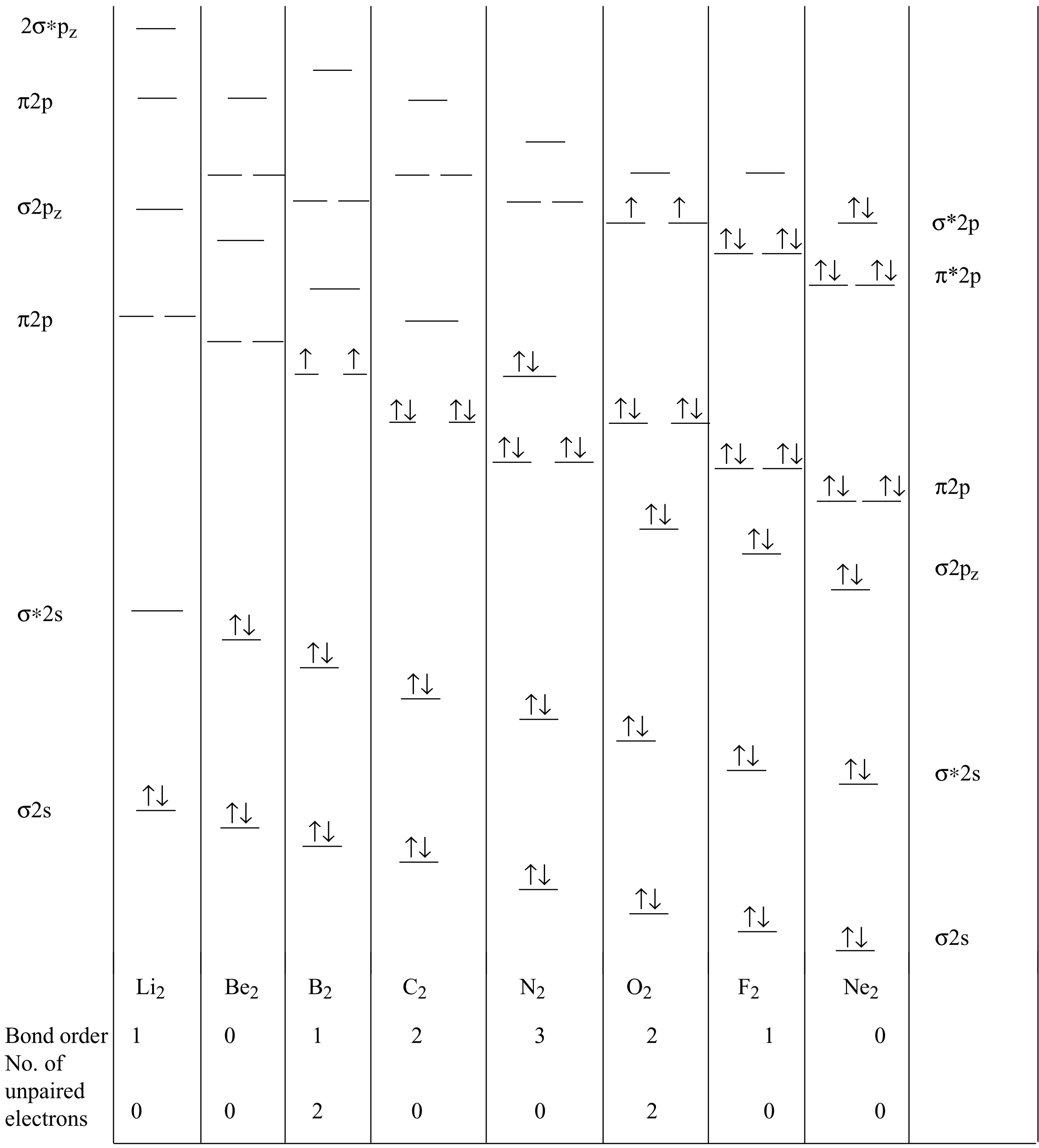


Fig 3.49 Energy levels of the homonuclear diatomic molecules of the second period

Similar to H_2 molecule, Li_2 molecule can also be converted into Li_2^+ and Li_2^- ions either by removing or by adding electrons resulting in the bond order in both cases to 0.5. The order of stability is $Li_2 > Li_2^+ > Li_2^-$. Li_2 is diamagnetic while Li_2^+ and Li_2^- are paramagnetic.

Be_2 Molecule: Be_2 has the same number of bonding and anti-bonding electrons and consequently a bond order of zero. Hence, like He_2, Be_2 is not a stable chemical species.

B_2 Molecule: B_2 molecule is an example in which the MO model has advantage over Lewis diagram. B_2 is found only in the gas phase and is paramagnetic. This behaviour can be explained if its two highest energy electrons occupy separate π MOs. Lewis diagram cannot account for the paramagnetic behaviour of this molecule.

B_2 is also a good example of the energy level shift caused by the mixing of s and p orbitals. In the absence of mixing, the σ2p orbital is expected to be lower in energy than the π2p orbitals and the resulting molecule would be diamagnetic. However, mixing of the σ2s and increase of the energy of σ2p MO lowers the energy of the σ2s orbitals and increase the energy of the σ 2p orbital to a higher level than the π orbitals, giving the order of energies shown in Fig 3.48. As a result, the last two electrons are unpaired in the degenerate π MOs and the molecule is paramagnetic. Its MO electronic configuration is KK*[$\sigma\ 2s^2\ \sigma^*2s^2\ \pi\ 2px^1\ \pi\ 2p_y^1$]. The bond order is 1.The bond dissociation energy of B_2 molecule is found to be 189 KJ mole^{-1} and bond length is 159 pm.

If an electron is removed from B_2 molecule, it has to be removed from bonding MO i.e., either π $2p_x$ or $\pi 2p_y$ to convert into B_2^+ ion. Therefore, the bond order decreases to 0.5 and becomes less stable than B_2. B_2^{2+} ion cannot exist because the bond order becomes zero due to removal of two electrons from two π bonding MOs. If an electron is added to B_2 molecule to convert it into B_2^- ion, the electron has to be added to $\pi 2p_x$ or $\pi 2p_y$ which are of equal energy. In such cases the bond order increases, bond length decreases and the stability of the B_2^- ion will be more than B_2 molecule. If a second electron is added to convert the B_2^- ion to B_2^{2-} ion it also enters into the π bonding MO. Therefore, the bond order increases to 2. Among B_2, B_2^+, B_2^- and B_2^{2-}, B_2, B_2^+ and B_2^- are paramagnetic while B_2^{2-} is diamagnetic.

Order of bond length $B_2^{2-} < B_2^- < B_2 < B_2^+$.

C_2 Molecule: The simple MO picture of C_2 predicts a doubly bonded molecule with all electrons paired, but with both **highest occupied molecular orbitals** (HOMO) having π symmetry. It is unusual because it has two π bonds and no σ bond. Its MO electronic configuration is KK* [$\sigma 2s^2 \sigma^* 2s \pi\ 2p_x^2\ \pi 2p_y^2$] The bond order of C_2 molecule is 2, its bond dissociation energy is 62 KJmol^{-1} and bond length is equal to 131pm. C_2 molecule can be converted into C_2^+ and C_2^{2+} ions by removing electrons and can be converted into C_2^- and C_2^{2-} ion by adding electrons. Their MO electronic configurations and bond orders are as follows.

$C_2^+ = KK^*[\sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2\ \pi 2p_y^1]$

Bond order 1.5 paramagnetic

$C_2^{2+} = KK^*[\sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^1\ \pi 2p_y^1]$

Bond order 1.0, paramagnetic

$C_2^- = KK^*[\sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2\ \pi 2p_y^2\ \sigma 2p_z^1]$

Bond order 2.5, paramagnetic

$C_2^{2-} = KK^*[\sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2\ \pi 2p_y^2\ \sigma 2p_z^2]$

Bond order 3.0, diamagnetic

Although C_2, C_2^+, C_2^{2+} and C_2^- are not common, the acetylide ion C_2^{2-} is well known particulary in compounds with alkali metals, alkaline earth metals and lanthanides.

Order of bond length $C_2^{2-} < C_2^- < C_2 < C_2^+ < C_2^{2+}$

Order of stability and bond energy

$C_2^{2+} < C_2^+ < C_2 < C_2^- < C_2^{2-}$

N_2 Molecule: The MO electronic configuration of N_2 molecule is KK*[$\sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2\ \pi 2p_y^2\ \sigma 2p_z^2$] and bond order is 3. The triple bond in N_2 molecule indicates that it must have a very high dissociation energy and shorter bond length which were found experimentally as 945.6 KJ mole^{-1} and 110 pm, respectively. It is diamagnetic. Atomic orbitals decrease in energy with increasing nuclear charge; as the effective nuclear charge increases, all orbitals are pulled to lower energies. The shielding effect and electron – electron interactions cause an increase in the difference between 2s and 2p orbital energies as Z increases from 5.7 eV for boron to 8.8 eV for carbon and 12.4 eV for nitrogen. As a result the σ2s and σ2p levels of N_2 interact (mix) less than the B_2 and C_2 levels, and the σ2p and π2p are very close in energy. The order of energies of these orbitals has been a matter of controversy. The photoelectron spectroscopy shows that the IE values of the $\sigma 2p_z$ and π2p electrons are about 15.6 and 16.7 eV, respectively, giving –15.6 and –16.7 eV as orbital energies and suggesting that sufficient s-p mixing occurs in this molecule to make $\sigma 2p_z$ level higher in energy than the π 2p orbitals.

If an electron is removed from N_2 molecule to convert it into N_2^+ ion, it has to be removed from one of the σ bonding MO. This causes the decrease in bond order to 2.5. Therefore, the bond length increases and stability decreases. The ion is paramagnetic. When N_2 is converted into N_2^- ion, an electron is added to one of the π anti-bonding MOs causing the decrease in the bond order to 2.5 so that the bond length increases and stability decreases. The N_2^- ion is also paramagnetic. Among N_2^+ and N_2^- ions, N_2^- ion is less stable due to the presence of electron in anti-bonding MO.

O_2 Molecule: The MO electronic congfiguration of O_2 molecule is KK* [$\sigma 2s^2\ \sigma^* 2s^2\ \sigma 2P_z^2\ \pi 2P_x^2\ \pi 2P_y^2\ \pi^* 2P_x^1\ \pi^* 2P_y^1$]. O_2 molecule is paramagnetic.This property as for B_2 cannot be explained by the traditional Lewis diagram [:Ö=Ö:] but is evident from the MO picture which assigns two electrons to the degenerate π*orbitals. O_2 molecule contain a double bond. Its bond dissociation energy is 494.6 KJ mol^{-1} and bond length is 120.07 pm.

If an electron is removed from O_2 molecule to convert it into O_2^+ ion, it has to be removed from one of the anti-bonding MOs. This causes the increase in bond order from 2 to 2.5. The O_2^+ ion is still paramagnetic since it has one more unpaired electron in one of the π*orbitals.

If an electron is added to O_2 molecule to convert it into superoxide ion, the electron has to be added to one of the anti-bonding MOs, which causes decrease in bond order from 2 to 1.5. Therefore, the bond length increases and stability decreases. O_2^- is still paramagnetic due to the presence of one unpaired electron in one of the π* orbitals. If one more electron is added to superoxide ion to covert into peroxide ion (O_2^{2-}), it should be added to another π* orbital. The bond order in peroxide is 1. Peroxide ion is diamagnetic. The MO electronic configurations and bond orders of different oxygen species are as follows.

$O_2 = KK*[\sigma 2s^2\ \sigma*2s^2 \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^2\ \pi*2p_x^1\ \pi*2p_y^1]$;

bond order is 2.0; paramagnetic; bond length 120.07 pm;

$O_2^+ = KK*[\sigma 2s^2 *2s^2 \sigma 2p_z^2\ \pi 2p_x^2 \pi 2p_y^2\ \pi*2p_x]$

bond order 2.5 paramagnetic; bond length112.3pm;

$O_2^- = KK*[\sigma 2s^2\ \sigma*2s^2 \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^2\ \pi*2p_x^2\ \pi*2p_y^1]$

bond order 1.5; bond length 128 pm; paramagnetic.

$O_2^{2-} = KK*[\sigma 2s^2\ \sigma*2s^2 \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^2\ \pi*2p_x^2\ \pi*2p_y^2]$;

bond order 1.0; bond length 1.49 pm; diamagnetic.

O – O bond length in O_2^- and O_2^{2-} is influenced by the cation.

F_2 Molecule: The MO electronic configuration of F_2 molecule $KK*[\sigma 2s^2\ \sigma*2s^2 \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^2\ \pi*2p_x^2\ \pi*2p_y^2]$ indicates that F_2 molecule is diamagnetic having a single F – F bond.

Ne_2 Molecule: All the molecular orbitals are filled, there are equal numbers of bonding and anti-bonding electrons and the bond order is therefore zero. The Ne_2 molecule is a transient species, if it exists at all.

3.7.8 Heteronuclear Diatomic Molecules

Heteronuclear diatomic molecules follow the same general bonding pattern as the homonuclear molecules described so far, but a greater nuclear charge on one of the atoms lowers its atomic energy levels and shifts the resulting MO levels. In dealing with heteronuclear molecules, it is necessary to have a way to estimate the energies of the atomic orbitals that may interact. For this, the orbital potential energies given in Table 3.13 are useful. Due to the difference in the electronegativities of the two atoms, the electrons in bonding MO spend more time near the more electronegative atom. On the other hand, the electrons in anti-bonding MOs are closer to the less electronegative atom.

NO Molecule: The nitrogen has 7 electrons while the oxygen has 8 electrons.The MO electronic configuration of NO molecule is

Bond order is (8 – 3)/2 = 2.5. The presence of unpaired electron in antibonding π MO makes it less stable. Bond

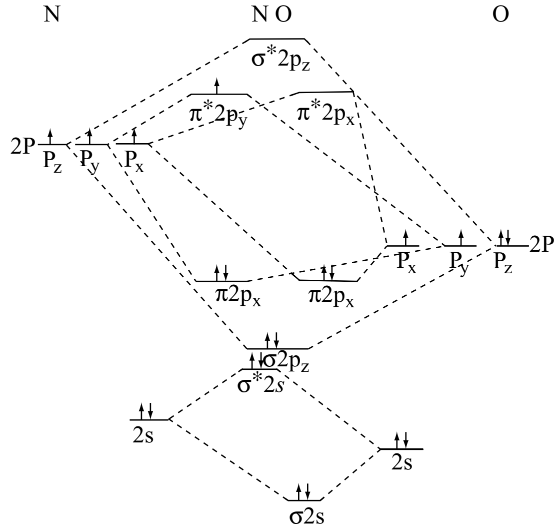


Fig 3.50 The molecular orbital diagram of NO

energy is 667.8 KJ mole^{-1}. It is paramagnetic. When NO molecule is converted in to NO^+ ion, the electron in π* MO is removed. Therefore, bond order increases, bond length decreases and becomes diamagneitc. The N – O bond length in NO^+ is 106 pm and is comparable with that of N – N bond length 110 pm in N_2.

CN^- ion: The CN^- ion has totally 14 electrons (6 from carbon, 7 from nitrogen and 1 due to negative charge) and is isoelectronic with N_2. The energy difference in s and p orbitals is less. Therefore, sp mixing takes place in the formation of CN^- ion. The MO electronic configuration of CN^- is $KK*[\sigma 2s^2\ \sigma*2s^2 \pi 2p_x^2\ \pi 2p_y^2\ \sigma 2p_z^2]$. Bond order is (8 – 2)/2 = 3.

CO Molecule: The molecular orbitals of CO are shown in Fig. 3.51.

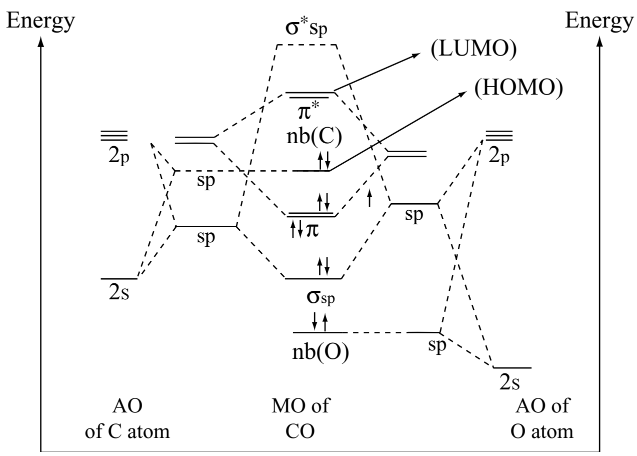


Fig 3.51 The molecular orbital diagram of CO

The orbital energies are lower for the more electronegative oxygen, so that the lower energy hybrid orbital of carbon is comparable in energy with the higher energy hybrid orbital of oxygen. These orbitals of comparable energy combine to give bonding and anti-bonding MOs. The very low energy hybrid orbital on oxygen and the high energy one on carbon are far apart in energy. Hence, even though their symmetries are the same, they are **non-bonding**. Rather than combining to give MOs, they remain as localized orbtials on 'O' and 'C', respectively. These are the orbitals occupied by the lone pairs on 'C' and 'O' directed away from the CO bond. Because the lone pair orbital on oxygen is close in energy to the oxygen atomic 2s, it is said to be mostly in s character; that is, it much like an oxygen 2s orbital. The lone-pair orbital on carbon, on the other hand, is close in energy to the carbon p-orbitals, so it is said to have a great deal of carbon p-character. With its high energy, the carbon, lone pair is the pair donated in forming dative bonds to other species such as BH_3 in $H_3B \leftarrow CO$ and also to metals in metal carbonyls. The electron cloud for the π orbitals is polarized towards the more electronegative oxygen atom (Fig 3.51). Actually because only one direction (z) is defined for the CO molecule, x and y are indistinguishable for the atomic and molecular orbitals. Consequently πp_x and πp_y together form a sheath having cylindrical symmetry. Although these bonds have axial symmetry, they are π, not σ, since there is node instead of a maximum in the electron density along the internuclear axis. The empty π^* orbitals are more localized on (polarized) carbon.

The MOs that will be of greatest interest for reactions between molecules are the **highest occupied molecular orbital (HOMO)** and the **lowest unoccupied molecular orbital (LUMO)** collectively known as **frontier orbitals** because they lie at the occupied – unoccupied frontier. The reaction chemistry of CO with transition metal cannot be explained by simple electronegativity arguments that place more electron density in the oxygen. If this is correct, the metal carbonyls should bond as M–O–C but the actual bonding order is M–C–O. The HOMO of CO is non-bonding orbital of carbon. The lone pair in this orbital forms a bond with a vacant orbital on the metal.

The ionization of CO gives CO^+ during which bond length decreases from 112.8 to 111.5 pm. This suggests that CO contains only a double bond and that ionization results in more electronic charge pulled into the inter nuclear region. A simple interpretation is that the MOs are considerably distorted towards the more electronegative oxygen atom and the ionization reduces the electronegativity difference.

3 .7.9 Isoelectronic Principle

The molecular species with the same number of atoms and the same total number of valence electrons are said to be isoelectronic and the isoelectronic principle states that such molecular species will have similar MOs and molecular structure. The molecular species involved may be molecules, anions or cations. The application of this principle enables some simple diatomic molecules to be likened to molecules of isoelectronic elements.

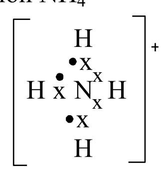
3.8 COORDINATE BOND OR DATIVE BOND

A covalent bond may also be formed by sharing a pair of electrons, providing both electrons by one of the two bonded atoms. In such a case, the bond is called as **dative bond.** It is just like a covalent bond, once it is formed, the two are not always distinguished in bond diagrams.

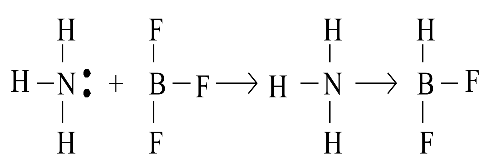
The atom providing the two electrons to make up the dative bond is known as the **donor**. It must, of course, have an 'unused' pair of electrons available, and such a pair is referred to as a **lone pair.** The atom sharing the pair of electron from the donor is known as the **acceptor.** The dative bond between atoms A and B is shown as $A \rightarrow B$, A being the donor and B the acceptor. This indicates, in a convenient way, the origin of the electrons making up the bond. The dative bond can also be shown as $\overset{\oplus}{A}-\overset{\ominus}{B}$. This method indicates the electrical charges which develop on atoms A and B as a result of dative bond formation. A, the donor, develops a positive charge by partly transferring two electrons to B; B, the acceptor, develops a corresponding negative charge. On this basis, the dative bond can be regarded as a covalent bond with a certain amount of ionic character and the term coionic, instead of dative, is intended to describe this state of affairs. Other terms which have been used are coordinate bond, semi-polar bond or semi-polar double bond. The use of dative bond is preferred because it was the term finally used by Sidgwick, who first developed the use of this type of bond.

The following structural formulae show the various ways in which dative bonds can occur in typical compounds.

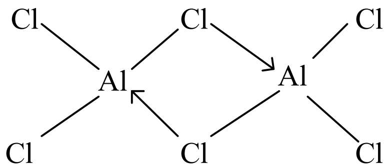
(i) The ammonium ion NH_4^+



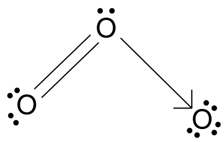
(ii) Combination of ammonia and boron trifluoride



(iii) Dimerisation of aluminium chloride



(iv) Formation of ozone



3.8.1 Characteristics of Coordinate Compounds

The properties of coordinate compounds are intermediate between the properties of ionic compounds and covalent compounds.

(i) Physical state: Coordinate compounds exist as gases, liquids and solids in ordinary conditions.

(ii) Melting points and Boiling points: The melting points and boiling points of coordination compounds are higher than covalent compounds but lower than ionic compounds.

(iii) Solubility: Coordination compounds are sparingly soluble in polar solvents like water but readily soluble in non-polar solvents.

(iv) Stability: Coorodination compounds are stable like covalent compounds but the addition compounds formed by dative bond are not very stable.

(v) Electrical conductivity: These compounds do not conduct electric current either in solutions or in fused state.

(vi) Reactivity: Coordinate compounds undergo molecular reactions which are slow.

(vii) Isomerism: The dative bond is directional in nature like covalent bond. Therefore, coordination compounds exhibit isomerism.

3.9 METALLIC BOND

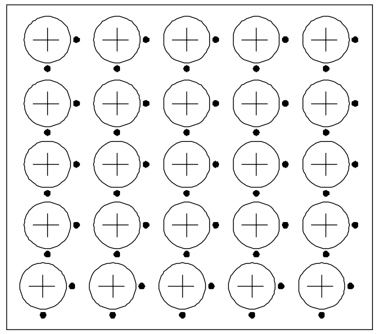
Metals have very distinctive properties. In particular, they are good conductors of heat and electricity; they are very opaque and have high shining nature i.e., lustre. They are generally hard, ductile and malleable. They have high melting and boiling points. Such properties cannot be accounted for normal ionic or covalent bonding, so the idea of special metal bond is necessary. The metal bond can be defined as **"the force that binds a metal atom to a number of electrons within its sphere of influence is known as a metal bond"**. Following are the main theories put forward to explain the metallic bond.

3.9.1 Electron Sea Model

This theory was first proposed by Drude and later developed by Lorentz. The main postulates of this theory are as follows.

(i) All the atoms in a metal have several unoccupied electron orbitals in their outer shells.

(ii) Generally the ionization energies of metals are less. So, they lose their outer electrons and form positive ions called metal nuclei.



(iii) These electrons delocalize and move throughout the metal in the vacant orbitals of the metal atoms. So the metal can be thought to be of metal cations immersed in a sea of electrons and hence this model is referred to as electron sea model.

(iv) Electrons pull the cations towards them from all directions and thus hold the cations leading to big force of attraction of several nuclei for several electrons. This gives the metal a stable structure.

Though this theory could explain the metallic properties, certain properties like resistance to electrical conductivity with increase in temperature in some metals while electrical conductivity increases with rise in temperatue in certain other cases and why some are virtually insulators could not be explained.

3.9.2 Valence Bond Theory

This theory proposed by Pauling considering that covalent bond with resonance is present in metals. For example in sodium metal crystal every sodium is surrounded by eight sodium atoms contributing 9 electrons for covalent bonding between sodium atoms. These are not sufficient to form electron pair bonds between all sodium atoms because 9 atoms require 16 electrons. This can be explained on the basis of resonance that at first a metal atom forms the covalent bond with one of its neighbouring atoms, then with another neighbouring atoms and so on. In this way, all the neighbouring atoms come close but the valence electrons remain free because of their delocalization between the differing neighbouring atoms. The process is continuous in whole of the crystal so that all

Table 3.14 Molecular orbital results for selected diatomic molecules

Molecule	Electronic configuration	Molecular orbital predictions: No. of outer electons: Bonding	Antibonding	Bond order	Bond length pm	Energy KJ mol^{-1}	Magnetic character
H_2^+	$\sigma 1s^1$	1	0	0.5	269.6	1.06	Para
H_2	$\sigma 1s^2$	2	0	1.0	257	432	Dia
H_2^-	$\sigma 1s^2 \sigma^* 1s^1$	2	1	0.5	—	—	Para
He_2^+	$\sigma 1s^2 \sigma^* 1s^1$	2	1	0.5	—	—	Para
He_2	$\sigma 1s^2 \sigma^* 1s^2 (KK^*)$	2	2	0	—	—	—
Li_2	$KK^*[\sigma 2s^2]$	2	0	1	267	105	Dia
Be_2	$KK^*[\sigma 2s^2 \sigma^* 2s^2]$	2	2	0	—	—	—
B_2	$KK^*[\sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^1 \pi 2p_y^1]$	4	2	1	159	289	Para
C_2	$KK^*[\sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2]$	6	2	2	131	627.9	Dia
N_2^+	$KK^*[\sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2 \sigma 2p_z^1]$	7	2	2.5	112	841	Para
N_2	$KK^*[\sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2 \sigma 2p_z^2]$	8	2	3	110	945.6	Dia
O_2^+	$KK^*[\sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* p_x^1]$	8	3	2.5	112	644	Para
O_2	$KK^*[\sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^1\ \pi^* 2p_y^1]$	8	4	2	121	494.6	Para
O_2^-	$KK^*[\sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^2\ \pi^* 2p_y^1]$	8	5	1.5	128	—	Para
O_2^{2-}	$KK^*[\sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^2\ \pi^* 2p_y^2]$	8	6	1.0	149	—	Dia
F_2	$KK^*[\sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* p_x^2\ \pi^* 2p_y^2]$	8	6	1.0	142	155	Dia
NO	$KK^*[\sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* p_x^1]$	8	3	2.5	—	667.8	Para
NO^+	$KK^*[\sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2]$	8	2	3.0	106		Dia

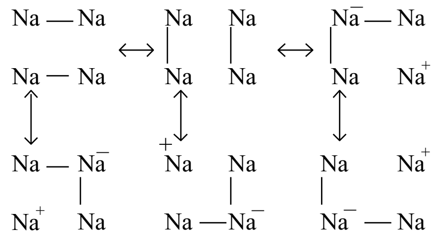


Fig 3.52 Resonance structure in sodium metal

the atoms in a metallic crystal are held very close. This is illustrated in Fig. 3.52 taking example of sodium metal by taking four sodium atoms.

Though only four atoms are shown in these pictures, the actual bonding includes all the atoms of the crystal and in three dimensional view. The actual structure of the metal is the resonance hybrid of different resonance structures. As the resonance always stabilizes the system, metals too are quite stable. Moreover as the number of resonance structures are large in number, the metals show sufficiently high stability.

To have resonance structures every metal atom should possess a vacant orbital in its valence shell, so that it can accept an electron from other atom and can form two bonds. Such a vacant orbital is termed as **metallic orbital.** Presence of such metallic orbitals is a characteristic of all metals. In diamond, carbon has no vacant orbital and forms four stable covalent bonds. Therefore, resonance or delocalization is not possible and hence is a non-conductor.

3.9.3 Band Theory

This model of the bonding in metals is based on the MO theory. We start with considering the sodium metal. If two sodium atoms come close to each other, the 3s orbital of one sodium atom can combine with 3s orbital in the second sodium atom to form two MOs of which, one is bonding MO with lower energy and the other is anti-bonding MO with higher energy. If a third sodium atom also come close to these two sodium atoms, three MOs are formed. In a similar way if n sodium atoms approach together, n MOs can be formed. The building up of such MOs can be represented as in Fig 3.53.

Depending upon different types of atomic orbitals which overlap, different energy bands gives a pictorial representation of the energy levels in metallic crystal. The arrangement of electrons in the different energy bands determines the characteristics of a metal. The energy bands formed from different atomic orbitals may overlap or be separated from each other. **The highest occupied energy band is called the valence band while the lowest unoccupied energy band is called the conduction band. The energy gap between the top of the valence band and the bottom of the conduction band is called the energy gap.**

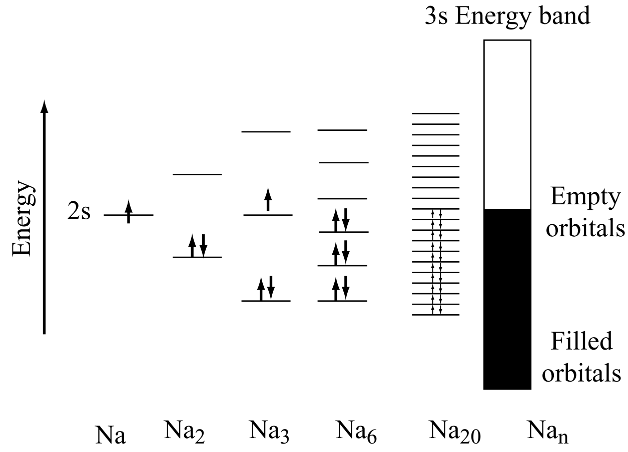


Fig 3.53 Formation of 3s energy bands in sodium metal

Eg. In the case of metals, the valence band may be half-filled or there may be an overlapping between the valence and conduction bands. This makes it possible for the electrons to go into the vacant bands and hence is responsible for electrical conductivity.

In sodium metal 1s atomic orbitals form 1s band, 2s atomic orbitals form 2s band, 2p atomic orbitals form 2p band and 3s atomic orbitals form 3s band. 1s, 2s and 2p bands are completely filled with electrons while 3s band is half-filled since 'n' orbitals contain 'n' electrons but each orbital can accomodate 2 electrons. The presence of electrons in the half-filled 3s band leads to high electrical conductance as a number of unoccupied energy levels are available in the valence band itself.

The electronic configuration of magnesium is $1s^2\ 2s^2\ 2p^6\ 3s^2$. Unlike sodium, 3s band is completely filled by the electrons as every 3s orbital of Mg contribute 2 electons. However, in Mg it has been found that the 3p energy band (formed by unoccupied 3p AOs of the Mg atoms) overlaps the 3s band. There is, therefore, no energy gap between the valence band and the conduction band and Mg is an excellent conductor. In the case of **insulators,** the energy gap is very large and therefore the vacant conduction band is not available to the electrons of the completely filled valence band. Several solids have electrical properties intermediate between those of metals and insulators. These are called **semiconductors.** The energy gap in these substances is very small and increase in temperature gives thermal energy for some of the electrons in the valence band to move into the conduction band. The energy gap values in KJ mole^{-1} for carbon group elements are diamond (511), silicon (111) and germanium (63).

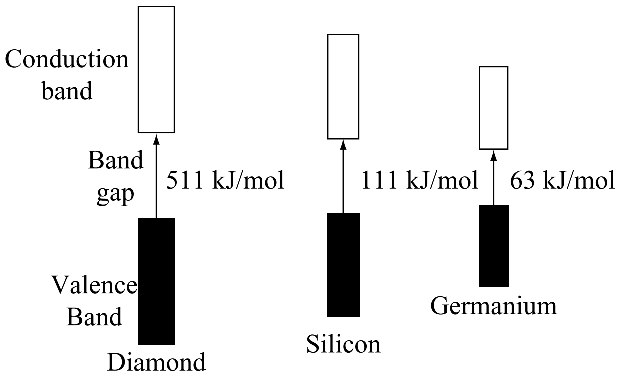
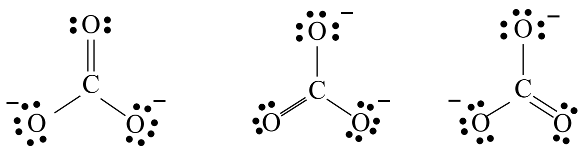


Fig 3.54 Energy bands in diamond, silicon and germanium

Diamond with a large band gap is an insulator. Silicon and germanium have fairly small band gaps and are called semiconductors. Their electrical conductivity is substantially greater than that of insulators but far less than that of conductors.

3.10 RESONANCE

One problem with Lewis diagrams is that more than one equivalent Lewis diagrams can be written for many molecules or ions. Consider for example, the carbonate ion CO_3^{2-} which can be represented by different but equivalent structures.



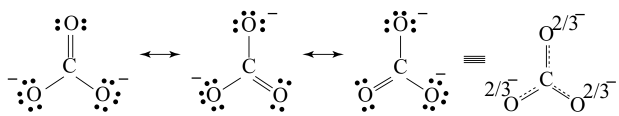
I II III

None of these alone can explain the important data about the carbonate ion. All the C – O bond lengths are equal and shorter than single bonds. None of the above structures, therefore, is correct. This can be explained by the **resonance theory**.

The different possible Lewis diagrams of a molecule or ion are called resonance structures. None of these structures can completely explain the physical and chemical properties of the substance. The actual molecule or ion will be better represented by a hybrid of these structures. Resonance structures are not structures for the actual molecule or ion; they exist only in theory and are hypothetical. As such, they can never be isolated. The actual structure which must be visualized does not consist of a mixture of the various possible structures. It is a single structure of its own, but as it cannot be written down simply on paper, it is convenient to think of it in terms of possible structures and is different from all the hypothetical structures called **resonance hybrid** structure. The individual structures are called **canonical forms.** Alternatively, using a terminology introduced by Ingold, the term **mesomeric forms** or **mesomeric structures** are used, and the conception is known as **mesomerism** (between the parts).

Various analogies have been suggested to facilitate the building up of a visual picture of what resonance means. Perhaps, the best is the idea of describing a rhinoceros in terms of a unicorn and a dragon. The rhinoceros, which has an actual existence, is thought of as a sort of resonance hybrid, interms of unicorns and dragons, which do not exist.

It must be emphasised that all the molecules in a resonance hybrid are alike, and that it is not just a mixture of different molecules. It is particularly important to distinguish between resonance and tautomerism, for they are easily confused. The latter may be regarded as the existence of two or more forms of a substance having different arrangement of atoms; the forms can sometimes be isolated. The possible structures contributing to resonance hybrid have the same arrangement of atoms but different arrangement of electrons; they can never be separated for they have no real existence. Resonance is represented by$\longleftrightarrow$ unlike the equilibrium by $\rightleftharpoons$. The resonance hybrid of the carbonate ion can be represented as follows.



The bonds in a resonance hybrid structure are represented by a combination of a solid line and a dashed line. This is to indicate that the bonds are something in between a single bond and double bond. The charge density will be equal on all oxygen atoms.

3.10.1 Resonance Energy

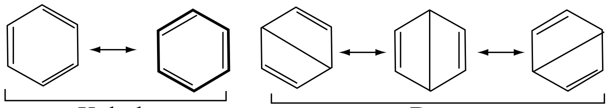
The resonance energy of a substance is the extra stability of the resonance hybrid compared with the most stable of the resonating structures. The resonance energy of carbonate ion is 176 KJ mol^{-1}. Resonance energy can be calculated by subtracting the calculated heat of formation of the CO_3^{2-} from the observed value of the heat of formation. In general

Resonance energy = Observed heat of formation – Calculated heat of formation.

3.10.2 Rules for Writing Resonance Structures

1. While writing the resonance structures, only the position of electrons must be moved but not the position of the nucleii of atoms.

2. All the resonance structures must be proper Lewis structures, for example we should not write structure in which carbon has five bonds.
3. All resonance structures must have the same number of unpaired electrons because the number of unpaired electrons decides the magnetic character and magnetic moment of the molecule.
4. Equivalent resonance structures contribute equally to the hybrid and a system described by them has large resonance stabilization.
5. More stable resonance structure contribute more to the resonance hybrid. For example, benzene has two Kekule and three Dewar structures. Dewar structures are less stable than Kekule structures since one of the π bonds is longer. Hence, Kekule structures contribute more to the resonance hybrid structure of benzene.



Kekule Dewar

6. Non-polar resonance structure having more number of covalent bonds (without charge separation) is more stable than polar resonance structure (charge separation) having less number of covalent bonds.

$$\underset{(I)}{CH_2=CH-CH=CH_2} \longleftrightarrow \underset{(II)}{\overset{+}{C}H_2-CH=CH-\overset{-}{\ddot{C}}H_2}$$

$$\longleftrightarrow \underset{(III)}{\overset{-}{\ddot{C}}H_2-CH=CH-\overset{+}{C}H_2}$$

The non-polar structure (I) is more stable than II and III because in I, formation of covalent bond lowers the energy.

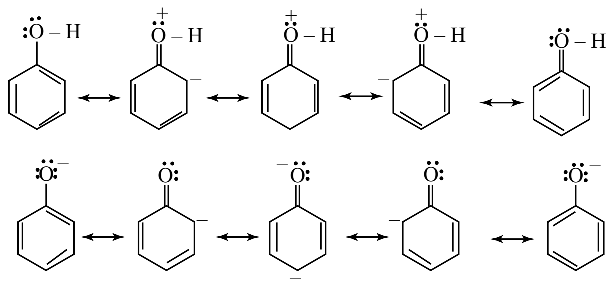
7. Resonance structures in which all the atoms have octet are especially stable and contribute more to the resonance hybrid. For example in the following, in structure IV the first carbon is having only six electrons while in structure V all the atoms are having octets.

$$\underset{IV}{\overset{+}{C}H_2-\ddot{\underset{\cdot\cdot}{O}}-CH_3} \longleftrightarrow \underset{V}{CH_2=\overset{+}{\underset{\cdot\cdot}{O}}-CH_3}$$

8. Separation of opposite charges requires energy. Therefore, the structures in which opposite charges are separated have greater energy (lower stability) than those that have no charge separation. Further increase in the distance betwen charges decreases the stability due to decrease in attraction between the charges.

$$CH_2=CH-\ddot{\underset{\cdot\cdot}{Cl}}: \longleftrightarrow \overset{-}{C}H_2-CH=\overset{+}{Cl}$$

9. Structures in which negative charge present on more electronegative atom and positive charge present on less electronegative atom are more stable than the structures in which negative charge present on less electronegative atom and positive charge present on more electronegative atom.



For example, in the resonance structures of phenol positive charge is present on more electronegative oxygen while negative charge is present on less electronegative carbon. In the resonance structures of phenoxide ion the negative charge is delocalized. Hence, the resonance hybrid of phenoxide ion is stable than that of phenol and hence phenol exhibits acidic character.

3.10.3 Examples of Resonance

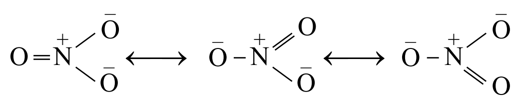
(i) Dinitrogen oxide: A resonance hybrid between

$$\underset{I}{N \overset{\leftarrow}{-} N=O} \longleftrightarrow \underset{II}{N\equiv N\rightarrow O}$$

The calculated bond lengths for various bonds involved in these structures are N = N, 120 pm, N≡N, 110 pm, N = O, 115 pm and N–O, 136 pm. The length of the molecule which is linear, is 231 pm, the probable bond lengths being N–N, 112 pm and N–O, 119 pm.

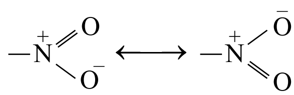
The dipole moment of nitrous oxide is very less so that the two structures contribute almost equally.

(ii) Nitrate ion: It is a resonance hybrid of the following structures.



The observed bond length is 121 pm. The calculated bond length for N – O is 136 nm and for N = O is 115 pm. The resonance energy is 185.3 kj mol^{-1}.

(iii) Nitro group: It is a resonance hybrid between

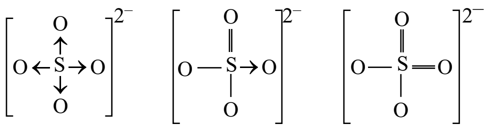


The N –O bond lengths in a variety of nitro compounds vary between 121 and 123 pm. A single bond between nitrogen and oxygen atoms would be expected to give a bond length of 136 pm and a double bond would be 115 pm.

1,4–dinitrobenzene has no dipole moment, which indicates that the dipole moments in the two nitrogroups must be equal in magnitude, but opposite in direction. The dipole moment of each nitrogroup must, infact, act in a direction which bisects the ONO angle. The dipole moments of each of the individual structures given above would act mainly along the direction of the dative bonds.

(iv) Sulphate ion: In the sulphate ion four oxygen atoms are arranged around the central sulphur atom almost tetrahedrally. The measured bond lengths are all equal 151 pm. The calculated bond length for single bonds between sulphur and oxygen atoms is 170 pm and for double bond length is 149 pm.

It is clear that there must be some double bond character about the bonds in the ion, and it is best represented as resonance hybrid between many various structures of which some typical ones are shown.



I II III

The stucutre (III) is probably the most important, but there are many similar ones because the double bonds may have alternative positions.

(v) Benzene: The properties of benzene can be accounted for, reasonably satisfactorily, by regarding it as a resonance hybrid between the two well-known Kekule structures with three possible Dewar structures making a small contribution as shown in section 3.10.2 rule 5.

The bond length of C – C bond would be expected to be 154 pm and that of C = C bond 134 pm. The actual bonds between carbon atoms in benzene are all alike, with a bond length of 139 pm.

The actual measured heat of formation of benzene is 5501 kJ mol^{-1} but the calculated heat of formation for Kekule structures is only 5339 kJ mol^{-1}. The resonance energy of benzene is therefore 162 kJ mol^{-1}. Also, a similar value is obtained from heat of hydrogenation for Kekule structures. The expected heat of hydrogenation for Kekule structures would be 372.9 kJ mol^{-1} i.e., three times the heat of hydrogenation of a C = C bond 129.3 kJ mol^{-1}. The measured heat of hydrogenation of benzene is, however, only 208.4 kJ mol^{-1}.The resonance energy, from the values, is 164.5 kJ mol^{-1}.

Representation of benzene as a resonance hybrid does not postulate the existence of true C=C bonds in benzene molecule, and this accounts for the lack of true unsaturation properties such as those shown by ethene.

Characteristics of Resonance:

(i) Resonance gives rise to identical bond lengths in molecules.
(ii) Resonance gives rise to extra stability to a molecule.
(iii) The greater the resonance energy of the molecule, the more is its stability.
(iv) More is the number of canonical forms and with more of canonical forms having same energy, more is the resonance energy and more is the stability of the molecule.
(v) Resonance hybrid structure is definite but canonical forms are imaginary and not real.

Bond order in Resonance hybrid structure:

The bond order i.e., number of bonds between two atoms in a resonance hybrid can be calculated as follows.

$$\text{Bond order} = \frac{\text{Total number of bonds around the central atom}}{\text{Total number of atoms around the central atom}}$$

For cyclic compounds, the bond order between two atoms in a ring can be calculated as follows

$$\text{Bond order} = \frac{\text{Total number of bonds in the ring}}{\text{Total number of atoms in the ring}}$$

3.11 POLAR MOLECULES: DIPOLE MOMENT

Whenever atoms with different electronegativities combine, the resulting molecule has polar bonds, with the electrons of the bond shifted more towards the electronegative atom. The greater the difference in electronegativity, the more the polar bond. As a result the bonds are polar, with positive and negative ends.

As a result of polarization the molecule possesses the **dipole moment.** A magnet has a magnetic moment equal to 'm*l*' where 'm' is the pole strength of the magnet and '*l*' is the distance between the poles. In a similar way, two equal and opposite, but separated, electrical charges constitute an electrical dipole moment measured by the charge multiplied by the distance between the charges. Mathematically, it is expressed as

Dipole moment μ = charge (Q) × distance between charges. Dipole moments are expressed in terms of Coulomb metre (cm) units or the older Debye unit, D being equal to 3.33564×10^{-30} cm. If charge is expressed in esu and distance in centimeter, dipole moment is represented in Debye unit D.

$$1\text{Debye} = 1 \times 10^{-18} \text{ esu cm}$$

Dipole moment is a vector quantity and is depicted by a small arrow with tail on the positive centre and head pointing towards the negative centre. For example, the dipole moment of HF may be represented as $\overset{\longleftrightarrow}{H–\ddot{F}}$: The shift in electron density is symbolized by crossed arrow ($+\!\!\longrightarrow$) above Lewis structure to indicate the direction of the shift. If the molecule has a very symmetric structure or if the polarities of different bonds cancel each other, the molecule as a whole may be non-polar even though the individual bonds are quite polar. This is due to vectorial property of the dipole moment.

3.11.1 Applications of Dipole moment

(a) Per cent ionic character: Using the dipole moment, the per cent ionic character of diatomic molecules can be calculated by using the formula.

$$\%\ \text{ionic character} = \frac{\propto_{\text{expt}}}{\propto_{\text{ionic}}} \cdot 100 \qquad (3.25)$$

μ_{expt} is the dipole moment determined experimentally while μ_{ionic} is the calculated dipole moment considering complete transfer of electrons.

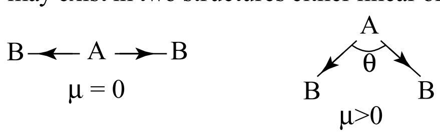
Illustrated Example: The experimental dipole moment for HC*l* is 1.03 D and its bond length is 1.27A°.
Calculate the per cent ionic character of HC*l*.
Solution: If the electron is completely transferred from hydrogen to chlorine, the calculated dipole moment will be

$$4.8\times10^{-10}\,\text{esu}\times1.27\times10^{-8}\,\text{cm} = 6.09\ \text{D}$$

$$\%\ \text{ionic character} = \frac{\propto_{\text{expt}}}{\propto_{\text{ionic}}} \cdot 100 = \frac{1.03}{6.09} \cdot 100 = 16.9\%$$

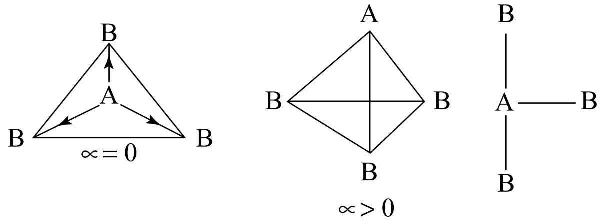
(b) Geometry of the Molecules: Dipole moment is the additive vector quantity acting in the direction of the chemical bond, and so the moment of the whole molecule is the vector sum of the constituent moments. A perfectly symmetrical molecule will therefore be non-polar, although it may contian polar linkages. This priniciple helps in the prediction of geometry of the molecules.

(i) AB_2 Type molecules: Triatomic molecule of the AB_2 type may exist in two structures either linear or angular.

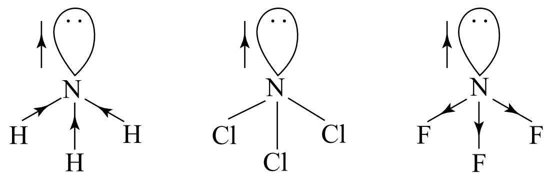


Since the linear molecule is symmetrical, its dipole moment is almost equal to zero, but the angular molecules have dipole moment and the dipole moment of the molecule depends on the angle θ. With increase in θ, dipole moment decreases. Molecules like CO_2, CS_2, etc which have dipole moment zero are linear while the molecules like H_2O, H_2S, NO_2 which have dipole moment greater than zero are angular in shape.

(ii) AB_3 Type molecules: AB_3 molecules have zero dipole moment if they have symmetric planar trigonal structure but have dipole moment greater than zero if they are in pyramidal or T shape.



Molecule like SO_3, BF_3, BCl_3, etc have zero dipole moment. Therefore, they have a symmetric planar trigonal structure. However, the molecues like NH_3, NF_3, PH_3, etc., have dipole moment and exist in a pyramidal shape. The molecules like ClF_3, $BrCl_3$, ICl_3, etc., exist in T shape which have dipole moment. If the central atom has one lone pair and dipole moment it will be in the pyramidal shape but if there are two lone pairs on the central atom and the molecule with dipole moment will be in T shape. In the pyramidal molecules, the dipole moment of the lone pair will be outward the central atom. Thus depending on the direction of bond moments sometimes they may have nearly zero dipole moment.

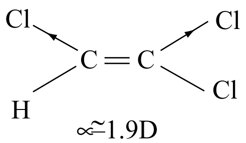


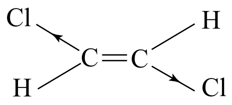
In ammonia the bond moments and dipole moment are acting in the same direction. In NCl_3 the bond moments are zero since the electronegativities of both nitrogen and chlorine are equal, but due to the dipole moment of lone pair on nitrogen it will have dipole moment less than ammonia. In NF_3 the dipole moment of lone pair and N – F bonds are acting in opposite direction. Therefore its dipole moment is nearly zero. Therefore, the dipole moments of these molecules will be in the order $NF_3 < NCl_3 < NH_3$.

(iii) AB_4 Molecules: AB_4 molecules have zero dipole moment if they have perfectly symmetric geometry. If AB_4 molecules have some dipole moment they have distorted geometries. For example, molecules like CH_4, CCl_4 $SiCl_4$ which are perfectly symmetric tetrahedral structure have zero dipole moment. But CH_3Cl, CH_2Cl_2, $CHCl_3$ have dipole moments due to which they have distorted tetrahedral structures with slightly different bond angles. The order of dipole moments of these molecules is

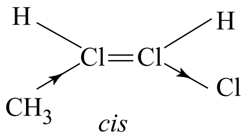
CCl_4	$< CHCl_3$	$< CH_2Cl_2$	$< CH_2Cl$
	1.01 D	1.60 D	1.87 D

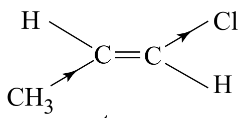
(c) Distinguishion between cis- and trans-isomers: Dipole moments are useful in identifying the cis- and trans-isomers. The trans-isomers usually possesses either zero dipole moment or very low value in comparison to the cis-form.


$\propto \simeq 1.9D$

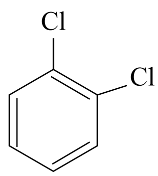

$\propto \simeq 0$

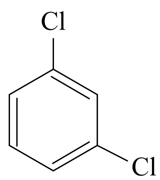
If the two groups have opposite inductive character, then the trans-isomer will have greater dipole moment.

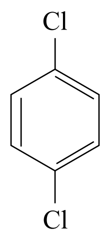

cis


trans

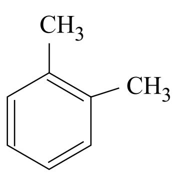
(d) Orientation in benzene ring: Dipole moment is useful to acertain the orientation of substituents. The paraisomers have zero dipole moment if both groups are similar and in plane. Orthoisomers have more dipole moment than metaisomers.

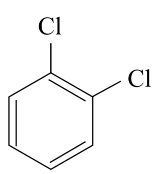

ortho
$\propto = 2.54$ D

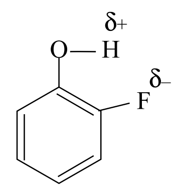

meta
$\propto = 1.48$ D


para
$\propto = 0$

In orthoisomers, experimental value of dipole moment is found different from theoritical value due to interaction between groups.


$\mu_{exp} = \mu_{theo}$


$\mu_{exp} < \mu_{theo}$


$\mu_{exp} > \mu_{theo}$

(e) Bond moment: The contribution of individual bonds in the dipole moment of a polyatomic molecule is termed bond moment. If the experimental dipole moment and bond angle of a triatomic molecule are known the individual bond moments can be calculated by using the formulae.

Observed dipole moment

$$\mu_{obs} = 2(\text{bond moment})(\cos \theta/2) \tag{3.26}$$

or observed dipole moment

$$\mu_{obs} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1 \mu_2 \cos\theta} \tag{3.27}$$

where θ is the bond angle.

Illustrated Example 3: The dipole moment of H_2S is 0.95 D. Find the bond moment if the bond angle is 97° (cos 48.5° = 0.662).

Solution

$\mu_{obs} = 2(\text{bond moment})(\cos\theta/2)$

$\therefore\ 0.95 = 2\ (\text{bond moment})\ (\cos 48.5°)$

$0.95 = 2\ (\text{bond moment})\ (0.662)$

μ_{S-H} Bond moment = 0.72D

3.12 HYDROGEN BONDING

Hydrogen atom is in bond with highly electronegative atoms like F, O or N. The hydrogen atom carries partial positive charge while the more electronegative atoms (F, O or N) carries negative charge due to difference in electronegativities. If hydrogen atom carrying partial positive charge come closer to the more electronegative atom (F, O or N) of another molecule of same compound or different compound. there exists some weak electrostatic attraction between positively charged hydrogen and negatively charged more electronegative atom (F, O or N). This **weak electrostatic attractive force which binds hydrogen atom of one molecule with more electronegative atom (F, O or N) of another molecule is known as hydrogen bond.**

Hydrogen bond is much weaker than a covalent bond. The strength of hydrogen bond ranges from 10 to 40 KJ mol^{-1}. It may also be noted that the bond length of hydrogen bond is larger than that of a covalent bond. For example, the covalent H – F bond length is 109 pm while the hydrogen bond length between F and H is 155 pm. To distinguish the hydrogen bond, it is written as a dotted line ex: (F.......... H – F). The tendency to form hydrogen bond increases rapidly from C – H through N – H and O – H to F – H with increase in electronegativity and it decreases in passing from O – H to S – H or from F – H to Cl – H with decrease in electronegativity. This shows that the tendency to form hydrogen bond increases with the increase in ionic character of the bond i.e., when the bond has the greatest polarity. Fluorine, with the highest electronegativity, forms the strongest hydrogen bond, and by far the greatest number of hydrogen bonds known are those which unite two oxygen atoms. Though nitrogen and chlorine have same electronegativity values (3.0), chlorine has less tendency to participate in hydrogen

Table 3.15 Enthalpy of dissociation of H-bonded pairs in the gas phase ΔH_{298} (A – H .... B) /KJ mol^{-1}

Weak		**Medium**		**Strong**	
HSH ....SH_2	7	FH .... FH	29	HOH .... Cl^-	55
NCH ....NCH	16	ClH .... OMe_2	30	$HCONH_2$....... OCH NH_2	59
H_2NH .... NH_3	17	FH .... OH_2	38	HCOOH ... OCHOH	59
MeOH ....OHMe	19			HOH ....F^-	98
HOH.......... OH_2	22			H_2OH^+ .... OH_2	151
				FH ..... F^-	169
				H COOH ..... F^-	~200

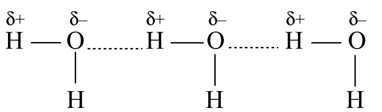
bond due to bigger size. For example, NH_3 shows hydrogen bonding while HC*l* does not. Enthalpy of dissociation of hydrogen bonded pair in the gas phase ΔH_{298} (A – H ...... B KJ mol^{-1}) are given in Table 3.15.

3.12.1 Types of Hydrogen Bond

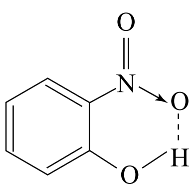
Hydrogen bonds are of two types.
(a) Intermolecular hydrogen bond
(b) Intramolecular hydrogen bond

(a) Intermolecular hydrogen bond: Intermolecular hydrogen bond is formed between two different molecules of the same or different substances. For example, hydrogen bond between the molecules of HF, H_2O or between H_2O and alcohols.

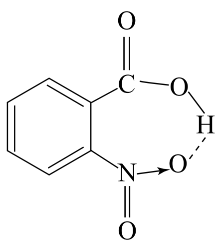
$$\overset{\delta+}{H}-\overset{\delta-}{F}\cdots\cdots\overset{\delta+}{H}-\overset{\delta-}{F}\cdots\cdots\overset{\delta+}{H}-\overset{\delta-}{F}$$



(b) Intramolecular hydrogen bond: Intramolecular hydrogen bond is formed between the hydrogen atom and the highly electronegative atom (F, O or N) present in the same molecule. Intramoelcular hydrogen bond results in the **chelation** (cyclisation) of the molecules. For example, the intramolecular hydrogen bond in onitrophenol and onitrobenzoic acid can be shown as follows



Orthonitrophenol



Orthonitrobenzoic acid

3.12.2 Influence of Hydrogen Bonding on the Properties

Hydrogen bonding has important effect on many physical properties such as melting point, boiling point, solubility etc., of the compounds.

(i) Physical state: Hydrogen bonding influences the physical state of the substances (solid, liquid or gas). For example, water is a liquid while H_2S is a gas. In water, oxygen is highly electronegative so that it forms hydrogen bonds. As a result the molecules of H_2O get associated with one another so that water exists as liquid. On the other hand, the electronegativity difference of atoms in H_2S (H = 2.1 and S = 2.5) is almost negligible. As a result H_2S is not associated and exists as a gas at room temperature. Similarly the existence of C_2H_5OH as liquid and C_3H_8 as gas at room temperature can be explained.

(ii) Melting points and Boiling points: The compounds containing hydrogen bonds have high melting and boiling points. It is due to hydrogen bonding the electrostatic attractive force in the molecule becomes large. Consequently, larger energy is required to separate these molecules before they can melt or boil. The melting and boiling points of the hydrides of groups 14, 15, 16 and 17 are given in Table 3.16.

It is clear from Table 3.16 that the melting and boiling points increase as the molecular mass increases in group 14. This is mainly due to the fact that the size of elements of group 14 increases, the number of electrons also increases. As a result, van der Waal's forces also increases and therefore melting and boiling points increase. However, hydrides of groups 15, 16 and 17 do not show this trend. In these groups, the melting and boiling points also increase with increase in molecular mass with the exception of first member. The first members NH_3 (group 15), H_2O (group 16) and HF (group 17) have abnormally, high melting and boiling points. The high melting and boiling points of the first member of each group is due to the intermolecular hydrogen bonding shown by these compounds.

Table 3.16 M.pts and B.pts (K) of the hydrides of groups 14, 15, 16 and 17

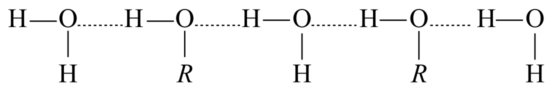
Hydride	M.pt	B.pt.
	Group – 14	
CH_4	89.0	111.5
SiH_4	88.0	161.2
GeH_4	108.0	183.0
SnH_4	23.0	221.0
	Group – 15	
NH_3	195.5	239.5
PH_3	138.0	185.0
AsH_3	159.0	218.0
SbH_3	184.0	256.0
	Group – 16	
H_2O	273.0	373.0
H_2S	190.3	211.2
H_2Se	209.0	231.0
H_2Te	222.0	271.0
	Group – 17	
HF	180.7	292.4
HC*l*	161.0	199.4
HBr	184.5	206.0
HI	222.2	237.0

The boiling points of HF and H_2O are maximum in the respective groups, the boiling point of NH_3 is greater than PH_3 and AsH_3 but less than SbH_3.. This is because the hydrogen bond in NH_3 is weaker when compared with H_2O and HF due to less polarity. The van der Waal's attractive forces dominate in SbH_3 due to bigger size.

The hydrogen bond in H–F is stronger than in H_2O due to greater polarity of H–F bond. Hence the boiling point of water is expected to be less than that of HF. But practically water has higher boiling point. To explain this it is suggested that because of two O–H bonds in H_2O molecule, the number of hydrogen bonds per water molecule are almost twice than those per HF molecule. Therefore, more energy is required to break greater number of hydrogen bonds in water. Secondly hydrogen fluoride exists as $(HF)_6$ even in vapour state i.e., all hydrogen bonds in HF are not broken before it is converted into vapour. But during the conversion of water into vapour all hydrogen bonds should be broken. This means that lesser energy is utilized to break hydrogen bonds in HF than in H_2O due to which the boiling point of water is higher than that of HF.

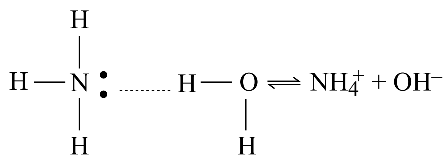
The compounds like H_2O which exist in liquid state due to association of molecules through hydrogen bond are called **associated liquids.** The compounds like CCl_4, $CHCl_3$, CS_2 etc. which exist in liquid state due to van der Waal's attractive forces are called **normal liquids.** Generally, associated liquids have more melting and boiling points than normal liquids.

(iii) Solubility: Hydrogen bond influences the solubility of several compounds in water. Several organic compounds like alcohols, amines, urea, sugar, glucose, etc. which are covalent dissolve in water due to hydrogen bonding. For example, alcohols are soluble in water due to hydrogen bonding as shown below.

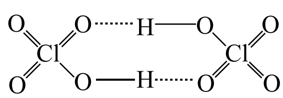


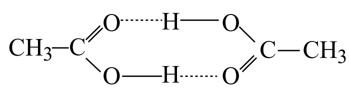
The association of molecules of different substances through hydrogen bonding is known as coassociation. Due to the coassociation of alcohols with water molecules, they are soluble in water. The OH group in alcohol is hydrophilic while the alkyl group is hydrophobic. The hydrophobic nature of alkyl group increases with increase in its size. Therefore, the solubility of alcohols decreases with increase in size of alkyl group. Methyl and ethyl alcohols are completely miscible with water in any proportions, but propyl and butyl alcohols are less soluble while higher alcohols are insoluble.

When ammonia is dissolved in water nitrogen atom in ammonia forms hydrogen bond with hydrogen atom of water through its lone pair but it does not readily form hydrogen bond with its hydrogen atom with oxygen atom of water. Because of this reason ammonia when dissolved in water forms NH_4OH and ionizes giving NH_4^+ and OH^- ions.



(iv) Abnormal Molecular Weights: Certain compounds like perchloric acid, carboxylic acids dimerise due to hydrogen bonding. For such compounds when molecular weights are determined using colligative properties, higher molecular weights (abonrmal molecular weights) will be obtained.


Perchloric acid


Acetic acid

(v) Properties of amines: The association and basic strength of amines are both explained interms of hydrogen bonding.

Primary and secondary amines are associated to some extent, though not greatly because nitrogen has less tendency to form hydrogen bonds. Tertiary amines are not associated at all for they have no hydrogen atom. Thus trimethyl amine, which is not associated, has a lower boiling point (4°C) than dimethyl amine (7°C) even though it has a high relative molecular mass.

In aqueous solution, amines react with water molecules as shown here under.

$$CH_3\text{–}NH_2 + H_2O \rightleftharpoons CH_3\text{—}\underset{\underset{H}{|}}{\overset{\overset{H}{|}}{N}}\text{—}H...O\text{—}H \rightleftharpoons (CH_3\text{–}NH_3)^+ + OH^-$$

The resulting solution will contain some CH_3NH_3OH molecules together with some $CH_3NH_3^+$ and OH^- ions. In a solution of a quarternery base, e.g., tetramethyl ammonium hydroxide $[(CH_3)_4N]OH$, however, there are no hydrogen atoms which could form hydrogen bonds. As a result the solution contain only $(CH_3)_4N^+$ and OH^- ions. This explains why quarternary bases are very much stronger than primary, secondary or tertiary amines. It was, infact, the marked basic strength or quarternary bases which first led Moore and Winmill (1912) to suggest the possible existence of hydrogen bonds.

(vi) Influence on the structure of compounds

(a) Ice: The crystal structure of ice shows a tetrahedral arrangement of water molecules.

The central oxygen atom is surrounded tetrahedrally by the oxygen atoms. All other oxygen atoms are arranged similarly. The arrangement of water molecules in ice is a very open structure and this explains the low density of ice. When ice melts, the structure breaks down and the molecules pack more closely together so that water has a high density. This breaking down process is not complete until a temperature of 4°C is reached. Again above 4°C due to expansion, density of water decreases with increase in temperature.

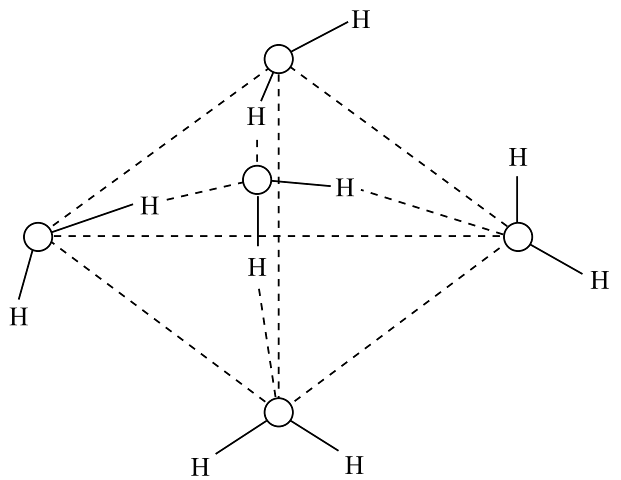


Fig 3.55 The crystal sturcture of ice

(b) $CuSO_4 \cdot 5H_2O$: Gentle heating of copper (II) sulphate –5 water will convert it into the monohydrate; it is infact efflorescent in hot, dry climates. A much higher temperature is required, to remove the final molecule of water of crystallization from the monohydrate. This suggests that one of the molecules of water of crystallization is different from the other four. Each Cu^{2+} ion is surrounded octahedrally by six oxygen atoms, four of them being in water molecules and two in sulphate ions. The extra water molecule is situated between two such octahedral groups. It is not attached directly to any Cu^{2+} ions; instead, it is linked by hydrogen bonds between the water and sulphate groups surrounding the Cu^{2+} ions. $CuSO_4{\cdot}5H_2O$ exist but $CuSO_4{\cdot}\ 5NH_3$ does not exist probably due to the fact that NH_3 molecules will not form hydrogen bonds so readily as H_2O molecules.

(c) Ammonium fluoride: In ammonium fluroide the radius ratio is 0.92 so that caesium chloride structure might be expected. Instead, ammonium fluoride crystallizes with wurtzite structure in which each ion is surrounded tetrahedrally by four F^- ions. The difference is due to the formation of hydrogen bonds between F^- ions and the hydrogen atoms of the ion. Such bonds lock the ions in position, but they are formed only in the fluoride because of the high electronegativity of fluroine.

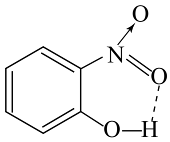
The crystal structure of many compounds is dominated by the effect of hydrogen bonds and numerous examples will be provided in the following chapters.

The hydrogen fluoride in liquid state has a zig-zag structure due to the presence of lone pairs on fluroine but not a linear structure.

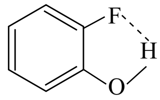
Influence of Intramolecular Hydrogen boindng:

2-nitrophenol boils at 214°C as compared with 290°C for 3- and 279°C for 4-nitrophenol. 2-nitrophenol is, moreover, volatile in steam and less soluble in water than the other two isomers. The intramolecular hydrogen bonding in 2-isomers prevents intermolecular bonding between two or more molecules. But in the 3- and 4-isomers, intra molecular bonding is not possible because of the size of the ring which would have to be formed. Intermolecular bonding therefore takes place and this cause, some degree of association, which accounts for the higher boiling points of the 3- and 4-isomers.

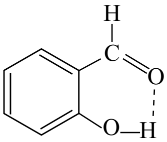
The low solubility of 2-nitrophenol may be explained in two ways. The formation of an internal hydrogen bond "suppresses" the hydroxylic character of the compound and this causes a lowering of solubility in water. In other words, the formation of an internal hydrogen bond prevents hydrogen bonding between 2-nitrophenol and water and this results in reduced solubility.



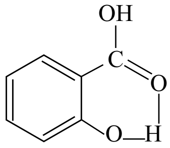
o-Nitrophenol



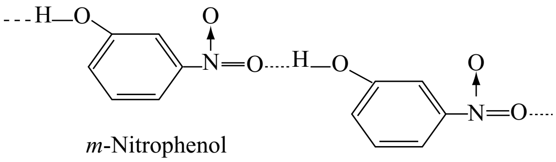
o-Flurophenol



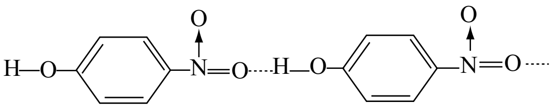
o-Hydroxy benzaldehdye
(or)Salicylaldehyde



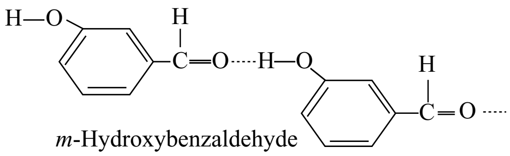
o-Hydroxy benzoic acid
(or)Salicylic acid



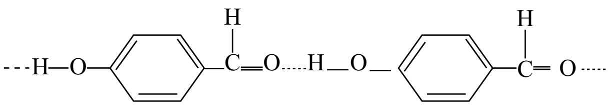
m-Nitrophenol



p-Nitrophenol



m-Hydroxybenzaldehyde



p-Hydroxybenzaldehyde

3.13 BOND PARAMETERS

3.13.1 Bond Order

Bond order is the number of bonds between two atoms in a bond. In Lewis description of covalent bond the bond order is the number of electron pairs shared between two atoms in a molecule. The bond order, for example in H_2 (one shared electron pair), in O_2 (with two shared electron pairs) and in N_2 (with three shared electron pairs, is 1,2 and 3, respctively. Similarly in CO (three shared electron pairs between C and O), the bond order is 3.

According to the MO theory the bond order is one half the difference between the number of electrons present in the bonding and anti-bonding orbitals.

In the resonance hybrid structure the bond order is related as follows.

$$\text{Bond order} = \frac{\text{Total no. of bonds arounds the central atom}}{\text{Total no. of atoms around the central atom}}$$

In cyclic compounds the bond order may be calculated as

$$\text{Bond order} = \frac{\text{Total no. of bonds in the ring}}{\text{Total no. of atoms in the ring}}$$

Isoelectronic molecules and ions have identical bond order; for example, F_2 and O_2^{2-} have bond order 1, N_2,CO and NO^+ have bond order 3.

In Lewis description of covalent bond, bond order is always an integer. But the bond orders calculated from the MO theory or in resonance hybrid structures may be fractional or integers. For example, the bond order in NO is 2.5 as per the MO theory. Similarly the bond order in BF_3, NO_3^-, CO_3^{2-} is 1.33, in benzene, SO_4^{2-} is 1.5 etc. as per resonance.

3.13.2 Bond Length

Bond length is the equilibrium distance between the nuclei of two bonded atoms in a molecule. The bonded atoms in a molecule always vibrate about their equilibrium positions so that the distance between the nuclei of these two atoms does not remain constant. It is an average value. Bond length is measured in angstroms (1×10^{-8} cm), nanometers (1×10^{-9} cm) or in picometers. (1×10^{-12} m). In covalent molecules the bond length is equal to the sum of the covalent radii of two atoms in bond. The bond lengths of some common bonds are given in Table 3.17.

Factors influencing the bond length:

(i) The magnitude of the bond length between the similar atoms in different molecules is generally same. **Eg:** "O–H" bond length in H_2O or H_2O_2 or C_2H_5O–H is the same and equal to 96 pm.

(ii) With increase in the bond order (number of bonds) the bond lengths decreases. **Eg:** C – C bond lengths in ethane, ethylene and acetylene with single, double and triple bonds, respectively are 154 pm, 134 pm and 120 pm.

(iii) The bond length also depends on the type of hybridization of the atom with which another atom is in bond.

Eg: C_{sp^3}–H in alkane = 110 pm
C_{sp^2}– H in alkene = 108 pm
C_{sp} – H in alkyne = 106 pm

(iv) Effect of ionic character: The sum of the normal covalent radii of two atoms A and B only gives the bond length A–B, when the bond is purely covalent or nearly so. In other cases the measured bond length is less than the sum of the covalent radii for the bonded atoms an effect which is attributed to the partial ionic character. In such cases the bond length may be calculated as

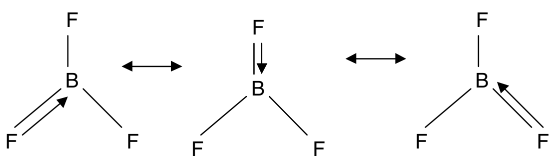
$A — B = r_A + r_B + C(\chi_A - \chi_B)$ where r_A and r_B are the covalent radii of A and B and χ_A and χ_B their respective electronegativities. The value of C depends on the type of atoms involved. For example, for bonds involving atoms

Table 3.17 Bond lengths of some common bonds

Bond	Bond length /pm	Molecule	Bond	Bond length /pm	Molecule
H – H	74	H_2	Si - C	189	$Si(CH_3)_4$
H – F	92	HF	Si – C*l*	208	$SiCl_4$
H – C*l*	127	HC*l*	Si – H	148	SiH_4
H – Br	141	HBr	Ge – C	198	$Ge(CH_3)_4$
B – Br	187	BBr_3	Ge – C*l*	208	$GeCl_4$
B – C	156	$B(CH_3)_3$	Ge – H	153	GeH_4
B – C*l*	175	BCl_3	Sn – C	218	$Sn(CH_3)_4$
B – F	130	BF_3	Sn – C*l*	231	$SnCl_4$
B – O	136	$B(OH)_3$	N – C	147	$N(CH_3)_3$
C – Br	194	CBr_4	N – F	137	NF_3
C – C	154	C_2H_6	N – H	101	NH_3
C = C	133	C_2H_4	P – C	184	$P(CH_3)_3$
C ≡ C	120	C_2H_2	P – C*l*	204	PCl_3
C – C*l*	177	CCl_4	P – F	155	PF_3
C – F	132	CF_4	P – H	142	PH_3
C – H	110	CH_4	P – O	160	P_4O_{10}
C – N	147	$N(CH_3)_3$	P = O	145	$POCl_3$
C ≡ N	115	HCN	As – C	196	$As(CH_3)_3$
C – O	141	$(CH_3)_2O$	As – C*l*	216	$AsCl_3$
C = O	122	$(CH_3)_2CO$	As – F	171	AsF_3
C – S	180	$(CH_3)_2S$	As – H	152	AsH_3
O – C*l*	170	OCl_2	O – F	141	OF_2
O – H	96	OH_2	S – C*l*	199	SCl_2
S – H	132	H_2S	S – F	158	SF_6
Se – H	146	SeH_2	S – O	141	SO_2F_2

of second period C = 0.08. For bonds between Si, P and S bonded to more electronegative atoms not belonging to first period C = 0.06. This equation is useful in many cases but it fails to account for all the measured bond lengths.

(v) Resonance effect: Due to resonance hybridization the normal bond lengths in several molecules will become less than the sum of the covalent radii. For example, the B – F bond length (130 pm) in BF_3 is less than the sum of covalent radii of boron (80 + 72 = 152 pm). This is because of the resonance hybridization of the following resonance structures due to back bonding.



The resonance effect on bond lengths is discussed in section 3.10.3.

(vi) Size of the atom: The bond length increases with increase in size of the atom. It is also clear from the table that the bond lengths for a given family increases with increase in atomic number.

(viii) Number of bonds: With increase in the number of bonds between two atoms bonds length decreases. Thus C ≡ C bond length is shorter than C = C bond which in turn is shorter than C – C bond i.e.,

C≡ C < C = C < C – C.

Simlarly, N ≡ N < N = N < N – N and O = O < O – O

3.13.3 Bond Energy

We have learnt that when a bond is formed between two atoms, energy is released. Obviously the same amount of energy will be required to break the bond. Bond energy can be defined as the **amount of energy required to break one mole of bonds of a particular type between two atoms in the gaseous state.** Bond energy is expressed in KJ mol^{-1}. For example, H – H bond energy in H_2 molecule is 435.8 kJ mol^{-1}.

The bond energy depends on the mode of cleavage of the bond. Diatomic molecule may be either homoatomic as

A_2 or heteroatomic as "AB". Then, the bond cleavage may occur in follwing ways.

$A:A_{(g)} \xrightarrow{\text{energy}} A\cdot_{(g)} + A\cdot_{(g)}$ Homolytic cleavage

$A:B_{(g)} \xrightarrow{\text{energy}} A\cdot_{(g)} + B\cdot_{(g)}$ Homolytic cleavage

Shared electron pair is equally divided.

$A:B \xrightarrow{\text{energy}} A^{+} + :B^{-}$ Heterolytic cleavage

$A:B \xrightarrow{\text{energy}} A\overset{-}{:} + B^{+}$ Heterolytic cleavage

Shared electron pair is unequally divided.

The energy required for the heterolytic cleavage of a bond is higher than that required for homolytic cleavage. The following example illustrates this.

$$C_2H_5 - Br \xrightarrow{\text{Homolytic cleavage}} (C_2H_5\cdot) + (\cdot Br);$$

Bond energy = 280.9 kJ mol^{-1}

$$C_2H_5 - Br \xrightarrow{\text{Heterolytic cleavage}} (C_2H_5^{+}) + (Br^{-});$$

Bond energy = 764.94 kJ mol^{-1}

In a polyatomic molecule, containing more than one covalent bond, the term bond energy can be interpreted in two ways. The C – H bond energy in methane, for example, might be taken as one quarter of the heat of formation of methane from its atoms i.e., one quarter of ΔH in the reaction.

$$\underset{\text{atom}}{C} + \underset{\text{atoms}}{4H} \rightarrow CH_4 \quad \Delta H = 1653.2 \text{ kJ mol}^{-1}$$

Such a bond energy is really **an average bond energy.** Its value for C – H is 413.3 kJ mol^{-1}.

Alternatively, the C – H bond energy in methane might be taken as the energy required to split one C – H bond in the molecule to form CH_3 and H as the heat of reaction D_{CH_3-H}.

$$CH_4 \longrightarrow CH_3 + H \qquad D_{CH_3-H} = 431.0 \text{ kJ mol}^{-1}$$
$$CH_3 \longrightarrow CH_2 + H \qquad D_{CH_2-H} = 364.0 \text{ kJ mol}^{-1}$$
$$CH_2 \longrightarrow CH + H \qquad D_{CH-H} = 523.0 \text{ kJ mol}^{-1}$$
$$CH \longrightarrow C + H \qquad D_{C-H} = 338.9 \text{ kJ mol}^{-1}$$

It will be seen that none of these bond dissociation energy values is equal to the average bond energy for C – H bond in methane (413.4 kJ mol^{-1}), but the sum of all four bond dissocation energies (1656.9 kJ mol^{-1}) is equal to 4×413.4 (1653.6) kJ.

Similarly the average bond energy for O – H bond in water is 461.2 KJ mol^{-1}, whereas the two bond-dissociation energies concerned are

$$H_2O \longrightarrow H + OH \qquad D_{H-OH} = 495.9 \text{ kJ mol}^{-1}$$
$$OH \longrightarrow H + O \qquad D_{H-O} = 426.5 \text{ kJ mol}^{-1}$$

as would be expected 2×461.2 is equal to 495.9 + 426.5. The bond dissociation enthalpies of some common bonds are given in Table 3.18.

Factors Infuencing the Bond energy:

(i) Number of bonds: As the number of bonds between similar atoms increases, the overall bond strength also increases. For example the C – C single, double and triple bond energies are 341.1, 610.7 and 827.6 kJ mol^{-1}, respectively. It should be noted that π bond is weaker than σ bond. The C – C double bond energy is not double to the C – C single bond energy because of π bond. The π bond energy will be about (827.6 – 610.7) = 215.9 kJ mol^{-1}.

(ii) Size of atom: Generally, elements of the same group form similar type of bonds. The bond energies of

Table 3.18 Bond dissociation enthalpies of some common bonds in KJ mol^{-1}

			Bond energies of single bonds			
H — H	C — C	N — N	O — O	S — S	F — F	F—H
435.8	347.0	163.02	146.0	266.9	137.7	568.2
	C — N	N — H	O — H	S — H	*Cl* — *Cl*	*Cl* — H
	290.0	388.7	464.0	366.1	243.5	431.9
	C — O				Br — Br	Br – H
	341.0				192.8	366.1
	C — H	Si — Si			I — I	I — H
	414	177.8			151.0	298.7
			Bond energies of multiple bonds			
C = C	C = N	N = N	C = O	O = O		
619.0	564.8	408.4	723.8	498		
C ≡ C	C ≡ O	N ≡ N				
836	1071	946.4				

similar type of bonds decreases down the group. For example, the bond energies of hydrogen halides are

H — F	H – C*l*	H – Br	H – I	
568.2	431.9	366.1	298.3	kJ mol^{-1}

(iii) Presence of lone pairs: As the number of lone pairs on bonded atoms increases, the bond energy decreases.

— C — C —	$-\ddot{N}-\ddot{N}-$	$-\ddot{\underset{..}{O}}-\ddot{\underset{..}{O}}-$	$:\ddot{\underset{..}{F}}-\ddot{\underset{..}{F}}:$
347	163.02	146.0	137.7

(iv) Effect of Resonance: Due to resonance hybridization bond energy increases, as the stability of resonance hybrid structure is more. To break all the bonds in a resonance hybrid structure the energy required will be more than the calculated values indicating the increase in bond energy. For example, the energy required to break all the bonds in benzene is (5501 kJ mol^{-1}) more than the calculated value (5339) from bond energies.

3.13.4 Bond Angle

Bond angle is the average angle between the orbitals containing bonding electron pairs around the central atom in a molecule.

The effect of various factors on bond angles have already been discussed in different parts of this chapter like VSEPR theory, hybridization, etc.

3.14 NON-BONDING INTERACTIONS BETWEEN MOLECULES

The forces between the molecules of a substance are known as **intermolecular forces.** Such forces exist in all states of matter and are responsible for many structural features and physical properties of matter. The **intramolecular forces** that exist within each molecule or polyatomic ion influences the chemical properties of the substance. Intermolecular forces are weak forces. The physical properties like boiling and melting points of substances reflect the strength of intermolecular forces operating among the molecules.

Intermolecular forces arise due to the following types of interactions. Ion-dipole, dipole-dipole, ion-induced dipole, dipole-induced dipole, dispersion forces and hydrogen bonding. The **dipole-dipole, dipole-induced dipole and dispersion forces are collectively called as van der Waal's forces.** Depending on the phase of the substance, the nature of chemical bonds and type of elements present more than one type of interaction may contribute to the total attraction between molecules.

3.14.1 Ion-dipole Interactions

Ion-dipole interactions are similar to ion-ion interactions except that they are more sensitive to distance (1/r^2 instead of 1/r) and tend to be somewhat weaker since the charges (q^+,q^-) comprising the dipole are usually considerably less than a full electronic charge.

Ion-dipole interactions attract an ion (either a cation or anion) and a polar molecule to each other. The charge density on the cation is higher than on anions, therefore, a cation interacts more strongly with dipoles than does an anion having same charge but with bigger size. Ion-dipole interactions are important in solutions of ionic compounds in polar solvents where solvated species such as [$Na(H_2O)_y$] and [$F(H_2O)_y$] (for solution of NaF in H_2O) exist. **Thus hydration of different ions is an example of ion-dipole interaction.** When an ionic compound like NaF dissolves, the water molecules act as a dielectric to keep the ions apart. Solvents like carbon tetrachloride, which is non-polar, lacks the ability to participate in ion dipole interaction. Therefore carbon tetrachloride is a poor solvent for ionic compounds as are most non-polar solvents.

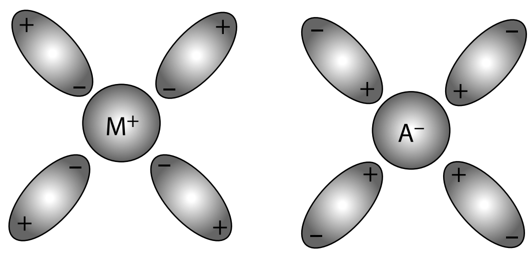


Fig 3.56 Ion-dipole interaction

3.14.2 Dipole-dipole Interactions

Dipole-dipole interactions depend upon the distance and orientation of the two dipoles. A favourable orientation of the two dipoles result in attractive dipole-dipole interactions. Dipole-dipole interactions tend to be even weaker than ion-dipole interactions and to fall off more rapidly with distance (1/r^3). Like ion-dipole forces, they are directional in the sense that there are certain preferred orientations and they are responsible for the association and structure of polar liquids.

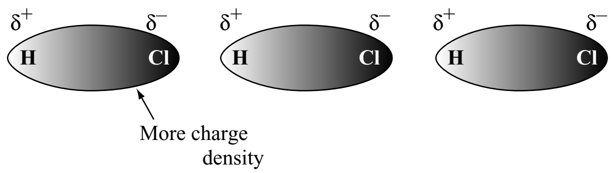


Fig 3.57 Dipole-dipole interaction between the HC*l* molecules

3.14.3 Ion-Induced Dipole Interactions

If a charged particle such as an ion, is introduced into the neighbourhood of an uncharged, non-polar molecule (e.g., an atom of a noble gas such as xenon), it will distort the electron cloud of the atom or molecule in much the same way that a charged cation can distort the electron cloud of a large, soft anion (Fajan's rules 3.4.3). The polarization of the neutral species will depend upon its inherent polarizability and on the polarizing field afforded by the charged ion $Z^{\pm}$. Similar interaction between a polar molecule and the induced dipole is called **dipole-induced dipole interaction.**

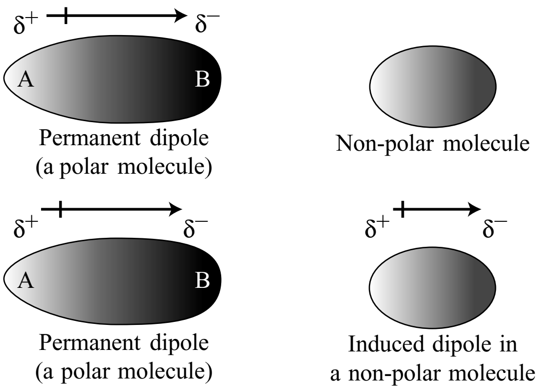


Fig 3.58 Induced dipole interaction

3.14.4 Instaneous Dipole-Induced Dipole interactions

In substances like O_2, N_2 or monoatomic gases such as He, Ne, Ar, etc. instantaneous dipoles are created. In such substances, the electrons are not confined in a particular molecular orbital / atomic orbital. These occupied molecular / atomic orbitals can be considered as "electron clouds". The shapes of these electron clouds can be altered by external forces. One such external force is the electric field of the electrons in nearby molecules. When two molecules / atoms approach each other closely as, in a liquid, the electron clouds of one molecule repel the electron clouds of the other. As a result, both molecules temporarily acquire small localized separation of charge called **instantaneous dipoles.** The deficiency of electron (positive charge) in one part of one molecule is attracted by the excess of electrons (negative charge) in part of another molecule. The attraction between induced dipoles is called **London dispersion forces.** Another way of looking at this phenomenon is to consider the electrons in two or more "non-polar" molecules as synchronizing their movement (at least partially) to minimize electron repulsion and maximize electron-nucleus attraction. Such attractions are extremely short ranged and weak, as are dipole-induced dipole forces.

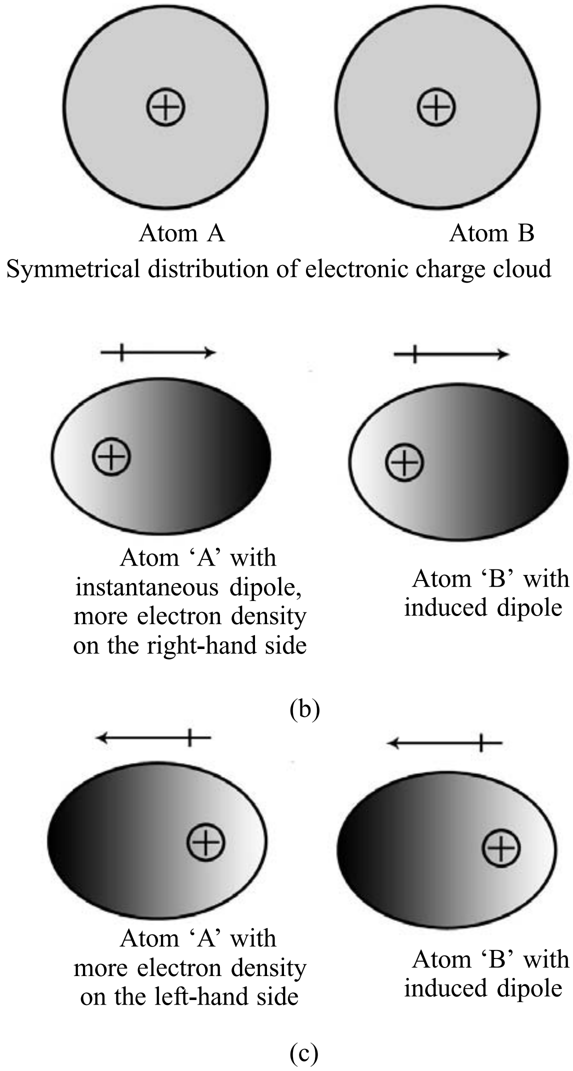
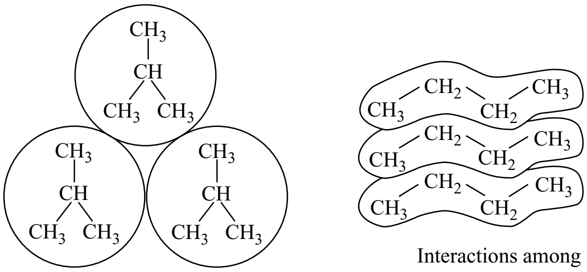


Fig 3.59 Dispersion forces or London forces between atoms

One important factor that determines the magnitude of van der Waal's forces is the relative polarizability of the electrons of the atoms involved. In the halogen family, for example, polarizability increases in the order F< C*l* < Br < I. Fluorine atoms show a very low polarizability because their electrons are very tightly held for they are close to the nucleus. Iodine atoms are large and hence are more easily polarized. Their valence electrons are far from the nucleus. Atoms with unshared pairs are generally more polarizable than those with only bonding pairs.

For example, fluorocarbons containing only carbon and fluorine have extraordinarily low boiling points when compared to hydrocarbon having same molecular weight. This is because of the very low polarizability of flourine atoms.



Interactions among spherical molecules
2-methylpropane
boiling point –10.2°C

Interactions among elongated molecules
butane
boiling point – 0.6°C

Fig 3.60 Comparison of intermolecular interactions for 2-methyl propane and butane

Since the van der Waal's interactions can act only through the parts of different molecules that are within a certain distance of each other, their three-dimensional shapes also play an important role. For example, the isomers butane and 2-methylpropane have different boiling points. 2-methylpropane is almost spherical, whereas butane is elongated. The molecules of butane have a greater surface area for interaction with each other than do the molecules of 2-methylpropane. The stronger interactions that are possible for butane are reflected in its boiling point, which is higher than the boiling point of 2-methylpropane.

Summary of chemical forces and interactions are given in Table 3.18. The forces are listed in order of decreasing strength from covalent and ionic bonds to the very weak London forces. Since London forces increase with increasing size and there is no limit to the size of molecules, these forces can become rather large. In general, however, they are very weak.

Table 3.19 Summary of chemical forces and interactions

Type of interaction	Strength	Energy-distance function
Covalent bond	very stong	complex, but comparatively long range
Ionic bond	very strong	1/r comparatively long range
Ion-dipole	strong	$1/r^2$ short range
Dipole-dipole	moderately strong	$1/r^3$ short range
Ion-induced dipole	weak	$1/r^4$ very short range
Dipole-induced dipole	very weak	$1/r^6$ extremely short range
London dispersion forces	very weak	$1/r^6$ extremely short range

The application of a knowledge of these forces to interpretation of chemical phenomena requires a certain amount of practice and chemical intention. For example, the boiling points of the noble gases are determined by London forces because no other forces are in operation. In a crystal of an ionic compound, however, although the London forces are still present they become negligible in comparison to the very strong ionic interactions to a first approximation.

KEY POINTS

- The force of attraction between atoms in a molecule is called **chemical bond**.
- Energy changes or the rearrangement of electrons are primarily responsible for the bond formation.
- The internal energy of a molecule is less than the sum of the internal energy of individual atoms in that atom and that difference is released outside as **bond energy**.
- In the formation of chemical bond there is always a decrease in potential energy and is the lowest at the inter atomic distance, i.e., the principle of minimum energy.

KOSSEL – LEWIS THEORY: OCTET THEORY

- **Kossel** proposed the electrovalent bond while **Lewis** proposed the covalent bond.
- Atoms combine to form molecules to decrease their internal energy and to acquire octet in their valence shell like inert gases. This is known as **octet rule**.
- Atoms acquire stable inert gas configuration in their valence shell by losing or by gaining or by sharing electrons when they combine with one another.

Lewis Dot Formulae

- **Lewis diagrams** provide a picture of bonding in molecule and ions in terms of the shared pairs of electrons and the octet rule.
- The molecules in which if all the atoms except hydrogen have not achieved octet are called electron **deficient molecules.** Ex. BF_3, $BeCl_2$, etc.

- The molecules in which the central atom is having an octet and if all the electron pairs are bond pairs are called **electron precise molecules**.
- The molecules, in which the central atom is having an octet but some are bond pairs and some are lone pairs are called **electron rich molecules**.
- Molecules whose central atom is having more than an octet of electrons are called **hypervalent molecules** also called **valence shell expansion** or **expanded octet** by utilizing the availability of valence shell d orbitals for bonding.

Octet Theory

- Octet theory could not explain the shape of molecules.
- Octet theory could not explain the relative stability of molecules being totally silent about the energy of molecule.

  Octet theory could not explain the formation of electron deficient and hypervalent molecules.

Formal Charge

- Formal charge is the apparent electronic charge of each atom in a molecule, based on Lewis diagrams.
- $$\text{Formal charge} = \begin{bmatrix}\text{No. of valence electrons} \\ \text{in a free atom of the element}\end{bmatrix} - \begin{bmatrix}\text{No. of unshared electrons} \\ \text{on the atom}\end{bmatrix} - \begin{bmatrix}\text{No. of bonds} \\ \text{to the atom}\end{bmatrix}$$

  (or) $F = [N_A - N_{LP} - N_{BP}]$
- Charge on the molecule or ion = sum of all the charges.
- Molecules having small formal charges are more stable.
- Charges on adjacent atoms are usually of opposite sign.
- More electronegative atoms should have negative charges rather than positive charges.
- Structure having formal charges of opposite sign should not be at longer distances.
- Structure in which there is largest difference in electronegativity between the nearest atoms is stable.

IONIC BOND

- The electrostatic attractive force between two oppositely charged ions formed by transfer of electrons from one atom to another is called an **ionic bond**.
- Ionic bond is formed between an atom of low ionization potential and an atom of more electron affinity with an electronegativity difference of about or greater than 1.7.
- An ionic bond having maximum ionic character is formed between caesium and fluorine.
- Favourable conditions for the formation of cation from an atom are (i) its ionization potential must be low, (ii) the ion should have low charge, (iii) the size of the atom should be large and (iv) the cation should have an octet (e.g., Ca^{2+}) rather than pseudoinert gas configuration (Zn^{2+}).
- Favourable conditions for the formation of anion are (i) electronegativity and electron affinity, should be high, (ii) small size of the atom and (iii) charge on the anion should be less.
- Large cation and small anion with less number of charges favour ionic bond formation with **more ionic character**.
- The bond formed between the atoms of more electronegative atoms will be due to electron sharing and it is a **covalent bond**.
- Elements in their lower oxidation state form ionic compounds while in the higher oxidation state form covalent compounds. Eg: $SnCl_2$ is ionic but $SnCl_4$ is covalent.

Covalent Character of Ionic Bond [Fajan's Rules]

- Increase in cationic size increases the ionic nature of the bond. Eg. $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$.
- Increase in anionic size favours the formation of covalent bond. E.g. CaF_2 is ionic while CaI_2 is more covalent.
- Highly charged cation or anion or both favour the covalent bond formation. E.g. NaCl is ionic while $AlCl_3$ is covalent.
- Cation with inert gas configuration favour the ionic bond formation while the cation with pseudoinert gas configuration favour the covalent bond formation Eg: $CaCl_2$ is ionic while $ZnCl_2$ is covalent.
- When the ionic bond is formed, the oppositely charged ions approach an equilibrium distance where the attractive forces overcome the repulsive forces.

Crystal Lattice Energy

- **Lattice energy** is the amount of energy released when one mole of solid is formed due to the attraction between oppositely charged ions which are at infinite distance.
- The amount of energy absorbed when one mole of solid crystal is broken into ions and the ions are separated to infinite distance.
- If lattice energy is more, the stability of the crystal is more and thus the strength of the ionic bond is more.

- For lattice energy to be high the cation and anion should be small in size, bond length should be less and they should have more charge.
- Lattice energies are calculated using Born–Haber cycle and Hess law of constant heat summation.
- Lattice energy of compound depends on both attractive and repulsive forces between the ions.
- Lattice energy may be calculated by using Born–lande equation or Born–Mayer equation or by Kapustinski equation.
- **Born–Lande equation** $U = -\dfrac{Z^+Z^-e^2AN}{4\pi \in R_e}\left[1-\dfrac{1}{n}\right]$

  where Z^+ and Z^- are the number of charges on ion, e is the charge on ion, A is called Madelung constant, R_e is the distance between the nearest oppositely charged ions and N is Avogadro number.
- **Born–Mayer equation** $U = -\dfrac{Z^+Z^-e^2AN}{r^+ + r^-}\left[1-\dfrac{\rho}{r^+ + r^-}\right]$
- ρ is the constant for ions with inert gas configuration and is about 0.35×10^{-10} m. r^+ and r^- are the radii of cation and anion respectively.
- **Kapustinski equation** $U = -\dfrac{n\,(Z^+Z^-)}{r^+ + r^-}\left[1-\dfrac{34.5}{r^+ + r^-}\right]\kappa$
- Where κ is a constant and its value is 1.21×10^5 KJ pm mol^{-1}, n is the number of ions in the formula considered, e.g., n = 2 for NaCl, 3 for CaF_2 etc. r_+ and r_- are the ionic radii.

Born–Haber Cycle

- Born–Haber cycle is useful for calculating lattice energy of ionic compounds.
- Born–Haber cycle is based on Hess's law of constant heat summation.

Coordination Number

- Number of oppositely charged ions surrounding a particular ion in an ionic crystal is called coordination number.
- Coordination number of Na^+ and Cl^- ions in sodium chloride crystal is 6.
- Coordination number of Cs^+ and Cl^- ions in caesium chloride is 8.
- Limiting radius ratio $= \dfrac{\text{radius of small ion}}{\text{radius of large ion}}$

  i.e. $\dfrac{\text{radius of cation}}{\text{radius of anion}}$ (Table 3.4)
- Unit cell is the smallest portion of the lattice which has the same structure as lattice and is extended in all directions.
- Total no. of ionic pairs (Na^+Cl^-) belonging to one unit cell of sodium chloride is 4.
- Total no. of ionic pairs (Cs^+Cl^-) belonging to one unit cell of caesium chloride is 1.
- An ion at a corner of cube will be shared by eight unit cells and the contribution of an ion to one unit cell is 1/8.
- An ion at the edge centre of a cubic cell will be shared by four unit cells and its contribution to one unit cell is 1/4.
- An ion at the face of a cubic cell is shared by two unit cells and its contribution to one unit cell is 1/2.
- An ion at the body centre of a cubic cell will be shared by only one unit cell and its contribution to one unit cell is 1.
- Sodium chloride crystal has **face centered cubic** lattice structure while caesium chloride crystal has **body centered cubic** lattice.

Characteristics of Ionic Compounds

- Melting and boiling points of ionic compounds are very high because to break the strong electrostatic attractive forces, high amount of energy is required.
- Ionic compounds are soluble in polar solvents like water but insoluble in non-polar solvents like benzene.
- When the ionic compounds are dissolved in polar solvents the ions get solvated which decreases the inter ionic forces and thus dissolves.
- Ionic compounds are non-conductors of electricity in solid state but conducts electricity in fused state or in aqueous solution state due to the free moving ions.
- Ionic compounds undergo electrolysis in fused state and in aqueous solution and hence they are called electrolytes.
- Ionic bonds are non-directional, hence they do not exhibit stereoisomerism.
- Ionic compounds shows the reactions of the constituent ions and the reactions are very fast.

Consequences of Lattice Energies

- The more the number of charges on the ions in an ionic compound the more is the lattice energy, thus having more melting points, more stability. E.g.: NaF has low m.p. (997°C) while MgO has high m,p. (2800°C) though NaF is more ionic than MgO.
- Increasing charge on ions results in increasing covalent character particularly when small cation combines with large anion due to which their m.p and b.p decreases. E.g.: silver halides having covalent character have low m.p and b.p (Table 3.5) than alkali metal halides.
- Ionic compounds having large difference in their ionic radii are more soluble, but the compounds having ions of equal size are least soluble.

- The solubilities of Group - II sulphates decreases from $MgSO_4$ to $BaSO_4$ down the group as the difference between the sizes of ions goes on decreasing while the solubilities of hydroxides and oxides increases from magnesium to barium as the difference in the sizes of ions increases.
- Ionic compounds dissolve in water when the sum of hydration energies of cation and anion of an ionic compound exceeds the lattice energy of that compound.
- Both hydration and lattice energies decreases with increase in the sizes of the cation and anion but the decrease in hydration energy is rapid than the lattice energy with increase in size of cation combined with large anion. Hence, solubility of ionic compounds containing large anion decreases compared with the compounds having small anion.
- If enthalpy of hydration is greater than the lattice energy of an ionic compound, the dissolution is exothermic. If enthalpy of solution for ionic compound is more, cooling effect takes place during the dissolution. This is because some work is done to pull apart the ions and increase in entropy in adiabatic process, at the expense of internal energy.
- If the enthalpy of solution is sufficiently positive and the favourable entropy may not be able to overcome it, the compound will be insoluble.
- Thermal stability of ionic compounds increases with increase in lattice energy.
- Thermal stability of ionic compounds containing anions like carbonate, sulphate, nitrate, superoxide, etc. increases with increase in the size of cation though lattice energy decreases.
- When the bigger anions like carbonate, sulphate, nitrate or super oxide decompose, they give small anion (O^{2-}). This when combined with small cation gives more lattice energy that compensate for the decomposition energy of the compound containing small cation and large anion but cannot compensate for the decomposition energy of the compound containing large cation and large anion because the lattice energy of large cation and small anion is also small.

$$\text{E.g.: } MgCO_{3(s)} \xrightarrow{\Delta} MgO_{(s)} + CO_{2(g)}$$

$$BaCO_{3(s)} \xrightarrow{\Delta} BaO_{(s)} + CO_{2(g)}$$

- The small Mg^{2+} ion when combined with small O^{2-} liberate more lattice energy and compensate for the energy required to decompose $MgCO_3$. Hence, the decomposition temperature of $MgCO_3$ is less but when the bigger Ba^{2+} ion combined with small O^{2-} liberate small amount of lattice energy. Hence, the decomposition temperature of $BaCO_3$ is large.
- Smaller anions stabilize the higher oxidation state of cations while bigger anions stabilize the lower oxidation states of cation because smaller non-metal atoms have more electronegativities and high electron affinities and liberate more lattice energies.
- Metal ions in higher oxidation states can oxidise the bigger anions because the bigger anions cannot hold the electrons strongly and act as reducing agents. E.g.: I^- acts as a reducing agent but F^- cannot.

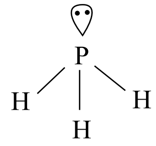
COVALENT BOND

- The concept of covalent bond was proposed by **Lewis**.
- Covalent bond is formed by **mutual sharing** of electrons between two atoms.
- Pure covalent bond (100 per cent) or pure ionic bond (100 per cent) is only an ideal situation. Even a covalent bond between similar atoms like H_2, Cl_2, etc. have some ionic character due to shifting of electrons in a fraction of a second.
- Covalent bonds formed between two atoms having difference in their electronegativities have **partial ionic character**.
- A covalent bond formed between two atoms by sharing one electron pair is a **single bond.** E.g.: H_2, F_2, Cl_2, HCl.
- If two pairs of electrons are shared between two atoms, the covalent bond formed is **a double bond**. E.g.: O_2, CO_2.
- If three electron pairs are shared between two atoms the covalent bond formed is a **triple bond**.
- **Covalence** is the number of electron pairs shared by an atom in the formation of a covalent compound.
- The molecules present in a covalent solid are attracted by weak van der Waals' forces. Hence, they have low m.pts and b.pts.
- Most of the covalent compounds are either gases or low melting solids.
- Polar covalent compounds like glucose, sugar, etc dissolve in polar solvents like water, alcohol, etc.
- Non-polar covalent substances like iodine, camphor dissolve in non-polar solvents like benzene, carbon tetrachloride, etc.
- Covalent compounds do not conduct electric current either in molten state or in their aqueous solution because of the absence of ions or free electrons.
- Covalent compounds are non-electrolytes.
- Covalent bonds are directional in nature; hence, covalent compounds exhibit isomerism.
- In chemical reactions covalent molecules participate which involves bond breaking and bond making. Hence, their reactions are slow.

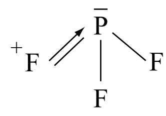
- Lewis theory cannot explain the shapes and bond angles in molecules.

Valence Shell Electron Pair Repulsion (VSEPR) Theory

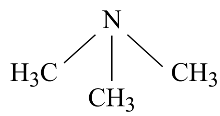
- VSEPR theory was proposed by **Sidgwick** and **Powell** and later developed by **Gillespy** and **Nyholm.**
- VSEPR theory predicts the shapes of molecules without reference to hybridization.
- The shapes of molecules depend on the number of electron pairs present on the central atom including lone pairs.
- A lone pair (LP) electrons occupy more space around the central atom than a bond pair (BP) of electrons because the lone pair electrons are attracted by only one nucleus while the bond pair electrons are under the attraction of two atoms sharing this electron pair.
- The order of repulsion between various electron pairs is LP-LP > LP-BP > BP-BP.
- Molecules having all bond pairs around the central atom have regular shapes as given in Table 3.8.
- The magnitude of repulsions between BPs and lone pair electrons depend on the electronegativity difference between the central atom and the other atoms bonded to it.
- With increase in the electronegativity of central atom bond angle increases, but with increase in electronegativity of the bonded atom bond angle decreases. This is because with increase in electronegativity of the central atom, the BP moves towards central atom causing more repulsion, but with increase in electronegativity of the bonded atom BP moves away from the central atom causing less repulsion.
- Double bonds cause more repulsion than single bonds and triple bonds cause more repulsion than double bonds.
- Repulsions between electron pairs in filled shells are greater than those between electron pairs in incompleted shells.
- Molecules having 5 and 7 electron pairs around the central atom possess asymmetric structures and the electron pairs with largest number of nearest neighbours will be located at a greater average distance from the central atom than the other electron pairs.
- As the number of lone pairs on the central atom increases distortion of the molecule also increases.
- A lone pair cause more repulsion (e.g., NO_2^-) than an unpaired electron (NO_2) and an unpaired electron cause more repulsion than in the molecule having all BPs. E.g., the order of bond angle in NO_2, NO_2^- and NO_2^+ will be $NO_2^- < NO_2 < NO_2^+$.
- If in a molecule the central atom is having vacant orbitals and the bonded atom has lone pair or vice versa delocalization of electrons from the atom having lone pair into the atom having vacant orbital causes the increase in bond angle and change in shapes of molecules.



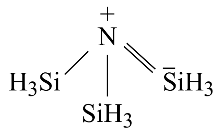
93.3°



104°



Pyramidal



Planar

- Molecules having 5 electron pairs adopt asymmetric trigonal bipyramid structure having two long axial bond lengths and three short equatorial bond lengths. The three equatorial bond angles are 120° while the axial to equatorial position are 90°.
- In the trigonal bipyramid structure, always the lone pair occupy the equatorial positions only.
- In TBP structure, always the double-bonded oxygen occupies the equatorial positions only.
- If there are different types of atoms in a TBP structure the more electronegative atom occupies the axial positions.
- In the octahedral structure having 6 electron pairs, if there are more than one LP, they occupy opposite corners as in XeF_4 and if there are double bonded oxygen atoms, they also occupy the opposite corners of octahedron as in XeO_2F_4.

Isoelectronic and Isostructural Species

- If Lewis diagrams of different species are identical, except for the identity of central atom and having same electronic structure are called **isoelectronic species**.
- Isoelectronic species have similar structures. E.g., $BO_3^{3-}, CO_3^{2-}, NO_3^-$ are isoelectronic and isostructural.

Ligand Close Packing

- For a series of molecules with the same central atom the non-bonded distances between the outer atoms are consistent, but the bond angles and bond lengths change.

Valence Bond Theory

- Valence Bond (VB) theory was proposed by Heitler and London and was later extended by Pauling and Slater.
- A covalent bond is formed by the overlap of the atomic orbitals of the two atoms.
- The greater the extent of overlap, the **stronger** is the bond formed.
- The **spin** of the two electrons present in the overlapping orbitals must be opposite.
- The imaginary line joining the two nuclei in a molecule is known as **internuclear axis** or **molecular axis**.
- The covalent bond formed with the linear overlap of atomic orbitals along the internuclear axis is known as a **σ-bond**.
- The covalent bond formed by the lateral overlap or sidewise overlap of the atomic orbitals perpendicular to internuclear axis is known as a **π-bond**.
- A **σ-bond** can exist independently but a **π-bond** cannot exist independently.
- A π-bond is formed only after the formation of a σ-bond.
- π-bond is weaker than a σ-bond.
- The shapes of molecules depend only on σ-bonds but not on π-bonds.
- The order of the strength of covalent bonds formed by overlapping of different types of orbitals will be in the order σ p - p > σ s - p > σ s - s > π p - p

Hybridization

- Hybridization was proposed by **Linus Pauling**.
- The inter-mixing of atomic orbitals of similar energies and their redistribution into an equal number of identical orbitals is known as hybridization.
- Orbitals of only one atom can participate in hybridization but orbitals of different atoms cannot participate in hybridization.
- The hybrid orbitals are symmetrically arranged around the nucleus so that the repulsion between them is less.
- The angle between any two hybrid orbitals in an atom must be equal, except in sp^3d and sp^3d^3.
- Only orbitals undergo hybridization but not electrons.
- The distribution of electrons in hybrid orbitals follow Pauli's exclusion principle and Hund's rule of maximum multiplicity.
- Hybrid orbitals always form σ-bonds only.
- For predicting the type of hybridization and the shapes of molecules refer to section 3.6.2.

Role of d-orbitals in Bonding

- The valence shell d-orbitals are far higher in energy and much more diffuse than the s- and p-orbitals and thus cannot contribute much to the bonding.
- When an atom is in bond with more electronegative atoms like fluorine or oxygen, positive charge develops on the central atom resulting the contraction of d-orbitals and decrease in energy, which can participate in bonding. Because of this reason elements can form compounds in higher oxidation states, utilizing d-orbitals, only with more electronegative elements but not with less electronegative elements. E.g.: SF_6 is formed but not SH_6.

Bents' Rule

- sp^3d hybrid orbitals may be considered to be combination of two sets of hybrid orbitals, one set of sp_xp_y (sp^2) and another set of $p_zd_z^2$ (pd).
- The pd hybrid orbitals make two linear bonds axially while the sp^2 hybrid orbitals form trigonal equatorial bonds.
- The sp^2 hybrid orbitals form stronger bonds and they are shorter than the weaker axial bonds.
- When different types of atoms are present in the -T.P structure more electronegative substituents "prefer" hybrid orbitals having less s-character, i.e., axial positions and more electropositive substituents 'prefer' hybrid orbitals having more s-character (Bent's rule).

Calculation of Per cent Character of Hybrid Orbitals

- There is a relation between hybridization and bond angles for s-p hybrid orbitals.
- For two or more equivalent orbitals, the per cent s-character or per cent p-character is given by the relationship.

$$\mathrm{Cos}\,\theta = \frac{S}{S-1} = \frac{P-1}{P}$$

  where θ is the angle between the equivalent orbitals and the s- and p-characters are expressed as decimal fractions.
- In the case of small atoms like N and O, the steric effects are more prominent, but in larger atoms in the respective groups such as P, As, Sb, S, Se and Te these effects are somewhat relaxed, allowing the reduced hybridization energy and more p-character in the bonding orbitals. Because of this reason only N and O are involved in sp^3 hybridization in NH_3 and H_2O but almost pure p-orbitals involve in bonding in larger atoms in their respective groups during the formation of hydrides.
- For the orbitals participating in different types of hybridization refer to Table 3.11.

MOLECULAR ORBITAL THEORY

- Molecular orbital theory was proposed by Hund and Mulliken.
- An electron in an atomic orbital is centred about a single nucleus but an electron in a molecular orbital (MO) is spread out over all the nuclei in a molecule.
- MO belongs to all the nuclei in a molecule.
- Mathematically an MO is a **linear combination of atomic orbitals (LCAO)** of two different atoms that approach one another.
- The number of MOs formed is equal to the no. of atomic orbitals that combined.
- Two atomic orbitals can be combined to yield two molecular orbitals; one bonding and another **anti-bonding** molecular orbitals.
- Bonding MO is at **lower energy** than the original atomic orbital.
- In the bonding MO most of the electron density is located in the region between the two nuclei. Both nuclei, are attracted to this electron density.
- Anti-bonding MO is at higher energy than the original atomic orbital and represented by an asterisk (*).
- In the anti-bonding MO only a small fraction of the electron density is in the region between the nuclei.
- If the overlapping of the two atomic orbitals takes place along their axes, the resulting MOs are known as σ-MOs.
- If the overlapping of the two atomic orbitals takes place side ways, the resulting MOs are known as π-MOs.
- π-MOs are formed by the overlapping of P_x-P_x and P_y-P_y orbitals if *z*-axis is selected as the molecular axis.
- MOs formed by the overlapping of 1s, 2s, $2p_x$, $2p_y$ and $2p_z$ orbitals can be arranged in order of increasing energies as follows.

$$\sigma 1s < \sigma^*1s < \sigma 2s < {}^*2s < \pi 2p_x = \pi 2p_y < \sigma 2p_z <$$

$$\pi^*2p_x = \pi^*2p_y < \sigma^*2p_z.$$

- This order is valid for Li_2 to N_2 and it is because of sp mixing of orbitals in these elements,
- For O_2 and F_2 the following sequence for the energies of MOs is observed.

$$\sigma 1s < \sigma^*1s < \sigma 2s < \sigma^*2s < \sigma 2p_z < \pi 2px = \pi 2p_y < \pi^*$$

$$2p_x = \pi^*2p_y < \sigma^*2p_z.$$

Electronic Configuration

- The number of electrons belonging to two atoms are counted and these are filled in the various MOs according to the following rules.
  (i) The MO in the lowest energy is filled first (**Auf bau principle**)
  (ii) An MO can accommodate a maximum of two electrons with opposite spins (**Pauli's exclusion principle**).
  (iii) The electrons present in bonding MOs are called bonding electrons (n_b) and the electrons in the anti-bonding MOs are called anti-bonding electrons (n_a).

Bond Order

- Bond order may be defined as half the difference between the number of bonding and anti-bonding electrons.

  Bond order $= \frac{1}{2}$ [No. of electrons in bonding MOs – No. of electrons in anti-bonding MOs]
- Bond order of 1, 2 or 3 corresponds to a single, double or triple bonds, respectively.
- Bond order may be fractional, integer, zero and rarely negative.
- Zero and negative bond order indicates that the species do not exist.
- The bond order of a molecule is directly proportional to its bond dissociation energy and stability and inversely proportional to bond length.
- The addition of an electron in the bonding MO makes the molecule more stable, whereas that in the anti-bonding MO makes it less stable.
- The removal of an electron from the bonding MO makes the molecule less stable, whereas from the anti-bonding MO makes it stable.
- Paramagnetism in a substance is due to the presence of unpaired electrons in its molecule.
- Molecules or ions having same number of electrons are called isoelectronic species, and they have same bond order and same MO electronic configuration.
- The highest occupied MO (HOMO) and the lowest unoccupied MO (LUMO) collectively are called as frontier orbitals.
- The electronic configurations and bonding properties of certain diatomic species can be seen in Table 3.14.
- **Heteronuclear diatomic molecules** also follow the same general bonding pattern as the homonuclear molecules, but greater nuclear charge on one of the atoms lowers its atomic energy levels and shifts the resulting MO levels lower.
- In heteronuclear diatomic molecules due to the difference in the electronegativities of the two atoms, the electrons in bonding MO spend more time near the more electronegative atom while the electrons in anti-bonding MOs are closer to the less electronegative atom.
- The order of the energies of different MOs follows the order of energies of MOs in more electronegative diatomic molecule. E.g., in CN^-, both carbon and

nitrogen are in the set of B, C, N, so the order of energies of MO will be as in B, C and N due to sp mixing but in NO they will follow O_2 molecule.
- CO molecule is of special case where HOMO is the nonbonding orbital of carbon.

COORDINATE BOND

- A bond formed by sharing a pair of electrons, providing both electrons by one of the bonded atoms is called a **dative bond** or a **coordinate covalent bond**.
- The atom providing two electrons to form a dative bond is called a **donor** and the atom sharing the pair of electrons from the donor is known as an acceptor.
- Dative bond may be represented with an arrow from the donor to acceptor as $A \rightarrow B$ or as $\overset{+}{A}-\overset{-}{B}$ with positive charge on donor and negative charge on acceptor.
- Donor should have atleast one lone pair of electrons and acceptor should have a vacant orbital.
- The properties of compounds containing dative bonds are intermediate to ionic and covalent compounds.
- Since the dative bond is directional in nature the compounds containing dative bonds exhibit stereoisomerism.
- Coordinate covalent compounds are non-electrolytes, do not conduct electricity because they do not ionize in water.
- Coordinate covalent compounds dissolve more in non-polar solvents like benzene, carbon tetrachloride, etc. than in polar solvents.

METALLIC BOND

- The force that binds a metal atom to a number of electrons within its sphere of influence is known as a metallic bond.
- According to the **electron sea model**, the metal atoms lose their outer electrons and convert into positive kernels.
- All the electrons coming from all atoms in a metal form as a sea of electrons in which the positive kernals are floating and the electrons delocalize, and move through out the metal in the vacant orbitals of metals.
- Electrons pull the cations towards them from all directions and thus hold the cations leading to a big force of attraction of several nuclei for several electrons which gives the metal a stable structure.
- **Valence bond theory** proposed by Linus Pauling is mainly based on resonance hybridization of several structures which gives stability for the metal.
- Every metal possess vacant orbitals in its valence shell called a **metallic orbital**, which can accept an electron from other atoms and can form more number of bonds.
- As per the theory electrical conductivity of metals is due to the change in position of +ve and –ve charges of resonance structures.
- According to the **MO theory** the orbitals of valence shell of all atoms combine to form a large number of MOs around all the atoms in metal crystals.
- Since large number of MOs having almost equal energy is formed, they are very close to each other and form as band of molecular orbitals.
- Depending upon different types of atomic orbitals which overlap, different energy bands are formed.
- The arrangement of electrons in the different energy bands determines the characteristics of a metal.
- The energy bands formed from different atomic orbitals may overlap or be separated from each other.
- The highest occupied energy band is called the **valence band** while the lowest unoccupied energy band is called the **conduction band.**
- In the case of metals, the valence band may be half-filled or there may be an overlapping between the valence and conduction bands which makes it possible for the electrons to go into the vacant bands and hence responsible for electrical conductivity.
- In the case of **insulators** the energy gap is very large and therefore the vacant conduction band is not available to the electrons of the completely filled valence band.
- In **semiconductors** the energy gap is intermediate between those of metals and insulators. The increase in temperature gives thermal energy for some electrons in valence band to move into the conduction band and hence their electrical conductivity increases with increase in temperature.

RESONANCE

- If it is possible to write different Lewis diagrams to the same molecules or ions, the different Lewis diagrams are called **resonance structures** or **canonical structures** and the actual structure will be a hybrid structure of all the resonance structures.
- Resonance structures are only hypothetical, and as such, they can never be isolated.
- Tautomerism is different from resonance and tautomeric forms exist actually and can be isolated sometimes.
- The resonance energy of a substance is the extra stability of the resonance hybrid compared with the most stable of the resonating structures.

Rules for Writing the Resonance Structures

- While writing the resonance structures only the position of electrons must be changed but not the position of nuclei of atoms.

- All the resonance structures must be proper Lewis structures.
- All resonance structures must have same number of unpaired electrons which decide the magnetic character of the molecule.
- More stable resonance structure contribute more to the resonance hybrid and equivalent resonance structures contribute equally to the hybrid.
- Nonpolar resonance structure having more number of covalent bonds (without charge separation) is more stable than polar resonance structures (charge separation) having less number of covalent bonds.
- Increase in the distance between charges in a resonance structure decreases the stability due to decrease in attraction between charges.
- Resonance structure in which negative charge present on more electronegative atom and positive charge present on less electronegative atom are more stable than the structures in which negative charge present on less electronegative atom and positive charge present on more electronegative atom.
- Resonance give rise to identical bond lengths in molecule and extra stability to the molecule.
- The more the resonance energy, the more is the stability of a molecule.
- The more the number of canonical forms and with more of canonical forms having same energy, the more is the resonance energy and more is the stability of the molecule.
- Bond order in a resonance hybrid structure is

$$\frac{\text{Total No. of bonds around the central atom}}{\text{Total No. of atoms around the central atom}}$$

- For cyclic compounds the bond order between two atoms in a ring is

$$\frac{\text{Total No. of bonds in the ring}}{\text{Total No. of atoms in the ring}}$$

DIPOLE MOMENT

- Whenever atoms with different electronegativities combine, the resulting molecule has polar bonds with the bond pair of electrons shifted more towards the more electronegative atom.
- Two equal and opposite but separated electrical charges in a molecule constitute an electrical dipole moment μ
- Dipolomoment μ = charge (q) $\times$ distance between charges (r)
- Dipole moment is expressed in terms of coulomb metre (cm units) or by Debye unit 'D' being equal to 3.33564×10^{-30} cm or 1 Debye = 1×10^{-18} esu-cm.

Dipole moment is useful in calculating the per cent ionic character of a polar covalent bond.

$$\%\text{ ionic character} = \frac{\propto_{\text{expt}}}{\propto_{\text{ionic}}} \cdot 100$$

- Dipole moment is an additive vector quantity acting in the direction of the chemical bond and so the dipole moment of a whole molecule is the vector sum of the constituent dipole moments.
- For AB_2 type molecules if dipole moment is zero, the molecules are linear but if they have any dipole moment greater than zero they are angular in shape.
- AB_3 type molecules having dipole moment zero, are planar triangular in shape, but if they have any dipole moment greater than zero then they have pyramidal structure or T-shape.
- AB_4 molecules having dipole moment zero have symmetrical tetrahedral structure but if they have any dipole moment greater than zero have distorted tetrahedral structure.
- Among geometrical isomers cis-isomers have dipole moment greater than zero while trans-isomers have dipole moment almost equal to zero.
- Among dichlorobenzenes para-isomers have zero dipole moment while orthoisomers have more dipole moment than metaisomers.
- The dipole moment of a molecule depends on the nature of bonds, direction, geometry and bond angles.
- Because dipole moment is a vector quantity, the bonds in a molecule may be polar but the molecule may be polar or non-polar.
- Strength of a covalent bond increases with increase in polarity.

HYDROGEN BOND

- The electrostatic force of attraction between a partially positive charged hydrogen atom of a molecule and a highly electronegative atom of the same molecule or different molecule is known as **hydrogen bond**.
- Hydrogen bond is formed when the hydrogen is bonded to a small highly electronegative atoms like F, O, N.
- Liquids in which molecules are associated through hydrogen bond are called **associated liquids.** E.g.: H_2O, HF, NH_3 alcohol.
- Liquid in which hydrogen bonds do not exist between their molecules are called normal liquids.
- The boiling points of associated liquids are greater than those of normal liquids.
- The energy of hydrogen bond is approximately 10–40 KJ mol^{-1} (3-10 k.Cal mol^{-1}) which is relatively small

when compared to the 418 KJ mol^{-1} (100 k.Cal mol^{-1}) for a covalent O-H bond.

- The bond length of hydrogen bond is 176 pm which is higher than the O-H bond length of 100 pm.
- Hydrogen bonding is of two types: (i) intermolecular hydrogen bonding and (ii) intramolecular hydrogen bonding.
- **Intermolecular hydrogen bonding** is between two different molecules of the same compound or different compounds. E.g.: HF, H_2O, alcohols.
- The compounds containing intermolecular hydrogen bonds have high m.pts and high b.pts because more energy is required to break the hydrogen bonds.
- The hydrogen bond in H-F is stronger than in H_2O, yet H_2O has a high b.pt than HF because in HF the hydrogen bond is so strong that it can exist in vapour phase also and exist as $(HF)_6$, i.e., the number of hydrogen bonds broken before it is converted into vapour are less. But during the conversion of water into vapour all the hydrogen bonds should be broken.
- The solubility of the compounds in water, which can form hydrogen bonds with water is more than those which cannot form hydrogen bonds. Alcohol which can form hydrogen bond with water is more soluble in water than ethers, aldehydes or ketones which cannot form hydrogen bonds.
- The solubility of alcohols in water decreases with increase in the size of alkyl group due to increase in hydrophobic nature of alkyl group.
- When ammonia is dissolved in water N-atom in ammonia form hydrogen bond with H-atom of water but not the H-atom of ammonia with O-atom of water.
- Due to hydrogen bonding molecules of certain compounds like perchloric acid, carboxylic acids dimerize and give abnormal molecular weights when determined by using colligative properties.
- Primary and secondary amine molecules can associate due to hydrogen bonding and have more b.pt than tertiary amine which cannot form hydrogen bond due to the absence of H-atoms.
- Quartenary bases like $[(CH_3)_4N]OH$ are stronger than primary, secondary and tertiary amines due to the absence of H-atoms in quarternary base.
- Due to the hydrogen bonding ice has an open structure in which O-atom in every water molecule is surrounded by 4 O-atoms tetrahedrally and joined by 4 hydrogen atoms two of which are in covalent bond while the other two are in hydrogen bonds.
- In several hydrated compounds some water molecules are present as hydrogen bonded water molecules. E.g., in $CuSO_4$ $5H_2O$ each Cu^{2+} ion is surrounded octahedrally by 6 O-atoms with 4 of them being in water molecules and 2 in SO_4^{2-} ions; the extra water molecule is situated between two such octahedral groups linked by hydrogen bonds.
- The crystal structure of several compounds is dominated by hydrogen bonds. E.g., the HF in liquid state have zigzag structure due to the presence of lone pairs on flourine atoms.
- The structure of NH_4F is expected as CsCl structure from the radius ratio (0.92) but it adopts wurtzite structure in which each NH_4^+ ion is surrounded tetrahedrally by four F^- ions due to hydrogen bonds between F^- ions and H atoms of NH_4^+ ion.
- The strength of hydrogen bond depends on the magnitude of polarity and size of the atoms and number of hydrogen bonds.
- Though electronegativities of nitrogen and chlorine are same (3.0) chlorine has the least tendency to form hydrogen bonds due to bigger size.
- **Intramolecular hydrogen bonding** is between two groups of the same molecule. E.g.: orthoisomers of nitrophenol, hydroxy benzaldehyde (salicylaldehyde), hydroxy benzoic acid (salicylic acid), chlorophenol, fluorophenol, nitroaniline, etc.
- Due to intramolecular hydrogen bonding the molecules of orthoisomers are not associated. Hence, their b.pts are less and are steam volatile.
- In meta- and paraisomers intramolecular hydrogen bonding is not possible due to increase in ring size. Hence, molecules of meta- and paraisomers are associated by intermolecular hydrogen bonding and hence have high m.pts and b.pts.
- Molecules of orthoisomers cannot form hydrogen bonds with water due to intramolecular hydrogen bonding but molecules of meta- and paraisomers can form hydrogen bonds with water molecules. Thus, orthoisomers are less soluble than meta- and paraisomers.

Bond Length

- Bond length is the distance between the nuclei of two covalently bonded atoms.
- The magnitude of the bond length between the similar atoms in different molecules is generally same. E.g., ‘OH’ bond length ion H_2O or H_2O_2 or C_2H_5OH is the same.
- With increase in the bond order (number of bonds) the bond lengths decreases, e.g., the C-C bond lengths in ethane, ethylene and acetylene with single, double and triple bonds, respectively are 154 pm, 134 pm and 120 pm.
- The bond length also depends on the type of hybridization of the atom with which another atom is in bond, e.g., C_{sp^3} - H (110 pm) > C_{sp^2} - H (108 pm) > C_{sp} - H (106 pm).

- In diatomic molecules the bond length is equal to the sum of the covalent radii of two atoms, if the bond is purely covalent. But in several cases the bond length is less than the sum of the covalent radii of the bonded atoms because of partial ionic character and in such cases the bond length may be calculated as

$$A - B = (r_A + r_B) + C(\chi_A - \chi_B)$$

where r_A and r_B are covalent radii of A and B while χ_A and χ_B are electronegativities of A and B, respectively. C is constant and its value range of 0.06–0.08 depends on the type of atoms in bonding.
- Due to resonance hybridization, the normal bond lengths in several molecules will become less than the sum of the covalent radii, e.g., in BF_3 the B - F bond length is shorter than the sum of covalent radii of boron and fluorine due to back bonding from fluorine to boron and resonance hybridization.
- The bond length increases with increase in size of the atom in similar compounds for the elements in a given family.

Bond Energy

- The energy required to break the bonds in one mole of diatomic molecules or the energy released when a bond is formed between two atoms to form one mole of substance is known as bond energy.
- The bond energies are expressed at room temperature.
- The bond energy depends on the type of cleavage of bond.
- The energy required for the **heterolytic cleavage** of bond is higher than required for **homolytic cleavage**.
- If the bond energy increases, stability of the bond increases and reactivity of molecule decreases.
- σ-bonds are stronger than π-bonds.
- As the number of bonds between two atoms increases, the bond strength increases.
- The bond energies of similar compounds in a group decreases down the group due to increase in bond length with increase in atomic size.
- With increase in the p-character in the hybrid orbitals the bond energy decreases as $sp > sp^2 > sp^3$.
- With increase in the number of lone pairs of electrons on bonded atoms, bond energy decreases as C-C > N – N > O – O.
- In multiatom molecule like CH_4 the bond energy is the average value of all bond dissociation energies.
- The heat of reaction can be estimated from bond energy values.

Non-bonding Interactions between Molecules

- At a given instant the asymmetry in the distribution of electrons around the nucleus of an atom cause distortion in the electron distribution in neighbouring atom or molecule itself due to which dipole moment develops.
- Attractive forces arising due to distortion of electron clouds under the influence of nuclei of atoms are known as **dispersion forces** or **London forces**.
- Attractive forces between the polar molecules are known as **Dipole - Dipole attractive forces**.
- The dipole moment produced in a non-polar molecule by a polar molecule is called **induced dipole moment**.
- The attraction between the permanent dipole and the induced dipole is called **dipole-induced dipole attractive force**.
- The induced dipole moment is due to the shift in the centre of gravity of the negative charge relative to the nuclear charge.
- Interaction between the dipolar molecules also include additive dipole - dipole induced interactions.
- Dipole - dipole, dipole - induced dipole and dispersion forces are collectively called as van der Waal's forces.
- The magnitude of van der Waal's forces is the relative polarizability of the electrons of the atom involved.
- Fluorocarbons containing only carbon and fluorine atoms have extraordinarily low boiling point when compared to hydrocarbons having same molecular weight. This is because of the very low polarizability of fluorine atoms.

PRACTICE

Multiple Choice Questions with Only One Answer

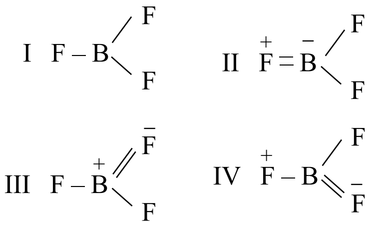
1. Which of the following molecules is adequately represented by a single Lewis structure?
(a) O_3 (b) NOCl
(c) SO_2 (d) N_2O

2. Nodal plane in the pi bonds
(a) a plane perpendicular to the molecular plane which contains C—C sigma bond
(b) a plane perpendicular to the molecular plane and bisecting the C—C sigma bond
(c) a plane parallel to the molecular plane
(d) the molecular plane

3. Assuming the bond direction to be z-axis, which of the overlapping of atomic orbitals of two atoms A and B will result in bonding (I) s orbital of A and p_x orbital to B, (II) s orbital of A and p_z orbital of B, (III) p_y orbital of A and p_z orbital of B and (IV) s orbitals of both A and B?
(a) I and IV (b) I and II
(c) III and IV (d) II and IV

4. Which is the correct sequence of decreasing O—O bond length in O_2, H_2O_2 and O_3?
(a) $H_2O_2 > O_3 > O_2$ (b) $O_2 > O_3 > H_2O_2$
(c) $O_2 > H_2O_2 > O_3$ (d) $O_3 > H_2O_2 > O_2$

5. Which of the following statements is correct about the molecular structure of boron trifluoride?



(a) All the structures contribute equally to the resonance hybrid.
(b) Structure I contributes maximum to the resonance hybrid.
(c) Structures II and IV contribute to a greater extent to the resonance hybrid.
(d) The B—F bond has been found to possess π-character.

6. Increasing order of dipole moments is given by
(a) $CF_4 < NH_3 < NF_3 < H_2O$
(b) $CF_4 < NH_3 < H_2O < NF_3$
(c) $CF_4 < NF_3 < H_2O < NH_3$
(d) $CF_4 < NF_3 < NH_3 < H_2O$

7. The common features among the species CN^-, CO and NO^+ is/are:
(a) isoelectronic and weak field ligands
(b) bond order three and isoelectronic
(c) bond order three and weak field ligands
(d) bond order two and π-acceptors

8. Amongst the species CO_3^{2-}, NH_4^+, PH_3, BCl_3, ClF_3, the isostructural pairs are:
(a) CO_3^{2-} and ClF_3 (b) NH_4^+ and PH_3
(c) CO_3^{2-} and BCl_3 (d) PH_3 and ClF_3

9. The geometrical shapes of XeF_5, XeF_6 and XeF_8^{2-}, respectively are:
(a) trigonal bipyramidal, octahedral and square planar
(b) square-based pyramidal, distorted octahedral and octahedral
(c) planar pentagonal, octahedral and square antiprismatic
(d) square-based pyramidal, distorted octahedral and square antiprismatic

10. Which of the following statements is/are true regarding ICl_2^- ion?
(a) I-atom is sp^3d hybridized.
(b) The three lone pairs occupy equatorial positions and two bond pairs the axial positions giving rise to linear
(c) ICl acts as Lewis acid and Cl^- ion acts as Lewis base.
(d) All of these.

11. CrO_2Cl_2 has
(a) tetragonal structure
(b) distorted tetrahedral structure
(c) square pyramidal structure
(d) octahedral structure

12. Consider the statements
I. Bond length in N_2^+ is 0.02 Å greater than in N_2.
II. Bond length in NO^+ is 0.09Å less than in NO.
III. O_2^{2-} has a shorter bond length than O_2.
Then, which of the following is correct?
(a) I and III (b) II and III
(c) I and II (d) I, II and III

13. In PO_4^{3-} the formal charge on each oxygen atom and the P—O bond order are respectively
(a) –0.75, 0.5 (b) –0.75, 1.25
(c) –0.75, 1.0 (d) –3, 1.25

14. PCl_5 in the solid state exists as PCl_4^- and PCl_6^- because
(a) solid PCl_5 is a conductor
(b) PCl_4^- and PCl_6^- have stable, symmetrical structures unlike PCl_5 that has an asymmetrical structure
(c) ion pairs are more stable than neutral molecules
(d) phosphorous belongs to V group in the periodic table

15. Regarding hybridization which is incorrect?
(a) BF_3, C_2H_4, C_6H_6 involves sp^2 hybridization
(b) BeF_2, C_2H_2 CO_2 involves sp hybridization
(c) NH_3, H_2O, CCl_4 involves sp^3 hybridization
(d) CH_4, C_2H_4, C_2H_2 involves sp^2 hybridization

16. The correct order of arrangement of bond length is:
(a) $F_2 > N_2 > Cl_2 > O_2$ (b) $Cl_2 > F_2 > O_2 > N_2$
(c) $O_2 > Cl_2 > N_2 > F_2$ (d) None of these

17. The Cl—C—Cl angle in 1, 1, 2, 2-tetrachloro ethene and tetrachloro methane respectively will be
(a) 109.5° and 90° (b) 120° and 109.5°
(c) 90° and 109.5° (d) 109.5° and 120°

18. Dipole moment is exhibited by
(a) 1,4-dichlorobenzene
(b) 1,2-dichlorobenzene
(c) trans 1,2-dichloroethene
(d) trans 1,2-dichloro-2-butene

19. Which one of the following about hydrogen bonding is false?
(a) It alters some of the physical properties of the compound.
(b) Not all H-compounds can have hydrogen bonding.
(c) The H-atom in a hydrogen bonded pair of atoms is equidistant from the two atoms.
(d) Hydrogen bond is a weak bond.

20. Arrange the following in the decreasing order of hydrogen bonding.
CH_4, CH_3 — OH, CH_3 — F, CH_3 — NH_2:
(a) CH_3, — OH > CH_3 — NH_2 > CH_3 — F > CH_4
(b) CH_3, — OH > CH_3 — F > CH_3 — NH_2 > CH_4
(c) CH_3, — F > CH_3 — OH > CH_3 — NH_2 > CH_4
(d) CH_4, > CH_3 — F > CH_3 — NH_2 > CH_3 — OH

21. The melting point of 2-nitrophenol is lower than that of 4-nitrophenol and this is because of
(a) intramolecular hydrogen bonding in 2-nitrophenol
(b) intramolecular hydrogen bonding in 4-nitrophenol
(c) intermolecular hydrogen bonding in 2-nitrophenol
(d) hydrogen bonding has no relation to the melting point

22. Which of the following is false?
(a) The bond formed between two non-metallic elements is covalent bond.
(b) The bond formed between two inert gas elements is van der Waal's bond.
(c) The bond formed between a metal and a non-metal is electrovalent bond.
(d) The bond formed between two metallic elements is covalent bond.

23. A, B, C are three substances. A does not conduct electricity in the solid or solution state. B conducts electricity both in the fused and solution states, while C conducts electricity only in the solution state.
Which of the following statement is false regarding A, B and C?
(a) A has polar covalent linkage.
(b) A has non-polar covalent linkage.
(c) B is ionic nature.
(d) C has polar covalent linkage.

24. Molecular shapes of SF_4, CF_4 and XeF_4 are:
(a) the same with 2, 0 and 1 lone pairs of electrons, respectively
(b) the same with 1, 1 and 1 lone pairs of electrons, respectively
(c) different with 0, 1 and 2 lone pairs of electrons, respectively
(d) different with 1, 0 and 2 lone pairs of electrons, respectively

25. $SnCl_4$ is a covalent liquid because
(a) electron clouds of the Cl^- ions are weakly polarized to envelop the cation
(b) electron clouds of the Cl^- ions are strongly polarized to envelop the cation
(c) its molecules are attracted to one another by strong Van der Waal's forces
(d) Sn shows inert-pair effect

26. The I_3^- ion has
(a) five equatorial lone pairs on the central I atom and two axial bonding pairs in a trigonal bipyramidal arrangement
(b) five equatorial lone pairs on the central I atom and two axial bonding pairs in a pentagonal bipyramidal arrangement
(c) three equatorial lone pairs on the central I atom and two axial bonding pairs in a trigonal bipyramidal arrangement
(d) two equatorial lone pairs on the central I atom and three axial bonding pairs in a pentagonal bipyramidal arrangement

27. The explanation of various intermolecular forces indicates
(a) the unusual (anomalous) behavior of H_2O, NH_3 and HF in terms of the relationship between molecular weight and bonding points is due to London forces
(b) ion-dipole forces account for the salvation energy which plays and important role in the dissolving of ionic solids
(c) for non-polar molecules in the liquid state an important force acting is gravitational attraction
(d) that London forces are due to very small permanent electric dipoles

28. Amongst NO_3^-, AsO_3^{3-}, CO_3^{2-}, ClO_3^-, SO_3^{2-} and BO_3^{3-} the non-polar species are:
(a) CO_3^{2-}, SO_3^{2-} and BO_3^{3-}
(b) AsO_3^-, ClO_3^- and SO_3^{2-}
(c) NO_3^-, CO_3^{2-} and BO_3^{3-}
(d) SO_3^{2-}, NO_3^- and BO_3^{3-}

29. Total number of electron pairs in PCl_5, PCl_4^+ and PCl_6^- are respectively:
(a) 20, 16, 24 (b) 24, 16, 20
(c) 16, 20, 24 (d) 24, 20, 16

30. Which of the following statements is not correct for NO_2, NO_2^+ and NO_2^-
(a) NO_2 is paramagnetic
(b) NO_2^+ is linear, NO_2 is bent, with bond angle slightly less than 120°
(c) NO_2^+ ion has the shortest and strongest bonds
(d) NO_2^- has longest and weakest bonds among these

31. The formal charges on the three atoms in O_3 molecule are:
(a) 0, 0, 0 (b) 0, 0, –1
(c) 0, 0, +1 (d) 0, +1, –1

32. Which of the following statements about LiCl and NaCl is wrong?
(a) LiCl has lower melting point that of NaCl.
(b) LiCl dissolves more in organic solvents, whereas NaCl does not.
(c) LiCl would ionize in water more than NaCl.
(d) Fused LiCl would be more conducting than fused NaCl.

33. The boiling points of methanol, water and dimethyl ether are respectively 65°C, 100°C and 34.5°C. Which of the following best explains these wide variations in b.p.?
(a) The molecular mass increases from water (18) to methanol (32) to diethyl ether (74).
(b) The extent of H-bonding decreases from water to methanol while it is absent in ether.
(c) The density of water 1.00 gm l^{-1}, methanol 0.7914 gm l^{-1} and that of diethyl ether is 0.7137 gm l^{-1}.
(d) The number of H-atoms per molecule increases from water to methanol to ether.

34. In one of the following molecules, the state of hybridization of the central atom is not the same as in others
(a) B in BF_3 (b) O in H_3O^+
(c) N in NH_3 (d) P in PCl_3

35. The correct order of the bond angle is
(a) $NH_3 > H_2O > PH_3 > H_2S$
(b) $NH_3 > PH_3 > H_2O > H_2S$
(c) $NH_3 > H_2S > PH_3 > H_2O$
(d) $PH_3 > H_2S > NH_3 > H_2O$

36. A diatomic molecule has a dipole moment of 1.2D. Its bond distance is 1.0Å. What fraction of electric charge 'e' exists on each atom?
(a) 12% of e (b) 18% of e
(c) 25% of e (d) 30% of e

37. Which of the following statements is not correct?
(a) In $CHCl_3$ molecule both dipole forces as well as dispersion forces exist.
(b) In H_2O molecule both hydrogen bonds as well as dispersion forces are present.
(c) In CCl_4 only dispersive forces exist.
(d) In salicylaldehyde both hydrogen bonds as well as dispersion forces are present.

38. In solid NH_3, each NH_3 molecule has 6 other NH_3 molecules as nearest neighbours, ΔH of sublimation of NH_3 at the melting point is 30.8 kJ/mole and the estimated ΔH of sublimation in the absence of hydrogen bonding is 14.4 kJ per mole. The strength of H-bond in solid NH_3 is
(a) 5.5 kJ/mole (b) 16.4 kJ/mole
(c) 2.7 kJ/mole (d) –8.7 kJ/mole

39. Consider the species NO_3^-, NO_2^+ and NO_2^- pick up the wrong statement:
(a) The hybrid state of N in all the species is not same.
(b) The shapes of both NO_2^+ and NO_2^- are bent while NO_3^- is planar.
(c) The hybrid state of N in NO_3^-, NO_2^- is same.
(d) The hybrid state of N in NO_2^+ is sp.

40. Calcium carbide gets hydrolyzed to form acetylene gas
I. Hybrid state of C does not change.
II. Bonding between the carbon atoms does not change.
III. Boiling point of carbon compound does not change.
(a) I and II are correct
(b) Only II is correct
(c) I and III are correct
(d) II and III are correct

41. Which of the following statements is correct?
(a) Non-bonding orbitals have the same energy as the bonding molecular orbitals
(b) Anti-bonding orbitals have higher energies than highest energy atomic orbitals from which they are formed
(c) Bonding orbitals have higher energies than the anti-bonding molecular orbitals
(d) All are correct

42. The bond order in peroxide ion and fluorine molecule is equal:
(a) these are isoelectronic
(b) their bond energies are nearly equal
(c) their bond lengths are nearly equal
(d) all of these

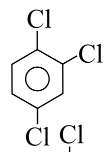
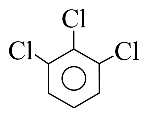
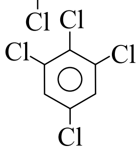
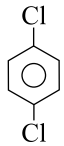
43. When N_2 changes to N_2^+, the N—N bond distance .... and O_2 changes to O_2^+, the O—O bond distance ....
(a) increases, decreases
(b) decreases, increases
(c) increases in both cases
(d) decreases in both cases

44. In the electronic structure of H_2SO_4, the total number of unshared electrons is:
(a) 20 (b) 16 (c) 12 (d) 8

45. A molecule may be represented by three structures having energies E_1, E_2 and E_3, respectively. The energies of these structures follow the order $E_3 < E_2 < E_1$, respectively. If the experimental bond energy of the molecule is E_0, the resonance energy is:
(a) $(E_1 + E_2 + E_3) - E_0$ (b) $E_0 - E_3$
(c) $E_0 - E_1$ (d) $E_0 - E_2$

46. The hybridization of the central atom will change when:
(a) CH_3^- combines with H^+
(b) H_3BO_3 combines with OH^-
(c) NH_3 forms NH_2^-
(d) H_2O combines with H^+

47. Which has maximum dipole moment

(a) 
(b) 
(c) 
(d) 

48. Select pair of compounds in which both have different hybridization but have same molecular geometry.
(a) BF_3, BrF_3 (b) ICl_2^-, $BeCl_2$
(c) BCl_3, PCl_3 (d) PCl_3, NCl_3

49. The correct order of increasing s-character (in percentage) in the hybrid orbitals of the following molecules/ions is:
I. CO_3^{2-} II. H_2S
III. NH_3 IV. CCl_4 V. $BeCl_2$
(a) II < III < IV < I < V (b) II < IV < III < V < I
(c) III < II < I < V < IV (d) II < IV < III < I < V

50. N_2 and O_2 are converted to monocations N_2^+ and O_2^+ respectively, which is a wrong statement now?
(a) In N_2^+, the N — N bond weakens
(b) In O_2^+ the O — O bond order increases
(c) In O_2^+, the paramagnetism decreases
(d) N_2^+ becomes diamagnetic

51. Among the following compounds, which has the maximum number of sp-hybridized C atoms?
(a) $(CN)_2$
(b) $CH_2 = C = CH - CN$
(c) $HC \equiv C - CH_2CH_2 - CH = C = CH_2$
(d) $HC \equiv C - CN$

52. Phosphorus shows a maximum covalency of:
(a) five (b) seven
(c) six (d) three

53. Which of following statements are not correct?
(a) Hybridization is the mixing of atomic orbitals.
(b) sp^2-hybrid orbitals are formed from two p-atomic orbitals and one s-atomic orbital.
(c) dsp^2-hybrid orbitals are all at 90° to one another.
(d) d^2sp^3-hybrid orbitals are directed towards the corners of regular octahedron.

54. Which of the following has a pyramidal shape?
(a) PCl_3 (b) SO_3
(c) CO_3^{-2} (d) NO_3^-

55. The compound in which carbon uses its sp^3 hybrid orbitals for bond formation is
(a) HCO_2H (b) $(H_2N)_2CO$
(c) HCHO (d) CH_3CHO

56. The hybridization of carbon involved in the C—C single bond in the molecule

$$CH \equiv C - CH = CH_2 \text{ is}$$

(a) $sp^3 - sp^2$ (b) $sp^3 - sp^3$
(c) $sp - sp^2$ (d) $sp^2 - sp^2$

57. Which of the following has been arranged in increasing order of size of the hybrid orbitals?
(a) $sp < sp^2 < sp^3$ (b) $sp^3 < sp^2 < sp$
(c) $sp^2 < sp^3 < sp$ (d) $sp^2 < sp < sp^3$

58. The shapes of PCl_4^+, PCl_4^- and $AsCl_5$ are respectively
(a) square planar, tetrahedral and see-saw
(b) tetrahedral, see-saw and trigonal bipyramidal
(c) tetrahedral, square planar and pentagonal bipyramidal
(d) trigonal bipyramidal, tetrahedral and square pyramidal

59. CO_2 is not isostructural with
(a) $HgCl_2$ (b) $SnCl_2$
(c) C_2H_2 (d) $ZnCl_2$

60. In the context of carbon, which of the following is arranged in the correct order of electro negativity?
(a) $sp < sp^2 > sp^3$ (b) $sp^3 < sp^2 > sp$
(c) $sp^2 < sp < sp^3$ (d) $sp^3 < sp > sp^2$

61. PCl_5 undergoes
(a) $sp^3 d_{z^2}$ hybridization
(b) $sp^3d_{x^2-y^2}$ hybridization
(c) sp^3d_{xy} hybridization
(d) sp^3d_{yz} hybridization

62. When 2s-2s, 2p-2p and 2p-2s orbitals overlap, the bond strength decreases in the order
(a) $p - p > s - s > p - s$
(b) $p - p > p - s > s - s$
(c) $s - s > p - p > p - s$
(d) $s - s > p - s > p - p$

63. Which of the following pairs are isostructural?
(a) CH_3^- and CH_3^+ (b) NH_4^+ and NH_3
(c) SO_4^{2-} and BF_4^- (d) NH_2^- and BeF_2

64. The shape of $XeOF_4$ is
(a) square pyramidal
(b) square antiprismatic

(c) distorted octahedral
(d) pentagonal bipyramidal

65. The O — N— O bond angle is maximum in
(a) NO_2^+ (b) NO_2
(c) NO_2^- (d) N_2O_3

66. The shapes of XeF_4, XeF_5^- and XeF_8^{2-} are
(a) octahedral, trigonal bipyramidal and square planar
(b) square pyramidal, pentagonal bipyramidal and octahedral
(c) square planar, planar pentagonal and square anti-prismatic
(d) see-saw, T-shaped and square pyramidal

67. The most likely arrangement of atoms in S_2Cl_2 is
(a) S — S — Cl — Cl (b) S — Cl — S — Cl
(c) S — Cl — Cl — S (d) Cl — S — S — Cl

68. Which of the following molecules have unequal bond lengths?
(a) NF_3 (b) SF_4
(c) SF_6 (d) XeF_4

69. Which of the following sets of characteristics lead to the increase in solubility of ionic substances?
(a) High dipole moment, strong attraction by an ion and large solvation energy
(b) Low dipole moment, weak attraction by an ion and high solvation energy
(c) High dipole moment, strong attraction by an ion and low solvation energy
(d) High dipole moment, weak attraction between ions and large solvation energy

70. Which of the following have undistorted octahedral structures?
I. SF_6 II. PF_6^-
III. SiF_6^{2-} IV. XeF_6
Select the correct answer using the codes given below;
(a) II, III and IV (b) I, III and IV
(c) I, II and III (d) I, II and IV

71. Sulphur reacts with chlorine in 1:2 ratio and forms X, Hydrolysis of X gives sulphur compound Y. What is the hybridization state of central atom in the anion of Y?
(a) sp^3 (b) sp^2
(c) sp^2 (d) sp^3d

72. In ICl_4^-, the shape is square planar. The number of bond pair-lone pair repulsion at 90° is
(a) 6 (b) 8 (c) 12 (d) 4

73. Arrange the following in the order of decreasing N — O bond length: NO_2^+, NO_2^-, NO_3^-
(a) $NO_3^- > NO_2^+ > NO_2^-$
(b) $NO_3^- > NO_2^- > NO_2^+$
(c) $NO_2^+ > NO_3^- > NO_2^-$
(d) $NO_2^- > NO_3^- > NO_2^+$

74. A molecule XY_2 contains two σ-, two π-bonds and one lone pair of electron in the valence shell of X. The arrangement of lone pair as well as bond pairs is
(a) square pyramidal (b) linear
(c) trigonal planar (d) unpredictable

75. Nodal planes of π-bond(s) in $CH_2 = C = C = CH_2$ are located in
(a) all are in molecular plane
(b) two in molecular plane and one in a plane perpendicular to molecular plane which contains C — C σ-bonds
(c) one in molecular plane and two in plane perpendicular to molecular plane which contains C — C σ-bonds
(d) two in molecular plane and one in a plane perpendicular to molecular plane which bisects C — C σ-bonds at right angle

76. Between the molecules FNO and FNO_2 one has dipole moment of μ = 1.81D and the other has μ = 0.47 D on the basis of VSEPR considerations assign the shapes and dipole moments to the two molecules.
(a) FNO is angular with 0.47 D dipole moment while FNO_2 is pyramidal with dipole moment 1.81 D.
(b) FNO is linear with 1.81 D dipole moment while FNO_2 is planar triangular dipole moment 0.47 D.
(c) FNO is angular with 1.81 D dipole moment while FNO_2 is planar trigonal shape with 0.47D dipole moment.
(d) FNO is angular with 1.81 D while FNO_2 is pyramidal with dipole moment 0.47 D.

77. In the following pairs select the pair with largest bond angle difference
(a) NH_3 and PH_3 (b) OF_2 and SF_2
(c) PF_4^- and PF_4^+ (d) ClF_3 and BF_3
(a) a (b) b (c) c (d) d

78. Which of the following is an example of a planar molecule having a net dipole moment?
(a) NH_3 (b) ClF_3 (c) XeO_3 (d) SO_3

79. The bond angle of H_2Se is best described as being
(a) Between 109° and 120°
(b) Greater than 120°
(c) Less than that in H_2S but not less than 90°
(d) Less than 90°

80. The bond dissociation energy will increase in the order of
(a) $N_2 < O_2 < F_2 < Cl_2$ (b) $F_2 < O_2 < Cl_2 < N_2$
(c) $F_2 < Cl_2 < O_2 < N_2$ (d) $Cl_2 < F_2 < O_2 < N_2$

81. In molecules of the type AX_2L_n where L represents lone pair 'n' its number then exists a bond between elements A and X. Then, XAX angle
(a) always decreases if 'n' increases
(b) always increases if 'n' increases
(c) will be maximum for n = 3
(d) will always be less than 180° if n = 0

82. Identify the correct statements
I. N_3^- is linear
II. ClF_3 has a dipole moment
III. SF_4 has distorted tetrahedral (see-saw shape)
IV. XeF_4 is tetrahedral in shape
(a) I, II, III and IV are correct
(b) Only I, II and III are correct
(c) Only I and II are correct
(d) Only III and IV are correct

83. Delocalization of charge may be expected in the species of formula

I. N_2H_4 II. NO_3^- III. HNO_3 IV. NH_4^+
(a) I, II and III only correct
(b) I and III only correct
(c) II, III and IV only correct
(d) II and III only correct

84. Which of the following possess two lone pair of electrons on the central atom and square planar in shape?
I. SF_4 II. XeO_4 III. XeF_4 IV. ICl_4
(a) I, II and III only correct
(b) II and IV only correct
(c) I and III only correct
(d) III and IV only correct

85. Which of the following molecules have a dative π-bond.
I. P_4O_{10} II. $(SiH_3)_3N$
III. P_4O_6 IV. N_2O_3
(a) I, II and III only correct
(b) II and IV only correct
(c) I and III only correct
(d) I and II only correct

86. Which of the following is the correct order of angle between two fluorine atoms linked to central atom in the given molecules?
I. $F_2C=O$ II. $N\equiv SF_3$
III. $IO_2F_2^-$ IV. $F_5I=O$
(a) I > II > III > IV (b) I > III > II > IV
(c) II > III > I > IV (d) III > II > I > IV

87. Arrange the following in the increasing order of their bond angles
I. NH_3 II. $N(CH_3)_3$
III. $N(SiH_3)_3$ V. NF_3
(a) I > II > III > IV (b) II > I > IV > III
(c) II > III > I > IV (d) III > II > I > IV

88. The electronic configuration of B_2 molecule is different from the electronic configuration of F_2 molecule is due to mixing of
(a) σls and σ2s orbitals
(b) σ2s and σ2p orbitals
(c) σ^*2s and σ2p orbitals
(d) σ^*2s and σ^*2p orbitals

89. Solid oxygen has a pale blue colour which is attributed to
(a) electronic transition from the singlet ground state to the triplet ground state
(b) electronic transitions from antibonding π* molecular orbitals (triplet state) to bonding $\sigma 2p_z$ (doublet) molecular orbitals
(c) electronic transitions from the antibonding π* molecular orbitals (triplet state) to excited antibonding $\sigma^* 2p_z$ molecular orbital (singlet state)
(d) electronic transitions from the triplet ground state to the singlet ground state

90. A sigma bond may be formed by the overlap of two atomic orbitals of atoms A and B. If the bond is formed along the *x*-axis which of the following overlaps is acceptable?
(a) s-orbital of A and p_z orbital of B
(b) p_x orbital of A and p_y orbital of B
(c) p_x orbital of A and p_z orbital of B
(d) p_x orbital of A and s-orbital of B

91. The way of writing Lewis structure of the cyanate ion OCN^- places one double bond between the carbon atom and the oxygen atom and another double bond between the carbon atom and nitrogen atom. What are the formal charges on the oxygen, carbon and nitrogen atoms, respectively for those structures?
(a) 0, 0, –1 (b) –1, 0, 0
(c) –1, + 1, –1 (d) –2, 1, 0

92. Which of the following is not a correct statement?
(a) Every AB_5 molecule does in fact have square pyramidal structure.
(b) Multiple bonds are always shorter than corresponding single bonds.
(c) The electron deficient molecules act as Lewis acids.
(d) The canonical structures have no real existence.

93. Most stable molecules
(a) have paired electrons and are diamagnetic
(b) have unpaired electrons and are para magnetic
(c) have paired electrons and are ferromagnetic
(d) have unpaired electrons and are ferromagnetic

94. The correct order of bond angle of NO_2^+, NO_2 and NO_2^- is
(a) $NO_2^+ < NO_2 < NO_2^-$
(b) $NO_2^+ < NO_2^- < NO_2$

(c) $NO_2^+ < NO_2 < NO_2^-$
(d) $NO_2^+ > NO_2 > NO_2^-$

95. Consider the following statements
I. Dispersion forces are evergreen and exist between all atoms, molecules and ions.
II. The extent of ion induced dipole interaction depends on the charge on ion.
III. Dry ice is held together by a network of C = O bonds.
IV. Among the hydrides of second period, decreasing order of boiling points is $HF > H_2O > NH_3 > CH_4$.
Correct statement(s) out of the above will be:
(a) I only (b) I and IV
(c) II and IV (d) I and II

96. The tetrahedral geometry of methane instead of square planar structure is supported by
(a) its dipole moment value
(b) its monochlorination
(c) its dichlorination
(d) its trichlorination

97. Which of the following is the correct order of the strength of hydrogen bonding in the given compound?
(a) $HF < NH_3$ (b) $H_2O > H_2O_2$
(c) $H_2O_2 > H_2O$ (d) $NH_3 > H_2O$

98. Which of the following statements is incorrect?
(a) A sigma bond has no free rotation around its axis.
(b) Two p-orbitals always overlap laterally.
(c) There can be more than one sigma bond between two atoms.
(d) All of these.

99. Which of the following conditions favours the formation of an ionic bond?
(a) Cation with low positive charge
(b) Cation with large radius
(c) Anion with small size
(d) All of these

100. The molecule MLE is planar with six pairs of electrons around 'M' in the valence shell. The value of 'E' is
(a) 6 (b) 2 (c) 4 (d) 3

101. Mark the incorrect statement in the following
(a) The bond order in species O_2, O_2^+ and O_2^- decreases as $O_2^+ > O_2 > O_2^-$.
(b) The bond energy in a diatomic molecule always increases when an electron is lost.
(c) Electrons in anti-bonding MO contribute to repulsion between two atoms.
(d) With increase in bond order, bond length decreases and bond strength increases.

102. Which of the following molecules has (have) a sp^2 hybridized atom?
I. CBr_4 II. $CH_2 = CH_2$
III. BF_3 IV. SO_2
(a) II and III (b) III and IV
(c) II, III and IV (d) II and IV

103. Which of the following about SF_4, SOF_4 and COF_2 molecules is correct?
(a) Equatorial FSF bond angle in SOF_4 will be less than in SF_4 molecule.
(b) Hybridization state of sulphur in SF_4 and SOF_4 molecules will be different.
(c) The bond angle FCO will be < 120° in molecule OCF_2.
(d) The axial FSF bond angle in SF_4 is 180°.

104. In which of the following pairs, both the species have the same hybridization?
I. SF_4, XeF_4 II. I_3^-, XeF_2
III. ICl_4^+, $SiCl_4$ IV. ClO_3^-, PO_4^{3-}
(a) I, II (b) II, III
(c) II, IV (d) I, II, III

105. Which of the following pairs have identical values of bond order?
(a) N_2 and O_2^{2-}
(b) NO^+ and N_2
(c) CN^- and O_2^-
(d) CO and O_2

106. A covalent bond is formed between
(a) an electronegative element and an electronegative element
(b) an electronegative element and an electropositive element
(c) an electropositive element and an electropositive element
(d) none of these

107. Which of the following pairs is paramagnetic according to MOT?
(a) O_2^- and O_2^{2-} (b) B_2 and C_2
(c) N_2^+ and O_2^+ (d) O_2^+ and H_2

108. Which of the following pairs of species have identical structures and shapes?
(a) NO_2^+ and NO_2^-
(b) PCl_5 and BrF_5
(c) XeF_4 and ICl_4^-
(d) XeF_2 and XeO_4

109. What would be the bond angle θ when the dipole moment is maximum for the triatomic molecule YXY?
(a) $\theta = 90°$ (b) $\theta = 120°$
(c) $\theta = 150°$ (d) $\theta = 180°$

110. BF_3 is a planar molecule while NF_3 is pyramidal because
(a) the N-F bond is more covalent than the B-F bond
(b) the N-atom has higher value of ionization energy than the B-atom
(c) the B-F bond is more polar than the N-F bond
(d) BF_3 has no lone pair of electrons but NF_3 has a lone pair of electrons

111. Which of the following states the correct relationship between the dissociation of N_2 and N_2^+?
(a) Dissociation energy of N_2 = dissociation energy of N_2^+
(b) Dissociation energy of N_2 > dissociation energy of N_2^+
(c) Dissociation energy of N_2 < dissociation energy of N_2^+
(d) Dissociation energy of N_2 can either be lower or higher than the dissociation energy of N_2^+

112. Which of the following set of species have planar structures?
(a) $I_3^-, CH_3^-, ClO_3^-, SiF_6^{2-}$
(b) $I_3^-, ICl_4^-, Al_2Cl_6, TeCl_4$
(c) $SCl_2, N_2O_5, SF_4, XeOF_4$
(d) $I_2Cl_6, XeF_2, BrF_4^-, XeF_5^-$

113. The boiling point of p-nitrophenol is higher than that of o-nitrophenol because
(a) p-nitrophenol has intermolecular hydrogen bonding while o-nitrophenol has intramolecular hydrogen bonding.
(b) p-nitrophenol has intramolecular hydrogen bonding while o-nitrophenol has intermolecular hydrogen bonding.
(c) —NO_2 group at p-position behaves in a different way than that of o-position.
(d) Hydrogen bonding exists in p-nitrophenol but no hydrogen bonding is present in o-nitrophenol.

114. In which species the electron pair geometry is the same as the molecular geometry?
(a) BeF_2 (b) PF_3
(c) SF_4 (d) IF_5

115. In H_2^+ ion
(a) one electron is bound to two protons
(b) two electrons are bound to two protons
(c) three electrons are bound to two protons
(d) none of these happens

116. Which of the following pairs have nearly identical values of bond energy?
(a) O_2 and H_2 (b) N_2 and CO
(c) F_2 and I_2 (d) O_2 and Cl_2

117. The molecular sizes of ICl and Br_2 are nearly the same, but the boiling point of ICl is about 39°C higher than that of Br_2. This is because
(a) the bond energy of I— Cl is greater than that of Br — Br
(b) the ionization energy of iodine is less than that of bromine
(c) ICl is polar while Br_2 is non-polar
(d) the size of iodine is greater than that of bromine

118. Which of the following factors is the most responsible for increase in boiling point as we move from He to Xe?
(a) Decrease in ionization energy
(b) Increasing in electronegativity
(c) Decrease in polarizability
(d) Increase in polarizability

119. Hydrogen bonding is exhibited by
(a) all substances containing H-atoms
(b) molecules in which hydrogen is bonded to F, O or N
(c) molecules in which one hydrogen is bonded to F and the other is bonded to Cl
(d) all substances containing H and O atoms

120. The density of water is greater than that of ice because of
(a) dipole-dipole interaction
(b) hydrogen bonding
(c) dipole-induced dipole interaction
(d) covalent bond formation

121. Which of the following is true?
(a) Bond order $\propto \dfrac{1}{\text{bond length}} \propto$ bond energy
(b) Bond order $\propto$ bond length $\propto \dfrac{1}{\text{bond energy}}$
(c) Bond order $\propto \dfrac{1}{\text{bond length}} \propto \dfrac{1}{\text{bond energy}}$
(d) Bond order $\propto$ bond length $\propto$ bond energy

122. Which of the following has been arranged in order of decreasing bond length?
(a) P — O > Cl — O > S — O
(b) P — O > S — O > Cl — O
(c) S — O > Cl — O > P — O
(d) Cl — O > S — O > P — O

123. During the formation of a molecular orbital from atomic orbitals, the electron density is
(a) minimum in the nodal plane
(b) maximum in the nodal plane
(c) zero in the nodal plane
(d) zero on the surface of the lobe

124. The oxygen molecule is paramagnetic because
(a) the bonding electrons outnumber the anti-bonding electrons in the molecular orbital

(b) it contains unpaired electrons in the antibonding molecular orbitals.
(c) it contains unpaired electrons in the bonding molecular orbitals.
(d) the number of bonding electrons equal that of the anti-bonding electrons in the molecular orbitals.

125. If a molecule MX_3 has zero dipole moment, the sigma bonding orbitals used by M (atomic number <2) are
(a) pure P (b) sp hybrid
(c) sp^2 hybrid (d) sp^3 hybrid

126. The volatility of HF is low because of
(a) its low polarizability
(b) the weak dispersion interaction between the molecules
(c) its small molecular mass
(d) its strong hydrogen bonding

127. Which of the following has the least dipole moment?
(a) NF_3 (b) $NC1_3$
(c) SO_2 (d) NH_3

128. Which of the following is the least polar?
(a) HF (b) HBr
(c) HI (d) HCl

129. Which of the following have been arranged in increasing order of bond order as well as bond dissociation energy?
(a) $O_2^{-2} < O_2^- < O_2^+ < O_2$
(b) $O_2^{-2} < O_2^- < O_2 < O_2^+$
(c) $O_2 < O_2^+ < O_2^{2-} < O_2^-$
(d) $O_2^+ < O_2^{2-} < O_2^- < O_2$

130. Orthonitrophenol is steam volatile but paranitrophenol is not because
(a) orthonitrophenol has intramolecular hydrogen bonding while paranitrophenol has intermolecular hydrogen bonding
(b) both ortho- and paranitrophenols have intramolecular hydrogen bonding
(c) orthonitrophenol has intermolecular hydrogen bonding and paranitrophenol has intramolecular hydrogen bonding
(d) Van der Waal's forces are dominant in orthonitrophenol

131. The maximum possible number of hydrogen bonds in which a water molecule can participate is
(a) four (b) three
(c) two (d) one

132. The overlapping powers (overlap integrals) of 2s, 2p, $2sp^2$, $2sp^3$ and 2sp orbitals are in the order
(a) $2sp > 2sp^2 > 2sp^3 > 2p > 2s$
(b) $2p > 2s > 2sp^3 > 2sp^2 > 2sp$
(c) $2sp^3 > 2sp^2 > 2sp > 2s > 2p$
(d) $2sp^3 > 2sp^2 > 2sp > 2p > 2s$

133. Increasing order of strength of hydrogen bonding in X…….. H — X for X among O, S, F, Cl and N is
(a) Cl < N < O < F (b) Cl < S < O < N
(c) S < Cl < N < O (d) Cl < S < O < F

134. The bond angle in NH_3 is
(a) Less than that in H_2O but greater than that in H_2S
(b) Greater than that in H_2O but less that that in H_2S
(c) Less than both those of H_2O and H_2S
(d) Greater than both those of H_2O and H_2S

135. In the following species, the one having a planar structure is
(a) BF_4^- (b) $SnCl_4$
(c) BrF_4^- (d) NH_4^+

136. H_2, Li_2, B_2 each has a bond order equal to 1, the order of their stability is
(a) $H_2 = Li_2 = B_2$ (b) $H_2 > Li_2 > B_2$
(c) $H_2 > B_2 > Li_2$ (d) $B_2 > Li_2 < H_2$

137. Which of the following facts given are correct?
I. Bond length order : $H_2^- = H_2^+ > H_2$
II. O_2^+, NO, N_2^- have same bond order of $2\frac{1}{2}$
III. Bond order can assume any value including zero.
IV. NO_3^- and CO_3^{2-} have same bond order for X — O bond (where X is the central atom).
(a) I, II and III (b) I and IV
(c) II and IV (d) I, II, III and IV

138. Which of the following have identical bond order?
I. CN^- II. O_2^-
III. NO^+ IV. CN^+
(a) I, III (b) I, II
(c) II, III (d) I, II, III

139. Among KO_2, AlO_2^-, BaO_2 and NO_2^+, unpaired electron is present in
(a) KO_2 only (b) NO_2^+ and BaO_2
(c) KO_2 and AlO_2^- (d) BaO only

140. Which is the correct order of size of O^-, O^{2-}, F and F^-
(a) $O^{2-} > F^- > O^- > F$ (b) $O^{2-} > F^- > F > O^-$
(c) $O^- > O^{2-} > F > F^-$ (d) $O^{2-} > O^- > F^- > F$

141. The geometry of ammonia molecule can be best described as
(a) nitrogen at one vertex of a regular tetrahedron, the other three vertices being occupied by the three hydrogen
(b) nitrogen at the centre of the tetrahedron, three of the vertices being occupied by three hydrogen
(c) nitrogen at the centre of equilateral triangle, three comers being occupied by three hydrogen
(d) nitrogen at the junction of a 'T' three open ends being occupied by three hydrogen

142. The compound with the highest degree of covalent character is

(a) NaCl (b) $MgCl_2$
(c) AgCl (d) CsCl

143. Which of the following is the correct electron dot structure of N_2O molecule?

(a) $:N = N = \ddot{O}:$ (b) $:N = N = \underset{\cdot\cdot}{\ddot{O}}:$

(c) $\ddot{N} = \ddot{N} = \ddot{O}:$ (d) $:N = \ddot{N} = \underset{\cdot\cdot}{\ddot{O}}:$

144. Metals possess lustre when freshly cut because

(a) they have a hard surface and light is reflected back
(b) their crystal structure contains ordered arrangement of their constituent atoms
(c) they contain loosely bound electrons which absorb the photons and then re emit
(d) they are obtained from the minerals on which light has been falling for years

145. Identify the correct statement about CH_2F_2

(a) $H\hat{C}H$ bond angle is equal to $F\hat{C}H$ bond angle.
(b) $H\hat{C}H$ bond angle is greater than $F\hat{C}H$ bond angle.
(c) $H\hat{C}H$ bond angle is less than $F\hat{C}H$ bond angle.
(d) bond angles are equal

146. The pair of species with similar shape is

(a) PCl_3, NH_3 (b) CF_4, SF_4
(c) $PbCl_2$, CO_2 (d) PF_5, IF_5

147. In which of the following compounds B — F bond length is shortest?

(a) BF_4^- (b) $BF_3 \leftarrow NH_3$
(c) BF_3 (d) $BF_3 \leftarrow N(CH_3)_3$

148. Which molecular geometry is least likely to result from a trigonal bipyramidal electron geometry?

(a) Trigonal planar (b) See-saw
(c) Linear (d) T-shaped

149. Give the correct order of initials T or F for the following statements. Use 'T' if statement is true and F if it is false.

I. The order of repulsion between different pair of electrons is $lp - lp > bp - lp > bp - bp$.
II. In general, as the number of lone pair of electrons on the central atom increases, value of bond angle from normal bond angle also increases.
III. The number of lone pair on O in H_2O is 2 while on N in NH_3 is 1.
IV. The structures of xenon fluorides and xenon oxyfluorides could not be explained on the basis of VSEPR theory.

(a) TTTF (b) FFTF (c) TFTT (d) TFFF

150. The correct increasing order of adjacent bond angle among BF_3, PF_3 and ClF_3 is

(a) $BF_3 < PF_3 < ClF_3$
(b) $PF_3 < BF_3 < ClF_3$
(c) $ClF_3 < PF_3 < BF_3$
(d) $BF_3 = PF_3 = ClF_3$

151. The bond angles of NH_3, NH_4^+ and NH_2^- are in the order

(a) $NH_2^- > NH_3 > NH_4^+$
(b) $NH_4^+ > NH_3 > NH_2^-$
(c) $NH_3 > NH_2^- > NH_4^+$
(d) $NH_3 > NH_4^+ > NH_2^-$

152. The compound MX_4 is tetrahedral. The number of XMX angles in the compound is

(a) three (b) four
(c) five (d) six

153. Out of $CHCl_3$, CH_4 and SF_4 the molecules that do not have a regular geometry are

(a) $CHCl_3$ only (b) $CHCl_3$ and SF_4
(c) CH_4 only (d) CH_4 and SF_4

154. Select the incorrect statement about N_2F_4 and N_2H_4

I. In N_2F_4, d-orbitals are contracted by electronegative fluorine atoms, but d-orbital contraction is not possible by H-atom in N_2H_4
II. The N — N bond energy in N_2F_4 is more than N — N bond energy in N_2H_4
III. The N —N bond length inN_2F_4 is more than that of in N_2H_4
IV. The N — N bond length in N_2F_4 is less than that of in N_2H_4

Choose the correct code

(a) I, II and III (b) I and III
(c) II and IV (d) II and III

155. Which combination will give the strongest ionic bond?

(a) Na^+ and F^- (b) Mg^{2+} and Cl^-
(c) Na^+ and O^{2-} (d) Mg^{2+} and O^{2-}

156. The HOMO in nitricoxide molecule is

(a) $\pi_{2py} = \pi_{2pz}$ (b) $\pi^*_{2py} = \pi^*_{2pz}$
(c) σ_{2px} (d) σ^*_{2px}

157. A simplified application of the MO theory to the hypothetical 'molecule' OF would give its bond order as

(a) 2 (b) 1.5 (c) 1.0 (d) 0.5

158. Which of the following compounds have the same number of lone pairs with their central atom?

I. XeF_5^- II. BrF_3

III. XeF_2 IV. H_3S^+ V. XeO_6^{4-}
(a) IV and V (b) I and III
(c) I and II (d) II, IV and V

159. Which of the following statements is correct?
(a) Low sublimation energy of non-metals favours the formation of ionic bond.
(b) High bond dissociation energies of metals favours the formation of ionic bond.
(c) Nitrogen due to stable half-filled configuration has lesser electronegativity value than its adjacent elements on either side.
(d) High lattice energy favours the formation of ionic compound.

160. If there is no sp mixing the bond order and magnetic character in C_2 molecule is
(a) 1 and paramagnetic
(b) 1 and diamagnetic
(c) 2 and paramagnetic
(d) 2 and diamagnetic

161. The strongest P— O bond is found in the molecule
(a) $F_3P = O$ (b) $Cl_3P = O$
(c) $Br_3P = O$ (d) $(CH_3)_3P = O$

162. F — As — F bond angle in AsF_3Cl_2 can be
(a) 90° and 180° only (b) 120° only
(d) 90° and 120° only (c) 90° only

163. Which statement is incorrect?
(a) Higher the polarization, higher will be relative solubility in non-polar solvent.
(b) Higher the polarization, higher will be the intensity of colour.
(c) Diamagnetic substances sometimes become coloured due to HOMO-LUMO transition.
(d) Higher the polarization in a metal oxide, higher will be the basic character.

164. If the ionic radii of K^+ and F^- are about 1.34Å each, then the expected values of atomic radii of 'K' and 'F' should be respectively
(a) 1.34 and 1.34 Å (b) 2.31 and 0.64 Å
(c) 0.64 and 2.31 Å (d) 2.31 and 1.34 Å

165. In which of the following pairs of species the central atom is not having the same hybridization?
(a) ClF_3O, ClF_3O_2 (b) $(ClF_2O)^+$, $(ClF_4O)^-$
(c) ClF_3, ClF_3O (d) $(ClF_4O)^-$, $(XeOF_4)$

166. Which compound has the smallest bond angle in each set?
I. OSF_2, $OSCl_2$, $OSBr_2$ (halogen-S-halogen angle)
II. $SbCl_3$, $SbBr_3$, SbI_3 (halogen-Sb-halogen angle)
III. PI_3, AsI_3, SbI_3 (halogen-central atom halogen angle)
(a) $OSBr_2$, $SbCl_3$, SbI_3
(b) $OSCl_2$, $SbBr_3$, AsI_3
(c) OSF_2, $SbCl_3$, SbI_3
(d) OSF_2, $SbCl_3$, PI_3

167. Arrange ClO_3^-, BrO_3^- and IO_3^- ions in the increasing order of their bond angles
(a) $ClO_3^- > BrO_3^- > IO_3^-$
(b) $IO_3^- > BrO_3^- > ClO_3^-$
(c) $ClO_3^- > IO_3^- > BrO_3^-$
(d) $BrO_3^- > ClO_3^- > IO_3^-$

168. Which of the following is incorrectly matched?
(a) SiF_4: can act as Lewis acid.
(b) Benzyne: All C-atoms are sp^2 hybridized.
(c) PBr_3: non-polar.
(d) CHF = C = CHF: Nodal planes of π-bonds are not lying in the same plane.

169. Which of the following molecule/species having minimum number of lone pairs on its central atom?
(a) BrF_3 (b) BrF_4^-
(c) XeF_5^+ (d) I_3^-

170. Classify the following statements into true and false statements and indicate them with T (true) and F (false).
S_1: Bond orbital which is symmetrical about the line joining the two nuclei is known as pi bond.
S_2: In KHF_2 the bonds present are ionic, covalent and coordinate covalent bonds.
S_3: In liquid NH_3, Bi can be reduced from 0 to –3 by using Na.
S_4: Li, Na, K can from nitrides by direct reaction with nitrogen.
(a) TFTF (b) FFTF
(c) TTFF (d) FTFT

171. In $[ICl_2]^-$ ion, the lone pairs present in iodine atom are separated at an angle of
(a) 90° (b) 120°
(c) 180° (d) 60°

172. In which compound do two ions composed of non-metal atoms combine to form an ionic compound?
(a) $Al_2(SO_4)_3$ (b) CH_3CN
(c) CH_3COOH (d) NH_4NO_3

173. The correct sequence of increasing melting points of $BeCl_2$, $MgCl_2$, $CaCl_2$, $SrCl_2$ and $BaCl_2$ is
(a) $BeCl_2 < SrCl_2, < CaCl_2 < MgCl_2 < BaCl_2$
(b) $BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2 < BaCl_2$
(c) $BeCl_2 < CaCl_2 < MgCl_2 < SrCl_2 < BaCl_2$
(d) $MgCl_2 < BeCl_2 < SrCl_2 < CaCl_2, < BaCl_2$

174. The electronic configurations of four elements L, P, Q and R are given below
$L = 1s^2 2s^2 2p^4$; $Q = 1s^2 2s^2 2p^6 3s^2 3p^5$
$P = 1s^2 2s^2 2p^6 3s^1$; $R = 1s^2 2s^2 2p^6 3s^2$
The formula of ionic compounds that can be formed between these elements are

(a) L_2P, RL, PQ, R_2O
(b) LP, RL, PQ, RQ
(c) P_2L, RL, PQ, RQ_2
(d) LP, R_2L, P_2Q, RQ

175. Among LiCl, $BeCl_2$, BCl_3 and CCl_4 the covalent bond character follows the order
(a) $LiCl > BeCl_2 > BCl_3 > CCl_4$
(b) $LiCl < BeCl_2 < BCl_3 < CCl_4$
(c) $LiCl > BeCl_2 > CCl_4 > BCl_3$
(d) $LiCl < BeCl_2 < BCl_3 > CCl_4$

176. Among NaF, NaCl, NaBr and NaI, NaF has highest melting point because
(a) it has maximum ionic character
(b) it has minimum ionic character
(c) it has associated molecules
(d) it has least molecular weight

177. Amongst LiCl, RbCl, $BeCl_2$ and $MgCl_2$ the compounds with the greatest and least ionic character respectively are
(a) LiCl and RbCl (b) RbCl and $BeCl_2$
(c) RbCl and $MgCl_2$ (d) $MgCl_2$ and $BeCl_2$

178. The correct order of decreasing dipole moment is
(a) $HF > SO_2 > H_2O > NH_3$
(b) $HF > H_2O > SO_2 > NH_3$
(c) $HF > NH_3 > SO_2 > H_2O$
(d) $H_2O > NH_3 > SO_2 > HF$

179. Knowing that $Na^+ > Mg^{2+}$ and $S^{2-} > Cl^-$ predict which compound will be least soluble in a polar solvent.
(a) MgS (b) Na_2S
(c) $MgCl_2$ (d) NaCl

180. Polarization may be called the distortion of the shape of an anion by an adjacently placed cation. Which of the following statements is correct?
(a) Minimum polarization is brought about by a cation of low radius.
(b) A large cation is likely to bring about a large degree of polarization.
(c) Maximum polarization is brought about by a cation of high charge.
(d) A small anion is likely to undergo a large degree of polarization.

181. The melting point of AlF_3 is 1040°C and that of SiF_4 is –77°C (it sublimes) because
(a) There is a very large difference in the ionic character of the A1 — F and Si — F bonds.
(b) In AlF_3, Al^{3+} interacts very strongly with the neighbouring F^- ions to give a three-dimensional structure but in SiF_4 no interaction is possible.
(c) The silicon ion in the tetrahedral SiF_4 molecule is not shielded effectively from the fluoride ions, whereas in AlF_3 and Al^{3+} ion is shielded on all sides.
(d) The attractive forces between the SiF_4 molecules are strong, whereas those between $A1F_3$ molecules are weak.

182. CaO and NaCl have the same crystal structure and approximately the same ionic radii. If 'u' is the lattice energy of NaCl, the approximate lattice energy of CaO is
(a) $\frac{u}{2}$ (b) u
(c) 2u (d) 4 u

183. In an ionic compound A^+X^- the degree of covalent bonding is greatest when
(a) A^+ and X^- ions are small
(b) A^+ is small and X^- is large
(c) A^+ and X^- ions are approximately of the same size
(d) A^+ and X^- ions are larger

184. Which of the following statement is correct?
(a) $FeCl_2$ is more covalent than $FeCl_3$
(b) $FeCl_3$ is more covalent than $FeCl_2$
(c) Both $FeCl_2$ and $FeCl_3$ are equally covalent
(d) $FeCl_2$ and $FeCl_3$ do not have any covalent character

185. For two ionic solids, CaO and KI, identify the wrong statement among the following
(a) Lattice energy of CaO is much larger than of KI.
(b) KI is soluble in benzene.
(c) CaO has a higher melting point.
(d) KI has a lower melting point.

186. Polarizing action of Cd^{2+} on anions is stronger that of Ca^{2+} because
(a) the charges of the ions are same
(b) their radii are same (Ca^{2+} = 0.104 nm, Cd^{2+} = 0.099 nm)
(c) the Ca^{2+} ion has a noble-gas electron configuration and the Cd^{2+} ion, an 18-electron configuration of its outer shell
(d) all are correct

187. Which set contain no ionic species?
(a) NH_4Cl, OF_2, H_2S
(b) CO_2, CCl_4, Cl_2
(c) BF_3, AlF_3, TlF_3
(d) I_2, CaO, CH_3Cl

188. Solubility of alkali metal fluorides increases down the group. Select the correct explanation for the given statement
(a) Hydration energy increases and lattice energies decreases down the group.
(b) Both energies decrease down the group but decrease in hydration energy is rapid.

(c) Both energies decrease down the group but decrease in lattice energy is rapid.
(d) Both energies increase down the group but increase in hydration energy is rapid.

189. An ionic bond can be formed between two atoms when
(a) one of them has a low ionization energy and the other a high electron affinity
(b) both the atoms have low values of ionization energy
(c) both the atoms have high values of ionization energy
(d) both the atoms have low values of electron affinity

190. The cohesive energy of an ionic crystal is the energy
(a) liberated during its formation from individual neutral atoms
(b) absorbed during its formation from individual neutral atoms
(c) liberated during the formation of positive ions
(d) absorbed during the formation of negative ions

191. According to Fajan's rules, covalent bond formation is favoured when there is a
(a) large cation and a small anion
(b) large cation and a large anion
(c) small cation and a small anion
(d) small cation and a large anion

192. Which of the following has been arranged in order of increasing covalent character?
(a) $KCl < CaCl_2 < AlCl_3 < SnCl_4$
(b) $SnCl_4 < AlC1_3 < CaCl_2 < KCl$
(c) $AlCl_3 < CaCl_2 < KCl < SnCl_4$
(d) $CaCl_2 < SnCl_4 < KCl < AlCl_3$

193. The radius of the Cl^- ion is 38 per cent larger than that of the F^- ion but the radius of the Br^- ion is only 6.5 per cent larger than that of the Cl^- ion. The relatively small difference in size between Cl^- and Br^- ions is due to the fact that
(a) the Br^- ion contains ten 3d electrons, which fail to shield the nuclear charge effectively
(b) the Br^- ion contains ten 3d electrons, which shield the nuclear charge effectively
(c) the Br^- ion contains six 4p electrons, which shield the nuclear charge effectively
(d) the Br^- ion contains ten 3d electrons and six 3p electrons, together they shield the nuclear charge effectively

194. Which of the following statement is correct in the context of Fajan's rule explaining covalency?
(a) The covalent character increases as the size of the cation decreases.
(b) The covalent character increases as the size of the anion increases.
(c) The covalent character is more pronounced in cations with non-noble gas configurations that as cation of similar size with noble gas configurations.
(d) All of these.

195. During the formation of an ionic bond the cation may achieve
(a) ns^2 or $ns^2\ np^6$ configuration
(b) Pseudonoble gas configuration
(c) Inert pair configuration
(d) All of these configurations

196. Both the lattice energy (ΔU_0) and hydration enthalpy (ΔH_h) of a binary salt are negative quantities. However, if ΔU_0 is more negative than ΔH_h then
(a) the salt will not dissolve in water
(b) salt will dissolve in water
(c) dissolution of salt in water is exothermic
(d) dissolution of salt in water is endothermic

197. The melting point of RbBr is 682°C while that of NaF is 988°C. The principal reason for this fact is
(a) The molar mass of NaF is smaller than that of RbBr.
(b) The bond in RbBr has more covalent character than the bond in NaF.
(c) The difference in electronegativity between Rb and Br is smaller than the difference between Na and F.
(d) The internuclear distance, $r_c + r_a$ is greater for RbBr than for NaF

Multiple Choice Questions with One or More than One Answer

1. Correct order of decreasing boiling points is
(a) HF > HI > HBr > HCl
(b) $H_2O > H_2Te > H_2Se > H_2S$
(c) $Br_2 > Cl_2 > F_2$
(d) $CH_4 > GeH_4 > SiH_4$

2. Which of the following statements are correct about sulphur hexafluoride?
(a) All S—F bonds are equivalent
(b) SF_6 is a planar molecule
(c) Oxidation number of sulphur is the same as number electrons of sulphur involved in bonding
(d) Sulphur has acquired the electronic structure of the gas argon

3. Which of the following process is/are associated with change of hybridization of the underlined compound?
(a) $\underline{Al}(OH)_3$ ppt dissolved in NaOH
(b) $\underline{B}_2H_6$is dissolved in THF
(c) $\underline{Si}F_4$ vapour is passed through liquid HF
(d) Solidification of $\underline{P}Cl_5$ vapour

4. Which combination of compounds, their geometry and hybridization is correct?
(a) XeF_4 — square planar, sp^3d^2
(b) BeF_3^- – trigonal planar, sp^2
(c) NH_2^- - linear, sp
(d) ClF_3 — T shape, sp^3 d

5. Which of the molecules has unpaired electrons in anti-bonding orbitals?
(a) CO (b) O_2^-
(c) NO (d) N_2^+

6. Select the correct statement(s) of the following
(a) NF_3 is a weaker base than NH_3 or NCl_3.
(b) $AlCl_3$ is largely covalent while AlF_3 is largely ionic.
(c) A triple bond between two atoms may be made up of two sigma and one π bonds.
(d) N^{3-} ion is more susceptible to polarization than O^{2-} ion.

7. In which of the following change(s) hybridization of the underlined atom is not affected?
(a) $\underline{P}H_3 + H^+ \longrightarrow PH_4^+$
(b) $\underline{B}F_3 + NaH \longrightarrow NaBH_4 + NaF$
(c) $H_3\underline{B}O_3 \longrightarrow HBO_2 + H_2O$
(d) $2H\underline{Cl}O_2 \longrightarrow HClO + HClO_3$

8. Which of the following is/are correct statement(s)?
(a) HF molecule forms two hydrogen bonds.
(b) Lower alcohols are soluble in water.
(c) Hydrogen bond is generally formed by covalent compounds.
(d) Hydrogen bond is not formed by non-polar covalent compounds.

9. Which of the following statement(s) are not correct?
(a) All C — O bonds in CO_3^{2-} are equal but not in H_2CO_3.
(b) All C — O bonds in HCO_2^- are equal but not in HCO_2H.
(c) C — O bond length in HCO_2^- is longer than C — O bond length in CO_3^{2-}.
(d) C — O bond length in HCO_2^- and C — O bond length in CO_3^{2-}

10. Which of the following is/are paramagnetic?
(a) B_2 (b) O_2
(c) N_2 (d) He_2^+

11. Intermolecular hydrogen bonding increases the enthalpy of vapourization of a liquid due to the
(a) decrease in the attraction between molecules
(b) increase in the attraction between molecules
(c) decrease in the molar mass of unassociated molecules
(d) increase in the effective molar mass of hydrogen-bonded molecules

12. Which of the following is/are correct statement(s)?
(a) Probability of finding the electron in bonding MO is more than combining atomic orbitals.
(b) Bonding MOs are formed when the same sign of orbitals overlap.
(c) *d-d* combination of atomic orbitals gives δ^- and δ^+ MOs.
(d) None of these.

13. According to MOT (Molecular Orbital Theory), the MOs are formed by mixing of atomic orbitals through LCAO (linear combination of atomic orbitals). The correct statement(s) about MOs is/are
(a) Bonding MOs are formed by additive wavefunctions of atomic orbitals.
(b) Anti-bonding MOs are formed by subtraction of wave functions of atomic orbitals.
(c) Non-bonding MOs do not take part in bond formation because they belong either to inner shells or of having same energy as those of atomic orbitals.
(d) Anti-bonding MOs provide stability to molecules while bonding MOs make the molecule unstable.

14. Depending on Lewis structure OF_2, O_2F_2 and S_2F_2 which of the following statements are correct?
(a) S_2F_2 has two possible isomeric structures, whereas O_2F_2 has only one.
(b) The bond angle in OF_2 is 103° but in OCl_2 is 111°.
(c) The O — O bond length in O_2F_2 is only 1.22 Å but that in H_2O_2 is 1.48 Å.
(d) The O — F bond length in O_2F_2 is 1.58 Å but in OF_2 is only 1.41 Å.

15. In which of the following pairs, both the species have the same state of hybridization?
(a) $[NO_3^-, NH_2^-]$ (b) $[NO_2^-, H_3O^+]$
(c) $[BF_3, CH_3^+]$ (d) $[PCl_4^+, NH_4^+]$

16. In which of the following set all the species are paramagnetic in nature?
(a) O_2, O_2^{2+}, N_2^{2-} (b) B_2, C_2, H_2
(c) O_2^-, O_2^+, O_2 (d) N_2^+, O_2^+, F_2^+

17. Which of the following reaction involves a decrease in bond length?
(a) $N_2 \longrightarrow N_2^+$ (b) $O_2 \longrightarrow O_2^+$
(c) $C_2 \longrightarrow C_2^{2-}$ (d) $O_2^+ \longrightarrow O_2$

18. Which of the following statements is incorrect?
(a) In O_2F_2 the O — O bond is shorter than the O — O bond in H_2O_2.
(b) In O_2F_2, the O — O bond is longer than the O — O bond in H_2O_2.

(c) H_2O_2 and O_2F_2 have similar structures; hence, the O — O bond of the two molecules is identical.
(d) O_2F_2 does not contain the peroxide bond (— O — O —).

19. The molecule/ion having last electron in the antibonding π molecular orbitals are/is
(a) O_2 (b) O_2^+
(c) Ne_2 (d) N_2^+

20. The correct statement about sulphur hexafluoride include
(a) There are 12 F—S—F 90° bond angles.
(b) S in SF_6 has an expanded octet.
(c) With H_2O, SF_6 can accept lone pair of electron with empty 3*d* atomic orbital and gets hydrolyzed.
(d) SF_6 has a distorted octahedral geometry.

21. Correct statements about CH_2F_2 molecule include
(a) All bond angles are equal.
(b) F—C—F bond angle is less than H—C—H bond angle.
(c) Dipole moment value is not equal to zero.
(d) H—C—H bond angle is greater than tetrahedral angle.

22. Which of the following statement(s) is/are not correct for the following compounds?

I. $SCl_2(OCH_3)_2$ II. $SF_2(OCH_3)_2$

(a) — OCH_3 groups in both cases occupy the same position.
(b) Cl-atom occupy equatorial position in case of (I) and F atoms occupy equatorial position in case of (II).
(c) Cl-atoms occupy axial position in case of (I) and F-atoms occupy equatorial position in case of (II).
(d) Cl- and F-atoms occupy either axial or equatorial position in case of (I) and (II), respectively.

23. Which of the following statement(s) is/are correct?
(a) The peroxide ion has a bond order 1 while the oxygen molecule has a bond order of 2.
(b) The peroxide ion has a weaker bond than the dioxygen molecule.
(c) The peroxide ion as well as the dioxygen molecules are paramagnetic.
(d) The bond length of the peroxide ion is greater than that of the dioxygen molecule.

24. Most ionic compounds have
(a) high melting points and low boiling points
(b) high melting points and non-directional bonds
(c) three dimensional network structures and are good conductors of electricity in the molten state
(d) high solubilities in polar solvents and low solubilities in non-polar solvent

25. Which of the following pairs have identical values of bond order?
(a) N_2^+ and O_2^+ (b) F_2 and Ne_2
(c) O_2 and B_2^{2-} (d) C_2^{2-} and N_2

26. Which of the following are paramagnetic?
(a) B_2 (b) O_2
(c) N_2 (d) He_2

27. Which of the following is correct?
(a) During O_2^+ formation, one electron is removed from the bonding MO.
(b) During O_2^+ formation, one electron is removed from anti-bonding MO.
(c) During O_2^- formation, one electron is added to the bonding MO.
(d) During C_2^- formation, one electron is added to the bonding MO.

28. Which of the following have (18, +2) electronic configuration?
(a) Pb^{2+} (b) Cd^{2+}
(c) Bi^{3+} (d) SO_4^{2+}

29. Which of the following statements is incorrect?
(a) O_2 is paramagnetic, O_3 is also paramagnetic.
(b) O_2 is paramagnetic, O_3 is diamagnetic.
(c) B_2 is paramagnetic, C_2 is also paramagnetic.
(d) Different observation is found in their bond lengths when $NO \longrightarrow NO^+$ and $CO \longrightarrow CO^+$.

30. Which of the following species are linear?
(a) N_3^- (b) I_3^-
(c) ICl^{2-} (d) ClO_2

31. Which of the following species have equal bond order?
(a) H_2^+ (b) H_2^-
(c) N_2 (d) O_2

32. Which of the following sets of compounds are isoelectronic and isostructural?
(a) $CO_2, CS_2, CNO^-, CN_2^{2-}$
(b) $ClO_3^-, BrO_3^-, XeO_3, SO_3^{2-}$
(c) $BF_4^-, NH_4^+, CH_4, SiH_4$
(d) $SO_2Cl_2, POCl_3, ClO_4, XeO_4$

33. Which of the following compound/compounds do not conduct electricity in their molten state?
(a) $MgCl_2$ (b) $BeCl_2$
(c) BeF_2 (d) MgF_2

34. In which of the following, the hybrid orbitals of the central metal atom have the same *s*-character?
(a) CH_4 (b) XeO_3
(c) $Ni(CO)_4$ (d) $[Ni(CN)_4]^{2-}$

35. The compound(s) which contain ionic, covalent and coordinate bonds is/are
(a) H_2SO_4 (b) NH_4Cl
(c) $K_4[Fe(CN)_6]$ (d) CaC_2

36. The molecules or ions which have bond pairs as well as lone pairs of electrons on the central atom?
(a) SF_4 (b) ClF_3
(c) XeF_2 (d) CO_3^{2-}

37. Select the correct statements
(a) NF_3 is weaker base than NH_3.
(b) NO^+ is more stable than O_2.
(c) $AlCl_3$ has higher m.pt. than AlF_3.
(d) $SbCl_3$ is more covalent than $SbCl_5$.

38. Which of the following relation is/are correct?
(a) Covalent character $\propto \dfrac{1}{\text{Dipole moment}}$
(b) Covalent character $\propto$ Pseudoinert gas configuration
(c) Ionic character $\propto$ inert gas configuration
(d) Ionic character $\propto \dfrac{1}{\text{Dipole moment}}$

39. $CuCl_2$ is more covalent. This can be justified on the basis of:
(a) VSEPR theory (b) Hybridization
(c) Fajan's rule (d) Hydration energy

40. Which of the following is/are correct relation?
(a) Bond energy $\propto$ (polarity of the bond)^1
(b) Bond energy $\propto$ (s-character of hybrid orbital)$^{-1}$
(c) Bond energy $\propto$ (atomic radius)$^{-1}$
(d) Bond energy $\propto$ (Bond order)^1

41. Which of the following are isostructural?
(a) N_3^- (b) I_3^-
(c) N_2O (d) NO_2

42. Which have fractional bond order?
(a) O_2^+ (b) I_3^-
(c) NO (d) H_2^+

43. In which of the following compounds having general formula X_2H_6 there is no X — X bond?
(a) B_2H_6 (b) C_2H_6
(c) I_2Cl_6 (d) Al_2Cl_6

44. A π-bond may be formed between two p_X orbitals containing one unpaired electron each when they approach each other approximately along
(a) X-axis (b) Y-axis
(c) Z-axis (d) any direction

45. The linear structure is assumed by
(a) $SnCl_2$ (b) NCO^-
(c) NO_2^+ (d) CS_2

46. Which of the following statements are correct about $[IOF_4]^-$ ion and $SeOCl_2$?
(a) In $[IOF_4]^-$ ion the lone pair and oxygen atoms occupy the opposite corners of the octahedron
(b) The angle OIF is less than that of FIF.
(c) $SeOCl_2$ molecule has a pyramidal shape.
(d) Angle ClSeO is smaller than that of Cl SeCl.

47. Which of the following are non-polar?
(a) SO_3 (b) BF_3
(c) CO_3^{2-} (d) NO_3^-

Passage Comprehension Questions

Passage-I

Study the following passage and answer the questions at the end of it.

According to MOT, two atomic orbitals overlap resulting in the formation of molecular orbitals. Number of atomic orbitals overlapping together is equal to the molecular orbitals formed. The two atomic orbitals thus formed by LCAO (linear combination of atomic orbital) in the same phase or in the different phase are known as bonding and anti-bonding molecular orbitals, respectively. The energy of bonding molecular orbital is lower than that of the pure atomic orbitals by an amount Δ. This is known as the stabilization energy. The energy of anti-bonding molecular orbital is increased by Δ. (destabilization energy).

1. Which among the following pairs contain both paramagnetic species?
(a) O_2^{2-} and N_2^- (b) O_2^- and N_2
(c) O_2 and N_2 (d) O_2 and N_2^-

2. How many nodal planes are present in σ (s and p) bonding molecular orbital?
(a) 0 (b) 1 (c) 2 (d) 3

3. Which of the following statements is not correct regarding bonding molecular orbitals?
(a) Bonding molecular orbitals possess less energy than the atomic orbitals from which they are formed.
(b) Bonding molecular orbitals have low electron density between the two nuclei.
(c) Electron in bonding molecular orbital contributes to the attraction between atoms.
(d) They are formed when the lobes of the combining atomic orbitals have the same sign.

4. The x-axis is the molecular axis, then π-molecular orbitals are formed by the overlap of
(a) $s + p_z$ (b) $p_x + p_y$
(c) $p_z + p_z$ (d) $p_x + p_x$

Passage-2

Read the following passage and answer the questions at the end of it.

The shapes of molecules can be predicated by VSEPR theory, hybridization and dipole moment. Total number of hybrid orbitals (H) on the central atom of a molecule can be calculated by using the following relation.

H = [Total no. of valence electron pairs (P) –3 × (no. of atoms surrounding the central atom, excluding hydrogen atoms)]

One can also calculate total no. of bond pairs (n) around central atom as

n = total number of atoms surrounding the central atom also, total no. of lone pairs (m) = H – n

Thus, VSEPR notation of a molecule can be written as AX_nE_m.

Where, A denotes central atom of the molecule. X denotes bond pairs on central atom of the molecule. E denotes lone pairs on central atom of the molecule. In a polar molecule, the net dipole moment of the molecule $\propto m$.

1. VSEPR notation of chlorine trifluoride molecule is
(a) AX_5 (b) AX_3
(c) AX_2E_3 (d) AX_3E_2

2. Some molecules are given below

CO_2	SO_2	H_2O
I	II	III

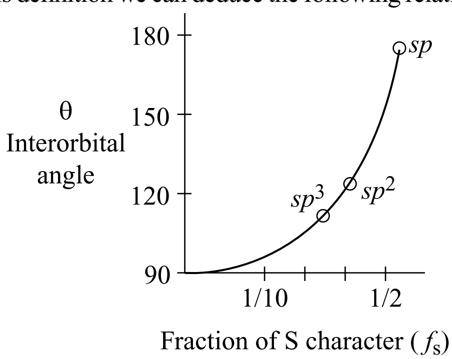
The correct increasing order of dipole moment of given species is
(a) I < II < III (b) II < II < I
(c) III < II < I (d) III < I < II

3. Total number of hybrid orbitals on central iodine in triiodide ion are
(a) 2 (b) 3 (c) 4 (d) 5

Passage-3

Departure From Normal Hybridization

Departure from normal hybridizations is quite prevalent. To begin with, we will define hybridization index (i) of a hybrid orbital as the superscript on p in the label. Thus, for sp^3, i = 3. From this definition we can deduce the following relationships.



$$f_s = \frac{1}{i+1}$$

$$f_p = \frac{1}{i+1}$$

$$i = \frac{f_p}{f_s}$$

$$\Sigma f_s = 1$$

$$f_s + f_p = 1$$

Finally, the relationship that generates the curve in the above figures is

$\cos\theta = \frac{-1}{1}$ where θ is the interorbital angle.

1. Bond angle in methane is 109°28′. The hybrid used in the formation of methane is
(a) sp^3 (b) sp^2
(c) $sp^{3.5}$ (d) $sp^{2.5}$

2. Bond angle of ammonia is 107.1°. What nitrogen hybrids are used in the formation of N-H bonds? (cos 107° = – 0.2923)
(a) sp^3 (b) $sp^{3.4}$
(c) $sp^{2.5}$ (d) sp^4

3. What type of nitrogen hybrid contains the non-bonding pair?
(a) sp^2 (b) $sp^{2.13}$
(c) sp^3 (d) $sp^{2.9}$

Passage-4

Bond length is the average distance between the nuclei of the two atoms held by a bond. This represents the inter nuclear distance corresponding to minimum potential energy for the system. Main factors which affect the bond length are given below:

I. Multiple bonds are shorter than corresponding single bonds.

II. Sometimes single bond distances are somewhat shorter than double of their respective covalent radii because bonds acquire some partial double bond character. This normally happens when one atom having vacant orbital and another atom containing lone pair. It is possible that it becomes shorter due to high ionic character in the covalent bond.

1. Which is not true about the N — N bond length among the following species?
I. H_2N-NH_2 II. N_2
III. $H_3\overset{+}{N}-\overset{+}{N}H_3$ IV. N_2O
(a) N—N bond length is shortest in II.
(b) N—N bond length in I is shorter than that in III.

(c) N—N bond length in III is shorter than that of in I.
(d) N—N bond length IV is intermediate between I and II.

2. In which of the following cases central atom-F bond has partial double bond character?
(a) NF_3 (b) CF_4
(c) PF_3 (d) OF_2

3. The C—Cl bond in vinyl chloride is …… and ... compared to C—Cl bond in ethyl chloride
(a) longer, weaker (b) shorter, weaker
(c) longer, stronger (d) shorter, stronger

Passage-5

According to molecular orbital theory all atomic orbitals combine to form molecular orbital by LCAO (linear combination of atomic orbitals) method. When two atomic orbitals have additive (constructive) overlapping, they form bonding molecular orbitals, overlap subtractively, higher energy antibonding molecular orbitals (AMO) are formed. Each MO occupies two electrons with opposite spin. Distribution of electron in MO follows Aufbau principle as well as Hund's rule. MO theory can successfully explain magnetic behaviour of molecules.

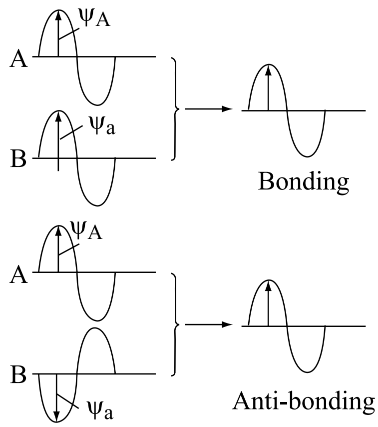
1. Which of the following is/are not paramagnetic?
(a) NO (b) B_2
(c) CO (d) O_2

2. Bond strength increases when
(a) bond order increases
(b) bond length increases
(c) antibonding electrons increases
(d) bond angle increases

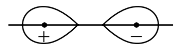
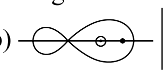
3. O_2^{2-} will have
(a) bond order equal to H_2 and diamagnetic
(b) bond order equal to H_2 but paramagnetic
(c) bond order equal to N_2 and diamagnetic
(d) bond order higher than O_2

Passage-6

According to molecular orbital theory when a pair of atomic orbitals combine they give rise to a pair of molecular orbitals. The number of molecular orbitals produced must always be equal to the number of atomic orbitals involved. The overlapping of orbitals means the overlapping of the wave of electron. Wave can overlap within the same phase or in the opposite phase, e.g.,



1. Which of the following combination gives bonding molecular orbitals?
(a) $\psi_A - \psi_B$ (b) $\psi_A + \psi_B$
(c) $\psi_A^2 + \psi_B^2$ (d) $\psi_A \times \psi_B$

2. Which of the following is bonding molecular orbital?
(a)  (b) 
(c) both of these (d) none of these

3. The probability of finding the electrons in bonding molecular orbital is equal to
(a) $\psi_A^2 - \psi_B^2$
(b) $\psi_A^2 + \psi_B^2 - 2\psi_A\ \psi_B$
(c) $\psi_A^2 + \psi_B^2 + 2\psi_A\ \psi_B$
(d) none of these

Passage-7

When hybridization involving *d*-orbitals are considered then all the five *d*-orbitals are not degenerate, rather $d_{x^2-y^2}, d_{z^2}$, and d_{xy}, d_{yz}, d_{zx} form two different sets of orbitals and orbitals of appropriate set is involved in the hybridization.

1. In sp^3d^2 hybridization, which set of d-orbitals is involved?
(a) $d_{x^2-y^2}, d_{z^2}$ (b) d_{z^2}, d_{xy}
(d) d_{xy}, d_{yz} (c) $d_{x^2-y^2}, d_{xy}$

2. In sp^3d^3 hybridization, which orbitals are involved?
(a) $d_{x^2-y^2}, d_{z^2} d_{xy}$ (b) d_{xy}, d_{yz}, d_{zx}
(d) $d_{x^2-y^2}, d_{xy} d_{xz}$ (c) d_{z^2}, d_{yz}, d_{zx}

3. Molecule having trigonal bipyramidal geometry and sp^3d hybridization, *d*-orbitals involved is
(a) d_{xy} (b) d_{yz} (d) $d_{x^2-y^2}$ (c) d_{z^2}

Passage-8

Covalent molecules formed by heteroatoms bound to have some ionic character. The ionic character is due to shifting of the electron pair towards *A* or *B* in the molecule *AB*. Hence, atoms acquire small and equal charge but opposite in sign. Such a bond which has some ionic character is described as polar covalent bond. Polar covalent molecules can exhibit dipolemoment. Dipole moment is equal to the product of charge separation, *q* and the bond length, '*d' for the bond. The unit of dipole moment is Debye. One Debye is equal to* 10^{-18} esu cm.

Dipole moment is a vector quantity. It has both magnitude and direction. Hence, dipole moment of molecules depends upon the relative orientation of the bond dipoles, but not on the polarity of bonds alone. A symmetrical structure shows zero dipole moment. Thus, dipolemoments help to predict the geometry of the molecules. Dipole moment values can be used to distinguish between *cis*- and *trans*-isomers; ortho-, meta- and para-forms of a substance, etc. The percentage of ionic character of a bond can be calculated by the application of the following formula:

% ionic charcter

$$= \frac{\text{Experimental value of dipole moment}}{\text{Theoretical value of dipole moment}} \cdot 100$$

1. Which are non-polar molecules?
(a) XeF_4 (b) SF_4
(d) NH_3 (c) H_2O

2. A diatomic molecule has a dipole moment of 1.2D. If the bond length is 1.0×10^{-8} cm, what fraction of charge does exist on each atom?
(a) 0.1 (b) 0.2
(d) 0.25 (c) 0.3

3. The dipole moment of NF_3 is very less than that of NH_3 because.
(a) number of lone pairs in NF_3 is much greater than in NH_3
(b) unshared electron pair is not present in NF_3 as in NH_3
(c) both have different shapes
(d) of different directions of moments of N—H and N—F bonds

Passage-9

Ionic compounds are soluble in water. They divide into ions in water and are hydrated with water molecules. In solid state, ionic compounds exhibit different crystal lattice structures like BCC, FCC, HCP, etc. In these structures each cation surrounds some anions and vice versa. The structure of ionic lattice can be determined by knowing coordination number.

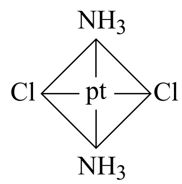
1. The unit cell of NaCl contains
(a) $4Na^+$ and $4Cl^-$ ions
(b) $3Na^+$ and $4Cl^-$ ions
(d) $4Na^+$ and $3Cl^-$ ions
(c) $2Na^+$ and $2Cl^-$ ions

2. Dielectric constants of some solvents are respectively $A = 25$, $B = 40$, $C = 55$, $D = 85$. Then, NaCl is highly soluble in
(a) *A* (b) *B*
(c) *C* (d) *D*

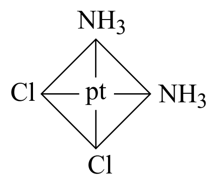
3. The limiting radius ratio of an ionic compound is 0.93. Then, the structure of ionic compound is
(a) Tetrahedral (b) Octahedral
(d) BCC (c) FCC

Passage-10

The platinum-chlorine distance has been found to be 2.32 Å in several crystalline compounds. This value applies to both of the compounds shown hereunder



(A)



(B)

Based on the above structures, answer the following questions assuming exact square planar structure.

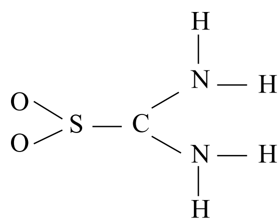
1. Cl—Cl distance in structure (A) is
(a) 2.32 Å (b) 4.64 Å
(d) 1.16 Å (c) 9.28 Å

2. Cl—Cl distance in structure (B) is
(a) 2.32 Å (b) 1.52 Å
(d) 2.15 Å (d) 3.28 Å

3. *A is*
(a) cis-isomer (b) trans-isomer
(c) chiral isomer (d) none of these

Passage-11

Thiourea S, S-dioxide $O_2SC(NH_2)_2$ has the following skeletal structure.



Based on the valence shell electron pair repulsion (VSEPR) model, predict geometries.

1. What is the geometry around the sulphur atom?
(a) Trigonal pyramidal
(b) Triangular planar
(c) T-shape
(d) Irregular tetrahedral

2. What is the geometry around the carbon atom?
(a) Trigonal pyramidal
(b) Triangular planar
(c) T-shape
(d) Irregular tetrahedral

3. What is the geometry around the N-atom?
(a) Trigonal pyramidal
(b) Triangular planar
(c) T-shape
(d) Irregular tetrahedral

Passage-12

The shapes of molecules can be predicted by VSEPR theory, hybridization and dipolemoment. There is a high degree of correlation between hybridization and shapes of molecules or the interorbital bond angles. If any depature in geometry of a molecule is observed on the basis of VSEPR theory then an apparent depature is observed in hybridization. The types of hybridization associated with various shapes are *sp*-linear; sp^2-trigonal; sp^3-tetrahedral.

In case of the departure observed the most easy way is following relationship.

For *sp* hybridization we have sp^1 hybrid characterized by

$$f_s = \frac{1}{i+1} \text{ or } f_p = \frac{i}{i+1} \text{ and } i = \frac{f_p}{f_s}$$

(in case of sp^3, the value of $i = 3$ and for sp^2 it is 2)

also for any orbital, fs + fp = 1 and all *s, p* hybrids of a given atom $\Sigma fs = 1.00$ must be satisfied. For finding the inter orbital angle between two equivalent (same f_s and same f_p) hybrid orbitals use the relation

$$\cos\theta = \frac{-1}{i}$$

To predict the geometrical shape of species containing only one central atom use the following method.

I. Total number of electron pairs

$$= \frac{(\text{Number of valence electrons}) \pm (\text{Total charge})}{2}$$

II. Number of bond electrons = (number of atom –1)

III. Number of electron pairs around central atom = [total number of electron pairs –3 (number of terminal atoms except hydrogen)]

IV. Number of lone pairs = [(number of central electron pairs) – (number of bond pairs)]

1. If we consider a molecule of H_2O

(H — O — H)

and we find that it consists of two bond pairs and two lone pairs. It was expected to be tetrahedral with a bond angle of 109.5° but it is V-shaped and has a bond angle of 104.5°. From this, predict the oxygen hybrids used in O—H.
(a) sp^3 (b) sp^4
(c) sp^5 (d) $sp^{2.8}$

2. The hybridization of P in PO_4^{3-} is the same as that of
(a) N in NO_3^- (b) S in SO_3
(c) I in ICl_2^+ (d) I in ICl_4^-

3. Select the molecules that contain non-bonding electrons on the central atom
(a) ICl_3 (b) ICl_3 and SF_3
(c) SF_4 and SO_2 (d) ICl_3, SF_4 and SO_2

Passage-13

Lewis concept of covalency of an element involved octet rule. Later on it was found that many elements in their compounds, e.g., BeF_2,BF_3, etc. have incomplete octet, whereas $PC1_5$,SF_6, etc. have expanded octet. This classical concept also failed in predicting the geometry of molecules. Modern concept of covalence was proposed in terms of valence bond theory. Hybridization concept along with valence bond theory successfully explained the geometry of various molecules but failed in many molecules. The geometry of such molecules was explained by VSEPR concept. Finally, molecular orbital theory was proposed to explain many other molecules.

1. Which are true statements among the following?
I. I_3^- has bent structure
II. $p\pi - d\pi$ bonds are present in SO_2
III. SeF_4 is see-saw in structures whereas ICl_3 is T-shaped
IV. XeF_2 and CO_2 have same shape
(a) II, III, IV (b) I, II, III, IV
(c) II, III, IV (d) I, III, IV

2. The bond angles in NO_2^+, NO_2 and NO_2^- are respectively
(a) 180°, 134°, 115° (b) 115°, 134°, 180°
(c) 134°, 180°, 115° (d) 115°, 180°, 130°

3. Which statements are correct about CO^+ and N_2^+ according to MO theory
I. Both have same configuration
II. Bond order for CO^+ and N_2^+ are 3.5 and 2.5
Ill. Bond order of CO^+ and N_2^+ are same

IV. During the formation of N_2^+ from N_2 bond length increases.

V. During the formation of CO^+ from CO, the bond length decreases.

(a) II, IV, V (b) I, II, IV, V
(c) I, III (d) I, II, III

Matching Types

1. Match the following List-I with II

List-I	List-II
(a) ICl_2^-	(p) Linear
(b) BrF_2^+	(q) Angular
(c) ClF_4^-	(r) Tetrahedral
(d) $AlCl_4^-$	(s) Square planar

2. Match the hybridization of central atom and geometry described in List-II with the species in List-I

List-I	List-II
(a) XeF_4	(p) sp^3d, see-saw geometry
(b) SF_4	(q) sp^3d^2, square planar
(c) SF_6	(r) sp^3d^3, distorted octahedral geometry
(d) XeF_6	(s) sp^3d^2,octahedral geometry

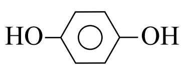
3. Match the following Column-I with Column-II

Column-I	Column-II
Compound	(Hybridisation of the central atom)
(a) SO_3	(p) sp
(b) I_3^-	(q) sp^2
(c) CO_2	(r) sp^3
(d) PO_4^{3-}	(s) sp^3d

4. Match the following List-I with List-II

List-I	List-II
(a) O_2 molecule	(p) Paramagnetic
(b) CO molecule	(q) One unpaired electron
(c) KO_2 molecule	(r) Bond order equal to N_2
(d) NO_2 molecule	(s) Angular

5. Match the following List-I with List-II

List-I	List-II
(a) PBr_3Cl_2	(p) Symmetric molecule
(b) 	(q) Asymmetrical structure
(c) PCl_3Br_2	(r) Non-zero dipole moment
(d) BF_3 molecule	(s) Zero dipole moment

6. Match the following Column-I with Column-II

Column-I	Column-II
(a) SiO_3^{2-}	(p) Planar triangular
(b) BF_3	(q) Non-polar
(c) CO_3^{2-}	(r) Bond order 1.33
(d) NO_3^-	(s) Resonance hybrid of three structure

7. Match the following List-I and List-II

List-I	List-II
(a) Solid PF_5	(p) Square planar
(b) Solid PCl_5	(q) Trigonal bipyramid
(c) Gas PCl_5	(r) Tetrahedral
(d) Gas PBr_5	(s) Octahedral
	(t) Trigonal planar

8. Match the following pair of molecules in List-I and with same property in List-II

List-I	List-II
(a) PCl_3F_2, PCl_2F_3	(p) Hybridization of central atom
(b) BF_3 and BCl_3	(q) Shape of molecule/ion
(c) CO_2 and CN_2^{2-}	(r) μ (dipole moment)
(d) C_6H_6 and $B_3N_3H_6$	(s) Total number of electrons

9. Match the following List-I with List-II

List-I	List-II
(a) N_3^-	(p) Trigonal bipyramidal geometry
(b) I_3^-	(q) sp hybridization
(c) O_3	(r) Formal charge on centre atom is +1
(d) $ICl_2^{\ominus}$	(s) Linear

10. Match the molecular species in List-I with their shape in List-II

List-I	List-II
(a) Linear shape	(p) CS_2
(b) sp hybridization	(q) XeF_2
(c) sp^3d hybridization	(r) C_2H_2
(d) CO_2 is isostructural to	(s) NCO^-

11. Match the following Column-I with Column-II

Column-I	Column-II
(a) Ionic bonds	(p) NH_4Cl
(b) Covalent bonds	(q) Non-directional
(c) Metallic bonds	(r) Diamond
(d) Coordinate bonds	(s) Gadolinium

12. Match the following

List-I	List-II
(a) BF_3	(p) sp^2 hybridization
(b) SO_2	(q) Bond angle 120°
(c) SO_3	(r) One lone pair
(d) CO_3^{2-}	(s) Planar triangular

13. Match the following List-I with List-II

List-I (Decreasing order)	List-II (Physical properties)
(a) NH_3, SbH_3, AsH_3, PH_3	(p) Dipolemement
(b) HF, HCl, HBr, HI	(q) Melting point
(c) SnH_4, GeH_4, SiH_4, CH_4	(r) Enthalpies of vapourization
(d) H_2O, H_2Te, H_2Se, H_2S	(s) Boiling point

14. Match the following List-I with List-II

List-I	List-II
(a) NaCl molecule	(p) Tetrahedral molecule
(b) H - Cl molecule	(q) More reactive than chlorine
(c) I - Cl molecule	(r) Non-directional bonds
(d) ClO_4^- ion	(s) Polar covalent bonds

15. Match the following List-I with List-II

List-I	List-II
(a) ClF_3 molecule is T-shaped	(p) Dipole moment is zero
(b) Bond order of O_3	(q) Similar to benzene
(c) Carbonate ion is stabilized	(r) VSEPR theory
(d) CO_2 is symmetrical	(s) Resonance theory

16. Match the following List-I with List-II

List-I	List-II
(a) Bond order of NO_3^-	(p) 2
(b) Bond order of SO_2	(q) Equal to the bond order of benzene
(c) Bond order of O_3	(r) 1.33
(d) Bond order of CO_2	(s) 1.5

17. Match the following List-I with List-II

List-I	List-II
(a) N_2, NH_3, O_2	(p) Paramagnetic
(b) K_2O, H_2O, Ag_2O	(q) Diamagnetic
(c) O_2, H_2^+, He_2^+	(r) Paramagnetic as well as diamagnetic
(d) Fe, Fe_3O_4	(s) Ferromagetic

18. Match the following List-I with List-II

List-I	List-II
(a) NH_4Cl	(p) Covalent bond
(b) HNC	(q) Ionic bond
(c) Liquid H_2O_2	(r) Hydrogen bond
(d) $CuSO_4$,$5H_2O$	(s) Coordinate bond

19. Match the reactions in Column-I with nature of the reactions/type of the products in Column-II

Column-I	Column-II
(a) $2O_2^- \rightarrow O_2 + O_2^{2-}$	(p) redox reaction
(b) $CrO_4^{2-} + H^+ \rightarrow$	(q) one of the products has trigonal planar
(c) $MnO_3^- + NO_2^- + H^+ \rightarrow$	(r) dimeric bridged tetrahedral metal ion
(d) $NO_3^- + H_2SO_4 + Fe^{2+} \rightarrow$	(s) disproportionation

20. Match the following List-I with List-II

List-I	List-II
(a) $XeO_3, XeF_2, ClF_3, PCl_5$	(p) All are sp^3d except one
(b) CH_4, XeO_3, ClO_4^-, $POCl_3$	(q) All are sp^3 except one
(c) $BF_4^-, SO_4^{2-}, CO_3^{2-}, POCl_3$	(r) All are sp^3
(d) $SO_2, SnCl_2, SO_3, SO_2Cl_2$	(s) All are sp^2 except one

21. Match the following List-I with List-II

List-I	List-II
(a) $CuSO_4$, $5H_2O$	(p) Ionic bond
(b) NH_4Cl	(q) Covalent bond
(c) NaOH	(r) Dative bond
(d) BF_3	(s) Hydrogen bond
	(t) Electron deficient bond

22. Match List-I (species) with List-II (bond orders) and select the correct answer

List-I	List-II
(a) N_2	(p) 1.0
(b) O_2	(q) 2.0
(c) F_2	(r) 2.5
(d) O_2^+	(s) 3.0
	(t) Electron deficient bond

23. Match the following List-I with List-II

List-I	List-II
(a) CO	(p) B.O. = 2.5
(b) O_2^+	(q) B.O. = 1.5
(c) F_2	(r) Diamagnetic
(d) O_2^-	(s) Paramagnetic

24. Match the following List-I with List-II

List-I	List-II
(a) I_3^-	(p) sp^2
(b) XeF_2	(q) sp^3d
(c) $SnCl_2$	(r) Linear
(d) SO_2	(s) V-shape

25. Match the following List-I with List-II

List-I	List-II
(a) PBr_3 (solid)	(p) sp^3
(b) PCl_2F_3	(q) sp^3d
(c) ICl_4^+	(r) Two types of bond length
(d) XeF_3^+	(s) Polar molecule

26. Match the following

Compound	Hybridization of central atom
(a) ClF_3	(p) sp^3
(b) XeO_2F_3	(q) sp^3d
(c) ClO_3^-	(r) sp^2
(d) BF_4^-	(s) sp^3d^2

27. Match the following List-I with List-II

List-I	List-II
(a) NH_4NO_3	(p) Contain covalent, coordinate and ionic bonds
(b) K_2CO_3	(q) Contain only covalent and ionic bonds
(c) $K[Pt(C_2H_4)Cl_3]$	(r) Contain trigonal planar ion
(d) $[Cu(NH_3)_4]SO_4$	(s) Contain tetrahedral ion
	(t) Contain square planar ion

ASSERTION AND REASON

Codes:

(a) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(b) Both **A** and **R** are true but **R** is not the correct explanation of **A**.
(c) **A** is true but **R** is false.
(d) **A** is false but **R** is true.

1. **Assertion (A):** Solubility of n-alcohol in water decreases with increase in molecular weight.
Reason (R): The hydrophobic nature of alkyl chain increases.
2. **Assertion (A):** The p-isomer of dichlorobenzene has higher melting point than o- and m-isomer.
Reason (R) : p-isomer is symmetrical and thus shows more closely packed structure.
3. **Assertion (A):** Bond order in a molecule can assume any value, positive or negative or integral or fractional including zero.
Reason (R): It depends upon the number of electrons in the bonding and anti-bonding orbitals.
4. **Assertion (A):** LiCl is predominantly a covalent compound.
Reason (R): Electronegativity difference between ‘Li’ and ‘Cl’ is too small.
5. **Assertion (A):** Salts of ClO_3^- and ClO_4^- are well known but those of FO_3^- and FO_4^- are non-existent.
Reason (R): F is more electronegative than ‘O’ while Cl is less electronegative than O.
6. **Assertion (A):** Bond energy of Cl-Cl bond is more than F-F bond.
Reason (R): Shorter the bond length, stronger the bond, more is the bond energy.
7. **Assertion (A):** Octet theory cannot account for the shape of the molecule.
Reason (R): Octet theory can produce relative stability and energy of a molecule.
8. **Assertion (A):** Carbonates and silicates are isostructural.
Reason (R): Carbon and silicon have the same number of valence shell electrons.
9. **Assertion (A):** The dipole moment of NF_3 is less than that of NH_3
Reason (R): The polarity of the N-F bond is less than that of the N-H bond.
10. **Assertion (A):** The boiling point of H_2O is greater than H_2S.
Reason (R): Both H_2O and H_2S contains hydrogen bond.
11. **Assertion (A):** Haber cycle is based on Hess’s law.
Reason (R): Lattice enthalpy can be calculated by Born-Haber cycle.
12. **Assertion (A):** As lattice energy increases, melting and boiling points of ionic compound increases.
Reason (R): As lattice energy increases, stability of ionic compound increases.
13. **Assertion (A):** In triangular bipyramidal hybridization a lone pair preferentially occupies equatorial position.
Reason (R): The most electronegative bonding partner/s preferentially occupy the axial positions in triangular bipyramidal stucture.
14. **Assertion (A):** Covalent bond is more stable than ionic bond.
Reason (R): An ionic bond is formed by transfer of electron from one atom to another atom.
15. **Assertion (A):** Antibonding molecular orbital posess higher energy relative to the constituent atomic orbitals.

Reason (R): The probability of finding the electrons in antibonding molecular orbitals is less than constituent atomic orbitals.

16. **Assertion (A):** The melting point of $ZnCO_3$ and $CdCO_3$ is found to be low in comparison to $CaCO_3$.
Reason (R): The polarizing power of Zn^{2+} and Cd^{2+} is much higher than the polarizing power of Ca^{2+}.

17. **Assertion (A):** Diethyl ether is insoluble in water.
Reason (R): Diethyl ether can form hydrogen bond with water.

18. **Assertion (A):** If there are two pairs of electrons with valence shell of the central atom the orbitals containing them will be oriented at 180° to each other.
Reason (R): The orbitals in CO_2 overlaps at 180° then the CO_2 molecule is linear.

19. **Assertion (A):** For π overlap the lobes of the atomic orbitals are perpendicular of the line joining the nuclei.
Reason (R): In π molecular orbitals ψ is zero along the internuclear axis.

20. **Assertion (A):** AgCl is more covalent than KCl.
Reason (R): Polarizing power of K^+ is more than the polarizing power of Ag^+ ions.

21. **Assertion (A):** In liquid water the hydrogen bonds are constantly swapping between molecules.
Reason (R): Water is one of the substance that is less dense as a solid that it is as a liquid.

22. **Assertion (A):** The dipole moment helps to predict whether a molecule is polar or non-polar.
Reason (R): The dipole moment helps to predict the geometry of molecules.

23. **Assertion (A):** Glucose is soluble in a polar solvent like water.
Reason (R): Ionic compounds dissolve in polar solvents.

24. **Assertion (A):** Bond order in carbon monoxide is three.
Reason (R): The HOMO of carbon monoxide is $\pi^*2py = \overset{*}{\pi}2p_x$

25. **Assertion (A):** Na^+ and Al^{3+} are isoelectronic but the magnitude of the ionic radius of Al^{3+} is less than that of Na^+.
Reason (R): The magnitude of the effective nuclear charge on the outer shell electrons in Al^{3+} is greater than that in Na^+.

26. **Assertion (A):** Fe^{3+} salts are less stable than Fe^{2+} salt.
Reason (R): Fe^{3+} ions are formed by loss of 's' and 'd' electrons while Fe^{2+} ions are formed by loss of only 's' electrons.

27. **Assertion (1):** H_2 molecule is more stable than HeH molecule.
Reason (R): The anti-bonding electron in the molecule destabilizes it.

Integer Answer Type

1. Covalency of sulphur in S_8 molecule is
2. No. of 109°28′ angles in $CHCl_3$ is
3. Total no. of electrons in anti-bonding orbitals of valency shell of oxygen molecule.
4. No. of electron pairs around the central atom in I_3^- ion is
5. Maximum possible electron pairs that can present around a central atom in a molecule is
6. The number of **(**$p\pi - d\pi$**)** bonds in XeO_4 is ……..
7. Total number of covalent bonds in C_3O_2 is ……
8. How many '*sp*' hybrid orbitals are there in allene C_3H_4?
9. The number of electrons present in valency shell of hydroxide ion is

Multiple Choice Questions with Only One Answer

1. b	**2.** d	**3.** d	**4.** a	**5.** d	**6.** d	**7.** b	**8.** c	**9.** d	**10.** d
11. b	**12.** c	**13.** b	**14.** b	**15.** d	**16.** b	**17.** b	**18.** b	**19.** c	**20.** a
21. a	**22.** d	**23.** a	**24.** d	**25.** b	**26.** c	**27.** b	**28.** c	**29.** a	**30.** b
31. d	**32.** c	**33.** b	**34.** a	**35.** a	**36.** c	**37.** d	**38.** b	**39.** b	**40.** a
41. b	**42.** d	**43.** a	**44.** b	**45.** b	**46.** b	**47.** b	**48.** b	**49.** a	**50.** d
51. c	**52.** c	**53.** c	**54.** a	**55.** d	**56.** c	**57.** a	**58.** b	**59.** b	**60.** d
61. a	**62.** b	**63.** c	**64.** a	**65.** a	**66.** c	**67.** d	**68.** b	**69.** d	**70.** c
71. a	**72.** b	**73.** b	**74.** c	**75.** c	**76.** c	**77.** d	**78.** b	**79.** c	**80.** c
81. c	**82.** b	**83.** c	**84.** d	**85.** d	**86.** d	**87.** d	**88.** b	**89.** d	**90.** d
91. d	**92.** a	**93.** a	**94.** d	**95.** d	**96.** c	**97.** c	**98.** d	**99.** d	**100.** b
101. b	**102.** c	**103.** c	**104.** c	**105.** b	**106.** a	**107.** c	**108.** c	**109.** a	**110.** d
111. b	**112.** d	**113.** a	**114.** a	**115.** a	**116.** b	**117.** c	**118.** d	**119.** b	**120.** b
121. a	**122.** b	**123.** c	**124.** b	**125.** c	**126.** d	**127.** a	**128.** c	**129.** b	**130.** a

131. a	**132.** a	**133.** a	**134.** d	**135.** c	**136.** c	**137.** d	**138.** a	**139.** a	**140.** a
141. b	**142.** c	**143.** a	**144.** c	**145.** b	**146.** a	**147.** c	**148.** a	**149.** b	**150.** c
151. b	**152.** d	**153.** b	**154.** b	**155.** d	**156.** b	**157.** b	**158.** c	**159.** d	**160.** c
161. a	**162.** a	**163.** d	**164.** b	**165.** b	**166.** c	**167.** a	**168.** c	**169.** c	**170.** a
171. b	**172.** d	**173.** b	**174.** c	**175.** b	**176.** d	**177.** b	**178.** b	**179.** a	**180.** c
181. b	**182.** d	**183.** b	**184.** b	**185.** b	**186.** c	**187.** b	**188.** c	**189.** a	**190.** a
191. d	**192.** a	**193.** a	**194.** d	**195.** d	**196.** a	**197.** d			

Multiple Choice Questions with More than One Answer

1. a, b, c	**2.** a, c	**3.** a, c, d	**4.** a, b, d	**5.** b, c
6. a, b, d	**7.** a, d	**8.** a, b, c, d	**9.** c, d	**10.** a, b, d
11. b, d	**12.** a, b, c	**13.** a, b, c	**14.** a, b, c	**15.** c, d
16. c, d	**17.** b, c	**18.** b, c, d	**19.** a, b	**20.** a, b
21. b, c, d	**22.** a, b, c, d	**23.** a, b, d	**24.** b, c, d	**25.** a, c, d
26. a, b	**27.** b, d	**28.** a, b, c	**29.** b, d	**30.** a, b, c
31. a, b	**32.** a, b, c	**33.** a, b, c	**34.** a, c	**35.** b, c
36. a, b, c	**37.** a, b	**38.** a, b	**39.** c, d	**40.** a, c, d
41. a, b, c	**42.** a, b, c, d	**43.** a, c, d	**44.** b, c	**45.** b, c, d
46. a, b, c	**47.** a, b, c, d			

Passage Comprehension Questions

Passage – 1	**1.** d	**2.** a	**3.** b	**4.** c
Passage – 2	**1.** d	**2.** a	**3.** d	
Passage – 3	**1.** a	**2.** b	**3.** b	
Passage – 4	**1.** c	**2.** c	**3.** d	
Passage – 5	**1.** c	**2.** a	**3.** a	
Passage – 6	**1.** b	**2.** b	**3.** c	
Passage – 7	**1.** a	**2.** b	**3.** d	
Passage – 8	**1.** a	**2.** c	**3.** d	
Passage – 9	**1.** a	**2.** d	**3.** c	
Passage – 10	**1.** b	**2.** d	**3.** b	
Passage – 11	**1.** b	**2.** b	**3.** a	
Passage – 12	**1.** b	**2.** c	**3.** d	
Passage – 13	**1.** a	**2.** a	**3.** b	

Match the Following Type Questions

1. a-p	b-q	c-s	d-r
2. a-q	b-p	c-s	d-r
3. a-q	b-s	c-p	d-r
4. a-p	b-r	c-p, q	d-p, q, s
5. a-q, s	b-p, s	c-q, r	d-p, s
6. a-p, q, r, s	b-p, q, r, s	c-p, q, r, s	d-p, q, r, s
7. a-q	b-r, s	c-q	d-r
8. a-p, q	b-p, q, r	c-p, q, r, s	d-p, q, r, s
9. a-q, r, s	b-p, r, s	c-r	d-p, r, s
10. a-p, q, r, s	b-p, r, s	c-q	d-p, q, r, s
11. a-p, q	b-p, r	c-q, s	d-p
12. a-p, q, s	b-p, r	c-p, q, s	d-p, q, s
13. a-q	b-p	c-q, r, s	d-p, r, s
14. a-r	b-q, s	c-q, s	d-p, s
15. a-r	b-q, s	c-p, r, s	d-p
16. a-r	b-p	c-q, s	d-p
17. a-r	b-q	c-p	d-s
18. a-p, q, s	b-p, s	c-p, r	d-p, q, r, s
19. a-p, s	b-r	c-p, q	d-p
20. a-p	b-r	c-p, q	d-s
21. a-p, q, r, s	b-p, q, r	c-p, q	d-q, r, t
22. a-s	b-q	c-p	d-r
23. a-r	b-p, s	c-r	d-q, s
24. a-q, r	b-q, r	c-p, s	d-p, s
25. a-p, s	b-q, r, s	c-q, r, s	d-q, r, s
26. a-q	b-q	c-p	d-p
27. a-p, r, s	b-q, r	c-p, t	d-p, s, t

Assertion and Reason

1. a	**2.** a	**3.** a	**4.** c
5. a	**6.** b	**7.** c	**8.** b
9. c	**10.** c	**11.** b	**12.** a
13. b	**14.** b	**15.** b	**16.** a
17. c	**18.** b	**19.** b	**20.** c
21. b	**22.** b	**23.** b	**24.** c
25. a	**26.** c	**27.** a	

Integer Type Question

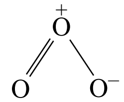
1. 2	**2.** 0	**3.** 4	**4.** 5	**5.** 8
6. 4	**7.** 8	**8.** 2	**9.** 8	

Answers

1 – 2. (Hint: each s atom share 2 electron pairs with two S atoms in S_8 molecule.)
2 – 0 (Hint: it is distorted tetrahedron.)
3 – 4 $\left(\sigma^* 2s^2 \ \ \pi 2p_x^1 \ \ \pi 2p_z^1\right)$
4 – 5 (3LP and 2 LP_
5 – 8 (maximum valency exhibited by os or Ru is 8)
6 – 4
7 – 8 (4σ and 4π bonds in 0 = C = C = C = 0)
8 – 2 (H_2C = C = CH_2 contain two sp hybrid orbitals)
9 – 8

Multiple Choice Questions with Only One Answer

2. In a π-bond the electron density will be above and below the plane of molecule. So, nodal plane will be in the plane of molecule.
3. If *z*-axis is molecular axis s-s and s-p_z overlaps form the bonds.
4. The bond orders in H_2O_2, O_3 and O_2 are 1, 1.5 and 2.
5. Between every B—F bond there is some π-dative bond character in the resonance hybrid.
6. CF_4 has no dipole moment due to symmetric tetrahedron. NF_3 have less dipole moment that NH_3 due to vectorial addition of bond moments with lone pair moment.
7. CN^-, CO and NO^+ are isoelectronic and bond order is 3.
8. CO_3^{2-} and BCl_3 are planar triangular.
9. XeF_5^+ have 6 EP 5 BP and one L.P. So square pyramid. XeF_6 have 7 EP 6 BP and one LP distorted octatedran. XeF_8^{2-} have 9 EP 8 BP one LP. LP is inert square antiprismatic.
10. lCl_2^- have 5 BP Trigonal bipyramid structure 3 LP at equatorial positions. Linear molecule.
11. CrO_2Cl_2 has a distorted tetrahedral structure due to two Cr ═ O and two Cr — Cl bonds.
12. When N_2 is converted to N_2^+ BO decreases from 3 to 2.5 so and length increases.
When NO is converted to NO^+ BO increases to 3.
When O_2 is converted to O_2^{2-} BO decreases from 2 to 1.
13. Three negative charges distributed on four oxygen atoms. So formal charge = 0.75. Bond order = no. of bonds/no. of atoms = 5/4.
14. PCl_5 has unsymmetric TBP structure but when converted to symmetric tetrahedral PCl_4^+ and symmetric octahedral PCl_6^- get stability.
15. Hybridization of carbon in CH_4 is sp^3, CH_2H_4 is sp^2 and C_2H_2 is sp.
16. Cl atom is larger than F so Cl — Cl bond is longer than F — F, O_2 and N_2 have double and triple bonds, respectively.
17. The carbon in 1, 1, 2, 2, ethane is sp^2 hybridized while in CCl_4 sp^3 hybridized.
18. o-dichlorobenzene has dipole moment.
19. H—X……H covalent bond is shorter than hydrogen bond.
20. Alcohols form stronger hydrogen bonds than amines.
21. Due to intramolecular hydrogen bond.
22. In metals there will be metallic bond.
23. A is non-polar. So do not ionize.
24. SF_4 see-saw shape with 1 LP, CF_4 tetrahedral 0 LP, XeF_4 square planar with 2 LP.
25. Fajan's rule. Smaller the cation with more no. of charges have more polarizing power.
26. I_3^- have same structure of lCl_2^- as in Q.No. 10.
28. AsO_3^{3-}, and ClO_3^- are pyramidal. So they have dipole moment NO_3^-, CO_3^2 and BO_3^{3-} are planar with zero dipole moment.
30. NO_2 is bent due to the presence of unpaired electron but bond angle is greater than 120°.
31.



32. LiCl and NaCl ionize completely in water but due to small size of Li^+ in fused state LiCl will have more conductivity.
34. B in BF_3 is sp^2 hybrid, but in other three molecules the central atoms are in sp^3 hybridization.
36. % charge $= \dfrac{1.2}{1 \cdot 4.8} \cdot 100 = 25\%$
37. Between salicylaldehyde molecules no hydrogen bonds exist due to intramolecular hydrogen bond.
38. NH_3 can form only one hydrogen bond due to the presence of one lone pair on nitrogen atoms.
39. NO_2^+ is linear and NO_2^- is bent.
41. Anti-bonding MOs have more energy than AOs from water formed.
42. F_2 and O_2^{2-} are isoelectronic.
43. When N_2 is converted into N_2^+ BO decreases, so bond length increases while in the conversion of O_2 to O_2^+ BO increases 2 to 2.5.
45. Resonance energy is the difference in internal energy of most stable structure (lowest internal energy) and resonance hybrid structure.
46. In H_3BO_3, B is sp^2 but when combines with OH^- changes to sp^3.
48. In lCl_2^-, is in sp^3d and in $BeCl_2$ Be is in sp hybridization but they are linear.
49. In $BeCl_2$, Be is *sp* (50 per cent *s* character); CO_3^{2-}, C is sp^2 (33.3 per cent s character) and in CCl_4, NH_3 and

H_2S, the central atoms are in sp^3 hybridization but with decrease in bond angle s-character of hybrid orbitals decreases.

53. In square planar structure (dsp^2) the diagonal orbitals are not at 90°.

66. XeF_4 has 6 EP of which 4 BP and 2LP. The 2LP occupy opposite comers of octahedron so square planar. XeF_5^- has 7 EP with 2 LP 5 BP. The two LP occupy opposite corners of pentagonal bipyramidal structure. XeF_8^{2-} has square antiprism structure.

69. More the solvation energy than the attractive force between ion, solubility is more. More the dipole moment of the structure it keeps the ions apart.

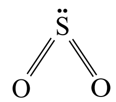
71. $S + 2Cl_2 \longrightarrow SCl_4$

$SCl_4 + 4H_2O \longrightarrow H_4SO_4 + 4HCl$

$H_4SO_4 \longrightarrow H_2SO_3 + H_2O$

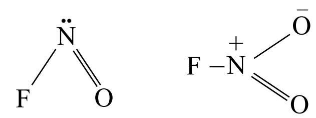
73. Bond order in NO_2^+ and NO_2 is 1.5 and in NO_3^- is 1.33. Due to positive charge on nitrogen the bond length in NO_2^+ is shorter than in NO_2^-.

74. Two BP and 1 LP will be arranged in trigonal planar structure. e.g.,



75. The terminal π-bonds will be perpendicular to the plane of molecule while the middle π-bond is in the plane of molecules.

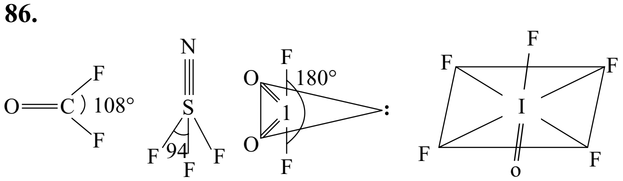
76. The structures are



77. ClF_3 is T shape, BF_3 planar triangle.

81. n cannot be 4 because to have n = 4. The valence shell should contain nine electrons to form AXL_4 · n can have a minimum value 3 as in XeF_2.

86.



Any FF angle in less than 90° due to repulsion by double bond

87. In $N(SiH_3)_3$ the structure is planar triangular due to back bonding. Bond angle is 120°. The remaining are pyramidal. More the electronegativity of bonded atom lesser the bond angle.

88. Orbitals having same symmetry mix up.

92. $\overset{-2}{C} = \overset{+1}{N} = \overset{0}{O}$

92. AB_5 molecule may have trigonal bipyramid or square pyramid structure.

96. If it is has square planar structure it should give geometrical isomers in dichlorination.

103. SF_4 and SOF have same TBP structure but lone pair repulsion is more than double-bonded oxygen and the axial FSF angle is less than 180 due to repulsion by LP.

122. In a period from left to right atomic size decreases.

126. Due to strong hydrogen bonds they do exist even in vapour phase and HF volatilizes without breaking hydrogen bonds.

136. B is smaller atom than Li so bond length is less stability is more.

145. More electronegative F atom draws the electron density from the C atom, so F—C—F bond angle decreases while HCH bond angle increases.

154. Neither N nor F contain *d* -orbitals. Further in N_2F_4 N—N bond is shorter than in N_2H_4 due to more *s*-character (Bent's rule).

155. More the number of charges on the ion stronger the attraction.

161. Due to strong π-back bonding in P—F bond electron donating capacity of P → O in π-dative bond increases (synergic effect).

162. As ClF_3 have trigonal bipyramid structure in which two F atoms are at axial and one F atom is at equatorial position.

163. Higher the polarization, covalent character increases.

166. With decrease in the size of bonded atom bond angle decreases and with increase in the size of central atom also bond angle decreases.

172. NH_4^+ and NO_3^-, all non-metal atoms.

173. Ionic character increases down the group with increase in electropositive character.

174. Valency of $L = 2$, $Q = 1$, $P = 1$, $Q = 2$.

175. In a period covalent character increases with increase in number of charges (Fajan's rule).

176. M.pt depends on lattice energy not on ionic character.

177. Fajan rule. Smaller the ion with more number of charges covalent character is more.

178. Dipolemoments of HF, H_2O, NH_3 and SO_2 are 1.86, 1.84, 1.46 and 1.6 D.

179. Fajan's rule. Cation with more number of +ve charges polarize the anion with more number of –ve charges. So, covalent character is more and is less soluble.

180. Fajan's rule

181. AlF_3 is ionic, while SiF_4 is covalent.

182. Lattice energy $= \dfrac{q_1 q_2}{r_c + r_a} = \dfrac{2 \cdot 2}{r_c + r_a}$

183. Fajan's rule.

184. Fajan's rule. More the number of charges on cation more the covalent character.

185. KI is ionic.

188. With increase in size of cation both lattice energy and hydration energies decreases, but decrease in lattice energy is more.

190. Individual neutral atoms converts into oppositely charged ions and attracted by each other to form ionic crystal.

192. With increase in the number of charges on cation, covalent character increases.

193. It is the middle row anomaly where sudden change appears in electronic configuration in penultimate shell.

195. Ex: In NaCl, $ZnCl_2$ and $PbCl_2$, Na^+, Zn^{2+} and Pb^{2+} have inert gas, pseudoinert gas and inert polar configurations, respectively.

196. If lattice energy is very high than hydration energy, salt is insoluble because hydration energy is not sufficient to break the lattice.

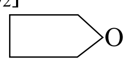
197. M.pt is directly proportional to lattice energy, which in turn is inversely proportional to $r_c + r_a$.

More Than One Answer Type Questions

1. (a), (b), (c)
B.pt depends on molecular size. HF and H_2O have more Bpt due to hydrogen bonding.

2. (a), (c)
SF_6 has symmetric octahedral structure with 12 electrons around central atom.

3. (a), (c), (d)
(a) $Al(OH)_3$ when dissolved in NaOH forms $Na[Al(OH)_4(H_2O)_2]$

(b) $\underset{sp^3}{B_2H_6}$+THF $\longrightarrow$  $\longrightarrow \underset{sp^3}{BH_3}$

(c) $SiF_4 + 2HF \longrightarrow \underset{sp^3d^2}{H_2SiF_6}$

(d) $\underset{\text{Vapour}}{2PCl_5} \longrightarrow \underset{\text{Solid}}{PCl_4^+ + PCl_6^-}$

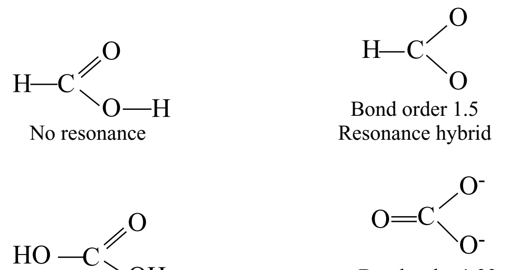
4. (a), (b), (c)
NH_2 is angular Sp^3

6. (a), (b), (c)
(a) F being more electronegative decreases the electron density on N.
(b) $AlCl_3$ covalent but AlF_3 ionic.
(c) Triple bond contain one sigma and two pi bonds.
(d) More the charge on anion more susceptible to polarization.

7. (a), (c)

1. no change	2. $sp^2 \to sp^3$
3. sp^2 to sp	4. no change

9. (c), (d)

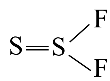


No resonance — Bond order 1.5 Resonance hybrid

No resonance — Bond order 1.33 Resonance hybrid

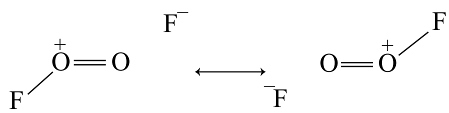
12. (a), (b), (c)
(a) In bonding MO electron density is localized between the nuclei.
(b) Only orbitals having same symmetry and energy can combine.
(c) MOs formed by d-orbital combination are δ-MOs.

14. (a), (b), (c)

(a) S_2F_2 may have F — S — S — F or 

(b) 

(c) Due to withdrawal of electron density by more electronegative fluorine from oxygen repulsion between lone pairs decreases and due to resonance hybridization O—O bond length decreases.

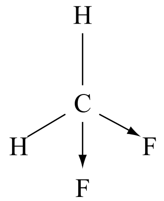


17. (b), (c)
Bond order increases in $O_2 \to O_2^{-1}, C_2 \to C_2^{2-}$ and $O_2^- \to O_2^+$ from 2 to 2.5, 2 to 3 and 1.5 to 2.

18. (b), (c), (d)
Refer to Q. No. 14.

20. (a), (b)
SF_6 has symmetric octahedral structure.

21. (b), (c), (d)



Since more electronegativity, F atom withdraws electron density from C—F bond, F—C—F angle decrease and H—C—F bond angle increases. Also, dipole moment cannot be zero.

22. (a), (b), (c), (d)
Bent's rule, More electronegative atom always occupy the axial positions in TBP structure.

25. (a), (c), (d)
O_2 and B_2^{2-} have bond order 2 and C_2^{2-} and N_2 have 3.

29. (a), (b), (d)
In both cases, N $\rightarrow$ NO^+ and CO $\rightarrow$ CO^+ bond length decreases.

31. (a) and (b)
Both H_2^+ and H_2^- have bond order 0.5.

32. (a), (b), (c)
The molecules or ions having same number total valence electrons are isoelectronic and they have similar structures.

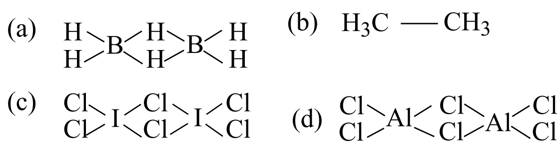
34. (a), (c)
In CH_4 and $Ni(CO)_4$ the central atoms are sp^3 hybridized but in $[Ni(CN)_4]^{2-}$ Ni is dsp^2. In XeO_3 as bond angle is less than 109°28' s character is also less.

38. (a), (b), (c)
(a) More ionic character more dipole moment Fajan's rule explains 2 and 3.

39. (c), (d)
Cu^+ has pseudo-inert gas configuration. Hydration energy and lattice energies are favourable for the formation of Cu^{2+}, Cu^+ disproportionate in water.

42. (a), (b), (c), (d)
Bond order in O_2^+, O_2^-, NO and H_2^+ are 2.5, 1.5, 2.5 and 0.5, respectively.

43. (a), (c), (d)



46. (a), (b), (c)
Lone pair and double bonded oxygen atoms occupy the opposite corners of octahedran. The LP has slightly greater repulsive effect than double-bonded oxygen. The IOF angle is 89°.

47. (a), (b), (c), (d)
All have symmetric planar triangular structure.

Passage Comprehension Questions

Passage-1

(a) O_2 contain two and N_2^- contain one unpaired electrons in π^* molecular orbitals.
(b) σs and σp bonding MOs do not contain nodal planes.
(c) In bonding MOs electron density is maximum between the nuclei.
(d) If X – axis is molecular axis p_x — p_x form σMO while $p_y - p_y$ and $P_z - P_z$ form π MOs.

Passage-2

(a) ClF_3 contain 3BP and 2LP. Hence AX_3E_2.
(b) Dipole moments of CO_2, SO_2 and H_2O are zero, 1.6 and 1.84 D.
(c) I is sp^3d hybridized.

Passage-3

(a) ClF_3 contain 3BP and 2LP. Hence AX_3E_2.
(b) Dipole moments of CO_2, SO_2 and H_2O are zero, 1.6 and 1.84 D.
(c) I is sp^3d hybridized.

Passage-4

(a) If bond angle is 109°28′ hybrid orbitals are sp^3.
(b) $i = \frac{1}{0.2923} = 3.4$
(c) fraction of s-character is $sp^{3.4}$ orbital

$$\frac{1}{1+3.4} = \frac{1}{4.4} = 3.4$$

Total contribution of each orbital

$$\frac{1}{4.4} + \frac{1}{4.4} + \frac{1}{4.4} + x = 1$$

(x is fraction of s-character of orbital having non-bonding electron pair)

$$x = 1 - \frac{3}{4.4} = 0.318$$

$$0.318 = \frac{1}{1+i}$$

$$\therefore\ i = 2.13$$

Passage-5

(a) Due to repulsions by positive charges N-N bond length increases in $H_2N—NH_2$
(b) Due to back bonding $P \rightleftarrows F$
(c) Due to mesomeric effect.

Passage-6

(a) As per LCAO, bonding MO is formed by addition of wave functions.

(b) Electron density is more between nuclei and no nodal plane.

(c) Probability of electrons in bonding MO is

$(\psi_A + \psi_B)2$

Passage-8

(a) XeF_4 square planar and non-polar.

(b) Fraction of charge $\frac{\propto_{expt}}{\propto_{cat}} = \frac{1.2}{4.8} = 0.25.$

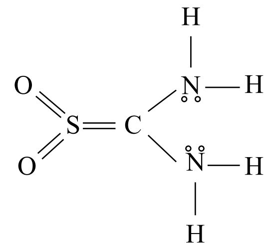
(c) In NF_3 N—F bond moments and lone pair moments are in opposite direction but in NH_3 in same direction.

Passage-9

2-More the dielectric constant more the polarity of the solvent so solubility of ionic compounds is also more.

Passage-11

The structure of compound can be written as



Passage-12

Calculating as in passage 3 we get *s*-character 20 per cent and *p*-character 80 per cent.

4 CHAPTER

Hydrogen and its Compounds

Inflammable air (hydrogen) is undoubtedly charged with abundance of principle of inflammability

T. Bergman

4.1 INTRODUCTION

Hydrogen is the lightest and simplest of all the elements. It can combine with almost all the elements in the periodic table except noble gases. It play an important role in the structure of the universe. It has been established from the spectra of the light emitted by the sun and other stars that hydrogen is the major constituent of these bodies. Moreover, it has been shown that the energy emitted by the sun and stars results from the fusion of hydrogen nuclei to form helium.

$$4\,{}^{1}_{1}H \longrightarrow {}^{4}_{2}He + 2\underset{\text{positron}}{{}^{0}_{+1}e} + \text{energy}$$

By using this reaction, the global concern related to energy can be overcome to a great extent by the use of hydrogen as a source of energy. Hydrogen is of greatest industrial importance.

4.2 POSITION OF HYDROGEN IN THE PERIODIC TABLE

Hydrogen is the first element of the periodic table. Its atomic number is 1 and its atomic weight is 1.008 amu. It has one proton in its nucleus and one electron in its K-shell or 1s energy level.

One of the most serious problems of the periodic classification of elements is the allotment of proper place to hydrogen. This is because of the fact that some of its properties resemble with those of alkali metals, whereas few other properties resemble with those of halogens. This dual behaviour of hydrogen can be explained on the basis of its electronic structure ($1s^1$). It can gain one electron to give H^- ion and thus resemble halogens. It can also lose its electron and give H^+ ion and thus resemble alkali metals.

Resemblances with alkali metals

- Similar to alkali metals hydrogen contain one electron in its outer most shell.
- Similar to alkali metals it loses its only electron and can convert into H^+ ion.

$$H \longrightarrow H^+ + e^-$$

$$Na \longrightarrow Na^+ + e^-$$

- Similar to alkali metals hydrogen is monovalent, electropositive in character and exhibit +1 oxidation state.
- When fused alkali metal salts (say Na*Cl*) are electrolysed alkali metals are liberated at cathode. Similar to this hydrogen is liberated at cathode during the electrolysis of water.
- Similar to alkali metals hydrogen is also a very good reducing agent.
- Similar to alkali metals hydrogen forms oxides, halides and sulphides.

  Though hydrogen is similar to alkali metals, it has some subtle points of differences with alkali metals.
- Inspite of hydrogen having one electron in its outer most orbital like alkali metals yet its ionization energy (1312 kJ mol^{-1}) is very high as compared to those of alkali metals, i.e., 520 kJ mol^{-1} for Li, 490 kJ mol^{-1} for Na, etc. However, ionization energy of hydrogen is quite close to halogens, i.e., 1681 KJ mole^{-1} for F, 1255 kJ mol^{-1} for C*l,* etc.
- Hydrogen does not possess metallic characteristics under normal conditions.

Resemblances with Halogens

- Similar to the electronic configuration of halogens ($ns^2\ np^5$) one electron short of inert gas configuration ($ns^2\ np^6$), the electronic configuration of hydrogen ($1s^1$) is also one electron short of helium configuration ($1s^2$).
- Similar to halogens hydrogen is a non-metal.
- Similar to halogens hydrogen exists as a diatomic molecule.
- Similar to halogens it gains one electron to from a negative ion such as H^-.
- Similar to halogens, hydrogen combines with metals and exhibit –1 oxidation state.
  Ex: Na*Cl*, NaH.
- Both hydrogen and halogens form similar compounds like CH_4, CCl_4; SiH_4, $SiCl_4$, etc.
- When fused metal hydrides such as CaH_2 are electrolysed, hydrogen is liberated at the anode. In this respect it resembles halogens which are also liberated at the anode when halides such as fused sodium chloride is electrolysed.
- Salts of both hydrogen and halogens are generally good electrolytes.

Though hydrogen resembles with halogens in many respects but at the same time it does differ from halogens. For example, hydrogen as well as halogens gain an electron to form negative ions. However, hydrogen can do this in reactions with highly electropositive metals, e.g., NaH,CaH_2. This is because of small size of 1s orbital of hydrogen, the introduction of the second electron in the 1s orbital (during the formation of H^- ion) produces interelectronic repulsion. As a result, the process is not very favourable and the size of H^- ion is 208 pm.

Loss of electron from hydrgen atom results in nucleus (H^+) of ~1.5×10^{-3} pm size. This is extremely small as compared to normal atomic and ionic sizes of 50 to 200 pm. As a consequence, H^+ does not exist freely and is always associated with other atoms or molecules.

From the above discussion, it is evident that hydrogen displays unique behaviour that it does resemble and at the same time differ from the elements of group I and group VII. Therefore, it is difficult to place it along with the elements of any of the above mentioned groups. Thus it is a prototype of the whole system and it must be kept isolated from others thereby confirming the view of Mendeleef.

4.3 OCCURRENCE

Dihydrogen (H_2) is the most abundant element in the universe (70% of the total mass of the universe). For example, 90 per cent of the mass of the sun is due to hydrogen. Thus, the sun, many stars and nebulae are considered to have been formed 12 to 15 billion years ago from a rotating mass of hydrogen gas that condensed under gravitational forces.

Hydrogen is the third most abundant non-metal on earth (after oxygen and silicon). Hydrogen in combined form accounts for about 15.4 per cent of the **atoms** in the earth's crust and oceans. Because it is the lightest element it occupies the ninth position in order of abundance by weight (0.9 per cent). In the combined form it occurs in water, plant and animal tissues, carbohydrates, proteins, hydrides including hydrocarbons and many other compounds.

4.4 ISOTOPES OF HYDROGEN

There are three isotopes of hydrogen with mass numbers 1, 2 and 3, each possessing an atomic number of one. Due to their comparatively large percentage mass differences, there is a greater difference in physical properties between the isotopes of hydrogen than between the isotopes of any other element. The atomic and physical properties of the three isotopes of hydrogen are summarized in Table 4.1.

The structures of the three isotopes of hydrogen are depicted in Fig. 4.1.

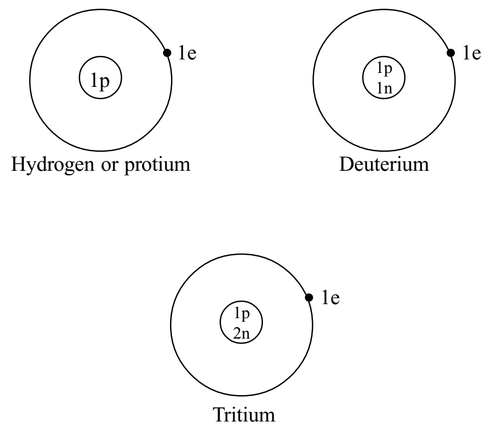


Fig 4.1 Isotopes of hydrogen

The nuclei of these three isotopes contain 0, 1 and 2 neutrons. As the atomic number of three isotopes of hydrogen is one, they will have similar chemical properties. However, their reaction rates will be different, because of the large mass ratio (property difference arising from the difference in mass is called isotopic effect). For example, a bond to a protium atom can be broken as much as 18 times faster than the bond to a deuterium atom. Thus the reaction of protium with chlorine is 13.4 times faster than deuterium because of lower activation energy. Similarly the addition of hydrogen to ethylene compounds is twice as fast as with deuterium at the same temperature. Protium is more rapidly adsorbed on the surface

Table 4.1 Atomic and physical properties of hydrogen, deuterium and tritium

Property	H_2	D_2	T_2
Relative atomic mass	1.007825	2.014102	3.016049
Nuclear spin quantum number		1	$\frac{1}{2}$
Radio activity	Stable	Stable	$\beta t_{1/2}$ =12.26y
Melting point /K	13.957	18.73	20.62
Boiling point /K	20.39	23.67	25.04
Heat of fusion / kJ mol^{-1}	0.117	0.197	0.250
Heat of vapourization / kJ mol^{-1}	0.904	1.226	1.393
Critical temperature / K	33.19	38.35	40.6 (calc)
Heat of dissociation / kJ mol^{-1} (at 298K)	435.88	443.35	446.9
Bond length (pm)	74.14	74.14	(74.14)
Bond energy / kJ mol^{-1}	436.0	443.4	446.9
Density / gL^{-1}	0.09	0.18	0.27
Covalent radius / pm	37	-	-
Ionic radius (H^-) pm	208		

than deuterium. Therefore it can be concluded that reactions of deuterium are slower than those of protium.This is because, the mass of deuterium is more than hydrogen, the D – D bond vibrational energies are more than H – H bond vibrational energies. This requires more bond dissociation energies.

Ordinary hydrogen is the most common isotope and constitutes 99.984 per cent of total hydrogen available in the nature. Deuterium constitutes only about 0.0156 per cent of total hydrogen occurring in nature. Tritium constitutes only 7×10^{-16} per cent of total natural hydrogen.

Tritium is formed in the upper atomosphere layers, by certain nuclear reactions.

$$^{14}_{7}N + ^{1}_{0}n \longrightarrow ^{12}_{6}C + ^{3}_{1}T$$

It can be produced artificially by nuclear transformations.

$$^{6}_{3}Li + ^{1}_{0}n \longrightarrow ^{4}_{2}He + ^{3}_{1}T$$

Tritium is radioactive isotope of hydrogen and its half-life period is 12.26 years.

$$^{3}_{1}H \longrightarrow ^{3}_{2}He + ^{0}_{-1}e \ (\beta\text{-decay})$$

The mass difference in hydrogen and deuterium and the radioactivity of tritium makes them used as tracers in the study of the reaction mechanism. Among the radioactive elements used as tracers, tritium is relatively cheap and also non-toxic since it emits low energy β-radiation and no γ-radiation.

4.5 TYPES OF HYDROGEN

(i) Nascent Hydrogen: The hydrogen at the moment of its production in contact with the substance to be reduced is called as nascent hydrogen. It is more reactive than the ordinary hydrogen gas. For example, hydrogen gas cannot reduce acidified solutions containing ferric, permanganate, dichromate or chlorate ions. But if a small zinc piece is added to such solutions, these ions are reduced to ferrous, manganous, chromic and chloride ions, respectively by the nascent hydrogen produced in solution.

A number of theories have been proposed for the superior activity of nascent hydrogen.

Initially it was considered that at the moment of its liberation, hydrogen exists as atoms which can react readily. The ordinary dihydrogen is less reactive because extra energy is required to break the H–H bond prior to chemical reaction. This theory failed in explaining why nascent hydrogen obtained by different sources has different reactivities. For example, chlorate cannot be reduced by the nascent hydrogen produced by the action of sodium amalgam and water but can be reduced by the nascent hydrogen produced by zinc and dilute sulphuric acid.

Another theory known as *energy association theory* proposes that some energy is associated with hydrogen at the time of its production and that energy makes the hydrogen more reactive. Since the energy associated with hydrogen in different chemical reactions will be different, the difference in the chemical reactivity of nascent hydrogen is also different.

This theory also explains the difference in the reducing powers of hydrogen produced by electrolysis using different metal chlorides. The higher the voltage required by the metallic cathode to liberate hydrogen, the higher the reactivity of hydrogen formed. Here again some of the extra energy supplied becomes associated with the hydrogen and enhances its chemical reactivity.

Another theory called **pressure theory** proposes that at the moment of production in the form of very minute bubble having high pressure, the nascent hydrogen is more reactive. The reactivity of ordinary hydrogen can also be increased by applying high pressure was shown by **Ipatiev**.

It may, however, be pointed out that none of the theories described so far is entirely satisfactory to explain the behaviour of nascent hydrogen.

(ii) Active hydrogen: Active hydrogen can be obtained by subjecting a stream of hydrogen gas at ordinary temperatures to silent electric discharge at a voltage of about 30,000 volts. It is highly reactive and can combine with lead and sulphur directly forming their hydrides at ordinary temperatures. It has been suggested that this active hydrogen exists either in the form of atoms or as H_3 molecules like ozone (O_3). But neither of these views has been finally accepted.

(iii) Atomic hydrogen: Atomic hydrogen can be prepared by passing molecular hydrogen through an electric arc set up between tungsten rods (Fig. 4.2). The temperature of the arc is maintained at 3500° – 4000°C. In order to get maximum formation of atomic hydrogen the pressure of the hydrogen gas lead into the arc should be close to the atmospheric pressure.

In the recent methods hydrogen containing mercury vapour is subjected to radiations emitted from a mercury vapour arc in a discharge tube. The radiation energy is first absorbed by the mercury vapour and then transferred from the mercury vapour to the hydrogen molecules which get dissociated into atoms.

$$Hg + h\nu \longrightarrow \overset{*}{Hg}$$
$$H_2 + \overset{*}{Hg} \longrightarrow 2H + Hg$$

Atomic hydrogen is highly unstable. Its half-life period is only 0.3 second. They immediately combine forming molecules liberating a large amount of heat.

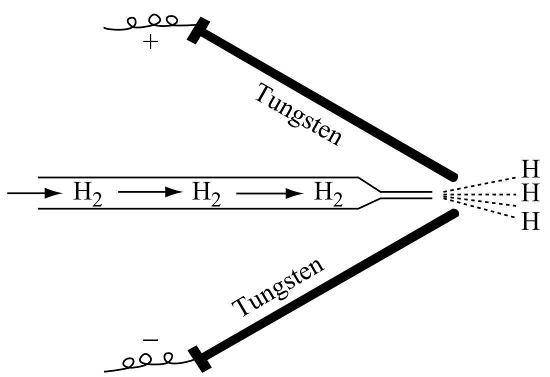


Fig 4.2 Electric arc method for the production of atomic hydrogen

$$2H \longrightarrow H_2 \quad \Delta H = -458.1 \text{ kJ mol}^{-1}$$

Atomic hydrogen combines with several metals and non-metals directly and acts as a strong reducing agent.

(iv) Ortho- and Parahydrogen: Similar to electron, the protons and neutrons in the nuclei also spin on their own axis. When the spin of the two nuclei is parallel in the hydrogen molecule (the resultant nuclear spin is 1) it is known as **orthohydrogen** but if the spin of the two nuclei is antiparallel in the hydrogen molecule (the resultant nuclear spin is zero) it is known as **parahydrogen**. Ortho- and parahydrogens cannot be called as allotropes but they may be called as nuclear spin isomers.

Ordinary hydrogen contain both forms of hydrogen but their proportions vary with temperature and one can convert into another. At room temperature ordinary hydrogen is a mixture of about 75 per cent ortho- and 25 per cent parahydrogen. At absolute zero, hydrogens exist only as pure paraform as the temperature increase parahydrogen converts into orthohydrogen. The proportion of orthoform cannot be increased more than 75 per cent even at very high temperatures.

The conversions parahydrogen to orthohydrogen and vice versa are normally very slow as they involve a forbidden transition between two energy states of different spin multiplicity. Introduction of atomic hydrogen or use of catalysts like O_2, NO, NO_2, Mn^{2+} (aq), Fe^{2+} (aq), Pt, Pd and charcoal accelerates the attainment of equilibrium. Most of these catalysts are paramagnetic. They either break the H–H bond, or at least weaken it or effect some sort of magnetic perturbations. Consistent with this, the rate of conversion of $p - H_2$ to $o - H_2$ is proportional to $[p - H_2]\,[\text{total } H_2]^{1/2}$ at high temperature.

Orthohydrogen, owing to the spins in same direction, has *more internal energy*. In the case of parahydrogen the spin get cancelled and so the molecules would have

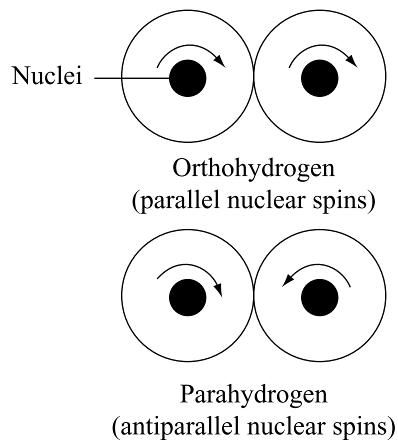


Fig 4.3 Ortho- and parahydrogens

Table 4.2 Certain physical properties of ortho- and parahydrogens and deuterium

	Ortho-hydrogen	Para-hydrogen	Ortho-deuterium	Normal D_2 (66.7% ortho)
M. P (K)	14.05	13.81	18.70	18.73
B. P (K)	20.45	20.29	23.63	23.67
Rotational specific heat (J mol^{-1} k^{-1})				
50 K	0.00	0.167		
100 K	0.305	6.276		
298 K	7.690	9.146		

less energy. Because of this reason at low temperature more stable paraform is formed and at high temperature orthohydrogen having more internal energy is formed.

Nuclear spin isomerism does not effectively influence the bond between the atoms and so the chemical properties of ortho- and parahydrogen (or any other element) are alike. Most physical properties are also little affected. However, the heat capacities of the two forms differ appreciably and so also do the thermal conductivities which are proportional to the heat capacities. Thus the thermal conductivity of parahydrogen is 50 per cent greater than that of orthohydrogen (this is used to estimate the o : p content). The melting and boiling points differ slightly. Note that the trend is reversed with D_2 and the difference also become smaller (increase of mass) For other elements, the differences are expected to be still smaller. Ortho- and parahydrogens differ in their band spectra.

Orthohydrogen, owing to the spins in the same direction, has **more internal energy**. In the case of parahydrogen the spins get cancelled and so the molecules would have less energy. Because of this reason, at low temperature more stable paraform is formed and at high temperature orthohydrogen having more internal energy is formed.

4.6 PREPARATION OF HYDROGEN

(a) Laboratory Methods

(i) Action of water on metals: This is essentially a process in which hydroxonium ions are reduced. Since other cations besides the hydrogen ion are hydrated in water and equations showing hydration look rather cumbersome, the simple cations will usually be written.

$$2H^+ + 2e^- \longrightarrow H_2$$

Metals with high negative electrode potentials (Group I A and II A) react with cold water. E.g.,

$$2Na + 2H_2O \longrightarrow 2NaOH + H_2$$

Magnesium reacts only slowly with water but vigorously with steam, the oxide rather than hydroxide is formed.

$$Mg + H_2O \longrightarrow MgO + H_2$$

(ii) Action of dilute acids on metals: This again is a reduction of hydrogen ions. All metals higher than hydrogen in the electrode potential series react with dilute hydrochloric acid and sulphruic acids to give hydrogen; the more negative the electrode potentials of the metal the more vigorous the reaction. Nitric acid and concentrated sulphuric acid must be avoided, since hydrogen appears as water in these instances and oxide of nitrogen and sulphur dioxide are evolved, respectively. Lead does not show much evidence of reaction with dilute hydrochloric or sulphuric acid since an insoluble layer of salt prevents appreciable action.

Hydrogen is normally prepared in the laboratory by reacting zinc with dilute hydrochloric acid (1vol of conc. HC*l* to 1 vol of water). The gas, however, is frequently contaminated with hydrogen sulphide and arsine from traces of zinc sulphide and arsenic in zinc.

$$Zn + 2H^+ \longrightarrow Zn^{2+} + H_2$$

Very pure zinc reacts only slowly with these acids and this phenomenon has been attributed to a high energy of activation for one or both of the reactions.

$$2H^+ + 2e^- \longrightarrow 2H$$

$$2H \longrightarrow H_2$$

The reaction can be catalyzed by either copper or platinum, which by themselves do not liberate hydrogen from acids, the hydrogen being evolved from non-zinc surface. Platinum is known to be a more effective catalyst for the recombination of hydrogen atoms.

(iii) Action of strong Alkali on Zinc and Aluminium: Zinc and more particularly aluminium, react with aqueous solutions of sodium and potassium hydroxides to give hydrogen.

$$Zn + 2OH^- + 2H_2O \longrightarrow \underset{\text{Zincate ion}}{[Zn\,(OH)_4]^{2-}} + H_2$$

$$2Al + 2OH^- + 6H_2O \longrightarrow \underset{\text{Aluminate ion}}{2[Al(OH)_4]^-} + 3H_2$$

(iv) Hydrolysis of ionic Hydrides: In this type of reaction a hydride is oxidized to hydrogen by loss of an electron;

it can equally well be regarded as an acid–base reaction in the Bronsted–Lowry sense. E.g.,

$$CaH_2 + 2H_2O \longrightarrow Ca(OH)_2 + 2H_2$$

$$\text{(or)} \quad H + H_2O \longrightarrow OH^- + H_2$$

(b) Commercial Methods

(i) Action of steam on coke (Bosch process) or hydrocarbons: The first process involving coke is carried out by passing steam over white hot coke at a temperature of about 1200°C. The reaction is endothermic and the temperature falls of 800°C. A blast of air is now turned on to raise the temperature

$$C + H_2O \longrightarrow \underbrace{CO + H_2}_{\text{water gas}} \text{(endothermic)}$$

$$2C + (O_2 + 4H_2) \longrightarrow \underbrace{2CO + 4N_2}_{\text{producer gas}} \text{(exothermic)}$$

The mixture of CO and H_2 is called water gas. The water gas is used for the synthesis of methanol and a number of hydrocarbons. So, it is also called **synthesis gas** or **Syn gas**. Nowadays, "syngas" is produced from sewage, sawdust, scrap wood, news papers, etc. The process of producing syngas from coal is called **coal gasification**.

The water gas now treated with more steam at a temperature of about 450°C in the presence of a special iron oxide or iron chromate catalyst; the carbon monoxide is oxidized to carbon dioxide and an extra molecule of hydrogen is formed

$$CO + H_2 + H_2O \rightleftharpoons CO_2 + 2H_2$$

This reaction is called **water- gas shift reaction.** Carbon dioxide is removed by scrubbing with sodium arsenite solution or water under pressure or by treatment with a hot solution of potassium carbonate.

The above method is now being replaced by a very similar process which uses cheap refinery hydrocarbons in place of coke. The first stage of the process is operated at a temperature of about 900°C with a nickel catalyst when a mixture of carbon monoxide and hydrogen (synthesis gas) is formed. The second stage (the water gas shift reaction) is exactly similar to the water gas / steam reaction above and the carbon dioxide is removed by the same technique. Traces of carbon monoxide still remaining are removed by absorption in an ammonical solution of copper (I) formate.

$$C_n H_{2n+2} + nH_2O \xrightarrow{900°C} nCO + (2n+1)H_2$$

$$nCO + nH_2O \longrightarrow nCO_2 + nH_2 \quad \text{(water gas shift reaction)}$$

(ii) Lane's Process: When superheated steam is passed over heated iron at about 1200 K, hydrogen is produced according to the following reaction

$$3Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$$

The Fe_3O_4 is reduced to iron by means of water gas ($CO + H_2$) at 1200 k.

$$Fe_3O_4 + 4CO \longrightarrow 3Fe + 4CO_2$$

$$Fe_3O_4 + 4H_2 \longrightarrow 3Fe + 4H_2O$$

In practice, first steam is passed for 10 – 15 minutes to oxidize iron into iron oxide which is reduced by passing water gas 20 – 30 minutes. In this way, the process is carried out alternatively.

(iii) Uyeno's process: By treating scrap aluminium with sodium hydroxide solution.

$$2Al + 2NaOH + 2H_2O \longrightarrow 2NaAlO_2 + 2H_2O$$

(iv) Electrolysis of water: Electrolysis of acidified or alkaline water using platinum or nickel electrodes gives hydrogen.

$$2H_2O_{(l)} \xrightarrow[\text{Traces of acid / base}]{\text{Electrolysis}} 2H_{2(g)} + O_{2(g)}$$

Highly pure (>99.95 per cent) hydrogen is obtained by the electrolysis of warm aqueous barium hydroxide solution between nickel electrodes.

(v) By-product, Hydrogen: Hydrogen is produced as a by-product during the cracking of hydrocarbons, and in the manufacture of chlorine and sodium hydroxide by electrolyzing a concentrated solution of sodium chloride. The reactions that takes place during electrolysis are

$$\text{at anode:} \quad 2Cl^-_{(aq)} \longrightarrow Cl_{2(g)} + 2e^-$$

$$\text{at cathode:} \quad 2H_2O_{(l)} + 2e^- \longrightarrow 2H_{2(g)} + 2OH^-_{(aq)}$$

The overall reaction is

$$2Na^+_{(aq)} + 2Cl^-_{(aq)} + 2H_2O_{(l)} \longrightarrow Cl_{2(g)} + H_{2(g)} + 2Na^+_{(aq)} + 2OH^-_{(aq)}$$

Presently ~77 per cent of the industrial hydrogen is produced from petrochemicals, 18 per cent from coal, 4 per cent from electrolysis of aqueous solutions and 1 per cent from other sources.

4.6.1 Physical Properties of Hydrogen

Hydrogen is a colourless, odourless gas without taste or smell. It can be liquified by compression and cooling in liquid nitrogen, followed by sudden expansion. Liquid hydrogen (b.p 20K) becomes solid at 14K. It is combustible gas. It is lighter than air and insoluble in water. Its other physical properties are given in Table 4.1.

4.6.2 Chemical Properties of Hydrogen

Hydrogen is not a very reactive element under normal conditions because H–H bond dissociation enthalpy is the highest for a single bond between two atoms of any element. It dissociates into atoms only to about ~0.081 per cent around 2000 K but at 5000 K it can dissociate to about 95.5 per cent. At room temperature it is relatively inert because of high H–H bond enthalpy. Since its electronic configuration is $1s^1$, it does combine with almost all elements either by loss of the only electron to give H^+ or by gaining one electron to form H^- and by sharing electrons to form single covalent bond.

(i) Reaction with halogens: Hydrogen reacts with halogens forming hydrogen halides (HX)

$$H_{2(g)} + X_{2(g)} \longrightarrow 2HX_{(g)} \ (X = F, Cl, Br, I)$$

The reaction with fluorine takes place even in dark and at 28 K explosively. Chlorine reacts only in the presence of light, i.e., sunlight or UV light; bromine can react at 573 K. The reaction with iodine is reversible and takes place at 713 K in the presence of platinum sponge catalyst. Even under these conditions the reaction never goes to completion.

(ii) Reaction with oxygen: Hydrogen burns in oxygen forming water. The reaction is highly exothermic.

$$2H_2 + O_2 \longrightarrow 2H_2O; \ \Delta H = -285.9 \text{ kJ mol}^{-1}$$

In the presence of Pd or Pt black catalyst, the reaction takes place at ordinary temperature.

(iii) Reaction with nitrogen: Hydrogen reacts with nitrogen forming ammonia.

$$3H_{2(g)} + N_{2(g)} \xrightarrow{673\text{ K, 200 atm}} 2NH_{3(g)}, \ \Delta H = -92.6 \text{ kJ mol}^{-1}$$

This reaction is the basis for the manufacture of ammonia by Haber's process.

(iv) Reaction with metals: Hydrogen combines with several metals at high temperatures to yield the corresponding metal hydrides (section 4.7).

$$H_{2(g)} + M_{(g)} \longrightarrow 2MH_{(s)} \ (M = \text{alkali metal})$$

$$H_{2(g)} + M_{(g)} \longrightarrow MH_{2(s)} + \ (M = \text{alkaline earth metal})$$

(v) Reduction properties: Hydrogen is a strong reducing agent. It reduces some metal ions in aqueous solution and oxides of less reactive metals than iron to their corresponding metals.

$$H_{2(g)} + Pd^{2+} \longrightarrow Pd_{(s)} + 2H^+_{(aq)}$$

$$yH_{2(g)} + M_xO_y \longrightarrow xM_{(s)} + yH_2O_{(l)}$$

(vi) Reaction with organic compounds: Hydrogen reacts with several organic compounds in the presence of catalysts to give useful hydrogenated products of commercial importance.

For example: (a) Hydrogenation of vegetable oils using nickel as catalyst gives edible fats (margarine and vanaspathi or vegetable ghee).

(b) Methanol is formed from CO on reaction with H_2.

$$2H_2 + CO \xrightarrow{\text{catalyst}} CH_3OH$$

(c) Hydroformylation of olefins yields aldehydes which further undergo reduction to give alcohols.

$$H_2 + CO + RCH = CH_2 \longrightarrow RCH_2CH_2CHO$$

$$H_2 + RCH_2CH_2CHO \longrightarrow RCH_2CH_2CH_2OH$$

(vii) Unsaturated hydrocarbons combine with hydrogen to form saturated hydrocarbons.

Example: $H_2C = CH_2 + H_2 \longrightarrow CH_3 - CH_3$

$$HC \equiv CH + 2H_2 \longrightarrow CH_3 - CH_3$$

4.6.3 Uses of Hydrogen

Till date, only small quantities of hydrogen were required as a fuel in the form of town gas and water gas, for filling balloons and in the oxyhydrogen blow lamp for welding. Nowadays, however, large quantities of the gas are employed. The utility of hydrogen can be understood under four heads.

(i) In the Synthesis of Industrial Chemicals

(a) Manufacture of ammonia: Nitrogen combines with hydrogen gas in the presence of finely divided activated iron catalyst at 450–500°C and 200 atmospheric pressure.

$$N_{2(g)} + 3H_{2(g)} \underset{}{\overset{\text{Fe cat}}{\rightleftharpoons}} 2NH_{3(g)}$$

This ammonia, in turn, is used in the manufacture of nitric acid, explosives, dyestuffs, nitrogenous fertilizers, etc.

(b) Manufacture of Hydrogen chloride and Hydrochloric acid: The hydrogen and chlorine are reacted together, and the product is hydrogen chloride which forms hydrochloric acid with water.

$$H_{2(g)} + Cl_{2(g)} \longrightarrow 2HCl_{(g)}$$

(c) Manufacture of Methanol: Methanol is obtained by reacting carbon monoxide with hydrogen at 400°C and

300 atmosphere pressure in the presence of zinc oxide and chromium (III) oxide.

$$CO + 2H_2 \xrightarrow[\Delta]{ZnO + Cr_2O_3} CH_3OH$$

(d) Production of vanaspathi ghee: Unsaturated oils like ground nut oil, cotton seed oil, etc are hydrogenated in the presence of catalyst (Ni, Pd, Pt or Raney nickel) at 5 atm pressure and 150–200°C temperature. This is called hardening of oils.

(e) Production of synthetic gasoline (Fischer–Tropsch Synthesis): Water gas mixed with half its volume of hydrogen, is passed over iron oxide catalyst to remove sulphur first. Finally it is passed over a cobalt catalyst at 200°C. A mixture of lower hydrocarbons is obtained. This is known as synthetic motor fuel or **synthetic petrol.**

$$CO_{(g)} + 3H_{2_{(g)}} \longrightarrow CH_{4_{(g)}} + H_2O_{(g)}$$

$$nCO + 2nH_{2_{(g)}} \longrightarrow C_nH_{2n_{(g)}} + nH_2O_{(g)}$$

$$nCO_2 + (2n+1)\,H_{2_{(g)}} \longrightarrow C_nH_{(2n+2)_{(g)}} + nH_2O_{(g)}$$

(ii) As a Reducing Agent in Metallurgy

Hydrogen is a good reducing agent. Metals which are having higher reduction electrode potentials can be readily produced by reduction with H_2. Hydrogen is used for producing pure metals from their oxide. Molybdenum and tungsten are obtained by this method of reducing their oxides at high temperatures.

$$MoO_3 + 3H_2 \longrightarrow Mo + 3H_2O$$

$$WO_3 + 3H_2 \longrightarrow W + 3H_2O$$

(iii) Use of Hydrogen Fuel

(a) Hydrogen gas is an important gaseous fuel. The heat of combustion of hydrogen is high (242 kJ mol^{-1}). It is not used singly as a fuel because (i) it is not available in the nature, (ii) Its production is costlier and (iii) its storage and transportation are very dangerous because it is highly inflammable. However, it is used as a fuel as a constituent of many fuel gases like coal gas, water gas and oil gas.

Coal gas: It is a mixture of hydrogen (45–55%), methane (25–35%) and carbon monoxide (4–11%). It is obtained by the destructive distillation of coal.

Water gas: It is a mixture of carbon monoxide (40–50%), hydrogen (45–50%) and small amounts of CO_2 and N_2. It is produced as explained in section 4.6 b(i).

Semiwater gas: It is a mixture of CO (25–28%); H_2 (10–12%); CO_2 (4–5%); N_2 (50–55%) and methane (1–2%). This gas is used as a fuel in the steel industry.

(b) Rocket fuel: Liquid hydrogen can be used as rocket fuel. For example, Saturn V, the rocket that took Neil Armstrong to the moon was powered by liquid hydrogen fuel. Oxygen (required for combustion) as well as hydrogen were carried in the rocket in separate tanks.

(c) Oxyhydrogen Torch: Hydrogen is burnt at a jet in an atmosphere of oxygen, producing an extremely hot flame called **oxyhydrogen flame.** This flame has a temperatue of about (2800°C). The apparatus in which oxyhydrogen flame is produced is known as **oxyhydrogen torch.** Some molecules of hydrogen dissociate as they pass through the electric arc.

The single atoms recombine upon collision with one another especially at the surface of the metal plates which are used for welding and release energy which was absorbed during dissociation in the electric arc.

$$2H_{(g)} \longrightarrow H_{2_{(g)}} + \text{heat}$$

Heat is also produced due to burning of hydrogen in oxygen.

$$2H_{2_{(g)}} + O_{2_{(g)}} \longrightarrow 2H_2O_{(g)} + \text{heat}$$

Heat liberated by both these reactions results in very high temperatures. Oxyhydrogen blow torch is used in cutting and welding of metals which melt at high temperatures, melting even the most infusible metals such as tungsten, tantalum and substances like quartz.

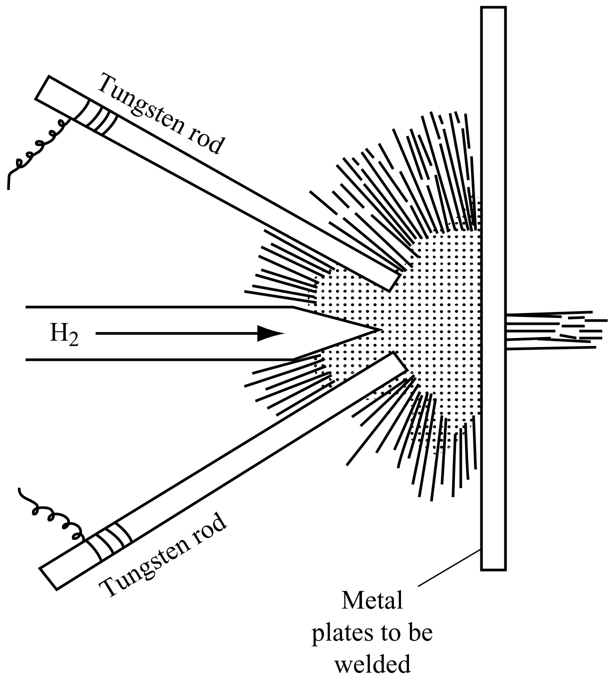


Fig 4.4 Atomic hydrogen blow torch

(d) Fuel cells: Electrical cells that are designed to convert the energy from the combusion of fuels like hydrogen and carbon monoxide or methane, directly

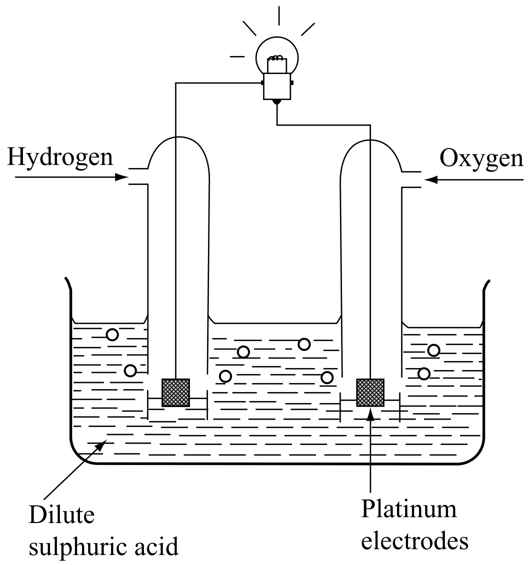


Fig 4.5 Hydrogen fuel cell

into electrical energy are called fuel cells. Hydrogen–oxygen fuel cells are useful devices for converting the energy of the reaction.

$$2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(l)}$$

The cell is shown in Fig. 4.5.

Hydrogen and oxygen gases are brought together over two electrodes, which can catalyze the reaction between them. The electrolyte can be an acid or an alkali, depending on the type of cell. The virtue of the fuel cell is that it produces electricity with only water as the by-product. The cells can operate with high power outputs and efficiencies greater than those of conventional methods of generating electricity like thermal or nuclear power stations. Fuel cells have been used by space vehicles, where they have the virtue of being reliable, efficient and of relatively small size compared to conventional electric cells. The electrode reactions are

$$\text{at anode:} \left[H_2(g) + 2OH^-(aq) \longrightarrow 2H_2O(l) + 2e^-\right] \times 2$$

$$\text{at cathode: } 2H_2O_{(l)} + O_{2(g)} + 2e^- \rightarrow 4OH^-_{(aq)}$$

$$\text{The overall reaction } 2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(l)}$$

The oritically, the efficiency of a fuel cell should be 100 per cent. So far, an efficiency of 60–70 per cent only has been achieved. Since the fuel cells are more efficient and do not pollute the atmosphere, attempts are made to produce them on a commercial scale.

(a) Motor fuel: One use of hydrogen that has been thoroughly researched in recent years is its potential use as a fuel. When hydrogen burns in air it produces significant amounts of energy, and it has the virtue of being pollution free. Cars have been built that use hydrogen as a fuel rather than petrol. It might be thought that carrying hydrogen around could be dangerous, but it can be absorbed by a number of metal alloys, which are perfectly safe.

(b) As a source of Atomic energy: In the sun, the stars and the hydrogen bomb energy is released by nuclear fusion. During fusion, atoms of lighter elements combine to form atoms of heavier elements.

(i) The energy of the sun and stars is due to fusion of hydrogen nuclei forming helium. In this fusion reaction, enormous amount of solar energy is generated and radiated to the earth and other planets. This reaction takes place at the temperatures that are of the order of many billion degrees. The fusion reactions that takes place in solar system are

$$^2_1H + ^2_1H \rightarrow ^4_2He \quad \Delta H = -23 \times 10^8 \text{ kJ mol}^{-1}$$

$$^2_1H + ^3_1H \rightarrow ^4_2He + ^1_0n \quad \Delta H = -17.2 \times 10^8 \text{ kJ mol}^{-1}$$

$$4(^1_1H) \rightarrow ^4_2He + 2^{0}_{+1}e \quad \Delta H = -26 \times 10^8 \text{ kJ mol}^{-1}$$

(ii) Thermonuclear reactions: A very large amount of energy is released in nuclear fusion reaction in which two or more lighter nuclei combine to form heavy nucleus. A large amount of energy is required to overcome the repulsion between positively charged nuclei to get them close enough to fuse and so for fusion reactions very high temperatures of over a million degrees are required. Thus, the nuclear reactions are termed as **"thermonuclear reactions"**. The fusion of two hydrogen nuclei is initiated by an atomic bomb which generates the required high temperature for the fusion reaction in the hydrogen bomb

$$^2_1H + ^2_1H \longrightarrow ^4_2He$$

Thus 23×10^8 kJ mol^{-1} of energy is released.

This reaction can be used as alternate sources of energy in the nuclear power plants. The use of hydrogen or deuterium in the place of uranium or plutonium as nuclear fuel in nuclear power plants has several advantages.

The fission products of uranium and plutonium are radioactive and harmful. Their disposal is of great problem. They should be stored for several years unitl their radioactivity becomes less harmful. If hydrogen is used as nuclear fuel the fusion product is non-radioactive helium gas. Further, hydrogen is available in large quantities in nature in the combined state like water, hydrocarbons, etc. But the abundance of uranium and plutonium in nature is very less and depleting.

The major hurdle in the usage of hydrogen as a source of nuclear energy is that the hydrogen fusion reactions are uncontrollable and this reaction takes place only at very high temperatures.

4.7 HYDRIDES

The binary compounds of hydrogen with other elements are called hydrides. Hydrogen combines with almost all elements in the periodic table except noble gases. The type of hydride formed by an element depends on its electronegativity and hence on the type of bond. Gibbs classified the hydrides, on the basis of their physical and chemical properties into four important types.

1. Saline or salt-like or electrovalent hydrides
2. Covalent or molecular hydrides
3. Interstitial or metallic hydrides
4. Polymeric or polynuclear hydrides

1. Saline or Salt-like or Electrovalent hydrides: These are formed by transfer of electrons from metals to hydrogen atoms. Only the group 1A and 2A metals are sufficiently electropositive to force the hydrogen atom to accept an electron. Hydrides of lithium, berylium and magnesium (LiH, BeH_2 and MgH_2) have significant covalent character. These hydrides are formed by heating the alkali metals and alkaline earth metals except (Be and Mg) in hydrogen at high temperatures.

(i) These are white crystalline solids.
(ii) They are stoichiometric compounds.
(iii) These hydrides decompose above 675 K except the hydrides of Li, Ca and Sr.
(iv) They are good conductors of electricity. When fused, liberate hydrogen at the anode on electrolysis.
(v) They are strong bases; when dissolved in water they from corresponding hydroxide and evolve hydrogen.
(vi) They are strong reducing agents especially at high temperatures.

$$SiCl_4 + 4NaH \longrightarrow SiH_4 + 4NaCl$$
$$PbSO_4 + 2CaH_2 \longrightarrow PbS + 2Ca(OH)_2$$
$$2CO + NaH \longrightarrow HCOONa + C$$

(vii) They form complex hydrides which are good reducing agents.

$$LiH + AlH_3 \longrightarrow LiAlH_4$$
$$2NaH + B_2H_6 \longrightarrow 2NaBH_4$$

(viii) They have high heat of formation showing their relative stability. The values of heat of formation in kJ mol^{-1} are given below.

LiH	NaH	KH	RbH	CsH	CaH_2	SrH_2	BaH_2
90.4	58.2	59.0	56.5	54.4	188.7	177.0	171.1

Stability decreases down the group due to decrease in lattice energy with increase in size of cation.

(ix) The density of these hydrides is greater than that of the metal from which they were formed since H^- ions occupy the holes of the lattice of the metal without distorting the metal lattice.
(x) Group I hydrides are more reactive than the corresponding group 2 hydrides and reactivity increases down the group.

2. Covalent or Molecular Hydrides: By far, the greatest number of hydrides come under this classification. These are formed by the p-block elements of the periodic table. These hydrides usually contains discrete covalent molecules between which only weak van dar Waal's attractive forces exist. So these hydrides are soft, have low melting and boiling points. They have low electrical conductivity. The bonds formed are mostly covalent in character but in some cases, e.g., HF etc., the bond may be partially ionic in nature.

These hydrides may be prepared either by direct combination of the elements or by hydrolysis of metal borides, carbides, nitrides, phosphides, etc. or by electrolytic reduction. The general formula of these hydrides is MH_{8-n} or MH_n where n is the group number of the element.

Depending on the relative number of electrons and bonds in their Lewis structures molecular hydrides are further classified into three types: (a) electron deficient, (b) electron precise and (c) electron-rich hydrides.

(a) Electron-deficient hydrides: These are formed by the elements of boron family. In these hydrides, the central atom do not have octet. They act as Lewis acids, i.e., electron acceptors.

(b) Electron-precise hydrides: These are generally formed by the elements of carbon family. In these hydrides, the central atom have an octet and all the electron pairs are bond pairs, e.g., CH_4, SiH_4, etc. They have a tetrahedral geometry.

(c) Electron-rich hydrides: These are formed by the elements of Group VA, VIA and VIIA elements. In these hydrides, the central atom have an octet and among the electron pairs present around the central atom some are bond pairs and some are lone pairs, e.g., NH_3, H_2O, HF, etc. They behave as Lewis bases i.e., electron donors. The presence of lone pairs on the highly electronegative atoms like N, O and F in these hydrides results in hydrogen bond formation between the molecules. This leads to association of molecules due to which their physical properties like m.pt and b.pt. changes.

Table 4.3 Boiling points of the hydrides of groups V, VI and VII elements

Group IV hydride	b.p °C	Group V hydride	b.p °C	Group VI hydride	b.p °C	Group VII hydride	b.p °C
CH_4	– 162	NH_3	– 33	H_2O	+100	HF	+19.4
SiH_4	– 111	PH_3	– 84.6	H_2S	– 59.6	HC*l*	– 85.0
GeH_4	– 88	AsH_3	– 55.1	H_2Se	– 42.6	HBr	– 67.1
SnH_4	– 52	SbH_3	– 17.0	H_2Te	– 1.8	HI	– 35.9

The stability of these hydrides in a particular group decreases with increasing atomic number. This is because of the decrease in E-H bond energy with increase in bond length due to increase in atomic size down the group. Some are so unstable in the presence of small traces of air, e.g., stannane (SnH_4). Some special methods are necessary for their preparation.

3. Metallic or Non-Stoichiometric (Or Interstitial) Hydrides: These are formed by d-and f-block elements of the periodic table. However, the metals of Mn, Co, Fe groups and some metals of chromium group do not form hydrides. This part of the perodic table is called **hydride gap.**

These are generally prepared by heating the metal with hydrogen under high pressure. At high temperatures, they decompose. This reaction can be used for the preparation of pure hydrogen.

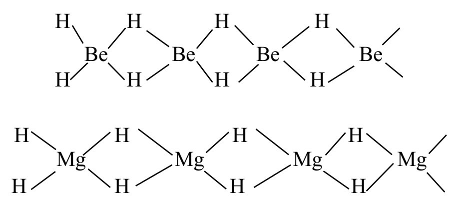
The properties of these hydrides are generally similar to those of the parent metals. They have metallic lustre, conduct electricity and have magnetic properties. Some expansion of the metal lattice occurs, since the density of the hydride is less than that of the parent metal. They are brittle due to the distortion of the crystal lattice of the metal.

No definite chemical formula can be allocated to these substances, i.e., they are non-stoichiometric hydrides. Their composition can be varied by changes in temperature and pressure. Examples are $TiH_{1.5-1.8}$, $ZrH_{1.3-1.75}$, $VH_{0.56}$, $PdH_{0.6-0.8}$, $NiH_{0.6-07}$, etc. These hydrides do not follow the law of constant composition. The uptake of hydrogen is reversible and can in all cases be removed by pumping at a sufficiently high temperature. These non-stoichiometric compounds may be regarded as **solid solutions**.

Formerly it was thought that in these hydrides, the hydrogen occupies the interstitial viods of the metal lattice, due to which the crystal structure is distorted without any change in the lattice type. Hence, they are named as **interstitial hydrides**. Metals which can form this type of hydrides can act as catalysts for hydrogenation reactions. The catalysts are thought to be effective through providing H atoms rather than H_2 molecules. It is not certain whether the hydrogen atoms present in the interstitial sites is in the form of hydrogen atoms or as H^+ ion with delocalized electrons, but they have strong reducing properties. Recent studies shows that except in the case of Ni, Pd, Co and Ac, the crystal structures of other hydrides are different from that of metal lattice. Among lanthanoids, **europium** and **ytterbium** form ionic hydrides EuH_2 and YbH_2.

Occlusion of Hydrogen by Palladium: When red hot palladium is cooled in the atmosphere of hydrogen first hydrogen is chemisorbed at the surface of Pd and at increased pressures it enters the metal lattice. Palladium absorbs about 100 times its own volume of hydrogen. This absorbed hydrogen is called **occluded hydrogen** and this phenomenon is known as **occlusion.** As the gas is abosrbed, the metallic conductivity falls till the material becomes a semiconductor at a composition of about $PdH_{0.5}$. Alloys of palladium such as Pd–Ag and Pd–Au occlude greater amount of hydrogen than the palladium metal.

4. Polymeric hydrides: These are formed by the elements having electronegativity that lies between 1.4 and 2.0. These are compounds which seem to be intermediate in structure between ionic and covalent hydrides and therefore sometimes referred to as intermediate hydrides. They are polymeric and it is thought that hydrogen atoms act as bridging unit, e.g., $(BeH_2)_n$, $(MgH_2)_n$, $(AlH_3)_n$, etc. The structures of $(BeH_2)_n$ and $(MgH_2)_n$ being represented as



Simple valence theories are inadequate for dealing with structures of this type but can be explained as 3c–2e bonds (Chapter 7).

These are good reducing agents. They form stable addition products with ethers and amines. They are highly reactive with water and lower alcohols.

4.8 WATER

Water has played a crucial part in the origin of life and it still has an essential role in maintaining plant and animal life. Plants depend on water for the transfer of nutrients and for photosynthesis owing to the presence of water in the cells and body fluids such as blood, human beings are approximately 60 per cent water. Nearly all the processes essential for life depend on reactions that takes place in an aqueous solution, be it the division of DNA in a cell, the digestion of food stuffs in the stomach or on the transport of oxygen around the body. Given the importance of water, it is not surprising that men and women can survive very much longer without food than they can without water.

Historically, the availability of water supplies has determined where villages, towns and cities are located. Nomadic peoples, and animals, may travel hundreds of miles over the course of a year following the seasonal variation in rainfall and a lack of good quality drinking water, and water for sanitation, brings deadly illnesses such as typhoid, cholera, etc.

All these factors and many more, make water a substance of great importance. From strictly a chemical point of view, the remarkable thing about water is the amount of hydrogen bonding there is, both in the solid (ice) and in the liquid. If it were not for the fact that hydrogen bonds are of intermediate strength (stronger than van der waal's bonds but weaker than ordinary ionic or covalent bonds) then life as we know could not exist and the world be without rivers, lakes and seas.

4.8.1 Physical Properties of Water

It is a colourless, odourless and tasteless liquid. Its physical properties are listed in Table 4.3 along with the physical properties of heavy water.

As already explained, the unusual properties of water in the condensed phase (liquid and solid states) are due to the presence of extensive hydrogen bonding between water molecules. This leads to high freezing point, high boiling point, high heat of vapourization and high heat of fusion in comparision to H_2S and H_2Se. In comparison to other liquids, water has high specific heat, thermal conductivity, surface tension, dipolemoment, dielectric constant, etc. The solubility of many salts is due to the polarity of the water molecules. The attraction of the negatively charged oxygen atoms allows them to congregate around positive ions. This is the process of hydration. The solubility of compounds in water is largely as a competition between the magnitude of the lattice energy of a solid and the hydration energies of its ions or molecules. Due to hydrogen bonding with polar molecules, even covalent componunds like alcohol and carbohydrates dissolve in water.

Table 4.4 Physical properties of H_2O and D_2O

Property	H_2O	D_2O
1. Molecular mass (g mol^{-1})	18.0151	20.0276
2. Melting point / K	273	276.8
3. Boiling point / K	373	374.4
4. Enthalpy of formation / kJ mol^{-1}	–285.9	–294.6
5. Enthalpy of vapourization (373k) /kJ mol^{-1}	40.66	41.61
6. Enthalpy of fusion / kJ mol^{-1}	6.01	–
7. Temperature of Max. density / K	276.98	284.2
8. Density (298 K) / g cm^{-3}	1.0000	1.1059
9. Viscosity / centipoises	0.8903	1.107
10. Dielectric constant /C^2 / N.m^2	78.39	78.06
11. Electrical conductivity (293K / ohm^{-1} cm^{-1})	5.7×10^{-8}	—
12. Ionic product K_w at 298 K	1×10^{-4}	3×10^{-15}
13. Solubility of Na*Cl* /100g of water at 298K	35.9	30.5

4.8.2 Chemical Properties of Water

Water reacts with a large number of substances. Its main properties are discussed here.

(i) Water is highly stable. It dissociates into H_2 and O_2 to an extent of 8.5 per cent at 2270K

(ii) Water under goes self-ionization, i.e., autoprotolysis.

$$H_2O + H_2O \longrightarrow H_3O^+ + OH^-$$

So, water can act both as Bronsted acid while dissolving alkalis and as Bronsted base when an acid is dissolved in it.

(iii) Reaction with metals: The way metals react with water can give us a measure of their reactivities. The most reactive metals are the alkali metals in group I. Each of them reacts with cold water, giving off hydrogen and leaving an alkaline solution.

$$2Na_{(s)} + 2H_2O_{(l)} \longrightarrow 2Na^+_{aq} + 2OH^-_{(aq)} + H_{2(g)}$$

The reactivity increases down the group. The group II metals react in a similar way but less violently (beryllium does not react at all).

Some of the transition metals react with hot water or steam to give an oxide. For example, if steam is passed over heated iron, hydrogen is released and an oxide is left.

$$3Fe_{(s)} + 4H_2O_{(l)} \longrightarrow Fe_3O_{4_{(s)}} + 4H_2O$$

Water plays an important part in the rusting of iron. In the reaction of metals with water, metals are oxidized.

(iv) Reactions with Non-metals: Elements such as carbon, sulphur and phosphorous normally do not react with water. Carbon in the form of coke, when it is white hot will react with steam, and the product is called water gas.

$$C_{(s)} + H_2O_{(g)} \longrightarrow CO_{(g)} + H_2O_{(g)}$$

Fluorine oxidizes water to O_2.

$$2F_{2_{(g)}} + 2H_2O_{(l)} \longrightarrow 4H^+_{(aq)} + 4F^-_{(aq)} + O_{2_{(g)}}$$

A solution of chlorine in water is known as **chlorine water.**

$$Cl_{2_{(g)}} + H_2O_{(l)} \longrightarrow HOCl_{(aq)} + H^+_{(aq)} + Cl^-_{(aq)}$$

The usefulness of chlorine water is due to the formation of hypochlorous acid which can act as oxidizing agent and is responsible for the antibacterial action of chlorine water and for its use as a bleach.

(v) Reaction with compounds: Several compounds that dissolve in water suffer hydrolysis.

(a) Metal oxides when dissolved in water forms alkalis, e.g.

$$Na_2O + H_2O \longrightarrow 2NaOH$$

(b) Non-metal oxides when dissolved in water forms acids, e.g.

$$SO_3 + H_2O \longrightarrow H_2SO_4$$

$$Cl_2O_7 + H_2O \longrightarrow 2HClO_4$$

$$N_2O_5 + H_2O \longrightarrow 2HNO_3$$

$$P_4O_{10} + 6H_2O \longrightarrow 4H_3PO_4$$

(c) Compounds which hydrolyze / ionize, give acidic solution.

$$AlCl_3 + 3H_2O \longrightarrow Al(OH)_3 + 3HCl$$

$$SiCl_4 + 2H_2O \longrightarrow SiO_2 + 4HCl$$

$$H_2S + H_2O \longrightarrow HS^- + H_3O^+$$

$$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$$

(d) Compounds which hydrolyze / ionize, give basic solution.

$$Na_2CO_{3_{(aq)}} + 2H_2O_{(l)} \longrightarrow 2Na^+_{(aq)} + 2OH^-_{(aq)} + H_2CO_3$$

$$NH_{3_{(aq)}} + H_2O_{(l)} \longrightarrow NH^+_{4(aq)} + OH^-_{(aq)}$$

(vi) Water is oxidized to oxygen during photosynthesis.

$$6CO_{2_{(aq)}} + 12H_2O_{(l)} \longrightarrow C_6H_{12}O_{6\,(aq)} + 6H_2O_{(l)} + 6O_{2_{(g)}}$$

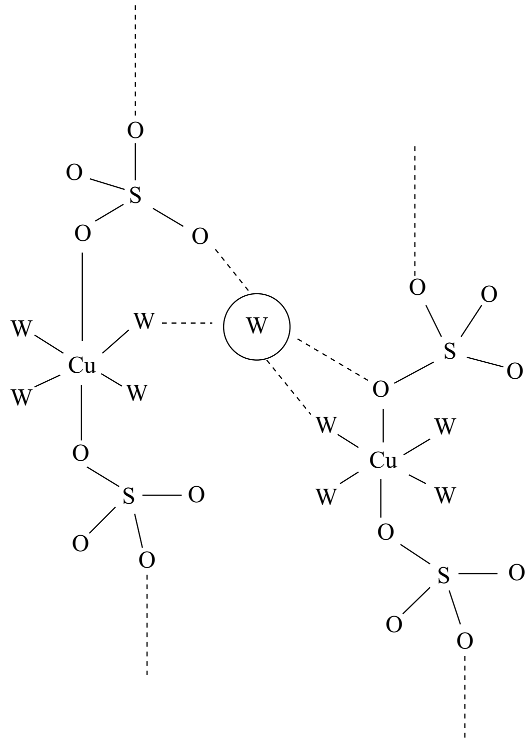


Fig 4.6 The crystal structure of copper (II) sulphate has chains of copper (II) ions bonded to four water molecules 'W' and the oxygen atoms of sulphate ions. The fifth water molecule, circled, lies between the chains. It is hydrogen bonded to oxygen atoms in both chains

(vii) Hydrate formation: During crystallization from their aqueous solutions, many salts crystallizes as hydrated salts. The association of water in these hydrated salts is of different types.

(i) In some compounds, water molecules exist as ligands and present as **coordinated water.** e.g., $[Cr(H_2O)_6]\ Cl_3$.

(ii) In certain compounds, water molecules exist as interstitial water, e.g., $BaCl_2 \cdot 2H_2O$.

(iii) In some compounds, water molecules exist in hydrogen bonds, e.g., $CuSO_4 \cdot 5H_2O$ exist as $[Cu(H_2O)_4]\ SO_4 \cdot H_2O$. In this compound, four water molecules exist as coordinated water and one water molecule exist as hydrogen bonded water as shown in Fig. 4.6.

4.8.3 Structure of Water

In the gas phase, water exists as discrete H_2O molecules. In water molecule oxygen is involved in sp^3 hybridization, in

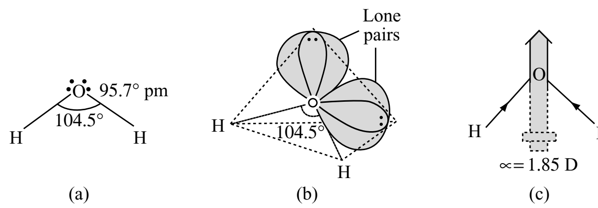


Fig 4.7 Structure of water molecule

which the four electron pairs are arranged tetrahedrally. Two of the electron pairs are lone pairs and the remaining two electron pairs are bond pairs. The shape of the molecule is V-shaped or bent molecule. The bond angle 104.5° is less than tetrahedral bond angle 109°28' due to lone pair bond pair repulsion. Its orbital overlap picture is shown in Fig. 4.8.

Ice has highly ordered three dimensional hydrogen bonded structure. X-ray analysis shows that each oxygen atom is surrounded tetrahedrally by four other oxygen atoms at a distance of 276 pm. Ice is crystalline and crystallized in the hexagonal form at 1 atmospheric pressure but at very low temperatures it condenses to cubic form. The water molecules line up in such a way as to maximize the amount of hydrogen bonding between them. This leaves a tremendous amount of space in the lattice. So, ice is less denser than water and floats on water. The open structure of ice is shown in Fig. 4.8.

Due to the open type structure with wide holes it can hold some other molecules of appropriate size interstitially. In winter season ice formed on the surface of a lake and provides thermal insulation which ensures the survival of the aquatic life. This fact is of great ecological significance.

4.9 HARD WATER

Depending upon the behaviour towards soap solution, water is categorized into two types namely soft water and hard water.

(i) The water that produces ready and permanent lather with soap solution is called **soft water.**

(ii) The water which does not produce lather with soap solution is called **hard water.**

Hardness of water is due to the dissolved salts like bicarbonates, chlorides and sulphates of magnesium and calcium. When the rain water passes through the ground, rocks gets contaminated with these salts. Soap is a mixture of sodium salts of higher fatty acids like stearic acid, palmitic acid, oleic acid, etc. and is soluble in water. If the soap reacts with soluble calcium or magnesium ions present in hard water precipitates as calcium or magnesium salts.

$$\underset{\text{Sodium stearate}}{2C_{17}H_{35}COONa} + M^{2+} \longrightarrow \underset{\text{Metal stearate}}{(C_{17}H_{35}COO)_2M\downarrow} + 2Na$$

(M = Ca or Mg)

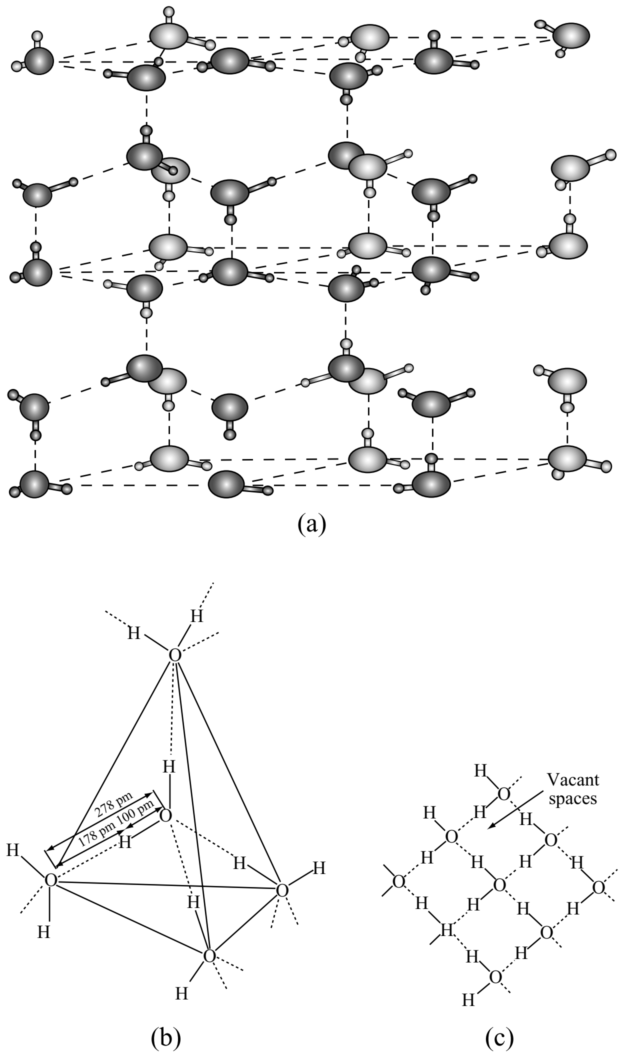


Fig 4.8 (a) Structure of ice, (b) Arrangement of H bonds and covalent bonds around oxygen in ice and (c) Open cage like structure of ice

Thus no lather is produced until all the calcium and magnesium ions have been precipitated. This leads to the consumption and, hence, wastage of lot of soap.

The hardness of water is two types.

(i) Temporary hardness: It is due to the presence of soluble calcium and magnesium bicarbonates. It is called temporary hard water because hardness can be removed simply by boiling the water. The bicarbonates of calcium and magnesium are formed in water due to dissolution of carbonates of calcium and magnesium in the presence of atmospheric carbon dioxide.

(ii) Permanent hardness: It is due to the presence of chlorides and sulphates of calcium and magnesium. It is called permanent hard water because it cannot be removed by boiling.

The process of removal of the calcium and magnesium (Ca^{2+} and Mg^{2+}) ions responsible for hardness of water is known as **softening of hard water.** The reagents used for softening of hard water are called **water softeners.**

4.9.1 Removal of Temporary Hardness

1. By boiling: Temporary hardness can be removed by simple boiling of the hard water in large boilers. During boiling the soluble Mg $(HCO_3)_2$ is converted into insoluble $Mg(OH)_2$ and $Ca(HCO_3)_2$ is converted into insoluble $CaCO_3$. The solubility product of $Mg(OH)_2$ is less than $MgCO_3$ · So, $Mg(OH)_2$ is precipitated instead of $MgCO_3$ but the solubility product of $Ca(OH)_2$ is larger than that of $CaCO_3$. So, in the case of $Ca(HCO_3)_2$, $CaCO_3$ is precipitated instead of $Ca(OH)_2$.

$$Ca\,(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 \downarrow + H_2O + CO_2$$

$$Mg(HCO_3)_2 \xrightarrow{\Delta} Mg(OH)_2 + 2CO_2$$

2. Clark's method: Temporary hardness can also be removed by the addition of calculated quantities of lime.

$$Ca\,(HCO_3)_2 + Ca(OH)_2 \longrightarrow 2CaCO_3 \downarrow + 2H_2O$$

$$Mg(HCO_3)_2 + 2Ca(OH)_2 \longrightarrow 2CaCO_3 \downarrow + Mg(OH)_2 \downarrow + 2H_2O$$

If excess of lime is added, the temporary hardness converts into permanent hardness due to soluble $Ca(OH)_2$ or converting into calcium bicarbonate by absorbing CO_2 from the atmosphere.

$$Ca(OH)_2 \longrightarrow Ca^{2+} + 2OH^-$$

$$Ca(OH)_2 + 2CO_2 \longrightarrow Ca(HCO_3)_2$$

4.9.2 Removal of Permanent Hardness

Permanent hardness cannot be removed by boiling. It can be removed by the following methods.

1. Washing soda method: In this method water is treated with calculated amount of washing soda (Na_2CO_3) which react with soluble calcium and magnesium salts in hard water, converting them into insoluble carbonates.

$$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaCl$$

$$MgSO_4 + Na_2CO_3 \longrightarrow MgCO_3 + Na_2SO_4$$

It also removes the temporary hardness.

$$Ca(HCO_3)_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaHCO_3$$

$$Mg(HCO_3)_2 + Na_2CO_3 \longrightarrow MgCO_3 + 2NaHCO_3$$

2. Calgon method (sequestration): Calgon is the commercial name of sodium hexametaphosphate. When calgon is added to hard water it reacts with calcium and magnesium ions making them sequestred (rendered ineffective) in the form of soluble complexes.

$$\underset{\text{Calgon}}{Na_2[Na_4(PO_3)_6]} + 2Ca^{2+} \longrightarrow Na_2[Ca_2(PO_3)_6] + 4Na^+$$

$$\underset{\text{Calgon}}{Na_2[Na_4(PO_3)_6]} + 2Mg^{2+} \longrightarrow Na_2[Mg_2(PO_3)_6] + 4Na^+$$

The complexed calcium and magnesium ions do not form any precipitate with soap. Therefore, water can easily form lather with soap.

3. Permutit method: Permutit is an artificial zeolite, **hydrated sodium alumino orthosilicate** $Na_2Al_2Si_2O_8.XH_2O$. It can be prepared by fusing sodium carbonate, alumina and silica together. The product is washed with water to remove soluble impurities leaving behind a porous mass of permutit. The permutit is loosely packed in a big tank over a layer of coarse sand (Fig. 4.9). To remove the hardness of water, it is allowed to percolate through this tank. Calcium and magnesium ions which cause hardness in water are replaced by sodium ions which do not cause hardness. Thus, water is softened. The chemical reactions that take place during softening of water are as given below.

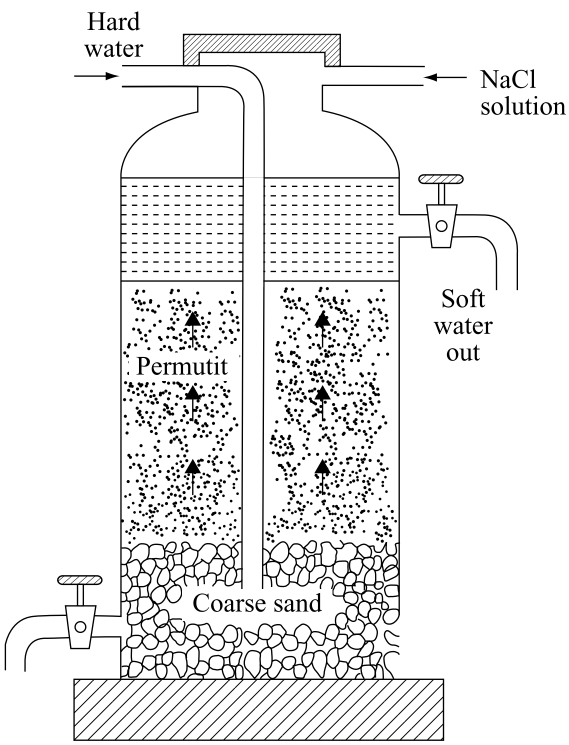


Fig 4.9 Permutit process for softening of hard water

$$Na_2Al_2Si_2O_8 \cdot xH_2O + Ca^{2+}_{(aq)} \longrightarrow CaAl_2Si_2O_8 \cdot xH_2O + 2Na^+_{(aq)}$$

$$Na_2Al_2Si_2O_8 \cdot xH_2O + Mg^{2+}_{(aq)} \longrightarrow MgAl_2Si_2O_8 \cdot xH_2O + 2Na^+_{(aq)}$$

As the process continues for sometime, the zeolite gets exhausted because the total sodium zeolite gets converted to calcium or magnesium zeolite. It may be regenerated by passing 10 per cent sodium chloride solution through it. The following reactions takes place.

$$CaAl_2Si_2O_8 \cdot xH_2O + 2Na^+ \longrightarrow Na_2\,Al_2\,Si_2O_8 \cdot xH_2O + Ca^{2+}$$

$$Mg\,Al_2Si_2O_8 \cdot xH_2O + 2Na^+ \longrightarrow Na_2Al_2Si_2O_8 \cdot xH_2O + Mg^{2+}$$

The regenerated sodium zeolite can be used again for softening of water.

4. Ion exchange resin method: Ion exchange resins are complex organic molecules having giant hydrocarbon framework. These ion exchange resins may contain **acidic or basic** groups. The ion exchange resins containing acidic groups are called **cation exchange resins** and the resins containing basic groups are called **anion exchange resins**. These are superior to zeolites because they can remove all types of cations as well as anions present in water. The resulting water is known as **demineralized or deionized water**.

The hard water is passed through a tank containing cation exchange resin. The calcium and magnesium ions and any other cations present in the water are replaced by H^+ ions from the resin.

$$2R - COOH + Ca^{2+} \longrightarrow (R - COO)_2Ca + 2H^+$$

$$2R - SO_3H + Mg^{2+} \longrightarrow (R - SO_3)_2Mg + 2H^+$$

The water coming out of the cation exchange resin is richer in H^+ ions and is acidic in nature. This water is then passed through another tank containing anion exchange resin which exchanges its OH^- ions with anions like Cl^-, SO_4^{2-}, etc. present in hard water.

$$R - NH_3OH + Cl^- \longrightarrow RNH_3Cl + OH^-$$

$$2R - NH_3OH + SO_4^{2-} \longrightarrow (RNH_3)_2SO_4 + 2OH^-$$

Thereafter, the H^+ ions (formed while coming out of cation exchange resin) combine with these OH^- ions to produce water and become neutral.

$$H^+ + OH^- \longrightarrow H_2O$$

Thus, water obtained by this method is free from all types of cations as well as anions and is called **deionized water**.

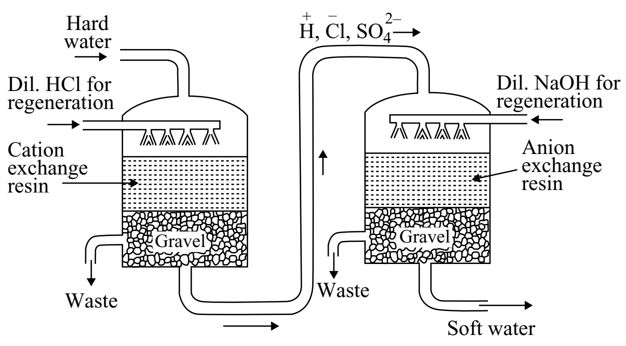


Fig 4.10 Ion exchange resin method for the removal of hardness of water

> Demineralized water is not useful for drinking purposes, because the drinking water should contain some minerals required to the body.

After use for sometime, the resins will be exhausted because all the H^+ ions in cation exchange resin are replaced by cations and all the OH^- ions in the anion exchange resin are replaced by anions. The exhausted cation exchange resin is regenerated by washing with moderately concentrated hydrochloric or sulphuric acid and the anion exchange resin is regnerated by washing with a moderately concentrated solution of sodium hydroxide

$$(R\,COO)_2Ca + 2HCl \longrightarrow 2RCOOH + CaCl_2$$

$$(RNH_3)_2SO_4 + 2NaOH \longrightarrow 2RNH_3OH + Na_2SO_4$$

4.9.3 Disadvantages of Hard Water

Water is chiefly used in industries for steam making, and its value for the purpose depending primarily on the amount and the chemical character of the mineral matter dissolved and suspended in it. The troubles in the boiler room practice caused by the mineral constituents matter are (1) formation of scale, (2) corrosion and (3) foaming.

When water is boiled in a boiler, while the water converts into steam, the mineral matter present in water deposits on the walls of the boiler. This is called scale formation. Due to these scales, fuel consumption increase. This is because they are poor conductors of heat and they also increase cost of boiler repairs.

The iron metal, used for making boilers, corrodes due to the action of hydrogen sulphide (present in ground waters) and the free acids. The constant corrosion of the boiler causes it to leak and makes it unfit for use.

Besides these disadvantages hard water also causes wastage of soap during washing purposes.

4.9.4 Measurement of Hardness of Water

The quantity of the calcium and magnesium salts present in a certain volume or weight of water measures the extent of hardness or degree of hardness. The results of the analysis of hard water are usually expressed in parts per million (ppm), the substances producing hardness being calculated as calcium carbonate.

> The degree of hardness of water is defined as the number of parts by weight of calcium carbonate (equivalent to various calcium and magnesium salts) present in a million parts (i.e., ppm) by weight of water)

Equivalents are given below

$$\begin{aligned} 100\text{g of } CaCO_3 &\equiv 162\text{g of } Ca(HCO_3)_2 \\ &\equiv 111 \text{ g of } CaCl_2 \equiv 136\text{g of } CaSO_4 \\ &\equiv 146\text{ g of } Mg\ (HCO_3)_2 \equiv 95\text{g of } MgCl_2 \\ &\equiv 120\text{ g of } MgSO_4 \end{aligned}$$

Illustrated Examples

4.1 A sample of hard water contains 12g of magnesium sulphate and 11.1 g of calcium chloride in 100kg water. What is the total hardness of water?

$$\begin{aligned} 12\text{g. of } MgSO_4 &\equiv 10\text{g of } CaCO_3 (\because 100CaCO_3 \\ &\equiv 120 \text{ of } MgSO_4) \\ 11.1\text{g. of } CaCl_2 &\equiv 10\text{g of } CaCO_3 (\because 100CaCO_3 \\ &\equiv 111 \text{ of } CaCl_2) \end{aligned}$$

$\therefore$ Total amount of $MgSO_4$ and $CaCl_2$ equivalent to $CaCO_3 = 20$g

20g of $CaCO_3$ is present in 100 kg water.

1kg of hard water contains 0.2g or 200 mg of $CaCO_3$.

$\therefore$ Total hardness of water is 200 ppm.

4.2 10 ml of standard soap solution (1ml = 0.001 g of $CaCO_3$) was required in titrating 50 ml of a sample of hard water to produce good lather. Calculate the degree of hardness.

Solution: 10 ml of soap solution = 0.01 g of $CaCO_3$ ($\therefore$ 1ml = 0.001g of $CaCO_3$)

50 ml of water = 50 g of water (taking density of water as 1g /cc)

50 g of hard water contains 0.01g of $CaCO_3$

1 kg of hard water contains 0.2g of $CaCO_3$

$\therefore$ Hardness of water is 200 ppm.

4.10 HEAVY WATER

Normal water is mainly protium oxide, H_2O. However, if protium atoms in normal water molecule are replaced completely by deuterium atoms, the resulting water is called **heavy water**.

4.10.1 Occurrence

(i) It is found on the leaves of weeping willows and banyan trees.

(ii) It is found to be present in the melt when large masses of ice are slowly evaporating as in the Himalayas.

(iii) **Washburn** and **Urey** discovered that water from some commercial electrolytic cells which had been working for a number of years that contain a high concentration of the heavy water.

4.10.2 Preparation of Heavy Water

1. Prolonged electrolysis of water: On electrolysis of water, protium is liberated about six times more readily than deuterium due to the following reasons.

(i) Hydrogen ion has a greater mobility than deuterium ion.

(ii) Hydrogen ions react more readily with the electron at the cathode to form atoms of hydrogen.

(iii) Atomic hydrogen combines more readily to form molecules.

(iv) O – D bond is stronger than O – H bond.

Therefore, as the process of electrolysis progressess, the deuterium content of the residual sample increases and by repeated electrolysis, water enriched with D_2O may be obtained.

The electrolytic cell is a steel vessel which itself acts as a cathode. The anode is a perforated cylindrical sheet of nickel. The gases obtained in each state are separately burnt and the water thus formed is returned to the previous stage. The concentration of heavy water gradually increases in the residual water usually and the whole process is completed in seven stages.

The electrolysis is carried out by taking 0.5M of NaOH solution as an electrolyte in the steel electrolytic cell. In the first stage, electrolysis is carried until the volume of electrolyte decreases to 1/6 of its original volume. Now, carbon dioxide gas is passed into the resulting solution to partly neutralize the alkali. It is then distilled and taken into another cell for the second stage of electroysis and the process is carried out upto seven stages. After the end of seven stages, 99 per cent D_2O is obtained.

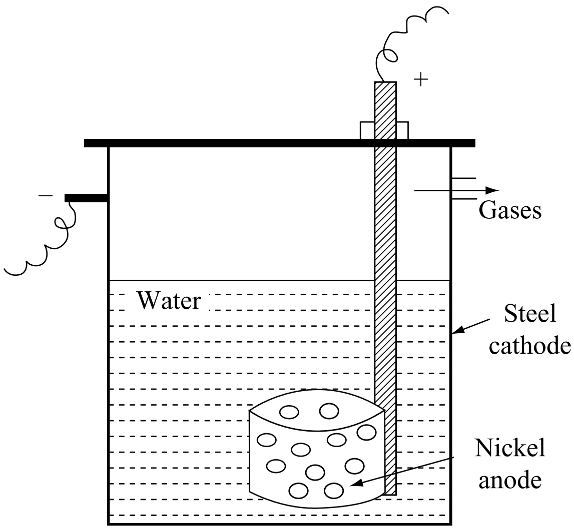


Fig 4.11 Electrolytic cell for preparation of heavy water

Table 4.5 Data of the products in different stages of electrolysis of water

State of electrolysis	Volume of electrolysed	Density of product	Per cent of D_2O in the product
1	2310 *l*	0.998	–
2	340 *l*	0.999	0.5
3	52 *l*	1.001	2.5
4	10.15 *l*	1.007	8.0
5	2 *l*	1.031	30.0
6	420 m*l*	1.098	93.0
7	82 m*l*	1.104	99.0

On account of increasing concentration of heavy water from the third stage, deuterium content of the gases also increase. These gases are burnt and the resulting mixture of H_2O and D_2O is added to the appropriate cell and subjected to electrolysis. Data of the stages are given in Table 4.5.

2. By Fractional Distillation of water: It can also be prepared by fractional distillation in about 13 meter long fractionating column at reduced pressure. The lighter fraction (H_2O) distills over first leaving behind the residue richer in heavy water. This method is being used for the production of heavy water in USA and NewZealand.

3. By exchange method: This is a new method developed recently. H_2S gas is passed through hot water where hydrogen atoms exchange with deuterium atoms from D_2O in the water, there by enriching the gas in D_2S (Fig. 4.12). On passing this enriched gas through cold water, the deuterium from the D_2S and the hydrogen from H_2O is again exchanged so the cold water becoming richer with D_2O.

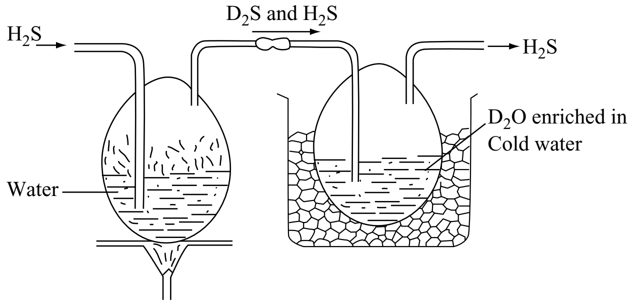


Fig 4.12 Preparation of heavy water by exchange reaction method

The process can be repeated and the water continuously enriched in D_2O.

4.10.3 Physical Properties of Heavy Water

(i) It is a colourless, odourless and tasteless liquid.
(ii) It is actually heavier than ordinary water. Its density is 1.1, i.e., its density is about 10 per cent higher than that of ordinary water, however, its maximum density is found to occur at 11.6°C instead of 4°C for ordinary water.
(ii) There are slight differences in the physical constants of heavy water as compared to ordinary water (Table 4.3).

Most of the differences in the physical constants (as listed in Table 4.3) stem from differences in mass and therefore, are fairly apparent.

(i) Higher viscosity of D_2O points to greater association in D_2O than in H_2O.
(ii) Conductances of strong electrolytes is less in D_2O because of the greater viscosity of the medium.
(iii) Ionic mobilites are less in D_2O than in H_2O due to greater viscosity of heavy water.
(iv) Ionic compounds are less soluble in heavy water as compared to ordinary water because D_2O has a lower dielectric constant.

4.10.4 Chemical Properties of Heavy Water

Chemical properties of heavy water are quite similar to those of ordinary water, however, reactions with heavy water proceed slowly as compared to those with ordinary water. A few important reactions are given below for comparing the behaviour of heavy water with ordinary water.

(i) Electrolysis: Electrolysis of acidified or alkaline solution of heavy water liberates deuterium at the cathode and oxygen at the anode.

$$\underset{}{2D_2O} \longrightarrow \underset{\text{Cathode}}{2D_2} + \underset{\text{Anode}}{O_2}$$

(ii) Reaction with metals: D_2O reacts with alkali and alkaline earth metals liberating deuterium

$$2Na + 2D_2O \longrightarrow D_2 + 2NaOD \quad \text{Sodium deuteroxide}$$

$$Ca + 2D_2O \longrightarrow D_2 + Ca(OD)_2 \quad \text{Calcium deuteroxide}$$

(iii) Reaction with metal oxides: When metal oxides are dissolved in heavy water, metal deuteroxides will be formed.

$$Na_2O + D_2O \longrightarrow 2NaOD$$

$$CaO + D_2O \longrightarrow Ca(OD)_2$$

(iv) Reaction with non-metal oxides: When non-metal oxides are dissolved in heavy water, deuteroacids or heavy acids will be formed.

$$P_2O_5 + 3D_2O \longrightarrow 2D_3PO_4 \text{ (Deuterophosphoric acid)}$$

$$SO_3 + D_2O \longrightarrow D_2SO_4 \text{ (Deuterosulphuric acid)}$$

$$N_2O_5 + D_2O \longrightarrow 2DNO_3 \text{ (Deuteronitric acid)}$$

(v) Reaction with metallic salts: Heavy water decomposes metal nitrides, phosphides, arsenides and carbides to from deuterocompounds.

$$Mg_3N_2 + 6D_2O \longrightarrow 3Mg(OD)_2 + 2ND_3 \text{ (Deuteroammonia)}$$

$$Ca_3P_2 + 6D_2O \longrightarrow 3Ca(OD)_2 + 2PD_3 \text{ (Deuterophosphine)}$$

$$Na_3As_2 + 6D_2O \longrightarrow 3NaOD + AsD_3 \text{ (Deuteroarsine)}$$

$$Al_4C_3 + 12D_2O \longrightarrow 4Al(OD)_3 + 3CD_4 \text{ (Deuteromethane)}$$

$$CaC_2 + 2D_2O \longrightarrow Ca(OD)_2 + C_2D_2 \text{ (Deuteroacetylene)}$$

The melting and boiling points of deuterocompounds are generally higher than the analogous hydrogen compounds. For example, the m.pt and b.pts of NH_3 are 195.2K and 239.8K, respectively while those of ND_3 are 199.5K and 242K, respectively.

(vi) Deuterolysis: Just as water hydrolyses certain inorganic salts, heavy water also reacts with certain salts. This interaction is known as deuterolysis.

$$AlCl_3 + 3D_2O \longrightarrow \underset{\substack{\text{Aluminium}\\\text{deuteroxide}}}{Al(OD)_3} + \underset{\substack{\text{Deuterium}\\\text{chloride}}}{3DCl}$$

(vii) Exchange reactions: A variety of hydrogen compounds undergo exchange reactions when treated with heavy water. Exchange reactions occur with great readiness if the bonds to the hydrogen are relatively easily broken, in particular if the bonds are ionic. Thus, hydrogen of HC*l* readily exchanges with deuterium atoms of D_2O to form DC*l*.

$$D_2O + HCl \longrightarrow DCl + HOD$$

Similarly, exchange reactions occurs if NH_4Cl is dissolved in D_2O presumably in a stepwise fashion.

$$NH_4Cl + D_2O \rightleftharpoons NH_3DCl + HOD$$

$$NH_3DCl + D_2O \rightleftharpoons NH_2D_2Cl + HOD$$

$$NH_2D_2Cl + D_2O \rightleftharpoons NHD_3Cl + HOD$$

$$NHD_3Cl + D_2O \rightleftharpoons ND_4Cl + HOD$$

Similarly when NaOH is dissolved in D_2O, there is a rapid ionic reaction leading to the formation of sodium deuteroxide.

$$NaOH + D_2O \rightleftharpoons NaOD + HOD$$

The hydrogen atoms of ammonia exchange with heavy water very slowly and those of methane do not exchange at all. However, deuteration of organic compounds (which do not yield hydrogen ions) can be catalyzed by the presence of hydrogenation catalysts such as nickel, palladium or platinum. For example, benzene when treated with deuterium oxide using nickel as a catalyst at a temperature of 200°C leads to the formation of pure hexadeuterobenzene.

$$C_6H_6 + 3D_2O \rightleftharpoons C_6D_6 + 3H_2O$$

(viii) Formation of Deuterates: Similar to hydrates, D_2O can associate with many compounds to form deuterates by crystallizing the solutions of salts in heavy water. Examples are $CuSO_4 \cdot 5D_2O; CoCl_2 \cdot 6D_2O; Na_2SO_4 \cdot 10D_2O$. However, the colour of deuterates of transition elements are lighter than those of hydrates. Also, deuterates have low vapour pressures than hydrates.

(ix) Physiological effects: D_2O appears to have germicidal effects. Tobacco seeds do not germinate in water containing high concentration of D_2O. Pure heavy water kills protozoa, tadpoles, small fish and mice when fed upon it. It retards growth of bacteria and, therefore, fermentation process are slower in D_2O than in ordinary water. It is found that while high concentrations are poisonous, very dilute solutions of D_2O have toxic effect stimulating vegetable growth.

4.10.5 Uses of Heavy Water

(i) As moderator for neutrons: Neutrons produced in the fission process are of extremely high energy and heavy water serves as a moderator to slow down the neutrons so that they may be readily captured by other fissionable nuclei and thereby promote the chain reaction.

(ii) For the preparation of deuterium: Deuterium required for making hydrogen bomb is produced either by the electrolysis of heavy water or by the action of sodium metal on heavy water.

(iii) As a tracer compound: D_2O is commonly used as a tracer for studying the reaction mechanisms in organic chemistry. For example, deuteration studies have helped to elucidate the mechanism of electrophillic aromatic substitution reactions.

D_2O is also very useful as a tracer in the study of metabolic processes.

(iv) The exchange reactions of D_2O are used to study the structures of certain oxyacids of phosphorous. E.g., H_3PO_2 is shown to be a monobasic acid while H_3PO_3 is a dibasic acid.

4.11 HYDROGEN PEROXIDE

Hydrogen peroxide was discovered by a French chemist J.L. Thenard in 1818. It is formed in the nature, in trace amounts, by the action of sunlight on water containing dissolved oxygen.

4.11.1 Preparation of Hydrogen Peroxide

(a) Laboratory Methods

1. Merck's process: To an ice cold 20 per cent solution of sulphuric acid, calculated amount of sodium peroxide is added in small amounts slowly with constant stirring.

$$Na_2O_2 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O_2$$

The sodium sulphate is removed by cooling when crystals of $Na_2SO_4 \cdot 10H_2O$ separate out. The product is always contaminated with sodium sulphate. A 30 per cent solution of H_2O_2 will be obtained.

2. From Barium peroxide: Barium peroxide is made into a thin paste (with ice-cold water) and added slowly to ice-cold dilute sulphuric acid.

$$BaO_2 + H_2SO_4 \longrightarrow BaSO_{4(s)} + H_2O_{2(l)}$$

Barium sulphate is allowed to settle down as a residue and is filtered off. The filtrate contain a dilute solution of H_2O_2. Anhydrous barium peroxide is not decomposed by dilute sulphuric acid on account of the formation of a layer of insoluble barium sulphate on the granules of peroxide. But the hydrated $BaO_2 \cdot 8H_2O$ is easily decomposed.

The hydrogen peroxide thus obtained may contain some heavy Ba^{2+} ions. These catalyze the decomposition of hydrogen peroxide. So, if phosphoric acid is used in the place of sulphuric acid complete precipitation of Ba^{2+} ions as barium phosphate takes place.

$$3BaO_2 \cdot 8H_2O + 2H_3PO_4 \longrightarrow Ba_3(PO_4)_2 + 24H_2O + 3H_2O_2$$

Alternatively a slow stream of carbon dioxide gas may be passed through the hydrated barium peroxide paste with ice-cold water. Then, H_2O_2 will be formed with precipitation of barium carbonate.

$$BaO_{2(s)} + CO_{2(g)} + H_2O_{(l)} \longrightarrow BaCO_{3(s)} + H_2O_{2(l)}$$

(b) Manufacture of Hydrogen peroxide: On a commercial scale, hydrogen peroxide can be prepared by the following methods.

1. Electrolytic process: In this method, hydrogen peroxide is manufactured by the electrolysis of 50 per cent sulphuric acid solution. The initial product in the course of electrolysis is peroxydisulphuric acid, formed by the electrolytic oxidation of sulphuric acid at anode.

$$2H_2SO_{4(aq)} \longrightarrow 2HSO^-_{4(aq)} + 2H^+_{(aq)}$$

$$2H_2SO_4^-(aq) \longrightarrow H_2S_2O_{8(aq)} + 2e^- \text{ at anode}$$

$$2H^+_{(aq)} + 2e^- \longrightarrow H_{2(g)} \text{ at cathode}$$

Peroxydisulphuric acid is taken out of the electrolytic cell and hydrolyzed with water to give hydrogen peroxide.

$$H_2S_2O_8 + 2H_2O \longrightarrow 2H_2SO_4 + H_2O_2$$

In the recent method equimolar mixture of sulphuric acid and ammonium sulphate is used in the place of 50 per cent sulphuric acid as an electrolyte for electrolysis.

$$(NH_4)_2SO_4 + H_2SO_4 \longrightarrow 2NH_4HSO_4$$

$$NH_4HSO_4 \rightleftharpoons NH_4SO_4^- + H^+$$

$$2NH_4SO_4^- \longrightarrow (NH_4)_2S_2O_8 + 2e^- \text{ at anode}$$

$$2H^+ + 2e^- \longrightarrow H_2 \text{ at cathode}$$

The electrolytic cell used in this method is shown in Fig. 4.13.

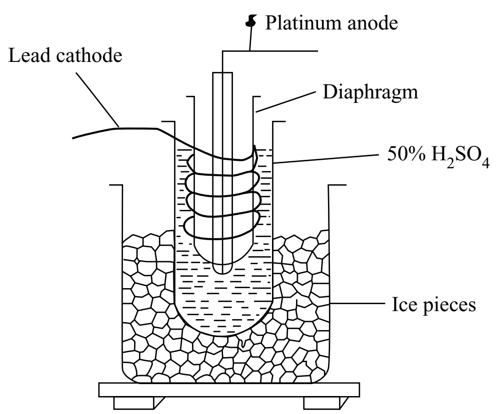
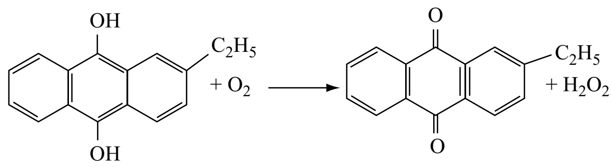


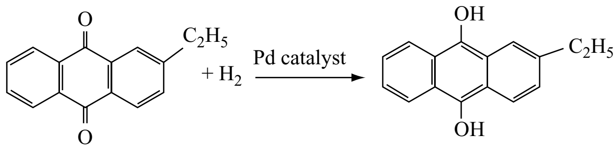
Fig 4.13 Manufacture of hydrogen peroxide by electrolytic process

2. Autooxidation process: Industrially, hydrogen peroxide is manufactured by the autooxidation of an organic compound 2-ethylanthraquinol. This is a cyclic process.

First 2-ethylanthraquinol is oxidized by air after dissolving in a mixture of benzene and cyclohexanol. Then, 2-ethylanthraquinone and hydrogen peroxide will be formed.



The 2-ethylanthraquinone can be used again. It is dissolved in benzene and hydrogen gas is passed through the solution in the presence of palladium catalyst. Then, it is reduced to 2-ethylanthraquinol.



In this process only hydrogen and atmospheric oxygen are used up to form hydrogen peroxide.

2-ethylanthraquinone is not consumed. So, this process is very cheap.

4.11.2 Concentration of Hydrogen Peroxide

The hydrogen peroxide obtained (about 1 per cent) should be concentrated stepwise carefully because H_2O_2 decomposes easily on heating. Rapid and uncontrolled decomposition of concentrated solution of hydrogen peroxide results in a dangerous explosion. The concentration of hydrogen peroxide is carried out in the following steps.

(i) Evaporation on a water bath: The dilute solution of H_2O_2 is carefully evaporated on a water bath preferably under reduced pressure using fractionating column. In this process about 30 per cent solution of H_2O_2 is obtained.

(ii) Dehydration in a vaccum dessicator: The above solution of H_2O_2 is taken in a vacuum dessicator containing concentrated sulphuric acid. The sulphuric acid absorbs water vapours. Thus about 90 per cent concentrated solution of H_2O_2 will be obtained.

(iii) Distillation under reduced pressure: The 90 per cent solution of H_2O_2 is subjected to distillation under reduced pressure (10–15 mm). Then water distills over 303–313K and 99 per cent pure hydrogen peroxide will be formed.

(iv) Fractional crystallization: Then the 99 per cent solution of H_2O_2 is cooled in a freezing mixture of solid carbon dioxide and ether. As a result of this, needle-like crystals of 100 per cent hydrogen peroxide separate out.

4.11.3 Storage of Hydrogen Peroxide

Hydrogen peroxide solutions are stored in plastic or wax coated glass bottles, because the rough glass surface and silica of glass accelerates the decomposition of H_2O_2. A small amount of phosphoric acid, glycerol, acetanilide, urea, sodium stannate or sodium phosphate is generally added which retard the decomposition of hydrogen peroxide. These are called **negative catalysts.** Solutions of H_2O_2 are fairly stable in acid medium or if alcohol or ether is present. Alkaline solutions are relatively unstable. Impurities like silica, alumina, manganese dioxide, metal powders like iron, manganese, etc., cause rapid decomposition. So, these are called positive catalysts for the decomposition of H_2O_2.

4.11.4 Strength of Hydrogen Peroxide

The strength of hydrogen peroxide may be expressed in different methods as given below.

(i) Molarity: It is the no. of moles of H_2O_2 per litre.

(ii) Normality: It is the no. of gram equivalents per litre.

(iii) Per cent by weight: It is the no. of grams of H_2O_2 present in 100 g of solution.

(iv) Volume strength: It is the no. of volumes of oxygen gas liberated at STP by 1 vol. of H_2O_2 solution. For example, 10 volume H_2O_2 solution means that 1 litre of this solution will give 10 litres of oxygen gas at STP. Hydrogen peroxide decomposes as per the equation.

$$2H_2O_{2_{(l)}} \longrightarrow 2H_2O_{(l)} + O_{2(g)}$$

From this equation it can be calculated that 2 moles of H_2O_2, i.e., 68gm (2×34) of H_2O_2 gives 1 mole of O_2 which occupies 22.4 *l* at STP. If only 34g H_2O_2 is present in one litre solution of H_2O_2, we can easily show that

1M = 2N = 3.4 % w/v = 11.2vol. (or) 0.893M = 1.786 N = 3.036 % w/v = 10 vol.

Commercial hydrogen peroxide sold in the market is called **perhydrol.** It is of about 30 per cent w/v or 100 vol. hydrogen peroxide.

4.11.5 Physical Properties of Hydrogen Peroxide

(i) Pure anhydrous hydrogen peroxide is a thick syrupy liquid. In thick layers, it has pale blue colour.

Table 4.6 Physical characteristics of H_2O_2

Physical property	Value
Melting point °C	–0.41
Boiling point °C (extrap)	150.2
Vapour pressure (25°C) / mm. Hg	1.9
Density of solid at –4.5°C /g cm^{-3}	1.6434
Density of liquid at 25°C /g cm^{-3}	1.4425
Viscosity (20°C) / centipoise	1.245
Dielectric constant at 25°C	70.7
Electrical conductivity at 25°C / Ω^{-1} cm^{-1}	5.1×10^{-8}
ΔH_f° / kJ mol^{-1}	–187.6
ΔG_f° / kJ mol^{-1}	–118.0
Dipolemoment	2.1D

(ii) It is more viscous, less volatile and denser than water. This indicates that H_2O_2 is more hydrogen bonded than water.

(iii) H_2O_2 decomposes easily on heating even before its boiling point at atmospheric pressure.
So, the estimated boiling point of H_2O_2 is about 152°C and melting point is 0.4°C. It boils at 85°C under a pressure of 68 mm. It forms prismatic crystals at –2°C.

(iv) It is completely miscible with water, alcohol and ether in all proportions.

(v) It is a very weak acidic substance; however, anhydrous H_2O_2 does not turn blue litmus to red.

Its important physical properties are given in Table 4.6.

4.11.6 Chemical Properties of Hydrogen Peroxide

Chemical properties of hydrogen peroxide can be classified as

1. Acidic character
2. Oxidation properties
3. Reduction properties
4. Addition reactions
5. Bleaching property

1. Acidic character: Pure hydrogen peroxide is a weak acid. It is neutral towards litmus. It can dissociate as

$$H_2O_2 + H_2O \longrightarrow H_3O^+ + HO_2^-$$

$$HO_2^- + H_2O \longrightarrow H_3O^+ + O_2^{2-}$$

So, it can form two types of salts called hydroperoxides and peroxides.

$$NaOH + H_2O_2 \longrightarrow NaHO_2 + H_2O$$

$$2NaOH + H_2O_2 \longrightarrow Na_2O_2 + 2H_2O$$

The metal peroxides like Na_2O_2 and BaO_2 are considered as the salts of H_2O_2.

$$Ba(OH)_2 + H_2O_2 \longrightarrow BaO_2 + 2H_2O$$

Hydrogen peroxide liberates CO_2 from Na_2CO_3

$$Na_2CO_3 + H_2O_2 \longrightarrow Na_2O_2 + H_2O + CO_2$$

2. Oxidation properties: Hydrogen peroxide has an interesting chemistry because of its ability to function as an **oxidizing agent** as well as **reducing agent** in both acid and alkaline media. The oxidation state of oxygen in H_2O_2 is –1, which is intermediate between the values for O_2 and H_2O. From the reduction potentials given below aqueous solutions of H_2O_2 should spontaneously disproportionate.

H_2O_2 can be oxidized to O_2 in which the oxidation state of oxygen increases from – 1 to 0. In this process, H_2O_2 acts as a reducing agent. H_2O_2 can be reduced to H_2O or OH^- in which the oxidation state of oxygen changes from –1 to –2. In this process, H_2O_2 acts as an oxidizing agent.

The standard reduction potentials while acting as oxidizing agent and reducing agent in both acid as well as alkaline media are given below.

As an oxidant:

$$\text{Acid medium } H_2O_2 + 2H^+ + 2e^- \longrightarrow 2H_2O \; E^\circ = +1.776V$$

$$\text{Alkaline medium } H_2O_2 + 2e^- \longrightarrow 2OH^- \; E^\circ = +0.878V$$

As a reductant:

$$\text{Acid medium } H_2O_2 \longrightarrow 2H^+ + O_2 + 2e^- \; E^\circ = -0.695V$$

$$\text{Alkaline medium } H_2O_2 + 2OH^- \longrightarrow 2H_2O + O_2 + 2e^- \; E^\circ = +0.076V$$

From the information given above it is known that H_2O_2 is a very powerful oxidizing agent and a poor reducing agent.

It will be noted that O_2 is always evolved when H_2O_2 acts as a reducing agent. In such reactions, experiments using ^{18}O in H_2O_2 show negligible exchange between H_2O_2 and H_2O and all the O_2 formed when H_2O_2 is used as reducing agent comes from the H_2O_2 implying that oxidizing agents do not break the O–O bond but simply remove electrons.

Typical reactions in which hydrogen peroxide functions as an oxidizing agent are given below.

2a Oxidation Properties in Acid Medium

(i) H_2O_2 oxidizes ferrous sulphate to ferric sulphate.

$$2FeSO_4 + H_2SO + H_2O_2 \longrightarrow Fe_2(SO_4)_3 + 2H_2O$$

(or)

$$2Fe^{2+} + H_2O_2 + 2H^+ \longrightarrow 2Fe^{3+} + 2H_2O$$

(ii) H_2O_2 oxidizes acidified potassium ferrocyanide to potassium ferricyanide.

$$2K_4[Fe(CN)_6] + H_2SO_4 + H_2O_2 \longrightarrow 2K_3[Fe(CN)_6] + K_2SO_4 + H_2O$$

(or)

$$2[Fe(CN)_6]^{4-} + H_2O_2 + 2H^+ \longrightarrow 2[Fe(CN)_6]^{3-} + 2H_2O$$

(iii) H_2O_2 liberates iodine from potassium iodide solution.

$$2KI + H_2SO_4 + H_2O_2 \longrightarrow K_2SO_4 + 2H_2O + I_2$$

(or)

$$2I^- + H_2O_2 + 2H^+ \longrightarrow 2H_2O + I_2$$

(iv) H_2O_2 oxidizes nitrite to nitrate, sulphite to sulphate and arsenite to arsenate.

$$KNO_2 + H_2O_2 \longrightarrow KNO_3 + H_2O$$

$$Na_2SO_3 + H_2O_2 \longrightarrow Na_2SO_4 + H_2O$$

$$Na_3AsO_3 + H_2O_2 \longrightarrow Na_3AsO_4 + H_2O$$

(v) H_2O_2 oxidizes sulphides to sulphates.

$$PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$$

(or)

$$S^{2-} + 4H_2O_2 \longrightarrow SO_4^{2-} + 4H_2O$$

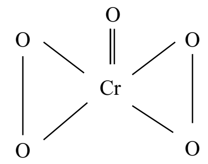
This property of H_2O_2 is useful in restoring the original white colour of the old lead paintings. The traces of hydrogen sulphide in the atmosphere blackens the lead paintings due to the formation of black lead sulphide. To restore the colour of lead paints, it is dipped in aqueous solution of H_2O_2. Then, black lead sulphide is oxidized to white lead sulphate.

(vi) A solution of chromic acid in sulphuric acid (i.e., acidified potassium dichromate solution) is oxidized to blue peroxide of chromium, CrO_5 which decomposes to give chromic salts and oxygen in acid medium. But the blue chromium peroxide is stable in organic layer like ether containing little pyridine to produce alkalinty. So in a test tube first potassium dichromate and sulphuric acid are taken, and to this mixture ether containing pyridine is added which forms a separate layer. Then if H_2O_2 is added, the chromium peroxide formed comes into organic layer and gets blue colour.

$$CrO^{2-}_{4(aq)} + 2H^+ + 2H_2O_{2(aq)} \longrightarrow \underset{\text{Blue chromium peroxide}}{CrO_5} + 3H_2O_{(l)}$$

$$4CrO_5 + 12H^+_{(aq)} \longrightarrow 4Cr^{+3}_{(aq)} + 6H_2O_{(l)} + 7O_{2(g)}$$

The chromium peroxide has a structure similar to a butterfly.



The CrO_5 is written as $CrO\,(O_2)_2$. It contains chromium in +VI oxidation state. It has 4 oxygen atoms involved in peroxy linkages. In ethereal solution containing pyridine and chromium peroxide, it forms Py (CrO_5).

2b Oxidation Reactions in Alkaline Medium

(i) In an alkaline medium, H_2O_2 oxidizes manganese sulphate to manganese dioxide.

$$MnSO_4 + H_2O_2 + 2NaOH \longrightarrow MnO_2 + Na_2SO_4 + 2H_2O$$

(or)

$$Mn^{2+} + H_2O_2 + 2OH^- \longrightarrow MnO_2 + 2H_2O$$

$$Mn^{2+} + H_2O_2 \longrightarrow Mn^{4+} + 2OH^-$$

(ii) In an alkaline medium, H_2O_2 oxidizes chromium (III) salts to chromates.

$$Cr_2(SO_4)_3 + 3H_2O_2 + 10NaOH \longrightarrow 2Na_2CrO_4 + 3Na_2SO_4 + 8H_2O$$

(or)

$$2Cr^{3+} + 3H_2O_2 + 10OH^- \longrightarrow 2CrO_4^{2-} + 8H_2O$$

(iii) H_2O_2 oxidizes formaldehyde to formic acid in alkaline medium in the presence of pyrogallol (1, 2, 3-trihydroxy benzene). In this reaction, H_2O_2 is reduced to H_2 (different from other oxidation reactions).

$$2HCHO + H_2O_2 + 2NaOH \longrightarrow 2HCOONa + 2H_2O + H_2$$

(iv) H_2O_2 oxidises benzene in the presence of ferrous sulphate to phenol.

$$C_6H_6 + H_2O_2 \xrightarrow{FeSO_4} C_6H_5OH$$

3a Reduction Reactions in Acid Medium

(i) H_2O_2 reduces acidified potassium permanganate solution to colourless manganous sulphate, due to which the pink colour of permanganate is discharged.

$$2KMnO_4 + 3H_2SO_4 + 5H_2O_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5O_2$$

(or)

$$2MnO_4^{2-} + 6H^+ + 5H_2O_2 \longrightarrow 2Mn^{2+} + 8H_2O + 5O_2$$

(ii) H_2O_2 reduces acidified potassium dichromate solution. As a result the orange red colour of $K_2Cr_2O_7$ changes to green due to the formation of chromium salt.

$$K_2Cr_2O_7 + 4H_2SO_4 + 3H_2O_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3O_2$$

(or)

$$Cr_2 + 8H^+ + 3H_2O_2 \longrightarrow 2Cr^{3+} + 7H_2O + 3O_2$$

(iii) H_2O_2 reduces manganese dioxide to manganous salt in acid medium.

$$MnO_2 + H_2SO_4 + H_2O_2 \longrightarrow MnSO_4 + 2H_2O + O_2$$
$$Mn^{4+} + H_2O_2 \longrightarrow Mn^{2+} + 2H^+ + O_2$$

(iv) H_2O_2 reduces ozone to oxygen.

$$H_2O_2 + O_3 \longrightarrow H_2O + 2O_2$$

(v) It reduces hypohalous acid to halide ion in acid medium.

$$H_2O_2 + HOCl \longrightarrow H_3O^+ + Cl^- + O_2$$

3b Reduction Reactions in Alkaline Medium

(i) H_2O_2 reduces potassium permanganate to manganese dioxide in alkaline medium.

$$2KMnO_{4(aq)} + 3H_2O_{2(aq)} \longrightarrow 2MnO_{2(s)} + 2H_2O_{(l)} + 2KOH_{(aq)} + 3O_{2(g)}$$

(ii) H_2O_2 reduces alkaline potassium ferricyanide to potassium ferrocyanide.

$$2K_3[Fe(CN)_6] + 2KOH + H_2O_2 \longrightarrow 2K_4[Fe(CN)_6] + H_2O_2 + O_2$$

(or)

$$2[Fe(CN)_6]^{3-} + 2OH^- + H_2O_2 \longrightarrow 2[Fe(CN)_6]^{4-} + 2H_2O + O_2$$

(iii) The oxides of metals are reduced to metals. For example, silver oxide is reduced to silver.

$$Ag_2O + H_2O_2 \longrightarrow 2Ag + H_2O + O_2$$

(iv) Lead dioxide is reduced to litharge (lead monoxide).

$$PbO_2 + H_2O_2 \longrightarrow PbO + H_2O + O_2$$

(v) H_2O_2 reduces halogens to halide ions.

$$Cl_2 + H_2O_2 + 2OH^- \longrightarrow 2Cl^- + 2H_2O + O_2$$

Because of this reaction hydrogen peroxide is used as an antichlor to remove the chlorine after bleaching.

(vi) It reduces hypohalites to halides.

$$NaOBr + H_2O_2 \longrightarrow NaBr + H_2O + O_2$$
$$CaOCl_2 + H_2O_2 \longrightarrow CaCl_2 + H_2O + O_2$$

4. **Addition reactions:** Hydrogen peroxide adds to alkenes to form glycols.

$$\underset{\text{Ethylene}}{CH_2 = CH_2} + H_2O \longrightarrow \underset{\text{Ethylene glycol}}{\begin{array}{l} CH_2OH \\ | \\ CH_2OH \end{array}}$$

Hydrogen peroxide also forms addition compounds with some inorganic salts.

Examples: $Na_2HPO_4 \cdot H_2O_2$; $(NH_4)_2SO_4 \cdot H_2O_2$; $H_2N\ CONH_2 \cdot H_2O_2$

The addition compound with urea is known as **hyperol.**

These addition compounds release hydrogen peroxide on the addtion of water.

5. **Bleaching property:** Hydrogen peroxide acts as a bleaching agent due to the release of nascent oxygen.

$$H_2O_2 \longrightarrow H_2O + O$$

Thus, the bleaching action of hydrogen peroxide is permanent and is due to oxidation. It oxidizes the colouring matter to a colourless product.

$$\text{colouring matter} + O \longrightarrow \text{colourless matter}$$

Hydrogen peroxide is used to bleach delicate materials like ivory, wool, feathers, etc. Ammonical solution of hydrogen peroxide bleaches the human hair to golden yellow colour.

4.11.7 Uses of Hydrogen Peroxide

H_2O_2 is widely used in various fields. This led to tremendous increase in the large-scale production of H_2O_2. Some of the uses are listed below.

(i) It is used as an antiseptic for washing the wounds, teeth and ears. As an antiseptic, it is sold as **perhydrol.**

(ii) It is used as a bleaching agent for delicate textiles, wool, ivory, silk and wood pulp.

(iii) In our daily life, H_2O_2 is used as hair bleach and as a mild disinfectant.

(iv) Highly concentrated solution of H_2O_2 is used as an oxidant for rocket fuel.

(v) Hydrogen peroxide solution is used to restore the colour of oil paintings which contain lead paints.

(vi) Recently, H_2O_2 is used in environmental chemistry such as in pollution control, e.g., treatment of domestic and industrial wastes, restoration of aerobic conditions to sewage wastes, oxidation of cyanides, etc.

(vii) It is used for preserving milk, wine and other liquors.

(viii) It is used as an antichlor in bleaching.

(ix) It is used in the manufacture of several compounds such as sodium perborates, percarbonates which are important constituents of high-quality detergents.

It is also used in the production of epoxides, propylene and polyurethanes.

(x) It is used as an oxidizing agent in the laboratory.

4.11.8 Detection of Hydrogen Peroxide

(i) Hydrogen peroxide turns the starch-iodide paper to blue. This is due to the liberated iodine from potassium iodide by H_2O_2.

(ii) When H_2O_2 is shaken with potassium dichromate solution in sulphuric acid and ether, an intense blue colour appears in ethereal layer.

(iii) Hydrogen peroxide gives orange colour with a solution of titanium dioxide (TiO_2) in concentrated sulphuric acid due to the formation of pertitanic acid.

$$TiO_2 + 2H_2SO_4 \longrightarrow Ti\,(SO_4)_2 + 2H_2O$$

$$Ti\,(SO_4)_2 + H_2O_2 + 2H_2O \longrightarrow H_2TiO_4 + 2H_2SO_4$$

4.11.9 Structure of Hydrogen Peroxide

The study of X-ray analysis and dipole moment values indicates that hydrogen peroxide has a non-planar structure. H_2O_2 has an open book structure, the two oxygen atoms lie on the spine of the book opened at an angle (dihedral angle). The two hydrogen atoms lie on the two cover pages of the book. The molecular dimensions in the gas phase and solid phase are listed in Table 4.7.

Table 4.7 Bond lengths and bond angles of H_2O_2 in gas and solid phase

Gas phase		Solid phase	
∠OOH	94° 48	∠OOH	= 101.54°
Dihedral angle	111° 30	Dihedral angle	= 90.20°
O – O bond length	147.5 pm	O – O bond length	= 145.8 pm
O – H bond length	94.8 pm	O – H bond length	= 98.98 pm

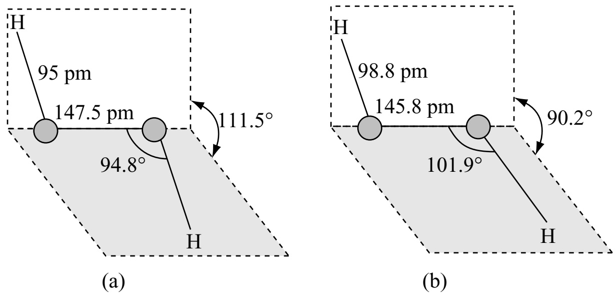


Fig 4.14 Structure of Hydrogen peroxide (a) gas phase (b) solid phase.

In the gas phase, the molecule adopts a skew configuration with a dihedral angle 111.5° as shown in Fig. 4.14. This is due to repulsive interaction of the O–H bonds with the lone pairs of electrons on each oxygen atom. The skew form persists in the liquid phase, but modified by hydrogen bonding. In the crystalline state at –163 °C, the dihedral angle is 90.2°.

KEY POINTS

- Hydrogen is the lightest and simplest of all the elements.

POSITION OF HYDROGEN IN THE PERIODIC TABLE

- Similar to alkali metals it can lose its 1s electron and and convert into H^+ ion.
- Similar to alkali metals, it contain one electron in its valence shell.
- Similar to alkali metals hydrogen is monovalent, electropositive in character and exhibit +1 oxidation state.
- Electrolysis of alkali metal halides liberate metals at cathode. Similar to this, hydrogen is liberated at cathode during electrolysis of water.
- Similar to alkali metals, hydrogen is also very good reducing agent.
- Similar to alkali metals, hydrogen forms oxides, halides and sulphides.
- Because its ionization potential is very high instead of losing electron it shares its electron with other elements and form mostly covalent compounds similar to halogens.
- Hydrogen can form H^- ion by gaining electron similar to halogens but only with highly electropositive elements like alkali metals, alkaline earth metals, aluminium, etc.
- Similar to halogens, hydrogen have electronic con figuration of one electron short of inert gas configuration.
- Similar to halogens, it is a non-metal and exist as a diatomic molecule.
- Electrolysis of fused metal halides liberate halogen at anode similar to which hydrogen is liberated at anode during the electrolysis of fused metal hydrides.
- The position of hydrogen in the periodic table is not certain because of its similarities with both alkali metals and halogens. It can be kept in both 1A group along and in VIIA group.

Occurrence

- Hydrogen is the most abundant (about 98%) element in the universe.
- Hydrogen is 9th most abundant element on the earth's surface.

Isotopes of Hydrogen

- Isotopes of hydrogen are ordinary hydrogen or protium (^1_1H), deuterium (^2_1H or D) and tritium (^3_1H).
- The difference in mass percentage of isotopes of hydrogen when compared with the isotopes of other elements is maximum due to which the properties of hydrogen isotopes differ much.
- Ordinary hydrogen is more reactive than deuterium.
- Ordinary water decompose six times faster than heavy water when electrolysed.
- Hydrogen and deuterium can be separated by gas diffusion method.
- Tritium is the lightest radioactive isotope with half-life period 12.26 years and emits β-particles during disintegration.
- Tritium is formed naturally by the bombardment of neutrons with nitrogen or by bombarding lithium with neutron in nuclear reactors (artificially).
- Deuterium and tritium are used as tracers in the study of the mechanism of complex reactions.

Types of Hydrogen

Nascent Hydrogen

- Hydrogen at the moment of its formation is known as **nascent hydrogen**.
- Ordinary hydrogen cannot reduce (decolourise) the acidified permanganate but the nascent hydrogen produced by adding zinc to acidified permanganate can reduce it.
- Ordinary hydogen cannot reduce acidified ferric chloride but nascent hydrogen can reduce it.
- Nascent hydrogen produced by the action of zinc with dilute HC*l* can reduce the chlorate to chloride but nascent hydrogen produced by the action of sodium amlagam with water cannot reduce.
- At the moment of production nascent hydrogen is present in a very small bubble where the internal pressure is very high due to which nascent hydrogen has more reactivity.
- By increasing pressure, reactivity of ordinary hydrogen can be increased.

Atomic Hydrogen

- Bond energy of H_2 molecule is very high and is 436 kJ mol^{-1}.
- At very high temperatures of about 2000k, H_2 molecules convert into atomic hydrogen by about 0.081 per cent which increases to 95.5 per cent at 5000K.
- The **life period** of atomic hydrogen is only 1/3 second.
- Atomic hydrogen is highly reactive and acts as a strong reducing agent.
- Conversion of atomic hydrogen into ordinary hydrogen is highly exothermic.
- Atomic hydrogen welding torch is used for cutting and welding the metal plates and rods.

Active Hydrogen

- Active hydrogen can be obtained by subjecting a stream of hydrogen gas at ordinary temperatures to silent electric discharge at a voltage of about 30,000 volts.
- Active hydrogen is highly reactive and combines with lead and sulphur directly at ordinary temperature.

Ortho- and Parahydrogen

- Hydrogen exists as **ortho-** and **parahydrogen**.
- If the nuclei in a molecule of hydrogen have **same spin** it is known as orthohydrogen but if they have **opposite spin** it is known as parahydrogen.
- Ortho- and parahydrogens are **spin isomers** and differ in physical properties like b.pts, specific heats, internal energies, band spectra, etc.
- The nuclear spin of orthoisomer is 1 while that of para is zero.
- At absolute zero, hydrogen is 100 per cent para but with increase in temperature it slowly converts into ortho.
- At ordinary temperatures the ratio between ortho- and parahydrogens is 3:1.
- It is possible to obtain pure parahydrogen but it is not possible to obtain a sample containing more than 75 per cent orthohydrogen.
- The **thermal conductivity** of parahydrogen is 50 per cent more than that of orthohydrogen.
- The m.pt of parahydrogen is 0.15K less than that of hydrogen containing 75 per cent of ortho hydrogen.

Preparation of Hydrogen

- Highly electropositive metals like alkali metals and alkaline earth metals react with water liberating H_2.
- In the laboratory, hydrogen is prepared by the action of zinc with dil. HC*l*, but pure zinc react only slowly because of high energy of activation.

- Pure hydrogen is not obtained by the action of metals on acids like HNO_3, H_2SO_4 and HCl because the hydrogen produced may be contaminated with oxides of nitrogen or sulphur and HCl vapours.
- Large quantities of hydrogen is prepared by passing steam over red hot coke and the process is known as **coal gasification.**

$$C + H_2O \longrightarrow CO + H_2$$

- Since the mixtrure of CO and H_2 (water gas) is used for the synthesis of methanol and a number of hydrocarbons, it is also called as **synthesis gas** or **syngas**.
- The water gas mixed with twice the amount of steam and passed over catalyst to convert CO to CO_2 and this reaction is called **water gas shift reaction**.

$$CO + H_2 + H_2O \xrightarrow[450^\circ C]{Fe_2O_3/Cr} CO_2 + 2H_2$$

- In **Lane's process** hydrogen is prepared by passing superheated steam over red hot iron.

$$3Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$$

- In **Uyeno's process,** hydrogen is prepared by treating scrap aluminium with sodium hydroxide solution.
- **Electrolysis** of acidified or alkaline water using Pt or Ni electrodes give hydrogen.
- **High pure hydrogen** is obtained by the electrolysis of warm $Ba(OH)_2$ solution between nickel electrodes.
- Hydrogen is produced as a by-product during the cracking of hydrocarbons and in the manufacure of Cl_2 and NaOH by the electrolysis of brine.

Preparation of Hydrogen

Method	Remark
1. Metal + water → Metal hydroxide + H_2	Na, K, Rb, Cs, Ca, Sr, Ba (cold)
2. Metal + steam → Metal oxide + H_2	Mg, Fe, Zn, Al.
3. Metal + acid → Salt + H_2	Metals above hydrogen in e.m.f. series. (pure H_2 is not obtained).
4. Metal + base → Salt + H_2	Amphoteric metals such as Be, Al, Zn, Sn.
5. Metal hydride + water → Metal hydroxide + H_2	NaH, CaH_2, etc.
6. Coke + steam → CO + H_2	Water gas
7. Hydrocarbons + steam $\xrightarrow{\text{catalyst}} CO + H_2$	
8. Electrolysis of acidulated or alkaline water → H_2	
9. By-product $2NaCl + 2H_2O \to 2NaOH + H_2 + Cl_2$ $CH_3 - CH_3 \to CH_2 = CH_2 + H_2$	In the manufacture of NaOH and Cl_2 by the electrolysis of brine and in cracking of petroleum

Properties of Hydrogen

- Hydrogen burns in oxygen forming water.The reaction between H_2 and O_2 without igniting is very slow at ordinary temperatures but can be catalyzed by Pd and platinum black.
- Hydrogen reacts with halogens forming hydrogen halides or hydraacids and the reactivity of halogens decreases in the order $F_2 > Cl_2 > Br_2 > I_2$.
- Hydrogen reduces several metal oxides and halides to their corresponding metals, e.g., CuO to Cu and $AgCl$ to Ag.
- Hydrogen combines with many unsaturated organic compounds at double and triple bonds converting them to saturated compounds.
- H_2 combines with CO forming methanol.
- Hydroformylation of olefins yields aldehydes.
- Hydrogen combines with several metals forming metal hydrides.
- Lithium hydride reacts with aluminium chloride forming **lithium aluminium hydride,** an important reducing agent.

Reactions of Hydrogen

1. $H_2 + Li \longrightarrow LiH$
2. $H_2 + Na \longrightarrow NaH$
3. $H_2 + Ca \longrightarrow CaH_2$
4. $H_2 + La \longrightarrow LaH_2$
5. $H_2 + U \longrightarrow UH_{2.87}$
6. $H_2 + CuO \longrightarrow Cu$
7. $H_2 + WO_3 \longrightarrow W$
8. $H_2 + Fe_3O_4 \longrightarrow Fe$
9. $H_2 + AgCl \longrightarrow Ag$
10. $H_2 + N_2 \longrightarrow NH_3$
11. $H_2 + Y_2 \longrightarrow 2HY$ (Y = F, Cl, Br or I)
12. $H_2 + CO \longrightarrow CH_3OH$

Uses of Hydrogen

- Hydrogen is used in industries for the (i) manufacture of **ammonia** by Haber's process, (ii) manufacture of **hydrochloric acid,** (iii) the manufacture of **methyl alcohol** and (iv) production of **vanaspathi** or **margarine** or **dalda** by the hydrogenation of vegetable oil and in the production of **synthetic petrol** by Fischer – Tropsch synthesis.
- Hydrogen is used as a **reducing agent** in the extraction of metals like molybdenum and tungsten.
- Hydrogen is used as a **fuel** in the form of water gas, coal gas, etc., which contain hydrogen.
- Oxyhydrogen blow torch produces very high temperatures of about 3000°C at which quartz and several metals can be melted or welded.
- Liquid hydrogen was used as a fuel in the rocket saturn V through which Neil Armstrong reached the moon.
- Though heat of combustion of hydrogen is very high it is not much used as a fuel and reducing agent because (i) preparation of hydrogen is costlier and (ii) storage and transportation of hydrogen is dangerous.
- The cells in which the chemical energy is directly converted into electrical energy are called **fuel cells** and their efficiency is very high.
- Hydrogen can be used in fuel cells for the production of electrical energy.
- Hydrogen can be used as a source of nuclear energy.
- When lighter nuclei like hydrogen, deuterium are fused to form heavier nuclei very high amount of energy will be released.
- To overcome the repulsion between the positively charged nuclei, the fusion reactions must be carried at high temperatures. Hence, they are known as **thermonuclear reactions.**
- The heat liberated during the fission bomb is used in the fusion of two isotopes of hydrogen in the hydrogen bomb.
- The energy coming from the sun and stars is due to nuclear fusion reaction between hydrogen nuclei forming helium.

HYDRIDES

- The binary compounds of hydrogen with other elements are called **hydrides** and they are of three types.

Saline or Ionic or Salt-like Hydrides

- These are formed with most of the s-block elements LiH, BeH_2 and MgH_2 have significant covalent character. These are prepared by heating s-block elements (except Be and Mg) in hydrogen at high temperatures.
- They have high melting points, insoluble in common non-aqueous solvents, react vigorously with water and acids liberating hydrogen. So, these are used as a source of hydrogen.
- Alkali metal hydrides have **rock-salt** structure.
- Thermal stability of alkali metal hydrides and alkaline earth metals decreases down the group.
- Electrolysis of these fused hydrides liberates hydrogen at anode.
- Densities of these hydrides are greater than that of the metal from which they are formed.
- Fires produced by saline hydrides cannot be extinguished by CO_2 as it get reduced by that hot metal hydrides. Only sand can extinguish the fire.
- Calcium hydride is called **hydrolith**, used in the laboratory to remove traces of water from solvents and inert gases such as N_2 and Ar.

Metallic or Interstitial Hydrides

- These are formed by d- and f-block elements and are formed by the presence of hydrogen atoms in the interstitial positions of the metal crystals.
- Metallic hydrides are non-stoichiometric compounds and show electric conduction.
- The properties of these hydrides are similar to those of parent metal and their densities are less than those of the parent metals.
- Some metals like Pt and Pd can accommodate a very large volume of hydrogen called **occluded hydrogen.**
- The hydrogen absorbed by the metals will be released when heated and this property is used in the purification of H_2, storing of H_2, and also for reduction and catalystic hydrogenation.
- In the chromium group only chromium form metallic hydride.
- In the periodic table, the region of groups 7–9 is referred to as the **hydride gap** because metals in these groups do not form metallic hydrides.

Molecular or Covalent Hydrides

- These are formed by p-block elements and consists of individual, discrete covalent molecules.
- The molecular formula of covalent hydrides can be written as MH_n or $MH_{(8-n)}$ where n = group number of the element in the short form periodic table.
- Covalent hydrides are named from the name of element and the suffix - ane. e.g., phosphane – PH_3; oxidane – H_2O and azane – NH_3.
- Molecular hydrides having less number of electrons for writing the conventional Lewis structure are called **electron-deficient hydrides,** e.g., B_2H_6.

- Molecular hydrides formed by the elements of IV or 14 group elements in which all the valence electrons are involved for bond formation are known as **electron-precise hydrides.**
- Molecular hydrides which contain more valence electrons, than the required for bond formation are called **electron-rich hydrides.** They contain lone pairs, e.g., NH_3, H_2O.
- Only weak van der Waal's attractive forces exists between the molecules of covalent hydrides. So they are volatile with low b.pts.
- The presence of lone pairs of electrons on highly electronegative like F, O, N in electron-rich hydrides result in hydrogen bond formation which leads to high b.pts, high permittivities and aggregates.

WATER

- Water has abnormally high freezing point, boiling point, heat of vapourisation, heat of fusion, etc., due to the association of molecules through hydrogen bonds.
- The high heat of vapourization and high heat capacity of water are responsible for moderation of the climate and body.
- Solubility of ionic compounds in water is due to its polar nature while the solubility of covalent compounds like alcohols, carbohydrates, urea, etc. is due to their ability to form hydrogen bonds with water.
- Under very high pressure and temperature water behaves like a non-polar solvent in which organic compounds are more soluble while inorganic salts are insoluble.
- Purest water is rain water. In water the ratio of hydrogen and oxygen is 2:1 by volume and 1:8 by weight.
- Dipole moment of water 1.85 D while dielectric constant is 78.39.
- Maximum density of water is at 4°C.
- Structurally ice have nine different forms, the one which is formed at atmospheric pressure is normal hexagonal form (I_n) but at very low temperatures it condenses as cubic form.
- Ice has lesser density than the water with which it is in equilibrium.
- The water molecules are joined together in three-dimensional network in which each oxygen atom is surrounded by four water molecules tetrahedrally and bond through four hydrogens, two by normal covalent bond and the other two by hydrogen bonds.
- Due to the three dimensional hydrogen bonded structure ice has got open spaces due to which its density is less than water and floats over water.
- Many salts crystallize from their aqueous solutions as hydrated salts and the water molecules associated with these salts are of five types: (i) coordinated water, (ii) hydrogen bonded water, (iii) lattice water, (iv) zeolite water and (v) clatharate water.
- In some hydrated salts water molecules are coordinated to metal ion (complex compounds).
  e.g. $[Ni(H_2O)_6]Cl_2$; $[Fe(H_2O)_6]Cl_3$; $[Cr(H_2O)_6]Cl_3$
- In some compounds water molecules are hydrogen bonded to oxygen containing anions. e.g., in $CuSO_4 \cdot 5H_2O$ four water molecules are coordinated to Cu^{2+} ion and one water molecule is in hydrogen bond with sulphate ions.

Reactions of Water	Remark
1. $H_2O + M \longrightarrow MOH + H_2$	with water
2. $H_2O + M \longrightarrow \text{Metal oxide} + H_2$	with steam and red hot condition
3. $H_2O + C \xrightarrow{\text{Red hot}} CO + H_2$	
4. $H_2O + F_2 \longrightarrow O_2$ and $O_3 + HF$	
5. $H_2O + Cl_2 \longrightarrow HOCl + HCl$	
6. $H_2O + \text{Metal oxides} \longrightarrow \text{Metal hydroxides}$	Na_2O, K_2O, CaO, BaO, etc.
7. $H_2O + \text{Non-metal oxides} \longrightarrow \text{oxoacids}$	N_2O_5, SO_3, P_4O_{10}, Cl_2O_7, etc.
8. $\text{Salts} + H_2O \longrightarrow \text{Hydrolysis}$	Li_3N, Mg_3N_2, AlN give NH_3 Ca_3P_2 give PH_3 CaC_2 give C_2H_4 BeC_2, Al_4C_3 give CH_4.

- In certain compounds water is present as interstitial water, i.e., it ocupies the interstices in the crystal lattice, e.g., $BaCl_2\ 2H_2O$.
- Water can act both as Bronsted acid and Bronsted base due to **autoprotolysis** or self-ionization and hence water is amphoteric.

$$H_2O + H_2O \longrightarrow H_3O^+ + OH^-$$

- Halogen oxidize water to oxygen. Solution of chlorine in water is called as **chlorine water.**
- Solutions of basic oxides in water are alkaline while the solutions of acidic oxides in water are acids.
- Salts undergo hydrolysis in water.
- Water reacts with carbon dioxide in the presence of sunlight and chlorophyll to give carbohydrates (photosynthesis).

HARD WATER

- Water which gives ready and permanent lather with soap is **soft water.**
- The water which does not give ready and permanent lather with soap is called **hard water**.
- Soap is sodium or potassium salt of oleic, pamitic or stearic acids.
- Hardness of water is due to the presence of bicarbonates, sulphates and chlorides of magnesium and calcium.
- The lather formed by normal sodium or potassium soap is coagulated by magnesium or calcium because they form insoluble salts.
- **Temporary hardness** is due to the presence of bicarbonates of magnesium and calcium.
- **Permanent hardness** is due to the presence of chlorides and sulphates of magnesium and calcium.
- Temporary hardness can be removed by boiling the water or by the addition of calculated amount of slaked lime (Clark's method) where soluble bicarbonates converts into insoluble carbonates.

$$M(HCO_3)_2 \xrightarrow{\Delta} MCO_3 + H_2O + CO_2 \ (M = Mg \text{ or } Ca)$$

$$Mg(HCO_3)_2 + 2Ca(OH)_2 \longrightarrow Mg(OH)_2 + 2CaCO_3 + 2H_2O$$

$$Ca(HCO_3)_2 + Ca(OH)_2 \longrightarrow 2CaCO_3 + 2H_2O$$

In the removal of temporary hardness by the addition of slaked lime, if more amount of slaked lime is added the water will get permanent hardness.

Removal of Permanent Hardness

- By adding **washing soda** both temporary and permanent hardness is removed due to conversion of soluble magnesium and calcium salts into insoluble magnesium and calcium carbonates.
- In **Permutit** method, permanent hardness of water is removed by exchanging the Ca^{2+} and Mg^{2+} ions with sodium ions present in artificially made zeolite.
- **Zeolite** is hydrated sodium alumino silicate $Na_2Al_2\ Si_2O_8 \cdot xH_2O$.
- Exhausted zeolite is regenerated by treating Mg or Ca zeolite with 10 per cent $NaCl$.
- Artificial zeolite can be prepared by fusing the quartz (SiO_2), soda ash (Na_2CO_3) and china clay ($Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O$)
- In **calgon process** permanent hardness is removed by using **sodium hexametaphosphate** which is known as **calgon**.
- Calgon reacts with Mg^{2+} and Ca^{2+} ions to form inactive complex anions which do not react with soap. This is known as **sequestration** (rendering active species into inactive species).
- In the **ion exchange method** all the cations and anions present in the hard water are removed.
- Resins are giant organic molecules, and the resins having acidic groups such as –COOH, $–SO_3H$ can exchange the cations with H^+ ions and are called **cation exchange resins** while the resins containing basic groups such as $–NH_3OH$ can exchange the anions with OH^- ions and are called **anion exchange resins.**
- Exhausted cation exchange resins can be regenerated by washing with moderate dilute acids while the exhausted anion exchange resins can be regenerated by treating with NaOH or Na_2CO_3 solution.
- The water obtained by passing through ion exchange resin is known as **deionized** water whose quality is equal to distilled water.
- If hard water is used in boilers, the heat resistant and corrosive scales are formed on the inner surface of boiler which causes the fuel wastage and corrosion of the metal by which the boiler is made.
- **Degree of hardness** is the amount of hardness in terms of calcium carbonate given in units of ppm.

$$\begin{array}{ccccccccc} 1CaCO_3 & \equiv & 1MgSO_4 & \equiv & 1MgCl_2 & \equiv & 1\ CaSO_4 & \equiv & 1CaCl_2 \\ 100\text{ ppm} & & 120\text{ ppm} & & 95\text{ ppm} & & 136\text{ ppm} & & 111\text{ ppm} \end{array}$$

HEAVY WATER

- It is **deuterium oxide**, discovered by **Urey**.
- Ordinary water contains 1 part of heavy water in 6000 parts of ordinary water.
- On large scale, heavy water is prepared by repeated and prolonged electrolysis of ordinary water containing N/2 NaOH.

- In the preparation of heavy water by electrolysis of ordinary water **cathode** is steel vessel and **anode** is cylindrical nickel sheet.
- When H_2S gas is passed through hot water the outcoming gas contains some D_2S due to exchange of H with D.
- When H_2S gas having more percentage of D_2S is passed in cold water repeatedly D_2O will be obtained due to exchange of D in D_2S to H in H_2O.
- Several physical properties like molecular weight, freezing point, b.pt, density, viscosity, specific heat, latent heat of vapourisation of heavy water are greater than ordinary water.
- The ionic product, dielectric constant, solubility of salts in heavy water when compared with ordinary water are less (Table 4.3).
- Since the dielectric constant of heavy water is less than ordinary water, the solubility of ionic compounds like $NaCl$, $BaCl_2$ are less in heavy water than in ordinary water.
- The difference in physical properties of heavy water from ordinary water is due to greater nuclear mass of deuterium and greater degree of association.
- The D–O bond is stronger than H–O bond, so that reactivity of heavy water is less than ordinary water.
- The ionic product of heavy water is 3.0×10^{-15} mole^2 lit^{-2} 25°C which is less than that of ordinary water.
- Electrolysis of heavy water gives deuterium at cathode and oxygen at anode.
- When metals like Na or Ca react with water form the corresponding metal deuteroxides NaOD and $Ca(OD)_2$ with the liberation of deuterium.
- Metal oxides when dissolved in heavy water, gives the bases containing metal deuteroxides.
- When non-metal oxides are dissolved in heavy water deutero acids will be formed, e.g., deuteronitric acid (DNO_3); deuterosulphuric acid (D_2SO_4).
- With nitrides heavy water gives deuteroammonia (ND_3) and with phosphides give deuterophosphine (PD_3).
- With carbides like CaC_2 gives deuteroacetylene (C_2D_2) and with Al_4C_3 gives deuteromethane (CD_4)
- Salts, which associated with heavy water molecules are called deuterates, e.g., $CuSO_4 \cdot 5D_2O$; $MgSO_4 \cdot 7D_2O$. Deuterates have low vapour pressure than hydrates.
- Hydrolysis of salts in heavy water is called **deuterolysis.**
- When heavy water is mixed with compounds containing normal hydrogen, the lighter isotope is exchanged by heavier isotope deuterium, e.g., NaOH to NaOD; NH_4Cl to NH_3DCl; HCl to DCl.
- Heavy water has different physiological action on plants and animals: (i) the growth of plants is less in heavy water, (ii) seeds cannot germinate in heavy water, (iii) in large doses heavy water is poisonous and (iv) aquatic animals die in heavy water.
- Heavy water is used as a **moderator** in nuclear reactors to slow down the fast moving neutrons.
- Heavy water is used as a tracer to study the mechanism of biochemical reactions, electrophil lic aromatic substitution mechanism, etc.
- The exchange reaction of heavier deuterium with lighter hydrogen atoms in heavy water is used to know the structure of oxoacids, e.g., H_3PO_2 is a monobasic acid and H_3PO_3 is a dibasic acid was decided by this reaction.
- Deuterium can be prepared either by the electrolysis of heavy water or by the reaction of sodium with heavy water required for making the hydrogen bomb.

HYDROGEN PEROXIDE

- Hydrogen peroxide can be considered as oxygenated water. It was first prepared by **Thenard.**
- Commercial name of H_2O_2 is **perhydrol** which is 30 per cent.
- When sunlight falls on water containing dissolved oxygen, little H_2O_2 will be formed.

Preparation of H_2O_2

- In the laboratory, H_2O_2 can be prepared by action of ice cold dilute acids on metal peroxides like Na_2O_2 or BaO_2.
- Anhydrous BaO_2 is not decomposed by dil. H_2SO_4 due to the formation of a layer of insoluble $BaSO_4$. So, hydrated $BaO_2 \cdot 8H_2O$ is used which easily decomposes.
- Phosphoric acid is preferred in the place of dil. H_2SO_4 because some Ba^{2+} ions present in solution catalyze the decomposition of H_2O_2 but H_3PO_4 completely precipitates Ba^{2+} ions.
- Commercially, H_2O_2 is manufactured by the electrolysis of 50 per cent H_2SO_4 or by autooxidation of 2-ethyl anthraquinol.
- In the electrolysis of 50 per cent H_2SO_4 or H_2SO_4 and $(NH_4)_2\ SO_4$ in 1 : 1 molar ratio is taken as electrolyte for the preparation of H_2O_2. Anode is platinum and cathode is lead.
- During the electrolysis of H_2SO_4, the anion oxidized at anode is HSO_4^-.
- Distillation of $H_2S_2O_8$ formed form the electrolysis of 50 per cent H_2SO_4 or $(NH_4)_2\ S_2O_8$ formed from the electrolysis of H_2SO_4 and $(NH_4)_2SO_4$ in 1 : 1 molar ratio gives H_2O_2.
- When 2-ethylanthraquinol is oxidized in air H_2O_2 and 2-ethylanthraquinone will be formed.

- 2-ethylanthraquinone can be reduced with H_2 in the presence of catalysts like Pt, Pd or Ni to 2-ethyl-anthraquinol.

Concentration of H_2O_2

- **Careful evaporation** under reduced pressure gives about 30 per cent H_2O_2.
- **Dehydration** in a vaccum dessicator containing conc. H_2SO_4 give about 90 per cent H_2O_2.
- Distillation under reduced pressure gives 99 per cent H_2O_2.
- Fractional crystallization using freezing mixtures gives 100 per cent needle like crystals of H_2O_2.

Storage of H_2O_2

- H_2O_2 decomposes rapidly on the rough surface.
- Urea, sodium stannate, acetanilide, sodium pyrophosphate acts as negative catalysts in the decomposition of H_2O_2.
- Dilute solutions of H_2O_2 are stable in the presence of acids, alcohols or ether (negative catalysts).
- The decomposition of H_2O_2 is accelarated by the presence of alkalis, sand, MnO_2, iron, manganese and alumina (positive catalysts).
- H_2O_2 is stored in plasitc bottle or glass containers coated inside with wax and in the presence of a stabilizing agents like glycerol, urea, sodium stannate, etc.

Strength of H_2O_2

- The strength of H_2O_2 is expressed in terms of volume strength, e.g., 10 vol, 20 vol, 30 vol, etc.
- 10 vol H_2O_2 means 1 ml H_2O_2 gives 10 c.c of O_2 at STP.
- The strength of perhydrol is 30 per cent w/v or 100 vol.
  10 vol H_2O_2 = 3.036% w/v = 1.786 N = 0.893M
  (or) 11.2 vol H_2O_2 = 3.4% w/v = 2N = 1 M

Physical Properties

- Pure H_2O_2 is a colourless, odourless, syrupy liquid. In thick layers it has light blue colour, miscible with water in all proportions.
- H_2O_2 is more hydrogen bonded than water. It is slightly acidic but cannot turn the blue litmus to red.
- The oxidation number of oxygen in H_2O_2 is 1. H_2O_2 decomposes to give O_2.

$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

- Decomposition of H_2O_2 is both oxidation and reduction (disproportionation) because in the oxygen liberated oxidation number increases from –1 to 0 and in another oxygen of H_2O oxidation number decreases from –1 to –2.

Chemical Properties

- H_2O_2 can function both as oxidizing and reducing agent.
- H_2O_2 is a very powerful oxidizing agent and poor reducing agent.
- While acting as a reducing agent O_2 gas will be evolved and always O_2 is liberated from H_2O_2 without breaking O–O bond.

Oxidation Reactions in Acid Medium

- H_2O_2 oxdizes PbS to $PbSO_4$; ferrous salts to ferric salts, liberates iodine from iodides; oxidizes nitrites to nitrates, sulphites to sulphates, arsenites to arsenates; potassium ferrocyanide to potassium ferricyanide.
- H_2O_2 gives blue chromium peroxide (CrO_5) with acidified dichromate which has a butterfly structure. It contain two peroxy bonds, four oxygen atoms in –1 oxidation state and one oxygen atom in –2 oxidation state while chromium is in +6 oxidation state.
- CrO_5 is unstable in acid medium converting into chromic salt liberating oxygen. It is stable in ether layer in the presence of pyridine forming blue layer. It is used for testing the presence of H_2O_2 or vice versa.
- H_2O_2 oxidizes mercury to mercuric oxide.

Oxidation Reactions in Alkaline Medium

- H_2O_2 oxidizes formaldehyde in the presence of pyrogallol (1, 2, 3% trihydroxy benzene) with the liberation of hydrogen converting formaldehyde to formic acid.
- Oxidation reactions of H_2O_2 are slow in acid medium but faster in alkaline medium.
- Due to the oxidation property, H_2O_2 acts as a mild bleaching agent and is used to bleach silk, wool, ivory, hair which are destroyed by the bleaching action of chlorine. Ammonical H_2O_2 bleaches human hair to golden yellow.

Reduction Properties of H_2O_2

- H_2O_2 reduce and decolourise acidified potassium permanganate, reduces halogens to halogen acids, reduces silver oxide to silver, potassium ferricyanide to ferrocyanide in basic medium, lead dioxide to yellow coloured lead monoxide (litharge) and ozone to oxygen.

Acidic Properties

- H_2O_2 reacts with alkalis and carbonates forming metal peroxides. It liberates CO_2 from carbonates. Because it is a weak dibasic acid, it forms two series of salts hydroperoxide and peroxides.

Formation of Addition Compounds

- H_2O_2 forms addition compounds $Na_2HPO_4 \cdot H_2O_2$; $(NH_4)_2SO_4 \cdot H_2O_2$; $H_2N\ CONH_2 \cdot H_2O_2$. The addition compound of H_2O_2 with urea is called hyperol. These addition compounds release H_2O_2 when dissolved in water.

Hydrogen Peroxide

Reaction	Remark
Preparation	
1. $Na_2O_2 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O_2$	Merck's process
2. $BaO_2 + H_2SO_4 \longrightarrow BaSO_4 + H_2O_2$	$BaSO_4$ insoluble
3. $BaO_2 + H_3PO_4 \longrightarrow Ba_3(PO_4)_2 + H_2O_2$	$Ba_3(PO_4)_2$ insoluble
4. $BaO_2 + CO_2 + H_2O \longrightarrow BaCO_3 + H_2O$	$BaCO_3$ insoluble
5. $\underset{50\%}{2H_2SO_4} \xrightarrow{\text{Electrolysis}} H_2S_2O_8 + H_2$	
$H_2S_2O_8 + 2H_2O \longrightarrow 2H_2SO_4 + H_2O_2$	
6. Electrolysis of equimolar mixture of H_2SO_4 and $(NH_4)_2\ SO_4$	
$(NH_4)_2SO_4 + H_2SO_4 \longrightarrow 2NH_4HSO_4$	
$2NH_4\ HSO_4 \xrightarrow{\text{Electrolysis}} (NH_4)_2S_2O_8 + H_2$	
$(NH_4)_2\ S_2O_8 + 2H_2O \longrightarrow 2NH_4HSO_4 + H_2O_2$	
Acidic Property:	
1. $H_2O_2 + NaOH \longrightarrow NaHO_2$	
2. $H_2O_2 + 2NaOH \longrightarrow Na_2O_2$	Metal peroxides are salts of H_2O_2
3. $H_2O_2 + Na_2CO_3 \longrightarrow Na_2O_2 + H_2O + CO_2$	
Oxidation Properties in acid medium:	
1. $H_2O_2 + FeSO_4 \longrightarrow Fe_2(SO_4)_3$	
2. $H_2O_2 + KI \longrightarrow I_2$	
3. $H_2O_2 + KNO_2 \longrightarrow KNO_3$	
4. $H_2O_2 + Na_2SO_3 \longrightarrow Na_2SO_4$	
5. $H_2O_2 + Na_3AsO_3 \longrightarrow Na_3AsO_4$	
6. $H_2O_2 + PbS \longrightarrow PbSO_4$	
7. $H_2O_2 + Cr_2O_7^{2-} \longrightarrow CrO_5$	
Oxidation reactions in alkaline medium	
1. $H_2O_2 + Mn^{2+} \longrightarrow MnO_2$	
2. $H_2O_2 + Cr^{3+} \longrightarrow CrO_4^{2-}$	
3. $H_2O_2 + HCHO \longrightarrow HCOOH + H_2$	Only reaction in which H_2O_2 is reduced to H_2
Reduction reactions in acid medium	
1. $H_2O_2 + \underset{\text{purple}}{KMnO_4} \longrightarrow \underset{\text{colourless}}{Mn^{2+}}$	
2. $H_2O_2 + \underset{\text{Orange red}}{K_2Cr_2O_7} \longrightarrow \underset{\text{green}}{Cr^{3+}}$	
3. $H_2O_2 + \underset{\text{Black}}{MnO_2} \longrightarrow Mn^{2+}$	
4. $H_2O_2 + O_3 \longrightarrow O_2$	
5. $H_2O_2 + HOCl \longrightarrow HCl$	

Reaction	Remark
Reduction reactions in alkaline medium	
1. $H_2O_2 + KMnO_4 \longrightarrow MnO_2$ (KMnO₄ Pink, MnO₂ Black)	
$H_2O_2 + K_3[Fe(CN)_6] \longrightarrow K_4[Fe(CN)_6]$	
$H_2O_2 + Ag_2O \longrightarrow Ag$ (Black)	
$H_2O_2 + PbO_2 \longrightarrow PbO$ (litharge)	
$H_2O_2 + Cl_2 \longrightarrow Cl^-$	
$H_2O_2 + NaOBr \longrightarrow NaBr$	
$H_2O_2 + CaOCl_2 \longrightarrow CaCl_2$ (CaOCl₂ Bleaching Powder)	
Addition Reaction	
1. $H_2O_2 + CH_2 = CH_2 \longrightarrow HOCH_2 - CH_2OH$	

Uses

- H_2O_2 is a harmless strong insecticide, so is used to clean the wounds.
- H_2O_2 is used as a mild bleaching agent to bleach silk, wool, ivory, etc.
- Conc. H_2O_2 is used as a rocket fuel.
- H_2O_2 is used in restoring the colour of old and spoiled lead paintings.

Tests for H_2O_2

- Turns starch iodide paper to blue due to the liberation of iodine from iodide by H_2O_2.
- Forms blue colour in ether layer when reacts with acidified potassium dichromate containing ether.
- Gives orange coloured pertitanic acid when reacts with acidified titanium dioxide.

$$Ti\,(SO_4)_2 + H_2O_2 + 2H_2O \longrightarrow H_2TiO_4 + 2H_2SO_4$$

Structure of H_2O_2

- H_2O_2 has a non-linear, non-planar structure with half-open book structure.
- The angle between O–H and O–O bonds is 94° 98'.
- The angle between the two hydrogen atoms which lie on the cover pages of open book (dihedral angle) is 111° 30'.
- The bond angles and bond lengths in H_2O_2 changes with the physical state of H_2O_2 and the values given above are for gaseous H_2O_2.
- When H_2O_2 changes from gaseous state to solid the dihedral angle decreases and reaches to 90° in crystalline state.

Multiple Choice Questions with Only One Answer

Single Answer Questions

1. The dipole moment of H_2O_2 is 2.1D while that of water is 1.84D. But water (H_2O) is a better solvent than that of H_2O_2 because
(a) Its dipole moment of H_2O_2
(b) It is less corrosive
(c) H_2O_2 gets ionized during chemical reactions
(d) H_2O_2 gets decomposed during chemical reactions

2. The H–O–O–H bond angles and O–H bond lengths are 101.9° and 98.8 pm, respectively in solid phase instead of 111°31' and 95 pm in gaseous phase of H_2O_2. This indicates that the structure of H_2O_2 in solid and gaseous phase are different. This is due to
(a) Intermolecular hydrogen bonding
(b) Intramolecular hydrogen bonding
(c) van der Waal's bonding
(d) All are true

3. Amongst the following, choose the correct statements.
I. Atomic hydrogen is obtained by passing hydrogen gas through electric arc.
II. Hydrogen gas will not reduce heated aluminium oxide.
III. Finely divided palladium absorbs large amount of hydrogen gas.
IV. Pure nascent hydrogen is best prepared by reacting Na with C_2H_5OH.
(a) I, II, III (b) II, IV
(c) I, III (d) All of these

4. If x and y are boiling point and dielectric constant of water, respectively and a and b are that of D_2O (heavy water) then
(a) $x > a, y < b$ (b) $x > a, y > b$
(c) $x < a, y < b$ (d) $x < a, y > b$

5. Polyphosphates are used as water softening agents because they
(a) form soluble complexes with anionic species
(b) precipitate anionic species
(c) form soluble complexes with cationic species
(d) precipitate cationic species

6. To an aqueous solution of $AgNO_3$ some NaOH (aq) is added, till a brown ppt is obtained. To this H_2O_2 is added dropwise. The precipitate turns black with the evolution of O_2. The black ppt is
(a) Ag_2O (b) Ag_2O_2
(c) AgOH (d) Ag

7. When KI in acidic medium was mixed in 50 mL H_2O_2 liberated I_2 requires 20 mL of 0.1N hypo. What is the strength of H_2O_2?
(a) 6.8 g/litre (b) 0.68 g/litre
(b) 680 g/litre (d) 68.0 g/litre

8. In the following reaction using isotopic ^{18}O in H_2O_2,
$2MnO_4^- + 3H_2O_2^{18} \rightarrow 2MnO_2 + 3O_2 + 2H_2O + 2OH^-$
isotopic oxygen goes
(a) Both with O_2
(b) Both with MnO_2
(c) Both with OH^-
(d) One with O_2 and one with MnO_2

9. The incorrect statements about hydrogen peroxide
I. It oxidizes potassium ferrocyanide to potassium ferricyanide in basic medium
II. It reduces potassium ferricyanide to potassium ferrocyanide in acidic medium.
III. It bleaches coloured matter to colourless matter by oxidation as well as reduction.
IV. It liberates iodine from KI solution which gives blue colour with amylose (a straight chain polymer containing many glucose units).
(a) I, II and III only (b) II, III and IV only
(b) I, II and IV only (d) I, III and IV only

10. There are three samples of H_2O_2 labelled as 10 V, 15 V, 20 V. Half-litre of each sample is mixed and then diluted with equal volume of water. Then, the volume strength of resultant solution is
(a) 7.5 (b) 1.339
(c) 5.6 (d) 15

11. What mass of CaO will be required to remove the hardness of 1000 litre of water containing 1.62g of calcium bicarbonate per litre?
(a) 56 g (b) 560 g
(c) 112 g (d) 1120 g

12. The hair dye available in the market generally contain two bottles one containing dye and other containing hydrogen peroxide. The bottles are mixed before applying the dye. The function of hydrogen peroxide is
(a) To dilute the dye
(b) To oxidize the dye to give desired colour
(c) To reduce the dye to give desired colour
(d) To acidify the dye solution

13. Identify the incorrect statement
I. H_2 is evolved at anode during the electrolysis of molten LiH.
II. All ionic hydrides react with water and liberate hydrogen gas.
III. Salts are more soluble in heavy water than water.
(a) I and II (b) III only
(c) II and III only (d) I only

14. The incorrect statements about the melting point of ice
I. Ice melts at a temperature lower than its usual melting point, when the pressure is increased. This is because ice is less denser than water.
II. Ice melts at a temperature higher than its usual melting point, when the pressure is increased. This is because ice is less denser than water
III. Ice melts at a temperature lower than its usual melting point, when the pressure in increased. This is because the chemical bonds break under pressure.
IV. Ice melts at a temperature lower than its usual melting point, when the pressure is decreased. This is because ice is not a true solid.
(a) I, II and III only (b) II, III and IV only
(c) I, II and IV only (d) I, III and IV only

15. Identify the correct order w.r.t hydrogen bonding.
(a) $H_2O^{16} > H_2O^{17} > H_2O^{18}$
(b) $H_2O^{17} > H_2O^{16} > H_2O^{18}$
(c) $H_2O^{18} > H_2O^{17} > H_2O^{16}$
(d) $H_2O^{16} = H_2O^{17} = H_2O^{18}$

16. The lithium ion (Li^+) and hydride ion (H^-) are isoelectronic ions. Which statement about these systems is true?
(a) Chemical properties of these ions are identical since they are isoelectronic
(b) Li^+ is a stronger reducing agent than H^-
(c) More energy is needed to ionize H^- than Li^+
(d) Radius of H^- is larger than that of Li^+

17. Alkali metals displace hydrogen from water forming bases due to the reason that
(a) They are far above the hydrogen in electrocheimcal series based on oxidation potential
(b) They are far below the hydrogen in electrochemical series based on oxidation potential.
(c) Their ionization enthalpy is less than that of other elements.
(d) They contain only one electron in their outermost shell.

18. When a small amount of solid calcium phosphide Ca_3P_2 is added to water, what are the most likely products ?
(a) Aqueous Ca^{2+} and OH^- ions and gaseous PH_3.
(b) Aqueous Ca^{2+} and OH^- ions and gaseous H_3PO_3.
(c) Solid CaH_2 and aqueous H_3PO_2.
(d) Solid CaO and gaseous PH_3.

19. If the water molecule had a linear rather than a bent structure, we would expect that
(a) both the boiling and freezing temperatures would be higher
(b) the boiling temperature would be higher but the freezing temperature would be lower
(c) both the boiling and freezing temperatures would be lower
(d) salts such as sodium chloride would be even more soluble in the linear structure than the bent structure

20. Select the correct statement
(a) H_2O_2 reduces MnO_4^- to Mn^{2+} in acid medium
(b) H_2O_2 reduces MnO_4^- to MnO_2 in basic medium
(c) H_2O_2 can be used to bleach blackened oil painting
(d) All are correct

21. 100 mL of 0.01M $KMnO_4$ oxidized 100 mL H_2O_2 in acidic medium. Volume of same $KMnO_4$ required in alkaline medium to oxidize 100 mL of same H_2O_2 will be (MnO_4^- changes to Mn^{2+} in acidic medium and to MnO_2 in alkaline medium)
(a) $\frac{100}{3}$ mL (b) $\frac{500}{3}$ mL
(c) $\frac{300}{5}$ mL (d) $\frac{200}{5}$ mL

22. In gaseous hydrogen peroxide, the dihedral angle between H-atoms is X° but in solid state it is Y°. The values of X and Y are respectively
(a) 94.8, 94.8 (b) 111.5, 90.2
(c) 90.2, 90.2 (d) 111.5, 111.5

23. When H_2O_2 is added to ice cold solution of acidified dichromate containing ether, the contents are shaken and allowed to stand
(a) a blue colour is obtained in ether due to formation of $Cr_2(SO_4)_3$
(b) a blue colour is obtained in ether due to formation of CrO_5
(c) CrO_3 is formed which dissolves in ether to give blue colour
(d) Chromyl chloride is formed

24. D_2O is preferred to H_2O as a moderator in nuclear reactors because
(a) D_2O slows down fast neutrons better
(b) D_2O has high specific heat
(c) D_2O is cheaper
(d) D_2O has high boiling point

25. A sample of water contains sodium chloride. It is
(a) hard water (b) soft water
(c) moderately hard (d) none

26. Deuteroammonia (ND_3) can be prepared
(a) by heating a solution of NH_4Cl in NaOD
(b) by the action of heavy water on magnesium nitride.
(c) by fractionation of ordinary ammonia
(d) by the exchange of hydrogen in ammonia by dissolving in heavy water

27. The reaction $Ag_2O + H_2O_2 \rightarrow 2Ag + H_2O + O_2$ takes place in
(a) basic medium
(b) acidic medium
(c) neutral medium
(d) both in acidic and basic medium

28. Which of the following statement is not correct?
(a) H_2O and D_2O differ in physical properties but resemble in chemical properties.
(b) Melting point and boiling point of D_2O are higher than those of H_2O.
(c) Ionic product and dielectric constant of D_2O are smaller than those of H_2O
(d) Strength of O–H bond in H_2O and O–D bond D_2O is equal.

29. Which of the following statements is wrong?
(a) Ordinary hydrogen is an equilibrium mixture of ortho- and parahydrogen.
(b) In orthohydrogen, spin of two nuclei is in the same direction.
(c) Ortho- and paraforms do not resemble in their chemical properties.
(d) In parahydrogen spin of nuclei is in the opposite direction.

30. Acidified potassium permanganate is dropped over sodium peroxide taken in a round bottom flask at room temperature, vigorous reaction takes place to produce .... and the sum of the coefficients of reactants and products are respectively
(a) Hydrogen peroxide; 15 and 15
(b) A mixture of hydrogen and oxygen; 20 and 15
(c) A colourless gas hydrogen; 15 and 20
(d) A colourless gas dioxygen; 15 and 21

31. By starting with 0.5 moles of sodium peroxide how many moles of dioxygen can be obtained by dropping excess of water on it?
(a) 0.5 mole (b) 1 mole
(c) 0.25 mole (d) 0.125

32. Electrolysis of aqueous Na*Cl* and NaH differs in
(a) formation of basic solution at the cathode by Na*Cl* only
(b) formation of basic solution at the cathode by NaH only
(c) formation of H_2 gas at cathode and anode both by Na*Cl* and NaH
(d) formation of NaOH at cathode and anode

33. 10 L of hard water required 0.56 g of lime (CaO) for removing hardness. Hence, temporary hardness in ppm (part per million 10^6) of $CaCO_3$is
(a) 100 (b) 200 (c) 10 (d) 20

34. In which of the properties listed below hydrogen does not show resemblance with halogen
I. Electropositive character
II. Electronegative character
III. Neutral nature of H_2O
IV. Atomicity
(a) I and III (b) I only
(c) II and III (d) III and IV

35. Which combination cannot be used for the preparation of hydrogen gas in the laboratory
I. Zn/conc. H_2SO_4
II. Zn/dil.HNO_3
III. pure Zn/dil.H_2SO_4
(a) I and II (b) I, II, III
(c) III only (d) I and III

36. Dipole moment of hydrogen peroxide is
(a) greater than that of water
(b) less than that of water
(c) equal to that of water
(d) unpredictable

37. When hard water percolates through the ion exchange resin with formula Res. COOH, it becomes free from
(a) all types of ions (b) Cl^- ions
(c) Ca^{2+} and Mg^+ ions (d) SO_4^{2-} ions

38. The stability of alkali metal hydrides is
(a) LiH > NaH > KH > RbH > CsH
(b) LiH < NaH < KH < RbH < CsH
(c) LiH = NaH = KH < RbH > CsH
(d) LiH < NaH > KH > RbH > CsH

39. H_2O_2 cannot be synthesized by
(a) addition of ice cold H_2SO_4 on BaO_2
(b) addition of ice cold H_2SO_4 on PbO_2
(c) areal oxidation of 2-ethyl anthraquinol
(d) electrolysis of $(NH_4)_2SO_4$ at a high density current

40. A mixture of ammonium sulphate and sulphuric acid in 1:1 molar ratio is electrolyzed using platinum electrodes. The products formed at anode and cathode are respectively
(a) H_2 and NH_4HSO_4 (b) H_2O_2 and H_2
(c) $(NH_4)_2S_2O_8$ and H_2 (d) H_2 and H_2O_2

41. An aqueous solution of compound (A) is weakly acidic in nature, it can form two series of salts, with TiO_2 in conc. H_2SO_4 it gives orange colour. Which of the following pair cannot give (A) in water?
(a) $H_2S_2O_8$ and KO_2 (b) $HClO_4$ and PbO_2
(c) H_2SO_5 and Na_2O_2 (d) BaO_2 and K_2O_2

42. Which of the following processes will produce hard water ?
(a) Saturation of water with $CaSO_4$
(b) Addition of Na_2SO_4 to water
(c) Saturation of water with Na_2CO_3
(d) Saturation of water with K*Cl*

43. The hydrides of alkali metals from Li to Cs share the following feature. Select the correct statement.
(a) All are white and crystalline in nature.
(b) None of the hydrides form polymeric covalent bridge bonds.
(c) Thermal stability of hydrides decreases from Li to Cs.
(d) All of the above.

44. The ionization constant of protium in water ($H_2O \rightleftharpoons H^+ + OH^-$) is 1×10^{-14} and that in heavy water ($D_2O \rightleftharpoons D^+ + OD^-$) is 3×10^{-15}. H_2O dissociates about
(a) three times as much as D_2O does
(b) thirty times as much as D_2O does
(c) 0.3 times as much as D_2O does
(d) 300 times as much as D_2O does

45. The order of the heats of fusion of T_2, D_2 and H_2 is
(a) $T_2 > D_2 > H_2$ (b) $H_2 > T_2 > D_2$
(c) $D_2 > T_2 > H_2$ (d) $D_2 = T_2 > H_2$

46. Which of the following statement is correct? NaO_2 has a
(a) graphite-like structure
(b) rock-salt (Na*Cl*)-like structure
(c) pyrite (FeS_2)-like structure
(d) fluorite-like structure

47. Acetone exhibits keto-enol tautomerism

$$CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-CH_3 \rightleftharpoons CH_5-\overset{\overset{\displaystyle OH}{|}}{C}=CH_2$$

Which of the following products is obtained when acetone is treated with an excess D_2O for a sufficient time in presence of small amount of dilute NaOH solutions?

(a) $CH_3-\overset{\overset{\displaystyle OD}{|}}{C}=CH_2$ (b) $CH_2D-\overset{\overset{\displaystyle OH}{|}}{C}=CH_2$

(c) $CD_3-\overset{\overset{\displaystyle O}{\|}}{C}-CD_3$ (d) $CH_3-\overset{\overset{\displaystyle OH}{|}}{C}=CD_2$

48. In which of the following compounds does deuteration take place easily on treating with D_2O
(a) $CH_3\ CH_2OH$ (b) $CH_3 - CH_3$
(c) $CH_3\ CH_2\ O\ CH_2\ CH_3$ (d) $CH_3\ CH_2\ Cl$

49. Which of the following statements is wrong?
(a) water is amphoteric
(b) water acts as an oxidizing agent
(c) water acts as a reducing agent
(d) sodium hydride is insoluble in water

50. Hydrogen molecule differ from chlorine molecule in which of the following property is correct
(a) Hydrogen molecule is polar while chlorine molecule is non-polar
(b) Hydrogen molecule can form intermolecular hydrogen bonds but chlorine molecule does not
(c) Hydrogen molecule cannot participate in coordination bond formation but chlorine molecule can
(d) Atomicity of both hydrogen and chlorine is same

51. Hydride ion reacts with water liberating hydrogen gas
$H^- + H_2O \longrightarrow H_2 + OH^-$
This reaction indicates that
(a) Hydride ion reduces water to hydrogen.
(b) Hydride ion oxidizes water to hydrogen.
(c) Hydride ion displaces H^+ ion from water.
(d) Hydride ion being stronger base than OH^- takes up H^+ ion from water.

52. A sample of water containing some dissolved table sugar and common salt is passed through an organic cation exchange resin. The resulting solution contain
(a) both sugar and common salt
(b) only sugar
(c) only pure water
(d) sugar, glucose and fructose

53. When formaldehyde is treated with hydrogen peroxide in the presence of alkaline pyrogallol, it the produces light. This is due to
(a) Reduction of hydrogen peroxide forming H_2 and chemiluminescence
(b) Oxidation of formaldehyde forming O_2 and chemiluminescence
(c) Oxidation of hydrogen peroxide forming H_2 and phosphorescence
(d) Reduction of formaldehyde forming O_2 and phosphorescence

54. The hydride having highest electric conductance is
(a) LiH (b) BeH_2
(c) CaH_2 (d) TiH_2

Multiple Choice Questions with One or More than One Answer Type Questions

1. Which of the following are correct?
(a) Ordinary water is electrolyzed more rapidly than D_2O.
(b) Reaction between H_2 and Cl_2 is much faster than D_2 and Cl_2.
(c) D_2O freezes at lower temperature than H_2O.
(d) Bond dissociation energy for D_2 is greater than H_2.

2. Select the correct statements.
(a) Ortho- and parahydrogens are different due to difference in their nuclear spins.
(b) Ortho- and parahydrogens are different due to difference in their electron spins.
(c) Parahydrogen has a lower internal energy than that of orthohydrogen.
(d) Parahydrogen is more stable at lower temperature.

3. H_2O_2 is "5.6 volume" then
(a) It is 1.7% weight by volume
(b) It is 1 N
(c) It is 1 M
(d) It is 5.6 M

4. Which of the following compounds gives H_2O_2 on action with acids?
(a) Na_2O_2 (b) BaO_2
(c) PbO_2 (d) K_2O_2

5. Hydrogen peroxide can be tested with
(a) A filter paper with a stain of black PbS
(b) An acidified solution of TiO_2
(c) KI + starch solution
(d) Acidified $K_2\ Cr_2O_7$ solution with little ether

6. Correct formula of exhausted permutit is /are
(a) $Na_2Al_2Si_2O_8 \cdot xH_2O$ (b) $MgAl_2Si_2O_8 \cdot xH_2O$
(c) $CaAl_2Si_2O_8 \cdot xH_2O$ (d) $K_2Al_2Si_2O_8 \cdot xH_2O$

7. Which of the following statements about H_2O_2 are true?
(a) H_2O_2 is used to clean oil paintings.
(b) H_2O_2 acts as oxidizing as well as reducing agent.
(c) Two hydroxyl groups in H_2O_2 lie in the same plane.
(d) It retains same structure in liquid and solid form with different bond parameters.

8. Which of the following statements are correct?
(a) The crystal lattice of ice is mostly formed by covalent as well as hydrogen bonds.
(b) The density of water increases when heated from 0°C to 4°C due to the change in the structure of the cluster of water molecules.
(c) Above 4°C, the thermal agitation of water molecules increases. Therefore, intermolecular distance increases and water starts expanding.
(d) The density of water increases from 0°C to a maximum at 4°C and the entropy of the system increases.

9. Atomic hydrogen is more reactive rather than molecular hydrogen. This is because
(a) Atomic hydrogen contain unpaired electron while molecular hydrogen contain paired electrons.
(b) Atomic hydrogen can participate in reactions directly while molecular hydrogen participates in reaction only after converting into atoms.
(c) The bond dissociation energy of molecular hydrogen is very high.
(d) Use of catalysts (often transition metals) makes dihydrogen more reactive either by reaction with it or by weakening H–H bond.

10. The correct statements among the following are
(a) H_2 is more rapidly adsorbed on solid surface than D_2.
(b) H–H bond dissociation energy is greater than D–D bond dissociation energy.
(c) Deuterated compound can be easily prepared by exchange reactions of deuterium with hydrogen using heavy water rather than D_2.
(d) The bond length is shortest in T_2 among the isotopes of hydrogen.

11. Which of the following statements are correct?
(a) Parahydrogen has lower internal energy than ortho- hydrogen.
(b) The hydrogen gas coming out of a tube packed with charcoal cooled to liquid air contain more parahydrogen.
(c) The difference in the properties such as b.pt, specific heat, thermal conductivity of ortho and parahydrogens is due to difference in their internal energy.
(d) Ortho-para conversions of hydrogen are catalyzed by certain transition metals and paramagnetic substances or ions.

12. The correct statements among the following is
(a) Ionic hydrides (e.g., NaH) have high melting points than the corresponding pure metals.
(b) The density of ionic hydrides is greater than that of the metal from which they are formed.
(c) The polymeric form of BeH_2 contain three centred two electron bonds and all the atoms are in the same plane.
(d) The density of metallic hydrides is less than the metal from which they are prepared.

13. Non-stoichiometric hydrides
(a) are solid solutions
(b) are formed by dissolving various amounts of H_2
(c) are brittle and finely powdered metal hydrides can be prepared
(d) can act as good catalysts

14. Among the following, the correct statement is
(a) Among lanthanides only europium and ytterbium form ionic hydrides similar to CaH_2.
(b) The elements of manganese, iron and cobalt group do not form hydrides.
(c) Metallic hydrides on heating converts into finely divided powder with liberation of hydrogen.
(d) Magnesium hydride is an intermediate hydride.

15. Regarding the softening of water, which of the following statements is/are correct
(a) Lime softening is used to remove temporary hardness.
(b) Calgon ‘sequester’ the magnesium and calcium ions present in hard water.
(c) Water softened by exchange resin method is pottable water.
(d) Washing soda removes both temporary and permanent hardness.

16. The correct statements among the following are
(a) The density of H_2O_2 is greater than H_2O.
(b) H_2O_2 is a better ionizing solvent than H_2O due to its high dipole moment.
(c) Boiling point of H_2O_2 is more than H_2O because of more hydrogen bonding in H_2O_2.
(d) On cooling water freezes earlier than H_2O_2.

17. Identify the correct statements (s) regarding hydrogen gas
(a) Unlike H_2, D_2 do not show spin isomerism.
(b) Orthohydrogen differs from parahydrogen w.r.t their b.pt, sp. heat and thermal conductivity.
(c) As the temperature is increased from absolute zero some of the para form gradually changes to ortho form.
(d) Parahydrogen has lower energy than ortho-hydrogen

18. In which of the following reaction(s) H_2O_2 can convert the underlined atom into the product with oxidation state = +6?
(a) $\underline{Cr}Cl_3 + H_2O_2 + OH^- \rightarrow$
(b) $K_2\underline{Cr_2}\,O_7 + H_2O_2 + H_2SO_4 \rightarrow$
(c) $\underline{SO_2} + H_2O_2 + OH^- \rightarrow$
(d) $H_2O_2 + Na_3\,\underline{AsO_3} + OH^- \rightarrow$

19. When zeolite which is hydrated sodium aluminium silicate is treated with hard water the sodium ions are exchanged with
(a) H^+ ions (b) Ca^{2+} ions
(c) SO_4^{2-} ions (d) Mg^{2+} ions

20. Which of the following statements(s) is/are correct?
(a) $Be_2C + H_2O \longrightarrow$ Marsh gas
(b) $Al_4C_3 + H_2O \longrightarrow$ gas is a content of CNG

(c) $CaC_2 + H_2O \longrightarrow$ gas used for welding purpose with O_2 gas

(d) $Ca_3P_2 + H_2O \longrightarrow$ gas used in Holme's signals with CaC_2

Comprehension Type Questions

Passage-I

Concentration of H_2O_2 is expressed in terms of volume strength, e.g., 10 volume, 15 volume, 20 volume, etc. H_2O_2 solution. It represents the volume of oxygen in mL obtained at NTP by the decomposition of 1 mL. of that H_2O_2 solution. For example, 20 volume of H_2O_2 solution means 1 mL of this solution on decomposition evolves 20 mL. of O_2 at NTP.

However, sometimes the concentration of H_2O_2 in a solution is expressed as percentage of H_2O_2 in solution (w/v). Thus 30 per cent solution of H_2O_2 means 30 grams of H_2O_2 are present in 100 mL of water.

1. Find the volume of oxygen gas liberated at STP, when 25 mL of '30 volumes' H_2O_2 is completely decomposed
(a) 30 mL (b) 900 mL
(c) 250 mL (d) 750 mL

2. Determine the volume of O_2 obtained at NTP by the decomposition of 6.8 g of H_2O_2
(a) 2.24 *l* (b) 22.4 *l*
(c) 224 *l* (d) 5 volume

3. Calculate the volume strength of 13.6 per cent solution of H_2O_2
(a) 10 volume (b) 20 volume
(c) 22.4 volume (d) 44.8 volume

Passage-2

Hydrogen peroxide is stable than H_2S_2, it is prepared by the electrolysis followed by hydrolysis of 1:1 sulphuric acid. It is also prepared by autooxidation process. It acts as an oxidizing agent as well as a reducing agent. The decomposition of hydrogen peroxide increases in the presence of metals and some metal oxides. The strength of H_2O_2 is expressed in 'volumes'.

1. 25 mL of hydrogen peroxide after acidification were added to excess of acidified potassium iodide solution. The iodine so liberated required 20 mL of 0.1 N sodium thiosulphate solution. Then, the normality of H_2O_2 is
(a) 0.05 (b) 0.06 (c) 0.08 (d) 0.5

2. When acidified potassium permanganate reacts with hydrogen peroxide, oxygen gas is liberated. Then, the correct statement is
(a) There is no O–O bond breaking take place
(b) There is O–O bond breaking take place
(c) O_2 liberated in this reaction from H_2O_2 as well as solvent used in the reaction
(d) O_2 liberated in this reaction from only solvent used in the reaction

3. Na_2O_2 is the salt of H_2O_2. It is used in qualitative analysis to identify the
(a) Ni^{2+} salts (b) Cr^{3+} salts
(c) Fe^{2+} salts (d) Pb^{2+} salts

Passage-3

Soap is the sodium salt of a long chain fatty acid. Sodium salt of stearic acid is soluble in water, whereas calcium and magnesium salts are insoluble, so when soap is added to soft water, it dissolves and forms lather readily. Temporary hardness in water can be easily removed by boiling. Permanent hardness can be removed by the addition of washing soda, Calgon process, ion exchange process.

1. Temporary hardness of water is due to the presence of
(a) $Ca(HCO_3)_2$ (b) $CaCl_2$
(c) $CaSO_4$ (d) $MgSO_4$

2. Calgon contains
(a) Ca^{2+} ion (b) Mg^{2+} ion
(c) Tetramer of phosphate anion
(d) Hexamer of metaphosphate anion

3. The degree of hardness of a sample of water containing 6 mg of magnesium sulphate (other salts are absent) per kilogram of water is
(a) 10 ppm (b) 5 ppm
(c) 15 ppm (d) 6 ppm

Matching Type Questions

1.

Column-I	Column-II
(a) $K_2Cr_2O_7 + H_2SO_4 + H_2O_2 \longrightarrow$	(p) Oxidizing property of H_2O_2
(b) $KMnO_4 + H_2SO_4 + H_2O_2 \longrightarrow$	(q) Reducing property of H_2O_2
(c) $NaOH + H_2O_2 \longrightarrow$	(r) One of the product has a tetrahedral geometry
(d) $PbS + H_2O_2 \longrightarrow$	(s) Liberated gas acts as a supporter of combustion

2.

Column-I (Type of hydride)	Column-II (Example)
(a) Ionic hydrides	(p) B_2H_6, CH_4, H_2O, SiH_4
(b) Covalent hydrides	(q) CuH, ZnH_2, CdH_2, HgH_2
(c) Metallic hydrides	(r) ScH_2, TiH_2, VH_2, ZrH_2
(d) Intermediate hydrides	(s) NaH, KH, CaH_2, SrH_2

3.

Column-I	Column-II
(a) Intermediate hydrides	(p) CaH_2, BaH_2
(b) Covalent hydrides	(q) CsH, RbH
(c) Salt-like hydrides	(r) AsH_3, SeH_2
(d) Interstitial hydrides	(s) TiH_n, PdH_n
	(t) $(BeH_2)_n$, MgH_2

4.

Column-I	Column-II
(a) Oxidant	(p) CsH
(b) Reductant	(q) Liquid hydrogen
(c) Bleaching agent	(r) H_2O
(d) Rocket fuel	(s) H_2O_2

Numerical Answer Type Question

1. The mass in grams of available oxygen per litre from a solution of H_2O, 10 mL of which when titrated with N/20 $KMnO_4$ solution required 25 mL in acid medium.
2. The no. of oxygen atoms involved in –1 oxidation state in chromium peroxide is
3. In the old oil painting basic lead carbonate was used as white pigment. This white colour slowly turns to black when exposed to air. These paintings are regenerated by washing with hydrogen peroxide to get white colour. In that process the no. of electrons involved in changing the black compound to white
4. The no. of lattice water molecules in hydrated barium chloride is
5. In solid ice the number of oxygen atoms bonded to oxygen atom of a water molecule through hydrogen atoms.
6. Total number of isotopic isomers of hydrogen peroxide will be equal to
7. How many of the following give different compounds having same molecular weight equal to heavy water on reaction with heavy water. Mg_3N_2, BeC_2, Al_4C_3, CaC_2, Ca_3P_2, AlN, SiF_4, $AlCl_3$?

Comprehension Type Question

Passage-1	**1.** d	**2.** a	**3.** d
Passage-2	**1.** c	**2.** a	**3.** b
Passage-3	**1.** a	**2.** d	**3.** b

Matching

1.	a-q, r, s	b-q, r, s	c-r, s	d-p, r
2.	a-s	b-p	c-r	d-q
3.	a-t	b-r	c-p, q	d-s
4.	a-r, s	b-p, r, s	c-s	d-q, s

Numerical Answer Type Questions

1. 2 **2.** 4 **3.** 8 **4.** 2
5. 4 **6.** 6 **7.** 4

Hydrogen and its Compounds Single Answer Quesitons

1. d	**2.** a	**3.** a	**4.** d	**5.** c	**6.** d
7. b	**8.** a	**9.** a	**10.** a	**11.** b	**12.** b
13. b	**14.** b	**15.** a	**16.** d	**17.** b	**18.** a
19. c	**20.** d	**21.** b	**22.** b	**23.** b	**24.** a
25. b	**26.** b	**27.** a	**28.** d	**29.** c	**30.** d
31. c	**32.** c	**33.** a	**34.** c	**35.** b	**36.** a
37. c	**38.** a	**39.** b	**40.** c	**41.** b	**42.** a
43. d	**44.** a	**45.** a	**46.** a	**47.** c	**48.** a
49. d	**50.** d	**51.** d	**52.** d	**53.** a	**54.** d

More than One Answer Questions

1. a, b, d	**2.** a, c, d	**3.** a, b
4. a, b, d	**5.** a, b, c, d	**6.** b, c
7. a, b, d	**8.** a, b, c, d	**9.** a, b, c
10. a, c	**11.** a, b, c, d	**12.** a, b, d
13. a, b, c, d	**14.** a, b, c, d	**15.** a, b, d
16. a, b, c, d	**17.** b, c, d	**18.** a, c
19. b, d	**20.** a, b, c, d	

Hints

1. H_2O_2 react with the compounds dissolved in it due to its oxidation, reduction, acidic and forming addition compound properties.
2. Due to intermolecular hydrogen bonding bond parameters changes.
3. When sodium reacts with ethyl alcohol some hydrocarbon gas is also formed along with H_2.
5. Polyphosphates form soluble complexes with Mg^{2+} and Ca^{2+} present in hard water.

6. $AgNO_3 + NaOH \longrightarrow AgOH + NaNO_3$

$$2AgOH \longrightarrow Ag_2O + H_2O$$

$$Ag_2O + H_2O_2 \longrightarrow \underset{\text{Black}}{2Ag} + H_2O + O_2$$

7. Normality of $H_2O_2 = \dfrac{20 \cdot 0.1}{50} = 0.04$ N

wt. of $H_2O_2 = V \times N \times E = 1 \times 0.04 \times 17 = 0.68$ g/lit

8. $H_2O_2 \longrightarrow 2H^+ + O_2 + 2e^-$

10. When equal volumes of 10 V, 15 V and 20 V are mixed the volume strength will be 15 V. Since the volume of solution is doubled the vol. strength is 7.5.

11. $\underset{162}{Ca(HCO_3)_2} + \underset{56}{CaO} \longrightarrow 2CaCO_3 + H_2O$

$Ca(HCO_3)_2$ present in 1000 lit = $1.62 \times 1000 = 1620$ g

$\therefore$ 560 g of CaO is required.

13. Salts are less soluble in heavy water than in water due to low dielectric constant.

14. According to Le chatelier's principle when pressure is increased equilibrium shifts in the direction of lesser volume.

15. Isotopic effect

18. $Ca_3P_2 + 3H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3 \uparrow$

19. If water molecule is linear it is non-polar and its dielectric constant will be zero. So, b.pt and m.pt decreases.

21. While $KMnO_4$ acts as oxidizing agent, in acid medium $5e^-$ and in alkaline medium $3e^-$ are involved. So, $100 \times 0.01 \times 5 = 0.05 \times 10$ acid medium

$KMnO_4$ H_2O_2

$V_2 \times 0.01 \times 3 = 0.05 \times 100$

$\therefore V_2 = \dfrac{500}{3}$ mL

26. $Mg_3N_2 + 6D_2O \longrightarrow 3Mg(OD)_2 + 2ND_3$

28. O–D bond is stronger than O–H bond.

30. $2KMnO_4 + 8H_2SO_4 + 5Na_2O_2 \longrightarrow K_2SO_4 + 5Na_2SO_4 + 2MnSO_4 + 8H_2O + 5O_2$

31. $2Na_2O_2 + 2H_2O \longrightarrow 4NaOH + O_2$

$\therefore$ 0.125 mole

33. 0.56 g CaO 1.0 g of $CaCO_3$

$\therefore$ Hardness of 1 litre = 100 ppm

35. Zn with conc. H_2SO_4 liberates SO_2 and with dil. HNO_3 liberates NO. Pure Zn do not liberate H_2 with dil. H_2SO_4 due to hydrogen over voltage.

37. Only cations are removed by Res. COOH.

38. Stability of alkali metal hydrides decreases with increase in the size of alkali metal ion due to decrease in lattice energy.

39. PbO_2 is a dioxide. It does not give H_2O_2.

42. Hardness is only due to Ca^{2+} and not due to Na^+ or K^+.

44. $\dfrac{1 \times 10^{-14}}{3 \times 10^{-15}} = 3.3$ times.

45. With increase in mass, heat of fusion increases.

48. With increasing ionic character exchange of H with D takes place easily.

50. H_2 cannot form coordinate bond due to absence of lone pair electrons.

52. When water containing salt and sugar is passed through cation exchange resin Na^+ ions are exchanged with H^+ ion and the water get acidic character. In acid medium inversion of sugar gives glucose and fructose.

54. Metallic hydrides have more electrical conductivity due to metallic property.

More than One Answer Questions

1. (1) O–D bond is stronger than O–H. So, ordinary water decompose rapidly on electrolysis, (2) the activation energy of D_2 and Cl_2 is more than H_2 and Cl_2 and (3) Freezing pt of D_2O is more than H_2O (4) D–D bond energy is greater than H–H.

9. Increase in the reactivity of H_2 in the presence of catalyst is not a reason for the more reactivity of atomic hydrogen.

10. Due to more rate of diffusion, H_2 will be adsorbed rapidly than D_2 on solid surface. Since O–D bond is polar while D–D bond is non-polar. So, exchange reactions with D_2O is easier. Atomic sizes of H_2, D_2 and T_2 are same.

11. (1) pairing of protons in para H_2 lowers internal energy. (2) At low temp. para H_2 is formed more. The remaining statements are also correct.

12. In BeH_2 due to sp^3 hydridization, in Be all atoms are not in the same plane.

15. Water softened by ion exchange resin is completely free from minerals and is not useful for drinking purpose.

16. H_2O_2 reacts with the compounds dissolved in it.

17. Both H_2 and D_2 show spin isomerism.

18. $CrCl_3$ is converted to CrO_4^{2-} and SO_2 is converted to SO_3 in which the oxidation state of Cr and S is +6 but no change in reaction with $K_2Cr_2O_7$ and in the case of Na_3AsO_4 product, formed from Na_3AsO_3 is +5.

Comprehension Type Questions

Passage-I

1. $25 \times 30 = 750$ mL

2. 68 g of H_2O_2 liberate 22.4 lit of O_2 at STP.
6.8 g of H_2O_2 liberates 2.24 lit of O_2 at STP.

3. Vol. strength of 3.4% = 11.2 v
$\therefore$ 13.6% = 44.8v

Passage-2

1. $\frac{20 \cdot 0.1}{25} = 0.08$ N.
2. $H_2O_2 \longrightarrow 2H^+ + O_2 + 2e^-$
   while acting as a reducing agent there is no O–O bond breaking.
3. Na_2O_2 oxides Cr (III) salts yellow chromates.

Passage-3

1. Temporary hardness is due to bicarbonates of Mg and Ca.
2. Calgon is sodium hexametaphosphate.
3. 120 g $MgSO_4 \equiv$ 100 g of $CaCO_3$
   $\therefore$ 6 mg/kg = 5mg/kg or 5 ppm

Numerical Answer Type Questions

1. Normality of $H_2O_2 = \frac{1}{20} \cdot \frac{25}{10} = 0.125$
   1 gm eq. contain __________ 16 g
2. **fig P (101)**
   4 oxygen atoms present in peroxy bonds are in –1 oxidation state.
3. $PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$
   The oxidation state of sulphur changes from –2 in PbS to +6 in $PbSO_4$. So, 8 electrons.
4. $BaCl_2 \cdot 2H_2O$
5. In ice, every water molecule is surrounded by four water molecules and thus 'O' atom is in bond with four other oxygen atoms (O – H …..O or O ……. H – O)
6. H–O–O–H, H–O–O–D, D–O–O–D,
   H–O–O–T, T–O–O–T, D–O–O–T
7. Mg_3N_2, AlN on hydrolysis give ND_3 having M.W. 20
   BeC_2, Al_4C_3 on hydrolysis give CD_4 having M.W. 20
   SiF_4 on hydrolysis give D_2SiF_6 and DF: DF have M.W. 21

5

CHAPTER

s-Block Elements
Group-IA (1): Alkali Metals

This is either a dissolution of the mutual involutions of the atoms, or a similar dispersion of the unsentient elements.

Marcus Aurelius

5.1 s-BLOCK ELEMENTS

The s-block elements are those in which the differentiating electron (last electron) enters into s-orbital. These elements do not possess any electron in (n-1) d orbitals, whereas their (n-1)s- and p-orbitals are completely filled up. The elements of Groups I A and IIA (Groups 1 and 2) of the periodic table constitute the s-block elements. The general electronic configuration of s-block elements is [noble gas] ns^1 for Group IA metals and [noble gas] ns^2 for Group II A metals.

5.2 GROUP-I: ALKALI METALS

The elements lithium (Li), sodium (Na), potassium (K), rubidium (Rb), caesium (Cs) and francium (Fr) constitute the Group IA. These elements are also called as the **alkali metals**. The ashes of the plants contain the compounds of sodium and potassium in large quantities. In Arabic, 'alkali' means **ash**. This ash when dissolved in water forms strongly basic solution due to the formation of hydroxides of sodium and potassium. So, the Group IA elements are called alkali metals (the basic character is referred to as alkaline character). Sodium is first used as a remedy to split headache. In Arabic, 'soda' means to "split apart". So it was named as sodium. The ashes of plants are rich with potassium carbonate. Potash means ashes of plants and hence it was named as potassium. Rubidium exhibits beautiful ruby colour in its spectral lines and hence it is named as rubidium. Caesium exhibits sky blue colour in its spectrum after the bright blue lines. So it was named as caesium, which means sky blue. At the suggestion of Madam curie, M.M. Perey named the last element as francium after her native land when she discovered it in 1939. Not much is known about francium since it is radioactive and all its isotopes are extremely short-lived. It is formed during the radioactive decay of actinium.

The alkali metals form a homogeneous group of extremely reactive elements which illustrate well the similarities and trends to be expected from the periodic classification, as discussed in Chapter 2. Their physical and chemical properties are readily interpreted in terms of their simple electronic configuration ns^1. Compounds of sodium and potassium have been known from ancient times and both elements are essential for animal life. They are also major items of trade, commerce and chemical industry. Lithium was first recognized as a separate element at the beginning of the nineteenth century but did not assume major industrial importance until about 50 years ago. Rubidium and caesium are of considerable academic interest but so far have few industrial applications.

5.3 OCCURRENCE

The alkali metals are highly reactive and hence do not occur in free state. Since these are most electropositive elements they occur in nature as cations in combination with most electronegative anions like chloride ions. Sodium occurs as sodium chloride in sea water and in rock salt and also as sodium nitrate in Chile salt petre. Sodium also occurs as carbonate (trona), sulphate (mirabilite) borate (boraxy kernite), etc. Potassium occurs as simple chloride (Sylvine), Carnalite $KCl \cdot MgCl_2 \cdot 6H_2O$. Lithium, rubidium and caesium are present in minute quantities in some alumino silicates.

Among the six elements, sodium and potassium have been the sixth and seventh most abundant elements in the earth's crust. Note that, although Na and K are almost equally abundant in the crystal rocks of the earth Na is 30 times as abundant as K in the oceans. This is partly because K salts with the larger anions tend to be less soluble than Na salts, likewise K is more strongly bound to complex silicates and alumino silicates in the soils. Again, K leached from rocks is preferentially absorbed and used by plants, whereas Na can proceed to the sea. The other alkali metals are much less common.

5.4 GENERAL CHARACTERISTICS OF ALKALI METALS

5.4.1 Physical Properties

1. Electronic configuration of Alkali metals: Electronic configurations of alkali metals are listed in Table 5.1.

From Table 5.1, it follows that atoms of alkali metals have one electron in s-orbital outside a noble gas core. Therefore, their general electronic configuration of the outer valency shell is ns^1 where n = 2 to 7. As all these elements have similar outer valency shell electronic configuration, they have similar physical and chemical properties described below.

The important atomic and physical properties of alkali metals are listed in Table 5.2.

Only first five members of alkali metals are discussed here because little is known about the last member which is a radioactive element. An attempt has been made to make a correlation between atomic structure and physical properties of alkali metals on the basis of the following two facts.

(i) Loose binding of s-electrons and
(ii) Size of alkali metal atoms and ions.

The various consequences of these two facts are described as follows.

2. Atomic and ionic Radii: The atoms and ions of alkali metals are largest in their corresponding periods. Atomic size increases from lithium to caesium. This is due to the presence of one extra shell of electron as we move down from one element to the other.

When an alkali metal loses its outer s-electron it gets converted into a positive ion. This results in the elimination of the outer shell. At the same time, the charge on the nucleus becomes higher than the number of electrons and therefore, electrons get attracted towards the nucleus more strongly. As a result of both these effects, a positive ion (cation) becomes smaller than the corresponding atom.

Atomic as well as ionic size got increased from Li to Cs because of the presence of an extra shell of electrons. Atomic volume (atomic wt/density) also gets increased on moving down the group from Li to Cs.

3. Physical state: These elements are silvery white, soft and light metals. These are highly malleable and ductile. The silvery lustre is present when freshly cut but fades rapidly as surface oxidation takes place. They have high electrical conductances. All these properties can be explained in terms of their structure.

At ordinary temperatures all the Group IA metals adopt body-centred cubic lattice structure, with a coordination number 8. Lithium at low temperatures forms hexagonal close packing structure with coordination number 12.

Group IA metals can contribute their only one valence electron to the metallic bond, the cohesive energy (the binding energy of the atoms or ions together) in the solid is very less. Further, decreases due to the diffuse nature of the outer bonding electron with increase in size on descending the group from Li to Cs. So the bonds are weaker, the cohesive energy decreases and the softness of the metals increases.

When light falls on these metals, the highly mobile electrons of the metallic lattice absorb energy and get excited. The excited electrons while coming to the ground state lose electromagnetic energy in the form of light. In other words the beam of light gets reflected from the surface of metal. Thus the light sets their electrons in oscillatory motion which gives the silvery lustre to the metals.

4. Density: Alkali metals have low density. The reason for this is that they have large atomic sizes. Density gradually increases on moving down the group from Li to Cs.

Table 5.1 Electronic configuration of alkali metals

Element	Atomic number	Electronic configuration	Electronic configuration of valency shell
Lithium	3	$1s^2\ 2s^1$	$2s^1$
Sodium	11	$1s^2\ 2s^2\ 2p^6\ 3s^1$	$3s^1$
Potassium	19	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$	$4s^1$
Rubidium	37	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 5s^1$	$5s^1$
Caesium	55	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^6\ 6s^1$	$6s^1$
Francium	87	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 4f^{14}\ 5s^2\ 5p^6\ 5d^{10}\ 6s^2\ 6p^6\ 7s^1$	$7s^1$

Table 5.2 Atomic and physical properties of alkali metals

Property	Lithium Li	Sodium Na	Potassium Rb	Rubidium K	Caesium Cs
Atomic number	3	11	19	37	55
Atomic mass (g mol^{-1})	6.94	22.991	39.10	85.47	132.91
Density at 20°C/gcm^{-3}	0.534	0.968	0.859	1.532	1.903
Melting point (K)	453	370.8	336.7	312	301.6
Boiling point (K)	1603	1156	1032	961	944
Specific heat at 273(K)	0.941	0.293	0.17	0.08	0.049
Atomic radius (pm)	152	186	227	248	265
Ionic radius M^+ (pm) (6- coordinate)	76	102	138	152	167
Heat of atomization $kJmol^{-1}$	162.22	108.58	89.6	82	78.2
ΔH_{fus}/kJ mol^{-1}	2.93	2.64	2.39	2.20	2.09
ΔH_{vap}/kJ mol^{-1}	148	108	89.6	82.0	78.2
Ionization energy/ kJ mol^{-1}	520.2	495.8	418.8	403.0	375.7
Electron affinity kJ mol^{-1}	59.8	52.9	46.36	46.88	45.5
Electronegativity	1.0	0.9	0.8	0.8	0.7
Flame colouration	Crimson red	Golden yellow	Lilac (Pale violet)	Red violet	Blue violet
λ/nm	670.8	589.2	766.5	780	455.5
Standard potentials E°/V for (M^+/M)	–3.045	–2.714	–2.925	–2.925	–2.923

However, potassium is lighter than sodium (anomaly). This is because that in potassium atom the penultimate shell is incompletely filled and also due to the bigger size, the interstitial spaces of potassium crystal lattice will become more.

5. Melting and boiling points: The melting and boiling points of alkali metals are very low due to the weak metallic bonding. With the increase in size of the metal atoms the metallic bond strength decreases. Thus, the melting and boiling points decreases down the group from Li to Cs.

6. Ionization enthalpy: Alkali metals have very low ionization enthalpies because of their larger atomic sizes. Further, as the atomic size increases on moving down the group, the outer electron gets farther and farther away from the nucleus and therefore, ionization enthalpy gets decreased on moving down from Li to Cs.

The second ionization enthalpy of alkali metals are fairly high because, the second electron is to be removed from the stable noble gas core.

7. Electropositive character or metallic character: All the alkali metals are highly electropositive in character because of their low ionization enthalpies. They can lose their valence electron easily. The tendency to lose electron is known as **metallic character or electropositive character.** The electropositive character increases down the group from Li to Cs because ionization enthalpy decreases down the group.

8. Formation of unipositive ions and oxidation states: All the alkali metals can convert into unipositive (M^+) ions by losing their only valence electron and exhibit an oxidation state +1. As the second ionization enthalpy is very large there is no possibility for the formation of M^{2+} ion. So, all alkali metals are univalent and form ionic compounds.

However, some degree of covalent character also occurs in certain cases, i.e., their vapours contain covalently bonded diatomic molecules Li_2, Na_2, etc. In certain organometallic compounds also they involve in covalent bonding, e.g., CH_3Li, C_2H_5Li, $C_6H_5CH_2Na$, etc. Strength of the covalent bond in alkali metals decreases down the group.

All the compounds of alkali metals are colourless and diamagnetic because in their ions all the electrons are paired. The colour of the compounds such as permanganates, dichromates, etc. is due to anion but not due to alkali metal ion.

9. Electronegativity: Alkali metals have low electronegativity values because they have more tendency to lose electron rather than to gain an electron. The electronegativity values decreases down the group from Li to Cs.

10. Effect of light: Alkali metals when irradiated with light emit electrons with ease due to low ionization enthalpies. This phenomenon is used in photoelectric cells, particularly caesium and potassium are used as electrodes in photoelectric cells.

11. Hydration energies: The alkali metal ions are highly hydrated. The smaller the ionic size, the higher the degree of hydration. Since the Li^+ ion is very small, it is heavily hydrated. Some water molecules touch the metal ion and bond to it forming a complex. These water molecules constitute the **primary shell** of water. Thus Li^+ is tetrahedrally surrounded by four water molecules using its four sp^3 hybrid orbitals as per the valence bond theory. The Na^+ and K^+ ions may also have fourfold primary hydration in aqueous solutions. In the heavier ions, particularly Rb^+ and Cs^+, six water molecules surround each metal ion octahedrally by using d-orbitals of their valence shell (sp^3d^2 hybridization) in the primary layer. A **secondary layer** of water molecules further hydrates the ions by the weak ion-dipole attractive forces. The secondary hydration decreases from lithium to caesium because the strength of ion-dipole attractive forces decreases with increase in distance due to increase in size of the metal ion.

The hydration of ions is an exothermic process. The energy released in the hydration of ions is termed as hydration energy. As the degree of hydration of M^+ ions decreases as we go down the group, the hydration energy of alkali metal ions gets decreased from Li^+ to Cs^+.

12. Ionic mobilities: All the simple salts dissolve in water, producing ions and consequently the solution conduct electricity. Since Li^+ ions are small, it might be expected that the solutions of lithium salts would conduct electricity better than the solutions of the same concentration of sodium, potassium, rubidium or caesium salts. The small ions should migrate more easily towards the cathode and thus conduct more than the larger ions. But in aqueous solutions the degree of hydration decreases from Li^+ to Cs^+ due to increase in size. Because of this reason the ionic radii of hydrated alkali metal ions also decreases from Li^+ to Cs^+. It implies that Li^+ ion which must be smallest is actually largest in size in water. So, it moves slowly in water. In contrast, Cs^+ is the least hydrated, and the radius of the hydrated Cs^+ ion is smaller than the radius of hydrated Li^+ and hence hydrated Cs^+ moves faster and conducts electricity more readily. So, the order of the electrical conductivity of alkali metal ions in aqueous solution is

$$Cs^+ > Rb^+ > K^+ > Na^+ > Li^+$$

13. Formation of hydrated salts: The crystallization of salts of alkali metals from the aqueous solutions results in the formation of hydrated salts. The decrease in hydration from Li^+ to Cs^+ is also shown in the crystalline salts. Nearly all the lithium salts are hydrated, commonly as trihydrates. In these hydrated Li salts Li^+ is coordinated to $6H_2O$ and the octahedra shares the faces forming chains. Many sodium salts are hydrated, e.g., $Na_2CO_3 \cdot 10H_2O$, $Na_2SO_4 \cdot 10H_2O$, etc. Few potassium salts and no rubidium or caesium salts are hydrated.

14. Flame colours: The alkali metals and their salts, when introduced in bunsen flame, impart characteristic colour to flame. Their characteristic colours, wavelengths and wave numbers are given in Table 5.4.

The heat from the burner excites one of the orbital electrons to a higher energy level. When excited electrons drops back to its original energy level it gives out the extra energy it obtained. The energy E is related to the frequency ν by the Einstein relationship

$$E = h\nu \text{ (h = Planck's constant)}$$

For alkali metals, the energy emitted appears as visible light, thus giving the characteristic flame colourations.

The colour actually arises from electronic transitions of short-lived species which are formed momentarily in the flame. The flame is rich in electrons and in the case of sodium the ions are temporarily reduced to atoms.

$$Na^+ + e^- \rightarrow Na$$

Table 5.3 Ionic mobilities and hydration energies of alkali metal ions

Ion	Ionic radius (pm)	Ionic mobility at infinite dilution	Approximate radius of hydrated ion (pm)	Approximate hydration number	Hydration terms ΔH° (kJ mol⁻)	ΔS°	ΔG°
Li^+	76	33.5	340	25.3	−544	−134	−506
Na^+	102	43.5	276	16.6	−435	−100	−406
K^+	138	64.5	232	10.5	−352	−67	−330
Rb^+	152	67.5	228	10.0	−326	−54	−310
Cs^+	167	68.0	228	9.9	−293	−50	−276

Table 5.4 Flame colours, wavelengths and wave number of alkali metals

Metal	Colour	Wave length (nm)	Wave number (cm^-)
Li	Crimson red	670.8	14908
Na	Golden yellow	589.2	16972
K	Lilac (or) pale violet	766.5	13046
Rb	Red-violet	780.0	12821
Cs	Blue-violet	455.5	21954

The colour of the flame is due to the electronic transition from one energy state to another. The sodium D-line (which is actually a doublet at 589.0 nm and 589.6 nm) arises from the electronic transition $3s^1 \rightarrow 3p^1$ in sodium atoms formed in the flame. The colours from different elements do not all arise from the same transition or from the same species. Thus the red line for lithium arises from a short-lived LiOH species formed in the flame.

Alkali metals can therefore be detected by the respective flame tests and can be determined by flame photometry or atomic absorption spectroscopy.

5.4.2 Chemical Properties

From the lower values of first ionization enthalpies and relatively higher values of second ionization energy, it follows that these elements are very reactive and tend to form M^+ ions. This tendency increases as we move down the group. So, the reactivity also increases down the group. The important chemical properties and their trends are discussed below.

1. Action of Air: All the alkali metals tarnish rapidly on exposure to air due to the formation of oxides, hydroxides and finally carbonates at the surface by the reaction with oxygen, moisture and carbon dioxide of air.

$$4Na + O_2 \rightarrow 2Na_2O$$
$$2Na_2O + 2H_2O \rightarrow 4NaOH$$
$$4NaOH + 2CO_2 \rightarrow 2Na_2CO_3 + 2H_2O$$

They burn vigorously in air forming oxides. Lithium shows exceptional behaviour in reacting with both oxygen and nitrogen forming oxide Li_2O as well as Li_3N. Other alkali metals reacts only with oxygen but not with nitrogen. Lithium forms monoxide, sodium forms peroxide and other alkali metals form super oxides.

2. Action of Water: The alkali metals react readily with water forming hydroxides and liberates hydrogen. For example

$$2Na + 2H_2O \rightarrow 2NaOH + H_2.$$

The reactivity with water increases on descending the group from lithium to caesium. With water, lithium reacts slowly, sodium vigorously, potassium violently, rubidium and caesium explosively. With potassium, rubidium and caesium the reaction is so vigorous that the evolved hydrogen catches fire spontaneously.

The large negative free energy of hydration (ΔG^o) of Li^+ ion (−506) indicates that in the reaction of lithium with water liberates more energy than the other alkali metals. So it might be expected that lithium should react with water vigorously when compared with other alkali metals. But it reacts with water gently while potassium which liberates less energy reacts with water violently and catch fire. This is because potassium having low melting point melts by the heat liberated during the reaction with water. The molten metal spread out and exposes a large surface area to water, so that it can react faster, gets even hotter and catches fire.

3. Reaction with Hydrogen: Alkali metals react with hydrogen to form ionic hydrides M^+H^-. The reaction of alkali metals with hydrogen decreases from Li to Cs.

$$2M + H_2 \rightarrow 2MH \text{ (M = Li, Na, K, Rb or Cs.)}$$

The alkali metal hydrides when dissolved in water yield back hydrogen.

$$MH + H_2O \rightarrow MOH + H_2$$

4. Reaction with Nitrogen: Lithium can react with nitrogen directly forming nitride.

$$6Li + N_2 \rightarrow 2Li_3N$$

Other alkali metals react with nitrogen directly, forming azides

$$2Na(s) + 3N_2(g) \rightarrow 2NaN_3(s)$$

All the alkali metals react with other elements of nitrogen family.

$$3M + P \rightarrow M_3P \text{ (Metal phosphides)}$$
$$3M + As \rightarrow M_3As \text{ (Metal arsenides)}$$
$$3M + Sb \rightarrow M_3Sb \text{ (Metal stibnides)}$$

5. Reaction with Oxygen: On burning in oxygen, lithium forms monoxide Li_2O (along with little peroxide), sodium forms the peroxide Na_2O_2 (and some super oxide), while potassium, rubidium and caesium form the superoxides. Under appropriate conditions pure compounds M_2O, M_2O_2 and MO_2 can be prepared. The principal combustion products of alkali metals are given in Table 5.5.

Now the question arises that, why lithium forms monoxide, sodium forms peroxide and potassium, rubidium and caesium forms superoxides mainly. This can be explained as follows.

As already seen in Chapter 3 (Chemical Bonding) a small anion has a more significant influence on the change in the lattice enthalpy as the cation size changes. The change in lattice enthalpy is relatively small when the parent compound has a large cation initially. When the cation is very big, the change in size of anion barely affects the variation of the lattice enthalpy. Therefore, with a given

Table 5.5 Principal combustion product of alkali metals (minor product)

Alkali metal	Oxide	Peroxide	Superoxide
Li	Li_2O	(Li_2O_2)	-
Na	(Na_2O)	Na_2O_2	(NaO_2)
K	-	-	KO_2
Rb	-	-	RbO_2
Cs	-	-	CsO_2

unstable polyatomic anion, the lattice enthalpy difference is more significant and favourable to decomposition when the cation is small than when it is large.

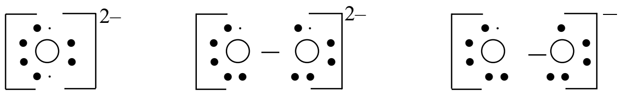
The difference in lattice enthalpy between MO and M_2O_2 or MO_2 is magnified by a larger charge on the cation as lattice enthalpy $\Delta H_L \propto Z^+Z^-/d$. As a result, thermal decomposition of peroxide or superoxide will occur at lower temperatures, if it contains a higher charged cation. Basing on these arguments now we can explain why lithium forms only monoxide, sodium forms peroxide while potassium rubidium and caesium forms superoxides.

Because the small Li^+ ion results in Li_2O having more favourable lattice enthalpy (in comparison to M_2O_2 or MO_2) than Na_2O, the decomposition reaction

$$M_2O_{2(s)} \rightarrow M_2O_{(s)} + \frac{1}{2}O_{2(g)}$$

is thermodynamically favourable for Li_2O_2 than for Na_2O_2. Similarly, the lattice enthalpy is more favourable for the formation of Na_2O_2 than for the formation of NaO_2. Likewise, we can explain the formation of superoxides by potassium, rubidium and caesium.

The structures of oxide, peroxide and superoxide ions can be represented as follows.



Oxide ion **Peroxide ion** **Superoxide ion**

The oxides and peroxides are colourless when pure. The superoxide ion contains one **unpaired electron** due to which the superoxides are **yellow or orange coloured** and are **paramagnetic**.

Alkali metals also react with other elements of the oxygen family forming metal sulphides, selenides and tellurides.

$$2M + S \rightarrow M_2S$$
$$2M + Se \rightarrow M_2Se$$
$$2M + Te \rightarrow M_2Te$$

6. Reaction with halogens: The alkali metals react readily with halogens to form halides. As the electropositive character gets increased from top to bottom in the group, the ease of formation of alkali metal halides gets increased from Li to Cs.

$2M + X_2 \rightarrow 2MX$ (M=Li, Na, K, Rb or Cs, X = F, C*l*, Br or I)

7. Solutions of alkali metals in liquid ammonia: All the alkali metals dissolve in liquid ammonia to form metastable solutions. When ammonia is pure and if there are no impurities evaporation of ammonia from these solution results in the recovery of unchanged alkali metals. The solubility of alkali metals in liquid ammonia is favoured by low metal lattice energy, low ionization energies and high cation solvation energy. The most striking physical properties of the solutions are their colour, electrical conductivity and magnetic susceptibility. The solutions all have the same blue colour when dilute, suggesting the presence of common coloured species. Gradual addition of alkali metal to liquid ammonia first gives blue-coloured solution, which changes to bronze-coloured phase that floats on the dark blue phase. At higher concentrations of alkali metal the solution becomes completely bronze coloured and metallic.

The properties like colour, electrical conductivity and paramagnetism of these solutions have been interpreted in terms of ionization of alkali metals to form alkali metal cations and electrons which are solvated by ammonia.

$$M \xrightarrow[\text{liquid ammonia}]{\text{dissolve in}} \underset{\text{solvated metal cation}}{[M(NH_3)_x]^+} + \underset{\text{solvated electron}}{[e(NH_3)_y]^-}$$

Ammoniated electrons are responsible for the blue colour while the electrical conductances are due to the both ammoniated cations as well as ammoniated electrons. The presence of large number of unpaired electrons is responsible for the paramagnetism. The magnetic susceptibilities correspond with those of calculated for the presence of one electron per metal atom.

The solutions occupy a much greater volume than the sum of volumes of the metal and solvent. So it is suggested that the electrons occupy cavities of radius 300–400 pm. Because of this increase in volume the density of the solutions decrease and hence floats.

As the solutions are made more concentrated the molar conductivity at first decreases reaching a minimum at about 0.05 molar. Again the molar conductances increases if the alkali metal concentration increases more than 0.05 M until it becomes saturated. The molar conductance of saturated solution is comparable to that of metal and they are diamagnetic. The variation of properties with concentration can best be explained in terms of the following three equilibria between five solute species M, M_2, M^+, M^- and e^-.

1. $M_{am} \rightleftharpoons M^+_{am} + e^-_{am}$; $K \sim 10^{-2}$ mol l^{-1}
2. $M^-_{am} \rightleftharpoons M_{am} + e^-_{am}$; $K \sim 10^{-3}$ mol l^{-1}
3. $(M_2)_{am} \rightleftharpoons 2M_{am}$; $K \sim 2 \times 10^{-4}$ mol l^{-1}

The subscript 'am' indicates that the species are solvated with ammonia molecules. At very low concentrations the first equilibrium predominates and the high ionic conductivity stems from the high mobility of the electron, which is some 280 times more than that of the cation. The M^-_{am} in the second equilibrium can be considered as an ion pair in which M^+_{am} and e^-_{am} are held together by coulombic forces. As the concentration increases, the second equilibrium begins to remove mobile electrons e^-_{am} as the complex M^-_{am} and the conductivity drops. Then M_{am} begins to dimerize to give $(M_2)_{am}$ in which the two electrons are paired and the solution becomes diamagnetic.

Reducing properties: Solutions of alkali metals in liquid ammonia are valuable as powerful and selective reducing agents. In the presence of impurities the solutions are unstable and forms amide.

$$M + NH_3 \rightarrow MNH_2 + \frac{1}{2}H_2$$

However, under anhydrous conditions and in the absence of catalytic impurities such as transition metal ions, solutions can be stored for several days with only little decomposition. Some reduction properties of solutions of alkali metals in liquid ammonia are as follows.

(i) It can reduce dioxygen to superoxide and peroxide.

$$O_2 \xrightarrow{e^-_{am}} O_2^- \xrightarrow{e^-_{am}} O_2^{2-}$$

(ii) Transition metal complexes can be reduced to unusally lower oxidation states with or without bond cleavage.

$$K_2[Ni(CN)_4] + 2K \xrightarrow{-33^\circ C/NH_3} K_4[Ni(CN)_4] \text{ i.e., Ni(0)}$$

$$[Pt(NH_3)_4]Br_2 + 2K \xrightarrow{-33^\circ C/NH_3} [Pt(NH_3)_4] \text{ i.e., Pt(0)}$$

$$Mn_2(CO)_{10} + 2K \xrightarrow{-33^\circ C/NH_3} 2K[Mn(CO)_5]; \text{ i.e., Mn(–1)}$$

$$Fe(CO)_5 + 2Na \xrightarrow{-33^\circ C/NH_3} Na_2[Fe(CO)_4] + CO \text{ i.e., Fe(–2)}$$

Solutions of alkali metals in liquid ammonia liberate hydrogen from protonic species.

$$RC \equiv CH + e^-_{am} \rightarrow RC \equiv C^- + \frac{1}{2}H_2$$

$$C_2H_5OH + e^-_{am} \rightarrow C_2H_5O^- + \frac{1}{2}H_2$$

$$NH_4^+ + e^-_{am} \rightarrow NH_3 + \frac{1}{2}H_2$$

These solutions react with many organic compounds and hydrides of non-metals. Some of the typical reactions are given below.

$$C_6H_5-O-C_6H_5 + Na \xrightarrow{NH_3} C_6H_5ONa + C_6H_6$$

$$C_6H_5-CO-C_6H_5 + 2Na \xrightarrow{NH_3} C_6H_5CH_2C_6H_5 + 2NaNH_2 + H_2O$$

$$2PH_3 + 2Na \xrightarrow{NH_3} 2NaPH_2 + H_2$$

Solutions of alkali metals in liquid ammonia have been developed as versatile reducing agents which effect reactions with organic compounds that are otherwise difficult or impossible. Aromatic systems are reduced smoothly to cyclic mono- or di-olefins and alkynes are reduced stereospecifically to *trans*alkenes (in contrast to Pd/H_2) which gives *cis*alkenes.

8. Reducing properties (Reduction potentials): Since the alkali metals have a great tendency to lose their s-electrons they act as **powerful reducing agents**. Reducing power is measured in terms of standard electrode reduction potentials. From Table 5.2 it can be seen that these elements have very low values of E°. These values indicate the strong reducing tendency of these metals.

From the ionization enthalpies of alkali metals, it can be seen that lithium has the highest value. Such a value indicates that it holds its electron most tightly, i.e., it has the least tendency to lose its 2s electron. In other words, it should be the poorest reducing agent among the alkali metals while the standard reduction potential values (E°) reveal that lithium should act as strong reducing agent. This discrepancy could be explained if we understand the fact that the ionization energy is the property of isolated atoms in the gaseous state while the reduction potential is concerned itself with the metal when it goes into solution.

The ionization energy involves only the following change.

$$M_{(g)} \rightarrow M^+_{(g)} + e^-$$

On the other hand, the reduction potential involves the process.

$$M_{(s)} \rightarrow M^+_{(aq)} + e^-$$

The process is thought to take place in three consecutive steps.

(i) $M_{(s)} \rightarrow M_{(g)}$

(ii) $M_{(g)} \rightarrow M^+_{(g)} + e^-$

(iii) $M^+_{(g)} + H_2O \rightarrow M^+_{(aq)}$ + Hydration energy

The energy required in step (i) is known as **sublimation energy**. This energy is about the same for all alkali metals.

The energy required for step (ii) is the **ionization energy**. This is the highest for lithium.

The energy liberated in step (iii) is known as **hydration energy**. Lithium ion has the highest heat of hydration as explained in physical properties of section 5.4.1 (11). The net result of energy liberated in all the three steps is larger for lithium. This explains the lowest reduction potential of lithium.

From the above discussion, it may be concluded that **the greater strength of lithium as reducing agent is due to its greater heat of hydration**.

9. Formation of alloys: The alkali metals are known to form alloys amongst themselves as well as with other metals. The alkali metals readily dissolve in mercury forming amalgams, and the process is highly exothermic.

10. Complex formation: In order to form complex compounds, a metal must possess the following characteristics.

(i) Small size
(ii) High effective nuclear charge
(iii) Tendency to accept electrons (i.e., presence of vacant orbitals)

Since alkali metals have none of these characteristics, they have little tendency to form complexes. However, due to the small size, lithium forms certain complexes. The complex forming tendency fall markedly down the groups as the atomic size increases.

5.5 PRINCIPLES OF EXTRACTION OF METALS

The alkali metals cannot be extracted by the usual methods because of the following difficulties.

(i) Alkali metals are themselves very strong reducing agents. So, a chemical reducing agent that can reduce the alkali metal compounds is not available. Therefore, alkali metals cannot be extracted by chemical reduction methods.
(ii) Alkali metals cannot be prepared from their aqueous solutions by using metal displacement method because the liberated metal immediately react with water forming metal hydroxide liberating hydrogen gas.
(iii) Alkali metals cannot be prepared by the electrolysis of their aqueous solutions because the discharge potential of alkali metal ions is more than that of H^+ ion. So, H^+ will be reduced at cathode preferential to alkali metal ions and hydrogen gas will be liberated at cathode.

Keeping in view of the above difficulties, the alkali metals are generally isolated by the electrolysis of the fused metal halides. But the melting points of the alkali metal halides are very high. These are further lowered by adding some other salt (colligative property).

5.6 USES OF ALKALI METALS

1. **Uses of Lithium**
   (i) It is used as a deoxidizer in the purification of copper and nickel.
   (ii) It is used in the manufacture of alloys. It increases the tensile strength and resistance to corrosion of magnesium and magnesium alloys. For example, lead lithium alloy is used in making bearings and electric cable sheathings.
   (iii) Its compounds are used in glass and pottery manufacture.
2. **Uses of Sodium**
   (i) Due to lightness and high thermal conductivity, it is used for filling exhaust valves of aeroplane engines.
   (ii) It is used as a catalyst in the production of artificial rubber, dyes and drugs.
   (iii) It is used for preparing large number of chemicals like sodium peroxide, sodium cyanide and sodamide.
   (iv) In inorganic chemistry, sodium amalgam is used as a reducing agent.
   (v) It is used in the manufacture of sodium vapour lamps.
   (vi) It is used in the detection of nitrogen, sulphur and halogens in organic compounds by Lassaigne's test.
   (vii) Na–Pb alloy is used in the manufacture of tetraethyl lead as an antiknocking agent for petrol.
3. **Uses of Potassium**
   (i) It is used in photoelectric cells.
   (ii) An alloy of potassium and sodium is used in liquid thermometers for measuring high temperatures.
   (iii) It is also used in synthesis of organic compounds.
4. **Uses of Rubidium**
   (i) Rubidium metal is used in photoelectric cells.
   (ii) It is used as a fuel in the ion propulsion engine.
   (iii) Rubidium carbonate is used in the preparation of the special glasses.
5. **Uses of Caesium**
   (i) Caesium metal is used in scintillation counters, in photoelectric cells and infrared detecting instruments.
   (ii) It is also used in ion propulsion engines for space travel and the conversion of heat to electricity.
   (iii) Caesium is also used in photoelectric cells.

5.7 COMPOUNDS OF ALKALI METALS: COMPARATIVE STUDY

The alkali metals form a complete range of compounds with all the common anions and have long been used to illustrate group similarities and trends. Some important compounds of alkali metals are discussed here.

5.7.1 Hydrides

All the alkali metals react with hydrogen to form ionic hydrides of the formula M^+H^-. These hydrides contain alkali metal as positive ion and the hydrogen as hydride ion H^-. This can be established on the liberation of hydrogen at the anode and the metals depositing at the cathode during the electrolysis of the fused hydrides.

Hydride ion is about the same size as fluoride ion and as with the alkali metal fluorides the lattice energy of the salt decreases more rapidly than the ionization enthalpy of the metal as we go down the group. Thus, ΔH is −91 kJ mol^{-1} for LiH and −50 kJ mol^{-1} for CsH.

The hydrides can be prepared by a direct combination of hydrogen with alkali metals.

$$2M + H_2 \longrightarrow 2MH$$

$$2Li + H_2 \xrightarrow{973-1073K} 2LiH$$

$$2Na + H_2 \xrightarrow{670K} 2NaH$$

All are colourless solids. The order and reactivity of the alkali metals with hydrogen is in the following order:

$$Li > Na > K > Rb > Cs$$

The ionic character of the bonds in these hydrides increases from LiH to CsH and their stability decreases in the same order. With increase in size of cation lattice energy decreases and hence the stability decreases. These hydrides act as strong reducing agents and liberate hydrogen with protonic solvents such as water, ammonia and ethyl alcohol with the formation of hydroxide, amide and ethoxide respectively showing that the hydride ion is a strong base.

$$MH + H_2O \rightarrow MOH + H_2$$
$$MH + NH_3 \rightarrow MNH_2 + H_2$$
$$MH + C_2H_5OH \rightarrow C_2H_5OM + H_2$$

The hydrides are frequently used as reducing agents, the product being a hydride or complex metal hydride depending on the conditions used, or the free element if the hydride is unstable. Illustrative examples using NaH are

$$2BF_3 + 6NaH \xrightarrow{200^\circ C} B_2H_6 + 6NaF$$

$$BF_3 + 4NaH \xrightarrow{C_2H_5OH/125^\circ C} NaBH_4 + 3NaF$$

$$AlBr_3 + 4NaH \xrightarrow{CH_3OCH_3} NaAlH_4 + 3NaBr$$

$$TiCl_4 + 4NaH \xrightarrow{400^\circ C} Ti + 4NaCl + 2H_2$$

Sulphur dioxide is uniquely reduced to dithionate and CO_2 to formate.

$$2SO_2 + 2NaH \rightarrow Na_2S_2O_4 + H_2$$
$$CO_2 + NaH \rightarrow HCOONa$$

Lithium also forms a complex hydride $Li[AlH_4]$, called lithium aluminium hydride which is useful reducing agent. It is made from lithium hydride in dry ether solution.

$$4LiH + AlCl_3 \longrightarrow Li[AlH_4] + 3LiCl$$

Lithium aluminium hydride is ionic and the $[AlH_4]^-$ ion is tetrahedral. $Li[AlH_4]$ is a powerful reducing agent and is widely used in organic chemistry, as it reduces carbonyl compounds to alcohols. It reacts violently with water, so it is necessary to use absolutely dry organic solvents, for example ethers which have been dried over sodium. $Li[AlH_4]$ will also reduce a number of inorganic compounds.

$$4BCl_3 + 3Li[AlH_4] \longrightarrow 2B_2H_6 + 3LiCl + 3AlCl_3$$
$$4PCl_3 + 3Li[AlH_4] \longrightarrow 2PH_3 + 3LiCl + 3AlCl_3$$
$$SiCl_4 + Li[AlH_4] \longrightarrow SiH_4 + LiCl + AlCl_3$$

Sodium tetrahydrido borate (sodium borohydride) $NaBH_4$ is another complex hydride. It is ionic, comprising tetrahedral $\left[BH_4\right]^-$ ions. It is obtained by heating sodium hydride with trimethylborate.

$$4NaH + B(OCH_3)_3 \xrightarrow{230-270^\circ C} Na\,[BH_4] + 3CH_3ONa$$

Other tetrahydridoborates for Group-IA and IIA metals, aluminium and some transition metals can be made from the sodium salt. These tetrahydridoborates are used as reducing agents, and the alkali metal compounds (particularly those of Na and K) are becoming increasingly used as they are much less sensitive to water than $Li[AlH_4]$. $Na[BH_4]$ can be crystallized from cold water and $K[BH_4]$ from hot water, so they have the advantage that they can be used in aqueous solutions. The others react with water.

$$\left[BH_4\right]^- + 2H_2O \longrightarrow BO_2^- + 4H_2$$

5.7.2 Nitrides

Except lithium all the other alkali metals do not directly combine with nitrogen to form nitrides. But they may be indirectly prepared.

Lithium nitride is formed when nitrogen is passed over lithium. The reaction takes place slowly at room temperature and more rapidly as the temperature increases.

$$6Li + N_2 \longrightarrow 2Li_3N$$

Sodium and potassium nitrides can be prepared (i) by the action of active nitrogen on a film of the metal.

(ii) by heating an intimate mixture of the metallic azide and the metal prepared by the evaporation of mixed solution of the metal and the azide in liquid ammonia.

Lithium nitride on heating in a current of hydrogen gas is reduced to lithium hydride. The reaction is reversible.

$$Li_3N + 3H_2 \rightleftharpoons 3LiH + NH_3$$

Lithium nitride forms double nitrides with certain other metallic nitrides. These may be solid solutions.

5.7.3 Oxides

When alkali metals are burnt in a limited supply of oxygen they form normal oxides of the formula M_2O. In excess of oxygen they also form M_2O, M_2O_2 and MO_2. They are termed as monoxides, peroxides and superoxides.

Lithium burns in oxygen giving lithium monoxide.

$$4Li + O_2 \longrightarrow 2Li_2O$$

When other metals burn in limited supply of oxygen monoxides are formed.

$$4Na + O_2 \longrightarrow 2Na_2O$$
$$4K + O_2 \longrightarrow 2K_2O$$

Oxides of any required type can, however, be prepared by dissolving the metal in ammonia and treating the solution with the required amount of oxygen.

All the alkali metal oxides are ionic compounds. These are strongly basic and highly soluble in water. They give strongly alkaline solution due to the formation of the hydroxides.

$$M_2O_{(s)} + H_2O_{(l)} \longrightarrow 2M^+_{(aq)} + 2OH^-_{(aq)}$$

The above reaction involve the abstraction of proton from water molecule, the oxide ion thus functions as strong base in the Bronsted–Lowry sense. Solubility of these oxides increases down the group from Li_2O to Cs_2O. There is a trend to increasing colouration with increasing atomic number, Li_2O and Na_2O are white when pure. K_2O is yellowish, Rb_2O is bright yellow and Cs_2O is orange.

5.7.4 Hydroxides

The alkali metal hydroxides are formed by dissolving their oxides in water.

$$M_2O + H_2O \longrightarrow 2MOH$$

Of all the hydroxides known, the alkali metal hydroxides are strongest bases. The basic strength of the hydroxide increases regularly from lithium to caesium hydroxide.

If the polarity of the M–O bond is more, the ionization becomes easier. The alkali metal hydroxides do not differ greatly in polarity (the electronegativities of alkali metals are nearly equal, see Table 5.2). Hence, the basic strength of alkali metal hydroxides depend mainly on the separation of hydroxide ion from cation. The greater the separation of cation and anion, i.e., the bond distance of M^+ and OH^- of a hydroxide, the greater is the ionization and hence more basic strength. In Group IA, the cation size increases with increasing atomic number of the metal. This causes increase in the ionic bond length of M^+ and OH^- ions. So, ionization becomes easier from LiOH to CsOH and thus basic character increases from LiOH to CsOH.

All the alkali metal hydroxides are white crystalline solids and deliquiscent in nature. Except lithium hydroxide, all are highly soluble in water and the solubility in water increases from LiOH to CsOH. Thermal stability of these hydroxides also increases from LiOH to CsOH. Lithium hydroxide tends to lose water on heating.

$$2LiOH \longrightarrow Li_2O + H_2O$$

Other alkali metal hydroxides do not decompose on heating. However, they sublime at about 673K and the vapours mainly consist of dimers $(MOH)_2$. The dissolution of the alkali metal hydroxides in water is exothermic due to the formation of hydrates and hydration.

5.7.5 Halides

Alkali metal halides can be prepared by the reaction of the appropriate oxide, hydroxide or carbonate with aqueous hydrohalic acid (HX).

$$M_2O + 2HX \longrightarrow 2MX + H_2O$$
$$MOH + 2HX \longrightarrow 2MX + H_2O$$
$$M_2CO_3 + 2HX \longrightarrow 2MX + H_2O + CO_2$$

All the alkali metals also react directly with halogens to form halides.

$$2M + X_2 \longrightarrow 2MX$$

In the above reactions M is any of the alkali metal (Li, Na, K, Rb, or Cs) and X is any of the halogen (F, C*l*, Br or I).

All the halides of alkali metals are colourless solids with high melting points and boiling points, but they turn yellow (NaC*l*), violet (KI), etc. owing to non-stoichiometric crystal defects.

(a) Stability: The stabilities of alkali metal halides can be compared by using their standard enthalpies of formation values ($\Delta_f H^o$).

From Table 5.6 it can be concluded that all the halides have negative enthalpies of formation. The $\Delta_f H^o$ values for fluorides become less negative as we go down the group whilst the reverse is true for $\Delta_f H^o$ for chlorides, bromides and iodides. For a given metal $\Delta_f H^o$ is always less negative from fluoride to iodide.

The order of $\Delta_f H^o$ for a given metal is

Fluoride > Chloride > Bromide > Iodide

Thus for any alkali metal, fluoride is most stable and iodide is least stable.

The order of $\Delta_f H^o$ for fluorides of alkali metals is

LiF > NaF > KF > RbF > CsF.

Thus, LiF is most stable while CsF is least stable. This trend is opposite to the trend predicted by Fajan's rules. This is because of the high lattice energy of LiF due to smaller size of both ions.

The order of $\Delta_f H^o$ for other halides (except fluorides) is not regular.

The stability of lithium halides is least while those of caesium halides is better.

The Born–Haber cycle (Chapter 3) may be recalled to explain these observations. With the same anion H_f of MX depends on $S_M + I_1 - U$ (sublimation, ionization and lattice energy, respectively). The variation of $S + I_1$ from Li to Cs is as follows.

	Li	Na	K	Rb	Cs
$S + I_1$ (KJ mol^{-1})	682	606	509	491	464

Now from Table 5.7 it can be seen that the lattice energies of the fluorides vary more rapidly (from Li to Cs) than does the sum of $S + I_1$. Lattice energy depends inversely on $(r_+ + r_-)$ and so the variation in lattice energy is greatest with smallest r^- (i.e., the F^- ion) and least when r^- is largest (I^- ion). For chlorides and iodides the sum $S + I_1$ decreases more rapidly than do the lattice energies. Thus in the fluorides, the lattice energy term predominates over others; this leads to a decreasing trend in the enthalpies of formation of LiF to CsF. For the large halides the lattice energies become smaller and less dominant. Greater ease of sublimation and ionization of the heavier alkali metals (i.e., lower values of S and I_1) drives ΔH_f to larger numerical values.

Table 5.6 Standard enthalpies of formation ($\Delta_f H^o$) for alkali metal halides (in kJ mol^{-1})

Alkali metal	Enthalpies of formation ($\Delta_f H^o$) kJ mol^{-1}			
	MF	**MCl**	**MBr**	**MI**
Li	–612	–398	–350	–271
Na	–569	–400	–360	–288
K	–563	–428	–392	–328
Rb	–549	–423	–389	–329
Cs	–531	–424	–395	–337

Table 5.7 Lattice enthalpies of group IA halides

Metal	Lattice enthalpies (kJ mol^{-1})			
	MF	**MCl**	**MBr**	**MI**
Li	–1035	– 845	– 800	– 740
Na	– 908	– 770	– 736	– 690
K	– 803	– 703	– 674	– 636
Rb	– 770	– 674	– 653	– 515
Cs	– 720	– 644	– 623	– 590

(b) Bond character: All the alkali metal halides are ionic compounds, but the per cent ionic character varies. This can easily be predicted by using Fajan's rules. Small cation will have more polarizing power and the polarizing power decreases with increase in the size of cation. Similarly, the polarizability of smaller anion is less and the polarizability increases with increase in size of the anion. So for a given halide ion ionic character increases with increase in the size of cation and for a given halogen ionic character decreases with increase in the size of halogen. The order of ionic character is

$$LiX < NaX < KX < RbX < CsX$$

$$MF > MCl > MBr > MI$$

This is confirmed from the observed and calculated dipole moments assuming that there is a net charge ± e (4.8×10^{-10} esu) on each ion. This assumption would be appropriate for a truly ionic compound between spherical ions. The dipole moments of such molecules would be equal to $e \times r_o \times 10^{18}$ where e is the fundamental charge in esu and r_o is the experimental value of the bond length. The observed dipole moments and the calculated dipole moments are given in Table 5.8.

From Table 5.8 it can be seen that the observed dipole moments are smaller than the calculated dipole moments. Thus the gaseous alkali metal halides do not consist of perfectly spherical ions, but instead the negative ion tends to distorted or polarized by the positive ion as shown below.

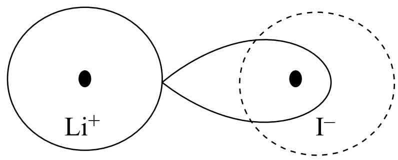


Table 5.8 The observed dipole moments and calculated dipole moments of some alkali metal halides

Alkali metal Halide	Bond distance r_o	$eq \times r_o$ (debyes)	μ observed (debyes)
LiBr	2.17	10.4	6.19
LiI	2.39	11.5	6.25
NaCl	2.36	11.3	8.5
KF	2.17	10.4	8.6
KCl	2.67	12.8	10.0
KBr	2.82	13.5	10.6
KI	3.05	14.6	11.1

Table 5.9 Melting and boiling points (K) of alkali metal halides (MX)

	Fluoride		Chloride		Bromide		Iodide	
M	**M.P.**	**B.P.**	**M.P.**	**B.P.**	**M.P.**	**B.P.**	**M.P.**	**B.P.**
Li	1116	1954	883	1655	820	1538	719	1463
Na	1268	1977	1081	1738	1028	1663	934	1573
K	1129	1775	1045	1680	1021	1653	950	1598
Rb	1048	1681	990	1654	955	1613	915	1573
Cs	955	–	918	1573	909	–	894	1553

Due to this reason the bond between alkali metal ion and halide ion gets some covalent character and the covalent character of the bond increases with increase in the difference between the observed and calculated dipole moments.

(c) Melting points: The melting points of the alkali metal halides can be explained basing on their lattice enthalpies. The greater the lattice enthalpy, the higher is the melting point. The m.pts and b.pts, are given in Table 5.9.

The trends in the melting and boiling points of the halides are also very interesting.

(i) For a given alkali metal, the melting points and boiling points are in the order.
Fluoride > Chloride > Bromide > Iodide

(ii) For a given halogen the melting and boiling points are in the order
Lithium < Sodium > Potassium > Rubidium > Caesium

(iii) For any particular halide ion (X^-), NaX has the highest melting and boiling point but not LiX. This can be largely rationalized in terms of the very small size of the Li^+ ion–a fact that leads to appreciable covalent character of lithium halides because of more polarizing power of small Li^+ ion with more charge density. Further due to small size of Li^+ ion there exists anion–anion repulsion resulting in the considerable increase of Li–X distance in the lithium salts than the expected from the sum of ionic radii. This results in appreciable lowering of the energy of the crystal, which, in turn, leads to lower the melting and boiling points.

Pauling made some corrections for the anion–anion repulsion and showed that the corrected melting and boiling points follow the expected regular trend. According to him, the equilibirum distance (R_0) between two ions in a crystal is given by

$$R_0 = (r_+ + r_-)\, F(\rho)$$

in which $F(\rho)$ is a function of the radius ratio $\rho = r^+/r^-$. It was shown that $F(\rho)$ becomes unity when $\rho = 0.75$. Pauling then corrected the lattice energy of the crystals for the radius-ratio effect. It was found that the corrected crystal energies are much higher for the lithium halides: about 33 kJ mol^{-1} for LiF, increasing to 63 kJ mol^{-1} in LiI. The corresponding corrections in melting and boiling points are also appreciable. The m.pt of LiF should be higher by about 264°C, the difference increasing to ~777K in LiI. The values of sodium halides show that they are only slightly affected by this radius-ratio effect, e.g., lattice energy of NaI changes by only 23kJ mol^{-1}, with a melting point correction by 180°C. For other halides, this influence is still smaller. The corrected melting points show a regular trend.

(d) Solubility: The solubility of a salt in water depends on its lattice energy and hydration energy. For a substance to dissolve, the hydration energy must be larger than the lattice energy. Conversely, if the hydration energy is less than the lattice energy the solid is insoluble. The hydration energies of alkali metal ions and halide ions are given in Table 5.10.

Table 5.10 Hydration enthalpies of alkali metal ions and halide ions (kJ mol^{-1})

Alkali metal ion	Hydration enthalpy kJ mol^{-1}	Halide ion	Hydration enthalpy kJ mol^{-1}
Li^+	– 506	F^-	– 513
Na^+	– 406	Cl^-	– 370
K^+	– 330	Br^-	– 339
Rb^+	– 310	I^-	– 293
Cs^+	– 276		

From Tables 5.7 and 5.10 we can compute the lattice and hydration energies and the difference between them for alkali metal halides. The hydration enthalpies of alkali metal halides can be obtained by adding the hydration enthalpies of both alkali metal ion and halide ions.

Similar values can also be computed for bromides and iodides. Their solubilities are as given in Table 5.11.

From these values it can be concluded that

(i) Among fluorides, the order of solubility is

LiF < NaF < KF < RbF < CsF

The low solubility of LiF is due to its very high lattice energy (due to small sizes of both cation and anion).

Table 5.11 Solubilities of alkali metal bromides and iodides

MBr	LiBr	NaBr	KBr	RbBr	CsBr
solubility	177	91	91	110	108
	(20.4)	(8.8)	(7.6)	(6.0)	
MI	LiI	NaI	KI	RbI	CsI
solubility	165	179	144	152	79

We can also find that the lattice enthalpy of LiF is greater than hydration energy.

(ii) In the case of lithium halides except LiF other halides are highly soluble in water because of the large hydration energy of small Li^+ ion.

(iii) Except fluorides, for the other halides the solubility decreases from NaX to KX and then increases to CsX. The solubility of potassium salts is less than sodium salts. This is because the difference in hydration and lattice energies of potassium salts is positive, whereas in the case of other metal halides is negative.

(iv) Lithium halides (except LIF) have significant covalent character and hence soluble in organic solvents like pyridine, alcohol, acetone, etc.

(e) Structure of crystal lattice: Cs*Cl*, CsBr and CsI have body centred cubic (bcc) structure in which the coordination number of each ion is 8. All the other alkali metal halides adopt face centred cubic structure in which the coordination number of each ion is 6. As per the radius-ratio value lithium must show a coordination number 4 theoretically. The anomalous behaviour of Li^+ ion has been attributed to more favourable lattice energy acquired by achieving a higher coordination number. Hence, the solubility of lithium halides is less.

(f) Conclusions: Small ions have strong electrostatic attraction for each other and also for water molecules, whereas large ions have weaker attraction for each other and also for water molecules. These factors work together to make compounds formed of two large ions or of two small ions less soluble than compounds containing one large ion and one small ion particularly when they have the same charge magnitude. For example, LiF with two small ions and CsI with two large ions are less soluble than LiI and CsF, with one large and one small ion. For the small ions, the larger lattice energies overcome the larger hydration

Table 5.12 Hydration and lattice enthalpies, the difference (hydration enthalpy-lattice enthalpy) and the solubilities of alkali metal fluorides

Metal fluoride	Hydration enthalpy kJ mol^{-1}	Lattice enthalpy kJ mol^{-1}	[Hydration enthalpy kJ mol^{-1}] – [Lattice enthalpy]	Solubility
LiF	– 1019	– 1035	+ 16	0.27 (0.1)
NaF	– 919	– 908	– 11	4.22 (1.0)
KF	– 843	– 803	– 40	92.3 (15.9)
RbF	– 823	– 770	– 53	130.6 (12.5)
CsF	– 789	– 720	– 69	367.0 (24.2)

Table 5.13 Hydration and lattice enthalpies, the difference (hydration enthalpy-lattice enthalpy) and the solubilities of alkali metal chlorides

Metal chloride	Hydration enthalpy kJ mol^{-1}	Lattice enthalpy kJ mol^{-1}	[Hydration enthalpy kJ mol^{-1}] – [Lattice enthalpy]	Solubility
LiCl	– 876	– 845	– 31	83.0 (19.6)
NaCl	– 776	– 770	– 6	36 (6.2)
KCl	– 700	– 703	+ 3	34.7 (4.8)
RbCl	– 680	– 674	– 6	91.0 (7.5)
CsCl	– 646	– 644	– 2	186.0 (11.0)

enthalpies and for the large ions the small hydration enthalpies allow the lattice energy to dominate. Inspite of less favourable enthalpy CsI is 15 times more soluble than LiF.

If two alkali metal halides are mixed, results in the formation of a new alkali metal halides one having more lattice enthalpy. For example if LiI and CsF are mixed, LiF and CsI will be formed because of the large lattice enthalpy of LiF.

$$LiI + CsF \longrightarrow CsI + LiF \quad \Delta H = -146 \text{ kJ mol}^{-1}$$

This is contrary to the simple electronegativity argument that the most electropositive and most electronegative elements form the most stable compounds. This can be explained basing on hard acid (least polarizable) combines with hard base (more polarizing power) and soft acid (more polarizable) combines with soft base (least polarizing power).

It should be noted that if an excess of hydrofluoric acid is used in the preparation of fluoride by adding HF to NaOH or KOH solid hydrogen fluorides will be formed.

$$NaOH + 2HF \longrightarrow NaHF_2 + H_2O$$
$$KOH + 2HF \longrightarrow KHF_2 + H_2O$$

The hydrogen fluoride ion contains one molecule of hydrogen fluoride, hydrogen bonded to the very electronegative fluoride ion.

$$(F - H \text{ ...... } F)^-$$

The hydrogen fluorides decompose to give normal fluorides and hydrogen fluoride.

$$NaHF_2 \longrightarrow NaF + HF$$
$$KHF_2 \longrightarrow KF + HF$$

5.7.6 Compounds with Carbon

Lithium on heating with carbon forms an ionic carbide Li_2C_2. Other alkali metals form similar carbides either by heating with acetylene or by passing acetylene through a solution of the metal in liquid ammonia.

$$Na + C_2H_2 \xrightarrow{\text{liquid NH}_3} NaHC_2 + \frac{1}{2}H_2$$
$$2Na + C_2H_2 \xrightarrow{\text{liquid NH}_3} Na_2C_2 + H_2$$

These carbides contain the carbide ion $(C \equiv C)^{2-}$ i.e., acetylide ion. These carbides on hydrolysis give acytelene and therefore are termed as acetylides.

$$Na_2C_2 + 2H_2O \longrightarrow 2NaOH + C_2H_2$$

5.8 SALTS OF OXO-ACIDS

The acids in which the acidic proton is on a hydroxyl group with an oxogroup attached to the same atom are called **Oxo-acids. E.g.:** H_2CO_3 [OC $(OH)_2$], sulphuric acid H_2SO_4 [O_2S $(OH)_2$]; nitric acid [O_2N (OH)], etc. The alkali metals form salts with all the oxo-acids. Comparative study of some important salts of oxo-acids are discussed here.

5.8.1 (a) Carbonates and Bicarbonates

All the alkali metals form carbonates and have the general formula M_2CO_3. These are formed by the reaction of carbondioxide with their oxides and hydroxides.

$$M_2O + CO_2 \longrightarrow M_2CO_3$$
$$2MOH + CO_2 \longrightarrow M_2CO_3 + H_2O$$

The bicarbonates of alkali metals on heating gives carbonates.

$$2MHCO_3 \longrightarrow M_2CO_3 + H_2O + CO_2$$

When excess carbondioxide is passed into the aqueous solution of carbonates, they convert into bicarbonate.

$$M_2CO_3 + H_2O + CO_2 \longrightarrow 2MHCO_3$$

The thermal stabilities of the alkali metal carbonates increases with increase in the size of the cation. Lithium carbonate decomposes on heating, to lithium oxide and carbon dioxide, whereas the carbonates of the other alkali metals are highly stable to heat.

$$Li_2CO_3 \longrightarrow Li_2O + CO_2$$

The exceptional behaviour of lithium carbonate to heat may be explained as follows.

As stated earlier the polarizing power of Li^+ ion, due to its large charge to radius ratio is more than that of any other alkali metal ion.

Further, the size of Li^+ ion is smaller when compared with CO_3^{2-} ion. So replacement of larger CO_3^{2-} ion by the smaller O^{2-} ion will lead to the increase in the lattice energy, hence favours decomposition (refer to consequences of lattice energy chapter, section 3.4.8).

Except lithium carbonate the other alkali metal carbonates are stable upto 1273 K, above which they melt and then are converted into oxides.

$$M_2CO_3 \xrightarrow{\Delta/>1273\text{K}} M_2O + CO_2.$$

Order of the thermal stability of alkali metal carbonates is $LiCO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$.

The carbonates are the salts of weak acid (carbonic acids and strong base (i.e., alkali metal hydroxide MOH). Hence they are hydrolyzed in water to give OH^- ion and thus basic in nature.

$$CO_3^{2-} + H_2O \longrightarrow HCO_3^- + OH^-$$

$$CO_3^{2-} + 2H_2O \longrightarrow H_2CO_3 + 2OH^-$$

Lithium carbonate is only sparingly soluble in water. Potassium carbonate is highly soluble. The solubility of alkali metal carbonate increases as the atomic number of alkali metal increases. The order of solubility of alkali metal corbonates is

$$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3.$$

(b) Bicarbonates: These can be obtained by passing carbon dioxide through a cold concentrated solution of the corresponding carbonate. E.g.

$$Na_2CO_3 + CO_2 + H_2O \longrightarrow 2NaHCO_3$$

Although the bicarbonates of several metals exist in aqueous solution, only those of the alkali metals can be obtained as solids and even these tend to decompose readily, e.g.

$$2KHCO_3 \xrightarrow{\Delta} K_2CO_3 + H_2O + CO_2$$

The heat of decomposition of bicarbonates increases with the increase in the size of cation. Hence the stability of bicarbonates will be in the order

$$LiHCO_3 < NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$$

The bicarbonates also hydrolyze in water to give OH^- ion and thus basic in nature.

$$HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^-$$

Bicarbonate solutions are less alkaline than carbonate solutions because only one OH^- ion is formed per one mole of bicarbonate while two OH^- ions are formed per mole of carbonate during their hydrolysis. Bicarbonates does not give colouration with phenolphthalein but gives yellow colour with methyl orange indicator. This is used in distinguishing the bicarbonate and carbonate and in the quantitative estimation of bicarbonate and carbonate in their mixture.

The bicarbonate ions tend to be held together in crystal structures by hydrogen bonding giving layers of polymeric anions. The structures of sodium and potassium bicarbonates provide interesting examples of hydrogen bonding. The potassium salt contains a dimeric anion of the structure as in Fig. 5.1a. In sodium salt, however, the bicarbonate anions form an infinite chain as shown in Fig. 5.1b.

The hydrogen bonds in each case are asymmetrical. Because of this infinite anion structure in sodium bicarbonate the solubility is very low in water which is the basis of the ammonia–soda process.

5.8.2 Nitrates and Nitrites

(a) Nitrates: The nitrates can be prepared by the action of nitric acid on the corresponding alkali metal hydroxide or carbonate.

$$MOH + HNO_3 \longrightarrow MNO_3 + H_2O$$

$$M_2CO_3 + 2HNO_3 \longrightarrow 2MNO_3 + H_2O + CO_2$$

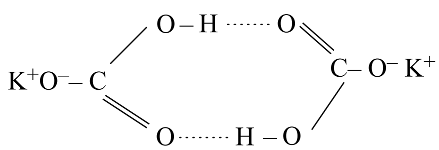
All the nitrates are very soluble in water. Solid $LiNO_3$ and $NaNO_3$ are deliquescent and because of this reason KNO_3 is used in preference to $NaNO_3$ in gun powder. Gun powder is a mixture of KNO_3, sulphur and charcoal.

KNO_3 is usually prepared from nitric acid and K_2CO_3. Formerly, it was made from $NaNO_3$.

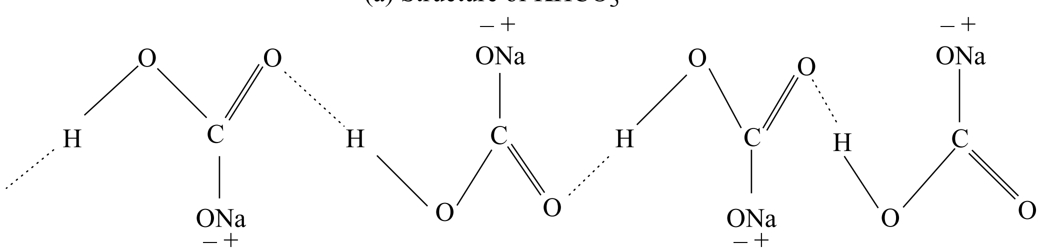
$$2HNO_3 + K_2CO_3 \longrightarrow 2KNO_3 + CO_2 + H_2O$$

$$NaNO_3 + KCl \longrightarrow KNO_3 + NaCl$$

Alkali metal nitrates have fairly low melting points. Their melting points are lowest among the most stable



(a) Structure of $KHCO_3$



(b) Structure of $NaHCO_3$

Fig 5.1 Hydrogen bonding in (a) $KHCO_3$ and (b) $NaHCO_3$

nitrates known. However, on strong heating they decompose into nitrites and at high temperatures to the oxide.

$$2NaNO_3 \xrightleftharpoons{500°C} 2NaNO_2 + O_2$$

$$4NaNO_3 \xrightleftharpoons{800°C} 2Na_2O + 5O_2 + 2N_2$$

Again $LiNO_3$, similar to Li_2CO_3 is least stable among alkali metal nitrates.

$$2LiNO_3 \rightleftharpoons Li_2O + 2NO_2 + \frac{1}{2}O_2$$

The reason for the least stability of $LiNO_3$ is same as that explained in the case of carbonates.

$LiNO_3$ is used for fireworks and red-coloured distress flares. Alkali metal nitrates are widely used to carry high temperature oxidations and also as a heat transfer medium.

(b) Nitrites: Nitrites are formed by heating alkali metal nitrates, e.g.

$$2NaNO_3 \longrightarrow 2NaNO_2 + O_2$$

Alkali metal nitrites can be made by chemical reduction of nitrates.

$$2NaNO_3 + C \longrightarrow 2NaNO_2 + CO_2$$

$$KNO_3 + Zn \longrightarrow KNO_2 + ZnO$$

They can also be prepared by the reaction of nitric oxide with a hydroxide.

$$2KOH + 4NO \longrightarrow 2KNO_2 + N_2O + H_2O$$

$$4KOH + 6NO \longrightarrow 4KNO_2 + N_2 + 2H_2O$$

Nitrites liberate brown fumes of NO_2 by reacting with dilute acids.

$$2NaNO_2 + 2HCl \longrightarrow 2NaCl + H_2O + NO_2 + NO$$

$$2NO + O_2 \longrightarrow 2NO_2$$

This reaction is used to identify the nitrites in the laboratory. Nitrites are used in the manufacture of organonitrogen compounds like azo dyes.

Nitrites are white, crystalline hygroscopic salts that are very soluble in water. When heated in the absence of air, they disproportionate.

$$5NaNO_2 \longrightarrow 3NaNO_3 + Na_2O + N_2$$

5.8.3 Sulphates

All the alkali metal sulphates can be prepared by the reaction of alkali metal oxides, hydroxides or carbonates with dilute sulphuric acid.

$$M_2O + H_2SO_4 \longrightarrow M_2SO_4 + H_2O$$

$$2MOH + H_2SO_4 \longrightarrow M_2SO_4 + 2H_2O$$

$$M_2CO_3 + H_2SO_4 \longrightarrow M_2SO_4 + H_2O + CO_2$$

All the alkali metal sulphates except lithium sulphate are water soluble. They also form double salts (alums) with trivalent metal ions like Al^{3+}, Fe^{3+}, Cr^{3+}, etc. Because of small size, lithium cannot form alums.

Alkali metal sulphates are reduced to the corresponding sulphides on heating with carbon.

$$M_2SO_4 + 4C \longrightarrow M_2S + 4CO.$$

5.9 ANOMALOUS PROPERTIES OF LITHIUM

Although lithium shows most of the characteristic properties of Group IA, it also shows some differences in many respects. This is mainly due to

(i) small size
(ii) high electronegativity
(iii) presence of only two electrons in the penultimate shell
(iv) absence of d-orbitals in the valence shell.

Lithium differs from the other alkali metals in the following respects.

1. Lithium is the hardest metal among the alkali metals. Its melting and boiling points are higher than those of the other alkali metals.
2. Lithium is the least reactive among the alkali metals towards various chemical reagents except hydrogen. For example, lithium reacts with water slowly whereas other alkali metals react with water violently and explosively.
3. Unlike the other alkali metals, lithium reacts with nitrogen directly forming nitride Li_3N.
4. Unlike the other alkali metals, lithium is not attacked by air easily and does not lose its lustre. So it can be stored by wrapping with wax coated paper but other alkali metals must be stored in hydrocarbons like kerosene.
5. When burnt in air or oxygen lithium forms only monoxide while other alkali metals form peroxides and superoxides.
6. Lithium reacts with hydrogen at higher temperatures (1000–1100 K) while other alkali metals react at lower temperatures (623–700K). Lithium hydride is more stable than the other alkali metal hydrides.
7. Lithium hydroxide decomposes on heating to form Li_2O while other alkali metal hydroxides do not decompose on heating but melts on heating.
8. Li_2O and LiOH are weakest bases among the oxides and hydroxides of alkali metals.
9. Salts of lithium like fluoride, phosphate, oxalate and carbonate are sparingly soluble in water while the corresponding salts of other alkali metals are freely soluble in water.

10. Lithium chloride is highly deliquescent and crystallizes as hydrated salt $LiCl \cdot 3H_2O$. Other alkali metal chlorides are not deliquescent and crystallizes from water as anhydrous salts.
11. Due to covalent character lithium halides except lithium fluorides, lithium nitrate are soluble in organic solvents, but the corresponding salts of other alkali metals do not dissolve in organic solvents.
12. Lithium carbonate decomposes on heating but other alkali metal carbonates do not decompose on heating.

$$Li_2CO_3 \longrightarrow Li_2O + CO_2$$

13. Lithium nitrate, on heating, decomposes giving Li_2O, NO_2 and O_2 while other alkali metal nitrates on heating converts into nitrites evolving oxygen.

$$2NaNO_3 \longrightarrow 2NaNO_2 + O_2$$

14. Lithium sulphate do not form alums but other alkali metal sulphates form alums.
15. Due to small size Li^+ ion is extensively hydrated in water resulting in low ionic mobility in water. So, its electrical conductance is less than the other alkali metal ions.
16. Lithium chloride undergoes hydrolysis in hot water though to a small extent but sodium and potassium chlorides do not hydrolyze at all.
17. Due to covalent character, the observed dipole moments of salts are lesser than the calculated dipole moments for several lithium salts.

5.10 DIAGONAL RELATIONSHIP OF LITHIUM WITH MAGNESIUM

Lithium differs from other alkali metals in many respects but shows close relationship with magnesium, the element of Group IIA, placed diagonally opposite to it in the periodic table. This type of behaviour is a case of diagonal relationship as explained in Chapter 2.

Resemblances of Lithium with Magnesium

1. The atomic radii of both lithium (134 pm) and magnesium (136.4 pm) are not much different.
2. Among alkali metals lithium is the hardest one and its hardness is comparable with that of magnesium.
3. Polarizing powers of Li^+ and Mg^{2+} ions are about the same.
4. Like magnesium but unlike other alkali metals, lithium is not easily attacked by air and can be melted without losing its lustre.
5. Like magnesium, lithium decomposes water slowly liberating hydrogen.
6. Both lithium and magnesium reacts with nitrogen directly forming nitrides Li_3N and Mg_3N_2.
7. When burnt in oxygen both lithium and magnesium forms only monoxides, e.g., Li_2O and MgO.
8. Hydroxides of both lithium and magnesium dissolve very slightly in water and they are less basic.
9. Fluorides, phosphates, oxalates and carbonates of both lithium and magnesium are sparingly soluble.
10. Chlorides of both lithium and magnesium crystallizes from their aqueous solutions as hydrated salts $LiCl \cdot 3H_2O$ and $MgCl_2 \cdot 6H_2O$. Also, both of these compounds are deliquescent.
11. Nitrates and carbonates of both lithium and magnesium decompose on heating similarly.

$$4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 4NO_2 + O_2$$
$$2Mg(NO_3)_2 \xrightarrow{\Delta} 2MgO + 4NO_2 + O_2$$
$$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$$
$$MgCO_3 \xrightarrow{\Delta} MgO + CO_2$$

5.11 COMPOUNDS OF SODIUM

5.11.1 Sodium Hydroxide

The solid sodium hydroxide is known as **caustic soda** because it breaks down the proteins of the skin and flesh to a pasty mass (caustic property). Aqueous sodium hydroxide is known as **soda lye**. Different methods used for the manufacture of sodium hydroxide are discussed here.

1. Causticizing process: This process is also known as **Gossage process**. But this method is not in use nowadays. But the principle is worth noting. In this process, milk of lime, $Ca(OH)_2$, is added to warm dilute solution (10%) of sodium carbonate at 80–85°C. The following reaction takes place.

$$Ca(OH)_2 + Na_2CO_3 \longrightarrow CaCO_3\downarrow + 2NaOH.$$

Calcium carbonate is slightly soluble and separates out as mud at the bottom. This process is called **causticization** or **caustication process**. Sodium hydroxide formed is about 98 per cent pure.

2. Nelson cell or Porous diaphragm process: Caustic soda is manufactured by electrolyzing solution of sodium chloride. The product obtained by this method is very pure and at the same time hydrogen and chlorine and obtained as valuable by-products.

When dissolved in water, sodium chloride is dissociated completely into sodium and chloride ions.

$$NaCl \longrightarrow Na^+ + Cl^-$$

Water also ionizes slightly into H^+ and OH^- ions.

$$H_2O \longrightarrow H^+ + OH^-$$

When an electric current is passed through an aqueous solution of sodium chloride, Na^+ and H^+ ions move towards the cathode while Cl^- and OH^- ions move towards the anode. But the discharge potential of Na^+ is more than that of H^+. Therefore, Na^+ ions remain in solution while H^+ ions are discharged at the cathode.

$$2H^+ + 2e^- \longrightarrow H_{2(g)}$$

As H^+ ions are discharged at the cathode, more H^+ ions are obtained by the dissociation of water, thereby accumulating OH^- ions.

The discharge potential of Cl^- is lower than that of OH^- ions. So, OH^- ions remain in solution while Cl^- ions are discharged at the anode.

$$2Cl^- \longrightarrow Cl_{2(g)} + 2e^-$$

When the solution containing Na^+ and OH^- is evaporated, the crystals of sodium hydroxide are obtained.

There is one difficulty in the above process that the chlorine gas evolved at the anode may react with sodium hydroxide and thus the whole labour may go wasted.

Hence the construction of the cell is done in such a manner that sodium hydroxide and chlorine produced do not come in contact with each other.

Nelson's cell consist of a perforated U-shaped steel tube which is lined inside with a thin lining of asbestos. This tube acts as the **cathode** and is suspended in a steel tank. A graphite rod serves as the anode. The anode is dipped in the brine flowing into the cathode tube as shown in Fig. 5.2.

When the electric current is passed through the brine solution percolating through the porous diaphragm, sodium ions penetrate through the asbestos lining and reach the cathode where hydrogen gas and OH^- ions are formed by the reduction of water. Sodium ions combine with hydroxyl ions to form sodium hydroxide. Hydrogen escapes through an opening at the top while sodium hydroxide is collected

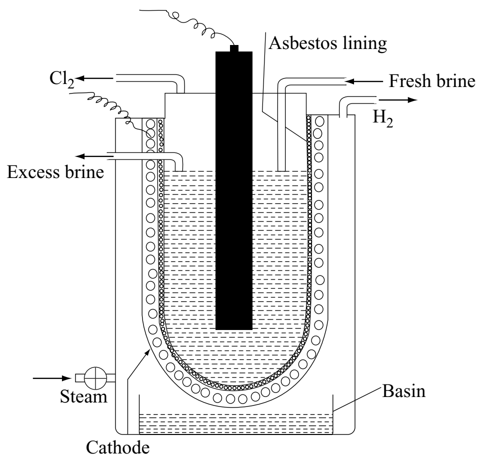


Fig 5.2 Nelson process for the manufacture of sodium hydroxide

in the outer tank from where it is run off at intervals. Chlorine is evolved at the anode and led out through the outlet.

The usual practice is to keep the space between the cathode and the outer tank full of steam. This keeps the reaction mixture hot and therefore reduces its resistance. This also keeps the pores of the diaphragm clean.

The concentration of sodium hydroxide prepared by Nelson cell is about 10–15 per cent containing some sodium chloride. When the sodium hydroxide solution is concentrated to 50 per cent, sodium chloride which is having low solubility particularly in the presence of sodium hydroxide, is almost completely separated. The filtered solution on evaporation gives the dry mass which is fused and casted into sticks or pellets.

3. Castner–Kellner process or Mercury cathode process: The Castner–Kellner cell consists of a large tank divided into three compartments by two slate partitions. These are not touching the bottom of the tank but are resting in the grooves produced at the bottom. A layer of mercury is placed at the bottom of the cell. The mercury in one compartment can flow into the other compartment but any solution placed in one compartment cannot flow into the other. The movement of mercury is kept controlled with the help of the eccentric wheel.

The two outer compartments are provided with graphite anodes and contain saturated brine solution. The central compartment contains dilute solution of caustic soda and also contain a series of iron rods which act as a cathode (Fig. 5.3).

When electricity is passed through Castner–Kellner cell, the following reactions occur.

(i) Outer compartments: When the electric current is passed, sodium chloride solution is electrolyzed in the two outer compartments. Chlorine is liberated at the anodes, whereas sodium is obtained at the mercury layer which acts as the cathode by induction. The liberated sodium dissolves in mercury to form sodium amalgam. Due to rocking motion of the cell, sodium amalgam travels to the central compartments, where it reacts with water to form 50 per cent sodium hydroxide solution of high purity, the reaction being catalyzed by the presence of iron grids. The mercury is then returned to the cell. The products are thus sodium hydroxide, chlorine and hydrogen.

Cathode				Anode
Na^+	$\longleftarrow$	$Na^+ Cl^-$	$\longrightarrow$	Cl^-
discharged				discharged

$$2Na^+ + 2e^- \longrightarrow 2Na$$

$$H_2O \longrightarrow H^+ + OH^-$$

$$2Cl^- \longrightarrow Cl_2 + 2e^-$$

$$2Na + Hg \longrightarrow Na_2Hg$$

$$Na_2Hg + 2H_2O \longrightarrow 2NaOH + H_2 + Hg$$

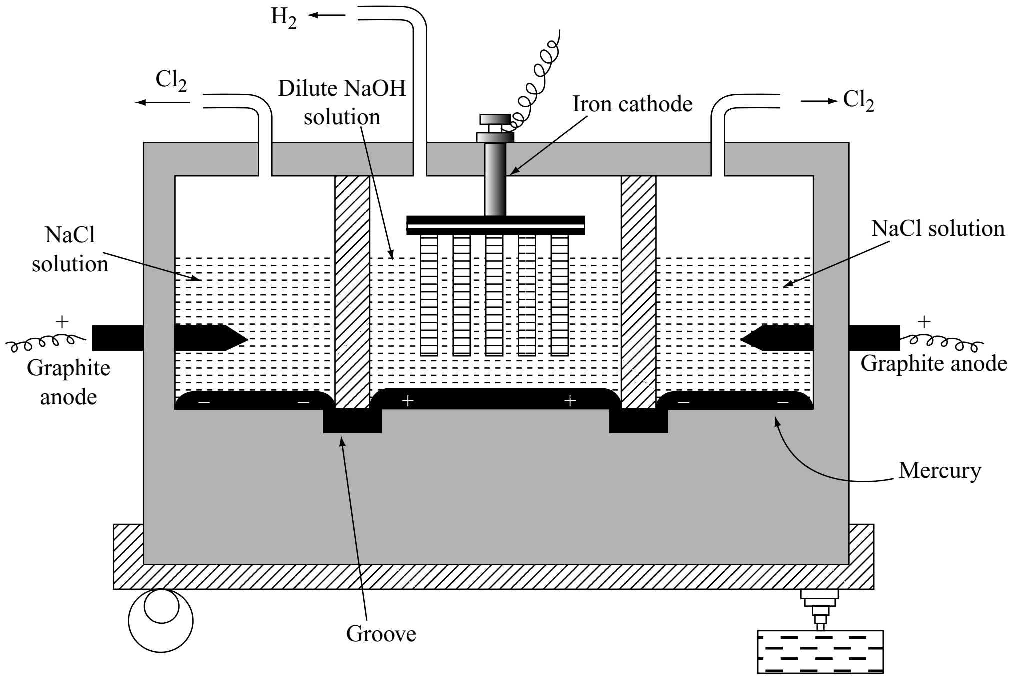


Fig 5.3 Castner–Kellner cell for the manufacture of sodium hydroxide

The mercury in the outer compartments acts as a cathode while in the middle compartment acts as an anode due to induction. Sodium is discharged in preference to hydrogen in the cell, since hydrogen has a high voltage at a mercury electrode. This amounts to saying that the discharge of hydrogen ions or the combination of hydrogen atoms to give molecules is difficult to achieve at mercury surface, i.e., mercury is a poor catalyst for either or both of these process. Since the sodium dissolves in mercury which is circulated through the cell, the formation of sodium hydroxide and hydrogen in the electrolytic cell is prevented.

As soon as the solution of caustic soda gets concentrated, it is withdrawn and concentrated to get solid caustic soda.

3. Kellner–Solvay process: It is also known as Solvay's trough cell. This cell has no compartments. It is a modified form of Castner–Kellner cell which has a number of graphite anodes and mercury cathode with no compartment. Mercury forms a layer of 1 cm thick in the cell. This flows across the bottom during electrolysis. The level of the brine in the cell is maintained constant throughout and it is made to flow through it in the same direction as the mercury.

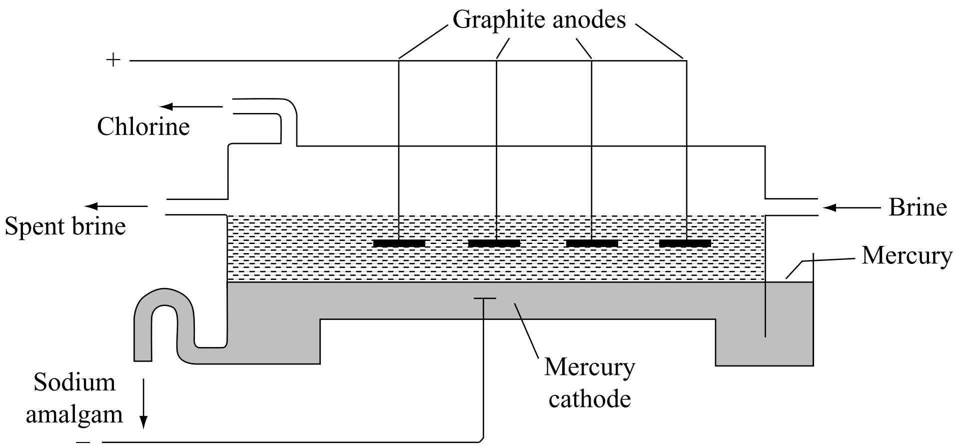


Fig 5.4 Kellner–Solvay process for the manufacture of sodium hydroxide

When electric current is passed through the cell, sodium ions get discharged at the mercury cathode to form sodium metal which dissolves in the mercury to form an amalgam. This flows out of the cell into an iron vessel containing water where it decomposes to give sodium hydroxide, hydrogen and mercury.

$$Hg_xNa_2 + 2H_2O \longrightarrow 2NaOH + xHg + H_2$$

The mercury liberated in the iron vessel is reused in the cell. The spent brine solution leaving the cell is first made saturated with sodium chloride solution and then used in the cell.

Physical properties

It is a deliquescent white crystalline solid in flakes, sticks, pellets or powder. It readily absorbs moisture and carbon dioxide from the atmosphere. So, it cannot be used as a primary standard. It freely dissolves in water with liberation of heat due to the formation of hydrates NaOH. xH_2O where x = 1, 2 or 7. It is soapy to touch. Its melting point is 591K. The solution of NaOH has a corrosive action on the skin as it attacks the proteins of the skin. It is therefore called as caustic soda.

Chemical Properties

(i) Action of heat: When heated at about 1573 K, it dissociates into its elements.

$$2NaOH \longrightarrow 2Na + H_2 + O_2$$

(ii) Action of water: When dissolved in water, it gives a strongly alkaline solution due to its complete ionization.

$$NaOH \longrightarrow Na^+ + OH^-$$

(iii) Basic character: Sodium hydroxide being a very strong alkali, reacts with almost all acids and acidic oxides.

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

$$NaOH + HNO_3 \longrightarrow NaNO_3 + H_2O$$

$$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$$

$$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$$

$$2NaOH + SO_2 \longrightarrow Na_2SO_3 + H_2O$$

$$2NaOH + SiO_2 \longrightarrow Na_2SiO_3 + H_2O$$

$$2NaOH + 2NO_2 \longrightarrow NaNO_3 + NaNO_2 + H_2O$$

Since sodium hydroxide reacts with silica of the glass of porcelain ware, it cannot be stored in the porcelain or glass ware.

(iv) Reaction with non-metals: Sodium hydroxide reacts with a number of non-metals. The products obtained depend on (1) nature of the non-metal, (2) concentration of NaOH and (3) temperature.

1. Boron reacts with NaOH forming borate liberating H_2.

$$2B + 6NaOH\ (\text{fused}) \longrightarrow 2Na_3BO_3 + 3H_2$$

2. Fused NaOH is reduced by carbon.

$$6NaOH + 2C \longrightarrow 2Na + 2Na_2CO_3 + 3H_2$$

3. Silicon when heated with concentrated NaOH solution liberates hydrogen forming sodium silicate.

$$Si + 2NaOH + H_2O \longrightarrow Na_2SiO_3 + 2H_2$$

4. Nitrogen do not react with NaOH but when white phosphorous is boiled with NaOH gives phosphine.

$$P_4 + 3NaOH + 3H_2O \longrightarrow 3NaH_2PO_2 + PH_3$$

5. When sulphur is boiled with NaOH solution sodium thiosulphate and sodium sulphide or polysulphides will be formed.

$$6NaOH + 4S \longrightarrow Na_2S_2O_3 + 2Na_2S$$

$$6NaOH + 12S \longrightarrow Na_2S_2O_3 + 2Na_2S_5$$

6. All halogens react with NaOH but give different products depending on the concentration of NaOH and temperature.

Flourine gives OF_2 with cold dilute NaOH and O_2 with hot concentrated NaOH.

$$\underset{\text{Cold and dil.}}{2NaOH} + 2F_2 \longrightarrow 2NaF + OF_2 + H_2O$$

$$\underset{\text{Hot and conc.}}{4NaOH} + 2F_2 \longrightarrow 4NaF + O_2 + 2H_2O$$

Chlorine, bromine and iodine react similarly with NaOH. In cold and dilute conditions, they form halides and hypohalites while in hot and concentrated conditions they form halides and halates.

$$\underset{\text{Cold and dil.}}{2NaOH} + X_2 \longrightarrow NaX + NaOX + H_2O$$

$$\underset{\text{Hot and conc.}}{6NaOH} + 3X_2 \longrightarrow 5NaX + NaXO_3 + 3H_2O$$

where X = Cl, Br or I.

The reaction of NaOH with P, S, Cl, Br, and I are called **disproportionation** reactions because in these reactions the non-metals undergo both oxidation and reduction.

(v) Reaction with metals: Generally electropositive metals do not react with bases. But the amphoteric metals like Zn, Al, Sn, Be, Ge, etc., can dissolve in sodium hydroxide liberating hydrogen.

$$Zn + 2NaOH \longrightarrow \underset{\text{Sod. zincate}}{Na_2ZnO_2} + H_2$$

$$2Al + 2NaOH + 2H_2O \longrightarrow \underset{\text{Sod. aluminate}}{2NaAlO_2} + 3H_2$$

$$Sn + 2NaOH + H_2O \longrightarrow \underset{\text{Sod. stannate}}{Na_2SnO_3} + 2H_2$$

$$Be + 2NaOH \longrightarrow \underset{\text{Sod. beryllate}}{Na_2BeO_2} + H_2$$

(vi) Reaction with compounds

1. **With ammonium salts:** Any ammonium salt on heating with sodium hydroxide liberates ammonia gas. In this reaction stronger base (NaOH) substitutes the weaker base NH_3.

$$NH_4Cl + NaOH \longrightarrow NaCl + H_2O + NH_3$$

This reaction is utilized to identify ammonium salts in the laboratory.

2. **Precipitation of hydroxides:** Sodium hydroxide precipitates many metals as their hydroxides from their salts. They may be divided mainly into three types.

(a) The hydroxides of the metal may be precipitated but insoluble by adding excess of sodium hydroxide.

$$FeSO_4 + 2NaOH \longrightarrow \underset{\text{grass green}}{Fe(OH)_2 \downarrow} + Na_2SO_4$$

$$FeCl_3 + 3NaOH \longrightarrow \underset{\text{reddish brown}}{Fe(OH)_3 \downarrow} + 3NaCl$$

$$NiSO_4 + 2NaOH \longrightarrow Ni(OH)_2 + Na_2SO_4$$

$$CuSO_4 + 2NaOH \longrightarrow Cu(OH)_2 \downarrow + Na_2SO_4$$

$$CrCl_3 + 3NaOH \longrightarrow Cr(OH)_3 \downarrow + 3NaCl$$

$$MnSO_4 + 2NaOH \longrightarrow Mn(OH)_2 \downarrow + Na_2SO_4$$

(b) Hydroxide of the metal may be precipitated but dissolves by adding excess of sodium hydroxide forming soluble complex salts.

$$AlCl_3 + 3NaOH \longrightarrow \underset{\text{white gelatinous}}{Al(OH)_3 \downarrow} + 3NaCl$$

$$Al(OH)_3 + NaOH \longrightarrow NaAlO_2 + 2H_2O$$

$$ZnSO_4 + 2NaOH \longrightarrow \underset{\text{white}}{Zn(OH)_2 \downarrow} + Na_2SO_4$$

$$Zn(OH)_2 + 2NaOH \longrightarrow Na_2ZnO_2 \downarrow + 2H_2O$$

$$SnCl_2 + 2NaOH \longrightarrow \underset{\text{white}}{Sn(OH)_2 \downarrow} + 2NaCl$$

$$Sn(OH)_2 + 2NaOH \longrightarrow Na_2SnO_2 + 2H_2O$$

$$BeCl_2 + 2NaOH \longrightarrow \underset{\text{white}}{Be(OH)_2 \downarrow} + 2NaCl$$

$$Be(OH)_2 + 2NaOH \longrightarrow Na_2BeO_2 + 2H_2O$$

(c) Hydroxides of the metal may be precipitated. The metal hydroxides precipitated are unstable and decompose into metal oxide. Only silver and mercury hydroxides decompose.

$$2AgNO_3 + 2NaOH \longrightarrow 2AgOH + 2NaNO_3$$

$$2AgOH \longrightarrow Ag_2O + H_2O$$

$$HgCl_2 + 2NaOH \longrightarrow Hg(OH)_2 + 2NaCl$$

$$Hg(OH)_2 \longrightarrow HgO + H_2O$$

(vii) Reactions with metal oxides

Generally metal oxides are basic in nature. So, they do not react with sodium hydroxide. But the metal oxides which are amphoteric in nature dissolve in sodium hydroxide forming corresponding salts.

$$Al_2O_3 + 2NaOH \longrightarrow \underset{\text{Sod. meta aluminate}}{2NaAlO_2} + H_2O$$

$$ZnO + 2NaOH \longrightarrow \underset{\text{Sod. Zincate}}{Na_2ZnO_2} + H_2O$$

$$SnO + 2NaOH \longrightarrow \underset{\text{Sod. stannite}}{Na_2SnO_2} + H_2O$$

$$SnO_2 + 2NaOH \longrightarrow \underset{\text{Sod. stannate}}{Na_2SnO_3} + H_2O$$

$$BeO + 2NaOH \longrightarrow \underset{\text{Sod. beryllate}}{Na_2BeO_2} + H_2O$$

$$PbO + 2NaOH \longrightarrow \underset{\text{Sod. plumbite}}{Na_2PbO_2} + H_2O$$

$$PbO_2 + 2NaOH \longrightarrow \underset{\text{Sod. plumbate}}{Na_2PbO_3} + H_2O$$

(viii) Reaction with carbon monoxide: When CO is passed over solid NaOH under high pressure at 150–200°C forms sodium formate.

$$NaOH + CO \xrightarrow[5-10\text{ atm}]{150-200°C} HCOONa$$

Uses of sodium hydroxide

Sodium hydroxide is used

(i) as a reagent in the laboratory
(ii) in the manufacture of soap, paper and rayon industries
(iii) in the manufacture of compounds like NaOX and $NaXO_3$
(iv) in petroleum refining
(v) in the mercirization of cotton
(vi) in the preparation of alumina, silicate glass, phosphates, etc.
(vii) to absorb SO_2 gas near electrical generators

5.11.2 Sodium Carbonate

The hydrated sodium carbonate $Na_2CO_3 \cdot 10H_2O$ is known as **washing soda** or **salt soda**. The anhydrous salt is known as **soda** or **soda ash**. Sodium carbonate is prepared mainly by three methods.

1. Le Blanc process, 2. Ammonia–Soda process and 3. Electrolytic process.

1. Le Blanc Process

This method is obsolete nowadays. It is important only for principle and the by-products. The raw materials used in the process are NaCl, concentrated sulphuric acid, lime stone and coke. The reactions that takes place during the process are as follows:

$$NaCl + H_2SO_4 \xrightarrow{\Delta} NaHSO_4 + HCl$$

$$NaHSO_4 + NaCl \longrightarrow Na_2SO_4 + HCl$$

$$Na_2SO_4 + 4C \xrightarrow{\Delta} Na_2S + 4CO$$

$$Na_2S + CaCO_3 \longrightarrow \underbrace{Na_2CO_3 + CaS}_{\text{Black ash}}$$

Brine is heated in large iron retorts with concentrated sulphuric acid. First, sodium bisulphate also known as **nitre cake** will be formed. This again reacts with concentrated sulphuric acid to form anhydrous sodium sulphate called **salt cake**. Salt cake on heating with coke and lime stone gives a mixture of sodium carbonate and calcium sulphide known as **black ash.** This is leached with water to remove calcium sulphide. The sodium carbonate dissolves and the insoluble impurities mainly consisting of CaS are left behind. The insoluble material is called sludge or alkali waste. The solution containing sodium carbonate is evaporated to get solid sodium carbonate.

2. Solvay's Ammonia–Soda process

Principle: When carbon dioxide gas is passed through a brine solution saturated with ammonia, ammonium bicarbonate will be formed.

$$2NH_3 + H_2O + CO_2 \longrightarrow (NH_4)_2CO_3$$

$$(NH_4)_2CO_3 + H_2O + CO_2 \longrightarrow 2NH_4HCO_3$$

Ammonium bicarbonate so formed interacts with sodium chloride to form sodium bicarbonate.

$$NaCl + NH_4HCO_3 \longrightarrow NaHCO_3 + NH_4Cl$$

Sodium bicarbonate, which is sparingly soluble in water, gets precipitated out. The precipitated sodium bicarbonate is filtered off and then calcinated to get sodium carbonate.

$$2NaHCO_3 \xrightarrow{\text{Calcination}} Na_2CO_3 + H_2O + CO_2$$

Raw materials: The raw materials are sodium chloride, ammonia and lime stone (for the supply of CO_2). When lime stone is heated carbon dioxide is obtained.

$$CaCO_3 \longrightarrow CaO + CO_2$$

The lime obtained as above can be used to recover ammonia from ammonium chloride on heating.

$$CaO + 2NH_4Cl \longrightarrow CaCl_2 + 2NH_3 + H_2O$$

Process: The flow sheet diagram of Solvay's ammonia–soda process is shown in Fig. 5.5. The process involves the following steps.

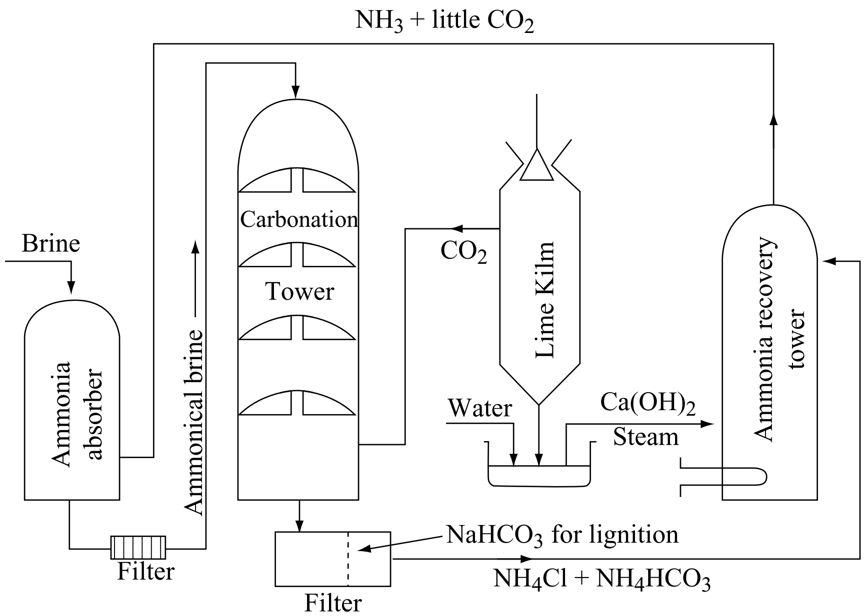


Fig 5.5 Flow sheet diagram of Solvay's ammonia–soda process for the manufacture of sodium carbonate

(i) Saturation of brine with ammonia: A 30 per cent solution of brine is admitted in a big tank. Then brine is pumped into the ammonia absorber. In this the brine solution is saturated by ammonia (obtained from ammonia recovery tower) mixed with little carbon dioxide. Any magnesium or calcium salts present as impurity in sodium chloride are precipitated as carbonates or hydroxide.

$$NH_3 + H_2O \longrightarrow NH_4OH$$

$$2NH_3 + H_2O + CO_2 \longrightarrow (NH_4)_2CO_3$$

$$MgCl_2 + 2NH_4OH \longrightarrow Mg(OH)_2 + 2NH_4Cl$$

$$CaCl_2 + (NH_4)_2CO_3 \longrightarrow CaCO_3 + 2NH_4Cl$$

After saturation, the liquid is pumped through the filter press to remove the precipitated, impurities, and then passed through the cooling pipes as shown in Fig. 5.5.

(ii) Carbonation: It is a tall tower (20 metres) fitted with perforated plates. Ammonical brine is allowed to enter a little above the middle of carbonating tower, whereas carbon dioxide gas from the lime kiln along with that obtained by heating sodium bicarbonate is admitted from the base of the tower. The downcoming brine meets an upward stream of carbon dioxide when small crystals of sodium bicarbonate appear.

(iii) Filtration: The solution containing crystals of sodium bicarbonate is then filtered with the help of rotary vacuum filter. The solid sodium bicarbonate deposits on its surface (which is scraped), whereas the filtrate is pumped to the top of ammonia recovery tower.

(iv) Calcination: The sodium bicarbonate obtained in step (iii) is heated strongly in a kiln to convert it into sodium carbonate

$$2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$$

The carbon dioxide gas is used in the carbonation tower along with the gas obtained by heating lime stone.

(v) Ammonia generator: The filtrate obtained in step (iii) contains ammonium salts such as ammonium bicarbonate and ammonium chloride. This filtrate is allowed to flow down the ammonia recovery tower, whereas slaked lime is entering a little above the middle of the tower and steam is admitted from the bottom.

$$NH_4HCO_3 \longrightarrow NH_3 + H_2O + CO_2$$

$$Ca(OH)_2 + 2NH_4Cl \longrightarrow 2NH_3 + CaCl_2 + 2H_2O$$

The mixture of ammonia and carbon dioxide is pumped to ammonia saturation tower where these are used for the saturation of brine. The calcium chloride is obtained as a by-product.

(iv) Lime kiln: In this the lime stone is heated to about 1273 K to obtain carbon dioxide and calcium oxide.

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

The carbon dioxide gas goes to the carbonation tower, whereas the lime is saturated with water in a tank known as slaker to form calcium hydroxide. The calcium hydroxide is pumped to the ammonia recovery tower.

From the above it is evident that ammonia can be recovered and sodium chloride and lime stone are the only raw materials consumed in the process. Calcium chloride is the only by-product. Further, the sodium carbonate manufactured by this method is pure and also cheap.

3. Electrolytic process

If carbon dixode under pressure is blown along with steam through the Nelson's cell which is being used for the manufacture of sodium hydroxide, the caustic soda produced will react with carbon dioxide to give sodium carbonate.

$$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$$

The solution is concentrated and crystallized to give crystals of sodium carbonate.

This method is preferred where electricity is cheap.

Physical properties: Sodium carbonate gives white crystalline solid. It is a decahydrate $Na_2CO_3 \cdot 10H_2O$. These crystals efflorescence on exposure to dry air giving the monohydrate $Na_2CO_3 \cdot H_2O$. This form is generally used as washing soda.

$$Na_2CO_3 \cdot 10H_2O \longrightarrow Na_2CO_3 \cdot H_2O + 9H_2O$$

When heated it changes into the anhydrous compounds which is stable towards heat. It is soluble in water with evolution of considerable amount of heat and the solution becomes alkaline due to hydrolysis.

$$Na_2CO_3 \longrightarrow 2Na^+ + CO_3^{2-}$$

$$CO_3^{2-} + 2H_2O \longrightarrow H_2CO_3 + 2OH^-$$

Chemical properties

(i) Action of acids: Acids decompose sodium carbonate with evolution of carbon dioxide.

$$Na_2CO_3 + HCl \longrightarrow NaHCO_3 + NaCl$$

$$NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$$

$$\text{(or) } Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$$

(ii) Action of carbon dioxide: When carbon dioxide gas is passed through the concentrated solution of sodium carbonate it converts into sodium bicarbonate.

$$Na_2CO_3 + H_2O + CO_2 \longrightarrow 2NaHCO_3$$

(iii) Reaction with silica: Fusion of silica with sodium carbonate gives sodium silicate called **water glass.**

$$Na_2CO_3 + SiO_2 \longrightarrow Na_2SiO_3 + CO_2$$

(iv) Reaction with sulphur and sulphur dioxide: When sulphur dioxide is passed through a solution of sodium carbonate it converts into sodium sulphite. If the solution also contains sulphur, sodium thiosulphate is formed.

$$Na_2CO_3 + SO_2 \longrightarrow Na_2SO_3 + CO_2$$

$$Na_2CO_3 + SO_2 + S \longrightarrow Na_2S_2O_3 + CO_2$$

(v) Reaction with Metal salts: Sodium carbonate precipitates several metals (except alkali metals) from their salt solutions as their normal carbonates or basic metal carbonates. More electropositive metals are precipitated as normal carbonates while less electropositive metals are precipitated as basic metal carbonates.

$$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaCl$$

$$BaCl_2 + Na_2CO_3 \longrightarrow BaCO_3 + 2NaCl$$

$$2MgCl_2 + 2Na_2CO_3 + H_2O \longrightarrow MgCO_3 \cdot Mg(OH)_2 + 4NaCl + CO_2$$

$$5ZnSO_4 + 5Na_2CO_3 + 4H_2O \longrightarrow [2ZnCO_3 \cdot 3Zn(OH)_2] \cdot H_2O + 5Na_2SO_4 + 3CO_2$$

$$2CuSO_4 + 2Na_2CO_3 + H_2O \longrightarrow Cu(OH)_2 \cdot CuCO_3 + CO_2 + Na_2SO_4$$

$$3(CH_3COO)_2Pb + 3Na_2CO_3 + H_2O \longrightarrow 2PbCO_3 \cdot Pb(OH)_2 + CO_2 + 6CH_3COONa$$

$$2AgNO_3 + Na_2CO_3 \longrightarrow Ag_2CO_3 + 2NaNO_3$$

Carbonates of metals like iron, aluminium, tin, etc., when formed are immediately hydrolyzed to hydroxides.

$$Fe_2(SO_4)_3 + 3Na_2CO_3 \longrightarrow Fe_2(CO_3)_3 + 3Na_2SO_4$$

$$Fe_2(CO_3)_3 + 3H_2O \longrightarrow 2Fe(OH)_3 + 3CO_2$$

The reaction of aluminium sulphate and sodium carbonate is used in foam fire extinguishers.

Uses

Sodium carbonate is used

(i) in the manufacture of glass, water glass and caustic soda

(ii) in laundries and washing purpose

(iii) in the softening of hard water

(iv) in petroleum refining process

(v) in the manufacture of ultramarine and artificial zeolites

(vi) in paper industry and in dyeing industry

(vii) in the laboratory as a reagent in qualitative and quantitative analysis

5.11.3 Sodium Bicarbonate

Sodium bicarbonate is also known as **baking soda**. In the laboratory it can be prepared by passing carbon dioxide into aqueous sodium carbonate solution.

$$Na_2CO_3 + H_2O + CO_2 \longrightarrow 2NaHCO_3$$

On large scale it is obtained in Solvay's ammonia–soda process.

Properties: It is sparingly soluble in water. Its solution is alkaline due to the hydrolysis of bicarbonate ion but to somewhat lesser extent than the carbonate ion.

$$NaHCO_3 \longrightarrow Na^+ + HCO_3^-$$

$$HCO_3^- + H_2O \longrightarrow H_2CO_3 + OH^-$$

Solid sodium bicarbonate on heating decomposes forming sodium carbonate and carbon dioxide.

$$2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$$

The metal salts which gives basic metal carbonates with sodium carbonate gives normal carbonates with sodium bicarbonate.

$$MgSO_4 + 2NaHCO_3 \longrightarrow MgCO_3 + Na_2SO_4 + H_2O + CO_2$$

$$ZnSO_4 + 2NaHCO_3 \longrightarrow ZnCO_3 + Na_2SO_4 + H_2O + CO_2$$

Uses

(i) It is used as an antacid in medicines to neutralize the acidity in stomach.

(ii) It is used in dry fire extinguishers.

(iii) It is used as a constituent of baking powders, sieldlitz powder and in effervescent drinks, baking powder is a mixture of potassium hydrogen tartrate, starch, sodium bicarbonate.

(iv) It is used as a mild antiseptic for skin infections.

5.11.4 Sodium Oxide

Sodium monoxide can be prepared by heating sodium metal in limited supply of oxygen.

$$4Na + O_2 \longrightarrow 2Na_2O$$

But it may contain little peroxide also. Pure Na_2O can be prepared by heating Na_2O_2,, NaOH, $NaNO_2$ or $NaNO_3$ with sodium metal.

$$Na_2O_2 + 2Na \longrightarrow 2Na_2O$$

$$NaOH + Na \longrightarrow Na_2O + \frac{1}{2}H_2$$

$$NaNO_2 + 3Na \longrightarrow 2Na_2O + \frac{1}{2}N_2$$

$$NaNO_3 + 5Na \longrightarrow 3Na_2O + \frac{1}{2}N_2$$

Pure Na_2O is also formed when sodium azide and sodium nitrite mixture is heated.

$$3NaN_3 + NaNO_2 \longrightarrow 2Na_2O + 5N_2$$

It is a white amorphous substance. It reacts with water violently forming sodium hydroxide with evolution of large amount of heat.

$$Na_2O + H_2O \longrightarrow 2NaOH$$

When Na_2O is heated in oxygen it converts into Na_2O_2

$$Na_2O + \frac{1}{2}O_2 \longrightarrow Na_2O_2$$

When heated to 400°C it disproportionates into sodium peroxide and sodium vapour.

$$2Na_2O \xrightarrow{400°C} Na_2O_2 + 3Na$$

When it reacts with liquid ammonia sodamide is formed

$$Na_2O + NH_3 \longrightarrow NaNH_2 + NaOH$$

It is used as a dehydrating agent and polymerizing agent in organic chemistry.

5.11.5 Sodium Peroxide

Sodium peroxide is prepared by first oxidizing sodium to sodium monoxide in a limited supply of oxygen (air) and then reacting this further to give sodium peroxide.

$$2Na + \frac{1}{2}O_2 \xrightarrow{heat} Na_2O \xrightarrow{1/2\,O_2} Na_2O_2$$

It is a pale yellow powder. Sodium peroxide reacts with moisture and carbon dioxide.

$$2Na_2O_2 + 2H_2O \longrightarrow 4NaOH + O_2$$

$$2Na_2O_2 + 2CO_2 \longrightarrow 4Na_2CO_3 + O_2$$

At low temperature when dissolved in water, in dilute solution gives H_2O_2 but in concentrated solution gives oxygen.

$$Na_2O_2 + 2H_2O \longrightarrow 2NaOH + H_2O_2$$

$$2Na_2O_2 + 2H_2O_2 \longrightarrow 4NaOH + O_2$$

With dilute acids it gives H_2O_2 at low temperatures.

$$Na_2O_2 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O_2$$

It reacts with carbon monoxide and carbon dioxide forming carbonates.

$$Na_2O_2 + CO \longrightarrow Na_2CO_3$$

$$Na_2O_2 + CO_2 \longrightarrow Na_2CO_3 + \frac{1}{2}O_2$$

The reaction with carbon dioxide is used in breathing apparatus for divers, firemen and in submarines.

Sodium peroxides oxidizes several compounds. For example, it oxidizes chromium (III) compounds to chromates, manganese (II) compounds to manganates, sulphides to sulphates, etc.

$$2CrCl_3 + 3Na_2O_2 + 4NaOH \longrightarrow 2Na_2CrO_4 + 6NaCl + 2H_2O$$

$$MnSO_4 + 2Na_2O_2 \longrightarrow Na_2MnO_4 + Na_2SO_4$$

$$Na_2S + 4Na_2O_2 \longrightarrow Na_2SO_4 + 4Na_2O$$

Sodium peroxide oxidizes benzoyl chloride to benzoyl peroxide which is used as a bleaching agent.

$$2C_6H_5\,COCl + Na_2O_2 \longrightarrow (C_6H_5CO)_2\,O_2 + 2NaCl.$$

Uses

It is used

(i) as an oxidizing agent in laboratory
(ii) for the production of oxygen under the name **oxone**
(iii) for the purification of air
(iv) for the preparation of benzoyl peroxide
(v) for bleaching the delicate articles like wool, silk, etc

5.11.6 Sodium Nitrate

Sodium nitrate occurs naturally as chile salt petre. It is also present in caliche. In the laboratory it can be prepared by the action of nitric acid on sodium hydroxide or sodium carbonate.

$$NaOH + HNO_3 \longrightarrow NaNO_3 + H_2O$$

$$Na_2CO_3 + 2HNO_3 \longrightarrow 2NaNO_3 + 2H_2O$$

Pure salt is obtained by crystallization.

Properties: It is a white, deliquescent, crystalline solid and highly soluble in water.

When heated to 500°C it converts to sodium nitrite.

$$2NaNO_3 \xrightarrow{500°C} 2NaNO_2 + O_2$$

But at 800°C it decomposes to sodium oxide, nitrogen and oxygen.

$$2NaNO_3 \xrightarrow{800°C} Na_2O + N_2 + \frac{5}{2}O_2$$

Chemical reduction with carbon or metals like Zn, Pb, etc. gives sodium nitrite.

$$2NaNO_3 + C \longrightarrow 2NaNO_2 + CO_2$$

$$NaNO_3 + Pb \longrightarrow NaNO_2 + PbO$$

When heated with sodium metal gives sodium monoxide.

$$2NaNO_3 + 10Na \longrightarrow 6Na_2O + N_2$$

Uses

(i) It is used as a nitrogenous fertilizer in agriculture.
(ii) It is used in the manufacture of HNO_3, $NaNO_2$ and KNO_3.

5.11.7 Sodium Nitrite

It is manufactured by the following methods.

(i) By heating sodium nitrate alone or reduction with lead or mixture of carbon and lime.

$$2NaNO_3 \longrightarrow 2NaNO_2 + O_2$$

$$2NaNO_3 + Pb \longrightarrow 2NaNO_2 + PbO$$

$$2NaNO_3 + C + Ca(OH)_2 \longrightarrow 2NaNO_2 + CaCO_3 + H_2O$$

(ii) Commercially sodium nitrite is manufactured by passing oxides of nitrogen into an aqueous solution of sodium hydroxide or sodium carbonate.

$$2NaOH + NO_2 + NO \longrightarrow 2NaNO_2 + H_2O$$

$$Na_2CO_3 + NO_2 + NO \longrightarrow 2NaNO_2 + CO_2$$

(iii) It can also be prepared by passing sulphur dioxide in a concentrated solution of sodium nitrate to which lime has been added.

$$NaNO_3 + CaO + SO_2 \longrightarrow NaNO_2 + CaSO_4$$

Properties

(i) It is a white crystalline substance, but ordinarily it is slightly yellow due to impurities. It is soluble in water (8.33% at 288 K).
(ii) On heating with ammonium chloride, nitrogen is evolved.

$$NH_4Cl + NaNO_2 \longrightarrow NaCl + NH_4NO_2$$

$$NH_4NO_2 \longrightarrow N_2 + 2H_2O$$

(iii) When heated with acids it gives off oxides of nitrogen.

$$NaNO_2 + HCl \longrightarrow NaCl + HNO_2$$

$$3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$$

In sodium nitrite the oxidation number of nitrogen is +III. It can increase or decrease its oxidation number. So, it can act as both oxidizing and reducing agent.

(iv) It oxidizes urea to nitrogen in acid medium.

$$H_2NCONH_2 + 2NaNO_2 + 2HCl \longrightarrow 2N_2 + 3H_2O + CO_2 + 2NaCl$$

(v) It oxidizes iodides to iodine in acid medium.

$$2NaNO_2 + 2KI + 4HCl \longrightarrow 2KCl + 2NaCl + 2NO + 2H_2O + I_2$$

(vi) It oxidizes stannous chloride, sulphur dioxide, etc.

$$SnCl_2 + 2NaNO_2 + 4HCl \longrightarrow SnCl_4 + 2NaCl + 2NO + 2H_2O$$

$$SO_2 + 2NaNO_2 + 2HCl \longrightarrow 2NaCl + H_2SO_4 + 2NO$$

(vii) If cobalt nitrite is treated with sodium nitrite and acetic acid, sodium cobalt-nitrite is formed.

$$Co(NO_3)_2 + 2NaNO_2 \longrightarrow Co(NO_2)_2 + 2NaNO_3$$

$$NaNO_2 + CH_3COOH \longrightarrow CH_3COONa + HNO_2$$

$$Co(NO_2)_2 + 2HNO_2 \longrightarrow Co(NO_2)_3 + H_2O + NO$$

$$Co(NO_2)_3 + 3NaNO_2 \longrightarrow Na_3[Co(NO_2)_6]$$

(viii) It reduces potassium permanganate, potassium dichromate, bromine, etc.

$$2KMnO_4 + 3H_2SO_4 + 5NaNO_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5NaNO_3$$

$$K_2Cr_2O_7 + 4H_2SO_4 + 3NaNO_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3NaNO_3$$

$$Br_2 + H_2O + NaNO_2 \longrightarrow 2HBr + NaNO_3$$

Uses

Nitrites are important in the manufacture of organonitrogen compounds like azodyes. It is used as a food preservative. Also, it used in both qualitative and quantitative analysis.

5.11.8 Sodium Chloride

It is the common salt mainly found in sea water and in land lakes. In the form of rock salt deposits it is known to exist. From these sources sodium chloride is obtained by evaporation by solar heat and wind.

Ordinary sodium chloride is slightly hygroscopic due to the presence of calcium and magnesium chloride. It is a typical compound whose **solubility changes very little with a change** in temperature. This is because its lattice enthalpy and hydration enthalpies are almost equal. It has a face centred cubic structure in its crystalline form.

Uses: It is an essential constituent in our diet. It is a starting material for the manufacture of sodium metal and chlorine gas (Down's process), caustic soda (Castner–Kellner process) and washing soda (Solvay's process) used as a preservative for meat and fish.

5.11.9 Sodium Sulphate $Na_2SO_4 \cdot 10H_2O$

It is also known as **Glauber's salt**. Anhydrous sodium sulphate (called salt cake) is manufactured by heating sodium chloride with sodium bisulphate.

$$NaCl + NaHSO_4 \longrightarrow Na_2SO_4 + HCl$$

When the salt cake is crystallized below 305 K, it is converted into Glauber's salt $Na_2SO_4 \cdot 10H_2O$.

Uses: Sodium sulphate is used in textile industry, medicines as purgative, manufacture of glass plates and sodium salts.

5.11.10 Sodium Ammonium Hydrogen Phosphate $Na(NH_4)HPO_4 \cdot 4H_2O$

It is also known as **microcosmic salt**. It is prepared by dissolving equimolecular quantities of disodium hydrogen phosphate and ammonium chloride in little water.

$$Na_2HPO_4 + NH_4Cl \longrightarrow Na(NH_4)HPO_4 + NaCl$$

It forms colourless, needle-like crystals. On heating, it gives a glassy mass of sodium metaphosphate.

$$Na(NH_4)HPO_4 \longrightarrow NaPO_3 + H_2O + NH_3$$

It is used in qualitative analysis for bead tests. When hot glassy bead of sodium meta phosphate is brought in contact with coloured substance and strongly heated a coloured bead is formed, if the substance contains Cu^{2+}, Ni^{2+}, Mn^{2+}, CO^{2+}, etc. It is particularly used for testing silica with which a cloudy bead containing floating properties of silica is obtained.

5.11.11 Sodium Amide (Sodamide) $NaNH_2$

It is prepared by passing a stream of dry NH_3 over metallic sodium at 575–675 K.

$$2Na + 2NH_3 \longrightarrow 2NaNH_2 + H_2$$

Properties

(i) It is a waxy solid which melts at 483 K. It reacts with water liberating ammonia.

$$NaNH_2 + H_2O \longrightarrow NaOH + NH_3$$

(ii) When heated with carbondioxide it forms cyanamide.

$$2NaNH_2 + 2CO_2 \longrightarrow Na_2CO_3 + H_2O + CNNH_2$$

(iii) With nitrous oxide, sodium azide is formed.

$$NaNH_2 + N_2O \longrightarrow NaN_3 + H_2O$$

(iv) On heating with charcoal it gives sodium cyanide.

$$NaNH_2 + C \longrightarrow NaCN + H_2$$

It is used in the manufacture of dyes and sodium cyanide.

5.11.13 Sodium Cyanide NaCN

(i) From sodamide: When sodamide is heated with red hot charcoal, first sodium cyanamide is formed which further react with carbon to give sodium cyanide.

$$2NaNH_2 + C \longrightarrow Na_2NCN + 2H_2$$

$$Na_2NCN + C \longrightarrow 2NaCN$$

(ii) From coal gas: Impure coal gas contain hydrocyanic acid, which is removed from the coal gas by passing in an alkaline ferrous sulphate solution, whereby sodium ferrocyanide is obtained. The later salt when heated with the metallic sodium in covered iron crucibles gives sodium cyanide.

$$NaOH + HCN \longrightarrow NaCN + H_2O$$

$$6NaCN + FeSO_4 \longrightarrow Na_4[Fe(CN)_6] + Na_2SO_4$$

$$Na_4[Fe(CN)_6] + 2Na \longrightarrow 6NaCN + Fe$$

(iii) From calcium cyanamide: It is manufactured by fusing crude calcium cyanamide (nitrolim) with carbon and common salt or sodium carbonate.

$$CaNCN + C + 2NaCl \longrightarrow 2NaCN + CaCl_2$$

$$CaNCN + C + Na_2CO_3 \longrightarrow 2NaCN + CaCO_3$$

(iv) It is manufactured by strong heating of sodium carbonate, powdered coal and finely divided catalyst (iron) in a current of nitrogen.

$$Na_2CO_3 + 4C + N_2 \xrightarrow{Fe} 2NaCN + 3CO$$

Properties

It is a colourless, crystalline, highly poisonous solid, soluble in water. It smells like bitter almonds due to hydrocyanic acid.

Its aqueous solution is alkaline due to hydrolysis.

$$NaCN + H_2O \longrightarrow NaOH + HCl$$

It forms number of complexes with number of transition metals

E.g.: $CdSO_4 + 4NaCN \longrightarrow Na_2[Cd(CN)_4] + Na_2SO_4$

$$AgNO_3 + 2NaCN \longrightarrow Na[Ag(CN)_2] + NaNO_3$$

With copper sulphate first gives cupric cyanide which dissociates into cuprous cyanide and cyanogen. The cuprous cyanide dissolves in excess of sodium cyanide due to the formation of complex.

$$CuSO_4 + 2NaCN \longrightarrow Cu(CN)_2 + Na_2SO_4$$

$$2Cu(CN)_2 \longrightarrow Cu_2(CN)_2 + (CN)_2$$

$$Cu_2(CN)_2 + 6NaCN \longrightarrow 2Na_3[Cu(CN)_4]$$

Sodium cyanide reacts with metallic silver and gold in the presence of air forming complex cyanides.

$$4Ag + 8NaCN + 2H_2O + O_2 \longrightarrow 4Na_2[Ag(CN)_2] + 4NaOH$$

$$4Au + 8NaCN + 2H_2O + O_2 \longrightarrow 4Na_2[Au(CN)_2] + 4NaOH$$

Uses: It is used in the extraction of silver and gold, used in the electroplating of silver, gold and other metals. As a reagent in the laboratory and in the synthesis of organic compounds.

5.12 IMPORTANT COMPOUNDS OF POTASSIUM

Potassium and potassium salts are not of much demand as cheaper sodium and its salts can be used for the same purpose. The preparation and properties of several compounds are almost similar to those of sodium salts. Here, we discuss about the compounds which differ either in preparation or in properties and those which are very important.

5.12.1 Oxides of Potassium

Potassium forms K_2O, K_2O_2, K_2O_3, KO_2 of which potassium monoxide and peroxide can be prepared exactly in a similar way as that of Na_2O and Na_2O_2. Their behaviour is also same as those of Na_2O and Na_2O_2.

K_2O_3 is obtained when oxygen is passed through liquid ammonia containing potassium.

$$4K + 3O_2 \xrightarrow{\text{Liquid } NH_3} 2K_2O_3$$

Potassium superoxide (KO_2) is prepared by burning potassium in excess of oxygen free from moisture.

$$K + O_2 \longrightarrow KO_2$$

Potassium superoxide is chrome yellow powder. When dissolved in water it forms hydrogen peroxide with evolution of oxygen.

$$2KO_2 + 2H_2O \longrightarrow 2KOH + H_2O_2 + O_2$$

It absorbs CO and CO_2

$$2KO_2 + CO \longrightarrow K_2CO_3 + O_2$$

$$4KO_2 + 2CO_2 \longrightarrow 2K_2CO_3 + 3O_2$$

It oxidizes sulphur forming potassium sulphate.

$$2KO_2 + S \longrightarrow K_2SO_4$$

Potassium superoxide is used as an oxidizing agent. It is used as air purifier in submarine, space capsules and in breathing masks as it produces oxygen and remove carbon dioxide.

5.12.2 Potassium Hydroxide

Solid potassium hydroxide is known as **caustic potash** while aqueous potassium hydroxide is known as **potash lye**. The preparation and properties are exactly similar to those of sodium hydroxide.

It is mainly used in making soft soaps. Alcoholic potassium hydroxide is a useful reagent in organic chemistry as it eliminates hydrogen halides from alkyl halides.

$$C_2H_5Br + KOH \xrightarrow{\text{alcohol}} CH_2 = CH_2 + KBr + H_2O$$

It is a better absorber for acidic oxides like CO_2, SO_2, gases etc. than sodium hydroxide because potassium salts are more soluble in water than sodium salts.

5.12.3 Potassium Carbonate

It is also called as **Pearl ash**. Solvay process cannot be used for the manufacture of potassium carbonate because potassium bicarbonate is highly soluble in water and does not crystallize like sodium bicarbonate. However, Le Blanc process can be used with modification, i.e., potassium chloride is used in the place of sodium chloride.

$$2KCl + H_2SO_4 \longrightarrow K_2SO_4 + 2HCl$$

$$K_2SO_4 + 4C + CaCO_3 \longrightarrow K_2CO_3 + CaS + 4CO$$

It is now manufactured by magnesia process (Precht’s process). In this process, carbon dioxide is passed into potassium chloride solution at 293 K in the presence of hydrated magnesium carbonate, $MgCO_3 \cdot 3H_2O$.

$$2KCl + 3(MgCO_3 \cdot 3H_2O) + CO_2 \longrightarrow 2(KHCO_3 \cdot MgCO_3 \cdot 4H_2O) + MgCl_2$$

The precipitate of potassium hydrogen magnesium carbonate is obtained by filtration and decomposed by boiling with water giving a precipitate of magnesium carbonate which can be reused.

$$2KHCO_3 \cdot MgCO_3 . 4H_2O \longrightarrow K_2CO_3 + 2MgCO_3 + 9H_2O + CO_2$$

The filtrate is evaporated and allowed to crystallize to get K_2CO_3.

It is a white solid (m.pt 1173 K), hygroscopic in nature and highly soluble in water. Its aqueous solution is alkaline in nature due to hydrolysis. It is quite thermally stable like sodium carbonate. It resembles sodium carbonate in its chemical reactions.

It is used in making hard glass, toilet soap and potassium compounds. It is also used as a drying agent in the laboratory.

5.12.4 Potassium Iodide

Preparation: Potassium iodide can be prepared as follows.

(i) In the laboratory it can be prepared by the action of hydroiodic acid on potassium hydroxide or potassium carbonate.

$$KOH + HI \longrightarrow KI + H_2O$$

$$K_2CO_3 + 2HI \longrightarrow 2KI + H_2O + CO_2$$

(ii) It can be manufactured by heating iodine with a hot and concentrated solution of potassium hydroxide. The resulting liquid is evaporated to dryness, and then ignited with powdered charcoal to change potassium iodate into potassium iodide.

$$3I_2 + 6KOH \longrightarrow 5KI + KIO_3 + 3H_2O$$

$$KIO_3 + 3C \longrightarrow KI + 3CO$$

This mass is extracted with water, filtered and evaporated to dryness.

(iii) Rubbing iodine with iron filings in the presence of water forms ferrous iodide, which again converts into ferrous ferric iodide. It is treated with potassium carbonate to precipitate iron as ferrous and ferric hydroxides

$$Fe + I_2 \longrightarrow FeI_2$$

$$3FeI_2 + I_2 \longrightarrow Fe_3I_8$$

$$4K_2CO_3 + Fe_3I_8 + 4H_2O \longrightarrow 8KI + 4CO_2 + Fe(OH)_2 + 2Fe(OH)_3$$

The solution is filtered and the filtrate is evaporated to crystallization.

Properties

(i) It forms white cubic crystals (m.pt 953K) soluble in water and alcohol.

(ii) It dissolves iodine forming tri-iodide, which is unstable and gives up extra iodine easily.

$$KI + I_2 \longrightarrow KI_3$$

(iii) When heated with concentrated sulphuric acid, first white fumes of HI will be liberated, which again oxidized to violet-coloured vapours of iodine by concentrated sulphuric acid.

$$2KI + 2H_2SO_4 \longrightarrow 2KHSO_4 + 2HI$$

$$2HI + H_2SO_4 \longrightarrow 2H_2O + SO_2 + I_2$$

(iv) If chlorine gas is passed through the solution of potassium iodide solution, iodine will be liberated.

$$2KI + Cl_2 \longrightarrow 2KCl + I_2$$

(v) Several oxidizing agents like $KMnO_4$, $K_2Cr_2O_7$, H_2O_2, etc., liberate iodine from potassium iodide.

$$2KMnO_4 + 8H_2SO_4 + 10KI \longrightarrow 6K_2SO_4 + 2MnSO_4 + 5I_2 + 8H_2O$$

$$K_2Cr_2O_4 + 7H_2SO_4 + 6KI \longrightarrow 4K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3I_2$$

$$H_2O_2 + H_2SO_4 + 2KI \longrightarrow K_2SO_4 + 2H_2O + I_2$$

(vi) With copper sulphate, first it forms unstable cupric iodide, which decomposes to cuprous iodide and iodine.

$$2CuSO_4 + 4KI \longrightarrow 2K_2SO_4 + 2CuI_2$$

$$2CuI_2 \longrightarrow Cu_2I_2 + I_2$$

(vii) It gives yellow precipitates of lead iodide with lead acetate and silver iodide with silver nitrate.

$$(CH_3COO)_2\,Pb + 2KI \longrightarrow PbI_2 \downarrow + 2CH_3COOK$$

$$AgNO_3 + KI \longrightarrow AgI \downarrow + KNO_3$$

(viii) It forms a red precipitate of HgI_2 with $HgCl_2$ which dissolves in excess of potassium iodide forming a complex potassium tetraiodido mercury (II). Its alkaline solution is known as **Nessler's reagent** and is used for detecting ammonia.

$$HgCl_2 + 2KI \longrightarrow HgI_2 \downarrow + 2KCl$$

$$HgI_2 + 2KI \longrightarrow K_2HgI_4$$

Uses: It is used

(i) as a laboratory reagent

(ii) in dissolving iodine and for preparation of tincture of iodine, throat pain, etc

(iii) in preparing Nessler's reagent used for the detection of ammonia

5.13 BIOLOGICAL IMPORTANCE OF SODIUM AND POTASSIUM

Sodium and potassium are the most common cations in biological fluids. Sodium ion is the major cation of extracellular fluids of animals and in blood plasma, including human beings which is known to activate certain enzymes in the animal body. Sodium ions are relatively harmless except when present in excess amounts in which case they may cause hypertension. The saline water containing excessive amounts of NaCl is harmful to plant and aquatic life because of the toxicity of Na^+ ions. These ions participate in the transmission of nerve signals. They also regulate flow of water across cell membranes and in transport of sugars, amino acids into the cells.

Though sodium and potassium have close resemblances in their chemical behaviour, surprisingly their biological functions are very different. The sodium and potassium ions, taken together constitute what is known as **Na^+-K^+ pump** or simply as **sodium pump**. There is transfer of Na^+ ions from the intracellular fluids to the extracellular fluids with the help of "carrier proteins". Simultaneously, it transfers K^+ ions from the extracellular fluids to the intracelluler fluids. Since in the each operation of the pump, the number of sodium ions pumped out of the cell is more than the number of K^+ ions that pumped into the cell. Thus the **interior** fluid of the cell acquires negative charge and the exterior of the cell acquires an excess of positive charge. This results in the development of electrical potential gradient across the cell membrane which is responsible for the transmission of nerve signals in animals. The **Na^+-K^+** pump also maintains the volume of the cell. Without the existence of **Na^+-K^+** pump in the cell the latter would have swelled in volume and ultimately burst.

Potassium ions are the most abundant cations within cell fluids, where they activate many enzymes that participate in oxidation of glucose to produce adenosine triphosphate (ATP).

A typical 70 kg adult contains about 90 g of Na^+ ions and 170g of K^+ ions. The daily requirement of sodium and potassium is about 2 g each.

KEY POINTS

- Except H and Fr, the other elements of Group 1A give strong bases by dissolving their oxides and hydroxides in water. So, they are known **as alkali metals**.
- Since these elements are highly reactive they do not occur in nature in a free state but occur as their compounds. The most abundant alkali metal is Na and least abundant metal is Cs. Francium is a radioactive metal.
- All these elements have ns^1 electronic configuration in their valence shell.

Physical Properties

- In a period alkali metal has **highest atomic size** and lowest density.
- Density increases from Li to Cs but K is lighter than Na because of the presence of vacant 3d-orbitals and also due to bigger atomic size of K large gaps between the atoms in crystalline structure of K occur.
- Due to larger size, their outer electron is weakly attracted by the nucleus and hence their ionization enthalpies are less and decrease down the group as the atomic size increases.
- Electropositive character increases down the group.
- Alkali metals exhibit a fixed oxidation state of +1.
- The second ionization enthalpy of alkali metals are very high because the second electron to be removed is from most stable inert gas configuration.
- The salts of alkali metals are colourless and diamagnetic because all electrons are paired.
- Alkali metals when irradiated with light emit electrons. This phenomenon is known as **photoelectric effect** and hence K and Cs are used as electrodes in photoelectric cells.
- **Hydration energies** of alkali metal ion decreases down the group due to increase in size.
- Every alkali metal ion have two layers of water molecules of hydration, primary and secondary layers. The water molecules in primary layer are coordinated water while in the secondary layer are in ion-dipole attractions.
- In the primary layer of Li^+ ion only four water molecules are coordinated to Li^+ tetrahedrally using its sp^3 hybrid orbitals. Similarly, Na^+ and K^+ ions may also have four-fold primary hydration in aqueous solution. Rb^+ and Cs^+ ions are surrounded by six water molecules octahedrally in the primary layer using sp^3d^2 hybrid orbitals.
- The secondary hydration decreases from Li^+ to Cs^+ since the ion-dipole attraction decreases with increase in size.
- Ionic mobilities increases from Li^+ to Cs^+ due to decrease in the size of hydrated ions.
- Tendency to form hydrated salts decreases from Li to Cs due to decrease in the attractive force towards water. Nearly all Li compounds are hydrated commonly trihydrates. In these hydrated salts Li^+ is coordinated

to $6H_2O$ molecules octahedrally sharing the faces forming chains. Many sodium salts and few potassium salts are hydrated. Rubidium and caesium do not form hydrated salts.

- Alkali metals give flame colours Li-crimson red, Na-golden yellow, K-lilac or pale violet, Rb-red violet and Cs-blue violet.
- The order of oxidation potentials is Li > Cs > Rb > K > Na.

Chemical Properties

- Chemical reactivity of alkali metals increases from Li to Cs.
- When exposed to air, alkali metals tarnish due to the formation of oxides, hydroxides and carbonates at the surface due to reaction with oxygen, moisture and CO_2 in air.
- When burnt in air Li forms monoxide as well as lithium nitride Li_3N. But other alkali metals react only with oxygen but not with nitrogen. Li forms monoxide, Na forms peroxide and other alkali metals forms super-oxides.
- Alkali metals react with water liberating hydrogen and the reactivity towards water increases from Li to Cs.
- Alkali metals react with hydrogen forming hydrides and the reaction of alkali metals with hydrogen decreases from Li to Cs. These alkali metal hydrides when dissolved in water yield hydrogen.
- Stability of superoxides increases from KO_2 to CsO_2 though lattice energy decreases with increase in size of cation. This is because when they decompose forming metal oxides which has more lattice enthalpy in K_2O (due to small size of K^+ and O^{2-}) that compensate the heat required to decompose KO_2 but in case of Cs_2O less lattice enthalpy (due to bigger Cs^+ ion) cannot compensate the heat required to decompose CsO_2.
- Pure oxides and peroxides are colourless but superoxides are coloured and paramagnetic due to the presence of unpaired electron.

Solutions of Alkali Metals in Ammonia

- All the alkali metals dissolve in ammonia to form metastable solutions. When ammonia is pure and if there are no impurities evaporation of these solutions yields alkali metals.
- Solutions of alkali metals in ammonia are blue coloured and have electrical conductivity, paramagnetic character and reducing properties due to the presence of ammoniated metal ions and ammoniated electrons.

$$M \xrightarrow{\text{Liquid } NH_3} [M(NH_3)_x]^+ + [e(NH_3)_y]^-$$

- The bronze-coloured solutions of alkali metals in ammonia have more volume than the sum of volumes of the metal and solvent probably due to occupation of electrons in the cavities of radius 300–400 pm. So, the density of solution decrease and floats.
- If the concentration of alkali metal in ammonia increases the blue colour changes to **bronze colour** and conductivity decreases reaching a minimum at 0.05 molar but again conductivity increases with increase in concentration and becomes equal to that of metal in saturated solution.
- At high concentrations dimerization of metal ions and pairing of electrons takes place due to which the solution becomes diamagnetic.
- Solutions of alkali metals in ammonia reduces O_2 to O_2^- and O_2^{2-}; transition metal complexes to unusual lower oxidation states, e.g., K_2 [Ni(NH)$_4$] to K_4 [Ni(NH)$_4$] i.e., Ni^{2+} to Ni (0) [Pt $(NH_3)_4$] Br_2 to [Pt$(NH_3)_4$] i.e., Pt^{2+} to Pt(0); $Mn_2(CO)_{10}$ to K [Mn(CO)$_5$] i.e., Mn (0) to Mn^-; Fe $(CO)_5$ to Na_2 [Fe(CO)$_5$] to Na_2 [Fe(CO)$_4$] i.e., Fe (0) to Fe^{-2}.
- Solutions of alkali metals in liquid ammonia liberate H_2 gas from protonic species like RC ≡ CH; C_2H_5OH, etc. **Complex-forming tendency** of alkali metals is very less because of their bigger size, less effective nuclear charge and less tendency to accept electrons.

Principles of Extraction

- Alkali metals cannot be extracted by the usual methods because (i) they themselves are strong reducing agents and strong chemical reducing agents which can reduce alkali metal compounds are not available and (ii) alkali metals cannot be prepared by the electrolysis of the aqueous solutions of their salts because the discharge potential of alkali metal ions is more than that of H^+ ion and hence H^+ ion will be reduced at cathode preferential to alkali metal ion and H_2 gas will be liberated at cathode.
- Alkali metals are generally prepared by the electrolysis of their fused metal halides after the addition of some other salt to decrease their m.pt.

Compounds of Alkali Metals

Hydrides

- All the alkali metals react with H_2 forming ionic hydrides of the formula M^+H^-. The reactivity of alkali metals towards hydrogen decreases from Li to Cs.
- The ionic character of these hydrides increases from Li to Cs and their stability decreases from LiH to CsH.

- These hydrides act as strong reducing agents and liberate H_2 with protonic solvents such as H_2O, NH_3, and C_2H_5OH with the formation of hydroxide, amide and ethoxide, respectively.
- NaH reduces several substances forming another hydride or complex hydride, e.g., BF_3 to B_2H_6 or $NaBH_4$ $AlBr_3$ to $NaAlH_4$; $TiCl_4$ to Ti.
- NaH reduces SO_2 to dithionate ($Na_2S_2O_4$) and CO_2 to formate (HCOONa).
- LiH forms $LiAlH_4$ with $AlCl_3$ which is used as a reducing agent in organic chemistry to reduce carbonyl compounds to alcohols, reduces several inorganic compounds, e.g., BCl_3 to B_2H_6 . PCl_3 to PH_3; $SiCl_4$ to SiH_4, etc.

Nitrides

- Lithium reacts with N_2 directly forming nitride Li_3N. Sodium forms azide.
- Sodium and potassium nitrides are prepared by the action of active nitrogen on metals or heating a mixture of metal azide and metal.
- Li_3N on heating with H_2 gives ammonia and LiH.

Oxides

- Alkali metals when burnt in oxygen, Li forms monoxide, Na, forms peroxide but others form super-oxides.
- Oxides of any required element can be prepared by dissolving the metal in ammonia and treating the solution with the required amount of O_2.
- All the alkali metal oxides are ionic, strongly basic and highly soluble in water forming strongly alkaline solutions of their hydroxides and solubility increases down the group.
- There is a trend of increasing colouration with increase in atomic number Li_2O and Na_2O are colourless or white, K_2O yellowish, Rb_2O bright yellow, Cs_2O is orange.

Hydroxides

- Alkali metal hydroxides are formed by dissolving their oxides in water, and these are the strongest bases. The strength of these hydroxides increases regularly from Li to Cs due to increase in electropositive character of metals.
- All the alkali metal hydroxides are white crystalline and deliquescent in nature. Except LiOH, all are highly soluble in water and solubility increases from LiOH to CsOH.
- Thermal stability of these hydroxides increases from LiOH to CsOH. LiOH tends to lose H_2O on heating converting into Li_2O.
- Except LiOH other alkali metal hydroxides do not decompose on heating but sublimes at about 673 K and their vapours contain $(MOH)_2$.
- The dissolution of alkali metal hydroxides in water is exothermic due to the formation of hydrates and hydration.

Halides

- Alkali metal halides are colourless solids with high m.pts and b.pts but they turn yellow (NaCl), violet (KI), etc. owing to non-stoichiometric crystal defects.
- The order of heat of formation for a given metal is F > Cl > Br > I. For any alkali metal, fluoride is most stable and iodide is least stable.
- The order of $\Delta_f H°$ for fluorides of alkali metal is LiF > NaF > KF > RbF > CsF. LiF is most stable and CsF is least stable.
- **Ionic character** of alkali metal halides is in the order

  LiX < NaX < KX < RbX < CsX

  MF > MCl > MBr > MI
- **M.pts** of halides for a given alkali metal will be in the order NaF > NaCl > NaBr > NaI.
- M.pt of alkali metal halides for a given halogen will be in the order.

  Li < Na > K > Rb > Cs.

  Low m.pts of lithium halides may be due to covalent character.
- The **solubility** of alkali metal fluorides is in the order

  LiF < NaF < KF < RbF < CsF
- The low solubility of LiF is due to its very high lattice energy because of small sizes of both Li^+ and F^-.
- Except fluorides, for the other halides solubility decreases from NaX to KX and then increases to CsX. The solubility of KX is less than NaX because the difference in hydration and lattice energies of potassium salts are positive, whereas in the halides of other alkali metals it is negative.
- Lithium halides (except LiF) have significant covalent character and hence soluble in organic solvents.
- CsCl, CsBr and CsI have a bcc structure in which coordination number (CN) of each ion is 8. All the other alkali metal halides have an fcc structure in which the CN of each ion is 6.
- Theoretically, the CN of Li^+ ion must be 4 but it abnormally adopts CN 6 which is attributed to more favourable lattice energy acquired by achieving a higher CN. Hence, the solubility of lithium halides is less.

- Mixing of two alkali metal halide solutions results in the formation of new alkali metal halide, the one having more lattice enthalpy.

$$LiI + CsF \longrightarrow CsI + LiF;\ \Delta H = -146\ KJ\ mol^{-1}$$

- If excess of HF is used in the preparation of fluoride by adding HF to NaOH or KOH, solid hydrogen fluoride $NaHF_2$ and KHF_2 will be formed which contain (F-H .... F)$^-$ ion.

Compounds of Carbon

- Lithium on heating with carbon forms an ionic carbide Li_2C_2. Other alkali metals react with acetylene or by passing acetylene through a solution of the metal in liquid ammonia.
- These carbides contain acetylide ion (C_2^{2-}) and on hydrolysis give acetylene.

Carbonates

- Alkali metals form carbonates of the type M_2CO_3 by passing CO_2 gas through their hydroxide solutions or by the reaction of CO_2 with their oxides.
- In the presence of excess CO_2 carbonates convert into bicarbonates.
- Thermal stability of alkali metal carbonates increases with increase in size of the cation though lattice energy decreases with increase in size of cation.
- The size of Li^+ is smaller when compared with CO_3^{2-} ion. So the replacement of larger CO_3^{2-} ion by the smaller O^{2-} will lead to the increase in the lattice energy hence favours the decomposition of Li_2CO_3.
- Except Li_2CO_3 other alkali metal carbonates are stable upto 1273 K above which they melt and then converted into oxides.
- Order of thermal stability of alkali metal carbonates is

$$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$$

- When dissolved in water they hydrolyze in water and their aqueous solutions are basic in nature.

$$CO_3^{2-} + 2H_2O \longrightarrow H_2CO_3 + 2OH^-$$

- Li_2CO_3 is only sparingly soluble in water and the solubility of alkali metal carbonates increases down the group.

$$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$$

Bicarbonates

- Bicarbonates of alkali metals (except Li) can be prepared in solid state. $LiHCO_3$ exist only in solution but cannot be prepared in solid state.
- The bicarbonates of alkali metals decompose on heating converting into carbonates. The heat of decomposition of bicarbonates increases with increase in the size of cation.

$$LiHCO_3 < NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$$

- The bicarbonates of alkali metals hydrolyze in water forming alkaline solution.

$$HCO_3^- + H_2O \longrightarrow H_2CO_3 + OH^-$$

- Bicarbonate solutions are less alkaline than carbonate solutions because more no. of OH^- ions are formed during the hydrolysis of carbonates (per mol).
- Bicarbonate does not give pink colour with phenolphthalein but carbonates can turn the phenolphthalein to pink and this test can be used to distinguish between carbonate and bicarbonate.
- Potassium bicarbonate dimerizes due to hydrogen bonding but $NaHCO_3$ form long chains. Because of this reason $NaHCO_3$ is very less soluble in water which is used as the basis of the ammonia–soda process.

Nitrates and Nitrites

- All the alkali metal nitrates are soluble in water. Solid $LiNO_3$ and $NaNO_3$ are deliquescent and because of this reason KNO_3 is used in preference to $NaNO_3$ in gunpowder, a mixture of KNO_3, sulphur and charcoal.
- Alkali metal nitrates have fairly low melting points and have lowest m.pts among the most stable nitrates. On strong heating they decompose into nitrites but at high temperatures to the oxide.

$$2NaNO_3 \xrightarrow{500^\circ C} 2NaNO_2 + O_2$$

$$4NaNO_3 \xrightarrow{800^\circ C} 2Na_2O + 5O_2 + 2N_2$$

- Similar to Li_2CO_3, $LiNO_3$ is also least stable among alkali metal nitrates and the reason is same as explained in the case of carbonate.

$$2LiNO_3 \longrightarrow Li_2O + 2NO_2 + \frac{1}{2}O_2$$

- $LiNO_3$ is used in fire works, red coloured distress flares and alkali metal nitrates are widely used to carry high temperature oxidations and also heat transfer medium.
- Nitrites of alkali metals can be prepared by heating their nitrates or by the reaction of NO with a hydroxide.
- Nitrites liberate brown fumes of NO_2 by reacting with acid.

Anomalous Properties of Lithium

- Anomalous properties of lithium are due to (i) its small size, (ii) high electronegativity, (iii) presence of only two electrons in its penultimate shell and (iv) absence of d-orbitals in its valence shell.

- Lithium is the hardest with high m.pt and b.pt among alkali metals.
- Except hydrogen, it is the least reactive metal among alkali metals.
- Unlike other alkali metals it reacts with nitrogen directly, can be stored by wrapping with wax-coated paper, whereas other alkali metals are stored under hydrocarbons like kerosene.
- On heating LiOH, Li_2CO_3 decomposes but other alkali metal hydroxides and carbonates do not decompose.
- Li_2O and LiOH are weak bases.
- Lithium salts like fluoride, phosphate, oxalate and carbonate are sparingly soluble while the corresponding salts of other alkali metals are freely soluble.
- LiCl is highly deliquescent and crystallize as a hydrated salt. $LiCl.3H_2O$ other alkali metal chlorides are not deliquescent and crystallize as anhydrous salts.
- Due to the covalent character of lithium halides (except LiF), they are soluble in organic solvents while corresponding salts of other alkali metals are not.
- $LiNO_3$ on heating decomposes giving Li_2O, NO_2 and O_2 while the nitrates of other alkali metals give nitrites (MNO_2) and O_2.
- Li_2SO_4 do not form alum but the sulphates of other alkali metals form alum.
- In aqueous solutions it has least ionic mobility due to extensive hydration because of its small size.

Diagonal Relationship of Lithium with Magnesium

- The similarities between lithium and magnesium are due to the similar ionic sizes and similar polarizing power.
- Both Li and Mg are harder and lighter than the other elements in their respective groups.
- Both Li and Mg react with water slowly and can react with N_2 directly forming nitrides Li_3N and Mg_3N_2.
- Oxides and hydroxides of both Li and Mg are weak bases, less soluble in water, decompose on heating.
- Li and Mg do not form peroxides.
- Carbonates and nitrates of Li and Mg decompose on heating similarly.
- Solid bicarbonates of both Li and Mg are not formed.
- Chlorides of both Li and Mg have covalent character and are soluble in organic solvents and they crystallize as hydrated salts.

Sodium Hydroxide

- Solid NaOH is known as **caustic soda** and aqueous NaOH is known as **soda iye**.
- In **Causticising process** NaOH is formed by the action of milk of lime on sodium carbonate.
- NaOH is manufactured by the electrolysis of brine in different electrolytic processes.
- In **Nelson's cell** method or porous diaphram method **cathode** is porous U-shaped steel vessel; **anode** is graphite rod separated by asbestos lining. **By-products** are chlorine and hydrogen.
- The **Castner–Kellner** method also known as **mercury cathode** method. In this method, the electrolytic cell contains three compartments.
  - (i) Mercury in the outer compartment acts as a cathode while in middle compartment acts as an anode due to induction.
  - (ii) Graphite rods in the outer compartments acts as an anode while the iron rods in the middle compartment acts as a cathode.
- Sodium liberated at mercury cathode in the outer compartments dissolve in mercury forming sodium amalgam which moves into middle compartment where it react with water at cathode forming NaOH, H_2 and Hg.
- Cl_2 gas is liberated at graphite anodes in the outer compartment.
- **Kellner–Solvay** cell is modified Castner–Kellner cell.
- NaOH is deliquescent, white crystalline solid. So, it absorbs moisture and CO_2 from air and hence cannot be used as a primary standard.
- Highly soluble in water with liberation of heat due to the formation of hydrates NaOH. nH_2O (n = 1, 2 or 7).
- NaOH dissociates the proteins of the skin into pasty mass and corrosive. Hence, it is called as **caustic soda**.
- NaOH is a strong base, neutralizes almost all acids and acidic oxides forming salts.
- Amphoteric metals like Zn, Al, Sn, Be, Ge, etc. can dissolve in NaOH liberating H_2 gas.
- F_2 liberates OF_2 with dilute NaOH and O_2 with conc. NaOH.
- Cl_2, Br_2 and I_2 gives **halides** and **hypohalites with** cold and dilute NaOH but with hot and concentrated NaOH gives **halides and halates**.
- When phosphorus is boiled with NaOH, phosphine gas is liberated with the formation of sodium hypophosphite.
- In the reactions of Cl_2, Br_2, I_2, S, and P with NaOH the oxidation numbers of non-metals are both increased and decreased. So, these reactions are **disproportionation reactions** or **redox reactions**.
- When Si is heated with NaOH sodium silicate is formed with the liberation of H_2.
- Carbon reduces the NaOH to Na and H_2 with the formation of Na_2CO_3.
- NaOH liberates ammonia from ammonium salts.
- Sodium hydroxide precipitates many metals from their salts as metal hydroxides. These are of three types.

(i) Metal hydroxides precipitates from their salts but insoluble in excess of NaOH, e.g., $Fe(OH)_2$, $Fe(OH)_3$, $Ni(OH)_2$, $Cu(OH)_2$, $Mn(OH)_2$.

(ii) Metal hydroxides are precipitated from their salts but the precipitates dissolve in excess of NaOH. These are amphoteric hydroxides like $Al(OH)_3$, $Sn(OH)_2$, $Zn(OH)_2$, $Cr(OH)_3$, $Pb(OH)_2$, $Be(OH)_2$.

(iii) Metal hydroxides are precipitated but are unstable and decompose into metal oxide and water, e.g., $Hg(OH)_2$ to HgO, AgOH to Ag_2O.

- Amphoteric metal oxides like Al_2O_3, ZnO, SnO, SnO_2, BeO, PbO, PbO_2, Cr_2O_3 dissolve in sodium hydroxide forming corresponding salts.
- When CO is passed over solid NaOH under high pressure at 150–200°C forms sodium formate.
- Sodium hydroxide is used, in the manufacture of soap, paper, rayon compounds like NaOX, $NaXO_3$, in petroleum refining, in mercerization of cotton, in the preparation of alumina, silica glass.

Reactions of NaOH

Reaction	Remark
I. Action of heat	
1. $NaOH \xrightarrow{1573K} Na + H_2 + O_2$	
II. Basic Character	
1. $NaOH + Acid \longrightarrow Salt + water$	All most all acids form salts
2. $NaOH + CO_2 \longrightarrow Na_2CO_3$	
3. $NaOH + SiO_2 \longrightarrow Na_2SiO_3$	
4. $NaOH + SO_2 \longrightarrow Na_2SO_3$	
5. $NaOH + NO_2 \longrightarrow NaNO_2 + NaNO_3$	Disproportionation
III. Reaction with non-metals	
1. $NaOH + B \longrightarrow Na_3BO_3 + H_2$	
2. $NaOH + Si \longrightarrow Na_2SiO_3 + H_2$	
3. $NaOH + C \longrightarrow Na_2CO_3 + H_2 + Na$	
4. $NaOH + P_4 \longrightarrow NaH_2PO_2 + PH_3$	Disproportionation
5. $NaOH + S \longrightarrow Na_2S + Na_2S_2O_3$	Disproportionation
6. $NaOH + F_2 \longrightarrow NaF + OF_2$	Cold and dilute condition
7. $NaOH + F_2 \longrightarrow NaF + O_2$	Hot and conc. condition
8. $NaOH + X_2 \longrightarrow NaX + NaOX$	Cold and dil. condition disproportionation
9. $NaOH + X_2 \longrightarrow NaX + NaXO_3$	Hot and dil condition disproportionation
IV. Reaction with metals	
1. $NaOH + Be \longrightarrow Na_2BeO_2 + H_2$	
2. $NaOH + Al \longrightarrow NaAlO_2 + H_2$	
3. $NaOH + Sn \longrightarrow Na_2SnO_2 + H_2$	
4. $NaOH + Pb \longrightarrow Na_2PbO_2 + H_2$	
5. $NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$	
V. Reaction with compounds	
1. Any ammonium salt + NaOH $\longrightarrow NH_3$	
2. Certain metal ions ppt as hydroxides insoluble in excess of NaOH.	Fe^{2+}, Fe^{3+}, Ni^{2+}, Cu^{2+}, Mn^{2+}
3. Amphoteric metal ions ppt as hydroxides but soluble in excess of NaOH	Be^{2+}, Al^{3+}, Sn^{2+}, Sn^{4+}, Pb^{2+}, Pb^{4+}, Zn^{2+}
4. Ag^+ and Hg^{2+} ions ppt as hydroxides and decompose to metal oxides.	
5. Amphoteric metal oxides dissolve in NaOH	BeO, Al_2O_3, SnO, SnO_2, PbO, PbO_2, ZnO
6. $NaOH + CO \xrightarrow[5-10\text{ atm}]{150-200°C} HCOONa$	

Sodium carbonate

- The hydrated sodium carbonate $Na_2CO_3 \cdot 10H_2O$ is known as **washing soda** while anhydrous Na_2CO_3 is known as **soda ash**.
- In **Le Blanc Process,** Na_2CO_3 is manufactured by heating the salt cake with coke and lime stone. The salt cake is obtained by heating NaCl with conc. H_2SO_4.
- Salt cake is reduced by coke to Na_2S which reacts with lime stone forming **black ash** (CaS and Na_2CO_3) from which Na_2CO_3 is leached.
- In **Solvay's process,** first brine is saturated with ammonia containing little CO_2 in ammonia saturation tower where impurities like Mg and Ca are removed in the form of $Mg(OH)_2$ and $CaCO_3$.
- In **Solvay's ammonia–soda process** sodium bicarbonate is precipitated by passing CO_2 into a solution of brine saturated with ammonia.
- The sodium bicarbonate on heating gives sodium carbonate.
- Ammonia can be recovered from the filtrate by boiling with lime and can be used again.
- In Nelson's cell method for the manufacture of NaOH, if a mixture of steam and CO_2 is used sodium carbonate will be obtained.
- $Na_2CO_3 \cdot 10H_2O$ is efflorescent and convert into monohydrate when exposed to air but converts into anhydrous compound on heating. Soluble in water and its aqueous solution is alkaline due to hydrolysis.
- With acids, it liberate CO_2 gas.
- With excess of CO_2, the carbonate converts into bicarbonate.
- When fused with SiO_2, it forms sodium silicate called **water glass**.
- When boiled with sulphur and sulphurdioxide, it forms hypo but with only SO_2 forms sodium sulphite.
- Except alkali metal carbonates, almost all other metal carbonates are insoluble. So, when Na_2CO_3 is added to soluble metal salts insoluble metal carbonates will be precipitated.
- With highly electropositive metal salts like $CaCl_2$, $BaCl_2$, etc. it forms precipitates of normal carbonates $CaCO_3$, $BaCO_3$, but with less electropositive metal salts gives basic metal carbonates

  e.g. (i) with Mg salts form $MgCO_3$ $Mg(OH)_2$.
  (ii) with Zn salts form $[2ZnCO_3 \cdot 3Zn(OH)_2]H_2O$
  (iii) with Cu salts form $CuCO_3 \cdot Cu(OH)_2$
  (iv) with Pb salts form $2PbCO_3 \cdot Pb(OH)_2$
- Carbonates of metals like iron (III), aluminium (III), etc. when formed hydrolyzes to hydroxides immediately liberating CO_2. This reaction is used in foam fire extinguishers.
- Sodium carbonate is used in the manufacture of glass, caustic soda, in softening hard water, in petroleum refining, in washing purposes and in laundries, in dyeing industries and in fire extinguishers.

Reactions of Sodium Carbonate

Reaction	Remark
1. $Na_2CO_3 + \text{Acid} \longrightarrow CO_2$	
2. $Na_2CO_3 + H_2O + CO_2 \longrightarrow NaHCO_3$	
3. $Na_2CO_3 + SiO_2 \longrightarrow Na_2SiO_3$	
4. $Na_2CO_3 + SO_2 \longrightarrow Na_2SO_3 + CO_2$	
5. $Na_2CO_3 + SO_2 + S \longrightarrow Na_2S_2O_3 + CO_2$	
6. $M^{2+} + Na_2CO_3 \longrightarrow MCO_3$	More electropositive metals e.g., Ca, Sr, Ba ppt as normal carbonates
7. $Na_2CO_3 + M^{2+} \longrightarrow$ Basic metal carbonate	Less electropositive metals e.g., Mg^{2+}, Pb^{2+}, Cu^{2+}, Zn^{2+}
8. $Na_2CO_3 + M^{3+} \longrightarrow M(OH)_3 + CO_2$	Fe^{3+}, Cr^{3+}, Al^{3+}

Sodium Bicarbonate

- Sodium bicarbonate is known as **baking soda**. In the laboratory it is prepared by passing CO_2 into sodium carbonate solution but industrially it is manufactured by Solvay's process.
- It is sparingly soluble in water and its aqueous solution is alkaline due to hydrolysis.
- On heating, it decomposes to give sodium carbonate.
- The metal salts which gives basic metal carbonate with sodium carbonate gives normal carbonates.
- Sodium bicarbonate is used, as an antacid in medicine, in dry fire extinguishers, in baking powders and as mild antiseptic for skin infections.

Sodium Oxide

- Pure sodium oxide is prepared by heating $NaNO_2$ or $NaNO_3$ with sodium metal or sodium azide with sodium nitrite.
- At 400°C, it disproportionates into Na_2O_2 and Na.
- With liquid ammonia it forms sodamide.

Sodium Peroxide

- It is obtained by heating sodium metal in excess of air.
- At low temperature when dissolved in water, in dilute solution gives H_2O_2 but in concentrated solution gives O_2. With dilute acids it gives H_2O_2.
- It absorbs CO and CO_2. In the reaction with CO_2 oxygen gas is liberated, so used in breathing apparatus, divers, firemen and in submarines.
- Na_2O_2 oxidizes several compounds, e.g., chromium (III) compounds to chromates, manganese (II) compounds to manganates, sulphides to sulphates, benzoyl chloride to benzoyl peroxide.
- Na_2O_2 is used for the production of O_2 under the name **Oxone**, for purification of air, to bleach delicate articles like wool, silk, etc.

Sodium Nitrate

- Sodium nitrate occurs naturally as chile salt petre. In the laboratory it is prepared by the action of HNO_3 on NaOH or Na_2CO_3.
- Reduction of $NaNO_3$ with Zn, Pb, etc. gives sodium nitrite. It is used as nitrogenous fertilizer in agriculture and in the manufacture of HNO_3, $NaNO_2$ and KNO_3.

Sodium Nitrite

- It is manufactured by heating sodium nitrate alone or reduction with lead or a mixture of lime and carbon.
- It can be manufactured by passing oxides of nitrogen NO and NO_2 into an aqueous solution of NaOH or Na_2CO_3.
- It is also formed when SO_2 is passed into a concentrated solution of $NaNO_3$ to which lime is added.
- When heated with NH_4Cl it gives N_2 gas.
- When heated with acids it gives of oxides of nitrogen (NO and NO_2).
- It can act both as oxidizing and reducing agent.
- It oxidizes urea to nitrogen in acid medium, liberates iodine from iodides in acid medium, oxidizes stannous chloride to stannic chloride, sulphur dioxide to sulphuric acid.
- If cobalt nitrite is treated with sodium nitrite and acetic acid, sodium cobaltinitrite $Na_3[Co(NO_2)_6]$ is formed.
- It reduces potassium permanganate, potassium dichromate, bromine, etc.
- It is used in the manufacture of azo dyes and as a food preservative.

Sodium Chloride

- It is mainly found in sea water and occurs as a rock salt.
- Ordinary sodium chloride is hygroscopic due to the presence of magnesium and calcium chlorides.
- Its solubility in water changes very little with change in temperature because its lattice energy and hydration energies are almost equal.

Sodium Sulphate

- $Na_2SO_4 \cdot 10H_2O$ is known as **Glauber's salt** and anhydrous Na_2SO_4 is known **as salt cake**, manufactured by heating NaCl with concentrated H_2SO_4.
- It is used in textile industry, medicines as purgative manufacture of glass plates and sodium salts.

Important Compounds of Potassium

Potassium Oxides

- Potassium forms K_2O, K_2O_2, K_2O_3 and KO_2 of which K_2O and K_2O_2 can be prepared exactly in a similar way as that of Na_2O and Na_2O_2. Their behaviour is also similar to Na_2O and Na_2O_2.
- K_2O_3 is obtained when O_2 is passed through liquid ammonia containing potassium.
- Potassium superoxide (KO_2) is prepared by burning potassium in excess of oxygen free from moisture.
- KO_2 is a chrome yellow powder, and when dissolved in water it forms H_2O_2 and O_2.
- It absorbs CO and CO_2 and liberates O_2.
- It oxidizes sulphur to potassium sulphate.
- It is used as an oxidizing agent, as air purifier in submarines, space capsules and in breathing masks as it produces oxygen and removes CO_2.

Potassium Hydroxide

- Solid KOH is known as **caustic potash** and aqueous KOH is known as **potash iye**. Its preparation and properties are similar to those of NaOH.
- It is mainly used in making of soft soaps. Alcoholic KOH is used in organic chemistry as it eliminates hydrogen halides from alkyl halides.
- It is better absorber for acidic oxides like CO_2, SO_2, etc. than NaOH because potassium salts are less soluble than sodium salts.

Potassium Carbonate: Pearl Ash

- It cannot be manufactured by Solvay's process because $KHCO_3$ is highly soluble in water and does not crystallize like $NaHCO_3$.
- It can be manufactured by Le Blanc's process by using KCl in the place of NaCl.
- It is now manufactured by magnesia process in which CO_2 is passed into KCl solution at 293 K in the presence of hydrated magnesium carbonate. Potassium hydrogen magnesium carbonate is precipitated which on decomposition with water gives $MgCO_3$ precipitate and K_2CO_3 solution from which K_2CO_3 can be prepared in solid state by evaporation.
- It resembles Na_2CO_3, used in making hard glass, toilet soaps and potassium compounds.

Potassium Iodide

- Potassium iodide can be prepared by the neutralization of KOH or K_2CO_3 with HI.
- It is manufactured by heating iodine with hot concentrated KOH solution. The resulting liquid contains KI and KIO_3. This is evaporated to dryness and heated with carbon to reduce KIO_3 to KI.
- It can also be prepared by treating ferrous ferric iodide (Fe_3I_8) formed by the action of iodine on iron fillings with potassium carbonate.

$$4K_2CO_3 + Fe_3I_8 + 4H_2O \longrightarrow 8KI + 4CO_2 + Fe(OH)_2 + 2Fe(OH)_3.$$

- It dissolves I_2 forming triodide, which is unstable and gives up extra I_2 easily.
- With conc. H_2SO_4 it gives violet coloured vapours of I_2 due to the oxidation of HI formed initially by conc. H_2SO_4.
- Several oxidizing agents like Cl_2, Br_2, $KMnO_4$, $K_2Cr_2O_7$, H_2O_2, etc. liberate I_2 from KI.
- With $CuSO_4$ first it gives cupric iodide (CuI_2) which decompose to cuprous iodide (white Cu_2I_2) and iodine.
- It gives yellow precipitates with lead and silver salts.
- If forms red precipitate of HgI_2 with $HgCl_2$ soluble in excess of KI due to the formation of complex K_2HgI_4. The alkaline solution of K_2HgI_4 is known as **Nessler's reagent** and is used for detection of ammonia.
- Tincture of iodine is a mixture of KI, I_2 and alcohol.

Reactions of Potassium iodide

Reaction		Remark
Preparation		
1. $KOH + HI \longrightarrow KI$		
2. $K_2CO_3 + HI \longrightarrow KI$		
3. $KOH + I_2 \longrightarrow KI + KIO_3$		
4 $KIO_3 + C \longrightarrow KI$		
Properties		
Reduction reactions		
1. $KI + H_2SO_4 \longrightarrow SO_2 + I_2$		
2. $KI + HNO_3 \longrightarrow NO_2 + I_2$		
3. $KI + KMnO_4 \longrightarrow Mn^{2+} + I_2$		
4. $KI + K_2Cr_2O_7 \longrightarrow Cr^{3+} + I_2$		
5. $KI + H_2O_2 \longrightarrow I_2$		
6. $KI + CuSO_4 \longrightarrow Cu_2 I_2 + I_2$		
Precipitation Reaction		
1. $KI + Ag^+ \longrightarrow AgI$	Yellow ppt	
2. $KI + Pb^{2+} \longrightarrow PbI_2$	Yellow ppt	
3. $KI + Hg^{2+} \longrightarrow HgI_2$	Red ppt soluble in excess KI	
$2KI + HgI_2 \longrightarrow K_2HgI_4$		

Biological Importance of Sodium and Potassium

- Sodium and potassium ions balance the negative charges associated with organic molecules present in cells.
- The ion transport activity is known as the **sodium pump**.
- Sodium ions can be pumped out of the cells but not potassium ions.
- The energy required for pumping out the sodium ions and taking of K^+ ions or the H^+ ions is provided by the hydrolysis of ATP to ADP.
- The presence of Na^+ and K^+ ions inside and outside the cell produce an electrical potential across the cell membrane.
- Sodium ions are associated with the movement of glucose into the cells.
- The potassium ions are essential for metabolism of glucose inside the cell, the synthesis of proteins and the activation of certain enzymes.

Multiple Choice Questions with Only One Answer

1. Cs+ ions impart violet colour to Bunsen flame. This is due to the fact that the emitted radiations are of
(a) High energy (b) Lower frequencies
(c) longer wavelengths (d) Zero wave number

2. $Na + Al_2O_3 \xrightarrow{\text{High temperature}} X \xrightarrow[\text{water}]{CO_2\text{in}} Y$; compound Y is
(a) $NaAlO_2$ (b) $NaHCO_3$
(c) $Al_2(CO_3)_3$ (d) Na_2CO_3

3. The correct order of solubility is
(a) $CaCO_3 < KHCO_3 < NaHCO_3$
(b) $KHCO_3 < CaCO_3 < NaHCO_3$
(c) $NaHCO_3 < CaCO_3 < KHCO_3$
(d) $CaCO_3 < NaHCO_3 < KHCO_3$

4. An aqueous solution of a halogen salt of potassium reacts with same halogen X_2 to give KX_3, a brown solution, in which halogen exist as X_3^- ion. X_2 acts as Lewis acid and X acts as Lewis base. Then, halogen X is
(a) Fluorine (b) Chlorine
(c) Bromine (d) Iodine

5. I. $Br_2 + 2NaOH \longrightarrow NaBr + X + H_2O$ (cold)
II. $Br_2 + 6NaOH \longrightarrow NaBr + Y + H_2O$ (hot)
The difference in the oxidation states of Br in X and Y is
(a) 6 (b) 4 (c) 2 (d) 0

6. The correct order of thermal stability of metal carbonate is
(a) $K_2CO_3 > Na_2CO_3 > Li_2CO_3$
(b) $Li_2CO_3 > K_2CO_3 > Na_2CO_3$
(c) $Na_2CO_3 > Li_2CO_3 > K_2CO_3$
(d) $K_2CO_3 > Li_2CO_3 > Na_2CO_3$

7. Lithium is similar to magnesium in many properties. This is because
(a) both have nearly the same size
(b) the ratio of the charge to size is nearly the same
(c) both have similar electronic configurations
(d) both have nearly the same N/P ratio

8. Which of the following is a correct statement about s-block compounds?
(a) Stability of metal chlorides decreases down the alkali metal group
(b) Lattice energy of M_2O_2 is greater than that of M_2O
(c) Stability of peroxide increases with the increase in size of alkaline earth metal ion
(d) The water of crystallization is greater in alkali metal salts than in alkaline earth metal salts

9. When excess of potassium superoxide is placed in a container containing CO_2, then
(a) the pressure of the container decreases
(b) the pressure of the container increases
(c) the pressure of the container remains constant
(d) the pressure of the container first increases and then decreases

10. The standard oxidation potential value of Li is greater than that of K. But K is more reactive towards water than Li, because
(a) The M.P. of 'K' is less than Li and the heat of reaction is sufficient to make it melt. So the molten metal spreads out, and exposes a larger surface to the water.
(b) The M.P. of 'Li' is less than 'K' and the heat of reaction is sufficient to make it melt. So the molten metal spreads out, and exposes a larger surface to the water.
(c) The E^0 value is more negative and ΔG is more positive for $Li^+ + e \longrightarrow Li$
(d) The E^0 value is more positive and ΔG is more negative for $K^+ + e \longrightarrow K$.

11. S_1; Alkali metals are generally extracted by electrolysis of their aqueous salts S_2; The electropositive character of alkali metals decreases with increase in atomic number S_3; Lithium is hardest element in alkali metals S_4; Potassium carbonate is prepared by Solvay's process
(a) T T F F (b) T F T F
(c) F T F T (d) F F T F

12. The magnitude of enthalpy of formation of alkali metal halides decreases in the order
(a) Fluoride > chloride > bromide > iodide
(b) Iodide > bromide > chloride > fluoride
(c) Bromide > iodide > fluoride > chloride
(d) Fluoride > chloride > iodide > bromide

13. Which of the following is arranged in the order of increasing radius in an aqueous solution
(a) $Cs^+ < Rb^+ < K^+ < Na^+ < Li^+$
(b) $Li^+ < Na^+ < K^+ < Rb^+ + Cs^+$
(c) $Li^+ < K^+ < Na^+ < Rb^+ < Cs^+$
(d) $Rb^+ < Na^+ < Li^+ < Cs^+ < K^+$

14. $KO_2 + CO_2 + H_2O \xrightarrow[CO_2]{\text{Excess}} A + O_2\uparrow$, Correct statement about 'A' is
(a) The anion present in crystalline A exists as a dimer due to hydrogen bonds
(b) A can't be isolated in solid state
(c) A is generally called washing soda
(d) A gives thermally unstable carbonate on heating

15. Mark the false statement
(a) The electropositive character of alkali metals decreases with increase in atomic number
(b) Lithium is a hard metal and cannot be cut with a knife
(c) Alkali metals are strong reducing agents
(d) Electronegativities of all alkali metals lie between 1.0 and 0.7

16. X reacts with NaOH (aqueous solution) to form Y and H_2. Aqueous solution of Y is heated to 323–333 K and on passing CO_2 into it, Na_2CO_3 and Z were formed. When Z is heated to 1200°C, Al_2O_3 is formed, X, Y and Z respectively are
(a) Al, $AlCl_3$, $NaAlO_2$
(b) Zn, Na_2ZnO_2, $Al(OH)_3$
(c) Al, $Al(OH)_3$, $AlCl_3$
(d) Al, $NaAlO_2$, $Al(OH)_3$

17. The conductivity increases from Li+ salts to Cs+ salts in an aqueous solution because of
(a) Increase in no. of water molecules surrounding the ion in their primary shell from Li^+ to Cs^+
(b) Decrease in no. of water molecules surrounding the ion in the secondary layer from Li^+ to Cs^+
(c) The presence of vacant d-orbitals in the valence shell of K^+, Rb^+ and Cs^+
(d) Decrease in no. of water molecules surrounding the ion in their primary shell from Li^+ to Cs^+

18. Which of the following statements is incorrect?
(a) NaOH solution dissolves Al_2O_3, forming soluble sodium aluminate
(b) NaOH solution dissolves HgO, forming soluble sodium mercurate
(c) NaOH solution dissolves SnO_2, forming soluble sodium stannate
(d) NaOH solution dissolves PbO_2, forming soluble sodium plumbate

19. The alkali-metal peroxides contain the $(\text{-O-O-})^{2-}$ ion. They are
(a) Diamagnetic and are oxidizing agents
(b) Paramagnetic and are oxidizing agents
(c) Paramagnetic and are colourless compounds
(d) Diamagnetic and are coloured compounds

20. KO_2 reacts with CO_2 to produce
(a) $K_2CO_3 + O_2$ (b) $K_2CO_3 + CO$
(c) K_2CO_3 (d) $K_2O + CO$

21. Which of the following statements is incorrect for group I metals?
(a) They all have one electron in the outer shell preceeded by a closed shell containing eight electrons.
(b) The compounds of group I metals are generally predominantly ionic and exists as high-melting solids in which as many ions of opposite charge surround each other as possible.
(c) Their compounds are generally water soluble and white, unless the anions happen to be coloured.
(d) As the atomic size increase, it becomes progressively less difficult to ionize the monovalent atom and the reactivity of the elements increases from lithium to caesium.

22. Which of the following is correct regarding solutions of sodium metal in liquid ammonia
(a) The solutions are generally blue, good conductors of electricity and conductivity decreases as temperature is raised
(b) The solutions are paramagnetic but paramagnetic character decreases as concentration of solution is increased
(c) When NH_4Cl solid is added to such a solution NaCl is precipitated
(d) All the above are correct

23. Solubility of NaOH in water
(a) increases with increase in temperature
(b) decreases with increase in temperature
(c) is not affected by a change in temperature
(d) first increases and then decreases with temperature

24. Sodium hydroxide cannot be used as a primary standard for acid base titration because
(a) it is corrosive and reacts with glass
(b) the dissolution of sodium hydroxide in water is highly exothermic, hence its concentration changes on dissolution
(c) it is hygroscopic and also react with atmospheric carbondioxide
(d) hydroxides cannot be used as primary standards

25. A metal from period 4 is added to water and a vigorous reaction takes place with the evolution of a gas. Which statements are correct?
I. Oxygen is evolved
II. Hydrogen is evolved
III. The resulting solution is acidic
IV. The resulting solution is basic
(a) I and III only (b) II and III only
(c) II and IV only (d) I and IV only

26. Li^+ and H^- ions have isoelectronic structures. Which of the following statements concerning these two ions is true?
(a) Li^+ is a strong reducing agent than H^-
(b) H^- is larger than Li^+
(c) More energy is needed in removing one electron from H^- than from Li^+
(d) The chemical properties of the two ions are identical since they have same electronic configuration

27. Which of the following statements regarding alkali metals is correct?
(a) The strength of alkali metal hydroxides decreases down the group
(b) The conductivity of aqueous alkali metal ions increase down the group
(c) Melting points of alkali metal halides follow the order MF < MCl < MBr < MI
(d) Melting points of LiX (X is halide) are more than those of the corresponding halides of Na

28. Which of the following is true?
(a) Li^+ being smallest in size and thus moves faster and hence a good conductor of electricity in aqueous solution among the elements of IA group
(b) $LiClO_4$ is more soluble than $CsClO_4$ in water
(c) CH_3NH_2 is a better Lewis base than CH_3CN
(d) Both 2 and 3 are correct

29. When hydrogen peroxide and alcohol are added to a concentrated solution of lithium hydroxide, a white precipitate of........is formed
(a) Li_2O (b) Li_2O_2 (c) LiH (d) $LiAlH_4$

30. Which of the following statements is true?
(a) Alkali metal hydride in aqueous medium is alkaline
(b) BeH_2 has less reducing tendency than MgH_2
(c) The melting point of LiCl, LiI, KF CsCl and NaCl can be arranged as LiI < LiCl < CsCl < NaCl < KF
(d) All the above

31. When silicon or boron is dissolved in caustic soda, hydrogen gas is liberated. It is
(a) an intermolecular redox reaction
(b) intramolecular redox reaction
(c) disproportionation reaction
(d) not a redox reaction

32. Potassium is produced by electrolyzing fused KCl in a cell similar to one used for Na but the cell must be operated
(a) at higher temperature since KCl has a higher m.pt than NaCl
(b) at lower temperature since KCl has a lower m.pt than NaCl
(c) at higher temperature since potassium is more electropositive with low m.pt
(d) at lower temperature since potassium is more electropositive with high m.pt

33. When a concentrated solution of ammonia is saturated with sodium chloride in the presence of pieces of dry ice, water cloud forms. This is
(a) due to the formation of solid Na_2CO_3
(b) due to the formation of solid $NaHCO_3$
(c) due to the formation of solid $(NH_4)_2CO_3$
(d) due to the formation of solid NH_4HCO_3

34. In the reaction $LiH + AlH_3 \longrightarrow LiAlH_4$, AlH_3 and LiH acts as
(a) Lewis acid and Lewis base
(b) Lewis acid and Lewis acid
(c) Bronsted base and Bronsted acid
(d) Arrhenius acid and Arrhenius base

35. Binary compound $A + S \longrightarrow B \xrightarrow{Al_2(SO_4)_3}$ Double salt. A is paramagnetic in nature and contains about 55 per cent of alkali metal. Thus, A is
(a) Na_2O_2 (b) K_2O_2
(c) RbO_2 (d) KO_2

36. The process that cause nerve signals in animals.
(a) Electrical potential gradient due to transfer of K^+ ions
(b) Electrical potential gradient due to transfer of Na^+ ions in $(Na^+–K^+)$ pumps
(c) Electrical potential gradient set up due to transfer of Ca^{2+} ions
(d) No nerve signal exists in Ca^{2+} ions

37. A mixture of Na_2CO_3 and K_2CO_3 is used as fusion mixture because
(a) it has lower m.pt than Na_2CO_3 and converts metal salts to carbonates which decompose to metal oxides
(b) it has higher m.pt than K_2CO_3 and converts metal salts to carbonates, which decompose to metal oxides
(c) it has lower melting point than both Na_2CO_3 and K_2CO_3 and converts the metal salts to carbonates, which decompose to metal oxides
(d) it has higher melting point than both Na_2CO_3 and K_2CO_3 and converts the metal salts to carbonates, which decompose to metal oxide.

38. $A + H_2O \longrightarrow NaOH$; $A + O_2 \xrightarrow{400°C} B \xrightarrow[\text{at } 25°C]{H_2O}$ $NaOH + O_2$ which of the following statement is false regarding B.
(a) B turns green chromic salt solution to yellow
(b) B can be used to purify the air in submarines
(c) B can be used as an oxidizing agent
(d) When crystallized from solution B is obtained as an anhydrous compound

39. When sodium hydroxide is added dropwise to a mixture of two compounds, a white precipitate appears but on adding excess of NaOH the amount of precipitate decreases. The possible compounds are
(a) $ZnSO_4$ and $FeSO_4$ (b) $CuSO_4$ and $Al_2(SO_4)_3$
(c) $AlCl_3$ and $MnCl_2$ (d) $ZnSO_4$ and $SnCl_2$

40. The species X and Y are obtained by the reaction of F_2 with cold and dilute solution of NaOH and X and Z are formed with hot concentrated solution of NaOH. The oxidation states of second most electronegative element in y and z are
(a) −2 and −2 (b) +2 and +1
(c) +2 and 0 (d) −2 and +2

41. Metal A reduces silica converting itself into B. B absorbs moisture and converts into C. When C is heated with the reduction product of silica liberates a gas. Then A, B, C and gas are
(a) Na, Na_2O, NaOH, H_2
(b) Na Na_2O_2, Na_2CO_3, O_2
(c) Na, Na_2O, Na_2CO_3, CO_2
(d) Na, Na_2O_2, NaOH, H_2

42. An alkali metal (A) reacts with CO_2 forming B. When milk of lime is added to the aqueous solution of B, C and D are formed. On heating C with white phosphorous a gas is liberated along with the formation of E. In these reactions A and E are
(a) Na, neutral salt (b) K, acidic salt
(c) Rb, basic salt (d) Li, mixed salt

43. A compound of sodium does not give CO_2 when heated but it gives CO_2 when treated with dilute acids. A crystalline compound is found to have 37.1 per cent Na and 1.6 per cent H_2. Hence the compound is
(a) $NaHCO_3 \cdot 10H_2O$ (b) $NaHCO_3 \cdot 5H_2O$
(c) $Na_2CO_3 \cdot 10H_2O$ (d) $Na_2CO_3 \cdot H_2O$

44. In industries caustic potash is used in the place of caustic soda to remove acidic oxides like SO_2, CO_2, etc. because
(a) Potassium salts are thermally stable than sodium salts
(b) Potassium salts are less soluble than sodium salts
(c) Caustic potash can react with them but not caustic soda
(d) Caustic potash is cheaper than caustic soda

45. LiF is less soluble in water than KF because
(a) LiF more is covalent than KF
(b) LiF has higher lattice energy than KF
(c) LiF has higher enthalpy of hydration than KF
(d) Li^+ ions are not extensively hydrated than K^+ ions

Multiple Choice Questions with One or More than One Answer

1. On heating to 400–500°C relatively stable hydrides and carbonates decompose. Which of the following will decompose when heated to 400–500°C?
(a) LiH (b) NaH
(c) Li_2CO_3 (d) Na_2CO_3

2. Identify the correct statements
(a) In aqueous solution, both NaI and KI dissolve iodine to form I_3^-
(b) In aqueous solution, both KI and RbI dissolve iodine to form I_3^-
(c) NaF has a lower melting point than NaI
(d) A solution of sodium metal in liquid NH_3 acts as a reducing agent due to the presence of NaH

3. Which one of the following statements are correct regarding alkali metals?
(a) They all have one loosely held valence electron in the outer shell
(b) They form univalent, ionic and colourless compounds
(c) Their oxosalts are unstable
(d) Their hydroxides and oxides are very strong bases

4. Which of the following can't be isolated in solid state?
(a) $LiHCO_3$ (b) $Ca(HCO_3)_2$
(c) $Mg(HCO_3)_2$ (d) NH_4HCO_3

5. When Cl_2 gas is passed through hot NaOH solution, oxidation number of Cl changes from
(a) −1 to 0 (b) 0 to −1
(c) 0 to +7 (d) 0 to +5

6. Which of the following statement regarding the oxides of alkali and alkaline earth metals is correct?
(a) The reactivity of K_2O towards water is more than that of Na_2O
(b) The oxides of alkaline earth metals are more basic than those of alkali metals
(c) MgO is used as a refractory material for lining of electric furnaces
(d) The milk of lime and lime water are two different solutions

7. Sodium nitrate decomposes above 800°C to give
(a) N_2 (b) O_2
(c) NO_2 (d) Na_2O

8. Which of the following statements are correct?
(a) Alkali metals are strong reducing agents but poor complexing agents
(b) Among Cs^+ and Mg^{2+}, Mg^{2+} form an acetate complex
(c) LiF and CsI have low solubility in water, whereas LiI and CsF are very soluble
(d) LiH has greater thermal stability than the other alkali metal hydrides, whereas Li_2CO_3 decomposes at a lower temperature than the other alkali metal carbonates

9. Which statement is/are correct?
(a) At normal temperature I A group metals adopt a body centred cubic type of lattice with a coordination number of eight
(b) At very low temperature lithium forms a hexagonal close packed structure with a coordination number of 12
(c) Lithium is softer than the other metals but is harder than lead
(d) LiO_2 an NaO_2 are yellow but KO_2 is colourless

10. Select the correct statements from the following
(a) Stability of peroxides and superoxides of alkali metals increases with an increase in the size of the metal ion
(b) NaOH does not form hydrated salt
(c) Increase in stability of carbonates of alkali metals is due to the stabilization of large anions by larger cation through lattice energy effects
(d) The low solubility of LiF is due to its high lattice energy, whereas low solubility of CsI is due to smaller energy effects

11. Which of the following metals are precipitated as hydroxides by the addition of sodium carbonate?
(a) $FeCl_3$ (b) $ZnCl_2$
(c) $CrCl_3$ (d) $AlCl_3$

12. Considering greater polarization in LiCl compared to that in NaCl which of the following statements, is/are correct?
(a) The m.pt of LiCl is lower than that of NaCl
(b) LiCl dissolves more in organic solvent than NaCl
(c) LiCl ionizes in water more than NaCl
(d) Fused LiCl would be more conducting than fused NaCl

13. Sodium when strongly heated in air forms a yellow compound (A) reacts with CO_2 to give (B). The correct statement about (B) is
(a) Its aqueous solution is basic in nature
(b) When B is fused with SiO_2 liberates CO_2
(c) It is used in textile industry
(d) When B reacts with SO_2, SO_3 is formed

Comprehension Type Questions

Passage-I

Lithium only forms monoxide when heated in oxygen. Sodium forms monoxide and peroxide in excess of oxygen. Other alkali metals form superoxide with oxygen, i.e., MO_2. The abnormal behaviour of lithium is due to small size. The larger size of nearer alkali metals also decides the role of formation of superoxides. The three ions are related to each other as follows:

$$O^{2-} \xrightarrow{1/2O_2} O_2^{2-} \xrightarrow{O_2} 2O_2^-$$

All the three ions abstract proton from water

1. Lithium does not form stable peroxide because
(a) of its small size
(b) d-orbitals are absent in it
(c) it is highly reactive and form superoxide in place of peroxide
(d) covalent nature towards water

2. Which anion is stable towards water
(a) O^{2-} (b) O_2^{2-}
(c) O_2^- (d) None of these

3. In hydrolysis of the alkali metal oxides, peroxides and superoxides water act as
(a) Bronsted acid (b) Bronsted base
(c) Lewis acid (d) Arrhenius

Passage-II

Alkali metals are chemically very reactive. The reactivity with water increases down the group. The alkali metal hydroxides are strong bases and their solubility and basic strength increases from LiOH to CsOH. All the alkali metals when heated with oxygen form different type of oxides and with hydrogen form ionic crystalline hydrides. With halogens, the reaction is vigorous and ionic crystalline halides are formed. They dissolve in liquid ammonia giving deep blue solutions when dilute, which are conducting, strongly reducing and paramagnetic in nature.

1. Which of the following alkali metal halides has lowest lattice energy?
(a) LiF (b) NaCl
(c) KBr (d) CsI

2. Amongst the alkali metal hydrides, the most stable one is
(a) LiH (b) NaH
(c) KH (d) RbH

3. Which of the following hydroxides is least stable to heat and decomposes to form the corresponding oxide
(a) KOH (b) NaOH
(c) CsOH (d) LiOH

Passage-III

Alkali metals readily react with oxyacids forming corresponding salts with evolution of hydrogen like

M_2CO_3, $MHCO_3$, MNO_3, M_2SO_4, etc.

They also dissolve in liquid NH_3 but without the evolution of hydrogen. The colour of its dilute solution is blue but when it is heated and concentrated then its colour becomes bronze

1. Among the nitrates of alkali metals which one can be decomposed to its oxide?
(a) $NaNO_3$ (b) KNO_3
(c) $LiNO_3$ (d) All

2. Among the carbonates of alkali metals which one has most stability?
(a) Cs_2CO_3 (b) Rb_2CO_3
(c) K_2CO_3 (d) Na_2CO_3

3. Which of the following statement about the sulphate of alkali metal is correct?
(a) Except Li_2SO_4 all sulphates of other alkali metals are soluble in water
(b) All sulphates of alkali metals except lithium sulphates forms alum
(c) The sulphates of alkali metals cannot be hydrolyzed
(d) All

Match the Following Type Questions

1. Match the following

Column-I	Column-II (property of the product)
(a) $AgNO_3 + NaOH$	(p) ppt. is soluble in excess of reagent
(b) $FeCl_3 + NaOH$	(q) ppt. is insoluble in excess of reagent
(c) $ZnSO_4 + NaOH$	(r) Oxide precipitate is obtained
(d) $AlCl_3 + NaOH$	(s) Hydroxide precipitate is obtained initially

2. Match the following

Column-I	Column-II
(a) $Na_2O + H_2O$	(p) H_2
(b) $Na_2O_2 + 2H_2O$	(q) O_2
(c) $2KO_2 + 2\ H_2O$	(r) NaOH
(d) $2Na + 2H_2O$	(s) H_2O_2

3. Match the following

Column-I	Column-II
(a) $KHCO_3$	(p) Exists in solid state
(b) $NaHCO_3$	(q) O_2
(c) $LiHCO_3$	(r) Hydrogen bonding
(d) NH_4HCO_3	(s) Dimeric anion

Numerical Type Questions

1. In the hydrated lithium ion the number of water molecules surrounding each lithium ion in its primary layer is.

2. Electrolysis of aqueous sodium chloride (x) forms three products A, B and C of which B and C are gases. B cannot react with A but C can react with A in cold condition to give x and y and in hot condition give x and z. The difference in the oxidation states of C in y and z is.

3. When white phosphorous is boiled with sodium hydroxide a gas is liberated with the formation of a compound in solution. The difference in the oxidation states of phosphorous in gaseous product and the compound in aqueous solution is.

4. How many of the following liberates hydrogen from sodium hydroxide?

Be, B, C, N, Mg, Al, Si, P, S, Sn, Pb.

5. In the reaction with sodium hydroxide, how many of the following substances are oxidized?

Be, B, C, Al, Si, P, S, Sn, Pb, NH_4^+, $FeCl_3$, $AlCl_3$, $SnCl_2$.

6. How many of the following substances are precipitated as hydroxides only when treated with sodium carbonate $FeSO_4$, $FeCl_3$, $MgCl_2$, $AlCl_3$, $MnSO_4$, $CrCl_3$.

7. How many of the following compounds liberate carbon dioxide gas from sodium carbonate.

$FeCl_3$, $MgCl_2$, $AlCl_3$, $CuSO_4$, $ZnSO_4$, $Pb(NO_3)_2$, $CaCl_2$, $BaCl_2$

Multiple Choice Questions with Only One Answer

1. a	**2.** d	**3.** d	**4.** d	**5.** b	**6.** a
7. b	**8.** c	**9.** b	**10.** a	**11.** d	**12.** a
13. a	**14.** a	**15.** a	**16.** d	**17.** b	**18.** b
19. a	**20.** a	**21.** a	**22.** d	**23.** b	**24.** c
25. c	**26.** b	**27.** b	**28.** d	**29.** a	**30.** d
31. a	**32.** b	**33.** b	**34.** a	**35.** d	**36.** a
37. c	**38.** d	**39.** c	**40.** c	**41.** a	**42.** a
43. d	**44.** b	**45.** b			

Multiple Choice Questions with One or More than One Answer

1. b, c	**2.** a, b	**3.** a, b, d
4. a, b, c	**5.** b, d	**6.** a, c, d
7. a, b, d	**8.** a, b, c, d	**9.** a, b
10. a, c, d	**11.** a, c, d	
12. a, b, d	**13.** a, b, c	

Comprehension Type Questions

Passage-I	**1.** 1	**2.** 4	**3.** 1
Passage-II	**1.** 4	**2.** 1	**3.** 4
Passage-III	**1.** 3	**2.** 1	**3.** 4

Matching Type Questions

1.	a-q, r, s	b-q, s	c-p, s	d-p, s
2.	a-r	b-r, s	c-q, s	d-p, r
3.	a-p, q, r, s	b-p, q, r	c-q	d-p

Numerical Type Questions

1. 4 **2.** 4 **3.** 4 **4.** 7 **5.** 9 **6.** 3 **7.** 6

HINTS AND SOLUTIONS

Hints to Problems for Practice

1. Violet colour radiations have more energy.

2. $6Na + Al_2O_3 \longrightarrow 2Al + 3Na_2O$

$Na_2O + H_2O \longrightarrow 2NaOH$

$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$

3. $CaCO_3$ is least soluble. Among $NaHCO_3$ and $KHCO_3$ $NaHCO_3$, is less soluble due to polymeric structure because of hydrogen bonding. $KHCO_3$ exist as a dimer only.

4. $KI + I_2 \longrightarrow KI_3$ (Brown solution)

5. X is NaOBr in which Br is in +1 oxidation state Z. Y is $NaBrO_3$ in which Br is in +5 oxidation state so the difference is 4.

6. Thermal stability of alkali metal carbonates increases down the group.

7. The charge density, i.e., charge $(\text{radius})^2$ of Li^+ and Mg^{2+} are nearly same and hence their polarizing powers are nearly same. So, they exhibit similarities in properties.

8. If the lattice energy liberated during the formation of an oxide by the decomposition is more, the stability of the compound is less. The lattice energy of oxides formed during the decomposition of peroxides decreases more down the group due to small size of anion, while the decrease in lattice energy of peroxide is small due to bigger size of peroxide ion.

9. $4KO_2 + 2CO_2 \longrightarrow 2K_2CO_3 + 3O_2$

As the number of gaseous molecules increases pressure increases.

10. Since the m.pt. of K is less it melts at low temperature and spreads out on the surface of water due to its less density giving more surface area for reaction.

11. Enthalpy of formation of alkali metal halides decreases with increase in size of the alkali metal ion.

12. The extent of hydration decreases with increase in the size of alkali metal ion.

13. $4KO_2 + 2CO_2 \longrightarrow 2K_2CO_3 + 3O_2$

$K_2CO_3 + H_2O + CO_2 \longrightarrow 2KHCO_3$

$KHCO_3$ exist as a dimer due to hydrogen bonding.

16. $\underset{X}{2Al} + 2NaOH + 2H_2O \longrightarrow \underset{Y}{2NaAlO_2} + 3H_2$

$2NaAlO_2 + 3H_2O + CO_2 \longrightarrow Na_2CO_3 + 2Al(OH)_3$

$\underset{Z}{2Al(OH)_3} \xrightarrow{1200°C} Al_2O_3 + 3H_2O$

17. Due to the decrease in extent of hydration down the group the size of the hydrated alkali metal ion decreases and thus conductivity increases.

18. HgO is insoluble in NaOH.

19. Peroxides are diamagnetic, colourless and act as oxidizing agents.

21. Lithium contains only two electrons in the penultimate shell.

23. Dissolution of NaOH in water is exothermic. So, the solubility decrease with increase in temperature.

25. The metal in period 4 may be either K or Ca which react with water liberating H_2 and forming basic solution of KOH or $Ca(OH)_2$.

26. H^- ion is larger than Li^+ ion.

27. For fluorides, m.pts decreases down the group. Ionic mobilities in aqueous solution increase down the group.

29. $2LiOH + H_2O_2 \longrightarrow Li_2O + 2H_2O + 1/2\ O_2$

30. (a) $H^- + H_2O \longrightarrow H_2 + OH^-$

(b) due to decrease in stability of hydrides down the group in periodic table.

(c) M.Pt $\propto$ lattice energy and % ion in character.

31. B and Si are oxidized to B^{3+} and Si^{4+} while H^+ ion in NaOH is reduced to H_2.

32. M.Pt of KC*l* is lesser than NaC*l*.

33. $NH_4OH + CO_2 \longrightarrow NH_4HCO_3$

$NH_4HCO_3 + NaCl \longrightarrow NaHCO_3 + NH_4Cl$

As $NaHCO_3$ is sparingly soluble it forms a cloud of white ppt.

35. $2KO_2 + S \longrightarrow K_2SO_4$

$K_2SO_4 + Al_2(SO_4)_3 + 24H_2O \longrightarrow K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

double salt.

37. A mixture of Na_2CO_3 and K_2CO_3 have low m.pt than both Na_2CO_3 and K_2CO_3. This on heating with metal salts, convert into their carbonates which decompose giving metal oxides on heating.

38. Sodium peroxide crystallizes from solution as $Na_2O_2 \cdot 8H_2O$

39. Both $AlCl_3$ and $MnCl_2$ gives white precipitate as $Al(OH)_3$ and $Mn(OH)_2$ by the addition of NaOH dropwise but by adding excess of NaOH $\cdot$ $Al(OH)_3$ dissolves due to the formation of $NaAlO_2$ but $Mn(OH)_2$ is insoluble in excess of NaOH.

40. Flourine with cold and dil NaOH forms NaF and OF_2 while with hot and conc. NaOH forms NaF and O_2. The second most electronegative element is O_2. Its oxidation states in the products are +2 in OF_2 and zero in O_2.

41. $\underset{A}{4Na} + SiO_2 \longrightarrow 2Na_2O + Si$

$\underset{B}{Na_2O} + H_2O \longrightarrow \underset{C}{2NaOH}$

$$2NaOH + Si + H_2O \longrightarrow Na_2SiO_3 + H_2$$

$$\underset{A}{4Na} + \underset{B}{3CO_2} \longrightarrow 2Na_2CO_3 + C$$

$$\underset{B}{Na_2CO_3} + \underset{C}{Ca(OH)_2} \longrightarrow \underset{D}{2NaOH} + CaCO_3$$

$$3NaOH + 3H_2O + P_4 \longrightarrow PH_3 + 3NaH_2PO_2$$

NaH_2PO_2 is a normal salt

More Than One Answer Questions

1. Only lithium compounds among alkali metal compounds decomposes on heating.
4. Lithium and alkaline earth metal carbonates exist only in aqueous solution and cannot be prepared in solid state.
5. When Cl_2 is passed into hot and conc. NaOH solution it converts into NaC*l* and NaC*l*O_3. So, the change in oxidation state is 0 to –1 in NaCl and 0 to +5 $NaClO_3$.
11. $4NaNO_3 \xrightarrow{800°C} 2Na_2O + 2N_2 + 5O_2$
    Small metal ions with more number of charges have more acidic character. Thus Fe^{3+}, Al^{3+} and Cr^{3+} liberate CO_2 from Na_2CO_3 and precipitate as hydroxides.
13. $2Na + O_2 \longrightarrow Na_2O_2$

$$2Na_2O_2 + 2CO_2 \longrightarrow 2Na_2CO_3 + O_2$$

$$Na_2CO_3 + 2H_2O \longrightarrow 2Na^+ + 2OH^- + H_2CO_3$$

$$Na_2CO_3 + SiO_2 \longrightarrow Na_2SiO_3 + CO_2$$

Numerical Type Questions

1. Since the outermost orbit in Li is 2nd orbit the Li^+ can accept 4 electron pairs from 4 water molecules.
2. Electrolysis of NaC*l* solution gives (a) NaOH (b) H_2 and (c) Cl_2.

   $2NaOH + Cl_2 \longrightarrow NaCl + NaOCl$
   Cold and dil.

   $6NaOH + 3Cl_2 \longrightarrow 5NaCl + NaClO_3$
   The difference in the oxidation states of Cl in NaOCl and NaC*l*O_3 is 4.
3. $P_4 + 3NaOH + 3H_2O \longrightarrow 3NaH_2PO_2 + PH_3$
   The difference in oxidation states of P in NaH_2PO_2 and PH_3 is 4.
4. Be, B, C, Al, Si, Sn and Pb liberate H_2 from NaOH.
5. Be, B, C, Al, Si, P, S, Sn, Pb undergo oxidation or disproportionation (redox) reactions in NaOH.
6. FeC*l*$_3$, $AlCl_3$ and $CrCl_3$ are precipitated as hydroxide.
7. $FeCl_3$, $AlCl_3$ are precipitated as hydroxides with liberation of CO_2.
   $MgCl_2$, $CuSO_4$, $ZnSO_4$ and $Pb(NO_3)_2$ precipitate as basic carbonates with liberation of CO_2.

6

CHAPTER

Group-II A (2) Alkaline Earth Metals

Atoms or systems into ruin hurled, and now a bubble burst, and now a world.....
Alexander Pope

6.1 INTRODUCTION

The Group-II A of the periodic table contains beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba) and radium (Ra). The oxides of these elements such as CaO (lime); SrO (strontia) and BaO (baryta) are thermally very stable and are basic in nature. Because of this reason the Group-II A metals are known as alkaline earth metals. The word **earth** was applied in old an days to a metallic oxide and because the oxides of calcium, strontium and barium produce alkaline solutions in water and therefore these metals came to known as **alkaline earth metals**. The term is now commonly applied to all the elements of Group-IIA. Radium corresponds to all the alkaline earth metals in its chemical properties but being radioactive, it is studied along with other radioactive elements.

Like the alkali metals, they are very reactive and hence never occur in nature in native state and react readily with many non-metals.

Electronic configuration

The electronic configuration of the alkaline earth metals are given in Table 6.1.

The electronic configurations show that for each element the neutral atom has two electrons after inert gas core and these electrons are in a completed s-subshell. Thus, the outer electronic configuration of each element is ns^2 where n is the number of the outer valence shell. It can be expected that the two electrons can be easily removed to give the inert gas configuration. Hence, these elements are all bivalent and tend to form ionic salts.

Table 6.1 Electronic configuration of alkaline earth metals

Element	Atomic number	Electronic configuration	Configuration of valence shell
Beryllium	4	$1s^2\ 2s^2$	$2s^2$
Magnesium	12	$1s^2\ 2s^2\ 2p^6\ 3s^2$	$3s^2$
Calcium	20	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$	$4s^2$
Strontium	38	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 5s^2$	$5s^2$
Barium	56	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^6\ 6s^2$	$6s^2$
Radium	88	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 4f^{14}\ 5s^2\ 5p^6\ 5d^{10}\ 6s^2\ 6p^6\ 7s^2$	$7s^2$

6.2 GENERAL CHARACTERISTICS OF ALKALINE EARTH METALS

Some important properties of alkaline earth metals are given in Table 6.2.

The variations in physical properties are not as regular as for the alkali metals because the elements of this group do not crystallize with the same type of metallic lattice. Be and Mg have a hexagonal close-packed structure. However, at high temperature Ca and Sr and at room temperature Ba have body centred cubic structures (similar to Group-1A) instead of expected hexagonal close-packed structure. This is probably at high temperature, the paired s-electron is excited to a 'd' level instead of a p-level and hence there is only one s-or p-electron per atom participating in metallic bonding.

1. Physical appearance: They are all silvery white metals as they are having a few valence electrons than the available orbitals. They exhibit greyish white lustre when freshly cut, but tarnish soon after their exposure to air. They are malleable and ductile but less than the alkali metals. Although they have been sufficiently soft yet less than the alkali metals as metallic bonding (bonding between atoms in its crystal lattice) in these elements has been stronger than in Group-IA elements.

2. Metallic properties: The alkaline earth metals are harder than the alkali metals. Hardness decreases with increase in atomic number. These show good metallic lustre and high electrical as well as thermal conductivity because the two s-electrons can easily move through the crystal lattice.

3. Atomic radius: The atoms of these elements are somewhat smaller than the atoms of the corresponding alkali metals in the same period. This is due to higher nuclear charge of those atoms which tends to draw the orbital electrons inwards. Due to the smaller atomic radius, the elements are harder and have higher melting points and higher densities than the elements of Group-IA. Atomic radius is seen to increase on moving down the group on account of the presence of an extra shell of electrons at each step.

4. Ionic radius: The ions are also large but smaller than those of the elements in Group-IA. This is due to the fact that the removal of two electrons in the formation of bivalent cations M^{2+} (Be^{2+}, Mg^{2+}, Ca^{2+}, Sr^{2+}, etc.) increases the effective nuclear charge which pulls the electrons inwards and thus reduces the size of the ions. The ionic radius is seen to increase on moving down the Group-II A.

5. Density: As the atomic number increases in this group, the density decreases upto calcium and then increases. They are denser than the alkali metals because they can be packed more tightly due to their greater charge and smaller radii. The densities do not show a regular trend with increasing atomic number because of the difference in crystal structure and variations in the rate of change of atomic weights as compared to atomic radii.

6. Melting and Boiling points: Both the melting and boiling points do not show regular trends because atoms adopt different crystal structures. They possess low melting and boiling points, however, higher than those of alkali metals because the number of bonding electrons in these elements is twice as great as for Group-IA, i.e., the metallic bonding in Group-II A elements is stronger than that in Group-IA elements.

7. Atomic volume: Due to the addition of an extra shell of electrons to each element from Be to Ra, the atomic volume increases from Be to Ra.

Table 6.2 Atomic and physical properties of alkaline earth metals

Property	Be	Mg	Ca	Sr	Ba	Ra
At. No	4	12	20	38	56	88
At. Wt	9.012	24.31	40.08	87.62	137.33	226.03
Electron configuration	$[He]2s^2$	$[Ne]3s^2$	$[Ar]4s^2$	$[Kr]5s^2$	$[Xe]6s^2$	$[Rn]7s^2$
Melting point (k)	1560	924	1124	1062	1002	973
Boiling point (k)	2745	1363	1767	1655	2078	(1973)
Density/g cm^{-3}	1.84	1.74	1.55	2.63	3.59	(5.5)
Metallic radius (pm)	112	160	197	215	222	–
Ionic radius (6 coord)/pm	27*	72	100	118	135	148
Ionization enthalpy (kJ mol^{-1})	(i) 899.4	737.7	589.8	549.8	502.9	509.3
	(ii)1757.1	1450.7	1145.4	1064.2	965.2	979.0
Electronegativity	1.5	1.23	1.04	1.0	0.97	0.97
Standard potential $E°/V(M^{2+}/M)$	–1.97	– 2.36	– 2.84	– 2.89	– 2.92	– 2.92
Hydration enthalpy kJ mol^{-1}	– 2494	– 1921	– 1577	– 1443	– 1305	–

* For beryllium, ionic radius is for 4 coordinats.

Element	Be	Mg	Ca	Sr	Ba	Ra
Atomic volume (c.c)	4.90	13.97	25.9	35.44	36.7	38.00
Atomic radii (pm)	112	160	197	215	222	–
Ionic radii of M^{2+} (pm)	31	65	99	113	135	140

8. Electronegativity: Group-II A elements have low electronegativity values, but have more than the corresponding alkali metals. They decrease down the group from Be to Ra. The relatively high value of Be is ascribed to its small size.

9. Ionization enthalpy: As the alkaline earth metals are having smaller size and greater nuclear charge than the alkali metals the electrons are more tightly held and hence the **first ionization energy would be greater than that of the alkali metals.**

Element	Be	Mg	Ca	Sr	Ba
IE_1 (eV)	9.3	7.6	6.1	5.7	5.2
IE_2 (eV)	18.2	15.0	11.8	11.0	10.1

The second ionization energy has been found to be nearly double than that of the first ionization energy.

This trend can be explained as follows.

(a) The first ionization energy means the energy needed to remove an electron from a neutral atom (M), whereas the second ionization energy means the energy needed to remove an electron from a positive ion (M^+), which of course, is difficult than the former case.

(b) After the removal of one electron, the effective nuclear charge gets increased and hence the remaining electron gets held even more tightly leading to a very high ionization energy.

It is interesting to note that the IE_2 of the alkaline earth metals is much higher than the IE_1, even then they are able to form M^{2+} ions. This anomaly can be attributed to their **high heat of hydration** in aqueous solution and **high lattice energy** in the solid state.

Metal ion	Be	Mg	Ca	Sr	Ba
Heat of hydration of M^{2+} ions $kJmol^{-1}$	–2494	–1921	–1577	–1443	–1305

The very high hydration energies of M^{2+} ions compensates the extra energy required for the second ionization energy.

10. Oxidation state: Because of the presence of the s-electrons in the outermost orbital, high heat of hydration of the dipositive ions or due to high lattice enthalpy values of M^{2+} ions in solid state and comparatively low ionization enthalpies of M^{2+} ions, the alkaline earth metals have been bivalent. Moreover, as the **bivalent** ions are having an inert gas configuration, it becomes very difficult to remove the third electron from the element and hence the oxidation states higher than II are not encountered. Further, the divalent ion is having no unpaired electron, hence their compounds are **diamagnetic** and **colourless** provided their anions have been also colourless. Thus, in brief, it can be said that the chemistry of these elements has been the chemistry of dipositive ions.

11. Nature of Bonding: Among Group-II A elements, since beryllium is having the highest ionization energy (i.e., least electropositive character), it is having the least tendency to form Be^{2+} ion. Thus its compounds with nitrogen, oxygen, sulphur and halogens (except fluorine) have been covalent, whereas the corresponding compounds of Mg, Ca, Sr and Ba are ionic.

12. Hydration energy: The hydration energy of the ions of Group-II A elements are four or five times greater than those of ions of Group-I A elements, because of their smaller size and increased charge. Hydration decreases from Be^{2+} to Ba^{2+}, because the size of ions increases. Because of this reason the solid compounds of Group-II A elements are more heavily hydrated than those in Group-I A elements, e.g., $MgCl_2 \cdot 6H_2O$; $CaCl_2 \cdot 6H_2O$; $BaCl_2 \cdot 2H_2O$, etc.

13. Ionic mobilities: With the decrease in the extent of hydration from Be^{2+} to Ba^{2+}, the ionic mobilities (i.e., ionic conductances) of these ions gets increased in the same order as the hydrated ionic size decreases.

14. Thermal and Electrical conductivity: Because of the presence of two loosely held valence electrons per atom which are free to move throughout the crystal structure, the alkaline earth metals act as good conductors of heat and eletricity.

Element	Be	Mg	Ca	Sr	Ba	Ra
Electrical resistivity (25°C) $\mu\Omega$ cm	3.7	4.68	3.42	13.4	34.0	(100)

15. Electropositive character: Because of their large size and comparatively low ionization enthalpies, the alkaline earth metals are strongly electropositive elements. However, these are not as strongly electropositive as the alkali metals and hence, unlike alkali metals, these elements fail to emit electron on exposure to light.

The alkaline earth metals are fairly strongly electropositive but the degree of electropositivity, i.e., the tendency to lose electrons is less than Group-I A elements. The electropositive character increases on moving from beryllium to barium. However, these are not as strongly electropositive as metals of Group-I A.

16. Heat of atomization: Heat of atomization decreases considerably from Be to Mg but the subsequent decrease is rather small. This reveals that the metal bond strength is maximum in the case of beryllium.

Element	Be	Mg	Ca	Sr	Ba	Ra
ΔH_{fus}/kJ mol^{-1}	15	8.9	8.6	8.2	7.8	(8.5)
ΔH_{vap}/kJ mol^{-1}	309	127.4	155	158	136	(113)
ΔH_f (monoatomic gas) kJ mol^{-1}	324	146	178	164	178	–

17. Electrode potentials: Just as in alkali metals the noble gas core shields the s-electrons from the direct attraction of the nuclear charge of alkaline earth metals. Therefore, the s-electrons are very loosely held.

From Table 6.2 it is evident that the first ionization energies of these elements are not particularly large but the second ionization energies of these elements are substantial. This suggests that the alkaline earth metals should have a tendency to form unipositive rather than bipositive ions. The following energetic relationships show that gaseous Ca^+ is scarcely stable with respect to gaseous Ca and Ca^{2+}.

$$Ca_{(g)} \longrightarrow Ca^+_{(g)} + e^- \quad \Delta H = 140 \text{ k.Cal}$$

$$Ca^{2+}_{(g)} + e^- \longrightarrow Ca^+_{(g)} \quad \Delta H = -274 \text{ k.Cal}$$

$$Ca_{(g)} + Ca^{2+}_{(g)} \longrightarrow 2Ca^+_{(g)} \quad \Delta H = -134 \text{ k.Cal}$$

But the situation is quite different when energetic relations between the aqueous ions and the solid metal are considered.

$$2Ca^+_{(g)} \longrightarrow 2\,Ca^+_{(aq)} \quad \Delta H \cong -152 \text{ k.Cal}$$

$$Ca_{(s)} \longrightarrow Ca_{(g)} \quad \Delta H = +142 \text{ k.Cal}$$

$$Ca^{2+}_{(aq)} \longrightarrow Ca^{2+}_{(g)} \quad \Delta H = +395 \text{ k.Cal}$$

$$Ca_{(g)} + Ca^{2+}_{(g)} \longrightarrow 2\,Ca^+_{(g)} \quad \Delta H = -134 \text{ k.Cal}$$

$$Ca_{(s)} + Ca^{2+}_{(aq)} \longrightarrow 2\,Ca^+_{(aq)} \quad \Delta H = +251 \text{ k.Cal}$$

Table 6.3 Standard oxidation potentials of alkaline earth metals

Element	Oxidation reaction	Standard oxidation potential (V)
Be	$Be \longrightarrow Be^{2+} + 2e^-$	1.97
Mg	$Mg \longrightarrow Mg^{2+} + 2e^-$	2.36
Ca	$Ca \longrightarrow Ca^{2+} + 2e^-$	2.84
Sr	$Sr \longrightarrow Sr^{2+} + 2e^-$	2.89
Ba	$Ba \longrightarrow Ba^{2+} + 2e^-$	2.92

From the relations it is clear that aqueous Ca^+ is energetically unstable with respect to $Ca^{2+}_{(aq)}$ and calcium metal. From the hydration energies it is clear that hydration energy of the bipositive ion is responsible for its stability in aqueous solution. Hence the oxidation potentials of alkaline earth metals can be considered to be sum of energies of the following reactions.

$$M_{(s)} \longrightarrow M^{2+}_{(g)} + 2e^-$$

ΔH_1 = Ionization potential + sublimation energy

$$M^{2+}_{(g)} + H_2O \longrightarrow M^{2+}_{(aq)}$$

ΔH_2 = Hydration energy

$$M_{(s)} + H_2O \longrightarrow M^{2+}_{(aq)} + 2e^- \quad \Delta H = \Delta H_1 + \Delta H_2$$

The oxidation potentials of alkaline earth metals are given in Table 6.3.

The data which partly explain these differences are given in Table 6.4.

Table 6.4 Energy changes for Electrode half-reactions

	Be	Mg	Ca	Sr	Ba
ΔH_{Sub} (Monoatomic gas) / kJ mol$^-$	324	146	178	164	178
$I_1 + I_2$ / kJmol^{-1}	2656.5	2188.4	1735.2	1614.0	1468.1
Hydration enthalpy kJ mol^{-1}	–2494	–1921	–1577	–1443	–1305
$\Delta H\left[M_{(s)} \rightarrow M^{2+}_{(aq)} + 2e^-\right]$	486.5	413.4	336.2	335	341.1

The oxidation potentials of alkaline earth metals are relatively high but lower than the alkali metals but there is a distinct difference between the lighter elements of the two groups. For example, the oxidation potentials of beryllium is +1.97 V compared to +3.05 V of lithium. However, as we go down the group there is a similarity between the alkali metals and the corresponding alkaline earth metals. Because the alkaline earth metals have an higher ionization energies compared to alkali metals, we could expect them to be less effective reducing agents than alkali metals because of their higher hydration energies. The energy released during the hydration of ions is more than sufficient to pull the electrons from the atom.

18. Flame colouration: Except beryllium and magnesium, the other alkaline earth metals and their compounds impart characterstic colours to the flame. Thus

Calcium	Strontium	Barium	Radium
Brick red	Crimson red	Apple green	Crimson red

The reason for imparting the colour to flame is that when elements or their compounds are put into a flame, the electrons absorb heat energy and excite to higher-energy levels. When they return to the ground state, they emit the absorbed energy in the form of radiations having particular wavelength in the visible region.

Beryllium and magnesium atoms are smaller and their electrons being strongly bound to the nucleus are not excited to higher-energy levels. Therefore, they do not give the flame test.

6.3 REACTIVITY OF ALKALINE EARTH METALS: CHEMICAL PROPERTIES

The chemistry of the elements is mainly decided by their electronic configuration, ns^2. Alkaline earth metals are strong electropositive elements and highly reactive like alkali metals but the reactivity of alkaline earth metals is less when compared with alkali metals. The high reactivity of alkaline earth metals is due to (i) low ionization energy, (ii) lower heat of atomisation and (iii) higher heat of hydration of their M^{2+} ion. The ionization potential value reveal the fact that they form M^{2+} ion readily. Except beryllium all other form ionic compounds predominantly. The reactivity of alkaline earth metals increases down the group, i.e., from Be to Ba. The reactivity of alkaline earth metals can be judged from the following chemical properties.

1. Action of Air: As the alkaline earth metals are less electropositive than alkali metals, they are only slowly oxidized on exposure to air. However, this tendency increases in the elements of Group-II A from Be to Ra due to an increase in the electropositive character.

$$2Ca + O_2 \longrightarrow 2CaO$$
$$2Sr + O_2 \longrightarrow 2SrO$$

When burnt in air, alkaline earth metals react with both oxygen and nitrogen in air forming a mixture of oxide and nitride. Beryllium is less reactive in bulk solid form and reacts at higher temperature but powdered beryllium is more reactive and burns brilliantly on ignition to give a mixture of oxide and nitride.

$$2Be + O_2 \xrightarrow{\text{heat}} 2BeO$$
$$3Be + N_2 \longrightarrow Be_3N_2$$

Further, the tendency of the metals to form higher oxides such as peroxide increases on moving down the group.

2. Action of Water: Beryllium do not react with water or steam even at high temperature. All the other alkaline earth metals react with water forming hydroxides with the liberation of hydrogen. Magnesium reacts only with boiling water, whereas other metals react even with cold water. e.g. ,

$$Mg + H_2O \longrightarrow MgO + H_2$$
$$Mg + 2H_2O \longrightarrow Mg(OH)_2 + H_2$$

The reaction with water becomes increasingly vigorous on moving down the group.

Ba > Sr > Ca > Mg > Be. Reactivity with water.

The inertness of Be and Mg towards water is ascribed to the formation of a protective thin layer of hydroxide on the surface of the metals.

3. Action of Acids: The alkaline earth metals stand high in electrochemical series and readily displace hydrogen from protonic acids and thus forming corresponding salts in solutions.

$$M + 2H^+ \longrightarrow M^{2+} + H_2 \text{ (M = alkaline earth metal)}$$

Conc. nitric acid renders the beryllium passive and so does not react with conc. HNO_3. This is due to the formation of a very thin oxide film on the surface of metals.

With the alkaline earth metals dil. or conc. HCl react similarly liberating H_2, dil. H_2SO_4 also liberate H_2 but conc. H_2SO_4 is reduced to SO_2. With nitric acid different products are formed under different conditions.

E.g.: $$Mg + \underset{\text{dil. or conc.}}{2HCl} \longrightarrow MgCl_2 + H_2$$

$$Mg + \underset{\text{dil.}}{H_2SO_4} \longrightarrow MgSO_4 + H_2$$

$$Mg + \underset{\text{conc.}}{2H_2SO_4} \longrightarrow MgSO_4 + 2H_2O + SO_2$$

$$Mg + \underset{\text{V. dil.}}{2HNO_3} \longrightarrow Mg(NO_3)_2 + H_2$$

$$Mg + \underset{\text{Conc.}}{4HNO_3} \longrightarrow Mg(NO_3)_2 + 2H_2O + 2NO_2$$

Only magnesium liberates hydrogen from dil. HNO_3

4. Action of Bases: Generally, metals do not react with bases. Be being amphoteric, metals react with bases liberating hydrogen but other alkaline earth metals do not react with bases.

$$Be + 2NaOH \longrightarrow Na_2BeO_2 + H_2$$

5. Action of Hydrogen: Except beryllium, all other elements of Group-IIA, when heated, combine with hydrogen to form hydrides MH_2. Magnesium hydride has a significant covalent character but the other hydrides are ionic.

$$Mg + H_2 \longrightarrow MgH_2$$
$$Ca + H_2 \longrightarrow CaH_2$$

6. Action of Carbon: When heated with carbon these elements form ionic carbides. Be forms methanide Be_2C

while other metals form MC_2 type acetylides. The acetylides have the sodium chloride structure with M^{2+} replacing Na^+ and C_2^{2-} replacing Cl^- ions.

7. Action of Nitrogen: These elements on burning in nitrogen form the nitrides of the type M_3N_2. The ease of formation of nitrides decreases down the group. They react with water liberating ammonia and form either metal oxide or hydroxide. E.g.,

$$3Ca + N_2 \longrightarrow Ca_3N_2$$
$$Ca_3N_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2NH_3$$

8. Action of Phosphorous: With phosphorous also these elements react similar to nitrogen forming phosphides of the type M_3P_2 which on hydrolysis in water liberates phosphine.

$$6Ca + P_4 \longrightarrow 2Ca_3P_2$$
$$Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$$

9. Action of Oxygen: All the elements of Group-II A excluding radium form oxides. Be and Mg form only monoxides. Ca, Sr and Ba forms peroxides also in excess of oxygen.

$$M + \frac{1}{2}O_2 \longrightarrow MO$$
$$M + O_2 \longrightarrow MO_2$$

10. Action of S, Se and Te: All the alkaline earth metals react with S, Se and Te forming sulphides, selenides and tellurides, respectively.

$$M + S \longrightarrow MS$$
$$M + Se \longrightarrow MSe$$
$$M + Te \longrightarrow MTe$$

11. Action of Halogens: All the alkaline earth metals react with halogens at appropriate temperatures to form halides of the type MX_2.

$$M + X_2 \longrightarrow MX_2$$

12. Formation of amalgams and alloys: All the alkaline earth metals form amalgams with mercury and alloys with other metals.

13. Complex formation: As the complex formation is favoured by the small size, high charged ion and suitable empty orbitals, alkaline earth metal ions, not having these characteristics are not having a significant tendency (although it is more than the alkali metals by virtue of their double charge) to form complexes. However, beryllium, because of its small size, is known to form a number of stable complexes, e.g., $[BeF_3]^-$, $[BeF_4]^{2-}$, $[Be(H_2O)_4]^{2+}$, $[Be(OH)_4]^{2-}$, etc.

It is to be noted that the coordination number of Be is at the maximum 4 and hence Be salts cannot have more than four molecules of water of crystallization because in Be no vacant d-orbitals are present, whereas magnesium can have a coordination number of six by employing 3d-orbitals as well as 3s- and 3p-orbitals.

The tendency to form complex gets decreased when we move down the group which is attributed to increase in the size of the ion. Thus, Mg^{2+} and Ca^{2+} ions exist as hexahydrates $[M(H_2O)_6]^{2+}$. The formation of complexes in solution forms the basis of volumetric estimation of Mg^{2+} and Ca^{2+} ions using standard solution of EDTA ethylene diamine tetraacetate (EDTA). Similarly, gravimetric estimation of Mg^{2+} ions with oxime (8-hydroxy quinoline) can be explained on the basis of formation of an insoluble complex.

14. Solubility in liquid ammonia: Like alkali metals Ca, Sr and Ba dissolve in liquid ammonia to give deep blue-black solutions. If the metal–ammonia solutions get evaporated, hexaammoniates $[M(NH_3)_6]^{2+}$ get formed. The ammoniates are good conductors of electricity. They gradually decompose to the corresponding amides, especially in the presence of catalysts.

$$[M(NH_3)_6]^{2+}_{(s)} \rightarrow M(NH_2)_{2(s)} + 4NH_{3(g)} + H_2(g)$$

The tendency for the formation of ammoniates gets decrease with increase in size of the metal atom, i.e., on moving down the group.

6.4 OCCURRENCE OF ALKALINE EARTH METALS

The Group-IIA elements are never found in the metallic form in nature because like the alkali metals they are active reductants and react readily with a variety of non-metals. The important minerals of various alkaline earth metals are given in Table 6.5.

6.5 PRINCIPLES OF EXTRACTION OF ALKALINE EARTH METALS

We know that alkaline earth metals are strong reducing agents. Therefore, they cannot be prepared in the metallic form by chemical reduction methods. But smaller quantities of these elements have been prepared by the reduction of their oxides with carbon or aluminium.

$$BeO + C \longrightarrow Be + CO$$
$$3BeO + 2Al \longrightarrow 3\,Be + Al_2O_3 + \text{heat}$$

Similar to alkali metals, alkaline earth metals cannot be prepared by the electrolytic reduction of their aqueous salt solutions. The reason for this is due to high electropositive nature of these metals, as soon as they are liberated at cathode, they react with water forming hydroxide and hydrogen. Thus, no metals are liberated at the cathode.

Table 6.5 Abundance and minerals of alkaline earth metals

Element	Abundance (ppm)	Minerals	
Be	6	Beryl	$3BeO \cdot Al_2O_3 \cdot 6SiO_2$
		Chrysoberyl	$BeO \cdot Al_2O_3$
		Phenacite	$2BeO \cdot SiO_2$ or Be_2SiO_4
Mg	29,900		
	(6th position in the order of abundance)	Magnesite	$MgCO_3$
		Dolamite	$MgCO_3 \cdot CaCO_3$
		Epsom salt	$MgSO_4 \cdot 7H_2O$
		Kieserite	$MgSO_4 \cdot H_2O$
		Carnalite	$KCl \cdot MgCl_2 \cdot 6H_2O$
		Talc	
		Kainite	$K_2SO_4 \cdot MgSO_4 \cdot MgCl_2 \cdot 6H_2O$
		Asbestos	$CaMg_3(SiO_3)_4$
		Sea water is the main commercial source of magnesium	
Ca	36,300		
	(3rd position in the abundance of order)	Lime stone, Chalk, Marble, Iceland spar	$CaCO_3$
		Gypsum	$CaSO_4 \cdot 2H_2O$
		Anhydrite	$CaSO_4$
		Phosphorite	$Ca_3(PO_4)_2$
Sr	300	Celestite	$SrSO_4$
		Strontianite	$SrCO_3$
Ba	250	Barytes or heavy spar	$BaSO_4$
		Whitherite	$BaCO_3$
Ra	1.3×10^{-6}	Pitch blende	U_3O_8
		Carnotite	$K_2 (UO_2)_2 (UO_4)_2 \cdot 5H_2O$

The alkaline earth metals are prepared by the electrolysis of their fused anhydrous salts containing alkali metal salts which are added to lower the melting points and to increase the conductivity of the electrolytic mixtures. e.g.,

(i) Beryllium is prepared by electrolyzing a mixture of fused beryllium chloride and sodium chloride.

(ii) Magnesium is prepared by electrolyzing a mixture of anhydrous carnallite (or anhydrous magnesium chloride) and potassium chloride in an iron cell (cathode) with a carbon anode.

6.6 COMPOUNDS OF ALKALINE EARTH METALS: A COMPARATIVE STUDY

6.6.1 Hydrides

Down the Group-II A, there is an increase in reactivity towards hydrogen. Beryllium and magnesium show only a slight tendency to react with hydrogen but calcium, strontium and barium react with hydrogen at high temperatures forming ionic hydrides of the type MH_2.

$$M + H_2 \longrightarrow MH_2 \text{ (M = Mg, Ca, Sr or Ba)}$$

However, beryllium and magnesium hydrides are prepared by the action of lithium aluminium hydride on their halides in ethereal solution.

$$2MgCl_2 + LiAlH_4 \longrightarrow 2MgH_2 + LiCl + AlCl_3$$

$$2BeCl_2 + LiAlH_4 \longrightarrow 2BeH_2 + LiCl + AlCl_3$$

Beryllium hydride is a covalent compound and magnesium hydride is partially ionic. The hydrides are all reducing agents and are hydrolyzed by water and dilute acids.

$$MH_2 + 2H_2O \longrightarrow M(OH)_2 + 2H_2$$

Beryllium and magnesium hydrides are polymeric. The polymeric solid $(BeH_2)_n$ contains hydrogen bridges between beryllium atoms. Be is bonded to four hydrogen atoms, and

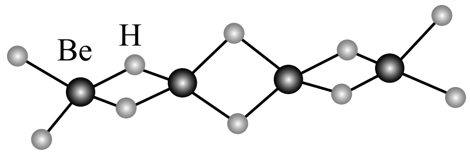


Fig 6.1 Polymeric structure of $(BeH_2)_n$ in solid state

the H atoms appear to be forming two bonds. Since Be has two valence electrons, and H only one, it is apparent that there are not enough electrons to form the usual electron pair bonds in which two electrons are shared between two atoms. Instead of this, **three-centre bonds** are formed in which a "banana shaped" molecular orbital (or three-centre bond) covers three atoms Be......H...... Be and contains two electrons (this is called a three-centre two-electron bond). The structure of $(BeH_2)_n$ is as shown in Fig. 6.1.

6.6.2 Nitrides

All the elements of Group-II A burn in nitrogen to form nitrides M_3N_2.

$$3Mg + N_2 \longrightarrow Mg_3N_2$$

The ease of formation of nitrides decreases down the group. All the nitrides are colourless solids.

Barium nitride can also be prepared by the decomposition of amides on heating.

$$3Ba(NH_2)_2 \longrightarrow Ba_3N_2 + 4NH_3$$

Magnesium nitride can also be prepared by heating magnesium hydride in a current of nitrogen.

$$3MgH_2 + 2N_2 \longrightarrow Mg_3N_2 + 2NH_3$$

All these nitrides are stable upto 1000°C. Addition of water causes hydrolysis and ammonia is liberated.

$$Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$$

On heating with hydrogen, nitrides are reduced.

$$Mg_3N_2 + 6H_2 \longrightarrow 3MgH_2 + 2NH_3$$

These nitrides are decomposed by carbon monoxide and carbon dioxide at 200°C.

$$Mg_3N_2 + 3CO \longrightarrow 3MgO + 3C + N_2$$
$$Mg_3N_2 + 3CO_2 \longrightarrow 3MgO + 3CO + N_2$$

The formation of magnesium nitride was used by Ramsay to remove nitrogen from air during the preparation of noble gases.

6.6.3 Oxides

All the elements of Group-II A excluding radium form oxides of MO and peroxides of MO_2. Beryllium forms monoxide only, which is predominantly covalent. The remaining oxides are highly stable ionic solids and have sodium chloride structure. Except BeO other alkaline earth metal oxides are usually obtained by thermal decomposition of the carbonates, MCO_3.

$$MCO_3 \longrightarrow MO + CO_2$$

Beryllium oxide is made by ignition of the metal or its compounds in oxygen. Alkaline earth metal oxides are also formed by heating their oxosalts such as $M(NO_3)_2$ and MSO_4 and also by heating $M(OH)_2$. The oxosalts are less stable to heat than the corresponding Group-IA salts because the metals and their hydroxides are less basic than those of Group-IA.

Alkaline earth metal oxides are extremely stable solids and their free energies of formation are highly negative and are in the range of –120 to –144 k.Cal mol^{-1} at 298 K as shown in Table 6.6.

Table 6.6 Thermodynamic properties of alkaline earth oxides at 298 K

	BeO	**MgO**	**CaO**	**SrO**	**BaO**
ΔH°_f k.Cal mol^{-1}	–143.1	–143.8	–151.9	–144.1	–133.4
ΔG°_f k.Cal mol^{-1}	–136.1	–136.1	–144.4	–133.8	–176.3

The oxides of alkaline earth metals except beryllium are ionic and have 6:6 sodium chloride structure. BeO is covalent and has 4:4 zinc sulphide (Wurtzite) structure. The radius ratio (M^{2+}/O^{2-}) values of BeO, MgO and CaO predicts a coordination number 4 for BeO and 6 for MgO and CaO, which are in accordance to the structure. But for SrO and BaO the radius ratio values predict a coordination number 8, though the structures found are of 6 coordinates. Crystals adopt the strcuture that has the most favourable lattice energy. Here the radius ratio concept is failed probably because the ionic radii are not known with great accuracy and they change with coordination numbers also, and ions are not necessarily spherical or inelastic.

Table 6.7 Radius ratios and coordination numbers of alkaline earth metal oxides

Oxide	**Radius ratio M^{2+}/O^{2-}**	**Predicted coordination number**	**Coordination number found**
BeO	0.22	4	4
MgO	0.44	6	6
CaO	0.56	6	6
SrO	0.81	8	6
BaO	0.96	8	6

The heavier oxides react with water to form hydroxides of the general formula $M(OH)_2$. Beryllium oxide is distinguished from other oxides by a large degree of covalency. It is polymeric and is therefore harder. Beryllium oxide is amphoteric and reacts slowly with strong acids.

$$BeO + 2HCl \longrightarrow BeCl_2 + H_2O$$

It also react with strong bases and enters into solution.

$$BeO + 2NaOH \longrightarrow Na_2BeO_2 + H_2O$$

The amphoteric character of BeO is due to strong polarizing power of smaller ionic size and relatively large charge to radius ratio.

BeO is insoluble in water. MgO is not very reactive CaO, SrO and BaO react exothermically with water forming hydroxides. Solubility in water increase from BeO to BaO, e.g.,

$$CaO + H_2O \longrightarrow Ca(OH)_2$$

These oxides have

(i) high melting points (BeO ~2500°C, MgO ~2800°C)
(ii) very low vapour pressure
(iii) very good conductors of heat
(iv) chemical inertness
(v) electrical insulators

These properties make them useful as refractory materials. Magnesium and calcium oxides are used in metallurgy as basic fluxes for the removal of acidic gangue especially silica in the extraction of the metals. Calcium oxide is used in the production of calcium carbide.

Peroxides: Peroxides are formed with increasing ease and increasing stability as the metal ion become larger. Barium oxide reacts with oxygen at 773 K to form the peroxide. Strontium peroxide can be formed in a similar way but this requires a high pressure (200 atm) and high temperature. Calcium peroxide may be obtained by the cautious dehydration of the hydrate obtained from hydroxide and hydrogen peroxide. Hydrated magnesium peroxide is prepared by an analogous reaction for use in tooth pastes. It cannot be dehydrated without decomposition.

The peroxides are white ionic solids having the peroxide $(O\text{-}O)^{2-}$ ion. The stability of peroxides increases as the size of the cation increases from magnesium to barium. These peroxides are considered as salts of the very weak acid hydrogen peroxide. The peroxides react with acids liberating hydrogen peroxide.

$$BaO_2 + 2HCl \longrightarrow BaCl_2 + H_2O_2$$

Beryllium do not form peroxide.

6.6.4 Hydroxides

Hydroxides are formed by the reaction of the oxides of alkaline earth metals with water.

$$MO + H_2O \longrightarrow M(OH)_2$$

The solubility of the hydroxides of alkaline earth metals in water increases with increase in atomic number of the cation. The solubility of hydroxides depend mainly on two facts.

(a) The lattice energy required to dissociate the components of hydroxide. This decreases from beryllium to barium.
(b) The hydration energy of cation M^{2+}. This also decreases from beryllium to barium as the size of cation increases.

Though both lattice and hydration energies are decreasing down the group the decrease in lattice energy is more rapid than the hydration energy and so their solubility increases on descending the group. $Be(OH)_2$ is soluble in solutions containing an excess of OH^- and is therefore amphoteric.

$$Be(OH)_2 + 2NaOH \longrightarrow Na_2[Be(OH)_4]$$

$Mg(OH)_2$ is weakly basic and is used as an antacid in medicines. Other hydroxides are strong bases. Basic strength of hydroxides increases from beryllium to barium. However, these hydroxides are less basic than those of alkali metal hydroxides because of higher ionization energies, smaller ionic sizes and double charge on ions.

6.6.5 Halides

Alkaline earth metals form halides of the type MX_2. They can be made directly by heating the metals with halogens

Table 6.8 Solubility of alkaline earth metals hydroxides

Property	Be	Mg	Ca	Sr	Ba
Ionic radius (pm)	31*	65	99	118	135
Hydration energy of M^{2+} kJ mol^{-1}	–2494	–1921	–1577	–1443	–1305
Solubility of $M(OH)_2$ mol lit^{-1} at 293K	2×10^{-6}	3×10^{-4}	0.022	0.066	0.23

*Four coordinate ratio.

or by the action of halogen acid on either of the metal or carbonate.

$$M + X_2 \longrightarrow MX_2$$
$$M + 2HX \longrightarrow MX_2 + H_2$$
$$MCO_3 + 2HX \longrightarrow MX_2 + H_2O + CO_2$$

When crystallized from solutions they form hydrated salts, and anhydrous $CaCl_2$, $SrCl_2$ and $BaCl_2$ can be prepared by heating the hydrated salts. Anhydrous beryllium halides cannot be obtained from materials made in aqueous solutions because the hydrated ion $[Be(H_2O)_4]^{2+}$ is formed, e.g., $[Be(H_2O)_4]Cl_2$ or $[Be(H_2O)_4]F_2$. Heating these compounds results in the formation of hydroxides or oxides.

$$[Be(H_2O)_4]\,Cl_2 \xrightarrow{\Delta} Be(OH)_2 + 2HCl + 2H_2O$$

Anhydrous beryllium chloride is made by the action of carbon tetrachloride on the oxide at 1073 K. This is a standard method for making the metal chlorides which cannot be prepared by thermal decomposition of hydrates isolated from aqueous solution.

$$2BeO + CCl_4 \xrightarrow{800°C} 2BeCl_2 + CO_2$$

The anhydrous beryllium halides are covalent. They have low melting points as well as the electrical conductivity in the fused state than the other alkaline earth halides. Further, beryllium halides are soluble in organic solvents suggesting the non-polar character of the compound. The small and highly charged beryllium ion polarizes neighbouring negative ions and degree of polarization is so large that the beryllium halides (except BeF_2) are covalently bonded molecules.

Beryllium chloride vapours contains $BeCl_2$ and $(BeCl_2)_2$, but in the solid state it is polymerized. Though the structure of $(BeCl_2)_n$ is similar to that of $(BeH_2)_n$, the bonding nature is different. The hydride is having three centred two electron bonds whereas the halides have halogen bridges, in which a halogen atom bonded to one beryllium atom uses a lone pair of electrons to form a coordinate bond to another beryllium atom.

Beryllium chloride also forms a stronger complex of formula $M_2[BeCl_4]$ with alkali metal chlorides, but these are decomposed by water. In aqueous media beryllium forms a stronger complex with fluoride than with chloride.

Ionic alkaline earth metal chlorides dissolve in water to produce neutral solutions of hydrated divalent metal ions and hydrated chloride ions.When their solutions are evaporated the following colourless hydrated salts are crystallized, $MgCl_2 \cdot 6H_2O$; $CaCl_2 \cdot 6H_2O$; $SrCl_2 \cdot 6H_2O$ and $BaCl_2 \cdot 2H_2O$. When covalent beryllium chloride comes in contact with water it hydrolyzes readily to form an acidic solution according to the equation.

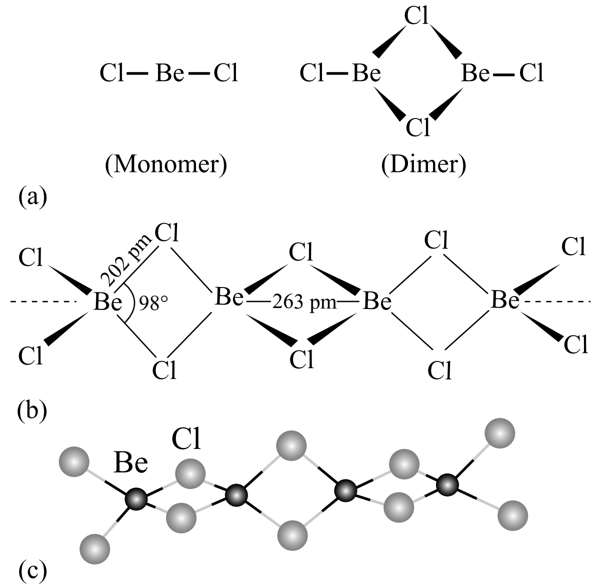


Fig 6.2 Structure of beryllium chloride (a) Monomer, (b) Dimer and (c) Polymer in solid state

$$BeCl_2 + 2H_2O \rightarrow Be(OH)_2 + 2HCl$$

The bromides and iodides are similar to the chlorides. The bromide and iodide of magnesium are appreciably soluble in organic solvents such as alcohols, ketones and ethers with which they form complexes.

The halides of alkaline earth metals are hygroscopic and form hydrates. Calcium chloride has a strong affinity for water and is a popular dehydrating agent.

Among the chlorides there is regular increase in the melting point as the size of the metal ion increases. Similarly, the conductivity of metal chlorides increases from beryllium to barium. This increase can be attributed to the increasing ionic character of chlorides. Thus, barium chloride is a stable, high melting solid and a conductor in the molten state. Chlorides, bromides and iodides of Mg, Ca, Sr, and Ba are ionic that have much lower melting points than fluorides.

Beryllium fluoride is very soluble in water owing to the solvation energy of Be in forming $[Be(H_2O)_4]^{2+}$. The other fluorides are all almost insoluble. The chlorides, bromides and iodides of Mg, Ca, Sr and Ba are readily soluble in water. The solubility decreases somewhat with increasing atomic number.

Ionic character of the halides of alkaline earth metals is in the order

$$BeX_2 < MgX_2 < CaX_2 < SrX_2 < BaX_2$$
$$MI_2 < MBr_2 < MCl_2 < MF_2$$

Solubility order

$$BeF_2 > MgF_2 > CaF_2 < SrF_2 < BaF_2$$

$$BeX_2 > MgX_2 > CaX_2 > SrX_2 > BaX_2$$
$$MF_2 < MCl_2 < MBr_2 < MI_2$$

6.6.6 Carbonates and Bicarbonates

The carbonates of the alkaline earth metals are salts of the weak carbonic acid H_2CO_3 and strong hydroxides $M(OH)_2$. The carbonates occur in nature as solid rock materials. These can be prepared by passing carbon dioxide into the aqueous solutions of alkaline earth metal hydroxides or by adding sodium or ammonium carbonate solution to the solution of a soluble salt of these metals. The carbonates are precipitated out.

$$M(OH)_2 + CO_2 \longrightarrow MCO_3 + H_2O$$
$$M^{2+} + CO_3^{2-} \longrightarrow MCO_3$$

All these carbonates decompose on heating.

$$MCO_3 \longrightarrow MO + CO_2$$

The thermal stability of carbonates increases from beryllium to barium.

Carbonate	$BeCO_3$	$MgCO_3$	$CaCO_3$	$SrCO_3$	$BaCO_3$
Decomposition temperature (K)	373	813	1173	1563	1633

Since $BeCO_3$ decomposes at low temperatures, it should be stored in an atmosphere of carbon dioxide.

The solubility of alkaline earth metal carbonates decreases as the atomic number of cation increases.

$$BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$$

All these carbonates are much more soluble in a solution of carbon dioxide than in water owing to the formation of the bicarbonate. The bicarbonates of alkaline earth metals cannot be prepared in solid state. They exist only in solution.

> Caves in lime stone regions often have stalactites growing down the roof, and stalagmites growing up from the floor. Water percolating through the limestone contains some calcium bicarbonate solution. The soluble bicarbonate decomposes slowly into the insoluble carbonate, and these results in the slow growth of the stalactites and stalagmites.

All the carbonates of alkaline earth metals are ionic, but $BeCO_3$ is unusual because it contains the hydrated ion $[Be(H_2O)_4]^{2+}$ rather than Be^{2+}. Calcium carbonate occurs as two different crystalline forms, **calcite** and **aragonite.** Calcite is the more stable form in which each Ca^{2+} ion is surrounded by six oxygen atoms from CO_3^{2-} ions. Aragonite is a metastable form. Its standard enthalpy of formation is about 5 kJ mol^{-1} higher than that of calcite. Though argonite is less stable and should convert into calcite, the high activation energy prevents this conversion. Aragonite can be prepared in the laboratory by precipitating from a hot solution. In aragonite, each Ca^{2+} ion is surrounded by 9 oxygen atoms, an unusual coordination number.

6.6.7 Sulphates

Sulphates of alkaline earth metals can be obtained by the action of dilute sulphuric acid on metals, oxides, hydroxides or carbonates.

$$M + H_2SO_4 \longrightarrow MSO_4 + H_2$$
$$MO + H_2SO_4 \longrightarrow MSO_4 + H_2O$$
$$M(OH)_2 + H_2SO_4 \longrightarrow MSO_4 + 2H_2O$$
$$MCO_3 + H_2SO_4 \longrightarrow MSO_4 + H_2O + CO_2$$

All the sulphates are white crystalline solids. Sulphates of Be, Mg, and Ca crystallize, as hydrated salts $BeSO_4 \cdot 4H_2O$, $MgSO_4 \cdot 7H_2O$ and $CaSO_4 \cdot 2H_2O$. Sulphates of Sr and Ba crystallize without water of crystallization.

The thermal stability of sulphates increases down the group.

Sulphate	**$BeSO_4$**	**$MgSO_4$**	**$CaSO_4$**	**$SrSO_4$**	**$BaSO_4$**
Decomposition temperature	773	1168	1422	1422	1647

Sulphates decompose on heating.

$$2MSO_4 \longrightarrow 2MO + 2SO_2 + O_2$$

When sulphates are heated with carbon, they are reduced to sulphides.

$$MSO_4 + 2C \longrightarrow MS + 2CO_2$$

The solubility of the sulphates in water decreases from Be to Ba. $BeSO_4$ and $MgSO_4$ are fairly soluble while $BaSO_4$ is completely insoluble.

6.6.8 Nitrates

All the alkaline earth metals form nitrates of the type $M(NO_3)_2$. They can be prepared by the reaction of nitric acid with oxides, hydroxides or carbonates.

$$MO + 2HNO_3 \longrightarrow M(NO_3)_2 + H_2O$$
$$M(OH)_2 + 2HNO_3 \longrightarrow M(NO_3)_2 + 2H_2O$$
$$MCO_3 + 2HNO_3 \longrightarrow M(NO_3)_2 + H_2O + CO_2$$

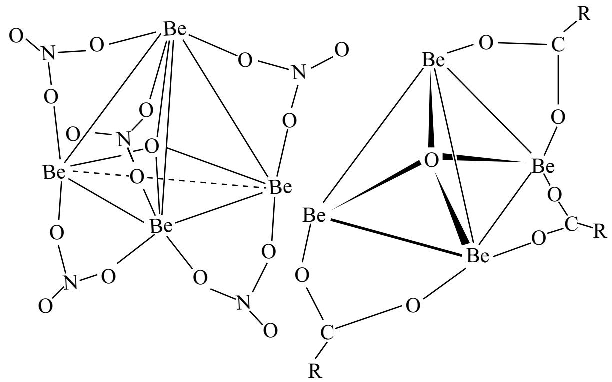


Fig 6.3 Structures of basic beryllium nitrate and basic carboxylates complexes $[Be_4O(RCO_2)_6]$. Only three of the six carboxylate groups are shown.

When crystallized from solution, they are crystallized as hydrated salts. Anhydrates cannot be prepared by heating hydrated nitrates as they decompose to metal oxide. Anhydrous nitrates can be prepared using liquid N_2O_4 and ethyl acetate. Beryllium forms basic beryllium nitrate, in addition to the normal salt.

$$BeCl_2 \xrightarrow{N_2O_4} Be(NO_3)_2 \cdot 2N_2O_4$$

$$\xrightarrow[\text{Vaccum}]{50°C} Be(NO_3)_2 \xrightarrow{125°C} [Be_4O\,(NO_3)_6]$$

Basic beryllium nitrate

Basic beryllium nitrate is covalent and has an unusual structure as shown in Fig. 6.3, Four Be atoms are arranged tetrahedrally. Six NO_3^- ions act as bridges between two Be atoms along 6 edges of the tetrahedron. The basic oxygen is at the centre of the tetrahedron. In this compound nitrate ion acts as a bidentate ligand.

All the nitrates decompose on heating liberating brown NO_2 gas along with oxygen.

$$2M(NO_3)_2 \longrightarrow 2MO + 4NO_2 + O_2$$

Thermal stability and solubility of the nitrates of Group-IIA elements follow the same order as in the case of carbonates and sulphates, but all nitrates are soluble.

6.6.9 Trends in the Solubilities and Thermal Stabilities of Compounds of Group-IIA Elements

It is interesting to note that the solubilities of oxides, hydroxides and fluorides of the elements of Group-IIA increases down the group while the solubilities of most of their salts decreases down the group. Another point to be noted is that thermal stabilities of most of their oxosalts like carbonates, sulphates, nitrates, etc. increases down the group, though their lattice energies decreases. As already explained in section 3.5.8 of Chapter 3, these trends depend on lattice enthalpies, hydration enthalpies and free energy changes. The hydration enthalpies of alkaline earth metal ion and the latice enthalpies of some of their compounds are given in Table 6.9.

The lattice energy is inversely proportional to the bond length in ionic solids.

$$\Delta H_L \propto \frac{1}{r_+ + r_-}$$

However, the hydration energy, with each ion being hydrated individually, is the sum of individual ion contributions.

$$\Delta_{hyd}\,H \propto \frac{1}{r_+} + \frac{1}{r_-}$$

In the expression for lattice energy, one small ion cannot make the denominator of the expression small by itself, so the lattice energy decreases more rapidly than hydration energy. For example, the decrease in lattice energy between MgO and BaO is about 803 kJ mol^{-1}, whereas the decrease in hydration energy between Mg^{2+} and Ba^{2+} (hydration energy of oxide ion being constant) is 616 kJ mol^{-1}. So, the hydration energy dominates the lattice energies and the compounds dissolve with increase in the size of cation down the group as in the case of oxides, hydroxides and fluorides.

But in the expression for lattice energy, if one ion is bigger, the denominator also becomes larger. Then the decrease in lattice energy is less rapid than the decrease in hydration energy and dominates the hydration energy. For example, the decrease in lattice energy between $MgCO_3$ and $BaCO_3$

Table 6.9 Hydration and lattice enthalpy values of compounds of group IIA elements

Metal ion	Hydration enthalpy kJ mol^{-1}	MO	MCO_3	MF_2	MI_2
Be^{2+}	–2494				
Mg^{2+}	–1921	–3923	–3178	–2906	–2292
Ca^{2+}	–1577	–3517	–2986	–2610	–2058
Sr^{2+}	–1443	–3312	–2718	–2459	
Ba^{2+}	–1305	–3120	–2614	–2367	

is about 564 kJ mol^{-1}, but the decrease in hydration energy between Mg^{2+} and Ba^{2+} (being the hydration energy of carbonate ions is same) is 616 kJ mol^{-1}. So, the solubility of the compound decreases with increase in the size of cation as in the case of carbonates, sulphates, nitrates, halides (except fluorides), etc. The sulphates of calcium, strontium and barium are insoluble, and the carbonates, oxalates, chromates and fluorides of the whole group are insoluble.

Thermal stability of carbonates, sulphates, nitrates or the oxosalts, which on decomposition give metal oxide as one of the product, can be explained basing on lattice energy of the metal oxide and its oxosalt. The lattice energy is more when metal ion combines with small oxide ion than combined with bigger oxoanion like CO_3^{2-}, SO_4^{2-}, NO_3^-, etc. The decomposition temperature depends on the difference in the lattice enthalpies of metal oxide and its oxosalt.

$$\Delta H° \approx \Delta_{decomp} H + H_L (MCO_{3(s)}) - H_L (MO_{(s)})$$

where Δ_{decomp} H is the enthalpy change for the gas phase decomposition of oxoanion. Because Δ_{decomp} H is large and positive, the overall reaction enthalpy is positive (decomposition is endothermic), but it is less positive if the lattice energy of the oxide is markedly greater than that of oxosalt because lattice enthalpy of oxosalt is less than that of metal oxide. It follows that the decomposition temperatures of oxosalts will be low that have relatively high lattice energies for oxides. Thus the compounds having small, highly charged cation, such as Be^{2+}, Mg^{2+}, etc. are unstable because small cation increases the lattice energy of its oxide more than that of an oxosalt. So, the thermal stability of oxosalts decreases with increase in the size of the cation.

6.7 ANOMALOUS BEHAVIOUR OF BERYLLIUM

Beryllium is the first member of Group-IIA, and differs from the remaining members of its group. This has been attributed to the following features:

(i) its small size
(ii) its high electronegativity
(iii) presence of only two electrons in its penultimate shell
(iv) absence of d-orbitals in its valence shell

The main points of differences are as follows.

1. Tendency to form ionic compounds: As the ionization energy of beryllium is highest among the Group-IIA elements, it will have the lowest tendency to form the Be^{2+} ion; consequently, it does not form ionic compounds readily.

2. Tendency to form covalent compounds: As the electronegativity of beryllium (1.5) is the highest among the Group-II A metals its compounds with non-metals such as oxygen, chlorine, nitrogen, etc. are more covalent in character than those of the corresponding compounds of magnesium, calcium, strontium and barium.

3. Reaction with water: Beryllium do not react with water even at red hot condition.

4. Oxidation potential: Beryllium is known to have the lowest oxidation potential among Group-IIA metals. This means that it will have the lowest electropositive character. This is evident from the nature of oxides and hydroxide. Both BeO and Be $(OH)_2$ are amphoteric, i.e., they dissolve in acids as well as in alkalis to form salts and beryllates, respectively.

$$BeO + 2HCl \longrightarrow BeCl_2 + H_2O$$

$$BeO + 2NaOH \longrightarrow Na_2BeO_2 + H_2O$$

The oxides and hydroxides of the remaining Group-IIA metals are basic in nature. The basic strength increases from Mg $(OH)_2$ to Ba $(OH)_2$.

BeO is covalent and has a 4:4 zinc sulphide (wurtzite) structure but all the others are ionic and have a 6:6 NaCl type structure.

5. Reaction with acids: Beryllium does not liberate hydrogen from acids so readily due to its comparatively lower oxidation potential. On the other hand, the remaining metals of Group-II A do.

6. Reaction with hydrogen: Beryllium does not react directly with hydrogen to form the hydride while the remaining metals of Group II do. BeH_2 and to some extent MgH_2 are covalent compounds while the hydrides of the remaining metals of Group-IIA are ionic in character.

7. Reaction with nitrogen: All the metals of Group-IIA burn in nitrogen to form nitrides M_3N_2. However, Be_3N_2 is volatile, whereas the other nitrides do not volatilize.

8. Reaction with carbon: Beryllium on heating with carbon forms Be_2C but not BeC_2. However, the remaining metals of Group-IIA form ionic carbides MC_2 by heating the metals or their oxides with carbon. When treated with water, beryllium carbide evolves methane while the remaining carbides evolve acetylene.

$$Be_2C + 2H_2O \longrightarrow 2BeO + CH_4$$

$$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + C_2H_2$$

9. Reaction with alkalis: Beryllium reacts with alkalis to liberate hydrogen and form beryllates, e.g.,

$$Be + 2NaOH \longrightarrow Na_2BeO_2 + H_2$$

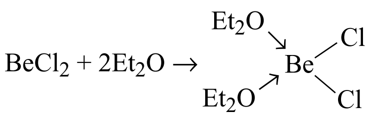
But other elements of Group-IIA do not react with alkalis.

10. Beryllium sulphide is insoluble in water while those of other metals are soluble in water.

11. Beryllium sulphate is soluble in water while those of other alkaline earth metals are sparingly soluble in water.

12. Formation of complexes: Anhydrous compounds of Be are predominantly two-covalent and they should be linear. In fact, linear molecules exist only in the gas phase. But in the solid state fourfold coordination is always achieved. There are several ways by which this can be achieved.

(i) Two ligands that have a lone pair of electrons may form coordinate bonds using the two unfilled orbitals of the valence shell of Be. Thus two F^- ions may coordinate to BeF_2, forming $[BeF_4]^{2-}$. If the solvent molecules are weak coordinators, $BeCl_2$ forms species of the type L_2BeCl_2 where L represents the weak coordinating solvent molecule. Thus, we have



The structure of L_2BeCl_2 is tetrahedral and the Be atom utilizes the four sp^3 hybrid orbitals. If, however, the solvent molecule is comparatively a stronger coordinator, $BeCl_2$ forms an ionic species of the type $[BeL_4]^{2+}\ 2Cl^-$ where L is NH_3 or H_2O. Thus, we have

$$BeCl_2 + 4H_2O \longrightarrow [Be(H_2O)_4]Cl_2$$

The water ligands in such salts are more strongly bound than is typical of other divalent cations. For instance, $[Be(H_2O)_4]Cl_2$ does not lose H_2O over strong dessicants such as P_2O_5. The stability of Be complexes with ligands containing nitrogen or other donors is lower than that of complexes possessing oxygen donor ligands. Thus $[Be(NH_3)_4]Cl_2$ is thermally stable but rapidly hydrolyzed to the tetraaqua ion.

The firm attachment of the H_2O molecule in $[Be(H_2O)_4]^{2+}$ cause a weakening of the OH bonds. This means that the aquaion is acidic, as shown in the reaction.

$$[Be(H_2O)_4]^{2+} \longrightarrow [Be(H_2O)_3(OH)]^+ + H^+$$

The aqueous solutions of beryllium salts are extensively hydrolyzed and are acidic. Hence, $[Be\ (H_2O)_4]^{2+}$ is an acid and $[Be(OH)_4]^{2-}$ is a base.

The tetrafluro beryllate ion $[BeF_4]^{2-}$ is formed in fluoride containing solutions, or in non-aqueous melts of acid fluorides such as NH_4HF_2. The tetrafluoro beryllate anion behaves in crystals much like SO_4^{2-}; thus, $PbBeF_4$ and $PbSO_4$ have similar crystal structures and solubilities.

(ii) The BeX_2 molecules may polymerize to form chains containing bridging halogen groups. For example, $(BeF_2)_n$, $(BeCl_2)_n$. Each halogen forms one normal covalent bond and uses a lone pair to form a coordinate bond.

(iii) $[Be\ (CH_3)_2]_n$ has essentially the same structure as $(BeCl_2)_n$ but the bonding in methyl compound is best regarded as three-centred two-electron bonds covering C–Be–C atoms.

The coordination of Be in these chains is not exactly tetrahedral. For instance, the internal Cl–Be–Cl angles in $(BeCl_2)_n$ are 98.2°, which means the Be $(\mu_2\text{–}Cl)_2$ Be units are somewhat elongated in the direction of the chain axis. In contrast, the C–Be–C angles in $[Be(CH_3)_2]_n$ are 114°. These distrotions from the ideal tetrahedral angle for a four-coordinate Be atom are dependent on the nature of the bridging groups and are related to the presence or absence of lone pairs on the bridging atoms.

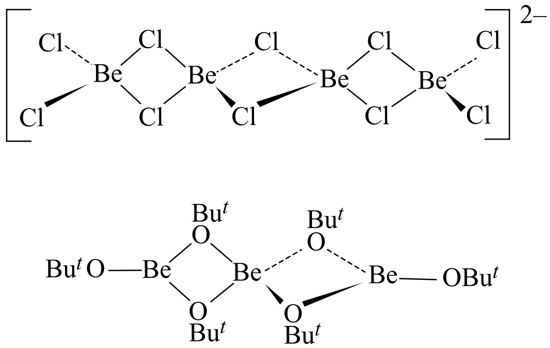


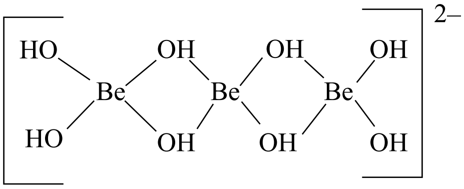
Fig 6.4 Polymeric structures of $(BeCl_2)_n$ and $[Be(OR)_2]_n$

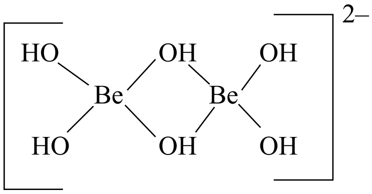
Alkoxides like $[Be(OR)_2]_n$ usually have associated structures with both μ_2-bridging and terminal OR groups. For example, $[Be(OCH_3)_2]_n$ is a high polymer, insoluble in hydrocarbon solvents. On the other hand, the tert-butoxy derivatives is less condensed, being only a trimer $[Be(O\text{-}t\text{-}Bu)_2]_3$ which is soluble in hydrocarbon solvents. Only when the alkoxide is bulky, monomers are obtained with two-coordinate Be.

(iv) Beryllium also achieves tetrahedral four coordination in compounds such as BeO and BeS, the structures which are often those of the corresponding Zn derivatives. Thus low-temperature BeO has the ZnO-wurtzite structure, whereas BeS adopts the ZnS-zinc blend structure. The structure of $Be(OH)_2$ is similar to that of $Zn(OH)_2$.

In water, beryllium salts are extensively hydrolyzed to give a series of hydroxocomplexes of unknown structure. They may be polymeric.

If an alkali is added to these solutions the polymers break down to give simple mononuclear tetrahedral beryllate ion $[Be(OH)_4]^{2-}$. Many beryllium salts contain the tetrahedral hydrated $[Be(H_2O)_4]^{2+}$ rather than simple Be^{2+}. Due to the formation of a complex, effective size of the ion increases in which charge is spreaded over a larger area making it stable. Thus stable ionic salts such as $[Be(H_2O)_4]SO_4$, $[Be(H_2O)_4](NO_3)_2$ and $[Be(H_2O)_4]Cl_2$ are known.





6.8 DIAGONAL RELATIONSHIP BETWEEN BERYLLIUN AND ALUMINIUM

Beryllium shows a strong similarity to aluminium. The two elements have the same electronegativity (Be 1.5; Al 1.5) and their charge/radius ratios are very similar, indicating, similar field strengths (Be 6.4, Al 6.0). Hence, beryllium resembles aluminium in several properties. Some of the similarities include.

(i) The standard oxidation potential of Be and Al are of the same order (Be = 1.97V, Al = 1.7V).

(ii) Beryllium atom is small and, therefore, has a high charge density. It has, therefore, a strong tendency to form covalent compounds. Aluminium, too, has a strong tendency to form covalent compounds.

(iii) Beryllium, like aluminium, is rendered passive on treatment with concentrated nitric acid.

(iv) Unlike alkaline earth metals but similar to aluminium, beryllium does not get readily attacked by dry air.

(v) Beryllium, like aluminium, reacts very slowly with dilute mineral acids due to the presence of an oxide layer.

$$Be + 2HCl \longrightarrow BeCl_2 + H_2$$

$$2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2$$

(vi) Both beryllium and aluminium react with caustic alkalis forming beryllates and aluminates, respectivley.

$$Be + 2OH^- \longrightarrow BeO_2^{2-} + H_2$$

$$2Al + 2OH^- + 2H_2O \longrightarrow 2AlO_2^- + 3H_2$$

(vii) Both beryllium and aluminium form carbides which on hydrolysis liberate methane.

$$Be_2C + 4H_2O \longrightarrow 2Be(OH)_2 + CH_4$$

$$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$$

(viii) Both form nitrides when heated in nitrogen which give ammonia by the reaction with water.

$$Be_3N_2 + 6H_2O \longrightarrow 3Be\,(OH)_2 + 2NH_3$$

$$AlN + 3H_2O \longrightarrow Al(OH)_3 + NH_3$$

(ix) Both form oxides which are amphoteric.

$$BeO + H_2SO_4 \longrightarrow BeSO_4 + H_2O$$

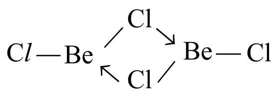
$$Al_2O_3 + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3H_2O$$

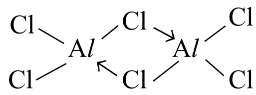
$$BeO + 2NaOH \longrightarrow Na_2BeO_2 + H_2O$$

$$Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + H_2O$$

(x) Both form halides by the direct reaction of metal and halogen which contain bridge bonds.

$$Be + Cl_2 \longrightarrow BeCl_2$$
$$2Al + 3Cl_2 \longrightarrow 2AlCl_3$$





Salts of both metals give hydrated ions in aqueous solution $[Be(H_2O)_4]^{2+}$ and $[Al(H_2O)_6]^{3+}$.

(xi) Ions of both metals (Be^{2+} and Al^{3+}) have strong tendency to form complexes. For example, Be^{2+} form tetrahedral $[BeF_4]^{2-}$ and $[Be(C_2O_4)_2]^{2-}$ while Al^{3+} forms octahedral $(AlF_6)]^{3-}$ and $[Al(C_2O_4)_3]^{3-}$.

(xii) Both Be and Al do not impart any colouration to the Bunsen flame.

6.9 IMPORTANT COMPOUNDS OF MAGNESIUM

6.9.1 Magnesium Oxide

Preparation: Magnesium oxide is mainly prepared by the calcination of magnasite. This is called **calcined magnesia**.

$$MgCO_3 \longrightarrow MgO + CO_2$$

It can also be prepared by heating magnesium hydroxide, nitrate or sulphate.

$$Mg(OH)_2 \longrightarrow MgO + H_2O$$

$$2Mg(NO_3)_2 \longrightarrow 2MgO + 4NO_2 + O_2$$

$$2MgSO_4 \longrightarrow 2MgO + 2SO_2 + O_2$$

Properties:

(i) It is a light weight infusible powder. Melting point is 3123 K.

(ii) It is sparingly soluble in water with which it combines slowly to form the hydroxide.

$$MgO + H_2O \longrightarrow Mg(OH)_2$$

(iii) It is basic in nature. It is easily soluble in acids but is unaffected by alkalis and ammonia.

$$MgO + 2HCl \longrightarrow MgCl_2 + H_2O$$

(iv) MgO dissolves in aqueous solution of magnesium chloride or bromide forming basic salts such as

$$MgBr_2 \cdot 9Mg(OH)_2 \cdot 5H_2O.$$

(v) MgO is reduced by carbon at the temperature of an electric arc furnace, giving magnesium carbide.

$$MgO + 3C \longrightarrow MgC_2 + CO$$

Uses: It is used

(i) in medicine as an antacid

(ii) When moistened with a saturated solution of magnesium chloride it sets into a hard mass called **Sorel's cement** whose composition is $MgCl_2 \cdot 5MgO \cdot xH_2O$. It is used in plaster casting, repairing porcelain ware and in filling teeth.

(iii) It is used as basic refractory material in metallurgical furnaces.

(iv) When mixed with asbestos it is used as an insulator for lagging steam pipes and boilers.

(v) Flocculent form of MgO is used as a rubber filler.

(vi) It is used as adsorbent in the manufacture of dynamite and in the vulcanization of rubber.

6.9.2 Magnesium Hydroxide

It occurs in the nature as mineral **brucite**

Preparation: It is obtained as a white precipitate when sodium hydroxide or ammonium hydroxide is added to a solution of soluble magnesium salt.

$$MgSO_4 + 2NaOH \longrightarrow Mg(OH)_2 + Na_2SO_4$$

On a large scale, it is manufactured by treating magnesium chloride of the potash industry with slaked lime.

$$MgCl_2 + Ca(OH)_2 \longrightarrow Mg(OH)_2 + CaCl_2$$

Properties

(i) It is a white powder.

(ii) It is very slightly soluble in water (19mg/lit at 291K) and is a strong electrolyte.

(iii) It dissolves in acids, but do not react with bases.

$$Mg(OH)_2 + 2HCl \longrightarrow MgCl_2 + 2H_2O$$

(iv) It decomposes at about 573 K forming the oxide. Magnesium hydroxide is soluble in ammonium chloride solution and is therefore not precipitated in Group III of qualitative analysis.

Uses

(i) In medicine in the form of suspension known as **milk of magnesia** that is used as an antacid.

(ii) In the extraction of sugar from molasses in sugar industry.

6.9.3 Magnesium Chloride $MgCl_2 \cdot 6H_2O$

It is present in sea water and the mineral carnalite $KCl.MgCl_2 \cdot 6H_2O$.

Preparation

(i) In the **laboratory** it is prepared by dissolving the metal, its oxide, hydroxide or carbonates in dilute hydrochloric acid and evaporating the solution to crystallize out as $MgCl_2 \cdot 6H_2O$.

(ii) On a **large scale,** it is manufactured by the fractional crystallization of fused carnalite. The mineral is fused and cooled to 449 K when almost whole of its potassium chloride separates out as a solid leaving $MgCl_2 \cdot 6H_2O$ in the fused state. When it is further cooled, remaining potassium chloride separates out leaving the hydrate of magnesium chloride $MgCl_2 \cdot 6H_2O$ in the fused state which crystallizes on cooling.

Anhydrous magnesium chloride cannot be prepared by heating the hydrated salt because the latter undergoes hydrolysis to form hydroxy chloride or hydroxide or oxide depending on the temperature.

$$MgCl_2 \cdot 6H_2O \xrightarrow{\Delta} MgCl_2 \cdot 2H_2O + 4H_2O$$

$$MgCl_2 \cdot 2H_2O \xrightarrow{\Delta} Mg(OH)Cl + HCl + H_2O$$

$$MgCl_2 \cdot 2H_2O \xrightarrow{\Delta} Mg(OH)_2 + 2HCl$$

$$Mg(OH)_2 \xrightarrow{\Delta} MgO + H_2O$$

Anhydrous magnesium chloride is prepared by the following methods.

(a) It is prepared by heating $MgCl_2 \cdot 6H_2O$ in vacuum or in a current of dry hydrogen chloride.

(b) It can be prepared by heating magnesium oxide with carbon in a current of chlorine gas.

$$MgO + C + Cl_2 \longrightarrow MgCl_2 + CO$$

(c) It can also be prepared by passing a mixture of CO and Cl_2 over magnesium oxide at 1023 K.

Properties: It is a colourless, deliquescent solid, soluble in water. When heated to about 473 K, it undergoes hydrolysis to form magnesium hydroxy chloride Mg(OH)Cl and hydrochloric acid.

$$MgCl_2 + H_2O \longrightarrow Mg(OH)Cl + HCl$$

When saturated solution of neutral magnesium chloride is treated with magnesium oxide, the resulting paste sets into a hard marble like mass. This mass is a mixture of the following compounds and is known as **Sorel's cement.**

$$\underset{\text{I}}{MgCl_2 \cdot 2Mg(OH)_2\ 4H_2O} \text{ and } \underset{\text{II}}{MgCl_2 \cdot 9Mg(OH)_2 \cdot 5H_2O}.$$

The second compound is stable and the first compound changes within a week to $MgCl_2 \cdot 3Mg(OH)_2 \cdot 8H_2O$.

Uses: In making Sorel's cement which is used as a dental filling, as a finish for plaster and for cementing glass with metals. It is used in lubricating cotton thread in spinning.

6.9.4 Magnesium Sulphate, $MgSO_4 \cdot 7H_2O$ Epsom Salt

It occurs in nature in the minerals Kieserite, $MgSO_4 \cdot H_2O$ in Stassfurt deposits and as epsomite, $MgSO_4 \cdot 7H_2O$ in certain gypsum deposits. It also occurs in certain mineral springs like Epsom bath and Seidlitz.

Preparation

(i) It is prepared by dissolving magnesium oxide or hydroxide or carbonate in dilute sulphuric acid and evaporating the solution gives heptahydrate $MgSO_4 \cdot 7H_2O$

$$MgO + H_2SO_4 \longrightarrow MgSO_4 + H_2O$$

$$Mg(OH)_2 + H_2SO_4 \longrightarrow MgSO_4 + 2H_2O$$

$$MgCO_3 + H_2SO_4 \longrightarrow MgSO_4 + H_2O + CO_2$$

(ii) It is also prepared by dissolving dolamite in boiling dilute sulphuric acid. Insoluble calcium sulphate is removed by filtration and the filtrate on evaporation and cooling yields colourless crystals of Epsom salt.

$$MgCO_3 \cdot CaCO_3 + 2H_2SO_4 \longrightarrow MgSO_4 + CaSO_4 + 2H_2O + 2CO_2$$

(iii) Anhydrous magnesium sulphate is obtained by heating any hydrate of magnesium sulphate to 478 K.

(iv) The Epsom salt is manufactured by dissolving the mineral Kieserite $MgSO_4 \cdot H_2O$ in boiling water and the resulting solution is crystallized when colourless crystals of $MgSO_4 \cdot 7H_2O$ separate out.

Properties: It is a colourless, efflorescent crystalline solid having a bitter taste. There are two crystal forms of the heptahydrate $MgSO_4 \cdot 7H_2O$. One is isomorphous with the rhombic crystals of zinc sulphate heptahydrate and the other is isomorphous with the monoclinic cyrstals of ferrous sulphate heptahydrate. Out of the seven water molecules, six are in coordination with Mg^{2+} ion and the seventh water molecule is in hydrogen bond with sulphate ion.

When heated to 423 K, it changes to monohydrate $MgSO_4 \cdot H_2O$ which at 473 K becomes $MgSO_4$ and on further heating it decomposes, evolving sulphur dioxide and oxygen.

$$MgSO_4 \cdot 7H_2O \xrightarrow[-6H_2O]{423K} MgSO_4 \cdot H_2O \xrightarrow[-H_2O]{473K} MgSO_4$$

$$2MgSO_4 \xrightarrow{\Delta} 2MgO + 2SO_2 + O_2$$

Uses: Epsom salt is used extensively

(i) as a purgative or laxative in medicine

(ii) in dyeing and tanning processes

(iii) in the manufacture of fire proofing fabrics

(iv) Platinized magnesium sulphate is used as a catalyst in the manufacture of sulphuric acid by Grillo's process.

6.9.5 Magnesium Carbonate, $MgCO_3$ (Magnesite)

Anhydrous magnesium carbonate is found as the mineral magnesite. In India, magnesite occurs in the chalk hills in the Salem district.

Preparation

(i) It is obtained as a white precipitate when sodium bicarbonate solution is added to a hot solution of magnesium salt.

$$MgSO_4 + 2NaHCO_3 \longrightarrow MgCO_3 + Na_2SO_4 + H_2O + CO_2$$

If sodium carbonate is used, basic magnesium carbonate is obtained.

$$2MgSO_4 + 2Na_2CO_3 + H_2O \longrightarrow MgCO_3 \cdot Mg(OH)_2 + CO_2 + 2Na_2SO_4$$

If the precipitation is carried out in the cold with dilute solutions, then **magnesia alba levis** having the approximate formula $MgCO_3 \cdot Mg(OH)_2 \cdot \frac{7}{2} H_2O$ is obtained. Hot concentrated solutions yield **magnesia alba ponderosa** having the approximate formula $MgCO_3 \cdot Mg(OH)_2 \cdot 3H_2O$. Magnesia alba is used in medicine as an antacid and laxative. It is also used in tooth powders and pastes and in silver polishes.

A clear solution of magnesium bicarbonate is obtained by passing carbon dioxide into a suspension of magnesium oxide. If the solution of the bicarbonate is crystallized at 325 K, heated with magnesium oxide crystals of magnesium carbonate trihydrate separate.

$$MgO + H_2O + 2CO_2 \longrightarrow Mg(HCO_3)_2$$

$$Mg(HCO_3)_2 + MgO \longrightarrow 2MgCO_3 + H_2O$$

The trihydrate $MgCO_3 \cdot 3H_2O$ has been found to be more reactive than the anhydrous salt and is used in the Engel-Precht process.

Properties: Magnesium carbonate is a white solid, very sparingly soluble in water. It, however, dissolves in the presence of excess of carbon dioxide forming magnesium bicarbonate, known as fluid magnesia. Magnesium carbonate is much more easily decomposed on heating than $CaCO_3$.

$$MgCO_3 \longrightarrow MgO + CO_2$$

Uses: It is used extensively

(i) in glass manufacture, inks, ceramics, fertilizers, insulation, industrial chemicals and rubber products

(ii) as a refractory material and for certain medicinal preparations

(iii) as a filler for paper, rubber and pigments

6.9.6 Magnesium Ammonium Phosphate $Mg(NH_4)PO_4 \cdot 6H_2O$

It is one of the least soluble compounds of magnesium. It is obtained as a white flocculent precipitate when a solution of disodium hydrogen phosphate Na_2HPO_4 is added to a solution of magnesium salt containing some ammonia and ammonium chloride.

$$MgCl_2 + Na_2HPO_4 + NH_4Cl + 6H_2O \longrightarrow Mg(NH_4)PO_4 \cdot 6H_2O + 2NaCl + HCl$$

The formation of white precipitate of magnesium ammonium phosphate is used in the detection and estimation of magnesium ions in a solution.

6.10 IMPORTANT COMPOUNDS OF CALCIUM

6.10.1 Calcium Oxide, CaO

Preparation: It is also known as **quick-lime** or **burnt-lime**. It is prepared on a large scale by burning lime stone in lime kilns.

$$CaCO_3 \xrightarrow{1273K} CaO + CO_2$$

Properties: It is a white solid. When CaO is heated in oxyhydrogen flame a brilliant white light known as **lime light** is emitted. It absorbs moisture from air.

$$CaO + H_2O \longrightarrow Ca(OH)_2$$

It readily absorbs acidic oxides from air.

$$CaO + CO_2 \longrightarrow CaCO_3$$
$$CaO + SO_2 \longrightarrow CaSO_3$$

CaO combines with solid acidic oxides at high temperatures.

$$CaO + SiO_2 \longrightarrow CaSiO_3$$
$$6CaO + P_4O_{10} \longrightarrow 2Ca_3(PO_4)_2$$

Uses

(i) It is used as a drying agent.

(ii) It is used in the manufacture of bleaching powder.

(iii) It is used in the manufacture of calcium carbide, cement, glass, lime mortar, etc.

(iv) It is used in the purification of sugar.

(v) It is used as basic refractory in the lining of furnaces.

(vi) It is used in water softening.

6.10.2 Calcium Hydroxide $Ca(OH)_2$

Preparation: It is obtained by the addition of water to quick lime. When water is added to quick lime, a hissing sound is heard and lumps of CaO crack and disintegrate into powder. This process is called **slaking of lime** and the product $Ca(OH)_2$ formed is called **slaked lime.**

$$CaO + H_2O \longrightarrow Ca(OH)_2 + Heat$$

Properties: Calcium hydroxide is a white amorphous powder. It is sparingly soluble in water. The dissolution of slaked lime in water is exothermic. Its solubility in water decreases with increase in temperature. The suspension of lime in water is called **milk of lime,** whereas the filtered and clear solution is known as **lime water.** Chemically, both milk of lime and lime water are calcium hydroxide.

When CO_2 is passed through lime water, first it turns milky which disappears by passing excess of CO_2 due to the formation of soluble bicarbonate.

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 \downarrow + H_2O$$
$$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$$

SO_2 gas also gives the similar reactions as CO_2 with lime water.

Milk of lime when treated with Cl_2 gives hypochlorite in cold condition and chlorate in hot condition along with chloride.

$$2Ca(OH)_2 + 2Cl_2 \xrightarrow{Cold} CaCl_2 + Ca(OCl)_2 + 2H_2O$$
$$6Ca(OH)_2 + 6Cl_2 \xrightarrow{Hot} 5CaCl_2 + Ca(ClO_3)_2 + 6H_2O$$

When chlorine gas is passed over dry slaked lime bleaching powder, a mixture of different products is formed.

$$3Ca(OH)_2 + 2Cl_2 \longrightarrow \underbrace{Ca(OCl)_2 + CaCl_2 + Ca(OH)_2 \cdot H_2O}_{\text{Bleaching powder}} + H_2O$$

Milk of lime liberates ammonia gas from amonium salts.

$$2NH_4Cl + Ca(OH)_2 \longrightarrow CaCl_2 + 2H_2O + 2NH_3 \uparrow$$

Uses: It is used

(i) for absorbing acid gases.

(ii) in the manufacture of bleaching powder and caustic soda

(iii) in the production of lime mortar for construction of buildings

(iv) in glass making, tanning industry and for purification of sugar

(v) for the preparation of NH_3 from NH_4Cl in Solvay's process

(vi) as lime water in laboratories

6.10.3 Calcium Chloride

Preparation: Calcium chloride forms hexahydrate $CaCl_2 \cdot 6H_2O$. It is obtained as a by-product in Solvay's process for the manufacture of sodium carbonate.

In the laboratory it is prepared by dissolving calcium carbonate in dilute hydrochloric acid. The solution on evaporating and cooling yields hexagonal crystals of $CaCl_2 \cdot 6H_2O$.

On heating $CaCl_2 \cdot 6H_2O$ to about 473 K it forms $CaCl_2 \cdot 2H_2O$ which finally becomes anhydrous $CaCl_2$ on further heating.

$$CaCl_2 \cdot 6H_2O \xrightarrow[-4H_2O]{473K} CaCl_2 \cdot 2H_2O \xrightarrow{\text{Fusion}} CaCl_2$$

The anhydrous calcium chloride is also called as fused calcium chloride.

Properties: Calcium chloride is extremely soluble in water. It dissolves in alcohol. It forms addition compounds with ammonia and methyl alcohol like $CaCl_2 \cdot 8NH_3$ and $CaCl_2 \cdot 4NH_3$.

Uses: Anhydrous calcium chloride is hygroscopic and deliquescent. Therefore, it is used as a drying agent in a dessicator. But it cannot be used to dry ethyl alcohol or ammonia as it forms addition compounds with them.

When mixed with ice the freezing point of water decreases to 255 K. So, it is used to melt ice and snow on roads. It is also used in making freezing mixtures and in refrigeration because it can give as low a temperature 218 K. Highly concentrated solutions of calcium chloride are used in liquid baths for heating purposes. For instance, a somewhat saturated solution of calcium chloride boils at 453 K.

6.10.4 Calcium Carbonate $CaCO_3$

Calcium carbonate is widely distributed in nature in huge deposits as limestone, chalk, marble and iceland spar.

In the laboratory it is prepared either by passing carbon dioxide through lime water or by adding calculated quantity of sodium carbonate to a solution of calcium chloride.

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$$

$$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaCl$$

It is a white solid, insoluble in water in all forms, but dissolves in the presence of carbon dioxide due to the formation of calcium bicarbonate.

$$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$$

Uses: It is used

(i) in the manufacture of cement, lime, washing soda and glass

(ii) as a building material

(iii) as a chalk in tooth powders and paste

(iv) as a flux in extraction of metals like iron when mixed with magnesium carbonate

6.10.5 Calcium Sulphate $CaSO_4 \cdot 2H_2O$

In nature it occurs as anhydrite $CaSO_4$ and gypsum $CaSO_4 \cdot 2H_2O$. Naturally occurring pure fine powder of calcium sulphate is known as **alabaster.**

Preparation: Calcium sulphate may be prepared by adding sodium sulphate to a soluble calcium salt. A precipitate of calcium sulphate is formed which is filtered and dried.

$$Ca^{2+} + SO_4^{2-} \longrightarrow CaSO_4$$

Calcium sulphate is sparingly soluble in water. Its solubility in water decreases with increase in temperature. Its solubility in water increases in the presence of ammonium sulphate due to the formation of the double salt $CaSO_4 \cdot (NH_4)_2SO_4 \cdot H_2O$.

When gypsum is heated, interesting observations are recorded. Initial heating alters its crystal structure. On further heating to 390 K it forms the hemihydrate known as **"plaster of paris"**. Above 473 K anhydrous calcium sulphate is formed. This is known as **dead burnt plaster** because it has no tendency to set into hard mass.

When heated above 1473 K it decomposes.

$$2CaSO_4 \longrightarrow 2\ CaO + 2SO_2 + O_2$$

When heated with carbon, the decomposition of calcium sulphate gets completed.

$$2CaSO_4 + C \longrightarrow 2CaO + 2SO_2 + CO_2$$

When calcium sulphate is heated with carbon at 1473 K, calcium oxide and sulphur dioxide are obtained. This reaction is reversible and shifts towards the right in the presence of silica and alumina. The function of these is to react with lime to form silicates and aluminates.

$$3CaSO_4 + CaS \longrightarrow 4CaO + 4SO_2$$

$$CaO + SiO_2 \longrightarrow CaSiO_3$$

$$CaO + Al_2O_3 \longrightarrow Ca(AlO_2)_2$$

The product thus obtained has the necessary constituents of cement. It has also the properties of cement. Thus, if a mixture of calcium sulphate and calcium sulphide is heated in a rotary kiln in the presence of silica and alumina, cement clinkers get collected at the bottom, whereas sulphur dioxide is obtained as a by-product which is used in the manufacture of sulphuric acid.

Uses: It is used

(i) in the manufacture of plaster of paris

(ii) in the manufacture of cement

(iii) in agriculture and for impregnating filter papers

6.10.6 Plaster of Paris $CaSO_4 \cdot \frac{1}{2} H_2O$

Plaster of paris is prepared by heating gypsum at 390 K. If the temperature of heating is above 390 K, setting properties of plaster of paris is partly destroyed.

$$CaSO_4\,2H_2O \longrightarrow CaSO_4 \cdot \frac{1}{2}\,H_2O + \frac{3}{2}\,H_2O$$

Plaster of paris, when mixed with water, forms, a plastic mass that sets into a hard solid in 5 to 15 minutes; hence, it is given the name plaster. As the setting is accompanied by a slight expansion in volume, it can take a sharp impression into which it is put. Therefore, it is used in making casts for statues, toys, etc. It is also used in surgery for plastering fractured parts of the body for preparing black board crayons.

The setting process involves two stages. In the first stage the plaster of paris converts into orthorhombic form gypsum (setting stage) and in the second stage orthorhombic gypsum changes into monoclinic form (hardening stage).

$$\underset{\text{plaster of paris}}{CaSO \cdot \frac{1}{2} H_2O} + \frac{3}{2} H_2O \xrightarrow{\text{Setting stage}} \underset{\text{orthorhombic gypsum}}{CaSO_4 \cdot 2H_2O}$$

$$\xrightarrow{\text{Hardening Stage}} \underset{\text{monoclinic gypsum}}{CaSO_4 \cdot 2H_2O}$$

The setting of plaster of paris is catalyzed by sodium chloride and is retarded by borax or alum. Addition of alum to plaster of paris gives hard setting. This mixture is known as **Keen's cement.**

Uses: It is used

(i) in making moulds and statues
(ii) plastering the fractured bones
(iii) in dentistry to fill the gaps
(iv) making black board crayons

6.10.7 Mortar

It is also known as lime mortar. It is an intimate mixture of 1 part of slaked lime, 3 parts of sand and water made into paste. This is used to bind the bricks firmly.

Setting of mortar is said to involve the following steps.

(i) Mortar loses water on account of evaporation.
(ii) Carbon dioxide is absorbed from the air converting into calcium carbonate which acts as a binding material.

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$$

(iii) Slaked lime reacts with silica forming calcium silicate which gives hardness.

$$Ca(OH)_2 + SiO_2 \longrightarrow CaSiO_3 + H_2O$$

(iv) The sand makes the mass porous so that the atmospheric CO_2 may freely penetrate inside the mass, thus facilitating the setting of mortar. The sand also prevents excessive shrinkage which might result in cracks.

Mortar becomes hard with time due to several chemical reactions taking place at the centre of the mass.

Mortar mixed with cement is called **cement mortar**. This is more harder than mortar and more water proof.

Mortar obtained by burning lime stone containing more than 10 per cent alumino silicates is called hydraulic mortar. This can set like cement on addition of water. This is used for bleaching purposes and as an antiseptic.

6.10.8 Cement

The cement was first introduced in 1824 by Joseph Aspidin, a mason of Leeds, England. He found that when a strongly heated mixture of lime stone and clay was mixed with water and allowed to stand, it hardened to a stone-like mass resembling portland rock which is a famous building stone of England. The name portland cement is since that time given to the product obtained by combining a material high in lime such as lime stone, with one which contains silica, iron oxide and alumina as chief constituents such as clay.

Composition of cement: The average composition is
CaO, 50–60%; SiO_2, 20–25%; Al_2O_3, 5–10%
MgO, 2–3%; Fe_2O_3, 1–2% and SO_3, 1–2%

The ratio of silica and alumina should be between 4 and 2.5 and the ratio of calcium oxide and the mixture of silica, alumina and ferric oxide should be close to 2.0. If lime is excess the cement cracks during setting. If lime is less than required by the above ratio, the cement is weak in strength.

The **raw materials** used for the manufacture of cement are lime stone and clay. These are powdered and made into slurry. It is introduced into the upper part of the rotary kiln which consists of an inclined steel rotating cylinder 150–200 feet long and 10 feet in diameter lined with bricks. The water evaporates at the upper end of the kiln. The dry material moves down and meets a coal gas flame which is at very high temperature. In the middle of the kiln the temperature is about 1000°C. Here the lime stone decomposes to give calcium oxide. At the lower end, the temperature rises to 1600°C. Here chemical reactions takes place between lime, alumina, ferric oxide and silica resulting in the formation of calcium aluminates and silicates.

The product is known as **cement clinker**. After cooling it is mixed with 2–3 per cent gypsum and ground to a fine

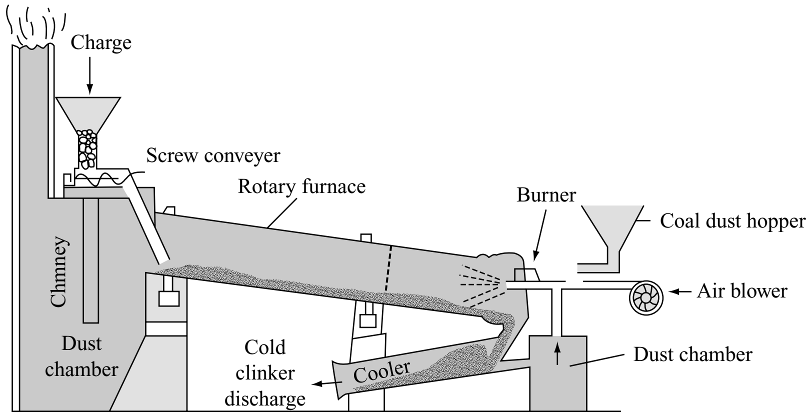


Fig 6.5 Manufacture of cement

powder. The function of gypsum is to slow down the setting of cement so that it gets sufficiently hardened. The main compounds present in the cement are

Dicalcium silicate $2CaO \cdot SiO_2$

Tricalcium silicate $3CaO \cdot SiO_2$

Tricalcium aluminate $3CaO \cdot Al_2O_3$

Similarly, magnesium oxide is changed to magnesium silicate, etc.

Setting of Cement: Cement sets into hard mass after being mixed with water. Though the exact mechanism of the setting of cement is not known, it is thought to be due to the hydration and hydrolysis of the different compounds present in cement primarily, the reactions involved are the hydration of calcium aluminates and calcium silicates which change into their colloidal gels at the same time, some calcium hydroxide and aluminium hydroxide are formed as precipitates due to hydrolysis. The gels soon begin to harden as a result of gradual dehydration. Finally, it becomes a hard silica gel containing interlocking or interspersed crystals of calcium hydroxide and other products of hydrolysis. These interlocking crystals impart extra strength to the hardened mass. This hardened mass contains $Ca(OH)_2$, rhombic hydrated dicalcium silicate, needle shaped hydrated tri-calcium silicate and a complex calcium sulphoaluminate.

The setting of cement is an exothermic reaction (due to hydration of compounds). Hence, cement structures must be cooled during setting; otherwise, uneven expansion of the mass due to heat of hydration is liable to produce cracks.

A mixture of cement with sand and crushed stones is called **concrete**. It is used in foundations and in the construction of bridges, dams and buildings.

6.11 BIOLOGICAL IMPORTANCE OF MAGNESIUM AND CALCIUM

Magnesium is an important constituent of chlorophyll. This absorbs the energy from light required for the process of photosynthesis. All enzymes that utilize ATP in phosphate transfer require magnesium as the cofactor. Magnesium ions are concentrated more in intracellular than in extracellular fluids in animal bodies. The presence of Mg^{2+} ions is a necessary requirement for the activation of phosphate transfer enzymes which participate in the energy releasing biochemical process occurring in animal body.

Furthers Mg^{2+} and Ca^{2+} ions are also responsible for the transmission of electrical impulses along the nerve fibre and the contraction of muscles.

Calcium ions are essential for the formation of bones and teeth. About 99 per cent of body calcium is present in bones and teeth. The enamel on teeth is also a double salt of calcium $3Ca_3(PO_4)_2 \cdot CaF_2$. It also plays important roles in maintaining rhythm of heart, clotting of blood, neuromuscular function, interneuronal transmission, cell membrane integrity, etc. The calcium concentration in plasma is regulated at about 100 mg L^{-1}. It is maintained by two hormones, calcitonin and parathyroid hormone. The substance present in bones is continuously solubilized and redeposited to the extent of 400 mg per day in man. All this calcium passes through the plasma.

KEY POINTS

- Be, Mg, Ca, Sr, Ba and Ra belongs to Group-II A and s-block of the periodic table.
- These elements are known as **alkaline earth metals** since these are abundant in earth and their oxides lime (CaO), strontia (SrO) and baryta (BaO) are alkaline. Hence, these elements are called alkaline earth metals.
- Radium is a radioactive metal.

General Chracteristics

- The variation in physical properties are not as regular as for the alkali metals because the elements of this group do not crystallize with the same type of metallic lattice. Be and Mg have hcp structure, and Ca and Sr have ccp structure.
- At high temperatures Ca and Sr, and at room temperature Ba have bcc structures (similar to Group-1A) instead of the expected hcp structure. This is probably at high temperature and the paired s-electron is the excited to a 'd' orbital instead of p-orbital and hence there is only one s- or p-electron per atom participating in metallic bonding.
- They are all white lustrous silvery white metals, malleable and ductile but less than alkali metals. This is because in these elements metallic bonding is stronger than in Group-IA elements since two electrons per atom participate in metallic bonding compared to one electron in alkali metals.
- Alkaline earth metals are harder than alkali metals and hardness do not show regular tendency with increase in atomic number.
- **Atomic radii** of these elements is less than those of corresponding alkali metals due to increased effective nuclear charge and hence these elements are harder, have higher m.pts and higher densities than alkali metals. Atomic radii increases down the group.
- Since the alkaline earth metal atoms possess smaller size and more nuclear charge they have more ionization energies than the corresponding alkali metals but decreases down the group due to increase in atomic size.
- The second ionization energies are almost double to the first ionization energies of alkaline earth metals.
- Since the hydration energies of M^{2+} ions are more than the sum of I_1 and I_2 of alkaline earth metals, their M^{2+} ions are easily formed. Hence, they do not exhibit +1 oxidation state.
- The hydration energies decreases from Be^{2+} to Ba^{2+} with increase in size.
- The M^{2+} ions of alkaline earth metals are colourless and diamagnetic because all the electrons in them are paired.
- The electropositive character of alkaline earth metals is less than the corresponding alkali metals from Be to Ba.
- Because of more attraction towards water molecules by alkaline earth metal ions their solid compounds are heavily hydrated than those of alkali metals, e.g., $MgCl_2 \cdot 6H_2O$, $CaCl_2 \cdot 6H_2O$, etc.
- Ionic mobilities decreases from Be^{2+} to Ba^{2+}.
- Standard oxidation potentials of these elements increases from Be to Ba and hence the reduction power increases from Be to Ba.
- Except Be and Mg the other alkaline earth metals give flame colouration as Ca-brick red; Sr - crimson red and Ba-apple green.
- Be and Mg atoms are smaller and their electrons being strongly bound to the nucleus are not excited to higher-energy levels. Therefore, they do not give the flame test.
- Alkaline earth metals are less reactive than the corresponding alkali metals due to small size, high charge and more ionization energies.
- The reactivity increases from Be to Ba as their ionization energies decreases with increase in atomic number.
- Since the alkaline earth metals are more reactive they do not occur in nature in native state but occur in the form of oxycompounds like carbonates, silicates, etc.

Reactivity of Alkaline Earth Metals

- The reactivity of alkaline earth metals increases down the group. Be mainly forms covalent compounds but other alkaline earth metals form mainly ionic componds.
- When burned in air these elements react with both oxygen and nitrogen forming oxides and nitrides.
- Except Be all the other Group-II A elements react with water forming hydroxides with the liberation of H_2. Mg reacts only with boiling water while other elements react even in cold condition but slowly.
- The inertness of Be and Mg towards water is ascribed to the formation of a protective thin layer of hydroxide on the surface of the metal.
- These elements readily displace hydrogen from protonic acids forming corresponding salts.
- Conc. HNO_3 render Be passive due to the formation of thin oxide film on the surface of the metal.
- Only Be react with alkalis forming beryllates with liberation of H_2, but other Group-II A elements do not react with bases.
- Except Be all the other elements directly combine with H_2 to form hydrides, MH_2. MgH_2 has some covalent character but the other hydrides are ionic.

- When heated with carbon these elements form ionic carbides. Be form methanide Be_2C, while other elements form acetylides.
- With nitrogen these elements form nitrides at high temperatures.
- On burning in nitrogen these elements form nitrides, M_3N_2. The ease of formation of nitrides decreases down the group. These nitrides react with water liberating ammonia.
- With phosphorous they form metal phosphides which on hydrolysis liberate phosphine gas.
- All the Group-II A elements except radium form oxides with oxygen. Be and Mg form only monoxides, whereas Ca, Sr and Ba also form peroxides in excess oxygen.
- All the Group-II A elements react with S, Se and Te forming sulphides, selenides and tellurides.
- All the Group-II A elements react with halogen forming halides of the type MX_2.
- These elements have less tendency to form complexes but have more tendency to form complexes than alkali metals. In these complexes, Be exhibit a maximum coordination number 4. So, Be salts cannot have more than four water molecules of crystallization. Mg can exhibit a coordination number 6 by utilizing the d-orbitals in its valence shell.
- Like alkali metals, Ca, Sr and Ba dissolve in liquid ammonia to give blue-black solutions.
- Unlike the solutions alkali metals in liquid ammonia, evaporation of the solutions of Group-II A metals in liquid ammonia gives hexaammoniates which gradually decompose to the corresponding amides. The tendency to form ammoniates decreases down the group.

Principle of Extraction of Alkaline Earth Metals

- Similar to alkali metals, alkaline earth metals cannot be prepared by the electrolytic reduction of their aqueous salt solution.
- Alkaline earth metals are prepared by electrolysis of their fused anhydrous salts containing alkali metal salts to lower the m.pt and to increase the condutivity of the electrolysis mixtures.

Reactivity of Alkaline Earth Metals

	Reaction	Remark
1.	**Action of air** **Exposed to air**	When alkaline earth metals are exposed to air, they tarnish due to the formation of basic metal carbonates
	$M + O_2 \longrightarrow MO$ $MO + H_2O \longrightarrow M(OH)_2$ $M(OH)_2 + CO_2 \longrightarrow MCO_3$	
	Burning in air	
	$M + O_2 \longrightarrow MO$ $M + N_2 \longrightarrow M_3N_2$	When burned in air, they react with both nitrogen and oxygen in air
2.	$M + H_2O \longrightarrow M(OH)_2 + H_2$	Be do not react with steam, Mg react with hot water and others react with cold water
3.	$M + 2HCl \longrightarrow MCl_2 + H_2$	All metals react with acids. Be becomes passive with conc. HNO_3.
4.	$Be + NaOH \longrightarrow Na_2\,[Be(OH)_4] + H_2$	Only Be react with bases because it is amphoteric
5.	$M + H_2 \longrightarrow MH_2$	Be and Mg forms polymeric covalent hydrides. Ca, Sr, and Ba form ionic hydrides
6.	$Be + C \longrightarrow Be_2C$ $M + C \longrightarrow MC_2$	Be forms methanide and others form acetylide
7.	$M + N_2 \longrightarrow M_3N_2$ $M + P_4 \longrightarrow M_3P_2$	All metals form nitrides and phosphides
8.	$M + O_2 \longrightarrow MO$	Normal oxides are formed by all metals
	$M + O_2 \longrightarrow MO_2$	Ca, Sr and Ba form peroxides with excess oxygen
	$M + X \longrightarrow MX$	X = S, Se, Te
9.	$M + X_2 \longrightarrow MX_2$	X = F, C*l*, Br, I
10.	$2M + NH_3 \longrightarrow M(NH_2)_2 + H_2$	All metals form amides at high temperature. Ca, Sr and Ba dissolve in liquid NH_3 forming blue-coloured solution.

COMPOUNDS OF ALKALI EARTH METALS

Hydrides

- The reactivity of alkaline earth metals towards hydrogen increases down the group.
- Except BeH_2, hydrides of other Group-II A elements can be prepared by heating the metal with hydrogen.
- BeH_2 and MgH_2 can be prepared by the action of $LiAlH_4$ on their halides.
- BeH_2 is covalent while MgH_2 is partially ionic. CaH_2 SrH_2 and BaH_2 are ionic. All these hydrides react with water liberating H_2.
- BeH_2 and MgH_2 are polymeric. $(BeH_2)_n$ contains 3c-2e hydrogen bridge bonds.

Nitrides

- All the elements of Group-II A burn in nitrogen forming nitrides M_3N_2. The ease of formation decreases down the group.
- Barium amide on heating decomposes to give Ba_3N_2.
- All these nitrides are stable upto 1000°C. Addition of water causes hydrolysis liberating ammonia.
- These nitrides on heating with hydrogen gives metal hydride and ammonia and decomposed by CO or CO_2 to metal oxide and nitrogen.

Oxides

- Be forms monoxide only which is predominantly covalent. The other Group-II elements form monoxides which are ionic and have NaCl structure.
- Except BeO the oxides of other Group-II elements are usually obtained by thermal decomposition of their carbonates. BeO is obtained by igniting the metal or its compounds in oxygen.
- The oxides of Group-II elements are extremely stable and their free energies of formation are highly negative.
- BeO has wurtzite structure with 4:4 coordination number. The radius-ratio values of SrO and BaO predict a coordination number 8 but they are found to have 6 coordinates, because this has the most favourable lattice energy.
- BeO is insoluble in water. MgO is not very reactive and CaO, SrO and BaO react exothermally with water forming hydroxides.
- BeO is amphoteric and react with acids and bases. The amphoteric character is due to strong polarizing power of small Be^{2+} ion with large charge to radius ratio.
- The oxides of Group-II A elements are useful refractory materials because of their high m.pts, low vapour pressure, high thermal conductivity, chemical inertness and electrical insulating property.
- MgO and CaO are used as basic fluxes in metallurgy. CaO is used in the production of calcium carbide.
- **Peroxides:** The tendency to form peroxides and their stability increases as the metal ion becomes bigger. Be do not form peroxide. These peroxides react with acids liberating H_2O_2.

Hydroxides

- Hydroxides are formed by the reaction of the oxides of alkaline earth metal with water.
- Solubility of these hydroxides in water increase down the group.
- $Be(OH)_2$ dissolves only in bases containing excess of OH^- ions.
- Basic character of these hydroxides increases down the group.
- $Mg(OH)_2$ is weakly basic and is used as an antacid in medicines.
- Hydroxides of Group-II A elements are less basic than those of Group-I elements because of less electropositive character.

Halides

- Group-II A elements form halides of the type MX_2. They can be obtained directly by heating the metals with halogens or by the action of halogen acid on either of the metal or carbonate.
- These halides crystallize from solution as hydrated salts. Anhydrous $CaCl_2$, $SrCl_2$ and $BaCl_2$ can be prepared by heating their hydrated salts.
- Anhydrous beryllium and magnesium halides cannot be prepared by heating their hydrated salt because the hydrated salts on heating gives metal oxides or hydroxides.
- Anhydrous $BeCl_2$ can be prepared by heating BeO in CCl_4. Anhydrous beryllium halides are covalent having low m.pts as well as low electrical conductivity in fused state.
- Beryllium halides are soluble in organic solvents suggesting the covalent character, which is due to polarizing power of small Be^{2+} ion on the neighbouring halide ions.
- $BeCl_2$ exist as a linear monomer above 1200°C but at low temperature the vapours contain dimers of $BeCl_2$. In the solid state, it has a polymeric $(BeCl_2)_n$ structure similar to $(BeH_2)_n$ but BeH_2 contain 3c-2e bonds, whereas $(BeCl_2)_n$ contain halogen bridges in which halogen atom bonded to one Be atom with coordinate bond and with another Be atom with covalent bond. Be atom is involved in sp hybridization in monomer,

sp^2 hybridization in dimer and sp^3 hybridization in polymer and the coordination number of Be changes from 2 to 3 to 4, respectively.

- $BeCl_2$ and BeF_2 forms 4-coordinate complexes like $[BeCl_4]^{2-}$ and $[BeF_4]^{2-}$.
- Fluoride and chloride of magnesium are ionic but its bromide and iodides have significant covalent character and are soluble in organic solvents.
- The halides of alkaline earth metals are hygroscopic. Due to the strong affinity towards water, anhydrous $CaCl_2$ is used as a dehydrating agent.
- Ionic character of the halides of alkaline earth metals is in the order of $BeX_2 < MgX_2 < CaX_2 < SrX_2 < BaX_2$ and $MI_2 < MBr_2 < MCl_2 < MF_2$.
- Order of solubility in water
  $BeF_2 > MgF_2 > CaF_2 < SrF_2 < BaF_2$
  $BeX_2 > MgX_2 > CaX_2 > SrX_2 > BaX_2$
  $MF_2 < MCl_2 < MBr_2 < MI_2$.

Carbonates and Bicarbonates

- These carbonates occur in nature as solid rock materials. They can be prepared by passing CO_2 into the aqueous solutions of alkaline earth metal hydroxides or by adding Na_2CO_3 or $(NH_4)_2CO_3$ solution to the solution of a soluble salt of these metals. The carbonate of alkaline earth metals will be precipitated out.
- All these carbonates decomposes on heating giving MO and CO_2.
- Thermal stability of these carbonates increases from Be to Ba
  $BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$.
  and decomposition temperature increases.
- Solubility in water decreases from $BeCO_3$ to $BaCO_3$.
  $BeCO_3 > MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$
- All these carbonates are much more soluble in a solution of CO_2 than in water owing to the formation of bicarbonates.
- The bicarbonates of alkaline earth metals cannot be prepared in solid state. They exist only in solution.
- All the carbonates of alkaline earth metals are ionic, but $BeCO_3$ is unusual because it contains the hydrated $[Be(H_2O)_4]^{2+}$ ion rather than Be^{2+} ion.
- $CaCO_3$ occur in two different crystalline forms **calcite** and **aragonite.** Calcite is a more stable form in which each Ca^{2+} ion is surrounded by six oxygen atoms of CO_3^{2-}, whereas in aragonite (a meta stable form) each Ca^{2+} ion is surrounded by nine oxygen atoms of CO_3^{2-} ion as an unusual coordination number.
- Though aragonite is less stable and should convert to calcite, the high activation energy prevents the conversion.

Sulphates

- Sulphates of alkaline metals can be obtained by the action of dil. H_2SO_4 on metals, oxides, hydroxides or carbonates.
- Sulphates of Be, Mg and Ca crystallize as hydrated salts $BeSO_4 \cdot 4H_2O$; $MgSO_4 \cdot 7H_2O$; $CaSO_4 \cdot 2H_2O$. Sulphates of Sr and Ba crystallize without water of crystallization.
- Thermal stability increases down the group similar to carbonates.
- Sulphates decompose into MO, SO_2 and O_2 on heating. But when heated with carbon they form sulphides.
- Solubility of the sulphates in water decreases from Be to Ba. $BeSO_4$ and $MgSO_4$ are fairly soluble while $BaSO_4$ is completely insoluble.

Nitrates

- The nitrates of alkaline earth metals can be prepared by the reaction of nitric acid with oxides, hydroxides or carbonates.
- When crystallized from solution they crystallize as hydrated salts, but anhydrous nitrates cannot be prepared by heating hydrated nitrates because they decompose to metal oxide.
- Anhydrous nitrates can be prepared by using N_2O_4 and ethyl acetate. Be forms unusual basic beryllium nitrate in addition to normal nitrate.
- In basic beryllium acetate 4 Be atoms are arranged tetrahedrally and $6NO_3^-$ ions act as bridges between two Be atoms along six edges of tetrahedron. The basic oxygen is at the centre of tetrahedron (Fig. 6.3). Similar structures occur $[Be_4O(O_2CR)_6]$ where $RCOO^-$ may be acetate, or any carboxylate ion.
- All nitrates decompose on heating into MO, NO_2 and O_2. Thermal stability and solubility of the nitrates of Group-II elements follow the same order as in the case of carbonates and sulphates but all nitrates are soluble.

Trends in the Solubilities and Thermal Stability of Compounds of Group-II A Elements

- The solubility of oxides, hydroxides and fluorides of the Group-II A elements increases down the group while the solubilities of most of their salts decreases down the group,
- Thermal stabilities of the oxosalts of the elements of Group-II increase down the group though their lattice energies decreases.
- The trends in solubilities and thermal stabilities of compounds depend on lattice enthalpies and hydration energies.

- Lattice energy is inversely proportional to the bond length in ionic solids.

$$\Delta H_L \alpha \frac{1}{r_+ + r_-}$$

- Hydration energy is related to each ion individually and for a compound it is the sum of individual contributions.

$$\Delta_{hyd} H \alpha \frac{1}{r_+} + \frac{1}{r_-}$$

- Lattice energy decreases rapidly with increase in the size of one ion if the other ion is smaller one but do not change much with increase in size of one ion if the size of other ion is very large.
- Though hydration energies decreases with increase in size of cation, it dominates the lattice energy when the anion is small because the decrease in lattice energy is more rapid than the decrease in hydration energy. So, the solubilities of the compounds containing smaller oxide, hydroxide and fluoride ions increases down the group with increase in size of cation.
- Since the decrease in lattice energy is less rapid than the decrease in hydration energy when anion is bigger one, the solubility of the compounds decreases down the group with increase in size of cation. Hence, the solubilities of compounds such as carbonates, sulphates, nitrates, phosphates, chlorides, bromides, iodides, etc. of alkaline earth metals decreases down the group with increase in size of cation.
- When oxosalts decompose one of the product is metal oxide.
- The lattice energy is more when metal ion combines with small oxide ion than combined with bigger oxoanion like CO_3^{2-}, SO_4^{2-}, NO_3^-, etc. The decomposition temperature depends on the difference in the lattice enthalpies of metal oxide and oxosalt and directly proportional to the difference in lattice enthalpies.
- When small cation like Be^{2+} combines with small oxide ion the lattice energy is more and the difference in lattice enthalpies of oxosalt and oxide is little. So, it decomposes at low temperatures because less energy is required to decompose.
- When bigger cation such as Ba^{2+} combines with small oxide ion, the lattice energy is less and the difference in lattice enthalpies of oxosalt and oxide is more. So, it decomposes at high temperatures because more energy (more difference in lattice enthalpies) is required to decompose.

Anomalous Behaviour of Beryllium

- The reasons for the anomalous behaviour of beryllium is attributed to (i) its small size, (ii) its high electronegativity, (iii) absence of d-orbitals in its valence shell and (iv) presence of only two electrons in its penultimate shell.
- Be compounds are predominantly covalent due to their high polarizing power and their salts are readily hydrolyzed.
- Be is not easily affected by dry air and does not decompose water, at ordinary temperature.
- Be is an amphoteric metal and dissolves in alkali solutions forming beryllates.
- $BeSO_4$ is soluble in water, whereas sulphates of Ca, Sr and Ba are insoluble.
- Be and its salts do not respond to flame test while Ca, Sr and Ba give characteristic flame colours.
- Be forms many complexes, while the heavier elements do not show much tendency to form complexes.
- Maximum covalency of Be is 4 while other alkaline earth metals can exhibit a covalency of 6.
- BeO is covalent and has 4:4 ZnS structure but all the others are ionic and have 6:6 NaCl structure.
- Be donot react with H_2 directly but others can react with H_2 directly.
- Be_3N_2 is volatile but the nitrides of others are not volatile.
- With carbon, Be forms methanide (Be_2C) while others form acetylides (MC_2).
- Beryllium halides polymerize to form chains containing bridging halogens groups, e.g., (BeF_2) and $(BeCl_2)_n$. But halides of other Group-II elements are ionic.

Diagonal Relationship between Be and Al

- Be and Al have the same electronegativity (Be 1.5; Al 1.5) and their charge/radius ratios are very similar indicating similar field strengths (Be 6.4, Al 6.0). So, Be and Al resemble in several properties.
- The standard oxidation potentials of both Be and Al are of nearly the same order (Be = 1.97V; Al = 1.7V).
- Since the polarizing power of both Be and Al are nearly the same, the covalent character of their compounds is also similar.
- Both Be and Al are rendered passive on treatment with conc. HNO_3.
- Unlike alkaline earth metals but like aluminium, Be does not get readily attacked by dry air.
- Both Be and Al reacts very slowly with dilute mineral acids due to the presence of oxide layer.
- Both Be and Al react with alkalis liberating H_2.
- Both Be and Al form carbides which on hydrolysis liberate methane.
- Both form nitrides when heated in nitrogen which give ammonia by the reaction with water.
- Both form oxides which are amphoteric.
- Halides of both Be and Al contain halogen bridge bonds.

- Both Be and Al do not impart flame coloration.
- Both Be^{2+} and Al^{3+} have a strong tendency to form complexes.

IMPORTANT COMPOUNDS OF MAGNESIUM

Magnesium Oxide

- Prepared by calcination of magnesite. It is called **calscined mangesia.**
- It can be prepared by heating $Mg(OH)_2$, $Mg(NO_3)_2$ or $MgSO_4$.
- It is sparingly soluble in water, less basic in nature and easily soluble in acids.
- It dissolves in aqueous solution of $MgCl_2$ or $MgBr_2$ forming basic salts such as $MgBr_2 \cdot 9Mg(OH)_2 \cdot 5H_2O$.
- It is used as an antacid in medicine. When moistened with saturated solution of $MgCl_2$, it sets into a hard mass called **Sorrell's cement** whose composition is $MgCl_2$. $5MgO \cdot xH_2O$. It is used in plaster casting, repairing porcelain ware and in filling teeth.
- It is used as a basic flux and basic refractory material in metallurgy.
- It is used as an adsorbent in the manufacture of dynamite and in vulcanization of rubber. Flocculent form of MgO is used as a rubber filler.

Magnesium Hydroxide

- It occurs in the mineral **brucite.** It is prepared by adding NaOH or NH_4OH to a solution of soluble magnesium salt and $Mg(OH)_2$ will be precipitated out.
- On large scale, it is manufactured by treating $MgCl_2$ of the potash industry with slaked lime.
- It is a white powder, slightly soluble in water, dissolves in acids and decomposes at 573 K to MgO.
- The suspension of Mg $(OH)_2$ in water is called **milk of magnesia** and is used as an antacid in medicine.

Magnesium Chloride $MgCl_2 \cdot 6H_2O$

- In the laboratory it is prepared by dissolving the metal, oxide, hydroxide or carbonate in dil. HCl and evaporating the solution.
- On large scale it is manufactured by fractional crystallization of fused carnallite.
- Anhydrous $MgCl_2$ cannot be prepared by heating hydrated $MgCl_2 \cdot 6H_2O$ because it hydrolyzes to form hydroxy chloride or hydroxide or oxide depending on the temperature.
- Anhydrous $MgCl_2$ can be prepared by (i) heating $MgCl_2 \cdot 6H_2O$ in vacuum in a current of dry HCl, (ii) heating a mixture of MgO and C in a current of Cl_2 gas and (iii) passing a mixture of CO and Cl_2 over MgO at 1023 K.
- It is colourless, deliquescent solid soluble in water.
- Used in making Sorell's cement and in lubricating cotton threads in spinning.

Magnesium Sulphate, $MgSO_4 \cdot 7H_2O$, Epsom Salt

- It occurs in the nature in the form of minerals kieserite $MgSO_4 \cdot H_2O$ and epsomite $MgSO_4 \cdot 7H_2O$.
- It is prepared by dissolving magnesium, or its oxide or hydroxide or carbonate in dil. H_2SO_4 and evaporating the solution which gives heptahydrate $MgSO_4 \cdot 7H_2O$.
- Dolamite when dissolved in dil. H_2SO_4 gives a mixture of $MgSO_4$ and $CaSO_4$. Insoluble $CaSO_4$ is removed by filtration and the filtrate on evaporation gives epsom salt.
- Anhydrous $MgSO_4$ is obtained by heating the hydrated salt at 473 K.
- Epsom salt is manufactured by dissolving kieserite in boiling water and crystallizing the resulting solution.
- It is colourless efflorescent crystalline solid having a bitter taste.
- There are two types of crystalline forms of $MgSO_4 \cdot 7H_2O$: one isomorphous with rhombic crystals of $ZnSO_4 \cdot 7H_2O$ and the other is isomorphous with monoclinic crystals of $FeSO_4 \cdot 7H_2O$. Out of the seven water molecules, six are coordinated to Mg^{2+} ion and the seventh water molecule is in hydrogen bond with SO_4^{2-} ion.
- It is extensively used as a purgative or laxative, in dyeing and tanning of leather and in the manufacture of fire proofing fabrics. Platinized $MgSO_4$ crystals are used as catalysts in the manufacture of H_2SO_4 by Grillo's process.

Magnesium Carbonate

- It occurs in nature as magnasite. It can be precipitated by adding sodium bicarbonate solution to a hot solution of a magnesium salt. If precipitation is carried in the cold condition, **magnesia alba levis** having the formula $MgCO_3 \cdot Mg(OH)_2$ is obtained. Hot concentrated solutions yields **magnesia alba ponderosa** having the approximate formula $MgCO_3 \cdot Mg(OH)_2 \cdot 3H_2O$. Magnesia alba is used in medicine as an antacid and laxative. Also, it is used in tooth powders and pastes and silver polishes.
- If the solution of the bicarbonate is crystallized at 325 K and heated with MgO, crystals of $MgCO_3 \cdot 3H_2O$ separate.
- The $MgCO_3 \cdot 3H_2O$ is more reactive than the anhydrous salt and is used in the Engel–Precht process.
- It is used in glass manufacture, inks, ceramics, fertilizers insulation process, industrial chemicals and rubber products, as a refractory material and for certain medicinal preparations and as a filler for paper, rubber and pigments.

Magnesium Ammonium Phosphate $Mg(NH_4)PO_4 \cdot 6H_2O$

- It is the least soluble compounds of magnesium and is formed as a white crystalline precipitate when a solution of Na_2HPO_4 is added to a solution of magnesium salt containing some ammonia and ammonium chloride.
- The formation of magnesium ammonium phosphate as a white precipitate is used in the detection and estimation of magnesium.

IMPORTANT COMPOUNDS OF CALCIUM

Calcium Oxide CaO

- It is also known as **quick lime** or **burnt lime**, prepared on large scale by burning lime stone in lime kiln.
- It is a white solid, when heated in oxyhydrogen flame a brilliant white light known as **lime light** is emitted.
- It absorbs moisture from air. So, it is used as a drying agent but cannot be used to dry acidic substances.
- It is used in the manufacture of bleaching powder, calcium carbide, cement, glass, lime mortar, etc. used in the purification of sugar, as a basic refractory material and basic flux in metallurgy.

Calcium Hydroxide $Ca(OH)_2$

- It is obtained by the addition of water to quick lime.
- When water is added to quick lime a hissing sound is heard and lumps of CaO crack and disintegrate into powder. This process is called **slaking of lime** and the product is called slaked lime.
- It is a white amorphous powder, sparingly soluble in water. Its solubility in water decreases with increase in temperature.
- The suspension of lime in water is called **milk of lime,** whereas the filtered and clear solution is called **lime water.**
- When CO_2 is passed through lime water, first it turns milky which disappears by passing excess of CO_2 due to formation of soluble bicarbonate.
- Milk of lime when treated with Cl_2 gives hypochlorite in cold condition and chlorate in hot condition along with chloride.
- When Cl_2 gas is passed over dry slaked lime, bleaching powder is formed.
- Milk of lime liberates ammonia gas from ammonium salts.
- It is used in the manufacture of bleaching powder, caustic soda, in the production of lime mortar, in glass making, tanning industry and for purification of sugar.

Calcium Chloride $CaCl_2 \cdot 6H_2O$

- It is obtained as a by-product in the Solvay's process for the manufacture of sodium carbonate.
- In the laboratory it is obtained by dissolving $CaCO_3$ in dilute HCl and crystallizing gives $CaCl_2 \cdot 6H_2O$.
- On heating $CaCl_2 \cdot 6H_2O$ at about 473 K forms $CaCl_2 \cdot 2H_2O$, which finally becomes anhydrous $CaCl_2$ on further heating. It is also called as **fused calcium chloride.**
- It is extremely soluble in water. It forms addition compound with ammonia, methyl alcohol like $CaCl_2 \cdot 8NH_3$ and $CaCl_2 \cdot 4NH_3$.
- Anhydrous $CaCl_2$ is hygroscopic and deliquescent, and hence used as a drying agent in dessicator but it cannot be used to dry ethylalcohol or ammonia as it forms addition compounds with them.
- When mixed with ice the freezing point of water decreases to 255K, so, it is used to melt ice and snow on roads. It is also used in making freezing mixtures and in refrigeration.
- Highly concentrated solutions of $CaCl_2$ are used in liquid baths for heating purposes.

Calcium Carbonate $CaCO_3$

- It is widely distributed in nature in huge deposits as lime stone, chalk, marble and iceland spar.
- In the laboratory it is prepared either by passing CO_2 through lime water or by adding calculated amount of sodium carbonate to a solution of $CaCl_2$.
- It is white solid, insoluble in water, but dissolves in the presence of CO_2 due to formation of calcium bicarbonate.
- Used in the manufacture of cement, lime, washing soda, glass, as chalk in tooth powders and pastes and as a flux in metallurgy.

Calcium Sulphate $CaSO_4 \cdot 2H_2O$

- In the nature it occurs in the form of **gypsum** $CaSO_4 \cdot 2H_2O$, anhydrous calcium sulphate $CaSO_4$ is known as **anhydrite.** The calcium sulphate occurring as a pure fine powder is known as **alabaster.**
- Gypsum is used in cement to retard setting time, in the manufacture of plaster of paris, in agriculture and to impregnate filter papers. It is also used in the production of SO_2 required in the manufacture of sulphuric acid.

Plaster of Paris $CaSO_4 \cdot \frac{1}{2}H_2O$

- It is prepared by heating gypsum at 390 K. If temperature of heating is above 390 K, setting properties of plaster of paris are partly destroyed and it is called **dead burnt.**

- It is hemihydrate and when mixed with water sets into hard mass which is known as setting of plaster of paris.
- During setting of plaster of paris little expansion takes place with liberation of little heat (exothermic).
- It sets into hard mass in two stages, the first stage is called **setting stage** during which it converts into **orthorhombic gypsum** and in the second stage known as **hardening stage** during which it converts into **monoclinic gypsum.** Setting of plaster of paris is due to hydration or uniting with water into another hydrate.
- It is used in making moulds, statues, in plastering the fractured bones and in making black board crayons.

Mortar

- It is also known as lime mortar. It is an intimate mixture of 1 part of slaked lime, 3 parts of sand and water made into a paste.
- Lime absorbs CO_2 from air and converts into $CaCO_3$ which acts as a binding material. Lime also reacts with silica forming calcium silicate which gives hardness. Sand makes the mass porous which prevents the crackings due to excessive shrinkage during drying.
- Mortar mixed with cement is called **cement mortar** which is more harder and more water proof.
- Mortar obtained by burning lime stone containing 10 per cent alumino silicates is called **hydraulic mortar** which can set like cement on addition of water. This is used for bleaching purposes and as an antiseptic.

Cement

- The name portland cement was given to it by Joseph Aspidin because when it is mixed with sand and water it hardens like the lime stone querried at Portland in England.
- The composition of cement is CaO, –50 to 60%; SiO_2, 20 to 25%; Al_2O_3, 5 to 10%; MgO, 2 to 3%; Fe_2O_3,1 to 2% and SO_3, 1 to 2%.
- The ratio between silica to alumina should be between 4 and 2.5 and the ratio of CaO to the mixture of SiO_2, Al_2O_3 and Fe_2O_3 should be close to 2.0.
- If lime is excess the cement cracks during setting but if it is less the cement will be weak.
- The raw materials for the manufacture of cement are lime stone and alumino silicates (clay, sand and shales).
- When the powdered raw materials are heated in a rotary kiln, sintered clinker will be obtained.
- The important compounds in the portland cement are dicalcium silicate (Ca_2SiO_4) 26 per cent, tricalcium silicate (Ca_3SiO_5) 51 per cent and tricalcium aluminate ($Ca_3\,Al_2O_6$) 11 per cent.
- The setting of cement by mixing with water is due to hydration of the molecules and their rearrangement.
- About 1 per cent gypsum is added to the cement to slow down the process of setting of the cement.

Biological Importance of Mg and Ca

- Mg^{2+} ions are concentrated in animal cells.
- Enzymes like "phosphohydrolases" and "phosphotransferases" contain Mg^{2+} ions. These enzymes participate in ATP reactions and release energy in the process. Mg^{2+} forms a complex with ATP.
- Mg^{2+} is a constituent of chlorophyll, the green pigment of plants.
- Ca^{2+} ion is present in bones and teeth as apetite [$Ca_3(PO_4)_2$]. Enamel of teeth is fluorapetite [$3Ca_3(PO_4)_2 \cdot CaF_2$].
- Ca^{2+} ions are necessary for clotting of blood, to maintain regular heart beating and for muscle contraction.

MULTIPLE CHOICE QUESTIONS WITH ONLY ONE ANSWER

1. X and Y are two metals. When burnt in air, X forms only oxide while Y forms oxide and nitride. The metals X and Y may be
   (a) Ca and Mg (b) Na and Mg
   (c) Li and Na (d) Na and K
2. Select the incorrect statement about alkaline earth metals.
   (a) Solubility of sulphates decreases down the group.
   (b) Solubility of hydroxides decreases down the group.
   (c) Thermal stability of carbonates increases down the group.
   (d) Basic nature of hydroxides increases down the group.
3. In polymeric $(BeCl_2)_n$, there are
   (a) three centre four-electron bonds
   (b) three centre three-electron bonds
   (c) two centre three-electron bonds
   (d) two centre two-electron bonds
4. A compound X on heating gives a colourless gas. The residue is dissolved in water to obtain Y. Excess CO_2 is bubbled through aqueous solution of Y and Z is formed. Z on gentle heating gives back X. The X is
   (a) $CaCO_3$ (b) $Ca(HCO_3)_2$
   (c) Na_2CO_3 (d) $NaHCO_3$
5. Metal M + air $\xrightarrow{\Delta}$ A $\xrightarrow{H_2O}$ B $\xrightarrow{HCl}$ White fumes; Metal M can be
   (a) Li, Mg or Al (b) Li, Al or K
   (c) Na, K or Mg (d) Li, Na or K

6. Alkaline earth metals form carbides of the type MC_2. But MgC_2 on heating converts into X which on hydrolysis gives
(a) Methane (b) Acetylene
(c) Propyne (d) Ether

7. A metal M readily forms water soluble sulphate and water insoluble hydroxide $M(OH)_2$. Its oxide MO is amphoteric, hard and has a high melting point. The alkaline earth metal M must be
(a) Mg (b) Be
(c) Li (d) Sr

8. $X + C + Cl_2 \xrightarrow[\text{of about 1000 K}]{\text{High temperature}} Y + CO; Y + 2H_2O \rightarrow$ $Z + 2HCl$. Compound Y is found in polymeric chain structure and is an electron-deficient molecule. Y must be
(a) BeO (b) $BeCl_2$
(c) BeH_2 (d) $AlCl_3$

9. The pair of substances that gives the same products on reaction with water is
(a) Mg and MgO (b) Sr and SrO
(c) Ca and CaH_2 (d) Be and BeO

10. Sort the odd one out
(a) $Ca(HCO_3)_2$ (b) $Mg(HCO_3)_2$
(c) $NaHCO_3$ (d) $LiHCO_3$

11. Which is the most stable among the following?
(a) $[Be(H_2O)_4]^{2+}$ (b) $[Mg(H_2O)_4]^{2+}$
(c) $[Ca(H_2O)_4]^{2+}$ (d) $[Sr(H_2O)_4]^{2+}$

12. The composition of Beryl is
(a) $Be_3Cr_2Si_6O_{18}$ (b) $Be_3Al_2Si_6O_{18}$
(c) Be_2SiO_4 (d) $3BeO.Al_2O_3$

13. Which of the following has lowest thermal stability?
(a) $BeCO_3$ (b) $MgCO_3$
(c) $CaCO_3$ (d) $BaCO_3$

14. The hydration energies of the Group-2 ions are four to five times greater than that of the Group-1 ions. This is due to their
(a) Smaller size and increased nuclear charge
(b) Greater size and decreased nuclear charge
(c) Smaller size and decreased nuclear charge
(d) Greater size and increased nuclear charge

15. Which of the following is arranged in the correct order of increasing thermal stability?
(a) $MgCO_3 < BaCO_3 < SrCO_3 < CaCO_3$
(b) $MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$
(c) $CaCO_3 < MgCO_3 < BaCO_3 < SrCO_3$
(d) $BaCO_3 < SrCO_3 < CaCO_3 < MgCO_3$

16. Calcium carbide is produced by heating
(a) CaO and CH_4 at 200°C
(b) CaO and C at 2000°C
(c) Ca and C at 1000°C
(d) CaNCN and C at 800°C

17. A metal X on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which on passing through $CuSO_4$ solution turns it deep blue. Y is
(a) $Mg(NO_3)_2$ (b) Mg_3N_2
(c) NH_3 (d) MgO

18. One of the following statements is incorrect.
(a) Elements of Group-II are good conductors of electricity and heat.
(b) Compounds of Group-II elements are diamagnetic in nature.
(c) The salts of Group-II elements are more heavily hydrated than those of elements of Group-I.
(d) Elements of Group-II are more electropositive than Group-I elements.

19. The order of decreasing polarity in the compounds CaO, CsF, KCl, MgO is
(a) CaO, CsF, KCl, MgO
(b) MgO, KCl, CaO, CsF
(c) KCl, CaO, CsF, MgO
(d) CsF, KCl, CaO, MgO

20. Alkaline earth metal compounds are less soluble in water than the corresponding alkali metal compounds because the former have
(a) Lower lattice energies
(b) Higher ionization enthalpies
(c) Higher covalent character
(d) Higher ionic character

21. Beryllium is placed above magnesium in the second group. Beryllium dust, therefore, when added to $MgCl_2$ solution will
(a) Have no effect
(b) Precipitate Mg metal
(c) Precipitate MgO
(d) Lead to dissolution of beryllium metal

22. Which of the following statements is false?
(a) Strontium decomposes water readily than beryllium.
(b) Barium carbonate melts at a higher temperature than calcium carbonate.
(c) Barium hydroxide is more soluble in water than magnesium hydroxide.
(d) Plaster of paris is obtained by partial oxidation of gypsum.

23. Identify the correct statement.
(a) Gypsum contains a lower percentage of calcium than plaster of paris.
(b) Gypsum is obtained by heating plaster of paris.
(c) Plaster of paris can be obtained by hydration of gypsum.
(d) Plaster of paris is obtained by partial oxidation of gypsum.

24. Which of the following on thermal decomposition yields a basic as well as an acidic oxide?
(a) $KClO_3$ (b) $CaCO_3$
(c) NH_4NO_3 (d) $NaNO_3$

25. A doctor by mistake administers a dilute $Ba(NO_3)_2$ solution to a patient for radiographic investigations. Which of the following should be the best to prevent the absorption of soluble barium and subsequent barium poisioning?
(a) $NaCl$ (b) Na_2SO_4
(c) Na_2CO_3 (d) NH_4Cl

26. The solubilities of sulphates of alkaline earth metals decrease down the group mainly due to decrease in
(a) Lattice energy of metal sulphate
(b) Entropy of solution of metal sulphates
(c) Interionic attraction
(d) Hydration energy of cations

27. 1 mole of a substance (X) was treated with an excess of water. 2 moles of readily combustible gas were produced along with solution which when reacted with CO_2 gas produced a white turbidity. The substance (X) could be
(a) Ca (b) CaH_2
(c) $Ca(OH)_2$ (d) $Ca(NO_3)_2$

28. In the alkaline earth metals, the electrons are more firmly held to the nucleus and hence
(a) atoms of alkaline earth metals are bigger than alkali metals
(b) ionization energy of alkaline earths is greater than alkali metals
(c) reactivity of alkaline earths is greater than alkali metals
(d) alkaline earths are less abundant in nature

29. A compound of calcium (X) is used in sugar industry for the purification of sugar. When exposed to an oxyhydrogen flame, it becomes incandescent and starts emitting white light, on treatment with CO_2 it forms a compound which can be decomposed to give back X. At a very high temperature, X is
(a) $CaCO_3$ (b) CaO
(c) $Ca(OH)_2$ (d) $CaSO_4$

30. Identify the correct statements in the compounds of IIA group.
I. Solubility of hydroxides increases from Ca to Ba
II. Solubility of carbonates decreases from Ca to Ba
III. Solubility of sulphates decreases from Ca to Ba
IV. Thermal stability of carbonates increases from Ca to Ba
(a) I, II and III only (b) I, II, III and IV
(c) II and IV only (d) I and IV only

31. A sodium salt on treatment with $MgCl_2$ gives a white precipitate only on heating. The anion of the sodium salt is
(a) HCO_3^- (b) CO_3^{2-}
(c) NO_3^- (d) SO_4^{2-}

32. Identify the correct order of solubility in 2nd group compounds of the modern periodic table.
I. $Ca(C_2O_4)_2 > Sr(C_2O_4)_2 > Ba(C_2O_4)_2$
II. $CaSO_4 < SrSO_4 < BaSO_4$
III. $Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$
IV. $CaF_2 < SrF_2 < BaF_2$
(a) III, IV only (b) I, III, IV only
(c) I, II only (d) I, IV only

33. Lattice energies of the oxides of Mg, Ca, Sr, Ba are in the order
(a) $BaO > SrO > CaO > MgO$
(b) $CaO > BaO > SrO > MgO$
(c) $MgO > SrO > CaO > BaO$
(d) $MgO > CaO > SrO > BaO$

34. When a saturated solution of magnesium sulphate is treated with NH_4Cl and NH_3, followed by the addition of disodium hydrogen phosphate, a white precipitate is formed. This on heating gives
(a) $Mg_2P_2O_7$ (b) $Mg_3(PO_4)_2$
(c) $Mg(NH_4)PO_4$ (d) $Mg(NH_4)HPO_4$

35. The solubility of the fluorides and hydroxides of alkaline earth metals increases on descending the group because the
(a) lattice energy of the compounds increases more rapidly than the hydration energy
(b) lattice energy of the compounds decreases more rapidly than the hydration energy
(c) size of the metals decreases on descending the group
(d) ionization energy of the metals increases on descending the group

36. On descending the Group-II, the ions of the corresponding metals becomes larger. Therefore, the
(a) lattice energy and the hydration energy of the compounds decreases but the decrease in lattice energy is small when anion is larger
(b) lattice energy and hydration energy of the compounds decreases but the decrease in lattice energy is small when anion is smaller
(c) lattice energy and hydration of the compounds decreases but the decrease in hydration is more when anion is smaller
(d) lattice energy and hydration of the compounds decreases but the decrease in hydration or lattice energy have no relation with size of anion

37. Alkaline earth metals are
(a) harder and have higher cohesive energies and m.pts than alkali metals
(b) softer and have lower cohesive energies and m.pts than alkali metals

(c) softer and have lower cohesive energies and higher m.pts than alkali metals
(d) harder and have higher cohesive energies and lower m.pts than alkali metals

38. Which of the following statements is incorrect?
(a) The ionization energy of Be is high and its compounds are covalent.
(b) The ionization energy of Be is low and its compounds are covalent.
(c) The compounds formed by Mg, Ca, Sr and Ba are predominantly divalent and ionic.
(d) The ionization energy of Be is low and its compounds are ionic.

39. $BeCl_2 + N_2O_4 \longrightarrow A \xrightarrow[\text{Vaccuum}]{50^\circ C} B \xrightarrow{125^\circ C} C.$
The compounds A, B and C are
(a) BeO, $Be(NO_2)_2$, $Be(NO_3)_2$
(b) BeO, $Be(NO_3)_2$, $Be(NO_2)_2$
(c) $Be(NO_3)_2 \cdot 2N_2O_4$, $Be(NO_3)_2$, $Be(NO_2)_2$
(d) $Be(NO_3)_2 \cdot 2N_2O_4$, $Be(NO_3)_2$, $[Be_4O(NO_3)_6]$

40. $[Be(H_2O)_4]Cl_2$ on heating gives
(a) O_2 (b) Be (c) $BeCl_2$ (d) $Be(OH)_2$

41. BeO dissolves in strongly basic solutions to form
(a) $Be(OH)_2$ (b) Be (c) Be_2O_3 (d) $[Be(OH)_4]^{2-}$

42. MgO and CaO are used for lining furnaces because
(a) They have high m.pts
(b) They are good conductors of heat
(c) They are electrical insulators
(d) All are correct

43. Calcium chloride is preferred over sodium chloride for clearing ice on roads particularly in very cold countries. This is because
(a) $CaCl_2$ is less soluble in water than NaCl.
(b) $CaCl_2$ is hygroscopic but NaCl is not.
(c) Eutectic mixture of $CaCl_2/H_2O$ freezes at –55°C while that of $NaCl/H_2O$ freezes at –18°C.
(d) NaCl makes the roads slippery but $CaCl_2$ does not.

44. Calcium cyanamide on hydrolysis gives a gas B which on oxidation with bleaching powder gives another gas C. When magnesium is heated in gas C and the resultant compound D on adding to water gives the same gas B. Then B, C and D are
(a) NH_3, N_2, Mg_3N_2
(b) N_2, NH_3, MgNH
(c) N_2, NH_3, $Mg(NO_3)_2$
(d) NH_3, NO, $Mg(NO_3)_2$

45. Alkaline earth metals form halides of the type MX_2. Which is false about them?
(a) They can be prepared by the direct reaction of metal and halogens.
(b) Beryllium halides are covalent while the halides of other elements are ionic.
(c) Their m.pts increases from BeX_2 to BaX_2.
(d) Except BeF_2, the solubility of other halides increases from BeX_2 to BaX_2.

46. Which one of the following statement is incorrect?
(a) The solubility of $CaSO_4$ increases with increase in temperature.
(b) Solubility of $CaSO_4$ increases in the presence of $(NH_4)_2SO_4$.
(c) $Ba(OH)_2$ is used for the preparation of standard alkali solutions for titration of acids.
(d) $BaCl_2$ is used as a laboratory reagent for the test of SO_4^{2-} radical.

47. Alkaline earth metals form dipositive ions instead of unipositive ions because
(a) The second ionization energy is almost double than the first ionization energy.
(b) Unipositive ions do not have stable configuration.
(c) The hydration energy of dipositive ion is more than that compensate the higher value of the second ionization energy.
(d) Unipositive ions are highly hydrated.

48. A metal salt solution forms a yellow/precipitate with K_2CrO_4 in acetic acid, a white precipitate with dil. H_2SO_4, but gives no precipitate with NaCl or NaI. The white precipitate obtained when Na_2CO_3 is added to the metal salt solution will contain
(a) $CaCO_3$ (b) $SrCO_3$ (c) $BaCO_3$
(d) Basic magnesium carbonate

49. To two different samples of magnesium sulphate A and B, sodium carbonate solution is added to sample A and sodium bicarbonate is added to sample B. In both cases there appears
(a) white precipitate of $MgCO_3$ in both A and B
(b) white precipitate of $MgCO_3$ in A and no precipitate in B
(c) white precipitate of $MgCO_3$ in A and $MgCO_3 \cdot Mg(OH)_2$ in B
(d) white precipitate of $MgCO_3 \cdot Mg(OH)_2$ in A and $MgCO_3$ in B

50. The solubility of gypsum in water can be increased by
(a) passing CO_2
(b) by increasing temperature
(c) adding ammonium sulphate
(d) adding ammonium chloride

Multiple Choice Questions with One or More Than One Answer

1. Identify the incorrect statement(s)
(a) Magnesium is lighter than calcium.
(b) The atomic radius of Mg is greater than that of Ca.

(c) Mg alloys are used in the construction of air crafts.
(d) Mg is used as a reducing agent.

2. Select the wrong statement(s)
(a) CaF_2 is soluble in water.
(b) $BaSO_4$ is soluble in water.
(c) Ba $(OH)_2$ is soluble in water.
(d) $MgSO_4$ is soluble in water.

3. Which of the following statements about the elements, Mg, Ca, Sr and Ba and their compounds is true?
(a) Solubility of the hydroxides in water increases with increasing atomic number.
(b) Thermal stability of the carbonates increases with increasing atomic number.
(c) All given elements except beryllium react with water or steam to give hydrogen.
(d) Metal chlorides are all liquids at room temperature.

4. Which statement is correct regarding the diagonal relationship between the Al and Be?
(a) BeO and Al_2O_3 are amphoteric in nature.
(b) Carbides of both elements on hydrolysis produce the same gas.
(c) Both can form a complex.
(d) Both have nearly similar m.p.

5. Identify the correct statements
(a) Beryllium liberates H_2 from HCl or H_2SO_4.
(b) Beryllium is rendered passive on treatment with conc. HNO_3.
(c) $BeCl_2$ has a polymeric structure in the solid state but exists as a dimer in the vapour state.
(d) $BeCl_2$ has a dimeric structure in the solid state but a polymeric structure in the vapour state.

6. Which of the following statements are correct?
(a) Beryllium forms covalent compounds due to small size and high charge of the Be^{2+} ion.
(b) The maximum coordination number of beryllium is six.
(c) The maximum coordination number of beryllium is four.
(d) Beryllium salts are extensively hydrolyzed.

7. Sodium sulphate is soluble in water but barium sulphate is sparingly soluble because
(a) The hydration energy of Na_2SO_4 is more than its lattice energy.
(b) The lattice energy of $BaSO_4$ is more than its hydration energy.
(c) The lattice energy has no role to play in solubility.
(d) The lattice energy of Na_2SO_4 is more than its hydration energy.

8. Choose the correct statement(s)
(a) $BeCO_3$ is kept in the atmosphere of CO_2 since it is least thermally stable.
(b) Be dissolves in an alkali solution forming $[Be(OH)_4]^{2-}$.
(c) BeF_2 forms complex ion with NaF in which Be goes with cation.
(d) BeF_2 forms complex ion with NaF in which Be goes with anion.

9. Which of the following statements is true regarding beryllium chloride?
(a) In solid state, beryllium chloride exists in the form of a chain structure.
(b) It readily dissolves in water and gets hydrolyzed to form basic solution.
(c) In vapour state, it exists as a dimer with bridged structure.
(d) Above 1200 K, it has a linear structure.

10. Which of the following statements is correct for anhydrous calcium chloride?
(a) It is prepared by heating hydrated calcium chloride above 533 K.
(b) It is used for drying alcohols and NH_3.
(c) It is used as a dehydrating agent to clear snow and ice on highways and pavements.
(d) When mixed in concrete it gives quicker initial setting and improves its strength.

11. Choose the correct statement.
(a) The solubility of Group-II salts depends upon lattice energy of solid and the hydration energy of ions.
(b) The solubility of Group-II chlorides and hydroxides have opposite trends.
(c) The solubilities of most Group-II salts decrease with an increase in atomic weight of corresponding metals.
(d) The solubilities of Group-II chlorides and hydroxides increases with molecular weight.

12. Which of the following statements are correct?
(a) Alkali metals are strong reducing agents but poor complexing agents.
(b) Among Cs^+ and Mg^{2+}, Mg^{2+} form an acetate complex.
(c) LiF and CsI have low solubility in water, whereas LiI and CsF are very soluble.
(d) LiH has greater thermal stability than the other alkali metal hydrides, whereas Li_2CO_3 decomposes at a lower temperature than the other alkali metal carbonates.

13. Which of the following statements are true?
(a) Magnesium hydroxide is a much more effective antacid than calcium or barium hydroxide.
(b) Group-I hydroxides are much more corrosive than Group-II hydroxides.

(c) The difference between the lattice enthalpies of the oxides and peroxides or oxides and carbonates of s-block elements decreases down the group resulting in decreased tendency for the peroxides and carbonates to decompose.
(d) Solubility of the carbonates of Group-2 elements increases in the presence of CO_2.

14. Which of the following statements are correct?
(a) Unlike magnesium chloride, calcium chloride can be obtained by heating $CaCl_2 \cdot 6H_2O$.
(b) Halides of elements of Group-2 with the exception of beryllium halides are ionic.
(c) The decrease in the solubility from $BeSO_4$ to $BaSO_4$ is primarily due to decrease in the hydration energy as one moves from Be^{2+} to Ba^{2+}.
(d) The increase in the solubility of chlorides and hydroxides of alkaline earth metals is primarily due to the decrease in lattice energy from Be to Ba salts.

15. Which of the following statements are correct?
(a) When one electron is removed from Group-2 metal the ratio of nuclear charge to the number of orbital electrons is increased so that the remaining electrons are more tightly held.
(b) The energy needed to remove the second electron from a Group-2 metal is nearly double that required for the first electron.
(c) When ionic compounds are formed by Group-2 metals, energy is released.
(d) When ionic compounds are formed by Group-2 metals, energy is absorbed.

16. Which of the following statements are correct for the compounds of Group-2 metals?
(a) The number of molecules of water of crystallization increases with the size of the metal ions.
(b) The number of molecules of water of crystallization increases as the size of the metal ion decreases.
(c) The number of molecules of water of crystallization decreases as the size of the metal ion increases.
(d) There is no relation to the size of the metal ion and the number of molecules of water of crystallization.

17. Which of the following is/are correct for negative lattice energy?
(a) Lattice energy of MgO is more than $MgCO_3$.
(b) Lattice energy of MgF_2 is less than MgO.
(c) Lattice energy of MgF_2 is more than MgO.
(d) Lattice energy of $MgCO_3$ is more than MgO.

18. Dry beds of ponds and lakes crack due to
(a) expanding property of $CaCO_3$
(b) clay contains sodium carbonate which has the property to expand when wet
(c) clay contains sodium bentonite which has the property to expand when wet.
(d) clay contains sodium adipate which has the pro perty to expand when wet

19. The compound MgCl does not exist because
(a) Mg has high charge and small size.
(b) Compounds of alkaline earth metals are less ionic than the corresponding alkali metals.
(c) Mg is more reactive than alkali metals.
(d) MgCl disproportionates into $MgCl_2$ and Mg.

20. During setting of plaster of paris which of the following process takes place?
(a) Hydration (b) Hydrolysis
(c) Crystalline state (d) Evolution of heat

21. Identify the correct statement
(a) The order of basic character of the following oxides is NiO < MgO < SrO < K_2O < Cs_2O.
(b) Be $(OH)_2$ is a stronger base than $Ba(OH)_2$.
(c) The order of the ionic character of the following chlorides is $BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2 < BaCl_2$.
(d) $Ca(OH)_2$ is more soluble in water than $Mg(OH)_2$.

22. Which of the following statements are correct?
(a) Alkaline earth metals are weaker reducing agents than alkali metals because of their comparatively high ionization energies.
(b) The reducing nature of alkaline earth metals follows the increasing order as Be < Mg < Ca < Sr < Ba.
(c) Compounds of Group-II A elements are colourless and diamagnetic due to the absence of unpaired electrons.
(d) Alkaline earth metals have high electrical and thermal conductivities as two s-electrons can move through the crystal lattice.

23. Property of the alkaline earth metal that increases with increase is their atomic number
(a) Solubility of their hydroxides
(b) Decomposition temperature of their carbonates
(c) Solubility of their sulphates
(d) Basic character of their oxides

24. Which of the following properties show a reverse trend on moving from magnesium to calcium within the Group-II?
(a) Density
(b) Solubility of sulphates
(c) Ionization
(d) Melting point of oxides

25. The metals that can form water soluble sulphates, water insoluble hydroxides and oxides which become inert on heating are
(a) Be (b) Mg
(c) Ca (d) Al

Comprehension Type Questions

Passage-I

$$\text{Metal} + \text{dil.HCl} \longrightarrow \text{A} \xrightarrow[\text{NH}_4\text{OH}]{\text{Na}_2\text{HPO}_4} \text{B} \quad \text{(white ppt).}$$

$$\text{A} \xrightarrow[\text{Heated}]{\text{HCl(g)}} \text{C} \xrightarrow[\text{presence of NaCl}]{\text{Electrolysis in}} \text{Metal (M)}$$

1. The compound A is
   (a) $CaCl_2 \cdot 2H_2O$ (b) $MgCl_2 \cdot 6H_2O$
   (c) $Na_2SO_4 \cdot 10H_2O$ (d) $CaSO_4 \cdot 2H_2O$
2. The compound B is
   (a) $Mg(NH_4)PO_4$ (b) $Ca_3(PO_4)_2 + NH_3$
   (c) $Na(NH_4)HPO_4$ (d) both (a) and (b)
3. The compound C and metal M are
   (a) NaCl, Na (b) $CaCl_2$, Ca
   (c) $MgCl_2$, Mg (d) $BeCl_2$, Be

Passage-II

2.2 g of a gas (P) which turned lime water milky was evolved when 8.4 g of a white solid (Q) was heated. Another gas (R) weighing 0.9 g was also evolved during this heating of white solid (Q). The residue (S) was left which on dissolving in water alkaline and gave a white precipitate (T) when excess of $BaCl_2$ was added. The precipitate gave brisk effervescence to give CO_2 when an acid was added.

1. The molecular weight of Q is
   (a) 84 (b) 100
   (c) 85 (d) 101
2. The bicarbonate 'M' is
   (a) More soluble than bicarbonate of K
   (b) Less soluble than bicarbonate of 'K'
   (c) More soluble than bicarbonate of Mg
   (d) Equally soluble as bicarbonate K
3. The compound 'T' is
   (a) $NaHCO_3$ (b) $BaCO_3$
   (c) Na_2CO_3 (d) K_2CO_3

Passage-III

Solubility depends on the lattice energy of the solid and the hydration energy of the ions. Some lattice energy values for Group-II compounds are much higher than the values for Group-I compounds, because of the effect of the increased charge on the ions in the Born–Lande equation. Taking any one particular negative ion the lattice energy decreases as the size of the metal increases. The hydration energy also decreases as the metal ions become larger. For a substance to dissolve, the hydration energy must exceed the lattice energy. Consider a related group of compounds, such as the chlorides of all the Group-II metals. On desending the group the metal ions become larger and so both the lattice energy and the hydration energy decreases. A decrease in lattice energy favours increased solubility, but a decrease in hydration energy favours decreased solubility. These two factors thus change in opposite directions, and the overall effect depends on which of the two has changed most.

1. The correct order of solubility is
   (a) $BeF_2 > CaF_2 > SrF_2 > BaF_2$
   (b) $BeF_2 < CaF_2 < SrF_2 < BaF_2$
   (c) $CaF_2 < SrF_2 < BaF_2 < BeF_2$
   (d) $CaF_2 > SrF_2 > BaF_2 > BeF_2$
2. The correct order of hydration energies is
   (a) $Be^{2+} < Ca^{2+} < Sr^{2+} < Ba^{2+}$
   (b) $Be^{2+} < Sr^{2+} < Ca^{2+} < Ba^{2+}$
   (c) $Be^{2+} > Sr^{2+} > Ca^{2+} > Ba^{2+}$
   (d) $Be^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$
3. The correct order of solubility is
   (a) $MgSO_4 < CaSO_4 < SrSO_4 < BaSO_4$
   (b) $MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$
   (c) $CaSO_4 > MgSO_4 > BaSO_4 > SrSO_4$
   (d) $MgSO_4 > SrSO_4 > CaSO_4 > BaSO_4$

Passage-IV

In the compounds of beryllium, polymerization may occur through bridging groups such as H, F, *Cl* or CH_3 giving chain polymers of the type $(BeF_2)_n$, $(BeCl_2)_n$ and $[Be(CH_3)_2]_n$. In $(BeCl_2)_n$. The bridge $Be(\mu_2 Cl)_2$. Be units are somewhat elongated in the direction of the chain axis. The distortions in the four coordinate Be atom are dependent on the nature of the bridging group and are released due to the presence of lone pairs on bridging atoms.

Alkoxides of beryllium $[Be(OR)_2]_n$ usually have associated structures with both μ_2-bridging and termial OR groups. For example, $[Be(OCH_3)_2]_n$ is a high polymer insoluble in hydrocarbon solvents. On the other hand, the tertiary-butoxy derivative is less condensed being only a trimar $[Be(O\text{-}t\text{-}Bu)_2]_3$. Only when the alkoxide is bulky are monomers obtained with two-coordinate Be.

1. The hybridization involved in beryllium in $[BeCl_2]_n$ is
   (a) sp (b) sp^2
   (c) sp^3 (d) dsp^2
2. The shape of $[BeCl_2]_n$ in solid state is
   (a) Polymeric regular tetrahedron
   (b) Polymeric irregular tetrahedron
   (c) Polymeric linear
   (d) Planar trigonal
3. The internal (bridge) *Cl*-Be-*Cl* angles in $[BeCl_2]_n$ is
   (a) $109°.28^1$ (b) Less than 109.28^1
   (c) More than $109° 28^1$ (d) 180°

4. The C-Be-C angle in $[Be(CH_3)_2]_n$ is
   (a) Equal to *Cl*-Be-C angle in $[Be(Cl_2)]_n$
   (b) Less than the *Cl*-Be-C angle in $[BeCl_2]_n$
   (c) Greater than the *Cl*-Be-C angle in $[BeCl_2]_n$
   (d) No relation to *Cl*-Be-C angle in $[BeCl_2]_n$
5. With bulky alkoxide groups the beryllium compound exists as monomers which have
   (a) linear structure with 180° bond angle
   (b) trigonal planar structure with 120° bond angle
   (c) tetrahedral structure with $109°28^1$ bond angle
   (d) distorted tetrahedral structure with more than $109°28^1$ in internal C-Be-C bond
6. The vapour phase $BeCl_2$ is
   (a) two coordinate with linear structure
   (b) three coordinate with planar triangular structure
   (c) four coordinate with regular tetrahedral structure
   (d) four coordinate with irregular tetrahedral polymeric structure

Passage-V

Magnesium is a valuable, light weight metal used as a structural material as well as in alloys, batteries and in chemical synthesis. Although magnesium is plentiful in earth's crust, it is mainly found in the sea water (after sodium). There is about 1.3 g of magnesium in every kilogram of sea water. The process for obtaining magnesium from sea water employs all three types of reactions, i.e., precipitation, acid–base and redox-reactions.

1. Precipitation reaction involves formation of
   (a) insoluble $MgCO_3$ by adding Na_2CO_3
   (b) insoluble $Mg(OH)_2$ by adding $Ca(OH)_2$
   (c) insoluble in $MgSO_4$ by adding Na_2SO_4
   (d) insoluble $MgCl_2$ by adding NaCl
2. Acid–base reaction involves reaction between
   (a) $MgCO_3$ and HCl (b) $Mg(OH)_2$ and H_2SO_4
   (c) $Mg(OH)_2$ and HCl (d) $MgCO_3$ and H_2SO_4
3. Redox reaction takes place (in the extraction of Mg)
   (a) In the electrolytic cell when fused $MgCl_2$ is subjected to electrolysis
   (b) When fused $MgCO_3$ is heated
   (c) When fused $MgCO_3$ is strongly heated
   (d) None of the above

Passage-VI

A halide of Be(X) sublimes on heating and is a bad conductor of electricity in molten state. From its aqueous solution it is difficult to obtain anhydrous salt. This halide of beryllium(X) is obtained by heating beryllium oxide with carbon tetrachloride at 800°C. This halide of beryllium forms a complex of the type $M_2[BeX_4]$.

1. What is the compound (X)?
   (a) $BeBr_2$ (b) BeF_2
   (c) $BeCl_2$ (d) any of these
2. What is the compound formed when $BeCl_2$ dissolves in water in cold conditions?
   (a) $Be(OH)_2$ (b) BeO
   (c) BeO_2 (d) $[Be(H_2O)_4]Cl_2$
3. Beryllium chloride forms $Na_2[BeCl_4]$ with NaCl because Be has
   (a) Vacant d-orbitals
   (b) Be^{2+} is small-sized and has high charge density
   (c) Both of these
   (d) $BeCl_2$ is a covalent solid

Match the Following Type Questions

Matching Type Questions

1. Column-I (Chemical Prop.)	Column-II (Metals)
(a) Metal sulphate $\xrightarrow{\Delta}$ metal oxide + SO_2 + O_2	(p) Ba
(b) Metal cation + H_2SO_4 $\longrightarrow$ white ppt.	(q) Sr
(c) Metal + NH_3 $\xrightarrow{\text{(liquid)}}$ blue solution	(r) Na
(d) MCl_2 + conc. H_2SO_4 $\longrightarrow$ white ppt.	(s) Mg

2. Match the following

List-I	List-II
(a) $BeCl_2$ (solid)	(p) Polymeric
(b) BeH_2 (solid)	(q) sp^3 hybridization
(c) BeO (solid)	(r) Amphoteric
(d) MgO (solid)	(s) Refractory material

3. Match the following

List-I	List-II
(a) CaO	(p) Refractory material
(b) $CaCl_2$	(q) Drying agent
(c) MgO	(r) Sorrell's cement
(d) $MgCl_2$	(s) Antacid

Numerical Type Questions

1. In epsom salt, the number of water molecules coordinated to Mg^{2+} is
2. In solid ionic BeO, the coordination number of Be is
3. No. of nitrate ions acting as bridges in basic beryllium nitrate is

4. In basic beryllium acetate, the number of acetate ions acting as bridges are

IInd Group Key

Only One Answer Questions

1. b	**2.** b	**3.** a	**4.** a	**5.** a	**6.** c	**7.** b
8. b	**9.** c	**10.** c	**11.** a	**12.** b	**13.** a	**14.** a
15. b	**16.** b	**17.** b	**18.** d	**19.** d	**20.** c	**21.** a
22. d	**23.** a	**24.** b	**25.** b	**26.** d	**27.** b	**28.** b
29. b	**30.** b	**31.** a	**32.** a	**33.** d	**34.** a	**35.** b
36. a	**37.** a	**38.** d	**39.** d	**40.** d	**41.** d	**42.** d
43. c	**44.** a	**45.** d	**46.** a	**47.** c	**48.** c	**49.** d
50. c						

More than One Answer Questions

1. a, b	**2.** a, b	**3.** a, b, c
4. a, b, c	**5.** a, b, c	**6.** a, c, d
7. a, b	**8.** a, b, d	**9.** a, c, d
10. a, c, d	**11.** a, b, c	**12.** a, b, c, d
13. a, b, d	**14.** a, b, c, d	**15.** a, b, c
16. b, c	**17.** a, b	**18.** c
19. d	**20.** a, c, d	**21.** a, c, d
22. a, b, c, d	**23.** a, b, d	**24.** a
25. a, b, d		

Comprehensive Type Questions

Passage I	1-b	2-a	3-c			
Passage II	1-a	2-b	3-c			
Passage III	1-c	2-d	3-d	4-d		
Passage IV	1-c	2-b	3-b	4-b	5-a	6-b
Passage V	1-b	2-c	3-a			
Passage VI	1-c	2-d	3-b			

Matching Type Questions

1. a-p, q, s	b-p, q	c-p, q, r, s	d-p,q
2. a-p, q	b-p, q	c-r, s	d-s
3. a-p, q	b-q	c-p, r, s	d-r

Numerical Type Questions

1. 6 **2.** 4 **3.** 6 **4.** 6

Hints

1. Na forms oxide but Mg can form both MgO and Mg_3N_2.
2. Solubility of hydroxides of Group-2 elements increases down the group.
3. Polymeric $(BeCl_2)_n$ contain one dative bond from $Cl \longrightarrow Be$ and one normal 2e covalent bond by each Cl atom with two Be atoms.
4. $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
   $CaO + H_2O \longrightarrow Ca(OH)_2$
   $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$
   $CaCO_3 + H_2O + CO_2 \longrightarrow CaHCO_3)_2 \xrightarrow{\Delta} CaCO_3 + H_2O + CO_2$
5. Li, Mg and Al react with N_2 in air forming metal nitrides which on hydrolysis gives NH_3 that gives white fumes with HCl.
6. MgC_2 on heating converts into Mg_2C_3 which on hydrolysis gives propyne.
7. $BeSO_4$ is water soluble but $Be(OH)_2$ is insoluble in water, BeO is amphoteric.
8. $BeO + C + Cl_2 \longrightarrow BeCl_2 + CO$
   $BeCl_2 + 2H_2O \longrightarrow Be(OH)_2 + 2HCl$
9. $Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2$
   $CaH_2 + 2H_2O \longrightarrow Ca(OH)_2 + 2H_2$
10. $NaHCO_3$ can be prepared into solid state but others exist only in solution.
11. Coordination number of Be is 4 but for other metals it is 6.
14. Smaller the size with more number of charges, more the attraction towards water molecules.
15. $CaO + 2C \xrightarrow{2000°C} CaC_2 + CO$
20. Compounds of 2nd group elements are lesser ionic and more covalent due to lesser electropositive character and more number of the charges on ions.
21. Be is less electropositive than Mg.
22. When gypsum converts into plaster of paris, mol. wt decreases due to loss of water, so percentage of calcium increases.
23. Na_2SO_4 forms insoluble $BaSO_4$.
31. $MgCl_2$ with bicarbonate ions forms soluble $Mg(HCO_3)_2$ which on heating forms insoluble $MgCO_3$.
32. Solubility of oxalates, hydroxides and fluorides increases from Ca to Ba but solubility of sulphates decreases.
33. As the size of cation increases, lattice energy decreases.
34. $MgSO_4 + NH_3 + Na_2HPO_4 \longrightarrow Mg(NH_4)PO_4 + Na_2SO_4$
    $2Mg(NH_4)PO_4 \longrightarrow Mg_2P_2O_7 + 2NH_3 + H_2O$
35, 36. Refer to consequences of lattice energies and solubility of compounds.
37. As the number of electrons participating in metalic bond increases bond strength increases and hence are harder and have high m.pts.
38. $BeCl_2 + N_2O_4 \longrightarrow Be(NO_3)_2 \cdot N_2O_4 \xrightarrow[\text{Vaccuum}]{50°C}$
    $Be(NO_3)_2 \xrightarrow{125°C} [Be_4O(NO_3)_6]$
44. $CaCN_2 + 3H_2O \longrightarrow CaCO_3 + 2NH_3$
    $2NH_3 + 3CaOCl_2 \longrightarrow 3CaCl_2 + 3H_2O + N_2$
46. $CaSO_4$ exhibits retrograde solubility with temperature.

48. $BaCl_2 + K_2CrO_4 \longrightarrow BaCrO_4 + 2KCl$
Calcium and strontium chromates are soluble in acetic acid.
$BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HCl$
$BaSO_4 + Na_2CO_3 \longrightarrow BaCO_3 + Na_2SO_4$
K_{sp} of $BaCO_3$ is less than $BaSO_4$.

49. Magnesium salts give basic magnesium carbonate with Na_2CO_3 and normal carbonate with $NaHCO_3$.

50. Solubility of $CaSO_4$ in water increases in the presence of $(NH_4)_2\ SO_4$ due to formation of double salt.

More than One Answer Type Questions

1. Calcium is less denser than magnesium and the radius of Mg is smaller than Ca.

2. CaF_2 and $BaSO_4$ are insoluble in water.

3. Solubility of hydroxides in water and thermal stability of carbonates of Group-2 elements increases down the group.

8. $BeCO_3$ is unstable and decomposes.

$$BeCO_3 \rightleftharpoons BeO + CO_2$$

To prevent the forward reaction, it should be stored in the atmosphere of CO_2. Be is amphoteric, dissolves in alkalis forming beryllate ion $[Be(OH)_4]^{2-}$, BeF_2 combines with NaF forming complex ion $[BeF_4]^{2-}$.

9. Refer to Fig. 6.2.

11. Solubility of a compound is more if hydration energy exceeds the lattice energy. Solubility of hydroxides increases while the solubility of chlorides decreases down the group.

12. Due to bigger size alkali metals have poor complex forming ability. Due to small size and more charge Mg^{2+} can form complex. LiF is less soluble due to more lattice energy while CsI is less soluble due to small hydration energies. But LiI and CsF are more soluble. Thermal stability of hydrides decreases with increase in the size of metal ion. Li_2CO_3 decomposes to Li_2O and CO_2.

13. $Ca(OH)_2$ and $Ba(OH)_2$ are strong alkalis and caustic in property. The difference between lattice enthalpies of the oxides and peroxides or oxides and carbonates is less when cation is smaller. Hence, peroxides or carbonates of smaller cation decompose easily because the energy liberated in the formation of oxide of small cation is more. As this difference increases the stability increases with increase in cation size.

17. With increase in the size of ion, lattice energy decreases, but increases with increase in the number of charges.

Comprehensive Type Questions

Passage-I dil $Mg + 2HCl$ (dil) $\longrightarrow MgCl_2 \cdot 6H_2O \xrightarrow[NH_4OH]{NaHPO_4} Mg(NH_4)PO_4$

$MgCl_2 \cdot 6H_2O \xrightarrow[\Delta]{HCl(g)} MgCl_2$ (anhydrous) $\xrightarrow{\text{Electrolysis}} Mg + Cl_2$

Passage-II $2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + 2CO_2$

84	9	22
8.4g	0.9g	2.2g

$Na_2CO_3 + 2H_2O \longrightarrow 2Na^+ + 2OH^- + H_2CO_3$

$Na_2CO_3 + BaCl_2 \longrightarrow BaCO_3 + 2NaCl +$

$NaHCO_3$ is less soluble than $KHCO_3$.

Passage-IV

1. Every Be in polymer is four coordinate, so sp^3 hybrid.

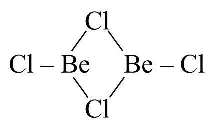
2. Due to elongation along the chain the tetrahedrons are distorted.

3. Due to elongation along the chain internal Cl-Be-Cl angle becomes less than the tetrahedral angle.

4. The C-Be-C angle is less than the Cl-Be-Cl angle because in $(BeCl_2)_n$ the repulsion between the lone pairs of Cl atoms which is absent in $[Be(CH_3)_2]_n$.

5. The monomers of Be compounds are linear with bond angle 180°.

6. The vapour phase $BeCl_2$ is



Numerical Questions

1. In $MgSO_4 \cdot 7H_2O$ six water molecules are coordinated to Mg^{2+} ion and one water molecule is in hydrogen bond with SO_4^{2-} ions.

2. Solid BeO is 4:4 coordinate solid.

3,4. In basic beryllium acetate and nitrate, the number of acetate or nitrate ions acting as bridges is six.

7 CHAPTER

p-Block Elements
Group-IIIA (13) Boron Family

The very word "experiment" refers to a situation where we can tell others what we have done and what we have learned.....

Niels Bohr

7.1 INTRODUCTION

The elements boron (B), aluminium (Al), gallium (Ga), indium (In) and thallium (Tl) constitute Group III A or 13 of the periodic table. They belong to p-block elements. The electronic configuration in their outer most orbit is $ns^2\ np^1$. These elements not only show marked similarities among them but also show a very wide variation in properties. Boron is a typical non-metal, aluminium is a metal but shows many chemical similarities to boron and the remaining elements are almost exclusively metallic in character. Although unipositive oxidation state is the characteristic one, for all the members of the group, the unipositive state occurs in compounds of all the elements except boron. In the case of thallium, the unipositive oxidation state is the stable one and infact it shows similarities to so many elements such as alkali metals, silver and mercury. Hence, it is nick named as **duck-billed platypus** among the elements. A feature of the chemistry of boron is the existence of large numbers of electron-deficient species which pose formidable problems in valence bond theory. These include not only the hydrides but also organic and metallic derivatives of the hydrides, the metal borides etc.

7.2 ABUNDANCE

Boron and aluminium of this family are considered to be familiar elements whereas gallium, indium and thallium are less familiar elements. Particularly aluminium is abundant in nature. The elements occur in nature in the following proportions in the crust of the Earth.

Boron $3 \times 10^{-40}\%$

Aluminium 8.13%

Gallium $1.5 \times 10^{-30}\%$

Indium $1 \times 10^{-50}\%$

Thallium 10^{-4} to $10^{-5}\%$

Aluminium is of course the most abundant of all the metals and the third most abundant of all the elements. The comparative scarcity of boron may be partially due to the ease with which the nuclei of its atoms are transmuted by natural bombardment process. Boron is well known, however, because of the existence of concentrated deposits of its compounds particularly in arid regions and because of the desirable properties of many of its compounds which have necessiated large-scale recovery of boron materials. Gallium, indium and thallium never found in concentrated deposits and until recently they were never recovered in sizable quantities.

7.3 OCCURRENCE

Borax $Na_2B_4O_7 \cdot 10H_2O$ or $Na_2\ [B_4O_5(OH)_4] \cdot 8H_2O$ is the principal source of boron. In India, it occurs in deserts and in the United States, at California. It occurs in hot springs and lakes in volcanic regions. Aluminium occurs mainly as bauxite (a hydrated oxide mineral), cryolite $Na_3[AlF_6]$ and also in the alumino silicate minerals such as mica and feldspar. Gallium, indium and thallium occur in traces in sulphide minerals. Gallium is also found in traces in bauxite.

7.4 ELECTRONIC CONFIGURATION

The electronic configuration of the elements of Group III A are listed in Table 7.1.

From Table 7.1, it follows that all the elements of Group III A have three electrons in their valence shell, two electrons in the s-orbital and one electron in the p-orbital, i.e., $ns^2\ np^1$.

Owing to the similarity in electronic configuration of their outermost shell, they closely resemble in their physical and chemical properties. However, it is to be noted that the penultimate shell of those five elements differ in the configuration; the penultimate shell of boron is having s^2, aluminium is having s^2p^6 and those of other three elements are having $s^2p^6d^{10}$ electrons. Thus, boron is expected to differ from aluminium and further boron and aluminium are expected to have some similar properties because penultimate shell of the B and Al has noble gas kernel and are different from other three elements.

7.5 PHYSICAL PROPERTIES

1. Atomic and ionic radii: The atomic and ionic radii are given in Table 7.2. Atomic and ionic sizes of Group III A elements do not increase regularly. The greater difference in the atomic radius between boron and aluminium is due to the fact that boron has lesser number of electrons (i.e., two electrons) in its inner shell than aluminium (i.e., eight electrons). Thus, in boron, the outer most electrons experience lesser shielding effect and greater nuclear attraction.

2. Density: As we move down in Group III A, density increases. However, boron and aluminium have comparatively low density. This can be attributed to their lower atomic weights as compared to the remaining elements. Ga is unusual because the liquid expands when it forms solid. Therefore, solid Ga is not denser than liquid Ga.

3. Melting point: The elements of Group III A do not show a regular change in their melting points with increase in atomic number. The melting point of boron is very high because it has the structure of giant covalent polymer in both solid and liquid states. The melting points decrease from B to Ga and then increase. Low melting point of Ga (29.8°) is attributed to the fact that it consists of only Ga_2 molecules; it remains liquid up to 2000°C and hence used in high-temperature thermometry.

Table 7.2 Atomic and ionic radii of Group III A elements

Element	Covalent radius (pm)	Metallic radius (pm)	Ionic radius (pm)	
			M^+	M^{3+}
B	80	(88.5)	–	27
Al	125	143	–	53.5
Ga	125	122.5	120	62
In	150	167	140	80
Tl	155	170	150	88.5

4. Boiling point: Boiling points of Group III A elements follow a regular decrease from boron to thallium. This shows that the strength of bonds holding the atoms in their liquid state decreases from boron to thallium. Note that the boiling point of Ga is in regular order with others whereas the melting point is not. The very low melting point is due to the unusual crystal structure, but the structure no longer exists in the liquid.

5. Heat of sublimation: It decreases regularly on moving down the group.

6. Ionization energies: The ionization energies increases as expected, i.e, first ionization energy < second ionization energy < third ionization energy. The sum of the first three ionization energies for each element is very high.

As the p-electron is less tightly held as compared to the s-electrons, the first ionization energy has been rather low in each case. The second and third ionization energies have been considerably higher.

Ionization energy decreases from boron to aluminium but does not change appreciably as we move to gallium, indium and thallium. Decrease in ionization energy from boron to aluminium is attributed to the

Table 7.1 Electronic configuration of Group III A elements

Element	Atomic number	Electronic configuration	Electronic configuration of valence shell
B	5	$1s^2\ 2s^2\ 2p^1$	$2s^2\ 2p^1$
Al	13	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$	$3s^2\ 3p^1$
Ga	31	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2 4p^1$	$4s^2\ 4p^1$
In	49	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^1$	$5s^2\ 5p^1$
Tl	81	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 4f^{14}\ 5s^2\ 5p^6\ 5d^{10}\ 6s^2\ 6p^1$	$6s^2\ 6p^1$

Table 7.3 Density, melting point, boiling point, electronegativity and heat of sublimation of Group III A elements

Element	Density (g cm^{-3})	Melting point (K)	Boiling point (K)	Heat of sublimation (kJ mol^{-1})	Electro negativity
B	2.35	2453	3923	564	2.0
Al	2.7	933	2740	324	1.5
Ga	5.9	303	2676	273	1.6
In	7.31	430	2353	241	1.7
Tl	11.85	576	1730	179	1.8

Table 7.4 Ionization energies of Group III A elements

	Ionization energies kJ mol^{-1}			
	Ist	**IInd**	**IIIrd**	**Sum of three**
B	801	2427	3659	6887
Al	577	1816	2744	5137
Ga	579	1979	2962	5520
In	558	1820	2704	5082
Tl	589	1971	2877	5437

increased size. Further, penultimate shell in both boron and aluminium has inert gas configuration (He configuration in the case of B and Ne configuration in the case of Al) whereas the penultimate shell in all the three remaining elements, i.e., gallium, indium and thallium has 18 electrons [$(n-1)s^2p^6d^{10}$]. The extra d-electrons fail to shield the nuclear charge effectively because shielding by electrons present in various orbitals has been found to be in the order s > p > d > f. Therefore, the outer electrons in the case of gallium, indium and thallium are held more tightly by the nucleus. Consequently, their atoms become smaller and thus their ionization energies become higher than expected. The ionization energy of thallium is further affected because of the poor shielding of 14 f-electrons present in the antipenultimate shell, and it is even more than that of Al, Ga and In.

For gallium and indium, the electronic configuration of the species left after the removal of three electrons is [Ar]$3d^{10}$ and [Kr] $4d^{10}$, respectively whereas for thallium, the species so formed has the configuration [Xe] $4f^{14}5d^{10}$. Thus, the fourth ionization energies of these three elements do not involve the removal of an electron from a noble gas configuration and the difference between the fourth and the third ionization energies is not nearly so large as for boron and aluminium.

7. Electronegativity: Among the Group III A elements, boron has the maximum electronegativity. It decreases from boron to aluminium as expected. However, from aluminium to thallium, it increases instead of decreasing in contrary to the expectation. This is again attributed to the poor shielding of d-electrons in gallium and indium and d- and f-electrons in thallium (Table 7.3).

8. Oxidation states: The Group III A elements contain three more electrons in their outer most orbit than the stable inert gas (with B and Al) or pseudo inert gas (with Ga, In and Tl) structures. Hence, a uniform +3 oxidation state is expected. This state is characteristic of all the elements. The electronic arrangement in the outer most orbit of these elements $ns^2\ np^1$ also suggests +1 oxidation states.

Outer electronic configuration

	ns	np			
Ground state	$\uparrow\downarrow$	$\uparrow$			+1
Excited state	$\uparrow$	$\uparrow$	$\uparrow$		+3

In thallium, +1 oxidation state is well known. A few unipositive indium compounds mostly halides are known, in the solid state. In boron, there are no evidences for +1 oxidation state. Though aluminium and gallium compounds are prepared at elevated temperatures, they yield most unstable unipositive compounds and the +1 oxidation state of these elements is of least important.

The stability of +1 oxidation state increases more and more when we move down the group from B to Tl. Thus, Tl(I) compounds are more stable than Tl(III) compounds. This is attributed to the inert pair effect. **The two s-electrons in the outer shell tend to remain paired and are not participating in compound formation**. This pair of electrons is called **inert pair** and the effect is called **inert pair effect**. The inert pair effect increases gradually in gallium, indium and thallium compounds. For example, Ga^+ compounds are unstable, In^+ compounds are moderately stable, whereas Tl^+ compounds are most stable. In fact, Tl(I) salts resemble alkali metals because thallium is having large size and low oxidation state. Some points of resemblence are as follows.

(i) TlOH is soluble in water yielding strong alkaline solution very similar to NaOH.
(ii) Tl (I) cyanide, perchlorate, sulphate, nitrate, phosphate and carbonate are stable and isomorphous with alkali metal salts.
(iii) TlF is having distorted NaCl-type structure whereas other thallous halides crystallize with CsCl structure.

(iv) Like alkali metals, thallium (I) is known to form alums, e.g., $Tl_2SO_4 \cdot Al_2(SO_4)_3\ 24H_2O$.

The stability of unipositive ions in aqueous solutions is consistent only in the case of thallium but for other elements the unipositive oxidation state disproportionate.

$$3M^+ \longrightarrow M^{3+} + 2M$$

Stability of +3 oxidation state decreases regularly as the atomic number increases from boron to thallium.

The stability of +1 oxidation state of Group III A elements should not be attributed strictly to inert pair effect. The inert pair effect explains what is happening, i.e., two electrons do not participate in bonding. The reason for not participating the two electrons in the outer most s-orbital in bonding can be explained basing on energy changes, i.e., excitation energy and the bond strength of the compound formed.

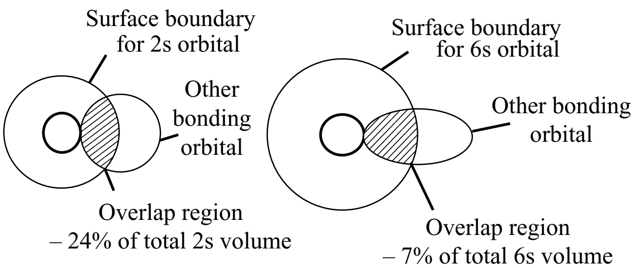


Fig 7.1 Comparison of overlappings between small (2s) orbital and large (6s) orbital with the orbital of another atom

The small sized 2s orbital of boron overlaps sufficiently with the orbital of another element (X) to yield strong M-X bonds that impart stability to the molecule. However, when we move downwards, the larger orbital (5s or 6s) is involved because of which overlapping is poor giving rise to lower bond energy of the M-X bond.

As the small energy of the M-X bond is not sufficient to compensate for the excitation energy of s-electrons, it follows that the larger elements show increasing tendency to form univalent compounds.

Only boron is sufficiently electronegative to show any tendency towards a negative oxidation state. In the borides of the most highly electropositive elements, boron presumably exists in the –3 oxidation state.

9. Nature of bonding: Boron never forms B^{3+} cation because the sum of the three ionization energies is very large. Further because of the very small size of B^{3+} ion (20 pm) in its ionic compounds, the tripositive boron ion will have much polarizing power on the adjacent atoms which results in the covalent character (Fajan's rules). Hence, in boron, +3 oxidation state is strictly covalent.

With other elements of Group III A, the +3 oxidation state is largely covalent. Tripositive cations are known in aqueous solutions for all the elements except boron. This is because of the fact that the hydration energies of tripositive cations overcome the ionization energies. Therefore, in aqueous solutions, they exist as hydrated cations and are greately hydrolysed in solution. For example, in the case of $AlCl_3$, the energy changes are as follows.

$$\Delta H_{\text{hydration}} \text{ for } Al^{3+} = -4665 \text{ kJ mol}^{-1}$$
$$\Delta H_{\text{hydration}} \text{ for } Cl^- = -381 \times 3 \text{ kJ mol}^{-1}$$

Total hydration energy = –5808 kJ mol^{-1}
Sum of the first three ionization energies = 5137 kJ mol^{-1}.

$$\text{for } Al \longrightarrow Al^{3+}$$

Thus, total hydration energy evolved (–5808 kJ mol^{-1}) more than offsets the ionization energy (5137 kJ mol^{-1}) required to convert Al to Al^{3+}.

The trivalent hydrated metal ions are having six molecules of water which are attached to them strongly giving an octahedral structure. These undergo hydrolysis to form acidic solutions. The strength of the metal oxygen bond in the hydrated ion would be able to weaken the O-H bonds. This causes some hydrolysis and protons get released giving acidic solution.

$$(H_2O)_5\,M \leftarrow O\begin{matrix}\diagup H \\ \diagdown H\end{matrix} \rightarrow [(H_2O)_5\,M \leftarrow O]^- + H^+ \quad (\text{with } O\text{–}H \text{ on the anion})$$

10. Electropositive character: Boron is a non-metal and its electropositive character is least. Aluminium is metal and is most electropositive. The remaining three elements gallium, indium and thallium are weakly metallic in nature and their electropositive character is less than that of aluminium and decreases from gallium to thallium.

The increase in electropositive character from boron to aluminium is ascribed to the increased size. The extra d^{10} electrons in case of gallium and indium whereas d^{10} and f^{14} electrons in the case of thallium do not shield the nuclear charge very effectively and these metals are, therefore, less electropositive. This is illustrated by the increase in ionization energy between aluminium and gallium even though the larger atom would be expected to have a lower value.

11. Electrode potentials: The standard electrode potentials E° for M^{3+}/M are given in Table 7.5.

The standard electrode potentials E° for M^{3+}/M become less negative from Al to Ga to In. As the free energy change $\Delta G = nFE°$ becomes more positive for the formation of the metal, the reaction $Al^{3+} + 3e^- \longrightarrow Al$ is not spotaneous. However, the reverse reaction $Al \longrightarrow Al^{3+} + 3e^-$ occurs spontaneously. As the standard potential

Table 7.5 Standard electrode potentials E°

-	M^{3+}/M (volts)	M^+/M (volts)
B	– 0.87*	-
Al	– 1.66	+ 0.55
Ga	– 0.56	– 0.79**
In	– 0.34	– 0.18
Tl	+ 1.26	– 0.34

* For $H_3BO_3 + 3H^+ + 3e^- \longrightarrow B + 3H_2O$
** Value in acidic solution.

becomes less negative down the group, so the reaction $M^{3+} \longrightarrow M$ occurs with less difficulty. Thus, the +3 oxidation state becomes less stable in aqueous solution on moving down the group. In a similar way, the increase in the stability of +1 oxidation state down the group is indicated by E° values for M^+/M. Thus, in thallium, +1 oxidation state is more stable than +3 oxidation state.

12. Complex formation: On account of their smaller size and greater charge, these elements have a much greater tendency to form complexes than the s-block elements. For example, the molecular trihalides or similar species of the elements of Group III A are still capable of accepting a pair of electrons and very large number of complexes such as $[BF_4]^-$, $[AlCl_4]^-$, $[GaCl_4]^-$ and $[InCl_4]^-$ are known. In boron, the second orbit is the outer most orbit and does not contain d-sub-shell but aluminium and other III A elements will contain d-subshells in their outer most orbits. Hence, aluminium and the heavier elements are not restricted to an octet of electrons in their valence shells. Hence, for these elements, coordination numbers higher than four may be found, e.g., $[AlF_6]^{3-}$ and $[TlF_6]^{3-}$. Thus, the covalence of boron is restricted to 4 only while the other elements can exhibit a covalence up to 6. Owing to this reason, boron cannot form complexes such as $[BF_6]^{3-}$.

13. Aqueous solution chemistry: All the tripositive aquo ions of these elements are acidic, that of aluminium the least and that of thallium the most so. Thus, aqueous solution of their salts are appreciably hydrolysed, and salts of weak acids (e.g., carbonates and cyanides) cannot exist in contact with water. In acidic solution, aluminium is present as the $[Al(H_2O)_6]^{3+}$ ion, as the acidity is decreased, polymeric hydrolysed species such as hydrated, $[Al_2(OH)_2]^{4+}$ and $[Al_7(OH)_{16}]^{5+}$ appear, then $Al(OH)_3$ is precipitated, and finally, in alkaline solutions, aluminate anions such as $[Al(OH)_6]^{3-}$ and $[Al(OH)_4]^-$ and polymeric species such as $[(HO)_3\ AlOAl\ (OH)_3]^{2-}$ are formed. The chemistry of gallium is broadly similar to that of aluminium in this respect, indium and thallium(III) hydroxides, however, are not amphoteric.

Redox potential data show that Al^{3+} (aq) is much less readily reduced than the other tripositive cations in aqueous solution. This, doubtlessly, arises partly from a more negative hydration free energy of the smaller Al^{3+} ion, but another important contributory factor is the increase in ionization energies between aluminium and gallium and between indium and thallium; there is relatively little variation in atomization enthalpies, and the overall variation in E° is, therefore, quite different from that in two preceding groups.

The E° value of Tl^{3+}/Tl^+, +1.26V indicates that it is a powerful oxidant. The value of E° is, however, very dependent upon the anion present, because Tl (I) resembles an alkali metal ion by forming a few stable complexes in aqueous solution (e.g., TlCl, unlike AgCl is not soluble in aqueous ammonia or potassium cyanide), whereas Tl (III) is very strongly complexed by a variety of anions. Thus, at unit chloride ion concentration, although TlCl is fairly insoluble, E° for the system $[TlCl_4]^-/TlCl$ is only +0.9 V. Iodide forms a more stable complex than chloride (soft acid-soft base relation) and at high iodide concentrations $[TlI_4]^-$ is a stable species even though $E^\circ_{Tl^{3+}/Tl^+}$ is much higher than $E^\circ_{½I_2/I^-}$ (+0.54V) and TlI is sparingly soluble. Thus, I_3^- $(I^- + I_2)$ in solid TlI_3 can under these conditions oxidize Tl (1) and bring about the reaction.

$$Tl^{I}I_3 + I^- \longrightarrow [Tl^{III}\ I_4]^-$$

In alkaline media, Tl(I) is also easily oxidized, as TlOH is soluble in water and hydrated $Tl_2O_3 \cdot 3H_2O$ or $2\ Tl(OH)_3$ is very sparingly soluble, with Ksp about 10^{-45}.

7.6 REACTIVITY OF GROUP III A ELEMENTS

Amorphous boron is reactive. It burns in air to yield the oxide and nitride and in the halogens forming trihalides. It reduces nitric and sulphuric acids and liberates hydrogen from alkalis. Crystalline boron is comparatively unreactive.

Aluminium is stable in air because of the formation of a protective oxide film. In the absence of this film, the metal rapidly gets oxidized and decomposes water. Gallium and indium are stable in air and are not attacked by water except when free oxygen present.

Thallium is a little more reactive and is superficially oxidized in air.

Reactivity of Group III A elements can be best illustrated by the following chemical properties.

1. Action of air: Pure boron is less reactive. It reacts with air only when heated.

$$4B + 3O_2 \longrightarrow 2B_2O_3$$

Aluminium reacts with air forming its oxide which protects it from further reaction.

$$4Al + 3O_2 \longrightarrow 2Al_2O_3;\ \Delta H = -3230,\ kJ\ mol^{-1}$$

The reaction is highly exothermic. When aluminium is in massive form, the extent of reaction is less because of less surface area. However, when finely powdered aluminium is kept in contact with liquid oxygen, owing to large surface area, the extent of reaction is more and thus the heat liberated is also more. Therefore, when aluminium is in massive form, it is not attacked by oxygen because of the formation of protective oxide layer but when it is exposed in powdered form, it explodes in contact with liquid oxygen.

Gallium and indium are stable in air whereas thallium forms an oxide layer.

Boron and aluminium when burnt in air react with both oxygen and nitrogen in air forming oxides and nitrides.

$$4B + 3O_2 \longrightarrow 2B_2O_3 \qquad 2B + N_2 \longrightarrow 2BN$$
$$4Al + 3O_2 \longrightarrow 2Al_2O_3 \qquad 2Al + N_2 \longrightarrow 2AlN$$

2. Reaction with water: Boron is not affected by water or steam. However, red hot boron decomposes steam liberating H_2.

$$2B + 3H_2O \longrightarrow B_2O_3 + 3H_2$$

Thermodynamically, aluminium should react with water, but owing to the presence of oxide layer on the surface, it do not react with cold water. However, it corrodes in salt water because the salts remove the oxide layer on the surface. Owing to this reason, aluminium vessels should not be kept filled with water overnight. Aluminium can decompose the steam.

$$2Al + 3H_2O \longrightarrow Al_2O_3 + 3H_2$$

Gallium and indium are attacked by water only in the presence of oxygen. Thallium is attacked by moist air to form TlOH.

$$4Tl + 2H_2O + O_2 \longrightarrow 4TlOH$$

3. Reaction with acids: Boron does not react with non-oxidizing acids such as HCl or with dil. H_2SO_4, but it reacts with strong oxidizing acids such as conc. HNO_3 or conc. H_2SO_4.

$$B + 3HNO_3 \xrightarrow{\Delta} H_3BO_3 + 3NO_2$$
$$2B + 3H_2SO_4 \xrightarrow{\Delta} 2H_3BO_3 + 3SO_2$$

The coherent protective oxide film present on the surface of aluminium metal prevents appreaciable reaction with dilute acids. Aluminium readily dissolve in dilute or concentrated hydrochloric acid. The reaction becomes more vigorous as the concentration of hydrochloric acid is increased.

$$2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2$$

Dilute sulphuric acid does not attack aluminium in cold condition because of insoluble oxide layer, whereas in hot condition, it liberates hydrogen. Concentrated sulphuric acid dissolves the metal easily and sulphur dioxide is liberated.

$$2Al + \underset{\text{dil}}{3H_2SO_4} \longrightarrow Al_2(SO_4)_3 + 3H_2$$
$$2Al + \underset{\text{Conc.}}{6H_2SO_4} \longrightarrow Al_2(SO_4)_3 + 6H_2O + 3SO_2$$

Very dilute nitric acid reacts slowly with impure aluminium forming aluminium nitrate and ammonium nitrate.

$$8Al + 30HNO_3 \longrightarrow 8Al(NO_3) + 3NH_4NO_3 + 9H_2O$$

Concentrated nitric acid makes aluminium passive because of the formation of a thin film of oxide layer on its surface.

Amalgamation with mercury or contact with solutions of salts of certain electropositive metals destroys the oxide film and permits further reaction.

Gallium also rendered passive with concentrated nitric acid. Indium and thallium react with acids forming salts.

4. Reaction with alkalis: Boron does not react with alkalis at low temperatures and resists attack by boiling concentrated aqueous NaOH or by fused NaOH up to 773K; however, at high temperatures, borates are formed.

$$2B + 6KOH \xrightarrow{T>773K} 2K_3BO_3 + 3H_2$$

Boron disolved by fusion mixtures such as Na_2CO_3/$NaNO_3$.

$$2B + 3Na_2CO_3 + 3NaNO_3 \xrightarrow{1173K} 2Na_3BO_3 + 3NaNO_2 + 3CO_2$$

Aluminium reacts with strong bases such as sodium or potassium hydroxide forming meta-aluminates or aluminates liberating hydrogen.

$$2Al + 2NaOH + 2H_2O \longrightarrow \underset{\text{Sodium meta aluminate}}{2NaAlO_2} + 3H_2\uparrow$$
$$2Al + 6NaOH \longrightarrow \underset{\text{Sodium aluminate}}{2Na_3AlO_3} + 3H_2$$

The meta aluminate ion in solution exists as the hydrated tetrahydroxo aluminate ion $[Al(OH)_4]^-_{(aq)}$ or $[Al(OH)_4\ (H_2O)_2]^-$. While the aluminate ion exists as $[Al(OH)_6]^{3-}$, in both cases, the coordination number of aluminium is 6.

Gallium also dissolves in alkali to give gallate. Indium and thallium are not affected by alkalis.

5. Reaction with metals: Boron reacts readily and directly with almost all metals at elevated temperatures except with the heavier members of groups 11–15. They are Ag, Au; Cd, Hg; Ga, In, Tl; Sn, Pb; Sb, Bi.

$$3Mg + 2B \longrightarrow Mg_3B_2$$

6. Reaction with non-metals: At high temperatures, boron reacts directly with all the non-metals except hydrogen, germanium, tellurium and the noble gases. Aluminium combines with most of the non-metals on heating. They do not react with hydrogen.

(i) With carbon, both boron and aluminium form carbides.

$$4B + C \xrightarrow{\Delta} B_4C$$

$$4Al + 3C \xrightarrow{\Delta} Al_4C_3$$

(ii) With nitrogen, both B and Al form nitrides.

$$2B + N_2 \xrightarrow{\Delta} 2BN$$

$$2Al + N_2 \xrightarrow{\Delta} 2AlN$$

(iii) With oxygen, both B and Al form oxides.

$$4B + 3O_2 \longrightarrow 2B_2O_3$$
$$4Al + 3O_2 \longrightarrow 2Al_2O_3$$

(iv) With sulphur, both B and Al form sulphides.

$$2B + 3S \longrightarrow B_2S_3$$
$$2Al + 3S \longrightarrow Al_2S_3$$

(v) With halogens, both B and Al form trihalides.

$$2B + 3X_2 \longrightarrow 2BX_3 \qquad X = F, Cl, Br \text{ or } I$$
$$2Al + 3X_2 \longrightarrow 2AlX_3$$

7. Reaction with ammonia: When heated with ammonia, both boron and aluminium form nitrides.

7.7 COMPOUNDS OF GROUP III A (13) ELEMENTS: A COMPARATIVE STUDY

7.7.1 Hydrides

These elements do not react directly with hydrogen. However, using indirect methods, some interesting hydrides of these elements have been prepared. These hydrides are important from the point of view of their structures.

Boron forms a number of stable covalent hydrides with general formulae B_nH_{n+4} and B_nH_{n+6}. The representative compounds of the two series are B_2H_6 and B_4H_{10}, respectively. They are often called as **boranes** by analogy with the alkanes.

Aluminium forms only one high molecular weight polymeric hydride $(AlH_3)x$ called **alane.** Recent structure investigation by x-ray and neutron diffraction studies show that it contains Al atoms surrounded octahedrally by six hydrogen atoms. The structure consists of $Al_2–H_2–Al_2$ bridges. On heating, it decomposes into its elements.

Gallium hydride GaH_3 is an unstable, volatile oil. Indium hydride $(InH_3)_x$ is an unstable polymeric solid whereas thallium hydride is extremely unstable.

Boron, aluminium and gallium form complex anionic hydrides by making use of the empty p-orbitals such as lithium borohydride $Li[BH_4]$, lithium aluminium hydride $Li[AlH_4]$ and lithium gallium hydride $Li[GaH_4]$. The formation of these hydrides is due to the hydride ion acting as *electron donor* XH_3 group behaves as *electron aceptor.*

$$BH_3 + H^- \longrightarrow BH_4^-$$

$$AlH_3 + H^- \longrightarrow AlH_4^-$$

$$GaH_3 + H^- \longrightarrow GaH_4^-$$

Electron acceptor   Electron donor

These complex hydrides are prepared as follows.

(i) $4LiH + AlCl_3 \xrightarrow{\text{ether}} Li[AlH_4] + 3LiCl$

(ii) $Li + Al + 2H_2 \xrightarrow[120-150°C]{\text{ether}} Li[AlH_4]$

Other complex hydrides can be prepared by similar methods. The stability of the complex hydrides increases when

(i) the extra electron can be completely stabilized by the anion.

(ii) the polarizing power of the central cation is maximum.

Thus, the alkali metal borohydrides are the most stable. Their stability increases with increasing size of the central cation. These complex hydrides, particularly $Li[AlH_4]$ and $Na[AlH_4]$, are extremely useful reducing agents. They are used to reduce aldehydes and ketones to alcohols and nitrites to amines.

7.7.2 Oxides

All the elements of Group IIIA form sesqui-oxides of the general formula M_2O_3. There is only spectroscopic evidence about the formation of Al_2O. Gallium and indium give only unstable monoxides Ga_2O and In_2O. Thallium forms most stable Tl_2O.

When strongly heated in air, boron burns with green flame to form acidic B_2O_3. Boron (III) oxide (B_2O_3) is commonly called boric oxide and is the anhydride of orthoboric acid H_3BO_3. It is prepared by heating orthoboric acid.

$$6H_3BO_3 \underset{+H_2O}{\overset{-H_2O}{\rightleftharpoons}} 2(HBO_2)_3 \underset{H_2O}{\overset{-H_2O \text{ Red hot}}{\rightleftharpoons}} 3B_2O_3$$

When strongly heated in air, aluminium burns with a brilliant white light, to form amphoteric Al_2O_3.

Gallium combines with oxygen to form white amphoteric oxide Ga_2O_3. It can also be prepared by the thermal decomposition of the nitrate or sulphate. Indium (III) oxide In_2O_3 is an yellow solid formed by thermal decomposition of its hydroxide, nitrate or sulphate. Thallium (III) oxide is a dark brown solid formed by the dehydration of $Tl(OH)_3$ when strongly heated in air.

B_2O_3 is acidic in nature, combines with basic metal oxides to form metaborates.

$$CoO + B_2O_3 \longrightarrow Co(BO_2)_2$$

B_2O_3 is an extremely stable solid and cannot be reduced by carbon at high temperatures. It dissolves in water forming boric acid.

$$B_2O_3 + 3H_2O \longrightarrow 2H_3BO_3$$

B_2O_3 reacts only with very strong acidic oxides such as P_2O_5 and SO_3 to give phosphates or sulphates respectively.

$$B_2O_3 + P_2O_5 \longrightarrow 2BPO_4$$
$$B_2O_3 + 3SO_3 \longrightarrow B_2(SO_4)_3$$

These reactions show the minimum basic character that B_2O_3 has. However, more recently, the above compounds are identified as double oxides rather than as salts. BPO_4 has silica-like structure, in which alternate silicon atoms have been replaced by boron and phosphorus and thus better considered a mixed oxide rather than a salt of cationic boron; nevertheless, it should be noted that aluminium phosphate has a similar structure.

Acidic character of B_2O_3 has been attributed to small size of born atom resulting in high positive charge density on atom. Owing to this, it pulls off electrons from water molecules resulting in the weakening of O-H bonds and facilitating the release of a proton giving acidic solution.

As the Al^{3+} and Ga^{3+} ions are relatively larger, their tendency to rupture the O-H bond by pulling off electrons becomes somewhat smaller. As a result of this, the oxides of these elements instead of being acidic becomes amphoteric. It means that the M-O and O-H bonds are nearly equal in polarity and hence their oxides are amphoteric in character.

Even though indium oxide In_2O_3 appears as amphoteric, it is mostly basic in nature. As the atomic size of indium is more than the first three elements, it must show more electropositive character; its properties also confirm the same.

In compact form, Tl_2O_3 is stable; however, in finely divided form when heated to 373K, it losses oxygen and converts into thallium monoxide Tl_2O. Tl_2O_3 is purely basic and will not show any acidic properties.

From the above discussion, it may be concluded that on moving down the group, there is a gradual change from acidic through amphoteric to basic character of the oxides. Boron oxide is purely acidic, aluminium oxide and gallium oxides are amphoteric and the last thallium oxide is purely basic in nature. This is due to the gradual change in electropositive character from boron to thallium.

In the lower oxide of thallium, Tl_2O, the thallium exhibits +1 oxidation state. It is more stable and it, like alkali metal oxides, dissolves in water to form a highly basic thallous hydroxide.

7.7.3 Hydroxides

The hydroxy compounds of boron and aluminium are formed when their oxides react with water. The low ionic radii and the high charges of the ions (+3) of the aluminim family account for the fact that the acidic properties of hydroxides become less pronounced with increasing ionic radius. B_2O_3, when hydrated, forms boric acid $B(OH)_3$ which is a weak acid ($pK_a = 9$) moderately soluble in water. In its reaction with water, it acts as a Lewis acid accepting OH^- ion from water.

$$B(OH)_3 + H_2O \longrightarrow [B(OH)_4]^- + H^+$$

In this reaction, boron again displays its tendency to accept a pair of electrons from the Lewis base OH^-. At lower concentrations, monomeric species such as $B(OH)_3$, $[B(OH)_4]^-$ are present whereas at higher concentrations (> 0.03M), these monomeric species get polymerized with the formation of polymeric species such as $[B_2O_3(OH)_4]^-$.

The hydroxide of aluminium $Al(OH)_3$ is amphoteric and dissolves both in acids and bases.

$$Al(OH)_3 + 3HCl \longrightarrow AlCl_3 + 3H_2O$$

$$Al(OH)_3 + 2NaOH \longrightarrow NaAlO_2 + 2H_2O$$

Practically, however, it acts as a base more than an acid and gives salts with acids. These salts contain the hydrated $[Al(OH)_6]^{3-}$ ion which shows more basic nature. $Al(OH)_3$ dissolves in sodium hydroxide to give sodium meta aluminate but reprecipitated by adding little acid or by passing carbon dioxide through the solution.

Gallium resembles aluminium in forming amphoteric oxide and hydroxide. The hydroxide is white gelatinous precipitate and forms gallate on reaction with alkali.

Indium and thallium sesquioxides are highly basic and do not form the respective hydroxides. They have practically no amphoteric character. Thallous hydroxide TlOH is a strong base. It differs from the trivalent hydroxide in that it is soluble in water. In this respect, it resembles the alkali metal hydroxides. When ever an element exists in several oxidation states, the lower oxidation state tends to be more basic.

7.7.4 Halides

Group III A elements form three types of halides viz mono halides, dihalides and trihalides, of which trihalides are much important for discussion whereas monohalides are somewhat less important.

Monohalides: Group III A elements form monohalides of the type MX in the gas phase at elevated temperatures.

$$AlCl_3 + 2Al \xrightarrow{\text{high temp.}} 3AlCl$$

These compounds are not very stable and are covalent except for Tl^+F^- which is ionic. Boron forms a number of stable polymeric monohalides $(BX)_n$.

$$B_2Cl_4 \xrightarrow{\text{Mercury discharge}} B_4Cl_4$$

$$B_2Cl_4 \xrightarrow{\text{slow decomposition}} B_8Cl_8,\ B_9Cl_9,\ B_{10}Cl_{10},\ B_{11}Cl_{11},\ B_{12}Cl_{12}$$

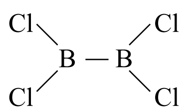
The compounds B_4Cl_4, B_8Cl_8 and B_9Cl_9 are crystalline solids and their structures have closed cage or polyhedron of boron atoms where each boron atom is bonded to three other boron atoms and one chlorine atom as boron has only three valence electrons, there are no enough electrons to form electron pair bonds and bonding probably involve multi-centre σ bonds covering all the boron atoms of the cage.

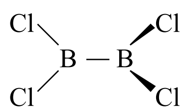
Gallium(I) chloride is produced by thermal decomposition of gallium trichloride at 1373K, but it has not been isolated. Pure indium(I) chloride can be isolated from the system $InCl_3$ - In.

Thallium(I) halides are stable compounds which in some ways resemble the silver(I) halides. Thus, thallium(I) fluoride is very soluble in water whereas the other halides are only sparingly soluble. The fluorides has a deformed sodium chloride structure, chloride and bromide have the caesium chloride structure, the iodide is dimorphic like mercury(II) iodide, existing in a red caesium chloride above 343K and an yellow layer lattice form below the temperature.

Dihalides: Boron forms dihalides of the formula B_2X_4. These decompose slowly at room temperature. B_2Cl_4 can be made as follows.

$$2BCl_3 + 2Hg \xrightarrow[\text{low pressure}]{\text{electric discharge}} B_2Cl_4 + Hg_2Cl_2$$





There is free rotation about the B-B bond, and in the gaseous and liquid states, the molecule adopts a non-eclipsed conformation. In the solid state the molecule is planar, because of crystal forces and ease of packing. Gallium and indium form dihalides.

$$GaCl_3 + Ga \longrightarrow GaCl_2 + GaCl$$

$$In + 2HCl_{(gas)} \longrightarrow InCl_2 + H_2$$

These are more properly written $Ga^+[GaCl_4]^-$ and contain M(I) and M(III) rather than divalent gallium and indium.

Trihalides: All the Group III A elements form trihalides of the type MX_3 (X = F, Cl, Br or I). However, $TlCl_3$ and $TlBr_3$ are unstable and tend to lose halogen, forming thallium(I) halide due to the oxidizing tendency of the +3 oxidation state of thallium coupled with reducing property of Cl^- and Br^-. In TlI_3, thallium exists in +1 oxidation state and contains $Tl^+\ I_3^-$ ions.

The decrease in stability of higher oxidation state as we go from fluoride to iodide is of course a general feature of the chemistry of all metals that show more than one oxidation state; for ionic compounds, it is easily explained by lattice energy accompanying an increase in oxidation state being greatest for the smallest anion.

All the boron halides BX_3 are covalent. Owing to the small size of boron atom, its halides are stable compounds. The fluorides of Al, Ga, In and Tl are ionic and have high melting points. The chlorides, bromides and iodides of these elements are largely covalent when anhydrous. Ionic character of halides of these elements increase with the increase in cation size. However, their covalent character decreases on moving from gallium and to thallium.

The trichlorides, tribromides and triiodides of aluminium, gallium and indium are all obtained by combination of the elements. They are relatively volatile and have layer lattices or lattices containing dimeric molecules, the vapours consist of dimeric molecules which are also present in solutions of the compounds in organic solvents.

Trihalides fume in moist air. They are readily soluble in water and undergo hydrolysis.

$$BCl_3 + 3H_2O \longrightarrow H_3BO_3 + 3HCl$$

$$AlCl_3 + 3H_2O \longrightarrow Al(OH)_3 + 3HCl$$

When water is dropped on to solid aluminium chloride, vigorous hydrolysis ensues, but in dilute aqueous solutions, $[Al\ (H_2O)_6]^{3+}$ and halide ions are present.

Lewis acid character of trihalides: In the trihalides of Group III A elements as the octet is not completed they can accept a pair of electrons. Hence, they act as Lewis acids. For example, AlF_3 forms the fluoroaluminate ion $[AlF_6]^{3-}$ by accepting three pairs of electrons from three F^- ions.

The comparison of relative strengths of trihalides of boron as Lewis acids has been done by measuring heats of the formation of compounds such as $H_3N \longrightarrow BX_3$

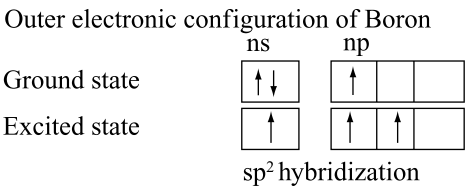
as well as by recording their infrared spectra. The power to accept lone pair of electrons has been found to increase in the following order.

$$BF_3 < BCl_3 < BBr_3$$

This order has been found to be reverse of what is expected on the basis of the electronegativities of halogens. This anomalous behaviour has been explained in terms of the tendency of the halogen atom by back bonding its electrons to the boron atom.

Boron trihalides are covalent compounds. In their formation, one s and two p-orbitals of boron atom undergo sp^2 hybridization, thus giving three sp^2 hybrid orbitals which are planar and inclined at 120° to one another. This sp^2 hybridization leaves one empty 2p orbital on boron. The three sp^2 hybrid orbitals of boron overlap with those of three halogen atoms giving rise to a planar triangular molecule. For example, the structure of BF_3 is shown in Fig 7.2.

Thus, in BF_3 molecule, one of the 2p orbitals of boron atom is vacant whereas each one of the three fluorine atoms has three completely filled 2p-orbitals, each having a lone pair of electrons. There occurs a lateral overlap between vacant p-orbital of boron and one completely filled 2p-orbital of fluorine. However, both the orbitals are 2p-orbitals. Therefore, an effective overlap takes place thus forming an additional pπ-pπ bond between the two atoms. It means that B-F acquires some double bond character. This type of bond formed by back donation of electron pair form fluorine to boron is known as back bonding. This has been illustrated in Figs 7.3 and 7.4.



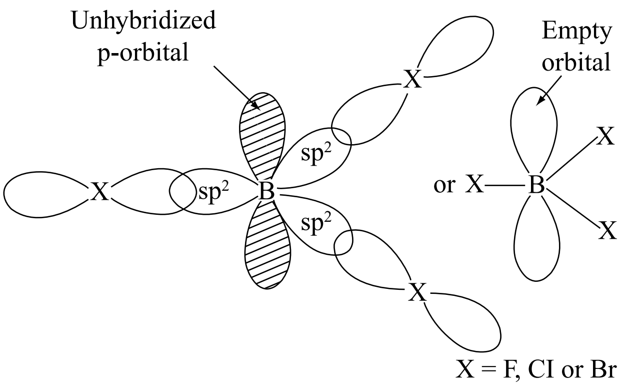


Fig 7.2 Structure of boron trihalides

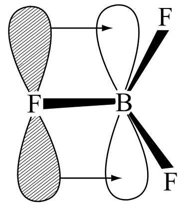


Fig 7.3 Formation of an additional pπ—pπ bond due to lateral overlap between vacant 2p orbital of boron and one completely filled 2p orbital of fluorine

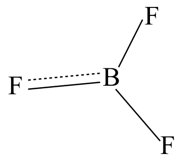


Fig 7.4 One B—F bond has some double bond character

As any one of the three fluorine atoms can participate in back bonding, the structure of BF_3 molecule may be assumed to be a resonance hybrid of the canonical structures in Fig 7.5.

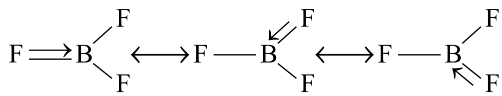


Fig 7.5 Various probable structures of BF_3 molecule involving back bonding

The electron deficiency of boron atom is partly made up because of the donation of electrons from fluorine to boron. The lateral overlap also takes place in other trihalides of boron between 2p orbital of boron and completely filled np orbital of halogen. However, the overlap is less effective because of the difference in the energies of overlapping orbitals, i.e., 2p of boron and np of halogen, which can be confirmed from the following discussion.

Even the weakest bases will form adducts with the trihalides of boron. Ethers, amines, phosphines, alcohols, anions, carbon monoxide and so on form adducts by the donation of an electron pair to boron. The rehybridization of boron that accompanies adduct formation results in a loss in B-X double-bond character, as shown in Fig 7.6.

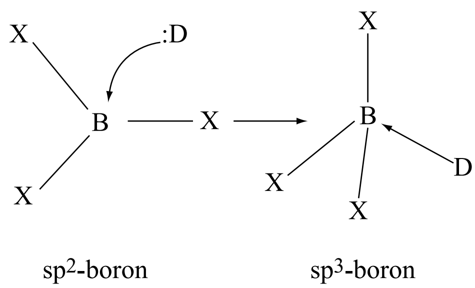


Fig 7.6 The reaction of trigonal trihalide of boron with a Lewis base (:D) to give tetrahedral adduct. The rehybridization of boron that is required when: D disrupts the B-X π bond in the BX_3 reactant

When the Lewis donor is trimethyl amine, the enthalpy change for adduct formation is more negative for BBr_3 and least for BF_3.

$$BX_3 + :N(CH_3)_3 \longrightarrow X_3B:N(CH_3)_3$$

We would expect that the higher electronegativity of fluorine should enhance the stability of the trimethyl amine adduct with BF_3. The least enthalpy of adduct formation with BF_3 indicates that the loss in B-X double-bond character upon rehybridization to form an adduct requires more energy. From this, we can conclude that the double bond character in the trihalides follows the order $BF_3 > BCl_3 > BBr_3$ a trend opposite to that expected from the electronegativities of the halides. Evidently, the π-bond system in BF_3 is especially strong because of effective overlap between the boron 2p and the fluorine 2p atomic orbitals – overlap because of the closely matched energies and sizes of the orbitals. The 3p and 4p atomic orbitals of Cl and Br have the proper symmetry for π overlap with the 2p atomic orbital of B in the compounds BX_3, but the π overlap is less effective because the energies and sizes of the π-donor orbitals (3p for chlorine and 4p for bromine) are not well matched to those of the π-acceptor (2p) orbital of boron.

From the above discussion, it can be seen that the tendency to form pπ—pπ bond is maximum in the case of BF_3 and decreases rapidly from BCl_3 to BBr_3. In other words, the tendency to accept electron pair increases from BF_3 to BBr_3, i.e., the strength of the trihalides as Lewis acids increases in the order

$$BF_3 < BCl_3 < BBr_3$$

Halides of aluminium and other members of the group have been also electron pair acceptors. However, this property gets decreased with the increase in the size of the cation. The Lewis acidities of the halides reflect the relative chemical hardness of the Group III A elements. Thus, towards a hard base (such as ethyl acetate, which is hard because of its O donor atoms), the Lewis acidities of the halides weaken as the softness of the acceptor element increases, so the Lewis acidities fall in the order.

Hard acids are Lewis acids which are small in size and whose electron charge cloud is not easily polarizable, e.g., light metal ions with high positive oxidation state.

Soft acids are Lewis acids which are comparatively larger in size and can be easily polarizable, e.g., heavy metal ions with low (or even zero) positive oxidation state.

Hard bases are Lewis bases which prefer to coordinate with hard acids. These are anions or neutral molecules which are not easily polarizable.

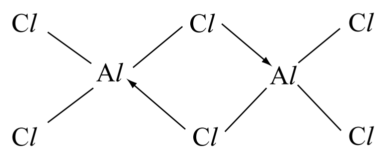
Soft bases are Lewis bases which prefer to coordinate with soft acids. These are anions or neutral molecules which are easily polarizable.

$$BCl_3 > AlCl_3 > GaCl_3$$

By contrast, towards soft Lewis base (such as dimethylsulfane, Me_2S which is soft because of its S atom), the Lewis acidities strengthen as the softness of the acceptor element increases.

$$GaX_3 > AlX_3 > BX_3 \ (X = Cl \text{ or } Br)$$

Structure: Boron trihalides are found only as monomers whereas the trihalides of aluminium, gallium and indium exist as dimers. The reason for this is that boron being small in size cannot coordinate with four large halide ions. However, now, a question arises that why BF_3 does not exist as dimer being the four small F atoms can surround the boron atoms. This can be understood easily that the rehybridization of boron that accompanies adduct formation results in the loss of resonance stability because of the loss in B-X double bond character. Aluminium, gallium and indium due to their large size can coordinate with four or more halide ions. Actually, they make use of vacant p-orbitals by another method, i.e., metal atoms complete their octet by forming dimers.



The structures of the halide dimers are almost exactly similar that of diborane but the bonding is different. In diborane, the hydrogen bridge bond is involved in three-centre two-electron bond whereas in the dimers of the halides, halogen bridges, in which a halogen atom bonded to one atom of Group III A element, use a lone pair of electrons to form coordinate bond to another atom of Group III A element. As the fluorides of the Group III A elements (except boron) are ionic compounds, they contain ion pairs not molecules.

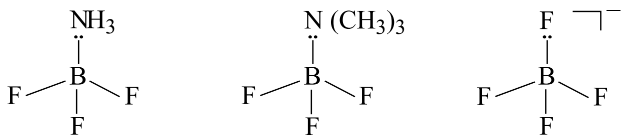
Bond lengths in boron halides: Both experimental and calculated bond lengths (the sum of covalent radii of boron atom and halogen atom) are given in Table 7.6. The covalent radius of boron atom is 80 pm.

From the values in Table 7.6, it can be noted that B-F bond length is significantly shorter than the sum of covalent radii of boron and fluorine when compared with the B-X bond lengths of BCl_3, BBr_3 and BI_3. This shortening of bond length clearly explains the effective overlapping of 2p-ortbitals of both boron and fluorine.

Table 7.6 Covalent radii, of halogen atom, calculated and experimental bond lengths in boron halides

Boron halide	Covalent radii of halogen (pm)	Calculated B-X bond length pm	Experimental B-X bond length pm	Difference
BF_3	72	152	131	21
BCl_3	99	179	174	5
BBr_3	114	194	187	7
BI_3	133	213	210	3

If the empty p-orbital on boron is filled with a lone pair of electrons from a donor molecule such as NH_3, $(CH_3)_3N$ or F^-, a tetrahedral molecule is formed.



The possibility for π bonding no longer exists and in $H_3N \longrightarrow BF_3$, $(CH_3)_3N \longrightarrow BF_3$ and $[BF_4]^-$, the B—F bond distances are 138 pm, 139 pm and 141 pm, respectively which are more than the experimental B—F bond length in BF_3. It can also be seen from the values in Table 7.6 that the difference in calculated and experimental bond distances of boron halides is decreasing from flourine to iodine indicating the decrease in effective overlapping.

7.8 ANOMALOUS BEHAVIOUR OF BORON

7.8.1 Differences of Boron from Other Elements

As usual, the first element boron of the Group III A shows different properties from the rest of the elements because of the following.

(i) Small size and high charge make the ion B^{3+} highly polarizing power. Thus, it does not exist. Almost all boron compounds are covalent.

(ii) Boron does not have d-orbitals. Thus, its coordination number is limited to four, whereas the other elements can have a coordination number of six.

(iii) Boron does not exhibit the **"inert pair"** effect.

(iv) Boron combines with metals forming borides whereas other elements do not combine. They form alloys with other metals.

(v) Boron cannot be attacked by non-oxidizing acids such as HCl whereas others are attacked.

(vi) Boron does not decompose water or steam whereas other elements of Group III A decompose hot water or steam.

(vii) Boron is non-metal and bad conductor of electricity but other elements are metals and good conductors of electricity.

(viii) Boron exhibits allotropy but others elements do not exhibit alltropy.

(ix) Boron never forms B^{3+} ion, but other elements can form M^{3+} ions.

(x) Boron forms a large number of volatile hydrides which are electron-deficient compounds whereas other elements form only one polymeric hydride. Thallium does not form hydride.

(xii) Boron halides are monomeric whereas the halides of the other elements are dimeric.

7.8.2 Similarities between Boron and Aluminium

(i) Electronic configuration: Both boron and aluminium have the same outer electronic configuration $ns^2\ np^1$.

(ii) Oxidation state: Both these elements exhibit +3 oxidation state. However, boron exhibits –3 oxidation state in metal borides.

(iii) Covalency: Both these elements form covalent compounds. However, aluminium may form electrovalent compounds with strong electron accepting groups or atoms.

(iv) Formation of oxides: Both these elements form similar sesquioxide of the type M_2O_3.

(v) Formation of chlorides: Both these elements form chlorides of the type MCl_3 when heated in a current of chlorine or by passing chlorine over the heated mixture of their oxides and charcoal.

$$2M + 3Cl_2 \longrightarrow 2MCl_3$$

$$M_2O_3 + 3C + 3Cl_2 \longrightarrow 2MCl_3 + 3CO$$

These chlorides are covalent and readily hydrolysed in water.

$$BCl_3 + 3H_2O \longrightarrow H_3BO_3 + 3HCl$$

$$AlCl_3 + 3H_2O \longrightarrow Al(OH)_3 + 3HCl$$

(vi) Formation of nitrides: Both the elements when heated with nitrogen or ammonia form nitrides.

$$2M + N_2 \longrightarrow 2MN$$

$$2M + 2NH_3 \longrightarrow 2MN + 3H_2$$

These nitrides undergo decomposition when heated with steam or sodium hydroxide liberating ammonia gas.

$$BN + 3H_2O \longrightarrow H_3BO_3 + NH_3$$

$$BN + 3NaOH \longrightarrow Na_3BO_3 + NH_3$$

$$AlN + 3H_2O \longrightarrow Al(OH)_3 + NH_3$$

$$AlN + NaOH + H_2O \longrightarrow NaAlO_2 + NH_3$$

(vii) Formation of sulphides: Both these elements react with sulphur at high temperature to form sulphides which undergo hydrolysis by water.

$$B_2S_3 + 6H_2O \longrightarrow 2H_3BO_3 + 3H_2S$$

$$Al_2S_3 + 6H_2O \longrightarrow 2\,Al(OH)_3 + 3H_2S$$

(viii) Formation of alkyl compounds: Both these elements form similar organic compounds with alkyl radicals.

(ix) Action with conc. H_2SO_4: Both these elements react with concentrated sulphuric acid to form sulphur dioxide

$$2B + 3H_2SO_4 \longrightarrow 2H_3BO_3 + 3SO_2\uparrow$$

$$2Al + 6H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 6H_2O + 3SO_2\uparrow$$

(x) Action with alkalis: They react with alkalis to evolve H_2.

$$2B + 6NaOH \longrightarrow 2Na_3BO_3 + 3H_2\uparrow$$

$$2Al + 2NaOH + 2H_2O \longrightarrow 2NaAlO_2 + 3H_2\uparrow$$

7.8.3 Dissimilarities of Boron and Aluminium

Boron and aluminium have more of difference than of similarity in their properties. This is due to the presence of two electrons in its next to outermost orbit of boron whereas there are eight electrons in the penultimate shell of aluminium. Boron differs from aluminium in the following respects.

(i) Boron is a typical non-metal whereas aluminium is a metal.

(ii) Boron exhibits allotropy whereas aluminium does not exhibit allotropy.

(iii) Crystalline boron is very hard whereas aluminium is sufficiently soft.

(iv) Aluminium is very good conductor of heat and electricity whereas boron is a bad conductor.

(v) Boron has very high melting point as compared with aluminium.

(vi) The maximum covalence shown by boron is 4 whereas aluminium shows a maximum covalence of 6.

(vii) Boron is not attacked even by steam whereas aluminium decomposes steam liberating hydrogen.

$$2Al + 3H_2O \longrightarrow Al_2O_3 + 3H_2$$

(viii) Dilute acids have no action on boron but concentrated. HNO_3 oxidizes it to boric acid. Aluminium evolves H_2 gas from dil. HCl and H_2SO_4 but concentrated. HNO_3 renders aluminium passive.

$$B + 3HNO_3 \longrightarrow H_3BO_3 + 3NO_2$$

$$2Al + 3H_2SO_4 \longrightarrow Al(SO_4)_3 + 3H_2$$

(ix) Boron dissolves in fused alkalis evolving hydrogen whereas aluminium reacts with hot alkali solutions.

$$2B + 6NaOH \longrightarrow 2Na_3BO_3 + 3H_2$$

$$2Al + 2NaOH + 2H_2O \longrightarrow 2NaAlO_2 + 3H_2$$

Borates are very stable as compared to the aluminates.

(x) Boron forms two types (B_nH_{n+4} and B_nH_{n+6}) of hydrides whereas aluminium does not form such hydrides.

(xi) The halides of boron are covalent in nature and get hydrolysed by water giving boric acid. Aluminium chloride in solution gives Al^{3+} ions.

(xii) Oxide and hydroxide of boron are acidic whereas of aluminium are amphoteric.

(xiii) Boron combines with metals to form borides, e.g., Mg_3B_2 but aluminium forms alloys only.

(xiv) Boron forms many covalent compounds as compared with aluminium.

7.8.4 Resemblance between Boron and Silicon: Diagonal Relationship

Boron shows more resemblences to silicon because of diagonal relationship as shown below.

(i) Non-metallic character: Boron and silicon are typical non-metals. They both have high melting points and are bad conductors of electricity.

(ii) Allotropy: Both these elements exhibit allotropy (amorphous and crystalline) crystalline forms are hard.

(iii) Density, atomic volume and electronegativity: The density and the atomic volumes of both the elements are low. Their electronegativities are almost similar (B = 2.0; Si = 1.8)

Their ionization energies (kJ mol^{-1}) are also almost similar (B = 801; Si-786).

(iv) Oxides: Boron and silicon burn in air or oxygen to form stable and acidic oxides B_2O_3, SiO_2. These oxides in association with water yield corresponding acids, boric acid and silicic acid. Both are weak acids.

$$B_2O_3 + 3H_2O \longrightarrow 2H_3BO_3$$
$$SiO_2 + H_2O \longrightarrow H_2SiO_3$$

(v) Extraction: Both these elements can be obtained by the reduction of their respective oxides with magnesium.

$$B_2O_3 + 3Mg \longrightarrow 2B + 3MgO$$
$$SiO_2 + 2Mg \longrightarrow Si + 2MgO$$

(vi) Carbides: The oxides of both these elements when fused with carbon form carbides; B_4C and SiC. These are very hard substances and are used as abrasives.

$$2B_2O_3 + 7C \longrightarrow B_4C + 6CO$$
$$SiO_2 + 3C \longrightarrow SiC + 2CO$$

(vii) Reaction with metals: Both combine with metals to form borides and silicides, which are decomposed by dilute acids to form volatile hydrides.

$$3Mg + 2B \longrightarrow Mg_3B_2$$
$$2Mg + Si \longrightarrow Mg_2Si$$
$$Mg_3B_2 + HCl \longrightarrow 3MgCl_2 + \text{Mixture of boron hydrides}$$
$$Mg_2Si + 4HCl \longrightarrow 2MgCl_2 + SiH_4$$

(viii) Reaction with halogens: Both boron and silicon form the halides with halogens. The fluorides of both are colourless fuming gases. The chlorides BCl_3 and $SiCl_4$ are liquids, which are readily hydrolysed by water to acids.

$$BCl_3 + 3H_2O \longrightarrow H_3BO_3 + 3HCl$$
$$SiCl_4 + 4H_2O \longrightarrow H_4SiO_4 + 4HCl$$

(ix) Hydrides: Both boron and silicon form a number of covalent hydrides.

$$4BCl_3 + 3LiAlH_4 \longrightarrow 3LiCl + 3AlCl_3 + 2B_2H_6$$
$$SiCl_4 + LiAlH_4 \longrightarrow LiCl + AlCl_3 + SiH_4$$

These hydrides are volatile, spontaneously inflammable and readily hydrolysed. These hydrides are known as boranes and silicoalkanes or silanes. Boron appears to act as a tetra-covalent element such as silicon in these hydrides.

Boroethane B_2H_6	Borobutane B_4H_{10}
Silicoethane Si_2H_6	Silicobutane Si_4H_{10}

(x) Action of alkalis: Both these elements and their oxides form borates and silicates with alkalis.

$$2B + 6\,NaOH \longrightarrow 2Na_3BO_3 + 3H_2$$
$$B_2O_3 + 2NaOH \longrightarrow \underset{\text{Sod. meta borate}}{2NaBO_2} + H_2O$$
$$Si + 2NaOH + H_2O \longrightarrow Na_2SiO_3 + 2H_2$$
$$SiO_2 + 2NaOH \longrightarrow Na_2SiO_3 + H_2O$$

(xi) Formation of esters: Both these elements form volatile esters of the type $B(OR)_3$ and $Si(OR)_4$ with alcohols.

$$B(OH)_3 + 3ROH \longrightarrow B(OR)_3 + 3H_2O$$
$$Si(OH)_4 + 4ROH \longrightarrow Si(OR)_4 + 4H_2O$$

7.9 BORON

Boron was known to the early Romans. It was used as a flux in metallurgy. In 1702, Hornberg prepared boric acid by the treatment of borax with sulphuric acid. In 1808, Davy isolated the boron by fusing boric acid with potassium. Its name was derived from Arabic word **burag** meaning white.

Occurrence: Boron occurs in combined state in nature. It is present about 10^{-3}% in the earth crust. The important ores of boron are

1. Borax or Tincal	$Na_2B_4O_7 \cdot 10H_2O$
2. Kernite or Rasorite	$Na_2B_4O_7 \cdot 4H_2O$
3. Boracite	$2Mg_3B_8O_{15}\ MgCl_2$
4. Colemanite	$Ca_2B_6O_{11} \cdot 5H_2O$
5. Panderinite	$Ca_2B_6O_{11} \cdot 3H_2O$
6. Borocalcite	$CaB_4O_7 \cdot 4H_2O$
7. Boric acid or Tuscany boric acid	H_3BO_3

All these ores are the compounds of boric acid.

Borax is one of the most important ores and found in Japan, Turkey, Tibet and Kashmir.

Extraction of Boron: The following two steps are involved for the extraction of boron from its minerals.

(i) Preparation of boron trioxide
(ii) Reduction of boron trioxide

(i) Preparation of boron trioxide

(a) From Borax: Borax is treated with calculated quantity of hydrochloric acid or sulphuric acid, when the sparingly soluble boric acid slowly separates out.

$$Na_2B_4O_7 + 2HCl \longrightarrow 2NaCl + H_2B_4O_7$$
$$H_2B_4O_7 + 5H_2O \longrightarrow 4H_3BO_3$$

The boric acid on heating converts into boron trioxide

$$2H_3BO_3 \xrightarrow{\Delta} B_2O_3 + 3H_2O$$

(b) From Colemanite: Colemanite is converted into borax by fusing or boiling with sodium carbonate. The insoluble calcium carbonate settles down and can be removed by filtration. The filtrate contains borax and sodium metaborate. The borax is crystallized from the mother liquor. The sodium metaborate present in mother liquor is treated with carbon dioxide gas to get borax again, which is changed into boric acid as mentioned earlier.

$$Ca_2B_6O_{11} + 2Na_2CO_3 \longrightarrow CaCO_3 \downarrow + Na_2B_4O_7 + 2NaBO_2$$

$$4NaBO_2 + CO_2 \longrightarrow Na_2B_4O_7 + Na_2CO_3$$

$$Na_2B_4O_7 + 2HCl \longrightarrow 2NaCl + H_2B_4O_7$$

$$H_2B_4O_7 + 5H_2O \longrightarrow 4H_3BO_3$$

The boric acid so obtained is converted into boron trioxide by heating.

$$2H_3BO_3 \longrightarrow B_2O_3 + 3H_2O$$

(ii) Reduction of boron trioxide

B_2O_3 can be reduced to boron by metals at high temperatures.

$$B_2O_3 + 3Mg \longrightarrow 2B + 3MgO$$

The boron obtained by magnesium is about 95–98% pure and is called **Moissan boron.** Other electropositive metals such as Li, Na, K, Be, Ca, Al and Fe can also be used for reduction of B_2O_3 but the product is generally amorphous and contaminated with refractory impurities such as metal borides.

Other methods

Massive crystalline boron (96%) can be prepared by reacting BCl_3 with zinc in a flow system at 900°C.

$$2BCl_3 + 3Zn \longrightarrow 2B + 3ZnCl_2$$

Electrolytic reduction of fused borates or tetra fluoroborates, e.g., KBF_4 in molten KCl/KF at 800°C gives boron. The process is comparatively cheap but yields only powdered boron of 95% purity.

The most effective method for the preparation of crystalline boron is the reduction of volatile boron compounds by hydrogen on a heated tantalum metal filament. Crystallinity improves with increasing temperatures. Amorphous products being obtained below 1000°C. Between 1000 and 1200°C, crystalline α and β rhombohedral modifications and above 1200°C tetragonal crystals will be obtained.

Thermal decomposition of boron hydrides when heated at temperatures up to 900°C gives amorphous boron whereas the thermal decomposition of BI_3 gives crystalline boron.

Boron can also be isolated from potassium fluoborate by heating it with potassium or magnesium.

$$KBF_4 + 3K \longrightarrow 4KF + B$$

$$2KBF_4 + 3Mg \longrightarrow 2KF + 3MgF_2 + 2B$$

The cooled mass is treated with hydrochloric acid to remove everything except dark brown amorphous powder of boron.

Properties: Boron exhibit allotropy. The allotropic forms of boron are of both amorphous and crystalline.

The crystalline boron contains clusters of boron atoms situated at the corners of a nearly regular icosahedraon. This basic B_{12} unit exists in many boron containing compounds

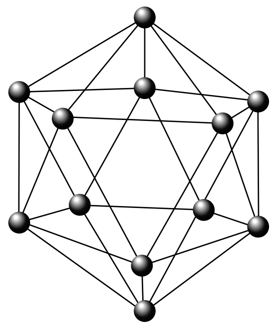


Fig 7.7 B_{12} icosahedral unit of boron

Amorphous boron is brown in colour and its density is 2.4 g·cm^{-3}. Its melting point is about 2475K but it volatilizes without melting at about 1875K. It is bad conductor of electricity. Crystalline boron is a blackish-grey, very hard and heat resistant. Its density is 3.3 cm^{-3}. It is also bad conductor of electricity. It is not effected by hot and concentrated nitric or sulphuric acid chemically. It is less active than amorphous form.

The chemical behaviour of boron is already discussed in Section 7.6.

Uses: It is used

(i) as a deoxidizer for metals in the form of calcium boride.
(ii) in making boron steels, which are very hard and used in atomic reactors as control rods.
(iii) as abrassive in the form of metallic borides.
(iv) as a rocket fuel in the form of boron hydrides.

Boron steel or boron carbide rods are used to control the nuclear reactions. Boron has a very high cross section to capture the neutrons. According to another concept, boron absorbs neutron to make the boron having even number of neutrons.

$${}^{10}_{5}B + {}^{1}_{0}n \rightarrow {}^{11}_{5}B$$

7.10 COMPOUNDS OF BORON

7.10.1 Boron Trioxide or Boric Oxide B_2O_3

Preparation: Boron troixide (boron sesqui-oxide) B_2O_3 is most common oxide of boron. It is formed when boron is burned in oxygen, but it is usually prepared by heating boric acid to redness.

$$4B + 3O_2 \longrightarrow 2B_2O_3$$

$$2H_3BO_3 \longrightarrow B_2O_3 + 3H_2O$$

It can also be prepared by dehydration of boric acid over phosphorous pentoxide at about 475K in vacuum.

Properties: It is a colourless transparent glassy mass. It has no definite melting point. It is hygroscopic. When exposed to the atmosphere, it absorbs moisture with which it combines become opaque and finally converting into boric acid.

$$B_2O_3 + H_2O \longrightarrow 2HBO_2 \text{ (Metaboric acid)}$$

$$HBO_2 + H_2O \longrightarrow H_3BO_3 \text{ (Orthoboric acid)}$$

Hence, it is also called as boric anhydride.

It is non-volatile even at red heat. It cannot be reduced to boron by carbon but the metals such as Na, K or Mg can reduce it into boron. It can displace other acids.

$$K_2SO_4 + B_2O_3 \longrightarrow 2KBO_2 + SO_3$$

$$CuSO_4 + B_2O_3 \longrightarrow Cu(BO_2)_2 + SO_3$$

The corresponding metaborates are colourled glassy bead and used for the identification of the metals which form coloured bead. This test is known as borax bead test.

It also exhibits feeble basic properties, as it forms unstable compounds such as $B_2(SO_4)_3$ and BPO_4 with SO_3 and P_2O_5.

$$B_2O_3 + 3SO_3 \longrightarrow B_2(SO_4)_3$$

$$B_2O_3 + P_2O_5 \longrightarrow 2BPO_4$$

Recently, the above compounds are identified as double oxides rather than as salts as already described in Section 7.7.2.

7.10.2 Boric Acids

Several boric acids are known (atleast in the form of salts); all these acids are derived from boron trioxide with different amounts of water.

Orthoboric acid	H_3BO_3	or	$B_2O_3 \cdot 3H_2O$
Metaboric acid	HBO_2	or	$B_2O_3 \cdot H_2O$
Pyroboric acid	$H_6B_4O_9$	or	$2B_2O_3 \cdot 3H_2O$
Tetraboric acid	$H_2B_4O_7$	or	$2B_2O_3 \cdot H_2O$

Salts also exist corresponding to more complex, condensed boric acids, e.g., $H_4B_6O_{11}$ and $H_2B_{10}O_{16}$. Of the free acids, orthoboric acid is the only important one, and of the salts **borax,** the most important boron compound, is sodium tetraborate.

Orthoboric acid: H_3BO_3

This is also known as simple boric acid and most important of the boric acids. It can be obtained from three different sources.

(i) From Borax: A hot and concentrated solution of borax is treated with calculated amount of sulphuric acid of hydrochloric acid. On cooling the solution, crystals of boric acid separate out.

$$Na_2B_4O_7 + H_2SO_4 + 5H_2O \longrightarrow Na_2SO_4 + 4H_3BO_3$$

(ii) From Colemanite: Colemanite is the most important source for the manufacture of boric acid. The powdered mineral is boiled with water and a stream of sulphurdioxide gas is passed through the solution when boric acid and calcium bisulphite are formed. On cooling, the hot solution boric acid crystallizes out.

$$Ca_2B_6O_{11} + 4SO_2 + 11H_2O \longrightarrow 2Ca(HSO_3)_2 + 6H_3BO_3$$

(iii) From Tuscany Soffioni: The jets of steam commonly known as soffioni, issuing from the ground in the volcanic regions of Tuscany contains traces of boric acid along with steam, carbon dioxide, hydrogen sulphide, ammonia and nitrogen. The boric acid is recovered by condensing the steam and concentrating the solution on cooling, the crystals of boric acid separates out.

It is also formed by the action of super heated water on boron nitride or boron sulphides.

$$2BN + 3H_2O \longrightarrow H_3BO_3 + 2NH_3$$

$$B_2S_3 + 6H_2O \longrightarrow 2H_3BO_3 + 2H_2S$$

Properties: It is a white crystalline substance, soft and soapy to touch. It is moderately souble in cold water. Its solubility increases with the rise in temperature. It is volatile in steam and imparts green colour to non-luminous flame.

On heating, it decomposes to form metaboric acid at 375 K, tetraboric acid at 435K and boron trioxide at red heat.

$$H_3BO_3 \xrightarrow{375K} HBO_2 \xrightarrow{435K} H_2B_4O_6 \xrightarrow{\text{red heat}} B_2O_3$$

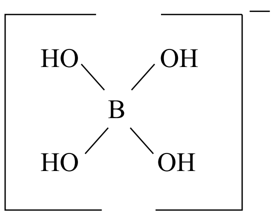
It is a very weak monobasic acid (pKa = 9.00). It does not liberate hydrogen ion but accepts a pair of electrons from OH^-, i.e., it behaves as Lewis acid not protonic acid.

$$H_3BO_3 + 2H_2O \longrightarrow [B(OH)_4]^- + H_3O^+$$

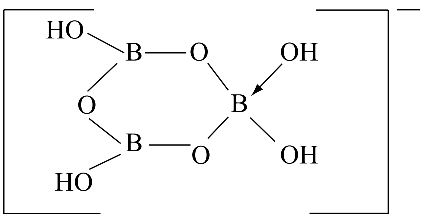
Metaborate ion $[B(OH)_4]^-$ is present at low concentration (< 0.025M). However, as the concentration of aqueous solution increases, the acidity increases (pKa = 6.84) because of the formation of polymeric metaborate $[B_3O_3(OH)_4]^-$

$$3H_3BO_3 \rightleftharpoons H_3O^+ + [B_3O_3(OH)_4]^-$$

The structures of metaborate ion and its polymeric form are given below.



Metaborate ion

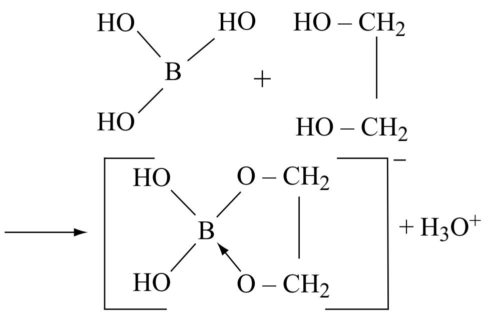


Polymeric borate ion

Being a weak acid, it cannot be titrated accurately with sodium hydroxide solution, because of the excessive hydrolysis of sodium metaborate $NaBO_2$.

$$H_3BO_3 + NaOH \longrightarrow NaBO_2 + 2H_2O$$

However, if titrations are carried out in the presence of polyhydroxy compounds such as mannitol, sorbitol, glucose or glycerol, boric acid acts as a much stronger acid and can be titrated using phenolphthalein as indicator.



If unsaturated diols are used only cis diols but not trans diols can form complexes and increase the strength of boric acid.

When boric acid is heated with ethyl alcohol, it gives ethyl borate vapours which burns with green edged flame.

$$H_3BO_3 + 3C_2H_5OH \longrightarrow B(OC_2H_5) + 3H_2O$$

This property is used for the detection of borate ion in the mixture. The mixture when treated with concentrated sulphuric acid gives boric acid.

Uses: It is used

(i) in making enamels, pottery, glazes and glass industry.
(ii) as antiseptic and in medicine (eyelotion).
(iii) as food preservative.

Structure: In the orthoborate ion BO_3^{3-}. boron atom is involved in sp^2 hybridization and is in bond with three OH groups in a planar triangular manner. Thus, the boric acid contains planar triangular $B(OH)_3$ units which are bonded together through hydrogen bonds into two-dimensional sheets.

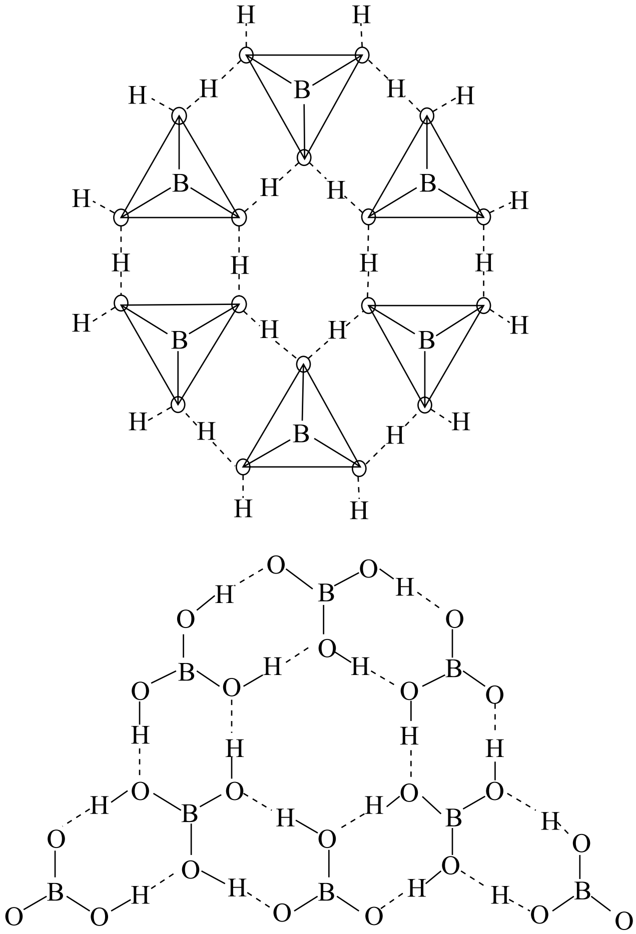


Fig 7.8 Hydrogen bonded structure of ortho boric acid

It is clear from Fig. 7.8 that each boron atom remains bonded to three oxygen atoms and each oxygen atom is bonded to hydrogen atom. Hence, hydrogen atoms act as bridge between the two oxygen atoms with a covalent bond on one side and a hydrogen bond on the other side of different BO_3^{3-} units.

7.10.3 Metaboric acid

It is prepared by heating orthoboric acid at 375K for some time.

$$H_3BO_3 \xrightarrow{375K} HBO_2 + H_2O$$

It is a white solid, stable below 475K, above which it decomposes to boric anhydride.

$$2HBO_2 \longrightarrow B_2O_3 + H_2O$$

It is soluble in water and forms ortho boric acid.

$$HBO_2 + H_2O \longrightarrow H_3BO_3$$

Metaborates are quite stable even at high temperature. It is clear from the above reactions that different metals form metaborates which are characterized by their different colours.

Sodium metaborate can be obtained by the action of sodium hydroxide with metaboric acid or borax.

$$NaOH + HBO_2 \longrightarrow NaBO_2 + H_2O$$

$$2NaOH + Na_2B_4O_7 \longrightarrow 4NaBO_2 + H_2O$$

Uses: Sodium metaborate is used in oxidizing and bleaching process. It is also used in tooth pastes and powders, cosmetics and soaps.

While naming some oxoacids and their salts, the prefixes ortho, pyro and meta are used to distinguish the acids differing in the content of water. These traditional names provide neither specific information on the number of oxygen atoms nor the number of hydrogen atoms, whether acidic or not. The acid which contain more water content is called ortho acid, whereas the same acid with less amount of water is called meta acid. For example,

(i) orthoboric acid H_3BO_3
meta boric acid HBO_2
(ii) orthophosphorus acid H_3PO_3
meta phosphorus acid HPO_2
(iii) ortho phosphoric acid H_3PO_4
meta phosphoric acid HPO_3

The term 'pyro' is used for the acids formed by condensation of acid molecules by losing water molecules on heating

$$2H_2SO_4 \xrightarrow{-H_2O} H_2S_2O_7 \text{ (Pyrosulphuric acid)}$$

$$2H_3PO_4 \xrightarrow{-H_2O} H_4P_2O_7 \text{ (Pyrophosphoric acid)}$$

$$4H_3BO_3 \xrightarrow{-3H_2O} H_6B_4O_9 \text{ (Pyroboric acid)}$$

Sometimes, these acids are polymerized. For example, polymerization of two HBO_2 molecules gives dimer, three HBO_2 molecules gives trimer and four HBO_2 molecules gives tetramer, which are also termed as meta acids. For example, the tetramer of metaboric acid is called tetrametaboric acid and its salt sodium tetra metaborate is called borax. Similarly, calgon is sodium hexametaphosphate.

7.10.4 Sodium Tetra-borate or Borax or Tincal $Na_2B_4O_7 \cdot 10H_2O$

Borax occurs naturally as tincal in certain dried up lakes of India, Tibet and California (the United States).

Preparation

(i) From Colemanite: The mineral is allowed to boil with concentrated solution of sodium carbonate.

$$Ca_2B_6O_{11} + 2Na_2CO_3 \longrightarrow 2CaCO_3 + Na_2B_4O_7 + 2NaBO_2$$

By filtration insoluble $CaCO_3$ is removed. Concentration of the remaining solution gives crystals of borax leaving behind sodium metaborate in the solution. A stream of carbon dioxide is passed through the mother-liquor for converting the metaborate into borax

$$4NaBO_2 + CO_2 \longrightarrow Na_2B_4O_7 + Na_2CO_3$$

The residual solution of sodium carbonate is reused for the treatment of a fresh quantity of colemanite.

(ii) From Tincal: When naturally occurring crude borax (Tincal) is dissolved in water, filtered, concentrated and crystallized pure borax would be obtained.

(iii) From Kernite (Rasorite): It constitutes the main source of borax. When the mineral is lixivated with hot water, the latter dissolves borax, leaving clay and sand. The solution is separated from the insoluble residue, concentrated and crystallized.

(iv) From boric acid: Small quantities of borax are obtained by neutralizing boric acid solution with soda ash.

$$4H_3BO_3 + Na_2CO_3 \longrightarrow Na_2B_4O_7 + 6H_2O + CO_2$$

Borax is sparingly soluble in cold water, but it is more soluble in hot water. 100 grams of water dissolves 3 grams of decahydrate at 10°C and 99.3 grams at 100°C. Depending on this, borax is crystallized from a saturated solution at different temperatures. If a saturated solution be allowed to crystallize at room temperature, **ordinary or prismatic borax $Na_2B_4O_7 \cdot 10H_2O$** separates out. However, if a saturated solution be allowed to crystallize above about 62°C, **octaheral or Jeweller's borax $Na_2B_4O_7 \cdot 5H_2O$** separates.

When heated, borax fuses, looses water and swells up into white porous mass, owing to the expulsion of the water. Finally, the borax melts to a clear glass called **borax glass** which is anhydrous borax. It is unstable in moist air as it absorbs moisture and converts into decahydrate.

Properties: It is sparingly soluble in cold water and fairly soluble in hot water. Its aqueous solution is alkaline in nature.

$$Na_2B_4O_7 \cdot 7H_2O \rightleftharpoons 4H_3BO_3 + 2NaOH$$

Borax on heating with ammonium chloride gives a mixture of boron nitride and oxide.

$$Na_2B_4O_7 + 2NH_4Cl \longrightarrow 2BN + B_2O_3 + 2NaCl + 4H_2O$$

When heated, it swells into a white mass by losing water and finally to a transparent glassy mass called **borax bead.** This bead gives characteristic coloured bead when again heated with a transition metal ion, because of the formation of metaborates.

$$Na_2B_4O_7\,10H_2O \longrightarrow Na_2B_4O_7 \xrightarrow{\text{heat}} \underbrace{NaBO_2 + B_2O_3}_{\text{glassy bead}}$$

$$CuO + B_2O_3 \xrightarrow{\Delta} Cu(BO_2)_2$$

Colour of meta borates of different metals is as

Cu	Fe	Co	Cr	Ni
Blue	Green	Blue	Green	Brown

Uses: Borax is used

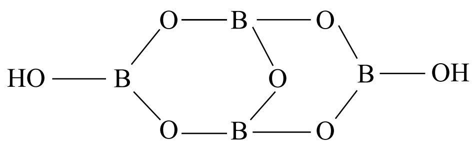
(i) in the manufacture of optical glass, soap, drying oils, glazes, enamels, paper, plastics and leather.
(ii) in the metallurgy, soldering and welding as flux.
(iii) in the form of aqueous solution as a buffer because it contains weak acid and its salt with strong base.

$$Na_2B_4O_7 + 7H_2O \longrightarrow 2Na[B(OH)_4] + 2H_3BO_3$$

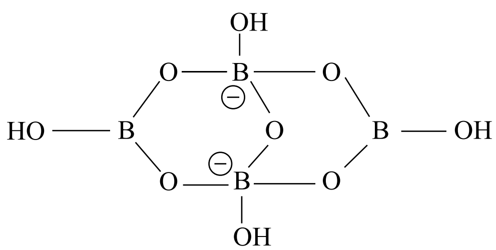
(iv) for stiffening candle wickes.
(v) in the preservation of food.
(vi) as water softening compound.
(vii) as an antiseptic, and eye wash under the name **boric lotion.**

Structure: The formation of borax can be explained as follows. First, four meta boric acid molecules polymerize to give tetraboric acid.

$$4HBO_2 \xrightarrow[-H_2O]{160°C} H_2B_4O_7$$



The two boron atoms on which there are no OH groups will accept lone pair from OH^- ions and convert into tetraborate ion and these two boron atoms are in tetrahedral structure whereas the remaining two boron atoms remain in planar triangular structure.



Tetra borate $[B_4O_5(OH)_4]^{2-}$ ion.

The remaining eight water molecules are associated with two sodium ions. Hence, the formula of borax is $Na[B_4O_5\,(OH)_4] \cdot 8H_2O$ and it is a metaborate.

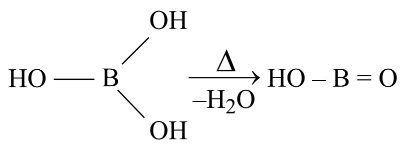
The observed bond lengths in boron oxygen compounds range from 120 pm for B—O in gaseous B_2O_3 molecule to around 155 pm. The mean value for triangular coordination is 136.5 pm and for tetrahedral coordination is 147.5 pm. In borax, the B—O bond length for planar BO_3 groups is usually close to 136 pm but for tetrahedral BO_4 groups the length increases to about 148 pm. This increase is very similar to that on going from BF_3 to $[BF_4]^-$ and suggest that in the planar grouping π bonding involving lone pairs of electrons from the oxygen atom occurs; this π bonding is necessarily lost in the tetrahedral group, in which a lone pair from the extra oxygen atom occupies the previously empty orbital on the boron atom. The number of BO_4 units being equal to the charge on the anion. Thus, $KB_5O_8 \cdot 4H_2O$ has one BO_4 and four BO_3, whereas $Ca_2B_6O_{11} \cdot 7H_2O$ has four BO_4 and two BO_3 groups.

Experimental evidences show that in aqueous media, species containing only three-coordinated boron are unstable and are rapidly converted into species containing some of the boron tetrahedrally coordinated.

7.10.5 Structures of Different Borates

Orthoboric acid on heating at 375K gives metaboric acid.

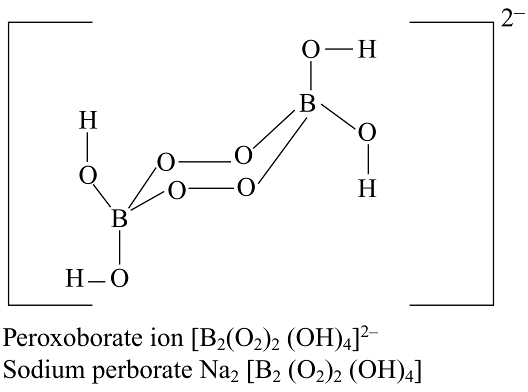
$$H_3BO_3 \xrightarrow{\Delta} HBO_2 + H_2O$$



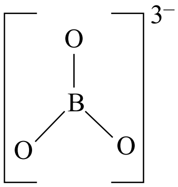
The orthoborates contain discrete BO_3^{3-} ions. The polymerization of metaborate ions forms a variety of polymeric chain and ring structures. The structures of different borate ions are given on page 7.20.

Peroxoborates

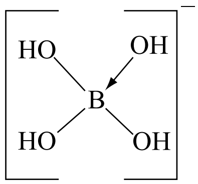
These contain peroxo bonds, used as brightners because they absorb UV light and emit visible light.



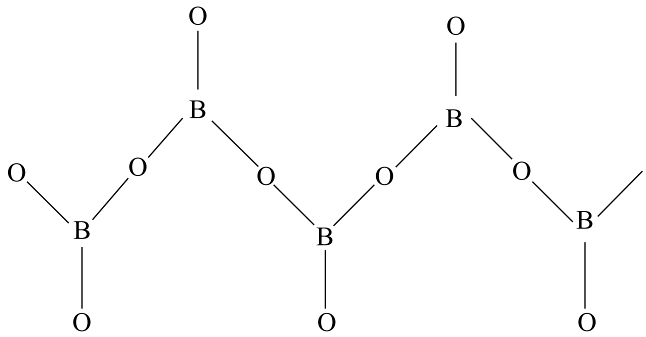
Peroxoborate ion $[B_2(O_2)_2\,(OH)_4]^{2-}$
Sodium perborate $Na_2\,[B_2\,(O_2)_2\,(OH)_4]$



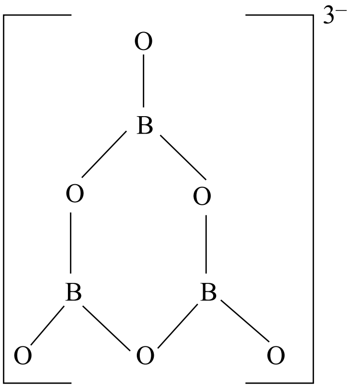
Orthoborate ion
e.g. $Mg_3(BO_3)_2$



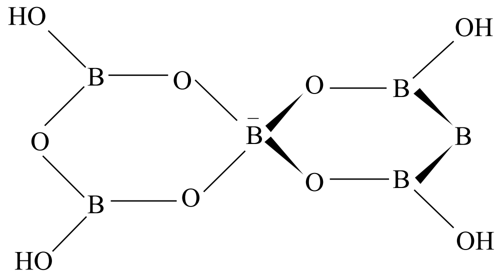
Tetrahedral
metaborate ion



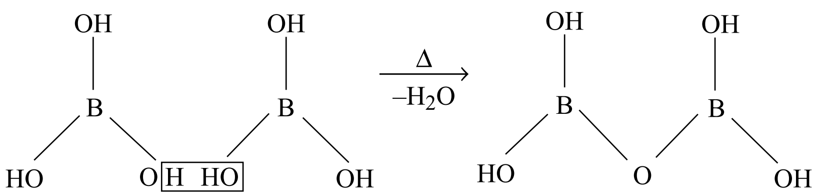
Metaborate chain
$[Ca(BO_2)_2]_n$



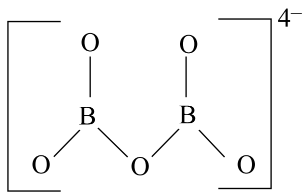
Triborate $[B_3O_6]^{3-}$
$K_3[B_3O_6]$



Complex metaborate K $[B_5O_6(OH)_4]$
spiro anion containing four triangular BO_3 units
and one tetrahedral BO_4 unit



Orthoboric acid → Pyroboric acid



Pyroborate or
diborate ion

Fig 7.9 Structures of borate ions

7.10.6 Boron Halides

Preparation: Boron combines with all the four halogens to form trihalides.

$$2B + 3X_2 \longrightarrow 2BX_3 \ (X = F, Cl, Br \text{ or } I)$$

However, BF_3 is generally prepared from fluorospar by heating with concentrated sulphuric acid and boron trioxide.

$$3CaF_2 + B_2O_3 + 3H_2SO_4 \longrightarrow 2BF_3 + 3CaSO_4 + 3H_2O$$

Other boron trihalides are prepared by heating boron trioxide with carbon in the presence of halogen.

$$B_2O_3 + 3C + 3X_2 \longrightarrow BX_3 + 3CO \ (X = Cl, Br \text{ or } I)$$

Properties: The physical states of these halides get progressively more condensed from fluoride to iodide. Thus, boron trifluoride is a gas; boron trichloride is a liquid; boron tribromide is a viscous liquid and triiodide is a crystalline solid. Therefore, their melting points and boiling points also increase with the increase in the atomic number of halogen $BI_3 > BBr_3 > BCl_3 > BF_3$.

All the trihalides of boron hydrolyses in water. The hydrolysis products of boron trifluoride are different from the other three boron halides. The degree of hydrolysis increases from BCl_3 to BI_3.

$$BCl_3 + 3H_2O \longrightarrow H_3BO_3 + 3HCl$$

$$BBr_3 + 3H_2O \longrightarrow H_3BO_3 + 3HBr$$
$$BI_3 + 3H_2O \longrightarrow H_3BO_3 + 3HI$$

The rapid hydrolysis supports other evidence that these halides are stronger Lewis acids than BF_3. BF_3 hydrolyses incompletely and form fluoborates. This is because the HF first formed reacts with the H_3BO_3.

$$4BF_3 + 12H_2O \longrightarrow 4H_3BO_3 + 12HF$$
$$12HF + 3H_3BO_3 \longrightarrow 3H^+ + 3[BF_4]^- + 9H_2O$$
$$\overline{4BF_3 + 3H_2O \longrightarrow H_3BO_3 + 3H^+ + 3[BF_4]^-}$$
$$BF_4^- + H_2O \longrightarrow [BF_3OH]^- + HF$$

The structures and their Lewis acidic characters are discussed already in Section 7.7.4.

Owing to the hydrolysis reactions discussed earlier and due to its Lewis acidic character, BF_3 is widely used to promote various organic reactions. Examples are

1. Ethers or alcohols + acids $\longrightarrow$ esters + H_2O or ROH
2. Alcohols + benzene $\longrightarrow$ alkylbenzenes + H_2O
3. Polymerization of alkenes and alkene oxides such as propylene oxide.
4. Friedel-Crafts-like acylation and alkylations.

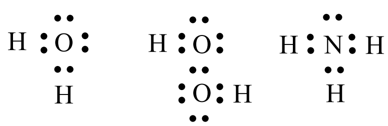
When boric acid is dissolved in aqueous HF forms fluoboric acid.

$$B(OH)_3 + 4HF \longrightarrow H_3O^+ + BF_4^- + 2H_2O$$

The fluoborate ion is tetrahedral and resembles the corresponding perchlorates in their solubilities and crystal structure. Like ClO_4^- and PF_6^-, the BF_4^- has low tendency to act as a ligand towards metal ions in aqueous solution. In non-aqueous media, there is evidence for complex formation.

7.10.7 Electron Deficiency and Acceptor Behaviour

If the structural formulae are represented by Lewis dot method for compounds such as H_2O, H_2O_2 and NH_3, each pair of electrons between the atoms show a covalent bond.



On observing these structural formulae, it is seen that the number of bonds are less than the number of atoms. Generally, in open chain compounds in which all the atoms are bonded with one another with a single bond and if the atoms at the two ends are not in bond formation, the number of bonds will be less by one, than the number of atoms. If the atoms at the two ends are also in bond to form a cyclic structure, then the number of bonds and atoms are equal. In general, except the cyclic molecules, the relationship between the number of atoms in a molecule and the number of bonds can be shown as follows.

$$Ne = 2\ Nb = 2\ (Na–1)$$

In the above equation, Ne is the number of bonding electrons, Nb is the number of bonds and Na is the number of atoms.

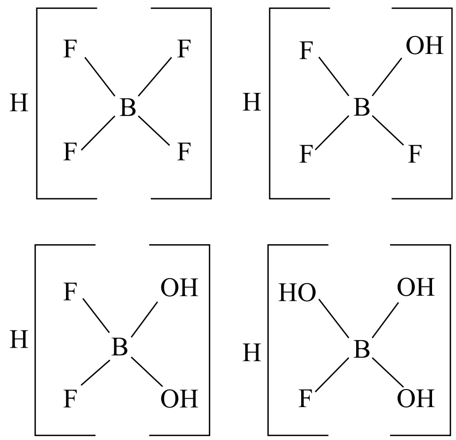
Boron the first element of the Group III A had only three electrons in its valence shell ($2s^2\ 2p^1$). Therefore, it can participate in the three bonds. Hence, it can be assumed theoretically that boron can form a compound of formula BH_3 with hydrogen. However, the simplest hydride of the boron is diborane. B_2H_6 containing two BH_3 groups having six valence electron, so the total number of bonding electrons in diborane are $(2 \times 3) + (6 \times 1) = 12$. These electrons are not in accordance with the formula $Ne = 2(Na-1)$.

Such compounds which do not satisfy the formula $Ne = 2$, $Nb = 2(Na-1)$, i.e., there are not enough electrons available than is needed for the formation of normal electron pair covalent bonds between all the atoms in the molecule and those compounds whose central atom has incomplete octet can be termed as **electron-deficient compounds**. The central atom of such molecules contain more stable low-energy atomic orbitals than the total number of electrons in the valence shell. Such empty low-energy atomic orbitals of an atom in the molecule possess a strong tendency of filling themselves through the process of a coordinate bond formation, i.e., they can readily accept a lone pair of electrons, if brought in contact with a molecule containing a donor atom such as nitrogen, oxygen, fluorine, and sulphur in NH_3, H_2O, HF and H_2S, respectively.

Electron-deficient compounds are more characteristic of the elements of Group III A which possess three valence etectrons only but have the tendency to be 4 covalent. This does not produce electron deficiency in all types of compounds as strong donor groups can impart 4 – covalence by the formation of coordinate linkages.

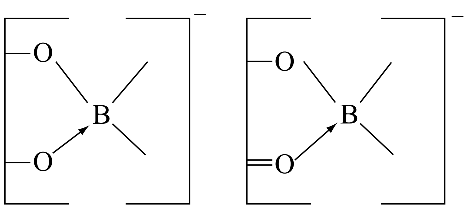
Owing to the acceptor strength of the boron atom, a variety of molecular addition compounds are known. These are formed most commonly between trihalides (especially boron trifluoride) boron alkyls or boron aryls and compounds containing donor nitrogen or donor oxygen. Some of these materials have been described in conjugation with the materials on hydrides of boron. Materials which are complexes in the more usual sense are the fluoborates, the chealated oxy-derivatives and perhaps the borohydrides are discussed in connection with the boron hydrides because of the close chemical relation.

The only stable halo complexes of boron are those containing fluorine. These are either completely fluo-compounds (e.g., HBF_4, H_2BF_5, $H_2B_2F_6$) or mixed aquafluo compounds (e.g., $(HO)_2$ HOB(F) B(F) BOH). Many of these compounds are poorly characterized or have unconfirmed compositions. Important fluoboric acids are the compounds of HBF_4 and those materials which are related to it as by successive substitution of hydroxyl groups for fluorides. These are related to each other as

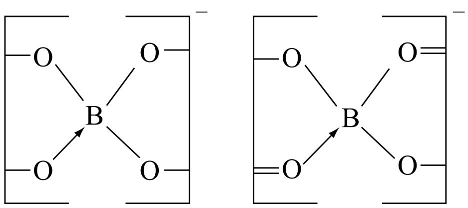


All these compounds are known and the compound $H[B(OH)_4]$ a substituted product of the last fluoride shown above is also known. The name fluoboric acid is properly restricted to the first compound HBF_4, the others are known as hydroxy fluoboric acids.

Chealated oxy-derivatives of boron containing chealate rings involving oxygen to boron. Either one or two chealate rings may be present, giving the fundamental structures.



Monochealate structures



Dichealate structures

Owing to this chealating ability, only the weak ortho boric acid acts as comparatively strong monobasic acid in the presence of polyhydroxy compounds such as glycerol or mannitol and several other chealating agents such as catechol, salicylic acid, β-diketones form chealate complexes.

Comparitively, few molecular addition compounds are known for the other Group III A elements and those which are known are also less stable than their boron analogues. Stability also decreases with increasing atomic weight, and the size of the Group III A element. Like boron halides, the analogous halides have the greatest tendency towards formation of compounds of this type. Nitrogen and oxygen are again the best donors. For the aluminium halides addition compounds with ammonia, the amines, a few cyanides, water, alcohols, ethers, aldehydes and ketones are known. Their composition are in general.

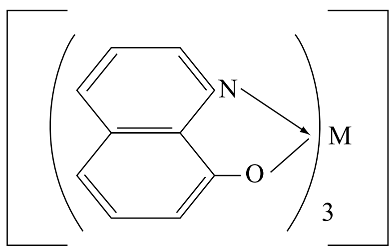
$$X_3Al \leftarrow NR_3 \text{ or } X_3Al \leftarrow OR_2$$
(R = hydrogen or other substituent)

Phosphine and hydrogen sulphide addition compounds of aluminium halides and alkyls are also preparable.

$$X_3Al \leftarrow PH_3;\ X_3Al \leftarrow SH_2;\ R_3Al \leftarrow OR_2$$

Fluoro complexes are more characteristic of aluminium and gallium. Chloro and bromo complexes are better characterized with heavier elements. Iodo complexes are almost unknown.

Chealated oxy compounds are almost 6-coordinate complexes. The β-diketones yield stable inner complexes with aluminium, gallium and indium. Polyhydric alcohols do not yield compounds comparable to those formed by boron. The complexes of dicarboxylic acids such as trioxalato and dioxalato salts are very common. Chealate rings involving oxygen and donor nitrogen characterize the 8-quinolinol and substituted 8-quinolinol chealates of these elements. The fundamental structure being



M = Al, Ga, In, TI

These are yellow crystalline compounds, which are insoluble in water but dissolve in chloroform to give solutions useful for the colorimetric determination of the metals.

7.10.8 Boron Hydrides

Boron does not form simple monomeric hydride, i.e., BH_3 although analogous boron trihalide BX_3 is well known and quite stable compound. This is due to the fact that hydrogen atoms in BH_3 have no free electrons to form pπ-pπ **back bonding** as in boron trifluoride and thus boron has incomplete octet. Actually, when BH_3 molecules come in contact with each other, they dimerize to form diborane B_2H_6.

The chemistry of the boron hydrides was started around 1920 because of the classical researches of **Alfred Stock**. Boron forms several hydrides which are somewhat analogous to the alkanes and so have been called **boranes**. These boranes can be classified into two series, namely B_nH_{n+4} and B_nH_{n+6}. Of these, latter are less stable and have low melting points.

(a) B_nH_{n+4} (where n = 2, 5, 6, 8, 10 and 18). These are quite stable and hence more important. These are commonly known as **boranes** or **Nidoboranes.**

(b) B_nH_{n+6} (where n = 4, 5, 6, 9, and 10). These are unstable and have low melting points. These are called as **hydroboranes** or **Arachano boranes.**

A convenient nomenclature for the boranes is to specify the number of boron atoms in the molecule by Greek prefix and the number of hydrogen atoms with an arabic number in parantheses. For example, B_5H_9 as pentaborane (9) and B_5H_{11} as pentaborane (11). The hydrogen number may be omitted where only one compound with the number of boron atoms in question is known, namely B_4H_{10} as tetraborane.

Among the numerous boranes, diborane (B_2H_6) is extensively studied. Its chemical behaviour is considered as representative of other boranes. Its structure is also very much useful in understanding the structure of electron-deficient compounds. Hence, its preparation, properties and structure are discussed in detail.

7.10.9 Diborane B_2H_6

(i) Diborane is prepared by treating boron trifluoride or boron trichloride with $LiAlH_4$ in diethyl ether.

$$4BF_3 + 3LiAlH_4 \longrightarrow 2B_2H_6 + 3LiF + 3AlF_3$$

$$4BCl_3 + 3LiAlH_4 \longrightarrow 2B_2H_6 + 3LiCl + 3AlCl_3$$

(ii) In the laboratory, it is conveniently prepared by the oxidation of sodium borohydride with iodine.

$$2NaBH_4 + I_2 \longrightarrow B_2H_6 + 2NaI + H_2$$

(iii) Industrially, diborane is produced by the reaction of BF_3 with sodium hydride or lithium hydride.

$$2BF_3 + 6NaH \xrightarrow{450\,K} B_2H_6 + 6NaF$$

$$2BF_3 + 6LiH \xrightarrow{450\,K} B_2H_6 + 6LiF$$

(iv) It can also be prepared by subjecting a mixture of boron trichloride and hydrogen to silent discharge.

$$2BCl_3 + 6H_2 \xrightarrow[\text{or Cu-Al catalyst at }450^\circ C]{\text{electric discharge}} B_2H_6 + 6HCl$$

Properties

Diborane is a colourless, highly toxic gas with a boiling point of 180 K. At low temperatures, in the absence of moisture and grease, it is stable. At ordinary temperatures, it converts into other hydrides of boron.

(i) **Action of air:** It burns or explodes in air with the-evolution of much heat.

$$B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O \quad \Delta_cH^\circ = -1976 \text{ kJ mol}^{-1}$$

Most of the higher boranes are also spontaneously flammable in air.

(ii) **Action of water:** It reacts with water to form boric acid and hydrogen.

$$B_2H_6 + 6H_2O \longrightarrow 2H_3BO_3 + 6H_2$$

(iii) **Action of alkalis:** It undergoes reaction with solutions of caustic alkalis to form borates with the evolution of hydrogen.

$$B_2H_6 + 6KOH \longrightarrow 2K_3BO_3 + 6H_2$$

$$B_2H_6 + 2KOH + 2H_2O \longrightarrow 2KBO_2 + 6H_2$$

(iv) **Reaction with halogen acids:** Diborane reacts scarcely with HCl, readily reacts with HBr at 90°C and with HI at 50°C (in the presence of their corresponding aluminium halides) to give halogen substituted diborane. On standing for sometime, they will be converted into trihalides.

$$6B_2H_6 \xrightarrow{AlBr_3\ 90^\circ C} 6BH_3 \cdot BH_2Br \longrightarrow 5BH_3 \cdot BH_3 \cdot 2BBr_3$$

(v) **Action of halogens:** Chlorine reacts with diborane violently. The reaction with bromine may be controlled to yield bromo-diborane. Iodine does not react at all.

$$B_2H_6 + Cl_2 \longrightarrow B_2H_5Cl + HCl$$

$$B_2H_6 + 3Cl_2 \longrightarrow 2BCl_3 + 3H_2$$

$$B_2H_6 + 6Cl_2 \longrightarrow 2BCl_3 + 6HCl$$

$$B_2H_6 + Br_2 \xrightarrow{100^\circ C} B_2H_5Br + HBr$$

(vi) **Action of ammonia:** Diborane reacts with ammonia. The products vary depending on the reaction conditions.

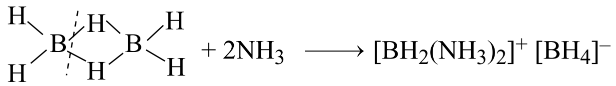
$$B_2H_6 + 2NH_3 \xrightarrow{120^\circ C} \underset{\text{Diammoniate of diborane}}{B_2H_6 \cdot 2NH_3}$$

$$3B_2H_6 \cdot 2NH_3 \xrightarrow{200^\circ C} \underset{\text{Borozole}}{B_3N_3H_6} + 12H_2$$

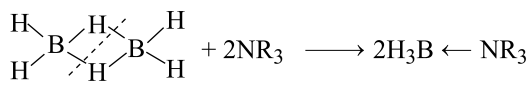
$$B_2H_6 + 2NH_3 \xrightarrow[1:2\text{proportion}]{200^\circ C} B_3N_3H_6$$

$$nB_3N_3H_6 \xrightarrow{200^\circ C} 3(BN)_n + 3nH_2$$

(vii) Lewis acid character: As diborane is electron-deficient molecule, it reacts with several molecules to form complexes (addition compounds). During the formation of these addition compounds, two different cleavage patterns have been observed, namely symmetric cleavage and asymmetric cleavage. In **symmetric cleavage,** B_2H_6 is broken symmetrically into two BH_3 fragments, each of which forms a complex with Lewis base.

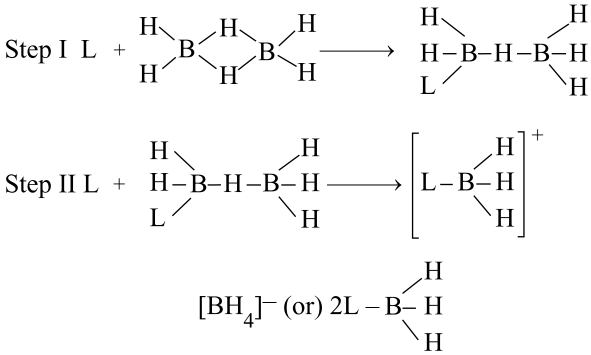


Unsymmetric cleavage



Symmetric cleavage

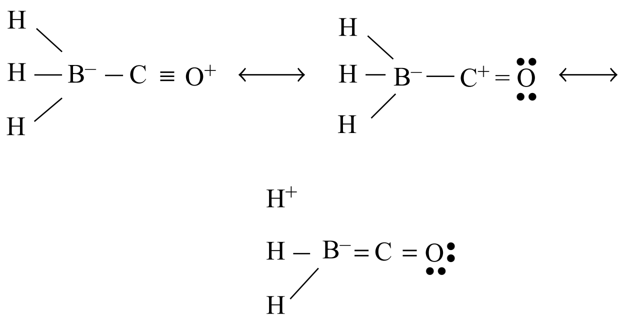
Cleavage of diborane is presumed to occur by a two-step mechanism.



In step II, the final product depends on the choice of L for the two B atoms. As most donor atoms are sufficiently electronegative (more than the H^- ion), the B atom containing L develops greater partial positive charge and favours the attack by the second L. This leads to unsymmetrical cleavage. However, if such attack causes steric congestion, the second L is expected to attack the other B atom, when symmetrical cleavage would occur. However, it is also possible that an unsymmetrical product formed initially undergoes rapid hydride ion transfer to give the symmetrical end product. It has been possible to isolate $[BH_2(H_2O)_2]^+[BH_4]^-$ at –130°C.

For $B_2H_6 \longrightarrow 2BH_3$, $\Delta H = 150\ kJ\ mol^{-1}$. With this value in mind, it is possible to compare the strength of BH_3 as a Lewis acid with the strengths of the boron alkyls and boron trihalides. As expected from electronegativity considerations, the acceptor properties of BH_3 towards simple bases (e.g., Me_3N) are intermediate between boron trihalides and boron trimethyl. However, BH_3 alone forms stable complexes with CO and PF_3. These ligands are good σ-donors (using lone pair electrons on C and P, respectively.

In its adducts with these ligands, BH_3 probably acts as an electron donor through "hyperconjugation", one may write the following canonical forms for the BH_3 co-adduct.



Many complexes of this kind exist. They are interesting partly because they are isoelectronic with 2,2-dimethyl propane $C(CH_3)_4$ stability trends, which indicate that BH_3 is a soft Lewis acid as illustrated by the reaction.

$$H_3B\text{–}N(CH_3)_3 + F_3B - S(CH_3)_2 \longrightarrow H_3B - S\,(CH_3)_2 + F_3B - N(CH_3)_3$$

In the above equation, BH_3 transfers to the soft S donor atom and the harder Lewis acid BF_3 combines with the hard N donor atom.

Some other reactions in which symmetric cleavage of B_2H_6 takes place while acting as Lewis acid are as follows.

(a) **Reaction with carbon monoxide:** It reacts with carbon monoxide under pressure.

$$B_2H_6 + 2CO \xrightarrow[20\ atm]{100°C} 2H_3B \cdot CO$$

(b) It reacts with ether or thioether to form addition compounds.

$$B_2H_6 + 2(CH_3)_2O \longrightarrow 2H_3B \leftarrow O(CH_3)_2$$

$$B_2H_6 + 2(CH_3)_2S \longrightarrow 2H_3B \leftarrow S(CH_3)_2$$

(c) It reacts with tertiary amines.

$$B_2H_6 + 2NR_3 \longrightarrow 2H_3B \leftarrow NR_3$$

(d) It reacts with pyridine.

$$B_2H_6 + 2C_5H_5N \longrightarrow 2H_3B \leftarrow NC_5H_5$$

(e) It combines with sodium amalgam forming an addition compound $B_2H_6 \cdot Na_2Hg$.

The complexes serve as source of diborane in chemical reaction as they can be stored or transported safely.

Unsymmetrical cleavage of this kind is generally observed when diborane react with strong sterically uncrowded bases at low temperatures. The steric repulsion is such that only two small ligands can attack one boron atom in the course of reaction.

The direct reaction of diborane and ammonia results in **unsymmetrical cleavage** leading to the formation of ionic product.

Trimethyl boron is a weaker Lewis acid than the boron trihalides or monoborane (BH_3 obtained by symmetrical cleavage of B_2H_6). Although there is no π bonding to be overcome when the planar molecule becomes tetrahedral in a donor acceptor complex, the electron-repelling effect of methyl groups hinders complex formation. No compound is formed by trimethyl boron with carbon monoxide and compounds with tertiary phosphines are much less stable than those with tertiary amines. As expected, steric factors may influence either donor or acceptor properties thus 2,6-dimethylpyridine does not react with trimethyl boron. The stability sequences of the donor acceptor complexes of boron compounds may be summarized as follows.

$(CH_3)_3N$ as donor; $BBr_3 > BCl_3 > BF_3 \sim BH_3 > B(CH_3)_3$

$(CH_3)_3P$ as donor: $BBr_3 > BCl_3 \sim BH_3 > BF_3 \sim B(CH_3)_3$

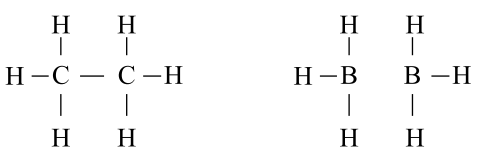
CO as donor: $BH_3 >> BF_3 \sim B(CH_3)_3$

Uses: Diborane is used

(i) as a catalyst in polymerization reactions.
(ii) for welding torches.
(iii) for preparing rocket fuels.
(iv) as a reducing agent in organic reactions.
(v) in the hydroboration reaction which is useful method for preparing hydrocarbons, alcohols, ketones etc.

Structure of Diborane

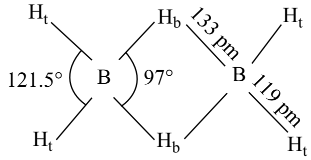
The boranes have been of interest from the point of view of their electronic structures, because they are electron deficient, i.e., enough electrons are not available to form the expected number of covalent bonds. For example, in diborane, there have been 12 electrons, three from each boron atom and six from hydrogens whereas ethane is having 14 such electrons. If diborane has to have structure such as ethane (analogue diborane) 14 electrons are necessary.



Ethane Diborane

In diborane, there are no sufficient electrons to form a covalent bond between two boron atoms. Then, the question arises how the two BH_3 groups combine to exist as dimer B_2H_6.

Electron diffraction results reveal that in B_2H_6 molecules having the coplanar BH_2 groups and that the two remaining, H atoms are lying centrally above and below the plane linking the BH_2 groups and disallow rotation between the two boron atoms. The structure may be represented as follows.



B-B bond length 177 pm
B-H_b bond length 133 pm
B-H_t bond length 119 pm
H_tBH_t bond angle 121.5°
H_bBH_b bond angle 97°

The four hydrogen atoms (represented by H_t) present in BH_2 groups are known as terminal hydrogen atoms. The remaining two hydrogens (represented by H_b) are called as bridge hydrogens because they link the two BH_2 groups in the molecule. These two bridge hydrogens lie in a plane perpendicular to the plane of the BH_2 groups. One of the bridge hydrogens lies above the plane and the other lies below the plane as shown in the above figure.

Generally, a covalent bond is formed by sharing a pair of electrons between the two atoms, one from each atom. If a covalent bond is formed by sharing a pair of electrons between more than two atoms, it can be shown as above between two boron atoms and one hydrogen atom; one pair of electron cloud is distributed (delocalized) over one hydrogen atom and two boron atoms which are on both sides to it. This electron pair cloud, with centrally situated hydrogen nucleus and two boron atoms, is known as **B-H-B hydrogen bridge bond**. There are two such hydrogen bridges in diborane. This bond is also known as **3-centre, 2-electron bond** abbreviated as **3c-2e bond**. This is also known as protonated bond as only protons remain in the bridge B-H-B bond. As one electron pair gets shared between three atoms instead of two, the strength of the bond would be only one-half of the strength of a normal 2-centre 2-electron (2c-2e) bond. It is equivalent to the bond order of 1/2 which is obtained in the resonance treatment.

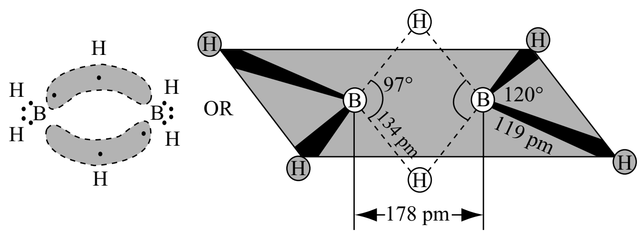


Fig 7.10 Structure of diborane

Nature of bonds in hydrogen bridges

Boron atoms undergoe sp^3 hybridization resulting in four equivalent orbitals. Three of these orbitals have one electron each and the fourth hybrid orbital is vacant. In forming the B-H-B bridge, sp^3 hybrid orbital with one electron from one boron atom, 1s orbital of one bridge hydrogen and the vacant sp^3 hybrid orbital of the second boron atom overlaps as shown in Fig 7.11.

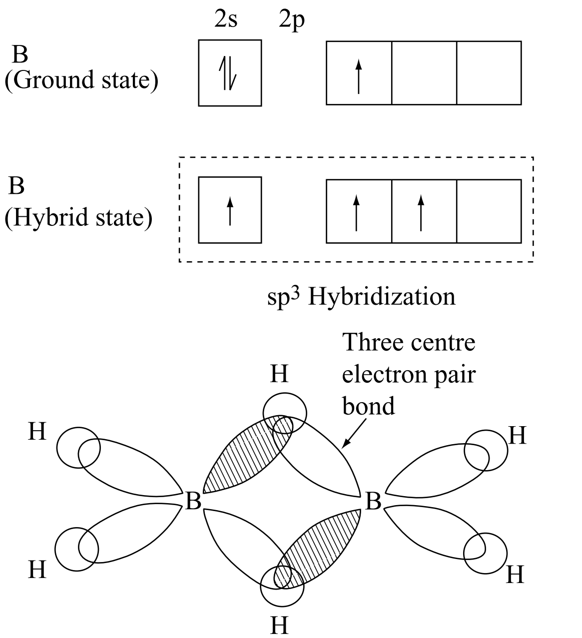


Fig 7.11 Hybridization in boron atom and orbital overlap

As molecular orbital formed in this B-H-B bridge bond appears like banana, it is also called as **banana bond** or **tau bond**.

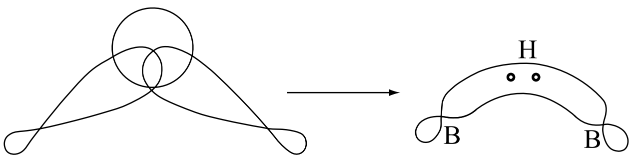


Fig 7.12 Formation of banana bond or B–H–B bridge bond

Evidences in support of bridge structure

The structure of diborane having hydrogen bridges linking the two BH_2 groups is supported by the following evidences.

(i) Specific heat measurements: The study reveals that the rotation of two BH_2 groups lying at the two ends against each other gets hindered. In ethane (H_3C-CH_3), two methyl groups get linked directly and the hinderence to this type rotation has been very little.

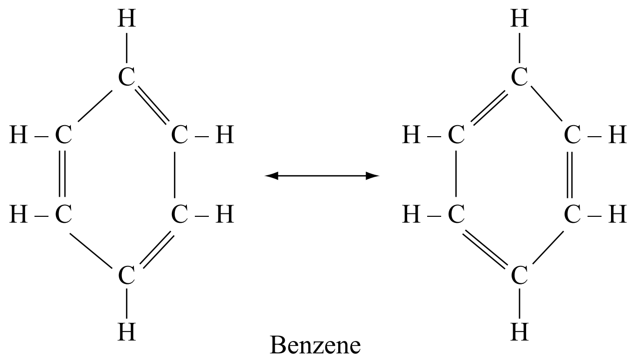
(ii) NMR and Raman Spectra: This study reveals that four hydrogen atoms have been of one type and remaining two of another type.

(iii) Diborane on methylation yields only $Me_4B_2H_2$: Two hydrogens which cannot be replaced by methyl groups have been different from the remaining four which have been replaced. These are the bridging hydrogen atoms.

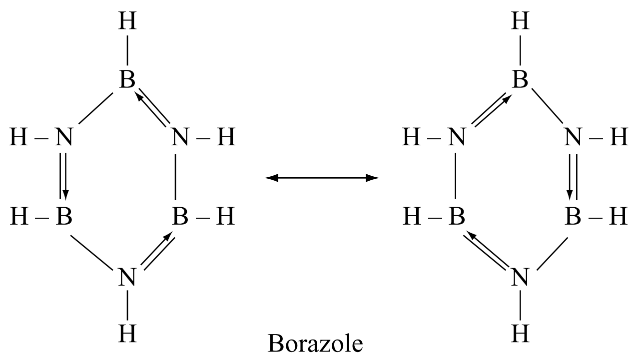
7.10.10 Borazole

The borazole formed by the reaction of diborane with ammonia has the structure similar to benzene. Therefore, it is also known as **inorganic benzene.** Borazole can be prepared by heating BCl_3 with NH_4Cl followed by reduction with $NaBH_4$.

$$3BCl_3 + 3NH_4Cl \xrightarrow{140^\circ C} B_3N_3H_3Cl_3 \xrightarrow{NaBH_4} B_3N_3H_6$$



Benzene



Borazole

The difference in the structures of benzene and borazole is that in benzene the π-bonds are covalent whereas in borazole the π bonds are dative bonds by the donation of lone pair on nitrogen atom being capable of contributing to either of two boron—nitrogen linkages.

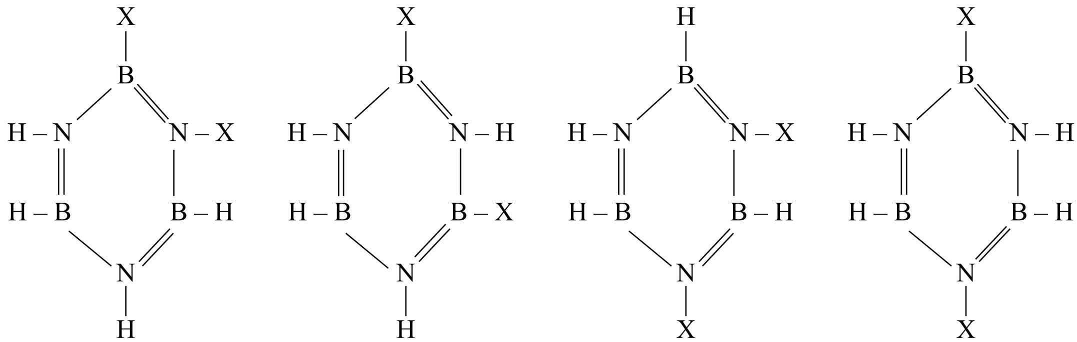
The molecular orbital description of borazole considers the formation of σ and π bonds in the ring. Owing to the difference in the electronegativity between boron and nitrogen, the π cloud in borazole is more with highly electronegative nitrogen than with boron. This partial localization weakens the π bonding in the ring.

In addition, as nitrogen retains some of its basic character and the boron the acidic character, polar compounds such as HCl can, therefore, attack the double bond between nitrogen and boron. The different electronegativities of boron and nitrogen tend to stabilize bonding to boron by electronegative substituents and to nitrogen by electro positive substituents. Thus, in contrast to benzene, borazole readily undergoes addition reactions.

If a criterion of aromaticity is taken to be the extent of delocalization of electrons in π-orbitals, it appears that the aromatic nature of borazole is significantly less than that of benzene. Moreover, it should be noted that the six electrons in the π-orbitals are derived from three nitrogen atoms and not from each of six atoms of the ring as in the case of benzene. Borazole hydrolyses in water whereas benzene does not

$$B_3N_3H_6 + 9H_2O \longrightarrow 3NH_3 + 3H_3BO_3 + 3H_2$$

Further with two similar substituents, benzene gives three isomers ortho, meta and para. Unlike benzene, borazole gives four isomers one ortho, two meta and one para. In the meta isomers, both similar substituents may be either on boron or on nitrogen. Comparison of the physical properties of borazole and benzene is given in Table 7.7.



ortho meta meta para

Table 7.7 Comparison of the physical properties of borazole and benzene

S. No.	Property	Borazole	Benzene
1.	Molecular mass	80.6	78
2.	Melting point	–58°C	6°C
3.	Boiling point	64.5°C	80°C
4.	Critical temperature	252	287
5.	Heat of vapourization at boiling point	29.26 kJ mol^{-1}	30.93 kJ mol^{-1}
6.	Troutans' constant	21.3	21
7.	Surface Tension at melting point dynes/cm	31.1	31.0
8.	Parachor	24	27
9.	Liquid density at boiling point g/mL	0.81	0.81
10.	Dipolemoment D	0	0
11.	Bond distance pm	-	
	B-N	144	
	B-H	120	
	N-H	102	
	C-C		139
	C-H		108

7.10.11 Borohydrides

The metal borohydrides contain tetrahedral complex ion $[BH_4]^-$. It is chemically tetrahydrido borate ion. The most important one is the sodium salt $Na[BH_4]$. It can be prepared by the reaction between trimethyl borate and sodium hydride.

$$4B(OCH_3)_3 + 4NaH \xrightarrow[\text{THF Solvent}]{250°C,\ \text{high P}} Na[BH_4] + 3Na\,[B(OCH_3)_4]$$

$Na[BH_4]$ has sodium chloride structure. Other metal borohydrides can be prepared by treating $Na[BH_4]$ with the appropriate metal chloride. Alkali metal borohydrides are white ionic solids and react with water with varying ease. Thus, $Li[BF_4]$ reacts with water violently, $Na[BH_4]$ may be recrystallized from cold water with only slight decomposition and $K\,[BH_4]$ is quite stable.

The beryllium, aluminium and transition metal borohydrides become increasingly covalent and volatile. In these, the $[BH_4]^-$ group acts as a ligand and forms covalent compounds with metal ions. One or more H atoms in a $[BH_4]^-$ act as a bridge and bond to the metal, forming a three-centre bond with two electrons shared by three atoms. The $[BH_4]^-$ ligand is unsusal in that it may form one, two or three such 3c-2e bonds to the metal ion. Thus, in $Al[BH_4]_3$ and $Zr\ [BH_4]_4$, each BH_4^- forms two hydrogen bridges whereas in $Be\ [BH_4]_2$, one of the each BH_4^- forms hydrogen bridges.

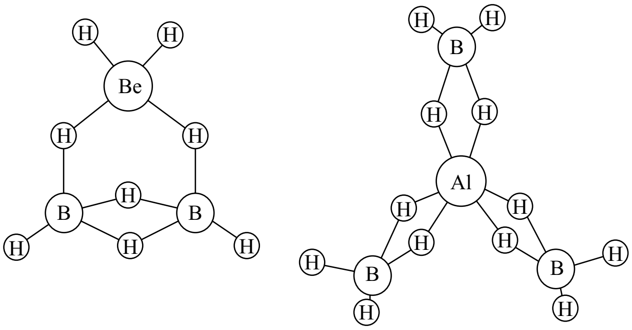


Fig 7.13 Structures of (a) $Be\ [BH_4]_2$ and (b) $Al\ [BH_4]_3$

7.10.12 Boron Nitride

The synthesis of boron nitride involves considerable technical difficulty. In the laboratory, it can be prepared by fusing borax with ammonium chloride.

$$Na_2B_4O_7 + 2NH_4Cl \longrightarrow 2BN + 2NaCl + B_2O_3 + 4H_2O$$

On large scale, it is prepared by the fusion of urea with boric acid. It is also formed by heating diborane with ammonia, boron in nitrogen or boron with ammonia.

$$B_2H_6 + 2NH_3 \xrightarrow{120^\circ C} B_2H_6 \cdot 2NH_3$$

$$3B_2H_6 \cdot 2NH_3 \longrightarrow 2B_3N_3H_6 + 12H_2$$

$$B_3N_3H_6 \xrightarrow{\text{above } 200^\circ C} 3(BN)_n + 3nH_2$$

$$2B + N_2 \longrightarrow 2BN$$

$$2B + 2NH_3 \longrightarrow 2BN + 3H_2$$

Pure boron nitride is obtained by the action of NH_3 on BCl_3.

$$BCl_3 + 6NH_3 \longrightarrow \underset{\text{Boron amide}}{B(NH_2)_3} + 3NH_4Cl$$

$$2B(NH_2)_3 \xrightarrow{\Delta} \underset{\text{Boronimide}}{B_2(NH)_3} + 3NH_3$$

$$B_2(NH)_3 \xrightarrow{\Delta} 2BN + NH_3$$

Properties: Boron nitride is colourless solid and good insulator. It melts at very high temperature, 3000°C. It is insoluble in water. However, it gets decomposed when heated in steam under pressure to yield ammonia.

$$2BN + 3H_2O \longrightarrow B_2O_3 + 2NH_3$$

It resists the attack of most reagents. Fluorine reacts with boron nitride quantitatively to BF_3 and N_2.

$$2BN + 3F_2 \longrightarrow 2BF_3 + N_2$$

When dissolved in HF, boron nitride forms $NH_4 \cdot BF_4$

$$BN + 4HF \longrightarrow NH_4BF_4.$$

Boron nitride is not attacked by alkalis. However, when fused with potassium carbonate, it yields potassium cyanate and potassium metaborate.

$$K_2CO_3 + BN \longrightarrow KCNO + KBO_2$$

Structure: Boron nitride consists planar sheets of atoms such as those in graphite. The planar sheets of alternating B and N atoms consists of edge-shared hexagons and as in graphite, the B-N distance with in the layer (145 pm) is much shorter than the distance between the sheets (333 pm). The difference between the structures

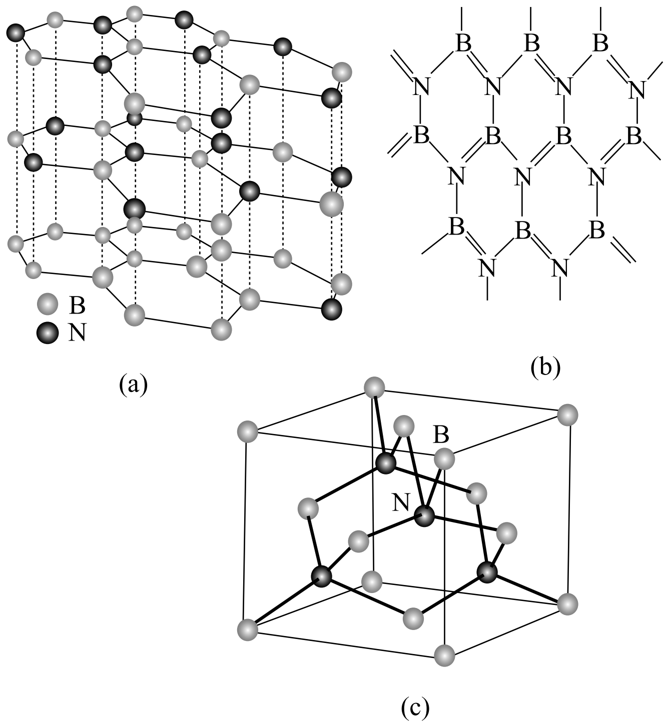


Fig 7.14 (a) The structure of layered hexagonal boron nitride. (b) Structure of single layer. (c) The sphalerite structure of cubic boron nitride

Table 7.8 Comparison of hexagonal boron nitride and graphite

	Inter layer distance/pm	Intra layer distance/pm	Density g cm^{-3}
BN(Hexagonal)	333	144.6	2.29
Graphite	335	142	2.255

of graphite and boron nitride, however, lies in the register of the atoms of neighbouring sheets in BN, the hexagonal rings are stacked directly over each other, with B and N atoms alternating in successive layers; in graphite, the hexagons are staggered. Molecular orbital calculations suggest that the stackings in BN stems form the partial positive charge on B and a partial negative charge on N. This charge distribution is consistent with the electronegativity difference of the two elements (B = 2.04; N = 3.04).

As with impure graphite, layered boron niride is a slippery material that is used as a lubricant. Unlike graphite, it is colourless electrical insulator, as there is a large gap between the filled and the vacant π bands. The size of the band gap is consistent with high electrical resistivity and lack of absorption in the visible spectrum.

Layered boron nitride changes into a denser cubic phase at high pressure and temperatures (60K bar and 2000°C). This phase is a hard crystalline analogue of diamond but, as it has lower lattice enthalpy, it has a slightly lower mechanical hardness. Cubic boron nitride is manufactured and used as an abrasive for certain high-temperature applications in which diamond cannot be used because it forms carbides with the materials being ground. The order of the hardness of different materials is

Diamond > Sphalerite cubic boron nitride > Boron carbide (B_4C) > Silicon carbide > Tungsten carbide > Alumina (Al_2O_3).

7.11 Important Compounds of Aluminium

7.11.1 Aluminium Hydride (Alane) AlH_3

It is prepared by the reaction of lithium hydride with aluminium chloride in dry ether.

$$3LiH + AlCl_3 \longrightarrow 3LiCl + AlH_3$$

Aluminium hydride is a white solid and it is stable up to 373K; above this temperature, it decomposes into its components. It reacts with lithium hydride in ether to form a powerful reducing agent, lithium aluminium hydride.

$$LiH + AlH_3 \longrightarrow LiAlH_4$$

It is an electron-deficient compound and acts as Lewis acid, e.g., $AlH_3 \cdot NH_3$ and $AlH_3 \cdot N(CH_3)_3$.

It hydrolyses violently to form aluminium hydroxide

$$AlH_3 + 3H_2O \longrightarrow Al(OH)_3 + 3H_2$$

It is an insoluble polymer $(AlH_3)_n$. The structure is a giant molecule of Al H_2 Al bridges (Fig 7.15) confirmed by X-ray and neutron diffraction studies.

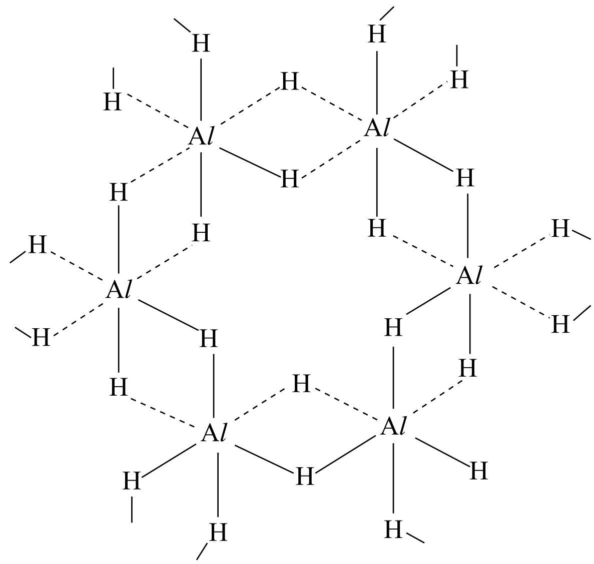


Fig 7.15 Structure of aluminium hydride $(AlH_3)_n$

7.11.2 Aluminium Oxide (Alumina) Al_2O_3

Preparation: It can be prepared by the thermal decomposition of aluminium salts.

$$Al_2(SO_4)_3 \longrightarrow Al_2O_3 + 3SO_3$$

$$4Al(NO_3)_3 \longrightarrow 2Al_2O_3 + 12NO_2 + 3O_2$$

$$2Al(OH)_3 \longrightarrow Al_2O_3 + 3H_2O$$

It can be prepared by heating the aluminium hydroxide precipitate obtained by the action of aluminium salts and ammonium hydroxide.

$$AlCl_3 + 3NH_4OH \longrightarrow Al(OH)_3 + 3NH_4Cl$$

Properties: Alumina is white amorphous powder with melting point 2325K and boiling point 3255K. It is amphoteric in nature

$$Al_2O_3 + 3H_2SO_4 \longrightarrow 2Al_2(SO_4)_3 + 3H_2O$$

$$Al_2O_3 + 3KOH \longrightarrow 2KAlO_2 + H_2O$$

Alumina is a very stable compound. It is not reduced to the metal. It is only reduced with great difficulty.

When it is heated with carbon in a current of nitrogen, it forms aluminium nitride which is hydrolysed by water.

$$Al_2O_3 + 3C + N_2 \longrightarrow 2AlN + 3CO$$

$$AlN + 3H_2O \longrightarrow Al(OH)_3 + NH_3$$

It reacts with carbon or calcium carbide at very high temperatures to form aluminium carbide which on hydrolysis gives methane.

$$2Al_2O_3 + 9C \longrightarrow Al_4C_3 + 6CO$$

$$2Al_2O_3 + 2CaC_2 \longrightarrow Al_4C_3 + 3CaO + 2CO$$

$$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$$

Uses

(i) It is used for the manufacture of aluminium and its salts.
(ii) It is used as a refractory material.
(iii) It is used as adsorbent in chromatography.
(iv) Fused alumina at 3275K in an arc furnace is known as **alundum** or artificial corrundum which is used as an abrasive.
(v) Fused mixture of alumina and lime is known as bauxite cement which is resistent to sea water and sets quickly.

Structure: The structural relations between the many crystalline forms of aluminium oxide and hydroxide are exceedingly complex but they are exceptional scientific interest and immense technological importance. The principal structural types are listed in Table 7.9. Al_2O_3 occurs naturally as the mineral corrundum contaminated with iron oxide and silica. Owing to its great hardness (Mohs 9), high melting point 2045°C, non-volatility, chemical inertness and good electrical insulating properties, it finds many applications in abrasives (including tooth paste), refractories and ceramics. Larger crystals, when coloured with metal-ion impurities are prized as gem stones.

Artificial gems are now produced by dropping finely powdered mixture of alumina, fluorspar and little colouring matter such as Cr_2O_3 for rubies through the centre of an oxyhydrogen flame. The fused mass is caught on a rod of alumina where it is deposited as a single crystal which may be cut and polished. If a mixture of alumina, ferric oxide (1.5%) and titanium dioxide (0.5%) is fused in a reducing flame, artificial sapphires are obtained.

In these precious stones, some Al^{3+} ions are replaced by transition metal ions. For example, in rubies, some Cr^{3+} ions are incorporated in the place of Al^{3+} ions and these metal ions coordinated by six oxygen atoms octahedrally. In the presence of these oxide ions, the d-orbitals split up into a group of three orbitals at lower energy (t_{2g}) and a group of two orbitals of higher energy (e_g). For the transition of electrons from one group of these orbitals (t_{2g}) to another group of ortbitals (e_g), energy is absorbed in the visible region. Therefore, they appear in the complimentary colour of the absorbed colour. Different transition metal ions produce different colours. (Refer to crystal field theory of coordination compounds chapter 14.5.)

Table 7.9 The main structural types of aluminium oxides and hydroxides

Formula		Mineral name idealized structure
α-Al_2O_3	Corrundum	Hcp O with Al in two-thirds of the octahedral sites.
α-$AlO(OH)$	Diaspore	Hcp O (OH) with chains of octahedra stacked in layers inter connected with H bonds. Al in certain octahedral sites.
α-$Al(OH)_3$	Bayerite	Hcp (OH) with Al in two thirds of the octahedral sites.
γ-Al_2O_3	–	ccp O defect spinel with Al in $21\frac{1}{3}$ of the 16 octahedral and 8 tetrahedral sites.
γ-$AlO(OH)$	Boehmite	ccp (OH) within layers; details uncertain.
γ-$Al(OH)_3$	Gibbsite (Hydrargillite)	ccp OH within layers of edge-shared $Al(OH)_6$; octahedra stacked vertically via H bonds.

Table 7.10 Composition of gem stones and their colours

Precious gem	Colour	Composition
1. Aqua marina	Bluish green, greenish	Be, Al Silicate
2. Fibrolite Jade	Blue white, green violet red orange	Be, Al Silicate Na, Al Silicate
3. Phenacite	Colourless as diamond	Be aluminate
4. Ruby	Red	Alumina with Cr^{3+}
5. Sapphire	Blue	Alumina with ($Fe^{2+/3+}Ti^{4+}$)
6. Emerald	Green	Alumina with (Cr^{3+}/V^{3+})
7. Amythest	Violet	Alumina with (Cr^{3+}/Ti^{4+})
8. Topaz	Yellow	Alumina with (Fe^{3+})

7.11.3 Aluminium Hydroxide $Al(OH)_3$

Preparation: It is prepared as a white gelatinous precipitate by the action of hydroxide solution with aluminium salt solution.

$$AlCl_3 + 3NH_4OH \longrightarrow Al(OH)_3 + 3NH_4Cl$$

Properties: The precipitate of aluminium hydroxide is amorphous and dries to a glassy solid.

It is amphoteric in behaviour .

$$Al(OH)_3 + 3HCl \longrightarrow AlCl_3 + 3H_2O$$

$$Al(OH)_3 + 3NaOH \longrightarrow 3Na_3AlO_3 + 3H_2O$$

$$Al(OH)_3 + NaOH \longrightarrow NaAlO_2 + 2H_2O$$

It dissolves in sodium hydroxide to form hydroxy aluminate ions, e.g.,

$$Al(OH)_3 + OH^- \longrightarrow [Al(OH)_4]^-$$

$$Al(OH)_3 + 3OH^- \longrightarrow [Al(OH)_6]^{3-}$$

Structure: It is a polynuclear complex. It has a layer structure such that each aluminium atom is associated with six OH groups and each OH group is associated with two aluminium atoms.

Uses: When aluminium hydroxide is partly dehydrated at 475–525K, it adsorbs moisture very rapidly. This alumina gel is used as a drying agent and as an adsorbent.

Aluminium hydroxide is used for water proofing cloth. The cloth is soaked in the solution of aluminium acetate and then subjected to the action of steam when colliodal aluminium hydroxide is precipitated in the pores.

It is also used as mordant in dyeing. Aluminium hydroxide is first precipitated on the fibres of cloth and the fabric is then dipped in the dye.

7.11.4 Aluminium Chloride $AlCl_3$

Preparation: It is prepared by the reaction of aluminium, or its oxide or hydroxide and hydrochloric acid.

$$2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2$$

$$Al_2O_3 + 6HCl \longrightarrow 2AlCl_3 + 3H_2O$$

$$Al(OH)_3 + 3HCl \longrightarrow AlCl_3 + 3H_2O$$

On crystallization, hydrated crystals $AlCl_3 \cdot 6H_2O$ are obtained. The anhydrous aluminium chloride cannot be prepared by heating hydrated crystals, because it decomposes on heating.

$$2AlCl_3 \cdot 6H_2O \longrightarrow 2AlCl_3 \cdot 3H_2O + 6H_2O$$

$$2AlCl_3 \cdot 3H_2O \xrightarrow{\Delta} 2Al(OH)_3 + 6HCl$$

$$2Al(OH)_3 \xrightarrow{\Delta} Al_2O_3 + 3H_2O$$

Anhydrous aluminium chloride can be prepared by the reaction of heated aluminium and chlorine or hydrogen chloride gas.

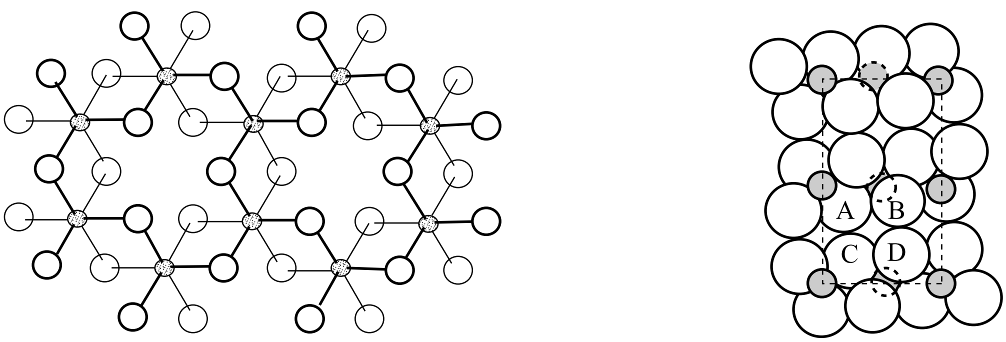
$$2Al + 3Cl_2 \longrightarrow 2AlCl_3$$

$$2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2$$

The vapours of anhydrous aluminium chloride are condensed in the receiver.

In Mac Affe method, anhydrous aluminium chloride can be prepared by heating a mixture of Al_2O_3 and coke in a current of dry chlorine gas.

$$Al_2O_3 + 3C + 3Cl_2 \longrightarrow 2AlCl_3 + 3CO$$



(a) (b)

Fig 7.16 (a) Part of the layer of $Al(OH)_3$ (idealized). The heavy and open circles represent OH groups above and below the plane of the Al atoms shaded. (b) Structure of $Al(OH)_3$ viewed in a direction parallel to the layers to illustrate the difference in packing of OH groups of different layers

Anhydrous aluminium chloride can also be prepared by the dehydration of hydrated aluminium chloride using thionyl chloride or by an ether such as 2,2-dimethoxy propane.

$$AlCl_3 \cdot 6H_2O + 6SOCl_2 \longrightarrow AlCl_3 + 6SO_2 + 12HCl$$

$$AlCl_3 \cdot 6H_2O + 6CH_3-\underset{\underset{OCH_3}{|}}{\overset{\overset{OCH_3}{|}}{C}}-CH_3 \longrightarrow AlCl_3 + 12CH_3OH + 6CH_3COCH_3$$

Properties: Anhydrous aluminium chloride is a white crystalline solid. It is deliquescent in nature when exposed to moist air, it gives fumes because of the formation of hydrogen chloride.

$$AlCl_3 + 3H_2O \longrightarrow Al(OH)_3 + 3HCl$$

On heating, it sublimes at 180°C and melts at 188°C under pressure (2 atm). Below 350°C, its vapour density corresponds to the formula Al_2Cl_6 whereas above 850°C, it is monomeric $AlCl_3$. It is covalent when anhydrous as it does not conduct current in fused state. It is soluble in organic solvents such as alcohol, ether and benzene. The dimeric formula is retained in non-polar solvents but is broken into $[Al(H_2O)_6]Cl_3$ on dissolution in water on account of heat of hydration as explained earlier in this chapter.

When dissolved in water, first, it changes to hydrated salt and then hydrolyses to give acidic solution.

$$Al_2Cl_6 + 12H_2O \longrightarrow 2AlCl_3 \cdot 6H_2O$$
$$AlCl_3 + 3H_2O \longrightarrow Al(OH)_3 + 3HCl$$

With ammonium hydroxide, it gives gelatinous white precipitate insoluble in excess of ammonium hydroxide.

$$AlCl_3 + 3NH_4OH \longrightarrow Al(OH)_3 + 3NH_4Cl$$

With sodium hydroxide, it gives gelatinous white precipitate soluble in excess of sodium hydroxide.

$$AlCl_3 + 3NaOH \longrightarrow Al(OH)_3 + 3NaCl$$
$$Al(OH)_3 + NaOH \longrightarrow NaAlO_2 + 2H_2O$$

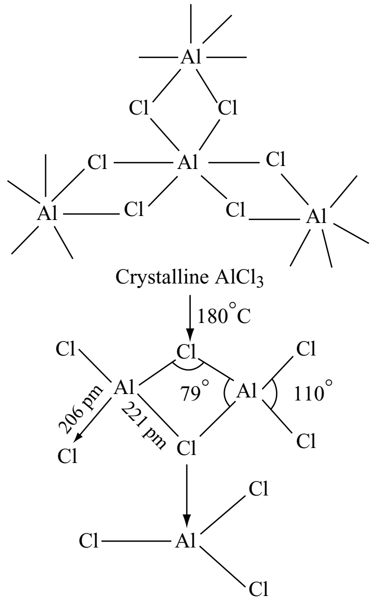
Uses

(i) It is used as a catalyst in Friedel-Craft's reaction.

(ii) It is used in the manufacture of gasoline by cracking of high boiling fractions of petroleum.

(iii) It is extensively used in the manufacture of dyes, drugs and perfumes.

Structure: Anhydrous aluminium chloride has **six-coordinate layer structure** in the solid phase and converts into **four coordinate** molecular dimers at its melting point. Above 850°C, it converts into tricoordinate monomeric $AlCl_3$.



Except AlF_3, all the trihalides of aluminium $AlCl_3$, $AlBr_3$ and AlI_3 exist as dimers.

There is evidence of bridging in the dimeric structure of various trihalides in which donor bonds are formed from halogen to metal giving the latter a covalency of four. The chlorine atoms are arranged tetrahedrally around each aluminium atom. It is observed that terminal halogen metal bond length is shorter. This may be due to the intramolecular pπ-dπ double bonding by accepting p-electrons from the terminal halogens in the vacant d-orbital of the metal.

7.11.5 Aluminium Sulphate

Preparation: It can be prepared by the reaction of aluminium metal or its oxide or hydroxide in dilute sulphuric acid.

$$2Al + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3H_2$$
$$Al_2O_3 + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3H_2O$$
$$2Al(OH)_3 + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 6H_2O$$

It is manufactured by the treatment of bauxite or china clay with boiling sulphuric acid.

$$Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 2SiO_2 + 5H_2O$$

In this process, silica remains as insoluble mass which is removed by filtration. The filtrate is concentrated to get its crystals $Al_2(SO_4)_3 \cdot 18H_2O$.

Properties: It is a white crystalline substance. It is soluble in water, the solution is acidic because of hydrolysis. On strong heating, it decomposes into alumina.

$$Al_2(SO_4)_3 \cdot 18H_2O \xrightarrow{\Delta} Al_2(SO_4)_3 + 18H_2O$$

$$Al_2(SO_4)_3 \longrightarrow Al_2O_3 + 3SO_3$$

Uses: It is used in purifying sewage water, dyeing, calico printing, foamite fire extinguishers, tanning of leather, water proof cloth and sizing of paper. It is also used in the manufacture of alums.

7.11.5 Potash Alum $K_2SO_4 \cdot Al_2\,(SO_4)_3 \cdot 24H_2O$

The alums are the double salts. The general formula of the alum is $M_2SO_4 \cdot M'_2(SO_4)_3 \cdot 24H_2O$ or $M{\cdot}M'(SO_4)_2 \cdot 12H_2O$ where M is monovalent cation or radical such as Na^+, K^+, Rb^+, Cs^+, Tl^+ and NH_4^+, and M′ is a trivalent cation Al^{3+}, Cr^{3+}, Fe^{3+}, Mn^{3+}, Co^{3+}, V^{3+}, Ti^{3+} etc.

Alums are named according to the metals constituting them, e.g.,

Potash iron alum	$K_2SO_4 \cdot Fe_2(SO_4)_3 \cdot 24H_2O$
Sodium aluminium alum	$Na_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$
Rubidium chromium alum	$Rb_2SO_4 \cdot Cr_2(SO_4)_3 \cdot 24H_2O$

Moreover, some common alums are named after one metal only, e.g.,

Potash alum	$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$
Chrome alum	$K_2SO_4 \cdot Cr_2(SO_4)_3 \cdot 24H_2O$
Manganic alum	$K_2SO_4 \cdot Mn_2(SO_4)_3 \cdot 24H_2O$
Ferric alum	$(NH_4)_2SO_4 \cdot Fe_2(SO_4)_3 \cdot 24H_2O$

All alums are isomorphous. In alums, every cation is surrounded by six water molecules octahedrally. Lithium cannot form alum because of its small size; it cannot be coordinated with six water molecules. The monovalent cations are larger in size hence the water molecules are too far from those to be chemically bound to the metal ion. The trivalent cation, being small, is chemically bound to the water molecules.

The common alum is potash alum.

Preparation: It is prepared by concentration of calculated quantities of potassium sulphate and aluminium sulphate solutions. On cooling, crystals of potash alum separate out.

$$K_2SO_4 + Al_2(SO_4)_3 + 24H_2O \longrightarrow K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$$

It is manufactured from the mineral alum stone or alunite $K_2SO_4 \cdot Al_2\,(SO_4)_3 \cdot 4Al(OH)_3$ by dissolving it in sulphuric acid when aluminium hydroxide changes into aluminium sulphate.

$$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 4Al(OH)_3 + 6H_2SO_4 \longrightarrow K_2SO_4 + 3Al_2(SO_4)_3 + 12H_2O$$

A calculated quantity of potassium sulphate is added to it and on crystallization, alum separates out.

$$[K_2SO_4 + 3Al_2(SO_4)_3 + 12H_2O] + 2K_2SO_4 + 60H_2O \longrightarrow 3K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$$

Alum shale (Al_2O_3 . $xSiO_2$. FeS_2) is also used for the manufacture of potash alum. It is first roasted in the excess of air when iron sulphide changes into ferrous sulphate and sulphuric acid. The sulphuric acid thus formed converts the alumina into aluminium sulphate.

$$2FeS_2 + 2H_2O + 7O_2 \longrightarrow 2FeSO_4 + 2H_2SO_4$$

$$Al_2O_3 \cdot xSiO_2 + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + xSiO_2 + 3H_2O$$

Ferrous sulphate is separated by fractional crystallization and the mother liquor is mixed with potassium sulphate when potash alum crystallizes out.

Properties: It is a white crystalline substance soluble in water. Its solution is acidic because of the hydrolysis of aluminium sulphate. On heating, it swells up because of the removal of water of crystallization and on further heating, it gives alumina.

Uses: Potash alum is used

(i) in the purification of drinking water.
(ii) as a styptic to arrest the bleeding.
(iii) as a mordant in dyeing.
(iv) in tanning of leather.
(v) in sizing the paper.

7.11.6 Pseudo Alums

The pseudo alums are the double salts of sulphates of bivalent and trivalent cations with 24 molecules of water of crystallization. These are not isomorphous with the true alums, e.g.,

Manganese aluminium pseudo alum

$$MnSO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$$

Ferrous aluminium pseudo alum

$$FeSO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$$

7.11.7 Ultramarines

These are double silicates of sodium and aluminium with sulphur. White ultramarine $Na_5Al_3Si_3SO_{12}$ is formed by fusing kaolin, soda ash, sulphur and some resin. It changes to green ultramarine $Na_5Al_3Si_3S_2O_{12}$ in the presence of air. Blue ultramarine $Na_5Al_3Si_3S_3O_{12}$ is obtained by further heating with sulphur and air. Violet form is obtained by heating in a current of chlorine. These substances are used in paints and laundry.

7.11.8 Lapislazuli

Lapislazuli is one of the rare minerals of beautiful blue colour. It is sodium aluminium silicate with some sulphur. It is used as a precious gem.

KEY NOTES

- Elements B, Al, Ga, In and Tl belong to III A Group and p-block of the periodic table.
- The outer electronic configuration of these elements is $ns^2\ np^1$.
- Boron has 2, aluminium has 8, gallium and indium have 18 electrons in their penultimate shells. Thallium has 18 electrons in the penultimate and 32 electrons in the antipenultimate shells.
- Owing to the difference in the electronic configuration in penultimate shells, there is difference in the properties of these elements, i.e., between B and Al; Al and Ga; and Ga, In and Tl.
- Aluminium is the third most abundant element and first most abundant metal in the earth's crust. Gallium, Indium and Thallium are less abundant in nature.
- Boron is non-metal and others are metals and the metallic character increases from B to Tl.

PHYSICAL PROPERTIES

- **Atomic and ionic sizes** of Group III A elements do not increase regularly. A large increase between B and Al is due to the lesser number of electrons in the penultimate shell of B. Al and Ga have almost equal size because of poor shielding capacity of 3d electrons in Ga. In and Tl are also almost equal in size because of lanthanide contraction in Tl.
- Density increases down the group. Ga has unusual expansion during solidification. Solid Ga is less denser than liquid Ga.
- Melting and boiling points of B are very high because of its giant covalent polymeric structure in both solid and liquid state. melting point decreases from B to Ga and then increases. Ga has abnormally low Melting point because it consists of Ga_2 molecules. Ga exist in liquid state in a wide range of temperatures 30–2000°C and hence used in high temperature thermometers.
- **Boiling points** decrease regularly from B to Tl. Though melting point of Ga is less, its boiling point is in regular order.
- **Ionization energies** decreases down the group. Large decrease between B to Al but then onwards the decrease is very little because of the poor shielding effect of d-electrons in Ga and In and d and f-electrons in Tl which results in the increase of effective nuclear charge.
- **Electronegativity** decreases from B to Al but then onwards there is little increase down the group instead of normal decrease which is attributed to the increased effective nuclear charge as said above.
- Group III A elements exhibit +1 and +3 **oxidation states**. The stability of +1 oxidation state increases down the group whereas the stability of +3 oxidation state decreases because of inert pair effect.
- The pair of electrons in ns orbital tend to remain paired and are not participating in compound formation. This pair of electrons is called **inert pair**.
- In +1 oxidation state, Tl behaves similar to alkali metals. Hence, it is nick named as **duck-gilled platypus**.
- In +1 oxidation state, these elements act as **reducing agents** and their reduction power decreases from B (+1) to Tl (+1) as the stability of +1 oxidation state increases.
- In **+3** oxidation state, they act as **oxidizing agents** and the oxidation power increases down the group because of the decrease in the stability +3 oxidation state because of inert pair effect.
- B never forms B^{3+} ion because the sum of three ionization energies ($I_1 + I_2 + I_3$) of B is very large. 'B' always form covalent compounds.
- AlF_3 and Al_2O_3 are ionic but other aluminium compounds are covalent in solid state, e.g., $AlCl_3$ is covalent in solid state; however, in solution, it ionizes because the hydration energy of small Al^{3+} ion and the sum of hydration energies and Ea, of three Cl^- ions exceeds the sum of three ionization energies of Al.
- Boron is non-metal. Aluminium is the most electropositive metal in the group. The electropositive character of Ga, In and T*l* is less than Al and decreases from Ga to Tl. This is again attributed to the increased effective nuclear charge because of poor shielding capacity of inner d electrons.
- The tendency to form complexes by these elements is more than the s-block elements. Boron exhibits a maximum covalence 4 because of the absence of d-orbitals in its valence shell but other elements exhibits a covalence up to 6 because of the availability of d-orbitals in their valence shells.

REACTIVITY OF GROUP III A ELEMENTS

- These elements do not react with air in cold condition. B burns in air forming B_2O_3. Al does not react with air because of the formation of protective oxide layer. However, when powdered Al is exposed to liquid oxygen, it explodes because the reaction between Al and O_2 is highly exothermic which takes place instantaneouly because of the availability of large surface area. Ga and In are stable in air but Tl forms an oxide layer.
- B and Al when burnt in air react with both O_2 and N_2 in air forming oxides and nitrides.
- Boron is not affected by water but red hot boron decomposes steam. Though the reaction between

Al and H_2O is thermodynamically possible, it does not react with water because of the presence of oxide layer. However, it corrodes in salt water because the oxide layer will be removed by salts. Owing to this reason, Al vessels should not be kept with water over night. Ga and In are attacked by water only in the presence of oxygen.

- Boron does not react with non-oxidizing acids but oxidized to boric acid with oxidizing acids such as conc. HNO_3.
- The oxide film present on the surface of Al prevents it to react with dil. acids but it readily react with dil. HCl and the reaction becomes vigorous with the increase in concentration of HCl. Dil. H_2SO_4 and dil. HNO_3 react with Al slowly. Conc. H_2SO_4 liberates SO_2 with Al but conc. HNO_3 renders the Al passive.
- If Al is amalgamated the, reactivity increases because of the destruction of oxide layer.
- B and Al dissolve in alkalis forming borates and aluminates with the liberation of H_2.
- Sodium meta aluminate exists as hydrated tetrahydroxo aluminates $[Al(OH)_4(H_2O)_2]^-$ and aluminate is $[Al(OH)_6]^{3-}$ in which Al exhibits a coordination number 6.
- Boron can react with highly electro positive metals such as Mg forming borides. Others form alloys with metals.
- With carbon, boron gives B_4C whereas aluminium gives Al_4C_3.
- Both B and Al react with N_2 forming nitrides BN and AlN which on hydrolysis give ammonia.
- Both B and Al give trihalides with halogens. Tl gives both trihalides and mono halides.

Reactions of Boron

Reaction	Remark
1. $B + O_2 \longrightarrow B_2O_3$	At high temperature
2. $B + S \longrightarrow B_2S_3$	At high temperature
3. $B + N_2 \longrightarrow BN$	At high temperature
4. $B + X_2 \longrightarrow BX_3$	X = F, C*l*, Br, I
5. $B + NaOH \longrightarrow Na_3BO_3 + H_2$	Fused with alkali
6. $B + NH_3 \longrightarrow BN + 3H_2$	
7. $B + M \longrightarrow M_xB_y$	Highly electropositive metals form borides.

Chemical reactivity of Group III A (except boron) elements

Reaction	Remark
Action of air 1. $M + O_2 \longrightarrow M_2O_3$	A*l* corrodes in air but corrosion is prevented by its oxide layer. When burnt in air, they form M_2O_3. T*l* forms some M_2O_3. Al also form, A*l*N. Powdered Al explodes in air because of highly exothermic reaction.
2. $M + H_2O \longrightarrow M_2O_3 + H_2$	Al does not react with pure H_2O because of oxide layer but corrodes easily in salt water. A*l* decomposes steam.
3. $M + acid \longrightarrow$	Al reacts with acids. Al becomes passive with conc. HNO_3.
4. $M + NaOH \longrightarrow NaMO_2 + H_2$	Only Al and Ga react with alkalis. $Na_3MO_3 + H_2$
5. $M + X_2 \longrightarrow MX_3$	All form trihalides; TlI_3 is $Tl^+ \left(I_3^-\right)$

COMPOUNDS OF GROUP IIIA ELEMENTS

Hydrides

- Boron forms a large number of hydrides called **boranes** with general formula B_nH_{n+4} and B_nH_{n+6}. Aluminium forms only one hydride called **alane** which have polymeric structure in which each Al is surrounded with six H atoms consisting Al H_2 Al bridges.
- Gallium and indium hydrides are unstable.
- Boron hydride and aluminium hydrides form complex hydrides such as $LiAlH_4$ $NaBH_4$, in which H^- ion acts as donor.

Oxides

- These elements form sesqui-oxides of the type M_2O_3. Ga and In form unstable but Tl form most stable monoxides Ga_2O, In_2O and Tl_2O.

- B_2O_3 is acidic Al_2O_3, Ga_2O_3 and In_2O_3 are amphoteric but Tl_2O_3 is basic. Acidic character decreases and basic character increases for the oxides down the group. Tl_2O_3 on heating converts into Tl_2O.

Hydroxides

- The hydroxy compounds of boron and aluminium are formed when their oxides react with water.
- The acidic character of the hydroxides decreases and basic character increases down the group.
- $B(OH)_3$ or H_3BO_3 is a weak acid, $Al(OH)_3$, $Ga(OH)_3$ are amphoteric $In(OH)_3$ is more basic than acidic but $Tl(OH)_3$ is completely basic. TlOH behaves almost similar to NaOH.

Halides

- Group III A elements form three types of halides viz., monohalides, dihalides and trihalides of which trihalides are much important.
- All the Group III A elements form trihalides of the type 'MX_3 (X = F, Cl, Br or I)'.
- $TlCl_3$ and $TlBr_3$ are unstable and tend to lose halogen forming mono halide because of the oxidizing tendency of the +3 oxidation state of thallium coupled with reducing property of Cl^- and Br^-.
- In TlI_3, thallium exists in +1 oxidation state containing Tl^+ and I_3^- ions.
- Stability of different halides for given element is fluoride > chloride > bromide > iodide.
- All boron halides are covalent. Fluorides of Al, Ga, In and Tl are ionic and have high melting points.
- Chlorides, bromide and iodides of Al, Ga, In and Tl are covalent when anhydrous, ionic character increases with the increase in the cation size.
- Trihalides fumes in air because of hydrolysis. The hydrogen halide due to reaction with moisture in air collects the water molecules, forms as droplets and appears as fumes.
- These trihalides are electron-deficient compounds and act as Lewis acids.
- The relative strengths of the trihalides of boron as Lewis acids is $BF_3 < BCl_3 < BBr_3$. This order is opposite to the order expected from electronegativities and is due to the back bonding of electrons from halogen to boron.
- The back bonding is effective in BF_3 and decreases rapidly from BCl_3 to BBr_3 because of the increasing difference in the energies of p-orbitals of boron and halogen participating in pπ-pπ back bonding.
- Boron trihalides can form adducts even with weakest bases such as ethers, amines, phosphines, alcohols, anions and CO by donating a lone pair to boron.
- The adduct formation with Lewis bases accompanies the rehybridization of boron resulting in the loss of B-X double bond character.
- The enthalpy change in the adduct formation with trimethyl amine is more for BBr_3 and least for BF_3 indicating the double bond character follows the order $BF_3 > BCl_3 > BBr_3$.
- From the experimental and calculated bond lengths (sum of the covalent radii of boron and halogen), it was found that the experimental value is less than the calculated value because of double bond character and the difference between calculated value and experimental value is more for BF_3 indicating the effective overlapping of 2p orbitals of both boron and fluorine.
- When BF_3 forms an adduct such as BF_4^- and $BF_3 \cdot NH_3$, the B-F bond length increases because of the change from double bond character to single bond character.
- Halides of other Group III A elements are also electron deficient and can act as Lewis acids but the Lewis acidic character decreases with the increase in the size of cation down the group.
- AlF_3 is ionic solid having high melting point. Other halides of Al are low melting volatile solids because AlF_3 is ionc but other halides of Al are covalent.
- Except fluorides, other halides of Al, Ga, and In exists as dimers. This is because boron being small atom cannot coordinate with four large halogen atoms.
- Though F atom is smaller, BF_3 does not dimerize. Owing to dimerization, there will be rehybridization of boron results in the loss of resonance stability because of the loss of B-X double bond character.

Anomalous Behaviour of Boron

- Owing to the absence of d-orbitals in the valence shell of boron, its coordination number is limited to 4 whereas other elements can have coordination number of 6.
- Boron does not exhibit inert pair effect.
- Boron can combine with metals to form metal borides in whereas it exhibit –3 oxidation state.
- Boron is not attacked by non-oxidizing acids such as HCl whereas others are attacked.
- Boron does not decompose water or steam but other Group III A elements decompose hot water or steam.
- Boron is non-metal and bad conductor of electricity but other elements are metals and good conductors of electricity.
- Boron exhibits, allotropy but other elements do not exhibit allotropy.
- Boron never forms B^{3+} ion but other elements can form M^{3+} ions.

- Boron forms a large number of volatile hydrides called boranes which are electron deficient but other elements form only one polymeric hydride. Thallium does not form hydride.
- B_2O_3 is acidic whereas similar oxides of other elements are either amphoteric or basic.
- Boron halides are monomeric whereas halides of other elements are dimeric.

Similarities between Boron and Aluminium

- Both B and Al have same outer electronic configuration.
- Both B and Al exhibit +3 oxidation state.
- Both B and Al form covalent compounds.
- Both form similar halides BCl_3 and $AlCl_3$ with chlorine and readily hydrolyse in water.
- Both form nitrides when heated in nitrogen which hydrolyses in water liberating ammonia.
- Both react with conc. H_2SO_4 liberating SO_2 gas.
- Both react with alkalis liberating H_2 gas.

Dissimilarities of Boron and Aluminium

- Boron is non-metal but aluminium is metal.
- Boron exhibits allotropy whereas aluminium does not exhibit allotropy.
- Crystalline boron is very hard whereas aluminium is sufficiently soft.
- Al is good conductor of heat and electricity whereas B is bad conductor.
- B has very high melting point compared to Al.
- Maximum covalency of B is 4 whereas that of Al is 6.
- Boron is not attacked even by steam whereas aluminium decomposes steam liberating H_2.
- Boron is not attacked by dilute acids but aluminium evolves H_2 from dil. H_2SO_4 and dil. HCl. Conc. HNO_3 oxidizes boron to boric acid but renders the aluminium passive.
- Borates are stable whereas aluminates are unstable.
- Boron forms a large number of hydrides called boranes but aluminium forms only one polymeric hydride alane.
- Oxide and hydroxide of boron are acidic whereas those of aluminium are amphoteric.
- Boron reacts with metals forming borides but aluminium forms alloys.

Diagonal Relationship of Boron with Silicon

- Both B and Si are non-metals having high melting points and are bad conductors of electricity.
- Both B and Si exhibit allotropy.
- The electronegativities and ionization energies of both B and Si are nearly equal.
- Oxides of both B and Si, B_2O_3 and SiO_2 are acidic and with water yield corresponding acids boric acid and silicic acid, which are weak acids.
- Both B and Si can be prepared by reduction of their oxides B_2O_3 and SiO_2 with Mg.
- Both B and Si form carbides B_4C and SiC which are very hard substances and are used as abrasives.
- Both B and Si react with metals forming borides and silicides which decompose by dilute acids to form hydrides.
- Both B and Si react with halogen forming halides which readily hydrolysed in water.
- Both B and Si form a number of covalent hydrides which are volatile, spontaneously inflammable and readily hydrolyses.
- Both these elements and their oxides form borates and silicates with alkalis.

BORON

- Borax on treating with HCl or H_2SO_4 gives boric acid which on heating gives B_2O_3.
- Colemanite when fused with sodium carbonate forms calcium carbonate, borax and sodium metaborate, sodium metaborate also converts into borax when CO_2 is passed through it.
- B_2O_3, obtained by heating boric acid, is reduced to get boron with Na, K, Mg etc.
- The boron obtained by the reduction of B_2O_3 with Mg is about 95–98% pure and is called **Moisson boron**.
- Reduction of BCl_3 with zinc gives crystalline boron.
- Thermal decomposition of boranes at about 900°C gives amorphous boron, but the thermal decomposition of BI_3 above 1000°C gives crystalline boron.
- Crystalline boron contains clusters of boron atoms as B_{12} unit arranged in a **icosahedron** structure.
- Boron steel or boron carbide rods are used to control nuclear reactions because boron nucleus has a very high cross section area to capture neutrons.

COMPOUNDS OF BORON

Boron Trioxide B_2O_3

- It is the anhydride of boric acid and can be prepared by heating boric acid. It can also be prepared by dehydration of boric acid over P_2O_5 at about 475K in vaccum.
- It is colourless transparent glassy mass, hygroscopic in nature; absorbs moisture from air, becomes opaque and finally converts into boric acid.

- When heated with transition metal salts, it forms metal metaborate having characteristic colour by which the metal can be detected. For example, $CuSO_4$ gives $Cu(BO_2)_2$ having blue colour.

Boric Acids

- Boron forms a large number of oxo acids such as ortho boric acid (H_3BO_3), metaboric acid (HBO_2), pyroboric acid ($H_6 B_4O_9$) and tetra boric acid ($H_2B_4O_7$).
- **Ortho boric acid** can be prepared by adding calculated amount of H_2SO_4 or HCl to borax.

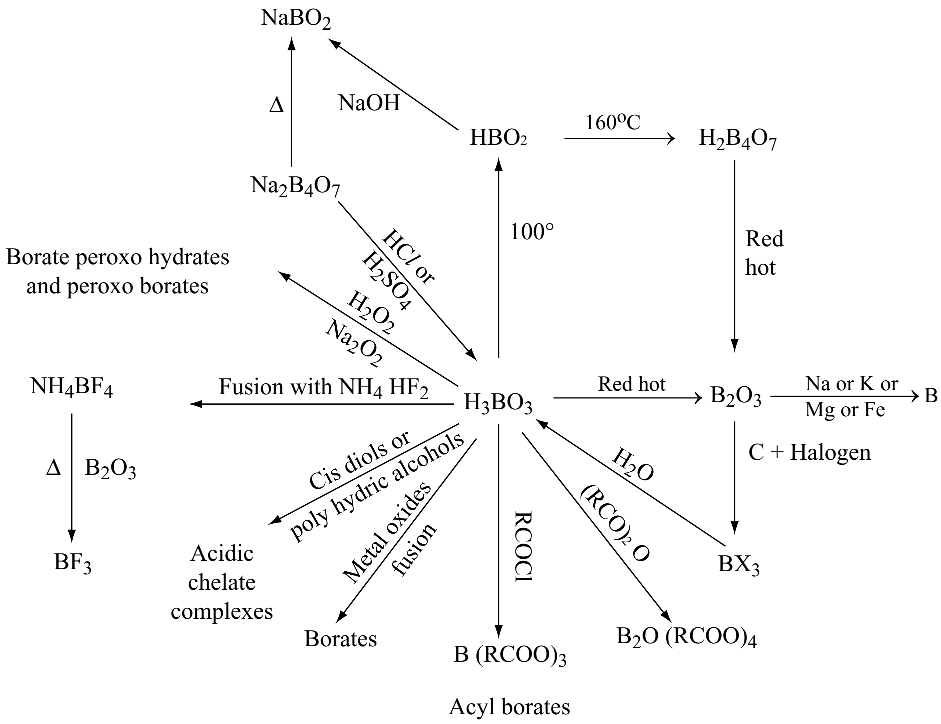
$$Na_2B_4O_7 + H_2SO_4 + 5H_2O \longrightarrow Na_2SO_4 + 4H_3BO_3$$

- When SO_2 gas is passed through a suspension of colemanite in water, boric acid is formed.

$$Ca_2B_6O_{11} + 4SO_2 + 11H_2O \longrightarrow 2Ca(HSO_3)_2 + 6H_3BO_3$$

- Boric acid is also formed by the action of super heated water on boron nitride or boron sulphide.
- It is white crystalline substance, soft, soapy to touch, moderately soluble in cold water.
- Solubility of boric acid increases with the increase in temperature.
- On heating at 375 K, it gives metaboric acid; at 435K, it gives tetraboric acid; and at red hot condition, it forms boron trioxide.
- Boric acid is **weak monobasic Lewis acid**. It does not liberate H^+ ion but accepts a pair of electrons from OH^-
- With increase in the concentration, acidity increases. At low concentration, it is present as meta borate ion $[B(OH)_4]^-$; however, as concentration increases, acidity increases because of the formation of polymeric metaborate ion $[B_3O_3(OH)_4]^-$
- It is a weak acid and cannot be titrated with NaOH solution. However, in the presence of polyhydroxy compounds such as mannitol, sorbitol, glucose, glycerol and ethylene glycol, boric acid acts as stronger acid and can be titrated with NaOH using phenolphthalein as indicator.
- When boric acid is heated with ethyl alcohol, it gives ethyl borate vapours which burns with green-edged flame. This is used for the detection of borate ion.
- In orthoborate ion, boron atom is involved in sp^2 hybridization and is in bond with three OH groups in a planar triangular manner.
- Boric acid contains planar triangular $B(OH)_3$ units which are bonded together through hydrogen bonds in two-dimensional sheets.
- Each boric acid molecule can form six hydrogen bonds and hydrogen atoms act as bridge between two oxygen atoms with a covalent bond on one side and a hydrogen bond on the other side of different BO_3^{3-} units.
- Borate ions are arranged in hexagonal rings and the number of hydrogen bonds in that hexagonal ring are six.

Summary of the reactions of Boric acid.



Metaboric Acid

- It is prepared by heating orthoboric acid.
- Metaboric acid when dissoved in water forms orthoboric acid.
- Metaborates are stable even at high temperatures.
- Metaboric acid on reaction with sodium hydroxide forms either sodium metaborate or borax.

Borax $Na_2B_4O_7 \cdot 10H_2O$

- Borax occurs naturally as tincol.
- It can be prepared from the mineral colemanite by boiling with concentrated solution of sodium carbonate.

$$Ca_2B_6O_{11} + 2Na_2CO_3 \longrightarrow 2CaCO_3 + Na_2B_4O_7 + 2NaBO_2$$

- The $NaBO_2$ also converts into borax by passing CO_2 into metaborate solution after separating $CaCO_3$ and $Na_2B_4O_7$.

$$4NaBO_2 + CO_2 \longrightarrow Na_2B_4O_7 + Na_2CO_3$$

- Naturally occuring tincol is dissolved in water, filtered, concentrated and crystallized to get borax.
- Small quantities of borax can be obtained by neutralizing boric acid with NaOH.
- Borax is sparingly soluble in cold water but highly soluble in hot water.
- Crystallization of borax from saturated solution at ordinary temperature gives **prismatic borax** $Na_2B_4O_7 \cdot 10H_2O$.
- Crystallization at 62°C from saturated solution gives **octahedral or Jeweller's borax** $Na_2B_4O_7 \cdot 5H_2O$.
- At red hot condition, it forms **borax glass** which is anhydrous and absorbs moisture converting into decahydrate.
- Borax on heating with ammonium chloride gives a mixture of **boron nitride** and **oxide**.

$$Na_2B_4O_7 + 2NH_4Cl \longrightarrow 2BN + B_2O_3 + 2NaCl + 4H_2O$$

- When heated, it swells into white mass finally to a transparent glassy mass called borax bead. This bead gives characteristic coloured bead when heated with a transition metal ion because of the formation of metaborate.
- In borax, two B atoms are in sp^3 and other two B atoms are in sp^2 hybridization having tetrahedral and planar triangular structures, respectively.
- The borax ion is $[B_4O_5(OH)_4]^{2-}$, the remaining eight water molecules are associated with the two sodium ions.
- The formula of borax is written as $Na_2\,[B_4O_5\,(OH)_4] \cdot 8H_2O$. It contain two six-membered heterocyclic rings, five B-O-B bridge bonds, 10 B-O bonds and four OH groups on four boron atoms.
  Borax is **metaborate**

Boron Halides

- BF_3 is prepared by the action of conc. H_2SO_4 on B_2O_3 and CaF_2

$$3CaF_2 + B_2O_3 + 3H_2SO_4 \longrightarrow 2BF_3 + 3CaSO_4 + 3H_2O$$

- Other boron trihalides can be prepared by heating B_2O_3 with carbon in the presence of halogen.
- BF_3 is gas, BCl_3 is a liquid, BBr_3 is viscous liquid and BI_3 is crystalline solid. Therefore, their melting points and boiling points also increase with them increase in the atomic number of halogen.
- All the trihalides of boron hydrolyse in water giving boric acid and hydrogen halide (HCl, HBr and HI).
- BF_3 on hydrolysis forms H_3BO_3, BF_4^-, $[BF_3OH]^-$, HF and H^+ ions because of incomplete hydrolysis and the reaction of H_3BO_3 with HF formed initially.

$$4BF_3 + 3H_2O \longrightarrow H_3BO_3 + 3H^+ + 3[BF_4]^-$$

$$BF_4^- + H_2O \longrightarrow [BF_3OH]^- + HF$$

- The degree of hydrolysis increases from BCl_3 to BI_3 which supports the other evidences for increasing Lewis acid character from BF_3 to BI_3.
- When boric acid is dissolved in aqueous HF forms fluoboric acid HBF_4 which is tetrahedral and has least tendency to act as a ligand.

Reactions of Boron Halides

Reaction boron trifluoride	Remark
1. $BF_3 + H_2O \longrightarrow H_3BO_3 + HF + BF_4^- + [BF_3OH]^-$	
2. $BF_3 + NH_4F \xrightarrow{\Delta} NH_4\,BF_4$	
3. $BF_3 + RMg\,Br \longrightarrow RBF_2, R_2BF, R_3B$	
4. $BF_3 + NH_3 \cdot NR_3 \longrightarrow F_3B \cdot NH_3, F_3B \cdot NR_3$	
5. $BF_3 + \text{Ether} \longrightarrow BF_3 \cdot OR_2$	
6. $BF_3 + NaBH_4 \xrightarrow{\text{Ether}} B_2H_6$	

	Remark
7. $BF_3 + CH_3CN \longrightarrow F_3B \cdot NC\,CH_3$	
8. $BF_3 \xrightarrow{Al_2Cl_6 \text{ or } Al_2Br_6} BCl_3$ or BBr_3	
9. $BF_3 \xrightarrow{NaH \text{ or } BF_4} B_2H_6$	
Reactions of BCl_3 and BBr_3 (BX_3; X = Cl or Br)	
1. $BX_3 + H_2 \xrightarrow{Heat} B$	
2. $BX_3 + H_2O \longrightarrow H_3BO_3$	
3. BX_3 + alcohol $\longrightarrow B(OR)_3$	
4. BX_3 + liquid $NH_3 \longrightarrow B(NH_2)_3$	
5. $BX_3 + C_6H_5NH_2 \xrightarrow[\Delta]{Benzene} B(NHC_6H_5)_3 \xrightarrow{HCl} B_3N_3Cl_3(C_6H_5)_3$	
6. $BX_3 + R_3N \longrightarrow X_3B \cdot NR_3$	
7. $BX_3 + NH_4Cl \xrightarrow[150°]{C_6H_5Cl} B_3N_3H_3Cl_3$	
8. $BX_3 + CH_3NH_2 \longrightarrow B_3N_3H_3(CH_3)_3$	

Boron Hydrides

- Boron forms two series of hydrides having general formula B_nH_{n+4} known as **boranes** and B_nH_{n+6} known as **hydroboranes**.
- Boranes are named by the number of boron atoms in the molecule by Greek prefix and the number of hydrogen atoms with arabic numbers in parentheses, e.g., pentaborane (9) for B_5H_9.
- **Diborane** is the simplest borane.
- Diborane is prepared by treating BF_3 or BCl_3 with $LiAlH_4$ in diethyl ether.
- Sodium borohydride on oxidation with I_2 gives diborane.

$$2NaBH_4 + I_2 \longrightarrow B_2H_6 + 2NaI + H_2$$

- On large scale, it is manufactured by the reaction of BF_3 with NaH or LiH.
- It can also be prepared subjecting a mixture of boron trichloride and hydrogen to silent electric discharge.
- It is a colourless, highly toxic gas, slowly changes into higher boranes.
- It burns or explodes in air with evolution of much heat.

$$B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O$$

- With water, it forms boric acid (H_3BO_3) and hydrogen.
- With alkalis, it forms borates with the evolution of hydrogen.

$$B_2H_6 + 6KOH \longrightarrow 2K_3BO_3 + 6H_2$$

$$B_2H_6 + 2KOH + 2H_2O \longrightarrow 2KBO_2 + 6H_2$$

- The reactivity of diborane with hydrogen halides increases from HCl to HI and forms halogen substituted diborane in the presence of corresponding aluminium halide, which will be converted into trihalides after some time.
- The reactivity of halogens with diborane decreases from Cl_2 to I_2. With Cl_2, it forms BCl_3; with Br_2, it forms B_2H_5Br at 100°C; and I_2 does not react with B_2H_6.
- With ammonia diborane forms diammoniate of diborane ($B_2H_6 \cdot 2NH_3$) at 120°C, borazole at 200°C and boron nitride above 200°C.
- As diborane is electron deficient, it reacts with several molecules to form complexes. During the addition of smaller sterically uncrowded Lewis bases such as NH_3, primary amine and secondary amine assymmetric cleavages take place.

$$B_2H_6 + 2NH_3 \longrightarrow [BH_2(NH_3)_2]^+ [BH_4]^-$$

- With **bulky Lewis bases, symmetric cleavage** takes place forming two equal BH_3 fragments each of which form complex with Lewis base.

$$B_2H_6 + 2(CH_3)_3N \longrightarrow 2(CH_3)_3N: \rightarrow BH_3$$

- With CO, ether, thioether, tertiary amines and pyridine, it forms adducts with diborane by symmetric cleavage such as $BH_3 \cdot CO$, $H_3B \cdot O(CH_3)_2$, $H_3B \cdot S(CH_3)_2$ and $H_3B \cdot NR_3$.
- With sodium amalgam diborane, it forms an addition compound $B_2H_6 \cdot Na_2Hg$.
- In diborane, two hydrogens are one type and the other four hydrogen atoms are another type which is confirmed by specific heat measurements, NMR and Raman spectra and methylation.
- Diborane on methylation yields only $Me_4B_2H_2$ which indicates that the two hydrogens which cannot be

replaced by CH_3 groups have been different from the other which are replaced.

- Diborane contain two **hydrogen bridge bonds**.
- In each hydrogen bridge bond in diborane, two electrons are delocalized on three atoms and hence it is known as **three-centred two-electron (3c-2e) bond** or (B-H-B) **hydrogen bridge bond** or **protonated bond**.
- The two hydrogen bridge bonds in diborane are present one above and the other below the plane of molecule whereas the other four hydrogen atoms and two boron atoms (Total 6) are in the same plane.
- In diborane, each boron is in **sp^3 hybridization**. In the B-H-B bridge bond one 'sp^3' orbital of one boron atom containing one electron, one 's' orbital of H atom and one vacant sp^3 hybrid orbital of another boron atom overlap forming molecular orbital which appear like banana. Therefore, these bonds are also called **banana bonds** or sometimes as **Tau bonds**.
- The angle H_tBH_t in diborane is 120 in the plane of molecule but 97.5° in the case of H_bBH_b.

Borazole

- Borazole formed by the reaction of diborane with ammonia has the structure similar to benzene. Therefore, it is also called **inorganic benzene**.
- The difference in the structures of benzene and borazole is that in benzene, the π bonds are covalent; however, in borazole, the π bonds are dative bonds by donation of lone pair on nitrogen.
- Benzene and borazole both are isoelectronic, non-polar. Though the B-N bonds in borazole are polar, the molecule is non-polar.
- Owing to the polarity of B-N bonds, borazole is more reactive than benzene. Generally, electrophile will be added to the more negative nitrogen whereas nucleophile will be added to the less electronegative boron during addition reactions.
- With two similar substituents, benzene gives three isomers ortho, meta and para; however, borazole gives four isomers one ortho, two meta and one para isomers. In meta isomer, both similar substitutes may be either on boron or on nitrogen.

Reaction of Diborane

Reaction	Remark
1. $B_2H_6 + O_2 \xrightarrow{\text{combustion}} B_2O_3 + H_2O$	
2. $B_2H_6 + H_2O \longrightarrow H_3BO_3 + H_2$	
3. $B_2H_6 + KOH \longrightarrow KBO_2 + H_2$ or $K_3BO_3 + H_2$	
4. $B_2H_6 \xrightarrow{\Delta}$ Higher boranes	
5. $B_2H_6 + Cl_2 \longrightarrow BCl_3 + HCl$	
$B_2H_6 + Br_2 \longrightarrow B_2H_5Br + HBr$	
$B_2H_6 + I_2 \longrightarrow$ No reaction	
6. $B_2H_6 + NH_3 \longrightarrow [H_2B(NH_3)_2]^+[BH_4]^- \xrightarrow{\Delta} B_3N_3H_6$ (borazole)	
$B_2H_6 + CH_3NH_2 \longrightarrow [H_2B(CH_3NH_2)_2]^+[BH_4^-]$	
$B_2H_6 + (CH_3)_2NH \xrightarrow{-42°C} H_3\bar{B} - {}^{+}NH(CH_3)_2 \xrightarrow{\Delta} (CH_3)_2NBH_2$	
$B_2H_6 + (CH_3)_3N \longrightarrow (CH_3)_3\overset{+}{N}-\bar{B}H_3$	
7. $B_2H_6 + CH_3OH \longrightarrow BH(OCH_3)_2 + B(OCH_3)_3 + H_2$	
8. $B_2H_6 + ROR \longrightarrow R_2O \rightarrow BH_3$	
$B_2H_6 + RSR \longrightarrow R_2S \rightarrow BH_3$	
9. $B_2H_6 + HCl \xrightarrow{AlCl_3} B_2H_5Cl$	
10. $B_2H_6 \xrightarrow{BCl_3 \text{ or } BBr_3} B_2H_5Cl$ or B_2H_5Br	
11. $B_2H_6 \xrightarrow{CO \text{ or } PF_3} H_3B \leftarrow CO$ or $H_3B \leftarrow PF_3$	
12. $B_2H_6 + LiR \longrightarrow LiBH_4 + BR_3$	
13. $B_2H_6 + (CH_3)_2PH \longrightarrow (CH_3)_2PH \cdot BH_3$	

Borohydrides

- Metal borohydrides contain tetrahedral $[BH_4]^-$ unit.
- The Be, Al and transition metal borohydrides are more covalent and volatile. In these, the $[BH_4]^-$ group acts as ligand and forms covalent compounds with metal ions in which one or more 'H' atoms of $[BH_4]^-$ group act as bridge between B and metal atoms forming 3c-2e bonds.
- In $Be(BH_4)_2$, the Be and B atoms are involved in sp^3 hybridization and have tetrahedral structure. Three tetrahedrons are joined through hydrogen bridges.

- From each BH_4^-, two hydrogen atoms are involved in 3c-2e hydrogen bridge bonds, one with another boron and the other with beryllium atom thus making total four bridge bonds two B-H-B and two B-H-Be bonds.
- $Al(BH_4)_3$ contain six B-H-Al bridge bonds but no B-H-B bridge bonds (Fig. 7.13).

Boron Nitride

- In the laboratory, boron nitride is formed by fusing borax with ammonium chloride.

$$Na_2B_4O_7 + 2NH_4Cl \longrightarrow 2BN + 2NaCl + B_2O_3 + 4H_2O$$

- On large scale, it is prepared by fusing urea with boric acid.
- It is also formed by heating diborane with ammonia or boron in nitrogen or boron with ammonia.
- Boron nitride is formed by the action of ammonia on BCl_3.

$$BCl_3 + 6NH_3 \xrightarrow{\Delta} B(NH_2)_3 + 3NH_4Cl$$

$$2B(NH_2)_3 \xrightarrow{\Delta} B_2(NH)_3 + 3NH_3$$

$$B_2(NH)_3 \xrightarrow{\Delta} 2BN + NH_3$$

- It decomposes when heated with steam under pressure to yield ammonia.
- With fluorine gives BF_3 but when dissolved in HF forms NH_4BF_4.
- Structure of boron nitride is similar to graphite having layered lattice structure. It is slippery in nature used as lubricant.
- Unlike graphite, it is colourless, electrical insulator as there is large gap between filled and vacant bands.
- The bond order in boron nitride is 1.33 where as in borazole is 1.5.
- Layered boron nitride changes into denser cubic phase at high pressure and temperatures which is hard analogous to diamond.

COMPOUNDS OF ALUMINIUM

Aluminium Hydride

- It is prepared by the reaction of LiH with $AlCl_3$ in dry ether.

$$3LiH + AlCl_3 \longrightarrow 3LiCl + AlH_3$$

- Stable up to 373K but decomposes into elements above 373K.
- It forms $LiAlH_4$ with LiH, a useful reducing agent.
- It is an electron-deficient compound, acts as Lewis acid and forms addition compounds with Lewis bases such as $AlH_3 \cdot NH_3$ and $AlH_3 \cdot N(CH_3)_3$.
- It hydrolyses in water forming $Al(OH)_3$ liberating H_2.
- It is polymeric giant molecule containing AlH_2Al bridges (Fig. 7.15).

Aluminium Oxide Al_2O_3

- It is prepared by thermal decomposition of $Al_2(SO_4)_3$, $Al(NO_3)_3$, $Al(OH)_3$ etc.
- It is white amorphous powder, amphoteric in nature and very stable compound.
- Alumina when heated with carbon or calcium carbide forms aluminium carbide which on hydrolysis liberate methane.
- It is used as refractory material and adsorbent in chromatography.
- Fused alumina at 3275k is called alundum used as abrasive.
- Fused mixture of alumina and lime is called bauxite cement.
- Rubies, saphires, topazs and emeralds are naturally occuring corrundum stones containing metal oxides as impurities.
- Artificial rubies and saphires can be prepared by dropping finely powdered mixture of alumina, fluorspar and little colouring matter (metal oxides) through the centre of an oxyhydrogen flame.
- In the precious stones, Al^{3+} ions are replaced by transition metal ions which are coordinated by six oxygen atoms octahedrally because of which splitting of d orbitals takes place and because of d-d transition of electrons in the presence of light, they get the colour.

Aluminium Hydroxide

- It is obtained as white gelatinous precipitate by the action of hydroxide solution with aluminium salt solution.
- It is amphoteric in nature, dissolves in sodium hydroxide to form hydroxy aluminate ions $[Al(OH)_4]^-$ or $[Al(OH)_6]^{3-}$.
- It is a **polynuclear complex** having layer structure such that each aluminium atom is associated with six OH groups and each OH group is associated with two aluminium atoms.
- Partly dehydrated aluminium hydroxide is called alumina gel which adsorbs moisture rapidly. Therefore, it is used as drying agent and adsorbent in chromatography.
- It is used in making water proof cloth, as mordant in mordant dyeing.

Aluminium Chloride

- Hydrated $AlCl_3 \cdot 6H_2O$ can be prepared by dissolving aluminium or Al_2O_3 or $Al(OH)_3$ in dilute HCl.

- Anhydrous aluminium chloride cannot be prepared by heating hydrated salt because it hydrolyses with its own water of crystallization forming $Al(OH)_3$ or Al_2O_3 depending on temperature.
- Anhydrous $AlCl_3$ can be prepared by heating aluminium with chlorine or HCl.
- In **Mac Affe method,** anhydrous $AlCl_3$ is prepared by heating a mixture of Al_2O_3 and coke in a current of dry Cl_2 gas.
- Anhydrous $AlCl_3$ can be prepared by the dehydration of hydrated $AlCl_3 \cdot 6H_2O$ with $SOCl_2$ or 2,2-dimethoxy propane.

$$AlCl_3 \cdot 6H_2O + 6SOCl_2 \longrightarrow AlCl_3 + 6SO_2 + 12HCl$$

$$AlCl_3 \cdot 6H_2O + 6(CH_3)_2C(OCH_3)_2 \longrightarrow AlCl_3 + 12CH_3OH + 6CH_3COCH_3$$

- It is deliquescent in nature when exposed to moist air gives fumes owing to the formation of HCl which collects the water molecules forming droplets which appear as fumes.
- It sublimes at 180°. Below 350°C, its vapour contains dimers Al_2Cl_6. Above 350°C, it converts into monomer.
- It is covalent when anhydrous does not conduct electric current in fused state. Soluble in organic solvents. The dimeric formula is retained in non-polar solvents but ionizes in water and exists as $[Al\ (H_2O)_6]^{3+}$ and Cl^- ions because of heat of hydration.
- With NH_4OH, forms white gelatinous precipitate insoluble in excess of NH_4OH; however, with NaOH, the white gelatinous precipitate formed dissolves in excess of NaOH.
- It is used as Friedel crafts catalyst and in manufacture of dyes, drugs etc.
- Anhydrous $AlCl_3$ has six-coordinated layer structure in the solid phase and converts into four coordinate molecular dimers at its melting point and above 850°C, it converts into tri-coordinate monomer $AlCl_3$.
- Except AlF_3, other trihalides of aluminium exist as dimers.
- In the dimer Al_2Cl_6 the bridge Al-Cl bonds are longer than terminal Al-Cl bonds.

Aluminium Sulphate

- It can be prepared by the reaction of aluminium or its oxide or hydroxide with dil H_2SO_4.
- It can be manufactured by treatment of bauxite or china clay with boiling H_2SO_4.

$$Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O + H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 2SiO_2 + 5H_2O$$

- It is a white crystalline solid soluble in water; the solution is acidic because of hydrolysis.
- On heating, it decomposes to Al_2O_3 and SO_3.
- It is a white crystalline solid soluble in water, dyeing, calico printing, foamite fire extinguishers, tanning of leather, water proof cloth and sizing of paper.

Alums

- Alums are double salts with general formula $M_2SO_4 \cdot M'_2(SO_4)_3 \cdot 24H_2O$ or $MM'(SO_4)_2 \cdot 12\ H_2O$ where M is monovalent cation such as Na^+, K^+, Rb^+, Cs^+, Tl^+ and NH_4^+ and M′ is trivalent cation such as Al^{3+}, Cr^{3+}, Fe^{3+}, Mn^{3+}, CO^{3+}, $V^{3+}$and Ti^{3+} etc.
- All alums are **isomorphous**. In alums, every cation is surrounded by six water molecules octahedrally.
- Lithium cannot form alum; because of its small size, it cannot be coordinated with six water molecules.
- The common alum is potash alum $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$.
- It is prepared by mixing equimolar solutions of K_2SO_4 and $Al_2(SO_4)_3$ followed by evaporating and crystallizaton gives alum.
- It is manufactured from the mineral alum stone or alunite or by dissolving in sulphuric acid and then adding required amount of K_2SO_4.

$$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 4Al(OH)_3 + 6H_2SO_4 \longrightarrow K_2SO_4 + 3Al_2(SO_4)_3 + 12H_2O$$

- Alum shale ($Al_2O_3 \cdot xSiO_2 \cdot FeS_2$) is also used in the preparation of alum.
- Alum shale on roasting converts into ferrous sulphate and sulphuric acid which converts Al_2O_3 to $Al_2(SO_4)_3$

$$2FeS_2 + 2H_2O + 7O_2 \longrightarrow 2FeSO_4 + 2H_2SO_4$$

$$Al_2O_3 \cdot xSiO_2 + 3H_2SO_4 \longrightarrow Al_2(SO_4)_3 + xSiO_2 + 3H_2O$$

- After the separation of $FeSO_4$, required amount of K_2SO_4 is added to get alum.
- It is white crystalline substance soluble in water; solution is acidic because of hydrolysis of aluminium sulphate.
- On heating, it swells up because of the removal of water of crystallization and on further heating, it gives alumina.
- Alum is used in the purification of drinking water, as a styptic in bleeding, as mordant in dyeing and in tanning of leather and sizing of paper.
- The double salts of sulphate of divalent and trivalent cations with 24 water molecules are called pseudo alums, e.g., $MnSO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$.

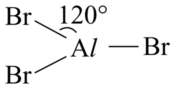
Ultramarines

- These are double silicates of sodium and aluminium with sulphur.
- **White ultramarine** $Na_5Al_3Si_3SO_{12}$ is obtained by fusing kaolin, soda ash, sulphur and some resin. It changes to green ultramarine $Na_5Al_3Si_3S_2O_{12}$ in the presence of air.
- **Blue ultramarine** $Na_5Al_3Si_3S_3O_{12}$ is obtained from green ultramarine by heating with sulphur and air.
- **Violet form** is obtained by heating blue ultramarine in a current of chlorine.
- Ultramarines are used in paints and laundry.
- **Lapis Lazuli** is one of the rare minerals of beautiful blue colour which is sodium aluminium silicate with some sulphur and is used as a precious gem.

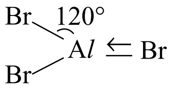
Single Answer Questions

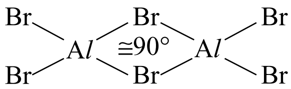
1. A colourless gas obtained by the electrolysis of NaCl in Nelson's cell is made to react with the halide of an element X using electric discharge. The resulting compound (A) has low melting point and boiling point. It can also act as weak reducing agent. The element X forms oxide and hydroxide of the formula X_2O_3 and X $(OH)_3$. X does not react with HCl. The resulting compound (A) is
(a) $AlCl_3$ (b) B_2H_6 (c) BCl_3 (d) Ca_2Cl_6
2. An alkali metal hydride NaH reacts with diborane in Y to give a tetrahedral compound Z which is extensively used as reducing agent in organic synthesis. The Y and Z in the above reaction are
(a) C_2H_6, C_2H_5Na (b) $C_2H_5OC_2H_5$, $NaBH_4$
(c) NH_3, $B_3N_3H_6$ (d) C_3H_8, C_3H_7Na
3. A compound of boron X reacts at very high temperature with NH_3 to give another compound Y which is isosteric with benzene. The compound Y is a colourless liquid and is highly light sensitive. Its melting point is–57°C. The compound (X) with excess of NH_3 and at still higher temperature gives boron nitride $(BN)_n$. The compounds X and Y are, respectively,
(a) BH_3, B_2H_6 (b) $NaBH_4$, C_6H_6
(c) B_2H_6, $B_3N_3H_6$ (d) B_4C_3, C_6H_6
4. Which of the following statements regarding elements of group 13 is not correct?
(a) The value of $E(M^{3+}/M)$ becomes less negative more and becomes positive on descending from Al to Tl.
(b) The acidic character of their oxides decreases on descending the group.
(c) The tendency of catenation decreases on descending the group.
(d) Hydrolysis of $SiCl_4$ followed by heating gives SiO_2.
5. Diborane is a Lewis acid forming addition compound $B_2H_6 \cdot 2NH_3$ with NH_3, a Lewis base. This
(a) is ionic and exists as $[BH_2(NH_3)_2]^+$ and BH_4^- ions.
(b) on heating, it is converted into borazine $B_3N_3H_6$ (called inorganic benzene).
(c) Both are correct.
(d) None is correct.
6. When the aluminium oxide is electrolysed in the industrial process for the production of aluminium metal, aluminium is produced at one electrode and oxygen gas is produced at the other. For a given quantity of electricity, what is the ratio of moles of aluminium to moles of oxygen gas?
(a) 1 : 1 (b) 2 : 1 (c) 2 : 3 (d) 4 : 3
7. Aluminium chloride catalyses certain reactions by forming carbocations with chloroalkanes as shown in the following equations.
$RCl + AlCl_3 \longrightarrow R^+ + AlCl_4^-$
The formation of $AlCl_4^-$ implies that
(a) $AlCl_3$ is covalent.
(b) The aluminium atom in $AlCl_3$ has an incomplete octet
(c) The chlorine atom in RCl has a vacant p-orbital
(d) $AlCl_4^-$ is tetrahedral
8. Which of the following properties describes the diagonal relationship between boron and silicon?
(a) BCl_3 is not hydrolysed whereas $SiCl_4$ can be hydrolysed.
(b) Both form oxides; B_2O_3 is amphoteric and SiO_2 is acidic.
(c) Both dissolve in cold and dilute nitric acid.
(d) Borides and silicides are hydrolysed by water.
9. The solubility of anhydrous $AlCl_3$ and hydrated $AlCl_3$ in diethyl ether are S_1 and S_2, respectively. Then,
(a) $S_1 = S_2$ (b) $S_1 > S_2$
(c) $S_1 < S_2$ (d) $S_1 < S_2$ but not $S_1 = S_2$
10. Borax is $Na_2B_4O_7 \cdot 10H_2O$. Consider the following statements about borax.
I. Two boron atoms have four B-O bonds whereas the other two have three B-O bonds
II. Each boron has one OH groups.
III. It is a salt of tetra boric acid.
IV. It is a cyclic meta borate having two six-membered rings. Select the correct statements.
(a) I, II, III (b) II, III, IV
(c) I, II, IV (d) I, II, III, IV
11. Which is not obtained when metal carbides react with H_2O?
(a) $Al_4C_3 + H_2O \longrightarrow CH \equiv CH$
(b) $CaC_2 + H_2O \longrightarrow CH \equiv CH$
(c) $Mg_4C_3 + H_2O \longrightarrow CH_3C \equiv CH$
(d) $Be_2C + H_2O \longrightarrow CH_4$

12. The structure of aluminium bromide is best represented as

(a) 

(b) $[AlBr_2]^+ + [AlBr_4]^-$

(c) 

(d) 

13. (i) $B_2O_3 + 6NaOH \longrightarrow (X) + 3H_2O$ (ii) $B_2O_3 + 2NaOH \longrightarrow (Y) + H_2O$ (X) and (Y) are

(a) Sodium borate and sodium meta borate.
(b) Both are sodium borates.
(c) Both are sodium meta borates.
(d) Sodium meta borate and sodium borate.

14. Which of the following statements regarding the structure of aluminium chloride is correct?

(a) All the bond angles Cl-Al-Cl and Al-Cl-Al in Al_2Cl_6 are identical.
(b) All the bond lengths Cl-Al in Al_2Cl_6 are identical.
(c) All the bond lengths Cl-Al as well as all the bond angles Cl-Al-Cl and Al-Cl-Al are equal.
(d) The bond lengths of terminal Al-Cl and bridged Al-Cl-Al have different values.

15. Which is true about the hydride of boron, that obtained by the reaction of BCl_3 and $LiAlH_4$?

(a) Its formula is BH_3.
(b) It is electron-rich compound.
(c) Its molecule contains only banana bonds.
(d) It contains three center and two center bonds.

16. In solid state, the $B(OH)_3$ units are

(a) Linked by hydrogen bonds and form three-dimensional sheets with trigonal symmetry.
(b) Linked by hydrogen bonds and form two-dimensional sheets with almost hexagonal symmetry.
(c) Very close to each other and are tightly held and so the crystals cannot be broken easily into small particles.
(d) Linked by hydrogen bonds intramolecularly and exist as discrete particle.

17. When borax is dissolved in water,

(a) Only $B(OH)_3$ is formed.
(b) Only $[B(OH_4)]^-$ is formed.
(c) Both $B(OH)_3$ and $[B(OH)_4]^-$ are formed.
(d) Both $B(OH)_3$ and B_2O_3 are formed.

18. Aluminium is more reactive than iron because its standard reduction potential is higher still aluminium is less easily corroded than iron because

(a) It has higher reducing power and forms a self-protective layer of Al_2O_3.
(b) It has higher reducing power and does not react with oxygen so easily.
(c) Al reacts with atmospheric carbon dioxide to form a self-protective layer of Al_2O_3.
(d) All of these.

19. Identify the correct statements regarding the structure of $Al(BH_4)_3$

(a) 'Al' is sp^3d^2 and 'B' is sp^3 hybridized.
(b) It has 6 3c-2e$^-$ bonds.
(c) It has 6 Al-H-B bonds.
(d) It has 6 2c-2e$^-$ bonds.

(a) only a, c, d (b) only a, b, c
(c) only a, b, d (d) a, b, c and d

20. $AlCl_3$ is more volatile than NaCl; this is because

(a) The $AlCl_3$ molecules are held by weak van der Waal's forces whereas the NaCl species are held by strong ionic forces in a giant lattice.
(b) $AlCl_3$ unlike NaCl is dimerized.
(c) Sodium is more metallic than aluminium.
(d) NaCl unlike $AlCl_3$ is a natural product.

21. Identify the incorrect statement about orthoboric acids.

(a) It has a layer structure in which planar BO_3 units are joined by hydrogen bonds.
(b) Orthoboric acid H_3BO_3 is a weak monobasic lewis acid.
(c) Above 370K, on heating, orthoboric acid forms metaboric acid and on further heating to red hot forming boric oxide.
(d) It is estimated by reacting borax with HCl using phenolphthalein as an indicator.

22. Borax structure contains

(a) Two BO_4 groups and two BO_3 groups
(b) Four BO_4 groups only
(c) Four BO_3 groups only
(d) Three BO_4 and one BO_3 groups

23. Select the incorrect statement about diborane

(a) B_2H_6 has three-centered two-electron bond.
(b) Each boron atom is sp^3 hybridized.
(c) H_tBH_t bond angle is 122°.
(d) All hydrogens in B_2H_6 lie in the same plane.

24. In the borax-bead test for transition metal compounds. the fused beads unique colour is due to the formation of

(a) glass-like metal metaborate bead
(b) opaque metal hexaborate bead
(c) B_2O_3
(d) glass-like metal orthoborate bead

25. Orthoboric acid contains

(a) triangular BO_3^{3-} units
(b) Pyramidal BO_3^{3-} units
(c) T-shaped BO_3^{3-} units
(d) Irregular tetrahedral BO_3^{3-} units

26. Diborane is instantly hydrolysed by water to give
(a) $B_2O_3 + H_2O_2$ (b) $H_3BO_3 + H_2$
(c) $H_3BO_3 + O_2$ (d) $H_3BO_3 + B_2O_3$

27. Which of the following reactions is not correct?
(a) $B_2H_6 + (CH_3)_3N \longrightarrow H_3B^- - N^+(CH_3)_3$
(b) $B_2H_6 + LiH \longrightarrow 2Li[BH_4]$
(c) $B_2H_6 + 6CH_3OH \longrightarrow 2(CH_3)_3B + 6H_2O$
(d) $B_2H_6 + 6H_2O \longrightarrow 2H_3BO_3 + 6H_2$

28. Al and Ga have nearly the same covalent radius because of
(a) Greater shielding effect of s-electrons of Ga atoms.
(b) Poor shielding effect of s-electrons of Ga atoms.
(c) Poor shielding effect of d-electrons of Ga atoms.
(d) Greater shielding effect of d-electrons of Ga atoms.

29. Anhydrous $AlCl_3$ is prepared by reaction between
(a) aluminium and concentrated HCl.
(b) heated aluminium and dry Cl_2(g) or HCl(g).
(c) aluminium and dilute HCl solution.
(d) aluminium carbide and dry HCl(g).

30. Which of the following statements about H_3BO_3 is not correct?
(a) It is strong tribasic acid.
(b) It is prepared by acidifying an aqueous solution of borax.
(c) It has a layer structure in which planar BO_3 units are joined by hydrogen bonds.
(d) It does not act as proton donor but acts as a Lewis acid by accepting hydroxyl ion.

31. Which of the following statements is incorrect?
(a) $B(OH)_3$ partially reacts with water to form H_3O^+ and $[B(OH)_4]^-$ and behaves like a weak acid.
(b) $B(OH)_3$ behaves like a strong monobasic acid in the presence of sugars, and this acid can be titrated against an NaOH solution using phenolphthalein as an indicator.
(c) $B(OH)_3$ does not donate a proton and hence it is not a Bronsted acid by accepting lone pair from OH^- ion.
(d) $B(OH)_3$ reacts with NaOH forming $Na[B(OH)_4]$.

32. Which of the following is not a characteristic of boranes?
(a) They undergo spontaneous combustion in air.
(b) Their combusion products are water and crystalline boron.
(c) They form borohydride complexes by reaction with alkali metal hydrides.
(d) They readily get hydrolysed by liberating H_2 gas.

33. In the boron family, boron has a very high melting point because of
(a) Strong binding forces in the covalent polymer.
(b) Strong van der Waal's forces between the boron atoms.
(c) Its ionic crystal structure.
(d) All of these.

34. Which of the following reactions is used to prepare BF_3?
(a) $B_2O_3 + CaF_2 + H_2SO_4(conc) \xrightarrow{heat}$
(b) $B_2O_3 + NH_4BF^4 \xrightarrow{heat}$
(c) $Na_2[B_4O_5(OH)_4] + HF + H_2SO_4(conc) \xrightarrow{heat}$
(d) All of these

35. $H_3BO_3 \xrightarrow{100^\circ C} A \xrightarrow{red\ heat} B$
(a) $A = H_2B_4O_7$, $B = HBO_2$
(b) $A = HBO_2$, $B = H_2B_4O_7$
(c) $A = H_2B_4O_7$, $B = B_2O_3$
(d) $A = HBO_2$, $B = B_2O_3$

36. What is correct statement among the following about the structure of borax molecule?
(a) All the four boron atoms are sp^3-hybridized.
(b) All the four boron atoms are sp^2-hybridized.
(c) Two of the four boron atoms are sp^3-hybridized and the other two are sp^2-hybridized.
(d) Out of the 10 water molecules generally shown as water of crystallization in its MF,4 molecules of H_2O are really involved in the structural formation.

37. The bond dissociation energy of B-F in BF_3 is 646 kJ mol^{-1} whereas that of C-F in CF_4 is 515 kJ mol^{-1}. The correct reason for higher B-F bond dissociation energy as compared to that of C-F is
(a) Stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4.
(b) Significant $p\pi$—$p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4.
(c) Lower degree of $p\pi$—$p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4.
(d) Smaller size of B-atom as compared to that of C-atom.

38. $2NH_3 + B_2H_6 \rightarrow A$
How many disubstituted (same substituents) isomers of A are possible?
(a) 3 (b) 2 (c) 4 (d) 6

39. Aluminium oxide exists in nature as gems with different colours. The reason for the difference in colour is that
(a) The oxidation states of aluminium in these gems are different.
(b) The extent of crystallinity in these gems is different.
(c) The Al-O bonding is different in these gem structure.
(d) There are different metal ions present as impurities in these gems.

40. A boron carbide rod is used in a nuclear reactor because boron (^{10}B) has a very
(a) low cross section area for capturing neutrons.
(b) high cross section area for capturing neutrons.

(c) low cross section area for removing positrons.
(d) high cross section for capturing protons.

41. Which of the following statements is incorrect?
(a) Boron carbide is used as an abrasive.
(b) Boron is used to increase the hardness of steel.
(c) Boron sesquioxide is used in making borosilicate glass.
(d) Orthoboric acid undergoes intramolecular hydrogen bonding.

42. Boron reacts with nitrogen at high temperature and high pressures forming boron nitride BN. Boron nitride is
(a) a slippery white solid with a layer structure similar to graphite
(b) a white solid with diamond-like structure
(c) a liquid and is structurally similar to silicon dioxide
(d) none of these

43. Which of the following statements is incorrect in regard to the B-F bond in BF_3?
(a) All the three B-F bond lengths are equal (1.30Å) and each of them is shorter than the sum of the covalent radii of Boron (0.80Å) and fluorine (0.72Å).
(b) The bond energy of the B-F bond is highest (646 kJ mol^{-1}).
(c) The unusually short bond length and high bond strength of the B-F bond is due to pπ–pπ interaction between the boron and fluorine atoms.
(d) The short length and high strength of the bonds is due to small size of B and F atoms.

44. Borazine is sometimes called inorganic benzene. Which of the following reactions is expected to give this compound

(a) $B_2H_6 + NH_3 \xrightarrow[\text{low tem.}]{\text{excess } NH_3}$

(b) $B_2H_6 + NH_3 \xrightarrow[\text{high tem.}]{\text{excess } NH_3}$

(c) $B_2H_6 + NH_3 \xrightarrow[\text{high tem.}]{\text{ratio } 2NH_3 : 1B_2H_6}$

(d) All of these

45. Which of the following statement is incorrect?
(a) All alums crystallize in the tetrahedral form and produce isomorphous series of double salts.
(b) All alums are double salts and have large quantities of water of crystallization.
(c) Aluminium sulphate is useful as a mordant in dyeing and printing.
(d) Alums are used in water purification, coagulation and tanning of leather.

46. Which of the following is correct statement?
(a) The hydroxide of aluminium is more acidic than that of boron.
(b) The hydroxide of boron is basic whereas that of aluminium is amphoteric.
(c) The hydroxide of boron is acidic whereas that of aluminium is amphoteric.
(d) The hydroxides of both boron and aluminium are amphoteric.

47. Which of the following statements is incorrect?
(a) Anhydrous aluminium chloride cannot be prepared by heating $AlCl_3 \cdot 6H_2O$ because of hydrolysis of $AlCl_3$ into $Al(OH)_3$
(b) $AlCl_3$ is a high-melting solid whereas AlF_3 is a low-melting volatile solid.
(c) Aluminium forms $[AlF_6]^{3-}$ ion but boron does not form $[BF_6]^{3-}$.
(d) Cryolite is added to alumina to lower the fusion temperature and to make the mass a good conductor of electricity.

48. The self-protective oxide film on aluminium can be removed by
(a) boiling aluminium with water
(b) amalgamating with mercury
(c) adding concentrated HNO_3
(d) reacting with chlorine

49. Boron nitride obtained by heating borazole is
(a) white solid with a diamond-like structure.
(b) slippery white solid with layered strucure similar to that of graphite.
(c) covalent liquid and is structurally similar to carbon monoxide.
(d) soft low-melting solid with rock-salt-like structure molecules with different structures.

50. Which of the following pairs contain molecules with different structures?
(a) Borazine and benzene
(b) Diborane and hydrazine
(c) NaCl and NiO
(d) Graphite and boron nitride

51. Aluminium chloride ionizes in an aqueous solution because
(a) its dissociation constant is low.
(b) it forms a dimer in an aqueous solution.
(c) it is a Lewis acid.
(d) its total hydration energy exceeds the ionization energy of the system.

52. In solid state,
(a) $AlCl_3$ exists as a close-packed lattice of Cl^- and Al^{3+} occupying tetrahedral holes.
(b) $AlCl_3$ exists as a close-packed lattice of Cl^- and Al^{3+} occupying trigonal holes.

(c) $AlCl_3$ exists as a close-packed layered lattice of Cl^- and Al^{3+} occupying octahedral holes.
(d) $AlCl_3$ exists as Al_2Cl_6.

53. In the structure of solid $AlCl_3$, each Al participates in
(a) four bridges (b) six bridges
(c) three bridges (d) twelve bridges

54. When borax is treated with mineral acids, the product formed is
(a) Boron oxide (b) Orthoboric acid
(c) Metaboric acid (d) Pyroboric acid

55. Aluminium vessels should not be washed with materials containing washing soda because
(a) Washing soda reacts with aluminium to form soluble aluminium compounds.
(b) Washing soda is expensive.
(c) Washing soda is easily decomposed.
(d) Washing soda reacts with aluminium to form insoluble chloride.

56. Borax is converted into crystalline boron by the following steps

Borax $\xrightarrow{X} H_3BO_3 \xrightarrow{\Delta} B_2O_3 \underset{\Delta}{\overset{Y}{\longrightarrow}} B$ X and Y are respectively

(a) HCl, Cu (b) HCl, C
(c) C, Al (d) HCl, Al

57. Aluminium hydroxide dissolves in excess of NaOH solution because of the formation of
(a) $[Al(H_2O)_4(OH)_2]^+$ (b) $[Al(H_2O)_3(OH)_3]$
(c) $[Al(H_2O)_2(OH)_4]^-$ (d) $[Al(H_2O)_6(OH)_3]$

58. Choose the correct sequence for the geometry of the given molecules Borazon, Borazole, $B_3O_6^{3-}$, Al_2Cl_6
[P stands for planar and NP stands for non-planar]
(a) NP, NP, NP, P (b) P, P, NP, NP
(c) NP, NP, NP, P (d) NP, P, P, NP

59. Which is not true about borax?
(a) It is a useful primary standard for titrating against acids.
(b) Borax forms basic buffer solution.
(c) Aqueous solution of borax can be used as buffer.
(d) It is made up of two six-membered heterocyclic rings.

60. Which of the following amines has little tendency to show the reaction $B_2H_6 + 2X \longrightarrow [BH_2(X)_2]^+ \left[BH_4^-\right]^-$
(a) NH_3 (b) CH_3NH_2
(c) $(CH_3)_2NH$ (d) $(CH_3)_3N$

61. BF_3 forms tetrahedral complex with the compounds which can donate lone pair of electrons. The bond length B-F in BF_3 is
(a) Equal to that existing in the tetrahedral complex.
(b) Greater than that existing in the tetrahedral complex.
(c) Smaller than that existing in the tetrahedral complex.
(d) Greater than or smaller than or equal to that existing in tetrahedral complex depending upon the nature of the electron-donor compound.

62. Alumina is insoluble in water because
(a) It is a covalent compound.
(b) It has high lattice energy and low heat of hydration.
(c) It has low lattice energy and high heat of hydration.
(d) Al^{3+} and O^{2-} ions are not excessively hydrated.

63. The non-metallic character of boron is indicated by
(a) reaction with metals forming borides
(b) liberation of hydrogen with alkali
(c) basic nature of its oxide
(d) reaction with metals forming alloys

64. In the reaction of aluminium with dilute sodium hydroxide solution, which of the following is not correct?
(a) Hydrogen gas will be liberated.
(b) Sodium meta aluminate is formed.
(c) The product exists as $Al(OH)_4^- \cdot 2H_2O$.
(d) Coordination number of aluminium in the product is 4.

65. Boron can be obtained by various methods but not by
(a) Thermal decomposition of B_2H_6.
(b) Pyrolysis of BI_3.
(c) Reducing BCl_3 with H_2.
(d) Electrolysis of fused BCl_3.

66. Identify the correct statement.
(a) BF_3 and BCl_3 are gases.
(b) BF_3 has partial double bond character because of π bonding and the molecule shows resonance.
(c) BF_3 is electron pair donor.
(d) BBr_3 is solid.

67. Alum helps in purifying water by
(a) Coagulating the mud particles.
(b) Making mud water soluble.
(c) Forming silicon complex with clay particles.
(d) Sulphate part which coming with dirt and remove it.

68. Boron trichloride on heating with ammonium chloride at 140°C gives a compound (A) which on reduction with sodium borohydride followed by hydrolysis gives another compound (B). Hence, compounds (A) and (B) are, respectively,
(a) Inorganic graphite and ammonia.
(b) Inorganic benzene and orthoboric acid.
(c) Trichloroborazine and orthoboric acid.
(d) diborane and orthoboric acid.

More than One Answer Type Questions

1. Which of the following statements is/are correct for B_2H_6?
(a) In diborane (B_2H_6), the electron deficiency is not associated with the bridging groups.

(b) The banana bond in the diborane molecule results from the overlap of two sp^3 orbitals of the B atoms (one from each) with 1s orbital of the bridged H-atom.
(c) In diborane, the banana bond of B-H-B bridge involves delocalization of the electrons over all these three atoms
(d) In diborane, the terminal B-H bond is a localized two-centred sigma bond

2. Which of the following product(s) may be formed by the reaction of diborane with ammonia at different temperatures?
(a) $(BN)_x$ (b) $B_2H_6 \cdot 2NH_3$
(c) $B_3N_3H_6$ (d) $B_2H_6 \cdot NH_3$

3. Aqueous solution of boric acid is treated with salicylic acid. Which of the following statements is/are incorrect for the product formed in the above reaction?
(a) No product will be formed because both are acids.
(b) Product is 4-coordinated complex and optically resolvable.
(c) Product is 4-coordinated complex and optically non-resolvable.
(d) There are two rings only which are five membered.

4. Borazine is called 'inorganic benzene' in view of its ring structure with alternate BH and NH groups. Which of the following statements is correct about borazine?
(a) Each B and N atom is sp^2 hybridized.
(b) Borazine satisfies the (4n+2) Huckel's rule.
(c) Like organic benzene, borazine does not give addition products.
(d) Borazine contains dative $p\pi$–$p\pi$ bond.

5. Which of the following methods can be used for the preparation of anhydrous aluminium chloride?
(a) Heating $AlCl_3 \cdot 6H_2O$.
(b) Heating a mixture of alumina and coke in a current of dry chlorine.
(c) Passing dry HCl gas over heated aluminium powder.
(d) Passing dry chlorine over heated aluminium.

6. Which of the following statements are correct?
(a) BH_3 is not a stable compound.
(b) All the B-H bonds in B_2H_6 are equal.
(c) The boron hydrides are readily hydrolysed.
(d) Boron hydrides are prepared by the action of dil. HCl on Mg_3B_2.

7. Hydrolysis products of BF_3 is/are
(a) H_3BO_3 (b) BH_4^-
(c) $[BF_3OH]^-$ (d) HF

8. Borax is
(a) orthoborate (b) metaborate
(c) pyroborate (d) tetraborate

9. Identify the correct statements regarding structure of diborane.
(a) There are two bridging hydrogen atoms.
(b) Each boron atom forms four bonds.
(c) The hydrogen atoms are not in the same plane.
(d) Each boron atom is in sp^3 hybridized state.

10. Which of the following are true about diborane?
(a) It has two bridging hydrogens and four terminal hydrogens.
(b) When methylated, the product is $Me_4B_2H_2$.
(c) The bridging hydrogen are in a plane perpendicular to the rest.
(d) All the six B–H bond distances are equal.

11. In which of the following molecules, vacant orbitals take part in hybridization?
(a) B_2H_6 (b) Al_2Cl_6
(c) H_3PO_3 (d) H_3BO_3

12. Which of the following statements are correct?
(a) When BF_3 is treated with excess of NaF in acidic aqueous solution gives $NaBF_4$.
(b) When BCl_3 is treated with excess of NaCl in acidic aqueous solution gives H_3BO_3.
(c) When BBr_3 is treated with $NH(CH_3)_2$ in a hydrocarbon solvent gives $B[N(CH_3)_2]_3$.
(d) When B_2H_6 is treated with BeH_2, the product formed is $Be(BH_4)_2$.

13. Select the correct statement(s)/reactions (s).
(a) The B-F bond length is greater in BF_3 compared to that of BF_3-NH_3.
(b) BF_3 is least hydrolysed among the boron trihalides.
(c) BF_3 is the weakest Lewis acid among the boron trihalides.
(d) $C_6H_6 + C_2H_5Cl \xrightarrow{BF_3} C_6H_5C_2H_5 + HCl$.

14. $Al_2(SO_4)_3 + NH_4OH \longrightarrow X$, X is:
(a) a white gelatinous precipitate.
(b) insoluble in excess of NH_4OH.
(c) soluble in excess of NaOH.
(d) amphoteric in nature.

15. Which of the following statements are true for H_3BO_3?
(a) It is mainly monobasic acid and a Lewis acid.
(b) It does not act as a proton donor but acts as an acid by accepting hydroxyl ions.
(c) It has a layer structure in which BO_3 units are joined by hydrogen bonds.
(d) It is obtained by treating borax with conc. H_2SO_4.

16. Among the following statements, the correct statements are
(a) The increasing Lewis acidity is $BF_3 < BCl_3 < AlCl_3$.
(b) When $BF_3 \cdot N(CH_3)_3$ reacts with BCl_3 the product formed is $BCl_3 \cdot N(CH_3)_3$.
(c) When $BH_3 \cdot CO$ reacts with BBr_3, the product formed is $BBr_3 \cdot CO$.
(d) BF_3 is formed by the reaction of B_2O_3 with CaF_2 and conc. H_2SO_4.

17. Pick out correct statement(s) regarding crystalline form of boron.
(a) All allotropic forms of boron contain icosahedral units with boron atoms at all 12 corners.
(b) In B_{12} unit, each boron atom is bonded to five boron atoms at a distance 1.77Å.
(c) α-rhombohedral form of boron consists of layers of icosahedra linked within each layer by three-centre B-B bonds and between layers B–B bonds.
(d) β-rhombohedral form consists of 12 B_{12} icosahedra arranged icosahedrally about a central B_{12} unit $(B_{12}(B_{12})_{12})$.

18. Which of the following statements is correct?
(a) The compound $B_2H_6 \cdot 2NH_3$ is ionic.
(b) On heating, it gives borazole $B_3N_3H_6$.
(c) A boron and nitrogen atom bonded together have the same number of valence electrons as two carbon atoms.
(d) Both benzene and borazole are planar and non-polar.

19. Which of the following statements is correct?
(a) Boric acid is a hydrogen-bonded substance.
(b) Boric acid combines with CuO to give metaborate in the borax bead test.
(c) Al_2O_3 is more acidic than B_2O_3.
(d) Al_2O_3 is amphoteric and B_2O_3 is acidic.

20. Which of the following statements is correct?
(a) B_2H_6 reacts with excess of ammonia at low temperature to form an ionic substance.
(b) B_2H_6 reacts with excess of ammonia at high temperature to form a white slippery solid called boron nitride.
(c) In boron nitride, the bond order is 1.5.
(d) Borazine is an inorganic benzene but has –ve and +ve charges in it on N and B, respectively.

21. Regarding boric acid, which of the following statements is correct?
(a) Boric acid has layered lattice structure.
(b) When excess of boric acid is added to a solution of acidic KHF_2 solution, the solution becomes alkaline.
(c) Boric acid on heating at 160°C forms pyroboric acid.
(d) Boric acid is a quite good lubricant.

22. Which of the following statements is/are correct?
(a) $NaBH_4$ is very much less rapidly hydrolysed in water than $NaAlH_4$.
(b) AlF_3 is insoluble in anhydrous HF but dissolves in the presence of KF.
(c) If BF_3 gas is passed through Na_3AlF_6, aluminium trifluoride is precipitated.
(d) The enthalpies of dimerization of aluminium alkyls decreases with increase in the size of alkyl groups.

23. Which of the following statements is correct?
(a) Trimethyl boron is a weak Lewis acid compared to boron halides.
(b) CO forms stable adduct with diborane than with BF_3.
(c) TlI_3 when added to aqueous sodium hydroxide gives Tl_2O_3 precipitate.
(d) If KI is added to TlI_3, the compound formed K $[TlI_4]$ contains Tl^{3+} ion.

24. Which is correct statement regarding borax?
(a) It contains eight water molecules as water of crystallization.
(b) In borax, all B-O bond lenghts are equal.
(c) Four boron atoms in borax are in bond with four OH groups of which two boron atoms carry negative charge.
(d) Borax is a based system of two six-membered rings.

25. Which of the following statements is/are correct?
(a) Borax is used for cleaning metals.
(b) Al_2O_3 is amphoteric while B_2O_3 is acidic.
(c) Boric acid on heating with CuO gives metaborate and borax bead test.
(d) Boron nitride is inorganic graphite.

26. Which of the following statement(s) is/are correct regarding the structure of borax?
(a) Number of B–B bonds are zero.
(b) Hybridization of each boron atom is sp^2.
(c) Number of B–O–B bonds are five.
(d) Borax contain two different types of *B-O* bonds

Comprehension Type Questions

Passage-I

The elements boron, aluminium, gallium, indium and thalium react with halogens forming halides. The boron halides exist as monomers whereas the halides of the other elements exists as dimers. The B–F bond length in BF_3 is shorter than the sum of the covalent radii of boron and fluorine. The halides of boron acts as lewis acids and their Lewis acidic character is in the order $BF_3 < BCl_3 < BBr_3 < BI_3$ which is opposite to the expected $BF_3 > BCl_3 > BBr_3 > BI_3$ depending upon electronegativities. AlF_3 cannot act as Lewis acid whereas $AlCl_3$ can act as Lewis acid. The fluorides of Al, Ga, In and Tl have high melting points (<950°C) whereas the chlorides, bromides and iodides have lower melting point. These halides hydrolyse in water. Thermal stability decreases with the increase in the atomic size of III Group element or halogen.

1. The electronegativity difference between boron and fluorine is about 2.5, BF_3 is covalent. This is due to

(a) Low polarizability of small F^- ion.
(b) High polarizability of small F^- ion.
(c) High polarizing power of small B^{3+} ion.
(d) BF_3 is an electron-deficient molecule.

2. Except BF_3, the fluorides of Al, Ga, In and Tl do not act as Lewis acids.
(a) In the halides of Al, Ga, In and Tl, the electron deficiency at central atom decreases because of back bonding by F.
(b) BF_3 is covalent whereas fluorides of third group elements are ionic.
(c) BF_3 is monomer whereas other halides are dimers.
(d) BF_3 is small molecule whereas others are bigger molecules.

3. TlI_3 can be prepared by adding iodine to TlI. In TlI_3, the oxidation state of Tl is
(a) +1 (b) +2 (c) +3 (d) both +1 and +3

4. The fluorides of Al, Ga, In and Tl have very high melting point (>950°C) whereas chlorides, bromides and iodides have low melting points. This is because
(a) Fluorides are monomers whereas other halides are dimers.
(b) Fluorides are more polar whereas other trihalides are less polar.
(c) Fluorides are ionic whereas other trihalides are polar covalent.
(d) Fluorides have high thermal stability while other trihalides have less thermal stability.

5. Which of the following statement is not correct?
(a) BF_3 exists as monomer because small boron atom cannot coordinate with four fluorine atoms.
(b) AlF_3 does not exist as dimer because it is ionic.
(c) Chlorides, bromides and iodides of boron exist as monomers in gaseous state.
(d) Solid or molten $AlCl_3$ is not a conductor of electricity but aqueous $AlCl_3$ is a good conductor.

6. Which of the hydrolysis reaction of the halides of III Group element is wrong?
(a) $AlCl_3 + 3H_2O \rightarrow Al(OH)_3 + 3HCl$
(b) $BCl_3 + 3H_2O \rightarrow H_3BO_3 + 3HCl$
(c) $BF_3 + H_2O \rightarrow H_3BO_3 + 3HF$
(d) $AlBr_3 + 3H_2O \rightarrow Al(OH)_3 + 3HBr$

Passage-II

A white precipitate (X) is formed when a mineral (A) is boiled with Na_2CO_3 solution. The precipitate is filtered and filtrate contains two compounds (Y) and (Z).

The compound (Y) is removed by crystallization and when CO_2 is passed through the filtrate obtained after crystallization, then (Z) changed to (Y). When compound (Y) is heated, it gives two compounds (Z) and (T). Compound (T) on heating with cobalt oxide produces blue coloured substance (S).

1. The mineral (A) is
(a) $Na_2B_6O_{11}$ (b) $CaCO_3$
(c) $Ca_2B_6O_{11} \cdot 5H_2O$ (d) B_2O_3

2. The compound (Y) in the filtrate when (A) is boiled with Na_2CO_3 is
(a) Na_2BO_2 (b) $Na_2B_4O_7$
(c) Na_3BO_3 (d) CaO

3. When cobalt oxide is heated with (Y), then a bead (S) is formed which is blue in colour. The bead (S) is
(a) $CoCO_3$ (b) $Co(BO_2)_2$
(c) CoO (d) B_2O_3

Passage-III

Aluminium is stable in air and water inspite of the fact that it is reactive metal. The reason is that a thin film of its oxide is formed on its surface which makes it passive for further attack. The layer is so useful that in industry, it is purposely deposites by an electrolytic process called anodizing. Reaction of aluminium with oxygen is highly exothermic and is called thermite reaction.

$$2Al(s) + \frac{3}{2}O_2(g) \longrightarrow Al_2O_3(s); \Delta H = -1670\,k = kJ.$$

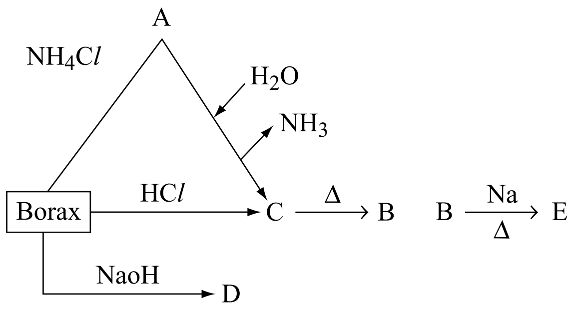
Thermite reaction finds applications in the metallurgical extraction of many metals from their oxides and for welding of metals. The drawback is that to start the reaction, high temperature is required for which an ignition mixture is used.

1. Anodizing can be done by electrolysing dilute H_2SO_4 with aluminium as anode. This results in:
(a) the formation of $Al_2(SO_4)_3$ on the surface of aluminium anode.
(b) the formation of oxide film (Al_2O_3) on the surface of aluminium anode.
(c) the formation of polymeric aluminium hydride film on the surface of aluminium anode.
(d) none of the above.

2. Which of the following reaction is not involved in thermite process?
(a) $3Mn_3O_4 + 8Al \rightarrow 9Mn + 4Al_2O_3$
(b) $Cr_2O_3 + 2Al \rightarrow 2Cr + Al_2O_3$
(c) $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$
(d) $B_2O_3 + 2Al \rightarrow 2B + Al_2O_3$

3. Which one of the following metals cannot be extracted using Al as a reducing agent?
(a) Na from Na_2O (b) Cr from Cr_2O_3
(c) W from WO_3 (d) Mn from Mn_3O_4

Passage-IV



1. The correct statement about the structure 'A' is
   (a) Each boron surrounds with three nitrogen atoms, whereas each N-atom surrounds with four boron atoms. In addition, both boron and nitrogen surrounds with 2σ bonds and 1π bond.
   (b) Each boron surrounds with four nitrogens whereas each N-atom surrounds with three-borons. In addition, both boron and nitrogen surround with 3σ bonds and 2π bonds.
   (c) Each boron surrounds with three nitrogen atoms, whereas each nitrogen surrounds with three borons and both boron and nitrogen surround with 3σ bonds and 1π bond.
   (d) Each boron surrounds with four nitrogen atoms, whereas each nitrogen surrounds with four borons and both boron and nitrogen surround with 3σ bonds and 1π bond.
2. The incorrect statement about 'C' is
   (a) It has trigonal planar units and held by H–bonds.
   (b) It can also be prepared by passing SO_2 gas through suspension of colemanite ore.
   (c) When the concentration of 'C' is very high, it exists as polymeric ion such as $[B_3O_3(OH)_4]^-$.
   (d) 'C' on heating gives $H_2B_4O_7$ at 100°C, HBO_2 at 160°C and B_2O_3 at high temperature.
3. The incorrect statement about 'B' is
   (a) Reduction of 'B' gives amorphous variety of boron with Na, Mg and K.
   (b) Electrolytic reduction of 'B' (B + MgO + MgF_2) gives amorphous boron.
   (c) Reduction of B gives crystalline variety of boron with Al. In addition, 'B' gives coloured bead with Al_2O_3.
   (d) B is a white hygroscopic solid.

Passage-V

Boron halides are electron-deficient compounds and acts as Lewis acids. In boron halides, there is double bond character between boron and halogen atom because of the back bonding. Even the weakest bases will form adducts with the trihalides of boron. The rehybridization of boron that accompanies adduct formation results in a loss of in B–X double bond character.

1. In the adduct formation of trimethyl amine with boron halide
   $$BX_3 + :N(CH_3)_3 \longrightarrow X_3B:N(CH_3)_3$$
   the enthalpy change is more negative in the case of
   (a) BF_3 (b) BCl_3
   (c) BBr_3 (d) All are equal
2. BF_3 exists as monomer but not dimer because
   (a) Boron cannot coordinate with four fluorine atoms to form dimer.
   (b) Dimerization on BF_3 leads to the rehybridization of boron that accompanies in a loss of B–X double bond character.
   (c) BF_3 is ionic compound.
   (d) of steric hinderence.
3. BCl_3 exists as monomer but not dimer because
   (a) boron cannot coordinate with four chlorine atoms to form dimer.
   (b) dimerization of BF_3 leads to the rehybridization of boron that accompanies in a loss of B–X double bond character.
   (c) BCl_3 is covalent compound.
   (d) Boron has no catenation power.

Passage-VI

Borax (X) is heated with conc. HCl or H_2SO_4. When a sparingly soluble compound (A) separates out. Compound A on heating gives compound (B). The compound (B) on heating with potassium gives an amorphous variety of an element with high melting point.

1. Borax on heating with NH_4Cl gives
   (a) $[B(OH)_2][NH_4]$ (b) BN
   (c) H_3BO_3 (d) B_2O_3
2. The compound (A) is
   (a) BH_3 (b) BSO_4
   (c) H_3BO_3 (d) $H_2B_4O_7$
3. The compound B and the element with high melting point is
   (a) B_2O_3 and B (b) $H_2B_2O_7$ and B
   (c) $B_2O + B$ (d) $B_2O_3 + H_4B_2O_7$

Passage-VII

Boric acid $B(OH)_3$ is weak monobasic acid reacts with alkali to form borates. The most common borate of boric acid is borax represented as $Na_2(B_4O_5(OH)_4) \cdot 8H_2O$ which is made up of two tetrahedral and two triangular units. On dissolution in water, these tetrahedral and triangular units are separated. Borax is useful primary standard for titration against acids.

1. The number of B–O–B linkage in borax is
   (a) 2 (b) 5
   (c) 4 (d) 6
2. Oxidation state of boron atom in borax is/are
   (a) +3 only
   (b) three atoms +3 and one atom +2
   (c) +2 only
   (d) two atoms +3 and two atoms +4
3. Number of equivalents of an acid required to neutralize aqueous solution of borax is
   (a) 4 (b) 1
   (c) 2 (d) 3

Matching Type Questions

Match the following List-I and List-II

1. List-I	List-II
(a) $AlCl_3$	(p) Lewis acid
(b) BCl_3	(q) Forms acidic solution
(c) B_2H_6	(r) Sublimes on heating
(d) Alum	(s) Swells on heating

2. List-I	List-II
(a) Borazole	(p) Lewis acid
(b) Diborane	(q) Dimer
(c) Aluminium chloride (vapour)	(r) Delocalization of electrons
(d) Boron trichloride	(s) Hydrolyses in water

3. List-I	List-II
(a) $B_2H_6 + NH_3 \longrightarrow$	(p) B_2H_6
(b) $2BF_3 + 6LiH \longrightarrow$	(q) Borazine
(c) Two electron three centre bond	(r) $AlCl_3$ (vapour)
(d) sp^3 hybrid orbitals	(s) Inorganic graphite

4. List-I	List-II
(a) H_3BO_3	(p) Hydrogen bonds
(b) $Na_2B_4O_7$	(q) Amphoteric
(c) Al_2O_3	(r) Basic
(d) TlOH	(s) Lewis acid

5. Column-I	Column-II
(a) $Na_2B_4O_7 + NH_4Cl \longrightarrow$	(p) B_2O_3
(b) $NaH + B_2O_3 \longrightarrow$	(q) $NaBO_2$
(c) $B + NaOH \longrightarrow$	(r) Na_3BO_3
(d) $Na_2B_4O_7 \xrightarrow{\Delta}$	(s) $NaBH_4$

6. Column-I	Column-II
(a) $B_3N_3H_6$	(p) Planar geometry
(b) $(BN)_x$	(q) No lone pair
(c) B_2H_6	(r) Non-polar molecule
(d) BF_3	(s) Non-planar geometry

7. Column-I	Column-II
(a) Solid $AlCl_3$	(p) Layered lattice
(b) Boron nitride	(q) Hexagonal fused rings
(c) Graphite	(r) sp^2 hybridization
(d) Borazole	(s) Electrical conductor

8. Column-I	Column-II
(a) Graphite	(p) Layered structure
(b) Boric acid	(q) Delocalization of electrons
(c) Borazole	(r) Electrical conductor
(d) Boron nitride	(s) Hydrogen bonds

9. Column-I	Column-II
(a) Borazine + $H_2O \longrightarrow$	(p) One of the products is gas
(b) Borax + $NH_4Cl \longrightarrow$	(q) Monobasic acid
(c) $BF_3 + NaH \xrightarrow{180^\circ C}$	(r) Graphite-like structure
(d) $B_2H_6 + H_2O \longrightarrow$	(s) One of the products turns mercurous nitrate paper black or can be used as a brightner in washing powder
	(t) Acidic oxide

Numerical Type

1. Tri alkyl aluminium molecules exist as dimers which contain 3c-2e bonds. The coordination number of bridged carbon is
2. Number of possible isomers for disubstituted borazole with similar substituents is
3. Number of 3c-2e bonds (hydrogen bridges) in $Be(BH_4)_2$ is,
4. Number of B–O–B bonds in sodium tetraborate is

5. In solid corrundum, the number of oxygen atoms coordinate to aluminium ion is
6. The number of hydrogen bonds that can be formed by each boric acid molecule is
7. The total number of molecules having three-centre two-electron bonds among the following is B_2H_6, Al_2Cl_6, $BeCl_2(s)$, $BeH_2(s)$, Al_2H_6, $[Be_2(CH_3)_2]_n$ C_2H_6, $Al_2(CH_3)_3$, $C_2(CH_3)_6$
8. Number of hexagonal rings in borax
9. In sodium tetraborate, the number of boron atoms involved in sp^3 hybridization is
10. In Jeweller's borax, the number of water molecules present in hydrated compound is
11. In $[B_4O_5(OH)_4]^{-2}$, the number of boron atoms having an octet of electrons is
12. The number of oxygen atoms in the plane of arrangement of aluminium atoms in corrundum is
13. In sodium aluminium hydroxide (sodium aluminate), the number of OH groups surrounding each aluminium atom is
14. In the polymeric aluminium hydride, each aluminium is surrounded by – number of 3c-2e bonds.
15. BCl_3 on reduction with $LiAlH_4$ gives a gas (X). The (X) when heated with ammonia forms an addition compound which on further heating to high temperature forms another compound (Y). How many dichloro derivatives will be formed by (Y)?
16. Borax contains some longer B–O and some shorter B-O bonds. The number of longer B–O bonds is
17. Number of tetrahedral boron atoms in colemanite is

Answers

1. b	**2.** b	**3.** c	**4.** c	**5.** c	**6.** d
7. b	**8.** d	**9.** b	**10.** d	**11.** a	**12.** d
13. a	**14.** d	**15.** d	**16.** b	**17.** c	**18.** a
19. d	**20.** a	**21.** d	**22.** a	**23.** d	**24.** a
25. a	**26.** b	**27.** c	**28.** c	**29.** b	**30.** a
31. c	**32.** b	**33.** a	**34.** a	**35.** d	**36.** c
37. b	**38.** c	**39.** d	**40.** b	**41.** d	**42.** b
43. d	**44.** c	**45.** a	**46.** c	**47.** b	**48.** b
49. b	**50.** b	**51.** d	**52.** c	**53.** b	**54.** b
55. a	**56.** d	**57.** c	**58.** d	**59.** b	**60.** d
61. c	**62.** b	**63.** a	**64.** d	**65.** d	**66.** b
67. a	**68.** c				

More than One Answer Type Questions

1. b, c, d	**2.** a, b, c	**3.** a, c, d
4. a, b, d	**5.** b, c, d	**6.** a, c, d
7. a, b, c, d	**8.** b, d	**9.** a, b, c, d
10. a, b, c	**11.** a, b	**12.** a, b, c, d
13. b, c, d	**14.** a, b, c, d	**15.** a, b, c, d
16. a, b, d	**17.** a, b, c, d	**18.** a, b, c, d
19. a, b, d	**20.** a, b, d	**21.** a, b, d
22. a, b, d	**23.** a, b, c, d	**24.** a, c, d
25. a, b, c, d	**26.** a, c, d	

Comprehensive Type Questions

Passage - I	**1.** c	**2.** b	**3.** a	**4.** c	**5.** a	**6.** c
Passage - II	**1.** c	**2.** b	**3.** b			
Passage - III	**1.** b	**2.** d	**3.** a			
Passage - IV	**1.** c	**2.** d	**3.** c			
Passage - V	**1.** c	**2.** b	**3.** a			
Passage - VI	**1.** b	**2.** c	**3.** a			
Passage - VII	**1.** b	**2.** a	**3.** c			

Matching Type Questions

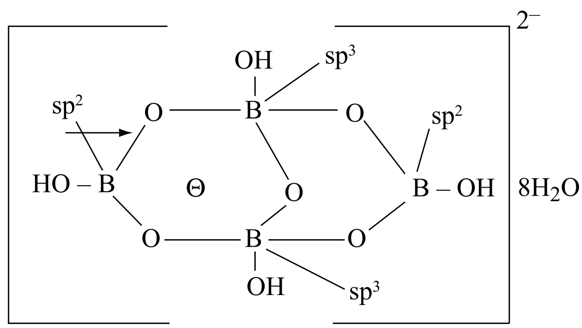
1. a - p, q, r	b - p, q	c- p, q	d - q, s
2. a - p, r, s	b - p, q, r, s	c- p, q, s	d- p, s
3. a - q, s	b - p	c - p	d - p, r
4. a - p, s	b - r	c - q	d - r
5. a - p	b - q, s	c - r	d - q, r
6. a - p, q, r	b - p, q, r	c - q, s	d - p, r
7. a - p	b - p, q, r	c - p, q, r, s	d - r
8. a - p, q, r	b - p, s	c - q	d - p, q
9. a - p, q, s	b - r, t	c - p	d - p, q

Numerical Questions

1. 5	**2.** 4	**3.** 4	**4.** 5	**5.** 6	**6.** 6
7. 5	**8.** 2	**9.** 2	**10.** 5	**11.** 2	**12.** 0
13. 6	**14.** 6	**15.** 4	**16.** 8	**17.** 4	

III Group Hints

1. In Nelson cell method, both H_2 and Cl_2 are liberated but H_2 is colourless and can be used to form a compound with boron halide to get B_2H_6 which is reducing agent.
9. $AlCl_3$ is covalent more soluble in organic solvents
10.



the sp^2 boron can participate in pπ-pπ back bonding with oxygen atoms and hence the six B–O bonds of two sp^2 boron-oxygen bonds are shorter.

11. Al_4C_3 on hydrolysis gives CH_4.

12. $AlBr_3$ exists as dimer.

14. Al_2Cl_6 have different Al–Cl bond lengths and bond angles.

16. Solid boric acid has layered lattice structure and in a layer boric acid units are held together by hydrogen bonds

17. $Na_2B_4O_7 + 7H_2O \longrightarrow 2NaOH + 4H_3BO_3$

18. Aluminium is less reactive because of the formation of protective oxide layer.

19. Refer to structure of $Al(BH_4)_3$.

20. $AlCl_3$ is covalent whereas NaCl is ionic.

21. H_3BO_3 is weak acid cannot be estimated with NaOH using phenolphthalien as indicator.

24. In borax bead test, metal oxides form metal metaborates.

27. Only four hydrogens can be substituted by methyl groups.

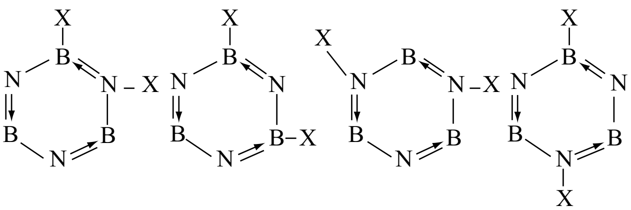
29. $Al_2O_3 + 3C + 3Cl_2 \longrightarrow 2AlCl_3 + 3CO$

32. Combustion of boranes give H_2O and B_2O_3

34. $B_2O_3 + 3CaF_2 + 3H_2SO_4 \longrightarrow 2BF_3 + 3CaSO_4 + 3H_2O$

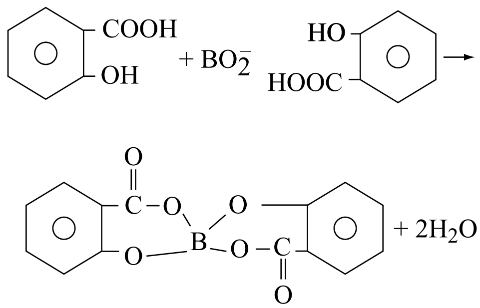
37. Owing to back bonding, the B–F bond has some double bond character and so high bond dissociation energy

38.



More than One Answer Hints

3.



$B(OH)_3 + H_2O \longrightarrow [B(OH)_4]^- + H$

$[B(OH)_4]^- \longrightarrow BO_2^- + 2H_2O$

It is optically resolvable because of asymmetric structure and it contains two six-membered rings.

4. Borazine easily participates in addition reaction with HCl because of polarity of B–N bond.

7. BF_3 partially hydrolyses in water.

$4BF_3 + 12H_2O \longrightarrow 4H_3BO_3 + 12HF$

$3H_3BO_3 + 12HF \longrightarrow 3HBF_4 + 9H_2O$

$4BF_3 + 3H_2O \longrightarrow H_3BO_3 + 3HBF_4$

$HBF_4 + H_2O \longrightarrow H[BF_3(OH)] + HF$

8. Borax is a sodium salt of tetrametaboric acid.

12. The F^- ion is chemically hard (non-polarizable) and fairly strong base. BF_3 is a hard and strong Lewis acid with high affinity for the F^- ion. Therefore F^- combines with BF_3 forming BF_4^-.

Excess F^- and acid prevent the hydrolysis of BF_3, BCl_3 hydrolyses vigorously in water rather than coordinate to aqueous Cl^-.

BBr_3 will undergo protolysis with the formation of B–N bond $BBr_3 + 6NH(CH_3)_2 \longrightarrow B[N(CH_3)_2]_3 + 3[NH_2(CH_3)_2]^+ Br^-$

$BeH_2 + B_2H_6 \longrightarrow Be(BH_4)_2$

16. Owing to back bonding, boron halides are weaker Lewis acids than aluminium halides. Moreover, owing to effective back bonding in BF_3, it is a weaker Lewis acid than BCl_3.

As BCl_3 is stronger Lewis acid than BF_3, BCl_3 substitutes the BF_3.

BBr_3 is a weaker Lewis acid than diborane. Therefore, no reaction

$B_2O_3 + 3CaF_2 + 3H_2SO_4 \longrightarrow 2BF_3 + 3CaSO_4 + 3H_2O$

18. $B_2H_6 . 2NH_3$ is ionic $[BH_2(NH_3)_2]^+ [BH_4]^-$
This on heating converts into borazole.
Owing to π dative bond from nitrogen to boron, each will have four electrons in their valence shells as in carbon.
Both C_6H_6 and $B_3N_3H_6$ are planar and non-polar. Though B–N bonds are polar the molecule is non-polar.

20. At low temperature, the ionic compound formed by B_2H_6 with ammonia (as in 7.10.12) converts to borazole on heating. At high temperature, it changes to boron nitride having structure similar to graphite. The bond order is 1.33. Owing to π-dative bond from N to B, there will be +ve and –ve charges on N and B.

Passage-I

1. Owing to small size and more number of charges on B^{3+} ion, it has more polarizing power. Hence, all boron compounds are covalent.

2. Fluorides of Al, Ga, In and Tl are ionic compounds.

3. TlI_3 consists of Tl^+ and I_3^- ions because Tl^{+3} can oxidize I^-.

4. Flourides are ionic with high melting points. However, other trihalides are covalent.
5. AlF_3 is ionic compound contains ions only.
6. $4BF_3 + 3H_2O \longrightarrow H_3BO_3 + 3HBF_4$
   $HBF_4 + H_2O \longrightarrow H[BF_3(OH)] + HF$

Passage-II

1. $Ca_2B_6O_{11} + 2Na_2CO_3 \longrightarrow 2CaCO_3 + Na_2B_4O_7 + 2NaBO_2$

   $Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + B_2O_3$

   $CaO + B_2O_3 \longrightarrow \underset{\text{Blue}}{Co(BO_2)_2}$

Passage-III

1. To prevent the action of air and salts on aluminium, its vessels are oxidized by taking as anode so that an oxide layer is formed on the surface called anodic oxidation.
2. Al is not used to prepare boron by aluminothermi process.
3. As Al is less electropositive than Na, it cannot reduce Na_2O.

Passage-IV

Borax $\xrightarrow{NH_4Cl}$ Boron nitride

$BN \xrightarrow{H_2O} H_3BO_3 + NH_3$

$H_3BO_3 \xrightarrow{100°C} HBO_2 \xrightarrow{160°C} H_4B_2O_7 \xrightarrow{\text{Red hot}} B_2O_3 \xrightarrow{Na} B$

Sol: $Na_2B_4O_7 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2B_4O_7$

$H_2B_4O_7 + 5H_2O \longrightarrow \underset{(A)}{4H_3BO_3}$

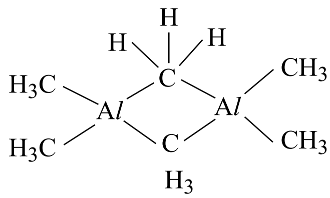
$2H_3BO_3 \xrightarrow{\Delta} \underset{(B)}{B_2O_3} + 3H_2O$

$B_2O_3 + 6K \longrightarrow \underset{(C)}{2B + 3K_2O}$

Passage-V

1. Back bonding in BBr_3 is weak and hence the energy consumed to break the back bonding in BBr_3 is least whereas it is maximum in BF_3.
2. If BF_3 has to dimerize the back π dative bond should break and sp^2 boron should convert into sp^3 boron which requires more energy.

Hints to Numerical Questions

1. 

Coordination number of bridged carbon is 5.
2. Refer to hint of Q. 38.
3. Refer to structure of Be $(BH_4)_2$.
4. Refer to structure of borax.
5. In solid Al_2O_3, each aluminium ion is surrounded by six oxide ions octahedrally.
6. Each boric acid contain three OH groups which can participate in two hydrogen bonds.
7. B_2H_6, $BeH_2(s)$, Al_2H_6 $[Be_2(CH_3)_2]_n$ $Al_2(CH_3)$ contain 3C-2e bonds.
8. Borax contain two heterocyclic B_3O_3 hexagonal rings.
9. In borax, two 'B' atoms are in sp^3 hybridization.
10. Jeweller's borax is $Na_2B_4O_7 \cdot 5H_2O$.
11. The two sp^3 hybrid boron atoms have octet in borax.
17. Number of tetrahedral boron atoms in a borate is equal to number of charges on the ion.

Hint to Matching

5. (a) $Na_2B_4O_7 + 2NH_4Cl \longrightarrow 2BN + B_2O_3 + 2NaCl + 4H_2O$

   (b) $4NaH + 2B_2O_3 \longrightarrow 2NaBH_4 + 3NaBO_2$

   (c) $2B + 6NaOH \longrightarrow 2Na_3BO_3 + 3H_2$

   (d) $Na_3B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + B_2O_3$

9. $B_3N_3H_6 + 9H_2O \longrightarrow 3NH_3 + 3H_3BO_3 + 3H_2$

   $Na_2B_4O_7 + 2NH_4Cl \longrightarrow 2BN + B_2O_3 + 4H_2O$

   $2BF_3 + 6NaH \xrightarrow{180°C} B_2H_6 + 6NaF$

   $B_2H_6 + 6H_2O \longrightarrow 2H_3BO_3 + 3H_2$

8

CHAPTER

Group-IV(A) (14) Carbon Family

Our proud theories are but temporary resting places of the mind on the unending road to knowledge

L. Rosenfeld

8.1 INTRODUCTION

The elements carbon, silicon, germanium, tin and lead constitute the Group IVA of the periodic table and belong to p-block elements. This group may be regarded in a way as the transitional group, for here we find that in the elements, the electropositive character as well developed as the electronegative. **The group is a transition between metals and non-metals.** The first three groups were distinctly metallic and the last three, the fifth, the sixth and the seventh, non-metallic. But the elements of this fourth group like carbon and silicon form very stable compounds with hydrogen as well as halogens. CH_4,CCl_4 and SiH_4, $SiCl_4$ which show a balance of the electrochemical character.

8.2 OCCURRENCE

Of these elements, carbon is found to occur in the free state in the form of coal, graphite and diamond. However, combined with oxygen and hydrogen it occurs in all living tissues belonging to plant or animal kingdom and in petroleum and coal deposits. It occurs as carbon dioxide in the atmosphere to the extent of 0.03 per cent and as carbonates in rock and minerals such as chalk, marble, limestone etc. carbon is a common constituent of all the organic matter while silicon is the main constituent of inorganic matter.

Silicon occurs, abundantly (28%) in the earth's crust as silica and silicates next to oxygen, Germanium is a rare element. Tin does not occur in the free nature. Its chief source is the ore cassiterite SnO_2, also known as tin stone. The chief ore of lead is galena, PbS which contains about 6.8 per cent of lead.

Electronic configuration: The electronic configurations of elements of Group IV A are shown in Table 8.1.

Table 8.1 Electronic configurations of elements of group IVA

Element	At. no.	Electronic configuration	Configuration of valency shell
Carbon	6	$1s^2\,2s^2\,2p^2$	$2s^2\,2p^2$
Silicon	14	$1s^2\,2s^2\,2p^6\,3s^2\,3p^2$	$3s^2\,3p^2$
Germanium	32	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}$ $4s^2\,4p^2$	$4s^2\,4p^2$
Tin	50	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}$ $4s^2\,4p^6\,4d^{10}\,5s^2\,5p^2$	$5s^2\,5p^2$
Lead	82	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}$ $4s^2\,4p^6\,4d^{10}\,4f^{14}\,5s^2\,5p^6$ $5d^{10}\,6s^2\,6p^2$	$6s^2\,6p^2$

The general outer electronic configuration of the Group IVA elements is $ns^2\,np^2$. The electronic configuration of Group IVA elements differ from one another in their penultimate shells. Thus carbon is having two, silicon 8 electrons in their penultimate shells (noble gas kernel), whereas the remaining elements possess 18 electrons in the penultimate shell (Pseudo inertgas configuration).

8.3 PHYSICAL PROPERTIES

Some important atomic and physical properties of Group IVA elements are given in Table 8.2.

1. Atomic and Ionic radius: Atomic radius increases with increase in atomic number. The same trend is observed with ionic radius in their +2 as well as +4 oxidation states.

Atomic radii increase down the group but the difference in the size between Si and Ge is less than might be otherwise expected. The reason for this is that the filling

Table 8.2 Physical properties of elements of group IVA

Property	C	Si	Ge	Sn	Pb
Abundance (ppm)	320	277,200	7	40	16
Atomic radius (pm)	77	117	122	141	146
Ionic radius (pm)					
M^{2+}	-	-	93	112	120
M^{4+}	15	41	53	71	85
Electronegativity	2.5	1.8	1.8	1.8	1.8
Boiling point (K)	5100	3553	3123	2896	2024
Melting point (K)	4373	1693	1218	505	600
Density (g cm^{-3})	3.52	2.42	5.36	7.30	11.34

of the 3d shell increases the nuclear charge but the 3d electrons shields the nuclear charge rather ineffectively. Also the small difference in size between Sn and Pb is because of the poor shielding effect of the 4f electrons.

2. Density: Carbon in the form of diamond has more density than silicon. With the exception of carbon, the density of the elements of this group increases with the increase in atomic number. This is attributed to the greater and steady increase of the atomic weight comparing to the increase in the atomic size of the elements in the group.

3. Melting points: The melting point of carbon is very high. The value decreases from carbon to tin but there is slight increase from tin to lead. The melting point depends upon the size of atoms. The smaller atoms are more closely packed and require greater amount of energy to overcome the force of attraction between them. As the atomic size increases down the group the decrease in the melting point in that order is expected. But the melting point of lead is more than that of tin but less than germanium.

4. Boiling points: The boiling point of carbon is exceedingly high. The value decreases gradually as we move down the group from carbon to lead.

The melting point and boiling point of Group IVA elements are higher than the corresponding Group IIIA elements. This is due to the formation of four covalent bonds on account of four valence electrons resulting in strong binding forces in between their atoms in solid as well as in liquid state. Carbon, silicon and germanium have diamond like structure. So they have high melting points because more energy is required to break more number of bonds but decreases with increase in atomic size. Tin and lead are metals but they do not use all the valence electrons in metallic bond.

5. Physical appearance: All the elements of Group IVA are solids having different colours. Carbon is black, silicon is light brown, germanium is greyish white whereas tin and lead are lustrous and silvery white appearance.

6. Ionization Energies: The first four ionization energies of the Group IVA elements are given in Table 8.3.

Table 8.3 Ionization energies of group IVA elements Ionization energies (kj mol^{-1})

Element	1st	2nd	3rd	4th
Carbon	1086	2354	4622	6223
Silicon	786	1573	3300	4409
Germanium	760	1534	3232	4351
Tin	707	1409	2943	3821
Lead	715	1447	3087	4081

Ionization energies decreases from carbon to tin but the decrease is not regular. There is large decrease between carbon and silicon, but from silicon onwards the decrease is very little and practically have nearly same ionization energies. This is because of the poor shielding effect of 3d and 4d electrons in germanium and tin. Lead is having a little more ionization energy than tin but less than germanium. This is because of the lanthanide contraction in the 14 lanthanide elements present before lead.

7. Electronegativity: Carbon is the most electronegative element in this group. Electronegativity decreases from carbon to silicon but remain constant in the remaining four elements. This is again due to the filling of d-orbitals in germanium and tin and also f electrons in the case of lead.

8. Metallic and non-metallic character: The elements in the Group IVA well illustrate the change from non-metallic character to metallic character with increasing atomic number. Carbon and silicon are non-metals, germanium has some metallic properties, while tin and lead are metals. The increase in metallic character is exhibited in the structures and appearance of the elements, in physical properties like malleability and electrical conductivity and in chemical properties like the increased tendency to form M^{2+} ions and the acidic or basic properties of the oxides and hydroxides.

9. Oxidation states: All the elements of Group IVA have ns^2 np^2 valence electronic configuration. So it is natural to expect for these elements either to lose these four electrons or gain four electrons to attain a stable configuration. These elements on account of their high ionization energies do not form stable M^{4+} ions. At the same time their inability to form tetranegative anion is due to their low electron affinity. The only alternative left is to share their four electrons to form covalent bonds. These elements form compounds in +2 as well as +4 oxidation states.

Outer electronic configuration

	ns	np	
Ground State	$\boxed{\uparrow\downarrow}$	$\boxed{\uparrow}\boxed{\uparrow}\boxed{\ }$	+ 2
Excited State	$\boxed{\uparrow}$	$\boxed{\uparrow}\boxed{\uparrow}\boxed{\uparrow}$	+ 4

As explained in the previous Chapter 7.5(8) the +2 oxidation state becomes increasingly stable on moving down the group i.e., from carbon to lead, while the stability of the +4 oxidation state decreases from carbon to lead (inert pair effect).

In +2 oxidation state these elements act as reducing agents. As the stability of +2 oxidation state increases from carbon to lead the reduction power in +2 oxidation state decreases from carbon to lead. For example carbon monoxide is a good reducing agent while PbO has no reduction power.

In +4 oxidation state these elements acts as oxidizing agents. As the stability of +4 oxidation decreases from carbon to lead due to inert pair effect, the oxidation power in +4 oxidation state also increases from carbon to lead. Stability order of oxidation states is

+2 oxidation state C < Si < Ge < Sn < Pb

+4 oxidation state C > Si > Ge > Sn > Pb

Order of the reduction power of Group IVA elements in +2 oxidation state is C > Si > Ge > Sn > Pb. In +2 oxidation state compounds of carbon to tin act as good reducing agents. From carbon to tin +4 oxidation state is more stable than +2. But in lead +2 oxidation state is more stable than +4 oxidation state. Oxidation power of Group IV elements in +4 oxidation state is in the order C < Si < Ge < Sn < Pb. Lead compounds in +4 oxidation state are strong oxidizing agents e.g., PbO_2.

10. Nature of bonding: All the compounds of Group IVA elements in the +4 oxidation state are covalent. This is in accordance with the Fajan's rule which may be stated that **the smaller the cation the greater is the amount of covalent character in its compounds**. The M^{4+} is smaller ion with more number of charges. So it will have more polarizing power and thus its compounds are covalent. Most of the compounds of carbon and silicon in +2 oxidation state are also covalent because of their non-metallic character and more electronegativity. However carbon tends to form C_2^{2-} ions as in carbides like CaC_2. The last three elements Ge, Sn and Pb tend to form M^{2+} ions. The tendency to form ionic compounds increases from Ge to Pb in +2 oxidation state.

11. Reluctance of heavier elements to involve p-orbitals for π bonding: In the Group IVA elements, only carbon can form double bonds or triple bonds between two carbon atoms. But other elements do not form such double bonds or triple bonds between the atoms of same element, although recently some compounds of silicon with Si = Si involving pπ-pπ bonding have been prepared. This is because in heavier elements as the atomic size increases the sigma bond length also increases which reduces or even excludes the sideways overlapping of p-orbitals. This is illustrated in Fig 8.1 by depicting the effect of the length of C–C, Si – Si, Ge – Ge and Sn – Sn sigma bonds on the sideways overlapping of p-orbitals of these elements.

A very small overlap implies a very weak pπ – pπ bond and if no overlap is possible, there is no formation of pπ – pπ bond at all.

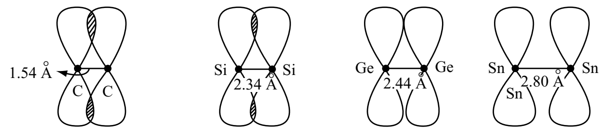


Fig 8.1 Effect of the length of sigma bond on the sideways overlapping of p-orbitals

12. Participation of d-orbitals in sigma bonding: Carbon has only s and p orbitals whereas other Group IVA elements have d-orbitals also in their valence shells. So Carbon can show a maximum covalency of 4 corresponding to sp^3 hybridization of its valence shell orbitals. In contrast the other elements of Group IVA show a covalency higher than 4 due to the involvement of valence d- orbitals also in hybridization. Thus while the covalence of C in CCl_4, CF_4 etc involving sp^3 hybridization is 4, that of Si in $[SiF_6]^{2-}$ involving sp^3d^2 hybridization is 6.

13. Participation of d-orbitals in π - bonding: Trimethyl amine is known to differ from trysilylamine being $(H_3Si)_3N$ is planar rather than pyramidal. Also $(H_3Si)_3N$ is weaker base than $(CH_3)_3N$. Disilylamine is also planar. These observations can be explained by supposing that nitrogen forms dative π bonds to the silicon atom as shown in Fig 8.2. We assume that the central nitrogen atom is sp^2 hybridized leaving a filled $2p_z$ orbital, which overlaps appreciably with an empty silicon $3d_{xz}$ (or $3d_{yz}$) orbital. Thus a dative pπ - dπ bond is established which provides additional bond strength in each Si – N linkage of the molecule. It is thus additional bond strength that stabilizes the NSi_3 skeleton in a planar configuration. In contrast for $N(CH_3)_3$, since carbon has no low lying d-orbitals, σ bonding alone determines the configuration at the AB_3E, carbon atom, which is pyramidal. As an interesting comparison, consider trisilyl phosphine $(H_3Si)_3P$ which is pyramidal. Evidently phosphorus is less stable than nitrogen to form a pπ → dπ dative bond to silicon.

In the vapour phase H_3SiNCO is linear (hydrogen atoms accepted). This can be explained by the formation of a pπ → dπ bond between nitrogen and silicon (H_3Si – N = C = O). The corresponding carbon compound (H_3C NCO) is not linear, since carbon has no vacant, low-lying d-orbitals. Interestingly H_3GeNCO is not linear in the gas phase. Evidently, effective pπ-dπ bonding occurs for Si – N but not Ge – N.

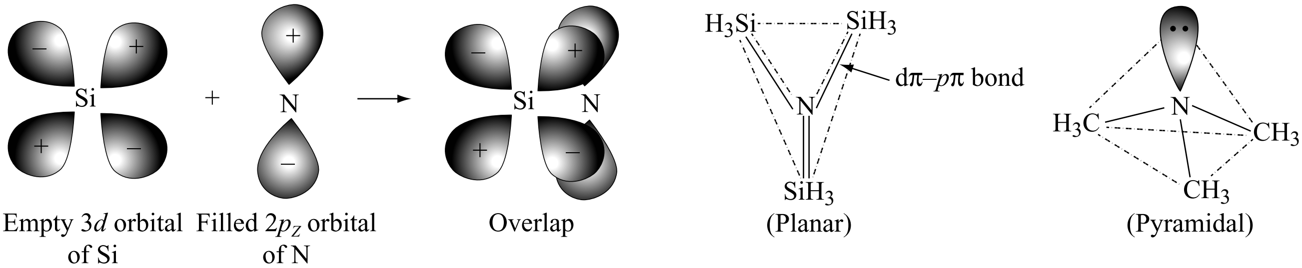


Fig 8.2 The formation of a dπ – pπ bond between silicon and nitrogen atoms in trisilylamine

The disilyl ethers $(R_3Si)_2O$, all have large angles at oxygen (140–180°) and both electronic and steric explanations have been suggested. Electronically overlap between filled oxygen pπ orbitals and empty silicon dπ orbitals would improve with increasing Si – O – Si angles, and might be most effective for linear Si – O – Si group. There may also be very strong steric factors favouring more linear structures especially for large R groups. For instance, the angle at oxygen is 180° for $(Ph_3Si)_2O$.

Silanols such as $(CH_3)_3SiOH$ are stronger protonic acids than their carbon analogues and form stronger hydrogen bonds. This is due to stabilization of the conjugate base anion by O(pπ) → Si(dπ) bond formation. A similar stabilization of the conjugate base anion can be invoked to explain the order of acidities (M = Si > Ge > C) in the series R_3MCOOH.

14. Catenation: Combining capacity of the atoms of the same element to form long chains, branched chains and cyclic compounds is called **catenation power**. An element can exhibit catenation only if it is minimum bivalent. Of all the elements carbon has the maximum catenation power. The reasons are

(i) its tetravalency
(ii) high C–C bond energy
(iii) absence of lone pairs on tetravalent carbon
(iv) absence of d - orbitals in its valence shell
(v) ability to form multiple bonds between carbon atoms.

Carbon is tetravalent and can form four covalent bonds. Carbon atoms form single bonds as well as double or triple bonds with other carbon atoms. Due to small size and absence of lone pairs of electrons C–C bond energy is very high and comparable with the bond energies between carbon and other elements.

H_3C-CH_3	H_2N-NH_2	$HO-OH$	$F-F$	Units
350	160	140	150	kJ mol^{-1}

From the M–M bond energies of second period elements it can be seen that there is profound drop between C – C and N – N bond energies. This difference is probably attributable to the effects of repulsion between non-bonding lone pairs. The result is that unlike carbon, nitrogen and oxygen have little tendency to catenation. It can also be seen that the bond energies decreases as we move down the group with increase in atomic size. In Table 8.4 it can be seen that there is steady decrease in the M–H bond strength. In general, with the exceptions of M–H bonds the strength of M–X bonds diminish less noticeably. Further due to the absence of d-orbitals in its valence shell, the compounds of carbon cannot be attacked by moisture in air and cannot be hydrolysed and thus they are stable when exposed to atmosphere. Because of all these reasons carbon exhibits maximum catenation power. The catenation power of Group IVA elements is in the order C >> Si > Ge > Sn > Pb.

Contrary to the situation in carbon chemistry, catenation in silicon compounds reaches its maximum in the halide rather than hydrides. This is because of additional back bonding from filled halogen pπ orbitals into the silicon dπ orbitals which thus synergically compensates for electron loss from silicon via σ bonding to the electronegative halogen. From Table 8.4 it can be seen that Si–F, Si–C*l* and Si–O bonds are stronger than corresponding bonds with carbon. This is thought to be due to the donation of electrons from F, C*l* or O to Si giving rise to pπ-dπ bonding.

15. Allotropy: The occurrence of the same element in two or more different physical forms having more or less

Table 8.4 Approximate average bond energies / kj mol^{-1}

M –	– M	– C	– H	– F	– C*l*	– Br	–I	– O
C	356	356	416	490	325	279	216	357.7
Si	226	360	323	596	400	325	248	452
Ge	188	255	289	471	339	281	216	
Sn	151	226	253	–	315	261	187	
Pb	98	130	205	411	308	–	–	

similar chemical properties but different physical properties is called **allotropy**. The different forms of the element are called allotropes and are formed due to the difference in the arrangement of atoms. Allotropes may be crystalline or amorphous.

Carbon has large number of allotropes.

(i) **Crystalline allotropes of carbon** are diamond, graphite and fullerenes.

(ii) **Amorphous allotropes** of carbon are coal, coke, wood charcoal, animal charcoal, lamp black, gas carbon, petroleum, coke, sugar charcoal.

Both diamond and graphite are thermodynamically unstable in the presence of air and, of the two allotropes, graphite is more stable than diamond. These facts can be seen from the equations.

$$C\ (Diamond) + O_{2(g)} \longrightarrow CO_{2(g)}\ \Delta H^0_{298} = -395.4\ kJ\ mol^{-1}$$

$$C\ (Graphite) + O_{2(g)} \longrightarrow CO_{2(g)}\ \Delta H^0_{298} = -393.4\ kJ\ mol^{-1}$$

The reason why diamond and graphite do not react with oxygen at room temperature and diamond does not spontaneously transform itself into graphite is because high activation energies oppose these reactions.

This type of allotropy is referred to as **monotropy** since at ordinary temperatures and pressures one form (graphite) is always thermodynamically more stable than the other (diamond).

Silicon also exhibit allotropy but there appear to be no allotropes of Si at ambient pressure but the 4 coordinate diamond type lattice of Si – I transforms to several other modifications at higher pressures of which distorted diamond Si – II, primitive hexagonal Si – V and eventually hexagonal close packed Si – VII. The structural sequence corresponds to a systematic increase in coordination number.

$$4(Si - I) \xrightarrow{8.8\,GPa} 6(Si - II) \xrightarrow{16\ GPa} 8\ (Si - V) \xrightarrow{40\,Gpa} 12\ (Si - VII)$$

$1GPa = 10\ K\ bar \approx 9869\ atm.$

Germanium forms brittle, grey white lustrous crystals with the diamond structure; it is a metalloid with similar electrical resistivity to Si at room temperature but with a substantially smaller band gap.

Tin exhibits allotropy, three crystalline forms being known with the transition temperature as shown below.

$$\underset{Grey}{\alpha - tin} \underset{}{\overset{13.2^oC}{\rightleftharpoons}} \underset{white}{\beta - tin} \overset{161^oC}{\rightleftharpoons} \gamma - tin \overset{273^oC}{\rightleftharpoons} liquid\ tin$$

Although β - tin should change into α-tin at a temeprature below 13.2°C the transformation only becomes rapid at about –50°C, unless some α-tin is added to catalyse the reaction. α-tin has the diamond like structure whereas both β-tin and γ-tin are more metallic (an approach to close packing of the atoms) Since α-tin has more open structure than β- and γ-tin its density is considerably less than the densities of the other two allotropes.

This type of allotropy in which two allotropes are equally stable at the transition temperature is referred as **enantiotropy**.

Lead exists only in one form which is metallic in nature.

The change over from non-metallic to metallic nature with increasing atomic number is thus reflected in the structures of the Group IVA elements.

8.4 CHEMICAL REACTIVITY

Elements of Group IVA are relatively unreactive but reactivity increases down the group. They are generally attacked by acids, alkalis and the halogens. Among these elements lead is less reactive than expected. This is partly due to the presence of oxide layer on the surface and partly due to hydrogen over voltage.

(i) **Action of air:** These elements are not attacked by air at ordinary temperature. Though the reaction of carbon with oxygen is highly exothermic it will not take place when carbon is kept in air. This is because the activation energy of reaction is very high. On heating all these elements combine with oxygen in the air.

$$C + O_2 \longrightarrow CO_2$$
$$Si + O_2 \longrightarrow SiO_2$$
$$Ge + O_2 \xrightarrow{\text{Red hot}} GeO_2$$
$$Sn + O_2 \longrightarrow SnO_2$$
$$2Pb + O_2 \longrightarrow 2PbO$$

(ii) **Action of water:** Water has no action on these elements. However at red hot condition, these elements react with steam to form hydrogen.

$$C + H_2O \longrightarrow CO + H_2O$$
$$Si + 2H_2O \longrightarrow SiO_2 + 2H_2$$
$$Sn + 2H_2O \longrightarrow SnO_2 + 2H_2$$

Lead is not affected by pure water except at the boiling point. It dissolves slowly in water containing dissolved oxygen. This dissolution of lead in water is called **plumbosolvency**.

$$2Pb + 2H_2O + O_2 \longrightarrow 2Pb(OH)_2$$

(iii) **Action of acids:** Carbon is not attacked by non-oxidizing acids. But carbon is oxidized to carbon dioxide by the oxidizing acids like concentrated sulphuric and nitric acids.

$$C + 2H_2SO_4 \longrightarrow 2H_2O + 2SO_2 + CO_2$$
$$C + 4HNO_3 \longrightarrow 2H_2O + 4NO_2 + CO_2$$

Silicon do not react with any acid except hydrofluoric acid. Dilute hydrochloric acid does not attack germanium but the germanium metal when heated in a current of HC*l* gas forms germanium chloroform and germanium tetrachloride. It also dissolves in hot concentrated H_2SO_4.

$$Ge + 3HCl \longrightarrow GeHCl_3 + H_2$$
$$Ge + 4HCl \longrightarrow GeCl_4 + 2H_2$$

Tin dissolves slowly in dilute HC*l* but quite rapidly in concentrated HC*l* liberating H_2. The slow reaction in dilute HC*l* is due to the high hydrogen over voltage at surface of the metal. Conc. H_2SO_4 in hot condition gives $SnSO_4$ and SO_2.

$$Sn + 2HCl \longrightarrow SnCl_2 + H_2$$
$$Sn + 2H_2SO_4 \longrightarrow SnSO_4 + 2H_2O + SO_2$$

Lead dissolves in hot concentrated hydrochloric acid forming chloroplumbous acid. The low solubility of $PbCl_2$ restricts the reaction.

$$Pb + 2HCl \longrightarrow PbCl_2 + H_2$$
$$PbCl_2 + 2HCl \longrightarrow H_2PbCl_4$$

Lead becomes passive with concentrated sulphuric acid due to the formation, insoluble lead sulphate on the surface. Germanium, tin and lead reacts with both dilute and concentrated nitric acid. Tin with dilute nitric acid gives ammonium nitrate and with concentrated nitric acid gives metastannic acid

$$4Sn + 10HNO_3 \longrightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$$
$$Sn + 4HNO_3 \longrightarrow H_2SnO_3 + 4NO_2 + H_2O$$

Lead with dilute HNO_3 gives NO while with concentrate HNO_3 gives NO_2.

$$3Pb + 8HNO_3 \longrightarrow 3Pb\,(NO_3)_2 + 4H_2O + 2NO$$
$$Pb + 4HNO_3 \longrightarrow Pb(NO_3)_2 + 2H_2O + 2NO_2$$

(iv) Action of Alkalis: Carbon is not attacked by the alkalis. But carbon reduces sodium hydroxide to the elements when fused together.

$$6NaOH + 2C \longrightarrow 2Na + 2Na_2CO_3 + 3H_2$$

The remaining elements react with alkalis forming their soluble alkali salts and liberate hydrogen.

$$Si + 2NaOH + H_2O \longrightarrow Na_2SiO_3 + 2H_2$$
$$Sn + 2NaOH + H_2O \longrightarrow Na_2SnO_3 + 2H_2$$
$$Pb + 2NaOH + H_2O \longrightarrow Na_2PbO_3 + 2H_2$$

(v) Reaction with metals: Carbon reacts with many metals directly at high temperatures forming carbides which are discussed latter.

Silicon also react with several metals forming silicides. Silicides have been reported for virtually all elements in group 1–10 except Be. No silicides are known for the metals in groups 11– 15 except copper. eg.,

$$2Mg + Si \longrightarrow Mg_2Si$$

Germanium, tin and lead being electropositive form alloys with other metals.

(vi) Reaction with non-metals: At high temperatures carbon reacts with many non-metals including hydrogen (in the presence of finely divided nickel catalyst) fluorine (but not with other halogens) oxygen, sulphur, silicon, boron. In these compounds carbon is known with all coordination numbers from 0-8 though compounds in which it is 3 or 4 coordinate are the most numerous. Some typical examples of compounds carbon with higher coordination number are given in Table 8.5.

Table 8.5 Compounds of carbon with higher coordination number

Coordination number	Example	Comment
5	$Al_2(CH_3)_6$	Alkyl bridged organo metallic
6	$C_2B_{10}H_{12}$	involving 3c-2e bonds Several
7	$(LiCH_3)_4$	Carboranes
8	Be_2C (antifluorite)	

Carbon reacts with oxygen in air or oxygen forming CO or CO_2 depending on the amount of oxygen available.

$$2C + O_2 \longrightarrow 2CO$$
$$C + O_2 \longrightarrow CO_2$$

Carbon in the form of white hot coke reacts with sulphur forming carbon disulphide.

$$C + 2S(\text{vapour}) \xrightarrow[\text{furnace}]{\text{Electric}} CS_2$$

Carbon reacts with fluorine forming carbon tetrafluoride

$$2C + 2F_2 \longrightarrow 2CF_4$$

With boron forms boron carbide and with silicon gives silicon carbide.

$$4B + C \longrightarrow B_4C$$
$$Si + C \longrightarrow SiC$$

Silicon in the massive crystalline forms is relatively unreactive except at high temperatures. Oxygen, water and steam all have little effect probably because of the formation of a very thin continuous, protective surface layer of SiO_2 a few atoms thick. Oxidation in air is not measurable below 900°C; between 950° and 1160° the rate of formation of vitreous SiO_2 rapidly increases.

$$Si + O_2 \longrightarrow SiO_2$$

At 1400° silicon reacts with nitrogen (in air) to give SiN and Si_3N_4.

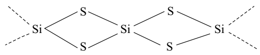
$$2Si + N_2 \longrightarrow 2SiN$$

$$3Si + 2N_2 \longrightarrow Si_3N_4$$

Sulphur vapour reacts with silicon at 600°C

$$Si + 2S \longrightarrow SiS_2$$

Silicon disulphide, as might be expected, a giant molecule unlike carbon disulphide which has the structure S = C = S. It crystallizes in the form of long silky needle a process consistent with chain like structure.



Phosphorous vapour reacts with silicon at 1000°C. Silicon reacts with all most all halogens and the reactivity with halogens decreases from F to I.

$$Si + 2F_2 \xrightarrow{\text{room temp}} SiF_4$$

$$Si + 2Cl_2 \xrightarrow{300°C} SiCl_4$$

$$Si + 2Br_2 \xrightarrow{:500°C} SiBr_4$$

$$Si + 2I_2 \xrightarrow{:500°C} SiI_4$$

Germanium is oxidized to GeO_2 in air at red heat and both H_2S and gaseous sulphur yield GeS_2. Cl_2 and Br_2 yield GeX_4 on moderate heating.

$$Ge + O_2 \xrightarrow{\text{red heat}} GeO_2$$

$$Ge + 2H_2S \longrightarrow GeS_2 + 2H_2$$

$$Ge + 2S \longrightarrow GeS_2$$

Tin reacts readily with Cl_2 and Br_2 in the cold and with F_2 and I_2 on warming SnX_4

$$Sn + 2X_2 \longrightarrow SnX_4 \ (X = F, Cl, Br \text{ or } I)$$

It reacts readily with heated sulphur and selenium to Sn^{2+} and Sn^{4+} chalcogenides depending on the proportions used and with tellurium to form SnTe.

$$Sn + S \longrightarrow SnS \qquad Sn + Se \longrightarrow SnSe$$

$$Sn + 2S \longrightarrow SnS_2 \qquad Sn + 2Se \longrightarrow SnSe_2$$

$$Sn + Te \longrightarrow SnTe$$

Finely divided Pb powder is pyrophoric but the reactivity of the metal is usually greatly diminished by the formation of thin, coherent, protective layer of soluble product such as oxide, oxocarbonate, sulphate or chloride. Lead reacts with fluorine at room temperature, with chlorine on heating to form PbF_2 and $PbCl_2$

$$Pb + F_2 \xrightarrow{\text{room temp}} PbF_2$$

$$Pb + Cl_2 \xrightarrow{\Delta} PbCl_2$$

Molten lead reacts with chalcogens to give PbS, PbSe, PbTe

$$Pb + X \longrightarrow PbX \ (X = S, Se \text{ or } Te)$$

Reducing property: Carbon reduces heated oxides of various elements. The products depend on the quantity of carbon used and the temperature. e.g.,

$$SnO_2 + 2C \longrightarrow Sn + 2CO$$

$$CO_2 + C \xrightarrow{\Delta} 2CO$$

$$Fe_2O_3 + 3C \xrightarrow{250-400°C} 2Fe + 3CO$$

Silicon is also used as deoxidizer in the manufacture of steel in the form of ferrosilicon. Tin also acts as good reducing agent.

8.5 IMPORTANT COMPOUNDS OF ELEMENTS OF GROUP IVA COMPARATIVE STUDY

8.5.1 Hydrides

All the elements of the groupIV form covalent hydrides. The number of the stable hydrides and the ease with which these are formed decreases as we move down the group. Carbon forms a large number of stable hydrides known as **hydrocarbons**. This is due to the strong tendency of the carbon atoms to form long chain i.e., catenation. The hydrides of carbon include alkanes, alkenes, alkynes, cycloalkanes and number of aromatic hydrocarbons, however, lies out side the scope of this chapter and is deal with in books devoted to organic chemistry.

Silicon forms a limited number of hydrides due to the less bond energy of Si - Si bond. The hydrides of silicon are known as **silanes**, which have the general formula Si_nH_{n+2}. Silicon hydrides (silanes) may be prepared by the action of (i) lithium aluminium hydride on silicon tetrachloride in ether, or (ii) by the action of HCl on magnesium silicide (mixture of silanes will be formed).

$$LiAlH_4 + SiCl_4 \longrightarrow LiCl_4 + AlCl_3 + SiH_4$$

$$Mg_2Si + H^+ \longrightarrow Mg^{2+} + SiH_4 + H_3Si - SiH_3 \text{ etc.}$$

Germanium forms a series of hydrides with formulae analogous to alkanes called **germanes**. The whole series upto and including the hydride containing six germanium atoms in a chain are known with certainity. They are named as germane GeH_4, digermane Ge_2H_6, trigermane Ge_3H_8 etc. They are prepared by the addition of alkali borohydride (MBH_4) to germanium dioxide GeO_2 in acid solution. Germane is also prepared by the action of $LiAlH_4$ on germanium tetrachloride.

Tin is known to form only two hydrides viz stannane SnH_4 and distannane Sn_2H_6. Stannane is prepared by the reduction of $SnCl_4$ with $LiAlH_4$ in ether at –30°C.

$$SnCl_4 + LiAlH_4 \longrightarrow SnH_4 + LiCl + AlCl_3$$

Lead forms only one hydride called plumbane PbH_4. It has been shown to exist by treating an alloy of magnesium and lead (containing $^{212}_{82}Pb$ which has radioactivity) with dilute acid. The detection of radioactivity in the gas phase shows that a volatile hydride of lead is formed.

Thermal stability: The thermal stability of the hydrides gets decreased from carbon to lead which is evident from their decomposition temperature.

Hydride	CH_4	SiH_4	GeH_4	SnH_4	PbH_4
Decomposition tem(°C)	800	450	285	150	0

With increase in the size of the central atom M the M–H bond length increases and the bond becomes weaker. The weak M–H bond in turn is because of large difference in sizes of the metal and hydrogen atom giving rise to poor overlapping and hence weak covalent bond. So the stabilities of these hydrides decreases as we descend from top to bottom in the group.

The stability of the higher hydrides of the elements of groups IVA also decreases from carbon to tin. This can be explained basing on the M–M bond strength and on the difference in the electronegativites of hydrogen and the elements of Group IVA. The values in Table 8.4 clearly shows the decrease in M-M bond strength. Further due to the similar electronegativites of carbon and hydrogen, the hydrogen is not able to withdraw electron from C–H bond in alkanes, with the result the C–C bonds do not get weakened on the other hand hydrogen is able to withdraw electrons from M–M bond in other hydrides because of high electronegativity of hydrogen than the electronegativities of the other elements of Group IV, which produces the weakening of M–M bonds.

$$\underset{2.5}{\overset{\delta-}{C}} \longrightarrow \underset{2.1}{\overset{\delta+}{H}} \qquad \underset{1.8}{\overset{\delta+}{Si}} \longrightarrow \underset{2.1}{\overset{\delta-}{H}}$$

Reducing character: As the thermal stability of hydrides gets decreases down the group, their ability to act as reducing agent gets increased. Thus unlike alkanes silanes are strong reducing agents and get readily hydrolysed by alkaline solution to liberate hydrogen.

$$SiH_4 + 2NaOH + H_2O \longrightarrow Na_2SiO_3 + 4H_2$$
$$Si_2H_6 + 4KOH + 2H_2O \longrightarrow 2K_2SiO_3 + 7H_2$$

However, it is important to observe that among these hydrides silanes are the least stable to hydrolysis with alkali. The order of stability of these hydrides towards alkali is as follows.

CH_4	>	GeH_4	>	SnH_4	>	SiH_4
Unattacked by alkali		Attacked by 33% alkali		Attacked by 30% alkali		Easily attacked even by water

It is not possible to explain the abnormality in trend on the basis of electronegativity of these elements. However it is possible to explain the more reactivity of silanes than alkanes on the basis of the polarity of Si—H bond and availability of d-orbitals with silicon atom.

8.5.2 Halides

The various facts about the halides of the elements Group IVA are as follows.

(i) The elements of Group IVA form tetrahalides of the general formula MX_4 except $PbBr_4$ and $PbI_4 \cdot PbBr_4$ and PbI_4 do not exist because of the inability of Br_2 and I_2 to oxidize Pb^{2+} to Pb^{4+}. In other words Pb^{4+} ion is a strong oxidizing agent while Br^- and I^- ions are high reducing agents. This results only in the formation of their divalent compounds.

(ii) All the tetrahalides are very volatile and covalent. The exceptions are SnF_4 and PbF_4 which are ionic and highly melting solids. SnF_4 sublimes at 978K, PbF_4 melts at 373K. The thermal stability of tetrahalides of different elements of Group IVA with a common halogen decreases with increase in atomic number.

$$CX_4 > SiX_4 > GeX_4 > SnX_4 > PbX_4$$

On the other hand the thermal stability and volatility of tetrahalides having a common central atom decreases with the increase in the molecular weight of tetrahalide.

$$MF_4 > MCl_4 > MBr_4 > MI_4$$

The above order can be ascribed to the decrease in the M–X bond energies from M–F to M–I as can be seen for C–X bonds in Table 8.4.

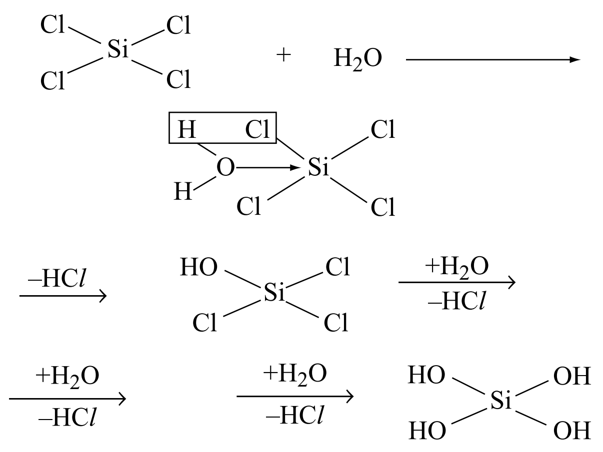
The fluorides of carbon are commonly known as fluorocarbons. These are most stable and inert. The mixed fluorochlorocarbons known as freons are volatile, non-toxic and non-corrosive. These are useful refrigerants.

(iii) Carbon halides are not hydrolysed by water because they cannot increase their coordination number beyond four due to the non-availability of d-orbitals. On the other hand the tetrahalides of the remaining elements are readily hydrolysed by water. The reason for this is due to the availability of vacant d-orbitals in their atoms and therefore they can easily increase their coordination number beyond four.

$$CCl_4 + H_2O \longrightarrow \text{No action}$$
$$SiCl_4 + 4H_2O \longrightarrow Si(OH)_4 + 4HCl$$

Hydrolysis of tetrahalides takes place in two stages.

The first step in the reaction is the donation of lone pair of electrons from the oxygen of water molecule to an empty 3d orbitals of silicon and forms a coordinate bond. Hydrogen chloride molecule is then eliminated in the second step. The stepwise formation of hydrolysed product of $SiCl_4$ is shown below.



The ultimate product due to stepwise hydrolysis is orthosilicic acid $Si(OH)_4$ or hydrated silicic acid $H_2SiO_3.H_2O$. With silicon tetrafluoride the reaction can proceed further and some hexafluorosilicic acid is formed as well.

$$SiF_4 + 4H_2O \longrightarrow Si(OH)_4 + 4HF$$

$$2SiF_4 + 4HF \longrightarrow 2H_2SiF_6$$

$$3SiF_4 + 4H_2O \longrightarrow Si(OH)_4 + 2H_2SiF_6$$

The carbon halides are not hydrolysed under normal conditions due to the absence of d-orbitals. However it is important to remember that empty orbitals having higher energy are always available with any atom and they can be used if sufficient energy is available for the reaction to occur. This is able to explain the hydrolysis of carbon tetrachloride with super heated steam especially in the presence of iron or copper

$$\underset{}{CCl_4} + \underset{\text{Steam}}{H_2O} \longrightarrow \underset{\text{Phosgene}}{COCl_2} + 2HCl$$

Lewis acidic character and complex formation: Except carbon tetrahalides, the tetrahalides of other Group IVA elements act as strong Lewis acids because of the availability of vacant d-orbitals in their valence shells. Carbon halides cannot act as Lewis acids because they cannot increase their coordination number beyond four due to the non-availability of d-orbitals in its valence shell. Because of this reason the tetra halides of Group IV elements except carbon can form complex ions such as $[SiF_6]^{2-}$, $[GeF_6]^{2-}$, $[SnCl_6]^{2-}$ by increasing their coordination number from four to six. $SiCl_4$ cannot form complex ion like $[SiCl_6]^{2-}$ because the silicon atom cannot be coordinated by bigger six chloride ions.

Dihalides: Carbon and silicon do not form dihalides whereas other elements form dihalides. The stability of the dihalides increases as we go down the group. This is due to the inert pair effect. Dihalides are ionic solids and have more melting points than the corresponding tetrahalides which are covalent. For example $SnCl_2$ and $PbCl_2$ are high melting solids whereas $SnCl_4$ and $PbCl_4$ are volatile liquids.

Charge transfer compounds: The tetrahalides are all colourless or white except for SnI_4 which is bright orange compound. The colour is associated with electrons being promoted from one energy level to another, and absorbing or emitting the energy difference between the two levels. This is common in the transition elements where there are often unfilled energy levels in the shells allowing promotion from one d-level to another. In the main groups the s and p shells are normally filled with electrons when a compound is formed. So such promotion is not possible. The promotion for example from 2p to 3p level involves so much energy that it would appear in the ultraviolet rather than the visible region. Thus the white colour is to be expected for the tetrahalides. The orange colour in SnI_4 is caused by the absorption of blue light, hence the reflected light contains a higher proposition of red and orange. The energy absorbed in this way causes transfer of an electron from iodine to tin (This corresponds to the temporary reduction of Sn(IV)). Since transferring an electron to another atom is transferring a charge. Such spectra are called charge transfer spectra. This occurs in SnI_4 because the atoms are close in the periodic table and have similar sizes and similar energy levels, but it does not occur with other tetrahalides. However, PbI_2 is also yellow in colour due to similar reasons.

Polymeric halides: Carbon forms a number of catenated halides, perhaps the best known being teflon or polytetrafluoroethylene. This is formed when tetrafluoroethylene is subjected to pressure and the polymer formed has chain lengths of several hundred atoms.

$$nC_2F_4 \xrightarrow{\text{Pressure}} -CF_2-CF_2-CF_2-CF_2-$$

Teflon is extremely useful because it is resistant to chemical attack and is good electrical insulator. It is also used as a coating for non-stick cooking utensils.

Silicon forms polymers $(SiF_2)_n$ and $(SiCl_2)_n$ by passing the tetrahalides over silicon. These polymers decompose on heating into low molecular weight polymers or oligomers of formula Si_nX_{2n+2}. Fluorides are known when n has values 1-14.

Germanium forms the dimer Ge_2Cl_6 whilst no catenated halides are known for tin and lead.

8.5.3 Oxides

All the elements of Group IVA form mainly two types of oxides, monoxides of the type MO and dioxides of the type MO_2.

Monoxides	CO	SiO	GeO	SnO	PbO
Dioxides	CO_2	SiO_2	GeO_2	SnO_2	PbO_2

Monoxides: All the members of Group IVA form monoxides. The SiO is formed only at elevated temperature and is very unstable. Among the monoxides CO is neutral, but GeO, SnO and PbO are amphoteric in nature. This is because the electropositive character increases from carbon to lead.

Stability of monoxides increases from CO to PbO because the stability of +2 oxidation state increases due to inert pair effect. Because of this reason monoxides can act as reducing agents but the reduction power decreases from CO to PbO. Carbon monoxide is a very good reducing agent while PbO has least reduction power. Reactivity of the monoxides also decreases from CO to PbO due to increase in the stability of +2 oxidation state.

Dioxides: All the members of Group IVA form dioxides. CO_2 is a gas and the rest all dioxides are crystalline solids with high melting points. Among the dioxides CO_2 and SiO_2 are acidic, GeO_2 distinctly acidic but less acidic than SiO_2. SnO_2 and PbO_2 are amphoteric. On moving down the group acidic character decreases, basic character increases for the dioxides. This is again due to increase in the electropositive character of the elements of Group IVA elements.

All the dioxides dissolve in alkalis giving carbonates, silicates, germanates, stannates and plumbates.

$$MO_2 + 2NaOH \longrightarrow Na_2MO_3 + H_2O$$

$$(M = C, Si, Ge, Sn \text{ or } Pb)$$

The germanates have complicated structures similar to silicates, but stannates and plumbates contain $[Sn(OH)_6]^{2-}$ and $[Pb(OH)_6]^{2-}$. The three oxides GeO_2, SnO_2 and PbO_2 are insoluble in acids except when complexing agents such as F^- or Cl^- present when complex ions such as $[GeF_6]^{2-}$, $[SnCl_6]^{2-}$ and $[PbCl_6]^{2-}$ are formed.

The stabilities of the dioxides decrease from carbon to lead due to increase in the inert pair effect. So the dioxides can act as oxidizing agents. The oxidation power of the dioxides is in the order $CO_2 < SiO_2 < GeO_2 < SnO_2 < PbO_2$. Thus CO_2 can oxidize only the highly electropositive metals like alkali metals and alkaline earth metals, but lead dioxide can oxidize even weak reducing agents like Cl^- etc. eg:

$$PbO_2 + 4HCl \longrightarrow PbCl_2 + 2H_2O + Cl_2$$

Structures: Carbon dioxide is a gas while other dioxides are solids. As discussed in chemical bonding (Chapter 3) CO_2 is linear molecule in which carbon atom is involved in sp hybridization and in double bond with each oxygen. It is a non-polar molecule with dipole moment zero. So between CO_2 molecules which are discrete, there exists only weak van der Waal's attractive forces due to which CO_2 is a gas.

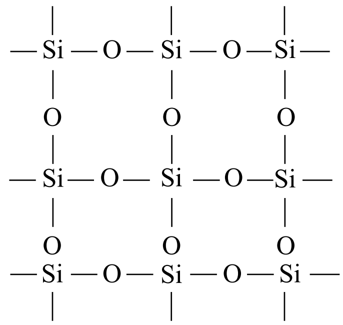


Fig 8.3 Structure of silica (SiO_2)

In analogy with CO_2 it is usually referred to silicon dioxide. This is not correct because the X-ray study of silica reveals that it forms giant molecules by linking together a number of tetrahedra in each of which silicon atom is bonded to four oxygen atoms together by sharing of the oxygen atoms. In SiO_2 every silicon atom is in sp^3 hybridization. The structure of silica is extended three dimensionally but for the sake of convenience it can be represented by a planar formula as given in Fig 8.3.

The dioxides of GeO_2, SnO_2 and PbO_2 normally adopt a TiO_2 structure with 6 : 3 coordination.

Other oxides: Carbon forms a sub-oxide C_3O_2 which is a gas (boiling point 6°C). It can be obtained by dehydration of malonic acid with phosphorus pentoxide at 150°C. It polymerizes at room temperature rapidly. The molecule is linear and its structure can be represented as

$$O = C = C = C = O$$

Lead also forms a mixed oxide Pb_3O_4 which is called red lead and is formed by heating the monoxide in air at 450°C. This is diamagnetic and hence contains only Pb(II) and Pb(IV). It is a mixed oxide of PbO and PbO_2.

8.6 CARBON

Some of the planets are thought to be surrounded by a layer of carbondioxide, and it is probable that all the deposits of carbon compounds in the earth originally came from carbondioxide in the atmosphere. Plants use atmospheric carbondioxide for growth and animals feed on plants. The decomposition of dead animals and vegetation, followed by geological upheavals which subjected the decaying matter to great temperatures and pressures for long periods of time in the absence of oxygen, almost certainly accounts of coal, oil and natural gas. The carbonate minerals were formed by the sedimentation of the shells of microscopic sea animals, followed by compression of the layers and the redistribution of the oceans. Pure carbon has three allotropes of which two diamond and graphite are found naturally.

8.6.1 Diamond

Diamonds occur naturally in igneous rocks, formed by crystallization of molten magma from deep within the earth. This igneous rock is blasted and allowed to weather; the light soluble material is then washed away and the heavy residue is pushed by water under pressure over a bed of grease to which the diamond adheres. Most rocks yield about 0.1g of diamond per tonne.

Diamond was first discovered in India and previously India was regarded as a diamond producing country. The famous diamonds like the **Kohinoor** (186 carats) and **Regent** or **pitt** (136.2 carats) which was studded in Napoleon state sword, were discovered near Krishna river of South India. But at the present time India does not produce many diamonds. However diamonds are mainly formed in South Africa, Belgium, Congo, Brazil, British Guiana etc. Brazil is famous for black diamonds. The largest diamond mined so far is **Cullinan diamond** weighed about 303.2 carats which was obtained from South Africa in January 1905.

In a diamond each carbon atom is in sp^3 hybridization and is linked to four carbon atoms by single bonds. Hence in diamond every carbon is surrounded by four other carbon atoms situated tetrahedrally and is covalently bonded to them. This results in the formation of giant molecule. The C – C bond distance in diamond is 154 pm and the CCC bond angle is 109° 28’. It is regarded as a three dimensional polymeric giant molecule.

Diamond is the hardest material so far known. In diamond each carbon is linked to four other carbon atoms by strong covalent bonds and more energy is required to break the bonds. So the melting point of diamond is very high (3600°C). Because of these three-dimensional strong covalent bonds diamond becomes the hardest material. Due to the close packing of atoms in diamond, it has maximum density 3.52 g cm^{-3} among the allotropes of carbon. Diamonds are transparent to light and X-rays and possess high refractive index 2.45. Because of this high refractive index when light falls on properly cut diamonds, total internal reflection takes place So diamonds have high shining nature, which gives them more value. The value of a diamond is expressed in terms of a mass unit called carat (1 carat = 200 mg).

Diamond is the purest form of carbon. When pure it is colourless but when impurities are present it has bright colours. Since all the valence electrons are participated in bonding there are no free moving electrons. Hence diamond is an insulator. It is chemically inert and is insoluble in any solvents, or acids or alkalis. When burnt in air at about 1000°C it converts into carbon dioxide, but on heating above 1500°C in vacuum it converts into graphite. This indicates diamond is chemically inert but

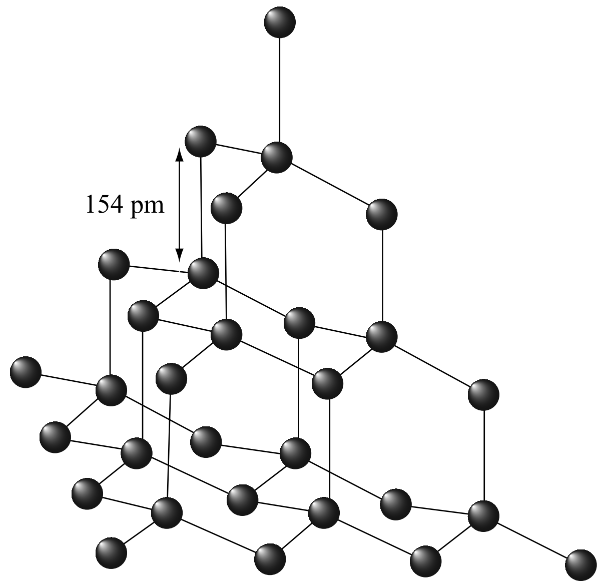


Fig 8.4 Structure of diamond

thermodynamically unstable than graphite. Diamond has the highest known **thermal conductivity** because this structure distributes thermal motion in three dimensions very effectively. Measurement of thermal conductivity is used to identify fake diamonds.

Uses

1. Diamonds are used as precious stones for jewellery of their ability to reflect and refract light.
2. Diamond dies are used for drawing thin wires.
3. Due to their hardness, black diamonds are used for making glass cutting, rock drilling agents and are used as abrasives.

8.6.2 Graphite

Graphite is widely distributed in different parts of the world. Large deposits have been found to occur in Siberia, Sri Lanka, Italy, Bohemia, Bavaria and U.S.A. In India it is found in small quantities at places like Coorg, Rajasthan and Orissa.

Artificial Graphite: Large amounts of artificial graphites are being manufactured at Niagara Falls in America by **Acheson process.** This process involves the heating of a mixture of sand and powdered anthracite (or coke) in an electric furnace by passing electricity through carbon rods embedded in the mass. By passing an alternating current, the heating is continued for 24 - 30 hours. Silicon carbide is first formed which decomposes at the very high temperature into silicon and graphite. The former volatilizes off, leaving behind the pure graphite.

$$SiO_2 + 3C \longrightarrow SiC + 2CO$$

$$SiC \longrightarrow Si + C_{Graphite}$$

Graphite is thermodynamically stable than diamond. Graphite has layered lattice structure. Every carbon in graphite is involved in sp^2 hybridization and is in bond with three nearest neighbours. The σ bond between neighbours with in the sheets are formed from the overlap of sp^2 hybrid orbitals, and the remaining perpendicular p-orbitals overlap to form π -bonds that are delocalized over the plane. Each layer is a hexagonal net of carbon atoms and may be regarded as a fused system of benzene rings. The bond order in graphite is 1.33 whereas in benzene is 1.5. The layers are widely separated from each other at a distance of 335 pm while the distance between the two nearest neighbours is 142 pm. The layers are held together by relatively weak Van der Waal's attractive forces and consequently the region between the planes is called **van der Waal's gap**. There are two different forms of graphite α- and β-forms. In normal or α- graphite the layers are arranged in the sequences ABAB .... with the third layer exactly above the first layer so that with carbon atoms in alternate layers vertically above each other. In β - (orthorhombohedral) graphite the staking sequence is ABC ABC. The two forms are interconvertible. Heating turns β into α, and grinding turns α into β. In both forms the inter layer distances and the C – C bond length with in the sheet are same.

Since the bonding between layers is weak, graphite cleaves easily between the layers which acounts for the remarkable softness of the crystals (> 1 on Moh's scale). Because of this reason the layers in graphite are slippery and can be used as a **solid lubricant** or **graphited oil** called **oil drog**. The wide spacing of layers in graphite also means that the atoms do not pack together to fill the space very

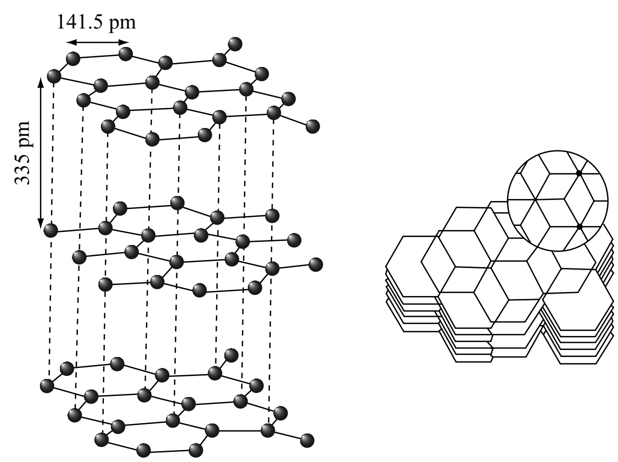


Fig 8.5 Structure of Graphite the rings are in register (one above the other) in alternate planes not adjacent planes

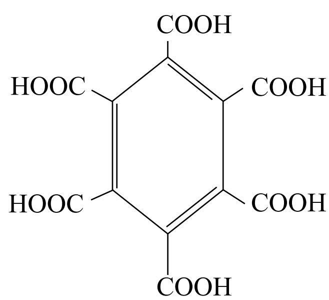
effectively. Thus the density of graphite (2.22gm cm^{-1}) is lower than of diamond (3.51 gm cm^{-1}).

Due to the presence of delocalized π electron in a layer graphite conducts electricity. Conduction can occur in a sheet, but not from one sheet to another under normal conditions. Its electrical conductivity perpendicular to the planes is low (5 ks cm^{-1} at 25°C) and increases with increasing temperature signifying that graphite is a semiconductor in that direction. The electrical conductivity is much higher parallel to the planes (30 ks cm^{-1} at 25°C) but decreases as the temperature is raised, indicating that graphite behaves as a metal, but precisely a semimetal (a semi metal is a substance in which two neighbouring bonds have zero density of states at their edge but zero band gap between them).

Graphite can act as either an electron donor or an electron acceptor towards atoms and ions that penetrate between the layers and give rise to an intercalation compounds. Thus K atoms reduce graphite by donating their valence electron to the empty orbitals of the π bond and the resulting K^+ ions penetrate between the layers so that they may penetrate either between alternate layers. The electrons added by potassium to the band increase the electrical conductivity of graphite.

When graphite is heated with a mixture of sulphuric and nitric acid **graphite bisulphates** are formed by the removal of electrons by π-bond and HSO_4^- ions penetrate between the sheets. In such cases also electrical conductivity increases than the pure graphite similar to p-type silicon by electron accepting dopants. Similar trend appears with involving atoms of Cl_2 or Br_2 also.

Though graphite is thermodynamically more stable than diamond, it is chemically more reactive than diamond. Like diamond, graphite is not attacked by dilute acids. However, chromic acid oxidizes it slowly to CO_2 concentrated nitric acid oxidizes it to graphitic acid $C_{11}H_4O_5$ which is an insoluble yellowish green substance. Alkaline permanganate oxidizes it to oxalic acid and mellitic acid $C_6(COOH)_6$.The structure of the latter compound is shown as below.



The formation of mellitic acid supports the hexagonal structure of graphite.

Uses

(i) When rubbed on paper, it leaves a black mark. So it is called black lead. Mixture of graphite and clay is used to make lead pencils.
(ii) Graphite is used as lubricant at high temperatures.
(iii) It is used in making electrodes, in painting the stoves and furnaces.
(iv) It is also used in making refractory crucibles used at high temperatures.
(v) It is used in electroplating and electrotyping.

8.6.3 Fullerenes

The discovery of the soccer-ball shaped C_{60} cluster in the 1980s created great excitement in the scientific community and in the popular press. Much of this interest undoubtedly stemmed from the fact than carbon is a common element and there had seemed little likelihood that new molecular carbon structures would be found.

When an electric arc is struck between carbon electrodes in an inert atmosphere a large quantity of soot is formed together with significant quantities of C_{60} and much smaller quantities of related **fullerenes** such as C_{70}, C_{76} and C_{84}. Some traces of fullerenes consisting of even number of carbon **atoms** upto 350 or above were also found. Fullerenes are the only pure form of carbon because they have smooth structure without having "dangling" bonds. This allotrope of carbon differs from diamond and graphite (which form lattice). Fullerenes form only discrete molecules. The C_{60} molecules look rather like a soccerball, and sometimes called a **Bucky ball** or **Buckminister fullerene**.

It consists of fused system of five- and six-membered rings. A six membered ring is fused with either six or five membered rings but a five membered ring can only fuse with six membered rings. C_{60} molecule contain twenty 6-membered rings and twelve 5-membered rings. All the carbon atoms are equal and they undergo sp^2 hybridization. Each carbon atom forms three sigma bonds with other three carbon atoms. The remaining electron at each carbon is delocalized in molecular orbitals which in turn give aromatic character to the molecule.

This ball shaped molecule has 60 vertices and each one is occupied by one carbon atom and it also contains both single and double bonds with C – C distances 143.5 pm and 138.3 pm respectively.

The reduced fullerenes with alkali metals like K_3C_{60}, Rb_2CsC_{60} acts as super conductors at about 18k and 33k respectively. Fullerenes are finding many applications in the production of carbon nanotubes. Fullerenes can also form complexes.

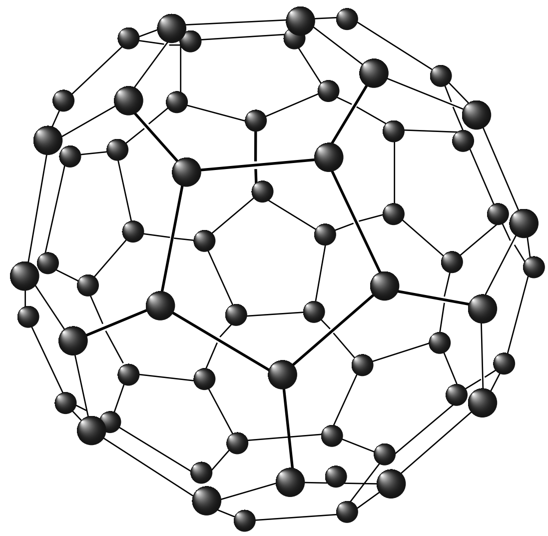


Fig 8.6 The structure of Buckminister fullerene (The molecule has the shape of a soccerball (Foot ball))

8.7 IMPORTANT COMPOUNDS OF CARBON

8.7.1 Oxides of Carbon

Carbon forms mainly two important oxides carbon monoxide and carbon dioxide.

(a) Carbon monoxide

Carbon monoxide, together with carbon dioxide, is formed when carbon or carbonaceous matter is burnt in a deficiency of air or oxygen. It is also produced when carbon dioxide is reduced by red hot carbon.

$$C + CO_2 \rightleftharpoons 2CO$$

It is also a constituent of water gas (a mixture of CO and H_2), producer gas (CO and N_2) and coal gas (CO, H_2, CH_4 and CO_2). All are important fuels.

$$C + H_2O \xrightarrow{\text{red heat}} CO + H_2$$

$$2C + \underbrace{O_2 + 4N_2}_{\text{air}} \longrightarrow 2CO + 4N_2 \text{ (producer gas)}$$

When heavy metal oxides like iron, zinc are heated with carbon, carbon monoxide will be formed

$$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO$$

$$ZnO + C \longrightarrow Zn + CO$$

Laboratory methods: In the laboratory it can be prepared by dehydrating formic acid with concentrated sulphuric acid.

$$HCOOH \xrightarrow[-H_2O]{\text{Conc. } H_2SO_4} CO$$

Dehydration of oxalic acid with concentrated sulphuric acid gives a mixture of CO and CO_2 gases

$$\begin{array}{l}COOH \\ | \\ COOH\end{array} \xrightarrow[-H_2O]{\text{Conc. } H_2SO_4} CO + CO_2$$

The CO_2 from this mixture can be removed by passing through caustic potash.

When potassium ferrocyanide crystals in powdered state is heated with concentrated H_2SO_4, CO is evolved. Dilute H_2SO_4 should never be used because it shall evolve highly poisonous gas HCN.

$$K_4[Fe(CN)_6] + 3H_2SO_4 \longrightarrow 2K_2SO_4 + FeSO_4 + 6HCN$$

$$6HCN + 12H_2O \xrightarrow{\text{dil}} 6HCOOH + 6NH_3$$

$$6NH_3 + 3H_2SO_4 \longrightarrow 3(NH_4)_2SO_4$$

$$6HCOOH \xrightarrow{\text{Conc. } H_2SO_4} 6CO + 6H_2O$$

$$\underset{\text{Conc}}{K_4[Fe(CN)_6] + 6H_2SO_4} \longrightarrow$$

$$2K_2SO_4 + FeSO_4 + 3(NH_4)_2SO_4 + 6CO$$

Properties: Carbon monoxide is a colourless, odourless gas which burns in air with blue flame, forming carbondioxide. It is exceedingly poisonous, combining with the haemoglobin in the blood more readily than oxygen so that normal respiration is impeded very quickly. When it is present as one volume in 800 volumes of air it leads to death. Ordinary gas masks are no protection against the gas, since it is not readily adsorbed by active charcoal. In the presence of air, a mixture of manganese (IV) oxide and copper (II) oxide catalytically oxidizes it to carbon dioxide and this mixed catalyst is used in the breathing apparatus worn by rescue teams in mine disasters.

It is a combustible gas but non-supporter of combustion. It burns with blue flame. It is a neutral oxide and is not decomposed by heat.

$$2CO + O_2 \longrightarrow 2CO_2$$

Carbon monoxide is a powerful reducing agent, being employed industrially in the extraction of iron and nickel.

$$Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$$

$$NiO + CO \longrightarrow Ni + CO_2$$
$$ZnO + CO \longrightarrow Zn + CO_2$$
$$PbO + CO \longrightarrow Pb + CO_2$$

It reduces Fehling solution to red cuprous oxide

$$2Cu(OH)_2 + CO \longrightarrow Cu_2O + 2H_2O + CO_2$$

It reacts with many transition metals forming volatile carbonyls. The formation of nickel carbonyl followed by its decomposition is the basis of the Mond's process for obtaining very pure nickel.

$$Ni + 4CO \xrightarrow{80°C} Ni(CO)_4 \xrightarrow{180°C} Ni + 4CO$$

Since all the valencies of carbon in carbon monoxide are not completely saturated, it forms addition compounds with sulphur it forms carbonyl sulphide and with chlorine in the presence of light to give carbonyl chloride (phosgene). The latter is an exceedingly poisonous gas.

$$CO + S \longrightarrow COS$$

$$CO + Cl_2 \longrightarrow COCl_2$$

Although carbon monoxide is not a true acid anhydride since it does not react with water to produce an acid, it reacts under pressure with fused sodium hydroxide to give sodium formate.

$$NaOH + CO \longrightarrow HCOONa$$

Formic acid is produced from the sodium salt by the addition of dilute hydrochloric acid.

With hydrogen under pressure and in the presence of a zinc oxide / chromium (III) oxide catalyst it react to give methanol; this reaction is of industrial importance.

$$CO + 2H_2 \xrightarrow{ZnO/Cr_2O_3} CH_3OH$$

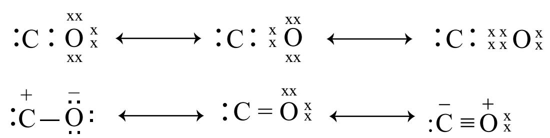
Carbon monoxide is readily absorbed by an ammoniacal solution of copper (I) chloride to give $CuCl \cdot CO \cdot 2H_2O$. It reduces an ammoniacal solution of silver nitrate to silver (black) in the absence of other reducing agents, this serves as a test for the gas. It can also be detected by its ability to reduce an aqueous solution of $PdCl_2$ to metallic Pd.

$$CO + PdCl_2 + H_2O \longrightarrow Pd + CO_2 + 2HCl$$

It can be estimated by reaction with iodine pentoxide, the iodine which is produced quantitatively being titrated with standard sodium thiosulphate solution

$$5CO + I_2O_5 \longrightarrow I_2 + 5CO_2$$

Structure: Carbon monoxide is considered to have a structure involving the three resonance structures.



The carbon atom of carbon monoxide forms sp hybrid orbitals, one of which contains lone pair of electrons and the other contains one electron which forms a σ bond with the p-orbital of oxygen. Out of the remaining two p-orbitals of carbon which donot participate in hybridization, the one which has a lone electron will participate in $p\pi - p\pi$ bond. With the vacant p-orbital of carbon the lone pair of electrons from one of the p-orbital of oxygen will form a dative bond.

The presence of triple bond between carbon and oxygen in CO is evident by the following facts.

(i) The bond length between carbon and oxygen (113 pm) corresponds to that of carbon-oxygen triple bond.

(ii) Carbon monoxide has only a slight dipolemoment inspite of large difference in the electronegativities of carbon and oxygen. The shift of electron pair from oxygen to carbon, has compensated the difference in the electronegativitiy values.

Because of this structure there develops some negative charge on carbon. So by donating the lone pair of electrons carbon always tries to lessen the negative charge. Hence carbon monoxide acts as a Lewis base (ligand) and can form coordinate bond with metals ($M \longleftarrow C \overset{\longleftarrow}{\equiv} O$) forming metal carbonyls.

Uses: It is used

(i) as a fuel in the form of water gas, producer gas etc.
(ii) in the manufacture of methanol, synthetic petrol.
(iii) in the extraction of iron as reducing agent.
(iv) in the refining of nickel by Mond's process.
(v) in the production of poisonous war gas like phosgene.

(b) Carbon dioxide

Atmospheric air contain by about 0.03 – 0.05 percent. It is liberated into the air due to the combustion of carbonaceous fossil fuels, fermentation of dead plants and animals and due to respiration of plants and animals. Plants absorb carbon dioxide during photosynthesis. The percentage of carbondioxide in the atmosphere remains constant by the operation of carbon cycle.

Preparation: In the laboratory it can be prepared by heating bicarbonates of all metals and carbonates of heavy metals except alkali metal carbonates

$$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2$$

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

It can also be prepared in the laboratory by the action of dilute acids on metal bicarbonates or carbonates.

$$NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$$

$$NaCO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$$

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$$

Generally in the laboratory carbondioxide is prepared by the action of dilute hydrochloric acid on marble chips using kipps apparatus. In this method dilute sulphuric acid should not be used because the insoluble calcium sulphate formed, forms as a layer on the surface of the marble chips which prevents the further reaction.

On large scale carbon dioxide is prepared by complete combustion of carbon in excess of air.

$$C + O_2 \longrightarrow CO_2$$

If any CO is formed it is also converted into CO_2 by burning.

$$2CO + O_2 \longrightarrow 2CO_2$$

On large scale carbon dioxide is obtained as a byproduct during the calcination of lime stone to produce burnt lime.

$$CaCO_3 \longrightarrow CaO + CO_2$$

During the fermentation of sugar present in the molasses into alcohol, a large amount of CO_2 is obtained as a by-product.

$$C_{12}H_{22}O_{11} + H_2O \longrightarrow \underset{\text{glucose \& fructose}}{2C_6H_{12}O_6} \longrightarrow 2C_2H_5OH + 2CO_2$$

Properties: It is a colourless, odourless gas with pungent taste. It is slightly soluble in water and the solubility of CO_2 in water increase with increase in pressure and decrease in temperature (Henry's law). It is heavier than air. It can be easily liquified by applying pressure. If the liquid carbon dioxide is allowed to rapid evaporation through a nozzle, it converts into white solid called **dry ice**. It is so called as it sublimes producing cooling effect. Solid CO_2 is used as a refrigerant under the name **drikold**. It provides cold as well as the inert atmosphere which helps in killing the undesirable bacteria.

It is neither combustible nor supporter of combustion. However metals like sodium, potassium, magnesium burns in CO_2.

$$3CO_2 + 4Na \longrightarrow 2Na_2CO_3 + C$$

$$CO_2 + 2Mg \longrightarrow 2MgO + C$$

So metal fires cannot be extinguised by CO_2. For extinguishing metal fires generally pyrene (CCl_4) is used because it easily vapourizes and because it is heavier than air it covers the fires so that the supply of oxygen will be cut off.

When carbondioxide is passed through a solution of carbonate, a bicarbonate is obtained.

$$Na_2CO_3 + H_2O + CO_2 \longrightarrow 2NaHCO_3$$

Carbon dioxide gives white precipitate with lime water, but the precipitate is soluble in excess of CO_2, when it forms soluble calcium bicarbonate.

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$$

$$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$$

When carbondioxide is passed through any base first carbonate is formed which converts into bicarbonate on passing excess of CO_2.

$$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$$

$$Na_2CO_3 + H_2O + CO_2 \longrightarrow 2NaHCO_3$$

Plants absorb CO_2 from air, water from soil and convert them into carbohydrates in the presence of sunlight. This process is called **photosynthesis**.

$$6CO_2 + 12H_2O \xrightarrow[\text{chlorophyll}]{\text{light}} C_6H_{12}O_6 + 6O_2$$

By this process plants make food for themselves as well as for animals and human beings.

Structure: The structure of CO_2 was already discussed in Chapter-3 (Chemical bonding 3.8.2). It is a linear molecule as shown below.

$$O = C = O$$

The structure of carbon dioxide is a point of discussion. The measured C–O bond length in CO_2 is 115 pm. The normal C = O and C ≡ O bond length are 121 pm and 110 pm respectively. The C–O bond (115 pm) in CO_2 is in between C = O and C ≡ O bond lengths. To explain this the following resonance was proposed.

$$:\ddot{O} = C = \ddot{O}: \leftrightarrow :\overset{-}{\underset{..}{\ddot{O}}} - C \equiv \overset{+}{O}: \leftrightarrow :\overset{+}{O} \equiv C - \overset{-}{\underset{..}{\ddot{O}}}:$$

I II III

It was proposed that due to resonance hybridization of the above canonical forms the bond length decreases to 115 pm. But this appears too far from the fact because the bond order in the resonance hybrid structure also remain 2 and is not changing from the normal structure O = C = O. Here the C – O bond length (121 pm) is calculated by summing up the covalent radii of double bonded carbon (=C) as 67 pm and double bonded oxygen (=O) as 55 pm equal to 122 pm. This may be true if carbon is in double bond on one side, but the other two are single bonds i.e., carbon is involved in 3σ bonds and 1π bond (sp^2 carbon). But in CO_2 carbon atom is double bonded on both sides involving only 2σ and 2π bonds i.e., sp carbon. So if the size of triple bonded carbon (≡ C– participating in 2σ and 2π bonds) 60 pm and the size of double bonded oxygen (=O) 55 pm are considered, then the sum of these values become equal to C–O bond length 115 pm. Recently it was also proposed that in CO_2 the oxygen atoms are also involved in sp^2 hybridization in addition to the sp hybrid carbon and hence the bond length is 116.3 pm.

Uses

(i) It is used in the manufacture of urea and in the preparation of aerated water and soft drinks.
(ii) Dry ice is used as a refrigerant for ice cream and frozen food.
(iii) Being heavy and non-supporter of combustion it is used as fire extinguisher.
(iv) It is used in the manufacture of white lead and sodium carbonate.
(v) Carbogen is a mixture of O_2 and CO_2 (5–10%). It is used for artificial respiration in the case of pneumonia patients and victims of CO poisoning.
(vi) It is also used to maintain inert atmosphere for certain chemical process.

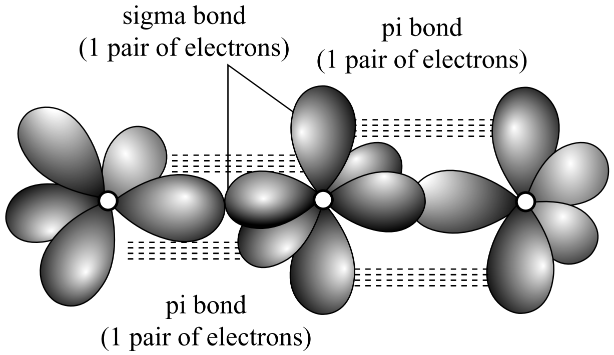


Fig 8.7 Structure of carbon dioxide

(c) Carbon suboxide

This is an evil-smelling gas and can be made by dehydrating propanedioic(malonic) acid, with phosphorus pentoxide.

$$3CH_2(COOH)_2 + P_4O_{10} \longrightarrow 3C_3O_2 + 4H_3PO_4$$

When heated to about 200°C, it decomposes into carbondioxide and carbon.

$$C_3O_2 \xrightarrow{\Delta} CO_2 + C$$

The molecule is thought to have a linear structure.

$$O = C = C = C = O$$

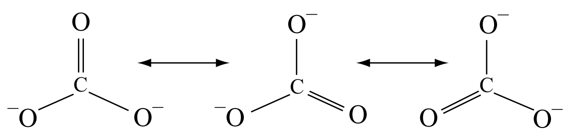
8.7.2 Carbonic Acid

When carbon dioxide dissolves in water the greater part of it is only loosely hydrated. This hydrated species is in equilibrium with carbonic acid, hydrogen ions, and hydrogen carbonate and carbonate anions.

$$CO_2 + aq \rightleftharpoons CO_{2(aq)} \rightleftharpoons H_2CO_3$$
$$\rightleftharpoons H^+ + HCO_3^- \rightleftharpoons 2H^+ + CO_3^{2-}$$

Although pure carbonic acid cannot be isolated, solid carbonates are plentiful and Group IA metals form solid hydrogen carbonates.

Carbonates contain discrete planar ions in which the three carbon oxygen bond lengths are identical, the ion is thus a resonance hybrid of containing equal contributions from the three forms.



Silicon does not form an analogous discrete silicate anion, SiO_3^{2-} because of the reluctance of the larger silicon atom to form a multiple bond with oxygen. Silicates contain polymeric silicate anions (to be discussed later) some of which contain the anion which approximates to the empirical formula SiO_3^{2-}.

Carbonates and hydrogen carbonates

Metallic carbonates are insoluble in water, except those of the Group IA metals. These last-named soluble carbonates can be obtained by saturating a solution of the alkali with carbon dioxide and then adding a second identical volume of the alkali.

$$OH^- + CO_2 \longrightarrow HCO_3^-$$

$$HCO_3^- + OH^- \longrightarrow CO_3^{2-} + H_2O$$

When a solution of an alkali metal carbonate is added to an aqueous solution of a salt it is possible to precipitate a carbonate, a basic carbonate or a hydroxide. The product obtained depends upon the size and charge of the metallic cation.

Precipitation of hydroxide: The addition of soluble carbonate to solutions containing aluminium, chromium (III) or iron (III) ions results in the precipitation of the hydroxide. Since aluminium, chromium (III) and iron (III) ions are highly charged and of small size they are strongly hydrated in solution and show an acid reaction. e.g.,

$$\left[Al(H_2O)_6\right]^{3+} + H_2O \rightleftharpoons \left[Al(H_2O)_5(OH)\right]^{2+} + H_3O^+$$

$$\left[Al(H_2O)_5(OH)\right]^{2+} + H_2O \rightleftharpoons \left[Al(H_2O)_4(OH)_2\right]^{+} + H_3O^+$$

The addition of soluble carbonate (the addition of carbonate anions which are strong bases) results in further ionization with the precipitation of hydrated aluminium hydroxide.

$$\left[Al(H_2O)_4(OH)_2\right]^{+} + CO_3^{2-} \rightarrow Al(H_2O)_3(OH)_3 + HCO_3^-$$

Precipitation of a basic carbonate: The addition of the soluble carbonate to solutions containing copper (II) or zinc ions results in the precipitation of the basic carbonate i.e., a mixture of the carbonate and the hydroxide. The copper (II) and zinc ions are larger and carry one unit of charge less than the cations discussed above, their solutions are thus less acidic and the ionization process is more difficult. It is therefore understandable why both carbonate and hydroxide are precipitated.

$$CuSO_4 + 2H_2O \longrightarrow Cu(OH)_2 + H_2SO_4$$
$$CuSO_4 + Na_2CO_3 \longrightarrow CuCO_3 + Na_2SO_4$$
$$H_2SO_4 + Na_2CO_3 \longrightarrow Na_2SO_4 + H_2O + CO_2$$

$$2CuSO_4 + 2Na_2CO_3 + H_2O \longrightarrow CuCO_3 \cdot Cu(OH)_2 + 2Na_2SO_4 + CO_2$$

$$MgSO_4 + 2H_2O \longrightarrow Mg(OH)_2 + H_2SO_4$$
$$MgSO_4 + Na_2CO_3 \longrightarrow MgCO_3 + Na_2SO_4$$
$$H_2SO_4 + Na_2CO_3 \longrightarrow Na_2SO_4 + H_2O + CO_2$$

$$2MgSO_4 + 2Na_2CO_3 + H_2O \longrightarrow Mg.CO_3 \cdot Mg(OH)_2 + 2Na_2SO_4 + CO_2$$

$$3ZnSO_4 + 6H_2O \longrightarrow 3Zn(OH)_2 + 3H_2SO_4$$
$$2ZnSO_4 + 2Na_2CO_3 \longrightarrow 2ZnSO_4 + 2Na_2SO_4$$
$$3H_2SO_4 + 3Na_2CO_3 \longrightarrow 3Na_2SO_4 + 3H_2O + 3CO_2$$

$$5ZnSO_4 + 5Na_2CO_3 + 4H_2O \longrightarrow [2ZnCO_3 \cdot 3Zn(OH)_2] H_2O + 5Na_2SO_4 + 3CO_2$$

Precipitation of Carbonate

The larger cations e.g., those of the Group IIA metals show little tendency to hydrate in solution and consequently are neutral or only very slightly acidic. The normal carbonate is precipitated in these cases.

$$Ba^{2+} + CO_3^{2-} \longrightarrow BaCO_3$$

Metallic ions that give rise to basic carbonates when they interact with carbonate ions generally are made to produce the normal carbonate on treatment with a soluble hydrogen carbonate (the bicarbonate ion is a weaker base than the carbonate ion because it carries one unit of negative charge less).

$$Cu^{2+} + 2HCO_3^- \longrightarrow CuCO_3 + H_2O + CO_2$$

Only the bicarbonates of the alkali metals can be isolated as solids and these decompose to give carbonate at about 70°C.

Soluble carbonate and bicarbonate can be estimated volumetrically by titration with standard hydrochloric acid, using methyl orange as indicator.

$$CO_3^{2-} + 2H^+ \longrightarrow H_2O + CO_2$$

$$HCO_3^- + H^+ \longrightarrow H_2O + CO_2$$

An M/20 solution of sodium carbonate has a pH of 11.5 and an M/10 solution of sodium hydrogen carbonate has a pH of 8.4, thus methyl orange (pH range 2.9–4.0) reacts with alkali (yellow pH > 4) but turns red in the presence of one drop of hydrochloric acid at the end point. When M/20 solution of sodium carbonate is titrated with acid as far as the bicarbonate stage the pH change will be from 11.5 to 8.4; phenolphthalein changes colour over this region, and thus a mixture of carbonate and bicarbonate can be estimated by using both indicators.

8.7.3 Carbides

The term carbide is generally applied to compounds in which carbon is bonded to elements of lower or approximately the same electronegativity. The definition excludes compounds in which oxygen, sulphur, phosphorus, nitrogen and the halogens are united with carbon. It is also usual to exclude the hydrides of carbon.

Carbides are conveniently classified in terms of the bonding present into salt-like (ionic), interstitial and covalent carbides.

(a) Ionic or salt-like carbides

The reactive metals i.e., those of Group IA and IIA form carbides which are colourless and non-conductors of electricity. These are prepared by heating the metals or their oxides or hydrides with carbon or with a source of carbon such as CO,CO_2, an aliphatic or an aromatic hydrocarbon. They are decomposed by water or acids with the formation of aliphatic hydrocarbons.

(i) Acetylides: These yield acetylene on hydrolysis and are believed to contain $(C \equiv C)^{2-}$ ion so termed as acetylides.

$$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + C_2H_2$$

Many of these carbides have a structure in someways resembling that of Na*Cl*, one cell axis being elongated to accomodate the $(C \equiv C)^{2-}$ ions. The coordination number of each ion is six, the same as in the structure of Na*Cl*, but the axis along with the acetylide ion lie is elongated.

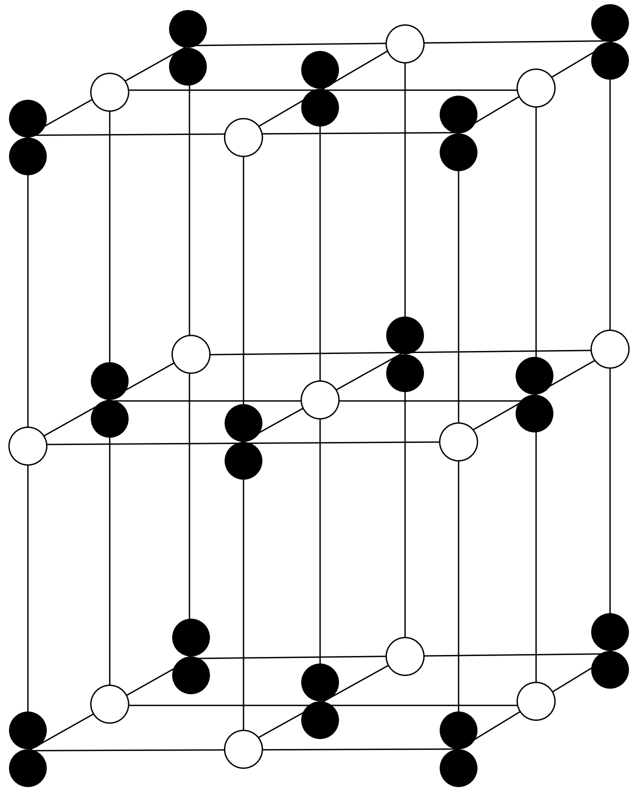


Fig 8.8 Structure of calcium carbide

(ii) Methanides: Beryllium and aluminium carbides give methane on hydrolysis with water, so termed as methanides.

$$Be_2C + 4H_2O \longrightarrow 2Be(OH)_2 + CH_4$$
$$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$$

These are considered as transition from true ionic to essential covalent carbides. The carbides of the lanthanides are considered to be ionic.

(iii) Allylides: Only example of this is the magnesium carbide which contain C_3^{4-} ion and yield propyne on hydrolysis.

$$\left(Mg^{2+}\right)_2 \left(C-C\equiv C\right)^{4-} + 4H_2O \longrightarrow 2Mg\left(OH\right)_2 + C_3H_4$$

(b) Covalent carbides: Compounds of carbon with elements of similar electronegativity are covalent in nature. Typical examples are silicon carbide (SiC) and boron carbide. Technically silicon carbide is called **Carborundum**. It is manufactured by heating a mixture of sand 54%, coke 34%, saw-dust 10% and salt 2% in an electric furnace.

$$SiO_2 + 3C \longrightarrow SiC + 2CO$$

Salt acts as flux and saw dust makes the mass porous. Chemically it is extremely inert. It resists the action of almost all the reagents and even hydrofluoric acid. However when it is fused with alkalis in the presence of air it forms soluble silicates and carbonates.

$$SiC + 4NaOH + 2O_2 \rightarrow Na_2SiO_3 + Na_2CO_3 + 2H_2O$$

When chlorine is passed over silicon carbide SiC at 875K silicon tetrachloride and carbon are formed.

$$SiC + 2Cl_2 \rightarrow SiCl_4 + C$$

The structure of silicon carbide is similar to that of diamond in which the silicon and carbon atoms each tetrahedrally surrounded by four of the other kind.

$$
\begin{array}{ccccccccccc}
 & | & & | & & | & & | & & | & \\
- & Si & - & C & - & Si & - & C & - & Si & - \\
 & | & & | & & | & & | & & | & \\
- & C & - & Si & - & C & - & Si & - & C & - \\
 & | & & | & & | & & | & & | & \\
- & Si & - & C & - & Si & - & C & - & Si & - \\
 & | & & | & & | & & | & & | & \\
- & C & - & Si & - & C & - & Si & - & C & - \\
 & | & & | & & | & & | & & | &
\end{array}
$$

Silcon carbide is as hard as diamond due to this structure. It is used as abrasive.

Boron carbide is also chemically inactive and have high melting point. It can be prepared by the reaction of B_2O_3 with carbon in an electric furnace.

$$2B_2O_3 + 7C \longrightarrow B_4C + 6CO$$

Its structure is immensely complicated. It is also very hard and is used as abrasive in making refractory crucibles and the shields which protect from radioactivity.

(c) Interstitial or metallic carbides

These are formed by many transition metals and are similar to the interstitial hydrides i.e., they are non-stoichiometric. Carbon atoms fit into the spaces between metal atoms in the lattice without too much distortion thus these carbides have very high melting points and are conductors of electricity. It has been shown that metals with an atomic radius of less than 130 pm cannot take up carbon atoms without causing distortion of the metal lattice. In these cases carbides with properties intermediate between ionic and interstitial are formed e.g., Fe_3C, Mn_3C and Ni_3C.

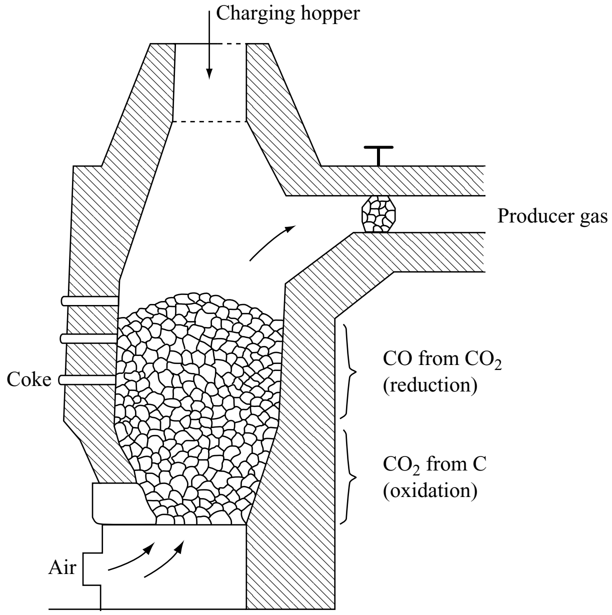


Fig 8.9 Gas producer

8.7.4 Gaseous fuels of Carbon

Any substance that liberates energy when burnt in oxygen is called fuel. A fuel should have the following characteristics to be a good fuel.

(i) It should have high calorific value.
(ii) It should leave very small amount of ash.
(iii) It should not liberate any irritating and offensive gases.
(iv) It should be available cheaper.
(v) It should have low ignition temperature.
(vi) It can be transported easily.

Generally fuels are three types (i) Solid fuels eg.coke, coal, wood, cow dung etc. (ii) Liquid fuels e.g., kerosene, petrol, diesel, alcohol, ether, benzene etc. (iii) Gaseous fuels e.g., coal gas, producer gas, water gas etc.

Gaseous fuels are more advantageous over the other because they have all the above characteristics. Also the gaseous fuels can be transported easily through pipes.

(a) Producer gas

Producer gas is essentially a mixture of carbon monoxide and nitrogen. It contains 33% CO and 64% N_2 and 3% of CO_2 and H_2. It is prepared in a furnace called gas producer. Hence it is called producer gas. Gas producer is shown in Fig 8.9.

Gas producer is a cylindrical furnace lined inside with fire bricks. While coke is introduced into this furnace through a hopper at the top, a blast of air is blown with the help of a pipe from the bottom of the producer. There is a provision for the removal of ash formed during the formation of producer gas in the producer. Following reactions takes place in the producer.

$$C + O_2 \longrightarrow CO_2\uparrow \qquad \Delta H = -394 \text{ KJ mol}^{-1}$$
$$CO_2 + C \longrightarrow 2CO\uparrow \qquad \Delta H = +163 \text{ KJ mol}^{-1}$$
$$2C + O_2 \longrightarrow 2CO\uparrow \qquad \Delta H = +231 \text{ KJ mol}^{-1}$$

Carbon burns at the bottom of the producer and forms carbondioxide. This carbon dioxide is reduced to carbon monoxide as it passes through the bed of red hot coke. The mixture of CO and N_2 thus formed is called producer gas. Producer gas is collected from the exit and along with it large amount of heat is also liberated. This should be used immediately. Hence producer gas is prepared and used on the spot while hot. Calorific value of producer gas is 5439.2 KJ mol^{-1}. Its calorific value is less because of the presence of large proportions of non-combustible nitrogen gas.

Uses

(i) Producer gas is used in gas engines.
(ii) It is used as an industrial fuel in the manufacture of steel and glass.
(iii) It is used in the manufacture of ammonia.

(b) Water gas

It is a mixture of carbon monoxide (40–50%) and hydrogen 45-50%.

The preparation of water gas is similar to that of producer gas. In the preparation of producer gas, coke is burnt in hot air whereas in the preparation of water gas, super heated steam is passed over white hot coke and the product obtained is a mixture of carbon monoxide and hydrogen.

$$C + H_2O \longrightarrow CO + H_2 \quad \Delta H = +121.22 \text{ KJ mol}^{-1}$$

This reaction is endothermic and the temperature of the coke comes down which favours a different reaction.

$$C + 2H_2O \longrightarrow CO_2 + 2H_2$$

To avoid the above reaction alternatively hot air and steam are introduced into the generator. When hot air is passed over the coke, CO_2 and N_2 are formed in this reaction and it is allowed to escape out.

$$C + O_2 \longrightarrow CO_2; \quad \Delta H = -394 \text{ KJ mol}^{-1}$$

The reaction being exothermic, the temperature increases. When the supply of air is stopped, steam is passed over red hot coke and now water gas is produced.

The generator used for the production of water gas is shown in Fig 8.10. The generator is filled with coke. It is provided with a coke inlet (hopper at the top) for introducing coke and two inlets at the bottom, one for steam and another for sending hot air into the generator. The upper part of the generator has an exit for collecting the gas. Alternatively hot air and steam are blown into the generator.

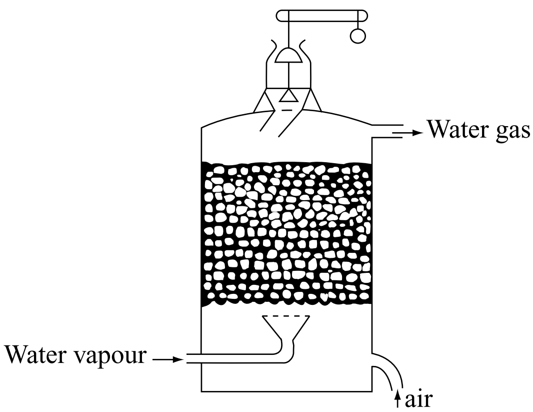


Fig 8.10 Water gas generator

The components of water gas burn with blue flame and hence is called blue gas. Because it is used for the synthesis of several compounds like methanol etc., it is also called as **synthesis gas** or **syn gas**. Since both CO and H_2 present in water gas are combustible, its calorific value is very high 13000 kJ m^3.

Uses

1. It is used as a fuel of high calorific value.

$$2CO + O_2 \longrightarrow 2CO_2; \quad \Delta H = -569 \text{ KJ mol}^{-1}$$

$$H_2 + \frac{1}{2} O_2 \longrightarrow H_2O; \quad \Delta H = -285 \text{ KJ mol}^{-1}$$

2. Large quantities of water gas is used for the manufacture of methyl alcohol in which a mixture of water gas and H_2 when passed over heated zinc oxide and chromium (III) oxide under pressure of 200 – 300 atmospheres produces methyl alcohol.

$$CO + 2H_2 \xrightarrow[Cr_2O_3]{ZnO+} CH_3OH$$

3. It is used in the manufacture of ammonia by Haber's process.

(c) Semiwater gas

It is a mixture of CO, H_2 and N_2. It is obtained by passing a mixture of air and steam in appropriate proportions over red hot coke. The composition of semiwater gas is 10–12% H_2, 25–28%, CO, 50–55% N_2 and 1–2% methane. Calorific value of semi water gas is 7524 KJm^{-3}.

Uses: Used as a fuel in steel industry and for generating power in internal combustion engines.

(d) Carburetted water gas

It is a mixture of water gas and hydrocarbons. The approximate composition of carburetted water gas is 30–40% H_2, saturated hydrocarbons 15–20%, unsaturated hydrocarbons 10–15%; CO 20 – 80%; CO_2 0.2% and nitrogen 2.5–5%.

(e) Coal gas

It is obtained by the destructive distillation of coal. The composition of coal gas is 40–50% H_2; 30–35% CH_4; 4% C_2H_2; 5–10% CO and 8% N_2.

Uses

(i) It is used in domestic and industrial operations.

(ii) It is used in providing inert atmosphere in certain chemical process.

(iii) It is used in illuminating, smelting of metals and alloys and as a reducing agent.

(f) Natural gas

It is obtained while digging for petroleum. The gas consists of chiefly methane. The approximate composition of natural gas is 85% CH_4; 9% C_2H_6; 3% C_3H_8 1% C_4H_{10} and 2% N_2. The composition may vary from place to place.

Uses

(i) Mainly used as fuel.

(ii) It is used for the production of hydrogen, carbon black and various petrochemicals.

(g) Liquefied Petroleum Gas (LPG)

It is generally isolated either from natural gas or from the cracking units of petroleum refineries. It mainly contains C_3 and C_4 hydrocarbons of the alkane, alkene series. It is liquified under high pressure. The main constituents are n-butane, isobutane, butene and propane.

Its calorific value is 55 KJ gm^{-1} or 29780 Kcal m^{-3}

Uses

(i) It is used as a domestic fuel, motor fuel, laboratory gas and a source of petrochemicals.

It is a colourless and odourless gas. So for finding the leakage and to avoid accidents it is mixed with little ethyl mercaptan or methyl mercaptan.

8.8 SILICON

Silicon is the second most abundant element occurring in the earth's crust (about 28% by weight) as the oxide, silica, in a variety of forms e.g., sand, quartz and flint, and as silicates in rocks and clays. It never occurs in free state.

Preparation

(i) 96–99% pure silicon is prepared by the reduction of quartzite or sand with high pure coke in an electric furnace. There must be excess of SiO_2 to prevent the formation of the carbide SiC.

$$SiO_2 + 2C \longrightarrow Si + 2CO$$

$$2SiC + SiO_2 \longrightarrow 3Si + 2CO$$

(ii) High pure silicon is made from $SiCl_4$ obtained by the chlorination of scrap Si or from $SiHCl_3$ (a by - product in silicone industry). These are purified and reduced with exceedingly pure Zn or Mg.

$$SiCl_4 + 2Mg \longrightarrow Si + 2MgCl_2$$

$$SiCl_4 + 2Zn \longrightarrow Si + 2ZnCl_2$$

The resulting spongy Si is melted and purified by zone refining method.

(iii) It is also prepared by the thermal decomposition of SiI_4 or SiH_4.

$$SiI_4 \longrightarrow Si + 2I_2$$

$$SiH_4 \longrightarrow Si + 2H_2$$

(iv) High pure silicon used in solar cells is made from Na_2SiF_6 which is a waste product from the phosphate fertilizer industry. It is reduced by metallic sodium. This reaction is highly exothermic and is self sustaining without the need of external fuel.

$$Na_2SiF_6 + 4Na \longrightarrow Si + 6NaF$$

Silicon exists in two allotropic modifications; the **amorphous silicon** and the **crystalline** and **adamantine silicon**. In the crystalline form silicon crystallizes in the diamond lattice with Si - Si bond distance of 235.17 pm. The so-called amorphous form consists of minute crystals of this form. Amorphous silicon is brownish powder having density 2.35 gm cm^{-3}. Crystalline silicon possesses a metallic lustre. It is very hard, scratches glass and has a density of 2.45 g cm^{-3}. It melts at 1693K and volatilizes at the temperature of electric arc. Chemically both amorphous and crystalline forms behave similarly which is already discussed.

8.9 COMPOUNDS OF SILICON

8.9.1 Silicon Dioxide (SiO_2)

It is simply called silica. Silica is widely distributed in nature and it is the common constituent of earth's crust. It exhibits allotropy. It is found in nature in different forms like crystalline and amorphous forms. Crystalline allotropes of silica are quartz, tridimite and cristobalite. Quartz is colourless when pure but the presence of impurities impart colour to the crystal. Coloured crystals are used as gems.

Pure silica is quartz. Quartz is known as rock crystal. It contains hexagonal prisms and ends in hexagonal pyramids. Quartz, tridimite and cristobalite, the three crystalline polymorphs have a low temperature α-form and a high temperature β-modification.

$$\begin{array}{ccc}
\beta\text{ - Quartz} & \xrightleftharpoons{846K} & \alpha\text{ - Quartz} \\
\text{Slow} \updownarrow 1140K & & \\
\beta\text{ - tridimite} & \xrightleftharpoons{393\text{–}533K} & \alpha\text{ - tridimite} \\
\text{Slow} \updownarrow 1743K & & \\
\beta\text{ - cristobalite} & \xrightleftharpoons[\text{fast}]{473\text{–}553K} & \alpha\text{ - cristobalite} \\
\text{Sluggish tends to vitrify} & 1986K & \\
\text{Liq.}SiO_2 & &
\end{array}$$

Sand is the crushed form of quartz. Sand stone is sand particles with iron oxide bound to them. Flint consists of amorphous silica associated with quartz. Kiesulghur is a siliceous rock composed of the remains (Kiesulghur) of minute germs.

Amorphous forms of silica are agate, onyx and jaspar. Silica is found in plants and animals.

Preparation

(i) It may be prepared by burning silicon in air or oxygen.

$$Si + O_2 \longrightarrow SiO_2$$

(ii) It may also be obtained by hydrolysis of silicon tetrachloride or silicon tetrafluoride.

$$SiCl_4 + 4H_2O \longrightarrow H_4SiO_4 + 4HCl$$

$$6SiF_4 + 4H_2O \longrightarrow H_4SiO_4 + 2H_2SiF_6$$

Orthosilicic acid obtained by the above reactions loses water at 1273K yielding pure silica.

$$H_4SiO_4 \longrightarrow SiO_2 + 2H_2O$$

(iii) Amorphous silica is also obtained by heating the gelatinous silicic acid to redness in a platinum dish. However the silicic acid for the reaction is obtained by hydrolysing sodium silicate with hydrochloric acid.

$$H_2SiO_3 \longrightarrow SiO_2 + H_2O$$

$$Na_2SiO_3 + 2HCl \longrightarrow H_2SiO_3 + 2NaCl$$

Properties

(i) It is insoluble in water. However it dissolves to a small extent when heated under pressure.

(ii) It is insoluble in all acids except hydrofluoric acid with which it reacts to form silicon tetrafluoride.

$$SiO_2 + 4HF \longrightarrow SiF_4 + 2H_2O$$

(iii) It dissolves in hot alkalis due to the formation of silicates.

$$SiO_2 + 2NaOH \longrightarrow Na_2SiO_3 + H_2O$$

(iv) At high temperatures, it reacts with metal oxides and give silicates.

$$SiO_2 + Na_2O \longrightarrow Na_2SiO_3$$
$$SiO_2 + CaO \longrightarrow CaSiO_3$$

(v) At high temperatures silica reacts with metal carbonates to form silicates and liberates carbon dioxide.

$$SiO_2 + Na_2CO_3 \longrightarrow Na_2SiO_3 + CO_2$$
$$SiO_2 + CaCO_3 \longrightarrow CaSiO_3 + CO_2$$

(vi) When heated with salts of volatile acids to a high temperature it, being a non-volatile substance, displace volatile acids from their salts.

$$Na_2SO_4 + SiO_2 \longrightarrow Na_2SiO_3 + SO_3$$
$$Ca_3(PO_4)_2 + 3SiO_2 \longrightarrow 3CaSiO_3 + P_2O_5$$

The second reaction is used in the extraction of phosphorus. These reactions reveal the acidic nature of silica.

(vii) When silica is heated with carbon in an electric furnace, carborundum or silicon graphite is formed

$$SiO_2 + 3C \longrightarrow SiC + 2CO$$

(viii) When heated to 1600°C silica melts to form quartz glass. Red hot quartz glass when plunged into water does not break. Hence it is used in making optical instruments.

Silica Gel: It is prepared by hydrolysing silicon tetrachloride or tetrafluoride or by mixing solutions of sodium silicate and ammonium sulphate or chloride or hydrochloric acid.

$$Na_2SiO_3 + 2HCl \longrightarrow 2NaCl + H_2O + SiO_2$$
$$(NH_4)SO_4 + Na_2SiO_3 \longrightarrow Na_2SO_4 + 2NH_3 + SiO_2 + H_2O$$

In either case, silica gel is obtained which is filtered and washed to remove the soluble impurities. It is dried by heating at 383K in an air oven.

Silica gel is of bluish white colour. Its composition is variable $SiO_2 \cdot xH_2O$.

When the gelatinous precipitate of silicic acid is heated it loses water and gets converted into dry silica gel which has the excellent capacity to take up a large amount of water.

Uses

(i) Silica is used as a building material.

(ii) Transparent quartz glass is used for making special glassware suitable for working with U.V. radiation.

(iii) Bricks made from a mixture of powdered sand, clay and lime stone are used for lining the furnace used in the manufacture of steel.

(iv) Coloured varieties of quartz are used as gems while the transparent varieties of quartz are used for the manufacture of lenses, optical instruments etc.

(v) Silica is used as an acid flux in the metallurgy.

(vi) Silica gel is used as an excellent drying agent in laboratory as well as in industry.

(vii) Since the dry silica gel adsorbs sulphur compounds, it is used for the removal of sulphur compounds from petroleum.

(viii) When silica gel is fused it converts into silica glass which has low thermal expansion coefficient and can withstand sudden changes in temperature. So it is used to make crucibles, dishes etc.

Silica Garden

When crystals of coloured salt e.g., salts of copper, cobalt and nickel are added to an undisturbed solution of sodium silicate, plant-like coloured shoots of insoluble silicates grow and rise up in the beaker. This plant-like coloured growth is called **silica garden**.

$$Na_2SiO_3 + CuSO_4 \longrightarrow CuSiO_3 + Na_2SO_4$$
$$Na_2SiO_3 + NiCl_2 \longrightarrow NiSiO_3 + 2NaCl$$

8.9.2 Silicones

Well over 1,00,000 organosilicon compounds have been synthesized. Of these during the past few decades, silicone oils, elastomers and resins have become major industrial products.

The silicones are polymeric organosilicon compounds containing oxygen-silicon linkages. The great thermal and chemical stability of the silicones derives from the strength of both of the Si – C and Si – O – Si linkages.

Preparation:

Preparation of silicones involves in two stages.

1. Preparation of organosilicon halides.
2. Hydrolysis and condensation of organosilicon halides.

1. Preparation of organo silicon halides:

(i) In the laboratory on small scale these are prepared by the reaction of silicon with organolithium, Grignard or organoaluminium reagents.

$$SiCl_4 + 4LiR \longrightarrow SiR_4 + 4LiCl$$
$$SiCl_4 + 3LiR \longrightarrow R_3SiCl + 3LiCl$$
$$SiCl_4 + 2LiR \longrightarrow R_2SiCl_2 + 2LiCl$$
$$SiCl_4 + LiR \longrightarrow RSiCl_3 + HCl$$

$$SiCl_4 + CH_3MgCl \longrightarrow CH_3SiCl_3 + MgCl_2$$
$$CH_3SiCl_3 + CH_3MgCl \longrightarrow (CH_3)_2SiCl_2 + MgCl_2$$
$$(CH_3)_2SiCl_2 + CH_3MgCl \longrightarrow (CH_3)_3SiCl + MgCl_2$$
$$(CH_3)_3SiCl + CH_3MgCl \longrightarrow (CH_3)_4Si + MgCl_2$$

(ii) Industrially these are prepared by the direct reaction of alkyl or aryl halides with fluidized bed of silicon in the presence of large amounts (10%) of copper catalyst.

$$Si + 2CH_3Cl \xrightarrow[280-300°C]{\text{Cu catalyst}} (CH_3)_2SiCl_2$$

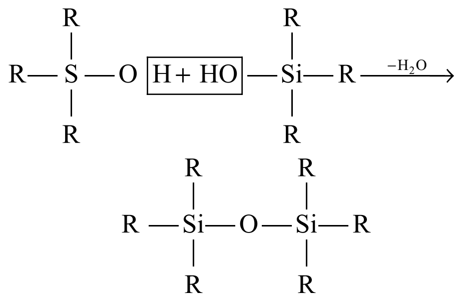
(iii) These are also prepared by the catalytic addition of Si-H to an alkene. The catalysts are BF_3, BCl_3 or $AlCl_3$. This method is not applicable for making the methyl and phenyl silanes.

$$Cl_3SiH + H_2C = CHR \longrightarrow Cl_3SiCH_2 — CH_2R.$$

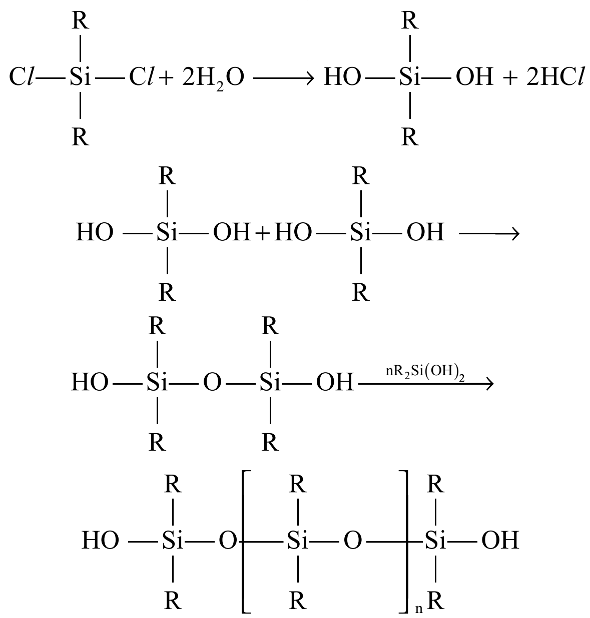
2. Hydrolysis and condensation of organo silicon halides:

Hydrolysis and condensation of trialkyl monochlorosilane R_3SiCl yields hexa-alkyl silicone. (or siloxane)

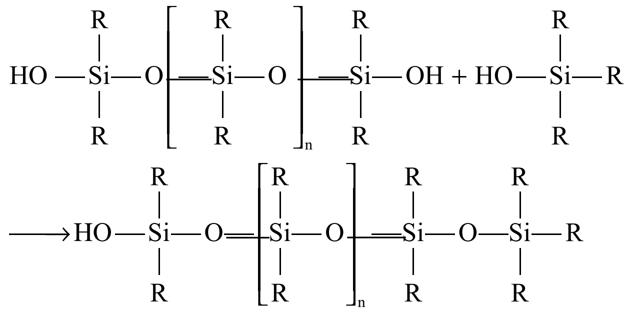
$$R_3SiCl + H_2O \longrightarrow R_3SiOH + HCl.$$



The dialkyl dichlorosilane R_2SiCl_2 on hydrolysis gives rise to straight chain polymers, since they have active OH group at each end of the chain. Polymerization continues and the chain increases in length.

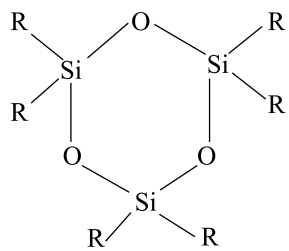


The chain growth can be controlled by the addition of R_3SiCl during hydrolysis and condensation. The R_3SiCl will block the end of the straight chain produced by R_2SiCl_2. Since there is no longer an active functional OH group at this end of the chain, it cannot grow any more at this end. Thus R_3SiCl is a chain terminating unit and the ratio of R_3SiCl and R_2SiCl_2 in the starting mixture will determine the average chain size.

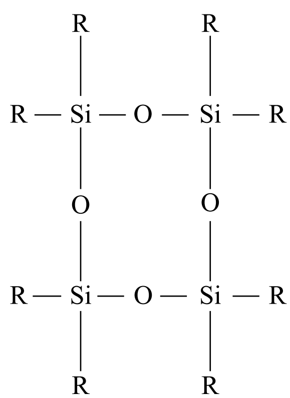


If some $RSiCl_3$ is added to the starting mixture it will give cross linkages between the chains because there is one more active functional OH group produces **cross linked silicones**.

Hydrolysis of R_2SiCl_2 under careful conditions can produce **cyclic silicones** with rings containing three, four, five or six silicon atoms. e.g.,

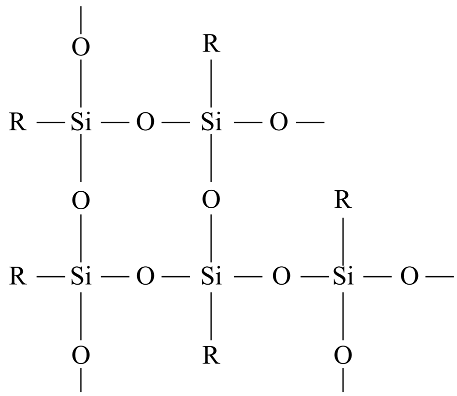


tris cyclo-dialkyl siloxane



tetrakis cyclo-dialkyl siloxane

Hydrolysis of alkyl trichlorosilane $RSiCl_3$ gives a very complex cross-linked polymer.



The starting materials for the manufacture of silicones are alkyl or aryl substituted chlorosilanes. For both economic and technical reasons commercial production of such polymers is almost entirely restricted to the methyl derivatives (and to the lesser extent the phenyl derivatives.) Silicones are fairly expensive but have many desirable properties. They were originally developed as electrical insulators, because they are more stable to heat than organic polymers and if they break down they do not produce conducting materials as carbon does. They are resistant to heat, oxidation and most chemicals due to strong Si-C and Si-O-Si bonds. They are strongly water repellant, good electrical insulators and have non-stick properties and antifoaming properties. Silicones may be made as fluids (oils), greases, emulsions, elastomers (rubbers) and resins.

Silicone fluids: Straight chain polymers of 20 to 500 units are used as silicone fluids. These liquids are used as dielectric insulating media, hydraulic oils and compressible fluids for liquid springs.

Silicone greases: The introduction of some phenyl groups and by thickening the methyl phenyl silicone oil with lithium soap gives silicone greases.

Silicone elastomers (Rubbers): Reinforced linear dimethyl polysiloxanes of exceedingly high molecular weight ($5 \times 10^5 - 10^7$) are the silicone rubbers. These rubbers with certain cross linking for every 100 - 1000 silicon atoms are very superior than any other synthetic or natural rubber in retaining their flexibility, elasticity and strength upto 250° and down to 100°C.

Silicone resins: These are prepared by the hydrolysis of phenyl substituted dichloro and trichloro silanes in toluene. The phenyl groups increase the heat stability, flexibility and processability of the resins. These are used in the insulation of electrical equipment and machinery and electronics as laminates for printed circuit boards. These are also used as high temperature paints and resinous release coating familiar on domestic cooking ware and industrial tyre moulds.

8.9.3 Silicates

Silicates are those substances which comprise SiO_4^{4-} units. In SiO_4^{4-} units the silicon is present in sp^3 hybridization. Now there are four lone electrons in the four sp^3 hybrid orbitals of silicon atom which form four bonds with four oxygen atoms yielding SiO_4 species. Every oxygen will take up one electron from some metal to complete its octet. Hence SiO_4^{4-} anion will be formed.

Since the silicon atom involves in sp^3 hybridization, it has tetrahedral structure as shown in Fig 8.10. The plane circles represent oxygen atom and the small shaded circle represents the silicon atom.

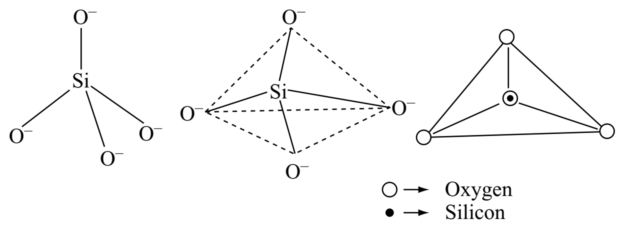


Fig 8.10 Tetrahedral structure of SiO_4^{4-} ion

From the above structures it is evident that silicon atom has its octet but each oxygen atom is still short of one electron to complete its octet. The oxygen atoms therefore have a tendency to obtain electrons from some other atom and thus acquire negative charge in this process. In some cases the oxygen atom of SiO_4 species tend to complete their octet by sharing electrons with other silicon atoms. The number of oxygen atoms which in this way form bridges to other silicon atoms may be one, two, three or all the four. In this way a number of complex silicates can be formed. The resulting silicate chains are largely charged anions because the unshared oxygen atoms carry negative charge due to gaining electrons from certain metal atoms to complete its octet. Thus metal cation such as Li^+, Na^+, K^+, Ca^{2+} and Al^{3+} are present in silicate minerals. These serve to hold the solid together by ionic attraction. The basic unit of structure is SiO_4 tetrahedron and these tetrahedron occur either singly or by sharing oxygen atoms in small groups, in small cyclic groups, in infinite chains or in infinite sheets. So according to such arrangement of tetrahedral SiO_4 units the silicates are classified as follows.

1. Orthosilicates: These silicates contain simple discrete SiO_4^{4-} **orthosilicate** anion and its corners are not shared. Different structures are formed depending on the coordination number adopted by the metal ion present in these silicates.
Eg: Zircon, $ZrSiO_4$, Phenacite Be_2SiO_4, Willemite Zn_2SiO_4. Forsterite Mg_2SiO_4.

2. Pyrosilicates: These contain condensed silicate anions i.e., those formed by combining two SiO_4 tetrahedra by

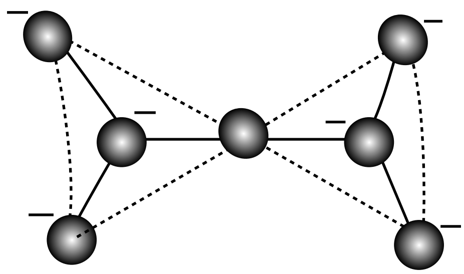


Fig 8.11 Structure of $[Si_2O_7]^{6-}$ unit in Island or Pyrosilicates

sharing one oxygen atom. These contain basic **pyrosilicate ion** $Si_2O_7^{6-}$ **e.g.,** Thortivetite ($Sc_2Si_2O_7$) hemimorphite $[Zn_4(OH)_2(Si_2O_7)$. H_2O

3. Chain Silicates: These are formed by combining of several SiO_4 tetrahedra linearly. These are mainly two types.

(a) Pyroxenes: These contain single strand chain of composition $\left(SiO_3\right)_n^{2-}$. In these silicates two oxygen atoms on two corners of each tetrahedron are shared with other tetrahedra. Due to the difference in the arrangement of the tetrahedra in space a variety of different structures are formed. **E.g.,** Spodumene Li*Al* $[(SiO_3)_2]$, enstatite $Mg_2[(SiO_3)_2]$ and diopsite $CaMg[(SiO_3)_2]$

(b) Amphiboles: These contain double-strand cross (linked chains or bonds. The basic unit in such silicates may be $(Si_2O_5)_n^{2n-}$ $(Si_4O_{11})_n^{6n-}$ $\left(Si_6O_{17}\right)_n^{10n-}$. Most common amphiboles are the asbestos minerals which contain structural unit $\left(Si_4O_{11}\right)_n^{6n-}$. In this structure some tetrahedra share two corners whilst others share three corners (with an average 2.5) **E.g.,** Tremolite $Ca_2Mg_5[(Si_4O_{11})_2]$ $(OH)_2$ and Crocidolites $Na_2Fe_3^{II}\,Fe_2^{III}\left[\left(Si_4O_{11}\right)_2(OH)_2\right.$. Amphiboles always contain OH groups which are attached to the metal ions.

4. Cyclic silicates: Cyclic silicates are formed by sharing two oxygen atoms per tetrahedron. The ring structure may be formed of general formula $(SiO_3)_n{}^{2-}$. Rings containing 3,4,6 and 8 tetrahedra units are known but with those with 3 and 6 are common. $Si_3O_9{}^{6-}$ unit is present in wollastonite $Ca_3Si_3O_9$ and in benitoite BaTi $[Si_3\,O_9]$. The $Si_6O_{18}{}^{12-}$ unit occurs in beryl $Be_3\,Al_2Si_6O_{18}$.

5. Sheet silicates: In these silicates SiO_4 tetrahedra are linked into infinite two dimensional networks as shown in Fig 8.14. In such silicates SiO_4 tetrahedra have three corners (Three bridge oxygen atoms per silicon) shared with three adjacent tetrahedra. The empirical formula of sheet silicate is $(Si_2O_5)_n^{2n}$. Many silicates have such sheet structures with the sheets are bound together by the cations which lie between them.

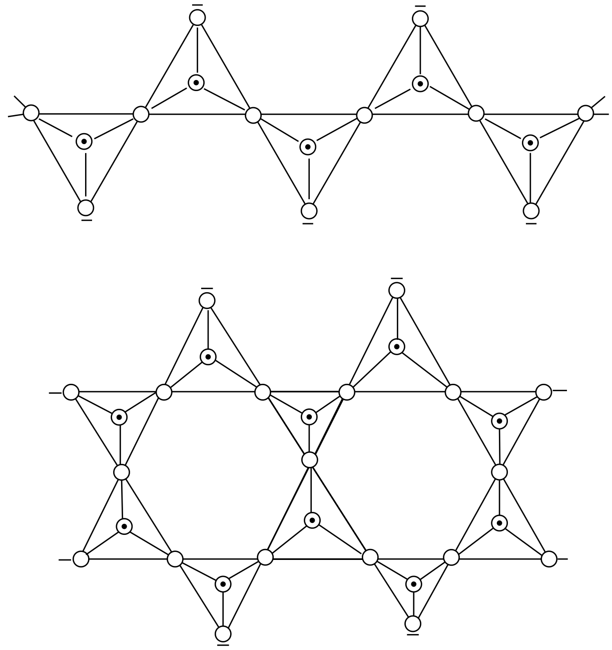


Fig 8.12 Structure of (a) linear silicate anion (SiO_3^{2-}) (b) Double chain silicate ($Si_4\,O_{11}^{6-}$)

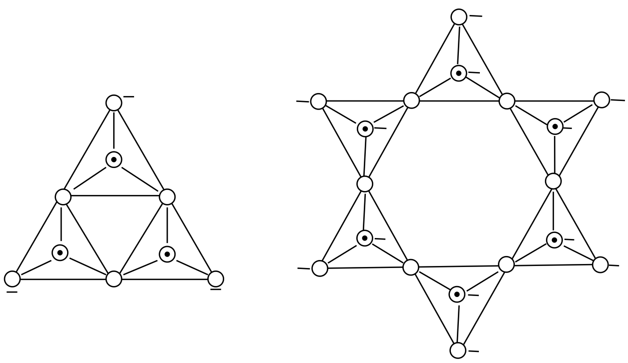


Fig 8.13 Structure of cyclic silicates

So they can be cleaved readily into thin sheets e.g., (1) Clay minerals (2) White asbestos (3) Micas (4) Montomorillonites

1. Clay Minerals:	Pyrophyllite	$Al_2\,(OH)_2\,[Si_2O_5]_2]$
	Talc	$Mg_3\,(OH)_2\,[(Si_2O_5)_2]$
2. White asbestos:	Chrystotite	$Mg_3\,(OH)_4\,[(Si_2O_5)]$
3. Micas:	Muscovite	$K\,Al_2\,(OH)_2\,[Al\,Si_3O_{10}]$
4. Montomorillonites:	Vermiculate	$Na_x\,(Mg\,Al\,Fe)_3\,(OH)_2\,[(SiAl)_2\,O_5)_2]\,H_2O$

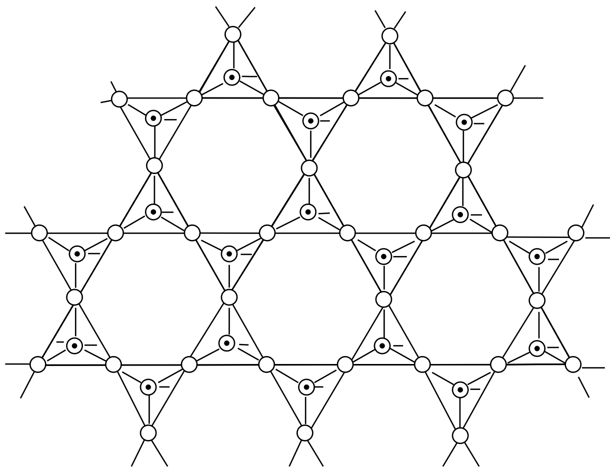


Fig 8.14 Structure of Sheet silicate

6. Three dimensional silicates: If all the oxygen atoms of a SiO_4 tetrahedron are shared with adjacent tetrahedra and if this process is repeated then an infinite three dimensional structure is formed. If silicon atom is not replaced by other metals, the silicate structure will have the neutral formula SiO_2 e.g., quartz, tridimite, crystobalite etc. These contain no metal ions. If some of the Si^{4+} ions are replaced by Al^{3+} and additional metals which are isomorphous results in the formation of three dimensional silicates. If one fourth of the Si^{4+} ions in SiO_2 are replaced by Al^{3+} gives a frame work $AlSi_3O_8$. To neutralize the negative charge on this unit there may be unipositive ions like K^+, Na^+ or dipositive ions like Ca^{2+} or Ba^{2+}. Replacement of one-quarter or one-half of the silicon atoms are quite common. Giving structures M^+ [Al Si_3 O_8] and M^{2+} [Al_2 Si_2 O_8]$^{2-}$. Three-dimensional silicates are three types.

(i) Feldspars: These are most important rock forming minerals. These constitute about two-thirds of the igneous rocks. They may be **orthoclare feldspars** K[$AlSi_3O_8$] and Ba[$Al_2Si_2O_8$] and plagiocare feldspars Na[$AlSi_3O_8$] and **anorthite.** They differ in the crystal structure due to the difference in the size of ions Na^+ and Ca^{2+} are smaller than K^+ and Ba^{2+} respectively.

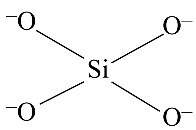
(ii) Zeolites: These are most important of three dimensional alumino silicates. Zeolite contain channels like honey comb structure. These channels are large enough to allow through them certain molecules which are smaller than the aperture and absorb in the cavities. Examples of such molecules are CO_2, NH_3 (gases), H_2O, C_2H_5OH (liquids) etc. Thus they act as molecular sieves to separate molecules of different sizes. Sodium zeolites like natrolite and permutit are used as water softners.

(iii) Ultramarines: Ultramarines are a group of sodium alumino silicates which have the frame work [$(SiAl)_n$.$(O_2)_n$] and balancing cations. Unlike zeolites ultramarines do not contain water. Further in addition to the metal ions introduced to balance the charges when Al^{3+} replaces Si^{4+}, ultramarines also contain some extra anions like Cl^-, S_2^{2-} and SO_4^{2-}. Some examples are.

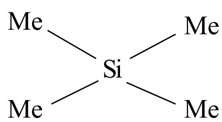
Ultramarine	Na_8 [$(AlSiO_4)_6$] S_2
Sodalite	Na_8 [$(AlSiO_4)_6$] Cl_2
Nosean	Na_8 [$(AlSiO_4)_6$] SO_4

Ultramarines are coloured compounds. The colour is due to the presence of sulphur as polysulphide and the variation in colour is due to the variation of the content of cations like Li^+, Na^+, K^+, Tl^+, Ag^+, Zn^{2+}, Mn^{2+}, Ba^{2+} etc.

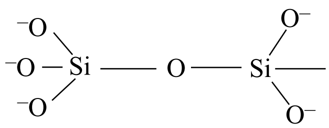
A comparative skeletons of frame work silicates and silicones are as follows.



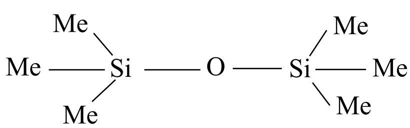
Ortho silicate



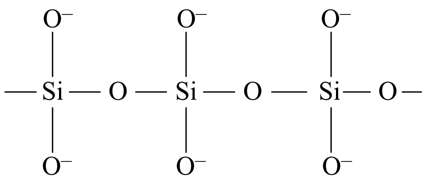
Tetra methyl silane



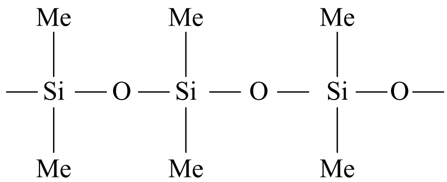
Disilicate (or) Pyrosilicate



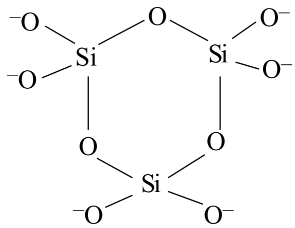
Hexamethyl disiloxane



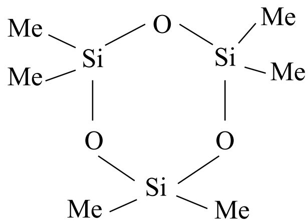
Pyroxenes



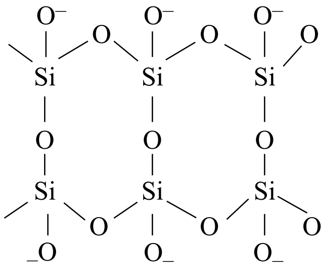
Polydimethyl siloxane



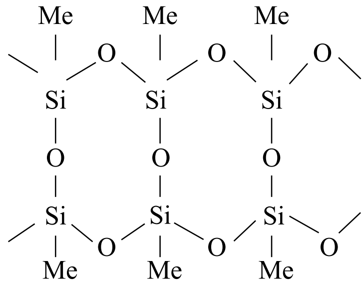
Cyclic metasilicates



Cyclic dimethyl siloxane



Amphiboles
Infinite sheet silicates
Frame work silicates



Methyl silsesquioxane
ladder polymer
Siloxenes
Silicone resins.

8.10 GLASS

Glass is a non-crystalline transparent super cooled mixture of silicates, one of which is always an alkali metal. So glass is man made artificial silicate considered as the **super cooled liquid**. Its composition varies with different varieties of glass. An approximate formula for ordinary glass may be given as

$M_2O \cdot M'O \cdot 6SiO_2$

Where M = Na or K and M′ may be Ba, Zn or Pb.

Raw materials: The following are the most commonly used substances.

(i) Sand: It is a source of silica.
(ii) Alkali metal salts such as Na_2CO_3, $NaNO_3$, K_2CO_3 and KNO_3.
(iii) Alkaline earth metal salts such as CaO, $CaCO_3$, $BaCO_3$ etc.
(iv) Oxides of heavy metals such as PbO, Pb_3O_4, ZnO, As_2O_3, TiO_2.
(v) Feldspar such as $Na_2O \cdot Al_2O_3 \cdot 6SiO_2$ or $K_2O \cdot Al_2O_3 \cdot 6SiO_2$
(vi) Oxides of non-metals such as B_2O_3 and P_4O_{10}.
(vii) Colouring materials: The following substances are used to get the required colour to the glass.

Fe_2O_3 - yellow; CoO - blue;
$FeO \cdot Cr_2O_3$- green; Cu_2O or SeO_2 - red;
MnO_2 - purple; CuO - light blue;
CdS - lemon yellow

(viii) **Cullet:** It is the crushed glass from imperfect or defective articles. It makes the melting easy and utilizes the waste.

Manufacture: The manufacture of glass involves four different operations as discussed below.

(i) Melting: The raw materials in proper proportion are finely ground to get homogeneous mixture called **batch**. The batch is then fused in a pot furnace or tank furnace at about 1400–1500°C. by using producer gas for about ten hours. The following chemical reactions takes place in the furnace.

$$Na_2CO_3 + SiO_2 \longrightarrow Na_2SiO_3 + CO_2$$

$$CaCO_3 + SiO_2 \longrightarrow CaSiO_3 + CO_2$$

Test samples of the molten mass are taken out from time to time till the opaque mass turns transparent on cooling and is free from gaseous bubbles.

(ii) Shaping: Various articles are made from the molten glass either by stamping with dies or by blowing from mouth or by compressed air such as bottle, flask and beakers etc.

(iii) Annealing: Glass being a semiconductor of heat, if cooled suddenly it becomes brittle and fragile. Therefore the article should be cooled slowly and the process is known as **annealing**. Annealing is done in a tunnel like oven called lehr. It is hot at one end and at room temperature at the other end. The articles are made to move slowly through the lehr and some time take a number of days to complete the journey. Different types of glasses have different annealing temperatures.

(iv) Finishing: After annealing, the glass articles are given finishing touch such as cleaning, trimming of edges, polishing.

Types of Glass: Glass has many types depending upon their composition and uses as given below.

(i) Soft Glass: Its formula is $Na_2O \cdot CaO \cdot 6SiO_2$. It is easily fusible and used for ordinary glass pots, tubes, plates, window panes etc

(ii) Hard glass: Its formula is K_2O. CaO. $6SiO_2$. It fuses with difficulty and is used for hard glass apparatus and bottles etc.

(iii) Flint glass: It is potassium and lead silicates. It has higher refractive index, density and transparency than ordinary glass. It is used for optical lenses, prisms, glass bulbs etc.

(iv) Jena glass: It is zinc and barium borosilicates. It is uneffected by heat, reagents, and jerks. It is used for making superior quality glass apparatus.

(v) Pyrex glass: It is sodium and aluminium borosiliates. Its qualities and uses are same as jena glass.

(vi) Crooke's glass: It is potassium, lead and cerium silicates. It absorbs ultraviolet rays harmful to eyes. It is used for spectacle lenses.

(vii) Safety glass: It consists of a thin layer of transparent plastic e.g., polyvinyl accetate resin held between the two thin layers of glass. The three layers are cemented by heat and pressure. It does not shatter under impact. It is used for automatic wind shields.

(viii) Quartz glass: This glass is obtained from silica. It has low coefficient of expansion and does not break even when plunged in water while red hot.

(ix) Ground glass:It is soft which has been ground by emery or turpentine.

(x) Water glass: It is sodium metasilicate Na_2SiO_3. It is soluble in water and hence known as water glass. It is used as a filler in soap manufacturer, as a fixative in calico-printing for preservation of eggs and for rendering flexible and timber fire proof.

8.11 TIN

The only ore of tin of any importance is cassiterite or tin stone, SnO_2 which is mined in Malaysia Nigeria, Indonesia and Bolivia.

Extraction: The extraction of tin from tin stone involves the following operations.

(i) Concentration: The ore is pulverized and washed with water to float away lighter impurities. The concentrated ore contains. ($FeWO_4$) and tin stone which are of the same density. Wolfram being magnetic is removed by electromagnetic method to increase the percentage of tin stone.

(ii) Roasting: The concentrated ore is roasted in a rotary furnace. During roasting sulphur and arsenic are removed as their oxides and iron and copper pyrites are converted into their oxides and sulphates.

(iii) Washing: The roasted ore is washed with water to remove water soluble impurities as sulphates of copper and ferrous. The washed ore is called **black - tin** and contains 60–70 % tin as oxide.

(iv) Smelting: The black tin is mixed with anthracite (carbon) lime stone, is added as flux to produce slag with the impurities.

$$SnO_2 + 2C \longrightarrow Sn + 2CO$$

$$CaCO_3 \longrightarrow CaO + CO_2$$

$$SnSiO_3 + CaO + C \longrightarrow CaSiO_3 + Sn + CO$$

(v) Purification: The metal thus obtained purified by the following stages.

(a) Liquation: First the metal is purified by liquation method by heating the metal on the sloping hearth of the reverboratory furnace when tin melts and flows down leaving the residue usually contains iron, copper arsenic and lead.

(b) Poling: The molten metal is further purified by stirring with poles of green wood. The hydrocarbon gases liberated from the green wood reduces the tin oxide present in it.

(c) Electrolytic refining: The metal may be purified by electrolysis. Impure tin metal blocks are taken as anode pure tin sheet is taken as cathode. The electrolyte is the hydroflurosilicic acid (H_2SiF_6), tin sulphate and sulphuric acid. During electrolysis the pure metal gets deposited on the cathode.

Properties:

Tin is silvery - white lustrous metal. melting point 505 K, boiling point 2900 K. It becomes brittle at 200°C and can be powdered. It produces a peculiar sound when it is bent. That cracking sound is known as **tin cry**. This is due to rubbing of crystals one above the other. It exists in three allotropic forms. Their transition temperatures are as follows.

$$\text{Grey tin} \underset{}{\overset{291K}{\rightleftharpoons}} \text{White tin} \overset{440K}{\rightleftharpoons} \text{Rhombic tin} \overset{505K}{\rightleftharpoons} \text{liquid}$$

	Grey tin	White tin	Rhombic tin
Sp. Gr	5.8	7.8	6.56
Crystal structure	Cubic	Tetragonal	Rhombic

Most common and stable allotrope is white tin. The conversion of white tin into grey tin in cold countries makes it brittle and powdered. This is termed as **tin disease** or **Tin pest** or **tin plague.**

Chemical reactivity of tin was already discussed earlier in this chapter.

Uses: It is used as a reducing agent, mordant in dying, for making purple of cassius used for colouring glass and pottery. It is used to tinning the copper and brass vessels. Tinning is a process of coating the inner walls of copper and brass. Organic acids present in food materials dissolves the copper present in copper and brass vessels. This causes the food poisoning.

The food materials are protected from poisoning by copper by depositing a thin layer of tin on the surface of copper and brass vessels. Tin foils are used for wrapping cigarettes, confectionary and for making tooth paste tubes. Tin amalgam is used in making mirrors.

8.12 COMPOUNDS OF TIN

8.12.1 Stannous Oxide

Preparation: It can be prepared by heating stannous oxalate in the absence of air.

$$\begin{array}{l} COO^- \\ | \quad Sn^{2+} \\ COO^- \end{array} \xrightarrow{\Delta} SnO + CO + CO_2$$

It can also be prepared by heating stannous hydroxide (obtained by adding sodium hydroxide to stannous chloride) or a mixture of solid stannous chloride and sodium carbonate in an atmosphere of carbon dioxide (air oxidizes it to stannic oxide).

$$Sn(OH)_2 \longrightarrow SnO + H_2O$$
$$SnCl_2 + Na_2CO_3 \longrightarrow SnO + CO_2 + 2NaCl$$

Properties: It is a dark grey or black powder. It burns in air with incandescence, forming stannic oxide.

$$2SnO + O_2 \longrightarrow SnO_2$$

It is an amphoteric oxide. It dissolves in acids forming stannous salts and also in alkali forming a white precipitate which dissolves in excess of the alkali forming sodium stannite.

$$SnO + 2HCl \longrightarrow SnCl_2 + H_2O$$
$$SnO + 2NaOH \longrightarrow Na_2SnO_2 + H_2O$$

The stannites are known only in aqueous solutions. These absorb oxygen from air forming stannates.

$$2Na_2SnO_2 + O_2 \longrightarrow 2Na_2SnO_3$$

8.12.2 Stannic Oxide

It occurs in nature as cassiterite or tin stone. It may be prepared by heating tin in air.

$$Sn + O_2 \longrightarrow SnO_2$$

It can also be prepared by heating metastannic acid obtained by the reaction of tin with concentrated nitric acid.

$$Sn + 4HNO_3 \longrightarrow H_2SnO_3 + 4NO_2 + H_2O$$
$$H_2SnO_3 \xrightarrow{\Delta} SnO_2 + H_2O$$

Properties: It is a white powder which is unreactive towards common acids. Sulphuric acid, however, reacts to give stannic sulphate $Sn(SO_4)_2$. When this is diluted, it reprecipitates oxide. It is readily attacked by alkalis forming stannates.

$$SnO_2 + 2KOH \longrightarrow K_2SnO_3 + H_2O$$

Uses: Stannic oxide finds use in polishing powders to glaze in ceramics and as pigments in pottery. A film of SnO_2 is put on air craft windows and prevents the window from frosting up.

8.8.3 Stannous Chloride ($SnCl_2$)

Preparation: It is prepared by dissolving tin in concentrated HCl followed by evaporation when $SnCl_2 \cdot 2H_2O$ is crystallized.

$$Sn + 2HCl \longrightarrow SnCl_2 + H_2$$

Anhydrous $SnCl_2$ cannot be prepared by heating hydrated salt because it undergoes hydrolysis giving white precipitate of basic chloride.

$$SnCl_2 \cdot 2H_2O \xrightarrow{\Delta} Sn(OH)Cl + HCl + H_2O$$

Anhydrous $SnCl_2$ can be prepared by heating tin in a current of dry HCl or by the distillation of excess of tin with mercuric chloride when mercury distills leaving behind anhydrous stannous chloride

$$Sn + 2HCl \longrightarrow SnCl_2 + H_2$$

$$Sn + HgCl_2 \longrightarrow SnCl_2 + Hg$$

Anhydrous salt is also prepared by heating tin and stannic chloride together

$$Sn + SnCl_4 \longrightarrow 2SnCl_2$$

Properties: It is transparent monoclinic crystals. It is soluble in water, alcohol and ether.

It combines with ammonia to form a number of double compounds depending upon the temperature of the reaction. These compounds are $SnCl_2 \cdot 2NH_3$ (formed at freezing mixture), $SnCl_2 \cdot NH_3$ (formed above 100°C) and $3SnCl_2 \cdot 2NH_3$ (formed above 100°C). Among these the last one is the most stable. When treated with water it hydrolyses forming a precipitate of basic chloride. However a clear solution is obtained in the presence of hydrochloric acid.

$$SnCl_2 + H_2O \rightleftharpoons Sn(OH)Cl + HCl$$

In acid medium stannous chloride acts as a strong reducing agent because Sn^{2+} is less stable than Sn^{4+}.

$$Sn^{2+} \longrightarrow Sn^{4+} + 2e^-$$

Mercuric chloride is first reduced to mercurous chloride and then to mercury.

$$2HgCl_2 + SnCl_2 \longrightarrow Hg_2Cl_2 + SnCl_4$$

$$Hg_2Cl_2 + SnCl_2 \longrightarrow 2Hg + SnCl_4$$

It also reduces ferric salts to ferrous salts, cupric salts to cuprous, auric salts to gold, nitric acid to hydroxyl amine.

$$2FeCl_3 + SnCl_2 \longrightarrow SnCl_4 + 2FeCl_2$$

$$2CuCl_2 + SnCl_2 \longrightarrow Cu_2Cl_2 + SnCl_4$$

$$2AuCl_3 + 3SnCl_2 \longrightarrow 2Au + 3SnCl_4$$

Decolourizes iodine. Therefore it can be estimated volumetrically using standard iodine solution.

$$Sn^{2+} + I_2 \longrightarrow Sn^{4+} + 2I^-$$

It decolourizes acidified potassium permanganate and reduces orange red dichromate to green chromic salt.

$$2MnO_4^- + 5Sn^{2+} + 16H^+ \longrightarrow 2Mn^{2+} + 5Sn^{4+} + 8H_2O$$

$$Cr_2O_7^{2-} + 3Sn^{2+} + 14H^+ \longrightarrow 2Cr^{3+} + 3Sn^{4+} + 7H_2O$$

It gives dark brown precipitate of stannous sulphide with H_2S

$$SnCl_2 + H_2S \longrightarrow SnS + 2HCl$$

With alkalis, it gives a white precipitate which dissolves in excess of alkali due to the formation of sodium stannite.

$$SnCl_2 + 2NaOH \longrightarrow Sn(OH)_2 + 2NaCl$$

$$Sn(OH)_2 + 2NaOH \longrightarrow Na_2SnO_2 + 2H_2O$$

Uses: It is used as a mordant in dyeing. It is used as a reducing agent in the laboratory. Also used in the manufacture of purple of cassius.

8.8.4 Stannic Chloride ($SnCl_4$)

Preparation: It may be prepared by passing chlorine over molten tin kept in a retort.

$$Sn + 2Cl_2 \longrightarrow SnCl_4$$

It can also be prepared by distilling tin with an excess of mercuric chloride.

$$2HgCl_2 + Sn \longrightarrow SnCl_4 + 2Hg$$

It may also be prepared by passing a mixture of dry chlorine and sulphur chloride vapour over heated stannic oxide.

$$2SnO_2 + 3Cl_2 + S_2Cl_2 \longrightarrow 2SnCl_4 + 2SO_2$$

Properties: It is a colourless fuming liquid boiling point 387K. It has unplesant odour. It is highly reactive in nature. It combines with water molecules forming hydrates containing 3,5,6, or 8 molecules of water of crystallization e.g., $SnCl_4 \cdot 3H_2O$; $SnCl_4 \cdot 5H_2O$; $SnCl_4 \cdot 6H_2O$ and $SnCl_4 \cdot 8H_2O$. The $SnCl_4 \cdot 5H_2O$ is called as **butter of tin or oxymuriate of tin**. With excess water it hydrolyses.

$$SnCl_4 + H_2O \longrightarrow Sn(OH)Cl_3 + HCl$$

$$Sn(OH)Cl_3 + 3H_2O \longrightarrow Sn(OH)_4 + 3HCl$$

If hydrochloric acid is added, hydrolysis will be prevented due to the formation of complex anion.

$$SnCl_4 + 2Cl^- \rightleftharpoons [SnCl_6]^{2-}$$

With ammonium chloride it forms ammonium chlorostannate, similarly with potassium chloride gives potassium chlorostannate.

$$2NH_4Cl + SnCl_4 \longrightarrow (NH_4)_2SnCl_6$$

$$2KCl + SnCl_4 \longrightarrow K_2SnCl_6$$

It forms double salts with ammonia $SnCl_4 \cdot 4NH_3$ which can be sublimed without decomposition. It also forms double salts with substances like N_2O_3, $NaCl$, PCl_5 etc.

Uses: It is used for fire-proofing cotton and also for increasing the weight of silk. Its pentahydrate is used as a mordant under the name butter of tin. Ammonium chlorostannate is also used as mordant under the trade name **pink salt**.

It is soluble in organic solvents and is not a conductor of electricity indicating that it is covalent compound. It is tetrahedral in shape like CCl_4.

8.13 LEAD

Lead is known since ancient times. In Rome lead pipes were used for conveying water and some compounds of lead were employed as paints and cosmetics. The symbol of lead is derived from the Latin word **plumbum**.

Occurrence: Lead occurs in free as well as in combined state in nature. Galena, PbS is the principle ore of lead. Its other ores are

Galena	–	PbS
Cerrusite	–	$PbCO_3$
Anglesite	–	$PbSO_4$
Lanarkite	–	$PbO.\ PbSO_4$

Extraction: Lead is extracted from the richer ores by self-reduction method and from the poorer ore by carbon reduction method.

(a) Self-Reduction Method

(i) Concentration: Finely powdered ore is concentrated by froth floatation process.

(ii) Roasting: The concentrated ore is roasted in a reverboratory furnace at 800 - 900K. Lead sulphide is partially oxidized to lead oxide and sulphate.

$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$

$$PbS + 2O_2 \longrightarrow PbSO_4$$

(iii) Reduction: At this stage the air supply is reduced, temperature is raised and more of the galena is added to the furnace. The galena reduces the oxide and sulphate to metallic lead (90%) and collected at the bottom.

$$PbS + 2PbO \longrightarrow 3Pb + SO_2$$

$$2PbS + PbSO_4 \longrightarrow 3Pb + 2SO_2$$

Thus, the process of roasting and smelting (reduction) are done in the same furnace at two different temperatures.

(b) Carbon-Reduction Method

(i) Concentration: It is concentrated as in the self-reduction method.

(ii) Roasting and sintering: The concentrated ore is mixed with lime and roasted in a sinterer in a blast of air at high temperature. The galena is oxidized to lead oxide.

$$2PbS + 2O_2 \longrightarrow 2PbO + 2SO_2$$

Since lime is more basic than PbO its presence prevents the formation of both $PbSO_4$ and $PbSiO_3$.

(iii) Smelting: The sintered ore its mixed with coke and lime and smelted in a blast furnace. During smelting, lead oxide is reduced to lead and silica is removed as calcium silicate slag.

$$PbO + C \longrightarrow Pb + CO$$

$$PbO + CO \longrightarrow Pb + CO_2$$

$$CaO + SiO_2 \longrightarrow CaSiO_3$$

$$PbSiO_3 + CaO \longrightarrow CaSiO_3 + PbO$$

The molten lead is drawn off from the tapping hole.

Purification: The lead obtained by the above methods contains the impurities of copper, silver, gold, tin, antimony, zinc etc. The following methods are employed for the purification of lead.

(i) Liquation: The impure lead is heated on sloped hearth. Lead having low melting point melts and flows down leaving behind the infusible impurities.

(ii) Softening: The impure metal is heated in a reverboratory furnace. Then air is passed to oxidize base metals (copper, iron, tin etc) that form a scum which is skimmed off.

(iii) Electrolytic refining: The lead further purified by electrolytic process. In this process impure lead blocks are taken as anode, pure metal is taken as cathode. The electrolyte consists of lead fluosilicate containing 8-10% hydrofluosilicic acid with a little gelatin. On passing current, a uniform and smooth deposit of pure lead is formed on the cathode.

Properties

It is bluish grey metal m.p. 600K, b.p.200K, sp gr 11.3. It is very soft metal, can be cut with a knife. It is highly malleable and ductile. It makes black mark on paper. It is poor conductor of electricity.

Lead is less reactive towards air due to the presence of oxide layer on the surface but when heated in air or oxygen it is oxidized to litharge (PbO) and red lead (Pb_3O_4)

$$2Pb + O_2 \longrightarrow 2PbO$$

$$6Pb + O_2 \longrightarrow 2Pb_3O_4$$

Action of water: Lead is not affected by air free water. It slowly decomposes steam at high temperature. It is readily dissolved by water containing dissolved oxygen to give lead hydroxide [$Pb(OH)_2$]. This solvent action of water on lead is known as **plumbosolvancy**.

This reaction is accelerated in the presence of nitrates, ammonium salts, organic acids and retreated in the presence of carbonates, sulphates and phosphates. Hence hard water has no action on lead. Since all the soluble lead compounds are very poisonous lead pipes should be used with caution for conveying drinking water.

Its chemical reactivity with acids, alkalis and non-metals was already discussed earlier in this chapter. It dissolves in acetic acid in the presence of air.

$$2Pb + 4CH_3COOH + O_2 \longrightarrow 2Pb(CH_3COO)_2 + 2H_2O$$

Uses: It is used for making water pipes and cable coverings. Lead chambers are used in sulphuric acid industry. Lead is used in lead accumulator. It is used in the manufacture of several useful compounds like white lead pigment, red lead

8.14 COMPOUNDS OF LEAD

8.14.1 Lead Monoxide PbO

It exists in two allotropic forms yellow powder called massicot and reddish yellow crystalline form called **litharge**. It is prepared by heating lead in air or by heating lead carbonate or nitrate.

$$2Pb + O_2 \xrightarrow{573K} \underset{\text{Massicot}}{2PbO} \xrightarrow{1150K} \underset{\text{Litharge}}{2PbO}$$

$$PbCO_3 \xrightarrow{\Delta} PbO + CO_2$$

$$2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$$

It is amphoteric in nature.

$$PbO + 2NaOH \longrightarrow Na_2PbO_2 + H_2O$$

When heated in oxygen it converts into red lead.

$$6PbO + O_2 \longrightarrow 2Pb_3O_4$$

Uses: It is used in making glass and ceramic glazes. It is also used in paints, pigments and varnishes.

8.14.2 Lead Dioxide PbO_2

It is prepared by the action of nitric acid on red lead.

$$Pb_3O_4 + 4HNO_3 \longrightarrow PbO_2 + 2Pb(NO_3)_2 + 2H_2O$$

In the laboratory, it is prepared by the oxidation of lead acetate with bleaching powder which contain some slaked lime.

$$Pb(CH_3COO)_2 + Ca(OH)_2 \longrightarrow Pb(OH)_2 + Ca(CH_3COO)_2$$

$$Pb(OH)_2 + CaOCl_2 \longrightarrow PbO_2 + CaCl_2 + H_2O$$

It can also be prepared by treating $PbCl_2$ with NaOCl and NaOH.

$$PbCl_2 + NaOCl + 2NaOH \longrightarrow PbO_2 + 3NaCl + H_2O$$

Properties: It is a chocolate brown powder, insoluble in water and dilute acids. It is amphoteric in nature. It liberates oxygen on heating.

$$2PbO_2 \xrightarrow{\Delta} 2PbO + O_2$$

It is a strong oxidizing agent. It oxidizes conc. HC*l* to chlorine.

$$PbO_2 + 4HCl \longrightarrow PbCl_2 + 2H_2O + Cl_2$$

It oxidizes and combines with sulphur dioxide.

$$PbO_2 + SO_2 \longrightarrow PbSO_4$$

It oxidizes Mn(II) salts to pink permanganic acid, which is used as a test for the detection of Mn(II) salts in qualitative analysis.

$$2MnSO_4 + 6HNO_3 + 5PbO_2 \longrightarrow 2HMnO_4 + 2PbSO_4 + 3Pb(NO_3)_2 + 2H_2O$$

It dissolves in alkalis. forming pulmbates.

$$PbO_2 + 2NaOH \longrightarrow Na_2PbO_3 + H_2O$$

Uses: It is used in match industry and in lead accumulators. It is also used as an oxidizing agent in the laboratories.

8.10.3 Red – Lead Pb_3O_4

It is prepared by heating litharge in air at 450° C.

$$6PbO + O_2 \xrightarrow{450^\circ C} 2Pb_3O_4$$

It is a scarlet crystalline powder insoluble in water. It is a mixed oxide made up with lead monoxide and lead dioxide ($PbO_2 \cdot 2PbO$). It reacts with concentrated acids as

$$Pb_3O_4 + 8HCl \longrightarrow 3PbCl_2 + 4H_2O + Cl_2$$

$$2Pb_3O_4 + 6H_2SO_4 \longrightarrow 6PbSO_4 + 6H_2O + O_2$$

$$Pb_3O_4 + 4HNO_3 \longrightarrow PbO_2 + 2Pb(NO_3)_2 + 2H_2O$$

When heated it decomposes giving O_2.

$$2Pb_3O_4 \longrightarrow 6PbO + O_2$$

Uses

It is used in match and glass industries. It is also used as a paint for iron and steel and for silver mirror. In laboratories it is used as an oxidizing agent.

8.14.4 Lead (II) Chloride

It is prepared by heating lead with chlorine gas or by adding HC*l* to soluble lead salt.

$$Pb + Cl_2 \longrightarrow PbCl_2$$

$$Pb(NO_3)_2 + 2HCl \longrightarrow PbCl_2 + 2HNO_3$$

It is a white crystalline solid, slightly soluble in cold water but soluble in hot water. It dissolves in concentrated HC*l* forming chloroplumbous acid.

$$PbCl_2 + 2HCl \longrightarrow H_2PbCl_4$$

8.14.5 Lead (IV) Chloride

It is prepared by dissolving PbO_2 in Conc. HCl and passing Cl_2 gas through the solution when dark brown solution of H_2PbCl_6 is obtained. It is then treated with NH_4Cl to get $(NH_4)_2PbCl_6$ which with H_2SO_4 gives yellow liquid of $PbCl_4$.

$$PbO_2 + 4HCl \longrightarrow PbCl_4 + 2H_2O$$

$$PbCl_4 + 2HCl \longrightarrow H_2PbCl_6$$

$$H_2PbCl_6 + 2NH_4Cl \longrightarrow (NH_4)_2PbCl_6 + 2HCl$$

$$(NH_4)_2PbCl_6 + H_2SO_4 \longrightarrow (NH_4)_2SO_4 + 2HCl + PbCl_4$$

It decomposes and liberates chlorine at room temperature.

$$PbCl_4 \longrightarrow PbCl_2 + Cl_2$$

8.10.6 White Lead $2PbCO_3 \cdot Pb(OH)_2$ (Basic Lead Carbonate)

It is prepared by exposing sheets of lead to the action of moisture, air, CO_2 and vapour of acetic acid.

$$2Pb + 2H_2O + O_2 \longrightarrow 2Pb(OH)_2$$

$$Pb(OH)_2 + 2CH_3COOH \longrightarrow Pb(CH_3COO)_2 + 2H_2O$$

$$Pb(CH_3COO)_2 + Pb(OH)_2 \longrightarrow [Pb\,(CH_3COO)_2 \cdot Pb(OH)_2]$$

$$3[Pb(CH_3COO)_2 \cdot Pb(OH)_2] + 4CO_2 + 2H_2O \longrightarrow 2[2PbCO_3 \cdot Pb(OH)_2] + 6CH_3COOH$$

The liberated acetic acid attacks the more lead. It can also be prepared by adding Na_2CO_3 to a solution of lead nitrate.

$$3Pb(NO_3)_2 + 2NaCO_3 + H_2O \longrightarrow 6NaNO_3 + CO_2 + [2PbCO_3 \cdot Pb(OH)_2]$$

It decomposes on heating.

$$2PbCO_3\ Pb(OH)_2 \xrightarrow{450°C} Pb_3O_4 + H_2O + CO + CO_2$$

Uses: It is extensively used as a white paint with linseed oil and possesses good covering power.

8.10.7 Lead Acetate $Pb(CH_3COO)_2$

It is prepared by dissolving litharge or lead carbonate in acetic acid.

$$PbO + 2CH_3COOH \longrightarrow Pb(CH_3COO)_2 + H_2O$$

$$PbCO_3 + 2CH_3COOH \longrightarrow Pb(CH_3COO)_2 + H_2O + CO_2$$

It is a white crystalline substance, sweet in taste and hence it is known as **lead sugar** or **sugar of lead** and soluble in water. On heating it decomposes forming acetone.

$$Pb(CH_3COO)_2 \longrightarrow PbO + CO_2 + CH_3COCH_3$$

It is used in dyeing and calico printing, in medicines as lead lotion and for the manufacture of pigments.

8.10.8 Tetra Ethyl Lead $Pb(C_2H_5)_4$

Tetra ethyl lead is a highly poisonous covalent liquid produced by reacting chloroethane vapour with sodium-lead alloy.

$$4C_2H_5Cl + 4Na/Pb \rightarrow Pb(C_2H_5)_4 + 3Pb + 4NaCl$$

It can also be prepared by the action of ethyl magnesium chloride with lead (II) chloride.

$$4C_2H_5MgI + 2PbCl_2 \rightarrow (C_2H_5)_4Pb + Pb + 4Mg\ ICl.$$

It is also formed when ethyl magnesium chloride is electrolysed in dry ether using lead anode.

It is a colourless, toxic, liquid, soluble in organic liquids but insoluble in water.

It is used as an antiknocking agent in petrol. On combustion first lead oxide is formed and then reduced to metallic lead which deposits inside the cylinders of the engine causing clogging of valves. To avoid this ethylene dibromide $C_2H_4Br_2$ is added along with tetra ethyl lead which removes the lead as non-toxic volatile $PbBr_2$.

KEY POINTS

- The elements carbon, silicon, germanium, tin and lead belong to Group – IVA and p-block of the periodic table.
- Carbon is a non-metal, silicon is chemically non-metal, Germanium is metalloid. Tin is metal with some non-metallic character, lead is purely metal.
- Group IVA is a transition between metals and non-metals and considered as **transition group** because in the elements of this group electropositive character as well developed as the electronegative character.
- The general outer electronic configuration of these elements is ns^2np^2.

Physical Properties

- **Atomic radius** increases from carbon to lead. The small difference in the atomic radii of silicon (117 pm) and germanium (122 pm) is due to the poor shielding capacity of d-electrons in the penultimate shell of germanium. Similarly the small difference in the atomic radii of tin (141 pm) and lead (146 pm) is due to the poor shielding capacity of the f-electrons in the anti-penultimate shell of lead.

- Except carbon density increases down the group. The carbon in the form of diamond has more density than silicon. Only lead is the densest element in the group.
- Melting point decreases from carbon to tin. Lead has more m.p than tin but less than germanium.
- Boiling point of carbon is exceedingly high. The Boiling point decreases gradually from carbon to lead.
- Carbon, silicon and germanium have **diamond like structure.** Hence they have high melting point. Tin and lead are metals but they do not use all the valence electrons in metallic bond due to inert pair effect.
- **Ionization energies** decreases from carbon to tin. There is large decrease between carbon and silicon but then onwards the decrease is little because of poor shielding of d-electrons in Ge and Sn and d- and f-electrons in Pb. Pb have a little more, I.E than Sn.
- **Electronegativity** decreases from C to Si but remains constant in the remaining elements. The reason is same as in ionization energies.
- These elements exhibit two types of oxidation states +2 and +4. From carbon to lead the stability of +2 oxidation state increases, while the stability of +4 oxidation state decreases due to **inert pair effect**.
- In +2 oxdiation state these elements act as **reducing agents** and their reduction power decreases from C to Pb due to increase in the stability of +2 oxidation state.
- In +4 oxidation state these elements act as oxidizing agents and their oxidation power increases from C to Pb due to decrease in the stability of +4 oxidation state.
- All the compounds of Group - IVA elements in +4 oxidation state are **covalent** (Fajan's rule: smaller the cation with more number of charges have more polarizing power). Most of the compounds of 'C' and 'Si' in +2 oxidation state are covalent. Ge, Sn and Pb tend to form M^{2+} ions and thus the tendency to form ionic compounds in +2 oxidation state increases from C to Pb.
- Carbon being a smaller atom the p-orbitals can participate in strong $p\pi - p\pi$ bond. As the size of atoms increases the distance between the parallel orbitals increases and cannot overlap. So $p\pi - p\pi$ bonding is absent in Si, Ge, Sn and Pb.
- The valence shell of carbon do not contain d-orbitals. So its maximum covalency is restricted to 4. But other elements can use the d-orbitals of their valence shell for σ-bonding and can exhibit a covalence upto 6 as in $[SiF_6]^{2-}$.
- The orbitals in the valence shell of silicon can also participate in $p\pi$-$d\pi$ bonding. For example $N(CH_3)_3$ is pyramidal but $N(SiH_3)$ is planar. Because the lone pair on N atom form π dative bond with the vacant d-orbital of Si. But trisilyl phosphine $(SiH_3)_3P$ is pyramidal because phosphorus is less able to form $p\pi \rightarrow d\pi$ dative than nitrogen to silicon.
- Due to $p\pi \rightarrow d\pi$ dative bond formation from 'N' atom to Si atom in H_3SiNCO and 'O' atom to 'Si' atom in $(R_3Si)_2O$ they become linear instead of angular.
- Due to $p\pi \rightarrow d\pi$ dative bond in silanols $(CH_3)_3SiOH$, resonance stabilization occurs in $(CH_3)_3Si\text{–}\overline{O}$ ion. Hence silanols are stronger acids than alcohols.
- **Catenation** is the combining capacity of the atoms of the same element to form long chains, branched chains and cyclic compounds. Catenation is not possible in the elements which exhibit valence less than 2.
- Of all the elements carbon has the maximum catenation power. The maximum catenation power of carbon is due to fact that C - C bond is as much strong as the bond formed by carbon atom with the atoms of other elements.
- Catenation power decreases from carbon to lead because of the decrease in the bond energy

$$C >> Si > Ge >> Sn > Pb.$$

- **Allotropy** is the occurrence of same element in two or more different physical forms having more or less similar chemical properties but different physical properties. The different forms of the element are called **allotropes**.
- Allotropy is due to the difference in the arrangement of atoms in solid state. Allotropes may be crystalline or amorphous.
- Carbon has large number of allotropes of which crystalline forms are diamond, graphite and fullerenes. Amorphous forms are coal, coke, wood charcoal, animal charcoal, lamp black, gas carbon, petroleum coke, sugar charcoal.
- At ordinary temperatures and pressures if one allotrope is thermodynamically more stable than the other, it is referred to as **monotropy**.
- Thermodynamically graphite is stable than diamond, so carbon exhibit monotropy.
- Silicon exhibit allotropy. At high pressures 4 coordinate diamond type lattice of Si – I changes to distorted - diamond Si – II, primitive hexagonal Si – V and eventually close packed Si – VII.
- Germanium forms brittle, grey white lustrous crystals with diamond structure.
- Tin has three crystalline allotropes with transition temperatures as shown below.

$$\underset{\text{Grey}}{\alpha\text{ - tin}} \xrightleftharpoons{13.2°C} \underset{\text{White}}{\beta\text{ - tin}} \xrightleftharpoons{161°C} \gamma\text{ - tin.}$$

α tin is less denser than β and γ - tin.

- The type of allotropy in which two allotropes are equally stable at the transition temperature is called **enantiotropy**. Tin exhibit enantiotropic allotropy. Lead do not exhibit allotropy.

Chemical Reactivity

- The elements of Group – IVA are relatively unreactive and the reactivity increases down the group partly.
- Lead is less reactive than expected partly because of the presence of oxide layer on the surface and partly due to hydrogen over voltage.
- These elements are not attacked by air at ordinary temperature. Though the reaction between carbon and oxygen is highly exothermic it will not take place in air because of high activation energy.
- Water has no reaction on these elements. At red hot temp. these elements decompose steam liberating H_2.
- Carbon is not attacked by non-oxidizing acids but oxidized to CO_2 by oxidizing acids like conc. H_2SO_4 and conc. HNO_3.
- Except with HF, silicon do not react with other acids.
- Carbon is not attacked by alkalis but other elements react with alkalis forming their soluble salts and liberate H_2.
  $Si + 2NaOH + H_2O \longrightarrow Na_2SiO_3 + 2H_2$
- Carbon and silicon react with metals forming carbides and silicides respectively. Ge, Sn and Pb forms alloys.
- Carbon burns in O_2 forming CO or CO_2 depending on temperature, forms CS_2 when heated with sulphur vapour in electric furnance, forms flurocarbon with F_2.
- Silicon forms SiO_2 when burnt in O_2, forms tetrahalides SiX_4 when heated with halogens.

COMPOUNDS OF GROUP IVA ELEMENTS

Hydrides

- All the Group IVA elements form covalent hydrides. The number of stable hydrides and the ease with which they are formed decreases as we move down the group.
- Hydrides of carbon are large in number and are known as **hydrocarbons.** Hydrides of silicon are limited and are known as **silanes**. Germanium hydrides are called **germanes**. Tin forms only two hydrides SnH_4 and Sn_2H_6. Lead forms only one hydride called **plumbane**.
- Thermal stability of the hydrides decreases from CH_4 to PbH_4 due to decrease in bond energy with increase in bond length with increase in atomic size.
- The stability of higher hydrides of these elements decreases from carbon to tin. This is because of the decrease in the M – M bond strength down the group and due to increased withdrawl of electron density by hydrogen.
  $\underset{2.5}{C} \longleftarrow \underset{2.1}{H} \qquad \underset{1.8}{Si} \longrightarrow \underset{2.1}{H}$
  This results in the weakening of M – M bond.
- As the thermal stability of hydrides gets decreased down the group their ability to act as reducing agent gets increased. Order of reduction power is $CH_4 < SiH_4 < GeH_4 < SnH_4 < PbH_4$.
- The reactivity of these hydrides towards alkalis is $CH_4 < GeH_4 < SnH_4 < SiH_4$. They hydrolyse in alkalis and liberate H_2.
- The more reactivity of silanes than hydrocarbons towards alkalis is due to polarity of Si – H and availability of d-orbitals with silicon atom.

Halides

- Elements of Group – IVA form tetrahalides of the type MX_4 except $PbBr_4$ and PbI_4.
- $PbBr_4$ and PbI_4 do not exist because of the inability of Br_2 and I_2 to oxidize Pb^{2+} to Pb^{4+}. In other words Pb^{4+} is strong oxidizing agent and Br^- and I^- are good reducing agents resulting the formation of $PbBr_2$ and PbI_2 only.
- Except SnF_4 and PbF_4 all the tetrahalides are covalent and volatile. SnF_4 and PbF_4 are ionic with high melting points.
- For a common halogen the stability of these halides is in the order $CX_4 > SiX_4 > Ge\,X_4 > SnX_4 > PbX_4$.
- With the same central atom the stability and volatility is in the order $MF_4 > MCl_4 > MBr_4 > MI_4$. This is because the decrease in bond energies from M – F to M – I.
- Fluorocarbons are most stable and chemically inert.
- Carbon halides do not hydrolyse in water. The halides of other elements hydrolyse in water because of the availability of d-orbitals in the valence shell
- SiF_4 on hydrolysis forms hexafluorosilicic acid
  $$3SiF_4 + 4H_2O \longrightarrow Si(OH)_4 + 2H_2SiF_6$$
- Except tetrahalides of carbon, tetrahalides of other elements acts as strong **Lewis acids** because of the availability of vacant d-orbitals in their valence shells. e.g., $[SiF_6]^{2-}$, $[GeF_6]^{2-}$, $[SnCl_6]^{2-}$.
- $SiCl_4$ cannot form complex ion $[SiCl_6]^{2-}$ because the silicon cannot be coordinated by bigger six chloride ions.
- Except carbon and silicon other elements of Group - IVA form dihalides of the type MX_2.
- Dihalides are ionic solids and have more melting points than the corresponding tetrahalides. e.g., $SnCl_2$ and $PbCl_2$ are high melting solids whereas $SnCl_4$ and $PbCl_4$ are volatile liquids.

- Except SnI_4 all other tetrahalides are colourless or white. SnI_4 is bright orange compound. The colour is due to charge transfer i.e., transfer of electron from iodine to tin.
- Carbon forms large number of polymeric halides. **Teflon** is polytetrafluoro ethylene used as good electrical insulator and in non-stick cooking utensils.
- Polymeric silicon halides are unstable. Ge forms only dimer Ge_2Cl_6. Tin and lead donot form catenated halides.

Oxides

- All the elements of Group - IVA form monoxides of the type MO and dioxides of the type MO_2.
- Among the monoxides SiO is formed only at elevated temperatures CO is neutral, GeO SnO and PbO are amphoteric.
- Stability of monoxides increases from CO to PbO but SiO is least stable.
- Monoxides act as reducing agents and the reduction power decreases from CO to PbO due to increase in stability down the group. Reactivity of monoxides also decreases from CO to PbO due to the same reason.
- Among the dioxides CO_2 is gas others are crystalline solids. CO_2 and SiO_2 are acidic, GeO_2 is distinctly acidic but less than SiO_2, SnO_2 and PbO_2 are amphoteric.
- All the dioxides dissolve in alkalis forming carbonates, silicates, germanates, stannates and plumbates.
  $MO_2 + 2NaOH \longrightarrow Na_2MO_3 + H_2O$
  (M = C, Si, Ge, Sn or Pb)
- GeO_2, SnO_2 and PbO_2 are insoluble in acids but dissolves in acids in the presence of complexing agents.
- **Stabilities** of dioxides decreases from CO_2 to PbO_2 due to inert pair effect. So the dioxides can act as oxidizing agents and the oxidation power increases from CO_2 to PbO_2.
- Because of non-polarity of linear CO_2 molecule only weak van der Waal's attractive forces exist between CO_2 molecules, so it is gas.
- SiO_2 has giant polymeric structure in which each silicon atom is shared by four oxygen atoms and each oxygen atom is shared by two silicon atoms (Fig 8.3) three dimensionally. Every silicon is sp^3 hybridized and have tetrahedral structure. So it is a solid. GeO_2, SnO_2 and PbO_2 are also solids having TiO_2 structure with 6:3 coordination.
- Carbon also form **suboxide** C_3O_2 which is linear and is obtained by dehydrating malonic acid. Lead forms mixed oxide Pb_3O_4.

Diamond

- Pure diamonds are colourless but the presence of impurities give colour. It is the densest allotrope of carbon. Its density is 3.5 g/cc.
- Diamonds glitter due to high refractive index 2.45.
- In diamond every carbon is in sp^3 hybridization and linked to four carbon atoms by single bonds. Hence in diamond every carbon is surrounded by four other carbon atoms tetrahedrally and covalently bonded to them resulting in the formation of **giant molecule**.
- The C – C bond length in diamond is 154 pm and CCC bond angle is 109º28'.
- To break all the bonds in the giant polymeric structure of diamond, large amount of energy is required hence diamond is the hardest material so far known and have very high melting point.
- Diamond has the highest known **thermal conductivity** because, this structure distributes thermal motion in three dimensions very effectively. But it is an **insulator** because there are no free moving electrons since all the valence electrons are participated in bonding. Fake diamonds can be identified by measuring the thermal conductivity.
- Diamond is chemically inert and insoluble in any solvent but on heating above 1500º C in vacuum it converts into graphite indicating diamond is chemically inert but thermodynamically unstable than graphite.
- Due to hardness black diamonds are used in making glass cutting, rock drilling tools and used as abrasives. Diamond dies are used for drawing thin wires.

Graphite

- Graphite can be manufactured by heating silica with carbon in an electric furnace. Silicon carbide formed initially dissociate to give silicon and graphite at the furnace temperature.
- Graphite has layered lattice structure. Every carbon in graphite is involved in sp^2 hybridization and in bond with three carbon atoms forming σ bonds.
- The p-orbitals which do not participate in hybridization are perpendicular to the plane of each layer and overlap laterally forming π bonds delocalized over the plane.
- Each layer is a hexagonal net of carbon atoms and may be regarded as a fused system of benzene rings, but the difference is that the **bond order** in benzene is **1.5** while in graphite is **1.33** since benzene ring has **three** π **– bonds** where as in graphite every ring has only **two** π **– bonds**.
- The distance between the layers is large 335 pm but the distance between the nearest neighbours in the

layer is 142 pm. Due to the large gap between layers the graphite is less denser than diamond.

- The layers in graphite are held together by weak van der Waal's forces due to which the layers are slippery in nature and can be cleaved easily. Also it can be used as a **lubricant**.
- Due to the presence of delocalized π electrons in a layer graphite conducts electricity and the conduction occurs in a layer, but not from one layer to another under normal conditions.
- Electrical conductivity in graphite perpendicular to the planes is low and increases with increasing temperature signifying graphite is **semiconductor** in that direction, but the electrical conductivity in planes is much higher and decreases with increase in temperature indicating it behaves as metal.
- Graphite exists in two forms α– and β–forms. In normal α – graphite the layers are arranged in the sequence AB AB AB .... with third layer exactly above the first layer, so that with carbon atoms in alternate layers vertically above each other. In β–form the layers are arranged in ABC ABC ABC ..... manner.
- Both forms are inter covertible. Heating β–form converts into α–form while grinding the α–form converts into β–form. In both forms the inter layer distances and C – C distance in a layer are equal.
- Chemically graphite is more reactive than diamond. Chromic acid oxidizes graphite slowly to CO_2. Conc. HNO_3 oxidizes it to graphitic acid $C_{11}H_4O_5$ and alkaline permanganate oxidizes it to oxalic acid, and melitic acid, (benzene hexacarboxylic acid) formation of which supports the hexagonal structure of graphite.
- Graphite is used in making lead pencils (graphite + clay), used as solid lubricant at high temperatures, in making electrodes, refractories.

Fullerenes

- Fullerenes are soccer shaped carbon clusters such as C_{60}, C_{76}, C_{84}. These are formed when electric arc is struck between carbon electrodes with significant quantities of C_{60}.
- Fullerenes containing even number of carbon atoms upto 350 or above are found.
- Fullerenes are the purest form of carbon because they have smooth structure without having **dangling bonds**.
- C_{60} is sometime referred as **Bucky ball** or **Buckminister** fullerene consists of fused system of 5 and 6-membered rings. A six-membered ring is fused with either six- or five-membered rings but a five-membered ring can only fuse with six-membered ring.
- C_{60} contain 20 six-membered rings and 12 five-membered rings.
- All the carbon atoms in C_{60} are equal and they undergo sp^2 hybridization. Each carbon forms three sigma bonds with other three carbon atoms. The remaining electron of each carbon is delocalized in molecular orbitals which in turn give **aromatic character** to the molecule.
- The ball shaped molecule has 60 vertices and each one is occupied by one carbon atom and it also contain single and double bonds with C – C distances 143.5 pm and 138.3 pm respectively.
- The reduced fullerenes with alkali metals like K_3C_{60}, $Rb_2Cs\ C_{60}$ act as **super conductors** at about 18K and 33K respectively.

COMPOUNDS OF CARBON

Carbon Monoxide

- It is formed in the incomplete combustion of carbon of carbonaceous fuels. Also it is formed when CO_2 is reduced by red hot carbon.
- It is a constituent of water gas ($CO + H_2$), producer gas ($CO + N_2$) and coal gas (CO, H_2, CH_4 and CO_2)
- When heavy metal oxides like iron, zinc are heated with carbon, CO is formed.

$$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO$$
$$ZnO + C \longrightarrow Zn + CO$$

- In the laboratory CO is prepared by the dehydration of formic acid or oxalic acid with conc. H_2SO_4.

$$HCOOH \xrightarrow{\text{Conc. } H_2SO_4} H_2O + CO$$
$$C_2H_2O_4 \xrightarrow{\text{Conc. } H_2SO_4} H_2O + CO + CO_2$$

  CO_2 from CO and CO_2 mixture can be removed by passing through caustic potash.

- Heating of potassium ferrocyanide crystals with conc. H_2SO_4 give CO. If dil. H_2SO_4 is used poisonous HCN will be evolved instead of CO
- It is a colourless, odourless gas, burns with blue flame, highly poisonous, combines with haemoglobin of the blood forming carboxy haemoglobin which loses the oxygen carrying capacity. Mixture of MnO_2 and CuO is used as catalyst in breathing apparatus used by rescue teams in mine disasters to oxidize CO to CO_2.
- It is a good reducing agent, reduces several metal oxides to their corresponding metals e.g., Fe_2O_3 to Fe, NiO to Ni, ZnO to Zn, PbO to Pb etc.
- It reduces Fehling solution to red cuprous oxide.
- It combines with several transition metals forming volatile metal carbonyls. The formation of $Ni(CO)_4$ followed by its decomposition is the basis of the Mond's process for refining of nickel.

- CO is an unsaturated compound because all the valencies of carbon are not satisfied and forms addition compounds e.g., It forms carbonyl sulphide (COS) with sulphur, carbonyl chloride (phosgene $COCl_2$), with chlorine sodium formate with NaOH, methyl alcohol with H_2 in the presence of ZnO/Cr_2O_3 as catalyst.
- The presence of CO can be detected by its reducing property of ammoniacal silver nitrate to black silver and an aqueous solution of $PdCl_2$ to metallic Pd.
- It can be estimated by estimating the I_2 liberated from I_2O_5 by CO.
- It is completely absorbed by ammoniacal solution of copper (I) chloride to give $CuCl \cdot CO \cdot 2H_2O$.
- In CO molecule, according to VB theory, between carbon and oxygen there are one sigma bond one π-bond and a third π dative bond from oxygen to carbon and is considered as the resonance hybrid of the following three structures.

$$:\overset{+}{C} - \underset{\cdot\cdot}{\overset{\cdot\cdot\,-}{O}}: \longleftrightarrow :C=\overset{\cdot\cdot}{O}: \longleftrightarrow :\overset{-}{C}\equiv\overset{+}{O}:$$

Carbon Dioxide:

- In the laboratory it can be prepared by heating any metal bicarbonate or carbonate (except alkali metal carbonates) or by the action of dilute acids on any metal bicarbonate or carbonate.
- Generally in the laboratory it is prepared by the action of dil. HCl on marble chips using Kipps apparatus. If dil. H_2SO_4 is used, the insoluble $CaSO_4$ forms as a layer on the marble chips and prevents the reaction.
- On large scale CO_2 is produced by complete combustion of carbon. It is also obtained as a by-product in the calcination of lime stone.
- It is colourless, odourless gas with pungent taste. It can be easily liquified by applying pressure. If the liquid CO_2 is allowed to rapid evaporation it converts into white solid called **dry ice**, used as a refrigerant under the name **drikold** or **cordice**.
- It is neither combustible nor supporter of combustion but metals like Na, K, Mg etc burn in CO_2. So metal fires cannot be extinguished by CO_2. To extinguish metal fires pyrene (CCl_4) is used.
- CO_2 turns the lime water milky due to the formation of insoluble $CaCO_3$ but the milkiness disappears when excess CO_2 is passed due to conversion of insoluble $CaCO_3$ to soluble $Ca(HCO_3)_2$. It is the test for detection of CO_2.
- When CO_2 is passed through any base first carbonate is formed which converts into bicarbonate on passing excess of CO_2.
- Plants absorb CO_2 from air, water from soil and convert them into carbohydrates in the presence of sunlight. This process is called **photosynthesis**.
- For structure refer 8.7.1b.
- It is used in the manufacture of urea, in the preparation of aerated water & soft drinks, in fire extinguishers in the manufacture of white lead & sodium carbonate.
- **Carbogen** is a mixture of O_2 and CO_2 (5 – 10%), used for artificial respiration in the case of pneumonia patients and victims poisoned by CO.

Carbonic Acid

- Carbonic acid (H_2CO_3) is formed when CO_2 is dissolved in water. The greater part of it is loosely hydrated which are in equilibrium with H_2CO_3, H^+ ions, HCO_3^- , CO_3^{2-} ions.

$$CO_2 + aq \rightleftharpoons CO_{2(aq)} \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^- \rightleftharpoons 2H^+ + CO_3^{2-}$$

- Carbonic acid is unstable cannot be isolated but solid carbonates are formed. Solid bicarbonates of alkali metals can only be prepared.
- Carbonates contain discrete planar ions in which all C – O bond lengths are equal due to resonance.
- Silicon does not form analogous silicate anion SiO_3^{2-} because larger silicon cannot form multiple bond with oxygen.
- Except alkali metal carbonates, other metallic carbonates are insoluble in water. When alkali metal carbonate solution is added to other metal salt solutions, a metal carbonates or a hydroxide or a basic carbonate precipitates. The product formed depends on the size and charge of metallic cation.
- The addition of soluble carbonate to solution containing Al^{3+}, Cr^{3+} or Fe^{3+} results in the precipitation of their hydroxides because these ions are small with more number of charges, they are hydrated in solution and show acid reaction.

$$[Al(H_2O)_6]^{3+} + 2H_2O \rightleftharpoons [Al(H_2O)_4(OH)_2]^+ + 2H_3O^+$$

$$[Al(H_2O)_4(OH)_2]^+ + CO_3^{2-} \longrightarrow \underset{\text{precipitate}}{[Al(H_2O)_3(OH)_3]} + HCO_3^-$$

- Addition of soluble carbonate to solutions of Cu^{2+} or Zn^{2+} ions results in the precipitation of basic carbonates $CuCO_3 \cdot Cu(OH)_2$ and $2ZnCO_3 \cdot 3Zn(OH)_2$. This is because the Cu^{2+} and Zn^{2+} are larger and carry less charge than Al^{3+}, Cr^{3+} and Fe^{3+} so are less acidic and the ionization process is more difficult.
- Large cations such as Group IIA metals are precipitated as normal carbonates because they are less hydrated in solution and thus are less acidic.

Carbides

- Carbides are the binary compounds of carbon with other elements having electronegativity less than carbon.
- Depending on the nature of bond, carbides are three types (i) ionic or salt like carbides (ii) covalent carbides and (iii) interstitial or metallic carbides.
- **Ionic carbides** are formed by Groups IA and IIA which are colourless and non-conductors. These are prepared by heating the metals or their oxides with carbon or with a source of carbon.
- Ionic carbides hydrolyse in water liberating a hydrocarbon gas.
- **Acetylides** are those which liberate acetylene on hydrolysis and contain acetylide ion. (C_2^{2-}) e.g., CaC_2
- **Methanides** are those which liberate methane gas on hydrolysis and contain methanide ion (C^{4-}) e.g., Al_4C_3, Be_2C.
- **Allylide** on hydrolysis gives allylene (C_3H_4). This contain C_3^{4-} ion. The only allylide is Mg_2C_3.
- **Covalent carbides** are formed by carbon with boron and silicon which have nearly same electronegativity with carbon.
- Silicon carbide, technically called as **carborrundum** is manufactured by heating a mixture of sand (54%), coke (34%), saw dust (10%) and salt (2%) in an electric furnace.
- It resists the action of almost all acids including HF but when fused with alkalis in air forms silicates and carbonates.
- When Cl_2 is passed over silicon carbide at 875K forms $SiCl_4$ and carbon.
- Structure of silicon carbide is similar to that of diamond in which Si and C atoms each tetrahedrally surrounded by four of the other kind, due to which it is as hard as diamond and used as abrassive.
- **Interstitial or metallic carbides** are formed by many transition metals and are similar to interstitial hydrides i.e., they are non-stoichiometric.
- Metals having radius less than 130 pm cannot form these carbides. In such cases carbides intermediate to ionic and interstitial are formed e.g., Fe_3C, Mn_3C and Ni_3C.

Fuel Gases

- **Fuel** is a substance that releases heat energy and light energy during combustion.
- Fuels are 3 types (i) solid fuels like coal, coke (ii) liquid fuels like kerosene, petrol (iii) gaseous fuels like natural gas, water gas etc.
- A good fuel is that which (i) is cheap (ii) easy to burn (iii) leaves no ash (iv) has high calorific value (v) can be transported easily.
- Gaseous fuels are advantageous over liquid, and solid fuels because they have above said qualities.

Producer gas

- It is mainly a mixture of CO 33% and N_2 64% remaining are CO_2 and H_2 etc. It is formed due to incomplete combustion of carbon.
- Its calorific value is less i.e., about 3762 kJ m^{-3} because of uncombustible N_2 gas, which act as **diluent.**
- When formed, producer gas will have high temperature and should be used immediately. It cannot be stored or transported.
- It is used in glass and steel industry, for illuminating purpose, in metallurgy and in the manufacture CH_3OH and NH_3.

Water gas

- Composition of water gas CO (40%), H_2 (50%) and the remaining are CO_2 and N_2.
- It is prepared by passing steam over red hot coke.
  $C + H_2O \longrightarrow CO + H_2 \quad \Delta H = +121.2$ kJ.
- Production of water gas is endothermic and hence steam and air are passed alternatively over coke.
- Its calorific value is 13,000 kJ m^{-3}. It burns with blue flame because both CO and H_2 burns with blue colour. So it is called **blue gas**. It is also called **synthesis gas** or **syn** gas because it is used for the synthesis of several compounds.
- It is used as a fuel in industry, in the manufacture of methyl alcohol, ammonia, and in the preparation of H_2 by Bosch process.
- Other fuel gases are **semiwater gas** (10 – 12% H_2, 25 – 28% CO, 50 – 55% N_2) **Carburetted water gas** (30 – 40% H_2, saturated hydrocarbons 15 - 20%, unsaturated hydrocarbons 10 – 15 %; CO 20 – 18%) **Coal gas** (40 – 50% H_2; 30 – 35% CH_4 ; 4% C_2H_6 5 – 10% CO and 8% N_2) **natural gas** [85% CH_4, 9% C_2H_6, 3% C_3H_8, 1% C_4H_{10} and 2% N) **LPG** (mainly contain C_3 and C_4 hydrocarbons)

SILICON

- Silicon is the second most abundant element occurring in the earth's crust (about 28% by wt). It never occurs in the free state but occurs as the oxide and silicates in rocks and clays.
- 96 – 99% pure silicon is prepared by the reduction of quarzite or sand with high pure coke in an electric furnace. Excess of SiO_2 should be used to prevent the formation of carbide SiC.

$$SiO_2 + 2C \longrightarrow Si + 2CO$$
$$2SiC + SiO_2 \longrightarrow 3Si + 2CO$$

- High pure silicon is made by the reduction of $SiCl_4$ or $SiHCl_3$ with Zn or Mg.
- It can also be prepared by thermal decomposition of SiI_4 or SiH_4.
- Silicon used in solar cells is made by the reduction of Na_2SiF_6 with sodium metal.

$$Na_2SiF_6 + 4Na \longrightarrow Si + 6NaF$$

- Silicon exists in two allotropric modifications amorphous and crystalline forms. The crystalline silicon has diamond like structure. Amorphous silicon contain minute crystals of this form.

COMPOUNDS OF SILICON

Silicon Dioxide

- Silica exhibits **polymorphism** and is the crushed form of quartz. **Sand stone** is sand particles bounded by iron oxide. **Flint** is amorphous silica associated with quartz. **Kiesulghur** is a siliceous rock composed of the remains of minute sea organisms.
- Crystalline silica exists as quartz, tridimite and crystobalite. Pure silica is quartz and contain **hexagonal prisms** and ends in hexagonal bipyramids.
- Quartz is known as **rock crystal**. It is transparent to U.V. light so used in making optical instruments.
- Quartz, tridimite and crystoballite are the three crystalline forms each having a low temperature. α - form and high temperature β - form.
- Amorphous forms of silica are **agate, jaspar** and **onyx**.
- Silica can be prepared by burning the silicon in air or by heating the hydrolysis product of $SiCl_4$ or SiF_4.

$$Si + O_2 \longrightarrow SiO_2$$
$$SiCl_4 + 4H_2O \xrightarrow{-4HCl} H_4SiO_4 \xrightarrow[-H_2O]{\Delta} SiO_2$$
$$3SiF_4 + 4H_2O \longrightarrow 2H_2SiF_6 + H_4SiO_4$$
$$\downarrow \Delta$$
$$2H_2O + SiO_2$$

- Acids except HF have no reaction with silica. Silica react with HF forming SiF_4 and H_2SiF_6.
- Silica dissolves in hot concentrated alkalis forming silicates.

$$SiO_2 + 2NaOH \longrightarrow Na_2SiO_3 + H_2O$$

- At high temperatures it reacts with metal oxides, carbonates and salts of non-volatile acids like sulphates and phosphates.

$$SiO_2 + Na_2O \longrightarrow Na_2SiO_3$$
$$SiO_2 + CaCO_3 \longrightarrow CaSiO_3 + CO_2$$
$$SiO_2 + Na_2SO_4 \longrightarrow Na_2SiO_3 + SO_3$$
$$6SiO_2 + 2Ca_3(PO_4)_2 \longrightarrow 3CaSiO_3 + P_4O_{10}.$$

- It is used, as acidic flux in metallurgy, in making acidic refractories, in making optical instruments and in building construction. Coloured quartz stones are used as precious stones.

Silicones

- Silicones are organosilicon polymers containing Si-O-Si bonds.
- Silicones are formed by the hydrolysis of alkyl or aryl substituted chlorosilanes.
- Silicones contain R_2SiO repeating unit. The empirical formula of silicone R_2SiO is analogous to that of ketone and hence are named silicones.
- Alkyl or aryl substituted chlorosilanes are prepared by reaction of RCl with silicon in the presence of metallic copper as catalyst.

$$RCl + Si \xrightarrow[370\ K]{Cu\ Powder} RSiCl_3 + R_2SiCl_2 + R_3SiCl$$

- Alkyl or aryl substituted chlorosilanes are also prepared by the reaction of Grignard reagent and silicon tetra chloride.
  $$RMgCl + SiCl_4 \longrightarrow \rightarrow RSiCl_3 + R_2SiCl_2 + R_3SiCl$$
  Product formed depends on the ratio of the reactants taken.
- Hydrolysis of substituted chlorosilanes yield corresponding silanols which undergo polymerization.
  $$R_2SiCl_2 + 2H_2O \longrightarrow R_2Si(OH)_2 + 2HCl$$
  $$2R_2Si(OH)_2 \longrightarrow HO - SiR_2 - O - SiR_2 - OH.$$
- Polymerization of dialkyl diols yield linear thermoplastic polymers. Properties of linear silicone depends on chain length. Chain growth can be controlled by the addition of R_3SiCl. If some $RSiCl_3$ is added it will give cross linkages between the chains forming cross linked silicones.
- Hydrolysis of R_2SiCl_2 under careful conditions can produce **cyclic silicones** by the elimination of water molecule from terminal OH groups.
- Hydrolysis of $RSiCl_3$ alone gives very complex cross linked polymer.
- Commercial silicone polymers are usually methyl derivatives and to a lesser extent phenyl derivatives.
- The great thermal and chemical stability of the silicones derives from the strength of both Si – C and Si – O – Si bonds.
- Silicones have chemical inertness, water repelling nature, heat resistance and good electrical insulating properties.

- Silicones are used as scalants, greases, electrical insulators and for water proofing of fabrics.
- Since silicones are biocompatible they are also used in making surgical and cosmetic implants.

Silicates

- Silicates are the metal derivatives of orthosilicic acid H_4SiO_4 or $Si(OH)_4$. The basic structural unit in the silicates is the SiO_4^{4-} tetrahedron. Depending on the number of corners (oxygen atoms) 0, 1, 2, 3, or 4 of SiO_4^{4-} tetrahedra shared various kinds of silicates are formed.
- Simple **ortho silicates** contain discrete SiO_4^{4-} units e.g., Zircon - $ZrSiO_4$, Phenacite Be_2SiO_4, Willimite Zn_2SiO_4.
- **Pyrosilicates** are formed when two SiO_4^{4-} tetrahedra share **one oxygen atom**. Pyrosilicate contain $(Si_2O_7)^{6-}$ unit. In pyrosilicate the bridging and terminal Si – O bonds have different bond lengths e.g., Thortevetite $Se_2Si_2O_7$, Hemimorphite $Zn_3Si_2O_7 \cdot Zn(OH)_2 \cdot 2H_2O$.
- When SiO_4 units share **two oxygen** atoms with each other **cyclic** or linear single chain silicates having the empirical formula $(SiO_3^{2-})_n$ are formed e.g., Beryl $Be_3Al_2Si_6O_{18}$.
- If two linear chains are cross linked by sharing two oxygen atoms in some tetrahedral units while in some others share three oxygen atoms, (on an average 2.5 oxygen atoms) double strand linear silicates having empirical formula $(Si_4O_{11})_n^{6n-}$ are formed. e.g., tremolite $Ca_2Mg_5 [(Si_4O_{11})_2] (OH)_2$ crocidolite $Na_2Fe^{II}{}_3Fe^{III}{}_2 [(Si_4.O_{11})_2] (OH)_2$. These are called **amphiboles** and they always contain hydroxyl groups attached to the metal ions.
- The pyroxenes and amphiboles cleave readily parallel to the chains forming fibres and hence are called **fibrous** minerals.
- **Asbestos** is formed from two different types of silicates amphiboles and sheet silicates.
- Two dimensional **sheet silicates** are formed by sharing **three oxygen atoms** of each SiO_4 tetrahedron. Sheet silicates contain $(Si_2O_5^{2-})$ units e.g., Talc – $[Mg(Si_2O_5)]_2 \cdot Mg(OH)_2$. Kaolin - $Al_2(OH)_4.Si_2O_5$. Sheet silicates are found mainly in clays.
- When all the four oxygen atoms of the SiO_4 tetrahedra are shared, three dimensional network silica will be formed. If a part of silicon of three dimensional network is replaced by Al^{3+} with incorporation of other cations such as Na^+, K^+ and Ca^{2+} to maintain the charge balance results in the formation of **three dimensional silicates**.
- Three dimensional silicates are three types (i) feldspars (ii) zeolites and (iii) ultramarines. Zeolites contain channel like honey comb structure. So these are used as molecular sieves to separate the molecules of different sizes. Sodium zeolite like natrolite and permutit are used as water softners.

Structural Units of Various Silicates

Type of silicate	Basic unit	No. of 'O' atoms shared
1. Orthosilicate	SiO_4^{4-}	0
2. Pyrosilicate	$Si_2O_7^{6-}$	1
3. Single strand chain silicates (pyroxenes)	$(SiO_3)_n^{2n-}$	2
4. Double strand chain silicates (amphiboles)	$(Si_2O_5)_n^{2n-}$	
	$(Si_4O_{11})_n^{6n-}$	2.5
	$(Si_6O_{17})_n^{10n-}$	
5. Cyclic silicates	$(SiO_3)_n^{2n-}$	
	$(Si_3O_9)^{6-}$	3
	$(Si_6O_{18})^{12-}$	6
6. Sheet silicates	$(Si_2O_5)_n^{2n-}$	3

GLASS

- Glass is man made artificial silicate.
- Ordinary glass is a mixture of sodium silicate, calcium silicate and silica with an approximate composition $Na_2SiO_3 \cdot CaSiO_3 \cdot 4SiO_2$ or $(Na_2O \cdot CaO \cdot 6SiO_2)$.
- Glass is super cooled liquid. It is amorphous and does not have sharp melting point. It contain non-crystalline silicates.
- Raw materials for the manufacture of glass are soda ash (Na_2CO_3). lime stone ($CaCO_3$) and sand (SiO_2). The homogeneous mixture of raw materials used for the production of glass is called **batch**.
- The broken glass pieces added to the batch in the production of glass is called **cullets**. The froth formed on the surface when the batch is heated at about 1700 K is known as **glass gall**.
- Addition of small amounts of transition metal compounds to the batch imparts colour to glass MnO_2 – violet; CoO – blue, CuO or Cr_2O_3 – green SeO_2 – red; CdS – lemon yellow.
- If the molten glass is suddenly cooled, it will be brittle and fragile but on cooling very slowly it becomes opaque.
- **Annealing** is a process of cooling the hot glass slowly from furnace temperature to room temperature by passing the glass articles through different zones of temperature.

- If most of the calcium in soft glass is replaced by lead, **flint** glass is obtained. It is used for making optical lenses and prisms. Its composition $Na_2SiO_3 \cdot K_2SiO_3$. $PbSiO_3 \cdot SiO_2$.
- **Pyrex glass** mainly contains sodium borosilicates and alumino borosilicate. Thermal expansion coefficient of pyrex glass is less and hence it can with stand the sudden changes in temperatures.
- **Crooke's glass** contain cerium oxide which checks the passage of U.V. light used by persons involved in welding process.
- **Safety glass** is layers of glass cemented with butyryl plastics by applying heat and pressure used in automobile wind shields.

TIN

- The only important ore of tin is **tinstone** or **cassiterite** SnO_2. It is concentrated first by levigation and then by magnetic separator to remove tungstates of Fe and Mn.
- The concentrated ore is roasted to remove impurities like S, As etc. as their volatile oxides and reduced with carbon to get the metal.

$$SnO_2 + 2C \longrightarrow Sn + 2CO$$

- Impure tin is purified by liquation method and then by poling methods. Finally it is purified by electrolytic refining using impure tin as anode, pure tin as cathode and solution containing H_2SiF_6, $Sn(SO_4)_2$ and H_2SO_4 as electrolyte.
- It becomes brittle at 200ºC and can be powdered. It produces a peculiar sound when it is bent. The cracking sound is known as **tin cry**. It exists in three allotropic forms.

$$\text{Grey Sn} \xrightleftharpoons{13.2°C} \text{White Sn} \xrightleftharpoons{160°C} \text{Rhombic Sn}$$

- Most stable allotrope of tin is white tin. Conversion of white tin into grey tin in cold countries makes it brittle. This is termed as **tin disease** or **tin pest** or **tin plague**.

COMPOUNDS OF TIN

Stannous Oxide (SnO)

- It can be prepared by the following methods.

$$Sn(OH)_2 \xrightarrow[CO_2]{\Delta} SnO + H_2O$$

$$SnCl_2 + 2NaOH \xrightarrow{\Delta} SnO + 2NaCl + H_2O$$

$$SnC_2O_4 \xrightarrow{\Delta} SnO + CO + CO_2$$

- It is dark grey or black powder insoluble in water but dissolves in both acids and bases showing its amphoteric nature.

Stannic Oxide (SnO_2)

- It is prepared as follows.

$$Sn + 4HNO_3 \longrightarrow H_2SnO_3 + 4NO_2 + H_2O$$

$$H_2SnO_3 \xrightarrow{\Delta} SnO_2 + H_2O$$

$$Sn + O_2 \xrightarrow{\Delta} SnO_2$$

- It is also amphoteric and dissolves in both acids and alkalis.
- It is used to glaze in ceramics and as pigments in pottery. A film of SnO_2 is put on air craft windows and prevents the window from frosting up.

Stannous Chloride

- It is prepared by dissolving tin in conc. HCl as $SnCl_2 \cdot 2H_2O$. Anhydrous $SnCl_2$ cannot be prepared by heating $SnCl_2 \cdot 2H_2O$ because it undergoes hydrolysis giving, white precipitate of basic chloride.

$$SnCl_2 \cdot 2H_2O \longrightarrow Sn(OH)Cl + HCl$$

- Anhydrous $SnCl_2$ can be prepared by heating tin in a current of dry HCl gas or by the distillation of excess of tin with Hg_2Cl_2 gives $SnCl_2$ as residue.

$$Sn + Hg_2Cl_2 \longrightarrow SnCl_2 + 2Hg$$

- It is soluble in water but on long standing the precipitate of $Sn(OH)Cl$ settles due to hydrolysis. Hydrolysis can be prevented by adding HCl.
- It is a strong reducing agent in the presence of HCl and reduces several substances. e.g., Hg_2Cl_2 to Hg; $FeCl_3$ to $FeCl_2$, $CuCl_2$ to $CuCl$, I_2 to I^-, decolourizes permanganate and reduces orange red dichromate to green chromic salt.
- It acts as a Lewis acid and form addition compounds like $SnCl_2 \cdot NH_3$, $SnCl_2 \cdot 2NH_3$.

Stannic Chloride ($SnCl_4$)

- It can be prepared by passing dry Cl_2 gas over tin or by distillation of $HgCl_2$ with Sn.

$$Sn + 2Cl_2 \longrightarrow SnCl_4$$

$$Sn + 2HgCl_2 \longrightarrow SnCl_4 + 2Hg$$

- It is colourless fuming liquid, soluble in benzene due to covalent nature forms hydrated salts with water $SnCl_4$. $3H_2O$, $SnCl_4 \cdot 5H_2O \cdot SnCl_4 \cdot 6H_2O$, $SnCl_4 \cdot 8H_2O \cdot SnCl_4 \cdot 5H_2O$ is called **butter of tin or oxymuriate of tin**.
- With excess of water it hydrolyses and on evaporation gives chlorostannic acid. $H_2SnCl_4 \cdot 6H_2O$.

$$SnCl_4 + 4H_2O \longrightarrow Sn(OH)_4 + 4HCl$$

- $SnCl_4 \cdot 5H_2O$ is used as mordant in dyeing. Its double salt $SnCl_4 \cdot 2NH_4Cl$ is used in calico printing.

LEAD

- Important ores of lead are (i) Galena - PbS (ii) Cerrusite $PbCO_3$ (iii) Anglisite $PbSO_4$ (iv) Lanarkite PbO. $PbSO_4$ Mainly extracted from galena.
- Galena is purified by froth floatation method and then roasted to convert into PbO. Some lime stone is added to prevent the formation of $PbSO_4$.

$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$
$$PbS + 2O_2 \longrightarrow PbSO_4$$
$$PbSO_4 + CaCO_3 \longrightarrow PbCO_3 + CaSO_4$$
$$PbCO_3 \longrightarrow PbO + CO_2$$

- The roasted ore on reduction with coke or CO in blast furnace gives lead.

$$2PbO + C \longrightarrow 2Pb + CO_2$$

- Lead is also extracted from galena by autoreduction or self reduction.

$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$
$$PbS + 2PbO \longrightarrow 3Pb + SO_2$$

- Crude lead is purified first by cooling the liquid metal to near the freezing point to remove first Cu and then Zn containing Ag or Au.
- Preferential oxidation removes As and Sn in the form of As_2O_3 and SnO_2 which float on the molten metal and can be skimmed off.
- Pure lead is prepared by electrolytic refining using impure lead as anode, pure lead as cathode and the electrolyte containing 1 – 15% $PbSiF_6$ and 5 – 10% H_2SiF_6.
- Dry air has no reaction on lead due to the presence of basic lead carbonate $2PbCO_3 \cdot Pb(OH)_2$ as protective layer.

$$2Pb + 2H_2O + O_2 \longrightarrow 2Pb(OH)_2 \xrightarrow{2CO_2} 2PbCO_3 + 2H_2O$$

- Lead is readily corroded in water containing dissolved oxygen forming $Pb(OH)_2$, soluble in water, rendering poisonous. This is known as **plumbosolvency**. Plumbosolvency is accelerated by the presence of nitrates and ammonium salts in water while the presence of carbonates, sulphates and phosphates retards the plumbosolvency due to the formation of a protective layer.
- With conc. HC*l* it forms chloroplumbous acid. dil HC*l* do not react with lead due to formation of insoluble $PbCl_2$ layer.

$$Pb + 2HCl \longrightarrow PbCl_2 + H_2$$
$$PbCl_2 + 2HCl \longrightarrow H_2PbCl_4$$

- Lead becomes passive with conc. H_2SO_4 due to the formation of insoluble $PbSO_4$ on the surface.
- Lead with dil. HNO_3 gives NO and with conc. HNO_3 gives NO_2.

$$3Pb + 8HNO_3 \longrightarrow 3Pb(NO_3)_2 + 4H_2O + 2NO$$
$$Pb + 4HNO_3 \longrightarrow Pb(NO_3)_2 + 2H_2O + 2NO_2$$

- Lead dissolves in alkalis liberating H_2.

$$Pb + 2NaOH + H_2O \longrightarrow NaPbO_3 + 2H_2$$

COMPOUNDS OF LEAD

Lead Monoxide (PbO)

- It exist in two allotropic forms yellow powder called **massicot** and reddish yellow crystalline form called **litharge**.

$$2Pb + O_2 \xrightarrow{573\text{ K}} \underset{\text{Massicot}}{PbO} \xrightarrow{1150\text{ K}} \underset{\text{Litharge}}{PbO}$$

- It is also formed by heating $Pb(NO_3)_2$. It is an amphoteric oxide reacts with both acids and bases. When heated in oxygen it converts into red lead.
- It is used in making glass and ceramic glazes, in paints and varnishes.

Lead Dioxide

- It is prepared by the oxidation of lead salt with bleaching powder containing slaked lime or with NaOC*l* containing NaOH.

$$Pb(CH_3COO)_2 + Ca(OH)_2 + CaOCl_2 \longrightarrow PbO_2 + Ca(CH_3COO)_2 + CaCl_2 + H_2O$$
$$PbCl_2 + NaOCl + 2NaOH \longrightarrow PbO_2 + 3NaCl + H_2O$$

- It is chocolate brown powder, insoluble in water and dilute acids. With conc. HC*l* gives chloroplumbic acid H_2PbCl_4.
- It dissolves in NaOH forming sodium plumbate Na_2PbO_3.
- It is a strong oxidizing agent oxidizes several substances.

$$PbO_2 + SO_2 \longrightarrow PbSO_4$$
$$PbO_2 + 4HCl \longrightarrow PbCl_2 + H_2O + Cl_2$$
$$2MnSO_4 + 6HNO_3 + 5PbO_2 \longrightarrow 2HMnO_4 + 2PbSO_4 + 3Pb(NO_3)_2 + 2H_2O$$

- Used in the manufacture of matches, as cathode in lead accumulator.

RED LEAD (Pb_3O_4)

- It is formed by heating litharge in air 450ºC.

$$6PbO + O_2 \longrightarrow 2Pb_3O_4$$

- It is a mixed oxide $PbO_2 \cdot 2PbO$, scarlet red crystalline powder insoluble in water.

- With dil. HNO_3 it forms $Pb(NO_3)_2$ and brown PbO_2 precipitate.

$$Pb_3O_4 + 4HNO_3 \longrightarrow PbO_2 + 2Pb(NO_3)_2 + 2H_2O$$

- It oxidizes conc. HC*l*

$$Pb_3O_4 + 8HCl \longrightarrow 3PbCl_2 + 4H_2O + Cl_2$$

- On heating with conc. H_2SO_4 it liberates O_2.

$$2Pb_3O_4 + 6H_2SO_4 \longrightarrow 6PbSO_4 + 6H_2O + O_2$$

- On heating converts into PbO giving off O_2.

Lead (II) Chloride

- It can be prepared by heating lead with Cl_2 or by adding HC*l* to lead nitrate.
- It is a white crystalline solid slightly soluble in cold water but soluble in hot water. Dissolves in conc. HC*l* forming chloro plumbous acid H_2PbCl_4.

Lead (IV) Chloride

- It is prepared by dissolving PbO_2 in conc. HC*l* and passing Cl_2 gas through the solution when a dark brown solution of H_2PbCl_6 is obtained. It is then treated with NH_4Cl to get $(NH_4)_2PbCl_6$ which with H_2SO_4 gives yellow liquid of $PbCl_4$.

$$PbO_2 + 6HCl \longrightarrow H_2PbCl_6 + 2H_2O$$

$$H_2PbCl_6 + 2NH_4Cl \longrightarrow (NH_4)_2\,PbCl_6 + 2HCl$$

$$(NH_4)_2PbCl_6 + H_2SO_4 \longrightarrow (NH_4)_2SO_4 + 2HCl + PbCl_4$$

- It fumes in moist air due to hydrolysis.

$$PbCl_4 + 2H_2O \longrightarrow PbO_2 + 4HCl$$

- At room temperature it decomposes forming $PbCl_2$ and Cl_2.

Single Answer Questions

1. An inorganic compound (A) made by two most occurring elements in the earth crust having polymeric tetrahedral network structure. With carbon, compound (A) produces a poisonous gas (B) which is the most stable diatomic molecule. Compound (A) and (B) will be
   (a) SiO_2, CO_2 (b) SiO_2, CO
   (c) SiC, CO (d) SiO_2, N_2
2. When oxalic acid is heated with conc. H_2SO_4, two gases produced are neutral and acidic in nature respectively. Potassium hydroxide absorbs one of the two gases. The product formed during this absorption and the gas which absorbed are respectively.
   (a) K_2CO_3 and CO_2 (b) $KHCO_3$ and CO_2
   (c) K_2CO_3 and CO (d) $KHCO_3$ and CO
3. A scarlet red compound (X) on treatment with conc. HNO_3 gives compounds (Y) and (Z) · (Z) with HC*l* produces a chloride compund (A) which can also be produced by treating (X) with conc HCl. Compounds (X), (Z) and (A) will be
   (a) Mn_3O_4, MnO_2, $MnCl_2$
   (b) Pb_3O_4, PbO_2, $PbCl_2$
   (c) Fe_3O_4, Fe_2O_3, $FeCl_2$
   (d) Fe_3O_4, Fe_2O_3, $FeCl_3$
4. In which of the following silicates, only two corners per tetrahedron are shared?
   I. Pyrosilicate II. Cyclic silicate
   III. Double chain silicate
   IV. Single chain silicate V. 3D silicate
   VI. Sheet silicate
   (a) I, II, III (b) IV and VI only
   (c) I and VI only (d) II and IV only
5. Select the false statement about silicates.
   (a) Cyclic silicate having three Si atoms contains six Si – O linkages.
   (b) $2\frac{1}{2}$ corner oxygen atoms per tetrahedron unit are shared in double chain silicate.
   (c) $(Si_2O_5)_n^{2n-}$ is the formula of double chain silicate.
   (d) SiO_4^{4-} units polymerize to form silicate because Si atom has less tendency to form π bond with oxgen.
6. The silicate anion in the mineral kaionite is a chain of three SiO_4 tetrahedra that share corners with adjacent tetrahedra. The mineral also contains Ca^{2+} ions, Cu^{2+} ions, and water molecules in a 1 : 1 : 1 ratio mineral is represented as:
   (a) $CaCuSi_3O_{10} \cdot H_2O$ (b) $CaCuSi_3O_{10} \cdot 2H_2O$
   (c) $Ca_2Cu_2Si_3O_{10} \cdot 2H_2O$ (d) none of these
7. $Si_2O_7^{6-}$ anion is obtained when.
   (a) no oxygen of a SiO_4 tetrahedron is shared with another SiO_4 tetrahedron.
   (b) one oxygen of a SiO_4 tetrahedron is shared with another SiO_4 tetrahedron.
   (c) two oxygen of a SiO_4 tetrahedron is shared with another SiO_4 tetrahedron.
   (d) three or all four oxygen of a tetrahedron are shared with other SiO_4 tetrahedron.
8. $PbCl_2$ is more ionic than PbO_2 because
   (a) The radius of Pb^{2+} is more than that of Pb^{4+}
   (b) Of inert pair effect
   (c) Chlorine is more electronegative than oxygen
   (d) Chlorine atom is smaller than oxygen atom
9. The hydrolysis of dimethyl dichloro silane gives
   (a) Silicones (b) Silicates
   (c) SiO_2 (d) Silicic acid

10. PbF_4 is good fluorinating agent because
(a) The Pb – F bond is very weak.
(b) It is a fluoride of a heavy metal.
(c) It is unstable and dissociates on warm, forming fluorine.
(d) Fluorine is highly electronegative.

11. Silicones are
(a) Synthetic polymers containing repeated R_2SiO units.
(b) Silicates with common SiO_4 unit.
(c) Ketones with silyl group (SiH_3) similar to alkyl, $(SiH_3)_2$ CO.
(d) Zircon (meso silicates).

12. In the carbon family, the elements other than carbon do not form pπ – pπ bonds, because the
(a) atomic orbitals are too small and diffused to undergo effective lateral overlap.
(b) atomic orbitals are too large and diffused to undergo effective lateral overlap.
(c) atomic orbitals are too large and less diffused to overlap linearly.
(d) atomic orbitals are too small to overlap both laterally and linearly.

13. Which of the following statements is incorrect in the context of diamond?
(a) Each C atom is tetrahedrally surrounded by four other carbon atoms, each at a distance of 154 pm.
(b) The tetrahedral units are linked together into a three-dimensional giant molecule.
(c) The unit cell has a body-centred cubic structure.
(d) Strong covalent bonds extend in all directions.

14. Even though thermodynamically favourable, the conversion of diamond into graphite does not occur normally since.
(a) the activation energy of the process is high.
(b) the activation energy of the process is low.
(c) the change of entropy is zero.
(d) the change of enthalpy is zero.

15. Calcium carbide on heating with nitrogen at 1100°C in an electric furnace produces
(a) calcium (b) calcium nitride
(c) calcium amide (d) calcium cyanamide

16. Solid CO_2 is produced as white snow by
(a) cooling the gas below its inversion temperature.
(b) cooling the gas below Boyle's temperature.
(c) expanding the gas at a high temperature.
(d) adiabatically expanding the compressed gas from the cylinder by allowing it to escape through a nozzle.

17. The correct order of the increasing carbon-oxygen bond length of CO, CO_3^{2-} and CO_2 is.
(a) $CO_3^{2-} < CO_2 < CO$ (b) $CO < CO_3^{2-} < CO_2$
(c) $CO < CO_3^{2-} < CO_2$ (d) $CO < CO_2 < CO_3^{2-}$

18. The silicate minerals are classified on the basis of the manner of linking of
(a) SiO_4^{2-} tetrahedral units
(b) SiO_4^{4-} tetrahedral units
(c) SiO_7^{2-} units
(d) $(SiO_3)_n^{4n-}$ triangular units

19. Which of the following minerals classified as an orthosilicate?
(a) $CaMg[(SiO_3)_2]$ (b) $Na_4Si_2O_7$
(c) $Ca_3[Si_3O_9]$ (d) $Zn_2[SiO_4]$

20. The length of the N - Si bond in $(SiH_3)_3N$ is shorter than what is normally expected for an N – Si single bond. This is due to
(a) $sp^2 - sp^2$ σ overlap between N and Si atoms.
(b) localized pπ - dπ bonding between the N atom and one of the three Si atoms.
(c) delocalized four-centred two-electron pπ-dπ bonding spread over the N atom and all the three Si atoms.
(d) localized pπ - pπ bonding between the N atom and one Si atom.

21. The melting point of AlF_3 is 104°C and that of SiF_4 is –77°C (it sublimes) because
(a) there is a very large difference in the ionic character of the Al – F and Si – F bonds.
(b) In AlF_3, Al^{3+} interacts very strongly with the neighbouring F^- ions to give a three dimensional structure but in SiF_4 no such interaction is possible.
(c) the silicon ion in the tetrahedral SiF_4 molecule is not shielded effectively from the fluoride ions whereas in AlF_3, then Al^{3+} ion is shielded on all sides.
(d) the attractive forces between the SiF_4 molecules are strong whereas those between the AlF_3 molecules are weak.

22. Sn(II) is a strong reducing agent than Pb(II) because
(a) Pb(II) is more stable than Sn(II)
(b) Pb(II) is less stable than Pb(IV)
(c) Sn(IV) forms more covalent compounds
(d) Pb(II) forms covalent compounds

23. The silicate anion in the kaionite is a chain of three SiO_4^{4-} tetahedra, that share corners with adjacent tetrahedra. The charge of the silicate anion is
(a) – 4 (b) – 8 (c) – 6 (d) – 2

24. The geometry with respect to the central atom of the following molecules are
$N(SiH_3)_3$; $(CH_3)_3N$; $(SiH_3)_3P$
(a) Planar, pyramidal, planar
(b) Planar, pyramidal, pyramidal

(c) Pyramidal, pyramidal, pyramidal
(d) Pyramidal, planar, pyramidal

25. $H_2C_2O_4(B) \xrightarrow{\Delta}$ gas (A) + gas (B) + liquid (C).
Gas (A) burns with a blue flame and is oxidized to gas (B)
Gas (A) + $Cl_2 \longrightarrow$ (D) $\xrightarrow{NH_3\Delta}$ (E)
A, B, C, and E are:
(a) CO_2, CO, H_2O, $HCONH_2$
(b) CO, CO_2, $COCl_2$, $HCONH_2$
(c) CO, CO_2, H_2O, NH_2CONH_2
(d) CO, CO_2, H_2O, $COCl_2$

26. The structure of silicon (IV) oxide belongs to the type
(a) ionic lattice
(b) macromolecular, with a layer structure
(c) molecular lattice, with Van der Waals' forces among the molecules
(d) macromolecular, with a non-layer structure

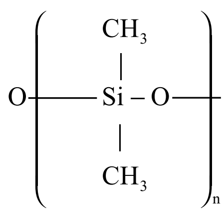
27. The correct order of decreasing ionic nature of lead dihalides is
(a) $PbF_2 > PbCl_2 > PbBr_2 > PbI_2$
(b) $PbF_2 > PbBr_2 > PbCl_2 > PbI_2$
(c) $PbF_2 < PbCl_2 > PbBr_2 < PbI_2$
(d) $PbI_2 < PbBr_2 < PbCI_2 < PbF_2$

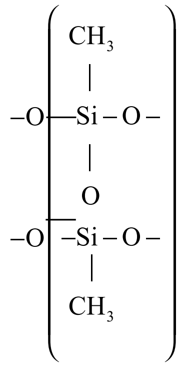
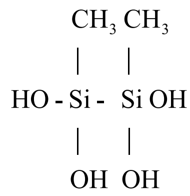
28. A student prepared a sample of silicon chloride by passing chlorine over heated silicon and collecting the condensed silicon chloride in a small specimen tube. He analysed the chloride by dissolving a known mass of it in water and titrating the solution with standard silver nitrate solution. The formula of the silicon chloride as obtained by this method was $SiCl_{2.6}$ as against a true formula $SiCl_4$.

Which of the following possible errors could have resulted in this wrong formula?
(a) The silicon chloride contained excess dissolved chlorine.
(b) More silicon chloride than the student supposed was actually used owing to inaccurate weighing.
(c) The small specimen tube was not dry.
(d) The reaction between the silicon and the chlorine stopped prematurely leaving some unreacted silicon in the reaction tube.

29. The dehydration of malonic acid $CH_2(COOH)_2$ with P_4O_{10} and heat give
(a) Carbon monoxide (b) Carbon sub oxide
(c) Carbon dioxide (d) All three

30. $CH_3Cl + Si \xrightarrow[570K]{Cu}$ A 'A' on hydrolysis followed by polymerization gives

(a) $Si(CH_3)_3 - Si(CH_3)_3$ (b) 

(c)  (d) 

31. Select incorrect statement
(a) Cyanamide ion $(CN_2)^{2-}$ is iso-electronic with CO_2 and has the same linear structure.
(b) Mg_2C_3 reacts with water to form propyne.
(c) CaC_2 has C_2^{2-} and it contains one sigma and two pi bonds.
(d) Al_4C_3 is an example of methanide carbide and it contains C_3^{4-}.

32. Name the structure of silicates in which three oxygen atoms of $[SiO_4]^{4-}$ are shared
(a) Pyrosilicate
(b) Sheet silicate
(c) Linear silicate
(d) Three dimensional silicate

33. Which of the following statements is not correct?
(a) The durability and inertness of silicones is due to the high bond enthalpy of Si - O bond.
(b) Silicones are used in water - proofing textiles.
(c) Silicone rubbers are excellent electrical insulators.
(d) The silicones always involve cross-linked between Si and O atoms.

34. $(COOH)_2 \xrightarrow{Conc.\ H_2SO_4}$ A + B + H_2O, A can be identified by using iodine pentoxide then incorrect statement about 'B' is
(a) Solid B is sublimated compound.
(b) B reacts with ammonia to form urea.
(c) Supercritical B is used as solvent.
(d) It is linear and neutral oxide.

35. Asbestos has composition as
(a) $CaO \cdot Al_2O_3 \cdot SiO_2 \cdot H_2O$
(b) $CaO \cdot 3MgO \cdot 4SiO_2$
(c) $3MgO \cdot 4SiO_2 \cdot H_2O$
(d) $Al_2O_3 \cdot SiO_2 \cdot 2H_2O$

36. How many oxygen atoms of SiO_4^{4-} units shared in the continuous 3D frame work silicates?
(a) 2 (b) 2 (c) 3 (d) 4

37. Trisilylamine $(SiH_3)_3$ N is
(a) Trigonal pyramidal and acidic
(b) Trigonal pyramidal and basic

(c) Trigonal pyramidal and neutral
(d) Trigonal planar and weakly basic

38. The inter layer distance in graphite is
(a) Very small, the layers being tightly packed.
(b) Many times greater than the covalent radius of carbon.
(c) Approximately 4½ times the covalent radius of Carbon.
(d) The same as the covalent radius of carbon.

39. In the equilibrium $C_{(s)}$ (diamond) $\rightleftharpoons$ $C_{(s)}$ (graphite) + heat (density of diamond and graphite are 3.5 and 2.3g/cm^3 respectively) the equilibrium will be shifted to the left at
(a) low temperature and very high pressure
(b) high temperature and low pressure
(c) high temperature and high pressure
(d) low temperature and low pressure

40. Carbon disulphide react with chlorine to give
(a) Carbon tetrachloride and sulphur monochloride
(b) Carbon tetrachloride and sulphur tetrachloride
(c) Carbon and sulphur dichloride
(d) Carbon and sulphur monochloride

41. In graphite which have several fused hexagonal rings of benzene the hybridization state of each carbon atom and the bond order of each carbon-carbon bond are respectively
(a) sp, 1.5 (b) sp^2, 1.5
(c) sp^2, 1.33 (d) sp^3, 1.5

42. Often ground-glass stopper gets stuck in the neck of a glass bottle containing an NaOH solution. The reason is that
(a) there are particles of dirt in between.
(b) a solid silicate is formed in between by the reaction of the SiO_2 of glass with NaOH.
(c) Solid Na_2CO_3 is formed in between by reaction of the CO_2 of air and NaOH.
(d) glass contain a boron compound which forms a precipitate with the NaOH solution.

43. The hybridization state of central atoms in the molecule $(CH_3)_3N$ and $(SiH_3)_3N$ and their shapes respectively are
(a) sp^3 with pyramidal shape in $(CH_3)_3N$ and sp^2 with planar triangular shape in $(SiH_3)_3N$.
(b) In both $(CH_3)_3N$ and $(SiH_3)_3N$, the hybridizaiton of N is sp^3 and both are pyramidal in shape.
(c) In both $(CH_3)_3N$ and $(SiH_3)_3$ N the hybridization of N is sp^2 and both are planar triangular in shape.
(d) sp^2 with planar triangular shape in $(CH_3)_3N$ and sp^3 with tetrahedral shape in $(SiH_3)_3N$.

44. The lone pair of electrons in the nitrogen atoms of the molecules $(CH_3)_3N$ and $(SiH_3)_3N$ are present respectively in the
(a) sp^3 orbital and sp^2 orbital
(b) p orbital and sp^2 - orbital
(c) sp^3 orbital and p$^-$ orbital
(d) p orbital in both cases

45. Which of the following statements is correct for silicon?
(a) forms molecular halides that are not hydrolysed.
(b) forms strong but unconjugated multiple bonds of the pπ- dπ variety, especially with O and N.
(c) does not undergo coordination number expansion.
(d) forms an oxide that has bonding similar to carborundum.

46. Which of the following pairs of ions represent in cyclic and double chain silicates?
(a) $Si_2O_7^{2-}$ and $(SiO_3)_3^{2n-}$
(b) $Si_3O_9^{6-}$ and $(Si_4O_n)_n^{6n-}$
(c) Si_2O_7 and $(Si_2O_5)_n^{2n-}$
(d) $Si_2O_7^{7-}$ and $(SiO_3)_3^{2n-}$

47. Carborundum on heating with caustic soda in the presence of air produces
(a) $Na_2SiO_3 + H_2$ (b) $Na_2SiO_3 + Na_2CO_3$
(c) $Na_2SiO_2 + H_2$ (d) $Na_2SiO_4 + O_2$

48. In $(Si_2O_5)_n^{2n-}$
(a) no oxygen of SiO_4^{4-} tetrahedron is shared with another tetrahedron.
(b) one oxygen of SiO_4^{4-} tetrahedron is shared with another tetrahedron.
(c) two oxygens of SiO_4^{4-} tetrahedron are shared with another SiO_4^{4-} tetrahedron.
(d) three oxygens of SiO_4^{4-} tetrahedron are shared with another SiO_4^{4-} tetrahedron.

49. $A \xrightarrow{\text{Red hot coke}} CO \xrightarrow{Cl_2} C \xrightarrow{H_2O}$ 2 HCl + A. The compounds A and C are
(a) CO_2, COCl_2 (b) CO, COCl_2
(c) C, CO_2 (d) CO_2, CO

50. The silicates having layer and sheet structures involves
(a) the discrete SiO_4^{4-} tetrahedra.
(b) the sharing of one oxygen atom between two SiO_4^{4-}tetrahedra.
(c) the sharing of two oxygen atoms of one SiO_4^{4-} tetrahedron with two other tetrahedra.
(d) the sharing of three oxygen atoms of one SiO_4^{4-} tetrahedron with three other tetrahedra.

51. An inorganic compound (X) on hydrolysis produces a gas which on treatment with sodium followed by its reaction with ethyl chloride forms another compound (Y). Compound (Y) on heating with Pd catalyst

gives (2Z) - pent - 2 - ene as major product. Hence the inorganic compound (X) is
(a) Tl_4C (b) BaC_2 (c) SiC (d) Mg_2C_3

52. Which of the following statement is correct:
(a) C – F bond is stronger than Si – F because C – F bond length is shorter than that of Si – F.
(b) C – F bond is weaker than Si – F bond because of less difference in electronegativity.
(c) Si – F bond is stronger than C – F bond because of double bond character due to back bonding from F to Si.
(d) Si – F bond is stronger than C – F bond due to more difference in electronegativities.

53. The wrong statement among the following
(a) Diamond is the best conductor of heat, so far known
(b) Graphite act as metallic conductor parallel to the layers but act as semiconductor perpendicular to the layers.
(c) C – C bond order in graphite is 1.5.
(d) SiO do not exist but CO is stable though stability of +2 oxidation state increase down the group because silicon cannot form $p\pi - p\pi$ bonds.

More than One Answer Type Questions

1. Which of the following statements are correct about silicones?
(a) They are formed by the hydrolysis of R_2SiCl_2.
(b) They are polymers made up of R_2SiO units.
(c) They are made up of SiO_4^{4-} units.
(d) They are macro molecules.

2. Which of the following statements(s) is/are correct?
(a) PbI_4 is not known because of the oxidizing power of Pb(+IV) and the reducing power of I^-.
(b) $SnCl_2$ does not hydrolyse.
(c) $SnCl_2$ is reducing agent.
(d) Tin (II) chloride is linear in shape.

3. Which of the following statements is/are correct?
(a) Trisilyl amine $(SiH_3)_3N$ is a weaker base than trimethyl amine $(CH_3)_3N$.
(b) Silicon has more tendency to form π bonds with oxygen than phosphorus.
(c) Structure of silicon is same as that of diamond.
(d) SiO_2 is a high melting solid.

4. When SO_2 is passed through sodium carbonate solution, then CO_2 gas is liberated. This is because
(a) CO_2 is more volatile than SO_2.
(b) CO_2 has a lower molecular weight than SO_2.
(c) SO_2 is a stronger acid than CO_2.
(d) CO_2 is a stronger acid than SO_2.

5. Identify A, B, C and D in the following reaction sequence.

$$\begin{array}{ccccc} Si + CH_3Cl & \xrightarrow[570\ k]{\text{Cu powder}} & (A) & \xrightarrow{H_2O} & (B) \\ & & \downarrow O_2 & & \downarrow \text{Polymerization} \\ & & (C) \xrightarrow[\text{electric arc}]{\text{Excess of coke}} & & (D)\ \text{Silicone} \end{array}$$

(a) The compound A is $(CH_3)_2\ SiCl_2$
(b) The compound C and D are SiO_2 and SiC respectively
(c) The compound B is $(CH_3)_2\ Si(OH)_2$.
(d) The compound B is $(CH_3)_2\ Si = O$

6. $SiO_2 + NaOH \longrightarrow A$

$SiO_2 \xrightarrow{H_2F_2} B \xrightarrow{H_2O} C + D$

Hence
(a) $A = Na_2SiO_3$
(b) $B = SiF_4$
(c) C and D are H_2SiO_3 and H_2SiF_6 respectively
(d) C and D are F_2 and H_2O_2 respectively

7. $C(OH)_4$ is unstable because a carbon atom cannot hold more than one – OH but $Si(OH)_4$ is a stable compound because
(a) Carbon can form strong $p\pi$ - $p\pi$ bonds with oxygen by eliminating water.
(b) Silicon atom due to bigger size cannot form strong $p\pi$ - $p\pi$ bonds with oxygen by eleminating water.
(c) Silicon can form strong $d\pi$ - $p\pi$ bond with lone pairs on oxygen.
(d) Silicon being more electronegative than carbon can hold four OH groups.

8. Which of the following is / are correct for group 14 elements?
(a) The stability of dihalides are in the order $CX_2 < SiX_2 < GeX_2 < SnX_2 < PbX_2$.
(b) The ability to form $p\pi$ - $p\pi$ multiple bonds among themselves increases down the group.
(c) The tendency for catenation decreases down the group.
(d) They all form oxides with the formula MO_2.

9. SiO_2 reacts with
(a) Na_2CO_3 (b) CO_2 (c) HF (d) HCl

10. Which of the following statements (S) is/ are true?
(a) The lattice structure of diamond and graphite are different.

(b) Graphite is an impure form of carbon while diamond is a pure form.
(c) Graphite is harder than diamond.
(d) Graphite is thermally more stable than diamond.

11. Which of the following statement is correct?
(a) Organo silicon polymers are known as silicones.
(b) Silicones have the general formula $(R_2SiO)_n$ where R is CH_3, C_2H_5, C_6H_5 etc.
(c) Hydrolysis of dialkyl dichlorosilane produces cross-linked silicone polymer.
(d) Hydrolysis of alkyl trichlorosilane produces cross-linked silicone polymer.

12. In which of the following all the atoms are in sp^3 hybridization?
(a) Diamond
(b) Carborundum
(c) Crystalline silicon
(d) Quartz

13. Among the following statements the correct statements(s) is/are
(a) Of all the elements carbon exhibit maximum catenation power.
(b) Silanes are less stable than hydrocarbons because – I effect of hydrogen, decreases the electron density in Si-Si bond.
(c) Silicon exhibit more catenation power in halides than in hydrides due to $p\pi$- $p\pi$ nature.
(d) CS_2 is a volatile liquid while SiS_2 is high melting solid due to polymeric structure.

14. Which among the following statements are correct?
(a) Aqua dag and oil dag are made up of graphite
(b) Graphite reacts with conc. HNO_3 to form mellitic acid $C_6(COOH)_6$.
(c) C_3O_2 is also toxic like CO.
(d) Zircon, $ZrSiO_4$ is a gemstone.

15. Which of the following carbides yield CH_4 gas on hydrolysis?
(a) Al_4C_3 (b) Be_2C
(c) Mg_2C_3 (d) Na_2C_2

16. Which of the following statements are correct?
(a) Lithium is the only alkali metal to form a stable nitride.
(b) With Me_3N as donor, the stability order of donor-acceptor complexes of boron compound is $BBr_3 > BCl_3 > BF_3 > BMe_3$.
(c) $NaBH_4$ is very much less rapidly hydrolysed by water than $NaAlH_4$.
(d) Magnesium silicides reacts with ammonium bromide in liquid ammonia to form silane.

17. Which of the following statement (s) are correct?
(a) The C – C bond order in graphite is 1.5.
(b) Graphite can act as electron donor or electron acceptor toward atom and ions that penetrate between the layers.
(c) The electrical conductivity of graphite is much similar to metals in the direction parallel to planes, but behaves like semi conductor in the direction perpendicular to planes.
(d) The oxidation products of graphite with alkaline permanganate support the hexagonal structure of graphite

18. Which of the following statements is/are correct?
(a) Aluminium carbide as well as beryllium carbide produce methane gas on treatment with water.
(b) On reacting with water calcium carbide produces acetylene while magnesium carbide gives propyne.
(c) Calcium carbide has a lattice similar to that of NaCl but the unit cell is elongated in one direction.
(d) Calcium carbide on heating with nitrogen form calcium imide.

19. Which of the following regarding fullerene (C_{60}) is/are correct?
(a) All carbon atoms are sp^2 hybridized.
(b) In fullerene the five membered rings and six membered rings are in 1 : 2 ratio.
(c) A five membered ring may be fused either with another five membered ring or with a six membered ring.
(d) It is aromatic.

20. Which of the following statements is incorrect?
(a) The CO_2 molecule behaves as a nonpolar molecule even though two of its resonating structures $\bar{O}-C\equiv\overset{+}{O}$ and $\overset{+}{O}\equiv C-\bar{O}$ are dipolar.
(b) Carbon dioxide is the anhydride of the unstable dibasic acid $O = C(OH)_2$.
(c) The CO_2 molecule is linear because the carbon atom utilizes its sp orbitals to form σ bonds.
(d) The carbon atom is sp^2 hybridized in the CO_2 molecule as well as its hydrate H_2CO_3.

21. Which of the following statement(s) is/are correct?
(a) Aqueous solution of sodium carbonate is alkaline because carbonate ion takes up proton and release OH^- ion from water.
(b) When sodium carbonate is added to the aqueous solutions of Al^{3+} and Fe^{3+} they are precipitated as their carbonates.
(c) If Na_2CO_3 solution is added to the aqueous solutions of Ca^{2+}, Sr^{2+}, and Ba^{z+} they are precipitated as their carbonates.

(d) Addition of Na_2CO_3 solution to the aqueous solutions of Mg^{2+}, Cu^{2+}, and Zn^{2+} precipitates them as their basic carbonates.

22. Fullerene can be prepared by
(a) the pulsed layer vapourization of graphite.
(b) vapourizing carbon by resistive heating.
(c) passing an arc discharge between carbon electrodes in a tube containing inert atmosphere at 100 torr.
(d) by heating gas carbon at very high temperatures and pressure.

23. $CaF_2 + H_2SO_4 \longrightarrow A + B$
$SiO_2 + B \longrightarrow C$
The compound C contains
(a) $[SiF_6]^{4-}$ (b) $[SiF_6]^{2-}$
(c) $[SiF_4]^{2-}$ (d) SiF_4

24. $SnCl_2$ was dissolved in sodium hydroxide to give a ..... (P) which dissolves in excess NaOH to give (Q) Q then reacts with $Bi(OH)_3$ to form a black substance (R). Hence
(a) Compound P = $Sn(OH)_2$
(b) Compound Q = Na_2SnO_2
(c) Substance R = Na_2SnO_3
(d) Substance R = Bi

25. Which of the following statements are correct?
(a) Boron carbide is an ionic carbide.
(b) In metallic carbides, carbon atoms occupy interstital positions in the crystal lattices of metals.
(c) Silicone rubber is not attacked by ozone.
(d) Dehydration of diakyldihydrosilane produces linear silicon.

26. Complete the reaction
$CS_2 + \text{aq NaOH} \longrightarrow - + - + H_2O$
(a) Na_2S (b) Na_2CS_3
(c) Na_2SO_3 (d) Na_2CO_3

27. Identify the correct statements from the following reactions:
I. $SnCl_2 + H_2O \longrightarrow X + HCl$
II. $SnCl_2 + 2NaOH \longrightarrow Y + 2NaCl$

$$\Big\downarrow \text{2NaOH (excess)}$$

$$Z + H_2O$$

III. $SnCl_2 + 2AuCl_3 \longrightarrow 3SnCl_4 + P$
IV. $SnCl_2 + 2HCl + I_2 \longrightarrow SnCl_4 + Q$
(a) X shows white turbidity and is basic.
(b) Y is white precipitate and Z is soluble in water.
(c) P is a metal.
(d) Q is an acid.

28. Which of the following reaction is/are correct?
(a) $3CS_2 + 6NaOH \longrightarrow Na_2CO_3 + 2Na_2CS_3 + 3H_2O$
(b) $SiCl_4 + LiAlH_4 \longrightarrow SiH_4 + AlCl_3 + LiCl$
(c) $SiO_2 + 4HF \longrightarrow SiH_4 + 2H_2O$
(d) $Na_2B_4O_7 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2B_4O_7$

29. Select the correct statement(s)
(a) Carbon has pronounced ability to form $p\pi$ - $p\pi$ multiple bonds to itself and to other elements like O an N.
(b) In diamond each carbon atom is linked tetrahedrally to four other carbon atoms by sp^3 hybridization.
(c) Graphite has planar hexagonal layers of carbon atoms held together by weak van der Waal's forces.
(d) Out of CO_2, SnO_2, PbO_2, CO_2, SiO_2 is amphoteric and PbO_2 is an oxidizing agent.

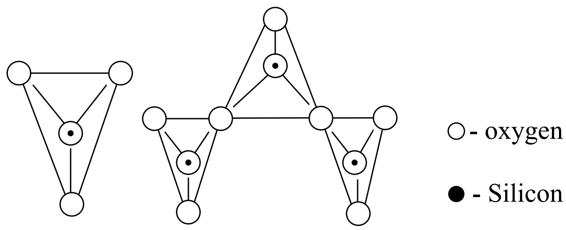
Comprehension Type Questions

Passage-I

Read the following passage and answer the questions followed by is.

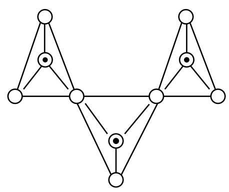
Silicates are a group of minerals which have the general formula SiO_2 the most common of which is quartz. Quartz is a frame work silicate with SiO_4 tetrahedra arranged in spirals. The spirals can turn in clockwise or anticlockwise direction a feature that results in there being two mirror images optically active varities of quartz.

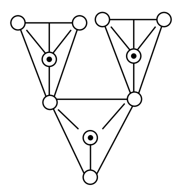
1. The following represents various silicate anions. Their formulae are respectively



(a) SiO_3^{2-}, SiO_7^{2-} (b) SiO_4^{4-} $Si_3O_{10}^{8-}$
(c) SiO_4^{2-}, $Si_3O_9^{2-}$ (d) SiO_3^{4-}, $Si_3O_7^{8-}$

2. $Si_3O_{10}^{8-}$ (having three tetrahedral units) is represented as

(a)


(b)


(c) Both (d) none

3. The silicate anion in the mineral kaionite is a chain of three SiO_4 tetrahedra that share corners with adjacent tetrahedra. The mineral also contain Ca^{2+} acid Cu^{2+} ions and water molecules in a 1 : 1 : 1 ratio. The mineral is represented as
(a) $CaCuSi_3O_{10} \cdot H_2O$
(b) $CaCuSi_3O_{10} \cdot 2H_2O$
(c) $Ca_2Cu_2Si_3O_{10} \cdot 2H_2O$
(d) $Ca_2CuSi_3O_{10} \cdot 4H_2O$

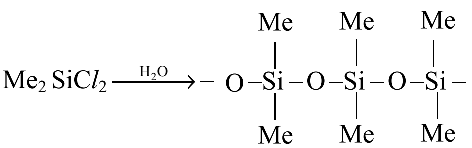
Passage-II

Read the following passage and answer the questions followed by it

Silicones are synthetic polymers containing repeated R_2SiO units. Since the empirical formula is similar to that of a ketone (R_2CO) the name silicone has been given to these materials silicones can be made into oils, rubbery elastomers and resins. They find a variety of applications because of their chemical inertness, water repelling nature, heat - resistance and good electrical insulating property.

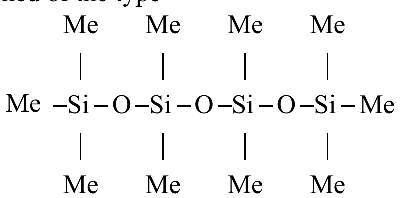
Commercial silicon polymers are usually methyl derivative and to a larger extent phenyl derivatives.

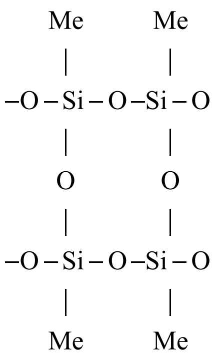
R_2SiCl_2 [R = methyl (MeE) or phenyl (ϕ)]



is mixed

1. If Me_3SiCl is mixed with Me_2SiCl_2 the silicones formed of the type

(a) 

(b) 

(c) Both
(d) Cross linked or three-dimensional silicones

2. If $MeSiCl_3$ alone is taken as starting material, the silicone formed is
(a) a linear silicone
(b) dimeric silicone
(c) cyclic silicone
(d) three dimensional silicone

3. If Me_2SiCl_2 alone is taken as starting material, the silicone formed is
(a) linear silicone (b) cyclic silicone
(c) cross linked silicone (d) both A and B

Passage-III

Although stoichiometry similarities exists between the compounds of carbon and those of the remaining elements of group IV A, there is no structural or chemical similarity between them. Ex: CO_2 is gas SiO_2 is solid. Reaction of the compounds of silicon and the lower elements of group donot give products analogous to those for carbon. For example the dehydration of alcohols gives alkene but the dehydration of silanols give different products accompanied by condensation.

In carbon, multiple bonds involve overlap of the $p\pi$ - $p\pi$ variety mulitple bonding for silicon and germanium usually arise form $p\pi$ - $d\pi$ component especially in bonds to O and N. It is important to note that this does not usually lead to conjugation as is so prevalent for carbon. The $p\pi$ – $d\pi$ bonding character decreases with increasing the atomic size down the group.

1. Among trisylylamine $(H_3Si)_3N$ and trimethyl amine $(CH_3)_3N$.
(a) $(H_3Si)_3N$ and $(CH_3)_3N$ both are pyramidal in shape and equally basic.
(b) $(H_3Si)_3N$ and $(CH_3)_3N$ both are pyramidal in shape but $(H_3Si)_3N$ is more basic than $(CH_3)_3N$.
(c) $(H_3Si)_3$ N is planar while $(CH_3)_3N$ is pyramidal and $(H_3Si)_3$ N is more basic than $(CH_3)_3N$.
(d) $(H_3Si)_3$ N is planar while $(CH_3)_3N$ is pyramidal and $(H_3Si)_3N$ is less basic than $(CH_3)_3$ N.

2. Among $(H_3Si)_3$ P and $(H_3Si)_3N$
(a) Both are pyramidal in which P and N are involved in sp^3 hybridization.
(b) Both are planar in which P and N are involved sp^2 hybridization.
(c) $(H_3Si)_3$ P is planar while $(H_3Si)_3N$ is pyramidal where P is involved in sp^2 and N is involved in sp^3 hybridization.
(d) $(H_3Si)_3$ P is pyramidal with P is involved in sp^3 hybridization while $(H_3Si)_3N$ is planar with N is involved in sp^2 hybridization.

3. Among H_2SiNCO and H_3CNCO
   (a) Both are linear
   (b) Both are angular
   (c) H_3Si NCO is linear while H_3CNCO is angular
   (d) H_3SiNCO is angular while H_3CNCO is linear

Passage-IV

Rocks, clays and soils are made up of silicates of aluminium, iron, magnesium and other metals. All silicates are made up of SiO_4 tetrahedral units in which Si is sp^3 - hybridized is surrounded by four oxygen atoms. The SiO_4 tetrahedra can be linked together in several different ways. Depending on the number of corners of the SiO_4 tetrahedra shared, various kinds of silicates are formed

1. A fibrous mineral which can withstand red hot flames without any damage is
   (a) Talc (b) Glass wool
   (c) Soap stone (d) Asbestos
2. Quartz watches contain
   (a) A crystal of quartz as an essential component
   (b) A coating of quartz on the outer body
   (c) Hands made of quartz
   (d) Silica coated on the number
3. Which of the following is not a crystalline form of silica?
   (a) Quartz (b) Tridymite
   (c) Cristobalite (d) Kieselguhr

Passage-V

Carbides are three types; ionic, covalent and interstitial. CaC_2 is one of the commercially important ionic carbide.

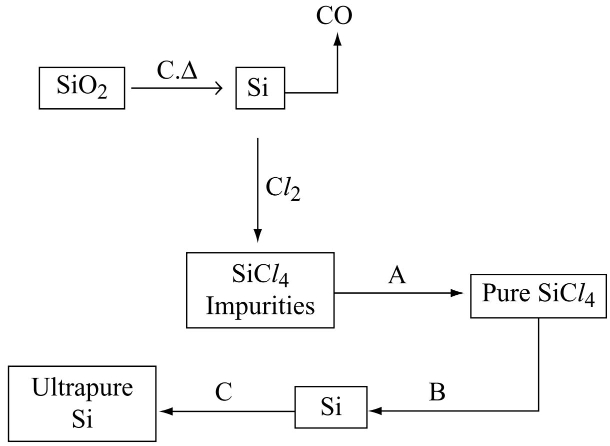
$$\underline{X} + C \xrightarrow{\Delta} CaC_2 + CO$$
$$CaC_2 + \underline{Y} \xrightarrow{1100°C} \underline{Z} + C$$
$$Z + H_2O \longrightarrow NH_3 + CaCO_3$$

'Z' is an important nitrogenous fertiliser

1. 'X' is
   (a) CaO (b) $CaCO_3$
   (c) $Ca(OH)_2$ (d) $CaCl_2$
2. Y and Z respectively are
   (a) N_2O and $Ca(CN)_2$ (b) $(CN)_2$ and $CaCN_2$
   (c) N_2 and $Ca(CN)_2$ (d) N_2 and $CaCN_2$
3. Covalent carbide among the following is
   (a) Mg_2C_3 (b) SiC
   (c) WC (d) Fe_3C

Passage-VI

Following flowsheet is used to prepare pure silicon. Answer the question based on it.



1. Step A is
   (a) fractional distillation (b) zone-refining
   (c) ultra-filtration (d) none of these
2. Step B involves
   (a) thermal decomposition
   (b) reduction with H_2
   (c) reduction with Na or Mg
   (d) hydrolysis
3. Pure quartz glass contains
   (a) SiO_2 (b) Si
   (c) SiO_2, B_2O_3 (d) SiO_2, Al_2O_3

Passage-VII

The elements like silicon, phosphorus, sulphur and chlorine which have empty d-orbitals are able to form pπ - dπ bonding in their various compounds. To obtain effective pπ - dπ overlap, the size of the d-orbital must be similar to the size of the p-orbital. On crossing a period (e.g., IIIrd period) in the periodic table, the nuclear charge is increased, the size of the atom and the size of the d-orbitals decreases form Si to Cl. The decrease in the size the 3d orbitals in this series of elements leads to progressively stronger pπ - dπ bonds. This helps (i) in determination of acidic character of various compound (ii) in understanding the shortening of bond length in various compounds (iii) in knowing the tendency of various oxoanions to under go polymerization etc.

Empty d-orbitals also help in back bonding (if some second period element having lone pairs is attached with atoms having empty d-orbitals) which decreases bond length than the expected and helps in delocalizing negative charge on anion.

1. Which of the following oxoanions have least tendency of polymerization?
   (a) SiO_4^{4-} (b) PO_4^{3-}
   (c) SO_4^{2-} (d) ClO_4^-

2. Which of the following statements is true on the basis of back bonding?
   (a) Si – O bond is stronger than C – O bond
   (b) Dimethyl ether acts as a better lewis base but not disilyl ether ($SiH_3 – O – SiH_3$)
   (c) $(CH_3)_3C – O – H$ is less acidic than $(CH_3)_3Si – O – H$
   (d) All of these
3. Which of the following statements is/are false?
   (a) NMe_3 and $N(SiMe_3)_3$ are not isostructural.
   (b) Methyl isocyanate ($CH_3 – N = C = O$) is bent but silyl isocyanate ($SiH_3 – N = C = O$) is linear.
   (c) In trisilyl amine $(SiH_3)_3N$ all N – Si bond lengths are identical but shorter than the expected N – Si bond length.
   (d) B–F bond length in BF_3 is greater than B–F bond length in $[NH_3 \cdot BF_3]$.

Comprehension Type Questions

Passage-VIII

Ionic character of metallic halides tend toward covalent nature as per Fajans rule. Such covalent halides behave as non-metal in their higher oxidation states. They exhibit property to hydrolyse to give oxy-acids of the element and corresponding halogen acid. For most non-metallic elements proceeds exceptionally in the way, keeping oxidation number of element and halide same in oxoacids.

Non-polar halides are immiscible in water, as they do not show hydrolysis, but halides of some elements with empty d-orbitals undergo hydrolysis. Stability of halides of higher state decreases down the group due to inert pair effect.

1. Which halide undergoes hydrolysis to give oxyacid of under lined elements?
   (a) $\underline{C}Cl_4$
   (b) $\underline{P}Cl_3$
   (c) $\underline{N}I_3$
   (d) $\underline{Sb}Cl_3$
2. Which one does not exist?
   (a) $PbCl_4$
   (b) PbF_4
   (c) PbI_4
   (d) $PbCI_2$
3. Which one is not correct?
   (a) $SiCl_4 + 4H_2O \longrightarrow Si(OH)_4 + 4HCl$
   (b) $SbCl_3 + 3H_2O \longrightarrow Sb(OH)_3 + 3HCl$
   (c) $AsCl_3 + 3H_2O \longrightarrow H_3AsO_3 + 3HCl$
   (d) $XeF_2 + H_2O \longrightarrow Xe + 2HF + \frac{1}{2}O_2$

Matching Type Questions

1.

Column-I	Column-II
(a) Sheet silicate	(p) $(SiO_3)_n^{2n-}$
(b) Pyroxene chain	(q) $(Si_4O_{11})_n^{6n-}$
(c) Pyro silicate	(r) 3 – corner oxygen atoms are shared
(d) Amphibole chain	(s) Non - planar

2.

Column-I	Column-II
(a) $(SiO_3)_n^{2n-}$	(p) Cyclic silicate
(b) Silicone	(q) Chain silicate
(c) $Si_2O_7^{6-}$	(r) Contain Si-O-Si bond
(d) $(Si_4O_{11})_n^{6n-}$	(s) Pyrosilicate

3.

Column-I	Column-II
(a) Orthosilicate	(p) Oxygen atoms shared < 2
(b) Pyrosilicate	(q) Oxygen atoms shared $\geq$ 2
(c) Single chain silicate	(r) Net charge = –2
(d) Ring silicate	(s) Net charge = –6

4.

Column-I	Column-II
(a) CO	(p) Neutral
(b) PbO_2	(q) Amphoteric
(c) GeO	(r) Reducing agent
(d) SnO	(s) Oxidizing agent

5.

Column-I	Column-II
(a) Cyclic silicates	(p) Tetrahedral hybridization
(b) Single chain silicate	(q) Si – O bonds are 50% ionic and 50% covalent
(c) Pyro silicates	(r) General formula is $(SiO_3)_n^{2n-}$
(d) Sheet silicates	(s) Two oxygen atoms per (two dimensional) tetrahedronare shared

Integer Type Questions

1. The total number of protons donated by one molecule of (H_3BO_3) boric acid is.
2. The covalency of silicon in hydrofluoro silicic acid is.
3. In the formation of cyclic silicones number of oxygen atoms belong to each silicon atom forming the ring is.
4. In cyclic silicate ion $Si_6O_{18}^{12-}$ the number of oxygen atoms shared

5. What is the coordination no.of Sn in crystalline layer structure of solid SnF_4?
6. Diamond is formed by the fusion of several carbon tetrahedrons in which carbon atoms form homocyclic rings. The number of carbon atoms in each ring is
7. How many of the following substances / molecules / ion have bond order 1.33?
   BF_3, Boron nitride, graphite,
   $$NO_3^{2-},\ SO_3^{2-},\ SO_4^{2-},\ PO_4^{3-}$$

Key IV Group

Single Answer Question

1. b	**2.** a	**3.** b	**4.** d	**5.** a	**6.** c	**7.** b
8. a	**9.** a	**10.** b	**11.** a	**12.** b	**13.** c	**14.** a
15. d	**16.** d	**17.** d	**18.** b	**19.** d	**20.** c	**21.** b
22. a	**23.** b	**24.** b	**25.** c	**26.** d	**27.** a	**28.** c
29. b	**30.** b	**31.** d	**32.** b	**33.** d	**34.** d	**35.** b
36. d	**37.** d	**38.** c	**39.** c	**40.** a	**41.** c	**42.** b
43. a	**44.** c	**45.** b	**46.** b	**47.** b	**48.** d	**49.** a
50. d	**51.** d	**52.** c	**53.** c			

More Than One Answer Type Questions

1. a, b, d	**2.** a, c	**3.** a, c, d
4. a, b, c	**5.** a, b, c	**6.** a, b, c
7. a, b	**8.** a, c, d	**9.** a, c
10. a, d	**11.** a, b, d	**12.** a, b, c, d
13. a, b, c, d	**14.** a, b, c, d	**15.** a, b
16. a, b, c, d	**17.** b, c, d	**18.** a, b, c
19. a, d	**20.** a, b, c	**21.** a, c, d
22. a, b, c	**23.** b	**24.** a, b, d
25. b, c, d	**26.** b, d	**27.** a, b, c, d
28. a, b, d	**29.** a, b, c, d	

Comprehensive Type Questions

Passage-I	**1.** b	**2.** c	**3.** c
Passage-II	**1.** a	**2.** d	**3.** d
Passage-III	**1.** d	**2.** d	**3.** c
Passage-IV	**1.** d	**2.** a	**3.** d
Passage-V	**1.** a	**2.** d	**3.** b
Passage-VI	**1.** a	**2.** c	**3.** a
Passage-VII	**1.** d	**2.** d	**3.** d
Passage-VIII	**1.** b	**2.** c	**3.** b

Matchings

1. a-q, r, s	b-p, s	c-s	d-p, s
2. a-p, q, r	b-r	c-r, s	d-q, r
3. a-p	b-p, s	c-q, r	d-q, r
4. a-p, r	b-q, s	c-q, r	d-q, r
5. a-p, q, r, s	b-p, q, r, s	c-p, q	d-p, q

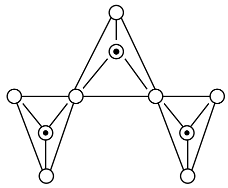
Integer Type Questions

1. 0	**2.** 6	**3.** 1	**4.** 6	**5.** 6	**6.** 6	**7.** 5

IV Group Hints

Single Answer Questions

1. The most abundant elements in the earth's crust are oxygen and silicon.They form SiO_2 which has polymeric tetrahedral network structure. This on heating with carbon forms SiC and CO.
2. Heating of oxalic acid produces CO and $CO_2 \cdot CO_2$ is absorbed by KOH forming K_2CO_3.
3. X is Pb_3O_4
   $Pb_3O_4 + 4HNO_3 \longrightarrow Pb(NO_3)_2 + PbO_2 + 2H_2O$
   $Pb(NO_3)_2 + 2HCl \longrightarrow PbCl_2 + 2HNO_3$
   $Pb_3O_4 + 8HCl \longrightarrow 3PbCl_2 + 4H_2O + Cl_2$

6. 

When three SiO_4 tetrahedrons are joined the silicate ion is $Si_3O_{10}^{8-}$. As the 8 negative charges are neutralized by cations Ca^{2+} and Cu^{2+} in 1 : 1 ratio. The formula of the silicate is

$$Ca_2Cu_2Si_3O_{10} \cdot 2H_2O.$$

8. Smaller the size (Pb^{4+}) with more number of charges covalent character is more.

20. Due to the $p\pi - d\pi$ bonding from $N \longrightarrow Si$ and due to resonance, the π bonding two electrons are delocalized and spread in three Si and one N atoms.

21. AlF_3 is ionic compound and between the Al^{3+} ion and F^- ions strong electrostatic attractive forces exist.

22. Sn^{2+} is less stable than Sn^{4+} but Pb^{2+} is more stable then Pb^{4+} due to inert pair effect.

25. $$\underset{}{H_2C_2O_4} \xrightarrow{\Delta} \underset{C}{H_2O} + \underset{A}{CO} + \underset{B}{CO_2}$$
   $2CO + O_2 \longrightarrow 2CO_2$
   $CO_2 + 2NH_3 \longrightarrow NH_2COONH_4 \xrightarrow{\Delta}$
   $NH_2CONH_2 + H_2O$

29. $CH_2(COOH)_2 \xrightarrow[-2H_2O]{P_4O_{10}} C_3O_2$

30. $2CH_3Cl + Si \xrightarrow[570K]{Cu} (CH_3)_2\,SiCl_2$
   Hydrolysis and polymerization of $(CH_3)_2\,SiCl_2$ gives linear polymer.

32. In pyrosilicate only one oxygen atom, in linear silicate two oxygen atoms, in three dimensional silicate all

four oxygen atoms of SiO_4^{4-} ions are shared. In sheet silicate only three oxygen atoms are shared.

34. CO_2 is acidic oxide.
$$\underset{}{(COOH)_2} \xrightarrow{\text{Cono.}H_2SO_4} \underset{A}{CO} + \underset{B}{CO_2} + H_2O$$
Solid CO_2 called dry ice sublimes. CO_2 with NH_3 form urea.

35. Composition of asbestos is $CaMg_3(SiO_3)_4$ or $CaO \cdot 3MgO \cdot 4SiO_2$

38. The inter layer distance in graphite is 335 pm where as the radius of carbon is 77 pm. So the interlayer distance is nearly $4\frac{1}{2}$ times of the radius of carbon.

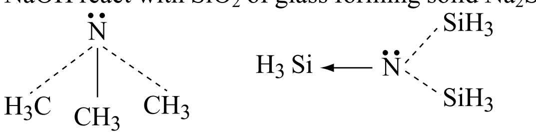
39. Graphite can be converted into diamond only at very high temperatures and pressures.

40. $CS_2 + 3Cl_2 \longrightarrow CCl_4 + S_2Cl_2$

41. In graphite every carbon is sp^2 hybridized and the bond order is 1.33.

42. NaOH react with SiO_2 of glass forming solid Na_2SiO_3.

43.



So N in $N(SiH_3)_3$ is in sp^2 hybridization and the molecule is planar triangular. Due to the absence of d - orbitals in carbon atom CH_3 group $p\pi$ - $d\pi$ bond is not possible hence pyramidal.

45. Silicon cannot from $p\pi$ - $p\pi$ bonds but can form $p\pi$ – $d\pi$ bonds with oxygen and nitrogen.

47. $SiC + 4NaOH + 2O_2 \longrightarrow$
$$Na_2SiO_3 + Na_2CO_3 + 2H_2O$$

49. $CO_2 \longrightarrow CO \xrightarrow{Cl_2} COCl_2 \xrightarrow{H_2O} CO_2 + 2HCl$

More than One Answer Type Questions

3. Due to the participation of lone pair of nitrogen in $p\pi$ – $d\pi$ bond with silicon, the lone pair is not available for donation. So it is weak base. Structure of crystalline silicon is similar to diamond SiO_2 is a high melting solid.
Silicon has less tendency to form $p\pi$ - $p\pi$ bonds with oxygen but can form $p\pi$ - $d\pi$ bonds with phosphorus easily.

7. $C(OH)_4$ easily lose two H_2O molecules and convert into stable CO_2 molecule which contain strong $p\pi$ - $p\pi$ bonds but silicon cannot form such bonds due to bigger size.

12. Diamond, carborrundum and crystalline silicon have same structure in which all the atoms are in sp^3 hybridization. In SiO_2 both silicon and oxygen atoms are in sp^3 hybridization but in oxygen two hybrid orbitals contain lone pairs.

14. Aquadag is graphite powder in water and oil dag is graphite powder in oil as lubricants. The remaining options are also correct.

16. Among alkali metals only lithium can react directly with nitrogen and form stable nitride.

In BMe_3 due to electron releasing effect the electron accepting capacity of B decreases and from halides back bonding decreases from flourine to bromine hence the lewis acid character is

$BBr_3 > BCl_3 > BF_3 > BMe_3$

In BH_4^- ion boron do not contain d-orbitals in the valence shell whereas in AlH_4^- contain d-orbitals in its valence shell
$$Mg_2Si + 4NH_4Br \longrightarrow 2MgBr_2 + SiH_4 + 4NH_3$$

17. The bond order in graphite is 1.33, other statements are correct. Due to large gap between layers electrical conductivity perpendicular to planes of layers depend on temperature and behave like semi metal.

With alkaline permanganate graphite gives oxalic acid and melitic acid. The formation of melitic acid $C_6(COOH)_6$ supports the hexagonal structure of graphite. In fullerene all Carbon atoms are sp^2 hybridized and is aromatic. C_{60} contain 20 six membered rings and twelve 5 membered rings. Five membered rings are fused only with 5 membered ring.

21. When sodium carbonate is added to the aqueous solution of Al^{3+} and Fe^{3+}: $Al(OH)_3$ and $Fe(OH)_3$ are precipitated instead of carbonate or Al^{3+} and Fe^{3+} are small in size with more charge, they hydrolyse in water producing H^+ ions which neutralize the carbonate ion forming hydroxides.
$$Al^{3+} + 3H_2O \longrightarrow [Al(OH)_3] + 3H^+$$
$$CO_3^{2-} + 2H^+ \rightarrow H_2O + CO_2$$

23. $CaF_2 + H_2SO_4 \longrightarrow CaSO_4 + 2HF$
$SiO_2 + 4HF \longrightarrow SiF_4 + 2H_2O$
$SiF_4 + 2HF \longrightarrow H_2SiF_6$

24. $SnCl_2 + 2NaOH \longrightarrow Sn(OH)_2 + 2NaCl$
$Sn(OH)_2 + 2NaOH \longrightarrow Na_2SnO_2 + H_2O$
$3Na_2SnO_2 + 2Bi(OH)_3 \longrightarrow 3Na_2SnO_3 + 2Bi + 3H_2O$

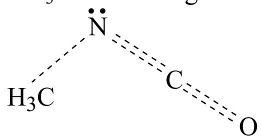
26. $3CS_2 + 6NaOH \longrightarrow (Na_2CO_3 + 2Na_2CS_3 + 3H_2)$

27. $X = Sn(OH)Cl$, $Y = Sn(OH)_2$, $Z = Na_2SnO_2$
$P = Au$, $Q = HI$

28. SiO_2 react with HF to form H_2SiF_6
$SiO_2 + 6HF \longrightarrow H_2SiF_6 + 2H_2O$
Hence a, b, d are correct

Passage-III

1. 4
2. $(SiH_3)_3$ P is pyramidal because pπ - dπ bonding P – π is not possible due to larger size. But $N(SiH_3)_3$ is planar triangular ( ) and nitrogen is in sp^2 hybridization.
3. H_3CNCO is angular in shape while H_3SiNCO is



$H_3Si \longleftarrow\!\!=\!\!=\; N = N = O$

Passage-V

$$CaO + 3c \xrightarrow{\Delta} CaC_2 + CO$$
$$CaC_2 + N_2 \xrightarrow{1100°C} CaNCN + C$$
$$Ca\,NCN + 3H_2O \longrightarrow CaCO_3 + 2NH_3$$

Passage-VI

$SiO_2 + 2C \longrightarrow Si + 2CO$
$Si + 2Cl_2 \longrightarrow SiCl_4$
$SiCl_4 + 2Mg \longrightarrow Si + 2MgCl_2$
$SiCl_4$ is liquid can be purified by fractional distillation.
Pure quartz glass contains only SiO_2.

Passage-VII

1. ClO_4^- do not polymerize whereas SiO_4^{4-}, PO_4^{3-} and SO_4^{2-} can polymerize.
2. Due to pπ - dπ bonding in Si $\leftleftharpoons$ O the Si – O bond gets double bond character and is stronger.
   As the lone pair on oxygen participate in pπ - dπ bonding in disilyl either, its basic character is less
   Due to resonance in $(CH_3)_3$ Si – $\bar{O}$ by participating in pπ - dπ bonding its stability increases.

$$CH_3\ Si\text{——}\bar{O} \longleftrightarrow (CH_3)_3\overline{Si} \longleftarrow\!\!\text{——}\ O$$

   So$(CH_3)_3$ SiOH is stronger acid than $(CH_3)_3COH$.
   In $H_3N \longrightarrow BF_3$ back bonding is absent.

Integer Questions

1. H_3BO_3 is not protonic acid.It will not give proton but release proton from water. So answer is zero.
2. In H_2SiF_6 covalency of silicon is 6.
3. In cyclic silicones each silicon is in bond with two oxygen atoms and each oxygen atom is in bond with two silicon atoms. So the share of each oxygen belonging to each silicon is 0.5. So the number of oxygen belonging to each silicon atom forming the ring is 1.
4. In $Si_6\,O_{18}^{12}$ six oxygen atoms are shared in the formation of cyclic silicate.
5. In solid SnF_4 each Sn atom is surrounded by six fluorine atoms.
6. In diamond the tetrahedral carbon atoms are fused in hexagonal rings.

9 CHAPTER

Group VA (15) Nitrogen Family

A Theory is a tool not a creed ….
J. J. Thomson

9.1 INTRODUCTION

Group VA of the periodic table consists of five elements, viz, **Nitrogen** (N), **Phosphorus** (P), **Arsenic** (As) **Antimony** (Sb) and **Bismuth** (Bi). After the name of the first element of the group, they are also named as nitrogen family. Collectively, these elements are termed as *Pnicogens* and their compounds are *Pnictides*. The name is derived from the Greek word **Pnigmos** which means suffocation. Nitrogen occurs as the diatomic gas N_2 which forms nearly 80 per cent by volume of the earth's atmosphere. In the combined form it is found as nitrates, constituent of proteins and nucleic acids. It has been an essential constituent of fertilizers, explosives and food-stuffs. These elements except phosphorus do not occur very abundantly in nature. Nitrogen constitutes only about 0.004 per cent phosphorus about 0.116 per cent, arsenic about 5×10^{-4} per cent, antimony about 1×10^{-4} per cent and bismuth about 1.8×10^{-5} per cent. Phosphorus ranks 10th in abundance.

Phosphorus is important because it is a constituent of phosphatic fertilizers and also needed for the formation of bones and teeth in the body. The remaining three elements, As, Sb and Bi, are less abundant and are found chiefly as sulphides and oxides.

9.2 OCCURRENCE

Nitrogen occurs in the free state to the extent of 78 per cent by mass or 80 per cent by volume. It occurs as chile salt petre $NaNO_3$ and Indian salt petre KNO_3.

Phosphorus is the 11th most abundant element in the earth's crust. The important sources of phosphorus are phosphate rocks like

Phosphorite	$Ca_3(PO_4)_2$
Fluorapatite	$3Ca_3(PO_4)_2 \cdot CaF_2$
Chlorapatite	$3Ca_3(PO_4)_2 \cdot CaCl_2$
Hydrapatite	$3Ca_3(PO_4)_2 \cdot Ca(OH)_2$

Arsenic occurs as

Realgar	As_4S_4	Orange-red colour
Orpiment	As_2S_3	Yellow

Antimony occurs mainly as stibnite Sb_2S_3 and in flue dust as Sb_2O_3.

Bismuth minerals are

Bismuth glance or bismuthinite	Bi_2S_3
Bismuth ocure or bismite	Bi_2O_3
Bismuthite	$BiO_2 \cdot CO_3$

Element	N	P	As	Sb	Bi
Abundance in earth's crust	19	11.2	1.8	0.2	0.008 ppm

Electronic Configuration

The electronic configurations of Group VA element are given in Table 9.1.

Thus all the elements possess ns^2np^3 electronic configuration. According to Hund's rule the three p-electrons must be in the unpaired state, i.e., $np_x^1\ np_y^1\ np_z^1$. Since the p-orbital is half-filled these elements are fairly stable and not very reactive.

9.3 PHYSICAL PROPERTIES

Some important atomic and physical properties of Group VA elements are given in Table 9.2.

1. Atomic size: Covalent radii of the atoms of the elements of Group VA do not increase regularly as we move down the group from nitrogen to bismuth. This is due to the poor shielding effect of d-electrons in As, Sb and Bi.

Table 9.1 Electronic configurations of group VA elements

Element	At. no	Electronic configuration	Configuration of valence shell
N	7	$1s^2\ 2s^2\ 2p^3$	$2s^2\ 2p^3$
P	15	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$	$3s^2\ 3p^3$
As	33	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^3$	$4s^2\ 4p^3$
Sb	51	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^3$	$5s^2\ 5p^3$
Bi	83	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 4f^{14}$ $5s^2\ 5p^6\ 5d^{10}\ 6s^2\ 6p^3$	$6s^2\ 6p^3$

Table 9.2 Atomic and physical properties of group VA elements

Property	N	P	As	Sb	Bi
Atomic radius (pm)	74	110	121	140	152
Ionic radius M^{3-}(pm)	171	212	222	245	
Ionic radius M^{3+}(pm)			92	108	
Melting point (k)	63	316	1090*	904	546
Boiling point (k)	77	552	906	1693	1775
Electronegativity	3.0	2.1	2.0	1.9	1.9
Density (gm cm^{-3})	0.88	1.82	5.7	6.7	9.8
Ionization energy (KJ mol^{-1})					
First	1403	1012	947	834	703
Second	2857	1897	1950	1590	1610
Third	4580	2910	2730	2440	2460
Latent heat of fusion (KJ mol^{-1})	0.33	0.63	21.32	20.06	10.87
Bond energy (KJ mol^{-1})	159	213.4	178	142.3	1046

* Arsenic sublimes at 878 K. So, its melting point is measured at 38.6 atm pressure.

Consequently, atoms with d^{10} inner shells have been smaller in size than would otherwise be expected. In the case of Bi, the inclusion of f^{14} inner shell (more poorly shielding) further affects the atomic size.

2. Density: Density increases from top to bottom in the elements of Group VA.

3. Physical State and Chemical Structure: Nitrogen is a gas, phosphorus is a soft, waxy and lustureless solid. Arsenic is a hard lustureless solid while antimony and bismuth are hard solids and have high characteristic metallic lustre. Nitrogen exists as triple bonded diatomic molecule, N_2, while phosphorus, arsenic and antimony exist as tetratomic molecules P_4, As_4 and Sb_4. Bismuth is monoatomic in gaseous state.

The existence of nitrogen as diatomic molecule is attributed to its ability to form strong pπ—pπ multiple bonds. So, it exists as a diatomic molecule containing triple bond (N≡N). The triple bond is constituted of one σ-bond and two π-bonds. The π bonds are formed by the lateral overlap of π-orbitals. The intermolecular forces between these small N_2 molecules are weak van der Waals' forces. So, nitrogen exists as a gas. Further due to high bond dissociation energy (945.4 KJ mol^{-1}) of the triple bond in nitrogen, it is apparently inactive under normal chemical reaction conditions. So, it behaves almost as an inert gas at room temperature.

A probable reason for the non-existence or instability of multiple bonded structures involving pπ—pπ bonding in heavier elements (P, As and Sb) is that the bigger length of the sigma bond in heavier element reduces or even excludes the sideway overlapping of p-orbitals. So phosphorus, arsenic and antimony forms P_4, As_4 and Sb_4 molecules in which the atoms are arranged in a tetrahedral manner e.g., yellow phosphorus P_4 molecule is shown in Fig. 9.1. The PPP angle is 60º and P—P distance is 221 pm.

4. Melting and Boiling Points: The melting points (except for Sb and Bi) and the boiling points both increase when we go down the group. The melting point of Bi has been usually low. Low melting point of Bi reveals that there is little possibility of the availability of the pair of electrons in s-orbital for bonding. The elements of this group are more

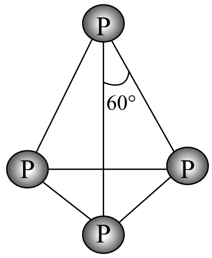


Fig 9.1 Structure of yellow phosphorus P_4 molecule

volatile than their immediate neighbours. This is due to the fact that these elements have five electrons in their valence shell. The increase in melting points and b. pts from nitrogen to arsenic is due to increase in their molecular sizes.

5. Ionization Energy: Ionization energy gets decreased regularly on descending the group. There is a large decrease between nitrogen and phosphorus. But from phosphorus to bismuth the ionization energy decreases slowly due to the poor shielding effect of d-electrons in arsenic and antimony and d- and f-electrons in bismuth.

6. Electronegativity: Electronegativity decreases gradually from nitrogen to bismuth indicating a gradual change from non-metallic character to metallic character.

7. Metallic and Non-metallic Character: The first two elements, nitrogen and phosphorus, are distinctly non-metals, the succeeding two elements arsenic and antimony are partly non-metals and partly metals, i.e., metalloids while the last element, bismuth is a metal. So, from top to bottom in the group gradually non-metallic character is decreasing and metallic character is increasing.

8. Oxidation States: From the valence shell electronic configuration ns^2np^3 of these elements, it is evident that these elements show oxidation states of –3, +3 and +5.

Outer Electronic Configuration

	ns	np	nd	
Ground state	↑↓	↑ ↑ ↑		–3 or +3
Excited state	↑	↑ ↑ ↑	↑	+5

The sign + or – for the oxidation state 3 depends on the electronegativity of the other element with which an element combines. The stability of the +3 oxidation state increases down the group while the stability of the +5 oxidation state decreases down the group due to inert pair effect. Apart from these oxidation states nitrogen exhibit a large number of oxidation states from –3 to +5 as exemplified below.

–3	–2	–1	–1/3	0	+1	+2	+3	+4	+5
NH_3	N_2H_4	NH_2OH	N_3H	N_2	N_2O	NO	N_2O_3	NO_2	N_2O_5

Nitrogen and phosphorus exhibits oxidation state +4 due to the ability of the lone pair of electrons in NH_3 and PH_3 to form dative bond with Lewis acids.

Since the electronegativities decrease as we move down the group towards heavier elements, the tendency to exhibit negative oxidation state decreases while the tendency to exhibit positive oxidation state increases. For example, nitrogen exhibits –3 oxidation state in almost all compounds except with fluorine and oxygen. But bismuth being a metal mainly exhibits +3 oxidation state.

9. Nature of bonding: In majority of cases of this group the bonds are covalent and this tendency gets increased with increasing oxidation states. Ionic bonds are formed by bismuth with the fluorine in BiF_3 which contain Bi^{3+} ion. In addition to BiF_3, the tripositive ion occurs in the salts of bismuth and antimony with strong acids as in $Bi(ClO_4)_3$ and $Sb_2(SO_4)_3$. But the Bi^{3+} and Sb^{3+} cations have been less stable in the presence of water and rapidly hydrolyzed to yield bismuthyl, BiO^+ and antimonyl SbO^+ ions.

Theoretically all the elements of this group may accept three electrons to yield the triply charged negative ion like nitride ion N^{3-} and the phosphide ion P^{3-} and thereby get noble gas configuration. Only small and more electronegative nitrogen atom and to some extent phosphorus atom may be able to form highly charged negative ion. The other members of this group have least tendency to form trinegative ion (M^{3-}) due to increase in size and decrease in electronegativity.

In most of their compounds the bonds formed are covalent as the difference in electronegativities is not sufficient to allow an ionic bond.

10. Participation of d-orbitals in Sigma Bonding: Nitrogen has no d-orbitals in its valence shell. So, it cannot form compounds like NCl_5. But all the other elements of Group VA contain d-orbitals in their valence shell which can participate in bonding. So they can form compounds like PF_5, PCl_5, etc. by using the d-orbitals in sigma bonding. Further, by using these vacant d-orbitals they can accept lone pairs of electrons and act as Lewis acids.

Ex: PCl_6^-, SbF_6^-, etc.

11. Participation of d-orbitals in π-Bonding:

(i) It is a common experience that trialkyl and triaryl phosphine oxides viz (RR′R″) PO are much more stable than the corresponding amine oxides. The bond dissociation energy of P → O bond is in the range of 500–600 KJ mol^{-1} compared to that of N → O bond which is in the range of 200–600 KJ mol^{-1}. This shows that P and O are linked together through more than one bond.

If the bonding between P and O is represented merely as P → O, then the greater the combined electronegativity electron attracting powers of (RR′R″), the lesser would

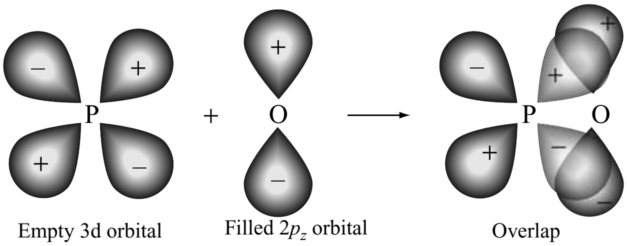


Fig 9.2 The formation of a dπ—pπ bond between P and O atoms in phosphine oxides

be the strength of P → O bond and the greater would be the length of P → O bond. The actual situation is just the reverse, i.e., the greater the combined electronegativity of (RR′R″) the greater is the strength of P → O bond and shorter is the length of P → O bond. The above facts can be explained only if we assume π-back bonding from fully filled pπ orbitals of oxygen to vacant dπ orbitals of phosphorus in phosphine oxide. Accordingly, the structure of the oxide would be as shown in Fig. 9.2.

Since nitrogen does not have any d-orbitals there would be no pπ—dπ bonding between N and O in the amine oxides and these compounds would, therefore, have a simple structure RR′R″N → O The π-back bonding in phosphine oxides thus readily accounts for their greater dissociation energy and therefore greater stability as compared to amine oxides.

It can be easily visualized that the greater the combined electronegativity of (RR′R″) the greater would be the partial positive charge created on P and hence the greater would be the flow of pπ electronic charge from O to P. This would impart greater double bond character to the bond between P and O resulting in the shortening of the P → O bond and thereby increasing the stability of the phosphine oxide.

(ii) The electronegativity difference between P and O is more than that in N and O. On the basis of this difference alone (i.e., in the absence of any π-back bonding) amine oxides should have been less polar than the corresponding phosphine oxides. But actually the situation is just the reverse. Phosphine oxides are less polar than the analogous amine oxides. For instance, the dipole moment of $(CH_3)_3NO$ is 5.0 Debye. The π-back bonding, i.e., pπ—dπ bonding in the P—O bond shifts back some of the charge to P and thus decreases the dipole moment of phosphine oxide. The P—O bond should thus be represented as P⇆O and not merely as P→O (the N—O bond would be represented merely as N→O). The dipole moment is the product of the amount of charges separated and the distance between the separated charges. On account of pπ—dπ back bonding, the distance between P and O and also the separated charges which are on P and O both decrease and hence there is a decrease in the dipole moment of phosphine oxide.

12. Catenation: The elements of VA group exhibit catenation power but to a less extent and it decreases from top to bottom in the group. This is due to gradual decrease in bond energies in the group. The catenation power in nitrogen is least because of the least stable N—N bond. The N—N bond energy is very small (160 KJ mol^{-1}). Compared to C—C bond energy due to the repulsion between non-bonding lone pairs. It is also less than even P—P bond energy (200.8 KJ mol^{-1}). The catenation in stable nitrogen compounds is thus restricted generally to two or three N atoms as, e.g., in N_2H_4, N_3^-, etc. Some very unstable compounds containing chains of upto eight nitrogen atoms have been synthesized, e.g., R—N = N—R, R—N = N—NR_2, R_2N—N = N—NR_2, RN = N—N(R)—N = NR and RN = N—N(R)—N = N—N(R)—N = N—R, where R is some alkyl or aryl radical or it may be hydrogen. From the single bond energies of N—N, P—P and As—As, phosphorus has maximum tendency to catenate among Group VA elements.

13. Covalency: Nitrogen may exhibit a maximum covalency of 4 (as in NH_4^+ ion) because it is having only four orbitals (one 2s and three 2p) available for bonding. Other elements are having vacant d-orbitals in their valence shells and hence they exhibit a covalency 5 and a maximum of 6 as in AsF_6^-, $SbCl_6^-$ and $Sb(OH)_6^-$.

14. Allotropy: All the elements show allotropy. Nitrogen has two solid forms known as α-nitrogen with cubic crystalline structure and β-nitrogen with hexagonal crystalline structure. The transition temperature is –238.5ºC (34.5K). Phosphorus exists in a number of allotropic forms such as white phosphorus, red phosphorus, scarlet phosphorus, metallic or α-black phosphorus and violet phosphorus. Arsenic exists in three allotropic forms, viz. grey, yellow and black. Similarly, antimony exists in three different forms. These are metallic forms, yellow or α-antimony and explosive antimony.

9.4 COMPOUNDS OF GROUP VA ELEMENTS: COMPARATIVE STUDY

9.4.1 Hydrides

All the elements of the group are known to form volatile hydrides of the formula MH_3. In addition to this, the first two elements (N and P) also form hydrides of the type M_2H_4, whereas nitrogen also forms hydrazoic acid HN_3.

The hydrides of the type MH_3 are discussed in detail.

Preparation: Hydrides of the type MH_3 can be prepared by the action of water or dilute acids on the binary metal compounds.

$$Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$$

$$Ca_3P_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2PH_3$$

$$Zn_3As_2 + 6HCl \rightarrow 3ZnCl_2 + 2AsH_3$$

$$Mg_3Sb_2 + 6HCl \rightarrow 3MgCl_2 + 2SbH_3$$

$$Mg_3Bi_2 + 6HCl \rightarrow 3MgCl_2 + 2BiH_3$$

The ease of formation of these hydrides decreases from NH_3 to BiH_3. The heat of formation of these hydrides decreases from NH_3 to BiH_3; infact, the heats of formation of NH_3 and PH_3 are negative but for the other hydrides the heat of formation is positive.

Properties

(i) All these hydrides are colourless gases. Their smell becomes more disagreeable as the atomic number increases from N to Bi.

(ii) These are posionous gases and the poisonous nature increases from NH_3 to BiH_3.

(iii) All these hydrides are covalent and the covalent character increases from NH_3 to BiH_3 as the electronegativity difference between hydrogen and Group VA elements decreases from N to Bi. Order of covalent character is $NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$.

(iv) Polarity of the molecules also decreases from NH_3 to BiH_3.

(v) Ammonia is highly soluble in water because it can form hydrogen bonding with water.

$$\begin{array}{ccccccc} & & H & & & & \\ & & | & & & & \\ \ldots H & - & N & \ldots H & - & O & \ldots \\ & & | & & & | & \\ & & H & & & H & \end{array}$$

As the polarity of the molecules decreases from NH_3 to BiH_3, solubility in water also decreases from NH_3 to BiH_3.

(vi) The ammonia molecules are associated through hydrogen bonding because of high polar character of N—H bond.

$$\begin{array}{cccccccccccc} & & H & & & H & & & H & & & H \\ & & | & & & | & & & | & & & | \\ H & - & N & \ldots H & - & N & \ldots H & - & N & \ldots H & - & N\ldots\ldots \\ & & | & & & | & & & | & & & | \\ & & H & & & H & & & H & & & H \end{array}$$

Table 9.3 Important hydrides of group VA elements

NH_3	Ammonia	N_2H_4	Hydrazine	HN_3
PH_3	Phosphine	P_2H_4	Diphosphine	(Hydrazoic acid)
AsH_3	Arsine			
SbH_3	Stibine			
BiH_3	Bismuthine			

As the polarity of M—H bond decreases from NH_3 to BiH_3, the tendency to form hydrogen bond also decreases. Except ammonia, in other hydrides of the Group VA elements the hydrogen bond is practically absent. So, ammonia can be liquified easily and its boiling point will be more than arsine but less than stibine.

The van der Waal's forces in bigger stibine molecules dominate the weaker hydrogen bonding in ammonia. The order of boiling points of the hydrides of Group VA elements is

	PH_3 <	AsH_3 <	NH_3 <	SbH_3 <	BiH_3
Boiling points (k)	185.5	210.6	238.5	254.6	256.2

But the order of melting points is

PH_3	$< AsH_3$	$< SbH_3$	$< NH_3$
139.5	156.7	185	195.2

(vii) **Basic character:** When ammonia is dissolved in water, there exists diequilibrium producing OH^- ions. So, it will act as a Arrhenius base.

$$NH_3 + H_2O \rightleftharpoons NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

But other hydrides cannot behave as Arrhenius bases as their solubility is less and cannot produce OH^- ions. However, as these hydrides have one lone pair of electrons on the central atom and therefore behave as Lewis bases. The Lewis basic character of these hydridess decreases from NH_3 to BiH_3. The hydrides of As, Sb and Bi have no basic character at all. PH_3 combines with HI forming PH_4I. The decrease in basic character of the hydrides is due to the decrease in availability of the lone pair of electrons as we go down the group. The size of the central atom increases as we move from N to Bi. As a result the electron charge cloud of the lone pair diffuses over larger volume and therefore the electron charge donor capacity (i.e., basic strength) of the elements decreases as we go down the group.

(viii) **Thermal stability:** Thermal stability of the hydrides of the elements of Group VA decreases progressively from NH_3 to BiH_3. As we move down

Table 9.4 Bond lengths, bond energies and bond angles of the hydrides of group VA elements

Hydride	Bond length (pm)	Bond energy KJ mol^{-1}	Bond angle
NH_3	101.7	389	107° 48′
PH_3	141.9	318	93° 36′
AsH_3	151.9	247	91° 48′
SbH_3	170.7	255	91° 18′

the group the size of the central atom increases and therefore, its tendency to form stable covalent bond with comparatively small hydrogen atom decreases consequently, the strength of M—H bond and hence the thermal stability goes on decreasing while moving from NH_3 to BiH_3. The largest atom bismuth forms least stable bond with hydrogen so that the hydride formed (BiH_3) is so unstable that it can be kept above 230 K.

(ix) **Reduction power:** All these hydrides can act as reducing agents. As we move from NH_3 to BiH_3, the reduction power increases due to decrease in thermal stability. Unstable compound will act as the strongest reducing agent while the stable hydride will become the least powerful reducing agent. The order of reduction power of these hydrides is $NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$.

(x) **Formation of Complexes:** By donating the lone pair present on the central atom of these hydrides, they can enter into complex formation with metal ions.

NH_3 can donate its lone pair easily and can form complexes with several metal ions such as Co, Ni, Cu and Zn groups, e.g., $[Cu(NH_3)_4]^{2+}$. Phosphine can enter into complex formation with metal ions which can reinforce the original coordinate bond by back bonding from π overlap of a filled d-orbital on the metal with an empty d-orbital on P as explained earlier in phosphine oxides of section 9.3(11).

The donor properties of the other hydrides are very weak and they have little or no tendency to form coordinate bonds.

(i) ***Structure and Donor Property*****:** All these hydrides are pyramidal in shape. In ammonia, nitrogen is involved in sp^3 hybridization The lone pair occupies one of the sp^3 hybrid orbital. Now, around the nitrogen atom in ammonia there are one lone pair and three bond pairs. Due to the repulsion between lone pair and bond pairs, the bond angle decreases to 107° 48′ from ideal tetrahedral angle 109° 28′. In other hydrides, the bond angles become close to 90°. The decrease in bond angles in other hydrides is due to the decreased tendency of the central atom to involve in sp^3 hybridization with increase in size, i.e., pure p-orbitals are utilized in M—H bonding.

If the three p-orbitals are used for M—H bonding, the lone pair must occupy a spherical s-orbital. This is larger, and less directional, and hence less effective for forming a coordinate bond. This means that any σ-bond will be very weak. In addition, the 4d- and 5d-orbitals are too large for effective π back bonding. So, the complex-forming ability decreases in AsH_3, SbH_3 and BiH_3.

(ii) ***Substitution reactions*****:** The hydrogen atoms of NH_3 may be substituted by groups like Cl_2 or alkyl groups such as $–CH_3$, etc. The ease of such substitution reactions decreases from NH_3 to BiH_3.

9.4.2 Halides

The elements of Group VA forms mainly two types of halides, trihalides of the type MX_3 and pentahalides of the type MX_5.

Trihalides

Preparation**:** The nitrogen trihalides can be prepared by the action of excess halogen on ammonia.

$$4NH_3 + 3F_2 \xrightarrow{Cu} NF_3 + 3NH_4F$$
$$NH_3 + 3Cl_2 \longrightarrow NCl_3 + 3HCl$$
$$2NH_3 + 3Br_2 \longrightarrow NBr_3 \cdot NH_3 + 3HBr$$
$$2NH_3 + 3I_2 \longrightarrow NI_3 \cdot NH_3 + 3HI$$

NBr_3 and NI_3 are known only as their unstable ammoniates, $NBr_3 \cdot 6NH_3$ and $NI_3 \cdot 6NH_3$.

The trihalides of other elements can be prepared by the direct reaction between elements using limited supply of halogens.

$$2M + 3X_2 \rightarrow 2MX_3$$
$$(M = P, As, Sb \text{ or } Bi \text{ and } X = Cl_2\ Br_2 \text{ or } I_2)$$

The PF_3 is best made by the action of CaF_2, ZnF_2 or AsF_3 on PCl_3. The trifluorides of As, Sb and Bi are prepared by the action of HF on the oxide M_2O_3 (direct reaction of M or M_2O_3 with F_2 gives MF_5).

Properties**:** The trihalides are predominantly covalent. BiF_3 is ionic and the other halides of Bi and SbF_3 are intermediate in character. Thus, all the trihalides of nitrogen and phosphorus and AsF_3, $AsCl_3$, $AsBr_3$, $SbCl_3$ and $SbBr_3$ are clearly volatile covalent molecular species. AsI_3, SbF_3 and BiX_3 have more extended interactions (close to ionic) in the solid state.

Stability**:** For a given element of Group VA, the stability of the halides will be in the order

Fluoride > Chloride > Bromide > Iodide

However, the stabilities of halides of different elements are far from the regular trend due to the difference in the structures and bond types. NCl_3, and the addition products of NBr_3 and NI_3 with ammonia are highly unstable and explodes.

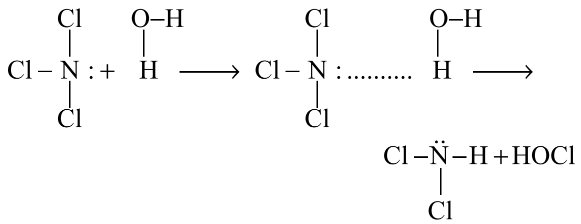
Hydrolysis**:** The trihalides typically hydrolyze with water but the products are different depending on the nature of element.

$$NCl_3 + 4H_2O \rightarrow NH_4OH + 3HOCl$$
$$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$$

$$AsCl_3 + 3H_2O \leftrightarrows H_3AsO_3 + 3HCl$$
$$SbCl_3 + H_2O \leftrightarrows SbOCl \downarrow + 2HCl$$
$$BiCl_3 + H_2O \leftrightarrows BiOCl \downarrow + 2HCl$$

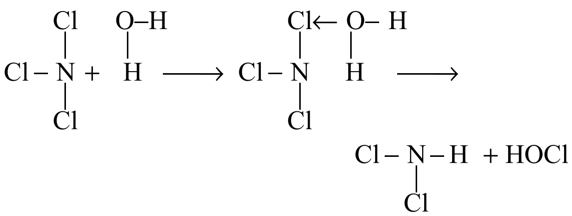
As already seen in the halides of the elements of Group IVA, during hydrolysis, the water molecule approaches the halide and donate a lone pair of electrons on oxygen to the central atom of the halide. Since in NF_3 there are no vacant d-orbitals either on nitrogen or on fluorine it do not hydrolyze in water. The hydrolysis of NCl_3 may proceed in one of the two methods proposed below.

In the first proposal, the hydrolysis is thought to proceed by a stepwise replacement of chlorine. The lone pair of electrons on nitrogen atom bonding to a hydrogen atom (carrying partial positive charge) of water molecule.



Similarly $NHCl_2 + H_2O \rightarrow NH_2Cl + HOCl$
$NH_2Cl + H_2O \rightarrow NH_3 + HOCl$

In another way it may be explained that water molecule may donate its lone pair on oxygen to the vacant d-orbital of chlorine and thus hydrolysis proceeds as follows.



Similarly $NHCl_2 + H_2O \rightarrow NH_2Cl + HOCl$
$NH_2Cl + H_2O \rightarrow NH_3 + HOCl.$

In PCl_3 both P and Cl atoms have vacant d-orbitals in their valence shells. Now, water molecule donates its lone pair on oxygen to P atom because it carries partial positive charge due to the less electronegativity of phosphorus than chlorine.

The hydrolysis reactions of As, Sb and Bi are reversible. These reactions clearly indicate the increase in electropositive (metallic) character from nitrogen to bismuth. Due to increase in electropositive character, the hydrolysis product $M(OH)_3$ distinctly will have basic character, and react with the acid HCl produced in the same reaction. Since antimony and bismuth have more metallic character, their hydrolysis reactions are only partial due to more basic character of $Sb(OH)_3$ and $Bi(OH)_3$.

PF_3, unlike the other trihalides of phosphorus, hydrolyzes only slowly with water, probably due to strong P—F bonds.

$$PF_3 + 3H_2O \rightarrow H_3PO_3 + 3HF$$

***Donor property*:** Similar to hydrides, the trihalides of Group VA elements have a lone pair on the central atom and can act as electron pair donors (Lewis bases and ligands) to form a coordinate bond. Though the electron density at phosphorus in PF_3 is less, due to withdrawal of electron density by more electronegative fluorine atoms, it acts as a strong donor, with metal atoms/ions which can reinforce the σ dative bond from phosphorus to metal atom/ion by the p-back bonding from a filled orbital on the metal to an empty d-orbital on P atom. NF_3 has little tendency to act as a donor molecule due to the absence of d-orbitals in the valence shell of N atom.

In the case of nitrogen halides, the tendency to act as Lewis base increases from NF_3 to NI_3

$$NF_3 < NCl_3 < NBr_3 < NI_3$$

This is due to the decrease in the electronegativity from fluorine to iodine. Also, the Lewis basic character decreases down the group from nitrogen to bismuth halides.
***Acceptor property*:** The trihalides also show acceptor properties and can accept an electron pair from another ion such as F^- forming complex ions such as $[SbF_5]^{2-}$ and $[Sb_2F_7]^-$. This tendency is exhibited by the trihalides of phosphorus and antimony especially fluorides and chlorides by using the vacant d-orbitals in their valence shells. Nitrogen trihalides do not exhibit such acceptor property due to the absence of vacant d-orbitals in its valence shell.

PCl_3 is the most important trihalide. It is widely used in organic chemistry to convert carboxylic acids to acid chlorides and alcohols to alkyl halides.

$$PCl_3 + 3RCOOH \rightarrow 3RCOCl + H_3PO_3$$
$$PCl_3 + 3ROH \rightarrow 3RCl + H_3PO_3$$

***Structures*:** All the trihalides have a pyramidal shape in which the central atom is involved in sp^3 hybridization. Around the central atom there will be one lone pair and three bond pairs. The bond angles of different halides of Group VA elements are given in Table 9.5.

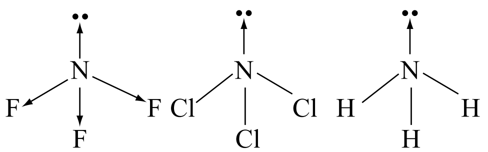
From the values in Table 9.5 it can be seen that the bond angle decreases for a particular halogen with increase in the size of the central atom. This is because that with increase in the size of the central atom the bond pair moves towards the halogen atom due to decrease in electronegativity of the central atom. So, the repulsion between bond pairs decreases and the bond angles decreases.

Table 9.5 Bond angles of the trihalides of group VA elements in gaseous state

Halide	Bond angle	Halide	Bond angle	Halide	Bond angle	Halide	Bond angle
NF_3	102.2°	PF_3	96.3°	AsF_3	96.2	SbF_3	87.3
		PCl_3	100°	$AsCl_3$	98.7	$SbCl_3$	99.5
		PBr_3	101°	$AsBr_3$	99.7	$SbBr_3$	97
		PI_3	102°	AsI_3	100.2	SbI_3	99.1

With the same central atom of a Group VA element, bond angle increases with increase in the size of halogen.

The dipole moment of NF_3 (0.234D) is only one-sixth of that NH_3 (1.47 D) presumably because N–F bond moments act in the opposite direction to that of the lone pair moment. NCl_3 will have more dipole moment than NF_3 because the N–Cl bonds are non-polar but the lone pair moment remains.



Pentahalides

Nitrogen do not form pentahalides because the second shell (valence shell) in nitrogen do not contain d-orbitals. The subsequent elements have suitable d-orbitals and form the following pentahalides.

PF_5	PCl_5	PBr_5	PI_5
AsF_5	$(AsCl_5)$		
SbF_5	$SbCl_5$		
BiF_5			

The pentahalides are prepared as follows

$$3PCl_5 + 5AsF_3 \rightarrow 3PF_5 + 5AsCl_3$$

$$PCl_3 + Cl_2 \rightarrow PCl_5$$

$$2As_2O_3 + 10F_2 \rightarrow 4AsF_5 + 3O_2$$

$$2Sb_2O_3 + 10F_2 \rightarrow 4SbF_5 + 3O_2$$

$$2Bi + 5F_2 \rightarrow 2BiF_5$$

The pentafluorides of As, Sb and Bi can be prepared by the direct reaction of F_2 with elements (As, Bi) or their oxides (As_2O_3, Sb_2O_3).

PF_5 is thermally stable and chemically reactive. PCl_5 is the most important pentahalide. Pentahalides hydrolyzes in water. Partial hydrolysis yield oxyhalides while complete hydrolysis yields the approximate -ic acid. Thus, PCl_5 gives $POCl_3$ if equimolar amounts are used but reacts violently with excess water.

$$PCl_5 + H_2O \rightarrow POCl_3 + 2HCl$$
$$PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl.$$

PCl_5 reacts with P_4O_{10} forming $POCl_3$ and with SO_2 forming thionyl chloride.

$$6PCl_5 + P_4O_{10} \rightarrow 10POCl_3$$

$$PCl_5 + SO_2 \rightarrow POCl_3 + SOCl_2$$

The pentahalides vary from gases such as PF_5 (b.p. 188 K) and AsF_5(b.p. 220 K) to solids such as PCl_5 (sublimes at 335 K) and BiF_5 (m.p.426 K). The gas phase pentahalides are trigonal bipyramidal. SbF_5 is a highly viscous liquid in which the molecules are associated through F-atom bridges. In solid SbF_5, these bridges result in a cyclic tetramer in which Sb (V) achieve a coordination number 6.

The trigonal bipyramid structure is not a regular structure. Electron diffraction methods indicate that in PF_5 some bond angles are 90° and others are 120° and the axial P–F bond lengths are 158 pm whilst the equatorial P–F bond lengths are 153 pm. But NMR studies indicate that all five P–F bond lengths are equal. This is because the electron diffraction method gives the instantaneous picture while NMR gives the picture averaged over several milliseconds. The axial and equatorial F atoms are thought to interchange their positions in less time than that needed to take the NMR. The interchange of axial and equatorial positions is called as pseudorotation. The mechanism of interchange of axial and equatorial of trigonal bipyramid is shown in Fig. 9.4.

PF_5 remains covalent and keeps the trigonal bipyramid structure in all the three physical states. PCl_5 is trigonal bipyramid in liquid and gas phase, but in solid state exists as $[PCl_4^+][PCl_6^-]$. In this case, the ionic contribution to the lattice enthalpy provides the driving force for the transfer of a Cl^- ion from one PCl_5 molecule to another. Another contributing factor may be the more efficient packing of PCl_4^+ and PCl_6^- units compared to the less efficient packing of PCl_5 units. In the solid state, PBr_5 exists as $[PBr_4]^+ Br^-$. Pentaiodide of phosphorus do not exist. The PI_5 reported maybe PI_3I_2 but certainly not $[PI_4^+] I^-$ as reported earlier.

Of the pentachlorides, PCl_5 and $SbCl_5$ are stable, whereas $AsCl_5$ is very unstable. This difference is a manifestation of the *alternation effect* or *middle row anomaly*. The relative instability of $AsCl_5$ when compared with PCl_5 and $SbCl_5$ is an example of the instability of the highest valency state of p-block elements following the completion of first (3d) transition series. This can be understood interms of incomplete shielding of the nucleus which leads to a *d-block contraction* and a consequent lowering of the energy of 4s-orbital in As and $AsCl_3$, thereby making it more difficult to promote one of the $4s^2$ electrons for the formation $AsCl_5$. The non-existence of

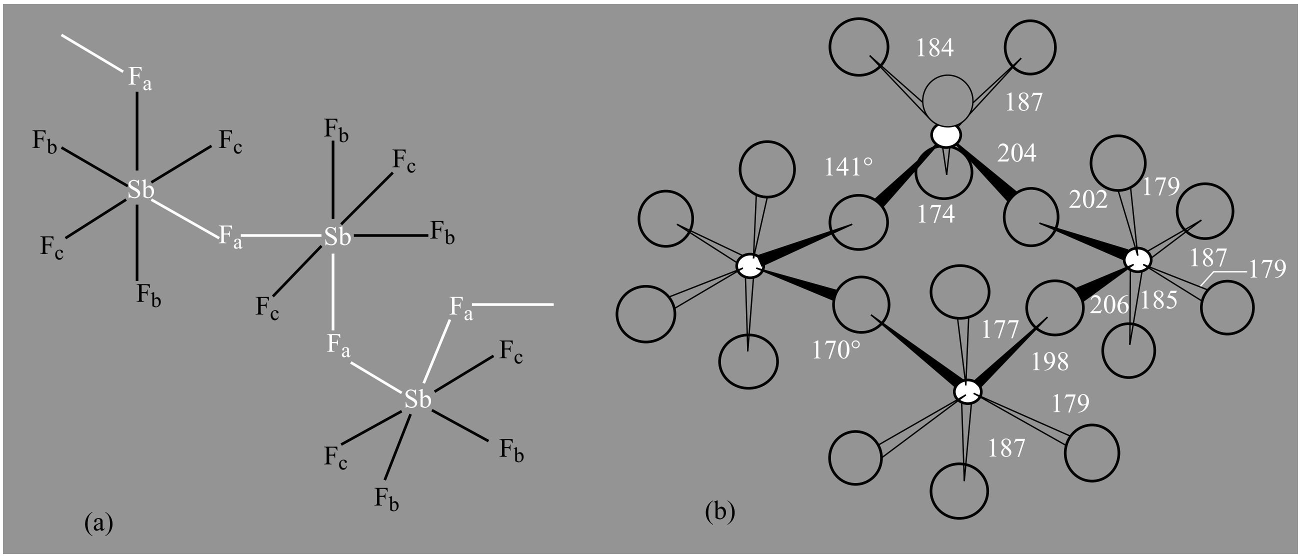


Fig 9.3 (a) The Cis bridged polymeric structure of liquid SbF_5 showing the three types of F atoms and (b) structure of tetrameric molecular unit of crystalline $(SbF_5)_4$ showing cis bridging of $4SbF_6$ octahedra (distances in pm)

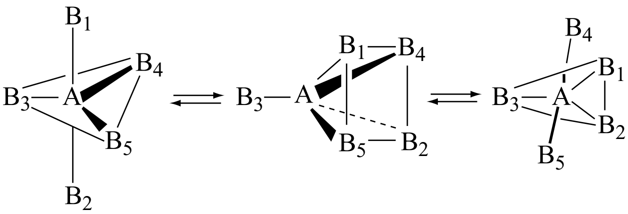


Fig 9.4 A simple mechanism that interchanges axial and equatorial atoms by passage through an sp internet

$BiCl_5$ likewise suggests that it is probably less stable than $SbCl_5$ due to analogous f-*block contraction* following the lanthanide elements.

The pentafluorides of phosphorus, arsenic, antimony and bismuth are strong Lewis acids. SbF_5 is a very strong Lewis acid. It is much stronger Lewis acid, e.g., than aluminium halides. When SbF_5 or AsF_5 is added to HF, a *superacid* is formed that is a more stronger acid than HF.

$$SbF_5\,(l) + 2HF\,(l) \rightarrow H_2F^+\,(\text{sol}) + SbF_6^-(\text{sol})$$

9.4.3 Oxides

All the elements of Group VA are known to form oxides. The important oxides of these elements are given in Table 9.6.

The following trends can be seen in the oxides

(i) Except the oxides of nitrogen and bismuth, the oxides of other elements exist as dimers.

Table 9.6 Important oxides of the elements of group VA

	Oxidation State of Element				
Element	**+1**	**+2**	**+3**	**+4**	**+5**
Nitrogen	N_2O	NO	N_2O_3	NO_2 (or) N_2O_4	N_2O_5
Phosphorus			P_4O_6	P_4O_8	P_4O_{10}
Arsenic			As_4O_6	As_4O_8	As_4O_{10}
Antimony			Sb_4O_6	Sb_4O_8	Sb_4O_{10}
Bismuth			Bi_2O_3		Bi_2O_5

(ii) Stability of the oxides in higher oxidation states gets decreased with increasing atomic number. Thus, N_2O_5 is appreciably stable while Bi_2O_5 is least stable.

(iii) In a given oxidation state the basic character of the oxides gets increased with increasing size of the central atom. Thus, nitrogen and phosphorus oxides (except N_2O and NO) are acidic, arsenic oxides are weakly acidic, antimony oxides are amphoteric and bismuth oxide is weakly basic. This trend is attributed to the size of the cation and change in the non-metallic character to metallic character from nitrogen to bismuth. N^{3+} being much smaller than Bi^{3+}, reacts very readily with water to yield acidic solution.

(iv) The acidic character of oxides of an element gets increased with the increase in the oxidation number of the element. Thus, N_2O and NO are neutral, N_2O_3

and NO_2 are weakly acidic while N_2O_5 is strongly acidic. In general pentoxides are more acidic than trioxides.

(v) The tetroxides of arsenic and antimony are the mixed oxides, e.g., Sb_2O_4 is $Sb^{III}\,Sb^{V}\,O_4$.

(vi) The solubility of the oxides in water decreases from nitrogen to bismuth.

(vii) When trioxides are dissolved in water the corresponding -ous acids will be formed. If pentoxides are dissolved in water, the corresponding -ic acids will be formed.

Trioxides

$$N_2O_3 + H_2O \rightarrow 2HNO_2$$

$$P_4O_6 + 6H_2O \rightarrow 4H_3PO_3$$

$$As_4O_6 + 6H_2O \rightarrow 4H_3AsO_3$$

Pentoxides

$$N_2O_5 + H_2O \rightarrow 2HNO_3$$

$$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$$

The oxides of antimony and bismuth are not soluble in water but they dissolve in alkalis showing the acidic character.

$$As_4O_6 + 12NaOH \rightarrow 4Na_3AsO_3 + 6H_2O$$

$$Sb_4O_6 + 12NaOH \rightarrow 4Na_3SbO_3 + 6H_2O$$

$$Bi_2O_3 + 6NaOH \rightarrow 2Na_3BiO_3 + 3H_2O$$

Similarly pentoxides of arsenic, antimony and bismuth dissolve in alkalis giving the corresponding salts showing their acidic nature.

$$As_4O_{10} + 12NaOH \rightarrow 4Na_3AsO_4 + 6H_2O$$

$$Sb_4O_{10} + 12NaOH \rightarrow 4Na_3SbO_4 + 6H_2O$$

$$Bi_2O_5 + 6NaOH \rightarrow 2Na_3BiO_4 + 3H_2O$$

These reactions show the decrease in acidic character of the oxides from nitrogen to bismuth.

The oxides in lower oxidation states (i.e., in +3) will act as reducing agents and their reduction power decreases from nitrogen to bismuth as the stability of +3 oxidation state increases due to inert pair effect.

The oxides in higher oxidation state (i.e., in +5) will act as oxidizing agents and their oxidation power should increase from nitrogen to bismuth as their stability in +5 oxidation state decreases from nitrogen to bismuth. But oxides of nitrogen (V) and Bi(V) are stronger oxidizing agents than the +5 oxidation states of the three intervening elements. The nitrogen is more electronegative than the elements immediately below it, accordingly the lighter nitrogen is generally less easily oxidized. Nitrogen is generally a good oxidizing agent in its positive oxidation states. Bismuth is much less electronegative, but favours the +3 oxidation state in preference to the +5 state on account of the inert pair effect.

9.4.4 Oxoacids

All the elements of Group VA form oxoacids. Nitrogen forms large number of oxoacids such as $H_2N_2O_2$, H_2NO_2, HNO_2, HNO_3, HOONO and $HOONO_2$.

Phosphorus also forms a large number of oxoacids such as H_3PO_2, H_3PO_3, $H_4P_2O_6$, H_3PO_4, $H_4P_2O_7$ and HPO_3 and H_3PO_5.

Arsenic forms two oxyacids, H_3AsO_3 and H_3AsO_4. Antimony forms only one oxyacid, H_3SbO_3, which is, however, unstable and exists only in solution. Bismuth gives one stable oxoacid acid namely metabismuthic acid, $HBiO_3$.

9.5 NITROGEN

Nitrogen was discovered in 1772 by Daniel Rutherford, a Scottish physician and chemist. Others who made important contributions towards its discovery have been Cavendish and Sheele who isolated nitrogen independently about the same time. The name nitrogen was derived from nitre a well known nitrogen compound.

Preparation: Nitrogen is conveniently prepared in the laboratory by the following methods

(i) *Heating of ammonium nitrite*: Ammonium nitrite on heating liberates nitrogen.

$$NH_4NO_2 \rightarrow N_2 + H_2O$$

Since the reaction is explosive, a solution containing equivalent amounts of ammonium chloride and sodium nitrite is warmed to get nitrogen.

$$NH_4Cl + NaNO_2 \rightarrow NaCl + 2H_2O + N_2\uparrow$$

(ii) *Oxidation of ammonia*: Nitrogen can be prepared by the oxidation of ammonia with red hot copper oxide, PbO, Cl_2, bleaching powder or sodium hypochlorite.

$$2NH_3 + 3CuO \rightarrow 3Cu + 3H_2O + N_2$$

$$2NH_3 + 3PbO \rightarrow 3Pb + 3H_2O + N_2$$

$$8NH_3 + 3Cl_2 \rightarrow 6NH_4Cl + N_2$$

$$2NH_3 + 3CaOCl_2 \rightarrow 3CaCl_2 + 3H_2O + N_2$$

$$2NH_3 + 3NaOCl \rightarrow 3NaCl + 3H_2O + N_2$$

(iii) When orange red crystals of ammonium dichromate are heated a violent reaction occurs which is accompanied by flashes of light and nitrogen gets liberated leaving behind a dark green residue of chromic oxide.

$$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} Cr_2O_3 + 4H_2O + N_2$$

(iv) Nitrogen can also be obtained by heating urea with an acidified solution of nitrite.

$$NaNO_2 + HCl \rightarrow HNO_2 + NaCl$$

$$NH_2CONH_2 + 2HNO_2 \rightarrow 2N_2 + CO_2 + 3H_2O$$

(v) Very pure nitrogen is prepared by heating sodium or barium azide.

$$2NaN_3 \rightarrow 2Na + 3N_2$$

(vi) ***From air***: Air is an important source of nitrogen where it is present in mixture chiefly with oxygen and carbon dioxide. Carbon dioxide is absorbed by caustic potash solution and oxygen is separated by passing the air on hot copper. The remaining part of the air which is mainly nitrogen is collected. It contains about 1 per cent impurities of inert gases.

Nitrogen is manufactured by fractional distillation of liquid air. Linde's process is used for the liquification of air. Boiling points of liquid air and liquid nitrogen are 90 K and 77.3 K, respectively. There is sufficient difference in the boiling points and they are separated by Claude's process.

Physical Properties

Nitrogen is a colourless, tasteless and odourless gas. It is sparingly soluble in water (2.3 volumes in 100 volumes at NTP). It can be liquified (b.p. 77.3 K) and solidifies (m.p 62.5 K at 86 mm pressure). It is slightly lighter than air. It is not poisonous but animals die in an atmosphere of nitrogen for want of oxygen. It is neither combustible nor a supporter of combustion.

Chemical Properties

Nitrogen is an inert diatomic gas (N≡N). The interatomic distance is 109.5 pm. It dissociates into atoms at very high temperature, hence it is chemically less reactive.

$$N_{2(g)} \rightleftharpoons 2N_{(g)} \; \Delta H = 940 \text{ KJ mol}^{-1}$$

Important chemical reactions of nitrogen are as follows:

(i) Nitrogen combines with several electropositive metals forming nitrides.

$$6Li + N_2 \rightarrow 2Li_3N$$

$$3Mg + N_2 \rightarrow Mg_3N_2$$

(ii) Nitrogen reacts with calcium carbide forming calcium cyanamide.

$$CaC_2 + N_2 \rightarrow CaNCN + C.$$

(iii) Nitrogen combines with hydrogen forming ammonia.

$$N_2 + 3H_2 \xrightleftharpoons[100-1000 \text{ atm}]{673-823K} 2NH_3$$

(v) Nitrogen reacts with oxygen, only at elevated temperatures (about 3273 K).

$$N_2 + O_2 \xrightleftharpoons{3273k} 2NO$$

9.6 HYDRIDES OF NITROGEN

9.6.1 Ammonia

Laboratory preparation

(i) Ammonia can be made in the laboratory by heating an ammonium salt with an alkali, e.g.,

$$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3$$

$$NH_4^+ + OH^- \rightarrow H_2O + NH_3$$

(ii) It is also obtained when an ionic nitride is hydrolyzed with water.

$$Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$$

$$AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3$$

(iii) Ammonia gas will be liberated when a nitrate or nitrite is heated with zinc or aluminium in alkaline medium.

$$[Zn + 2NaOH \rightarrow Na_2ZnO_2 + 2H] \times 4$$
$$NaNO_3 + 8H \rightarrow NaOH + 2H_2O + NH_3$$

$$4Zn + 7NaOH + NaNO_3 \rightarrow 4Na_2ZnO_2 + 2H_2O + NH_3$$

$$[Zn + 2NaOH \rightarrow Na_2ZnO_2 + 2H] \times 3$$
$$NaNO_2 + 6H \rightarrow NaOH + H_2O + NH_3$$

$$3Zn + 5NaOH + NaNO_2 \rightarrow 3Na_2ZnO_2 + H_2O + NH_3$$

$$[Al + NaOH + H_2O \rightarrow NaAlO_2 + 3H] \times 8$$
$$[NaNO_3 + 8H \rightarrow NaOH + 2H_2O + NH_3] \times 3$$

$$8Al + 5NaOH + 3NaNO_3 + 2H_2O \rightarrow 8NaAlO_2 + 3NH_3$$

$$[Al + NaOH + H_2O \rightarrow NaAlO_2 + 3H] \times 2$$
$$NaNO_2 + 6H \rightarrow NaOH + H_2O + NH_3$$

$$2Al + NaOH + H_2O + NaNO_2 \rightarrow 2NaAlO_2 + NH_3$$

(i) When urea is heated with sodium hydroxide, ammonia gas will be evolved.

$$NH_2CO\,NH_2 + 2NaOH \rightarrow Na_2CO_3 + 2NH_3$$

(ii) Hydrolysis of cyanamide gives ammonia.

$$CaCN_2 + 3H_2O \rightarrow CaCO_3 + 2NH_3$$

(iii) Heating of ammonium compounds also gives ammonia.

$$(NH_4)_2SO_4 \xrightarrow{\Delta} NH_3 + NH_4HSO_4$$

$$NH_4H_2PO_4 \xrightarrow{\Delta} NH_3 + HPO_3 + H_2O$$

Manufacture of Ammonia

The two important sources for the manufacture of ammonia have been ammonical liquor and air. The latter has been actually a source of nitrogen which gets converted into ammonia either by Haber's synthetic process or by cyanamide process.

(i) ***From Ammonical Liquor*****:** When coal is subjected to destructive distillation three fractions, viz.(a) coal gas, (b) ammonical liquor and (c) coaltar will be obtained. The ammonical liquor is a concentrated solution of ammonium salts. Milk of lime is added to the ammonical liquor and steam is blown through the mixture. The mixture of steam and ammonia evolved is allowed to bubble through sulphuric acid. Ammonium sulphate formed gets crystallized out. It is largely used as a fertilizer.

$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$$

In order to get liquor ammonia, ammonia evolved is purified and bubbled through water under pressure when we get a concentrated solution of ammonia.

(ii) ***Haber's Process*****:** Ammonia is manufactured on a large scale by the synthetic method from nitrogen and hydrogen elements

$$N_2 + 3H_2 \rightleftharpoons 2NH_3;\ \Delta H = -93.63\ \text{kJ}$$

In this process, nitrogen obtained from the air and hydrogen obtained from steam or petroleum are reacted together at a high pressure. The above reaction reveals that (a) it is a reversible reaction, (b) the forward reaction is exothermic and (c) the reaction is accompanied by a decrease in volume of the system, i.e., the number of molecules of the products are less than the total number of reactant molecules.

The reaction occurs slowly. The formation of ammonia in larger amounts is favoured by the following conditions according to Le Chatelier's principle.

(i) As the reaction is exothermic, low temperatures are suitable.

(ii) As the reaction is accompanied by a decrease in volume of the system during the forward reaction, high pressures are favourable.

(iii) A catalyst, if present, increases the rate of formation of NH_3.

In practice, a temperature of 725 to 775 K; 200 to 1000 atmospheres pressure and a solid comprising of finely divided iron as catalyst and molybdenum as promoter are used. As an alternative iron as a catalyst promoted by small amounts of mixture of potassium and aluminium oxide (K_2O and Al_2O_3) can be used.

The flow chart for the manufacture of ammonia is shown in Fig. 9.5.

The nitrogen and hydrogen used in the Haber's process must be very pure. Further, the unreacted nitrogen, and hydrogen must be recycled.

The gases (N_2 and H_2) are dried by passing over solid sodium hydroxide and mixture of nitrogen and hydrogen in 1:3 ratio by volume is compressed to 200–300 atmospheres. Then these gases are passed into a catalytic chamber and the incoming gases are preheated in the heat exchanger. They react in the presence of the catalyst at 725–775 K. Under these conditions, about 10 per cent ammonia is formed. It is separated from the unreacted N_2 and H_2 by condensation in the condenser. The unreacted gases are mixed with the initial amounts of the gas mixture.

(iii) ***Cyanamide Process*****:** Cyanamide process is another important commercial method for the preparation of NH_3. In this process finely divided calcium carbide reacts with N_2 gas in a cylindrical electric furnace at 1273–1373 K. Finely powdered anhydrous calcium chloride or fluoride is added to serve as a catalyst. Then a mixture of calcium cyanamide and graphite known as *nitrolime* is formed.

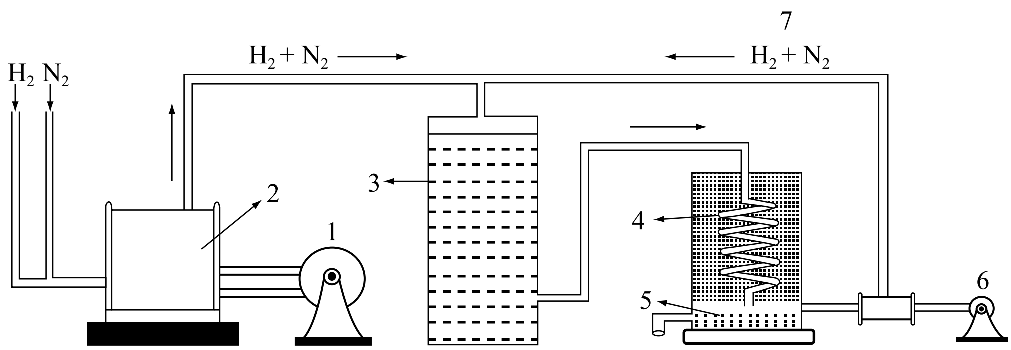


Fig 9.5 Synthesis of NH_3 – Haber's process. 1. Compressor 2. Purifier and drier 3. Catalytic chamber, Heat exchanger 4. Condenser 5. Ammonia liquid 6. Recycling pump 7. Unreacted N_2 and H_2

$$CaC_2 + N_2 \xrightarrow{1273-1378k} \underbrace{CaCN_2 + C}_{\text{Nitrolim}}(\text{graphite})$$

Calcium cyanamide is hydrolyzed with superheated steam at 453K to get NH_3

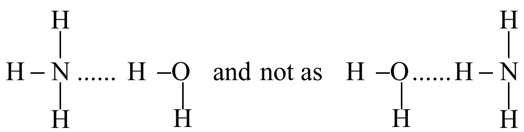
$$CaCN_2 + H_2O \longrightarrow CaO + H_2NCN$$
$$H_2NCN + H_2O \longrightarrow H_2NCONH_2$$
$$H_2NCONH_2 + H_2O \longrightarrow 2NH_3 + CO_2$$
$$CaO + CO_2 \longrightarrow CaCO_3$$
$$CaCN_2 + 3H_2O \longrightarrow CaCO_3 + 2NH_3$$

(iv) ***As a By-product*:** In the purification of bauxite by Serpeck's process, bauxite is mixed with coke and heated in the presence of nitrogen when it gives aluminium nitride which gives ammonia on hydrolysis.

$$Al_2O_3 + 3C + N_2 \rightarrow 2AlN + 3CO$$
$$AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3.$$

Physical Properties

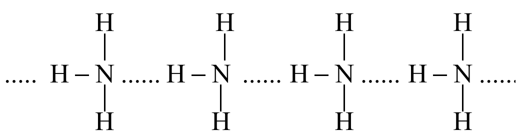
It is a colourless pungent gas, m.p 195K, b.p 240K. It is highly soluble in water (1300 volumes gas in 1 volume of water at 273K.) The solubility of ammonia in water is due to the formation of intermediate compounds with water as



This is indicated by the formation of NH_4OH. Aqueous ammonia (46.2%) hydrated ammonia $H_3N \cdot HOH$ (52.4%) and ammonium and hydroxyl ions (1.4%).

$$NH_3 + H_2O \rightleftharpoons NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

It is lighter than air (sp. gr. 0.5963). It liquefies into a colourless liquid (b.p. 240K) and freezes to white snowy crystals (m.p. 195K). Latent heat of vapourization of ammonia is 1380 J. Ammonia molecules are associated due to hydrogen bonding.



Due to this hydrogen bonding ammonia has higher m.p, b.p, latent heat of fusion and surface tension with respect to those of other hydrides of this group.

Chemical Properties

***Stability*:** It is quite a stable gas, decomposes only at red hot condition or when electric sparks are passed through it.

$$2NH_3 \rightarrow N_2 + 3H_2$$

***Combustion*:** Although it shows no reaction with air, it burns with pale green flame in oxygen and nitrogen and steam are produced.

$$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$$

***Oxidation*:** Ammonia is a reducing agent. Chlorine, bleaching powder and sodium hypochlorite being strong enough oxidizing agents to react with the gas at room temperature to produce nitrogen.

$$2NH_3 + 3Cl_2 \rightarrow N_2 + 6HCl\ (NH_4Cl \text{ formed with excess})$$
$$2NH_3 + 3CaOCl_2 \rightarrow N_2 + 3CaCl_2 + 3H_2$$
$$2NH_3 + 3NaOCl \rightarrow N_2 + 3NaCl + 3H_2$$

Ammonia is oxidized to nitrogen when passed over heated copper (II) oxide or lead (II) oxide

$$2NH_3 + 3CuO \rightarrow 3Cu + 3H_2O + N_2$$
$$2NH_3 + 3PbO \rightarrow 3Pb + 3H_2O + N_2$$

In the presence of platinum catalyst ammonia is oxidized to nitric oxide. This is the first stage of the industrial method for making nitric acid.

$$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$$

***Reaction with Halogens*:** Ammonia reacts with halogens. When halogen is excess nitrogen trihalides are formed. NBr_3 and NI_3 are formed as addition products $NBr_3 \cdot 6NH_3$ and $NI_3 \cdot 6NH_3$ which explodes on rubbing or on shock.

$$NH_3 + 3X_2 \rightarrow NX_3 + 3HX\ (K = F, Cl, Br \text{ or } I)$$

But if ammonia is excess the halogens oxidize the ammonia to nitrogen

$$2NH_3 + 3X_2 \rightarrow N_2 + 6HX.$$

***Basic Character*:** Ammonia readily dissolves in water and the solution turns red litmus to blue. A solution of ammonia in water is commonly called as *ammonium hydroxide*, although NH_4OH molecules are not present. The solution, which contains ammonia hydrogen bonded to water is best represented as an equilibrium between $NH_{3(aq)}$ and NH_4^+ and OH^- ions.

$$NH_{3(aq)} + H_2O \rightleftharpoons NH_4^+ + OH^-$$

$$K_{298} = \frac{[NH_4^+][OH^-]}{[NH_{3(aq)}]} = 1.81 \times 10^{-5}\ \text{mol dm}^{-3}$$

The low value of dissociation constant means that an aqueous solution of ammonia is a weak base, and forms salts with acids. Ammonium salts generally resemble the corresponding alkali metal salts in solubility and structure; they are, however, thermally unstable, e.g.,

$$NH_4Cl \xrightarrow{\Delta} NH_3 + HCl$$

$$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$$

$$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$$

***Reaction with Metals*:** The metals of the Groups IA and IIA are sufficiently electropositive to react with ammonia on heating to give ionic amides, e.g.,

$$2Na + 2NH_3 \rightarrow 2NaNH_2 + H_2$$

***Reaction with Metal Salts*:** Ammonium hydroxide when added to metal salts, several metals are precipitated as their hydroxides. Some metal hydroxide precipitates are insoluble even if excess of ammonium hydroxide is added. But some metal hydroxide precipitates dissolve if excess of ammonium hydroxide is added due to the formation of complexes. These reactions are useful to separate the metal ions from their mixtures.

The metal ions which are precipitated as their hydroxides but are insoluble in excess of ammonium hydroxide are Fe^{2+}, Fe^{3+}, Mn^{2+}, Al^{3+}

$$FeSO_4 + 2NH_4OH \rightarrow \underset{\text{Dirty green}}{Fe(OH)_2} \downarrow + (NH_4)_2SO_4$$

$$FeCl_3 + 3NH_4OH \rightarrow \underset{\text{Reddish brown}}{Fe(OH)_3} \downarrow + 3NH_4Cl$$

$$AlCl_3 + 3NH_4OH \rightarrow \underset{\text{White gelatinous}}{Al(OH)_3} \downarrow + 3NH_4Cl$$

$$MnSO_4 + 2NH_4OH \rightarrow Mn(OH)_2 \downarrow + (NH_4)_2SO_4$$

The metal ions which are precipitated as their hydroxides but dissolve in excess of ammonium hydroxide are Cu^{2+}, Cd^{2+}, Ag^{+}, Zn^{2+}, Co^{3+}, etc.

$$CuSO_4 + 2NH_4OH \rightarrow \underset{\text{Blue}}{Cu(OH)_2} \downarrow + (NH_4)_2SO_4$$

$$Cu(OH)_2 + (NH_4)_2SO_4 + 2NH_4OH \rightarrow \underset{\text{Dark blue}}{[Cu(NH_3)_4]SO_4} + 4H_2O$$

$$CdSO_4 + 2NH_4OH \rightarrow Cd(OH)_2 + (NH_4)_2SO_4$$

$$Cd(OH)_2 + (NH_4)_2SO_4 + 2NH_4OH \rightarrow [Cd(NH_3)_4]\ SO_4 + 4H_2O.$$

$$ZnSO_4 + 2NH_4OH \rightarrow Zn(OH)_2 + (NH_4)_2SO_4$$

$$Zn(OH)_2 + (NH_4)_2SO_4 + 2NH_4OH \rightarrow [Zn(NH_3)_4]SO_4 + 4H_2O$$

$$AgNO_3 + NH_4OH \rightarrow AgOH \downarrow + NH_4NO_3$$

$$2AgOH \rightarrow Ag_2O + H_2O$$

$$Ag_2O + 2NH_4NO_3 + 2NH_4OH \rightarrow 2\ [Ag(NH_3)_2]\ NO_3 + 4H_2O$$

$$NiCl_2 + 2NH_4OH \rightarrow Ni(OH)_2 + 2NH_4Cl$$

$$Ni(OH)_2 + 2NH_4Cl + 4NH_4OH \rightarrow [Ni(NH_3)_6]\ Cl_2 + 6H_2O$$

$$CoCl_3 + 3NH_4OH \rightarrow Co(OH)_3 + 3NH_4Cl.$$

$$Co(OH)_3 + 3NH_4Cl + 3NH_4OH \rightarrow [Co(NH_3)_6]\ Cl_3 + 6H_2O$$

***Reaction with Mercury Salts*:** Ammonia forms a white precipitate with mercuric chloride due to the formation of amidomercuric chloride

$$HgCl_2 + NH_4OH \rightarrow NH_2HgCl + NH_4Cl + H_2O$$

With mercurous chloride it forms a black precipitate

$$Hg_2Cl_2 + 2NH_4OH \longrightarrow \underbrace{NH_2HgCl + Hg}_{\text{black}} + NH_4Cl + 2H_2O$$

***Reaction with Nessler's Reagent*:** When potassium iodide is added to mercuric chloride first a red precipitate is formed which dissolves in excess of potassium iodide due to the formation of complex potassium tetraiodido hydrargate (II)

$$HgCl_2 + 2KI \rightarrow HgI_2 \downarrow + 2KCl$$

$$HgI_2 + 2KI \rightarrow K_2[HgI_4]$$

Alkaline solution of K_2HgI_4 is called *Nessler's reagent.* This gives a brown precipitate with ammonia due to the formation of amidomercuric oxymercuric iodide, also called iodide of Milon's base.

$$2K_2HgI_4 + 3KOH + NH_3 \rightarrow \underset{\text{Brown}}{H_2NHgOHgI} \downarrow + 7KI + 2H_2O$$

Uses

Ammonia is used

(i) For the preparation of nitrogenous fertilizers like ammonium sulphate, urea, calcium ammonium nitrate, etc.
(ii) as a refrigerant in ice plants
(iii) in the manufacture of sodium carbonate by Solvay's process, and in ammonia saturation tower
(iv) in the preparation of rayon and artificial silk, in the form of tetraammine copper (II) sulphate called *Schweitzer's reagent* as solvent for cellulose acetate
(v) in the manufacture of nitric acid by Ostwald's process
(vi) in the preparation of explosives like ammonium nitrate, *Amatol* (80% NH_4NO_3 + 20% TNT) and *Ammonal* ($NH_4\ NO_3$ + aluminium powder)
(vii) Liquid ammonia is a useful solvent for both ionic as well as covalent compounds
(viii) in the laboratory as a reagent and for removing grease oils
(ix) in dry cleaning

Tests for Ammonia

(i) It turns the moist red litmus to blue and moist turmeric to brown.
(ii) It gives dense white fumes when a glass rod dipped in concentrated HCl is exposed to it due to the formation of NH_4Cl.
(iii) With Nessler's reagent it gives brown precipitate.
(iv) When a filter paper dipped in mercuric chloride is exposed to ammonia it turns to grey.
(v) When ammonia gas is passed through a solution of copper sulphate, a deep blue colour is formed.

The structure of ammonia is already discussed in Chapter 3 (Chemical Bonding).

9.6.2 Hydrazine

The oxidation number of nitrogen in hydrazine is –2. It is isoelectronic with nitrogen amine in which the oxidation number of nitrogen is –1. These are formally related to NH_3 by the replacement of a H atom by an NH_2 group in hydrazine or an OH group in Hydroxylamine. Both compounds are liquids at room temperature.

Preparation

Hydrazine is mostly prepared by *Raschig's process* in which ammonia is oxidized by sodium hypochlorite in dilute aqueous solution

$$NH_3 + NaOCl \rightarrow NH_2Cl + NaOH \text{ (fast)}$$
$$2NH_3 + NH_2Cl \rightarrow NH_2NH_2 + NH_4Cl \text{ (slow)}$$

This is formally a redox reaction, but is more cutricate than simple electron transfer because it proceeds through the formation of the intermediate NH_2Cl. Once formed, the NH_2Cl is attacked by the nucleophile NH_3 which results in the displacement of Cl^- and the formation of N—N bond. A side reaction is formed between chloramine and hydrazine to form ammonium chloride and nitrogen

$$N_2H_4 + 2NH_2Cl \rightarrow N_2 + 2NH_4Cl$$

The reaction is catalyzed by heavy metal ions present in solution. So, distilled water free from metal ion rather than tap water must be used. Glue or gelatin must be added to mask the remaining metal ions by complexation. By using excess ammonia, the reaction between chloramine and hydrazine can be decreased. The use of dilute solution of the reactants is necessary to minimize another side reaction.

$$3NH_2Cl + 2NH_3 \rightarrow N_2 + 3NH_4Cl$$

A 2 per cent solution of hydrazine can be made by this method. The solution is concentrated either by distillation or by adding H_2SO_4 to precipitate the salt hydrazine sulphate $N_2H_4 \cdot H_2SO_4$. The precipitate is removed and treated with an alkali to get hydrazine hydrate. Thus on distillation under reduced pressure over an alkaline dehydrating agent like barium oxide or solid sodium hydroxide yields free hydrazine

$$N_2H_4 + H_2SO_4 \rightarrow N_2H_4 \cdot H_2SO_4$$
$$N_2H_4 \cdot H_2SO_4 + 2NaOH \rightarrow N_2H_4 \cdot H_2O + Na_2SO_4 + H_2O$$
$$N_2H_4 \cdot H_2O + BaO \rightarrow N_2H_4 + Ba(OH)_2$$

Physical Properties

Hydrazine is a covalent liquid which is colourless and fuming in air, smells like ammonia. It is hygroscopic, strongly associated through hydrogen bonding.

Chemical Properties

Pure hydrazine burns in air with evolution of large amount of heat.

$$N_2H_4 + O_2 \rightarrow N_2 + 2H_2O \quad \Delta H = -621 \text{ KJ mol}^{-1}$$

Due to this, hydrazine and its alkylated derivatives unsymmetrical dimethyl hydrazine. $(CH_3)_2NNH_2$ (UDMH) and CH_3NHNH_2 mixed with N_2O_4 are used as rocket fuels.

Basic nature: It is a weak base and reacts with acids forming two series of salts because it acts as diacid base. The salts are white ionic crystalline solids and are soluble in water

$$N_2H_4 + HX \rightarrow N_2H_5^+ + H^-$$
$$N_2H_4 + 2HX \rightarrow N_2H_6^{2+} + 2X^-$$

Action of sodium: It reacts with sodium in an inert atmosphere to form sodium hydrazide.

$$2N_2H_4 + 2Na \rightarrow 2NaN_2H_3 + H_2$$

Reaction with nitrous acid: With nitrous acid it forms hydrazoic acid, N_3H.

$$N_2H_4 + HNO_2 \rightarrow N_3H + 2H_2O$$

Reducing property: When dissolved in water (in neutral or basic solutions) hydrazine and its salts are powerful reducing agents. They are used to produce silver and copper mirrors and to precipitate the platinum metals. It also reduces I_2 and O_2 and salts of silver and gold.

$$PtCl_4 + N_2H_4 \rightarrow Pt + N_2 + 4HCl$$
$$4AuCl_3 + 3N_2H_4 \rightarrow 4Au + 3N_2 + 12HCl$$
$$4AgNO_3 + N_2H_4 \rightarrow 4Ag + N_2 + 4HNO_3$$
$$N_2H_4 + 2I_2 \rightarrow 4HI + N_2$$
$$N_2H_4 + 2O_2 \rightarrow 2H_2O + N_2$$

$$N_2H_4 + 2CuSO_4 \rightarrow 2Cu + N_2 + 2H_2SO_4$$

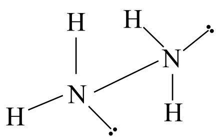
In acid solutions it acts as a mild reducing agent, though powerful reducing agents can reduce N_2H_4 to NH_3, thus N_2H_4 to be oxidized.

$$N_2H_4 + Zn + 2HCl \rightarrow 2NH_3 + ZnCl_2$$

Hydrazine may act as an electron donor. The nitrogen atoms have a lone pair of electrons which can form coordinate bonds to metal ions such as Ni^{2+} and CO^{2+}.

It is mostly used as a rocket fuel, as a reducing agent and as a reagent in organic chemistry.

Structure: Electron diffraction and infrared data indicate that the structure of hydrazine is related to that of ethane. Each N atom is tetrahedrally surrounded by one N, two H atoms and a lone pair. The two halves of the molecule are rotated 95° about the N—N bond and adopt a *gauche* (non-eclipsed) configuration. The N—N bond length is 145 pm and the bond energy is about 160 KJ mol^{-1}.



9.6.3 Hydroxylamine

Preparation

(i) ***By the reduction of nitric oxide***: On passing nitric oxide gas through a reducing mixture of granulated tin and concentrated hydrochloric acid, Hydroxylamine hydrochloride will be formed which remains in solution.

$$2NO + 6\,[H] \rightarrow 2NH_2OH$$

$$NH_2OH + HCl \rightarrow NH_2OH \cdot HCl$$

The excess tin is removed by passing H_2S as sulphide after it is dissolved completely and the solution is filtered and evaporated to dryness. The residue of Hydroxylamine hydrochloride on distillation with sodium methoxide in methyl alcohol will yield free Hydroxylamine.

$$NH_2OH \cdot HCl + CH_3ONa \rightarrow NH_2OH + NaCl + CH_3OH$$

(ii) ***By reduction of sodium nitrite***: When sodium nitrite is reduced by sulphur dioxide in the presence of sodium carbonate, Hydroxylamine will be formed.

$$NaNO_2 + SO_2 + NaHSO_3 + 2H_2O \rightarrow 2NaHSO_4 + NH_2OH$$

(iii) ***By electrolytic reduction of nitric acid***: Electrolysis of a mixture of 50 per cent nitric acid and 50 per cent sulphuric acid using lead electrodes in cathode compartment gives hydroxylamine sulphate and sulphuric acid. The excess sulphuric acid is removed by precipitating it with baryta water as $BaSO_4$.

$$HNO_3 + 2H \rightarrow HNO_2 + H_2O$$

$$HNO_2 + 4H \rightarrow NH_2OH + H_2O$$

(iv) ***By reduction of ethylnitrate***: When ethylnitrate is reduced with tin and hydrochloric acid, it gives Hydroxylamine hydrochloride.

$$C_2H_5NO_3 + 6H \rightarrow NH_2OH + C_2H_5OH + H_2O$$

Properties

At room temperature, hydroxylamine exists as colourless, deliquescent crystals, melting point 306 K, b.p 328–330K at 22 mm. It is readily soluble in water or alcohol but only slightly soluble in ether and benzene.

Basic nature: Hydroxylamine like hydrazine is a weaker base than ammonia. Its aqueous solution is basic due to hydrolysis.

$$NH_2OH + H_2O \rightleftharpoons (HO\ NH_3)\ OH$$

$$\rightleftharpoons HO\ NH_3^+ + OH^-$$

In aqueous solution it acts as a proton acceptor

$$NH_2OH + HCl \rightarrow HON^+H_3 + Cl^-$$

Stability: It is extremely unstable. Even at room temperature it slowly decomposes evolving nitrogen, nitrous oxide and ammonia.

$$2NH_2OH \rightarrow 3H_2O + NH_3 + N_2$$

In both acidic and basic medium hydroxylamine acts as a good reducing agent. For example, hydroxylamine hydrochloride in acid solution reduces ferric salts to ferrous salts and nitric acid to nitric oxide.

$$2NH_2OH \rightarrow HON = NOH + 4H^+ + 4e^-$$

$$HONH_3Cl + HNO_3 \rightarrow 2NO + 2H_2O + HCl$$

In basic medium hydroxylamine precipitates metallic gold from auricchloride, cuprous oxide from Fehling's solution and metallic silver from ammonical solution of silver salts. It reduces mercuric chloride first to mercurous chloride and then to the metal. In all these reactions, the hyponitrous acid initially formed decomposes and the final products are nitrous oxide and water.

$$2NH_2OH + 4OH^- \rightarrow N_2O + 5H_2O + 4e^-$$

$$\underset{\text{(as complex)}}{4Cu^{2+}} + 2NH_2OH + 8OH^- \rightarrow \underset{\text{Red ppt}}{2Cu_2O} + N_2O + 7H_2O$$

$$\underset{\text{(as complex)}}{4Ag^+} + 2NH_2OH + 4OH^- \rightarrow \underset{\text{silver mirror}}{4Ag} + N_2O + 5H_2O$$

$$4HgCl_2 + 2NH_2OH + 4OH^- \rightarrow \underset{\text{White ppt.}}{2Hg_2Cl_2} + N_2O + 5H_2O + 4Cl^-$$

$$2Hg_2Cl_2 + 2NH_2OH + 4OH^- \rightarrow \underset{\text{Black ppt.}}{4Hg} + N_2O + 5H_2O + 4Cl^-$$

Hydroxylamine reduces halogens to hydraacids.

$$2NH_2OH + 2Cl_2 \rightarrow N_2O + H_2O + 4HCl$$

It reduces bromates and iodates to bromides and iodides respectively.

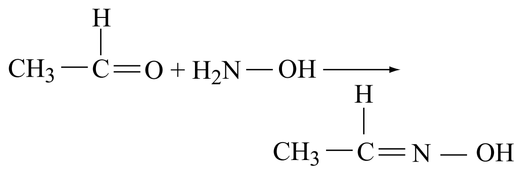
$$12NH_2OH + 4NaBrO_3 \rightarrow 4NaBr + 18H_2O + 3O_2 + 6N_2$$

***Oxidizing property*:** Hydroxylamine acts as an oxidizing agent in both acidic and basic medium. In strong acidic medium, it oxidizes stannous chloride to stannic chloride, sulphur dioxide to ammonium sulphate, etc. In basic medium it oxidizes ferrous hydroxide to ferric hydroxide quantitatively, it also oxidizes sodium arsenite to sodium arsenate.

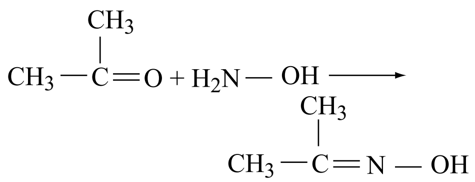
$$2Fe(OH)_2 + NH_2OH + H_2O \rightarrow 2Fe(OH)_3 + NH_3$$

$$Na_3AsO_3 + NH_2OH \rightarrow Na_3AsO_4 + NH_3$$

***Reaction with carbonyl compounds*:** Hydroxylamine reacts with aldehydes and ketones forming the corresponding oximes.



aldoxime.



ketoxime

Uses

(i) as a reagent in organic chemistry for the identification of carbonyl compounds

(ii) It is used for preparing cyclohexanone oxime which is first converted into caprolactum and then polymerized to nylon-6.

(iii) for the preparation of oximes which are used as important analytical reagents for the determination of metalions

9.6.4 Hydrazoic Acid

***Preparation*:** It can be prepared by the action of nitrous acid on hydrazine or its salts.

$$NH_2\,NH_2 + HNO_2 \rightarrow N_3H + 2H_2O$$

(iii) Hydrazoic acid can be conveniently prepared by passing nitrous oxide over fused sodamide at 190ºC when sodium azide is formed. Free hydrazoic acid will be formed when dry sodium azide is warmed with dilute sulphuric acid.

$$2NH_3 + 2Na \longrightarrow 2NaNH_2 + H_2$$

$$2NaNH_2 + N_2O \xrightarrow{190^\circ} NaN_3 + NaOH + NH_3$$

$$3NaNH_2 + NaNO_3 \xrightarrow{175^\circ} NaN_3 + 3NaOH + NH_3$$

$$2NaN_3 + H_2SO_4 \longrightarrow Na_2SO_4 + 2N_3H$$

***Properties*:** It is a poisonous, highly explosive liquid with an offensive smell, m.p 193K and b.p. 236K. It dissolves readily in water.

It explodes violently by dissociating into its constituents with the evolution of a large amount of heat.

$$2N_3H \rightarrow 3N_2 + H_2 \quad \Delta H = -51.7 \text{ KJ}$$

***Reduction*:** It is reduced to ammonia or hydrazine by sodium amalgam

$$N_3H + 4H_2 \rightarrow 3NH_3$$

$$2N_3H + 5H_2 \rightarrow 3NH_2 \cdot NH_2$$

***Oxidation*:** Hydrazoic acid can be oxidized to nitrogen and water by oxidizing agents like acidified permanganate. The reaction with ceric ion is quantitative.

$$2N_3H + O \rightarrow 3N_2 + H_2O$$

$$2Ce^{4+} + 2N_3H \rightarrow 2Ce^{3+} + 3N_2 + 2H^+$$

The oxidation of hydrazoic acid with nitrous acid is also quantitative under certain conditions.

$$N_3H + HNO_2 \rightarrow N_2 + N_2O + H_2O$$

***Action of acids*:** Acids like hydrochloric acid, hydroiodic acid, etc. decompose hydrazoic acid with the evolution of nitrogen.

$$N_3H + 2HCl \rightarrow NH_3 + N_2 + Cl_2$$

***Acidic nature*:** It is a slightly stronger acid than acetic acid. The aqueous solution readily dissolves iron, zinc and copper with the formation of azide, nitrogen and ammonia.

$$Zn + 3HN_3 \rightarrow Zn(N_3)_2 + N_2 + NH_3$$

Silver, lead and mercurous azides are insoluble and may be obtained by adding a soluble salt of the metal to a solution of sodium azide.

$$N_3H + AgNO_3 \rightarrow AgN_3\downarrow + HNO_3$$

Azides of many metals are known, those of heavy metals are generally explosive. Silver, lead and mercurous azides explode violently when struck. Lead azide, $Pb(N_3)_2$, is used in the explosive industry in detonation caps. It is preferred to mercury fulminate because it deteriorates on storage.

Azides of electropositive metals are not explosive, infact, decompose smoothly and quantitatively when heated to 300º or higher. For example

$$2NaN_{3(s)} \rightleftharpoons 2Na_{(l)} + 3N_{2(i)}$$

Azide ion also functions as a ligand in complexes of transition metals. In general N_3^- behaves rather than like a halide ion and is commonly considered as pseudohalide although the corresponding pseudohalogen $(N_3)_2$ is not known.

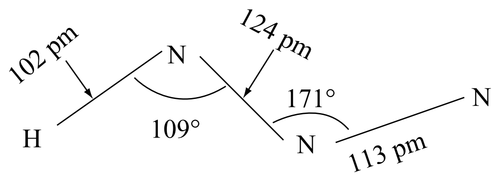
***Structure*:** The X-ray analysis shows that azide ion itself is symmetrical and linear (N—N 116pm) and its electronic structure may be represented in valence bond theory as

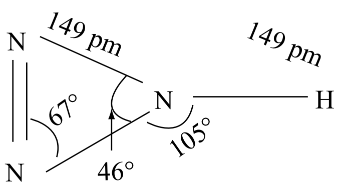
$$:\overset{-}{\ddot{N}}=\overset{+}{N}=\overset{-}{\ddot{N}}: \longleftrightarrow :N\equiv\overset{+}{N}-\overset{2-}{\underset{..}{\ddot{N}}}: \longleftrightarrow :\overset{2-}{\underset{..}{\ddot{N}}}-\overset{+}{N}\equiv N:$$

In covalent azides, the symmetry is lost as is evident from the structure of HN_3 and CH_3N_3. In such covalent azides, the electronic structure is a resonance hybrid of the following structures.

$$R:\ddot{N}=\overset{+}{N}=\overset{-}{\ddot{N}}: \longleftrightarrow R:\overset{-}{\ddot{N}}:\ \overset{+}{N}\equiv N:$$

In the gas phase hydrazoic acid's three N'atoms are almost collinear and the angle HNN is 109º and the two N—N distances are appreciably different. The structure of isomeric cyclotriazene is also given below for comparison.





9.7 OXIDES OF NITROGEN

9.7.1 Nitrous Oxide or Nitrogen (I) Oxide (N_2O)

***Preparation*:** In nitrous oxide nitrogen is present in +1 oxidation state. It can be readily prepared by heating ammonium nitrate or a mixture of sodium nitrate and ammonium sulphate to prevent the explosion of ammonium nitrate.

$$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$$

Nitrous oxide was first prepared by Priestly. When nitric oxide is passed over moist iron filings N_2O is formed.

$$2NO + H_2O + Fe \rightarrow N_2O + Fe(OH)_2$$

It can also be prepared by reducing nitric acid with stannous chloride and hydrochloric acid.

$$4SnCl_2 + 2HNO_3 + 8HCl \rightarrow 4SnCl_4 + N_2O + 5H_2O$$

***Properties*:** It is a colourless gas with pleasant odour producing mild laughing hysteria. So it is called a *laughing gas* and also used as an anaesthetic agent for minor operations.

It decomposes above 823K yielding oxygen and nitrogen.

$$2N_2O \rightarrow 2N_2 + O_2$$

Therefore, it readily supports combustion acting as a source of oxygen. Non-metals like carbon, sulphur, phosphorus, etc. burn in N_2O.

$$C + 2N_2O \rightarrow CO_2 + 2N_2$$
$$S + 2N_2O \rightarrow SO_2 + 2N_2$$
$$P_4 + 10N_2O \rightarrow P_4O_{10} + 10N_2$$

Metals are also oxidized to their corresponding oxides when heated in N_2O

$$2Na + N_2O \rightarrow Na_2O + N_2$$
$$Mg + N_2O \rightarrow MgO + N_2$$
$$Cu + N_2O \rightarrow CuO + N_2$$

A mixture of nitrous oxide and hydrogen in equal volumes explodes with a violent reaction.

$$N_2O + H_2 \rightarrow N_2 + H_2O$$

It is also notable that N_2O (like N_2 itself) can act as a ligand by displacing H_2O from the aquocomplexes, e.g., $[Ru(NH_3)_5(H_2O)]^{2+} + N_2O_{(aq)} \rightarrow [Ru(NH_3)_5(N_2O)]^{2+} + H_2O$.

***Uses*:** It is used as a propellent gas. A mixture of nitrous oxide and oxygen is used as an anaesthetic agent in dental and other minor surgical operations. It is also used as aerating agent for "Whipped" ice -cream.

***Structure*:** It is linear and a symmetrical molecule with a very small dipole moment explained by its being a reasonance hybrid of the following two structures each of which would be highly polar in opposite sense. It is proposed that there is delocalization of charge.

$$:\overset{-}{\ddot{N}}=\overset{+}{N}=\ddot{O}: \longleftrightarrow :N\equiv\overset{+}{N}-\overset{-}{\underset{..}{\ddot{O}}}:$$

The bond lengths in the linear molecule have been respectively 112 pm and 119 pm for N—N and N—O bonds.

$$N \ \underline{\ 112\text{ pm}\ }\ N\ \underline{\ 119\text{ pm}\ }\ O$$

9.7.2 Nitric Oxide (NO)

Preparation: It can be prepared by the direct reaction between nitrogen and oxygen at about 3000ºC.

$$N_2 + O_2 \rightarrow 2NO$$

It can be prepared by the oxidation of ammonia with oxygen above 773K in the presence of platinum gauze (Ostwald's process).

$$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$$

In the laboratory it can be prepared by the action of 8N nitric acid on copper.

$$3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$$

The liberated gas may contain NO_2 and N_2O. These are separated by passing the mixture through ferrous sulphate solution. Ferrous sulphate absorbs NO due to the formation of a brown complex. When the solution is heated pure nitric oxide will be liberated.

Pure sample of the gas may be obtained by the reduction of potassium nitrate by heating with ferrous sulphate acidified with sulphuric acid.

$$2HNO_3 + 6Fe^{2+} + 6H^+ \rightarrow 2NO + 6Fe^{3+} + 4H_2O$$

Properties: It is a colourless gas which is slightly soluble in water. It is a neutral oxide. It is an odd electron molecule having a paramagnetic character.

It immediately combines with oxygen in air converting into brown NO_2 gas.

$$2NO + O_2 \rightarrow 2NO_2$$

It is the most stable oxide of nitrogen. On heating, it starts decomposing at 1175K.

$$2NO \rightarrow N_2 + O_2$$

It is combustible and supports the combustion of only boiling sulphur and vigorously burning phosphorus. Burning sulphur and feebly burning phosphorus gets extinguished because at that temperature NO do not break up and do not supply oxygen for burning of sulphur and phosphorus.

$$S + 2NO \rightarrow SO_2 + N_2$$

$$P_4 + 10NO \rightarrow P_4O_{10} + 5N_2$$

Red hot copper decomposes the gas thereby forming nitrogen and copper oxide.

$$2Cu + 2NO \rightarrow 2CuO + N_2$$

Oxidizing properties: NO can act as an oxidizing agent and gets reduced to nitrogen but sometimes ammonia or nitrous oxide (N_2O) are also formed. It oxidizes hydrogen to water, sulphurous acid to sulphuric acid and hydrogen sulphide to sulphur.

$$2NO + 2H_2 \xrightarrow{\text{sparking}} N_2 + 2H_2O$$

$$2NO + 5H_2 \xrightarrow{\text{Pt black}} 2NH_3 + 2H_2O$$

$$2NO + H_2SO_3 \longrightarrow N_2O + H_2SO_4$$

$$2NO + H_2S \longrightarrow N_2O + H_2O + S.$$

Stannous chloride reduces nitric oxide to Hydroxylamine.

$$[SnCl_2 + 2HCl \longrightarrow SnCl_4 + 2H] \times 3$$

$$2NO + 6H \longrightarrow 2NH_2OH$$

$$\overline{3SnCl_2 + 6HCl + 2NO \longrightarrow 3SnCl_4 + 2NH_2OH}$$

Nitric oxide directly combines with halogens forming nitrosyl halides.

$$2NO + X_2 \rightarrow 2NOX \ (X = F, Cl \text{ or } Br)$$

Reducing property: As nitric oxide can be readily oxidized, it can act as a good reducing agent. It reduces acidified potassium permanganate and itself gets oxidized to nitric acid.

$$[2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)] \times 3$$

$$[2NO + H_2O + 3(O) \rightarrow 2HNO_3] \times 5$$

$$\overline{6KMnO_4 + 9H_2SO_4 + 10NO \rightarrow 3K_2SO_4 + 6MnSO_4 + 10HNO_3 + 4H_2O}$$

Iodine also oxidizes nitric oxide to nitric acid.

$$3I_2 + 4H_2O + 2NO \rightarrow 6HI + 2HNO_3.$$

Concentrated nitric acid is able to oxidize nitric oxide to nitrogen dioxide in accordance with the following reversible equation.

$$2HNO_3 + NO \rightleftharpoons 3NO_2 + H_2O$$

This reaction explains why concentrated nitric acid reacts with metals to form NO_2, whereas dilute nitric acid yields NO. With concentrated nitric acid the reaction takes place in the forward direction, but in the presence of water, e.g., with dilute acid, it takes place backwards and with moderately strong nitric acid both the gases are formed.

Reaction with ferrous sulphate: It gets absorbed by a cold solution of ferrous sulphate with the formation of dark brown nitrosyl complex $[Fe(H_2O)_5(NO)]^{2+}$. The complex is stable only at low temperature and decompose on heating with evolution of nitric oxide.

$$[Fe(H_2O)_5(NO)]^{2+} \xrightleftharpoons{\text{Heat}} Fe^{2+} + NO + 5H_2O$$

This reaction is used for its purification and separation from other gases.

The complex $[Fe(H_2O)_5(NO)]^{2+}$ contain nitrosonium ion (NO^+) and Fe^+. The magnetic moment of the species is 3.9 BM which shows the presence of three unpaired electrons.

Uses

(i) It is used in the manufacture of nitric acid.
(ii) It is used as a catalyst in the manufacture of sulphuric acid by Lead chamber's process.
(iii) It is used in the detection of oxygen to distinguish it from N_2O.

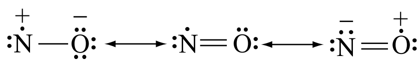
Structure: Nitric oxide molecule is having a total of 11 electrons (five from nitrogen and six from oxygen) in the valence shells of nitrogen and oxygen atoms. Its paramagnetism reveals the presence of odd number of electrons but its properties have been different from other odd electron molecules in the following aspects.

(i) It is colourless in the gaseous state and becomes brown when it comes in contact with air and is blue in the liquid state.
(ii) It is somewhat less active chemically.
(iii) It does not dimerize under ordinary condition.

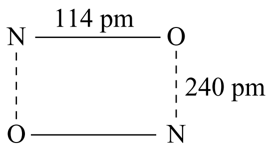
Because of these unusual properties it is regarded that the unpaired electron in nitric oxide gets spread over the whole molecule and is generally depicted with double bond as well as three electron bond between the atoms as shown below

$$:\mathrm{N} \overset{\cdots}{=} \mathrm{O}:$$

Three electron bonds get formed between atoms of similar electronegativity (N and O here) and may be regarded as resonance hybrid of the following structures



As a three-electron bond has been about half as a covalent bond, the bond length in NO would be in between that of double and triple bond. The bond length in the monomer has been 114 pm. In the liquid state it dimerizes to some extent.



The liquid has a low dielectric constant. The molecule has been very stable. So, the association to N_2O_2 at ordinary temperature has been exothermic.

9.7.3 Nitrogen Sesquioxide or Dinitrogen Trioxide or Nitrogen (III) Oxide N_2O_3

Preparation: It is obtained as a blue liquid when equimolar mixtures of NO and NO_2 are cooled to about –20ºC.

$$NO_{(g)} + NO_{2(g)} \rightleftharpoons \underset{\text{Blue}}{N_2O_{3(l)}}$$

It can be prepared by distilling a mixture of 60 per cent nitric acid with arsenious oxide or starch and condensing the vapours in U-tube dipped in the freezing mixture.

$$As_4O_6 + 4HNO_3 + 4H_2O \rightarrow 4H_3AsO_4 + 2N_2O_3$$

It is also obtained by the action of 5N nitric acid on metallic copper.

$$3Cu + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$$
$$Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2$$
$$4Cu + 12HNO_3 \longrightarrow 4Cu(NO_3)_2 + 6H_2O + \underbrace{2NO + 2NO_2}_{\downarrow \; 2N_2O_3}$$

or $\quad 2Cu + 6HNO_3 \rightarrow 2Cu(NO_3)_2 + 3H_2O + N_2O_3$

Properties: Solid nitrogen sesquioxide forms blue crystals, m.p. 393K. Thermal dissociation starts on melting.

$$\underset{\substack{\text{Blue}\\\text{liquid}}}{N_2O_3} \xrightarrow{\text{Room Temp}} \underbrace{NO + NO_2}_{\text{Brown gas}}$$

When dissolved in water it forms nitrous acid and with alkalis it forms nitrites.

$$N_2O_3 + H_2O \rightarrow 2HNO_2$$
$$N_2O_3 + 2NaOH \rightarrow 2NaNO_2 + H_2O$$

So, theoretically it is considered as an anhydride of nitrous acid.

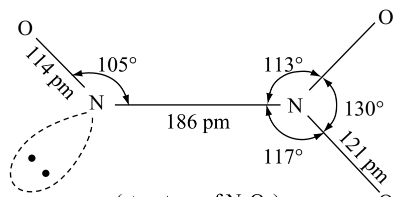
It is also absorbed by sulphuric acid forming nitrosyl sulphuric acid.

$$N_2O_3 + 2H_2SO_4 \rightarrow 2SO_2(OH)NO_2 + H_2O$$

Structure: Nitrogen tracer studies have shown rapid exchange between NO and NO_2 consistent with the equilibrium.

$$N_2O_3 \rightleftharpoons NO_{(g)} + NO_{2(g)}$$

The solid is believed to have two forms: an unstable one of structure ONONO and the other with long N—N bond.



(structure of N_2O_3)

Structure of N_2O_3 is of two forms:

Asymmetrical form Symmetric form

9.7.4 Dinitrogen Tetroxide $\rightleftharpoons$ Nitrogen Dioxide, Nitrogen (IV) Oxide

Preparation: Nitrogen dioxide is obtained by the action of nitric oxide with oxygen.

$$2NO + O_2 \rightarrow 2NO_2$$

In the laboratory it is conveniently prepared by heating lead nitrate.

$$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$$

It can be prepared by heating concentrated nitric acid with copper turnings.

$$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2$$

It can also be obtained by the action of nitric acid on sodium nitrite.

$$NaNO_2 + 2HNO_3 \rightarrow NaNO_3 + H_2O + 2NO_2$$

Properties: It is a reddish brown gas having pungent smell. With change in temperature, it associates or decomposes.

$$\underset{\text{Colourless gases}}{2NO + O_2} \xrightleftharpoons{770K} \underset{\text{Reddish brown gas}}{2NO_2} \xrightleftharpoons{410K} \underset{\text{Gas begins to fade in colour}}{2NO_2}$$

$$\xrightleftharpoons{300K} \underset{\text{Yellow liquid}}{N_2O_4} \xrightleftharpoons{200K} \underset{\text{Colourless liquid}}{N_2O_4}$$

It clearly indicates that with decrease in temperature association of NO_2 molecules to N_2O_4 increases.

It dissolves in nitric acid forming *fuming nitric acid*. It is highly poisonous.

When dissolved in water it forms a mixture of nitrous and nitric acids.

$$2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$$

So, it is considered as a mixed anhydride of nitrous and nitric acids. When passed through alkalis it disproportionate forming nitrites and nitrates.

$$2NO_2 + 2NaOH \rightarrow NaNO_2 + NaNO_3 + H_2O$$

When dissolved in warm water it gives nitric acid and nitric oxide.

$$3NO_2 + H_2O \rightarrow 2HNO_3 + NO$$

Supporter of combustion: It is not combustible but supports the combustion of brightly burning phosphorus, magnesium ribbon or glowing charcoal.

$$C + 2NO_2 \rightarrow CO_2 + 2NO$$
$$P_4 + 10NO_2 \rightarrow P_4O_{10} + 10NO$$
$$Mg + NO_2 \rightarrow MgO + NO$$

The heat liberated during the brightly burning phosphorus or magnesium is sufficient to break the NO_2 so that oxygen released will make the phosphorus and magnesium continue to burn. But the heat liberated during the burning of sulphur is not sufficient to break NO_2. So, burning sulphur extinguishes in NO_2.

Oxidation properties: Because of its tendency to give nascent oxygen, NO_2 can act as a good oxidizing agent.

$$NO_2 \rightarrow NO + O$$

It oxidizes sulphur dioxide to sulphuric acid, hydrogen sulphide to sulphur, carbon monoxide to carbon dioxide and liberates iodine from iodides.

$$SO_2 + H_2O + NO_2 \rightarrow H_2SO_4 + NO$$
$$H_2S + NO_2 \rightarrow H_2O + NO + S$$
$$CO + NO_2 \rightarrow CO_2 + NO$$
$$2KI + 2NO_2 \rightarrow 2KNO_2 + I_2$$

It oxidizes ferrous sulphate to ferric sulphate and the NO formed in the reaction combines with ferrous sulphate forming a brown complex.

$$2FeSO_4 + H_2SO_4 + NO_2 \rightarrow Fe_2(SO_4)_3 + H_2O + NO$$
$$FeSO_4 + NO \rightarrow \underset{\text{Brown complex}}{FeSO_4 \cdot NO}$$

Several metals like copper, lead, tin, etc. When heated in NO_2 oxidizes to their corresponding oxides.

$$Pb + NO_2 \rightarrow PbO + NO$$
$$Cu + NO_2 \rightarrow CuO + NO$$

Concentrated sulphuric acid absorbs NO_2 forming nitrosyl hydrogen sulphate.

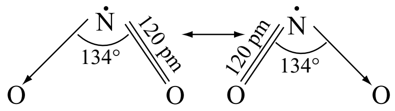
$$2NO_2 + H_2SO_4 \rightarrow NOHSO_4 + HNO_3$$

Reducing Property: NO_2 reduces more powerful oxidizing agents, for example potassium permanganate.

$$\begin{aligned} 2KMnO_4 + 3H_2SO_4 &\longrightarrow K_2SO_4 + 2H_2SO_4 \\ &\qquad + 3H_2O + 5(O) \\ \left[2NO_2 + H_2O + O \right.&\longrightarrow \left. 2HNO_3\right] \times 5 \\ \hline 2KMnO_4 + 3H_2SO_4 + 10NO_2 + 2H_2O &\longrightarrow K_2SO_4 \\ &\qquad + 2H_2SO_4 + 10HNO_3 \end{aligned}$$

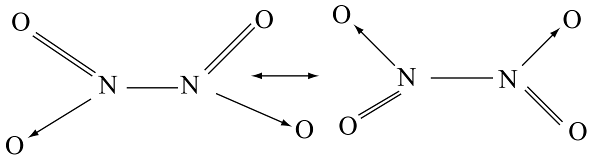
Uses: It is used in the manufacture of nitric acid and as a catalyst in the manufacture of sulphuric acid by lead chamber process.

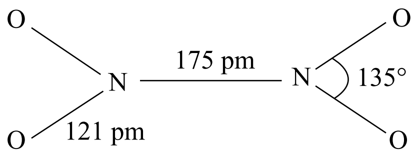
Structure: NO_2 is an odd electron molecule with 17 valence electrons (5 from nitrogen and 12 from two oxygen atoms). It has an unpaired electron and is paramagnetic in nature and since both the N—O bond lengths are equal it is considered as a resonance hybrid of the following structures



Nitrogen dioxide

When NO_2 dimerizes it becomes colourless and diamagnetic indicating the unpaired electron participate in N—N bond. The N—N bond is very weak and hence N_2O_4 is unstable.





9.7.5 Dinitrogen Pentoxide N_2O_5

It is prepared by dehydrating concentrated nitric acid with phosphorus pentoxide.

$$4HNO_3 + P_4O_{10} \xrightarrow{-10^\circ C} N_2O_5 + 4HPO_3$$

It is also formed when ozone is passed over liquid N_2O_4.

$$N_2O_4 + O_3 \rightarrow N_2O_5 + O_2$$

It can be prepared by passing dry chlorine gas over dry silver nitrate.

$$4AgNO_3 + 2Cl_2 \rightarrow 4AgCl + 2N_2O_5 + O_2$$

Properties: It is a highly deliquescent, light sensitive, colourless crystalline solid. It becomes yellow by raising the temperature on account of partial decomposition into brown NO_2.

$$2N_2O_5 \rightarrow 4NO_2 + O_2$$

When dissolved in water it gives nitric acid. It is the true anhydride of nitric acid.

$$N_2O_5 + H_2O \rightarrow 2HNO_3$$

With alkalis it forms nitrates.

$$N_2O_5 + 2NaOH \rightarrow 2NaNO_3 + H_2O$$

It reacts violently as an oxidizing agent towards many metals, non-metals and organic substances.

$$N_2O_5 + Na \rightarrow NaNO_3 + NO_2$$

$$N_2O_5 + NaF \rightarrow NaNO_3 + FNO_2$$

$$N_2O_5 + I_2 \rightarrow I_2O_5 + N_2.$$

When dissolved in H_2SO_4 or HNO_3, it ionizes as follows

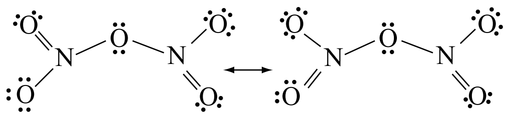
$$N_2O_5 \rightleftharpoons NO_2^+ + NO_3^-$$

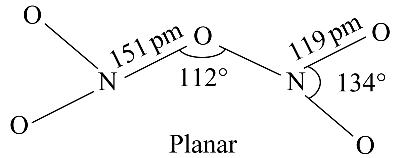
With sodium chloride it forms sodium nitrate and nitronium chloride.

$$N_2O_5 + NaCl \rightarrow NaNO_3 + NO_2Cl$$

This reaction indicates the presence of NO_2^+ and NO_3^- ions in N_2O_5 and so it is also called as *nitronium nitrate*.

Structure: X-ray diffraction studies show that solid N_2O_5 consists of an ionic array of linear NO_2^+ (N—O bond length 115.4 pm) and planar NO_3^- (N—O bond length 124 pm). In the gas phase and in solution (CCl_4, $CHCl_3$ $POCl_3$, etc.) the compound is molecular. The structure is not well established but may be $O_2N—O—NO_2$ with central N—O—N angle close to 112º.





9.8 OXOACIDS OF NITROGEN

Nitrogen forms numerous oxoacids, though several are unstable in the free state and are known only in the aqueous solution or as their salts. Of these, by far the most stable is nitric acid and this compound together with its salts, the nitrates, is a major product of the chemical industry. The different oxoacids of nitrogen are given in Table 9.7.

9.8.1 Nitrous Acid HNO_2

The free existence of nitrous acid is not very definite and exists only as an aqueous solution. However, there is spectroscopic evidence that nitrous acid can exist in the vapour phase.

Preparation: Solutions of nitrous acid can be easily prepared by acidifying solutions of nitrites with mineral acids.

$$NaNO_2 + HCl \rightarrow NaCl + HNO_2$$

$$2KNO_2 + H_2SO_4 \rightarrow K_2SO_4 + 2HNO_2$$

Table 9.7 Oxoacids of nitrogen

Name	Formula		Structural formula
1	Hyponitrous acid	$H_2N_2O_2$	HON = NOH
2	Hydronitrous acid (or) Nitroxylic acid	$H_4N_2O_4$	$(HO)_2N-N(OH)_2$
3	Nitrous acid	HNO_2	HO—N = O
4	Peroxonitrous acid	HNO_3	H—O—O—N = O
5	Nitric acid	HNO_3	$H-O-N(=O)\rightarrow O$
6	Peroxonitric acid	HNO_4	$H-O-O-N(=O)\rightarrow O$

Salt free aqueous solutions of nitrous acid can be made by choosing the acid which can form insoluble salt

$$Ba(NO_2)_2 + H_2SO_4 \rightarrow BaSO_4 + 2HNO_2$$

$$AgNO_2 + HCl \rightarrow AgCl + HNO_2$$

Nitrous acid is also formed when N_2O_3 is dissolved in water.

$$N_2O_3 + H_2O \rightarrow 2HNO_2$$

Properties: It is a weak acid, Ka = 4.5×10^{-4}. Aqueous solutions of nitrous acid are unstable and decompose rapidly when heated according to the equation

$$3HNO_2 \rightarrow HNO_3 + H_2O + 2NO$$

Thus in aqueous solution, nitrous acid undergoes auto oxidation.

Oxidation properties: It is a good oxidizing agent in acid medium due to the following reactions

$$H-O-N=O + H^+ \rightarrow H-\overset{+}{\underset{\underset{H}{|}}{O}}-N=O \rightarrow NO^+ + H_2O$$

$$NO^+ + e^- \rightarrow NO$$

(i) It oxidizes hydrogen sulphide to sulphur with evolution of brown NO_2 fumes.

$$H_2S + 2HNO_2 \rightarrow S + 2H_2O + 2NO$$

$$2NO + O_2 \rightarrow 2NO_2.$$

(ii) It oxidizes several reducing agents like sulphur dioxide to sulphuric acid, stannous chloride to stannic chloride, potassium iodide to iodine and acidified ferrous sulphate to ferric sulphate.

$$SO_2 + 2HNO_2 \rightarrow H_2SO_4 + 2NO$$

$$SnCl_2 + 2HCl + 2HNO_2 \rightarrow SnCl_4 + 2NO + 2H_2O$$

$$2KI + H_2SO_4 + 2HNO_2 \rightarrow K_2SO_4 + 2NO + I_2 + 2H_2O$$

$$2FeSO_4 + H_2SO_4 + 2HNO_2 \rightarrow Fe_2(SO_4)_3 + 2NO + 2H_2O$$

Sodium arsenite is oxidized to sodium arsenate.

$$Na_3AsO_3 + 2HNO_2 \rightarrow Na_3AsO_4 + 2NO + H_2O$$

Reduction Properties: It also acts as a good reducing agent due to the following reaction

$$HNO_2 + H_2O \rightarrow HNO_3 + 2H^+ + 2e^-$$

It decolourises acidified potassium permanganate, orange red dichromate to green chromic sulphate, bromine to hydrobromic acid and hydrogen peroxide to water.

$$\begin{array}{l} 2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2H_2SO_4 + 3H_2O + 5(O) \\ [HNO_2 + (O) \rightarrow HNO_3] \times 5 \\ \hline 2KMnO_4 + 3H_2SO_4 + 5HNO_2 \rightarrow K_2SO_4 + 2H_2SO_4 + 5HNO_3 + 3H_2O \end{array}$$

$$\begin{array}{l} K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3(O) \\ [HNO_2 + O \rightarrow HNO_3] \times 3 \\ \hline K_2Cr_2O_7 + 4H_2SO_4 + 3HNO_2 \rightarrow K_2SO_4 + 2H_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3HNO_3 \end{array}$$

$$\begin{array}{l} Br_2 + H_2O \longrightarrow 2HBr + (O) \\ HNO_2 + (O) \longrightarrow HNO_3 \\ \hline Br_2 + H_2O + HNO_2 \longrightarrow 2HBr + HNO_3 \end{array}$$

$$\begin{array}{l} H_2O_2 \longrightarrow H_2O + (O) \\ HNO_2 + (O) \longrightarrow HNO_3 \\ \hline H_2O_2 + HNO_2 \longrightarrow HNO_3 + H_2O \end{array}$$

Reaction with ammonia: It decomposes ammonia, yielding nitrogen.

$$NH_3 + HNO_2 \longrightarrow NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$$

Reaction with compounds containing $-NH_2$ group: It decomposes urea and other aliphatic amines giving nitrogen.

$$NH_2CONH_2 + 2HNO_2 \rightarrow CO_2 + 2N_2 + 3H_2O$$

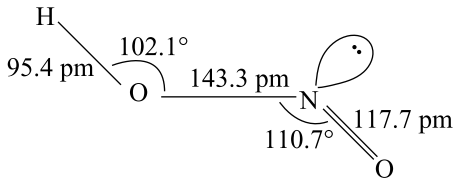
$$C_2H_5NH_2 + HNO_2 \rightarrow C_2H_5OH + N_2 + H_2O$$

Reaction with aromatic amines: At low temperatures, it reacts with aromatic amines to yield diazonium salts.

$$C_6H_5NH_2 \cdot HCl + HNO_2 \rightarrow C_6H_5N=N—Cl + 2H_2O$$

Uses: It is mainly used in the manufacture of azodyes.

Structure: Microwave spectroscopy shows that the gaseous compound is predominantly in the trans-planar configuration



9.8.2 Nitric Acid HNO_3

It is the most important and useful oxoacid of nitrogen. In the laboratory, it is prepared by heating a nitrate (sodium nitrate or potassium nitrate) with concentrated sulphuric acid.

$$NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$$

Manufacture of Nitric Acid

(a) Birkland and Eyde Process

This method is used at places where electric power is cheap. In this method, atmospheric nitrogen and oxygen can be made to combine under the influence of an electric arc (temperature 3000–3500ºC) to give nitric oxide.

$$N_2 + O_2 \xrightleftharpoons{\text{Electric arc}} 2NO \quad \Delta H = +180.7\text{ KJ}$$

The nitric oxide is then oxidized to nitrogen dioxide by the oxygen which is then absorbed in water to give nitric acid.

$$2NO + O_2 \rightarrow 2NO_2$$

$$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$$

The formation of NO from N_2 and O_2 is reversible reaction and endothermic. So, the yield of nitric oxide, however, is extremely small (only about 1% by the time it is oxidized to NO_2) and hence the method has now become practically obsolete.

(b) Ostwald's Process

Nitric acid is now almost invariably prepared by the catalytic oxidation of ammonia. The ammonia obtained by Haber's process is mixed with 10 volumes of air and passed over a platinum–rhodium gauze at 1023–1173K when ammonia is oxidized to nitric oxide.

$$4NH_3 + 5O_2 \xrightarrow{\text{Pt}} 4NO + 6H_2O; \Delta H = -21.5\text{k. Cal}$$

The nitric oxide is then oxidized to nitrogen dioxide which is cooled to about 323 K and absorbed in water (in the presence of air) to give nitric acid.

$$2NO + O_2 \rightleftharpoons 2NO_2$$

$$4NO_2 + 2H_2O + O_2 \longrightarrow 4HNO_3$$

If nitrogen dioxide is not dissolved in the presence of air, the following reaction occurs

$$3NO_2 + H_2O \rightarrow 2HNO_3 + NO$$

The yield of nitric acid becomes less. The nitric oxide is immediately oxidized to nitrogen dioxide which is again absorbed by water. The acid formed is about 61 per cent concentrated.

Concentration of nitric acid: The crude HNO_3 is concentrated in three stages to get crystals of pure HNO_3.

(i) Distillation of 61 per cent nitric acid gives 68 per cent nitric acid

(ii) 68 per cent nitric acid is mixed with conc H_2SO_4 and distilled again when 98 per cent acid is obtained.

(iii) 98 per cent nitric acid is cooled in a freezing mixture. Crystals of pure HNO_3 separates out.

Properties: Nitric acid is a colourless fuming liquid, m.p. 231.5K, b.p. 359K, with decomposition. Its density at 288K is 1.5241. The pure nitric acid, like that of water, undergoes autoionization giving nitrate ion and nitronium ion.

$$2HNO_3 \rightleftharpoons NO_2^+ + NO_3^- + H_2O$$

At 233 K the percentage dissociation of pure nitric acid ionized is to an extent of 3.5. When the pure nitric acid is diluted the increased concentration of water shifts the equilibria to the left and water molecules replace nitric acid molecules as proton acceptors.

The aqueous solution having 68 per cent of nitric acid by weight forms azeotropic mixture (constant boiling mixture) which boils at 393.5 K. Dilute nitric acid cannot be concentrated beyond 68 percent by boiling.

Pure nitric acid is colourless but it develops a yellow colour due to the photochemical decomposition to brown nitrogen dioxide.

$$4HNO_3 \rightarrow 2H_2O + 4NO_2 + O_2$$

The brown NO_2 can be removed by passing air through warm acid. Fuming nitric acid may be prepared by dissolving an excess of nitrogen dioxide in concentrated nitric acid. It is brown in colour. Because of its strong acidic and oxidation properties it can dissolve several substances like metals, non-metals and compounds. Hence, it is also called as *aqua fortis*.

Chemical properties: Chemical properties of nitric acid can be studied under the following headings

I. Acid properties

II. Nitration properties

III. Oxidation properties

(i) *Acid Properties*: Nitric acid is a very strong acid. In aqueous solution its ionization is almost complete.

$$HNO_3 + H_2O \rightleftharpoons H_3O^+ + NO_3^-$$

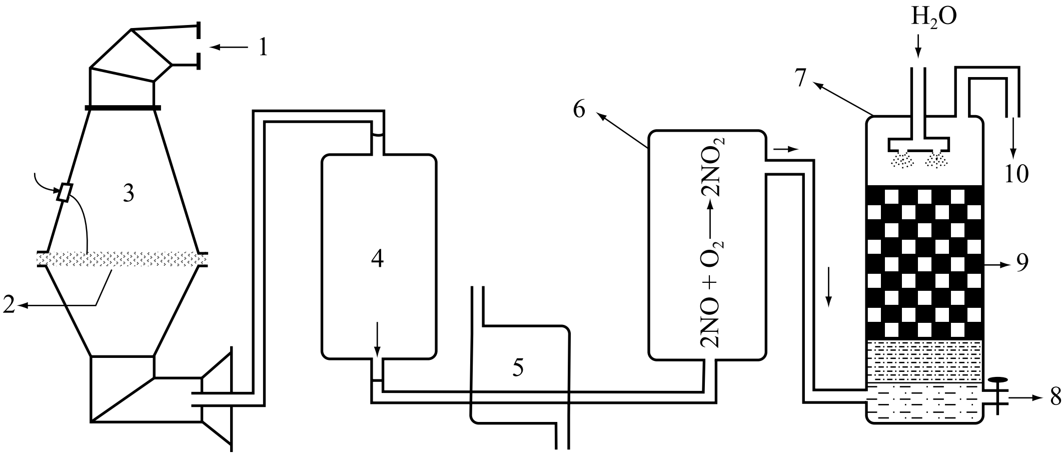


Fig 9.6 Manufacture of HNO_3 – Ostwald's process 1. Ammonia, oxygen mixture 2. Platinum guaze 3. Catalytic chamber 4. Aluminium or chrome steel vessel (cooling tower) 5. Boiler 6. Oxidation chamber 7. Absorption tower 8. 61% HNO_3 9. Bricks checker works 10. Unreacted NO mixed with O_2

Thus it reacts with metal oxides, hydroxides, carbonates and bicarbonates forming salts.

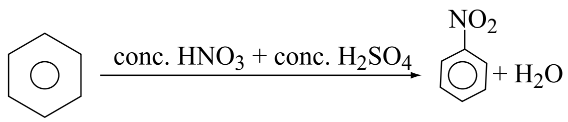
$$CaO + 2HNO_3 \rightarrow Ca(NO_3)_2 + H_2O$$

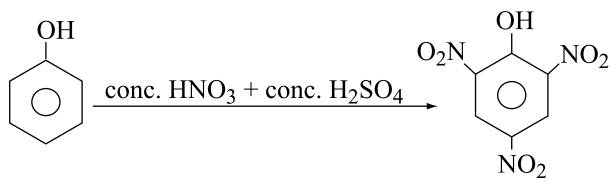
$$NaOH + HNO_3 \rightarrow NaNO_3 + H_2O$$

$$Na_2CO_3 + 2HNO_3 \rightarrow 2NaNO_3 + H_2O + CO_2$$

$$NaHCO_3 + HNO_3 \rightarrow NaNO_3 + H_2O + CO_2$$

(ii) *Nitration Properties*: Nitric acid in the presence of sulphuric acid reacts with aromatic compounds forming nitrocompounds, the process being known as nitration. Thus benzene and phenol when nitrated with a mixture of concentrated nitric and sulphuric acids give nitrobenzene and trinitrophenol (picric acid), respectively.



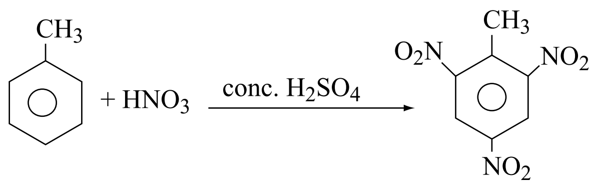


A mixture of nitric acid and conc. sulphuric acid reacts with glycerine to form nitro glycerine, an explosive substance. It is an ester.

$$\begin{array}{l} CH_2OH \\ | \\ CHOH \\ | \\ CH_2OH \end{array} + 3HNO_3 \xrightarrow{H_2SO_4} \begin{array}{l} CH_2ONO \\ | \\ CH\ ONO \\ | \\ CH_2ONO \end{array}$$

Cotton when reacted with concentrated nitric acid in the presence of sulphuric acid forms cellulose nitrate known as *gun cotton*.

Toluene when reacted with concentrated nitric acid and concentrated sulphuric acid forms trinitro toluene (TNT), a valuable explosive



Action on proteins: Nitric acid attacks proteins giving a yellow nitrocompound called *xantho protein*. Hence, nitric acid stains the skin and renders yellow. This is used as a delicate test of protein.

(iii) *Oxidation Properties*: Standard potential data imply that the NO_3^- ion is a moderately strong oxidizing agent. However, its reactions are generally slow in dilute acid solution. In dilute solution HNO_3 is fully ionized.

$$HNO_3 \rightleftharpoons H^+ + NO_3^-$$

But in concentrated solution NO_3^- is protonated (due to lesser ionization). Protonation of the oxygen atom makes the N—O bond breaking easily.

$$H-O-NO_2 + H^+ \longrightarrow H-\underset{\substack{|\\ H}}{\overset{+}{O}}-NO_2 \longrightarrow H_2O + NO_2^+$$

$$NO_2^+ + e^- \longrightarrow NO_2$$

So the reactions are faster at low pH, i.e., at higher concentration of acid. A sign of this oxidizing character is the yellow colour of concentrated acid, which indicates its instability with respect to decomposition.

$$4HNO_3 \rightarrow 2H_2O + 4NO_2 + O_2$$

The oxidation properties of nitric acid may be divided into

(a) Oxidation of non-metals
(b) Oxidation of compounds
(c) Oxidation of metals

(iv) *Oxidation of Non-Metals and Metalloids*: Concentrated nitric acid oxidizes the solid non-metals and metalloids to their respective oxoacids. For example, it oxidizes phosphorus, sulphur, selenium, tellurium, iodine, arsenic, antimony and tin to phosphoric acid (H_3PO_4), sulphuric acid (H_2SO_4), selenious acid (H_2SeO_3), tellurous acid (H_2TeO_3), iodic acid (HIO_3) arsenic acid (H_3AsO_4), antimonic acid (H_3SbO_4) and metastannic acid (H_2SnO_3), respectively.

Writing the Balanced Equation

(i) When concentrated nitric acid is used, it acts as an oxidizing agent according to the reaction

$$2HNO_3 \rightarrow H_2O + 2NO_2 + (O)$$

(ii) Write the reactant and product of the substance to be oxidized. For example, sulphur is oxidized to sulphuric acid.

$$S \rightarrow H_2SO_4$$

(iii) Find the difference between reactant and product and write them in this equation. Here the difference between S and H_2SO_4 is 'H_2O' and three O'atoms on reactant side (left side). Now write the reaction by adding these to the reactant (sulphur).

$$S + H_2O + 3(O) \rightarrow H_2SO_4$$

(iv) Multiply the reaction (i) with 3 to get three O atoms.

$$[2HNO_3 \rightarrow H_2O + 2NO_2 + (O)] \times 3$$

$$\text{i.e., } 6HNO_3 \rightarrow 3H_2O + 6NO_2 + 3(O)$$

(v) Add the two reactions (iii) and (iv) to get the final reaction.

$$\begin{array}{c} 6HNO_3 \rightarrow 3H_2O + 6NO_2 + 3(O) \\ S + H_2O + 3(O) \rightarrow H_2SO_4 \\ \hline 6HNO_3 + S \rightarrow H_2SO_4 + 6NO_2 + 2H_2O \end{array}$$

(i) Phosphorus to phosphoric acid.

$$\begin{array}{l} P_4 + 6H_2O + 10(O) \longrightarrow 4H_3PO_4 \\ 20HNO_3 \longrightarrow 10H_2O + 20NO_2 + 10(O) \\ \hline P_4 + 20HNO_3 \longrightarrow 4H_3PO_4 + 20NO_2 + 4H_2O \end{array}$$

(ii) Arsenic to arsenic acid.

$$\begin{array}{l} As_4 + 6H_2O + 10(O) \longrightarrow 4H_3AsO_4 \\ 20HNO_3 \longrightarrow 10H_2O + 20NO_2 + 10(O) \\ \hline As_4 + 20HNO_3 \longrightarrow 4H_3AsO_4 + 20NO_2 + 4H_2O \end{array}$$

(iii) Antimony to antimonic acid.

$$\begin{array}{l} Sb_4 + 6H_2O + 10(O) \longrightarrow 4H_3SbO_4 \\ 20HNO_3 \longrightarrow 10H_2O + 20NO_2 + 10(O) \\ \hline Sb_4 + 20HNO_3 \longrightarrow 4H_3SbO_4 + 20NO_2 + 4H_2O \end{array}$$

(iv) Selenium to selenious acid.

$$\begin{array}{l} Se + H_2O + 2(O) \longrightarrow H_2SeO_3 \\ 4HNO_3 \longrightarrow 2H_2O + 4NO_2 + 2(O) \\ \hline Se + 4HNO_3 \longrightarrow H_2SeO_3 + 4NO_2 + H_2O \end{array}$$

(v) Tellurium to tellurous acid.

$$\begin{array}{l} Te + H_2O + 2(O) \longrightarrow H_2TeO_3 \\ 4HNO_3 \longrightarrow 2H_2O + 4NO_2 + 2(O) \\ \hline Te + 4HNO_3 \longrightarrow H_2TeO_3 + 4NO_2 + H_2O \end{array}$$

(vi) Tin to metastannic acid.

$$\begin{array}{l} Sn + H_2O + 2(O) \longrightarrow H_2SnO_3 \\ 4HNO_3 \longrightarrow 2H_2O + 4NO_2 + 2(O) \\ \hline Sn + 4HNO_3 \longrightarrow H_2SnO_3 + 4NO_2 + H_2O \end{array}$$

(vii) Iodine to iodic acid.

$$\begin{array}{l} I_2 + H_2O + 5(O) \longrightarrow 2HIO_3 \\ 10HNO_3 \longrightarrow 5H_2O + 10NO_2 + 5(O) \\ \hline I_2 + 10HNO_3 \longrightarrow 2HIO_3 + 10NO_2 + 4H_2O \end{array}$$

(viii) Carbon is oxidized to carbon dioxide. Since carbonic acid is unstable, CO_2 is liberated.

$$\begin{array}{l} C + 2(O) \longrightarrow CO_2 \\ 4HNO_3 \longrightarrow 2H_2O + 4NO_2 + 2(O) \\ \hline C + 4HNO_3 \longrightarrow CO_2 + 2H_2O + 4NO_2 \end{array}$$

(b) *Oxidation of Compounds*

Several compounds are oxidized by dilute or concentrated nitric acid. The reactions of nitric acid while acting as oxidizing agent in dilute and concentrated conditions are

(i) $2HNO_3 \text{ (dil)} \rightarrow H_2O + 2NO + 3(O)$
(ii) $2HNO_3 \text{ (Conc)} \rightarrow H_2O + 2NO_2 + (O)$

Balanced chemical equations for oxidation of different compounds by nitric acid can also be written in a similar way as explained in the oxidation of non-metals and metalloids. But when dilute nitric acid is used the reaction (i) and while concentrated nitric acid is used the reaction (ii) given above should be used.

Dilute nitric acid can oxidize hydrogen sulphide to sulphur, sulphur-dioxide to sulphuric acid, liberates iodine from iodides, oxidizes ferrous sulphate to ferric sulphate.

$$\begin{array}{l} 2HNO_3 \longrightarrow H_2O + 2NO + 3(O) \\ [H_2S + O \longrightarrow H_2O + S] \times 3 \\ \hline 3H_2S + 2HNO_3 \longrightarrow 3S + 2NO + 4H_2O \end{array}$$

(ii) Sulphur dioxide to sulphuric acid.

$$\begin{array}{l} 2HNO_3 \longrightarrow H_2O + 2NO + 3(O) \\ [SO_2 + H_2O + (O) \longrightarrow H_2SO_4] \times 3 \\ \hline 3SO_2 + 2HNO_3 + 2H_2O \longrightarrow 3H_2SO_4 + 2NO. \end{array}$$

(iii) Liberates iodine from potassium iodide.

$$\begin{array}{l} 2HNO_3 \longrightarrow H_2O + 2NO + 3(O) \\ [2KI + 2HNO_3 + (O) \longrightarrow 2KNO_3 + I_2 + H_2O] \times 3 \\ \hline 6KI + 8HNO_3 \longrightarrow 6KNO_3 + 4H_2O + 2NO + 3I_2. \end{array}$$

(iv) Ferrous sulphate to ferric sulphate.

$$\begin{array}{l} 2HNO_3 \longrightarrow H_2O + 2NO + 3(O) \\ [2FeSO_4 + H_2SO_4 + (O) \longrightarrow Fe_2(SO_4)_3 + H_2O] \times 3 \\ \hline 6FeSO_4 + 3H_2SO_4 + 2HNO_3 \longrightarrow 3Fe_2(SO_4)_3 + 4H_2O \\ \qquad + 2NO. \end{array}$$

The NO formed in the above reaction combine with $FeSO_4$ to form a brown complex $[Fe(H_2O)_5(NO)]^{2+}$, which is the basis for the brown ring test for nitrates.

(v) Dilute HNO_3 can liberate bromine from HBr and iodine from HI.

$$\begin{array}{l} 2HNO_3 \longrightarrow H_2O + 2NO + 3(O) \\ [2HBr + (O) \longrightarrow H_2O + Br_2] \times 3 \\ \hline 2HNO_3 + 6HBr \longrightarrow 4H_2O + 2NO + 3Br_2 \end{array}$$

$$\begin{array}{l} 2HNO_3 \longrightarrow H_2O + 2NO + 3(O) \\ [2HI + (O) \longrightarrow H_2O + I_2] \times 3 \\ \hline 2HNO_3 + 6HI \longrightarrow 4H_2O + 2NO + 3I_2. \end{array}$$

The compounds which are oxidized by dilute nitric acid can also be oxidized by conc. HNO_3 but the reactions should be written and balanced by using the equation

$$2HNO_3 \rightarrow H_2O + 2NO_2 + (O).$$

(vi) Stannous chloride reduces the nitric acid to hydroxylamine in the presence of HCl.

$$\begin{array}{l} [SnCl_2 + 2HCl \longrightarrow SnCl_4 + 2H] \times 7 \\ 2HNO_3 + 14H \longrightarrow NH_2OH + NH_3 + 5H_2O \\ NH_3 + HNO_3 \longrightarrow NH_4NO_3 \\ \hline 7SnCl_2 + 14HCl + 3HNO_3 \longrightarrow 7SnCl_4 + NH_2OH \\ \qquad + NH_4NO_3 + 5H_2O \end{array}$$

(vii) Cane sugar is oxidized by conc. HNO_3 to oxalic acid.

$$\begin{array}{l} [2HNO_3 \longrightarrow H_2O + 2NO_2 + (O)] \times 18 \\ C_{12}H_{22}O_{11} + 18(O) \longrightarrow 6(COOH)_2 + 5H_2O \\ \hline C_{12}H_{22}O_{11} + 36HNO_3 \longrightarrow 6(COOH)_2 + 36NO_2 + 23H_2O \end{array}$$

(c) *Oxidation of Metals*

Nitric acid reacts with metals differently. The metals are oxidized to their corresponding positive ions, whereas nitric acid is reduced to NO_2, NO, N_2O, NH_2OH or NH^+_4 depending upon

(i) the nature of metal
(ii) concentration of the acid and
(iii) temperature

In the case of metals which exhibit variable oxidation states such as mercury, tin and iron form -*ous* salts with dilute nitric acid while excess concentrated nitric acid forms -*ic* salts. Ion-electron equations for these reactions can be written as

$HNO_3 + H^+ + e^- \rightarrow H_2O + NO_2$

$HNO_3 + 3H^+ + 3e^- \rightarrow 2H_2O + NO$

$2HNO_3 + 8H^+ + 8e^- \rightarrow 5H_2O + N_2O$

$HNO_3 + 6H^+ + 6e^- \rightarrow 2H_2O + NH_2OH$

$HNO_3 + 9H^+ + 8e^- \rightarrow 5H_2O + NH_4^+$

or $2HNO_3 + 8H^+ + 8e^- \rightarrow 3H_2O + NH_4NO_3$

These equations must be considered as approximate, indicating what is considered to be the principal reactions.

Reduction of nitric acid with nascent hydrogen occurs in the case of the metals which are above hydrogen in the electrochemical series. Metals below hydrogen in the electrochemical series are not capable of liberating hydrogen and are oxidized by nitric acid to their corresponding basic oxides which dissolve in nitric acid to form nitrates.

$2HNO_3 \text{ conc.} \rightarrow H_2O + 2NO_2 + (O)$
$2HNO_3 \text{ dil.} \rightarrow H_2O + 2NO + 3(O),$

Metal + O → Metal oxide

Metal oxide + nitric acid → Metal nitrate + water

More the change in the oxidation state of nitric acid, the weaker is its oxidation nature. As already explained oxidation power of HNO_3 increases with increase in concentration due to protonation. 6 per cent nitric acid is considered as very dilute, 20 per cent nitric acid as dilute and above 60 per cent nitric acid is considered as concentrated nitric acid.

Reactions of Metals with Very Dilute Nitric Acid

Aluminium, zinc, iron and tin, reduces the very dilute nitric acid to ammonium nitrate.

$$\left[Zn + 2HNO_3 \longrightarrow Zn(NO_3)_2 + 2H\right] \times 4$$
$$\underline{2HNO_3 + 8H \longrightarrow NH_4NO_3 + 3H_2O}$$
$$4Zn + 10HNO_3 \longrightarrow 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$$

Iron, tin and magnesium are divalent metals like zinc and give similar reactions.

$$4Fe + 10HNO_3 \rightarrow 4Fe(NO_3)_2 + NH_4NO_3 + 3H_2O$$
$$4Sn + 10\ HNO_3 \rightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$$

For trivalent aluminium the reaction is as follows

$$\left[Al + 3HNO_3 \longrightarrow Al(NO_3)_3 + 3H\right] \times 8$$
$$\underline{\left[2HNO_3 + 8H \longrightarrow NH_4NO_3 + 3H_2O\right] \times 3}$$
$$8Al + 30HNO_3 \longrightarrow 8Al(NO_3)_3 + 3NH_4NO_3 + 9H_2O$$

Magnesium and manganese react with very dilute nitric acid to liberate hydrogen. The conditions must, however, be very carefully controlled and the very dilute acid should be about 1 per cent. More concentrated acid gives oxides of nitrogen.

$$Mg + 2HNO_3 \rightarrow Mg\ (NO_3)_2 + H_2$$
$$Mn + 2HNO_3 \rightarrow Mn\ (NO_3)_2 + H_2$$

Reactions of metals with dilute nitric acid

(i) Most of the metals except noble metals dissolve in dilute nitric acid forming metal nitrate with evolution of NO.

(ii) Tin reduces the dilute nitric acid to ammonium nitrate.

(iii) Iron and zinc reduces the dilute nitric acid to N_2O.

(iv) Iron and mercury form metal nitrates in their lower oxidation states.

With Metals above Hydrogen in Electrochemical Series

Lead

$$\left[Pb + 2HNO_3 \longrightarrow Pb(NO_3)_2 + 2H\right] \times 3$$
$$\underline{\left[HNO_3 + 3H \longrightarrow 2H_2O + NO\right] \times 2}$$
$$3Pb + 8HNO_3 \longrightarrow 3Pb(NO_3)_2 + 4H_2O + 2NO$$

Zinc

$$\left[Zn + 2HNO_3 \longrightarrow Zn(NO_3)_2 + 2H\right] \times 4$$
$$\underline{2HNO_3 + 8H \longrightarrow N_2O + 5H_2O}$$
$$4Zn + 10HNO_3 \longrightarrow 4Zn(NO_3)_2 + N_2O + 5H_2O$$

Iron

$$\left[Fe + 2HNO_3 \longrightarrow Fe(NO_3)_2 + 2H\right] \times 4$$
$$\underline{2HNO_3 + 8H \longrightarrow N_2O + 5H_2O}$$
$$4Fe + 10HNO_3 \longrightarrow 4Fe(NO_3)_2 + N_2O + 5H_2O$$

Tin

$$\left[Sn + 2HNO_3 \longrightarrow Sn(NO_3)_2 + 2H\right] \times 4$$
$$\underline{2HNO_3 + 8H \longrightarrow NH_4NO_3 + 3H_2O}$$
$$4Sn + 10HNO_3 \longrightarrow 4Sn(NO_3)_2 + NH_4NO_3 + 3H_2O$$

With Metals Below Hydrogen in Electrochemical Series

Copper

$$2HNO_3 \longrightarrow H_2O + 2NO + 3(O)$$
$$\left[Cu + (O) \longrightarrow CuO\right] \times 3$$
$$\underline{\left[CuO + 2HNO_3 \longrightarrow Cu(NO_3)_2 + H_2O\right] \times 3}$$
$$3Cu + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$$

Silver

$$2HNO_3 \longrightarrow H_2O + 2NO + 3(O)$$
$$\left[2Ag + (O) \longrightarrow Ag_2O\right] \times 3$$
$$\underline{\left[Ag_2O + 2HNO_3 \longrightarrow 2AgNO_3 + H_2O\right] \times 3}$$
$$6Ag + 8HNO_3 \longrightarrow 6AgNO_3 + 4H_2O + 2NO$$
$$3Ag + 4HNO_3 \longrightarrow 3AgNO_3 + 2H_2O + NO$$

Mercury

$$2HNO_3 \longrightarrow H_2O + 2NO + 3(O)$$
$$\left[2Hg + (O) \longrightarrow Hg_2O\right] \times 3$$
$$\underline{\left[Hg_2O + 2HNO_3 \longrightarrow Hg_2(NO_3)_2 + H_2O\right] \times 3}$$
$$6Hg + 8HNO_3 \longrightarrow 3Hg_2(NO_3)_2 + 4H_2O + 2NO$$

Reactions of Metals with concentrated nitric acid

Most of the metals dissolve in concentrated nitric acid forming metal nitrate with evolution of NO_2 gas. Mercury forms mercuric nitrate. Tin is converted into metastannic acid.

With Metals above Hydrogen in Electrochemical Series

Zinc

$$
\begin{array}{c}
Zn + 2HNO_3 \longrightarrow Zn(NO_3)_2 + 2H \\
[HNO_3 + H \longrightarrow H_2O + NO_2] \times 2 \\
\hline
Zn + 4HNO_3 \longrightarrow Zn(NO_3)_2 + 2NO_2 + 2H_2O
\end{array}
$$

Lead

$$
\begin{array}{c}
Pb + 2HNO_3 \longrightarrow Pb(NO_3)_2 + 2H \\
[HNO_3 + H \longrightarrow H_2O + NO_2] \times 2 \\
\hline
Pb + 4HNO_3 \longrightarrow Pb(NO_3)_2 + 2H_2O + 2NO_2
\end{array}
$$

Tin

$$
\begin{array}{c}
[2HNO_3 \longrightarrow H_2O + 2NO_2 + (O)] \times 2 \\
Sn + H_2O + 2(O) \longrightarrow H_2SnO_3 \\
\hline
Sn + 4HNO_3 \longrightarrow H_2SnO_3 + H_2O + 4NO_2
\end{array}
$$

With Metals below Hydrogen in Electrochemical Series

Copper

$$
\begin{array}{c}
2HNO_3 \longrightarrow H_2O + 2NO_2 + (O) \\
Cu + O \longrightarrow CuO \\
CuO + 2HNO_3 \longrightarrow Cu(NO_3)_2 + H_2O \\
\hline
Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2
\end{array}
$$

Silver

$$
\begin{array}{c}
2HNO_3 \longrightarrow H_2O + 2NO_2 + (O) \\
2Ag + (O) \longrightarrow Ag_2O \\
Ag_2O + 2HNO_3 \longrightarrow 2AgNO_3 + H_2O \\
\hline
2Ag + 4HNO_3 \longrightarrow 2AgNO_3 + 2H_2O + 2NO_2 \\
Ag + 2HNO_3 \longrightarrow AgNO_3 + H_2O + NO_2
\end{array}
$$

Mercury

$$
\begin{array}{c}
2HNO_3 \longrightarrow H_2O + 2NO_2 + (O) \\
Hg + O \longrightarrow HgO \\
HgO + 2HNO_3 \longrightarrow Hg(NO_3)_2 + H_2O \\
\hline
Hg + 4HNO_3 \longrightarrow Hg(NO_3)_2 + 2H_2O + 2NO_2
\end{array}
$$

Metals which become passive with conc. HNO_3

Metals like beryllium, aluminium, chromium, iron, cobalt and nickel become passive in concentrated nitric acid. Conversion of reactive species into chemically inactive species is called passivity. The passivity of these metals by concentrated nitric acid is due to the formation of a thin oxide layer on the surface of the metal which prevents further reaction with the reagent. Because of this reason in certain countries aluminium vessels are used for carrying nitric acid.

Metals which do not React with Nitric Acid

Noble metals like gold, platinum, iridium, rhodium, etc. do not react with nitric acid. However, these metals dissolve in aquaregia. Aquaregia is a mixture of 3 parts of concentrated hydrochloric acid and 1 part concentrated nitric acid. The enhanced ability to dissolve metals shown by aquaregia is due to the enhanced oxidation power of nitric acid in the presence of Cl^- ions which can form complexes with metal ions.

$$Au + 4H^+ + NO_3^- + 4Cl^- \rightarrow AuCl_4^- + NO + 2H_2O$$

$$3Pt + 16H^+ + 4NO_3^- + 18Cl^- \rightarrow 3PtCl_6^{2-} + 4NO + 8H_2O$$

Gold

$$
\begin{array}{c}
2HNO_3 \longrightarrow H_2O + 2NO + 3(O) \\
2Au + 6HCl + 3(O) \longrightarrow 2AuCl_3 + 3H_2O \\
2AuCl_3 + 2HCl \longrightarrow 2H[AuCl_4] \\
\hline
2Au + 2HNO_3 + 8HCl \longrightarrow 2H[AuCl_4] + 4H_2O + 2NO
\end{array}
$$

$$\text{or } Au + HNO_3 + 4HCl \rightarrow H[AuCl_4] + 2H_2O + NO$$

Platinum

$$
\begin{array}{c}
[2HNO_3 \longrightarrow H_2O + 2NO + 3(O)] \times 2 \\
[Pt + 4HCl + 2(O) \longrightarrow PtCl_4 + 2H_2O] \times 3 \\
[PtCl_4 + 2HCl \longrightarrow H_2[PtCl_6]] \times 3 \\
\hline
3Pt + 4HNO_3 + 18HCl \longrightarrow 3H_2[PtCl_6] + 8H_2O + 4NO
\end{array}
$$

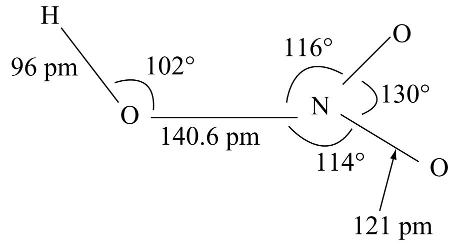
Uses

Nitric acid is used

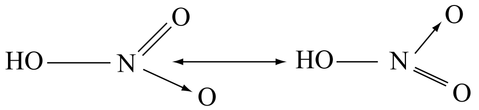
(i) in the manufacture of fertilizers like basic calcium nitrate [$CaO \cdot Ca(NO_3)_2$]
(ii) in the preparation of explosives like TNT and nitroglycerine
(iii) in the laboratory as nitration mixture along with conc. H_2SO_4
(iv) in the preparation of perfumes, dyes and medicines
(v) in the oxidation of cyclohexanol or cyclohexanone to adipic acid, p-xylene to terephthallic acid
(vi) in the preparation of cellulose nitrate that is used as artificial silk
(vii) in the manufacture of nitrates such as silver nitrate, sodium nitrate, etc.

Structure

The nitric acid molecule is planar in the gas phase with the dimensions shown.



It is supposed to be the resonance hybrid of the following two resonance structures



Test for Nitrate Ion

***Brown ring test*:** An aqueous solution of the nitrate is taken in a test tube. Then it is mixed with freshly prepared ferrous sulphate solution. To this mixture concentrated sulphuric acid is added carefully along the edges of the test tube. A brown ring appears at the junction of the solution and concentrated sulphuric acid.

The nitrate ions are reduced by the ferrous ions forming nitric oxide (NO) which then combines with ferrous sulphate forming a brown complex.

$$NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$$

$$2HNO_3 \rightarrow H_2O + 2NO + 3(O)$$

$$2FeSO_4 + H_2SO_4 + (O) \rightarrow Fe_2(SO_4)_3 + H_2O$$

$$[Fe(H_2O)_6]\ SO_4 + NO \rightarrow [Fe\ (H_2O)_5(NO)]\ SO_4.$$

***Ammonia Test*:** When nitrates are heated with zinc or aluminium metal in alkaline medium, ammonia gas having a characteristic smell will be liberated.

$$\left[Zn + 2NaOH \longrightarrow Na_2ZnO_2 + 2H\right] \times 4$$

$$\underline{NaNO_3 + 8H \longrightarrow NaOH + 2H_2O + NH_3}$$

$$NaNO_3 + 7NaOH + 4Zn \longrightarrow 4Na_2ZnO_2 + 2H_2O + NH_3$$

$$\left[Al + NaOH + H_2O \longrightarrow NaAlO_2 + 3H\right] \times 8$$

$$\underline{\left[NaNO_3 + 8H \longrightarrow NaOH + 2H_2O + NH_3\right] \times 3}$$

$$3NaNO_3 + 5NaOH + 2H_2O + 8\ Al \longrightarrow 8NaAlO_2 + 3NH_3$$

Action of Heat on some nitrogen compounds

$$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} \underset{\text{green}}{Cr_2O_3} + 4H_2O + N_2$$

$$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$$

$$NH_4NO_2 \longrightarrow N_2 + 2H_2O$$

Except $LiNO_3$ the other alkali metal nitrates on heating converts into nitrites liberating oxygen.

$$2NaNO_3 \rightarrow 2NaNO_2 + O_2$$

Heavy metal nitrates on heating decompose to metal oxide, brown NO_2 gas and O_2 gas.

$$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$$

Mercury and silver nitrates on heating gives metals as residue.

$$Hg(NO_3)_2 \rightarrow Hg + 2NO_2 + O_2$$

$$2AgNO_3 \rightarrow 2Ag + 2NO_2 + O_2$$

9.9 PHOSPHORUS

In 1669, Brand discovered phosphorus. Since it glowed in dark, it is named phosphorus (meaning light bearing). Its elementary nature was proved by Lavoiser in 1977. It occurs in combined state in nature as phosphates, e.g.,

Phosphorite	$Ca_3(PO_4)_2$
Apatite	$3Ca_3(PO_4)_2\ CaF_2$
Chlorapatite	$3Ca_3(PO_4)_2\ CaCl_2$

Phosphorus is an essential constituent of bones and also present in blood, brain and other parts of the body. Phosphorus is present in the soil as phosphates. In milk and eggs, it is present as phosphoproteins.

***Preparation*:** Phosphorus is now usually manufactured by the reduction of phosphate rock or bone ash with sand and coke in an electric furnace.

$$Ca_3(PO_4)_2 + 3SiO_2 \rightarrow 3CaSiO_3 + P_2O_5$$

$$2P_2O_5 + 10C \rightarrow 10CO + P_4$$

A mixture of calcium phosphate, sand and coke is introduced in the furnace (Fig. 9.7) through hooper. The mixture is heated to 1500–1800 K by carbon electrodes. The above reactions takes place and the vapours of phosphorus are condensed under water. The slag escapes through the exit.

Phosphorus obtained in this process is impure. It is melted by heating with potassium dichromate and sulphuric acid to oxidize the impurities. Phosphorus is washed by water and filtered through canvas. It is then cast in the form of rods. Extra pure phosphorus is obtained by distillation.

***Allotropes of phosphorus*:** Phosphorus exists in different allotropic forms such as white or yellow phosphorus, red phosphorus, scarlet phosphorus, metallic or α-black phosphorus, β-black phosphorus and violet phosphorus. Out of these white, red and black are the important forms.

***White phosphorus*:** White or yellow phosphorus is prepared by rapidly cooling the vapours. In pure state, it is the white waxy solid. When exposed to air, it gives a light yellow colour and therefore called yellow phosphorus. Below 1075K, its vapour density corresponds to P_4 (Fig. 9.8).

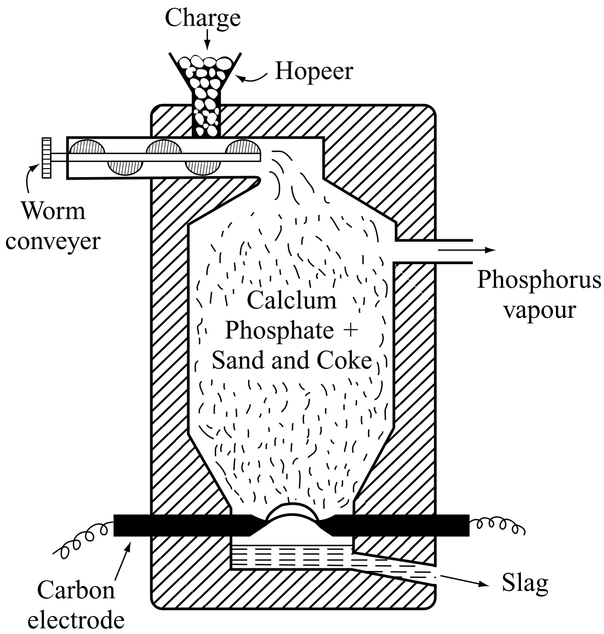


Fig 9.7 Modern electrothermic method for the manufacture of phosphorus

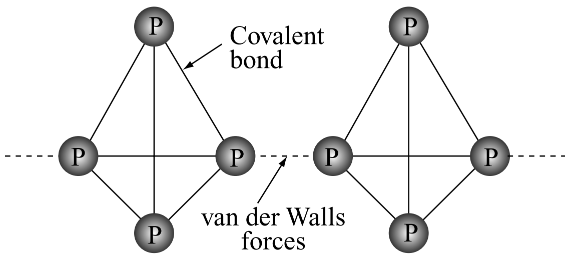


Fig 9.8 Structure of white phosphorus

Above 1075K it dissociates to P_2 and at 1975K its formula is P_2. Its melting point is 317.1K and boiling point is 563K. Due to slow oxidation in dark it glows. This phenomenon is known as phosphorescence.

The P_4 molecule is tetrahedral in shape, in which each P atom is in bond with three other P atoms. Such a structure requires bond angles of 60º. In as much as the smallest inter orbital angle available using only s- and p-orbitals is 90º (pure p-orbitals). The smaller bond angle in P_4 molecule must be accomplished either through the introduction of d-character or through the use of bent bonds. There has been much discussion of the nature of the bonds in P_4 molecule. Although the use of pd^2 hybrids (theoretical bond angle 66º 26′) would result in the least strain it seems that the promotion of two 3p-electrons to 3d-orbitals is unlikely on energetic grounds, and the bonds are largely p in character. The P—P bond length is 221 pm. In the P_2 molecule in the vapour of red P≡P bond length is 189.5 pm.

In spite of the ring strain, the P_4 molecule is stable relative to P_2 or the non-existent P_8. Neverthless, the molecule is quite reactive. It can be stored under water, but it burns readily in oxygen. Thermodynamically, it is the least stable allotrope of phosphorus.

***Metallic or α-black phosphorus*:** It is obtained as black phosphorus by dissolving red phosphorus in fused lead or bismuth at 675 K in a sealed tube for a long time and cooling. Lead or bismuth are separated by dissolving in dilute nitric acid. It is very stable, chemically inert and cannot be oxidized by air unless heated very strongly. It is a non-conductor.

β-*Black Phosphorus*: It is obtained by heating white phosphorus at 473K and at very high pressure (4000 atms). It can also be prepared by heating white phosphorus at 520–650K for 8 days in the presence of mercury as a catalyst with a seed of black phosphorus.

This is thermodynamically the *most stable* allotrope of phosphorus. It is inert and has a layer structure. Each P atom is in bond with three atoms. PPP angle 99º and the P—P bond length 218 pm. Distance between adjacent layers is 388pm. It is a *semiconductor* of electricity.

***Scarlet Phosphorus*:** It resembles red phosphorus in its physical properties and it is chemically reactive as white phosphorus but slowly oxidized in air.

Violet phosphorus is a bad conductor of electricity and does not oxidize in air.

***Red Phosphorus*:** It is made by heating white phosphorus at 675K for several hours in an inert atmosphere using iodine as a catalyst. It consists of long chains of phosphorus atoms which are covalently bonded forming a giant molecule. This structure requires large amount of energy to break the more number of bonds. Due to this highly polymerized structure, red phosphorus is less reactive and less volatile.

Properties

(i) ***Combustion*:** It burns in air forming oxides.

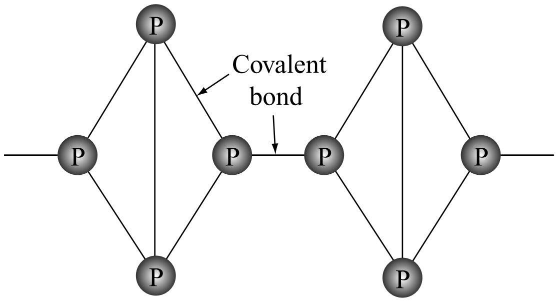
$$P_4 + 3O_2 \rightarrow P_4O_6$$
$$P_4 + 5O_2 \rightarrow P_4O_{10}$$

(ii) ***Reaction with caustic alkali*:** White phosphorus reacts with hot solutions of caustic alkalis in an inert atmosphere yielding phosphine.

$$P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$$

(iii) ***Reaction with the halogens*:** Phosphorus forms first trihalides and then the pentahalides. This reaction is more vigorous with the white phosphorus than with red phosphorus.

$$P_4 + 6Cl_2 \rightarrow 4PCl_3$$
$$P_4 + 10Cl_2 \rightarrow 4PCl_5$$



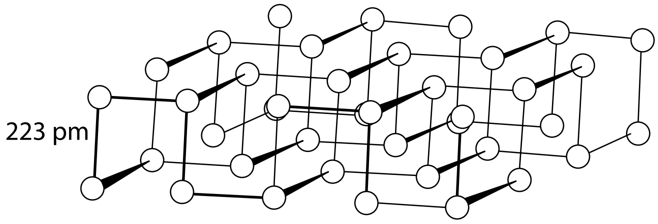


Fig 9.9 Structures of red and black phosphorus

(iv) ***Reaction with sulphur***: Phosphorus reacts with sulphur with an explosive violent reaction to form large number of sulphides like P_2S_3, P_2S_5, P_4S_3 and P_4S_7. The sulphide P_4S_3 has been used in "**strike any where"** matches.

(v) ***Reaction with metals***: Phosphorus combines with a number of metals forming metal phosphides.

$$12Na + P_4 \rightarrow 4Na_3P$$

$$6Mg + P_4 \rightarrow 2Mg_3P_2$$

(vi) ***Reducing Properties***: It acts as a powerful reducing agent because it can be easily oxidized. It reduces nitric acid to nitrogen dioxide, sulphuric acid to sulphurdioxide and reduces copper sulphate to copper and silver nitrate to silver.

$$P_4 + 20HNO_3 \rightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$$

$$P_4 + 10H_2SO_4 \rightarrow 4H_3PO_4 + 10SO_2 + 4H_2O$$

$$P_4 + 3CuSO_4 + 6H_2O \rightarrow Cu_3P_2 + 2H_3PO_4 + 3H_2SO_4$$

$$Cu_3P_2 + 5CuSO_4 + 5H_2O \rightarrow 8Cu + 5H_2SO_4 + 2H_3PO_4$$

$$P_4 + 20AgNO_3 + 16H_2O \rightarrow 4H_3PO_4 + 20HNO_3 + 20Ag$$

(vii) ***With oxidizing agents***: When treated with oxidizing agents such as potassium chlorate or potassium nitrate, it forms explosive mixtures.

(viii) ***As ligand*** **(*complexing agent*)**: It acts as a ligand. The first example of complex involving η' mode of the red-brown compound is $[Fe\,(CO)_4\,\mu_3P_4]$. Later, several complexes like $[RhCl\ \eta^2\text{– }P_4)\ (PPh_3)_2]$ are prepared.

(x) ***Phosphorescence***: When white phosphorus is kept in air it undergoes spontaneous slow oxidation giving greenish glow in dark. This phenomenon of glowing of phosphorus in the dark is known as phosphorescence. The main product of slow oxidation has been found to be phosphorus trioxide.

The glow appearing in phosphorescence disappears in the presence of vapours of ether, essential oils, etc.

Uses

(i) It is used in the preparation of rat poisons.

(ii) It is used in war for producing fire bombs and smoke bombs.

(iii) It is used in the production of phosphor bronze which is hard, tenacious and not corroded by water.

(iv) Red phosphorus is used in match box industry instead of yellow phosphorus which was in use earlier and has been discarded because of its poisonous nature.

The match stick tips contain red phosphorus, (to ignite) sulphur or antimony sulphide for burning potassium dichromate or potassium chlorate (to supply oxygen), glass pieces (to produce friction and gum. People working in match box industries will be attacked by a disease called *Phossy Jaw* due to inhalation of phosphorus vapours.

(v) Red phosphorus is used in the preparation of HBr and HI.

(vi) Radioactive phosphorus (^{32}P) is used in the treatment of leukaemia and other blood disorders.

(vii) It is used in the manufacture of compounds like hypophosphites (medicine), phosphorus chlorides employed in industry, calcium phosphide used in Home's signals and ortho phosphoric acid.

9.10 HYDRIDES OF PHOSPHORUS

Phosphorus forms many hydrides such as PH_3, P_2H_4, P_4H_2, P_5H_2, P_9H_2, etc. But phosphine is the most important which is a gas, P_2H_4 (diphosphine) is a liquid and other hydrides are solids.

9.10.1 Phosphine PH_3

Preparation: It is prepared by the following methods.

(i) By the hydrolysis of metal phosphides.

$$Na_3P + 3H_2O \rightarrow 3NaOH + PH_3$$

$$Ca_3P_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2PH_3$$

(ii) By the reaction of phosphonium iodide.

$$PH_4I + NaOH \rightarrow NaI + H_2O + PH_3$$

Table 9.8 Comparison between white and red phosphorus

Property	White phosphorus	Red phosphorus
1 Colour	White or pale yellow	Reddish violet
2 Physical state	Solid	Solid
3 Smell	Garlic smell	No smell
4 Phosphorescence	Emits light in dark	Does not show phosphorescence
5 Melting point.	317.1K.	885K
6 Boiling point.	560K	Very high
7 Solubility	Soluble in CS_2	Insoluble in CS_2
8 Physiological action	Highly poisonous	Non-poisonous
9 Ignition temp.	318K	533K
10 Action of air	Forms oxide (P_2O_3) at ordinary temp.	Forms oxide on heating
11 Action of alkalis	Forms phosphine	
12 Chemical reactivity	Very reactive	Less reactive
13 Action of Cl_2	Forms tri- and pentachlorides at ordinary temperatures	Forms PCl_3 and PCl_5 on heating
14 Solubility in water	Insoluble	Insoluble
15 Stability	Unstable	Stable

(iii) By heating orthophosphorus acid.

$$4H_3PO_3 \rightarrow 3H_3PO_4 + PH_3$$

(iv) In the laboratory, it is prepared by boiling white phosphorus and concentrated solution of sodium hydroxide in an inert atmosphere of carbon dioxide, hydrogen, oil gas, coal gas, etc.

$$P_4 + 3NaOH + 3H_2O \rightarrow 3NaH_2PO_2 + PH_3$$

Phosphine evolved is bubbled through water. As soon as it comes to the surface, it spontaneously ignites due to the presence of diphosphine (P_2H_4) impurity which come off in the form of vortex rings (Fig. 9.10). Pure gas may be obtained by using alcoholic solution of potassium hydroxide instead of solution of sodium hydroxide.

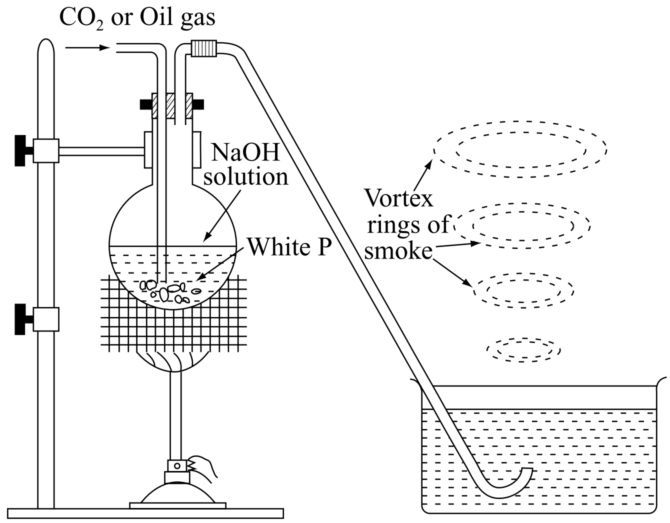


Fig 9.10 Preparation of phosphine

Properties

(i) It is a colourless, poisonous gas with a rotten fish odour. It is soluble in water, alcohol and ether. It condenses to a colourless liquid (b.p. 188K) and solidified to a white solid (m.p. 139.5K). It is poisonous in nature.

(ii) ***Thermal stability***: It is thermally less stable than ammonia. It decomposes at 715K to its elements.

$$4PH_3 \rightarrow P_4 + 6H_2$$

It explodes when it comes in contact with oxidizing agents, e.g., chlorine gas, nitric acid, etc.

$$4PH_3 + 8O_2 \rightarrow P_4O_{10} + 6H_2O$$

$$PH_3 + 4N_2O \rightarrow H_3PO_4 + 4N_2$$

(iii) ***Action of chlorine***: It reacts with chlorine to form phosphorus trichloride or phosphorus pentachloride.

$$PH_3 + 3Cl_2 \rightarrow PCl_3 + 3HCl$$

$$PH_3 + 4Cl_2 \rightarrow PCl_5 + 3HCl$$

(iv) ***Action of metallic salts***: Phosphine is a strong reducing agent. Metallic salts are reduced to phosphides or to the metals.

$$3CuSO_4 + 2PH_3 \rightarrow Cu_3P_2 + 3H_2SO_4$$

$$6AgNO_3 + PH_3 + 3H_2O \rightarrow 6Ag + 6HNO_3 + H_3PO_4$$

(v) ***Basic nature*****:** It is a much weaker base than ammonia. It reacts with acids to form phosphonium salts (like ammonium salts).

$$PH_3 + HX \rightarrow PH_4X \text{ (HX = halogen acid)}$$

But these salts on hydrolysis give phosphine gas (unlike ammonium halides)

$$PH_4X \xrightarrow{H_2O} PH_3 + HX$$

(vi) ***Addition Compounds*****:** It forms addition compounds with anhydrous $AlCl_3$ and $SnCl_4$.

$$AlCl_3 + 2PH_3 \rightarrow AlCl_3 \cdot 2PH_3$$

$$SnCl_4 + 2PH_3 \rightarrow SnCl_4 \cdot 2PH_3$$

When phosphine is passed through cuprous chloride solution in HCl, it forms an addition compound.

$$Cu_2Cl_2 + 2PH_3 \rightarrow 2CuCl \cdot PH_3$$

Ligand nature**:** Like nitrogen in ammonia, phosphorus atom has a lone pair of electrons and can act as a ligand forming several coordination complexes. E.g., $[Cr(CO)_2(PH_3)_4]$ $[Co(CO)_2(NO)(PH_3)]$, $[Ni(PF_3)_2(PH_3)_2]$. PH_3 is also a π-acceptor ligand similar to PPh_3. It accepts lone pairs of electrons from metal for pπ—dπ back bonding.

Uses

(i) ***Smoke Screens*****:** Calcium phosphide is used in making of smoke screens. Phosphine thus obtained catches fire to yield the required smoke.

(ii) ***Holme's Signal's*****:** The spontaneous inflammability of phosphine in the presence of P_2H_4 is made use of in making *Holme's* signals. A mixture of calcium phosphide and calcium carbide is taken in a container. Two holes are made to the container and thrown into the sea when required. Water enters the container and decompose the phosphide and carbide to produce phosphine and acetylene, respectively. Phosphine evolved catches fire and lights up acetylene. Burning gases serve as a signal to the approaching ships.

Structure**:** The electronegativity of phosphorus and hydrogen is the same, i.e., 2.1. Hence, there is a covalent bonding between the elements. Like ammonia (NH_3), phosphine also has a pyramidal shape and the bond angle of HPH is 93° indicating the tendency of sp^3 hybridization of phosphorus atom is less and nearly pure p-orbitals participating in bonding. P—H bond length is 142 pm.

9.10.2 Diphosphine or Phosphorus Dihydride P_2H_4

Preparation**:** It is prepared along with phosphine by the action of water on crude calcium phosphide which contains Ca_2P_2.

$$Ca_2P_2 + 4H_2O \rightarrow 2Ca(OH)_2 + P_2H_4.$$

The diphosphine is separated by passing the mixture of gases through a freezing mixture when diphosphine gets separated as a liquid and phosphine remains as a gas.

It is a colourless liquid. It decomposes on exposure to light to give phosphine and red phosphorus.

$$3P_2H_4 \rightarrow 4PH_3 + P_4.$$

It do not exhibit basic character.

Its structure is as follows.

$$\begin{array}{ccccccc} H & - & \ddot{P} & - & \ddot{P} & - & H \\ & & | & & | & & \\ & & H & & H & & \end{array}$$

9.11 HALIDES OF PHOSPHORUS

Phosphorus forms trihalides and pentahalides with almost all halogens. Of these, phosphorus trichloride and phosphorus pentachloride are important.

9.11.1 Phosphorus Trichloride: PCl_3

Preparation**:** On a small scale, it is prepared by the action of thionyl chloride on phosphorus.

$$P_4 + 8SOCl_2 \rightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$$

In the laboratory, it is prepared by heating white phosphorus in a current of limited supply of dry chlorine.

$$P_4 + 6Cl_2 \rightarrow 4PCl_3$$

PCl_5 formed, if any, is removed by distilling the phosphorus trichloride over white phosphorus.

Properties**:** It is a colourless liquid (b.p. 347K) having a pungent odour.

(i) It fumes in moist air and reacts violently with water.

$$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl.$$

(ii) It reacts with organic compounds containing hydroxyl groups (–OH) such as acetic acid (CH_3COOH), alcohol (C_2H_5OH), etc.

$$3C_2H_5OH + PCl_3 \rightarrow 3C_2H_5Cl + H_3PO_3$$

$$3CH_3COOH + PCl_3 \rightarrow 3CH_3COCl + H_3PO_3$$

(iii) It behaves like an unsaturated compound, combines with chlorine, oxygen and sulphur.

$$PCl_3 + Cl_2 \rightarrow PCl_5$$

$$2PCl_3 + O_2 \rightarrow 2POCl_3$$

$$PCl_3 + S \rightarrow PSCl_3$$

(iv) It also act as a reducing agent when treated with SO_3, conc. H_2SO_4 and sulphur chloride.

$$PCl_3 + SO_3 \rightarrow POCl_3 + SO_2$$

$$PCl_3 + 2H_2SO_4 \rightarrow SO_3HCl + SO_2 + 2HCl + HPO_3$$

$$3PCl_3 + S_2Cl_2 \rightarrow PCl_5 + 2PSCl_3$$

(v) It reacts with finely divided metals when hot.

$$12Ag + 4PCl_3 \rightarrow 12AgCl + P_4$$

$$6Na + PCl_3 \rightarrow 3NaCl + Na_3P$$

Structure of PCl_3 is pyramidal and was already discussed earlier in this chapter.

9.11.2 Phosphorus Pentachloride, PCl_5

Preparation

(i) It may be prepared by the action of sulphuryl chloride on phosphorus or phosphorus trichloride.

$$P_4 + 10SO_2Cl_2 \rightarrow 4PCl_5 + 10SO_2$$

$$PCl_3 + SO_2Cl_2 \rightarrow PCl_5 + SO_2$$

(ii) In the laboratory, it is prepared more conveniently by the action of an excess of chlorine on phosphorus trichloride.

$$PCl_3 + Cl_2 \rightarrow PCl_5$$

Properties: It is a yellowish white crystalline compound, having a sharp odour. It melts at 318K under pressure. However, it sublimes on heating at 433K.

(i) ***Dissociation***: When heated, it dissociates into phosphorus trichloride and chlorine.

$$PCl_5 \rightleftharpoons PCl_3 + Cl_2$$

(ii) ***Hydrolysis***: It undergoes hydrolysis violently by water giving phosphoryl chloride with insufficient water or phosphoric acid with excess of water.

$$PCl_5 + H_2O \rightarrow POCl_3 + 2HCl$$

$$POCl_3 + 3H_2O \rightarrow H_3PO_4 + 3HCl$$

$$PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$$

(iii) ***With hydroxy Compounds:*** With hydroxycompounds, PCl_5 reacts forming chloroderivatives replacing the hydroxyl group.

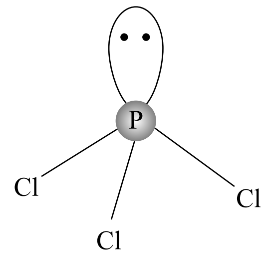
$$C_2H_5OH + PCl_5 \rightarrow C_2H_5Cl + POCl_3 + HCl$$

$$CH_3COOH + PCl_5 \rightarrow C_2H_5COCl + POCl_3 + HCl$$

$$SO_2(OH)_2 + PCl_5 \rightarrow SO_2Cl_2 + 2POCl_3 + 2HCl$$

(iv) ***Action of metals***: When heated with finely divided metals, yields the corresponding halides.

$$2Cu + PCl_5 \rightarrow 2CuCl + PCl_3$$



Pyramidal structure of PCl_3

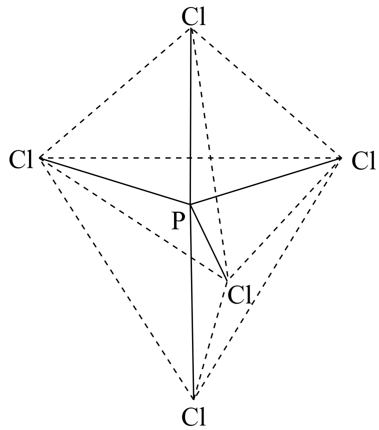


Fig 9.11 Structures of PCl_3 and PCl_5

$$2Ag + PCl_5 \rightarrow 2AgCl + PCl_3$$

$$Sn + 2PCl_5 \rightarrow SnCl_4 + 2PCl_3$$

Uses: In organic chemistry, it is used in the detection and determination of the number of –OH groups and to replace –OH group with chlorine atom.

Structure was already discussed earlier in this chapter.

9.12 OXIDES OF PHOSPHORUS

Phosphorus forms three oxides

(i) Phosphorus trioxide P_4O_6 or Phosphorus III oxide
(ii) Phosphorus tetroxide P_4O_8 or Phosphorus IV oxide
(iii) Phosphorus Pentoxide P_4O_{10} or Phosphorus V oxide.

9.12.1 Phosphorus Trioxide, P_4O_6

Preparation: Phosphorus trioxide is obtained by controlled oxidation of P_4 in an atmosphere of 75 per cent O_2 and 25 per cent N_2 at 90 mm Hg and 323K followed by distillation of the product from the mixture.

$$P_4 + 3O_2 \rightarrow P_4O_6$$

Properties: It is a white waxy solid, melting at 296.8K and boiling at 447K. It is soluble in benzene, chloroform and carbon disulphide. At low temperature it takes up oxygen from the air to form P_4O_{10} but inflames if slightly warmed.

$$P_4O_6 + 2O_2 \rightarrow P_4O_{10}$$

It reacts with cold water by slowly forming phosphorus acid.

$$P_4O_6 + 6H_2O \rightarrow 4H_3PO_3$$

With hot water it gives phosphoric acid and phosphine.

$$P_4O_6 + 6H_2O \rightarrow 3H_3PO_4 + PH_3$$

It reacts with chlorine and bromine by forming oxyhalides.

$$\underset{}{P_4O_6} + 5Cl_2 \longrightarrow \underset{\substack{\text{Phosphorus}\\\text{oxychloride}}}{2POCl_3} + \underset{\substack{\text{Metaphosphorus}\\\text{oxychloride}}}{2PO_2Cl}$$

P_4O_6 can also act as a ligand-forming complexes like $[\{Ni(CO)_3\}_4(P_4O_6)]$.

9.12.2 Phosphorus Tetroxide

It is obtained by heating phosphorus in a limited supply of air at 563K.

$$4P + 4O_2 \rightarrow P_4O_8$$

Another method is by heating P_4O_6 in a sealed tube at 713K.

$$4P_4O_6 \rightarrow 3P_4O_8 + P_4\ \text{(red)}$$

Phosphorus tetroxide sublimes and is collected, whereas red phosphorus remains behind.

Properties: Phosphorus tetroxide is a white solid which sublimes on heating. When it reacts with water, it forms phosphorus H_3PO_3 and phosphoric acids.

$$P_4O_8 + 6H_2O \rightarrow 2H_3PO_3 + 2H_3PO_4$$

9.12.3 Phosphorus Pentoxide

It is prepared by burning phosphorus in excess of air or oxygen.

$$P_4 + 5O_2 \rightarrow P_4O_{10}$$

This may contain some P_4O_6. This can be purified by heating with phosphorus at 663K in a rapid stream of ozonized oxygen when P_4O_{10} sublimes and its vapours are condensed.

Properties: It is white crystalline solid which sublimes on heating. When pure it is colourless but the garlic odour of a common sample is due to the presence of P_4O_6. It is readily soluble in water.

(i) It reacts with water to form a mixture of phosphoric acids whose composition depends upon the quantity of water and other conditions.

$$P_4O_{10} \xrightarrow{H_2O} \underset{\substack{\text{Metaphos-}\\\text{phoric acid}}}{(HPO_3)_n} \xrightarrow{H_2O} \underset{\substack{\text{Triphosp-}\\\text{horic acid}}}{H_5P_3O_{10}} \xrightarrow{H_2O} \underset{\substack{\text{Pyropho-}\\\text{sphoric acid}}}{H_4P_2O_7} \xrightarrow{H_2O} \underset{\substack{\text{Ortho}\\\text{phosphoric}\\\text{acid.}}}{H_3PO_4}$$

(ii) It has great affinity towards water. So, it is used as a drying and dehydrating agent. It removes water molecule from many organic and inorganic compounds.

$$4HNO_3 + P_4O_{10} \longrightarrow 2N_2O_5 + 4HPO_3$$

$$2\,H_2SO_4 + P_4O_{10} \longrightarrow 2SO_3 + 4HPO_3$$

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow[-6H_2O]{P_4O_{10}} C$$

$$C_{12}H_{22}O_{11} \xrightarrow[-11H_2O]{P_4O_{10}} 12C$$

$$(C_6H_{10}O_5)_n \xrightarrow[-5nH_2O]{P_4O_{10}} 6nC$$

(iii) It is reduced to phosphorus on strong heating with carbon.

$$P_4O_{10} + 10C \rightarrow P_4 + 10CO$$

(iv) It forms phosphates when fused with basic salts.

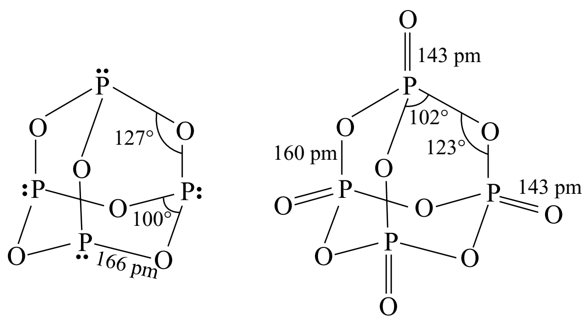
$$6CaO + P_4O_{10} \rightarrow 2Ca_3(PO_4)_2$$

It cannot be used to dry the basic substances such as CaO, NH_3 because they form salts with P_4O_{10}.

9.12.4 Structures of Phosphorus Oxides

In all these oxides, the four phosphorus atoms are arranged in a tetrahedral manner.

In these molecules, six oxygen atoms act as bridges between the tetrahedrally arranged phosphorus atoms along the edges. The molecular structure has a tetrahedral symmetry and comprises 4 fused 6-membered P_3O_3 heterocycles each with a chair conformation as shown in Fig. 9.12. These contain six P—O—P bridge bonds. In P_4O_8, two oxygen atoms are bonded to phosphorus atoms



P_4O_6 P_4O_{10}

Fig 9.12 Structures of P_4O_6 and P_4O_{10}

by coordination bonds (i.e., one oxygen atom with each phosphorus atom). In P_4O_{10}, every phosphorus atom has an additional oxygen atom which is linked by the coordinate bond.

Measurements of the P—O bond lengths shows that the bridging bonds on the edges are 160 pm in all the three oxides which is equal to single P—O bond length. The bonds on the corners are much shorter (140 pm) than a single bond and are in fact double bonds. These double bonds are different in origin from the 'usual' double bonds such as that in alkenes, which arises from $p\pi - p\pi$ overlap. With one electron coming from each carbon atom, the second bond in P = O is formed by $p\pi - d\pi$ bonding. A p-orbital containing a lone pair on the oxygen atom overlaps laterally with an empty d-orbital on the phosphorus atom. Thus, it differs from the double bond in ethene in two respects.

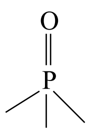
1. A p-orbital overlaps a d-orbital rather than p with p.
2. Both electrons come from one atom and hence the bond is a dative bond.

The P—O bond lengths in all the three oxides are nearly equal but there is a large variation in bond angles.

9.13 OXOACIDS OF PHOSPHORUS

The oxoacids of phosphorus are more numerous than those of any other element, and the number of oxoanions and oxosalts is probably exceeded by only those of silicon. Many are of great importance technologically and their derivatives are vital in many biological process. Fortunately, the structural principles covering this extensive array of compounds are very simple and can be stated as follows.

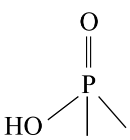
(i) All P atoms in the oxoacids and oxoanions are 4-coordinate and contain at least one P—O unit. In all these oxoacids and oxoanions P atom is involved in sp^3 hybridization having tetrahedral geometry.



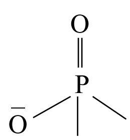
(1)

(ii) All P atoms in the oxoacids have at least one P—OH group (2a) and this often occurs in the anions also; all such groups are ionizable as proton donors (2b).

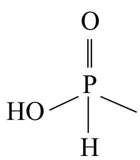
(iii) Some species also have one (or more) P—H group (3); such bonded H atoms are not ionizable.



2a

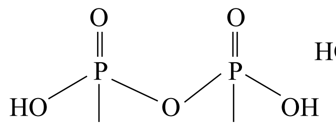


2b

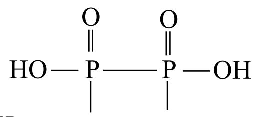


3

(iv) Catenation is by P—O—P links (4a) or via direct P—P bonds (4b) with the former and both open chain (linear) and cyclic species are known but only corner sharing of tetrahedra occurs, never edge or face-sharing.

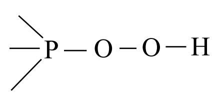


4a

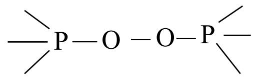


4b

(v) Peroxo compounds feature either



or

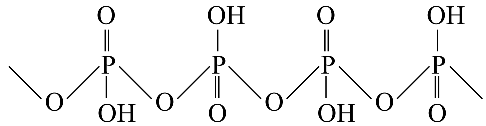


It follows from these structural principles that each P atom is 5-covalent. However, the oxidation state of P atom is +5 only when it is directly bound to four oxygen atoms; the oxidation number is reduced by 1 each time a P—OH is replaced by a P—P bond and by 2 each time a P—OH is replaced by a P—H.

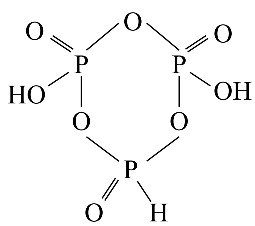
All the oxoacids and their salts containing P—H groups will act as reducing agents. Basicity of an oxoacid of phosphorus is equal to the number of P—OH groups. Some examples of phosphorus oxoacids are listed in Table 9.9. It will be seen that the numerous structural types and the variability of oxidation states pose several problems of nomenclature which offer a rich source of confusion in the literature.

Metaphosphoric do not exist as a monomer. It may exist in the form of polymetaphosphoric acids or cyclic metaphosphoric acids like cyclo di, tri, tetra, etc. metaphosphoric acids.

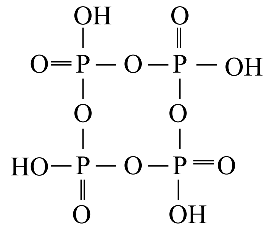
The structures of polymetaphosphoric acid, cyclotrimetaphosphoric acid and cyclo tetrametaphosphoric acid is shown here under.



Polymetaphosphoric acid

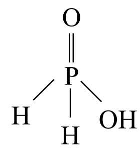
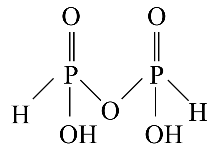
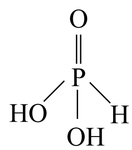
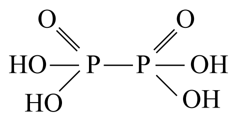
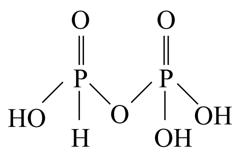
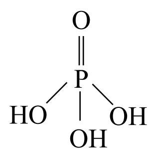
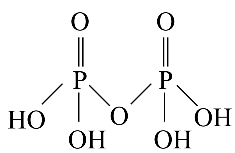
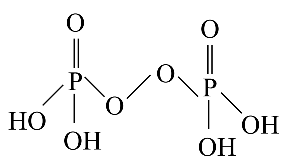


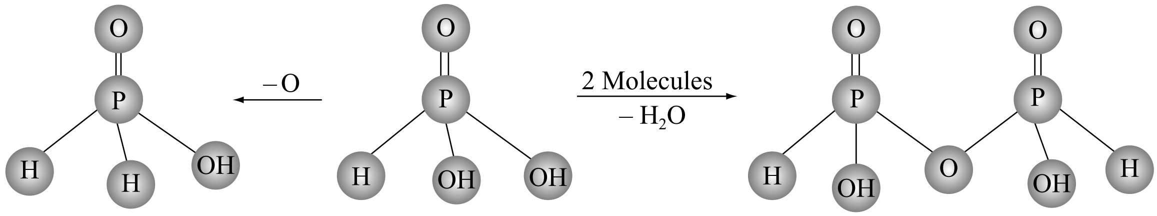
Cyclotrimeta phosphoric acid



Cyclotetrameta phosphoric acid

Table 9.9

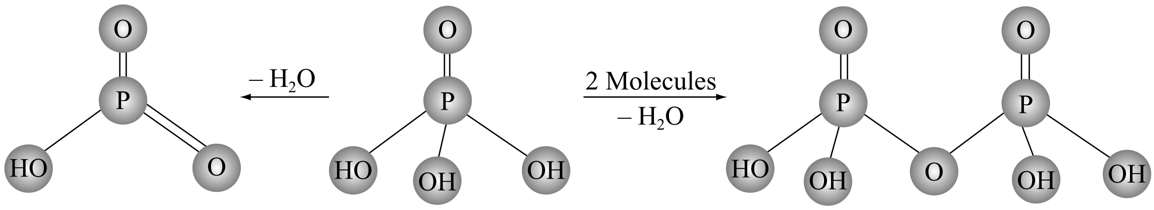
S. no.	Name	Formula	Structure	State	Basicity
1	Hypophosphorus acid (Phosphinic acid)	H_3PO_2		+1	1
2	Diphosphorus or pyrophosphorus acid (Diphosphinic acid)	$H_4P_2O_5$		+3	2
3	Orthophosphorus acid	H_3PO_3		+3	2
4	Hypophosphoric acid Diphosphoric (IV) acid	$H_4P_2O_6$		+4	4
5	Isohypophosphoric acid (Diphosphoric (III, V) acid	$H_4P_2O_6$		+5	3
6	Orthophosphoric acid	H_3PO_4		+5	3
7	Pyrophosphoric acid (Diphosphoric acid)	$H_4P_2O_7$		+5	4
8	Metaphosphoric acid	HPO_3		+5	
9	Peroxodiphosphoric acid	$H_4P_2O_8$		+5	4



Phosphinic or Hypophosphorus acid H_3PO_2 (O.N. of P = +1) (monobasic)

Phosphonic or Phosphorus acid H_3PO_3 (O.N. of P = +3) (dibasic)

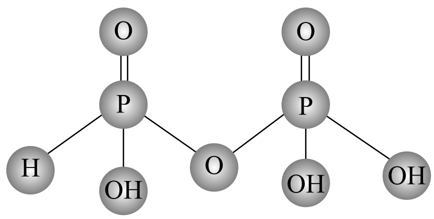
Pyrophosphorus acid $H_4P_2O_5$ (O.N. of P = +3) (dibasic)



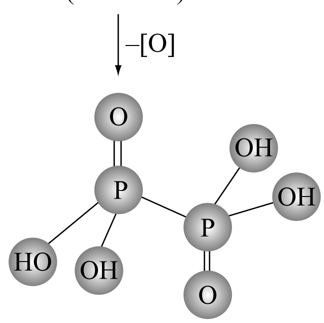
Metaphosphoric acid HPO_3 (O.N. of P = +5) (monobasic)

Orthophosphoric acid H_3PO_4 (O.N. of P = +5) (tribasic)

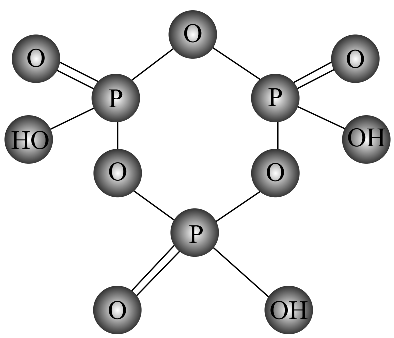
Pyrophosphoric acid $H_4P_2O_7$ (O.N. of P = +5) (tetrabasic)



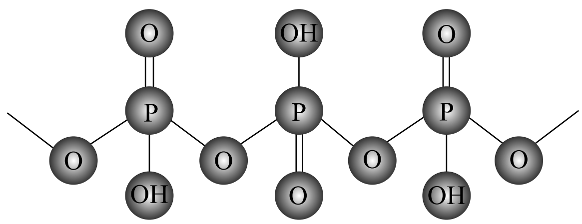
Iso hypophosphoric acid $H_4P_2O_6$ (tribasic)



Hypophosphoric acid $H_4P_2O_6$ (tetrabasic)



Cyclotrimetaphosphoric acid, $(HPO_3)_3$



Polymetaphosphoric acid, $(HPO_3)_n$

Fig 9.13

9.13.1 Hypophosphorus Acid H_3PO_2

Preparation

(i) It can be prepared by the oxidation of phosphine by iodine and water.

$$PH_3 + 2I_2 + 2H_2O \rightarrow H_3PO_2 + 4HI$$

(ii) When white phosphorus is boiled with barium hydroxide, barium hypophosphite and phosphine will be formed. Excess of barium hydroxide is removed by passing CO_2 and the filtrate is evaporated to get barium hypophosphite. Its solution when treated with calculated quantity of sulphuric acid gives hypophosphorus acid.

$$2P_4 + 3Ba(OH)_2 + 6H_2O \rightarrow 3Ba(H_2PO_2)_2 + 2PH_3$$

$$Ba(H_2PO_2)_2 + H_2SO_4 \rightarrow BaSO_4 + 2H_3PO_2$$

(iii) White phosphorus on boiling with hot solution of an alkali gives phosphine and the salt of hypophosphite.

$$P_4 + 3KOH + 3H_2O \rightarrow 3KH_2PO_2 + PH_3$$

Free acid may be obtained by decomposing the alkali salt with calculated quantity of dilute H_2SO_4 and extracting with absolute alcohol.

$$KH_2PO_2 + H_2SO_4 \rightarrow KHSO_4 + H_3PO_2$$

Properties: It is a colourless crystalline solid soluble in water, m.p. 299.3K.

(i) ***Acidic Character***: It is a strong monobasic acid and forms only one series of salt MH_2PO_2.

$$H_3PO \rightleftharpoons H_2PO_2^- + H^+ \; Ka = 10^{-2}.$$

Two other hydrogen atoms are not ionizable. Active metals like zinc, magnesium, etc. dissolve in H_3PO_2 with the evolution of hydrogen.

$$2H_3PO_2 + Mg \rightarrow Mg(H_2PO_2)_2 + H_2$$

(ii) ***Action of heat***: When heated strongly, it undergoes disproportionation to yield phosphine and phosphoric acid.

$$2H_3PO_2 \rightarrow PH_3 + H_3PO_4$$

(iii) ***Action of nascent hydrogen***: It is reduced to phosphine.

$$H_3PO_2 + 4[H] \rightarrow 2H_2O + PH_3$$

(iv) ***Reducing property:*** Due to the presence of two P—H bonds it acts as a strong reducing agent, oxidizing itself to phosphoric acid.

$$H_3PO_2 + 2H_2O \rightarrow H_3PO_4 + 4H^+ + 4e^-$$

(a) It reduces mercuric salt first to mercurous salt and then to mercury.

$$4HgCl_2 + H_3PO_2 + 2H_2O \rightarrow 2Hg_2Cl_2 + 4HCl + H_3PO_4$$

$$2Hg_2Cl_2 + H_3PO_2 + 2H_2O \rightarrow 4Hg + 4HCl + H_3PO_4$$

(b) It reduces solutions of the salts of Au, Ag, Pt and Bi to metals.

$$4AuCl_3 + 3H_3PO_2 + 6H_2O \rightarrow 4Au + 3H_3PO_4 + 12HCl$$

$$4AgNO_3 + H_3PO_2 + 2H_2O \rightarrow 4Ag + H_3PO_4 + 4HNO_3$$

$$PtCl_4 + H_3PO_2 + 2H_2O \rightarrow Pt + H_3PO_4 + 4HCl$$

$$4BiCl_3 + 3H_3PO_2 + 6H_2O \rightarrow 4Bi + 3H_3PO_4 + 12HCl$$

(c) It reduces chlorine and iodine to the corresponding hydraacids.

$$H_3PO_2 + 2Cl_2 + 2H_2O \rightarrow H_3PO_4 + 4HCl$$

$$H_3PO_2 + 2I_2 + 2H_2O \rightarrow H_3PO_4 + 4HI$$

(d) It reduces copper sulphate to cuprous hydride.

$$4CuSO_4 + 3H_2PO_2 + 6H_2O \rightarrow 2Cu_2H_2 + 3H_3PO_4 + 4H_2SO_4$$

Uses

(i) It is added to certain pharmaceutical preparations, E.g., ferrous iodide syrup to stop the oxidation of the ingredients.

(ii) Some of its salts like sodium, potassium, calcium, etc. are used as nerve tonics and in the treatment of lung diseases.

9.13.2 Phosphorus Acid H_3PO_3

Preparation

(i) It is prepared by dissolving phosphorus trioxide in cold water.

$$P_4O_6 + 6H_2O \rightarrow 4H_3PO_3$$

(ii) It is also prepared by the hydrolysis of phosphorus trichloride.

$$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$$

Hydrolysis of phosphorus trichloride is better done by addition of phosphorus trichloride to anhydrous oxalic acid.

$$PCl_3 + 3H_2C_2O_4 \rightarrow H_3PO_3 + 3CO + 3CO_2 + 3HCl$$

An advantage of this reaction is that all the products are gases except H_3PO_3.

Properties: It is a white deliquescent crystalline compound, m.p., 343.6K. It is extremely soluble in water.

(i) ***Action of heat***: On heating it disproportionates forming phosphoric acid and phosphine.

$$4H_3PO_3 \xrightarrow{200^\circ C} 3H_3PO_4 + PH_3$$

(ii) ***Acidic nature***: It is a strong dibasic acid, though it contain three hydrogens one is in P—H bond.

$$H_3PO_3 \rightleftharpoons H_2PO_3^- + H^+ \; K_1 = 1 \times 10^{-2}$$

$$H_2PO_3^- \rightleftharpoons HPO_3^{2-} + H^+ \; K_2 = 2 \times 10^{-7}$$

It forms two series of salts like NaH_2PO_3 and Na_2HPO_3 known as primary phosphites and secondary phosphites Na_2HPO_3. Primary phosphites are acid salts because they contain one more ionizable –OH group but secondary phosphites are normal salts.

(iii) ***Reducing Property***: It is a reducing agent though less than H_3PO_2 and acts as reducing agent according to the reaction

$$H_3PO_3 + H_2O \rightarrow H_3PO_4 + 2H^+ + 2e^-$$

It precipitates the heavy metals from solutions of their salts.

$$2HgCl_2 + H_3PO_3 + H_2O \rightarrow Hg_2Cl_2 + H_3PO_4 + 2HCl$$

$$CuSO_4 + H_3PO_3 + H_2O \rightarrow Cu + H_3PO_4 + H_2SO_4$$

$$2AgNO_3 + H_3PO_3 + H_2O \rightarrow 2Ag + H_3PO_4 + 2HNO_3$$

$$2AuCl_3 + 3H_3PO_3 + 3H_2O \rightarrow 2Au + 3H_3PO_4 + 6HCl$$

(iv) Phosphorus acid is oxidized by air in the presence of iodine (catalyst) forming phosphoric acid.

(v) It is oxidized to phosphoric acid with several oxidizing agents like potassium permanganate, iodine, etc.

$$\begin{array}{l} 2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 \\ \qquad\qquad\qquad\qquad + 3H_2O + 5(O) \\ [H_3PO_3 + (O) \longrightarrow H_3PO_4] \times 5 \\ \hline 2KMnO_4 + 3H_2SO_4 + 5H_3PO_3 \rightarrow K_2SO_4 + 2MnSO_4 \\ \qquad\qquad\qquad\qquad + 3H_2O + 5H_3PO_4 \end{array}$$

$$H_3PO_3 + H_2O + I_2 \rightarrow H_3PO_4 + 2HI$$

(vi) It reacts with phosphorus pentachloride to form phosphorus oxychloride.

$$H_3PO_3 + 3PCl_5 \rightarrow PCl_3 + 3POCl_3 + 3HCl$$

9.13.3 Hypophosphoric Acid, $H_4P_2O_6$

Preparation: It is obtained on exposing phosphorus to limited supply of moist air along with phosphoric and phosphorus acids. This mixture of acids is treated with lead acetate to get the precipitate of lead hypophosphate. This is decomposed by passing hydrogen sulphide gas to get free acid.

$$Pb_2P_2O_6 + 2H_2S \rightarrow 2PbS + H_4P_2O_6$$

In another method, if sodium acetate is added to the above mixture of hypophosphoric, phosphoric and phosphorus acids the sparingly soluble disodium hypophosphate $Na_2H_2P_2O_6 \cdot 6H_2O$ separates, from which free acid can be obtained by acidification.

Properties: It is a colourless crystalline solid. It is a dihydrate, m.p. 343K. It is hydrolyzed in water to a mixture of phosphorus and phosphoric acids.

$$H_4P_2O_6 + H_2O \rightarrow H_3PO_3 + H_3PO_4$$

It is hygroscopic in nature. It is a tetrabasic acid. It decomposes on heating above its melting point.

$$H_4P_2O_6 \rightarrow H_3PO_3 + HPO_3$$

It is not a reducing agent. It is not reduced by zinc and sulphuric acid to phosphine. The hypophosphates (except alkali metal salts) are very sparingly soluble.

9.13.4 Orthophosphoric Acid, H_3PO_4

Preparation: It is the best known oxoacid of phosphorus. It is also known as phosphoric acid.

(i) It is prepared by dissolving P_4O_{10} in water and the solution is boiled to convert metaphosphoric acid into orthophosphoric acid.

$$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$$

(ii) In the laboratory it is best prepared by the oxidation of red phosphorus with 50 per cent nitric acid in the presence of iodine flakes which act as a catalyst. The role of iodine in this reaction is not clear. However, the reaction may takes place in the following manner.

$$\begin{array}{ll} [P_4 + 6I_2 \longrightarrow 4PI_3] & \\ [PI_3 + 3H_2O \longrightarrow H_3PO_3 + 3HI] & \times 4 \\ [H_3PO_3 + 2HNO_3 \longrightarrow H_3PO_4 + 2NO_2 + H_2O] & \times 4 \\ [2HI + 2HNO_3 \longrightarrow 2H_2O + 2NO_2 + I_2] & \times 6 \\ \hline P_4 + 20HNO_3 \longrightarrow 4H_3PO_4 + 20NO_2 + 4H_2O & \end{array}$$

Manufacture

(i) On a large scale, it is prepared by heating calcium phosphate with calculated amount of dil. H_2SO_4 for several hours and allowed the mixture to stand so that calcium sulphate settles down. The clear supernatant syrup is separated. It is about 8.5 per cent and is used as such.

$$Ca_3(PO_4)_2 + 3H_2SO_4 \rightarrow 3CaSO_4 + 2H_3PO_4$$

(ii) Another convenient method involves preparation of phosphorus pentoxide and dissolving it in water.

$$2Ca_3(PO_4)_2 + 6\ SiO_2 \rightarrow 6CaSiO_3 + P_4O_{10}$$

$$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$$

It is also formed by the hydrolysis of phosphorus pentachloride.

$$PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$$

Properties: It is a transparent deliquescent crystalline solid, m.p. 315.3K. It absorbs water and gives a colourless syrupy liquid.

(i) ***Action of heat***: When heated at 523k it loses one water molecule to give pyrophosphoric acid, which on further heating at 873K loses another water molecule forming metaphosphoric acid.

$$2H_3PO_4 \xrightarrow[-H_2O]{523\,k} H_4P_2O_7 \xrightarrow[-H_2O]{873\,k} 2HPO_3$$

(ii) ***Acidic nature***: It is a tribasic acid and ionizes according to the following equations

$$H_3PO_4 \rightleftharpoons H_2PO_4^- + H^+ \qquad K_1 = 7.5 \times 10^{-3}$$

$$H_2PO_4^{2-} \rightleftharpoons HPO_4^{2-} + H^+ \qquad K_3 = 6.2 \times 10^{-8}$$

$$HPO_4^2 \rightleftharpoons PO_4^{3-} + H^+ \qquad K_3 = 1.1 \times 10^{-12}$$

The first dissociation constant is much greater than the second or third indicating that first ionization takes place considerably while second ionization takes place only very slightly, whereas the third ionization is very small. With alkalis it gives three types salts, viz. (i) primary phosphates e.g., NaH_2PO_4, (ii) secondary phosphates, e.g., Na_2HPO_4 and (iii) tertiary phosphates, e.g., Na_3PO_4. The primary and secondary phosphates are acid salts while the tertiary salt is a normal salt.

Primary phosphates on heating converts into metaphosphates.

$$NaH_2PO_4 \xrightarrow{\Delta} NaPO_3 + H_2O$$

Secondary phosphates on heating converts into pyrophosphates.

$$2Na_2HPO_4 \xrightarrow{\Delta} Na_4P_2O_7 + H_2O$$

Normal salt is not affected by heating.

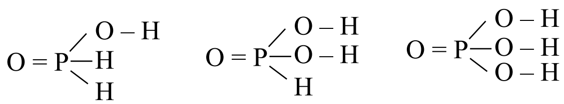
If ammonium ion is present in the salt, it behaves like hydrogen.

$$Na(NH_4)HPO_4 \xrightarrow{\Delta} NaPO_3 + NH_3 + H_2O$$

$$Na(NH_4)_2PO_4 \xrightarrow{\Delta} NaPO_3 + 2NH_3 + H_2O$$

$$2Mg(NH_4)PO_4 \xrightarrow{\Delta} Mg_2P_2O_7 + 2NH_3 + H_2O$$

Among the three oxoacids of phosphorus (H_3PO_2, H_3PO_3 and H_3PO_4) acidity actually decreases as $H_3PO_2 > H_3PO_3 > H_3PO_4$. This is contrary to the expectation from the number of unprotonated oxygen atoms alone. This trend can be explained basing on the structures shown.



H_3PO_2
pKa = 1.244

H_3PO_3
pKa_1 = 1.257
pKa_2 = 6.7

H_3PO_4
pKa_1 = 2.15
pKa_2 = 7.20
pKa_3 = 12.37

Acidity aroses from the –O–H groups, one in H_3PO_2, two in H_3PO_3 and three in H_3PO_4. The inductive effect of the single oxygen atom acts on the single –O–H bond in H_3PO_2 but it is distributed over two- and three –O–H bonds in H_3PO_3 and H_3PO_4, respectively. Accordingly, acidity decreases from H_3PO_2 to H_3PO_4.

(iii) ***Reaction with Salts***: It gives yellow precipitate with silver nitrate and white precipitate with barium chloride.

$$3AgNO_3 + H_3PO_4 \rightarrow Ag_3PO_4 + 3HNO_3$$

$$3BaCl_2 + 2H_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6\,HCl$$

(iv) It liberates HBr and HI from bromides and iodides. This reaction is preferred for the preparation of HBr and HI in the place of H_2SO_4 because H_3PO_4 cannot oxidize HBr and HI but H_2SO_4 oxidizes them to Br_2 and I_2.

$$3NaBr + H_3PO_4 \rightarrow Na_3PO_4 + 3HBr$$

$$3KI + H_3PO_4 \rightarrow K_3PO_4 + 3HI$$

(v) ***Reaction with magnesium salt***: Magnesium salts give a white precipitate with phosphoric acid in the presence of ammonium chloride and ammonium hydroxide.

$$MgSO_4 + NH_4Cl + H_3PO_4 \rightarrow Mg(NH_4)\,PO_4 + H_2SO_4 + HCl$$

This reaction is utilized for the detection of Mg^{2+} ions in qualitative analysis.

Uses: It is used

(i) in the preparation of HBr and HI
(ii) in the manufacture of phosphorus, phosphates and phosphatic fertilizers
(iii) in medicines

9.13.5 Pyrophosphoric Acid, $H_4P_2O_7$

Preparation

(i) It may be prepared by heating orthophosphoric acid at 523–533K.

$$2H_3PO_4 \rightarrow H_4P_2O_7 + H_2O$$

(ii) Pure acid can be prepared by heating an equimolar mixture of orthophosphoric acid crystals and phosphorus oxychloride.

$$5H_3PO_4 + POCl_3 \longrightarrow 3H_4P_2O_7 + 3HCl$$

Properties: It is colourless crystalline solid. Melting point is 334 K. It is soluble in water.

(i) ***Action of heat***: when heated at 873 K it converts into metaphosphoric acid.

$$H_4P_2O_7 \xrightarrow{875K} 2HPO_3 + H_2O$$

(ii) ***Reaction with water***: On boiling with water, it is reconverted into orthophosphoric acid.

$$H_4P_2O_4 + H_2O \longrightarrow 2H_3PO_4$$

(iii) ***Acidic nature***: It is a tetrabasic acid and forms two series of salts, $Na_2H_2P_2O_7$ and $Na_4P_2O_7$. The dissociation constants for the four stages are

$K_1 = 1 \times 10^{-1}$, $K_2 = 3.2 \times 10^{-2}$, $K_3 = 1.7 \times 10^{-6}$, $K_4 = 6.0 \times 10^{-9}$.

9.13.6 Metaphosphoric Acid

Preparation

(i) It is prepared by treating phosphorus pentoxide with small amount of water at 273K.

$$P_4O_{10} + 2H_2O \rightarrow 4HPO_3$$

(ii) This is also prepared by heating orthophosphoric acid or pyrophosphoric acids at 873K.

$$H_3PO_4 \xrightarrow{873\text{ k}} HPO_3 + H_2O$$

$$H_4P_2O_7 \xrightarrow{873\text{ k}} 2HPO_3 + H_2O$$

(ii) It can also be prepared by heating secondary ammonium orthophosphate strongly.

$$(NH_4)_2HPO_4 \rightarrow HPO_3 + 2NH_3 + H_2O$$

Properties: As it is a transparent glassy solid, it is also called as **glacial phosphoric acid**. It is a deliquiscent solid (m.p. 311.4 K) soluble in water. On boiling with water it converts into orthophosphoric acid.

$$HPO_3 + H_2O \rightarrow H_3PO_4$$

It reacts with silver nitrate forming a white precipitate.

$$AgNO_3 + HPO_3 \rightarrow AgPO_3 + HNO_3$$

With barium chloride it gives a white precipitate which is insoluble in acetic acid.

$$BaCl_2 + 2HPO_3 \rightarrow Ba(PO_3)_2 + 2HCl$$

It is a monobasic acid, and forms salts with alkalis. Its salts are called metaphosphates. Sodium metaphosphate when heated with transition metal salts forms coloured mixed phosphate. This reaction is used in qualitative analysis for the detection of coloured ions.

Metaphosphoric acid and its salts exist as polymers.

Uses

(i) It is used in the manufacture of dental cements.

(ii) It is used as a dehydrating agent.

(iii) Its polymers $(NaPO_3)_6$ is used in modern water purification methods.

Metaphosphoric acid coagulates the white of an egg.

9.13.6 Peroxophosphoric Acids

These acids are derived from hydrogen peroxide by the replacement of one and two hydrogen atoms by H_3PO_3 group. Thus

$$\begin{array}{lll}
O-H & O-H_2PO_3 & O-H_2PO_3 \\
| & | & | \\
O-H & O-H & O-H_2PO_3 \\
H_2O_2 & H_3PO_5 & H_4P_2O_8
\end{array}$$

Phosphorus forms two peroxophosphoric acids, i.e., peroxomonophosphoric acid H_3PO_5 and peroxodiphosphoric acid $H_4P_2O_8$.

Peroxomonophosphoric acid is obtained by the action of phosphorus pentoxide and 30 per cent hydrogen peroxide (perhydrol) at low temperature. Peroxodiphosphoric acid is obtained by the action of hydrogen peroxide and phosphorus pentoxide.

9.14 FERTILIZERS

Plants require about 16 elements for their growth (besides sun light and water) which are called plant nutrients. Out of these carbon, hydrogen and oxygen are derived from water and air. The remaining 13 elements are supplied by the soil and may be classified as

(i) ***Primary (essential) nutrients***: Nitrogen, phosphorus and potassium.

(ii) ***Secondary nutrients***: Calcium, magnesium and sulphur.

(iii) ***Micronutrients***: Zinc, boron, manganese, copper, iron, molybdenum and chlorine.

Since these elements are exhausted after repeated cultivation, these nutrients in the form of their compounds have to be added in order to make up for their deficiency. These compounds are known as *fertilizers*. Thus, "Fertilizers are the substances which are added to the soil in order to make up the deficiency of essential elements required by the plants".

Functions of Primary Nutrients

(i) ***Nitrogen***: It is highly essential for rapid growth of plants and hence it improves the yields of crops. It also raises protein content of the crops and thus add to their food value.

(ii) ***Phosphorus***: Experiments have shown that phosphates promote early growth as well as early maturity. It helps in blooming and seed formation, increases resistance to frost and increases yield and quality of the products.

(iii) ***Potassium***: It counteracts undesirable effects due to excessive supply of other nutrients especially nitrogen. It helps in the development of healthy root system which helps the plants to obtain regular supply of nutrients from the soil. It is needed for the production of albuminoids and helps in the formation of sugar starch, cellulose, increase vigour and disease resistance and makes stalks stronger.

Essential Characteristics of a Good Fertilizer

(i) The essential elements present in it must be readily available to the plants.

(ii) It should be soluble in water, cheap, stable for a long time and able to maintain to correct pH (7–8) of the soil.

(iii) It should not be harmful or injurious to plants.

Types of Fertilizers: Fertilizers are two types.

(a) ***Natural manures***: These manures are obtained from plants and animals. These include leaves, barks, seeds, ashes, dung bones, horns, blood, hair stock, leather waste, fish scrap, night soil, guano, sewage sludge, animal excreta, etc.

(b) ***Artificial manures or Fertilizers***: These are various chemical compounds which may occur naturally but can be synthesized artificially. They are of three types.

(i) Nitrogenous fertilizers

(ii) Phosphatic fertilizers

(iii) Potash fertilizers

9.14.1 Nitrogenous Fertilizers

These are nitrogen-containing compounds. The important nitrogenous fertilizers are discussed here.

1. Ammonium sulphate: It is obtained by the following methods.

(i) From ammonical liquor obtained from dry distillation of coal: When the coal is subjected to destructive distillation, the nitrogen present in the coal (plant material) gets converted into ammonium salts and collects along with other volatile matter as *ammonical liquor*. The ammonium salts present in the liquor are ammonium carbonate $(NH_4)_2CO_3$, ammonium sulphide $(NH_4)_2S$, ammonium chloride NH_4Cl and ammonium sulphate $(NH_4)_2SO_4$. To recover ammonia, this liquor is mixed with lime and heated in stills by steam when the salts decompose evolving ammonia.

$$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3$$

$$(NH_4)_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2H_2O + 2NH_3$$

$$(NH_4)_2CO_3 + Ca(OH)_2 \rightarrow CaCO_3 + 2H_2O + 2NH_3$$

$$NH_4HS \rightarrow NH_3 + H_2S$$

Ammonia coming off from the salts along with steam is passed through 60 per cent sulphuric acid when crystals of ammonium sulphate separate out.

$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$$

(ii) (a) ***Ammonia obtained from Haber's process***: The synthetic ammonia obtained from Haber's process is also converted into ammonium sulphate by absorbing it in sulphuric acid.

(b) ***Sindri process***: Ammonia obtained by Haber's process is passed into a suspension of finely powdered gypsum $(CaSO_4)$ in water in the presence of carbon dioxide.

$$2NH_3 + CO_2 + CaSO_4 \cdot 2H_2O \rightarrow (NH_4)_2SO_4 + CaCO_3 + H_2O$$

Calcium carbonate is removed by filtration. The solution is concentrated and crystallized.

Ammonium sulphate contains theoretically over 25 per cent ammonia. When it is added to soil the ammonia is first converted into nitrite and nitrate ions by the nitrifying bacteria present in the soil. The nitrite and nitrate ions can be taken up easily by plants from the soil. Ammonium sulphate should not be used too frequently as this renders the soil acidic. This is particularly so when the soil is poor in lime which if present can neutralize acidity to some extent. It should be used after the seeds have germinated.

2. Calcium ammonium nitrate (CAN): It is also known as *nitrolime stone*. Ammonia synthesized by Haber's process is partly converted into nitric acid by Ostwald's process using platinum or platinum–rhodium catalyst. Nitric acid and ammonia are then made to react with each other to give ammonium nitrate. This ammonium nitrate contains nitric acid which is neutralized by adding calcium carbonate. On cooling, grains of calcium ammonium nitrate separate out.

$$NH_3 + HNO_3 \longrightarrow NH_4NO_3$$

$$CaCO_3 + 2HNO_3 \longrightarrow Ca(NO_3)_2 + H_2O + CO_2.$$

$$CaCO_3 + NH_3 + 3HNO_3 \longrightarrow \underbrace{Ca(NO_3)_2 + NH_4NO_3}_{\text{CAN.}} + H_2O + CO_2$$

It is hygroscopic. To prevent the absorption of water from atmosphere, the grains are coated with a thin layer of soap stone. Ammonium nitrate is an explosive. The lime stone present in it prevent it from exploding. It is superior to ammonium sulphate because it is more soluble in water and the lime present in it prevents the development of acidity in soil.

3. Basic calcium nitrate: It is manufactured by the action of nitric acid on calcium carbonate (lime stone) and mixing the calcium nitrate so obtained with calcium oxide on heating and cooling, basic calcium nitrate crystallizes out.

$$CaCO_3 + 2HNO_3 \rightarrow Ca(NO_3)_2 + H_2O + CO_2$$

$$Ca(NO_3)_2 + CaO \rightarrow \underset{\text{Basic calcium nitrate}}{Ca(NO_3)_2.\ CaO}$$

It is readily soluble in water and basic in nature. It is easily assimilable and gives best results of all the artificial nitrogenous fertilizers.

4. Calcium cyanamide: It is manufactured by heating powdered calcium carbide at 1275K in an atmosphere of nitrogen in an electric furnace. A mixture of calcium cyanamide and graphite is obtained which is sold in the market under the trade name nitrolim.

$$CaC_2 + N_2 \rightarrow CaNCN + C$$

When it is applied to the soil, it undergoes a series of changes giving successively cyanamide, urea, ammonia and finally the nitrates which are assimilable by plants.

$$CaNCN + H_2O \rightarrow CaO + H_2NCN$$

$$H_2NCN + H_2O \rightarrow H_2N\ CONH_2$$

$$H_2N\ CONH_2 + H_2O \rightarrow 2NH_3 + CO_2$$

$$CaO + CO_2 \rightarrow CaCO_3$$

It is a base-producing fertilizer and can be applicable to acid soils, and its basic character is three times as high as that of sodium nitrate. The intermediate products are toxic in nature and therefore cyanamide should be applied at least two or three weeks before planting and should be well mixed in soil. It is a slow fertilizer and its effects on fertility are of more permanent nature.

5. Urea: It is manufactured as described below.

(i) It is manufactured by reacting liquid ammonia and gaseous carbon dioxide at 135ºC and about 220 atm pressure to yield ammonium carbamate, some of which decomposes simultaneously to give urea.

$$2NH_3 + CO_2 \rightarrow \underset{\text{Amm. carbamate}}{NH_2\ COONH_4}$$

$$NH_2COONH_4 \rightarrow NH_2CONH_2 + H_2O$$

The undecomposed ammonium carbamate is processed separately under reduced pressure.

In the modern manufacturing plants urea is produced using highly economical methods. Hydrogen required for the manufacture of ammonia and carbon dioxide required for reacting with ammonia are both obtained from **naphtha**. For this purpose crude naphtha from oil refineries is subjected to partial combustion in specially designed burners to give a mixture of hydrogen and carbon monoxide. The mixture is made to react with steam in the presence of a catalyst (usually a mixture of Fe, Cr and Co) when carbon monoxide is oxidized to carbon dioxide.

$$\underbrace{CO + H_2}_{\text{From naphtha}} + \underset{\text{Steam.}}{H_2O} \xrightarrow{\text{Catalyst}} CO_2 + 2H_2$$

The hydrogen, obtained as above, is converted into ammonia by Haber process. The carbon dioxide is regenerated and made to react with ammonia to get urea.

Apart from the nitrogenous fertilizers described so far other nitrogen containing compounds which can be used as fertilizers are sodium nitrate, calcium nitrate, ammonium nitrate, ammonium chloride and ammonium sulphato nitrate.

9.14.2 Phosphatic Fertilizers

The phosphate rock which is primarily tricalcium phosphate $Ca_3(PO_4)_2$ and apatite $CaF_2 \cdot 3Ca_3(PO_4)_2$ are too little soluble in water and thus cannot be absorbed by the plants. Therefore these are converted into soluble calcium dihydrogen phosphate or other compounds which are soluble and thus serve as good fertilizers. The important phosphatic fertilizers are discussed here.

1. ***Calcium Superphosphate or Superphosphate of lime***: It is a mixture of calcium dihydrogen phosphate $[Ca(H_2PO_4)_2]$ and gypsum $(CaSO_4 \cdot 2H_2O)$. It is manufactured by treating calcium phosphate (phosphorite rock or bone ash) with calculated quantity of sulphuric acid.

$$Ca_3\ (PO_4)_2 + 2H_2\ SO_4 + 5H_2O \longrightarrow \underbrace{Ca\ (H_2PO_4)_2 \cdot H_2O + 2Ca\,SO_4 \cdot 2H_2O}_{\text{Superphosphate of lime}}$$

The composition of superphosphate of lime may change and need not be always represented by this formula.

The plant used for the preparation of superphosphate of lime is shown in Fig. 9.14. The phosphate rock is ground to a fine powder. It is charged into the cast iron mixture. A calculated quantity of conc. H_2SO_4 (chamber acid 60% acid) is added to it. The reaction mixture is stirred with blades present in the mixture. When the reaction had started, the charge is dumped into one of the dens, D_1, and D_2 through either the valve V_1 or the valve V_2. The reaction is allowed to take place for 24–36 hours in the dens.

A large quantity of heat is produced which dries the products and a dry mixture of primary calcium phosphate and gypsum is obtained. The carbonate and the fluoride that the phosphate rock usually contains are decomposed by H_2SO_4 liberating CO_2 and hydrogen fluoride gases. They escape through the outlet at the top. While escaping, the gases make the mass porous. The final product is a hard mass (due to the presence of gypsum). The hard mass is crushed to a fine powder and sold in the market under the name *superphosphate of lime.* It contains 16 per cent of the available phosphorus as P_2O_5.

2. ***Triple superphosphate***: It is an excellent phosphatic fertilizer and is being manufactured by adding phosphoric acid to calcium phosphate.

$$Ca_3(PO_4)_2 + 4H_3PO_4 \rightarrow 3Ca(H_2PO_4)_2$$

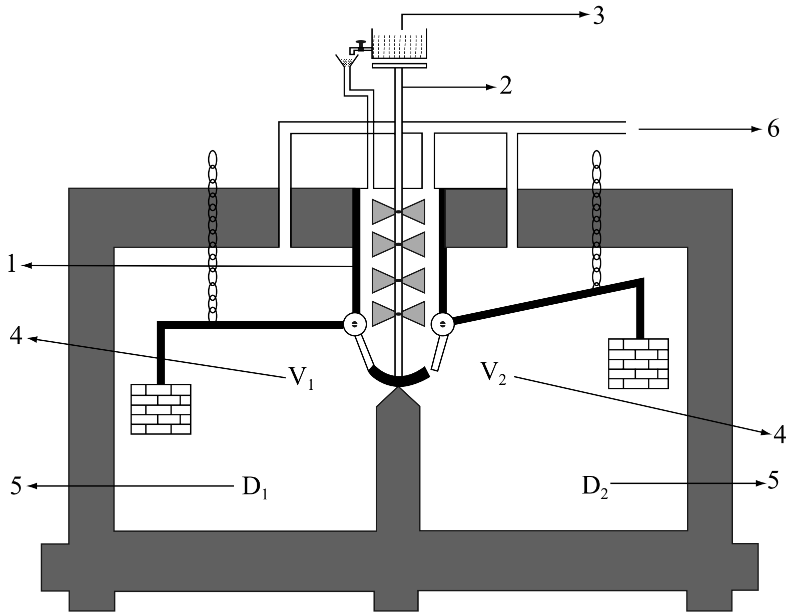


Fig 9.14 Manufacture of superphosphate of lime 1. Mixer 2. Stirrer 3. Chamber acid 4. Valves 5. Brick work Dens 6. Waste gases

In order to manufacture triple superphosphate, a weighed quantity of rock phosphate is reacted with 54 per cent phosphoric acid in a mixture. The reaction product is allowed to move over a belt conveyor, where it sets to a solid mass of triple superphosphate.

Triple superphosphate contains 45 per cent available phosphorus as P_2O_5 which is about three times more concentrated than superphosphate.

3. ***Phosphatic (Thomas) Slag***: It is prepared by grinding the basic slag obtained as a by-product in the manufacture of steel. It is a double salt of calcium phosphate and calcium silicate $[Ca_3(PO_4)_2 \cdot CaSiO_3]$.

It contains 14.22 per cent of P_2O_5 and about 40 per cent of lime. It decomposes in soil in due course of time giving a steady supply of phosphoric acid for the plant growth.

4. ***Nitrophosphate***: It is prepared by the action of dilute nitric acid on calcium phosphate (phosphatic rock).

$$Ca_3(PO_4)_2 + 4HNO_3 \rightarrow Ca(H_2PO_4)_2 + 2Ca(NO_3)_2$$

9.14.3 Potash Fertilizers

Potassium Nitrate, KNO_3

It is a chief potash fertilizer. It occurs in the soil and may be obtained from sodium nitrate and potassium chloride.

$$NaNO_3 + KCl \rightarrow NaCl + KNO_3$$

It is an excellent fertilizer because it contains both potassium and nitrogen as nitrate.

Potassium chloride and potassium sulphate are also used as potash fertilizers.

9.14.4 NPK Fertilizers

These are also known as mixed or complex fertilizers. These are prepared by mixing nitrogenous, phosphatic and potash fertilizers in suitable proportions. It is observed that these fertilizers produce much better results.

KEY POINTS

- Nitrogen, phosphorus, arsenic, antimony and bismuth belongs to the VA group and p-block of the periodic table.
- The general outer electronic configuration of these elements is ns^2np^3.
- Most of the available nitrogen on the earth is present as gas in the atomosphere and is about 78 per cent by volume.
- Nitrogen is inert at room temperature because it requires very high energy (225 k cal mol^{-1} or 945.4 kJ mol^{-1}) to break the triple bond in N_2 molecule ($N \equiv N$).
- Because of chemical inertness, nitrogen occurs in the elemental state as well as in the form of compounds like chile salt petre ($NaNO_3$), Indian salt petre (KNO_3) but other elements occur only in the combined state.

- Among Group-VA elements, phosphorus is the most abundant element in the earth's crust. Important sources of phosphorus are phosphate rock [$Ca_3(PO_4)_2$] and flour apatite [$3Ca_3(PO_4)_4 \cdot CaF_2$].

Physical Properties

- **Atomic size** of Group-VA elements do not increase regularly. As usual this is due to the poor shielding effect of d- and f-electrons in the penultimate and antipenultimate shells of As, Sb and Bi.
- **Physical state:** N_2 is a gas while others are solids. Nitrogen is diatomic, whereas phosphorus, arsenic and antimony are tetraatomic. Bismuth is monoatomic in gaseous state.
- Since nitrogen can form strong pπ–pπ multiple bonds it exists as diatomic molecule. With increase in the size of atoms, the tendency to form pπ–pπ bonds decreases due to the increase in the distance between the parallel p-orbitals to be overlapped. So phosphorus, arsenic and antimony forms P_4, As_4 and Sb_4 molecules in which the atoms are arranged in tetrahedral manner.
- Both the melting points (except for Sb and Bi) and boiling points increase down the group. The low melting point of Bi is due to non-participation of ns^2 (inert pair) in metallic bond.
- These elements are more volatile than their immediate neighbours because of the stable half-filled electronic configuration. The increase in melting points and boiling points from N_2 to As_4 is due to increase in their molecular sizes.
- **Ionization energies** decreases down the group. There is a large decrease between N and P, but thereafter the decrease is little. This is again due to poor shielding effect of d-electrons in As and Sb and d- and f-electrons in Bi.
- **Electronegativity decreases** gradually from N to Bi indicating a gradual change from non-metallic character to metallic character. N and P are non-metals, As and Sb are metalloids while Bi is a metal.
- These elements exhibit −3, +3 and +5 **oxidation states**.
- Nitrogen exhibit a wide range of oxidation states as given in the following compounds.
- NH_3 N_2H_4 NH_2OH N_3H N_2 N_2O NO N_2O_3 N_2O_4 N_2O_5.
  −3 −2 −1 −1/3 0 +1 +2 +3 +4 +5
- Due to more electronegativity, N can exhibit −3 oxidation state and the tendency to exhibit −3 oxidation state decreases while the tendency to exhibit +3 oxidation state increases down the group due to decrease in electronegativity.
- N being more electronegative can gain three electrons to form N^{3-} ions in the ionic compounds such as Ca_3N_2. The tendency to convert in M^{3-} ion decreases down the group while the tendency to convert into M^{3+} ion increases. Hence, Bi can form compound BiF_3 in which Bi is present as Bi^{3+} ion. In most of the compounds the bond formed is **covalent**.
- Nitrogen has no d-orbitals in its valency shell, so cannot form compounds like NCl_5. But other elements can form compounds like PF_5, PCl_5, etc., by using **d-orbitals in sigma bonds**. Further, by using the vacant d-orbitals they also act as Lewis acids, e.g., PCl_6^-, SbF_6^-, etc.
- Phosphorus can also use its d-orbitals for π-bonding. For example, the trialkyl or triaryl phosphine oxides $R_3P–O$ are much more stable than the corresponding amine oxides R_3NO. In $R_3P \rightarrow O$, oxygen can donate its lone pair into the vacant d-orbital of phosphorus forming a pπ → dπ bond. Because of the double bond character in $R_3P \rightleftarrows O$, the P–O bond is shorter and stronger than the normal P–O single bond.
- Further due to the pπ–dπ back bonding the P–O bond in $R_3P \rightarrow O$ is less polar than the N–O bond in $R_3N \rightarrow O$ which is contrary to the expected as per electronegativity difference. The dipole moment of P–O bond in phosphine oxides is less than the N–O bond in amine oxides due to the pπ – dπ back bonding resulting in the decrease in P–O bond length and charges.
- In the elements of Group VA, phosphorus has the maximum catenation power. The catenation power of nitrogen is less because N–N bond is weaker due to the repulsion between lone pairs on small 'N'atoms. Catenation power decrease from P to As. Sb and Bi do not exhibit catenation power.
- The maximum covalency of nitrogen is 4 because it cannot expand more than the octet due to the absence d-orbitals in its valency shell. Other elements can exhibit a covalency 5 and a maximum upto 6 by using the vacant d-orbitals in their valence shells.
- All the elements of Group VA exhibit allotropy. N_2 has two solid forms: α-N_2 with cubic crystalline structure and β-N_2 with hexogonal crystalline structure. Phosphorus has a large number of allotropes, viz. white, red, black, scarlet red and violet varieties. Arsenic exists in three allotropic forms: grey, yellow and black. Similarly, antimony also has three allotropic forms which are metallic.

COMPOUNDS OF GROUP-VA ELEMENTS

Hydrides

- Elements of Group VA forms MH_3 type hydrides where M is the VA group element. Nitrogen and phosphorus can also form other hydrides like N_2H_4, N_3H and P_2H_4.

- MH_3 type hydrides can be prepared by the action of water or dilute acids on the binary compounds such as Mg_3N_2, Ca_3P_2 Zn_3As_2, Mg_3 Sb_2 and Mg_3 Bi_2.

  e.g. $Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$

- The ease of formation of these hydrides and their heats of formation decreases from NH_3 to BiH_3.
- All these are colourless, volatile, covalent hydrides with poisonous nature. They have disagreeble odour.
- The covalent character of these hydrides increases from NH_3 to BiH_3 due to decrease in difference of electronegativities between hydrogen and Group VA element.
- **Solubility** of these hydrides decreases from NH_3 to BiH_3 due to decrease in polarity. NH_3 is highly soluble in water because it can form hydrogen bonds with water molecules.
- Since NH_3 can form hydrogen bonds between its molecules it can be liquified easily and its boiling point is more than PH_3 and AsH_3 but less than SbH_3. The order of boiling points is $PH_3 < AsH_3 < NH_3 < BiH_3$ but the order of melting point. is $PH_3 < AsH_3 < SbH_3 < NH_3$.
- When ammonia is dissolved in water it behaves as **Arrhenius base**.

  $NH_3 + H_2O \rightleftharpoons NH_4OH \rightleftharpoons NH_4^+ + OH^-$

- But other hydrides cannot behave as Arrhenius bases. However, all these hydrides behave as **Lewis bases** by donating the lone pair on the central atom.
- The **Lewis base character** of these hydrides decreases from NH_3 to BiH_3 because of the decrease in tendency to donate the lone pair as the ns^2 electron cloud is distributed in a larger area with increase in the atomic size.
- **Thermal stability** of the hydrides decreases from NH_3 to BiH_3 due to the decrease in M–H bond strength with increase in M–H bond length because of the increase in atomic size.
- As the thermal stability is decreasing from NH_3 to BiH_3 their **reduction power** increases from NH_3 to BiH_3.
- These hydrides can enter into complex formation by donating their lone pair and the tendency to donate the lone pair decreases from NH_3 to BiH_3.
- All the hydrides are **pyramidal** in shape. 'N'atom in NH_3 is in sp^3 hybridization. The bond angle in NH_3 is $107^0 48^1$. But the bond angles in other hydrides are close to 90° indicating that pure p-orbitals are participating in bonding and the lone pair remains in the ns-orbital.
- The lone pair present in s-orbital has a non-directional nature and hence less effective for forming a coordinate bond. So, complex forming ability decreases in SbH_3 and BiH_3.
- The hydrogen atoms of NH_3 may be substituted by groups like Cl_2 or an alkyl group like CH_3. The ease of substiution decreases from NH_3 to BiH_3.

Halides

- The elements of Group VA forms mainly two types of halides: trihalides of the type MX_3 and pentahalides of the type MX_5.
- Nitrogen trihalides can be prepared by the action of excess halogen on ammonia.

  $NH_3 + 3X_2 \longrightarrow NX_3 + 3HX\,(H = F, Cl, Br\ or\ I)$

- NF_3 is a gas, NCl_3 is an unstable explosive liquid and NBr_3 and NI_3 are known only as their unstable ammoniates $NBr_3 \cdot 6NH_3$ amd $NI_3 \cdot 6NH_3$.
- Trihalides of other elements can be prepared by the direct reaction between elements using limited supply of halogens.
- All trihalides except BiF_3 are covalent. BiF_3 is an ionic solid.
- For a given element of Group-VA, the order of stability is in the order: Fluoride > Chloride > Bromide > Iodide.
- Trihalides of different elements do not show a regular trend due to difference in the structures and bond types.
- These trihalides hydrolyze in water but the products are different.

$$NCl_3 + 4H_2O \longrightarrow NH_4OH + 3HOCl$$
$$PCl_3 + 3H_2O \longrightarrow H_3PO_3 + 3HCl$$
$$AsCl_3 + 3H_2O \rightleftharpoons H_3AsO_3 + 3HCl$$
$$SbCl_3 + H_2O \rightleftharpoons SbOCl + 2HCl$$
$$BiCl_3 + H_2O \rightleftharpoons BiOCl + 2HCl$$

- NF_3 do not hydrolyze in water since there are no d-orbitals in the valence shell of the nitrogen or fluorine.
- The hydrolysis reactions of $AsCl_3$, $SbCl_3$ and $BiCl_3$ are reversible which indicate the increase in electropositive character from N to Bi.
- These trihalides can act as electron pair donors (Lewis bases and ligands) by donating the lone pair on central atom.
- Contrary to electronegativity, PF_3 acts as a strong donor due to the $p\pi$–$d\pi$ back bonding from F $\rightarrow$ P. Since N has no vacant d-orbital in its valence shell, this back bonding is absent and hence NF_3 has least donor character. The order of donor character in nitrogen halides is $NF_3 < NCl_3 < NBr_3 < NI_3$.
- Trihalides except nitrogen trihalides can also act as electron pair acceptors (Lewis acid) using vacant d-orbitals in their valence shells.

- All the trihalides are **pyramidal in shape** in which the central atom is involved in sp^3 hybridization.
- **Bond angles** decreases for a particular halogen with increase in the size of the central atom but with same central atom bond angle increases with increase in the size of halogen.
- **Pentahalides** can be prepared by heating the elements in excess of halogen or by heating the trihalides in halogen.
- Pentahalides also hydrolyze in water forming the corresponding 'ic' acids

$$\text{e.g., } PCl_5 + 4H_2O \longrightarrow H_3PO_4 + 5HCl$$

- PF_5 is thermally stable and chemically reactive, remain covalent and keeps the trigonal bipyramid structure in all the three physical states.
- PCl_5 is a trigonal bipyramid in liquid and gas phase but in solid state exists as $[PCl_4^+]$ $[PCl_6^-]$ ions which have symmetric structures and can be packed efficiently compared to assymetric TBP PCl_5 molecules.
- PBr_5 exists as $[PBr_4^+]$ Br^- in solid state. PI_5 do not exist.
- Among pentachlorides PCl_5 and $SbCl_5$ are stable but $AsCl_5$ is very unstable due to alteration effect or middle row anomaly.
- Pentafluorides of P, As, Sb and Bi are **strong Lewis acids.**

Oxides

- All the elements of Group VA mainly form three types of oxides: trioxides of the type M_2O_3, tetroxides of the type M_2O_4 and pentaoxides of the type M_2O_5.
- Except the oxides of nitrogen and bismuth, the oxides of other elements exist as dimers.
- Stability of the oxides in higher oxidation states gets decreased with increasing atomic number.
- In a given oxidation state, acidic character decreases while basic character increases with increasing size of the central atom, e.g., N_2O_3 and P_4O_6 are acidic, As_4O_6 and Sb_4O_6 are amphoteric and Bi_2O_3 is basic.
- In the oxides of same elements acidic character of oxides increases with increase in oxidation number e.g., N_2O and NO are neutral, N_2O_3 and NO_2 are weakly acidic and N_2O_5 is strongly acidic. Solubility of the oxides in water decreases from nitrogen to bismuth.
- When trioxides are dissolved in water, the corresponding 'ous' acids are formed while the pentoxides form 'ic' acids. E.g., N_2O_3 gives nitrous acid while N_2O_5 gives nitric acid.
- Oxides of antimony and bismuth are insoluble in water but dissolve in alkalis showing the acidic character.
- Oxides in lower oxidation state act as reducing agents and their reduction power decreases from N to Bi due to increase in the stability of +3 oxidation state because of inert pair effect.
- Oxides in the higher oxidation states act as oxidizing agents and their oxidation power should increase from N_2O_5 to Bi_2O_5 due to the decrease in the stability of +5 oxidation state (inert pair effect). But N_2O_5 and Bi_2O_5 are stronger oxidizing agents than those present between them. The stronger oxidizing power of N_2O_5 is due to its more electronegativity.

Oxyacids

- All the elements of Group VA form oxoacids. Nitrogen and phosphorus form a large number of oxoacids. Arsenic forms two oxoacids, H_3AsO_3 and H_3AsO_4. Antimony forms only one oxoacid (H_3SbO_3) which is stable only in solution. Bismuth gives only one stable oxoacid, metabismuthic acid $HBiO_3$.

Nitrogen

- **Preparation:** N_2 is prepared by the following methods.

$$NH_4NO_2 \xrightarrow{\text{Heat}} N_2 + 2H_2O\,(\text{explosive})$$

$$NH_4Cl + NaNO_2 \xrightarrow{\text{Heat}} N_2 + NaCl + 2H_2O$$

$$\underset{\text{orange red}}{(NH_4)_2Cr_2O_7} \xrightarrow{\Delta} N_2 + 4H_2O + \underset{\text{green}}{Cr_2O_3}$$

- **Oxidation** of ammonia with CuO, PbO, Cl_2, $CaOCl_2$ or NaOCl gives nitrogen.
- It can also be prepared by the action of nitrous acid on urea or by heating sodium or barium azide.
- Commercially, N_2 is prepared from air by fractional distillation of liquid air.

Ammonia

- Any ammonium salt on heating with any base liberate ammonia.

$$NH_4Cl + NaOH \longrightarrow NH_3 + NaCl + H_2O$$

- Hydrolysis of metal nitrides give ammonia.

$$Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$$

- Reduction of nitrates and nitrites with zinc or aluminium or Devarda's alloy in alkaline medium gives ammonia.
- Hydrolysis of calcium cyanamide also gives ammonia.
- On a large scale, ammonia is manufactured by Haber's process by the direct reaction between nitrogen and hydrogen. The favourable conditions for the manufacture of ammonia are (i) optimum temperature

725–775 K, (ii) high pressure 200–1000 atmospheres and (iii) catalyst finely divided iron powder mixed with molybdenum powder or mixture of K_2O and Al_2O_3 as promoter.

- In the cyanamide process, ammonia is manufactured by the hydrolysis of nitrolim obtained by passing air over calcium carbide.

$$CaC_2 + N_2 \xrightarrow{1000°C} \underbrace{CaNCN + C}_{\text{Nitrolim}}$$

$$CaNCN + 3H_2O \longrightarrow CaCO_3 + 2NH_3$$

- Ammonia is obtained as a by-product in the purification of bauxite by Serpeck's process.

$$Al_2O_3 + 3C + N_2 \longrightarrow AlN + 3CO$$
$$AlN + 3H_2O \longrightarrow Al(OH)_3 + NH_3$$

- The solubility of ammonia in water is due to hydrogen bonding with N atom of NH_3 and H atom of H_2O but not H-atom of NH_3 and O atom of H_2O.
- It burns in air with a pale green flame forming N_2 and H_2O.
- NH_3 can be oxidized by Cl_2, bleaching powder, sodium hypochlorite at room temperature and with CuO or PbO.
- When ammonia is heated in the presence of Pt catalyst with air, it is oxidized to NO.
- Ammonia react with halogens. When halogens are in excess nitrogen trihalides are formed but if ammonia is excess it is oxidized to N_2.

$$NH_3 + 3X_2 \longrightarrow NX_3 + 3HX$$
$$2NH_3 + 3X_2 \longrightarrow N_2 + 6HX$$

- Aqueous solution of ammonia is called ammonium hydroxide and it is a weak base and forms salts with almost all acids which resemble alkali metal salts in solubility and structure, but thermally unstable.
- Metals of Group IA and IIA dissolve in liquid ammonia but when heated forms ionic amides, e.g., $NaNH_2$ with liberation of H_2.
- When ammonium hydroxide is added to metal salts, several metals are precipitated as their hydroxides.
- The metal ions which are precipitated as their hydroxides but are insoluble in excess of ammonium hydroxide are Fe^{2+}, Fe^{3+}, Mn^{2+} and Al^{3+}.
- The metal ions which are precipitated as their hydroxides but dissolve in excess of ammonium hydroxide are Cu^{2+}, Cd^{2+}, Ag^{+}, Zn^{2+}, Ni^{2+}, Co^{3+} etc. (For reactions, refer section 9.4.1 of the text.)
- Ammonia forms a white precipitate with mercuric chloride due to the formation of amido mercuric chloride (NH_2HgCl) and black precipitate with mercurous chloride due to the formation of Hg and NH_2HgCl.
- Alkaline solution of K_2HgI_4 is called Nessler's reagent which gives a brown precipitate with ammonia due to the formation of amido mercuric oxymercuric iodide, also known as iodide of Milon's base.
- Ammonia is used in the manufacture of fertilizers, as refrigerant in ice plants, in the manufacture of Na_2CO_3 by Solvay's process, in the preparation of rayon and artificial silk in the form of tetrammine copper (II) sulphate called **schweitzer's** reagent as solvent for cellulose, in the manufacture of nitric acid and explosives like **amatol** (80% NH_4NO_3 + 20% TNT) and ammonal (NH_4NO_3 + aluminium powder).

Ammonia

Reaction	Remark
Preparation	
1. $NH_4Cl + NaOH \rightarrow NaCl + H_2O + NH_3$	Any ammonium salt + any base
2. $Mg_3N_2 + 6H_2O \rightarrow Mg(OH)_2 + NH_3$	Hydrolysis of metal nitrides
3. $4Zn + 7NaOH + NaNO_3 \rightarrow 4Na_2ZnO_2 + 2H_2O + NH_3$	Reduction of nitrates and nitrites with Zn or Al in alkaline medium
$3Zn + 5NaOH + NaNO_2 \rightarrow 3Na_2ZnO_2 + H_2O + NH_3$	
$8Al + 5NaOH + 3NaNO_3 + 2H_2O \rightarrow 8NaAlO_2 + 3NH_3$	
$2Al + NaOH + NaNO_2 + H_2O \rightarrow 2NaAlO_2 + NH_3$	
4. $CaCN_2 + 3H_2O \rightarrow CaCO_3 + 2NH_3$	Hydrolysis of cyanamide
5. $(NH_4)_2SO_4 \xrightarrow{\Delta} NH_3 + NH_4HSO_4$	Action of heat on ammonium salts
$NH_4H_2PO_4 \xrightarrow{\Delta} NH_3 + HPO_3 + H_2O$	
6. $N_2 + 3H_2 \underset{200-300\text{ atm}}{\overset{450-475°C}{\rightleftharpoons}} 2NH_3$	Haber's process of manufacture. Using Fe as a catalyst and Mo as a promoter.

Properties

1.	$2NH_3 \xrightarrow{\Delta} N_2 + 3H_2$	
2.	$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$	
3.	$2NH_3 + 3Cl_2 \rightarrow N_2 + 6HCl$	
4.	$2NH_3 + 3CaOCl_2 \rightarrow N_2 + 3H_2O + 3CaCl_2$	
5.	$2NH_3 + 3NaOCl \rightarrow N_2 + 3H_2O + 3NaCl$	
6.	$2NH_3 + 3CuO \rightarrow 3Cu + 3H_2O + N_2$	
7.	$2NH_3 + 3PbO \rightarrow 3Pb + 3H_2O + N_2$	
8.	$4NH_3 + 5O_2 \xrightarrow[700°C]{Pt} 4NO + 6H_2O$	
9.	$NH_3 + 3X_2 \rightarrow NX_3 + 3HX$	X = F, Cl, Br, I and X is excess
10.	$2NH_3 + 3X_2 \rightarrow N_2 + 6HX$	NH_3 is excess, X = F, Cl, Br, I
11.	$2NH_3 + 2Na \rightarrow 2NaNH_2 + H_2$	
12.	Metal salt + $NH_4OH \rightarrow$ Metal hydroxide + ammonium salt	Metal hydroxides precipitate and insoluble in excess of ammonia also in excess of NaOH, Fe^{2+}, Fe^{3+}, Mn^{2+}
13.	Metal salt + $NH_4OH \rightarrow$ Metal hydroxide + ammonium salt	Metal hydroxide precipitate and dissolve in excess of ammonia but insoluble in NaOH, Cu^{2+}, Cd^{2+}, Ag^+, Ni^{2+}, Co^{3+}
14.	Metal salt + $NH_4OH \rightarrow$ Metal hydroxide + ammonium salt	Metal hydroxide precipitates and is insoluble in excess of NH_4OH but soluble in excess of NaOH eg. Al^{3+}
15.	Metal salt + NaOH $\rightarrow$ Metal hydroxide + ammonium salt	Metal hydroxide precipitates and dissolves in excess of both NH_4OH and NaOH, Zn^{2+}
16.	$HgCl_2 + NH_4OH \rightarrow NH_2HgCl + NH_4Cl + H_2O$	White ppt.
17.	$Hg_2Cl_2 + 2NH_4OH \rightarrow NH_2HgCl + Hg + NH_4Cl + H_2O$	Black ppt.
18.	$2K_2HgI_4 + 3KOH + NH_3 \rightarrow NH_2HgO\ HgI + 7KI + 2H_2O$	Brown ppt, test for ammonia

Hydrazine

- It can be prepared by **Raschig's** process in which ammonia is oxidized by NaOCl in dilute aqueous solution in the presence of glue.

$$NH_3 + NaOCl \longrightarrow NH_2Cl + NaOH$$
$$2NH_3 + NH_2Cl \longrightarrow N_2H_4 + NH_4Cl$$

- In the absence of glue the chloramine formed as an intermediate react with hydrazine to form NH_4Cl and N_2. This reaction is catalyzed by heavy metal ions present in solution. Glue masks these metal ions by complexation.
- Hydrazine is a covalent colourless liquid and fumes in air, smells like ammonia, hygroscopic and strongly associated through hydrogen bonding.
- Hydrazine burns in air with evolution of large amount of heat. Hence, hydrazine or its alkyl derivatives are used as rocket fuels.
- It is a weak diacidic base, forming two series of salts.
- It reacts with sodium in an inert atmosphere to form sodium hydrazide.

$$2N_2H_4 + 2Na \longrightarrow 2NaN_2H_3 + H_2$$

- With nitrous acid it forms hydrazoic acid.

$$N_2H_4 + HNO_2 \longrightarrow N_3H + 2H_2O$$

- It is a good reducing agent and reduces several metal salts to the corresponding metals, e.g., $PtCl_4$ to Pt, $AuCl_3$ to Au, $AgNO_3$ to Ag, I_2 to HI, $CuSO_4$ to Cu.
- Hydrazine may act as an electron donor (Lewis base and ligand) and form complexes with metal ions such as Ni^{2+} and Co^{2+}.
- Its structure is similar to ethane adopting a **gauche** configuration with lone pairs in gauche position.

Hydroxylamine

- It can be prepared by the reduction of NO with tin and conc. HCl or by the reduction of $NaNO_2$ with SO_2in the presence of Na_2CO_3 or by the electrolytic reduction of nitric acid.
- It is a weak base, extremely unstable, decomposes to N_2, NO and NH_3.
- It is a good reducing agent. In acid medium it reduces ferric salts to ferrous salts. In basic medium it reduces $AuCl_3$ to Au, Fehling's solution to Cu_2O and ammonical solution of silver salts to silver. It reduces $HgCl_2$ first to Hg_2Cl_2 and then to the metal. In all these reactions, the hyponitrous acid initially formed decomposes and the final products are N_2O and H_2O.
- Hydroxylamine reduces halogens to hydraacids, bromates to bromides and iodates to iodides.

- Hydroxylamine can also act as an oxidizing agent, can oxidize $SnCl_2$ to $SnCl_4$, SO_2 to ammonium sulphate in strong acid medium. In basic medium it oxidizes Fe $(OH)_2$ to Fe $(OH)_3$, and Na_3AsO_3 to Na_3AsO_4.
- It forms oximes with carbonyl compounds.

Hydrazoic Acid

- It can be prepared by the action of HNO_2 on hydrazine $N_2H_4 + HNO_2 \rightarrow N_3H + 2H_2O$
- It is also formed when N_2O is passed over fused sodamide at 190°C when NaN_3 is formed from which N_3H is formed by the addition of dil. sulphuric acid.
- It is a poisonous, highly explosive liquid, explodes violently by dissociating into its constituents with evolution of a large amount of heat.
- It is reduced to ammonia or hydrazine by sodium amalgam.
- It can be **oxidized** to N_2 and H_2O by oxidizing agents like acidified permanganate and ceric ion. Nitrous acid is oxidized, quantiatively to N_2, N_2O and H_2O.
- Acids decompose hydrazoic acid to NH_3 and N_2.
- It is acidic and dissolves iron, zinc and copper with the formation of metal azides and ammonia ($H_2$is not liberated).
- Azides of Ag, Pt and Hg_2^{2+} are insoluble in water. These can be prepared by adding a soluble salt of these metals to a solution sodium azide.
- Azides of Ag, Pb and Hg_2^{2+} **explode** violently when struck, so they are used in detonator caps.
- Azides of **electropositive metals** are not explosive but decompose smoothly when heated or on shock.
- Azide ion acts as a **ligand** in complexes of transition metals. It is commonly, considered as a **psuedohalide** though the corresponding pseudohalogen $(N_3)_2$ is not known.
- Hydrazoic acid is **angular** in shape but azide is **linear.**

OXIDES OF NITROGEN

Nitrogen (I) Oxide or Nitrous Oxide

- It is prepared by heating ammonium nitrate by reduction or nitric acid with $SnCl_2$ and HCl.
- It is colourless gas with pleasant odour producing mild laughing hysteria, so it is also called **laughing gas** and used as an anaesthetic agent for minor operations.
- It decomposes above 823K yielding oxygen and nitrogen, so readily supports combution. It oxidizes non-metals and metals to their corresponding oxides.
- It can also act as a ligand.
- It is a linear asymmetrical molecule with a very small dipole moment and it is also a resonance hybrid of the following structures each of which is polar in opposite sense but it has small dipole moment due to delocalization of charge. N – N bond is shorter than N – O bond.

$$:\overset{-}{\ddot{N}} = \overset{+}{N} = \ddot{O}: \rightleftharpoons :N \equiv \overset{+}{N} - \overset{-}{\ddot{O}}:$$

Nitric Oxide, NO

- It is prepared by the direct reaction between N_2 and O_2 at 3000°C or by the oxidation of ammonia at 773K in the presence of Pt catalyst or by the action of dilute nitric acid on copper.
- NO can be completely absorbed by $FeSO_4$ forming a brown complex and on heating again it gives pure NO.
- It is a colourless, neutral, paramagnetic oxide. It is an odd electron molecule and readily combines with oxygen forming a brown NO_2 gas.
- It is combustible and supports combustion of only boiling sulphur and vigorously burning phosphorus. But burning sulphur or feebly burning phosphorus gets extinguished in NO because at that temperature NO do not break and do not supply oxygen for burning S or P.
- It can act as an **oxidizing agent**, and oxidizes hydrogen to water, sulphurous acid to sulphuric acid and H_2S to S.
- $SnCl_2$ reduces NO to Hydroxylamine.
- NO forms addition compounds with halogens (NOX).
- It can also act as a **reducing agent**, reduces acidified $KMnO_4$ and itself gets oxidized to HNO_3.
- NO reduces I_2 to HI, HNO_3 to NO_2.
- It can also act as a **ligand** and form complexes with transition metals.
- Its structure is a resonance hybrid of the following structures

$$:\dot{N}-\ddot{O}: \leftrightarrow :\dot{N}=\ddot{O}: \leftrightarrow :\overset{-}{\ddot{N}}=\overset{+}{\dot{O}}:$$

- It has a **three-electron bond**. In liquid state it dimerizes to some extent.

Dinitrogen Trioxide or Nitrogen Sesqui Oxide

- It is formed as a blue liquid when equimolar mixture of NO and NO_2 are cooled to about –20°C.
- It can also be prepared by distilling a mixture of 60 per cent HNO_3 with As_2O_3 or starch and condensing the vapours or by the action of 5N nitric acid on copper.
- Solid N_2O_3forms blue crystals but dissociate into NO and NO_2 on melting. When dissolved in water it forms nitrous acid and hence it is considered as an anhydride of nitrous acid.
- It exists in two forms having symmetric and asymmetric structures.

Dinitrogen Tetroxide or Nitrogen (IV) Oxide

- It is formed by the action of oxygen on NO or by heating lead nitrate or by heating copper with conc. HNO_3. It can also be obtained by the action of nitric acid on sodium nitrite.
- It is a reddish brown gas having a pungent smell. In liquid state it is colourless due to dimerization, but in gaseous state it is brown in colour due to dissociation of dimer into monomer.
- When dissolved in water it forms a mixture of nitrous and nitric acid, so it is called a mixed anhydride of nitrous and nitric acid.
- It disproportionates in alkalis forming nitrites and nitrates.
- It can act as an **oxidising agent.** Oxidizes SO_2 to H_2SO_4, H_2S to S, CO to CO_2, ferrous sulphate to ferric sulphate and liberates I_2 from iodide.
- It also oxidizes several metals like Cu, Pb and Sn to their corresponding oxides.
- It can also act as a **reducing agent**, e.g., reduces acidified permanganate.
- It is an odd electron molecule and is paramagnetic. The N–O bond lengths are equal because of the resonance hybrid of two (p. 9.22) structures.
- When NO_2 dimerizes it becomes colourless and diamagnetic indicating the unpaired electron participates in N–N bond. The N–N bond is weak and hence N_2O_4 is unstable.

Dinitrogen Pentoxide

- It is prepared by dehydrating conc. HNO_3 with P_4O_{10}. It is also formed by the action of O_3 on N_2O_4 or chlorine on silver nitrate.
- It is a colourless crystalline solid and becomes yellow by raising the temperature on account of partial decomposition into brown NO_2.
- When dissolved in water it gives nitric acid. It is the true anhydride of nitric acid. With alkalis it forms nitrates.
- It is a strong oxidizing agent towards many metals, non-metals and organic substances.
- When dissolved in H_2SO_4 or HNO_3 it ionizes as NO_2^+ and NO_3^-. With sodium chloride it forms sodium nitrate and nitronium chloride indicating the presence of NO_2^+ and NO_3^- ion in N_2O_5. So, it is also called as **nitronium nitrate**.
- The structure is not well established but may be $O_2N - O - NO_2$ with N – O – N angle close to 112°.

Nitrous Acid, HNO_2

- Nitrous acid do not exist freely, but exists only in solution.
- It is prepared by the addition of dilute mineral acids to nitrites like $NaNO_2$ preferably to barium nitrite. It can also be prepared by dissolving N_2O_3 in water.
- It is a **weak acid**, unstable and decompose rapidly into HNO_3, NO and H_2O. It undergoes autooxidation.

$$3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$$

- It is a good **oxidising agent** as it oxidizes H_2S to S with liberation of NO_2. It oxidizes SO_2 to H_2SO_4, $SnCl_2$ to $SnCl_4$, liberates iodine from iodide, oxidizes ferrous sulphate to ferric sulphate and arsenite to arsenate. In these reactions it is reduced to NO.
- It also acts as a **reducing agent** decolourises acidified $KMnO_4$, reduces orange red dichromate to green chromic sulphate, Br_2 to HBr, H_2O_2 to H_2O and itself gets oxidized to HNO_3.
- It combines with ammonia forming NH_4NO_2 which on heating decomposes to N_2 and H_2O.
- With aliphatic primary amines and urea it liberates N_2. Aliphatic primary amines are converted into alcohols.
- At low temperatures it reacts with aromatic amines forming diazonium salts.
- Nitrite ion is angular in shape with a bond angle of 110.7.

Nitric Acid, HNO_3

- It is called as **'aqua fortis'** because it can dissolve several substances.
- In the laboratory it can be prepared by the action of conc. H_2SO_4 on metal nitrate.
- In **Birkland and Eyde** process, conc. HNO_3 is manufactured form air. The atmospheric nitrogen and oxygen are made to combine to give NO by producing electric arcs in them. The NO by combining with oxygen gives NO_2 which dissolves in water in the presence of oxygen.

$$N_2 + O_2 \longrightarrow 2NO$$
$$2NO + O_2 \longrightarrow 2NO_2$$
$$4NO_2 + 2H_2O + O_2 \longrightarrow 4HNO_3$$

- **Ostwald's process** is the modern method in which HNO_3 is manufactured from ammonia. First NH_3 oxidized to NO in the presence of Pt–Rh catalyst and then the process is the same as above. About 61 per cent HNO_3 will be obtained.

$$4NH_3 + 5O_2 \xrightarrow{Pt} 4NO + 6H_2O$$

- While dissolving NO_2 in water if O_2 is not present the yield of nitric acid is less because some NO escapes out.

$$3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$$

- Distillation of 61 per cent HNO_3 gives 68 per cent azeotropic mixture of HNO_3 with water. This is distilled again in the presence of conc. H_2SO_4 to get 98 per cent

HNO_3. Fractional crystallization gives 100 per cent pure HNO_3 crystals.

- It is a colourless fuming liquid. Like water it can undergo autoionization giving NO_3^- and NO_2^+ ions.
- Pure nitric acid is colourless, but develops yellow colour due to dissolving NO_2 formed by the photochemical decomposition. The brown colour can be removed by passing air through warm acid.
- It is a **strong acid** and combines with almost all bases forming salts. But with ammonia, the salt formed (NH_4NO_3)decomposes on heating to N_2O and H_2O.
- Nitric acid in the presence of sulphuric acid reacts with aromatic compounds forming nitrocompounds and the process is known as **nitration**. Several nitrocompounds like TNT, nitroglycerine, nitrocellulose (gun cotton) are explosives.
- Nitric acid reacts with protiens forming a yellow coloured xanthoprotein. Hence, HNO_3 stains the skin and renders yellow. This is used as a delicate test of protein known as **xanthoproteic test**.
- **Oxidation properties:** NO_3^- ion is a moderately strong oxidizing agent. The reactions are slow in dilute solutions because it is completely ionized but in concentrated solution NO_3^- is protonated (due to lesser ionization. Protonation of the oxygen atoms makes the N – O bond breaking easy and hence the reactions are faster.

$$\underset{\text{dil}}{HNO_3} \longrightarrow H^+ + NO_3^-$$

$$\underset{\text{conc}}{H-O-NO_2} + H^+ \longrightarrow H-\overset{+}{\underset{\underset{H}{|}}{O}}-NO_2 \longrightarrow H_2O + NO_2^+$$

$$NO_2^+ + e^- \longrightarrow NO_2$$

- **Oxidation of non-metals and metalloids:** Conc. HNO_3 oxidizes solid non-metals and metalloids to their respective 'ic' acids. For example, P_4 to H_3PO_4, S to H_2SO_4, Se to H_2SeO_3, Te to H_2TeO_3, I_2 to HIO_3, Sb to H_3SbO_4, Sn to metastannic acid (H_2SnO_3) and C to CO_2. Conc. HNO_3 acts as an oxidizing agent as per the reaction:

$$2HNO_3 \longrightarrow H_2O + 2NO_2 + (O)$$

So, in all the above reactions conc. HNO_3 is reduced to NO_2.

- **Oxidation of compounds:** When **dil. HNO_3** act as an oxidizing agent it is reduced to **NO** but when **conc. HNO_3** act as an oxidizing agent it is reduced to **NO_2**.
- Dil. HNO_3 oxidizes H_2S to S, SO_2 to H_2SO_4, iodide to iodine and ferrous sulphate to ferric sulphate.
- $SnCl_2$ reduces nitric acid to Hydroxylamine and ammonium nitrate. Conc. HNO_3 oxidizes cane sugar to oxalic acid.
- **Oxidation of Metals**
  Metals above hydrogen in the electrochemical series liberate H_2 which reduces the HNO_3 to different products depending on the concentration of HNO_3, nature of metal, temperature, etc.
- Metals below hydrogen in the electrochemical series are oxdized by nitric acid. When dil. HNO_3 is used nitric acid is reduced to NO while with conc. HNO_3 it is reduced to NO_2.
- The more the change in the oxidation state of nitrogen, the weaker the oxidation power of nitric acid.
- Very dil. HNO_3 is about 6 per cent, dil. HNO_3 is about 20 per cent and conc. HNO_3 is above 60 per cent.
- Al, Zn, Fe and Sn reduces the **very dilute nitric acid** to **ammonium nitrate**.
- **Mg** and **Mn** can only liberate H_2 from **very dil. HNO_3** (about 1%).
- **Reactions of Metals with dil. HNO_3**
  (i) Most of the metals except noble metals dissolve in dil. HNO_3 forming metal nitrate with evolution of NO.
  (ii) Tin reduces the dil. HNO_3 to ammonium nitrate.
  (iii) Fe and Zn reduces the dil. HNO_3 to N_2O.
  (iv) Fe and Hg form metal nitrates in their lower oxidation states.

Nitric acid

Reactions	Remark
1. ***Oxidation of non-metals and Metalloids*** $C \rightarrow CO_2$ $S \rightarrow H_2SO_4$ $P_4 \rightarrow H_3PO_4$ $I_2 \rightarrow HIO_3$ $Se \rightarrow H_2SeO_3$ $Te \rightarrow H_2TeO_3$ $As \rightarrow H_3AsO_4$	Concentrated nitric acid oxidizes the non-metals and metalloids Nitric acid will be reduced to NO_2

$Sb \rightarrow H_3SbO_4$
$Sn \rightarrow H_2SnO_3$

2. ***Oxidation of compounds***

$H_2S \rightarrow S$
$SO_2 \rightarrow H_2SO_4$
$KI \rightarrow I_2$
$FeSO_4 \rightarrow Fe_2(SO_4)_3$
$HBr \rightarrow Br_2$
$HI \rightarrow I_2$
$7SnCl_2 + 14HCl + 3HNO_3 \rightarrow 7SnCl_4 + NH_2OH + NH_4NO_3 + 5H_2O$
$C_{12}H_{22}O_{11} + 36HNO_3 \rightarrow 6(COOH)_2 + 36NO_2 + 23H_2O$

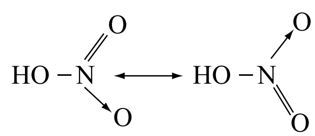
Dilute HNO_3 oxidizes the compounds and itself is reduced to NO. The compounds which are oxidized by dil. HNO_3 can also be oxidized by conc. HNO_3 but itself is reduced to NO_2.

3. ***Oxidation of Metals***

Concentration of HNO_3	**Metal**	**Main products**
Very dil. HNO_3	Mg, Mn	H_2 + Metal nitrate
	Fe, Zn, Sn, Al	NH_4NO_3 + Metal nitrate
Dil. HNO_3	Pb, Cu, Ag	NO + Metal nitrate
	Fe, Zn	N_2O + Metal nitrate
	Sn	$NH_4NO_3 + Sn(NO_3)_2$
Conc. HNO_3	Zn, Pb, Cu, Ag	NO_2 + Metal nitrate
	Sn	$NO_2 + H_2SnO_3$ (Metastannic acid)

***** Hg gives $Hg_2(NO_3)_2$ and NO with dil HNO_3 and $Hg(NO_3)_2$ and NO_2 with conc. HNO_3.**

- **Reactions of metals with Conc. HNO_3**
  (i) Most of the metals except noble metals dissolve in conc. HNO_3 with evolution of NO_2.
  (ii) Hg forms mercuric nitrate.
  (iii) Sn is converted to metastannic acid (H_2SnO_3) (for reactions refer the properties of nitric acid, section 9.6.2)
- Be, Al, Cr, Fe, Co and Ni become **passive** in conc. HNO_3. The passivity of these metals by conc. HNO_3 is due to the formation of an oxide layer on the surface. So aluminium vessels can be used to carry nitric acid.
- **Metals which do not react with HNO_3:** Noble metals like Au, Pt, Ir, Rh, etc. do not react with HNO_3, but they dissolve in aquaregia. **Aquaregia** is a mixture of 3 parts of conc. HCl and 1 part of conc. HNO_3. The reason for dissolution of these metals in aquaregia is the enhanced oxidation power of nitric acid in the presence of Cl^- ions which can form complexes with metals.
- Nitric acid is used in the manufacture of explosives like TNT, nitroglycerine, amatol, etc. fertilizers, perfumes, dyes and medicines. It is used in the oxidation of cyclohexanol or cyclohexanone to adipic acid. p-xylene to terephthallic acid, in the preparation of cellulose nitrate that is used as artificial silk.
- Nitric acid molecule is planar in the gas phase, supposed to be a resonance hybrid of following structures.



- Nitrate ion can be detected by the brown ring test. It evolves ammonia gas when heated with zinc or aluminium or Devarda's alloy in alkaline medium.
- **Action of heat on some nitrogen compounds**

$$(NH_4)_2Cr_2O_7 \longrightarrow Cr_2O_3 + 4H_2O + N_2$$
$$NH_4NO_2 \longrightarrow N_2 + 2H_2O$$
$$NH_4NO_3 \longrightarrow N_2O + 2H_2O$$

- Except $LiNO_3$, other alkali metal nitrates on heating converts into nitrites liberating oxygen.
- Heavy metal nitrates on heating gives metal oxide, NO_2 and O_2.
- Mercury and silver nitrates on heating gives metals as residue, NO_2 and O_2 gases.

PHOSPHORUS

- It is manufactured by reduction of phosphate rock or bone ash with sand and coke in an electric furnace.

$$Ca_3(PO_4)_2 + 3SiO_2 \longrightarrow 3CaSiO_3 + P_2O_5$$
$$2P_2O_5 + 10C \longrightarrow 10CO + P_4$$

- **Allotropy:** Phosphorus exists in several allotropic forms such as white or yellow phosphorus, red phosphorus, scarlet phosphorus, metallic or α- black phosphorus, β-black phosphorus and violet phosphorus.
- **White or yellow phosphorus** is prepared by cooling the vapours in pure state. It is a white waxy solid when exposed to air, gives a light yellow colour and so called as yellow phosphorus. Its molecular formula is P_4. Due to slow oxidation in dark it glows, which is known as **phosphorescence.**
- P_4 molecule is **tetrahedral** in shape. Every P atom is in bond with three other P atoms, bond angle 60°. No. of P-P bonds is six. Due to the ring strain P_4 molecule is highly reactive and burns in air. Hence it is stored under water. It contains 4 three-membered rings. Thermodynamically it is the least stable allotrope.
- **Metallic or** α-black phosphorus is obtained by dissolving red phosphorus in fused lead or bismuth for long time and cooling. It is stable, chemically inert and is a non-conductor.
- **β-black phosphorus** is obtained by heating white P at 473K at very high pressure. It is the thermodynamically most stable allotrope of phosphorus. It has a layered structure. Each P atom is in bond with 3P atoms, PPP angle is 99°. It is a semiconductor.
- **Red phosphorus** is made by heating white P at 675K in an inert atmosphere using iodine as a catalyst. It consists of long chains of P atoms which are covalently bonded forming a giant molecule. This structure requires large amount of energy to break the more number of bonds and hence it is less reactive and less volatile.
- Phosphorus burns in air forming P_4O_6 and P_4O_{10}.
- When boiled with alkalis liberate phosphine gas forming sodium hypophosphite due to disproportionation.
- It combines with halogens forming trihalides and pentahalides, with metals form metal phosphides.
- It can act as a **reducing agent** as it reduces conc. HNO_3 to NO_2, conc. H_2SO_4 to SO_2, $CuSO_4$ to Cu and $AgNO_3$ to Ag.
- P_4 molecules can also act as **ligand** by donating the lone pair electrons on P atoms.
- Red phosphorus is used in the match box industry. The match stick tips contain red P, sulphur or antimony sulphide, $K_2Cr_2O_7$ or $KClO_3$, glass pieces and gum. People working in match industries will be attacked by a disease called **Phossy Jaw** due to inhalation of phosphorus vapours.
- White phosphorus is soluble in carbon disulphide but red P is insoluble. So, they can be separated by using CS_2.

Phosphine

- It can be prepared by the hydrolysis of metal phosphides, or by heating phosphonium iodide with NaOH, or by heating orthophosphorus acid. In the laboratory it is mainly prepared by boiling white phosphorus with NaOH.
- Phosphine ignites spontaneously due to the presence of diphosphine (P_2H_4) as impurity which ignite PH_3 that come out as smoke in the form of rings called **vortex rings**.
- It is a colourless, poisonous gas with rotten fish odour.
- It is thermally less stable than ammonia, decomposes at 715K into elements. It explodes, when it comes in contact with oxidizing agents such as Cl_2 gas, HNO_3, etc.
- With Cl_2 it forms PCl_3 or PCl_5.
- Phosphine is a strong **reducing agent**. Metal salts are reduced to metal phosphides or to the metals. $CuSO_4$, is reduced Cu_3P_2 and $AgNO_3$ is reduced to Ag.
- It can act as an electron pair donor, i.e. **Lewis base or ligand but weaker than ammonia,** forms addition compounds with anhydrous **$AlCl_3$, $SnCl_4$ and Cu_2Cl_2**.
- Calcium phosphide is used in **smoke screens**. Mixture of calcium phosphide and calcium carbide is used in **Holme's signals**.

Halides of Phosphorus

- PCl_3 can be prepared by the action of thionyl chloride ($SOCl_2$) on phosphorus or by the action of chlorine on phosphorus.
- With moist air or with water it hydrolyzes giving phosphorus acid.
- PCl_3 reacts with compounds containing -OH groups.
- PCl_3 gives $POCl_3$ with oxygen and $PSCl_3$ with sulphur.
- PCl_3 acts as a reducing agent when treated with SO_3, conc. H_2SO_4 and sulphur chloride.

$$PCl_3 + SO_3 \longrightarrow POCl_3 + SO_2$$

$$PCl_3 + 2H_2SO_4 \longrightarrow SO_3HCl + SO_2 + 2HCl + HPO_3$$

$$3PCl_3 + S_2Cl_2 \longrightarrow PCl_5 + 2PSCl_3$$

- With finely divided metals PCl_3 react in hot condition.

$$12Ag + 4PCl_3 \longrightarrow 12AgCl + P_4$$

$$6Na + PCl_3 \longrightarrow 3NaCl + Na_3P$$

- PCl_5 can be prepared by the action of sulphuryl chloride on phosphorus or phosphorus trichloride or by the action of Cl_2 on PCl_3.
- Reactions of PCl_5 are summarized as follows.

Reactions of Phosphorus Pentachloride

1. $PCl_5 + H_2O \rightarrow POCl_3 + 2HCl$
   $PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$

2. $6PCl_5 + P_4O_{10} \rightarrow 10POCl_3$
3. $PCl_5 + SO_2 \rightarrow POCl_3 + SOCl_2$
4. $PCl_5 + C_2H_5OH \rightarrow C_2H_5Cl + POCl_3 + HCl$
5. $PCl_5 + CH_3COOH \rightarrow C_2H_5COCl + POCl_3 + HCl$
6. $2PCl_5 + H_2SO_4 \rightarrow SO_2Cl_2 + 2POCl_3 + 2HCl$
7. $PCl_5 + 2Cu \rightarrow Cu_2Cl_2 + PCl_3$
8. $PCl_5 + 2Ag \rightarrow 2AgCl + PCl_3$
9. $2PCl_5 + Sn \rightarrow SnCl_4 + 2PCl_3$

Oxides of Phosphorus

- P_4O_6 is obtained by controlled oxidation of P_4 in an atmosphere of 75 per cent O_2 and 25 per cent N_2. It is a white waxy solid soluble in benzene, chloroform and CS_2.
- It reacts with cold water slowly forming H_3PO_3 but with hot water gives H_3PO_4 and PH_3.
- It reacts with Cl_2 and Br_2 forming oxyhalides.

$$P_4O_6 + 4Cl_2 \longrightarrow \underset{\text{Phosphorus oxychloride}}{2POCl_3} + \underset{\text{Metaphosphorus oxychloride}}{2PO_2Cl}$$

- It can also act as a ligand by forming complexes like $[\{Ni(CO)_3\}_4P_4O_6]$.
- P_4O_{10} is formed by burning phosphorus in excess of air. It is a white crystalline solid and sublimes on heating. When pure it is colourless but the garlic smell is due to the presence of P_4O_6.
- It reacts with water to form a mixture of phosphoric acids depending on the amount of water and other conditions.

$$P_4O_{10} \xrightarrow{H_2O} (HPO_3)n \xrightarrow{H_2O} H_5P_3O_{10} \xrightarrow{H_2O} H_4P_2O_7 \xrightarrow{H_2O} H_3PO_4$$

- It is a good drying and dehydrating agent and removes water molecules from several compounds, e.g., HNO_3 to N_2O_5; H_2SO_4 to SO_3; carbohydrates to carbon; formic acid to CO; oxalic acid to CO and CO_2; acetic acid to acetic anhydride; amides to cyanides, etc. It cannot be used to dry the basic substances such as CaO, NH_3 because they form salts with P_4O_{10}.

  In P_4O_6 and P_4O_{10}:
  (i) The four phosphorus atoms are arranged in a tetrahedral manner.
  (ii) Six oxygen atoms act as bridges between P atoms along the edges.
  (iii) There are six P-O-P bridge bonds.
  (iv) There are 4 six-membered P_3O_3 heterocyclic rings.
- In P_4O_6 every 'P'atom is surrounded by three oxygen atoms while in P_4O_{10} every 'P' atom is surrounded by four oxygen atoms.
- In P_4O_{10} the fourth oxygen atom in every 'P' atom is in double bond with one sigma dative bond from P to O (P→O) and the other π dative bond from P←O (dπ–pπ). Hence the terminal P–O bonds are shorter than bridge bonds.

  P_4O_6 contains 12 P–O bonds while P_4O_{10} contains 16 P–O bonds.
- The P–O–P bond angles within the molecule or between the molecules are not equal.

Oxoacids of Phosphorus

- Phosphorus forms a large number of oxoacids. Structural principles covering these oxoacids are
  (i) All P atoms in the oxoacids and oxoanions are of **4 coordinates** and contain at least one P – O unit.
  (ii) In all the oxoacids and oxoanions, P – atom is involved in sp^3 hybridization.
  (iii) All P atoms in the oxoacids have at least one P–OH group. All such groups are ionizable as proton donors and responsible for acidic character.
  (iv) Some oxoacids or oxoanions may have one or more P–H group; such bonded H atoms are not ionizable and are responsible for reduction properties of oxoacids and oxoanions.
  (v) Catenation is by P–O–P links or via direct P–P bonds. The oxoacids containing P–O–P links may be either open chain (linear) or cyclic species. Only corner sharing occurs, never edge sharing.
  (vi) Peroxocompounds may be either –P–O–O–H or –P–O–O–P–.
  (vii) The oxidation state of P in an oxoacid is 5 when it is directly bound to four oxygen atoms and the oxidation number decreases by 1 each time when a P–OH is replaced by a P–P bond and by 2 each time a P–OH is replaced by a P–H (for structural aspects, refer to Table 9.9).

Hypophosphorus Acid, H_3PO_2

- It can be prepared by the oxidation of phosphine with I_2 and water or by the addition of acid to a salt of hypophosphite, preferably H_2SO_4 to Ba $(H_2PO_2)_2$. The hypophosphites required in this method can be obtained by boiling white phosphorus in the corresponding alkali.
- It is a colourless crystalline solid soluble in water. It is a **strong monobasic acid** as active metals like Mg and Zn dissolve in the acid with evolution of H_2.
- On heating it **disproportionates** into H_3PO_4 and PH_3 and with nascent hydrogen it is reduced to PH_3.
- It is a good **reducing agent**, itself gets oxidized to H_3PO_4.
- It reduces $HgCl_2$ first to Hg_2Cl_2 and then to Hg, reduces the solution of salts of Au, Ag, Pt and Bi metals,

reduces Cl_2 to HCl and I_2 to HI. It also reduces $CuSO_4$ to Cu_2H_2.
- In all these reactions the fundamental reaction for the reduction property of H_3PO_2 is

$$H_3PO_2 + 2H_2O \longrightarrow H_3PO_4 + 4H^+ + 4e^-$$

Phosphorus Acid

- It is prepared by dissolving P_4O_6 in water or by the hydrolysis of PCl_3.
- It is a white crystalline solid, highly soluble in water.
- On heating disproportionates into H_3PO_4 and PH_3.
- It is a strong dibasic acid and forms two series of salts: (i) acidic salts like NaH_2PO_3 and (ii) normal salts like Na_2HPO_3.
- It can act as a **reducing agent** though less than H_3PO_2. The fundamental reaction for the reduction property is

$$H_3PO_3 + H_2O \longrightarrow H_3PO_4 + 2H^+ + 2e^-$$

- It reduces $HgCl_2$to Hg_2Cl_2, $CuSO_4$ to Cu, $AgNO_3$ to Ag and $AuCl_3$ to Au. It also reduces the acidified permanganate, iodine, etc.
- It reacts with PCl_5 forming PCl_3, $POCl_3$ and HCl.

Hypophosphoric Acid

- It is formed when phosphorus is exposed to limited supply of moist air along with some phosphorus and phosphoric acids.
- It can also be prepared by the addition of sodium acetate to a mixture of hypophosphoric, phosphoric and phosphorus acids, resulting in the precipitation of disodium hypophosphate $Na_2H_2P_2O_6 \cdot 6H_2O$ from which the free acid can be obtained by acidification.
- It is a colourless crystalline solid. It is a **dihydrate.** It hydrolyzes in water forming a mixture of H_3PO_3 and H_3PO_4.
- It is a **tetrabasic acid.** It decomposes on heating to H_3PO_4 and PH_3.
- It cannot act as a reducing agent and cannot be reduced by zinc and H_2SO_4 to phosphine.

Orthophosphoric Acid, H_3PO_4

- It is prepared by dissolving P_4O_{10} in water, or by the oxidation of red phosphorus with 50 per cent HNO_3 using I_2 as a catalyst.
- On a large scale, it is manufactured by heating $Ca_3(PO_4)_2$ with calculated amount of dil. H_2SO_4. The insoluble $CaSO_4$ settles down while the supernatant syrup is separated.
- It is also obtained by the hydrolysis of PCl_5.
- It is a transparent deliquescent crystalline solid. It absorbs water and gives a syrupy liquid.
- On heating at 523 K gives pyrophosphoric acid which on further heating at 873 K gives metaphosphoric acid and at red hot conditions gives P_4O_{10}.
- It is a tribasic acid, forms three series of salts: (i) primary phosphates (acidic salts), e.g., NaH_2PO_4, (ii) secondary phosphates (acidic, e.g., Na_2HPO_4 and (iii) tertiary phosphates (normal), e.g., Na_3PO_4.
- Primary phosphates on heating converts into metaphosphates while secondary phosphates on heating converts into pyrophosphates.
- If ammonium ion is present in the salt it behaves like H^+ ion i.e., Na $(NH_4)_2$ PO_4 on heating gives metaphosphate and $Na_2NH_4PO_4$ on heating gives pyrophosphate.
- It gives a yellow precipitate of Ag_3PO_4 with $AgNO_3$ and white precipitate of $Ba_3(PO_4)_2$ with $BaCl_2$.
- It liberates HBr and HI from bromides and iodides, so it is preferred for the preparation of HBr and HI in the place of H_2SO_4 which can oxidize HBr to Br_2 and HI to I_2.
- In the presence of NH_4Cl it gives a white precipitate with magnesium salts due to the formation of $Mg(NH_4)PO_4$. This reaction is used for the detection of Mg^{2+} ions.

Pyrophosphoric Acid

- It is prepared by heating H_3PO_4 at 523 K or by heating equimolar mixture of H_3PO_4 crystals and $POCl_3$.

$$5H_3PO_4 + POCl_3 \longrightarrow 3H_4P_2O_7 + 3HCl$$

- When dissolved in water it converts into H_3PO_4. On heating to 873 K it converts into metaphosphoric acid.
- It is a tetrabasic acid and forms two series of salts, $Na_2H_2P_2O_7$ and $Na_4P_2O_7$.

Metaphosphoric Acid

- It is prepared by treating P_4O_{10} with a small amont of water at 273 K or by heating H_3PO_4 or $H_4P_2O_7$ at 873 K.
- It is also obtained by heating secondary ammonium hydrogen orthophosphates strongly.

$$(NH_4)_2HPO_4 \longrightarrow HPO_3 + 2NH_3 + H_2O$$

- It is a transparent glassy solid, so also called as **glacial phosphoric acid.** It is the most concentrated form of phosphoric acid.
- It forms white precipitates of $AgPO_3$ and $Ba(PO_3)_2$ with $AgNO_3$ and $BaCl_2$, respectively.

- It is a monobasic acid and forms only one type of salt with alkalis.
- Metaphosphoric acid and its salts exist as polymers. Its polymer $(NaPO_3)_6$ is used in modern water purification methods.
- Metaphosphoric acid coagulates the white of egg.

Fertilizers

Ammonium Sulphate

- The ammonical liquor obtained during the distillation of coal is heated with lime to liberate ammonia gas from various ammonium compounds present in it. The liberated ammonia is passed into 60 per cent H_2SO_4 to get $(NH_4)_2SO_4$.
- Ammonium sulphate is also manufactured by passing ammonia gas obtained in **Haber process** into sulphuric acid.
- In **Sindri process** ammonium sulphate is manufactured by passing ammonia into a suspension of powdered gypsum in water through which a stream of CO_2 is also passed.

$$2NH_3 + CO_2 + H_2O \longrightarrow (NH_4)_2CO_3$$

$$CaSO_4 + (NH_4)_2CO_3 \longrightarrow CaCO_3 + (NH_4)_2SO_4$$

- Ammonium sulphate contains 24–25% ammonia. Ammonia cannot be utilized directly by plants. Ammonia is converted into nitrates by nitrifying bacteria which are taken up easily by plants. Frequent usage of ammonium sulphate makes the soil acidic.

Calcium Cyanamide, CaNCN (Nitrolim)

- When calcium carbide is heated with nitrogen, a mixture of calcium cyanamide and graphite called **nitrolim** will be formed.
- In soil it is converted into urea which then decomposes into ammonia. Ammonia is finally converted into nitrates by nitrifying bacteria.

$$Ca\,CN_2 + H_2O + CO_2 \longrightarrow CaCO_3 + \underset{\text{(cyanamide)}}{CN \cdot NH_2}$$

$$CNNH_2 + H_2O \longrightarrow H_2NCONH_2 \text{ (urea)}$$

$$H_2N\,CONH_2 + H_2O \longrightarrow 2NH_3 + CO_2$$

Calcium Ammonium Nitrate, $Ca(NO_3)_2 \cdot NH_4NO_3$ (CAN)

- A part of ammonia obtained in Haber's process is converted into nitric acid by Ostwald process and made to react with the remaining part of ammonia to get ammonium nitrate. The solution of ammonium nitrate when treated with lime stone, the excess nitric acid present reacts with calcium carbonate by forming calcium nitrate.

Urea

- It is manufactured by treating ammonia and carbon-dioxide under 200 atmospheric pressure. Ammonium carbamate is formed as an intermediate product, which changes to urea.

$$CO_2 + 2NH_3 \longrightarrow NH_2\,COONH_4$$

$$NH_2COONH_4 \longrightarrow NH_2CONH_2 + H_2O$$

Urea contains about 46.6 per cent nitrogen.

Superphosphate of Lime or Calcium Superphosphate

- Superphosphate of lime is a mixture of calcium dihydrogen phosphate (primary phosphate) and gypsum. Its composition may be represented as $Ca(H_2PO_4)_2 \cdot 2\,CaSO_4 \cdot 2H_2O$.
- Superphosphate of lime is manufactured by treating calcium phosphate (phosphorite or bone ash) with conc. H_2SO_4.

$$Ca_3(PO_4)_2 + 2H_2SO_4 + 4H_2O \longrightarrow Ca(H_2PO_4)_2 + 2CaSO_4 \cdot 2H_2O + \text{Heat}$$

- The gases liberated during the manufacture of superphosphate of lime are CO_2 and HF due to the impurities $CaCO_3$ and CaF_2 in the phosphorite mineral.
- Superphosphate of lime is soluble in water but calcium phosphate is insoluble in water. Plants can absorb only soluble compounds.
- **Triple superphosphate** is obtained by the action of phosphoric acid on phosphate mineral.

$$Ca_3(PO_4)_2 + 4H_3PO_4 \longrightarrow 3Ca(H_2PO_4)_2$$

- It contains three times the amount of available P_2O_5 in comparison to superphosphate to lime. This do not contain gypsum.
- **Nitrophosphate is** obtained by heating phosphorite mineral with nitric acid. It is an useful fertilizer.

$$Ca_3(PO_4)_2 + 4HNO_3 \longrightarrow Ca_3(H_2PO_4)_2 + 2Ca(NO_3)_2$$

- **Thomas slag** obtained as a by-product during the production of steel can be used as a fertilizer. It contains calcium silicate and calcium phosphate.

EXERCISE-I

1. Select the correct statements about the hydrolysis of BCl_3 and NCl_3

(a) NCl_3 is hydrolyzed and gives HO*Cl* but BCl_3 is not hydrolyzed

(b) Both NCl_3 and BCl_3 on hydrolysis gives HCl
(c) NCl_3 on hydrolysis gives HO*Cl* but BCl_3 gives HCl
(d) Both NCl_3 and BCl_3 on hydrolysis gives HOCl

2. Consider the reactions

I. $Zn + conc.\ HNO_3\ (hot) \rightarrow Zn\ (NO_3)_2 + (X) + H_2O$

II. $Zn + dil.\ HNO_3\ (cold) \rightarrow Zn\ (NO_3)_2 + (Y) + H_2O$

Compounds X and Y are respectively

(a) N_2O, NO (b) NO_2, N_2O
(c) N_2,N_2O (d) NO_2, NO

3. In a molecule of phosphorus (V) oxide, there are
(a) 4 P—P, 10 P—O and 4 P = O bonds
(b) 12 P—O and 4 P = O bonds
(c) 2 P—O and 4 P = P bonds
(d) 6 P—P, 12 P—O and 4 P = P bonds

4. The number of sigma bonds in P_4, P_4O_6 and P_4O_{10} is
(a) 12, 6, 16 (b) 6, 16, 12
(c) 6, 12, 16 (d) 16, 16, 12

5. $(X) \xrightarrow{KOH} Y$ (gas turns red litmus to blue + Z
$Z \xrightarrow{Zn+KOH} Y$
$(X) \xrightarrow{\Delta}$ gas (support combustion)
Identify (X) to (Z)
(a) $X = NH_4NO_2$ $Y = NH_3$, $Z = KNO_2$
(b) $X = (NH_4)_2\ Cr_2O_7$, $Y = NH_3$, $Z = Cr_2O_3$
(c) $X = (NH_4)_2\ SO_4$, $Y = NH_3$, $Z = K_2\ SO_4$
(d) $X = NH_4NO_3$ $Y = NH_3$ $Z = KNO_3$

6. The pair that act as both oxidizing and as well as reducing agent is
(a) NO, SO_3 (b) NO_2, H_2O_2
(c) CO_2, SO_2 (d) N_2O_5, O_3

7. In the balanced stoichiometric equation of
'x'Zn + 'y' $HNO_3 \rightarrow x\ Zn\ (NO_3)_2 + NH_4NO_3 + zH_2O$,
the x and y are
(a) 2, 4 (b) 6,4 (c) 4,8 (d) 4,10

8. 60 mL of a mixture of nitrous oxide and nitric oxide was exploded with excess of hydrogen. If 38 mL of N_2 was formed, the volume (mL) of N_2O in the mixture is
(a) 44 (b) 24 (c) 16 (d) 8

9. In which of the following reactions does HNO_2 act as an oxidizing agent?
(a) $2KI + H_2SO_4 + 2HNO_2 \rightarrow K_2SO_4 + 2\ NO + I_2 + 2H_2O$
(b) $SnCl_2 + 2HCl + 2HNO_2 \rightarrow SnCl_4 + 2NO + 2H_2O$
(c) $2FeSO_4 + H_2SO_4 + 2HNO_2 \rightarrow Fe_2(SO_4)_3 + 2NO + 2H_2O$
(d) All of these

10. Which of the following reaction shows the correct sequence of Ostwald process in the manufacture of nitric acid?

(a) $S + O_2 \rightarrow SO_2 \xrightarrow{O_2} SO_3 \xrightarrow{H_2O} H_2SO_4 \rightarrow$,
$NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO$

(b) $N_2 + O_2 \xrightarrow[\text{high pressure}]{\text{low temp}} 2NO + heat$
$\xrightarrow[\text{catalyst}]{O_2} + NO_2 \xrightarrow{H_2O} HNO_3$

(c) $4NH_3 + 5O_2 \xrightarrow[\text{Catayst}]{750-900^\circ C} 4NO + 6H_2O + hea$
$\xrightarrow{O_2} NO_2 \xrightarrow{H_2O} HNO_3$

(d) All of these

11. NH_4Cl (s) is heated in a test tube. Vapours are brought in contact with red litmus paper, which changes to blue and then to red. It is because of:
(a) Formation of NH_4OH and HCl
(b) Formation of NH_3 and HCl
(c) Greater diffusion of NH_3 than HCl
(d) Greater diffusion of HCl than NH_3

12. It has been claimed that NH_4NO_3 fertilizer can be rendered unexplodable by a process that involves additives such as diammonium hydrogen phosphate $(NH_4)_2HPO_4$. Analysis of such a "desensitized" sample of NH_4NO_3 showed the mass percent of nitrogen to be 33.81 per cent. Assuming that the mixture contains only NH_4NO_3 and $(NH_4)_2HPO_4$, what is the mass per cent of $(NH_4)_2HPO_4$?
(a) 4.6 (b) 8.63 (c) 12.4 (d) 6.84

13. The ease of hydrolysis of trichlorides of Group-15 elements decreases in the order
(a) $NCl_3 > PCl_3 > AsCl_3 > SbCl_3 > BiCl_3$
(b) $PCl_3 > NCl_3 > AsCl_3 > SbCl_3 > BiCl_3$
(c) $AsCl_3 > NCl_3 > PCl_3 > SbCl_3 > BiCl_3$
(d) $SbCl_3 > BiCl_3 > PCl_3 > NCl_3 > AsCl_3$

14. The bond angle in PH_3 is less than the bond angle in PF_3. This is attributed to
(a) Enhanced repulsion due to the presence of double bond in PF_3
(b) Increased bond pair-bond pair repulsion due to multiple bond
(c) Both 1 and 2
(d) Displacement of electron cloud in P–F bond towards F in PF_3

15. In N_2O, the N—N distance corresponds to
(a) N = N bond (b) N ≡ N bond
(c) N—N bond (d) Intermediate of N = N and N ≡ N

16. A deep brown gas is formed by mixing two colourless gases which are
(a) NO_2 and O_2 (b) N_2O and NO
(c) NO and O_2 (d) NH_3 and HCl

17. Bones glow in the dark because
(a) They contain shining material
(b) They contain red phosphorus
(c) White phosphorus undergoes slow combustion in contact with air
(d) White phosphorus changes to red

18. The high reactivity and high volatility of white phosphorus is because
(a) It contains tetrahedrally arranged P_4 units
(b) It contain equilateral triangles
(c) The van der Waals forces of attraction are weak
(d) Angle strain is large

19. The total number of lone pairs of electrons present, the total number of single bonds formed and number of bonds formed by each 'P' atom in a P_4 molecule respectively are
(a) 2, 4, 4 (b) 6, 3, 6 (c) 4, 6, 3 (d) 8, 5, 5

20. Copper metal on treatment with dil. HNO_3 produce a gas 'A'.'A' when passed through acidic solution of stannous chloride a nitrogen-containing compound 'B' is obtained. 'B' on reaction with nitrous acid produces a gas 'C'. Then, 'C' is
(a) NO (b) NO_2 (c) N_2O (d) N_2

21. Arrange the following in decreasing order of bond angle: NH_3, PH_3, NF_3, PF_3
(a) $NH_3 > NF_3 > PF_3 > PH_3$
(b) $NF_3 > PF_3 > NH_3 > PH_3$
(c) $NF_3 > NH_3 > PH_3 > PF_3$
(d) $PH_3 > PF_3 > NH_3 > NF_3$

22. $P_4O_{10} + PCl_5 \rightarrow ?$
(a) $POCl_3$ (b) P_4O_6 (c) PO_2Cl (d) PCl_3

23. Both NF_3 and NCl_3 are covalent but they differ in the extent of hydrolysis because:
(a) NF_3 is more stable than NCl_3 and hydrolysis product of NF_3, that is, HFO, does not exist
(b) dipole moment of NF_3 is greater than that of NCl_3
(c) electronegativity of F is greater than that of Cl
(d) Cl can expand its octet by using d-orbitals

24. $d\pi - p\pi$ bonding is shown in
(a) NO_3^-, NO_2^-, N^{3-}, CN^- (b) PF_3, P_2O_5, PO_4^{3-}
(c) NH_3, PH_3, BiH_3 (d) CO, NO, CO_2, NO_2

25. Which of the following are incorrect about P_4O_6 and P_4O_{10}?
(a) There are two types of P—O bond lenghths in P_4O_{10} but only one type in P_4O_6
(b) The P—O single bond in P_4O_6 is longer than in P_4O_{10}
(c) The P—O single bond in P_4O_{10} is longer than in P_4O_6
(d) P_4O_6 can show Lewis base character but P_4O_{10} cannot

26. An element was burnt in limited supply of air to give oxide 'A', which on treatment with water gives an acid 'B'. Acid 'B' on oxidation gives acid 'C', which gives a yellow precipitate with $AgNO_3$ solution. A is
(a) SO_2 (b) NO_2 (c) P_4O_6 (d) SO_3

27. Which of the following statements regarding P_4O_{10} is not correct?
(a) Each P atom is bonded to four oxygen atoms
(b) P atoms are arranged tetrahedrally with respect to each other
(c) Each P—O bond has identical bond length
(d) Each 'P' is bonded to one O atom with considerable $p\pi - d\pi$ back bonding

28. The bond angles of NH_3, NH_4^+ and NH_2^- are in the order
(a) $NH_2^- > NH_3 > NH_4^+$ (b) $NH_4^+ > NH_3 > NH_2^-$
(c) $NH_3 > NH_2^- > NH_4^+$ (d) $NH_3 > NH_4^+ > NH_2^-$

29. The incorrect statement amongst the following is
(a) Black phosphorus is thermodynamically the most stable allotrope of phosphorus
(b) Fe(III) is thermodynamically more stable than Fe(II)
(c) Graphite is thermodynamically more stable than diamond
(d) White phosphorus is kinetically least stable allotrope of phosphorus and so is graphite in comparison to diamond

(**Note:** All comparisions are in their respective standard state.)

30. Nitrous oxide can be made by the careful thermal decomposition of molten NH_4NO_3 at about 210ºC.

$$NH_4NO_3 + \xrightarrow{\Delta} N_2O + 2H_2O$$

Now, choose the correct statements
(a) Thermal decomposition of $N^*H_4NO_3$ gives N N* O
(b) Thermal decomposition of $NH_4N^*O_3$ gives N N* O
(c) Thermal decomposition of $N^*H_4NO_3$ and $N^*H_4N^*O_3$ give N N*O
(d) Thermal decomposition of $N^*H_4NO_3$ and $NH_4N^*O_3$ give NN*O

31. Which of the following reactions is not giving products as shown?
(a) Hypo $\xrightarrow{\Delta}$ $Na_2S_5 + Na_2SO_3$
(b) Microcosmic salt $\xrightarrow{\Delta}$ $NaPO_3 + NH_3 + H_2O$
(c) Ammonium dichromate $\xrightarrow{\Delta}$ $N_2 + Cr_2O_3 + H_2O$
(d) $Na_2HPO_3 \xrightarrow{\Delta} PH_3 + Na_2PO_4 + Na_2HPO_4$

32. Zinc gives H_2 gas with dil. H_2SO_4 and dil. HC*l* but not with dil. HNO_3 because
(a) NO_3^- ion is reduced in preference to hydronium ion
(b) dil. HNO_3 is a weaker acid than dil. H_2SO_4 and dil. HC*l*.
(c) dil. HNO_3 acts as a reducing agent
(d) zinc is more reactive than H_2

33. Which of the following compounds is stable towards heat?
(a) NaH_2PO_4 (b) Na_2HPO_4
(c) $Cu(OH)_2$ (d) Na_3PO_4

34. Consider the ionic reaction:

$$HC_2O_4^- + PO_4^{-3} \rightleftharpoons HPO_4^{2-} + C_2O_4^{2-}$$

Identify the incorrect statement(s)
(a) The above equilibrium mixture can act as a buffer (both acids have nearly the same dissociation constant)
(b) $HC_2O_4^-$ and $C_2O_4^{-2}$ are conjugate acid–base pairs
(c) The P—O bond order in PO_4^{-3} is 1.75
(d) The C—O bond order in $C_2O_4^{2-}$ is 1.50

35. The kinetically most stable allotrope of phosphorus is
(a) White phosphorus (b) Red phosphorus
(c) Black phosphorus (d) Yellow phosphorus

36. $X \xleftarrow{H_2O} H_4P_2O_7 \xrightarrow{825k} Y$
In the above sequence of reactions, x and y are respectively
(a) H_3PO_4, H_3PO_4 (b) HPO_3, H_3PO_4
(c) H_3PO_4, HPO_3 (d) HPO_3, HPO_3

37. A colourless salt gives a white ppt (soluble in ammonium acetate) and a brown coloured pungent gas on reaction with conc. H_2SO_4. The salt is
(a) $Ba(NO_3)_2$ (b) $Pb(NO_3)_2$
(c) $NaNO_3$ (d) NH_4NO_3

38. Match the products in the following reaction

List-I	List-II
(a) $PCl_5 + H_2O \xrightarrow{1:1}$	(a) $N_3P_3Cl_6 + 12HCl$
(b) $PCl_5 + \underset{\text{excess}}{H_2O} \longrightarrow$	(b) $[PCl_4]^+ [AlCl_4]^-$
(c) $PCl_5 + AlCl_3 \longrightarrow$	(c) $H_3PO_4 + HCl$
(d) $PCl_5 + NH_4Cl \longrightarrow$	(d) $POCl_3 + HCl$

(a) 1-d, 2-c, 3-b, 4-a (b) 1-b, 2-c, 3-d, 4-a
(c) 1-d, 2-a, 3-b, 4-c (d) 1-c, 2-b, 3-d, 4-a

39. N_2O reacts with red-hot copper to give
(a) N_2 (b) NO_2 (c) N_2O_3 (d) Cu_3N_2

40. In the formation of the dimer N_2O_4 from two molecules of NO_2, the odd electron on each of the nitrogen atoms of the NO_2 molecules gets paired to form a
(a) weak N—N bond, and all the four N-O bonds become non-equivalent
(b) strong N—N bond, and all the four N-O bonds become equivalent
(c) weak N—N bond, two N-O bonds become equivalent and the other two N-O bonds become non-equivalent
(d) weak N—N bond, and all the four N—O bonds become equivalent

41. Which of the following statements is correct in the context of the N_2O molecule?
(a) The N—N bond is longer than the N—O bond
(b) The N—N bond is as long as the N—O bond
(c) The N—N bond has no π character
(d) The N—N bond is shorter than the N—O bond

42. Gold and platinum react with aquaregia to form soluble complexes. The complexes with oxidation number +3 and +4, respectively, are
(a) $H_2[AuCl_5]$ and $H_2[PtCl_6]$
(b) $H[AuCl_4]$ and $H_2[PtCl_6]$
(c) $H[AuCl_4]$ and $H_4[PtCl_8]$
(d) $H_3[AuCl_6]$ and $H[PtCl_5]$

43. Arrange the following in the increasing order of their bond angles
I. NH_3 II. $N(CH_3)_3$
III. $N(SiH_3)_3$ IV. NF_3
(a) I > II > III > IV (b) II > I > IV > III
(c) II > III > I > IV (d) III > II > I > IV

44. The cyclotrimetaphosphoric acid is
(a) $(HPO_3)_3$ and contains 9σ-bonds
(b) $H_3P_3O_6$ and contains 12σ-bonds
(c) $(HPO_3)_3$ and contains 15σ-bonds
(d) $H_3P_3O_9$ and contains 18σ-bonds

45. Platinum metal (Pt) dissolves in aquaregia but not in concentrated HCl or HNO_3 because
(a) HCl oxidizes Pt in the presence of HNO_3
(b) HNO_3 reacts with HCl to form chlorine which attacks Pt
(c) HNO_3 oxidizes Pt which is followed by formation of chlorocomplex
(d) HCl and HNO_3 together give O_2 that oxidizes Pt

46. Which of the following compounds does not give nitrogen on heating?
(a) NaN_3 (b) $(NH_4)_2SO_4$
(c) NH_4NO_2 (d) NH_4ClO_4.

47. Select the incorrect statement
(a) Nitrous oxide supports combustion more vigorously than air.
(b) Phosphorus pentoxide dehydrates nitric acid by forming nitrogen pentoxide.
(c) Reactivity order of various allotropic forms of phosphorus is white > red > black.
(d) Red phosphorus changes to white phosphorus on heating with I_2 at 250ºC and high pressure.

48. The species CO, CN^- and NO^+, which are isoelectronic with N_2, are much more reactive than N_2 because
(a) the bonds in these species are partly polar, whilst in N_2 they are not
(b) the bonds in these species are purely covalent
(c) these species behave as Lewis acids
(d) these species have higher bond energies

49. Nitrous oxide is obtained by the thermal decomposition of
(a) molten ammonium nitrate
(b) microcosmic salt

(c) sodium azide
(d) ammonium carbonate

50. Which of the following complexes is responsible for the brown colour of the ring formed in the ring test for the nitrates?
(a) $[Fe(CN)_5NO]^{3+}$ (b) $[Fe(H_2O)_4NO]^{2+}NO^+$
(c) $[Fe(CN)_5NO]^{2+}$ (d) $[Fe(H_2O)_5\ NO]^{2+}$

51. Platinum reacts with aquaregia to produce chloroplatinic acid represented by the formula
(a) $H_4\ [PtCI_6]$
(b) $H[PtCI_4]$
(c) $H_2[PtCI_6]$
(d) $H_2[PtCl_4]$

52. In P_4O_{10}
(a) the second bond in P = O is formed by $p\pi - d\pi$ back bonding
(b) P = O is formed by $p\pi - p\pi$ bonding
(c) P = O is formed by $d\pi - d\pi$ bonding
(d) P = O is formed by $\sigma p - p$ back bonding

53. $BiCl_3$ on hydrolysis forms a white ppt of
(a) Bismuth acid
(b) Bismuth oxychloride
(c) Bismuth pentachloride
(d) Bismuth hydride

54. The decrease in catenation power from nitrogen to bismuth among V group elements (except in P) is due to
(a) decrease in bond formation energy
(b) lack of ability to form strong $p\pi - p\pi$ bonds
(c) increase in metallic character
(d) their ability to react with other elements

55. Choose the correct increasing order of acidic strength
(a) $NO < CO_2 < NO_2 < N_2O_5$
(b) $N_2O_5 > NO_2 > CO_2 > NO$
(c) $CO_2 < NO < NO_2 < N_2O_5$
(d) $N_2O_5 > NO_2 > NO > CO_2$

56. What is the gas produced when dilute HNO_3 is added to zinc metal?
(a) N_2 (b) NO (c) NO_2 (d) N_2O

57. Arrange the following in order of decreasing N–O bond length in NO_2^+, NO_2^-, NO_3^-.
(a) $NO^-_3 > NO_2^+ > NO_2^-$ (b) $NO_3^- > NO_2^- > NO_2^+$
(c) $NO_2^+ > NO_3^- > NO_2^-$ (d) $NO_2^- > NO_3^- > NO_2^-$

58. Which of the following equations is not correctly written?
(a) $P_4 + 20HNO_3 \rightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$
(b) $C + 4HNO_3 \rightarrow H_2CO_3 + 4NO_2 + H_2O$
(c) $I_2 + 10HNO_3 \rightarrow 2HIO_3 + 10NO_2 + 4H_2O$
(d) $1/8S_8 + 6HNO_3 \rightarrow H_2SO_4 + 6NO_2 + 2H_2O$

59. Which pair of the following halide produces oxoacids of pnicogen on hydrolysis?
(a) NCl_3, PCl_3 (b) PCl_3, $AsCl_3$
(c) PCl_3, $BiCl_3$ (d) NF_3, NCl_3

60. When zinc reacts with very dilute nitric acid, the oxidation state of nitrogen changes from
(a) +V to +I (b) +V to –III
(c) +V to + V (d) +V to +III

61. Nitrogen compound 'A' is produced by the oxidation of HCN by O_2 using a silver catalyst.
Then the incorrect statement about A is
(a) It is a linear molecule
(b) It is a pseudohalogen
(c) It undergoes disproportionation in basic solution
(d) It can't be prepared by the action of Cu^{2+} and CN^-

62. The NO molecule
(a) Often act as a one-electron donor in contrast to most ligands which donate two electrons
(b) Often act as a three-electron donor in contrast to most ligands which donate two electrons
(c) is a coloured paramagnetic oxide
(d) is a colourless diamagnetic oxide

63. The azide ion has
(a) 20 outer electrons and is isoelectronic with Br_2O
(b) 18 outer electrons and is isoelectronic with NO_2^-
(c) 16 outer electrons and is isoelectronic with CO_2
(d) 14 outer electrons and is isoelectronic with H_2O_2

64. PCl_5 reacts with NH_4Cl to form
(a) Phosphonium chloride and NH_3
(b) Phosphonitrile chloride polymer $(NPCl_2)_n$
(c) Phosphorus triammine and Cl_2
(d) Phosphorus pentammine and N_2

65. Which of the following allotropic form of phosphorus is the most stable, least reactive, has a graphite-like structure and is a good conductor of electricity?
(a) White phosphorus
(b) Red phosphorus
(c) Black phosphorus
(d) Scarlet phosphorus

66. Which of the following statements regarding N_2O_4 is not correct?
(a) The molecule of N_2O_4 is planar
(b) The molecule of N_2O_4 contains a weak N—N bond
(c) In liquid N_2O_4, NOCl acts as a base
(d) The dipole moment of N_2O_4 is zero

67. Which of the following statement is not correct?
(a) Pure nitric acid is a colourless liquid
(b) Nitric acid with water forms an azeotropic mixture containing 68 per cent of acid
(c) Nitric acid when mixed with concentrated H_2SO_4 produces nitronium ion
(d) Covalent nitrates are more stable than ionic nitrates

68. Of the following statements regarding hydrazine which is not correct?
(a) It is a monofunctional base
(b) Anhydrous N_2H_4 is a fuming colourless liquid

(c) The structure of hydrazine is similar to ethane with a non-eclipsed structure
(d) Aqueous solution of hydrazine is a powerful reducing agent in basic solution

69. There is a very little difference in acid strength in the series H_3PO_4, H_3PO_3 and H_3PO_2 because
(a) Phosphorus in these acids exists in different oxidation states
(b) Number of unprotonated oxygen atoms responsible for increase of acidity due to inductive effect remains nearly the same
(c) Phosphorus is not a highly electronegative element
(d) Phosphorus oxides are less basic

70. A hydride (X) of Group-15 element is distinctly basic and has unexpectedly high boiling point. It reacts with NaOCl to give another hydride (X) which is a strong reducing agent and is used in organic analysis. X and Y are
(a) PH_3, P_2H_4 (b) NH_3,N_2H_4
(c) AsH_3, As_2H_4 (d) NH_3, NH_2Cl

71. $P_4 \xrightarrow{\text{limited oxygen}} x \xrightarrow{O_2} y + z$. In this sequence, y and z are.
(a) $POCl_3$, PO_2Cl_2 (b) $POCl_3$, PCl_3
(c) PCl_3, PO_2Cl (d) $POCl_3$, PCl_5

72. What is not true about $POCl_3$?
(a) The molecule does not contain any $p\pi - p\pi$ bond
(b) The molecule contains four sigma bonds
(c) The molecule contains one $p\pi - d\pi$ bond
(d) The molecule does not contain any $p\pi - d\pi$ bond

73. What is not produced when $NI_3 \cdot NH_3$ is rubbed against the hard surface?
(a) I_2 (b) N_2 (c) NH_3 (d) NH_4I

74. Which of the following is not correct?
(a) White and red phosphorus react with chlorine at room temperature
(b) White phosphorus is metastable, while red phosphorus is stable
(c) White phosphorus is lighter than red phosphorus
(d) White phosphorus is highly poisonous while red phosphorus is not

75. The nitrate which when heated gives off a gas or a mixture of gases which cannot relight a glowing splinter is
(a) Sodium nitrate
(b) Ammonium nitrite
(c) Lead nitrate
(d) Silver nitrate

76. An inorganic salt (A) on heating produces a colourless and odourless gas (X) which is neutral to litmus. Gas (X) when passed into another vessel containing lime stone and excess of coke at 1273 K produces a well-known fertilizer, nitrolim. Hence, salt (A) may be
(a) NH_4NO_3 (b) $Ba(N_3)_2$
(c) $LiNO_3$ (d) $Pb(NO_3)_2$

77. One mole of P_4O_{10} reacts with six moles of water to give finally four moles of orthophosphoric acid. Which of the following oxoacid of phosphorus is not formed in the intermediate stage during the course of reaction?
(a) Tripolyphosphoric acid ($H_5P_3O_{10}$)
(b) Tetrametaphosphoric acid ($H_4P_4O_{12}$)
(c) Tetrapolyphosphoric acid ($H_6P_4O_{13}$)
(d) Pyrophosphoric acid ($H_4P_2O_7$)

78. Bi^{3+} ion readily get hydrolyzed in water to give bismuthyl ion.

$$Bi^{3+} + H_2O \rightleftharpoons BiO^+ + 2H^+$$

This equilibrium can be moved in the forward direction by
(a) making the solution alkaline
(b) making the solution acidic
(c) adding Na_2SnO_2
(d) adding KI solution

79. A gas (X) is formed on heating copper turnings with dil. HNO_3. The gas (X) when passed into an acidic solution of stannous chloride a nitrogen-containing compound (y) is formed. (y) on treating with HNO_2 liberates a gas (z). Gas Z is
(a) NO (b) N_2 (c) NO_2 (d) N_2O

More than One Answer Type Questions

1. When a mixture of NH_4Cl, NH_4OH and Na_2HPO_4 was added to a solution containing Mg^{2+}, a white precipitate (A) was formed. When A was heated strongly, the residue B was obtained. A and B are
(a) $Mg(NH_4)\,PO_4$ (b) $Mg_2P_2O_7$
(c) $Mg_3(PO_4)_2$ (d) MgO

2. Which of the following statements are correct?
(a) If an electron is removed by oxidizing NO, the nitrosonium ion NO^+ is formed
(b) In NO^+, the bond order is 3.0
(c) The N–O bond length contracts from 1.15A in NO to 1.06A in NO^+
(d) NO is a coloured gas due to the presence of an unpaired electron

3. Which of the following statements is correct in the context of the NO_3^- ion?
(a) The structure of the NO^-_3 ion is planar triangular
(b) All the three oxygen atoms are equivalent
(c) The bond order of each N–O bond is 3/2
(d) The bond order of each N–O bond is 4/3

4. NH_3 can be obtained by
(a) Heating of NH_4NO_3 or NH_4NO_2
(b) Heating of NH_4Cl or $(NH_4)_2CO_3$
(c) Heating of NH_4NO_3 with NaOH
(d) Reaction of AIN or Mg_3N_2 or $CaCN_2$ with H_2O

5. Which of the following statement(s) is /are correct?
(a) Nitrogen (IV) oxide in basic aqueous solution disproportionates.
(b) Sodium nitrate and lead nitrate each on heating to 800ºC decomposes liberating both NO_2 and O_2 gases.
(c) Ammonia and sodium hypochlorite react in dilute aqueous solutions in the presence of gelatin and liberate nitrogen gas.
(d) Hydrogen peroxide is produced on a large scale by the oxidation of 2-ethyl anthraquinol.

6. Pyrophosphorus acid, $H_4P_2O_5$
(a) Contains P in +5 oxidation state
(b) Is a dibasic acid
(c) Is oxidizing in nature
(d) Contains one P—O—P bond

7. Which of the following are correct statements?
(a) Solid PCl_5 exists as tetrahedral $[PCl_4]^+$ and octahedral $[PCl_6]^-$ ions
(b) Solid PBr_5 exists as $[PBr_4]^+$ Br^- ions
(c) Solid N_2O_5 exists as NO_2^+ NO_3^- ions
(d) Oxides of phosphorus P_2O_3 and P_2O_5 exists as monomers

8. In the dark brown ring test for nitrate ions
(a) The colour is due to charge transfer spectra
(b) Iron has a formal +2 oxidation state and NO has no charge
(c) The complex species can be represented as $[Fe^{I}(H_2O)_5 NO^+] SO_4$
(d) The dark brown colour is due to NO_2 evolved in the reaction

9. Which of the following statements are correct?
(a) NO_2 acts as an oxidizing agent
(b) N_2O_5 with alkalis gives nitrates
(c) Moist NH_3 gas is dried over $CaCl_2$
(d) Action of NH_3 on Cl_2 gives NH_4Cl stable product

10. Ammonia on reaction with hypochlorite anion can form
(a) NO (b) NH_2Cl (c) N_2H_4 (d) HNO_2

11. Identify the correct statements
(a) The hydrolysis products of $BiCl_3$ are BiOCl and HCl
(b) Red P_4 is less volatile than white P_4
(c) Reducing properties of hypophosphorus acid is due to O–H bonds
(d) Hypophosphoric acid has one P—O—P bond

12. In the following statements, select the correct statement
(a) $N(CH_3)_3$ has a pyramidal structure
(b) $N(SiH_3)_3$ shows planar arrangement
(c) SiC is highly volatile
(d) SiO_2 is called silane

13. Nitrogen (I) oxide is produced by
(a) thermal decomposition of ammonium nitrate
(b) disproportionation of N_2O_4
(c) thermal decomposition of ammonium nitrite
(d) interaction of hydroxylamine and nitrous acid

14. Which of the following will favour the formation of NH_3 by Haber's process?
(a) Increase of temperature
(b) Increase of pressure
(c) Addition of catalyst
(d) Addition of promoter

15. Select the correct statement(s) about the compound NO $[BF_4]$
(a) It has 5σ and 2π bonds
(b) Nitrogen–oxygen bond length is higher than in nitric oxide (NO)
(c) It is a diamagnetic species
(d) B—F bond length in this compound is lower than in BF_3

16. Select the correct statement(s)
(a) NF_3 is a weaker base than NH_3
(b) NO^+ is more stable than O_2^+
(c) $AlCl_3$ has a higher melting point than AlF_3
(d) $SbCl_3$ is more covalent than $SbCl_5$

17. Pick out the correct statements
(a) In PCl_5, P atom is involved in sp^3d hybridization and has trigonal bipyramidal geometry
(b) The structure of PCl_5 is symmetric
(c) PCl_5 can accept a lone pair and thus can act as Lewis acid
(d) In PCl_5 the axial chlorine atoms are closer to the central P atom than equatorial atoms

18. Pick out the correct statements
(a) Red phosphorus consists of a chain structure and black phosphorus has a layered structure
(b) Nitrogen shows a little tendency for catenation, because N—N single bond is very strong
(c) The maximum number of covalent bonds formed by nitrogen is four since it has no d-orbitals in its valence shell
(d) The Group-15 elements do not form M^{5+} ions, but +5 oxidation state is obtained in their compounds through covalent bonding

19. Which of the following statements are incorrect?
(a) The solid N_2O_5 is ionic and is represented by NO_2^+ NO_3^- (nitronium nitrate).
(b) The nitronium ion (NO_2^+) is isoelectronic with CO_2 and both have similar V-shaped structures.
(c) The nitrite ion (NO_2^-) has a T-shaped structure.
(d) The NO_2^- ion has a pyramidal structure.

20. When PbO_2 reacts with conc. HNO_3 at high temperature, the gases evolved is/are
(a) NO_2 (b) O_2 (c) N_2 (d) N_2O

21. Which of the following statements are correct for the P_4O_6 molecule?
(a) The four phosphorus atoms are arranged in a tetrahedral form
(b) The six oxygen atoms are situated along the edges of a tetrahedron
(c) Each oxygen atom is bonded to two adjacent phosphorus atoms
(d) The structure of P_4O_6 is derived from that of PCl_3

22. Which of the following molecules have a dative bonding ($p\pi - d\pi$)?
(a) P_4O_{10} (b) $(SiH_3)_3N$ (c) PF_3 (d) N_2O_5

23. The statement (s) that are true among the following are
(a) PBr_5 exists as $[PBr_4]^+ [PBr_6]^-$
(b) Nitrite is both oxidizing and reducing agent
(c) Bi_2O_3 is only basic and not acidic and is insoluble in alkali
(d) The oxidation state of Fe in brown ring complex is +1

24. Which of the following order(s) is/are incorrect?
(a) $H_3PO_4 > H_3PO_3 > H_3PO_2$ (reducing character)
(b) $N_2O < NO < N_2O_3 < N_2O_5$ (oxidation state on nitrogen atom)
(c) $NH_3 > PH_3 > AsH_3 > SbH_3$ (basicity)
(d) $SbH_3 > NH_3 > AsH_3 > PH_3$ (reducing character)

25. Which of the following statement(s) is/are correct?
(a) NH_3 is oxidized to NO_2 by oxygen at 800ºC in the presence of a catalyst platinum.
(b) Nitric acid on standing slowly turns yellow or brown.
(c) Colloidal sulphur is formed when H_2S gas is passed through nitric acid solution.
(d) N_2O_3 gas dissolves in water giving a pale blue solution.

26. Which of the following statements is/are correct regarding ammonia?
(a) it is oxidized to NO with oxygen at 700ºC in the presence of platinum
(b) it gives black precipitate with calomel
(c) it can be dried by P_2O_5, H_2SO_4 and $CaCl_2$
(d) it gives white fumes with HCl

27. Which of the following statement(s) are not correct?
(a) The covalency of N in HNO_3 is +5
(b) HNO_3 in gaseous state has a trigonal planar structure
(c) The oxidation state of N in HNO_3 is +4
(d) Gold dissolves in HNO_3 to form gold nitrate

28. Which of the following metal nitrates give oxides on heating?
(a) Li (b) Mg (c) Na (d) Ca

29. Select the correct statement (s)
(a) In P_4O_{10} all ‘P—O’ bonds do not have the same bond length.
(b) In PF_5, all ‘P—F’ bond lengths are same
(c) Interhalogen compounds are more reactive than halogens (except F_2)
(d) In P_4O_{10} all P–O–P bond angles are not equal

30. PH_3 can be obtained by heating
(a) white phosphorus with hot concentrated alkali
(b) heating phosphinic acid (H_3PO_2)
(c) heating phosphorus acid (H_3PO_3)
(d) heating phosphoric acid (H_3PO_4)

31. Generally in a group the oxidation power in higher oxidation state increases but Vth group compounds of N(V) and Bi(V) are stronger oxidizing agents than the +5 oxidation state of the three intervening elements because
(a) Nitrogen is a more electronegative element
(b) Bismuth is a less electronegative element
(c) Bismuth favours the +3 oxidation state in preference to the +5 state on account of the inert pair effect
(d) Nitrogen in +5 oxidation state is more stable

32. Which of the following statement(s) regarding nitrogen sesqui oxide is(are) correct?
(a) Nitrogen sesquioxide is stable only in the liquid state it dissociates in the vapour state
(b) Nitrogen sesquioxide is neutral oxide
(c) Nitrogen sesquioxide contains a weak N—N bond
(d) Nitrogen sesquioxide exists in two different crystalline forms

33. Identify the correct statement(s)
(a) P_4O_{10} is used as a drying agent
(b) P_4O_{10} contains $d\pi - p\pi$ back bonding
(c) In P_4O_{10} each P atom is bonded to three oxygen atoms
(d) P_4O_{10} hydrolyzes violently in water forming phosphorus acid

34. Correct statements about phosphoric acid include that
(a) its basicity is three
(b) at 600ºC it forms metaphosphoric acid
(c) it contains two P = O bonds
(d) on gentle heating it forms pyrophosphoric acid whose basicity is +4

35. Which of the following statements are correct?
(a) an acidified solution of potassium permanganate oxidizes nitric oxide to nitrate ion
(b) The reaction $2HNO_3 + NO \rightarrow 3NO_2 + H_2O$ completely moves in the forward direction with conc. HNO_3
(c) The action of concentrated HNO_3 on metals produces NO_2 because the reaction $2HNO_3 + NO \rightleftharpoons 3NO_2 + H_2O$ lies far towards right side

(d) The action of dilute HNO_3 on metals produce NO because the reaction
$HNO_3 + NO \rightleftharpoons 3NO_2 + H_2O$ lies far towards left side

36. Which of the following statements are not correct?
(a) The HNH bond angle in NH_3 is greater than HAsH bond angle in AsH_3
(b) NO though an odd electron molecule, is diamagnetic in liquid state
(c) NO is reduced to N_2O when heated over copper
(d) NO is reduced to Hydroxylamine by hydrogen in the presence of Pt black

37. Which of the following statements are correct?
(a) Nitrogen sesquioxide reacts with concentrated acids forming nitrosyl salts.
(b) NO combines with halogens forming nitrosyl halides.
(c) Nitrogen sesquioxide exists in two different forms symmetrical and asymmetrical forms.
(d) Nitrogen tetroxide is a planar molecule with N—N bond length equal to the expected N—N single bond.

38. Which of the following statements are not correct
(a) Nitronium ion is NO^+
(b) In the reaction between metal and nitric acid, the more dilute the acid, the lesser the reduction of NO^-_3 ion
(c) Non-metals are oxidized to their highest oxoacids with conc. nitric acid
(d) Nitric acid is a non-planar molecule with no resonating structures

39. Which of the following statements are correct?
(a) The ionization energy of NO is greater that of N_2
(b) The second ionization energy of nitrogen is less than that of oxygen
(c) The odd electron in NO is more easily removed from the antibonding π-MO to form the NO^+ ion
(d) The electronegativity of nitrogen is greater than that of oxygen

40. Which of the following statements are correct about the reaction between the copper metal and dilute HNO_3?
(a) The principal reduction product is NO gas
(b) Cu metal is oxidized to Cu^{2+}(aq) ion which is blue in colour
(c) NO is paramagnetic and has one unpaired electron in antibonding molecular orbital
(d) NO reacts with O_2 to produce NO_2 which is linear in shape

41. Which of the following compounds are explosive?
(a) NF_3 (b) NH_4NO_3
(c) NCl_3 (d) $NI_3 \cdot NH_3$

42. What is correct about the product formed by heating phosphoric acid at 200ºC?
(a) It is a tetrabasic acid
(b) It contains P—O—P bond
(c) It forms only two series of salts
(d) Heating the product to 600ºC gives metaphosphoric acid

43. When PH_3 is passed through copper sulphate solution then copper phosphide is obtained. This is because
(a) PH_3 is highly unstable
(b) PH_3 shows acidic behaviour
(c) P^{3-} ion substitute SO_4^{2-} ion
(d) PH_3 is a strong ligand

44. White phosphorus (P_4) has
(a) 4 equilateral triangles
(b) 6 P—P bonds
(c) its PPP angle as 109º28'
(d) Tetrahedral structure

45. The hybridization of central nitrogen in azide ion is same as that of
(a) N_2O (b) C_2H_2 (c) C_6H_6 (d) CO_2

46. $\underset{\text{white}}{P_4} \xrightarrow[H_2O]{NaOH} A + B$

$B \xrightarrow{O_2} C + H_2O$

which of the following is correct for the above reactions?
(a) Compound A is NaH_2PO_2
(b) Compound B is PH_3
(c) Compound B is H_3PO_3
(d) Compound C is H_3PO_4

47. Which of the following are correct statements?
(a) N_3H is a weak acid
(b) N_3^- ion can act as a ligand
(c) N_3H do not liberate hydrogen with metals
(d) Hybridization of nitrogen in N_3^- is sp

48. Which of the following gives nitrogen on heating?
(a) $Ba(N_3)_2$ (b) $(NH_4)_2Cr_2O_7$
(c) $NaNO_2 + NH_4Cl$ (d) $CuO + NH_3$

49. Which of the following reactions are possible?
(a) $Zn + NaOH + NaNO_3 \rightarrow Na_2ZnO_2 + NH_3 + H_2O$
(b) $Al + NaOH + NaNO_2 \rightarrow NaAlO_2 + N_2 + H_2O$
(c) $N_2H_4 + HNO_2 \rightarrow N_3H + 2H_2O$
(d) $NH_3 + NaOCl \rightarrow N_2H_4 + NaCl + NH_4Cl$

50. Which of the following tests can be used to detect ammonia in the laboratory?
(a) its characteristic pungent odour
(b) forming dense white fumes of NH_4Cl with the glass rod dipped in a bottle of conc. HCl
(c) forming yellow–orange–brown precipitate with Nessler's reagent
(d) burning in air to give a pale blue flame

51. Among the following which will not react with NaOH or which is not an acid salt?
(a) NaH_2PO_2 (b) Na_2HPO_3
(c) NaH_2PO_3 (d) $NaHCO_3$

52. Which of the following ammonium salts do not liberate ammonia gas on heating?
(a) $(NH_4)_2CO_3$ (b) $(NH_4)_2Cr_2O_7$
(c) NH_4ClO_4 (d) $H_2NCOONH_4$

53. In liquid N_2O_4
(a) NOCl act as an acid
(b) $NaNO_3$ act as a base
(c) The reaction between NOCl and $NaNO_3$ in liquid N_2O_4 is a redox reaction
(d) The N – N bond is longer in N_2O_4 than in N_2H_4

Comprehension Type Questions

Passage-I

The following flow diagram represents the industrial preparation of nitric acid from ammonia

$$NH_3 + O_2 \xrightarrow[900°C]{(A)} NO \xrightarrow[air]{(B)} C \longrightarrow HNO_3 + NO$$

1. Which line of entry describes the underlined reagents, products and reaction conditions?

	A	B	C
(a)	Catalyst	Room Temp. (25ºC)	NO_2
(b)	Catalyst	Room Temp (25ºC)	N_2O
(c)	Catalyst	High pressure	NO_2
(d)	High pressure	Catalyst	N_2O_3

2. Formation of HNO_3 when (C) is dissolved in H_2O takes place through various reactions. Select the reaction not observed in this step.
(a) $NO_2 + H_2O \rightarrow HNO_3 + HNO_2$
(b) $HNO_2 \rightarrow H_2O + NO + NO_2$
(c) $NO_2 + H_2O \rightarrow HNO_3 + NO$
(d) $NO_2 + H_2O \rightarrow HNO_3 + N_2O$

3. Immediately after the formation of NO the temperature of catalyst must be decreased, otherwise which of the following reactions takes place?
(a) $2NO \rightarrow N_2 + O_2$
(b) $2NO + O_2 \rightarrow 2NO_2$
(c) $NO + NO_2 \rightarrow N_2O_3$
(d) $NO + NH_3 \rightarrow N_2 + H_2O$

Passage-II

When white phosphorus was boiled with caustic soda a gas, 'A' and the compound 'B' were obtained. When the gas 'A' was passed through $AgNO_3$ solution, a black precipitate 'C' was obtained.

1. The oxidation number of phosphorus in A and B are
(a) +3, +3 (b) –3, –3 (c) –3, +1 (d) +3, +5

2. The correct statement regarding the reaction between white phosphorus and caustic soda is:
(a) The reaction can occur when red phosphorus is used instead of white phosphorus.
(b) It is a disproportination reaction.
(c) The reaction cannot occur when KOH is used instead of NaOH.
(d) The phosphorus gets reduced from 0 to –1 in the reaction.

3. The product 'C' is
(a) Ag_3P (b) Ag_3PO_4
(c) $Ag_4P_2O_7$ (d) Ag

Passage-III

There are some deposits of nitrates and phosphates in earth's crust. Nitrates are more soluble in water. Nitrates are difficult to reduce under the laboratory conditions but microbes do it easily. Ammonia forms a large number of complexes with transition metal ions. Hybridization easily explains the ease of sigma donation capability of NH_3 and PH_3. Phosphine is a flammable gas and is prepared from white phosphorus.

1. Among the following, the correct statement is
(a) Phosphates have no biological significance in humans.
(b) Between nitrates and phosphates, phosphates are less abundant in earth's crust.
(c) Between nitrates and phosphates, nitrates are less abundant in earth's crust.
(d) Oxidation of nitrates is possible in soil.

2. Among the following, the correct statement is
(a) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional
(b) Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional
(c) Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional
(d) Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

3. Which one of the following metal cannot liberate hydrogen gas with dil. HNO_3?
(a) Iron (b) Copper (c) Tin (d) All

4. Bond order in NO^{3-} is equal to
(a) Bond order in O_3 (b) Bond order in CO_3^{2-}
(c) Bond order in NO^- (d) Bond order in NO

Passage-IV

When a vapour consisting of a mixture of a colourless gas X and a brown gas Y, at atmospheric pressure is gradually heated from 25ºC, its colour is found to deepen at first and then to fade as the temperature is raised above 160ºC. At 600ºC, the vapour is almost colourless consisting of a gas Z and O_2 but its colour deepens when the pressure is raised at this temperature.

1. The colourless gas X in the mixture is
(a) Bromine
(b) Pure nitrogen dioxide
(c) Dinitrogen tetroxide
(d) Nitric oxide

2. The brown gas Y in the mixture is
(a) Nitric oxide (b) Nitrous oxide
(c) Dinitrogen tetroxide (d) Nitrogen dioxide

3. The gas Z in the above reaction is
(a) Nitrous oxide (b) Nitric oxide
(c) Dinitrogen tetroxide (d) Nitrogen dioxide

Passage-V

1. Read the passage and answer the following questions.

The principal factors responsible for the difference between nitrogen and phosphorus group chemistry are those responsible for the C to Si differences namely (a) the diminished ability of the second-row element to form pπ–pπ multiple bonds and (b) the possibility of utilizing the lower lying 3d-orbitals.

The first explains features such as the fact that nitrogen forms esters (O = NOR) whereas phosophorus gives $P(OR)_3$. Nitrogen oxides and oxoacids all involve multiple bonds, whereas phosphorus oxides have single P—O bonds in P_4O_6 and phosphoric acid is $PO(OH)_3$ in contrast to NO_2 (OH).

The utilization of d-orbitals has three effects. First, it follows some $p\pi - d\pi$ bonding as in $R_3P{=}O$ or $R_3P = CH_2$. The amine oxides, R_3NO, have only a single canonical structure ($R_3N^+ –O^–$) and are chemically reactive while P – O bonds are shorter than expected from the sum of single bond radii indicating multiple bonding and are very strong about 500 KJ $mol^{–1}$. Second, there is the possibility of expansion of the valence shell, whereas nitrogen has a covalency maximum of four. Thus, we have compounds such as PF_5, $P(C_6H_5)_5$, $P(OCH_3)_6^-$ and PF_6^-.

Notice that for many of the five coordinate species especially of phosphorus, the energy difference between the trigonal bipyramidal and square pyramidal configuration is small and such species are usually stereochemically non-rigid.

When higher coordination numbers occur for the elements in the +III oxidation state, as in $[SbF_5]^{2–}$, the structures take the form of a square pyramid.

Divalent nitrogen and other elements in compounds such as $N(C_2H_5)_3$, $P(C_2H_5)_3$ and As $(C_2H_5)_3$ have lone pairs and act as donors. There is a profound difference in their donor ability towards transition metals. This follows from the fact that although NR_3 has no low lying acceptor orbitals, the others do have such orbitals namely the empty d-orbitals. These can accept electron density from filled metal d-orbitals to form $d\pi - d\pi$ bonds.

1. The compound NF_3 has no donor properties at all, but PF_3 forms numerous complexes with metals for example $Ni(PF_3)_4$ because
(a) In NF_3 the lone pair is present in 2s-orbital while in PF_3 the lone pair is present in 3s-orbital.
(b) In NF_3 the electron density at nitrogen is less due to high electronegativity.
(c) In PF_3 the electron density at phosphorus is more due to back bonding of the electrons of fluorine.
(d) Both 2 and 3.

2. The complexes formed by transition elements with trialkyl phosphine are more stable than formed with trialkyl ammonia because
(a) alkyl phosphines are stronger ligands while trialkyl ammonia are weaker ligands.
(b) There is a multiple bond formation between metal ion and phosphorus due to the presence of vacant d-orbitals in phosphorus.
(c) Since nitrogen in ammonia is more electronegative the bond between nitrogen and transition metal ion will be ionic in nature.
(d) The donation capacity of 2s-electrons in nitrogen is less than that of 3s-electrons in phosphorus.

3. The structures of $[SbF_5]^{2–}$ and SbF_5 are
(a) both trigonal bipyramidal
(b) both square pyramidal
(c) $[SbF_5]^{2–}$ squared pyramidal while SbF_5 is trigonal bipyramidal
(d) $[SbF_5]^{2–}$ is trigonal bipyramidal while SbF_5 is square pyramidal

4. Among $R_3P{=}O$ and $R_3N^+–O^–$ the amine oxide ($R_3N^+–O^–$) is reactive while $R_3P{=}O$ is stable because
(a) $R_3N^+–O^–$ is unstable because of no resonance structures
(b) In R_3P—O, the P—O bond is stronger due to multiple bonding
(c) R_3P—O has more number of resonance structures
(d) All are correct

5. Phosphorus forms compounds of the type $P(OR)_3$ but will not form compounds of the type O = POR while nitrogen can form compounds of the type O = NOR but not $N(OR)_3$

(a) because small-sized nitrogen is involved in strong $p\pi - p\pi$ bonding.
(b) Because of high electronegativity nitrogen can form a strong bond with oxygen.
(c) Because of the absence of d-orbitals in the valence shell of nitrogen.
(d) Because maximum covalency of nitrogen is four.

6. The compounds like RN = NR can exist but RP = PR is less stable and try to convert into cyclic oligomers of the type $(RP)_n$ because
(a) In the cyclic structures there exist single bonds between phosphorus atoms.
(b) The π-bond in HP = PH is weaker due to bigger size of phosphorus atoms.
(c) Cyclic structures have more stability.
(d) Both 1 and 2.

Passage-VI

Nitrogen forms the largest number of oxides as it is capable of forming stable multiple bonds with oxygen. They range from N_2O (O.S. of nitrogen is +1) through NO, N_2O_3 to N_2O_5 (O.S. of nitrogen is + 5). The following points are important regarding the study of oxides of nitrogen.

(a) All oxides of nitrogen except N_2O_5 are endothermic as a large amount of energy is required to dissociate the stable molecules of oxygen and nitrogen.
(b) The small electronegativity difference between oxygen and nitrogen make N—O bond easily breakable to give oxygen and hence oxides of nitrogen are said to be better oxidizing agent.
(c) Except N_2O_5 all are gases at ordinary temperature. N_2O_3 is stable only at lower temperature (253 K).
(d) Except N_2O and NO which are neutral oxides, all are acidic oxides which dissolve in water forming corresponding oxoacids.
(e) They are also good examples for illustrating the concept of resonance.

1. The gas which is acidic in nature is
(a) NO (b) N_2O (c) NO_2 (d) Both 1 and 3
2. Which of the following statements is not correct for the oxides of nitrogen?
(a) Dinitrogen trioxide dissolves in potassium hydroxide forming potassium nitrite
(b) Aqueous solution of nitrogen dioxide behaves both as a reducing agent and as an oxidizing agent
(c) Nitrous oxide is fairly soluble in cold water and turns blue litmus to red
(d) Nitrogen dioxide is an odd electron molecule and in the liquid phase, exists in equilibrium with the dimer N_2O_4
3. Identify the incorrect statement
(a) In N_2O_4 the N – N bond length is longer than the usual N – N single bond distance
(b) NO_2 molecule is angular with N – O distance equal to intermediate distance between a single and double bond
(c) N_2O is a linear molecule and has a small dipole moment
(d) None of these

Passage-VII

The 15th group elements form halides of the type MX_3. The halides are predominantly covalent in nature and have a pyramidal shape. Due to the presence of an electron pair on the central atom, the halides like the hydrides exhibit Lewis base character.

1. Trihalides of the 15th group which are not readily hydrolyzed are
(a) PCl_3; NCl_3 (b) NF_3; PF_3
(c) SbF_3; $SbCl_3$ (d) NF_3; NCl_3
2. The products of hydrolysis of NCl_3 and PCl_3 are respectively
(a) NOCl; H_3PO_3 (b) NH_3; H_3PO_3
(c) NH_3, PH_3 (d) $NOCl_3$; PH_3
3. The trihalide of nitrogen which will show very little Lewis base nature is
(a) NI_3 (b) NCl_3 (c) NBr_3 (d) NF_3

Passage-VIII

$$PCl_5 + SO_2 \longrightarrow A + B$$
$$A \xrightarrow{H_2O\,(3\,moles)} C + HCl$$
$$C \xrightarrow{Red\ hot} D + H_2O$$

1. The compound 'A' is
(a) $POCl_3$ (b) PCl_3 (c) $SOCl_2$ (d) SO_2Cl_2
2. The compound 'D' is
(a) P_4O_6 (b) P_4O_{10} (c) SO_3 (d) SO_2
3. The compound 'B' is
(a) $SOCl_2$ (b) SO_2Cl_2 (c) $POCl_3$ (d) PCl_3

Passage-IX

Aquaregia a 3:1 mixture (by volume) of concentrated hydrochloric acid and nitric acid and was developed by the alchemists as a means to "dissolve" gold. The process is actually a redox reaction with the following simplified chemical equation.

$$Au_{(s)} + NO_3^- + Cl^-_{(aq)} \rightleftharpoons AuCl^-_{4(aq)} + NO_{2(g)}$$

Gold is too noble to react with nitric acid. However, gold does react with aquaregia because the complex ion $AuCl^-_4$ forms

1. What are the oxidizing and reducing agents in the dissolution of gold in aquaregia?
(a) Cl and Au (b) HNO_3 and Au
(c) Cl^- and Au (d) NOCl and Au

2. The function of HCl is to provide Cl^-. What is the purpose of the Cl^- for the above reaction?
(a) Cl^- is an oxidizing agent
(b) Cl^- is a reducing agent
(c) Cl^- is a complexing agent
(d) Cl^- is a catalyst

3. The shape of the complex formed in the above process is
(a) Tetrahedral (b) Octahedral
(c) Square planar (d) Pyramidal

Passage-X

PCl_5 in solid salt exists as PCl_4^+ and PCl_6^- and also in some solvents it undergoes dissociation as

$$2PCl_5 \rightleftharpoons PCl^+{}_4 + PCl_6^-$$

1. The geometry and hybridization of PCl_4^+ is
(a) Tetrahedral, sp^3 (b) Octahedral, sp^3d^2
(c) Trigonal bipyramid, sp^3d (d) See-saw, sp^3d

2. The geometry and hybridization of PCl_6^- is
(a) octahedral, sp^3d^2 (b) tetrahedral, sp^3
(c) square planar bipyramid, sp^3
(d) see-saw, sp^3d

3. Which statement is wrong?
(a) In PCl_5 all the P–Cl bonds are of same energy
(b) PCl_5 has no lone pair of electrons
(c) PCl_5 is a white solid which melts at 167ºC
(d) PCl_5 gives white fumes with moist air

Passage-XI

Phosphorus forms oxides of the type P_4O_6, P_4O_8, P_4O_7 and P_4O_{10}. Phosphorus pentoxide is the most important oxide and is quite common. It is dimeric and has the formula P_4O_{10} not P_2O_5. The oxides P_4O_7, P_4O_8 and P_4O_9 are very uncommon. In these oxides, the second bond in P = O is formed by $p\pi - d\pi$ back bonding. Phosphorus forms two series of oxoacids, phosphoric series of acids and phosphorus series of acids. In the former, the oxidation state of P is (+V) and compounds have oxidizing properties while in the latter, the acids contain P in the oxidation state (+III) and which are reducing agents. In all of these P is four coordinate and tetrahedrally surrounded whereever possible $p\pi - d\pi$ back bonding gives rise to P = O bonds.

A very large number of polyphosphoric acids and their salts, the polyphosphates arise by polymerizing acidic $[PO_4]$ units forming isopoly acids.

1. Consider the following oxide
I. P_4O_6 II. P_4O_8 III. P_4O_9
The oxides which yield H_3PO_3 as one of the products of hydrolysis
(a) I, II (b) II, III
(c) I, III (d) all of these

2. Consider the following oxyocids of phosphorus
I. H_3PO_2 II. $H_4P_2O_6$
III. $H_4P_4O_{12}$ IV. $H_6P_4O_{13}$
Acids which can act as reducing agents are
(a) I only (b) IV only (c) I, III (d) I, II, IV

3. Consider the following acids
I. H_3PO_2 II. H_3PO_3 III. H_3PO_4
The order of acidic strength is
(a) I > II > III (b) III > II > I
(c) I = II = III (d) III > II = I

Passage-XII

An inorganic compound (A) that is transparent like glass is a strong reducing agent. It hydrolyzes in water to give white turbidity (B). Aqueous solution of A gives white ppt(C) with NaOH which is soluble in excess NaOH. (A) reduces I_2 and give chromyl chloride test

1. A will be
(a) $Sn(OH)_2$ (b) $SnCl_2$
(c) $SnCl_4$ (d) Na_2CO_3

2. Its hydroxide react with NaOH and produce a compound. What will be the compound and how many moles of NaOH it will consume?
(a) Sn(OH)Cl, 2 (b) $SnCl_4$, 1
(c) Na_2SnO_2, 2 (d) $SnCl_2$, 2

3. A reduces I_2 and gives chromyl chloride test. What will be the product?
(a) $SnCl_2$ + HI (b) $SnCl_4$ + HI
(c) Sn(OH)Cl + HCl (d) none of these

Matching Type Questions

1. Match the following List-I with List-II

List-I	List-II
(a) N_2O	(p) Contains N – N bond
(b) N_2H_4	(q) Acidic
(c) N_3H	(r) Basic
(d) N_2O_4	(s) Reducing agent
	(t) Acts as a ligand

2. Match the following List-I with List-II

List-I	List-II (One of the product)
(a) $NaNO_3 \xrightarrow{500^\circ}$	(p) NO_2
(b) $Pb(NO_3)_2 \xrightarrow{\Delta}$	(q) N_2
(c) $NaN_3 \xrightarrow{\Delta}$	(r) O_2
(d) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$	(s) Cr_2O_3

3. Match the following Column-I with Column-II

Column-I	Column-II
(a) PH_3	(p) Acidic in nature
(b) HNO_2	(q) Oxidizing agent
(c) H_3PO_3	(r) Reducing agent
(d) N_2O_5	(s) Basic in nature

4. Match the following Column-I with Column-II

Column-I	Column-II
(a) $3CuO + 2NH_3 \xrightarrow{\Delta}$	(p) PH_3
(b) $PH_4I + NaOH \xrightarrow{\Delta}$	(q) Metal
(c) $Ba(N_3)_2 \xrightarrow{\Delta}$	(r) H_3PO_4
(d) $2H_3PO_2 \xrightarrow{\Delta}$	(s) Nitrogen

5. Match the reactions in Column-I with their corresponding products in Column-II

Column-I (Reaction)	Column-II (Products)
(a) $NH_3 + NaOCl \xrightarrow[\text{glue}]{\Delta}$	(p) N_2H_4
(b) $NH_4NO_2 + NaOH \xrightarrow{\Delta}$	(q) N_2
(c) $NH_4NO_3 + Ca(OH)_2 \xrightarrow{\Delta}$	(r) NH_3
(d) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$	(s) H_2O

6. Match the following

Column-I	Column-II
(a) Para magnetic	(p) N_2O_3
(b) Two pπ-pπ bonds	(q) NO_2
(c) Brown gas	(r) NO
(d) Acidic in nature	(s) N_3H

7. Match the following

List-I	List-II
(a) N_2O	(p) Neutral
(b) NO	(q) Oxidizing agent
(c) NO_2	(r) Odd electron molecule
(d) P_4O_6	(s) Reducing agent

8.

Column-I	Column-II
(a) NO_2	(p) Disproportionate in alkaline medium
(b) N_2O_3	(q) Acidic
(c) Cl_2O	(r) Bleaching agent
(d) SO_2	(s) Can act as both oxidizing and reducing agent

9. Match the following List-I with List-II

List-I	List-II
(a) $Pb + HNO_3(\text{dil.}) \rightarrow$	(p) One of the product N-atom exhibit sp^2 hybridization
(b) $Cu + HNO_3$ (very dil.) $\rightarrow$	(q) NO
(c) $Zn + HNO_3$ (conc.) $\rightarrow$	(r) NO_2
(d) $Fe + HNO_3$ (dil.) $\rightarrow$	(s) N_2O

10. Match the following Column-I with Column-II

Column-I	Column-II
(a) Hypophosphoric acid	(p) All hydrogens are ionizable in water
(b) Pyrophosphoric acid	(q) Lewis acid
(c) Boric acid	(r) Monobasic acid
(d) Hypophosphorus acid	(s) sp^3 hybridized central atom

11. Match the following List-I with List-II

Molecules	Geometry of molecule
(a) Solid PCl_5	(p) Tetrahedral
(b) Solid PBr_5	(q) Trigonal bipyramidal
(c) Solid PF_5	(r) Octahedral
(d) Gas PCl_5	(s) Planar triangle

12. Match the following

Column-I	Column-II
(a) Thomas slag	(p) CaNCN + C
(b) Nitrolim	(q) Fertilizer
(c) Phosphorite	(r) $Ca_3(PO_4)_2 + CaSiO_3$
(d) Bones	(s) $Ca_3(PO_4)_2$

13. Match Column-I with Column-II

Column-I	Column-II
(a) $NaBH_4$	(p) Covalent linkage
(b) PCl_5(s)	(q) Ionic linkage
(c) PBr_5(s)	(r) Coordinate linkage
(d) P_4O_{10}	(s) sp^3 hybridization

14. Match the compounds in Column-I with their corresponding characteristics in Column-II.

Column-I	Column-II
(a) Nitrogen (II) oxide	(p) Gas at room temperature
(b) Nitrogen pentoxide	(q) Dimerize on cooling
(c) Nitrogen (IV) oxide	(r) Contain ionic bonds in solid state
(d) Nitrogen trioxide	(s) Can be absorbed by $FeSO_4$
	(t) Paramagnetic

Integer Answer Type Questions

1. A typical air bag contains approximately 260 g of NaN_3. The number of moles of nitrogen produced when the azide is detonated
2. No. of electrons involved in reduction per one nitric acid molecule during reduction of very dilute nitric acid to ammonium nitrate with zinc is
3. What will be the n-factor of the reactant in the following reaction?

$$(NH_4)_2\,Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$$

4. No. of dative bonds (including σ and π) in P_4O_{10} molecule
5. The number of sp^3 atoms in cyclotrimetaphosphoric acid is… … . .
6. In the reduction of very dilute nitric acid by zinc the change in oxidation number of nitrogen is
7. A colourless salt (X) on heating with NaOH gave a gas 'Q' which can also be obtained when Mg_3N_2 is treated with H_2O. When the reaction of X with NaOH is complete the resultant solution on treatment with $FeSO_4$ and conc. H_2SO_4 gave a brown-coloured ring (R) between the two layers. (X) on strong heating forms (S) and (T) which reacts together forming a dibasic acid (U) that can exist as cis- and trans- isomers. The number of atoms present in (X) is
8. The number of six membered heterocyclic rings in P_4O_{10}
9. The number of P–O–P bonds in P_4O_6
10. The number of P–P bonds in P_4 molecule
11. The number of atoms in largest possible heterocyclic ring in P_4O_{10} is
12. The number of oxygen atoms in the largest possible heterocyclic ring in P_4O_{10} is

Single Answer Questions

1. c **2.** b **3.** b **4.** c **5.** d **6.** b **7.** d
8. c **9.** d **10.** c **11.** c **12.** b **13.** a **14.** c
15. d **16.** c **17.** c **18.** d **19.** c **20.** c **21.** a
22. a **23.** d **24.** b **25.** c **26.** c **27.** c **28.** b
29. b **30.** b **31.** a **32.** a **33.** d **34.** c **35.** c
36. c **37.** b **38.** a **39.** a **40.** d **41.** d **42.** b
43. d **44.** c **45.** c **46.** b **47.** d **48.** a **49.** a
50. d **51.** c **52.** a **53.** b **54.** a **55.** a **56.** d
57. b **58.** b **59.** b **60.** b **61.** d **62.** b **63.** c
64. b **65.** c **66.** c **67.** d **68.** a **69.** b **70.** b
71. a **72.** d **73.** c **74.** a **75.** b **76.** b **77.** a
78. a **79.** d

More than One Answer Questions

1. a, b **2.** a, b, c **3.** a, b, d
4. b, c, d **5.** a, c, d **6.** b, d
7. a, b, c **8.** a, c **9.** a, b
10. b, c **11.** a, b **12.** a, b
13. a, d **14.** b, c, d **15.** a, c
16. a, b **17.** a, c **18.** a, c, d
19. b, c, d **20.** a, b **21.** a, b, c
22. a, b, c **23.** b, d **24.** a, d
25. b, c, d **26.** a, b, d **27.** a, c, d
28. a, b, d **29.** a, c, d **30.** a, b, c
31. a, c **32.** a, c, d **33.** a, b
34. a, b, d **35.** a, b, c, d **36.** b, c, d
37. a, b, c **38.** a, b, d **39.** b, c
40. a, b, c **41.** b, c, d **42.** a, b, c, d
43. c **44.** a, b, d **45.** a, b, d
46. a, b, d **47.** a, b, c, d **48.** a, b, c, d
49. a, b **50.** a, b, c **51.** a, b
52. b, c, d **53.** a, b, d

Comprehensive Type Questions

Passage I **1.** a **2.** d **3.** a
Passage II **1.** c **2.** b **3.** a
Passage III **1.** c **2.** c **3.** d **4.** b

Passage IV	**1.** c	**2.** d	**3.** b		
Passage V	**1.** d	**2.** b	**3.** c	**4.** d	**5.** a
Passage VI	**1.** c	**2.** c	**3.** d		
Passage VII	**1.** b	**2.** b	**3.** d		
Passage VIII	**1.** a	**2.** b	**3.** a		
Passage IX	**1.** b	**2.** c	**3.** c		
Passage X	**1.** a	**2.** a	**3.** a		
Passage XI	**1.** d	**2.** a	**3.** a		
Passage XII	**1.** b	**2.** d	**3.** b		

Matching Type Questions

1.	a-p, t	b-p, r, s, t	c-p, q, s	d-p, q, s
2.	a-r	b-p, r	c-q	d-q, s
3.	a-r, s	b-p, q, r	c-p, r	d-p, q
4.	a-q, s	b-p	c-q, s	d-p, r
5.	a-p	b-r, s	c-r, s	d-q, s
6.	a-q, r	b-p, s	c-q	d-p, q, s
7.	a-p, q	b-p, b, r, s,	c-q, r, s	d-s
8.	a-p, q, s	b-s	c-p, q, r, s	d-q, r, s
9.	a-q	b-p	c-p, r	d-p, s
10.	a-p, s	b-p, s	c-q, r	d-r, s
11.	a-p, r	b-p	c-q	d-q
12.	a-q, r	b-p, q	c-s	d-s
13.	a-p, q, r, s	b-p, q, r, s	c-p, q, s	d-p, r, s
14.	a-p, q, s, t	b-r	c-p, q, t	d-p

Integer Questions

1. 6	**2.** 4	**3.** 6	**4.** 8	**5.** 9	**6.** 8	**7.** 9
8. 4	**9.** 6	**10.** 6	**11.** 8	**12.** 3		

Hints

Multiple Choice Questions with Only One Answer

1. $NCl_3 + 3H_2O \rightarrow NH_3 + 3HOCl$

$BCl_3 + 3H_2O \rightarrow H_3BO_3 + 3HCl$

2. I. $Zn + 4HNO_3 \rightarrow Zn(NO_3)_2 + 2NO_2 + 2H_2O$

II. $4Zn + 10HNO_3 \rightarrow 4Zn(NO_3)_2 + N_2O + 5H_2O$

$NH_4NO_3 \xrightarrow{KOH} NH_3 + KNO_3$

$KNO_3 \xrightarrow{Zn + kOH} K_2ZnO_2 + H_2O + NH_3$

$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

5. N_2O supports combustion

8. $\underset{a}{N_2O} + \underset{b}{2NO} + 3H_2 \longrightarrow 2N_2 + 3H_2O$

$$a + \frac{b}{2} \longrightarrow 38$$

$$\frac{a}{2} + \frac{b}{2} = \frac{60}{2}$$

$$\Rightarrow \frac{a}{2} = 8 \quad \text{or} \quad a = 16$$

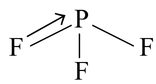
12. $NH_4NO_3 \longrightarrow x \quad (NH_4)_2\,HPO_4 \longrightarrow 100 - x$

$$\therefore x\left(\frac{28}{80}\right) + (100 - x)\left(\frac{28}{132}\right) = 33.81$$

$\therefore (NH_4)_2\ HPO_4 = 8.63\%$

13. Down the group metallic character increases. During hydrolysis the base produced by the metals participate in reaction with acid formed in the same reaction. So, hydrolysis becomes difficult down the group.

14. In PF_3, the lone pair of F atom participate in $p\pi$–$d\pi$ bonds due to which double bond character arises. A



15. Due to resonance hybridization of the following structures the N—N distance will become intermediate to N—N double bond and triple bond length

$$N \equiv N \longrightarrow O \longleftrightarrow \overset{-}{N} = \overset{+}{N} = O$$

17. Bones contain $Ca_3(PO_4)_2$. While phosphorus in bones undergoes slow oxidation in contact with air, light emitted is called phosphorescense

20. $3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$

$3SnCl_2 + 6HCl + 2NO \rightarrow 3SnCl_4 + 2NH_2OH$

$NH_2OH + HNO_2 \rightarrow N_2O + 2H_2O$

21. Since F is more electronegative the bond pair move away from nitrogen in NF_3 than in NH_3. When compared to NF_3 in PF_3 F attracts the bond pair more.

22. $P_4O_{10} + 6PCl_5 \rightarrow 10POCl_3$

23. In NF_3 both N and F atoms have no d-orbitals in their valence shells.

26. $P_4 + 3O_2 \rightarrow P_4O_6$

$P_4O_6 + 6H_2O \rightarrow 4H_3PO_3$

$2H_3PO_3 + O_2 \rightarrow 2H_3PO_4$

$H_3PO_4 + 3AgNO_3 \rightarrow \underset{\text{Yellow}}{Ag_3PO_4\downarrow} + 3HNO_3$

33. $NaH_2PO_4 \xrightarrow{\Delta} NaPO_3 + H_2O$

$Na_2HPO_4 \xrightarrow{\Delta} Na_4P_2O_7 + H_2O$

$Cu(OH)_2 \xrightarrow{\Delta} CuO + H_2O$

$Na_3PO_4 \xrightarrow{\Delta}$ No change.

34. The P—O bond order in PO_4^{3-} is 1.25

36. $2H_3PO_4 \xrightarrow{H_2O} H_4P_2O_7 \xrightarrow{825} 2HPO_3 + H_2O$

37. $2Pb(NO_3)_2 + 2H_2SO_4 \rightarrow 2PbSO_4$ White $\downarrow + 4NO_2 + 2H_2O + O_2$

$PbSO_4 + 2CH_3COONH_4 \rightarrow Pb(CH_3COO)_2 + (NH_4)_2SO_4$ Soluble.

38. $PCl_5 + H_2O \xrightarrow{1:1} POCl_3 + 2HCl$

$PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$

$PCl_5 + AlCl_3 \rightarrow [PCl_4]^+ [AlCl_4]^-$

$3PCl_5 + 3NH_4Cl \rightarrow N_3P_3Cl_6 + 12HCl.$

39. $Cu + N_2O \rightarrow CuO + N_2$

40. The N—N bond is weaker and hence N_2O_4 easily converts into NO_2.

41. In the resonance structures of N_2O $N \equiv N \rightarrow O$ and $N \rightleftarrows N = O$ the N—N bond order is more than N—O bond order.

43. $N(SiH_3)_3$ is planar triangular due to back bonding. In NF_3 due to the shifting bond pairs towards more electronegative F bond angle decreases.

45. Oxidation power of HNO_3 is enhanced by complex-forming ability of Cl^- ion in aquaregia.

47. White phosphorus changes to red phosphorus on heating at 250º C and high pressure using I_2 as a catalyst but not red phosphorus to white phosphorus.

48. N_2 is non-polar but other species are polar.

49. $NH_4NO_3 \rightarrow N_2O + 2H_2O$

52. In the formation of P = O, a σ dative bond by donating lone pair on P to oxygen and oxygen form a π dative bond by donating its lone pair to the vacant d-orbital of phosphorus.

53. $BiCl_3 + H_2O \rightarrow BiOCl + 2HCl$

54. N—N bond energy is less due to repulsion by lone pairs present on nitrogen atoms which are close to one another due to small size.

55. NO is a neutral oxide. When compared with carbon dioxide nitrogen oxides are more acidic due to more non-metallic character. In the oxides of same element acidic character increase with increase in oxidation number.

57. The bond orders in NO_3^-, NO_2^- and NO_2^+ are 1.33, 1.5 and 1.5, respectively but due to positive charge on NO_2^+ attraction on bond pairs increases bond length actually decreases

58. H_2CO_3 is unstable, so CO_2 is liberated

59. $PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$

$AsCl_3 + 3H_2O \rightarrow H_3AsO_3 + 3HCl$

60. In very dilute HNO_3 Zn reduces two moles of nitric acid to one mole of NH_4NO_3. So, the oxidation state +5 in HNO_3 changes to –3 in ammonia.

61. Oxidation of HCN with oxygen using silver as a catalyst produces cyanogen which can also be prepared by the action of CN^- with Cu^{2+}

$$Cu^{2+} + 2CN^- \rightarrow Cu(CN)_2$$

$$2Cu(CN)_2 \rightarrow Cu_2(CN)_2 + (CN)_2$$

62. NO often acts as a three-electron donor. For example, when it reacts with carbonyl compounds three CO molecules can be substituted by two NO molecules.

63. N_3^- has (3 × 5) + 1 = 16 outer electrons and is isoelectronic with CO_2 which contain

(2 × 6) + 4 = 16 outer electrons

64. $3PCl_5 + 3NH_4Cl \rightarrow N_3P_3Cl_6 + 12HCl$

66. The substance which increases the concentration of a cation in a solution is called acid

$$N_2O_4 \rightarrow NO^+ + NO_3^-$$

$$NOCl \rightarrow NO^+ + Cl^-$$

67. Ionic nitrates are more stable than covalent nitrates, the remaining statements are correct.

68. N_2H_4 contains two lone pairs on two nitrogen atoms and can act as a bifunctional base or diacidic base.

69. The acidic strength of an oxoacid depends on the inductive effect of unprotonated oxygen atoms. If the number of unprotonated oxygen atoms increases inductive effect increases and hence the acidic character. Since all the three acids H_3PO_2, H_3PO_3 and H_3PO_4 have the same number of unprotonated oxygen atoms their acidic strength is also nearly the same.

70. $NH_3 + NaOCl \xrightarrow{\text{glue}} NH_2Cl + NaOH$

$2NH_3 + NH_2Cl \rightarrow N_2H_4 + NH_4Cl$

71. $P_4 + 3O_2 \xrightarrow{\text{limited } O_2} P_4O_6$

$P_4O_6 + 4Cl_2 \rightarrow 2POCl_3 + 2PO_2Cl.$

72. $POCl_3$ molecule contains pπ – dπ bond.

73. $8NH_3 \cdot NI_3 \rightarrow 5N_2 + 9I_2 + 6NH_4I$

74. White phosphorus reacts with Cl_2 directly but not red P.

75. $NH_4NO_3 \rightarrow N_2O + 2H_2O$.
Since no oxygen is liberated it will not rekindle the glowing splinter. But other nitrates on heating give O_2.

Multiple Choice Questions with One or More than One Answer

1. $Mg^{2+} + NH_4OH + NH_4Cl + Na_2HPO_4 \rightarrow Mg(NH_4)PO_4 + 2Na^+ + NH_4Cl + H_2O$

5. $2NO_2 + 2NaOH \rightarrow NaNO_2 + NaNO_3 + H_2O$
$4NaNO_3 \rightarrow 2Na_2O + 2N_2 + 5O_2$
$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$.
NH_3 and NaOCl react to produce hydrazine. But not N_2

7. Solid PCl_5 contains PCl_4^+ and PCl_6^- ions
Solid PBr_5 contains PBr_4^+ and Br^- ions
Solid N_2O_5 contains NO_2^+ and NO_3^- ion
Phosphorus oxides are dimers of P_4O_6 and P_4O_{10}

8. The brown colour of brown ring is due to the formation of the complex $[Fe(H_2O)_5NO]^{2+}$ in which electron transfer takes place from NO to Fe^{2+} is called charge transfer.

15. $NO[BF_4]$ contains NO^+ and BF_4^- ions. There are one sigma bond in NO and four sigma bonds in BF_4^- ions. When NO is converted into NO^+ ion, bond order increases to 3 and bond length decrease.

16. In NF_3 the electron density on nitrogen decreases due to withdrawl by more electronegative fluorine and hence weaker base than NH_3 in NO^+ the bond order is 3 while in $\overset{+}{O_2}$ bond order is 2.5.

20. $PbO_2 + 4HNO_3 \rightarrow Pb(NO_3)_2 + 2H_2O + O_2$
$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$

26. $4NH_3 + 5O_2 \xrightarrow{\text{Pt cat}} 4NO + 6H_2O$

$$Hg_2Cl_2 + 2NH_3 \longrightarrow \underbrace{Hg + NH_2HgCl + NH_4Cl}_{\text{Black}}$$

$NH_3 + HCl \longrightarrow NH_4Cl$ white fumes

21. Covalency is the number of electron pairs shared. So, in HNO_3 covalency of nitrogen is 4 structure of HNO_3.as in the oxidation state of N in HNO_3 is +5. Gold is insoluble in nitric acid but dissolves in aquaregia.

28. $LiNO_3$, $Mg(NO_3)_2$ and $Ca(NO_3)_2$ on decomposition gives metal oxides NO_2 and O_2 but $NaNO_3$ gives $NaNO_2$ and O_2.

30. P_4 (white) $+ 3NaOH + 3H_2O \rightarrow 3NaH_2PO_2 + PH_3$

$$2H_3PO_2 \xrightarrow{\Delta} H_3PO_4 + PH_3$$
$$4H_3PO_4 \xrightarrow{\Delta} 3H_3PO_4 + PH_3$$

31. Nitrogen being a more electronegative element can take up the electrons easily and hence in +5 oxidation state acts as a good oxidizing agent. Statement 3 is correct.

Comprehensive Type Questions

Passage-I

$$4NH_3 + 5O_2 \xrightarrow[900^\circ C]{\text{Pt catalyst}} 4NO + 6H_2O.$$
$$2NO + O_2 \longrightarrow 2NO_2$$
$$3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$$

Passage-III

1. Since nitrates are soluble in water and also due to absorption by plants they exist in a low level in the earth crust.
2. NH_3 is a better donor than PH_3 because the electron pair is distributed in a smaller area.
3. Except Mg and Mn all the other metals do not liberate H_2 from HNO_3.
4. Bond order in both NO_3^- and CO_3^{2-} is 1.33.

Passage-V

1. In PF_3 the electron density at phosphorus increases due to back bonding from fluorine to phosphorus. So, PF_3 acts as a strong ligand.
2. In trialkyl phosphine complexes there is a synergic effect due to the back π bonding from metal to the vacant 'd' orbitals of phosphorus of trialkyl phosphines.
3. In SbF_5^{2-} there are six electron pairs arranged in an octahedral manner, but one electron pair is lone pair. So, it has a square pyramidal structure, whereas SbF_5 has the original bipyramid structure.
4. In $R_3P{=}O$ there is back bonding from oxygen to phosphorus which is absent in $R_3\overset{+}{N}-\overset{-}{O}$ $R_3P{=}O$ and have more number of resonate structures also.
5. The small nitrogen can form strong $p\pi - p\pi$ bond with oxygen but phosphorus cannot. So, phosphorus forms single bonds in cyclic compounds.

Passage-VIII

$$PCl_5 + 5O_2 \longrightarrow \underset{A}{POCl_3} + \underset{B}{SOCl_2}$$
$$POCl_3 + 3H_2O \longrightarrow H_3PO_4 + 3HCl$$
$$\underset{C}{4H_3PO_4} \xrightarrow{\text{Red hot}} \underset{D}{P_4O_{10}} + 6H_2$$
$$\underset{D}{P_4O_{10}} + 6PCl_5 \longrightarrow 10POCl_3$$

Integer Type Questions

1. $2NaN_3 \longrightarrow 2Na + 3N_2$

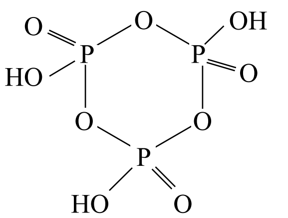
130 3 moles

260 6 moles

2. Zinc reduces very dilute nitric acid to ammonium nitrate. Two moles of HNO_3 converts into NH_4NO_3 in which N in +5 oxidation state changes to –3. So, 8 electrons are involved for two molecules. For one molecule, therefore, there are 4 electrons.

3. In P_4O_{10}, each phosphorus is in double bond with one oxygen of which one is a sigma dative bond from $P \rightarrow O$ and a π dative bond from O to P. So, there are 4σ dative bonds and 4π dative bonds in $4P = O$ bonds (total 8).

5. 3'P' atoms, 3 'O' atoms in ring and 3 'O' atoms in OH groups are in sp^3 hybridization.



7. The compound X is NH_4NO_3. Number of atoms in NH_4NO_3 is 9. The reactions are as follows

$NH_4NO_3(X) + NaOH \rightarrow NaNO_3 + H_2O + NH_3(g)$

$Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3\ (g)$

$NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$

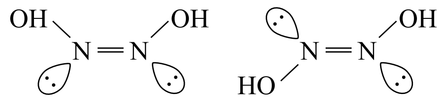
$2HNO_3 \rightarrow H_2O + 2NO + 3(O)$

$6FeSO_4 + 3H_2SO_4 + 3(O) \rightarrow 3\ Fe_2(SO_4)_3 + 3H_2O$

$[Fe\ (H_2O)_6]SO_4 + NO \rightarrow \underset{\text{Brown complex (R.)}}{[Fe(H_2O)_5\ NO]\ SO_4} + H_2O$

$$NH_4NO_3 \xrightarrow{\Delta} \underset{S}{N_2O} + \underset{T}{2H_2O}$$

$$N_2O + H_2O \longrightarrow H_2N_2O_2$$



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CHAPTER 10

Group VIA (16) Oxygen Family

Quickly o! Dame fire that I may burn sulphur, the cure of ills
The Odyssey of Homer

10.1 INTRODUCTION

Group VIA consists of five elements namely oxygen (O), sulphur (S), selenium (Se), tellurium (Te) and polonium (Po). The first four elements are non-metallic in character and collectively called as *chalogens* (meaning ore-forming elements) because many metals occur in nature as oxides, sulphides or selenides. The last element polonium is a radioactive element with a very short half-life and therefore, very little is known about the chemistry of this element.

10.2 OCCURRENCE

Oxygen is the most abundant of the elements. It makes up 46.6 per cent of the earth's crust in the form of oxides and oxo salts. It occurs to 21 per cent by volume in air and to ~ 86 per cent by weight in oceans. Sulphur is less abundant and makes up ~ 0.04 – 0.03 per cent of the earth's crust. It occurs in the native state in considerable amount and also in the form of metal sulphides and metal sulphates. Selenium occurs in the trace quantities (6×10^{-5} %) and accompanies sulphide ores as metal selenides. Tellurium and polonium are comparatively rare and constitute 10^{-7} and 10^{-4} per cent in the earth's crust, respectively.

Electronic Configuration

The electronic configurations of elements of group VIA are given in Table 10.1.

From the Table 10.1, it can be seen that all these elements have $ns^2\ np^4$ electronic configuration. In three p-orbitals, four electrons are arranged as $p_x^2\ p_y^1\ p_z^1$. Thus, two half-filled p-orbitals are available which are responsible for chemical bonding with other elements.

Further like other groups, the first element oxygen is not having any d-orbitals, while all the other elements of this group are having vacant d-orbitals and hence oxygen is different from other elements of the group in several respects.

10.3 GENERAL PROPERTIES

Some important atomic and physical properties of Group VIA elements are given in Table 10.2.

1. ***Atomic radii***: In group VIA, there occurs an increase in size with increasing atomic number due to the effect of extra shells of electrons being added, this outweighs the effect of increased nuclear charge. As usual in p-block elements, there is large increase in size from first element to

Table 10.1 Electronic configurations of elements of group VIA

Element	Atomic number	Electronic configuration	Configuration of valence shell
Oxygen	8	$1s^2\ 2s^2\ 2p^4$	$2s^2\ 2p^4$
Sulphur	16	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$	$3s^2\ 3p^4$
Selenium	34	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^4$	$4s^2\ 4p^4$
Tellurium	52	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^4$	$5s^2 5p^4$
Polonium	84	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}$ $4f^{14}\ 5s^2\ 5p^6\ 5d^{10}\ 6s^2\ 6p^4$	$6s^2\ 6p^4$

Table 10.2 Some important atomic and physical properties of the elements of group VIA

Property	O	S	Se	Te	Po
Atomic weight	15.999	32.054	78.96	127.6	210
Melting point(K)	55	392(Monoclinic)	490(Grey)	725(Metallic)	520
Boiling point(K)	90	718	958	1260	1235
Heat of atomization (KJ mol^{-1})	284.64	238.98	208.32	196.14	–
Density in solid state (g cm^{-3})	1.14	2.07	4.19	6.24	(9.2)
Atomic radius (pm)	66	104	116	135	
Ionic radius (pm) of divalent ion. E^{2-}	140	184	198	221	–
Atomic Volume (c.c)	14.0	15.5	16.5	20.5	22.7
Ionization energy (KJ mol^{-1})	1314	1000	941	869	813
Electron gain enthalpy (Δeg H KJ)	–141	–200	–195	–190	–174
Electronegativity	3.50	2.44	2.48	2.01	1.76

second element i.e., from oxygen to sulphur but from then onwards the increase in size is little. This is due to poor shielding effect of d-electrons in selenium and tellurium and d-and f-electrons in polonium.

2. *Ionic radius*: In the formation of anions in this group, two electrons are being added to an atom. Therefore the effective nuclear charge is reduced and hence the electron cloud expands. Thus the negative ions are bigger than the corresponding atoms.

3. *Density*: With the increase in the atomic number, there is corresponding increase in the size of the atoms and ions and hence the atomic volume increases. But at the same time, the atomic mass also increases which outweighs the increase in atomic volume. So the density in the solid state increases with increase in atomic number.

4. *Physical State*: The first member of this group i.e., oxygen is a gas where as others are solids. This is due to the fact that oxygen exist as small diatomic molecules between which the van der Waal's forces are weak. Sulphur, and selenium exist as staggered 8 atoms rings or zig-zag chains. However, the tendency to exist in 8-atom rings become greatest with sulphur and gets decreased as we go down the group. Tellurium has hexagonal rings.

Existence of oxygen as a diatomic molecule is ascribed to the ability of small oxygen atom to form stable pπ–pπ multiple bond while sulphur and other elements of the group are not capable of pπ–pπ overlap due to bigger atomic sizes. On the other hand, the special stability of the eight-membered chain in sulphur is because of the sp^3 hybridization of sulphur atom which is involving both bonding and non-bonding pairs of electrons. The great ability of sulphur, selenium and tellurium (and not that of oxygen) to form homoatomic chains gets reflected in single bond energy values between the similar atoms e.g., O—O 145.8KJ mol^{-1} and S—S 267 KJ mol^{-1}.

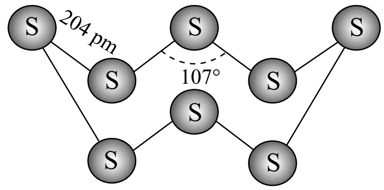


Fig 10.1 Puckered ring structure of sulphur

5. *Melting and boiling points*: Melting and boiling points increase steadily down the group VIA. This trend is consistent with the increasing van der Waal's forces in the non-polar molecules with increase in molecular sizes.

There is a large difference in melting and boiling points between oxygen and sulphur. This has been attributed, in part to the fact that oxygen exists in the form of O_2 molecule while sulphur in S_8 molecules. Further, the melting and boiling points of polonium are less than those of tellurium. This has been attributed to the diminished availability of s-electron pair.

6. *Ionization energies*: The elements of group VIA elements have high values of ionization energies. This reveals that the formation of positive ions is extremely difficult and the elements are predominantly non-metallic in character. However, the ionization energy decreases down the group as the size of the atom increases. Due to this metallic character increases while ability to act as oxidizing agent decreases down the group.

7. *Electronegativity*: Electronegativity decreases gradually from oxygen to polonium. This decreasing value also indicates change from non-metallic to metallic character.

8. *Metallic and non-metallic character*: The metallic character gets increased with increase in atomic number. For example oxygen and sulphur are strongly non-metallic. Selenium is also non-metal, tellurium is lesser non-metallic with some metallic character, where as polonium is metal.

9. ***Oxidation States***: From Table 10.1, it is evident that all the elements of group VIA have ns^2 np^4 electronic configuration in their valence shell. Being very nearer to the noble gases, they tend to attain stable inert gas configuration by gaining two electrons and form chalcogenide ions like O^{2-}, S^{2-}, Se^{2-} and Te^{2-} or by sharing the two electrons thus forming two covalent bonds. But oxygen is most electronegative in this group. Therefore, most of the oxides are ionic and contain O^{2-} ions, this giving an oxidation state of –2. However, sulphur, selenium and tellurium being less electronegative than oxygen form compounds with the most electropositive elements which are not more than 50 per cent ionic. Hence there is less chance of formation of S^{2-}, Se^{2-} and Te^{2-} ions. These elements also form covalent compounds having two covalent bonds e.g., H_2O, Cl_2O, SCl_2, H_2S etc.

As the electronegativities and ionization energies decreases down the group VIA, the tendency of the elements to exhibit –2 oxidation state decreases from sulphur to polonium. However, they exhibit +2 oxidation state.

As the electronegativity of oxygen is very high, second only to fluorine, it shows only a negative oxidation state. It does not show positive oxidation state, except in the compounds of fluorine. So oxygen exhibit –2 oxidation state in almost all compounds except in oxygen fluorides, peroxides and super oxides. It exhibit +1 in O_2F_2, +2 in OF_2, –1 in peroxides and –1/2 in super oxides. In the case of remaining elements the electronegativities decrease down the group. Therefore the tendency to exhibit positive oxidation state become more common down the group. From the electronic configuration ns^2np^4 it is evident that these elements might exhibit three positive oxidation states +2, +4 and +6. The +2 and +4 oxidation states are expected when only the p-electrons are involved while +6 state is expected when s – as well as p – electrons are involved

	Outer electronic configuration			Oxidation state
Ground state	↑↓	↓↑ ↑ ↑	□ □ □ □	–2 or +2
Ist excited state	↑↓	↑ ↑ ↑	↑ □ □ □	+4
IInd excited state	↑	↑ ↑ ↑	↑ ↑ □ □	+6

In the case of oxygen, its second shell is limited to 8 electrons because there are no d – orbitals in this shell. Further too much energy is required to excite an electron into higher shell therefore, it is never more than divalent.

The remaining elements, S, Se, Te and Po have vacant d – orbitals in their valence shells. Therefore they can also show +4 and +6 oxidation states.

The ability to exist in +6 and +4 oxidation state decreases down the group. This is due to gradual increase in atomic radius and decrease in electronegativity. In the case of last element polonium, +4 oxidation state is more evident than +6 oxidation state due to inert pair effect.

10. ***Nature of bonding***: Oxygen and sulphur form ionic compounds in –2 oxidation state with most electropositive metals of Groups IA and IIA. Majority of the compounds of all these elements are covalent in almost all the oxidation states except in the compounds of alkali and alkaline earth metals.

11. ***Participation of d-orbitals in sigma bonding***: As explained in chapter 3 (Chemical Bonding 3.8.4), the size and energy of the d-orbitals are sensitive functions of both the oxidation state (positive charge developed on the atom) and the electronic configuration. When sulphur is in bond with highly electronegative fluorine atoms, the positive charge developed on sulphur atom will be more due to which they contract and their energy decreases due to coming close to the nucleus. Then their energies are comparable with the energies of p-orbitals and can participate in hybridization like sp^3d or sp^3d^2 which can participate in bonding as in SF_4, SF_6 etc. Similarly the d-orbitals in selenium and tellurium also can participate in bonding in the formation of SeF_4, TeF_4, $SeCl_6$, $TeCl_6$ etc.

12. ***Participation of d – orbitals in π – bonding***: For sulphur particularly, as in other third-row elements, there is multiple dπ–pπ bonding, but little if any pπ–pπ bonding. The short S-O distances in SO_4^{2-} and related compounds (where s and p-orbitals are used for σ – bonding) is a result of multiple dπ–pπ bond character. This dπ–pπ bonding arises from the flow of electrons from filled pπ orbitals on 'O' atoms to empty dπ orbitals on S atoms.

13. ***Catenation***: Sulphur has a strong tendency to catenation, equalled or exceeded only by carbon. Sulphur forms compounds for which there are no known oxygen, selenium, tellerium analogs. Oxygen exists as diatomic gaseous molecules S, Se and Te exist as S_8, Se_8 and Te_8 rings or chains. The tendency of catenation is maximum in sulphur and decrease down the group. Sulphur shows chain formation specially the compounds of the type polysulphide ions S_n^{2-}, polythionate ions $O_3S - S_n - SO_3$ and compounds of the type X S_n X where X = H, Cl, CN or NR_2

14. ***Allotropy***: All the elements in Group VIA exhibit allotropy. Ozone is considered as the allotrope of oxygen. Sulphur exists in a large number of allotropic forms such as rhombic (or) α sulphur, monoclinic (or) β– sulphur, γ–sulphur, plastic sulphur, sulphur flowers, engel sulphur etc. Selenium has six forms, of which four are red varieties and two grey varieties. Of these six forms, the most stable allotropic form is the metallic grey variety form. Tellurium do not exhibit allotropy. It has only one crystalline form. This is isomorphous with the grey form of selenium. But the allotrope of tellurium has stronger metallic interactions. Polonium also exists in two forms α – and β – forms, both are metallic.

10.4 CHEMICAL REACTIVITY OF THE ELEMENTS

Similar to nitrogen, oxygen, the first member of the group, inspite of its high electronegativity, relatively inactive gas at room temperature. This is ascribed to the fact that in oxygen molecule the oxygen atoms get linked together by a double bond which being shorter (121 Pm) than the oxygen – oxygen single bond, becomes very strong (dissociation energy = 490.31 KJ mol^{-1}). However, oxidation reactions (reactions of oxygen) becomes highly exothermic, once initiated, they proceed vigorously without further application of heat. Important chemical reactions are given as follows.

1. *Reaction with metals*: Oxygen, sulphur and selenium react with metals to form oxides, sulphides and selenides respectively. For instance

$$4Na + O_2 \longrightarrow 2Na_2O$$
$$Cu + S \longrightarrow CuS$$

2. *Reaction with non-metals*: Oxygen and sulphur react with non-metals to form oxides and sulphides respectively e.g.,

$$S + O_2 \longrightarrow SO_2$$
$$C + O_2 \longrightarrow CO_2$$
$$C + 2S \longrightarrow CS_2$$

3. *Reaction with hydrogen*: Oxygen and sulphur react with hydrogen to form oxide and sulphide respectively e.g.,

$$2H_2 + O_2 \longrightarrow 2H_2O$$
$$H_2 + S \longrightarrow H_2S$$

4. *Action of air*: Sulphur, selenium and tellurium burn in air to form dioxides SO_2, SeO_2 and TeO_2.

5. *Reactions with halogens*: Sulphur, selenium and tellurium combine energetically with fluorine to yeild hexa fluorides XF_6 and with chlorine to give tetrachlorides XCl_4.

6. *Action of acids*: Sulphur, selenium and tellurium get attacked by hot nitric and sulphuric acids but not by non-oxidizing acids like HCl and HF.

$$2H_2SO_4 + S \longrightarrow 2H_2O + 3SO_2$$
$$6HNO_3 + S \longrightarrow 2H_2O + H_2SO_4 + 6NO_2$$

$$2H_2SO_4 + Se \longrightarrow 2H_2O + 2SO_2 + SeO_2$$
$$4HNO_3 + Se \longrightarrow 2H_2O + H_2SeO_3 + 4NO_2$$

$$2H_2SO_4 + Te \longrightarrow 2H_2O + 2SO_2 + TeO_2$$
$$4HNO_3 + Te \longrightarrow H_2O + H_2TeO_3 + 4NO_2$$

7. *Action of alkalis*: Sulphur and selenium get attacked by hot concentrated alkalis to form sulphides and selenides which if treated with excess of sulphur and selenium yield polysulphides and polyselenides respectively.

$$4S + 6KOH \longrightarrow K_2S_2O_3 + 2K_2S + 3H_2O$$
$$K_2S + 4S \longrightarrow K_2S_5$$

10.5 COMPOUNDS OF GROUP VIA ELEMENTS: COMPARATIVE STUDY

10.5.1 Hydrides

All the elements of this group form volatile bivalent hydrides of the type H_2R. Oxygen form another hydride hydrogen peroxide. Sulphur also form the other hydrides H_2S_2 and H_2S_n where n = 2 to 10.

***Preparation*:** Hydride of oxygen, H_2O is obtained by burning hydrogen in the atmosphere of oxygen. On the other hand, hydrides of S, Se and Te are best prepared by the action of acids on metal sulphides, selenides and tellurides. However, H_2Po is prepared only in trace quantities by dissolving magnesium foil plated with Po in 0.2 N HCl.

$$2H_2 + O_2 \longrightarrow 2H_2O$$
$$FeS + H_2SO_4 \longrightarrow FeSO_4 + H_2S$$
$$K_2Se + H_2SO_4 \longrightarrow K_2SO_4 + H_2Se$$

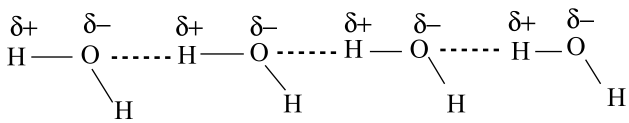
Properties

(i) With the exception of water, all the hydrides are colourless, poisonous gases having unpleasant odours. Water on the other hand, is a well-known colourless, non-toxic liquid.

(ii) ***Volatility*:** The volatility increases markedly from H_2O to H_2S and then decreases. This can be seen from their boiling points given below.

Hydride	H_2O	H_2S	H_2Se	H_2Te
B.Pt (K).	373	212.25	231.5	271.2

The abnormally high boiling point of water has been attributed to hydrogen bonding which results in association of a number of water molecules. Hydrogen bond formation in H_2S is always weak. Therefore, H_2S gas is non-associated and is highly volatile. However, volatile nature decreases from H_2S to H_2Se and H_2Te due to the increasing molecular weight of the hydrides.



(iii) All these hydrides are covalent and the covalent character increases from H_2O to H_2Te as the electronegativity difference between hydrogen and chalcogen decreases from oxygen to tellurium. Order of covalent character is $H_2O < H_2S < H_2Se < H_2Te$.

(iv) Polarity of the molecules also decreases from H_2O to H_2Te.

(v) ***Acidic nature***: Aqueous solutions of hydrides of the group behave as weak diprotic acid which dissociates to varying degree to give H^+ ions.

$$H_2S \rightleftharpoons H^+ + HS^-$$

$$HS^- \rightleftharpoons H^+ + S^{2-}$$

The acidic strength increases from H_2O to H_2Te. This is evident from dissociation constants given below.

Hydride	H_2O	H_2S	H_2Se	H_2Te
Ka_1	1×10^{-14}	1×10^{-7}	1×10^{-4}	2.3×10^{-3}

The increasing trend in acidic strength can be explained in two ways, but cannot be explained on the basis of electronegativities.

(a) On the basis of fall in dissociation energies: When the size of atom gets increased the bond between hydrogen and the chalcogen becomes weakened and hence the release of proton gets facilitated. This is also according to the increasing bond length between hydrogen and chalcogen (O to Te).

(b) On the basis of charge densities of anions (conjugate base): With increase in the size of the anion charge distribution takes place in larger area and hence the charge density decreases. The charge densities of these anions tend to follow the order

$$\xrightarrow[\text{Increasing size. Decreasing charge density}]{O^{2-} > S^{2-} > Se^{2-} > Te^{2-}}$$

With decrease in charge density, stability of the anion increases. Thus oxide ion (O^{2-}), being the smallest, exhibits the maximum resistance to dissociation i.e., it is having the greatest affinity for proton and hence acts as a strongest base. Thus water is the weakest acid while H_2Te is the strongest acid among these hydrides.

(vi) ***Thermal stability***: Thermal stability of these hydrides has been found to decrease with rise in molecular weight. Thus water undergoes dissociation only at 2073k where as tellurium hydride undergoes decomposition even at ordinary temperatures. As we move down the group, the size of the chalcogen atom increases. So the bond energy decreases with increase in bond length. Hence thermal stability decreases from H_2O to H_2Te.

(vii) ***Reduction power***: Owing to the decrease in the thermal stability from H_2O to H_2Po the reducing character increases accordingly.

(viii) ***Structure***: All these hydrides are angular in shape. From the table 10.3 it is evident that bond angles in H_2S, H_2Se and H_2Te are much smaller than in H_2O. This trend has been explained on the basis of VSEPR theory by taking into account the large size and lower electronegativity of S, Se and Te than that of oxygen. However, this can be explained in the following two ways.

(a) On the basis of decreasing electronegativity of the central atom: When we proceed from H_2O to H_2Te, the bonding electrons in EH_2 get drawn further away from the element E due to the decreasing electronegativity of E. This gives rise to decrease in bond pair – bond pair repulsion and thereby allowing the closing up of bond angles. This also gives rise to a decrease in the lone-pair – bond pair repulsions.

(b) On the basis of decreasing tendency of sp^3 hybridization: In water, the central atom, oxygen is sp^3 hybridized; the slight distortion from the normal tetrahedral bond angle of 109º 28´ is because of lone pair of electrons. In other elements, the electronegativities decrease and sp^3 hybridization gets less and less distinct with the increasing size of the electron clouds. In other words if we calculate the per cent character by using bond angles as explained in chapter 3 (chemical bonding 3.8.6) it will be nearer to 100 per cent p. So we can expect pure p orbitals are involved in bonding.

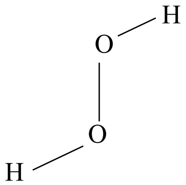
Hydrides of the formula H_2R_2: The oxygen and sulphur only form hydrides of the type H_2R_2 (viz H_2O_2, H_2S_2). These two hydrides behave differently. H_2O_2 is fairly stable except in the presence of certain impurities like Cu, Pt etc where as H_2S_2 is comparatively less stable and undergoes decomposition readily to yield sulphur and hydrogen sulphide. The decomposition has been accelerated by OH^- ions.

Table 10.3 Important comparative properties of hydrides of elements of group VIA

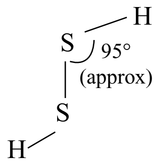
Property	H_2O	H_2S	H_2Se	H_2Te
1. Melting point (k)	273	187.5	207.3	221.9
2. Boiling point (k)	373	212.25	231.5	271.5
3. Bond angle	104.5º	92.5º	90º	90º
4. Bond length H—X(pm)	95	130	145	172
5. Dissociation constant as an acid at 298k	1×10^{-14}	1×10^{-7}	1×10^{-4}	2.3×10^{-3}
6. Heat of formation	–242	–20.2	85.8	154.4
7. Heat of vaporization (KJ. mol^{-1})	40.8	18.7	19.9	24.0

H_2O_2 acts as a strong oxidizing agent while H_2S_2 does not.

Due to hydrogen bonding, H_2O_2 is highly associated while H_2S forms discrete molecules. Both hydrides are having similar non-planar structures as shown Fig 10.2.



Dihedral angle 111.5° gas
O–O bond length 147 pm



Dihedral angle ≈90°
S–S bond length 205 pm
S–H bond 133 pm length

Fig 10.2 Structures of H_2O_2 and H_2S_2

10.5.2 Halides

The elements of Group VIA have been known to form many halides which are listed in Table 10.4.

As fluorine is more electronegative than oxygen, its compounds with oxygen are known as **fluorides**. Thus, it is more correct to write F_2O as OF_2 and should be named as oxygen difluoride. But oxygen is more electronegative than chlorine, bromine and iodine. Therefore the compounds of these halogens with oxygen are known as oxides. For instance, Cl_2O_7 has been named as chlorine heptoxide. The compounds of oxygen with halogens have been discussed in detail elsewhere. However, the rest of the important halides have been described as below.

Important Conclusions from the Table 10.4

(i) All elements except oxygen form hexafluorides i.e., among halogens only with fluorine exhibits the maximum valency of six which is ascribed to its small size and high electronegativity. The more electronegative fluorine only bring the S, Se or Te into its highest oxidation state.

(ii) When the size of the halogen atom gets increased, the maximum coordination number of the elements of this group gets decreased. Thus no hexa chlorides, bromides or iodides are known. Only three tetrabromides and two tetraiodides are known.

(iii) Although S_2Cl_2 and Se_2Cl_2 are known, yet the corresponding Te_2Cl_2 and Po_2Cl_2 are not known. This is because of weaker Te—Te and Po—Po bonds as they are very large atoms. Further chlorine being more electronnegative will withdraw electrons from Te or Po thereby making Te—Te and Po—Po bonds weaker.

(iv) Thermal stability of these halides has been found to follow the order F > Cl > Br > I.

(v) All the elements except selenium, form dichlorides and dibromides. The dihalides are unstable.

***Hexafluorides*:** Amongst hexahalides, hexafluorides are the most stable compounds. Fluorine reacts with S, Se and Te to form hexafluorides SF_6, SeF_6, TeF_6, respectively

$$S + 3F_2 \longrightarrow SF_6.$$

All the hexafluorides are colourless gases having low boiling points. The low boiling point reveals a high degree of covalency.

The reactivity gets increased from SF_6 to TeF_6. Hence SF_6 is extremely stable and inert both thermally and chemically. SF_6 can be heated to 500ºC without decomposition and is unattacked by most metals, P, As etc. even when heated. It is also unreactive towards high-pressure steam presumably. It is due to the kinetic factors since the gas phase reaction is exothermic

$$SF_6 + 3H_2O \longrightarrow SO_3 + 6HF \quad \Delta H = -460 \text{ kj mol}^{-1}$$

The low reactivity of SF_6, particularly toward hydrolysis, which contrasts with the very high reactivity of SF_4 (since average bond energy in SF_6 is slightly less that in SF_4) is mainly due to kinetic factors, not to thermodynamic stability, since the reaction of SF_6 with H_2O to give SO_3 and HF would be decidedly favorable (ΔG = –460 kJ mol^{-1}). The fluorine atoms surround the central sulphur atom so

Table 10.4 Halides of group VIA elements

Element	Hexahalides	Tetrahalides	Dihalides	Mono halides	Others
Oxygen	—	—	OF_2, Cl_2O	O_2F_2	ClO_2 Cl_2O_6 Cl_2O_7
Sulphur	SF_6	SF_4 SCl_4	SF_2, SCl_2	S_2F_2, S_2Cl_2 S_2Br_2	
Selenium	SeF_6	SeF_4 $SeCl_4$ $SeBr_4$		Se_2Cl_2 Se_2Br_2	
Tellurium	TeF_6	TeF_4, $TeCl_4$, $TeBr_4$, TeI_4	$TeCl_2$ $TeBr_2$ TeI_2		
Polonium	—	$PoCl_4$ $PoBr_4$ PoI_4	$PoCl_2$ $PoBr_2$		

well that an attacking water molecule (or hydroxide ion) cannot get its electron near enough to the sulphur to start making a bond.

SeF_6 is quiet reactive while TeF_6 is completely hydrolysed by water

$$TeF_6 + 6H_2O \longrightarrow 6HF + H_6TeO_6$$

The trend in the reactivity of the hexafluorides can be ascribed to (a) the increasing size of the central atom which permit higher coordination number and (b) the increasing polarity of the E—F bond. The increasing metallic character of the element is able to make the central atom more accessible to nucleophillic attack by water molecules.

Only TeF_6 adds F^- ions to form $[TeF_7]^-$ and $[TeF_8]^{2-}$ ions. TeF_6 combines with Lewis bases such as amines forming eight coordinated adducts $[(CH_3)_2N]_2TeF_6$.

***Uses*:** Of all the hexafluorides, only SF_6 has been industrially important as a gaseous insulator in high voltage equipments. Its use is ascribed to its inert character and favourable dielectric characteristics.

***Structure*:** The hexafluorides get formed by sp^3d^2 hybridization and, therefore possess **octahedral structure**. The formation of SF_6 is illustrated as an example.

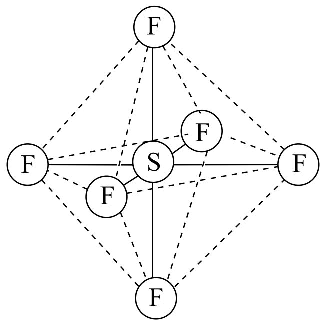


Fig 10.3 Structure of SF_6

***Tetrahalides*:** All these elements form tetrafluorides. However, these cannot be obtained by direct combination of the elements with fluorine because it is not possible to arrest the reaction at tetrafluoride stage. As fluorine is very reactive the end product is always hexafluoride. However, it is possible to prepare tetrafluoride by indirect methods. For example

$$S + 4CoF_3 \longrightarrow SF_4 + 4CoF_2$$

$$SeCl_4 + 4AgF \longrightarrow SeF_4 + 4AgCl$$

$$TeO_2 + 2SeF_4 \longrightarrow TeF_4 + 2SeOF_2$$

SF_4 is a gas, SeF_4 is liquid while TeF_4 is solid. All these fluorides are most stable to thermal decomposition. than the other tetrahalides. They are all strong fluorinating agents. They are strong Lewis acids because they accept electrons and change over to octahedral form. SF_4 reacts with fluorine to form SF_6

$$SF_4 + F_2 \longrightarrow SF_6$$

$$SeF_4 + 2F^- \longrightarrow SeF_6^{2-}$$

Tetrachlorides of the elements of Group VIA (except oxygen) are known. All the tetrachlorides are obtained by direct combination with chlorine

$$S + 2Cl_2 \longrightarrow SCl_4$$

$$Se + 2Cl_2 \longrightarrow SeCl_4$$

Lower electronegativity and greater polarity make $TeCl_4$ more stable than SCl_4 and $SeCl_4$.

Just like tetrafluorides, tetrachlorides (especially $TeCl_4$) react with HCl to form complexions

$$TeCl_4 + 2HCl \longrightarrow H_2[TeCl_6]$$

Selenium, tellurium and polonium are known to form tetrabromides by the direct combination of elements. The stability of the tetrabromides increases down the group as the polarity of bonds increases from $SeBr_4$ to $PoBr_4$. $SeBr_4$ is unstable and dissociates into $SeBr_2$ and Br_2

$$SeBr_4 \longrightarrow SeBr_2 + Br_2.$$

Only tellurium and polonium are known to form tetraiodides. Sulphur and selenium do not form tetraiodides because the bonds S—I and Se—I would be expected to be nearly nonpolar with iodine atom as positive. This along with atomic size difference make these bonds weaker. Although the bonds are not very polar in TeI_4, but are between the atoms of nearly the same radius, this makes the bonds quite stable.

All the tetrahalides hydrolyses in water SF_4 rapidly decomposes in the presence of moisture

$$SF_4 + 2H_2O \longrightarrow 4HF + SO_2$$

$$SCl_4 + 4H_2O \longrightarrow H_4SO_4 + 4HCl$$

$$H_4SO_4 \xrightarrow{\Delta} H_2SO_3 + H_2O$$

All the tetrahalides acts as Lewis bases and forms adducts with Lewis acids like $AlCl_3$ e.g., $SCl_4 \cdot AlCl_3$.

***Structure*:** Tetrahalides have trigonal bipyramidal structures with one position occupied by lone pair which are formed by sp^3d hybridization.

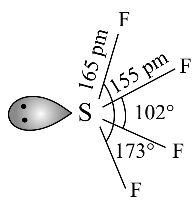


Fig 10.4 Structure of SF_4

***Dihalides*:** Except selenium, all the other elements form stable dichlorides and dibromides. However, these elements do not form diiodides. Dihalides of selenium are not stable (middle row anomaly). Amongst the dihalides SCl_2 is best known dihalide and it may be prepared by saturating sulphur monochloride with chlorine. $TeBr_2$ is obtained by direct combination of the elements. SF_2 can be prepared by fluorinating gaseous SCl_2 with activated KF or with HgF_2 at 150º.

SCl_2 is a dark cherry red liquid with a foul odour. All the dihalides form bent molecules in which the VIA element uses sp^3 hybrid orbitals for bonding and for accommodating the two lone pairs of electrons. Due to the repulsion between lone pairs amongst themselves; and between the lone pairs and bond pairs, the tetrahedral bond angle gets distorted

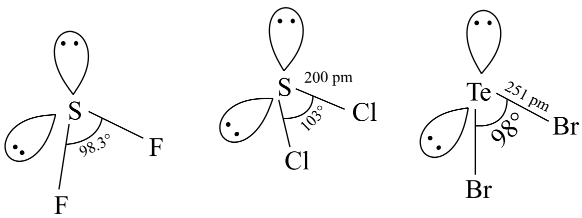


Fig 10.5 Structures of SF_2, SCl_2 and $TeBr_2$

***Monohalides*:** Amongst dimeric monohalides, the best characterized are S_2F_2, Se_2Cl_2 and Se_2Br_2. These get hydrolysed slowly and tend to undergo disproportionation.

$$2S_2F_2 + 2H_2O \longrightarrow 4HF + SO_2 + 3S$$

$$2SeCl_2 \longrightarrow Se_2Cl_2 + Se.$$

The structures of monohalides are similar to hydrogen peroxide. The bond angle 104º is due to sp^3 hybridization distorted by two lone pairs.

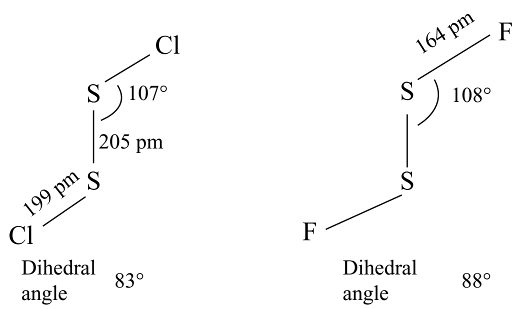
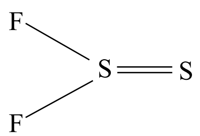


Fig 10.6 Structures of S_2Cl_2 and S_2F_2

S_2F_2 is an unstable compound. It exists in two isomeric forms, the second arrangement being



Oxygen Fluorides

Oxygen fluoride may be prepared by passing fluorine into 2 per cent NaOH solution.

$$2F_2 + 2NaOH \longrightarrow 2NaF + H_2O + OF_2.$$

It occurs as a pale yellow poisonous gas (b.p. 128K). It is thermally stable at 25ºC. It is non-explosive due to its negative free energy of formation. It dissolves slowly in water

$$OF_2 + H_2O \longrightarrow O_2 + 2HF.$$

It acts as a strong oxidizing agent and liberates other halogens from their salts or acids; its aqueous solution is able to convert Cr^{3+} to chromate

$$OF_2 + 4HX \longrightarrow 2X_2 + 2HF + H_2O$$

It is used as a rocket fuel.

The structure of oxygen difluoride is similar to any other difluoride with sp^3 hybridized oxygen

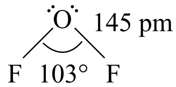


Fig 10.7 Structure of OF_2

Dioxygen difluoride resembles those of dimeric monohalides e.g., S_2Cl_2. O_2F_2 is prepared by direct union of O_2 and F_2. It is prepared by passing a silent electric discharge through a mixture of fluorine and oxygen at very low temperature (i.e., at liquid air temperature) and at low pressure.

$$F_2 + O_2 \xrightarrow[\text{discharge}]{\text{Silent electric}} O_2F_2$$

O_2F_2 is a brown gas. And it is a powerful oxidizing and fluorinating agent but less reactive than F_2. Structurally it belongs to M_2X_2 type halogen compounds of VIA group elements. The O_2F_2 has an open book structure.

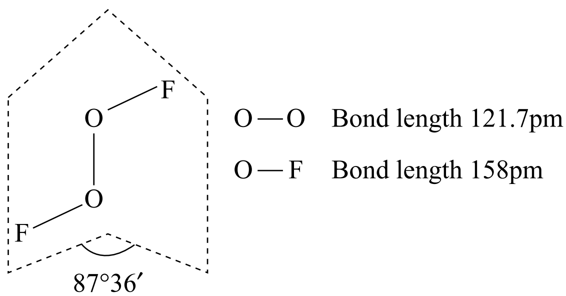


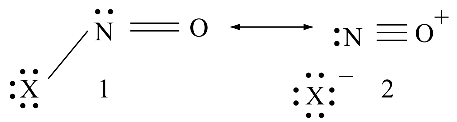
Fig 10.8 Structure of O_2F_2

O—O bond length 121.7pm
O—F bond length 158pm.

It is notable that the remarkably short O—O distance in O_2F_2 is 121.7pm while in H_2O_2 is 147.5 pm. Similarly another point to be noticed is that O—F bond length in O_2F_2 is 157.5pm while in OF_2 is 140.5pm.

In O_2F_2 fluorine atoms withdraw electron density from oxygen atoms, so that the repulsion between lone pairs decreases and oxygen atoms come close together. Therefore the O—O distance in O_2F_2 is smaller than H_2O_2. Also the extreme length of O—F bond in O_2F_2 (compare with 140.5pm in OF_2) and shortness of O—O bond length in O_2F_2 (Compare with 147.5pm in H_2O_2) suggest an anology with nitrosyl halides.

The nitrosyl halide molecules are non linear with bond angles 110º, 113º and 117º in NOF, NOCl and NOBr, respectively, and the N—O bond length is close to 114pm as in NO. This bond length and the fact that N—X bond lengths (152, 197 and 213pm) are considerably greater than normal single bond lengths suggest that these molecules are not adequately represented by the simple structure (1), but there may be resonance with the ionic structure (2).



Support for this polar structures comes from dipole moments of pure NOCl and NOBr (2.19D and 1.90D respectively), much greater than the values (around 0.3D) estimated for structure (1).

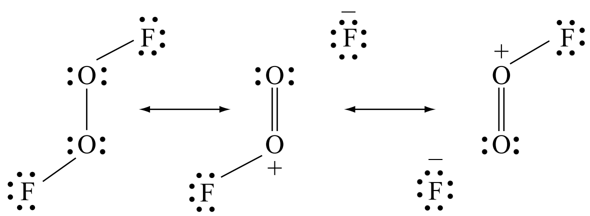


Fig 10.9 Resonating structures of O_2F_2

10.5.3 OXIDES

The elements of group VIA are known to form a variety of oxides which are listed in Table 10.5.

Some salient features of these oxides are as follows:

(i) As we go down the group VIA, the oxides of these elements (in the same oxidation state) becomes less acidic. For example, SO_2 is an acidic oxide whereas TeO_2 is an amphoteric oxide

(ii) Oxides of sulphur have been found to be more stable than the corresponding oxides of other elements. SO is formed only by sulphur. Also S_2O_7 is only formed by sulphur. SO_2 and SO_3 are more stable than the corresponding oxides of the other elements.

(iii) Acidic character of oxides of a particular element increases with the increase in oxidation number of the central element.

(iv) For any element, dioxide is more stable than trioxide.

(v) Dioxides can act both as Lewis acids and Lewis bases because they contain vacant d-orbitals and also lone pair of electrons in their valence shell.

(vi) Dioxides can act both as oxidizing and reducing agents because in dioxides the chalcogen is in +4 oxidation state and can increase or decrease its oxidation state.

(vii) Reduction power of dioxides decreases from SO_2 to PoO_2 while oxidation power increases from SO_2 to PoO_2.

(viii) Trioxides can act only as oxidizing agents. Oxidation power of trioxides should increases from SO_3 to SeO_3. But SO_3 acts as strong oxidizing agent in acid medium protonation makes the S—O bond breaking easier.

***Monoxides*:** Except selenium, all the elements of group VIA form monoxides. Sulphur monoxides S_2O is produced by passing electric discharge through sulphur dioxide and sulphur-until recently it was incorrectly formulated as SO. TeO and PoO are produced by heating $TeSO_3$ and $PoSO_3$ respectively.

$$TeSO_3 \longrightarrow TeO + SO_2$$

$$PoSO_3 \longrightarrow PoO + SO_2$$

Sulphur monoxide is very unstable and decomposes to SO_2 and polymeric oxide. Its structure is S—S—O with S—S and S—O bond lengths of 188pm and 146pm respectively and S—S—O bond angle 118º.

***Dioxides*:** Ozone is considered as the dioxide of oxygen (OO_2). The dioxides of S, Se, Te and Po are formed when these elements are burnt in air. These oxides differ from one another considerably in their properties and structures. SO_2 and SeO_2 are acidic oxides and they form corresponding -ous acids H_2SO_3 and H_2SeO_3 when

Table 10.5 Oxides of S, Se, Te and Po

Oxidation state	S	Se	Te	Po	Name
+1	S_2O	–	TeO	PoO	Monoxide
+ 4	SO_2	SeO_2	TeO_2	PoO_2	Dioxide
+ 6	SO_3	SeO_3	TeO_3	–	Trioxide
+7	S_2O_7	–	–	–	Heptoxide

dissolved in water. TeO_2 and PoO_2 are insoluble in water. However, these dissolve in concentrated acids or bases indicating amphoteric character of these oxides.

The dioxides of all the elements can be prepared by burning these elements in air. Alternatively SO_2 can be prepared by heating metal sulphides in air

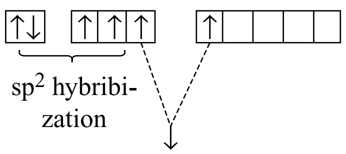
$$4FeS_2 + 11O_2 \longrightarrow 2Fe_2O_3 + 8SO_2$$
$$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$$

Selenium and tellurium dioxides can be prepared by treating the metals with hot nitric acid to form H_2SeO_3 and 2 TeO_2 HNO_3 and then heating these to remove water and nitric acid.

SO_2 can be readily condensed to a liquid which is used as non-aqueous solvent. It acts as a mild reducing agent in acidic solution but a strong reducing agent in basic solution. However, SeO_2 is an important oxidizing agent rather than a reducing agent. The reason is that sp^2 hybridization is less characteristic of selenium than that of sulphur. SeO_2 can oxidize SO_2 to SO_3

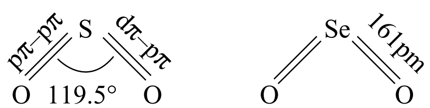
$$2SO_2 + SeO_2 \longrightarrow Se + 2SO_3.$$

Structures: SO_2 and SeO_2 are angular in shape in gaseous state, involving sp^2 hybridization of sulphur and selenium. Sulphur and selenium in Ist excited state have the following electronic configuration.



These two electrons form two π (pπ–pπ and pπ–dπ) bonds with oxygen

The O—S—O bond angle in SO_2 gets reduced from 120 to 119.5° due to the presence of a lone pair of electrons on sulphur.



S – O bond length 143pm

Angle not determined

Fig 10.10 Structure of SO_2 and SeO_2

At room temperature, SeO_2 is a white volatile solid. In solid state SeO_2 has polymeric zig-zag structure. The chains are not planar.

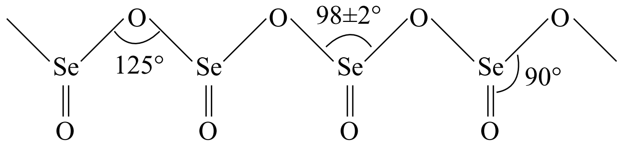


Fig 10.11 Structure of Solid SeO_2

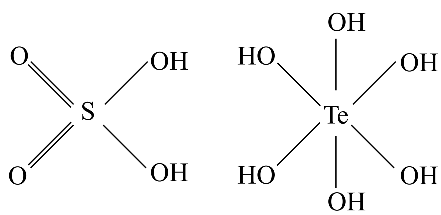
The Se – O bridge bond length is 178 pm and Se = O bond length 173 pm. TeO_2 and PoO_2 both crystalize in ionic form.

Trioxides: Sulphur, selenium and tellurium form the trioxides of the general formula $MO_3 \cdot SO_3$ is made by catalytic oxidation of SO_2 with air. SeO_3 is obtained by passing silent electric discharge on Se and $O_2 \cdot TeO_3$ is made by heating telluric acid H_6TeO_6.

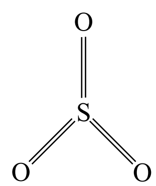
The trioxides are all acidic in nature. However, the solubility of TeO_3 is less than that of SO_3 and SeO_3. This is due to lesser tendency of tellurium to form multiple bonds to oxygen and lower electronegativity, thereby causing greater polarity in Te—O bond. Due to this, TeO_3 is polymeric and has low solubility in water.

$$SO_3 + H_2O \longrightarrow H_2SO_4$$
$$SeO_3 + H_2O \longrightarrow H_2SeO_4$$
$$TeO_3 + 3H_2O \longrightarrow H_6TeO_6$$

The strength and stability of the oxoacids gets decreased from H_2SO_4 to H_6TeO_6. The strength of an oxoacid depends on the inductive effect of the unprotonated oxygen atoms. With increase in the number of unprotonated oxygen atoms the acidic character increases. Due to hydration of telluric acid the number of unprotonated oxygen atoms decrease in telluric acid.



Structure: In the gaseous state SO_3 is planar triangular molecule which involves sp^2 hybridization of the sulphur atom. Its formation can conveniently represented as below.



Thus SO_3 molecule is having three S—O σ bonds (sp^2—p overlap) and three S—O π bonds (one pπ—pπ and two pπ–dπ overlap). The O—S—O bond angle has been of 120° and all the S—O bonds have been 143pm long. The equivalence of all S—O bonds indicates that SO_3 is a resonance hybrid of the following structures.

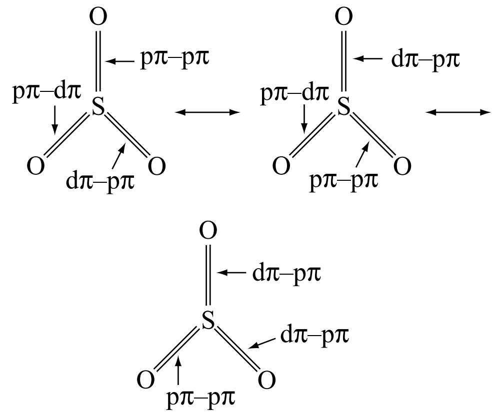


Fig 10.12 Structure of gaseous SO_3

The structure of solid sulphur trioxide is complex. It is having either cyclic trimer (α-form) structure or an infinite helical chain made up of linked SO_4 tetrahedron (β- and ω-forms). Each of the tetrahedron is able to share two oxygens.

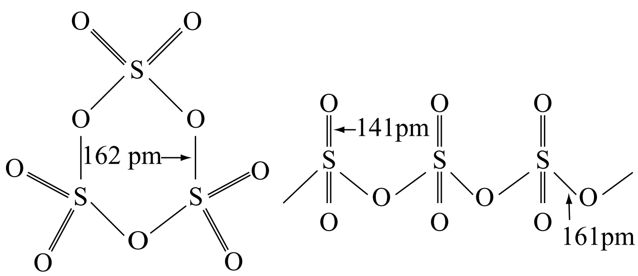


Fig 10.13 Structure of Solid SO_3

Like SO_3 it is at least dimorphic. Although the cyclic modification forms mixed crystals with S_3O_9 and tetrameric S_4O_{12}.

10.5.4 Oxoacids

Sulphur, selenium and tellurium form a variety of oxoacids. The more important are listed in Table 10.6.

-ic acids are stronger than -ous acids. -ic acid act as oxidizing agents while -ous acids can act both as oxidizing and reducing agents. Oxidation power of -ic acids increases from H_2SO_4 to H_6TeO_6. Also oxidation power of -ous acids increases from H_2SO_3 to H_2TeO_3 while reduction power decreases from H_2SO_3 to H_2TeO_3. Acidic character of both ous and ic acids decreases down the group.

Table 10.6 Important oxoacids of sulphur, selenium and tellurium

Sulphur	Selenium	Tellurium
Sulphurous acid H_2SO_3	Selenous acid H_2SeO_3	Tellurous acid H_2TeO_3
Sulphuric acid H_2SO_4	Selenic acid H_2SeO_4	Telluric acid H_6TeO_6
Thiosulphuric acid $H_2S_2O_3$		
Pyrosulphuric acid $H_2S_2O_7$		
Dithionic acid $H_2S_2O_5$		
Permonosulphuric acid H_2SO_5		
Perdisulphuric acid $H_2S_2O_8$		

10.6 OXYGEN

Oxygen is the most essential constituent of air. It occurs in free and combined state in nature. It is present in water (89% by weight) and earth's crust (about 47%) in combined state. Air contains about 21 per cent by volume (23% by weight). This percentage remains constant by the operation of the highly complex process termed photosynthesis, whereby green plants exposed to sunlight utilize carbon-dioxide (one product of respiration and combustion) and water to build up carbohydrate molecules, at the same time releasing oxygen. It is an essential ingredient in all living matter (together with carbon, hydrogen and nitrogen) and is of prime importance in respiration and combustion processes. Although only slightly soluble in water, enough oxygen dissolves to support marine life.

Oxygen can be obtained in the laboratory in a variety of ways

(i) By the thermal decomposition of the oxides of metals, low in the electrode potential series:

The oxides of most metals are thermally stable. However, the oxides of metals with very low electrode potentials e.g., Silver and mercury are thermally unstable

$$2Ag_2O \longrightarrow 4Ag + O_2$$
$$2HgO \longrightarrow 2Hg + O_2$$

The latter reaction is of historical interest, being the method used by Priestly to obtain oxygen in 1774.

(ii) By the thermal decomposition of higher oxides

So-called higher oxides such as Pb_3O_4 and PbO_2 decompose thermally into the lower oxide and oxygen

$$2Pb_3O_4 \longrightarrow 6PbO + O_2$$
$$2PbO_2 \longrightarrow 2PbO + O_2$$
$$3MnO_2 \longrightarrow Mn_3O_4 + O_2.$$

(iii) By the decomposition of peroxides.

Hydrogen peroxide is readily decomposed into water and oxygen by catalysts such as finely divided metals and manganese dioxide

$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

Solid peroxides such as sodium peroxide evolve oxygen on the addition of water

$$2Na_2O_2 + 2H_2O \longrightarrow 4NaOH + O_2$$

(iv) By the thermal decomposition of salts containing anions rich in oxygen.

Examples of salts that decompose thermally to give oxygen include nitrates, permanganates, dichromates and chlorates

$$2KNO_3 \longrightarrow 2KNO_2 + O_2$$
$$2KMnO_4 \longrightarrow K_2MnO_4 + MnO_2 + O_2$$
$$4K_2Cr_2O_7 \longrightarrow 4K_2CrO_4 + 2Cr_2O_3 + 3O_2$$
$$2KClO_3 \longrightarrow 2KCl + 3O_2$$

(v) By heating $KMnO_4$ and $K_2Cr_2O_7$ with concentrated H_2SO_4

$$4KMnO_4 + 6H_2SO_4 \longrightarrow 2K_2SO_4 + 4MnSO_4 + 6H_2O + 5O_2$$

$$2K_2Cr_2O_7 + 8H_2SO_4 \longrightarrow 2K_2SO_4 + 2Cr_2(SO_4)_3 + 8H_2O + 3O_2$$

(vi) By the addition of H_2O_2 to acidified permanganate

$$2KMnO_4 + 3H_2SO_4 + 5H_2O_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5O_2$$

***Manufacture*:** On large scale oxygen is prepared in one of the following methods.

(a) ***Brin's process*:** Oxygen was formerly prepared on a large scale by this method. Barium oxide (BaO) is heated at 500ºC in air. It is converted into barium peroxide (BaO_2). Decomposition of barium peroxide at 800ºC liberates oxygen.

(b) ***By electrolysis*:** Electrolysis of aqueous solutions of alkalis and acids results in the discharge of hydrogen at the cathode and oxygen at the anode. Pure water also decomposes in the same way but, since pure water is very poor conductor of electricity, hydrogen and oxygen are only discharged slowly.

(c) ***From air*:** Industrially, oxygen is obtained from the air by first removing carbon dioxide and water vapour. The remaining gases are then liquified and fractionally distilled to give nitrogen and oxygen.

***Properties*:** Oxygen is a colourless and odourless diatomic gas. It liquifies at –183ºC and freezes at –219ºC. Chemically it is very reactive, forming compounds with all other elements except the noble gases and apart from the halogens and some unreactive metals, these can be made to combine directly with oxygen under the right conditions. The action of a silent electrical discharge upon oxygen gives ozone in concentrations upto 10 per cent.

In addition to its importance in normal respiration and combustion processes, oxygen is used in oxyacetylene welding and cutting, in the manufacture of many metals particularly steel and in hospitals, high altitude flying and in mountaineering. The combustion of fuels, e.g., kerosene in liquid oxygen provides the tremendous thrust required to put satellites into space.

10.7 OXIDES

The binary compounds of oxygen with other elements are called oxides. The compounds of oxygen and fluorine (OF_2, O_2F_2) are not called as oxides but are called as oxygen fluorides as fluorine is more electronegative than oxygen.

Oxides may be prepared by directly combining the elements with oxygen under right conditions or by heating metal carbonates and hydroxides. The oxides of some non metals and weak metals can be obtained by oxidation of the element with nitric acid e.g.,

$$C + 4HNO_3 \longrightarrow CO_2 + 4NO_2 + 2H_2O$$
$$Sn + 4HNO_3 \longrightarrow SnO_2 + 4NO_2 + 2H_2O$$

Classification of oxides

Any realistic attempt to classify the diverse range of oxides requires a knowledge of their structures and their mutual reactions.

(a) ***Classification basing on structure*:** The oxides of most metals are essentially ionic and consist of a regular array of positive and negative ions, their arrangement being dictated by the radius ratio criterion and the necessity for maintaining overall electrical neutrality. For instance magnesium oxide adopts the sodium chloride structure in which each magnesium ion is surrounded symmetrically by six oxide ions and vice versa, and sodium monoxide Na_2O has a structure in which the coordination numbers of the sodium and oxide ions are four and eight respectively.

The oxides of non-metals and weak metals tend to be covalent and structures ranging from those of discrete molecules e.g., carbon dioxide (O = C = O) right through to three-dimensional giant molecules e.g., silicon dioxide [empirical formula $(SiO_2)n$] are observed.

(b) Classification of oxides on the basis of oxygen: Irrespective of whether an oxide is ionic or covalent the oxides are classified into four varieties depending on the amount of oxygen in an oxide.

1. *Normal oxides*: These contain oxygen permitted by the normal valency of the element. These involve bonds only between the element and oxygen e.g., H_2O, MgO, Al_2O_3 etc.

2. *Polyoxides*: These contain more oxygen than permitted by the normal valency of the element. These are again three types.

(i) *Peroxides*: These involve bonds between atoms of oxygen in addition to bonds between the element and oxygen typical examples are Na_2O_2, BaO_2, H_2O_2.

(ii) *Superoxides*: These also involve bonds between oxygen atoms e.g., KO_2, RbO_2, CsO_2.

(iii) *Dioxides*: These are oxides of an element in the higher oxidation state which can form stable oxide in lower oxidation state MnO_2, PbO_2 etc.

3. *Sub Oxides*: These contain less oxygen than permitted by the normal valency of the element. These involve bonds between atoms of the element in addition to bonds between the element and oxygen. Typical examples are

$O = C = C = C = O$	$\bar{N} = \overset{+}{N} = O$
Carbon suboxide	Nitrous Oxide

4. *Mixed Oxides*: These contain two simpler oxides of the same element e.g., red lead $Pb_3O_4(2PbO + PbO_2)$, ferrosoferric oxide $Fe_3O_4(FeO + Fe_2O_3)$, dinitrogen trioxide N_2O_3 $(NO + NO_2)$. On treating the mixed oxides of the metals with acids they form two compounds corresponding to the two oxides.

$$Fe_3O_4 + 8HCl \longrightarrow FeCl_2 + 2FeCl_3 + 4H_2O$$
$$Pb_3O_4 + 4HNO_3 \longrightarrow 2Pb(NO_3)_2 + PbO_2 + 2H_2O$$

(c) *Classification basing on mutual reactions of oxides:* There is a strong tendency for a metallic oxide to react with a non-metallic oxide to form a salt; this tendency can be given a quantitative meaning by employing tabulated values of standard free energy changes (quoted at 25ºC and 1 atmosphere pressure).

The standard free energy changes for reactions involving a selection of metallic oxides and a common non-metallic oxide are given below.

	ΔG^0_{298} kJ mol^{-1}
$K_2O + CO_2 \longrightarrow K_2CO_3$	– 356
$BaO + CO_2 \longrightarrow BaCO_3$	– 218
$CaO + CO_2 \longrightarrow CaCO_3$	– 130
$MgO + CO_2 \longrightarrow MgCO_3$	– 67

From the above figures it is apparent that the basic tendencies of the metallic oxides decrease in the order.

$$K_2O > BaO > CaO > MgO$$

For a number of reactions involving a common metallic oxide and a variety of non-metallic oxides, the following free energy changes are obtained.

	ΔG^0_{298} kJ mol^{-1}
$CaO + SO_3 \longrightarrow CaSO_4$	– 347
$CaO + N_2O_5 \longrightarrow Ca(NO_3)_2$	– 272
$CaO + CO_2 \longrightarrow CaCO_3$	– 130
$CaO + SiO_2 \longrightarrow CaSiO_3$	– 67
$CaO + H_2O \longrightarrow Ca(OH)_2$	– 59

From these figures the following order of decreasing acidic tendencies of the non-metallic oxides are observed.

$$SO_3 > N_2O_5 > CO_2 > SiO_2 > H_2O$$

Metallic and non-metallic oxides can be arranged in a series according to their basic and acidic tendencies. Towards middle of this table these tendencies will begin to merge and it is not therefore, surprising that some oxides can show both basic and acidic properties under different conditions e.g., zinc oxide. Such oxides are said to be amphoteric.

$ZnO + Na_2O \longrightarrow Na_2ZnO_2$ (Zinc oxide showing acidic property)

$ZnO + CO_2 \longrightarrow ZnCO_3$ (Zinc oxide showing basic property)

Using this system of classification, water would be regarded as an amphoteric oxide. It is possible to arrive at five possible classification of oxides on the basis of their acid-base properties.

1. *Very basic oxides*: Oxides which react readily with amphoteric oxides and very readily with acidic oxides e.g., Na_2O, K_2O, CaO etc.

2. *Moderately basic oxides*: Oxides which scarcely react with amphoteric oxides, but readily react with acidic oxides. e.g., MgO, FeO, Fe_2O_3, CuO etc.

3. ***Amphoteric oxides*:** Oxides which react with very basic oxides and acidic oxides e.g., Al_2O_3, ZnO, SnO, BeO, H_2O etc.

4. ***Acidic Oxides*:** Oxides which react with amphoteric oxides as well as basic oxides of both types e.g., SO_2, N_2O_5, CO_2, SiO_2 etc.

5. ***Neutral oxides*:** Oxides which do not react with any others e.g., N_2O, NO, CO etc.

The boundaries between these five classes are not rigid and not all the predicted reactions occur easily, neverthless this classification eliminates the need for such terms as "higher oxide" which convey little meaning.

Using structure and acid-base properties to describe oxides, MgO is a "normal basic oxide", BaO is a "very basic oxide", aluminium oxide is a "normal amphoteric oxide" and SO_2 is a "normal acidic oxide". These names are precise and convey information about the structure and reactions of each oxide.

10.8 OZONE

In 1785 von Marum observed a characteristic odour when oxygen gas is subjected to silent electric discharge. In 1840 Schonbein suggested that the odour might be due to the formation of a new gas. Ozone (OZO means odour in Greek) Soret determined the molecular formula as O_3 to the gas.

Ozone is considered as an allotrope of oxygen. Ozone is too reactive to remain for long in the atmosphere at sea level, but at a height of about 20 kilometers it is formed from atmospheric oxygen by the energy of sunlight. This ozone layer protect the earth's surface from an excessive concentration of ultraviolet radiation. Very small concentrations of ozone are formed in the vicinity of electrical machinery.

Preparation

When a slow dry stream of oxygen is passed through a silent electrical discharge upto 10 per cent conversion to ozone occurs

$$3O_2 \longrightarrow 2O_3 \qquad \Delta H^0_{298} = +142 \text{ KJ mol}^{-1}$$

Since ozone is endothermic compound it is necessary to use a silent electrical discharge, otherwise sparking would generate heat and decompose it. The apparatus used for this production of O_3 is known as *ozonizer*. Two kinds of ozonizers are commonly used in the laboratory.

(a) ***Siemen's Ozonizer*:** The ozonizer consists of two coaxial glass tubes sealed on one side (Fig 10).

The innerside of the innertube and the outersides of the outer tube are coated with tin foils. These tin foils are connected to the powerful induction coil. Cold, dry oxygen gas is passed through the annular space from one end. Oxygen undergoes silent electric discharge partially and ozone is formed (about 10% conversion takes place) ozone, oxygen mixture is known

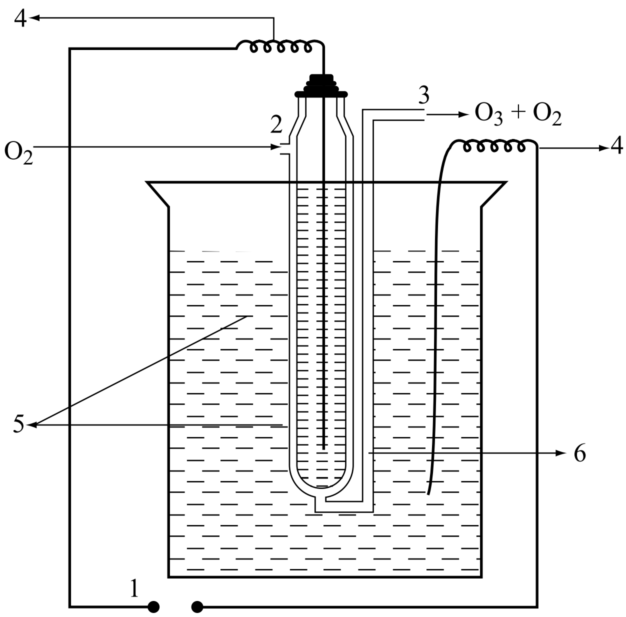


Fig 10.15 Brodie's ozonizer 1. Induction coil 2. Dry oxygen 3. Ozonized oxygen 4. Cu wire 5. Dilute H_2SO_4 6. Annular space

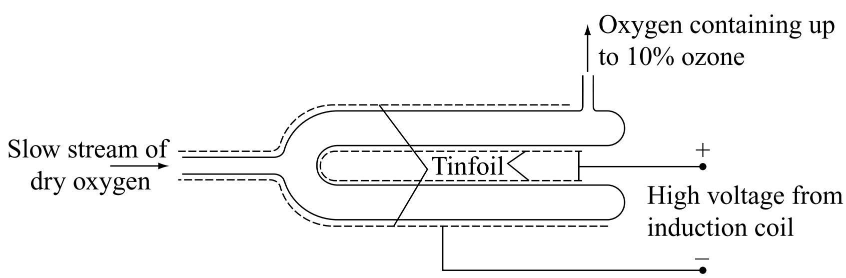


Fig 10.14 Siemen's Ozonizer

as ozonized oxygen. It is collected from the other end of the apparatus. O_3 is separated from O_2 if only required, by condensing the gases when O_3 liquifies first, leaving O_2 gas behind. It is removed and recycled to get more O_3.

(b) ***Brodie's Ozonizer*****:** Brodie's ozonizer also works in a similar manner to Siemen's ozonizer. The two ozonizers differ in their structures

In Brodie's ozonizer dilute sulphuric acid is used in place of tin foils to produce silent electric discharge as in Siemen's ozonizer. Two platinum or copper wires are immersed into the sulphuric acid one inside the Brodie's ozonizer and the other outside the ozonizer and then connected to a high power induction coil. Pure dry oxygen is passed through the annular space and gets converted into ozonized oxygen under the influence of silent electric discharge. If required O_3 and O_2 are separated by the condensation process.

(c) ***Commercial Preparation*****:** For the preparation of ozonized air on a large scale Siemen-Halske ozonizer is used (Fig 10.16). It is madeup of a cast iron box containing a number of aluminum rods which are enclosed in glass cylinders around which water is circulated to maintain temperature low. Each aluminium rod rests on a glass plate and therefore insulated from the box.

When air enters the ozonizer from near the base it raises up the annular space between the cylinders and the aluminium rods. The rods are charged to a high potential of 8000 to 10000 Volts whereas the cast iron box gets earthed. The escaping air is having about 5 per cent of ozone.

(d) ***Electrolytic Method*****:** The electrolysis of acidulated water with a high current density and platinum anode produces a gas at the anode which is having up to 95 per cent ozone.

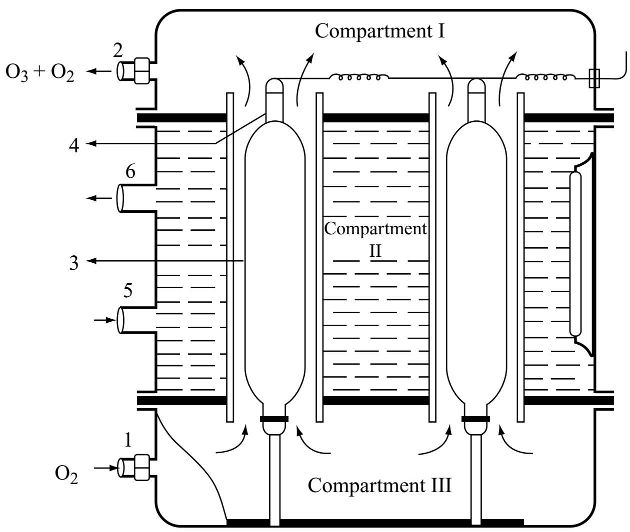


Fig 10.16 Siemen – Halske's Ozonizer 1. Dry air 2. Ozonized oxygen 3. Glass cylinders 4. Aluminium rod 5. Water inlet 6. Water outlet

(e) ***Thermal Method*****:** Ozone can also be prepared by heating oxygen to 2773K and then cooling it to get O_3.

(f) ***Chemical Method*****:** Fluorine reacts with water at very low temperature liberating a mixture of ozonized oxygen.

$$2F_2 + 2H_2O \longrightarrow 4HF + O_2$$
$$3F_2 + 3H_2O \longrightarrow 6HF + O_3$$

Properties

Physical Properties**:** Ozone has a characteristic smell and in small concentrations it is harmless. However, if concentration raises above about 100 ppm breathing becomes uncomfortable and it causes headache.

(i) When pure it is a pale blue gas; liquid ozone is a darker blue and solid ozone is violet black.
(ii) It is heavier than air.
(iii) It is slightly soluble in water but more soluble in turpentine oil, glacial acetic acid or carbon tetrachloride.
(iv) It is diamagnetic and endothermic.

Chemical Properties

Decomposition**:** Ozone is thermodynamically unstable with respect to oxygen, since its decomposition into oxygen results in the liberation of heat (ΔH is negative) and an increase in entropy (ΔS is positive since triatomic molecule is broken down into simpler diatomic oxygen). These two factors reinforce each other, resulting in a large negative free energy change (ΔG) for its conversion into oxygen.

$$\Delta G = \Delta H - T\Delta S$$

It is not really surprising, therefore, that high concentrations of ozone can be dangerously explosive.

$$2O_3 \longrightarrow 3O_2, \Delta H = 284 \text{ Kj mol}^{-1}$$

Oxidation Properties

Ozone is a strong oxidizing agent. The oxidation reactions of O_3 can be divided into two categories.

(A) Reactions in which oxygen is evolved i.e., one oxygen atom per ozone molecule is utilized.

(B) No oxygen is evolved i.e., all the oxygen atoms in ozone are consumed.

(A) *Examples of O_3 reactions in which O_2 is liberated*

(i) Black lead sulphide is oxidized to white lead sulphate

$$PbS + 4O_3 \longrightarrow PbSO_4 + 4O_2$$

(ii) Iodine is liberated from potassium iodide solution.

$$2KI + H_2O + O_3 \longrightarrow 2KOH + I_2 + O_2$$

(iii) Halogen acids are oxidized to the corresponding halogens e.g., hydrochloric acid is oxidized to chlorine.

$$2HCl + O_3 \longrightarrow Cl_2 + H_2O + O_2$$

(iv) Green potassium manganate is oxidized to purple permanganate.

$$2K_2MnO_4 + H_2O + O_3 \longrightarrow 2KMnO_4 + 2KOH + O_2$$

(v) Nitrites are oxidized to nitrates

$$NaNO_2 + O_3 \longrightarrow NaNO_3 + O_2$$

(vi) Ferrous sulphate is oxidized to ferric sulphate.

$$2FeSO_4 + H_2SO_4 + O_3 \longrightarrow Fe_2(SO_4)_3 + H_2O + O_2$$

(vii) Sodium sulphite is oxidized to sodium sulphate

$$Na_2SO_3 + O_3 \longrightarrow Na_2SO_4 + O_2$$

(viii) Sodium arsenite is oxidized to sodium arsenate

$$Na_3AsO_3 + O_3 \longrightarrow Na_3AsO_4 + O_2$$

(ix) Potassium ferrocyanide is oxidized to potassium ferricyanide

$$2K_4[Fe(CN)_6] + H_2O + O_3 \longrightarrow 2K_3[Fe(CN)_6] + 2KOH + O_2$$

(x) Alkaline KI is oxidized to potassium iodate and periodate

$$KI + 3O_3 \longrightarrow KIO_3 + 3O_2$$
$$KI + 4O_3 \longrightarrow KIO_4 + 4O_2$$

(xi) H_2S is oxidized to sulphur

$$H_2S + O_3 \longrightarrow H_2O + S + O_2.$$

Oxidation of Non-metals

Most non metals like iodine, sulphur, phosphorous and arsenic are oxidized to their corresponding ic acids

$$I_2 + H_2O + 5O_3 \longrightarrow 2HIO_3 + 5O_2$$
$$S + H_2O + 3O_3 \longrightarrow H_2SO_4 + 3O_2$$
$$P_4 + 6H_2O + 10O_3 \longrightarrow 4H_3PO_4 + 10O_2$$
$$As_4 + 6H_2O + 10O_3 \longrightarrow 4H_3AsO_4 + 10O_2.$$

Dry iodine is oxidized to yellow powder I_4O_9

$$2I_2 + 9O_3 \longrightarrow I_4O_9 + 9O_2.$$

Oxidation of Metals

Silver metal is blackened in ozone due to alternate oxidation of the metal and reduction of the metal oxide.

$$2Ag + O_3 \longrightarrow Ag_2O + O_2$$
$$Ag_2O + O_3 \longrightarrow 2Ag + 2O_2$$

When ozone is passed into mercury, it gets oxidized to mercurous oxide which dissolves in mercury. Then mercury loses its upper meniscus and non sticking nature to the glass

$$2Hg + O_3 \longrightarrow Hg_2O + O_2$$

The loss of metallic lustre and the meniscus of mercury metal in the presence of ozone and as a result making mercury stick to the glass surface is called *tailing of mercury.*

(B) *Examples of reactions where O_3 is completely consumed*

(i) SO_2 is oxidized to SO_3

$$3SO_2 + O_3 \longrightarrow 3SO_3$$

(ii) $SnCl_2$ is oxidized to $SnCl_4$ in the presence of HCl

$$3SnCl_2 + 6HCl + O_3 \longrightarrow 3SnCl_4 + 3H_2O$$

Ozone is the fourth strongest oxidizing agent. The others are fluorine, atomic oxygen and oxygen fluoride OF_2. Due to high value of oxidation potential (–2.07V) it acts as a strong oxidizing agent.

***Reactions with peroxides*:** Ozone reacts with barium peroxide and hydrogen peroxide to liberate oxygen

$$BaO_2 + O_3 \longrightarrow BaO + 2O_2$$
$$H_2O_2 + O_3 \longrightarrow H_2O + 2O_2.$$

Ozone reduces Ag_2O to silver

$$Ag_2O + O_3 \longrightarrow 2Ag + 2O_2$$

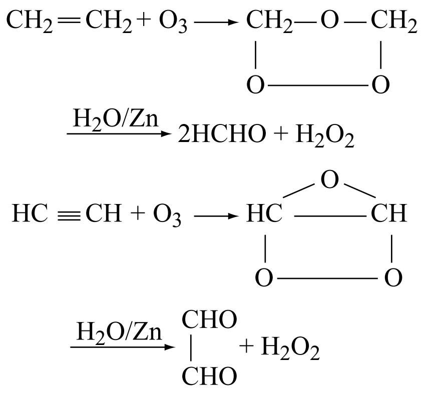
The reactions of ozone with peroxides and silver oxide are examples of *mutual reduction reactions.* These are so called because ozone as well as the reactant (H_2O_2, BaO_2 Ag_2O) lose one oxygen atom each. Thus both the reactants are reduced as loss of oxygen is reduction.

Bleaching Property

It is a good bleaching agent on account of its oxidizing action on organic matter like rubber, corks oil, ivory, flour, starch, waxes, wood pulp etc.

Addition Reactions

With unsaturated organic compounds containing double bonds, it forms addition compounds called *ozonides*. The ozonides decomposed by water or dilute acids forming carbonyl compounds. The total reaction is called ozonolysis.



The identification of carbonyl compounds so obtained helps to locate the position of the double bond in the molecule of the original unsaturated compound.

Uses

(i) Ozone is used as germicide and disinfectant for sterilization of water and for improving the atmosphere of crowded places like underground railways, mines, cinema halls etc.
(ii) It is used in the manufacture of artificial silk and camphor.
(iii) It is used to identify the position of unsaturation and number of double bonds or triple bonds.
(iv) It is used in the manufacture of potassium permanganate by oxidation of potassium manganate.

Tests for Ozone

(i) It can be detected by its characteristic rotten fishy smell.
(ii) It turns the starch iodide paper blue.
(iii) It can be identified by its tailing effect of mercury.
(iv) It turns the white shining silver to black.
(v) It converts the benzidine paper to brown and tetramethyl or ethyl base paper to violet.

Formula of Ozone

We know that ozone is prepared by passing silent electric discharge through oxygen and gives oxygen on heating. This indicates that ozone is made up of oxygen atoms and nothing else.

Soret's experiment: Soret enclosed the equal volumes of ozonized oxygen in two flasks of equal capacity over mercury. The flasks are provided with graduated necks. In one of the flasks he introduced turpentine oil and heated the second flask. A decrease in volume was observed in the first flask because the ozone was absorbed by turpentine oil. An increase in volume occurs in the second flask because of the decomposition of ozone to oxygen. The decrease in volume was found to be double than the increase i.e., if the increase is one volume, the decrease is double (= 2 volumes).

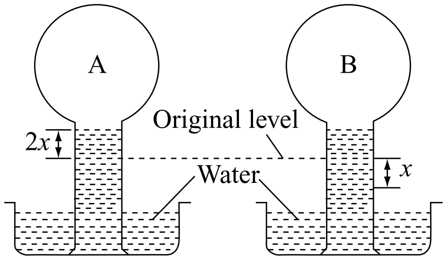


Fig 10.17 Soret's experiment

Moreover, the decrease in volume (= volumes) with turpentine oil would be directly the volume of ozone present in ozonized oxygen. This increases by one volume on heating when ozone (2 vol) decomposes to yield oxygen i.e., the volume of oxygen obtained = 2 vol + 1 vol = 3 vol. It indicates that two volumes on decomposition by heating give three volumes of oxygen.

Ozone	$\longrightarrow$	O_2
2Vol		3 Vol
2molecules		3 molecules
1 molecule		$\frac{3}{2}$ molecules or 3 atoms

On applying Avogadro's law 2 molecules of ozone gives 3 molecules of oxygen or 1 molecule of ozone gives 3/2 molecules or 3 atoms of oxygen. So formula of ozone should be O_3.

Structure of Ozone

Ozone is a planar molecule. The bond length is 128 pm. It is intermediate between a O—O single bond length (148pm) and double bond length (121pm) ozone is thus regarded as the resonance hybrid of the following canonical form.

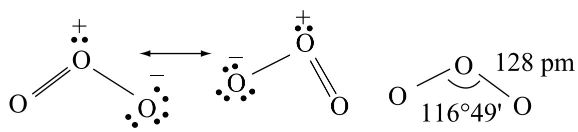


Fig 10.18 Structure of ozone

The central oxygen atom in the molecule is involved in sp^2 hybridization. Two of the sp^2 hybrid orbitals form two O—O σ bonds and the third sp^2 hybrid orbital contains the lone pair of electrons. The p orbital which is not participated in the hybridization on central oxygen atom participate in π–bonding. The bond order in ozone is 1.5.

10.9 SULPHUR

Sulphur occurs with many metals as sulphides, if they are insoluble in water e.g: the sulphides of zinc, lead, copper and mercury. Many sulphate minerals occur including anhydrite $CaSO_4$, gypsum $CaSO_4 \cdot 2H_2O$ and Epsom $MgSO_4 \cdot 7H_2O$ other important sources of sulphur are crude oil and natural gas, from which it is extracted as hydrogen sulphide.

Sulphur occurs in the free state in Japan and underground in Texas and Lousiana, where it was discovered by Frasch in 1867.

Frasch Process**:** In order to extract this underground sulphur, three concentric pipes are sunk deep into the ground (Fig 10.19). Superheated water at 170ºC is forced down the outer pipe into the sulphur which is melted. Compressed air blown through the inner pipe now forces the sulphur as a liquid to the surface, where it is allowed to solidify. Sulphur of about 99.5 per cent purity is obtained by this process.

Sulphur from waste products**:** Alkali waste of the Leblanc process mainly contains calcium sulphide. It is suspended in water and treated with carbon dioxide.

$$CaS + H_2O + CO_2 \longrightarrow CaCO_3 + H_2S$$

Hydrogen sulphide so liberated is allowed to burn in an insufficient supply of air when sulphur gets deposited.

$$2H_2S + O_2 \longrightarrow 2S + 2H_2O.$$

10.9.1 The Allotropy of Sulphur

Unlike oxygen, which is a discrete molecule, two atoms being united by a double bond, sulphur atoms show a marked reluctance to double bond with themselves and the two main allotropes of sulphur contain S_8 molecules in which single bonds unite sulphur atoms into a puckered octagonal ring. The high molecular weight of these S_8 structural units explains why sulphur, unlike oxygen, is a solid. The bond angles of 107º are consistent with the simple VSEPR theory. Sulphur exists in several allotropic forms. The important allotropes are discussed here.

Rhombic Sulphur**:** This is the form of sulphur normally encountered and consists of S_8 structural units packed together to give crystals. Fairly large crystals can be obtained by allowing a solution of powdered roll sulphur in carbon disulphide to evaporate slowly; they are yellow, transparent and have density 2.06 g cm^{-3}.

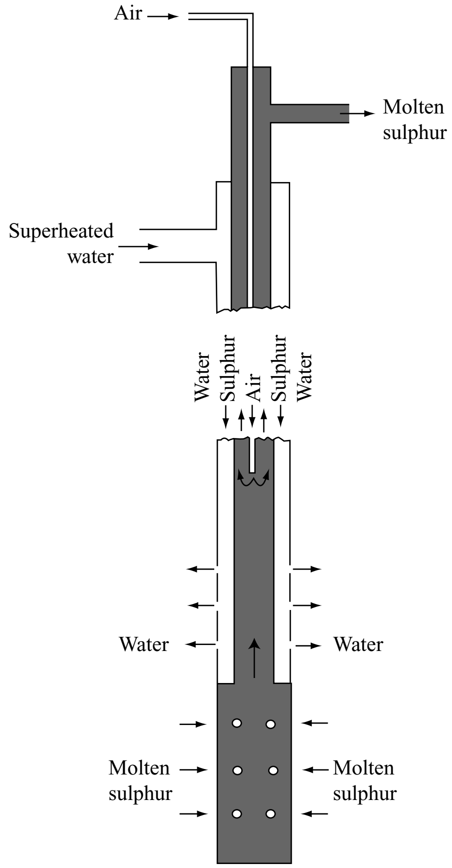


Fig 10.19 Frasch process for extraction of sulphur

Monoclinic Sulphur**:** This form of sulphur is formed when molten sulphur is allowed to crystallize above 95.6ºC. Like rhombic sulphur it consists of S_8 structural units but these are arranged differently in the crystal lattice. The temperature of 95.6ºC is the transition temperature for sulphur, below this temperature rhombic sulphur is the more stable allotrope and above it monoclinic sulphur is the more stable allotrope of the two forms. This type of allotropy, in which a definite transition point exists where two forms becomes equally stable is called *enantiotropy.*

$$\text{Rhombic sulphur} \rightleftharpoons \text{Monoclinic sulphur}$$

Crystals of monoclinic sulphur are amber – yellow in colour and have a density of 1.96 gm cm^{-3}. As they gradually change over into rhombic sulphur below 95.6ºC, each crystal retains its overall shape but changes into a mass of small rhombic crystals. Both rhombic and monoclinic sulphurs are soluble in CS_2.

Amorphous Sulphur: A number of forms of sulphur which possess no regular crystalline form can be obtained when sulphur is liberated in chemical reactions, e.g., by the action of dilute hydrochloric acid on a solution of sodium thiosulphate.

$$Na_2S_2O_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2 + S$$

Plastic Sulphur: This is obtained as an amber – brown soft and elastic solid, by pouring nearly boiling sulphur into cold water. It consists of a completely random arrangement of chains of sulphur atoms which when stretched, align themselves parallel to each other. On standing, it slowly changes over into rhombic sulphur, as the chains of sulphur atoms break and reform the S_8 cyclic units.

10.9.2 Action of Heat on Sulphur

Both rhombic and monoclinic sulphur melts to a yellow liquid. Owing to the conversion of rhombic to monoclinic sulphur, and also to possible variations in the percentage of allotropes of liquid sulphur formed, the melting points are not sharp; rhombic sulphur melts at approximately 113ºC and monoclinic sulphur at approximately 119ºC. As the temperature rises the colour of the liquid darkens with increase in viscosity until it nearly black and it becomes viscous. At about 200ºC the viscosity begins to fall and at its boiling point of 444ºC the liquid is again mobile. When sulphur vapour comes in contact with a cool surface, it sublimes to give a pale yellow liquid.

There is still some doubt concerning a complete explanation of these observations, but a recent theory runs as follows; as the sulphur melts the S_8 rings begin to open and it is possible that other ring systems containing possibly six and four sulphur atoms form. It is known, however, that sulphur chains begin to form and reach their maximum chain length at 200ºC, corresponding to the maximum viscosity of liquid sulphur. The decrease in viscosity of liquid sulphur that occurs above 200ºC is explained as being due to the break down of these long chains and the reformation of S_8 rings. Sulphur vapour contains S_8 rings, together with smaller fragments such as S_6, S_4 and S_2. At very high temperatures atomic sulphur is formed.

10.9.3 Chemical Properties of Sulphur

Sulphur combines with most metals when heated, finely divided metals such as magnesium and aluminium (which are high up in the electrode potential series) reacting with considerable vigour, e.g.,

$$Mg + S \longrightarrow MgS$$

Non metals that combine with sulphur directly include fluorine, chlorine, oxygen and carbon; hydrogen combines reversibly to a slight extent when passed through molten sulphur near its boiling point.

$$S + 3F_2 \longrightarrow SF_6$$
$$2S + Cl_2 \longrightarrow S_2Cl_2$$
$$S + O_2 \longrightarrow SO_2$$
$$C + 2S \longrightarrow CS_2$$
$$H_2 + S \longrightarrow H_2S$$

Sulphur is oxidized by concentrated nitric and sulphuric acids; with hot concentrated solutions of alkalis a sulphide and sulphite are formed, which react with more sulphur to form polysulphides and a thiosulphate respectively.

$$S + 6HNO_3 \longrightarrow 2H_2O + H_2SO_4 + 6NO_2$$
$$S + 2H_2SO_4 \longrightarrow 2H_2O + 3SO_2$$
$$3S + 6OH^- \longrightarrow 2S^{2-} + SO_3^{2-} + 3H_2O$$

followed by

$$\underset{\text{Polysulphide ion}}{S^{2-} + nS \longrightarrow S_{n+1}^{2-}} \qquad n = 1 \text{ to } 8$$

$$\underset{\text{Thiosulphate ion}}{SO_3^{2-} + S \longrightarrow S_2O_3^{2-}}$$

Uses: Sulphur is used

(i) in the manufacture of SO_2, H_2SO_4, CS_2, matches, gun powder, fire works etc.
(ii) as a disinfectant for houses and for destroying bacteria fungi, insects etc.
(iii) for vulcanizing rubber and in the manufacture of sulphur dyes.
(iv) in medicines for internal as well as external applications.

10.10 COMPOUNDS OF SULPHUR

10.10.1 Hydrogen Sulphide, H_2S

Hydrogen sulphide occurs in great quantities in the natural gas deposits of France and Canada and these sources are of major importance to the sulphuric acid industry. Very small quantities of the gas are released from rotten eggs.

Preparation: It is usually prepared in the laboratory by the action of dilute hydrochloric acid on iron (II) sulphide in a kipps apparatus.

$$FeS + 2HCl \rightarrow FeCl_2 + H_2S.$$

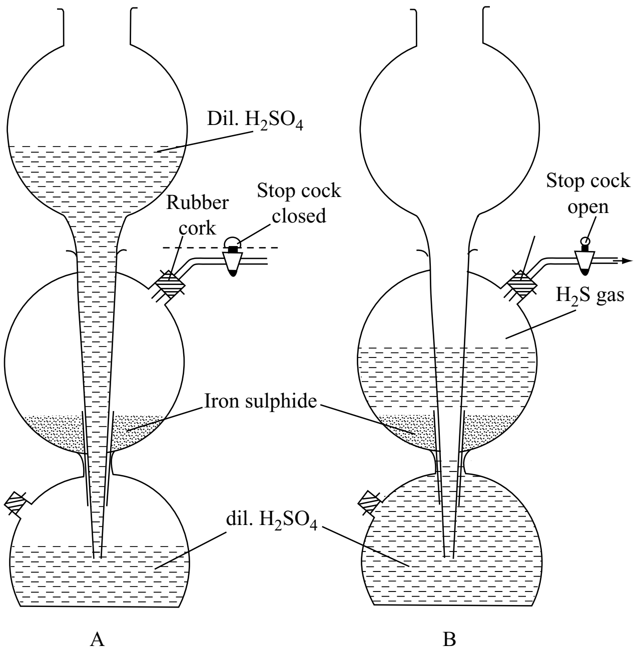


Fig 10.20 Preparation of hydrogen sulphide

Since iron (II) sulphide always contains some uncombined iron, the gas is contaminated with free hydrogen. A purer gas can be obtained by warming antimony (III) sulphide with concentrated hydrochloric acid

$$Sb_2S_3 + 6HCl \longrightarrow 2SbCl_3 + 3H_2S.$$

Physical Properties

(i) Unlike water, hydrogen sulphide is a gas, since the electronegativity of the sulphur atom is insufficient to participate in hydrogen bonding.
(ii) It is slightly heavier than air, moderately soluble in cold water, solubility decreases with increase in temperature.
(iii) It is an extremely poisonous substance (as little as 1 part per 1000 parts of air is fatal) but fortunately its rotten eggs smell becomes intolerable long before the fatal concentration is reached.
(iv) It can be easily liquified by pressure. The b.p 213k, freezes to a transparent solid at 187.4k.

Chemical Properties

1. *Thermal decomposition*: On heating, hydrogen sulphide decomposes into hydrogen and sulphur. Dissociation begins at 583k and is completed at 1043k.

$$H_2S \rightleftharpoons H_2 + S$$

2. *Combustibility*: It is combustible and burns with a blue flame to give sulphur in limited supply of air and sulphur dioxide in excess of air.

$$2H_2S + O_2 \longrightarrow 2H_2O + 2S$$
$$2H_2S + 3O_2 \longrightarrow 2H_2O + 2SO_2$$

3. *Reducing properties*

(i) It reacts with dissolved oxygen in water, so that a saturated solution of hydrogen sulphide will slowly go cloudy after standing for some days.

$$2H_2S + O_2 \longrightarrow 2H_2O + 2S$$

(ii) It reduces halogens into corresponding hydrogen halides

$$X_2 + H_2S \longrightarrow 2HX + S \quad (X_2 = F_2, Cl_2, Br_2 \text{ or } I_2)$$

(iii) It reduces sulphur dioxide to sulphur

$$SO_2 + 2H_2S \longrightarrow 2H_2O + 3S$$

(iv) It reduces ferric chloride to ferrous chloride

$$2FeCl_3 + H_2S \longrightarrow 2FeCl_2 + 2HCl + S$$

(v) It reduces H_2O_2 and O_3

$$H_2O_2 + H_2S \longrightarrow 2H_2O + S$$
$$O_3 + H_2S \longrightarrow H_2O + O_2 + S$$

(vi) It reduces the oxidizing acids like H_2SO_4 and HNO_3

$$H_2SO_4 + H_2S \longrightarrow 2H_2O + SO_2 + S$$
$$2HNO_3 + H_2S \longrightarrow 2H_2O + 2NO_2 + S$$

(vii) It reduces acidified dichromate and permanganate solutions to chromium (III) and manganese (II) ions respectively.

$$K_2Cr_2O_7 + 4H_2SO_4 + 3H_2S \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3S$$

$$2KMnO_4 + 3H_2SO_4 + 5H_2S \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5S.$$

Note that in the reactions above, the hydrogen sulphide is always oxidized to sulphur.

Acid Properties

Hydrogen sulphide has no effect on litmus paper when dry, but once it becomes wet it will turn blue litmus paper a wine – red colour showing that it functions as a weak acid in solution. Ionization in water takes place in two stages, the second occuring only to a very minute extent.

$$H_2O + H_2S \rightleftharpoons H_3O^+ + HS^-$$
$$H_2O + HS^- \rightleftharpoons H_3O^+ + S^{2-}$$

In the presence of hydroxyl ions, a much stronger proton acceptor than water molecules, the ionization of H_2S is much more extensive, thus the reaction with sodium hydroxide solution produces the hydrosulphide i.e., NaHS and Na_2S.

$$NaOH + H_2S \longrightarrow NaHS + H_2O$$
$$NaOH + NaHS \longrightarrow Na_2S + H_2O.$$

Even so the reaction do not go to completion i.e., the alkali metal hydrosulphides and sulphides show an alkaline reaction in solution. Thus demonstrating the presence of OH^- ions.

Precipitation Reactions

Hydrogen sulphide is used in qualitative analysis for precipitating the sulphides of many metals, in practice the precipitation is done under controlled conditions so that two groups of sulphides can be distinguished (analysis Groups II and IV which are in no way related to Groups II and IV of the periodic classification).

Group II – the sulphides of mercury (II), lead, bismuth, copper (II), cadmium, arsenic, antimony and tin.

Group IV – the sulphides of nickel, cobalt, manganese and zinc.

The sulphides of Group II have smaller solubility product than those of Group IV i.e., they are less soluble in water. In order to ensure that the Group II sulphides precipitate but the Group IV sulphides do not, the sulphide ion concentration must be decreased to an even smaller value than it already is in a saturated solution of H_2S. This is achieved by carrying out the precipitation in the presence of acid, the additional H_3O^+ ions shifting the two equilibria (shown below) to the left and thus decreasing $[S^{2-}]$.

$$H_2O + H_2S \rightleftharpoons H_3O^+ + HS^-$$
$$H_2O + HS^- \rightleftharpoons H_3O^+ + S^{2-}$$

Both equilibria shifted to the left by the extra H_3O^+ ions. The points made above are best considered by using an example.

Illustrated Example

Show that both copper (II) sulphide and nickel sulphide precipitate when hydrogen sulphide is passed through molar solutions of copper (II) and nickel salts, but that only copper (II) sulphide precipitates when the solution is made 0.3 molar with respect to hydrochloric acid. The solubility products of CuS and NiS are approximately 10^{-36} and 10^{-21} mol^2 dm^{-6} respectively.

(a) In a neutral solution the sulphide ion concentration is approximately 10^{-15} mol dm^{-3}

$$\left[Cu^{2+}\right]\left[S^{2-}\right] = 10^{-36}\ \text{mol}^2\ \text{dm}^{-6}$$

$$\left[Cu^{2+}\right] = 10^{-36}/10^{-15} = 10^{-21}\,\text{mol dm}^{-3}$$

$$\left[Ni^{2+}\right]\left[S^{2-}\right] = 10^{-21}\ \text{mol}^2\ \text{dm}^{-6}$$

$$\left[Ni^{2+}\right] = 10^{-21}/10^{-15} = 10^{-6}\ \text{mol dm}^{-3}$$

The minimum concentration of Cu^{2+} and Ni^{2+} necessary for precipitation to occur (10^{-21} and 10^{-6} mol dm^{-3} respectively) is considerably less than the concentrations of these ions present (molar solutions), therefore both sulphides precipitate.

(b) In 0.3 molar solution of hydrochloric acid, a saturated solution of H_2S has a sulphide ion concentration of approximately 10^{-22} mol dm^{-3}

$$\left[Cu^{2+}\right]\left[S^{2-}\right] = 10^{-36}\ \text{mol}^2\ \text{dm}^{-6} \quad \left[Cu^{2+}\right] = 10^{-36}/10^{-22} = 10^{-14}\,\text{mol dm}^{-3}$$

$$\left[Ni^{2+}\right]\left[S^{2-}\right] = 10^{-21}\ \text{mol}^2\ \text{dm}^{-6} \quad \left[Ni^{2+}\right] = 10^{-21}/10^{-22} = 10\ \text{mol dm}^{-3}$$

The minimum concentrations of Cu^{2+} and Ni^{2+} necessary for precipitation to occur (10^{-14} and 10 mol dm^{-3}) are now very much larger, and only in the case of copper (II) sulphide can precipitation occur, i.e., the minimum concentration of Ni^{2+} ions needed for precipitation to occur is ten times higher than the concentration present.

The illustrated example shows clearly that the acidity of the solution can influence the precipitation of metallic sulphides. However, the problem is seldom as simple as this, and often has to consider the possibility of complex formation. Even more serious is the fact that widely divergent values of solubility products can be found in the literature.

To ensure the complete precipitation of the Group IV sulphides the solution is made alkaline (generally by adding ammonium hydroxide solution in the presence of ammonium chloride). The presence of the OH^- ensures a greater sulphide ion concentration than the present in a neutral solution of H_2S (the OH^- ions combine with H_3O^+ ions formed when the H_2S ionizes to form water and more H_2S ionizes as a result).

$$H_2O + H_2S \rightleftharpoons H_3O^+ + HS^-$$

$$H_2O + HS^- \rightleftharpoons H_3O^+ + S^{2-}$$

$$+$$

$$OH^-$$

$$\downarrow\uparrow$$

$$2H_2O$$

Both equilibria shifted this way by the addition of OH^- ions.

The formation of a black precipitate of lead sulphide, formed when H_2S comes in contact with a filter paper dipped in lead acetate solution, is a convenient *test* for the gas.

$$Pb^{2+} + H_2S \longrightarrow PbS + 2H^+$$

Metal Sulphides which are not precipitated: The sulphides of Na, K, Mg, Ca, Sr, Ba, Al Cr are soluble in water and are not precipitated in both acidic and alkaline media.

Reaction with ammonium hydroxide: When H_2S gas is passed through ammonium hydroxide yellow ammonium sulphide is formed.

$$2NH_4OH + H_2S \longrightarrow (NH_4)_2S + 2H_2O$$

$$(NH_4)_2S + nH_2S \longrightarrow (NH_4)_2S_{n+1} + nH_2$$

Uses

(i) In the laboratory it is an important reagent in the qualitative analysis for the detection of metal ions.

(ii) It is used in the preparation of metal sulphides which are used in paint industry.

(iii) It is also used as a reducing agent.

Tests

(i) It can be identified by its offensive rotten eggs smell.

(ii) It turns lead acetate paper black.

(iii) It gives purple (violet) colour with sodium nitroprusside.

***Structure* of H_2S is angular in shape:**

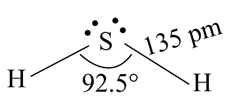


Fig 10.21 Structure of H_2S

10.10.2 Other Hydrides of Sulphur

Sulphur forms a number of hydrides which contain catenated sulphur atoms such as H_2S_2, H_2S_3, H_2S_4 etc. These are yellow oils which readily decompose into H_2S and free sulphur. They can be represented by the general formula

$$H-(S)_n-H.$$

The first member of the series (n=2) is the sulphur analogue of H_2O_2 and decomposes in a similar manner

$$2H_2S_2 \longrightarrow 2H_2S + 2S$$

$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

10.10.3 Oxides of Sulphur

Sulphur forms large number of oxides of which more important are sulphur dioxide and sulphur trioxide.

3A *Sulphur Dioxide*

Preparation

(i) Sulphur dioxide is formed, together with a little sulphur trioxide, when sulphur is burnt in air or oxygen.

$$S + O_2 \longrightarrow SO_2$$

(ii) In the laboratory it is readily generated by reacting a sulphite or bisulphite with dilute acid.

$$Na_2SO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + SO_2$$

$$2NaHSO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O + 2SO_2$$

(iii) It can also be obtained by heating concentrated sulphuric acid with sulphur, copper, silver etc.

$$S + 2H_2SO_4 \longrightarrow 2H_2O + 3SO_2$$

$$Cu + 2H_2SO_4 \longrightarrow CuSO_4 + 2H_2O + SO_2$$

$$2Ag + 2H_2SO_4 \longrightarrow Ag_2SO_4 + 2H_2O + SO_2$$

(iv) *Industrially* it is produced as a by-product during the roasting of sulphide ores and by other methods.

$$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$$

$$4FeS_2 + 11O_2 \longrightarrow 2Fe_2O_3 + 8SO_2$$

$$2CaSO_4 + C \longrightarrow 2CaO + 2SO_2 + CO_2.$$

Physical Properties

(i) Sulphur dioxide is a colourless dense gas with a choking smell.

(ii) It is highly soluble in water –1volume of water at 0ºC dissolves about 80 volumes of the gas.

(iii) It boils at 263k, liquifies under three atmospheres at 20ºC and is used as fumigant.

(iv) It is heavier than air.

Chemical Properties

(a) ***Combustibility*****:** It is neither combustible nor supporter of combustion. But burning magnesium and potassium continue to burn in its atmosphere.

$$3Mg + SO_2 \longrightarrow 2MgO + MgS$$
$$4K + 3SO_2 \longrightarrow K_2SO_3 + K_2S_2O_3$$

(b) ***Thermal decomposition*****:** On heating, it decomposes at 1473k producing sulphur trioxide and sulphur.

$$3SO_2 \longrightarrow 2SO_3 + S$$

(c) ***Acidic Nature*****:** Sulphur dioxide dissolve readily in water forming a solution of sulphurous acid, but any attempt to isolate this acid by evaporation results in sulphur dioxide being recovered.

$$SO_2 + H_2O \rightleftharpoons H_2SO_3 \rightleftharpoons H^+ + HSO_3^- \rightleftharpoons 2H^+ + SO_3^{2-}$$

It reacts very readily with sodium hydroxide solution, forming sodium sulphite, which then reacts with more sulphur dioxide to form the bisulphite.

$$2NaOH + SO_2 \longrightarrow Na_2SO_3 + H_2O$$
$$Na_2SO_3 + H_2O + SO_2 \longrightarrow 2NaHSO_3$$

In its reaction with water and alkalis, the behaviour of SO_2 is very similar to that of CO_2. For example it also turns the lime water milky which disappears by passing excess of SO_2

$$Ca(OH)_2 + SO_2 \longrightarrow CaSO_3 + H_2O$$
$$CaSO_3 + H_2O + SO_2 \longrightarrow Ca(HSO_3)_2$$

So this reaction cannot be used to distinguish between CO_2 and SO_2.

It substitutes less acidic and more volatile CO_2 from carbonates and bicarbonates.

$$Na_2CO_3 + SO_2 \longrightarrow Na_2SO_3 + CO_2$$
$$NaHCO_3 + SO_2 \longrightarrow NaHSO_3 + CO_2$$

It combines with several basic metal oxides.

$$BaO + SO_2 \longrightarrow BaSO_3$$

Addition Reactions

Sulphur dioxide combines with chlorine in the presence of charcoal (which acts as catalyst) to give sulphuryl chloride.

$$SO_2 + Cl_2 \longrightarrow SO_2Cl_2$$

It is catalytically oxidized to sulphur trioxide by oxygen in the presence of vanadium (V) oxide.

$$2SO_2 + O_2 \longrightarrow 2SO_3$$

It combines with lead dioxide to form lead sulphate

$$PbO_2 + SO_2 \longrightarrow PbSO_4$$

Oxidation Properties

Sulphur dioxide can also act as an oxidizing agent particularly with stronger reducing agents.

(i) H_2S is oxidized by SO_2 to sulphur.

$$2H_2S + SO_2 \longrightarrow 2H_2O + 3S$$

(ii) Burning metals when introduced into SO_2 they continue to burn and themselves are oxidized.

$$3Mg + SO_2 \longrightarrow 2MgO + MgS$$
$$4K + 3SO_2 \longrightarrow K_2SO_3 + K_2S_2O_3$$
$$3Fe + SO_2 \longrightarrow 2FeO + FeS$$

(iii) It oxidizes CO to CO_2.

$$2CO + SO_2 \longrightarrow 2CO_2 + S$$

(iv) Hydrogen burns in SO_2 at 1273k.

$$SO_2 + 2H_2 \longrightarrow S + 2H_2O$$

(v) It oxidizes stannous chloride to stannic chloride and mercurous chloride to mercuric chloride in the presence of HCl.

$$2SnCl_2 + SO_2 + 4HCl \longrightarrow 2SnCl_4 + 2H_2O + S$$
$$2Hg_2Cl_2 + SO_2 + 4HCl \longrightarrow 4HgCl_2 + 2H_2O + S.$$

Reducing Properties

Sulphur dioxide behaves as a reducing agent in the presence of water. However, these reactions are best regarded as redox reactions involving the sulphite ion and are discussed in section 10.11.1.

Bleaching action**:** In the presence of moisture SO_2 acts as a bleaching agent. The bleaching property of SO_2 is due to its reducing nature. During bleaching of the coloured compounds to colourless compounds it converts into sulphuric acid.

$$SO_2 + 2H_2O \longrightarrow H_2SO_4 + 2H$$
$$\text{Coloured compund} + H \longrightarrow \text{Colourless compound}$$

The bleaching action of sulphurdioxide is temporary and reversible. The bleached matter when exposed to air, it regains its colour due to oxidation.

Uses

(i) It is mainly used in the manufacture of sulphuric acid.
(ii) It is used to bleach the delicate articles like wool, silk and straw.
(iii) It is used in petroleum refining and sugar industry.
iv) It is used as refrigerant in the form of liquid SO_2.
(v) It is used as antichlor for removing chlorine from the textiles after bleaching.

***Structure*:** Its structure is discussed in 10.5.3.

10.10.3(B) Sulphur Trioxide SO_3: *Preparation*

Sulphur trioxide is prepared

(i) By heating ferric sulphate

$$Fe_2(SO_4)_3 \longrightarrow Fe_2O_3 + 3SO_3$$

Heating of ferrous sulphate gives a mixture of SO_2 and SO_3.

$$2FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$$

(ii) By dehydration of sulphuric acid: Sulphur trioxide is prepared by the dehydration of concentrated sulphuric acid distilled with phosphorous pentoxide.

$$2H_2SO_4 + P_4O_{10} \longrightarrow 2SO_3 + 4HPO_3$$

(iii) By the direct combination of SO_2 and O_2. A mixture of dry sulphur dioxide and oxygen is passed over heated spongy platinum or platinized asbestos and the sulphur trioxide produced is collected in an ice-cooled receiver.

The reaction is exothermic and the optimum working temperature is 673–723k. It has been found that 98 per cent of sulphur dioxide is converted.

$$2SO_2 + O_2 \rightleftharpoons 2SO_3; \quad \Delta H = -2 \times 94.6\,kJ$$

***Physical Properties*:** Sulphur trioxide exists in three allotropic forms.

(i) α-Sulphur trioxide: It is a colourless liquid that solidifies in the form of prismatic needle like crystals. Its m.p is 290k and b.p. is 318k. Its specific gravity is 1.9229 (at 298K).
This form is prepared by repeated fractionation of sulphur trioxide formed by any of the methods given above. The molecular weight of this substance as determined by freezing point depression is about 80.

(ii) β-Sulphur trioxide: The α-form, when allowed to stand at 10ºC for some time, changes slowly into the β-form. This form solidifies to give asbestos-like needles at 335.5k. Its molecular weight has been close 160 which corresponds to the formula S_2O_6 on being at a temperature between 50ºC and 100ºC, it changes into α-Variety.

(iii) γ-Sulphur trioxide: It is produced when β-form is completely dried. It sublimes without melting under ordinary temperature.

Chemical Properties

(i) Chemically, all the three forms have been almost identical, although α-form has been comparatively more active than the rest. Any of these forms dissolves in water readily to form sulphuric acid and evolving much heat.

(ii) Action of heat: SO_3 on being heated decomposes into sulphur dioxide and oxygen. Decomposition gets completed at 1273k.

$$2SO_3 \longrightarrow 2SO_2 + O_2$$

(iii) Action of water: It is having great affinity for water and dissolves in it producing hissing sound and yielding a mist of fine droplets of sulphuric acid. It fumes strongly in moist air.

$$SO_3 + H_2O \longrightarrow H_2SO_4$$

(iv) Formation of oleum: Sulphur trioxide on dissolving in concentrated sulphuric acid yields oleum (fuming sulphuric acid or pyrosulphuric acid)

$$H_2SO_4 + SO_3 \longrightarrow H_2S_2O_7$$

(v) Acidic nature: It is an acidic oxide. It is the anhydride of sulphuric acid. It reacts with alkalis, carbonates or basic oxide to give sulphates.

$$CaO + SO_3 \longrightarrow CaSO_4$$

(vi) With hydrochloric acid: Sulphur trioxide forms an addition product, chlorosulphonic acid.

$$SO_3 + HCl \longrightarrow ClSO_2 \cdot OH$$

(vii) Oxidation property: Sulphur trioxide acts as powerful oxidizing agent. For example, it oxidizes hydrogen bromide to bromine.

$$SO_3 + 2HBr \longrightarrow H_2O + Br_2 + SO_2.$$

Uses

Sulphur trioxide is used

(i) in the manufacture of sulphuric acid and oleum.
(ii) for drying gases.

***Structure*:** Structure is already discussed in 10.5.3.

10.11 OXOACIDS OF SULPHUR

Sulphur form numerous oxoacids. The oxoacids of sulphur may be divided into four series, depending on their structural similarities. These series of acids are

(i) Sulphurous acid series
(ii) Sulphuric acid series
(iii) Thionic acid series
(iv) Peroxo acid series.

Each of these acids may have more than one structure. The structures of these given in Table 10.7 are one of the several possible resonating or tautomeric forms. Oxoacids with S—S linkages are known as thioacids. Such analogous thioacids are not known for selenium and tellurium.

In all the oxoacids, the basicity is 2 which is equal to the number of –OH groups present in a molecule of acid.

In these acids, in general, it can be taken that those sulphur atoms that are linked to one or more oxygen atoms undergo sp^3 hybridization. The necessary excited states of electrons in the sulphur are as follows.

Outer electronic configuration of sulphur

1st excited state (assumed to be present in +III or +IV oxidation state.)

$$\underbrace{\boxed{\uparrow\downarrow}\quad\boxed{\uparrow}\boxed{\uparrow}\boxed{\uparrow}}_{sp^3\text{ hybridization}}\quad\boxed{\uparrow}\boxed{}\boxed{}\boxed{}\boxed{}$$

2nd excited state (assumed to be present in +VI oxidation state.)

$$\underbrace{\boxed{\uparrow}\quad\boxed{\uparrow}\boxed{\uparrow}\boxed{\uparrow}}_{sp^3\text{ hybridization}}\quad\boxed{\uparrow}\boxed{\uparrow}\boxed{}\boxed{}\boxed{}$$

10.11.1 Sulphurous Acid H_2SO_3

An aqueous solution of sulphurous acid, H_2SO_3, is obtained when sulphur dioxide is passed into water, the solution contains H^+, HSO_3^- and SO_3^{2-} ions together with free sulphur dioxide.

Although pure sulphurous acid does not exist, sulphites of the alkali metals e.g., Na_2SO_3, can be obtained as solids. Hydrogen sulphites e.g., $NaHSO_3$ also exist in solution, but when attempts are made to isolate them, two hydrogen sulphite ions condense with elimination of water, and a pyrosulphite is deposited.

$$2NaHSO_3 \rightleftharpoons \underset{\text{Sod. pyrosulphite}}{Na_2S_2O_5} + H_2O$$

$$2HSO_3^- \rightleftharpoons S_2O_5^{2-} + H_2O$$

The above reaction is reversible, since pyrosulphite gives the reactions of hydrogen sulphites (bisulphites) in aqueous solution.

***Reducing properties*:** Sulphurous acid is reducing agent. Half reaction where in itself is oxidized is as under

$$H_2SO_3 + H_2O \longrightarrow SO_4^{2-} + 4H^+ + 2e^-$$

All reducing properties of sulphur dioxide in aqueous solution have been infact those of sulphurous acid.

Its important reducing properties are as follows.

(i) Reduction of halogens to hydracids.

$$H_2SO_3 + 2H_2O \longrightarrow H_2SO_4 + 2H$$
$$2H + X_2 \longrightarrow 2HX \qquad (X_2 = F_2, Cl_2, Br_2 \text{ or } I_2)$$
$$\overline{SO_2 + 2H_2O + X_2 \longrightarrow H_2SO_4 + 2HX}$$

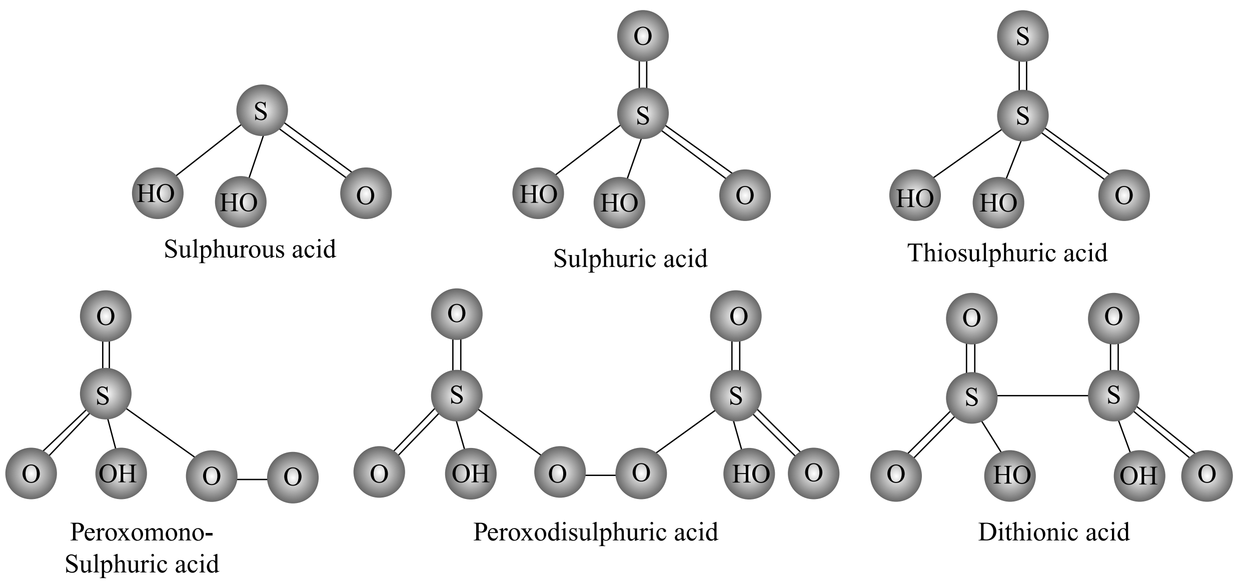
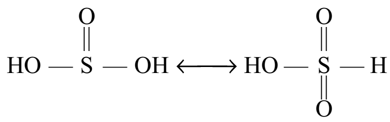
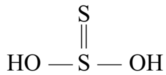
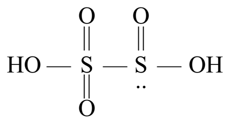
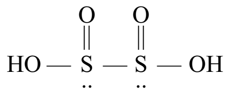
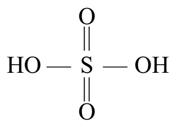
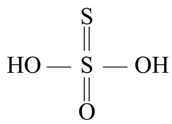
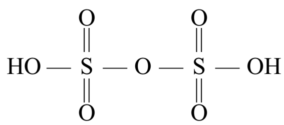
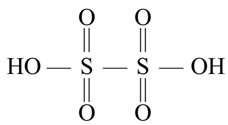
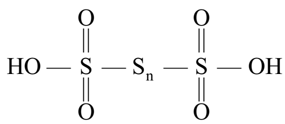
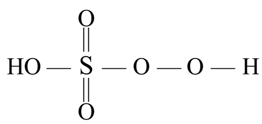
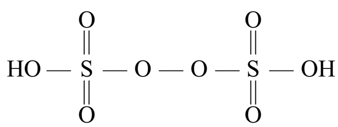


Fig 10.22 Structures of oxoacids of sulphur

Table 10.7 Oxo acids of sulphur
T = Terminal sulphur; C = Central sulphur; Av = Average oxidation state of S

Name of the oxoacid	Molecular formula	Structural formula	Oxidation State of oxoacid		
			T	C	Av.
I *Sulphurous acid series*					
(a) Sulphurous acid	H_2SO_3		–	+IV	+IV
(b) Thiosulphurous acid	$H_2S_2O_2$		–II	+IV	+I
(c) Disulphurous or Pyrosulphurous acid	$H_2S_2O_5$		+III	+V	+IV
(d) Dithionous acid	$H_2S_2O_4$		+III	+III	+III
II *Sulphuric acid series*					
(a) Sulphuric acid	H_2SO_4		–	+VI	+VI
(b) Thiosulphuric acid	$H_2S_2O_3$		–II	+VI	+II
(c) Disulphuric or Pyrosulphuric acid	$H_2S_2O_7$		+VI	+VI	+VI
III *Thionic acid series*					
(a) Dithionic acid	$H_2\ S_2O_6$		+V	+V	+V
(b) Polythionic acid	$H_2S_{n+2}O_6$		+V	O	$\frac{10}{n+2}$
(a) Peroxomono sulphuric acid or Caro's acid	H_2SO_5			+VI	+VI
(b) Peroxodisulphuric acid or Perdisulphuric acid or Marshall's acid	$H_2S_2O_8$		+VI	+VI	+VI

(ii) Reduces ferric salts to ferrous salts

$$H_2SO_3 + H_2O \longrightarrow H_2SO_4 + 2H$$
$$\underline{Fe_2(SO_4)_3 + 2H \longrightarrow 2FeSO_4 + H_2SO_4}$$
$$Fe_2(SO_4)_3 + SO_2 + 2H_2O \longrightarrow 2FeSO_4 + 2H_2SO_4$$

(iii) Reduces acidified potassium permanganate

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$$
$$[H_2SO_3 + H_2O \longrightarrow H_2SO_4 + 2H] \times 5$$
$$\underline{[2H + (O) \longrightarrow H_2O] \times 5}$$
$$2KMnO_4 + 5SO_2 + 2H_2O \longrightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4$$

(iv) Reduces orange red dichromate to green chromic sulphate

$$K_2Cr_2O_7 + 4H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3(O)$$
$$[SO_2 + 2H_2O \longrightarrow H_2SO_4 + 2H] \times 3$$
$$\underline{[2H + (O) \longrightarrow H_2O] \times 3}$$
$$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$$

This reaction is used as a convenient test for the detection of the gas and also for distinguishing from carbon dioxide.

(v) It reduces iodates to iodine.

$$2KIO_3 + H_2SO_4 \longrightarrow K_2SO_4 + 2HIO_3$$
$$[SO_2 + 2H_2O \longrightarrow H_2SO_4 + 2H] \times 5$$
$$\underline{2HIO_3 + 10H \longrightarrow I_2 + 6H_2O}$$
$$2KIO_3 + 5SO_2 + 4H_2O \longrightarrow K_2SO_4 + 4H_2SO_4 + I_2$$

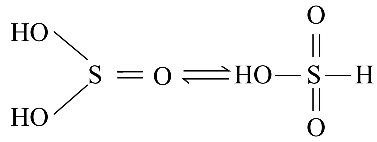
***Acidic properties*:** Sulphurous acid ionizes in two steps

$$H_2SO_3 \rightleftharpoons H^+ + [HSO_3]^- \qquad K_1 = 1.72 \times 10^{-2}$$
$$[HSO_3]^- \rightleftharpoons H^+ + SO_3^{2-} \qquad K_2 = 6.24 \times 10^{-2}$$

The first dissociation has been fairly strong but the second one has been weak. Generally forms two types of salts sulphites and bisulphites.

***Structure*:** It is regarded doubtful. If sulphurous acid gets formed when sulphur dioxide is dissolved in water, although the solution obtained is acidic in nature, its salts – sulphites and bisulphites – have been, however, well known and the latter has been predominant in dilute solution.

Sulphurous acid and bisulphites have been reducing agents due to the tautomerism.



Like P—H bonds in phosphorous acid, the S—H bonds in sulphurous acid must make it a reducing agent and this would also explain the formation of HSO_3^- ion.

The sulphite ion is pyramidal in shape with one of the positions occupied by a lone pair. The sulphur atom is involved in sp^3 hybridization.

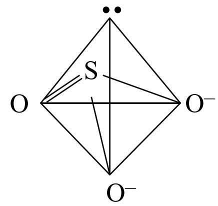


Fig 10.23 Structure of SO_3^{2-} ion

As all the S—O bonds in sulphite ion are identical (139 pm) the structure of sulphite ion is considered as a resonance hybrid of the following three resonance structures.

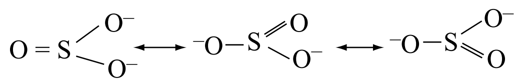


Fig 10.24 Resonance structures of SO_3^{2-} ion

10.11.2 SULPHURIC ACID H_2SO_4

Sulphuric acid is the most important acid used in chemical industry. In the latter middle ages, it was referred as *Oil of Vitriol*. Because of its wide applications, it was also called as King of chemicals. The acid is manufactured by the following process.

Essentially the manufacture of this acid involves the conversion of sulphur dioxide into sulphur trioxide which is then dissolved in water to form sulphuric acid.

Sources of Sulphur Dioxide

(i) By burning of sulphur.
(ii) From anhydrite by heating with carbon.
(iii) By roasting of iron pyrites.
(iv) By roasting sulphide minerals as byproduct.
(v) By burning hydrogen sulphide obtained from oil refining.
(vi) Isolation of sulphur from natural gas. Hydrogen sulphide from natural gas and from crude oil is extracted and burnt in air to form SO_2, this then reacted with more H_2S to form sulphur.

$$2H_2S + 3O_2 \longrightarrow 2H_2O + 2SO_2$$
$$2H_2S + SO_2 \longrightarrow 2H_2O + 3S.$$

Sulphur dioxide is converted into sulphuric acid by "the lead chambers process" and by "the contact process".

(a) ***Lead chamber's process***: This method is known as lead chambers process because sulphuric acid is manufactured in the chambers made with lead. Lead is cheaper and becomes passive with concentrated sulphuric acid. In this process SO_2 is oxidized to SO_3 in the presence of NO_2 as catalyst.

$$2SO_2 + O_2 \xrightarrow[Cat^-]{NO_2} 2SO_3$$
$$2NO + O_2(air) \longrightarrow 2NO_2$$
$$SO_3 + H_2O(Steam) \longrightarrow H_2SO_4$$

Mechanism proposed by Berzelius is as follows

$$2SO_2 + 2NO_2 \longrightarrow 2SO_3 + 2NO$$
$$2NO + O_2 \longrightarrow 2NO_2$$

The NO and NO_2 act as oxygen carriers i.e., they are able to transfer oxygen from the air to sulphur dioxide. But this mechanism could not explain the formation of chamber's acid i.e., nitroso sulphuric acid HSO_4NO. Hence it is suggested by Davy and Lunge that the nitroso acid is formed as an intermediate during the formation of sulphuric acid.

$$NO + NO_2 \longrightarrow N_2O_3$$
$$2SO_2 + N_2O_3 + H_2O + O_2 \longrightarrow 2HSO_4 \cdot NO$$
$$2HSO_4NO + H_2O \longrightarrow 2H_2SO_4 + NO + NO_2$$

It should be noted that nitroso sulphuric acid is formed if insufficient steam is present but does not prove that it is formed as an intermediate product under normal working conditions.

(b) ***Contact process:*** In contact process SO_2 is catalytically oxidized with atmospheric air to SO_3

$$2SO_2 + O_2 \xrightarrow[\Delta]{Catalyst} 2SO_3$$

The pure SO_3 formed is absorbed in 98 per cent conc H_2SO_4 to get oleum or pyrosulphuric acid $H_2S_2O_7$. Oleum is diluted with water to to get sulphuric acid of desired concentration.

$$SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$$
$$H_2S_2O_7 + H_2O \longrightarrow 2H_2SO_4$$

If SO_3 is directly passed into water much heat will be liberated due to the great affinity of SO_3 towards water. Hence the solution boils and forms a mist of corrosive nature. Moreover the reaction is inefficient.

The oxidation of SO_2 to SO_3 in the presence of a catalyst is a reversible reaction.

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)} \qquad \Delta H = -189\ KJ.$$

Since the reaction is reversible and exothermic, a high yield of the trioxide will be favoured reasonably at low temperature (Le Chatelier's principle). But the temperature must not be too low because the rate of reaction will become too slow at low temperatures. A temperature of about 450ºC is the optimum temperature and a conversion rate of about 95 to 98 percent is achieved.

The reaction is also attended by a decrease in volume, hence an increase in pressure should increase the equilibrium yield of sulphur trioxide and also the rate of reaction. Since both yield and reaction rate are quite satisfactory at ordinary pressure it has been decided that the extra expenditure of making pressure equipment would not be justified.

The rate of formation of SO_3 is enhanced by using a catalyst. Different catalysts are in use. The name of the catalyst and the name of the corresponding process are given in Table 10.8. All the catalysts used in the contact process are susceptible for poisoning and therefore, the gases must be used extremely pure. In modern plants excess of oxygen is used in the gaseous mixture since air contains only 21 per cent by volume, the decrease in volume (ΔV air) is much less than that implied by the stoichiometric equation.

The flow chart for the manufacture of sulphuric acid is given in Fig. 10.25.

SO_2 is produced in **pyrite burners** or **sulphur burners**. The dust particles present in the gases are removed in **dusting tower** then the gases are cooled in **cooling pipes**. Then the gases are purified by the water sprinkled in **scrubbing towers**. After drying the gases in a **drying tower**, they are passed into **arsenic purifier** to remove arsenic oxide by reacting with gelatinous ferric hydroxide present in it. The gases are tested for their purity in a test box. If the gas contain any impurities, they should be recycled. The pure gases are heated to 673–723K, then passed into the **catalytic chamber** packed with the catalyst V_2O_5. Then SO_2 is

Table 10.8 Catalysts in contact process

S. no	Catalyst used	Name of the process
1.	Platinized asbestos	Baundische process
2.	Vanadium pentoxide	Baundische process
3.	Finely divided platinum deposited on $MgSO_4$	Grillo's process
4.	A mixture of Fe_2O_3 and cupric oxide. (CuO)	Mannheim process

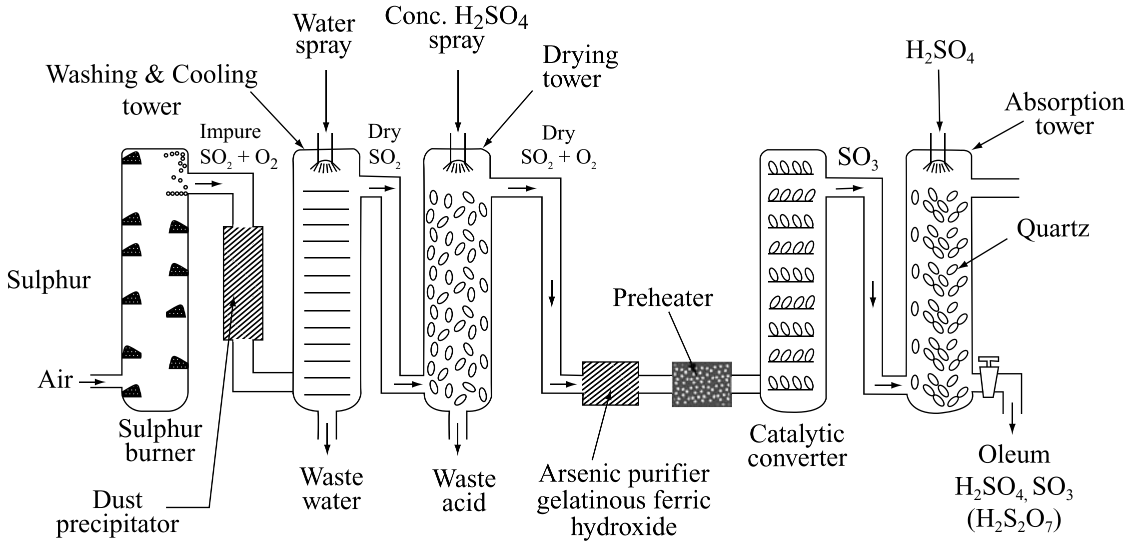


Fig 10.25 Flow diagram for the manufacture of sulphuric acid

converted into SO_3. The SO_3 thus obtained is absorbed in conc H_2SO_4 to form oleum $H_2S_2O_7$. The oleum is diluted with water to get the acid of required concentration.

Contact process is more advantageous because, the H_2SO_4 is extremely pure and concentrated. Further the impurities can be tested and the reactants can be recycled.

Physical Properties

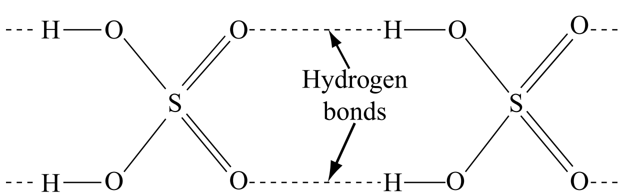
(i) Pure sulphuric acid is a colourless viscous syrupy liquid. It contains about 98.3 per cent sulphuric acid. Its specific gravity is 1.84 gm cm^{-3}.

(ii) It boils at 511k and freezes at 283.5k.

(iii) In the absence of water it will not turn blue litmus red nor will it react with metals to form hydrogen.

(iv) It decomposes on boiling to form sulphur trioxide and steam and a constant boiling mixture (azeotropic mixture) is formed containing 98.3 per cent acid.

(v) The high boiling point and viscosity are presumably due to the presence of hydrogen bonding which link the molecules into larger aggregates.



(vi) Pure sulphuric acid is covalent.

(vii) It is highly soluble in water. It has great affinity towards water. So much heat is liberated when it is dissolved in water due to the formation of various hydrates such as $H_2SO_4 \cdot H_2O$; $H_2SO_4 \cdot 2H_2O$; $H_2SO_4 \cdot 3H_2O$; $H_2SO_4 \cdot 4H_2O$ etc.

(viii) Dilution of concentrated sulphuric acid must be carried by adding concentrated sulphuric acid slowly to cold water with constant stirring. Water should not be added to concentrated sulphuric acid for dilution because the solution boils and forms fog and mist due to the much heat liberated.

Chemical Properties

These are most conveniently discussed under several headings.

(a) ***Acidic Properties***

(i) Concentrated sulphuric acid reacts violently with water to give a solution which exhibits the properties of strong acid. The ionization of sulphuric acid in water takes place in two stages i.e., it is dibasic; the first ionization is virtually complete, where as the second takes place to the extent of about 10 per cent

$$H_2SO_4 + H_2O \rightleftharpoons H_3O^+ + HSO_4^-$$

(virtually complete)

$$HSO_4^- + H_2O \rightleftharpoons H_3O^+ + SO_4^{2-}$$

(about 10% Complete)

An aqueous solution of the acid therefore contains H_3O^+ ions and very many more HSO_4^- and SO_4^{2-} ions. In the presence of OH^- ions which are stronger proton acceptors than the neutral water molecules more extensive ionization of the HSO_4^- ions takes place and SO_4^{2-} ions predominate.

$$H_2SO_4^- + OH^- \rightleftharpoons H_2O + SO_4^{2-}$$

(equilibrium to the right)

This explains why the interaction of sodium hydroxide solution and dilute sulphuric acid produce sodium sulphate and why special conditions are needed to crystallize the bisulphate

$$NaOH + H_2SO_4 \longrightarrow NaHSO_4 + H_2O$$
$$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$$

(ii) It decomposes carbonates and bicarbonates liberating carbon dioxide.

$$Na_2CO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2$$
$$NaHCO_3 + H_2SO_4 \longrightarrow NaHSO_4 + H_2O + CO_2.$$

(iii) It liberates hydrogen gas with electropositive metals.

$$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$$
$$Fe + H_2SO_4 \longrightarrow FeSO_4 + H_2.$$

(iv) It displaces more volatile acids from their metal salts. A less volatile acid can substitute a more volatile acid from a salt. Since concentrated sulphuric acid is less volatile acid and can substitute several volatile acids from their metal salts, this reaction is the basis for the detection of certain anions in qualitative analysis. It can substitute HCl from chlorides, nitric acid from nitrates, phosphoric acid from phosphates, oxalic acid from oxalates, H_2S from sulphides, acetic acid from acetates, hydrogen fluoride from fluorides etc. e.g.,

$$2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$$
$$2NaNO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + 2HNO_3$$

(b) ***Dehydrating Properties*:** Concentrated sulphuric acid has such an affinity for water that it will remove it from mixtures and compounds with evolution of much heat. Because of this reason when sulphuric acid falls on the skin produces burning sensation due to the heat evolved during absorption of water from the body.

Gases are dried in the laboratory by passing them through a wash bottle containing concentrated sulphuric acid. The gases like ammonia which are basic in nature and combine with sulphuric acid cannot be dried using concentrated sulphuric acid. When hydrogen is dried using concentrated sulphuric acid, explosion takes place because hydrogen burns due to the heat liberated during drying.

Many organic compounds loose the elements of water when treated with the concentrated sulphuric acid. Thus formic acid gives carbon monoxide, oxalic acid gives an equimolar mixture of carbon monoxide and carbon dioxide, carbohydrates are chared to black due to formation of carbon.

$$HCOOH \xrightarrow[-H_2O]{\text{Conc } H_2SO_4} CO$$
$$\begin{array}{l} COOH \\ | \\ COOH \end{array} \xrightarrow[-H_2O]{\text{Conc } H_2SO_4} CO + CO_2$$
$$\underset{\text{glucose}}{C_6H_{12}O_6} \xrightarrow[-6H_2O]{\text{Conc } H_2SO_4} 6C$$
$$C_{12}H_{22}O_{11} \xrightarrow[-11H_2O]{\text{Conc } H_2SO_4} 12\,C$$

The dehydration of organic compounds is of industrial importance e.g., diethyl ether is made by dehydrating ethanol and the insecticide DDT is made by dehydrating a mixture of chloro benzene and trichloroacetaldehyde (chloral).

(c) ***Oxidizing Properties*:** Hot concentrated sulphuric acid functions as an oxidizing agent, although less effectively than concentrated nitric acid. A number of reduction products of sulphuric acid can be formed, their relative proportions in any one particular reaction depends upon such factors as concentration of the acid, strength of the reducing agent present and temperature. These possible reduction products are shown schematically below

$$H_2SO_4 \xrightarrow{-[O]} (H_2SO_3) \longrightarrow H_2O + SO_2$$
$$H_2SO_4 \xrightarrow{-4[O]} H_2S$$

Which may be followed by oxidation of the hydrogen sulphide to sulphur.

Oxidation of Non-metals: Non metals such as carbon and sulphur reduce sulphuric acid, when heated, to sulphur dioxide, they themselves being oxidized to their dioxides.

$$C + 2H_2SO_4 \longrightarrow 2H_2O + CO_2 + 2SO_2$$
$$S + 2H_2SO_4 \longrightarrow 2H_2O + 3SO_2$$

Oxidation of metals: With dilute sulphuric acid metals above hydrogen in the electrode potential series liberate hydrogen. But with concentrated sulphuric acid it is possible to obtain SO_2, H_2S and S in addition to the metallic sulphate; further reaction between the metalic sulphate and hydrogen sulphide can give rise to the significant amounts of the metallic sulphides. These reactions are therefore quite complex; however, in general it can be said that SO_2 is the main product when metals below the electrode potential series are used e.g., copper, but that increasing amounts of H_2S are formed with more reactive metals e.g., zinc.

The following equations, therefore, do no more than indicate the main reactions that takes place. Metals below hydrogen in electrode potential series, like copper, silver, mercury, do not liberate hydrogen from dilute sulphuric acid but oxidized in the presence of air.

$$2Cu + 2H_2SO_4 + O_2 \longrightarrow 2CuSO_4 + 2H_2O$$

Zinc with dilute sulphuric acid liberates hydrogen gas (acid property) but gives different products with concentrated sulphuric acid.

$$\underset{\text{dil}}{Zn + H_2SO_4} \longrightarrow ZnSO_4 + H_2$$

$$Zn + \underset{98\ \%}{2H_2SO_4} \longrightarrow ZnSO_4 + H_2O + SO_2$$

$$4Zn + \underset{90\ \%}{5H_2SO_4} \longrightarrow 4ZnSO_4 + 4H_2O + H_2S$$

$$ZnSO_4 + H_2S \longrightarrow ZnS + H_2SO_4$$

(**d**) **Oxidation of compounds:** As already explained, since concentrated sulphuric acid is less volatile it substitutes the volatile acids from several salts. But some of them will be oxidized by conc H_2SO_4. For example when concentrated sulphuric acid is added to halides, HF is liberated from fluorides, HCl is liberated from chlorides but HBr and HI liberated initially from bromides and iodides will be oxidized by conc H_2SO_4 to Br_2 and I_2 respectively.

$$CaF_2 + H_2SO_4 \longrightarrow CaSO_4 + 2HF$$

$$2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$$

$$2KBr + H_2SO_4 \longrightarrow K_2SO_4 + 2HBr$$

$$2HBr + H_2SO_4 \longrightarrow 2H_2O + SO_2 + Br_2$$

$$2KI + H_2SO_4 \longrightarrow K_2SO_4 + 2HI$$

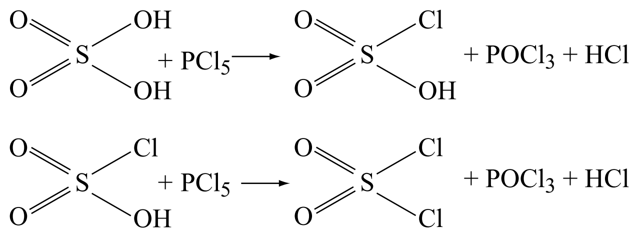
$$2HI + H_2SO_4 \longrightarrow 2H_2O + SO_2 + I_2$$

$$8HI + H_2SO_4 \longrightarrow 4H_2O + H_2S + 4I_2$$

These reactions explain why the free halogen is always liberated together with hydrogen halide when bromides and iodides are treated with concentrated sulphuric acid.

(e) ***Miscellanious reactions***

(i) ***Action of PCl_5*:** Sulphuric acid reacts with PCl_5 in two steps first forming chloro sulphonic acid $SO_2(OH)Cl$ then sulphuryl chloride in the second. This reaction proves the presence of two –OH groups in sulphuric acid.



(ii) ***Reaction with potassium ferrocyanide*:** On heating potassium ferrocyanide with dilute sulphuric acid liberates poisonous HCN but with concentrated sulphuric acid liberates carbon monoxide.

$$K_4[Fe(CN)_6] + 3H_2SO_4 \longrightarrow$$
$$2K_2SO_4 + FeSO_4 + 6HCN$$

$$K_4[Fe(CN)_6] + 6H_2SO_4 + 6H_2O \longrightarrow$$
$$2K_2SO_4 + FeSO_4 + 6CO + 3(NH_4)_2SO_4$$

(iii) ***Reaction with potassium chlorate*:** When potassium chlorate is heated with concentrated sulphuric acid, chlorine dioxide is evolved with explosion.

$$3KClO_3 + 3H_2SO_4 \longrightarrow$$
$$3KHSO_4 + HClO_4 + 2ClO_2 + H_2O$$

(iv) Sulphuric acid reacts with certain metal salts to form insoluble metal sulphates.

$$BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HCl$$

$$Pb(NO_3)_2 + H_2SO_4 \longrightarrow PbSO_4 + 2HNO_3.$$

(v) H_2SO_4 absorbs SO_3 forming oleum or fuming sulphuric acid or pyrosulphuric acid

$$H_2SO_4 + SO_3 \longrightarrow H_2S_2O_7$$

H_2SO_4 can be dehydrated with phosphorous pentoxide to get SO_3

$$H_2SO_4 + P_2O_5 \longrightarrow SO_3 + 2HPO_3$$

(**b**) ***Reaction with organic compounds*:** Alkenes undergo addition reactions with concentrated sulphuric acid, if the addition products are diluted with water and heated, alcohols are produced. This reaction is of industrial importance e.g.,

$$\underbrace{CH_3 - CH_2 - CH = CH_2}_{\text{But}-1-\text{ene}} + H_2SO_4 \longrightarrow$$

$$CH_3 - CH_2 - \underset{\underset{OSO_3H}{|}}{CH} - CH_3$$

$$\underset{\displaystyle |\atop OSO_3H}{CH_3-CH_2-CH-CH_3}+H_2O\longrightarrow$$

$$\underset{\text{Butan-2-ol}}{\underset{\displaystyle |\atop OH}{CH_3-CH_2-CH-CH_3}}+H_2SO_4$$

Sulphuric acid also reacts with long chain hydrocarbons with the elimination of water e.g.,

$$C_{18}H_{30}+H_2SO_4\longrightarrow \underset{\text{a sulphonic acid}}{C_{18}H_{29}SO_3H}+H_2O$$

The sodium salts of sulphonic acids are used in detergent preparations e.g., sodium benzene dodecyl sulphonate $C_{18}H_{29}SO_3Na$ is the main surface active constituent of Tide, Daz, Surf and Omo.

***Uses*:** Sulphuric acid is used on an enormous scale.

(i) Most of the sulphuric acid is used in the manufacture of fertilizers e.g., superphosphate of lime and ammonium sulphate.

(ii) It is used in industries for the manufacture of paints—titanium (IV) oxide and lithophone.

Synthetic fibres—rayon and plastics

Acid production—Hydrochloric and hydrofluoric acids

Metallurgy—Pickling steel and sulphates for electrolysis

Detergents—mostly alkyl-aryl sulphonates

Petroleum refining—mostly extraction of alkenes

Organic chemicals—Dyestuffs and explosives and drugs.

When used in the manufacture of dye stuffs and explosives its purpose is to enhance the nitrating property of nitric acid.

Pickling is a process of the removal of basic metal oxides, hydroxide and carbonates present on the metal surface by dipping in dilute sulphuric acid before the metals are galvanized, painted, enameled or electroplated.

Sulphuric acid is used in lead storage batteries. During the discharging of lead storage batteries sulphuric acid is consumed

$$Pb + PbO_2 + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O$$

***Structure*:**

Sulphuric acid and sulphate ion have tetrahedral geometry in which sulphur is involved in sp^3 hybridization.

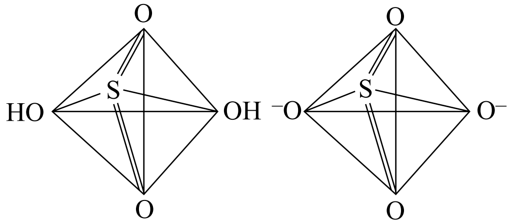
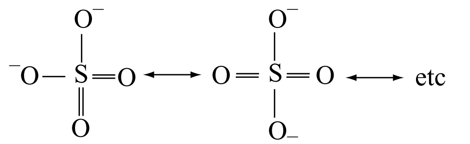


Fig 10.26 Structure of H_2SO_4 and SO_4^{2-} ion

X-ray analysis of the ionic crystals containing SO_4^{2-} ions has shown that SO_4^{2-} ion has tetrahedral shape and all the S—O bonds are equal. This indicates the SO_4^{2-} is a resonance hybrid of several resonance structures.



10.12 CHLORO DERIVATIVES OF SULPHUROUS AND SULPHURIC ACIDS

10.12.1 Thionyl Chloride $SOCl_2$ – A Derivative of Sulphurous Acid

Thionyl chloride is made by reacting sulphur dioxide (or a sulphite) with phosphorous pentachloride, the liquid product then being separated by distillation.

$$SO_2 + PCl_5 \longrightarrow SOCl_2 + POCl_3$$

It is hydrolysed by water, like most covalent halides and on warming sulphur dioxide and hydrogen chloride are expelled.

$$SOCl_2 + H_2O \longrightarrow SO_2 + 2HCl$$

It is mainly used in organic chemistry for the replacement of an hydroxyl group by the chlorine atom e.g., the conversion of carboxylic acids to acid chlorides and alcohols to chloroalkanes.

$$CH_3COOH + SOCl_2 \longrightarrow CH_3COCl + SO_2 + HCl$$

$$C_2H_5OH + SOCl_2 \longrightarrow C_2H_5Cl + SO_2 + HCl$$

The thionyl chloride is pyramidal with the lone pair electrons occupying the fourth tetrahedral position.

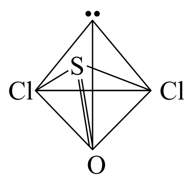


Fig 10.27 Structure of $SOCl_2$

10.12.2 Chlorosulphonic Acid $HOSO_2Cl$ – A Derivative of Sulphuric Acid

This is obtained by reacting hydrogen chloride with fuming sulphuric acid, the water produced being absorbed by the excess of sulphuric acid.

$$H_2SO_4 + HCl \longrightarrow HOSO_2Cl + H_2O$$

Chlorosulphonic acid is a colourless liquid which fumes in moist air owing to hydrolysis; with much water, it readily convertes back again into sulphuric acid.

$$HOSO_2Cl + H_2O \longrightarrow H_2SO_4 + HCl$$

Its molecule has a tetrahedral shape like that of sulphuric acid, the only difference being the replacement of one OH group by the chlorine atom.

10.12.3 Sulphuryl Chloride (Dichloro-Sulphonic Acid) SO_2Cl_2 – A Derivative of Sulphuric Acid

When concentrated sulphuric acid is reacted with phosphorous pentachloride, both OH groups are replaced by chlorine atoms to produce sulphuryl chloride.

$$H_2SO_4 + 2PCl_5 \longrightarrow SO_2Cl_2 + 2POCl_3 + 2HCl$$

It can be made by the direct combination of SO_2 and Cl_2 in the presence of charcoal as catalyst.

$$SO_2 + Cl_2 \longrightarrow SO_2Cl_2$$

Sulphuryl chloride is a colourless liquid but its boiling point 342k is much lower than that of chlorosulphonic acid, since the possibility of hydrogen bonding is ruled out. Like chloro sulphonic acid it is readily hydrolysed by water.

$$SO_2Cl_2 + 2H_2O \longrightarrow H_2SO_4 + 2HCl.$$

The molecule of sulphuryl chloride is tetrahedral as expected.

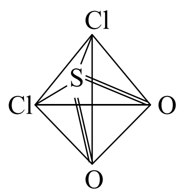


Fig 10.28 Structure of SO_2Cl_2

10.13 SODIUM THIOSULPHATE $Na_2S_2O_3 \cdot 5H_2O$

The anhydrous salt $Na_2S_2O_3$ is simply called as *sodium thiosulphate*, but the hydrated sodium thiosulphate $Na_2S_2O_3 \cdot 5H_2O$ is known as *hypo*. It is a salt of thio sulphuric acid which has never been isolated. They contain the thiosulphate ion which can be regarded as being derived from the sulphate ion by replacement of one oxygen atom by sulphur.

Preparation

(i) In the laboratory it is prepared by boiling alkaline or neutral sodium sulphite with "flowers of sulphur".

$$Na_2SO_3 + \underset{\text{excess}}{S} \xrightarrow{\text{boiling}} Na_2S_2O_3$$

The unreacted excess sulphur is filtered off and the filtrate is concentrated to the crystallization point to get $Na_2S_2O_3$. $5H_2O$ crystals.

(ii) When sulphur is boiled with sodium hydroxide, sodium thiosulphate and sodium sulphide or sodium penta sulphide will be formed.

$$6NaOH + 4S \longrightarrow Na_2S_2O_3 + 2Na_2S + 3H_2O$$

$$6NaOH + \underset{\text{excess}}{12S} \longrightarrow Na_2S_2O_3 + 2Na_2S_5 + 3H_2O$$

The sodium penta sulphide also converts into hypo when exposed to air.

$$2Na_2S_5 + 3O_2 \xrightarrow{\text{heating in air}} 2Na_2S_2O_3 + 6S$$

(iii) Sodium sulphide solution can also react with SO_2 to form $Na_2S_2O_3$.

$$2Na_2S + 3SO_2 \longrightarrow 2Na_2S_2O_3 + S.$$

(iv) *Spring's reaction*: Hypo is formed when a mixture of sodium sulphite and sodium sulphide is treated with calculated amount of iodine

$$Na_2S + Na_2SO_3 + I_2 \longrightarrow Na_2S_2O_3 + 2NaI.$$

(v) When SO_2 gas is passed through the waste liquor obtained in the Leblanc process for the manufacture of sodium carbonate which contain Na_2CO_3, Na_2SO_3 and Na_2S, hypo will be obtained.

$$2Na_2S + Na_2SO_3 + 3SO_2 \longrightarrow 3Na_2S_2O_3$$

$$2Na_2S + Na_2CO_3 + 4SO_2 \longrightarrow 3Na_2S_2O_3 + CO_2$$

This method is used for the production of hypo industrially.

(vi) When SO_2 gas is passed through the solution of Na_2CO_3 containing sulphur, sodium thiosulphate will be formed.

$$Na_2CO_3 + SO_2 \longrightarrow Na_2SO_3 + CO_2$$

$$Na_2SO_3 + S \longrightarrow Na_2S_2O_3.$$

Properties

Physical Properties

(i) It is a colourless crystalline, *efflorescent* substance.
(ii) It is highly soluble in water. It is one of the substances that can form supersaturated solution.

Chemical Properties

(i) ***Action of heat***: Hypo loses all the water molecules of water of crystallization when heated to about 488K. But on heating above 498k decomposes into sodium sulphate and sodium pentasulphide.

$$4Na_2S_2O_3 \longrightarrow 3Na_2SO_4 + Na_2S_5$$

On strong heating it decomposes to give H_2S, SO_2 and S.

(ii) ***Reaction with dilute acids***: With dilute mineral acids (HCl or H_2SO_4) hypo decomposes evolving SO_2 gas and forming colloidal sulphur.

$$Na_2S_2O_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2 + S$$

(iii) ***Reaction with AgNO$_3$***: When hypo reacts with $AgNO_3$ two kinds of reactions takes place.

When silver nitrate is excess first a white precipitate forms which slowly changes to black solid (Ag_2S).

$$Na_2S_2O_3 + 2AgNO_3 \longrightarrow \underset{\text{White}}{Ag_2S_2O_3}\downarrow + 2NaNO_3$$

$$Ag_2S_2O_3 + H_2O \longrightarrow \underset{\text{Black}}{Ag_2S}\downarrow + H_2SO_4$$

When hypo is excess the white precipitate formed in the begining dissolves by reacting with excess of hypo due to the formation of complex compound sodium argento thiosulphate.

$$Na_2S_2O_3 + 2AgNO_3 \longrightarrow \underset{\text{White}}{Ag_2S_2O_3}\downarrow + 2NaNO_3$$

$$3Na_2S_2O_3 + Ag_2S_2O_3 \longrightarrow 2Na_3\left[Ag(S_2O_3)_2\right]$$

(iv) ***Reaction with silver halides***: Silver halides (AgCl, AgBr, AgI) which are insoluble in water dissolve in the presence of hypo due to formation of complex.

$$AgBr + 2Na_2S_2O_3 \longrightarrow Na_3\left[Ag(S_2O_3)_2\right] + NaBr$$

This reaction is utilized in photography for fixing or developing the photographic reel. The black and white photographic reels are coated with AgBr. After taking the photograph the unreacted AgBr on the photographic film is removed by washing with hypo.

(v) ***Reaction with halogens***: Chlorine and bromine oxidizes the hypo to sodium sulphate or sodium hydrogen sulphate

$$Na_2S_2O_3 + Cl_2 + H_2O \longrightarrow Na_2SO_4 + S + 2HCl$$

$$Na_2S_2O_3 + 4Cl_2 + 5H_2O \longrightarrow 2NaHSO_4 + 8HCl.$$

This reaction is used in removing excess chlorine in certain industrial processess. For example in textile industry to remove the chlorine after the cloth is bleached with bleaching powder. So hypo is also called as *antichlor.*

Bromine also reacts with hypo in a similar way. Iodine oxidizes hypo quantitatively to sodium tetrathionate.

$$2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$$

This reaction is used in volumetric analysis for iodiometric estimations of several substances.

(vi) ***Reaction with ferric chloride***: When hypo is added to excess ferric chloride, a violet colour appears and remain permanent but when ferric chloride is added to excess of hypo violet colour appears and disappears due to reduction of ferric ion by hypo.

$$\underset{\text{excess}}{2FeCl_3} + 3Na_2S_2O_3 \longrightarrow \underset{\text{Violet}}{Fe_2(S_2O_3)_3} + 6NaCl$$

$$2FeCl_3 + \underset{\text{excess}}{2Na_2S_2O_3} \longrightarrow 2FeCl_2 + Na_2S_4O_6 + 2NaCl$$

(vii) ***Reaction with copper sulphate***: First cupric thiosulphate will be formed. This will be reduced to cuprous thiosulphate and then dissolves in excess of hypo due to the formation of complex.

$$CuSO_4 + Na_2S_2O_3 \longrightarrow CuS_2O_3 + Na_2SO_4$$

$$2CuS_2O_3 + Na_2S_2O_3 \longrightarrow Cu_2S_2O_3 + Na_2S_4O_6$$

$$3Cu_2SO_3 + 2Na_2S_2O_3 \longrightarrow Na_4\left[Cu_6(S_2O_3)_5\right]$$

(viii) ***Reaction with gold Chloride***: First the auric chloride is reduced to aurous chloride which dissolves in excess of hypo due to the formation of complex.

$$AuCl_3 + 2Na_2S_2O_3 \longrightarrow AuCl + Na_2S_4O_6 + 2NaCl$$

$$AuCl + 2Na_2S_2O_3 \longrightarrow Na_3\left[Au(S_2O_3)_2\right] + NaCl.$$

(x) ***Reaction with calcium and barium chlorides***: When hypo is added to calcium and barium chlorides they form calcium and barium thiosulphates. Calcium thiosulphate is soluble but barium thiosulphate is insoluble and precipitates.

$$CaCl_2 + Na_2S_2O_3 \longrightarrow CaS_2O_3 + 2NaCl$$

$$BaCl_2 + Na_2S_2O_3 \longrightarrow \underset{\text{White ppt.}}{BaS_2O_3}\downarrow + 2NaCl$$

Uses

Sodium Thiosulphate Is Used

(i) as antichlor to remove excess of chlorine from bleached textiles.
(ii) in photography as a fixing agent.
(iii) as a reagent in iodometric and iodimetric titrations for the estimations of several substances.
(iv) in the extraction of gold and silver.
(v) in medicine as an antiseptic.

Structure

The thiosulphate ion is regarded as derived from the sulphate ion by replacement of one oxygen atom by sulphur.

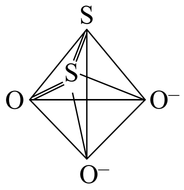


Fig 10.29 Structure of $S_2O_3^{2-}$ ion

X-ray analysis of the ionic crystals having $S_2O_3^{2-}$ ion has shown that $S_2O_3^{2-}$ has tetrahedral shape. This shape results from sp^3 hybridization of sulphur atom (central atom).

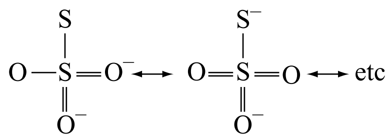
The structure of SO_3^{2-} ion as shown above reveals that the two sulphur atoms in SO_3^{2-} ion are not equivalent in bonding and hence are not interchangeable. The presence of two different types of S-atoms in $S_2O_3^{2-}$ ion could be verified by the fact that a thiosulphate is prepared from sulphite and radioactive sulphur (e.g., ^{35}S)

$$SO_3^{2-} + {}^{35}S \longrightarrow \left[{}^{35}S - SO_3\right]^{2-}$$

It gets decomposed by acid to give exclusively radio active ^{35}S and a non-radioactive aqueous solution of SO_3^{2-} ions arising from SO_2 evolved.

$$\left[{}^{35}S - SO_3\right]^{2-} + 2H^+ \longrightarrow {}^{35}S + H_2O + SO_2$$

Similar to sulphate ion, the thiosulphate ion also may have the following resonance structures.



In the above resonance structures since the terminal sulphur atom is getting a formal charge –1, its oxidation state is considered as –II and the oxidation state of central sulphur is considered as +VI.

KEY POINTS

- Oxygen, sulphur, seleniun, tellurium and polonium belong to the VIA group and p-block of the periodic table.
- The first four elements are collectively called as chalcogens since many metals occur as oxides and sulphides. Chalcogen means ore forming.
- Polonium was a radioactive element with very short half-life and therefore very little is known about the chemistry of it.
- Oxygen is the most abundant element in the earth's crust in the form of oxides and oxosalt, sulphur is less abundant (~0.04–0.03%) selenium in trace amounts while tellurium and polonium are rare.
- Their general outer electronic configuration is ns^2np^4.

General Properties

- Atomic radius increases from oxygen to polonium but not regularly. The reason is as usual in the p-block elements.
- **Ionization energies** decreases from oxygen to polonium. These elements have high values of ionization energies indicating that the formation of positive ions is extremely difficult and the elements are predominantly non-metallic in character.
- **Electronegativity** decreases gradually from oxygen to polonium.
- Density increases from oxygen to polonium.
- Oxygen is a gas others are solids. Oxygen being small in size can form strong pπ–pπ bonds and exist as diatomic molecule O_2. pπ–pπ bonding is not possible in the bigger atoms of S, Se and Te as explained in Vth group. Sulphur, selenium and tellurium exist as S_8, Se_8 and Te_8 molecules which are 8 atoms rings or zig-zag chains.
- The special stability of eight membered ring or chain of sulphur is because of the sp^3 hybridization of S atom. S_8 ring has puckered ring or crown shape in which 4 S atoms are in one plane while the other 4 atoms are in another plane. SSS bond angle is 107° while S–S bond length is 204 pm.
- The m.pts and b.pts increases down the group from oxygen to tellurium. There is large difference in the m.pt. and b.pt between oxygen and sulphur due to large difference in their molecular sizes. Polonium has low m.pt and b.pt than Te. This is due to inert pair effect.
- All the elements except oxygen exhibit –2, + 2, +4 and +6 **oxidation states**. Since oxygen is second most electronegative element next to fluorine, oxygen never exhibit positive oxidation states except in the

compounds of fluorine. Oxygen exhibit +1 oxidation state in O_2F_2, +2 in OF_2, – 1 in H_2O_2 and –1/2 in super oxides like KO_2. Except these in all other cases it exhibit only –2 oxidation state. The ability to exist in +6 and +4 oxidation state decreases down the group. This is due to increase in atomic size and decrease in electronegativity.

- Order of **electron affinity** of these elements is O < Te < Se < S.
- Oxygen and sulphur form ionic compounds in –2 oxidation state with more electropositive metals. Majority of the compounds of these elements are covalent.
- When sulphur, selenium and tellurium are in bond with more electronegative atom fluorine, the **d-orbitals** of valence shell can participate in sigma bonding due to the decrease in their size and energy, because of the development of positive charge on the central atom. So compounds like SF_6, SeF_6, TeF_6 SF_4, etc can be formed, but cannot form compounds like SH_6 as the electronegativity of hydrogen is less and cannot develop positive charge on central atom.
- Sulphur has a tendency to form **dπ-pπ bonding** but has little tendency to form **pπ-pπ** bonding. For example in SO_4^{2-} ion and related compounds multiple dπ-pπ bonding occurs due to the flow of electrons from filled pπ orbitals on oxygen to dπ orbitals on s atoms.
- Sulphur has a strong tendency to **catenation.** Oxygen has least catenation tendency due to weak O–O bond because of the repulsion between non bonding electron pairs. Catenation power decreases down the group due to decrease in M–M bond strength with increase in bond length as the atomic size increases.
- All the elements exhibit **allotropy.** O_3 is considered as the allotrope of oxygen. Sulphur has large number of allotropes such as rhombic sulphur, monoclinic sulphur, γ- sulphur, plastic sulphur etc. Selenium has six allotropes of which 4 are red varieties and 2 are grey varieties. Among these the most stable allotrope is grey sellenium. Tellurium exists in only one crystalline form which is isomorphous with grey selenium. Polonium also have two allotropes α- and β-forms.

Chemical Reactivity

- Similar to nitrogen oxygen is relatively less reactive at room temperature though it is more electronegative element. This is because of the double bond in O_2 which requires more bond dissociation energy.
- Oxygen, sulphur and selenium react with metals forming oxides, sulphides and selenides respectively.
- Oxygen and sulphur react among themselves and also with several other non metals forming oxides and sulphides.
- Sulphur, selenium and tellurium burn in air to form dioxides SO_2, SeO_2 and TeO_2 respectively.
- Sulphur, selenium and tellurium combine energetically with fluorine to yield hexafluorides XF_6 and with chlorine to give tetrahalides XCl_4.
- Sulphur, selenium and tellurium are oxidized by hot conc nitric acid and conc H_2SO_4 but not by non - oxidizing acids such as HCl and HF.
- When boiled with alkalis sulphur and selenium get attacked to form sulphides and selenides which on treating with excess of sulphur and selenium yield polysulphides and polyselenides respectively.

Hydrides

- Elements of Group VIB form hydrides of the type H_2M. Oxygen and sulphur form other hydrides H_2O_2 and H_2S_2. Sulphur also form polyhydrides H_2S_n where n = 2 to 10.
- H_2O can be prepared by the direct reaction between H_2 and O_2. The hydrides of S, Se and Te are prepared by the action of dilute acids on metal sulphides, selenides and tellurides H_2Po can be prepared by dissolving magnesium foil plated with Po in 0.2 N HCl.
- Except water all the other hydrides are colourless, poisonous gases with unpleasant odour.
- Water is liquid due to hydrogen bonding. Volatility increase from H_2O to H_2S and then decreases. Order of volatility $H_2S > H_2Se > H_2Te > H_2O$. Order of B.pts is $H_2S < H_2Se < H_2Te < H_2O$.
- All these hydrides are **covalent** and covalent character increases from H_2O to H_2Te. Polarity of the molecules and dipolemoments decreases from H_2O to H_2Te.
- **Acidic character** increases from H_2O to H_2Te. This is due to decrease in bond dissociation energy H – X and increase in the stability of conjugate base HX^- due to delocalization of charge with increase in size of HX^-.
- Thermal stability of the hydrides decreases from H_2O to H_2Po due to the increase in bond length and decrease in bond energy due to increase in atomic size $H_2O > H_2S > H_2Se > H_2Te > H_2Po$
- Owing to the decrease in the thermal stability from H_2O to H_2Te, the reducing character increases accordingly $H_2O < H_2S < H_2Se < H_2Te$.
- All the hydrides have angular shape. The HMH bond angle in water is 104° 31’ but in other hydrides it is almost equal to 90°. In H_2O oxygen is involved in sp^3 hybridization but in other hydrides pure p-orbitals are participated in the bonding.

Halides

- Elements of Group VIA form monohalides of the type H_2X_2; dihalides of the type MX_2; tetrahalides of the type MX_4 and hexahalides of the type MX_6 (M = S, Se, Te; X = halogen). The oxidation states of S, Se and Te in monohalides is +1, in dihalides +2, in tetra halides is +4 and in hexahalides is +6.
- Since the electronegativity of fluorine is greater than oxygen, the compounds of fluorine and oxygen are called **oxygen fluorides** rather than oxides of fluorine.

Important Conclusions About the Halides

(i) All elements form only hexafluorides. This is due to its small size and high electronegativity which can bring the S, Se and Te into their highest oxidation state. Hexachlorides, hexabromides and hexaiodides are not formed because of the bigger size of halogen atoms, six halogen atoms cannot coordinate with S, Se or Te. atoms.

(ii) Though S_2Cl_2 and Se_2Cl_2 are known, the corresponding Te_2Cl_2 and Po_2Cl_2 are not known because of the weaker Te–Te and Po–Po bonds, as they are very large atoms. The Te–Te and Po – Po bonds are further become weak due to withdrawl of electron density by more electronegative halogen.

(iii) Thermal stability of these halides has been found to follow the order F > Cl > Br > I.

(iv) All the elements form dichlorides and dibromides. The dihalides are unstable.

- **Hexafluorides:** All the hexafluorides are colourless gases having low B.pts showing that they have high degree of covalent character. Reactivity of hexafluorides increases from SF_6 to TeF_6, SF_6 is extremely stable and inert both thermally and chemically. It do not hydrolyse even in steam. This is because though the S–F bonds in SF_6 are weak the F-atoms surrounding S atom will not allow the water molecule to donate its lone pair to 'S' atom due to steric crowding.
- All the dihalides are angular in shape in which central atom is involved in sp^3 hybridization. Due to repulsion by lone pairs the tetrahedral bond angle gets distorted. The bond angle decreases with decrease in the size of halogen atom and increases with increase in the size of the central atom.
- **Monohalides:** Only S_2F_2, Se_2Cl_2 and Se_2Br_2 are best characterized. These get hydrolysed slowly and tend to undergo disproportionation.
- Structures of monohalides are similar to H_2O_2 with open book structure. S_2F_2 is an unstable compound and exist in two isomeric forms.
- **Oxygenfluorides:** OF_2 can be prepared by passing F_2 gas into cold and dil (2%) NaOH solution. It is a pale yellow poisonous gas. It is non-explosive due to its negative free energy of formation. It dissolves in water and liberates oxygen.
- OF_2 is a strong **oxidizing agent** and liberates halogens from their salts or acids. It can oxidize Cr^{3+} to chromate. OF_2 is angular in shape in which 'O' atom is involved in sp^3 hybridization. FOF bond angle is 103°.
- O_2F_2 is prepared by passing silent electric discharge through a mixture of O_2 and F_2 at low temperatures and pressures.
- O_2F_2 is a brown gas and it is a powerful oxidizing agent but less than F_2.
- O_2F_2 structure is similar to that of H_2O_2.
- O – O bond length in O_2F_2 is shorter than in H_2O_2 and the O – F bond length in O_2F_2 is longer than in that of OF_2. This is due to the resonance hybrid of the three structures.

Oxides

- Elements of Group VIA form a variety of oxides but the dioxides and trioxides are important.
- Acidic character decreases while basic character increases down the group for these oxides e.g., SO_2 is acidic, TeO_2 is amphoteric.
- As the size of the Se and Te atoms increases the reactivity SeF_6 and TeF_6 increases and TeF_6 is more reactive than SeF_6.
- SF_6 is used as gaseous insulator in high voltage equipments.
- All hexafluorides have octahedral structure in which the central atom is sp^3d^2 hybridized.
- **Tetrahalides:** Tetrafluorides are formed by all the elements and these are prepared by using fluorinating agents on the elements or on their compounds.
- SF_4 is a gas SeF_4 is liquid while TeF_4 is solid. The tetrafluorides are thermally more stable than the other tetrahalides.
- Tetrachlorides of all the Group VIA elements except oxygen are known. Tetrachlorides can be prepared by the direct reaction between the chalcogen and Cl_2.
- $TeCl_4$ is more stable than SCl_4 and $SeCl_4$ because of more polarity (ionic character).
- Se, Te and Po form tetrabromides by the direct combination of elements and their stability increases down the group due to increase in ionic character.
- Only TeF_4 and PoI_4 are known but SI_4 and SeI_4 are not known because S – I and Se – I bonds are almost non - polar and due to difference in atomic size make

the bonds weaker. Though Te – I bond is not very polar equal size of the atoms makes the bonds quite stable.

- All tetrahalides hydrolyse in water.
- All the tetrahalides act as **lewis bases** and the tetrafluorides and tetrachlorides can also act as **lewis acids.**
- Tetrahalides have trigonal bipyramid structure with one equatorial position is occupied by lone pair which are formed by sp^3d hybridization. The molecules have see-saw shape.
- **Dihalides:** Except selenium all the other elements form stable dichlorides and dibromides. Di-iodides are not formed by these elements. Dihalides of selenium are not stable (middle row anamoly).
- SCl_2 is prepared by saturating sulphur monochloride with chlorine. $TeBr_2$ can be prepared by the reaction between the elements. SF_2 can be prepared by fluorinating SCl_2 with activated KF or with HgF_2 at 150°.
- SCl_2 is a dark red liquid with foul odour.

Oxides

- Oxides of sulphur are more stable than the corresponding oxides of other elements. SO_2 and SO_3 are more stable than the oxides of the other elements.
- Acidic character of the oxides of the same element increases with increase in oxidation number. Trioxides are more acidic than the corresponding dioxide.
- For any element dioxide is more stable than trioxide.
- Dioxides can act as Lewis acid and Lewis bases because they contain vacant d - orbitals as well as lone pair in their valence shell.
- Dioxides can act both as oxidizing and reducing agents because in dioxides the chalcogen is in +4 oxidation state and can increase or decrease its oxidation state.
- Reduction power of dioxides decreases from SO_2 to PoO_2 while oxidation power increases from SO_2 to PoO_2.
- Trioxides can act only as oxidizing agents. Oxidation power of trioxides should increase from SO_3 to SeO_3 but SO_3 acts as strong oxidizing agent in acid medium because protonation makes the S – O bond breaking easier.
- **Dioxides** can be prepared by burning these elements in air.
- SO_2 and SeO_2 are acidic oxides and they form corresponding 'ous' acids H_2SO_3 and H_2SeO_3 when dissolved in water TeO_2 and PoO_2 are insoluble in water but they dissolve in concentrated acids and bases indicating their amphoteric nature.
- The SO_2 and SeO_2 are angular in shape in which the central atom is involved in sp^2 hybridization. The molecules contain one pπ-pπ and one pπ-dπ bonds. Solid SeO_2 has polymeric zig-zag chain structure, which is not planar. TeO_2 and PoO_2 are crystalline ionic compounds.
- Sulphur, selenium and tellurium forms trioxides of the type MO_3. All the trioxides are acidic.
- Solubility of TeO_3 in water is less than SO_3 and SeO_3 because of its less tendency to form multiple bonds with oxygen and lower electronegativity causing greater polarity in Te-O bond resulting in the polymeric structure.
- The strength and stability of the oxoacids gets decreased from H_2SO_4 to H_6TeO_6. Due to hydration of telluric acid the number of unprotonated oxygen atoms decrease resulting in the decrease in its acidic character.
- SO_3 is planar triangular molecule in gaseous state. sulphur atom is involved in sp^2 hybridization. It is non polar molecule with bond angle 120°. It contain one pπ-pπ and two pπ-dπ bonds.
- Solid SO_3 may either cyclic trimer (α-form) or an infinite helical chain made up of linked SO_4 tetrahedra.
- Cyclic SO_3 contain 3 S-O-S bonds 12 S-O bonds. The S-O bonds in the ring are longer than the S-O bonds outside.

Oxoacids

- Sulphur, selenium and tellurium form a variety of oxoacids. 'ic' acids are stronger than 'ous' acids. 'ic' acids can act only as oxidizing agents while 'ous' acids can act both as oxidizing and reducing agents. Oxidation power of 'ic' acids increases from H_2SO_4 to H_6TeO_6. Also oxidation power of 'ous' acids increases from H_2SO_3 to H_2TeO_3 while reduction power decreases from H_2SO_3 to H_2TeO_3.
  Acidic character of both 'ous' and 'ic' acids decreases down the group.

Oxygen

- In the laboratory oxygen can be prepared by any one of the following methods.
  (i) Heating the thermally unstable metal oxides like HgO and Ag_2O which decompose into metal and oxygen on heating.
  (ii) Thermal decomposition of higher oxides such as Pb_3O_4, PbO_2 and MnO_2 into lower oxides forming PbO, PbO and Mn_3O_4 liberating O_2.
  (iii) Decomposition of H_2O_2 and metal peroxides gives oxygen.
  (iv) Compounds like alkali metal nitrates, $KClO_3$, $KMnO_4$ and $K_2Cr_2O_7$ gives oxygen on heating.
  (v) Heating the $KMnO_4$ and $K_2Cr_2O_7$ with conc H_2SO_4 gives O_2. On large scale it is manufactured by the electrolysis of alkaline or acidified water or by fractional distillation of liquid air.

- In the Brin's process first BaO is heated in air at 500°C to get BaO_2 which on further heating at 800°C decompose to give BaO and O_2.

Oxides

Classification of oxides basing on the basis of oxygen content

1. **Normal Oxides:** These contain oxygen permitted by the normal valency of the element e.g., H_2O, MgO, Al_2O_3 etc.
2. **Polyoxides:** These contain more oxygen than permitted by the normal valency of the element. These are again three types.
   - **(a) Peroxides:** These involve bonds between oxygen atom in addtion to the bonds between the element and oxygen e.g., H_2O_2, Na_2O_2, BaO_2.
   - **(b) Superoxides:** These also involve bonds between oxygen atoms e.g., KO_2, RbO_2, CsO_2.
   - **(c) Dioxides:** These are oxides of an element in the higher oxidation state which can form stable oxides in lower oxidation states MnO_2 PbO_2 etc.
3. **Sub oxides:** They contain less oxygen than permitted by normal valency of the element. These involve the bonds between the atoms of the element in addition to bonds between the element and oxygen e.g C_3O_2, N_2O etc.
4. **Mixed oxides:** These contain simpler oxides of the same element e.g Red lead Pb_3O_4 ($2PbO + PbO_2$); Magnetite (Fe_3O_4) and dinitrogen trioxide N_2O_3 ($NO + NO_2$) On treating the mixed oxides of the metals with acids form two compounds corresponding to the two oxides e.g., Fe_3O_4 gives both ferrous and ferric salts.

Classification based on mutual reactions of oxides

1. **Very basic oxides:** Oxides which react readily with amphoteric oxides and very readily with acidic oxides e.g., Na_2O, CaO.
2. **Moderately basic oxides:** Oxides which scarcely react with amphoteric oxides but react readily with acidic oxide e.g., MgO, FeO, Fe_2O_3, CuO etc.
3. **Amphoteric oxides:** Oxides which react with very basic oxides and acidic oxides e.g., Al_2O_3, ZnO, SnO, BeO, H_2O etc.
4. **Acidic oxides:** Oxides which react with amphoteric oxide as well as basic oxides of both types. e.g., SO_2, N_2O_5, CO_2, SiO_2.
5. **Neutral oxides:** Which do not react with any other e.g., CO NO, N_2O etc.

Ozone

- Ozone is considered as an allotrope of oxygen. It is formed from atmospheric oxygen by the energy of sunlight. This ozone layer protects the life on the earth from UV light.
- Ozone can be prepared by the action of silent electric discharge on a slow dry stream of oxygen. Since ozone is an endothermic substance sparking would produce heat and decompose it.
- In **Siemen's ozonizer** silent electric discharge is produced by passing high voltage from induction coil through tin foils.
- In **Brodies ozonizer** silent electric discharge is produced by passing current through dilute H_2SO_4 using Pt or Cu wires immersed into it.
- In the above two methods about 10 per cent conversion takes place and it is seperated by condensing, during which O_3 liquifies first.
- Commercially ozone is prepared in Siemen-Halske method.
- Electrolysis of acidulated water with high current density using platinum anode produces-about 95 per cent O_3 at anode.
- Ozone is also formed thermally by heating O_2 at 2773 K.
- When fluorine is passed through water, F_2 can oxidize the water to O_2 and O_3.
- It is poisonous in large amounts but harmless in small concentration. It has characteristic smell of rotten fish. It is pale blue in gaseous state, dark blue in liquid state and violet black in solid state.
- It is a good oxidizing agent and oxidizes PbS to $PbSO_4$; liberates iodine from iodide, oxidizes hydrogen halides to halogens (except HF), green potassium manganate (K_2MnO_4) to purple potassium permanganate ($KMnO_4$), nitrite to nitrate, sulphite to sulphate, arsenite to arsenate, ferrous sulphate to ferric sulphate, potassium ferrocyanide to potassium ferricyanide.
- Ozone oxidizes alkaline KI to potassium iodate or potassium periodate, moist iodine to iodic acid; dry iodine to yellow powder I_4O_9.
- Ozone oxidizes silver metal to silver oxide and again reduces the silver oxide to silver. So due to alternate oxidation reduction white shining silver becomes black.
- When ozone is passed into mercury, it gets oxidized to mercurous oxide which dissolves in mercury. Then mercury loses its upper meniscus and non sticking nature of the glass. This is known as **tailing of mercury**.
- In all the above oxidation reactions only one atom of ozone is utilized in the oxidation while the remaining two oxygen atoms liberate out as O_2.
- In the oxidation of SO_2 to SO_3 and $SnCl_2$ to $SnCl_4$ all the three oxygen atoms in ozone are completely consumed.

- The reaction of ozone with peroxides can be considered as mutual reduction because both O_3 and peroxides lose oxygen.
- Ozone is a good bleaching agent on account of its oxidizing action on organic matter like rubber, corks, oil, ivory etc.
- Ozone forms **ozonides** by the addition to unsaturated organic compounds containing double bonds and triple bonds which on hydrolysis gives carbonyl compounds. The total reaction is called **ozonolysis**.
- Ozone is used as germicide and disinfectant for sterilizing the water and for improving the atmosphere of crowded places. Ozone and cyanogen mixture is used as rocket fuel.
- It can be detected by the following tests.
  (i) It turns starch iodide paper to blue (ii) it turns the white shining silver to black (iii) it converts the benzidine paper to brown or tetra methyl or ethyl base to violet.
- In O_3, the central oxygen atom is in sp^2 hybridization. It is angular in shape with O–O bond length 121 pm and bond angle 116°48'. It is a resonance hybrid of two structures. The O-O bond order in O_3 is 1.5.

Sulphur

- Sulphur from underground surface of earth is extracted by Frasch process. It is also prepared from alkali waste of Leblanc process and from hydrogen sulphide.
- Sulphur has large number of allotropes α-sulphur or rhombic sulphur or octahedral sulphur, β-sulphur or monoclinic sulphur or prismatic sulphur, γ-monoclinic sulphur, χ-sulphur or plastic sulphur; amorphous sulphur.
- The **most stable** allotrope at room temperature is rhombic sulphur. It is insoluble in water but soluble in organic solvents like benzene, alcohol, ether etc.
- Monoclinic sulphur is stable above 368.5K (95.5°C). At 368.5K both rhombic and monoclinic sulphurs are at equilibrium and this temperature is known as **transition temperature** of sulphur. Also these are known as enantiotropic allotropes.
- Density of rhombic sulphur (2.06 gm cm^{-3}) is more than that of monoclinic sulphur (1.96 gm cm^{-3}).
- When nearly boiling sulphur is poured into cold water soft and elastic amber - brown solid known as plastic or χ - sulphur is formed. It consists of completely random arrangement of chains of sulphur which when stretched align themselves parallel to each other. On standing it slowly converts into rhombic sulphur.
- α –, β – and γ – sulphurs contain S_8 rings which are puckered rings or with crown shape.
- Rhombic sulphur melts at approximately 113°C and monoclinic sulphur at approximately 119°C. It becomes viscous as the temperature rises upto 200°C and the viscosity again decreases abouve 200° and becomes mobile at its boiling point 444°C.
- It was proposed that the S_8 rings open forming chains reach their maximum chain length at 200°C corresponding to the maximum viscosity of liquid sulphur. Above 200°C the long chains break down and reform S_8 rings together with smaller fragments such as S_6, S_4 and S_2 and at high temperature atomic sulphur is formed.

Hydrogen Sulphide H_2S

- In the laboratory it is prepared by the action of H_2SO_4 on FeS pure H_2S is obtained by the action of HCl on Sb_2S_3.
- It burns with blue flame to give sulphur in limited supply of air and sulphur dioxide in excess of air.
- It is a good reducing agent and itself oxidizes to sulphur. It reduces halogens to hydrogen halides, sulphur dioxide to sulphur, ferric chloride to ferrous chloride, reduces H_2O_2, O_3, oxidizing acids like H_2SO_4 and HNO_3, decolourizes acidified permanganate, reduces orange red dichromate to green chromic sulphate.
- It is a weak dibasic acid, form two series of salts hydrosulphides and sulphides.
- The metals that are precipitate as sulphides in acid medium are mercury (II), lead, bismuth, copper (II) cadmium, arsenic, antimony and tin.
- The metals ions that are precipitated as sulphides in alkaline medium are nickel, cobalt, manganese and zinc.
- The metal sulphides which are not precipitated are of Na, K, Ca, Sr, Ba, Al, Cr. These sulphides are water soluble.
- With ammonium hydroxide H_2S forms yellow ammonium sulphide $(NH_4)_2\,S_n$.

Sulphur Dioxide

- It is formed when sulphur is burnt in air or oxygen. In the laboratory it is readily prepared by reacting bisulphite or sulphite with dilute acid. It can also be obtained by heating conc. H_2SO_4 with S, Cu or Ag.
- Industrially it is produced as a by-product during roasting of sulphide ores or by heating gypsum with carbon.
- Sulphur dioxide is a colourless dense gas with a smell of burning sulphur, highly soluble in water, can be liquified easily. It is heavier than air.
- It is neither combustible nor supporter of combustion but burning Mg or K burns in SO_2. When potassium burns in SO_2 the products are K_2SO_3 and $K_2S_2O_3$.

- It is acidic oxide, forms sulphurous acid when dissolved in water.

$$SO_2 + H_2O \rightleftharpoons H_2SO_3 \rightleftharpoons H^+ + HSO_3^- \rightleftharpoons 2H^+ + SO_3^{2-}$$

- With bases it forms two series of salts bisulphites e.g., $NaHSO_3$ and sulphites Na_2SO_3.
- With lime water it behaves similar to CO_2. First it turns lime water milky due to the formation of insoluble $CaSO_3$ which disappears on passing excess of SO_2 due to the conversion of insoluble $CaSO_3$ to soluble $Ca(HSO_3)_2$.
- It substitutes less acidic more volatile CO_2 from carbonates and bicarbonates forming sulphites and bisulphites respectively. It combines with basic metal oxides to form metal sulphites.
- It forms addition compounds. With Cl_2 it forms SO_2Cl_2, with PbO_2 forms $PbSO_4$ and with O_2 gives SO_3.
- It can act both as **oxidizing** and **reducing agent**.
- It oxidizes H_2S to S, Mg to MgO and MgS, potassium to K_2SO_3 and $K_2S_2O_3$, iron to FeO and FeS; CO to CO_2. It oxidizes $SnCl_2$ to $SnCl_4$ and Hg_2Cl_2 to $HgCl_2$.
- The reduction properties are best regarded as redox reactions involving the sulphite ion discussed later.
- SO_2 bleaches several substances in the presence of moisture and its bleaching property is due to reduction. The bleaching action of SO_2 is temporary and reversible. The bleached matter when exposed to air, regains its colour due to oxidation.

Sulphur Trioxide

- It is prepared by heating ferric sulphate. Heating of **ferrous sulphate** gives a mixture of SO_2 and SO_3 gases leaving behind Fe_2O_3 as residue. It is also obtained by the dehydration of conc H_2SO_3 with P_4O_{10} or by the oxidation of SO_2 with oxygen in the presence of catalyst such as platinized asbestos.
- It exists in three allotropic forms. α-, β- and γ - forms.
- α- SO_3 is a colourless liquid that solidifies in the form of prismatic crystals. β - form consists of asbestos like needles. Its molecular weight is about 160 corresponds to the formula S_2O_6. γ - SO_3 is produced when β - form is completely dried.
- α - form is more reactive than the rest. All three forms dissolve in water readily to form H_2SO_4 evolving much heat.
- On heating it decomposes to SO_2 and O_2. When dissolved in conc H_2SO_4 forms oleum (fuming sulphuric acid) $H_2S_2O_7$.
- It is an acidic oxide. It is the **anhydride** of sulphuric acid, reacts with alkalis, carbonates or basic oxides to give sulphates.
- With HCl it forms an addition product chlorosulphonic acid $ClSO_3H$.
- It is a good **oxidizing agent**.

Oxoacids of Sulphur

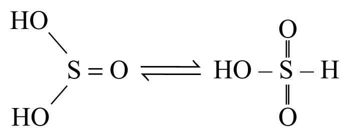
- The different oxoacids, their structural formula, oxidation numbers are given in Tabel 10.7. Of these acids the sulphurous acid and sulphuric acid are important.

Sulphurous Acid H_2SO_3

- Its aqueous solution is obtained by passing SO_2 into water. It is a weak dibasic acid forms two series of salts solid sulphites and bisulphites in solutions only. Attempts to prepare solid bisulphites results into condensation of two HSO_3^- ions into pyrosulphite ion $S_2O_5^{2-}$ Pyrosulphites when dissolved in water gives back again bisulphite ion.
- Sulphurous acid acts as a reducing agent where in itself oxidized to sulphuric acid.

$$H_2SO_3 + H_2O \longrightarrow SO_4^{2-} + 4H^+ + 2e^-$$

- All reducing properties of SO_2 in aqueous solution have been infact those of sulphurous acid.
- It reduces halogens to hydracids, ferric sulphate to ferrous sulphate, reduces acidified permanganate, orange red dichromate to green chromic sulphate, reduces iodates to iodine.
- It exists in tautomeric forms.



- Sulphite ion is pyramidal in shape with one of the positions occupied by lone pair. Sulphur atom is involved in sp^3 hybridization. All S – O bond length are identical due to resonance.

Sulphuric Acid H_2SO_4 (Oil of Vitriol)

- Sulphuric acid is manufactured by two process namely chambers process and contact process.
- In **Chamber's process** the chambers are made with lead because it is cheaper and passive towards sulphuric acid. In this process SO_2 is oxidized to SO_3 in the presence of **oxides of nitrogen** as catalysts. In the presence of oxides of nitrogen and water SO_2 will be converted into **nitrososulphuric acid** also known as **chamber's acid** ($NOHSO_4$) which hydrolyses in water to give sulphuric acid.

- In the manufacture of sulphuric acid, SO_2 is produced either by burning sulphur or by roasting iron pyrites. In this process the catalyst is V_2O_5. The arsenic impurity in SO_2 is removed by passing over ferric hydroxide in the form of ferric arsenate. The SO_3 will be absorbed into con H_2SO_4 to get oleum or pyrosuphuric acid $H_2S_2O_7$. Conc H_2SO_4 is prepared by adding oleum to water to get the desired strength.
- On addition of water to conc H_2SO_4 much heat is evolved and temperature raises to about 120°C due to the formation of hydrates such as $H_2SO_4 \cdot H_2O$, $H_2SO_4 \cdot 2H_2O$ etc. Density is about 1.84 g/cc.
- Sulphuric acid is a strong dibasic acid so it forms two types of salts bisulphates and sulphates. Due to high affinity toward water it acts as powerful dehydrating agent. It dehydrates formic acid to CO, oxalic acid to CO and CO_2, carbohydrates to carbon. Burning sensation of skin by conc H_2SO_4 is due to dehydration. Charing of wood, paper, cloth etc. are due to dehydration.
- Sulphuric acid is a good oxidizing agent. A number of reduction products of H_2SO_4 can be formed, their relative proportions in any one particular reaction depend on concentration of the acid, strength of the reducing agent and temperature. The possible reduction products given are the main products.
- Conc H_2SO_4 oxidizes C to CO_2, S to SO_2, P_4 to H_3PO_4.
- With dil H_2SO_4 metals above hydrogen in the electrochemical series liberate H_2. But with conc H_2SO_4 the product may be SO_2, H_2S and S in addition to metallic sulphate. Further reaction with metal sulphate and H_2S can give metal sulphides. For example zinc with dil H_2SO_4 gives H_2; with 98 per cent H_2SO_4 gives SO_2 and with 90 per cent first gives H_2S which in turn react with $ZnSO_4$ to give ZnS.
- Metals below hydrogen in the electrochemical series like copper, silver, mercury do not liberate hydrogen from dil H_2SO_4, but oxidized to the corresponding metal sulphate in the presence of air, but with conc H_2SO_4 gives SO_2.
- Since conc H_2SO_4 is less volatile it substitutes the volatile acids from several salts e.g., HF from CaF_2, HCl from chlorides, HNO_3 from nitrates, H_2S from sulphides, CO_2 from carbonates etc.
- Some of the acids liberated will be oxidized by conc H_2SO_4. For example, HBr and HI liberated initially from bromides and iodides will be oxidized by conc. H_2SO_4 to Br_2 and I_2 respectively.
- With PCl_5 conc H_2SO_4 reacts in two steps first forming chlorosulphuric acid ($ClSO_2OH$) and sulphuryl chloride (SO_2Cl_2) indicating the presence of two – OH groups in H_2SO_4.
- When potassium ferrocyanide crystals are heated with conc H_2SO_4 the products formed are K_2SO_4, $FeSO_4$, $(NH_4)_2SO_4$ and CO. If dil H_2SO_4 is used in this reaction poisonous HCN is liberated instead of CO.
- When potassium chlorate is heated with conc H_2SO_4 the products formed are Cl_2O (chlorine dioxide) and $HClO_4$ (perchloric acid).
- Sulphuric acid is called **king of chemicals** because it has many applications. Sulphuric acid is used in the **lead storage batteries**.
- **Pickling** is an industrial process of removing the layers of basic oxides from metal surfaces like Fe and Cu by dipping in dil H_2SO_4 before they are electroplated, enamelled, galvanized or soldered.

Sodium Thiosulphate

- The anhydrous $Na_2S_2O_3$ is simply called **sodium thiosulphate** but the hydrated sodium thiosulphate $Na_2S_2O_3 \cdot 5H_2O$ is called **hypo**. It is a salt of thiosulphuric acid which never been isolated.
- In the laboratory it is prepared by boiling alkaline or neutral sodium sulphite with flowers of sulphur.
- When sulphur is boiled with sodium hydroxide, sodium thiosulphate and sodium sulphide or sodium pentasulphide will be formed. The sodium pentasulphides also converts into hypo when exposed to air.
- Hypo is also formed when SO_2 react with sodium sulphide.

$$2Na_2S + 3SO_2 \rightarrow 2Na_2S_2O_3 + S$$

- In **spring's reaction** hypo is formed when a mixture of sodium sulphite is treated with calculated amount of iodine.

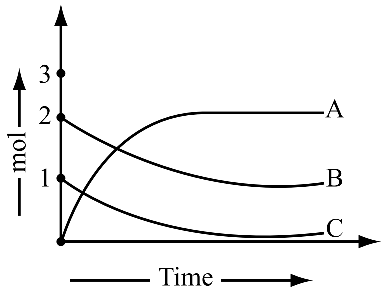
$$Na_2S + Na_2SO_3 + I_2 \rightarrow Na_2S_2O_3 + 2NaI$$

- When SO_2 gas is passed through the waste liquor obtained in the Leblanc process for the manufacture of Na_2CO_3 which contain Na_2CO_3, Na_2SO_3 and Na_2S, hypo will be obtained.
- Hypo is also formed when SO_2 is passed through the solution of Na_2CO_3 containing sulphur.
- It is colourless, crystalline **efflorescent** substance. It has a tendency to form **supersaturated solution**.
- When heated it becomes anhydrous at about 438K but on heating above 498K it decomposes to Na_2SO_4 and Na_2S_5. On strong heating it decomposes to give H_2S, SO_2 and S.
- With dilute acids (HCl or H_2SO_4) it decomposes evolving SO_2 gas forming colloidal sulphur (difference from sulphites).
- When excess silver nitrate reacts with hypo first a white precipitate of $Ag_2S_2O_3$ forms which slowly changes to black Ag_2S precipitate.

- When excess of hypo reacts with $AgNO_3$ the white precipitate $Ag_2S_2O_3$ formed in the begining dissolves in excess of hypo due to the formation of complex $Na_3[Ag(S_2O_3)_2]$.
- The insoluble silver halides AgCl, AgBr and AgI also dissolve in excess of hypo due to the formation of complex $Na_3[Ag(S_2O_3)_2]$.
- Chlorine and bromine oxidizes hypo to $NaHSO_4$ or Na_2SO_4 and S, iodine quantitatively oxidizes hypo to sodium tetra thionate ($Na_2S_4O_6$).
- When hypo is added to excess of ferric chloride a violet colour appears and remain permanent. When ferric chloride is added to excess of hypo violet colour appears and disappears due to reduction of ferric ion by hypo. Violet colour is due to the formation of complex $Fe_2\,(S_2O_3)_3$.
- When hypo is added to copper sulphate first cupric thiosulphate will be precipitated which then reduced to cuprous thiosulphate ($Cu_2S_2O_3$) by excess of hypo and then dissolves due to the formation of complex $Na_4\,[Cu_6\,(S_2O_3)_5]$
- Hypo reduces auric chloride to aurous chloride (AuCl) which dissolves in excess of hypo due to the formation of complex $Na_3[Au(S_2O_3)_2]$.
- When hypo is added to $BaCl_2$ a white precipitate of barium thiosulphate BaS_2O_3 is formed but calcium salts do not give precipitate because CaS_2O_3 is soluble.
- Hypo is used as antichlor in textile industry, in photography as a fixing agent due to its ability to form complex with silver ion, in the laboratory as a reagent in iodometric and iodimetric estimations and in the extraction of silver and gold.
- Structure of S_2O_3 ion is tetrahedral and similar to SO_4^{2-} ion but one oxygen atom is replaced by sulphur atom and it is in –2 oxidation state while the central sulphur is in +6 oxidation state with an average oxidation number +2.

Multiple Choice Questions With Only One Answer

1. Compounds A and B are treated with dil HCI separately. The gases liberated are y and z respectively. y turns acidified dichromate paper green while z turns lead acetate paper black. So A and B compounds are respectively.
   (a) Na_2SO_3, Na_2S (b) NaCl, Na_2CO_3
   (c) Na_2S, Na_2SO_3 (d) Na_2SO_3, K_2SO_4
2. PCl_5 and excess of sulphuric acid are allowed to react. The main product formed along with $POCl_3$ and HCl is
   (a) $ClSO_2OH$ (b) $SOCl_2$
   (c) SO_2Cl_2 (d) S_2Cl_2
3. For the reaction involving the conversion of sulphur (IV) oxide to sulphur (VI) oxide, the change in number of moles vs time is shown by different curves in the figure for different substances A, B and C are respectively



   (a) SO_3, O_2, SO_2 (b) SO_3, SO_2, O_2
   (c) SO_2, SO_3, O_2 (d) O_2, SO_2, SO_3
4. The structure of thionyl chloride is
   (a) linear.
   (b) planar triangular, with the sulphur atom at the centre.
   (c) tetrahedral, with one corner of the tetrahedron occupied by a lone pair of electrons and the sulphur atom at the centre.
   (d) square planar, with the sulphur atom at the centre.
5. In H_2SO_4, there are
   (a) Two sp^3 hybridized centres and $2(p\pi–d\pi)$ bonds
   (b) Three sp^3 hybridized centres and $2(p\pi–d\pi)$
   (c) Four sp^3 hybrized centres and $2(p\pi–d\pi)$ bonds
   (d) Only one sp^3 hybridized centre and $2(p\pi–d\pi)$ bonds
6. A yellow powder reacted with F_2 to form a colourless gas X which is used as gaseous insulator in high power generators. It does not get hydrolysed. Another compound is obtained by reaction of sulphur dichloride with NaF. It can be easily hydrolysed and has see-saw shape. X and Y respectively are
   (a) Agl, AgBr (b) SF_6, SF_4
   (c) SF_4, SF_6 (d) SCl_4, SCl_6
7. $SF_4 \xrightarrow{\text{Hydrolysis}} A + B$

   $A + H_2S \longrightarrow C$ The 'C' is
   (a) Solid sulphur (b) Colloidal sulphur
   (c) SO_3 (d) Gaseous sulphur
8. An oxide of a non-metal has the following properties:
   I. It acts both as a proton-donor as well as a proton acceptor
   II. It reacts readily with basic and acidic oxides
   III. It oxidizes Fe at its boiling point
   IV. It is a poor conductor of electricity. The oxide is
   (a) H_2O (b) SO_2 (c) NO_2 (d) CO_2

9. It is possible to obtain oxygen from air by fractional distillation because
(a) Oxygen and nitrogen are placed in different groups of the periodic table
(b) Oxygen is more reactive than nitrogen
(c) Oxygen has a higher boiling point than nitrogen
(d) Oxygen has a lower density than nitrogen

10. The increasing order of acid character of the oxides; CO_2, N_2O_5, SiO_2 and SO_3 follows the sequence
(a) $CO_2 > N_2O_5 > SiO_2 > SO_3$
(b) $SiO_2 < CO_2 < N_2O_5 < SO_3$
(c) $SO_3 < N_2O_5 < CO_2 < SiO_2$
(d) $N_2O_5 > SO_3 < SiO_2 < CO_2$

11. What is the correct relationship between the pH values of isomolar solutions of sodium oxide (pH_1), sodium sulphide (pH_2), sodium selenide (pH_3) and sodium telluride (pH_4)?
(a) $pH_1 < pH_2 < pH_3 < pH_4$
(b) $pH_1 > pH_2 > pH_3 > pH_4$
(c) $pH_1 < pH_2 < pH_3 \approx pH_4$
(d) $pH_1 > pH_2 \approx pH_3 > pH_4$

12. Which of the following statements regarding the manufacture of H_2SO_4 by contact process is not true
(a) Sulphur is burnt in air to form SO_2
(b) SO_2 is catalytically oxidized to SO_3
(c) SO_3 is dissolved in water to get 100 per cent sulphuric acid
(d) H_2SO_4 obtained by contact process is of higher purity than that obtained by lead chamber process

13. The oxidation states of sulphur atoms in the anions SO_3^{2-}, $S_2O_4^{2-}$ and $S_2O_6^{2-}$ follow the order
(a) $S_2O_4^{2-} < SO_3^{2-} < S_2O_6^{2-}$
(b) $S_2O_3^{2-} < S_2O_4^{2-} < S_2O_6^{2-}$
(c) $S_2O_4^{2-} < S_2O_6^{2-} < SO_3^{2-}$
(d) $S_2O_6^{2-} < S_2O_4^{2-} < S_2O_3^{2-}$

14. Which of the following statements is incorrect for the SO_4^{2-} ion?
(a) It is tetrahedral
(b) All the S—O bond lengths are equal, and shorter than expected single bond length
(c) It contains four σ bonds between S and the O atoms, two π bonds delocalized over the S and four O atoms and all the S—O bonds have a bond order 1.5
(d) It is square planar

15. Which of the following statements are incorrect?
(a) SO_3 is a stronger oxidizing agent and more acidic than SO_2
(b) Selenium forms only two oxoacids i.e., selenous acid (H_2SeO_3) and selenic acid (H_2SeO_4)
(c) The acidic strength and oxidizing power of oxo acids is greater in +6 oxidation state than in +4 oxidation state
(d) The thermal stability of oxides of group 16 elements decreases in the order $SO_2 > SeO_2 > TeO_2 > PoO_2$

16. Identify the correct sequence of increasing number of π – bonds in the structure of the following molecules
I. $H_2S_2O_6$ II. $H_2S_2O_3$ III. $H_2S_2O_5$
(a) I, II and III (b) II, I and III
(c) II, III and I (d) I, III and II

17. Which of the following statements are correct about sulphur hexaflouride
I. All S—F bonds are equivalent
II. SF_6 is a planar molecule
III. The oxidation number of sulphur is the same as the number of electrons it uses in bonding
IV. Sulphur has aquired the electronic structure of the inert gas argon
(a) I, II, III only correct (b) I, III only correct
(c) II, IV only correct (d) I, III, IV only correct

18. The addition of concentrated sulphuric acid to potassium iodide is not suitable for making hydrogen iodide because
I. The reaction is a redox reaction instead of double decomposition reaction
II. Hydrogen iodide is contaminated by reduction products of the sulphuric acid
III. Hydrogen iodide is oxidized to iodine
IV. Sulphuric acid is too weak to displace hydrogen iodide from its salt
(a) I, II and III only correct
(b) I and III only correct
(c) II and IV only correct
(d) I and II only correct

19. $SF_4 + BF_3 \rightarrow$ (A) the compound 'A' is
(a) $[SF_5]\,[BF_2]^+$ (b) $[SF_3]^+\,[BF_4]^-$
(c) SF_6 (d) S_2F_4

20. H_2S is less stable than H_2O because
(a) the bonding orbitals of sulphur are larger and more diffuse than those of oxygen, and hence they overlap less effectively with the 1s orbital of the hydrogen atom.
(b) the bonding orbitals of sulphur are smaller and more diffuse than those of oxygen, and hence they overlap less effectively with the 1s orbital of the hydrogen atom.
(c) the bonding orbitals of sulphur are smaller and less diffuse than those of oxygen, and hence they overlap less effectively with the 1s orbital of the hydrogen atom.
(d) H_2O molecules form H bonds but H_2S molecules do not.

21. The structure of the SO_3 molecule in the gas phase contains
(a) only σ bonds between sulphur and oxygen.
(b) σ bonds and $p\pi$–$p\pi$ bond between sulphur and oxygen.
(c) σ bonds and $p\pi$–$d\pi$ bond between sulphur and oxygen.
(d) σ bonds, $p\pi$–$p\pi$ bond and $p\pi$–$d\pi$ bond between sulphur and oxygen.

22. Identify the incorrect statement.
(a) Concentrated H_2SO_4 cannot convert Cu into Cu^{2+}.
(b) Concentrated H_2SO_4 absorbs water.
(c) The first proton of H_2SO_4 dissociates readily.
(d) The second proton of H_2SO_4 dissociates much less readily and so the solutions of hydrogen sulphates are acidic.

23. Cyclic trimeric structure of SO_3 contain:
(a) Six S = O bonds and three S—O—S bonds
(b) Three S = O bonds and six S—O—S bonds
(c) Six S = O bonds and two S—O—S bonds
(d) Three S = O bonds and three S—O—S bonds

24. Consider the following reactions:
I. $Na_2S_2O_4 + \text{dil HCl} \xrightarrow{\Delta}$
II. $Na_2S_2O_3\ (aq.) + I_2 \xrightarrow{\Delta}$
III. $SO_2 + H_2S \xrightarrow{\Delta}$
IV. $H_2S + H_2O_2 \xrightarrow{\Delta}$
The reaction(s) which give(s) yellow turbidity is/are
(a) II, III and IV (b) III and IV only
(c) I, III and IV (d) I, II, III and IV

25. Which of the following acids are dibasic and do not have an identical non-metal bond?
(a) $H_2S_2O_8$ (b) $H_2S_2O_4$
(c) $H_4P_2O_6$ (d) $H_2N_2O_2$

26. Oxidation state of terminal sulphur in $S_2O_3^{2-}$ is
(a) +2 (b) –2 (c) –4 (d) +6

27. Which of the following compound is obtained by passing O_3 gas into Hg metal?
(a) HgO (b) Hg_2O (c) Both (d) HgS

28. Acidic $K_2S_2O_8$ can oxidize MnO_2 to MnO_4^- and acidic $K_2S_2O_8$ and acidic MnO_2 oxidize I^-, Br^-, Cl^- to I_2, Br_2 and Cl_2 respectively from the given data, the sequence that represents the correct order of increasing oxidation ability is
(a) $I_2 > K_2S_2O_8 > Br_2$
(b) acidic $K_2S_2O_8 >$ acidic $MnO_2 > Cl_2$
(c) $K_2S_2O_8 > I_2 > Br_2$
(d) $Cl_2 > K_2S_2O_8 > Br_2$

29. The pair that act as both oxidizing and as well as reducing agent is
(a) NO,SO_3 (b) NO_2,H_2O_2
(c) CO_2, SO_2 (d) N_2O_5, O_3

30. Which of the following reactions does not occur?
(a) $H_2SO_4 + HNO_3 \longrightarrow HSO_4^- + NO_2^+ + H_2O$
(b) $KIO_3 + SO_2 + H_2O \longrightarrow I_2 + KHSO_4 + H_2SO_4$
(c) $NaHSO_3 + Na_2CO_3 \longrightarrow Na_2SO_3 + H_2O + CO_2$
(d) $2SO_{2(g)} \rightarrow SO^{2+} + SO_3^{2-}$

31. A mixture of Na_2CO_3 and Na_2SO_3 is treated with dil H_2SO_4 in a setup such that the gaseous mixture emerging can pass through a solution of $BaCl_2$ and then through one of $K_2Cr_2O_7$ acidified with dil H_2SO_4. Which of the following will you observe?
(a) The $BaCl_2$ solution remain unaffected and acidified dichromate solution turns green.
(b) The $BaCl_2$ solution gives white precipitate and acidified dichromate solution remains unaffected.
(c) The $BaCl_2$ solution gives a white precipitate and acidified dichromate solution turns green.
(d) Both the solution remain unaffected.

32. Which of the following is correct?
(a) S_3O_9 – contains no S – S linkage
(b) $S_2O_6^-$ contains – O – O linkage
(c) $(HPO_3)_8$ – contains P – P linkage
(d) $S_2O_8^{2-}$ contains S – S linkage

33. Sodium thiosulphate solution is treated with $AgNO_3$ solution. This causes the precipitation of white substance which changes rapidly to black through yellow, orange and brown colours. The white and black precipitates are respectively:
(a) Ag_2SO_4; Ag_2S (b) $Ag_2S_2O_3$; Ag
(c) $Ag_2S_2O_3$; Ag_2S (d) $Na_3[Ag(S_2O_3)_2]$; Ag_2S.

34. S – S bond is present in:

$H_2S_2O_6$	$H_2S_2O_8$	$H_2S_2O_7$	$H_2S_2O_5$
I	II	III	IV

(a) I, IV (b) II, IV
(c) I, II (d) III, IV

35. The statement that is *not* true about SF_4 is
(a) It is fluorinating agent for organic compounds
(b) It has sp^3 hybridization of S and a tetrahedral shape
(c) The hybridization in SF_4 is sp^3d and a see-saw shape
(d) SF_4 is a Lewis base

36. Choose the correct statement:
(a) Basicity of phosphorous acid is three
(b) Perbromic acid is having one peroxy linkage
(c) β-SO_3 is having cyclic structure
(d) Borazine is having zero dipolemoment

37. Select from each set the molecule or ion having the smallest bond angle

I. NH_3, PH_3 or AsH_3

II. O_3^+, O_3

III. NO_2^-, O_3

IV. X-S-X angle in $SOCl_2$ and SOF_2

(a) NH_3, O_3^+, O_3, $SOCl_2$

(b) PH_3, O_3^+, NO_2^-, SOF_2

(c) AsH_3, O_3, NO_2^-, SOF_2

(d) AsH_3, O_3^+, O_3, SOF_2

38. When SO_2 is passed through acidified $K_2Cr_2O_7$ solution

(a) The solution turned blue

(b) The solution is decolourized

(c) SO_2 is reduced

(d) Green $Cr_2(SO_4)_3$ is formed

39. Which of the following statement is/are false for polythionic acid series

(a) The average oxidation state of S-atom increases with decrease in number of S atoms

(b) The absolute oxidation state of S-atom increases with decrease in number of S-atoms

(c) The average oxidation state of S-atom decreases with increase in number of S-atoms

(d) The absolute oxidation state of S-atoms remain constant with increase or decrease in number of S-atoms

40. SO_3 + conc. HCl $\rightarrow$ (A) (A) on hydrolysis gives

(a) H_2SO_3 (b) H_2SO_4

(c) H_2SO_5 (d) $H_2S_2O_8$

41. Consider the following bond angles:

α = O–O–O in ozone

β = P–P–P in P_4 (white)

γ = N–N–N in azide anion (N_3^-)

δ = C–C–C in diamond

Then,

(a) $\alpha + \beta = \gamma$ (b) $\beta + \delta > \gamma$

(c) $\frac{\delta}{\beta} > \frac{\gamma}{\alpha}$ (d) $\gamma - \alpha = \alpha - \beta$

42. Which of the following statements regarding sulphuric acid is incorrect?

(a) When heated, concentrated H_2SO_4 decomposes before boiling

(b) As an oxidizing agent its equivalent weight is 49

(c) It acts as dehydrating agent against wood, starch and paper

(d) It explodes with $KMnO_4$ crystals

43. In the reaction of hypo with dil acids

(a) One sulphur is oxidized

(b) One sulphur oxidized, one sulphur reduced

(c) One sulphur is reduced

(d) Sulphur neither oxidized nor reduced.

44. In sulphur trioxide trimer, each sulphur atom is bonded to

(a) Four oxygen atoms (b) Three oxygen atoms

(c) Two oxygen atoms (d) Two sulphur atoms

45. Which of the following statements regarding sulphurous acid is not correct

(a) In alkaline solution reducing action of sulphurous acid is due to sulphite ion

(b) In acidic solution reducing action of sulphurous acid is due to SO_2

(c) The structure of sulphurous acid is tetrahedral including the lone pair of electrons on sulphur

(d) Sulphurous acid acts as a stronger reducing agent in acid medium than in alkaline medium

46. A pale yellow precipitate and a gas with pungent odour are formed on warming dil.hydrochloric acid with a solution containing:

(a) sulphate ion (b) sulphide ion

(c) thiosulphate ion (d) sulphite ion

47. When an equimolar mixture of Na_2S and Na_2SO_3 is treated with excess iodine, the products formed are

(a) $Na_2S_2O_3$ and NaI (b) Na_2SO_4,S and HI

(c) NaI and S (d) $Na_2S_4O_6$ and NaI

48. In which of the following reaction change of oxidation state of Br is more than unity

I. $H_2SO_4 + 2KBr \longrightarrow$

II. $KBr + H_3PO_4 \longrightarrow$

III. $Cl_2 + KBr + OH^- \longrightarrow$

IV. $Br_2 + NaClO_3 \longrightarrow$

(a) I, II and III (b) III and IV

(c) II, III and IV (d) I and IV

49. Liquid oxygen and liquid nitrogen are allowed to flow between the poles of an electromagnet. Choose the correct observation

(a) Both will be attracted but to opposing poles

(b) Both will be attracted to the same pole

(c) Liquid oxygen will be attracted and liquid nitrogen will be repelled

(d) Liquid oxygen will be attracted but liquid nitrogen unaffected

50. Urea reacts with SO_3 in the presence of H_2SO_4 produce

(a) NH_2SO_3H (Sulphamic acid)

(b) NH_2OH

(c) NH_2CSNH_2 (Thio urea)

(d) $(NH_4)_2CO_3$

51. Sodium pyrosulphate, $Na_2S_2O_7$ can be made by heating

(a) $NaHSO_4$ strongly

(b) $NaHSO_3$ strongly

(c) a mixture of $Na_2S_2O_3$ and SO_2
(d) a mixture of Na_2SO_3 and excess of sulphur

52. When $KHSO_4$ is added to a concentrated solution of H_2SO_4, the acidity of the solution
(a) increases (b) decreases
(c) remain constant (d) cannot be predicted

53. When ferric chloride is added to excess hypo solution a violet colour appears and disappears immediately. The disappearence of violet colour is due to
(a) formation of ferric sulphide
(b) conversion into colourless complex
(c) reduction of ferric chloride to ferrous chloride
(d) The violet complex is unstable and dissociate immediately

54. Hypo is used to remove excess chlorine from fabric in textile industry where chlorine oxidizes hypo to a single product. In this reaction the change in the oxidation state of terminal sulphur of hypo is
(a) 4 units (b) 6 units
(c) 8 units (d) no change

55 A silver foil is turned black in the presence of ozone involves
(a) oxidation reaction (b) reduction reaction
(c) both oxidation and reduction
(d) neither oxidation nor reduction

56 The structure of ozone involves
(a) delocalized three – centred σ bonding
(b) delocalized three – centred π bonding
(c) delocalized three – centred σ as well as π bonding
(d) localized π-bonding

57 O_2F_2 is unstable yellow orange solid and H_2O_2 is colourless liquid, both have O – O bond. O – O bond lengths in H_2O_2 & O_2F_2 are respectively.
(a) 1.22A° 1.48A° (b) 1.48A°, 1.22A°
(c) 1.22A°, 1.22A° (d) 1.48A° 1.48A°

58. Which of the following statement is incorrect regarding sulphur compounds?
(a) $H_2S_2O_8$ and H_2SO_5 both have one O – O linkage
(b) In SF_6 all S—F bonds are equivalent
(c) $SCl_2(OCH_3)_2$ and $SF_2(OCH_3)_2$;–OCH_3 groups in both cases occupy the equatorial position
(d) In $H_2S_2O_3$ oxidation number of sulphur is +6 and –2

59. Which does not possess sulphur with oxidation number +6?
(a) Caro's acid (b) Marshall's acid
(c) Oleum (d) Thiosulphurous acid

60. Which of the following statement (s) is incorrect?
(a) Oxoanions of sulphur have little tendency to polymerize compared with the phosphates and silicates
(b) In pyrosulphurous acid $H_2S_2O_5$, the oxidation states of both the sulphur atoms are not same and these are +IV and +II
(c) Conc. HNO_3 oxidizes both sulphur and carbon to H_2SO_4 and CO_2 respectively
(d) Most metal oxides are ionic and basic in nature while non-metallic oxides are usually covalent and acidic in nature

Multiple Choice Questions with One or More than One Answers

1. Which of the following statements regarding thiosulphate ion is/are correct?
(a) shape of thiosulphate ion is tetrahedral
(b) the two sulphur atoms in thiosulphate ion are equivalent
(c) there is S–S bond in thiosulphate ion
(d) with I_2 thiosulphate ion gives tetrathionate ion

2. Which of the following statements about sulphur is/are correct?
(a) sulphur exists as octaatomic S_8 molecule with puckered ring structure
(b) In S_8 molecule each sulphur atom undergoes sp^3–hybridization involving both bonding and non bonding pairs of electrons
(c) There are two single covalent bonds and two lone pairs of electrons associated with each S atom in S_8 molecule
(d) The S—S—S bond angle in S_8 molecule in $109°28^1$

3. Which among the following statements are correct?
(a) Carbon monoxide is neutral where as SO_3 is acidic
(b) Potassium oxide is basic where as nitrous oxide is acidic
(c) Aluminium and zinc oxides are amphoteric
(d) Sulphur trioxide is acidic where as phosphorus pentoxide is basic

4. Identify the correct statement about ozone.
(a) It is thermodynamically unstable with respect to oxygen
(b) The conversion of O_2 to O_3 is endothermic
(c) In the convertion of O_2 to O_3, there is decrease in entropy
(d) Ozone is diamagnetic in nature

5. When a compound X reacts with ozone in aqueous medium a substance Y is produced. Ozone also reacts with Y in alkaline medium and produces compound Z which can act as an oxidizing agent. Then X, Y and Z are
(a) X = HI, Y = I_2 and Z = IO^-_3
(b) X = KI, Y = I_2 and Z = IO^-_3

(c) $X = KI, Y = I_2$ and $Z = IO^-_4$
(d) $X = HI, Y = I_2$ and $Z = IO^-_4$

6. When radioactive sulphur is added to an alkaline sodium sulphite solution, radioactive thiosulphate ion is formed. Upon adding Ba^{2+}, a precipitate of BaS_2O_3 is formed. The precipitate is filtered, dried and then treated with acid, producing solid sulphur, SO_2 gas and water. The correct set of products is (*indicates radioactive sulphur)
(a) $SO_2 + S_8$ (b) $SO_2 + S^*_8$
(c) $S^*O_2 + S_8$ (d) $S^*O_2 + S^*_8$

7. Which of the following oxyacid(s) of sulphur have S—S bond?
(a) $H_2S_2O_8$ (b) $H_2S_2O_4$
(c) $H_2S_2O_5$ (d) $H_2S_2O_6$

8. Select the *correct* statements about $Na_2S_2O_3 \cdot 5H_2O$
(a) It is also called as hypo
(b) It is used in photography to form complex with AgBr
(c) It can be used as antichlor
(d) It is used to remove stains of I_2

9. In which of the following S—S link is present?
(a) Caro's acid (b) Dithionic acid
(c) Thiosulphuric acid (d) Chlorosulphonic acid

10. Which of the following statements is incorrect?
(a) In O_2F_2, the O—O bond is shorter than the O-O bond in H_2O_2
(b) In O_2F_2, the O-O bond is longer than the O-O bond in H_2O_2
(c) H_2O_2 and O_2F_2 have similar structures, hence the O-O bond of the two molecules are identical
(d) O_2F_2 does not contain the peroxide bond (—O—O—)

11. Which of the following statements are ***true*** about sodium thiosulphate, $Na_2S_2O_3$?
(a) It is used in the estimation of iodine
(b) It gives a white precipitate with $AgNO_3$ which turns black on standing.
(c) It is used to remove the unexposed AgBr from photographic films
(d) It contains, ionic, covalent and coordinate covalent bonds.

12. The correct statement about sulphur hexafluoride include
(a) There are 12 F—S—F 90º bond angles
(b) S in SF_6 has an expanded octet
(c) With H_2O, SF_6 can accept lone pair of electron in the empty 3d atomic orbital and gets hydrolysed
(d) SF_6 has a distorted octahedral geometry

13. Identify the correct statement(s):
(a) Ozone is a powerful oxidizing agent as compared to O_2
(b) Ozone reacts with KOH and gives an orange coloured solid KO_3
(c) There is a decrease in volume when ozone decomposed to form O_2
(d) The decomposition of O_3 to O_2 is exothermic

14. Which of the following statement(s) is/are correct?
(a) Marshall's acid on partial hydrolysis gives caro's acid & sulphuric acid
(b) conc H_2SO_4 on reaction with potassium chlorate gives chlorine dioxide
(c) Iodine solution is decolourized by sodium thiosulphate solution
(d) Aqueous solution of SO_2 liberates iodine from an iodate solution and decolourizes iodine solution

15. Which of the following statements is correct?
(a) The hydrogen halides can be prepared by the action of a nonvolatile, nonoxidizing acid on halide salts.
(b) Sulphuric acid reacts with F^- and Cl^- to give HF and HCI, but it oxidizes Br^- and I^- into Br_2 and I_2 respectively.
(c) Phosphoric acid converts NaBr and Nal into HBr and HI respectively but H_2SO_4 oxidizes both of them.
(d) Concentrated sulphuric acid converts all the halide ions into corresponding hydrogen halides.

16. Which of the following statements are correct?
(a) SO_3 is anhydride of sulphuric acid
(b) In the solid state SO_3 has plane triangular structure
(c) At room temperature SO_3 is a gas
(d) SO_3 is a powerful oxidizing agent when hot

17. Which of the following are correct?
(a) $SF_4 + BF_3 \rightarrow [BF_2^+]^+ SF_5]^-$
(b) $SF_4 + PF_5 \rightarrow [SF_3]^+[PF_6]^-$
(c) $SF_4 + CsF \rightarrow [Cs]^+[SF_5]^-$
(d) $BF_3 + LiF \rightarrow [Li]^+ [BF_4]^-$

18. Which of the following species have zero dipolemoment?
(a) SO_3 (b) CS_2 (c) SO_4^{2-} (d) SO_3^{2-}

19. Which of the following statments is/are correct?
(a) Sulphur exhibit enantiotropic allotropy
(b) Rhombic sulphur is stable at room temperature
(c) Both rhombic and monoclinic sulphur are soluble in CS_2
(d) Both rhombic and monoclinic sulphur contain S_8 molecule

20. Oxygen is evolved when
(a) Permanganate is heated
(b) Sodium peroxide reacts with water
(c) Ammonium nitrate is heated
(d) Sodium nitrate is heated

21. Which of the following statements are correct?
(a) Sulphurous acid is a white crystalline solid

(b) Sulphurous acid is a strong reducing agent
(c) The S—O bond in SO_2 is smaller than the expected value
(d) It is a polar molecule

22. When conc H_2SO_4 is added to NaCl solid, then HCl gas is liberated because
(a) H_2SO_4 is a stronger acid than HCl
(b) H_2SO_4 is an oxidizing agent
(c) HCl is more volatile than H_2SO_4
(d) HCl is a small molecule

23. Sodium thiosulphate can be prepared by
(a) boiling Na_2SO_3 solution with sulphur in acidic medium
(b) boiling an aqueous solution of NaOH with sulphur
(c) neutralizing H_2SO_4 with NaOH
(d) boiling Na_2SO_3 with sulphur in alkaline medium

24. Gas A + $Na_2CO_3 + H_2O \longrightarrow$ B + C (gas)

$$B + Na_2CO_3 \longrightarrow D + C\ (gas) + H_2O$$

$$D + S \xrightarrow{\Delta} E$$

Which of the following is/are correct?
(a) A is SO_2 (b) D is Na_2SO_3
(c) C is CO_2 (d) E is $Na_2S_2O_3$

25. Which of the following statements is/are correct?
(a) Ozone oxidizes iodide ion to iodine in neutral condition
(b) Ozone oxidizes iodide ion to iodate or periodate ions in alkaline medium
(c) Ozone oxidizes iodine to iodic acid in the presence of moisture
(d) Ozone oxidizes iodine to iodine pentoxide in dry condition

26. Identify the correct statement regarding formation of oleum during synthesis of sulphuric acid in which H_2SO_4 and SO_3 react to give oleum $H_2S_2O_7$?
(a) It is an exothermic reaction
(b) It is a neutralization reaction
(c) H_2SO_4 acts as Bronsted acid and SO_3 as Lewis base in the reaction
(d) H_2SO_4 acts as a Lewis base and SO_3 as a Lewis acid in the reaction

Comprehension Type Questions

Passage-I

Read the Passage and Answer the Questions Followed

Sulphur forms hexahalides, tetrahalides, dihalides and monohalides. Sulphur forms only hexafluoride but not hexachlorides, hexabromides and hexaiodides. Sulphur halides tend to hydrolyse easily. Sulphur hexafluoride is an exception.

Of the oxohalides, the most important are those of sulphur especially sulphur dichloride oxide (Thionyl chloride) $SOCl_2$ and sulphur dichloride dioxide (sulphuryl chloride) SO_2Cl_2. These also hydrolyse in water.

1. SF_6 do not hydrolyse in water because
(a) Due to strong S—F bonds which cannot be broken easily
(b) Because of steric hinderance of six fluorine atoms surrounding sulphur H_2O molecules cannot approach sulphur
(c) Due to double bond character of S—F bonds because of back bonding from fluorine to sulphur
(d) All the above

2. Thionyl chloride is dissolved in water. Which of the following statement is wrong about the solution?
(a) The solution will give white precipitate with baryta water soluble in dil. HCl
(b) The solution turns orange dichromate to green
(c) The solution turns the lead acetate paper to black
(d) The solution is acidic in nature

3. Sulphuryl chloride is dissolved in water. Which of the following statement is wrong about the solution
(a) Gives white ppt with barium chloride insoluble in any acid
(b) The solution contain two different types of acids a monobasic and a dibasic acid
(c) The solution can decolourize the permanganate
(d) The oxidation states of the elements in SO_2Cl_2 donot change when dissolved in water

4. In $SOCl_2$ and SO_2Cl_2.
(a) The oxidation states of sulphur is same
(b) The hybridization states of sulphur is same
(c) The shapes of both $SOCl_2$ and SO_2Cl_2 are same
(d) The *ClSCl* angle in both $SOCl_2$ and SO_2Cl_2 is same

5. Which halide of sulphur that undergoes disproportionation in water?
(a) SCl_4 (b) SF_2
(c) S_2Cl_2 (d) Both SF_2 and S_2Cl_2

Passage-II

A yellow powder X is burnt in a stream of fluorine to obtain a colourless gas Y which is thermally stable and chemically inert. Its molecule has octahedral geometry. Another colourless gas 'Z' with same constituent atoms as that of 'Y' is obtained when sulphur dichloride is heated with sodium fluoride. Its molecule has trigonal bipyramidal geometry

1. The yellow powder 'X' is
   (a) $K_2Cr_2O_7$ (b) $FeCl_3$
   (c) K_2CrO_4 (d) S
2. The colourless gas 'Y' is
   (a) SF_4 (b) SF_6 (c) NaF (d) S_2F_2
3. The colourless gas 'Z' is
   (a) SF_4 (b) SF_6 (c) S_4F_4 (d) NaF

Passage-III

A student accidentally found a packet in the lab. He was tempted to find what it is. He therefore subjected it to many reactions but only few gave him inferences. The substance (X) in the packet gave the following observations

(i) gave brown coloured gas (Y) with conc. HNO_3
(ii) dissolved in sodium sulphite on heating to form a colourless solution (Z)
(iii) (Z) on reaction with iodine decolourized it
(iv) (Z) on heating with sulphuric acid gave sulphur as turbidity and liberated a gas (Q)

1. The brown coloured gas (Y) is
   (a) N_2O_3 (b) NO_2 (c) SO_2 (d) Br_2
2. The colourless solution (Z) when subjected to experiment (IV) gives gas (Q). Hence (z) and gas Q are respectively.
   (a) $Na_2S_2O_3$ and SO_3 (b) $Na_2S_2O_3$ and SO_2
   (c) $Na_2S_4O_6$ and SO_3 (d) $Na_2S_4O_6$ and SO_2
3. Hence from all these observations obtained when substance (X) is subjected at experiments (i) to (iv) it can be concluded that (X) is
   (a) H_2SO_4 (b) H_2SO_3 (c) SO_3 (d) S

Passage-IV

Sulphur forms more than a dozen of oxoacids. They are mainly classified into four types such as sulphurous, sulphuric, thionic and peroxy acids. In these acids, there are hypo, thio, peroxo, thioperoxo and pyro links present. Accordingly, these acids have been named. The molecular structure of all oxyacids of sulphur have symmetry except persulphuric acid.

All oxyacids of sulphur are dibasic in nature irrespective of the type of links present in them and hybridization of sulphur atom in all of these is sp^3. However, these compounds vary in their acidic nature, mainly due to difference in the oxidation state of sulphur. As oxidation state of sulphur increases the acidic nature of the oxyacid increases. Based upon the information given answer the following questions.

1. Most acidic among the following acids is
   (a) Thiosulphurous (b) Hyposulphurous
   (c) Pyrosulphurous (d) All are equally acidic
2. S—S or S = S bond is absent in which of the following acids?
   (a) $H_2S_2O_2$ (b) $H_2S_2O_3$ (c) $H_2S_2O_6$ (d) $H_2S_2O_7$
3. Which of the following formulae of oxoacids of sulphur is not possible?
   (a) $H_2S_2O_8$ (b) $H_2S_4O_6$ (c) $H_2S_4O_5$ (d) H_2SO_5

Matching Type Questions

1. Match the following

List-I	List-II
(a) $S_2O_3^{2-}$	(p) S is in sp^3 hybridization
(b) SO_4^{2-}	(q) Contain S—S bond
(c) SO_5^{2-}	(r) Oxidizing agent
(d) $S_2O_4^{2-}$	(s) Complexing agent
	(t) Reducing agent

2. Match the following Column-I with Column-II

Column-I	Column-II
(a) SO_2	(p) Acidic nature
(b) SO_3	(q) Oxidizing agent
(c) O_3	(r) Reducing agent
(d) O_2^{2-}	(s) Bleaching agent

3. Column-I	Column-II
(a) Dithionous acid	(p) S—O—S bond is not present
(b) Thiosulphuric acid	(q) All 'S' atoms in the molecule has oxidation state +3
(c) Caro's acid	(r) Acidic strength of –OH groups present in the molecule is different
(d) Pyrosulphurous acid	(s) At least one S atom has oxidation state +5 in molecule

4. Match the following

List-I	List-II
(a) SO_2	(p) Reducing agent
(b) O_3	(q) Oxidizing agent
(c) SO_3	(r) Bleaching agent
(d) H_2S	(s) Acidic in nature in aqueous solution

5. Match the following Column-I with Column-II

Column-I	Column-II
(a) Thiosulphate ion	(p) Contain S—O—S bond
(b) Dithionite ion	(q) Both S-atoms in same oxidation state
(c) Pyrosulphite ion	(r) Both S-atoms in different oxidation state
(d) Pyrosulphate ion	(s) May be oxidized

6. Match the following Column-I with Column-II

Column-I	Column-II
(a) Dihydrogen hypophosphate ion	(p) All hydrogens are ionizable
(b) Hydrogen pyrophosphite ion	(q) Number of acidic hydrogens present is two
(c) Pyrosulphurous acid	(r) sp^3 hybridized central atom
(d) Hydrogen peroxymonosulphate ion	(s) Overall non planar structure
	(t) Ion having peroxy linkage

7. Match the following

List-I	List-II
(a) $O_3 + H_2O_2 \longrightarrow$	(p) Blue
(b) $O_3 + \text{starch KI} \longrightarrow$	(q) Tailing
(c) $O_3 + Hg \longrightarrow$	(r) HIO_3
(d) $O_3 + I_2 + H_2O \longrightarrow$	(s) $O_2 + H_2O$

8.	Column-I Elements	Column-II Covalency
	(a) 'O' atom in O_3	(p) One
	(b) 'S' atom in hypo sulphorous acid	(q) Two
	(c) 'N' atom in $NO^+$and N_2O_4	(r) Three
	(d) 'N' atom in hydrazoic acid	(s) Four

9. Match the oxoacids of sulphur in column-I with their characteristics in Column-II

Column-I	Column-II
(a) $H_2S_2O_5$	(p) Dibasic
(b) $H_2S_2O_4$	(q) S—O—S bond
(c) $H_2S_4O_6$	(r) S—S bond with same oxidation state of sulphur
(d) $H_2S_2O_7$	(s) S—S bond with different oxidation state of sulphur
	(t) Atleast one sulphur is in +6 oxidation state

Numerical Type Questions

1. The number of pπ–dπ bonds in cyclic trimer of sulphur trioxide is
2. No. of moles of ozone that should be passed through hydrogen peroxide solution to get 224 lit of oxygen at STP is… … … … … … … .
3. The number of sp^3 hybrid atoms in the product formed by oxidation of hypo with iodine
4. The oxidation state of the product when KBr and $KBrO_3$ react in acid medium is
5. How many of the following can react with bases? Mn_2O_7, ZnO, Al_2O_3, BeO, SeO_2, NO, BrO_5, PbO, As_2O_3
6. In the balanced reaction

$$xS + yHNO_3 \longrightarrow zH_2SO_4 + mNO_2 + nH_2O$$

The value of $\frac{y^2 - x^2}{z + m}$ is

7. When fluorine react with hydrogen sulphide a product (x) of sulphur is formed. The difference in the oxidation state of sulphur in H_2S and the in the product (x) is

VI Group Key

Single Answer Type Questions

1. a	**2.** a	**3.** b	**4.** c	**5.** b	**6.** b	**7.** b
8. a	**9.** c	**10.** b	**11.** b	**12.** c	**13.** a	**14.** d
15. d	**16.** c	**17.** b	**18.** a	**19.** b	**20.** a	**21.** d
22. a	**23.** a	**24.** c	**25.** a	**26.** b	**27.** b	**28.** b
29. b	**30.** d	**31.** b	**32.** a	**33.** c	**34.** a	**35.** b
36. d	**37.** c	**38.** d	**39.** b	**40.** b	**41.** c	**42.** b
43. b	**44.** a	**45.** d	**46.** c	**47.** d	**48.** b	**49.** c
50. a	**51.** a	**52.** a	**53.** c	**54.** c	**55.** c	**56.** c
57. b	**58.** c	**59.** d	**60.** b			

More than One Answer Questions

1. a, c, d	**2.** a, b, c	**3.** a, c	**4.** a, b, c, d
5. b, c	**6.** b	**7.** b, c, d	**8.** a, b, c, d
9. b, c	**10.** b, c, d	**11.** a, b, c	**12.** a, b
13. a, b, d	**14.** a, b, c, d	**15.** a, b, c	**16.** a, d
17. b, c, d	**18.** a, b, c	**19.** a, b, c, d	**20.** a, b, d
21. b, c, d	**22.** c	**23.** b, d	**24.** a, b, c
25. a, b, c	**26.** a, b, d		

Comprehensive Type Questions

Passage – I	**1.** b	**2.** c	**3.** c	**4.** b	**5.** d
Passage – II	**1.** d	**2.** b	**3.** a		
Passage – III	**1.** b	**2.** b	**3.** d		
Passage – IV	**1.** c	**2.** d	**3.** c		

Match the Type Questions

1.	a-p, q, s, t	b-p, s	c-p, r	d-p, q, t
2.	a-p, q, r, s	b-p, q	c-q, r, s	d-q, r, s
3.	a-p, q	b-p	c-p, r	d-p, r, s
4.	a-p, q, r, s	b-p, q, r	c-q, s	d-p, s
5.	a-r, s	b-q, s	c-r, s	d-p, q
6.	a-p, q, r, s	b-r, s	c-p, q, r, s	d-p, r, s, t
7.	a-s	b-p	c-q	d-r
8.	a-p, q, r	b-s	c-r, s	d-q, r, s
9.	a-p, s	b-p, r	c-p, r, s	d-p, q, t

Integer Type Questions

1. 6 **2.** 5 **3.** 4 **4.** 0 **5.** 9 **6.** 5 **7.** 8

VI Group Hints

1. $Na_2SO_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2$

$Na_2S + 2HCl \longrightarrow 2NaCl + H_2S$

Both SO_2 and H_2S reduces acidified dichromate to green chromic sulphate. H_2S turns lead acetate to black PbS but SO_2 has no effect

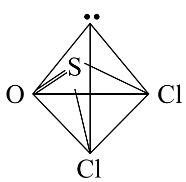
2. $\underset{\text{excess}}{PCl_5} + H_2SO_4 \longrightarrow ClSO_3H + POCl_3 + 2HCl$

$\underset{\text{excess}}{2PCl_5} + H_2SO_4 \longrightarrow SO_2Cl_2 + 2POCl_3 + 2HCl$

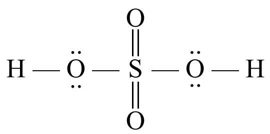
3. $2SO_2 + O_2 \longrightarrow 2SO_3$

Concentration of SO_3 increases and concentrations of SO_2 and O_2 decreases with time

4. Tetrahedral



5.

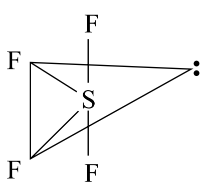


One 'S' and two 'O' atoms are in sp^3 hybridization. Two 'O' atoms are in $p\pi$–$d\pi$ bonds with S

6. $S + 3F_2 \longrightarrow SF_6$

SF_6 is used gaseous insulator

$SCl_2 + 4NaF \longrightarrow SF_4 + 2NaCl + Na_2S$



$SF_4 + 4H_2O \longrightarrow H_4SO_4 + 4HF$

7. $H_4SO_4 \longrightarrow H_2SO_3 + H_2O$

$H_2SO_3 + 2H_2S \longrightarrow 3H_2O + 3S$ (Colloidal)

8. $H_2O \rightleftharpoons H^+ + OH^-$

Both basic and acidic oxides dissolve in water forming hydroxides and oxoacids. Water is poor conductor

$Na_2O + H_2O \longrightarrow 2NaOH$

$Na_2S + 2H_2O \longrightarrow 2NaOH + H_2S$

$Na_2Se + 2H_2O \longrightarrow 2NaOH + H_2Se$

$Na_2Te + 2H_2O \longrightarrow 2NaOH + H_2Te$.

Acidic character of chalcogen hydrides increases down the group. So pH decreases.

12. Due to more affinity towards water much heat is liberated when SO_3 is dissolved in water. So the solution boils and form fog.

14. SO_4^{2-} is tetrahedral.

15. SO_2 is more stable than other oxides but in the case of other oxides stability increases due to increase in polarity.

18. $H_2SO_4 + 2KI \longrightarrow K_2SO_4 + 2HI$

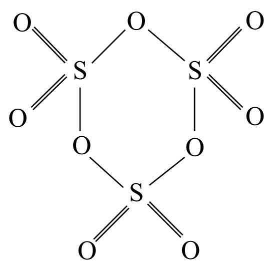
$H_2SO_4 + 2HI \longrightarrow 2H_2O + SO_2 + I_2$

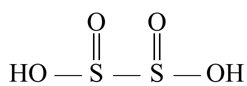
So the HI formed will be contaminated with SO_2 and I_2

22. Conc H_2SO_4 oxidizes copper to Cu^{2+} ion

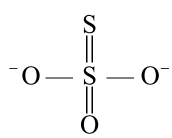
$Cu + 2H_2SO_4 \longrightarrow CuSO_4 + 2H_2O + SO_2$

23. 6 S = O and 3 S—O—S bonds



25.  HO — N = N — OH.

26. The structure of SO_3^{2-}



Due to resonance the terminal sulphur gets negative charge and hence it is in –2 oxidation state.

27. $2Hg + O_3 \longrightarrow Hg_2O + O_2$

28. Acidic $K_2S_2O_8$ can oxidize MnO_2 to MnO_4^-.

31. When a mixture of CO_2 and SO_2 are passed through $BaCl_2$ solution they are completely absorbed by $BaCl_2$ forming $BaCO_3$ and $BaSO_3$. As SO_2 is not coming out $K_2Cr_2O_7$ is unaffected.

37. I. Among the Vth group hydrides NH_3, PH_3, AsH_3, AsH_3 have lowest bond angle as the bond angle decreases down the group.
II. Due to positive charge bond angle in O^+_3 is less than O_3.
III. has lesser bond angle than O_3.
IV. Due to more electronegativity of F bond pair move away from central atom and bond angle decreases.

40. $SO_3 + HCl \longrightarrow ClSO_3H$

$ClSO_3 + H_2O \longrightarrow H_2SO_4 + HCl$

41. $\alpha = 116.8 \quad \beta = 60 \quad \gamma = 180 \quad \delta = 109.28'$

42. Equivalent of H_2SO_4 depend on the reduction product of H_2SO_4.

43. $Na_2S_2O_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2 + S$

Oxidation state of terminal sulphur increases from –2 to 0 oxidation state of central sulphur decreases from +6 to +4.

45. H_2SO_3 is stronger reducing agent in alkaline medium than in acid medium because it can give electrons easily in alkaline medium.

$SO_3^{2-} + 2OH^- \longrightarrow SO_4^{2-} + H_2O + 2e^-$

$H_2SO_3 + H_2O \longrightarrow H_2SO_4 + 2H^+ + 2e^-$

47. $Na_2S + Na_2SO_3 + I_2 \longrightarrow Na_2S_2O_3 + 2NaI$
(Springs reaction)

$Na_2S_2O_3 + \underset{\text{excess}}{I_2} \longrightarrow Na_2S_4O_6 + 2NaI$

48. $2H_2SO_4 + 2KBr \longrightarrow K_2SO_4 + SO_2 + H_2O + Br_2$
change is 2 untis

$2KBr + H_3PO_4 \longrightarrow K_3Po_4 + 3HBr$ No change

$Cl_2 + 2KBr \longrightarrow 2KCl + Br_2$

$Br_2 + 2OH^- \longrightarrow \underset{-1}{Br^-} + \underset{+1}{BrO^-} + H_2O$

Change is 2 units

$\underset{0}{Br_2} + 2NaClO_3 \longrightarrow 2Na\underset{+5}{BrO_3} + Cl_2$

Change is 5 units

49. Oxygen is paramagnetic but nitrogen is diamagnetic.

50. $2NH_2CONH_2 + 2H_2SO_4 \longrightarrow H_2NSO_3 + NH_4HSO_4 + CO_2$

51. $2NaHSO_4 \xrightarrow{\Delta} Na_2S_2O_7 + H_2O$

52. When $KHSO_4$ is dissolved in H_2SO_4 solution the H^+ concentration increases due to ionization of $KHSO_4$.

54. $Na_2S_2O_3 + 5H_2O + 4Cl_2 \longrightarrow 2NaHSO_4 + 8HCl$

The oxidation state of terminal sulphur changes from –2 to +6

55. $2Ag + O_3 \longrightarrow Ag_2O + O_2$

$Ag_2O + O_3 \longrightarrow 2Ag + 2O_2$

More than One Answer Questions

1. The two sulphur atoms in thiosulphate are not equivalent and they are in different oxidation states. The remaining statements are correct.

2. The S—S—S bond angle in S_8 molecule is 107º due to repulsion by lone pairs. Other statements are correct.

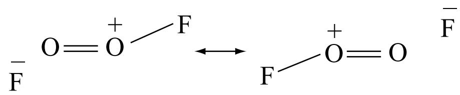
3. 1. CO neutral SO_3 acidic
2. K_2O basic, N_2O neutral
3. Al_2O_3 and ZnO are amphoteric
4. both SO_3 and P_4O_{10} are acidic

4. O_3 is endothermic substance and thermodynamically unstable with respect to O_2. As 3 molecules of O_2 convert into 2 molecules of O_3 entropy decreases O_3 is diamagnetic.

5. O_3 oxidizes KI to iodine in neutral medium but in alkaline medium O_3 oxidizes KI to KIO_3 or KIO_4.

6. When Na_2SO_3 is boiled with radioactive S*, the radioactive isotope is added as terminal sulphur and during the reaction with acids again the terminal radioactive S* will be precipitated.

10. In O_2F_2 the O—O bond is shorter than that in H_2O_2 due to resonance.



Though O_2F_2 and H_2O_2 has similar structures the O—O bond in O_2F_2 has double bond character as shown above, but H_2O_2 contain O—O bond.

14. (a) $H_2S_2O_8 + H_2O \longrightarrow H_2SO_4 + H_2SO_5$

(b) $3KClO_3 + 3H_2SO_4 \longrightarrow 3KHSO_4 + HClO_4 + 2ClO_2 + H_2O$

(c) $2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI.$

(d) $SO_2 + 2H_2O + I_2 \longrightarrow H_2SO_4 + 2HI.$

15. Conc H_2SO_4 can oxidize Br^- and I^- to Br_2 and I_2 respectively but cannot oxidize F^- and Cl^-. Other statements are correct.

16. In solid state SO_3 exist as either cyclic or chain forms in which sulphur is in sp^3 hybridization. At room temperature SO_3 is a solid.

18. SO_3^{2-} is pyramidal in shape and is polar with some dipolemoment.

19. If two allotropes exist in equilibrium at a particular temperature it is called enantiotropy. Rhombic and monoclinic sulphurs are enantiotropic allotropes. Rhombic sulphur is stable at room temperature.

20. NH_4NO_3 on heating gives N_2O not O_2.

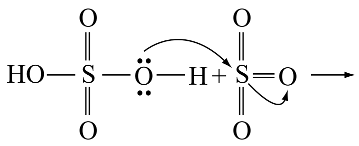
21. Pure sulphurous acid cannot be isolated. It exist only in solution.

22. Less volatile H_2SO_4 substitute more volatile HCl.

24. $$\underset{A}{2SO_2} + Na_2CO_3 + H_2O \longrightarrow 2\underset{B}{NaHSO_3} + \underset{C}{CO_2}$$

$$2\underset{B}{NaHSO_3} + Na_2CO_3 \longrightarrow 2\underset{D}{Na_2SO_3} + \underset{C}{CO_2} + H_2O$$

$$\underset{D}{Na_2SO_3} + S \longrightarrow \underset{E}{Na_2S_2O_3}$$

26. 

$$HOSO_2 - \overset{\overset{H}{|}}{\underset{+}{O}} - SO_3^- \longrightarrow HO - SO_2 - OSO_2OH$$

CHAPTER 11

Group VIIA (17) Halogens

God give me strength to face through it stay me.
Thomas Huxby

11.1 INTRODUCTION

This group includes the highly electronegative elements, namely, fluorine, chlorine, bromine and iodine besides the artificially produced astatine (atomic number 85). The first four elements fluorine, chlorine, bromine and iodine are known as the *halogens* (Greek, halos = sea salts, genes = born) because the salts of chlorine, bromine and iodine are found in sea water. Astatine is a newly discovered radioactive element and is highly unstable, e.g., ^{211}At ($t_{1/2}$ = 7.2h) or ^{211} At ($t_{1/2}$ = 8.3h). Much less is known about the chemistry of astatine.

All the halogens are too reactive to occur free in nature. They only occur in the combined state.

On account of their close resemblance and gradual gradation in their both physical and chemical properties, the halogens constitute a well marked group.

ELECTRONIC CONFIGURATIONS

The electronic configurations of halogens are listed in Table 11.1.

From Table 11.1, it is evident that atoms of these elements are characterized by an identical electronic configuration ns^2 np^5 in their valence shell. As the halogens are having one electron less than the adjacent noble gas configuration, they have a strong tendency to attain a noble gas configuration either by gaining an electron from an electropositive metal atom or by sharing an electron with an atom of an element whose attraction for the electrons is too high to permit complete transfer. Hence the halogens are amongst the most reactive elements.

The halogen atoms can also share their unpaired electrons among themselves to get the noble gas configuration. Hence all halogens exist as diatomic molecules X_2 and because they can gain an electron to get noble gas configuration and they easily convert into X^- ions.

Like the first elements of the other groups, only fluorine is not having a d–orbital, it is expected to differ markedly from the other members of the family. As the atoms of halogens are having similar configuration in their valence shell, they resemble closely in their properties than in any other groups except IA and IIA.

Table 11.1 Electronic configurations of halogens

Element	Atomic number	Electronic configuration	Valence shell configuration
Fluorine	9	$1s^22s^22p^5$	$2s^22p^5$
Chlorine	17	$1s^22s^22p^63s^23p^5$	$3s^23p^5$
Bromine	35	$1s^22s^22p^63s^23p^63d^{10}4s^24p^5$	$4s^2\ 4p^5$
Iodine	53	$1s^22s^22p^63s^23p^63d^{10}4s^24p^64d^{10}\ 5s^25p^5$	$5s^25p^5$
Astatine	85	$1s^22s^22p^63s^23p^63d^{10}4s^24p^6\ 4d^{10}4f^{14}$ $5s^25p^65d^{10}6s^26p^5$	$6s^26p^5$

11.2 PHYSICAL PROPERTIES

Some important atomic and physical properties are summerized in Table 11.2.

The densities given for F, Cl and Br are given in the liquid state at the temperature given in parenthesis. The density given for iodine is in solid state. The electrode potentials E°/(V) given are for the half – cell reaction $X_{2(g)} + 2e^- \rightarrow 2X^-_{(aq)}$.

1. *Atomic and ionic sizes*: The atomic or ionic radius of halogens gets increased gradually from fluorine to iodine as the principal quantum number of the outermost shell gets increased.

2. *Physical state*: All the halogens exist as diatomic covalent molecules F_2, Cl_2, Br_2 and I_2 and they are non-polar. Between these non-polar discrete molecules there exist only weak Van der Waal's attractive forces which increases with increase in the size of molecules from F_2 to I_2. Therefore fluorine having smaller molecules is a gas. Chlorine is also a gas, the Van der waals forces between bigger Br_2 molecules are sufficient to bring them together to form liquid. Iodine is a solid due to stronger Van der Waal's forces between the larger I_2 molecules.

3. *Melting and Boiling points*: As explained above since the Van der Waal's attractive forces are increasing from F_2 to I_2, the melting and boiling points also increases from F_2 to I_2.

4. *Atomic volume and density*: As expected, these increase regularly from fluorine to iodine.

5. *Ionization Energies*: The ionization energies of all halogens are all very high. This indicates these have little tendencies to lose electrons. The ionization energy decreases gradually from fluorine to iodine because the size of atom increase from fluorine to iodine. The value for iodine is comparatively so low that it can lose an electron and forms I^+ ion.

It is interesting to note that first ionization energies of Cl, Br and I are less than that of H. **Now the question arises why H^+ exists whereas simple X^+ ions are not formed**. The reason is that H^+ is very small and the ionization energy is compensated in the form of a high lattice energies. However, certain compounds are known in which I^+ is stabilized by forming complexes with Lewis bases e.g., $[I\ (Pyridine)_2]^+NO_3^-$.

6. *Electronegativity*: Halogens have very high values of electronegativity. Fluorine has the highest electronegativity (4). The electronegativity decreases from fluorine to iodine. This trend indicates a decrease in the non-metallic character from fluorine to iodine.

7. *Non-metallic character*: All halogens are non-metals. The non-metallic character decreases from fluorine to iodine. The last element iodine has a metallic lustre and has a tendency to form I^+ ion by losing an electron (cf alkali metals). For instance ICl and ICN ionize giving I^+ ion in each case.

$$ICl \longrightarrow I^+ + Cl^-$$

$$ICN \longrightarrow I^+ + CN^-$$

Enough evidence is available for the existence of I^{3+} ion. For example, iodine acetate ionizes in the following manner.

$$I(CH_3COO)_3 \rightleftharpoons I^{3+} + 3CH_3COO^-$$

8. *Bond Energy*: The bond energy of the X_2 molecules decreases because the atoms become larger as increased

Table 11.2 Atomic and physical properties of halogens

Property	F	Cl	Br	I	At
Atomic mass (g mol^{-1})	19.00	35.45	79.90	126.90	210
Covalent radius (pm)	64	99	114	133	–
Ionic radius x$^-$(pm)	133	184	196	220	–
Ionization enthalpy (KJ mol^{-1})	1680	1256	1142	1008	–
Electron gain enthalpy (KJ mol^{-1})	–333	–349	–325	–296	–
Electronegativity	4	3.2	3.0	2.7	2.2
Δ_{Hyd} H[X] KJ mol^{-1}	515	381	347	305	–
Melting point (K)	54.4	172	265.8	386.6	–
Boiling point (K)	84.9	239	332.5	458.2	
Density (g cm^3)	1.5(85)	1.66(203)	3.19(273)	494(293)	
X - X bond length (pm)	143	199	228	266	
Bond dissociation enthalpy (KJ mol^{-1})	158.8	242.6	192.8	151.1	
E°/(V)	2.87	1.36	1.09	0.54	

size causes less effective overlap of orbital. This trend can be seen in Cl_2, Br_2 and I_2. However, the bond energy in F_2 is exceptionally low and even less than bromine and nearly equal to iodine. Different explanations are given for this low bond energy of fluorine.

According to *Mulliken*, in Cl_2, Br_2 and I_2 there is possibility of multiple bonding involving d-orbitals. This accounts for the increased bond energy of Cl_2 and Br_2. But in F_2 there are no d-orbitals in its valence shell, and hence no possibility of p—d overlap in F_2. Therefore no multiple bonding in F_2 is possible which results in low F—F energy. But this explanation is not so correct because the d-orbitals are, in general, too large and too high in energy to overlap with the s and p-orbitals. Also p-orbital of one atom may overlap with a d-orbital of another atom only laterally to form π – overlap.

According to *Coulson's* explanation fluorine atoms are small and the two atoms are closer (F—F bond length 143 pm) to each other. Then appreciable inter nuclear repulsions and large lone pair – lone pair electronic repulsions in the fluorine molecule are possible. These repulsions weakens the bond and hence the bond energy is less. Because of this reason the F—F bond length (143 pm) is longer than the sum of the covalent radii of two fluorine atoms (64 + 64 = 138 pm). These repulsions in other halogen molecules are relatively negligible because of the larger atomic sizes and larger inter atomic distances.

Of the two above explanations, i.e., Mulliken and Coulson's explanations, Coulson's explanation is simpler and widely accepted.

9. *Electron Affinity*: As the halogen atoms are having seven electrons in their outermost orbit, they can readily accept one electron to get the stability and thus they have high electron affinity values. However, the electron affinity of fluorine is less than chlorine, though it is more electronegative than chlorine. This is because of the repulsion between the electrons present in the relatively compact 2p – sublevel of fluorine and the electron being added. So to overcome this repulsion some energy is consumed from the energy liberated. Hence the liberated energy becomes less. The decreasing order of electron affinity for halogens has been as follows

$$Cl > F > Br > I.$$

10. *Colour*: All the halogens are coloured. Fluorine is light yellow, chlorine is yellowish green, bromine is orange red and iodine is deep violet in gaseous state but blackish brown in solid state. The colour deepens with the rise in atomic number from fluorine to iodine. The colour is due to the absorption of energy in the visible light by the electrons to excite from HOMO to LUMO (Frontier orbitals). Thus they appear in the complimentary colour of the absorbed colour. The energy absorbed depends on the size of the atom. Greater the size of atom, smaller is the excitation energy. Thus the excitation energy decreases from fluorine to iodine. Fluorine due to its smaller size requires a large amount of excitation energy and therefore absorbs violet (high energy) and hence appears yellow. On the other hand gaseous molecules of I_2 absorb yellow light (lower energy) and appear violet. Similarly the yellowish green colour of chlorine and orange red colour of bromine can be accounted.

The colour of iodine in different solvents varies. The colour of iodine solutions in aliphatic hydrocarbons or CCl_4 is bright violet, those in aromatic hydrocarbons are pink or reddish brown and those in stronger donor solvents such as alcohols, ethers or amines are deep brown. This variation can be understood in terms of a weak donor-acceptor interaction leading to complex formation between the solvent (donor) and I_2 (acceptor) which alters the optical transition energy. Thus referring to the conventional M.O energy diagram for I_2 (or other X_2) as shown in Fig 11.1 the violet colour of I_2 vapour can be seen to arise as a result of the excitation of an electron from the HOMO (the anti-bonding π level) into the LUMO (the antibonding σ level). In non-coordinating solvents such as aliphatic hydrocarbons or their fluoro- or chloro- derivatives the transition energy (and hence the colour) remains essentially unmodified. However, in electron donor solvents L the vacant antibonding σ^* P_z orbital of I_2 acts as an electron acceptor thus weakening the I–I bond and altering the energy of electronic transitions.

Consistent with this: (i) the solubility of iodine in the donor solvents tends to be greater than in the non-donor solvents (ii) brown solutions frequently turn violet on being heated, and brown again on cooling, due to the ready dissociation and reformation of the complex and (iii) addition of a small amount of a donor solvent to a violet solution turns the colour brown.

11. *Oxidation States*: As fluorine is the most electronegative element it always exhibit only negative oxidation state but never exhibit positive oxidation state. Further due to the absence of d-orbitals in its valence shell, it never exhibit higher oxidation states. So fluorine always exhibit a fixed oxidation state –1 only. The remaining halogens in their ground state exhibit either –1 or +1 depending on the electronegativity of the other element with which they combine. They also exhibit +III, +V and +VII oxidation states in 1st, 2nd and 3rd excited states respectively.

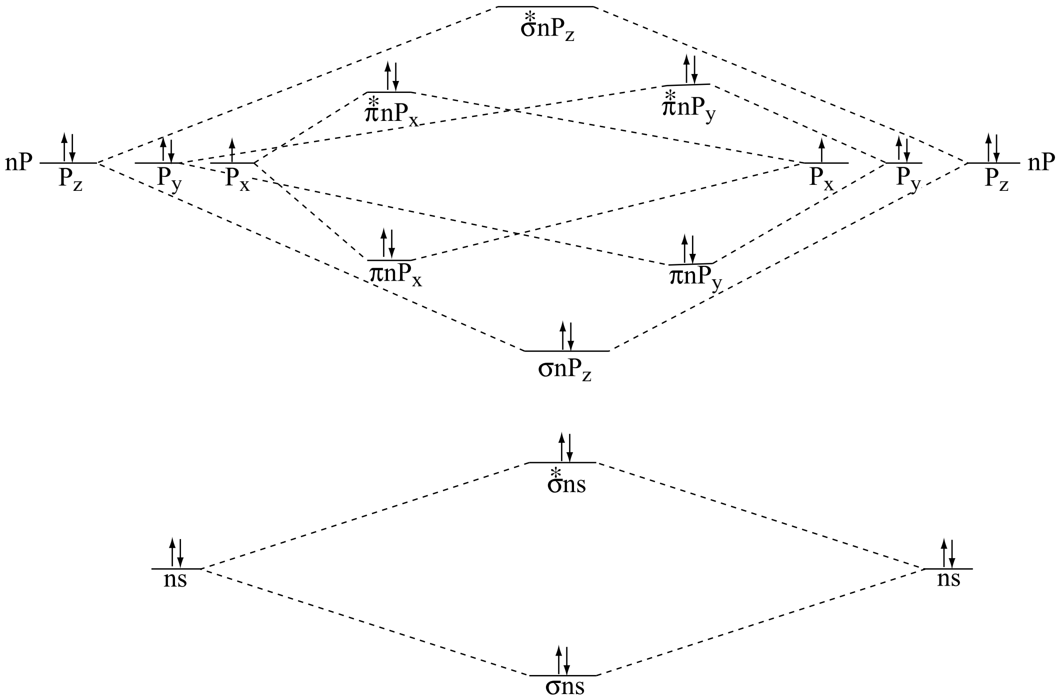


Fig 11.1 Molecular orbital energy diagram for diatomic halogen molecules

	Outer electronic configuration of halogen			Oxidation state
	ns	np	nd	
Ground state	↓↑	↓↑ ↓↑ ↑	☐☐☐☐☐	−1 or +1
1st excited state	↓↑	↑↓ ↑ ↑	↑ ☐☐☐☐	+III
2nd excited state	↑↓	↑ ↑ ↑	↑ ↑ ☐☐☐	+V
3rd excited state	↑	↑ ↑ ↑	↑ ↑ ↑ ☐☐	+VII

Apart from these oxidation states chlorine and bromine exhibit unusual oxidation states +IV in ClO_2 and BrO_2 and +VI in Cl_2O_6 and BrO_3 respectively.

It is interesting to note that bromine compounds in +VII oxidation state are rare. Br_2O_7 do not exist. This is again due to the **middle row anomaly.** Bromine is having $3s^2\ 3p^6\ 3d^{10}$ configuration in its penultimate shell which has weak screening effect of d electrons. Therefore the energy required to promote an electron from s–orbital to the vacant d – orbital is higher than in the case of chlorine. This energy is not realized in reactions and therefore, bromine does not exhibit an oxidation state of +7.

12. *Oxidation Power*: All halogens act as good oxidizing agents. The oxidizing power decreases from fluorine to iodine. Although, the electron affinity of chlorine is higher than fluorine, the former acts as a weaker oxidizing agent than the latter. It is possible to explain this anomaly on the basis of the fact that the strength of an oxidizing agent (i.e., its oxidation potential) is dependent not only on the electron affinity but also on several other energy terms. The following steps are involved in the reduction (oxidizing strength) of a halogen molecule in its standard state to the hydrated halide ion $\left[\frac{1}{2}X_2 + e^- \longrightarrow X^-_{(aq)}\right]$ in aqueous solution.

(i) $\frac{1}{2}X_{2(s)} \longrightarrow \frac{1}{2}X_{2(l)}$; Heat of fusion ΔH_{fus}

(ii) $\frac{1}{2}X_{2(l)} \longrightarrow \frac{1}{2}X_{2(g)}$; Heat of vapourization ΔH_{Vap}

(iii) $\frac{1}{2}X_{2(g)} \longrightarrow X_{(g)}$; Heat of dissociation ΔH_{diss}

(iv) $X_{(g)} + e^- \longrightarrow X^-_{(g)}$; Electron affinity ΔEa,

(v) $X^-_{(g)} + H_2O \longrightarrow X^-_{(aq)}$; Heat of hydration ΔH_{hyd}

The above changes i.e., reduction of $\frac{1}{2}X_2$ to $X^-_{(aq)}$ can be represented in the form of Born–Haber cycle.

$$\frac{1}{2}X_{2(s)} \xrightarrow{\frac{1}{2}\Delta H_{fus}} \frac{1}{2}X_{2(l)} \xrightarrow{\frac{1}{2}\Delta H_{vap}} \frac{1}{2}X_{2(g)}$$

$$\xrightarrow{\frac{1}{2}\Delta H_{diss}} X_{(g)} \xrightarrow{Ea_1} X^-_{(g)} \xrightarrow{\Delta H_{hyd}} X^-_{(aq)}$$

Now as the processe of converting a solid into liquid, a liquid into gas and a molecular halogen to atomic halogen always takes place with the **absorption of energy**., their heat values i.e., ΔH_{fus}, ΔH_{vap} and ΔH_{diss} would always **positive.** On the other hand, formation of a negative ion and the process of hydration takes place with the release of energy, their heat values i.e., **electron affinity** and ΔH_{hyd} would always negative. Consequently, from Hess's law it is possible to calculate the net energy (E) which is a measure of **oxidation potential** and hence the strength of the oxidizing agent from the following expression.

$$E = \frac{1}{2}\Delta H_{fus} + \frac{1}{2}\Delta H_{vap} + \frac{1}{2}\Delta H_{diss} - E_{a1} - \Delta H_{hyd}$$

As fluorine and chlorine are gases at room temperature, ΔH_{fus} and ΔH_{vap} are not involved, there by omitted. Also, in the case of bromine which is liquid at ordinary temperature, ΔH_{fus} is not involved and hence omitted. The values of E calculated as above for the various halogens are given in Table 11.3.

Hence fluorine having the highest value of E acts as the strongest oxidizing agent. The net higher value of E of fluorine is attributed to its low heat of dissociation and high heat of hydration which have more than compensated the lower value of electron affinity. Hence in short, it is the oxidation potential and not the electronaffinity alone which is able to control the strength of an oxidizing agent.

Although Br_2 and I_2 are having **low dissociation energies** than chlorine, they act as **poor oxidizing agents** than Cl_2. This is ascribed to their **small electron affinities** and **smaller hydration energies**.

11.3 CHEMICAL REACTIVITY OF HALOGENS

Halogens are all highly reactive elements. They can react with metals as well as non-metals and other substances. The order of their reactivity can be given in general as.

$$F >> Cl > Br > I$$

The decreasing order of reactivity in the halogens can be perceived in the conditions in which they react with different substances.

1. *Reaction with water*: From the action of halogens on water two observations can be noted clearly. First, their solubility in water decreases from F_2 to I_2. Second, their chemical reaction with water provides an illustration for the gradation in the properties as the atomic weight of the halogen increases.

Fluorine reacts with water very vigorously and oxidizes the water to ozonized oxygen.

$$2F_2 + 2H_2O \longrightarrow 4HF + O_2$$
$$3F_2 + 3H_2O \longrightarrow 6HF + O_3$$

Chlorine dissolves in water giving a solution known as *Chlorine water*. A freshly prepared solution of chlorine in water contains HCl and HOCl.

$$Cl_2 + H_2O \longrightarrow HCl + HOCl$$

HOCl being unstable, dissociates to give nascent oxygen.

$$HOCl \longrightarrow HCl + (O)$$

Table 11.3 Values of energy changes in the oxidation reaction of halogens ($KJ.\ mol^{-1}$)

S. No.	Halogen	ΔHfus (Fusion)	ΔHVap (Vapourization)	ΔHdiss (Bond energy)	Ea_1 Electron affinity	ΔHhyd (Hydration energy)	Overall energy change E
1	Fluorine	–	–	$+\frac{158.8}{2}$	– 333	– 513	– 766.4
2	Chlorine	–	–	$+\frac{242.6}{2}$	– 349	– 370	– 597.3
3	Bromine	–	+30	$+\frac{192.8}{2}$	– 325	– 339	– 537.4
4	Iodine	+15.52	+41.94	$+\frac{151.1}{2}$	– 296	– 274	– 437.02

The overall reaction can be written as

$$Cl_2 + H_2O \longrightarrow 2HCl + (O)$$

Because of this reaction chlorine water has acidic property as well as bleaching property.

Bromine is less soluble in water. When bromine is allowed to stand in water, it slowly dissolves giving bromine water similar to chlorine water. Bromine water also act as an oxidizing agent and bleaching agent but less powerful than chlorine water.

$$Br_2 + H_2O \longrightarrow HBr + HOBr$$

Iodine is sparingly soluble in water. The free energy in the reaction of iodine with water is positive. So iodine cannot react with water, but the reverse reaction is spontaneous.

$$H_2O + I_2 \longrightarrow 2HI + (O) \qquad \Delta G = +439 \text{ KJ}$$

Air (oxygen) can oxidize iodide ions to iodine.

Thus we can also see that the solubility of halogens in water decreases from F_2 to I_2.

2. *Reaction with alkalis*: Fluorine reacts with cold and dilute alkalis forming oxygen difluoride, but with hot and concentrated alkalis liberates oxygen gas.

$$2F_2 + \underset{\text{Cold \& dil}}{2NaOH} \longrightarrow 2NaF + H_2O + OF_2$$

$$2F_2 + \underset{\text{Hot \& conc}}{4NaOH} \longrightarrow 4NaF + 2H_2O + O_2$$

Chlorine, bromine and iodine react with alkalis differently. With cold and dilute alkalis they give halide and hypohalites while with hot and concentrated alkalis they give halide and halates.

$$\underset{\text{Cold \& Dil}}{2NaOH + X_2} \longrightarrow \underset{\text{Halide}}{NaX} + \underset{\text{hypohalite}}{NaOX} + H_2O; \; X_2 = Cl_2, Br_2 \text{ or } I_2$$

$$\underset{\text{Hot \& Conc}}{6NaOH + 3X_2} \longrightarrow \underset{\text{Halide}}{5NaX} + \underset{\text{halate}}{NaXO_3} + 3H_2O; \; X_2 = Cl_2, Br_2 \text{ or } I_2$$

In these reactions the oxidation state of halogen both increases and decreases. So these reactions are disproportionation reactions.

3. *Reaction with non-metals*: Halogens react with a number of non-metals like S, P, As etc directly. The reactivity again decreases as we move down the group.

(a) ***Reaction with hydrogen*:** All the halogens react with hydrogen to yield their respective hydrogen halides (HX)

$$H_2 + X_2 \longrightarrow 2HX \qquad X_2 = F_2, Cl_2, Br_2 \text{ or } I_2$$

Fluorine reacts violently with hydrogen even in dark at –245ºC, chlorine also react violently but in the presence of light only, bromine reacts non-explosively at about 300ºC. The reaction of iodine with hydrogen takes place at about 440ºC in the presence of platinum as catalyst. Moreover the reaction with iodine is reversible and incomplete.

(b) ***Reaction with boron*:** All halogens react with boron forming boron trihalides.

$$2B + 3X_2 \longrightarrow 2BX_3 \qquad X_2 = F_2, Cl_2, Br_2 \; \& \; I_2$$

(c) ***Reaction with carbon*:** Only fluorine can react with carbon directly. Other halogen cannot combine with carbon directly. The compounds of fluorine and carbon are called fluorocarbons.

$$C + 2F_2 \longrightarrow CF_4$$

(d) ***Reaction with silicon*:** All halogens react with silicon directly forming silicon tetrahalides

$$Si + 2X_2 \longrightarrow SiX_4 \qquad X_2 = F_2, Cl_2, Br_2 \text{ or } I_2$$

(e) ***Reaction with phosphorous*:** Phosphorous react with all halogens forming trihalides of the type PX_3 and pentahalides (except pentaiodide) of the type PX_5

$$2P + 3X_2 \longrightarrow 2PX_3 \qquad X_2 = F_2, Cl_2, Br_2 \text{ or } I_2$$

$$2P + 5X_2 \longrightarrow 2PX_5 \qquad X_2 = F_2, Cl_2, Br_2$$

(f) ***Reaction with sulphur*:** Fluorine and chlorine react with sulphur

$$S + 3F_2 \longrightarrow SF_6$$

$$2S + Cl_2 \longrightarrow S_2Cl_2$$

(g) ***Reaction with halogens*:** The halogens react with one another among themselves forming inter halogen compounds which are discussed separately.

(h) Halogens do not react with nitrogen and oxygen directly.

4. *Reaction with inert gases*: Fluorine react with xenon forming xenon fluorides XeF_2, XeF_4 and XeF_6.

5. *Reaction with metals*: Fluorine reacts with almost all metals including noble metals forming ionic fluorides. Chlorine also react with several metals but slowly with noble metals. Bromine and iodine do not react with gold and platinum and some less active metals.

6. *Reaction with halides*: Fluorine, being the strongest oxidizing agent, can displace the remaining three halogens from their halides.

$$2NaCl + F_2 \longrightarrow 2NaF + Cl_2$$

$$2KBr + F_2 \longrightarrow 2KF + Br_2$$

$$2KI + F_2 \longrightarrow 2KF + I_2$$

Chlorine can displace bromine from bromides and iodine from iodides but cannot displace fluorine from fluorides.

$$2KBr + Cl_2 \longrightarrow 2KCl + Br_2$$

$$2KI + Cl_2 \longrightarrow 2KCl + I_2$$

Bromine can displace iodine from iodides but cannot displace fluorine from fluorides and chlorine from chlorides.

$$2KI + Br_2 \longrightarrow 2KBr + I_2$$

Iodine cannot displace any other halogen from their halides. These reactions clearly indicate the decrease in oxidation power of halogens from fluorine to iodine.

11.4 ANOMALOUS BEHAVIOUR OF FLUORINE

Similar to the first member of other groups in the periodic table, fluorine being the first member of the VII A group is different considerably from the other halogens. This is because of the following characteristics of fluorine.

(a) small size
(b) high electronegativity
(c) non-availability of d-orbitals in its valence shell
(d) Low F—F bond dissociation energy

The various consequences of the above peculiarities of fluorine are given below.

1. It exhibits only one oxidation state i.e., –1. This is because of its most electronegativity.
2. The maximum covalency of fluorine is one. The reason is that it cannot expand its valence shell beyond the octet as no vacant d-orbital is available in its valence shell. Other halogens can exercise covalencies upto 7 due to the presence of vacant d-orbitals in their valence shells.
3. The electronaffinity of fluorine is less than chlorine, though it is more electronegative than chlorine. This is because of its small size.
4. Because of its high electronegativity and small size fluorine can form hydrogen bonding. Therefore, under ordinary conditions HF is liquid while other hydrogen halides are gases.
5. Because of its small size and high electronegativity maximum number of fluorine atoms can attach to the central atom. Thus fluorine is able to bring about the maximum coordination number in other elements. e.g., SF_6, IF_7, $[SiF_6]^{2-}$ etc.
6. Because of the low F—F bond dissociation energy, fluorine is the most reactive among halogens.
   (a) It is able to displace all the other three halogens from their halides.
   (b) It combines with hydrogen violently even at very low temperatures and in dark.
   (c) It is able to decompose water violently and liberates ozonized oxygen.
   (d) It directly react with carbon, where as other halogens do not react even under drastic conditions.
   (e) It react with alkalis in a different manner.

$$\underset{\text{Cold \& dil}}{2NaOH} + 2F_2 \longrightarrow 2NaF + H_2O + OF_2$$

$$\underset{\text{Cold \& dil}}{2NaOH} + X_2 \longrightarrow 2NaX + NaOX + H_2O$$

$$\underset{\text{Hot \& conc}}{4NaOH} + 2F_2 \longrightarrow 4NaF + 2H_2O + O_2$$

$$\underset{\text{Hot \& conc}}{6NaOH} + 3X_2 \longrightarrow 5NaX + NaOX_3 + 3H_2O$$

$$(X_2 = Cl_2, Br_2 \text{ or } I_2)$$

   (f) With most of the other elements it forms strong bonds and thus form stable compounds.
7. ***Hydrofluoric acid*** is a weak acid whereas other halogen acids (HCl, HBr and HI) are strong acids.
8. ***Solubility of fluorides*** is just opposite to that of other halides; AgF is soluble in water whereas other silver halides are insoluble; CaF_2 is insoluble whereas the other halides of calcium are soluble.
9. ***Fluorides are having maximum ionic character*** than the other halides. Thus only AlF_3 and SnF_4 are ionic whereas other aluminium and tin (IV) halides are covalent. This is because of the large differences in the electronegativities of Al (1.5) or tin (1.8) and fluorine (4.0).
10. ***Fluorides have been found to be more stable than*** the corresponding chlorine compounds e.g., UF_6 is more stable than UCl_6; NF_3 is non-explosive whereas NCl_3 is violently explosive.
11. Fluorine is not able to form any **polyhalide** whereas other halogens form polyhalides e.g., I^-_3, Br^-_3 and Cl^-_3 ions.
12. The fluoride ion (F^-) can form the anion HF^-_2 because of hydrogen bonding $(F\ldots . H—F)^-$. Hence it is possible to explain the existence of KHF_2 written as $K^+(F\ldots .H—F)^-$ while $KHCl_2$ or $KHBr_2$ are not known.

11.5 RESEMBLENCES OF FLUORINE WITH OXYGEN

Fluorine and its compounds have been found to resemble more with oxygen than the other halogens. Few such properties are given as follows.

(i) Both fluorine and oxygen react directly with carbon; other halogens do not react with carbon.
(ii) Fluorine and oxygen form such stable compounds in which the other element is not exhibiting the

higher valencies than the usual ones. For example, sulphur shows higher valency in SF_6 and SO_3. The parallel compounds SCl_6 does not exist at ordinary temperatures.

(iii) The heat of formation of their hydrides has been close (ΔH_f for HF and H_2O are −161 and 241.6 KJ mol^{-1}) than those of hydrides of other halogens.

(iv) The boiling point of HF (292 k) has been nearer to water as compared to the hydrides of other halogens.

(v) HF and water have been weak acids as compared to other hydrides of the family.

11.6 FLOURINE

For the first time fluorine was discovered by Scheele. Fluorine was very reactive element and so it is not found in the nature in native state but widely found in the combined state as fluorides. Its chief minerals have been

1. **Fluorospar:** CaF_2
2. **Cryolite:** Na_3AlF_6 (or) $3NaF \cdot AlF_3$
3. **Fluorapatite:** $CaF_2 \cdot 3Ca_3(PO_4)_2$

Small quantities of fluorine are present as fluorides in the soil, river water plants, bones and teeth of animals.

11.6.1 Difficulties Encountered in the Isolation of Fluorine

The isolation of fluorine from its minerals was a huge challenge in chemistry. Many attempts were made unsuccessfully to isolate fluorine for over six decades. The main obstacles in its isolation are

(i) Extreme reactivity of fluorine: Fluorine has been so reactive that it attacks the materials of all vessels in which it is prepared.

(ii) Fluorine is a strongest oxidizing agent: Since fluorine itself is the strongest oxidizing agent chemical oxidizing agents that can oxidize fluorides to fluorine are not available. So fluorine cannot be prepared by chemical oxidation methods.

(iii) Greater Stability and non-conducting nature of hydrofluoric acid: Hydrofluoric acid is quite stable, highly poisonous and corrosive in nature. The anhydrous hydrofluoric acid is covalent and is non conductor.

(iv) Electrolysis of aqueous solutions containing fluorides including HF do not liberate fluorine at anode, instead ozonized oxygen is liberated at the anode.

Finally Henry Moissan achieved, on 26 june 1886, the success of preparation of fluorine by the electrolysis of a cooled solution of KHF_2 in anhydrous liquid HF. This led to the award of Nobel prize to him in 1906.

Moissan Method

In Moissan's method a mixture of anhydrous HF and KF in 12:1 mole proportions is electrolysed at 250k in a platinum-iridium U-tube. The electrodes are made with platinum irridium alloy.

$$2KF + H_2F_2 \longrightarrow 2KHF_2$$

$$KHF_2 \xrightarrow{\text{Fused at } 70-100^\circ C} K^+ + H^+ + 2F^-$$

$$2H^+ + 2e^- \longrightarrow H_2 \quad \text{at cathode}$$

$$2F^- \longrightarrow F_2 + 2e^- \quad \text{at anode}$$

Different methods are used for the preparation of fluorine. Commercially several types of electrolytic cells are used for this purpose. All these methods utilize the same principle of Moissan's method in the isolation of fluorine.

Modern Method

A typical fluorine gas generator is shown in Fig 11.2. It consists of a mild-steel pot (cathode) containing the electrolyte $KF \cdot 2HF$ which is kept at 353–373k either by a heating jacket when the cell is quiescent or by cooling system when the cell is working. The anode consists of a central rod of compacted, ungraphitized carbon and the product gases are kept separate by a skirt or diaphram dipping below the electrolyte surface. The temperature is automatically controlled as is the level of the electrolyte by controlled addition of make-up anhydrous HF.

Earlier, graphite was used as anode. But graphite anodes should not be used, since graphite reacts with

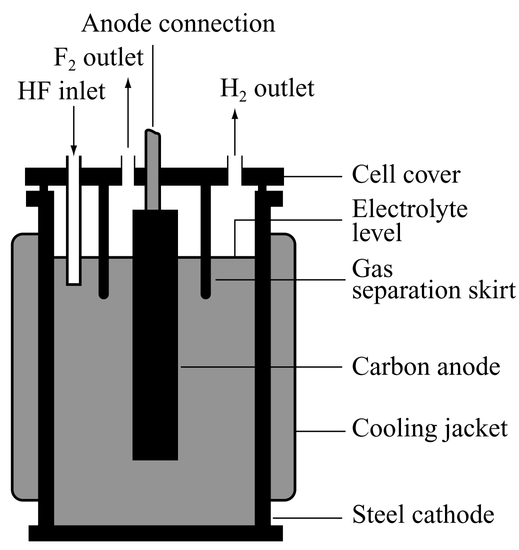


Fig 11.2 Electrolytic fluorine generating cell

fluorine forming an interstitial compound due to invasion of fluorine atoms in between the layers of graphite which approximates to CF. This stops graphite gradually from conducting. Eventually an explosion may occur. Ungraphitized carbon is used to avoid this. So anodes are preferably made from powdered petroleum coke impregnated with copper, and the valves are made of monel metal or nickel with teflon packing.

Physical Properties

Fluorine is a pale yellow coloured gas with a pungent smell. It is heavier than air and is poisonous in nature. It can be condensed to an yellow liquid (b.p. 86k) or to yellow crystals (m.p. 53k). It is diamagnetic in nature.

Chemical Properties

Fluorine is highly reactive element and reacts with large number of substances.

1. *Reaction with water*: Fluorine reacts with water producing oxygen and ozone.

$$2F_2 + 2H_2O \longrightarrow 4HF + O_2$$
$$3F_2 + 3H_2O \longrightarrow 6HF + O_3$$

2. *Reaction with alkalis*: Fluorine liberates OF_2 with cold and dilute alkali, and oxygen gas with hot and concentrated alkali

$$\underset{\text{Cold \& dil}}{2\,NaOH} + 2F_2 \longrightarrow 2NaF + H_2O + OF_2$$

$$\underset{\text{Hot \& conc.}}{4\,NaOH} + 3F_2 \longrightarrow 4NaF + 2H_2O + O_2$$

If excess of fluorine is passed through the alkali, ozonized oxygen and HF may be formed.

3. *Reaction with non-metals*: Fluorine reacts with almost all non-metals directly except nitrogen and oxygen.

$$C + 2F_2 \longrightarrow CF_4$$
$$Si + 2F_2 \longrightarrow SiF_4$$
$$2P + 3F_2 \longrightarrow 2PF_3$$
$$2P + 5F_2 \longrightarrow 2PF_5$$
$$S + 3F_2 \longrightarrow SF_6$$
$$2B + 3F_2 \longrightarrow 2BF_3$$

Hydrogen reacts with fluorine even at very low temperatures (at 21–23k) and in dark resulting HF.

$$H_2 + F_2 \longrightarrow 2HF \qquad \Delta H = -536 \text{ KJ mol}^{-1}$$

The heat of the reaction shows that the fluorine has great affinity towards hydrogen and thus F_2 is highly reactive.

Fluorine also reacts with the remaining three halogens to form interhalogen compounds which are discussed later.

e.g., : ClF, ClF_3, BrF_3, BrF_5, IF_5, IF_7

4. *Reaction with Metals*: Fluorine reacts with almost all metals including noble metals forming metal fluorides.

$$2Na + F_2 \longrightarrow 2NaF$$
$$Mg + F_2 \longrightarrow MgF_2$$
$$2Al + 3F_2 \longrightarrow AlF_3$$
$$2Ag + F_2 \longrightarrow 2AgF$$

Copper and mercury do not react with fluorine because the CuF_2 and HgF_2 formed initially on the surface of metal prevents the further reaction.

Metals which exhibit variable valency form compounds in the higher oxidation state with fluorine.

$$2Fe + 3F_2 \longrightarrow 2FeF_3$$

5. *Reaction with halides*: Fluorine being the strongest oxidizing agent oxidizes all the other halides to corresponding halogens and displaces from their compounds.

$$2KCl + F_2 \longrightarrow 2KF + Cl_2$$
$$MgBr_2 + F_2 \longrightarrow MgF_2 + Br_2$$
$$2KI + F_2 \longrightarrow 2KF + I_2$$

6. *Reaction with inert gases*: Heavier inert gases like Kr and Xe react with fluorine to form compounds like XeF_2, XeF_4 and XeF_6. e.g.,

$$Xe + 2F_2 \longrightarrow XeF_6.$$

7. *Oxidation properties*: It is a strong oxidizing agent and oxidizes several compounds.

$$2KHSO_4 + F_2 \longrightarrow K_2S_2O_8 + 2HF$$
$$KClO_3 + H_2O + F_2 \longrightarrow KClO_4 + 2HF$$
$$KIO_3 + H_2O + F_2 \longrightarrow KIO_4 + 2HF$$
$$H_2S + 4F_2 \longrightarrow SF_6 + 2HF$$

8. *Reaction with ammonia*: When fluorine is excess ammonia gives NF_3 but if ammonia is excess it is oxidized to nitrogen.

$$NH_3 + 3F_2 \longrightarrow NF_3 + 3HF$$
$$2NH_3 + 3F_2 \longrightarrow N_2 + 6HF$$

9. *Reaction with hydrocarbons*: It reacts with hydrocarbons explosively and forms fluorinated hydrocarbons

$$CH_4 \xrightarrow[-HF]{F_2} CH_3F \xrightarrow[-HF]{F_2} CH_2F_2$$
$$\xrightarrow[-HF]{F_2} CHF_3 \xrightarrow[-HF]{F_2} CF_4$$

10. ***Reaction with methyl alcohol***: Fluorine reacts with methyl alcohol forming CF_3OF a very strong oxidizing agent

$$CH_3OH + 4F_2 \xrightarrow{AgF_2} CF_3OF + 4HF$$

Uses: Fluorine has several uses mostly in the form of its compounds.

(i) It is used as a rocket fuel.

(ii) Its compounds like NaF and AlF_3 are used as insecticides.

(iii) SF_6 is used in high voltage electricity.

(iv) Just like DDT (a chloro compound), DDFT (a fluoro compound) is used as insecticide and fungicide.

(v) Its carbon compounds freons (chlorofluoro carbons) are used as refrigirents and a polymer of tetrafluoroethylene called teflon is used in making non sticking frying pans and laboratory ware.

(vi) The uranium isotopes ^{235}U and ^{238}U are separated by the diffusion of their gaseous fluorides $^{235}UF_6$ and $^{238}UF_6$.

(vii) HF is used in etching of the glass. Silica (SiO_2) present in glass reacts with HF and gives hydrofluosilicic acid (H_2SiF_6) that is soluble.

$$SiO_2 + 4HF \longrightarrow SiF_4 + 2H_2O$$
$$SiF_4 + 2HF \longrightarrow H_2SiF_6$$

11.7 CHLORINE

Chlorine does not occur in uncombined state in nature. It is however, widely found in the combined state in the form of chlorides of various metals. The most important chloride is sodium chloride which is found in sea water (2.5% w/v) lakes and rocks. Some other metal chorides which occur in the nature are

Carnalite	$KCl \cdot MgCl_2 \cdot 6H_2O$
Horn silver	AgCl
Sylvine	KCl

Preparation

Chlorine was first prepared by Scheele by oxidizing HCl with MnO_2. This method can also be used as laboratory method of preparation.

Laboratory Methods: It is possible to prepare chlorine by oxidation of hydrochloric acid by any of the following oxidizing agents.

(i) Manganese dioxide

$$MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$$

(ii) Potassium permanganate or **dichromate**

$$2KMnO_4 + 6HCl \longrightarrow 2KCl + 2MnCl_2 + 3H_2O + 5(O)$$
$$10\,HCl + 5(O) \longrightarrow 5H_2O + 5Cl_2$$
$$\overline{2KMnO_4 + 16\,HCl \longrightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2}$$

$$K_2Cr_2O_7 + 8HCl \longrightarrow 2KCl + 2CrCl_3 + 4H_2O + 3(O)$$
$$6HCl + 3(O) \longrightarrow 3H_2O + 3Cl_2$$
$$\overline{K_2Cr_2O_7 + 14\,HCl \longrightarrow 2\,KCl + 2CrCl_3 + 7H_2O + 3Cl_2}$$

Because of these reactions potassium permanganate or potassium dichromate cannot be used as oxidizing agents in quantitative analysis in hydrochloric acid medium.

(iii) Bleaching powder or a hypochlorite like sodium hypochlorite (NaOCl)

$$CaOCl_2 + 2HCl \longrightarrow CaCl_2 + H_2O + Cl_2$$
$$NaOCl + 2HCl \longrightarrow NaCl + H_2O + Cl_2$$

(iv) Lead dioxide or red Lead

$$PbO_2 + 4HCl \longrightarrow PbCl_2 + 2H_2O + Cl_2$$
$$Pb_3O_4 + 8HCl \longrightarrow 3PbCl_2 + 4H_2O + Cl_2$$

(v) In the place of hydrochloric acid a mixture of common salt and manganese dioxide is heated with concentrated sulphuric acid is commonly used

$$2\,NaCl + MnO_2 + 3H_2SO_4 \longrightarrow$$
$$2NaHSO_4 + MnSO_4 + 2H_2O + Cl_2$$

Manufacture

Chlorine is manufactured by any of the following methods.

1. ***Decon's process***: In this method chlorine is manufactured by the oxidation of hydrochloric acid by air in the presence of cupric chloride as catalyst and heated to 673K.

$$4HCl + O_2 \underset{673\,K}{\overset{Cu\,Cl_2}{\rightleftharpoons}} 2H_2O + 2Cl_2$$

The catalytic action cupric chloride could be explained as follows.

$$2CuCl_2 \xrightarrow{\text{high temp.}} Cu_2Cl_2 + Cl_2$$
$$2Cu_2Cl_2 + O_2 \longrightarrow 2Cu_2OCl_2$$
$$Cu_2OCl_2 + 2HCl \longrightarrow 2CuCl_2 + H_2O$$

2. ***Weldon's process***: In this process concentrated hydrochloric is heated with pyrolusite mineral (MnO_2)

$$MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$$

The manganese chloride present in the waste liquor is converted into a product that can be used to oxidize the hydrochloric acid instead of using fresh MnO_2. The waste liquor containing $MnCl_2$ is treated with excess of lime and air is blown in the heated mixture. The following reactions takes place

$$MnCl_2 + Ca(OH)_2 \longrightarrow Mn(OH)_2 + CaCl_2$$
$$2Mn(OH)_2 + O_2 \text{ (air)} \longrightarrow 2MnO_2 + 2H_2O$$
$$MnO_2 + Ca(OH)_2 \longrightarrow CaMnO_3 + H_2O$$

The calcium manganite settles down as a dark coloured mud called *Weldon mud*. This is used for the oxidation of hydrochloric acid

$$CaMnO_3 + 6HCl \longrightarrow CaCl_2 + MnCl_2 + 3H_2O + Cl_2$$

3. *Nitrosyl chloride process*: When common salt is heated with concentrated nitric acid a mixture of chlorine and nitrosyl chloride will be evolved.

$$3NaCl + 4HNO_3 \longrightarrow 3NaNO_3 + NOCl + 2H_2O + Cl_2$$

The gaseous mixture is oxidized with oxygen

$$2NOCl + O_2 \longrightarrow 2NO_2 + Cl_2$$

The gases are liquiefied and chlorine is distilled out. Nitrogen dioxide is absorbed in water in the presence of air to form nitric acid which can be used again.

$$4NO_2 + 2H_2O + O_2 \longrightarrow 4HNO_3$$

4. *Electrolytic process*: At present chlorine is manufactured by the electrolytic process described below.

If an electric current is allowed to pass through brine (NaCl Solution) Na^+ ions move towards the cathode where hydrogen and OH^- are formed by reduction of water. Sodium ions react with OH^- ions to form sodium hydroxide. The chloride ions on the other hand get directed towards the anode where by they lose one electron each and get liberated as chlorine atoms which immediately form chlorine molecules and get liberated as chlorine gas

$$NaCl \longrightarrow Na^+ + Cl^- \qquad \text{ionization}$$
$$\left.\begin{aligned} 2H_2O + 2e^- &\longrightarrow H_2 + 2OH^- \\ Na^+ + OH^- &\longrightarrow NaOH \end{aligned}\right\} \text{ at cathode}$$
$$2Cl^- \longrightarrow Cl_2 + 2e^- \qquad \text{at anode}$$

If sodium hydroxide and chlorine so produced come in contact, they will react again.

$$2NaOH + Cl_2 \longrightarrow NaCl + NaOCl + H_2O$$

So the cell must be constructed such that sodium hydroxide and chlorine once produced do not come in contact with each other.

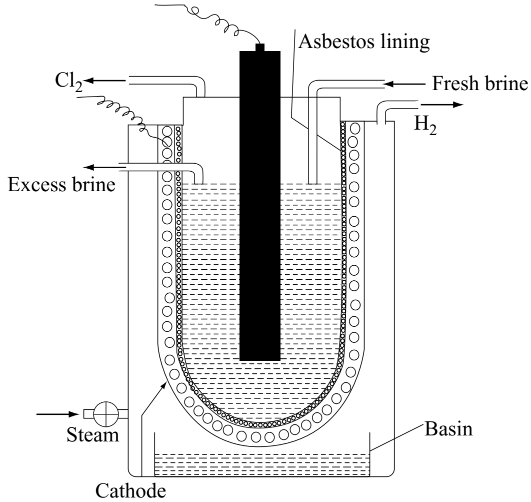


Fig 11.3 Electrolytic chlorine generating Nelsons cell

Based on the above principle there are number of electrolytic cells. The one commonly used has been *Nelson's* cell (Fig 11.3) which consists of a U – shaped perforated steel vessel which acts as cathode. It is lined inside with asbestos and suspended in a big iron tank. A carbon anode is dipped into brine.

When an electric current is passed chlorine is liberated at the anode and escape through the outlet above and compressed into steel cylinders. Sodium ions penetrate through the asbestos, reach the cathode at which it combines with OH^- ions formed from steam to give sodium hydroxide. Sodium hydroxide produced would be collected in the outer tank while hydrogen is drawn off through the exit at the top and collected as a valuble by – product.

5. *Pure Chlorine*: It is possible to prepare pure chlorine by heating dry auric chloride or platinic chloride in a hard glass tube

$$2AuCl_3 \xrightarrow{463k} 2Au + 3Cl_2$$
$$PtCl_4 \xrightarrow{873k} Pt + 2Cl_2$$

Physical Properties

(i) It is an yellowish green gas having pungent odour. It is heavier than air (about 2 1/2 times as heavy as air)
(ii) It is poisonous in nature. It causes headache when inhaled in small quantities. It may prove fatal when it is taken in large quantities.
(iii) It dissolves in water forming chlorine water and on cooling deposits greenish – yellow crystals $Cl_2 \cdot 8H_2O$.
(iv) It can be easily liquified by cooling under pressure to a yellow liquid (b.p. 238.4k) and freezes at 172.4k to give pale yellow solid.

Chemical Properties

1. *Reaction with water*: Chlorine dissolves in water. This solution is known as chlorine water. Freshly prepared solution of chlorine water contains HCl and HOCl.

$$Cl_2 + H_2O \longrightarrow HCl + HOCl$$

Chlorine water is strong oxidizing agent because hypochlorous acid is unstable and decomposes into HCl and nascent oxygen.

$$HOCl \longrightarrow HCl + (O)$$

It also act as a bleaching agent due to the liberation of nascent oxygen. Colourled substance + Nascent oxygen → colourless oxidized product.

The bleaching action of chlorine is permanent and irreversible.

2. *Reaction with alkalis*: Chlorine react with cold and dilute alkalis producing chlorides and hypochlorites.

$$2Ca(OH)_2 + 2Cl_2 \longrightarrow CaCl_2 + Ca(OCl)_2 + 2H_2O$$

With hot and concentrated alkalis it gives chlorides and chlorates.

$$6Ca(OH)_2 + 6Cl_2 \longrightarrow 5CaCl_2 + Ca(ClO_3)_2 + 6H_2O$$

Chlorine when passed over dry slaked lime gives bleaching powder.

$$2Ca(OH)_2 + 2Cl_2 \longrightarrow Ca(OCl)_2 + CaCl_2 + 2H_2O$$

3. *Reaction with metals*: Active metals combine with Cl_2 to give respective metal chlorides e.g.,

$$2Na + Cl_2 \longrightarrow 2NaCl$$

$$Mg + Cl_2 \longrightarrow MgCl_2$$

$$2Al + 3Cl_2 \longrightarrow 2AlCl_3$$

If a metal can form more than one chloride, the chloride in the highest stable oxidation state, is produced by direct combination of the elements

e.g., $Cu + Cl_2 \longrightarrow CuCl_2$

$$2Fe + 3Cl_2 \longrightarrow 2FeCl_3$$

4. *Reaction with non-metals*: Chlorine react with almost all non metals directly except with carbon, nitrogen and oxygen.

$$H_2 + Cl_2 \xrightarrow{h\nu} 2HCl$$

$$2B + 3Cl_2 \longrightarrow 2BCl_3$$

$$P_4 + 6Cl_2 \longrightarrow 4PCl_3$$

$$P_4 + 10Cl_2 \longrightarrow 2PCl_5$$

$$2S + Cl_2 \xrightarrow{\Delta} S_2Cl_2$$

$$Si + 2Cl_2 \longrightarrow SiCl_4$$

5. *Reaction with other halides*: Chlorine can displace bromine from bromides and iodine from iodides.

$$2KBr + Cl_2 \longrightarrow 2KCl + Br_2$$

$$2KI + Cl_2 \longrightarrow 2KCl + I_2$$

It should be noted that bromides and iodides must be ionic in nature.

6. *Reaction with ammonia*: If ammonia reacts with excess of chlorine, nitrogen trichloride is formed but if ammonia is excess ammonia will be oxidized to nitrogen.

$$NH_3 + \underset{\text{excess}}{3Cl_2} \longrightarrow NCl_3 + 3HCl$$

$$\underset{\text{excess}}{8NH_3} + 3Cl_2 \longrightarrow N_2 + 6NH_4Cl$$

7. *Addition reactions*: Chlorine forms addition compounds with SO_2, CO, and NO in the presence of sun light.

$$SO_2 + Cl_2 \xrightarrow{h\nu} SO_2Cl_2 \quad \text{(Sulphuryl Chloride)}$$

$$CO + Cl_2 \longrightarrow COCl_2 \quad \text{(Carbonyl Chloride or phosgene)}$$

$$2NO + Cl_2 \longrightarrow 2NOCl \text{ (Nitrosyl Chloride)}$$

8. *Oxidation properties*: Chlorine oxidizes several substances

(i) Ferrous sulphate to ferric sulphate

$$2FeSO_4 + H_2SO_4 + Cl_2 \longrightarrow Fe_2(SO_4)_3 + 2HCl$$

(ii) Sulphite to sulphate

$$Na_2SO_3 + H_2O + Cl_2 \longrightarrow Na_2SO_4 + 2HCl$$

(iii) Hydrogen sulphide to sulphur

$$H_2S + Cl_2 \longrightarrow 2HCl + S.$$

9. *Reaction with organic compounds*:

(i) With saturated hydrocarbons gives substitution products e.g.,

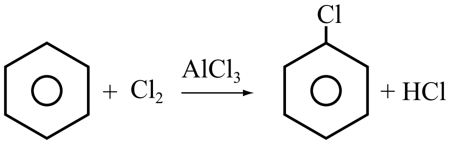
$$CH_4 + Cl_2 \xrightarrow[-HCl]{h\nu} CH_3Cl \xrightarrow[-HCl]{Cl_2/h\nu} CH_2Cl_2 \xrightarrow[-HCl]{Cl_2/h\nu} CHCl_3 \xrightarrow[-HCl]{Cl_2/h\nu} CCl_4$$

(ii) Unsaturated hydrocarbons like alkenes and alkynes readily undergo addition reactions e.g.,

$$CH_2 = CH_2 + Cl_2 \xrightarrow{\text{Room Temp.}} \overset{\displaystyle Cl}{\overset{|}{CH_2}} - \overset{\displaystyle Cl}{\overset{|}{CH_2}}$$

$$HC \equiv CH \xrightarrow[\text{Room Temp.}]{Cl_2} \overset{\displaystyle Cl}{\overset{|}{HC}} = \overset{\displaystyle Cl}{\overset{|}{CH}} \xrightarrow[\text{Room Temp.}]{Cl_2} H - \underset{\displaystyle Cl}{\underset{|}{\overset{\displaystyle Cl}{\overset{|}{C}}}} - \underset{\displaystyle Cl}{\underset{|}{\overset{\displaystyle Cl}{\overset{|}{C}}}} - H$$

(iii) In the presence of Lewis acids like $AlCl_3$ Chlorine react with benzene to give chloro-benzene



10. ***Affinity for Hydrogen*****:** If lighted hydrocarbons like wax candle or turpentine is introduced into a jar containing chlorine they continue to burn. The chlorine removes the hydrogen from the hydrocarbon (an oxidation) and leaves carbon behind.

$$\underset{\text{Wax}}{C_{10}H_{22(S)}} + 11Cl_{2(g)} \longrightarrow 10C_{(S)} + 22\,HCl_{(g).}$$

$$\underset{\text{Turpentine}}{C_{10}H_{16}} + 8Cl_2 \longrightarrow 10\,C + 16\,HCl$$

Uses

Chlorine is used

(i) In the sterilization of water because it kills bacteria
(ii) in the extraction of gold and platinum.
(iii) in chemical industries for the manufacture of solvents ($CHCl_3$, CCl_4 ethylene dichloride etc), refrigerants (Freons), insecticides (DDT, BHC), plastics (PVC, rubber)
(iv) as a bleaching agent for wood pulp, rayon, cotton or linen
(v) in the manufacture of poisonous gases like mustard gas ($ClCH_2\ CH_2S\ CH_2\ CH_2Cl$); phosgene ($COCl_2$); tear gas ($CCl_3NO_2$)
(vi) in the manufacture of bleaching powder, chlorates, hypochlorites, hydrochloric acid etc.

11.8 BROMINE

It is never found in free state in nature. It occurs generally in the form of bromides of sodium, potassium, magnesium and calcium. Sea water contains these salts of bromine to the extent of 0.068 per cent. The important compounds of bromine are, bischofite ($MgCl_2 \cdot 6H_2O$) tachydrite ($CaCl_2 \cdot 2MgCl_2 \cdot 12H_2O$) and carnalite $MgCl_2 \cdot KCl \cdot 6H_2O$. All of these contain 0.2 per cent bromine as bromides. The mother liquor left after the crystallization of potassium chloride from carnalite contain some bromo carnalite $KBr\ MgBr_2\ 6H_2O$. Bromargyrite is a mineral AgBr.

Preparation

Laboratory Methods

(i) When chlorine gas is passed through the solution of potassium or magnesium bromides, bromine is liberated.

$$2KBr + Cl_2 \longrightarrow 2KCl + Br_2$$

$$MgBr_2 + Cl_2 \longrightarrow MgCl_2 + Br_2$$

(ii) By heating a mixture of potassium bromide, manganese dioxide and concentrated sulphuric acid also gives bromine.

$$2KBr + MnO_2 + 3H_2SO_4 \longrightarrow 2KHSO_4 + MnSO_4 + 2H_2O + Br_2$$

(iii) Bromine is liberated when hydrochloric acid is added to a mixture of potassium bromide and potassium bromate

$$5KBr + KBrO_3 + 6HCl \longrightarrow 6KCl + 3H_2O + 3Br_2$$

Manufacture

Most of the worlds supply of bromine is obtained from sea water (bromide content about 1 part in 15000). The sea water is first acidified (to prevent appreciable hydrolysis of chlorine and bromine) and then treated with chlorine. The bromide ions are oxidized to bromine which is then expelled as a vapour by passing air through the liquid mixture. The air containing a very small concentration of bromine is reduced back again to bromide ions and the air recycled to pick up more bromine which is reduced to bromide and added to the aqueous solution. By this means a solution containing a high concentration of bromide ion is obtained. Treatment of this solution with chlorine results in the displacement of bromine in high concentration, which can be distilled off on heating the solution. Dilute sulphuric acid is produced in the process and is used to treat the sea water prior to addition of chlorine.

$$2Br^- + Cl_2 \longrightarrow 2Cl^- + Br_2 \text{ (low concentration)}$$

$$Br_2 + SO_2 + 2H_2O \longrightarrow 2H_2SO_4 + 2HBr$$

(high concentration)

$$2HBr + Cl_2 \longrightarrow 2HCl + Br_2$$

The evolved bromine is blown out by means of a current of air. The bromine coming out along with air is passed into sodium carbonate solution. The bromine reacts with sodium carbonate forming sodium bromide and sodium bromate.

$$3Na_2CO_3 + 3Br_2 \longrightarrow 5NaBr + NaBrO_3 + 3CO_2$$

The solution containing bromide and bromate is distilled with sulphuric acid to recover bromine.

$$[2NaBr + H_2SO_4 \longrightarrow Na_2SO_4 + 2HBr] \times 5$$
$$2NaBrO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + 2HBrO_3$$
$$[HBrO_3 + 5HBr \longrightarrow 3Br_2 + 3H_2O] \times 2$$
$$10NaBr + 2NaBrO_3 + 6H_2SO_4 \longrightarrow 6Na_2SO_4 + 6Br_2 + 6H_2O$$
$$5NaBr + NaBrO_3 + 3H_2SO_4 \longrightarrow 3Na_2SO_4 + 3Br_2 + 3H_2O$$

Physical Properties

(i) It is a reddish–brown, heavy, mobile liquid (sp. gr 3.19).
(ii) Its b.p. is 301.5k. It freezes to give a yellowish–brown crystalline solid at 265.7K.
(iii) It is soluble in water but more soluble in organic solvents such as alcohol, ether, chloroform and carbon disulphide.
(iv) Bromine water when cooled to 273k forms bromine hydrate $Br_2 \cdot 10H_2O$.
(v) It is having an irritating smell and is poisonous in nature. It is corrosive and attacks the skin producing sores.
(vi) Bromine turns the starch yellow.

Chemical Properties

The chemical properties of bromine are almost similar to those of chlorine.

Uses
It is used
(i) as germicide.
(ii) in the manufacture of bromides which find use in medicines and photography (AgBr).
(iii) as an oxidizing agent.
(iv) in the preparation of ethylene dibromide an additive to petrol.
(v) in making dyes, tear gas, photographic films, plates and papers.
(vi) as a laboratory reagent.

11.9 IODINE

Iodine does not occur free in nature. In the combined state it is found as iodides and iodates. Traces of iodine compounds are found in plants, animals and in certain minerals and sea water. Its main sources are.
(i) Sea-weeds (Laminaria species). These species contain iodides. Their ashes contain about 0.5 per cent iodine in the form of iodides.
(ii) *Caliche* the crude chile salt petre contains about 0.2 per cent of sodium iodate.

Laboratory Preparation

(i) Laboratory method for preparation of iodine is very similar to that of chlorine and bromine. When an iodide is heated with manganese dioxide and concentrated sulphuric acid, the vapours of iodine coming out are condensed as solid upon the walls of a cold dish or beaker.

$$2NaI + MnO_2 + 3H_2SO_4 \longrightarrow 2NaHSO_4 + MnSO_4 + 2H_2O + I_2$$

(ii) It can also be prepared by adding chlorine water or bromine water to iodides.

$$2NaI + Cl_2 \longrightarrow 2NaCl + I_2$$
$$2NaI + Br_2 \longrightarrow 2NaBr + I_2$$

(iii) By adding any oxidizing agent like hydrogen peroxide, potassium permanganate, potassium dichromate copper sulphate etc. to potassium iodide solution.

$$2KI + H_2SO_4 + H_2O_2 \longrightarrow K_2SO_4 + 2H_2O + I_2$$
$$2KMnO_4 + 8H_2SO_4 + 10KI \longrightarrow 6K_2SO_4 + 2MnSO_4 + 8H_2O + 5I_2$$
$$K_2Cr_2O_7 + 7H_2SO_4 + 6KI \longrightarrow 4K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3I_2$$
$$CuSO_4 + 2KI \longrightarrow CuI_2 + K_2SO_4$$
$$2CuI_2 \longrightarrow Cu_2I_2 + I_2$$

Manufacture

1. ***From caliche*****:** The mother liquor left after the crystallization of nitre from crude chile salt petre is having some iodine as sodium iodate. It is made to treat with calculated quantity of sodium bisulphite when iodine is precipitated because of the reduction of sodium iodate.

$$[NaIO_3 + 3NaHSO_3 \longrightarrow 3NaHSO_4 + NaI] \times 5$$
$$NaIO_3 + NaHSO_4 \longrightarrow Na_2SO_4 + HIO_3$$
$$[NaI + NaHSO_4 \longrightarrow Na_2SO_4 + HI] \times 5$$
$$HIO_3 + 5HI \longrightarrow 3H_2O + 3I_2$$
$$6NaIO_3 + 15NaHSO_3 \longrightarrow 9NaHSO_4 + 6Na_2SO_4 + 3H_2O + 3I_2$$

or $$2NaIO_3 + 5NaHSO_3 \longrightarrow 3NaHSO_4 + 2Na_2SO_4 + H_2O + I_2$$

If $NaHSO_3$ present in excess, HI is produced due to reduction of iodine by sodium bisulphite.

$$NaHSO_3 + I_2 + H_2O \longrightarrow NaHSO_4 + 2HI$$

The precipitated iodine is filter–pressed and dried. The commercial iodine is also having some ICl, IBr and ICN as impurities. These can be removed by distilling iodine over potassium iodide and resublimation.

$$KI + ICl \longrightarrow KCl + I_2$$

$$KI + IBr \longrightarrow KBr + I_2$$

$$KI + ICN \longrightarrow KCN + I_2$$

2. ***From sea weeds***

(i) Sea weeds, particularly of the laminaria variety, are collected, dried and burnt in shallow pits so as to avoid any loss of iodine. The ash known as *kelp* contains about 0.4–1.3% iodine as iodates in addition to chlorides and sulphates.

(ii) The ash is treated with hot water to dissolve out salts. On concentration, the sulphates and chlorides separate out from the solution.

(iii) The mother liquor having iodides is mixed with manganese dioxide and concentrated sulphuric acid in a cast iron retort with lead covers and connected to stone ware receivers known as aludels. When heated, iodine is liberated which sublimes and condenses in the aludels.

Physical Properties

(i) It is a dark violet solid with shining nature i.e., metallic lustre.

(ii) It sublimes on heating yielding violet vapours with irritating smell.

(iii) It melts at 387K and boils at 457K. Sp.gr 4.98.

(iv) It is slightly soluble in water, but much more soluble in water in the presence of potassium iodide due to the formation tri-iodide.

$$KI + I_2 \longrightarrow KI_3$$

The solution behaves as a simple mixture of potassium iodide and free iodine.

(v) Iodine dissolves more in organic solvents like $CHCl_3$, CCl_4 and CS_2

Chemical Properties

Chemically iodine behave similar to chlorine and bromine but its chemical reactivity and oxidation power is less than chlorine and bromine.

1. ***Thermal dissociation*****:** On heating it starts dissociating at 973 k and is complete at 1973K.

$$I_2 \rightleftharpoons 2I$$

2. ***Reaction with alkalis*****:** With cold dilute alkali it forms iodide and hypoiodite.

$$2NaOH + I_2 \longrightarrow NaI + NaOI + H_2O$$

With hot concentrated alkalis it forms iodide and iodates.

$$6NaOH + 3I_2 \longrightarrow 5NaI + NaIO_3 + 3H_2O$$

3. ***Reaction with metals*****:** It combines with metals forming iodides.

$$2Na + I_2 \longrightarrow 2NaI$$

$$Mg + I_2 \longrightarrow MgI_2$$

$$2Al + 3I_2 \longrightarrow 2AlI_3$$

The metals which can form more than one iodide form iodides in their stable lower oxidation state. The iodides formed in the higher oxidation state are unstable because iodide ion have reduction property.

$$2CuI_2 \longrightarrow Cu_2I_2 + I_2$$

$$PbI_4 \longrightarrow PbI_2 + I_2$$

4. ***Reaction with non-metals*****:** It combines with several non-metals directly.

$$H_2 + I_2 \xrightleftharpoons{440°C} 2HI$$

$$2P + 3I_2 \longrightarrow 2PI_3$$

5. ***Oxidation properties*****:** Like bromine, iodine acts as an oxidizing but it is a less powerful oxidizing agent than chlorine and bromine.

(i) $H_2S + I_2 \longrightarrow 2HI + S$

(ii) $SO_2 + 2H_2O + I_2 \longrightarrow H_2SO_4 + 2HI$

(iii) $NaNO_2 + H_2O + I_2 \longrightarrow NaNO_3 + 2HI$

(iv) $Na_2SO_3 + H_2O + I_2 \longrightarrow Na_2SO_4 + 2HI$

(v) $Na_3AsO_3 + H_2O + I_2 \longrightarrow Na_3AsO_4 + 2HI$

(vi) $2FeSO_4 + H_2SO_4 + I_2 \longrightarrow Fe_2(SO_4)_3 + 2HI$

6. ***Reaction with ammonia*****:** Iodine combines with liquor ammonia to form nitrogen tri-iodide, a black explosive powder.

$$2NH_3 + 3I_2 \longrightarrow NI_3\,NH_3 + 3HI$$

Suppose if we rub 1g of iodine with liquor ammonia in a glass mortar a dark brown precipitate of nitrogen triiodide is obtained. When it is sprinkled on the floor of the class room it soon gets dried up. As the class goes out, mild explosions would be heard because of decomposition of nitrogen tri-iodide causing confusion among the students.

$$8NI_3 \cdot NH_3 \longrightarrow 5N_2 + 9I_2 + 6NH_4I$$

7. ***Reaction with chlorates and bromates*****:** Iodine cannot displace chlorine and bromine from chlorides and bromides, respectively but it displaces them from their oxo salts.

$$2KClO_3 + I_2 \longrightarrow 2KIO_3 + Cl_2$$

$$2KBrO_3 + I_2 \longrightarrow 2KIO_3 + Br_2$$

This is because in chlorides and bromides, the chlorine and bromine are in negative oxidation states (–I). A more electronegative element can substitute a less electronegative atom since iodine is less electronegative than chlorine and bromine, it cannot substitute them from their halides. But in chlorates and bromates, chlorine and bromine are in positive oxidation states (+V). Iodine is more electropositive than chlorine and bromine. So it can substitute chlorine and bromine, from their oxosalts.

8. *Reaction with sodium thiosulphate*: Iodine oxidizes hypo quantitatively to sodium tetrathionate.

$$2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$$

This reaction is used in iodometric estimations using starch as indicator.

9. *Reaction with strong oxidizing agents*: Iodine can be oxidized to iodic acid with concentrated nitric acid.

$$I_2 + 10HNO_3 \longrightarrow 2HIO_3 + 10NO_2 + 4H_2O$$

Ozone oxidizes moist iodine to iodic acid but oxidizes the dry iodine to I_4O_9 an yellow powder.

$$I_2 + H_2O + 5O_3 \longrightarrow 2HIO_3 + 5O_2$$

$$2I_2 + 9O_3 \longrightarrow I_4O_9 + 9O_2$$

Chlorine water oxidizes iodine to iodic acid.

$$I_2 + 5Cl_2 + 6H_2O \longrightarrow 2HIO_3 + 10HCl$$

10. *Reaction with mercuric oxide*: Iodine reacts with freshly precipitated mercuric oxide forming hypoiodous acid.

$$2HgO + 2I_2 + H_2O \longrightarrow HgO \cdot HgI_2 + 2HOI$$

11. *Reaction with alcohols and carbonyl compounds*: Alcohols and carbonyl compounds containing α-methyl group react with iodine in alkaline medium forming iodoform an yellow precipitate.

$$CH_3\,CH_2OH + 6KOH + 3I_2 \longrightarrow CHI_3 + HCOOK + 5KI + 5H_2O$$

This reaction is used for distinguishing alcohols and carbonyl compounds containing α-methyl group from the others.

12. *Reaction with starch*: It gives dark blue colour with starch.

Uses

Important uses of iodine are

(i) It is used as a reagent in the laboratory and in volumetric analysis.

(ii) It is used in the manufacture of iodides (used in medicine and photography) iodoform, iodole which are used as an antiseptic and dyestuffs.

(iii) It is used in the form of tincture of iodine and iodex as disinfectant, antiseptic and analgesic.

Tincture of iodine is a mixture of potassium iodide iodine dissolved in alcohol (rectified spirit or methylated spirit).

(iv) Deficiency of iodine causes a disease goitre. For good health commonsalt is mixed with sodium iodide.

(v) To cure thyroid radioactive iodine is used.

11.10 COMPOUNDS OF HALOGENS

11.10.1 Hydrides

Preparation: *Direct Synthesis* All the halogens react with hydrogen to form hydrides.

$$H_2 + X_2 \longrightarrow 2HX$$

Affinity of halogens for hydrogen decreases from fluorine to iodine. Thus, fluorine combines explosively with hydrogen even in cold and dark, chlorine also combines explosively but in the presence of light, bromine combines non-explosively but on heating only and iodine combines slowly on heating in the presence of platinum catalyst. The reaction with iodine is reversible and incomplete.

(ii) Reaction of an ionic halide with concentrated sulphuric acid: HF and HCl can be prepared by the action of concentrated sulphuric acid on ionic fluoride or chloride (Displacement of more volatile acid by a less volatile acid).

$$CaF_2 + H_2SO_4 \longrightarrow CaSO_4 + 2HF$$

$$NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl \quad \text{(in the cold)}$$

$$NaCl + NaHSO_4 \longrightarrow Na_2SO_4 + 2HCl \text{ (on the heating)}$$

Similar methods cannot be usefully employed to obtain hydrogen bromide and hydrogen iodide, since these hydrides are reducing agents and are readily oxidized by concentrated sulphuric acid to free halogens.

$$KX + H_2SO_4 \longrightarrow KHSO_4 + HX$$

$$2HX + H_2SO_4 \longrightarrow 2H_2O + SO_2 + X_2$$

Where X = Br or I

HBr and HI can be prepared by heating the phosphoric acid with bromides and iodides.

$$3KBr + H_3PO_4 \longrightarrow K_3PO_4 + 3HBr$$

$$3KI + H_3PO_4 \longrightarrow K_3PO_4 + 3HI.$$

(iii) The action of water on phosphorus trihalides Phosphorus trihalides readily hydrolyses in water to give orthophosphorus acid and hydrogen halides.

$$PX_3 + 3H_2O \longrightarrow H_3PO_3 + 3HX \quad (X = Cl,\ Br \text{ or } I)$$

HBr is obtained by dropping bromine into a mixture of red phosphorous and a little water. PBr_3 is formed initially and is then hydrolysed.

$$2P + 3Br_2 \longrightarrow 2PBr_3$$

$$PBr_3 + 3H_2O \longrightarrow H_3PO_3 + 3HBr$$

Since iodine is a solid a slight variation on this method is used for obtaining HI. Water is dropped into a mixture of red phosphorous and iodine.

$$2P + 3I_2 \longrightarrow 2PI_3$$

$$PI_3 + 3H_2O \longrightarrow H_3PO_3 + 3HI$$

(iv) HCl, HBr and HI can be prepared by passing H_2S or SO_2 gas through the aqueous solutions of Cl_2, Br_2 or I_2.

$$X_2 + H_2S \longrightarrow 2HX + S$$

$$X_2 + SO_2 + H_2O \longrightarrow 2HX + H_2SO_4$$

Industrial production

Industrial production of HF is same as that of laboratory method in which HF is obtained by the action of concentrated sulphuric acid on fluorspar (CaF_2).

HCl is obtained industrially by burning hydrogen in chlorine, It is also available in considerable quantities as a by-product formed during many organic chlorination reactions e.g., chlorination of hydrocarbons. A small amount is still obtained by the reaction of sodium chloride with concentrated sulphuric acid, since the sodium sulphate produced in this process is used in the manufacture of glass.

A solution of hydrogen chloride in water is called hydrochloric acid.

Properties

1. *Physical state*: HCl, HBr and HI are gases but HF is a liquid with b.p 292K. The high melting and boiling points of HF are due to hydrogen bonding.

Table 11.4 Some physical properties of hydrogen halides

	HF	HCl	HBr	HI
M.P (k)	189.5	158.8	184.4	222
B.P (k)	292.5	187.9	205.9	237.9
Liquid range (1 atm) K.	376	302.1	294.5	288.9
Density gcm^{-3}	1.002	1.187	2.603	2.85
	(273k)	(115k)	(189k)	(226k)

2. *Polar covalent nature*: All the halides are essential covalent compounds with some ionic character. The ionic character increases with increasing electronegativity of the halogen. Thus the order of ionic character decreases in the following order.

$$HF > HCl > HBr > HI.$$

The presence of predominant covalent character in these hydrides is indicated by

(i) Their low m.p. and b.pts and

(ii) These are bad conductors of electricity.

They dissolve in non-ionizing solvents such as toluene and trichloromethane, preserving their covalent character e.g., a solution of hydrogen chloride in toluene is a non conductor of electricity and will not affect blue litmus paper.

3. *Thermal stability*: The H—X bond strength gets decreased from HF to HI as is evidenced from the values of their heat of formation and bond dissociation energies.

Thus HF is most stable where as HI is least stable This is evident from their decomposition reaction: HF and HCl are stable up to 1200ºC. HBr dissociates slightly and HI dissociates considerably (20%) at 440ºC. The decreasing stability of the hydrogen halides can also be seen from the values of dissociation energy of the H—X bond. The strength of the bond is related to H—X bond length. When the size of the halogen atom increases the bond length increases and the strength decreases.

4. *Reducing properties*: The strength of hydrogen halides as reducing agents increases from HF to HI. This is because of the decrease in the stability of H-X from H-F to HI. Thus HF is a poor reducing agent, HCl is a mild reducing agent can be oxidized by strong oxidizing agents such as MnO_2, $KMnO_4$ etc but cannot be oxidized by conc H_2SO_4. HBr is a stronger reducing while HI is a very strong reducing agent. HBr and HI can be oxidized by conc H_2SO_4. Even the dilute solution of HI on exposure to air slowly acquires a brown colour due to the liberation of iodine by the action of atmospheric oxygen.

$$4HI + O_2 \longrightarrow 2H_2O + 2I_2$$

In addition to relative stability explanation of HX, it is also possible to explain their reducing action on the basis of the increasing size of the halide ion from F^- to I^-.

In the smallest F^- ion, the electron which is to be removed during oxidation has been closest to the nucleus and therefore most difficult to be removed. Hence HF is a poor reducing agent. On the other hand in the largest I^- ion the electrons are **less tightly held** and be readily lost. Thus HI is the strongest reducing.

5. *Relative strength as acids*: When perfectly dry. They do not have any action on litmus, but in aqueous solution they function as acids and hence are able to turn blue litmus red. The relative strength of acids gets increased in the order

Table 11.5 Some physical properties of hydrogen halides

	HF	HCl	HBr	HI
Dielectric constant ε	83.6 (273k)	9.28 (178k)	7.0 (188k)	3.39 (223k)
Dipolemoment μ/D	1.86	1.11	0.788	0.382.
Electrical conductivity Tk/Ohm^{-1} cm^{-1}	~10^6 (273)	~10^9 (188)	~10^9 (188)	~10^{10} (223)

Table 11.6 Heat of formation and bond parameters of hydrogen halides

	HF	HCl	HBr	HI
ΔH^o_f (298k) KJ mol^{-1}	–271.12	–92.31	–36.40	26.48
ΔH_{diss} (H–X) kj mol^{-1}	573.98	431.62	362.50	294.58
H–X Bond length (pm)	91.7	127.4	141.4	160.9

given below which is in accordance with their dissociation constants Ka.

HF < HCl < HBr < HI (increasing acidic strength)

	HF	HCl	HBr	HI
Ka	7×10^{-4}	7×10^{8}	7×10^{10}	7×10^{11}

In aqueous solution they get ionized and act as acids. Thus HCl, HBr and HI are strong acids. It is interesting to note that HF is the weakest acid in water although it is most polar and more ionic in nature than the other hydrides.

This anomaly can be explained on the basis that the aspect of ionization (i.e., acidic character) of H^+ is not a one step process.

$$HX_{(aq)} \longrightarrow H^+_{(aq)} + X^-_{(aq)};\ \Delta H$$

But involves many steps as given below.

(i) $HX_{(aq)} \longrightarrow HX_{(g)}$; ΔH_1 (Heat of dehydration)

(ii) $HX_{(g)} \longrightarrow H_{(g)} + X_{(g)}$; ΔH_2 (Heat of dissociation)

(iii) $H_{(g)} \longrightarrow H^+_{(g)} + e^-$; ΔH_3 (Ionization potential)

(iv) $X_{(g)} + e^- \longrightarrow X^-_{(g)}$; ΔH_4 (Electron affinity)

(v) $H^+_{(g)} + H_2O \longrightarrow H^+_{(aq)}$; ΔH_5 (Heat of hydration)

(vi) $X^-_{(g)} + H_2O \longrightarrow X^-_{(aq)}$; ΔH_6 (Heat of hydration)

Table 11.7 Physical properties of hydrogen halides

Property	HF	HCl	HBr	HI
Enthalpy dehydration	48	18	21	23
Enthalpy dissociation	574	428	363	295
Ionization energy $H \to H^+$	1311	1311	1311	1311
Electron affinity of X	–338	–355	–331	–302
Enthalpy hydration H^+	–1091	–1091	–1091	–1091
Enthalpy hydration X^-	–513	–370	–339	–394
Total ΔH	–18	–68	–75	–167
TΔS	51	56	59	62
ΔG = ΔH–TΔS	–69	–124	–134	–229

Note that the terms ΔH_2 and ΔH_5 are common to all HX.

The steps (i), (ii) and (iii) are endothermic where as the other steps are exothermic. Hence the heat of ionization ΔH i.e acid strength would be as follows.

$$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3 - \Delta H_4 - \Delta H_6$$

The ΔH value is minimum in the case of HF showing its minimum acidic character. The values involved in this process are given in Table 11.7.

The weakest acidic character of HF is because of the following three factors:

(a) The strong H–F bond which needs very high amount of energy to dissociate (ΔH_2).
(b) The HF molecules are associated and exists as liquid due to hydrogen bonding and further more hydrated in aqueous solution due to high polarity. Hence ΔH_1 is also very high.
(c) The electron affinity of fluorine is low.

The only factor that increases the ionization of HF has been very high heat of hydration energy of small F^- ions. (ΔH_6).

But this value, although very high, cannot compensate the above three values.

The weakest acidic character of hydrofluoric acid may also be ascribed to the most negative and hence

Table 11.8 Boiling points (k) of azeotropes and their composition

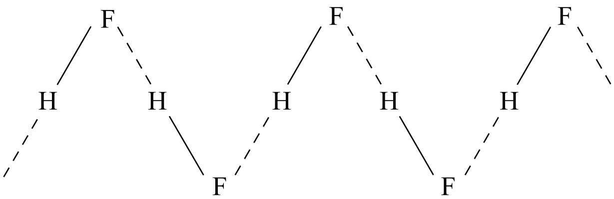
Acid	BP(k)	BP of azeotrope with water	Wt % of acid
HF	292.5	393	36.0
HCl	189	382	20.2
HBr	206	399	48.0
HI	237.5	400	57.0

unfavourable entropy change (ΔS) that accompanies its dissociation. The **negative entropy is probably because of the formation of a well ordered covering of water molecules around the highly hydrated fluoride ion**.

6. ***Azeotropic mixtures*****:** The aqueous solution of different halogen acids are forming azeotropic mixtures with maximum boiling points because of a negative deviation from Raoult's law.

7. ***Abnormal properties of hydrofluoric acid***

(i) It is a liquid due to association of molecules through hydrogen bonds but others are gases. In liquid state it has zig-zag structure.



(ii) Its viscosity and b.p are high.

(iii) HF is unique in forming acid salts having hydrogen difluoride ion $[HF_2]^-$. This is because in aqueous solution it exist as H_2F_2 and ionizes to give H^+ and $[HF_2]^-$ ions Hence it can form salts like KHF_2. No such acid salts are known for other halogen acids.

(iv) It does not yield precipitate with silver nitrate because AgF is soluble but other halogen acids give precipitates with silver nitrate.

(v) It gives white precipitate of BaF_2 with $BaCl_2$ but other halogen acids do not give precipitates with $BaCl_2$.

(vi) It is extremely stable and cannot be oxidized.

(vii) It reacts with glass and silica but other halogen acids cannot react with glass and silica.

Tests for Chloride, Bromide and Iodide Ions

(a) ***Action of concentrated sulphuric acid*****:** Ionic halides evolve a sharp–smelling gas which fumes in moist air (the halogen hydride) when treated with a little concentrated sulphuric acid. A brown colour of bromine is also apparent from a bromide and a purple colour of iodine from an iodide.

(b) ***Reaction with silver nitrate solution*****:** An aqueous solution of a chloride solution forms a white precipitate of silver chloride when mixed with silver nitrate solution.

$$Cl^- + Ag^+ \longrightarrow AgCl.$$

Under similar conditions a bromide forms a pale yellow precipitate of silver bromide and an iodide a yellow precipitate of silver iodide.

Silver chloride readily dissolves in dilute ammonia solution; silver bromide is only sparingly soluble and silver iodide is almost insoluble in ammonia solution.

(c) ***Displacement reactions*****:** An aqueous solution of a bromide is oxidized to free bromine when treated with chlorine water. If a small quantity of carbon tetrachloride is added to the mixture bromine is extracted into the organic layer and the brown colour of bromine is easily visible.

A similar reaction with an aqueous solution of an iodide produces a purple colour of free iodine in the organic layer.

11.10.2 Hydrochloric Acid

Stephen Hales (1727) noticed that a gas very soluble in water was made by heating sulphuric acid with sal ammoniac (ammonium chloride) and J. priestly about 1772, collected the gas over mercury, and gave it the name **marine acid air**. It was later renamed **muriatic acid** and when the elementary nature of chlorine had been established, the name *hydrochloric acid* was given to it.

Preparation

Laboratory method**:** In the laboratory hydrochloric acid is prepared by heating sodium chloride with concentrated sulphuric acid. As the gas is heavier than air, it is collected by an upward displacement of air.

$$NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl \text{ (in cold)}$$
$$2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl \text{ (in hot)}$$

Hydrogen chloride gas can be dried by passing it through conc H_2SO_4. Quick lime or phosphorus pentoxide cannot be used for this purpose as they react with the gas.

$$CaO + 2HCl \longrightarrow CaCl_2 + H_2O$$
$$P_4O_{10} + 3HCl \longrightarrow POCl_3 + 3HPO_3$$

Manufacture**:** Industrially hydrogen chloride is obtained by burning hydrogen in chlorine which are obtained as by products during the manufacture of caustic soda by electrolysis of sodium chloride.

Industrially it is also available in considerable quantities as a by-product formed during many organic chlorination reactions e.g., the chlorination of hydrocarbons. A small amount is still obtained by reacting sodium chloride with concentrated sulphuric acid during the manufacture of sodium carbonate by Leblanc process.

Physical Properties

(i) It is colourless, pungent smelling gas having acidic taste.

(ii) It fumes in moist air. It has more affinity towards water so it collects the water molecules in air and form as small droplets which appear as white fumes.

(iii) It is heavier than air.
(iv) With water it yields a constant boiling mixture (having 22.2 per cent of the acid and boiling at 383K). It means that dilute solutions of the acid cannot be concentrated by boiling beyond 22.2 per cent.
(v) It gets easily liquified to a colourless liquid (b.p. 190k) and frozen to a white crystalline solid (m.p. 160k).

Chemical Properties

(i) It is neither combustible nor supporter of combustion.
(ii) Although it is a typical acid yet perfectly dry gas does not affect litmus. In moist state or in solution, it is able to turn blue litmus to red.
It is able to react with metals, their oxides, hydroxides or carbonates to form chlorides e.g.,

$$Zn + 2HCl \longrightarrow ZnCl_2 + H_2$$
$$NaOH + HCl \longrightarrow NaCl + H_2O$$
$$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$$
$$NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$$

(iii) **Stability:** As it is quite stable, it gets oxidized only by very strong oxidizing agents such as MnO_2, $KMnO_4$, $K_2Cr_2O_7$, PbO_2, Pb_3O_4 etc.
(iv) **Action of silver nitrate and lead acetate:** When treated with silver nitrate, it yields a white precipitate which is insoluble in nitric acid but soluble in ammonium hydroxide.

$$HCl + AgNO_3 \longrightarrow AgCl + HNO_3$$
$$AgCl + 2NH_4OH \longrightarrow [Ag(NH_3)_2]Cl + 2H_2O$$

With lead acetate, it yields a white precipitate which is soluble in hot water.

$$Pb(CH_3COO)_2 + 2HCl \longrightarrow PbCl_2 + 2CH_3COOH$$

(v) **Action of halogens:** chlorine could be liberated from hydrochloric acid by fluorine but no other halogen can do it.

$$2HCl + F_2 \longrightarrow 2HF + Cl_2$$

(vi) **Action with ammonia:** With ammonia it gives dense white fumes of ammonium chloride.
(vii) **Aquaregia:** It is a mixture of 3 parts of concentrated HCl and 1 part of conc HNO_3. It is used to dissolve noble metals such as gold and platinum. Due to the complex forming ability of Cl^- ions the oxidation power of nitric acid will be enhanced.

$$Au + 4H^+ + NO_3^- + 4Cl^- \longrightarrow AuCl_4^- + NO + 2H_2O$$
$$3Pt + 16H^+ + 4NO_3^- + 18Cl^- \longrightarrow 3PtCl_6^{2-} + 4NO + 8H_2O$$

(viii) **Action on salts:** Hydrochloric acid decomposes salts of weaker acids such as carbonates, bicarbonates, sulphides, sulphites, thiosulphates and nitrites. It has been used as a test for these acid radicals.

$$Na_2SO_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2$$
$$Na_2S_2O_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2 + S$$
$$Na_2S + 2HCl \longrightarrow 2NaCl + H_2S$$
$$2NaNO_2 + 2HCl \longrightarrow 2NaCl + NO + NO_2 + H_2O$$

Uses
Hydrochloric acid is used
(i) In the manufacture of chlorides and chlorine.
(ii) In the, textile, dyeing and leather tanning industries.
(iii) In the form of aquaregia to dissolve metals like gold and platinum.
(iv) In cleaning the iron sheets during tin plating galvanization.
(v) For extraction of glue from animal tissues and bones.

11.10.3 Oxides

Halogens do not combine directly with oxygen. However, it is possible to prepare binary halogen–oxygen compounds by indirect methods. Twelve such oxides are positively known which are listed in Table 11.9.

As fluorine is more electronegative than oxygen, compounds of fluorine with oxygen are called oxygen fluorides which are already discussed in Chapter 10 (10.5.2) on the other hand, as oxygen is more electronegative than chlorine, bromine and iodine, it is correct to call their oxygen derivatives as oxides but not halides.

Important features of binary halogen – oxygen compounds
(i) Chlorine is known to form the largest number of oxides whereas iodine forms the least.
(ii) The halogen – oxygen bonds are mainly covalent because of the small difference in electronegativity between the halogens and oxygen.
(iii) Polarity of the halogen – oxygen bond gets increased down-wards i.e., from Cl—O to I—O
(iv) Iodine oxides are the most stable followed by that of chlorine whereas bromine oxides are the least stable

Table 11.9 Oxides of halogens with oxidation numbers

Fluorine	Chlorine	Bromine	Iodine
O_2F_2 –1	Cl_2O + 1	Br_2O + 1	
OF_2 –2	ClO_2 + 4	BrO_2 + 4	I_2O_5 + 5
	Cl_2O_6 + 6	BrO_3 + 6	
	Cl_2O_7 + 7		

Iodine-oxygen bond is stable because of greater polarity of the bond, the chlorine – oxygen bond is stable because of multiple bond formation involving the d-orbitals of the chlorine atom. Bromine being in between the two, lacks both these characteristics (middle row anomaly) and hence form the least stable oxides.

(v) The oxides in the higher oxidation states have been more stable than that in the lower state.

In addition to the oxides listed in the Table 11.9 iodine is also known to form I_2O_4 and I_4O_9 which are ionic and salt-like in character.

$$I_2O_4 = IO^+IO^-_3$$
$$I_4O_9 = I^{3+}(IO_3)^-_3$$

Here iodine exhibits a +3 oxidation state in cations while +5 oxidation state in the anions given above.

11.11.4 Monoxides

Cl_2O and Br_2O can be prepared by treating the freshly prepared mercuric oxide with Cl_2 or Br_2 at 273k.

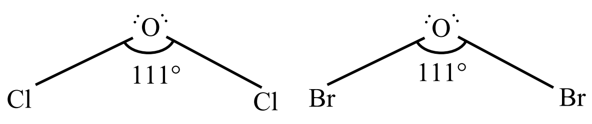
$$2HgO + 2Cl_2 \longrightarrow HgO\cdot HgCl_2 + Cl_2O$$
$$2HgO + 2Br_2 \longrightarrow HgO\cdot HgBr_2 + Br_2O$$

Cl_2O is an orange coloured gas, Br_2O is a dark brown liquid. When these are dissolved in water, they form hypohalous acids. So they are considered as the anhydrides of their hypohalous acids.

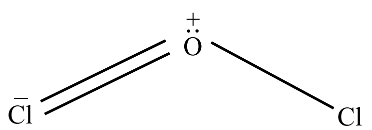
$$Cl_2O + H_2O \longrightarrow \underset{\text{Hypochlorous acid.}}{2HOCl}$$
$$Br_2O + H_2O \longrightarrow \underset{\text{Hypobromous acid.}}{2HOBr}$$

They act as oxidizing agents

Structure**:** Both are angular in shape. Oxygen atom is involved in sp^3 hybridization.



The bond angles are larger than tetrahedral angle due to the participation of lone pair of electrons in oxygen in π-dative bond with the vacant d-orbitals of chlorine, which causes the opening of the bond angle.



11.11.5 Dioxides

Chlorine dioxide can be prepared by reacting chlorine with sodium chlorite or silver chlorate

$$2NaClO_2 + Cl_2 \longrightarrow 2NaCl + 2ClO_2$$
$$2AgClO_3 + Cl_2 \longrightarrow 2AgCl + 2ClO_2 + O_2$$

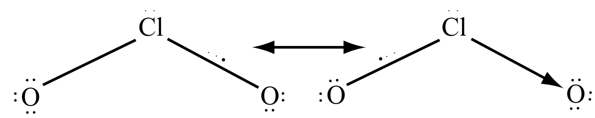
It can also be prepared by heating potassium chlorate with conc H_2SO_4 or with oxalic acid.

$$3KClO_3 + 3H_2SO_4 \longrightarrow 3KHSO_4 + HClO_4 + 2ClO_2 + H_2O$$
$$2KClO_3 + 2H_2C_2O_4 \xrightarrow{H_2O,\,90} 2ClO_2 + 2\,CO_2 + K_2C_2O_4 + 2H_2O$$

It is an yellow gas which explodes violently. It acts as a powerful oxidizing agent and chlorinating agent. Its bleaching power is 30 times more than that of Cl_2. So it is used as bleaching agent for paper pulp and textiles and in water treatment. It reacts with alkalis to yield chlorites and chlorates and hence can be considered as a mixed anhydride of chlorous ($HClO_2$) and chloric ($HClO_3$) acids.

$$2ClO_2 + 2NaOH \longrightarrow NaClO_2 + NaClO_3 + H_2O.$$

ClO_2 has bent structure with Cl—O bond length 149 pm and O—Cl—O angle 117º. It is considered to have a coordinate bond and a 3-electron bond coupled with a usual two electron covalent bond. Its structure is regarded as a resonance hybrid of the following structures.



The shorter Cl—O bond length (149 pm) than the usual (172 pm) is attributed to the presence of 3-electron bond. The presence of unpaired electron in the molecule is able to explain the paramagnetic nature, marked colour and reactivity of ClO_2. However, although ClO_2 is an odd electron molecule yet it is having no tendency to dimerize (difference from NO_2) as the odd electron is less localized on the central atom chlorine (because of the strong electronegativity of oxygen, it is near to oxygen).

Bromine dioxide (BrO2)**:** It may be prepared by passing an electric discharge through a mixture of bromine and oxygen gases at low temperatures. It is stable only below –40ºC. With alkalis it gives bromides and bromates.

$$6BrO_2 + 6NaOH \longrightarrow NaBr + 5NaBrO_3 + 3H_2O$$

11.11.6 HIGHER OXIDES

1. ***Dichloro hexoxide (chlorine hexoxide)*** Cl_2O_6 can be prepared by the action of ozone on chlorine dioxide.

$$2ClO_2 + 2O_3 \longrightarrow Cl_2O_6 + 2O_2$$

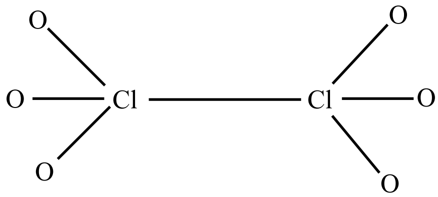
It is a reddish oily liquid, unstable explosive liquid. With alkalis it forms a mixture of chlorates and perchlorates. So it is considered as mixed anhydride of chloric ($HClO_3$) and perchoric ($HClO_4$) acids.

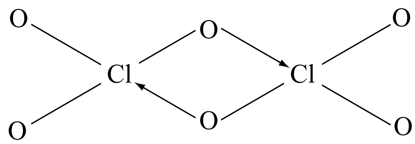
$$Cl_2O_6 + 2NaOH \longrightarrow NaClO_3 + NaClO_4 + H_2O$$

From the molecular weight determination, it follows that in the liquid state it exists as Cl_2O_6 (diamagnetic) while in the gaseous state it exists as ClO_3 (paramagnetic).

$$Cl_2O_6 \underset{}{\overset{\Delta}{\rightleftharpoons}} 2ClO_3$$

The structure of Cl_2O_6 is not known. Possible structures are as follows.





With HF and N_2O_4 it reacts as follows

$$Cl_2O_6 + HF \rightleftharpoons FClO_2 + HClO_4$$

$$2Cl_2O_6 + N_2O_4 \longrightarrow 2ClO_2 + 2[NO_2]^+[ClO_4]^-$$

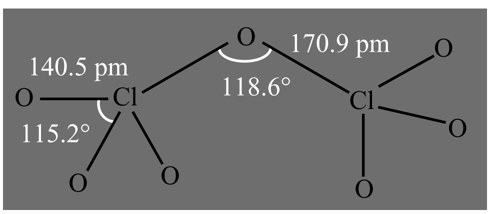
2. ***Dichloro heptoxide (chlorine heptoxide) Cl2O7***: It is prepared by dehydrating pure perchloric acid over phosphorous pentoxide.

$$2HClO_4 + P_2O_5 \longrightarrow Cl_2O_7 + 2HPO_3$$

Chlorine heptoxide, the only exothermic oxide of chlorine (the remaining oxides of chlorine are endothermic) has been the most stable chlorine oxide which is not so strong an oxidizing agent as the other oxides of chlorine. It reacts with water to form perchloric acid. It is a true anhydride.

$$Cl_2O_7 + H_2O \longrightarrow 2HClO_4$$

Cl_2O_7 molecule is having two ClO_3 units which are linked by an oxygen atom; each chlorine gets tetrahedrally linked to four oxygen atoms.



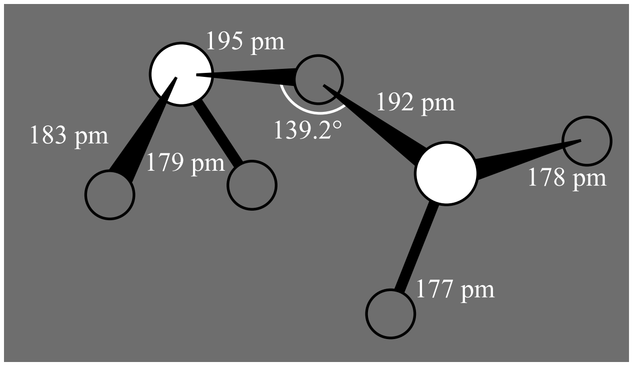
3. ***Iodine pentoxide. I2O5***: It is the only true oxide of iodine which is prepared by the dehydration of iodic acid at 195ºC.

$$2HIO_3 \xrightarrow{195^\circ C} I_2O_5 + H_2O$$

It is a white solid and is the most stable of all the oxides. However it gets decomposed into elements when heated above 300ºC. It dissolves in water to form back iodic acid, hence it may be considered as an anhydride of iodic acid. It acts as a strong oxidizing agent and oxidizes CO, H_2S and HCl.

$$I_2O_5 + 5CO \longrightarrow I_2 + 5CO_2$$

The structure of I_2O_5 is



I_4O_9 is formed by the action of O_3 on I_2 or by the dehydration of HIO_3 with P_2O_5 or H_3PO_4

$$2I_2 + 9O_3 \longrightarrow I_4O_9 + 9O_2$$

Dehydration of HIO_3 with conc H_2SO_4 gives I_2O_4. The structures of these oxides are not known, but as explained earlier I_2O_4 is probably $IO^+ \; IO^-_3$ and I_4O_9 is probably $I^{3+}(IO_3)^-_3$.

11.11 OXOACIDS OF HALOGENS

Halogens form four series of oxoacids with formulae HXO, HXO_2, HXO_3 and HXO_4 although many of these could be known only in solution or as salts. Fluorine forms only one oxoacid HOF. They are listed in Table 11.10.

11.11.1 Hypohalous Acids

The hypohalous acids HOF, HOCl, HOBr and HOI are all known and the halogen has the oxidation state (+1).

HOF was first made in 1968 using matrix isolation technique. When F_2 and H_2O are trapped in an unreactive matrix of solid nitrogen and photolysed, HOF is formed. Recently HOF is prepared by passing F_2 over ice at 0ºC and removing the product into a cold trap.

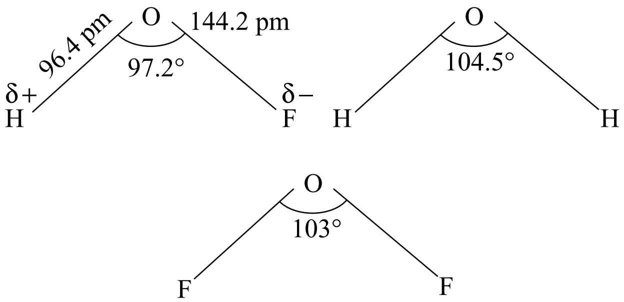
$$F_2 + H_2O \overset{-40^\circ C}{\rightleftharpoons} HOF + HF$$

Table 11.10 Oxoacids of halogens

Oxidation state	Fluorine	Chlorine	Bromine	Iodine	Name of acid	Name of anion
+1	HOF	HClO	HBrO	HIO	Hypohalous	Hypohalite
+3	—	$HClO_2$	—	—	Halous	Halite
+5	—	$HClO_3$	$HBrO_3$	HIO_3	Halic	Halate
+7	—	$HClO_4$	$HBrO_4$	HIO_4	Perhalic	Perhalate

HOF is unstable, and decomposes on its own to HF and O_2 It is a strong oxidizing agent and oxidizes H_2O to H_2O_2. HOF should be stronger acid than HOCl.

HOF is a non-linear molecule



The HOF bond angle (97.2º) is the smallest known bond angle at an unrestricted 'O' atom. It has been suggested that this arises due to attraction between positive charge on hydrogen and negative charge on fluorine, and bring the two atoms nearer. The negative charge on F is intermediate between those estimated for F in HF and OF_2 and this emphasizes the strictly formal nature of the +1 oxidation state for F in HOF.

HOCl, HOBr, HOI have been thermally unstable and therefore have not been isolated in the pureform. They are known only in aqueous solution. These are prepared by shaking freshly precipitated HgO in water with the desired halogen.

$$2HgO + 2X_2 + H_2O \longrightarrow \underset{\text{Precipitate}}{HgO \cdot HgX_2} + 2HOX$$

The salts of these acids are known as hypohalites. All hypohalous acids and hypohalites are unstable and tend to disproportionate at elevated temperatures.

$$3OX^- \longrightarrow \underset{\text{Halide}}{X^-} + \underset{\text{Halate}}{XO_3^-}$$

11.11.2 Halous Acids

The only halous acid definitely known is chlorous acid $HClO_2$ in which chlorine is in +3 oxidation state. It can be prepared by the action of sulphuric acid on barium chlorite.

$$Ba(ClO_2)_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HClO_2$$

It exist only in solution. It is a weak acid but is stronger than HOCl. Its salts are called chlorites. They are stable in alkaline solution even when boiled but in acid solution they disproportionate particularly when heated.

$$3HClO_2 \longrightarrow 2HClO_3 + HCl.$$

11.11.3 Halic Acids

$HClO_3$ and $HBrO_3$ are known only in solution. HIO_3 can be prepared in solid state. $HClO_3$ and $HBrO_3$ are readily obtained by the action of their barium salts with sulphuric acid. Chlorate or bromate of barium are obtained by treating hot concentrated $Ba(OH)_2$ with Cl_2 or Br_2

$$6Ba(OH)_2 + 6Cl_2 \longrightarrow 5BaCl_2 + Ba(ClO_3)_2 + 6H_2O$$

$$Ba(ClO_3)_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HClO_3$$

Similarly $HBrO_3$ can be prepared, however HIO_3 may be prepared by the action of concentrated nitric acid on iodine

$$10HNO_3 + I_2 \longrightarrow 2HIO_3 + 4H_2O + 10NO_2$$

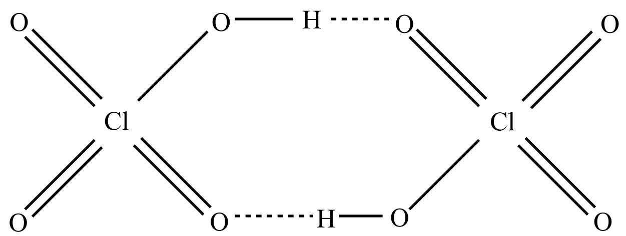
11.11. 4 Perhalic Acids

Perchloric acid and periodic acids and their salts are well known. Perbromates are prepared recently and are not common. Perchloric acid can be prepared first by electrolytic oxidation of sodium chlorate and on heating the product $NaClO_4$ formed with conc H_2SO_4

$$NaClO_3 + H_2O \xrightarrow{\text{electrolysis}} NaClO_4 + H_2$$

$$2NaClO_4 + H_2SO_4 \longrightarrow Na_2SO_4 + 2HClO_4$$

Perchloric acid exist as dimer due to hydrogen bonding.



Most of the perchlorates are highly soluble except those of potassium, rubidium, caesium and ammonium. Magnesium perchlorate is very hygroscopic and is a very effective desiccant called *anhydrone*. $KClO_4$ is used in fireworks and flares. Perchlorate ion has least tendency to act as ligand to form complexes.

Perbromic acid and perbromates are prepared in 1968 by (i) electrolytical method (ii) by oxidation with XeF_2 (iii) by oxidation of bromate in alkali by molecular fluorine. $HBrO_4$ is stable in solution upto a concentration of 6M. The perbromate has been a some what stronger oxidant than perchlorate and periodate but its oxidizing reactions are sluggish.

Several series of *periodic acids* and *periodates* are known. Most important of these include meta periodic acid HIO_4 and *para periodic acid* H_5IO_6 and their salts. Periodates can be made by oxidizing I_2 or I^- in aqueous solution commercially. They are made by oxidizing iodates in alkaline solution either with Cl_2 or electrolytically.

$$IO_3^- + 6OH^- + Cl_2 \longrightarrow IO_6^{5-} + 3H_2O + 2Cl^-$$

$$IO_3^- + 6OH^- \xrightarrow[-2\text{ electrons}]{\text{Eletrolysis}} IO_6^{5-} + 3H_2O$$

The common form of periodic acid is $HIO_4 \cdot 2H_2O$ or H_5IO_6. This is called paraperiodic acid which exist as white crystals. At 100ºC it loses water under reduced pressure yielding periodic acid

$$2H_5IO_6 \xrightarrow[-4H_2O]{100^\circ C} 2HIO_4 \xrightarrow{200^\circ C} I_2O_5 + O_2 + H_2O$$

Structure and Properties of Oxoacids

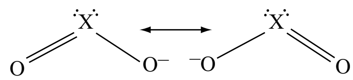
The structures of oxoanions of halogens are given in Table 11.11 the central atom in all the oxoanions is halogen. The oxidation state of halogen in these oxyanion changes gradually in each series for a particular halogen increases in the order +I, +III, +V and VII. The central halogen atom is involved in sp^3 hybridization. Therefore the structure of all these oxoanions have tetrahedron as the base structure, the number of oxygen atoms bonded to the central halogen increases through the series. The structural parameters of oxoanions of chlorine are given in Table 11.12.

Hypohalite ion have no resonance structures. Structures of halite, halate, and perhalate ions are resonance hybrid structures of 2, 3, and 4 resonance structures respectively as shown below. The X—O bond orders are 1, 1.5, 1.67 and 1.75 respectively for XO^-, XO_2^-, XO_3^- and XO_4^-

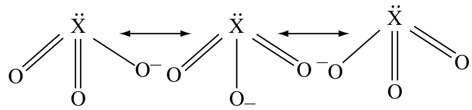
$X - O^-$ no resonance

Hypohalite ion

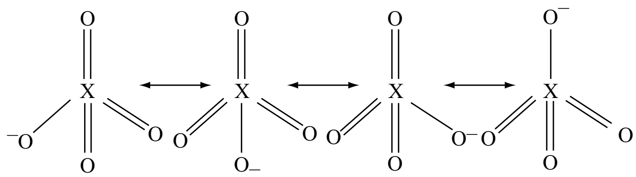
Bond order 1



Halite ion Bond Order 1.5



Halate ion Bond Order 1.67



Perhalate ion Bond Order 1.75

Table 11.11 Structures of oxoanions of halogens.

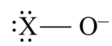
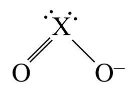
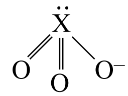
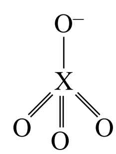
Oxidation state of halogen	Electronic state	Outer ns	Electronic np	Configuration nd	Oxoanion formed	Hybridization	Shape	Structure
+ I	Ground state	[↓↑]	[↓↑][↓↑][↑]	[ ][ ][ ][ ][ ]	Hypohalite	sp^3	Linear	
+ III	1st excited state	[↓↑]	[↓↑][↑][↑]	[↑][ ][ ][ ][ ]	Halite	sp^3	Angular	
+V	2nd excited state	[↑↓]	[↑][↑][↑]	[↑][↑][ ][ ][ ]	Halate	sp^3	Pyramidal	
+VII	3rd excited state	[↑]	[↑][↑][↑]	[↑][↑][↑][ ][ ]	Perhalate	sp^3	Tetrahedral	

Table 11.12 Structural parameters of oxoacids of chlorine

No	Oxoacid	Oxoanion	Cl—O bond length (pm)	∠OClO in oxoanion	Cl—O Bond energy KJ mol–1
1	$HClO$	ClO^-	170	—	209
2	$HClO_2$	ClO_2^-	164	111º	245
3	$HClO_3$	ClO_3^-	157	106º	244
4	$HClO_4$	ClO_4^-	145	109º28′	364

***Thermal Stability*:** Thermal stability of both the acids and their anions get increased with the increasing oxidation state of the halogen. For example the stability and bond strength of the oxychloride anion gets increased in the order

$$ClO^- < ClO_2^- < ClO_3^- < ClO_4^-$$

This can be attributed to the fact that with increasing number of oxygen atoms in the series, the number of resonance structures increases due to which the charge distribution takes place. Also the bond order increases due to which X—O bond strength increases.

***Oxidizing Power*:** As the stability of the anion increased from ClO^- to ClO_4^- the oxidizing power would decrease from ClO^- to ClO_4^-.

Order of oxidation power $Cl—O^- > ClO_2^- > ClO_3^- > ClO_4^-$.

***Relative Acidity*:** The order of acidic strength of oxoacids of chlorine is $HOCl < HClO_2 < HClO_3 < HClO_4$. Their ionization constants are as follows

HOCl	$HClO_2$	$HClO_3$	$HClO_4$
10^{-7}	10^{-2}	10^3	10^8

As the number of unprotonated oxygen atoms around chlorine atom increases, the +ve charge developed on chlorine atom also increases due to the inductive effect of more electronegative oxygen atoms. This causes the increase in the ionic character of O—H bond due to which acidic strength increases

Alternatively, it is possible to explain the relative strength of their conjugate bases. Due to delocalization of charge with increase in the number of resonance structures, the stability of conjugate bases increases from ClO^- to ClO_4^-. So the conjugate basic strength decreases in the order

$$ClO^- > ClO_2^- > ClO_3^- > ClO^-_4$$

If the conjugate base is weak the acid is stronger. So the acidic strength of the acids will be in the order

$$HOCl < HClO_2 < HClO_3 < HClO_4$$

For oxoacids of different halogens in the same oxidation state can be explained in a similar way. The order of the acidic strength of hypohalous acids is

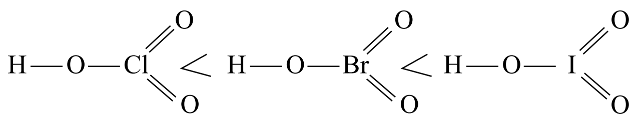
$$HOF > HOCl > HOBr > HOI.$$

As the electronegativity of halogen decreases from F to I the polarity of the H—O bond also decreases as oxygen can attract electron density from both H and halogen resulting the decrease in ionization.

$$\underset{\text{Unknown}}{H \rightarrow O \rightarrow F} \qquad \underset{2.9 \times 10^{-8}}{H \rightarrow O \leftarrow Cl}$$

$$\underset{5 \times 10^{-9}}{H \rightarrow O \leftarrow Br} \qquad \underset{\sim 10^{-11}}{H \rightarrow O \leftarrow I}$$

But in the case of halic acids of different halogens the acidic character should increase in the order



This is expected because the unprotonated oxygen atoms should withdraw more electron density from halogen atom with decrease in its electronegativity and thus developing more positive charge on halogen atom. Thus the positive charge developed on the halogen atoms must be in the order Cl < Br < I. So the ionic character of O—H group must be more in HIO_3 and least in $HClO_3$. Hence the expected order of relative acidic strength is as shown above. But surprisingly it was found that both chloric and bromic acids are strong acids of nearly equal strength with pka [≤] 0 where as iodic acid is slightly weaker with pk_a 0.804 ie. ka 0.157. In concentrated solutions of HIO_3, the iodate ions formed by deprotonation react with undissociated acid according to equilibrium.

$$IO_3^- + HIO_3 \rightleftharpoons [H(IO_3)_2]; \ k = 41 \text{ mol}^{-1}$$

In the case of perhalic acids due to bigger size and more positive charge developed on iodine, it allows more coordination number. So HIO_4 in aqueous solution exist as hydrated H_5IO_6 causing the decrease in number of unprotonated oxygen atoms. So its acidic strength becomes less than perchloric acid.

The modes of thermal decomposition of the halates and their complex oxidation reduction chemistry reflect the interplay of both thermodynamic and kinetic factors.

On the one hand thermodynamically feasible reactions may be sluggish, whilst on the other hand, traces of catalyst may radically alter the course of the reaction. In general, for a given cation, thermal stability decrease in the sequence

Iodate > Chlorate > Bromate

The oxidizing power of the halate ions in aqueous solutions as measured by their standard reduction potentials decreases in the order

Bromate ≥ Chlorate > Iodate

but the rate of reactions follow the order

Iodate > Bromate > Chlorate

The E° values of the reaction

$$XO_4^- + 2H^+ + 2e^- \longrightarrow XO_3^- + H_2O$$

indicate that BrO_4^- is stronger oxidant than ClO_4^- and lO_4^-

$$BrO_4^- \ (+1.76V) > IO_4^- \ (1.64\,V) > ClO_4^- \ (1.23V)$$

Generally one may expect a steady decrease in pπ – dπ bond strength from O – Cl to O – I (size increase and hence the stability sequence must be $HClO_4$ > $HBrO_4$ > HIO_4. The rmodynamic data indicates that perbromate is less stable than either perchlorate or periodate.

	$KClO_4$	$KBrO_4$	KIO_4
$\Delta_f H^o$ (kJ mol^{-1})	–432	–288	–461
$\Delta_f G^o$ (kJ mol^{-1})	–302	–174	–349

The weaker Br – O bond in BrO_4^- is due to middle row anomaly. In aqueous solution IO_4^- (4-coordinate) converts into $H_4IO_6^-$ (6–coordinate). Due to this ΔH° becomes negative since two extra I – O bonds are formed. Greater order in the 6 – coordinate makes ΔS° also negative. This causes ΔG° also less negative. So KIO_4 acts as stronger oxidizing agent than $KClO_4$, though KIO_4 is thermodynamically more stable than $KClO_4$.

11.12 BLEACHING POWDER

Bleaching powder is a mixture of calcium hypochlorite [$Ca(OCl)_2 \cdot 3H_2O$] and basic calcium chloride [$CaCl_2 \cdot Ca(OH)_2 \cdot H_2O$] It is manufactured by passing dry chlorine gas over dry slaked lime.

$$2Ca(OH)_2 + 2Cl_2 \longrightarrow Ca(OCl)_2 + CaCl_2 + 2H_2O$$
$$CaCl_2 + Ca(OH)_2 + H_2O \longrightarrow CaCl_2 \cdot Ca(OH)_2 \cdot H_2O$$
$$3Ca(OH)_2 + 2Cl_2 \longrightarrow \underbrace{Ca(OCl)_2 \cdot CaCl_2 \cdot Ca(OH)_2 \cdot H_2O}_{\text{Bleaching Powder.}} + H_2O$$

Bleaching powder can be manufactured by two methods.

(1) Hasenclover's method and (2) Beckmann's method.

(1) *Hasenclover's Method*: Hasenclover's plant consists horizontal iron cylinders attached to each other as shown in Fig 11.4. Each cylinder is fitted with a shaft fitted with blades

Dry slaked lime is introduced through the hopper at the top and Cl_2 from the bottom. In these cylinders lime and Cl_2 come in contact with each other to form bleaching powder.

The Backmann's plant consists of a vertical cast iron tower. Chlorine and air are introduced into the plant at the bottom separately. Slaked lime is dropped through the

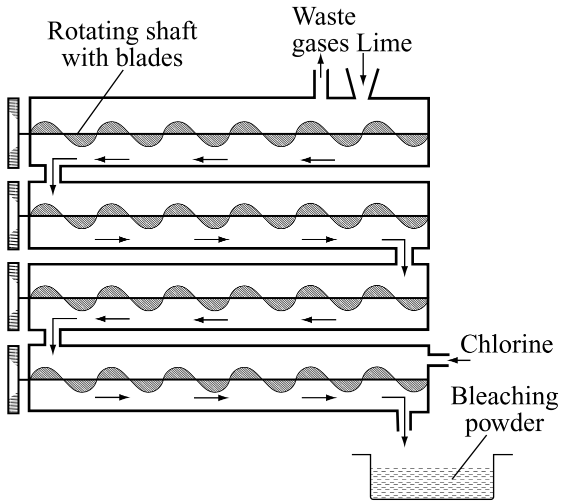


Fig 11.4 Hasenclover's plant

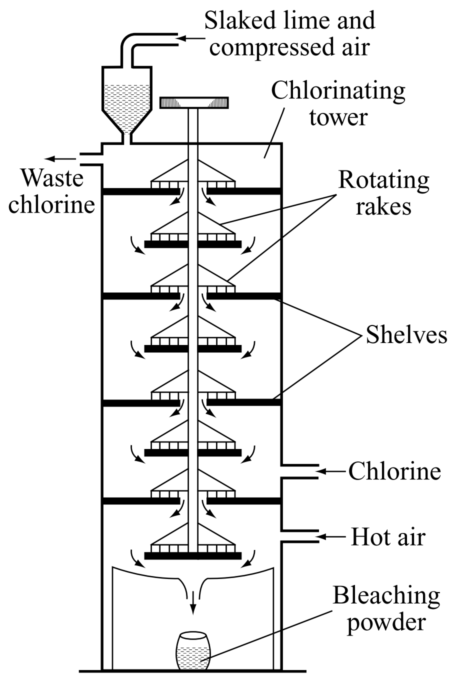


Fig 11.5 Beckmann's plant

hopper at the top of the tower. Unused chlorine and air escape through the outlet at the top below the hopper. In the tower there are several shelves on which rotating rollers are present. Slaked lime added into the tower moves down by the rotating rollers. The lime moving downwards will react with the chlorine gas moving upwards. The reactants i.e., slaked lime and chlorine which are moving in opposite direction and react with each other. This is known as counter current principle. The bleaching powder is collected at the bottom. The current of hot air drives away the unused Cl_2.

Physical Properties

(i) It is a yellowish white powder with a strong smell of chlorine.
(ii) It is soluble in water, the insoluble part is the unreacted lime present in it.

Chemical Properties

The amount of chlorine absorbed is never so complete as is represented by the equation quoted above. The constitution of bleaching powder has been much discussed. Under very favourable conditions slaked lime can be saturated with no more than 43.5 per cent of available chlorine, and the facts corresponds with the formula ascribed to W.Odling viz $Ca\begin{cases}Cl\\OCl\end{cases}$ and for long accepted.

It has now been shown (Bunn, Clark and Clifford 1935) by microscopic examination by X-ray examination of slaked lime during the process of chlorination, that ordinary bleaching powder is a mixture of calcium hypochlorite, $Ca(OCl)_2 \cdot H_2O$ and a non-deliquescent basic chloride $CaCl_2 \cdot Ca(OH)_2 \cdot H_2O$ in approximately molecular proportions. The first products of the chlorination are the basic chloride and a basic hypochloride,

$Ca(OCl)_2 \cdot 2Ca(OH)_2$. This latter on further chlorination forms the hypochlorite.

Here the chemical properties are explained by taking Odling formula.

1. *Autooxidation*: When bleaching powder is stored for longer periods it slowly undergoes autooxidation forming calcium chloride and calcium chlorate.

$$6CaOCl_2 \longrightarrow 5CaCl_2 + Ca(ClO_3)_2$$

2. *Reaction with water*: Bleaching powder dissolves in cold water and ionizes as follows

$$CaOCl_2 \longrightarrow Ca^{2+} + Cl^- + ClO^-$$

In hot water the hypochlorite ion undergoes disproportionation, giving chloride and chlorate ions.

$$3ClO^- \longrightarrow 2Cl^- + ClO_3^-$$

3. *Reaction with dilute acids*: If small amounts of dilute acid is added to bleaching powder finally liberates oxygen.

$$2CaOCl_2 + H_2SO_4 \longrightarrow CaSO_4 + CaCl_2 + O_2$$

Due to the liberation of nascent oxygen bleaching powder is used as oxidizing agent and bleaching agent.

If excess dilute acid is added chlorine gas will be liberated.

$$CaOCl_2 + H_2SO_4 \longrightarrow CaSO_4 + H_2O + Cl_2$$

The amount of chlorine liberated by the addition of excess dilute acid to bleaching powder is called **available chlorine**. The quality of bleaching powder depends on the available chlorine. A good quality bleaching powder contains 35–38 per cent of available chlorine. The bleaching powder stored for longer periods loses the available chlorine due to conversion into $CaCl_2$ and $Ca(ClO_3)_2$ by autooxidation.

4. *Reaction with carbon dioxide*: When CO_2 gas is passed over bleaching powder also liberate chlorine gas.

$$CaOCl_2 + CO_2 \longrightarrow CaCO_3 + Cl_2$$

5. *Effect of a catalyst*: Bleaching powder decomposes in the presence of cobalt chloride catalyst liberating oxygen.

$$2CaOCl_2 \xrightarrow{CoCl_2} 2CaCl_2 + O_2$$

6. *Oxidizing properties*: It oxidizes several substances

$$PbO + 2CaOCl_2 \longrightarrow PbO_2 + 2CaCl_2$$

$$CaOCl_2 + 2FeSO_4 + H_2SO_4 \longrightarrow Fe_2(SO_4)_3 + CaCl_2 + H_2O$$

$$CaOCl_2 + KNO_2 \longrightarrow CaCl_2 + KNO_3$$

$$3CaOCl_2 + 2NH_3 \longrightarrow 3CaCl_2 + 3H_2O + N_2$$

$$CaOCl_2 + 2KI + 2HCl \longrightarrow CaCl_2 + 2KCl + H_2O + I_2$$

$$CaOCl_2 + Na_3AsO_3 \longrightarrow CaCl_2 + Na_3AsO_4$$

7. *Reaction with organic compounds*: Bleaching powder oxidizes ethyl alcohol, acetaldehyde and acetone to chloroform. In this reaction the organic compounds under goes oxidation, chlorination and hydrolysis

$$CaOCl_2 + H_2O \longrightarrow Ca(OH)_2 + 2(Cl)$$

$$CH_3CH_2OH + CaOCl_2 \longrightarrow CaCl_2 + CH_3CHO + H_2O$$

$$CH_3CHO + 6Cl \longrightarrow CCl_3CHO + 3HCl$$

$$2CCl_3\ CHO + Ca(OH)_2 \longrightarrow 2CHCl_3 + Ca(HCOO)_2$$

With acetone.

$$CH_3\ CO\ CH_3 + 6Cl \longrightarrow CCl_3CO\ CH_3 + 3HCl$$

$$2CCl_3\ CO\ CH_3 + Ca(OH)_2 \longrightarrow 2CHCl_3 + Ca(CH_3COO)_2$$

Uses: Bleaching powder is used

(i) in the sterilization of water.
(ii) as a bleaching agent for cotton and paper pulp.
(iii) in the preparation of chloroform.
(iv) as oxidizing agent and as a chlorinating agent.

11.13 INTERHALOGEN COMPOUNDS

The binary compounds formed by halogens amongst themselves are known as inter-halogen compounds. Generally the halogens exist in dihalogen species X_2. In addition to these dihalogen species, all halogens combine with one another to form dihalogen species having different halogen atoms XY with all possible combinations. There are many compounds in which a less electronegative halogen atom is bound to three, five or seven more electronegative halogen atoms to form stable molecules. The known interhalogen compounds are listed in Table 11.13.

The interhalogen compounds are represented as XY, XY_3 XY_5 and XY_7 where Y is always halogen with more electronegativity. There are never more than two different halogens in any known interhalogen compound. It should be noted that the interhalogen compounds of the type XY_n where 'n' is odd number 1, 3, 5 and 7. Because 'n' is always odd, all the interhalogen compounds are diamagnetic having all valency electrons present either as shared bonding or unshared pairs.

In naming the interhalogen compound it is regarded as the halide of the more electropositive element. For example ClF is named as chlorine monofluoride and ICl_3 as iodine trichloride.

The coordination number i.e., the number of halogen atoms around the other halogen atom will be decided by the radius radio and electronegativity difference between the two different halogen atoms combining.

$$\text{Radius ratio} = \frac{\text{Radius of larger halogen}}{\text{Radius of smaller halogen}}$$

XY and XY_3 type interhalogen compounds are generally formed when the electronegativity difference, and radius ratio values are small. XY_5 and XY_7 type interhalogen compounds are formed if the electronegativity difference, and radius ratio values are large.

General Methods of Preparation. All the interhalogen compounds can be prepared by direct reaction between the halogens or by the action of a halogen on a lower interhalogen. The product formed depend on the conditions.

$$Cl_2 + F_2 \xrightarrow[\text{1:1 Vol.}]{200^\circ C} 2ClF$$

$$Cl_2 + 3F_2 \xrightarrow[\text{excess } F_2]{300^\circ C} 2ClF_3$$

$$I_2 + Cl_2 \xrightarrow[\text{(equimolar).}]{\text{liquid}} 2ICl$$

$$I_2 + \underset{\text{excess}}{3Cl_2} \xrightarrow{\text{liquid}} 2ICl_3$$

$$Br_2 + \underset{\text{diluted with } N_2}{3F_2} \longrightarrow 2BrF_3$$

$$Br_2 + 5F_2 \xrightarrow{\text{excess } F_2} 2BrF_5$$

$$I_{2(g)} + 5F_2 \xrightarrow{20^\circ C} 2IF_5$$

$$I_{2(g)} + 7F_2 \xrightarrow{250-300^\circ C} 2IF_7$$

Properties

1. ***Stability***: The most stable of the XY type compounds are ClF, ICl. BrCl is substantially stable while IBr is detectably stable and dissociates into the constituent elements at 25ºC BrF is unstable and disproportionates into Br_2 and BrF_3.

The bond strengths of interhalogen compounds are clearly related to the difference in the electronegativity between the constituent halogens. As the electronegativity difference increases the ionic character of the bond increases as expected on the basis of Pauling's theory. Further the tendency to form interhalogen compounds having more bonds as BrF_5, IF_5 and IF_7 will depend on the electronegativity of the central atom. Only iodine form IF_7, IF_5 or ICl_3. The *lower oxidation states* of halogens where the less electronegative halogen is present as central atom are unstable and disproportionates.

Table 11.13 Interhalogen compounds (Bond energies KJ mol^{-1})

Difference in electronegativity	XY	Bond energy	XY3	Bond energy	XY5	Bond energy	XY7	Bond energy
1.38	IF	277.8	IF_3	~272	IF_5	207.8	IF_7	231
1.28	BrF	248.6	BrF_3	201.2	BrF_5	187		
0.95	ClF	252.5	ClF_3	172.4	ClF_5	~142		
0.43	ICl	207.9	$(ICl_3)_2$					
0.33	BrCl	215.9						
0.10	IBr	175.3						

$$3BrF \longrightarrow Br_2 + BrF_3$$

$$5IF \longrightarrow 2I_2 + IF_5$$

The tendency of disproportionation is common among the lower fluorides of bromine and iodine. It can be explained on the basis of the behaviour of iodine fluorides. Both IF and IF_3 tend to disproportionate. The IF cannot be isolated due to this reason even though I – F bond strength is more than any other inter halogen compound. This is because fluorine being stronger oxidizing agent, will try to oxidize iodine into its higher oxidation state. In IF_5 and IF_7 because more number of bonds are formed, the IF disproportionates. The IF_7 though more stable is reactive and acts as a strong fluorinating agent than IF_5 because the bond strength is less due to the steric factors and resistance to high oxidation state of iodine. Similarly bromine fluoride disproportionates but BrF_3 and BrF_5 are stable.

Bromine trifluoride autoionizes in liquid state

$$2BrF_3 \longrightarrow BrF_2^+ + BrF_4^-$$

2. ***Lewis acid–base character and solvent property*:** Interhalogens are non aqueous ionizing solvents. The solvent properties of BrF_3 have been investigated. It dissolves number of halides salts

$$CsF_{(s)} + BrF_{3(l)} \longrightarrow CsBrF_4 \text{ (soln)}$$

$$SbF_{5_{(s)}} + BrF_{3(l)} \longrightarrow SbBrF_8 \text{ (soln)}$$

Bromine trifluoride is a useful solvent for ionic reactions that must be carried out under highly oxidizing conditions.

3. ***Reactions with alkenes*:** Like halogens, the interhalogen compounds will be added to the double bond in alkenes

$$CH_2 = CH_2 + BrCl \longrightarrow \underset{Br}{CH_2} - \underset{Cl}{CH_2}$$

4. ***Physical state*:** The interhalogen compounds are covalent liquids or gases or solids due to the small electronegativity difference and the melting and boiling points increase as the difference in electronegativities increases. (Ref. Table 11.14).

5. ***Reactivity*:** The interhalogen compounds are more reactive than halogens except F_2 because X—Y bond in interhalogens is weaker than the X—X bond in the halogens. Also the interhalogen compounds are polar and the polarity increases with the increase in the electronegativity difference of the two halogens while halogens are non-polar.

6. ***Oxidation property*:** All the interhalogens are oxidizing agents. Since ClF_3 is thermodynamically more stable than fluorine, it is expected as a weaker fluorinating agent than fluorine. But it is stronger fluorinating agent towards many elements and compounds. There is no relation to the thermodynamic stability and the rates of oxidation reactions of interhalogens. Thus ClF_3 and BrF_3 are much more strong fluorinating agents than BrF_5, IF_5 and IF_7 . IF_5 is a mild fluorinating agent that can be handled in glass apparatus. The interhalogen compounds fluorinate many metal oxides, metal halides and metals.

$$3ClO_2 + 4BrF_3 \longrightarrow 3ClF_4 + 2Br_2 + 3O_2$$

Table 11.14 Physical properties of interhalogen compounds

Interhalogen compound	Physical state	ΔH^of (298k) KJ mol^{-1}	M.P°C	B.P °C	X–Y Bond length pm.	X–Y energy KJ mol^{-1}	Dipolemoment D
ClF	Colourless gas	−56.5	−155.6 100.1	−100.1	162.81	252.5	0.881
BrF	Pale brown	−58.6	−33	disproportionate	175.6	248.6	1.29
IF	Unstable	−95.4		Disprop	190.9	~277	–
BrCl	Red brown gas	+14.6	–66	Dissoc	213.8	215.1	0.57
ICl	Ruby red Crystals	–35.3	27.2	97–100	232.07	207.7	0.65
IBr	Black Crystals	–10.5	41	~116	248.5	175.4	1.21
ClF_3	Colourless gas/liq	–164(g)	–76.3	11.8	–	174	1.885
BrF_3	Straw coloured liquid	–301	8.8	125.8	–	202	1.19
IF_3	Yellow solid	–485(g)	Calc	–	–	–	275 calc.–
I_2Cl_6	Bright yellow solid	–89.3(s)	101.(16 atms)	–	–	–	–
ClF_5		–255	–103	–13.1	–	154	–
BrF_5		–429	–60.5	41.3		187	1.51
IF_5		–843	9.4	104.5	–	269	2.18
IF_7		–962	6.5	4.8	–	232	0

7. ***Hydrolysis*****:** The interhalogen compounds are hydrolysed to form halide and oxyhalide ions. The oxyhalide is generally formed from the larger halogen. The oxidation state of the larger halogen in inter halogen does not change during hydrolysis.

$$BrF_5 + 3H_2O \longrightarrow 5HF + HBrO_3$$

$$ICl + H_2O \longrightarrow HCl + HIO$$

8. ***Ionization*****:** The conductivity of some interhalogens indicates a small degree of self ionization in the liquid state. For example ICl ionizes as

$$2ICl \rightleftharpoons I^+ + ICl_2^-$$

Structures

The structures of interhalogen compounds are given in Table 11.15. The central halogen atom in all these interhalogens is less electronegative halogen. The oxidation state of central halogen increases in the order +I, +III, +V and +VII in XY, XY_3, XY_5 and XY_7 respectively.

1. All XY type inter halogens are linear and their structural discriptions are given Table 11.15.
2. In XY_3 type molecules central halogen is in sp^3d hybridization around the central halogen. There will be total five pairs of electrons of which two are lone pairs and three bond pairs. These five pairs are arranged in trigonal bipyramidal structure. The two lone pairs occupy the equatorial positions. So these molecules have T-shape.

   ICl_3 exist as dimer. I_2Cl_6 in which iodine atom of one ICl_3 gets one extra electron pair from chlorine atom of another ICl_3, so that each iodine atom have 6 electron pairs around it. Each iodine atom is involved in sp^3d^2 hybridization. The six electron pairs are arranged in octahedral manner, of which two lone pairs occupy the opposite corners of the octahedron. All the atoms are arranged in the same plane, so that two squares are sharing an edge.

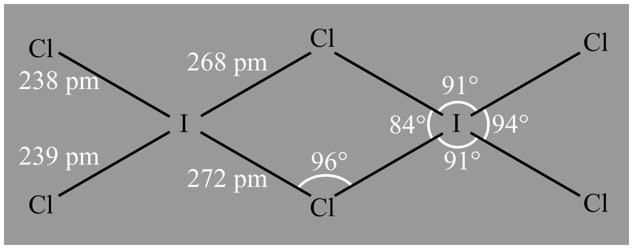


Fig 11.7 Structure of $(ICl_3)_2$

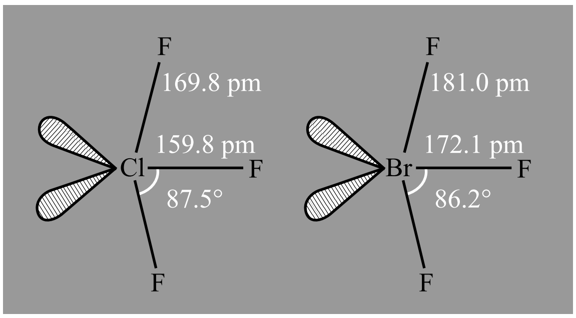
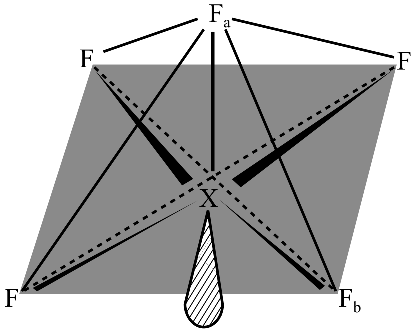


Fig 11.6 Structures of XY_3 Type interhalogens

3. In XY_5 type interhalogens the central halogen is in sp^3d^2 hybridization around which there are six electron pairs i.e., one lone pair and five bond pairs. These are arranged in octahedral manner. The lone pair occupy one corner of the octahedron due to which the molecules have square pyramidal structure. In BrF_5 and IF_5 the Br and I atoms are just below the base of the pyramid because of the repulsion between the lone pair and bond pairs.
4. The only compound of the type XY_7 is IF_7 in which iodine atom is in sp^3d^3 hybridization. The molecule is pentagonal bipyramidal structure. All I—F bond lengths approximately equal 182.5 pm.

Table 11.15 Structural discriptions of Interhalogen compounds

Oxidation state of halogen	Electronic state	Outer electronic configuration ns	np	nd	Inter halogen formed	Hybridization	Shape
+ I	Ground state	↓↑	↓↑ ↓↑ ↑		XY		Linear
+ III	1st excited state	↓↑	↑↓ ↑ ↑	↑	XY_3	sp^3d	T–Shape
+V	2nd excited state	↓↑	↑ ↑ ↑	↑ ↑	XX_5	sp^3d^2	Square pyramid
+VII	3rd excited state	↑	↑ ↑ ↑	↑ ↑ ↑	XX_7	sp^3d^3	Pentagonal bipyramid



	ClF_5	BrF_5		IF_5	
	(gas)	(gas)	(cryst)	(gas)	(cryst)
$X–F_b$/pm	~172	177.4	178	186.9	189
$X–F_a$/pm	~162	168.9	168	184.4	186
$LF_a–X–F_b$	~90° (assumed)	84.8°	84.5°	81.9°	80.9°

Fig 11.8 Structures of BrF_5 and IF_5

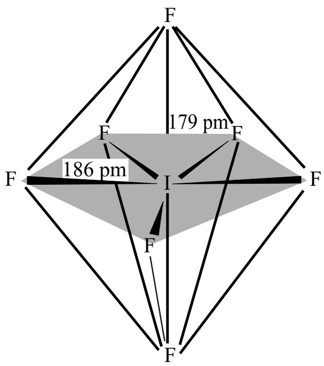


Fig 11.9 Structure of IF_7

The five equatorial F atoms not quite planar.

Applications of Interhalogen Compounds

(i) Iodine monochloride is used in the estimation of iodine number which is a measure of the unsaturation in acids and fats.

$$-CH = CH - + ICl \longrightarrow -\underset{\displaystyle |}{\overset{\displaystyle I}{}}\!HC - \overset{\displaystyle Cl}{\underset{}{|}}CH-$$

(ii) Iodine monochloride is used to iodinate organic compounds through chlorination which may occur depending upon the experimental conditions.

$$\text{Salicylic acid} \begin{cases} \xrightarrow{ICl} \text{Chlorination} \\ \xrightarrow[\text{Nirobenzene.}]{ICl} \text{Iodinnation} \end{cases}$$

(iii) Liquid BrF_3 and ClF_3 have been used as fluorinating agents for the preparation of number of metal fluorides.

(iv) BrF_3, ClF_3 and ICl have been used for the preparation of polyhalides

$$NaBr + ICl \longrightarrow NaBrICl$$

(v) Liquid BrF_3 has been used to carry out various types of reactions and new compounds such as BrF_2SbF_6, $(BrF_2)_2SnF_6$ have been prepared from reactions in this solvent.

(vi) BrF_5 is used as an oxidizer for propellents.

11.14 PSEUDOHALOGENS

The term 'pseudohalogen' was first applied by Birkenbach and Kellarman to certain univalent electronegative inorganic radicals which show a resemblance to the halogens in their chemical properties.

Table 11.16 Pseudohalogens and pseudohalide ions

Pseudohalogen		Pseudohalide ion	
Name	**Formula**	**Name**	**Formula**
Cyanogen	$(CN)_2$	Cyanide	CN^-
Thiocyanogen	$(SCN)_2$	Thiocyanate	SCN^-
Selenocyanogen	$(SeCN)_2$	Selenocyanate	$SeCN^-$
Tellurocyanogen	$(TeCN)_2$	Tellurocyanate	$TeCN^-$
Azido carbondisulphide	$(SCSN_3)_2$	Azidothiocarbonate	$SCSN_3^-$
Oxycyanogen	$(OCN)_2$	Cyanate	OCN^-
		Azide	N_3^-

Oxycyanogen is not available in the free state.

A pseudohalogen can be defined as "a molecule consisting of more than two electronegative atoms which, in free state, resemble the halogens". The pseudohalogens give rise to anions which resemble the halide ions in their behaviour.

Brown and his co-workers suggested the name "halogenoids" instead of psudohalogens. Later Welden and Audrieth defined halogenoids.

"Any univalent chemical aggregate composed of two or more electronegative atoms which shows in the free state certain characteristics of halogens and which combine with hydrogen to form an acid and with silver to form salt insoluble in water".

Some important pseudohalogens and their pseudohalide ions are listed in Table 11.16.

Similarities between Pseudohalogens and Halogens

(i) Both halogens and pseudohalogens are volatile in their free state except the polymeric thiocyanogen.

(ii) Both halogens and pseudohalogens are isomorphous in their structures in their free and solid states.

(iii) Like halogens, pseudohalogens also combine with many metals to form salts e.g., NaCN, NaSCN etc.

(iv) Like the halides of silver, mercury (I) and lead (II) the pseudohalides of these metals are insoluble in water.

(v) Like halogens, the pseudohalogens also combine with hydrogen to form monobasic acids. However, these acids are very weak compared to halogen acids.

(vi) Like halogens, the pseudohalogens also combine among themselves to give inter pseudohalogen compound e.g., CNN_3 etc and also with halogens to give halogen halogenoid e.g., ICN.

(vii) The halogenoid ions will form complexes with metals as do the halide ions e.g., $K_2[Hg(SCN)_4]$,$Na[Au(CN)_2]$

(viii) Like halogens the pseudohalogens can also be prepared in the free state by oxidizing their hydra acids or their salts or by thermal decomposition of higher valent compounds.

(ix) Several compounds of halogens and pseudohalogens are similar e.g., ICN and ICl, $Pb(SCN)_4$ and $PbCl_4$

(x) The reactions of pseudohalogens are very much similar to those of halogens

$$(CN)_2 + 2OH^- \longrightarrow CN^- + OCN^- + H_2O$$

$$(SCSN_3)_2 + 2OH^- \longrightarrow SCSN_3^- + OSCSN_3^- + H_2O$$

$$Cl_2 + 2OH^- \longrightarrow Cl^- + OCl^- + H_2O$$

(xi) Like halogens pseudohalogens will be added at the double or triple bonds in unsaturated hydrocarbons

$$CH_2 = CH_2 + (SCN)_2 \longrightarrow \underset{\substack{|\\ SCN}}{CH_2} - \underset{\substack{|\\ SCN}}{CH_2}$$

$$CH_2 = CH_2 + Cl_2 \longrightarrow \underset{\substack{|\\ Cl}}{CH_2} - \underset{\substack{|\\ Cl}}{CH_2}$$

(xii) Some lead compounds of pseudohalogens are decomposed in a similar way to the lead halides.

$$Pb(SCN)_4 \longrightarrow Pb(SCN)_2 + (SCN)_2$$

$$PbCl_4 \longrightarrow PbCl_2 + Cl_2$$

(xiii) The hydracids of halogens and pseudohalogens react with oxidizing agents in similar manner.

$$MnO_2 + 4HSCN \longrightarrow Mn(SCN)_2 + 2H_2O + (SCN)_2$$

$$MnO_2 + HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$$

From the above discussion, it is clear that halogens and pseudohalogens are very close in their behaviour. The preparation, properties and structures of some important pseudohalogens are described below.

Cyanogen

Cyanogen is among most important pseudohalogens and was first prepared by Gay-Lussac by heating mercuric or silver cyanide or by the reaction of the cupric ion with cyanide ion in aqueous solution.

$$2Hg(CN)_2 \longrightarrow Hg_2(CN)_2 + (CN)_2$$

$$2Cu^{2+} + 4CN^- \longrightarrow Cu_2(CN)_2 + (CN)_2$$

It can also be prepared by the gas-phase oxidation of HCN

$$2HCN + (O) \longrightarrow (CN)_2 + H_2O$$

Properties: It is a colourless poisonous gas with b.p –25ºC. It burns with violet coloured flame forming CO_2 and N_2

$$C_2N_2 + 2O_2 \longrightarrow 2CO_2 + N_2$$

On heating, the pure gas above 600ºC, it polymerizes to so called paracyanogen. It reacts with alkalis like halogens to give cyanide and cyanate

$$(CN)_2 + 2KOH \longrightarrow KCN + KCNO + H_2O$$

The structure of cyanogen may be represented as

$$N \equiv C - C \equiv N$$

The C—N bond distance is 116 pm which is normal for triple bond but the C—C bond distance 137 pm which suggests that some double bond character. Hence it was assumed that cyanogen exists in the following resonating structure.

$$N{\equiv}C-C{\equiv}N \leftrightarrow \bar{N}=C=C=\overset{+}{N} \leftrightarrow \overset{+}{N}=C=C=\bar{N}$$

11.15 POLYHALIDES

When halide ions associate with molecules of halogens or interhalogens univalent ions are formed. These are known as *polyhalides ions* and the compounds of these are known as *polyhalides.*

Generally polyhalides are formed by the large cations like alkali metal ion, an alkaline earth metal ion, a coordination complex ion such as $[Co(NH_3)_6]^{3+}$ etc. Polyhalide formation is more common in iodides than the other halides. The tendency of polyhalide formation increases with decreasing lattice energy in the original halide. Hence chlorides form less polyhalides, bromides form more where as iodides commonly enter into polyhalide formation. Polyhalides do not exhibit isomerism but due to the difference in the crystal structure $CsIBr_2$ exist in two forms having different colours.

Classification of polyhalides

(i) Type X–n: Examples are PbI_3, TlI_3, CsI_3, KI_3 etc in which polyhalide contain same type of halogen atoms.

(ii) Type XX–n: These are common e.g., IBr_2^-, ICl_2^-, $ClBr_2^-$ etc. These polyhalide ions contain two different types of halogen atoms.

(iii) Type X X′ X″n: These are very few eg $CsFIBr_3$, $RbFICl_3$, KClIBr.

Table 11.17 Types of polyhalide salts. (M represents a large univalent cation like Cs^+, $(CH_3)_4N^+$ etc.)

Type X–n	Type XX′n		Type X X′ X″n
MBr_3	$MClF_2$,	$MClF_4$	MIBrF
MI_3	$MIBr_2$	$MBrF_4$	$MIFCl_3$
MI_5	$MICl_2$,	$MBrF_6$	MIClBr
MI_7	$MBrCl_2$,	MIF_4	
MI_9	$MClBr_2$,	MIF_6	
	$MICl_4$		

In addition to the above polyhalide anions, polyhalide cations like BrF^+_2, ICl^+_2 are also known which are formed in the unusual dissociation of interhalogens XX_3' and XX_5'

$$ICl_3 \longrightarrow ICl_2^+ + Cl^-$$

$$2BrF_3 \longrightarrow BrF_2^+ + BrF_4^-$$

$$2IF_5 \longrightarrow IF_4^+ + IF_6^-$$

Preparation

1. Polyhalide can be prepared by the direct addition of halogen to halide

$$\text{Halide} + \text{Halogen} \longrightarrow \text{Polyhalide}$$

$$CsI + I_2 \longrightarrow CsI_3$$

$$KI + 2Cl_2 \longrightarrow KCl_4$$

2. By the direct addition of interhalogen compond to halogen.

$$\text{Halide} + \text{Inter halogen} \longrightarrow \text{Polyhalide}$$

$$NaBr + IBr \longrightarrow NaIBr_2$$

$$NH_4Br + IBr \longrightarrow NH_4IBr_2$$

3. They can also be prepared by the displacement of one halogen by another or by displacement by one interhalogen with another interhalogen.

$$CsBr_3 + I_2 \longrightarrow CsIBr_2 + IBr$$

$$CsICl_2 + 2IBr \longrightarrow CsIBr_2 + 2ICl$$

4. Polyhalides can be prepared by metathesis with another polyhalide.

$$RbCl + HICl_2 \longrightarrow RbICl_2 + HCl$$

Properties: Polyhalides are generally coloured ranging from yellow to blue-black. They have low melting points. They dissolve in liquids having high dielectric constants like water, ethanol acetone, etc. Polyhalides are insoluble in liquids having low dielectric constant except the free halogens and interhalogens. Dissociation in water is generally followed by hydrolysis.

Stability: Polyhalides are typically ionic compounds though they tend to decompose on heating. The products of decomposition are governed by lattice energy of the products. Because lattice energy of halide with smaller halide ion will be more, smaller halogen atom remain in metal halide.

$$RbICl_2 \longrightarrow RbCl + ICl$$

For a given type of polyhalide, the thermal stability increases as the size of the unipositive cation increases

$$Na^+ < K^+ < NH^+_4 < Rb^+ < Cs^+$$

The stability for polyhalide ion for a given cation increases in the following order.

$$[BrI]^- < [FIBr]^- < [ClBrCl]^- < [I_3]^- < [BrIBr]^- < [ClICl]^-$$

Fluorine cannot form polyhalide ion like F_3^- due to the absence of d-orbitals in its valence shell.

Replacement reactions: A more electronegative halogen can replace a less electronegative halogen from polyhalide but the central halogen atom is not replaceable. Thus chlorine can replace bromine from IBr_2^- to give ICl_2^- but cannot replace iodine atom.

$$IBr_2^- + Cl_2 \longrightarrow ICl_2^- + Br_2$$

Action of halogens: A halogen may add directly to the polyhalide Ex: Cl_2 can be added to ICl_2^- to give ICl_4^-

$$ICl_2^- + Cl_2 \longrightarrow ICl_4^-$$

Oxidation property: ClF_4^- and BrF_6^- are powerful oxidizing agents which explode in organic solvents.

Structures

The polyhalide ions of the type X_3^-, $[XX'_2]^-$ are linear with the central atom involved in sp^3d hybridization, and three lone pairs occupying the equatorial positions of trigonalbipyramid structure, e.g.,

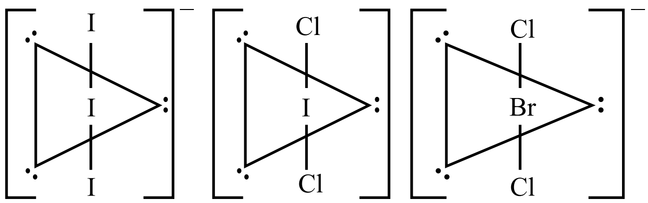


Fig 11.10 Structures of $[XX_2]^-$ type polyhalides

The $[XX'_4]^-$ type polyhalide ions have square planar structures. The central atom is involved in sp^3d^2 hybridization. The two lone pairs occupy the opposite corners of the octahedron, e.g.,

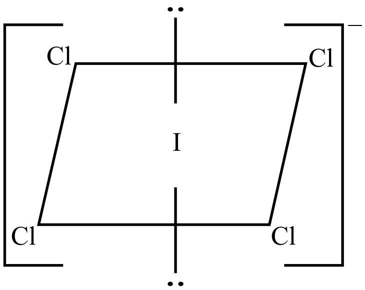


Fig 11.11 Structure $[XX_4]^-$ polyhalide

Note: These structures can be easily predicted by using VSEPR theory and (V + M – c + a)/2 rule.

The I_5^-, I_7^- and I_9^- ions are rather loose aggregates. In the tetramethyl ammonium salt, I_5^- has been shown to have the structure depicted.

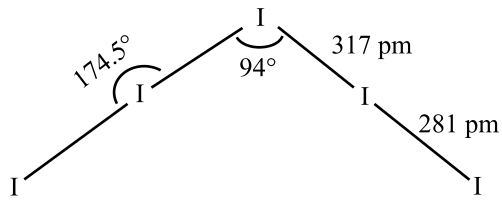
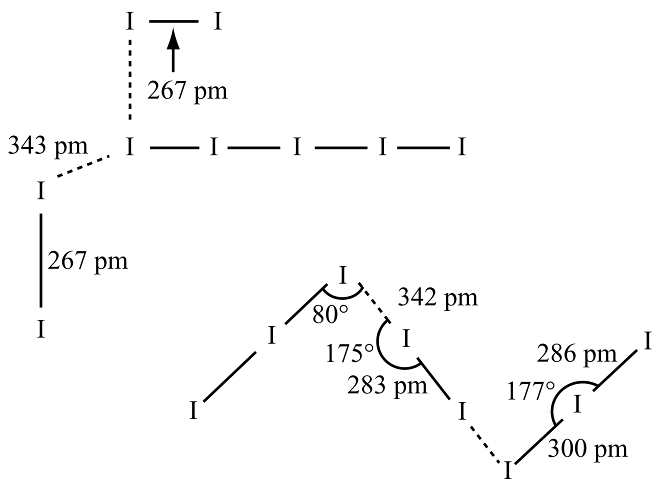


Fig 11.12 Structure of I_5^-

The short I – I distance is greater than in I_2 molecule which is 267 pm but the long I – I distance indicates very weak bonding and the ion can be considered to consist of two I_2 molecules fairly weakly coordinated to I^-

In I_7^- ion two I_2 molecules combine with I_3^- ions

In I_8^{2-} ion two I_3^- ions joined with I_2 molecule



KEY POINTS

- The elements Fluorine (F), Chlorine (Cl), Bromine (Br), Iodine (I) and Astatine (At) belong to VII A group and p block of the periodic table.
- Except astatine the other elements are called as halogens because they are mainly, produced from sea water (Halos = salt, gene = product)
- Fluorine occurs mostly as fluorospar CaF_2, Fluorapetite $3Ca_3(PO_4)_2 \cdot CaF_2$. Fluorine is called **super halogen** because of its high electronegativity.
- Chlorine, bromine and iodine occurs in sea water as salts e.g., NaCl, $MgBr_2$, $NaIO_3$.
- The general outer electronic configuration of elements is $ns^2\ np^5$.
- Halogens can attain stable inert gas configuration by accepting an electron, hence they are very reactive and occur only in the combined state.

Physical Properties

- Atomic and ionic sizes of halogens gets increased gradually from fluorine to iodine.
- All the halogen exist as **diatomic covalent molecules,** F_2, Cl_2, Br_2 and I_2 and they are non polar.

- With increase in size of molecules from F_2 to I_2 Van der Wall's attractive forces also increases. Hence F_2 and Cl_2 are small molecules and exist as gases, bromine is liquid and iodine is solid.
- M.pts and B.pts increases from F_2 to I_2 due to increase in the Van der Wall's attractive forces. Density also increases from fluorine to iodine.
- **Ionization energies** of all halogens are very high indicating their non-metallic character and decreases from F to I.
- Electronegativity decreases from fluorine to iodine. Non-metallic character decreases from F to I. Iodine has metallic lustre and has a tendency to form I^+.
- Bond length increases from F_2 to I_2. Order of bond energy is Cl-Cl > Br-Br > F-F > I-I.
- According to **Mulliken** the high bond energies of Cl_2, Br_2 are due to multiple bonding extra to the strong σ bond multiple bonding is possible involving d-orbitals, where as in F_2 this is not possible due to the absence of d-orbitals in its valency shell.
- According to **Coulson,** the low bond dissociation energy of fluorine is due to the repulsion between the lone pairs of electrons on the two smaller fluorine atoms which are at a smaller bond distance. The most widely accepted explanation about the low bond energy of F_2 is Coulson's theory.
- The order of electron affinities is in the order Cl > F > Br > I. The electron affinity of fluorine is less than chlorine though it is most electronegative due to its small size and repulsion between newly added electron and the electrons already present in its small 2p orbital.
- All the halogens are coloured F_2 is light yellow, Cl_2 is light green, Br_2 is reddish brown and I_2 is black in solid but violet in gaseous state. The intensity of the colour increases from F_2 to I_2. The colour is due to absorption of energy in the visible light for the excitation of electron from HOMO to LUMO.
- The energy absorbed decreases with the increase in the size of atom. F_2 being small in size requires large amount of energy therefore absorbs violet and hence appears in complimentary colour yellow, gaseous molecules of I_2 absorb yellow light and hence appear violet.
- When iodine is dissolved in aliphatic hydrocarbons or CCl_4 it is bright violet, those in aromatic hydrocarbons are pink or reddish brown and those in stronger donor solvents such as alcohols, ethers of amines are deep brown. This is because in donor solvents. I_2 accepts the electron pair from solvent molecule into its LUMO i.e., $^*\sigma P_z$ and thus weakening I-I bond altering the energy of electronic transition. In non-coordinating solvents the electronic transition is from HOMO to LUMO due to which violet colour appears.
- Solubility of iodine in the donor solvents tends to be greater than in the non donor solvents.
- Except fluorine all the other halogens exhibit –1, +3, +5 and +7 oxidation states. Halogens exhibit –1 or +1 in the ground state, +3 in the 1st excited state, +5 in the 2nd excited state and +7 in the third excited state. Fluorine exhibit a fixed valency 1.
- The compounds formed by halogens with most electropositive metals are ionic. If the metal ions are small in size and have more charge, then their compounds are covalent according to Fajan's rule. Compounds with non metals are covalent.
- Oxidation power decreases from fluorine to iodine. Though the electron affinity of fluorine is less than chlorine, the oxidation power of fluorine is more than chlorine. It is the strongest oxidizing agent because of its low bond dissociation energy and high heat of hydration.

Chemical Reactivity of Halogens

- Fluorine is the most reactive element. The reactivity of halogens decreases from fluorine to iodine.
- Fluorine oxidizes water forming ozonized oxygen and HF.
- Chlorine dissolves in water and the solution is called chlorine water. Chlorine water contain HCl and HOCl. Chlorine water is a good oxidizing and bleaching agent due to the liberation of nascent oxygen from HOCl. Bromine is less soluble in water than chlorine. Bromine water is less powerful oxidizing and bleaching agent than chlorine water.
- The free energy change in the reaction of water and iodine is positive. So the reaction is not spontaneous but the HI can be oxidized to iodine and water by atmospheric oxygen.
- With cold dilute alkali fluorine forms OF_2 and fluoride, but with hot concentrated alkali liberates oxygen. Other halogens (Cl_2, Br_2 and I_2) with cold and dilute alkalis formes halide and hypohalite while with hot and concentrated alkali from halide and halates. Except fluorine other halogens disproportionate in alkalis.
- Halogens combines with **hydrogen** forming hydrogen halides. With hydrogen flourine react violently even in dark at –245°C. Cl_2 react in the presence of light violently. Br_2 react only at 300°C while the reaction with I_2 is possible at about 440°C in the presence of catalyst like platinum and further it is a reversible reaction, never goes to completion.
- **Boron** react with halogens forming **trihalide**. Only fluorine can react with carbon directly. Other halogens cannot react with carbon. All halogens react with

silicon forming silicon tetrahalides, with **phosphorous** form trihalide and pentahalides. Fluorine with **sulphur** forms SF_6 chlorine forms S_2Cl_2. Halogens react among themselves react to form interhalogen compounds.

- **Oxidation power of** halogens decreases from fluorine to iodine. Fluorine displaces all the other three halogens from their halides. Chlorine can displace bromine and iodine from bromides and iodides, bromine displaces iodine from iodide but iodine cannot displace other three halogens from their halides.

Anomalous Behaviour of Fluorine

- Fluorine shows some differences in the properties from the other halogens because (i) of its small size (ii) high electronegativity (iii) absence of d-orbitals in its valency shell and (iv) low F.F bond energy.
  (i) fluorine exhibits only one oxidation state i.e., –1 (ii) its maximum covalency is only one. (iii) its electron affinity is less than chlorine though it is more electronegative than chlorine. (iv) it can form hydrogen bonds. (v) it can bring about the maximum coordination number in other elements because of its small size and high electronegativity. (vi) because of its low bond dissociation energy it reacts with hydrogen explosively, can decompose water to oxygen and ozone, can combine with carbon directly, can react with alkali liberating OF_2 in cold and dilute condition and O_2 in hot and concentrated condition.
- Its hydracid HF is weak acid and exist in liquid state.
- Solubility of fluorides is just opposite to that of other halides e.g., AgF soluble but other silver halides are insoluble CaF_2 is insoluble but other calcium halides are highly soluble.
- Fluorine compounds are more stable than the corresponding compounds of other halides.

Fluorine

- Fluorine occurs in the nature in the form of fluorospar CaF_2, cryolite Na_3AlF_6 or $3NaF \cdot AlF_3$ and fluorapetite $CaF_2 \cdot 3Ca_3(PO_4)_2$.
- In the extraction of fluorine the main difficulties are
  (i) Fluorine is so reactive it attacks almost all materials in which it is prepared.
  (ii) Fluorine itself is the strongest oxidizing agent, So it cannot be prepared by chemical oxidation methods because a chemical oxidizing agent that can oxidize fluoride ion is not available.
  (iii) HF is quite stable, highly poisonous and the anhydrous HF is covalent and non conductor.
  (iv) Electrolysis of aqueous HF liberates O_2 and O_3 at anode instead of F_2.
- Fluorine was prepared by Moissan by the electrolysis of a cooled solution of KHF_2 in anhydrous HF using Pt-Ir alloy as electrodes and to made the electrolytic cell.
- In the preparation of fluorine graphite rods cannot be used as anodes because the fluorine liberated at anode invade between the layers due to which electrical conductivity decreases gradually and explosion may takes place.
- In the preparation of fluorine the anodes are made with ungraphitized carbon impregnated with copper and the valves are made with monel metal or nickel with teflon packing.
- It is a pale yellow gas, can be condensed to an yellow liquid.
- Fluorine combines with almost all non metals except nitrogen and oxygen directly.
- Fluorine also reacts with almost all metals including noble metals. Metals which exhibit variable valency form compounds in the higher oxidation state. Copper and mercury do not react with fluorine because the CuF_2 and HgF_2 formed initially on the surface of metal prevents the further reaction.
- Fluorine reacts with heavier inert gases like Kr and Xe to form compounds like XeF_2, XeF_4, XeF_6 etc.
- It is a strong oxidizing agent oxidizes all the other three halide (Cl^-, Br^-, I^-) to halogens, bisulphate to persulphate, chlorate to perchlorate, iodate to periodate, H_2S to SF_6.
- With ammonia when fluorine is excess NF_3 is formed but if ammonia is excess it is oxidized to N_2.
- Fluorocarbons called freons are used as refrigerants and polymer of tetrafluoro ethylene called teflon is used in making non stick frying pans and laboratory ware.
- The uranium isotopes ^{235}U and ^{238}U are separated by the diffusion of their gaseous fluorides $^{235}UF_6$ and $^{238}UF_6$.
- HF is used in etching of the glass. Silica present in glass reacts with HF and gives hydroflurosilicic acids.

Chlorine

- Chlorine was prepared by Scheele by oxidizing HCl with MnO_2. It can be prepared by oxidizing conc HCl with oxidizing agents such a potassium permanganate or potassium dichromate, lead dioxide, red lead, bleaching powder, sodium hypochlorite.
- Chlorine is manufactured in **Decon's** process by the oxidation of conc. HCl with air using cupric chloride as catalyst.
- In Weldon process chlorine is manufactured by the conc HCl with MnO_2. When $MnCl_2$ is treated with lime and air, calcium manganite ($CaMnO_3$) will be settles as dark coloured mud called Weldon mud with which HCl can be oxidized to Cl_2.

- Chlorine is obtained as a by-product during the manufacture of, sodium metal by Down's process, magnesium metal by the electrolysis of magnesium chloride and in the production of sodium hydroxide by Nelson's cell process or Castner-Kellner process or Kellner Solvay process.
- Pure chlorine can be prepared by heating dry auric chloride or platinic chloride.
- It is an yellowish green gas having pungent odour, poisonous in nature, soluble in water, crystallizes from solution as $Cl_2 \cdot 8H_2O$.
- The solution of chlorine in water is called chlorine water that contains HCl and HOCl. It acts as a strong oxidizing agent and bleaching agent.
- With cold dilute alkalis it gives chloride and hypochlorites and with hot concentrated alkalis gives chloride and chlorates.
- With dry slaked lime it gives bleaching powder.
- It reacts with several metals forming their chlorides. The metals which exhibit variable valency form chlorides in their highest oxidation state.
- It reacts with almost all non-metals except carbon, nitrogen and oxygen directly.
- With ammonia when chlorine is excess NCl_3 is formed but when ammonia is excess it oxidized to nitrogen.
- It forms addition compounds. With CO to give $COCl_2$ (carbonyl chloride or phosgene) with SO_2 gives sulphuryl chloride (SO_2Cl_2) and with NO gives nitrosyl chloride NOCl.
- It is a strong oxidizing agent and oxidizes several substances e.g., Ferrous sulphate to ferric sulphate, sulphite to sulphate, H_2S to sulphur.
- It has great affinity towards hydrogen and burns in hydrocarbons like $C_{10}H_{22}$, turpentine ($C_{10}H_6$) forming HCl and carbon.
- It is used (i) in the extraction of gold and silver, (ii) sterilizing the drinking water, (iii) in the manufacture of polymers like PVC, neoprene rubber (iv) in the manufacture of solvents like $CHCl_3$, CCl_4, ethylene dichloride (v) manufacture of bleaching powder, poisonous gases like mustard gas ($ClCH_2\ CH_2S\ CH_2\ CH_2\ Cl$), phosgene ($COCl_2$), tear gas ($CCl_3NO_2$).

Bromine

- Bromine is prepared by passing chlorine gas through the solutions of KBr or $MgBr_2$.
- On heating a mixture of KBr and MnO_2 with conc H_2SO_4 gives bromine. Acidification of KBr and $KBrO_3$ also gives bromine.
- Bromine is manufactured mainly from sea water which contains bromides of metals like Na, K, Mg etc.
- It is a reddish brown, heavy, mobile liquid, soluble in water but more soluble in organic solvents such as alcohol, ether, $CHCl_3$ and CCl_4. Cooling of aqueous solution of bromine gives crystals of $Br_2{\cdot}10H_2O$.
- Its chemical properties are almost similar to those of chlorine.

Iodine

- Iodine is mainly present in certain sea-weeds (laminaria species) as iodides and in caliche as $NaIO_3$.
- In the laboratory it is prepared by heating a mixture of sodium iodide and MnO_2 with conc H_2SO_4.
- It can also be prepared by adding any oxidizing agent such as H_2O_2, $KMnO_4$, $K_2Cr_2O_7$, $CuSO_4$ to KI solution.
- It is mainly manufactured from $NaIO_3$ present in caliche. This on heating with sodium bisulphite part of it is reduced to iodide which react with the remaining part of iodate liberating iodine.
- $2NaIO_3 + 5NaHSO_3 \longrightarrow 3NaHSO_4 + 2Na_2SO_4 + H_2O + I_2$
- If $NaHSO_3$ present in excess, HI is produced due to reduction by $NaHSO_3$.
- The impurities such as ICl, IBr, ICN etc can be removed by adding KI.
- The sea weeds are dried, burnt in shallow pits. The ash formed is called kelp from which the sulphates and chlorides are separated by dissolving in water followed by crystallization. This mother liquor on heating with MnO_2 liberates iodine.
- It is a dark violet solid having some metallic lustre, sublimes on heating yielding violet vapours, slightly soluble in water but more soluble in the water in the presence of KI due to the formation of KI_3 which behaves as a simple mixture of KI and I_2. It is highly soluble in organic solvents such as $CHCl_3$, CCl_4 and CS_2.
- With cold dilute alkalis it gives iodide and hypoiodite and with hot concentrated alkalis it gives iodide and iodate.
- It combines wih several metals forming iodides. The metals which can form more than one iodide form the iodide in lower oxidation state because of the reducing property of iodide ion.
- It combines with several non metals directly forming compounds such HI, PI_3 etc.
- It is a good oxidizing agent but less powerful oxidizing agent than the other halogens. It oxidizes H_2S to S; SO_2 to H_2SO_4; nitrite to nitrate, sulphite to sulphate aresenite to arsenate, ferrous sulphate to ferric sulphate.
- With ammonia when iodine is excess forms $NI_3\ NH_3$ which is an explosive powder. This explodes forming N_2, I_2 and NH_4I.

- I_2 can substitute Cl from $KClO_3$ and Br from $KBrO_3$ because I_2 is more electropositive than Cl_2 and Br_2.
- I_2 quantitatively oxidizes sodium thiosulphate (hypo) to sodium tetrathionate. This reaction is used in iodometric and iodimetric estimations using starch as indicator.
- Iodine is oxidized to HIO_3 by conc HNO_3 and with ozone in the presence of water. But ozone oxidizes the dry iodine to I_4O_9. Chlorine water also oxidizes iodine to iodic acid.
- Iodine reacts with freshly precipitated mercuric oxide forming hypoiodous acid.
- Iodine gives blue colour with starch.
- Iodine is used in the manufacture of iodoform, iodole which are used as an antiseptic and analgesic.
- Tincture of iodine is a mixture of KI + I_2 + alcohol (rectified spirit or methylated spitric).
- Deficiency of iodine causes a disease called goitre. For good health common salt is mixed with sodium iodide. To cure thyroid radioactive iodine is used.

Hydrides

- Hydrogen halides can be prepared by the direct reaction between halogens and hydrogen. The reactivity of halogen's towards hydrogen decreases from F_2 to I_2.
- HF and HCl can be prepared by heating an ionic fluoride or chloride with conc H_2SO_4. But HBr and HI cannot be prepared by similar method because the HBr and HI are oxidized by conc H_2SO_4 to Br_2 and I_2 respectively. HBr and HI can be prepared by heating the bromides and iodides with phosphoric acid.
- Hydrogen halides are also formed by the hydrolysis of phosphorous trihalides in water.
- HCl, HBr and HI are formed by passing H_2S gas through the aqueous solutions of Cl_2, Br_2 or I_2.
- Industrially HF is produced by the action of conc H_2SO_4 on CaF_2, HCl is produced by burning H_2 in Cl_2. Considerable amounts of HCl is formed as a by - product during many chlorination reactions such as chlorination of hydrocarbons. The solution of HCl in water is called hydrochloric acid.
- HCl, HBr and HI are **gases** but HF is a liquid due to association of molecules by hydrogen bonding.
- **Polarity** and **dipolemoments** decreases from HF to HI. When dissolved in water they ionizes and behave as acids but when dissolved in non-ionizing solvents such as toluene they behave as covalent compounds, do not ionize, do not conduct electricity and will not affect the blue litmus.
- **Thermal stability** decreases from HF to HI due to decrease in bond energy with increase in bond length as the atomic size increases from F to I.
- The hydrogen halides acts as reducing agents and the reduction power increases from HF to HI, because of the decrease in thermal stability or electron holding capacity of halogens decreases from F to I.
- Acidic character increases from HF to HI. Though HF is more polar, it acts as a weak acid because of high H - F bond energy, hydration energy and low electron affinity of fluorine and due to association of molecules by hydrogen bonding.
- The aqueous solutions of different halogen acids form azeotropic mixtures because of more attraction towards water molecules.
- HF is unique in forming acid salts having difluoride ion $[HF_2]^-$. This is because in aqueous solution HF exists as H_2F_2 and ionizes to gives H^+ and $[HF_2]^-$ ions.
- HF do not give precipitate with silver salts because AgF is soluble but other hydrogen halides gives precipitates with $AgNO_3$.
- HF form white precipitate of BaF_2 when added to $BaCl_2$ but other hydrogen halides do not give precipitates with $BaCl_2$.
- HF is extremely stable and cannot be oxidized. HF reacts with silica or glass but other hydrogen halides do not react with silica or glass.

Hydrochloric Acid

- In the laboratory hydrochloric acid is prepared by heating sodium chloride with conc H_2SO_4.
- Industrially it is manufactured by burning hydrogen in chlorine.
- It is obtained as a by product in several process as explained 11.11.1.
- It is a colourless, pungent smelling gas having acidic taste.
- It has more affinity towards water and collects the water molecule when exposed to moist air and form small droplets which appear as white fumes.
- With water it forms an azeotropic mixture with a composition of 22.2 per cent acid. Hence 100 per cent concentrated acid cannot be prepared by distillation.
- Perfectly dry gas has no effect on litmus but in moist state or in solution it turns the blue litmus to red.
- With several metals it liberate H_2. The oxides, hydroxides, carbonates and bicarbonates of metals react with hydrochloric acid forming their chlorides.
- It is highly stable and can be oxidized only by very strong oxidizing agents such as MnO_2, $KMnO_4$, $K_2Cr_2O_7$, PbO_2, Pb_3O_4 etc.
- With $AgNO_3$ it forms white curdy precipitate soluble in ammonia due to the formation of complex $[Ag(NH_3)_2]Cl$.

- With a solution of lead salt it give white precipitate of $PbCl_2$ soluble in hot water but crystallizes on cooling.
- With ammonia it gives dense white fumes of NH_4Cl.
- Hydrochloric acid decomposes salts of weaker acids such as carbonates, bicarbonates, sulphides, sulphites, thiosulphates and nitrites.

Oxides

- Chlorine forms large number oxides where as iodine forms the least.
- The halogen - oxygen (X-O) bonds are mainly covalent because of the small difference in electronegativity between halogen and oxygen.
- Polarity of X-O bond increases from Cl-O to I-O.
- Iodine oxides are most stable followed by that of chlorine whereas bromine oxides are least stable. I-O bond is stable because of more polarity of the bond while the Cl-O bond is stable because of multiple bonds involving the d-orbitals of chlorine atom. Due to middle row anomaly Br-O bond is least stable.
- Oxides in the higher oxidation state are more stable than in lower oxidation state.
- The higher oxides of I_2O_4 and I_4O_9 are ionic and salt like in character

$$[I_2O_4 = IO^+ IO_3^- \text{ and } I_2O_9 = I^{3+}(IO_3)_3^-].$$

- Here iodine exhibits a +3 oxidation state in cation and +5 oxidation state in the anions.

(A) Monoxides

- Cl_2O and Br_2O can be prepared by treating the freshly prepared mercuric oxide with Cl_2 or Br_2 at 273K.
- Cl_2O is an orange coloured gas, Br_2O is a dark brown liquid and forms corresponding hypohallous acids HOCl and HOBr when dissolved in water. They act as oxidizing acids.
- Both Cl_2O and Br_2O are angular in shape. Oxygen atoms are involved in sp^3 hybridization. Bond angle in both Cl_2O and Br_2O is 111°. The larger bond angles than tetrahedral angles is due to the participation of lone pair of electrons on oxygen in π dative bond with the vacant d-orbitals of chlorine which causes the opening of bond angle.

(B) Dioxides

- ClO_2 can be prepared by reacting chlorine with sodium chlorite or silver chlorate. It can be also prepared by heating $KClO_3$ with conc H_2SO_4 or with oxalic acid.
- ClO_2 is a powerful oxidizing agent and chlorinating agent. It bleaches 30 times more than Cl_2.
- ClO_2 reacts with alkalis to yield chlorites and chlorates and hence considered as mixed anhydride of chlorous ($HClO_2$) and chloric ($HClO_3$) acids.
- It is an odd electron molecule, paramagnetic in nature, angular in shape, containing a 3 electron bond coupled with usual two electron covalent bond.
- Though ClO_2 is an odd electron molecule it has least tendency to dimerize, as the unpaired electron is less localized on the central atom due to strong electronegativity of oxygen.
- BrO_2 can be prepared by passing an electric discharge through a mixture of Br_2 and O_2 at low temperatures.
- With alkalis it gives bromites and bromates.

(C) Higher Oxides

- Cl_2O_6 can be prepared by the action of ozone on ClO_2.
- It is a reddish oily liquid, unstable explosive liquid. With alkalis it forms a mixture of chlorates and perchlorates. So it is considered as mixed anhydride of chloric ($HClO_3$) and perchloric ($HClO_4$) acids.
- It is considered that ClO_3 and Cl_2O_6 are in equilibrium. Its structure is not established correctly.
- With HF and N_2O_4 it reacts as follows.

$$Cl_2O_6 + HF \longrightarrow FClO_2 + HClO_4$$

$$2Cl_2O_6 + N_2O_4 \longrightarrow 2ClO_2 + 2[NO_2]^+ [ClO_4]^-$$

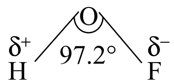
- Cl_2O_7 can be prepared by the dehydration of pure perchloric acid with P_4O_{10}.
- The only exothermic oxide of Cl_2 is Cl_2O_7. It is the most stable oxide of chlorine. It gives perchloric acid with water. It is the true anhydride of $HClO_4$.
- In Cl_2O_7 two ClO_3 units are linked through an oxygen at angle 118.6°. Each chlorine gets tetrahedrally linked with four oxygen atoms. Bridge Cl-O bond lengths are longer than terminal Cl-O bond lengths.
- I_2O_5 is only true oxide of iodine which can be prepared by dehydration of iodic acid.
- It is a white solid and is the most stable of all oxides. But decomposes into elements at 300°C. When dissolved in water is gives iodic acid. So it is considered as the anhydride of iodic acid. It is a strong oxidizing acid and oxidizes CO, H_2S, HCl etc.

Oxoacids of Halogens

- HOF, HOCl, HOBr and HOI are known and the halogen is in +1 oxidation state.
- HOF is formed by passing F_2 over ice at 0°C and removing the product into a cold trap.

$$F_2 + H_2O \longrightarrow HF + HOF$$

- HOF is unstable and decomposes on its own to HF and O_2. It is a strong oxidizing agent and oxidizes H_2O to H_2O_2.
- HOF is **non linear (angular)** molecule with a bond angle 97.2° which is smallest known bond angle at sp^3 hybrid 'O' atom due to attraction between +ve charge on 'H' atom and –ve charge on 'F' atom.



- HOCl, HOBr and HOI are thermally unstable, cannot be isolated in pure state. They can be prepared by shaking freshly precipitated HgO in water with desired halogen.

$$2\,HgO + 2\,X_2 + H_2O \longrightarrow HgO\,HgX_2 + 2\,HOX$$

- The hypohalous acids and their salts hypohalides are unstable and tend to disproportionate at elevated temperatures.

$$3\,OX^- \longrightarrow 2X^- + XO_3^-$$

- The only **halous acid** known is chlorous acid which can be prepared by the action of sulphuric acid on barium chlorite.
- It exist only in solution and disproportionate on heating to $HClO_3$ and HCl.
- $HClO_3$ and $HBrO_3$ exist in solution. HIO_3 can be prepared in solid state. $HClO_3$ and $HBrO_3$ can be prepared by the action of sulphuric acid on barium chlorate or barium bromate.
- HIO_3 can be prepared by the oxidation of iodine with conc HNO_3.
- Perchloric acid can be prepared by adding conc H_2SO_4 to $NaClO_4$. $NaClO_4$ can be obtained by electrolytic oxidation of $NaClO_3$.
- Perchloric acid exist as dimer due to hydrogen bonding.
- Meta periodic acid HIO_4, para periodic acid H_5IO_6 and their salts are made by oxidizing I_2 or I^- in aqueous solution.
- Structural aspects and bond parameters of oxoacids of chlorine and their anions are given in Table 11.11 and 11.12.
- Orders of different oxoanions for corresponding properties are as follows.

Order of bond order $ClO^- < ClO_2^- < ClO_3^- < BrO_4^-$
Order of bond energy $ClO^- < ClO_2^- < ClO_3^- < BrO_4^-$
Order of bond length $ClO^- > ClO_2^- > ClO_3^- > BrO_4^-$
Thermal stability $ClO^- < ClO_2^- < ClO_3^- < BrO_4^-$
Oxidation power $ClO^- > ClO_2^- > ClO_3^- > BrO_4^-$
Basic character $ClO^- > ClO_2^- > ClO_3^- > BrO_4^-$
Acidic character of
oxo acids $HOCl < HClO_2 < HClO_3 < HClO_4$

- For oxo acids of different halogens in the same oxidation state is HOF > HOCl < HOBr > HOI. This is due to decrease in electronegativity of halogen.
- The order of acidic character of halic acids is

$$HClO_3 \approx HBrO_3 < HIO_3.$$

- In aqueous solution HIO_4 is weaker acid than $HClO_4$ because HIO_4 converts into H_5IO_6 due to hydration which decrease the number of unprotonated oxygen atoms.
- Thermal stability of halates is in the order

  iodate > bromate > chlorate.
- The oxidation power of halate ions is

  Bromate > Chlorate > Iodate. But the rate of reactions follow the order, Iodate > Bromate > Chlorate.
- The oxidation power and rate of reactions for perhalate ions follow the order $BrO_4^- > IO_4^- > ClO_4$.

Bleaching Powder

- **Bleaching powder** is a mixture of calcium hypochlorite. [$Ca(OCl)_2$ $3H_2O$] and basic calcium chloride [$CaCl_2 \cdot Ca(OH)_2 \cdot H_2O$].
- Bleaching powder is manufactured by passing dry chlorine gas over dry slaked lime.
- Bleaching powder is manufactured in Hasenclover's method or in Beckmann's plant.
- The Odling formula of bleaching powder is Ca(Cl)OCl, but now it has been shown that it is a mixture of calcium hypochlorite and basic calcium chloride. Here the chemical properties are explained by taking Odling formula.
- Bleaching powder is an yellowish white powder having smell like chlorine. It is soluble in water, the residue being unreacted lime.
- In cold water it ionizes into Ca^{2+}, Cl^- and ClO^- ions but in hot water ClO^- ion disproportionates to Cl^- and ClO_3^-.
- With insufficient (less amount) acids bleaching powder forms hypochlorous acid which gives nascent oxygen on decomposition. So it acts as oxidizing and bleaching agent.
- With excess dilute acids bleaching powder liberates chlorine. The amount of chlorine liberated by the action of excess acids in bleaching powder is called available chlorine. The quality of bleaching powder is decided depending on available chlorine. Good quality of bleaching powder contain 35–38 per cent **available chlorine**.

- Bleaching powder liberates Cl_2 by absorbing CO_2 but liberates oxygen in the presence of $CoCl_2$ catalyst.
- On long standing it undergoes auto oxidation (dis proportionation) and converts into calcium chloride and calcium chlorate. On long standing the quality of bleaching powder decreases due to the decrease in available chlorine because of auto-oxidation.
- Bleaching powder oxidizes lead salts to lead dioxide.
- Bleaching powder reacts with ethyl alcohol, acetaldehyde and acetone forming chloroform.
- Bleaching powder is used (i) in the bleaching of wood pulp, cloth (ii) for sterilization of drinking water (ii) in the manufacture of chloroform used in medicine as an anaesthetic agent.

Inter Halogen Compounds

- The binary compounds formed by halogens among themselves are known as **interhalogen compounds**.
- Inter halogen compounds are four types XY; XY_3; XY_5 and XY_7 where Y is always lighter halogen with more electronegativity.
- An interhalogen compound never contain more than two different types of halogens. All inter halogen compounds are diamagnetic. XY and XY_3 type compounds are formed when the electronegativily difference and radius ratio values are small. XY_5 and XY_7 type compounds are formed if the electro-negativity difference and radius ratio values are large.
- Inter halogen compounds can be prepared by the direct reaction between different halogen by maintaining suitable conditions.
- More the electronegativity difference more stronger the inter halogen bond X-Y because ionic character of the bond also increases.
- The lower oxidation states of halogens where less electronegative halogen is present as central atom, are unstable and disproportionates e.g - BrF disproportionates to Br and BrF_3; IF disproportionaes to I and IF_5.
- Inter halogen compounds are more reactive than halogens except fluorine because X-Y bond in interhalogen compounds is weaker than the X-X bond in halogens. Also interhalogen compounds are polar while halogens are non-polar.
- All the interhalogens are strong oxidizing agents.
- Interhalogen compounds hydrolyse in water forming halide and oxyhalide ions. The oxyhalide is generally formed from the larger halogen. The oxidation state of larger halogen does not change during hydrolysis.
- **XY type** inter halogen compounds are **linear** in shape.
- In **XY3 type** inter halogen compounds X is involved in sp^3d hybridization and have trigonal bipyramidal geometry of electron pairs, with two lone pairs occupying the equatorial positions. So the XY_3 type inter halogen compounds have T shape and the bond angles are less than 90° due to repulsion by lone pairs.
- In **XY5 type** inter halogen compounds X is in sp^3d^2 hybridization and have octahedral shape with one corner occupied by a lone pair. So XY_5 type inter halogen compounds have square pyramidal structure in which the eight YXY bond angles are less than 90°.
- In XY_7 type interhalogen compounds X is involved in sp^3d^3 hybridization and have pentagonal bipyramidal structure.
- ICl_3 exist as dimer I_2Cl_6 in which around each iodine four chlorine atoms are arranged in square planar structure while the two lone pairs occupy the opposite corners of the octahedron.

Pseudohalogens

- A molecule consisting of more than two electronegative atoms which in free state resemble the halogens is called **pseudohalogen**. The pseudohalogens give rise to anions which resemble the halide ions in their behaviour are called **pseudohalide ions**.
- Pseudohalide ion is also defined as any univalent chemical aggregate composed of two or more electronegative atoms which behave similar to halogens and which combine with hydrogen to form an acid and with silver salt insoluble in water.
- Examples of pseudohalide ions are CN^-, SCN^-, $SeCN^-$, $TeCN^-$, $SCSN_3^-$, OCN^-, N_3^- and examples for pseudohalogens are $(CN)_2$, $(SCN)_2$, $(SeCN)_2$, $(TeCN)_2$, $(SCSN_3)_2$, $(OCN)_2$.
- Similarities between halogens and pseudohalogens are
  (i) Both are volatile, isomorphous in their structures and can form pseudo interhalogen compounds like pseudohalogens.
  (ii) Like halogens pseudohalogens also combine with metals to form salts like NaCN, NaSCN etc.
  (iii) Like the halides of Ag, Hg(I) and Pb(II) the pseudo halides of these metals are insoluble in water.
  (iv) Like halogens pseudohalogens also combine with hydrogen to form monobasic acids which are weak acids.
  (v) Similar to halogens pseudohalogens can also be prepared in the free state by oxidizing their hydracids or their salts.
  (vi) Like halogens pseudohalogens also dispropor-tionate in alkalis eg: $(CN)_2$ disproportionate into CN^- and CNO^- in alkalis.
  (vii) Similar to halogens, pseudohalogens also add at the double or triple bonds in unsaturated hydrocarbons.

Polyhalides

- The univalent ions formed by the association of a halide ion with the molecules of halogens or inter halogens are known as **polyhalide ions** and the compounds of these are known as **polyhalides**.
- Polyhalides are formed by the bigger cations like alkali metal ions or alkaline earth metal ions or with coordinate complex ions.
- Polyhalides are formed more by iodides than the other halides.
- The tendency to form polyhalide increases with decreasing the lattice energy in the orginal halide.
- Polyhalides do not exhibit isomerism.
- X_n^- type polyhalides are PbI_3, TlI_3, CsI_3, KI_3 etc in which polyhalide ion contain same type of halogen atoms XX_n^- contain two different halogens e.g.: IBr_2^-, ICl_2^-, $ClBr_2^-$.
- XX_n^- contain two different types of halogens.
- $XX'X_n''$ contain, two or three different type of halogens. These are less common e.g.: Cs $FIBr_3$, $RbFICl_3$ KClIBr.
- Polyhalides can be prepared by the direct addition of halogen to halide or a halide and inter halogen compound.
- Polyhalides are generally coloured, with low melting points. They are typically ionic compounds.
- On heating a polyhalide the metal halide with small halide ion which gives more lattice energy will be formed. For example on heating $RbICl_2$ gives RbCl and ICl.
- For a given polyhalide ion the order of stability increases with increase in size of cation
  $Na^+ < K^+ < Rb^+ < Cs^+$.
- For a given metal ion the stability of polyhalide increases in the order.

$[Br\ I]^- < [F\ l\ Br]^- < [ClBrCl]^- < [I_3]^- < [BrIBr]^- < [ClICl]^-$

- Fluorine cannot form polyhalide ion like F_3^-.

Single Answer Questions

1. Which of the following is correct order of acidic strength?
   (a) HOCI > HOBr > HOI
   (b) HOBr > HOCI > HOI
   (c) HOI > HOBr > HOCI
   (d) HOBr > HOI > HOCI
2. F_2 and CI_2 are soluble in water because
   (a) F_2 and CI_2 are small molecules and can be accommodated in the voids of H_2O
   (b) F_2 reacts with H_2O giving HF and O_2 and CI_2 gives HOCl and HCl
   (c) F_2 reacts with H_2O giving HOF while Cl_2 gives HCI and HOCl
   (d) F_2 reacts with H_2O to form HOF whereas CI_2 reacts with H_2O to give HOCI
3. A greenish yellow gas reacts with an alkali metal hydroxide to form a halate which can be used in fire works and safety matches, the gas and halate respectively are
   (a) Br_2, $KBrO_3$ (b) Cl_2, $KCIO_3$
   (c) l_2, $NaIO_3$ (d) Cl_2, $NaCIO_3$
4. When dry chlorine is passed through silver chlorate heated to 90ºC, then which of the oxides of chlorine is obtained?
   (a) CIO_2 (b) CI_2O
   (c) CI_2O_3 (d) CI_2O_5
5. When any chloride is heated with conc. H_2SO_4, then HCl is obtained while when any bromide is heated with conc. H_2SO_4, then Br_2 is obtained and not HBr. This is because
   (a) HCI is a gas while HBr is a liquid
   (b) HCI is a liquid while HBr is a gas
   (c) HBr is a stronger reducing agent than HCI
   (d) Br_2 is a liquid and HCI is a gas.
6. The radius of the *CI–* ion is 38 per cent larger than that of the *F–* ion but the radius of the *Br–* ion is only 6.5 per cent larger than that of the *CI–* ion. The relatively small difference in size between *CI–* and *Br–* ions is due to the fact that
   (a) the Br^- ion contains ten 3d electrons, which fail to shield the nuclear charge effectively.
   (b) the Br^- ion contains ten 3d electrons, which shield the nuclear charge effectively.
   (c) the Br^- ion contains six 4p electrons, which shield the nuclear charge effectively.
   (d) the Br^- ion contains ten 3d electrons and six 3p electrons, together they shield the nuclear charge effectively.
7. In spite of having small dissociation energies, bromine and iodine are weaker oxidizing agents than chlorine due to their
   (a) smaller electron affinities and greater hydration energies.
   (b) smaller electron affinities and smaller hydration energies.
   (c) greater electron affinities and greater hydration energies.
   (d) greater electron affinities and smaller hydration energies.
8. A mixture of Nal and $NalO_3$ is treated with dilute H_2SO_4. The iodine-containing product formed is
   (a) HIO_3 (b) $NaIO_4$ (c) I_2 (d) HIO_4
9. The structure of OF_2 is
   (a) tetrahedral with one corner occupied by a lone pair of electrons.

(b) tetrahedral with two corners occupied by two lone pairs of electrons.
(c) square planar with two corners occupied by two lone pairs of electrons.
(d) pentagonal bipyramid with three corners occupied by three lone pairs of electrons.

10. Which of the following arrangements gives the correct order of increasing basic character of the conjugate bases of the oxoacids of chlorine?
(a) $ClO_3^- < BrO_4^- < ClO_2^- < ClO^-$
(b) $BrO_4^- < ClO_3^- < ClO^- < ClO_2^-$
(c) $ClO^- < ClO_2^- < ClO_3^- < BrO_4^-$
(d) $BrO_4^- < ClO_3^- < ClO_2^- < ClO^-$

11. Iodine is obtained commercially from chile salt petre. Which of the following reactions is used in the process?
(a) $IO_3^- + 5I^- + 6H^+ \longrightarrow 3I_2 + 3H_2O$
(b) $2I^- + H_2O_2 \longrightarrow I_2 + 2OH^-$
(c) $2IO_3^- + 5HSO_3^- \longrightarrow 3HSO_4^- + 2SO_4^{2-} + H_2O + I_2$
(d) $2KI + Cl_2 \longrightarrow I_2 + 2KCl$

12. The ions of oxyacids of halogens are stabilized due to
(a) pπ–dπ bonding between full 2p orbitals on oxygen with empty d orbitals on the halogen atom
(b) pπ–pπ bonding between full p orbitals on the halogen atom
(c) pπ–pπ bonding between full 2p orbitals on oxygen with p orbitals on the halogen atom
(d) dπ–dπ bonding between d orbitals on oxygen with full 2p orbitals on the halogen atom

13. Which pair of products are formed when fluorine is passed through the aqueous solution of potassium chlorate
(a) $KFO_3 + Cl_2$ (b) $KF + O_2 + Cl_2$
(c) $KHF_2 + Cl_2O$ (d) $KClO_4 + H_2F_2$

14. The main impurities present in commercial iodine are ICl, IBr, ICN etc. Which of the following reagent is generally used for its purification
(a) Sodium thiosulphate
(b) Potassium bisulphate
(c) Potassium iodide
(d) Potassium iodate

15. Select correct statement regarding behaviour of HF as non-aqueous solvent
(a) HCl behaves as an acid and HF as a base
(b) $HClO_4$ behaves as a base and HF as an acid
(c) HNO_3 and H_2SO_4 behave as base and HF as an acid
(d) All are correct statements

16. Which of the following represents the correct order of increasing pKa values of the given acids
(a) $HClO_4 < HNO_3 < H_2CO_3 < B(OH)_3$
(b) $HNO_3 < HClO_4 < B(OH)_3 < H_2CO_3$
(c) $B(OH)_3 < H_2CO_3 < HClO_4 < HNO_3$
(d) $HClO_4 < HNO_3 < B(OH)_3 < H_2CO_3$

17. A dark violet solid (X) reacts with NH_3 to form a mild explosive which decomposes to give a violet coloured gas. (X) also reacts with H_2 to give an acid (Y). (Y) can also be prepared by heating its salt with H_3PO_4 (X) and (Y) are
(a) Cl_2, HCl (b) SO_2, H_2SO_4
(c) Br_2, HBr (d) I_2, HI

18. Kelp is a name given to
(a) Mother liquor left after the crystallization of KCl from carnalite
(b) Mother liquor left after the crystallization of sodium nitrate from caliche
(c) Ash left after burning dried sea weeds in shallow pits
(d) Ash left after burning leguminous variety plants in shallow pits

19. $HCl \underset{4}{\overset{1}{\rightleftharpoons}} Cl_2 \underset{3}{\overset{2}{\rightleftharpoons}} KClO_3$
In the above sequence of reactions, the step 2 and 3 require following reagents respectively?
(a) NaOH, I_2 (b) HCI, I_2
(c) KOH, I_2 (d) KOH, O_2

20. Which of the following is incorrectly matched with its property?
(a) Electron affinity – Cl > F > Br > I
(b) Bond dissociation energy –$Cl_2 > F_2 > Br_2 > I_2$
(c) M.P and B.P – $F_2 < Cl_2 < Br_2 < I_2$
(d) Intensity of colour –$F_2 < Cl_2 < Br_2 < I_2$

21. Incorrect statement of following:
(a) Higher oxidation states of halogens are found in inter halogen compounds, oxides and oxoacids
(b) Density of halogens decreases with increase of atomic numbers
(c) Frequency (v) of excitation energy of halogens decreases down the group
(d) In halogens (except fluorine) some multiple bonding may be present according to Mulliken

22. The oxidation states of second most electronegative atom in the products obtained on passing F_2 gas through dil NaOH
(a) 0, –2 (b) 0, +2
(c) –2, +2 (d) –2 only

23. When chlorine reacts with hot and con alkali, it disproportionates to chloride and oxo anion. Which of following is correct regarding that anion?
(a) It is the weakest base in aq. solution
(b) Central atom is sp^2 hybridized
(c) It possesses least bond angle among oxo anions of chlorine
(d) It possesses three σ bond pairs and three π bond pairs

24. Following energy changes may be observed in reactions in which halogens acts as oxidizing agents
(a) ΔH_{fusion} (b) $\Delta H_{vapourization}$
(c) $\Delta H_{dissociation}$ (d) Electron affinity
(e) $\Delta H_{hydration}$

Which is incorrect regarding energy changes?

(a) Fluorine –c, d, e (b) Chlorine –b, c, d, e

(c) Bromine –b, c, d, e (d) Iodine –a, b, c, d, e

25. Which of the following is incorrect regarding isolation of fluorine

(a) Aqueous hydrogen fluoride is non-conductor of electricity

(b) HF cannot be readily oxidized to F_2

(c) Fluorine reacts with all materials used in its preparation

(d) Reduction potential of fluorine is very high

26. $Cl_2 + F_2$ (excess) $\xrightarrow{573\text{ K}}$ A (inter halogen compound)

Which of following is correct regarding A?

(a) It is non-polar molecule

(b) It is pyramidal molecule with 2 LP and sp^2 hybridization

(c) It is trigonal planar with 2LP on axial position and sp^3d hybridization

(d) It is trigonal bipyramidal with 2LP on equatorial position and sp^3d hybridization

27. Consider A, B, C in which case(s) O_2 evolves

A. $CaOCl_2 + H_2SO_4$ (insufficient)

B. $CaOCl_2 + H_2SO_4$ (Excess)

C. F_2 + con. NaOH

(a) Only A (b) A & B only

(c) A & C only (d) All A, B, C

28. $2KX + X_2' \longrightarrow 2KX' + X_2$

X_2 and X_2' can not be respectively

(a) F_2, Cl_2 (b) I_2, Br_2

(c) I_2, Cl_2 (d) Br_2, Cl_2

29. Which of the following reaction is incorrect?

(a) $H_2S + 4F_2 \longrightarrow SF_6 + 2HF$

(b) $2Fe + 2Cl_2 \longrightarrow 2FeCl_2$

(c) $2KHSO_4 + F_2 \longrightarrow K_2S_2O_8 + 2HF$

(d) $NH_3 + 3F_2 \longrightarrow NF_3 + 3HF$

30. Which of the following reaction is not possible?

(a) $CaOCl_2 \xrightarrow{CoCl_2} CaCl_2 + O_2$

(b) $PbO \xrightarrow[+H_2O]{+CaOCl_2} PbO_2$

(c) $2CCl_3CHO + Ca(OH)_2 \longrightarrow 2CHCl_3 + Ca(HCOO)_2$

(d) $CaOCl_2 \xrightarrow{\text{auto - Oxidation}} CaCl_2 + O_2$

31. Which one of the following arrangements does not truly represent the property indicated against it?

(a) $ClO_2^- > BrO_4^- > ClO_3^- \longrightarrow$ bond angle

(b) $OCl^- > ClO_2^- > ClO_3^- > BrO_4^- \longrightarrow$ Cl—O bond energy

(c) $OCl^- > ClO_2^- > ClO_3^- > BrO_4^- \longrightarrow$ Cl—O distance

(d) $OCl^- < ClO_2^- < ClO_3^- < BrO_4^- \longrightarrow$ Chlorine oxidation state

32. ICl_3 exist as dimer I_2Cl_6. In this molecule

(a) All atoms are in the same plane with two chlorine bridges

(b) All atoms are in the same plane with I—I bond

(c) The two I atoms and terminal Cl atoms are in one plane while the remaining two Cl atoms are one above and below the plane

(d) It contains two 3 centre two electron bonds

33. At ordinary temperature and pressure, among halogens, the chlorine is a gas, bromine is a liquid and iodine is a solid. This is because

(a) The specific heat is in the order $Cl_2 > Br_2 > I_2$

(b) Intermolecular forces among molecules of chlorine are the weakest and those in iodine are the strongest

(c) The order of density is $I_2 > Br_2 > Cl_2$

(d) The order of stability is $Cl_2 > Br_2 > I_2$

34. Arrange ClO_3^-, BrO_3^- and IO_3^- ions in the increasing order of their bond angles

(a) $ClO_3^- > BrO_3^- > IO_3^-$

(b) $IO_3^- > BrO_3^- > ClO_3^-$

(c) $ClO_3^- > IO_3^- > BrO_3^-$

(d) $BrO_3^- > ClO_3^- > IO_3^-$

35. The correct order of increasing thermal stability

(a) HI < HBr < HF < HCl

(b) HI < HCl < HF < HBr

(c) HI < HBr < HCl < HF

(d) HF < HBr < HCl < HI

36. $Cl_2(g) + Ba(OH)_2 \longrightarrow X(aq) + BaCl_2 + H_2O$

$X + H_2SO_4 \longrightarrow Y + BaSO_4$

$Y \xrightarrow[\Delta > 365\text{ K}]{\Delta} Z + H_2O + O_2$

Y and Z are respectively:

(a) $HClO_4, ClO_2$ (b) $HClO_3, ClO_2$

(c) $HClO_3, Cl_2O_6$ (d) $HClO_4, Cl_2O_7$

37. O_2F_2 is an unstable yellow orange solid and H_2O_2 is colourless liquid, both have O—O bond. O—O bond length in H_2O_2 & O_2F_2 is respectively

(a) 1.22°*A*, 1.48°*A* (b) 1.48°*A*, 1.22°*A*

(c) 1.22°*A*, 1.22°*A* (d) 1.48°*A*, 1.48°*A*

38. The stability of interhalogen compounds:

(a) $BrF_3 > ClF_3 > IF_3$ (b) $BrF_3 > IF_3 > ClF_3$

(c) $ClF_3 > BrF_3 > IF_3$ (d) $ClF_3 > IF_3 > BrF_3$

39. Cl_2 cannot be prepared from

(a) MnO_2 + HCl(conc) (b) $KMnO_4$ + HCl(conc)

(c) O_3 + HCl (d) KCl + Br_2

40. Which of the following reactions show oxidizing property of chlorine?
(a) $FeSO_4 + Cl_2 + H_2SO_4 \longrightarrow Fe_2(SO_4)_3 + HCl$
(b) $Na_2SO_3 + Cl_2 + H_2O \longrightarrow Na_2SO_4 + 2HCl$
(c) $Na_2S_2O_3 + 4Cl_2 + 5H_2O \longrightarrow 2NaHSO_4 + 8HCl$
(d) All of these

41. Which of the following is the correct representation of the physical properties of halogens?
(a) Low ionization energy, high electron affinity and high electronegativity.
(b) High ionization energy, low electron affinity and high electronegativity.
(c) High ionization energy, high electron affinity and low electronegativity.
(d) High ionization energy, high electron affinity and high electronegativity.

42. Though ClO_2 is an odd-electron molecule, it does not undergo dimerization because
(a) the odd electron is delocalized
(b) the odd electron is not delocalized
(c) It is a bent molecule (V-shaped)
(d) the O—Cl bonds are shorter than expected for a single bond

43. Which of the following statements is incorrect regarding the structure of the ClO_2 molecule?
(a) The ClO_2 molecule is angular with O—Cl—O bond angle being 118°
(b) The two Cl—O bonds lengths are equal.
(c) Both Cl—O bond lengths are greater than expected for a single Cl—O bond.
(d) Both Cl—O bond lengths are shorter than expected for a single Cl—O bond.

44. Which of the following is correctly arranged in increasing order of hydration energy?
(a) $ClO_4^- < ClO_3^- < ClO^- < ClO_2^-$
(b) $ClO_4^- < ClO_3^- < ClO_2^- < ClO^-$
(c) $ClO_4^- < ClO^- < ClO_2^- < ClO_3^-$
(d) $ClO^- < ClO_2^- < ClO_3^- < ClO_4^-$

45. HBr can be prepared by the action of following on KBr
(a) H_2SO_4 (b) H_2O_2 (c) HNO_3 (d) H_3PO_4

46. The colour of halogens progressively deepens from fluorine to iodine because
(a) halogens of higher atomic number absorb light of longer wavelength since the difference in energy between the frontier orbitals decreases as the atomic number increases
(b) fluorescence and phosphorescence become more intense as the atomic number of halogen increases
(c) the standard electrode potential increases from I_2 to F_2
(d) halogens of higher atomic number absorb light of shorter wavelength since the difference in energy between the frontier orbitals increases as the atomic number increases

47. Which of the following pairs of halogens have approximately identical bond energy
(a) F_2 and Cl_2 (b) Cl_2 and Br_2
(c) F_2 and Br_2 (d) F_2 and I_2

48. The standard reduction potentials of the halogens are in the order
(a) $F_2 > Cl_2 > I_2 > Br_2$
(b) $Cl_2 > F_2 > I_2 > Br_2$
(c) $I_2 > Br_2 > Cl_2 > F_2$
(d) $F_2 > Cl_2 > Br_2 > I_2$

49. Among the following which acts as the strongest acid in reactions where HNO_3 behaves like a base.
(a) HI (b) H_3PO_3 (c) HF (d) HIO_4

50. Arrange CCl_4, $AlCl_3$, PCl_5 and $SiCl_4$ according to ease of hydrolysis
(a) $CCl_4 < SiCl_4 < PCl_5 < AlCl_3$
(b) $AlCl_3 < CCl_4 < PCl_5 < SiCl_4$
(c) $CCl_4 < AlCl_3 < PCl_5 < SiCl_4$
(d) $CCl_4 < AlCl_3 < SiCl_4 < PCl_5$

51. The increasing order of the dipolemoment of halogen acids is
(a) HI < HBr < HCl < HF
(b) HF < HCl < HBr < HI
(c) HBr < HI < HCl < HF
(d) HCl < HF < HBr < HI

52. When $KClO_3$ is heated in the absence of catalyst the products formed are
(a) $KClO_4$ and $KClO_3$ (b) KCl and O_2
(c) $KClO_4$ and KCl (d) $KClO_4$ and Cl_2

53. Which of the following statement is false?
(a) The shape of H_5IO_6 is octahedral
(b) The common form of periodic acid $HIO_4 \cdot 2H_2O$ or H_5IO_6 is called paraperiodic acid
(c) On strong heating H_5IO_6 gives I_2O_7
(d) Chlorine substitutes iodine from iodic acid

54. Which reaction yields the greatest quantity of chlorine from a given quantity of hydrochloric acid?
(a) Warming conc HCl with MnO_2
(b) Warming conc HCl with PbO_2
(c) Mixing conc HCl with $KMnO_4$
(d) Heating bleaching powder with HCl

55. Which of the following is the correct order of decreasing oxidizing power of perhalates during conversion to halates?
(a) $ClO_4^- > BrO_4^- > IO_4^-$
(b) $BrO_4^- > ClO_4^- > IO_4^-$
(c) $IO_4^- > BrO_4^- > ClO_4^-$
(d) $BrO_4^- > IO_4^- > ClO_4^-$

56. The formula of bromic acid is $HBrO_3$ while that of dysprosium oxide is Dy_2O_3. The formula of dysprosium bromate is
(a) Dy_2BrO_3 (b) $Dy(BrO_3)_3$
(c) Dy_3BrO_3 (d) $Dy_2(BrO_3)_3$

57. In which of the following reaction change of oxidation state of Br is more than unity
I. $H_2SO_4 + 2KBr \rightarrow$ II. $KBr + H_3PO_4 \rightarrow$
III. $Cl_2 + KBr + OH^- \rightarrow$ IV. $Br_2 + NaClO_3 \rightarrow$
(a) I, II, III (b) III, IV
(c) II, III, IV (d) I, IV

58. Bleaching powder ($CaOCl_2$) reacts with iodide ion as

$$ClO^- + 2I^- + 2H^+ \longrightarrow I_2 + Cl^- + H_2O$$

0.6 gm sample of bleaching powder requires 34 mL of 0.1 M $Na_2S_2O_3$ to titrate the liberated iodine. The % of Cl in the sample is
(a) 40.2 (b) 20.0 (c) 5.15 (d) 45.1

59. Potassium iodide reacts with iodine in aqueous solution to form the tri-iodide ion

$$KI\,(aq) + I_2\,(s) \longrightarrow KI_3\,(aq)$$

What would happen if CCl_4 is added in the reaction mixture?
(a) The KI would tend to dissolve in the CCl_4 layer
(b) The I_2 would tend to dissolve in the CCl_4 layer
(c) Both KI_3 and I_2 would dissolve in the CCl_4 layer
(d) Neither KI_3 nor I_2 would dissolve in the CCl_4 layer

More than One Answer Type Questions

1. Which are correct statements?
(a) All halogens form oxoacids
(b) All halogens show –1, +1, +3, +5, +7 oxidation states
(c) Hydrofluoric acid is a dibasic acid and attacks glass
(d) Oxidizing power is in order $F_2 > Cl_2 > Br_2 > I_2$

2. Select correct statement:
(a) ClO_2 and Cl_2O are used as bleaching agents for paper pulp and textiles
(b) OCl^- (hypohalites) salts are used as detergent
(c) OCl^- disproportionates in alkaline medium
(d) BrO^-_3 is oxidized to Br_2 by Br^- in acidic medium

3. Which of the following reactions are correct
(a) $CsBr_3 \rightleftharpoons Cs^+ + Br_3^-$
(b) $I_4O_9 \rightleftharpoons I^{3+} + 3(IO_3)^-$
(c) $AgBrO_3 \rightleftharpoons Ag^+ + BrO_3^-$
(d) $I_2O_4 \rightleftharpoons I^+ + IO_4^-$

4. Identify the correct statements:
(a) Fluorine is a super halogen
(b) Iodine shows basic nature
(c) AgF is insoluble in water
(d) SCN^- is pseudohalide

5. Which of the following properties of the elements chlorine, bromine and iodine increase with increasing atomic number?
(a) Ionization energy
(b) Ionic radius
(c) Bond energy of the molecule X_2
(d) Enthalpy of vaporization

6. $Cl_2O_6 + NaOH \rightarrow ?$
(a) $NaClO_4$ (b) NaOCl
(c) $NaClO_2$ (d) $NaClO_3$

7. Which one of the following arrangements truely represent the property indicated against it?
(a) $Br_2 < Cl_2 < F_2$ bond energy
(b) $Br_2 < Cl_2 < F_2$ electronegativity
(c) $Br_2 < Cl_2 < F_2$ oxidizing power
(d) $Br_2 < Cl_2 < F_2$ electron affinity

8. In which of the following reaction, change of oxidation state of Br is unity
(a) $H_2SO_4 + 2KBr \rightarrow$ (b) $KBr + H_3PO_4 \rightarrow$
(c) $Cl_2 + KBr + OH^- \rightarrow$ (d) $Br_2 + NaClO_3 \rightarrow$

9. Among the interhalide species IF_2^-, IF_3, IF_4^-, IF_5 and IF_7, incorrect statement is/are
(a) All iodine centres are either sp^3d or sp^3d^2 hybridized
(b) The minimum angular separation between fluorine atoms is 60º
(c) The anionic species are both isoelectronic and isostructural to XeF_2 and XeF_4
(d) There is no species having a single lone pair of electrons

10. The species that undergo disproportionation in an alkaline medium are
(a) Cl_2 (b) MnO_4^{2-}
(c) NO_2 (d) ClO_4^-

11. In which of the following pairs of species the central atom has same hybridization?
(a) ClF_3O, ClF_3O_2 (b) $(ClF_2O)^+$, $(ClF_4O)^-$
(c) ClF_3, ClF_3O (d) $(ClF_4O)^-$, $(XeOF_4)$

12. Which of the following statements are true regarding ICl_3?
(a) It exist as dimer $(ICl_3)_2$ or I_2Cl_6
(b) In liquid state it conducts electricity
(c) It has a planar structure
(d) All I-Cl bonds are identical

13. Oxygen gas is not evolved when ozone reacts with
(a) SO_2 (b) acidified $SnCl_2$
(c) Hg (d) KI

14. Which of the following is correct order with respect to rate of oxidation of many molecules or ions by halogen oxoanions

(a) $ClO_4^- < ClO^-_3 < ClO_2^- \approx ClO^- \approx Cl_2$
(b) $BrO_4^- < BrO^-_3 \approx BrO^- \approx Br_2$
(c) $IO_4^- < IO_3^- < I_2$
(d) $ClO_4^- < BrO_4^- < IO_4^-$

15. Which of the following is correct order with respect to acidic character of oxoacids of halogens

(a) $HOCl < HClO_2 < HClO_3 < HClO_4$
(b) $HOCl > HOBr > HOI$
(c) $HClO_3 \approx HBrO_3 > HIO_3$
(d) $HClO_4 > HBrO_4 > HIO_4$

16. Which of the following statements are correct?

(a) Anhydrous HF is slightly ionized and is therefore a poor conductor of electricity
(b) A mixture of KF and HF increases the conductivity of HF
(c) HF is stronger acid than HCl because it is more ionic
(d) Gaseous HF is very toxic

17. Which of the following will displace the halogen from the solution of the halogen compound?

(a) I_2 added to $NaClO_3$ solution
(b) Cl_2 added to KBr solution
(c) F_2 added to KCl solution
(d) Br_2 added to NaI solution

18. Which of the following statements are not correct?

(a) In a lesser ionic solvent, the HF has maximum acid strength among hydrohalic acids
(b) Acid HNO_3 behaves as a base in liquid HF
(c) Perchloric acid behaves as a base in liquid HF
(d) The fluoride acceptor compounds in HF acts as an acid

19. Which of the following statements are correct?

(a) The chlorine atom in all its oxoacids is in sp^3 hybridization
(b) All the oxoacids of chlorine except perchloric acid disproportionate on heating
(c) Chlorous acid is weaker acid than hypochlorous acid
(d) Chlorites are stronger oxidizing agents than perchlorates

20. Which of the following statements is/are true?

(a) I^- is weaker base than F^-
(b) NH_2OH is weaker base than NH_3
(c) OH^- is stronger base than NH_2^-
(d) F_3C^- is stronger base than Cl_3C^-

21. Which of the following halides is most acidic?

(a) PCl_5 (b) $SbCl_3$
(c) $BiCl_3$ (d) CCl_4

22. Which of the following reactions are correct?

(a) $2Cl_2 + 2HgO \longrightarrow Cl_2O + HgO \cdot HgCl_2$
(b) $Cl_2 + 2AgClO_3 \longrightarrow 2ClO_2 + O_2 + 2AgCl$
(c) $2HClO_4 \xrightarrow[\Delta]{P_4O_{10}} Cl_2O_7 + H_2O$
(d) $HClO_4 + HClO_3 \xrightarrow{P_4O_{10}} Cl_2O_6 + Cl_2O_4 + H_2O$

23. When $KClO_3$ is heated with conc HCl

(a) a mixture of Cl_2 and ClO_2 is liberated
(b) a mixture of Cl_2 and Cl_2O_7 is liberated
(c) The mixture of gases liberated is called euchlorine.
(d) The product acts as strong bleaching agent

24. $KClO_3$ on reaction with SO_2 gives

(a) ClO_2 (b) $HClO_4$
(c) $KHSO_4$ (d) SO_2Cl_2

25. Which of the following are the characteristics of interhalogen compounds?

(a) They are more reactive than halogens
(b) They are quite unstable but none of them is explosive
(c) They are covalent in nature
(d) They have low boiling points and are highly volatile

26. Identify the correct statements regarding bleaching powder

(a) Smell of Cl_2 is observed when placed in moist air
(b) When KI is added to its aq. solution in acetic acid I_2 is liberated
(c) When CO_2 is passed through solution white precipitate is obtained
(d) When its aqueous solution is heated with ethyl alcohol a product of anaesthetic use is obtained

27. Which of the following statements/(s) is/are correct?

(a) ClO_2 disproportionate in NaOH
(b) Cl_2O_6 is a stronger oxidizing agent
(c) ClO_2 is an odd electron molecule and dimerizes to Cl_2O_4
(d) Bond angle in Cl_2O is greater than in ClO_2

Comprehension Type Questions

Passage-I

Covalent compounds formed when two halogens react are termed interhalogen compounds. The actual product obtained depend on the relative concentrations of the halogens reacting.

The reaction of the interhalogen compounds are generally not largely different from those of the halogens. Type X-Y compounds resemble the free halogen. But as the X-Y bond is weaker than the Y-Y bond hence X-Y compounds are generally more reactive than the Y_2 molecule.

1. It is known that Cl_2 reacts with water to form HCl and HOCl. So when ICl hydrolysed then
   (a) Only HC*l* is formed
   (b) Only HI is formed.
   (c) HOC*l* and HI are formed
   (d) HOI and HC*l* are formed.
2. What the products formed when BrF**5** is reacted with NaOH?
   (a) $NaFO_3$ and NaBr formed.
   (b) $NaBrO_3$ and NaF are formed.
   (c) NaF and NaBr are formed.
   (d) NaBrF is formed along with F_2
3. Iodine pentoxide on heating with dry HCl gives
   (1) ICl_3 (2) Cl_2 (3) ICl_5 (4) ICl
   correct code.
   (a) 1, 2 (b) 1, 3 (c) 1, 2, 3, 4 (d) 1, 2, 3

Passage-II

Study the following passage and answer the questions.

Halogens in their reactions among themselves and with a variety of other elements give rise to a large number of compounds. Halogen derivatives and interhalogen compounds represent major type of halogen compounds.

A "black and white" photographic film contains a coating of silver bromide on a support such as cellulose acetate. During the developing process unexposed AgBr is washed away by complexation of Ag(I) by sodium thiosulphate solution. These washings are often disposed off as waste. However metallic silver can be recovered from them by adding cyanide, followed by zinc. The most reactive halogen, fluorine, reacts with other halogens Cl_2, Br_2 and I_2 under controlled conditions giving tetra-atomic, hexa-atomic and an octa-atomic molecule, respectively. Chlorine acts as a strong oxidizing agent and can oxidize hypo and chlorine dioxides react with alkali

1. The shape of IF_5 is
   (a) Trigonal bipyramid with lone pair at equatorial position
   (b) Trigonal bipyramid with lone pair at axial position
   (c) Tetragonal pyramid with lone pair at one corner
   (d) Octahedral with lone pair at one corner
2. The chemical equation for the oxidation of sodium thiosulphate to an ion containing the highest oxidation state of sulphur by chlorine

(a) $4Cl_2 + S_2O_3^{2-} + 5H_2O \longrightarrow 8Cl^- + 2SO_4^{2-} + 10H^+$

(b) $2Cl_2 + S_2O_3^{2} + 3H_2O \longrightarrow 4Cl^- + 2SO_4^{2-} + 6H^+$

(c) $4Cl_2 + S_2O_3^{2-} + 4H_2O \longrightarrow 8Cl^- + S_2O_8^{2-} + 8H^+$

(d) $4Cl_2 + S_2O_3 + 3H_2O \longrightarrow 8Cl^- + 2SO_5^{2-} + 6H^+$

3. Chlorine dioxide reacts with sodium hydroxide. What are the oxidation states of chlorine in the products?
   (a) –1 and +1 (b) +3 and +5
   (c) –1 and +3 (d) –1 and +5
4. The photochemical reaction that occurs when light falls on $AgBr_{(s)}$ coated on a film
   (a) $AgBr \xrightarrow{h\upsilon} Ag^+ + Br^-$
   (b) $5AgBr \xrightarrow{h\upsilon} AgBr_3 + Br_2$
   (c) $2AgBr \xrightarrow{h\upsilon} 2Ag + Br_2$
   (d) $2AgBr + 3O_2 \xrightarrow{h\upsilon} 2AgBrO_3$
5. The compound present in the washings of the photographer films while developing them is
   (a) $Na[Ag(S_2O_3)]$ (b) $Na[Ag(S_2O_3)_2]$
   (c) $Na_3[Ag\ (S_2O_3)_2]$ (d) $Na_2[Ag(S_2O_3)_3]$
6. The reactions involved in the recovery of silver from the washings of photographic films after developing are

(a) $Ag[(S_2O_3)_2]^{3-} + 2CN^- \longrightarrow [Ag(CN)_2] + 2S_2O_3^{2-}$

$2[Ag(CN)_2]^- + Zn \longrightarrow [Zn(CN)_4]^{2-} + 2Ag\downarrow$

(b) $Ag[(S_2O_3)^-] + CN^- \longrightarrow AgCN + S_2O_3^{2-}$

$2AgCN + Zn \longrightarrow 2Ag\downarrow + Zn(CN)_2$

(c) $[Ag(S_2O_3)_2]^{3-} + 2CN^- \longrightarrow [Ag(CN)_2]^- + 2S_2O_3^{2-}$

$[Ag(CN)_2^-]^- + Zn \longrightarrow [Zn(CN)_2]^- + 2Ag$

(d) $[Ag(S_2O_3)_2]^{2-} + 4CN^- + Zn \longrightarrow [Zn(CN)_4]^{2-} + 2S_2O_3^{2-} + Ag$

Passage-III

Bleaching powder is a mixed salt of hydrochloric acid and hypochlorous acid. It has the formula, $CaOCl_2 \cdot H_2O$. It is manufactured by the action of chlorine on dry slaked lime at 40ºC. There is also a view that bleaching powder is a mixture of calcium hypochlorite and basic calcium chloride $[Ca(OCl)_2 + CaCl_2 \cdot Ca\ (OH)_2 \cdot H_2O]$.

The amount of chlorine obtained from a sample of bleaching powder by the treatment with excess of dilute acids or CO_2 is called available chlorine. A good sample of bleaching powder contains 35–38 per cent of available chlorine. On long standing, it undergoes auto-oxidation and the amount of available chlorine decreases.

The estimation of available chlorine is done volumetrically by (a) iodometric method or by (b) arsenite method.

In textile industry, the cotton cloth is mainly bleached with the help of bleaching powder.

1. Maximum percentage of available chlorine on the basis of $CaOCl_2 \cdot H_2O$ formula is:
   (a) 35 (b) 40
   (c) 45 (d) 49
2. The percentage of available chlorine in commercial samples of bleaching powder is usually between 33–38%. The low value is due to:
   (a) Incomplete reaction between slaked lime and Cl_2 during its formation
   (b) Impurities present in the original slaked lime
   (c) Decomposition of bleaching powder when kept in air
   (d) All of the above
3. 3.55 g of bleaching powder when treated with acetic acid and excess of KI liberated iodine which required 60 mL of 0.5 N sodium thiosulphate solution. The percentage of available chlorine in the sample is:
   (a) 30.0 (b) 25.0
   (c) 20.0 (d) 35.0

Passage-IV

One halogen combines with another halogen to form some other compounds called inter halogen compounds. These are more reactive than halogens (except fluorine). These compounds undergo hydrolysis to form hydra acids and oxy acids. IF is unstable compound which undergoes decomposition to form IF_5 and I_2

1. Inter halogen compounds are more reactive than halogens (Except F_2) because
   (a) Inter halogen compounds contains two halogens
   (b) X – X′ bond in inter halogen compounds is weaker than X—X bond in halogens
   (c) Inter halogen compounds are soluble in water
   (d) Inter halogen compounds has vital force
2. ICl on hydrolysis gives
   (a) HCl, HOI (b) HI, HOCl
   (c) HCl, HOI, HOCl (d) HI only
3. Identity the wrong statement about ClF_3
   (a) ClF_3 is inter halogen compound
   (b) Shape of ClF_3 is T-shape
   (c) All Cl—F bonds lengths are same
   (d) *Cl*—F axial bond length is greater than the Cl-F equatorial bond length

Matching Type Questions

1. Column-I	Column-II
(a) Halogen gas is liberated when a halide treated with MnO_2 and conc. H_2SO_4	(p) F_2
(b) Halogen displaces when halide is treated with H_2SO_4 (conc)	(q) Cl_2
(c) Halogen can be prepared by chemical method	(r) Br_2
(d) Oxidizing agent	(s) I_2

2. Match the following Column-I with Column-II

Column-I	Column-II
(a) Euchlorine	(p) Berthelot's salt
(b) $KClO_3$	(q) Oxidizer
(c) Anhydrone	(r) $ClO_2 + Cl_2$
(d) Desiccant	(s) $Mg(ClO_4)_2$

3. Match the following Column-I with Column-II

Column-I	Column-II
(a) HI > HBr > HCl	(p) Acidic nature
(b) HCl > HBr > HI	(q) Boiling point
(c) $Cl^- > Br^- > I^-$	(r) Dipolemoment
(d) $NH_3 > PH_3 > AsH_3$	(s) Thermal stability
	(t) Basic nature

4. Match the following Column-I with Column-II

Column-I	Column-II
(a) Cl_2 + cold & dil NaOH	(p) Disproportionation takes place
(b) Cl_2 + Hot & conc NaOH	(q) Chloride ion (Cl^-) is produced
(c) $HOCl \xrightarrow{\Delta}$	(r) Chlorate ion (ClO_3^-) is produced
(d) $ClO_2 + NaOH \longrightarrow$	(s) Hypochlorite ion (ClO^-) is produced
	(t) Chlorite ion (ClO_2^-) is produced

5. List-I	List-II
(a) NO_2	(p) Paramagnetic
(b) NO	(q) Dimerizes on cooling
(c) Cl_2O	(r) Disproportionate either in cold or hot water
(d) ClO_2	(s) Act both oxidizing and reducing agent

Numerical Type Questions

1. The number of atoms on which the negative charge is distributed in perchlorate ion is
2. No. of pπ-dπ bonds in perchlorate ion is
3. Reaction of Br_2 with Na_2CO_3 in aqueous solution gives sodium bromide and sodium bromate with evolution of CO_2. The number of sodium bromide molecules involved in the balanced chemical equation is
4. When ClO_2 is dissolved in alkali two products are formed. The difference in the oxidation state of Cl in products is
5. Average oxidation state of chlorine in bleaching powder is

VI Group Key

Single Answer Type Qeustions

1. a	**2.** b	**3.** b	**4.** a	**5.** c	**6.** a	**7.** b
8. c	**9.** b	**10.** d	**11.** c	**12.** a	**13.** d	**14.** c
15. c	**16.** a	**17.** d	**18.** c	**19.** c	**20.** b	**21.** b
22. c	**23.** c	**24.** b	**25.** a	**26.** d	**27.** c	**28.** a
29. b	**30.** d	**31.** b	**32.** a	**33.** b	**34.** a	**35.** c
36. b	**37.** b	**38.** a	**39.** d	**40.** d	**41.** d	**42.** a
43. c	**44.** b	**45.** d	**46.** a	**47.** d	**48.** d	**49.** c
50. d	**51.** a	**52.** c	**53.** d	**54.** d	**55.** d	**56.** b
57. b	**58.** b	**59.** b				

More than One Answer Questions

1. a, c, d	**2.** a, b, c	**3.** a, b, c	**4.** a, b, d
5. b, d	**6.** a, d	**7.** b, c	**8.** a, b
9. a, b, d	**10.** a, c	**11.** a, c, d	**12.** a, b, c
13. a, b	**14.** a, b, c	**15.** a, b, c, d	
16. a, b, d	**17.** a, b, c, d	**18.** a, b, d	**19.** a, b, d
20. a, b, d	**21.** a	**22.** a, b, c	**23.** a, c, d
24. a, b, c	**25.** a, b, c	**26.** a, b, c, d	**27.** a, b

Comprehensive Type Questions

Passage – I	**1.** d	**2.** b	**3.** d			
Passage – II	**1.** d	**2.** a	**3.** b	**4.** c	**5.** c	**6.** a
Passage – III	**1.** d	**2.** a	**3.** a			
Passage – IV	**1.** b	**2.** a	**3.** c			

Matching Type Questions

1.	a-q, r, s	b-r, s	c-q, r, s	d-p, q, r, s
2.	a-q, r	b-p, q	c-s	d-s
3.	a-p, q	b-r, s	c-t	d-r, s, t
4.	a-p, q, s	b-p, q, r	c-p, q, r	d-p, r, t
5.	a-p, q, r, s	b-p, q, s	c-r	d-p, r

Integer Type Questions

1. 5 **2.** 4 **3.** 5 **4.** 2 **5.** 0

VII Group Hints

1. As the electronegativity of halogen decreases ionic character of OH group decreases

3. $3Cl_2 + 6KOH \longrightarrow 5KCl + KClO_3 + 3H_2O$, $KClO_3$ is used in fire works.

4. $2AgClO_3 + Cl_2(dry) \longrightarrow 2AgCl + 2ClO_2 + O_2$

6. Due to the poor shielding effect of 3d electrons the size of Br^- ion nearly same as that of Cl^- ion.

8. $NaI + H_2SO_4 \longrightarrow NaHSO_4 + HI$

$$NaIO_3 + H_2SO_4 \longrightarrow NaHSO_4 + HIO_3$$

$$5HI + HIO_3 \longrightarrow 3I_2 + 3H_2O$$

10. Acidic character of oxyacids of chlorine increases with increase in the oxidation number of chlorine. So the basic character of their conjugate bases decreases in the same order.

11. Chile salt petre contains $NaIO_3$ from which iodine is obtained by the following reaction.

$$2NaIO_3 + 5NaHSO_4 \longrightarrow 3NaHSO_4 + 2Na_2SO_4 + H_2O + I_2$$

13. Fluorine oxidizes potassium chlorate to potassium perchlorate.

14. ICl, IBr, ICN etc. reat with KI liberating I_2 converting into KCl, KBr, KCN etc respectively.

15. As a non-aqueous solvent HF ionizes as

$$2HF \longrightarrow HF_2^- + H^+$$

HF_2^- exist as $[F—H \ldots F]^-$ ion, HNO_3 and H_2SO_4 can be protonated, thus acts as bases.

17. $2NH_3 + \underset{(X)}{3I_2} \longrightarrow 3HI + NH_3 \cdot NI_3$ (explosive)

$$8NH_3 \cdot NI_3 \longrightarrow 5N_2 + 9I_2 + 6NH_4I$$

$$I_2 + H_2 \longrightarrow \underset{(Y)}{2HI}$$

19. Cl_2 in KOH disproportionate forming KCl and $KClO_3$. As I_2 is more electropositive than Cl_2, I_2 displaces Cl_2 from $KClO_3$.

20. Order of bond energy of halogens is $Cl_2 > Br_2 > F_2 > I_2$

22. $2NaOH + 2F_2 \rightarrow 2NaF + OF_2 + H_2O$
The oxidation state of oxygen in OF_2 is +2 and in H_2O is –2

23. With hot conc alkali Cl_2 forms $KClO_3$. The ClO_3^- ion is pyramidal in shape and have lesser bond angle than ClO_4^- (tetrahedral) due to lone pair repulsion. Also lesser than in ClO_2^- ion

24. Cl_2 is a gas. No need of $\Delta H_{vapourization}$

25. HF ionizes in water good conductor of electricity.

26. With excess F_2, chlorine forms ClF_3 which has trigonal bipyramid structure in which two lone pairs occupy equatorial positions so that ClF_3 have T-shape.

27. (A) $CaOCl_2 + H_2SO_4$(insufficient)

$$\longrightarrow CaCl_2 + CaSO_4 + 2HOCl$$

$$2HOCl \longrightarrow 2HCl + O_2$$

(B) $CaOCl_2 + H_2SO_4$(excess)

$$\longrightarrow CaSO_4 + H_2O + Cl_2$$

(C) $4NaOH + 2F_2 \longrightarrow 4NaF + 2H_2O + O_2$

29. Cl_2 forms $FeCl_3$ with iron but not $FeCl_2$ because it is a good oxidizing agent and oxidizes Fe to its higher oxidation state Fe^{3+}

30. $6CaOCl_2 \xrightarrow{\text{auto oxidation}} 5CaCl_2 + Ca(ClO_3)_2$

31. The bond orders in ClO^-, ClO_2^-, ClO_3^- and ClO_4^- are 1, 1.5, 1.67 and 1.75 due to resonance. So bond energy increases from ClO^- to $ClO^-{}_4$

32. I_2Cl_6 has the following structure.

Around each I atom there are six electron pairs of which two are lone pairs and occupy the opposite corners of octahedron. So two square planar structures share an edge. The bridge I – Cl bonds are different from terminal I – Cl bonds. There is no I – I bond. It contain 3 centred 4-electron I—Cl—I bonds.

34. As the electronegativity of the central atom decreases the bond pairs move away from the central atom. So bond angles decreases due to repulsion by lone pair (All the three ions have pyramidal shape with one lone pair).

35. HF to HI bond length increases, bond energy decreases.

36. $$6Cl_{2(g)} + 6Ba(OH)_2 \longrightarrow \underset{(X)}{Ba(ClO_3)_2} + 5BaCl_2 + 6H_2O$$

$$\underset{(X)}{Ba(ClO_3)_2} + H_2SO_4 \longrightarrow BaSO_4 + \underset{(Y)}{2HClO_3}$$

$$\underset{(Y)}{4HClO_3} \xrightarrow{D} \underset{Z}{4ClO_2} + 2H_2O + O_2$$

37. O—O bond in O_2F_2 is shorter than in H_2O_2. It is 1.48A° in H_2O_2 and 1.22A° in O_2F_2.

38. Inter halogen bond strength increases with increase in the difference of electronegativity. I—F bond is more stronger and thus IF_3 must be more stable but it is very unstable and dispropotionate because fluorine oxidizes the iodine to higher oxidation state forming IF_5 or IF_7.

43. Due to the delocalization of electron on to the three atoms the Cl—O bond length decreases.

44. With increase in size of the ion hydration energy decreases.

45. Frontier orbitals are highest occupied MO (HOMO) and Lowest unoccupied MO (LUMO). The energy difference between these decreases with increase in atomic number.

47. Bond energies of F_2 and I_2 are 158.8 and 151.1 KJ mol^{-1}

49. Refer to the Hint to Q. 15

50. PCl_5 can easily expand its coordination number to a maximum covalence six. Carbon cannot expand its coordination number due to the absence of d-orbitals in its valence shell.

52. When $KClO_3$ is heated in the absence of catalyst it disproportionate

$$4KClO_3 \longrightarrow 3KClO_4 + KCl$$

53. In iodic acid iodine is in positive oxidation state +5 More electropositive element can substitute less electropositive element. Cl_2 being less electropositive than I_2, it cannot substitute I_2 from HIO_3.

55. Due to middle row anomaly perbromates are less stable and stronger oxidizing agent. When compared with I—O bond, Cl—O bonds are stronger. So periodate ion is stronger oxidizing agent than perchlorate.

58. $I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

Moles of I_2 liberated = 1.7 m mole

moles of OCl = 1.7 m mole

moles of Cl in bleaching powder = 3.4 m mole = 0.12 gm Cl

$$\%\ Cl = \frac{0.12}{0.6} \cdot 100 = 20$$

More than One Answer Questions

1. Fluorine do not exhibit variable valence. Fluorine can form oxoacid HOF. Hydrofluoric acid exist as H_2F_2 and can react with glass.

6. $Cl_2O_6 + 2NaOH \longrightarrow NaClO_3 + NaClO_4 + H_2O$

9. In IF_2^- and IF_3, I atom is in sp^3d hybridization In IF_4^- and IF_5, I atom is in sp^3d^2 and in IF_7 I atom is in sp^3d^3 hybridization. The IF_2^- and IF_4^- are isoelectronic and isostructural with XeF_2 and XeF_4 respectively.

11. In ClF_3O and ClF_3O_2, Cl is in sp^3d hybridization.

In Cl F_2O^+, Cl is in sp^3 and in ClF_4O^- , Cl is in sp^3d^2 hybridisation.

In ClF_3 and ClF_3O, Cl is in sp^3d hybridization.

In ClF_4O^- and $XeOF_4$, the Cl and Xe atoms are in sp^3d^2 hybridization.

14. Order of oxidation power of perhalates is $BrO_4^- > IO_4^- > ClO_4^-$ For reason refer to hint of Q. No 55 (single answer).

21. PCl_5 can take up Cl^- ion converting into PCl_6^- ion having symmetric structure

23. $2KClO_3 + 4HCl \longrightarrow 2KCl + 2ClO_2 + Cl_2 + 2H_2O$

24.
$$KClO_3 + 3SO_2 + 3H_2O \longrightarrow KCl + 3H_2SO_4$$
$$3KClO_3 + 3H_2SO_4 \longrightarrow 3KHSO_4 + 3HClO_3$$
$$2HClO_3 \longrightarrow 2ClO_2 + H_2O + O$$
$$HClO_3 + O \longrightarrow HClO_4$$
$$\overline{4KClO_3 + 3SO_2 + 2H_2O \longrightarrow 3KHSO_4 + KCl + 2ClO_2 + HClO_4}$$

26. $CaOCl_2 + H_2O \longrightarrow Ca(OH)_2 + Cl_2$

$Cl_2 + 2KI \longrightarrow I_2 + 2KCl$

$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$

$CHCl_3$ is formed.

Comprehensive Type Questions (Hints)

Passage-I

1. When ICl react with water H^+ ion combine with more electronegative Cl and OH^- ion combine with less electronegative I
2. $BrF_5 + 6NaOH \longrightarrow 5NaF + NaBrO_3 + 3H_2O$
3. $I_2O_5 + 10HCl \longrightarrow ICl_3 + ICl_5 + Cl_2 + 5H_2O$

Passage-II

1. IF_5 has octahedral shape with one corner occupied by lone pair
2. $Na_2S_2O_3 + 4Cl_2 + 5H_2O \longrightarrow 2NaHSO_4 + 8HCl$
3. $2ClO_2 + H_2O \longrightarrow \underset{+3}{HClO_2} + \underset{+5}{HClO_3}$
4. $2AgBr \xrightarrow{hu} 2Ag + Br_2$
6. From the washings of photographic films after developing the silver is recovered as follows.

$$AgBr + 2Na_2S_2O_3 \longrightarrow Na_3\left[Ag(S_2O_3)_2\right] + NaBr$$

$$Na_3\left[Ag(S_2O_3)_2\right] + 2NaCN \longrightarrow Na\left[Ag(CN_2)\right] + 2Na_2S_2O_3$$

$$2Na\left[Ag(CN_2)\right]Zn \longrightarrow Na_2\left[Zn(CN)_4\right] + 2Ag$$

Passage-III

1. From the molecular formula $CaOCl_2 \cdot H_2O$ the approximate percent of Cl_2 is 49
2. In the manufacture of bleaching powder there remains some unreacted lime and bleaching powder is a mixture of calcium hypochlorite $[Ca(OCl)_2 \cdot 3H_2O]$ and basic calcium chloride $[CaCl_2 \cdot Ca(OH)_2 \cdot H_2O]$
3. wt of $Cl_2 = \dfrac{60 \cdot 0.5 \cdot 35.5}{1000} = 1.065$
4. 55 g of bleaching powder contain 1.065 g of Cl_2 $\therefore$ $\% = 30$

12

CHAPTER

Group 18 Noble Gases

We live immersed at the bottom of a sea of elemental air… …

Evangelista Torricelli

12.1 INTRODUCTION

Sometimes it seems that the most honour goes to those people who are furthest removed from the nasty practicalities of life, so it is with the gases helium, neon, argon, krypton, xenon and radon. These gases once thought to be so extremely idle that they had to belong to the nobility. The hurly burly of chemical reactions was thought to be above their station in life. Hence they were given the alternative name of the *inert gases*, these elements are also called as *zero group elements* because of their position in the periodic table. It is preferred to call them as *noble* gases because like *noble metals*, they occur in the free state and have least reactivity. These gases are also called as *rare gases* because they occur in traces in the nature and as *aerogens* because all of them occur in atmosphere with the exception of radon which is obtained from radio-active disintegration of radium.

12.2 POSITION IN THE PERIODIC TABLE

Mendeleef compiled his periodic table in 1869, the noble gases were not known till that time. He did not leave any vacant spaces for them. Ramsay, the discoverer of these gases introduced a new group for them in the periodic table and referred to as zero group. This group occupies an intermediate position between the highly electronegative halogens (VII A Group) and the most-electro-positive alkali metals (IA Group) in the periodic table as shown in Table 12.1.

The change from the high electronegative character of halogens to the high electropositive character of alkali metals is sudden, since every period starts with an electropositive element and ends with electronegative element.

It appeared anomalous as the electronegative character of an element suddenly changes into electropositive character. The addition of noble gas elements (with zero oxidation state) in between these groups Group IA (with oxidation state +1) and Group VIIA (with oxidation state –1) served as a bridge between the two characters of the elements. It is for this reason that these elements are placed at the end of every period.

Table 12.1 Position of noble gases in the periodic table

Electronegative halogen	Zero group	Electropositive alkali metal
–	$_{2}$He	$_{3}$Li
$_{9}$F	$_{10}$Ne	$_{11}$Na
$_{17}$Cl	$_{18}$Ar	$_{19}$K
$_{35}$Br	$_{36}$Kr	$_{37}$Rb
$_{53}$I	$_{54}$Xe	$_{55}$Cs
$_{85}$At	$_{86}$Rn	$_{87}$Fr

Electronic Configuration

All the noble gases except helium have s^2p^6 electronic configuration for the outermost (valence) shell. Helium has $1s^2$ configuration. As these have completely filled s and p orbitals in the valency shell, they behave as inert elements.

12.3 DISCOVERY OF NOBLE GASES

Discovery of all the noble gases took more than a century (1795–1898). For the exhaustive studies conducted on noble gases Ramsay and Rayleigh were awarded the noble prize. These elements were not discovered at any single point of time but were isolated after sustained research over a long period.

Table 12.2 Electronic configuration of noble gases

Element	Atomic number	Electronic configuration	Valence shell electronic configuration
He	2	$1s^2$	$1s^2$
Ne	10	$1s^2\ 2s^2\ 2p^6$	$2s^2\ 2p^6$
Ar	18	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$	$3s^2\ 3p^6$
Kr	36	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$ $4s^2\ 4p^6$	$4s^2\ 4p^6$
Xe	54	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$ $4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^6$	$5s^2\ 5p^6$
Rn	86	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$ $4s^2\ 4p^6\ 4d^{10}\ 4f^{14}$ $5s^2\ 5p^6\ 5d^{10}\ 6s^2\ 6p^6$	$6s^2\ 6p^6$

The discovery of the noble gases was a feat in itself. In 1868 there occurred a total solar eclipse and the entire chromosphere was observed by P.J.C. Janssen, J.N Lockyer and Frankland. They found a bright yellow line near D_1 and D_2. This was named as D_3. They realized that these lines in the spectrum belong to a new element which they named as helium from a Greek word ‘helios’ meaning the sun, since he believed it to exist only on the sun and not on the earth. Twenty-one years later, Hillebrand obtained a gas by heating clevite or monazite (uranium minerals) and supposed to be nitrogen. In 1895 Ramsay examined the spectrum of the gas and observed a new bright yellow line not coinciding D_1 and D_2 lines of sodium. He sent a sample of the gas to Crookes, a specialist in spectroscopy, who identified the yellow line to be identical with that of helium i.e., D_3 line. In this way Ramsay proved the existence of helium on earth.

In 1985 H. Cavendish in his classic work on the composition of air noted that after repeatedly sparking a sample of air with an excess of O_2, there was a small residue of gas which he was able to remove by the chemical means. About 1/120th part of air remained unreactive. He could not further characterize this component of air. This observation remained without proper recognition for over a century. In 1891 Lord Rayleigh found the density (gL^{-1}) of the atmospheric nitrogen (isolated from air by passing purified air repeatedly over red hot copper) is higher than the chemical nitrogen (obtained from chemical compounds like urea or ammonium nitrite). He suggested that the reason for this difference is due to the presence of heavier gas in the atmospheric nitrogen. He also observed that this heavier gas is inert with an atomic weight of 40. This gas was named as argon (argon – Lazy). Later it was found that this ‘argon’ gas was a mixture of several gases.

After the discovery of helium and argon, the problem of their position in the periodic table arose. In 1896 Juliot Thomson suggested the desirability of inserting a new group of six elements with atomic weights 4, 20, 36, 54, 86, 132 and 212 and named as zero group. This idea made Ramsay and Travers very enthusiastic about the discovery of remaining four elements.

In 1898 Ramsay and Travers carried out systematic fractional evaporation of 16 litres of liquid argon under reduced pressures. The first fraction was found to contain a new element neon (means new). Its vapour density was found to be 10.1 and hence atomic weight 20.2. Its spectrum established its identity as a new element. The final fraction gave spectra of two new elements from the spectroscopic examination and named them krypton (hidden) and xenon (stranger) respectively. The vapour densities of krypton and xenon were found to be 40 and 64 and hence atomic weights were 80 and 128 respectively.

In 1900 Dorn discovered the last member of the zero group as a disintegration product of radium and named it radon.

$$^{226}_{86}Ra \longrightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$$

Occurrence

Due to inert nature, the noble gases always occur in the free state. The atmosphere is the most important source of the noble gases in which they are present in the proportions given in Table 12.3. Helium is present up to 2 percent in the natural gas found in the United States of America and Canada. This is the most important source of helium.

Table 12.3 The discovery and occurrence of noble gases in air

Element	Year of discovery	Discovered by	% by Volume in the air	% by weight in the air
Helium	1895	Lockeyer, Frankland	5×10^{-4}	3.7×10^{-5}
Neon	1898	Ramsay & Travers	1.8×10^{-3}	1×10^{-3}
Argon	1894	Ramsay & Rayleigh	9.32×10^{-1}	1.285
Krypton	1898	Ramsay & Travers	1.5×10^{-3}	2.8×10^{-4}
Xenon	1898	Ramsay & Travers	8.7×10^{-6}	4×10^{-5}
Radon	1900	Friedrich Dorn		

In small quantity helium with neon is present in the minerals of the radioactive elements uranium and thorium e.g., clevite, uranite, monazite, etc. The gas is present in the state of solid solution and given off on heating to 1300k in vacuum or on treatment with an acid. In India, monazite is found in large quantity at the coast of travancore in Kerala. Helium, neon and argon are also found in certain radioactive spring waters as dissolved gas.

12.4 ISOLATION OF NOBLE GASES

Air is the chief source of all the noble gases, therefore, all these gases are isolated from air by removing oxygen nitrogen, carbon dioxide, water vapour, dust particles etc. This can be achieved either by physico-chemical methods or by physical methods based on fractional distillation of liquid air.

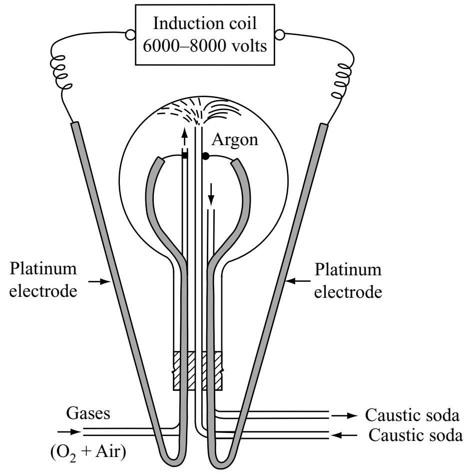


Fig 12.1 Separation of the mixture of noble gases from atmosphere

12.4.1 Physico-Chemical Methods

It consists of two steps:

(i) The first step involves the separation of noble gas mixture from air by using chemical methods.

(ii) The second step involves the separation of different noble gases from one another by adopting a physical method based on adsorption over activated charcoal developed by Dewar.

(i) ***Separation of noble gas mixture from air***

(a) ***Ramsay-Rayleigh's first method***: In this method, pure and dry air free from carbon dioxide (absorbed by soda lime) is passed over heated copper when oxygen of the air reacts with it forming copper oxide. Now the residual air is passed over heated magnesium when nitrogen of the air combines to form magnesium nitride.

$$2Cu + O_2 \longrightarrow CuO$$

$$3Mg + N_2 \longrightarrow Mg_3N_2$$

This process is repeated several times, so that, nitrogen and oxygen are completely removed, leaving behind the mixture of noble gases. Since it takes a very long time for completion, a modified method was developed.

(b) ***Ramsay-Rayleigh's second method***: The principle of this method was laid down by Cavendish and the technique was much improved by Ramsay and Rayleigh.

In this method a mixture of air and oxygen (1:11 by volume) is passed into the glass globe about 50 litre capacity fitted with two heavy platinum electrodes and subjected under the influence of an electric discharge 6000–8000 volts. Two glass tubes are introduced for the circulation of concentrated solution of sodium hydroxide while the third glass tube is for introduction of a mixture of air and oxygen Nitrogen of the air combines with oxygen to form oxides of nitrogen which are absorbed by sodium hydroxide circulating in the flask to form sodium nitrite and sodium nitrate.

$$N_2 + O_2 \longrightarrow 2NO$$

$$2NO + O_2 \longrightarrow 2NO_2$$

$$2NO_2 + 2NaOH \longrightarrow NaNO_2 + NaNO_3 + H_2O$$

The remaining gas contains a mixture of noble gases with small traces of oxygen. When this is passed through a solution of alkaline pyrogallol, oxygen is absorbed leaving behind a mixture of noble gases.

(c) ***Fischer–Ring's method***: In this method dry and pure air is passed over a mixture containing 90% calcium carbide and 10% calcium chloride in an iron tube heated to 1073k. Oxygen and nitrogen gases are removed by the following reactions.

$$CaC_2 + N_2 \longrightarrow CaNCN + C$$

$$C + O_2 \longrightarrow CO_2$$

$$2CaC_2 + 3CO_2 \longrightarrow 2CaCO_3 + 5C$$

$$2C + O_2 \longrightarrow 2CO$$

Carbon monoxide produced, if any, is removed by passing the gases over heated copper oxide when CO is oxidized into CO_2 which is absorbed in caustic soda.

$$CuO + CO \longrightarrow Cu + CO_2$$

$$CO_2 + 2KOH \longrightarrow K_2CO_3 + H_2O$$

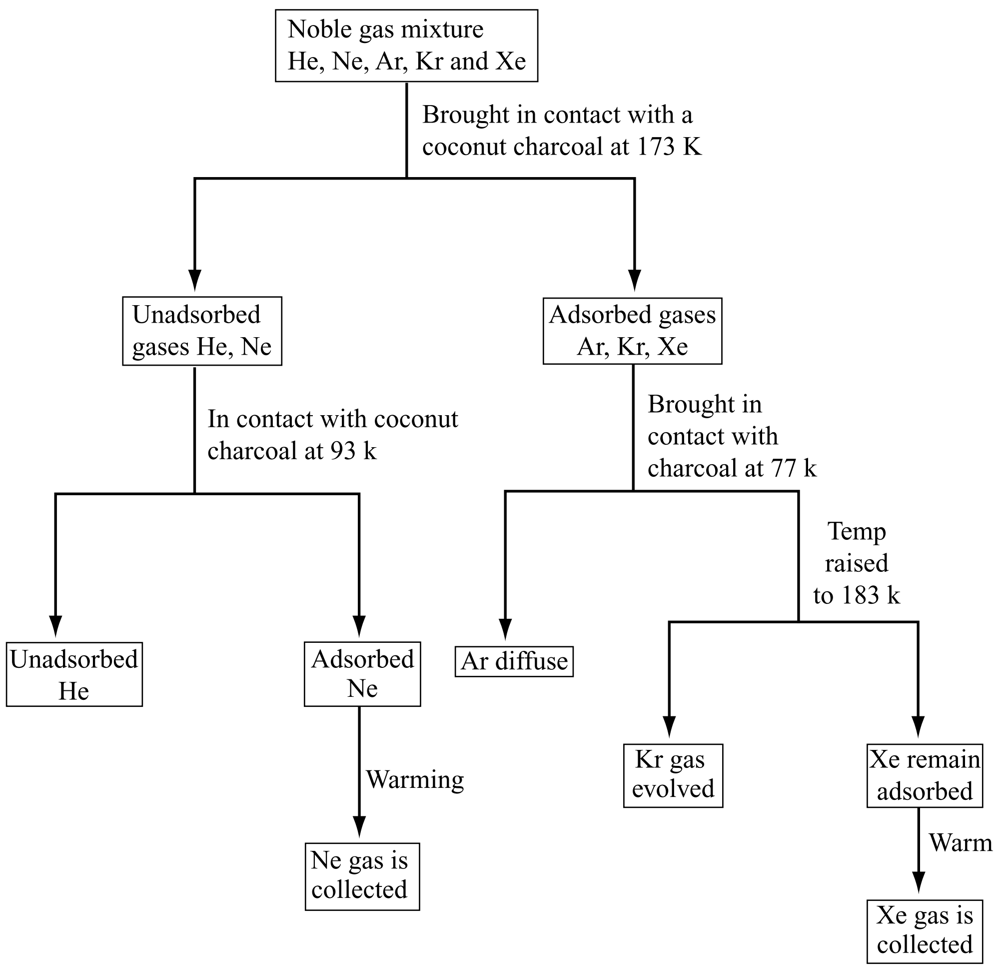
The remaining mixture contains all the noble gases. It is then dried over conc. sulphuric acid and finally over phosphorous pentoxide before the mixture of the gases is collected.

(ii) ***Separation of noble gases*:** The individual constituents of the noble gas mixture are separated by Dewar's method. This method is based on differential adsorption of noble gases by coconut charcoal. Except helium all the noble gases (from neon to xenon) can be adsorbed over activated charcoal. Adsorption of these gases on the charcoal increases with increase in atomic weight due to increased van der Waal's forces. Adsorption on the charcoal also depends on the temperature. Lower the atomic weight of the noble gas, lower is the temperature needed to adsorb it, i.e., noble gases with less atomic weights get more adsorbed at lower temperatures.

In this method the mixture of noble gases is passed into Dewar's flask at 173k and allowed to remain there for about half an hour. The unadsorbed gases (He and Ne) are then removed through a vacuum pump and passed into another flask at 93k. Here neon is completely adsorbed and helium is obtained in the free state. The first charcoal containing argon, krypton and xenon is kept in contact with another charcoal bulb cooled at the liquid air temperature, argon diffuses into the other charcoal which may be obtained by heating. The temperature of first charcoal still containing krypton and xenon is raised to 183k when krypton is set free while xenon remains adsorbed in the charcoal. Xenon which remained on the first charcoal is released by warming and is collected separately.

The following flow sheet will illustrate the above method of separation of noble gases.

Flow Chart of Separation of Noble Gases



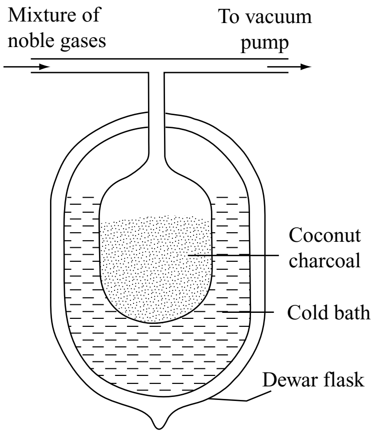


Fig 12.2 Dewar's charcoal method for the separation of noble gases

12.4.2 Separation of Noble Gases from Liquid Air

This method is employed when large excess of noble gases are required. This is based on the fact that the boiling points of various constituents of liquid air widely differ from each other. The boiling points of the various components present in liquid air at 760 mm pressure are given Table 12.4.

Due to the difference in the boiling points the following fractions collect at different stages.

(i) Helium and neon mixed with gaseous nitrogen
(ii) Argon will be above liquid oxygen
(iii) Krypton and xenon present in liquid oxygen

The apparatus used for the fractionation of liquid air is called Clauds apparatus which is shown in Fig. 12.3.

The various noble gases in the above fractions are separated as follows.

(i) ***Separation of helium and neon*****:** The first fraction containing He, Ne and N_2 is cooled by passing through a spiral tube immersed in liquid nitrogen. Most of the nitrogen undergoes liquification. Here if any nitrogen is remained in this mixture it is removed by passing over heated calcium carbide

$$CaC_2 + N_2 \xrightarrow{1073k} CaNCN + C$$

The resulting mixture is then chilled in liquid hydrogen (20k) when neon gas changes to liquid neon, while helium remains as a gas. The two can be easily separated by pumping out the helium gas.

(ii) ***Separation of Argon*****:** The portion containing argon, krypton and xenon is first cooled in a bath containing liquid nitrogen when oxygen, krypton and xenon being less volatile get liquified while argon remains as a gas. The gas is then separated. The gas thus obtained still contains a little oxygen. It is freed from oxygen by passing over heated copper, when argon in pure state is obtained.

The liquid portion containing oxygen, krypton and xenon is subjected to fractional distillation by which these gases are separated.

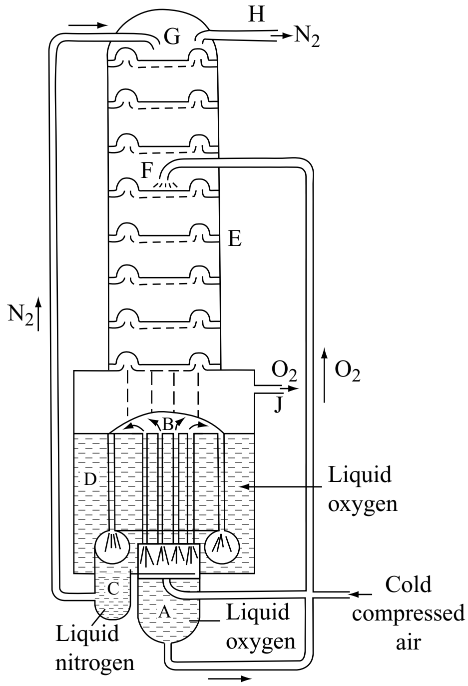


Fig 12.3 Fractionation of liquid air

Table 12.4 Boiling points (in K) of components in air

Gas	B.pt-(K)
He	4.22
Ne	27.0
N_2	77.20
Ar	87.10
O_2	90.02
Kr	119.8
Xe	164.9

12.5 PHYSICAL PROPERTIES OF NOBLE GASES

Some atomic and physical properties of noble gases are given in Table 12.5.

Table 12.5 Some atomic and physical properties of noble gases

Property	He	Ne	Ar	Kr	Xe	Ra
Atomic number	2	10	18	36	54	86
Atomic weight	4.00	20.0	39.91	83.7	131.3	222
Atomic volume (liq) cm^{-3}	31.8	16.8	28.5	32.2	42.9	50.9
van der Waal's radius pm	120	160	191	200	220	–
Melting point k	0.9	24.4	83.6	115.8	161.0	202.0
Boiling point k	4.22	27.0	87.1	119.8	164.9	211.0
Critical temp k	5	53	156	210	288	–
Critical pressure (atm)	2.26	26.9	50.0	54.3	58.2	62.4
Density. g Cm^{-3} at 273 k I atm press.	0.1785	0.8999	1.784	3.743	5.896	9.96
Density of liquid	0.126	1.20	1.40	2.6	3.06	44
Ionization energy KJ mol^{-1}	2372	2080	1521	1351	1170	1037
Heat of fusion KJ mol^{-1}	0.14	0.33	1.17	1.43	2.30	–
Heat of vapourization KJ mol^{-1} at b.p	0.09	1.8	6.3	9.7	13.7	18.0
Thermal conductivity at O°C 1J 51 m^{-1} k^{-1}	0.1418	0.0461	0.0169	0.00874	0.00506	–
Solubility in water at 20°C cm^{-3} kg^{-1}	8.61	10.5	33.6	59.4	108.1	230

(i) All the noble gases are colourless, odourless and tasteless gases.

(ii) All the noble gases are monoatomic. The mono atomic nature is confirmed by the value (γ = Cp/Cv) being close to 1.66.

(iii) Melting and boiling points: It is seen that boiling as well as the melting points of noble gases are very low. This is due to very weak inter molecular (van der Waals) forces of attraction in these cases. The melting and boiling points of helium are the lowest of any known element. Further, it is seen that the boiling points and melting points, in most cases increase regularly when we move down the group. This is due to increase in the magnitude of the van der Waal's forces with increase in the size of the noble gas atoms.

All the gases, except radon, **solidify** at 2° to 9° below their normal boiling points. Helium is most difficult to solidify only when it is cooled to about 1 k under a pressure of 25 atm. Liquid Helium is therefore used as an ideal cooling agent for research at low temperatures.

As the first member of this unusual group. He has, of course, a number of unique properties. Among these is the astonishing transition from so called He-I to He-II which occurs at 2.2k (the λ – point temperature), when liquid He is cooled by continuous pumping. He–I is a normal liquid but at the transition the specific heat increases abruptly by a factor of 10, the thermal conductivity by the order of 10^6 and the viscosity becomes zero. Hence He–II is called *super fluid.* If the bottom of a suitable container is dipped into a bath of He–II and cooled, liquid He flows upwards without friction over the edge of the container until the levels inside and outside are equal. It is believed that He–II consists of two components one a superfluid with zero viscosity and entropy, and the second one of a normal fluid. At absolute zero the total He converts into He–II. No satisfactory explanation for this phenomenon is available.

(iv) Liquifaction: The noble gases are liquified with great difficulty. This is indicated by their very low critical temperatures. This is again because of weak van der Waals or inter molecular forces existing in these gases. Heat of vapourization is also low. However, van der Waals forces seem to increase with increase in atomic size as is evident from increase in critical temperatures on moving from helium to xenon. The heat of vapourization also increase in the same order.

(v) Solubility in water: These gases are only slightly soluble in water (8 to 40 mL per litre at 25 ºC). The solubility in general increases with increasing the molar mass. The solubility of noble gases in water is due to dipole-induced dipole interaction between polar water molecules and the non-polar noble gas atoms. As the size of noble gas atom increases polarizability of the inert gas atom by polar water molecule increases. So solubility of inert gases increases with increase in atomic number of inert gases.

(vi) The ionization energies and electron affinities: The ionization energies of these elements are quite high. The values however, decrease with increasing molar mass, as shown in Table 12.5. **The electron affinities are close to zero.**

12.6 COMPOUNDS OF NOBLE GASES

The noble gases have most stable electronic configurations. Till recent times, their chemical compounds were almost unknown. In fact on account of their high ionization energies and zero electron affinities, they were not expected to be reactive at all. However, recent researches have shown that under certain conditions these elements can form a few compounds.

12.6.1 Hydrates

The noble gases (Ar, Kr and Xe) form compounds with water at 273k and high pressure. These compounds are known as hydrates. The compounds like $Ar \cdot 6H_2O$, $Kr \cdot 6H_2O$ and $Xe \cdot 6H_2O$ have been obtained and studied. The noble gases get polarized under the influence of water molecules (strong dipole) i.e., acquire a weak induced dipole of their own. This causes dipole-induced dipole interaction between the noble gases and water molecules, yielding the hydrate compounds. This property increases from helium to xenon, thus $Xe \cdot 6H_2O$ is most stable. This is also the reason that solubility of noble gases in water increases from helium to xenon.

12.6.2 Clathrates

Noble gases form a number of compounds in which *the gases are trapped within cavities of crystal lattices of certain organic and inorganic substances*. Such compounds are called as *clathrates* (Latin: clathrate = enclosed or protected by cross bars), these are also referred to as *Cage compounds*.

When o-dihydroxy benzene [quinol $C_6H_4(OH)_2$] or urea [$H_2N\ CONH_2$] is crystallized from aqueous solution in the presence of noble gases [Ar, Kr or Xe] under a pressure of 10–40 atmospheres, the atoms of noble gases get trapped within the lattice of the organic molecule. The crystal structure with cavity is called the *host* while the atom entrapped in it is called *guest.* The crystals are quite stable and can persist for several years but decompose on melting or dissolution in a suitable solvent.

X-ray studies show that somewhat spherical cage of about 400pm diameter is formed by three quinol molecules being bound with hydrogen bonds as shown in Fig 12.4.

Hence the ratio of quinol to entrapped atom or molecule is usually 3:1. Because all the cavities may not be filled at a time hence the ratio may be higher i.e., 3:0.74 or 3:0.88. But if the entrapped atom or molecule is too small (i.e., helium or neon) they are able to escape easily from the cavity. Therefore helium and neon do not form clathrate compounds with quinol.

Because helium and neon do not form clathrate compounds with quinol, they can be separated from the mixture of noble gases of argon, krypton and xenon. Clathrates of argon, krypton and xenon provide convenient means of handling and transporting of isotopes of these gases i.e., Kr-85 and Xe clathrates provide a source of β- and γ- radiations respectively.

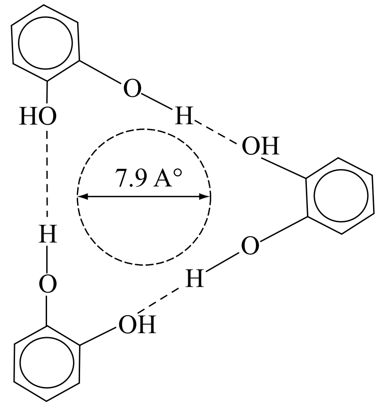


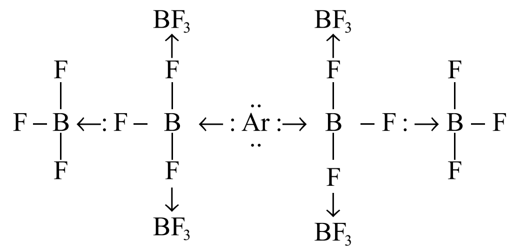
Fig 12.4 Clathrate compound of noble gas

12.6.3 Coordination Compounds

In 1935 Booth and Wilson reported that argon forms a number of unstable compounds with BF_3 molecule as $Ar \cdot nBF_3$ (Where n = 1, 2, 3, 6, 8 or 16). In these compounds Argon donates a pair of electrons to boron atom which has only six electrons in its outermost orbit. The structure of these molecules may be represented as follows.

$$:\ddot{\underset{\cdot\cdot}{Ar}}: \rightarrow BF_3 \qquad F_3B \leftarrow :\ddot{\underset{\cdot\cdot}{Ar}}: \rightarrow BF_3 \qquad F_3B \leftarrow :\underset{\substack{\cdot\cdot \\ \downarrow \\ BF_3}}{\ddot{Ar}} : \rightarrow BF_3$$

The formation of compounds like Ar. $8BF_3$ or Ar. $16BF_3$ has been explained by assuming that fluorine in BF_3 might also donate an electron pair to boron atom as shown below.



Structure of $Ar \cdot 8BF_3$ molecule

In the same way Ar. $16BF_3$ may be shown. All these compounds dissociate on heating above their melting points. Krypton does not form such compounds.

12.6.4 Chemical Compounds of Noble Gases

Noble gases are not very reactive because they have high ionization energies and their electron affinities are almost zero. Both these factors can be explained on the basis of their electronic configuration. At each noble gas the electronic sublevel is just complete (ns^2np^6) and their next available sub-level is of much higher energy. The electron distribution is spherically symmetrical and this particular configuration resists change or disturbance of any kind. Therefore all noble gases are extremely unreactive as compared to other elements. However, in recent years a fairly good number of compounds of these gases have been reported which are discussed now.

In 1962, Bartlett prepared a solid compound O_2PtF_6 The X-ray analysis showed that the compound is ionic and can be represented as $O_2^+PtF_6^-$. This would imply that platinum hexafluoride is a very strong oxidizing material and could even attract electron from molecular oxygen. Since the ionization energy of oxygen molecule is 1165 KJ mol^{-1} and is very close to the ionization energy of xenon atom (1170 KJ mol^{-1}) and as their molecular diameters are also similar (Xe ~ 400 pm, O_2 ~ 400 pm). Bartlett had every reason to believe that xenon would also be susceptible to oxidation with platinum hexafluoride. Therefore, he proceeded to mix xenon and platinum hexafluoride (both gases) at room temperature and obtained on his first attempt an orange yellow crystalline compound of composition $XePtF_6$ as he had predicated.

$$Xe(g) + PtF_6\,(g) \longrightarrow Xe^+\,[PtF_6]^-_{(g)}$$

Since this discovery, a large number of other noble gas compounds have been prepared. A summary of these compounds is given in Table 12.6.

Table 12.6 Known compounds of noble gases

	Xenon		Krypton
$Xe[PtF_6]$	$XeF_2 \cdot 2SbF_5$	$MXeF_7$	KrF_2
XeF_2	$XeOF_2$	M_2XeF_8	KrF_4?
XeF_4	$XeOF_4$		
XeF_6	$XeOF_6 \cdot 2SbF_5$		
	XeO_3		
XeO_4			
XeO_2F_2			
$XeF_6 \cdot BF_3$			

Before we discuss the compounds of noble gases, it will be useful if we consider some questions which are essential for the understanding of our later discussion on these compounds.

1. Which elements of this family are expected to be more reactive and why? We may expect only the heavier noble gases which have low ionization energy values to show some reactivity as a result of electron loss. Ionization energies of noble gases are given in Table 12.5. From the ionization energy values, it is obvious that we may expect radon to go into chemical reactions with greater ease. Since radon is radioactive with isotopes having short half life (about 3.8 days only) the little about its ability to form chemical compounds is known. Xenon forms the largest number of compounds as compared to any other noble gas. Krypton tends to form compounds but with difficulty as compared to xenon. The lighter noble gases helium, neon and argon which have very high ionization energies are expected to show very little reactivity. So far experimental evidence also suggests that no chemical compounds of these gases have been reported.
2. Why most of these compounds involve only fluorine and oxygen? Any chemical reactivity shown by these gases may be attributed to their tendency to lose electrons. For this reason, the combining atom must be highly electronegative.

Xenon forms stable compounds only with the most electronegative elements fluorine (E.N = 4.0) and oxygen (E.N = 3.5) or with highly electronegative groups such as $OSeF_5$ and $OTeF_5$ that contain both oxygen and fluorine. Relatively uncommon bonds of xenon are known with chlorine (E.N = 3.0) and nitrogen (E.N = 3.0).

The ability of compound formation by xenon under different conditions are discussed here.

12.6.5 Xenon Fluorides

The fluorides of xenon can be obtained by direct reaction but conditions need to be carefully controlled if these are to be produced individually in pure form.

(a) ***Xenon difluoride XeF_2***

Preparation**:** It is best prepared in the laboratory in high yields and purity by heating xenon and fluorine in the molar ratio 2:1. Excess xenon must be taken to avoid the formation of any xenon tetrafluoride. The reaction is carried out in a nickel or monel vessel at 400ºC. The reaction products are quenched to room temperature and XeF_2 is isolated in vacuum sublimation.

$$Xe + F_2 \xrightarrow{400^\circ C} XeF_2.$$

It may be prepared by simple exposure to sunlight a mixture of xenon and fluorine in a dry pyrex glass vessel at room temperature.

In another method it is prepared by the reaction of xenon with O_2F_2 at –118ºC.

$$Xe + O_2F_2 \longrightarrow XeF_2 + O_2$$

Properties: It is a colourless, transparent crystalline solid, melting point 129ºC. It is less volatile than xenon hexafluoride and can be stored in nickel containers for a long time when it is free from impurities. It dissolves in anhydrous HF and the solution does not conduct electricity.

It oxidizes water, hydrogen, iodine and HCl.

$$2XeF_2 + 2H_2O \longrightarrow 2Xe + 4HF + O_2$$
$$XeF_2 + H_2 \longrightarrow Xe + 2HF$$
$$XeF_2 + I_2 \xrightarrow{BF_3} Xe + 2IF$$
$$XeF_2 + 2HCl \longrightarrow Xe + 2HF + Cl_2$$

It is rapidly hydrolysed by aqueous solution of a base.

$$2XeF_2 + 4OH^- \longrightarrow 2Xe + 4F^- + 2H_2O + O_2$$
$$2XeF_2 + 2H_2O \longrightarrow 2Xe + 4HF + O_2$$

It forms addition compounds with metal fluorides (as NbF_5, TaF_5, RuF_5, RhF_5, OsF_5, IrF_5 and PtF_5) and non-metal fluorides (as PF_5, AsF_5 and SbF_5). These compounds may be represented as

$[XeF]^+ [MF_6]^-$	XeF_2 and MF_5 in the ratio 1:1
$[Xe_2F_3]^+ [MF_6]^-$	XeF_2 and MF_5 in the ratio 2:1
$[XeF]^+ [M_2F_{11}]^-$	XeF_2 and MF_5 in the ratio 1:2

It is a versatile mild fluorinating agent and will difluorinate olefins to 1, 1- and 1, 2-difluoro olefins.

$$CH_2 = CH_2 + 2XeF_2 \longrightarrow \underset{|}{\overset{F}{}}\!\!CH_2 - \overset{F}{\underset{|}{}}\!\!CH_2 + CH_3 - CHF_2 + 2Xe$$

It reacts with nitric oxide to give nitrosyl fluoride and nitrogen dioxide to give nitronium fluoride.

$$2NO + XeF_2 \longrightarrow 2NOF + Xe$$
$$2NO_2 + XeF_2 \longrightarrow 2NO_2F + Xe.$$

(b) *Xenon tetrafluoride XeF_4*

Preparation: It is best prepared by heating a mixture of xenon and fluorine in 1:5 volume at 400ºC under 6 atm pressure in a nickel vessel.

$$Xe + 2F_2 \longrightarrow XeF_4$$

It can also be prepared by passing an electric discharge in a reaction vessel maintained at –80ºC containing xenon and fluorine in the molar ratio 1:2 at a pressure 2–15 mm.

Properties: It is a white crystalline solid, easily sublimes, melting point 117.1ºC. It is less volatile than xenon hexafluoride and dissolves in trifluoro acetic acid. Its properties are similar to xenon difluoride except that it is rather stronger fluorinating agent and acts as a good oxidizing agent.

$$XeF_4 + 2H_2 \longrightarrow Xe + 4HF$$
$$XeF_4 + 4HCl \longrightarrow Xe + 4HF + 2Cl_2$$
$$3XeF_4 + 4BCl_3 \longrightarrow 3Xe + 4BF_3 + 6Cl_2$$
$$XeF_4 + 2Hg \longrightarrow Xe + 2HgF_2$$

When XeF_4 is heated with excess of xenon, XeF_2 is formed.

$$XeF_4 + Xe \longrightarrow 2XeF_2$$

It oxidizes iodide to iodine.

$$XeF_4 + 4I^- \longrightarrow Xe + 4F^- + 2I_2$$

It disproportionates in water and forms highly explosive solid compound XeO_3

$$6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$$

It forms addition compounds with SbF_5 having composition $XeF_4 \cdot 2SbF_5$. Other Lewis acids like PF_5, AsF_5, NbF_5, RuF_5, OsF_5 also form similar addition compounds.

It is fairly strong fluorinating agent and fluorinates several compounds and metals.

$$2CH_2 = CH_2 + XeF_4 \longrightarrow \underset{|}{\overset{F}{}}\!\!CH_2 - \overset{F}{\underset{|}{}}\!\!CH_2 + CH_3 - CHF_2 + Xe$$
$$XeF_4 + Pt \longrightarrow Xe + PtF_4$$
$$XeF_4 + 4NO \longrightarrow Xe + 4NOF$$
$$XeF_4 + 4NO_2 \longrightarrow Xe + 4NO_2F$$

(c) *Xenon hexafluoride XeF_6*

Preparation: It is best prepared by reacting xenon and fluorine in the ratio 1:20 in a nickel vessel. The reaction is carried out at low temperatures and at 50 atms. pressure.

$$Xe + 3F_2 \xrightarrow[50\ atm]{1:20} XeF_6$$

It can also be prepared by the oxidation of XeF_4 with O_2F_2 at a temperature below –80º

$$XeF_4 + O_2F_2 \longrightarrow XeF_6 + O_2$$

Properties: It is a crystalline solid but its vapours have greenish yellow colour, melting point 49.5ºC. It can be stored is a nickel vessel, but rapidly attacks glass and silica. It is more volatile than XeF_2 and XeF_4. It is stronger oxidizing and stronger fluorinating agent.

It is soluble in liquid hydrogen fluoride which is a good conductor of electricity due to the formation of ions.

$$XeF_6 + HF \longrightarrow XeF_5^+ + HF_2^-$$

It oxidizes H_2, HCl and NH_3

$$XeF_6 + 3H_2 \longrightarrow Xe + 6HF$$
$$XeF_6 + 6HCl \longrightarrow Xe + 6HF + 3Cl_2$$
$$XeF_6 + 2NH_3 \longrightarrow Xe + 6HF + N_2$$

It hydrolyses in water forming xenon oxyfluorides and finally highly explosive xenon trioxide.

$$XeF_6 + H_2O \longrightarrow XeOF_4 + 2HF$$
$$XeOF_4 + H_2O \longrightarrow XeO_2F_2 + 2HF$$
$$XeO_2F_2 + H_2O \longrightarrow XeO_3 + 2HF.$$

It also react with silica and hence it cannot be stored in glass vessels.

$$2XeF_6 + SiO_2 \longrightarrow 2XeOF_4 + SiF_4$$
$$2XeOF_4 + SiO_2 \longrightarrow 2XeO_2F_2 + SiF_4$$
$$2XeO_2F_2 + SiO_2 \longrightarrow 2XeO_3 + SiF_4.$$

Similar to XeF_2 and XeF_4, it also acts as strong fluorinating agent and undergoes addition reactions with Lewis acids e.g.,

$$XeF_6 + AsF_5 \longrightarrow [XeF_5]^+ [AsF_6]^-$$
$$XeF_6 + BF_3 \longrightarrow [XeF_5]^+ [BF_4]^-$$
$$XeF_6 + 2NOF \longrightarrow [XeF_8]^{2-} [NO]_2^{2+}$$

It reacts with alkali metal fluorides (except LiF) to form hepta fluoroxenates.

$$XeF_6 + NaF \longrightarrow NaXeF_7$$

It hydrolyses in strongly alkaline solutions.

$$2XeF_6 + 16OH^- \longrightarrow 8H_2O + 12F^- + XeO_6^{4-} + Xe + O_2$$

12.6.6 Xenon Oxides

(a) ***Xenon Trioxide***

Preparation: It is formed either by the hydrolysis of xenon tetrafluoride or xenon hexafluoride.

$$2XeF_4 + 3H_2O \longrightarrow Xe + XeO_3 + 6HF + F_2$$
$$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$$

Properties: It is a white non-volatile compound. It is stable in aqueous solution and is soluble in water. It explodes violently when dry (like TNT). It is a strong oxidizing agent.

$$6Pu^{3+} + XeO_3 + 6H^+ \longrightarrow 6Pu^{4+} + Xe + 3H_2O$$

It reacts with sodium hydroxide and undergoes disproportionation. Xenon gas is evolved and sodium perxenate, $Na_4XeO_68H_2O$ can be recovered from the solution.

$$2XeO_3 + 4NaOH + 6H_2O \longrightarrow Xe + O_2 + Na_4XeO_6 \cdot 8H_2O$$

Sodium perxenate when dissolved in acid solution decomposes to xenon trioxide.

$$2Na_4XeO_6 + 8HCl \longrightarrow 8NaCl + 2XeO_3 + 4H_2O + O_2$$

(b) ***Xenon Tetroxide XeO_4***

It can be prepared by the reaction of barium perxenate with anhydrous sulphuric acid.

$$Ba_2XeO_6 + 2H_2SO_4 \longrightarrow XeO_4 + 2BaSO_4 + 2H_2O$$

It is not a stable compound as xenon trioxide. It decomposes to xenon and oxygen.

$$XeO_4 \longrightarrow Xe + 2O_2$$

12.6.7 Structures of Xenon Compounds

According to the valence bond approach, hybridization takes place in the central xenon atom in which the s, p and d orbitals of the valency shell participate. The atomic number of xenon is 54 and its electronic configuration is

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^6$

From this electronic configuration that under normal conditions, it fails to form any chemical compound. However, under excited conditions it can form three series of compounds depending upon the state of excitation.

Outer electronic configuration of xenon

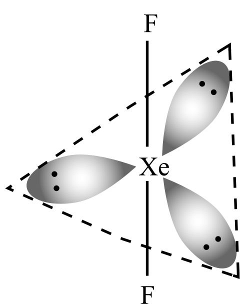
	5s	5p	5d
Xe is ground state	↑↓	↑↓ ↑↓ ↑↓	☐ ☐ ☐ ☐ ☐
Xe is Ist excited state	↑↓	↑↓ ↑↓ ↑	↑ ☐ ☐ ☐ ☐
Xe in IInd excited state	↑↓	↑↓ ↑ ↑	↑ ↑ ☐ ☐ ☐
Xe in IIIrd excited state	↑↓	↑ ↑ ↑	↑ ↑ ↑ ☐ ☐
Xe in IVth excited state	↑	↑ ↑ ↑	↑ ↑ ↑ ↑ ☐

The structures of various xenon compounds are summarized in Table 12.7.

Table 12.7 The structures of various xenon compounds

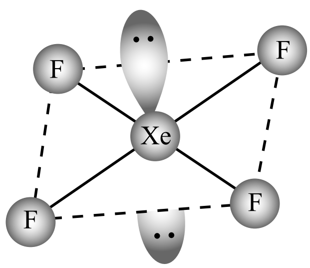
Compound	Total No. of EP $\left(\frac{V + M - c + a}{2}\right)$	No. of BP	No. of LP	Type of hybridization	Shape	Bond angle	Remark
XeF_2	5	2	3	sp^3d	linear	180°	3LP occupy equatorial positions of TBP structure
XeF_4	6	4	2	sp^3d^2	square planar	90°	2LP occupy opposite corners of octahedron
XeF_6	7	6	1	sp^3d^3	Distorted octahedron or capped octahedron		1LP protrudes into one of the triangular faces
XeO_3	4	3	1	sp^3	Pyramidal	103°	
XeO_4	4	4	–	sp^3	Tetrahedral	$109^{o}28^{1}$	
$XeOF_2$	5	3	2	sp^3d	T – Shape		
XeO_2F_2	5	4	1	sp^3d	see – saw		
XeO_3F_2	5	5	–	sp^3d	Trigonal bipyramidal		
$XeOF_4$	6	5	1	sp^3d^2	Square Pyramidal		1 LP and oxygen atom occupy opposite corner of octaedron
XeO_2F_4	6	6	–	sp^3d^2	Octahedral		Both O atoms are at opposite corners of octahedron
$[XeF_3]^+$	5	3	2	sp^3d	T – shape		
$[XeF_5]^+$	6	5	1	sp^3d^2	Square pyramid		
$[XeOF_5]^-$	7	6	1	sp^3d^3	Distorted octahedron		
$[XeF_8]^{2-}$	9	8	1	sp^3d^4	Square anti prismatic		Lone pair inactive.

Structures of Xenon Compounds



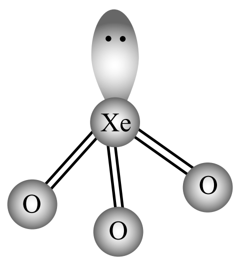
XeF_2

Linear molecule (3 lone pairs occupy the equatorial position in trigonal bipyramidal)

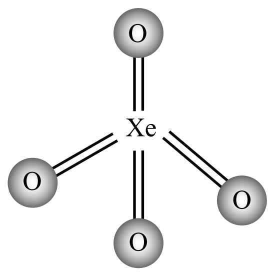


XeF_4

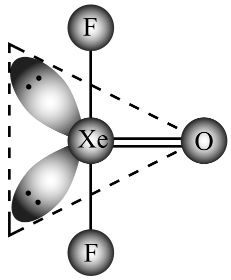
Square planar. Two LP occupy opposite corners of octahedron.



XeO_3
Pyramidal

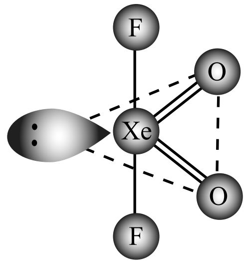


XeO_4
Tetrahedral



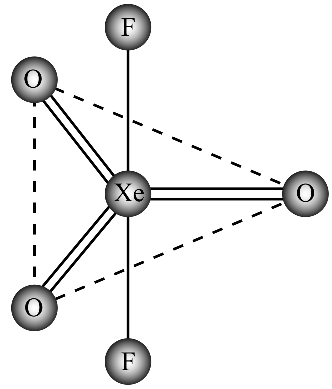
$XeOF_2$
T-shaped

Double-bonded oxygen and two lone pairs occupy equatorial positions

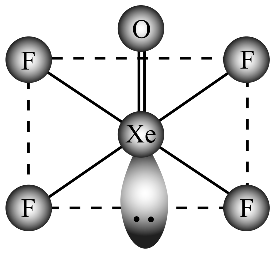


XeO_2F_2
Distorted trigonal bipyramidal

See – saw
Double-bonded oxygen and one lone pairs occupy equatorial positions

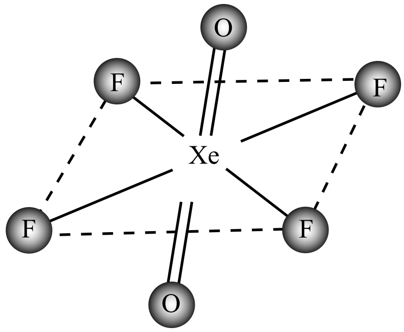


XeO_3F_2
Trigonal bipyramidal. The double-bonded oxygens occupy equatorial positions

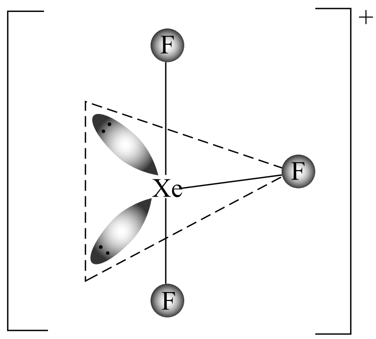


$XeOF_4$
Square pyramidal

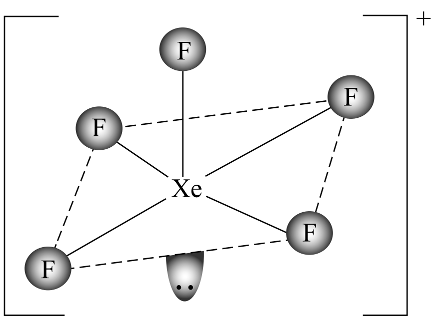
Lone pair and oxygen occupy opposite corners of octahedron



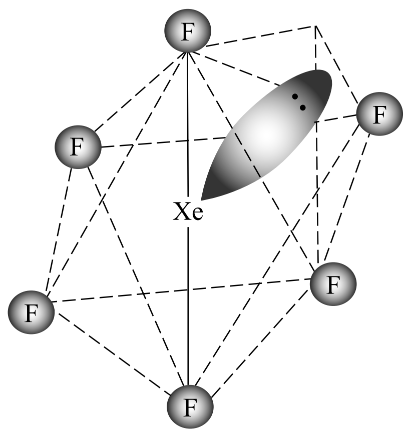
XeO_2F_4
Octahedral
Two oxygen atoms occupy the opposite corners of octahedron



$[Xe\,F_3]^+$ ion
Two lone pairs occupy equatorial positions



$[Xe\,F_5]^+$
one lone pair occupies one corner of octahedron



XeF_6
Distorted octahedron or capped octahedron. One lone pair protrudes into one triangular face of octahedron

12.6.8 Uses of Noble Gases

(a) ***Helium***

(i) It is used in filling the balloons and air ships because it is lighter and non combustible gas.

(ii) Unlike nitrogen, helium is not soluble in blood, under pressure. Hence, a mixture of 80% helium and 20% oxygen is used instead of ordinary air for respiration in deep sea diving. If air is used, the nitrogen present in the air dissolves in blood and would give a painful sensation called *bends* or *Caisen sickness* due to its evaporation as they come out.

(iii) A mixture of helium and oxygen can be taken by asthama patients because it is lighter than a mixture of N_2 and O_2 and diffuse easily through narrow respiratory tracts.

(iv) Liquid helium is used as cryogenic liquid to provide low temperatures for studying various phenomena occuring near absolute zero.

(v) Helium is used in gas thermometers.

(vi) On account of its lightness (density = 2 while that of air = 14.4) helium is used for inflating the tyres of big aeroplanes.

(vii) It is used in nuclear reactors as heat transfer agent.

(viii) Helium is used for providing an inert atmosphere in a number of metallurgical operations in the preparation of reactive metals such as magnesium.

(ix) Helium is used for providing inert atmosphere during the welding of magnesium, aluminium and stainless steel since these metals are attacked by oxygen and nitrogen at red hot temperatures, welding is carried in an atmosphere of helium.

(b) ***Neon***

(i) Neon has the property of giving an orange red glow in a discharge tube at 2 mm of pressure. It is extensively used in glow lamps (known as neon tubes) for advertising purposes. Different colours can be obtained when mixed with mercury vapour or with argon.

(ii) The glow of neon lamps is visible even through fog and mist and hence, are used as signal lights and as beacon lights for safe air navigation.

(iii) It is used in safety devices, relays and rectifiers as it has capacity for carrying high voltage for protecting electric instruments such as television sets, radio photographs, production of sound movie, warning signals and volta meters.

(c) ***Argon***

(i) A mixture of argon and mercury vapour is used in fluorescent tubes.

(ii) It is used in filling electric bulbs, Geiger counter tubes, thermionic tubes and other discharge tubes.

(iii) It is also used in creating inert atmosphere in rectifiers and in welding of aluminium and stainless steel.

(d) ***Krypton***

(i) It is used in minors caps lamps.

(ii) $_{85}Kr$ is used in electronic tubes for voltage regulations.

(iii) $_{85}Kr$ is used for the measurement of the thickness of the metal sheets and joints.

(e) ***Xenon***

(i) It is used in high speed photographic flash bulbs.

(ii) Liquid xenon is used for the detection of neutral mesons and gamma photons in the bubble chamber.

(f) ***Radon***

(i) Radon is used in the preparation of ointments in the treatment of cancer.

(ii) It is also used in locating the defects in steel casting.

(iii) Radon is used as a substitute for X-rays in industrial radiography.

THE NOBLE GASES

Summary Points

- Helium, neon, argon, krypton, xenon and radon are the **zero group** elements of the periodic table.
- Since these elements are chemically inert due to the presence of stable ns^2np^6 electronic configuration in their outermost orbit they are called **inert gases**.
- Since these elements are participating in chemical reactions but less reactive like noble metals, now-a-days they are called as **noble gases**.
- Because their occurrence is less in nature they are also called **rare gases.** Because these are prepared from air they are also called as **aerogens**.
- In the periodic table these elements are placed as zero group in between the highly electronegative halogens which exhibit –1 oxidation state and highly electropositive alkali metals which exhibit +1 oxidation state. Thus they show a change from the highly electronegative character of halogen to zero electronegative character of noble gases and again starts with highly electropositive character of alkali metals as they occupy intermediate position.
- Except helium all the other elements have ns^2 np^6 outer electronic configuration but helium has $1s^2$ electronic configuration.

Discovery and Occurrence

- Discovery of all the noble gases took about more than a century. Helium was discovered in the chromosphere of sun during the total solar eclipse in 1869 by Janssen, Lockyer and Frankland. The spectrum of helium is similar to that of sodium which contain D_1 and D_2 in the yellow region with an extra D_3 line in the spectrum of He.

- The discovery of Neon, Argon, Krypton and Xenon is mainly due to the work of Ramsay, Rayleigh, Travers etc. They separated these gases from air by fractional distillation of liquid air. Ramsay and Rayleigh were awarded noble prize for their work in the discovery of noble gases.
- Radon was discovered by Dorn as a disintegration product of radium ${}^{226}_{88}Ra \longrightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$.
- Helium occurs about 2% in the natural gas found in Canada etc. Helium was also found as occluded in the radioactive minerals. Helium is present in gases around sun and stars. Argon is found to be present in certain spring waters. Argon is the most abundant inert gas in the air.The inert gas radon do not present in air.

Isolation of Noble Gases

- Noble gases are separated and isolated from air either by physico-chemical method or physical method.
- In physico-chemical method, the mixture of noble gases from air are separated by removing N_2, O_2 CO_2 etc using chemical methods and then the noble gases are separated from one another using physical method.
- In **Ramsay-Rayleigh's first method:** (i) CO_2 is removed by passing through soda lime or caustic potash (ii) O_2 is removed by passing the air over finely divided red hot copper where O_2 react with copper forming CuO and thus removed (iii) N_2 is removed by passing over hot magnesium ribbon where N_2 converts into Mg_3N_2.
- In **Ramsay-Rayleigh second method:** Electric arcs are produced in air and oxygen in 1:9 ratio to convert the nitrogen into its oxides which are absorbed in NaOH. CO_2 is also absorbed by NaOH. Excess O_2 is removed by passing through alkaline pyrogallol.
- In **Fischer-Ringe's method:** Both N_2 and O_2 from air are removed by passing over hot mixture of 90% CaC_2 and 10% $CaCl_2$. where N_2 reacts with CaC_2 forming CaNCN and graphite. The carbon formed react with oxygen in the air forming CO and CO_2. The outcoming gas is now passed over hot cupric oxide where any CO converts into CO_2 which is removed by passing through NaOH. Moisture is removed by passing over conc H_2SO_4 or P_4O_{10}.
- The individual noble gases are separated by Dewar charcoal adsorption method. The adsorption capacity of different noble gases on charcoal depends (i) at low temperatures the adsorption capacity of noble gases increases with increase in the atomic weight (ii) adsorption capacity also depends on temperature and is inverse proportional to temperature.
- When a mixture of inert gases are passed over activated coconut charcoal at 173K, heavier Ar, Kr and Xe gases will be adsorbed while lighter helium and neon gases comes out. When the mixture of He and Ne is passed over activated charcoal at 93K. Ne will be adsorbed while He comes out.
- The charcoal on which Ar, Kr and Xe adsorbed is kept in contact with another charcoal at 77K, the lighter Ar diffuses into the charcoal at 77K leaving behind Kr and Xe.
- If the temperature of charcoal on which Kr and Xe are adsorbed at 173K is raized to 183K, Kr only comes out leaving behind Xe which can be collected later by heating.
- In the **physical method** noble gases are separated by fractional distillation of liquid air using the difference in the boiling points of different gases.
- Liquid air is separated into three fractions by fractional distillation (i) He, Ne along with N_2 (ii) Argon and (iii) Krypton xenon and liquid oxygen.
- From the first fraction N_2 is removed by passing over CaC_2, the remaining mixture of He and Ne is passed through a condensor cooled in liquid hydrogen at 20K. Ne condenses, He comes out.
- From the second fraction most of the oxygen is removed by passing through pipes cooled in liquid nitrogen. The remaining oxygen is removed by passing over heated copper.
- Krypton, xenon and oxygen can be separated by fractional distillation depending on the large difference in their b.pts.

Physical Properties

- All the noble gases are colourless, odourless, tasteless gases. The solubility of noble gases in water is very less and increases from He to Xe. Atomic radii increases from He to Rn.
- Density, b.pts, heat of vapourization increases from He to Xe.
- All the noble gases are monoatomic gases. Only weak van der Waal's forces exist between the atoms of noble gases. As the atomic size increases from He to Xe van der Waals forces also increases from He to Xe and hence b.pts increases.
- Since noble gas atoms have stable outer ns^2 np^6 electronic configuration, their ionization energies are very high. Also their electron affinities are negative i.e., electron gain enthalpies are positive.
- The liquification of noble gases is very difficult because weak van der Waal's attractive forces existing between them Ar, Kr and Xe can be liquified easily when compared to He and Ne because of their higher atomic weights.

- Helium condenses as He-I at 4.2 K and He-II at 2.2K. He-I is normal liquid but at the transition the specific heat increases abruptly, thermal conductivity increases and viscosity becomes zero, hence He-II is called **super fluid**.
- When a beaker is immersed in He-II, the liquid He flows upwards without friction over the edge of the container until the levels inside and outside are equal. At absolute zero the total He converts into He-II.

Compounds of Noble Gases

Hydrates

- With water Ar, Kr and Xe form hydrates at 273K and high pressure e.g., $Ar \cdot 6H_2O$, $Kr \cdot 6H_2O$; and $Xe \cdot 6H_2O$. These hydrates are formed due to dipole (water) induced dipole interactions. The stability of hydrates increases from Ar to Xe due to increase in the polarizability.

Clathrates

- Noble gases form a number of compounds in which the gases are trapped within cavities of crystal lattice of certain organic and inorganic substances called **clathrates** or **cage compounds**. The gas trapped is called guest while crystal structure with cavity is called **host**.
- When clathrates are melted or dissolved the trapped inert gas comes out. He and Ne cannot form clathrates because of their small size and thus they can be separated from others.

Coordination Compounds

- Argon forms large number of coordination compounds with BF_3 having formula Ar. nBF_3 where n = 1, 2, 3, 6, 8 or 16. In these compounds Ar donates a pair of electrons to boron in BF_3 atom which is electron deficient compound.

Chemical Compounds

- In 1960 Bartlett prepared the first ever chemical compound of inert gas $Xe^+[PtF_6]^-$. Bartlett prepared a solid compound O_2PtF_6 by reacting PtF_6 with molecular oxygen. Then Bartlett got the idea that PtF_6 can also react with xenon since the ionization energies of O_2 and Xe are nearly same and prepared Xe[Pt F_6].
- Inert gases form chemical compounds mainly with fluorine and oxygen because they are strongly electronegative elements. Further among inert gases xenon forms more number of compounds because it has lowest ionization energy among noble gases.

Xenon Fluorides

- XeF_2 can be prepared by heating a mixture of Xe and F_2 in 2:1 ratio in nickel vessel at about 400°C. It can also be prepared by exposing a mixture of Xe and F_2 to sunlight by taking in a pyrex glass vessel at room temperature. It is also obtained by the reaction of xenon with O_2F_2 at –118°C. It is a colourless, transparent, crystalline solid. M.Pt 129°C. It is less volatile than XeF_6, dissolves in anhydrous. HF and the solution does not conduct electricity.
- It oxidizes water to HF and O_2, H_2 to HF, I_2 to IF and HCl to HF and Cl_2. It is a good fluorinating agent and fluorinate alkenes, reacts with NO forming NOF and NO_2 forming NO_2F.
- **Xenon tetrafluoride** is best prepared by heating a mixture Xe and F_2 in 1 : 5 volume at 400° at a pressure of 6atm in a nickel vessel. It can also be prepared by passing electric discharge in mixture of Xe and F_2 in 1 : 2 molar ratio at –80°C.
- It is a white crystalline solid, easily sublimes, m.pt. 117.1°C. Its properties are similar to XeF_2 except that it is rather stronger flourinating agent and acts as a good oxidizing agent. It oxidizes H_2 to HF : HCl to HF and Cl_2; BCl_3 to BF_3 and Cl_2, Hg to HgF_2; iodide to iodine.
- It disproportionates in water and form highly explosive solid XeO_3. It also form addition compounds with SbF_5 having composition XeF_4. 2 SbF_5. Similar compounds are also formed by other Lewis; acids like PF_5, AsF_5, NbF_5, RuF_5 etc.
- It is strong fluorinating agent, fluorinates several compounds and metals. For example, Pt to PtF_4, NO to NO_2F etc.
- **Xenon hexafluoride** is best prepared by reacting xenon and fluorine in the ratio 1 : 20 in a nickel vessel at low temperatures and at 50 atms pressure. It is also formed by the oxidation of XeF_4 with O_2F_2 at a temperature below –80°C.
- XeF_6 is a crystalline solid, m.pt. 49.5°C, more volatile than XeF_2 and XeF_4. It is stronger oxidizing and stronger fluorinating agent.
- XeF_6 dissolves in liquid HF which is a good conductor of electricity due to the formation of XeF_5^+ and HF_2^- ions.
- It oxidizes H_2, HCl and NH_3. It hydrolyses in water forming xenon oxyfluorides and finally to a highly explosive XeO_3.
- XeF_6 also react with silica forming xenon oxyfluorides and finally converts to XeO_3. So XeF_6 cannot be stored in glass vessels.
- XeF_6 hydrolyses in strongly in alkaline solutions forming perxenate ion XeO_6^{4-}.

Xenon Oxides

- XeO_3 is formed either by hydrolysis of XeF_4 or XeF_6. XeO_3 is a white non-volatile compound, stable in aqueous solution, explodes violently when dry. It is a strong oxidizing agent oxidizes Pu^{3+} to Pu^{4+}.
- XeO_3 reacts with NaOH and disproportionate to Xe and sodium perxenate Na_4XeO_4. $8H_2O$. Sodium perxenate decomposes to XeO_3 when dissolved in acid solution.
- XeO_4 is prepared by the reaction of barium perxenate with anhydrous H_2SO_4.
- XeO_4 is unstable and decomposes into Xe and O_2. For structural details refer to Table 12.7 and for uses refer to 12.7.8.

Single Answer Questions

1. Noble gases are sparingly soluble in water due to
(a) Dipole-dipole interactions
(b) Dipole-induced dipole interactions
(c) Induced dipole-induced dipole interactions
(d) Hydrogen bonding

2. The ease of liquefaction of noble gases decreases in the order
(a) He > Ne > Ar > Kr > Xe
(b) Xe > Kr > Ar > Ne > He
(c) Kr > Xe > He > Ar = Ne
(d) Ar > Kr > Xe > He > Ne

3. The first compound of noble gas prepared by Bartlett was
(a) $XeOF_4$ (b) $Xe^+[PtF_6]^-$
(c) XeF_4 (d) XeF_6

4. Helium-oxygen mixture is used by deep sea divers in preference to nitrogen-oxygen mixture because
(a) Helium is much less soluble in blood than nitrogen
(b) Nitrogen is much less soluble in blood than helium
(c) Due to high pressure under the deep sea nitrogen and oxygen react to give poisonous nitric oxide
(d) Nitrogen is highly soluble in water

5. Which of the following statements about noble gases is false?
(a) They are used to provide inert atmosphere in many chemical reactions
(b) They are only sparingly soluble in water
(c) They form diatomic molecules
(d) Some of them are used to fill discharge tubes used for advertising signs

6. Gradual addition of electronic shells in the noble gases causes a decrease in their
(a) Ionization enthalpy (b) Atomic radius
(c) Boiling point (d) Density

7. Helium is used in balloons instead of hydrogen because it is
(a) Lighter than hydrogen
(b) Incombustible
(c) More abundant than hydrogen
(d) Radioactive and easily detected

8. Which of the following statements is not correct?
(a) Helium has the lowest boiling point among the noble gases
(b) Argon is used in electric bulbs
(c) Krypton is obtained during radioactive disintegration
(d) Xe forms XeF_6

9. XeF_6 dissolves in anhydrous HF to give a good conducting solution which contains
(a) H^+ and $XeF^-{}_2$ ions (b) $HF^-{}_2$ and $XeF^+{}_5$ ions
(c) $HXeF^+{}_6$ and F^- ions (d) H_2F^+ and $XeF^-{}_7$ ions

10. $XeF_2 + PF_5 \longrightarrow (A^+B^-)$

$XeF_4 + SbF_5 \longrightarrow (C^+D^-)$ then the correct statement about the equations is
(a) The geometry of A^+ and C^+ are same
(b) The geometry of A^+ and D^- are same
(c) The geometry of B^- and D^- are same
(d) The geometry of C^+ and B^- are same

11. O_2 can be converted into O_2^+ by using
(a) KF (b) F_2
(c) PtF_6 (d) XeF_6

12. Which of the following elements is unique and has two liquid phases?
(a) He (b) Ne
(c) Ar (d) Kr

13. $XeF_6 + BF_3 \rightarrow ?$
(a) $[XeF_5]^+[BF_4]^-$
(b) $[XeF_7]^-[BF_2]^+$
(c) $XeF_4 + 2BF_3$
(d) $2XeF_2 + BF_3$

14. The increase in boiling points of the noble gases from He to Xe is due to
(a) Decrease in ionization enthalpy
(b) Increase in polarization
(c) Increase in atomic volume
(d) Increase in electron gain enthalpy

15. The idea which prompted Bartlett to prepare first ever compound of noble gas was
(a) High bond energy of Xe–F
(b) Low bond energy of F–F in F_2 molecule
(c) Ionization enthalpy of molecular oxygen and xenon are almost similar
(d) none of the above

16. The atomic weight of noble gases is determined with the help of the following relationship
(a) Atomic weight = equivalent weight × valency
(b) Atomic weight = equivalent weight/valency
(c) Atomic weight = 2 × vapour density = molecular weight
(d) Atomic weight = valency/equivalent weight

17. The heats of vapourization of noble gases vary in the order:
(a) He > Ne > Ar > Kr > Xe > Rn
(b) He < Ne < Ar < Kr < Xe < Rn
(c) Xe < Kr < Ne < He < Rn < Ar
(d) He < Ne ≈ Ar > Kr < Xe < Rn

18. If ionization enthalpy for hydrogen atom is 13.6 eV, then ionization enthalpy of He^+ will be:
(a) 54.4 eV (b) 6.8 eV (c) 13.6 eV (d) 24.5 eV

19. In XeF_4 molecule, the two lone pairs of electrons on Xe atom occupy which of the following positions on the square planar structure?
(a) Two adjacent corners on the planar square
(b) Two diagonally opposite corners on the planar square
(c) One corner of the planar square and one trans position
(d) Any two adjacent positions

20. The density of nitrogen gas prepared from air is slightly greater than that of nitrogen prepared by a chemical reaction from a compound of nitrogen due to the presence of the following in arial nitrogen
(a) Argon
(b) CO_2
(c) Some nitrogen molecules analogous to O_2
(d) Greater amount of nitrogen molecules derived from ^{15}N isotope

21. Welding of magnesium can be done in an atmosphere of
(a) Xe (b) He (c) Kr (d) O_2

22. The reason that only xenon fluorides are known but the corresponding chlorides have not been reported is
(a) High bond energy of Xe-F bond and low dissociation energy of F_2 molecule
(b) Smaller bond energy of Xe-Cl bond and larger bond dissociation of Cl_2 molecule
(c) Both 1 and 2 (d) None of the above

23. The first ionization energy of Ar is less than that of Ne. An explanation of this fact is that
I. The effective nuclear charge experienced by a valence electron in Ar is much larger than that in Ne
II. The effective nuclear charge experienced by a valence electron in Ar is much smaller than that in Ne
III. The atomic radius of Ar is larger than that of Ne
IV. The atomic radius of Ar is smaller than that of Ne
(a) I and III (b) I and II
(c) II and III (d) III and IV

24. Which of the following contains minimum number of lone pairs around Xe atom?
(a) XeF_4 (b) XeF_6
(c) $XeOF_2$ (d) XeF_2

25. Liquids flow from a higher to a lower level. Which of the following liquids can climb up the walls of the glass vessel in which it is placed?
(a) Mercury (b) Liquid N_2
(c) Liquid He II (d) Water

26. Structure of XeF_5^+ ion is
(a) Trigonal bipyramidal (b) Square pyramidal
(c) Octahedral (d) Pentagonal

27. D_3 line observed in the yellow region of the sun's spectrum is due to
(a) Na (b) Ne (c) Kr (d) He

28. In analogy to $O_2^+[PtF_6]^-$ a compound $N_2^+[PtF_6]^-$ will not be formed because
(a) The ionization enthalpy of N_2 gas is higher than that of O_2 gas
(b) The ionization enthalpy of N_2 gas is lower than that of O_2 gas
(c) The ionization enthalpy of N_2 gas is higher than that of N atom
(d) None of these

29. $[SbF_5]$ reacts with XeF_4 to form an adduct. The shapes of cation and anion in the adduct are respectively
(a) Square planar, trigonal bipyramidal
(b) T– Shaped, octahedral
(c) Square pyramidal, octahedral
(d) Square planar, octahedral

30. Xenon tetrafluoride, XeF_4 is
(a) tetrahedral and acts as a fluoride donor with SbF_5
(b) Square planar and acts as fluoride donor with PF_5
(c) Square planar and acts as fluoride donor with NaF
(d) see – saw shape and acts as a fluoride donor with AsF_5

31. $MF + XeF_4 \longrightarrow A$ (M^+ is alkali metal cation) the hybridization of central atom of A and its shape
(a) sp^3d, TBP
(b) sp^3d^3, distorted octahedral
(c) sp^3d^3 pentagonal planar
(d) No such compound is formed

32. Helium II is the most extraordinary liquid with
(a) Zero viscosity and very high heat conductivity
(b) Zero Viscosity and low heat conductivity
(c) Very high Viscosity and zero heat conductivity
(d) Very high Viscosity and very high heat conductivity.

33. XeO_4 contains
(a) 4 π bonds and the remaining 4 electron pairs form a tetrahedron
(b) 3π bonds and the remaining 5 electron pairs form a trigonal bipyramid
(c) 4 π bonds and the remaining 4 electron pairs form a square planar structure
(d) 4π bonds and the remaining 4 electron pairs form pyramid.

34. In which of the following pairs of noble gases there is large difference in van der Waals radii
(a) Kr and Xe (b) He and Ne
(c) Ne and Ar (d) Ar and Kr

35. XeF_4 reacts with NO_2 to produce
(a) $NO_2F + Xe$ (b) $N_2 + Xe$
(c) $N_2O + XeO_3$ (d) $NO_2F_2 + XeO_3$

36. XeF_6 has
(a) a tetrahedral structure with lone pair
(b) a trigonal bipyramidal structure with two lone pairs
(c) a capped octahedral structure with one lone pair
(d) a capped octahedral structure with two lone pairs

37. The fluoride of xenon with zero dipolemoment.
(a) XeF_6 (b) XeO_3
(c) XeF_4 (d) $XeOF_2$

38. Helium is used in low temperature gas thermometry because of its
(a) high thermal expansion coefficient
(b) high transition temperature.
(c) low boiling point and near-ideal behaviour
(d) it behave like a real gas

More than One Answer

1. Complete hydrolysis of XeF_2 gives
(a) Xe (b) O_2 (c) HF (d) $XeOF_2$

2. XeO_3 is formed by the hydrolysis of
(a) XeF_4 (b) XeF_6
(c) $XeOF_4$ (d) XeO_2F_2

3. Which of the following statements is/are correct?
(a) Ionization energy of Xe is almost equal to that of O_2
(b) The noble gas compound that was prepared for the first time is $Xe[PtF_6]$
(c) XeO_3 is an explosive solid
(d) Among the fluorides of Xenon, XeF_6 has lower boiling point

4. $XeOF_4$ is prepared by
(a) partial hydrolysis of XeF_4
(b) partial hydrolysis of XeF_6
(c) reaction of XeO_3 with XeF_6
(d) reaction of oxygen with XeF_4

Comprehension Type Questions

Passage-I

Except radon, all noble gases are present in the atmosphere throughout the universe in the free state. He is also found as natural gas and in various minerals containing radioactive elements. Ne, Ar, Kr and Xe are isolated from liquid air by repeated fractionation. The individual noble gases are then separated by adsorption over coconut charcoal.

1. Study of solar spectrum revealed the presence of
(a) H_2 and He (b) H_2 and Ar
(c) H_2 and Ne (d) Ar and Xe

2. Helium is found in radioactive minerals because
(a) It is a radioactive gas
(b) It reacts with radioactive elements
(c) It is formed by the disintegration of the radioactive elements present in minerals like clevite, monazite and remains enclosed within them
(d) none of the above

3. The gas which is not adsorbed on activated coconut charcoal is
(a) Xe (b) Ne (c) He (d) Ar

Passage-II

The noble gases have closed-shell electronic configuration and are monoatomic gases under normal conditions. The low boiling points of the lighter noble gases are due to weak dispersion forces between the atoms and the absence of other interatomic interactions.

Xenon reacts directly with fluorine and gives compound from oxidation states II to VIII are known. XeF_4 and XeF_6 are violently hydrolysed by water to give stable aqueous solution up to 11 M of XeO_3

1. $XeF_6 + NaNO_3 \longrightarrow A.$
The compound A is
(a) $XeOF_4$ (b) $XeOF_2$
(c) XeO_3 (d) XeO_3F_2

2. The reaction between XeF_6 and XeO_3 produce
(a) $XeOF_4$ (b) XeO_2F_2
(c) XeO_3F_2 (d) $XeOF_2$

3. Which of the following statement is correct?
(a) XeF_6 oxidizes HCl to chlorine.
(b) XeF_4 oxidize platinum to PtF_6
(c) XeF_6 with NOF forms $[NO]_2^+ [XeF_8]^{2-}$
(d) XeF_4 on heating with Xe gives XeF_2

Matching Type Questions

1. Match the following Column-I with Column-II

Column-I	Column-II
(a) XeF_4	(p) One lone pair on central atom
(b) $XeOF_4$	(q) sp^3d^2-hybridization
(c) $XeOF_2$	(r) two lone pairs on central atom
(d) XeO_3	(s) sp^3d-hybridization

2. Match the following which may have more than one matching (+5 marks for correct answer and zero for wrong answer)

List-I	List-II
(a) XeO_3	(p) sp^3 hybridization
(b) XeO_4	(q) sp^3d^2 hybridization
(c) XeF_6	(r) sp^3d^3 hybridization
(d) XeF_4	(s) Tetrahedral

Single Answer Questions

1. b	**2.** b	**3.** b	**4.** a	**5.** c	**6.** a
7. b	**8.** c	**9.** b	**10.** c	**11.** c	**12.** a
13. a	**14.** c	**15.** c	**16.** c	**17.** b	**18.** a
19. b	**20.** a	**21.** b	**22.** c	**23.** c	**24.** b
25. c	**26.** b	**27.** d	**28.** a	**29.** b	**30.** b
31. c	**32.** a	**33.** a	**34.** b	**35.** a	**36.** c
37. c	**38.** c				

More than One Answer Questions

1. a, b, c **2.** a, b, c, d **3.** a, b, c, d **4.** b, c

Comprehensive Type Questions

Passage-I	**1.** a	**2.** c	**3.** c
Passage-II	**1.** a	**2.** b	**3.** a

Matching Type Questions

1. a-q, r	b-p, q	c-r, s	d-p
2. a-p	b-p, s	c-r	d-q

Hints

1. Water is polar and induces polarity in the inert gas atoms.
2. With increase in size of atoms van der Wall's attractive forces increases. So liquification becomes easier.

9. $XeF_6 + HF \longrightarrow XeF_5^+ + HF_2^-$

10. $XeF_2 + PF_5 \longrightarrow XeF^+ + PF_6^-$

$XeF_4 + SbF_5 \longrightarrow XeF_3^+ + SbF_6^-$

Both PF_6^- and SbF_6^- have octahedral structures

11. PtF_6 oxidizes O_2 to O_2^+ and form compound $O_2^+[PtF_6]^-$

12. Helium exists in two liquid phases He—I and He—II

13. $XeF_6 + BF_3 \longrightarrow XeF_5^+ + BF_4^-$

16. Inert gases are monoatomic gases, so
At. wt. = 2 × V.D = M.wt.

17. $13.6 \times Z^2$ i.e., $13.6 \times 4 = 54.4$ ev

21. Welding of Mg should be carried in inert atmosphere because it burns in air forming MgO and Mg_3N_2

29. $XeF_4 + SbF_5 \longrightarrow [XeF_3^+] + [SbF_6]^-$

XeF_3^+ is T shaped and SbF_6^- is octahedral.

30. XeF_4 has 6 electron pairs around Xe atom of which 2 electron pairs occupy opposite corners of octahedron. So XeF_4 is square planar and act as fluoride donor as above.

31. $NaF + XeF_4 \longrightarrow NaXeF_5$

XeF_5^- has 7 electron pairs arranged in pentagonal bipyramid. The two lone pairs occupy the two axial positions. So it has pentagonal planar structure.

32. XeO_4 has tetrahedral structure with four Xe = O bonds.

34. van der Wall's radii of inert gases 120, 160, 191, 200 and 200pm for He, Ne, Ar, Kr and Xe respectively.

36. XeF_6 has capped octahedral structure.

13 CHAPTER

The d- and f-Block Elements

(Transition and Innertransition Elements)

I was sufficiently interested to pursue the subject
Alexander Fleming

d-BLOCK ELEMENTS (TRANSITION ELEMENTS)

13.1 INTRODUCTION

The element in which the differentiating electron enters into d-orbital are called d-block elements. There are four series of elements in the periodic table in which the, 3d, 4d, 5d and 6d shells are filled by electrons. These four series of elements together are known as d-block elements and are present between the s- and p-block elements in the periodic table. These elements often called as transition elements because they show transition from highly metallic nature of s-block elements which typically form ionic compounds to the non-metallic nature of p-block elements which form largely covalent compounds.

Transition elements exhibit some typical characteristic properties some of which are

(i) Variable oxidation states
(ii) Para and ferromagnetic properties
(iii) Formation of coloured hydrated ions and compounds
(iv) Catalytic activity
(v) Complex forming ability
(vi) Alloy forming ability
(vii) Metallic character
(viii) Formation of interstitial compounds

Zinc, cadmium and mercury do not exhibit most of the above characteristic properties of transition elements. Typically the transition elements have an incompletely filled d-level. The elements of zinc group have a d^{10} configuration. So they do not exhibit the characteristic properties of transition elements. As per this statement, the coinage metals copper, silver, gold cannot be treated as transition elements as they also have d^{10} configuration. So a broader definition is adopted to include such elements as the elements which have an incompletely filled d-sublevel either in its elemental state or in any of its chemically significant oxidation state. Thus Cu^{2+} with $3d^9$, Ag^{2+} with $4d^9$ and Au^{3+} with $5d^8$ configurations could be included in the transition elements. The elements scandium, yttrium, lanthanum and actinium which are the initial elements in their respective series also show some differences markedly in their properties with other transition elements such as their compounds are diamagnetic and colourless. However these elements, since they have partly filled d-subs shells, are considered as transition elements. Zinc, cadmium and mercury are not considered as transition elements because the d-orbital's of the penultimate shells of these elements are completely filled in their elemental state as well as in any oxidation state. However, the initial (Sc, Y, La and Ac) and terminal (Zn, Cd and Hg) elements of the respective transition series are termed as 'non-typical transition elements' while the other transition elements are termed as typical transition elements. But both typical and non typical transition elements are studied as d-block elements in order to maintain a rational classification of elements. Here the terms 'd-block elements' and 'transition element' are used synonymously.

13.2 ELECTRONIC CONFIGURATIONS

The general outer electronic configurations of these elements is (n-1) d^{1-10} ns^{1-2}. However, this generalization has several exceptions because of very little energy difference between (n-1)d and ns orbital's. Furthermore half and completely filled sets of orbitals are relatively more stable due to spherical symmetry and more number

Table 13.1 Outer electronic configurations and lattice structures of transition elements

1st Series

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Z	21	22	23	24	25	26	27	28	29	30
4s	2	2	2	1	2	2	2	2	1	2
3d	1	2	3	5	5	6	7	8	10	10
Lattice	hcp	hcp	bcc	bcc	x	bcc	ccp	ccp	ccp	x
structure	(bcc)	(bcc)	(ccp)	(hcp)	(hcp)	(hcp)				(hcp)

2nd Series

	Y	Zr	Nu	Mo	Tc	Ru	Rh	Pd	Ag	Cd
Z	39	40	41	42	43	44	45	46	47	48
5s	2	2	1	1	1	1	1	0	1	2
4d	1	2	4	5	6	7	8	10	10	10
Lattice	hcp	hcp	bcc	bcc	hcp	hcp	ccp	ccp	ccp	x
structure	(bcc)	(bcc)								ccp

3rd Series

	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
Z	57	72	73	74	75	76	77	78	79	80
6s	2	2	2	2	2	2	2	1	1	2
5d	1	2	3	4	5	6	7	9	10	10
Lattice	hcp	hcp	bcc	bcc	hcp	hcp	ccp	ccp	ccp	x
structure	(ccp)	(bcc)								

4th Series

	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Uub
Z	89	104	105	106	107	108	109	110	111	112
7s	2	2	2	2	2	2	2	2	1	2
6d	1	2	3	4	5	6	7	8	10	10

bcc = body centered cubic, hcp = hexagonal close packed
ccp = cubic close packing x = a typical metal structure

of columbic exchange energies. Due to this factor certain elements have anomalous electronic configurations. The outer electronic configurations and the lattice structures of d-block elements are given in Table 13.1.

13.3 GENERAL CHARACTERISTICS

There are more horizontal similarities in the properties of the transition elements in contrast to the main group elements. However some group similarities also exist. We shall first discuss the general characteristics and their trends in the horizontal rows (particularly 3d row) and then consider some group similarities.

13.3.1 Physical Properties

1. Atomic and ionic radii: The variation in atomic and ionic radii in transition series is not similar as in the case of s- and p-block elements with increase in atomic number in a given series. The atomic radii and ionic radii generally decrease on moving from left to right in a period due to increase in atomic number. But the decrease in the radii across a transition series is not uniform. Further the decrease is very small when compared to the decrease in s- and p-block elements. This is because the new electron enters a d-orbital each time the nuclear charge increased by unity. It may be recalled that the shielding effect of a d electron is poor when compared with the shielding effect of s- and p-orbital's. Because of this reason the net electrostatic attraction between the nuclear charge and the outer most electron increases but the increase in electrostatic attraction is not as much as the electron enters into outer orbit. This is due to the electron entered into the d-sub shell of penultimate shell which partially shield and partially allow the increased nuclear attraction. So the decrease in atomic radii within a series is quite small. The atomic radii of the elements from chromium to copper are very close to one another. From iron onwards pairing of electrons starts. Between the paired electrons there will be some repulsive forces. As a result, the increased electrostatic attractive force will be opposed by the repulsive forces of electron pairs. So the atomic radii do not alter much on moving from chromium to copper. As the number of electron pairs increases repulsive forces between electrons also increases which are greater than the attractive forces due to increased nuclear charge. As a result the atomic and ionic radii of the terminal elements of each d-series increase.

When the atomic sizes of one series compared with those of the corresponding elements in the other series, there is an increase from the first (3d) to the second (4d) series of the elements but the radii of the third (5d) series are virtually the same as those of the corresponding members of the second series. This trend is also observed in ionic radii. This is because in the third transition series in between lanthanum and hafnium, there are the fourteen lanthanide elements in which lanthanide contraction takes place.

Table 13.2 Metallic radii (Pm) of transition elements

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
162	147	134	128	127	126	125	124	128	134
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
180	160	146	139	136	134	134	137	144	151
La*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
187	159	146	139	137	135	135.5	138.5	144	151

* 14 Lanthanoide elements

The lanthanide contraction cancels almost exactly the normal increase in size on descending the group of transition elements. The metallic radius of Hf (159 pm) and ionic radius of Hf^{4+} (71 pm) are actually smaller than the corresponding values of Zr (160 pm) and Zr^{4+} (72 pm). The metallic and ionic radii of other 2nd and 3rd transition series elements in a group have almost equal values. Hence they have similar lattice energies, solvation energies and ionization energies. Thus the differences in properties of 1st transition series and 2nd transition series are much greater than the differences between the 2nd and 3rd transition series. The effect of the lanthanide contraction is less pronounced towards the right of the d-block. However the effect still shows to some extent in the p-block elements also.

2. Densities: The atomic sizes (molar volumes) of transition elements are much lower than those of the s- and p-block elements of the neighbouring groups. As a result of decrease in molar volume there is corresponding increase in density. Accordingly the densities of the transition elements are quite high. Most of these elements have densities greater than 59 cm^{-1}. Scandium, titanium and yttrium with densities 3.00, 4.50 and 4.5g cm^{-3} respectively are the only exceptions.

In a series from left to right density increases up to the last but one element. But the last element in every series have lesser density than the elements before them. This is because of increase in the atomic size.

In a group of transition elements the density increases from top to bottom. The increase in density from the first element to second element in a group is little when compared with large increase in density from the second element to third element. This is because of similar sizes of the elements of second and third transition series elements in group due to lanthanide contraction, but the atomic masses of third transition series elements are almost double to those of second transition series elements in a group. So the densest elements are found in third transition series. Infact osmium and iridium are the densest elements.

Table 13.3 Physical properties of transition elements

1st Series

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Density (g cm^{-3})	3	4.5	6.11	7.14	7.43	7.874	8.9	8.908	8.95	7.14
	1.3	1.5	1.6	1.8	1.5	1.8	1.8	1.8	1.9	1.6
M.P^{0}C	1539	1667	1915	1900	1244	1535	1495	1485	1083	419.5
B.P^{0}C	2748	3285	3350	2690	2060	2750	3100	2920	2570	907
ΔH fus/KJ mol^{-1}	15.77	18.8	17.5	28	–13.4	13.8	16.3	17.2	13	7.28
ΔH vap/KJmol^{-1}	332.71	425	459.7	590	221	340	382	375	307	114.2

2nd Series

	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
Density g cm^{-3}	4.5	6.51	8.57	10.28	11.5	12.37	12.39	11.99	10.49	8.65
	1.2	1.4	1.6	1.8	1.9	2.2	2.2	2.2	1.9	1.7
M.P^{0}C	1530	1857	2468	1620	2200	2282	1960	1552	961	320.8
B.P^{0}C	3264	4200	4758	4650	4567	4050 extrap	3760	2940	2155	765
ΔH fus/KJ mol^{-1}	11.5	19.2	26.8	28	23.8	~25.5	21.6	17.6	11.1	6.4
ΔH vap/KJmol^{-1}	367	567	680.2	590	585	–	494	362	25.8	100

3rd Series

	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
Density g cm^{-3}	6.17	13.28	16.65	19.3	21	22.59	22.56	21.45	19.32	13.534
	1.1	1.3	1.5	1.7	1.9	2.2	2.2	2.2	2.4	1.9
M.P^{0}C	920	2222 or 2467	2980	3422	3180	3045	2443	1769	1067	–38.9
B.P^{0}C	3420	4450	5534	(5500)	(5650)	5025 extrap	4550	4170	2808	357
ΔH fus/KJ mol^{-1}	8.5	(25)	24.7	(35)	34	31.7	26.4	19.7	12.8	2.3
ΔH vap/KJmol^{-1}	402	571	758.2	824	704	738	612	469	343	59.1

3. Metallic Character: In the d-block elements the ns electron configuration remains almost constant (except in the case of anomalous electronic configurations) but the penultimate shell of electrons is expanding. Thus they have many physical and chemical properties in common. Thus all transition elements are metals. They are therefore good conductors of electricity and heat, have metallic lustre and hard, strong and ductile.

4. Lattice Structures: These elements exhibit all the three types of structures face centered cubic (fcc) hexagonal close packed (hcp) and body centered cubic (bcc). The VIII and IB group metals are soft and are more ductile. This is because of their crystal structure of fcc type. The other metals have hcp or bcc structure. The fcc type crystal lattices will contain more planes where deformation can occur. So these elements are softer and ductile.

5. Melting and Boiling Points: The transition metals have very high melting and boiling points. The high melting and boiling points of transition metals are attributed to the stronger forces that bind their atoms together. This can be explained as follows.

According to valence bond theory, the presence of one or more unpaired d-electrons contributes to higher interatomic forces on account of covalent bonding and therefore, to high melting points. In each series the melting points of these metals rise to maximum value and then decreases with increase in atomic number. The number of unpaired electrons increases upto manganese and then decreases to zero in zinc due to pairing of electrons. So the maximum melting points can be expected in manganese group, but they have low melting points than the elements in the adjacent groups. This is because Mn and Tc have stable half filled d^5 configuration due to which the d-electrons are reluctant to participate in covalent bonding. Of all the elements tungsten have maximum melting point because more number of d-electrons participate in bonding (note the tungsten has no anomalous electronic configuration unlike chromium and molybdenum). In each series the last element has lowest melting points and boiling points because in these elements the d-sub level is completely filled and do not participate in bonding.

Table 13.4 Melting points and boiling points of d-block elements (°C)

Sc	**Ti**	**V**	**Cr**	**Mn**	**Fe**	**Co**	**Ni**	**Cu**	**Zn**
1541	1660	1887	1857	1244	1535	1495	1456	1083	419.5
2831	3287	3377	2672	1962	2750	2870	2732	2567	907
Y	**Zr**	**Nb**	**Mo**	**Tc**	**Ru**	**Rh**	**Pd**	**Ag**	**Cd**
1522	1852	2468	2617	2200	2310	1966	1552	962	321
3338	4377	4742	4612	4567	3900	3727	3140	2212	765
La	**Hf**	**Ta**	**W**	**Re**	**Os**	**Ir**	**Pt**	**Au**	**Hg**
921	2230	2996	3422	3180	3054	2410	1772	1064	–38.9
3457	5197	5534	5657	5627	5027	4130	3827	2808	357

The trends in melting points and boiling points can also be explained in terms of band theory (Chapter 3). In a simplified approach, we may understand the initial rise in melting point and boiling point as more and more electrons occupy the lower energy bonding levels in the metallic bond produced primarily by the d-orbitals. The bonding energy levels are completely filled in metals having five d-electrons per atom. Later, the electrons will enter energy levels in the bond which are non bonding or anti bonding in nature. So this will weaken the net bonding interaction among metal atoms. However, the sample trend expected in this manner is subject to many alterations from a number of different factors. Some of these are common for most atomic properties. Proceeding horizontally along any transition series, effective nuclear charge increases steadily from one element to the next owing to poor screening by the d-electrons. Consequently the (n–1) d-orbitals are gradually lowered in energy—the orbitals contract and the electrons occupying them ultimately become a part of core electrons as one reaches group 12 (Zn, Cd, ….). The participation of (n–1)d and ns valence electrons in bonding, the metal atoms is thus greatly influenced. The melting point and boiling point both show a maxima at group 5. Vanadiun ($3d^3 4s^2$), effectively uses all five electrons per atom to fill the lower half (bonding) of the band of energy levels. The situation gets complicated in later elements due to several factors: (i) the separation between (n-1)d and ns orbitals gradually increases; (ii) exchange energy increases for the d^5 configuration (iii) electron-electron repulsion also increases to force some electrons to occupy the upper anti bonding region in the energy band (against to aufbau principle). This weakens the net bonding interaction as we observe in the very low melting point of Mn (and to a smaller extent in Tc).

In Tc, and also in Re the inter-electron repulsion decrease owing to larger size of the orbitals. So the decrease is not great. (iv) Along the third transition series the increase in effective nuclear charge is further enhanced by the very poorly screening 4f electrons.

Moving vertically down the group we observe that both melting point and boiling point remarkably rise from 3d to 5d, indicating stronger metal-metal bonding. It appears that the extent of overlap between valence-shell d-orbitals increase as 3d–3d < 4d–4d < 5d–5d. This trend is reverse of the trend observed in groups 1 and 2. It is likely that the d-orbitals can provide better overlap through their longer spatial distribution and lower inner-electron repulsion.

6. Electrical Conductivity and Metallic Lustre: The good electrical conductivity and metallic luster of these elements is due to the delocalization of ns electrons over the entire crystal lattice. The electrical conductivity and metallic luster also indicates the presence of metallic bonding along with covalent bonding.

7. Ionization Enthalpies: The ionization enthalpies of transition elements are fairly high due to their small size. The first ionization potentials lie between the values of those of s- and p-block elements. This shows that the d-block elements are less electro positive than s-block elements (alkali and alkaline earth metals).

In a given transition series ionization enthalpies increases from left to right with increase in atomic number due to an increase in nuclear charge which accompanies the filling of the inner d-orbitals. The first three ionization enthalpies of the transition elements are given in Table 13.5.

The increase in ionization enthalpies across a series is not so pronounced as in the case of s- and p-block elements and the shielding effect created due to the expansion of (n-1) d-orbital oppose each other. But due to the poor shielding effect of d-electrons the nuclear attraction reaching the ns electrons is less. On account of these effects the ionization enthalpies increase rather slowly on moving in a period of the first transition series.

There are some certain irregularities in the first ionization enthalpies of 3d series elements. They are not much important for their chemical behaviour. However these irregularities can be accounted as follows. The energies of 3d and 4s orbitals are very close and are sensitive to the nuclear charge. The conversion of a neutral metal atom by removing an electron alters the energies of 3d and 4s orbitals. In the unipositive ion the energy of 3d orbital becomes lesser and hence the remaining electron in unipositive ion may come into 3d orbital. Due to this reorganization of electron coulombic exchange energies increases. With increase in atomic number, the number of d-electrons increases and also due to transfer of s electron into d-orbital in unipositive ions coulombic exchange energies also increases.

Along any series, electrons are successively added to the $(n-1)d$ orbital while the first ionization involves removal of an electron from the ns level. The steady increase in effective nuclear charge along the series (the d-electrons have poor shielding ability) contracts the sizes of the metal atoms and leads to an overall increase in the first ionization energy. But after removing the first electron, due to reshuffling of ns electron into $(n-1)d$ orbital coulombic exchange energy increases. This compensate some ionization energy and hence ionization energies do not change much in a transition series. From the Table 13.6 it can be seen that increase in coulombic exchange energy is maximum at d^3s^2 (V) and d^8s^2 (Ni). So their ionization energies are less than their adjacent elements. Though we expect more first ionization enthalpies for chromium (increase in coulombic exchange energy zero but nuclear charge increases), it does not occur because the removal of first electron gives the stable half-filled d^5 configuration.

Table 13.5 The first three ionization enthalpies of first transition series elements $\Delta i\ H^0/KJ\ mol^{-1}$

	Sc	**Ti**	**V**	**Cr**	**Mn**	**Fe**	**Co**	**Ni**	**Cu**	**Zn**
I_1	631	656	650	653	717	762	758	736	745	906
I_2	1235	1309	1414	1592	1509	1561	1644	1752	1958	1734
I_3	2393	2651	2828	2990	3260	2962	3243	3402	3556	3829
	Y	**Zr**	**Nb**	**Mo**	**Tc**	**Ru**	**Rh**	**Pd**	**Ag**	**Cd**
I_1	616	660	664	685	702	711	720	805	731	868
I_2	1181	1267	1382	1558	1471	1617	1743	1875	2072	1631
I_3	1980	2218	2416	2621	2850	2747	2997	3176	3361	3616
	La	**Hf**	**Ta**	**W**	**Re**	**Os**	**Ir**	**Pt**	**Au**	**Hg**
I_1	538	642	761	770	760	840	880	870	890	1007
I_2	1067	1440	1500	1700	1260	1600	1680	1791	1980	1810
I_3	1850	2250	2100	2300	2510	2400	2600	2800	2899	3300

Table 13.6

Elec. config of M	Exchange energy	Elec. config of M^+	Exchange energy	Increase in exchange energy	Elec. config of M^{2+}	Exchange energy	Loss in exchange energy
d^1s^2	o	d^2s^0	E	E	d^1	o	E
d^2s^2	E	d^3s^0	3E	2E	d^2	E	2E
d^3s^2	3E	d^4s^0	6E	3E	d^3	3E	3E
d^5s^1	10E	d^5s^0	10E	o	d^4	6E	4E
d^5s^2	10E	d^6s^0	10E	o	d^5	10E	o
d^6s^2	10E	d^7s^0	11E	E	d^6	10E	E
d^7s^2	11E	d^8s^0	13E	2E	d^7	11E	2E
d^8s^2	13E	d^9s^0	16E	3E	d^8	13E	3E
$d^{10}s^1$	20E	$d^{10}s^0$	20E	o	d^9	16E	4E
$d^{10}s^2$	20E	$d^{10}s^1$	20E	o	d^{10}	20E	o

The electronic configuration of palladium is $4d^{10}\ 5s^0$. The removal of first electron decreases the coulombic exchange energy by about 4E. So first ionization energy of palladium is greater than its adjascent elements.

The second ionization energies of the d-block elements present a comparatively regular trend since most of them involve removal of an electron from the d^n configuration. The corresponding loss in exchange energy is easily calculated. We can see that $M^+(g)$ ions having d^5 and d^{10} configurations undergo maximum loss in exchange energy on loosing one more electron. Accordingly we expect that the second ionization energies of corresponding metals will be high. The values of I_2 in Table 13.5 support this conclusion for all three rows in the d-block.

The trend in the third ionization enthalpies is not complicated by the 4s orbital factors and shows the greater difficulty of removing an electron from the d^5 (Mn^{2+}) and d^{10} (Zn^{2+}) ions. We can see the third ionization enthalpy of iron (Fe^{2+}) is less than those of manganese (Mn^{2+}) and cobalt (Co^{2+}). This is again due to the formation of Fe^{3+} ion with stable half-filled d^5 configuration. The very high third ionization energies of nickel, copper and zinc indicate that the formation of oxidation state higher than two is very difficult for these elements.

Ionization enthalpies cannot be generalized for stability of the oxidation states.

8. Standard Electrode Potentials of M^{2+}/M: The metals have a tendency to lose electrons and convert into positive ions which go into solution when placed in water or solution containing its own ions.

$$M_{(s)} \longrightarrow M^{n+}_{(aq)} + ne^-$$

The magnitude of the electrode potential E set up between the two depends on the particular metal, the number of electrons involved, the activity of the ions in solution and the temperature. E^0 is the standard electrode potentials measured under standard condition of temperature and with unit activity. It is constant for a particular metal. These terms are related as

$$E = E^0 + \frac{RT}{nF} \ln \frac{1}{a_{M^{n+}}}$$

Where R is gas constant, T is the absolute temperature, $a_M{}^{n+}$ is the activity of ions in solution, n is the valency of the ion and F is the faraday. The $a_{M^{n+}}$ is generally replaced by the concentration of ions in solution. The single electrode potential will be measured with reference to standard hydrogen electrode ($E^0 = 0$). The standard electrode potentials are also known as redox potentials and the species involved in the redox reaction are called redox couple. The redox potentials will be affected by

1. Sublimation energy of the solid metal
2. Ionization energy of a gaseous metal atom
3. Hydration energy of a gaseous ion

These are best considered in terms of Born-Haber cycle.

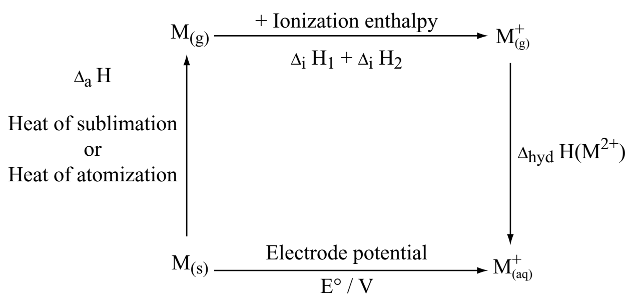


Table 13.7 Thermochemical data (KJ mol^{-1}) for the 1st transition series elements and the standard electrode potentials for the reduction of M^{2+} to M

Redox couple	Half reaction	$\Delta_a H(H)$	$\Delta_i H_i$	$\Delta_i H_2$	$\Delta_{hyd} H(M^{2+})$	E^0/V
Ti^{2+}/Ti	$Ti^{2+} + 2e \rightarrow Ti$	469	661	1310	–1866	–1.63
V^{2+}/V	$V^{2+} + 2e \rightarrow V$	515	648	1370	–1895	–148
Cr^{2+}/Cr	$Cr^{2+} + 2e \rightarrow Cr$	398	653	1590	–1925	–0.90
Mn^{2+}/Mn	$Mn^{2+} + 2e \rightarrow Mn$	279	716	1510	–1862	–1.18
Fe^{2+}/Fe	$Fe^{2+} + 2e^- \rightarrow Fe$	418	762	1560	–1998	–0.44
Co^{2+}/Co	$Co^{2+} + 2e \rightarrow Co$	427	757	1640	–2079	–0.28
Ni^{2+}/Ni	$Ni^{2+} + 2e^- \rightarrow Ni$	431	736	1750	–2121	–0.25
Cu^{2+}/Cu	$Cu^{2+} + 2e^- \rightarrow Cu$	339	745	1960	–2121	+0.34
Zn^{2+}/Zn	$Zn^{2+} + 2e^- \rightarrow Zn$	130	908	1730	–2059	–0.76

The standard reduction potentials of first row transition elements are fairly negative except in the case of copper the values of E^0 becomes less negative across the series. This is because of the increase in the sum of the first and second ionization enthalpies. It is interesting to note that the values of E^0 for Mn, Ni and Zn are more negative than expected from the trend. The more negative E^0 for Mn and Zn are related to the stability of half filled d^5 configuration in Mn^{2+} and completely filled d^{10} configuration of Zn^{2+} the more negative E^0 value of nickel is due to more hydration energy of Ni^{2+} ion.

Copper is unique in the 3d series having a positive E^0 value. This accounts its inability to liberate hydrogen from acids. Only oxidizing acids like conc. HNO_3 and conc. H_2SO_4 react with copper, the acids being reduced.

The transition metals are sufficiently electropositive to react with mineral acids liberating hydrogen. A few have low standard electrode potentials and remain unreactive or noble like gold and platinum. Some more metals are unreactive because of the formation of protective film of oxide on the surface. For example, chromium is so unreactive that it can be used as a protective non-oxidizing metal. Transition metals should act as good reducing agents. However they are less powerful reducing agents than alkali and alkaline earth metals and the metals of IIIrd group. This is due to less tendency to convert into ions because of their high ionization enthalpies, high sublimation energies and low hydration energies. The low hydration energies are due to the comparatively large size and low charge density.

9. Standard Electrode Potentials of M^{3+}/M^{2+}: The standard electrode potentials (E^0/V) for M^{3+}/M^{2+} are as follows:

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
–2.08	–0.37	–0.26	–0.41	+1.57	+0.77	+1.97			

The very low value of scandium is due to the stability of Sc^{3+} which has noble gas configuration. Zinc have very high value because of the removal of third electron from stable d^{10} configuration of Zn^{2+}, requires more energy. Similarly manganese also have comparatively higher value, shows that Mn^{2+} is having stable d^5 configuration. Comparatively low value for iron is due to the extra stability of Fe^{3+} having stable d^5 configuration.

Stability of Oxidation States

'Stability' is a relative term, meaningful only when referred to a given set of conditions or environment or both. Thus compounds containing Cu(I) are thermally stable, but in aqueous solution Cu^+ readily disproportionate into Cu^0 and Cu(II). In presence of cyanide ion, however, Cu(I) can be stabilized in aqueous solution.

Again the stability mentioned above means that the species is thermodynamically more stable. Redox data informs us only about the thermodynamic feasibility of a process which may or may not actually take place at an appreciable rate. Thus the permanganate ion is expected to oxidize water in acid medium ($E^0_{MnO_4^-/Mn^{2+}} = 1.51$). But the reaction is so slow that it may be neglected in volumetric analysis. We shall briefly discuss the influence of acidity and nature of ligands on the thermodynamic stability of oxidation states.

Since acidity of a metal ion increases with the increase of oxidation state, high oxidation states are favoured by alkaline media. An aquo ion may be considered to form an equilibrium of the following type

$$\left[M(H_2O)_6\right]^{n+} \underset{H^+}{\overset{OH^-}{\rightleftharpoons}} \left[M(OH)_6\right]^{(6-n)-}$$

Removal of an electron i.e., oxidation is now easier from the hydroxo species which is negatively charged than from the aquo ion which is positively charged. This explains why many transition metal hydroxides are readily oxidized by atmospheric oxygen, oxo species like chromate

(VI), manganate (VI) and ferrate (VI) are all formed by oxidative fusion of the lower valent compounds in alkalime medium. Anodic oxidation in alkaline solution is also used in several cases e.g., in the preparation of manganates, permanganates and ferrates.

The hydroxides of transition metal in higher oxidation state are often less soluble than the corresponding hydroxides in lower oxidation states. Thus solubility product of $Mn(OH)_2 \simeq 10^{-13}$ and that of $Mn(OH)_3 \simeq 10^{-36}$. Hence in alkaline medium Mn^{2+} is much more incompletely precipitated then Mn^{3+} and the potential of Mn (III) –Mn (II) couple is largely depressed showing stabilistation of Mn (III)

$$Mn^{3+} + e \rightleftharpoons Mn^{2+}\ eq \qquad E^0 = +1.51V$$

$$Mn(OH)_3 + e \rightleftharpoons Mn(OH)_2 + OH^- \quad E^0 = +1.51V$$

Same is the case with Fe(III)–Fe(II) couple complex formation by a ligand may have pronounce effect on the stability of a particular oxidation state relative to that in, say water. Here again, problem arises to separate kinetic and equilibrium stabilities. Kinetic stabilization of a particular oxidation state may be attained through for example by (a) use of bulky ligands to prevent reactions by steric hinderance and (b) insolubility of complex. Here we shall try to generalize regarding the thermodynamic stability of an oxidation state.

It is easy to guess that low oxidation states will be stabilized with ligands having reducing power or in reducing solvents. The ligand field stabilization gained through the electron configurations in different oxidation states and the ease of delocalization of charge with π-acceptor ligands also contribute significantly in the stabilization of different oxidation states. Several factors influence the stability of a particular oxidation state and a single generalization is difficult to arrive.

The stability of higher oxidation states is mostly influenced by the ligands that combine with them.

(i) Small, hard ligands with strong σ-donor properties like fluoride and oxide stabilize the higher oxidation states as explained later in oxidation states.

(ii) Anions containing a central atom in a high oxidation state also stabilize high oxidation states of some transition metals e.g., $M^{I}_{9}\left[Cu^{III}(TeO_6)_2\right] \cdot xH_2O$ when the cation combine with an anion, the hydrated water molecules around the cation are released in solution and thus entropy increases. Large anions with high charge are more efficient in this respect.

(iii) Some organic ligands which by themselves stable towards oxidation by oxidizing agents are known to stabilize high oxidation states: e.g., pyridine, pyridine-2-carboxylate, dipyridil, o-phenanthroline etc.

The stability of higher oxidation state in a redox couple increases with decrease in the reduction potential as shown below.

Couple	**E^0(volt)**
$\left[Fe(H_2O)_6\right]^{3+} + e^- \rightleftharpoons \left[Fe(H_2O)_6\right]^{2+}$	0.77
$[FeF_6]^{3-} + e^- \rightleftharpoons [FeF_6]^{4-}$	0.40
$\left[Fe(CN)_6\right]^{3-} + e^- \rightleftharpoons \left[Fe(CN)_6\right]^{4-}$	0.36
$\left[Fe(oxinate)_3\right] + e^- \rightleftharpoons \left[Fe(oxinate)_3\right]^-$	–0.20

Stabilization of a lower oxidation state in comparison to water is marked by an increase in the standard reduction potential of the couple above the aqueous potential as show below.

Couple	**E^0(volt)**
$\left[Fe(H_2O)_6\right]^{3+} + e \rightleftharpoons \left[Fe(H_2O)_6\right]^{2+}$	0.77
$\left[Fe(dipy)_3\right]^{3+} + e \rightleftharpoons \left[Fe(dipy)_3\right]^{2+}$	0.96
$\left[Fe(o\text{-}phen)_3\right]^{3+} + e \rightleftharpoons \left[Fe(o\text{-}phen)_3\right]^{2+}$	1.12

The electron configurations and spin states of an ion play an important role in the stabilization of oxidation states. This will be discussed in the next chapter (Chapter 14).

A redox predominance diagram shows the predominant (thermodynamically most stable) oxidation state and chemical form of an element at any given potential. In this type of diagram, elements with no redox chemistry (such as argon) will predominate at all potentials, and will cover the entire diagram. Among those with more than one thermodynamically stable oxidation state, more strongly oxidizing stable chemical forms are confirmed to higher regions of redox predominance diagrams. Correspondingly reducing stable chemical forms will be low in the predominance diagrams.

From the Fig 13.1 it can be seen that most strongly oxidizing is the ferrate ion FeO_4^{2-} and the most strongly

+6 oxidation state	$FeO_4^{2-} + 8H^+ + 3e^- \rightleftharpoons Fe^{3+} + 4H_2O$ $E^\circ = +2.20$	Fe_3O_4 2.20
+3 oxidation state	$Fe^{3+} + e^- \rightleftharpoons Fe^{2+} \quad E^\circ = +0.77V$	Fe^{3+} 0.77
+2 oxidation state	$Fe^{2+} + 2e^- \rightleftharpoons Fe(s) \quad E^\circ = +0.77V$	Fe^{2+} –0.44
		Fe

Fig 13.1 Redox predominance diagram for the element iron, with the corresponding redox reactions

reducing is iron metal Fe. The horizontal line with potential value separates two chemical forms of the element in question. Above the E^0 shown at horizontal line, the (more strongly oxidizing) chemical from above line is predominant form; below that E^0; the (less oxidizing) chemical form below the line is the predominant form.

We can note that the thermodynamically stable forms of the elements are arranged in the diagrams so that higher oxidation states of elements occurs higher in redox predominance diagrams. Thus iron (VI), in the chemical form FeO_4^{2-} is the most stable form in a solution of pH 0 and a total iron concentration of 1M above a potential of +2.20 V. As the potential is lowered, the stable oxidation state drops, first to +3, then to +2, then to 0, in elemental iron.

Redox reactivity may be predicted from the redox predominance diagram. Two species that touch each other or overlapping redox predominance regions have a potential at which they can coexist and will not tend to react with each other. Thus FeO_4^{2-} and Fe^{3+} have predominance areas that touch at +2.20 V; this means that they are equilibrium with each other at a total iron concentration of 1M at pH 0.

Two species that have non-overlapping predominance areas are expected to react with each other to give products that are stable in the presence of each other. This redox reaction will generate a standard cell emf E^0 that equals the numerical gap between their predominance regions. Thus FeO_4^{2-} will oxidize either Fe^{2+} or Fe metal. The cell emf generated by the reaction with Fe^{2+} will be 2.20 – 0.77 = +1.43 V. In this case, the only possible product of each species will be the form of iron between these two Fe^{3+}.

The balanced redox reaction is obtained by combining the reduction half-reaction with the oxidation half reaction.

$$FeO_4^{2-} + 8H^+ + 3e^- \longrightarrow Fe^{3+} + 4H_2O$$

$$Fe^{2+} \longrightarrow Fe^{3+} + e^-$$

The balanced overall redox reaction must not contain electrons. All the electrons produced by the oxidation half-reaction are consumed by the reduction half-reaction. The balanced overall reaction is

$$FeO_4^{2-} + 8H + 3Fe^{2+} \longrightarrow 4Fe^{3+} + 4H_2O \qquad E^0 = +1.43V$$

When a reaction is carried in aqueous solution, water is a potential reactant. Above an E^0 of + 1.229 V, water is oxidized to oxygen

$$2H_2O \longrightarrow 4H^+(aq) + O_2 + 4e^- \quad E^0_{redn} = 1.229V$$

The rate of this reaction is generally slow unless an over potential of about 0.5 – 0.6 V is also present. So, oxidizing agents with predominance regions confined to above +1.8V placed in water (at pH 0) will decompose rapidly, releasing O_2, which is in fact the fate of iron (VI) species FeO_4^{2-} under standard conditions.

If a reducing agent is introduced that has a predominance area entirely below 0.00V(standard conditions), the hydrogen ion in water will be reduced to H_2 by half reaction

$$2H^+ (aq, activity = 1) + 2e^- \longrightarrow H_2 \text{ (pressure = atm)}$$
$$E^0 = 0.00 \text{ V}$$

This reaction may also be slow unless an additional over potential is provided, so rapid reaction is expected if E^0 is below –0.6V. Consequently the species of an element under consideration will react rapidly with water and decompose, if its predominance range does not overlap the short-term predominance (stability) range of water, –0.6 to +1.8V.

For long term, look the narrower predominance range for water of 0.0 to +1.229V is more appropriate. Compounds or ions with predominance regions not overlapping these ranges at the appropriate pH will not persist in aqueous reactions, but may be important in non-aqueous or solid-state reactions. Hence the most important oxidation states of the elements (at pH 0) will be those that fall within this stability range of the most common solvent water.

10. Reactivity: The transition elements are less reactive this is because as already pointed

1. High sublimation energy
2. High ionization enthalpies
3. Low heat of hydration of their ions.

They have rather a tendency to remain unreactive or ‘noble’. This tendency is very well pronounced in platinum and gold. The chemical reactivity of transition metals vary widely as explained already in standard electrode potentials. The 3d series metals except copper are relatively more reactive and are oxidized by 1M H^+. But the rate of evolution of hydrogen have been found to be very slow in some of the cases. For example, titanium and vanadium, in practice are passive to dilute non oxidizing acids at room temperature. The E^0 values given in Table 13.7 also indicate the decreasing tendency to convert into divalent cations across the series.

The E^0 values for the redox couple M^{3+}/M indicate that Mn^{3+} and Co^{3+} ions are the strongest oxidizing agents in aqueous solutions. The Ti^{2+}, V^{2+} and Cr^{2+} ions are strong reducing agents and will liberate hydrogen from acids

$$2Cr^{2+} + 2H^+ (aq) \rightarrow 2Cr^{3+} (aq) + H_2$$

13.4 TYPICAL CHARACTERISTIC PROPERTIES OF TRANSITION ELEMENTS

13.4.1 Variable Oxidation States

One of the most important property of transition elements is that these elements exhibit variable valency i.e., exist in several different oxidation states. In this aspect these elements distinguishes from non-transition elements like s-block elements and the p-block elements which exhibit variable valency to a lesser extent. All transition elements except the first and the last members of each series show variable oxidation states. Further the oxidation states change in units of one e.g., Fe^{2+} and Fe^{3+}; Cu^{+} and Cu^{2+} etc whereas in p-block elements oxidation states normally differ by two units.

In transition elements the energies of ns and (n – 1)d orbitals are very close and differ only slightly. Hence the ns and (n–1)d electrons can participate in bonding. Because of this reason the transition elements exhibit variable oxidation states. The different oxidation states shown by 3d series elements are given in Table 13.8.

The minimum oxidation state that can be exhibited by a transition metal is equal to the number of ns electrons present in it. Generally all transition elements have two electrons in their ns orbital. So the minimum oxidation state exhibited by a transition metal is 2. But due to anomalous electronic configuration chromium and copper exhibit +I oxidation state because they have only one electron in their ns orbital.

The maximum oxidation state that can be exhibited by a transition metal is equal to the sum of the electrons present in both ns and (n-1)d sub-shells. The elements which exhibit the maximum number of oxidation states occur in near the middle of the series. For example, in the first transition series mangnese exhibits all the oxidation states from +2 to +7. Also in first transition series the element that exhibits highest oxidation state +7 is manganese. But in 2nd and 3rd transition series ruthenium and osmium exhibit highest oxidation state +8 in their compounds RuO_4, and OsO_4.

The stabilities of different oxidation states of a transition metal can be explained basing on the electronic configurations. The ions having fully filled and exactly half filled d-sub level and those having octet in their outer shell

Table 13.8 Oxidation states of the first transition series Metals. Most common oxidation states are underlined

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Ele. Config ns	2	2	2	1	2	2	2	2	1	2
(n–1)d	1	2	3	5	5	6	7	8	10	10
				+1					+1	
	+2	+2	+2	$\underline{+2}$	$\underline{+2}$	$\underline{+2}$	$\underline{+2}$	$\underline{+2}$	$\underline{+2}$	$\underline{+2}$
	$\underline{+3}$	+3	+3	$\underline{+3}$	+3	$\underline{+3}$	+3	+3	+3	
		$\underline{+4}$	+4	+4	$\underline{+4}$	+4	+4	+4		
			$\underline{+5}$	+5	+5		+5			
				$\underline{+6}$	+6	+6				
					$\underline{+7}$					

are stable. Thus Sc^{3+}, Ti^{4+}, V^{5+} are more stable because they have octet in their outer shells.

Chromium has two stable oxidation states +3 and +6. The stability of +6 oxidation state of chromium is due to the presence of octet in its outer shell. The stability of +3 oxidation state of chromium cannot be explained basing on electronic configuration but can be explained basing on thermodynamic data. The high lattice energy in solid state and the high hydration energy in its aqueous solution (table 13.7) makes the Cr^{3+} ion stable. Chromium in +1 oxidation state should be stable because it is having stable half filled $3d^5$ configuration but it is very unstable, because of its anomalous electronic configuration and unfavourable electrode potentials and thermodynamic values.

Manganese has three stable oxidation states +2, +4 and +7. The stability of +2 oxidation state can be explained by the extra stability of half filled d-sub shell while the stability of +7 oxidation state can be explained in a similar way as in the earlier elements. Again the stability of +4 oxidation state can be explained in a similar way as in the stability of +3 oxidation state of chromium.

Among these first five elements, the correlation between electronic structure and minimum and maximum oxidation states in simple compounds is complete. In the highest oxidation state of these first five elements all the ‘s’ and ‘d’ electrons are being used for bonding. Thus the properties depend only on the size and valency, and consequently show some similarities with elements of the main group elements in similar oxidation states. For example SO_4^{2-} and CrO_4^{2-} are iso structural as are $SiCl_4$ and $TiCl_4$.

Once the d^5 configuration is exceeded i.e., in the last five elements the tendency for all the d electrons to participate in bonding decreases. When the 4s electrons are removed, the 3d orbital contracts more and close to the nucleus. So the 3d electrons are stabilized and thus require more energy to remove a further electron to form a tripositive ion. Hence after manganese which does exhibit the oxidation number +7, the dipositive oxidation state is stable but in the case of iron, +3 oxidation state is stable, because of the extra stability of half filled $3d^5$ electronic configuration with the increase in the effective nuclear charge, the d-electrons have been stabilized and infact become core electrons.

The situation in the case of copper is interesting where the $3d^{10}$ is complete many copper (I) compounds are unstable in aqueous solution and undergo disproportion

$$2Cu^{+} \rightarrow Cu + Cu^{2+}$$

The marked instability of the Cu^{+} ion in water, in spite of its d^{10} configuration with high columbic exchange energy, may be largely attributed to the smaller size and higher charge density of the Cu^{2+} ion. This makes the hydration energy of Cu^+ much lower than that for Cu^{2+}. In fact the energy of interaction of water dipoles with the Cu^+ ion should be still less if we expect that the Cu^+ ion is 2-coordinate in water also as it is in a number of complexes like the diammine $[Cu(NH_3)_2]^+$. The Cu^{2+} ion, on the other hand is six coordinate, with four stronger and two weaker interactions from water molecules (However, on passing to silver, we shall find that this trend may be reversed making Ag^{2+} highly unstable).

Thus the stability of Cu^{2+} (aq) rather than Cu^+ (aq) is due to the much more negative hydration energy ($\Delta_{hya}H^O$) of Cu^{2+} (aq) than Cu^+ (aq) which more than compensate for the second ionization enthalpy of Cu.

The stability of Cu^{2+}(aq) rather than Cu^+(aq) is due to the much more negative hydration energy ($\Delta_{hyd}H^0$) of Cu^{2+}(aq) than Cu^+(aq) which more than compensate for the second ionization enthalpy of Cu.

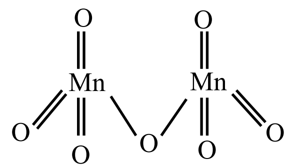
Stability of higher oxidation states: Generally higher oxidation states of elements are achived by most electronegative elements fluorine and oxygen. The highest oxidation states are achieved by titanium in TiX_4 (tetra halides) vanadium in VF_5 and chromium in CrF_6. Manganese do not form simple halides but its compound MnO_3F is known. The ability of fluorine to stabilize the highest oxidation state is due to either higher lattice energy as the case of CoF_3 or higher bond energy terms for the higher covalent compound e.g., VF_5 and CrF_6.

Though VF_5 is stable other vanadium halides undergo hydrolysis in aqueous solution to give oxohalides VOX_3. Some fluorides of certain metals are unstable in lower oxidation states because fluorine being a strong oxidizing agent always forms the compounds in higher oxidation state. On the other hand copper (II) iodide is unstable because Cu^{2+} oxidizes I^- to iodine.

Stable halides of the 3d series metals are given in Table 13.9.

Oxygen can also stabilize the higher oxidation states in the oxides of transition metals. The highest oxidation state in the oxides is equal to the group number up to manganese group (3 to 7). Beyond group 7, no higher oxides of iron above Fe_2O_3 are known, though ferrates (VI) $(FeO_4)^{2-}$ are formed in alkaline media but they readily decompose to Fe_2O_3 and O_2. Besides the oxides their oxocations like V^{5+} in VO_2^+. V^{4+} in VO^{2+} and Ti^{4+} in TiO^{2+}. The ability to stabilize the higher oxidation state is more for oxygen than fluorine. For example manganese do not form MnF_7 but forms Mn_2O_7. This is because oxygen can form multiple bonds with metals.

Nature of bonding: Generally transition metals form ionic compounds in the lower oxidation states but covalent compounds in their higher oxidation states. For example Mn^{2+} compounds are ionic but Mn^{7+} compounds are covalent. $FeCl_2$ is ionic but anhydrous $FeCl_3$ is covalent. In Mn_2O_7 two MnO_4 tetrahedrons are joined through one oxygen bridge



V^{5+}, Cr^{6+}, Mn^{5+} and Mn^{7+} ions also known to form tetrahedral $[MO_4]^{n-}$ ions.

Properties: The stability of lower oxidation state (mainly +2) increases from left to right in a transition series while the stability of higher oxidation states goes on decreasing. The transition metals in the lower oxidation state acts as reducing agents while in higher oxidation state acts as oxidizing agents. Oxidation power in the higher oxidation state increases from left to right. Thus the order of oxidation power is $Ti^{4+} < V^{5+} < Cr^{6+} < Mn^{7+}$. The reduction power in the lower oxidation state decreases from left to right in a series. For example the order of reduction power in +2 oxidation state is $Ti^{2+} > V^{2+} > Cr^{2+} > Mn^{2+}$. Fe^{2+} is stronger reducing agent than Mn^{2+} because Mn^{2+} cannot be easily oxidized due to its stable half-filled $3d^5$ configurations, but the oxidation product Fe^{3+} of Fe^{2+} is stable due to the same stable $3d^5$ configurations.

Table 13.9 Stable halides of 3d series metals

Oxidation state	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
+1									
+2	TiX_2^C	VX_2	CrX_2	MnX_2	FeX_2	CoX_2	NiX_2	CuX_2^b	ZnX_2
+3	TiX_3	VX3	CrX_3	MnF_3	FeX_3^a	CoF_2		CuX^c	
+4	TiX_4	VX_4^a	CrX_4	MnF_4					
+5		VX_5	CrF_5						
+6			CrF_6						

X = F, Cl, Br, I; X^a = F, Cl, Br; X^b = F, Cl X^C = Cl, Br, I.

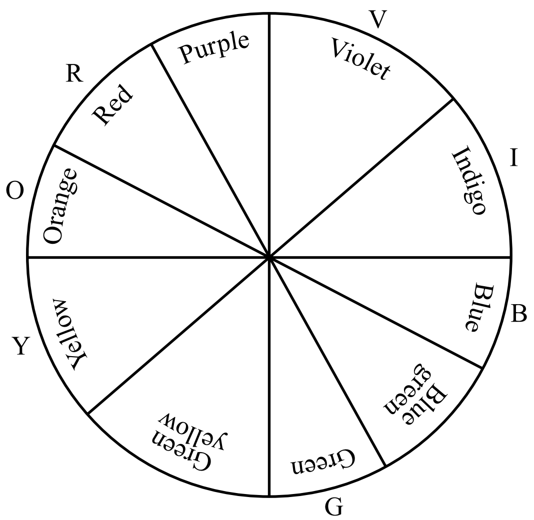
Nature of Oxides: In the oxides of the same transition metal, basic nature decreases and acidic nature increases with increases in oxidation number. In the lower oxidation states, form basic oxides but in higher oxidation state form acidic oxides e.g.,

CrO	MnO	basic
Cr_2O_3	MnO_2	amphoteric
CrO_3	Mn_2O_7	acidic

The second and third row transition elements also exhibit similar oxidation states but not exactly. For example in the iron group (VIII) the second and third row elements show a maximum oxidation state (+8) compared with +6 for iron. An interesting feature is that the higher oxidation states of 4d and 5d. series of elements are generally more stable than those of the elements of 3d series. For example molybdenum and technecium of 4d series and tungsten and rhenium of 5d series form ions like MoO_4^{2-}, TcO_4^- ReO_4^- which are stable and in which the transition elements concerned show their maximum oxidation state. The corresponding oxidation states of chromium and manganese namely chromates and permanganates are less stable and act as oxidizing agents. This is an opposite trend to the p-block elements where the stability of higher oxidation state decreases down the group (due to inert pair effect).

13.4.2 Colour

When light passes through a material, some light radiations might be absorbed by the substance. If the absorption occurs in the visible region of the spectrum, the transmitted light is coloured with the complementary colour of the absorbed colour of light. The complementary colours can be identified using Munshel colour wheel.



The colour absorbed and their wavelengths, their complementary colours are given in Table 13.10.

Table 13.10 Absorbed colours, their wavelengths and complementary colours

Colour absorbed	Approx. wavelength of the absorbed colour (nm)	Complementary colour of the absorbed colour
Violet	400–450	Yellow green
Blue	450–490	Yellow
Green blue	480–490	Orange
Blue green	490–500	Red
Green	500–560	Purple
Yellow green	560–575	Violet
Yellow	575–590	Blue
Orange	590–625	Green blue
Red	625–750	Blue green

The majority of transition metal ions are coloured, the colour and the number of 3d electrons present in some hydrated ions are given in Table 13.11.

Notice that the hydrated Sc^{3+} and Zn^{2+} ions have respectively none and ten 3d electrons these ions are colourless. The hydrated Cu^+ ion having $3d^{10}$ electrons is also colourless. It clearly shows the relation between the unpaired electrons and colour of the transition metal ions.

A complete theory of colour is very complex but, put simple, it is due to the movement of electrons from one d-level to another. Since the five separate 3d orbitals are oriented differently in space an electron (or electrons) which is close to a ligand will be repelled and hence the energy of such orbitals will be raised relative to others. The degeneracy of the 3d levels is therefore destroyed; this is represented pictorially for the copper (II) ions (Fig. 13.2).

It will be seen from the Fig 13.2 that two of the 3d levels are raised in energy relative to the other three, the energy difference being ΔE where, by the Plank's equation on

$$\Delta E = h\nu$$

Radiation of frequency given by the above equation will rise an electron from the lower 3d level to the upper one, and in the case of transition metal ions this radiation is part of the visible light. Hydrated copper (II) ions are blue

Table 13.11 The colours of some transition metal ions

$Sc^{3+}(aq)$	Colourless	$3d^0$	$Fe^{3+}(aq)$	Yellow	$3d^5$
$Ti^{3+}(aq)$	Purple	$3d^1$	$Fe^{2+}(aq)$	Green	$3d^6$
$V^{3+}(aq)$	Green	$3d^2$	$Co^{2+}(aq)$	Pink	$3d^7$
$Cr^{3+}(aq)$	Violet	$3d^3$	$Ni^{2+}(aq)$	Green	$3d^8$
$Mn^{3+}(aq)$	Violet	$3d^4$	$Cu^{2+}(aq)$	Blue	$3d^9$
$Mn^{2+}(aq)$	Pink	$3d^5$	$Zn^{2+}(aq)$	Colourless	$3d^{10}$

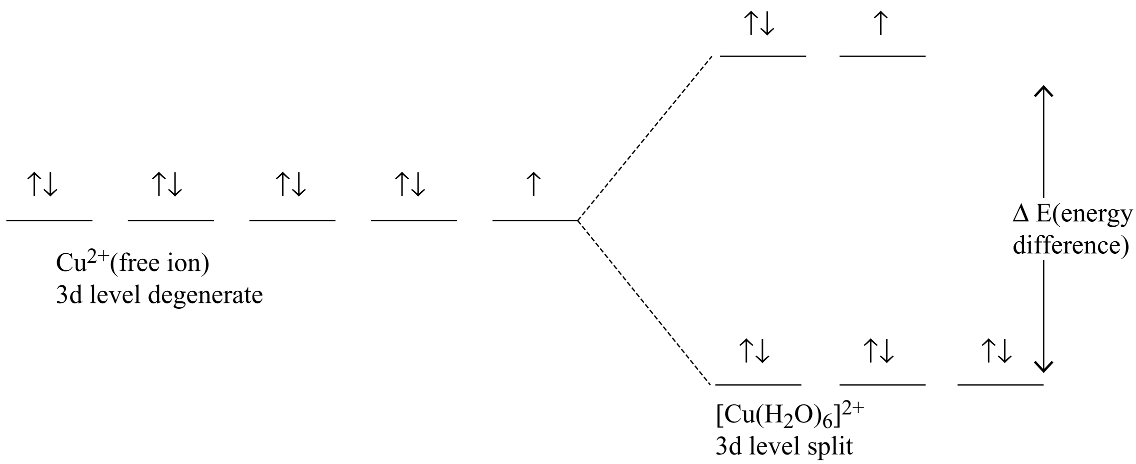


Fig 13.2 Splitting of d-orbitals in hydrated Cu (II) ion in water.

because red light of the appropriate frequency is absorbed (while light minus red gives blue).

The colour of a particular transition metal ion depends upon the nature of the ligands (either neutral molecules such as water which contain lone pairs or negative ions such as the chloride ion) bonded to the ion. The pale blue hydrated copper (II) ion changes to dark blue in the presence of sufficient ammonia and to green if sufficient chloride ions are added. Copper (II) chloride solution is therefore either blue or green, depending upon the relative concentration of water molecules and chloride ions.

$$[Cu(H_2O)_6]^{2+} + 4Cl^- \rightleftharpoons [CuCl_4]^{2-} + 6H_2O$$

In practice an aqueous solution of copper (II) chloride contain a variety of complex ions containing both water molecules and chloride ions, but in dilute solution the hydrated complex predominate and in large concentrations of chloride ions the chloride complex will be the predominating species.

Similarly hydrated cobalt (II) ions are pink but in the presence of sufficient chloride ions a blue complex is formed $[CoCl_4]^{2-}$.

From the above discussion it is clear that the light radiation absorbed will have the energy equal to ΔE i.e., the difference in the energy between the two sets of d-orbitals. This ΔE depends on the nature of metal ion, the nature of ligands and several other factors. If the ligands repels the d-orbitals more, the ΔE also increase and hence the energy of the light radiation required for excitation of electron, changes and hence colour also changes. because of this reason, the colour of Cu^{2+} ion changes to deep blue colour in the presence of excess of ammonia and green colour in the presence of excess of chloride ions.

13.4.3 Magnetic Properties

Most of the transition elements and their compounds are paramagnetic and much of our understanding of transition metal chemistry has been derived from magnetic data. So it is necessary to explain the salient facts and principles of magnetism from chemical view point.

Origin of magnetic moment: All the magnetic properties of substances in bulk are ultimately determined by the electrical properties of the subatomic particles, electrons and nucleons. Because the magnetic effects due to nucleons and nuclei are some 10^{-3} times of those due to electrons, they do not have detectable effect on magnetic phenomena of direct chemical significance.

We know that an electric current travelling in a loop of wire produces magnetic character. The flow of electric current is nothing but the flow of electrons. Similar to this the movement of electrons in an atom also produce magnetic character. This magnetic character of electrons is due to their spin angular momentum and also due to orbital angular momentum The magnetic properties of any individual atom or ion will result from the combination of these two properties, i.e., the spin moment and orbital moment of the electron.

Types of Magnetism: When a substance is placed in a magnetic field (between the poles of magnets) the magnetic field produced by the electrons in that substance interacts with the externally applied magnetic field. It is interesting to note that when the various substances are placed between the poles of a magnet (i.e., in a magnetic field) they do not behave in a similar way i.e., they show different behaviours which are known as magnetic behaviours. These are classified as diamagnetism, paramagnetism, ferromagnetism, antiferromagnetism and ferrimagnetism.

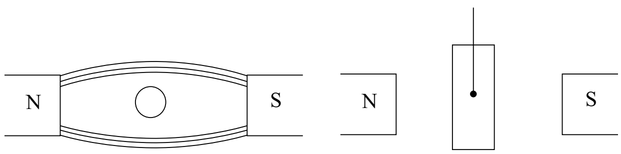
Table 13.12 Different types of magnetism and characteristic feature

Type of magnetism	Sign of χ_H	Magnitude of χ_H in cgs units	Dependence of χ_H on H	Origin
Diamagnetisn	–	$1–500 \times 10^{-6}$	Independent	Electron charge
Paramagnetism	+	$0–10^{-2}$	Independent	Spin & orbital motion of electrons in individual atoms
Ferromagnetism	+	$10^{-2}–10^{6}$	Dependent	Cooperative interaction between magnetic moments of individual atoms
Ferrimagnetism	+	$0–10^{-2}$	May be dependent	

Of these the last three are of rare occurrence and will therefore not be considered in detail.

The various magnetic behaviours differ in regard to (i) sign and magnitude of χ (ii) temperature of χ (iii) field strength dependence of χ.

Diamagnetism: If the magnetic lines of force are repelled by the substance, the field (B) in the substance is lesser than the applied field (H) (i.e., B< H). The magnetic lines of force tend to avoid such substances. It implies that it is hard for the magnetic lines to pass through a diamagnetic material than through vacuum. Then, if permitted, such substances move from stronger part if the field to the weaker part of the magnetic field.

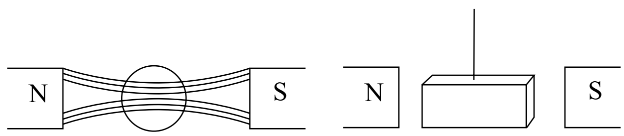


Diamagnetism is a property of all forms of matters. All substances contain at least some, if not all, electrons in closed shells. In closed shells all the electrons are paired, so their electron spin moments and orbital moments of individual electrons balance one another, so there is no net magnetic moment directly proportional to the strength of the field is induced. The electron spins have nothing to do with this induced moment; they remain tightly coupled together antiparalled pairs. However, the planes of the orbitals are tipped slightly so that a small net orbital moment is set up in the opposite direction to the applied field. It is because of this opposition the diamagnetic substances repel external magnetic field.

Even an atom with permanent magnetic moment will have diamagnetic behaviour in opposition to the paramagnetism when placed in a magnetic field provided only that the atom should contain one or more closed shells of electrons. Thus the net paramagnetism measured is slightly less than the free paramagnetism because some of the latter is cancelled by the diamagnetism.

Since diamagnetism is usually several orders (1-100 times) of magnitude weaker than paramagnetism, substances with unpaired electrons almost always have a net paramagnetism. Another important feature of diamagnetism is that its magnitude does not change with temperature. This is because the moment induced depends only on the sizes and shapes of the orbitals in the closed shells and these are not temperature dependent.

Paramagnetism: The substances which when placed in a magnetic field allow the magnetic lines of force to pass through them rather than through vacuum i.e., the field (B) in the substance is greater than the applied field (H) (B > H) are called paramagnetic substance. This phenomenon is know as paramagnetism. A para magnetic substance tends to set itself with its length parallel to the magnetic field. This paramagnetic substance is attracted into the magnetic field



Paramagnetism is due to the presence of unpaired electrons in the atoms, or ions or molecules of the substance. The greater the number of unpaired electrons, greater will be paramagnetism shown by the substance.

In substances containing one or more unpaired electrons, the magnetic field caused by the unpaired electrons are not mutually cancelled since each of the unpaired electrons has equal magnetic moment and thus some permanent and definite value of resultant magnetic moment is obtained. Thus resultant magnetic moment which is a combination of spin and orbital magnetic moments induced by the externally applied magnetic field. Such a substance experiences attraction on a magnetic field i.e., it shows paramagnetic behaviour.

Ferromagnetism it is a special case of paramagnetism. In ferromagnetic substance the strength (B) is very greater than H (i.e., B >> H). In ferromagnetic substance the lines of force are attracted very strongly and the substance may even be magnetized permanently. In these substances the magnetic moments of the individual atoms are aligned in the same direction. Ferromagnetism is exhibited by some transitional metals and their compounds e.g., Metals Fe, Co, Ni, compounds Fe_3O_4 (magnetite), $FeWO_4$ (wolframite). It is strongest in iron among the iron triad. Ferromagnetism in crystalline substance disappears in the solution form of the substance.

The magnetic property is generally expressed in Bohr magnetons (μ_{BH}) one Bohr magneton is equal to $9.273 \cdot 10^{-21}$ erg gauss^{-1} and is obtained from the relation

$$1\ \text{BM} = \frac{eh}{4\pi m_e}$$

Where e = Charge on electron h = plancks constant; and m_e = mass of electron.

Effective Magnetic Moment: The effective magnetic moment can be obtained by adding the magnetic moments due to the orbital motion (i.e., orbital magnetic moment μ_L) and that due to the spin motion (i.e., spin magnetic moment μ_s) of the electron given by

Orbital magnetic moment $\propto_L = g_L\sqrt{L(L+1)}$

And spin magnetic moment $\propto_S = g_s\sqrt{S(S+1)}$

Here L is the total orbital angular momentum, S is total spin angular momentum g_L and g_s are the Lande splitting factors for orbital and spin magnetic moments sine for $\mu_L = 1$ and μ_s, $g_s = 2$ then μ_L and μ_s are given by

$$\propto_L = \sqrt{L(L+1)}$$
$$\propto_s = 2\sqrt{S(S+1)} = \sqrt{4S(S+1)}$$

The μ_{BM} is given by

$$\propto_{BM} = \propto_L + \propto_S = \sqrt{L(L+1)+4S(S+1)}\quad \text{BM}$$

In many compounds of the 3d series metals, the orbital angular momentum due to orbital motion of unpaired electron is small because the orbital motion of these electrons quenched by the surrounding species in compounds or in solution. So the magnetic moment due to orbital motion can be ignored therefore the observed magnetic moment can be ascribed to the spin of the unpaired electrons only. Then the spin only magnetic moment.

$$\propto_S = \sqrt{4S(S+1)}\ \text{BM or}\ \sqrt{n(n+2)}\ \text{BM}$$

Where n = number of unpaired electrons and S is the sum of the spin of all the unpaired electrons having parallel spin

$$S = \frac{1}{2}\cdot\ n = \frac{n}{2}$$

Table 13.13 Magnetic properties of first row transition metal ions

Ion	Electronic configuration	μ calculated		μ_{expt}
		μ_{S+L}	μ_S	
Ti^{3+}	d^1	3.01	1.73	1.7–1.8
V^{3+}	d^2	4.49	2.83	2.8–3.1
V^{2+}, Cr^{3+}	d^3	5.21	3.87	3.7–39
Cr^{2+}, Mn^{3+}	d^4	5.5	4.9	4.8–4.9
Mn^{2+}, Fe^{3+}	d^5	5.92	5.92	5.7–6.0
Fe^{2+}, Co^{3+}	d^6	5.5	4.9	5.0–5.6
Co^{2+}	d^7	5.21	3.87	4.3–5.2
Ni^{2+}	d^8	4.49	2.83	2.9–3.9
Cu^{2+}	d^9	3.01	1.73	1.9–2.1

The formula $\mu_s = \sqrt{n(n+2)}$ is called spin only formula.

The values of magnetic moments calculated from the spin only formula for first row transition metal ions are given in Table 13.13. Also the magnetic moments calculated by using $\propto = \propto_L + \propto_s$ and the experimental magnetic moments of first row transition ions are also listed in Table 13.13.

Thus we can see from the Table 13.13 that for ions Mn^{2+} and Fe^{3+} (d^5 ions) the magnetic moment due to orbital angular momentum is largely suppressed or quenched by the metal ion. In some cases, for example, Co^{2+} the observed value for μ is higher than calculated by the spin only formula. This suggests, there is also an orbital contribution to μ_s. In the second and third row transition elements the orbital contribution is significant and so L must be included in the formula $\mu_{(S+L)}$.

13.4.4 Catalytic Activity

The catalytic activity of transition metals and their compounds is associated with their variable oxidation states. Typical catalysts are vanadium (V) oxide (Contact process) finely divided iron (Haber process) and nickel (Catalytic hydrogenation).

Catalysts at a solid surface involves the formation of bonds between reactant and the surface catalysts atoms (The first transition series elements have 3d electrons in addition to the 4s electrons which can be utilized in bonding); this has the effect of increasing the concentration of the reactants at the catalyst surface and also of weakening the bonds in the reactant molecules (the activation energy is lowered) then the reaction takes place between the reactant molecules at lowered activation energy. When once the reaction is complete the products desorbs from the surface of the catalyst giving space for the absorption of reactant molecules again.

Transition metal ions function as catalysts by changing their oxidation states e.g., iron (III) ions catalyse the reaction between iodide and persulphate ions.

$$2I^- + S_2O_3^{2-} \longrightarrow I_2 + 2SO_4^{2-}$$

A plausible, though possibly simplified explanation of this catalysis might be

$$2Fe^{3+} + 2I^- \longrightarrow 2Fe^{2+} + I_2$$

$$2Fe^{2+} + S_2O_8^{2-} \longrightarrow 2Fe^{3+} + 2SO_4^{2-}$$

It is known that both the above reactions can take place and it would be expected that two reactions between ions of opposite charge would be faster than one reaction between ions of the same type of charge.

Another example is the use of vanadium pentoxide as catalyst in the manufacture of sulphuric acid by contact process. In this the vanadium (V) oxide is first converted into vanadium (IV) oxide during the reaction but vanadium (V) oxide will be formed at the end of the process as shown below

$$SO_2 + V_2O_5 \longrightarrow SO_3 + V_2O_4$$

$$2SO_3 + V_2O_4 + O_2 \longrightarrow 2VOSO_4 \text{(Vanadyl sulphate)}$$

$$2VOSO_4 \longrightarrow V_2O_5 + SO_3 + SO_2$$

$$3SO_2 + V_2O_5 + O_2 \longrightarrow 2SO_3 + SO_2 + V_2O_5$$

So the reaction is

$$2SO_2 + O_2 \rightarrow 2SO_3$$

Some of the important catalytic activates of transition elements are given here

(i) $TiCl_4$ is used as a Ziegler – Natta catalysts in the production of polymers
(ii) MnO_2 acts as a catalyst in the decomposition of $KClO_3$. Manganese actate is used as catalyst in the oxidation of acetaldehyde to acetic acid by air
(iii) Cr_2O_3 is used as catalyst in the production of methyl alcohol from water gas
(iv) $FeCl_3$ is used as a catalyst in the production of CCl_4 from CS_2 and Cl_2. Also iron powder is used as catalyst in the manufacture of ammonia by Haber's process
(v) Cobalt chloride is used as catalyst in the production of synthetic petrol
(vi) Nickel is used as catalyst in the hydrogenation of oils
(vii) Cupric chloride is used as catalyst in the production of chlorine by Deacon's process.

13.4.5 Complex Formation

A complex ion is one that contains a central ion linked to other atoms, ions or molecules which are called ligands. If the ligands are easily removed the complex is said to be unstable and if they are difficult to remove the complex is a stable one. Among the transition metals the permanganate and chromate anions respectively MnO_4^- and CrO_4^{2-} are considered to be stable complexes and can be represented formally as MnO_4^- and CrO_4^{2-} (the Chemistry of the complex compounds is dealt separately in chapter 14). The transition metals form a large number of complex compounds. This is due to the following reasons

(i) Comparatively small sizes of the metal ions
(ii) Their high ionic charges
(iii) Availability of d-orbital's for bond formation.

13.4.6 Interstitial Compounds

Transition metals form interstitial compounds in which the small atoms like hydrogen, carbon and nitrogen are accommodated in the lattice of the transition metal crystal lattices. Some expansion of the lattice occures since the density of the hydride is less than that of the parent metal. No definite chemical formula can be assigned to these substances since they are non-stoichiometric e.g., TiC, Mn_4N, Fe_3H, $VH_{0.56}$ and $TiH_{1.7}$ etc. The formulas quoted do not of course, correspond to any normal oxidation state of the metal. Because of the nature of their composition they are referred to as interstitial compounds. They are generally have the following properties.

(i) They have high melting points higher than those of pure metals
(ii) They are very hard, for example boron carbide is as hard as diamond
(iii) They retain metallic character i.e., electrical conductivity
(iv) They are chemically inert

Sometimes stoichiometric compounds may becomes non stoichiometric compounds at higher temperature and therefore become coloured e.g., ZnO is white in cold, and yellow when hot.

13.4.7 Alloy Formation

An alloy is a homogeneous mixture of a metal with other metals or non metals or metalloids which has physical properties of the parent metal.

Pure metals have poor mechanical properties; therefore they are not used as such. The properties of the metals can be modified by converting them into alloys. Alloys are one such class of solid solutions. The properties like malleability, ductility, toughness, resistance to corrosion of metals will be modified so that they suit for industry.

Preparation of Alloys

1. Fusion method: This method involves fusing together the components of the alloy in proper proportion. In case the component metals are having different melting points, the metal with higher melting point is first melted and lumps of the other metal having lower melting point are then added to the melt. Brass an alloy of copper (m.p 1089°C) and zinc (m.p 419°C) is prepared by this method

2. Reduction method: In this method a suitable compound generally oxide of one metal is reduced in the presence of other constituent metal of the alloy. Aluminium bronze is prepared by reducing alumina in the presence of copper in electric furnace. Stainless steel is prepared by the reduction of iron chromite ore $FeCr_2O_4$.

3. Electrodeposition method: In this method alloys are prepared by the simultaneous deposition of the different component metals from the electrolytic solution having their salts solution mixture by passing current. For example Brass is obtained by the electrolysis of mixed solutions of copper and zinc cyanides dissolved in potassium cyanide.

4. Powder metallurgy or Compression method: In this method an intimate mixture of two or more component metal powders is subjected to great pressure at a temperature below the melting point of the alloy. Because of diffusion and melting, the tiny particles of one metal diffuse into another producing alloy. In this method alloys having proper structure are obtained and raw materials are not wasted e.g., Alloys involving metals like Mo and W.

5. Quenching: In this method iron–carbon alloy is produced. Iron is heated in charcoal to red hot condition and then suddenly plunged into oil or water. Steel from this method is tougher and its property depends on the percentage of carbon.

6. Amalgamation: Alloys of mercury with metals are called amalgams. These are prepared by grinding the metals with mercury. Amalgamation is used in the extraction of gold.

Table 13.14 Some important alloys and their uses

S. no	Name of the alloy	Composition of the alloy	Common uses
1	Invar	64% Fe, 35% Ni, Mn and C in trace amounts	It has low temperature coefficient used for making pendulum rods
2	Nichrome	60% Ni,25% Fe, 15% Cr	Used in heating elements of fire stoves and furnaces
3	Manganese Steel	12-15% Mn, remaining steel	Hard and tough pulverizing stones, helmets
4	Stainless steel	11.5% Cr, 2% Ni, remaining steel	Resist corrosion. House hold utensils
5	Tungsten steel	14-20% W, 3-8% Cr, remaining steel	High speed tools
6	Alnico	60% Fe, 12% Al, 20% Ni	For making permanent magnets
7	Chrome vanadium steel	1% Cr; 0.15% V, remaining steel	High tensile strength springs and shafts
8	Nickel steel	96-98% Fe; 2-4% Ni	Hard and rust proof cables automobile spare parts, gears, armor plates
9	Magnalium	85-99% Al, 15-1% Mg	Light and hard used in making balance beams, aeroplane parts or motor parts
10	Electron	95% Mg, 5% Zn	Aeroplane body and spare parts
11	Duralumin	95% Al, 4% Cu, 0.5%Mg, 0.5%Mn	Tough and light. Used in making air ships
12	Aluminium bronze	10-12% Al, 90-88% Cu	Used in making utensils, cheap jewellery, photoframes, currency coins
13	γ-alloy	1.5% Mg, 2% Ni, 4% Cu, 92.5% Al	Very light, tough, resist corrosion. Used in making parts of aeroplane
14	Type metal	60-80% Pb; 13-30% Sb, 3-10% Sn	Used in sharply defined castings
15	Wood's metal	50% Bi; 25% Pb; 12.5% Sn, 12.5% Cd	Used to automatic fire alarms; sprinklars systems.
16	Devarda's alloy	50% Cu; 45% Al, 5% Zn.	Used to reduce nitrates and nitrites to ammonia
17	Solder	50% Sn; 47.5% Pb, 2.5% Sb	Used in soldering
18	German silver	50-60% Cu; 10-30% Ni, 20-30 % Zn	Used in spoons, forks and utensils
19	Bell metal	80% Cu; 20% Sn.	Used in bells
20	Bronze	75-90% Cu; 10-25% Sn	Used in utensils, coins and statues
21	Gun metal	88% Cu; 10% Sn; 2% Zn	Used in bearing; guns
22	Brass	60-80% Cu, 20-40%	Used in machinery parts, utensils
23	Perm alloy	78% Ni, 21% Fe + C	Electromagnets

Classification of alloys: Alloys are classified in to three types

- **(i) Ferrous alloys:** These contain iron as the main constituent metal e.g., invar, nichrome, stainless steel, nickel steel, molybdenum steel, tungsten steel, ferrosilicon etc.
- **(ii) Non-ferrous alloys:** These do not contain iron e.g., brass, bronze, magnallium etc.
- **(iii) Amalgams:** These contain mercury as the constituent metal. several metals form amalgams e.g., Na_2Hg, AlHg etc.

Effect of alloying: Alloys are manufactured because they have properties more suitable for certain applications than do the simple metals. Alloys are harder and stronger than a pure metal. Alloys are prepared

- (i) For decreasing the melting points e.g., fuse wire (Wood's metal)
- (ii) For increasing tensile strength e.g., steels
- (iii) For increasing castabilty e.g., type metal
- (iv) For increasing corrosion resistance e.g., stainless steel
- (v) For modifying colour e.g., aluminium bronze having gold colour
- (vi) For changing thermal and electrical conductivity e.g., nichrome
- (vii) For increasing hardness e.g., gold is mixed with copper

13.5 TRENDS IN PHYSICAL AND CHEMICAL PROPERTIES OF 1ST, 2ND AND 3RD SERIES

In a given group, elements of the second and third series have been found to resemble each other but differ from that of the first series. Some important aspects are discussed here.

1. Atomic and ionic radii: The second row elements are larger than the first row elements. Because of the lanthanide contraction, the radii of the third row are almost the same as those for the second row. But in the first group in the d-block show the increase in size Sc $\rightarrow$ Y$\rightarrow$ La $\rightarrow$ Ac. However in the subsequent groups there is an increase in radius of 10 to 12 pm between the first and second member, but almost there is no increase between the second and third elements. This trend is shown in atomic radii and ionic radii. This is because of lanthanide contraction.

2. Density: The densities of the second row are high and those of third row elements are even higher than those of first row elements. This is because of the decrease in size due to lanthanide contraction and with the increase in mass due to increase in the number of nucleons.

3. Irregular electronic configuration: In the first transition series, only two elements namely chromium and copper have irregular electronic configurations. In the second series seven elements namely Nb, Mo Tc, Ru, Rh, Pd and Ag have irregular electronic configurations. In the third series, only two elements namely Pt and Au have irregular configurations. The irregular configurationns of Cr, Cu, Mo, Ag and Au can be explained on the basis of special stability associated with half filled and completely filled d-orbitals but others have got no reasonable explanation.

4. Reactivity of Metals: The reactivity of the transition metals decrease in going from 1st series to the 3rd series. Thus the elements of the 1st series are much more reactive than those of the second and third series.

As the size of atoms does not change much with increasing nuclear charge, Their ionization energies go on increasing but this is off set by small ions having high solvation energies. This tendency to noble character is more pronounced for platinum metals (Ru, Rh, Pd, Os, Ir, Pt) and gold.

5. Oxidation states: As we proceed from the 1st transition series to 3rd series, the lower oxidation states becomes progressively less stable and higher ones more stable. The +2 and +3 states are important for all the first row transition elements. Simple ions M^{2+} and M^{3+} are common with the 1st row but are less common for 2nd and 3rd row elements which have few ionic compounds. Similarly the 1st row elements form larger number of extremely stable complexes such as $CrCl_6^{3-}$ and $[Co(NH_3)_6]^{3+}$. Such complexes are not formed by Mo, W of Cr group and Rh or Ir of Co group.

The higher oxidation states of the 2nd and 3rd row elements are more important and much more stable than those of 1st row elements. Thus the chromate CrO_4^{2-} is strong oxidizing agent but molybdate MoO_4^{2-} and tungstate WO_4^{2-} are stable. Similarly, the permanganate (MnO_4^-) ion is a strong oxidizing agent but pertechnate TeO_4^- and perrhenate ReO_4^- ions are stable.

Some compounds exist in higher oxidation states which have no counterparts in the 1st row. For example WCl_6, ReF_7, RuO_4, OsO_4 and PtF_6. Maximum oxidation state in 1st series is exhibited by Fe group elements. Fe shows a maximum oxidation state of +6 in FeO_4^{2-} but Ru and Os exhibit maximum oxidation state of +8 in their tetroxide MO_4.

6. Complexes: The elements of 1st series rarely show coordination number (CN) greater than 6. CN of 7 and 8 are uncommon in 1st row but are much more common in the early members of the 2nd and 3rd rows.

For example in $Na_3[Zr F_7]$, the $[ZrF_7]^{3-}$ is a pentagonal bipyramid and in $(NH_4)_3$ $[ZrF_7]$ it is capped trigonal prism. In these two complexes CN is 7. In $Cu_2[ZrF_8]$ the Zr is at the centre of square antiprism where the CN is 8.

1st Transition series elements are able to form more stable complexes with N, O and F donors while those of 2nd

and 3rd series are able to form more stable complexes with P, S and X atoms.

The four coordinate complexes of the 1st series of transition elements are mainly tetrahedral, whereas that of 2nd and 3rd series are usually square planar.

7. Abundance: The ten 1st row elements are reasonably common and make up 6.79% of the earths crust. The remaining transition elements are mostly very scarce: even though the abundance of Zr is 162ppm, La 31ppm, V is 31 ppm and Nb is 20 ppm. The 20 elements in the 2nd and 3rd series elements together make up only 0.025% of the earth's crust. Tc does not occur in nature.

8. Magnetism: It is possible to explain the magnetic behaviour of the transition elements of the 1st series very easily whereas the interpretation of the magnetic behaviour of the 2nd and 3rd series elements is quite complex.

The 1st row elements form low spin complexes in the presence of strong but form high spin complexes in the presence of weak ligands (discussed later in chapter 14).

The spin only formula gives reasonable agreement relating the observed moment of 1st row metal complexes to the number of unpaired electrons. For the 2nd and 3rd row elements, the orbital contribution is significant and in addition spin-orbit coupling may occur. Thus the spin only formula is no longer valid and more complicated equations must be used. The 2nd and 3rd transition series also show extensive temperature dependant paramagnetism. This is explained by the spin orbit coupling removing degeneracy from the lowest energy level in the ground state.

9. Metal–metal bonding and cluster compounds: Metal-Metal (M-M) bonding occurs not only in the metals themselves but also in some compounds. M-M bonding is quite rare in 1st row transition elements. It occurs only in few carbonyl compounds such as $Mn_2(CO)_{10}$, $Fe_2(CO)_9$ etc and in carboxylate complexes such as chromium (II) acetate $Cr_2(CH_3COO)_4(H_2O)_2$ and in solid nickel dimethyl glyoxime.

f-BLOCK ELEMENTS (INNER TRANSITION ELEMENTS)

13.6 INTRODUCTION

The elements in which the differentiating electron enters (n-2)f – orbitals are called inner transition elements because in these elements, the differentiating electron enters into the orbital one more inner to the transition elements and also they form one more transition series inside a transition series. These are also called f-block elements because the differentiating electron enters the f-orbital. As there are seven f-orbitals in a given shell, it means that 14 electrons can be occupied in a given f-block i.e., a particular f-block series should have fourteen elements. It is possible to subdivide f-block elements into two series depending upon the nature of f-orbitals (4f or 5f) filled. The fourteen elements in which the differentiating electron enters into 4f orbitals are called lanthanides or lanthanons or lanthanoids and the other 14 elements in which the differentiating electron enters into 5f orbital are called actinoides or actinons or actinoids.

The lanthanides resemble one another more closely than do the members of ordinary transition elements in any series. They have only one stable oxidation state and their chemistry provides an excellent opportunity to observe the effect of small changes in size and nuclear charge along a series of otherwise similar elements. The chemistry of the actinoids is on the other hand much more complicated. The complication is due to the variable oxidation states in these elements and partly because their radioactivity creates special problems in their study, the two series will be considered separately here.

13.7 THE LANTHANOIDS

13.7.1 Discovery and Occurrence of the Lanthanides

As early as 1794 the Swedish chemist Gadolin announced the discovery of an oxide which he named 'yttria' and it was not until about fifty years later that this so called substance was broken down into three portions named 'yttria', 'erbia' and 'terbia'. Over the years further separations were achieved including the isolation of an oxide called lutetia. Of course the names of the lanthanides as we now know them are obtained by changing the ending –a in the oxide into-um e.g., yttria changes yttrium.

The major source of the lanthanides is monazite sand which is composed chiefly of the phosphates of thorium, cerium, neodymium and lanthanum; the phosphate portion of monazite contains small traces of other lanthanaide ions and the only lanthanaide that does not occur naturally is promethium, which is made artificially by nuclear reactions.

13.7.2 Electronic Configuration

The Electronic configurations of lanthanum and 14 lanthanides are given in table 13.15 the names, symbols, electronic configurations of some ionic states and atomic and ionic radii are also given Table 13.15 the symbol Ln is used commonly for all lanthanoids.

Lanthanum is the first member of the third transition series, and it has one 5d and two 6s electrons; the next element cerium also retain two 6s and one 5d electrons but one electron enters into f-orbital. The third element praseodymium, while still retaining two 6s electrons has 3 electrons in 4f orbits and none in the 5d orbital. The filling up of the 4f-orbitals is regular with but one exception; the

Table 13.15 Electronic configuration, atomic and ionic radii of lanthanoids and their ions

Atomic number	Name	Symbol	Ln	Ln^{2+}	Ln^{3+}	Ln^{4+}	Radii / pm Ln	Radii / pm Ln^{3+}
57	Lanthanum	La	$5d^1 6s^{2-}$	$5d^1$	$4f^0$		187	106
58	Cerium	Ce	$4f^1\ 5d^1\ 6s^2$	$4f^2$	$4f^1$	$4f^0$	183	103
59	Praseodymium	Pr	$4f^3\ 6s^2$	$4f^3$	$4f^2$	$4f^1$	182	101
60	Neodymium	Nd	$4f^4\ 6s^2$	$4f^4$	$4f^3$	$4f^2$	181	99
61	Promethium	Pm	$4f^5\ 6s^2$	$4f^5$	$4f^4$		181	98
62	Samarium	Sm	$4f^6\ 6s^2$	$4f^6$	$4f^5$		180	96
63	Europium	Eu	$4f^7\ 6s^2$	$4f^7$	$4f^6$		199	95
64	Gadolinium	Gd	$4f^7\ 5d^1\ 6s^2$	$4f^7\ 5d^1$	$4f^7$		180	94
65	Terbium	Tb	$4f^9\ 6s^2$	$4f^9$	$4f^8$	$4f^7$	178	92
66	Dysprosium	Dy	$4f^{10}\ 6s^2$	$4f^{10}$	$4f^9$	$4f^8$	177	91
67	Holmium	Ho	$4f^{11}\ 6s^2$	$4f^{11}$	$4f^{10}$		176	89
68	Erbium	Er	$4f^{12}\ 6s^2$	$4f^{12}$	$4f^{11}$		175	88
69	Thullium	Tm	$4f^{13}\ 6s^2$	$4f^{13}$	$4f^{12}$		174	87
70	Ytterbium	Yb	$4f^{14}\ 6s^2$	$4f^{14}$	$4f^{13}$		173	86
71	Lutetium	Lu	$4f^{14}\ 5d^1\ 6s^2$	$4f^{14}\ 5d^1$	$4f^{14}$	–	–	–

Only electrons outside [Xe] core indicated.

element europium has the outer electronic configuration $4f^7\ 5s^2\ 5p^6\ 5d^1\ 6s^2$ and the next element gadolinium has the extra electron in the 5d orbital i.e., its outer electronic configuration is $4f^7\ 5s^2\ 5p^6\ 5d^1\ 6s^2$. This is because half filled f-sub shell give some extra stability. The element ytterbium has a full complement of 4f electrons ($4f^{14}\ 5s^2\ 5p^6\ 5d^1\ 6s^2$). Except for lanthanum, cerium, gadolinium and lutetium (the 1st, 2nd, 8th and 15th members) which have a single 5d electron, the lanthanides do not have electrons in the 5d orbitals.

13.7.3 Atomic and Ionic Radii

In the lanthanoid series with increase in atomic number, the nuclear charge increase by one unit and one electron is added. The new electrons are added to the same 4f-sub-shells. The shielding effect of one 4f electron on another 4f electron is quite poor even than one d-electron by another with the increase in nuclear charge. This is because of the more diffused shapes of the f-orbitals. Hence with increase in nuclear charge, the effective nuclear charge experienced by the valence electrons increases. As a result there will be contraction in the sizes of the atoms of the successive elements. The contraction in size from one element to the next element in the lanthanide series is fairly small. But the cumulative effect over 14 lanthanides from Ce to Lu is about 20 pm. This cumulative effect of the contraction of the atomic size of lanthanoid series is called lanthanide contraction. There is dramatic increase in the atomic radii of europium. This is believed to be due to the difference in metallic bonding which contribute only two electrons for metallic bonding.

Consequences of Lanthanoid Contraction

(i) Due to lanthanoid contraction, the crystal structure and other properties of the 14 lanthanoids and their compounds become very close.

(ii) As a consequence of lanthanoid contraction it becomes difficult to separate them from the mixture that occur naturally.

(iii) Due to lanthanoid contraction the sizes of Zr and Hf become equal and hence the properties of both these elements and their compounds are almost similar. Because of this reason these two elements occur together and their separation from one another is also very difficult.

(iv) Due to lanthanide contraction the atomic sizes of post lanthanide elements (elements after lanthanum) in the 3rd transition (5d) series are almost become equal to the corresponding 2nd transition (4d) series elements, which are just above them. Therefore the 4d and 5d transition elements are more closer in properties than those exhibited by the elements of 3d and 4d transition elements.

13.7.4 Oxidation States

The lanthanides exhibit a principle oxidation state of +3 in which the M^{3+} ion contains an outer shell containing eight electrons and an underlying layer containing upto fourteen 4f electrons. The +3 ions of lanthanum, gadolinium and lutetium, which contain respectively an empty, a half-filled and a completely filled 4f level are especially stable. Cerium can exhibit an oxidation state of +4 in which it has the same electronic configuration as +3 lanthanum, i.e., an empty 4f level; similarly +4 terbium exists which has the same electronic configuration as +3 gadolinium i.e., a half-filled 4f level.

$$\text{La(III)}\,4f^0\,5s^2\,5p^6 \quad \text{Ce(IV)}4f^0\,5s^2\,5p^6\,(\text{empty 4f level})$$
$$\text{Gd(III)}4f^7\,5s^2\,5p^6 \quad \text{Tb(IV)}4f^7\,5s^2\,5p^6\,(\text{half-filled 4f level})$$

Further support of the idea that an empty, a half filled and a completely filled 4f level confers some extra stability on a particular oxidation state is available. Thus +2 europium is isoelectronic with +3 gadolinium i.e., a half-filled 4f level, and +2 ytterbium is isolelectronic with +3 lutetium.

$$\text{Gd(III)}\,4f^7\,5s^2\,5p^6 \quad \text{Eu(II)}\,4f^7\,5s^2\,5p^6\,(\text{half-filled 4f level})$$
$$\text{Lu(III)}\,4f^{14}\,5s^2\,5p^6 \quad \text{Yb(II)}\,4f^{14}\,5s^2\,5p^6\left(\begin{matrix}\text{Completely}\\ \text{filled 4f level}\end{matrix}\right)$$

However, other factors are involved since Pr(IV), Nd(IV), Tb(IV), Dy (IV), Sm(II) and Tm(II) exists which contain one, two, seven, eight, six and thirteen 4f electrons.

Although a few lanthanides exhibit an oxidation state of +4 while some other show an oxidation state +2, the +3 oxidation state is the most stable state in every case. For instance cerium (IV) is strongly oxidizing and samarium (II) is strongly reducing

$$Ce^{4+} + Fe^{2+} \longrightarrow Ce^{3+} + Fe^{3+}$$
$$2Sm^{2+} + 2H_2O \longrightarrow 2Sm^{3+} + 2OH^- + H_2$$

Note that in the above examples both Ce^{4+} and Sm^{2+} are converted into the +3 ions Ce^{3+} and Sm^{3+} respectively.

13.7.5 Colour

These metals are silvery white in colour but their ions have colours in both crystalline state and in aqueous solutions. The colours of these ions show a definite periodicity. The colour of the ion is directly connected to the number of unpaired electrons present in 4f-subshell. The colour of the element with $4f^n$ configuration is similar to that lanthanide elements with $4f^{14-n}$ configuration. This is illustrated in Table 13.16.

Absorption spectra of compounds of M^{3+} ions in these cases show very sharp peaks. These are due to internal transition in partly filled 4f-shell electrons and are called f-f transition. Since the 4f-orbitals are shielded from external perturbation, so the complexing agents have little effect on the spectral bands of these lanthanide ions.

Lanthanide ions having completely vacant half filled or completely filled 4f-orbitals are colourless e.g., $La^{3+}(4f^o)$, $Gd^{3+}(4f^7)$ and Lu^{3+} $(4f^{14})$ions are colourless. Dipositive and tetra positive ions which are isoelectronic with tripositive ions have different colours.

13.7.6 Magnetic Properties

Due to the presence of unpaired electrons in the 4f orbitals, all the lanthanide ions except those of La^{3+}, Lu^{3+}, Yb^{2+} and Ce^{4+} show paramagnetic character. In the case of lanthanides 4f electrons are well shielded and cannot participate in bond formation so they are well shielded from the quenching effect of the environments. So the magnetic moments of lanthnides are calculated by taking into consideration spin and orbital contributions a more complex formula is used.

$$\propto = \sqrt{L(L+1) + 4S(S+1)}$$

Where L is the total orbital angular momentum and S is total spin angular momentum. This is shown in Fig 13.3.

It can be seen from the figure that La^{3+} is diamagnetic (due to f^0), the value of magnetic moments increases from La to neodymium which has maximum value. Then decrease is observed for Sm (μ = 1.47). Magnetic moment values again increase in dysprosium and holmium have maximum values. These again fall and reach zero in Lu which is diamagnetic (due to f^{14}).

Table 13.16 Colours of some lanthanide ions

La^{3+}	Colourless	$4f^0$	Gd^{3+}	Colourless	$4f^7$
Ce^{3+}	Colourless	$4f^1$	Tb^{3+}	Pale-pink	$4f^8$
Pr^{3+}	Green	$4f^2$	Dy^{3+}	Yellow	$4f^9$
Nd^{3+}	Red Violet	$4f^3$	Ho^{3+}	Pink-yellow	$4f^{10}$
Pm^{3+}	Pink-yellow	$4f^4$	Er^{3+}	Pink	$4f^{11}$
Sm^{3+}	Yellow	$4f^5$	Tm^{3+}	Pale green	$4f^{12}$
Eu^{3+}	Pale pink	$4f^6$	Yb^{3+}	Colourless	$4f^{13}$
			Lu^{3+}	Colourless	$4f^{14}$

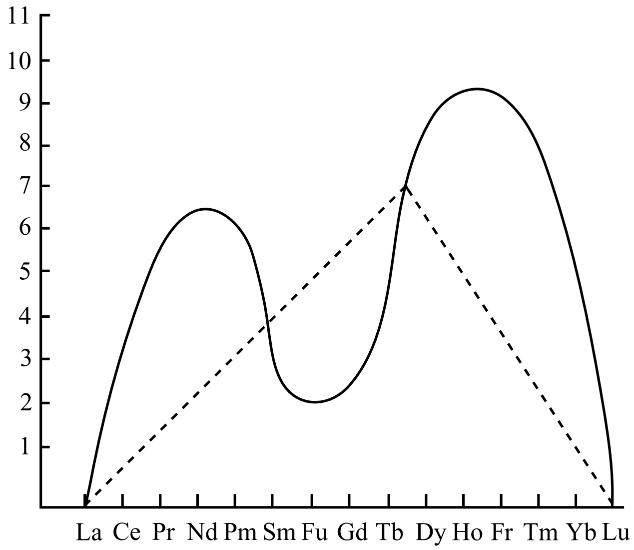


Fig 13.3 Magnetic moments

13.7.7 General Characteristics

1. **Action of air:** All the lanthanoids are silvery white soft metals and tarnish rapidly in air.
2. **Hardness:** They are hard metals. The hardness increases with increasing atomic number. Samarium is steel hard.
3. **Melting and boiling points:** Their melting points and boiling points range between 1000 to 1200k. Samarium has high melting point 1623k.
4. **Lattice structure:** They have typical metallic structure and are good conductors of heat and electricity.
5. **Density and other physical properties** change smoothly except for Eu and Yb and occasionally for Sm and Tm.
6. **Ionization energies:** The first ionization energies of the lanthanoid are around 1200kj mol^{-1}, the second about 1200 mol^{-1} comparable with those of calcium. The third ionization energies indicate that the coulombic exchange energies of completely vacant, half-filled and completely filled orbitals give more stability as in 3d series elements. Thus lanthanum, gadolinium and lutetium have low value of third ionization energy.

13.7.8 Chemical Reactivity

The earlier members of the lanthanoid series are similar in chemical behaviour with calcium, but with increase in atomic number, the latter elements behave similar to aluminium. The electrode potentials for the half reaction

$$Ln^{3+}_{(aq)} + 3e^- \rightarrow Ln_{(s)}$$

Are in the range of –2.2 to 2.4V except for Eu which has a value of –2.0V

Reaction with hydrogen: On gently heating in hydrogen gas lanthanoids from non stochiometric hydrides approaching LnH_2 and LnH_3 in composition.

Reaction with carbon: When lanthanoids are heated with carbon carbides Ln_3C, Ln_2C_3 and LnC_2 are formed.

Reaction with nitrogen: They react with nitrogen especially when warmed to form corresponding nitrides LnN.

Reaction with oxygen: On heating in oxygen lanthanoids form oxides of the Ln_2O_3. These with water form hydroxides $Ln(OH)_3$. The hydroxides are definite compounds but not just hydrated oxides. They are basic like alkaline earth metal oxides and hydroxides.

Due to lanthanoid contraction the decrease in size of Ln^{3+} ions from La^{3+} to Lu^{3+} increases the covalent character i.e., ionic character decreases between Ln^{3+} and OH^- ions in hydroxides of lanthanoids. So basic character decreases from $La(OH)_3$ to $Lu(OH)_3$. Similarly the basic character of the oxides also decreases from La_2O_3 to Lu_2O_3.

Reaction with non metals: Lanthanides also react with non-metals such as halogens, sulphur, phosphorus and silicon to form corresponding compounds.

Reaction with acids: They liberate hydrogen from dilute acids

Electropositive character: Their very low negative electrode potentials indicate that they are strongly electropositive in character. Thus they act as reducing agents.

Complex formation: Although the lanthanide ions have high charge (+3), their large size (85–103 pm) imparts them low charge density. So they have less tendency to form complexes. Their complexes with unidentate ligands are few, however complexes with a few, chelating ligands such as β- diketones, oximes and EDTA are fairly common.

The tendency to form complexes and stability tends to increase with increasing atomic number. This property finds use in the separation of lanthanoids from one other. However this order becomes reverse in the case of hydrated ions, Ln^{3+}(aq) i.e., the tendency to form complexes decrease among the hydrated lanthanoid ions with increase in atomic number.

With a specific ligand the order of complex formation for the Ln^{2+}, Ln^{3+} and Ln^{4+} ions is $Ln^{2+} > Ln^{3+} > Ln^{4+}$.

13.7.9 Separation and Extraction of the Lanthanoids

In view of the very similar properties of this group of elements, it is only comparatives recently that methods of separation have been worked out which have enabled the very pure elements to be made commercially available.

Due to lanthanoid contraction the size decreases from La^{3+} to Lu^{3+}. As the ionic radii decreases along the series the ability to form complex ions increases and this is the basis of their separation on an exchange column.

A solution containing +3 lanthanoid ion is placed at the top of a cation exchange column and the lanthanoid ions release an equivalent amount of hydrogen ions from the column.

$$M^{3+} + \underset{\substack{\text{cation}\\\text{exchange}}}{3HR} \rightarrow M(R)_3 + 3H^+$$

A citrate buffer solution (which complexes with the lanthanoid ions) is slowly run down the column and the cations partition themselves between the column itself and the moving citrate solution. Since the similar ions show a greater preference for complexing with the citrate solution, these ions at first to emerge from the column. By the correct choice of conditions the lutetium ion Lu^{3+}, emerges first from the column followed by the cations ytterbium, thulium, erbium etc., in order of increasing radius.

Lanthanum, cerium, praseodymium, neodymium and gadolinium may be obtained by reduction of their trichlorides with calcium at about 1000°C e.g.,

$$2PrCl_3 + 3Ca \rightarrow 3CaCl_2 + 2Pr$$

Terbium, dysprosium, holmium, erbium and thulium are obtained by a similar process except that their trifluorides are used, since their trichlorides are too volatile e.g.,

$$2HoF_3 + 3Ca \rightarrow 3CaF_2 + 2Ho$$

Europium, samarium and ytterbium are obtained by chemical reduction of their trioxides.

13.7.10 Uses

1. Lanthanoids are used in production of alloy steels for plates and pipes.
2. An alloy of lanthanoids is **misch metal** which consists of about 95 per cent lanthanoid, 5 per cent iron, traces of S,C, Ca and Al. A magnesium based alloy of misch metal is used to produce bullets, shells and lighter flint.
3. Mixed oxides of lanthanoids are employed as catalysts in petroleum cracking, hydrogenation, oxidation etc.
4. Many lanthanide oxides are used as phosphors in television screens and similar fluorescing surfaces.
5. Ceric sulphate is used as a good oxidizing agent in the laboratory.
6. Neodymium and praseodymium oxides are used for making coloured glasses for goggles. CeO_2 is used in gas mantles.

13.8 ACTINOIDS

13.8.1 Introduction

The elements with atomic numbers 89-103 in which the extra electron is entering into 5f-orbitals are known as 5f series or actinoides or actinons or actinoids. The actinoides constitute the second inner transition elements. The name 'actinide' is derived from actinium which is prototype of actinoides. It is important to note that only first four elements of this series namely actinium, thorium, protactinium and uranium are found in nature. The other elements are obtained artificially by nuclear bombardment. These are known as transuranium elements. All the actinoids are radioactive elements and the earlier members have relatively long half lives, the latter ones have half life values ranging from a day to 3 minutes e.g., for lawrencium (z = 103). The latter members can be prepared only in nanogram amounts which renders their study more difficult.

13.8.2 Electronic Configuration of Actinoides

General electronic configuration of the actinoides may be represented as [Rn] $5f^{1-14}\ 6d^{0-1}\ 7s^2$. In these atoms 5s, 4p, 5d, 6s, 6p and 7s are completely filled. The electron distribution is not yet clear and opinions differ. This is because in these elements the energy difference between 5f and 6d orbitals is small. Thus in these elements electrons may occupy the 5f and 6d orbital or sometimes both. Spectroscopic, chemical and other data shows that 5f level with increasing atomic number i.e., from plutonium onwards fills in a regular way except with minor changes where it is possible to attain a half filled or fully filled shell but upto plutonium i.e., in Th, Pa, U, and Np the electrons may occupy either 5f or 6d orbitals. The electronic configurations of actinoides are given Table 13.17 from the table it can be seen curium (at no 96) where the 5f shell attain half and lawrencium (at no 103) where the 5f shell attain fully filled, the electron is pushed into 6d shell.

13.8.3 Ionic Size

It can be noted from the Table 13.17 that for both +3 and +4 ions there is contraction in the size analogous to lanthanoid contraction. This is known as actinoid contraction. Although the actinoid contraction parallels initially with that of lanthanoid contraction, the elements from curium onwards are similar than might be expected. This is probably due to the poorer shielding by 5f electrons in actinoides than 4f electrons in lanthanoids.

13.8.4 Oxidation States

In the actinoides, the energies of 5f, 6d and 7s orbitals are almost equal. So the electrons in all of these orbitals may participate in bonding, unlike lanthanides. Hence some elements if not all exhibit variable oxidation states. The actinoids in

Table 13.17 Electronic configurations, atomic and ionic radii of actinoids

Atomic number	Name	Symbol	Electronic configuration M	Electronic configuration M^{3+}	Electronic configuration M^{4+}	Radii/pm M^{3+}	Radii/pm M^{4+}
89	Actinium	Ac	$6d^1\ 7s^2$	$5f^0$	–	111	
90	Thorium	Th	$6d^2\ 7s^2$	$5f^1$	$5f^0$	–	99
91	Protactinium	Pa	$5f^2\ 6d^1\ 7s^2$	$5f^2$	$5f^1$	–	96
92	Uranium	U	$5f^3 6d^1\ 7s^2$	$5f^3$	$5f^2$	103	93
93	Neptunium	Np	$5f^4\ 6d^1\ 7s^2$	$5f^4$	$5f^3$	101	92
94	Plutonium	Pu	$5f^6\ 7s^2$	$5f^5$	$5f^4$	100	90
95	Americium	Am	$5f^7\ 7s^2$	$5f^6$	$5f^5$	99	89
96	Curium	Cm	$5f^7\ 6d^1\ 7s^2$	$5f^7$	$5f^6$	99	88
97	Berkelium	Bk	$5f^9\ 7s^2$	$5f^8$	$5f^7$	98	87
98	Californium	Cf	$5f^{10}\ 7s^2$	$5f^9$	$5f^8$	98	86
99	Einstenium	Es	$5f^{11}\ 7s^2$	$5f^{10}$	$5f^9$	–	–
100	Fermium	Fm	$5f^{12}\ 7s^2$	$5f^{11}$	$5f^{10}$	–	–
101	Mendelevium	Md	$5f^{13}\ 7s^2$	$5f^{12}$	$5f^{11}$	–	–
102	Nobelium	No	$5f^{14}\ 7s^2$	$5f^{13}$	$5f^{12}$	–	–
103	Lawrencium	Lr	$5f^{14}\ 6d^1\ 7s^2$	$5f^{14}$	$5f^{13}$	–	–

Table 13.18 Oxidation states of actinium and actinoids

Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
3		3	3	3	3	3	3	3	3	3	3	3	3	3
	4	4	4	4	4	4	4	4						
		5	5	5	5	5								
			6	6	6	6								
				7	7									

general show +3 oxidation state, but the elements in the first half exhibit the higher oxidation states. The maximum oxidation state increases from +4 in Th to +5, +6 and +7 respectively in Pa, U and Np, but decreases in succeeding elements.

Since in the first half of the actinoides, the energy required for the promotion of electron from 5f to 6d is less, the lower actinoids show more higher oxidation states. But in the second half of actinoides, contraction of 5f orbital takes place due to increase in nuclear charge and come close to the nucleus. Due to this the energy required for promotion of electron from 5f to 6d becomes more. So the higher actinoides show only lower oxidation states. The actinoids resemble the lanthanoids in having more compounds in +3 states than in +4 state. However +3 and +4 ions tend to hydrolyze.

Important ionic species in which these elements are found are M^{3+}, M^{4+}, MO_2^+ MO_2^{2+}. In +6 oxidation state the actinide ions are no longer simple. The oxygenated ions are formed due to high charge density e.g., UO^{2+}, UO_2^{2+}, and NpO_2^{2+} are covalent cations. These oxocations are stable, in acid and aqueous solutions.

Maximum tendency to form complexes is observed in M^{4+} ions and decreases in others in the following order

$$M^{4+} > MO_2^{2+} > M^{3+} > MO_2^+$$

The higher complex formation tendency of MO_2^{2+} ion may be attributed to high concentration of charge on metal.

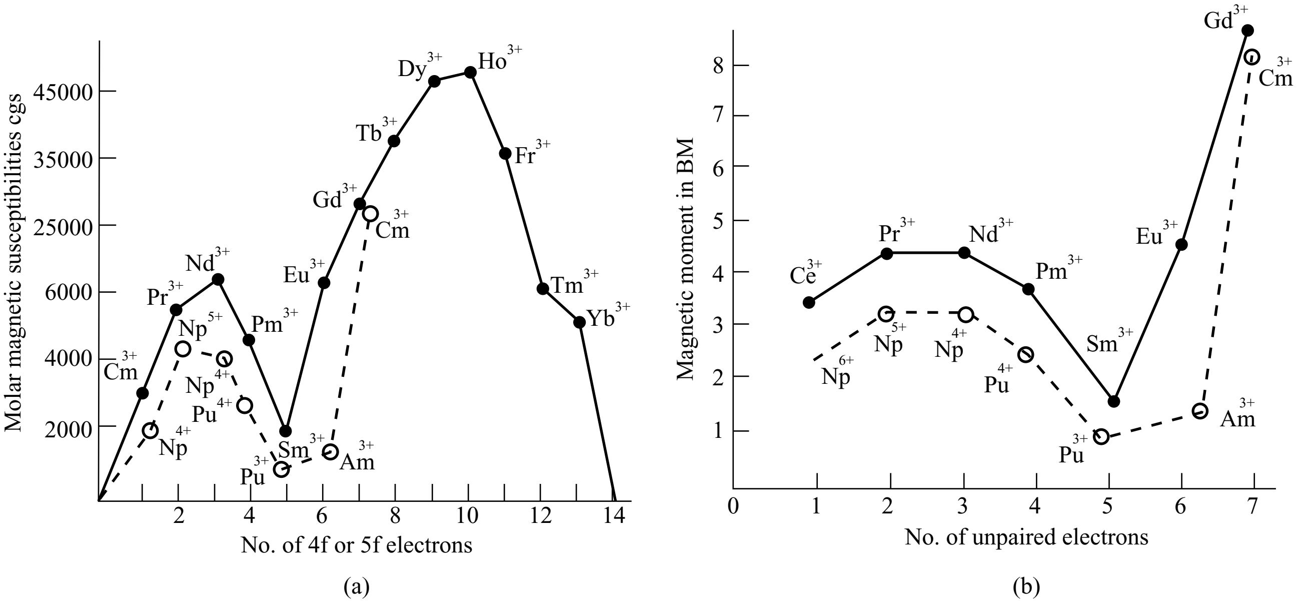


Fig 13.4 (a) Molar magnetic susceptibilities (b) Magnetic moments of lanthanides and actinoides

13.8.5 Magnetic Properties

A comparison of the plots of molar subscriptions of tripositive lanthanide and actinoide ions against the number of 4f or 5f electrons reveals that there are remarkable similarities between two plots.

The plot of lanthanide ions has two humps while that of actinoide ion has only one. In both cases the first humps is however, at the identical place. The moments of lanthanide ions agree closely with the theoritical predictions but those of the transuranic ions are somewhat lower than expected. This is because the 5f electrons of the transuranic ions are less effectively screened from the crystal field, which quenches the orbital contribution, than are the 4f electrons of i.e., lanthanide, ion.

13.8.6 General Characteristics

Lattice Structure: The actinoid metals are all silvery in appearance but display a variety of structure. The structural variability is obtained due to irregularities in metallic radii which are far greater than lanthanoids.

Chemical Reactivity: The actinoids are highly reactive metals, especially when finely divided. The action of boiling water on them for example, gives a mixture of oxide and hydride and combination with most non-metals takes place at moderate temperatures. Hydrochloric acid attacks all metals but most are slightly affected by nitric acid owing to the formation of protective oxide layers; alkalies have no action.

Ionization Energies: It is evident from the behaviour of the actinoids, though not accurately known, but are lower than those of the early lanthanoids. This is because that 5f orbitals are more diffused out and less penetrated into the inner core of electrons. So the 5f electrons are more effectively shielded from the nuclear attraction than the 4f electrons of the corresponding lanthanoids. Due to this reason the outer electrons are less firmly held and are available for bonding in the earlier actinoids.

From the discussion made so far it is evident that the earlier actinoids differ from the lanthanoids with respect to different characteristics. Even then the earlier actinoids resemble the lanthanoids in showing close similarities with each other in the series and in gradual variation in properties, which do not entail change in oxidation state. The lanthanoid and actinoid contraction also play an important role in the properties of post lanthanoid and post actinoid elements but lanthanoid contraction is more important because the chemistry of elements succeeding the actinoid are much less known.

13.9 COMPARSION OF LANTHANIDES AND ACTINOIDES

Similarities

(i) Both exhibit +3 oxidation state.
(ii) (n-2)f orbitals are progressively filled in both series.

(iii) Contraction of atomic and ionic sizes are observed in both series.

(iv) Due to f-f transition, the absorption spectra of the elements of the both series give sharp line- like band spectra.

(v) The nitrates, per chlorates and sulphates of trivalent elements of both the series are available.

(vi) The carbonates and hydroxides of trivalent of both the series are insoluble.

(vii) Members of both the series show ion exchange behaviour.

Differences

Lanthanoids	Actinoids
1. The chemistry of all the members of this series are similar due to large energy difference between 4f and 5d sub-levels.	1. Considerable variation is observed in the chemistry of these elements due to very small energy difference between 5f and 6d sub levels.
2. These have more nuclear binding energy.	2. Comparatively these have low nuclear binding energy.
3. These elements exhibit a maximum oxidation state +4 e.g., Ce^{4+}.	3. These elements show higher oxidation states such as +4, +5, +6, and +7.
4. 4f electrons have better shielding effect.	4. 5f electrons have lesser shielding effect than 4f electrons.
5. Paramagnetic properties can be easily explained.	5. Paramagnetic properties of these elements are difficult to explain.
6. Complex forming tendency of these elements is less. Complexes with π bonding ligands are not known.	6. Complex forming tendency is very high and form complexes even with π bonding ligands.
7. Except Pm all the other lanthanoids are non-radioactive.	7. All the elements in this series are radioactive.
8. These elements form less basic compounds.	8. These elements form compounds which are more basic.
9. They show no tendency to form oxycations.	9. They form oxycations like UO_2^{2+}, UO^+, NpO_2^+ PuO_2^+ PuO_2 etc.

d–BLOCK ELEMENTS KEY POINTS

- d-block elements are those in which differentiating electron enters, into d-orbital. In the d-block elements the outer most as well as the penultimate shells of the atoms are incompletely filled with electrons. In the d-block elements the penultimate shell is being expanded from eight to eighteen electrons due to the addition of 10 electrons to d-orbital.
- d-block elements are called transition elements because they show transition from metallic nature of the s-block elements to the non-metallic nature of the p-block elements and also ionic nature of the compounds of s-block elements to the covalent nature of the compounds of p-block elements.
- Zinc, cadmium and mercury of IIB (or group 12) are not considered as transition metals because they do not exhibit characteristic properties of transition elements.
- Characteristic properties of transition elements are (i) variable valency (ii) para and ferromagnetic properties (iii) formation of coloured hydrated ions or compounds (iv) catalytic activity, (v) complex forming ability (vi) alloy forming ability, (vii) metallic character (viii) formation of interstitial compounds.
- Transition elements are those which have incompletely filled or partially filled d-orbitals either in the elemental form or in their significant oxidation states.
- Since Zn, Cd and Hg contain completely filled d-orbitals they are not considered as transition elements. These are termed as 'non-typical transition elements'. Both typical and non-typical transition elements are studied as d-block elements.
- The general outer electronic configuration of d-block elements is $(n-1)d^{1-10}\ ns^{1\text{ or }2}$. Some d-block elements have anomalous electronic configurations which give stability to the atoms due to spherical symmetry and more number of coulombic exchange energies.
- Transition elements show more horizontal similarities than group similarities in contrast to main group elements.

- In a transition series from left to right atomic and ionic radius decreases slowly due to poor shielding effect of d-electrons.
- Due to the repulsion between paired electrons, the attractive force due to increased effective nuclear charge will be reduced and hence the atomic and ionic radii of terminal elements of each series (Zn, Cd and Hg) increases.
- In the same group of transition elements atomic and ionic radii increases from the first element to the second element but the atomic sizes of second and third elements (except in III A or 3rd group) are almost equal due to lanthanide contraction.
- In a series from left to right density increases up to the last but one element. But the last element in every series have lesser density than the element before them due to increase in the atomic size.
- In a transition group density increases from top to bottom. The increase in density from the first element to second element in a group is little when compared with increase in density from the second element to third element is very large. Most densest elements are found in 3rd transition series. This is because of lanthanide contraction.
- All the transition elements are metals and are good conductors of electricity and heat, have metallic lustre and are hard, strong, ductile and malleable.
- The transition elements exhibit all the three types of crystal structures fcc, hcp and bcc. The VIII and IB group metals are more soft and ductile because they have fcc crystal structure which contain more number of planes where deformation can occur.
- In addition to the normal metallic bonding, the unpaired electrons present in d-orbitals of transition elements contributes to higher inter atomic forces on account of covalent bonding and therefore they have high m.pts.
- The last elements in every series have low b.pt because they do not contain unpaired electrons. The elements of manganese group also have low m.pts than the elements in the adjacent groups because they have stable half-filled d^5 configuration due to which the d-electrons are reluctant to participate in covalent bonding of all the elements tungsten has maximum m.pt.
- Due to high b.pts the enthalpies of atomization than the corresponding elements of the 1^{st} series, metal-metal bonding occurs in the compounds of heavy transition elements.
- The ionization energies of transition elements are fairly high and their 1^{st} IE values are intermediate to s- and p-block elements. In a transition series ionization enthalpies increases from left to right as the effective nuclear charge increases due to poor shielding effect of d electrons, but the increase is slow.
- As the energy difference between 3d and 4s orbitals is little, after removing the first electron from 4s-orbital, the remaining electron in 4s orbitals comes into 3d-orbital due to decrease in its energy. Due to reorganization of electron, stability increases as the number of coulumbic exchange energies increases, and hence the second IE will become high.
- Chromium has low 1st IE though nuclear charge increases because the removal of electron does not alter the d-configuration.
- In every series the last element have highest ionization energies because of more effective nuclear charge.
- The second IE values of Cr and Cu are higher than those of their neighbours because Cr^+ and Cu^+ ions have extra stable d^5 and d^{10} configurations.
- The standard reduction potentials M^{2+}/M of 1^{st} row transition elements are fairly negative except in the case of copper and becomes less negative across the series.
- The more negative E^o for M^{2+}/Mn and Zn^{2+}/Zn are related to the stability of half-filled d^5 configuration in Mn^{2+} and completely filled d^{10} configuration of Zn^{2+}. The more negative E^o value of Ni^{2+}/Ni is due to more hydration energy of Ni^{2+} ion.
- The positive E^o value Cu^{2+}/Cu accounts its inability to liberate H_2 from acids. A few transition metals have low standard electrode potentials and remain unreactive or noble like gold and platinum.
- The less reduction power and less reactivity of transition metals is due to their high ionization energies, high sublimation energies and low hydration energies. The low hydration energies are due to comparatively large size and low charge density.
- The standard electrode potential of Sc^{3+}/Sc^{2+} is more negative because of the more stability of Sc^{3+} ion.

13.4.1 Variable Oxidation States

- The transition elements exhibits variable oxidation states because (n–1)d electrons can also participate in bonding due to less energy difference between (n–1)d and ns orbitals.
- The oxidation states of transition elements changes in units of one whereas in p-block elements oxidation states normally differ by two units.
- The minimum oxidation state exhibited by a transition elements is equal to the number of electrons in ns orbital.
- The maximum oxidation state that can be exhibited by a transition element is equal to the total number of electrons present in both ns and (n–1) d orbitals.

- Due to anomalous electronic configuration chromium and copper can exhibit a minimum oxidation state +1.
- In the 3d series maximum oxidation state increases upto Mn and then decreases from Fe onwards due to pairing of electrons. In 3d series Mn exhibit the maximum oxidation state +7.
- In 4d series a stable maximum oxidation state exhibited by technicium and an unstable maximum oxidation state +8 is exhibited by ruthenium. In 5d series a stable maximum oxidation state +8 is exhibited by osmium.
- The transition metal ions having fully filled and exactly half filled d-sublevel and those having octet in their outer shell are stable.
- The stabilities of Cr^{3+} and Mn^{4+} ions is due to the high lattice energy in solid state and high hydration energy in their aqueous solutions. Fe^{3+} ion is more stable than Fe^{2+} ion because of stable half-filled $3d^5$ electronic configuration in Fe^{3+}.
- In the last 5 elements of 3d series 3d electrons are stabilized and require more energy for their removal because the 3d orbital contracts more and come nearer to the nucleus with increase in nuclear charge. Thus in the last five elements the dipositive oxidation state becomes more stable (except Fe^{3+}).
- The stability of $Cu^{2+}_{(aq)}$ rather than $Cu^{+}_{(aq)}$ is due to the much more negative hydration energy of $Cu^{2+}_{(aq)}$ than $Cu^{+}_{(aq)}$ which more than compensate the second IE of Cu.
- Higher oxidation states of transition elements are achieved with most electronegative elements fluorine and oxygen. The ability of fluorine to stabilize the highest oxidation state is due to either higher lattice energy as in the case of CoF_3 or more bond energy terms for the higher covalent compounds e.g., VF_5 CrF_6 etc.
- The VF_5 is stable but other vanadium halides hydrolyse in water to form oxohalides VOX_3. Fluorides of certain metals are unstable in lower oxidation state because fluorine being strong oxidizing agent always form compounds in higher oxidation state. But bromides and iodides of certain elements in higher oxidation state are unstable because Br^- and F^- reduce the metal ion in higher oxidation state which act as oxidizing agent.
- Oxygen also stabilize the higher oxidation states in the oxides of transition elements. The highest oxidation state in the oxides of transition elements is equal to group number upto Mn group (3 to7).
- The ability to stabilize the higher oxidation state is more for oxygen than fluorine e.g., Mn do not form MnF_7 but forms Mn_2O_7 because oxygen can form multiple bonds with metals.
- Transition metals form ionic compounds in the lower oxidation states but form covalent compounds in their higher oxidation states, e.g., $FeCl_2$ is ionic but $FeCl_3$ is covalent. In Mn_2O_7 two MnO_4 tetrahedrons are joined through one oxygen bridge.
- The stability of lower oxidation state (mainly +2) increase from left to right in a transition series while the stability of higher oxidation state goes on decreasing.
- The transition elements in their lower oxidation state act as reducing agents while in higher oxidation state act as oxidizing agents. Oxidation power in the higher oxidation state increases from left to right in a series while the reduction power in lower oxidation state decreases from left to right in a series.
- In the oxides of the same transition metal basic nature decreases and acidic nature increases with increase in oxidation number e.g., CrO and MnO are basic, Cr_2O_3 and MnO_2 are amphoteric, CrO_3 and Mn_2O_7 are acidic.
- The higher oxidation states of 4d and 5d series of elements are generally more stable than those of the elements of 3d series a trend opposite to p-block elements where the stability of higher oxidation state decreases down the group (due to inert pair effect).

13.4.2 Colour

- Transition metals form coloured ions or compounds due to the partially filled d-orbitals.
- In the presence of solvent molecules (in solutions) or ligands (in complexes) or counter ions (in crystals) the d-orbitals split into two sets.
- The electrons in transition metal ions which occupy one set of d-orbitals having lower energy can be excited to another set of d-orbitals having higher energy by absorbing energy from visible light.
- Since the energy difference (ΔE) is small between the two sets of d-orbitals the light in the visible region only is absorbed by the electron during its excitation.
- The colour of the transition metal ion is due to d-d excitation or d-d transition of the electron. During d-d excitation the electron absorb one colour in the visible light and thus it appears in the complimentary colour of the absorbed light.
- The number of electrons undergoing d-d transition and the energy difference between the two sets of orbitals decide the colour. The colour of a particular transition metal ion. e.g., copper (II) ion which is blue in aqueous solution changes to dark blue in the presence of sufficient ammonia and to green if sufficient chloride ions are added.
- ΔE depends on the nature of metal ion, the nature of ligands and several other factors. Some metal ions exhibit different colours in different oxidation states.
- Transition metal ions having completely vacant d-orbitals (Sc^{3+}, Y^{3+}, La^{3+}, Ti^{3+}, Ti^{4+}, Zr^{4+} $H6^{4+}$) and

completely filled d-orbitals (Zn^{2+}, Cd^{2+}, Hg^{2+}, Cu^{+}, Ag^{+}, Au^{+}) are colourless.

- The colour of CrO_4^{2-}, MnO_4^{-} ions though d-orbitals in chromium and manganese are completely vacant, due to charge transfer phenomenon.

13.4.3 Magnetic Properties

- Paramagnetic substances are those which attract the external magnetic field or which move from a weaker to stronger part of the magnetic field. Paramagnetic substances contain unpaired electrons.
- Diamagnetic substances are those which repel the external magnetic field or which tend to move away from stronger part to weaker part of the magnetic field. Diamagnetic substances are those which contain all paired electrons.
- Ferromagnetic substances are those in which the spins of all unpaired electrons and thus their magnetic moments are aligned in the same direction. Iron, cobalt and nickel are ferromagnetic substances.
- The magnetic property of a substance is due to the spin angular momentum and orbital angular momentum of two electron, and can be calculated by using the formula

$$\propto_{S+L} = \sqrt{4S(S+1)+L(L+1)} \quad \text{BM}$$

Where S is the sum of spin quantum numbers of electrons and L is the sum of orbital angular momentum quantum numbers.

- The units of magnetic moment are Bohr Magnetrons (BM)

one $\text{BM} = eh/\pi 4\, m_e = 9.273 \times 10^{-21}$ erg Gauss^{-1}
where m_e = Mass of electron, h = Planck's constant.

- In many compounds of 3d series metals, the magnetic moment due to orbital motion of electron is neglected because the orbital motion of these electrons is quenched by two surrounding species in compounds or solution. Then the spin only magnetic moment can be calculated by using

$$\propto_s = \sqrt{4S(S+1)} \quad \text{BM} \qquad \text{or} \qquad \propto_s = \sqrt{n(n+2)} \quad \text{BM}$$

- The experimental magnetic moments of certain transition metal ions are more than the calculated value due to some contribution of magnetic moment due to orbital motion.

Catalytic Activity

- Transition elements and their compounds acts as good catalysts in industries and biological systems.
- The catalytic activity of transition metals is due to their variable valency or to form coordination compounds.
- In the manufacture of H_2SO_4 for the oxidation of SO_2 to SO_3, V_2O_5 acts as catalyst by changing its oxidation state from V^{5+} to V^{4+} while oxidizing SO_2 to SO_3 and then again converted into V^{5+} from V^{4+} by oxygen.
- In certain cases the catalysts provide suitable surface where the reactant molecules adsorb and form unstable intermediate by the incompletely filled d-orbitals during which activation energy of reactant molecules decreases and can participate in the reaction.
- Certain examples of transition metal catalysis are
  (i) Iron powder mixed with molybdenum powder is used as catalyst in the manufacture of ammonia by Haber's process.
  (ii) Platinized asbestos is used as catalyst for the oxidation of SO_2 to SO_3 in the manufacture of H_2SO_4.
  (iii) Platinum-iridium mixture is used as catalyst for the oxidation of NH_3 to NO in the manufacture of HNO_3 by Ostwald's process.
  (iv) Raney nickel is used as catalyst in the hydrogenation reactions.
  (v) Fenton's reagent ($FeSO_4 + H_2O_2$) is used as catalyst in the oxidation of primary alcohols to aldehydes.

Interstitial Compounds

- Compounds formed by the occupation of small atoms like H, B, C and N in the interstitial spaces of the metal crystals are called interstitial compounds.
- Interstitial compounds are non-stoichiometric compounds and do not contain definite composition.
- Non-stoichiometric compounds are hard, brittle and have high m.pts and b.pts.
- Since the non-metal atoms occupy the holes of the metal crystal, the metal crystal lattice is not altered but the lattice expands. Hence the non-stoichiometric compounds are less denser than the pure metals.
- Hydrogen always occupies the tetrahedral holes while carbon and nitrogen tend to occupy larger octahedral holes.
- Non-stoichiometry occurs in the oxides and sulphides of transition elements which show more than one oxidation state, e.g., $Fe_{0.82}O$ to $Fe_{0.94}O$.
- Interstitial compounds show electrical conductivity and have intense colour. Some stoichiometric compounds become non-stoichiometric compounds at high temperature and become coloured, e.g., ZnO is white when cold but yellow when hot.

Alloys

- An intimate mixture of a metal with other metals (or) metalloids or non-metals having physical properties similar to that of the metal is called an alloy.

- Alloys are prepared to modify properties like malleability, toughness, resistant to corrosion.
- Alloys can be prepared by melting a mixture of metals or the components taken in proper proportion and then cooling, e.g., Brass
- When a mixture of dilute solutions of complex compounds of the respective metals are electrolysed, alloy will be formed by the simultaneous deposition of metals at cathode. Electrolysis of Cu and Zn cyanides give brass.
- Simultaneous reduction of two metal oxides give alloy. When iron chromite ore is reduced stainless steel will be formed.
- If the fine powders of two or more metals are mixed and compressed at high temperature, alloy will be formed.
- Sudden cooling of red hot steel by dipping in oil or water is called quenching. Quenched steel is tougher and its property depends on the percentage of carbon.
- Alloys are three types (i) Ferrous alloys (ii) Non-ferrous alloys and (iii) amalgams.
- In ferrous alloys one of the constituent metal should be iron, e.g., Stainless steel, cast iron, invar, nichrome are examples of ferrous alloys.
- Non-ferrous alloys do not contain iron, e.g., Brass, german silver, bell metal, gun metal etc are examples of non-ferrous alloys.
- Amalgams contain mercury as one of the constituent metal. Several metals form amalgams, e.g., Na_2Hg, AlHg etc.
- Alloys are prepared to modify the mechanical properties e.g.,
  (i) For decreasing the m.pts e.g., fuse wire (wood's metal).
  (ii) to increase the tensile strength e.g., steels.
  (iii) to increase castability e.g., type metal.
  (iv) to increase the corrosion resistance e.g., stainless steel.
  (v) to modify colour, e.g., aluminium bronze having gold colour.
  (vi) to change thermal and electrical conductivity e.g., nichrome.
  (vii) to increase hardness e.g., gold mixed with copper.

f-BLOCK ELEMENTS

- The f-block elements are those which have partly filled f orbitals of antipenultimate (3rd to the outermost) i.e., (n–2) f energy shells in their elementary or ionic state.
- f-block elements are also called as inner transition elements because they form a transition series with in the transition series. Inner transition elements are two series. (i) Lanthanoid series and (ii) Actinoid series.

Lanthanoid Series

- In the lanthanoid series the differentiating electron enters into the 4f-subshell. This series include 15 elements i.e., $_{57}La$ to $_{71}Lu$. Since $_{57}La$ ($4f^0\ 5d^1\ 6s^2$) and $_{71}Lu$ ($4f^{14}\ 5d^1\ 6s^2$) do not contain partly filled subshell, they should not be considered as lanthanoids but they closely resemble lanthanum and hence are considered together.
- All these elements contain same number of outer electrons ($5d^0\ 6s^2$) while they differ only in the number of 'f' electrons which are deep-seated and cannot participate in reactions. Hence they show similar properties.
- Gadolinium and ytterbium have anomalous electronic configuration where half-filled, completely filled f-orbitals are possible.
- The lanthanoid elements having electron in 5d orbital are lanthanum, cerium, galolinium and lutecium.
- The atomic radii of lanthanoids and the ionic radii of Ln^{3+} ions decreases from lanthanum to lutecium. The cumulative decrease in atomic and ionic radii is called lanthanide contraction. The lanthanide contraction is more pronounced in Ln^{3+} ion than in lanthanide atoms.
- The dramatic increase in the atomic size of europium compared to its adjacent elements is attributed to the difference in the metallic bonding that europium contributes only two electrons to the metallic bonding.
  The consequences of lanthanide contraction are
  (i) the crystal structure and other properties of the lanthanoids and their compounds become very close.
  (ii) The separation of the elements from the mixture that occur naturally becomes difficult.
  (iii) The atomic size of Zr and Hf become equal and hence the properties of both these elements and their compounds are almost similar. So they occur together in nature and their separation from one another is very difficult.
  (iv) Due to lanthanide contraction the atomic sizes of post lanthanide elements in 5d series are almost equal to the corresponding 4d series elements which are just above them. So the properties of 4d and 5d series elements are more close.
- The lanthanide contraction is due to poor shielding of one of 4f electrons by another in the same sub shells.
- The principal oxidation state exhibited by all these elements is $+3(Ln^{3+})$ in aqueous solutions and in their solid compounds.
- Lanthanum, gadolinium and lutecium exhibit only +3 oxidation state because they contain stable completely vacant or exactly half-filled or completely filled 4f sub-shell respectively in their oxidation state.

- Europium and ytterbium also exhibit +2 oxidation state because they contain stable half-filled and completely filled 4f sub shell respectively in their +2 oxidation state.
- Cerium and ytterbium also exhibit +4 oxidation state because they contain stable completely vacant and exactly half-filled 4f subshells in their +IV oxidation states.
- In all lanthanides +3 oxidation state is most stable oxidation state. So Eu^{2+}, Yb^{+2} ions will act as reducing agents while Ce^{4+} and Tb^{4+} ions will act as oxidizing agents to convert into stable +3 ions.
- Many trivalent lanthanoid ions are coloured both in solid state and in aqueous solutions due to the presence of incompletely filled f-subshell where internal transition, i.e., f-f transition takes place. Absorption spectra of compounds of M^{3+} ions in these cases show a very sharp lines because the 4f-orbitals are shielded from external perturbation by the complexing agents.
- The colour of the ions with $4f^n$ configuration is similar to that lanthanide ions with $4f^{14-n}$ configuration. i.e ions having same number of unpaired electrons have similar colours.
- Lanthanide ions having completely vacant, half-filled and completely filled f orbitals are colourless, e.g., La^{3+} ($4f^o$); Gd^{3+} ($4f^7$) and Lu^{3+} ($4f^{14}$) ions are colourless. Dipositive and tetrapositive ions which are iso-electronic with tripositive ions have different colours.
- Due to the presence of unpaired electrons in the 4f orbitals, all the lanthanide ions except La^{3+}, Lu^{3+}, Yb^{2+} and Ce^{4+} show paramagnetic character.
- Since the 4f electrons are well shielded, in lanthanides, quenching effect of the environments on 4f electrons is absent. So their magnetic moment is due to both spin and orbital contributions.
- The first and second ionization energies of lanthanoids are comparable to those of calcium ($IE_1 = 600$ KJ mol^{-1} and $IE_2 = 1200$ KJ mol^{-1}). The third ionization energy of lanthanoid is abnormally low in the case of lanthanum, gadolinium and lutetium in which stable completely filled f level configurations are possible respectively.
- The chemical behaviour of earlier members of the lanthanoids is similar to calcium while that of later elements behave like aluminium.
- Due to lanthanide contraction the hydroxides of earlier lanthanoids are ionic, while the hydroxides of later lanthanoids are covalent (Fajan's rule). Thus basic character of hydroxides decreases from lanthanum to lutetium.
- Lanthanoids form non-stoichiometric hydrides with hydrogen, carbides of the type Ln_3C, Ln_2C_3 and LnC_2 with carbon, nitrides of the type LnN with nitrogen, oxides of the type Ln_2O_3 with oxygen.
- Lanthanoids liberate hydrogen from acids.
- Lanthanoids are mainly used in the production of alloy steels for plates and pipes. The misch metal is an alloy consisting of lanthanoid metal (95%) and iron (5%) along with traces of S, C, Ca and Al. It is used in magnesium based alloy to produce bullets, sheets and lighter flint. Mixed oxides of lanthanoids are used as catalysts in petroleum cracking. Some lanthanoid oxides are used as phosphors in television screens and similar fluoresing surfaces.

ACTINOIDS

- The 15 elements from actinium (z = 89) to Lawrencium (z = 103) constitute the actinoids. Actinium is not the member of the actinoid series, however due to its close resemblence with the members of the series of all these elements are called actinoids.
- Since in these elements the number of electrons in the outermost as well as in the penultimate shell remain same, all the actinoids resemble one another closely.
- Actinoids unlike lanthanoids exhibit variable valency since 5f electrons can also participate in bonds as the 5f orbitals are more diffused outside the energies of 5f, 6d and 7s electrons are nearly equal.
- With increase in atomic number in actinoids the contraction of 5f orbital takes place and the 5f electrons cannot participate in bonds. So the later actinoids cannot exhibit variable valency but exhibit only +3 oxidation state.
- Similar to lanthanoid contraction there is actinoid contraction in actinoids which is again due to the poor shielding effect of 5f electrons.
- In +6 oxidation state the actinoid form oxocations such as UO^{4+}, UO_2^{2+} NpO_2^{2+} are formed due to high charge density. These are stable in acid and aqueous solutions.
- Maximum tendency to form complexes is observed in M^{4+} ions and decreases in others in the following order.

$$M^{4+} > MO_2^{2+} > M^{3+} > MO_2^{1}$$

The higher complex formation tendency of MO_2^{2+} ion may be due to high concentration of charge on metal atom.
- The first four elements of the actinoid series namely actinium, thorium, protactinium and uranium occur in nature. Actinium and protactinium occur in all uranium ores as decay products of ^{235}U. Thorium is mostly obtained from monazite sand while uranium is mostly obtained from pitch blend and carnotite. The elements beyond uranium in the actinoid series do not

occur in nature but are made artificially by nuclear transmutations. These are all synthetic and called transuranic elements.

d-BLOCK ELEMENTS

Single Answer Questions

1. Which of the following is an incorrect statement?
   (a) The lowest oxide of a transition metal is acidic whereas the highest one is usually basic
   (b) Transition metal usually exhibits higher oxidation states in fluorides and oxides
   (c) Transition metal halides become more covalent with the increasing oxidation state of the transition metal and are more susceptable to hydrolysis
   (d) The highest oxide of a transition metal is acidic whereas the lowest one is usually basic
2. The noble character of platinum and gold is favoured by
   (a) High enthalpies of sublimation, high ionization energies and low enthalpies of solvation
   (b) High enthalpies of sublimation, low ionization energies and low enthalpies of solvation
   (c) Low enthalpies of sublimation, high ionization energies and low enthalpies of solvation
   (d) High enthalpies of sublimation, high ionization energies and high enthalpies of solvation
3. Which of the following statements is incorrect?
   (a) The compounds formed by the corresponding 3d transition metals in lower valence states are ionic but those in higher valence states are covalent
   (b) The 4d and 5d transition metals form less ionic compounds than the 3d transition metals
   (c) The compounds formed by the 3d transition metals are less ionic than the corresponding compounds formed by the 4d and 5d transition metals
   (d) The ionization energies of 3d, 4d and 5d transition metals are greater than those of group 1 and 2 metals
4. Which of the following statement is correct?
   (a) The purple colour of the permanganate ion does not arise from d-d transition but from charge transfer
   (b) The purple colour of the permanganate ion arises due to d-d transition
   (c) The purple colour of permanganate ion arises due to both d-d transition and charge transfer
   (d) For charge transfer to occur in the permanganate ion the difference in energy between Mn^{7+} and O^{2-} should be very high
5. Which of the following statements is incorrect in the context of paramagnetic materials?
   (a) A paramagnetic material placed in a magnetic field attracts magnetic lines of force towards it
   (b) A paramagnetic material moves from a weaker to a stronger part of the magnetic field
   (c) Paramagnetism arises as a result of unpaired electron spins in the atom
   (d) It is difficult for magnetic lines of force to travel through a paramagnetic material than through a vaccum
6. The magnetic moment ($\propto$) of a transition metal can provide information about
   (a) The number of unpaired electrons present in the metal atom
   (b) The occupied orbitals of the metal atom
   (c) The structure of the molecule and the complex formed by the metal atom
   (d) All of these
7. Which of the following statement is correct?
   (a) The second row elements have smaller radii than the corresponding third row ones
   (b) Because of the lanthanide contraction the radii of the third row elements are almost the same as those of the first row elements
   (c) Because of the lanthanide contraction, the radii of the third row elements are almost the same as those of the second row elements
   (d) Because of the lanthanide contraction the separation of second row elements from one another is easier
8. The first ionization enthalpies of the element of the first transition series (Ti $\rightarrow$ Cu)
   (a) Increase as the atomic number increases
   (b) Decrease as the atomic number increases
   (c) Do not show any change as the addition of electrons takes places in the inner (n-1)d orbitals
   (d) Increase form Ti to Mn and then decrease from Mn to Cu
9. The atomic radii of transition elements in a row are
   (a) Smaller than those of s-block as well as p-block elements
   (b) Greater than those of s-block as well as p-block elements
   (c) Smaller than those of s-block but greater than those of p-block
   (d) Greater than those of s-block but smaller than those of p-block
10. E° values for the couples Cr^{3+}/Cr^{2+} and Mn^{3+}/Mn^{2+} are –0.41 and +1.51 volts respectively. These values suggest that
    (a) Cr^{2+} acts as a reducing agent whereas Mn^{3+} acts as an oxidizing agent
    (b) Cr^{2+} is more stable than Cr^{3+} state
    (c) Mn^{3+} is more stable than Mn^{2+}
    (d) Cr^{2+} acts as an oxidizing agent whereas Mn^{3+} acts as a reducing agent

11. Of the ions Zn^{2+}, Ni^{2+} and Cr^{3+} [At. Nos. –Zn = 30, Ni = 28, Cr = 24]
(a) Only Zn^{2+} is colourless and Ni^{2+} and Cr^{3+} are coloured
(b) All three are colourless
(c) All three are coloured
(d) Only Ni^{2+} is coloured and Zn^{2+} and Cr^{3+} are colourless

12. Which of the following statement is false?
(a) With fluorine vanadium can form VF_5
(b) With chlorine vanadium can form VCl_5
(c) Vanadium exhibits highest oxidation state in oxohalides $VOCl_3$, $VOBr_3$ and fluoride VF_5
(d) With iodine vanadium cannot form VI_5 due to oxidation power of V^{5+} and reducing nature of I^-

13. The pair of compounds in which metals are in the highest possible oxidation state is
(a) $[Fe(CN)_6]^{4-}, [Co(CN)_6]^{3-}$
(b) CrO_2Cl_2, MnO_4^-
(c) TiO_2, MnO_2
(d) $[Co(CN)_6]^{3-}, MnO_3$

14. Among the following series of transition metal ions the one where the metal ion have $3d^2$ electronic configuration is
(a) $Ti^{3+}, V^{2+}, Cr^{3+}, Mn^{4+}$
(b) $Ti^{3+}, V^{4+}, Cr^{3+}, Mn^{7+}$
(c) $Ti^{4+}, V^{3+}, Cr^{2+}, Mn^{3+}$
(d) $Ti^{2+}, V^{3+}, Cr^{4+}, Mn^{5+}$

15. Four successive members of the first row transition elements are listed below with their atomic numbers. Which one of them is expected to have the highest third ionization enthalpy?
(a) Vanadium (Z = 23)
(b) Manganese (Z = 25)
(c) Chromium (Z = 24)
(d) Iron (Z = 26)

16. A transition metal exists in its highest oxidation state. It is expected to behave as
(a) A chelating agent
(b) A central metal in a coordination compound
(c) An oxidizing agent
(d) A reducing agent

17. When K_2CrO_4 is added to $CuSO_4$ solution, there is formation of $CuCrO_4$ as well as $CuCr_2O_7$. Formation of $CuCr_2O_7$ is due to
(a) basic nature of $CuSO_4$ solution
(b) acidic nature of $CuSO_4$ solution
(c) $CuSO_4$ oxidizes CrO_4^{2-} to $Cr_2O_7^{2-}$
(d) this is the typical property of $CuSO_4$

18. Cu is transition element because
(a) Cu has $3d^{10}$ configuration
(b) Cu^+ has $3d^9$ configuration
(c) Cu^{2+} has $3d^9$ configuration
(d) Cu^+ has $3d^{10}$ configuration

19. The m.pt of transition elements is very high. Due to
(a) Involvement of greater no. of electrons in (n-1)d and ns electrons in the inter atomic metallic bonding
(b) Non-involvement of greater no. of electrons in (n-1)d and ns electrons in the inter atomic metallic bonding
(c) Involvement of greater no. of electrons in 'ns' electrons in the inter atomic metallic bonding
(d) Non-involvement of greater no. of electrons in 'ns' electrons in the inter atomic metallic bonding

20. Which of 3d series transition element exhibit the largest number of oxidation states
(a) Sc (b) V
(c) Mn (d) Fe

21. The observed magnetic moment of Co^{2+} is much higher than the calculated magnetic moment by $\sqrt{n(n+2)}$ formula due to
(a) The spin angular momentum
(b) The presence of more number of unpaired electrons than expected
(c) The consideration of orbital angular momentum along with spin angular momentum of electrons.
(d) The presence of more no. of paired electrons

22. The catalyst used in the reduction of perdisulphate to sulphate with iodide ion is
(a) Fe^{3+} (b) Fe^{2+}
(c) Both 1 and 2 (d) Cr^{2+}

23. A transition element M forms the oxides MO, M_2O_3, MO_3, and M_2O_7, which of the following statement about these oxides is true?
(a) MO is most acidic
(b) M_2O_3 is the most likely to be a strong oxidizing agent
(c) MO_3 is the most acidic
(d) M_2O_7 is the one that cannot be a reducing agent

24. An element of 3d-transition series shows two oxidation states x and y, differ by two units then
(a) compounds in oxidation state x are ionic if $x > y$
(b) compounds in oxidation state x are ionic if $x < y$
(c) Oxidation state has no relation to the nature of bond
(d) compounds in oxidation state y are covalent if $y < x$

25. Amongst the following species, maximum covalent character is exhibited by
(a) $FeCl_2$ (b) $ZnCl_2$
(c) $HgCl_2$ (d) $CdCl_2$

26. Arrange following ions in the decreasing order of their magnetic moment
I. V^{4+} II. Mn^{4+} III. Fe^{3+} IV. Ni^{2+}
[At nos. V = 23 Mn = 25 Fe = 26 Ni = 28]
(a) II > III > I > IV (b) III > IV > II > I
(c) III > II > IV > I (d) I < IV < III < II

27. Which of the following pairs of transition metal ions are the stronger oxidizing agents in aqueous solutions
(a) V^{2+} and Cr^{2+} (b) Ti^{2+} and Cr^{2+}
(c) Mn^{3+} and Co^{3+} (d) V^{2+} and Fe^{2+}

28. Which of the following statements is incorrect?
(a) The melting points and boiling points are generally of the 3rd transition elements > 2nd transition elements > 1st transition elements
(b) The ionization energies of the 3rd transition elements are greater than the corresponding 2nd transition elements
(c) The atomic radii of the 3rd transition element after Hf is almost the same as that of the 2nd transition element right above it
(d) Down the group always for transition elements the atomic radii increases

29. Lanthanides like Eu and Yb can form hydrides like EuH_2 and YbH_2. These hydrides are
(a) Ionic hydrides (b) Covalent hydrides
(c) Metallic hydrides (d) Intermediate hydrides

30. Lanthanoid contraction implies
(a) Decrease in density (b) Decrease in mass
(c) Decrease in ionic radii (d) Decrease in radioactivity

31. If the lanthanoid element with xf electrons has a pink colour, then the lanthanoid with (14-x)f electrons will have the colour as
(a) Blue (b) Red
(c) Green (d) Pink

32. The electronic configuration of actinoides cannot be assigned with degree of certainty because of
(a) Overlapping of inner orbital's
(b) Free moment of electrons over all the orbital's movement
(c) Small energy difference between 5f and 6d levels
(d) None of the above

33. Which of the following statement is not correct?
(a) $La(OH)_3$ is less basic than $Lu(OH)_3$
(b) In lanthanide series ionic radius of Ln^{3+} ions decreases
(c) La is actually an element of transition series rather than lanthanide series
(d) Atomic radii of Zr and Hf are same because of lanthanide contraction

34. Across the lanthanide series, the basic character of the lanthanide hydroxides
(a) Increases
(b) Decreases
(c) First increases and then decreases
(d) First decreases and then increases

35. The reason for the stability of Gd^{3+} ion is
(a) 4f sub shell-half-filled
(b) 4f sub shell-completely filled
(c) Possesses the general electronic configuration of noble gases
(d) 4f sub shell empty

36. The +3 ion of which one of the following has half filled 4f sub shell?
(a) La (b) Nd (c) Gd (d) Ac

37. Lanthanide for which +2 and +3 oxidation states are common is
(a) La (b) Nd (c) Ce (d) Eu

38. Misch metal is
(a) An alloy of Al
(b) A mixture of chromium and $PbCrO_4$
(c) An alloy of lanthanoid metals
(d) An alloy of copper

39. More number of oxidation states are exhibited by the actinoides than by the lanthanides. The main reason for this is
(a) Greater metallic character of the lanthanoids than that of the corresponding actinoids
(b) More active nature of the actinoids
(c) More energy difference between 5f and 6d orbital's than between 4f and 5d orbital's
(d) Lower energy difference between 5f and 6d orbital's than between 4f and 5d orbital's

40. Identify the incorrect statement among the following:
(a) Shielding power of 4f electrons is quite weak
(b) There is a decrease in the radii of the atoms or ions while proceeding from La to Lu
(c) Lanthanoid contraction is the accumulation of successive shrinkages
(d) As a result of lanthanoid contraction, the properties of 4d series of the transition elements have no similarities with 5d series of elements.

41. Oxide of metal cation which is not amphoteric
(a) Al^{3+} (b) Cr^{3+} (c) Mn^{2+} (d) Zn^{2+}

42. Metal–metal bonding is more frequent in 4d or 5d series than in 3d series. This is due to
(a) Their greater enthalpies of atomization
(b) The large size of the orbital's which participates in the metal-metal bond formation
(c) Their ability to involve both ns and (n-1)d electrons in the bond formation
(d) The comparable size of 4d and 5d series elements.

43. Among the TiF_6^{2-}, CoF_6^{3-}, Cu_2Cl_2 and $NiCl_4^{2-}$ the colourless species are
(a) CoF_6^{3-} and $NiCl_4^{2-}$
(b) TiF_6^{2-} and CoF_6^{3-}
(c) Cu_2Cl_2 and $NiCl_4$
(d) TiF_6^{2-} and Cu_2Cl_2

44. The sum of the first three ionization energies of the lanthanoids Ce, Eu, Gd, Yb, Lu is
(a) Ce > Eu > Gd > Yb > Lu
(b) Yb > Lu > Eu > Gd > Ce

(c) Yb > Eu > Lu > Gd > Ce
(d) Yb > Eu > Gd > Lu > Ce

45. The basic character of the monoxides of the transition series is the following order
(a) CrO > VO > FeO > TiO
(b) TiO > VO > CrO > FeO
(c) VO > CrO > TiO > FeO
(d) TiO > FeO > CrO > VO

46. Hg_2^{2+} ion exists but Cd_2^{2+} ion does not exist due to
(a) Large ionization enthalpy of Hg
(b) Large ionization enthalpy of Cd
(c) Weak metallic bonds in Hg
(d) less reactivity of Hg^+ than Cd^+ ion

47. The ionization energy of the 1st transition series
(a) Decreases across the period
(b) First increases then decreases across the period
(c) Constantly increases across the period and then from Cu to Zn it increases with small variations
(d) increases then remains same.

48. AgF, AgCl, NaCl, NaBr, NaI are colourless but AgBr and AgI are coloured because
(a) Ag^+ polarizes Br^- and I^- and not able to polarize Cl^- and F^-
(b) AgBr has unpaired electron
(c) AgBr has defects
(d) All of these

49. The oxidation potential of the transition elements are lower than the oxidation potential of the s-block elements because
(a) The hydration energy of transition metal cations are high
(b) The ionization energy of the transition elements are higher than that of the corresponding s-block elements
(c) Ionization energy of transition elements are lower than that of the corresponding s-block elements
(d) The melting point of s-block elements are low

50. The reactivity of copper is low because of its
(a) high enthalpy of sublimation and low ionization energy
(b) high enthalpy of sublimation and high ionization energy
(c) low enthalpy of sublimation and high ionization energy
(d) low enthalpy of sublimation and low ionization energy

51. The melting point of copper is higher than that of zinc because
(a) copper has a bcc structure
(b) the atomic volume of copper is higher
(c) the d electrons of copper are involved in metallic bonding.
(d) the s as well as d electrons of copper are involved in metallic bonding

52. The ionization energy of copper is higher than that of potassium though both have a 4s configuration because the d electrons in copper
(a) form a poor shield, making copper smaller
(b) form a poor shield, making copper bigger
(c) are strongly shielded, making copper smaller
(d) are strongly shielded making copper bigger

53. The metallic radius of gold is almost identical with that of silver because of
(a) transition metal contraction
(b) the same crystal structure of silver and gold
(c) the high electropositive character of gold in comparison to silver
(d) the effect of lanthanide contraction

54. Which of the following ions is coloured?
(a) $_{57}La^{3+}$ (b) $_{63}Eu^{3+}$
(c) $_{64}Gd^{3+}$ (d) $_{71}Lu^{3+}$

55. Which of the following pairs is expected to exhibit the same colour
(a) $_{58}Ce^{3+}$, $_{67}Ho^{3+}$ (b) $_{60}Nd^{3+}$, $_{68}Er^{3+}$
(c) $_{61}Pm^{3+}$ $_{69}Tm^{3+}$ (d) $_{63}Sm^{3+}$, $_{70}Yb^{3}$

56. Which of the following pairs is expected to form colourless compound?
(a) $_{57}La^{3+}$, $_{59}Pr^{3+}$ (b) $_{57}La^{3+}$, $_{71}Lu^{3+}$
(c) $_{60}Nd^{3+}$ $_{61}Pn^{3+}$ (d) $_{63}Eu^{3+}$, $_{65}Tb$

57. Cerium can show the oxidation state of +4 because
(a) It resembles alkali metals
(b) It has very low value of ionization energy
(c) of its tendency to attain noble gas configuration of xenon
(d) of its tendency to attain f^0 configuration.

More than One Answer Questions

1. Correct order of melting point in transition element is
(a) W > Mo > Cr (b) Cr > Fe > Mn
(c) Cu > Au > Ag (d) V > Cr > Ti

2. Cu^+ ion is not stable in aqueous solution because
(a) $Cu^{2+}_{(aq)}$ is more stable than $Cu^+_{(aq)}$
(b) $Cu^{2+}_{(aq)}$ can form interstitial compounds in aqueous solution
(c) The equilibrium constant for
$Cu^+_{(aq)} \rightleftharpoons Cu^{2+}_{(aq)} + Cu$ is very high
(d) The equilibrium constant for
$Cu^+_{(aq)} \rightleftharpoons Cu^{2+}_{(aq)} + Cu$ is very low

3. Many transition metals form interstitial compounds. The characteristics of these interstitial compounds are
(a) They have high melting points, higher than those of pure metals

(b) They are very hard
(c) They retain metallic conductivity
(d) They are chemically more reactive than the pure metals

4. Which of the following statements are correct about Zn, Cd and Hg?
(a) They exhibit high enthalpies of atomization as the d-sub shell is full
(b) Zn, Cd do not show variable valency while Hg shows +1 and +2 oxidation states
(c) Compounds of Zn, Cd and Hg are paramagnetic in nature
(d) Zn, Cd and Hg are called soft metals

5. Which of the following is/are characteristic (s) of the transition elements in the series from scandium to zinc?
(a) The formation of coloured cations
(b) The presence of at least one unpaired electron in a d-orbital of a cation
(c) The ability to form complexe ions
(d) The possession of an oxidation state of +1.

6. Which of the following are correct about actinoids?
(a) They show large number of oxidation states
(b) They show actinoid contraction
(c) They form oxo-cations
(d) All of them are radioactive

7. An aqueous solution of $[Ti(H_2O)_6]^{3+}$ ion has a mild violet colour of low intensity. Which of the following statements is/are correct?
(a) The ion absorbs visible light in the region of 500 nm
(b) The colour is due to d-d transition of electron between two sets of d-orbitals having difference in energy
(c) The low colour intensity is due to less difference in the energy of two sets of d-orbitals
(d) The transition is the result of metal ligand back bonding.

8. When NH_4VO_3 is heated
(a) NH_3 is liberated
(b) N_2 is formed
(c) V_2O_5 is formed
(d) NH_4VO_2 is formed

9. Which of the following is correct trend in the m.pt order
(a) V > Mn > Ni
(b) V > Ni > Mn
(c) V > Cr > Ti > Sc
(d) Co > Ni > Cu > Zn

10. Which of the following statements are correct?
(a) Transition elements show variable oxidation states due to the participation of valence electrons
(b) p-block elements show variable oxidation states due to the participation of valence electrons
(c) Transition elements show variable oxidation states due to the participation of both valence and penultimate shell electrons
(d) The stability of higher oxidation state decrease in p-block elements but increase in transition element while descending in a group

11. Transition elements act as good catalysts because
(a) Presence of partially filled d-orbitals
(b) Presence of vacant d-orbitals
(c) Transition elements show variable oxidation state
(d) Easy inter convertibility of oxidation states due to low oxidation and reduction potential

12. Which of the following represent the correct sequence of indicated property?
(a) $Sc^{3+} > Fe^{3+} > Mn^{3+}$; Stability of +3 oxidation state
(b) $Mn^{2+} < Ni^{2+} < Co^{2+} < Fe^{2+}$; magnetic moment
(c) FeO > CoO > NiO; basic character
(d) Sc < Ti < V < Cr number of oxidation states

13. Among the following the correct statement is/are
(a) In 6th (VI A) transition group, the lower oxidation state is favoured by heavier elements (Mo and W) when compared to chromium
(b) Cu^{2+} is more stable than $Cu^{+}{}_{(aq)}$
(c) E^0 of Zn^{2+}/Zn is more negative than expected because of extra stability of d^{10} configuration
(d) In 3d-series Mn^{3+} and Co^{3+} are the strongest oxidizing agents when only +3 oxidation state is considered

14. Which of the following characterstics of lanthanides is correct?
(a) Reducing property of lanthanides decreases from La to Lu.
(b) Complex forming ability of lanthanides increases from La to Lu.
(c) Basic character of the oxides and hydroxides increases from La to Lu.
(d) Lanthanides form stable compounds only in +3 oxidation state.

15. Which of the following statements are correct
(a) of the two ions Cu^{2+} and Zn^{2+} the paramagnetic ion is Cu^{2+}
(b) Of the two ions Fe^{2+} and Fe^{3+} the ion having higher magnetic moment is Fe^{3+}
(c) The type of magnetism exhibited by $[Mn(H_2O)_6]^{2+}$ ion is paramagnetism
(d) The magnetic moment of Fe^{2+} is identical to that of Co^{3+}

16. Which of the following statements concerning lanthanide elements is correct?
(a) Lanthanides are separated from one another by an exchange method
(b) ionic radii of trivalent lanthanides steadily increases with increase in the atomic number
(c) all lanthanides are dense metals
(d) more characteristic oxidation state of lanthanide elements is +3

17. The correct statements among the following are
(a) Cu^{2+} is stronger oxidizing agent than Cr^{2+}
(b) Co^{2+} is stronger reducing agent than V^{2+}
(c) Cr^{2+} can be easily oxidized than Ni^{2+}
(d) Co^{3+} can be reduced easily than Cr^{3+}

Comprehension Type Questions

Passage–I

f-block elements exhibit different oxidation states, colour, complex formation like properties. The size of lanthanides decreases due to poor screening effect of 4f electrons. It is called lanthanide contraction. Lanthanide hydroxides are basic in nature.

1. Ce^{3+}, La^{3+}, Pm^{3+} and Yb^{+3} have ionic radii in the increasing order as
(a) $La^{3+} > Ce^{3+} > Pm^{3+} > Yb^{3+}$
(b) $Yb^{3+} < La^{3+} < Ce^{3+} < Pm^{3+}$
(c) $La^{3+} < Ce^{3+} < Pm^{3+} < Yb^{3+}$
(d) $Yb^{3+} < Pm^{3+} < La^{3+} < Ce^{3+}$

2. The colour of lanthanide ion is due to
(a) f-f transitions in partially filled f-orbitals
(b) f-f transitions in completely filled f-orbitals
(c) both
(d) none

3. Which of the following hydroxide of lanthanide element has more basic nature
(a) Ce (b) Lu
(c) Gd (d) all are same

d-Block Elements key

Single Answer Question

1. a	**2.** a	**3.** c	**4.** a	**5.** d	**6.** d	**7.** c
8. a	**9.** c	**10.** a	**11.** a	**12.** b	**13.** b	**14.** d
15. b	**16.** c	**17.** b	**18.** c	**19.** a	**20.** c	**21.** c
22. a	**23.** d	**24.** b	**25.** c	**26.** c	**27.** c	**28.** d
29. a	**30.** c	**31.** d	**32.** c	**33.** a	**34.** b	**35.** a
36. c	**37.** d	**38.** c	**39.** d	**40.** d	**41.** c	**42.** a
43. d	**44.** c	**45.** b	**46.** a	**47.** c	**48.** a	**49.** b
50. b	**51.** d	**52.** a	**53.** d	**54.** b	**55.** b	**56.** b
57. d						

More than One Answer Questions

1. a, b, c	**2.** a, c	**3.** a, b, c	**4.** b, d
5. a, b, c	**6.** a, b, c, d	**7.** a, b, c	**8.** a, c
9. b, c, d	**10.** b, c, d	**11.** a, c, d	**12.** a, c, d
13. b, c, d	**14.** a, b, d	**15.** a, b, c, d	**16.** a, c, d
17. a, c, d			

Comprehensive Type Questions

Passage: I **1.** a **2.** a **3.** a

d-Block elements Hints

1. For a transition metal, the oxide in lower oxidation state is basic while higher oxidation state is acidic.

2. Due to high enthalpy of sublimation, high ionization energies and enthalpies of solvation, the reactivity of gold and platinum is less and become noble metals.

3. Due to more effective nuclear charge (because of poor shielding effect of d and f electrons) the compounds of 4d and 5d metals are less ionic than those of 3d metals.

10. Lesser and negative reduction potential indicate Cr^{2+} is reducing agent. Higher positive reduction potential indicate Mn^{3+} is stronger oxidizing agent.

11. Zn^{2+} has $3d^{10}$ configurations so colourless.

13. The highest oxidation states of Cr and Mn are +6 and +7 which are present in CrO_2Cl_2 and MnO^-_4 respectively.

15. Mn^{2+} have stable half-filled $3d^5$ configuration. So third ionization energy is very high.

16. In higher oxidation state a transition metal act as oxidizing agent.

17. $CuSO_4$ solution is acidic due to hydrolysis in water forming $Cu(OH)_2$ and H_2SO_4. In acid medium chromates convert into dichromates.

19. In transition element the ns and (n-1) d electrons also participate in inter atomic metallic bonding. As the number of bonds are more m.pts are also high.

25. Zn^{2+}, Cd^{2+} and Hg^{2+} have pseudo inert gas configuration, but due to lanthanide contraction and more effective nuclear charge $HgCl_2$ is more covalent.

27. Mn^{3+} and Co^{3+} are stronger oxidizing agents because
1. Mn^{2+} is stable than Mn^{3+} due to stable half filled d^5 configuration.
2. in aqueous solution Co^{2+} is more stable than Co^{3+}.

28. In d-block elements though atomic radius increases from 3d to 4d element, the radii of 4d and 5d elements are almost equal due to lanthanide contraction.

31. Lanthanoid element having same number of unpaired electrons have same colours xf and (14-x)f configurations have same number of unpaired electrons.

32. Because of the small energy difference between 5f, 6d and 7s orbitals electron transition between these orbitals can takes place easily.

33. Due to lanthanide contraction covalent character of lanthanide compounds increases from La to Lu. So basic character of their oxides and hydroxides decreases from La $(OH)_3$ to $Lu(OH)_3$.

37. Europium has $4b^76s^2$ configuration. So it exhibit +2 oxidation state due to the stability of half-filled $4f^7$ configuration in Eu^{2+}. Also it exhibit the common oxidation state +3 of lanthanoids.

39. Due to small energy difference between 5f and 6d orbitals 5f electrons can also participate in bonding so actinoids exhibit variable valency.

40. Due to lanthanoid contraction the atomic sizes of 4d and 5d series elements in a group have same atomic sizes and hence their properties are also nearly same.

42. Due to the metal–metal bonding they have greater enthalpies of atomization.

43. In TiF_6^{2-} the titanium exist as Ti^{4+} with d^0 configuration while in Cu_2Cl_2, Cu^+ ion has d^{10} configuration.

45. With increase in atomic number in a transition series effective nuclear charge increases, ionic size decreases, covalent character of oxides increases so basic character decreases.

47. The ionization energy of the 1st transition series elements consistently increases from Sc to Zn with small variation.

48. The colour of AgBr and AgI is due to polarization of Br^- and I^- by the Ag^+ ion having pseudo inert gas configuration. Hence charge transfer takes place.

49. The oxidation potentials depend on the sublimation enthalpies, ionization energies and hydration energies. Transition elements have high sublimation enthalpies, high ionization energies and low hydration energies.

50. In zinc d-orbitals are completely filled and cannot participate in metallic bonding but in copper the d-electron can also participate in metallic bonding.

54. In $La^{3+}(4f^0)$ and $Lu^{3+}(4f^{14})$ there is no possibility of f-f transition also in 4f $(4f^7)$. There is no possibility of f-f transition due to spin forbidden rule. So they are colourless.

56. Ce^{4+} by losing four electrons get the $4f^0$ configuration.

More than One Answer (Hints)

1. In a series with increase in number of unpaired electrons strength of metallic bond increases, thus m.pts increases but Mn having stable d^5 configuration have low m.pt than Cr and Fe. Second and third series elements have more m.pt because of metal–metal bonding.

2. The hydration energies and lattice energies are favourable for the formation of Cu^{2+} ion.

6. Actinoids exhibit variable valency as their 5f electrons can also participate in bonding. Other statements are also correct.

8. $2NH_4VO_3 \rightarrow 2NH_3 + V_2O_5 + H_2O$

10. In transition elements the stability of higher oxidation state increases down the group.

13. Sc^{3+} $(3d^0)$; Fe^{3+} $(3d^5)$; Mn^{3+} $(3d^4)$ d^0 and d^5 configurations give more stability. Mn^{2+}, Ni^{2+}, Co^{2+}, Fe^{2+} have 5, 2, 3 and 4 unpaired electrons respectively.

15. Due to lanthanide contraction ionization energies increases from La to Lu. So reduction property of lanthanides decreases from La to Lu. For all lanthanides the stable oxidation state is +3.

CHAPTER 14

Coordination Compounds

Let us learn to dream, then perhaps shall find truth....
F.A.Kekule

14.1 INTRODUCTION

It is difficult to state when the first metal complex was discovered perhaps the earliest one on the record is Prussian blue the $KCN \cdot Fe(CN)_2$ $Fe(CN)_3$ or $KFe[Fe(CN)_6]$ which was obtained by the artist's colour maker Diesbach, in Berlin at the beginning of the eighteenth century, however, the date usually cited is that of the discovery of hexammine cobalt (III) chloride $CoCl_3 \cdot 6NH_3$by Tassaret (1798). This discovery marks the real beginning of coordination chemistry because the existence of a compound with a unique properties of $CoCl_3 \cdot 6NH_3$ stimulated considerable interest in and research on similar systems.

14.1.1 What Is a Coordination Compound?

When two or more independent, stable compounds combine, either physically or chemically, addition compounds are formed. These are two types

1. **Double salts or lattice compounds:** The compounds formed by the addition of two stable independent compounds exist only in solid state and ionizes completely in water are called double salts or lattice compounds e.g.,

(i) KCl and $MgCl_2 \cdot 6H_2O$ form $KCl \cdot MgCl_2 \cdot 6H_2O$

(ii) K_2SO_4 and $Al_2(SO_4)_3$ form $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$(Alum)

(iii) $(NH_4)_2SO_4$ and $FeSO_4$ form $(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O$ (Mohr's salt)

These compounds when dissolved in water behave like a mixture of two compounds and give reactions of the ions present in the individual compounds.

2. **Coordination compounds or complex compounds** The compounds formed by the addition of two stable independent compounds, exist in both solid and solution, do not ionize completely and do not give tests for all the ions present in them. Such compounds are called coordination compounds.The properties of such compounds are totally different from the individual compounds from which it is prepared. e.g.,

(i) $[Cu (NH_3)_4]SO_4$
(ii) $K_4[Fe(CN)_6]$
(iii) $K_3[Fe(CN)_6]$
(iv) $[Co(NH_3)_6] Cl_3$
(v) $K[Ag(CN)_2]$
(vi) $K_2[Ni(CN)_4]$

14.1.2 Characteristics of Coordination Compounds

(i) Coordination compounds are formed by the chemical union of apparently stable compounds. For example copper sulphate reacts with ammonia to form $CuSO_4 \cdot 4NH_3$.

(ii) The coordination compound is a new chemical species different from the compound from which it is formed. For example potassium ferrocyanide contain the anion $[Fe(CN)_6]^{4-}$ which is present in neither KCN nor in $Fe(CN)_2$ from which it is formed.

(iii) Complex species (molecule or ion) retain its identity in solution.

(iv) Several physical and chemical properties of the coordination compounds are different from the compounds from which it is formed. Physical properties like colour, electrical conductivity, optical activity, solubility are different for the complex compound and the compounds from which it is formed.

The chemical properties also completely changes. For example $Fe(OH)_2$ does not precipitate upon the addition of OH^- to a solution of $[Fe(CN)_6]^{4-}$ or $[Fe(EDTA)]^{2-}$ whereas $Fe(OH)_2$ will be precipitated by the addition of OH^- to any soluble compound of Fe^{2+} Similarly Cl^- does not precipitate Ag^+ in $[Ag(NH_3)_2]^+$, NH_3in $[Cr(NH_3)_6]^{3+}$ or $[Co(NH_3)_6]^{3+}$ shows no reaction with Nessler's reagent.

Coordination compounds are known by various names such as complex compounds (because the formation of these compounds could not be explained with the theories of bonding known by that time) Werner's complexes, and additive compounds.

14.2 THEORIES OF FORMATION OF COORDINATION COMPOUNDS

Earlier interesting, but wholly unsuccessful, explanations offered by Bloomstrand, Jorgensen and others are no more than historical importance and need not be discussed.

In 1893, Alfred Werner at the age of only 26 proposed what is now commonly referred to a Werner's coordination theory. This Werner's greatest contribution to coordination chemistry came as a flash of inspiration (at 2 O'clock in the morning in a dream) and crystallized into a comprehensive theory. It is said by noon of the day following this dream Werner had completed the revolutionary paper (A. Werner Z. anorg. chem. 3, 267; 1893) which summarized these views on structures and properties and opened an entirely new field for investigation. Werner devoted himself to proving the postulates of his original theory. It is significant that every postulate was verified experimentally not only by Werner but by many other working independently. The Nobel award in 1913 was a fitting recognition of Werner's contribution.

14.2.1 Werner's Coordination Theory

Werner's coordination theory has been a guiding principle of inorganic chemistry and in the concept of valence. The important postulates of Werner's theory are as follows.

(i) Metals possess two types of valencies (a) primary or ionizable valence and (b) secondary or non-ionizable valence. In modern terminology (a) corresponds to oxidation state and (b) to coordination number.

(ii) Primary valencies are those which a metal exercise in the formation of its simple salts. In the formation of simple salts always metal cations combine with negative ions. So primary valencies are always satisfied only by negative ions.

(iii) Secondary valencis can be satisfied by either neutral molecules or by negative ions. The number of primary valencies of a metal ion is equal to its oxidation state. The number of secondary valencies of a metal is called coordination number (The coordination number for different metals may vary from 2-10 but 90 per cent of coordination compounds have coordination number either 4 or 6).

(iv) In every case all the primary valencies and secondary valencies should be completely satisfied otherwise a complex compound cannot be formed.

(v) Primary valencies are non-directional in nature but secondary valencies are directional in nature. Due to the directional nature of secondary valencies complex compounds exhibit stereoisomerism.

14.2.2 Experimental Evidence in Support of Werner's Theory

A large number of complexes were studied by Werner and other workers. Werner used molar conductance values to determine the number of ions per molecule thus assigning groups to the secondary valencies. Chemical methods like precipitation of Cl^- by $AgNO_3$ to determine the number of ionizable Cl^- ions per molecules were also used.

Table 14.1 Number of chloride ions precipitated as AgCl

Complex	No of Cl^- ions precipitated	Molar conductance Ohm^{-1}
$CoCl_3 \cdot 6NH_3$	3	430
$CoCl_3 \cdot 5NH_3$	2	250
$CoCl_3 \cdot 4NH_3$	1	100
$CoCl_3 \cdot 3NH_3$	0	0

Table 14.2 Molar conductivity of platinum (IV) complexes

Complex	Molar conductivity ohm^{-1}	Number of ions indicated
$PtCl_4 \cdot 6NH_3$	523	5
$PtCl_4 \cdot 5NH_3$	404	4
$PtCl_4 \cdot 4NH_3$	229	3
$PtCl_4 \cdot 3NH_3$	97	2
$PtCl_4 \cdot 2NH_3$	0	0
$PtCl_4 \cdot NH_3 \cdot KCl$	109	2
$PtCl_4 \cdot 2KCl$	256	3

Werner's postulates can be explained by using the formation of chloramine complexes of cobalt (III) which are given in Table 14.1. The addition of a solution of silver nitrate to a freshly prepared solution of $CoCl_3 \cdot 6NH_3$ results in the immediate precipitation of all these chloride ions.

The same experiment with $CoCl_3 \cdot 5NH_3$, $CoCl_3 \cdot 4NH_3$ and $CoCl_3 \cdot 3NH_3$ causes instant precipitation of 2, 1 and 0 chloride ions.

Another kind of experiment provides useful information about the number of ions present in solution of different complexes. The greater the number of ions in a solution, the greater is the electrical conductivity of the solution. Therefore a comparison of the conductivities of the solutions containing the same concentrations of coordination compounds permits an estimate of the number of ions in each complex compound. The results in Table 14.1 and 14.2 shows that as the number of ammonia molecules in the compounds decreases the number of ions also falls to zero.

Now by using these observations the Werner used dotted lines (……) to represent the primary valencies and solid lines to represent the secondary valencies. The chloramine complexes of cobalt (III) can be written as follows.

According to the theory the first member of the series $CoCl_3 \cdot 6NH_3$ is designated as I. The primary valence of cobalt (III) is 3. The three chloride ions saturate the primary valencies of cobalt; the ions that neutralize the charge of the metal ion use the primary valence. The ammonia molecules use secondary valence. Here Co (III) is already surrounded by six ammonias so that the chloride ions cannot be accommodated in secondary valencies and hence are farther from the metal complex, conducts current equivalent to four ions

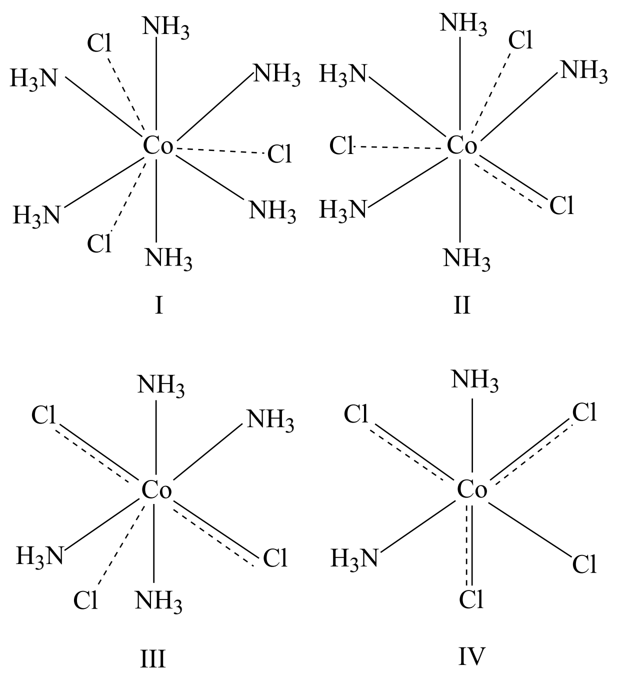


Fig 14.1 Werner's representation chloramine complexes of cobalt (III)

and the chloride ions are readily precipitated by Ag^+ as silver chloride.

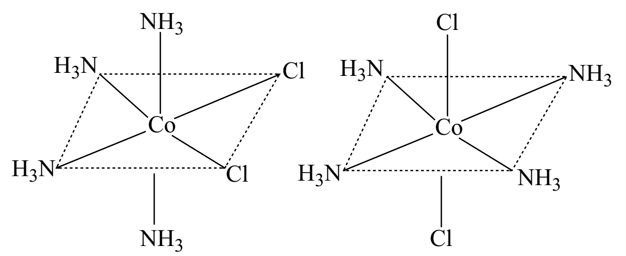
Returning now to theory one finds that Werner represented $CoCl_3 \cdot 5NH_3$ as II. He did this in accord with postulate (iv) which states that both the primary and secondary valence tend to be satisfied. In $CoCl_3$ there are only five ammonia molecules to satisfy the secondary valencies. Therefore, one chloride ion must serve the dual function of satisfying both a primary and a secondary valencies. Werner represented the bond between such a ligand and the central metal by a combined dotted-solid line such a chloride is not readily precipitated from the solution by Ag^+.

Extension of this theory to the next member of the series $CoCl_3 \cdot 4NH_3$ requires the formula III. Two chloride ions satisfy both primary and secondary valencies; hence they are firmly held to the cobalt (III). In solution the compound therefore dissociates into two ions one cation and a complex ion. The fourth one $CoCl_3 \cdot 3NH_3$ do not yield any Cl^- ion in solution because all the three chloride ions are in secondary valence.

Postulate (V) of Werner's theory deals specifically with the stereochemistry of metal complexes. Werner successfully demonstrated that complexes in which the secondary valencies of metal ion is 6, the species satisfying the secondary valencies are situated at positions symmetrically equidistant from the central metal atom. Werner showed that the two compounds (a violet and a green) of the composition $CoCl_2.4NH_3$ can be attributed to the existence of cis and trans isomers. The groups satisfying the secondary valencies are at the corners of an octahedron.

Thus Werner's theory not only provides an explanation of isomerism, but also predicts the existence of isomers of types which were not previously observed.

The Werner's contribution is unique one. And one must appreciate the remarkable work done by Werner. The fundamental postulates by Werner have been found to be as valid today as it was proposed 110 years ago, despite the tremendous advances in theory, the remarkable increase in the number of coordination compounds and enormous data of such compounds.



cis-isomer (violet) tran-isomer (green)

Fig 14.2 Geometrical isomers of $CoCl_3 \cdot 4NH_3$

14.2.3 Definition of the Terms used in Coordinate Compounds

1. Central atom/ion: In a coordination compound, the atom/ion to which a fixed number of ions/groups are bound in definite geometrical arrangement around it, is called the central atom or ion. For example in chloramine cobalt (III) complexes, cobalt (III) ion is the central ion.

2. Ligands: The ions or molecules which satisfy the secondary valencies of metal ion/atom in Werner's complexes are called ligands. These may be simple ions such as Cl^-, small molecules such as H_2O or NH_3, larger molecules such as $NH_2\,CH_2\,CH_2\,NH_2$ or $N\,(CH_2CH_2NH_2)_3$ or even macromolecules such as proteins.

3. Coordination number: The number of secondary valencies of the metal atom/ion in a complex compound is called coordination number. Usually abbreviated as CN. For example in the chloroammine cobalt (III) complexes explained earlier the CN of Cobalt (III) is 6. CN of different metals varies form 2 to 10, but the most common CNs are 4 and 6, but may be 2 or 8 or an odd number in rare cases.

The CN is previously considered to be a fixed number for a particular metal, but many complexes are known in which the same metal ion has more than one CN. The maximum CN of elements in the 2^{nd} row of elements of the periodic table is 4, for elements in the 3^{rd} and 4^{th} row, it is 6 and for the elements in the 5^{th} or 6^{th} row, 6 or 8 coordination numbers are more commonly seen and in some cases it is 10. Some examples are given in Table 14.3.

Table 14.3 Coordination numbers of some metal ions

Metal ion	CN	Metal ion	CN	Metal ion	CN
Ag^+	2	Cu^{2+}	4, 6	Os^{3+}	6
Au^+	2, 4	Zn^{2+}	4	Ir^{3+}	6
Ti^+	2	Pb^{2+}	4	Au^{3+}	4
Cu^+	2, 4	Pt^{2+}	4	Pt^{4+}	6
V^{2+}	6	Sc^{3+}	6	Pd^{4+}	6
Fe^{2+}	6	Cr^{3+}	6	Nb^{5+}	7
Co^{2+}	4, 6	Fe^{3+}	6	Mo^{5+}	8
Ni^{2+}	4, 6	Co^{3+}	6	Hg^{2+}	4

It is important to note that coordination number of the central atom/ion is determined only by the number of σ bonds formed by the ligand with the central atom/ion. π bonds, if formed between the ligand and the central atom/ion are not counted for this purpose.

4. Coordination sphere: Representing the complex compounds according to Werner's method occupy more space on the paper. So they are simplified by representing the central metal atom/ion and the ligands attached to it are enclosed in square bracket and is collectively termed as coordination sphere or inner sphere: The ionizable groups are written outside the square bracket and are called counter ions or outer sphere or ionizable sphere. So the chloramine cobalt (III) complexes studied in Werner's theory can be written as shown in Table 14.4.

Table 14.4 Representation of Werner's complexes

Complex	Coordination sphere	Counter ion
I $[Co(NH_3)_6]\,Cl_3$	$[Co(NH_3)_6]^{3+}$	Cl^-
II $[Co(NH_3)_5\,Cl]\,Cl_2$	$[Co(NH_3)_5\,Cl]^{2+}$	Cl^-
III $[Co(NH_3)_4\,Cl_2]\,Cl$	$[Co(NH_3)_4\,Cl_2]^+$	Cl^-
IV $[Co(NH_3)_3\,Cl_3]$	$[Co(NH_3)_3\,Cl_3]$	

5. Coordination polyhedron: The spatial arrangement of the ligand atoms which are directly attached to the central atom /ion defines a coordination polyhedron about the central atom/ion. The most common coordination polyhedra are octahedral, square planar and tetrahedral. These are discussed later.

6. Oxidation number: The oxidation number of the central atom in a complex is the charge it would carry if all the ligands are removed along with the electron pairs that are shared with the central atom.

7. Homoleptic and hetroleptic complexes: Complexes in which a metal is bound to only one kind of ligands e.g., $[Co(NH_3)_6]^{3+}$ are known as homoleptic complexes. The complexes in which a metal atom/ion is surrounded by more than one kind of ligands e.g., $[Co(NH_3)_3Cl_3]$ are known as heteroleptic complexes.

14.3 ELECTRONIC INTERPRETATION OF CO-ORDINATION

Werner's theory was satisfactory at the time it was propounded, when the theory of the electronic structure of matter was unknown. To make the Werner theory compatible with the electronic concept of valence, different theories were proposed of which important theories are discussed here.

14.3.1 Sidgwick Theory

Sidgwick and Lowry gave an electronic interpretation of Werner's theory.

According to this theory the electron transfer corresponds to the electro valence of the metal or formation of electrovalent bonds i.e., primary valencies of Werner. The secondary or non-ionizable valencies of the metal corresponds to the coordinate covalent bond in the complex.

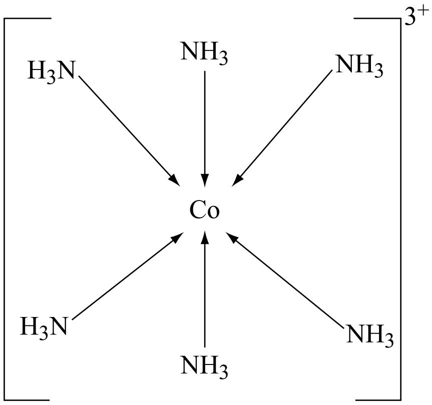


Fig 14.3 Sidgwick's dative bonds in $[Co(NH_3)_6]^{3+}$

All neutral molecules or anions which act as ligands were capable of being coordinated to central metal ion have atoms with at least one unshared lone pair of electrons in their valence shells. Thus the linkage of these coordinating groups to metal ion by secondary valencies arises due to the donation of the electron pairs to the central metal by these groups. Bonds of this type are termed as coordinate bonds. These bonds have been indicated by arrows. For example the structure of $[Co(NH_3)_6]^{3+}$ ion may be written as shown in Fig 14.3. Some examples are given in Table 14.5.

Sidgwick had suggested that the metal ion tends to accept electron pairs from ligands until the total number of electrons of the metal in the complex is equal to that of the next inert gas. The total number of electrons of the central metal in a complex including those gained by bonding is called **effective atomic number**.

The Effective atomic number (EAN) of a metal complex is obtained by subtracting the number of electrons lost by the metal in its ion formation from the atomic number (Z) and then adding the number of electrons gained through coordination.

EAN of a metal in a given complex =

Z- (number of electrons lost by the metal) + (number of electrons gained by the metal through coordination)

From the Table 14.5 it is evident that tendency to attain an inert gas configuration is a significant factor but not necessary condition for complex formation. Some complexes of chromium and platinum are equally stable, yet the effective atomic number is not equal to inert gas configuration. For complex formation, it is essential to produce a symmetrical structure, i.e., tetrahedral, square planar, octahedral, without any consideration of the number of electrons involved.

14.3.2 Defects in Sidgwick's Theory

Sidgwick's electronic interpretation of coordination is not entirely satisfactory. Some objections are.

(i) The donation of electron pairs to a central cation would produce an improbable accumulation of negative charge in the metal atom/ion.

(ii) Another difficulty presented by this concept is the fact that the electron pair, available for donation in H_2O, NH_3 and many other neutral molecules is $2s^2$ pair. These electrons have no bonding characteristics and to excite them to a higher energy level would require more energy than is usually available in bond formation.

(iii) This theory fails to predict the type of metal orbital's which may be involved in bonding.

(iv) This theory could not explain the magnetic behaviour, colour and spectra of complexes.

Table 14.5 Effective atomic number of some metals in complexes

Atom	Atomic number	Complex	No. of e^- lost (or) oxidation estate	No. of electrons gained by coordination	EAN
Fe	26	$[Fe(CN)_6]^{4-}$	2	12	26 – 2 + 12 = 36 [Kr]
Co	27	$[Co(NH_3)_6]^{3+}$	3	12	27 – 2 + 12 = 36 [Kr]
Ni	28	$[Ni(Co)_4]$	0	8	28 – 0 + 8 = 36 [Kr]
Cu	29	$[Cu(NH_3)_4]^{2+}$	2	8	29 – 2 + 8 = 35
Cu	29	$[Cu(CN)_4]^{3-}$	1	8	29 – 1 + 8 = 36 [Kr]
Pd	46	$[Pd(NH_3)_6]^{4+}$	4	12	46 – 4 + 12 = 54 [Xe]
Pt	78	$[PtCl_6]^{-2}$	4	12	78 – 4 + 12 = 86[Xe]
Cr	24	$[Cr(NH_3)_6]^{3+}$	3	12	24 – 3 + 12 = 33
Fe	26	$[Fe(CN)_6]^{3-}$	3	12	26 – 3 + 12 = 33
Ni	28	$[Ni(NH_3)_6]^{2+}$	2	12	28 – 2 + 12 = 38
Pd	46	$[PdCl_4]^{2-}$	2	8	46 – 2 + 8 = 52
Pt	78	$[Pt(NH_3)_4]^{2+}$	2	8	78 – 2 + 8 = 84

(v) This theory fails to explain satisfactorily the geometrical shapes of the complexes.

(vi) Generally metals are electropositive. This theory could not explain how such electropositive metals accept electron pairs from ligands?

Because of the above drawbacks, Sidgwick's theory was replaced by other theories which provided better theoretical justification and could explain the known facts about the chemistry of coordination compounds. The three main theories of bonding of coordination compounds have been (i) valence bond theory (ii) crystal field theory (iii) molecular orbital theory. However here only the first two theories have been described.

14.4 VALENCE BOND THEORY

In 1931 Pauling introduced a new theory which was based on the revolutionary idea of hybridization. His theory was able to account for and even predicts the geometrical shapes of many complexes.

The formation of a complex between a metal and the ligands, according to this theory, is assumed to follow the postulates given below.

1. The metal atom first loses requisite number of electrons to form the ion. The number of electrons lost is numerically equal to the oxidation state of the metal.
2. The central metal ion in the complex makes available a number of vacant orbitals for the formation of coordination bonds with suitable ligands.
3. The number of vacant orbitals made available for this purpose is equal to coordination number of the central metal ion.
4. The appropriate atomic orbitals (s, p and d) of the metal hybridize to give a set of equivalent orbitals of definite geometry such as linear, tetrahedral, square planar, octahedral and so on.

The types of hybridization that involved for different geometries of the complexes are given Table 14.6.

5. The non-bonding electrons of the metal occupy the inner orbitals. These electrons are arranged according to Hund's rule, when the complex is formed by weak ligands like H_2O, F^-, Cl^- etc. But if ligands are strong like NH_3, CO, CN^-, etc. The electrons are rearranged against to Hund's rule vacating certain d-orbitals that can participate in hybridization.
6. In the presence of weak ligands the d-orbitals belonging to nd (d-orbital's of outermost shell) if inner d-orbital's are not vacant, participate in hybridization and the complex is called outer orbital complex or spin free complex or high spin complex. But in the presence of strong ligands the d-orbitals belonging to (n-1)d (d-orbitals of penultimate shell) participate in hybridization and the complex is called inner orbital complex or spin paired complex or low spin complex (spin decreases due to pairing of electrons by rearrangement).
7. Each ligand has at least one σ-orbital containing a lone pair of electrons.
8. The vacant hybrid orbital of the metal ion overlap with the filled orbitals of the ligand to form metal ligand coordinate covalent bond.
9. The magnetic properties of the complex compound depend on the number of unpaired electrons. Depending on the number of unpaired electrons (which can be determined experimentally) the geometry of the complex can be predicted.

The details of some common complex ions are given in Table 14.7.

14.4.1 Important Aspects Regarding the Complexes of 1st Row Transition Metals

Titanium: Solution containing the $[Ti(H_2O)_6]^{3+}$ ion can be prepared by reduction of titanium (IV) compounds in acid solution with zinc e.g., a solution of titanium (IV) chloride a colourless solution (d^0) is reduced to titanium (III) ion, which is purple and unstable in the presence of air and readily reverts to the +4 oxidation state.

Table 14.6 Hybridization and shapes of complex species

Coordination number	Type of hybridization	Orbitals involved in hybridization	Shape
2	sp	s, p_z	Linear
3	sp^2	s, p_z, p_x or p_y	Planar trigonal
4	sp^3	s, p_x, p_y, p_z	Tetrahedral
4	dsp^2	$d_{x^2-y^2}$, s, p_x, p_y	Square planar
5	sp^3d (or) dsp^3	s, p_x, p_y, p_z, $d_z{}^2$	Trigonal bipyramid
6	sp^3d^2 or d^2sp^3	s, p_x, p_y, p_z, $d_{x^2-y^2}$, d_{z^2}	Octahedral

Complex	Atomic No. of central atom/ion	Oxi state of central atom/ion	Type of ligand	Outer electronic configuration of central atom/ion in complex				Type of hybridization	Shape	No. of unpaired electrons	Magnetic character	Magnetic moment	Type of complex
				3d	4s	4p	4d						
$[Cr(H_2O)_6]^{3+}$	24	+III	Weak	[↑][↑][↑][ ][ ]	[ ]	[ ][ ][ ]	[ ][ ][ ][ ][ ]	d^2sp^3	Octahedral	3	Para	3.87	Spin free, inner orbital
$[Cr(CN)_6]^{3-}$	24	+III	Strong	[↑][↑][↑][ ][ ]	[ ]	[ ][ ][ ]	[ ][ ][ ][ ][ ]	d^2sp^3	Octahedral	3	Para	3.87	Spin free, inner orbital
$[Cr(CO)_6]$	24	0	Strong	[↑↓][↑↓][↑↓][ ][ ]	[ ]	[ ][ ][ ]	[ ][ ][ ][ ][ ]	d^2sp^3	Octahedral	0	Dia	0	Spin paired, inner orbital
$[MnBr_4]^{2-}$	25	+II	Weak	[↑][↑][↑][↑][↑]	[ ]	[ ][ ][ ]		sp^3	Tetrahedral	5	Para	5.91	Spin free
$[Mn(H_2O)_6]^{2+}$	25	+II	Weak	[↑][↑][↑][↑][↑]	[ ]	[ ][ ][ ]	[ ][ ][ ][ ][ ]	sp^3d^2	Octahedral	5	Para	5.91	Spin free, outer orbital
$[Mn(CN)_6]^{4-}$	25	+II	Strong	[↑↓][↑↓][↑][ ][ ]	[ ]	[ ][ ][ ]		d^2sp^3	Octahedral	1	Para	1.73	Spin paired inner orbital
$[Fe(CO)_6]^{2-}$	26	–II	Strong	[↑↓][↑↓][↑↓][↑↓][↑↓]	[ ]	[ ][ ][ ]	[ ][ ][ ][ ][ ]	sp^3d^2	Octahedral	0	Dia	0	Spin paired outer orbital
$[Fe(CO)_5]$	26	0	Strong	[↑↓][↑↓][↑↓][↑↓][ ]	[ ]	[ ][ ][ ]		dsp^3	Trigonal bipyramid	0	Dia	0	Spin paired inner orbital
$[FeCl_4]^{2-}$	26	+II	Weak	[↑↓][↑][↑][↑][↑]	[ ]	[ ][ ][ ]		sp^3	Tetrahedral	4	Para	4.9	Spin free
$[Fe(H_2O)_6]^{2+}$	26	+II	Weak	[↑↓][↑][↑][↑][↑]	[ ]	[ ][ ][ ]	[ ][ ][ ][ ][ ]	sp^3d^2	Octahedral	4	Para	4.9	Spin free outer orbital
$[Fe(CN)_6]^{4-}$	26	+II	Strong	[↑↓][↑↓][↑↓][ ][ ]	[ ]	[ ][ ][ ]		d^2sp^3	Octahedral	0	Dia	0	Spin paired inner orbital
$[FeF_6]^{3-}$	26	+III	Weak	[↑][↑][↑][↑][↑]	[ ]	[ ][ ][ ]	[ ][ ][ ][ ][ ]	sp^3d^2	Octahedral	5	Para	5.91	Spin free outer orbital
$[Fe(CN)_6]^{3-}$	26	+III	Strong	[↑↓][↑↓][↑][ ][ ]	[ ]	[ ][ ][ ]		d^2sp^3	Octahedral	1	Para	1.73	Spin paired inner orbital
$[Co(CO)_4]^{-}$	27	–I	Strong	[↑↓][↑↓][↑↓][↑↓][↑↓]	[ ]	[ ][ ][ ]		sp^3	Tetrahedral	0	Dia	0	Spin paired
$[CoCl_4]^{2-}$	27	+II	Weak	[↑↓][↑↓][↑][↑][↑]	[ ]	[ ][ ][ ]		sp^3	Tetrahedral	3	Para	3.87	Spin free
$[Co(H_2O)_6]^{2+}$	27	+II	Weak	[↑↓][↑↓][↑][↑][↑]	[ ]	[ ][ ][ ]	[ ][ ][ ][ ][ ]	sp^3d^2	Octahedral	3	Para	3.87	Spin free outer orbital

(Continued)

(Cont.)

Complex	Atomic No. of central atom/ ion	Oxi state of central atom/ion	Type of ligand	Outer electronic configuration of central atom/ion in complex	Type of hybridization	Shape	No. of unpaired electrons	Magnetic character	Magnetic moment	Type of complex
$[CoF_6]^{3-}$	27	+III	Weak	[↑↓][↑][↑][↑][↑] [ ] [ ][ ][ ] [ ][ ][ ][ ][ ]	sp^3d^2	Octahedral	4	Para	4.9	Spin free outer orbital
$[Co(NH_3)_6]^{3+}$	27	+ III	Strong	[↑↓][↑↓][↑↓][ ][ ] [ ] [ ][ ][ ]	d^2sp^3	Octahedral	0	dia	0	Spin paired inner orbital
$[Co(C_2O_4)_3]^{3-}$	27	+ III	Weak	[↑↓][↑↓][↑↓][ ][ ] [ ] [ ][ ][ ]	d^2sp^3	Octahedral	0	Dia	0	Spin paired inner orbital
$[Ni(CO)_4]$	28	0	Strong	[↑↓][↑↓][↑↓][↑↓][↑↓] [ ] [ ][ ][ ]	sp^3	Tetrahedral	0	Dia	0	Spin paired
$[NiCl_4]^{2-}$	28	+II	Weak	[↑↓][↑↓][↑↓][↑][↑] [ ] [ ][ ][ ]	sp^3	Tetrahedral	2	Para	2.83	Spin free
$[Ni(CN)_4]^{2-}$	28	+II	Strong	[↑↓][↑↓][↑↓][↑↓][ ] [ ] [ ][ ][ ]	dsp^2	Square planar	0	Dia	0	Spin paired inner orbital
$[Ni(H_2O)_6]^{2+}$ $[Ni(NH_3)_6]^{2+}$	28	+II	Weak or Strong	[↑↓][↑↓][↑↓][↑][↑] [ ] [ ][ ][ ] [ ][ ][ ][ ][ ]	sp^3d^2	Octahedral	2	Para	2.83	Spin free outer orbital
$[Cu(CN)_4]^{3-}$	29	+I	Strong	[↑↓][↑↓][↑↓][↑↓][↑↓] [ ] [ ][ ][ ]	sp^3	Tetrahedral	0	Dia	0	
$[Cu(NH_3)_4]^{2+}$	29	+II	Strong	[↑↓][↑↓][↑↓][↑↓][↑] [ ] [ ][ ][↑] [ ][ ][ ][ ][ ]	dsp^2	Square planar	1	Para	1.73	Inner orbital
$[CuCl_4]^{2-}$	29	+II	weak	[↑↓][↑↓][↑↓][↑↓][↑] [ ] [ ][ ][ ] [ ][ ][ ][ ][ ]	sp^3	Tetrahedral	1	Para	1.73	
$[Zn(H_2O)_4]^{2+}$ $[Zn(NH_3)_4]^{2+}$	30	+II	weak or Strong	[↑↓][↑↓][↑↓][↑↓][↑↓] [ ] [ ][ ][ ] [ ][ ][ ][ ][ ]	sp^3	Tetrahedral	0	Dia	0	
$[Ag(NH_3)_2]^+$ $[Ag(CN)_2]^+$	47	+I	weak or Strong	4d [↑↓][↑↓][↑↓][↑↓][↑↓] 5s [ ] 5p [ ][ ][ ]	sp	Linear	0	Dia	0	
$[Au(CN)_2]$	79	+I	weak or Strong	5d [↑↓][↑↓][↑↓][↑↓][↑↓] 6s [ ] 6p [ ][ ][ ]	sp	Linear	0	Dia	0	
$[PdCl_4]^{2-}$	46	+II	weak or Strong	4d [↑↓][↑↓][↑↓][↑↓][ ] 5s [ ] 5p [ ][ ][ ]	dsp^2	Square planar	0	Dia	0	Spin paired inner orbital
$[PtCl_4]^{2-}$ $[Pt(NH_3)_4]^{2+}$	78	+II	weak or Strong	5d [↑↓][↑↓][↑↓][↑↓][ ] 6s [ ] 6p [ ][ ][ ]	dsp^2	Square planar	0	Dia	0	Spin paired inner orbital
$[PdCl_6]^{2-}$ $[Pd(NH_3)_6]^{2+}$	46	+IV	weak or Strong							
$[Pt(Cl_6)]^{2-}$ $[PtCl(NH_3)_6]^{4+}$	78	+IV	weak or Strong	(n–1)d [↑↓][↑↓][↑↓][ ][ ] ns [ ] np [ ][ ][ ]	d^2sp^3	Octahedral	0	Dia	0	Spin paired inner orbital

Vanadium $[V(H_2O)_6]^{2+}$ can be obtained by shaking a solution of ammonium vanadate in dilute sulphuric acid with zinc amalgam. Solutions containing hydrated V^{2+} ions have lavender colour and are strongly reducing and react with water evolving hydrogen, can only be preserved in the presence of a reducing agent such as zinc amalgam and dilute acid.

Chromium: Chromium in +3 oxidation state form a great number of complexes. In all these complexes the 3d-orbitals contain only 3 electrons. Pairing of electrons (rearrangement of electrons) does not takes place even the ligand is strong. So in all its complexes it forms only inner orbital complexes but spin free or high spin complexes. This is because pairing of electrons starts only when the number of electrons exceeds three. In the formation of $[Cr(CO)_6]$ the 4s electrons also rearrange in the 3d orbitals.

Manganese: Manganese in +II oxidation state can form both 4 coordinate and 6 coordinate complexes. The 4 coordinate complexes are tetrahedral while the 6 coordinate complexes are octahedral. In the six coordinate complexes electron pairing takes place in the presence of strong ligands forming inner orbital or spin paired complexes but in the presence of weak ligands rearrangement of electrons does not takes place and only outer orbital or spin free complexes will be formed.

$[Mn(CN)_6]^{4-}$ is very unstable and can be easily oxidized to $[Mn(CN)_6]^{3-}$. Mn^{3+} in aqueous solution is unstable and can act as oxidizing agent but can be stabilized by complex formation.

Iron: Iron form large number of complexes in +II and also in +III oxidation states, all of which are octahedral. In the presence of strong ligands electron rearrangement takes place giving spin paired or inner orbital complexes and in the presence of weak ligands rearrangement of electrons does not takes place forming spin free or outer orbital complexes.

Iron (III) do not form complexes with ammonia. In acid solution NH_3 molecule cannot act as a ligand because it is protonated by donating its lone pair to H^+ ion. So it can form complexes either in neutral solution or in alkaline solution. In low concentration, metal ions will be precipitated by the hydroxide ion produced by ammonia in aqueous solution.

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

In the presence of excess of ammonia, the hydroxide precipitates are dissolved with the formation of complexes Eg. Zn^{2+}, Cu^{2+}, Ni^{2+}, Ag^+ etc.

The extent of which a hydroxide or ammonia complex is formed with a cation depends not only upon the concentration of ammonia but also upon the stability of the complex that is produced. Cations such as iron (III) aluminium (III) always precipitate as hydroxide with ammonia because the stabilities of ammonia complexes are insufficient to put the hydroxide into solution.

Cobalt: For cobalt the most stable oxidation state is +II in aqueous solution where it exist as $[Co(H_2O)_6]^{2+}$. It is stable only in the absence of complexing agents like NH_3 and CN^-. Cobalt (II) complexes containing NH_3 and CN^- ligands are readily oxidized to Co(III) complexes. In solution cobalt (II) salts are pink, the colour being due to the presence of $[Co(H_2O)_6]^{2+}$ ions.

When present as an integral part of a complex, compounds containing cobalt in +3 oxidizing state are considerably more stable i.e., they are not such powerful oxidizing agents. Many Co^{3+} complexes exist and they are all octahedral in shape. The $[Co(CN)_6]^{3-}$ ion is of interest in that it is stable in the presence of water unlike $[Co(CN)_5]^{3-}$ which contains Co^{2+}; indeed $[Co(CN)_5]^{3-}$ attacks water liberating hydrogen and itself oxidized to $[Co(CN)_6]^{3-}$.

In $[Co(C_2O_4)_3]^{3-}$ though oxalate ion is weaker ligand than water pairing of electrons takes place and it is a spin paired, inner orbital complex with diamagnetic character.

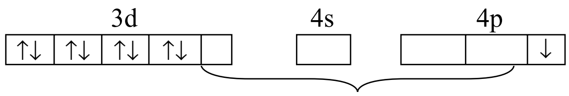
Indeed all known octahedral complexes including $[Co(H_2O)_6]^{3+}$ have diamagnetic ground states except for $[Co(H_2O)_3F_3]$ and $[CoF_6]^{3-}$ which are paramagnetic with four unpaired electrons.

Nickel: All 6 coordinate complexes of Ni^{2+} are octahedral and outer orbital complexes. They are paramagnetic due to the presence of two unpaired electrons. Even in the presence of strong ligands though electron pairing takes place only one 3d orbital will be vacated, which is not useful for d^2sp^3 hybridization.

Copper: All copper complexes in +1 oxidation state have tetrahedral structure and are diamagnetic.

Copper in +2 oxidation state form many complexes with ammonia and amines such as $[Cu(H_2O)_5NH_3]^{2+}$, $[Cu(H_2O)_4(NH_3)_2]^{2+}$; $[Cu(H_2O)_3(NH_3)_3]^{2+}$ and $[Cu(NH_3)_6]^{2+}$. This is because of the distortion of its octahedral complex and makes the 5th and 6th bonds (along z-axis) long and weak.

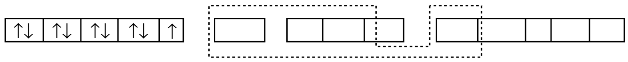
In $[Cu(NH_3)_4]SO_4$, the $[Cu(NH_3)_4]^{2+}$ ion is square planar in structure, in contrast to the expected tetrahedral geometry. Depending on the geometry it was proposed that in $[Cu(NH_3)_4]^{2+}$, Cu^{2+} is involved in dsp^2 hybridization. For this purpose to vacate one d-orbital the unpaired electron present in 3d is promoted to up orbital



dsp^2 hybridization

The above configuration indicates that the unpaired electron is present in higher energy 4p-orbital. So it can be easily removed oxidizing Cu^{2+} to Cu^{3+}. But this is not happening. To explain this Huggin suggested as sp^2d hybridization.

Cu^{2+} in $[Cu(NH_3)_4]^{3+}$



sp^2d hybridization

The halide complexes of copper (II) show two different stereochemistries. In $(NH_4)_2[CuCl_4]$ and $[Et\ NH_3]_2\ [CuCl_4]$ the $[CuCl_4]^{2-}$ ion is square planar but in $Cs_2[CuCl_4]$ the $[CuCl_4]^{2-}$ is tetrahedral in structure. Tetrahedral $[CuCl_4]^{2-}$ ions are orange and square planar. $[CuCl_4]^{2-}$ ions are yellow.

Zinc: All 4 coordinate zinc complexes are tetrahedral and diamagnetic and 6 coordinate zinc complexes have octahedral structure. Examples for 6 coordinate complexes of zinc is $[Zn(en)_3]^{2+}$.

The transition metals in 2nd and 3rd transition series always form inner orbital complexes i.e, spin paired complexes irrespective of the strength of ligand.

14.4.2 Defects in Valence Bond Theory

Pauling's theory explained the chemistry of coordination compounds satisfactorily but failed to explain some of the aspects.

1. Although it provides a satisfactory pictorial representation of the complex qualitatively, it does not provide quantitative interpretation of the stability of the complexes.
2. V.B theory could not explain the colour and absorption spectra of complexes.
3. It could not explain why certain complexes are more labile than others.
   Labile complexes are those in which one ligand can be easily displaced by another ligand. On the other hand inert complexes are those in which displacement of ligands is slow.
4. It does not account for the relative energies of different structures.
5. It fails to explain why at one time the electron must be arranged against the Hund's rule while at other times the electronic configuration is not disturbed.
6. It fails to provide any satisfactory explanation for the existence of inner orbital and outer orbital complexes.
7. Sometimes the theory needs the transfer of electron from lower energy level (e.g., 3d) to higher energy level (4p) which is very much unrealistic in the absence of energy supplier.
8. Electron spin resonance reveals that in Cu (II) complexes the electron is not in 4p level and the complex is square planar.

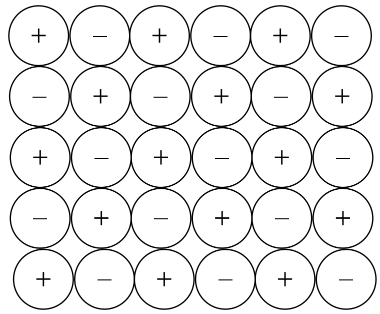


Fig 14.4 Arrangement of sodium and chloride ions in one layer of NaCl lattice

14.5 CRYSTAL FIELD THEORY

Crystal field theory (CFT) was originally proposed by Bethe and Van Vleck for explaining the optical properties of crystalline solids.

For example in sodium chloride crystal every Na^+ ion is surrounded by six Cl^- ions octahedrally which produce a negative charge field around sodium ion.

This negative charge field of Cl^- ions interacts with the negative charge of the electrons present in the outermost shell of sodium ion i.e., the p-orbitals of sodium ion. Due to this reason the energy of these orbitals increases. Since all the three p-orbitals p_x, p_y and p_z are equally repelled by the $6Cl^-$ ions arranged octahedrally around Na^+ ion along x, y and z axes, the energy of three p-orbitals also increases equally.

But in transition metal compounds the outermost orbitals of the transition metal ions are d-orbitals. Due to the difference in the orientation of d-orbitals, the repulsion by the negative charge field around the transition metal ion is also different on d-orbitals. This causes the development of energy differences between d-orbitals.

This concept was extended mainly by Orgel to explain about the formation of complexes. This theory starts with assumption that as far as transition metal complexes are concerned, the central metal ion can be regarded as a point charge i.e., a charge concentrated into a very small volume, and the surrounding ligands be they anions or neutral molecules containing lone pair of electrons can likewise be represented as point charges. The bonding between the central metal ion and the surrounding ligands is assumed to be ionic.

The five d-orbitals in an isolated gaseous metal atom/ion have same energy i.e., they are degenerate. This degeneracy is maintained if spherically symmetrical field of negative ends of dipolar molecules such as NH_3, H_2O in a complex, it becomes asymmetrical and the degeneracy is lifted due to the difference in the orientation of different d-orbitals. It results in splitting of the d-orbitals energies. The pattern of splitting depends upon the nature of the crystal field. The splitting of d-orbitals energies and its effects infact, form the basis of the crystal field treatment of coordination compounds. Let us explain this splitting in different crystal fields.

14.5.1 Crystal Field Splitting of d-orbitals in Octahedral Coordination Compounds

For convenience let us assume that the six ligands are positioned symmetrically along the cartesian axes, with the metal atom at the origin. As the ligands approach, first there is an increase in the energy of d-orbitals relative to that of the free ion just as would be the case in a spherical field. Crystal field theory considers what effect the approach of these six ligands (along x,y and z axes) have on the 3d orbitals of the transition metal ion. The orientations in space of the five 3d-orbitals (which in the free ion are degenerate) are shown, the approaching ligands being denoted by the letter L (Fig 14.5).

When ligands approach along the x,y and z axes, electrons in the 3d orbitals will be repelled but as can be seen from the Fig 14.5 the effect will be greater for the $3d_{z^2}$ and $3d_{x^2-y^2}$ orbitals since these two orbitals have lobes lying along the line of approaching ligands. The net result is that the energy of the $3d_{z^2}$ and $3d_{x^2-y^2}$ orbitals is raised relative to the energy of the $3d_{xy}$, $3d_{xz}$ and $3d_{yz}$ orbital i.e., the degeneracy of the 3d orbitals is now destroyed since

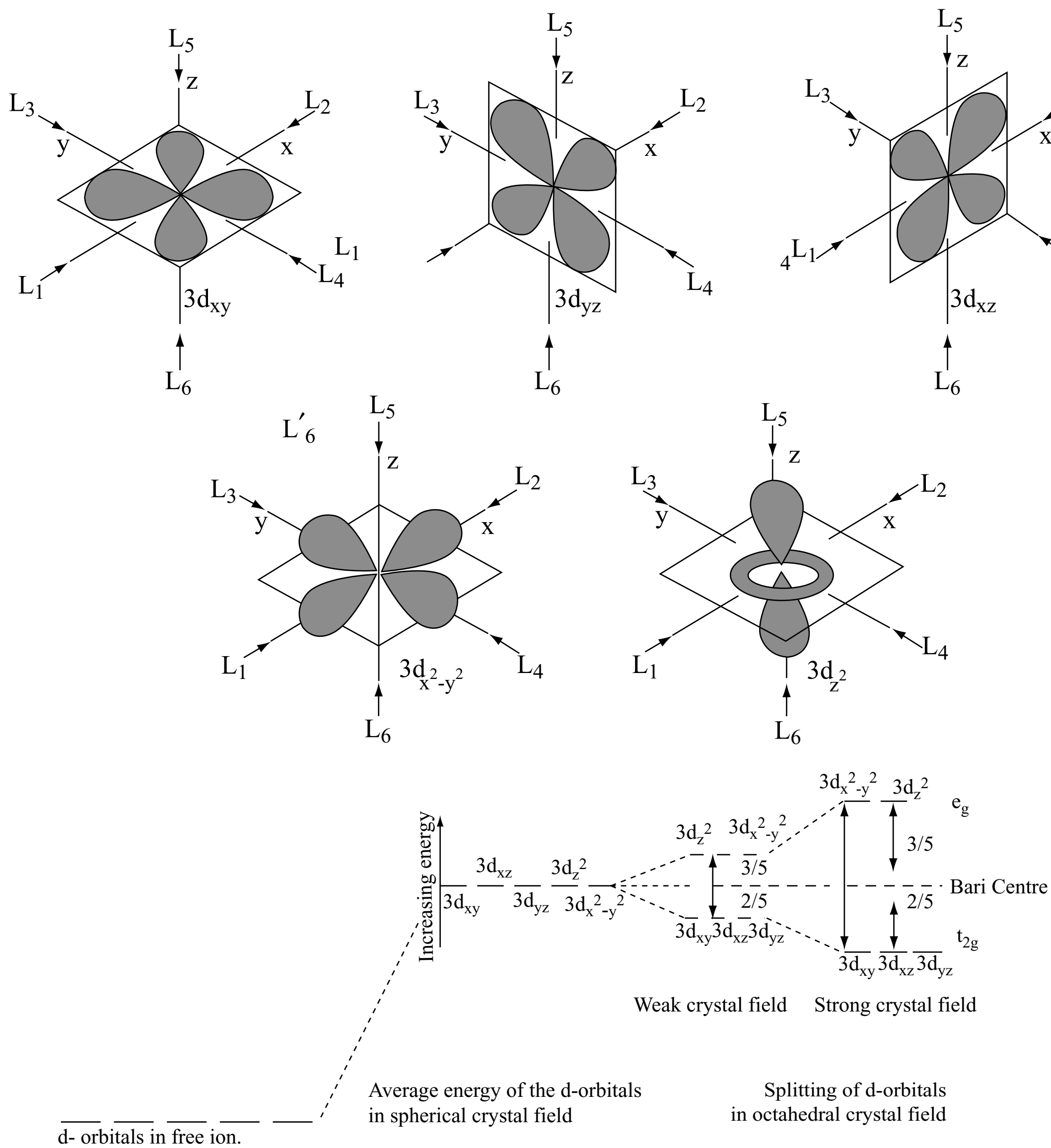


Fig 14.5 The splitting of the d-orbitals when a transition metal ion is surrounded octahedrally by six ligands

their relative orientations in space preclude the possibility of equal intersection with the surrounding ligands; the splitting of the 3d levels is shown in Fig 14.5.

The 3d levels are split into an upper group of two (doubly degenerate and labeled e_g) and a lower group of three (trebly degenerate and labeled t_{2g}); the splitting of the levels is represented by the symbol Δ. If we reckon the zero of energy as the state of affairs that would obtain if each of the five 3d orbitals had interacted equally with the six ligands, then each of the upper two orbitals is raised by 3/5 Δ (or 6/5 Δ collectively) while each of the lower three orbitals is lowered by 2/5 Δ (or 6/5 Δ collectively). As the diagram shows the degree of splitting depends upon the strength of the crystal field.

The crystal field splitting is thus the energy difference between t_{2g} and e_g orbitals. It is denoted by the symbol Δ_o or 10 Dq (the subscript 'o' indicated octahedral complexes) the Δ is frequently measured in terms of a parameter Dq the magnitude of splitting is arbitrarily set as 10 Dq i.e., 4Dq for each unpaired electron.

In an octahedral complex that contain one d-electron (For example $[Ti(H_2O)_6]^{3+}$) that electron will reside in the d orbital of lowest energy. In fact the electron goes into a t_{2g} orbital that has an energy 0.4 Δ_o (2/5 Δ_o) or 4 Dq less than that of the hypothetical degenerated d orbitals which results in crystal field splitting. Thus the complex will be $0.4\Delta_o$ or 4 Dq more stable than the simple electrostatic model predicts. In simple terms we can say that the d electron, and hence the whole complex, has a lower energy as a result of the placement of the electron in a t_{2g} d orbitals that is as far from the ligands as possible. The 0.4 Δ_o or 4Dq is called the crystal field stabilization energy (CFSE) for the complex.

For each electron entering into e_g orbital the destablizing energy is $0.6\Delta_0$ or 6Dq. In other words the stabilization energy assigned for an electron in e_g orbital is −0.6 Δ_o or −6 Dq. Working on this basis it is possible to calculate crystal field stabilization energies for various metal ions in octahedral complexes.

In d^2 and d^3 coordination entities, the d electrons occupy the t_{2g} orbitals singly in accordance with the Hund's rule. For d^4 ions, two possible patterns of electron distribution arise (i) the fourth electron may enter an e_g orbital (of higher energy) or (ii) it may pair an electron in the t_{2g} orbitals. The actual configuration adopted is decided by the relative values of Δ_o and P. 'P' represents the energy required for electron pairing in a single orbital.

If Δ_o is less than P ($\Delta_o < P$) as in the case of weak field situation or high spin situation (in the presence of weak ligands), the fourth electron enters one of the e_g orbitals giving the configuration $t_{2g}^3\,e_g^2$. If now a fifth electron is added to a weak field coordination entity the configuration becomes $t_{2g}^3\,e_g^2$.

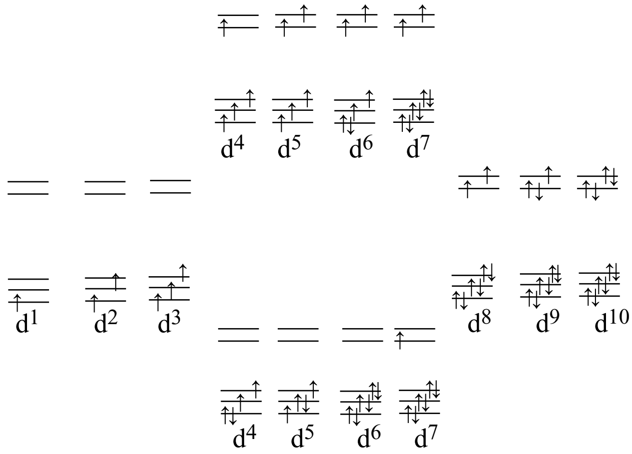


Fig 14.6 Weak and strong field arrangement of d electrons

When $\Delta_o > P$ as in the case of strong field or low spin situation, and pairing will occur in the t_{2g} level with the eg. level remaining unoccupied in entities of d^1 to d^6 ions.

Also columbic exchange energies (due to exchange of electron positions with parallel spin) plays a role in the stabilization of complex. Only the electron with parallel spin can change their positions either in t_{2g} or e_g orbitals but not from t_{2g} to e_g or vice versa (due to energy difference). The CFSEs are given in table 14.8. Weak and strong field arrangements of d electrons are shown in fig 14.6.

The relationship between the difference between the t_{2g} and e_g energy levels (Δ_o) the pairing energy (P) and the exchange energy (E) determines the orbital configuration of the electrons. The configuration with the lower total energy is the ground state for the complex. For example the change in exchange energy (E) from high spin to low spin is zero for d^5 ions. But for d^6 ions the exchange energies increases from high spin to low spin.

However two new pairs are formed as shown in fig 14.6. So the electronic arrangement in d^6 ions takes place depending on the overall energy change.

Experimentally it was found that all aqua complexes are high spin complexes except $[Co(H_2O)_6]^{3+}$. In $[Co(H_2O)_6]^{3+}$ Δ_0 and p values are nearly same. So it is a low spin complex and is the only low-spin aqua complex.

14.5.2 Crystal Field Splitting in Tetrahedral Coordination Compounds

A regular tetrahedron is related to cube. Assume the metal ion at the centre of cube and the four ligands are at

Table 14.8 CFSE for weak field and strong field ligands

	Weak field ligands				Strong field			
d^n	Electronic configuration	CFSE Δ_o	P	E	Electronic configuration	CFSE	P	E
d^1	$t_{2g}^1 e_g^0$	–0.4	–	–	$t_{2g}^1 e_g^0$	–0.4	–	–
d^2	$t_{2g}^2 e_g^0$	–0.8	–	–E	$t_{2g}^2 e_g^0$	–0.8	–	E
d^3	$t_{2g}^3 e_g^0$	–1.2	–	–3E	$t_{2g}^3 e_g^0$	–1.2	–	–3E
d^4	$t_{2g}^3 e_g^1$	–0.6	–	–3E	$t_{2g}^4 e_g^0$	–1.6	+P	–3E
d^5	$t_{2g}^3 e_g^2$	0	–	–4E	$t_{2g}^5 e_g^0$	–2.0	+2P	–4E
d^6	$t_{2g}^4 e_g^2$	–0.4	+ P	–4E	$t_{2g}^6 e_g^0$	–2.4	+3P	–6E
d^7	$t_{2g}^5 e_g^2$	–0.8	+2P	–5E	$t_{2g}^6 e_g^1$	–1.8	+3P	–6E
d^8	$t_{2g}^6 e_g^2$	–1.2	+3P	–7E	$t_{2g}^6 e_g^2$	–1.2	+3P	–7E
d^9	$t_{2g}^6 e_g^3$	–0.6	+4P	–7E	$t_{2g}^6 e_g^3$	–0.6	+4P	–7E
d^{10}	$t_{2g}^6 e_g^4$	0	+5P	–8E	$t_{2g}^6 e_g^4$	0	+5P	–8E

the alternate corners of the cube which makes a regular tetrahedron. The t_{2g} orbitals (d_{xy}, d_{yz} and d_{xz}) are directed towards the edge centres of the cube while the e_g ($d_{x^2-y^2}$ and d_{z^2}) are directed to the face centres and are away from the ligands which are at the corners of the cube. So the repulsion by the ligands on d-orbitals will be less by about 2/3 times. (Mathematically because the d-orbitals are away from ligands) compared to the octahedral field in which the ligands directly coincide the d-orbitals. Also the number of ligands in tetrahedral complexes are only 4 while in that of octahedral complex is 6. So there is further decreases in repulsion (due to less number of ligands) by about 4/6 i.e., 2/3 times in tetrahedral complexes than in octahedral complexes. Thus the tetrahedral crystal field splitting energy Δt is roughly $2/3 \times 2/3 = 4/9$ of the octahedral crystal field splitting energy Δ_0. The Δ_t value is always much smaller than the octahedral splitting energy Δ_0 and never energetically favourable to pair electrons and all tetrahedral complexes are high spin complexes.

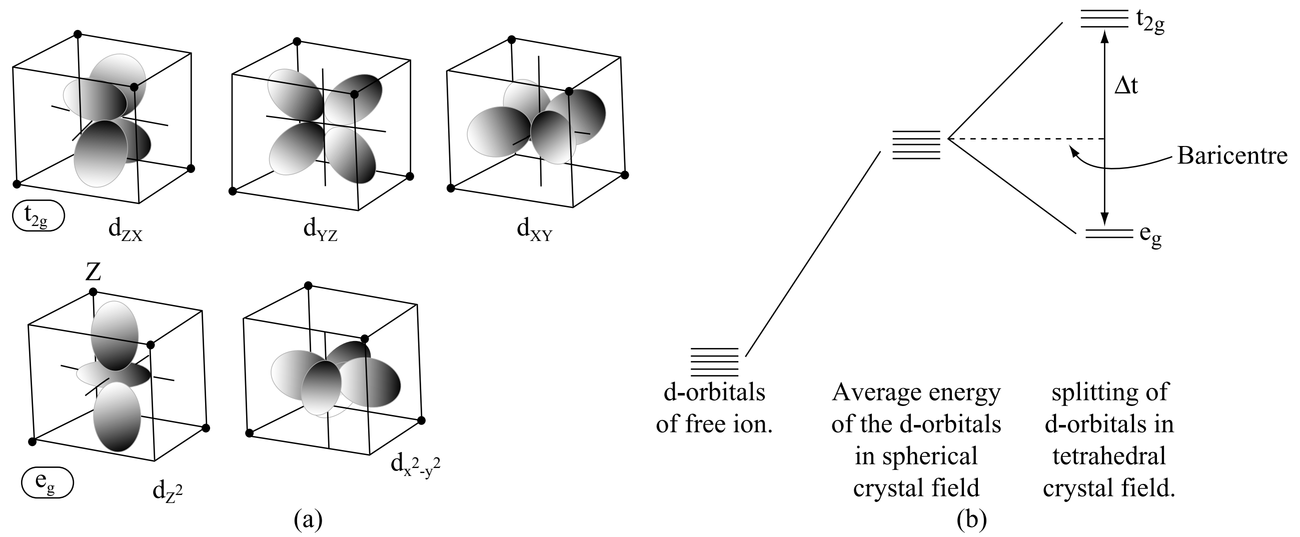


Fig 14.7 (a) The effect of tetrahedral crystal field on a set of d-orbitals is to split them into two sets; the e_g orbitals lie lower in energy than t_{2g} orbitals (b) d-orbital splitting in tetrahedral crystal field

It should be noted that the edge centres of the cube are near to the corner of cube than its face centre. So the d-orbitals (t_{2g}) which are at edge centres are repelled more by the ligands when compared to the d-orbitals (e_g) at face centres. Thus the energy of e_g orbitals will be less than the bari centre. This crystal field splitting is the opposite way round that in octahedral complexes.

The t_{2g} orbital's are 0.4 Δ_t above the weighted average (Bari centre) of the two groups and the e_g orbitals are 0.6 Δ_t below the weighted average (Bari centre).

14.5.3 Crystal Field Splitting in Square Planar Complexes

According to crystal field theory square planar complexes are formed by the removal of two ligands along the z axis in an octahedral complex. When the two ligands along the z-axis are removed the repulsion of these ligands on the d-orbital's having z component also decreases. So they are stabilized. The d_{z^2} orbital is more stabilized than $d_{x^2-y^2}$ and d_{yz} and d_{xz} are more stabilized than dxy. The d-orbitals splitting in a square planar complex will be as shown in Fig 14.8.

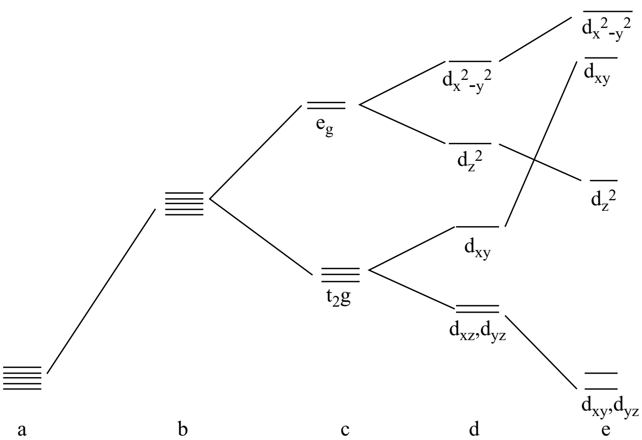


Fig 14.8 (a) d-orbital's of free ion (b) average energy of the d-orbital's in spherical crystal field (c) splitting of d-orbital's in octahedral field (d) tetragonal distortion i.e., the ligands along z axis are moving away from the metal ion (e) d-orbital splitting in square planar complex

14.5.4 Determination of Δ

As already seen that Δ is the energy difference between two sets of d-orbital t_{2g} and e_g. This can be measured experimentally. When light is passed through the solution of complex, it absorbs the radiation whose energy is equal to the excitation of electron from lower energy group of d-orbital to higher energy group orbital. For example the absorption spectrum of $[Ti(H_2O)_6]^{3+}$ shows a strong band corresponding to a wave length of 5000A^0. This represents an energy change of 238.9Kj mol^{-1} which can be calculated as follows. The relationship between Δ and the frequency of light absorbed is given by the usual expression.

$$\Delta = h\nu \text{ or } h\ \frac{c}{\lambda}$$

Where h is Planck's constant (6.626×10^{-34} Js), C is the velocity of light (3×10^{-8}m) and is λ the wavelength of light absorbed.

$$\therefore \Delta = \frac{6.626\times10^{-34}\times3\times10^{8}}{5000\times10^{-10}}\text{ J atom}^{-1}$$

$$\text{or } \Delta = \frac{6.626\times10^{-34}\times3\times10^{8}\times6.023\times10^{23}}{5000\times10^{-10}\times1000}\text{ KJ mol}^{-1}$$

$$= 239.4\text{ KJ mol}^{-1}$$

Thus the Δ value for different complexes can be determined.

14.5.5 Factors Affecting Δ

(i) Geometry of the complex: In octahedral complexes the splitting energy of d-orbitals has been found to be more than twice than in tetrahedral complexes for the same metal ion and ligands as already discussed.

(ii) Nature of the ligands: Δ Value of a complex depend on the nature of ligand. Different ligands with same metal ion produce different values due to difference in the strength of repulsion shown on d-orbital. We can compare the values of Δ produced by various ligands with same ion and then arranged in the order of their strength in the form of a series called spectrochemical series.

Weak field ligands

$$I < Br^- < S^{-2} < \underline{S}CN^- < Cl^- < N\underline{O}^-_2 < N^{3-} < F^- < OH^-$$
$$< C_2O_4^{2-} < H_2O < \underline{N}CS^- < CH_3CN < Py < NH_3 < en < bipy$$
$$< phen < \underline{N}O^-_2 < PPh_3 < \underline{C}N^- < CO$$

Strong field ligands

(The donor atom in an ambidentate ligand is underlined).

The ligand which lies on left side to H_2O are considered as weak field ligands while the ligands that lies to the right side of H_2O are considered as strong field ligands.

(iii) Charge on the metal ion: As stated earlier that crystal field theory is based on electrostatic model, a metal with higher charge attracts the ligands nearer to it, thus causing more splitting of d-orbital than an ion with lesser charge. So Δ value will be more. For example Δ_o for hexa aqua complexes of Cr^{2+} and Cr^{3+} are

164.6KJ mol^{-1} and 217.1KJ mol^{-1}. Similarly Δ_o is more for Fe^{3+} in $[Fe(CN)_6]^{3-}$ than for Fe^{2+} in $[Fe(CN)_6]^{4-}$

(iv) Position of the metal in transition series: For analogous coordination compounds within a group values differ the general trend being 3d < 4d < 5d. Thus while going from Cr to Mo or Co to Rh the Δ_0 increase by 50%. As a consequence of this, complexes of the second and third transition series have a greater tendency to be low spin as compared to the first transition series. For example the Δ_0 values for $[Co(NH_3)_6]^{3+}$, $[Rh(NH_3)_6]^{3+}$ and $[Ir(NH_3)_6]^{3+}$ are 274.6 kJ mol^{-1}, 406.0kJ mol^{-1} 498.5kJ mol^{-1}. The interaction down a group reflects the larger size of the 4d and 5d orbitals compared with the compact 3d orbitals and the consequent stronger interactions with the ligands.

14.5.6 Application of Crystal Field Theory

1. Colour: As already explained in 13.4.2 most of complexes of transition metals give absorption bands in the visible region i.e., they are coloured. According to the crystal field theory the colour is due to d-d transitions. In the presence of ligands only the d-orbitals split up into two sets t_{2g} and e_g. The mechanism of light absorption in coordination compounds is that photons of appropriate energy can excite the electron from lower energy d-orbitals to higher energy d-orbitals. So the complex will appear in the complimentary colour of the absorbed radiation.

For example the hydrated copper (II) sulphate $CuSO_4 . 5H_2O$ is blue coloured but the anhydrous copper (II) sulphate is colourless. In hydrated compound the water molecules act as ligands due to which d-orbital splitting takes place where as the d-orbitals are degenerate in the anhydrous $CuSO_4$.

Since the degree of splitting of the 3d levels depends upon the particular ligands themselves the variation in colour of ions of a particular transition metal is explained. For example hydrated cobalt (II) ions are pink, but in the presence of sufficient chloride ions a blue complex $[CoCl_4]^{2-}$ is formed. The pale blue colour of hydrated copper (II) ion changes to dark blue in colour in the presence of sufficient ammonia and to green if sufficient chloride ions are added. Similarly the green colour of aqueous $[Ni(H_2O)_6]^{2+}$ turns blue when ammonia is added giving $[Ni(NH_3)_6]^{3+}$, violet $[Cr(H_2O)_6]^{3+}$ gives bright blue $[Cr(H_2O)_6]^{2+}$ upon reduction. Hydrated Sc^{3+}, Ti^{4+}, etc are colourless as all these metal ions have vacant d-orbital and d-d transitions of same (n-1)d are not possible. Similarly Zn^{2+}, Cd^{2+}, Hg^{2+}, Cu^+, Ag^+ and Au^+ are also colourless because in these metals ions the d-orbitals are completely filled and thus d-d transitions are not possible.

But Cr^{6+} and Mn^{7+} form oxo anions $Cr_2O_7^{2-}$, CrO_4^{2-} and MnO_4^- which are strongly coloured. These colours are not due to d-d transitions, but due to different phenomenon known as charge transfer phenomenon i.e., due to the excitation of lone pair of electrons in lower energy level of oxygen to higher vacant energy levels of metal ion. The complete discussion of this phenomenon is beyond the scope of this book.

Generally the gems are naturally occurring corundum stones. When pure they are colourless. But when they contain certain transition metal oxide as impurity they have bright colours. For example ruby is aluminum oxide containing about 0.5 – 1% Cr^{3+} ions (with d^3 configuration). These chromium ions are randomly distributed in position normally occupied by Al^{3+} ions. Now we consider these chromium (III) species as octahedral chromium (III) complexes surrounded by six oxide ions which are incorporated into the alumina lattice: d-d transitions at these centres gives rise to the colour. In emerald Cr^{3+} ions occupy octahedral sites in the mineral beryl ($Be_3Al_2Si_6O_{18}$). The absorption bands shift to longer wavelength namely yellow-red and blue causing emerald to transit in the green region. Similarly due to the presence of different transition metals different coloured gems are formed (refer to 7.9.2).

2. Magnetic Properties: The crystal field theory is successful in calculating the number of unpaired electrons in high spin and low spin complexes. With the help of number of unpaired electrons, one can find the value of spin only moment by applying the formula $\propto_s = \sqrt{n(n+2)}$ BM. From the knowledge of unpaired electrons and spin only moment it is possible to know two important aspects of complexes

(i) The valence state of the metal ion in an octahedral complex can be calculated.

(ii) If there are more than three 3d electrons, the nature of bonding in a complex may be known.

These two points may be understood by studying the following examples

(a) the magnetic moment of $Hg[Co(CNS)_4]$ is 4.5B.M indicating that cobalt is in the bivalent state (Co^{2+} with d^7)and the bonding belongs to spin-free type corresponding to n = 3

(b) The magnetic moment of $\left[Fe(diph)_{3.}\right](ClO_4)_3$ is 2.2 B.M. From this value it is evident that iron is in the trivalent state $\left(Fe^{3+} \text{ with } d^5 \text{ config}\right)$ and the bonding belongs to spin paired type which corresponds to n = 1

(c) In the complexes of iron (i) $\left[Fe(H_2O)_5 NO\right]^{2+}$ (ii) $\left[Fe(CN)_5)NO\right]^{3+}$ (iii) $\left[Fe(H_2O)_5 NO\right]^{3+}$ and (iv) $\left[Fe(CN)_5 NO\right]^{2+}$ the magnetic moments are different from those expected. In (i) and (ii) iron is in +2 oxidation state while in (iii) and (iv) it is in +3 oxidation state. The distribution of d-electrons in weak field and strong field complexes of iron (II & III) are as shown in Fig 14.9.

In weak field iron (II) and in weak or strong field iron (III), low energy t_{2g} orbitals are not completely filled. The unpaired electrons present in high energy anti bonding MO of NO molecule (Refer to Chapter 3) will be transferred to low energy vacant t_{2g} orbitals of iron in (i), (iii) and (iv) complexes due to which the number of unpaired electrons decreases to 3, 4 and 0 in (i) (iii) and (iv), complexes.

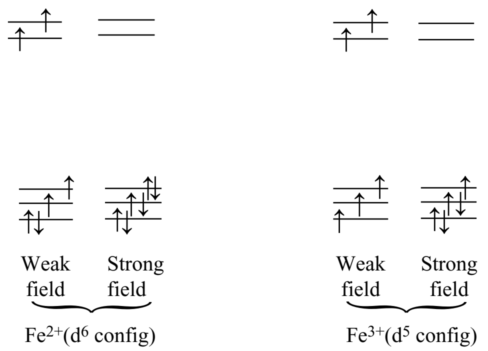


Fig 14.9 Distribution of electrons in Fe^{2+} and Fe^{3+} in weak and strong field complexes

Correspondingly the magnetic moments also decreases to 3.87 BM, 4.9 BM and 0 respectively in (i), (iii) and (iv) complexes. The oxidation states of iron also changes from +2 to +1 in (i) and +3 to +2 in (iii) and (iv). But in the case of complex (ii) since there is no vacancy in low energy t_{2g} orbitals electron transfer from NO to iron has to takes place into high energy e_g orbitals which is not possible. So the magnetic moment and oxidation state of iron does not change.

Recent studies on the kinetics and mechanism of the reaction of NO with Fe(II) (aq) and EPR and Mössbauer spectra of the reaction product $[Fe(H_2O)_5(NO)]^{2+}$, suggest that the complex ion is best described by the presence of high spin Fe(III) coordinated by NO^- i.e., as $[Fe^{III}(H_2O)_5(NO^-)]^{2+}$ The observed spin-state (S = 3/2) is explained on the basis of "antiferromagnotic coupling of Fe(III) with NO^- i.e., an arrangement in which the two unpaired electrons on NO^- have spins opposite to those of the five d-elecrons in iron(III). Displacement of a coordinated water molecule from $[Fe^{II}(H_2O)_6]^{2+}$ by NO in the rate - determining step is followed by a rapid intramolecular charge redistribution process leading to formal oxidation of Fe^{II} to $Fe^{III}$and reduction of NO to NO^-. The IR spectrum shows a single peak at 1810 cm^{-1}. This value is close to wave numbers found around 1777 cm^{-1} in the Fe^{III}-edta-NO complex, which is known to find as Fe^{III}-NO^-. The assignment is further supported by EPR and Mössbauer spctra.

Wanat A,; Schneppersiepr.T.; Stochel.G-, Van Eldik.R.; Bill.E.; Weighardt..K.; Inorganic Chemistry, 2002, 41, 4-10

3. Stabilization of complexes: The crystal field theory explains why certain oxidation states are preferentially stabilized by coordination to specific ligands. The electron configuration and spin states of an ion play vital role in the stabilization of oxidation states. The case of Co(III) stabilized by ligands like ethylenediamine may be taken as an illustration.

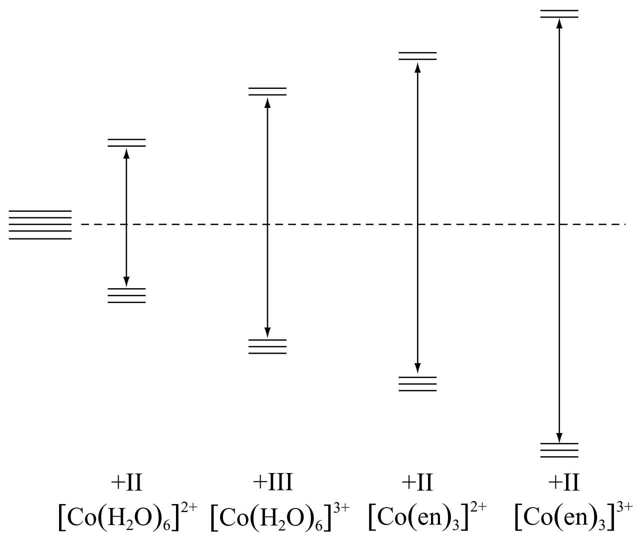


Fig 14.10 Relative crystal field splitting for Co(II) and Co(III) ions; not to scale

$[Co(H_2O)_6]^{2+}$ and $[Co(en)_3]^{2+:}$ $t_{2g}^5e_g^2$; CFSE relative spherical field is $-0.8\Delta_o$ or $-8Dq$; $[Co(H_2O)_6]^{3+}$ and $[Co(en)_3]^{2+}$; $t_{2g}^6e_g^0$; CFSE relative to spherical field = $-2.4\Delta_o$ or +24 Dq.

CFSE increases in strong field complexes due to increase in Δ and also due to increase in the number of electrons in low energy t_{2g} orbitals. This also increases the number of exchange energies. For example in Co(II) complexes with weak ligands have $-0.8\Delta_o$ and –5E (exchange energies) in t_{2g}^5 e_g^2 configuration, but increases to $-1.8\Delta_o$ and –6E in t_{2g}^6 e_g^1 configuration with strong ligands. The electron present in higher energy e_g (in strong field) can be removed easily when compared to remove an electron from lesser energy t_{2g} (in weak field). So Co^{2+} can be oxidized easily to Co^{3+} (in the presence to strong ligands such as CN^-, NH_3 etc. Therefore Co^{2+} is more stable in the presence of weak ligand like H_2O and unstable in the presence of strong ligands.

The stabilization of the higher oxidation state may thus be ascribed to (i) greater stabilization in the low spin state and also (ii) greater σ acceptor capacity of the metal in the higher oxidation state. However, this does not permit us to draw a generated conclusion regarding the nature of ligands stabilizing high oxidation states.

Stabilization of low oxiditation states of a metal implies that the valence electrons of the metal have been stabilized for some reason whatsoever. Thus (i) large saturated ligands like the iodide and sulphide which are good reducing agents (fluorides and oxides were poor reducing agents) are expected to stabilize low oxidation states; the difference in electrostatic energy between the higher and lower oxidation states in combination with these ligands are also expected to be small. (ii) Secondly, ligands with π-acceptor capacity provide a secondary mode for the "drainage" of electron density from the metal to the ligand, thereby stabilizing the low oxidation states. This is discussed later in the bonding with π-acceptor ligands such as CO, NO, PR_3, RNC etc.,

4. Stereochemistry of Complexes: The common geometries found in complex compounds are tetrahedral, square plannaer and octahedral. Unless opposed by severe steric or electronic factors, the octahedral coordination, using the largest number of available sigma bonding orbitals on the metal, is the most stable one. Such complexes are also favoured by higher CFSE than tetrahedral complexes. However, if we compare such trends in a series of ions where all other factors contributing to ΔH^o vary uniformly, the effect of the difference between the CFSE's may be appreciated to some extent.

CFSE in both tetrahedral and octahedral fields may be expressed in terms of Δ_o (using the relation $\Delta_t = \frac{4}{9}\Delta_o$). Thus for a d^1 ion in tetrahedral field (e^1), the CFSE is $\frac{3}{5}\Delta_t = \frac{3}{5}\times\frac{4}{9}\Delta_o = 0.266\ \Delta_o$. The values of CFSE thus obtained for all the d^0 to d^{10} configuration may be similarly calculated and are tabulated as follows.

If appears that the d^3 and d^8 (high spin) configurations will gain high CFSE on passing from tetrahedral to octahedral coordination over tetrahedral. In fact, tetrahedral chromium (III) complexes are virtually unknown.

The d^4 and d^9 configurations are next to gain a higher CFSE by adopting octahedral coordination, for example Mn(III) and Cu(II). However, these configuration gain further stabilization through distortion.

Ions with d^1, d^2, d^6 and d^7 configuration will gain little additional CFSE on passing from tetrahedral to octahedral coordination. Hence, these ions may be expected to form many tetrahedral complexes. Thus V(III) (d^2), unlike Cr(III) forms tetrahedral VX_4^- species (X = Cl, Br, I); Co(II) (d^7), unlike Ni(II) (d^8), similarly forms a large number of tetrahedral complexes $[CoX_4]^-$ or $[CoL_2X_2]$ with the unidentate ligands such as Cl, Br, I, SCN and OH^- and neutral ligands of group 15 donor atom. Ligand polarizability is an important factor determining which sterochemistry is adopted, the more polarizable ligands favouring the tetrahedral form since fewer of them are required to neutralize the metal's cationic charge. Thus if ligand is pyridine, replacement of Cl^- by I^- makes the stable form tetrahedral and if ligand is phosphine or arsine the tetrahedral form is favoured irrespective of X.

The most obvious distinction between the octahedral and tetrahedral compounds is that in general the former are pink to violet in colour whereas the latter are blue, as exemplified by the well-known equilibrium

$$\underset{\text{Pink}}{[Co(H_2O)_6]^{2+}} + 4Cl^- \rightleftharpoons \underset{\text{Blue}}{[CoCl_4]^{2-}} \cdot 6H_2O$$

Complexes of cobalt(II) are less common than those of cobalt (III) but, lacking any configuration comparable in stability with the t_{2g}^6 of Co^{3+} they show a greater diversity of types and are more labile. Even $[Co(H_2O)_6]^{3+}$ is low spin but it is such a powerful oxidizing agent that it is unstable in aqueous solution. Thus in the presence of weak ligands complexes of Co^{2+} are stable while in the presence of strong ligands complexes of Co^{3+} are stable. This is because of more CFSE associated with the t_{2g}^6 configuration in the presence of strong ligands.

d^0, d^5 (high spin) and d^{10} configurations have no crystal field stabilization either in tetrahedral or octahedral environment.

We have seen how the degeneracy of the five d- orbitals is removed in a crystal field to provide greater stability by way of crystal field stabilition energy (CFSE). Further removal of the degeneracy of the d-orbitals can some times occur due to unsymmetrical electron distribution in t_{2g} and e_g levels resulting in additional stabilization of the system.

	d^1	d^2	d^3	d^4	d^5	d^6	d^7	d^8	d^9	d^{10}
CFSE Δ_o	0.4	0.8	1.2	0.6	0	0.4	0.8	1.2	0.6	0
CFSE Δ_t	0.6	1.2	0.8	0.4	0	0.6	1.2	0.8	0.4	0
$\Delta_t = \frac{4}{9}\Delta_o$	0.267	0.533	0.356	0.178	0	0.267	0.533	0.356	0.178	0

This leads to the distortion in the perfectly symmetric geometrics. This is knows as Jahn-Teller distortion.

For example consider $[Cu(H_2O)_6]^{2+}$ ion. In an octahedral field, the d^9 ion has an electron distribution $t^6_{2g}\ e_g^3$. Since the e_g levels are degenerate, it is possible to arrange the electrons in two alternate ways.

Case-I: $d^1_{x^2-y^2} d^2_{z^2}$, The ligands along the x and y axes are screened only by one electron from the electrostatic attraction of Cu^{2+} ion than the two ligands on the z axis by two electrons. Hence the four ligands in the xy plane will be drawn more closely to the cation than the other two on the z-axis. This gives rise to a tetragonal distortion of the octahedron which is elongated along the z-axis (z-out).

Case-II: $d^2_{x^2-y^2} d^1_{z^2}$, The four ligands in the xy plane will be more screened from the electrostatic attraction of the Cu^{2+} ion by the two electrons than the two ligands on the z-axis by one electron. Hence two axial ligands will be drawn more closely than the other four and we shall get a tetragonally distorted octahedron with contraction along z-axis i.e., flattened (z-in). For copper(II) compounds, most experimental data indicate an elongated octahedron with two long and four short distances.

Following the same line of argument, we may expect Jahn-Teller distortion in complexes of metal ions having unsymmetrical electron distribution in e_g or t_{2g} level. Since the e_g group of orbitals lie directly along the lines of approach of the ligands in an octahedral field, the distortion is much pronounced with unsymmetrical electron occupancy of the e_g level. Thus the Jahn-Teller distortions with ions of the following configuration in addition to the $t^6_{2g}\ e^3_g$ configuration discussed above may be expected.

$t^3_{2g}\ e^1_g$: High Spin Cr(II) and Mn(III)

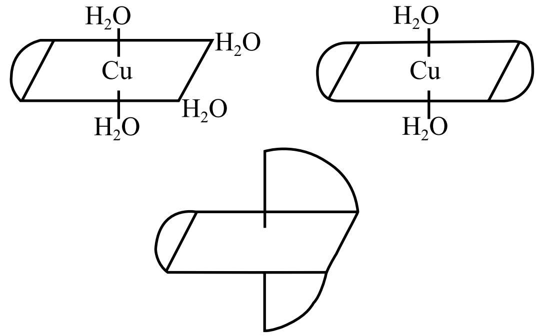
$t^6_{2g}\ e_g^1$: Low Spin Co(II) and Ni(III)

Except with low spin Co(II) there are ample evidences to confirm the expected distortion. Indeed some Cu(II) compounds undergo such a high amount of distortion that the coordination around the metal appears virtually square planar.

Jahn-Teller distortions may also be expected when an octahedral complex has 1, 2, 4 or 5 electrons in the t_{2g} level. Since these orbitals are not oriented directly along the lines of approach of the ligands, the spliting of these orbitals consequent to distortion would be naturally very small and difficult to detect.

In chelated compounds or ions susceptible for Jahn-Teller distortion, the effective span between coordinating atoms may oppose elongation and the consequent distortion of the complex. For example the trichelate complex $[Cu(en)_3]^{2+}$ is extremely unstable.

This may be related to the strain developed in the complex when it tends to elongate the axial bonds due to



Jahn-Teller effect. This causes sretching the chelate ligand joining, axial positions with equatorial positions in the coordination octahedron, thereby straining the molecule.

The crystal field theory explains that copper(II) forms square planar complexes than tetrahedral or octahedral complexes. The reason for this is that copper(II) possesses a much higher CFSE (–12.28 Dq) in a square planar configuration than octahedral (CFSE = –6 Dq) or tetrahedral configuration (CFSE = –2.56 Dq).

The CFSE also explains that the square planar complexes of 4d and 5d metal ions form more readily than 3d ions.

Failures of Crystal Field Theory

(i) In CFT model metal – ligand bonding is taken to purely electrostatic which is quite unrealistic. It provides no account of partial covalent character of the metal – ligand bonds.

(ii) The theory cannot explain for the relative strengths of ligands. For example it provides no account why negative OH^- has smaller splitting effect than neutral H_2O.

(iii) CFT considers only the metal ion d-orbitals without giving any consideration to the other metal orbitals such as s, p_x p_y or p_z and ligand π – orbitals.

(iv) CFT is unable to consider the possibility of the existence of π-bonding in complexes despite of its frequent existence in complexes of metals in low or high oxidation.

(v) CFT keeps the total emphasis on the metal orbital without consideration to the ligand orbitals therefore all properties dependent upon the ligand orbitals and their interactions with metal orbitals are not at all explained.

14.6 TYPES OF LIGANDS

Any atom, ion or molecule which is capable of donating a pair of electrons to the metal atom is called a coordinating group or ligand. In a ligand, the particular atom which actually donates the electron pair is called donor atom.

4.6.1 Classification Based on the Number of Donor Atoms Present in the Ligands

Ligands may be classified according to the number of electron pairs donated or the number of positions they occupy around the metal ion. As per the number of electron pairs donated they are named as mono, bi, tri etc., dentate ligands.

(a) Monodentate ligands: If a ligand contains only one atom i.e., it is capable of forming only one coordinate bond to the central metal atom, it is known as monodentate or unidentate ligand eg. Cl^-, Br^-, H_2O, NH_3 etc.

(b) Polydentate ligands: When a ligand have two or more donor atoms which may simultaneously co-ordinate to a metal atom, it is termed as a multi or polydentate ligand. Depending upon the number of donor sites, these ligands may be referred to as bi-dentate (two donor atoms), tridentate (three donor atoms), tetra- dentate (four donor atoms), pentadentate (five donor atoms), hexadentate (six donor atoms) and so on.

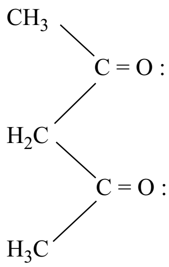
1. Bidentate ligands: Certain examples for bidentate ligands their molecular formula, their abbreviations and their IUPAC names are given below

$$\begin{array}{l} CH_2-\ddot{N}H_2 \\ | \\ CH_2-\ddot{N}H_2 \end{array}$$

Ethylene diamine
(en)
1, 2 – ethane–diamine.

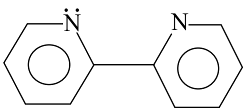
$CH_3\,CO\,CH_2\,CO\,CH_3$
Acetyl acetone
(ac ac)
2, 4– Pentanedione

or

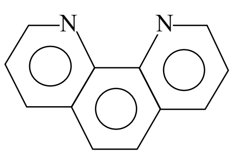


$$\begin{array}{l} CH_3-C=N-OH \\ \quad\ \ | \\ CH_3-C=N-OH \end{array}$$

Dimethyl glyoxime (DMG)
Butane dioxime



2, 2 - bipyridine
(bipy)
2, 2 - bipyridine



1, 10 - Phenanthroline
or phenanthroline
(phen, o-phen)1,10 – diaminophenanthroline

$$\begin{array}{l} COO^- \\ | \\ COO^- \end{array}$$

Oxalate
(ox)

Bidentate ligands are again two types (a) symmetric bidentate ligands and (b) asymmetric bidentate ligands.

If both coordinating atoms in a bidentate ligand are same as given in above examples, they are known as symmetric bidentate ligand, but if the two coordinating atoms in a bidentate ligand are different it is called asymmetric bidentate ligand.

$$\begin{array}{l} CH_2\text{-}NH_2 \\ | \\ CH_2\text{-}NH_2 \end{array} \qquad \begin{array}{l} CH_2\text{-}NH_2 \\ | \\ CO\text{-}O^-: \end{array}$$

Symmetric bidentate ligand — Unsymmetric bidentate ligand
(e.g., given is glycinate ion)

The symmetric and unsymmetric bidentate ligands are usually represented as (AA) and (AB) respectively where A and B denote the donor atoms.

(ii) Tridentate ligands contain 3 donor atoms e.g.,

$NH_2\,CH_2\,CH_2\,NH\,CH_2\,CH_2\,NH_2$
Diethylene triamine
(dien)
2,2-diaminodiethyl amine
or 1,4,7-triazaheptane

$$\begin{array}{l} CH_2\text{-}NH_2 \\ | \\ CH\text{-}NH_2 \\ | \\ CH_2\text{-}NH_2 \end{array}$$

1,2,3 propane triamine

(iii) Tetradentate ligands contain 4 donor atoms e.g.,

$NH_2\,CH_2\,CH_2\,NH\,CH_2\,CH_2\,NH\,CH_2\,CH_2\,NH_2$
Triethylene tetraamine (trien)
1, 4, 7, 10 – tetraazadecane (IUPAC name)

$$\begin{array}{l} NH_2CH_2\,CH_2\,N\,CH_2\,CH_2\,NH_2 \\ \qquad\qquad\quad | \\ \qquad\qquad CH_2\,CH_2\,NH_2 \end{array}$$

β, β′, β″ – *triaminotriethylamine (tren)*
β, β′, βo″ – *tris*(2 - *aminoethyl) amine. (IUPAC name)*

(iv) Pentadentate ligands contain 5 donor atoms e.g.,

$NH_2CH_2CH_2NHCH_2CH_2NHCH_2CH_2NHCH_2CH_2NH_2$

Tetraethylene pentamine

1, 4, 7, 10, 13 pentaazatridecane (IUPAC name)

(v) Hexadentate ligands contain 6 donor atoms e.g.,

$$\begin{array}{ccc} {}^-OOC\,H_2C & & CH_2\,COO^- \\ & NCH_2\,CH_2N & \\ {}^-OOC\,H_2C & & CH_2\,COO^- \end{array}$$

Ethylenediaminetetraacetate (EDTA)

1, 2-ethanediyl (dinitrilo) tetraacetate

Flexidentate Character of Polydentate Ligands

Polydentate ligands are said to have flexidentate character if they do not use all its donor atoms to get coordinated to the metal ion. An interesting example of such polydentate ligands is ethylene diamine tetraacetic acid. This ligand generally acts as a hexadentate ligand but it acts as a pentadentate ligand e.g., $[Cr^{III}(HEDTA)(OH)]^-$ $[Cr^{III}(HEDTA)(Br)]^-$ and as tetradentate ligand e.g.,

[$Pd^{II}(H_2EDTA)$]. Another polydentate ligand having flexidentate character is sulphate ion which acts as a monodentate e.g., $[Co^{III}(NH_3)_5\ SO_4]^+$ and bidentate e.g., $[Co^{III}(en)_2\ SO_4]^+$ respectively.

4.6.2 Types of Monodentate Ligands

Monodentate ligands may be further classified basing on the charge they carry

(i) Neutral monodentate ligands: Examples of these are H_2O, NH_3, CO, CS, NO, etc.

(ii) Positive monodentate ligands: Examples are NO^+. H_2N. N^+H_3 etc.

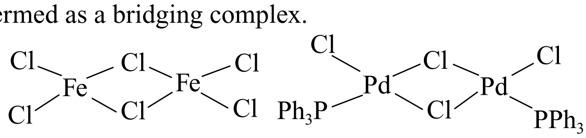
(iii) Negative monodentate ligands: Examples are F^-, Cl^-, OH^-, CN^-, SCN^-, NO_2^- etc.

Ambidentate ligands: Certain ligands contain two or more donor atoms but in forming complexes they use only one donor atom to attach themselves to the metal ion at a given time. Such ligands are known as ambidentate ligands e.g.,

Ambidentate ligands		Donor atoms
Thiosulphate ion	$S_2O_3^-$	S or O
Selenocyanide ion	$SeCN^-$	Se or N
Cyanate ion	NCO^-	N or O
Thiocyanate ion	SCN^-	S or N
Cyanide ion	CN^-	C or N
Nitrite ion	NO_2^-	N or O

π - electron donor ligands: Unsaturated organic compounds like alkenes or alkynes can donate the electron pair in their **π-bonds**. They do not contain lone pairs e.g., $CH_2 = CH_2$, $HC \equiv CH$, benzene etc.

Bridged ligands: in some cases monodentate ligands may simultaneously coordinate with two or more metal atoms under these circumstances the ligands acts as a bridge between different metal atoms and is therefore called as bridging ligands and the resulting compound is termed as a bridging complex.



Some examples of bridged ligands are

$OH^-, F^-, Cl^-, NH_2^-, O_2^-, SO_4^{2-}, CO, etc$

4.6.3 Classification Based on π-Donor and π-Acceptor Property of the Ligand

π-donor ligand: π-donor ligand contains extra lone pairs present in orbitals of π-symmetry around M-L axis e.g., Cl^-, Br^-, OH^-, H_2O etc. In lewis acid – base terminology, π–donor ligands are π-bases. These will form a π–dative bond from ligand to metal $(M \rightrightarrows L)$ extra to the normal σ dative bond. Due to this, π-donor ligands decreases Δ_0 (this can be explained basing on MO theory which is beyond the scope of this book.)

π-acceptor ligand: A π-acceptor ligand is a ligand that has vacant π orbitals that are available for occupation. In lewis acid–base terminology, a π-acceptor ligand is a π-acid. Typically the π-acceptor orbitals are vacant antibonding orbitals on the ligand (usually the LUMO) as in CO, N_2 which are higher in energy than the metal d-orbitals. The π orbital of CO, for instance has the correct symmetry for overlap with the metal t_{2g} orbitals, so CO can act as a π-acceptor ligand. Phosphines (PR_3) are also able to accept π-electron density and also act as π acceptors. The π-acceptor ligands increase Δ_0.

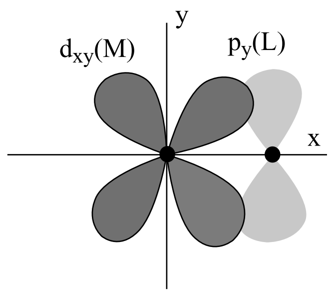


Fig 14.11 The overlap that may occur between a ligand p-orbital perpendicular to the M-L axis and a metal d_{xy} orbital

Increasing order of Δ_0.

π-donor < weak π donor < no effects < π acceptor

Representative ligands that match these classes are

π donor	weak π donor	No effects	π acceptor
I^-, Br^-, Cl^-, F^-	H_2O	NH_3	PR_3, CO

Notable examples where the effect of σ bonding dominates include amines NR_3, CH^-_3 and H^-, none of which has orbitals of symmetry of an appropriate energy and thus there are neither π-donor nor π-acceptor ligands.

The π-acceptor ligands are weak σ donor to Lewis acids such as BF_3 and $AlCl_3$, forms strong complexes with transition elements, since the drift of π-electron density from M to C tends to make the ligands more negative and so enhances its σ donor power. This is known as synergic effect. The preexisting negative charge on CN^- increases its σ donor propensity but weaken its effectiveness as a π acceptor. It is thus possible to rationalize many chemical observations by noting that effectiveness as a π donor decreases in the sequence $CN^- > RNC > \overset{+}{N}O \approx CO$ where as$_+\pi$ effectiveness as acceptor follows the reverse sequence NO > CO >> RNC > CN^-. By implication, back donation into antibonding CO orbital weakens the CO bond and this manifest in the slight increase in inter atomic distance from 112.8 pm in free CO to ~ 115 pm in many complexes.

14.7 BONDING IN ORGANOMETALLIC COMPOUNDS

Organo metallic compounds are those compounds which contain one or more metal – carbon bonds. It may be noted that all the compounds containing carbon and a metal atom are not organo metallic. We must use this term for compounds which contain at least one M-C bond. For example sodium ethoxide or sodium acetate can not be considered as organometallic compounds because in these compounds sodium is in bond with oxygen but not with carbon (C_2H_5ONa and CH_3COONa). On the other hand the compounds n-butyl lithium, tetra ethyl lead, trialkyl aluminium etc are called organo metallic & because in these compounds there is metal-carbon bond.

Historically, the first organometallic compound of a d-block element (platinum) was prepared by W.C. Zeise. In 1927, which is also popularly known as Ziege's-salt. Now a large number of oraganometallic compounds of d-and f block elements are known.

Ziese's salt $K[PtCl_3(C_2H_4)]$ is as shown below

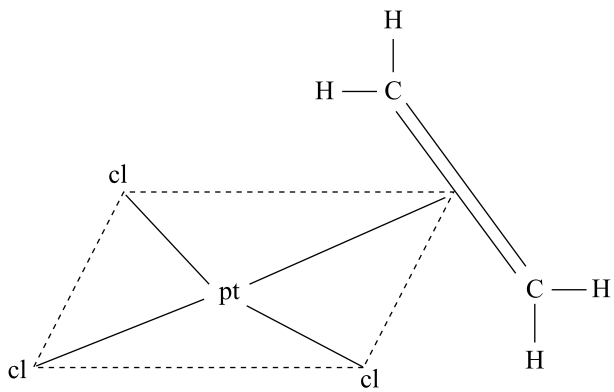


Fig 14.12 Structure of zeise's salt

In this structure the three Cl atoms and the middle point of ethylene double bond form a square plane. The platinum atom is present in the centre of the square and C=C double bond of ethylene molecule is perpendicular to plane containing Pt and Cl atoms.

It is written as $K[Pt\ Cl_3\ (\eta^2\text{-}C_2H_4)]$. In these cases the number of carbon atoms bound to the metal (Hapticity) is indicated by the Greek letter η (eta) followed by a number. In this case Pt is bound to two carbon atoms (dihapto) and therefore, it is written as η^2

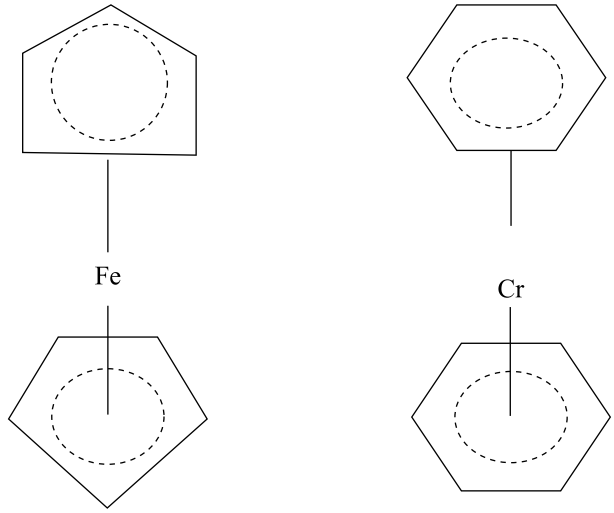


Fig 14.13 Structures (a) Ferrocene (b) dibenzene chromium

Ferrocene or bis (η^5-cyclopentadienyl) iron(II), $[Fe(\eta^5\text{-}C_5H_5)_2]$ (cyclopentadiene is pentahapto ligand. Its structure is given in Fig 14.13.

The structure of ferrocene is regarded as sandwich structure in which iron atom is sandwiched between two C_5H_5 organic rings. The planes of the rings are parallel so that all the carbon atoms are at the same distance from the iron atom.

The discovery of ferrocene was a land mark in the advancement of modern organometallic chemistry. For explaining its stability, structure and bonding, the new concept of π–bonding of carbocyclic rings to the metal atom was involved. This also lead to the preparation of a large number of organometallic compounds. These compounds are collectively called metallocenes.

14.7.1 Metal Carbonyls

Metal carbonyl compounds are also considered as organometallic compounds in which carbon monoxide (monohapto) acts as ligand. All the carbonyl compounds are homoleptic complexes.(which contain only one type of ligands). Some of these are $[V(CO)_6]^-$, $[Cr(CO)_6]$, $[Mo(CO)_6]$, $[W(CO)_6]$, $[Mn_2(CO)_{10}]$,$[Fe(CO)_5]$ $[Fe_2(CO)_9]$, $[Co_2(CO)_8]$,$[Ni(CO)_4]$ etc

Structure of Metal Carbonyls

Homoleptic binary metal carbonyls have simple well defined structures. Some of these have simple structures for example hexa carbonyl chromium (0) is octahedral, penta carbonyl iron (0) is trigonal bipyramidal, tetra carbonyl nickel (0) is tetrahedral. In all these cases the effective atomic number is equal to their nearest inert gas configuration. Some metals form bridge carbonyls. Some of the bridged metal carbonyls contain metal-metal bond, some may not. This depends on the total number of electrons around the central metal atom.

If the total number of electrons around metal atom i.e EAN is equal to nearest inert gas configuration metal –metal bond does not takes place in bridged (binuclear) complexes. If the EAN is not equal to nearest inert gas configuration then metal-metal bond takes place in the binuclear carbonyls but for calculation of EAN only one electron of metal-metal bond should be considered. For example in $Co_2(CO)_8$ the EAN is 35 calculated as follows

At. No of Co	=	27
No of electrons gained by coordination by 3CO molecules 3×2	=	6
No of electrons contributed by the bridging CO molecule one from each CO	=	2
EAN		35

In $Fe_2(CO)_9$ the EAN is

At. No of iron	=	26
No of electrons gained by coordination by 3CO molecules 3×2	=	6
No of electrons contributed by the bridging CO molecules one from each CO	=	3
EAN	=	35

So in $[Co_2(CO)_8]$ $Fe_2(CO)_9$ there is Co-Co, Fe-Fe bond and by adding one electron from this bond to the above gives the nearest inert gas configuration 36. The structures of some metal carbonyls are given in Fig 14.14.

Bonding in Metal Carbonyls

The molecular orbital diagram of CO molecules is shown in Fig 14.15. The highest occupied orbital is on carbon which is close in energy, to the carbon p-orbitals, so it is said to have great deal of carbon p-character with it high energy, the carbon lone pair is the pair donated in forming dative bonds with other speices like metals in metal carbonyls. The empty $\overset{*}{\pi}$ orbitals are more localized (polarized) on carbon.

When CO acts as a ligands the highest occupied molecular orbital of carbon donates the lone pair in forming dative bond. This orbital has less donating capacity due to the more electronegative character of oxygen which

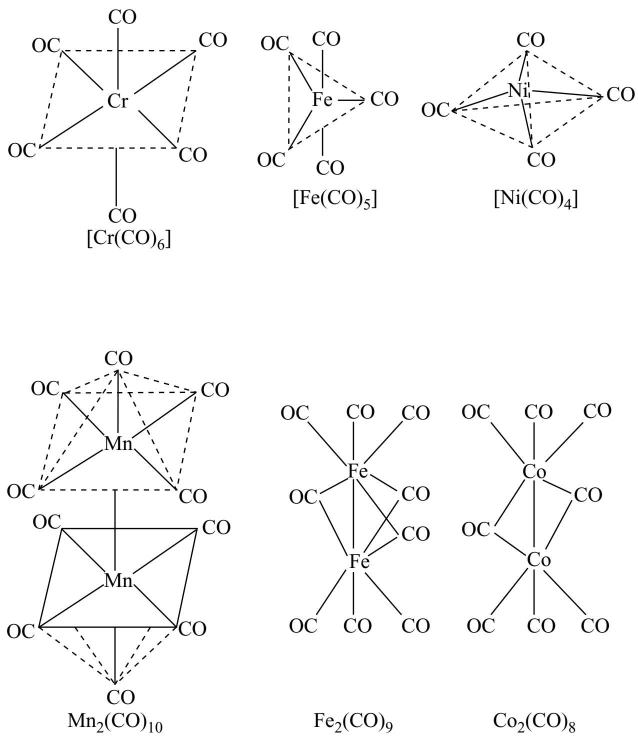


Fig 14.14 Structurres of some metal carbonyls

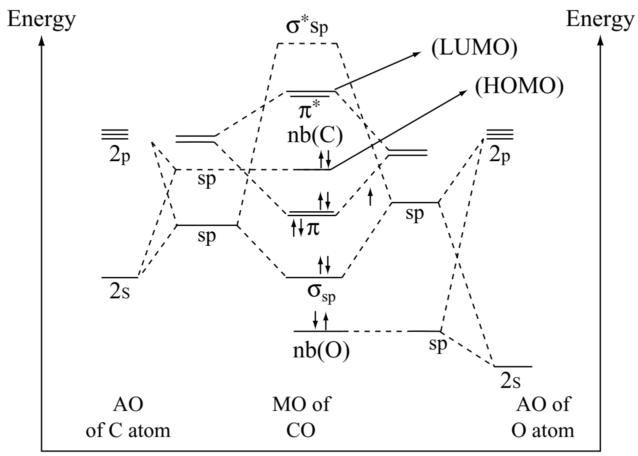


Fig 14.15 Molecular orbital diagram of CO

pulls the electron density towards it. So M ← CO bond is weak. Now there is a π –overlap involving donation of electrons from filled metal d-orbitals into vacant antibonding π molecular orbitals of CO (LUMO). This results in the formation of M → C π bond. This is also called back donation or back bonding and the CO is called π -acceptor ligand. Due to this the electron density at carbon increases and thus the donation in M ← C bond increases and strengthens the bond between CO and the metal. This effect is known as **synergic effect** because the drift of metal electrons into CO orbital will tend to make the CO as a whole negative and hence to increases its basicity via the σ orbital of carbon, at the same time the drift of electrons to the metal in the σ bond tends to make the CO positive, thus enhancing the strength of the π-orbitals. Thus upto a point the effects of σ bond formation strengthen the π bonding and vice versa. The bonding in metal carbonyls is shown in Fig 14.16. In these figures, the shaded orbitals represents filled orbitals.

According to the above discussion of bonding, as the extent of back bonding from M to antibonding MO of CO increases the M-C bond becomes stronger and the C-O bond becomes weaker due to decrease in bond order in C-O. Thus the multiple bonding should be evidenced by shorter M-C and longer C-O bonds. The bond length in CO itself is 112.8pm while the bond lengths in metal carbonyl molecules are 115pm. Moreover, when changes are made, that should increase the extent of M–C back bonding, the C-O bond length increases more. Thus if some CO groups are replaced by ligands with low or negligible back accepting ability, those CO groups that remain must accept dπ electrons from the metal to a greater extent in order to prevent the accumulation of charge on the metal atom. Thus the C-O bond length increases for $[Cr(CO)_6]$ when three CO ligands are replaced by amine groups which have essentially no ability to back – accept. Similarly when we go from $Cr(CO)_6$ to the isoelectronic $[V(CO)_6]^-$, when more negative charge must be taken from the metal atom, more back bonding takes place. So C-O bond length in $[V(CO)_6]^-$ will be more than in $[Cr(CO)_6]$. Conversely a change that would to inhibit the shift of electrons from the metal to CO π orbitals such as placing a positive charge on the metal should cause the less back bonding due to which the decrease in C-O bond order is also less and thus C-O bond length is less. With increase in number of positive charges on the metal ion back bonding decreases thus the back bonding in $[Fe(CO)_6]^{2+}$ is less than in $[Mn(CO)_6]^+$. So the C-O bond length in $[Fe(CO)_6]^{2+}$ is shorter than in $[Mn(CO)_6]^+$.

In the free CO, the electrons are polarized towards the more electronegative oxygen. For example the electrons in the π orbitals are concentrated nearer to the oxygen atom than to carbon. The presence of transition metal cation tends to reduce the polarization in the C-O bond by attracting the bonding electrons

$$C^{\delta+} \equiv O^{\delta-} \quad M^{n+} \leftarrow C^{\delta+} \equiv O^{\delta-}$$

The consequence is that the electrons in the positively charged complex are more equally shared by the carbon and the oxygen giving rise to a stronger bond and lesser C-O bond length (since back donation from M^{n+} is less).

For a molecule of the type $[M(CO)_6]$ having three pairs in d-orbitals of metals, the number of lone pairs on metal available for back bonding to each CO ligand is only ½ i.e., ½ of a π bond for each M-C pair, if the dπ electrons are fully used. If three ligands (L) which are incapable of accepting dπ electrons, incorporated into the complex each of the three CO group has access to 1/3 of the three pairs (i.e one lone pair for each M-C pair of dπ electrons. Thus if the dπ electrons enter fully into the back bonding, each M-C pair will have a full π-bond. Thus if we assume full participation by the metal dπ electrons in each case the M-C bond order go from 1.5 in $M(CO)_6$ to 2.0 in cis $ML_3(CO)_3$ where L is a ligand without π acceptor orbitals e.g., aliphatic amino nitrogen atom. The C-O bond orders in these two cases should then be respectively 2.5 and 2.0. So the C-O bond length in $ML_3(CO)_3$ is longer than in $M(CO)_6$.

From the above discussion it can be easily understood that with increase in the number of dπ electron pairs of metal the C-O bond order decreases and the C-O bond length increases. For example $[Cr(CO)_6]$, $[Fe(CO)_5]$ and

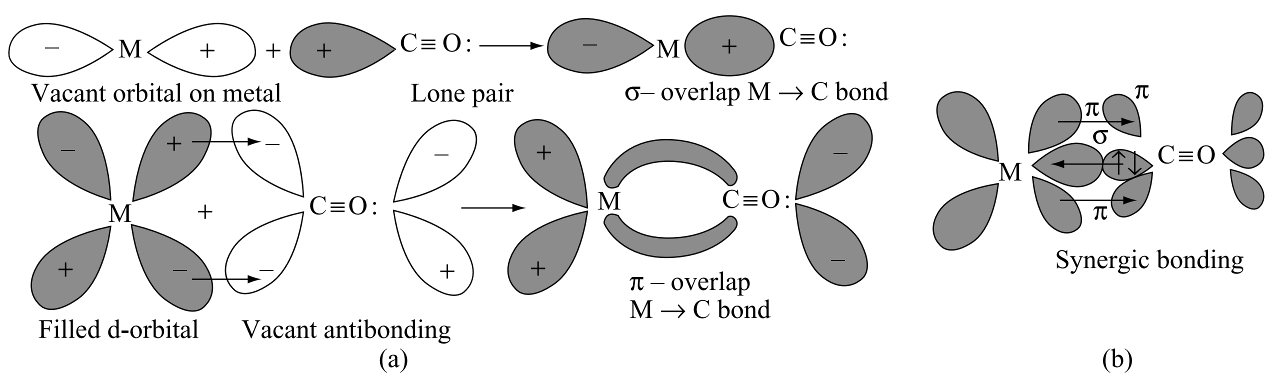


Fig 14.16 (a) Bonding in metal carbonyls (b) Synergic effect in bonding of metal carbonyls

$[Ni(CO)_4]$ the dπ electron pairs available are 3,4 and 5 for back bonding in Cr, Fe and Ni respectively, but the number of CO ligands that can accept these electron pairs is 6, 5 and 4 in $[Cr(CO)_6]$ $[Fe(CO)_5]$ and $[Ni(CO)_4]$. So the share of dπ electrons from Cr, Fe and Ni for back donation to each CO in these complexes will be 3/6, 4/5 and 5/4. Hence the C-O bond length in these complexes will be in the order

$$Cr(CO)_6 < Fe(CO)_5 < Ni(CO)_4$$

The vibrational stretching frequency of the CO bond υ_{co} is directly related to its bond order. The vibrational stretching frequency of a group is written as υ_{group}. If the bond order i.e., bond strength of a group is more, its IR absorption band due to the stretching vibration of that group would occur at higher frequency. Thus the IR spectra can serve as an excellent tool to determine the bond order of CO and thereby study the π back bonding in metal carbonyls. IR absorption for free CO occurs at about 2250 cm^{-1} whereas the absorption for ligated CO occurs at a considerably lower frequency between 2220 – 1700 cm^{-1}. This establishes a lower bond order or higher bond length for ligated CO this is due to transfer of electronic charge from metal to π orbitals of CO (M$\rightarrow$L back bonding). The IR absorption data for some carbonyls is given in Table 14.9.

Table 14.9 IR absorption data of some carbonyls

Complex	υ_{CO} cm^{-1}	Complex	υ_{CO} cm^{-1}
$[Ti(CO)_6]^{2-}$	1748	$[Fe(CO)_4]^{2-}$	1790
$[V(CO)_6]^{-}$	1859	$[Co(CO)_4]^{-}$	1890
$[Cr(CO)_6]$	2000	$[Ni(CO)_4]$	2060
$[Mn(CO)_6]^{+}$	2100		
$[Fe(CO)_6]^{2+}$	2204		

Table 14.10 Bond orders in metal carbonyls determined from spectroscopic data

Metal carbonyl	M-C Bond order	C-O Bond order
$Ni(CO)_4$	1.33	2.64
$[Co(CO)_4]^{-}$	1.89	2.14
$[Fe(CO)_4]^{2-}$	2.16	1.85

In the discussion made so far we have considered the carbonyls (complexes in which only CO is acting as ligands and mixed carbonyls in which some CO ligands are replaced by electron donating ligand without π-acceptor character. If in the mixed carbonyl, the other ligand is also π acceptor, it would compete with the CO ligands for gaining the metal dπ electron charge. If that ligand is a better π-acceptor than CO, most of the metal dπ electron charge would be utilized for π-back bonding with that other ligand and very little dπ electron charge would get transferred to antibonding π^* orbital's of CO. This results in less decrease in CO bond order and hence more bond stretching frequency. The reverse would happen if the other ligand is a poorer π acceptor character than CO. For example in $[(CH_3)_3P\ Ni(CO)_3]$, $(CH_3)_3P$ is a weaker π-acceptor than F_3P in $[F_3P\ Ni(CO)_3]$ so the C–O bond stretching frequency in $[F_3P\ Ni(CO)_3]$ is higher (1990 cm^{-1}) than in $[(CH_3)_3P\ Ni(CO)_3]$ (1980 cm^{-1}).

There are several compounds of trivalent phosphorous and arsenic which can act as π-acceptor ligands. This is because both P and As have vacant d-orbitals which are of proper symmetry to overlap with the d-orbitals of the central metal atom. The overlap facilitates shifting of the electron charge from the metal to the ligand through π back bonding.

The availability of d-orbitals of P or As to the d-orbitals of the metal is however, influenced by the electronegativity of the group attached to P or As. For example in PF_3, F being highly electronegative, would polarize the sigma electron charge cloud from P towards to F. Thereby reducing σ donor character of P. However the decrease in electron density at P by F atoms makes it stronger π - acceptor ligand. But in the case of $P(C_2H_5)_3$, this effect is just opposite. Hence $[Ni(PF_3)_4]$, $[Ni(PF_3)_3(CO)]$, $[Ni(PF_3)_2(CO)_2]$…… $Ni(CO)_4$ can be easily prepared. But it is not possible to get beyond the $\{Ni[P(C_2H_5)_3]_2(CO)_2\}$. The IR absorption frequencies of the vibration C-O bond stretching are given in Table 14.11.

Table 14.11 IR Absorption frequencies of some mixed carbonyls

Metal carbonyl	IR Absorption frequency υ_{Co} cm^{-1}
$Mo(CO)_6$	2050, 1900
Cis $(Py)_3Mo(CO)_3$	1883, 1723
Cis $(Ph_3P)_3Mo(CO)_3$	1890, 1730
Cis $(Ph_3As)_3Mo(CO)_3$	1934, 1835
Cis $(Ph_2PCl)_3Mo(CO)_3$	1980, 1880
Cis $(Ph_3PCl_2)_3Mo(CO)_3$	2010, 1890
Cis $(PCl_3)_3Mo(CO)_3$	2030, 1910

It should be noted that υ_{CO} is proportional to the CO bond order and that any increase in M-C bond order due to π bonding would lead to an equivalent decrease in the CO bond order or vice versa.

It can be seen from the table 14.11 the unsubstituted $Mo(CO)_6$ absorb around 2050 and 1900 cm^{-1}. If three CO groups are substituted by a better σ donor ligand diene which is incapable of forming π bonds with the metal due to lack of d-orbitals in N atoms, the availability of electron pairs to the remaining CO groups on metal is more resulting in the back donation increases for each CO group. So the extent of π back bonding in M-C is more in [Mo(diene) $(CO)_3$] than in $Mo(CO)_6$ the bond order of M-C is, therefore, higher and the bond order of CO is lower in the former than in the latter carbonyl.

Pyridine (Py) is a weaker σ donor, than diene and has π orbitals to accept dπ electron charge from the metal. Therefore, although the M-C bond order is higher in $[Mo(Py)_3 (CO)_3]$ than in $Mo(CO)_6$, it is lower than the bond of M-C in [Mo(diene) $(CO)_3$]. Similarly the CO bond order is lower in $[Mo(Py)_3 (CO)_3]$ than in $Mo(CO)_6$ but is higher than the bond order of CO in [Mo(diene) $(CO)_3$]. These arguments are supported by the IR absorption frequencies of CO in $[Mo(Py)_3 (CO)_3]$, $Mo(CO)_6$ and [Mo(diene) $(CO)_3$].

Phosphines and arsines having the same group attached to P and As, cause approximately the same extent of increase in M-C bond order and similar extent of decrease in CO bond order indicating that π acceptor capabilities of these ligands are of similar magnitude.

If a ligand such as Ph_3P, which is a weaker π acceptor than CO, partially replaces the COs from a binary carbonyl complex, the remaining COs present in the carbonyl get more dπ electron charge density in their π orbitals hence, the CO bond order in such mixed carbonyls would be lower than in the corresponding pure carbonyls. This is evident from the relative positions of υ_{CO} observed for such carbonyls Table 14.11.

If the Ph groups of Ph_3P are substituted by more electron attracting Cl groups the υ_{CO} in the resulting mixed carbonyls goes on increasing Table 14.11. This is due to the fact that the presence of electron – attracting groups in the phosphine ligand would facilitate the transfer of dπ electrons from the metal to P so that more and more of dπ electrons charge is used up in M-P back bonding and less and less of dπ electron charge is available for M-C back bonding in the mixed carbonyl with each replacement of Ph of triphenyl phosphine with Cl. Thus the CO bond order in mixed phosphine carbonyls varies in the order

$[(Ph_3P)_3\ Mo(CO)_3] < [(Ph_2PCl)_3\ Mo(CO)_3] < [(PhPCl_2)_3\ Mo(CO)_3] < [(PCl_3)_3\ Mo(CO)_3]$

Attempts have been made to arrange PX_3, AsX_3, SbX_3 ligands in the order of their π-accepting tendency using C-O bond stretching frequency in $LNi(CO)_3$. Based on the stretching frequencies the π-acceptor tendency of common phosphorus ligands is in the order.

$PF_3 > PCl_3 > P(OAr)_3 > P(OR)_3 > P(Ar)_3 > PR(Ar)_2 > PR_2Ar > PR_3$

Isocyanide (RNC) group is another π-acceptor ligand similar to CO since N:R is isoelectronic with O. thus an isocyanide can enter similar type of σ - π synergic bonding as it happened in CO. However, the extent of π bonding is less in isocyanide complexes than in metal carbonyls as is evinced by IR absorption spectra. υ_{NC} decreases upto about 100 cm^{-1} in ligated isocyanides than in free isocyanides. Isocyanides are stronger π donors than CO and in several isocyanide complexes, the π bonding is of minor importance compared to σ bonding as in $[Ag(CNR)_4]^+$ $[Fe(CNR)_6]^{2+}$ and $[Mn(CNR)_6]^{3+}$. However the isocyanide ligands are capable of extensive back acceptance of π electrons from the metal atoms in their lower oxidation states. This is indicated by their ability to form complexes such as $Cr(CNR)_6$ and $Ni(CNR)_4$ in which both Cr and Ni are in zero oxidation state.

The worth study of the another π-acceptor ligand is nitric oxide (NO) which is very similar to CO molecule in forming complexes with transition metals. However, there is one major difference in their structures is that NO molecule contain one additional electron in its π molecular orbital. The odd electron in π molecular orbital of NO molecule (bond order 2.5) is quite readily lost to produce NO^+ (bond order 3) this is evinced by the IR spectra of free NO(υ_{NO} ~ 1800 cm^{-1}) with those of salts containing NO^+ ion υ_{NO^+} = 2200-2300 cm^{-1} large number of complexes of NO are reported to have NO^+ ion (nitrosonium ion) as the ligand which appears to act as a 3-electron donor. Most of such complexes obey EAN or inert gas rule if we assume that NO ligand is a 3-electron donor. These nitrosonium complexes are formed in the following steps.

1. Initially during its complexation with the metal, NO loses one π electron to the metal reducing its valence by one unit and getting itself converted to NO^+

$$M + NO \rightarrow M^- + :N \equiv \overset{+}{O}:$$

2. In the second step, NO^+ which is isoelectronic with CO, donates a pair of electrons to the metal to get coordinated to the metal

$$M^- + :N \equiv \overset{+}{O}: \rightarrow \overset{2-}{M} - \overset{+}{N} \equiv \overset{+}{O}$$

In this way NO in its nitrosonium complexes acts as a 3-electron donor.

The $\overset{2-}{M} \leftarrow \overset{+}{N} \equiv \overset{+}{O}$ bond would, evidently be highly polar hence, the electron charge from the filled d π orbitals of the metal would get transferred to π molecular orbitals of NO^+

$$\overset{2-}{M} \leftarrow \overset{+}{N} \equiv \overset{+}{O} \xrightarrow{\pi\ \text{back bonding}} \overset{-}{M} \rightleftarrows N = \overset{+}{O}$$

The IR absorption frequency of NO^+ in nitrosonium (NO^+) complexes falls between 1600-1900 cm^{-1} which is much lower than the absorption frequency of NO^+ in its ionic compound (containing NO^+ as free cation) which falls between 2200-2300 cm^{-1}. It is much higher compared to similar lowering of υ_{CO} in case of metal carbonyls establishing there by that π back bonding in nitrosonium complexes is much greater than in metal carbonyls.

As in metal carbonyls the presence of negative charge on the nitrosonium complexes increases the extent of π back bonding as is obvious from the following IR absorption data.

Complex	Absorption frequency of NO^+ group
$[Fe (NO) (CN)_5]^{2-}$	1944 cm^{-1}
$[Mn (NO) (CN)_5]^{3-}$	1730 cm^{-1}
$[V (NO) (CN)_5]^{3-}$	1530 cm^{-1}

The complex formed between NO and Fe^{2+} (aq) which is dark brown in colour that employed as ring test for the detection of NO_3^- ion has been established to be a nitrosonium complex. The brown colour is due to the charge transfer bands occurring in the visible region.

Also there are several complexes of NO in which the π electron of NO remains localized on NO itself and is not transferred to M, the ligand would act as a 2-electron σ donor through its nitrogen atom and the metal complex formed would be paramagnetic. Several complexes such as $[Mn (NO)(CN)_5]^{2-}$ and $[Cr (NO) (CN)_5]^{3-}$ are paramagnetic and at one time these were thought to be derived from NO. But it was now been established that the unpaired electron in most of such complexes is located on the metal rather than on the ligand NO.

There are examples in which first NO accept electron from metal ion M^{n+} and convert into NO^- ion which then donate an electron pair through its N to the resulting $M^{(n+1)}$ to form a σ coordinate bond $M \rightarrow N^+O^-$: In this way nitric oxide in these complexes acts effectively as 1-electron donor. E.g $[Co(NH_3)_5 (NO)]^{2+}$ and $[Co(CN)_5(NO)]^{3-}$.

In complexes such as Fe $(NO)_4$,NO exists in both forms $[(NO^+)_3 Fe(NO^-)]$. Its IR spectrum shows absorption in both regions i.e., 1600–1900 cm^{-1} and 100–1200 cm^{-1}.

14.8 NOMENCLATURE OF INORGANIC COMPLEXES

The nomenclature system outlined below is the one recommended by the inorganic committee of the international union of pure and applied chemistry (IUPAC) and later modified by a similar committee in 1972. the following rules are to be observed in naming the coordination compounds and ions.

1. Order of naming a coordination compound: If the complex is ionic, the name of positive ion is written first followed by the full name of the negative ion. Note that the ion names are separated by a space and the number of each type of ion is not indicated e.g.,

(a) $K_4[Fe(CN)_6]$ In this compound the cation is K^+ ion. So its name should be started with potassium and ended with the name of complex ion.

(b) $[Co(NH_3)_6]Cl_3$ in this compound the cation is complex ion, so its name should be started with the name of complex ion and ended with the name of anion chloride ion.

If a complex is non-ionic, it is given a one word name.

2. Naming of ligands

(a) Naming of neutral ligands: For neutral ligands which are less common, the names of free molecules are used as such e.g.,

PH_3 – Phosphine	CH_3NH_2 – methyl amine
PPh_3 – Triphenyl phosphine	CH_2-NH_2 / CH_2-NH_2 ethylene diamine

However for some of more common neutral ligands, special names are used e.g.,

H_2O – aquo or aqua	NO – nitrosyl
NH_3 – ammine	CS – thiocarbonyl
CO – carbonyl	NS – thionitrosyl

(b) Naming of the negative or anionic ligands: When the name of anion ends in –ide, -te or ate, the final –e is replaced by -O giving –ido, -ito and – ato respectively

SO_3^{2-} – Sulphito	NH_2^- amido
SO_4^{2-} – Sulphato	NH^{2-} Imido
CO_3^{2} – Carbonato	NO_3^- nitrate
CH_3COO – Acetato	N_3^- azido
C_2O_4 – Oxalato	$S_2O_3^{2-}$ thiosuphato.
S^{2-} – Sulphido or thio	

Some exceptions to this rule are

F^- – Fluro or flurido	OH^- – Hydroxo or hydroxido
Cl^- – Chloro or chlorido	O^2 – Oxo or Oxido
Br^- – Bromo or Bromido	O_2^2 – Peroxo or Peroxido
I^- – Iodo or Iodido	O_2^- – Superxo or Superoxido
CN^- – Cyano or cyanido	O_2^{2-} – Perhydroxo or Perhydroxido

Note: In the case of these anions the exceptions allowed previously for historically known bigands given first are now withdrawn. The later names are recommended now by IUPAC

Anionic ligands derived from organic compounds by loss of protons, but which are not normally acidic and, hence, have no established anion name, are named by adding the ending – ato to the compounds name (dropping a final 'e' if present) e.g.,

$$CH_3CO\,CH_2CO\,CH_3 \xrightarrow{-H^+} CH_3CO\,CH^-CO\,CH_3$$

Acetyl acetone → acetyl acetonato

$$\begin{matrix} CH_2\text{–}C\text{=}NOH \\ CH_2\text{–}C\text{=}NOH \end{matrix} \xrightarrow{-H^+} \begin{matrix} CH_3\text{–}C\text{=}N\text{–}O^- \\ CH_3\text{–}C\text{=}N\text{–}OH \end{matrix}$$

Dimethylglyoxime → Dimethylglyoximato

(c) Naming the positive or cationic ligands: The names of positive ligands should be ended with –'ium' e.g.,

$NH_2\overset{+}{N}H_3$ – Hydrazinium; NO^+ – Nitrosylium.

3. Order of naming the ligands: All ligands (negative, positive or neutral) are arranged alphabetically without any preference order. Numerical prefixes (di, tri etc) are not considered in determining that order.

4. Numerical prefixes to indicate number of ligands: The number of simple ligands is given by Greek prefixes di, tri, tetra, penta, hexa etc are used for 2, 3, 4, 5, 6 etc respectively. The term mono is not used.

For more complicated ligands (which contain di, tri, etc in the name itself) the terms bis, tris, tetrakis etc are used and the ligands are enclosed in parenthesis in both formula and name.

5. Naming of the bridging ligands:

(i) The complexes having two or more metal atoms are called polynuclear complexes. In naming such complexes, the bridging group is indicated in the formula of the complex by separating it from the rest of the complex by adding the prefix μ before its use. Two or more bridging groups of the same kind are indicated by di- μ, tri- μ …..etc.

(ii) If the number of nuclear atoms bound by one bridging group exceeds two, the number shall be indicated by adding a subscript numeral in the μ i.e., μ_3, μ_4…etc..

(iii) If more than one kind of bridging groups are present, the μ is repeated before each of which is different.

(iv) If two different central elements are bound by bridging groups, the order given above cannot always be used, and the complex should be noted " from end to end" with the bridging groups named between.

6. Naming of ambidentate ligands: The ligands which may be coordinated to the central metal ion through either of the two donor atoms are known as ambidentate ligands. The ligands are named by special names such as thiocyanate for –SCN and isothiocyanate for –NCS, or the symbol of the element – coordinated with the metal ion is written after the name of the ligand thiocyanato – S for a thiocyanate coordinated to the metal ion through sulphur atom and thiocyanato –N for a thiocyanate group coordinated through nitrogen . Some other examples are

–ONO	Nitro - O or nitrito
$–NO_2$	Nitro - N or nitro
–SCN	Thiocyanato - S or thiocyanato
–NCS	Thiocyanato - N or isothiocyanato
–CN	Cyanido - C or cyanido
–NC	Cyanido - N or isocyanido

7. Naming of central metal ion (i) The central atom of each complex is placed first in the formula for the complex but comes last in the name i.e., after all ligands attached to it have been named. Thus the contents of coordination sphere are specified: First the ligands, then the central metal ion followed by a roman numeral in parenthesis to indicate its oxidation state.

(ii) The name of anionic complex is ended in **ate** but if it is named as an acid, it is ended in ic. For neutral and cationic complexes the name of metal is used without any characteristic ending.

In the case of elements whose symbols are taken from Latin names, the names of the metal ion should be written in latin language and should be ended with 'ate' when the complex ion is an ion, but if the complex ion is cation, the name of metal should be written as English name as such

(iii) In complexes containing metal to metal bonds the prefix 'bi' is used.

8. Oxidation States of the central ion: The oxidation states of central metal atom/ion is designated by Roman numeral in parenthesis (such as II, III IV) after the name of the metal ion.

9. Naming of geometrical isomers: Geometrical isomers are named either by using the prefixes cis or trans before the name of complex ion (i.e., before the names of ligands

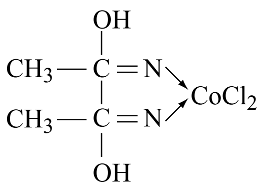
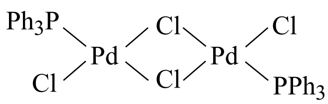
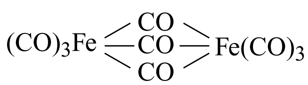
Element	Latin name	Symbol	Name in anionic complex
Sodium	Natrum	Na	
Potassium	Kalium	K	
Tin	Stannum	Sn	Stannate
Lead	Plumbum	Pb	Plumbate
Antimony	Stibnum	Sb	Stibnamate
Iron	Ferrum	Fe	Ferrate
Copper	Cuprum	Cu	Cuprate
Silver	Argentum	Ag	Argentate
Gold	Aurum	Au	Aurate
Mercury	Hydrargium	Hg	Hydrargate
Tungsten	Wolfram	W	Wolframate

in the complex ion. They can also be represented with the numbers for 1, 2 or 3, 4 for cis in square planar complexes. In the case of octahedral complexes the adjacent position numbers for cis and opposite position numbers for trans octahedral complexes can also be given.

10. Naming of optical isomers: Dextro and laevorotatory optically active compounds are designated by (+) and (–) or by d- and l- respectively

Some examples are given below.

w

(i)	$[Co(NH_3)_6]^{3+}$	Hexaammine cobalt (III) ion
(ii)	$[CrCl(H_2O)(en)_2]$	Aquachloridobis(ethylenediamine)chromium(III)
(iii)	$[Co(NH_3)_5Cl]^{2+}$	Pentamminechloridocobalt (III) ion
(iv)	$K[Co(CN)(CO)_2NO]$	Potassiumdicarbonylcyanidonitrosyl cobalt(0)
(v)	$[Co(NO_2)_3(NH_3)_3]$	Triamminetrinitrocobalt (III)
(vi)	$[Co(NH_3)_3\ NO_2Cl(CN)]$	Triamminechlorido cyanidonitrocobalt (III)
(vii)	$[CoCl_2(NH_3)_4]Cl$	Tetramminedichloridocobalt (III) chloride
(viii)	$[Co(NH_3)_4\ SO_4]\ NO_3$	Tetramminesulphatocobalt (III) nitrate
(ix)	$[Co(NH_3)_4\ (H_2O)Br](NO_3)_2$	Tetrammineaquabromidocobalt (III) nitrate
(x)	$[Pt(NH_3)_4Cl_2]^{2+}\ [PtCl_4]^{2-}$	Tetramminedichloridoplatinum (IV) tetrachloridoplatinate (II)
(xi)	$[NiCl_4]^{2-}$	Tetrachloridonickelate (II) ion
(xii)	$[Mn(H_2O)_6]^{2+}$	Hexaaquamanganese (II) ion
(xiii)	$[Co(en)_3]^{3+}$	Tris(ethylenediamine) cobalt (III) ion
(xiv)	$K_3[Fe(C_2O_4)_3]$	Potassium tri(oxalato)ferrate (III)
(xv)	$[Fe(CN)_6]^{4-}$	Hexacyanidoferrate(II) ion
(xvi)	$K\ [BF_4]$	Potassium tetrafluoridoborate (III) ion
(xvii)	$[Co(ONO)_6]^{3-}$	Hexanitritocobaltate (III) ion
(xviii)	$K_3[Fe(CN)_6]$	Potassium hexacyanidoferrate (III)
(xix)	Na_3AlF_6	Sodium hexafluoridoaluminate (III)
(xx)	$Hg[Co\ (CNS)_4]$	Mercury tetrathiocyanatocobaltate (II)
(xxi)	$K[AgF_4]$	Potassium tetrafluoridoargentate (III)
(xxii)	$K_2[Zn(OH)_4]$	Potassium tetrahydroxidozincate (II)
(xxiii)	$Na[Au(CN)_2]$	Sodium dicyanidoaurate (I)
(xxiv)	$Na_2\ [SiF_6]$	Sodium hexafluoridosilicate (IV)
(xxv)	$K_3[Cu(CN)_4]$	Potassium tetracyanidocuprate (I)
(xxvi)	$[Pt\ (NH_3)_4]\ [PtCl_4]$	Tetrammineplatinum(II) tetrachloridoplatinate (II)
(xxvii)	$Na\ [BH_4]$	Sodium tetrahydridoborate (III)
(xxviii)		Dichloridodimethylglyoxime - cobalt (II)
(xxix)	$[(NH_3)_5Cr - OH - Cr\ (NH_3)_5]Cl_5$	Pentaamminechromium (III) - μ-hydroxopentamminechromium (III) chloride
(xxx)		Chloridotriphenylphosphinepalladium (II) - μ - dichloridotriphenyl - phosphine palladium (II)
(xxxi)		Tri - μ - carbonyl bis (tricarbonyl) iron (0)

14.9 ISOMERISM IN COORDINATION COMPOUNDS

The word "isomerism" is coined from two Greek words; Iso – same and *meso* – parts. Two or more compounds having the same molecular formula but different properties are called isomers and the phenomenon is called isomerism.

Isomers can be broadly classified into two major categories (A) Structural isomerism (B) Stereoisomerism.

14.9.1 Structural Isomerism

The isomers which have same molecular formula but different structural arrangement of atoms or groups of atoms around the central metal ion are called structural isomers. Different types of structural isomerism are discussed here.

(i) Ionization isomerism: Compounds which have same molecular formula but give different ions in solutions are called ionization isomers. This type of isomerism occurs when the counter ion in a coordination compound is itself a potential ligand. Some examples are given below.

(1a) $[Co(NH_3)_5Br]SO_4 \rightarrow [Co(NH_3)_5Br]^{2+} + SO_4^{2-}$
Red Violet

(1b) $[Co(NH_3)_5SO_4]Br \rightarrow [Co(NH_3)_5SO_4]^{+} + Br^{-}$

These two can be distinguished as

1 (i)(a) will give white precipitate with $BaCl_2$ but (i)(b) do not give ppt with $BaCl_2$

(b) will have more electrical conductivity than (i)(b)

(ii)(a) $[Co(NH_3)_4Cl_2]NO_2 \rightarrow [Co(NH_3)_4Cl_2]^{+} + NO_2^{-}$

(b) $[Co(NH_3)_4Cl(NO_2)]Cl \rightarrow [Co(NH_3)_4Cl_2(NO_2]^{+} + Cl^{-}$

(ii)(a) do not give ppt with $AgNO_3$ while (ii)(b) gives ppt with $AgNO_3$

(iii)(a) $[Co(NH_3)_5NO_3]SO_4 \rightarrow [Co(NH_3)_5NO_3]^{2+} + SO_4^{2-}$

(b) $[Co(NH_3)_5SO_4]NO_3 \rightarrow [Co(NH_3)_5SO_4] + NO_3^{-}$

(iii)(a) will give white ppt with $BaCl_2$ but (iii)(b) do not give ppt with $BaCl_2$ and also (iii)(a) will have more electrical conductivity than (iii)(b).

2. Hydrate isomerism: The compounds which have the same molecular formula but differ in the number of water molecules present as ligands or as molecules of hydration are called hydrate isomers. This type of isomerism is similar to ionization isomerism and may occur inside and outside the coordination sphere as a coordinated group or a water of hydration. For example there are three isomers for $CrCl_3 \cdot 6H_2O$.

(i) $[Cr(H_2O)_6]Cl_3$ It is violet in colour, it does not lose any H_2O molecules on dehydration with conc $H_2SO_4 \cdot$ $3Cl^-$ ions are precipitated with $AgNO_3$ as AgCl. Total number of ions formed in solution is 4 (one complex ion and $3Cl^-$ ions)

(ii) $[Cr(H_2O)_5Cl]\ Cl_2 \cdot H_2O$ It is light green in colour. It loses one H_2O molecule on dehydration with conc $H_2SO_4 \cdot 2Cl^-$ ions are precipitated with $AgNO_3$ as AgCl. Total number of ions formed in solution is 3(one complex ion and $2Cl^-$ ions)

(iii) $[Cr(H_2O)_4Cl_2]\ Cl \cdot 2H_2O$: It is dark green in colour. It loses two water molecule on dehydration with conc H_2SO_4. One Cl^- ion is precipitated with $AgNO_3$ as AgCl. Total number of ions formed in solution is 2 (one complex ion and one Cl^- ion)

Due to the difference in the numer of ions in solution their electrical conductivity order is

$$[Cr(H_2O)_6]Cl_3 > [Cr(H_2O)_5Cl]Cl_2 \cdot H_2O > [Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$$

Other hydrate isomers are

$$[Co(en)_2(H_2O)Cl]Cl_2 \text{ and } [Co(en)_2Cl_2]Cl{\cdot}H_2O$$
$$[Co(NH_3)_4(H_2O)Cl]Cl_2 \text{ and } [Co(NH_3)_4Cl_2]Cl{\cdot}H_2O$$
$$[Co(Py)_2(H_2O)_2Cl_2]Cl \text{ and } [Cr(Py)_2H_2O{\cdot}Cl_3]{\cdot}H_2O$$

3. Coordination isomerism: This type of isomerism occurs in compounds containing both cationic and anionic coordination entities and the isomers differ in the distribution of ligands in the coordination entity of cationic and anionic parts e.g.,

$$[Co(NH_3)_6][Cr(CN)_6]$$
$$[Cr(NH_3)_6][Co(CN)_6]$$

Coordination isomers in the above example are also possible due to partial exchange of ligands

$$\left[Co(NH_3)_5(CN)\right]\left[Cr(CN)_5(NH_3)\right]$$
$$\left[Co(NH_3)_4(CN)_2\right]\left[Cr(CN)_4(NH_3)_2\right]$$
$$\left[Co(NH_3)_2(CN)_4\right]\left[Cr(CN)_2(NH_3)_4\right]$$
$$\left[Co(NH_3)(CN)_5\right]\left[Cr(CN)(NH_3)_5\right]$$

This type of isomerism is also shown by compounds in which the metal ion is the same in both cationic and anionic complexes.

$$\left[Cr(NH_3)_6\right]\left[Cr(CN)_6\right]$$
$$\left[Cr(NH_3)_5(CN)\right]\left[Cr(CN)_5(NH_3)\right]$$
$$\left[Cr(NH_3)_4(CN)_2\right]\left[Cr(CN)_4(NH_3)_2\right]$$

The examples for the coordination compounds with coordination number 4

$$\left[Cu\left(NH_3\right)_4\right]^{2+}\left[PtCl_4\right]^{2-}$$
$$\left[Cu\left(NH_3\right)_3Cl\right]\left[PtCl_3\left(NH_3\right)\right]$$
$$[Cu(NH_3)Cl_3][PtCl(NH_3)_3]$$
$$[CuCl_4][PtCl_4]$$

Another example with coordination number 4 is

$$\left[Pt\left(NH_3\right)_4\right]\left[PtCl_4\right]$$
$$\left[Pt\left(NH_3\right)_3Cl\right]\left[Pt\left(NH_3\right)Cl_3\right]$$

It can be noted that in the coordination compounds with CN 6 exhibiting coordination isomerism, the total number of isomers is 6, if the central metal ions in both coordination entities are different, but if both coordination entities have same type of metal ion the total no. of coordination isomers are only 3. Similarly in the coordination compounds with CN 4 showing coordination isomerism the total number of isomers is 4 if central metal ions are different whereas if the central metal ions in both cationic and anionic part of coordination entities is same, the total number of isomers is only 2.

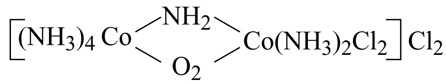
Other examples are

$[Co(NH_3)_6][Cr(C_2O_4)_3]$ and $[Cr(NH_3)_6][Co(C_2O_4)_3]$
$[Co(en)_3][Cr(CN)_6]$ and $[Cr(en)_3][Co(CN)_6]$

Sometimes the same metal which appears in both cation and anion may have different oxidation states e.g.,

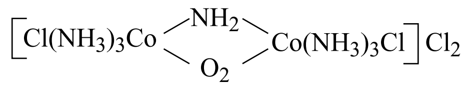
$$[Pt^{II}(NH_3)_4][Pt^{IV}Cl_6] \text{ and } [Pt^{IV}(NH_3)_4Cl_2][Pt^{II}Cl_4]$$

Coordination position isomerism: This type of isomerism occurs in poly nuclear complexes in which coordination groups may be present in the same number but may arrange themselves differently with respect to the different metal ions present i.e., it is a result of the exchange of ligands between the different metal ions present. An example of this is.



and

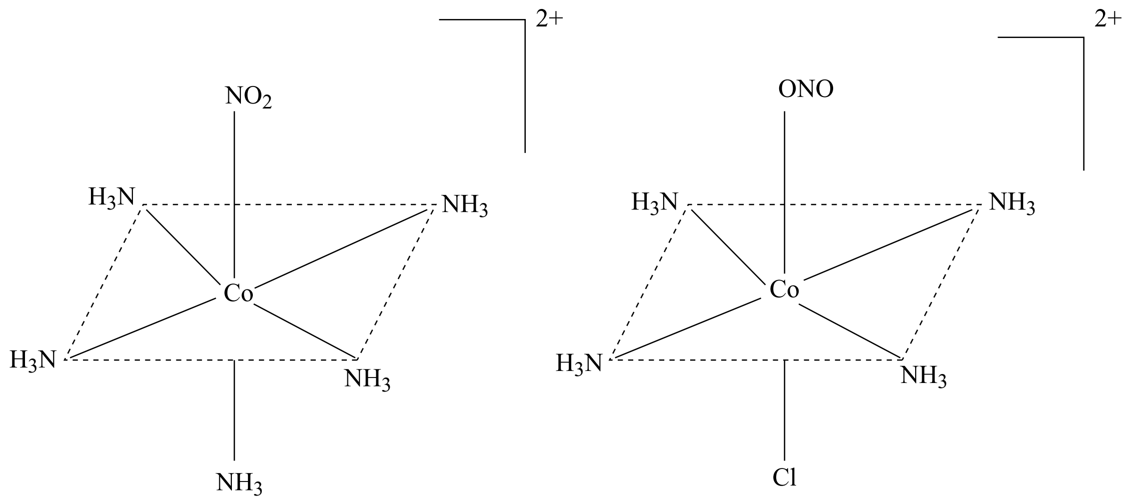
unsymmetric form.



Symmetric form.

Thus ammonia molecules and chloride ions are differently placed relative to the two cobalt ions.

4. Linkage isomerism: The compounds which have the same molecular formula but differ in the mode of attachment of ligand to the metal atom or ion are called linkage isomers i.e., this type of isomerism occurs in the complexes containing ambidentate ligands. For example, in NO_2^- ion, the nitrogen atom as well as the oxygen atom can donate their lone pairs e.g.,



$[Co(NH_3)_5(NO_2)]Cl_2$

tran-isomer (green)

Pentaamminenitrocobalt (III) chloride (Red)

Pentaamminenitrocobalt (III) chloride (Yellow brown)

Fig 14.17 Linkage isomers of pentamminenitrocobalt (III) chloride

Other example of ambidentate ligands which form linkage isomers are

-CN Cynido (through C)

-SCN Thiocyanato (through S)

-NC Isocynido (through N)

-NCS Isothiocyanato (through N)

5. Polymerization isomerism: This term is used to describe compounds which have the same stoichiometric composition but whose molecular compositions are multiples of the simplest stoichiometric arrangement: e.g.,

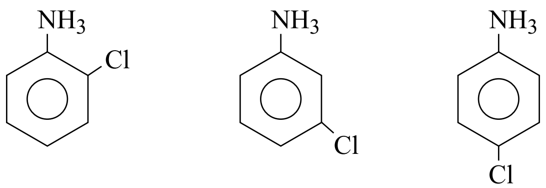
$$\text{I}[Pt(NH_3)_2Cl_2]$$
$$\text{II}[Pt(NH_3)_4][PtCl_4]$$
$$\text{III}[Pt(NH_3)_3Cl]_2[PtCl_4]$$

In the above example, the second and third compounds may said to be polymers of the first. Actually, in the strict sense of the word, these are not examples of polymerization. Polymerization is usually interpreted as meaning of multiplication of simple molecules. Here we have difference in arrangements as well as a multiplication of molecular weight.

Another example illustrating polymerization isomerism is

$$[Cr(NH_3)_3(CNS)_3]$$
$$[Cr(NH_3)_5(CNS)][Cr(CNS)_6]_2$$

6. Ligand Isomerism: Many ligands are themselves capable of existing in isomeric states. When such ligands are associated to complexes, the complexes are isomers of each other. Such isomers are known as ligand isomers and the phenomenon is known as ligand isomerism. For example $[Co(en)_2Cl$ (Chloro aniline)$]^{2+}$ exists in these isomers because chloroaniline may be o–, m–, p–



o-Chloroaniline m-Chloroaniline p-Chloroaniline

Diamino propane is another ligand which can exhist both as, 1, 2- diamino propane (pn) and 1, 3 diamino propane (tn)

NH_2 NH_2 CH_2-CH-CH_2	NH_2 NH_2 CH_2-CH-CH_2
1, 2-diamino propane (pn)	1, 3-diamino propane (tn)

When pn and tn are associated into complexes, the so obtained complexes are isomers of each other. $[Co(pn)_2Cl_2]^+$ and $[Co(tn)_2Cl_2]^+$

14.9.2 Stereo Isomerism

When the same molecular formula represent two or more compounds which differ in the spatial arrangement of atoms or groups, then such compounds are called stereo isomers. The phenomenon is known as stereo isomerism (Stereo – occupying space). It is further classified into two types.

(i) Geometrical isomerism (or Cis-trans isomerism)

(ii) Optical isomerism (or mirror image isomerism)

14.9.3 Geometrical Isomerism or Cis-Trans Isomerism

(a) In cis isomers the two identical ligands occupy the positions adjacent to each other

(b) In trans – isomers, the two identical ligands occupy the positions diagonal to each other.

Geometrical isomers have different physical and chemical properties.

14.9.4 Geometrical Isomerism in Complexes of Coordination Number 4

The complexes having coordination number 4 adopt tetrahedral or square planar geometry. The geometrical isomerism is not possible in tetrahedral geometry because any two positions are adjacent to each other. However, square planar complexes show geometrical isomerism.

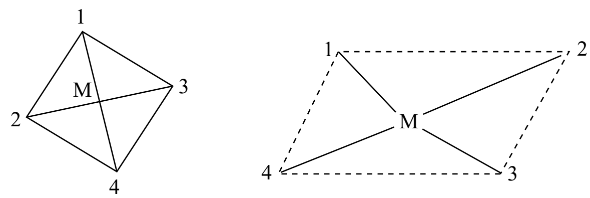


Fig 14.18 Tetrahedral and square planar entities

In square planar complexes the positions 1,2; 2,3; 3,4 and 1,4 are cis with respect to each other while the positions 1,3 and 2,4 are trans to each other.

Square planar complexes of the type MA_2X_2, MA_2XY or $MABX_2$ can exist in cis trans isomers. Here A and B are neutral ligands like H_2O, NH_3, Py etc. while X and Y are anionic ligands like NO_2^-, SCN^-, Cl^-, Br^-, I etc.

(i) $[Pt(NH_3)_2Cl_2]$ exists in cis and trans forms as in Fig 14.19

The cis isomer is used in medicine as an antitumor agent called cisplatin

(ii) $[Pt(Py)_2(NH_3)Cl]$ exists in cis and trans forms as in Fig 14.20

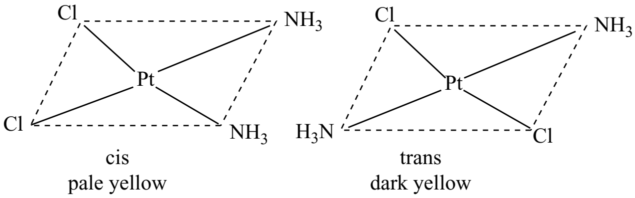


Fig 14.19 Cis and trans isomers of [Pt $(NH_3)_2$ Cl_2]

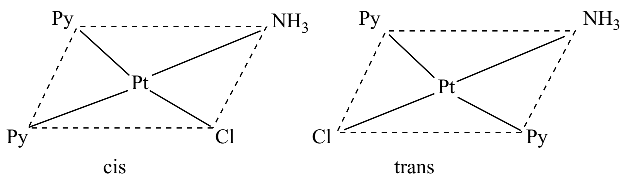


Fig 14.20 Cis and trans isomers of [Co $(Py)_2$ (NH_3) Cl]

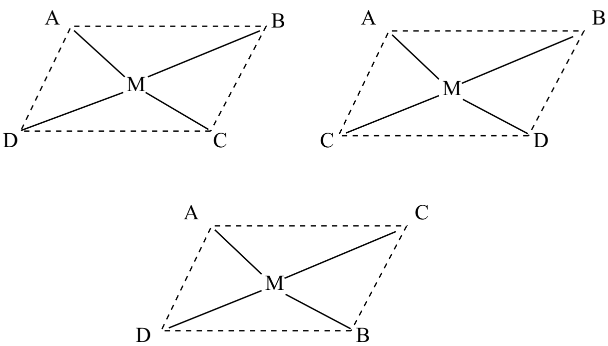


Fig 14.21 Geometrical isomers of [M (ABCD)]

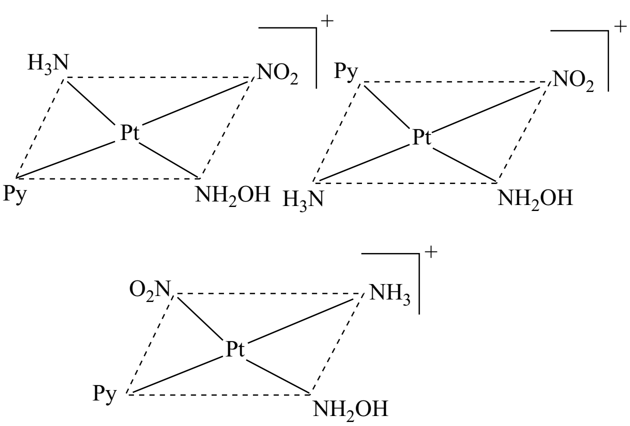


Fig 14.22 Geometrical Isomers of $[Pt(NH_3)\ (NH_2OH)\ (Py)(NO_2)]^+$

Square planer complexes of the type M ABCD show three isomers. The structures of these isomers can be written by fixing the position of one ligand (say A) and placing the other ligands B, C and D trans to it.

For example the complex $[Pt\ (NH_3)\ (NH_2OH)\ (Py)\ (NO_2)]^+$ exists in three geometrical isomers as represented in Fig 14.22.

(iv) Geometrical isomerism cannot occur in complexes of the type MA_4, MA_3B or MAB_3 because all possible spatial arrangements for any of these complexes will be exactly equivalent.

(v) Chelate rings involving bidentate ligands are formed only in cis form because the chelating ligand is too small to span the trans position. The distance across the two trans positions is too large for all but very large ligands can span to trans positions and synthesis with such large rings is difficult.

(vi) The square planar complexes containing unsymmetrical bidentate ligands such as $[M\ (AB)_2]$ also exhibit geometrical isomerism. For example, the complex $[Pt\ (gly)_2]$ where gly = $NH_2CH_2COO^-$ (glycinate ion) exists in cis and trans form

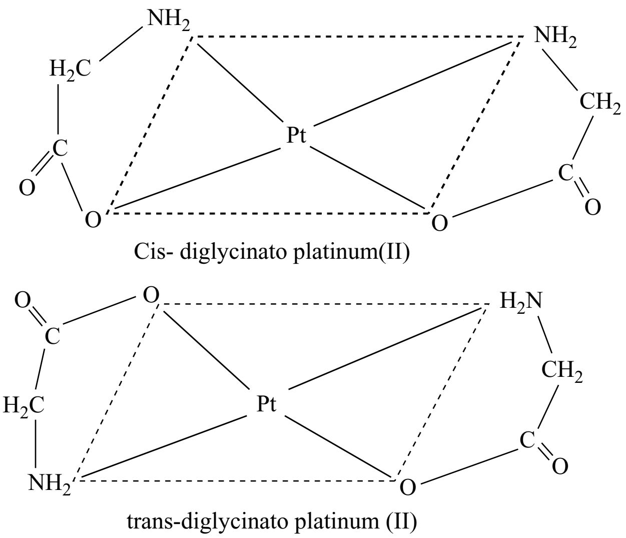
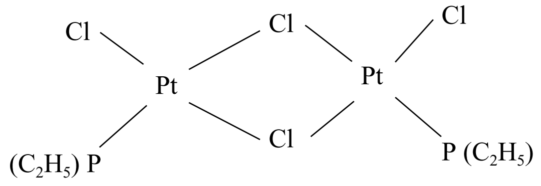


Fig 14.23 Geometrical isomers of $[Pt\ (gly)_2)]$

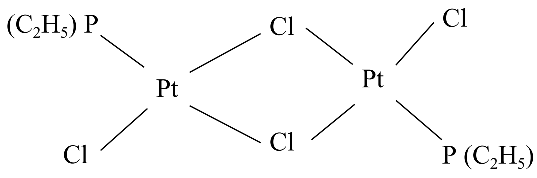
(vii) Geometrical isomerism is also shown by bridged binuclear complexes of the type $M_2A_2X_4$. For example the complex $[PtCl_2\ P(C_2H_5)_3]_2$ exhibits in geometrical isomers Fig 14.24.

14.9.5 Geometrical Isomerism in Complexes of Coordination Number 6

The complexes having coordination number 6 adopt octahedral geometry



Cis form



Trans form

Fig 14.24 Geometrical isomers of [Pt Cl_2 $P(C_2H_5)_3]_2$

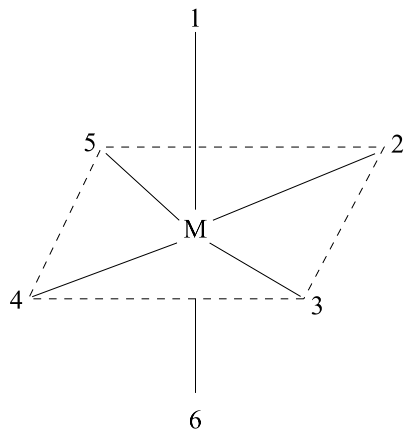


Fig 14.25 Octahedral entity coordination

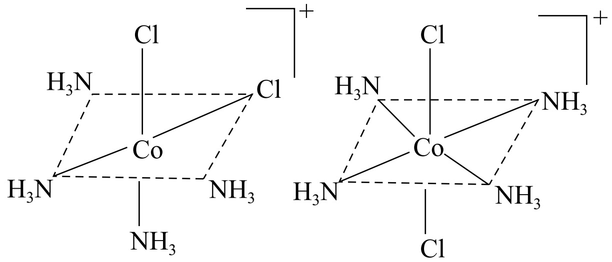
In the octahedral complexes, positions 1,6; 2,4 and 3,5 are trans while the others such as 1,2; 1,3; 1,4; 1,5; 2,5; 2,3; 3,4; 4,5; 2,6; 3,6; 4,6 and 5,6 are cis to each other

(i) The octahedral complexes of the type MA_6, MA_5B or MAB_5 type do not exhibit geometrical isomerism

(ii) The octahedral complexes of the type MA_4B_2, MA_2B_4 and MA_4BC etc exhibit geometrical isomerism . Some common examples are

$$[Co(NH_3)_4\ Cl_2]^+;\ [Pt\ (NH_3)_4\ Cl_2]^{2+}$$

(iii) Octahedral complexes of the type MA_3B_3 also exist in two geometrical isomers. When the three ligands



cis trans

Fig 14.26 Geometrical isomers of $[Co(NH_3)_4Cl_2]$

(with same donor atoms) are on the same triangular face of the octahedron, the isomer is called facial or **fac** isomer. When the three ligands are on the same equatorial plane the other in a plane bisecting the molecule, this isomer is called **meridional** or **mer** isomer. In a simple way in facial isomer the three ligands are at the corners of a triangular face while in meridional isomer, the three ligands are at the corners of a square plane as shown below.

(iv) Octahedral complexes having bidentate ligands of the type $[M(AA)_2\ X_2]$ and $[M(AA)_2\ XY]$ can also exist cis

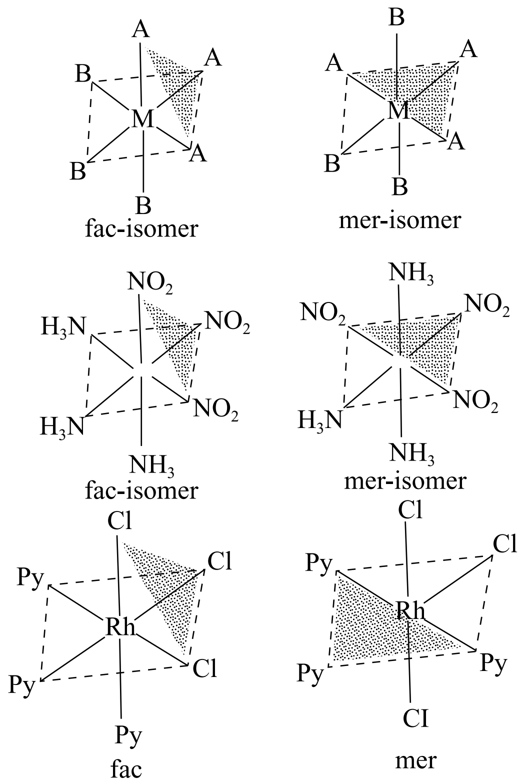
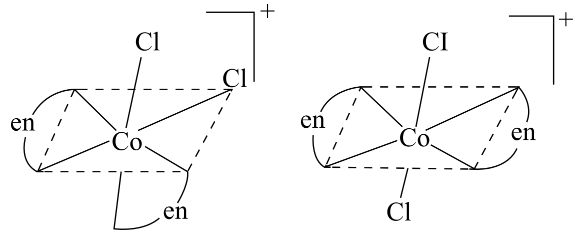


Fig 14.27 Facial (fac) and merdional isomers of MA_3B_3 type complexes



cis trans

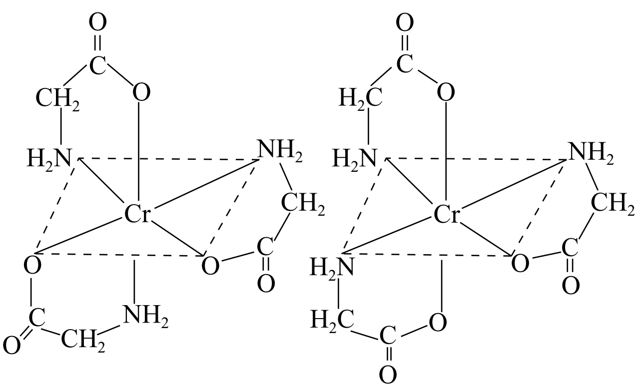
Fig 14.28 Geometrical isomers of [Co $(en)_2$ $Cl_2]^+$

and trans isomers where AA represents a symmetrical bidentate ligand such as ethylene diamine (en) oxalate ion (ox). For example an octahedral complex [Co $(en)_2$ $Cl_2]^+$ exist as two isomers.

(v) Octahedral complexes having six different ligands of the type M(ABCDEF) would exhibit geometrical isomerism. These isomers may be written by fixing a ligand at one position and then placing the other ligands trans to it. Theoritically, 15 different isomers are possible for such type of complexes. The only compound of this type that has been prepared is [Pt (Py) (NH_3) (NO_2) (Cl) (Br) (I)]. It has been possible to isolate only three different forms of this kind.

(vi) The complexes containing unsymmetrical bidentate ligands also show geometrical isomerism . For example the complex triglycinato chromium (III) [Cr $(gly)_3$] where gly is glycinate ions $H_2NCH_2COO^-$ exists in cis and trans forms as shown in Fig 14.29.

There are very large number of other geometrical isomers known but we have considered only simple examples.



Cis-form(fac) Trans-form(mer)

Cis triglycinato chromium (III) Trans-triglycinatochromium (III)

Fig 14.29 Geometrical isomers of [Cr $(gly)_3$]

14.9.6 Optical Isomerism

There are certain substances which can rotate the plane of polarized light. These are called optically active substances. These isomers which rotate the plane of polarized light equally but in opposite direction are called optically active isomers. These are also called enantiomers. The isomer which rotates the plane of polarized light to the right is called dextrorotatory designated as 'd' and the one which rotates the plane of polarized light to the left is called laevorotatory designated as (*l*).A1:1 mixture of d and *l* isomers gives a net zero rotation is called recemic mixture. The d and *l* isomers are mirror image compounds, a non-super imposable on each other and do not possess the plane of symmetry. These optical isomers also possess chirality (handedness) The essential condition for a substance to show optical activity is that the substance should not have a plane of symmetry in its structure. The optical isomers have identical physical and chemical properties. They differ only in the direction in which they rotate the plane of polarized light.

14.9.7 Optical Isomerism in 4-Coordination Compounds

In tetrahedral and square planer complexes of the type MA_4, MA_3B and MAB_3, stereoisomerism is not possible. The reason for this is that all the possible arrangements of the ligands around the central metal ion M, are exactly equivalent.

14.9.8 Optical Isomerism in Square Planar Complexes

Optical isomerism is rarely observed in square planer complexes because they have all the four ligands and the central metal ion in the same plane and hence possess a plane or axis of symmetry. Thus these complexes cannot exhibit optical isomerism even if all the four ligands

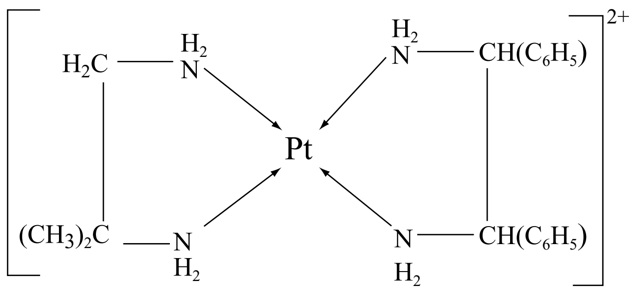


Fig 14.30 Isobutelenediamine mesostilbenediamine platinum (II) cation

are different. However Mills and Quibell in 1935 were successful in preparing a 4 coordinate complex of Pt (II) viz isobutylenediamine mesostillbene diamine platinum (II) cation [Pt (NH_2CH (C_6H_5) CH (C_6H_5) NH_2) (NH_2 CH_2C $(CH_3)_2$ $NH_2]^{2+}$. It has square planar structure and shows optical activity.

14.9.9 Tetrahedral Complexes

(i) Optical isomerism is expected in an asymmetric molecule with tetrahedral structure when the central atom is surrounded by four different groups [M (ABCD)]. Two optical isomers (mirror-image isomers) of tetrahedral complex. [As (CH_3) (C_2H_5) (S) $(C_6H_4COO)]^{2+}$ was reported in 1963 but it could not be resolved.

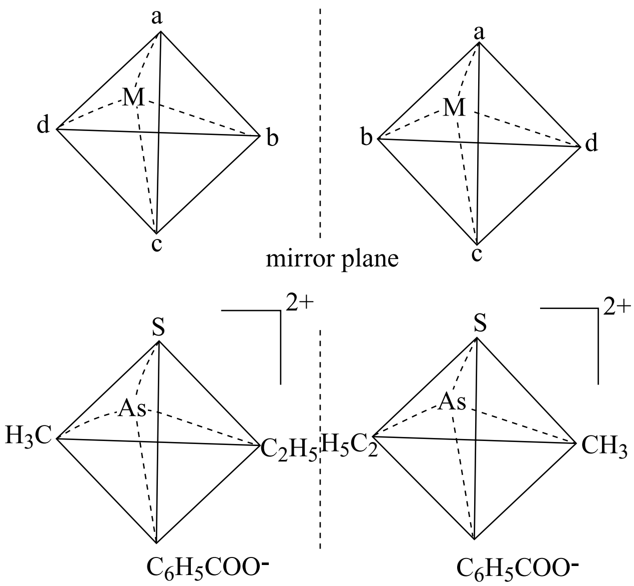


Fig 14.31 Optical isomers of [As (CH_3) (C_2H_5) (S) $(C_6H_5COO)]^{2+}$

(ii) However compounds containing two unsymmetric bidentate ligands are easy to resolve into optical isomers and known for complexes of Be (II), Zn (II) and B(III). The compounds bis benzoylacetonato beryllium (II) exhibits the following two enantiomarphs.

As this complex has no centre or plane of symmetry the two forms are not super imposable on each other (Fig 14.32).

14.9.10 Optical Isomerism in 6-Coordination Compounds

This type of isomerism is more common with examples having coordination number 6 than in coordination number 4. Optical isomerism is very common in the following type of octahedral complexes.

1. Octahedral complexes containing only monodentate ligands

(i) [$MA_2B_2C_2$] type complex has two optical isomers as shown in the Fig. 14.33

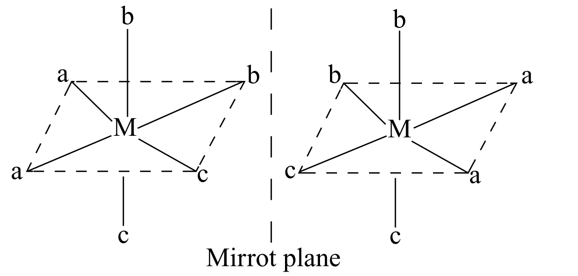


Fig 14.33 Two optical isomers of an octahedral complex of [$MA_2B_2C_2$]

(ii) In the complex [M ABCDEF] containing six different ligands the central atom is asymmetrically disposed. For each of the fifteen that are possible for such compounds, the total optical isomers are 30 because each of which would exist in d- and l- forms
e.g., [Pt (Py) (NH_3) (NO_2) (Cl) (Br) (I)]

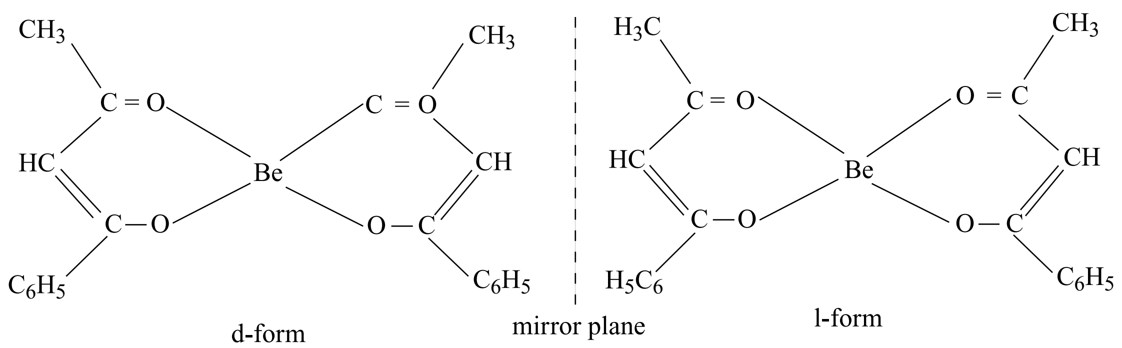


Fig 14.32 Optical isomers of bisbenzoylacetanato beryllium (II)

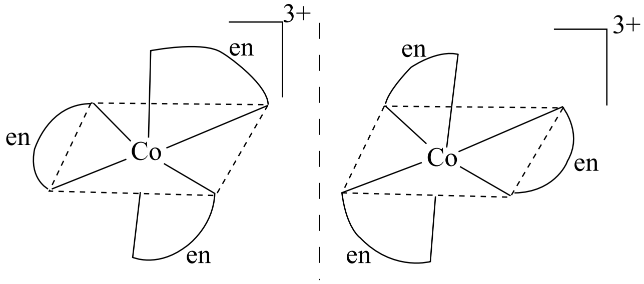


Fig 14.34 Optical isomers of $[Co\,(en)_3]^{3+}$

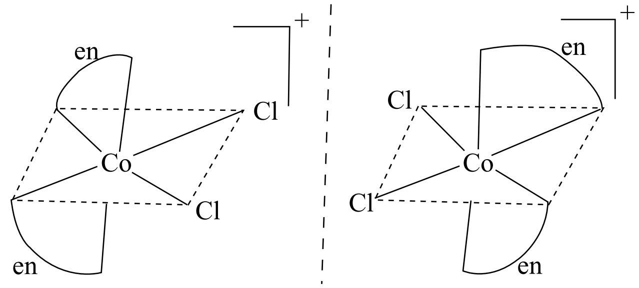


Fig 14.35 Optical isomers of $[Co\,(en)_2\,Cl_2]^+$

2. **Octahedral complexes containing only symmetrical bidentate chelating ligands**
   (i) Complexes of the type $M(AA)_3$ (where AA is symmetrical bidentate ligand) such as $[Co\,(en)_3]^{3+}$, $[Cr\,(ox)_3]^{3-}$ and $[Pt\,(en)_3]^{4+}$ etc. exist as optical isomers.
   (ii) The complexes of the type $[M(AA)_2\,(BB)]$ having different types of symmetrical bidentate ligands also exist in two optical isomers e.g., $[Co\,(en)_2\,(CO_3)_2]^+$ and $[Co\,(en)_2\,(C_2O_4)]^+$
3. **Octahedral complexes containing monodentate and symmetrical bidentate chelating ligands**
   (i) The complexes of the type $\left[M\left(AA\right)_2 B_2\right]^{n\pm}$ and $\left[M\left(AA\right)_2 BC\right]^{n\pm}$ (where AA= bidentate chelating ligand and B and C are monodatate ligands) show optical isomerism e.g., $[Co\,(en)_2\,Cl_2]^+$ and $[Co\,(en)_2\,Cl\,(NO_2)]^+$ etc.

cis $[Co(en)_2\,Cl_2]^+$ isomer has no plane of symmetry so exhibit optical isomerism but trans $[Co(en)_2\,Cl_2]^+$ has plane of symmetry hence cannot exhibit optical isomerism.

The examples for the type $[M(AA)_2\,BC]$ which can exhibit optical isomerism are $[Co\,(en)_2\,(NH_3)(Cl)]^{2+}$, $[Ru\,(Py)\,(C_2O_4)_2\,(NO)]^-$

   (ii) The complexes of the type $[M(AA)B_2C_2]^{n+}$ exist in three forms; one is optically active and two are inactive or symmetrical e.g., $[Co(en)\,(NH_3)_2\,Cl_2)^+$

In the Fig 14.36 I and II have plane of symmetry, so do not exhibit optical isomerism but the III and its mirror image IV are not super imposable because they do not have plane of symmetry.

4. **Octahedral complexes containing polydentate ligands**

Octahedral complexes containing hexadentate ligands such as ethylenediamine tetracetate (EDTA) also show optical isomerism. For example $[Co\,(EDTA)]^-$ exist as two optical isomers Fig 14.37.

14.10 CALCULATION OF NUMBER OF ISOMERS

When the number of different ligands increased there have been several possible isomers. There have been several schemes for calculating the maximum number of isomers for each case. One approach to tabulating isomers is shown in Fig 14.38 and table 14.12 the notation <ab>

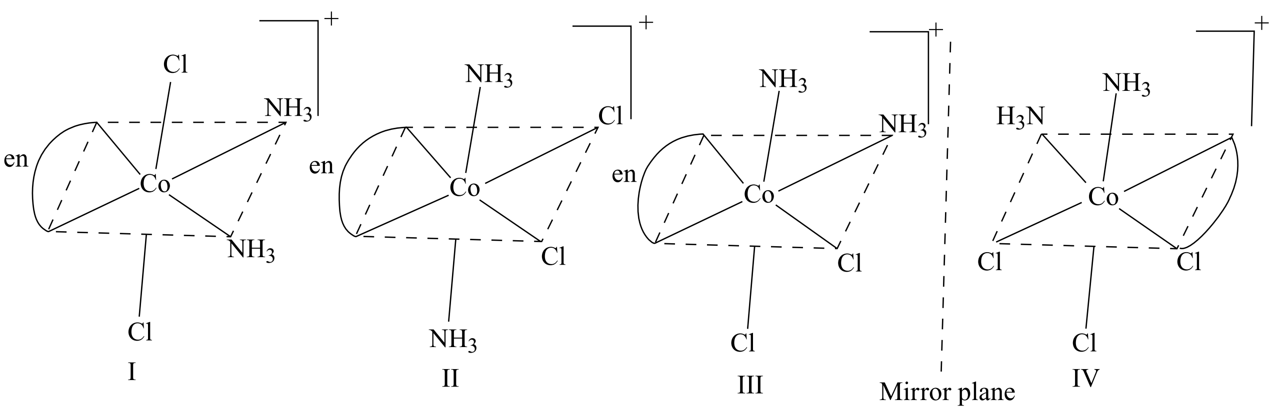


Fig 14.36 Optical isomers of $[Co\,(en)\,(NH_3)_2\,Cl_2)^+$

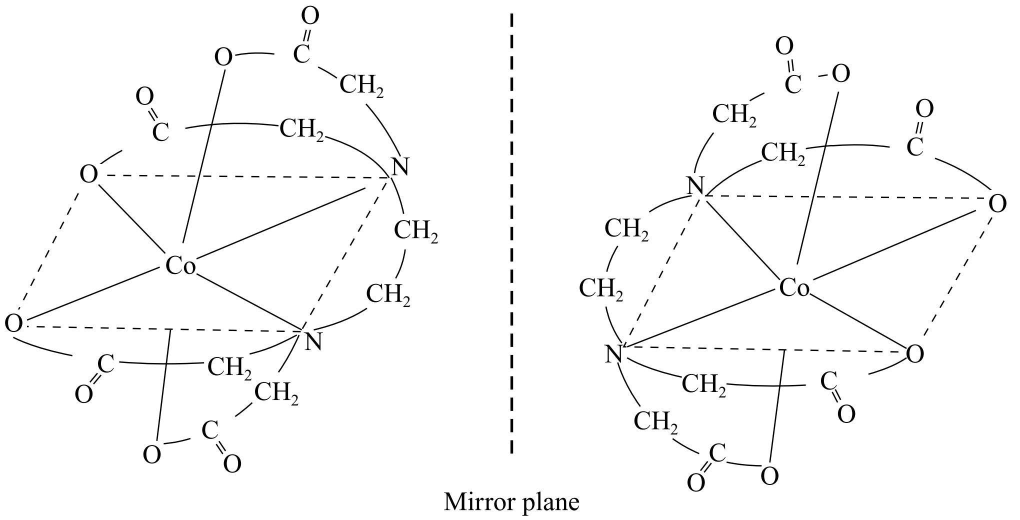


Fig 14.37 Optical isomers of $[Co\ (EDTA)]^{-}$

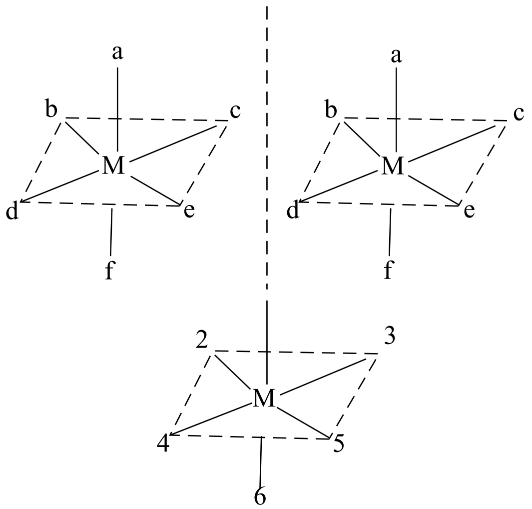


Fig 14.38 [M<ab><cd><ef>] isomers and the octahedral numbering system

Table 14.12 [M abcdef] isomers

	A	B	C
1	ab	ab	ab
	cd	ce	cf
	ef	df	de
2	ac	ac	ac
	bd	be	bf
	ef	df	de
3	ad	ad	ad
	bc	be	bf
	ef	cf	ce
4	ae	ae	ae
	bc	bf	bd
	df	cd	cf
5	af	af	af
	bc	bd	be
	de	ce	cd

indicates that a and b are trans to each other with M the metal ion and a,b,c,d,e and f monodentate ligands. The [M<ab><cd><ef>] isomers of [Pt (Py) (NH_3) (NO_2) (Cl) (Br) (I)] are examples shown in Fig 14.38 the six octahedral positions are commonly numbered as in the figure with positions 1 and 6 in axial positions 1 and with 2 through 5 in counter clock wise order as viewed from the 1 position.

If the ligands are completely scrambled rather than limited to the trans pairs shown in Fig 14.38 there are 15 different diastereo isomers (different structures that are not mirror images of each other) each of which has an enantiomer (mirror image). This means that a complex with 6 different ligands in an octahedral complex can have 30 different isomers! Isomers of [M abcdef] are given in Table 14.12.

Each of the 15 entries represents an isomer and its entiomer for a total 30 isomers. Each entry lists the trans pairs of ligands; for example C3 represents the two enantiomers of [M<ad><bf><ce>].

Finding the number and identity of the isomers of a complex is primarily a matter of systematically listing the possible structures and then checking for identical species, and chirality. The method suggested by Bailar uses a list of isomers. One trans pair such as <ab> is held constant the second

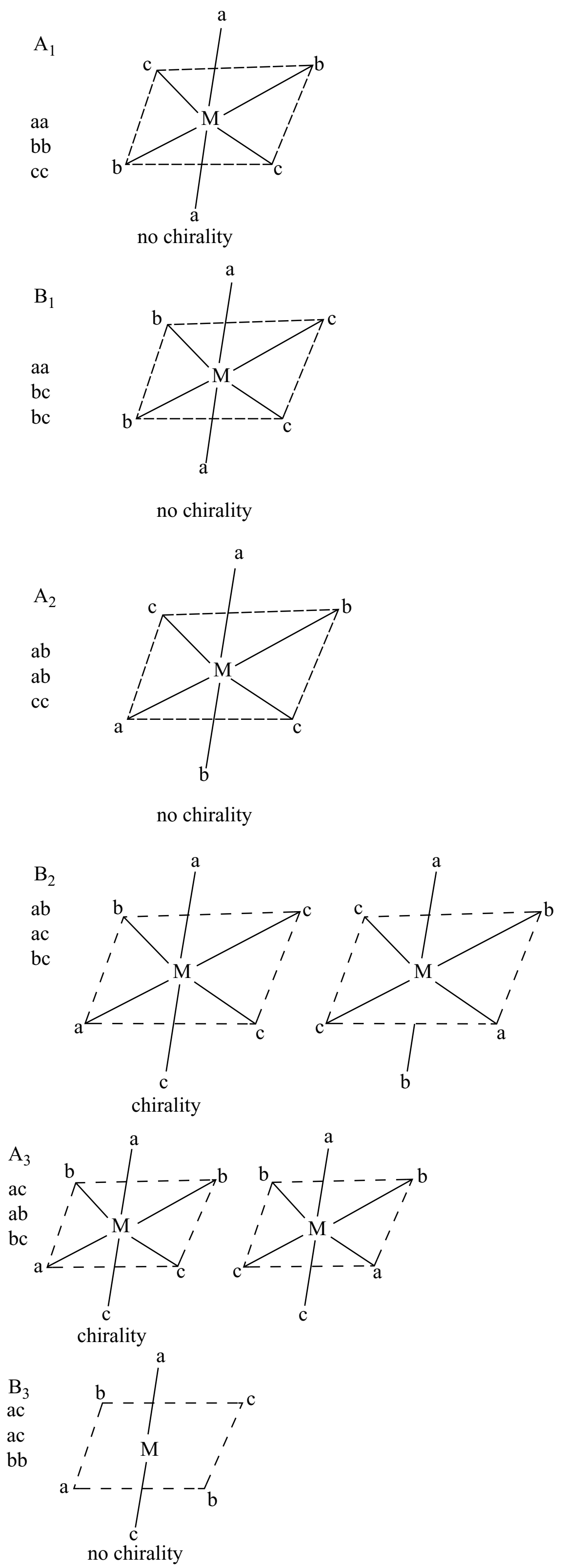


Fig 14.39 Sterco isomers of $Ma_2b_2c_2$

pair has one component constant and the other systematically changes and the third pair is whatever is left over. Then the second component of the first pair is changed and the process is continued. The results are given Table 14.12.

Each isomer (A1, A2 ……) has the trans pairs listed A1 is shown in Fig 14.38 each isomer also has a mirror image (enantiomer)

The same approach can be used for chelating ligands, with limits on the location of the ring. For example a normal bidentate chelate ring cannot connect trans positions.

Example

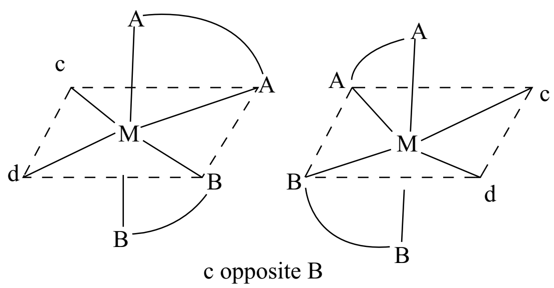
The isomers of $Ma_2b_2c_2$ can be found by this method. In each row the first pair of ligands is held constant (<aa>, <ab> and <ac> in rows1, 2 and 3 respectively. In column B one component of the second pair is traded for a component of the third pair (For example, in row 2 <ab> and <cc> becomes <ac> and <bc>)

Once all the trans arrangements are listed, drawn and checked for chirality, we can check for duplicates; in this case A_3 and B_2 are identical. Overall, there are four non chiral isomers and one chiral pair for a total of six.

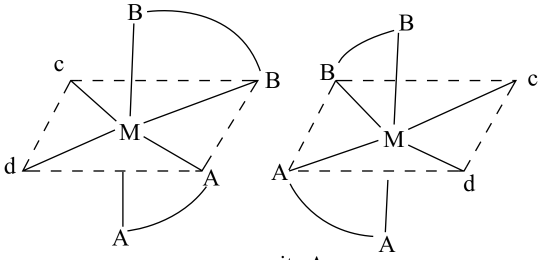
After listing all the isomers without this restriction those that are sterically impossible can be quickly eliminated and the others checked for duplicates and then for enantiomers are calculated. Table 14.13 lists the number of isomers and enantiomers for many general formulas.

Example

A methodical approach is important in finding isomers. AA and BB must be in cis positions because they are linked in the chelate ring. For [M(AA) (BB) cd] We first try c and d in Cis positions one A and one B must be trans to each other.



c opposite B
d opposite A
The mirror image is different.
So there is a chiral pair



c opposite A
d opposite B
The mirror image is different.
So there is a chiral pair

Fig 14.40 Stereoisomers of [M (AA) (BB) cd]

Then trying c and d in transpositions where AA and BB are in the horizontal plane.

The mirror images are identical, so there is only one isomer. There are two chiral pairs and one individual isomer, for a total of five isomers.

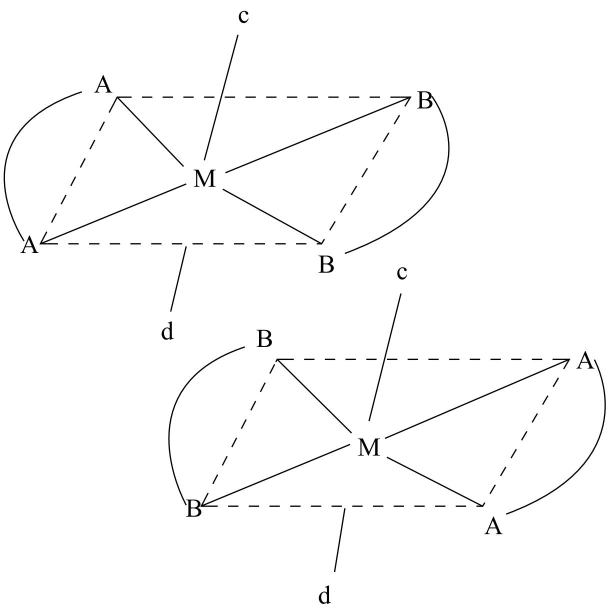


Fig 14.41

Table 14.13 Number of possible isomers for specific complexes

Formula	Number of stereoisomer's	Pairs of enantiomers
Ma_6	1	0
Ma_5b	1	0
Ma_4b_2	2	0
Ma_3b_3	2	0
Ma_4bc	2	0
Ma_3bcd	5	1
Ma_2bcde	15	6
Mabcdef	30	15
$Ma_2b_2c_2$	6	1
Ma_2b_2cd	8	2
Ma_3b_2c	3	0
M(AA)(BC)de	10	5
M(AB)(AB)cd	11	5
M(AB)(CD)ef	20	10
$M(AB)_3$	4	2
M(ABA)cde	9	3
$M(ABC)_2$	11	5
M(ABBA)cd	7	3
M(ABCBA)d	7	3

Note: Capital letters represent chelating ligands and small letters represent monodentate ligands

14.11 STABILITY OF COORDINATION COMPOUNDS

Most of the complexes are highly stable. The interaction between metal ion and ligand may be regarded as Lewis acid-base reaction. If the interaction is strong, the complex formed would be thermodynamically more stable. Stability constants are represented by equilibrium constants. Let us consider the equilibrium reaction between metal ion (M^{a+}) and n ligands (L^{x-}) to form complex $[ML_n]^{b+}$

$$M^{a+} + nL^{x-} \rightleftharpoons [ML_n]^{b\pm}$$

Where a^+, x^- and $b^\pm$ are the charges on the metal, ligand and complex respectively. Charge balance requires that

$$\left(a^+\right) + n\left(x^-\right) = \left(b^\pm\right)$$

The stability constant for the above general reaction may be written as

$$K = \frac{\left[ML_n^{b\pm}\right]}{\left[M^{a+}\right]\left[L^{x-}\right]^n}$$

More the numerical value of K, more thermodynamically stable is the complex.

The reaction proceeds through stepwise addition of monodentate ligand. In solution the metal ion may exist as aqua complex. For convenience sake the charges on metal ion, ligand and complex are omitted.

$$\left[M(H_2O)_n\right] + L \rightleftharpoons \left[M(H_2O)_{n-1}L\right]H_2O$$

$$K_1 = \frac{\left[M(H_2O)_{n-1}L\right]}{\left[M(H_2O)_n\right][L]}$$

Since concentration of water is very large and essentially remains constant, the stability constant K_1 (equilibrium constant) is written without considering the concentration of water.

$$\left[M(H_2O)_{n-1}L\right]L \rightleftharpoons \left[M(H_2O)_{n-2}L_2\right] + H_2O$$

$$K_2 = \frac{\left[M(H_2O)_{n-2}L_2\right]}{\left[M(H_2O)_{n-1}L\right][L]}$$

Similarly ML_n is formed in different steps as

$$[M(H_2O)_{n-2}L_2] + L \rightleftharpoons [M(H_2O)_{n-3}L_3] + H_2O$$

to

$$[M(H_2O)L_{n-1}] + L \rightleftharpoons [ML_n] + H_2O$$

$$K_n = \frac{[ML_n]}{\left[M(H_2O)L_{n-1}\right][L]}\ {}^{n}$$

The overall reaction is

$$[M(H_2O)_n] + nL \rightleftharpoons [ML_n] + nH_2O$$

$$\beta_n = \frac{[MLn]}{[M(H_2O)_n][L]^n}$$

Where β_n is called as overall stability constant while K_1, K_2, …….K_n are called stepwise formation or stability constants.β_n is related to stepwise stability constants as $\beta_n = K_1 \times K_2 \times K_3 \ldots\ldots \times K_n$

β_n is used to determine the thermodynamic functions. As an example let us study the reaction of aqueous Cu (II) with NH_3

$$\left[Cu(H_2O)_4\right]^{2+} + NH_3 \rightleftharpoons \left[Cu(H_2O)_3NH_3\right]^{2+} + H_2O$$

$$K_1 = \frac{\left[Cu(H_2O)_3(NH_3)^{2+}\right]}{\left[Cu(H_2O)_4^{2+}\right][NH_3]} = 1.66\times10^4$$

$$\left[Cu(H_2O)_3NH_3\right]^{2+} + NH_3 \rightleftharpoons \left[Cu(H_2O)_2(NH_3)_2\right]^{2+} + H_2O$$

$$K_2 = \frac{\left[Cu(H_2O)_2(NH_3)_2^{2+}\right]}{\left[Cu(H_2O)_3(NH_3)^{2+}\right][NH_3]} = 3.16\times10^3$$

$$\left[Cu(H_2O)_2(NH_3)_2\right]^{2+} + NH_3 \rightleftharpoons \left[Cu(H_2O)(NH_3)_3\right]^{2+} + H_2O$$

$$K_3 = \frac{\left[Cu(H_2O)(NH_3)_3^{2+}\right]}{\left[Cu(H_2O)_2(NH_3)_2^{2+}\right][NH_3]} = 8.31\times10^2$$

$$\left[Cu(H_2O)(NH_3)_3\right]^{2+} + NH_3 \rightleftharpoons \left[Cu(NH_3)_4\right]^{2+} + H_2O$$

$$K_4 = \frac{\left[Cu(NH_3)_4^{2+}\right]}{\left[Cu(H_2O)(NH_3)_3^{2+}\right][NH_3]} = 1.51\times10^2$$

The overall reaction is

$$\left[Cu(H_2O)_4\right]^{2+} + 4NH_3 \rightleftharpoons \left[Cu(NH_3)_4\right]^{2+} + 4H_2O$$

$$\beta = \frac{\left[Cu(NH_3)_4^{2+}\right]}{\left[Cu(H_2O)_4^{2+}\right][NH_3]^4} = 6.58\times10^{12}$$

The overall stability constant value is equal to the product of the values of K_1, K_2, K_3 and K_4.

There is steady decrease in K values from K_1 to K_4 as the number of ligands increases. The reasons are (1) statistical factors (2) increased steric hindrance as the number of ligands increases if they are more bulky than the H_2O molecules (3) columbic factors due to charged ligands

The values of K for a few reactions are given below

$$Ag^+ + 2NH_3 \rightleftharpoons \left[Ag(NH_3)_2\right]^+ \quad K = 1.6\times10^7$$

$$Ag^+ + 2CN^- \rightleftharpoons \left[Ag(CN)_2\right]^- \quad K = 5.4\times10^{18}$$

$$Cu^{2+} + 4NH_3 \rightleftharpoons \left[Cu(NH_3)_4\right]^{2+} \quad K = 4.5\times10^{21}$$

$$Cu^{2+} + 4CN^- \rightleftharpoons \left[Cu(CN)_4\right]^{2+} \quad K = 2.0\times10^{27}$$

$$Co^{3+} + 6NH_3 \rightleftharpoons \left[Co(NH_3)_6\right]^{3+} \quad K = 5.0\times10^{33}$$

The values indicate that CN^- ion is a stronger ligand than NH_3 molecule because the stability constants for cyanide complexes are very large in comparison to corresponding ammine complexes.

14.11.1 Factors Affecting the Stability of a Complex

A very large amount of information on the stabilities of metal complex is now available. This permits an assessment of the various factors that influence the stability of a metal complex. These factors are summarized briefly here. First the stability of a complex obviously depends on the nature of the metal and of the ligand.

1. With reference to the metal the following factors are important
   - **(i) Size of the metal ion:** Smaller the size of the metal ion, the more stable are the metal complexes. If we consider the bivalent metal ions, then stability of their complexes (irrespective of the ligands) increases with decrease in the ionic radius of the central metal ion. The stability order of the complexes of the borderline metal ions (First transition series) with a given ligand.

$$Mn^{2+} < Fe^{2+} < Co^{2+} < Ni^{2+} < Cu^{2+} > Zn^{2+}$$

This order is known as **Irwing-william's series.** Chelating ligands did not follow this order. Also in the case of copper the order is not correct because this cannot form strong fifth and sixth coordinate complexes. The increasing stability in this series from Mn to Cu is a measure of inherent acidity of the metal (largely due to decreasing size)

   - **(ii) Charge on the central metal ion:** In general the greater the charge density on the central ion, the greater is the stability of its complexes. In other words, the greater the charge and smaller the size of an ion i.e., the larger the charge/radius ratio of an ion; the greater is the stability of its complexes. For example Fe^{3+} ion carries higher charge than Fe^{2+} ion but their size is about the same. So charge density is higher on Fe^{3+} than on Fe^{2+} ion. The complexes of Fe^{3+} ion are therefore more stable.

$$Fe^{3+} + 6CN^- \longrightarrow \left[Fe(CN)_6\right]^{3-} \quad K = 1.2\times10^{31}$$

$$Fe^{2+} + 6CN^- \longrightarrow \left[Fe(CN)_6\right]^{4-} \quad K = 1.8\times10^{6}$$

   - **(iii) Crystal field effects:** The crystal field stabilization energy (CFSE) plays an important role in the stability of transition-metal complexes. For analogous coordination compounds with in a group Δ values differ. The general trend being 3d < 4d < 5d. Thus while going from Cr to Mo or Co to Rh the Δ_0 increases by ~50%. As a consequence of this complexes of the second and third row transition series have greater stability as compared to the first transition series. For example the $\Delta_0$values for $[Co(NH_3)_6]^{3+}$, $[Rh(NH_3)_6]^{3+}$ and $[Ir(NH_3)_6]^{3+}$ are 274.6, 406.0 and 498.5 KJ mol^{-1} respectively. Because of these values the stability order of these complexes is $[Co(NH_3)_6]^{3+} < [Rh(NH_3)_6]^{3+} < [Ir(NH_3)_6]^{3+}$.
   - **(iv) Class 'a' and class 'b' metals:** The more electropositive metals for example, the metals of groups 1 and 2, the lanthanides and the early members of the transition series (groups 3 to 6) belong to class 'a' which have relatively a few electrons beyond an inert gas core. These form their most stable complexes with ligands in which the donor atom is N, O or F.

The less electropositive metals like heavy metals such as Pt, Pd, Hg, Rh, Ag, Ir, Au, Pb etc having relatively full d-orbitals belongs to class 'b'. These metals form stable complexes with ligands in which the donor atom is one of the heavier elements in the N, O or F families. It is believed that the stability of the complexes of the class 'b' metals results from an important covalent contribution to metal-ligand bonds and from the transfer of electron density from the metal to ligand via π - bonding

2. With reference to the role of the ligands in determining the stability of metal complexes, the following factors are important
   - **(i) Basic Strength:** The greater the base strength of a ligand, the greater is the tendency of the ligand to form stable complexes with class 'a' metals

A complex can be considered as an acid, containing "metal" in the place of H^+ and another species frequently an anion. A conjugate base of weak acid such as HCN is strong and also act as strong ligand. For example the order of acidic character of hydrogen halides is HF < HCl < HBr < HI. Thus the basic character in turn ligand strength of their anion will be in the order $F^- > Cl^- > Br^- > I^-$

   - **(ii) Size and charge of ligands:** For anionic ligands the higher the charge and the smaller the size, the more stable is the complex formed. Thus F^- ion form more stable complex than does Cl^- ion.
   - **(iii) Chelate effect:** Chelating ligands in general form more stable complexes than their monodentate analogs. This is known as the **chelate effect.** It is explained in terms of favourable entropy for chelation process.

In a qualitative way one can understand the favourable entropy. The replacement of a coordinated water molecule by either an NH_3 or 'en' molecule should be about equally probable. The replacement of a second water molecule by the other amine group in the coordinated 'en', however, is much more probable than its replacement by a free NH_3 molecule from solution, since the 'en' is already tied to the

Table 14.14 The effect of chelation on the stability of complexes

Complex	β_1	β_2	β_3	β_4	β_5	β_6
$\left[Ni(NH_3)_6\right]^{2+}$	5×10^2	6×10^4	3×10^6	3×10^7	1.3×10^8	1×10^8
$\left[Ni(en)_3\right]^{2+}$	5×10^7	1.1×10^{14}	4×10^{18}			
$\left[Ni(dien)_2\right]^{2+}$	6×10^{10}	8×10^{18}				
$\left[Ni(trien)(H_2O)_2\right]^{2+}$	2×10^{14}					

metal ion and the free end of the molecule is in the immediate vicinity of the H_2O, it will replace. Thus the formation of $[Ni(en)(H_2O)_4]^{2+}$ is more probable than the formation of the less stable $[Ni(NH_3)_2(H_2O)_4]^{2+}$.

Another way to visualize the more favourable entropy change is to realize that if one bidentate molecule enters into coordination entity two monodentate ligands are freed due to which entropy increases.

Ter,quadri and other polydentate ligands can replace 3, 4 or more monodentate ligands respectively to form even more stable complexes. The stability constants for some complexes of Ni^{2+} with polydentate ligands are listed in Table 14.14.

To find the effect of chelation compare β values that are markedly similar for example for the first two complex β_2 & β_1, β_4 & β_2 β_6 & β_3 should be compared

(iv) Chelate ring size: The stability of the complex ion has been observed to be dependent on the number of atoms in the ring. In general it has been observed that for ligands that do not contain double bonds and those that form five-membered metal-chelate rings give the most stable products. Ligands that contain **double bonds** form very stable metal complexes containing six-membered rings. Chelate rings containing either four atoms or more then six atoms have been observed but they are relatively unstable and uncommon.

(v) Steric strain: Because of steric factors large bulky ligands form less stable metal complexes than do analogous smaller ligands. For example (en) form more stable complexes than $(CH_3)_2NCH_2CH_2CH_2N(CH_3)_2$. The strain is some times due to the geometry of the ligand coupled with the stereochemistry of the metal complex. For example the straight chain 'trien' can form more stable complexes with Cu^{2+} than does the branched-chain amine $\left[N(CH_2CH_2NH_2)_3\right]$ having same number of donor atoms. Straight chain ligand can adopt the preferred square planar geometry but the branched chain ligand cannot.

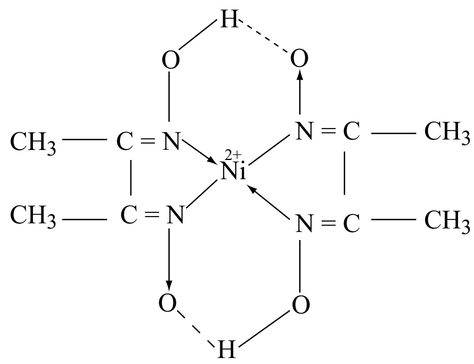


Fig 14.42 Structure of nickel dimethyl glyoxime

(vi) Hydrogen bonds: In addition to the chelate formation if intra molecular hydrogen bonds can be formed between the ligands, the stability increases further. For example the formation of hydrogen bonds in nickel-dimethyl glyoxime complex increases its stability

3. Environmental factors: Temperature and pressure. The complexes having volatile ligands (e.g., water, ammonia, ethylene diamine, carbon monoxide etc) are less stable at elevated temperature and commonly undergo decomposition on heating e.g., the hydrates lose water, the $\left[Co(NH_3)_6\right]Cl_3$ lose ammonia, the $\left[Cr(en)_3\right]Cl_3$ lose ethylene diamine (en) molecules and $(Ni(CO)_4)$ decompose into metal and carbon monoxide.

Secondly the effect of temperature is the transformation of certain complexes from one another. For example Ag_2HgI_4 is reversibly transformed at 45°C from a red to yellow form

$$\underset{\text{Red}}{AgHg\left[AgI_4\right]} \xrightarrow{45°C} \underset{\text{Yellow}}{Ag_2\left[HgI_4\right]}$$

The effect of pressure is the same as that of concentration

4. Concentration factors: The green species of $\left[CuCl_4\right]^{2-}$ complexes exist in the solid state but when dissolved in water, a pale blue hydrate of copper (II) ion gets formed which on addition of excess of chloride ion regenerates the green complex

$$\underset{\text{Green}}{\left[CuCl_4\right]^{2-}} \underset{4Cl^-}{\overset{X\ H_2O}{\rightleftharpoons}} \underset{\text{Blue}}{\left[Cu\left(H_2O\right)_4\right]^{2+}}$$

Similarly the blue $\left[Co\left(CNS\right)_4\right]^{2-}$ complex of cobaltous ion of the type undergoes the change as given below

$$\left[Co\left(CNS\right)_4\right]^{2-} \underset{4CNS^-}{\overset{6H_2O}{\rightleftharpoons}} \left[Co\left(H_2O\right)_6\right]^{2+}$$

5. Nature of the counter ion: If ions like $CN^-, SCN^-, Cl^-, Br^-, C_2O_4^{2-}$ and NO_3^- present as counter ions outside the coordination entity, they have a tendency to enter the coordination sphere substituting the ligands already coordinated to the metal

6. Role of pH: It is general rule that complexation of a metal ion by a ligand increases with decreasing hydrogen ion concentration, that is, with increasing pH.

14.12 APPLICATIONS OF COORDINATION COMPOUNDS

The ability of metal ions to form complexes with a variety of molecular species with different physico chemical properties has been used in many ways. In the recent years the complexes and the complex formation methods have been finding extensive uses. Some of these are given below.

1. Estimation of hardness of water: Hardness of water is estimated by simple titration with EDTA. The Ca^{2+} and Mg^{2+} ions form stable complexes with EDTA. The selective estimation of these ions can be done due to differences in the stability constants of calcium and magnesium complexes.

2. In water treatment: complex compounds like sodium metaphosphate are used to remove Ca^{2+} and Mg^{2+} ions from hard water by forming complexes with these ions this prevents the scale formation in boilers.

3. In Mordant dyeing: Mordants are insoluble substances which are uniformly deposited in the fibers to be dyed. These mordants can then attach to the molecules of the dyes by complex formation and help in fixing the dye to the fibers in a stable form. The important mordants used are $Fe\left(OH\right)_3$ and $Al\left(OH\right)_3$.

4. In Photography: In black and white photography to remove the unreacted AgBr after taking a photo, hypo is used as a fixing agent. Hypo forms a soluble complex with silver bromide $Na_3\left[Ag\left(S_2O_3\right)_2\right]$ complex.

5. In Medicine: Cis-chloroplatini, cis $\left[Pt\left(NH_3\right)_2Cl_2\right]$ is very good antitumor agent, causing regression of both fast and slow–growing cancers. This complex form diaquo complex which reacts with nitrogens of DNA bases, leading to changes in the DNA structure responsible for the anti-cancer effect.

The compounds like citric acid and cortisone which are present in body form complexes with lead, copper etc and prevent metabolism. Thus excess of copper and iron are removed by the chelating ligands D – pencilamine and desferrioxime B via the formation coordination. Compound EDTA is used in the treatment of lead poisoning.

6. In metallurgical process: Complex formation techniques are also used for the extraction of metals such as gold and silver. For example, silver is extracted from its ores by the cyanide process. In this process, silver passes into solution with formation of complex Na $[Ag(CN)_2]$

$$Ag_2S + 4NaCN \rightleftharpoons 2Na\,[Ag(CN)_2] + Na_2S$$

The solution containing the silver complex is removed and treated with zinc dust when silver precipitates out

$$2\,Na\,[Ag(CN)_2] + Zn \longrightarrow Na_2\,[Zn\,(CN)_4] + 2Ag.$$

7. In Electroplating: The coordination compounds of silver and gold are used as constituents of electroplating baths for controlled delivery of metal ions for reduction. For example, in electroplating bath for silver plating $K[Ag(CN)_2]$ is used as an electrolyte. Similarly, $K_3[Cu(CN)_4]$ is used for copper plating and K $[Au(CN)_2]$ is used in gold plating.

8. In catalysis: Certain coordination compounds act as catalysts for different reaction. For example penta carbonyl cobalt (II) acts as catalyst in the hydrogenation of alkenes. During hydrogenation Co(II) is oxidized to Co (III).

Organometallic compounds are also used as catalysts.

As homogeneous catalysis: Organometallic compounds are used as homogeneous catalysts for a variety of reactions in solution. For example Wilkinson's catalyst $[RhCl(PPh_3)_3]$ or chloro tris (triphenyl phosphine) rhodium (I) is used as a homogeneous catalyst in hydrogenation of alkenes.

As heterogeneous catalysis: organo metallic compounds are also used as heterogeneous catalysts in many reactions. For example Ziegler-Natta catalyst (solution of $TiCl_3$ and Trialkyl aluminium) is used as a catalyst for polymerization of olefins.

9. Purification of Nickel: Nickel forms a volatile complex $[Ni(CO)_4]$ at low temperature of about 50-80°C which decomposes again at 180°C into Ni and carbon monoxide. This reaction is used in the purification of nickel by Mond's process.

10. In qualitative analysis: In qualitative analysis complex formation is used in the detection of several anions and cations. Some are given as follows.

Identification of anions

(i) Sulphide ion can be identified by using sodium nitroprusside solution which gives purple colour with sulphide

$$S^{2-} + [Fe(CN)_5\,NO]^{2-} \rightarrow [Fe\,(CN)_5\,NOS]^{4-}$$

(ii) When neutral or slightly alkaline solution of thiosulphate is treated with the nickel ethylenediamine nitrate reagent forms crystalline violet precipitate is formed.

$$[Ni(en)_3]^{2+} + S_2O_3^{2+} \rightarrow [Ni\,(en)_3]\,S_2O_3$$

Identification of cations

(i) Potassium ferrocyanide can be used for the identification of iron, copper, zinc etc.

$3[Fe(CN)_6]^{4-} + 4Fe^{3+} \rightarrow Fe_4[Fe(CN)_6]_3$ Prussian blue ppt.

$[Fe(CN)_6]^{4-} + 2Cu^{2+} \rightarrow Cu_2[Fe(CN)_6]$ Chocolate brown ppt

$[Fe(CN)_6]^{4-} + 2Zn^{2+} \rightarrow Zn_2[Fe(CN)_6]$ White ppt.

(ii) Potassium ferricyanide can be used for the identification iron, copper, cobalt

$2[Fe(CN)_6]^{3-} + Fe^{+2} \rightarrow Fe_3\,[Fe(CN)_6]_2$ Turnbulls blue ppt.

$[Fe(CN)_6]^{3-} + Fe^{3+} \rightarrow Fe\,[Fe(CN)_6]$ Brown colouration.

$2[Fe(CN)_6]^{3-} + 3Cu^{2+} \rightarrow Cu_3[Fe(CN)_6]_2$ Green ppt

$2[Fe(CN)_6]^{3-} + 3Co^{2+} \rightarrow Co_3[Fe(CN)_6]_2$ Red ppt

(iii) Sodium hexanitrocobaltate (III) can be used for the identification of potassium

$$3K^+ + [Co(NO_2)_6]^{3-} \rightarrow K_3[Co(NO_2)_6]$$

(iv) Alkaline solution of potassium tetraiodido hydrargate (II) (Nessler's reagent) can be used for the identification of ammonium ion.

$$NH_4^+ + 2[HgI_4]^{2-} + 4OH^- \rightarrow \underset{\text{Brown ppt}}{H_2N\cdot HgO\cdot HgI} + 7I^- + 3H_2O$$

(v) Complexing agent, dimethyl glyoxime is used for the identification of nickel and palladium. Nickel dimethylglyoxime complex is cherry red precipitate while palladium complex is yellow precipitate.

(vi) The complex formation is used for separation and identification of cations since there is a change of colour in the solution.

$$\underset{\text{blue}}{Cu^{2+}} + 4NH_3 \rightarrow \underset{\text{dark blue}}{[Cu(NH_3)_4]^{2+}}$$

$$\underset{\text{pale green}}{Fe^{2+}} + 6CN^- \rightarrow \underset{\text{yellow}}{[Fe(CN)_6]^{4-}}$$

$$\underset{\text{Green}}{Ni^{2+}} + 6NH_3 \rightarrow \underset{\text{Blue}}{[Ni(NH_3)_6]^{2+}}$$

Separation of cations

The ligand cyanide ion can be used for the separation of copper and cadmium in the second group. The complexes formed have different stabilities

$$CuSO_4 + 2KCN \rightarrow Cu(CN)_2 + K_2SO_4$$

$$2Cu\,(CN)_2 \rightarrow Cu_2(CN)_2 + (CN)_2$$

$$Cu_2(CN)_2 + 6KCN \rightarrow K_3\,[Cu\,(CN)_4]$$

$$CdSO_4 + 2KCN \rightarrow Cd(CN)_2 + K_2SO_4$$

$$Cd(CN)_2 + 2KCN \rightarrow K_2\,[Cd(CN)_4]$$

Separation of IIA and IIB Groups

Elements of IIA Group can be separated from those of IIB by using yellow ammonium sulphide. IIB group cations form soluble complexes while those of IIA do not form.

$$As_2S_3 + 3(NH_4)_2\,S \rightarrow 2(NH_4)_3AsS_3$$

$$As_2S_5 + 3(NH_4)_2\,S \rightarrow 2(NH_4)_3AsS_4$$

$$Sb_2S_3 + 3(NH_4)_2\,S \rightarrow 2(NH_4)_3SbS_3$$

$$Sb_2S_5 + 3(NH_4)_2\,S \rightarrow 2(NH_4)_3SbS_4$$

$$SnS + (NH_4)_2\,S \rightarrow (NH_4)_2SnS_2$$

$$SnS_2 + (NH_4)_2S \rightarrow (NH_4)_2SnS_3$$

Selective dissolution of a precipitate from a mixture one can be separated from another. For example if ammonia is added to a mixture of AgCl and $PbCl_2$, AgCl dissolves while $PbCl_2$ is insoluble and can be separated

$$AgCl + 2NH_3 \rightarrow [Ag(NH_3)_2]Cl$$

Quantitative Analysis

Many organic complexing ligands are used for the quantitative precipitation and determination of metal ions since they are often insoluble in aqueous solution if the complex is unstable at the drying temperatures, they can be ignited to metal oxides. Some examples are

(i) 8- Hydroxy quinoline (oxime) is used – for the determination of magnesium, zinc, copper, cadmium, lead, indium with the compositions $M(C_9H_6ON)_2$; for the determination of aluminium, iron, bismuth, gallium with the composition $M\,(C_9\,H_6\,ON)_3$.

(ii) α- Benzoin oxime (cupron) is used for the determination of copper with the composition $Cu(C_{14}\,H_{11}O_2N)$ it can also be used for the determination of molybdenum.

(iii) α- Nitroso – β- napthol is used for the determination of cobalt Co $(C_{10}H_2O_2N)_3$ in the presence of nickel.

In volumetric analysis: EDTA is used for the determination of most of the cations titrimetrically (complexometric titrations). The indicators used are also complexing agents whose metal complexes have different colours from the reagent. These complexometric titrations are used for the determination of hardness of water.

14.12.1 Biological Importance

Many biological important natural compounds exist as coordinated complex. For example, chlorophyll, vitamin B_{12} and haemoglobin

Haemoglobin: Haemoglobin is the red pigment in the red-blood cells (erythrocytes). It is a globular protein consisting of four polypeptides chains arranged tetrahedral. The prosthetic non-protein group is "heme". This plus the "globin" is together haemoglobin. Haemoglobin contains one porphyrin ring and one iron atom.

Porphyrins bind their pyrrole nitrogen (four) atoms to ferrous iron (Fe^{2+}) to form Fe complexes. These complexes are called **hemes.** Metallophorphyrin containing ferric ion (Fe^{3+}) is called **hematin.** The other two co-ordinate linkages besides the four linkages with nitrogen atoms in porphyrin ring will be formed perpendicular to the plane of the heme. Of these the fifth linkage to Fe^{2+} is to N-atom from histidine (an amino acid present in globin) the sixth position around Fe^{2+} is occupied by O_2 molecule or H_2O molecule. The basic structure is shown in Fig 14.43.

During respiration haemoglobin combines with oxygen forming oxy haemoglobin

$$Hb + O_2 \rightarrow HbO_2$$

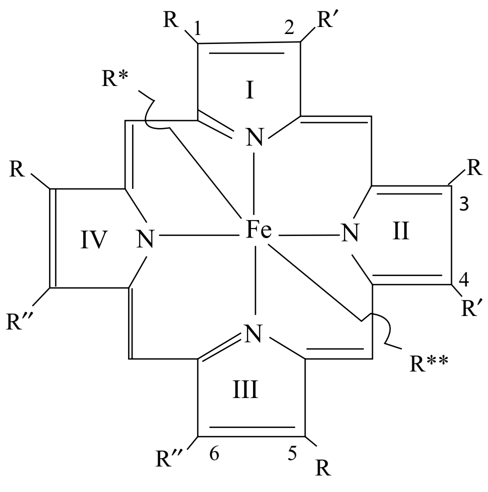


Fig 14. 43 Basic structure of Heme-b R = CH_3; R′ = CH = CH_2; R″ = CH_2CH_2 COOH; R*-Imidazole group of histidine residue; R**-Unoccupied or O_2 or other ligands

This process is not oxidation but oxygenation. Iron still remains in +2 state only. In oxyhaemoglobin Fe^{2+} is diamagnetic. In deoxyhaemoglobin Fe^{2+} is paramagnetic. Porphyrin is planar and rigid. The size of Fe^{2+} is increased by 28% when it changes from diamagnetic to paramagnetic state (61 pm to 78 pm) during respiration. In deoxyhaemoglobin Fe^{2+} is too big to fit into the centre of porphyrin and Fe^{2+} is situated 70 to 80 pm above the ring. This distorts the bonds around Fe^{2+}.

Chlorophyll: Chlorophyll is the green pigment present in leaves of plants. It is two types namely "chlorophyll a" and "chlorophyll b". These are the characteristic pigments of all the photosynthetic organisms. The chlorophylls have tetrapyrrole ring structure similar to that present in "heme" but Mg^{2+} ion is at the centre instead of Fe^{2+} ion. The "chlorophyll a" absorbs light of wavelength 4300 A^0 (violet blue) and 6800 A^0 (red). The green colour is due the transmitted light. The absorbed light is used in photosynthesis in the plants.

In photosynthesis CO_2 and H_2O vapour of air chemically combine under the influence of sunlight in the presence of chlorophyll in the green plants

$$6CO_2 + 6H_2O \xrightarrow[\text{sunlight}]{\text{Chlorophyll}} C_6H_{12}O_6 + 6O_2$$

The basic structure of chlorophyll is shown in fig 14.44.

Vitamin B_{12} (Cyanocobalamine): The vitamin B_{12} molecule has two characteristic components. The first part is nucleotide like structure and the second part a corrin ring system. The corrim ring system resembles porphyrins containing 4 pyrrole rings but two rings are directly joined rather than through =C- bridge. Coordinate to the four inner nitrogen atoms of the corrin ring is Co^{3+} ion is through the nitrogen atom of the amide group of the nucleotide. The sixth coordination position is occupied by cyanide (-CN) group. The vitamin B_{12} is called cyanocobalamin.

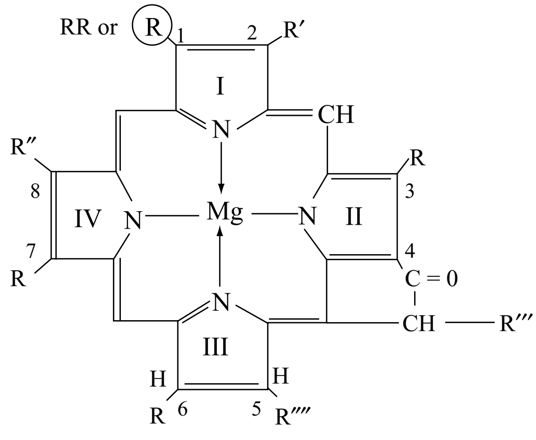


Fig 14. 44 Structure of a R = CH_3; Ⓡ = CHO in chlorophyll b R′ = C_2H_5; R″ = CH = CH_2; R‴ = $COOCH_3$; R⁗ = $CH_2CH_2COOC_{20}OH_{39}$

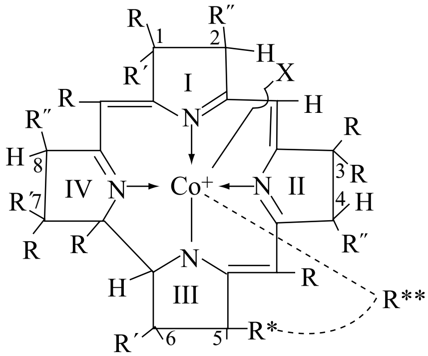


Fig 14. 45 Basic structure of vitamin B_{12} (Corrin ring part is shown) R = CH_3; R′ = CH_2CONH_2; R″ = CH_2CH_2 $CONH_2$; R*–R** = α – 5, 6-dimethyl Benzimidazole ribo-nucleotide

Vitamin B_{12} maintain normal bone marrow function for producing erthrocytes (RBC). The lack of enough cobalt leads to nutritional type anaemia excess of cobalt produces of erythrocytes causing polycythemia. The basic structure of vitamin B_{12} is shown in Fig 14.45.

Enzymes are the catalysts of biological systems, which are complex compounds. They not only control the rate of reactions but by favoring certain geometries in the transition state can lower the activation energy for the formation of one product rather than another.

KEY POINTS

Double Salts and Complex Compounds

- Compounds formed by the addition of two or more stable compounds exist only in solid state but ionize completely in solution and give test for all the ions present in it are called **double salts** e.g., carnalite KCl. $MgCl_2 \cdot 6H_2O$, potash alum $K_2SO_4 \cdot A1_2(SO_4)_3 \cdot 24H_2O$
- Compounds formed by the addition of two are more stable independent compounds exist in both solid state and in solution, and do not ionize in solution do not give tests for all the ions present in them are called **complex compounds** e.g., $K_3[Fe(CN)_6]$ potassium ferricyanide contains the anion $[Fe(CN)_6]^{3-}$ Which is not present in either KCN or $Fe(CN)_3$ from which it is formed.
- The physical properties such as colour, electrical conductivity, optical activity, solubility of the complex species are distinctly different from the substances from which they are formed.
- The chemical properties of complex compounds also completely changes e.g., $Fe(OH)_3$ does not precipitate by the addition of OH^- to a solution of $[Fe(CN)_6]^{4-}$ or $[Fe(EDTA)]^{2-}$ where as $Fe(OH)_2$ will be precipitated by the addition of OH^- ion to any soluble compound of Fe^{2+}.

Werner's Theory and Terms

- **Alfred Werner** first proposed the theory how complexes are formed for which he won the Noble prize.
- Metals possess two types of valencies:
  (i) Primary valencies which are ionizable
  (ii) Secondary valencies which are not ionizable
- The number of **primary valencies** are equal to the **oxidation number** of the metal. Primary valencies can be satisfied only by **negative ions.**
- Secondary valencies can be satisfied by negative ions or neutral molecules or rarely by positive ions. Every metal has a fixed number of secondary valencies known as **coordination number.** The ions or molecules that satisfy secondary valencies are known as **ligands** and the metal ion is known as **central ion.**
- Secondary valencies are directional in nature and hence coordination compounds exhibit stereoisomerism.
- The coordination number is previously considered to be a fixed number for a particular metal, but many complexes are known in which the same metal ion has more than one C.N. The C.N. of a metal atom/ion is determined only by the number of a bonds formed by the ligand.
- The central metal atom/ion and the ligands attached to it are enclosed in square brackets and is collectively termed ad **coordination sphere** or **inner sphere.** The ionizable sphere.
- The spatial arrangement of the ligand atoms around the central metal atom/ion gives the **coordination polyhedra.** The common are octahedral, square planar and tetrahedral.
- The **oxidation number** of the central atom in a complex is defined as the charge it would carry if all the ligands are removed along with the electron pairs that they shared with the central atom.

- Complexes in which a metal is bound to only one kind of ligands are known as **homoleptic complexes** while those surrounded by more than one kind of ligands are known as **heteroleptic complexes.**

Sidgwick Theory

- Transition metal ions are **Lewis acids.** Ligands are **Lewis bases.** Transition metal ions form complex compounds by accepting the lone pair of electrons from lignads. The bond present between the transition metal ion and ligand is **co-ordinate covalent bond.** Transition metal ion should contain vacant orbitals to accept the electron pairs from ligands. Ligand should contain at least one **lone pair** of **electrons.**
- **Effective atomic numbers (EAN)** is the total number of electrons present around central ion in a complex.
  EAN = [Atomic No. of metal] – [Number of electrons lost in the formation of its ion] + [Number of electrons gained from ligands]
- Sidgwick proposed that a complex is stable if the EAN of the metal ion in a complex is equal to the atomic number of the nearest inert gas.
- **Drawbacks of Sidgwick Theory:**
  (i) The donation of electron pairs to a central cation would produce an improbable accumulation of negative charge on the metal atom/ion.
  (ii) Several ligands such as H_2O, NH_3 the electron pair to be donated is of 2s pair which have no bonding characteristics.
  (iii) This theory could not explain the magnetic behaviour, colour, geometrical shapes of complexes and also the type of metal orbital involved in bonding.

Valence Bond (VB) Theory

- Valence bond theory, to explain bonding in complexes, was proposed by **Linus Pauling.** The main postulates of VB there are as follows.

Coordination number	Type of hybridization	Molecular geometry
2	sp	linear
3	sp^2	trigonal planar
4	sp^3	tetrahedral
4	dsp^2	square planar
5	sp^3d or dsp^3	Trigonal bipyramid
6	sp^3d^2 or d^2sp^3	octahedral

(i) The central atom loses a requisite number of electrons from the cation. The number of electrons lost is equal to the valence of the resulting cation.
(ii) The central cation makes available a number of vacant orbitals equal to its coordination number for the formation of dative bonds with the ligands.
(iii) The cation orbitals hybridize to form new set of equivalent hybrid orbitals with definite directional characteristics.
(vi) The non-bonding metal electrons occupy the inner d-orbitals and do not participate in the hybridization.
(v) In the presence of strong ligands such as CN^-, NO, CO the *d* electrons are rearranged vacating some d-orbitals (when the number of d-electrons are more than 3 only) which can participate in hybridization.
(vi) In the presence of weak ligands such as F^-, Cl^-, H_2O etc the d-electrons are not rearranged.
(vii) The d-orbitals involved in the hybridization may be either (n-1)d orbitals or outer d-orbitals. The complexes formed by the involvement of (n - l) d orbitals in hybridization are called **inner orbital complexes** or **low spin complexes.** The complexes formed by the involvement of d - orbitals of outer orbit are called outer orbital complexes or high spin complexes.
(viii) Each ligand contains a lone pair of electrons. A dative bond is formed by the overlap of a vacant hybrid orbital of metal ion and a filled orbital of ligand.
(ix) The complex will be paramagnetic, if any unpaired electrons present, otherwise diamagnetic.
(x) The number of impaired electrons in a complex gives out the geometry of the complex or vice versa.

(For Examples refer to Table 14.7)

Some Important Points

- In acid solution NH_3 molecule cannot act as a ligand because it is protonated by donating its lone pair to H^+ ion, so always NH_3 form complexes either in neutral or in alkaline solution.
- In low concentration of NH_3 metal ions are precipitated as their hydroxide but dissolves in excess of NH_3 due to formation of complex e.g., Zn^{2+}, Cu^{2+}, Ni^{2+}, Ag^+ etc. The formation of complex depends on the concentration of NH_3 and the stability of the complex formed.
- Cation of iron (III) and aluminum (III) always precipitate as hydroxide with ammonia because the stabilities of their ammonia complexes are insufficient to dissolve the hydroxide.

- Cobalt in +2 oxidation is stable in aqueous solution but unstable in the presence of strong ligands such as CN^-, NH_3 and converts easily to Co^{3+}
- In $[Co(C_2O_4)_3]^{3-}$ through oxalate ion is weaker ligand than water, pairing of electrons takes place and it is a spin paired, inner orbital complex with diamagnetic character.
- All six coordinate complexes of Ni^{2+} are octahedral and are outer orbital complexes with paramagnetic character.
- All copper complexes in +1 oxidation state have tetrahedral structure and are diamagnetic.
- Six coordinate complexes of copper (II) are unstable because the ligands along the z-axis are at longer distances than in *xy* plane in the octahedral structure due to distortion because of unequal distribution of electrons.
- The four coordinate complexes of copper (II) are square planar. For this structure to have *dsp*^{2} hybridization one electron from 3d orbital is promoted to 4p orbital.
- The halide complexes such as $[CuCl_4]^{2-}$ have square planar structure when the counter ion is NH_4^+ ion and tetrahedral structure if the counter ion is Cs^+ ion. Tetrahedral $[CuCl_4]^{2-}$ is orange and square planar $CuCl_4^{2-}$ is yellow in colour.
- The transition metals of 2nd and 3rd rows always form inner orbital complexes i.e., spin paired complexes irrespective of the strength of ligand.

Defects in VB Theory

- VB theory could not explain the colour and absorption spectra of complexes. It could not explain the stability of complexes. It could not explain why the electrons in d-orbitals rearrange in the presence of certain ligands against to Hund's rule while in the presence of other ligands the electronic configuration is not disturbed. It fails to provide satisfactory explanation for the existence of inner orbital and outer orbital complexes. Some times the VB theory needs the transfer of electron from lower energy level (3d) to the higher energy level (4P) which is very unrealistic in the absence of energy supplier.

Crystal Field Theory

- Crystal field (CFT) was originally proposed by Bethe and **Vanvleck** for explaining the optical properties of crystalline solids and it was extended by Orgel to explain the formation of complexes.

Crystal Field Splitting of d-Orbitals in Octahedral Complexes

- The five d-orbitals in an isolated gaseous metal atom/ion have same energy i.e, they are degenerate.
- When ligands approach along the X,Y and Z axes to form octahedral complex the $d_{x^2-y^2}$ and d_{z^2} orbitals are repelled more than d_{xy},d_{yz} and d_{xz} orbitals. The. net result is that the energy or $d_{x^2-y^2}$ and d_{z^2} orbitals is raised relative to the energy of d_{xy}, d_{yz} and d_{xz} orbitals i.e., the degeneracy of d-orbitals is destroyed and split up into two sets as shown in Fig 14.5. The group of d -orbitals d_{xy}, d_{yz} and d_{xz} having lower energy are designated and labeled t_{2g} and the group of d-orbitals $d_{x^2-y^2}$ and d_{z^2} having more energy are designated and labeled e_g.
- The energy difference between the two sets of d-orbitals t_{2g} and e_g is represented by Δ_o, the subscript o represents the octahedral field.
- For each electron entering into t_{2g} orbital the decrease in energy is about 0.4 Δ_o (2/5 Δ_o) and the complex will be 0.4 Δ_o more stable. This 0.4 Δ_o is called **the crystal field stabilization energy** (CFSE).
- For each electron entering into e_g orbital, the destabilizing energy is 0.6 Δ_o.
- If Δ_o is less than pairing energy as in the case of weak field (weak ligands) the fourth electron enters into one of the e_g orbitals. when Δ_o is more than pairing energy as in the case of strong field (strong lignds) pairing will occur in the level.
- Coulombic exchange energies also play a role in the stabilizaton of complex.
- The relationship between the difference in the t_{2g} and e_g energy levels. Δ_o, the pairing energy (p) and the exchange energy (E) determines the orbital configuration of electrons and are given in Table 14.8.

Crystal Field Splitting in Tetrahedral Complexes

- The repulsion by ligands on d-orbitals is less by about 2/3 times compared to the octahedral field because the ligands are directed away from the d-orbital, Also the repulsion of 4 ligands in tetrahedral complex is less by about another 2/3 times compared to the repulsion by 6 ligands in octahedral field. Thus the tetrahedral crystal field splitting energy Δ_t is approximately 4/9 of the octahedral crystal field splitting energy Δ_o.
- The Δ_t value is always much smaller than Δ_o and never energetically favourable to pair electrons and **all tetrahedral complexes are high spin complexes.**
- In tetrahedral field the t_{2g} orbitals are near to the ligands compared to e_g orbitals and hence the t_{2g} set of orbitals become the higher energy group while the e_g set of orbitals becomes lower energy group. So for each electron entering into t_{2g} orbitals the stability of the complex decreases by about $0.4\Delta_t$.

- The value of Δ can be determined experimentally by measuring the wavelength of the radiation absorbed whose energy is equal to Δ for the excitation of electron from lower energy group of d-orbitals to higher energy group of orbitals.

$$\Delta = h\nu \text{ or } hc/\lambda$$

- Where h is Planck's Constant, c is the velocity of light and λ is the wavelength of light absorbed.

Factors Affecting Δ

- The value is more in octahedral splitting of d-orbitals than in tetrahedral splitting for the same metal ion and ligands.
- Different ligands with same metal ion produce different values due to difference in the strength of repulsion shown on d-orbitals. The arrangement of various ligands in the order of their strength with same metal ion is called spectro chemical series.

Weak Field Ligands

$I^{-1} < Br^- < \underline{S}CN^- < Cl^- < N\underline{O}^-{}_2 < N^{3-} < F^- < OH^- < C_2O_4^{2-} < H_2O <$
$\underline{N}CS < CH_3CN < Py < NH_3 < en < bipy < Phen < \underline{N}O_2^- < PPh_3 < CN^- < CO$

Strong Field Ligands

- The donor atom in ambidentate ligand is underlined
- A metal ion with higher charge attracts the ligands near to it thus causing more Δ.
- In a transition group Δ value increases by ~50% from 3d series element to 4d series element and another ~30 % from 4d series element to 5d series element. This is because of increase in the size of 4d and 5d orbitals compared with the compact 3d orbitals and the consequent stronger interactions with the ligands.

Application of Crystal Field Theory

- **Colour:** The colour of the transition metal compounds is due to the *d-d* transition. The coordination compounds absorb radiation of appropriate energy for the excitation of the electrons from lower energy d-orbital to higher energy d-orbitals. So the complex will appear in the complementary colour of the absorbed radiation.
- Since the degree of splitting of 3d-levels depends upon the strength of ligands the colour of a particular transition metal ion also vary with the change in the type of ligand with which it is bound.
- The oxoanions such as $Cr_2O_7^{2-}$, CrO_4^{-2} and MnO_4^- are coloured not due to d-d transitions but due to different phenomenon known as **charge transfer** phenomenon.
- **Magnetic properties** The crystal field theory is successful in calculating the number of unpaired electrons in high spin and low spin complexes. From thë knowledge of unpaired electrons and spin only moment it is possible to know (i) the valence state of the metal ion and (ii) the nature of bonding in a complex that contain more than three 3d electrons. For example, the magnetic moment 4.5 BM of $Hg[Co(CNS)_4]$ indicate that cobalt is in bivalent state (Co^{2+}' with d^7) and the bonding belongs to spin free type corresponding to n = 3.
- The magnetic moment of $[Fe(diph)_3](ClO_4)_3]$ is 2.2 BM indicating that iron is in the trivalent state (Fe^{3+} with d^5 config) and bonding belongs to spin paired type corresponding n = 1.
- In the complexes (i) $[Fe(H_2O)_5NO]^{2+}$ (ii) $[Fe(CN)_5NO]^{-3}$ (iii) $[Fe(H_2O)_5NO]^{3+}$ and (iv) $[Fe(CN)_5NO]^{2-}$ the magnetic moments are different from those expected. In (ii) iron is in +2 oxidation state but in the complexes (i), (iii) and (iv) the oxidation state of iron changes from +2 to +1 in (i) and +3 to +2 in (iii) and (iv) due to transfer of electron from NO to iron atom.
- **Stability of complexes:** The CFT explains why certain oxidation states are preferentially stabilized by coordination to specific ligands. For example water stabilizes Co^{2+} where as ammonia stabilizes Co^{3+} because in the presence strong ligands such as NH_3 the electronic configuration is $t_2g^6\ e_g^1$ for Co^{2+} and the electron present in higher energy e_g^1 can be easily removed so that it is oxidized to Co^{3+}.
- Stereochemistry of the complexes can be explained by CFT e.g., copper (II) forms square planar complexes because it possess a much higher CFSE (–12.28 Dq) or tetrahedral complex (CFSE = 22.56 Dq)
- **Draw backs of CFT:** CFT is based on mainly electrostatic attraction between metal and ligand which is quite unrealistic. It do not consider about the partial covalent character of the metal-ligand bonds.
- CFT could not explain for the relative strengths of ligands e.g., why negative OH^- is weaker ligand than neutral H_2O.
- CFT did not considered the metal orbital (except d orbitals) such as s, p_x p_y, p_z and ligand π-orbitals and unable to consider the existence of π bonding in complexes.

Types of Ligands

Classification Based on the Number of Donor Atoms

- Monodentate ligands contain only one donor atom ex. Cl^-, Br^-, H_2O, NH_3 etc.

- Polydentate ligands contain two or more donor atoms. Bidentate ligands contain two donor atoms e.g., ethylene diamine ($H_2N—CH_2—CH_2—NH_2$) acetyl acetone ($CH_3C\ddot{O}CH_2C\ddot{O}CH_3$). Bidentate ligands are two types symmetric bidentate ligands like ethylene diamine and asymmetric bidentate ligands such as glycinate ion ($H_2NCH_2COO^-$).
- Tridentate ligands contain three donor atoms e.g., dien (diethylene triamine) $H_2NCH_2CH_2NHCH_2CH_2NH_2$. Similarly tetradentate ligands contain four donor atoms, pentadentate ligands contain five donor atoms, hexadentate ligands contain six donor atoms. EDTA is an example to hexadentate ligand.
- If a polydentate ligand do not use all its donor atoms to coordinate to metal ion, it is called **flexidentate** ligand.

Types of Monodentate Ligands

- Monodentate ligands are three types depending on the charge they carry (i) neutral e.g., NH_3,H_2O (ii) Negative e.g., Cl^-, F^-, CN^- and (iii) Positive e.g, NO^+_2, $H_3^+NNH_2$
- Monodentate ligands which contain two or more atoms, but in forming complexes if they use only one donor atom they are called ambidentate ligands. e.g., SCN^- NO^-_2 etc.
- Unsaturated organic compounds like alkenes or alkynes can donate the electron pair in their π bonds and these are called π electron donors e.g., $CH_2 = CH_2$, $HC \equiv CH$ etc.
- Some monodentate ligands may simultaneously coordinating with two or more metal atoms. Such ligands are called bridging ligand and the complex is called **bridging** complex e.g., dimer of $FeCl_3$, Fe_2Cl_6 and polymer of $PdCl_2$.

Classification Based on π-Donor and π-Acceptor Property

- π-donor ligand contain extra lone pair in orbital of π-symmetry around M — L axis and form an extra π dative bond with metal ion M ← L with a normal σ dative bond.
- π- acceptor ligand contain a vacant π-orbitals and can form an extra π-dative bond by accepting a lone pair from metal ion to the normal σ dative bond from ligand $M \rightleftarrows L$
- Increasing order of Δ_0

  π Donor < weak π donor < no π effect < π acceptor
- Examples for such ligands. π donors I^-, Br^-, Cl^-, F^-; weak π donor H_2O; no π effects NH_3 and π acceptors PR_3, CO
- π acceptor ligands are weak σ donors to Lewis acids such as BF_3 and $AlC1_3$ but form strong complexes with transition elements since the drift of π electron density from M → L tends to make the ligand more negative and so enhances its σ donor power and this is known as synergic effect.
- The order of the strength of π acceptor ligands is CN^- < RNC << CO < NO^+ · CN^- is a weak π acceptor but σ donor ligand due to the presence of negative charge

Bonding in Organometallic Compounds

- Organometallic compounds are those which contain one or more **metal-carbon bonds.** The first organometallic compound Ziese salt $K[PtCl_3(C_2H_4)]$ was prepared by W.C. Ziese.
- Ferrocene or bis (cyclopentadienyl) iron [Fe(η^5 - $C_5H_5)_2$] is sandwich compound. Iron atom is present between two C_5H_5 rings. The planes of the rings are parallel so that the carbon atoms are at the same distance from iron atom.
- **Metal carbonyls** are considered as organometallic compounds in which carbon monoxide acts as ligands.
- Homoleptic binary metal carbonyls have simple well defined structures e.g., $Cr(CO)_6$ is octahedral; $Fe(CO)_5$ is trigonal bipyramid and $Ni(CO)_4$ is tetrahedral.
- Some metals form bridged carbonyls. Some of the bridged carbonyl contain M—M bond and some may not. If the EAN of metal is equal to nearest inert gas configuration M—M bond does not takes place in bridged bond (binuclear) carbonyl. If the EAN of metal is not equal to the nearest inert gas configuration M—M bond occurs in the binuclear complexes.
- In metal carbonyls CO donates the lone pair present in HOMO i.e., non-bonding MO of carbon which has less donating capacity due to more electronegative character of oxygen So M→ CO bond is weak. Now the metal atom forms a π -dative bond with CO by donating its lone pair into the LUMO of CO i.e., π anti-bonding MO of CO
- Due to this synergic effect the $M \rightleftarrows C$ bond becomes stronger but because the lone pair, of metal atom goes into π antibonding MO the bond order in CO decreases while the C—O bond length increases from the normal C—O bond length 112.8 pm to ~115 Pm.
- In isoelectronic species $[Ti(CO)_6]^{2-}$ $[V(CO)_6]^-$, $Cr(CO)_6$ and $[Fe(CO)_6]^{2+}$ back donation from metal to CO increases with increase in negative charges and decreases with increase in positive charges as the negative charge makes the metal to back donate the pairs easily while the positive charge inhibit the metal

to back donate. So the order of C—O and M—C bond lengths will be

$$C—O: [Ti(CO)_6]^{2-} > [V(CO)_6]^- > Cr(CO)_6 > [Mn(CO)_6]^+ > [Fe(CO)_6]^{2+}$$

$$M—O: [Ti(CO)_6]^- < [V(CO)_6]^- < Cr(CO)_6 < [Mn(CO)_6]^+ < [Fe(CO)_6]^{2+}$$

- The strength and bond length of M—C and C—O also depend on the number of available lone pair electrons per CO molecule. For example in a complex like $M(CO)_6$ having three lone pairs in metal d-orbitals, if some CO ligands can be substituted by certain other ligands L which are without π -acceptor orbitals such as NH_3 or aliphatic amino nitrogen like $ML_3(CO)_3$. Comparing these two $M(CO)_6$ and $ML_3(CO)_3$ in $M(CO)_6$ three electron pairs are back donated to six CO molecules where as in $ML_3(CO)_3$. Three electron pairs are back donated to three CO molecules. Therefore the C—O bond order in these two cases should then be respectively 2.5 and 2.0. So the C—O bond length in $ML_3(CO)_3$ is longer than in $M(CO)_6$.
- In $Cr(CO)_6$,$Fe(CO)_5$ and $Ni(CO)_4$ the number of lone pairs available are 3,4 and 5 respectively. So the share of dπ electrons from Cr, Fe and Ni for back donation to each CO in these complexes will be 6/3, 4/5 and 5/4. Hence the C—O bond length in these complexes will be in the order $[Cr(CO)_6] < [Fe(CO)_5] < [Ni(CO)_4]$.

NOMENCLATURE OF COORDINATION COMPOUNDS

- The basic rule for writing the name of coordination compounds are as follows.
- The positive ion is named first followed by the negative ion. Name of the non-ionic or neutral complex must be written in word.
- Names of the neutral ligands should be written as such. Exceptions are H_2O-aquo; NH_3-ammine; CO-carbonyl; NO-nitrosyl; CS-thiocarbonyl; NS-thionitrosyl.
- Names of the negative ligands must be ended with 'O'. e.g., Cl - Chlorido, Br - Bromido, I - Iodido, F - Flurido, $CH_3COO \sim$ - acetato, CO_3^{2-} - carbonato, $C_2O_4^{2-}$ - oxalato, NO_3^- - nitro, NO_3^- - nitrato, S^{2-} - sulphido, SO_4^{2-} - sulphato, NH_2^- - amido, NH^{2-} - imido, OH^- - hydroxido, O_2^{2-} - peroxido etc.
- The names of the positive ligands should be ended with ium e.g., NO_2^+ - nitronium; $N_2H_5^+$ - hydrazinium.
- If the same complex compound contains different ligands their names should be written in alphabetical order.
- Prefixes di, tri, tetra, penta, hexa are used before the ligands to mention their number.
- If the same ligand can donate lone pairs from more than one centre. They are named as bidentate, tridentate, etc depending on the number of lone pairs donated.
- The number of bidentate, tridentate etc ligands is mentioned with bis, tris, tetrakis etc, if they already contain di, tri etc in their names.
- If the complex ion is +ve then the name of metal is written normally. If the complex ion is -ve ion the name of the metal should be ended with ate. If the complex ion is anion and symbol of the metal is taken from latin language, their names should be taken from latin language. e.g., cuprum-cuprate; ferrum-ferrate, stannum-stannate; plumbum-plumbate; argentum- argentate; aurum-aurate; hydrorgium-hydrorgate; wolfram-wolframate.
- The oxidation number of the central metal ion should be mentioned in the roman numerals in the parenthesis immediately after the name of the metal ion.
- While naming the ambidentate ligands. The ligands are named after point of attachment e.g., SCN^-, thiocyanate or thiocynato-S and for NCS as isothiocyanato or thiocyanato-N.
- If the complex contain two or more metal atoms, the bridging ligands are indicated by prefix μ before the name of such ligands and are separated by hyphen (–).
- Water molecules of crystallization are indicated after the name of the complex. Arabic numerals are used to indicate the number of such molecules.
- Geometrical isomers are named either by using the prefixes cis for adjacent (90° apart) positions and trans for opposite (180° apart) positions before the name of the ligands or by numbering system.
- Dextro and leavo rotatory optically active compounds are designated either by (+) and (–) or by d- and l- respectively.

Structural Isomerism

- Compounds having same composition but yield different ions in solution are called **ionization isomers**. Ionization isomers are formed by the interchange of the position of ligands with the counter ion which itself can act as a ligand. Ionization isomers can be distinguished by using conductance measurements or by chemical tests, e.g., $[Co(NH_3)_5C1]SO_4$ and $[Co(NH_3)_5SO_4]Cl$.
- Hydrate isomerism is due to the difference in the position of water molecules in a complex as ligand and hydrated molecules e.g., $CrCl_3.6H_2O$ has three isomers.
  (i) $[Cr(H_2O)_6]Cl_3 \cdot$ violet (3 ionic Cl^-)
  (ii) $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ green (2 ionic Cl^-)
  (iii) $[Cr(H_2O)_4Cl_2]Cl \cdot 2H_2O$ dark green (1 ionic Cl^-)

- Due to the difference in the number of ions formed they can be distinguished either by electric conductance or by the number of AgCl moles precipitated.
- **Coordination isomerism** occurs when the compound contain both cationic and anionic complex ions. There may be an exchange of ligands between the two, giving rise to isomers known as coordination isomers e.g., $[Cr(NH_3)_6][Co(CN)_6]$. If both cationic and anionic complex ions are hexacoordinate and the metal ions are different, the number of coordination isomers possible are 6 but if the two complex ions contain same metal ions the number of possible coordination isomers are 3. Similarly for four coordinate complex compounds containing both cationic and anionic complex ions the possible number of coordination isomers are 4 for different metal ions and 2 for same metal ion.
- **Linkage or structural isomerism** occur when more than a single atom can act as donor in a monodentate ligand i.e., ambidentate ligand e.g., Thiocyanate ion (SCN^-) may coordinate either through the sulphur or through the nitrogen $[Cr(NH_3)_5(SCN)]Cl_2$; $[Cr(NH_3)_5(NCS)]Cl_2$.
- **Polymerization isomerism** is the term used to describe the compounds which have the same stoichiometric composition but the molecular compositions are multiples of the simplest stoichiometric arrangements.

Stereo Isomerism

- When the same molecular formula represents two or more compounds which differ in the spatial arrangement of atoms or groups then such compounds are called stereo isomers and this phenomenon is known as stereoisomerism. It is of two types (i) Geometrical isomerism and (ii) Optical isomerism.

Geometrical Isomerism

- In cis isomers the two identical ligands occupy the positions adjacent to each other. In trans isomers the two identical ligands occupy the position diagonal to each others.
- Tetrahedral complexes do not exhibit geometrical isomerism because any two positions are adjacent to each other.
- Square planar complexes exhibit geometrical isomerism. MA_4,MA_3B and MAB_3 type square planar complexes do not exhibit geometrical isomerism. MA_2B_2 and MA_2BC type square planar complex exhibit geometrical isomerism.
- Square planar complexes of the type MABCD show three isomers the structures of these isomers can be written by fixing the position of one ligand (say A) and placing the other ligands B, C and D trans to it.
- Square planar complexes containing unsymmetric bidentate ligands such as $[M(AB)_2]$ also exhibit geometrical isomerism.
- The octahedral complexes of the type MA_6,MA_5B, MAB_5 do not exhibit geometrical isomerism.
- Octahedral complexes of the type MA_3B_3 exists in two geometrical isomers. The isomer with three same donor atoms on the same triangular face of the octahedron is called **facial** or **fac isomer** and the isomer with the three same donor atoms on the same equatorial plane is called **meridional** or **mer isomer.** In facial isomer the donor atoms are at the corners of a triangular face while in meridional isomer the three donor atoms are at the corners of the square plane.
- Octahedral complexes having bidentate ligands of the type $[M(AA)_2X_2]$ and $[M(AA)_2XY]$ also exhibit geometrical isomerism. AA represents a symmetrical bidentate ligand.
- Octahedral complex having six different ligands would exists in 15 geometrical isomers.
- $[M(AB)_3]$ type complex containing unsymmetric bidentate ligand exhibit geometrical isomerism and the isomers are facial (cis) and meridional *(trans).*

Optical Isomerism

- Tetrahedral complex containing four different types of ligands exhibit optical isomerism but the MA_4, MA_3B, MA_2B_2, MA_2BC tetrahedral complexes do not exhibit optical isomerism.
- Square planar complexes do not exhibit optical isomerism but only one square planar complex isobutelene diamine-mesostilbene diamine platinum (II) cation can exhibit optical isomerism.
- Octahedral complexes of the type $MA_2B_2C_2$ has five geometrical isomers of which one isomer with both *A* are *cis, B* are *cis* and *C* are *cis* to each can exhibit optical isomerism.
- Each of the 15 isomers of the complex MABCDEF will exhibit optical isomerism.
- Octahedral complex of the type $M(AA)_3$ containing three symmetrical bidentate ligands exhibit optical isomerism but not geometrical isomerism.
- Octahedral complexes of the type $[M(AA)_2b_2]$ and $[M(AA)_2bc]$ show geometrical isomerism of which cis isomers exhibit optical isomerism.
- Octahedral complexes containing hexa dentate ligands like EDTA also exhibit optical isomerism.

Stability of Coordination Compounds

- Stability constants are represented by equilibrium constant. For example the equilibrium reaction between metal ion (M^{a+}) and ligands L to form the complex $[ML_n]^{a+}$

$$M^{a+} + nL \rightleftharpoons [M^{a+}L_n]^{a+}$$

- The stability constant for the above general reaction may be written as

$$K = \frac{[ML_n^{a+}]}{[M^{a+}][L]^n}$$

- More the numerical value of K more thermo dynamically stable is the complex. The reaction proceeds through stepwise addition of monodentate ligand. If the equilibrium constants for each stepwise formation constant are represented as k_1, k_2, $k_3 \ldots k_n$ then the overall stability constant β_n can be calculated as ($\beta_n = k_1 \times k_2 \times k_3 \ldots k_n$)

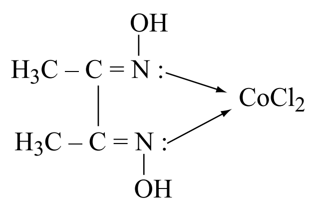
Factors Affecting the Stability of Complex

- **Size of the metal ion:** Smaller the size of the metal ion, the more stable are the metal complexes. The stability order of the complexes of the border line metal ions (first transition series with a given ligand is $Mn^{2+} < Fe^{2+} < Co^{2+} < Ni^{2+} < Cu^{2+} > Zn^{2+}$). This order is known as **Irwing-William series.**
- **Charge on the central metal ion:** More the charge density on the central ion, greater is the stability of its complexes e.g., $[Fe(CN)_6]^{3-}$ is more stable than $[Fe(CN)_6]^{4-}$
- **Crystal field effects:** Increase in the CFSE increases the stability of the complexes. The general trend for analogous complexes within a group is 3d < 4d < 5d...
- **Class '*a*' and class 'b' acceptors:** The more electropositive metals of groups 1 and 2, the lanthanides and early members of the transition series (Groups 3 to 6) belongs to **Class** *'a'* and form most stable complexes with ligands in which the donor atom is N, O and F.
- The less electropositive metals like heavy metals such as Pt, Pd, Hg, Rh, Ag, Ir, Au, Pb etc having relatively full d-orbitals belong to **Class** *b* and form stable complexes with the ligands in which the donor atom is heavier member of N, O or F families.
- Greater the **basic strength** of ligands, greater the stability of the complex $F^- > Cl^- > Br^- > I^-$
- For anionic ligands higher the charge and smaller the size, the more stable is the complex formed. F^- form stable complex than Cl^-.
- Chelating ligands form more stable complexes and it is called **chelate effect.** The reason is that when a polydenatate ligand replaces mono dentate ligands to form chelate complex entropy increases. Polydentate ligands that do not contain double bonds form more stable complex if the ring is **five membered ring.** If it contain double bonds then chelate having **six membered ring** is stable.
- **Steric strain:** Because of steric factors, large bulky ligands form less stable metal complexes than do analogous smaller ligands.
- **Hydrogen bonds:** In addition to the chelate formation if intra-molecular hydrogen bonds can be formed between the ligands of the same complex, stability increases further e.g., hydrogen bonds in nickel-dimethyl glyoxime increases its stability.
- The complexes having volatile ligands such as H_2O, NH_3, CO etc are less stable at elevated temperature.
- **Concentration factors:** Concentration of the ligand also play an important role in the stability of complex. $[CuCl_4]^{2-}$ is green but when dissolved in water changes to pale blue hydrate of Cu^{2+} ion again regain green colour on adding excess of Cl^- ions.

If the counter ion also act as a donor it can enter into the coordination entity substituting the ligands already coordinated to the metal.

Single Answers Type Questions

1. The correct IUPAC name of the complex is



(a) dichlorido dimethyl glyoximato cobalt (II)
(b) bis (dimethyl glyoxime) dichlorocobalt (II)
(c) dimethyl glyoxime cobalt (II) chloride
(d) dichlorido dimethyl glyoxime-N,N-cobalt(II)

2. Which of the following complex do not exhibit optical isomerism?

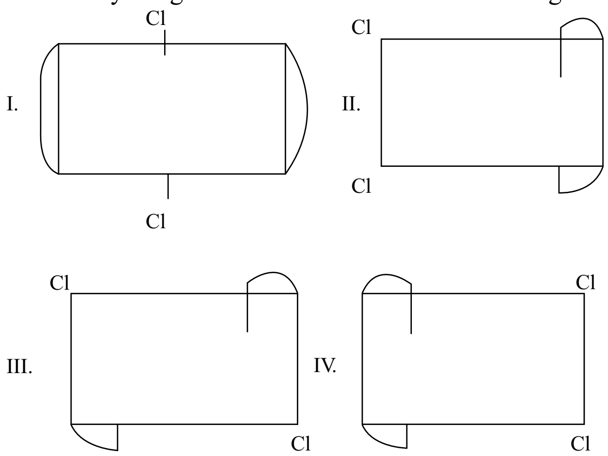
(a) $[Pt(Br)(Cl)(I)(NO_2)(C_5H_5N)(NH_3)]^-$

(b) $Cis-[Co(en)_2Cl_2]^+$

(c) $[Co(en)(NH_3)_2Cl_2]^+$

(d) $[Cr(NH_3)_4Cl_2]^+$

3. Identify the geometrical isomers in the following



(a) I with II (b) II with IV
(c) I with II and IV (d) None of these

4. Complex/complex ion 'X' is the most stable amongst following. Then 'X' must be
(a) Fe $(CO)_5$ (b) $[Fe(CN)_6]^{3+}$
(c) $[Fe (C_2O_4)]^{3-}$ (d) $[Fe (H_2O)_6]^{3+}$

5. From the stability constant (hypothetical values) given below predict which is the strongest ligand
(a) $Cu^{2+} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+} K = 4.5 \times 10^{11}$
(b) $Cu^{2+} + 4CN^- \rightleftharpoons [Cu(CN)_4]^{2-} K = 2.0 \times 10^{27}$
(c) $Cu^{2+} + 2en \rightleftharpoons [Cu(en)_2]^{2+} K = 3 \times 10^{15}$
(d) $Cu^{2+} + 4H_2O \rightleftharpoons [Cu(H_2O)_4]^{2+} K = 9.5 \times 10^{8}$

6. Consider the following isomerisms
I. Ionization II. Hydrate
III. Coordination IV. Geometrical V. Optical
Which of the above isomerisms are exhibited by $[Cr(NH_3)_2(OH)_2 CI_2]^{-1}$
(a) I and V (b) II and III
(c) III and IV (d) IV and V

7. Which one of the following can show optical isomerism?
(a) $FeSO_4 \cdot 7H_2O$ (b) $K_3[Cr(C_2O_4)_3]$
(c) $K_3[Fe(CN)_6]$ (d) $[Cr(NH_3)_6]Cl_3$

8. In the complex $FeK_2[Fe(CN)_6]$
(a) Both Fe atoms are in the same oxidation state
(b) Both Fe atoms are in different oxidation state
(c) The coordination number of iron is 4
(d) The complex is a high spin complex

9. The structure of $K[PtCl_3(C_2H_4)]$ and hybridization of Pt respectively are
(a) Square planar sp^3d^2 (b) square planar dsp^2
(c) tetrahedral sp^3 (d) octahedral d^2sp^3

10. Which of the following statement is incorrect?
(a) The stability constant of $[Co(NH_3)_6]^{3+}$ is greater than that of $[Co(NH_3)_6]^{2+}$
(b) The cyano complexes are more stable than those formed by halide ions
(c) The stability of halide complexes follows the order $I^- < Br^- < CI^- < F^-$
(d) The stability constant of $[Cu(NH_3)_4]^{2+}$ is greater than that of $[Cu(en)_2]^{2+}$

11. Which of the following is an outer orbital complex?
(a) $[Fe(CN)_6]^{4-}$ (b) $[Mn(CN)_6]^{4-}$
(c) $[Cr(NH_3)_6]^{3+}$ (d) $[Ni(NH_3)_6]^{2+}$

12. On treatment of 100 ml of 0.1 M solution of complex $CoCl_3 \cdot 6H_2O$ with excess of $AgNO_3$, 4.305g of AgCl was obtained. The complex is
(a) $[Co(H_2O)_3C1_3] \cdot 3H_2O$
(b) $[Co(H_2O)_4C1_2]C1 \cdot 2H_2O$
(c) $[Co(H_2O)_5C1]CI_2 \cdot H_2O$
(d) $[Co(H_2O)_6]Cl_3$

13. The magnetic moment of $[MnX_4]^{2-}$ is 5.9 BM. The geometry of the complex ion is (X = monodentate halide ion)
(a) Tetrahedral (b) Square planar
(c) Both possible (d) None of these

14. Which of the following is correctly matched?
(a) $[Fe(CN)_6]^{4-}$ and $[Fe(CN)_6J^3 \rightarrow$ both are Octahedral and diamagnetic with Fe,d^2sp^3 hybridized
(b) $Ni(CO)_4$ and $[Ni(CN)_4]^2 \rightarrow$ both are tetrahedral and diamagnetic with Ni,sp^3 hybridized
(c) Ni $(CO)_4$ and $[Co(CO)_4]^- \rightarrow$ both are tetrahedral and diamagnetic
(d) $[Fe(H_2O)_6]^{3+}$ and $[Cr(H_2O)_6]^{3+} \rightarrow$ both are paramagnetic and metal is $d^2\ sp^3$ hybridized

15. Increasing order of EAN of the metals in $[Ni(CN)_4]^{2-}$ $[Fe(CN)_6]^{3-}$ and $[Cu(CN)_4]^{3-}$ is
(a) $[Ni(CN_4]^{2-} < [Fe(CN)_6]^{3-} < [Cu(CN)_4]^{3-}$
(b) $[Ni(CN)_4]^{2-} < [Fe(CN)_6]^{3-} = [Cu(CN)_4]^{3-}$
(c) $[Ni(CN)_4]^{2-} < [Cu(CN)_4]^{3-} < [Fe(CN)_6]^{3-}$
(d) $[Cu(CN)_4]^{3-} < [Fe(CN)_6]^{3-} < [Ni(CN)_4]^{2-}$

16. Select correct statement:
(a) $[Fe(CN)_6]^{4-}$ and $[Fe(H_2O)_6]^{2+}$ can be distinguished by magnetic moment
(b) $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$ both are diamagnetic
(c) $[Fe(CN)_6]^{3-}$and $[Co(NH_3,)_6]^{2+}$ have equal magnetic moment
(d) All are correct statements

17. EAN of the elements (*) are equal in:
(a) $[\overset{*}{Ni}(CO)]_4$, $[\overset{*}{Fe}(CN)_6]^{2-}$
(b) $[*Ni(en)_2]^{2+}$, $[Fe*(H_2O)_6]^{2+}$
(c) $[*Co(CN)_6]^{3-}$ $[Fe*(CN)6]^{3-}$
(d) $[*Ni(en)_2]^{2+}$, $[*Fe(H_2O)_6]^{3+}$

18. On adding excess of HC1 to a light pink coloured aqueous solution of cobalt (II) chloride, a deep blue coloured solution is formed; this is due to
(a) transformation of octahedral $[Co(H_2O)_6]^{2+}$ into tetrahedral $[CoCl_4]^{2-}$
(b) transformation of $[Co(H_2O)_6]^{2+}$ into $[CoCl_6]^{4-}$
(c) formation of undissolved $CoCl_2$,
(d) formation of a Complex of Co (III) in the redox reaction.

$$[Co(H_2O)_6]^{2+} + H^+ + Cl^- \longrightarrow [Co(H_2O)_5Cl]^{2+} + H_2O + \frac{1}{2}H_2$$

19. The magnetic moment of $[NiX_4]^{2-}$ ion is found to be zero. Then ion is (X = monodentate anionic ligand)
(a) sp^3 hybridized (b) spd^2 hybridized
(c) dsp^2 hybridized (d) d^2sp hybridized

20. 0.1M solution of which of the following compounds will have the lowest freezing point?
(a) $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$
(b) $[Co(Cl)(NH_3)_5]Cl_2$
(c) $K_3[Fe(CN)_6]$
(d) $[CrCl_2(NH_3)]Cl$

21. Zn (II) ion first gives a white precipitate with NaOH which dissolves in excess of NaOH. This is due to the formation of a complex. The oxidation state of zinc in this complex will be
(a) Zero (b) +II
(c) +IV (d) +VI

22. In the following pairs find the pair in which first compound do not form complex with NH_3 and second give coloured complex with NH_3
(a) $CoCl_2$, $ZnSO_4$ (b) AgI, $CuSO_4$
(c) $NiCl_2$, $CuSO_4$ (d) $Cd(NO_3)_2$, $ZnSO_4$

23. For which of the following types of ions the number of unpaired electrons in octahedral complexes fixed at the same number as in the free ion no matter how weak or strong the crystal field is?
(a) d^3 (b) d^4 (c) d^5 (d) d^6

24. The formula for the compound tris(eithane-1, 2-diamine) cobalt (III) sulphate is
(a) $[Co(en)_3]SO_4$
(b) $[Co(SO_4)(en)_3]$
(c) $[Co(en)_3(SO_4)_2]$
(d) $[Co(en)_3]_2(SO_4)_3$

25. Which of the following is incorrectly matched?

Complex	Number of unpaired electrons
(a) $[FeF_6]^{3-}$	5
(b) $[Cr(en)_3]^{2+}$	2
(c) $[Co(NH_3)_6]^{3+}$	0
(d) $[Mn(H_2O)_6]^{2+}$	5

26. Which of the following is incorrectly matched?

Column-I	Column-II	Column-III
(a) $[Cr(CO)_6]$	Paramagnetic	Octahedral, sp^3d^2
(b) $[Fe(CO)_5]$	Paramagnetic	Trigonal bipyramid, sp^3d
(c) $[Co(CO)_4]$	Diamagnetic	Tetrahedral, sp^3
(d) $[Ni(CO)_4]$	Diamagnetic	Square planar, dsp^2

27. Ni^{2+} ion can be estimated by using dimethyl glyoxime and forms a cherry-red precipitate. The complex is stabilized by
(a) Ionic bonds
(b) Coordinate covalent bonds
(c) Dative π^- bonds
(d) Hydrogen bonds

28. Dimethyl glyoxime forms a square planar complex with Ni^{2+}. This complex should be
(a) Diamagnetic
(b) Paramagnetic having 1 unpaired electrons
(c) Paramagnetic having 2 unpaired electrons
(d) Ferromagnetic

29. Which of the following will show optical isomerism?

cis - $[Co(NH_3)_2(en)_2]^{3+}$ (I)	trans - $[IrCl_2(C_2O_4)_2]^{3-}$ (II)
$Rh(en)_3^{3+}$ (III)	Cis- $[Ir(H_2O)_3Cl_3]$ (IV)

(a) I, II, III only correct (b) II, IV only correct
(c) I, III, IV only correct (d) I, III only correct

30. Which of the following statements is correct?
(a) $[CoF_6]^{3-}$ and $[Co(NH_3)_6]^{3+}$ both are paramagnetic complexes.
(b) $[CoF_6]^{3-}$ and $[Co(NH_3)_6]^{3+}$ both are high spin complexes.
(c) $[CoF_6]^3$ is octahedral while $[Co(NH_3)_6]^{3+}$ has a pentagonal pyramid shape.
(d) $[CoF_6]^{3-}$ is outer orbital complex while $[Co(NH_3)_6]^{3+}$ is inner orbital complex.

31. A solution containing 0.319 g of $CrCl_3 \cdot 6H_2O$ was passed through a cation exchange resin in the acid form, and the acid liberated was titrated with a standard solution of NaOH. This required 28.5 cm^3 of 0.125 M NaOH. The correct formula of the Cr(III) complex is

(a) $[Cr(H_2O)_6]Cl_3$
(b) $[CrCl(H_2O)_5]Cl_2 \cdot H_2O$
(c) $[CrCl_2(H_2O)_5]Cl_2H_2O$
(d) $[CrCl_3(H_2O)_3] \cdot 3H_2O$

32. Which statement is incorrect?
(a) $[NiBr_2(PEt_3)_2]$ and $[Cu(NH_3)_4](NO_3)_2]$ have same geometries but different magnetic properties.
(b) $[Co(NH_3)_5C1]C1_2$ and $K_4[Cr(CN_6)] \cdot 3H_2O$ both are inner orbital complexes but the former is diamagnetic and later one is paramagnetic with two unpaired electrons.
(c) $[Co(NH_3)_2Cl_2(en)]^+$ can have total five stereoisomeric forms.
(d) Complex $[Pt(NH_3)_4][PtCl_6]$ can show coordination isomerism.

33. The oxidation state of central metal ion and magnetic moment of Brown ring complex is
(a) +1 and 2.8 BM (b) +l and 3.87 BM
(c) +1 and 4.8 BM (d) +2 and 4.89BM

34. The geometry of $[PtCl_4]^{2-}$ and $[Pt(CN)_4]^{2-}$ are
(a) Tetrahedral, square planar
(b) Square planar, tetrahedral
(c) Tetrahedral, tetrahedral
(d) Square planar, square planar

35. The correct name for the complex $[Cr(NH_3)_6][Co(C_2O_4)_3]$ is
(a) Hexaamminechromium (III) trioxalatocobalt (III)
(b) Hexaamminechromate (III) trioxalatocobaltate (III)
(c) Hexaamminechromium (III) trioxalatocobaltate (III)
(d) Hexaamminechromate (III) trioxalatocobalt (III)

36. In coordination compounds, the hydrate isomers differ
(a) In the number of water molecules of hydration
(b) In the number of water molecules present as ligands
(c) Both in (l) and (2)
(d) In their coordination number of the metal atom.

37. Amongst the following ions which one has the highest paramagnetism?
(a) $\left[Cr\left(H_2O\right)_6\right]^{3+}$ (b) $\left[Fe\left(H_2O\right)_6\right]^{2+}$
(c) $\left[Cu\left(H_2O\right)_6\right]^{2+}$ (d) $\left[Zn\left(H_2O\right)_6\right]^{2+}$

38. Which one of the following statements is incorrect?
(a) Greater the stability constant of a complex ion, greater is its stability
(b) Greater the charge on the central metal ion, greater is the stability of the complex
(c) Greater is the basic character of the ligand, the greater is the stability of the complex
(d) Chelate complexes have low stability constants.

39. Which of the following is a correct Irving - Williams order?
(a) $Mn^{2+} < Fe^{2+} < Co^{2+} < Ni^{2+}$
(b) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$
(c) $Fe^{2-} < Mn^{2+} < Ni^{2+} < Co^{2+}$
(d) $Co^{2+} < Mn^{2+} < Fe^{2+} < Ni^{2+}$

40. Which of the following statement is incorrect?
(a) Overall stability constant of a complex is denoted by β_n
(b) β_n is equal to the product of $K_1, K_2, K_3 \ldots\ldots\ldots K_n$ formation constants
(c) $K_1, K_2, K_3 \ldots\ldots K_n$ are called stepwise stability or formation constants
(d) $\beta_n = K_1 + K_2 + K_3 + \ldots\ldots + K_n$

41. Ammonia forms the complex ion $[Cu[NH_3)_4]^{2+}$ with copper ions in alkaline solutions but not in acidic solutions. What is the reason for it?
(a) In acidic solutions, protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available.
(b) In alkaline solutions insoluble $Cu(OH)_2$ is precipitated which is soluble in excess of any alkali.
(c) Copper hydroxide is an amphoteric substance.
(d) In acidic solutions hydration protects copper ions.

42. The no. of σ and π-bonds in $Fe_2(CO)_9$ respectively are:
(a) 22σ and 15 π (b) 23 σ and 15 π
(c) 22σ and 16 π (d) 15 σ and 8 π

43. What is the ratio of uncomplexed to complexed Zn^{2+} ion in a solution i.e., 10M NH_3, if the stability constant of $[Zn(NH_3)_4]^{2+}$ is 3×10^9
(a) 3.3×10^9 (b) 3.3×10^{-14}
(c) 3.3×10^{-14} (d) 3×10^{-9}

44. An octahedral complex of molecular formula $Co(en)_2(NCS)_2Br$ reacts with ethylene diamine but fails to respond to $AgNO_3$ test. Hence the complex is
(a) trans-bis(ethylenediamine) dithiocyanato (N) cobalt(III)bromide
(b) trans-bromidobis(ethylenediamine) thiocyanato (N) cobalt (III) thiocyanate
(c) cis-bromido bis(ethylenediamine)thiocyanato (N) cobalt(III) thiocyanate
(d) cis bis(ethylenediamine) dithiocyanato(N) cobalt (III)bromide

45. Which of the following is low spin to strong field ligands?
(a) $d^2_{xy}d^2_{xz}d^1_{yz}$
(b) $d^2_{xy}d^1_{yz}d^1_{xz}d^1_{x^2-y^2}d^1_{z^2}$

(c) $d_{xy}^2 d_{xz}^1 d_{yz}^1 d_{x^2-y^2}^1$

(d) $d_{xy}^1 d_{xz}^1 d_{yz}^1 d_{x^2-y^2}^2$

46. A six coordinate complex of formula $CrCl_3 \cdot 6H_2O$ has green colour. A 0.1 M solution of the complex when treated with excess of $AgNO_3$ gave 28.7 g of white precipitate. The formula of the complex would be:

(a) $[Cr(H_2O)_6)]Cl_3$

(b) $[CrCl(H_2O)_5]Cl_2 \cdot H_2O$

(c) $[CrCl_2(H_2O)_4]Cl \cdot 2H_2O$

(d) $[Cr(H_2O)_3 Cl_3]$

47. Which of the following is arranged in order of increasing stability of the metal complexes? (Oxidation states are given in parenthesis)

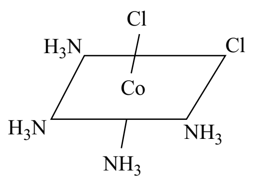
(a) Ag (I) < Pt (II) < Cr (III) < Co (III)

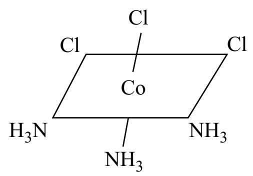
(b) Co (III) < Cr (III) < Pt (II) <Ag (I)

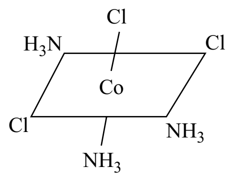
(c) Cr (III) < Pt (II) < Ag (I) < Co (III)

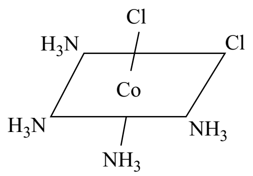
(d) Pt (II) < Co (III) < Ag (I) < Cr (III)

48. Consider the following arrangements of the octahedral complex ion $[Co(NH_3)_4 Cl_2]^+$

I. 

II. 

III. 

IV. 

Which of the following statements is incorrect regarding these structures

(a) I and II are enantiomers

(b) II and III are cis and trans isomers respectively

(c) III and IV are trans and cis isomers respectively

(d) II and IV have identical structures

49. Tetrahedral complexes of the types $[Ma_4]$ and $[Ma_3b]$ (where M = metal, a, b = achiral ligands) are not able to show optical isomerism because

(a) These molecules/ions possess C_n axis of symmetry

(b) These molecules/ions possess a plane of symmetry and hence are achiral

(c) These molecules possess a centre of symmetry

(d) These molecules/ions have non superimposable mirror images

50. A complex whose IUPAC name is not correctly written is

Complex	Name
1. $Fe(C_5H_5)_2$	Bis (η^5-cyclopentadienyl) iron II
2. $Cr(C_6H_6)_2$	Bis (phenyl) chromium (O)
3. $[CoCl_2(H_2O)_4]2H_2O$	Tetra aquodichlorido cobalt (II)-2-water
4. $[Zn(NCS)_4]^{2-}$	Tetra thioscyanato -N-Zincate (II)

51. Which of the following statements is incorrect regarding the stereo isomerism of the complexes noted below (where M stands for a metal and a, b, c and d are achiral ligands)?

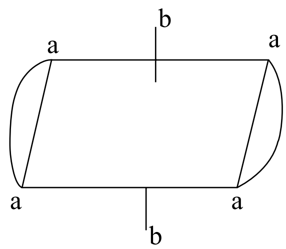
(a) In the cis and trans isomers of the octahedral complex (Ma_2b_4) the two ligands occupy the position 1,2 and 1,6 respectively

(b) The facial and meridional isomers of the octahedral complex Ma_3b_3 possess three identical ligands (a or b) at positions 1,2,3 and 1,2,6 respectively

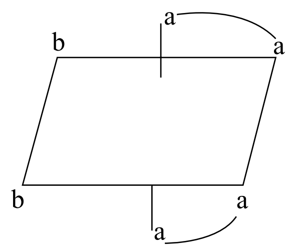
(c) The octahedral complex M (abcd $\overline{ee}$) where $\overline{ee}$ is a bidentate ligand should have 12 stereo isomers

(d) The facial and meridional isomers of Ma_3b_3 octahedral complex are both optically active

52. Which of the following statements is correct regarding the chirality (optical isomerism) of the cis and trans isomers of the type M $(aa)_2b_2$ (M stands for a metal a and b are achiral ligands and aa is bidentate ligand)?



trans



cis

(a) The trans form is achiral and optically inactive while the cis form is chiral and exists in two enantiomeric forms

(b) Both the cis and trans forms are achiral and optically inactive

(c) The trans form is chiral and exists in two enantiomeric forms while the cis form is achiral and optically inactive

(d) Both the cis and trans forms are chiral and each exists in the enantiomeric forms

53. The tris (ethylene diamine) cobalt (III) cation $[Co(en)_3]^{3+}$ can have

(a) Three stereo isomers, all chiral and optically active

(b) Two chiral stereo isomers (enantiomers)

(c) Three stereo isomers all achiral

(d) Two stereo isomers both achiral

54. The total number of structural and stereoisomers of $[Co(NH_3)_4(CN)_2]NO_3$ is
(a) 4 (b) 6 (c) 8 (d) 10

55. The number of linkage isomers for the compound $K_2[Cu(CN)_2(NCS)_2]$ are (including it)
(a) 4 (b) 6 (c) 16 (d) 9

56. A octahedral complex of cobalt contains $5NH_3$ groups one chloride and bromide group. This complex gives pale yellow precipitate with excess of $AgNO_3$. Then the ionization isomer of that complex is
(a) $[Co(NH_3)_5Cl]Br$ (b) $[Co(NH_3)BrCl]$
(c) $[Co(NH_3)_4Cl_2]$ (d) $[Co(NH_3)_4ClBr]$

57. IUPAC name of Zeise's salt is
(a) Potassium trichloroethylene platinum (II)
(b) Potassium trichloroethylene platinate (II)
(c) Potassium trischloroethylene platinate (II)
(d) Potassium trichloroethylene platinate (0)

58. The magnetic moments of complexes given below are in the order:
I. $[Ni(CO)_4]$ II. $[Mn(CN)_6]^{4-}$
III. $[Cr(NH_3)_6]^{3+}$ IV. $[CoF_6]^{3-}$
(a) I > II > III > IV (b) I < II < III < IV
(c) IV > II > I > III (d) IV < II < I < III

59. In $[Cr(C_2O_4)_3]^{3-}$ the isomerism shown is
(a) ligand (b) opticaI
(c) geometrical (d) ionization

60. If the coloumbic exchange energy for every electron pair 'E' then the exchange energies for high spin and low spin d^6 ions in an octahedral complex according to valence bond theory are
(a) 10 E, 10 E (b) 4E, 6 E
(c) 6E, 4E (d) 10E, 6E

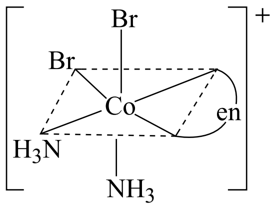
61. Aqueous solution of Ni^{2+} contains $[Ni(H_2O)_6]^{2+}$ and its magnetic moment is 2.83 BM. When ammonia is added in it, comment on the magnetic moment of solution:
(a) It will remain same
(b) It increases from 2.83 BM
(c) It decreases from 2.83 BM
(d) It cannot be predicted theoretically

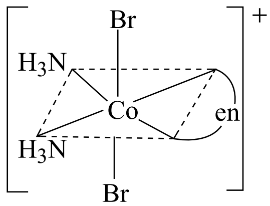
62. The total possible coordination isomers for the following compounds respectively are:
$[Co(en)_3][Cr(C_2O_4)_3]$ $[Cu(NH_3)_4][CuCl_4]$
$[Ni(en)_3][Co(NO_2)_6]$
(a) 4,4,4 (b) 2,2,2 (c) 2,2,4 (d) 4,2,4

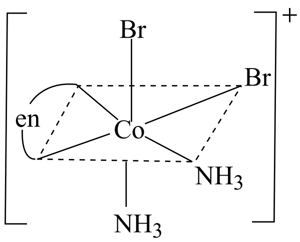
63. $CoCl^{2-}_{4(aq)}$ is blue in colour while $[Co(H_2O)_6]^{2+}_{(aq)}$ is pink. The colour of reaction mixture
$Co(H_2O)^{2+}_{6(aq)} + 4Cl^-_{(aq)} \rightleftharpoons CoCl_4^{2-} + 6H_2O_{(l)}$ is blue at room temperature while it is pink when cooled hence
(a) reaction is exothermic
(b) reaction is endothermic
(c) equilibrium will shift in backward direction on adding water to reaction mixture
(d) equilibrium will shift in forward direction on adding water to reaction mixture

64. In which of the following metal carbonyls the M - C bond is strongest
(a) $[Cr(CO)_6]$
(b) $[Cr(CO)_5(CH_3NH_2)]$
(c) $[Cr(CO)_4(CH_3NH_2)_2]$
(d) $[Cr(CO)_3(CH_3NH_2)_3]$

65. Three arrangements are shown for the complex $[CoBr_2(NH_3)_2(en)]$. Which one is wrong statement?

I) 

II) 

III) 

(a) 1 and II are geometrical isomers
(b) II and III are optically active isomers
(c) I and III are optically active isomers
(d) II and III are geometrical isomers

66. The number of ions formed when bis (ethane-1,2-diamine) copper (II) sulphate is dissolved in water
(a) 2 (b) 1 (c) 3 (d) 4

67. Which of the following correct increasing order of acidity?
(a) $[Na(H_2O)_6]^+ < [Mn(H_2O)_6]^{2+} < [Ni(H_2O)_6]^{2+} < [Sc(H_2O)_6]^{3+}$
(b) $[Sc(H_2O)_6]^{3+} < [Mn(H_2O)_6]^{2+} < [Na(H_2O)_6]^{2+} < [Ni(H_2O)_6]^{2+}$
(c) $[Mn(H_2O)_6]^{2+} < [Na(H_2O)_6]^+ < [Ni(H_2O)_6]^{2+} < [Sc(H_2O)_6]^{3+}$
(d) $[Na(H_2O)_6]^+ < [Ni(H_2O)_6]^{2+} < [Mn(H_2O)_6]^{2+} < [Sc(H_2O)_6]^{3+}$

68. Which of the following carbonyl is less stable?
(a) $Mn_2(CO)_{10}$ (b) $V(CO)_6$
(c) $Ni(CO)_4$ (d) $Fe(CO)_5$

69. A solution of concentrated aqueous ammonia is added dropwise to 1 mL of a dilute aqueous solution of copper (II) nitrate until a total of 1 mL of the ammonia solution has been added. What observations can be made during this process?

(a) The colourless copper (II) nitrate solution turns blue and yields a dark blue precipitate
(b) The colourless copper (II) nitrate solution yields a white precipitate which turns dark blue upon standing
(c) The light blue copper (II) nitrate solution yields a precipitate which redissolve to form a dark blue solution
(d) The light blue copper (II) nitrate solution turns dark blue and yields a dark blue precipitate

70. Which of the following coordination compounds would exhibit optical isomerism
(a) pentaamminenitrocobalt (III) iodide
(b) diamminedichloroplatinum (II)
(c) trans - dicyano bis - (ethylene diamine) chromium (III) chloride
(d) tris (ethylene diamine) cobalt (III) bromide

71. The compound that is not olefinic organometallic is
(a) $K[C_2H_4 \cdot PtCl_3] \cdot 3H_2O$ (b) $Be(CH_2)_2$
(c) $K[C_2H_4 \cdot PtCl_3]$ (d) $C_4H_4Fe\ (CO)_3$

72. Which one of the following statements is not correct?
(a) The complexes $[Ni(CO)_4]$ and $[NiCl_4]^{2-}$ differ in the state of hybridization
(b) The complexes $[Ni(CO)_4]$ and $[NiCl_4]^{2-}$ differ in the magnetic properties
(c) The complexes $[Ni(CO)_4]$ and $[NiCl_4]^{2-}$ differ in the primary valencies of nickel
(d) The complexes $[Ni(CO)_4]$ and $[NiCl_4]^2$ have nickel in the different oxidation state

73. Mixture (X) of 0.02 mole of $[Co(NH_3)_5SO_4]Br$ and 0.02 mole of $[Co(NH_3)_5\ Br]SO_4$ was prepared in 2 litre of solution

1 litre of mixture (X) + excess of $AgNO_3 \rightarrow Y$

1 litre of mixture (X) + excess of $BaCl_2 \rightarrow Z$

Number of moles of Y and Z respectively are
(a) 0.01,0.02 (b) 0.02,0.01
(c) 0.01,0.01 (d) 0.02,0.02

74. Which of the following is an outer orbital complex?
(a) $[Fe(CN)_6]^{4-}$ (b) $[Mn(CN)_6]^{4-}$
(c) $[Cr(NH_3)_6]^{3+}$ (d) $[Ni(NH_3)_6]^{2+}$

75. Which of the following statement is not correct?
(a) bis (glycinato) zinc II is optically active.
(b) $[NiCl_4]^{2-}$ and $[PtCl_4]^{2-}$ have different shapes
(c) $[Ni(CN)_4]^{4-}$ is square planar complex
(d) $[Ni(CN)_4]^{3-}$ and $[Ni(CO)_4]$ have same magnetic moment

76. Which of the following statement is incorrect?
(a) Co-ordination compounds and complexes are synonymous terms
(b) Complexes always give ion in the solution
(c) Complexes may give ions in the solution or may not give ions in the solution
(d) Complex ion does not dissociate into its component parts even in the solution

77. Which of the following pairs of complexes are isomeric with each other but their aqueous solutions exhibit different molar conductivities?
(a) $[PtCl_2(NH_3)_2Br_2]$ and $[PtBr_2(NH_3)_4]Cl_2$
(b) $[CoCl_2(NH_3)_4]NO_2$ and $[CoCl(NO_2)(NH_3)_4]Cl$
(c) $[Co(NO_2)\ (NH_3)_5]Cl_2$ and $[Co(ONO)\ (NH_3)_5]C1_2$
(d) $[CoBr(NH_3)_5]SO_4$ and $[Co(SO_4)\ (NH_3)_5]Br$

78. In the isoelectronic series of metal carbonyl, the CO bond strength is expected to increase in the order:
(a) $[Mn(CO)_6]^+ < [Cr(CO)_6] < [V(CO)_6]^-$
(b) $[V(CO)_6]^- < [Cr(CO)_6] < [Mn(CO)_6]^+$
(c) $[V(CO)_6]^- < [Mn(CO)_6]^+ < [Cr(CO)_6]$
(d) $[Cr(CO)_6] < [Mn(CO)_6]^+ < [V(CO)_6]^-$

79. Which is not true about metal carbonyls?
(a) Here CO acts as a Lewis base as well as Lewis acid
(b) Here metal acts as Lewis base as well as Lewis acid
(c) Here $d\pi$ - $p\pi$ back bonding takes place
(d) Here $d\pi$ - $p\pi$ back bonding takes place

80. The IUPAC name for $K_2[Cr(CN)_2O_2(O_2)\ NH_3]$ is
(a) Potassium amminedicyanotetraoxochromium (III)
(b) Potassium amminedicyanodioxygendioxochromate (IV)
(c) Potassium amminedicyanoperoxo superoxochromate (III)
(d) Potassium amminedicynidoxidoperoxidochromate (VI)

81. Which of the following is true about the complex $[PtCl_2(H_2O)\ (NH_3)]$?
(a) It exhibits geometrical isomerism
(b) It is paramagnetic complex
(c) Its geometry is tetrahedron
(d) Platinum is sp^3 hybridized

82. A metal complex of coordination number six having three different types of ligands a, b and c of composition $Ma_2b_2c_2$ can exist in several stereo isomeric forms; the total number of such isomers is:
(a) 3 (b) 5 (c) 7 (d) 6

83. Complex compound $[Cr(NCS)(NH_3)_5]\ [ZnCl_4]$ will be:
(a) colourless and diamagnetic
(b) green coloured and diamagnetic
(c) green coloured and shows coordination isomerism
(d) diamagnetic and shows linkage isomerism

84. An aqueous solution of titanium chloride, when subjected to magnetic measurement, measured zero magnetic moment. Assuming the octahedral complex in aqueous solution, the formula of the complex is:
(a) $[Ti(H_2O)_6]Cl_2$ (b) $[Ti(H_2O)_6]Cl_4$
(c) $[TiCl_3(H_2O)_3]$ (d) $[TiCl_2(H_2O)_4]$

85. Which of the following statement is not true for the reaction given below?

$$[Cu(H_2O)_4]^{2+} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+} + 4H_2O$$

(a) It is a ligand substitution reaction
(b) NH_3 is a relatively strong field ligand while H_2O is a weak field ligand
(c) During the reaction, there is a change in colour from light blue to dark blue
(d) $[Cu(NH)_4]^{2+}$ has a tetrahedral structure and is paramagnetic

86. Give the correct initials T or F for following statements. Use T if statement is true and F if it is false
I. Co(III) is stabilized in presence of weak field ligands, while Co(II) is stabilized in presence of strong field ligand
II. Four coordinated complexes of Pd(II) and Pt (II) are diamagnetic and square planar
III. $[Ni(CN)_4]^{4-}$ ion and $[Ni(CO_4)]$ are diamagnetic tetrahedral and square planar respectively
IV. Ni^{2+} ion does not form inner orbital octahedral complexes
(a) TFTF (b) TTTF
(c) TTFT (d) FTFT

87. Which of the following has largest number of isomers?
(a) $[Co(en)_2Cl_2]^+$ (b) $[Co(NH_3)_5Cl]^{2+}$
(c) $[Ir(PR_3)_2(NH_3)_2]$ (d) $[Ru(NH_3)_2Cl_2]^+$

88. Which of the following statement is true?
(a) In $[PtCl_2(NH_3)_2]^{2+}$ the cis form is optically inactive while trans form is optically active
(b) ln $[Fe(C_2O_4)_3]^{3-}$, geometrical isomerism does not exist while optical isomerism exists
(c) Mabcd, square planar complexes show both optical as well as geometrical isomerism
(d) In Mabcd tetrahedral complex, optical isomerism cannot be observed

89. The following complexes are given:
I. trans-$[Co(NH_3)_4Cl_2]^+$
II. cis-$[Co(NH_3)_2(en)_2]^{3+}$
III. trans-$[Co(NH_3)_2(en)_2]^{3+}$
IV. $NaCl_4^{2-}$
V. TiF_6^{2-}
VI. CoF_6^{3-}
Choose the correct Code:
(a) I, II are optically active, III is optically inactive
(b) II is optically active, I, III are optically inactive
(c) IV, V are coloured and VI is colourless
(d) IV is coloured and V, VI are colourless

90. Which of the following will have three stereoisomeric forms?
I. $[Cr(NO_3)_3(NH_3)_3]$ II. $K_3[Co(C_2O_4)_3]$
III. $K_3[CoCl_2(C_2O_4)_2]$ IV. $[CoBrCl(en)_2]$
(a) III and IV (b) I, III and IV
(c) IV only (d) All four

91. A coordination complex of type MX_2Y_2 (M-metal ion; X, Y-monodentate ligands), can have either a tetrahedral or a square planar geometry. The maximum number of possible isomers in these two cases are respectively:
(a) 1 and 2 (b) 2 and 1
(c) 1 and 3 (d) 3 and 2

92. $[CoCl_2(NH_3)_4]^+ + Cl^- \longrightarrow [CoCl_3(NH_3)_3] + NH_3$
If in this reaction two isomers of the product are obtained, which is true for the initial (reactant) complex:
(a) Compound is in cis-form
(b) Compound is in trans-form
(c) Compound is in both (cis and trans) form
(d) can't be predicted

93. Which of the following isomerisms is not exhibited by $[Cr(NH_3)_2(H_2O)_2Cl_2]Cl$
(a) Ionization (b) Geometrical
(c) Hydrate (d) Optical

94. Consider the following complex: $[Co(NH_3)_5CO_3]ClO_4$
The coordination number, oxidation number, number of d-electrons and number of unpaired d-electrons on the metal are respectively
(a) 6,3,6,0 (b) 7,2,7,1
(c) 7,1,6,4 (d) 6,2,7,3

95. The hybridization and geometry of Wilkinson's catalyst is
(a) sp^3, tetrahedral (b) dsp^2, square planar
(c) sp^3d^2, octahedral (d) d^2sp^3, octahedral

96. The magnetic moment of $K_2[Cu(CN)_4]$ is
(a) 1.73 BM (b) 2.83 BM
(c) 3.89BM (d) 0

97. Metal carbonyl with longest carbon-oxygen bond length is;
(a) $[Mn(CO)_6]^+$ (b) $[Fe(CO)_5]$
(c) $[Cr(CO)_6]$ (d) $[V(CO)_6]^-$

98. A $[M(H_2O)_6]^{2+}$ complex typically absorb at around 600 nm. It is allowed to react with ammonia to form a new complex $[M(NH_3)_6]^{2+}$ that should have absorption at
(a) 800 nm (b) 580 nm (c) 620 nm (d) 320 nm

99. Mark out the correct statement(s):
(a) $[Fe(CN)_6]^{3-}$ is more stable than $[Fe(CN)_6]^{4-}$
(b) $[Fe(CN)_6]^{3-}$ is less stable than $[Fe(CN)_6]^{4-}$
(c) $[Fe(CN)_6]^{3-}$ gets easily oxidized to $[Fe(CN)_6]^{4-}$ in alkaline medium
(d) $[Fe(CN)_6]^{3-}$ is reduced to $[Fe(CN)_6]^{4-}$ in acid medium by H_2O_2

100. Potassium diamminedicyanosulphatosuperoxo platinate (IV) is
(a) $K[Pt(CN)_2(O_2)(SO_4)(NH_3)_2]$
(b) $K_4[Pt(NH_3)_2(CN)_2(SO_4)(O_2)]$
(c) $K_2[Pt(NH_3)_2(CN)_2(SO_4)(O_2)]$
(d) $K_3[Pt(NH_3)_2(CN)_2(SO_4)(O_2)]$

101. Which of the following pair the EAN of central metal atom is not same?
(a) $[Fe(CN)_6]^{3-}$and $[Fe(NH_3)_6J^{3+}$
(b) $[Cr(NH_3)_6]^{3+}$ and $[Cr(CN)_6]^{3-}$
(c) $[FeF_6]^{3-}$ and $[Fe(CN_6)]^{3-}$
(d) $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$

102. Effective atomic number of Co $(CO)_4$ is 35 and hence is less stable. It attains stability by:
(a) Oxidation of Co (b) Reduction of Co
(c) Dimerization (d) Both 2 and 3

103. Which statement about coordination number of a cation is true?
(a) Most metal ions exhibit only a single characteristic coordination number
(b) The coordination number is equal to the number of ligands bonded to the metal atom
(c) The coordination number is determined solely by the tendency to surround the metal atom with the same number of electrons as one of the rare gases
(d) For most cations, the coordination number depends on the size, and charge of the cation

104. The IUPAC name of the Wilkinson's catalyst $[RhCl(PPh_3)_3]$ is:
(a) Chlorotris (triphenylphosphine) rhodium (I)
(b) Chlorotris (triphenylphosphine) rhodium (IV)
(c) Chlorotris (triphenylphosphine) rhodium (0)
(d) Chlorotris (triphenylphosphine) rhodium (VI)

105. The structure of $K[PtCl_3(C_2H_4)]$ and hybridization of Pt respectively are:
(a) square planar, sp^2d^2 (b) square planar, dsp^2
(c) tetrahedral, sp^3 (d) octahedral, d^2sp^3

106. Which one of the following pairs of isomers and types of isomerism are correctly matched?
I. $[Co(NH_3)_5(NO_2)]Cl_2$ and $[Co(NH_3)(ONO)]Cl_2$ ……………….(Linkage)
II. $[Cu(NH_3)_4][PtCl_4]$ and $[Pt(NH_3)_4]$ $[CuCl_4]$ ………(coordination)
III. $[PtCl_2(NH_3)_4]Br_2$ and $[PtBr_2(NH_3)_4]Cl_2$ ………… (ionization)

Select the correct answer using the codes given below:
(a) II and III (b) I, II and III
(c) I and III (d) I and II

107. For the complex ML_2, stepwise formation constants for
$M + L \rightleftharpoons ML$
$ML + L \rightleftharpoons ML_2$
are 4 and 3. Hence, overall stability constant for
$M + 2L \rightleftharpoons ML_2$ is
(a) 12 (b) 7 (c) 1.33 (d) 0.75

108. Trioxalato aluminate (III) and tetrafluoro-borate (III) ions are:
(a) $[Al(C_2O_4)_3]^{3-}$, $[BF_4]^{3-}$
(b) $[Al(C_2O_4)_3]^{3+}$, $[BF_4]^{3+}$
(c) $[Al(C_2O_4)_3]^{3-}$, $[BF_4]$-
(d) $[Al(C_2O_4)_3]^{2-}$, $[BF_4]^{2-}$

109. Which complex is likely to show optical activity:
(a) Trans-$[Co(NH_3)_4Cl_2]^+$
(b) $[Cr(H_2O)_6]^{3+}$
(c) Cis-$[Co(NH_3)_2(en)_2]^{3+}$
(d) Trans-$[Co(NH_3)_2(en)_2]^{3+}$

110. The IUPAC name of the red coloured complex $[Fe(C_4H_7ON_2)_2]$ obtained from the reaction of Fe^{2+} and dimethyl glyoxime
(a) bis (dimethyl glyoxime) ferrate (II)
(b) bis (dimethyl glyoximato) iron (II)
(c) bis (2,3-butanediol dioximato) iron (II)
(d) bis (2,3-butanedione dioximato) iron (II)

111. The molar ionic conductances of the octahedral complexes.
I. $PtCl_4 \cdot 5NH_3$ II. $PtCl_4 \cdot 4NH_3$
III. $PtCl_4 \cdot 3NH_3$ IV. $PtCl_4 \cdot 2NH_3$
(a) I < II < III < IV (b) IV < III < II < I
(c) III < IV < II < I (d) IV < III < I < II

112. $[Co(en)_3]^{3+}$ ion is expected to show
(a) Two optically active isomers: d and *l* forms
(b) Three optically active isomers: d, *l* and meso forms
(c) Four optically active isomers: cis, d and *l* isomers and trans d and *l* isomers
(d) None of these

113. Which of the following polymerization isomers of the compound having empirical formula $Cr(NH_3)_3(NO_2)_3$ has the lowest molecular mass?
(a) $\left[Cr(NH_3)_4(NO_2)_2\right]^+\left[Cr(NH_3)_2(NO_2)_4\right]^-$
(b) $\left[Cr(NH_3)_6\right]^{3+}\left[Cr(NO_2)_6\right]^{3-}$
(c) $\left[Cr(NH_3)_5(NO_2)\right]^{2+}\left[Cr(NH_3)(NO_2)_5\right]^{2-}$
(d) All have equal molecular mass

114. The two compounds pentaammine sulphato cobalt (III) bromide and pentaammine sulphato cobalt(III). chloride represent

(a) Linkage isomerism
(b) Ionization isomerism
(c) Coordination isomerism
(d) No isomerism

115. In the complex K Fe $[Fe(CN)_6]$:

(a) Both iron atoms are in same oxidation state.
(b) Both iron atoms are in different oxidation state.
(c) The coordination number of iron is 3.
(d) The complex is a high spin complex.

116. Among the following compounds whose optical activity does not depend on the orientation of the ligands around the metal cation

I: $[CoCl_3(NH_3)_3]$
II: $[Co(en)_3]Cl_3$,
III: $[Co(C_2O_4)_2,(NH_3)_2]^-$
IV: $[CoCl_2(NH_3)_2(en)]^+$

(a) II and III (b) I, II, III
(c) II and IV (d) Only II

117. $[PdCl_2(PMe_3)_2]$ palladium and nickel belong to same transition group how many total isomers are possible for analogous paramagnetic complex of Ni(II)

(a) zero (b) 1 (c) 2 (d) 4

118. Which of the following statement is not correct?

(a) bis (glycinato) zinc II is optically active.
(b) $[NiCl_4]^{2-}$ and $[PtCl_4]^{2-}$ have different shapes
(c) $[Ni(CN)_4]^{2-}$ is square planer complex
(d) $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$ have same magnetic moment

119. The pair in which both species have same magnetic moment [spin only]

(a) $[Cr(H_2O)_6]^{2+}$, $[CoCl_4]^{2-}$
(b) $[Cr(H_2O)_6]^{2+}$,$[Fe(H_2O)_6]^{2+}$
(c) $[Mn(H_2O)_6]^{2+}$,$[Cr(H_2O)_6]^{2+}$
(D) $[CoCl_4]^{2-}$, $[Fe(H_2O)_6]^{2+}$

120. The magnetic momentum of $K[Fe(CN)_5(NO)]$ is

(a) 1.73 BM. (b) 2.84 BM.
(c) 6.9 BM. (d) zero

121. Which of the following statements are correct?

(a) $[Fe(CN)_6]^{4-}$ is diamagnetic but $[Fe(CN)_6]^{3-}$ is paramagnetic
(b) Fe^{3+} ions always form tetrahedral complexes
(c) In a compound with an octahedral structure, the d_{xy} and d_{yz} orbitals of a metal ion should be vacant
(d) $[Fe(H_2O)_6]^{3+}$ is less paramagnetic than $[Fe(CN)_6]^{3-}$

122. Select correct statement

(a) $[Fe(CN)_6]^{4-}$ and $[Fe(H_2O)_6]^{2+}$ can be distinguished by magnetic moment
(b) $[Fe(CN)_6]^{4-}$ and$[Ni(CO)_4]$ both are diamagnetic
(c) $[Fe(CN)_6]^{3-}$ and $[Co(NH_3)_6]^{2+}$ have equal magnetic moment
(d) All are correct statements

123. The relative stability of the octahedral complexes of Fe (III) over Fe (II) with the bidentate ligands

I. $HO-CH_2-CH_2-OH$
II. $HO-CH_2-CH_2-NH_2$
III. $H_2N-CH_2-CH_2-NH_2$
IV. $H_2N-CH_2-CH_2-SH$

(a) I > II > III > IV (b) II > I > IV > III
(c) III > II > IV > I (d) IV > I > III > II

124. If 'M' is the element of actinoid series, the degree of complex formation decreases in the order.

(a) $M^{4+} > M^{3+} > MO_2^{2+} > MO_2^+$
(b) $MO_2^+ > MO_2^{2+} > M^{3+} > M^{4+}$
(c) $M^{4+} > MO_2^{2+} > M^{3+} > MO_2^+$
(d) $MO_2^{2+} > MO_2^+ > M^{4+} > M^{3+}$

125. According to IUPAC nomenclature system, which of the following is correct name of the complex compound given

$Na_3[Co(NO_2)(CN)_2(ONO)O_2(O_2)]$; = 4.899 BM
Where NO_2 and O_2 written separately are different ligands

(a) Sodium dicyanodinitrodiperoxocobaltate (III)
(b) Sodium dicyanodinitroniumdisuperoxocobaltate (III)
(c) Sodium dicyanonitronitritodioxygenperoxo cobaltate (III)
(d) Sodium dicyanonitronium nitrodioxygen peroxocobaltate (II)

126. The number of isomers for the complex $[Co(en)Cl_2Br_2]^-$ are

(a) 2 (b) 3
(c) 4 (d) 6

127. Fe^{3+} (aq) with thiocyanate ion SCN^- forms a deep red complex ion $[Fe(SCN)]^{2+}$ which have formation constant 1×10^{-3} $mol^{-1}L$. If the equilibrium concentrations of Fe^{3+} (aq) and SCN^- (aq) are 0.01 M and 3×10^{-4} M respectively. Calculate the concentration of complex ion

(a) 6×10^{-6} (b) 3×10^{-9}
(c) 1.8×10^{-10} (d) 8.0×10^{-7}

128. Which statement is correct regarding crystal field theory

(a) There is no interaction between metal orbitals and ligand orbitals
(b) In most transition metal complexes, either six or four ligands surround the metal because they give spherically symmetrical field

(c) Trans-dicyanobis – (ethylenediamine) chromium (III) Chloride
(d) Tris (ethylenediamine) cobalt (lll) bromide

4. Which of the following complexes have four different isomers only?
(a) $[Co(en)_2Cl_2]Cl$ (b) $[Co(en)_2(NH_3)_2]Cl$
(c) $[Co(PPh_3)_2 (NH_3)_2 Cl_2]Cl$ (d) $[Co(en)_3]Cl_3$

5. Among the following the correct statements are
(a) Invar is an alloy of Fe, Co, Ni used for making clock pendulums
(b) $[Fe(CN)_6]^{4-}$ is diamagnetic and low spin complex where as $[Fe(H_2O)_6]^{3+}$ is high spin complex as well as paramagnetic
(c) $[Cu (NH_3)_4]^{2+}$ is paramagnetic and coloured
(d) $[Pt (NH_3)_2 Cl_2]$ is square planar complex

6. Which among the following are correct?
(a) $K_3 [Mn(C_2O_4)_3]$ $\mu = 2.82BM$
(b) $[Ni(CO)_4]$ $\mu = 0.00BM$
(c) $[Ni(CN)_4]^{2-}$ $\mu = 5.90BM$
(d) $[FeCl_4]^{2-}$ $\mu = 1.73BM$

7 The complex [Mabcdef] can exhibit
(a) Optical isomerism
(b) Geometrical isomerism
(c) Structural isomerism
(d) Polymerization isomerism

8. Which of the following statements is not true about the complex ion $[Cr Cl (NO_2) (en)_2]^+$
(a) It has two geometrical isomers Cis and Trans
(b) Cis and Trans forms are not diastereomers to each other
(c) Only the Cis isomer displays optical activity
(d) It has three optically active isomers: d,l and meso forms

9. The correct statements among the following are
(a) The complexes $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ differ in the magnetic properties
(b) The complexes $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ differ in the geometry
(c) The complexes $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ differ in primary valencies of nickel
(d) The complexes $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ differ in the state of hybridization

10. Among the following, the ions having same magnetic moment are
(a) $[Ti(H_2O)_6]^{3+}$ (b) $[Sc(H_2O)_6]^{3+}$
(c) VO^{2+} (d) $[Mn(H_2O)_6]^{+3}$

11. In which of the following the central atom is present in its highest oxidation state?
(a) $[Fe(CN)_6]^{4-}$ (b) $[NiCl_4]^{2-}$
(c) $[Cr(H_2O)_6]^{3+}$ (d) $[Sc(H_2O)_6]^{3+}$

12. The coloured complex ions among the following are
(a) $[Ti(H_2O)_6]^{3+}$ (b) $[Zn(OH)_4]^{2-}$
(c) $[CuCl_3]^-$ (d) $[Cu(NH_3)_4]^{2+}$

13. The true statements among the following
(a) Hexacyanoferrate (II) ion has four unpaired electrons in 3d orbital
(b) Tetra cyanonickelate (II) ion has square planar geometry
(c) IUPAC name of $[Zn(OH)_4]^{2-}$ ion is tetrahydroxy zinc ion
(d) The coordination number of Cr in $[Cr(NH_3)_2(en)_2]^{3+}$ is 6

14. Find out the correct statements among the following
(a) The oxidation state of iron in sodium nitroprusside $Na_2[Fe(CN)_5(NO)]$ is +II
(b) $[Ag (NH_3)_2]^+$ is linear in shape
(c) In $[Fe(H_2O)_6]^{3+}$ Fe is $d^2 sp^3$ hybridized
(d) In $[Ni(CO)_4]$ the oxidation state of Ni is zero

15. Identify the correct statements given below
(a) In ferrocyanide ion, the effective atomic number is 36
(b) Chelating ligands are atleast bidentate ligand
(c) $[Cr Cl_2(CN)_2 (NH_3)_2]^-$ and $[Cr Cl_3 (NH_3)_3]$ both have d^2sp^3 hybridization
(d) As the number of rings in complex increases, stability of complex (chelate) also increases

16. Which of the following statements are true?
(a) The π-bond between metal and carbonyl carbon reduces the bond order of C-O from triple bond in CO towards double bond
(b) The pair of compounds $[Cr(H_2O)_6] Cl_3$ and $[CrCl_3(H_2O)_3] \cdot 3H_2O$ show hydrate isomerism
(c) dz^2 orbital of the central metal atom /ion is used dsp^2 hybridization
(d) Facial and meridional isomers are associated with $[Ma_3b_3]^{n+}$ type complex compound, both are optical inactive

17. Regarding metal carbonyls which of the following statements are correct.
(a) In $Mn_2(CO)_{10}$ bond order of Mn-Mn is 0
(b) In $Fe_3(CO)_{12}$ number of Fe-Fe bonds are 3
(c) In $Fe_2(CO)_9$ all bond lengths are same
(d) $Fe(CO)_5$ is diamagnetic

18. A complex is represented as $CoCl_3 \cdot xNH_3$. Its 0.1 molal in water shows $\Delta T_f = 0.558^0$. if the cryoscopy and ebullioscopy constants for H_2O are 1.86 and 0.52 respectively. Identify the correct versions assuming 100% ionization of the complex.
(a) Hexaammine cobalt (lll) chloride is the IUPAC name for the complex
(b) The above solution will boil between 100°C to 101°C

(c) The central metal ion is d^2sp^3 hybridized
(d) The coordination sphere is incapable of exhibiting geometrical isomerism

19. Identify the correct statements.
(a) Stability order in halide complexes of a given more electropositive metal is $I^- > Br^- > Cl^- > F^-$
(b) $[Cu(NH_3)_4]^{2+}$ ion has higher ease of formation than $(Cu(en)_2)$
(c) Stable EDTA complexes of alkaline earth metal ion form primarily because of entropy effects
(d) Alkyl substitution in H_2O increase the tendency to act as ligand in complex formation

20. Given chemical formula and name, which are correctly matched?
(a) $K[Pt(NH_3)Cl_5]$ – Potassium amminepentachloro platinate (IV)
(b) $[Ag(CN)_2]^-$ - dicyanoargentate (I) ion
(c) $K_3[Cr(C_2O_4)_3]$ – Potassium trioxalato chromate (III)
(d) $Na_2[Ni(EDTA)]$ – Sodium ethylene diamine tetracetato nickel (I)

21. The compounds having colour due to charge transfer
(a) $K_2Cr_2O_7$ (b) $KMnO_4$
(c) $[Fe(H_2O)_5NO]SO_4$ (d) Cu_2O

22. The correct statements about $Mn_2(CO)_{10}$ is/are
(a) There is Mn-Mn bond
(b) There are two bridged CO molecules
(c) There are three bridged CO molecules
(d) Each Mn surrounds with 5 CO molecules

23. The compounds that gives white precipitates with $AgNO_3$
(a) $[Co(NH_3)_6]Cl_3$ (b) $[Co(NH_3)_3Cl_3]$
(c) $K_2[Pt(en)_2Cl_2]$ (d) $[Fe(en)_3]Cl_3$

24. Which of the following statements are correct?
(a) The stability constant of $[Co(NH_3)_6]^{3+}$ is greater than that of $[Co(NH_3)_6]^{2+}$
(b) The cyano complexes are far more stable than those formed by halide ions
(c) The stability of halide complexes follow the order $I < Br < Cl$
(d) The stability constant of $[Cu(NH_3)_4]^{2+}$ is greater than that of $[Cu(en)_2]^{2+}$

25. Regarding the complex $[Cr\,Cl_3(OH)_2(NO_3)]^{3-}$ ion the correct statements are
(a) It has three geometrical isomers
(b) Only one space isomer is optically active and remaining are inactive
(c) There are total four space isomers
(d) The magnetic moment of the complex ion is 3.89 BM

26. Which of the following statements are false?
(a) In $[PtCl_2(NH_3)_4]^{2+}$ complex ion the cis form is optically active while trans-form is optically inactive
(b) In $[Fe(C_2O_4)_3]^{3-}$ geometrical isomerism does not exist while optical isomerism exists
(c) In $[Mabcd]^{n+}$ tetrahedral complexes optical isomerism cannot be observed
(d) In square planar complexes, optical isomerism can be observed

27. When Na_2S is added to sodium nitropusside solution, the correct observations made are
(a) Violet colour is produced
(b) A complex $Na_4[Fe(CN)_5NOS]$ is formed
(c) $[Fe(CN)_5NOS]^{4-}$ is formed
(d) $Na_2[Fe(CN)_5NOS]$ is formed

28. Among the following the diamagnetic species are
(a) $\left[Fe(CN)_6\right]^{4-}$ (b) $[Cu(NH_3)_4]^{2+}$
(c) $\left[Ti(H_2O)_6\right]^{3+}$ (d) $\left[Ni(en)_2\right]^{2+}$

29. The compounds having metal atom in zero oxidation state are
(a) $Ni(CO)_4$ (b) $Cr(CO)_6$
(c) $Fe_2(CO)_9$ (d) $Fe(CO)_5$

30. In the following reactions
I. $FeSO_4 + NO + H_2O \longrightarrow X$
II. Sodium nitroprusside + $Na_2S \longrightarrow Y$
(a) Products X and Y both are paramagnetic
(b) In the reaction I change of oxidation state of central atom occurs while in reaction (II) there is no change in oxidation state
(c) Hybridization of central atom in reaction II is not changed
(d) Magnetic moment of compound X is $\sqrt{15}$

31. In the formation of which of the following complexes the charge transfer takes place
(a) $\left[Fe(H_2O)_5NO\right]^{2+}$ (b) $\left[Fe(CN)_5NO\right]^{3-}$
(c) $\left[Fe(H_2O)_5NO\right]^{3+}$ (d) $\left[Fe(CN)_5NO\right]^{2-}$

32. A d-block element forms octahedral complex but its magnetic moment remains same either in strong field or in weak field ligand. Which of the following statements are correct about this complex?
(a) Element always form colourless compound
(b) Number of electrons in t_{2g} orbital's are higher than in e_g orbitals
(c) It can have either d^3 or d^8 configuration
(d) It can have either d^7 or d^8 configuration

33. For which of the following d^n configuration of octahedral complexes cannot exist in both high spin and low spin forms
(a) d^3 (b) d^5 (c) d^6 (d) d^8

(c) Generally the crystal field splitting energy (Δ) for different ligands is in the order shown down. N- donors < O-donors < C-donors
(d) Only CO among different ligands can form back bonding (π bond form metal to ligand)

129. Which of the following complex is sp^3 hybridized and paramagnetic?
(a) $Ni(CO)_4$ (b) $[Ni(CN)_4]^{2-}$
(c) $[NiCl_4]^{2-}$ (d) $Cu(NH_3)_4]^{2+}$

130. Choose correct statements regarding the following complexes
$x = [Cr(NH_3)_6]^{3+}$; $y = [Ni(CN)_4]^{2-}$
(a) x is diamagnetic and y is paramagnetic
(b) x undergoes d^2sp^3 hybridization and y follows dsp^2 hybridization
(c) x is paramagnetic and exhibit sp^3d^2 hybridization
(d) y is paramagnetic and exhibits sp^3 hybridization

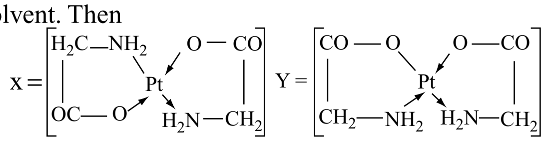
131. Co^{2+} (aq) + SCN^- (aq) complex(x)
Ni^{2+} (aq) + Dimethyl glyoxime complex (y)
The coordination number of cobalt and nickel in complexes (x) and (y) are four. The IUPAC names of the complexes (x) and (y) are respectively
(a) tetrathiocyanato-S-cobalt (II) and bis (dimethyl glyoximate) nickel (II)
(b) tetrathiocyanato-S-cobaltate (II) and bis (dimethyl glyoximato) nickel (II)
(c) tetrathiocyanato-S-cobaltate (II) and bis (dimethyl glyoximato) nickel (II)
(d) tetrathiocyanato-S-cobaltate (III) and bis (dimethyl glyoximato) nickel (II)

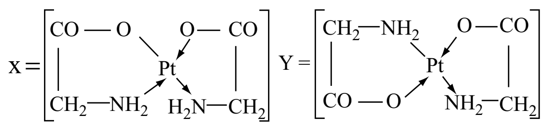
132. Consider the following coordination compounds
(i) $Ni(CO)_4$ (ii) $[Co(CO)_4]^-$ (iii) $Fe(CO)_4]^{2-}$
The stretching frequency of M - C bond (which is directly proportional to bond strength follows the order):
(a) (ii) > (iii) > (i) (b) (iii) > (ii) > (i)
(c) (i) > (ii) > (iii) (d) (i) > (iii) > (ii)

133. $K_2[PtCl_4] + 2Kgly \longrightarrow X + 4KCl$

$[Pt(NH_3)_4](NO_3)_2 + 2Kgly \longrightarrow Y + 2KNO_3$

Here Kgly is potassium glycinate and 'X' is soluble in polar solvent while 'Y' is soluble in non-polar solvent. Then

(a) 

(b) 

(c) Both X and Y are in cis form
(d) Both X and Y are in trans form

134. Identify the correct order of wavelength of light absorbed for the following complex ions

I. $[Co(H_2O)_6]^{3+}$ II. $[Co(CN)_6]^{3-}$

III. $[CoI_6]^{3-}$ IV. $[Co(NH_3)_6]^3$

(a) III > I > II > IV (b) III > I > IV > II
(c) II > IV > I > III (d) I > III > IV > II

135. Select the correct order of C-O bond length in the following complexes.

(a) $[Mo(CO)_3(PF_3)_3] < [Mo(CO)_3(PCl_3)_3] < [Mo(CO)_3(PMe_3)_3]$

(b) $[Mo(CO)_3(PMe_3)_3] > [Mo(CO)_3(PF_3)_3] > [Mo(CO)_3(PCl_3)_3]$

(c) $[Mo(CO)_3(PCl_3)_3] > [Mo(CO)_3(PMe_3)_3] > [Mo(CO)_3(PF_3)_3]$

(d) $[Mo(CO)_3(PF_3)_3] > [Mo(CO)_3(PCl_3)_3] > [Mo(CO)_3(PMe_3)_3]$

Coordination Compounds

Multiple Choice Questions with One or More than One Answers

1. Select the correct statements
(a) $[Ni(en)_3]^{2+}$ is less stable than $[Ni(NH_3)_6]^{2+}$
(b) Increase in stability of the complexes due to presence of multidentate cyclic ligand is called macro cyclic effect
(c) A complex ion that exchanges ligands slowly is said to be non-labile or inert
(d) For a given ion the ligand, greater the charge on the metal ion, greater the stability

2. Among the following the chelate complexes are
(a) Bis (dimethlyglyoximate) nickel (II)
(b) Potassiumethylenediaminetetracyanato chromate (III)
(c) Tetrammine dicyanato cobalt (lll) nitrate
(d) trans-diglycinato palladium (ll)

3. The coordination compounds which exhibit optical isomerism
(a) Pentaammine nitrocobalt (lll) iodide
(b) Diamminechloro platinum (II)

34. Select the correct statements among the following.
(a) Chelation effect is maximum for five and six membered rings
(b) Greater the charge on the central metal cation, greater the value of (CFSE)
(c) In complex ion $\left[CoF_6\right]^{3-}$ ion is weak ligand so that $\Delta_0 < P$ and it is low spin complex
(d) $\left[CoCl_2(NH_3)_2(en)\right]^+$ Complex ion will have four different isomers

35. Among the following the false statements are.
(a) In $\left[CoBrCl(en)_2\right]^+$ geometrical isomerism exist, while optical isomerism does not exist
(b) Potassium aquadicyanosuperoxo chromate (III) is the IUPAC name of $K_2\left[Cr(CN)_2O_2(O_2)(H_2O)\right]$
(c) There are 3 geometrical and 15 stereo isomers possible for $\left[Pt(NO_2)(NH_3)(NH_2OH)(Py)\right]^+$ and $\left[PtBrCl(NO_2)_2(NH_3)(Py)\right]$ respectively
(d) Cis and Trans forms are not diastereomers to each other

36. The number of nonchiral and chiral isomers for the complex $\left[Ma_2b_2c_2\right]^{\pm n}$ in which a, b, c are monodentate ligands.
(a) 4, 2 (b) 6, 2 (c) 6, 4 (d) 6, 3

37. The complex $\left[Fe(H_2O)_5NO\right]^{2+}$ is formed in ring test for nitrate when freshly prepared $FeSO_4$ solution is added to aqueous solution of $NaNO_3$. The complex formed by charge transfer in which
(a) Fe^{2+} changes to Fe^{3+} and NO changes to NO^+
(b) Fe^{3+} changes to Fe^{2+} and NO^+ changes to NO^+
(c) Fe^{2+} changes to Fe^{1+} and NO changes to NO^+
(d) Fe^{3+} changes to Fe^+ and NO changes to NO

38. Among the following the correct statements are
(a) $[Ag(NH_3)_2]^+$ is linear with sp hybridized Ag^+ ions
(b) $NiCl_4^{2-}$, CrO_4^{2-} and MnO^-_4 have tetrahedral geometry
(c) $[Cu(NH_3)_4]^{2+}$, $[Pt(NH_3)_4]^{2+}$ and $[Ni(CN)_4]^{2-}$ have dsp^2-hybridization of the metal ion.
(d) $Fe(CO)_5$ has trigonal bipyramidal structure with dz^2sp^3 hybridized ion.

39. Among the following pairs that show coordination isomerisms are
(a) $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$
(b) $[Co(NH_3)_3(H_2O)_2Cl]Br_2$ and $[CO(NH_3)_3(H_2O)ClBr]Br$
(c) $[Pt(NH_3)_4Cl_2]Br_2$ and $[Pt(NH_3)_3Br_2]Cl_2$
(d) $[Co(NH_3)_6[Cr(C_2O_4)_3]$ and $[Cr(NH_3)_6[Co(C_2O_4)_3]$

40. Regarding tetramminedithiocyanato –S Cobalt – (III) tris (oxalato) Cobaltate (III) the correct statements are
(a) Formula of the complex is $[Co(SCN)_2(NH_3)_4][Co(Ox)_3]$
(b) It is a chelating complex and show linkage isomerism
(c) it shows optical isomerism
(d) it shows geometrical isomerism

41. Which of the following statements are correct?
(a) In metal carbonyl complexes $d_{c\text{-}o}$ increases compared to that of C-O molecule
(b) The pair of compounds $[Cr(H_2O)_6]Cl_3$ and $[CrCl_3(H_2O)_3]\cdot 3H_2O$ show hydrate isomerism
(c) dz^2 orbital of central atom/ion is used in dsp^2 hybridization
(d) Facial and meridional isomers associated with Ma_3b_3 type complex compound both are optically active.

42. Which of the following isomerism is exhibited by $[Cr(NH_3)_2(H_2O)_2Cl_2]Cl$
(a) Ionization (b) Geometrical
(c) Hydrate (d) Optical

43. The correct statements among the following
(a) $MnCl^-_4$ ion has tetrahedral geometry and paramagnetic
(b) $[CuCl_4]^{2-}$ has square planar geometry and is paramagnetic
(c) $[Mn(CN)_6]^{2-}$ ion has octahedral geometry and is paramagnetic
(d) $[Ni(Ph_3P)_2Br_3]$ has trigonal bipyramidal geometry and one unpaired electron

44. The correct statements about the complex ion $[Cr(en)_2Cl_2]$ are
(a) It has two geometrical isomers – cis and trans
(b) Both the cis and trans isomers display optical activity
(c) Only the cis isomer displays optical activity
(d) Only the cis isomer has non-super imposable mirror image

45. Find out the false statements among the given below
(a) $[Cu(CN)_4]^{3-}$ has tetrahedral geometry and dsp^2 hybridization
(b) $[Ni(CN)_6]^{4-}$ is octahedral and Ni has d^2sp^3 hybridization
(c) $[ZnBr_4]^{2-}$ is tetrahedral and diamagnetic
(d) $[Cr(NH_3)_6]^{3+}$ has octahedral geometry and sp^3d^2 hybridization

46. The paramagnetic moment is nearly 1.73 BM for the complex is

(a) $[Ni(dmg)_2]$ (b) $[Cu(H_2O)_4]^{2+}$

(c) $[Cu(dmg)_2]$ (d) $[CoCl_4]^{2-}$

47. Iron cannot form complexes with ammonia because

(a) In alkaline solution iron salts are precipitated as their hydroxides which are insoluble in excess of alkali

(b) In acid solutions NH_3 cannot form complex because of protonation it changes to NH_4^+ ion which have no donor site

(c) In the presence of ammonia iron nitrides are formed

(d) In solution aqua complexes of iron salts are produced which protects the ions of iron

48. The correct stability orders are

(a) $[Cu(NH_3)_4]^{2+} < [Cu(en)_2]^{2+} < [Cu(trien)]^{2+}$

(b) $[Fe(H_2O)_6]^{3+} < [Fe(NO_2)_6]^{3-} < [Fe(NH_3)_6]^{3+}$

(c) $[Co(H_2O)_6]^{3+} < [Rh(H_2O)_6]^{3+} < [Ir(H_2O)_6]^{3+}$

(d) $[Cr(NH_3)_6]^{+} < [Cr(NH_3)_6]^{2+} < [Cu(NH_3)_6]^{3+}$

49. Which of the following will have two stereo isomeric forms only?

(a) $[Cr(NO_3)_3(NH_3)_3]$ (b) $K_3[Fe(C_2O_4)_3]$

(c) $[CoCl_2(en)_2]^+$ (d) $[CoBrCl(Ox)_2]^{3-}$

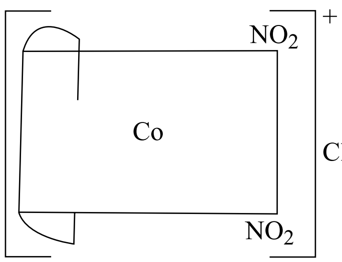
50. In Complexes of gold, the coordination number of gold may be

(a) 6 (b) 4 (c) 5 (d) 2

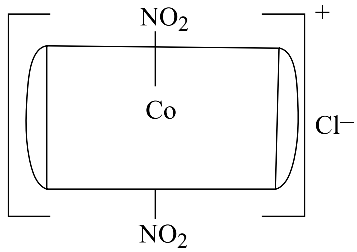
Comprehensions

Passage-1

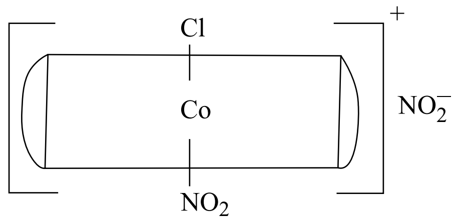
Following experiment has been given to identify isomer. Read the experiment and answer the questions A compound $Co(en)_2(NO_2)_2Cl$ has been prepared in a number of isomeric forms. One form undergoes no reaction with $AgNO_3$ or (en), and is optically inactive. A second form reacts with $AgNO_3$ but not with (en) and is optically inactive. A third form is optically active and reacts with both $AgNO_3$ and (en) complexes are?



(a) Cis - bis [ethylene diamine) dinitrocobalt (III) chloride



(b) Trans - bis (ethylene diamine) dinitro cobalt (III) chloride



(c) Trans - chloro nitro bis (ethylene diamine) cobalt (III) nitrite

1. First form of the complex is

(a) A (b) B (c) C (d) Any of these

2. One of the complex turns KI-starch paper blue in acidic medium. The complex is

(a) C (b) B (c) A (d) All of these

3. Second form of the complex is

(a) A (b) B (c) C (d) Any of these

4. Third form of the complex is

(a) A (b) B (c) C (d) Any of these

Passage-2

Complex compounds are molecular compounds which retain their identities even when dissolved in water. They do not give all the simple ions in solution but instead furnish complex ions with complicated structures. The complex compounds are often called coordination compounds because certain groups called ligands are attached to the central metal ion by coordinate or dative bonds. Coordination compounds exhibit isomerism, both structural and stereoisomerism. The structure, magnetic property, colour and electrical properties of complexes are explained by various theories.

1. Arrange the following compounds in order of their molar conductance:

I. $K[Co(NO_2)_4(NH_3)_2]$

II. $[Cr(ONO)_3(NH_3)_3]$

III. $[Cr(NO_2)(NH_3)_5]_3[Co(NO_2)_6]_2$

IV. $Mg[Cr(NO_2)_5(NH_3)]$

(a) II< I< IV< III (b) I< II< III< IV
(c) II < I <III < IV (d) IV < III < II < I

2. The oxidation number and coordination number of chromium in the following complex is $[Cr(C_2O_4)_2(NH_3)_2]^{-1}$
(a) O.N. = +4, C.N. = 4 (b) O.N. = –1, C.N. = 6
(c) O.N.= –1, C.N. = 4 (d) O.N. = +3, C.N. = 6

3. In which of the following pairs, both the complexes have the same geometry?
(a) $[Ni(Cl)_4]^{2-}$, $[Ni(CN)_4]^{2-}$
(b) $[CoF_6]^{3-}$, $[Co(NH_3)_6]^{3+}$
(c) $[Ni(CO)_4]$, $[Ni(CN)_4]^{2+}$
(d) $[Cu(NH_3)_4]^{2+}$, $[NiCl_4]^{2-}$

Passage-3

Valence bond theory successfully explains the magnetic behaviour of complexes. The substances which contains unpaired electrons are paramagnetic and paramagnetic character increases as the number of unpaired electrons increases. Magnetic moment of a complex can be determined experimentally and by using formula $\sqrt{n(n+2)}$ and we can determine the number of unpaired electrons in it. This information is important in writing electronic structure of complex which in turn also useful in deciding the geometry of complex.

1. There are four complexes of Ni. Select the complexes which will be attracted by magnetic field?
$[Ni(CN)_4]^{2-}$ (I) $[NiCl_4]^{2-}$ (II) $Ni(CO)_4$ (III) $[Ni(H_2O)_6]^{2+}$ (IV)
(a) I only (b) II and IV only
(c) II, III and IV (d) II and III

2. The magnetic moment of complexes given below are in the order
$Ni(CO)_4$ (I) $[Mn(CN)_6]^{4-}$ (II) $[Cr(NH_3)_6]^{3+}$ (III) $[CoF_6]^{3-}$ (IV)
(a) I > II > III > IV (b) I < II < III < IV
(c) IV > II > I > III (d) IV < II < I < III

3. Which of the following are diamagnetic?
I. $K_4[Fe(CN)_6]$ II. $K_3[Cr(CN)_6]$
III. $K_3[Co(CN)_6]$ IV. $K_2[Ni(CN)_4]$
(a) I, II & IV (b) I, III & IV
(c) II & III (d) I & IV

Passage-4

An excess of aqueous potassium cyanide (KCN) is added to a solution of 0.02 moles of ferrous sulphate (Iron (II) sulphate)

$$Fe^{2+} + 2CN^- \rightarrow Fe(CN)_2$$

The mixture is boiled in a fume cupboard and the precipitate of iron(II) cyanide then dissolves to form $[Fe(CN)_6]^{4-}$ ions.

$$Fe(CN)_2 + 4CN^- \rightarrow [Fe(CN)_6]^{4-}$$

Chlorine is bubbled through this solution and the $[Fe(CN)_6]^{4-}$ ions are converted to $[Fe(CN)_6]^{3-}$ ions.

$$Cl_2 + 2[Fe(CN)_6]^{4-} \rightarrow 2Cl^- + 2[Fe(CN)_6]^{3-}$$

Crystals of potassium hexacyanoferrate (III), are obtained from this solution and purified by crystallization.

1. What mass of iron (II) sulphate crystals, $FeSO_4.7H_2O$ should be weighed to obtain 0.02 moles of iron (II) sulphate? [Fe = 56).
(a) 1.66 g (b) 2.78 g
(c) 3.04g (d) 5.56 g

2. What is the minimum number of moles of potassium cyanide which must be added to the 0.02 moles of iron (II) sulphate to obtain the maximum yield of potassium hexacyanoferrate (II) $K_4[Fe(CN)_6]$
(a) 0.02 (b) 0.04 (c) 0.08 (d) 0.12

3. The most likely reason for boiling the mixture in a fume cupboard is that
(a) Potassium cyanide is a covalent compound and would vapourize rapidly on boiling the mixture.
(b) Steam would be prevented from escaping into the laboratory.
(c) Iron (II) sulphate is less likely to be oxidized by atmospheric oxygen to iron (III) sulphate.
(d) Some hydrogen cyanide is formed by hydrolysis and would be given off on boiling the mixture.

4. The chlorine is prepared from hydrochloric acid. Before the chlorine is bubbled through the solution, any traces of hydrogen chloride must be removed. This can best be done by passing the chlorine through wash bottles containing
(a) water
(b) aqueous potassium iodide
(c) aqueous sodium hydroxide
(d) aqueous chlorine

Passage-5

Complex compounds can exhibit isomerism like structural and sterio. The complexes containing coordination number 1,2,3 cannot exhibit geometrical isomerism. Geometrical and optical isomerism is common in the complexes having coordination number 4 and 6. Ma_3b_3 shows facial and meridonial isomerism

1. Number of geometrical isomers possible for complex $[Pt(NH_3)(Py)(Cl)(Br)]$ is
(a) 2 (b) 3 (c) 1 (d) 4

2. The number of unpaired electrons present in $\left[Co\left(NH_3\right)_6\right]^{3+}$ and $\left(CoF_6\right)^{3-}$ are
   (a) 4 and 4 (b) 0 and 4
   (c) 4 and 0 (d) 2 and 4
3. $[Co(NH_3)_5NO_2]C1_2$ and $[Co(NH_3)_5ONO]C1_2$ are
   (a) Linkage isomers
   (b) Ionization isomers
   (c) Coordinate isomers
   (d) Geometrical isomers

Passage-6

A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms (X) and (Y).

The form (X) reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous ammonia, whereas (Y) gives a pale yellow precipitate soluble in concentrate ammonia

1. Choose the true statement regarding compound X and Y
   (a) Both compounds X and Y have different magnetic moment (spin only)
   (b) Both complexes X and Y have same magnetic moment (spin only)
   (c) Magnetic moment of both X and Y is calculated by assuming they are low spin complex
   (d) Both compounds X and Y have different primary valence
2. The hybridization of compounds X and Y is respectively
   (a) sp^3d^2, d^2sp^3 (b) sp^3d^2; sp^3d^2
   (c) d^2sp^3,d^2sp^3 (d) d^2sp^3, sp^3d^2
3. Choose the correct statement regarding the complex X and Y
   I. Both complexes have same Van't Hoff factor with respect to any solvent
   II. Both complexes have exactly same conductivity in H_2O
   III. In both the complexes NH_3 is involved only to secondary valence
   IV. In both the complexes electronic configuration of t_{2g} is same
   (a) I, II and III (b) I, II, III and IV
   (c) II, III and IV (d) I, III and IV

Passage-7

A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms (X) and (Y).

The form (X) reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous ammonia, whereas (Y) gives a pale yellow precipitate soluble in concentrate ammonia

(i) 'X' is
   (a) $[Cr(NH_3)_4Cl_2]Br$ (b) $[Cr(NH_3)_4BrCl]$ CI
   (c) $[Cr(NH_3)_6]Cl$ (d) $[Cr(NH_3)_6]Br$

(ii) 'Y' is -
   (a) $[Cr(NH_3)_4Cl_2]$ Br (b) $[Cr(NH_3)_4BrCl]$ CI
   (c) $[Cr(NH_3)_6]Cl$ (d) $[Cr(NH_3)_6]Br$

(iii) 'X' and 'Y'are
   (a) Linkage isomers
   (b) Ionization isomers
   (c) Coordinate isomers
   (d) Ligand isomers

Passage-8

In carbon monoxide the C-O bond length is 112.8 pm. Weakening of the C-O bond increases in bond length is found in complexes containing CO as ligand with C-O distance is approximately 115 pm for many carbonyls

While it is possible to formulate the bonding in terms of a resonance hybrid of following structures

$$M^+ - C^+ \equiv O \leftrightarrow M = C = O$$

The molecular orbital formulation is more accurate. There is first a dative overlap of the filled carbon orbital and a second dative overlap of filled dπ or pπ metal orbital with an empty anti bonding pπ orbital of the CO. This bonding mechanisms synergic since the drift of metal electrons into CO orbital will tend to make the CO as a whole negative and hence to increase-its basicity via the σ orbital of carbon, at the same time the drift of electrons to the metal in the σ bond tend to make the CO positive, this enhancing the acceptor strength of the orbitals.

As the extent of back-donation from 'M'to CO increases the M-C bond becomes stronger and the CO bond becomes weaker. Thus the multiple bonding should be evidenced by shorter MC and longer CO bonds as compared MC single bonds and CO triple bonds respectively.

If some CO groups are replaced by ligands with low or negligible back accept ability those CO groups that remain, must accept electrons from the metal to a greater extent in order to prevent the accumulation of negative charge on the metal atom which causes the shortening of M-C bond increase in C-O bond lengths. Similarly when we go from $Cr(CO)_6$ to the is electronic $[V(CO)_6]^-$ when more negative charge must be taken from the metal atoms further the M-C bond length decreases while C-O bond length increases. Conversely a change that would tend to inhibit the shift of electrons from metal to CO π orbital's. Such as placing a positive charge on the metal should cause the increase in M-C bond length and decrease in C-O bond length

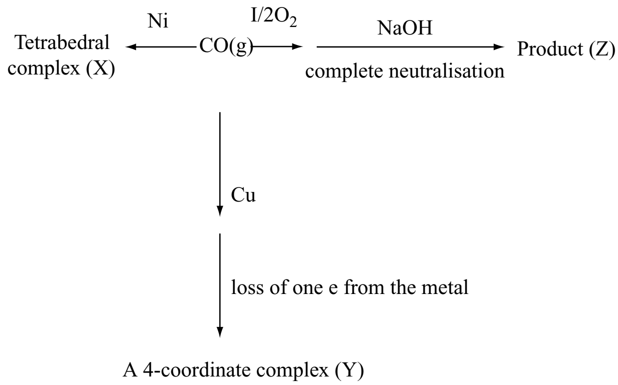
For molecules of the type $ML_3(CO)_3$ where 'L' has negligible back accepting ability, each C-O group shares two metal *d* orbital from which it receives electrons with the ligand L. In $M(CO)_6$ each d orbital is contributing equally to all six ligands. Since there are three dπ orbitals each containing one electron pair. There can be at least 1/2 of a π bond for each M - C pair, If the dπ electrons are fully used now in the case where 'L' represents a ligand incapable of accepting dπ electrons, each of the three CO groups has access to 1/3 of the three pairs of d π electrons enter fully into the back bonding, each M-C pair will have a full dπ bond. Thus we assume full participation by the metal dπ electrons in each case, the M-C bond orders go from 1.5 in M $(CO)_6$ to 2.0 in ML_3 $(CO)_3$ where L is a ligand atom without π acceptor orbitals. The C-O bond order in these two cases should then be respectively 2.5 and 2.0.

1. Which of the following is correct increasing order of C-O bond length of the carbonyls given below:
   I. $Cr(CO)_6$
   II. $[V(CO)_6]^-$
   III. $[Mn(CO)_6]^+$
   IV. $[Cr(CH_3NH_2)_3(CO)_3]$
   (a) IV < II < I < III (b) I < II < III < IV
   (c) II > IV > I > III (d) IV < I < II < III
2. Among $Fe(CO)_5$, $[Fe(CO)_4\ (PEt_3)]$ and $[Fe\ (CO)_4\ (PPh_3)]$ the one having longest M-C bond
   (a) Fe $(CO)_5$ (b) [Fe $(CO)_4$ (PEt_3)]
   (c) [Fe $(CO)_4$ (PPh_3)]
   (d) All have equal M-C bond lengths
3. N_2 is iso electronic with CO but N_2 is a weaker π -acceptor than CO because
   (a) In N, σ and π levels are very close together in energy and they are all symmetric
   (b) The CO levels are farther apart in energy and skewed toward C
   (c) The geometric overlap for CO is better than
   (d) All are correct

Passage-9

Various physical and chemical methods are used to distinguish between isomeric complexes. Most common of them are magnetic moment determination in solid states and molar conductance and colligative properties measurements of complexes in dilute solution, $AgNO_3$ test, $BaCl_2$ test etc. beside IR and electronic spectral studies which are beyond the scope of the level of junior college students. While molar conductance of an electrolytic complex depends upon number of ions, charges carried by them and speed of ions, colligative properties depend only on the number of particles (molecules or ions) dissolved in a given amount of solvent. Metal complexes are generally coloured due to d - d transition. For d-d transition the central metal in a complex which is generally a transition metal ion must have incomplete d-orbital. Amongst the lakhs of metal complexes reported so far metal carbonyls are of special importance. They are formed by the interaction of carbon monoxide with metal in atomic state. Metal carbonyls are stabilized by back-bonding which exerts synergic effect i.e., sum of the bond energies of two double bonds (M = C = O) is more than bond energies of one M - O single bond and one C-O triple bond ($M - C \equiv O$). In aqueous solution a metal ion exists in the form of aqua complexes.

1. Which of the following can't help in distinguishing between cis-$[Co(en)_2(NH_3)Br]SO_4$ and trans-$[Co(en)_2(NH_3)(SO_4)]$ Br isomeric pair.
   (a) Reacting them with ethylenediamine (en)
   (b) Chemical tests like $BaCl_2$ test and $AgNO_3$ test
   (c) Colligative property measurement
   (d) Molar conductance measurement
2. Carbon monoxide gas is subjected to following reaction to give the products X, Y and Z.



   The C - O bond length in the marked products and in CO itself marked as R are in the order
   (a) Z > Y > X > R (b) Z > X > Y > R
   (c) X = Y > R > Z (d) Z > R > Y > X
3. Which one of the following pairs of ions will both impart colour in aqueous solution?
   (a) $[Sc^{3+}, Ti^{3+}]$ (b) $[V^{3+}, Cu^{2+}]$
   (c) $[Ca^{2+}, Mn^{4+}]$ (d) $[Ni^{2+}, Zn^{2+}]$

Passage-10

The following ionization constants are determined in three separate experiments

I. $[Ag(S_2O_3)_3]^{5-} \rightleftharpoons [Ag\ (S_2O_3)_2]^{3-} + S_2O_3^{2-};\ K_1 = 2.0\times10^{-5}$ mole dm^{-3}

II. $[Ag(S_2O_3)_2]^{3-} \rightleftharpoons [Ag\ (S_2O_3)_2]^{-} + S_2O_3^{2-};\ K_2 = 3.3\times10^{-5}$ mole dm^{-3}

III. $[Ag(S_2O_3)^{-}] \rightleftharpoons Ag^{+} + S_2O_3^{2-};\ K_3 = 1.5\times10^{-9}$ mole dm^{-3}

1. Which of the following is the most stable?
   (a) $[Ag(S_2O_3)_3]^{5-}$ (b) $[Ag(S_2O_3)_2]^{3-}$
   (c) $[Ag(S_2O_3)]^{-}$ (d) $S_2O_3^{2-}$
2. In which of the equilibrium system, the concentration of $S_2O_3^{2-}$ is maximum
   (a) I (b) II
   (c) III (d) same conc in all
3. The complex formation constant for the reaction
   $Ag + 2S_2O_3^{2-} \rightleftharpoons [Ag(S_2O_3)_2]^{3-}$
   (a) 3.3×10^{-5} (b) 1.5×10^{-9}
   (c) 4.95×10^{-14} (d) 2×10^{13}

Passage-11

Ruby crystals have a deep red colour and are well known for their use in Jewellery. Not many people know that the heart of the first laser, built in 1960 by Maiman was a big ruby crystal. The red colour of ruby originates from the absorption of light by chromium atoms that are incorporated in colourless aluminum oxide crystals by replacing aluminum

1. Indicate what will happen with the ruby, when kept in a balance where magnet is kept directly under the side with the ruby
   (a) The magnet attracts the ruby (the ruby moves down)
   (b) The magnet has no influence on the ruby (the ruby does not move)
   (c) The magnet repels the ruby (the ruby moves up)
   (d) The magnet has an oscillating effect on the ruby (the ruby moves up and down)
2. In the rubies the number of oxygen atoms coordinating to chromium ion is
   (a) 4 (b) 6 (c) 2 (d) 8
3. The magnetic moment of each chromium ion in the ruby is
   (a) 0 (b) 2.814 (c) 3.85 (d) 5.90

Passage-12

A complex compound of cobalt contains five NH_3 molecules, one nitro group and two chloride ions for one Co atom. One mole of this compound produces three mole ions in aq. solution. One mole reacting with excess of $AgN0_3$ solution, two moles of AgCl get precipitated.

1. The formula of the complex compound is
   (a) $[Co(NH_3)_4NO_2Cl][(NH_3)Cl]$
   (b) $[Co(NH_3)_5Cl][Cl(NO_2)]$
   (c) $[Co(NH_3)_5NO_2]Cl_2$
   (d) $[Co(NH_3)_5NO_2Cl_2]$
2. The type(s) of isomerism(s) shown by the complex compound is(are)
   (a) Geometrical (b) Ionization
   (c) Linkage (d) Both (2) and (3)
3. The IUPAC name of the complex compound is
   (a) penta ammine nitro cobalt (II) chloride
   (b) penta ammine nitro cobalt (III) chloride
   (c) penta amine nitrotocobalt (II) chloride
   (d) penta amine chloride cobalt (II) nitrite

Passage-13

A double salt consisting of the formula $Cu(NH_3)Cl_y \cdot zH_2O$ had a molecular mass 277.5 g/mol. 1.388 g of salt gave 2.87 g of silver chloride and the same amount when boiled with excess of sodium hydroxide liberated NH_3 which neutralizes 10 mL of 1 M HC1.

1. The number of chlorine atoms present per molecule of the complex is
   (a) 1 (b) 2
   (c) 3 (d) 4
2. The number of NH_3 liberated per 1.388 g of the complex salt is
   (a) $17X\ 10^{-2}$ g (b) $0.17\ X\ 10^{-2}$ g
   (c) 1700 g (d) 1.7g
3. The formula of the complex is
   (a) $Cu(NH_3)_2Cl_4 \cdot 2H_2O$
   (b) $Cu(NH_3)_4Cl_4 \cdot 2H_2O$
   (c) $Cu(NH_3)_4Cl_3 \cdot 2H_2O$
   (d) $Cu(NH_3)_4Cl_2 \cdot 2H_2O$

Passage-14

A metal complex having composition $Cr(NH_3)Br_2I$ was isolated in two forms (X) and (Y). Form (X) reacts with $AgNO_3$ to give a pale yellow precipitate which is partially soluble in excess of NH_4OH whereas (Y)gives a greenish yellow precipitate which is insoluble in NH_4OH

1. Which of the following statement is correct for X and Y.
   (a) The formula of (X) and (Y) are $[Cr(NH_3)_4IBr]Br$ and $[Cr(NH_3)_3IBr]Br$ respectively.
   (b) The formula of (X) and (Y) are $[Cr(NH_3)_4IBr]Br$ and $[Cr(NH_3)_4Br_2]I$ respectively.
   (c) The formula of (X) and (Y) are both $[Cr(NH_3)_4I]BrI$
   (d) The formula of (X) and (Y) are $[Cr(NH_3)_2IBr_2]BrI$
2. Both the (X) form and (Y) form show
   (a) linkage isomerism
   (b) coordination isomerism
   (c) ionization isomerism
   (d) none of these

3. (a) (X) – cis form optically inactive
(Y) – cis form optically active
(b) (X) – cis form optically inactive
(Y) – trans form optically active.
(c) The cis and trans forms of both X and Y are optically active.
(d) The cis and trans form of both X and Y are optically inactive.

Comprehensions

Passage-15

VB theory describes the bonding in terms of hybridized orbitals of the central metal atom or ion. The theory mainly deals with the geometry (i.e., shape and magnetic properties) The salient features of the theory are

1. The central metal atom loses a requisite number of electrons to form the ion. The number of electrons lost is the valence of the resulting cation. In some cases, the metal atom does not lose electrons.
2. The cental metal atom/ion makes available a number of empty s, p and d-atomic orbitals equal to its coordination number. These vacant orbitals hybridized together to form hybrid orbitals which are same in the number as the atomic orbitals hybridizing together. These hybrid orbitals are vacant, equivalent in energy and have definite geometry.

1. $[Cu(NH_3)_4]^{+2}$ has hybridization
(a) sp^3 (b) dsp^2
(c) sp^2d (d) none of these
2. Which complex has zero magnetic moment (spin only)?
(a) $[NiCl_4]^{-2}$ (b) $[Ni(CN)_4]^{2-}$
(c) $[Cr(CN)_6]^{3+}$ (d) all of these
3. Which of the cyano complex would exhibit the lowest value of paramagnetic behaviour
(a) $[Cr(CN_6)]^{-3}$ (b) $Co(CN)_6]^{-3}$
(c) $[Fe(CN)_6]^{-3}$ (d) $[Mn(CN)_6]^{-3}$

Matching Type Question

Match the following which may have *more* than one matching

1. Match the following

List-I	List-II
(a) $K_3[Cr(C_2O_4)_3]^{3-}$	(p) Ionization isomerism
(b) $[Co(en)_2NO_2Cl]Cl$	(q) Linkage isomerism
(c) $[Co(en)(NO_2)_2Cl_2]^-$	(r) Geometrical isomerism
(d) $[Pt(NH_3)(C_5H_5N)(Cl)(Br)]$	(s) Optical isomerism

2.

Column -I	Column -II
(a) $Cr(en)_3]^{3+}$	(p) Paramagnetic
(b) $[Mn(en)_6]^{3-}$	(q) $\mu = \sqrt{15}$ BM
(c) $[Co(H_2O)_6]^{3-}$	(r) Two unpaired electrons
(d) $[Fe(CN)_6]^{3-}$	(s) Low Spin complex

3. Match the complexes in Column-I with their properties listed in column-II. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column-I	Column-II
(a) $[Co(NH_3)_4(H_2O)_2]Cl_2$	(p) Geometrical isomers
(b) $[Pt(NH_3)_2Cl_2]$	(q) Paramagnetic
(c) $[Co(H_2O)_5Cl]Cl$	(r) Diamagnetic
(d) $[Ni(H_2O)_6]Cl_2$	(s) Metal ion with +2 oxidation state

4.

Column-I	Column-II
(a) $[Ni(H_2O)_6]Cl_3$	(p) d^2sp^3 hybridization
(b) $[Co(CN)_2(NH_3)_4]OC_2H_5$	(q) Ionization isomerism
(c) $[IrCl_6]^{3-}$	(r) $\mu = 2.83$ BM
(d) $[PtCl_2(NH_3)_4]Br_2$	(s) $\Delta_0 < P$

5. Match the complexes in Column-I with their properties listed in Column-II

Column-I	Column-II
(a) $[Cr(en)_2Cl_2]^+$	(p) Optical isomerism
(b) $[Fe(CN)_6]^{3-}$	(q) d^2sp^3 hybridization
(c) $[Ni(NH_3)_6]^{2+}$	(r) Paramagnetic
(d) $[CoF_6]^{3-}$	(s) sp^3d^2 hybridization

6. Match the following

(a) A compound which cannot form isomers	(p) $[Pt(NH_3)_2NOCl]$
(b) A compound in which metal involve six orbital's in Hybridization	(q) $[Pt(NH_3)_4ClBr]SO_4$
(c) Complex can show ionization isomerism	(r) $[Pt(NH_3)_3Cl]Br$
(d) A compound which can form cis and trans isomers	(s) $[PtNH_3Cl_5]^-$

7. Match the following Column-I with Column-II

Column-I	Column-II
(a) $[Co(NO_2)_2(H_2O_2)_2(NH_3)_2]NO_3$	(p) Number of Stereo isomer's = 6
(b) $[Ni(en)_3]Br_3$	(q) Linkage isomerism

Column-I	Column-II
(c) $[Co(NH_3)_3(py)_3]Br_2$	(r) Ionization isomerism
(d) $[Pt(en)(SCN)_2](NO_3)_2$	(s) Optical isomerism
	(t) Geometrical isomerism

8. Column-I	**Column-II**
(a) $[Mn(H_2O)_6]^{2+}$	(p) Paramagnetic
(b) $[Mn(CN)_6]^{3+}$	(q) Diamagnetic
(c) $[Co(NH_3)_6]^{3+}$	(r) High spin complex (or) outer orbital complex
(d) $[Ni(NH_3)_6]^{2+}$	(s) Low spin complex (or) inner orbital complex

9. Match the following:

List-I	List-II
(a) $CoCl_3 \cdot 4NH_3$	(p) A non electrolyte
(b) $CoCl_3 \cdot 5NH_3$	(q) Contains a dispositive coordination ion
(c) $PtCl_2 \cdot 2NH_3$	(r) 1 mole of the compound gives, with an excess of $AgNO_3$ solution, a quick precipitation of 1 mole of AgCl
(d) $PtCl_4 \cdot 2NH_3$	(s) Shows cis-trans isomerism

10. Column-I (Pair of complex compounds)	**Column-II (Property which is different in given pair)**
(a) $[Ni(CO)_4]$ and $K_2[Ni(CN)_4]$	(p) Magnetic moment
(b) $[Cu(NH_3)_4]SO_4$ and $K_3[Cu(CN)_4]$	(q) Oxidation no. of central metal
(c) $K_2[NiCl_4]$ and $K_4[Ni(CN)_4]$	(r) Geometry
(d) $K[NiCl_4]$ and $K_2[PtCl_4]$	(s) EAN of central metal

11. Match the following Column-I with Column-II

Column-I	Column-II
(a) $[Cr(gly)_3]^0$	(p) Low spin complex
(b) $[CoBr_2Cl_2(SCN)_2]^{3-}$	(q) High spin complex
(c) $[PtCl_4]^{2-}$	(r) Optical isomerism
(d) $Na[PtBrCl(NO_2)(NH_3)]$	(s) Geometrical isomerism

12. Match the following Column-I with Column-II

Column-I	Column-II
(a) Sodium nitroprusside	(p) $\mu = 0$ BM
(b) Brown ring complex	(q) Octahedral
(c) Complex of Ag formed during the extraction	(r) $\mu = \sqrt{15}$ BM
(d) Potassium ferrocyanide	(s) NO^+ ligand

13. Match the following Column-I with Column-II

Column-I	Column-II
(a) $Ni(CO)_4$	(p) Octahedral paramagnetic
(b) $[Ni(CN)_4]^{2-}$	(q) Square planar diamagnetic
(c) $[Ni(NH_3)_6]^{2-}$	(r) Tetrahedral Diamagnetic
(d) $[NiCl_4]^{2-}$	(s) Tetrahedral paramagnetic

14. Match the following List-I with List-II

List-I Pair of complex compounds	List-II Property which is same in a given pair
(a) $Ni(CO)_4$ and $[NiF_4]^{2-}$	(p) No. of unpaired electrons
(b) $[Cu(NH_3)_4]C1_2$ and $Na_2|Ni(CN)_4]$	(q) Hybridization of central atom ion
(c) $[Cr(NH_3)_6]^{3+}$ and $[Co(NH_3)_6]^{3+}$	(r) Shows isomerism
(d) $[Cr(NH_3)_4(NO_2)_2]Br$ and $[Co(NH_3)_4(NO_2)]Br$	(s) Structure of complex

15. Column-I	Column-II
(a) $[Co(en)_2Cl_2]Br$	(p) Linkage isomerism
(b) $[Co(NH_3)_3Cl_3]$	(q) Optical isomerism
(c) $[Ni(H_2NCH_2COO)_2]$	(r) Geometrical isomerism
(d) $[Co(NH_3)_5(NO_2)]Cl_2$	(s) Ionization isomerism

16. Match the following Column-I with Column-II

Column-I	Column-II
(a) $K_3[Fe(CN)_5(CO)]$	(p) Complex having lowest bond length of CO ligand
(b) $K[PtCl_3(C_2H_4)]$	(q) Follow rule of EAN
(c) $Na[Co(CO)_4]$	(r) Complex involved in synergic bonding
(d) $[V(CO)_6]$	(s) Complex having highest bond length of CO ligand

17.

Column-I	Column-II
(a) $[Pt(NH_2)_3Cl_2]$	(p) Ionization isomerism
(b) $[Co(en)_3]^{3-}$	(q) Geometrical isomerism
(c) $[Co(NH_3)_3(NO_2)_3]$	(r) Optical isomerism
(d) $[Co(NH_3)_5Br]Cl$	(s) Facial-meridional isomerism

18. Match the following-I with Column-II Coordination compounds

Column-I	Column-II
(a) $[Fe(CN)_6]^{4-}$	(p) Paramagnetic
(b) $[Fe(H_2O)_6]^{2+}$	(q) Diamagnetic
(c) $[Cu(NH_3)_6]^{2+}$	(r) Inner orbital complex
(d) $[Ni(CN)_6]^{4-}$	(s) Outer orbital complex

19. Match the following Column-I with Column-II

Column-I (Pair of complexes)	Column-II (Property which is similar in given pair)
(a) $[Fe(CN)_6]^{3-}$ and $[Co(NH_3)_6]^{3+}$	(p) Magnetic moment
(b) $[Fe(H_2O_6)]^{2+}$ and $[Fe(CN)_6]^{4-}$	(q) Geometry
(c) $[Ni(CN)_4]^{4-}$ and $[Ni(CO)_4]$	(r) Hybridization
(d) $[Ni(H_2O)_6]^{2+}$ and $[NiCl_4]^{2-}$	(s) Number of d-electrons

20.

Column-I	Column-II
(a) $[Ni(CN)_4]^{2-}$	(p) sp^3
(b) $CuCl_5^{2-}$	(q) dsp^2
(c) $AuCl_4^-$	(r) sp^3dz
(d) ClO_4^-	(s) $d_{x^2-Y^2}$

21. Match the following Column-I with Column-II

Column-I	Column-II
(a) $[Mn(CN)_6]^{3-}$	(p) Diamagnetic
(b) $[MnCl_4]^{2-}$	(q) Paramagnetic
(c) $[Ni(CN)_4]^{2-}$	(r) d^2sp^3
(d) $[NiCl_4]^{2-}$	(s) sp^3

22. Match the following

Complexes	No. of geometrical isomers
(a) $[PtBrCl(NH_3)_2]$	(p) 2
(b) $[CoCl_2(NH_3)_2]^+$	(q) 3
(c) $[PtBrCl(NH_3)(CH_3NH_2)]$	(r) 4
(d) $[Co(NCS)_3(NH_3)_3]$	(s) 0

23. Match the complex with the oxidation state of its cobalt ion

List-I	List-II
(a) $[Co(NCS)(NH_3)_5]SO_3$	(p) +3
(b) $Na[Co(CO)_4]$	(q) 0
(c) $Na_4[Co(S_2O_3)_3]$	(r) –1
(d) $Co_2(CO)_8$	(s) +2

Matching Type Question

24. Matching

In Column-I some complexes are given and in column-II the shape of the complex and the d-orbital involved in hybridization are given. Match the complex in Column-I with the property given Column-II

Column-I	Column-II
(a) $[CuCl_4]^{3-}$	(p) orbital involved is $d_{x^2-Y^2}$
(b) $[Cu(CN)_4]^{2-}$	(q) orbital involved is d_{z^2}
(c) $[FeF_6]^{3-}$	(r) orbital involved is d_{xy}
(d) $[Ni(CO)_4]$	(s) Tetrahedral
	(t) square planar

25. Match the Column-I with Column-II

Column-I	Column-II
(a) $[Co(NH_3)_4(H_2O)_2]Cl_2$	(p) Geometrical isomer
(b) $[Pt(NH_3)Cl_2]$	(q) Paramagnetic
(c) $[Co(H_2O)_5 Cl]Cl$	(r) Diamagnetic
(d) $[Ni(H_2O)_6]Cl_2$	(s) Metal ion with +2 oxidation state
	(t) Give ppt. with $AgNO_3$

26. Match the complexes in Column-1 with corresponding characteristics in Column-II

Column-I (Complexes)	Column-II (Characteristics)
(a) $Ni(CO)_4$	(p) Tetrahedral
(b) $[NiCl_4]^{2-}$	(q) Square planar
(c) $[Pt(en)Cl_2]$	(r) Can exhibit geometrical isomerism
(d) $[Pd(gly)_2]$	(s) Primary valency of central metal atom or ion = 2
	(t) Secondary valency of the central metal atom ion = 4

Numerical Type Questions

1. Number of sterio isomers are possible for a complex ion $[Co(NH_3)_2(H_2O)_2ClBr]^+$ is
2. Number of unpaired electrons at centrons at central atom of $[Co(C_2O_4)_3]^{3-}$ ion is

3. Co-ordination number of Fe in heamoglobin is
4. A metal complex of coordination number six having three different types of ligands a, b and c of composition $Ma_2b_2c_2$ can exist in several stereo isomeric forms; the total number of such isomers is:
5. No. of ions produced by dissolving $PtCl_4.2NH_3$ in aqueous medium is
6. $FeSO_4$ reacts with excess KCN solution to form a complex then the no. of unpaired d-electrons present in the central atom of that complex is
7. No. of possible stereo isomers for the octahedral complex ion $[ML_1L_2(L_3)_2(L_4)_2]^{M+}$
8. The difference in the number of coulombic exchange pairs for high spin and low spin d^5 ion in an octahedral complex according to crystal field theory is
9. The number of atoms involved in sp^3 hybridization in product formed by the oxidation of hypo with I_2 is
10. In the reaction $\left[CoCl_2(NH_3)_4\right]^- + Cl^- \longrightarrow \left[CoCl_3(NH_3)_3\right] + NH_3$, if the reactant complex is cis, how many isomers of product complex are formed?
11. Find out maximum number of geometrical isomers theoritically possible in a octahedral complex of formula $Pt(en)(NO_2)(Cl)(Br)(I)$
12. The possible no of coordination isomers of the complex $\left[Cr(NH_3)_6\right]\left[Co(CN)_6\right]$ is
13. The possible no of coordination isomers of the complex $\left[Pt(NH_3)_4\right]\left[PtCl_4\right]$ is
14. Number of unpaired electrons in K [Ag $(CN)_2$] is

Single Answer Questions

1. d	**2.** d	**3.** c	**4.** c	**5.** b	**6.** d	**7.** b
8. a	**9.** b	**10.** d	**11.** d	**12.** d	**13.** a	**14.** c
15. a	**16.** d	**17.** a	**18.** a	**19.** c	**20.** a	**21.** b
22. b	**23.** a	**24.** d	**25.** b	**26.** c	**27.** d	**28.** a
29. d	**30.** d	**31.** a	**32.** c	**33.** b	**34.** d	**35.** c
36. c	**37.** b	**38.** d	**39.** a	**40.** d	**41.** a	**42.** a
43. b	**44.** c	**45.** a	**46.** b	**47.** a	**48.** a	**49.** b
50. b	**51.** d	**52.** a	**53.** b	**54.** c	**55.** a	**56.** b
57. b	**58.** b	**59.** b	**60.** d	**61.** a	**62.** d	**63.** b
64. d	**65.** b	**66.** a	**67.** a	**68.** b	**69.** c	**70.** d
71. d	**72.** a	**73.** c	**74.** d	**75.** a	**76.** b	**77.** d
78. b	**79.** d	**80.** c	**81.** a	**82.** d	**83.** c	**84.** b
85. d	**86.** d	**87.** a	**88.** b	**89.** b	**90.** a	**91.** a
92. a	**93.** a	**94.** a	**95.** b	**96.** a	**97.** b	**98.** b
99. a	**100.** a	**101.** d	**102.** d	**103.** d	**104.** a	**105.** b
106. b	**107.** a	**108.** c	**109.** c	**110.** a	**111.** b	**112.** a
113. d	**114.** d	**115.** b	**116.** d	**117.** b	**118.** c	**119.** b
120. d	**121.** a	**122.** d	**123.** c	**124.** c	**125.** c	**126.** c
127. b	**128.** a	**129.** c	**130.** b	**131.** b	**132.** b	**133.** b
134. b	**135.** a					

More than One Answer Questions

1. b, c, d	**2.** a, b, d	**3.** d	**4.** b
5. b, c, d	**6.** b	**7.** a, b	**8.** b, d
9. a, b, d	**10.** a, c	**11.** d	**12.** a, d
13. b, d	**14.** a, b, d	**15.** a, b, c, d	**16.** a, b, d
17. b, d	**18.** b, d	**19.** c, d	**20.** a, b, c
21. a, b, c, d	**22.** a, d	**23.** a, d	**24.** a, b, d
25. a, d	**26.** a, c, d	**27.** a, b, c	**28.** a, d
29. a, b, c, d	**30.** b, c, d	**31.** a, c, d	**32.** b, d
33. a, d	**34.** a, b, d	**35.** a, b, c, d	**36.** a
37. c	**38.** a, b, c, d	**39.** a, d	**40.** b, c, d
41. a, b	**42.** b, c, d	**43.** a, c, d	**44.** a, c, d
45. a, b, d	**46.** b, c	**47.** a, b	**48.** a, c, d
49. a, b	**50.** b, d		

Comprehension

Passage – 1	**1.** c	**2.** a	**3.** b	**4.** a
Passage – 2	**1.** a	**2.** d	**3.** b	
Passage – 3	**1.** b	**2.** b	**3.** b	
Passage – 4	**1.** d	**2.** d	**3.** d	**4.** d
Passage – 5	**1.** b	**2.** b	**3.** a	
Passage – 6	**1.** b	**2.** c	**3.** d	
Passage – 7	**1.** b	**2.** a	**3.** b	
Passage – 8	**1.** c	**2.** b	**3.** d	
Passage – 9	**1.** c	**2.** a	**3.** b	
Passage – 10	**1.** a	**2.** a	**3.** b	
Passage – 11	**1.** a	**2.** b	**3.** c	
Passage – 12	**1.** c	**2.** d	**3.** b	
Passage – 13	**1.** d	**2.** a	**3.** a	
Passage – 14	**1.** b	**2.** c	**3.** d	
Passage – 15	**1.** b	**2.** b	**3.** b	

Match Type Questions

1.	a-s	b-p, r, s, q	c-q, r, s	d-r
2.	a-p, q	b-p, s, r	c-p, q	d-p, s
3.	a-p, q, s	b-p, r, s	c-q, s	d-q, s
4.	a-r, s	b-p, q	c-p	d-p, q
5.	a-p, q, r	b-q, r	c-r, s	d-r, s
6.	a-s	b-q, s	c-q, r	d-p, q
7.	a-p, q, r, s, t	b-s	c-t	d-q, r
8.	a-p, r	b-p, s	c-q, s	d-p, r
9.	a-r, s	b-q	c-p, s	d-p, s
10.	a-q, r, s	b-p, q, r, s	c-p, q, s	d-p, r, s
11.	a-q, r	b-q, r, s	c-p	d-p, s
12.	a-p, q, s	b-q, r, s	c-p	d-p, q
13.	a-r	b-q	c-p	d-s
14.	a-q, s	b-q, r, s	c-q, r, s	d-q, r, s
15.	a-q, r, s	b-r	c-r	d-p, s

16.	a-q, r, s	b-q	c-q, r	d-p, r
17.	a-q, r, s	b-q, r, s	c-q, s	d-p
18.	a-q, r	b-p, s	c-p, s	d-p, s
19.	a-q, r	b-q, s	c-p, q, r, s	d-p
20.	a-q, s	b-r	c-q, s	d-p, s
21.	a-q, r	b-q, s	c-p	d-q, s
22.	a-p	b-p	c-q	d-p
23.	a-p	b-r	c-s	d-q
24.	a-s	b-p, t	c-p, q	d-s
25.	a-p, q, s, t	b-p, r, s	c-q, s, t	d-q, s, t
26.	a-p, t	b-p, s, t	c-q, s, t	d-p, r, s, t

Numerical Question

1. 6	**2.** 0	**3.** 6	**4.** 6	**5.** 0	**6.** 0	**7.** 8
8. 0	**9.** 4	**10.** 2	**11.** 6	**12.** 6	**13.** 2	**14.** 0

Coordination Compounds (Hints)

Single Answer Questions

1. Dimethy glyoxime is neutral ligand. So the name dichlorodimethyl glyoxime –N,N-Cobalt (II) is correct

4. Due to chelate effect $[Fe(C_2O_4)_3]^{3-}$ is stable.

5. More the stability constant more the stability of complex. So stronger ligands form stable complexes

10. $[Co\ (NH_3)_6]^{3+}$ is more stable than $[Co(NH_3)_6]^{2+}$

11. All hexa coordinate complexes of nickel are outer orbital complexes

12. 0.01 mol of complexes giving 0.03 moles of AgCl. So the formula of complex is $[Co(H_2O)_6]Cl_3$

13. 5.9 Bm indicates the presence of 5 unpaired electrons in Mn^{2+} ion. So it will have tetrahedral structure.

14. $[Fe(CN)_6]^{4-}$ is diamagnetic while $[Fe(CN)_6]^{3-}$ is paramagnetic. $[Ni(CO)_4]$ is tetrahedral while $[Ni(CN)_4]^{2-}$ is square planar

15. $Ni(CO)_4$ and $[Co(CO)_4]^-$ are isoelectronic and tetrahedral. $[Fe(H_2O)_6]^{3+}$ is outer orbital complex with sp^3d^2 hybridization

18. In the presence of Cl^- ligand cobalt forms tetrahedral $[CoCl_4]^{2-}$ complex. Tetrahedral complexes have dark colours compared to octahedral.

19. Zero magnetic moment for Ni^{2+} is possible when all the d-electrons are paired and thus have dsp^2 hybridization

20. $FeSO_4(NH_4)_2SO_4 \longrightarrow Fe^{2+} + 2SO_4^{2-} + 2NH_4^+$
Total 5 ions are formed. More the number of particles more the depression in freezing point.

22. $[Cu(en)_2]^{2+}$ is more stable than $[Cu(NH_3)_4]^{2+}$ due to chelating effect

23. for d^3 configuration whether the ligand is weak or strong pairing does not take place

25. Cr in +2 oxidation state contain d^4 configuration. So in the presence of strong ligand pairing takes place.

27. The complex of nickel with dimethyl glyoxime is stabilized by hydrogen bonds extra to the chelating effect.

29. In the complexes containing two bidentate ligands only cis forms show optical isomerism. The complex containing three bidentate ligands show only optical isomerism.

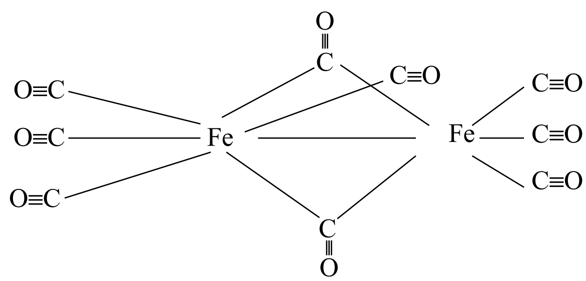
31. 0.319 of $CrCl_3 \cdot 6H_2O$ = 0.0012 moles
The acid liberated = 0.00356 moles
So the No . of H^+ ion formed = $\dfrac{0.00356}{0.0012} = 3$
The number of positive charges on cation = 3

32. $[Co\ (NH_3)_2\ Cl_2\ en]^+$ can have only four stereoisomer of three geometrical of which one is optically active.

33. All tetra coordinate complexes of platinum are inner orbital complexes with square planar structure

40. $\beta n = K_1 \times K_2 \times K_3 \times .... \times K_n$

42.



43. $\dfrac{[Zn(NH_3)_4]^{2+}}{[Zn^{2+}][NH_3]^4} = 3 \cdot 10^9$

$$\frac{[Zn(NH_3)_4]^{2+}}{[Zn^{2+}]} = 3\times10^9\times10^4 = 3\times10^{13}$$

$$\frac{[Zn^{2+}]}{[Zn(NH_3)_4} = \frac{1}{3\times10^{13}} = 3.3\times10^{-14}$$

44. A bidentate ligand can substitute two monodentate ligands which are cis to each other. Since the complex is not responding to $AgNO_3$ the Br^- ion is in coordination sphere.

46. 0.1H complex is giving 0.2 moles of AgCl indicating two Cl^- ions are in ionization sphere.

47. With increase in the number of changes on metal ion the stability of the complex increases. For the ions having same charge stability increases with decrease in size of metal ion.

48. I and II are mirror images; but they have plane of symmetry so they are not enantiomers.

50. The IUPAC name of Cr $(C_6H_6)_2$ is bis [η^6 benzene chromium].

51. The facial and meridional isomers do not exhibit optical isomerism.

56. Since the complex is giving pale yellow precipitate with $AgNO_3$ the Br^- ion is ionization sphere. It's ionization isomer should be $[Co(NH_3)_5Br]Cl$

58. The number of unpaired electrons in the complexes $[Ni(CO)_4]$ $[Mn(CN)_6]^{4-}$, $[Cr(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ are 0, 1, 3 and 4 respectively.

60. Electronic config. for high spin and low spin d^6 ions in octahedral complex as per VB Theory.

High spin

↑↓	↑	↑	↑	↑

10E – P

Low spin

↑↓	↑↓	↑↓		

6E – 3P

61. All six coordinate. So addition of ammonia does not change the magnetic moment.

64. In all the complexes the number of lone pairs available for back bonding in Cr atom is 3. CH_3NH_2 ligand is not a π-acceptor ligand. So as the number of CH_3NH_2 increases the number of lone pairs available for each CO molecule from metal atom increases and hence the M-C bond strength increases while C-O bond strength decreases.

66. $[Cu(en)_2]SO_4 \rightarrow [Cu(en)_2]^{2+} + SO_4^{2-}$

67. Smaller the cation with more number of charges has more polarizing power. Hence more acidic.

68. In vanadium the electron pairs available for back bonding are less and further its EAN is not equal to inert gas config

69. $$\underset{}{Cu(NO_3)_2} + 2NH_4OH \longrightarrow \underset{\text{Blue}}{Cu(OH)_2\downarrow} + 2NH_4NO_3$$

$$Cu(OH)_2 + 2NH_4NO_3 + 2NH_4OH \longrightarrow \underset{\text{Dark blue}}{[Cu(NH_3)_4](NO_3)_2} + 4H_2O$$

71. $C_4H_4Fe(CO)_3$ do not contain alkenes

75. bis (glycinate) zinc (ii) has plane of symmetry. So is not optically active

76. Neutral complexes do not give ions

77. $[Co(Br)(NH_3)_5]SO_4$ and $[Co(SO_4)(NH_3)_5]Br$ are ionization isomers but their molar conductivities are different due to difference in the number of charges or ions.

78. When number of electron pairs available from metal atom per each CO molecule is equal the ion carrying negative charge back donate easily thus decreasing C-O bond strength, while in the ion carrying positive inhibit the back donation

82. $Ma_2b_2c_2$ type complex have five geometrical isomers of which one is optically active. So Total isomers are six.

83. Cr^{3+} complexes are generally green in colour and by exchanging NCS^- ion with Cl^- ion it can exhibit coordination isomerism.

91. Tetrahedral complex of the type Mx_2y_2 do not exhibit any isomerism. So exist as one isomer

Mx_2y_2 type square planar complexes can exist in cis and trans isomers.

92. Cis – isomer can produce two isomers fac and mer by the substitution of one NH_3 by Cl^- but not trans.

95. Wilkinson catalyst is $[Rh\,Cl(PPh_3)_3]$. It is square planar with dsp^2 hybridizition.

97. The number of lone pairs available per CO molecule in $Fe(CO)_5$ is 4/5 where as in other cases is 3/6. If the number of lone pairs donated by metal atom increases the M – O bond order increases and C-O bond order decreases.

98. When weak ligand (H_2O) is substituted by strong ligand (NH_3) Δ_O increases. So radiation having shorter wave length will be absorbed so 580 nm (320 nm is u.v region)

102. $Co(CO)_4$ gets the nearest inert gas config either by reduction or by dimerization, so that it gets stability.

103. Several metal ions exhibit different coordination numbers which depend on their size and the charge

107. Overall stability constant $\beta_n = K_1 \times K_2$. So $4 \times 3 = 12$

114. In both complex ions are same but the counter ions only differ.

116. In $[Co(en)_3]Cl_3$ any change in the arrangement of ligands give same configuration because bidentate ligand can donate only to adjacent (cis) positions but not trans

117. To be paramagnetic Ni^{2+} should form outer orbital's complex with sp^3 hybridization. So only one isomer with the formula $[Ni\,Cl_2(PMe_3)_2]$

118. In $[Ni(CN)_4]^{4-}$ the oxidation state of Ni is zero. So it will form tetrahedral complex with sp^3 hybridization after rearranging the 4s electrons in 3d orbitals

119. Both Cr^{2+} and Fe^{2+} have same number of unpaired electrons (4)

120. In $K_2[Fe(CN)_5(NO)]$, due to transfer of electron from NO to iron, its oxidation state is +2 in which all the electrons are paired

123. All are chealate complexes the order of strength of donor atoms will be N > O > S with Fe^{3+}

124. The higher the charge on the metal ion smaller is the ionic size and more is the complex forming ability. Thus the degree of complex formation decreases in the order

$$M^{4+} > MO_2^{2+} > M^{3+} > MO_2^{+}$$

The higher tendency of complex formation of MO_2^{2+} as compared to M^{3+} is due to higher charge density on metal atom M in MO_2^{+}

$Na_3 [Co (NO_2)(CN)_2(NO_2)(O_2)(O_2)]$
$3 \times 1 + 3 + a - 2 + b + c + d$
$a + b + c + d = -4$
now a and b must have –1, or +1
and c and d must have –2, –1, 0
by hit and trial method we get
$a = -1$ or 0 $b = 0$ or -1 $c = -2$ or -1
$d = -1$ or -2 or 0
Hence the name of complex is
Sodiumdicyamonitronitritodioxygen peroxo superoxo cobaltate (III)

127.
Solutions:
$= 0.01 \times 3 \times 10^{-4} \text{ x } 10^{-3}$
$= 0.03 \times 10^{-7}$
$x = 3 \times 10^{-9}$

128.
Ans: a
Solution: Crystal field theory is based mainly on electrostatic interaction between metal ion and ligands

In six and four coordinate complexes d-orbital splitting takes place due to the loss of spherically symmetric field

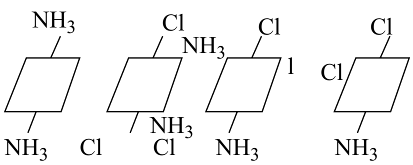
131.
Sol: $X = [Co(SCN)_4]^{2-}$; $Y[Ni(dmg)_2]$

132.
Solution: Due to increase in charge density on central metal, electron transfer through back bonding increases which increases M - C bond strength and thus its stretching frequency

133.
Solution: Since cis form is having dipole moment it can dissolve in polar solvent where as trans form is non polar hence dissolve in non polar solvent.

Coordination Compounds

More than One Answer Questions Hints

4. 

6. In $K_3 [Mn (C_2O_4)_3]$ Mn is in +3 oxidation state with four unpaired electrons

In $Ni(CO)_4$ the $4s^2$ electrons are paired in 3d so the it has no unpaired electrons

7. M abcdef has 15 geometrical isomers each of which in -4rn have optical isomer

9. $[NiCl_4]^{2-}$ is tetrahedral in which Ni^{2+} is in sp^3 hybridization with two unpaired electrons
$[Ni(CN)_4]^{2-}$ is square planar in which Ni^{2+} is in dsp^2-hybridization with zero unpaired electrons.

10. In $[Ti(H_2O)_6]^{3+}$ and in VO^{2+} both Ti^{3+} and V^{4+} have one unpaired electron only.

11. The maximum oxidation state of Sc is only +3 other elements can exhibit higher oxidation states

17. In $Mn_2 (CO)_{10}$ there is Mn-Mn bond
$Fe_3(CO)_{12}$ contain 3 Fe-Fe bonds
$Fe_2 (CO)_9$ contain different type of bonds
In $Fe(CO)_5$ all electrons are paired.

18. From the values the given compound is $[Co(NH_3)_5Cl]Cl_2$

19. When polydentate EDTA substitues monodentate ligands entropy increases. Alkyl groups with have electron releasing effect due to which donor capacity of oxygen increases

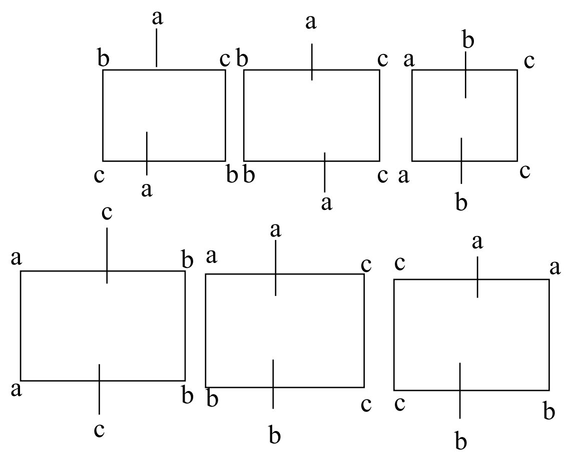
30. $X = [Fe(H_2O)_5 NO] SO_4$ and $Y = Na_4[Fe(CN)_5 NOS]$
There is change in the oxidation state of iron in X due to electron transfer from NO to Fe^{2+}. No change in hybridization in the reaction II. $[Fe(H_2O)_5 NO]SO_4$ contain 3 unpaired electrons

31. In $[Fe (CN)_5 NO]^{3-}$ the iron is in +2 oxidation state with 6 electrons filled in lower energy t_{2g} orbitals. So electron transfer is not possible

32. Pairing of electrons starts only when the number of electrons is more than there. So metal ions with d^3 configuration always form high spin complex and does not change with field strength. For d^8 configuration octahedral complexes are always outer orbital complexes as sufficient d-orbitals are not available even if pairing takes place in strong field.

35. The Cis isomer of $[Co BrCl (en)_2]^+$ exhibit optical isomerism the IUPAC name of the complex is correct
For square planar Mabcd complex have 3 geometrical isomers while Mabcdef type octahedral complex have 15 geometrical isomers.

36.



40. The anionic part of complex $[Co(ox)_3]^{3-}$ exhibits optical isomerism and also it is a chelate complex cationic part of complex $[Co(SCN)_2(NH_3)_4]$ can exist in Cis-trans isomers.

41. In dsp^2 hybridization the d orbital participating in hybridization is $d_{x^2-y^2}$ not d_{z^2}

45. $[Cu(CN)_4]^{3-}$ has tetrahedral geometry with sp^3 hybridization all six coordinate complexes of N_i^{2+} are outer orbital complexes will sp^3d^2 hybridization.
All Cr^{3+} complexes are inner orbital complexes with d^2sp^3 hybridization as there are only 3 unpaired electrons

46. Cu^{2+} has d^9 configuration with one unpaired electron

48.
1. Chelate complexes are more stable and as the number of rings increases stability increases.
2. NO_2^{2-} is stronger ligand than NH_3
3. In a transition group, stability increases down the group due to increase in effective nuclear charge for the metal ions with same charge.
4. For the same metal ion stability of the complex increases with increase in the number of charges

50. Gold exhibit coordination numbers 2 and 4, examples $[Au(CN)_2]^-$ and $[AuCl_4]^-$

Comprehension Type Question

Passage-1

1. First form of compound is not responding to $AgNO_3$. So Cl^- is inside the coordination sphere. Also it is not reacting with en indicating, it is in trans position to NO_2.
2. NO_2^- ion present outside the coordination sphere oxidizes KI to iodine.
3. Second form react with $AgNO_3$. So Cl^- ion is outside the coordination sphere. Since it is optically active it should be trans isomer.
4. Third form reacts with $AgNO_3$ indicating the Cl^- is outside the coordination sphere. Reaction with en indicates that the NO_2 groups are in cis position.

Passage-2

1. More the number of ions or the ions with more number of charges have more conductances
2. $K[Co(NO_2)_4(NH_3)_2] \rightarrow K_+^+[Co(NO_2)_4(NH_3)_2]^-$ S
3. $[Cr(ONO)_3(NH_3)_3] \rightarrow$ No ions
4. $[Cr(NO_2)(NH_3)_5]_3[CO(NO_2)_6]_2$
$\rightarrow 3[Cr(NO_2)(NH_3)_5]^{2+} + 2[Co(NO_2)_6]^{3-}$
5. $Mg[Cr(NO_2)_5(NH_3)] \rightarrow Mg^{2+} + [Cr(NO_2)_5(NH_3)]^{2-}$
6. Both $[CoF_6]^{3-}$ $[Co(NH_3)_6]^{3+}$ have octahedral geometry.

Passage-3

1. Both $NiCl_4^{2-}$ and $[Ni(H_2O)_6]^{2+}$ have two unpaired electrons and are paramagnetic
2. No of unpaired electrons
$$Ni(CO)_4 = 0; [Mn(CN)_6]^{4-} = 1$$
$$[Cr(NH_3)_6]^{3+} = 3;\ [CoF_6]^{3-} = 4$$
3. I, III and IV do not contain unpaired electrons

Passage-4

1. Mwt of $FeSO_4.\ 7H_2O = 278$.
$\therefore$ 0.02 moles = 5.56
2. $Fe^{2+} + 6CN^-$ $[Fe(CN)_6]^{4-}$
0.02 0.02*6 $\therefore$ 0.12
3. $KCN + H_2O \rightarrow KOH + HCN$
HCN is highly poisonous
4. HCl dissolves in aqueous chlorine solution but chloride is evolved out as already the solution is saturated with Cl_2

Passage-6

4. Due to the difference in the sizes of Cl^- and Br^- ions the conductivities of
$[Cr(NH_3)_4Cl_2]Br$ and $[Cr(NH_3)_4ClBr]Cl$

Passage-7

In metal carbonyls back bonding occurs as $M \rightleftarrows C \equiv O$. The electron pair back donated by metal goes into antibonding MO of carbon monoxide, thus decreases the bond order and increases the bond length. This effect depends on the number of electron pairs available on metal atom for back bonding. Negative charge enhances the back bonding while positive charge decreases the back bonding.

Passage-8

1. Both isomers give same number of ions (particles). So colligative properties cannot help in distinguishing the

i. $\underset{X}{Ni(CO)_4} \xleftarrow{Ni} \underset{R}{CO} \xrightarrow{1/2O_2} CO_2 \xrightarrow{NaOH} \underset{Z}{Na_2CO_3}$

$$\begin{array}{l} Cu \\ \downarrow \text{loss of one } e^- \\ Cu^+ \\ \downarrow \\ \underset{(y)}{[Cu(CO)_4]^+} \end{array}$$

In $Ni(CO)_4$ five electron pairs on Ni are available for back donation.
$[Cu(CO)_4]^+$ also contain five electron pairs but with positive charge.

Passage-10

1. Dissociation is least in the case of $[Ag(S_2O_3)]^-$. So it is most stable.
2. As K_1 is more dissociation of $[Ag(S_2O_3)_3]^{5-}$ is more. So concentration of $S_2O_3^{2-}$ is more in its equilibrium system.
3. Complex formation constant for the reaction

$$Ag^+ + 2S_2O_3^{2-} \rightleftarrows [Ag(S_2O_3)_2]^{3-}$$

$$= \frac{1}{K_2 * K_3} = 2*10^{13.}$$

Passage-11

1. Cr^{3+} contain 3 unpaired electrons and is paramagnetic.
2. Each Cr^{3+} ion is surrounded by six oxygen atoms octahedral.
3. Magnetic moment is 3.85 due to three unpaired electrons.

Passage-13

1. From the data the number of Cl^-ions present per one indiums are 4

$$\frac{2.77.5*2.87}{1.385} = 573.AgCl \equiv 4 \text{ moles}$$

2. No. of moles of ammonia liberated = 0.01 mole or $17*10^{-2}$ g.
3. The number of moles of ammonia present in complex

Solution

= 4.899 BM so no. of unpaired electrons is 4 electronic configuration of cobalt ion having four unpaired electrons must be $3d^4\ 4s°$. Thus Co must be in +3 oxidation state O_2 can have a charge zero in dioxygen (O_2), –1 in superoxide or –2 in peroxide

NO_2, can have charge zero in nitrogen dioxide (NO_2), +l in nitronium ion or –1 in nitrite ion so for the complex.

CHAPTER 15

Metallurgy

Gold is for the mistress, Silver for the maid
Copper for the craftsman, Cunning at his trade;
"Good" said the Baron, sitting in his hall
"But iron is the master of them all."
Kipling

15.1 OCCURRENCE OF METALS

In nature metals are found either in native (free state) or in combined state i.e., in the form of stable compounds. Most of the metals occur in the earth's crust in the form of stable insoluble compounds and a few of the metals are found in sea water in the form of soluble salts. The metals like silver, gold, platinum etc. which resist the attack of water and other chemical reagents occur in the native state. The natural materials in which the metals or their compounds occur in the earth are called minerals. All minerals are not suitable for the extraction of metals commercially. The minerals from which the metals can be extracted profitably are called ores. Thus all ores are minerals but all minerals are not ores. For example, clay and bauxite are the important minerals of aluminium but aluminium can be profitably extracted only from bauxite. Hence bauxite is an ore of aluminium while clay is mineral.

Generally the minerals are associated with a lot of impurities like sand, clay rock etc. These impurities are called gangue or matrix. This type of impurities changes from ore to ore and place to place where the ore occurs. The selection of the ore for the extraction of metal depends on

(i) Percentage of the metal in the ore
(ii) Chemical nature of the ore
(iii) Expenditure involved in the extraction of metal
(iv) Cost of the metal
(v) Value of the by-products

15.2 CLASSIFICATION OF THE ORES

Ores may be grouped into different classes depending upon their chemical nature some important ores and their composition are given in Table 15.1.

Table 15.1 Classification of ores

Type of ore	Ore or mineral	Composition
Oxide ore	Corrundum	Al_2O_3
	Diaspore	$Al_2O_3 \cdot H_2O$
	Bauxite	$Al_2O_3 \cdot 2H_2O$
	Gibbsite	$Al_2O_3 \cdot 3H_2O$
	Cuprite (or) Ruby Copper	Cu_2O
	Zincite	ZnO
	Haematite	Fe_2O_3
	Magnatite	Fe_3O_4
	Pyrolusite	MnO_2
	Chromite	$FeCr_2O_4$
	Cassiterite or Tin stone	SnO_2
	Pitch blend	U_3O_8
Sulphide ores	Copper pyrites (or) Chalco pyrities	$CuFeS_2$ Or $Cu_2S \cdot Fe_2S_3$
	Copper Glance	Cu_2S
	Zinc blend	ZnS
	Galena	PbS
	Cinnabar	HgS
	Iron pyrites	FeS_2

Table 15.1 (Cont.)

Type of ore	Ore or mineral	Composition
	Argentite or silver glance	Ag_2S
	Pentlandite	(NiFe)S
Sulphate ores	Epsomite	$MgSO_4 \cdot 7H_2O$
	Kieserite	$MgSO_4 \cdot H_2O$
	Gypsum	$CaSO_4 \cdot 2H_2O$
	Anhydrite	$CaSO_4$
	Celestite	$SrSO_4$
	Barytes (or) Heavy spar	$BaSO_4$
	Anglesite	$PbSO_4$
	Lanarkite	$PbO \cdot PbSO_4$
Carbonate ores	Magnesite	$MgCO_3$
	Dolamite	$MgCO_3 \cdot CaCO_3$
	Limestone	$CaCO_3$
	Calamine	$ZnCO_3$
	Stronianite	$SrCO_3$
	Whitherite	$BaCO_3$
	Cerussite	$PbCO_3$
	Siderite or sepathic iron	$FeCO_3$
	Malachite	$CuCO_3 \cdot Cu(OH_2)$
	Azurite	$2CuCO_3 \cdot Cu(OH)_2$
Halide ores	Horn Silver	AgCl
	Rock Salt	NaCl
	Sylvine	KCl
	Carnalite	$KCl \cdot MgCl_2 \cdot 6H_2O$
	Fluorspar	CaF_2
	Cryolite	Na_3AlF_6 or $3NaF \cdot AlF_3$
Silicate ores	Asbestos	$CaSiO_3 \cdot 3MgSiO_3$
	Beryl	$3BeO \cdot Al_2O_3 \cdot 6SiO_2$ Or $Be_3Al_2Si_6O_{18}$
	Chrsoberyl	$BeO \cdot Al_2O_3$
	Willemite	Zn_2SiO_4
	Feldspar	$K_2O \cdot Al_2O_3 \cdot 6SiO_2$ (Or) $KAlSi_3O_8$
	Potash Mica	$K_2O \cdot 3Al_2O_3 \cdot 6SiO_2 \cdot 2H_2O$
	Zircon	$ZrSiO_4$
	Talc	$Mg_2(Si_2O_5)_2 \cdot Mg(OH)_2$
Phosphate ores	Phosphorite	$Ca_3(PO_4)_2$
	Fluorapetite	$CaF_2 \cdot 3Ca_3(PO_4)_2$
	Chlorapetite	$CaCl_2 \cdot 3Ca_3(PO_4)_2$
	Hydrapetite	$Ca(OH)_2 \cdot 3Ca_3(PO_4)_2$
Nitrate ores	Chile salt Petre	$NaNO_3$
	Bengal salt Petre	KNO_3

15.3 EXTRACTION OF METALS

The extraction methods of metals from their ores involve the following three main stages

1. Concentration of the ore
2. Extraction of crude metal
3. Purification of metal

15.3.1 Concentration of the Ore (or) Ore Dressing

Before subjecting the ore to metallurgical process, it is necessary to remove the unwanted impurities known as gangue or matrix. It should be removed mechanically so that the concentration of the ore increases. The process of removal of gangue is known as ore dressing or concentration of the ore. If the gangue is present in large quantity in the ore, the flux should be added which causes the increase in fuel expenditure. Also the size of the furnace must increase. In order to reduce these difficulties and to extract the metal with less expenditure, the ore must be concentrated. The different methods used in the ore dressing are illustrated here.

1. Hand Picking: This method is used when the ore can be separated in sufficient degree of purity simply by hand picking. The gross lumps of rocks may be removed from the ore by simply picking up by hand and these are then broken away with hammer.

2. Gravity Concentration Methods: These methods are based upon the difference in specific gravity of the ore particles and impurities.

(i) Washing with water: The ore is powdered and then washed with running stream of water. The heavy ore particles settle down rapidly while the lighter gangue particles are washed away. The heavier ore particles settled down are collected behind the riffles kept at the bottom of the stream of water.

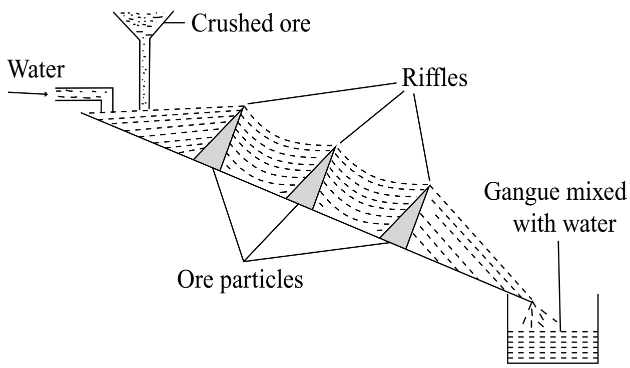


Fig 15.1 Washing with water

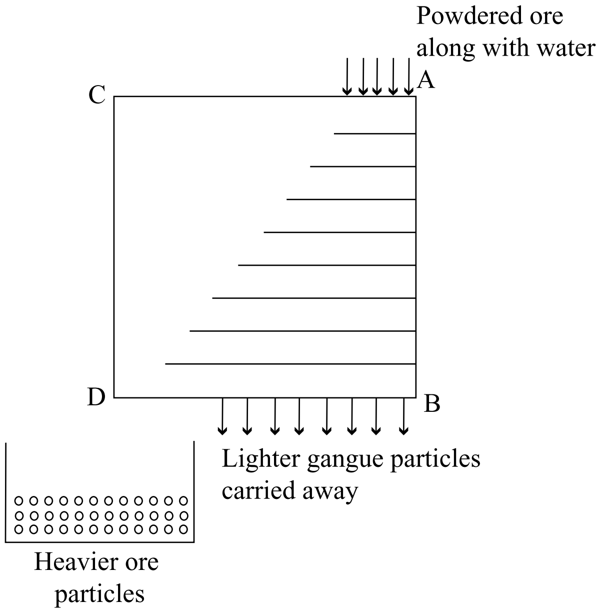


Fig 15.2 Wilfley table method

(ii) Wilfley Table Method: Wilfley table is a wooden table on which long wooden strips called chats (Or) riffles are fixed. The table is kept moving or vibrating continuously mechanically

The table is arranged on standing floor so that the height of AB side will be little more than CD side. Then table is provided with an inlet for dropping powered ore suspended in a stream of water on the top of the table and is allowed to flow across the table in a direction at right angles to the slope. The heavier ore particles are obstructed by the riffles while the lighter impurities pass over and carried away by the stream of the water by the jerks of the table. The ore particles collect behind the riffles move to left side by jerks of the table and are collected. For example gold is separated from sand by this method.

(iii) Hydraulic Classifier Method: Hydraulic classifier is a conical reservoir provided with hopper at the top and pipe at the bottom.

Finely powdered ore is dropped into conical reservoir (hopper). A powerful stream of water is introduced through the pipe provided at the bottom. The lighter gangue particles are carried away by water while the heavier ore particles are accumulated at the top of the cone. The conical shape of hopper reduces the velocity of water and this prevents the ore particles from being carried away along with the stream of water.

3. Magnetic Separation: This process is used if one of the ore or impurities is magnetic. For example tin stone

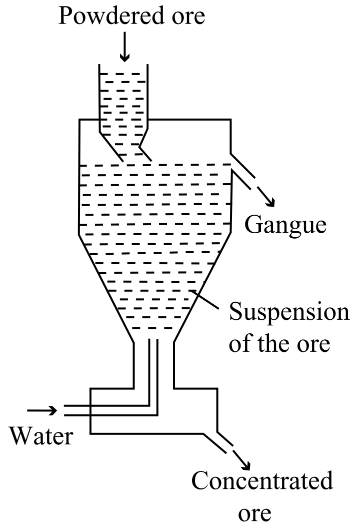


Fig 15.3 Hydraulic classifier

is separated from magnetic wolfram by this method. Similarly magnetic oxide of iron (Fe_3O_4) can be separated from pyrolusite which is manganese ore.

The pulverized ore is allowed to fall on a rubber belt that moves horizontally on two rollers one of which is magnetic. When the ore moves to the end of magnetic roller the magnetic particles are attracted and fall nearer the magnetic roller while non-magnetic particles fall away from the roller.

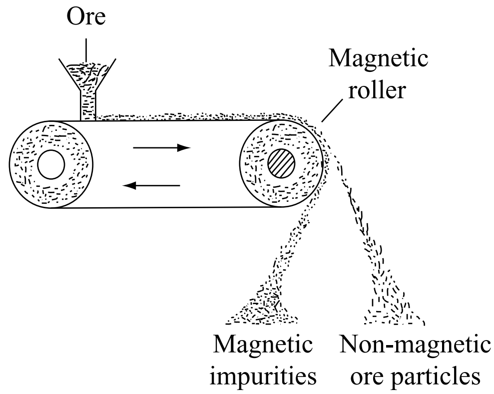


Fig 15.4 A magnetic separation

4. Liquation: This method depends on the difference in the melting points of ore and gangue. If the melting point of the ore is less than the melting point of gangue, then the impure ore is heated on an open hearth with slanting position. The ore melts first and flows down. For example stibnite can be separated from impurities by this method.

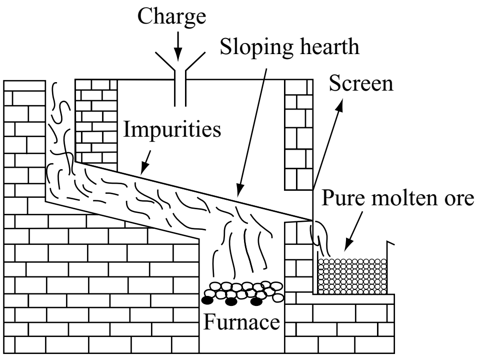


Fig 15.5 Liquation in reverberatory furnace

5. Froth Flotation Process: Generally this method is used to purify the sulphide ores. The process is based on the principle of preferential wetting of the solid surface by various liquids.

The finely powdered ore is made into slurry with water to which small quantities of pine oil (or) eucalyptus oil as froather and sodium ethyl xanthate as collector (cresols and aniline are also added as collectors) along with lime (or) sodium carbonate.

The water is agitated violently with compressed air. When froth is formed at the water interface, the sodium ethyl xanthate dissociates into positively charged sodium ion and negatively charged xanthate ion. The xanthate ion forms a layer on the particles by reacting with the metal ion of the ore particle which is insoluble in water. The hydrocarbon part of this layer repels water and so the ore particles rise to the surfaces along with the froth. The stony (gangue) particles are wetted by the water and remain in water below the foam. The froth containing the mineral is skimmed off over the top and squeezed through the canvas bag.

In some conditions, certain substances known as depressors or activators are used in froth flotation process. Depressors do not allow the froth to stick on the ore particles while activators will make the frothers to stick easily to the ore particles.

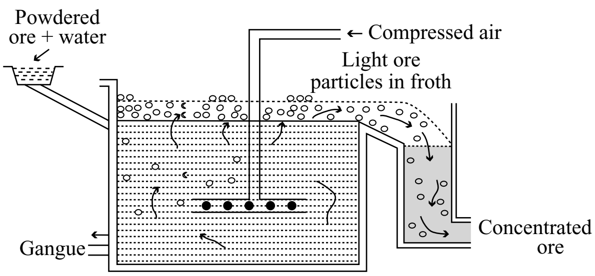


Fig 15.6 Froth flotation process

If sodium cyanide is added to galena containing zinc sulphide, the sodium cyanide acts as a depressor for zinc sulphide and makes it to sink down in water. Zinc sulphide react with sodium cyanide forming a complex $Na_2[Zn(CN)_4]$ on the surface of ZnS particles due to which ethyl xanthate is prevented to form as a layer on the surface of ZnS. So it is wetted with water and prevented coming into the froth, while PbS' comes into froth because it cannot react with NaCN.

If copper sulphate is added it acts as an activator for zinc sulphide and makes it to float. This type of separation of two ores from gangue is called differential flotation method.

6. Leaching: It is a chemical method for the concentration of the ore. In this process the powdered ore is dissolved selectively in certain acids, bases or other suitable reagents. Thus impurities remain un-dissolved as sludge. The solution of ore is filtered and the ore is recovered by precipitation or crystallization. Following examples illustrate the process of leaching.

(i) Leaching alumina from bauxite: The principal ore of aluminium is bauxite. It contains SiO_2, iron oxides and titanium (iv) oxide as impurities. The ore is concentrated by digesting the powdered ore with 45 per cent caustic soda solution at 473-523k and 35-36 bar pressure. Al_2O_3 dissolves in water due to the formation of soluble sodium metaluminate. SiO_2 also dissolves as sodium silicate

$$Al_2O_3 \cdot 2H_2O_{(s)} + 2NaOH_{(aq)} + H_2O \longrightarrow \underset{\text{Soluble}}{2Na\,[Al(OH)_4]_{(aq)}}$$

The solution is filtered to remove insoluble impurities. Next aluminium hydroxide is precipitated from the strongly alkaline solution. This is done by bubbling CO_2, an acidic oxide which lowers the pH. At this stage solution may be seeded with freshly prepared aluminium hydroxide precipitate.

$$Na[Al(OH)_4]_{(aq)} + CO_{2(g)} \longrightarrow Al(OH)_3 + NaHCO_3$$

The sodium silicate remains in the solution. The $Al(OH)_3$ precipitate is filtered, dried and heated strongly to get purified Al_2O_3

$$2Al(OH)_3 \xrightarrow{1470K} Al_2O_{3(s)} + 3H_2O_{(g)}$$

(ii) Other Examples: Sulphide ore of silver (Ag_2S – argentite) is also concentrated by leaching. The ore is treated with dilute solution of sodium cyanide

$$Ag_2S_{(s)} + 4NaCN\,(aq) \rightarrow 2Na\,[Ag(CN)_2] + Na_2S_{(aq)}$$

Silver and gold ores containing native metal are leached with dilute solution of sodium cyanide or potassium cyanide in the presence of air.

$$4Ag + 8NaCN + 2H_2O + O_2 \longrightarrow 4Na\left[Ag(CN)_2\right] + 4NaOH$$
$$4Au + 8NaCN + 2H_2O + O_2 \longrightarrow 4Na\left[Au(CN)_2\right] + 4NaOH$$

From the solution the metal is recovered by treatment with more electropositive metal (zinc or aluminium) by displacement.

$$2Na\left[Ag(CN)_2\right] + Zn \longrightarrow Na_2\left[Zn(CN)_4\right] + 2Ag$$
$$2Na\left[Au(CN)_2\right] + Zn \longrightarrow Na_2\left[Zn(CN)_4\right] + 2Au$$

15.3.2 Fluxes and Slags

Though the ores are concentrated by various methods described in ore dressing or concentration of the ore, still there remains some earthy matter (gangue). The gangue is not fusible at the ordinary temperatures of furnaces and hence cannot be removed in molten state. There are certain substances which when mixed with ore and on heating in furnaces they combine with gangue to form an easily fusible material which is not soluble in the molten metal. Such substances are called **fluxes** and the easily fusible mass formed by combining the flux with gangue is called **slag**. Since the slag is lighter than the molten metal it floats over the surface of molten metal and can be easily removed. Fluxes are of two types

(a) Acidic Fluxes: Silica is the main acidic flux employed in metallurgical operations. In nature it occurs in the form of sand, sand stone, quartz etc. in sufficient purest form. Acidic fluxes are employed when the gangue is basic. For example, in the extraction of copper from pyrites, silica is used as a flux to remove iron oxide

$$\underset{\text{gangue}}{FeO} + \underset{\text{flux}}{SiO_2} \longrightarrow \underset{\text{slag}}{FeSiO_3}$$

Similarly lime may be removed with silica

$$\underset{\text{gangue}}{CaO} + \underset{\text{flux}}{SiO_2} \longrightarrow \underset{\text{slag}}{CaSiO_3}$$

(b) Basic Fluxes: If the ore is associated with acidic impurities such as SiO_2, P_2O_5 etc then employed fluxes must be basic. The common basic fluxes used are calcium oxide and magnesium oxide which can be obtained by decomposing lime stone, magnesite respectively

$$\underset{\text{gangue}}{SiO_2} + \underset{\text{flux}}{CaO} \longrightarrow \underset{\text{slag}}{CaSiO_3}$$
$$\underset{\text{gangue}}{SiO_2} + \underset{\text{flux}}{MgO} \longrightarrow \underset{\text{slag}}{MgSiO_3}$$
$$\underset{\text{gangue}}{P_4O_{10}} + \underset{\text{flux}}{6CaO} \longrightarrow \underset{\text{slag}}{2Ca_3(PO_4)_2}$$

For example to remove silica impurity in haematite ore, lime stone is used as flux The slag must have the following characteristics

(i) It should be easily fusible and must flow
(ii) It should have less density than metal and hence it must float over the molten metal
(iii) It must be a poor conductor of heat which prevents the super heating of the metal
(iv) Most of the impurities must be soluble in slag

It should be noted that the content of acidic oxides in a slag is higher than that of basic oxides, the slag is called acidic. If the contents of basic oxides are greater than the content of acidic oxides the slag is known as basic. If acidic and basic oxides are in equal parts the slag is known as neutral. Example

Basic slag	$(CaO)_4\,SiO_2$
Neutral slag	$CaO\ SiO_2$
Acidic slag	$(CaO)_2\,(SiO_2)_3$

$$\text{Basicity} = \frac{M_{CaO}}{M_{SiO_2}}$$

In order to control the process and to regulate the smelting conditions, slags are analyzed in the course of metallurgical operations.

Slags are used (1) in road surfacing (2) in the manufacture of cement (3) in the manufacture of fertilizers (4) in the construction of buildings.

Thomas slag used as a fertilizer is calcium phosphate $Ca_3(PO_4)_2$

15.3.3 Preliminary Treatment of the Concentrated Ore

The process of extraction of metal from the concentrated ore depends upon the nature of the ore as well as the nature of impurities present in the ore. Before the concentrated ore is subjected to final metallurgical operations in order to get the metal in the free state, the preliminary chemical treatment may be necessary. The objective of this preliminary chemical treatment is

(i) to get rid of impurities which would cause difficulties in the later stages; and

(ii) to convert the ore into oxide of the metal because it is easier to reduce an oxide than the carbonate or sulphide.

1. Calcination: Calcination is the process in which the ore is subjected to the action of heat at high temperature in the absence of air. In calcination, the hydrated oxide ores or carbonate ores are converted into metallic oxide by heating them at high temperatures. Metal hydroxides expel water while carbonates expel carbon dioxide. Other impurities like carbon, sulphur, phosphorous, arsenic etc if present are also removed as their volatile oxides. In calcination the ore becomes porous and dry

$$Fe_2O_3 \cdot xH_2O \longrightarrow Fe_2O_3 + xH_2O$$
$$ZnCO_3 \longrightarrow ZnO + CO_2$$
$$PbCO_3 \longrightarrow PbO + CO_2$$
$$CaCO_3 \cdot MgCO_3 \longrightarrow CaO + MgO + 2CO_2$$
$$CuCO_3 \cdot Cu(OH)_2 \longrightarrow 2CuO + H_2O + CO_2$$

Calcination is usually carried in reverboratory furnace

2. Roasting: In roasting the ore is heated below its melting point in the presence of air either alone or after addition of other materials

Roasting is generally carried in reverboratory furnace or in blast furnace. During roasting volatile impurities like S,P,As,Sb, etc get oxidized and escape out as volatile gases. The sulphide ores convert into their oxides. The ore becomes porous and the moisture will be removed. Roasting may be conveniently divided into the following groups.

(i) Oxidizing Roasting: In this type of roasting, the ores are converted into their oxides. The impurities like sulphur, phosphorous, arsenic, antimony etc escapes out as their volatile oxides. This type of roasting is done for copper pyrites, zinc blende galena etc.

$$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$$
$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$

(ii) Blast Roasting: In this process the oxidation is carried out by blast of hot air. This process is applied in the case of galena and copper pyrites

(iii) Sulphatizing Roasting: In this method of roasting, the ore converts into a soluble metal sulphate. For example CuS to $CuSO_4$ or ZnS to $ZnSO_4$

$$CuS + 2O_2 \longrightarrow CuSO_4$$
$$PbS + 2O_2 \longrightarrow PbSO_4$$
$$ZnS + 2O_2 \longrightarrow ZnSO_4$$

(iv) Chloridizing Roasting: In this method, the metal or its ore is converted into chloride by heating the ore with sodium chloride in the presence of air

$$Ag_2S + 2NaCl \longrightarrow 2AgCl + Na_2S$$

3. Smelting: Smelting is a process in which the metal or its sulphide is separated in a molten condition from impurities. Generally in smelting, the ore is melted with a flux and often with a reducing agent.

Flux combines with the gangue forming slag. The mixture of the ore, flux and fuel or reducing agent heated in a furnace is called charge.

In the extraction of iron the charge is haematite (ore) lime stone (flux) and coke (fuel)

$$C + O_2 \longrightarrow CO_2$$
$$C + CO_2 \longrightarrow 2CO$$
$$Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$$

The lime stone decomposes giving CaO which combines with impurities like silica phosphorus pentoxide to form slag.

$$CaCO_3 \longrightarrow CaO + CO_2$$
$$CaO + SiO_2 \longrightarrow CaSiO_3$$
$$6CaO + P_4O_{10} \longrightarrow 2Ca_3(PO_4)_2 \text{ slag.}$$

In the extraction of copper, silica is used as flux which combines with FeO forming $FeSiO_3$

Slag

$$2CuFeS_2 + O_2 \longrightarrow Cu_2S + 2FeS + SO_2$$
$$2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$$
$$2FeO + SiO_2 \longrightarrow \underset{\text{slag}}{FeSiO_3}$$

15.3.4 Extraction of the Crude Metal

Generally the metals in their ores are present in positive oxidation states. Extraction of the metal from their ores is nothing but the reduction of their ores.

$$M^{n+} + ne^- \longrightarrow M$$

Thus the concentrated ore is converted into a form which is suitable for reduction. Oxides are easier for reduction therefore the ores are converted into oxides by one of the method as explained already.

The process of extracting the metal by heating its ore with suitable reducing agent is called pyrometallurgy. Reduction of the roasted/calcinated/smelted ore is carried by various process.

1. Thermal reduction
2. Chemical reduction
3. Autoreduction
4. Hydrometallurgy

5. Electrochemical or Electrolytic reduction
6. Amalgamation method

1. Thermal reduction: This method is used for the metal oxides which are unstable at high temperatures. For example when cinnabar is heated at high temperatures in a shaft furnace, mercury metal will be obtained as vapour which on condensation gives mercury liquid.

$$2HgS + 3O_2 \longrightarrow 2HgO + 2SO_2$$
$$2HgO \longrightarrow 2Hg + O_2$$

2. Chemical reduction: The concentrated ore can be reduced to metal with a suitable chemical reducing agent. The various chemical reduction methods are as follows

(i) Carbon reduction: Carbon is the cheapest reducing agent and is available in the nature directly as coke. Carbon in the form of coke or carbon monoxide is used as reducing agent. The oxide of metal obtained from nature or by calcinations or by roasting is reduced by coke or carbon monoxide

$$MO + C \longrightarrow M + CO\uparrow$$
$$MO + CO \longrightarrow M + CO_2\uparrow$$

Metals are obtained in molten state.

$$ZnO + C \xrightarrow{950^\circ C} Zn + CO$$
$$SnO_2 + 2C \xrightarrow{1300^\circ C} Sn + 2CO$$
$$Fe_2O_3 + 3CO \xrightarrow{700^\circ C} 2Fe + 3CO_2$$

(ii) Reduction with water gas: In certain cases water gas can be used as a reducing agent. For example, nickel oxide can be reduced with water gas

$$2NiO + \underbrace{CO + H_2}_{\text{Water gas}} \xrightarrow{300^\circ C} 2Ni + CO_2 + H_2O$$

(iii) Reduction with Hydrogen: Although hydrogen is a very good reducing agent it is not much used in the extraction of metals because it is not available as such in the nature. Its preparation is costlier. Further its storage and transportation is dangerous because it easily catch fire and explodes. However where other methods are not applicable hydrogen is used as reducing agent. For example molybdenum and tungsten are extracted by the reduction with hydrogen because in carbon reduction method they yield carbides

$$MoO_3 + 3H_2 \xrightarrow{\text{Red Hot}} Mo + 3H_2O$$
$$WO_3 + 3H_2 \xrightarrow{850^0 C} W + 3H_2O$$
$$In_2O_3 + 3H_2 \xrightarrow{\text{High Temp}} 2In + 3H_2O$$

(iv) Reduction with more electropositive metals: More electropositive metals can reduce less electropositive metals. So by using highly electropositive metals like Na, K, Mg, Ca Al etc certain metals are extracted by the reduction of their oxides or halides

$$B_2O_3 + 6Na \longrightarrow 2B + 3Na_2O$$
$$B_2O_3 + 3Mg \longrightarrow 2B + 3MgO$$
$$SiO_2 + 2Mg \longrightarrow Si + 2MgO$$
$$Rb_2O_3 + 3Mg \longrightarrow 2Rb + 3MgO$$
$$TiCl_4 + 2Mg \longrightarrow Ti + 2MgCl_2$$
$$UCl_4 + 2Ca \longrightarrow U + 2CaCl_2$$

The method in which aluminium is used as reducing agent is called **Goldschmit alumino thermy process** or simply **Gouldshmit thermite process**. This method is mainly used for the extraction of chromium and manganese and for thermite welding of railway tracks.

$$Fe_2O_3 + 2Al \longrightarrow 2Fe + Al_2O_3$$
$$Cr_2O_3 + 2Al \longrightarrow 2Cr + Al_2O_3$$
$$3Mn_3O_4 + 8Al \longrightarrow 9Mn + 4Al_2O_3$$

These reactions are highly exothermic, so the metals are obtained in molten state. The reaction with iron oxide is used in welding rails or heavy parts of the machinery without removing them from their position

A mixture of haematite (3 parts) and aluminium powder (1 part), is called thermit, is taken in a crucible lined inside with magnetite. This is covered with ignition mixture consisting of aluminium powder and barium peroxide. Magnesium ribbon is inserted in this mixture which serves as fuse. When the magnesium ribbon is lighted the ignition mixture catches fire. Aluminium powder reduces iron oxide to iron and this being exothermic reaction the iron

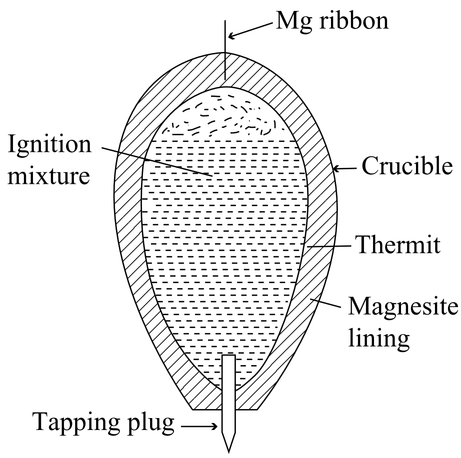


Fig 15.7 Alumino thermic process

will present in the molten state. This can be allowed to flow through the tapping plug between the joints to be welded. On solidification the broken pieces join together.

3. Auto reduction (or) Air reduction: Ores of certain metals are sometime reduced without using any additional reducing agent. Ores of copper, lead, mercury are reduced without any external reducing agent, during the extraction of the metal. Their sulphide ores are at first partially roasted so that some oxide is formed. This oxide is now reduced by the remaining sulphide ore resulting in the formation of metal. This process is called **auto reduction** or **air reduction**

$$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$$
$$2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$$

$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$
$$PbS + 2PbO \longrightarrow 3Pb + SO_2$$

$$2HgS + 3O_2 \longrightarrow 2HgO + 2SO_2$$
$$2HgO + HgS \longrightarrow 3Hg + SO_2$$

4. Hydrometallurgy: Hydrometallurgy is the treatment of the ore by suitable chemical reagent to bring the metal into solution followed by the extraction of the metal by use of proper precipitating agent or by electrolysis. For example the ores of silver when agitated with a dilute solution of sodium cyanide, the silver goes into solution as sodium argentocyanide complex from which the silver is precipitated by the addition of zinc

$$4Ag + 8NaCN + 2H_2O + O_2 \longrightarrow 4Na\left[Ag(CN)_2\right] + 4NaOH$$

$$AgCl + 2NaCN \longrightarrow Na\left[Ag(CN)_2\right] + NaCl$$

$$2Na\left[Ag(CN)_2\right] + Zn \longrightarrow Na_2\left[Zn(CN)_4\right] + 2Ag$$

The copper oxide obtained by calcination of its carbonate ores or by roasting its sulphide ores is leached with dilute sulphuric acid

$$CuO + H_2SO_4 \longrightarrow CuSO_4 + H_2O$$

The copper can be recovered by treating the above solution with scrap iron

$$CuSO_4 + Fe \longrightarrow FeSO_4 + Cu$$

5. Amalgamation method: This method is used for the extraction of noble metals like gold and silver from the native ores. When the powdered ore is treated with mercury, the metal converts into its amalgam. The metal is then recovered by distillation.

6. Electrochemical or Electrolytic reduction: The extraction of metals from ores electrochemically is confined to the more electropositive metals like Na, K, Mg, Ca, Al etc. Except for aluminium, halides are employed since oxo acid salts would decompose thermally at the high temperatures needed to operate the cells and give infusible metallic residues. Fused salts must be used since any water present would lead to selective discharge of hydrogen. Often halide impurities are added to lower the melting point and thus economize on the electrical power needed to keep the salt molten. The reactions taking place in the electrolytic cell may be represented as

$$MX \longrightarrow M^+ + X^-$$
$$M^+ + e^- \longrightarrow M \text{ Reductions at cathode}$$
$$X^- \longrightarrow X + e^- \text{ Oxidation at anode}$$
$$X + X \longrightarrow X_2$$

Sodium metal is obtained by the electrolysis of NaCl mixed with small amounts of $CaCl_2$. Similarly magnesium is extracted by the electrolysis of fused magnesium chloride mixed with a small amount of KCl and $CaCl_2$.

15.4 THERMODYNAMIC PRINCIPLES OF METALLURGY

The reduction of the metallic oxide usually involves heating it with some reducing agent such as carbon, carbon monoxide, hydrogen or some other metal (pyrometallurgy). The reducing agent combines with oxygen of the metal oxide

$$M_XO_y + yC \longrightarrow xM + yCO$$

Some metal oxides get reduced easily while others are reduced with difficulty. Some oxides are reduced at relatively low temperature while others are reduced at high temperatures. Thermodynamic considerations play an important role in deciding the temperature and the choice of reducing agent in the thermal reduction process during metallurgy (Pyrometallurgy).

A reaction carried out in a closed system takes place with a decrease in free energy. While metallurgical process may or may not be carried out under these conditions, nevertheless no great inaccuracy is introduced by assuming closed system conditions to apply; the relevant equation is

$$\Delta G = \Delta H - T\Delta S$$

Gibbs energy ΔG^0, is related to equilibrium constant K through $\Delta G^0 = -RT \ln K$ and a negative value of ΔG^0 correspondence $K > 1$. It should be noted that equilibrium is rarely attained in commercial processes because in many such process involve dynamic stages. For example, reactants and products are in contact only for short times. Furthermore, even a process at equilibrium for which $K < 1$ can be viable if the product is removed out of the reaction

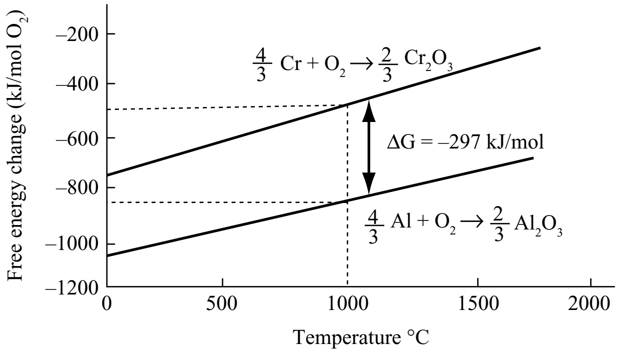


Fig 15.8 The free energy change of formation of aluminium oxides and chromium (III) oxide as a function of temperature

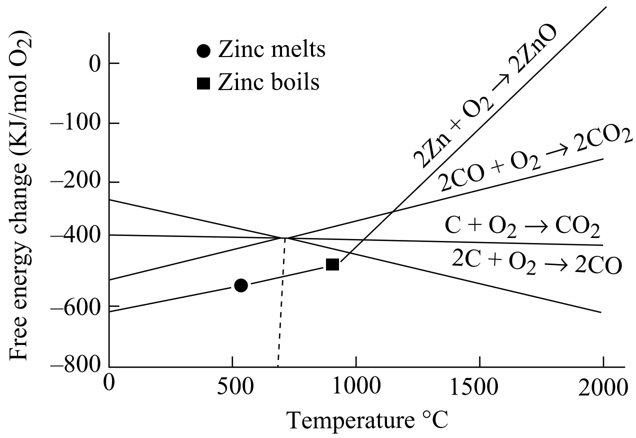


Fig 15.9 The variation of the free energies of formation of four systems as a functions of temperature

chamber and the reaction continues. In principle, we also need to consider rates when judging whether a reaction is feasible in practice, but reactions are often fast at high temperature and thermodynamically favorable reactions are likely to occur.

It is possible to calculate the changes in enthalpy ΔH and in entropy ΔS for chemical reactions from a wide range of experimental data. When corrections are made for variations in ΔH and ΔS with temperature (these are small for a larger change in ΔS when a liquid boils, since there is a large increase in disorder) it becomes possible to calculate free energy changes at a series of temperature. These values are conveniently plotted on a graph, the method first used by Ellingham and are called **Ellingham diagrams**. These diagrams refer to reactions involving one mole of oxygen at one atmosphere pressure. Fig 15.8 shows the variation of free energies of formation of aluminium oxide and chromium oxide as a function of temperature.

The free energy formation of aluminium oxide is more negative than that of chromium (III) oxide at all temperatures, so that aluminium should be capable of reducing the latter to chromium. This is easily seen by considering the two individual reactions.

$$4/3\ Al + O_2 \longrightarrow 2/3\ Al_2O_3;\ \Delta G = -827\ KJ\ mol^{-1}\ of\ O_2$$
$$4/3\ Cr + O_2 \longrightarrow 2/3\ Cr_2\ O_3;\ \Delta G = -540\ KJ\ mol^{-1}\ of\ O_2$$

(The free energy changes are for a temperature of 1000°C)

Free energy changes can be treated algebraically just like heats of reaction, thus by subtraction

$$4/3\ Al + 2/3\ Cr_2\ O_3 \longrightarrow 4/3\ Cr + 2/3\ Al_2O_3\ \Delta G = -287 KJ$$

Since there is a decrease in free energy the reaction is thermodynamically possible. The fact that it does not take place at low temperatures is because reactions between solids generally involve high energies of activation.

It is therefore clear that aluminium and magnesium find industrial application as reducing agents because their oxide formation is attended by a large decrease in free energy.

The reducing action of carbon and carbon monoxide can be readily be understood in similar manner. Fig 15.9 shows the variation of free energies of formation of carbon monoxide, carbon dioxide and zinc oxide with temperature.

There are three curves for carbon, corresponding to complete oxidation of carbon to carbon dioxide, partial oxidation to carbon monoxide and oxidation of carbon monoxide to the dioxide. The three curves pass through a common point at about 710°. Thus the free energies of formation of carbon dioxide from carbon monoxide and carbon dioxide from carbon are identical at this temperature.

1. $C + O_2 \longrightarrow CO_2\ \Delta G = X\ KJ\ mol^{-1}$ of O_2
2. $2C + O_2 \longrightarrow 2CO\ \Delta G = X\ KJ\ mol^{-1}$ of O_2
3. $2CO + O_2 \longrightarrow 2CO_2\ \Delta G = X\ KJ\ mol^{-1}$ of O_2

Subtracting one equation from the other (1 and 3) and rearranging the following result is obtained

$$CO_2 + C \longrightarrow 2CO\ \Delta G = 0$$

i.e., an equilibrium is set up. The following conclusions can be made from the figure

- For temperature at which C/CO line lies below the metal oxide line, carbon can be used to reduce the metal oxide and itself is oxidized to carbon monoxide
- For temperature at which C/CO_2 line lies below the metal oxide line, carbon can be used to achieve the reduction, but is oxidized to carbon dioxide
- For temperature at which the CO/CO_2 line lies below the metal oxide line, carbon monoxide can reduce the metal oxide to the metal and is oxidized to carbon dioxide.

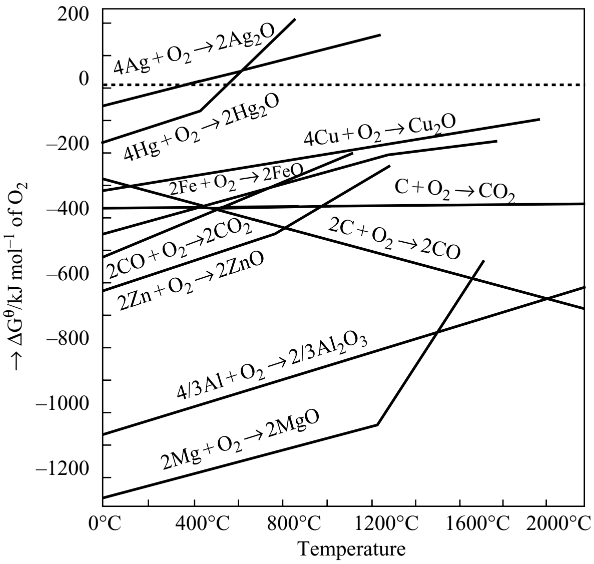


Fig 15.10 Ellingham diagram for some oxides

Fig 15.10 shows an Ellingham diagram for all the metals shown in the diagram, even magnesium and calcium could be accomplished by pyrometallurgy, heating with a reducing agent. Ellingham diagrams helps us in predicting the feasibility of a thermal reduction of an ore. The criterion of feasibility is that at a given temperature, change in Gibbs energy for the reaction must be negative.

15.4.1 The Ellingham Diagram for Oxides Show the Following Features

(i) Ellingham diagram normally consist of plots ΔG^0 VsT for the formation of oxides of elements i.e., for the reaction

$$2x\ M_{(s)} + O_{2(g)} \longrightarrow 2M_xO_{(s)}$$

(ii) The graphs for metal to metal oxide all slope upwards because the change in Gibbs energy becomes less negative with increase in temperature. This is due to the factor that, ΔS is negative for the reaction and hence $T\Delta S$ becomes more and more negative with increase in temperature.

(iii) Each plot follows a straight line unless there is some change in phase (such as solid to liquid or liquid to gas) which results in large change in entropy and hence changes the slope of line. For example Mg-MgO line changes slope at 1120°C. Similarly Zn-ZnO line there is sudden change in slope due to melting.

(iv) When temperature is increased a point will be reached when the line crosses $\Delta G = O$ line. Below this temperature the ΔG^0 of oxide formation is negative and hence the oxide is stable. Above this temperature ΔG^0 of the oxide formation is positive and hence the oxide becomes unstable and decomposes on its own into metal and oxygen. Theoretically all metal oxides can be decomposed into metal and oxygen if sufficiently high temperature can be attained. In practice, the oxides of mercury, silver and gold are the only oxides which can be decomposed at temperatures which are easily attainable.

In a number of processes, one metal is used to reduce the oxides of other metals. A metal can reduce oxides of other metals which lie above it in the Ellingham diagram because in such a process Gibbs free energy change becomes negative by an amount equal to the difference between that two graphs at that particular temperature.

However there are severe practical limitations. It is interesting to note that the value of carbon as reducing agent is due to the marked increase in disorder that takes place when carbon (an ordered solid) reacts with one mole of oxygen to give two moles of carbon monoxide, the net effect is an extra mole of gas and hence an increase in disorder (an increase in entropy) capable of reducing most metallic oxides to metals. Efforts to produce aluminium by pyrometallurgy (most notably in Japan where electricity is expensive) were frustrated by the volatility of Al_2O_3 at the very high temperatures required. A difficulty of different kind is encountered in the pyrometallurgical extraction is that the formation of metal carbides at high temperature instead of metal.

15.4.2 Limitations of Ellingham Diagram

Thermodynamic conclusions derived from Ellingham diagram about what will reduce a given compound have the following limitations:

1. The conclusions of Ellingham diagrams are based in the assumption that reactants and products are in equilibrium which is often not true.
2. These conclusions are derived on the basis of thermodynamic concepts only and kinetics of the reaction are not taken into consideration. Thus on the basis of Ellingham diagram we can find out whether a reaction is feasible or not but cannot predict the rate of reaction

15.5 REFINING OR PURIFICATION OF METALS

The crude metal extracted by various metallurgical operations is usually associated with impurities. The impurities largely consist of

1. Other metals formed due to simultaneous reduction of their respective oxides that are presented in the ore.

2. Non-metals like silicon or phosphorus formed by the reduction in the furnace
3. The substances which are added to the furnace as flux, fuel etc and the slag formed

The impure metal is purified by different methods depending upon the nature of impurities to be removed and also on the nature of metal. Various methods employed are as follows.

1. Distillation: Volatile metals can be separated from non-volatile impurities by distillation. Metals like zinc, mercury and cadmium having low boiling points are purified by this method. Pure metal vapourizes and separately condensed in a receiver while the non-volatile impurities are left in the retort.

2. Liquation: Fusible (easy melting) metals like tin etc may be purified by this method. Impure metal is heated on a sloping hearth of a reverboratory furnace in absence of air. Fused metal flows down and collected leaving behind infusible impurities.

3. Poling: If the metals contain their own oxides as impurities, the poling method is used for purification. Metals containing their oxides are melted first and then the anthracite powder is sprinkled on the molten metal. Then it is agitated with poles of green wood. The hydrocarbon gases liberated from these poles and the carbon reduces the oxides to the metal. Copper and tin are purified by this method.

4. Oxidation processes: oxidation process are employed for refining of the metals when the impurities have more affinity towards oxygen than the metal. If air is passed over or through the metal, the impurities only get oxidized. The oxidized impurities are removed from the surface by skimming. Sometimes, the oxide of the metal itself is added which provides the oxygen to the impurities. Thus copper oxide is added to copper and iron oxide is added to steel. The various oxidation processes employed for different metals are oxidizing fusion, pudding process, bessemarization, cupellation etc.

(i) Oxidizing fusion: Metals like gold, silver, bismuth are purified by this method. The impure metal is melted in a crucible after addition of oxidizing flux (e.g., Borax, potassium nitrate). The oxidized impurities form a slag which can be removed to give pure metal.

(ii) Pudding process: In the preparation of cast iron from pig iron, the reverboratory furnace is lined with haematite. The pig iron is melted on this hearth. When the molten metal is agitated with iron rods, the impurities in the pig iron are oxidized by the haematite lining. The impurities like carbon, sulphur, phosphorous are oxidized to volatile oxides, silicon converts into silicate slag.

(iii) Bessemerization: In this method the molten metal containing impurities is taken in a Bessemer converter and compressed air is passed through it from the bottom. Then the impurities will get oxidized. For example, when air is passed through the molten pig iron, some of the impurities are oxidized to volatile gaseous oxides and some are removed as slag.

(iv) Cupellation: This method is used when silver containing lead as impurity. The metal is melted in crucibles made of bone ash over which a current of air is blown. Lead is oxidized to litharge (PbO) and is swept away by the current of air.

(v) Electrolytic refining: It is most widely used method for refining of a metal. Metals of highest purity can be obtained by this method. An acidified aqueous solution of a suitable salt of the metal is taken as the electrolyte in a cell. Impure metal is taken as anode while a thin plate of pure metal is taken as cathode. On passing the current of suitable voltage (depending upon the metal to be refined) metal cation of electrolyte goes to cathode and deposits there as pure metal. An equivalent amount of metal goes into solution from anode to maintain the concentration of the electrolyte as such

The impurities in anode left behind. Out of these, the less electropositive impurities go on collecting near the anode in the form anode mud while the more electropositive impurities go into the solution and remain in the electrolyte but not deposited at cathode because of unsuitable voltage for them. The anode mud formed during the refining of copper may contain valuable metals like gold and silver. Copper, silver, aluminium etc are purified by this method.

5. Zone refining: This is one of the most modern methods used for the purification of metals. The principle involved

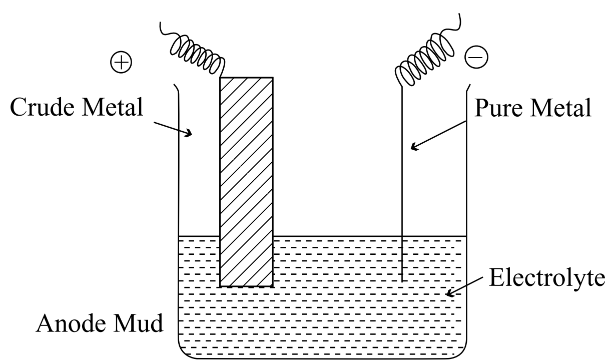


Fig 15.11 Electrolytic refining

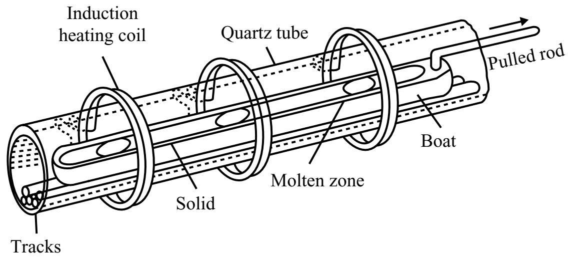


Fig 15.12 Cage zone melting technique

in this method is fractional crystallization. The method is capable of decreasing the concentration of impurities to less than one part per billion and is based on the principle that an impure metal on solidification will separate crystals of the pure metal. The impurities are left behind in the molten part of the metal. It is shown diagrammatically in the Fig 15.16.

A long tube made of suitable material is taken which is surrounded by a series of induction coils which supply heat for melting the metal in narrow zones. The impurities which are more soluble in liquid remain in the high temperature zone and are swept to one end of the ingot and finally discharged.

In this method the impure metal is taken as a rectangular bar on a movable platform. The whole arrangement is sealed in a vessel under high vacuum. When current is passed through the coil, corresponding currents are induced in the bar, as a result of which the bar melts due to which heat is produced. A part of the metal melts as the induction coil moves to the other end. The metal at the end melts while the previously molten metal solidifies. Impurities moves into the molten end of the metal. If this process is carried repeatedly, pure metal is obtained. The impurities moves into the molten metal because of their more soluble tendency in molten metal than in solid metal So, the metal crystallizes in pure condition.

6. Vapour phase refining: It is of two methods.

(a) Mond's process: This method is applied for the purification of nickel. Nickel when heated in a stream of carbon monoxide at about 60–80°C forming a volatile nickel carbonyl.

$$\mathrm{Ni + 4CO \xrightarrow{60-80^\circ C} Ni(CO)_4}$$

The impurities present in the impure nickel left behind as solid. These nickel carbonyl vapours are passed through a tower known as decomposer maintained at a temperature of about 180° having granulated nickel. Here the nickel carbonyl decomposes and the nickel deposits on the granulated nickel

$$\mathrm{Ni\,(CO)_4 \xrightarrow{180^\circ C} Ni + 4CO}$$

The carbon monoxide can be used again and the nickel thus obtained is about 99.8 per cent pure.

(b) Van Arkel – deBoer Method: Small amounts of very pure titanium or zirconium metal can be prepared by this method. Impure metal is heated in an evacuated vessel with iodine. TiI_4 or ZrI_4 are formed which vapourizes leaving behind impurities. The gaseous MI_4 is decomposed on a white hot tungsten filament.

$$\underset{\text{Impure}}{\mathrm{Zr + 2I_2}} \xrightarrow{870\text{K}} \underset{\substack{\text{Tungsten}\\ \text{Filament}}}{\mathrm{ZrI_4}} \xrightarrow{2075} \mathrm{Zr + 2I_2}$$

$$\underset{\text{Impure}}{\mathrm{Ti + 2I_2}} \xrightarrow{525\text{K}} \underset{\substack{\text{Tungsten}\\ \text{Filament}}}{\mathrm{TiI_4}} \xrightarrow{1675} \mathrm{Ti + 2I_2}$$

7. Chromatographic Method: This method is based on difference in extent of adsorption of different components of a mixture on an adsorbent. In column chromatography, a suitable adsorbent such as alumina (Al_2O_3) is packed in glass tube having a stop-cock near the bottom (Fig 15.13). This constitutes the stationary phase. The mixture to be separated is dissolved in some suitable solvent and added to the column. The different components of the mixture get adsorbed to different extent. Some suitable solvent called eluent is then added to the column. The eluent constitutes the mobile phase. The weakly adsorbed components reach the bottom of the column first. It is followed by more strongly adsorbed components. Thus different components of the mixture reach the bottom one by one and in this way get separated. This method is suitable for purification of those elements which are available in small quantities and the impurities are not much different in chemical behaviour from the element to be purified. Lanthanides (rare earth elements are purified by this technique).

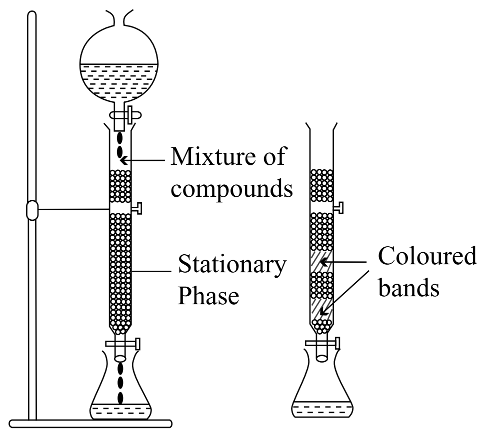


Fig 15.13 Column chromatography

15.6 REFRACTORY MATERIALS

Metallurgical operations are carried at very high temperatures (1000–2000°C) in furnaces. So the furnaces should be constructed with such material which can withstand high temperatures. Such materials are called refractory materials.

Refractory materials are those which do not melt or soften or crumble into powder at high temperatures. They should withstand sudden changes in temperatures they are not affected by the slags formed during the extraction of metals. These are used in the form of bricks for the internal linings of furnaces. Refractory materials are of three types.

(i) **Acidic refractory materials:** Silica, quartz, sand stone etc are used in making acidic refractories.
(ii) **Basic refractory materials:** Magnetite, lime stone, dolomite etc are used in making basic refractories.
(iii) **Neutral refractory materials:** Graphite, chromites, bone ash etc are used in making neutral refractories.

15.7 FURNACES USED IN METALLURGY

The device used in heating operations of the ore is called furnace. The furnaces used in metallurgical operations are mainly of two types:

(i) Fuel furnaces
(ii) Electric furnaces.

Fuel Furnaces: Any substance which liberates heat energy while burning in air is called fuel. Fuel may be solid, liquid or gas. In fuel furnaces like blast furnace, reverboratory furnace, Bessemer converter etc, solid fuel is used, open hearth furnace uses the gaseous fuel.

(a) Blast Furnace: It is widely used furnace. It is mainly used for reduction by smelting. It is a tall structure of about 25–60m height and 6–9m in diameter. It is made up with steel lined inside with fire bricks. The furnace may be cooled by a water jacket when required.

The different parts of the furnace and their functions may be listed as

(i) the mouth through which charge is fed
(ii) the body in which different temperatures are maintained
(iii) the boshes – the widest part of the furnaces
(iv) the hearth – where the molten metal and slag separate as distinct liquids
(v) The chimney through which waste gases goes out.

The mouth is arranged with cup and cone arrangement, sometimes a double cup and cone arrangement through which the charge is fed. Charge is a mixture to be reduced, oxide ore, reducing agent (usually coke) and the flux.

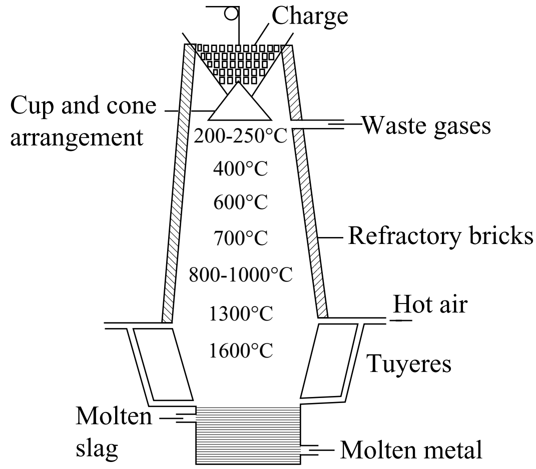


Fig 15.14 Blast furnace

The cup and cone arrangement prevents the escaping of hot gases of the furnace and thus prevents the loss of heat. The fuel burns in the middle portion and a blast of hot air is passed through a series of nozzles called tuyeres in the lower middle part of the furnace.

There is a tap hole near the bottom of the furnace to take out the molten metal and another tap hole is provided at little height to take out the slag which floats over the surface of the molten metal. There is an outlet (chimney) at the top of the furnace for the waste gases.

As the charge moves down form top, it undergoes various chemical reactions. The fuel burns by the hot air blown and the temperature is regulated in such a way that the temperature increases gradually from 200°C at the top to 1500°C at the bottom as the charge moves down. The portion near the bottom is called zone of fusion and slag formation where the temperature is 900–1200°C and the upper zone of reduction 500–900°C.

The metal oxide is reduced in zone of reduction, molten metal slides down, collects at bottom and is tapped out. The impurities combine with the flux to form slag which, being lighter, floats over the molten metal and tapped out separately.

Oxides of Pb, Cu, Fe are commonly reduced by smelting in blast furnace. However, for smelting of Cu and Pb ores, a smaller and a modified model is used.

(b) Reverboratory Furnace: It is another most widely used furnace in metallurgical operations. It can be used for calcinations, roasting as well as for purification process. It consists mainly three parts.

(i) **Fire place:** In which fuel is burnt.
(ii) **Hearth** in which the substance to be heated is kept over a bed.
(iii) **Chimney** through which waste gases escapes out.

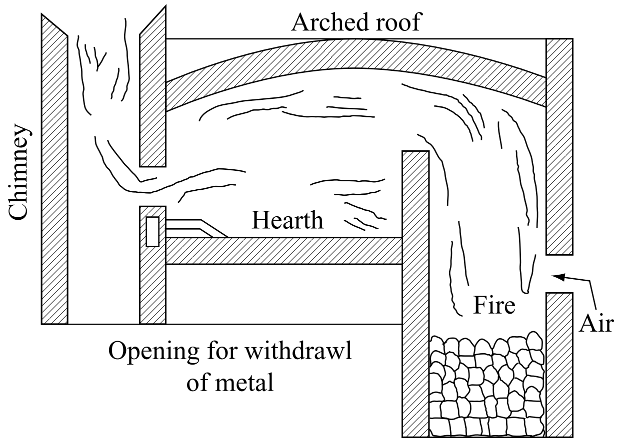


Fig 15.15 Reverberatory fumance

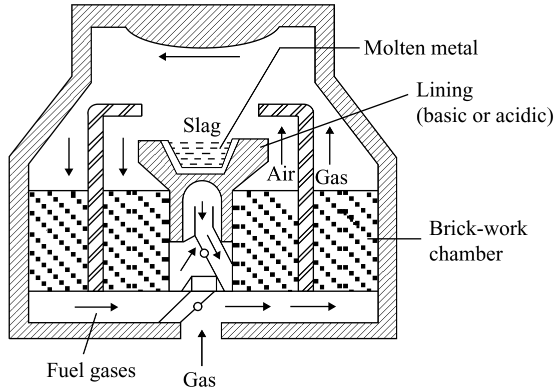


Fig 15.16 Open hearth furnace

The reverberatory furnaces are constructed with fire bricks as shown in the Fig 15.15. The fire place, hearth and chimney are separated. The roof of the hearth is in concave shape so as to reverberate the radiations from the fire place to the bed. Suitable fuel is burned in the fire place. Air is blown from the side door. The hot flames rise to the ceiling and reflect back on to the charge taken on the hearth where the ore gets heated. Chemical reactions occur on the hearth. The waste gases go out through the chimney.

As the fuel does not come in contact with the heated substances, the furnaces can be used for oxidizing and reducing purpose both. Further the metal will not be contaminated by the ashes. Reverboratory furnace is used for calcination and roasting and some times for smelting.

(c) Open Hearth furnace (Regenerative Furnace): It is a gaseous fuel furnace. The open hearth furnace is also known as regenerative furnace because the heat content of the fuel gases and waste gases are utilized to their maximum limit. It is used for the removal of acidic impurities that are present in the ore. It is made up of fire bricks lined inside. The fire bricks used may be acidic to remove basic impurities or basic to remove acidic impurities. The furnace has four chambers. Air and fuel gas (generally producer gas) are passed separately through two heated chambers as shown in Fig 15.16 both mix above the hearth and produce large amount of heat due to combustion. Waste gases are passed through the two chambers of the other side so that these two chambers also get heated waste cool gases are now led to chimney.

Air and fuel gas mixture is now passed through other two chambers by reversing the process. This goes on alternatively and continuously for some time bringing the temperature of almost 1600°C. Impurities get oxidized and form slag or escape. This process is economical as the hot waste gases are reutilized.

This furnace is commonly used in the manufacture of steel.

Bessemer Converter

It is named after Henry Bessemer who devised it in 1856. It is a pear shaped furnace made of steel plates it is used for auto-reduction and for removing the impurities in the form of their volatile oxides. The process carried is called Bessemerization Bessemer converter is lined inside with either acidic (SiO_2) or basic (CaO) or (MgO) fire bricks. It is mounted on trunions which enables to be tilted to any position. Bessemer converter is used in the production of copper and steel. To convert the impurities into their volatile oxides blast of air is blown through tuyers fitted below the middle of the furnace. The charge in the form of molten mass, called matte in introduced. The desired reactions take place, molten metal quickly drops below the tuyeres so that it escapes being oxidized by the blast of air.

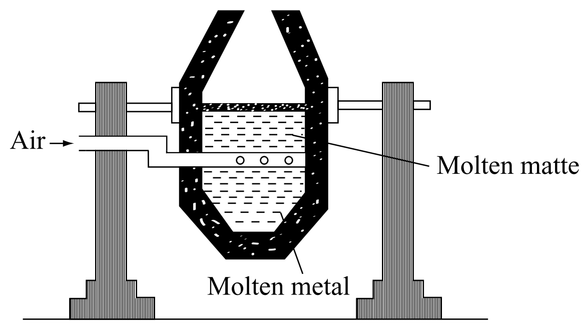


Fig 15.17 Bessemer converter

Electric Furnaces

Furnaces in which heating is carried by electricity are called electric furnaces. Where electricity is cheap and product of

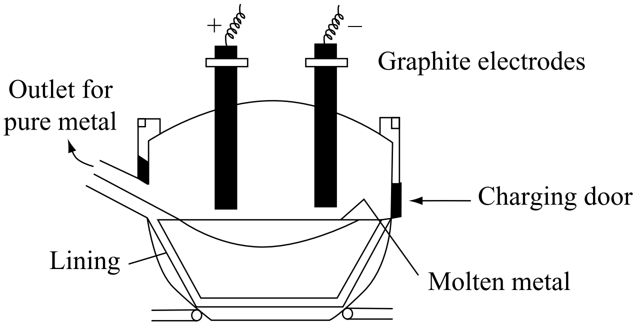


Fig 15.18 Heroult's furnace

quality is desired, an electric furnace is very suitable. Two commonly used electric furnaces are Heroult's furnace and electric muffle furnace.

(i) Heroult's Furnace: It consists of a crucible made of steel and lined inside with dolomite or magnetite which acts as a flux. A temperature as high as 4000°C can be produced by striking an electric arc between two graphite electrodes. The furnace is used for purification of some metals and in the production of high quality alloy steels.

Due to the electric arcs produced by passing electric current, the charge on the hearth melts due to high temperature and the impurities combine with flux to form slag which floats over the molten metal and removed.

(ii) Electric Muffle Furnace: It is a box type furnace with a refractory lining which is heated around electrically. Sufficient high temperatures can be attained in this furnace. Purification of silver by cupellation process can be done in the furnace by keeping the cupel in the muffle.

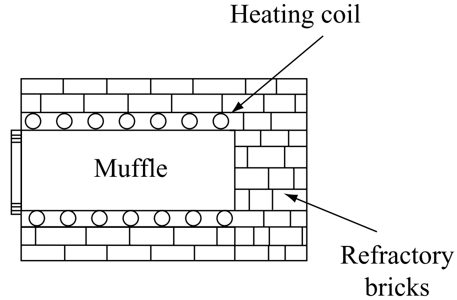


Fig 15.19 Muffle furnace

Table 15.2 Extraction of metals

Electrode process	Standard electrode potential (volts)	Main occurrence	Main method of extraction	Equation for extraction
Li, Li^+	−3.04	Spodumene $LiAl(SiO_3)_2$	Electrolysis of fused LiCl with KCl added	
K, K^+	−2.92	Carnallite $KCl \cdot MgCl_2 \cdot 6H_2O$	Electrolysis of fused KCl with $CaCl_2$ added	
Ba, Ba^{2+}	−2.90	Witherite $BaCO_3$ Barytes $BaSO_4$	Electrolysis of fused $BaCl_2$	
Sr, Sr^{2+}	−2.89	Strontianite $SrCO_3$ Celestine $SrSO_4$	Electrolysis of fused $SrCl_2$	
Ca, Ca^{2+}	−2.87	Limestone $CaCO_3$ Gypsum $CaSO_4$	Electrolysis of fused $CaCl_2$ and CaF_2	All involve electrolytic reduction: $M^{n+} + ne^- \longrightarrow M$
Na, Na^+	−2.71	Rock salt NaCl Chile salpetre $NaNO_3$	Electrolysis of fused NaCl with $CaCl_2$ added	
Mg, Mg^{2+}	−2.37	Carnallite $KCl \cdot MgCl_2 \cdot 6H_2O$ Magnesite $MgCO_3$	Electrolysis of fused $MgCl_2$ with KCl added	
Be, Be^{2+}	−1.70	Beryl $3BeO \cdot Al_2O_3.6SiO_2$	Electrolysis of fused BeF_2 with NaF added	
Al, Al^{3+}	−1.66	Bauxite $Al_2O_3 \cdot 2H_2O$ Silicate rocks	Electrolysis of Al_2O_3 in molten Na_3AlF_6	
Mn, Mn^{2+}	−1.18	Pyrolusite MnO_2 Hausmannite Mn_3O_4	Reduction of oxide with Al or C	$3Mn_3O_4 + 8Al \longrightarrow 9Mn + 4Al_2O_3$

Table 15.2 (Cont.)

Electrode process	Standard electrode potential (volts)	Main occurrence	Main method of extraction	Equation for extraction
Ti, Ti^{4+}	−0.95	Ilmenite $TiO_2 \cdot FeO$ Rutile TiO_2	Reduction of $TiCl_4$ with Mg or Na	$TiCl_4 + 2Mg \longrightarrow Ti + 2MgCl_2$
Zn, Zn^{2+}	−0.76	Zinc blende ZnS Calamine $ZnCO_3$	Reduction of ZnO with C or electrolysis of $ZnSO_4$	$ZnO + C \longrightarrow Zn + CO$
Cr, Cr^{3+}	−0.74	Chromite $FeO \cdot Cr_2O_3$	Reduction of Cr_2O_3 with Al	$Cr_2O_3 + 2Al \longrightarrow 2Cr + Al_2O_3$
Fe, Fe^{2+}	−0.44	Magnetite Fe_3O_4 Haematite Fe_2O_3	Reduction of oxides with CO	$Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$
Co, Co^{2+}	–0.28	Smaltite $CoAs_2$	Reduction of Co_3O_4 with Al	$3Co_3O_4 + 8Al \longrightarrow 9Co + 4Al_2O_3$
Ni, Ni^{2+}	–0.25	Millerite NiS	Reduction of NiO with CO	$NiO + 5CO \longrightarrow Ni(CO)_4 + CO_2$ $Ni(CO)_4 \longrightarrow Ni + 4CO$
Sn, Sn^{2+}	–0.14	Cassiterite SnO_2	Reduction of SnO_2 with C	$SnO_2 + 2C \longrightarrow Sn + 2CO$
Pb, Pb^{2+}	–0.13	Galena PbS	Reduction of PbO with C	$PbO + C \longrightarrow Pb + CO$
Bi, Bi^{3+}	+0.32	Bismuth glance Bi_2S_3 Bismuthite Bi_2O_3	Reduction of Bi_2O_3 with C	$Bi_2O_3 + 3C \longrightarrow 2Bi + 3CO$
Cu, Cu^{2+}	+0.34	Copper pyrites $CuFeS_2$ Cuprite Cu_2O	Partial oxidation of sulphide ore	$2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$
Ag, Ag^{+}	+0.80	Argentite Ag_2S Occurs as metal	Special methods involving use of sodium cyanide	$Ag_2S + 4NaCN \longrightarrow 2NaAg(CN)_2 + Na_2S$ $2NaAg(CN)_2 + Zn \longrightarrow 2Ag + Na_2Zn(CN)_4$
Hg, Hg^{2+}	+0.85	Cinnabar HgS	Direct reduction of HgS by heat alone	$HgS + O_2 \longrightarrow Hg + SO_2$
Pt, Pt^{2+}	+1.20	Occurs as metal Sperrylite $PtAs_2$	Thermal decomposition of $(NH_4)_2PtCl_6$	$(NH_4)_2PtCl_6 \longrightarrow Pt + 2NH_4Cl + 2Cl_2$
Au, Au^{2+}	+1.50	Occurs as metal	Special methods involving use of sodium cyanide	Similar to that for silver $2NaAu(CN)_2 + Zn \longrightarrow 2Au + Na_2Zn(CN)_4$

OCCURENCE AND PRINCIPLES OF EXTRACTION OF SOME METALS

15.8 IRON

Iron is the second most abundant metal occurring in the earth's crust. It is a reactive metal and does not occur in the free state. In combined state it occurs as oxides, carbonate and sulphides.

15.8.1 Common Ores of Iron

(i)	Haematite	Fe_2O_3 (red oxide of iron)
(ii)	Magnetite	Fe_3O_4 (Magnetic oxide of iron)
(iii)	Limonite	$2Fe_2O_3 \cdot 3H_2O$ (Hydrated oxide of iron)
(iv)	Siderite (or) sepathic iron	$FeCO_3$
(v)	Iron pyrites	FeS_2

The iron pyrites FeS_2 is not considered to be an important ore of iron, but it is used to produce sulphur dioxide for the manufacture of sulphuric acid.

15.8.2 Extraction of Cast Iron

The extraction of cast iron involves the following steps:

(i) Ore Dressing: Generally cast iron is extracted from haematite or magnetite. There is no need of concentration for these ores. Only in few cases, the crushed ore is concentrated by gravity separation process in which it is washed with water to remove clay, sand etc.

(ii) Calcinations or roasting: The concentrated ore is heated strongly in the presence of limited supply of air in a reverboratory furnace. The following changes may occur in the furnace.

$$2Fe_2O_3 + 3H_2O \longrightarrow 2Fe_2O_3 + 3H_2O$$

$$FeCO_3 \longrightarrow FeO + CO_2$$
$$4FeO + O_2 \longrightarrow 2Fe_2O_3$$

If there is any FeO, it would combine with silica and goes into slag. So it should be converted into Fe_2O_3 by roasting. The impurities such as sulphur, phosphorus, and arsenic are converted to their gaseous oxides which are volatile and escape. The entire mass becomes porous which helps in the reduction process at a later stage.

Smelting: The roasted ore, coke (free from sulphur) and lime stone are mixed in 8 : 4 : 1 parts by weight. This charge is fed into the blast furnace. A blast of hot air is sent into the furnace through tuyers. The added coke serves as a fuel as well as reducing agent, while the lime serves as flux. The temperature range in the furnace is 500–2000°k – various reactions takes place in the blast furnace in different temperature zones as described below.

(i) Zone of combustion: At the base, coke burns to produce CO_2 which starts rising upward during the reaction. The reaction is exothermic and heat produced rises the temperature to about 1800k. this region is called combustion zone.

$$C + O_2 \longrightarrow CO_2 \ \Delta H = -393.4KJ$$

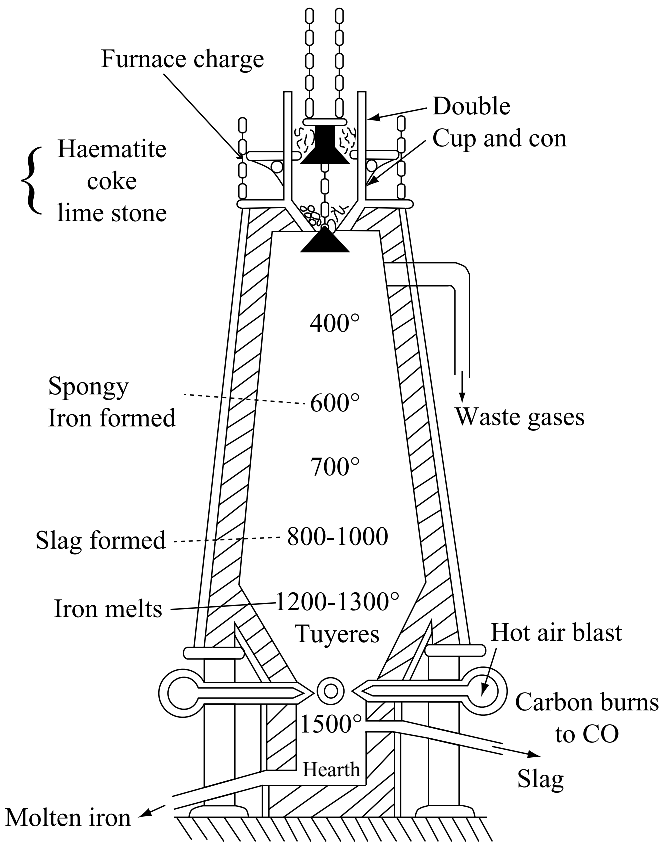


Fig 15.20 Blast furnace

15.8.3 Thermodynamics of the Reduction of Iron Oxides

In the blast furnace iron oxide is reduced to iron. One of the main reduction step in this process is

$$FeO_{(S)} + C_{(S)} \longrightarrow Fe_{(S)} + CO_{(g)}$$

The above reaction can be written in two steps as

(i) $FeO_{(S)} \longrightarrow Fe_{(S)} + ½\, O_{2(g)}\ \Delta G_{FeO,\,Fe}$

(ii) $C_{(S)} + ½\, O_2 \longrightarrow CO_{(g)}\ \Delta G_{c,\,co}$

By rearranging and subtracting these two reactions (i&ii) we get the main reaction for which the net Gibbs energy change becomes

$$\Delta G_{c,co} + \Delta G_{FeO,\,Fe} = \Delta_r G$$

The reaction will takes place if $\Delta_r G$ is negative. From the Ellingham diagram (Fig 15.10) it can be seen that the line C $\rightarrow$ CO goes downward. So above 1073k the C, CO line comes below the Fe, FeO line [$\Delta G_{c,\,co} < \Delta G_{Fe,FeO}$]. Hence in this range coke reduces the FeO and itself oxidizes to CO. But at relatively low temperatures (below 1073 k) CO $\rightarrow CO_2$ line is below Fe $\rightarrow$ FeO line. Therefore CO reduces Fe_2O_3 and Fe_3O_4 to Fe below 1073K.

(iii) Zone of fusion: As carbon dioxide rises upwards it comes in contact with coke present in the charge falling down and converts into carbon monoxide

$$CO_2 + C \longrightarrow 2CO;\ \Delta H = +163.2KJ$$

This is an endothermic reaction and therefore the temperature is lowered to about 1570k. The iron produced in the upper part of blast furnace melts here. Any oxides of iron, if present are reduced by hot coke to iron

$$FeO + C \longrightarrow Fe + CO$$
$$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO$$

(iv) Zone of slag formation: This is the middle portion of the furnace where the temperature is about 1270k. Here the lime stone decomposes into lime (CaO) and carbon dioxide. The lime thus produced (flux) combines with the silica (gangue) forming slag.

$$CaCO_3 \longrightarrow CaO + CO_2$$
$$CaO + SiO_2 \longrightarrow \underset{\text{Slag}}{CaSiO_3}$$

The oxides of phosphorus if present also combines with lime forming calcium phosphate which comes into the slag.

$$6CaO + P_4O_{10} \longrightarrow 2Ca_3(PO_4)_2$$

This slag containing calcium silicate and calcium phosphate is called **Thomas slag**.

(v) Zone of reduction: It is the top portion of the furnace where the temperature ranges 500-875k. Here the oxides of iron are reduced by carbon monoxide to iron. So this region is called zone of reduction.

$$3Fe_2O_3 + CO \longrightarrow 2Fe_3O_4 + CO_2$$

$$Fe_3O_4 + CO \longrightarrow 3FeO + CO_2$$

$$FeO + CO \longrightarrow Fe + CO_2$$

The spongy iron produced in the zone of reduction moves slowly and melts in the fusion zone. At the lower part where the temperature is high, the main reaction is

$$FeO + C \longrightarrow Fe + CO$$

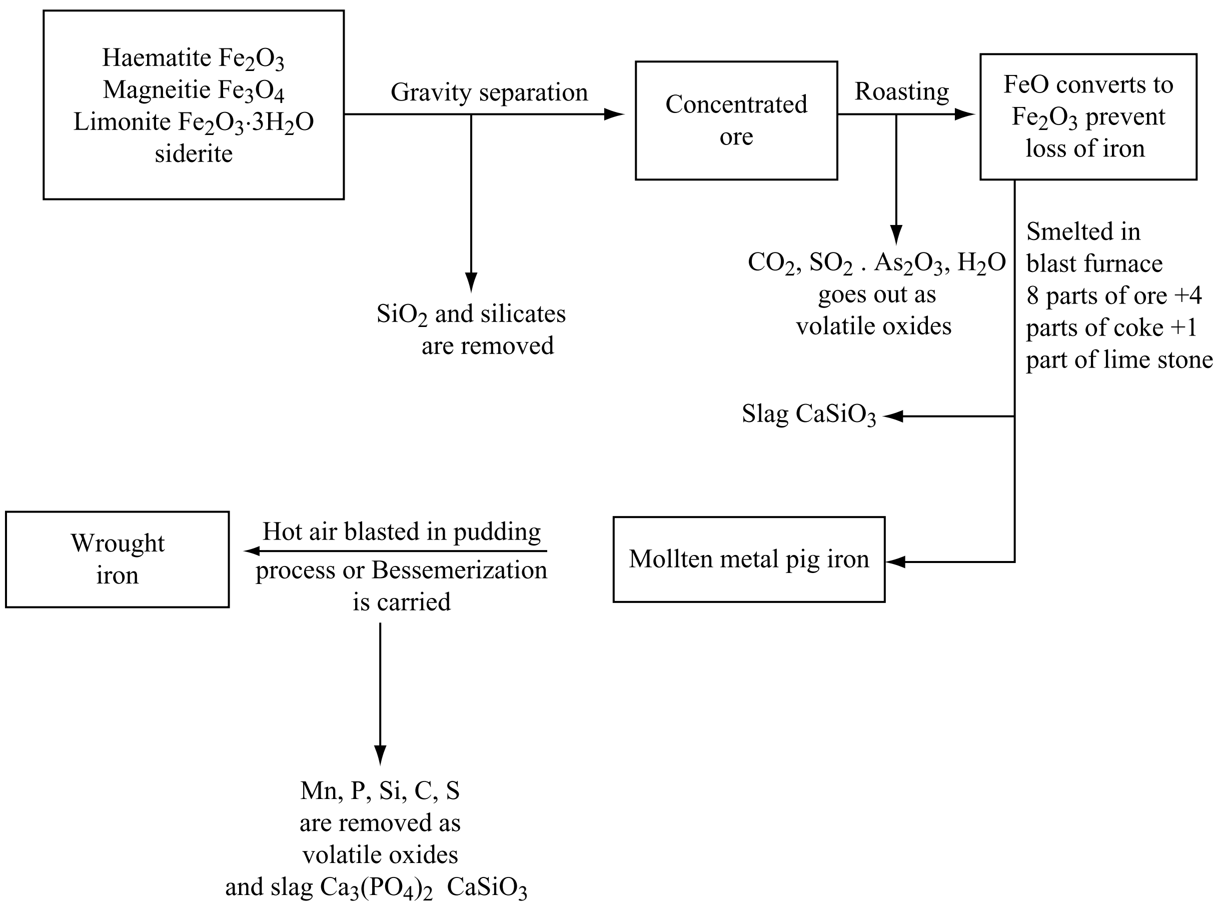
In the lower part of furnace where the temperature is very high the oxides of silicon, phosphorous and manganese are also reduced to the corresponding elements and they dissolve in molten iron along with carbon. The iron thus formed in the blast furnace is called **pig iron**. This contains about 93 per cent iron, 5 per cent carbon and the remaining are silicon, phosphorous, manganese etc as impurities.

The pig iron taken out of the blast furnace is heated again with scrap iron and coke in the presence of blast of hot air. Then the percentage of carbon decreases to 3-3.5 per cent. This is called **cast iron**. It expands slightly on solidifying and therefore reproduces the shape of the mould.

Cast iron is two types: (i) **White cast iron:** This is formed by sudden cooling of the pig iron. It contains carbon in the form of cementite Fe_3C (ii) **Grey cast iron:** it is formed by the slow cooling of the pig iron. It contains carbon in the form of graphite.

The cast iron is extremely hard but brittle. The melting point is about 1473 k.

Flow Chart for the Extraction of Iron



15.8.4 Commercial Forms of Iron

There are three commercial forms of iron which differ from each other mainly in their carbon content.

(i) Cast iron or pig iron: It contains about 2–5 per cent carbon along with impurities such as sulphur, silicon, phosphorus, manganese etc. It is the least pure form of iron. It is brittle and is very difficult to weld. It is resistant to corrosion and is used for sewage pipes.

(ii) Steel: It contains 0.1 to 1.5 per cent carbon and other impurities. There are many varieties of steel depending on the amount of carbon present in it.

(a) Mild steels: These contain low percentage of carbon. Such steels show the properties of wrought iron along with elasticity and hardness.

(b) Hard steels: These contain high percentage of carbon. They are hard and brittle.

(c) Special steels: Steel mixed with small amounts of nickel, cobalt, chromium, tungsten, molybdenum manganes etc acquires special properties. Such steels are called alloy steels.

(iii) Wrought iron: It is the purest form of iron and contains carbon and other impurities less than 0.2 per cent it is malleable and can be easily welded this type of iron permanently. It is used for making wires chains, electromagnets etc.

15.8.5 Preparation of Wrought Iron

It is the purest form of iron. It is obtained by refining of cast iron by puddling process.

The cast iron is heated with haematite (Fe_2O_3) so that the impurities are oxidized.

$$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO$$
$$2Fe_2O_3 + 3Si \longrightarrow 4Fe + 3SiO_2$$
$$Fe_2O_3 + 3Mn \longrightarrow 2Fe + 3MnO$$
$$MnO + SiO_2 \longrightarrow MnSiO_3 \text{ (slag)}$$
$$5Fe_2O_3 + 6P \longrightarrow 10Fe + 3P_2O_5$$

As the impurities decreases the melting point of the metal increases and it becomes a semi-solid mass. This is taken out and hammered to squeeze out the impurities. Wrought iron thus produced contains about 0.2–0.5 per cent carbon and traces of P and Si its melting point is above 1400°C.

15.8.6 Preparation of Steel

Steel is manufactured from cast iron by reducing its carbon content from about 5 per cent to between 0.2 and 1.5 per cent depending upon the quality of steel to be prepared.

1. Bessemer Process: In this method molten cast irons is poured into a pear shaped Bessemer converter lined inside

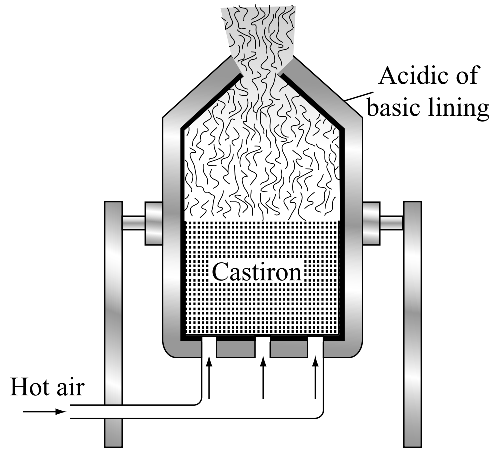


Fig 15.21 Bessemer converter

with silica bricks. It is mounted on trunions so that it can be tilted as one desires. A blast of air or oxygen diluted with steam or carbon monoxide is blown through the inlets at the bottom.

Carbon and various other impurities present in cast iron are oxidized and form slag.

$$2Mn + O_2 \longrightarrow 2MnO$$
$$Si + O_2 \longrightarrow SiO_2$$
$$MnO + SiO_2 \longrightarrow \underset{\text{Slag}}{MnSiO_3}$$

It may be noted that the furnace is lined with silica (acidic lining) if the iron contains basic impurities like manganese. However if iron contains acidic impurities like phosphorus and sulphur, then the lining must be basic (eg CaO or MgO). The phosphorus is oxidized to P_2O_5 which may combine with CaO to form calcium phosphate.

$$4P + 5O_2 \longrightarrow P_4O_{10}$$
$$P_4O_{10} + 6CaO \longrightarrow 2Ca_3(PO_4)_2$$

The slag formed is called **Thomas slag** containing calcium phosphate. This is used as a fertilizer.

When whole of the carbon is oxidized, and flames of CO stop, the air blast is cut off, a calculated amount of carbon is added in the form of spiegeleisen (an alloy with 15–20 per cent Mn, 60 per cent C and the rest iron) Manganese reduces the loss of iron. This process is rapid and inexpensive but the steel obtained is not of high quality. It is difficult to control the process and also the quality of steel is variable.

2. Open Hearth Process: This gives steel of very high quality. The open hearth furnace is a shallow saucer-like hearth. The pig iron is taken into the hearth. It contains two sets of brick work chambers (checkers) on each side. The furnace is made of cast iron. It is lined either by dead

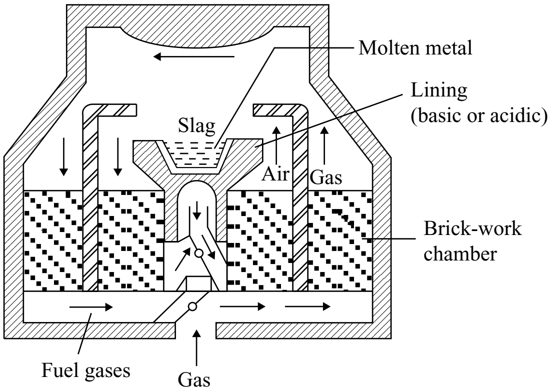


Fig 15.22 Open hearth furnace

burnt magnesium oxide or by silica. A mixture of fuel gas (producer gas or coal gas) and air is burnt over the surface of pig iron. A temperature of about 1600ºC is maintained in the furnace the fuel gas and air are introduced into the furnace through one set of checkers and exhaust gases pass out through the other set. The passage of the fuel gas and air is now reversed i.e., passed from the second set, so that heat economy is possible.

The charge taken on the open hearth furnace consists of pig iron, scrap iron, iron ore (haematite) and lime stone. The changes in the open-hearth furnace are carried out slowly. The quality of the steel is analysed from time to time. Iron ore is added till the carbon content is at the desired level. To prepare the alloys requisite quantity of ferromanganese or other alloys are added. The steel and slag are collected from the holes in the hearth separately. This method is advantageous over Bessemer process because the composition of steel can be controlled and gives steel of good quality. It has also greater fuel economy.

3. Oxygen Blowing Process: In this process the liquid iron from the blast furnace is charged into converter (Fig 15.23). Scrap steel is added to this and a jet of oxygen is blown through a retractable steel lance into or over the surface of liquid metal. During this, the impurities are oxidized which on the addition of lime form slag. The slag is removed by tilting the converter. When steel of desired composition is obtained, the oxygen is stopped and the molten steel is poured into for casting ingots.

4. The Electric Arc Process: This is the recent widely used method for the manufacture of high quality alloy steels such as stainless steel. In this method a charge of scrap steel is fed into the furnace and is melted by striking electric arc between the adjustable graphite electrodes. The furnace has acidic or basic lining depending upon the nature of the impurities present in the scrap. The electric

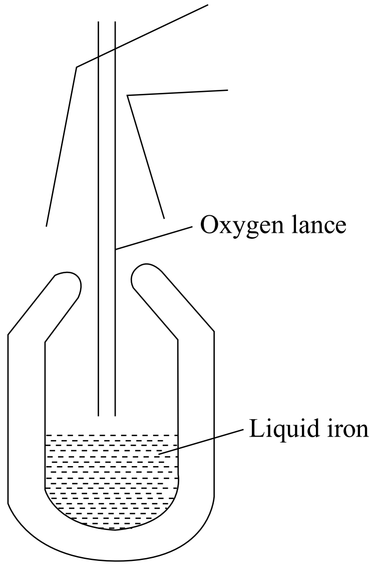


Fig 15.23 Oxygen top Blowing process

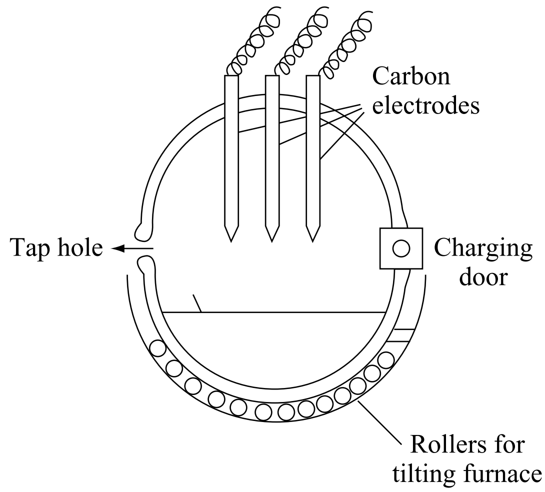


Fig 15.24 Electric arc process

arc furnaces generally produce between 25–50 tones of steel at a time.

This method is widely used in manufacture of alloys and other high quality steels such as stainless steel and high speed cutting steel.

5. The High Frequency Induction Process: In this method, a charge of alloy scrap of known composition together with iron is fed into the furnace. Alternating current at 500–2000Hz per second is passed through the insulated water cooled copper coils. The resulting magnetic field sets up steady current which produces heat. The

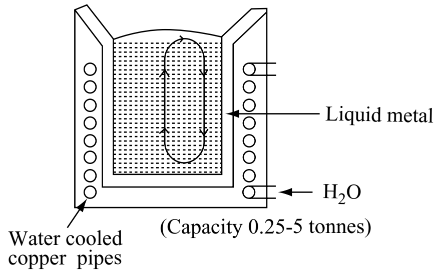


Fig 15.25 High frequency induction process

circulation of the metal caused by these currents produces strong stirring effect.

This induction furnace produces high quality alloy steels containing tungsten, vanadium, chromium, manganese, cobalt, molybdenum and nickel etc for making ball bearings, magnets, dies and tool steel etc.

15.8.7 Heat Treatment of Steel

The hardness of steel can be increased or decreased by heat treatment as discussed below.

1. **Quenching:** If a steel article is heated to redness and then suddenly cooled by plunging into water or some oil, the steel becomes hard and brittle. This treatment is called quenching or hardening of steel.
   This phenomenon can be explained as follows. Above 850°C the structure is face-centred cubic and the carbon is present in solid solution. Rapid cooling does not allow sufficient time for the carbon to separate as cementite, Fe_3C, and instead carbon is deposited; the presence of carbon atoms makes it impossible for the iron to adopt a true body-centred structure and distorted modification which is hard but brittle is the result.
2. **Tempering:** When the quenched steel is heated to a temperature of about 550k and kept at that temperature for some time and then cooled slowly, the steel obtained is quite hard but less brittle. This treatment is called **tempering**.

   During tempering the surface of the steel takes on certain colours, being yellow at 500k and bule at 570k. These colours are due to the formation of very thin oxide film on the surface of the steel, the thickness of the film determining the final colour. Watch springs are blue, since tempering at 570k removes all brittleness and produces an extremely tough and springy steel.

   During 'tempering' microscopically small areas of cementite are formed and thus helps to relieve the strain present in 'quenched steel' without significantly altering the hardness of steel.
3. **Annealing:** If the quenched steel is heated to a temperature below red hot and then allowed to cool slowly, it becomes soft. This process is called annealing.
4. **Case hardening:** When the mild steel is heated with charcoal and then plunged into oil. Then a thin film of hardened steel is formed on the surface of the mild steel. This is called case hardening. This steel is resistant to wear and tear.
5. **Nitriding:** Production of hard iron nitride coating on the surface of steel is called nitriding. This is made by heating steel in the atmosphere of dry ammonia at 500-600°C for about 3–4 days when a hard coating of iron nitride is formed on the surface of steel.

15.8.8 Passive Iron

When a piece of iron is dipped in concentrated nitric acid, a reaction takes place which stops after some time completely. The iron does not appear to undergo any change in appearance. But it becomes inactive. We know that ordinary iron liberates hydrogen from dilute acids and displaces copper from copper sulphate. However this iron does not liberate hydrogen from dilute acids and cannot displaces copper from copper sulphate solution. The iron in this form is said to be passive and the process is called passivation of iron. The loss of chemical activity of an active species is called passivity. The passivity of iron is due to the formation of thin insoluble and invisible iron oxide film on the surface which prevents its further reactions. The passivity can be removed by scratching, scrubbing or by dissolving out in iodine solutions. The passivity of iron is important in preventing corrosion of iron.

15.8.9 Rusting of Iron

When exposed to air and moisture, iron gets rapidly oxidized to hydrous oxide known as rust of iron. Although the composition of rust varies some what, it consists of mainly of hydrated iron oxide $2Fe_2O_3 \cdot xH_2O$ together with small quantity of iron (II) carbonate.

Theory of rusting: If an electrolyte is in contact with two metals which are themselves in contact, the three essential elements of an electrochemical cell are present and corrosion takes place. For instance, rusting takes place at the exposed surfaces of scratched tinplate, the electrolyte being the moisture in the air.

Pure iron out of contact with any other metal will still corrode by an electrolytic mechanism, for it found that iron exposed to a higher concentration of oxygen will accept electrons from iron where the oxygen concentration is less.

Lower oxygen concentration (–ve pole)

$$2Fe \longrightarrow 2Fe^{2+} + 4e^-$$

Higher oxygen concentration (+ve pole)

$$O_2 + 2H_2O + 4e^- \longrightarrow 4OH^-$$

At some point on the surface of the iron (between the regions acting as negative and positive poles) iron (II) hydroxide is formed by diffusion of the iron (II) and hydroxyl ions, this is then oxidized to rust (hydrated iron (III) oxide). Rusting is accelerated if the moisture contains a strong electrolyte, since more current can flow through the tiny electrochemical cells e.g., sea spray and the sodium chloride used to thaw ice on roads in the winter can play havoc with unprotected car chassis.

The fact that rust is formed some way between the positive and negative poles explains why paint must be entirely stripped off slightly corroded areas, for it may well be that the iron under the paint has corroded appreciably.

Protection against rusting can be afforded by connecting a metal with a higher negative electrode potential than iron to the structure e.g., underground steel pipes have been protected from corrosion by connecting easily renewable magnesium rods along them at specific points. Since magnesium has a higher negative electrode potentials than iron it will pass into solution as ions (and thus corrode) leaving the steel structure intact.

15.9 COMPOUNDS OF IRON

Iron shows oxidation states of +3 (the most stable) and +2 (reducing). There is also an unstable oxidation state of +6 which is powerfully oxidizing. The presence of five singly occupied 3d orbital (half filled) in the Fe^{3+} ion and only four singly occupied 3d orbital in the Fe^{2+} ion is often cited as an explanation of the easy oxidation of Fe^{2+} to Fe^{3+}

Iron (III) Compounds

15.9.1 Iron (III) Oxide

Iron (III) oxides occurs naturally as haematite; it can be prepared as rust coloured solid by heating iron (III) hydroxide or iron (II) sulphate or by burning iron pyrites.

$$2Fe(OH)_3 \longrightarrow Fe_2O_3 + 3H_2O$$
$$2FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$$
$$4FeS_2 + 11O_2 \longrightarrow 2Fe_2O_3 + 8SO_2$$

It is a deep red powder insoluble in water it is armphoteric

$$Fe_2O_3 + 2NaOH \longrightarrow 2NaFeO_2 + H_2O$$
$$Fe_2O_3 + Na_2CO_3 \longrightarrow 2NaFeO_2 + CO_2$$
$$Fe_2O_3 + 6HCl \longrightarrow 2FeCl_3 + 3H_2O$$

It is reduced to iron by H_2 and CO

$$Fe_2O_3 + 3H_2 \longrightarrow 2Fe + 3H_2O$$
$$Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$$

When heated to 1300°C it converts to magnetic oxide

$$6Fe_2O_3 \longrightarrow 4Fe_3O_4 + O_2$$

It is used as a red pigment under the name vanation red and also as pigment (yellow, red and purple forms can be obtained by a variety of methods). It is used as an abrasive polishing powder under the name **Jeweller's Rouge**.

Naturally occurring iron (III) oxide has a similar structure to that of aluminium oxide (corundum) i.e., it is ionic.

15.9.2 Iron (III) Hydroxide $Fe_2O_3 \cdot xH_2O$

Iron (III) hydroxide, which is best represented as iron (III) oxide associated with an indefinite number of molecules of water, is precipitated as a rust coloured gelatinous solid when aqueous solutions containing hydroxyl and iron (III) ions are mixed. It reacts readily with dilute acids to give iron (III) salts but since it also reacts with concentrated solutions of alkalis to give ferrates (III) e.g., $NaFeO_2$, it must be regarded as amphoteric.

Colloidal iron (III) hydroxide is obtained as a rust coloured solution when a few drops of iron (III) chloride solution are added to a large volume of boiling water

$$FeCl_3 + 3H_2O \longrightarrow Fe(OH)_3 + 3HCl.$$

15.9.3 Magnetic Oxide Fe_3O_4

This oxide occurs naturally as the ore magnetite which is magnetic. It is a black solid and is formed when steam is passed over heated iron.

$$3Fe + 4H_2O \longrightarrow Fe_3O_4 + 4H_2$$

When reacted with acids it gives a mixture of iron (II) and iron (III) salts (ratio 1:2) and this is consistent with a structure containing Fe^{2+} and Fe^{3+} ions (ratio 1:2) with enough oxide ions to maintain electrical neutrality.

15.9.4 Iron (III) Halides

Iron (III) fluoride, chloride and bromide can be made by heating iron in the presence of the halogen; pure iron (III) iodide cannot be isolated. $FeCl_3$ and $FeBr_3$ are dark red covalent solids which in the vapour state exist as dimerized molecules Fe_2Cl_6 and Fe_2Br_3 respectively similar to that of aluminium chloride. Both compounds dissociate on heating to give the iron (II) halide and halogen and both dissolve very readily in water to give the yellow $[Fe(H_2O)_6]^{3+}$ ion and the halide ion.

It is not really surprising that iron (III) iodide cannot be obtained in the pure state, since a solution containing iron (III) ions readily oxidizes iodide ions to iodine.

$$2Fe^{3+} + 2I^- \longrightarrow 2Fe^{2+} + I_2$$

The instability of FeI_3 with respect to FeI_2 and iodine means that iodine is not sufficiently powerful oxidizing agent to convert Fe^{II} to Fe^{III}; this behaviour is very much in line with the more extensive dissociation of iron (III) bromide into iron (II) bromide on heating as compared with the corresponding chloride.

15.9.5 Ferric Chloride $FeCl_3$

This is the most important ferric halide. It is known in anhydrous and hydrated forms.

Preparation:

Anhydrous ferric chloride can be prepared by passing dry chlorine gas over heated iron fillings.

$$2Fe + 3Cl_2 \longrightarrow 2FeCl_3$$

Hydrated ferric chloride is obtained by the action of hydrochloric acid on ferric carbonate, ferric hydroxide or ferric oxide

$$Fe_2(CO_3)_3 + 6HCl \longrightarrow 2FeCl_3 + 3H_2O + 3CO_2$$
$$Fe\ (OH)_3 + 3HCl \longrightarrow FeCl_3 + 3H_2O$$
$$Fe_2O_3 + 6HCl \longrightarrow 2FeCl_3 + 3H_2O$$

Crystallization from solution gives hydrated compound $FeCl_3 \cdot 6H_2O$. Anhydrous $FeCl_3$ cannot be prepared by heating hydrated salt, because it decomposes on heating.

$$2FeCl_3 \cdot 6H_2O \xrightarrow{\Delta} 2FeCl_3 \cdot 3H_2O + 6H_2O$$
$$2FeCl_3 \cdot 3H_2O \xrightarrow{\Delta} 2Fe(OH)_3 + 6HCl$$
$$2Fe(OH)_3 \xrightarrow{\Delta} Fe_2O_3 + 3H_2O$$

Anhydrous $FeCl_3$ can be prepared from hydrated salt by using some dehydrating agents such as thionyl chloride or an ether like 2,2-dimethoxy propane

$$FeCl_3 \cdot 6H_2O + 6SOCl_2 \longrightarrow FeCl_3 + 6SO_2 + 12\ HCl$$

$$FeCl_3 \cdot 6H_2O + 6CH_3 \cdot \underset{\substack{|\\ OCH_3}}{\overset{\substack{OCH_3\\ |}}{C}} \cdot CH_3 \rightarrow FeCl_3 + 12CH_3OH + 6CH_3COCH_3$$

Properties: It is a dark red brown deliquescent solid. It sublimes at 300°C and its vapour density corresponds to dimeric formula Fe_2Cl_6. The dimer dissociates at high temperatures to $FeCl_3$. The dissociation into $FeCl_3$ is complete at 750°C, above which it breaks into ferrous chloride and chlorine.

$$Fe_2Cl_6 \xrightarrow{750°C} 2FeCl_3 \xrightarrow{\text{above } 750°C} 2FeCl_2 + Cl_2$$

Anhydrous ferric chloride is covalent, soluble in non-polar solvents like ether, alcohol etc. Its aqueous solution is acidic in nature due to hydrolysis.

$$FeCl_3 + 3H_2O \longrightarrow Fe(OH)_3 + 3HCl$$

Anhydrous $FeCl_3$ is a good Lewis acid and form addition compounds with Lewis bases like NH_3

$$FeCl_3 + 6NH_3 \longrightarrow FeCl_3 \cdot 6NH_3$$

Ferric chloride acts as a good oxidizing agent.

(i) It oxidizes H_2S to S, $SnCl_2$ to $SnCl_4$, SO_2 to H_2SO_4 and liberates iodine from iodides

$$2FeCl_3 + H_2S \longrightarrow 2FeCl_2 + 2HCl + S$$
$$2FeCl_3 + SnCl_2 \longrightarrow 2FeCl_2 + SnCl_4$$
$$2FeCl_3 + SO_2 + 2H_2O \longrightarrow 2FeCl_2 + H_2SO_4 + 2HCl$$
$$2FeCl_3 + 2KI \longrightarrow 2FeCl_2 + 2KCl + I_2$$

With ammonium hydroxide and sodium hydroxide it forms a reddish-brown precipitate of ferric hydroxide insoluble in excess of NH_4OH or NaOH

$$FeCl_3 + 3NH_4OH \longrightarrow Fe(OH)_3 + 3NH_4Cl$$
$$FeCl_3 + 3NaOH \longrightarrow Fe(OH)_3 + 3NaCl$$

With potassium or ammonium thiocyanate solutions ferric chloride solution gives blood red colouration due to formation of complex salt.

$$FeCl_3 + NH_4CNS \longrightarrow Fe\ (SCN)\ Cl_2 + NH_4Cl$$
$$\text{Or } FeCl_3 + 3NH_4CNS \longrightarrow Fe\ (SCN)_3 + 3NH_4Cl$$

With potassium ferrocyanide it forms a blue complex known as Prussian blue.

$$4FeCl_3 + 3K_4[Fe(CN)_6] \longrightarrow \underset{\text{Prussian blue}}{Fe_4[Fe(CN)_6]_3} + 12KCl$$

Structure: $FeCl_3$ have a very interesting structural chemistry of its own. In the vapour at low temperatures the compound is in the form of Fe_2Cl_6 molecules. When these condense to form a crystal a radical rearrangement takes place and instead of a finite molecule, in which each iron atom is attached to four chlorine atoms, there is an infinite 2-dimensional layer in which every Fe is bonded to six Cl atoms. When this crystal dissolves in a non-polar solvent such as CS_2 the Fe_2Cl_6 molecules reform, but in polar solvents such as ether simple tetrahedral molecules $(C_2H_5)_2O \rightarrow FeCl_3$ are formed. The hexa hydrate consists of octahedral $[FeCl_2\ (H_2O)_4]^+$ and Cl ions and H_2O molecules which are not attached to metal atoms. The structural chemistry of $FeCl_3$ can be shown as follows.

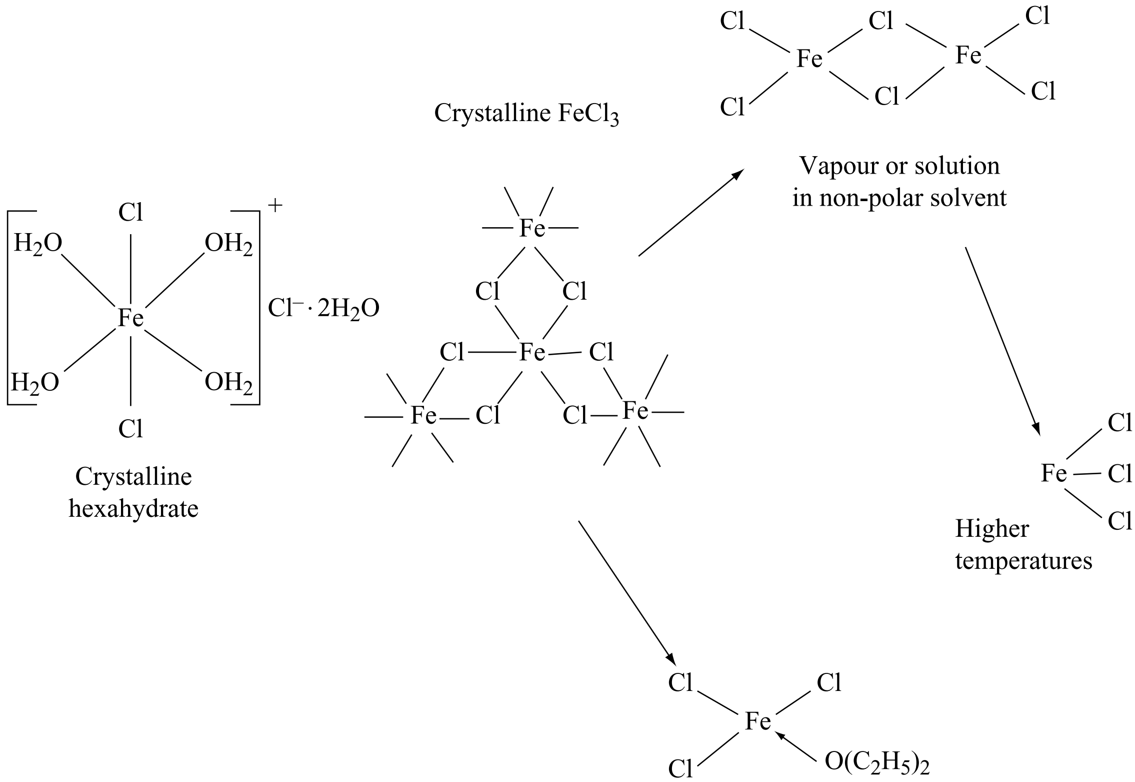


Fig 15.26 Structural Chemistry of $FeCl_3$

15.9.6 Iron (III) Sulphate $Fe_2(SO_4)_3$

Iron (III) sulphate can be obtained by heating an aqueous acidified solution of iron (II) sulphate with hydrogen peroxide (oxidizing agent) or by oxidation with nitric acid.

$$2FeSO_4 + H_2SO_4 + H_2O_2 \longrightarrow Fe_2(SO_4)_3 + 2H_2O$$

$$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \longrightarrow Fe_2(SO_4)_3 + 2NO + 4H_2O$$

It is also obtained by dissolving ferric hydroxide and sulphuric acid

$$2Fe(OH)_3 + 3H_2SO_4 \longrightarrow Fe_2(SO_4)_3 + 6H_2O$$

Properties: The solution of ferric sulphate is yellow with brown but on crystallization gives hydrate $Fe_2(SO_4)_3 \cdot 9H_2O$. The anhydrous sulphate is white.

The aqueous solution of ferric sulphate is readily hydrolyzed to brown basic salts of indefinite composition

$$Fe_2(SO_4)_3 + 2H_2O \longrightarrow Fe_2(SO_4)_2\,(OH)_2 + H_2SO_4$$

On heating it dissociate liberating SO_3

$$Fe_2(SO_4)_3 \longrightarrow Fe_2O_3 + 3SO_3$$

Iron (III) sulphate forms alums, the best known being ammonium iron (III) sulphate $(NH_4)_2SO_4 \cdot Fe_2(SO_4)_3 \cdot 24H_2O$

Iron (II) Compounds

15.9.7 Iron (II) Oxide FeO

Iron (II) oxide can be obtained as a black solid by heating iron (II) oxalate (the carbon monoxide provides a reducing atmosphere and prevents aerial oxidation to iron (III) oxide)

$$FeC_2O_4 \longrightarrow FeO + CO + CO_2$$

The oxide is made by the above method is pyrophoric becoming incandescent when sprinkled into the air. It has the sodium chloride structure, indicating that it is ionic, but the crystal lattice is some what deficient in iron (II) ions and it is non-stoichiometric. If heated in an inert atmosphere to a high temperature and allowed to cool slowly it tends to disproportionate into iron and magnetic oxide of iron.

$$4FeO \longrightarrow Fe + Fe_3O_4$$

15.9.8 Iron (II) Hydroxide $Fe(OH)_2$

Iron (II) hydroxide when pure is a white solid, but it is normally obtained as a green gelatinous precipitate by adding sodium hydroxide solution to an aqueous solution containing iron (II) ions.

$$Fe^{2+} + 2OH^- \longrightarrow Fe(OH)_2$$

The green colour is due to the presence of some iron (III) hydroxide produced either by aerial oxidation or by there being some iron (III) ions in the iron (II) solution. Prolonged standing in air results in complete conversion into iron (III) hydroxide, essentially hydrated iron (III) oxide.

Iron (II) hydroxide dissolves readily in dilute acids to produce the green hydrated ion $[Fe(H_2O)_6]^{2+}$; Since it also reacts with concentrated sodium hydroxide solution, it must be regarded as being some what amphoteric.

15.9.9 Iron (II) Halides

Anhydrous iron (II) chloride, $FeCl_2$ can be made by passing a stream of dry hydrogen chloride over the heated metal, the anhydrous fluoride is made in a similar way using a stream of dry hydrogen fluoride. Note that the action of either chlorine or fluorine on the heated metal produces the iron (III) halide.

Both these halides dissolve in water from which the hydrates $FeF_2 \cdot 8H_2O$ and $FeCl_2 \cdot 6H_2O$ can be crystallized; the hydrated salts can also be made by the action of dilute hydrofluoric acid or dilute hydrochloric acid on the metal.

Iron (II) fluoride and iron (II) chloride, whether anhydrous or hydrated are predominantly ionic solids.

Iron (II) bromide is obtained by heating iron in the presence of bromine vapour, an excess of metal is used to prevent the formation of any iron (III) bromide. Iron (II) iodide is made by a similar method from iron and iodine but since iron (III) iodide does not exist, it is not necessary in this case to use an excess of the metal.

$FeCl_2 \cdot 6H_2O$ has trans - $[FeCl_2(H_2O)_4]$ units. Vapours at about 1000°C contain Fe_2Cl_4 and $FeCl_2$, but above 1300°C contain normal $FeCl_2 \cdot FeCl_2$ is oxidized on heating in air. Hydrogen is evolved on heating in steam.

$$12FeCl_2 + 3O_2 \longrightarrow 2Fe_2O_3 + 8FeCl_3$$
$$3FeCl_2 + 4H_2O \longrightarrow Fe_3O_4 + 6HCl + H_2$$

15.9.10 Iron (II) Sulphate $FeSO_4 \cdot 7H_2O$ (Green Vitriol)

It can be prepared by dissolving scrap iron in dilute sulphuric acid

$$Fe + H_2SO_4 \longrightarrow FeSO_4 + H_2O$$

It can also be prepared by dissolving iron (II) carbonate and iron (II) sulphide in dilute H_2SO_4

$$FeCO_3 + H_2SO_4 \longrightarrow FeSO_4 + H_2O + CO_2$$
$$FeS + H_2SO_4 \longrightarrow FeSO_4 + H_2S$$

It is manufactured by the slow oxidation of iron pyrites in the presence of air and moisture.

$$2FeS_2 + 2H_2O + 7O_2 \longrightarrow 2FeSO_4 + 2H_2SO_4$$

The free sulphuric acid is removed by the addition of scrap iron. On crystallizations green heptahydrate $FeSO_4 \cdot 7H_2O$ is deposited.

Properties: Hydrated iron (II) sulphate is a green crystalline compound where as anhydrous iron (II) sulphate is white or colourless. When exposed to air it turns to brownish yellow due to formation of basic ferric sulphate

$$4FeSO_4 + 2H_2O + O_2 \longrightarrow 4\ Fe\ (OH)\ SO_4$$

On heating it loses water of crystallization at 300°C and on strong heating breaks up to form ferric oxide with the evolution of SO_2 and SO_3

$$FeSO_4 \cdot 7H_2O \longrightarrow FeSO_4 + 7\ H_2O$$
$$2FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$$

Its aqueous solution is slightly acidic due to hydrolysis.

$$FeSO_4 + 2H_2O \longrightarrow Fe(OH)_2 + H_2SO_4$$

Reduction property: It is a good reducing agent.

(i) Reduces acidified permanganate

$$2KMnO_4 + 8H_2SO_4 + 10\ FeSO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 8H_2O$$

(ii) Reduces orange red dichromate to green chromic sulphate

$$K_2Cr_2O_7 + 7H_2SO_4 + 6FeSO_4 \longrightarrow K_2SO_4 + Cr_2\ (SO_4)_3 + 3Fe_2(SO_4)_3 + 7H_2O$$

(iii) Reduces gold chloride to gold

$$AuCl_3 + 3FeSO_4 \longrightarrow Fe_2(SO_4)_3 + FeCl_3 + Au$$

(iv) Reduces mercuric chloride to mercurous chloride.

$$6HgCl_2 + 6FeSO_4 \longrightarrow 3Hg_2Cl_2 + 2Fe_2(SO_4)_3 + 2FeCl_3$$

A solution of iron (II) sulphate contains the pale green $[Fe(H_2O)_6]^{2+}$ ions; in the presence of nitric oxide one molecule of water is displaced from the complex and replaced by NO to give $[Fe(H_2O)_5\ NO]^{2+.}$ This complex ion is dark brown and the above reaction is the basis of the 'brown ring' test for ionic nitrates.

It reacts with potassium cyanide (excess) forming potassium ferrocyanide

$$FeSO_4 + 2KCN \longrightarrow Fe(CN)_2 + K_2SO_4$$
$$Fe(CN)_2 + 4KCN \longrightarrow K_4[Fe(CN)_6]$$

With potassium ferrocyanide it forms blue precipitate called. Turn bull's blue.

$$FeSO_4 + K_3[Fe(CN)_6] \longrightarrow KFe[Fe(CN)_6] + K_2SO_4$$
$$3FeSO_4 + 2K_3[Fe(CN)_6] \longrightarrow Fe_3[Fe(CN)_6]_2 + 3K_2SO_4$$

Turn bull's blue has been shown to be identical with Prussian blue and it is likely that iron (II) ions first reduce the hexacyano ferrate (III) to hexacyano ferrate (II)

$$Fe^{2+} + [Fe^{III}(CN)_6]^{3-} \longrightarrow Fe^{3+} + [Fe(CN)_6]^{4-}$$

Followed by

$$K^+ + [Fe^{II}(CN)_6]^{4-} + Fe^{3+} \longrightarrow \underset{\substack{\text{Turn bull's blue} \\ \text{(or)} \\ \text{Prussian blue.}}}{K\,Fe^{III}[Fe^{II}(CN)_6]}$$

Iron (II) sulphate is one of the cheapest industrial chemical and is used to make iron (III) oxide (used as pigment), Prussian blue (pigment) and inks.

$FeSO_4 \cdot 7H_2O$ is isomorphous with Epsom salt $MgSO_4 \cdot 7H_2O$, and white vitriol $ZnSO_4 \cdot 7H_2O$ in which six water molecules are coordinated to metal ion and one water molecule is in hydrogen bond with sulphate ion.

15.9.11 Iron (II) Ammonium Sulphate $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ Mohr's Salt

This double salt is obtained by crystallizing a solution containing equivalent amounts of iron (II) sulphate and ammonium sulphate. Unlike hydrated iron (II) sulphate it neither effloresces nor oxidize on exposure to the atmosphere. It is used as primary volumetric standard, especially for standardizing potassium permanganate solutions.

15.10 COPPER

Copper is known to man since ancient times Its symbol 'Cu' comes from roman name cuprum.

Occurrence: Copper metal occurs in the elemental form to little extent. It occurs mostly in the combined state as its oxy compounds or sulphur compounds. The important minerals are.

Cuprite or ruby copper Cu_2O
Copper glance or chalocite Cu_2S
Copper pyrites or chalcopyrites $CuFeS_2$
Malachite (Green) $CuCO_3 \cdot Cu(OH)_2$
Azurite (blue) $2CuCO_3 \cdot Cu(OH)_2$

Extraction of copper

In the Ellingham diagram (Fig 15.10) one can see that $\Delta G°$ Vs T graph for Cu_2O is almost at the top. Therefore it is quite easy to reduce Cu_2O directly to metal by heating with coke. The lines of C-CO and C-CO_2 are much lower than the line of Cu, Cu_2O particularly at temperatures' above 500k. however, most of the ores of copper are sulphide ores these sulphide ores are roasted/smelted to get oxides which can then be easily reduced to metallic copper using coke.

$$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$$
$$Cu_2O + C \longrightarrow 2Cu + CO$$

Copper metal is extracted from copper pyrites ($CuFeS_2$) in the following steps

1. **Concentration:** The ore is concentrated by froth floatation process.
2. **Roasting:** The concentrated ore is heated in the presence of excess of air in a reverberatory furnace. Volatile impurities sulphur, phosphorous, arsenic, antimony etc are expelled as their volatile oxides.

$$S + O_2 \longrightarrow SO_2$$
$$P_4 + 5O_2 \longrightarrow P_4O_{10}$$
$$As_4 + 5O_2 \longrightarrow As_4O_{10}$$

A mixture of sulphides of copper and iron is obtained. The sulphides are partially oxidized to the respective oxides.

$$2CuFeS_2 + O_2 \longrightarrow Cu_2S + 2FeS + SO_2$$
$$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$$
$$2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$$

3. **Smelting:** The roasted ore is mixed with little coke and sand (silica) and smelted in a blast furnace. A blast of air is necessary for the combustion of coke is blown through the tuyeres present at the bottom of the furnace. The oxidation of sulphides of copper and iron will be completed further. A slag of iron silicate is formed according to the following reactions.

$$2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$$
$$FeO + SiO_2 \longrightarrow FeSiO_3$$
$$Cu_2O + FeS \longrightarrow Cu_2S + FeO$$

Copper has greater affinity to sulphur than to oxygen and iron has more affinity to oxygen than sulphur. So Cu_2S and FeO are formed.

The molten mass obtained from the blast furnace mostly contains Cu_2S and a little FeS. This product is called matte.

4. **Bessemerization:** The matte is transferred to a Bessemer converter lined inside with silica. It is fitted with pipes known as tuyeres through which air blast is admitted near the bottom. The following reactions take place in the Bessemer converter

$$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$$
$$2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$$

The iron oxide forms the slag with silica

$$FeO + SiO_2 \longrightarrow FeSiO_3$$

The cuprous oxide reacts with the remaining cuprous sulphide to form copper (auto reduction)

$$2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$$

After the completion of reaction the molten copper is poured off the Bessemer converter. As it cools it gives up the sulphur dioxide in it which comes out in the form of bubbles which appears like blisters on the surface of copper, so it is called **blister copper** which is 98 per cent pure.

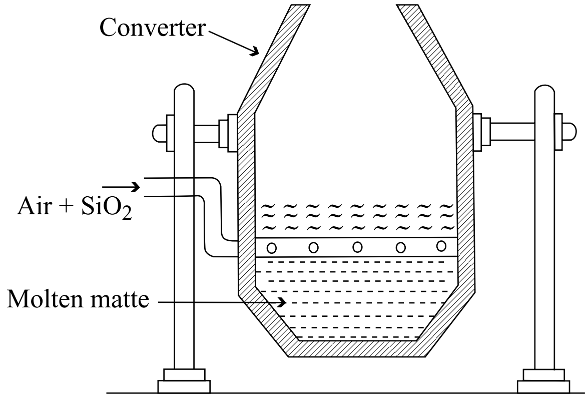


Fig 15.27 Bessemer converter for copper

5. Purification

(i) Poling: The blister copper is now purified by melting it in a reverboratory furnace where it is exposed to an oxidizing atmosphere. The impurities are either converted into slag or expelled as volatile oxides. During this process the molten mass is stirred with long poles of green wood. This process is known as poling. The reducing gases evolved from the wood prevent the oxidation of copper.

(ii) Electrolytic refining: The impure copper metal is made into blocks. They are suspended into lead-lined tanks containing copper (II) sulphate solution. Thin plates of pure copper serves as cathode. The cathode plates are coated with graphite on electrolysis, pure copper is deposited at the cathode.

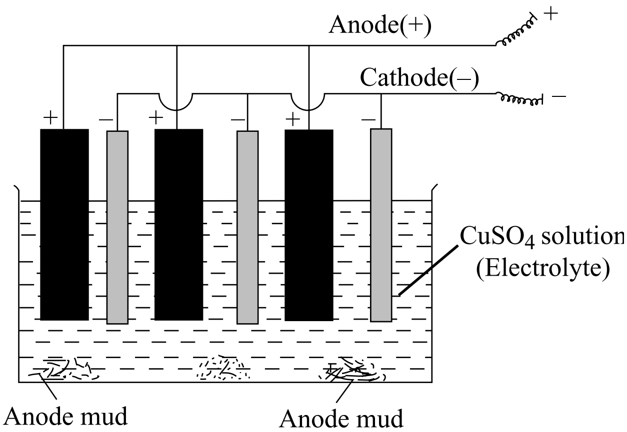


Fig 15.28 Electrolytic cell on an industrial scale

The copper obtained is almost 100 per cent pure. The impurities fall under anode as anode mud. The anode mud contains valuable metals like gold and silver.

Hydrometallurgical Process

Big heaps of copper sulphide ores are exposed to air & rain. If necessary water is sprinkled. After long periods the copper sulphide is slowly oxidized to copper sulphate. The liquor flowing from the bottom of the heaps is run into pans. To this scrap iron is added which displaces copper.

$$2Cu_2S + 5O_2 \longrightarrow 2CuSO_4 + 2CuO$$
$$CuSO_4 + Fe \longrightarrow Cu + FeSO_4$$

The carbonate ores (malachite and azurite) are calcined in a reverboratory furnace. The carbonate decomposes to form the oxide and the impurities either volatile or are oxidized.

$$CuCO_3 \cdot Cu(OH)_2 \xrightarrow{\Delta} 2\,CuO + CO_2 + H_2O$$

Copper is obtained from this oxide either by reduction with carbon or by displacement with scrap iron after leaching with dilute sulphuric acid.

$$CuO + C \longrightarrow Cu + CO$$
$$Cu_2O + C \longrightarrow 2Cu + CO$$
$$CuO + H_2SO_4 \longrightarrow CuSO_4 + H_2O$$
$$CuSO_4 + Fe \longrightarrow Cu + FeSO_4$$

Properties: Copper has a melting point of 1083°C and a density of 8.92 g cm^{-3}, it is an attractive golden coloured metal, being very malleable and ductile, and its electrical and thermal conductivities are second only to those of silver. The metal is slowly attacked by moist air and its surface gradually becomes covered with an attractive green layer of basic copper carbonate.

$$2Cu + H_2O + CO_2 + O_2 \longrightarrow CuCO_3 \cdot Cu(OH)_2$$

At about 300°C, it is attacked by air or oxygen and a black coating of copper (II) oxide forms on its surface. At a temperature of about 1000°C copper (I) oxide is formed instead. Copper is attacked by sulphur vapour with the formation of copper (I) sulphide

$$2Cu + S \longrightarrow Cu_2S$$

Halogens react with copper forming copper (II) halide (except iodine which forms copper (I) iodide)

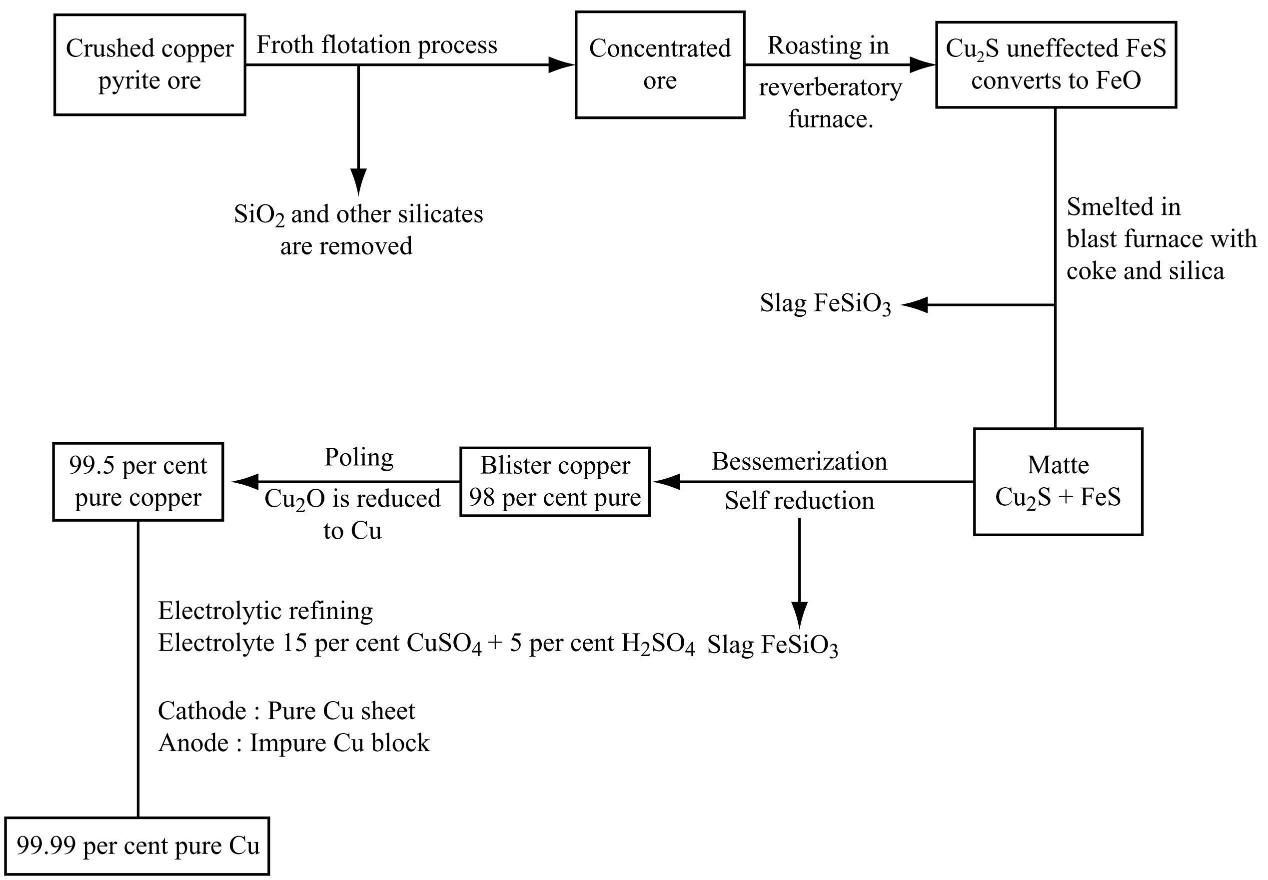
$$Cu + X_2 \longrightarrow CuX_2 \text{ (X = F, Cl or Br)}$$

The metal is not attacked by water or steam and dilute non-oxidizing acids such as hydrochloric acid and dilute sulphuric acid. However these acids react with copper in the presence of air.

$$2Cu + 2H_2SO_4 + O_2 \longrightarrow 2CuSO_4 + 2H_2O$$

Boiling concentrated hydrochloric acid attacks the metal with the evolution of hydrogen, a surprising result in view of the positive electrode potential of copper. This

Flow Chart for Extraction of Copper from Copper Pyrite



phenomenon is due to the formation of the $[CuCl_2]^-$ ion which drives the equilibrium to the right.

$$2Cu + 2H^+ \rightleftharpoons 2Cu^+ + H_2$$

$$+$$

$$4Cl^- \text{ (from HCl)}$$

$$\uparrow\downarrow$$

$$2[CuCl_2]^-$$

Hot concentrated sulphuric acid attacks the metal and so too does dilute and concentrated nitric acid; the equations below do not more than indicated the main reactions, since there are significant side reactions.

$$Cu + \underset{\text{Conc}}{2H_2SO_4} \longrightarrow CuSO_4 + 2H_2O + SO_2$$

$$3Cu + \underset{\text{dil}}{8HNO_3} \longrightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$$

$$Cu + \underset{\text{Conc}}{4HNO_3} \longrightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2$$

Copper plates are slowly attacked by acetic acid (vinegar) in presence of air to form basic copper acetate called **verdigris**

$$2Cu + 2CH_3COOH + O_2 \longrightarrow \underset{\text{Basic copper acetate (Vedrigris)}}{Cu(CH_3COO_2)\cdot Cu(OH)_2}$$

Copper dissolves in aqueous ammonia in the presence of air or oxygen with the formation of complex

$$2Cu + 8NH_3 + 2H_2O + O_2 \rightarrow 2[Cu\,(NH_3)_4]\,(OH)_2$$

This reaction is utilized in the removal of ammonia present in nitrogen (prepared from ammonia) by passing over moist copper chips.

Copper can displace silver, gold and platinum from their salt solutions.

$$2AgNO_3 + Cu \rightarrow Cu\,(NO_3)_2 + 2Ag$$

$$2AuCl_3 + 3Cu \rightarrow 3CuCl_2 + 2Au$$

Uses: copper is useful for the windings of dynamos and for conveying electrical power; its resistance to chemical attack and its high thermal conductivity make it a useful metal for the construction of condensers for chemical plants and car radiators. Brass is an alloy of copper and zinc, is used for making cartridge containers, head lamp reflectors and the working parts of watches and clocks. Bronze an alloy of copper and tin is used for fabricating bearings and ships fittings. Phosphor bronze, which contains some phosphorous is used for watch springs and galvanometer suspensions. Finely divided copper is used as an industrial catalyst, in the oxidation of methanol to formaldehyde.

15.11 Compounds of Copper

Copper exhibits oxidation states of +2 (the most common) and +1 (only stable in aqueous solution in the form of stable complex ion). A few compounds containing copper (III) are known e.g., K_3CuF_6, but this oxidation state of the metal is unimportant.

15.11.1 Copper (I) Compounds

In aqueous solutions hydrated copper (I) ion is unstable and disproportionation into the copper (II) ion and copper i.e., undergoes self-oxidation and self-reduction. This is indicated by the standard redox potentials for the systems Cu^+/Cu and Cu^{2+}/Cu which are given below

$$Cu^+ + e \rightarrow Cu \qquad E^0 = +0.52V$$
$$Cu^{2+} + e \rightarrow Cu^+ \qquad E^0 = +0.16V$$
Or $$2Cu^+ \rightarrow Cu + Cu^{2+} \qquad E^0_{total} = +0.36V$$

The positive e.m.f of the above cell reaction implies that hydrated copper (I) ions are thermodynamically unstable in solutions with respect to copper and hydrated copper (II) ions. The value of the equilibrium constant for this disproportionation reaction has been estimated to be in the order of 10^6 litre mol^{-1} at 25°C

$$2Cu^+ \rightleftharpoons Cu + Cu^{2+}$$
$$K = \frac{[Cu^{2+}]}{[Cu^+]^2} = 10^6 \text{ dm}^3 \text{ mol}^{-1}$$

i.e., the concentration of hydrated Cu^+ ions in solution is extremely minute. The equilibrium can be shifted to the left by adding anions which precipitate out an insoluble copper (I) compound e.g., I^- ions precipitate insoluble CuI, or by adding a substance which forms a more stable complex ion with Cu^+ than with Cu^{2+} e.g., ammonia.

Although the chemistry of copper (I) is largely that of its water insoluble compounds and of its stable complexes, other copper (I) compounds are perfectly stable in the absence of moisture e.g., copper (I) sulphate Cu_2SO_4.

15.11.2 Copper (I) Oxide Cu_2O

Copper (I) oxide is obtained as red solid by the reduction of an alkaline solution of copper (II) sulphate. Since the addition of alkali to a solution of a copper (II) salt would result in the precipitation of copper (II) hydroxide, the copper (II) ions are complexed with tartrate ions; under these conditions the Cu^{2+} ions are present in such low concentration that the solubility product of copper (II) hydroxide is not exceeded.

The experimental procedure is as follows: a solution of copper (II) sulphate is added to an alkaline solution of sodium potassium tartrate when a deep blue solution is obtained, the colour being due to the presence of copper (II)-tartrate complex. The solution is warmed with a solution of glucose (reducing agent) when red deposit of copper (I) oxide is obtained.

Copper (I) oxide reacts with dilute sulphuric acid on warming to give a solution of copper (II) sulphate and a deposit of copper, i.e., disproportionation occurs.

$$Cu_2O + 2H^+ \rightarrow Cu^{2+} + Cu + H_2O$$

It dissolves in concentrated hydrochloric acid with the formation of the $[CuCl_2]^-$ complex ion.

Copper (I) oxide is a covalent solid, each oxygen atom being surrounded by four copper atoms and each copper atom lying midway between two oxygen atoms (4:2 coordination's).

15.11.3 Copper (I) Chloride CuCl

This is a white solid which is insoluble in water. It can be obtained by boiling a solution of copper (II) chloride, excess of copper turnings and concentrated hydrochloric acid. Copper (I) is present in this solution as the $[CuCl_2]^-$ complex ion.

$$\begin{array}{l} Cu + Cu^{2+} \rightleftharpoons 2Cu^+ \\ \qquad\qquad\quad + \\ \qquad\qquad\quad 4Cl^- \\ \qquad\qquad\quad \uparrow\downarrow \\ \qquad\qquad\quad 2[CuCl_2]^- \end{array}$$

On pouring the solution into air free distilled water copper (I) chloride precipitates as a white solid. It must be rapidly washed, dried and sealed in the absence of air, since a combination of air and moisture oxidizes it to copper (II) chloride.

Copper (I) chloride is essentially covalent in structure is similar to that of diamond; i.e., each copper atom is surrounded tetrahedrally by four chlorine atoms and each chlorine atom is surrounded tetrahedrally by four copper atoms, in the vapour phase dimeric and trimetric species are present, i.e., $(CuCl)_2$ and $(CuCl)_3$. It is soluble in water in the presence of entities such as Cl^-, $S_2O_3^{2-}$ and NH_3 with which it forms complex ions eg $[Cu(NH_3)_2]^+$.

Copper (I) chloride is used in conjunction with ammonium chloride as a catalyst in the dimerization of ethyne to but-1-ene-3-yne (vinyl acetylene), which is used in the production of synthetic rubber. In the laboratory a mixture of copper (I) chloride and hydrochloric acid is used for converting benzene diazonium chloride into chloro benzene (Sandmeyer reaction)

$$C_6H_5N_2^+Cl^- \xrightarrow{CuCl\,/\,HCl} C_6H_5Cl + N_2$$

An ammonical solution of copper (I) chloride solution absorbs carbon monoxide and precipitates copper (I) carbide when ethane is bubbled through it. The latter reaction can evolves ethane when treated with dilute acid.

15.11.4 Copper (II) Compounds

This is the most common oxidation state of copper and in aqueous solution copper (II) salts are blue, the colour being due to the presence of $[Cu(H_2O)_6]^{2+}$ ions.

15.11.5 Copper (II) Oxide

Copper (II) oxide may be obtained as black solid by heating either copper (II) carbonate (actually a basic salt) or copper nitrate.

$$CuCO_3 \rightarrow CuO + CO_2$$
$$2Cu(NO_3)_2 \rightarrow 2CuO + 4NO_2 + O_2$$

On heating to a temperature of about 800°C it decomposes into copper (I) oxide and oxygen. It readily reacts with dilute mineral acids on warming, with the formation of copper (II) salts. and also easily reduced to copper on heating in a stream of hydrogen.

$$4CuO \rightarrow 2Cu_2O + O_2$$
$$CuO + 2H^+ \rightarrow Cu^{2+} + H_2O$$
$$CuO + H_2 \rightarrow Cu + H_2O$$

The detection of carbon and hydrogen in organic compounds can be achieved by heating with dry copper (II) oxide, when carbon dioxide and water are formed.

15.11.6 Copper (II) Hydroxide

Copper (II) hydroxide is precipitated as a blue green gelatinous solid when an aqueous solution of a copper (II) salt is made alkaline with sodium hydroxide solution. It can be filtered and dried at 100° C to the composition $Cu(OH)_2$; however, heating the unfiltered suspension to about 80°C results in decomposition into copper (II) oxide and water.

Copper (II) hydroxide is readily soluble in an aqueous solution of ammonia with the formation of the intensely blue $[Cu(NH_3)_4]^{2+}$ Ion.

$$\begin{array}{c} 2Cu(OH)_2 \rightleftharpoons Cu^{2+} + 2OH^- \\ + \\ 4NH_3 \\ \uparrow\downarrow \\ [Cu(NH_3)_4]^{2+} \end{array}$$

It reacts readily with dilute acids to give copper (II) salts but, since the freshly precipitated solid is also slightly soluble in sodium hydroxide solution, it must be considered to be somewhat amphoteric.

$$Cu(OH)_2 + 2H^+ \longrightarrow Cu^{2+} + 2H_2O$$
$$Cu(OH)_2 + 2OH^- \longrightarrow [Cu(OH)_4]^{2-}$$

15.11.7 Copper (II) Chloride

The anhydrous copper (II) chloride is a dark brown coloured solid. This can be prepared by passing chlorine over heated copper. It is predominantly covalent and a layer structure in which each copper atom is surrounded by four chlorine atoms at a distance of 230pm and to more at a distance of 295 pm.

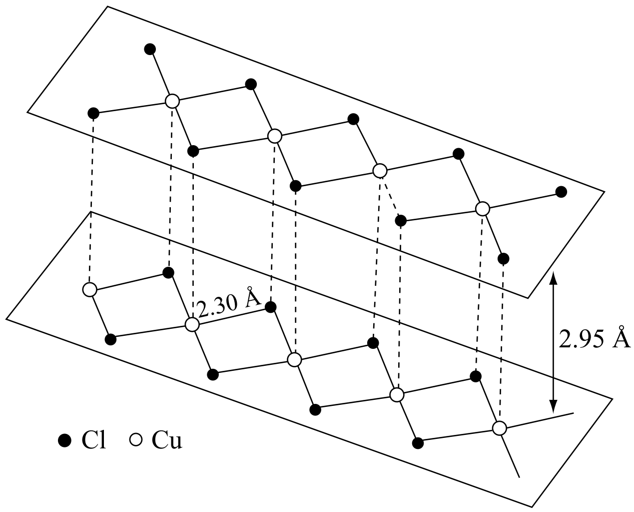


Fig 15.29 The layer structure of an hydrous Copper (II) chloride (two layers only shown)

Copper (II) chloride is very soluble in water; a concentrated aqueous solution is dark brown, the colour being due to the presence of complex ions such as $[CuCl_4]^{2-}$ but with dilution the colour changes to green and then to blue. These colour changes are due to the successive replacement of chloride ions in the complexes by water molecules, the final colour being that of the $[Cu(H_2O)_6]^{2+}$ ion. The dihydrate $CuCl_2 \cdot 2H_2O$ which is a green solid, can be obtained by crystallizing the solution.

15.11.8 Copper (II) Sulphate (Blue Vitriol) $CuSO_4 \cdot 5H_2O$

In the laboratory copper (II) sulphate may be prepared by reacting copper (II) oxide or copper (II) hydroxide or copper (II) carbonate with dilute sulphuric acid. The solution is heated to obtain a saturated solution and the blue solid penta hydrate separates on cooling (a few drops of dilute sulphuric acid are generally added before heating in order to prevent hydrolysis)

$$CuO + H_2SO_4 \longrightarrow CuSO_4 + H_2O$$
$$Cu(OH)_2 + H_2SO_4 \longrightarrow CuSO_4 + 2H_2O$$
$$CuCO_3 + H_2SO_4 \longrightarrow CuSO_4 + H_2O + CO_2$$

On an industrial scale copper (II) sulphate is obtained by forcing air through a hot mixture of copper and dilute sulphuric acid.

$$2Cu + 2H_2SO_4 + O_2 \longrightarrow 2CuSO_4 + 2H_2O$$

Properties: It is a blue crystalline compound and is fairly soluble in water. It is an efflorescent substance turns to pale blue powder $CuSO_4 \cdot 3H_2O$. When heated the copper (II) sulphate penta hydrate loses four of its water molecules of crystallization at about 100°C; the fifth molecule of water is lost at a temperature of about 250°C.

$$CuSO_4 \cdot 5H_2O \xrightarrow{100°C} CuSO_4 \cdot H_2O + 4H_2O \xrightarrow{250°C} CuSO_4 + 5H_2O$$

Anhydrous copper sulphate (white) regains its blue colour when moistened with a drop of water (a test for moisture or water), the anhydrous salt decomposes into copper (II) oxide and sulphur trioxide on really strong heating.

$$CuSO_4 \xrightarrow{720°C} CuO + SO_3$$
$$\downarrow$$
$$SO_2 + \frac{1}{2}O_2$$

Action of NH_4OH: With ammonia solution first it forms a blue precipitate of copper (II) hydroxide which dissolves in excess of ammonia solution due to the formation of complex

$$CuSO_4 + 2NH_4OH \longrightarrow Cu(OH)_2 + (NH_4)_2SO_4$$
$$Cu(OH)_2 + (NH_4)_2SO_4 + 2NH_4OH \longrightarrow [Cu(NH_3)_4]SO_4 + 4H_2O$$

The blue complex is known as **Schweitzer's reagent** which is used for dissolving cellulose acetate in the manufacture of artificial silk, rayon.

Reaction with alkalis: With alkalis it forms blue precipitate of copper (II) hydroxide

$$CuSO_4 + 2NaOH \longrightarrow Cu(OH)_2 + Na_2SO_4$$

Reaction with potassium iodide: When potassium iodide is added to copper (II) sulphate first cupric iodide is formed which decomposes to give white cuprous iodide and iodine

$$[CuSO_4 + 2KI \rightarrow CuI_2 + K_2SO_4] \times 2$$
$$2CuI_2 \rightarrow Cu_2I_2 + I_2$$
$$\overline{2CuSO_4 + 4KI \rightarrow Cu_2I_2 + 2K_2SO_4 + I_2}$$

Reaction with potassium cyanide: Potassium cyanide reacts with copper (II) sulphate similar to potassium iodide. First cupric cyanide is formed which decomposes to give cuprous cyanide and cyanogen gas. Cuprous cyanide dissolves in excess of potassium cyanide due to the formation of complex potassium cuprous cyanide.

$$[CuSO_4 + 2KCN \rightarrow Cu(CN)_2 + K_2SO_4] \times 2$$
$$2Cu(CN)_2 \rightarrow Cu_2(CN)_2 + (CN)_2$$
$$Cu_2(CN)_2 + 6KCN \rightarrow 2K_3[Cu(CN)_4]$$
$$\overline{2CuSO_4 + 10KCN \rightarrow 2K_3[Cu(CN)_4] + 2K_2SO_4 + (CN)_2}$$

Reaction with potassium ferrocyanide: Copper (II) salts give chocolate brown precipitate with potassium ferrocyanide. It is a test for the detection of Cu^{2+} ions

$$2CuSO_4 + K_4[Fe(CN)_6] \longrightarrow Cu_2[Fe(CN)_6] + 2K_2SO_4$$

Reaction with H_2S: When H_2S is passed through copper sulphate solution, a black precipitate of copper sulphide is formed, soluble in hot dilute nitric acid.

$$CuSO_4 + H_2S \longrightarrow CuS + H_2SO_4$$
$$3CuS + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO + 3S$$

Reaction with potassium thiocyanate: Cupricthiocyanate is formed

$$CuSO_4 + 2KCNS \longrightarrow Cu(SCN)_2 + K_2SO_4$$

If SO_2 is passed through the solution a white precipitate of cuprous thiocyanate is formed.

$$2CuSO_4 + 2KSCN + SO_2 + 2H_2O \longrightarrow Cu_2(SCN)_2 + K_2SO_4 + 2H_2SO_4$$

More electropositive metals like iron, zinc etc can substitute copper from copper (II) sulphate

$$CuSO_4 + Fe \longrightarrow Cu + FeSO_4$$
$$CuSO_4 + Zn \longrightarrow Cu + ZnSO_4$$

Copper (II) sulphate forms double salts with alkali metal sulphates eg $K_2SO_4 \cdot CuSO_4 \cdot 6H_2O$ and with ammonium sulphate $(NH_4)_2SO_4 \cdot CuSO_4 \cdot 6H_2O$.

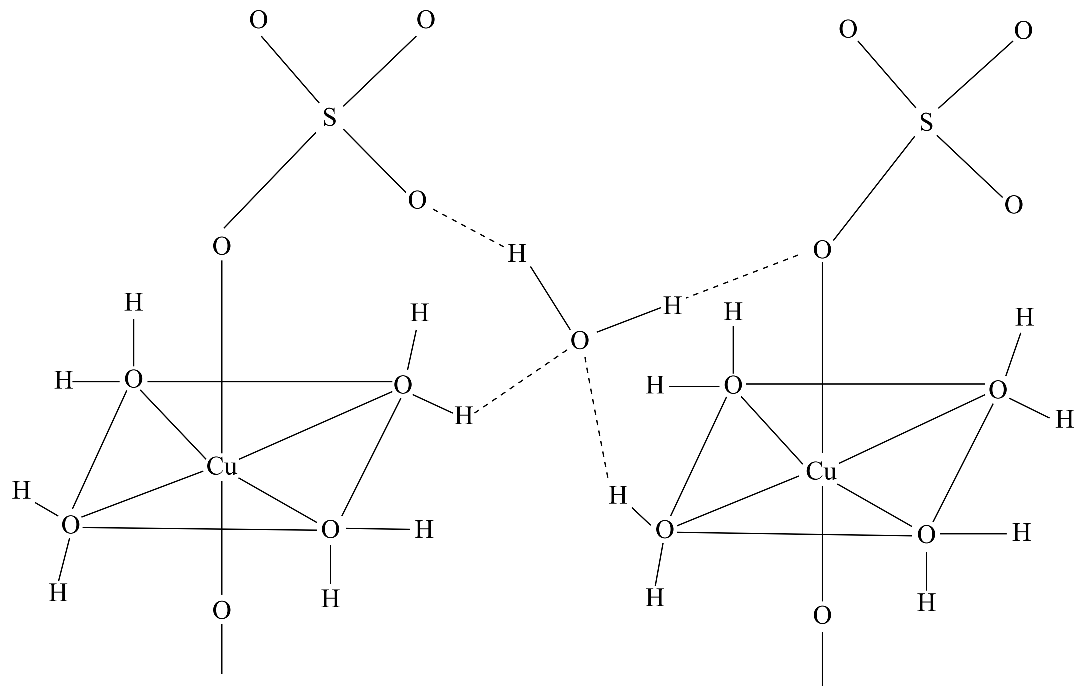


Fig 15.30 The partial structure of Copper (II) sulphate pentahydrate

Reaction with sodium thio sulphate: Copper (II) sulphate reacts with hypo first to give cupric thiosulphate which is reduced to white precipitate of cuprous thiosulphate. This dissolves in excess of sodium thiosulphate to form a complex

$$CuSO_4 + Na_2S_2O_3 \longrightarrow CuS_2O_3 + Na_2SO_4$$
$$2CuS_2O_3 + Na_2S_2O_3 \longrightarrow Cu_2S_2O_3 + Na_2S_4O_6$$
$$3Cu_2S_2O_3 + 2Na_2S_2O_3 \longrightarrow Na_4\,[Cu_6(S_2O_3)_5]$$

In the solid pentahydrate each copper (II) ion is surrounded by four water molecules arranged at the corners of a square the fifth and sixth octahedral positions are occupied by oxygen atoms from sulphate anions and the fifth water molecule is held in position by hydrogen bonding.

Uses: It is used as a fungicide and germicide. Bordeaux mixture consisting copper sulphate and lime is sprayed on plants and crops to kill moulds and fungus. It is also used as a laboratory reagent particularly in the preparation of Fehling's solution. It is also used in electroplating, electrotyping. Its tetraammine complex Schweitzer's reagent is used in the manufacture of artificial silk or rayon.

15.12 SILVER

Occurrence

Silver occurs in nature in the combined as well as in native state. The important ores of silver are

(i) Argentite or silver glance : Ag_2S
(ii) Horn silver or chlorargirite : AgCl
(iii) Pyrargyrite or ruby silver : $Ag_2S \cdot Sb_2S_3$
(iv) Prousite : $Ag_3As\ S_3$

Extraction of silver

Silver metal is extracted from the argentite by cyanide process. This process is also known as Mac Arthur Forrest process. The various steps involved in the process are as follows

(i) Concentration of the ore: The ore is crushed to a fine powder in ball mills. The finely powered ore is concentrated by froth floatation process. The ore is suspended in water. Suitable oil like pine oil and a little sodium carbonate is added to it. The suspension is agitated by current of air. The ore particles are wetted more by oil and so go with the froth. The concentrated ore is separated from the froth.

(ii) Treatment with sodium cyanide: The concentrated ore is treated with dilute solution (about 0.5 per cent) of sodium cyanide for several hours. The solution is continuously agitated by passing a current of air. The silver gradually passes into solution as a complex sodium argentocyanide.

$$Ag_2S + 4NaCN \rightleftharpoons 2\,\underset{\text{Sodiumdicyanoargentate(I)}}{Na[Ag(CN)_2]} + Na_2S$$

Sodium sulphide formed is oxidized to sodium sulphate by the air blown into the solution. This forces the above reaction to go to completion.

$$2Na_2S + H_2O + O_2 \longrightarrow Na_2S_2O_3 + 2NaOH$$
$$Na_2S_2O_3 + 2NaOH + 2O_2 \longrightarrow 2Na_2SO_4 + H_2O$$

The other ores like chlorargirite or the ores containing native silver can also be treated with sodium cyanide solution to get the silver into solution.

$$AgCl + 2NaCN \longrightarrow NaAg(CN)_2] + NaCl$$
$$4Ag + 8NaCN + 2H_2O + O_2 \longrightarrow 4Na[Ag(CN)_2] + 4NaOH.$$

(iii) Precipitation of silver: The above solution filtered to remove insoluble impurities. It is then treated with zinc dust. Silver being less electropositive gets displaced by more electropositive zinc and is precipitated.

$$2Na[Ag(CN)_2] + Zn \longrightarrow Na_2\,[Zn(CN)_4] + 2Ag$$

(iv) Fusion: The silver that is precipitated is filtered off. Then it is fused with a flux (borax or potassium nitrate) to oxidize any zinc present as impurity.

(v) Purification: The silver obtained in fusion is purified by electrolytic process to get very pure metal. Impure silver containing Zn, Cu, Au etc as impurities is made as the anode in the electrolytic bath containing $AgNO_3$ with a little HNO_3. A thin sheet of pure silver is made cathode. On electrolysis pure silver go into solution, while gold if at all present, falls down as anode mud. The reactions talking place in the cell may be represented as

$$Ag^+ + e^- \longrightarrow Ag \text{ at cathode}$$
$$Ag \longrightarrow Ag^+ + e^- \text{ at anode.}$$

Extraction from argentiferrous lead

Galena, a lead ore contains small amount of silver sulphide. The lead extracted from such galena contain very small amounts of silver. This lead is called argentiferrous lead. The recovery of silver from argentiferrous lead is made (i) by Pattinson's process (or) (ii) by Parke's process

(i) Pattisons process: This method is based on the fact that lead-silver system forms an eutectic mixture with 2.6 per cent silver melting at 303°C whereas pure lead melts at 327°C.

Molten argentiferrous lead is allowed to cool slowly to solidify the crystals of pure lead until the silver content of the mixture has risen to 2.6 per cent The crystals of pure lead are removed and the alloy rich in silver content is then subjected to cupellation to remove the remaining lead.

(ii) Parke's process: This method is based on the following points

(a) Molten lead and molten zinc are immiscible.
(b) Silver is more soluble in molten zinc than in molten lead.
(c) Molten lead forms the bottom layer while molten zinc forms the upper layer.
(d) Zinc-silver alloy will solidify first when temperature decreases, being it has more melting points.
(e) Zinc being volatile, can be separated from silver by distillation.

Argentiferrous lead is mixed with zinc dust and heated to melting point of zinc. Zinc dissolves silver and forms the upper layer. When temperature is decreased the zinc-silver alloy crystallized while the lead still remaining as liquid. The solid zinc-silver alloy is separated by using perforated laddles. This may contain a little lead also. This is distilled using iron retorts. Zinc distills over leaving behind silver containing little lead. This is further refined first by cupellation method and then by electrolytic method.

Properties: Silver has a melting point 960°C and a density of 10.5g cm^{-3}. It is a white lustrous metal and is very malleable and ductile; its thermal and electrical conductivities exceed those of copper and, indeed, it is the best conductor known. The metal is resistant to attack by air and moisture, although the presence of hydrogen sulphide results in the familiar black stain of silver sulphide. Steam and dilute non-oxidizing acids are without effect on the metal; however, it is attacked by hot concentrated sulphuric acid and cold dilute nitric acid, with the formation of silver (I) Ag^+ ions

$$2Ag + 2H_2SO_4 \longrightarrow Ag_2SO_4 + 2H_2O + SO_2$$
$$3Ag + 4HNO_3 \longrightarrow 3AgNO_3 + 2H_2O + 2NO$$
$$Ag + 2HNO_3 \longrightarrow AgNO_3 + H_2O + NO_2$$

Silver is not attacked by alkalis but dissolves in alkali metal cyanides due to formation of complex.

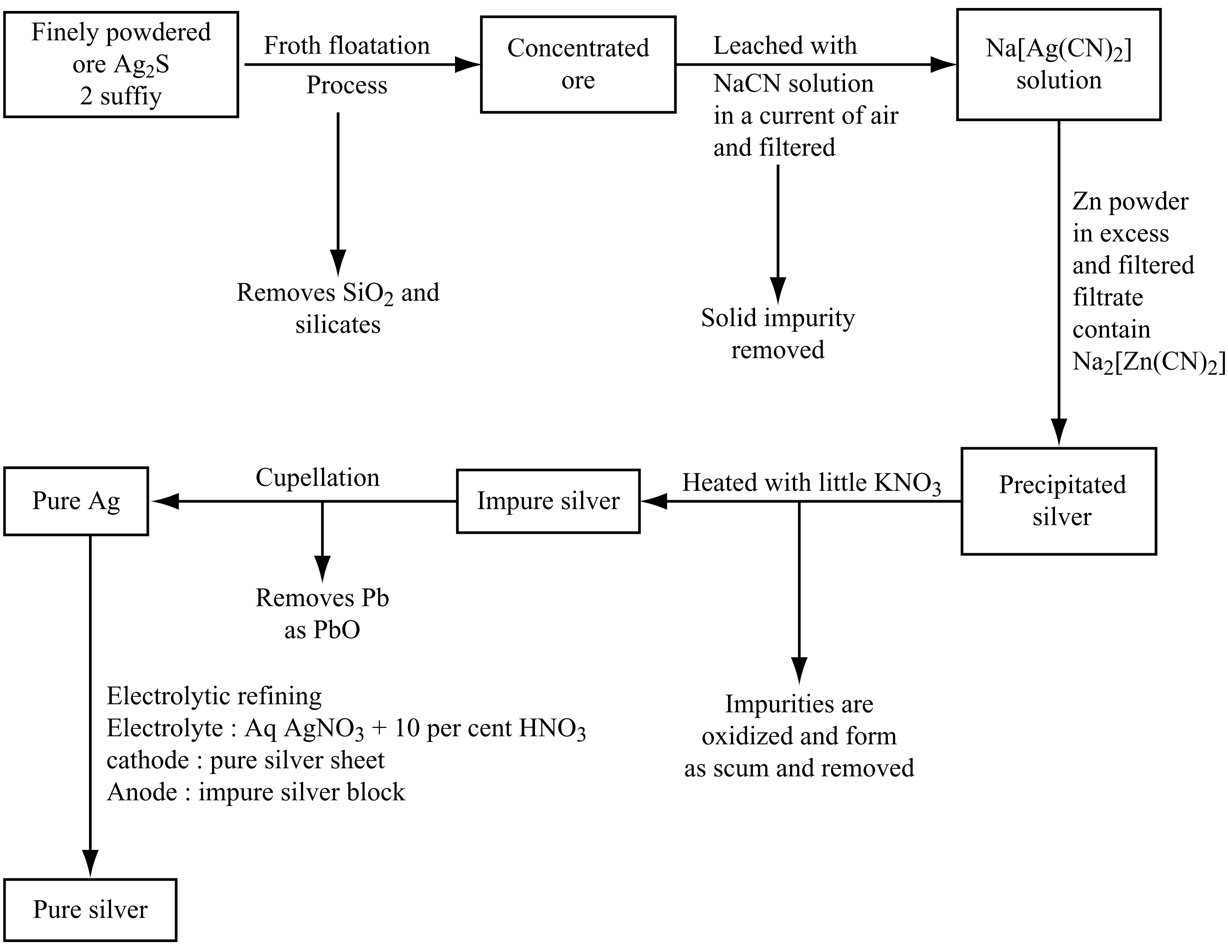
Silver is used in making ornaments, silver ware and silver plating the articles of base metals. It is used in the preparation of silver salts used in photography, silver mirroring, medicines etc.

15.13 COMPOUNDS OF SILVER

15.13.1 Silver Oxide

The addition of sodium hydroxide solution to a solution of a silver salt results in the precipitation of brown silver

Flow Chart for the Extraction of Silver by Mac-Arthur - Forrest Process



oxide. It is possible that the hydroxide is formed initially and then decomposes into the oxide and water

$$2Ag^+ + 2OH^- \longrightarrow 2AgOH$$

$$2AgOH \longrightarrow Ag_2O + H_2O$$

Silver oxide turns moist red litmus blue. i.e., in the presence of water it produces OH^- ions. The moist silver oxide is used in the organic chemistry for the conversion of alkyl halides into the corresponding alcohols.

$$2RI + Ag_2O + H_2O \longrightarrow 2ROH + 2AgI$$

The use of dry silver oxide results in the formation of an ether

$$2RI + Ag_2O \longrightarrow R\text{-}O\text{-}R + 2AgI$$

The silver oxide is readily soluble in ammonia solution forming the $[Ag(NH_3)_2]^+$ complex

$$Ag_2O + H_2O + 4NH_3 \longrightarrow 2[Ag(NH_3)_2]^+ + 2OH^-$$

This complex is readily reduced to silver by aldehydes and reducing sugars e.g., glucose, and this reaction is used in testing for aldehyde groups. The solution containing the complex should be washed away immediately after use, since explosions occurred with this reagent. It has been suggested that explosive silver nitride Ag_3N is formed on standing.

Silver oxide is thermally unstable and gentle heating is sufficient to cause decomposition into silver and oxygen

$$2Ag_2O \longrightarrow 4Ag + O_2$$

15.13.2 Silver Nitrate (Lunar Caustic) $AgNO_3$

Commercially silver nitrate is called as lunar caustic. It is prepared by dissolving silver in warm dilute nitric acid and then crystallizing the solution.

$$3Ag + 4HNO_3 \longrightarrow 3AgNO_3 + 2H_2O + NO$$

Properties: It forms large colourless rhombic crystals. Its melting point is 482k. It is not hygroscopic, highly soluble in water. Solubility increases with increasing temperature. The aqueous solutions are susceptible to decomposition by light.

Action of heat: At 723K it decomposes to silver nitrite

$$2AgNO_3 \xrightarrow{723K} 2AgNO_2 + O_2$$

But at red hot condition (980k) it decomposes to metallic silver, oxide of nitrogen and oxygen

$$2AgNO_3 \xrightarrow{980K} 2Ag + 2NO_2 + O_2$$

Silver nitrate is also decomposed by organic matter such as glucose, paper, skin and cork. It has very caustic and destructive effect on organic tissues. It leaves a white stain (usually black) like the moon luna on skin and hence it is called **Lunar caustic**. Since it is susceptible for decomposition by light it is stored in amber coloured bottles.

Reaction with halides: When treated with halides, it gives precipitates of corresponding halides

$AgNO_3 + NaCl \longrightarrow NaNO_3 + AgCl\downarrow$ (white curdy ppt)

$AgNO_3 + KBr \longrightarrow KNO_3 + AgBr\downarrow$ (Pale yellow ppt)

$AgNO_3 + KI \longrightarrow KNO_3 + AgI\downarrow$ (yellow ppt)

Reactions with oxo salts: Solutions of carbonate, sulphates, phosphates also give corresponding silver salts which precipitates

$2AgNO_3 + Na_2CO_3 \longrightarrow 2NaNO_3 + Ag_2CO_3$ (White ppt)

$2AgNO_3 + Na_2SO_4 \longrightarrow 2NaNO_3 + Ag_2SO_4$ (white ppt)

$2AgNO_3 + K_2CrO_4 \longrightarrow 2KNO_3 + Ag_2CrO_4$ (Brick red ppt)

$3AgNO_3 + Na_3PO_4 \longrightarrow 3NaNO_3 + Ag_3PO_4$ (Yellow ppt)

Reaction with hydrogen sulphide: With H_2S silver nitrate forms black precipitate of silver sulphide.

$2AgNO_3 + H_2S \rightarrow 2HNO_3 + Ag_2S$ (Black ppt)

Reaction with thiosulphate: The reaction of silver nitrate with hypo is two types. When silver nitrate is excess first white precipitate forms which slowly changes to yellow, orange and finally black.

$$2AgNO_3 + Na_2S_2O_3 \rightarrow \underset{\text{White ppt}}{Ag_2S_2O_3\downarrow} + 2NaNO_3$$

$$2Ag_2S_2O_3 + H_2O \rightarrow \underset{\text{Black ppt}}{Ag_2S\downarrow} + H_2SO_4$$

When hypo is excess the white precipitate formed initially, dissolves in excess of hypo due to formation of complex.

$$2AgNO_3 + Na_2S_2O_3 \rightarrow \underset{\text{White ppt}}{Ag_2S_2O_3\downarrow} + 2NaNO_3$$

$$Ag_2S_2O_3 + 3Na_2S_2O_3 \rightarrow 2Na_3[Ag(S_2O_3)_2]$$

Reaction with iodine: Silver nitrate reacts with iodine in two ways.

$$\underset{\text{excess}}{6AgNO_3} + 3I_2 + 3H_2O \rightarrow AgIO_3 + 5AgI + 6HNO_3$$

$$5AgNO_3 + \underset{\text{excess}}{3I_2} + 3H_2O \rightarrow HIO_3 + 5AgI + 5HNO_3$$

Reaction with sodium hydroxide: It forms yellow precipitate of silver oxide

$$2AgNO_3 + 2NaOH \rightarrow Ag_2O + 2NaNO_3 + H_2O$$

Reaction with ammonium hydroxide: Silver nitrate reacts with ammonium hydroxide to form silver oxide which further dissolves to form complex

$$2AgNO_3 + 2NH_4OH \rightarrow Ag_2O + 2NH_4NO_3 + H_2O$$

$$Ag_2O + 4NH_4OH \rightarrow 2[Ag(NH_3)_2]OH + 3H_2O$$

Reaction with potassium cyanide: Initially white precipitate will be formed which dissolves in excess of KCN due to formation of complex

$$AgNO_3 + KCN \rightarrow AgCN + KNO_3$$

$$AgCN + KCN \rightarrow K[Ag(CN)_2]$$

Reduction to Ag: More electropositive metals like Cu, Zn etc and reducing agents like PH_3, AsH_3, SbH_3 etc reduces silver nitrate to silver

$$2AgNO_3 + Cu \rightarrow Cu(NO_3)_2 + 2Ag$$

$$2AgNO_3 + Zn \rightarrow Zn(NO_3)_2 + 2Ag$$

$$6AgNO_3 + PH_3 + 3H_2O \longrightarrow 6Ag + 6HNO_3 + H_3PO_3$$

Reaction with ethyne: An ammoniacal solution of silver nitrate reacts with ethyne forming a precipitate of silver carbide Ag_2C_2 which explodes if struck when dry

$$C_2H_2 + 2[Ag(NH_3)_2]^+ \rightarrow Ag_2C_2 + 2NH_4^+ + 2NH_3$$

The action of dilute nitric acid on silver carbide regenerates pure ethyne and this reaction may be used to purify this hydrocarbon e.g., to separate it from alkanes and alkenes

Uses: Silver nitrate is used

(i) As a laboratory reagent and in volumetric analysis.
(ii) In the production of light sensitive plates, films and papers.

(iii) In the manufacture of marking inks (indelible ink) and hair dyes.
(iv) As a caustic in surgery under the name lunar caustic.
(v) In medicines in nervous diseases.
(vi) For the preparation of silver halides used in photography.
(vii) In silvering of mirrors.

Silvering of Mirrors

The process of depositing a uniform and thin layer of silver on a clean glass surface is called silvering of mirror. It is used for making looking mirrors, concave mirrors and reflecting surfaces. The process is based on the reduction of ammoniacal solution by formaldehyde, tartrate, glucose etc.

$$2AgNO_3 + 2NH_4OH \rightarrow Ag_2O + 2NH_4NO_3 + H_2O$$

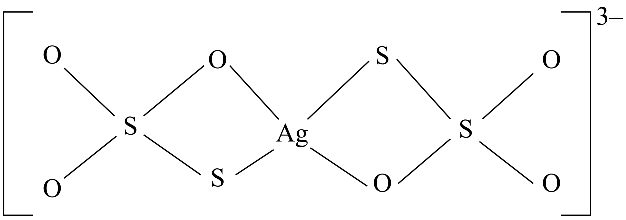
$$Ag_2O + HCHO \rightarrow \underset{\substack{\text{Silver}\\ \text{mirror}}}{2Ag} + HCOOH$$

15.13.3 Silver Thiosulphate

Silver ion forms complex with thiosulphate ion $[Ag(S_2O_3)_2]^{3-}$. Also it can form complexes with ammonia, cyanide ion etc. The ammonia and cyanide complexes are $[Ag(NH_3)_2]^+$ and $[Ag(CN)_2]^-$. Among these complexes thiosulphate complex is of importance in the "fixing" process in photography while the cyanide complex is used during the extraction of silver. The stability of these three complexes decreases in the order.

$$[Ag(CN)_2]^- > [Ag(S_2O_3)_2]^{3-} > [Ag(NH_3)_2]^+$$

The structure of silver thiosulphate complex is as follows



The formation of this silver thiosulphate is utilized to remove the unchanged silver bromide on a photographic film or plate.

$$AgBr + 2Na_2S_2O_3 \longrightarrow Na_3[Ag(S_2O_3)_2] + NaBr$$

15.14 GOLD

Gold is nearly always found as the free element, often in association with copper and silver; it in obtained in significant amounts during the extraction of lead from its ores and the electrolytic refining of copper.

One method of extraction of the metal involves treatment of the pulverized ore with an aerated solution of sodium cyanide and treatment of the cyanide complex $[Au(CN)_2]^-$ with zinc to liberate the free metal *cf* the cyanide process used in the extraction of silver.

$$4Au + 8NaCN + 2H_2O + O_2 \longrightarrow 4Na[Au(CN)_2] + 4NaOH.$$

$$2Na[Au(CN)_2] + Zn \longrightarrow Na_2[Zn(CN)_4] + 2Au.$$

The mixture of gold and silver obtained by this method can be separated by electrolytic methods.

Properties: Gold has a melting point of 1063°C and a density of 19.3 g cm^{-3}. It is extremely malleable and ductile. It is one of the most unreactive elements. It is not attacked by air, water or steam and the common mineral acids. But it dissolves in aquaregia. It is thought that the gold is first oxidized by nitric acid to Au^{3+} which is then removed by complexing with the chloride ions

$$Au + 6HNO_3 \rightleftharpoons Au^{3+} + 3NO_3^- + 3H_2O + 3NO_2$$

$$Au^{3+} + 4Cl^- \longrightarrow [AuCl_4]^-$$

In view of its lack of chemical reactivity and its attractive bright yellow colour, gold is much used in Jewellery. Since pure metal is rather soft it is generally hardened by the incorporation of other metals such as silver and copper. Gold is sold by the carat which signifies the number of parts by weight of gold in 24 parts by weight of the alloy, thus pure gold is 24 carat.

15.14.1 Compounds of Gold

Gold exhibits oxidation states of +1 and +3 but all its compounds are thermally unstable and yield free metal on gentle heating. The solution chemistry of gold is essentially that of its complex ions, typical complexes including $[Au(CN)_2]^-$ which contains gold (I) and $[AuCl_4]^-$ which contains gold (III).

The standard redox potentials for the system Au^{3+}/Au and Au^+/Au are gives below; they indicate that Au^+ is unstable in solution with respect to Au^{3+} and Au.

$$Au^+ + e^- \longrightarrow Au \qquad E^O = +1.68V$$

$$Au^{3+} + 2e^- \longrightarrow Au^+ \qquad E^O = +1.29V.$$

or $$3Au^+ \longrightarrow 2Au + Au^{3+} \qquad E^O_{total} = +0.39V.$$

The positive e.m.f of the cell reaction implies that Au^+ ions disproportionate into Au^{3+} ions and Au in solution as explained in the case of disproportionation of Cu^+ into Cu and Cu^{2+} ion aqueous solution. This disproportionation reaction is only prevented from occurring in aqueous solution by adding substances which form more stable complexes with gold (I) than with gold (III) e.g., CN^- ions which form the $[Au(CN)_2]^-$ complex ion.

The action of fluorine and chlorine on gold at moderate temperatures results in the formation of the gold (III) halide AuX_3

$$2Au + 3X_2 \longrightarrow 2AuX_3$$

The addition of water results in the hydrolysis of these halides to give hydrated gold (III) oxide $Au_2O_3 \cdot xH_2O$. This oxide dissolves in hydrochloric acid to give gold (III) chloride and then the complex ion $[AuCl_4]^-$. It dissolves in alkaline solutions to give the $[Au(OH)_4]^-$ ion so much be considered to be amphoteric.

$$Au_2O_3 + 6HCl \longrightarrow 2AuCl_3 + 3H_2O$$

$$AuCl_3 + Cl^- \longrightarrow [AuCl_4]^-$$

$$Au_2O_3 + 3H_2O + 2OH^- \longrightarrow 2[Au(OH)_4]^-$$

Gold (III) chloride can be reduced to gold by several reducing agents.

$$2AuCl_3 + 3SnCl_2 \longrightarrow 2Au + 3SnCl_4$$

When reduced with stannous chloride the product stannic chloride formed hydrolyses in water forming colloidal stannic hydroxide which adsorbs the gold particles. This colloidal solution of stannic hydroxide adsorbed by gold particles is called purple of cassius, which is used for colouring of glass and pottery.

15.15 ZINC

Zinc is not found in native state because it is highly reactive metal. The main ores are

1. Zinc blende ZnS
2. Calamine $ZnCO_3$
3. Willimite Zn_2SiO_4
4. Zincite ZnO
5. Franklinite $ZnO \cdot Fe_2O_3$.

Extraction

The principle ore is zinc blende. The following are the metallurgical steps.

1. **Concentration:** The ore is powdered and concentrated by froth floatation process
2. **Roasting:** The concentrated ore is roasted in the presence of excess of air. Then zinc sulphide is converted into zinc oxide.

$$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$$

Some zinc sulphide may be oxidized to zinc sulphate. Roasting at high temperature is carried to decompose any zinc sulphate is formed

$$ZnS + 2O_2 \rightarrow ZnSO_4$$

$$2ZnSO_4 \rightarrow 2ZnO + 2SO_2 + O_2$$

If the ore is calamine, it is calcined to get zinc oxide

$$ZnCO_3 \rightarrow ZnO + CO_2$$

(II) Reduction: The oxide is then mixed with carbon and heated in retorts.

$$ZnO + C \rightarrow Zn + CO$$

The reduction is carried in different methods

(a) Belgian Process: Circular or elliptical retorts of fireclay closed at one end and open at the other are employed in this process of heating. The retorts are filled with a mixture of 2 parts of ore and 1 part coal. The temperature is raised and the carbon monoxide which evolves during the process of reduction burns with a blue blame at the end of clay adapter and gives an indication of the process. After some time, at higher temperature the cadmium present along with zinc begins to volatilize giving brown flames. At this stage, an iron condenser is placed at the end and the metal condensed. The metal deposited is taken out and the process is again continued, fresh charges also being introduced in the furnace from time to time.

This process is continuous. The roasted ore is mixed with coke and made into small pellets. These are fed in the vertical retorts made of retractory material and heated externally by burning producer gas. Zinc vapour along with carbon monoxide is led to the condenser by applying sucction and forcing producer gas from below. Molten zinc is tapped off from the condensers from time to time. The residue moves downwards and is worked out from the bottom of the plant.

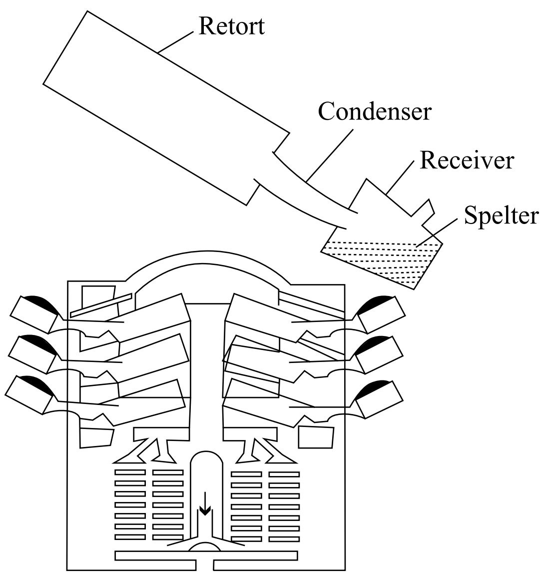


Fig 15.31 Belgian process

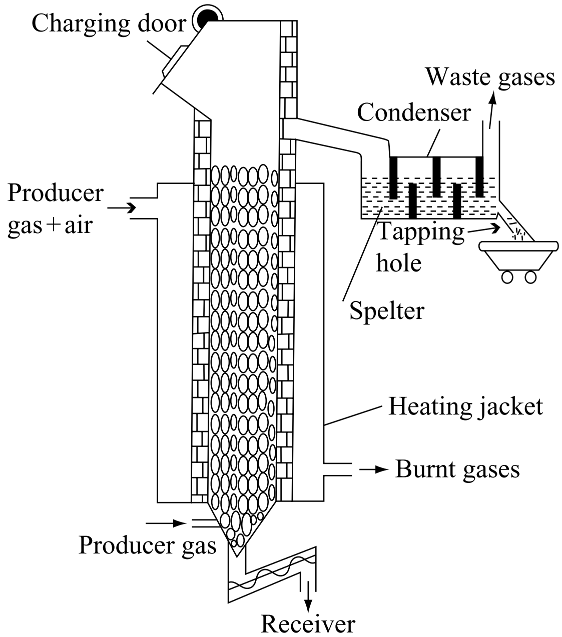


Fig 15.32 Vertical retort process

(iv) Purification: The impure zinc obtained as above is known as spelter, contains about 97.8 per cent zinc, the rest being mostly lead and some iron, cadmium and arsenic. This is purified by the following methods.

(a) Distillation: The separation of impurities is possible because of the difference in their boiling points. Iron at 3000°and lead at 1600°. These are higher than the boiling points of zinc (907°C) and cadmium (767°C). When therefore, distillation is carried around 950–1000°C only zinc and cadmium distil over. From this sample zinc is separated by passing through a condenser at about 800°C when the more volatile cadmium passes out.

(b) Electrolytic refining: Impure zinc is made as anode while a plate of zinc is made of cathode. The electrolyte is zinc sulphate containing a small amount of dilute sulphuric acid. On passing current, zinc from the electrolyte is deposited at the cathode while an eqivalent amount of zinc from anode goes into the electrolyte. Pure zinc is obtained at cathode.

Special Types of Zinc

(i) Zinc dust: It is prepared by melting zinc and then atomizing it with blast of air.

(ii) Granulated zinc: It is prepared by pouring molten zinc into cold water.

Zinc is a white lustrous metal. It adopts a somewhat distorted hexagonal close - packed structure (each atom surrounded by twelve nearest neighbours, the distortion being indicative of some degree of covalent bonding and hence of weak non-metallic properties.

It is a fairly reactive metal and when exposed to moist air for any length of time a protective layer is formed on its surface. This is the oxide initially, but over a period of time the basic carbonate is formed. Dilute non-oxidizing acids such as hydrochloric acid and sulphuric acid attack the metal with the formation of its +2 ion and the liberation of hydrogen e.g.,

$$Zn + 2H^+ \rightarrow Zn^{2+} + H_2$$

However pure metal is resistant to attack by these acids since they exhibit a high over potential to discharge of hydrogen (this appears to be an activation energy effect). Concentrated sulphuric acid attacks the metal with the formation of sulphur dioxide; oxides of nitrogen are formed by the action of both dilute and concentrated nitric acid.

Zinc is chemically more reactive due to its more negative electrode potential and hot zinc will displace hydrogen from steam.

$$Zn + H_2O \rightarrow ZnO + H_2$$

Non-metals such as oxygen, sulphur and halogens combine directly with zinc on heating.

Zinc reacts with sodium hydroxide solution liberating hydrogen

$$Zn + 2OH^- + 2H_2O \rightarrow [Zn(OH)_4]^{2-} + H_2$$

Uses: Appreciable quantities of zinc are used to protect iron and steel from corrosion. Galvanized iron is made by dipping the article into a bath of molten zinc. Small articles like screws, nuts and bolts are treated by process known as **sheradizing**; this involves heating them in a rotating drum with zinc dust. An electroplating process is used for applying a thin yet long-lasting film to metal objects.

15.16 COMPOUNDS OF ZINC

15.16.1 Zinc Oxide

It is prepared by igniting the carbonate or nitrate or hydroxide obtained by precipitating from zinc sulphate

$$ZnCO_3 \rightarrow ZnO + CO_2$$

$$2Zn(NO_3) \rightarrow 2ZnO + 4NO_2 + O_2$$

$$Zn(OH)_2 \rightarrow ZnO + H_2O$$

Zinc oxide formed by burning zinc in air, condenses as wooly material which is known as **Philosopher's Wool**

$$2Zn + O_2 \rightarrow 2ZnO$$

Zinc oxide is a white solid and is considered to be essentially covalent; it adopts the diamond structure in which each zinc atom is surrounded tetrahedrally by four oxygen atoms and each oxygen atom is likewise surrounded by four zinc atoms. On heating, it develops a yellow colour and this phenomenon is associated with the loss of some oxygen from the lattice leaving it with an excess negative charge. This excess negative charge electrons can be moved through the lattice on the application of a potential difference and thus this oxide is a semiconductor. Zinc oxide returns to the former colour (white) on cooling, since the oxygen, which was lost on heating returns to the crystal lattice.

It is amphoteric oxide reacts with both acids and bases.

$$ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$$

$$ZnO + 2NaOH \rightarrow Na_2ZnO_2 + H_2O.$$

When heated with cobalt nitrate, a green –coloured product is obtained which may be regarded as the solid solution of cobalt zincate $CoZnO_2$ in zinc oxide

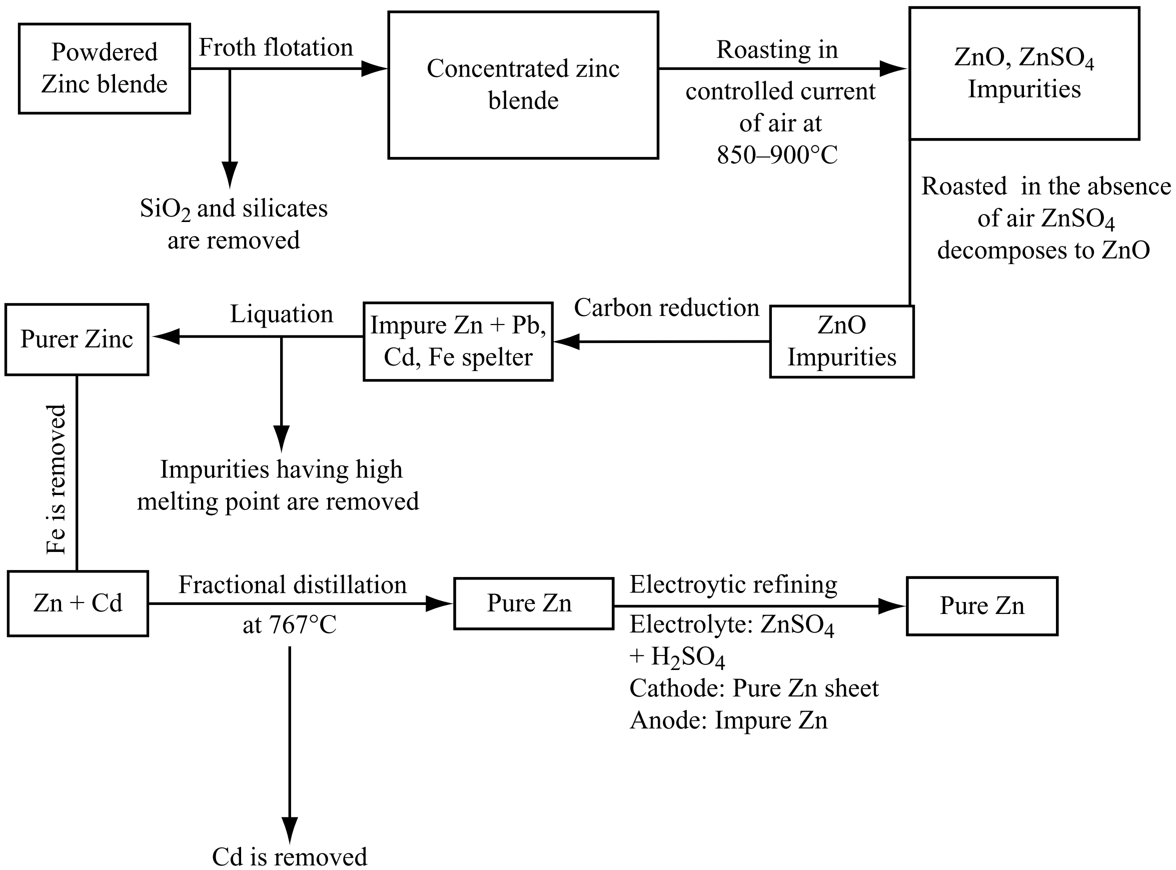
$$2Co(NO_3)_2 \rightarrow 2CoO + 4NO_2 + O_2$$

$$CoO + ZnO \rightarrow CoZnO_2$$

This solid solution is sold under the name of Rinmann's green to be used as a pigment

Zinc oxide is the important compound of zinc; it is used as a white pigment, as a filler in rubber and as a component in various glazes, enamels and antiseptic ointments. In combination with chromium (III)oxide, it is used as a catalyst in the manufacture of methanol from synthesis gas (water gas).

Flow Chart for the Extraction of Zinc



15.16.2 Zinc Chloride

It is obtained by dissolving zinc metal or its oxide, hydroxide or carbonate in dilute hydrochloric acid; partial evaporation followed by crystallization results in the separation of white hydrated chlorides $ZnCl_2 \cdot 2H_2O$

$$Zn + 2HCl \rightarrow ZnCl_2 + H_2$$

$$ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$$

$$Zn(OH)_2 + 2HCl \rightarrow ZnCl_2 + 2H_2O$$

$$ZnCO_3 + 2HCl \longrightarrow ZnCl_2 + H_2O + CO_2$$

Anhydrous salt cannot be prepared from these crystals by merely heating, since a basic chloride is formed during the decomposition.

$$Zn + Cl_2 \rightarrow ZnCl_2$$

$$2ZnO + Cl_2 \rightarrow 2ZnCl_2 + O_2$$

$$Zn + HgCl_2 \rightarrow ZnCl_2 + Hg$$

Zinc chloride is very deliquescent and extremely soluble in water. 100gms of water dissolve 330gms of the anhydrous salt at 10°C. It is also soluble in alcohols and acetone and this indicates that it is having appreciable covalent character.

Since zinc chloride has a remarkable power of absorbing water, it is used for drying gases, and also as a condensing reagent where it is desired to eliminate molecules of water (for example in the preparation of phenolphthalein from phenol and phthalic anhydride).

Zinc chloride is used as a flux in soldering and as a timber preservative. Both uses depend upon the ability of the compound to behave as a Lewis acid i.e., to accept an electron pair. In soldering it is essential to remove the oxide film on the surfaces to be joined at the temperature employed zinc chloride melts and removes the oxide film by forming covalent bonded complexes with the oxygen atoms. When it is applied to timber, Zinc chloride forms covalent bonds with the oxygen atoms in the cellulose molecules. The timber is therefore coated with a layer of zinc chloride which like all zinc compounds, is toxic to any living matter. Zinc chloride solution cannot be filtered through paper since it reacts with cellulose.

It is also used for weighting cotton goods and for making parchment paper and fire board. An oxychloride ($ZnCl_2$ ZnO) formed by mixing syrupy zinc chloride solution with zinc oxide sets to a hard mass and, therefore, is used as a dental filling.

15.16.3 Zinc Sulphate (White Vitriol) $ZnSO_4 \cdot 7H_2O$

It can be prepared by reacting the appropriate oxide, hydroxide or carbonate or directly zinc metal with dilute sulphuric acid.

$$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$$

$$ZnO + H_2SO_4 \rightarrow ZnSO_4 + H_2O$$

$$Zn(OH)_2 + H_2SO_4 \rightarrow ZnSO_4 + 2H_2O$$

$$ZnCO_3 + H_2SO_4 \rightarrow ZnSO_4 + H_2O + CO_2.$$

Evaporation and crystallization gives white crystalline solid

Properties: It is a colourless crystalline solid, efflorescent in nature, freely soluble in water.

Action of heat: On heating the following changes takes place at different temperatures.

$$ZnSO_4 \cdot 7H_2O \xrightarrow[\text{below } 70^\circ C]{\text{above } 39^\circ C} ZnSO_4 \cdot 6H_2O \xrightarrow{\text{above } 70^\circ C}$$
$$ZnSO_4 \cdot H_2O \xrightarrow{\text{above } 280^\circ C}$$

$$ZnSO_4 \xrightarrow{800^\circ C} ZnO + SO_2 + \frac{1}{2}O_2$$

Reaction with NaOH: With NaOH, zinc sulphate first gives white precipitate which dissolves in excess NaOH due to the formation of sodium zincates

$$ZnSO_4 + 2NaOH \rightarrow Na_2SO_4 + Zn(OH)_2$$
$$Zn(OH)_2 + 2NaOH \rightarrow Na_2ZnO_2 + 2H_2O$$

Reaction with ammonium hydroxide: First it gives white precipitate which dissolves in excess of ammonium hydroxide due to the formation of complex

$$ZnSO_4 + 2NH_4OH \rightarrow Zn(OH)_2 + (NH_4)_2SO_4$$

$$Zn(OH)_2 + (NH_4)_2SO_4 + 2NH_4OH \rightarrow [Zn(NH_3)_4]SO_4 + 4H_2O$$

Reaction with sodium carbonate: Basic zinc carbonate will be precipitated

$$4ZnSO_4 + 4Na_2CO_3 + 3H_2O \rightarrow ZnCO_3 \cdot 3Zn(OH)_2 + 4Na_2SO_4 + 3CO_2$$

Reaction with sodium hydrogen carbonate, normal carbonate will be precipitated

$$ZnSO_4 + 2NaHCO_3 \rightarrow ZnCO_3 + Na_2SO_4 + H_2O + CO_2.$$

Reaction with H_2S: Gives white precipitate of zinc sulphide soluble in hot dilute HCI

$$ZnSO_4 + H_2S \rightarrow ZnS\downarrow + H_2SO_4$$

$$ZnS + 2HCl \rightarrow ZnCl_2 + H_2S.$$

Reaction with potassium ferroyanide. Gives white precipitate of zinc ferrocyanide.

$$\underset{\text{white ppt.}}{2ZnSO_4 + K_4[Fe(Cn)_6] \rightarrow Zn_2[Fe(CN)_6] \downarrow + 2K_2SO_4}$$

Chemically zinc chloride and zinc sulphate shows the similar reactions since the solutions of both contain Zn^{2+} ions.

Uses: It is used

(i) As mordant in dyeing and calico priming.
(ii) In zinc plating and electrolytic refining of zinc.
(iii) As eye lotion.

In the manufacture of lithopone, a white pigment.

$$ZnSO_4 + BaS \rightarrow \underbrace{BaSO_4 + ZnS}_{\text{Lithophone.}}$$

15.17 MERCURY

Mercury is extracted from the mineral cinnabar. Mercury is readily extracted from the concentrated ore by heating it in air mercury vapour is evolved which is condensed to liquid.

$$HgS + O_2 \rightarrow Hg + SO_2$$

Commercial mercury is impure and contains lead and less frequently zinc and tin. It is purified by distillation under reduced pressure.

Properties: Mercury is a dense silvery – coloured metal and the only one that is liquid at room temperature, It freezes at –39°C and boils at 357°C. Although it has very small vapour pressure, the vapour is exceedingly poisonous and any mercury accidentally spit should immediately be dusted with powdered sulphur. It dissolves many metals including sodium, zinc, tin, silver and gold, to form amalgams, but does not attack iron, cobalt, nickel hence it can be stored in iron bottles. Pure mercury is not attacked by air. At ordinary temperatures Mercury (I) oxide is slowly formed in the region of about 350°C but at slightly higher temperatures it decomposes into mercury and oxygen; other non-metals that combine directly with mercury include halogens and sulphur.

In view of its high positive electrode potential, mercury is not attacked by dilute non-oxidizing acids. Nitric acid attacks the metal, with the liberation of oxides of nitrogen; excess of concentrated nitric acid tends to give nitrogen dioxide and mercury (II) nitrate but with dilution and in the presence of excess of mercury, nitric oxide and mercury (I) nitrate tend to predominates.

Hot concentrated sulphuric acid attacks the metal with liberation of sulphur dioxide; excess acid tends to give mercury (II) sulphate and excess of mercury tends to give mercury (I) sulphate.

$$4Hg + O_2 \xrightarrow{\sim 350°C} 2Hg_2O$$

$$2Hg_2O \xrightarrow{\text{above } 350°C} 4Hg + O_2$$

$$Hg + X_2 \rightarrow HgX_2$$

$$Hg + S \rightarrow HgS.$$

$$\underset{\text{excess}}{6Hg} + \underset{\text{dil}}{8HNO_3} \rightarrow 3Hg_2(NO_3)_2 + 4H_2O + 2NO$$

$$Hg + \underset{\text{excess conc}}{4HNO_3} \rightarrow Hg(NO_3)_2 + 2H_2O + 2NO_2$$

$$\underset{\text{excess}}{2Hg} + \underset{\text{conc}}{2H_2SO_4} \rightarrow Hg_2SO_4 + 2H_2O + SO_2$$

$$Hg + \underset{\text{excess conc.}}{2H_2SO_4} \rightarrow HgSO_4 + 2H_2O + SO_4$$

Uses: Mercury is used

(i) In the ammeters, barometers, electrical switches and in mercury arc lights.
(ii) Sodium and zinc amalgams are used in laboratory as reducing agents and tin amalgam is used in dental filling.
(iii) Mercury fulminate is a shock-sensitive explosive which is used in detonator.
(iv) The major uses of mercury compounds are in agriculture and in horticulture, e.g., organomercury compounds are used as fungicides and as timber preservatives.

15.17.1 Compounds of Mercury

Mercury exhibits oxidation states +2 and +1, in the latter, two mercury atoms are united by a single covalent bond as in the $[Hg\text{-}Hg]^{2+}$ ion. Mercury compounds are more covalent than those of zinc and cadmium compounds and, indeed, covalent nature is the rule rather than exception. Mercury compounds form stable complexes with variety of ligands, indeed, they are generally many orders of magnitude more stable than the corresponding complexes of zinc and cadmium.

15.17.2 Mercury (II) Compounds

Mercury (II) oxide: When mercury is heated for a long time at about 350°C, mercury (II) oxide will be formed. In this method red variety of mercury (II) oxide is formed. Yellow mercury (II) oxide may be obtained by adding sodium hydroxide solution to a solution of a mercury (II) salt

$$HgCl_2 + 2NaOH \longrightarrow Hg(OH)_2$$

$$Hg(OH)_2 \longrightarrow HgO + H_2O$$

Red and yellow varieties differ only in particle size. Particles of yellow variety are smaller in size. Yellow form changes to red form on heating to 400°C. On strong heating it decomposes to mercury and oxygen

$$2HgO \xrightarrow{\Delta} 2Hg + O_2$$

It is used as a red pigment in oil paints and mild antiseptic in ointments.

15.17.3 Mercury (II) Chloride (Corrosive Sublimate) $HgCl_2$

Mercury (II) chloride forms rapidly when chlorine is brought into contact with mercury

$$Hg + Cl_2 \longrightarrow HgCl_2$$

On large scale it is prepared by heating a mixture of $HgSO_4$, NaCl and MnO_2. MnO_2 prevents the formation of mercurous chloride

$$HgSO_4 + 2NaCl \xrightarrow{MnO_2} HgCl_2 + Na_2SO_4$$

It sublimes on heating. It is therefore known as corrosive sublimate. It is a white solid and readily dissolves in organic solvents such as ethanol and ether, thus indicating that it is predominantly covalent. Although only sparingly soluble in cold water, it dissolves readily on warming but the solutions so obtained is only a very feeble electrolyte and it has been estimated that about 99 per cent of the dissolved mercury (II) chloride solution is readily reduced to mercury (I) chloride and then to mercury by the addition of tin (II) chloride solution and this is a convenient test for mercury (II) ion

$$2HgCl_2 + SnCl_2 \longrightarrow SnCl_4 + \underset{\text{White ppt}}{Hg_2Cl_2}$$

$$Hg_2Cl_2 + SnCl_2 \longrightarrow SnCl_4 + \underset{\text{Black}}{2Hg}$$

It is highly poisonous, acting by coagulating protein in living cells; the antidote is the immediate administration of the white of an egg which provides an alternative protein material for it to act upon. In very dilute solution mercury (II) chloride is sometimes used as an antiseptic e.g., for sterilizing the hands.

15.17.4 Mercury (II) Iodide

This compound precipitates as a yellow solid which then turns rapidly red when the correct amount of potassium iodide solution is added to a solution of mercury (II) chloride.

$$HgCl_2 + 2KI \longrightarrow HgI_2 + 2KCl$$

On heating to about 127°C the red form changes into a yellow variety which has a different crystal structure (polymorphism). On cooling again the change is reversed and this change in crystal structure may be accelerated by scratching the material with a glass rod.

$$\underset{\text{Red}}{HgI_2} \underset{\text{below 127°C}}{\overset{\text{above 127°C}}{\rightleftharpoons}} \underset{\text{Yellow}}{HgI_2}$$

Mercury (II) iodide readily dissolves in an excess of potassium iodide solution with the formation of the $[HgI_4]^{2-}$ complex ion

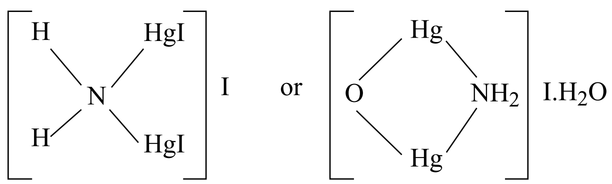
$$HgI_2 + 2KI \longrightarrow K_2[HgI_4]$$

Salts containing this complex anion include. Ag_2HgI_4 (yellow) and Cu_2HgI_4 (red) which are both insoluble.

Nessler's reagent is a solution of K_2HgI_4 to which some potassium hydroxide has been added. It reacts with traces of ammonia to give a yellow colouration; larger quantities of the gas produces a brown precipitate.

$$2K_2HgI_4 + NH_3 + 3KOH \longrightarrow H_2NHgOHgI + 7KI + 2H_2O$$

The brown precipitate is called **iodide of Milon's base**. Chemically it is called as amidomercuric oxymercuric iodide and its probable composition is



Mercury (I) Compounds

15.17.5 Mercury (I) Chloride (Calomel) Hg_2Cl_2

Mercury (I) chloride precipitates as a white solid when solutions containing mercury (I) and chloride are mixed

$$Hg_2^{2+} + 2Cl^- \rightarrow Hg_2Cl_2$$

It may also be obtained by subliming a mixture of mercury (II) chloride and mercury.

$$Hg_2Cl_2 + Hg \longrightarrow Hg_2Cl_2$$

It sublimes on heating and then decomposes

$$Hg_2Cl_2 \longrightarrow HgCl_2 + Hg$$

The action of ammonia solution on mercury (I) chloride gives black deposit of mercury and a compound containing mercury (II) i.e., disproportionate occurs

$$Hg_2Cl_2 + 2NH_3 \longrightarrow Hg + HgNH_2Cl + NH_4Cl.$$

The compound $HgNH_2Cl$ contains the mercury-nitrogen covalent bond, the formation of which assists in moving the disproportionate reaction from left to right. This reaction is a convenient test for the mercury (I) ion.

Mercury (I) chloride is employed in the calomel electrode which is used as reference standard electrode in redox potential measurements.

15.18 ELECTROCHEMICAL PRINCIPLES OF METALLURGY

So far we have seen the application of thermodynamics to pyrometallurgy. In these pyrometallurgical techniques reduction of metal ions are carried in solution or molten state. But in electrochemical process metal ions are reduced by electrolysis. These methods are based on electrochemical principles which could be understood through the equation

$$\Delta G^0 = -nFE^0$$

Here n is the number of electrons and E^0 is the electrode potential of the redox couple formed in the system. More reactive metals have large negative values of the electrode potential. So their reduction is difficult.

In some electrolysis, the M^{n+} ions are discharged at negative electrodes (cathodes) and deposited there. The material should be selected keeping in mind the reactivity of the metal produced. Some times flux is added to the electrolyte for making molten mass more conducting and to decrease the melting point of the metal salt.

Some examples of extraction of metals illustrating the application of electrochemical principles of metallurgy are discussed here.

15.18.1 Sodium

Sodium was first isolated by Davy in 1807 by the electrolysis of fused sodium hydroxide.

Occurrence: It does not occur in free in nature because it is very reactive metal. In combined state it is found as

(i) Sodium chloride in the seas, lakes and as rock salt
(ii) Sodium nitrate or Chile salt petre $NaNO_3$
(iii) Sodium sulphate or glauber's salt $Na_2SO_4 \cdot 10H_2O$
(iv) Sodium carbonate is an efflorescence on the soil
(v) Borax or tincol $Na_2B_4O_7 \cdot 10H_2O$

Extraction

(I) Castner's Process: In the Castner's process, sodium is prepared by the electrolysis of fused caustic soda the electrolysis is carried out just above 313°C, melting point of caustic soda. Sodium hydroxide ionizes.

$$4NaOH \longrightarrow 4Na^+ + 4OH^-$$

At cathode $\quad 4Na^+ + 4e^- \longrightarrow 4Na$

At anode $\quad 4OH^- \longrightarrow 2H_2O + O_2 + 4e^-$

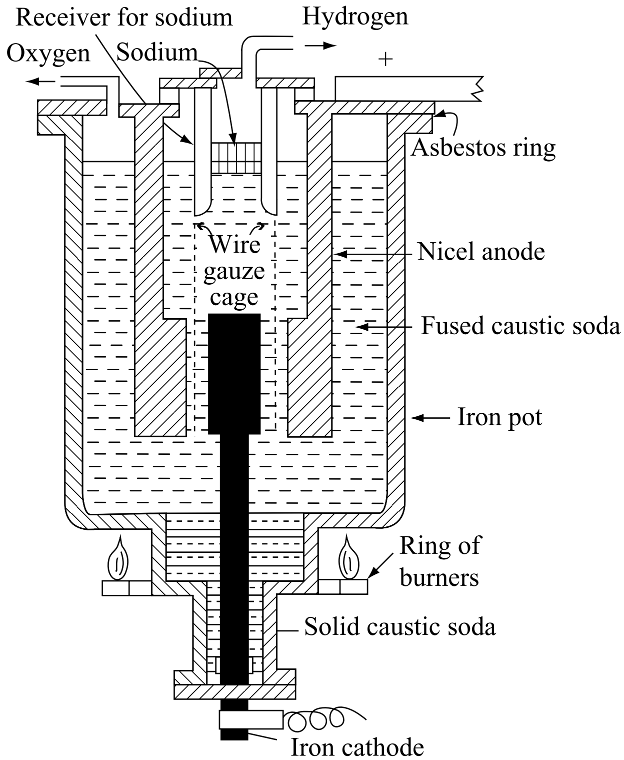


Fig 15.33 Extraction of sodium, Castner's process

The electrolytic cell used in the process is shown in fig 15.33.

Fused sodium hydroxide is taken in a cylindrical iron tank. The electrolyte is fused by heating to 330°C. with a ring of burners. An iron rod is introduced into the fused electrolyte from the bottom of the tank. This functions as cathode. A hallow nickel cylindrical anode surrounds the cathode. The two electrodes are separated by a wire gauze. During the process of electrolysis, the sodium ions migrate to the cathode and get deposited. Since the molten sodium is lighter than molten sodium hydroxide, it rises to the surface. The wire gauze prevent the passage of the liberated sodium to the anode. Sodium is separated by perforated ladles from time to time.

Down's Process

Nowadays sodium is manufactured by the electrolysis of fused sodium chloride. The pure sodium chloride melts at 803°C. At this temperature, separation of sodium ion from fused electrolyte is difficult so the melting point of the electrolyte is brought down to a temperature between 500 to 600°C by the addition of calcium chloride or potassium

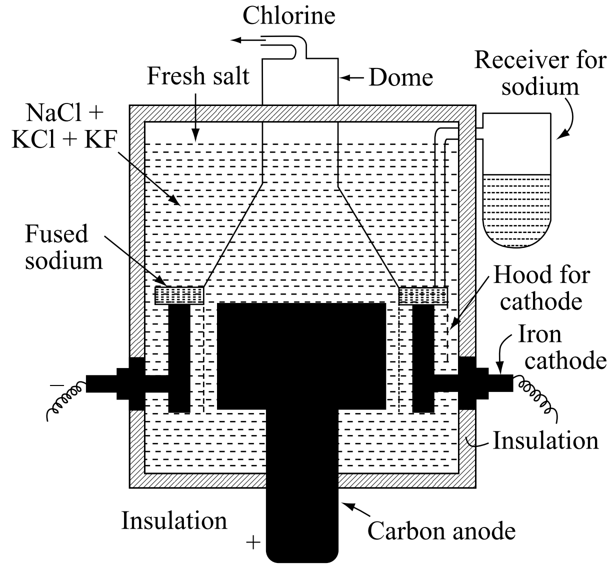


Fig 15.34 Extraction of sodium – Down's process

chloride and potassium fluoride in suitable proportions. There are three advantages

(i) The m.p is decreased, so the fuel wastage is reduced
(ii) At lower temperatures, the vapour pressure of sodium is less and hence the probability catching fire in air is minimized.
(iii) At lower temperatures loss of metal due to its dissolution in fused electrolyte is also reduced. Moreover the process of electrolysis, takes place smoothly and yields are improved. The reactions that takes place in the cell can be written as follows.

Ionization $NaCl \longrightarrow Na^+ + Cl^-$

At cathode $Na^+ + e^- \longrightarrow Na$

At cathode $2Cl^- \longrightarrow Cl_2 + 2e^-$

In spite of the possibility of calcium impurity in this method, the metal obtained is very pure.

The electrolytic cell in the Down's process consists of an iron or a steel tank. It is lined inside with fire bricks and acid resistant materials. A graphite rod introduced from the bottom of the tank acts as anode.

The anode is surrounded by a ring shaped iron cathode. A conical connecting hood is placed above separates the anode and the cathode. The wire gauze prevents the passage of sodium liberated at cathode to anode and also the mixing up of Cl_2 formed at anode, with sodium.

15.18.2 Magnesium

Magnesium is distributed in nature in combined form. We find it in sea water, in animal body, in the leaves (as chlorophyll) etc.

Occurrence

Magnesite	$MgCO_3$
Dolomite	$MgCO_3 \cdot CaCO_3$
Carnallite	$KCl \cdot MgCl_2 \cdot 6H_2O$
Kieserite	$MgSO_4 \cdot H_2O$
Epsomite	$MgSO_4 \cdot 7H_2O$
Kainite	$K_2SO_4 \cdot MgSO_4 \cdot MgCl_2 \cdot 6H_2O$
Asbestos	$CaMg_3 (SiO_3)_4$
Sea water	

contains $MgCl_2$ and $MgSO_4$

Extraction

Magnesium is obtained on a large scale by the electrolysis of either fused anhydrous magnesium chloride or magnesium oxide.

Electrolysis of fused magnesium chloride: Fused magnesium chloride is obtained by the action of chlorine over a mixture of magnesium oxide and carbon heated to red hot condition. When fused carnalite $KCl \cdot MgCl_2 \cdot 6H_2O$ is employed, this substance requires to be rendered anhydrous before electrolysis. Four water molecules are readily removed, but the elimination of the last two molecules leads to the following reaction.

$$MgCl_2 \cdot 2H_2O \longrightarrow MgO + 2HCl + H_2O$$

Excess of hydrogen chloride, however, checks this reaction and the final stages of dehydration are carried out in a stream of dry HCl gas at 350°C. The reactions that takes place are.

Ionization $MgCl_2 \longrightarrow Mg^{2+} + 2Cl^-$

At cathode $Mg^{2+} + 2e^- \longrightarrow Mg$

At anode $2Cl^- \longrightarrow Cl_2 + 2e^-$

The electrolytic bath consists of rectangular iron vessel. The optimum bath temperature is 700°C, as is sufficiently

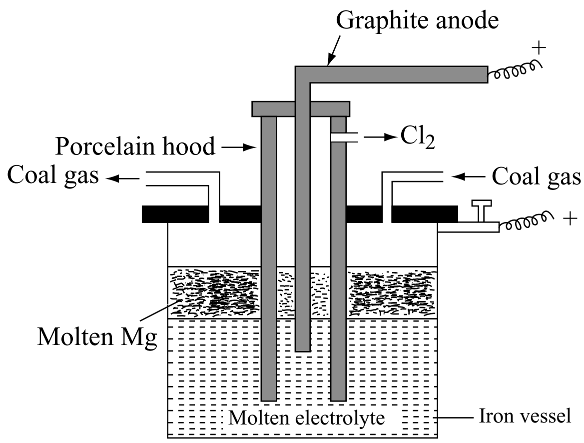


Fig 15.35 Electrolysis of molten magnesium chloride

above the melting point of the metal to allow its separation in liquid form. The anodes of carbon or graphite are lead covered and have a porcelain hood which dips into the molten electrolyte, the submerged part being protected by water – cooled tubes, which cause a crust of solidified electrolyte to form a protection against corrosion by fluorides when such salts are used as fluxes. The chlorine evolved is removed through pipes in these hoods and utilized in the magnesium chloride plant. The iron vessel containing the electrolyte acts as the cathode. Magnesium collects in the form of drops which coalesce and rise to the surface, being protected from the attack of chlorine by the intervening hoods. The air in the apparatus is displaced by some inert gas like coal gas or hydrogen. The metal separated from the electrolyte by perforated laddles is 99.9 percent purity; it is further refined by remelting in a flux of anhydrous magnesium chloride and sodium chloride.

By Electrolysis of Magnesia

By a process like that used for aluminium, magnesium may be obtained by the electrolysis of magnesia (MgO) dissolved in fused mixture of magnesium, barium and sodium fluorides.

The electrolytic bath is a steel tank. The iron cathodes are projected from below into the electrolyte while the carbon anodes are suspended from above. The temperature

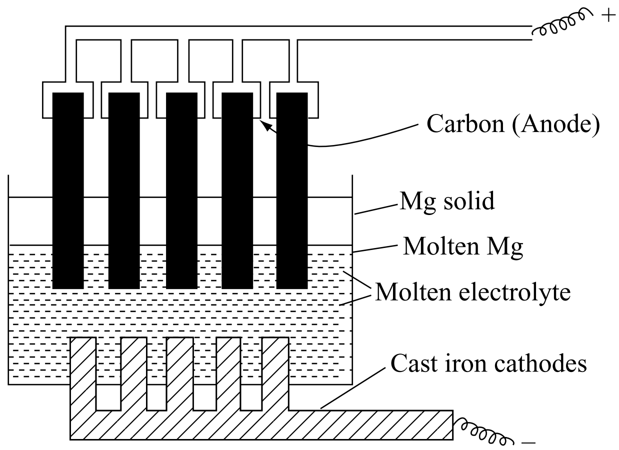
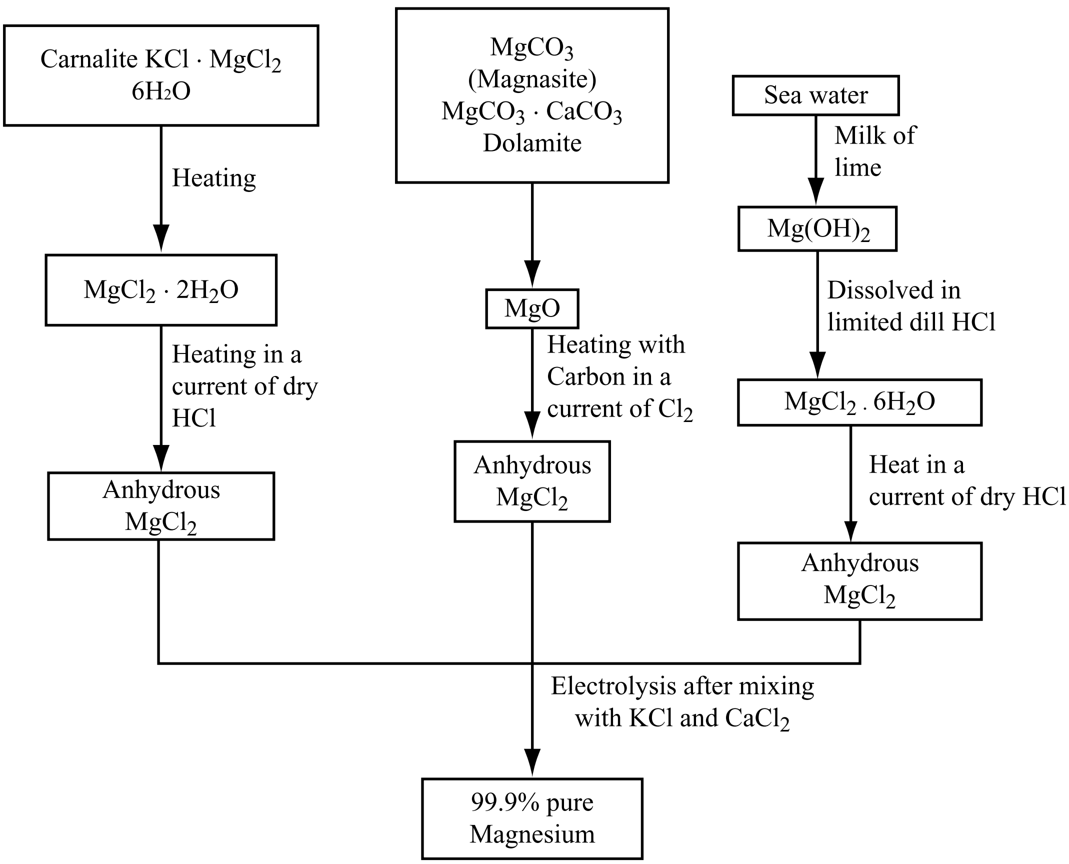


Fig 15.36 Preparation of magnesium from magnesia

Flow Chart for the Extraction of Magnesium



of the bath is about 950°C. A layer of the solidified electrolyte soon forms over the surface of the fused mass and this protects the metal from being oxidized. The magnesium metal collects under the protective layer and is removed periodically.

15.18.3 Aluminium

Though aluminium is the third most abundant element. It does not occur freely in nature.

Occurrence

Oxide ores	Corrundum	Al_2O_3
	Diaspore	$Al_2O_3 \cdot H_2O$
	Bauxite	$Al_2O_3 \cdot 2H_2O$
	Gibbsite	$Al_2O_3 \cdot 3H_2O$
Fluoride ore	Cryolite	Na_3AlF_6 or $3NaF \cdot AlF_3$
Basic sulphate ore	Alunite	$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 4Al(OH)_3$
Basic phosphate ore	Turquoise	$AlPO_4 \cdot Al_2O_3 \cdot H_2O$
Silicate ores	Felspar	$K_2O \cdot Al_2O_3 \cdot 6SiO_2$ (Or) $KAl\ Si_3O_8$
	Kaolin	$Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O$

Extraction

The metallurgy of aluminium involves three stages.

1. Refining of Bauxite: Bauxite has the appearance of red or brown clay containing the oxides of iron, silicon and titanium as impurities. They must be removed to get a good quality of aluminium. Purification method depends on the nature of impurities present in bauxite. If bauxite contains iron oxide as impurity it can be purified either by Baeyer's process or by Hall's process. When bauxite contains silica as impurity it is purified by Serpeck's process.

Baeyer's Process: Powdered bauxite is first roasted to convert ferrous oxide to ferric oxide. It is then digested with concentrated sodium hydroxide solution at 423k for few hours in an autoclave. Aluminium oxide reacts with sodium hydroxide and forms sodium meta –aluminates solutions while the impurity (Fe_2O_3) remains unaffected and hence removed by filtration.

$$Al_2O_3 \cdot 2H_2O + 2NaOH \longrightarrow 2NaAlO_2 + 3H_2O$$

The filtrate containing meta-aluminates is diluted with water and a crystal of $Al(OH)_3$ is added which acts as a seeding agent. Sodium meta-aluminates' undergoes hydrolysis and forms aluminium hydroxide.

$$NaAlO_2 + 2H_2O \rightarrow NaOH + Al(OH)_3 \downarrow$$

$Al(OH)_3$ is filtered off and washed. The filtrate containing caustic soda is concentrated and used again

Hall's Process: In this method bauxite ore is fused with sodium carbonate and sodium meta aluminate formed is extracted with water. Sodium meta aluminates' dissolves in water leaving behind the impurities. These can be removed by filtration.

$$Al_2O_3 \cdot 2H_2O + Na_2CO_3 \rightarrow 2NaAlO_2 + 2H_2O + CO_2$$

The solution is heated to about 50-60°C and a current of carbon dioxide is passed through when hydrated alumina separates out.

$$2NaAlO_2 + 3H_2O + CO_2 \rightarrow 2Al(OH)_3 + Na_2CO_3.$$

Serpeck's process: This method is employed to purify the bauxite which contains silicon dioxide as impurity. Powdered bauxite is mixed with coke and heated to 1800°C in a current of nitrogen. Aluminium nitride is formed while silicon dioxide is reduced to silicon which escapes out as vapour.

$$SiO_2 + 2C \xrightarrow{1800°C} Si\uparrow + 2CO\uparrow$$

$$Al_2O_3 + 3C + N_2 \xrightarrow{1800°C} AlN + 3CO$$

Aluminium nitride formed in the above reaction is subjected to hydrolysis to get aluminium hydroxide

$$AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3$$

2. Calcination of aluminium hydroxide: The aluminium hydroxide by the above three process is washed and then calcined at 1500°C in a rotary furnace similar to that compared in cement industry.

$$2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$$

Calcinations of aluminium hydroxide is carried because for the production of aluminium, pure non-hygroscopic anhydrous alumina is required.

Electrolysis of fused alumina

The alumina (purified above) is dissolved in molten cryolite ana is electrolysed in an iron tank lined inside with carbon. The composition of the electrolytic solution is Na_3AlF_6(80–85%) CaF_2(5–7%) AlF_3(5–7%) Al_2O_3(2–8%). The molten cryolite (Na_3AlF_6) decreases the melting point to about 1173k and also increases the electrical conductivity. This method is known as **Hall – Heroult's process** because it was developed by Hall and Heroult. The process

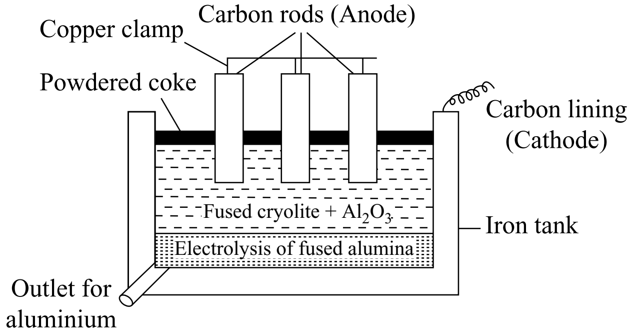


Fig 15.37 Electrolysis of fused alumina

of electrolysis is carried in an iron tank lined inside with carbon, which acts as the cathode. The anode consists of a number of carbon rods which dip in the molten electrolyte. The electrolyte is covered with a layer of powdered coke.

On passing electric current electrolysis occurs

At Cathode: $Al^{3+}(\text{melt}) + 3e^- \rightarrow Al.$

At anode: $O^{2-} \rightarrow O + 2e^-$

$C + O \rightarrow CO$

$CO + O \rightarrow CO_2$

The aluminium is liberated at the cathode and gets collected at the bottom of tank, from where it is removed periodically. The oxygen liberated at anode. combines with the carbon of the anode to produce carbon monoxide. CO either burns to CO_2 or escape out. For each kg of aluminium produced about 0.5kg of the carbon anode is burnt away. Therefore anodes need to be replaced periodically.

Refining of Aluminium: The aluminium metal obtained by the electrolysis of alumina is 99 per cent pure. It can be further refined by Hoop's electrolytic method. The process is carried out in an iron tank inside with carbon.

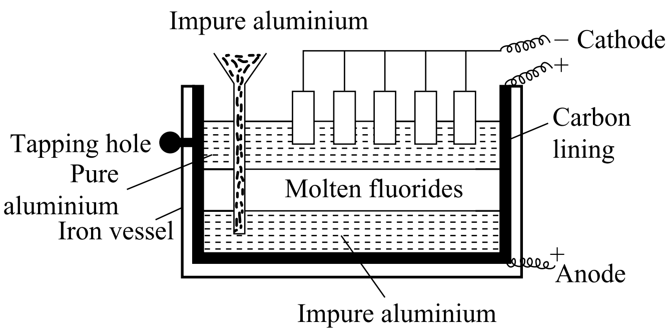


Fig 15.38 Purification of aluminium by Hoop's cell

The electrolyte is three layers of molten liquids having different densities.

(i) The top layer consists of pure aluminium having carbon electrodes dipping in it. This layer serves as cathode.

(ii) The middle layer has mixture of fluorides of sodium, barium and aluminium in the molten state. This acts as an electrolyte.

(iii) The bottom layer consist of impure aluminium which along with carbon lining acts as the anode.

On passing electric current, aluminium ions from the middle layer are discharged at the cathode as pure aluminium. The pure aluminium is removed from time to time from the tapping hole. An equivalent amount of aluminium from the bottom layer moves into the middle layer leaving behind the impurities. Thus this method gives completely pure aluminium.

15.19 POTASSIUM DICHROMATE $K_2Cr_2O_7$

Preparation: It is manufactured from chromite ore ($FeCr_2O_4$) The production of potassium dichromate from chromite ore involves the following steps.

1. **Conversion of chromite of sodium chromate:** The chromite ore is fused with sodium hydroxide or sodium carbonate in the presence of air

$$4FeCr_2O_4 + 16NaOH + 7O_2 \longrightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8H_2O$$

$$4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \longrightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$$

2. **Conversion of sodium chromate to sodium dichromate:** The yellow solution of sodium chromate is filtered and acidified with dilute sulphuric acid giving its dichromate

$$2Na_2CrO_4 + H_2SO_4 \longrightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$$

On cooling sodium sulphate crystallizes out as $Na_2SO_4 \cdot 10H_2O$ and is removed. The resulting solution contains sodium dichromate.

3. **Conversion of sodium dichromate into potassium dichromate:** Potassium dichromate is prepared by mixing a hot concentrated solution of sodium dichromate and potassium chloride in equimolar proportions.

$$Na_2Cr_2O_7 + 2KCl \longrightarrow K_2Cr_2O_7 + 2NaCl.$$

Sodium chloride being the least soluble precipitates out from the hot solution and is removed by filtration orange crystals of potassium dichromate separate out from mother liquor on cooling.

Properties: It is an orange red crystalline solid having melting point 670 k. It is moderately soluble in cold water but readily soluble in hot water on heating it decomposes to potassium chromate, chromic oxide and oxygen.

$$4K_2Cr_2O_7 \xrightarrow{\Delta} 4K_2CrO_4 + 2Cr_2O_3 + 3O_2$$

Action of alkalis: In alkaline medium orange colour of dichromate changes to yellow due to conversion into chromate ions

$$\underset{\text{orange}}{K_2Cr_2O_7} + 2KOH \longrightarrow \underset{\text{yellow}}{2K_2CrO_4} + H_2O$$

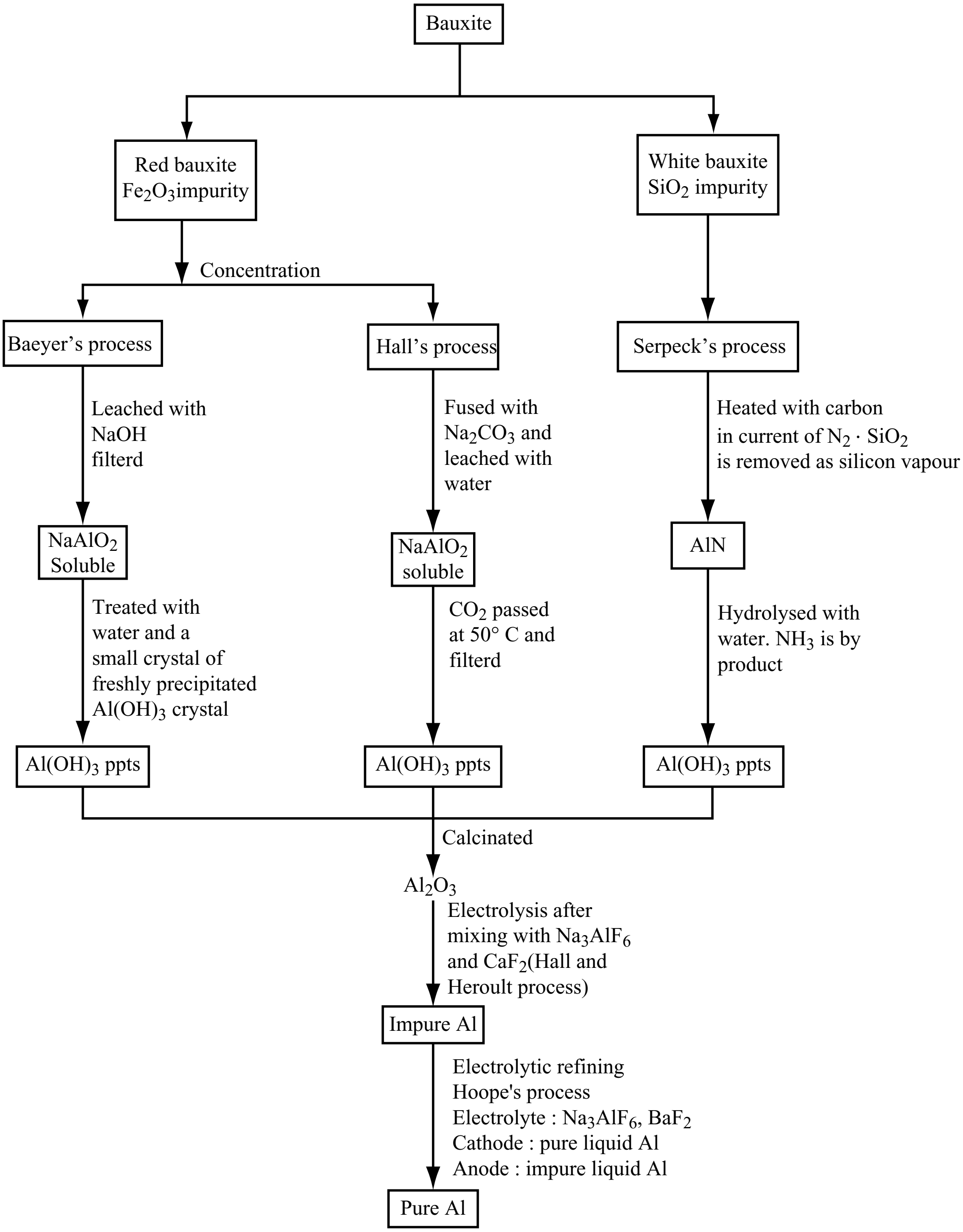
or $\quad Cr_2O_7^{2-} + 2OH^- \longrightarrow 2CrO_4^{2-} + H_2O$

on acidifying the yellow solution, the colour again changes to orange red due to reversible reaction

$$2K_2CrO_4 + H_2SO_4 \longrightarrow K_2Cr_2O_7 + K_2SO_4 + H_2O$$

(or) $\quad 2CrO_4^{2-} + 2H^+ \longrightarrow Cr_2O_7^{2-} + H_2O$

Flow Chart for the Extraction of Aluminum



Thus in a pH range 2–6 the dichromate and chromate ions exist in equilibrium which are inter convertible by changing the pH of the solution.

$$2CrO_4^{2-} + 2H^+ \underset{\text{alkali}}{\overset{\text{acid}}{\rightleftharpoons}} 2HCrO_4^- \underset{\text{alkali}}{\overset{\text{acid}}{\rightleftharpoons}} Cr_2O_7^{2-} + H_2O$$

Oxidation properties: The dichromate act as a good oxidizing agent in acidic medium. In the presence of dilute sulphuric acid $K_2Cr_2O_7$ liberates nascent oxygen and therefore acts as an oxidizing agent

$$K_2Cr_2O_7 + 4H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3(O)$$

$$\text{or } Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O \quad E^O = 1.33\text{ V.}$$

So as per the above reactions equivalent weight of potassium dichromate is M.W/6 = 294/6 = 49.

With concentrated sulphuric acid it gives chromic anhydride

$$K_2Cr_2O_7 + 2H_2SO_4 \longrightarrow 2CrO_3 + 2KHSO_4 + H_2O.$$

Both $Na_2Cr_2O_7$ and $K_2Cr_2O_7$ are oxidizing agents but $K_2Cr_2O_7$ is preferred since $Na_2Cr_2O_7$ is hygroscopic while $K_2Cr_2O_7$ is not and can be used as primary standard. Some of the oxidation reactions of potassium dichromate are as follows.

(i) It liberates iodine from potassium iodide

$$K_2Cr_2O_7 + 4H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3(O)$$

$$\underline{[2KI + H_2SO_4 + (O) \longrightarrow K_2SO_4 + H_2O + I_2] \times 3}$$

$$K_2Cr_2O_7 + 7H_2SO_4 + 6KI \longrightarrow 4K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3I_2$$

(ii) It oxidizes ferrous salts to ferric salts

$$K_2Cr_2O_7 + 4H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3(O)$$

$$\underline{[2FeSO_4 + H_2SO_4 + (O) \longrightarrow Fe_2(SO_4)_3 + H_2O] \times 3}$$

$$K_2Cr_2O_7 + 7H_2SO_4 + 6FeSO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3Fe_2(SO_4)_3 + 7H_2O$$

(iii) It oxidizes hydrogen sulphide to sulphur

$$K_2Cr_2O_7 + 4H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3(O)$$

$$\underline{[H_2S + (O) \longrightarrow H_2O + S] \times 3}$$

$$K_2Cr_2O_7 + 4H_2SO_4 + 3H_2S \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3S$$

(iv) It oxidizes sulphur dioxide to sulphuric acid or sulphites to sulphates.

$$K_2Cr_2O_7 + 4H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3(O)$$

$$\underline{[SO_2 + H_2O + (O) \longrightarrow H_2SO_4] \times 3}$$

$$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$$

$$K_2Cr_2O_7 + 4H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3(O)$$

$$\underline{[Na_2SO_3 + O \longrightarrow Na_2SO_4] \times 3}$$

$$K_2Cr_2O_7 + 4H_2SO_4 + 3Na_2SO_3 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3Na_2SO_4 + 4H_2O.$$

(v) It oxidizes concentrated hydrochloric acid and evolves chlorine.

$$K_2Cr_2O_7 + 8HCl \longrightarrow 2KCl + 2CrCl_3 + 4H_2O + 3(O)$$

$$\underline{[2HCl + O \longrightarrow H_2O + Cl_2] \times 3}$$

$$K_2Cr_2O_7 + 14HCl \longrightarrow 2KCl + 2CrCl_3 + 7H_2O + 3Cl_2$$

Because of thus reaction potassium dichromate cannot be used as oxidizing agent in volumetric analysis in hydrochloric acid medium. Some extra dichromate will be consumed by chloride ions.

It also oxidizes chlorides to chlorine, bromides to bromine, nitrites to nitrates etc.

Chromyl chloride test: When potassium dichromate is heated with concentrated sulphuric acid and a soluble metal chloride (e.g., NaCl) orange red vapours of chromyl chloride are evolved

$$K_2Cr_2O_7 + 4NaCl + 6H_2SO_4 \longrightarrow 2KHSO_4 + 4NaHSO_4 + 3H_2O + 2CrO_2Cl_2.$$

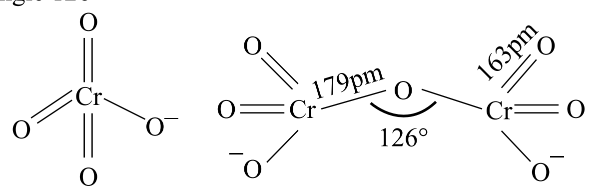
The orange red chromyl chloride vapours when passed into sodium hydroxide turns to yellow colour due to the formation sodium chromate, which gives yellow precipitate with lead acetate

$$CrO_2Cl_2 + 4NaOH \longrightarrow Na_2CrO_4 + 2NaCl + 2H_2O$$

$$Na_2CrO_4 + Pb(CH_3COO)_2 \longrightarrow \underset{\text{yellow}}{PbCrO_4} \downarrow + 2CH_3COONa$$

Structure of Chromate and Dichromate Ions

The chromate ion has tetrahedral structure in which four oxygen atoms around chromium atoms are oriented in tetrahedrons manner. The dichromate ion consists of two tetrahedrons sharing one corner with Cr–O–Cr bond angle 126°



In chromate ion all the Cr–O bond lengths are equal due to resonance. But in dichromate ion the bridge Cr–O bond lengths are longer than two terminal Cr–O bond lengths. All the terminal Cr–O bond lengths in dichromate ion are equal due to resonance

Uses: Potassium dichromate is used

(i) in dyeing and calico printing

(ii) in chrome tanning in leather industry

(iii) for the preparation of large number of chromium compounds such as chrome yellow [$PbCrO_4$], chrome red [$PbCrO_4 \cdot PbO$], chrome alum [$K_2SO_4 \cdot Cr_2(SO_4)_3 \cdot 24H_2O$], zinc yellow [$ZnCrO_4$] etc.

(iv) as volumetric reagent in laboratory for the estimation of ferrous ions, iodide ions etc.
(v) Chromic acid (a mixture of $K_2Cr_2O_7$ and Conc H_2SO_4) is used in cleaning the laboratory ware
(vi) In Organic chemistry as an oxidizing agent.

15.20 POTASSIUM PERMANGANATE $KMnO_4$

Preparation: It is manufactured from the mineral pyrolusite (black in colour). The manufacture involves the following steps.

(i) Conversion of pyrolusite ore to potassium manganate. The pyrolusite is fused with caustic potash or potassium carbonate in the presence of air or oxidizing agent such as potassium nitrate or potassium chlorate to give a green mass due to the formation or potassium manganate.

$$2MnO_2 + 4KOH + O_2 \xrightarrow{\Delta} \underset{\text{Green pot. manganate}}{2K_2MnO_4} + 2H_2O$$

$$2MnO_2 + 2K_2CO_3 + O_2 \xrightarrow{\Delta} 2K_2MnO_4 + 2CO_2$$

$$MnO_2 + 2KOH + KNO_3 \xrightarrow{\Delta} K_2MnO_4 + KNO_2 + H_2O$$

(ii) Oxidation of potassium manganate to potassium permanganate: The green mass is extracted with water resulting in green solution. The solution is then treated with a current of chlorine or ozone or carbon dioxide to oxidize potassium manganate to potassium permanganate. The solution is concentrated and the dark purple crystals of potassium permanganate separate out.

$$2K_2MnO_4 + Cl_2 \longrightarrow 2KMnO_4 + 2KCl$$

$$2K_2MnO_4 + O_3 + H_2O \longrightarrow 2KMnO_4 + 2KOH + O_2$$

$$3K_2MnO_4 + 2CO_2 \longrightarrow 2KMnO_4 + 2K_2CO_3 + MnO_2 \downarrow$$

Electrolytic Method: Potassium mangante can be alternatively oxidized by electrolytic method. The potassium manganate solution is taken in an electrolytic cell which contains iron cathode and nickel anode. The potassium manganate solution is taken in anodic compartment while dilute alkali solution is added in the cathode compartment. When current is passed, the manganate ion is oxidized to permanganate ion at anode and hydrogen is liberated at cathode.

$$2K_2MnO_4 \rightleftharpoons 2K^+ + MnO_4^{2-}$$

$$\text{at anode} \quad \underset{\text{green}}{MnO_4^{2-}} \longrightarrow \underset{\text{purple}}{MnO_4^-} + e^-$$

$$\text{at cathode} \quad \underset{\text{from water}}{2H^+} + 2e^- \longrightarrow H_2$$

Properties: It is a dark violet crystalline solid having metallic luster melting point 523k. It is moderately soluble in water forming purple solution.

Action of heat: When heated strongly it decomposes giving oxygen.

$$2KMnO_4 \longrightarrow K_2MnO_4 + MnO_2 + O_2$$

$$2K_2MnO_4 \longrightarrow \underset{\text{Pot.manganite}}{2K_2MnO_3} + O_2$$

Action of conc H_2SO_4: With concentrated sulphuric acid it forms highly explosive Mn_2O_7

$$2KMnO_4 + H_2SO_4 \longrightarrow K_2SO_4 + Mn_2O_7 + H_2O$$

$$2Mn_2O_7 \xrightarrow{\text{explodes}} 4MnO_2 + 3O_2$$

Action of alkalis: On heating with alkalis potassium permanganate changes into potassium manganate and oxygen is evolved

$$4KMnO_4 + 4KOH \longrightarrow 4K_2MnO_4 + 2H_2O + O_2.$$

Oxidizing character: Potassium permanganate is a powerful oxidizing agent in neutral, alkaline or acidic solutions because it liberates nascent oxygen as

Neutral solution: MnO_2 is formed

$$2KMnO_4 + H_2O \longrightarrow 2KOH + 2MnO_2 + 3O$$

or $$2MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$$

So equivalent weight of $KMnO_4$ in neutral medium is M.W/3 = 158/3= 52.67.

Alkaline medium: Manganate ions are formed

$$2KMnO_4 + 2KOH \longrightarrow 2K_2MnO_4 + H_2O + O$$

Potassium manganate is further reduced to MnO_2 in the presence of a reducing agent.

$$[K_2MnO_4 + H_2O \longrightarrow MnO_2 + 2KOH + (O)] \times 2$$

Therefore the complete reaction is

$$2KMnO_4 + 2H_2O \longrightarrow 2MnO_2 + 2KOH + 3(O)$$

or $$2MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$$

This is the same as in neutral medium, and so equivalent weight of $KMnO_4$ in alkaline medium is also MW/3 = 158/3 = 52.67

Acid Medium: Mn^{2+} ions are formed.

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$$

or $\quad MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

However $KMnO_4$ is used in acidic medium quite frequently in the laboratory. The important oxidation reactions are as follows.

Oxidation reactions in acidic medium

(i) Oxidizes ferrous salts to ferric salts

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$$
$$[2FeSO_4 + H_2SO_4 + (O) \longrightarrow Fe_2(SO_4)_3 + H_2O] \times 5$$
$$\overline{2KMnO_4 + 10FeSO_4 + 8H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 8H_2O}$$

(ii) Oxidizes acidified potassium iodide to iodine

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 \times 3H_2O + 5(O)$$
$$[2KI + H_2SO_4 + (O) \longrightarrow K_2SO_4 + H_2O + I_2] \times 5$$
$$\overline{2KMnO_4 + 8H_2SO_4 + 10KI \longrightarrow 6K_2SO_4 + 2MnSO_4 + 8H_2O + 5I_2}$$

(iii) Oxidizes H_2S to sulphur

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$$
$$[H_2S + (O) \longrightarrow H_2O + S] \times 5$$
$$\overline{2KMnO_4 + 3H_2SO_4 + 5H_2S \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5S}$$

(iv) Oxidizes sulphur dioxide to sulphuric acid

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$$
$$\left[SO_2 + H_2O + (O) \longrightarrow H_2SO_4\right] \times 5$$
$$\overline{2KMnO_4 + 5SO_4 + 2H_2O \longrightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4}$$

(v) Oxidizes nitrite to nitrate

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$$
$$[NaNO_2 + (O) \longrightarrow NaNO_3] \times 5$$
$$\overline{2KMnO_4 + 3H_2SO_4 + 5NaNO_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5NaNO_3}$$

(vi) Oxidizes oxalate to carbon dioxide

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$$
$$[H_2C_2O_4 + O \longrightarrow H_2O + 2CO_2] \times 5$$
$$\overline{2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 10CO_2}$$

(vii) Oxidizes hydrogen halides to halogens (X_2) X = Cl, Br or I

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$$
$$[2HX + (O) \longrightarrow H_2O + X_2] \times 5$$
$$\overline{2KMnO_4 + 3H_2SO_4 + 10HX \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5X_2}$$

Oxidation Reactions in neutral medium

(i) Oxidizes hydrogen sulphide to sulphur

$$2KMnO_4 + H_2O \longrightarrow 2KOH + 2MnO_2 + 3O$$
$$[H_2S + O \longrightarrow H_2O + S] \times 3$$
$$\overline{2KMnO_4 + 3H_2S \longrightarrow 2KOH + 2MnO_2 + 2H_2O + 3S}$$

(ii) Oxidizes manganous sulphate to manganese dioixide

$$2KMnO_4 + H_2O \longrightarrow 2KOH + 2MnO_2 + 3(O)$$
$$[MnSO_4 + H_2O + O \longrightarrow MnO_2 + H_2SO_4] \times 3$$
$$2KOH + H_2SO_4 \longrightarrow K_2SO_4 + 2H_2O$$
$$\overline{2KMnO_4 + 3MnSO_4 + 2H_2O \longrightarrow 5MnO_2 + K_2SO_4 + 2H_2SO_4}$$

(iii) Oxidizes sodium thiosulphate to sulphate and sulphur

$$2KMnO_4 + H_2O \longrightarrow 2KOH + 2MnO_2 + 3O$$
$$[Na_2S_2O_3 + O \longrightarrow Na_2SO_4 + S] \times 3$$
$$\overline{2KMnO_4 + 3Na_2S_2O_3 + H_2O \longrightarrow 2MnO_2 + 3Na_2SO_4 + 2KOH + 3S}$$

Oxidation reactions in alkaline medium

(i) Oxidizes iodides to iodates in alkaline medium

$$2KMnO_4 + H_2O \longrightarrow 3MnO_2 + 2KOH + 3O$$
$$KI + 3O \longrightarrow KIO_3$$
$$\overline{2KMnO_4 + KI + H_2O \longrightarrow 2MnO_2 + 2KOH + KIO_3}$$

(ii) Dilute alkaline permanganate is called Baeyer's Reagent oxidizes alkenes to glycols

$$\begin{matrix} CH_2 \\ \| \\ CH_2 \end{matrix} + H_2O + O \longrightarrow \begin{matrix} CH_2OH \\ | \\ CH_2OH \end{matrix}$$

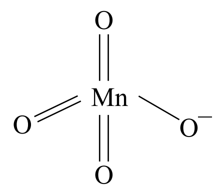
Ethyleneglycol

Uses: It is used

(i) as an oxidizing agent in the laboratory and industry
(ii) as a disinfectant for well water
(iii) in qualitative and quantitative analysis
(iv) Alakaline potassium permanganate is used in organic chemistry under the name Baeyer's reagent.

Structure

MnO_4^- ion is tetrahedral and all Mn–O bonds are covalent



All Mn–O bond lengths in MnO_4^- ion are equivalent due to resonance hybridization.

KEY POINTS

- Generally the metals and non-metals occur in nature in the form of compounds in which metals exists in positive oxidation states (M^{n+}) and non-metals exists in negative oxidation state (E^{n-}).
- Extraction of metals means reduction of the positive metal ions while the extraction of non-metals means oxidation of negative non-metal ions.
- A few metals like noble metals (Ag, Au pt etc) having least electropositive character occur in nature in native (free) state.
- The compounds of metals that occur in nature along with rocky and other impurities are called **minerals**.
- The rocky, sandy and siliceous impurities associated with minerals are called **gangue** or **matrix**.
- The mineral from which a metal can be extracted commercially (profitably) is called ore.
- Depending on the type of anion present in the ore, they are divided into different types (Refer the Table 15.1).
- Selection of a mineral as an ore depends on (i) percentage of the metal in the ore (ii) chemical nature of the ore (iii) Expenditure involved in the extraction of metal (iv) cost of the metal (v) value of by-products.

Concentration of the Ore or Ore Dressing

- The process of removal of gangue and increasing the percentage of ore is called ore dressing or concentration of the ore.
- **Hand picking** is the process of removal of sand and stones by hand e.g., The sand and stones from haematite are removed by hand.
- The **levigation method** depends upon the difference in specific gravity of the ore particles and impurities
- **Wilfley method** also depend on the difference in specific gravity method.
- **Magnetic separation** method depends on the difference in the magnetic properties of ore and impurities.
- If the ore or impurity is magnetic and the other is non-magnetic they can be separated by magnetic separation method.
- When the powdered ore is poured on a travelling belt which passes over a magnetic roller the ore and impurities are separated into different piles.
- The magnetic wolframite can be separated from cassiterite by magnetic separation method.
- The magnetic ores like magnetite (Fe_3O_4), chromite ($FeCr_2O_4$) are concentrated by magnetic separation method.
- **Liquation** method depends on the difference in the melting points of the ore and gangue.
- When an impure ore is heated on a sloped hearth the ore having low melting point melts and flows down leaving behind the impurities having high melting point.
- Stibnite is purified by liquation method.
- **Froth floatation** method is used for the concentration of sulphide ores.
- Pine oil or eucalyptus oil is added to form froth and are called frothers.
- The potassium ethyl xanthate or potassium amyl xanthate is added to enhance the non-wettability of the mineral particles are called **collectors**.
- Cresols and aniline which are added to stabilize the froth are called **froth stabilizers**.
- Froth floatation method depends on wettability of gangue particles and non-wettability of ore particles which are wetted by oil.
- The ore particles are collected into the froth while the gangue particles sink down.
- The substances which are added to prevent certain type of particles from forming the froth are called **depressants**.
- When sodium cyanide is added to a mixture of ZnS and PbS, NaCN forms a layer of zinc complex $Na_2[Zn(CN)_4]$ on the surface of ZnS and prevents the ZnS coming into froth. Thus sodium cyanide acts as depressant for ZnS and allows the PbS to form the froth.
- Copper sulphate acts as **activator** for ZnS and makes it to float.
- Separation of two different sulphide ores by adding a depressant is called differential flotation method.
- Leaching is a chemical method of concentration of ore. The ore is dissolved in certain acids, bases or suitable reagents while the impurities remain undissolved.
- Alumina containing iron oxide is concentrated by leaching method using NaOH or Na_2CO_3 in which alumina dissolves but iron oxide is unaffected.
- Ores of silver and gold are concentrated by leaching with sodium cyanide.

Fluxes and Slags

- Flux is a chemical substance that can react with infusible gangue to form a fusible mass called **slag**.
- Slag is a chemical compound formed by the reaction between flux and gangue

$$\text{Flux} + \text{gangue} \longrightarrow \text{Slag}$$

- **Acidic fluxes** are employed when the gangue is basic e.g., sand, sand stone, quartz

Gangue		Flux		Slag
FeO	+	SiO_2	$\longrightarrow$	$FeSiO_3$
CaO	+	SiO_2	$\longrightarrow$	$CaSiO_3$

- Basic fluxes are used when the gangue is acidic

Gangue		Flux		Slag
SiO_2	+	CaO	$\longrightarrow$	$CaSiO_3$
SiO_2	+	MgO	$\longrightarrow$	$MgSiO_3$

- Slag should be easily fusible and must flow.
- The density of slag is less than the molten metal.
- Slag is poor conductor of heat which prevents the super heating of metal.
- Slags are used in road surfacing, in the manufacture of cement, in the manufacture of fertilizers and in the construction of buildings.
- Thomas slag used as fertilizer is calcium phosphate $Ca_3(PO_4)_2$ and calcium silicate $CaSiO_3$.

Preliminary Treatment of the Concentrated Ore

- The concentrated ore must be converted into a form which is suitable for reduction to the metal.
- To convert the hydroxide or carbonate or sulphide ores to oxide ores they are subjected to different processes.
- **Calcination** is a process of strong heating the ore in the absence of air. So that part of the ore is removed in the form of volatile gas.
- Hydroxides and carbonates are subjected to calcination

$$2Al(OH)_3 \longrightarrow Al_2O_3 + 3H_2O$$
$$CaCO_3 \longrightarrow CaO + CO_2$$
$$ZnCO_3 \longrightarrow ZnO + CO_2$$

- Calcination is generally carried in reverberatory furnace.
- **Roasting** is a process of strongly heating the ore below to is melting point in the presence of air.
- In **oxidizing roasting** the sulphide ores are converted into oxides

$$2ZnS + 3O_2 \longrightarrow 2ZnO + 2SO_2$$
$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$
$$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$$

- **Sulphatizing** roasting is carried under controlled conditions where sulphide ores converted into sulphates which on strong roasting again converted into oxides

$$ZnS + 2O_2 \xrightarrow{650°C} ZnSO_4$$
$$2ZnSO_4 \xrightarrow{\Delta} 2ZnO + 2SO_2 + O_2.$$

- In **chloridizing roasting** sulphide ores are converted into chlorides

$$Ag_2S + 2NaCl + 2O_2 \xrightarrow{600°C} 2AgCl + Na_2SO_4$$

- Roasting is generally carried out in reverberatory furnace.
- Smelting is a process of the separation of the molten metal or molten sulphide ore from the impurities.
- **Smelting** is carried in blast furnace.
- Generally in smelting the ore is smelted with a flux and often with a reducing agent.
- The furnaces used for the preliminary treatment of the ore are made with refractory bricks covered outside with steel sheets. The different furnaces are (i) shaft furnace (ii) reverberatory furnace (iii) muffle furnace (iv) electric furnace (v) blast furnace and (vi) open hearth furnace.
- Reverberatory furnace contain three parts (1) fire box (2) hearth (3) chimney.
- The heat content present in the waste gases going out cannot be used repeatedly so the efficiency of the furnace is less.
- Reverberatory furnace is used in metallurgy of copper, steel and lead.
- **Blast furnace** is a cylindrical furnace about 100 feet height and 25 feet in diameter at the widest part. Blast furnace is constructed with wrought iron lined inside with fire bricks.
- The blast furnace consists of (i) mouth (ii) chimney (iii) body (iv) bosches (v) hearth.
- The mouth is provided with cup and cone arrangement to prevents the loss of heat.
- The waste gases goes out through chimney.
- The body is the longest part of the furnace where different zones of temperatures are maintained at which different types of reactions takes place.
- Bosches is widest part of the furnace at which nozzels called tuyers are present, through which hot air is blown.
- In the hearth the products are collected in the molten state.
- **Open hearth** furnace also known as regenerative furnace utilizes the heat content of fuel gases and waste gases to maximum limit.

Extraction of the Crude Metal

- Extraction of the metal from their ores is nothing but the reduction of their ores.
- Reduction of the ore is carried by various processes (i) chemical reduction (ii) Auto reduction (or) self reduction (iii) Hydrometallurgy or displacement method (iv) Electrolytic reduction.
- **Chemical reduction methods** are two types (i) carbon reduction and (ii) reduction using a reducing agent other than carbon.

- In the carbon reduction method carbon in form of coal, coke, charcoal and carbon monoxide are used as reducing agents

$$SnO_2 + 2C \longrightarrow Sn + 2CO$$
$$ZnO + C \longrightarrow Zn + CO$$
$$Fe_2O_3 + CO \xrightarrow{823\text{ K}} 2FeO + CO_2$$
$$FeO + CO \xrightarrow{1123\text{ K}} Fe + CO_2$$

- Chemical reduction by reducing agents other than carbon are of several types
  Reduction with (i) water gas (ii) hydrogen (iii) metals (iv) complex compounds (v) auto reduction

- Nickel is extracted by reducing the nickel oxide with water gas

$$2NiO + \underbrace{CO + H_2}_{\text{Water gas}} \xrightarrow{573\text{ K}} 2Ni + CO_2 + H_2O$$

- The oxides of heavier metals like molybdenum, tungsten etc are reduced by hydrogen.

$$MoO_3 + 3H_2 \longrightarrow Mo + 3H_2O$$
$$WO_3 + 3H_2 \longrightarrow W + 3H_2O$$

- Metals like chromium, manganese are obtained by reduction of their oxides with aluminium in **Gould Schmidt alumino thermi** process

$$Cr_2O_3 + 2Al \longrightarrow 2Cr + Al_2O_3$$
$$3Mn_3O_4 + 8Al \longrightarrow 9Mn + 4Al_2O_3$$
$$Fe_2O_3 + 2Al \longrightarrow 2Fe + Al_2O_3$$

- The reduction of haematite with aluminium is used in thermite welding.
- More electropositive metals like magnesium and calcium are used as reducing agents in the extraction of some elements

$$TiO_2 + 2Mg \longrightarrow Ti + 2MgO$$

- In **Auto reduction** or **air reduction** method a part of the ore is oxidized while the remaining part of the ore reduces the oxidized ore.
- Mercury, copper and lead are extracted by auto reduction method

$$HgS + O_2 \longrightarrow Hg + SO_2 \qquad \text{or}$$
$$2HgS + 3O_2 \longrightarrow 2HgO + 2SO_2$$
$$2HgO \longrightarrow 2Hg + O_2$$
$$\text{also} \quad 2HgO + HgS \longrightarrow 3Hg + SO_2$$
$$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$$
$$Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2$$
$$2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$$
$$PbS + 2PbO \longrightarrow 3Pb + SO_2$$

- Auto reduction is carried either ion Bessemer converter or in the pierce smith converter.
- More electro positive metals such as Na, Mg, Al reduces the oxides and halides of less electro positive metals such as $B_2O_3, SiO_2, TiCl_4, UCl_4$ etc.
- **Hydrometallurgy** is the treatment of the ore by a suitable chemical reagent to bring the metal into solution followed by the displacement of the metal by use of more electropositive metals.
- Silver and gold are extracted by hydrometallurgy or displacement method

$$4M + 8NaCN + 2H_2O + O_2 \longrightarrow 4Na[M(CN)_2] + 4NaOH$$
$$2Na[M(CN)_2] + Zn \longrightarrow Na_2[Zn(CN)_4] + 2M$$
$$(M = Ag \text{ or } Au)$$

- Highly electropositive metals of group 1 and 2 are extracted by electrolytic reduction method of their suitable fused salt.
- Electrolysis is carried after adding a suitable electrolyte to the fused chlorides of metals of group 1 and 2 to lower their melting point.
- The reactions taking place in the electrolytic cell may be

$$MX \longrightarrow M^+ + X^-$$
$$2X^- \longrightarrow X_2 + 2e^- \text{ (at anode)}$$
$$M + e^- \longrightarrow M \qquad \text{(at cathode)}$$

Refining of Crude Metals

- The crude metals extracted by various methods may contain impurities like the unreduced metal oxides and sulphides, the non-metals like silicon or phosphorus formed by the reduction in furnace, other metals formed due to simultaneous reduction of their oxides present in the ore and the substances which are added as flux, fuel etc, the slag formed.
- **Liquation** method is used to purify the metals having low melting points like tin from impurities having high melting points.
- When the impure metal having low melting point is heated on a sloped hearth, the metal melts and flows down leaving behind the impurities having high melting point.
- **Poling method** is used to purify the metal containing its own oxide as impurity.

- When the molten metal containing its own oxide as impurity is stirred with the poles of green wood, the hydrocarbon gases liberated from the green wood reduces the metal oxide.
- Copper and tin and purified by poling method.
- Cupellation method is used to purify silver containing lead.
- The silver containing lead when heated in a crucible made with bone ash, lead is selectively oxidized to litharge, a part of it was blown out and the remaining part will be absorbed by the crucible.
- **Distillation method** is used to purify the volatile metals like Zn, Hg etc from non-volatile impurities.
- **Electrolytic refining** is used to get pure metals like Cu, Ag, Au, Pb, Zn, Al etc.
- In the electrolytic refining method pure metal strip is taken as cathode, impure metal blocks are taken as anode and a suitable salt of the metal is taken as electrolyte. Pure metal deposits on the cathode leaving the impurities at anode known as anode mud. High pure metals can be obtained in this method.
- **Zone refining** method is used to get ultrapure metals.
- Germanium, silicon, boron, gallium and indium are purified by zone refining method.
- Zone refining method is based on the principle of fractional crystallization i.e. the impurities are more stable in the melt than in the pure metal.
- In the zone refining method when the molten impure metal in long boat is moved from high temperature the pure metal solidifies leaving the impurities in molten metal.
- Zone refining method is used to produce semiconductors like germanium.
- **Vapour phase refining** is used to purify nickel in Mond's process.
- When impure nickel is heated in a stream of carbon monoxide it forms volatile nickel carbonyl $Ni(CO)_4$ at low temperatures (330 – 350 k) which again decomposes at 450–470K

$$Ni + 4CO \xrightarrow{330-350\ K} Ni(CO)_4$$

$$Ni(CO)_4 \xrightarrow{450-470\ K} Ni + 4CO$$

- **Van Arkel** process is vapour phase refining in which when impure metal is heated with iodine vapour, the metal forms volatile metal iodide. The metal iodide vapour when heated over a tungsten filament at high temperature pure metal is obtained by decomposition.
- Zirconium is purified by Van Arkel method

$$Zr + 2I_2 \xrightarrow{870\ K} ZrI_4(\text{Vapour})$$

$$ZrI_4 \xrightarrow[\text{Tungsten filament}]{2075\ K} Zr + 2I_2$$

- **Chromatographic** method is based on the difference in the rate of adsorption of different substances in a mixture on an adsorbent.
- When a solution containing a mixture of substances is passed through a column packed with an adsorbent like Al_2O_3, the components in the mixture are adsorbed at different parts.
- Separation of bands of different elements by using a suitable solvent is known as **elution**.
- The solvent used for elution of adsorbed substance is called eluent.
- Chromatographic method is particularly suitable when the elements are in minute quantities and the impurities are not very much different in their chemical behaviour from the element of be purified.

Thermodynamic Principles of Metallurgy

- Ellingham diagrams helps us in predicting the feasibility of a thermal reduction of an ore.
- The criterion of feasibility is that at a given temperature change in Gibbs energy for the reaction must be negative.
- Ellingham digram consists of plots of $\Delta G°$ VsT for the formation of oxides of elements i.e, for the reaction

$$2xM(s) + O_2(g) \longrightarrow 2M_xO(s)$$

- The graphs for metal to metal oxide all slope upwards because the change in Gibbs energy becomes less negative with increase in temperature. This is because ΔS is negative for the reaction and hence $T\Delta S$ becomes more and more negative with increase in temperature.
- Each plot follows a straight line unless there is some change in phase (such as solid to liquid or liquid to gas) which results in large change in entropy and hence change the slope of line.
- When temperature is increased a point will be reached when the line crosses $\Delta G = 0$ line. Below this temperature $\Delta G°$ of oxide formation is negative, above which it is unstable since $\Delta G°$ is positive.
- The element below in Ellingham diagram can reduce the element above it but not in Vice Versa.
- For temperature at which C/CO line lies below the metal oxide line, carbon can be used to reduce the metal oxide and itself is oxidized to carbon monoxide.
- For temperature at which C/CO_2 line lies below the metal oxide line, carbon can be used to achieve the reduction, but is oxidized to carbon dioxide.
- For temperature at which the CO/CO_2 line lies below the metal oxide line, carbon monoxide can reduce the metal oxide to the metal and is oxidized to carbon dioxide.
- The conclusions of Ellingham diagrams are based on the assumption that reactants and products are in equilibrium which is not correct in several cases.

- The conclusions of Ellingham diagrams are derived on the basis of thermodynamic concepts only and kinetics of the reaction are not taken into consideration. So we can predict whether a reaction is feasible or not but cannot predict the rate of reaction.

IRON

- **Important ores**: Haematite (Fe_2O_3); Limonite ($2Fe_2O_3 \cdot 3H_2O$); Magnetite (Fe_3O_4); iron pyrites (FeS_2) and copper pyrites ($CuFeS_2$).
- Iron is mainly extracted from haematite by carbon reduction method.
- The concentrated ore (8 parts) is mixed with desulphurized coke (4 parts) and lime stone (1 part) is smelted in a blast furnace. Since the blast furnace has different temperatures at different zones, different reactions takes place at different zones.
- **Zone of combustion** (1500–1600°C) is near the bottom and little above the layers where the coke burns.

$$C + O_2 \longrightarrow CO_2$$
$$CO_2 + C \longrightarrow 2CO$$
$$2C + O_2 \longrightarrow 2CO$$

- **Zone of fusion** (1200–1500°C) is just above zone of combustion. Here the iron produced in upper zones melts and collected in the hearth while slag formed and floats over molten iron and prevent its oxidation by hot air blown.
- **Zone of slag formation** (800–1200°C): Here lime stone decomposes

$$CaCO_3 \longrightarrow CaO + CO_2$$

Lime combines with silica forming slag

$$CaO + SiO_2 \longrightarrow CaSiO_3$$

- In lower part of the blast furnace where the temperature is high certain impurities are reduced by carbon and get mixed with iron.

$$MnO_2 + 2C \longrightarrow Mn + 2CO$$
$$Ca_3(PO_4)_2 \longrightarrow 3CaO + P_2O_5$$
$$P_2O_5 + 5C \longrightarrow 2P + 5CO$$
$$SiO_2 + 2C \longrightarrow Si + 2CO$$

- **Zone of reduction** (300–400°C): Here heamatite is reduced to iron

$$2Fe_2O_3 + CO \longrightarrow 2Fe_3O_4 + CO_2 (300 - 400°C)$$
$$Fe_3O_4 + CO \longrightarrow 3FeO + CO_2 (680 - 700°C)$$
$$FeO + CO \longrightarrow Fe + CO_2 (700 - 800°C)$$

- The iron obtained in the zone of reduction is known as sponge iron.
- Iron collected in the hearth of blast furnace is called pig iron or cast iron.

Pig Iron or Cast Iron

- It is hard and brittle due to the presence of more carbon content.
- It expand on solidification, does not rust easily, cannot be welded, tempered and magnetized.
- The pig iron heated with scrap iron and coke in a blast of air is called cast iron. This contain 3–3.5 per cent carbon.

Wrought Iron

- It is prepared from pig iron by puddling process which involved the mixing of pig iron with haematite and puddling with steel rods

$$3C + Fe_2O_3 \longrightarrow 2Fe + 3CO$$
$$3Mn + Fe_2O_3 \longrightarrow 2Fe + 3MnO$$
$$3Si + 2Fe_2O_3 \longrightarrow 4Fe + 3SiO_2$$
$$MnO + SiO_2 \longrightarrow MnSiO_3.$$

- It is soft, malleable, ductile, can be tempered, magnetized, forged and welded.

Steel

- In Bessemer process it is prepared by flowing air through molten pig iron taken in Bessemer converter Impurities like C, Si, P and Mn and oxidized which are removed as slag

$$Si + O_2 \longrightarrow SiO_2$$
$$2Mn + O_2 \longrightarrow 2MnO$$
$$MnO + SiO_2 \longrightarrow MnSiO_3$$
$$4P + 5O_2 \longrightarrow P_4O_{10}$$
$$P_4O_{10} + 6CaO \longrightarrow 2Ca_3(PO_4)_2$$

- The basic slag containing $Ca_3(PO_4)_2$ is known as **Thomas slag** used as phosphatic fertilizer.
- In **Open hearth process** molten pig iron is heated in the hearth of reverberatory furnace in the presence of air. Impurities are oxidized and react with the lining of the hearth forming slag.
- The difference between pig iron, wrought iron and steel is only in carbon content and impurities.
- Cast or pig iron contains 4.3 per cent carbon with other impurities like Si, P, S, and Mn.

- Wrought iron is the purest form of iron containing 0.15–0.25 per cent carbon with less than 0.5 per cent of other impurities.
- Steel contain less than 2 per cent carbon.
- Different steels are made by mixing it with different metals like Ni, Mn, Cr, W etc.
- The steel obtained in open hearth furnace is of high quality.
- In the open hearth furnace the composition of steel can be controlled, iron ore, scrap iron and low grade pig iron can be used.
- In open hearth furnace since no blast of air is used there is no loss of iron due to slag formation.
- In open hearth furnace since the heat content present in the waste gases is used repeatedly with more economic.

Heat Treatment of Steel

- **Quenching** or **hardening** of steel is a process of heating steel to red hot and then suddenly plunging in cold water or oil. The steel becomes hard and brittle.
- **Tempering** is a process of heating the quenched steel at about 550 k for some time and then cooled slowly Tempering makes the steel hard but less brittle.
- **Annealing** is a process of heating quenched steel below red hot condition and then allowed to cool slowly. Annealing makes the steel soft.
- **Case hardening** is a process of heating mild steel in charcoal and then plunging into oil. Then a thin film of hardened steel is formed on the surface.
- Production of hard iron nitride coating on the surface of steel is called **nitriding**.
- Nitriding is carried by heating steel in the atmosphere of dry ammonia at 500–600°C for about 3–4 days.
- When iron is dipped in concentrated nitric acid, it becomes passive due to the formation of thin film of iron oxide on the surface.
- Passive iron cannot reduce copper sulphate and cannot liberate hydrogen from acids.
- Passive iron can be made active by scratching or scrubbing or by dissolving in iodine solution or by heating in charcoal.

COMPOUNDS OF IRON

Iron (III) Oxide

- It occurs naturally as heamatite and can be prepared by heating Fe $(OH)_3$ or FeS_2 or $FeSO_4$.
- Heating $FeSO_4$ gives Fe_2O_3, SO_2 and SO_3.
- It is amphoteric on heating at 1300°C converts into magnetic oxide (Fe_3O_4).
- It is used as red pigment under the name vanatian red. Also used to polish the Jewellary under the name Jeweller's rouge.
- Magnetic oxide contains Fe^{2+} and Fe^{3+} ion in 1:2 ratio.

Ferric Chloride

- FeF_3, $FeCl_3$ and $FeBr_3$ can be prepared by heating iron with halogen.
- FeI_3 cannot be prepared because Fe^{3+} act as oxidizing agent and I^- act as reducing agent.
- Anhydrous $FeCl_3$ can be prepared by passing dry chlorine gas over heated iron fillings.
- Hydrated ferric chloride is prepared by dissolving $Fe_2(CO_3)_3$, $Fe(OH)_3$ or Fe_2O_3 in dil HCl. It crystallizes as $FeCl_3 \cdot 6H_2O$.
- Anhydrous $FeCl_3$ cannot be prepared from hydrated salt by heating since it hydrolyses with its own water of crystallization forming $Fe(OH)_3$ or Fe_2O_3 depending on temperature.
- Anhydrous $FeCl_3$ can be prepared from hydrated salt using dehydrating agent such as thionyl chloride or an ether like 2, 2– dimethoxy propane $(CH_3)_2C(OCH_3)_2$
- Anhydrous $FeCl_3$ is covalent soluble in organic solvents. Its aqueous solution is acidic in nature due to hydrolysis. It fumes in moist air.
- It is a good oxidizing agent oxidizes H_2S to S; $SnCl_2$ to $SnCl_4$, SO_2 to H_2SO_4 and liberates I_2 from KI.
- It gives reddish brown precipitate of $Fe(OH)_3$ with NaOH or NH_4OH, insoluble in excess of reagent.
- It gives blood red colour with NH_4SCN or KSCN due to formation of Fe (SCN) Cl_2 or Fe $(SCN)_3$.
- With $K_4[Fe(CN)_6]$ it forms a blue complex $Fe_4[Fe(CN)_6]$ known as Prussian blue.
- Solid anhydrous $FeCl_3$ has layered lattice structure in which each iron atom is surrounded by six chlorine atoms
- In vapour phase it exist as dimer Fe_2Cl_6 but becomes monomer at high temperature.
- In $FeCl_3 \cdot 6H_2O$ four H_2O molecules and two Cl^- ions coordinate to Fe^{3+} in octahedral manner. The two Cl^- ions are in trans position $[Fe(H_2O)_4Cl_2]Cl \cdot 2H_2O$.

Iron (II) Oxide FeO

- It is obtained as a black solid by heating ferrous oxalate since CO produced in the reaction provide a reducing atmosphere.
- It has sodium chloride structure. It is non-stoichiometric deficient of Fe^{2+} ions.
- It is stable at high temperature (in inert atmosphere) but disproportionate to Fe and Fe_3O_4 on cooling to room temperature.

Iron (II) Halides

- FeF_2 and $FeCl_2$ can be prepared by passing a stream of dry HF and HCl over the heated metal respectively. $FeBr_2$ can be prepared by heating excess of iron with bromine. FeI_2 can be made by direct heating of iron with iodine.

Iron (II) Sulphate $FeSO_4 \cdot 7H_2O$ Green Vitriol

- Ferrous sulphare can be prepared by dissolving iron metal or $FeCO_3$ or FeS in dilute H_2SO_4.
- It is manufactured by slow oxidation of iron pyrites in the presence of air and moisture

$$2FeS_2 + 2H_2O + 7O_2 \longrightarrow 2FeSO_4 + 2H_2SO_4$$

- Hydrated $FeSO_4 \cdot 7H_2O$ is green in colour while anhydrous $FeSO_4$ is colourless or white. When exposed to air it turns to brownish yellow due to formation of basic ferrous sulphate $Fe(OH)SO_4$.
- At high temperature it decomposes liberating SO_2 and SO_3 gases leaving behind brown Fe_2O_3.
- It is a good reducing agent, reduces acidified $KMnO_4$, acidified $K_2Cr_2O_7$ quantitatives. It reduces $AuCl_3$, to gold, $HgCl_2$ to Hg_2Cl_2.
- In aqueous solution with nitric oxide form brown complex $[Fe(H_2O)_5NO]^{2+}$.
- With $K_3[Fe(CN)_6]$ it forms blue precipitate $Fe_3[Fe(CN)_6]_2$ known as Turn Bulls blue.
- Prussian blue and Turn Bulls blue are found to be same and is proposed that when Fe^{2+} ion is added to $[Fe(CN)_6]^{3-}$ first Fe^{2+} is oxidized to Fe^{3+} by $[Fe(CN)_6]^{3-}$ and converts itself to $[Fe(CN)_6]^{4-}$.
- Mohr's salt is a double salt $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ It resists the oxidation by air than $FeSO_4$.

COPPER

- Important ores (i) Cuprite (Cu_2O) (ii) Malachite $CuCO_3 \cdot Cu(OH)_2$ (iii) Azurite $2CuCO_3 \cdot Cu(OH)_2$ (iv) Copper pyrites or Chalcopyrites ($CuFeS_2$) (v) Copper glance (Cu_2S).
- Copper is extracted either by pyrometallurgical process or by hydrometallurgical process (wet process).

Pyrometallurgy

- The ore is concentrated by froth floatation method and then roasted to remove. As, Sb, S etc as their volatile oxides.
- The roasted ore is smelted in a reverboratory furnace after mixing with silica as flux to remove part of iron as ferrous silicate.
- The molten ore $Cu_2S \cdot FeS$ is called **matte**.
- From the matte, copper is extracted in Bessemer converter by self reduction or auto reduction method by blowing a blast or sand and air from the middle portion of Bessemer converter. Iron is removed in the form of ferrous silicate.
- Copper metal is purified first by poling to remove copper oxide and then finally by electrolytic refining Electrolyte is 3.5 per cent $CuSO_4$ + 15 per cent H_2SO_4. Cathode is pure copper and anode is impure copper.

Hydrometallurgy

- The powdered ore is exposed to air and water, copper sulphide is oxidized to copper sulphate. Some iron sulphate and sulphuric acid are also formed.
- The non-sulphide ores are powdered and leached with dilute sulphuric acid

$$CuCO_3 \cdot Cu(OH)_2 + 2H_2SO_4 \longrightarrow 2CuSO_4 + CO_2 + 3H_2O$$

- Copper is precipitated by adding scrap iron to copper sulphate solution.
- When copper is exposed to moist air a green coating of basic copper carbonate is formed on the surface.
- In the presence of air copper dissolves in aqueous ammonia forming blue complex

$$2Cu + 4NH_4OH + O_2 \longrightarrow 2[Cu(NH_3)_4](OH)_2 + 3H_2O$$

COMPOUNDS OF COPPER

Copper (I) Compounds

- In aqueous solution Cu^+ is unstable and disproportionate to Cu and Cu^{2+}

$$2Cu^+ \rightleftharpoons Cu + Cu^{2+}$$

The equilibrium constant is very high (10^6 dm^3 mol^{-1}). The equilibrium can be shifted to the left by adding anions which precipitate Cu^+ or by adding a substance that can form more stable complex with Cu^+ such as NH_3, CN^- etc.
- **Copper (I) oxide** can be obtained as red solid by reduction of alkaline $CuSO_4$ complexed with tartarate ions.
- Cu_2O dissolves in conc HCl with formation of $[CuCl_2]^-$ complex ion.
- Cu_2O is covalent, each oxygen atom is surrounded by four copper atoms and each copper atom lying midway between two oxygen atoms.
- The red colour of Cu_2O is due to charge transfer.
- **Copper (I) chloride** can be prepared by boiling a solution of $CuCl_2$ with excess of copper in concentrated hydrochloric acid which gives $[CuCl_2]^-$ complex ion. This on dilution with water precipitates Cu_2Cl_2.

- Cu_2Cl_2 is essentially covalent, similar in structure to diamond in which each Cl atom is surrounded by four Cu atoms tetrahedrally and each Cu atom is surrounded by four Cl atoms tetrahedrally.
- Cu_2Cl_2 is insoluble in water but dissolves in the presence of complexing agents such as $Cl^-, S_2O_3^{2-}$ and NH_3.
- A mixture of Cu_2Cl_2 and NH_4Cl is used as catalyst in the dimerization of ethyne to get vinyl acetylene.
- Ammonical Cu_2Cl_2 absorbs carbon monoxide and precipitates copper (I) carbide when ethyne is bubbled through it.

Copper (II) Compounds

- Copper (II) oxide can be obtained as black solid by heating $CuCO_3$ or $Cu(NO_3)_2$.
- At 800°C CuO decomposes to Cu_2O and O_2.
- Copper (II) chloride is dark brown solid, can be prepared by passing chlorine gas over heated copper.
- It is covalent and have layered structure in which each copper atom is surrounded by four chlorine atoms in the same layer and two chlorine atoms between the layers.
- The four Cu – Cl distances in a layer are shorter than the Cu – Cl distances between the layers.
- The colour of $CuCl_2$ in aqueous solution changes from Brown to Green to Blue due to different number of Cl^- ions and H_2O molecules in complex.
- Crystallization from water gives green $CuCl_2 \cdot 2H_2O$.

Copper Sulphate $CuSO_4 \cdot 5H_2O$ Blue Vitriol

- It is prepared by dissolving CuO or $CuCO_3 \cdot Cu(OH)_2$ in dilute H_2SO_4. It is obtained by dissolving scrap copper in dilute H_2SO_4 in the presence of air. It is also obtained by slow oxidation of $CuFeS_2$.
- It is efflorescent and turns to pale blue $CuSO_4 \cdot 3H_2O$ when exposed to air. It becomes white anhydrous $CuSO_4$ on heating which can regain blue colour by absorbing moisture. This is used to test the presence of moisture.
- With KCN first it gives cupric cyanide $Cu(CN)_2$ which decomposes into cuprous cyanide $Cu_2(CN)_2$ and cyanogen $(CN)_2$. The $Cu_2(CN)_2$ then dissolves in excess of KCN due to the formation of complex $K_3[Cu(CN)_4]$.
- With ammonium hydroxide it forms blue coloured complex $[Cu(NH_3)_4]^{2+}$ known as **Schweitzer's reagent** used as solvent in the manufacture of artificial silk rayon.
- With potassium iodide first it forms CuI_2 which decomposes to Cu_2I_2 and I_2.
- In the presence of potassium thiocyanate, it is reduced by SO_2 and precipitated as cuprous thiocyanate (CuSCN).
- A mixture of $CuSO_4$ and milk of lime is known as **Bordeaux mixture** used to kill moulds and fungi.

SILVER

- **Important ores** are (i) Horn silver or chlorargirite (AgCl); (ii) Argentite or silver glance (Ag_2S) (iii) pyrargirite or ruby silver ($3Ag_2S \cdot Sb_2S_3$).
- Silver is extracted by Mac Arthur-Forrest cyanide process i.e. hydrometallurgy.
- The sulphide is concentrated by froth flotation process.
- The powdered concentrated ore is leached with 0.5 per cent NaCN solution

$$Ag_2S + 4NaCN \rightleftharpoons 2Na[Ag(CN)_2] + Na_2S$$

To prevent the back ward reaction air is blown to convert Na_2S into $Na_2S_2O_3$ and Na_2SO_4.
- The metallic silver and AgCl present in the ore also goes into solution by leaching with NaCN solution as $Na[Ag(CN)_2]$.
- From the complex cyanide solution silver is precipitated by adding more electropositive metals like Al or Zn.
- Silver is purified by electrolytic refining. Electrolyte is $AgNO_3$ solution containing 1 per cent HNO_3, anode is impure silver and cathode is pure silver.

Silver Nitrate

- It is prepared by dissolving silver in dilute HNO_3.
- It decomposes on heating leaving silver metal as residue

$$2AgNO_3 \longrightarrow 2Ag + 2NO_2 + O_2.$$

- It also decomposes in a similar way in the presence of sun light or when it falls on the skin or cloths. It is caustic in property and forms black stains of silver on skin, so it is known as Lunar caustic.
- With chloride it gives white curdy precipitate AgCl soluble in ammonia, with bromide forms pale yellow precipitate AgBr partially soluble in ammonia and with iodide gives yellow precipitate AgI insoluble in ammonia.
- With K_2CrO_4 gives red precipitate of Ag_2CrO_4 and with Na_3PO_4 gives yellow precipitate of Ag_3PO_4.
- With excess iodine it gives AgI and HIO_3 while with excess $AgNO_3$ gives AgI and $AgIO_3$.
- With ammonia forms a complex $[Ag(NH_3)_2]NO_3$ known as Tollen's reagent.
- It is used in the preparation of indelible ink and in silvering of mirrors.

Silver Bromide

- It is yellowish white solid insoluble in water but dissolves in KCN and hypo.
- AgBr is coated on photographic and x-ray films as it is photosensitive.

$$2AgBr \xrightarrow{\text{light}} 2Ag + Br_2$$

- The unreacted AgBr on photographic films is removed as a soluble complex with hypo during developing.

$$AgBr + 2Na_2S_2O_3 \longrightarrow Na_3[Ag(S_2O_3)_2] + NaBr$$

GOLD

- Gold occurs mainly in native state as alluvial gold (in sand, mud or gravel in the beds of river).
- The concentrated ore is leached with 0.5 per cent KCN solution in the presence of atmospheric oxygen. Then gold converts into soluble complex.

$$4Au + 8KCN + 2H_2O + O_2 \longrightarrow 4K[Au(CN)_2] + 4KOH$$

From the solution gold is precipitated by adding more electropositive metals like Al or Zn.

- High pure gold is prepared by electrolytic refining. Electrolyte is acidified $AuCl_3$ solution, anode is impure gold and cathode is pure gold.
- Gold dissolves in aquaregia.
- Gold (III) chloride or auric chloride $AuCl_3$ can be prepared by passing chlorine gas over finely divided gold.
- It decomposes to AuCl and Cl_2 on heating.
- Purple of cassius is a colloidal solution of gold obtained by reduction of $AuCl_3$ with $SnCl_2$ used in gold lining of ceramics.

ZINC

- Important ores are (i) zinc blende (ZnS) (ii) calamine ($ZnCO_3$) (ii) willimite ($ZnSiO_4$) (iv) zincite (ZnO).
- The ore is concentrated by froth flotation method and then roasted to convert into oxide.
- The oxide on reduction with carbon gives zinc.
- Since the reduction of ZnO with carbon is endothermic, reduction is carried at high temperature (1400°C) and to prevent the backward reaction excess of carbon is used.
- Zinc is first purified by distillation and then by electrolytic refining. Electrolyte is acidified $ZnSO_4$ solution, anode is impure zinc and cathode is pure zinc.

Zinc Oxide (zinc white) ZnO

- It is obtained by heating. zinc in air or by heating zinc nitrate or zinc carbonate or by roasting zinc sulphide.
- Zinc oxide formed by burning zinc in air condenses as wooly material which is known as philosopher's wool.
- It is white when cold but yellow when hot.
- It is amphoteric oxide and dissolves in both acids and bases.
- When heated with cobalt nitrate it forms green mass cobalt zincate ($CoZnO_2$) known as Rinmann's green.

Zinc Sulphide

- It occurs in nature as ZnS.
- It can be prepared by passing H_2S gas in to the aqueous solutions of zinc salts.

$$ZnSO_4 + H_2S \longrightarrow ZnS + H_2SO_4$$

- It is phosphorescent and is used as luminous paint on watches.
- A mixture of ZnS + $BaSO_4$ is known as lithophone used as white paint.

Zinc Sulphate $ZnSO_4 \cdot 7H_2O$ White Vitriol

- It is prepared by dissolving zinc or ZnO or $ZnCO_3$ in dilute H_2SO_4.
- It is an efflorescent substance.
- With NaOH first gives white precipitate which dissolves in excess of NaOH due to formation of $Na[Zn(OH)_2]$.
- With NH_4OH first gives white precipitate which dissolves in excess of NH_4OH due to formation of complex $[Zn(NH_3)_4]^{2+}$.
- With sodium carbonate it forms basic carbonate $ZnCO_3 \cdot 3Zn(OH)_2$ but with sodium bicarbonate gives normal carbonate.
- With $K_4[Fe(CN)_6]$ it forms white precipitate of $Zn_2[Fe(CN)_6]$.

Zinc Chloride

- It is prepared by dissolving zinc or ZnO or $ZnCO_3$ in dilute HCl. Crystallization from solution gives $ZnCl_2 \cdot 2H_2O$.
- Anhydrous $ZnCl_2$ cannot be prepared by heating since it hydrolyses with its own water of crystallization forming Zn(OH)Cl.
- Anhydrous $ZnCl_2$ can be prepared by heating zinc with chlorine gas or $HgCl_2$.
- It is highly deliquescent highly soluble in water and also soluble in alcohol and acetone indicating its covalent nature.
- It is a very good dehydrating agent.
- It can accept electron pairs and can form bonds with OH groups present in cellulose of cotton and timber

so it can form a coating on timber being toxic preserve the timber. Because of this reason it cannot be filtered through paper.

MERCURY

- The only important ore of mercury is cinnabar HgS.
- When concentrated ore is roasted in shaft furnace in excess of air, vapours of mercury formed are condensed as liquid which is about 99.5 per cent.

$$HgS + O_2 \longrightarrow Hg + SO_2.$$

- Mercury is purified by distillation under reduced pressure.
- Dil HCl and dil H_2SO_4 has no reaction with mercury.
- With dil HNO_3, $Hg_2(NO_3)_2$ and with conc HNO_3, $Hg(NO_3)_2$ are formed.
- With conc H_2SO_4 it gives $HgSO_4$.
- Iron, cobalt, nickel, platinum do not form amalgams with mercury.

Mercuric Oxide (HgO)

- On heating mercury at about 623 k for long time red variety of HgO is formed

$$2Hg + O_2 \longrightarrow 2HgO$$

- It can also be prepared by heating mercuric nitrate.
- Yellow variety of HgO is obtained by adding sodium hydroxide solution to mercuric chloride.
- Red and yellow varieties differ only in particle size particles of yellow variety are smaller in size. Yellow form changes to red form on heating to 673 k.
- On strong heating it decomposes to Hg and O_2.
- It is used as red pigment in oil paints and mild antiseptic ointments.

Mercurous Chloride, Hg_2Cl_2, Calomel

- It can be prepared by the action of dil HCl or NaCl on $Hg_2(NO_3)_2$.
- It is also formed by subliming a mixture of $HgCl_2$ and Hg in iron vessel.
- It sublimes on heating and then decomposes to Hg and $HgCl_2$.
- With ammonia it forms black precipitate of Hg and NH_2HgCl.
- It is used as purgative in medicine and in making standard electrode.

Mercuric Chloride $HgCl_2$, Corrosive Sublimate

- It can be prepared by heating mercury with chlorine.
- On large scale it is prepared by heating a mixture of $HgSO_4$, NaCl and $MnO_2 \cdot MnO_2$ prevents the formation of mercurous chloride.
- It is sparingly soluble in cold water but is soluble in hot water readily soluble in organic solvents due to covalent nature.
- It is reduced first to Hg_2Cl_2 and then to Hg with $SnCl_2$.
- With ammonia it forms black precipitate of NH_2HgCl.

Mercuric Iodide HgI_2

- It is prepared by adding slowly aqueous solution of potassium iodide to an aqueous solution of mercuric chloride

$$HgCl_2 + 2KI \longrightarrow HgI_2 + 2KCl.$$

- Below 400k it exists as red solid while above 400 k it exist as yellow solid.
- It dissolves in excess of potassium iodide forming soluble complex $K_2[HgIu]$. Its alkaline solution known as Nessler's reagent is used for the detection of ammonium ion

$$HgI_2 + 2KI \longrightarrow K_2[HgI_4]$$

Potassium Dichromate $K_2Cr_2O_7$

- It is prepared by fusing chromite with caustic soda or sodium carbonate in the presence of air

$$4FeCrO_4 + 16NaOH + 7O_2 \longrightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8H_2O$$

$$4FeCrO_4 + 8Na_2CO_3 + 7O_2 \longrightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8CO_2$$

- Acidification of Na_2CrO_4 converts into $Na_2Cr_2O_7$ which converts into $K_2Cr_2O_7$ by heating with KCl.
- $K_2Cr_2O_7$ is preferred to $Na_2Cr_2O_7$ in volumetric analysis because $Na_2Cr_2O_7$ is hygroscopic while $K_2Cr_2O_7$ is not.
- It is orange red in colour, moderately soluble in cold water but appreciably soluble in hot water.
- It exist as K_2CrO_4 in alkaline medium but exist as $K_2Cr_2O_7$ in acid medium. In a pH range 2–6 both exist in equilibrium.
- On heating it decomposes into K_2CrO_4, Cr_2O_3 and oxygen.
- With conc H_2SO_4 it gives chromic anhydride CrO_3.
- It is a good oxidizing agent and oxidizes several compounds, it self reduced to chromic salt, Equivalent weight is M.W/6.
- It oxidizes $FeSO_4$ to $Fe_2(SO_4)_3$; H_2S to S; SO_2 to H_2SO_4; iodide to iodine.
- When solid dichromate is heated with soluble ionic chloride and conc H_2SO_4 orange red oily vapours of chromyl chloride (CrO_2Cl_2) will be liberated. These vapours give yellow solution with NaOH due to formation of Na_2CrO_4. This gives yellow precipitate with lead salt. This total reactions are called chromyl chloride test for chloride ion.

- Chromate ion is tetrahedral while dichromate ion contain two tetrahedrons joined through oxygen.
- In CrO_4^{2-} all Cr–O bond lengths are equal due to resonance but in $Cr_2O_7^{2-}$ the six terminal Cr–O bond lengths are equivalent. In both CrO_4^{2-} and $Cr_2O_7^{2-}$ chromium is involved in d^3s hybridization.

Potassium Permanganate $KMnO_4$

- When pyrolusite (black MnO_2) is fused with KOH or K_2CO_3 in presence of air or oxidizing agents like KNO_3 green mass K_2MnO_4 will be formed.
- The K_2MnO_4 can be converted into $KMnO_4$ by oxidation with Cl_2 or O_3 or by passing CO_2 into K_2MnO_4 solution. K_2MnO_4 can also be oxidized to $KMnO_4$ electrolytically.
- On heating it liberates oxygen converting itself to K_2MnO_4 and K_2MnO_3.
- With conc H_2SO_4 it forms explosive Mn_2O_7.
- It is a good oxidizing agent in acid neutral and basic media.
- In acid media it is reduced to Mn^{2+} involving 5 electron change Eq. Wt = M.W/5.
- In neutral and basic media it is reduced to MnO_2 involving 3 electron change Eq. Wt = M.W/3.
- In acid medium it oxidizes $FeSO_4$ to $Fe_2(SO_4)_3$, oxalic acid CO_2, H_2S to S, SO_2 to H_2SO_4, KI to I_2.
- In alkaline medium it oxidizes KI to KIO_3.
- $K_2Cr_2O_7$ and $KMnO_4$ both oxidize HCl to Cl_2 and hence oxidation reactions involving $K_2Cr_2O_7$ or $KMnO_4$ cannot be carried in HCl acid medium.
- MnO_4^- ion is tetrahedral and all Mn–O bonds are covalent and equal in length due to resonance. Manganese is involved in d^3s hybridization.

Single Answer Type Questions

1. Identity x, y, z for the following metallurgical process.
Metal sulphide $\xrightarrow{X}$ Metal oxide $\xrightarrow{Y}$ Impure metal $\xrightarrow{Z}$ Pure metal
(a) Roasting, smelting, electrolysis
(b) Roasting, calcination, smelting
(c) Roasting, auto-reduction, Bessemerization
(d) None of the above is correct

2. One of the processes used for concentration of ores is froth floatation process. This process is generally used for concentration of sulphide ores. Sometimes in this process we add NaCN as a depressant. NaCN is generally added in case of ZnS and PbS minerals. What is the purpose of addition of NaCN during the process of froth floatation?
(a) NaCN causes reduction by precipitation
(b) A soluble complex is formed by reaction between NaCN and ZnS while PbS forms froth.
(c) A soluble complex is formed by reaction between NaCN and PbS while ZnS forms froth
(d) A precipitate of $Pb(CN)_2$ is produced while ZnS remain unaffected.

3. Which of the following processes is used to get pure metal from impure zinc known as spelter?
(a) Liquation
(b) Poling
(c) Fractional distillation
(d) None Of these

4. When $K_2Cr_2O_7$ is heated with concentrated H_2SO_4 and soluble chloride such as KCl
(a) Cl^- ion is oxidized to Cl_2 gas
(b) $Cr_2O_7^{2-}$ ion is reduced to green Cr^{3+} ion
(c) Red vapours of CrO_2Cl_2 is evolved
(d) $CrCl_3$ is formed

5. The correct statement of the following is.
(a) FeI_3 is stable is aqueous solution
(b) An acidified solution of K_2CrO_4 gives yellow precipitate on mixing with lead acetate
(c) The species $[CuCl_4]^{2-}$ exists but $[CuI_4]^{2-}$ does not
(d) Both copper (I) and copper (II) salts are known in aqueous solution.

6. Pick out the correct statement of the following
(a) The stability of either of $HgCl_2$ and $SnCl_2$ is not affected when present simultaneously in aqueous solution.
(b) Both $Cu(OH)_2$ and $Fe(OH)_2$ are soluble in aqueous NH_3
(c) Copper (I) salts are not known is aqueous solution
(d) white precipitate of $Zn(OH)_2$ is obtained on adding excess of NaOH to aqueous $ZnSO_4$

7. First four ionization ionization energes of Ni and Pt are given below.

	$(IE_1 + IE_2)$	$(IE_3 + IE_4)$
Ni	2.49×10^3 kj mol^{-1}	8.8×10^3 kj mol^{-1}
Pt	2.66×10^3 kj mol^{-1}	6.7×10^3 kj mol^{-1}

From the data it can be concluded that
(a) Ni (II) Compounds are thermodynamically more stable than Pt (II) Compounds
(b) Pt (IV) compounds are thermodynamically more stable than Ni compounds
(c) Both are correct
(d) none of these is correct

8. Zinc is extracted from ZnS by the
(a) calcinations of ZnS followed by hydrogen reduction at 400°C

(b) calcinations of ZnS followed by carbon monoxide reduction
(c) roasting of ZnS followed by aluminium reduction at 1200°C in a muffle furnace
(d) roasting of ZnS followed by carbon reduction at 1200°C in a smelter

9. In which of the following reactions potassium ferrocyanide produced?
(a) The reaction between $K_3[Fe(CN)_6]$ and an excess of $FeSO_4$
(b) The reaction between $FeSO_4$ and an excess of KCN
(c) The reaction between $Fe_2(SO_4)_3$ and an excess of KCN
(d) The reaction between Fe_2O_3 and excess of KCN

10. Of the following reduction processes

$$A : Fe_2O_3 + C \rightarrow Fe$$
$$B : ZnO + C \rightarrow Zn$$
$$C : Ca_3(PO_4)_2 + C \rightarrow P_4$$
$$D : PbO + C \rightarrow Pb$$

Correct processes are
(a) A, B, C and D (b) B, C
(c) A, B, C, D (d) B, D

11. Consider the following steps $Cu_2S \xrightarrow{\text{roast in air}} A \xrightarrow{\text{roast without air}} B$
Which is not the correct statement?
(a) It is self reduction
(b) It involves disproportionate ion $Cu_2S \rightarrow Cu + CuS$
(c) A is a mixture of Cu_2O and Cu_2S and B is a mixture of Cu and SO_2
(d) Conversion of A to B take place in Bessemerization

12. Consider the following metallurgical processes
1. Heating metal with CO and distilling the resulting volatile carbonyl (b.p.43°C) and finally decomposing at 150°C to 200°C to get the pure metal
2. Heating the sulphide ore in air until a part is converted to oxide and then further heating in the absence of air to let the oxide react with unchanged sulphide.
3. Electrolyzing the molten electrolyte containing approximately equal amounts of the metal chloride and $CaCl_2$ to obtain the metal

The process used for obtaining sodium, nickel and copper are respectively
(a) 1, 2 and 3 (b) 2, 3 and 1
(c) 3, 1 and 2 (d) 2, 1 and 3

13. $FeCr_2O_4$ (Chromite) is converted to Cr by following steps

$$\text{chromite} \xrightarrow{\text{I}} Na_2CrO_4 \xrightarrow{\text{II}} Cr_2O_3 \xrightarrow{\text{III}} Cr$$

I, II and III are

I	II	III
(a) Na_2CO_3/air, Δ	C	C
(b) NaOH/air, Δ	C, Δ	Al, Δ
(c) NaOH/air, Δ	C, Δ	Mg, Δ
(d) Conc H_2SO_4, Δ	NH_4Cl, Δ	C, Δ

14. Which is not correct statement
(a) Casseterite, chromites and pitch blende are concentrated by hydraulic washing
(b) Pure Al_2O_3 is obtained from the bauxite ore by leaching in the Beayer's process
(c) Sulphide ore is concentrated by calcination method
(d) Roasting can convert sulphide into oxide or sulphate and part of sulphide may also act as a reducing agent.

15. XCl_2 (excess) + $YCl_2 \longrightarrow XCl_4 + Y\downarrow$;

$$YO \xrightarrow[>400°]{\Delta} \frac{1}{2}O^2 + Y, \text{ore of Y would be :}$$

(a) Siderite (b) Cinnabar
(c) Malachite (d) Hornsilver

16. Give the correct order of initials T and F for following statements. Use 'T' if statement is true and 'F' if it is false
I. Every mineral is an ore but every ore is not a mineral
II. Slag is product formed during extraction of metal by combination of flux and impurities
III. Highly pure metals can be obtained by zone refining
IV. Carnallite is an ore of magnesium and sodium
(a) TTTF (b) FTTF
(c) FTTT (d) TFTF

17. Froth floatation process used for the concentration of sulphide ore:
(a) is based on the difference in wettability of different minerals
(b) uses sodium ethyl xanthate, $C_2H_5OCS_2Na$ as collector
(c) uses NaCN as depressant in the mixture of ZnS and PbS when ZnS forms soluble complex and PbS forms froth
(d) All are correct statements

18. In the leaching of Ag_2S with NaCN, a stream of air is also passed. It is because of:
(a) reversible nature of reaction between Ag_2S and NaCN is prevented
(b) to oxidize Na_2S formed into Na_2SO_4 and sulphur
(c) Both 1 and 2
(d) None of the above

19. In the extraction of aluminium
Process (X): applied for red bauxite to remove iron oxide (chief impurity)

Process (Y): (Serpeck's process): applied for white bauxite to remove 'Z' (chief impurity) then, process X and impurity Z are:
(a) X = Hall and Heroult's process and Z = SiO_2
(b) X = Baeyer's process and Z = SiO_2
(c) X = Serpeck's process and Z = iron oxide
(d) X = Baeyer's process and Z = iron oxide

20. The electrolysis of pure alumina is not feasible because:
(a) it is bad conductor of electricity and its fusion temperature is high
(b) it is volatile in nature
(c) it is decomposed when fused
(d) it is amphoteric

21. Blister copper is refined by stirring molten impure metal with green logs of wood because such as a wood liberates hydrocarbon gases (like CH_4). This process X is called …….. and the metal contains impurities of 'Y' is …….
(a) X = Cupellation, Y = Cu_2O
(b) X = Polling, Y = Cu_2O
(c) X = Polling, Y = CuO
(d) X = Cupellation, Y = CuO

22. Select correct statement:
(a) The decomposition of an oxide into oxygen and metal vapour entropy increases
(b) Decomposition of an oxide is an endothermic change
(c) To make $\Delta G°$ negative, temperature should be high enough so that $T\Delta S° > \Delta H°$
(d) All are correct statements

23. If a metal has low oxygen affinity then purification of metal may be carried out by
(a) Liquation (b) Distillation
(c) Zone refining (d) Cupellation

24. In zone refining method, the molten zone
(a) Consists of impurities only
(b) Contain impurity than the original metal
(c) Contains the purified metal
(d) Moves to either side

25. Formation of metallic copper from the sulphide ore in the commercial thermo-metallurgical process essentially involves which one of the following reaction
(a) $Cu_2S + \frac{3}{2}O_2 \rightarrow Cu_2O + SO_2$; $CuO + C \rightarrow Cu + CO$
(b) $Cu_2S + \frac{3}{2}O_2 \rightarrow Cu_2O + SO_2$; $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$
(c) $Cu_2S + 2O_2 \rightarrow CuSO_4$; $CuSO_4 + Cu_2S \rightarrow 3Cu + 2SO_2$
(d) $Cu_2S + \frac{3}{2}O_2 \rightarrow Cu_2O + SO_2$; $Cu_2O + CO \rightarrow 2Cu + CO_2$

26. High temperature (>1000°C) electrolytic reduction is necessary for isolating.
(a) Al (b) Cu (c) C (d) F_2

27. Addition of high proportions of manganese makes steel useful in making rails or rail roads, because manganese
(a) Gives hardness
(b) Helps formation of oxide of iron
(c) Can remove oxygen and sulphur
(d) Can show highest oxidation state of +7

28. A piece of steel is heated until redness and then plunged into cold water or oil. This treatment of iron makes it
(a) Soft and malleable
(b) Hard but not brittle
(c) More brittle
(d) Hard and brittle

29. Froth floatation process for concentration of ores is an illustration of the practical application of
(a) Adsorption
(b) Absorption
(c) Coagulation
(d) Sedimentation

30. Railway wagon axles are made by heating iron rods embedded in charcoal powder. This process is known as
(a) Sheradizing
(b) Annealing
(c) Tempering
(d) Case hardening

31. Gold Schmidt thermite process is used for
(a) Welding of broken iron pieces
(b) Converting iron into steel
(c) Extraction of sulphur
(d) Reduction of metallic oxide by magnesium

32. In the froth floatation process for the purification of ores, the ore particles float because
(a) They bear electrostatic charge
(b) They are insoluble
(c) They are light
(d) Their surface is not easily wetted by water

33. By fusing dry silver chloride with sodium carbonate, we get pure
(a) Na (b) Chlorine
(c) CO_2 (d) Ag

34. Carbon cannot be used in the reduction of Al_2O_3 because
(a) The enthalpy of formation of CO_2 is more than that of Al_2O_3

(b) Pure carbon is not easily available
(c) The enthalpy of formation of Al_2O_3 is very high
(d) It is an expensive proporition

35. A sulphide ore is generally roasted to the oxide before reduction, because
(a) The enthalpy of formation of CO_2 is more than that of CS_2
(b) A metal sulphide is generally more stable than the metal oxide
(c) No reducing agent is found suitable for reducing a sulphide ore
(d) A sulphide ore cannot be reduced at all

36. In a mixture of PbS, ZnS and FeS each component is separated from other by using the reagents in the following sequence in froth floatation process.
(a) Potassium ethyl xanthate, KCN
(b) Potassium ethyl xanthate, KCN, NaOH, $CuSO_4$ acid
(c) KCN, $CuSO_4$, acid
(d) None of these

37. When hydrated $MgCl_2 \cdot 6H_2O$ is strongly heated
(a) MgO is formed
(b) Anhydrous $MgCl_2$ is formed
(c) Mg(OH) HCl is formed
(d) Mg(OH) Cl is formed

38. Which of the following statements is incorrect?
(a) Mercurous chloride can be obtained by heating mercuric chloride with mercury
(b) Mercurous chloride exists in a dimeric form Hg_2Cl_2 and it contains $[Hg\text{-}Hg]_2^{2+}$ ions
(c) Mercurous chloride contains the ion Hg^+ and not Hg_2^{2+}
(d) In mercurous chloride two Hg^+ ions get bonded using their 6s orbital's to form $[Hg\text{-}Hg]^{2+}$

39. Which of the following reactions is involved in the extractions of Cu by Bessemer Process?
(a) $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$
(b) $Cu_2S + O_2 \rightarrow 2Cu + SO_2$
(c) $Cu_2O + C \rightarrow 2Cu + CO$
(d) $Cu_2O + Zn \rightarrow ZnO + 2Cu$

40. Which of the following chemical changes is incorrect?
(a) $Hg^{2+} + Cu \rightarrow Hg + Cu^{2+}$
(b) $HgI_2 + 2I^- \rightarrow [HgI_4]^{2-}$
(c) $HgI_2 + 2Cl^- \rightarrow HgCl_2 + 2I^-$
(d) $Hg + Hg^{2+} \rightarrow 2Hg_2^{2+}$

41. AgCl dissolves in an excess of NH_3, KCN and $Na_2S_2O_3$ solution respectively producing complex ions represented by the following chemical formulae
(a) $Ag(NH_3)_2^+, Ag(CN)_2^{2-}$ and $Ag_2(S_2O_3)_4^{3-}$
(b) $Ag(NH_3)_2^{2+}, Ag(CN)_2^-$ and $Ag(S_2O_3)_2^{3-}$
(c) $Ag.2NH_3, Ag \cdot KCN, AgCl \cdot S_2O_3$
(d) $Ag(NH_3)_2^+, Ag(CN)_2^- \cdot Ag_2(S_2O_3)_2^{3-}$

42. Which of the following reaction sequences is correct for the extraction of gold?
(a) $AuCN + Na_2SO_4 \rightarrow A \xrightarrow{Zn} gold$
(b) $Au + NaCN + H_2O + O_2 \rightarrow A \xrightarrow{Zn} gold$
(c) $Au_2S_3 + NaCN + H_2O + O_2 \rightarrow A \xrightarrow{Mg} gold$
(d) $Au + NaCN + H_2O + O_2 \rightarrow A \xrightarrow{Zn(CN)_4^{2-}} gold$

43. The $Cr_2O_7^{2-}$ ion is made up of
(a) Two tetrahedral chromate units (CrO_4) linked by an oxygen atom
(b) Two square planar chromate units linked by an oxygen atom
(c) Two CrO_4^{2-} tetrahedral ions having Cr – Cr link
(d) Two CrO_4^{2-} tetrahedral ions which are linked by Cr atom

44. Which of the following is an incorrect statement?
(a) In a redox reaction in acidic medium $KMnO_4$ produces Mn^{2+} ions
(b) In a redox reaction in strongly alkaline medium $KMnO_4$ produces MnO_4^{2-} ions
(c) In a redox reaction in neutral medium $KMnO_4$ produces Mn^{2+} ions
(d) In a redox reaction in acidic medium $KMnO_4$ produces Mn_2O_7 ions

45. Which is the incorrect statement?
(a) Ore bauxite is purified by alkaline treatment, the Baeyer process and then dissolved in molten cryolite and reduced electrolytically in the fused salt system to get metal.
(b) Sulphide ore of some of the less electropositive metals like Hg, Cu, Pb etc are heated in air, a part of these is changed into oxide or sulphate then that reacts with the remaining part of the sulphide ore to give its metal and SO_2.
(c) Metallic silver is dissolved from its ore in dilute NaCN solution in presence of air and the solution so obtained is treated with scrap zinc when silver metal is precipitated.
(d) Magnesium is precipitated as $Mg(OH)_2$ from sea water by treatment with lime and then it dissolved in dilute HCl and reduced electrolytically in aqueous medium to get magnesium metal.

46. Collectors are the substances which help in attachment of an ore particle to air bubble in froth. A popular collector used industrially is
(a) Sodium ethyl xanthate
(b) Sodium xanthate

(c) Sodium pyrophosphate
(d) Sodium nitroprusside

47. Zone refining is based on the principle of
(a) Fractional distillation
(b) Fractional crystallization
(c) Partition coefficient
(d) Chromatographic separation

48. During initial treatment, preferential wetting of ore by oil and gangue by water takes place in
(a) Levigation (gravity separation)
(b) Froth floatation
(c) Leaching
(d) Bessemerization

49. Silica is added to roasted copper ore during extraction in order to remove
(a) Cuprous sulphide
(b) Ferrous oxide
(c) Ferrous sulphide
(d) Cuprous oxide

50. Match List-I with List-II

List-I (Property)	List-II (Element/compound)
I. Explosive	A: Cu
II. Self-reduction	B: Fe_3O_4
III. Magnetic material	C: $Cu(CH_3COO)_2 \cdot Cu(OH)_2$
IV. Verdigris	D: $Pb(NO_3)_2$

(a) I-A, II-B, III-C, IV-D (b) I-D, II-A,III-B,IV-C
(c) I-D, II-B, III-A, IV-C (d) I-C,II-A, III-B, IV-D

51. The principal reaction in the zone of heat absorption of blast furnace employed in the metallurgy of iron is
(a) $C + O_2 \rightarrow CO_2$
(b) $2C + O_2 \rightarrow 2CO$
(c) $CaCO_3 + SiO_2 \rightarrow CaSiO_3 + CO_2$
(d) $Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$

52. In which of the following pair of metals, both are extracted from their respective ores by carbon reduction method?
(a) [Zn, Cu] (b) [Fe, Cu]
(c) [Sn, Zn] (d) [Al, Ag]

53. Poling process is used for
(a) The removal Cu_2O from Cu
(b) The removal of Al_2O_3 from Al
(c) The removal Fe_2O_3 from Fe
(d) All the above

54. The method of electrolytic refining is most unsuitable for the extraction of
(a) Aluminium (b) copper
(c) Mercury (d) silver

55. The method of cupellation is used to separate silver from
(a) iron (b) lead
(c) cobalt (d) zinc

56. The process of converting hydrated alumina into anhydrous alumina is called
(a) roasting (b) calcination
(c) dressing (d) smelting

57. Ferrous sulphate on heating gives:
(a) SO_2 and SO_3 (b) SO_2 only
(c) SO_3 only (d) SO_2 and O_2

58. Complex formation or cyanide method is used for the extraction of:
(a) Cu (b) Fe
(c) Hg (d) Ag

59. The chemical process in the production of steel from haematite ore involves:
(a) Reduction
(b) Oxidation
(c) Reduction followed by oxidation
(d) Oxidation followed by reduction

60. Which of these industrial processes does not involve oxidation or reduction?
(a) Formation of iron from the reaction of iron ore and carbon monoxide
(b) Electrolysis of molten aluminium ore
(c) Purification of zinc ore by roasting with pure oxygen
(d) The addition of NaOH to digest aluminium oxides

61. Magnesium is manufactured by the electrolysis of fused magnesium chloride using
(a) a nickel cathode and a graphite anode
(b) a nickel container as cathode and an iron
(c) an iron container as cathode and a graphite anode
(d) a graphite rod acts as an anode and a lead plate as a cathode

62. During the extraction of aluminium by the Hall-Heroult process
(a) A graphite – lined steel tank serves as cathode and a graphite rod acts as an anode
(b) A graphite – lined steel tank serves as anode and a graphite rod acts as cathode
(c) A graphite rod acts as an anode and another as a cathode
(d) A graphite rod acts as an anode and a lead plate as a cathode

63. $ZnSO_4$ react with an excess of KCN solution to produce
(a) The complex $[Zn(CN)_4]^{2-}$ which has tetrahedral structure
(b) The complex $[Zn(CN)_2]^-$ which has linear structure
(c) The complex $[Zn(CN)_4]^{2-}$ which has square planar structure
(d) The complex $[Zn(CN)_6]^{4-}$ which has octahedral structure

64. The formulae of the soluble complexes formed during the extraction of Ag and Au by the cyanide process are respectively
(a) $Na_2[Ag(S_2O_3)_4]$ and $H[AuCl_4]$
(b) $Na[Ag(CN)_2$ and $Na[Au(CN)_2]$
(c) $Na_2[Ag(CN)_4]$ and $Na_4[Au_6(CN)_5]$
(d) $Na_3[Ag(CN)_2]$ and $Na_2[Au(CN)_4]$

65. Which of the following statement is correct?
(a) Pig iron or cast iron hard and brittle
(b) The type of iron that contains maximum amount of iron is pig iron.
(c) The type of iron that contains maximum amount of carbon is wrought iron.
(d) The hardness of steel increases as its carbon content decreases

66. Which process of purification is represented by the following equation?

$$\underset{\text{(impure)}}{Ti + 2I_2} \xrightarrow{250°C} TiI_4 \xrightarrow{1400°C} \underset{\text{(pure)}}{Ti + 2I_2}$$

(a) Cupellation (b) Polling
(c) Van Arkel process (d) Zone refining

67. The role of calcination in metallurgical process is
(a) To remove moisture
(b) To decompose carbonate into oxide
(c) To remove organic matter
(d) All of these

68. Which of the following beneficiation process is used for bauxite $Al_2O_3 \cdot 2H_2O$?
(a) Froth flotation
(b) Leaching
(c) Liquation
(d) Magnetic separation

69. Which of the following acts as activator in froth flotation process?
(a) Ethyl xanthate
(b) KCN
(c) Copper sulphate
(d) $K_4[Fe(CN)_6]$

70. Which of the following can be reduced by Zn?
(a) Na $[Ag(CN)_2]$
(b) Na $[Au(CN)_2]$
(c) (a) and (b) both
(d) None of these

71. Which of the following cannot be reduced by Al?
(a) Fe_2O_3 (b) Cr_2O_3
(c) MnO_2 (d) MgO

72. In the metallurgy of iron, when limestone is added to the blast furnace the calcium ions end up in
(a) Calcium silicate (slag)
(b) Gangue
(c) Metallic calcium
(d) Calcium carbonate

73. The metallurgical process in which a metal is obtained in free state is called.
(a) Smelting (b) Roasting
(c) Calcination (d) Froth flotation

74. Which of the following is not an ore?
(a) Bauxite (b) Malachite
(c) Zinc blende (d) Pig iron

75. If the temperature needed for carbon to reduce an oxide is too high for economic practical purpose, then reduction can be affected by using another highly electropositive element which is
(a) Al (b) Mg
(c) Na (d) All of these

76. The process for removing layers of basic oxides from metal before electroplating is called
(a) Galvanizing (b) anodizing
(c) pickling (d) poling

77. The ultra pure metal is obtained by
(a) Calcination (b) sublimation
(c) Zone refining (d) None of these

78. Pure zirconium is obtained by.
(a) Van Arkel method (b) Hydrometallurgy
(c) Zone refining (d) None of these

79. The molten material obtained after treatment of copper pyrites in the blast furnace has the composition of
(a) Cu_2S (b) $Cu_2S + FeS$
(c) $Cu_2S + FeO$ (d) $Cu_2S + Cu_2O$

80. Gold is mainly extracted by
(a) Carbon reduction
(b) Self reduction
(c) Electrolytic method
(d) Mc Arthur – Forrest process

81. In which of the following heat is absorbed?
(a) $C + O_2 \rightarrow CO_2$
(b) $C + \frac{1}{2}O_2 \rightarrow CO$
(c) $CO_2 + C \rightarrow 2CO$
(d) $CO_2 + \frac{1}{2}O_2 \rightarrow CO_2$

82. Which of the following act as depressant in froth floatation process?
(a) NaCN (b) Pine Oil
(c) NaOH (d) o-Cresol

83. Which of the following reaction is not taking place during the extraction of Ag from Ag_2S by cyanide process?
(a) $Ag_2S + CN^- \rightarrow [Ag(CN)_2]^- + S^{2-}$
(b) $Zn + 2[Ag(CN)_2]^- \rightarrow [Zn(CN)_4]^{2-} + 2Ag\downarrow$
(c) $Cu + 2[Ag(CN)_2]^- \rightarrow [Cu(CN)_4]^{2-} + 2Ag\downarrow$
(d) $S^{2-} + O_2 \rightarrow SO_4^{2-} + S + S_2O_3^{2-}$

84. A black mineral on roasting breaks up into two compounds A and B with the liberation of gas C. When air is passed through the molten mixture of A and B, B converts into oxide that can be reduced by air. The mineral is
(a) Chalocite (b) Feldspar
(c) Chalcopyrite's (d) Pyrargirite

85. A black coloured compound (A) in solid state is fused with KOH and $KClO_3$ and the mixture extracted with water to obtain a green colour solution (B). On passing Cl_2 through the solution the colour changes to pink with a black residue (C). Which of the following is correct?
(a) The pink colouration is decolourized by acidified ferrous sulfate solution
(b) The black residue is same as compound (A)
(c) When O_3 gas is passed through the green colour solution it changes to pink
(d) All of the above

86. Gold smiths use borax while making gold jewellery. They heat the mixture of a gold sample and borax on a flame, through which air is passed using a blow pipe because
(a) Heat and air convert the impurities into their oxides that react with molten borax forming a slag which can be separated from gold.
(b) Gold oxide formed by heat and air gets reduced by borax to pure gold
(c) Air increases the heating temperature at which gold dissolves in borax and on cooling gets recrystallized in pure form
(d) Borax reduces the hardness of gold at high temperature so that it can be stretched to form jewellery easily

87. The function of fluorspar in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AIF_6) is
(a) That of catalyst
(b) To lower the temperature of the melt and improve the conductivity of the cell
(c) In the presence of cryolite which forms a melt with lower melting temperature
(d) In the presence of cryolite which forms a melt with higher melting temperature

88. Which of the following statement is correct?
(a) Mercurous compounds are paramagnetic
(b) Mercurous compounds are ferromagnetic
(c) Mercurous compounds are diamagnetic
(d) Mercurous compounds are electromagnetic

89. Mercury can be extracted from cinnabar
(a) By roasting the concentrated ore in air at 600°C followed by cooling
(b) By heating the ore with scrap iron
(c) By heating the ore with quick lime
(d) All of these

90. Which of the following statements is wrong?
(a) Mercury forms two types of oxides HgO and Hg_2O
(b) Mercury (II) oxide is thermally stable even at high temperature
(c) Mercury (II) oxide is thermally unstable and readily decomposes into mercury and oxygen on being heated above 400°C
(d) Mercury (II) sulphide is precipitated from Hg^{2+} solution by passing H_2S in the presence of high concentration of hydrogen ions

91. Hg_2Cl_2 reacts with liquid ammonia to produce a black precipitate consisting of
(a) $Hg(NH_2)Cl$
(b) Hg and $Hg(NH_2)Cl$
(c) Hg and HgCl
(d) $Hg(NH_3)_2\ Cl_2$ and $HgCl_2$

92. Which of the following statement is incorrect?
(a) Reaction of Zn with dil H_2SO_4 becomes faster when zinc is treated with a drop of $CuSO_4$ Solution
(b) Zinc hydroxide dissolves both in ammonium hydroxide and sodium hydroxide solution
(c) Zn^{2+} is precipitated by H_2S as ZnS in an alkaline medium but not in a strong acid medium
(d) The mercurous ion is written as Hg_2^{2+} and the cuprous ions written as Cu_2^{2+}

93. Chloroauric acid $H[AuCl_4]$ reacts with NaOH solution at room temperature to produce
(a) $Au(OH)_3$ (b) Au_2O_3
(c) Au (d) AuOH

94. When mild steel is heated to a high temperature and suddenly plunged into cool water, it become hard and brittle. The process is called
(a) Hardening (b) annealing
(c) Quenching (d) Temperature

95. Which of the following statement is correct
(a) Cast iron cannot be welded
(b) Wrought iron can not be welded easily
(c) Steel can be welded easily
(d) Cast iron can be welded with phosphorous

96. Which of the following statements is correct in the context of magnetization
(a) Cast iron cannot be permanantly by magnetized
(b) Wrought iron is easy to magnetize but the magnetization is not permanent
(c) Steel can be permanently magnetized
(d) All of these

97. Which of the following statements is correct?
(a) FeI_3 does not exist in a pure state because Fe^{3+} oxidizes I^-
(b) Fe(III) Complexes are less stable than those if Fe(II)

(c) Fe^{+3} has a d^6 configuration
(d) Fe^{2+} reacts with the SCN^- ion to produce $[Fe(SCN)(H_2O)_5]^{2+}$

98. $KMnO_4$ is manufactured on large scale by
(a) Fusing MnO_2 with KOH and then oxidizing the fused mixture with KNO_3
(b) Fusing MnO_2 with Na_2CO_3 in the presence of O_2
(c) Fusing MnO_2 with KOH and KNO_3 to form K_2MnO_4 which is then electrolytically oxidized in an alkaline solution
(d) Fusing MnO_2 with KNO_3 and then acidifying the fused mixture

99. Which of the following statements is correct?
(a) Quenched steel is mild steel
(b) Nitriding is heating iron in atmosphere of N_2
(c) Stainless steel is produced by heating wrought iron in molten chromium
(d) Mild steel is obtained by annealing

100. Which of the following statements regarding the metallurgy of magnesium using electrolytic method is not correct?
(a) Electrolyte is anhydrous magnesium chloride containing a little NaCl and NaF
(b) NaCl and NaF in the electrolyte act as flux
(c) Electrolysis is done in the atmosphere of coal gas
(d) Air tight iron pot act as a cathode

101. During the manufacture of steel from cast iron by open hearth process, a neutral gas was evolved which reacts with a metal X to form a complex compound 'Y'. Cation of metal X gives following tests

$X^{n+} + SCN \rightarrow$ blood red precipitate

$X^{n+} + CH_3COO^- \rightarrow$ deep red colouration

$X^{n+} + K_4(Fe(CN)_6) \rightarrow X_4[Fe(CN)_6]_n$

The compound Y was found to have EAN = 36 which of the following statements is/false?
(a) The evolution of neutral gas was accompanied by the formation of MnO, SiO_2 and SO_2
(b) The magnetic moment in Y is zero
(c) The bond energy of the bond in the neutral gas increases when it forms Y
(d) All of the above

102. Which of the following is not the product of reaction $HMnO_4 \xrightarrow{\Delta} \ldots + \ldots ?$
(a) MnO_2 (b) H_2O (c) O_2 (d) MnO

103. When $KMnO_4$ is treated with concentrated H_2SO_4 in cold a dark brown oily compound (X) is formed which decomposes explosively on heating. The compound (X) is
(a) MnO_2 (b) Mn_2O_7
(c) K_2MnO_4 (d) MnO

104. The metal extracted by leaching with a cyanide is
(a) Na (b) Ni
(c) Ag (d) Ku

105. Given reagents are HCI, NaOH, $ZnCI_2$, Na_2CO_3, NH_4CI and Zn. Which of these reagent(s) can intensify Hydrolysis of $FeCI_3$ when added to its solution
(a) NaOH, Na_2CO_3 (b) NaOH, NH_4CI, HCI
(c) Na_2CO_3, NaOH + HCI (c) NH_4CI, HCI, $ZnCI_2$

106. Given below is extraction of a particular type of metal

$$\text{(A) Sulphide of (A)} + NaCI \rightarrow \underset{\text{white}}{B \downarrow} + Na_2S \ldots.(i)$$

$$B + Na_2S_2O_3 \rightarrow (C) + NaCI \ldots.(ii)$$

$$C + Cu \rightarrow A \downarrow + Na_3[Cu(S_2O_3)_2] \ldots.(iii)$$

The nitrate of (A) when treated with NH_4OH gives a brown precipitate (D) which dissolved in excess of the reagent forming (E). Based on the above observations select the incorrect option from the following:
(a) E is $[Ag(NH_3)_2]NO_3$
(b) Metal A can be extracted by hydrometallurgy
(c) Compound C is a linear complex
(d) Compound E can oxidize ketones

107. Consider $FeSO_4 \xrightarrow{\Delta} A + B + C$; (B) and (C)are gases (A) is red brown solid. (B) can be oxidized to (C). (B) also turns acidified $K_2Cr_2O_7$ solution green. (A) dissolves in HCI to give deep yellow solution $(D) \cdot (D)$ gives blue colour (E) with potassium ferrocyanide. Identify A, B, D and E
(a) $Fe_2O_3, SO_2, FeCl_3, Fe(CN)_2$
(b) $Fe_2O_3, SO_3, FeCl_3, KFe^{III}[Fe^{II}(CN)_6]$
(c) $Fe_2O_3, SO_2, FeCl_3, KFe^{III}[Fe^{II}(CN)_6]$
(d) $FeO, SO_2, FeCl_2, KFe^{III}[Fe^{II}(CN)_6]$

One or More than One Correct Type Questions

1. Which of the following statements is/are correct about the making of blue prints?
(a) Ferric oxalate or citrate is reduced to ferrous salt on being exposed to light
(b) Ferrous salts give blue colour with $K_3[Fe(CN)_6]$
(c) Ferric salts give blue colour with $K_3[Fe(CN)_6]$
(d) white lines are obtained in place of the ink drawing on a deep blue back ground

2. Which of the following statements is/are correct?
(a) Anhydrous ferric chloride can be obtained by heating hydrated ferric chloride
(b) A solution of ferric oxalate in dilute H_2SO_4 will decolourize $KMnO_4$

(c) Ferric salts are more stable than ferrous salts.
(d) The Mohr's salt is resistant to oxidation by atmospheric oxygen

3. Which of the following complex ions that zinc can form
(a) $[Zn(CN)_6]^{4-}$
(b) $[Zn(CN)_4]^{2-}$
(c) $[Zn(NH_3)_4]^{2-}$
(d) $[Zn(CNS)_4]^{2-}$

4. Which of the following statements is/are true?
(a) Both Hg^{2+} and Hg_2^{2+} ions show the divalency
(b) The ionization potentials of group 12 metals are fairly greater then those of coinage metals yet more reactive than the latter.
(c) Ionization potential of Hg is smaller than that of Cd
(d) Zn, Cd and Hg exhibit positive oxidation potentials.

5. In which of the following pairs, both the minerals are oxides?
(a) Sylvine, saltpetre (b) Casseterite, litharge
(c) Siderite, corrundum (d) Cuprite, tinstone

6. Amphoteric nature of aluminium is employed in which of the following process for extraction of aluminium?
(a) Baeyer's process (b) Hall's process
(c) Serpeck's process (d) Dow's process

7. The disadvantage of carbon reduction method are:
(a) high temperature needed which is expensive and requires the use of a blast furnace
(b) Many metals combine with carbon forming carbides
(c) Carbon combines with oxygen to form poisonous CO
(d) Carbon cannot be used with highly electropositive metals

8. Roasting of copper pyrites is done:
(a) to remove moisture
(b) to oxidize free sulphur
(c) to decompose pyrite into Cu_2S and FeS
(d) to remove volatile organic impurities

9. Which of the following is a correct statement?
(a) Calamine is the ore of zinc
(b) Pyrolusite is the ore of manganese
(c) Cassiterite is the ore of tin
(d) Calcite is the ore of calcium

10. Which of the following pair consists of ore of the same metal?
(a) Bauxite, Limonite
(b) Haematite, Siderite
(c) Cinnabar, Cassiterite
(d) Galena, Cerrusite

11. Froth floatation:
(a) is physical method of separating mineral from the gangue
(b) is a method to concentrate the ore depending on the difference in wettability of gangue and the ore
(c) is used for the sulphide ores
(d) is a method in which impurities sink to the bottom

12. Which of the following reaction(s) occur during calcination?
(a) $CaCO_3 \rightarrow CaO + CO_2$
(b) $4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$
(c) $2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$
(d) $CuS + CuSO_4 \rightarrow 2Cu + 2SO_2$

13. During the production of iron and steel
(a) The oxide ore is primarily reduced to iron by solid coke according to the reaction
$2Fe_2O_3 + 3C \rightarrow 4Fe + 3CO_2$
(b) The oxide ore is reduced by the carbon monoxide according to the reaction
$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
(c) Major silica impurities are removed as calcium silicate slag by addition of a fluxing agent limestone
(d) The silicate slag is used in manufacturing cement

14. Which of the following reduction reactions are actually employed in commercial extraction of metals?
(a) $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$
(b) $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$
(c) $2Na[Au(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Au$
(d) $Cu_2S + Pb \rightarrow Cu + PbS\downarrow$

15. Which of the following are true for electrolytic extraction of aluminium?
(a) Cathode material contains graphite
(b) Anode material contains graphite
(c) Cathode reacts away forming CO_2
(d) Anode reacts away forming CO_2

16. Aluminothermi process used for the spot welding of large iron structures is based upon the fact that:
(a) As compared to iron, aluminium has greater affinity for oxygen
(b) As compared to aluminium iron has greater affinity for oxygen
(c) Reaction between aluminium and oxygen is endothermic
(d) Reaction between iron oxide and aluminium is exothermic

17. Which of the following metals is/are extracted from its oxide by air reduction method?
(a) Cu (b) Hg (c) AI (d) Pb

18. In the process of extraction of gold
Roasted gold ore $+ CN^- + H_2O \xrightarrow{O_2} [X] + OH^-$
$[X] + Zn \rightarrow [Y] + Au$
Identify the complexes [X] and [Y]
(a) $X = [Au(CN)_2]^-$

(b) $Y = [Zn(CN)_4]^{2-}$

(c) $Y = [Zn(CN)_6]^{4-}$

(d) $X = [Au(CN)_4]^{3-}$

19. The process(es) by which lighter earthy impurities are freed from the heavier particles using water is/are.
(a) gravity separation (b) levigation
(c) hydraulic washing (d) leaching

20. Hydrated $FeCl_3(FeCl_3 \cdot 6H_2O)$ cannot be converted to anhydrous $FeCl_3$ on heading but forms a compound X. When hydrated ferric chloride is treated with Y, then an hydrous $FeCl_3$ is formed along with SO_2 · Here
(a) X = FeOCI (b) Y = $SOCI_2$
(c) X = Fe_2O_3 (d) Y = SO_2Cl_2

21. A collector is
(a) Collect impurities form surface of ore for example pine oil
(b) Collect impurities from the bottom of the sulphide ore for example alkyl xanthate
(c) Adsorb themselves on polar groups to grains of ores and thus derive them on the surface to pass on into the froth, for example ethyl xanthate
(d) Collect ore particles, form precipitate for example lime

22. Which of the following statement is/are correct?
(a) The changes in ΔG that occur when one mole of oxygen is used may be plotted against temperature for a number of reactions of metals to form their oxides.Such graph is known as Ellingham diagram
(b) Ellingham diagram helps as in predicting the feasibility of thermal reduction of an ore
(c) At a given temperature, as a criterion of feasibility of reaction, change in Gibbs free energy must be positive
(d) Thermodynamic considerations play an important role in pyrometallurgy

23. Calcination and roasting processes of ores to their oxides are beneficial
(a) to convert ores into porous form so that their reduction becomes easier
(b) as volatile impurities like P. As, Sb, S are removed
(c) as organic impurities are removed.
(d) as the ores are converted into oxide form which makes the reduction easier

24. In the equation.
$M + CN^- + H_2O_2 \rightarrow [M(CN)_2]^- + 4OH^-$ 'M' can be
(a) Au (b) As (c) Sb (d) Ag

25. Auto reduction process is used in extraction of.
(a) Cu (b) Hg (c) Al (d) Fe

26. Which of the following process(es) is/are used for purification of bauxite ore?
(a) Hall's process (b) Serpeck's process
(c) Baeyer's process (d) Mond's process

27. Metals which are extracted by smelting process
(a) Pb (b) Fe (c) Zn (d) Mg

28. Which of the following is/are not correct statement (s)?
(a) Spelter is impure Zn (b) Spelter is impure Fe
(c) Spelter is pure Zn (d) Spelter is impure Al

29. Which of the following processes are used to get ultra pure metals?
(a) Van Arkel method (b) Zone refining
(c) Cupellation (d) Poling

30. Which of the following metals cannot be obtained by electrolysis of their aqueous solutions of their salts?
(a) Ag (b) Mg (c) Al (d) Cu

31. Which of the following ions decolourize $KMnO_4$ without evolving any gas.
(a) H_2O_2 (b) Sn^{2+} (c) SO_3^{2-} (d) Cl^-

32. What happens when malachite, $Cu(OH)_2 \cdot CuCO_3$ is heated in a furnace?
(a) Cupric oxide is formed
(b) Cuprous oxide is formed
(c) Carbon dioxide is formed
(d) It is dehydrated without decomposition

33. Which of the following oxides can be reduced by carbon?
(a) Al_2O_3 (b) SnO_2
(c) PbO (d) Fe_2O_3

34. In blast furnace temperature higher than 1000°C are found in.
(a) Zone of fusion
(b) Zone of combustion
(c) Zone of slag combustion
(d) Zone of reduction

35. Carnallite on electrolysis gives
(a) Ca (b) Cl_2 (c) Mg (d) CO_2

36. Which of the following process involve the extraction of copper from malachite process by hydrometallurgy?
(a) $CuCO_3 \cdot Cu(OH)_2 \rightarrow 2CuO + H_2O + CO_2$
(b) $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$
(c) $CuSO_4 + Hg \rightarrow HgSO_4 + Cu$
(d) $CuSO_4 + Fe \rightarrow FeSO_4 + Cu$

37. Which of the following is/are correct statements(s)?
(a) Nickel is purified by Mond's process
(b) $Ni(CO)_4$ is a volatile compound
(c) Silicon can be refined by Van Arkel process
(d) Ni $(CO)_4$ is a non-volatile compound

38. Which of the following processes is/are not used in extraction of magnesium?
(a) Fused salt electrolysis
(b) Aqueous solution of salt electrolysis

(c) Thermite reduction
(d) Self reduction

39. Which of these industrial processes involve oxidation or reduction?
(a) Formation of iron from the reaction of iron ore and carbon monoxide
(b) Electrolysis of molten aluminium ore
(c) Purification of zinc ore by roasting with pure oxygen
(d) The addition of NaOH to digest aluminium oxides

40. Zn and Ag can be separated from each other by
(a) Distillation
(b) Heating with conc. NaOH
(c) Treating with dil. HNO_3
(d) Treating with conc. H_2SO_4

41. Which of the following is/are correct definitions?
(a) Refractory means capable of enduring high temperature without damage
(b) Galvanized steel is the steel coated with zinc to improve its corrosion resistance
(c) A stalactite is a $CaCO_3$ deposit on the ceiling of a limestone cavern formed over long period of time
(d) A stalagmite is a deposit similar to a stalactite but on the floor of a lime stone cavern

42. Which of the following statement is/are correct?
(a) In metallurgy of PbS, $CuSO_4$ is an example of activator while KCN is used as a depressant
(b) In metallurgy of galena NaCN depresses the floatation property of ZnS and FeS_2
(c) The conversion of small pieces of substance into larger ones in known as sintering
(d) When Al_2O_3 contians high percentage of Fe_2O_3, purification is done by acidic leaching

43. Pick up the correct statements.
(a) Asbestos and willemite are silicate minerals
(b) Anglesite and barites are sulphate minerals
(c) Sylvine and fluorspar are halide minerals
(d) Calamine and calcite are minerals of calcium

44. Which of the following statements are correct
(a) Cu^{2+} is a stronger oxidizing agent than Cr^{2+}
(b) Co^{2+} is a stronger reducing agent than V^{2+}
(c) Cr^{2+} can be easily oxidized than Ni^{2+}
(d) Co^{3+} can be reduced easily than Ti^{3+}

45. Which of the following compounds are coloured due to charge transfer spectrum
(a) $K_2Cr_2O_7$ (b) $KMnO_4$
(c) $[Fe(H_2O)_5(NO)]SO_4$ (d) SnO_2

46. A (Sulphide ore) $+$ NaCN $\underset{}{\overset{\text{air}}{\rightleftharpoons}}$ B(Complex) $+ Na_2S \xrightarrow{\text{oxygen}} Na_2SO_4$ then B is
(a) Paramagnetic
(b) diamagnetic
(c) Linear complex
(d) Co ordination number of central atom is 4

47. After partial roasting, the sulphide of copper can be reduced by
(a) Carbon (b) Self-reduction
(c) Cyanide process (d) Heating

48. Which of the following statement (s) is/are correct with reference to $Fe^{2+}$and Cu^{2+}?
(a) Both react with NaOH
(b) Cu^{2+} gives blue ppt of $Cu(OH)_2$ whereas, Fe^{2+} gives $Fe(OH)_2$ [green ppt]
(c) Cu^{2+} gives chocolate brown ppt whereas Fe^{2+} gives sky blue ppt with $K_4[Fe(CN)_6]$
(d) Both have four unpaired lelectrons

49. Which of the following statement(s) is/are correct?
(a) $ZnSO_4$ is unstable towards heat
(b) $Zn(OH)_2$ decomposes on heating while NaOH does not
(c) $ZnCO_3$ and Li_2CO_3 both are thermally unstable
(d) $ZnSO_4$ is insoluble in strong alkalies

50. Which of the following ores can be concentrated by froth floatation process
(a) Zinc blende (b) Copper pyrites
(c) Argentite (d) Horn silver

51. Which of the following statements is/are correct
(a) Liquation is applied when the metal has low m.pt than that of impurities
(b) Presence of carbon in steel makes it hard due to formation of Fe_3C called cementite
(c) Less reactive metals like Hg, Pb and Cu are obtained by autoreduction of their sulphides or oxide ores
(d) Sodium and magnesium are purified by electrolytic process

52. Which of the following reactions in the blast furnace are endothermic.
(a) $C_{(s)} + O_{2(g)} \rightleftharpoons CO_{2(g)}$
(b) $CO_{2(g)} + C_{(s)} \rightleftharpoons 2CO_{(g)}$
(c) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$
(d) $Fe_2O_{3(s)} + 3CO_{(g)} \rightleftharpoons 2Fe_{(l)} + 3CO_{2(g)}$

53. The advantages of using carbon to reduce a number of oxides and other compounds are
(a) easy availability of coke
(b) low cost of using carbon
(c) tendency of carbon to show reduction of oxides
(d) presence of carbon lowers the melting point of the oxides.

54. An aqueous solution of $CuSO_4$ is added to KCN solution. This results in the formation of a poisonous gas (A) and a soluble complex (B). When gas (A) is passed into caustic soda solution a compound (C) is formed, which contain carbon as well as oxygen. Identify the correct statements.

(a) Compound(C) is sodium cyanate
(b) Gas (A) is cyanogen
(c) Compound (B) is $K_3\,[Cu(CN)_4]$
(d) Gas (A) is pseudohalogen

55. In which of the following pair(s) the minerals are converted into metals by self-reduction process?

(a) Cu_2S, PbS (b) PbS, HgS
(c) PbS, ZnS (d) Ag_2S, Cu_2S

56. Which of the following radical (s) decolourizes $KMnO_4$ without evolving any gas?

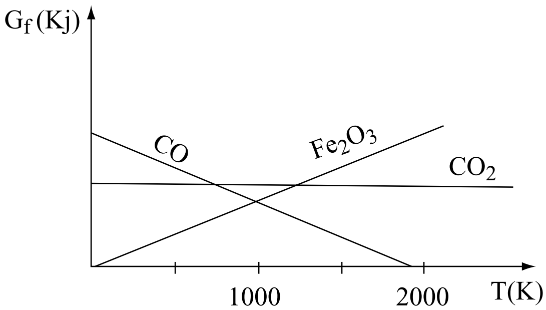
(a) SO_4^{2-} (b) Sn^{2+}
(c) SO_3^{2-} (d) Ag^+

Comprehensions Passage-I

Metals are extracted from their ores by a wide variety of techniques. The most common ores are oxides. (MnO_2, Al_2O_3, SnO_2), sulfides (PbS, ZnS), Chlorides(NaCl, KCl, $CaCl_2$, $MgCl_2$), and phosphates ($Ca_3(PO_4)_2$). Most metals are obtained by direct treatment of their ores with chemical agents, but the extraction of certain metals requires electrolysis. An example of the former type of process is the extraction of iron from its oxide, described by the following equation.

$$Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$$

The relative case of extraction of a metal from its oxide can be estimated using the Ellingham diagram, which is shown in figure-1. This diagram plots the free energies of formation of various oxides per mole of consumed oxygen as a function of absolute temperature.



Electrolytic extraction proceeds in three steps (see equations). In the first step, the crystalline lattice of the ore is thermally disintegrated to form a liquid containing free metal cations. In the second step, the metal cations are stabilized by solvating them with some thermally stable non-aqueous solvent (NAS). In the final step, an applied electric potential reduces the cations to neutral atoms.

$$\text{Step} - 1\ \text{Ore(s)} \rightarrow M^{n+}(l)$$
$$\text{Step} - 2\ M^{n+}(l) + \text{NAS} \rightarrow M^{n+}(\text{solvated})$$
$$\text{Step} - 3\ M^{n+}(\text{solvated}) + ne^- \rightarrow M$$

Extraction of a metal is usually proceeded by enrichment of the ore. Some ores can be concentrated after pulverization by the use of specific collectors such as salts of organic acids and bases, which make the ores hydrophobic and thus separable from hydrophilic admixtures. Other enrichment techniques include density separation and magnetic separation.

1. If for a certain ore the enthalpies of step-1 and 2 were 248.50 and –250.25 kj, respectively. Which of the following would determine the rate of extraction?

(a) The lattice energy of the ore
(b) The solvation energy of the metal cations
(c) The magnitude of the electric potential used
(d) The size of the metal cations.

2. Which of the following best explains why the free energy of formation of Fe_2O_3 become less negative as the temperature increases?

(a) The free energy of formation is independent of the absolute temperature
(b) Entropy drops as a result of the consumption of oxygen
(c) At low temperature, the free energy of formation become less dependent on the enthalpy of formation
(d) As entropy increases, the free energy of formation increases.

3. When ferric oxide is reduced to obtain iron metal, CO_2 or CO can be produced. The production of CO_2.

(a) Is directly proportional to the temperature
(b) Is inversely proportional to the temperature
(c) Is independent of the temperature
(d) requires cooling of the C

Passage-2

Much of the world supply of platinum group metals is derived from the residues recovered from the electrolytic refining of copper and nickel. The residues when heated with aquaregia, the gold platinum and palladium go into solution. This is filtered off and to the filtrate on adding ferrous sulphate solution gold is precipitated. The platinum and palladium remain in solution.

1. The solubility of the gold, platinum and palladium in aquaregia is due to the formation of

(a) AuCl, $PtCl_2$, $PdCl_2$

(b) $AuCl_3$, $PtCl_4$, $PdCl_4$
(c) $HAuCl_4$, H_2PtCl_6, H_2PdCl_4
(d) $HAuCl_4$, H_2PtCl_4, H_2PdCl_4

2. The role of ferrous sulphate in the precipitation of gold is
(a) Oxidant (b) Reluctant
(c) Complexing agent (d) Substituent

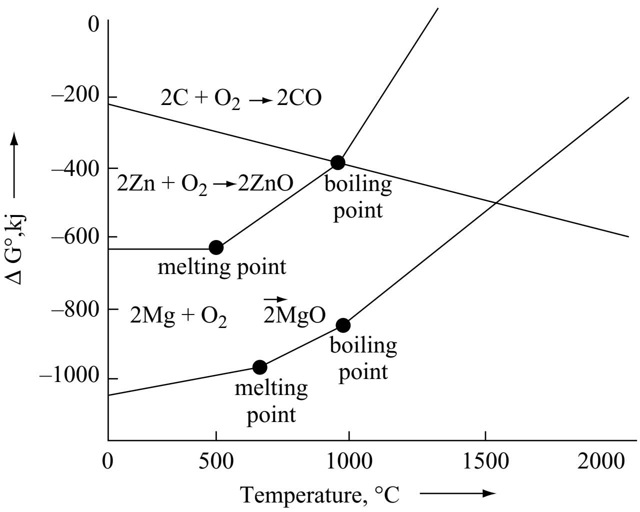
3. The structures of the compounds of platinum and palladium remained in the solution are
(a) Square planar & square planar
(b) Square planar & Tetrahedral
(c) Tetrahedral & Square planar
(d) Octahedral and square planar

4. The spin only magnetic moments of the compounds of gold, platinum and palladium are
(a) 0, 0, 0 (b) 1.73, 1.73, 1.73
(c) 1.73, 0, 2.85 (d) 0, 2.85

5. In all these compounds the d orbital's involved in hybridization are
(a) Outer orbital's
(b) Inner orbital's
(c) Outer orbital's in gold and inner orbital's in platinum & palladium
(d) Inner orbital's in gold and outer orbital's in platinum & palladium

Passage-3

The Ellingham diagram for zinc, magnesium and carbon converting into corresponding oxides is shown below:



1. At what temperature, zinc and carbon have equal affinity for oxygen?
(a) 1000°C (b) 1500°C
(c) 500°C (d) 1200°C

2. To make the following reduction process spontaneous, temperature should be:

$$ZnO + C \rightarrow Zn + CO$$

(a) 1000°C (b) > 1100°C
(c) < 500°C (d) < 1000°C

3. At 1100°C, Which reaction is spontaneous to a maximum extent?
(a) $MgO + C \rightarrow Mg + CO$
(b) $ZnO + C \rightarrow Zn + CO$
(c) $MgO + Zn \rightarrow Mg + ZnO$
(d) $ZnO + Mg \rightarrow MgO + Zn$

Passage-4

Magnesium is a valuable, light weight metal used as a structural material as well as in alloys, batteries, and in chemical synthesis. Although magnesium is plentiful in Earth's crust, it is mainly found in the sea water (after sodium). There is about 1.3g of magnesium in every kilogram of sea water. The process for obtaining magnesium from sea water employs all three types of reactions, i.e., precipitation, acid-base, and red ox-reactions.

1. Precipitation reaction involves formation of:
(a) insoluble $MgCO_3$ by adding Na_2CO_3
(b) insoluble $Mg(OH)_2$ by adding $Ca(OH)_2$
(c) insoluble in $MgSO_4$ by adding Na_2SO_4
(d) insoluble $MgCI_2$ by adding NaCI

2. Acid-base reaction involves reaction between:
(a) $MgCO_3$ and HCI
(b) $Mg(OH)_2$ and H_2SO_4
(c) $Mg(OH)_2$ and HCI
(d) $MgCO_3$ and H_2SO_4

3. Redox reaction takes place (in the extraction of Mg):
(a) In the electrolytic cell when fused $MgCI_2$ is subjected to electrolysis
(b) When fused $MgCO_3$ is heated
(c) When fused $MgCO_3$ is strongly heated
(d) None of the above

Passage-5

$$FeCr_2O_4 + NaOH + air \rightarrow (A) + Fe_2O_3$$
$$(A) + (B) \rightarrow Na_2Cr_2O_7$$
$$Na_2Cr_2O_7 + X \xrightarrow{\Delta} Cr_2O_3$$
$$Cr_2O_3 + Y \xrightarrow{\Delta} Cr$$

1. Compounds (A) and (B) are:
(a) Na_2CrO_4, H_2SO_4
(b) $Na_2Cr_2O_7$, HCI
(c) Na_2CrO_5, H_2SO_4
(d) $Na_4[Fe(OH)_6]$, H_2SO_4

2. (X) and (Y) are:
(a) C and Al (b) Al and C
(c) C in both (d) Al in both

3. Na_2CrO_4 and Fe_2O_3 are separated by:
   (a) dissolving in conc. H_2SO_4
   (b) dissolving in NH_3
   (c) dissolving in H_2O
   (d) dissolving in dil. HCI

Passage-6

A white substance 'X' when heated in a test tube, produces a colourless odorless gas leaving a residue, yellow when hot and white when cold. The residue was dissolved in dil. HCI, made alkaline with NH_4Cl and NH_4OH and H_2S gas was passed through it a white ppt 'Y' was formed. It was dissolved in dil. HCI to give 'Z' which on treatment with $K_4[Fe(CN)_6$ gave bluish white precipitate

1. 'X' is
   (a) ZnO (b) $ZnSO_4$, $7H_2O$
   (c) Cu_2CI_2 (d) $ZnCO_3$
2. Y and Z are respectively
   (a) $ZnCO_3$, $ZnSO_4$ (b) ZnS, $ZnCl_2$
   (c) $ZnCI_2$, $Zn(OH)_2$ (d) ZnO, ZnS
3. Bluish white precipitate formed is
   (a) $Zn_2[Fe(CN)_6]$ (b) $K_2[Zn(CN)_4]$
   (c) $Zn_3[Fe(CN)_6]_2$ (d) None of these

Passage-7

Among the various ores of metal (M) (Sulphide, carbonates, oxides, hydrated or hydroxides) two ores[X] and [Y] show the following reactively.

(i) [X] on calcination gives a black solid (S), carbon dioxide and water.
(ii) [X] dissolved in dil. HCI on reaction with KI gives a white ppt. (P) and iodine.
(iii) [Y] on roasting gives metal (M) and a gas (G_1) (a) which turns acidified $K_2Cr_2O_7$ solution green.
(iv) [Y] on reaction with dil. HCI gives a white precipitate and another gas (G_2) which turns lead acetate solution black and also reacts with gas (G_1) to precipitate colloidal in presence of moisture.

The M, gives greenish blue flame.

1. Which of the following statements is correct about [Y]?
   (a) [Y] is converted to metal (M) by self reduction.
   (b) Carbonate extract of [Y] gives yellow ppt, with suspension of $CdCO_3$.
   (c) [Y] is chalcocites or chalcopyrites
   (d) All of these
2. The white ppt. (P) is of.
   (a) Cu_2l_2 (b) Cul_2
   (c) $K_2[Cul_4]$ (d) None

Passage-8

There are many methods which can be used for the manufacture of steel. One such process is Bessemer process. Here the impurities in pig iron are completely oxidized in the presence of hot blast i.e. virtually obtaining wrought iron. This is then mixed with spiegeleisen in order to obtain steel. The process is carried out in Bessemer converter lined with silica bricks. The molten pig iron is introduced in the converter and a blast of hot air is blown through it from the bottom keeping the mouth of the converter vertically upwards. When the blue flame suddenly dies out then required amount of spiegeleisen is added. The blast is continued for a moment to ensure complete mixing

1. What is spiegeleisen?
   (a) Mixture of wrought iron and cast iron
   (b) Alloy of iron, manganese and carbon
   (c) SiO_2 and iron
   (d) Fe_2O_3 and FeO
2. The acidic flux added is
   (a) MnO (b) SiO_2
   (c) FeO (d) CO_2
3. Stainless steel contains
   (a) 14 per cent Co
   (b) 14–20 per cent W and 3–8 per cent Cr
   (c) 11 per cent Cr and 2 per cent Ni
   (d) 36 per cent Ni

Passage-9

An ore 'A' on roasting with sodium carbonate in presence of air gives two compounds 'B' and 'C'. The solution of 'B' in conc. HCI on treatment with $K_4[Fe(CN)_6]$ gives blue colour of D. The aqueous solution of 'C' on treatment with conc. H_2SO_4 gives a yellowish orange compound 'E'. 'E' when treated with KCl gives an orange red compound 'F' which is used as oxidizing agent. The solution of 'F' on treatment with oxalic acid and then with excess of potassium oxalate gives blue compound G.

1. A and B are respectively
   (a) $FeCr_2O_4$, Fe_2O_3 (b) $FeCr_2O_4$, Na_2CrO_4
   (c) $FeCr_2O_4$, CO_2 (d) $F_2Cr_2O_4$, Cr_2O_3
2. 'F' is
   (a) $Na_2Cr_2O_7$ (b) $K_2Cr_2O_7$
   (c) K_2CrO_4 (d) Na_2CrO_4
3. D and G are respectively
   (a) $Fe_3[Fe(CN)_6]_4, Cr^{3+}$
   (b) $Fe_4[Fe(CN)_6]_3, [Cr(C_2O_4)_3]^{3-}$
   (c) $Fe_4[Fe(CN)_6]_3, CO_2$
   (d) $FeCI_3, CrO_5$

Passage-10

A black mineral 'A' on heating in presence of air gives a pungent smelling gas 'B', 'A' on reaction with dil H_2SO_4 gives a gas 'C' and a solution of compound 'D'. On passing gas 'C' into aqueous solution of 'B', a white turbidity is obtained. The aqueous solution of 'D' on reaction with $K_3[Fe(CN)_6]$ gives a blue compound 'E'

1. 'A' is
   (a) FeO (b) CuO
   (c) FeS (d) MnO_2
2. B and C are respectively
   (a) CO_2, SO_2 (b) SO_2, H_2S
   (c) H_2S, CO (d) O_2CO_2
3. D and E are respectively
   (a) $FeSO_4, Fe_3[Fe(CN)_6]_2$
   (b) $FeCI_2, Fe[Fe(CN)_6]$
   (c) $Fe_2O_3, K_4[Fe(CN)_6]$
   (d) $FeO, Fe_2[Fe(CN)_6]_3$

Passage-11

A compound (A) having formula M_2O_3 on reaction with dilute HCI gave compounds (B) and (C). Compound (C) on heating with conc. HCl gave compound (B) and liberated a greenish yellow coloured gas (X) which liberated Br_2 and I_2 from bromides and iodides. When fluorine gas was passed through the compound (B) in aqueous solution then the same coloured gas was obtained. Compound (C) when heated with potassium hydroxide and air gave a green mass (D). When gas (X) was passed through the solution of (D), purple coloured needle like crystals were obtained (E). (E) is a powerful oxidizing agent which is decolourized by alkenes in alkali medium

1. The oxide M_2O_3 must be an oxide of
   (a) Fe (b) Cr (c) Ti (d) Mn
2. The compound (C) must be
   (a) MnO (b) MnO_2
   (c) MnO_3 (d) MnO_4
3. The compound (E) must be
   (a) K_2MnO_4
   (b) $KMnO_2$
   (c) $KMnO_4$
   (d) K_2MnCl_2

Passage-12

(i) $\text{Metal}(M) \xrightarrow[\text{in dil.}H_2SO_4]{\text{Dissolves}} \underset{(A)}{\text{pale green soln.}} \xrightarrow[\text{alkali}]{\text{add}}$ whitish precipitate quickly turning brown and dissolves in HCI giving yellow solution
(B)

(ii) $\text{Black substance} \xrightarrow[\text{air with }Na_2CO_3]{\text{heated in}} \underset{(C)}{\text{green mass}} \xrightarrow[\text{with water}]{\text{extracted}} \underset{(D)}{\text{green soln.}} \xrightarrow[\text{water}]{\text{chlorine}} \underset{(E)}{\text{pink soln.}}$

1. The green coloured substance formed is
   (a) K_2MnO_4 (b) Na_2MnO_4
   (c) $KMnO_4$ (d) $NaMnO_4$
2. The metal (M) in reaction series (i) is
   (a) Mn (b) Fe (c) Sn (d) Hg
3. The yellow coloured solution in series (i) and the compound (C) in series (ii) are
   (a) $FeCl_3$ and MnO_4^{2-} (b) $FeCl_2$ and MnO_4^-
   (c) $FeSO_4$ and Mn (d) $Fe_2(SCN)_3$ and MnO_4^{2-}

Passage-13

Chiufen, the old mining town located within the hills in the north – east Taiwan, is a place where you can really experience. Taiwan's historical legacy. It was the site of one of the largest gold mines in Asia. Accordingly, Chiufen is often referred to as gold capital of Asia. The compound KCN is traditionally used to extract gold from ore. Gold dissolves in cyanide solutions in the presence of air to form $Au(CN)_2^-$, Which is stable in aqueous solution.

$$Au_{(s)} + CN^-_{(aq)} + O_{2(g)} + H_2O_{(I)} \rightleftharpoons Au(CN)^-_{2(aq)} + OH^-_{(aq)}$$

Aquaregia; a 3:1 mixture (by volume) of cone. HCl and HNO_3 was developed by the alchemists as a means to dissolve gold. The process is actually a redox reaction.

$$Au_{(s)} + NO_{3(aq)} + Cl^-_{(aq)} \rightleftharpoons AuCl^-_{4(aq)} + NO_{2(g)}$$

Gold is too noble to react with nitric acid. However, gold does react with aquaregia because the complex $AuCl_4^-$ forms

$$Au^{+3}_{(aq)} + 3e^- \rightarrow Au_{(s)} \qquad E^0 = 1.5V \rightarrow 1$$

$$AuCl^-_{4(aq)} + 3e^- \rightarrow Au_{(s)} + 4Cl^-_{(aq)} \qquad E^0 = IV \rightarrow 2$$

1. How many grams, approximately, of NaCN are needed to extract 20g of gold from ore? (Au = 197)
   (a) 20g (b) 6.5g (c) 10g (d) 8g
2. Calculate the formation constant, approximately, of $AuCl_4^-$ at 25°C.
   (a) 10^5 (b) 10^{25} (c) 10^{12} (d) 10^{42}

3. The function of HCl is to provide Cl^-. What is the purpose of the Cl^- in the above reaction. Select your choice from the following
   (a) it is an oxidizing agent
   (b) it is a reducing agent
   (c) it is a complexing agent
   (d) it is a catalyst

Passage-14

A light green salt (A) on heating gives black residue (B) and 2 acidic oxides (C) and (D). The salt (A) gives white ppt. with aq $BaCl_2$ solution which does not dissolve in HCl or HNO_3. The gas C also gives E with $BaCl_2$ solution

1. The salt A will be
   (a) $CrSO_4$ (b) $FeSO_4$ (c) $BaSO_4$ (d) $CuSO_4$
2. White ppt which does not dissolve in conc. HCl or HNO_3 will be
   (a) $BaSO_3$ (b) $BaCO_3$ (c) $BaSO_4$ (d) BaC_2O_4
3. Black residue is
   (a) $FeSO_4$ (b) Fe_2O_3
   (c) $Fe_2(SO_4)_3$ (d) none of these

Matching Type Questions

1. Column-I	Column-II
(a) $MgCl_2 \cdot 6H_2O \xrightarrow[\text{dryHCl(g)}]{\text{heat}} MgCl_2 + 6H_2O$	(p) Calcination
(b) $4Au + 8NaCN + 2H_2O + O_2(\text{air}) \rightarrow 4Na[Au(CN)_2] + 4NaOH$	(q) Smelting
(c) $CuFeS_2 + 2H_2SO_4 \rightarrow CuSO_4 + FeSO_4 + 2H_2S$	(r) Leaching
(d) $Fe_3O_4 + 4CO \rightarrow Fe + 4CO_2$	(s) Hydrometallurgy

2. Match the following column-I with column-II

Column-I	Column-II
(a) Poling	(p) Titanium
(b) Cupellation	(q) Copper
(c) Electro-refining	(r) Silver
(d) Van Arkel method	(s) Tin

3. Match the following Column-I with Column-II

Column-I	Column-II
(Metal)	(process involved in commercial extraction from their ore)
(a) Pb	(p) Bessemerization
(b) Cu	(q) Roasting
(c) Zn	(r) Pyrometallurgy
(d) Fe (steel)	(s) Self-reduction method

4. Match the following List-I with List-II

List-I	List-II
(a) Carbon oxide	(p) Cu
(b) Liquation	(q) Ag
(c) Cupellation	(r) Ni
(d) Hydrometallurgy	(s) Sn

5. Match the following one or more match are possible

List-A	List-B
(a) $NH_4(WO_4)$	(p) Reduction by H_2
(b) NiO	(q) Goldsmidt thermite process
(c) Fe_2O_3	(r) Van Arkel process
(d) ZrI_4	(s) Mond' process
	(t) Carbon

6. Match the following.

Column-I	Column-II
(a) Metal extraction from galena	(p) Leaching with complexing agent or an alkali
(b) Metal extraction silver	(q) Electrolytic reduction/refining
(c) Metal extraction from haematite	(r) carbon reduction process
(d) Metal extraction from bauxite	(s) Self reduction process.

7. Match the methods given in List-II for the refining of metals given in List-I

List-I	List-II
(a) Zinc	(p) Poling
(b) Mercury	(q) Distillation
(c) Copper	(r) Liquation
(d) Silver	(s) Electrolytic refining
	(t) Cupellation

8. Match the following in column-I with in column-II

Column-I	Column-II
(a) Copper pyrites	(p) carbon reduction
(b) Galena	(q) Self reduction
(c) Silver glance	(r) Froath floatation
(d) gold	(s) Mac-Arthur Forrest process

9. Match the compounds in List-I with their properties in List-II

List – I	List-II
(a) K_2MnO_4	(p) Transition element in +6 state
(b) $KMnO_4$	(q) Oxidizing agent in acid medium
(c) $K_2Cr_2O_4$	(r) Manufactured from pyrolusite ore
(d) K_2CrO_4	(s) Manufactured from chromite ore

10. Match the following column-I with column-II

Column-I	Column-II
(a) Non-stoichiometric	(p) Fe_3O_4
(b) Ferric hydroxide is heated below 140°C	(q) $Fe(OH)_2$
(c) Sodium ferrite is hydrolyzed to give	(r) FeO
(d) Does not give red colour with KCNS	(s) Fe_2O_3

11. Column-I	Column-II
(a) Kipps apparatus waste	(p) $(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O$
(b) Green coloured compound	(q) $Cu(OH)_2 \cdot CuCO_3$
(c) Leaves (s) brown residue on heating	(r) $FeSO_4$
(d) Leaves (s) black residue on heating	(s) $CuCl_2 \cdot 2H_2O$
	(t) Castner's process

12. Column-I	Column-II
(a) soluble in a concentrated NH_3 solution	(p) Ag_2S
(b) soluble in excess KCN Solution	(q) $Cu(OH)_2$
(c) Soluble in excess hypo solution	(r) AgBr
(d) soluble in conc. HCl	(s) AgCl

13. List-I	List-II
(a) Sodium cyanide	(p) Depressent
(b) sodium hydroxide	(q) Leaching agent
(c) potassium ethyl xanthate	(r) Collector
(d) calcium carbonate	(s) Flux

14. Match the following column-I with column-II

Column-I	Column-II
(a) Asbestos	(p) Calcium
(b) Willimite	(q) Lead
(c) Anglesite	(r) Silicate
(d) Diaspore	(s) Aluminium
	(t) Sulphate

Numerical Type Questions

1. Find the no. of gaseous species (including water vapours) obtained on heating basic lead carbonate, $PbCO_3 \cdot Pb(OH)_2$ at 450°C
2. Iodate ions (IO_3^-) can be reduced to iodine by iodide ions. How many moles of iodine are produced for every mole of iodate ions consumed in the reaction.
3. The number of chloride ions surrounding each Cu^{2+} ion in solid copper (II) chloride is.
4. The number of chloride ions surrounding each Fe^{3+} ion in solid anhydrous ferric chloride is
5. The number of chloride ions coordinated to Fe^{3+} ion in hydrated $FeCl_3 \cdot 6H_2O$
6. The number of water molecules coordinated to Fe^{3+} in hydrated $FeCl_3 \cdot 6H_2O$
7. Solid $CuCl_2$ has layered lattice structure and each Cu^{2+} ion is surrounded some chloride ions in the same layer and by some chloride ions of adjacent layers. So all the Cu-Cl bond lengths are not equal. The number of short bonds is
8. Among the following the number of compounds having layered lattice structure in solid state
   (a) Graphite
   (b) Boron nitride
   (c) solid anhydrous ferric chloride
   (d) solid anhydrous aluminium chloride
   (e) solid cupric chloride
   (f) Black phosphorous
   (g) diamond
   (h) Borazole.
9. Number of oxygen atoms surrounding each Cu^{2+} ion in solid $CuSO_4 \cdot 5H_2O$

Single Answer Type Questions

1. a	**2.** b	**3.** c	**4.** c	**5.** c	**6.** c	**7.** c
8. d	**9.** b	**10.** a				
11. b	**12.** c	**13.** b	**14.** c	**15.** b	**16.** b	**17.** d
18. c	**19.** b	**20.** a				
21. b	**22.** d	**23.** d	**24.** b	**25.** b	**26.** a	**27.** a
28. d	**29.** a	**30.** d				
31. a	**32.** d	**33.** d	**34.** c	**35.** a	**36.** b	**37.** a
38. c	**39.** a	**40.** c				
41. d	**42.** b	**43.** a	**44.** d	**45.** d	**46.** a	**47.** b
48. b	**49.** b	**50.** b				
51. c	**52.** c	**53.** a	**54.** c	**55.** b	**56.** b	**57.** a
58. d	**59.** c	**60.** d				
61. c	**62.** a	**63.** a	**64.** b	**65.** a	**66.** c	**67.** d
68. b	**69.** c	**70.** c				
71. d	**72.** a	**73.** a	**74.** d	**75.** d	**76.** c	**77.** c
78. a	**79.** b	**80.** d				
81. c	**82.** a	**83.** c	**84.** c	**85.** d	**86.** a	**87.** b
88. c	**89.** a	**90.** b				
91. b	**92.** d	**93.** a	**94.** c	**95.** a	**96.** d	**97.** a
98. c	**99.** c	**100.** b				
101. c	**102.** d	**103.** b	**104.** c	**105.** a	**106.** d	**107.** c

More than One Answer Questions

1. a, b, d	**2.** b, c, d	**3.** b, c, d	**4.** a, b
5. b, d	**6.** a, b	**7.** a, b, d	**8.** a, b, c, d

9. a, b, c, d	**10.** b, d	**11.** a, b, c, d	**12.** a, c, d
13. b, c, d	**14.** b, c	**15.** a, b, d	**16.** a, d
17. a, b, d	**18.** a, b	**19.** a, b, c	**20.** b, c
21. c	**22.** a, b, d	**23.** a, b, c, d	**24.** a, d
25. a, b	**26.** a, b, c	**27.** a, b, c	**28.** b, c, d
29. a, b	**30.** b, c	**31.** b, c	**32.** a, c
33. b, c, d	**34.** a, b	**35.** b, c	**36.** a, b, d
37. a, b, c	**38.** b, c, d	**39.** a, b, c	**40.** a, b
41. a, b, c, d	**42.** a, b, c	**43.** a, b, c	**44.** a, c, d
45. a, b, c	**46.** b, c	**47.** b, d	**48.** a,b
49. a, b, c	**50.** a, b, c	**51.** a, b, c	**52.** b, c, d
53. a, b, c	**54.** a, b, c, d	**55.** a, b	**56.** b, c

Passage – I	**1.** c	**2.** b	**3.** c		
Passage – II	**1.** c	**2.** b	**3.** d	**4.** a	**5.** b
Passage – III	**1.** a	**2.** b	**3.** d		
Passage – Iv	**1.** b	**2.** c	**3.** a		
Passage – V	**1.** a	**2.** a	**3.** c		
Passage – VI	**1.** d	**2.** b	**3.** a		
Passage – VII	**1.** a	**2.** a			
Passage – VIII	**1.** b	**2.** b	**3.** c		
Passage – IX	**1.** a	**2.** b	**3.** b		
Passage – X	**1.** c	**2.** b	**3.** a		
Passage – XI	**1.** d	**2.** b	**3.** c		
Passage – XII	**1.** b	**2.** b	**3.** a		
Passage – XIII	**1.** c	**2.** b	**3.** c		
Passage – XIV	**1.** b	**2.** c	**3.** b		

Match the Type Questions

1. a-p	b-r, s	c-r, s	d-q
2. a-q, s	b-r	c-q, r, s	d-p
3. a-q, r, s	b-p, q , r, s	c-q, r	d-p, r
4. a-p, r, s	b-s	c-q	d-p, q
5. a-p	b-p, s	c-q, t	d-r
6. a-r, s	b-p, q	c-r	d-p, q
7. a-q, r, s	b-q	c-p, s	d-s, t
8. a-p, q, r	b-p, q, r	c-r, s	d-s
9. a-p, q, r	b-q, r	c-p, q, s	d-p, q, s
10. a-p, r, s	b-s	c-s	d-q, r
11. a-r	b-p, q, r, s	c-p, r	d-q
12. a-q, r, s	b-p, q, r, s	c-q, r, s	d-q, s
13. a-p, q	b-q	c-r	d-s
14. a-p, r	b-r	c-q, t	d-s

Integer Type Questions

1. 3 **2.** 3 **3.** 6 **4.** 6 **5.** 2 **6.** 4 **7.** 4
8. 6 **9.** 6

Hints to Single Answer Type Questions

2. NaCN forms a complex $Na_2[Zn(CN)_4]$ on the surface of ZnS, ore particles so that it is not adsorbed by potassium ethyl xanthate. Hence it remain in solution. PbS comes into froth

3. Zinc is purified by fractional distillation to remove the impurities iron and cadmium

4. When $K_2Cr_2O_7$, a soluble chloride and conc H_2SO_4 are heated orange red oily vapours of chromyl chloride are liberated

5. Since Cu^{2+} acts as oxidizing agent and I^- acts as reducing agent $[CuI_4]^{2-}$ is not stable, but Cu^{2+} cannot oxidize Cl^- ions

6. copper (I) salts are stable only in solid state but disproportionate into Cu^{2+} and Cu in aqueous solution.

7. As the sum of first four ionization energies are less for platinum it is more stable in +4 oxidation state than Ni^{4+}

10. All the four elements can be extracted by carbon reduction method

14. Sulphide ores are concentrated by roasting

15. Cinnabar HgS on heating first converts to HgO which is unstable at high temp and decomposes to Hg and O_2

17. $Ag_2S + 4NaCN \rightarrow 2Na[Ag(CN]_2 + Na_2S$
By blowing air Na_2S is oxidized to Na_2SO_4 preventing the backward reaction

18. Bauxite containing Fe_2O_3 is purified by Baeyer's process, while that containing silica as impurity is purified by Serpeck's process

27. Quenching makes the steel hard and brittle.

32. $AgCl + Na_2CO_3 \rightarrow Ag_2CO_3 + 2NaCl$
$2Ag_2CO_3 \rightarrow 4Ag + 2CO_2 + O_2$
$\Delta H_{CO_2} = -393 KJ\ mol^{-1}$ $\Delta H_{CS_2} = +89 KJ\ mol^{-1}$

36. KCN depress ZnS and FeS by forming complexes. Only PbS comes into froth. After removing PbS if NaOH is added it removes complex of zinc from the surface of ZnS ore particles so that they come into froth. If $CuSO_4$ and acid is added it acts as activator for ZnS to come into froth

37. $MgCl_2 \cdot 6H_2O \xrightarrow{\Delta} MgCl_2 \cdot 2H_2O + 4H_2O$
$MgCl_2 \cdot 2H_2O \xrightarrow{\Delta} Mg(OH)Cl + HCl + H_2O$
$Mg(OH)Cl \xrightarrow{\Delta} MgO + HCl$

38. Mercury in +1 oxidation state always exist as Hg_2^{2+} with metal –metal bonding $[Hg\text{-}Hg]^{2+}$

44. Mn_2O_7 is formed with conc H_2SO_4

45. Magnesium cannot be extracted from aqueous solution by electrolysis as H_2 is liberated at cathode instead of Mg

48. Ferrous oxide combines with silica forming $FeSiO_3$ slag

50. $CaCO_3 + Heat \rightarrow CaO + CO_2$
$CaO + SiO_2 \rightarrow CaSiO_3$

58. First haematite and oxides of impurities are reduced to corresponding elements. In the purification of iron the impurities are oxidized

75. Pickling is a process of the removal basic metal oxides, hydroxides and carbonate from the surface of metal by dipping in dil H_2SO_4 before galvanization, electroplating, enameling painting

84. Chalcopyrites $CuFeS_2$ which breaks up into Cu_2S, FeS and SO_2 during roasting. When air is passed Cu_2S converts into Cu_2P which is reduced by Cu_2S.

85. $\underset{\text{Black}}{MnO_2} + 2KOH + KNO_3 \rightarrow \underset{\text{green}}{K_2MnO_4} + KNO_2 + H_2O$
$2K_2MnO_4 + H_2O + O_3 \rightarrow 2KMnO_4 + 2KOH + O_2$

89. HgO is unstable at high temperatures and decomposes mercury metal and oxygen

91. $HgCl_2 + 2NH_3 \rightarrow NH_2.Hg \cdot Cl + Hg + NH_4Cl$

92. Pure zinc react with dil H_2SO_4 slowly due to hydrogen over voltage but reacts fastly in the presence of $CuSO_4$
$Zn(OH)_2$ dissolves in both NaOH and NH_4OH due to formation of Na_2ZnO_2 and $[Zn(NH_3)_4]^{2+}$

93. $H[AuCl_4] + 4NaOH \rightarrow Au(OH)_3 + 4NaCl + H_2O$

97. Fe^{3+} is an oxidizing agent and I^- is a reducing agent

98. $KNO_3 \rightarrow KNO_2 + (O)$
$MnO_2 + 2KOH + (O) \rightarrow K_2MnO_4 + H_2O$
$K_2MnO_4 + H_2O \xrightarrow{\text{Electrolysis}} KMnO_4 + KOH + H_2$

100. NaCl and NaF increases the conductivity and decreases the melting point of anhydrous magnesium chloride

More than One Answer Type Questions

1. The oxalate or citrate ion reduces ferric salt to ferrous salt in the presence of light (photochemical reduction) which gives blue colour with potassium ferricyanide (Turn bull's blue) so white lines are obtained in place of the ink drawing on a deep blue back ground.
2. Oxalate ion can reduce the acidified permanganate ferric ion is more stable than ferrous ion due to stable $3d^5$ configuration
Mohr's salt is less susceptible for oxidation
5. Cassiterite is SnO_2; Litharge is PbO
Cuprite is Cu_2O, tinstone is SnO_2
6. In Baeyer's process and Hall's process red bauxite is leached with NaOH and Na_2CO_3 respectively in which amphoteric nature of aluminium is utilized
7. 1 Calamine $ZnCO_3$ 2. Pyrolusite-MnO_2
3. Cassiterite –SnO_2 4. Calcite – $CaCO_3$
10. 2 Haematite Fe_2O_3; Siderite $FeCO_3$
4 Galena PbS Cerrusite $PbCO_3$

31. $Fe_2O_3 + 3CO \xrightarrow[\Delta]{\text{blastfurnace}} 2Fe + 3CO$
$ZnO + C \xrightarrow{1200°C} Zn + CO$
$2Ca_3(PO_4)_2 + SiO_2 + 10C \xrightarrow{\Delta} 6CaSiO_3 + 10CO + P_4$
$MgO + C \xrightarrow{2000°C} Mg + CO$

32. $Cu(OH)_2 \cdot CuCO_3 \rightarrow 2CuO + CO_2 + H_2O$

40. Zinc having low boiling point distill over on boiling leaving behind Ag. If boiled with NaOH, Zinc dissolves due to formation of Na_2ZnO_2 but silver is unaffected.

47. Partial roasting gives Cu_2O. The remaining Cu_2S reduces the oxidized ore.
$Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$

49. 1. $2ZnSO_4 \rightarrow 2ZnO + 2SO_2 + O_2$
2. $Zn(OH)_2 \rightarrow ZnO + H_2O$
3. $ZnCO_3 \rightarrow ZnO + CO_2$
$Li_2CO_3 \rightarrow Li_2O + CO_2$
4. $ZnSO_4 + 2NaOH \rightarrow Zn(OH)_2 + Na_2SO_4$
$Zn(OH)_2 + 2NaOH \rightarrow Na_2ZnO_2 + 2H_2O$

Comprehensive Type Questions

Passage-I

1. As the enthalpies of liquifaction and salvation are almost equal, the rate of extraction depends on the magnitude of the electric potential used.
2. As solid Fe_2O_3 is formed by consuming gaseous O_2 entropy decreases and hence ΔG becomes less negative.
3. The ΔG formation of CO_2 does not depend on absolute temperatures

Passage-2

1. Gold, platinum and palladium dissolves in aquaregia forming $HAuCl_4$, H_2PtCl_6 and H_2PdCl_4
2. Fe^{2+} reduces the Au^{3+} to Au
3. H_2PtCl_6 is octahedral and H_2PdCl_4 square planar

4,5. All the four coordinate complexes of 2nd and 3rd transistion series elements are low spin inner orbital complexes.

Passage-4

1. Extraction of magnesium from sea water is called Dow process in which Mg is precipitated from sea water as $Mg(OH)_2$ by adding slaked lime $[Ca(OH)_2]$
2. The $Mg(OH)_2$ precipitate is dissolved in HCl
3. Magnesium is extracted by the electrolytic reduction of Mg^{2+} ion by subjecting the fused anhydrous $MgCl_2$ to electrolysis

Passage-5

$$4FeCr_2O_4 + 16NaOH + 7O_2 \longrightarrow 8Na_2CrO_4 + 2Fe_2O_3 + 8H_2O$$

$$\underset{A}{2Na_2CrO_4} + \underset{B}{H_2SO_4} \longrightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$$

$$Na_2Cr_2O_7 + C \xrightarrow{\Delta} Na_2CO_3 + Cr_2O_3 + CO$$

$$Cr_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Cr$$

Passage-6

$$\underset{(X)}{ZnCO_3} \rightarrow ZnO + CO_2$$

$$ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$$

$$ZnCl_2 + H_2S \rightarrow \underset{(Y)}{ZnS} + 2HCl$$

$$ZnS + 2HCl \rightarrow \underset{(Z)}{ZnCl_2} + H_2S.$$

$$2ZnCl_2 + K_4[Fe(CN)_6 \rightarrow Zn_2[Fe(CN)_6]$$

Passage-7

X maybe malachite or azurite

$$CuCO_3 \cdot Cu(OH)_2 \xrightarrow{\Delta} \underset{\text{black.}}{CuO} + H_2O + CO_2$$

$$CuCl_2 + 2HCl \rightarrow CuCl_2 + H_2O$$

$$CuCl_2 + 2KI \rightarrow CuI_2 + 2KCl$$

$$2CuI_2 \rightarrow Cu_2I_2 + I_2$$

Y is sulphide ore which on roasting gives SO_2 gas which can act both oxidizing and reducing agent.

Passage-9

$$\underset{(A)}{4FeCr_2O_4} + 16NaOH + 7O_2 \rightarrow \underset{(C)}{Na_2CrO_4} + \underset{(B)}{2Fe_2O_3} + 8H_2O$$

$$\underset{(B)}{Fe_2O_3} + 6HCl \rightarrow 2FeCl_3 + 3H_2O$$

$$4FeCl_3 + 3K_4[Fe(CN)_6] \rightarrow \underset{(D)}{Fe_4[Fe(CN)_6]_3} + 12KCl$$

$$\underset{(C)}{2Na_2CrO_4} + H_2SO_4 \rightarrow \underset{(E)}{Na_2Cr_2O_7} + Na_2SO_4 + H_2O$$

$$\underset{(E)}{Na_2Cr_2O_7} + 2KCl \rightarrow \underset{(F)}{K_2Cr_2O_7} + 2NaCl$$

Oxalic acid reduces the potassium dichromate to Cr^{3+} and then forms a complex $K_3[Cr(C_2O_4)_3]$ with potassium oxalate crystals.

Passage-11

$$\underset{(A)}{Mn_2O_3} + 2HCl \rightarrow \underset{(B)}{MnCl_2} + \underset{(C)}{MnO_2} + H_2O$$

$$\underset{(C)}{MnO_2} + 4HCl \rightarrow \underset{(B)}{MnCl_2} + 2H_2O + \underset{(X)}{Cl_2}$$

$$2KBr + \underset{(X)}{Cl_2} \rightarrow 2KCl + Br_2$$

$$MnCl_2 + F_2 \rightarrow MnF_2 + \underset{(X)}{Cl_2}$$

$$\underset{(C)}{MnO_2} + 2KOH + (O) \rightarrow \underset{(D)}{K_2MnO_4} + H_2O$$

$$\underset{(D)}{2K_2MnO_4} + Cl_2 \rightarrow \underset{(E)}{2KMnO_4} + KCl$$

Passage-12

(1) $$Fe + H_2SO_4 \rightarrow \underset{\substack{\text{green} \\ (A)}}{FeSO_4} + H_2$$

$$FeSO_4 + 2NaOH \rightarrow Fe(OH)_2 + Na_2SO_4$$

$$4Fe(OH)_2 + 2H_2O + O_2 \rightarrow 4Fe(OH)_3$$

$$Fe(OH)_3 + 3HCl \rightarrow FeCl_3 + 3H_2O$$

(2) $$MnO_2 + Na_2CO_3 + (O) \rightarrow \underset{C\,\&\,D}{Na_2MnO_4} + CO_2$$

$$2Na_2MnO_4 + Cl_2 \rightarrow \underset{E.}{NaMnO_4} + NaCl$$

Passage-13

1. $4Au + 8NaCN + 2H_2O + O_2 \rightarrow 4Na[Au(CN)_2] + 4NaOH$

   $4 \times 197 \quad 8 \times 49$

   $197 \qquad 98$

   $\therefore$ for 20 gm approximately 10

2. $Au^{3+}{}_{aq} + 4Cl^-{}_{aq} \rightleftarrows [AuClu]^-{}_{aq}$

   At equilibrium E = 0 and K = Keq = K formation
   The above reaction can be written as.

$Au_{(S)} + 4cl_{(eq)} \rightarrow AuClu_{(eq)} \qquad E° = -1V$

$Au^{3+}{}_{eq} + 3e^- \rightarrow Au_{(S)} \qquad E° = 1.5V$

$Au^{3+}{}_{(eq)} + 4Cl^-{}_{(eq)} \rightleftarrows [AuClu]^-{}_{(eq)} \quad E° = 0.5$

Using nearest equation.

$E = E° - \frac{0.059}{n} 10g\,Keq$

$O = 0.5 - \frac{0.059}{n} 10g\,Keq$

$Keq = 10^{25.}$ (approximately).

16 CHAPTER

Qualitative Analysis

16.1 INTRODUCTION

Qualitative analysis deals with the identification of inorganic substance or a mixture of substances. Generally inorganic substances dissociate in water to give positive and negative ions. The positive ions are also called as **basic radicals** and the negative ions are called as **acid radicals**. The tests carried for the identification of different ions can be divided into three types

1. Preliminary tests
2. Tests for anions
3. Tests for cations

16.2 PRELIMINARY TESTS

16.2.1 Colour

The colour of the substance gives the indication of the presence of certain cations and anions. Generally the transition metals form coloured compound. From the colour of the substance, the presence of certain cations can be predicted. Some of the commonly occurring compounds are listed below.

Red: $Pb_3O_4, As_2S_3, HgO, Sb_2S_3, CrO_3, Cu_2O$ $K_3[Fe(CN)_6]$; dichromates are orange red; permanganates and chrome alums are reddish purple.

Pink: Hydrated salts of manganese and of cobalt.

Yellow: CdS, As_2S_3, SnS_2, PbI_2, HgO (precipitated) $K_4[Fe(CN)_6]\cdot 3H_2O$; Normal chromates, ferric chloride and nitrate.

Green: $Cr_2O_3, Hg_2I_2, Cr(OH)_3$; ferrous salts i.e., $FeSO_4\cdot 7H_2O; FeSO_4\cdot (NH_4)_2SO_4\cdot 6H_2O; FeCl_2\cdot 4H_2O$ nickel salts, $CrCl_3\cdot 6H_2O, CuCl_2\cdot 2H_2O, CuCO_3, K_2MnO_4$

Blue: Anhydrous cobalt salts; hydrated cupric salts; Prussian blue.

Brown: PbO_2, CdO, Fe_3O_4, SnS, Fe_2O_3, and $Fe(OH)_3$ (reddish brown)

Black: PbS, CuS, CuO, HgS, FeS, MnO_2, Co_3O_4, CoS, NiS, Ag_2S.

The colour of the solution obtained when the substance is dissolved in water or in dilute acids should be noted. As this may often give valuable information. The following colours are shown by the ions (the cations are usually hydrated) present in the dilute solution

Blue: Cupric copper
Green: Nickel, Ferrous iron, chromic chromium
Yellow: Chromates, ferrocyanides, ferric iron
Orange-red: Dichromates
Purple: Permanganates
Pink: Cobalt, manganese

16.2.2 Action of Heat

Place a small quantity of the substance in a dry ignition tube so that none of it remain adhering to the sides and heat cautiously; the tube should be held in an almost horizontal position. The temperature is gradually raised.

Observation	Inference
(a) The substance changes colour	
(i) Yellow when hot white when cold	ZnO and many Zn Salts
(ii) Yellowish-brown when hot, yellow when cold	SnO_2 or Bi_2O_3
(iii) Red to black when hot, brown when cold	Fe_2O_3
(b) Sublimate is formed	
(i) White sublimate	$HgCl_2$, $HgBr_2$, Hg_2Cl_2 ammonium halides, As_2O_3, Sb_2O_3
(ii) Blue black sublimate	Iodine
(c) A gas or vapour is evolved	
(i) Water is evolved; Test with litmus paper	Compounds with water of crystallization (often accompanied by change in colour) ammonium salts, acid salts, hydroxides
The water is alkaline	Ammonium salts
The water is acid	Readily decomposed salts of strong acids
(ii) Oxygen is evolved (rekindles glowing splint).	Nitrates, chlorates, perchlorates, bromates, iodates, peroxides, permanganates.
(iii) Nitrous oxide (rekindles glowing splint) and steam are evolved	Ammonium nitrate
(iv) Reddish brown fumes	Nitrates or nitrites of heavy metals.
(v) CO_2 is evolved (lime water rendered turbid)	Carbonates, bicarbonates
(vi) Ammonia is evolved (odour; turns red litmus paper blue, turns mercurous nitrate paper black)	Ammonium salts
(vii) SO_2 is evolved (odour smell of burning sulphur, turns $K_2Cr_2O_7$ paper green)	Normal and acid sulphites
(viii) H_2S is evolved (Rotten eggs smell, turns lead acetate paper black.)	Acid sulphides

16.2.3 Flame Colour Test

Place a small quantity of the substance on a watch glass moisten with a little concentrated hydrochloric acid and introduce a little of the substance on a clean platinum wire into the base of the non luminous Bunsen flame. Since chlorides are more volatile, the substance is mixed with concentrated hydrochloric acid to convert the metal salt into metal chloride.

Flame colour	Inference
Golden Yellow	Sodium
Violet (lilac)	Potassium
Carmine red	Lithium
Brick red	Calcium
Crimson red	Strontium
Apple green (Yellow green)	Barium
Bluish green	Copper
Green flashes	Zinc or Manganese

16.2.4 Charcoal Cavity Test

Heat a little substance in a small cavity scooped in a charcoal block using a blow pipe. The salt is heated in the charcoal cavity directly or after mixing with sodium carbonate or fusion mixture ($Na_2CO_3 + K_2CO_3$)

The sodium carbonate or fusion mixture converts the metallic salt into a carbonate which decomposes into metal oxides on heating.

The metal oxides

(i) the metal oxides are left as coloured residues, the colour of the residue being characteristic of the basic radical present (or)

(ii) the metal oxides undergo reduction to metallic state by the reducing action of carbon or charcoal resulting in the formation metallic beads of easily fusible (or)

(iii) the metal so formed volatilizes in the form of vapours which burn in air to form oxide. These get in the form of the layer or crust around the cavity (incrustation)

Observation	Inference
(i) White infusible and incandescent when hot	BaO, SrO, CaO, MgO (residue is alkaline to litmus paper) Al_2O_3, ZnO (residue not alkaline to litmus paper)
(ii) Incrustation with out metal	
(a) white, yellow when hot	ZnO
(b) white, garlic odour	As_2O_3
(iii) Incrustation with metal	
(a) White incrustation; brittle metal	Sb
(b) Yellow incrustation; brittle metal	Bi
(c) Yellow incrustation, malleable metal, marks paper	Pb
(iv) Metal without incrustation	
(a) Grey metallic particles attracted by magnet	Fe, Ni, Co
(b) Malleable beads	Ag and Sn (white), Cu (red flakes)

Cobalt nitrate Test: This test is performed if white infusible residue is obtained in charcoal cavity test. The salts of aluminium, magnesium, zinc, calcium etcyields oxides which are white in colour. This residue is heated with cobalt nitrate solution in a reducing flame. Cobalt oxide formed as a result of decomposition of cobalt nitrate combines with metallic oxides to form coloured compounds which are characteristic cations.

$$2Co(NO_3)_2 \longrightarrow 2CoO + 4NO_2 + O_2$$

$$CoO + ZnO \longrightarrow CoZnO_2$$

Green residue (Rinmann's green)

$$CoO + Al_2O_3 \longrightarrow Co(AlO_2)_2$$

Blue residue (Thenard Blue)

$$CoO + MgO \longrightarrow CoMgO_2$$

Pink residue

16.2.5 Borax Bead Test

When borax is heated, it first loses its water of crystallization and then decomposes to give a clear and transparent bead consisting of boric anhydride and sodium metaborate.

$$Na_2B_4O_7 \cdot 10H_2O \xrightarrow{\Delta} Na_2B_4O_7 + 10H_2O$$

$$Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + B_2O_3$$

Boric anhydride being less volatile displaces more volatile acidic anhydrides from their salts to form metaborates which possess characteristic colours

$$B_2O_3 + CuSO_4 \longrightarrow Cu(BO_2)_2 + SO_3$$

In some cases meta borates of two different colours are produced in oxidizing and reducing flames

$$B_2O_3 + CuSO_4 \longrightarrow Cu(BO_2)_2 + SO_3$$

green when hot and blue
when cold in oxidising flame.

$$2Cu(BO_2)_2 + C \longrightarrow 2CuBO_2 + B_2O_3 + CO$$

Colourless

In certain cases the bead becomes even opaque in the reducing flame. This happens when too much of the metal borate is present and its reduction to metallic state is possible.

$$2CuBO_2 + C \longrightarrow 2Cu + B_2O_3 + CO$$

Reddish brown when hot.

The colours produced by different cations are shown in Table 16.1.

Table 16.1 Colour of the borax bead with different cations

Colour of bead in oxidizing flame		Colour of bead in reducing flame		Metal
Hot	**Cold**	**Hot**	**Cold**	
(i) Green	Blue	Colourless	Opaque red	Copper
(ii) Yellowish-brown or red	Yellow	Green	Green	Iron
(iii) Dark yellow	Green	Green	Green	Chromium
(iv) Violet (amethyst)	Violet (amethyst)	Colourless	Colourless	Manganese
(v) Blue	Blue	Blue	Blue	Cobalt
(vi) Brown	Brown	Grey or Black	Opaque	Nickel

16.2.6 Microcosmic Salt Bead Test

The bead is produced similarly to the borax bead except that microcosmic salt is used. The colourless transparent bead contains sodium metaphosphate

$$Na(NH_4)HPO_3 \rightleftharpoons NaPO_3 + H_2O + NH_3$$

This combines with metallic oxides forming orthophosphates which are often coloured.

The colours of the metal phosphate beads are same as in the borax bead test given in Table 16.1.

16.3 TESTS FOR ANIONS

The tests for anions can be grouped into three types

1. The anions which respond to dilute acid.
   CO_3^{2-}, SO_3^{2-}, S^{2-}, CH_3COO^-
2. The anions which respond to conc. H_2SO_4
   Cl^-, Br^-, I^-, NO_3^-
3. The anions which does not respond to dil or conc H_2SO_4, SO_4^{2-}

16.3.1 Anions which Respond to Dilute Acid

Treat a small amount of the substance in a small test-tube with 2 mL of 2N hydrochloric acid or sulphuric acid and note the reaction taking place

Observation	Inference
1. Colourless gas is evolved with effervescence. Gas is odourless and produces turbidity when passed into lime water	CO_2 from carbonate or bicarbonate

Observation	Inference
2. Colourless gas is evolved, with suffocating odour; turns filter paper moistened with acidified $K_2Cr_2O_7$ solution to green	SO_2 from sulphite
3. Colourless gas with rotten eggs smell; gives above test and turns the filter paper dipped in lead acetate to black	H_2S gas from sulphides
4. Smell of vinegar on warming	Acetic acid from acetates

16.4 CONFIRMATORY TESTS AND REACTIONS OF ANIONS WHICH RESPOND WITH DILUTE ACID

16.4.1 Reactions of Carbonate Ion

1. All normal carbonates except those of alkali metal carbonates and ammonium carbonate are insoluble in water. The bicarbonates of Ca, Sr, Ba, Mg and possibly of iron exist only in solution. The bicarbonates of the alkali metals are soluble in water but are less soluble than the corresponding normal carbonates.

2. With dilute hydrochloric acid: When dil HCl or dil H_2SO_4 is added to a salt containing carbonate or bicarbonate brisk effervescence will be evolved due to evolutions of CO_2 by the decomposition of carbonate or bicarbonate. The gas liberated when passed into lime water turns it milky due to the formation of calcium carbonate. But passing more, the white precipitate formed will be dissolved again.

$$Na_2CO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2 \uparrow$$

$$NaHCO_3 + H_2SO_4 \longrightarrow NaHSO_4 + H_2O + CO_2 \uparrow$$

$$\underset{\text{Lime water}}{Ca(OH)_2} + CO_2 \longrightarrow \underset{\text{milky}}{CaCO_3 \downarrow} + H_2O$$

$$CaCO_3 + CO_2 + H_2O \longrightarrow Ca(HCO_3)_2$$

3. Barium Chloride or Calcium Chloride Solutions: White precipitate of barium carbonate $BaCO_3$ or calcium carbonate $CaCO_3$ with solutions of normal carbonates. The precipitate is soluble in dil HCl or dil HNO_3

$$BaCl_2 + Na_2CO_3 \longrightarrow BaCO_3 + 2NaCl$$

$$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaCl$$

$$BaCO_3 + 2HCl \longrightarrow BaCl_2 + H_2O + CO_2$$

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$$

4. Silver Nitrate Solution: White precipitate of silver carbonate Ag_2CO_3 with solution of normal carbonates soluble in ammonia solution and in nitric acid. The precipitate become yellow upon addition of excess of the reagent and is partly decomposed on boiling with water into brown silver oxide Ag_2O and carbon dioxide.

5. Test for Bicarbonate: If sodium bicarbonate solution is added to magnesium sulphate, no precipitate is formed because magnesium bicarbonate is soluble in water. But on boiling the solution a white precipitate appears due to the formation of magnesium carbonate $MgCO_3$. Solutions of carbonates give white precipitate of magnesium carbonate in cold condition with magnesium sulphate solution.

$$2NaHCO_3 + MgSO_4 \xrightarrow{\text{Cold}} Mg(HCO_3)_2 + Na_2SO_4$$

$$Mg(HCO_3)_2 \xrightarrow{\Delta} MgCO_3 + H_2O + CO_2$$

$$MgSO_4 + Na_2CO_3 \xrightarrow{\text{Cold}} MgCO_3 + Na_2SO_4$$

6. Test with Phenolphthalein: Carbonate solutions turns the phenolphthalein to pink while bicarbonate solutions cannot.

16.4.2 Reactions of Sulphite Ion SO_3^{2-}

1. Solubility: Only the sulphites of the alkali metals and ammonia are soluble in water; the sulphites of other metals are either difficultly soluble or insoluble in water. The bisulphites of the alkali metals are soluble in water.

The bisulphites of the alkaline earth metals exist only in solution.

2. Dilute Hydrochloric or Dilute Sulphuric Acid: Decomposes more rapidly on warming with evolution of sulphur dioxide which may be identified

(i) By its suffocating odour of burning sulphur

(ii) It turns the filter paper dipped in acidified potassium dichromate to green

$$Na_2SO_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2$$

$$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$$

3. Barium Chloride or Calcium Chloride Solution: White precipitate of the sulphite $BaSO_3$ or $CaSO_3$ readily soluble in dilute hydrochloric acid. On standing, the precipitate is slowly oxidized to the sulphate and is then insoluble in dilute mineral acids. This change is rapidly effected by warming with bromine water or a little concentrated nitric acid or with hydrogen peroxide

$$Na_2SO_3 + BaCl_2 \longrightarrow BaSO_3 + 2NaCl$$

4. Lime Water: When sulphur dioxide gas is passed through a white precipitate of $CaSO_3$ is formed.

The precipitate dissolves by passing excess of SO_2 due to the formation of the soluble calcium bisulphite $Ca(HSO_3)_2$

$$Ca(OH)_2 + SO_2 \longrightarrow CaSO_3 + H_2O$$

$$CaSO_3 + SO_2 + H_2O \longrightarrow Ca(HSO_3)_2$$

5. Potassium Permanganate Solution: Decolourized owing to the reduction of permanganate by sulphite

$$2KMnO_4 + 5H_2SO_3 \longrightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4 + 3H_2O$$

6. Potassium Dichromate Solution: A green solution containing chromic salt is produced due to reduction by sulphite

$$K_2Cr_2O_7 + 3H_2SO_3 + H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O$$

7. Silver Nitrate Solution: White crystalline precipitate of silver sulphite soluble in excess of the sulphite solution forming the Complex salt, sodium argento sulphite $Na[Ag(SO_3)]$. The precipitate is also soluble in ammonia solution and dilute nitric acid. On boiling the solution of the complex salt or an aqueous suspension of the precipitate, grey metallic silver is precipitated.

$$Na_2SO_3 + 2AgNO_3 \longrightarrow 2NaNO_3 + Ag_2SO_3$$

$$Ag_2SO_3 + Na_2SO_3 \longrightarrow 2Na[AgSO_3]$$

$$2Na[AgSO_3] \longrightarrow Na_2SO_4 + SO_2 + 2Ag$$

16.4.3 Reactions of Sulphide Ion S^{2-}

1. Solubility: The acid, (H_2S) normal and poly sulphides of the alkali metals are soluble. The normal sulphides of alkaline earth metals are difficultly soluble but slowly changes to hydrosulphides in the presence of water. Sulphides of aluminium, chromium and magnesium are completely hydrolyzed in water.

$$Na_2S + H_2O \rightleftharpoons NaHS + NaOH$$

$$NaHS + H_2O \rightleftharpoons H_2S + NaOH$$

$$2CaS + 2H_2O \rightleftharpoons Ca(SH)_2 + Ca(OH)_2$$

$$Al_2S_3 + 6H_2O \rightleftharpoons 2Al(OH)_3 + 3H_2S$$

2. Dilute Hydrochloric Acid: H_2S gas with rotten eggs smell is evolved, turns the filter paper moistened with lead acetate to black

$$Na_2S + 2HCl \rightarrow 2NaCl + H_2S$$

$$Pb(CH_3COO)_2 + H_2S \rightarrow PbS + 2CH_3COOH$$

3. Potassium Dichromate and Potassium Permanganate Solutions: Hydrogen sulphide, like sulphur dioxide, is a good reducing agent, reduces (i) acidified potassium permanganate (ii) acidified potassium dichromate and (iii) iodine solution.

$$2KMnO_4 + 3H_2SO_4 + 5H_2S \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5S$$

$$K_2CrO_7 + 4H_2SO_4 + 3H_2S \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3S$$

$$H_2S + I_2 \longrightarrow 2HI + S$$

4. Silver Nitrate Solution: Black precipitate of silver sulphide Ag_2S, insoluble in cold, but soluble in hot dilute nitric acid

$$Na_2S + 2AgNO_3 \longrightarrow Ag_2S + 2NaNO_3$$

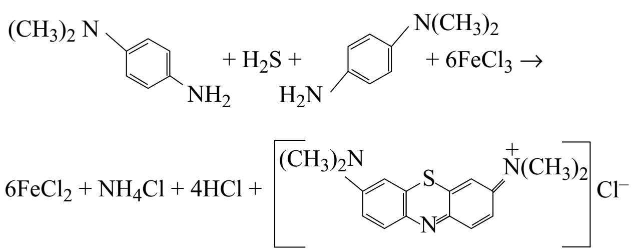
$$3Ag_2S + 8HNO_3 \longrightarrow 6AgNO_3 + 4H_2O + 2NO + 3S$$

5. Barium Chloride Solution: No precipitate
6. Sodium Nitroprusside Solution: $Na_2[Fe(CN)_5NO]$

In alkaline medium gives purple colour. No reaction with solutions of H_2S or with free gas.

$$Na_2S + Na_2[Fe(CN)_5(NO)] \rightarrow Na_4[Fe(CN)_5NOS]$$

7. Methylene Blue Test: para-Aminodimethyl aniline is converted by ferric chloride and hydrogen sulphide in strongly acid solution into water – soluble dyestuff methylene blue.



16.4.4 Reactions of Acetate Ion CH_3COO^-

1. Solubility: All normal acetates are readily soluble in water. Silver and mercury acetates are sparingly soluble. Some basic acetates i.e., those of iron, aluminium and chromium are insoluble in water.
2. Dilute sulphuric acid: Acetic acid, easily recognized by its vinegar-like odour is evolved on warming

$$CH_3COONa + H_2SO_4 \rightarrow NaHSO_4 + CH_3COOH$$

This reaction is also given by concentrated sulphuric acid.
3. Rubbing Test: When acetates are rubbed with oxalic acid after moistening, smell of vinegar will appear due to the formation of acetic acid.

$$2CH_3COONa + H_2C_2O_4 \longrightarrow 2CH_3COOH + Na_2C_2O_4$$

4. Ethyl alcohol and concentrated sulphuric acid

When acetates are heated with conc. H_2SO_4 and ethyl alcohol, a pleasant fruity smell of apple perceives due to the formation of ester, ethyl acetate.

$$CH_3COOH + C_2H_5OH \xrightarrow{\text{Conc } H_2SO_4} CH_3COOC_2H_5 + H_2O$$

If the reaction is carried with iso-amyl alcohol, the ripened banana smell will be perceived.

5. Silver Nitrate solution: A white crystalline precipitate of silver acetate CH_3COOAg is produced in concentrated solution in cold. The precipitate is more soluble in boiling water (at 80°C) and readily soluble in dilute ammonia solution.

$$CH_3COONa + AgNO_3 \longrightarrow CH_3COOAg + NaNO_3$$

6. Ferric chloride solution: Deep red colouration, due to ferric acetate $Fe(CH_3COO)_3$ is produced with neutral solutions of acetates. On diluting and boiling the red colour solution, the iron is precipitated as basic ferric acetate $[Fe(OH)_2 \cdot CH_3COO]$. The colouration is destroyed by dilute hydrochloric acid.

$$FeCl_3 + 3CH_3COONa \longrightarrow Fe(CH_3COO)_3 + NaCl$$

$$Fe(CH_3COO)_3 + 2H_2O \longrightarrow Fe(OH)_2 \cdot CH_3COO + 2CH_3COOH$$

16.5 ANIONS WHICH RESPOND TO CONCENTRATED SULPHURIC ACID

Treat a small amount of the substance in a test tube with 2-3mL of concentrated sulphuric acid and note the reaction taking place.

Observation	Inference
1. Colourless gas evolved with pungent smell and fumes in the air, white fumes of NH_4Cl in contact with glass rod moistened with NH_3 solution	HCl from chloride
2. Gas evolved with pungent odour, reddish colour and fumes with moist air	HBr and Br_2 from bromides
3. Violet vapour evolved accompanied by pungent acid fumes and often SO_2 and even H_2S	HI and I_2 from iodides
4. Pungent acid fumes, often coloured brown by NO_2; colour deepens upon addition of copper turnings	HNO_3 and NO_2 from nitrate

CONFIRMATORY TESTS AND REACTIONS OF ANIONS WHICH RESPOND WITH CONCENTRATED SULPHURIC ACID REACTIONS

16.5.1 Chloride Ion Cl^-

1. Solubility: Most chlorides are soluble in water. Hg_2Cl_2, AgCl, $PbCl_2$ (this is sparingly soluble in cold water but readily soluble in hot water) CuCl, BiOCl. SbOCl and Hg_2OCl_2 are insoluble in water.

2. Concentrated Sulphuric Acid: On warming complete decomposition occurs evolving HCl gas which is recognized

(i) by its pungent odour and the production of white fumes consisting of fine drops of HCl acid, on blowing across the mouth of the tube

(ii) by the formation of white clouds of ammonium chloride when a glass rod moistened with ammonia solution is held near the mouth of the vessel and

(iii) by its turning blue litmus paper red

$$NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl$$

3. Manganese Dioxide and Concentrated Sulphuric Acid: If the solid chloride is mixed with MnO_2 and Conc H_2SO_4 and heated gently chlorine gas is evolved which is identified by

(i) its suffocating odour

(ii) yellowish green colour

(iii) its bleaching of moistened litmus paper

(iv) turning starch-iodide paper to blue. The HCl formed initially is oxidized to chlorine

$$MnO_2 + 4HCl \longrightarrow MnCl_2 + Cl_2 + 2H_2O$$

4. Silver Nitrate Solution: White curdy precipitate of silver chloride AgCl insoluble in water and in dilute nitric acid but soluble in dilute ammonia solution and in potassium cyanide and sodium thiosulphate solution. Silver chloride is reprecipitated from the ammoniacal solution by the addition of dilute nitric acid.

$$NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$$

$$AgCl + 2NH_3 \longrightarrow [Ag(NH_3)_2]Cl$$

$$[Ag(NH_3)_2]Cl + 2HNO_3 \longrightarrow AgCl + 2NH_4NO_3$$

5. Chromyl Chloride Test: If the solid chloride is heated after mixing with solid potassium dichromate and conc H_2SO_4, a reddish oily vapours of chromyl chloride CrO_2Cl_2 will be evolved. If these gases are passed into sodium hydroxide solution forms a yellow, solution of sodium chromate. To this solution if lead acetate is added, yellow precipitate of $PbCrO_4$ will be formed

$$K_2Cr_2O_7 + 4NaCl + 6H_2SO_4 \longrightarrow 2CrO_2Cl_2 + 2KHSO_4 + 4NaHSO_4 + 3H_2O$$

$$CrO_2Cl_2 + 4NaOH \longrightarrow Na_2CrO_4 + 2NaCl + 2H_2O$$

$$Na_2CrO_4 + Pb(CH_3COO)_2 \longrightarrow \underset{\text{Yellow ppt}}{PbCrO_4} + 2CH_3COONa$$

The following points should be noted while carrying chromyl chloride test

(i) This test must not be carried out in the presence of chlorates because of the danger of forming explosive chlorine dioxide
(ii) Bromides and iodides give rise to the free halogens which yield colourless solution with sodium hydroxide. If the ratio of iodide to chloride exceeds 1:15, the chromyl chloride formation is largely prevented and chlorine is evolved due to the reaction of iodine with chromic acid. So chromyl chloride test fails in the presence of bromides and iodides.
(iii) The chlorides of mercury owing to their slight ionization do not respond to this test.
(iv) only partial conversion to CrO_2Cl_2 occurs with the chlorides of lead, silver, antimony and tin. So they do not give reliable chromyl chloride test.
(v) Nitrates interfere with this test since nitrosyl chloride is formed but they can be reduced to ammonium salts.
(vi) Chromyl chloride test must be carried in dry test tube. If water is present the chromyl chloride vapors hydrolyses and do not liberate out

$$CrO_2Cl_2 + H_2O \longrightarrow CrO_3 + 2HCl$$

16.5.2 Reaction of Bromide Ion Br^-

1. Solubility: AgBr, Hg_2Br_2 and Cu_2Br_2 are insoluble in water. $PbBr_2$ is sparingly soluble in cold, but more soluble in boiling water.

2. Concentrated Sulphuric Acid: With solid bromide a reddish brown solution is first formed and reddish brown vapors of bromine accompany the hydrogen bromide (Fuming in moist air) which is evolved; the reactions accelerated by warming. The bromine is produced by the oxidation of the hydrogen bromide initially formed by the sulphuric acid

$$KBr + H_2SO_4 \longrightarrow KHSO_4 + HBr$$

$$2HBr + H_2SO_4 \longrightarrow Br_2 + 2H_2O + SO_2$$

3. Manganese Dioxide and Concentrated Sulphuric Acid: When a mixture of solid bromide, precipitated manganese dioxide and concentrated sulphuric acid is warmed, reddish brown vapors of bromine are evolved, which is recognized

(i) by its powerful irritating odour
(ii) by its bleaching of litmus paper
(iii) by turning the starch paper to orange red

$$2KBr + MnO_2 + 3H_2SO_4 \longrightarrow 2KHSO_4 + MnSO_4 + 2H_2O + Br_2$$

4. Silver Nitrate Solution: Curdy pale yellow precipitate of silver bromide AgBr, sparingly soluble in dilute, but readily soluble in concentrated ammonia solution. The precipitate also soluble in potassium cyanide and sodium thiosulphate solution but insoluble in dilute nitric acid.

$$AgBr + 2NH_3 \longrightarrow [Ag(NH_3)_2]Br$$

5. Chlorine water: If chlorine water and CS_2 or CCl_4 or $CHCl_3$ are added to a solution of bromide, the organic layer will get red colour. This is due to the liberation of bromine by chlorine and dissolution in organic layer

$$2KBr + Cl_2 \longrightarrow 2KCl + Br_2$$

If excess of chlorine water is added, the bromine is converted into yellow bromine monochloride or into colourless hypobromous acid, bromic acid and a pale yellow or colourless solution results.

$$Cl_2 + Br_2 + 2H_2O \longrightarrow 2HOBr + 2HCl$$

$$5Cl_2 + Br_2 + 6H_2O \longrightarrow 2HBrO_3 + 10HCl$$

$$2KBr + HOCl + HCl \longrightarrow 2KCl + Br_2 + H_2O$$

$$Br_2 + 5HOCl + H_2O \longrightarrow 2HBrO_3 + 5HCl$$

16.5.3 Reactions of Iodide Ion I^-

1. Solubility: Solubility of the iodides are similar to those of the chlorides and bromides. Silver, mercurous, mercuric, cuprous and lead iodides are the least soluble salts.

2. Concentrated Sulphuric Acid: With solid iodide iodine is liberated; on warming violet vapours are evolved which turn starch paper blue. Some hydrogen iodide is also formed. HI reduces the sulphuric acid to SO_2, H_2S and S, the relative proportions of which depend upon the concentration of the reagents

$$KI + H_2SO_4 \longrightarrow KHSO_4 + HI$$

$$2HI + H_2SO_4 \longrightarrow I_2 + 2H_2O + SO_2$$

$$6HI + H_2SO_4 \longrightarrow 3I_2 + 4H_2O + S$$

$$8HI + H_2SO_4 \longrightarrow 4I_2 + 4H_2O + H_2S$$

If manganese dioxide is added to the mixture, only iodine is evolved.

$$2KI + MnO_2 + 3H_2SO_4 \rightarrow MnSO_4 + 2H_2O + 2KHSO_4 + I_2$$

3. Silver Nitrate Solution: Yellow, curdy precipitate of silver iodide AgI, readily soluble in KCN and in $Na_2S_2O_3$

Slightly soluble in concentrated ammonia and insoluble in dilute nitric acid.

$$KI + AgNO_3 \longrightarrow KNO_3 + AgI$$

4. Lead Acetate Solution: Yellow precipitate of lead iodide PbI_2 soluble in much hot water forming colourless solution and yielding golden-yellow plates (sprangles) on cooling

$$Pb(CH_3COO)_2 + 2KI \longrightarrow PbI_2 + 2CH_3COOK$$

5. Chlorine Water: When chlorine water is added to a solution of iodide, the solution turns brown due to liberation of iodine. This on shaking with few drops of CS_2 or CCl_4 or $CHCl_3$, the iodine dissolves in organic layer giving violet colour.

$$2KI + Cl_2 \rightarrow 2KCl + I_2$$

If excess chlorine water is added, the iodine is oxidized to colourless iodic acid.

$$5Cl_2 + I_2 + 6H_2O \longrightarrow 2HIO_3 + 10HCl$$

$$2KI + HOCl + HCl \longrightarrow 2KCl + I_2 + H_2O$$

$$I_2 + 5HOCl + H_2O \longrightarrow 2HIO_3 + 5HCl$$

6. Mercuric Chloride Solution: Scarlet precipitate of mercuric iodide HgI_2 soluble in excess of potassium iodide solution

$$HgCl_2 + 2KI \longrightarrow HgI_2 + 2KCl$$

$$HgI_2 + 2KI \longrightarrow K_2[HgI_4]$$

16.5.4 Reactions of Nitrate Ion NO_3^-

1. Solubility: All nitrates are soluble in water. The nitrates of mercury and bismuth yield basic salts on treatment with water, these are soluble in dilute nitric acid.

2. Concentrated Sulphuric Acid: When nitrates are heated reddish brown vapours of nitrogen dioxide will be evolved

$$NaNO_3 + H_2SO_4 \longrightarrow NaHSO_4 + HNO_3$$

$$4HNO_3 \longrightarrow 2H_2O + 4NO_2 + O_2$$

3. Concentrated Sulphuric Acid and Copper Turnings: On heating the nitrates with conc H_2SO_4 and copper turnings the reddish brown vapors of nitrogen dioxide are evolved immediately. The solution acquires blue colour due to the formation of copper nitrate.

$$Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2$$

$$3Cu + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$$

$$2NO + O_2 \longrightarrow 2NO_2$$

4. Brown Ring Test: This test is carried with sodium carbonate extract after neutralizing with dilute sulphuric acid. To this freshly prepared solution of ferrous sulphate is added and then concentrated sulphuric acid is added slowly down the side of the test-tube so that the acid forms a layer beneath the mixture. A brown ring will be formed at the junction of the two liquids. The brown ring is due to the formation of the compound $[Fe(NO)(H_2O)_5]SO_4$. On shaking and warming the mixture the brown colour disappears, nitric oxide is evolved and a yellow solution of ferric sulphate remains.

$$NaNO_3 + H_2SO_4 \longrightarrow NaHSO_4 + HNO_3$$

$$6FeSO_4 + 2HNO_3 + 3H_2SO_4 \longrightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$$

$$[Fe(H_2O)_6]SO_4 + NO \longrightarrow [Fe(H_2O)_5NO]SO_4 + H_2O$$

Note: Bromides and iodides interfere because of the liberation halogens. This test is not trust worthy in the presence of chromates, sulphites, thiosulphates, iodates, cyanides, thiocyanates, Ferro- and ferricyanides. They must be removed by adding excess of Ag_2SO_4 before carrying the test.

Nitrites react similar to nitrates. If nitrite is present along with nitrate ion, it must be decomposed by adding sulphamic acid or urea.

5. Aluminium or Zinc and Sodium Hydroxide Solution (Ammonia Test): When the solution of nitrate is boiled with zinc dust or gently warmed with aluminium powder and sodium hydroxide solution ammonia gas will be evolved due to the reduction of nitrate ion. Excellent results will be obtained by using Devarda's alloy (45% Al, 5% Zn and 50% Cu) The reduction is due to the nascent hydrogen produced in the reaction.

$$NaNO_3 + 4Zn + 7NaOH \longrightarrow 4Na_2[ZnO_2] + 2H_2O + NH_3$$

$$3NaNO_3 + 8Al + 5NaOH + 2H_2O \longrightarrow 8Na[AlO_2] + 3NH_3$$

Note: Nitrites also give similar reaction

16.6 REACTIONS OF ANIONS WHICH DO NOT RESPOND TO DILUTE OR CONCENTRATED ACID

Reactions of Sulphate Ion SO_4^{2-}

1. Solubility: The sulphates of barium, strontium and lead are practically insoluble in water. Calcium and mercuric sulphates are slightly soluble. Most of the remaining sulphates are soluble, some basic sulphates, such as those of mercury, bismuth and chromium are insoluble in water, but they dissolve in dilute hydrochloric or nitric acid.

2. Barium chloride, lead acetate and silver nitrate solutions: The solutions of sulphate will give precipitate with the solutions of barium chloride, lead acetate or silver nitrate

$$Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2NaCl$$

$$Na_2SO_4 + Pb(CH_3COO)_2 \longrightarrow PbSO_4 + 2CH_3COONa$$

$$Na_2SO_4 + 2AgNO_3 \longrightarrow Ag_2SO_4 + 2NaNO_3$$

Barium sulphate is insoluble in mineral acids, lead sulphate is soluble in hot concentrated solution of ammonium acetate and in sodium hydroxide solution.

$$PbSO_4 + 2CH_3COONH_4 \longrightarrow Pb(CH_3COO)_2 + (NH_4)_2SO_4$$

$$PbSO_4 + 4NaOH \longrightarrow \underset{\text{sod.plumbite}}{Na_2PbO_2} + Na_2SO_4 + 2H_2O$$

16.7 SODIUM CARBONATE EXTRACT

Many salts of anions are water insoluble. To get the anions into solution they have to be converted into the sodium salt by double decomposition. Some times some coloured cations also interfere with the tests for anions. For example the blue coloured Cu^{2+} ion may mask the brown ring test for nitrate ion. So sodium carbonate extract is best suited to remove such ions

$$CuSO_4 + Na_2CO_3 \longrightarrow CuCO_3 + Na_2SO_4$$

Since all carbonates other than those of the alkali metals are insoluble, the cation is removed in the residue as carbonate while the anions remain in solution as the sodium salt. The solution after removing the insoluble residue is known as sodium carbonate extract. This contains much free sodium carbonate along with the sodium salt of the anion which should be destroyed with suitable acid before testing for an anion.

1. Generally the halide ions present in sodium carbonate extract are tested with silver nitrate since they form insoluble silver halides in water. Before the test is carried the sodium carbonate extract must be neutralized with dilute nitric acid but not with hydrochloric or sulphuric acid since the chloride or sulphate ions form insoluble salts with silver ions.
2. Nitrate ion present in sodium carbonate extract is tested with brown ring test. Before the test is carried the sodium carbonate extract must be neutralized with dilute sulphuric acid.
3. Sulphate ion present in sodium carbonate extract is tested with barium chloride solution. So before the test is carried the sodium carbonate extract must be neutralized with dilute hydrochloric acid.

16.8 TESTS FOR CATIONS

The cations are divided into several groups based on the solubilities of their salts. Thus all ions which form insoluble chlorides can be precipitated by the addition of dilute hydrochloric acid. From a mixture of various cations only a few ions which form insoluble chlorides could be precipitated thus leaving the other ions in solution. These ions form an **analytical group** and hydrochloric acid which precipitates the ions is the **group reagent**. On such basis the cations have been grouped into six groups. The ions are precipitated in the order of these groups from the mixture. When the reagents for the successive groups are added, the precipitates being removed before the next group reagent is added.

Table 16.2 Classification of cations into groups

Group no	Group reagent	Cations	Precipitate	Solubility product Ksp
I	Cold dil HCl	Pb^{2+}	$PbCl_2$ white	1×10^{-4}
		Hg_2^{2+}	Hg_2Cl_2 white	1.2×10^{-10}
		Ag^+	AgCl white	1.0×10^{-10}
II	Cold dil HCl and H_2S	Sub group II A		
		Hg^{2+}	HgS Black	1.0×10^{-47}
		Pb^{2+}	PbS Black	3.4×10^{-28}
		Cu^{2+}	CuS Black	3.5×10^{-42}
		Cd^{2+}	CdS Yellow	7.1×10^{-28}
		Bi^{3+}	Bi_2S_3 Dark brown	1.6×10^{-72}
		Sub group II B		
		As^{3+}	As_2S_3 Yellow	
		Sb^{3+}	Sb_2S_3 Red orange	
		Sn^{2+}	SnS Brown	8×10^{-29}
		Sn^{4+}	SnS_2 yellow	
III	NH_4Cl + NH_4OH	Al^{3+}	$Al(OH)_3$ White	1×10^{-15}
		Fe^{2+}	$Fe(OH)_2$ Green	1×10^{-14}
		Fe^{3+}	$Fe(OH)_3$ Red brown	1.1×10^{-36}
		Cr^{3+}	$Cr(OH)_3$ Bluish green	6.7×10^{-31}
IV	NH_4Cl + NH_4OH + H_2S	Mn^{2+}	MnS Pale Pink	7×10^{-16}
		Zn^{2+}	ZnS White	7×10^{-26}
		Co^{2+}	CoS Black	1.9×10^{-27}
		Ni^{2+}	NiS Black	1.1×10^{-27}
V	NH_4Cl + NH_4OH + $(NH_4)_2CO_3$	Ca^{2+}	$CaCO_3$ White	1.2×10^{-8}
		Sr^{2+}	$SrCO_3$ White	1.6×10^{-9}
		Ba^{2+}	$BaCO_3$ White	8.1×10^{-9}
VI	None	Mg^{2+}	_	

16.9 REACTIONS OF GROUP – I CATIONS

Ist Group contains three cations Ag^+, Hg_2^{2+} and Pb^{2+}

Group Reagent: Dilute hydrochloric acid.

These cations precipitate, when slightly excess of dilute HCl is added, as their chlorides, because the solubility product of these chlorides is less than the solubility product of all other chlorides which thus remain in solution.

$$Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2 \downarrow + 2HNO_3$$

$$AgNO_3 + HCl \rightarrow AgCl \downarrow + HNO_3$$

$$Hg_2(NO_3)_2 + 2HCl \rightarrow Hg_2Cl_2 \downarrow + 2HNO_3$$

1. A slightly excess of hydrochloric acid is needed for the following reasons
   (a) to ensure the precipitation of all three cations
   (b) to reduce the solubility of the precipitated chlorides through the common-ion effect
   (c) to prevent precipitation of basic salts such as SbOCl and BiOCl.
2. A large excess of hydrochloric acid should be avoided since precipitation of lead chloride may be prevented because of the formation of soluble complex ion $[PbCl_4]^{2-}$
3. Barium chloride and sodium chloride are insoluble in concentrated hydrochloric acid
4. Lead chloride is much soluble than silver chloride or mercurous chloride and is not completely precipitated in Group I. Hence the dissolved $PbCl_2$ is precipitated in Group II as lead sulphide. Hence lead is included in both group I and group II.

16.9.1 Separation and Identifications of Group-I Cations

1. Boil the mixture of $PbCl_2$, Hg_2Cl_2 and AgCl precipitates with water, $PbCl_2$ dissolves in hot water. Filter the solution. The filtrate contain $PbCl_2$ while the precipitate contain Hg_2Cl_2 and AgCl.
2. Add ammonia to the residue containing AgCl and Hg_2Cl_2 · AgCl dissolves by forming complex $[Ag(NH_3)_2]$ Cl. Filter the solution. The residue contains Hg and NH_2HgCl.
3. The residue is dissolved in aquaregia to get mercuric chloride.

16.9.2 Reactions of Lead Ion Pb^{2+}

1. Dilute hydrochloric acid: White precipitate of lead chloride $PbCl_2$ formed only in cold and not too dilute solution.

$$Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2 + 2HNO_3$$

The precipitate is soluble in hot water but precipitates again when cooled. It is also soluble in concentrated hydrochloric acid and in concentrated alkali chloride solutions owing to the formation of complex compounds. These are decomposed in dilution with water with the separation of lead chloride.

$$PbCl_2 + HCl \rightleftharpoons H[PbCl_3]; PbCl_2 + 2HCl \rightleftharpoons H_2[PbCl_4]$$

2. Potassium Iodide Solution: Yellow precipitate of lead iodide PbI_2 moderately soluble in boiling water to yield a colourless solution from which it separates on cooling in golden plates. It is also soluble in excess of potassium iodide solution forming a complex salt which is decomposed on dilution with deposition of lead iodide.

$$Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$$

$$PbI_2 + 2KI \rightarrow K_2[PbI_4]$$

3. Dilute Sulphuric Acid: White precipitate of lead sulphate insoluble in excess acid, but soluble in a concentrated solution of ammonium acetate due to formation of lead acetate

$$Pb(NO_3)_2 + H_2SO_4 \rightarrow PbSO_4 + 2HNO_3$$

$$PbSO_4 + 2CH_3COONH_4 \rightarrow Pb(CH_3COO)_2 + (NH_4)_2SO_4$$

4. Potassium Chromate Solution: Yellow precipitate of lead chromate $PbCrO_4$ insoluble in acetic acid and in ammonia solution but soluble in alkali hydroxide and in nitric acid

$$Pb(NO_3)_2 + K_2CrO_4 \rightarrow PbCrO_4 + 2KNO_3$$

$$PbCrO_4 + 4NaOH \rightarrow Na_2PbO_2 + Na_2CrO_4 + 2H_2O$$

5. Hydrogen Sulphide: Black precipitate of lead sulphide PbS. The precipitate is often red in the hydrochloric acid; this is due to the initial formation of lead sulpho-chloride Pb_2SCl_2 which is decomposed on dilution and by passage of hydrogen sulphide forming black lead sulphide

$$Pb(NO_3)_2 + H_2S \rightarrow PbS + 2HNO_3$$

$$2PbCl_2 + H_2S \rightarrow Pb_2SCl_2 + 2HCl$$

PbS is soluble in hot dilute nitric acid

$$3PbS + 8HNO_3 \rightarrow 3Pb(NO_3)_2 + 4H_2O + 2NO + 3S$$

The black lead sulphide turns to white lead sulphate on treatment with hydrogen peroxide

$$PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$$

6. Sodium Hydroxide Solution: White precipitate of lead hydroxide $Pb(OH)_2$ soluble in excess of the reagent to form sodium plumbite.

$$Pb(NO_3)_2 + 2NaOH \rightarrow Pb(OH)_2 + 2NaNO_3$$

$$Pb(OH)_2 + 2NaOH \rightarrow Na_2[PbO_2] + 2H_2O$$

16.9.3 Reactions of the Mercurous Ion Hg_2^{2+}

1. Dilute Hydrochloric Acid: White precipitate of mercurous chloride (Calomel) Hg_2Cl_2 insoluble in hot water and in cold dilute acids, but soluble in aquaregia.

$$Hg_2(NO_3)_2 + 2HCl \rightarrow Hg_2Cl_2 + 2HNO_3$$

The precipitate becomes black when ammonia solution is poured over it due to the production of amino-mercuric chloride (infusible white precipitate) and finely divided mercury (black), this black mixture is soluble in aquaregia with the formation of mercuric chloride.

$$Hg_2Cl_2 + 2NH_3 \rightarrow Hg(NH_2)Cl + Hg + NH_4Cl$$

$$Hg + 2HNO_3 + 2HCl \rightarrow HgCl_2 + 2H_2O + 2NO_2$$

$$Hg(NH_2)Cl + 2HCl \rightarrow HgCl_2 + NH_4Cl$$

2. Potassium Iodide Solution: Yellowish-green precipitate of mercurous iodide Hg_2I_2 which yields soluble potassium mercuri – iodide $K_2[HgI_4]$ and black finely divided mercury with excess of reagent.

$$Hg_2(NO_3)_2 + 2KI \rightarrow Hg_2I_2 + 2KNO_3$$

$$Hg_2I_2 + 2KI \rightarrow K_2[HgI_4] + Hg$$

3. Potassium Chromate Solution: Brown amorphous precipitate of mercurous chromate Hg_2CrO_4 in the cold, which is converted into red crystalline form on boiling.

$$Hg_2(NO_3)_2 + K_2CrO_4 \rightarrow Hg_2CrO_4 + 2KNO_3$$

4. Hydrogen Sulphide: Immediate black precipitate of mercuric sulphide HgS and mercury.

$$Hg_2(NO_3)_2 + H_2S \rightarrow HgS + 2HNO_3 + Hg$$

5. Sodium Hydroxide Solution: Yellow precipitate of mercurous oxide Hg_2O, insoluble in excess of precipitant.

$$Hg_2(NO_3)_2 + 2NaOH \rightarrow Hg_2O + 2NaNO_3 + H_2O$$

6. Ammonia Solution: Black precipitate, consisting of mercuric amino salt and finely divided mercury.

$$2Hg_2(NO_3)_2 + 4NH_3 + H_2O \rightarrow O\begin{matrix} Hg \\ \\ Hg \end{matrix}NH_2 \cdot NO_3 + 2Hg + 3NH_4NO_3$$

7. Stannous Chloride Solution: Grey Finely – divided mercury is obtained with excess of the reagent.

$$Hg_2(NO_3)_2 + SnCl_2 + 2HCl \rightarrow 2Hg + SnCl_4 + 2HNO_3$$

16.9.4 Reactions of Silver Ion Ag+

1. Dilute Hydrochloric Acid: White, curdy precipitate of AgCl which darkens on exposure to light. The precipitate is insoluble in water but is soluble in dilute ammonia solution owing to the formation of the complex ion $[Ag(NH_3)_2]^+$. It is precipitated from the ammoniacal solution by the addition of dilute nitric acid; silver iodide is precipitated with potassium iodide solution.

$$AgNO_3 + HCl \rightarrow AgCl + HNO_3$$

$$AgCl + 2NH_3 \rightarrow [Ag(NH_3)_2]Cl$$

$$[Ag(NH_3)_2]Cl + 2HNO_3 \rightarrow AgCl + 2NH_4NO_3$$

$$[Ag(NH_3)_2]Cl + KI \rightarrow AgI + KCl + 2NH_3$$

2. Potassium Iodide Solution: Yellow precipitate of silver iodide AgI insoluble in concentrated ammonia solution but readily soluble in solutions of potassium cyanide and of sodium thiosulphate.

$$AgNO_3 + KI \rightarrow AgI + KNO_3$$

$$AgI + 2KCN \rightarrow K[Ag(CN)_2] + KI$$

$$AgI + 2Na_2S_2O_3 \rightarrow Na_3[Ag(S_2O_3)_2] + NaI$$

3. Potassium Chromate solution: Red precipitate of silver chromate Ag_2CrO_4, insoluble in dilute acetic acid but soluble in dilute nitric acid and ammonia solution.

$$2AgNO_3 + K_2CrO_4 \rightarrow Ag_2CrO_4 + 2KNO_3$$

4. Hydrogen Sulphide: Black precipitate of silver sulphide Ag_2S, insoluble in water and in ammonia solution but soluble in hot dilute nitric acid.

$$2AgNO_3 + H_2S \rightarrow Ag_2S + 2HNO_3$$

$$3Ag_2S + 8HNO_3 \rightarrow 6AgNO_3 + 4H_2O + 2NO + 3S$$

5. Sodium Hydroxide Solution: Brown precipitate of silver oxide Ag_2O insoluble in excess of the precipitant

$$2AgNO_3 + 2NaOH \rightarrow Ag_2O + 2NaNO_3 + H_2O$$

6. Ammonia Solution: White precipitate at first which quickly passes into brown silver oxide Ag_2O soluble in excess of the precipitant

$$2AgNO_3 + 2NH_4OH \rightarrow Ag_2O + 2NH_4NO_3 + H_2O$$

$$Ag_2O + 4NH_3 + H_2O \rightarrow 2[Ag(NH_3)_2]OH$$

16.10 REACTIONS OF GROUP – II CATIONS

Group – II Contains Hg^{2+}, Pb^{2+}, Cu^{2+}, Cd^{2+}, Bi^{3+}, As^{3+}, Sb^{3+} and Sn^{2+} or Sn^{4+}.

Group Reagent: Dilute hydrochloric acid and H_2S gas. H_2S is a weak electrolyte and ionization is suppressed in the presence of hydrochloric acid due to common ion effect of H^+ ion.

$$H_2S \rightleftharpoons 2H^+ + S^{2-}$$

$$HCl \rightarrow H^+ + Cl^-$$

In that low concentration of sulphide ions the ionic product of group-II cation and sulphide ion exceeds the solubility product of their sulphides so they will be precipitated whereas the ionic products of the other cations and sulphide ion are still less than their solubility products so they remain in solution.

Note if oxidizing agents such as permanganate, chromate, nitrate, sulphite or Fe^{3+} are present, They oxidize the hydrogen sulphide to sulphur. So sometimes even if there is no cation of group II in the solution, yellow turbidity of sulphur appears in group II.

When chloride ions are present in large amounts, they form stable complexes with lead, cadmium and stannic ions. This may result in the incomplete precipitation of these ions.

$$HgCl_2 + H_2S \rightarrow HgS + 2HCl$$
$$2Pb(CH_3COO)_2 + H_2S \rightarrow PbS + 2HCl$$
$$2BiCl_3 + 3H_2S \rightarrow Bi_2S_3 + 6HCl$$
$$CuCl_2 + H_2S \rightarrow CuS + 2HCl$$
$$CdCl_2 + H_2S \rightarrow CdS + 2HCl$$
$$2AsCl_3 + 3H_2S \rightarrow As_2S_3 + 6HCl$$
$$2SbCl_3 + 3H_2S \rightarrow Sb_2S_3 + 6HCl$$
$$SnCl_2 + H_2S \rightarrow SnS + 2HCl$$
$$SnCl_4 + 2H_2S \rightarrow SnS_2 + 4HCl$$

16.10.1 Separation of Group II Cations into Sub Groups II A and II B

The precipitates of Group II cations when treated with sodium hydroxide and little yellow ammonium sulphide the subgroup II B cation sulphides dissolve due to conversion into thio salts.

$$As_2S_3 + 3(NH_4)_2S_2 \rightarrow S + 2(NH_4)_3AsS_4$$
$$Sb_2S_3 + 3(NH_4)_2S_2 \rightarrow S + 2(NH_4)_3SbS_4$$
$$SnS_2 + (NH_4)_2S_2 \rightarrow S + (NH_4)_2SnS_3$$
$$SnS + (NH_4)_2S_2 \rightarrow S + (NH_4)_2SnS_2$$

The residue will contain Group II A and the filtrate contain Group II B cations.

16.10.2 Separation of Group II A Cations

The residue insoluble in NaOH and yellow ammonium sulphide will contain HgS, PbS, Bi_2S_3, CuS and CdS. They are separated and tested.

1. HgS is insoluble in dilute HNO_3 but the PbS, Bi_2S_3, CuS and CdS dissolve in hot dilute nitric acid with the formation of nitrate.

$$3PbS + 8HNO_3 \rightarrow 3Pb(NO_3)_2 + 4H_2O + 2NO + 3S$$
$$3CuS + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO + 3S$$
$$3CdS + 8HNO_3 \rightarrow 3Cd(NO_3)_2 + 4H_2O + 2NO + 3S$$
$$Bi_2S_3 + 8HNO_3 \rightarrow 2Bi(NO_3)_2 + 4H_2O + 2NO + 3S$$

The insoluble mercuric sulphide is then converted into mercuric chloride by dissolving in aquaregia and then tested for mercuric ion.

$$3HgS + 2HNO_3 + 6HCl \rightarrow 3HgCl_2 + 3S + 2NO + 4H_2O$$

2. The above filtrate containing nitrates of lead, copper, cadmium and bismuth is treated with dilute sulphuric acid. Only lead precipitates as lead sulphate leaving behind the other metal sulphates in solution.

$$Pb(NO_3)_2 + H_2SO_4 \rightarrow PbSO_4 + 2HNO_3$$

This lead sulphate is separated and dissolved in ammonium acetate.

$$PbSO_4 + 2CH_3COONH_4 \rightarrow Pb(CH_3COO)_2 + (NH_4)_2SO_4$$

This solution used for testing lead ion.

3. The filtrate containing the sulphates of Bi, Cu and Cd when treated with ammonia solution. Bismuth will be precipitated as $Bi(OH)_3$ and the copper and cadmium remain in solution in the form of complexes.

$$Bi_2(SO_4)_3 + 6NH_4OH \rightarrow 2Bi(OH)_3 + 3(NH_4)_2SO_4$$
$$CuSO_4 + 4NH_4OH \rightarrow [Cu(NH_3)_4]SO_4 + 4H_2O$$
$$CdSO_4 + 4NH_4OH \rightarrow [Cd(NH_3)_4]SO_4 + 4H_2O$$

The $Bi(OH)_3$ is separated by filtration and dissolved in dilute hydrochloric acid to test for bismuth ion.

4. The filtrate is used for testing copper and cadmium ions.

16.10.3 Reactions of the Mercuric Ion Hg^{2+}

1. Hydrogen Sulphide: Initially white, then yellow, brown and finally a black precipitate of mercuric sulphide HgS. In all cases, excess of hydrogen sulphide gives the black mercuric sulphide. The white precipitate is the chloro sulphide $Hg_3S_2Cl_2$ (Or $HgCl_2$, 2HgS) which is decomposed by hydrogen Sulphide.

$$3HgCl_2 + 2H_2S \rightarrow Hg_3S_2Cl_2 + HCl$$
$$Hg_3S_2Cl_2 + H_2S \rightarrow 2HCl + 3HgS$$
$$HgCl_2 + H_2S \rightarrow HgS + 2HCl$$

The black HgS dissolves in aquaregia.

$$3HgS + 2HNO_3 + 6HCl \rightarrow 3HgCl_2 + 2NO + 3S + 4H_2O$$

2. Stannous Chloride Solution: A white precipitate of mercurous chloride Hg_2Cl_2 is first obtained which is reduced by excess of the reagent to grey-black metallic mercury.

$$2HgCl_2 + SnCl_2 \rightarrow SnCl_4 + Hg_2Cl_2$$
$$Hg_2Cl_2 + SnCl_2 \rightarrow SnCl_4 + 2Hg$$

3. Sodium Hydroxide Solution: Gives yellow mercuric oxide

$$HgCl_2 + 2NaOH \rightarrow Hg_2O + 2NaCl + H_2O$$

4. Ammonia Solution: White precipitate of aminomercuricchloride $Hg(NH_2)Cl$ known "as infusible white precipitate" for it volatilizes without melting

$$HgCl_2 + 2NH_3 \rightarrow Hg(NH_2)Cl + NH_4Cl$$

5. Potassium Iodide Solution: Red precipitate of mercuric iodide HgI_2 Soluble in excess of the precipitant owing to the formation of the complex salt potassium mercuri-iodide $K_2[HgI_4]$

$$HgCl_2 + 2KI \rightarrow HgI_2 + 2KCl$$
$$HgI_2 + 2KI \rightarrow K_2[HgI_4]$$

16.10.4 Reactions of Bismuth Ion Bi^{3+}

1. Hydrogen Sulphide: Brown precipitate of bismuth sulphide Bi_2S_3, insoluble in cold dilute acids and in ammonium sulphide but soluble in hot dilute nitric acid

$$2Bi(NO_3)_3 + 3H_2S \longrightarrow Bi_2S_3 + 6HNO_3$$
$$Bi_2S_3 + 8HNO_3 \longrightarrow 2Bi(NO_3)_3 + 4H_2O + 2NO + 3S$$

2. Sodium Hydroxide Solution: White precipitate of bismuth hydroxide $Bi(OH)_3$ in the cold, very slightly soluble in excess of the reagent, soluble in acids. It becomes yellow on boiling due to partial dehydration.

$$Bi(NO_3)_3 + 3NaOH \longrightarrow Bi(OH)_3 + 3NaNO_3$$
$$Bi(OH)_3 \longrightarrow BiO \cdot OH + H_2O$$

If H_2O_2 is added to the solution containing white or yellowish white precipitate in suspension, brown bismuthic acid $HBiO_3$ is formed.

$$BiO \cdot OH + H_2O_2 \longrightarrow HBiO_3 + H_2O$$

3. Ammonia Solution: A white basic salt of variable composition is precipitated. The precipitate is insoluble in excess of the reagent (distinction from copper and cadmium)

4. Potassium Iodide Solution: Dark brown precipitate of BiI_3 readily soluble in excess of reagent to give yellow solution of the complex $K[BiI_4]$. This on dilution first gives a precipitate of the trioxide and then an orange coloured precipitate of the basic bismuthiodide (BiO) I

$$Bi(NO_3)_3 + 3KI \longrightarrow BiI_3 + 3KNO_3$$
$$BiI_3 + KI \longrightarrow K[BiI_4]$$
$$BiI_3 + H_2O \rightleftharpoons (BiO)I + 2HI$$

5. Sodium Stannite Solution: Black precipitate of finely divided bismuth in cold

$$2Bi(NO_3)_3 + 6NaOH + 3Na_2[SnO_2] \longrightarrow 2Bi + 3Na_2[SnO_3] + 6NaNO_3 + 3H_2O$$

6. Water: When a solution of a bismuth salt is poured into a large volume of water, a white precipitate of the corresponding basic salt is produced, which is soluble in dilute mineral acids

$$Bi(NO_3)_3 + H_2O \rightleftharpoons (BiO)NO_3 + 2HNO_3$$
$$2Bi(NO_3)_3 + \underset{\text{large excess}}{3H_2O} \rightleftharpoons \underset{\text{Basic salt}}{(BiO)_2(OH)NO_3} + 5HNO_3$$

$$BiCl_3 + H_2O \rightleftharpoons (BiO)Cl + 2HCl$$

16.10.5 Reactions of Cupric Ion Cu^{2+}

1. Hydrogen Sulphide: A black precipitate of cupric sulphide soluble in hot dilute nitric acid is formed.

$$CuSO_4 + H_2O \longrightarrow CuS + H_2SO_4$$
$$3CuS + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO + 3S$$

2. Sodium Hydroxide Solution: Blue precipitate of cupric hydroxide insoluble in excess of NaOH but turns to black on heating due to conversion into cupric oxide.

$$CuSO_4 + 2NaOH \longrightarrow \underset{\text{Blue}}{Cu(OH)_2 \downarrow} + Na_2SO_4$$

$$Cu(OH)_2 \longrightarrow \underset{\text{Black}}{CuO \downarrow} + H_2O$$

3. Ammonia Solution: Pale blue precipitate of basic salt soluble in excess of the precipitant with the formation of a deep blue solution containing the complex salt, tetra-ammine copper (II) sulphate

$$2CuSO_4 + 2NH_3 + 2H_2O \longrightarrow CuSO_4 \cdot Cu(OH)_2 + (NH_4)_2SO_4$$

$$CuSO_4 \cdot Cu(OH)_2 + (NH_4)_2 SO_4 + 6NH_3 \longrightarrow 2[Cu(NH_3)]SO_4 + 2H_2O$$

4. Potassium Ferrocyanide Solution: Chocolate brown precipitate of cupric ferrocyanide from neutral or acidic solutions. It is insoluble in dilute acids, but dissolves in aqueous ammonia and decomposed by alkalis

$$2CuSO_4 + K_4[Fe(CN)_6] \longrightarrow Cu_2[Fe(CN)_6] + 2K_2SO_4$$

5. Potassium Cyanide Solution: Yellow precipitate of cupric cyanide $Cu(CN)_2$ which dissociate to cuprous cyanide and cyanogens.The cuprous cyanide dissolves in excess of the reagent forming potassium cupro cyanide complex

$$2CuSO_4 + 4KCN \longrightarrow 2Cu(CN)_2 + 2K_2SO_4$$

$$2Cu(CN)_2 \longrightarrow Cu_2(CN)_2 + (CN)_2$$

$$Cu_2(CN)_2 + 6KCN \longrightarrow 2K_3[Cu(CN)_4]$$

6. Potassium Iodide Solution: Cupric iodide CuI_2 is first precipitated; this immediately decomposes into white cuprous iodide and free iodine

$$CuSO_4 + 2KI \longrightarrow CuI_2 + K_2SO_4$$

$$2CuI_2 \longrightarrow Cu_2I_2 + I_2$$

7. Potassium or Ammonium Thiocyante Solution: Black precipitate of cupric thiocyante $Cu(SCN)_2$ which upon adding sulphurous acid solution changes into white insoluble cuprous thiocyanate.

$$CuSO_4 + 2NH_4SCN \longrightarrow Cu(SCN)_2 + (NH_4)_2 SO_4$$

$$2Cu(SCN)_2 + H_2SO_3 + H_2O \longrightarrow 2CuSCN + 2HSCN + H_2SO_4$$

16.10.6 Reactions of Cadmium Ion Cd^{2+}

1. Hydrogen Sulphide: Yellow precipitate of cadmium sulphide CdS from acidified solution is formed. It is soluble in hot dilute nitric acid and also in hot dilute sulphuric acid (distinction from copper) but insoluble in potassium cyanide (difference from copper). Precipitation does not occur in strongly acid solution owing to the reversibility of the reaction .

$$CdSO_4 + H_2S \rightleftharpoons CdS + H_2SO_4$$

2. Sodium Hydroxide Solution: White precipitate of cadmium hydroxide $Cd(OH)_2$ insoluble in excess of the reagent.

$$CdSO_4 + 2NaOH \longrightarrow Cd(OH)_2 + Na_2SO_4$$

3. Ammonia Solution: White precipitate of cadmium hydroxide, soluble in excess of the precipitating reagent due to the formation of complex

$$CdSO_4 + 2NH_4OH \longrightarrow Cd(OH)_2 + (NH_4)_2 SO_4$$

$$Cd(OH)_2 + (NH_4)_2 SO_4 + 2NH_3 \longrightarrow [Cd(NH_3)_4]SO_4 + 2H_2O$$

4. Potassium Cyanide Solution: White precipitate of cadmium cyanide, soluble in excess of the reagent to yield the complex $K_2[Cd(NH_3)_4]$. A sufficiently high concentration of cadmium ions is produced by the dissociation of the complex ions to give precipitate of yellow cadmium sulphide with hydrogen sulphide.

$$CdSO_4 + 2KCN \longrightarrow Cd(CN)_2 + K_2SO_4$$

$$Cd(CN)_2 + 2KCN \longrightarrow K_2[Cd(CN)_4]$$

$$K_2[Cd(CN)_4] + H_2S \longrightarrow CdS + 2KCN + 2HCN$$

The marked difference in the values of the instability constants of the complex ions $[Cd(CN)_4]^{2-}$ and $[Cu(CN)_4]^{3-}$ serves as the basis for separation and detections of the cadmium ions in the presence of copper (II) ion.

5. Ammonium Thiocyanate Solution: No precipitate (distinction from copper).

16.10.7 Separation of Group IIB Cations

The filtrate after treating the precipitates of Group II cations with NaOH and yellow ammonium sulphide will contain thiosalts of arsenic antimony and tin. These on acidifying will be precipitated as sulphides.

$$2(NH_4)_3AsS_4 + 6HCl \longrightarrow As_2S_5 + 6NH_4Cl + 3H_2S$$

$$2(NH_4)_3SbS_4 + 6HCl \longrightarrow Sb_2S_5 + 6NH_4Cl + 3H_2S$$

$$(NH_4)_2SnS_3 + 2HCl \longrightarrow SnS_2 + 2NH_4Cl + H_2S$$

$$(NH_4)_2SnS_2 + 2HCl \longrightarrow SnS + 2NH_4Cl + H_2S$$

The separation and identification of these cations will be carried as follows.

1. When concentrated HCl is added only As_2S_5 is unaffected while Sb_2S_5 and SnS_2 dissolve forming chlorides

$$Sb_2S_5 + 6HCl \longrightarrow 2SbCl_3 + 3H_2S + 2S$$

$$SnS_2 + 4HCl \longrightarrow SnCl_4 + 2H_2S$$

2. The above filtrate is divided into two portions and used for testing antimony and bismuth
3. As_2S_5 or As_2S_3 precipitate is converted into arsenate by ammonical hydrogen peroxide.

$$As_2S_5 + 16NH_3 + 20H_2O_2 \longrightarrow 2(NH_4)_3AsO_4 + 5(NH_4)_2SO_4 + 12H_2O$$

$$As_2S_3 + 12NH_3 + 14H_2O_2 \longrightarrow 2(NH_4)_3AsO_4 + 3(NH_4)_2SO_4 + 12H_2O$$

The arsenate is readily identified by the formation of yellow precipitate with ammonium molybdate

$$(NH_4)_3AsO_4 + 12(NH_4)_2MoO_4 + 24HNO_3 \rightarrow (NH_4)_3AsO_4 \cdot 12MoO_3 \downarrow + 24NH_4NO_3 + 12H_2O$$

4. To one portion of the filtrate when ammonia and large excess of oxalic acid are added more stable tin oxalate and less stable antimony oxalate complexes will be formed. If H_2S is passed into this solution antimony will be precipitated.

$$H_2C_2O_4 + 2NH_4OH \longrightarrow (NH_4)_2C_2O_4 + 2H_2O$$

$$SbCl_3 + 3(NH_4)_2C_2O_4 \longrightarrow (NH_4)_3[Sb(C_2O_4)_3] + 3NH_4Cl$$

$$(NH_4)_3[Sb(C_2O_4)_3] \longrightarrow 3NH_4 + [Sb(C_2O_4)_3]^{3-}$$

$$[Sb(C_2O_4)_3]^{3-} \longrightarrow Sb^{3+} + 3C_2O_4^{2-}$$

$$2Sb^{3+} + 3H_2S \longrightarrow Sb_2S_3 + 6H^+$$

$$SnCl_4 + 3(NH_4)_2C_2O_4 \longrightarrow (NH_4)_2[Sn(C_2O_4)_3] + 4NH_4Cl$$

5. To the second portion of the filtrate if metallic iron is added, it reduces the $SnCl_4$ to $SnCl_2$. The $SnCl_2$ is identified by the usual reaction with $HgCl_2$ solution. The $FeCl_2$ produced has no action on $HgCl_2$

$$SnCl_4 + Fe \longrightarrow SnCl_2 + FeCl_2$$

$$2HgCl_2 + SnCl_2 \longrightarrow \underset{\text{White ppt}}{Hg_2Cl_2} + SnCl_4$$

$$Hg_2Cl_2 + SnCl_2 \longrightarrow \underset{\text{grey}}{2Hg} + SnCl_4$$

16.11 REACTIONS OF GROUP III CATIONS

Group-III contains Al^{3+}, Cr^{3+}, Fe^{2+} and Fe^{3+} ions

Group reagent: $NH_4OH + NH_4Cl$

Ammonium hydroxide is weak electrolyte while ammonium chloride is strong electrolyte, ionizes completely. So the ionization of ammonium hydroxide decreases in the presence of ammonium chloride due to common ion effect of NH_4^+ ion

$$NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

$$NH_4Cl \longrightarrow NH_4^+ + Cl^-$$

In that low concentration of OH^- ion the ionic product of OH^- ion and the third group cation exceeds the solubility product of their hydroxides. So they will be precipitated as their hydroxides. The solubility product of ferrous hydroxide is more, so complete precipitation of iron is not possible, if it is present in the form of ferrous iron. So before proceeding to III group analysis the solution is boiled with few drops of concentrated nitric acid to oxidize the ferrous iron to ferric iron.

$$FeCl_3 + 3NH_4OH \longrightarrow Fe(OH)_3 + 3NH_4Cl$$

$$AlCl_3 + 3NH_4OH \longrightarrow Al(OH)_3 + 3NH_4Cl$$

$$CrCl_3 + 3NH_4OH \longrightarrow Cr(OH)_3 + 3NH_4Cl$$

The separation and identification of these precipitates is carried out as follows.

1. When the precipitate is boiled with NaOH and H_2O_2, the $Al(OH)_3$ dissolves due to the conversion into $NaAlO_2$ and $Cr(OH)_3$ also dissolves due to oxidation to soluble yellow sodium chromate $Na_2CrO_4 \cdot Fe(OH)_3$ remains undissolved.

$$Al(OH)_3 + NaOH \longrightarrow NaAlO_2 + 2H_2O$$

$$2Cr(OH)_3 + 3H_2O_2 + 4NaOH \longrightarrow 2Na_2CrO_4 + 8H_2O$$

If lead acetate solution is added to the above solution lead chromate will be precipitated.

$$Pb(CH_3COO)_2 + Na_2CrO_4 \longrightarrow PbCrO_4 + 2CH_3COONa$$

Aluminium is identified by precipitation as $Al(OH)_3$ by boiling in the presence of NH_4Cl.

$$NaAlO_2 + NH_4Cl + H_2O \longrightarrow Al(OH)_3 + NaCl + NH_3$$

2. The precipitate left after boiling with NaOH and H_2O_2 will contain ferric hydroxide. When acidified with dil HCl, the $Fe(OH)_3$ converts into $FeCl_3$. It is used to study the reactions of iron.

16.11.1 Reactions of Aluminium Ion Al^{3+}

1. Ammonia Solution: White gelatinous precipitate insoluble in excess of ammonia solution.

$$Al_2(SO_4)_3 + 6NH_4OH \longrightarrow 2Al(OH)_3 + 3(NH_4)_2SO_4$$

2. Sodium Hydroxide Solution: White precipitate of aluminium hydroxide $Al(OH)_3$ soluble in excess of the reagent with the formation of sodium meta aluminates

$$Al_2(SO_4)_3 + 6NaOH \longrightarrow 2Al(OH)_3 + 3Na_2SO_4$$

$$Al(OH)_3 + NaOH \rightleftharpoons NaAlO_2 + 2H_2O$$

The reaction is reversible and any reagent which will decrease the hydroxyl ions concentration sufficiently should cause the reaction to proceed from right to left with consequent precipitation of aluminium hydroxide. Addition of ammonium chloride or by the addition of acid; in latter case a large excess of acid causes the precipitated hydroxide to redissolve

$$NaAlO_2 + NH_4Cl + H_2O \longrightarrow Al(OH)_3 + NH_3 + NaCl$$

$$NaAlO_2 + HCl + H_2O \longrightarrow Al(OH)_3 + NaCl$$

$$Al(OH)_3 + 3HCl \longrightarrow AlCl_3 + 3H_2O$$

3. Sodium Carbonate Solution: White precipitate of aluminium hydroxide, soluble in excess of the precipitant.

$$Al_2(SO_4)_3 + 3Na_2CO_3 \longrightarrow Al_2(CO_3)_3 + 3Na_2SO_4$$

$$Al_2(CO_3)_3 + 6H_2O \longrightarrow 2Al(OH)_3 + 3H_2CO_3$$

$$2Al(OH)_3 + Na_2CO_3 \longrightarrow 2NaAlO_2 + CO_2 + 3H_2O$$

4. Cobalt Nitrate Solution Test: Aluminium compounds when heated with sodium carbonate upon charcoal in the blowpipe flame give a white infusible solid, which glows when hot. If the residue with one or two drops of cobalt nitrate solution is heated a blue infusible mass (Thenard's blue or cobalt metaaluminate) is obtained.

$$2Al_2O_3 + 2Co(NO_3)_2 \longrightarrow 2Co(AlO_2)_2 + 4NO_2 + O_2$$

16.11.2 Reactions of Chromium (III) Ion Cr^{3+}

1. Ammonia Solution: Grey-green to grey-blue gelatinous precipitate of chromium hydroxide [$Cr(OH)_3$], slightly soluble in excess of the precipitate in the cold forming a violet or pink solution containing complex chrome ammines; upon boiling the solution chromium hydroxide is precipitated

$$CrCl_3 + 3NH_3 + 3H_2O \longrightarrow Cr(OH)_3 + 3NH_4Cl$$

$$Cr(OH)_3 + 6NH_3 \longrightarrow \left[Cr(NH_3)_6\right](OH)_3$$

2. Sodium Hydroxide Solution: Precipitate of chromic hydroxide, readily soluble in acids and also in excess of the precipitant in the cold forming a green solution containing sodium chromite $NaCrO_2$. Upon boiling the latter solution hydrolysis occurs and the chromium is almost quantitatively reprecipitated as the hydroxide.

$$Cr(OH)_3 + NaOH \longrightarrow NaCrO_2 + 2H_2O$$

If hydrogen peroxide is added to the sodium chromite solution a yellow solution of sodium chromate is produced which may be identified by the "per chromic acid" reaction

$$2Na\left[CrO_2\right] + 3H_2O_2 + 2NaOH \longrightarrow 2Na_2CrO_4 + 4H_2O$$

To this if acid hydrogen peroxide and amyl alcohol are added a blue colour is acquired by the organic layer due to the formation of blue chromium peroxide.

$$H_2Cr_2O_7 + 4H_2O_2 \rightleftharpoons 2CrO_5 + 5H_2O$$

To the sodium chromate solution if little acetic acid and barium chloride is added, a yellow precipitate of barium chromate if formed

$$Na_2CrO_4 + BaCl_2 \longrightarrow BaCrO_4 \downarrow + 2NaCl$$

16.11.3 Reactions of Ferrous Ion Fe^{2+}

1. Sodium Hydroxide Solution: White precipitate of ferrous hydroxide $Fe(OH)_2$ in the complete absence of air, insoluble in excess, but soluble in acids. Upon exposure to air ferrous hydroxide is rapidly oxidized. Under ordinary conditions it appears as a dirty green of grass green precipitate; the addition of hydrogen peroxide changes immediately to ferric hydroxide.

$$FeSO_4 + 2NaOH \longrightarrow Fe(OH)_2 + Na_2SO_4$$

$$4Fe(OH)_2 + 2H_2O + O_2 \longrightarrow 4Fe(OH)_3$$

2. Ammonia Solution: Precipitation of ferrous hydroxide occurs as above. In the presence of ammonium chloride solution the common ion effect of the ammonium ions lower the concentration of the OH^- ions to such an extent that the solubility product of ferrous hydroxide Fe $(OH)_2$ is not attained and precipitation does not occur

$$FeSO_4 + 2NH_4OH \longrightarrow Fe(OH)_2 + (NH_4)_2SO_4$$

3. Potassium Cyanide Solution: Yellowish brown precipitate of ferrous cyanide $Fe(CN)_2$ soluble in excess of the precipitant forming a yellow solution of potassium ferrocyanide $K_4[Fe(CN)_6]$

$$FeSO_4 + 2KCN \longrightarrow Fe(CN)_2 + K_2SO_4$$
$$Fe(CN)_2 + 4KCN \longrightarrow K_4[Fe(CN)_6]$$

4. Potassium Ferrocyanide Solution: In the complete absence of air white precipitate of potassium ferrous ferro cyanide $K_2Fe[Fe(CN)_6]$ is produced. Under ordinary atmospheric conditions a pale blue precipitate is obtained; partial oxidation to potassium ferric ferro cyanide, $KFe^{III}[Fe^{II}(CN)_6]$ occurs

5. Potassium Ferricyanide Solution: A dark blue precipitate is produced. This formerly termed as Turnbull's blue and was formulated as potassium ferrous ferricyanide $K\,Fe^{II}\left[Fe^{III}(CN)_6\right]$. It is now considered to be identical with Prussian blue i.e., potassium ferric ferrocyanide $KFe^{III}\left[Fe^{II}(CN)_6\right]$

$$FeSO_4 + K_3\left[Fe(CN)_6\right] \longrightarrow KFe\left[Fe(CN)_6\right] + K_2SO_4$$

The precipitate is decomposed by sodium or potassium hydroxide solution, ferric hydroxide being precipitated.

6. Ammonium Thiocyanate Solution: No colouration with pure ferrous salts

16.11.4 Reaction of the Ferric Ion Fe^{3+}

1. Ammonia Solution: Reddish brown, gelatinous precipitate of ferric hydroxide $Fe(OH)_3$ insoluble in excess of the reagent but soluble in acids.

$$FeCl_3 + 3NH_4OH \longrightarrow Fe(OH)_3 + 3NH_4Cl$$

The solubility product of ferric hydroxide is so small $\left(1.1\times10^{-36}\right)$ that complete precipitation takes place even in the presence of ammonium salts.

2. Sodium Hydroxide Solution: Reddish-brown precipitate of ferric hydroxide insoluble in excess of the reagent

3. Hydrogen Sulphide: Reduced to ferrous salt in the presence of acid with separation of sulphur

$$2FeCl_3 + H_2S \longrightarrow 2FeCl_2 + 2HCl + S$$

4. Potassium Ferrocyanide Solution: Intense blue precipitate of potassium ferric ferrocyanide (Prussian blue)

$$FeCl_3 + K_4\left[Fe(CN)_6\right] \rightarrow KFe[Fe(CN)_6] + 3KCl$$

The precipitate is insoluble in dilute hydrochloric acid but dissolves in oxalic acid solution, in concentrated hydrochloric acid and also in a large excess of potassium ferrocyanide solution with the production of a blue solution.

5. Potassium Ferricyanide Solution: A brown colouration is produced, due to ferric ferricyanide $Fe[Fe(CN)_6]$ (Distinction from ferrous salts)

$$FeCl_3 + K_3[Fe(CN)_6] \longrightarrow Fe[Fe(CN)_6] + 3KCl$$

Addition of H_2O_2 or little stannous chloride to the above solution turns to blue due to reduction of ferricyanide to ferrocyanide which then reacts with ferric ions

6. Ammonium Thiocyanate Solution: Deep red colouration The colouration was formally attributed to undissociated ferric thiocyanate but is now known to be due to the ferri-thiocyanate ion $[Fe(SCN)]^{2+}$. The colour is discharged by mercuric chloride.

$$Fe^{3+} + SCN \rightleftharpoons [Fe(SCN)]^{2+}$$
$$2[Fe(SCN)]^{2+} + HgCl_2 \rightleftharpoons 2Fe^{3+} + Hg(SCN)_2 + 2Cl$$

Distinctive tests for ferrous and ferric salts

Reagent	Ferrous salt	Ferric salt
Potassium ferrocyanide solution	White or pale-blue precipitate	Deep-blue precipitate (Prussian blue)
Potassium ferricyanide solution	Deep blue precipitate	No precipitate, brown colouration
Ammonium thiocyanate solution	No colouration in complete absence of ferric salts	Deep red colouration discharged by mercuric chloride solution
Ammonia solution	White or greenish white precipitate	Reddish brown precipitate

16.12 REACTIONS OF GROUP IV CATIONS

Group – IV contains Zn^{2+}, Mn^{2+}, Co^{2+} and Ni^{2+} ions

Group reagent $NH_4Cl + NH_4OH + H_2S$

Hydrogen sulphide is a weak electrolyte. But in the presence of ammonium hydroxide due to neutralization of H^+ ions, ionization of H_2S increases

$$H_2S \rightleftarrows 2\,H^+ + S^{2-}$$
$$NH_4OH \rightleftarrows NH_4^+ + OH^-$$

So the ionization of H_2S increases producing more number of S^{2-} ions. In that high concentration of S^{2-} the ionic product of S^{2-} ions and group IV cations increases the solubility products of their sulphides. So they will be precipiated as their sulphide.

$$MnCl_2 + H_2S \longrightarrow MnS + 2HCl$$
$$ZnCl_2 + H_2S \longrightarrow ZnS + 2HCl$$
$$CoCl_2 + H_2S \longrightarrow CoS + 2HCl$$
$$NiCl_2 + H_2S \longrightarrow NiS + 2HCl$$

The separation of NiS, CoS, MnS and ZnS is based on the following facts

1. When the above precipitates are boiled with dilute HCl, the MnS and ZnS dissolve while CoS and NiS are left behind

$$MnS + 2HCl \longrightarrow MnCl_2 + H_2S$$
$$ZnS + 2HCl \longrightarrow ZnCl_2 + H_2S$$

2. The solution containing $MnCl_2$ and $ZnCl_2$ on treatment with NaOH, the $Zn(OH)_2$ and $Mn(OH)_2$ will be precipitated but the $Zn(OH)_2$ react with excess of NaOH and converts into soluble sodium zincate $Na_2[ZnO_2]$

$$MnCl_2 + 2NaOH \longrightarrow Mn(OH)_2 + 2NaCl$$
$$ZnCl_2 + 2NaOH \longrightarrow Zn(OH)_2 + 2NaCl$$
$$Zn(OH)_2 + 2NaOH \longrightarrow Na_2[ZnO_2] + 2H_2O$$

After separation, the reactions of zinc and manganese are studied.

3. The precipitate left after treatment with dilute HCl will contain CoS and NiS. This on heating with aquaregia or sodium hypochlorite solution and dilute HCl. CoS and NiS dissolves

$$3CoS + 2HNO_3 + 6HCl \longrightarrow 3CoCl_2 + 2NO + 4H_2O + 3S$$
$$3NiS + 2HNO_3 + 6HCl \longrightarrow 3NiCl_2 + 2NO + 4H_2O + 3S$$

This solution is divided into two portions to carry the tests for cobalt nickel separately.

To one portion, to test for cobalt ion if potassium nitrite is added, a yellow precipitate of potassium cobaltinitrite will be formed (for reactions refer to reactions of Co^{2+} ion) If ammonium thiocyanate is added, a blue solution of cobalt thiocyanate complex will be formed.

Nickel can be identified by the formation of red precipitate in another portion with dimethyl glyoxime in a solution just alkaline with ammonia.

16.12.1 Reactions of Zinc Ion Zn^{2+}

1. Sodium Hydroxide Solution: White gelatinous precipitate of zinc hydroxide readily soluble in excess of the reagent wih the formation of sodium zincate. The precipitate also dissolves in dilute acids

$$ZnSO_4 + 2NaOH \longrightarrow Zn(OH)_2 + Na_2SO_4$$
$$Zn(OH)_2 + 2NaOH \longrightarrow Na_2ZnO_2 + 2H_2O$$

2. Ammonia Solution: White precipitate of zinc hydroxide readily soluble in excess of the reagent and in solution of ammonium salts owing to the formation of complex. No precipitation of zinc hydroxide occurs by ammonia solution in the presence of ammonium chloride is due to lowering of the hydroxyl ion concentration to such a value that the solubility product of $Zn(OH)_2$ is not attained.

$$ZnSO_4 + 2NH_4OH \longrightarrow Zn(OH)_2 + (NH_4)_2SO_4$$
$$Zn(OH)_2 + 4NH_3 \longrightarrow [Zn(NH_3)_4](OH)_2$$

3. Hydrogen Sulphide: Partial precipitation of zinc sulphide but complete precipitation in alkaline medium

$$ZnSO_4 + H_2S \rightleftharpoons ZnS + H_2SO_4$$
$$Na_2ZnO_2 + H_2S \rightleftharpoons ZnS + 2NaOH$$

4. Potassium Ferrocyanide Solution: White precipitate of zinc ferrocyanide $Zn_2 [Fe(CN)_6]$ which is converted by excess of the reagent into the less soluble zinc potassium ferrocyanide $Zn_3K_2[Fe(CN)_6]_2$

$$2ZnSO_4 + K_4[Fe(CN)_6] \longrightarrow Zn_2[Fe(CN)_6] + 2K_2SO_4$$
$$3Zn_2[Fe(CN)_6] + K_4[Fe(CN)_6] \longrightarrow 2Zn_3K_2[Fe(CN)_6]$$

5. Cobalt Nitrate Test: The zinc hydroxide precipitate when heated with little cobalt nitrate forms a green mass (called Rinmann's green) consisting largely of cobalt zincate $CoZnO_2$ is obtained

$$2Zn(OH)_2 + 2Co(NO_3)_2 \longrightarrow 2Co\,ZnO_2 + 4NO_2 + 2H_2O + O_2$$

16.12.2 Reactions of Manganous Ion Mn^{2+}

1. Sodium Hydroxide Solution: White precipitate of manganous hydroxide $Mn(OH)_2$ insoluble in excess of the reagent The precipitate rapidly oxidized on exposure to air becoming brown, the brown compound is either manganese hydroxide $Mn(OH)_3$ or hydrated manganese dioxide $MnO_2 \cdot XH_2O$

$$MnCl_2 + 2NaOH \longrightarrow Mn(OH)_2 + 2NaCl$$
$$4Mn(OH)_2 + 2H_2O + O_2 \longrightarrow 4Mn(OH)_3$$
$$2Mn(OH)_2 + O_2 \longrightarrow 2MnO(OH)_2$$

2. Ammonia Solution: Partial precipitation of white manganous hydroxide $Mn(OH)_2$ soluble in solutions of ammonium salts. No precipitation occurs in the presence of ammonium salts owing to the lowering of the hydroxyl ion concentration and consequent failure to achieve the solubility product of $Mn(OH)_2$. On exposure to air brown manganic hydroxide or hydrated manganese dioxide is precipitated from ammonia solution

$$MnCl_2 + 2NH_4OH \longrightarrow Mn(OH)_2 + 2NH_4Cl$$

3. Lead Dioxide and Concentrated Nitric Acid: On boiling a dilute solution of a manganous salt (free from chlorides or hydrochloric acid), with lead dioxide and a little concentrated nitric acid and allowing the solution to settle, the supernatant liquid acquires a violet-red (or purple) colour due to permanganic acid

$$2MnSO_4 + 5PbO_2 + 6HNO_3 \longrightarrow 2HMnO_4 + 3Pb(NO_3)_2 + 2PbSO_4 + 2H_2O$$

The same colour due to permanganic acid is also obtained by heating manganous salt with ammonium or potassium persulphate in the presence of silver nitrate as catalyst or with sodium bismuthate or with potassium periodate

$$2MnSO_4 + 5(NH_4)_2S_2O_8 + 8H_2O \longrightarrow 2HMnO_4 + 10NH_4HSO_4 + 2H_2SO_4$$

$$2MnSO_4 + 5NaBiO_3 + 16HNO_3 \longrightarrow 2HMnO_4 + 5Bi(NO_3)_3 + NaNO_3 + 2Na_2SO_4 + 7H_2O$$

$$2MnSO_4 + 5KIO_4 + 3H_2O \longrightarrow 2HMnO_4 + 5KIO_3 + 2H_2SO_4$$

16.12.3 Reactions of Cobaltous Ion Co^{2+}

1. Sodium Hydroxide Solution: A blue basic salt is precipitated in the cold; upon warming with excess of alkali the basic salt is converted into pink cobaltous hydroxide. This when exposed to air changes to brownish-black cobalt hydroxide $Co(OH)_3$.$Co(OH)_2$ cannot be precipitated from ammonia solutions or from solutions containing ammonium salts owing to the formation of complex.

$$Co(NO_3)_2 + NaOH \longrightarrow Co(OH)NO_3 + NaNO_3$$

$$Co(OH)NO_3 + NaOH \longrightarrow Co(OH)_2 + NaNO_3$$

$$2Co(OH)_2 + H_2O + O \longrightarrow 2Co(OH)_3$$

2. Ammonia Solution: Precipitates blue basic salt as with NaOH, readily soluble in excess of the precipitant and in solutions of ammonium salts

$$Co(NO_3)_2 + NH_4OH \longrightarrow Co(OH)NO_3 + NH_4NO_3$$

$$4Co(OH)NO_3 + 28NH_3 + 6H_2O + O_2 \longrightarrow 4[Co(NH_3)_6](OH)_3 + 4NH_4NO_3$$

3. Potassium Cyanide Solution: Reddish-brown precipitate of cobaltous cyanide soluble in excess of the reagent to form a brown solution containing potassium cobaltocyanide. It slowly assumes yellow colour due to oxidation to potassium cobalticyanide

$$Co(NO_3)_2 + 2KCN \longrightarrow Co(CN)_2 + 2KNO_3$$

$$Co(CN)_2 + 4KCN \longrightarrow K_4[Co(CN)_6]$$

$$4K_4[Co(CN)_6] + 2H_2O + O_2 \rightarrow 4K_3[Co(CN)_6] + 4KOl$$

4. Potassium Nitrite Solution: Yellow precipitate of potassium cobaltinitrite is formed when potassium nitrite is added in excess to a concentrated solution of cobalt salt acidified with acetic acid.

$$Co(NO_3)_2 + 2KNO_2 \longrightarrow Co(NO_2)_3 + 2KNO_3$$

$$Co(NO_3)_2 + 2KNO_2 + 2CH_3COOH \longrightarrow Co(NO_2)_3 + 2CH_3COOK + NO + H_2O$$

$$Co(NO_2)_3 + 3KNO_2 \longrightarrow K_3[Co(NO_2)_6]$$

$$Co(NO_3)_2 + 7KNO_2 + 2CH_3COOH \longrightarrow K_3[Co(NO_2)_6] + 2KNO_3 + 2CH_3COOK + NO + H_2O$$

5. Ammonium Thiocyanate Solution: Blue solution due to the formation of cobalt-thiocyanate ion $[Co(SCN)_4]^{2-}$

$$Co(NO_3)_2 + 4NH_4SCN \longrightarrow (NH_4)_2[Co(SCN)_4] + 2NH_4NO_3$$

16.12.4 Reactions of Nickel Ion Ni^{2+}

1. Sodium Hydroxide Solution: Green precipitate of $Ni(OH)_2$, insoluble in excess of reagent is formed.

$$NiSO_4 + 2NaOH \rightarrow Ni(OH)_2 + Na_2SO_4$$

2. Ammonia Solution: Green precipitate of basic salt soluble in excess of the reagent forming complex nickel ammine complex having dark blue colour

$$NiCl_2 + NH_4OH \longrightarrow Ni(OH)Cl + 2NH_4Cl$$

$$Ni(OH)Cl + 7NH_3 + H_2O \longrightarrow [Ni(NH_3)_6](OH)_2 + NH_4Cl$$

No precipitation occurs with ammonia solution in the presence of ammonium salts due to less OH^- ion concentration

3. Hydrogen Sulphide: Only partial precipitation of NiS takes place in neutral solution but complete precipitation occurs in alkaline medium

$$NiCl_2 + H_2S \rightleftharpoons NiS + 2HCl$$

4. Potassium Cyanide Solution: Initially green precipitate of $Ni(CN)_2$ is formed which dissolves in excess of KCN due to the formation of complex

$$NiCl_2 + 2KCN \longrightarrow Ni(CN)_2 + 2KCl$$

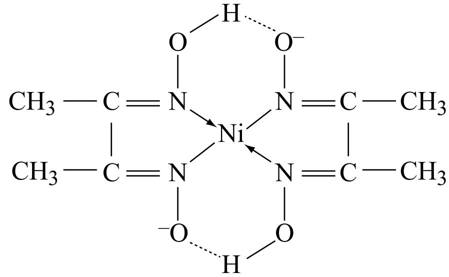
$$Ni(CN)_2 + 2KCN \longrightarrow K_2[Ni(CN)_4]$$

5. Potassium Nitrite Solution: No precipitate is produced in the presence of acetic acid (difference from cobalt)

6. Dimethylglyoxime Reagent: Red precipitate of nickel dimethyl glyoxime in solutions just alkaline with ammonia solution

$$\begin{array}{l} CH_3 - C = NOH \\ \qquad | \qquad\qquad + NiSO_4 + 2NH_4OH \longrightarrow \\ CH_3 - C = NOH \end{array}$$

$$+ (NH_4)_2SO_4 + 2H_2O$$



16.13 REACTIONS OF GROUP – V CATIONS

16.13.1 Separation of group – V Cations

V Group contains Ca^{2+}, Sr^{2+}, Ba^{2+} ions
Group Reagent $NH_4Cl + (NH_4)_2CO_3 + NH_4OH$
Ammonium carbonate is a weak electrolyte and its ionization decreases in the presence of NH_4Cl a strong electrolyte due to common ion effect

$$(NH_4)_2CO_3 \rightleftharpoons 2NH_4^+ + CO_3^{2-}$$
$$NH_4Cl \longrightarrow NH_4^+Cl^-$$

In that low concentration of carbonate ion the ionic product of carbonate ion and group V cation exceeds the solubility product of their carbonates. So they are precipitated as carbonates. But the ionic product of carbonate ion and magnesium ion is less than the solubility product of magnesium carbonate ($MgCO_3$). So magnesium ion remains in solution.

$$CaCl_2 + (NH_4)_2CO_3 \longrightarrow CaCO_3 + 2NH_4Cl$$
$$SrCl_2 + (NH_4)_2CO_3 \longrightarrow SrCO_3 + 2NH_4Cl$$
$$BaCl_2 + (NH_4)_2CO_3 \longrightarrow BaCO_3 + 2NH_4Cl$$

The separation and identification of the ions can be carried as follows:

1. The carbonates of the fifth group cations are soluble in acetic acid due to conversion into acetates.

$$CaCO_3 + 2CH_3COOH \longrightarrow Ca(CH_3COO)_2 + H_2O + CO_2$$
$$SrCO_3 + 2CH_3COOH \longrightarrow Sr(CH_3COO)_2 + H_2O + CO_2$$
$$BaCO_3 + 2CH_3COOH \longrightarrow Ba(CH_3COO)_2 + H_2O + CO_2$$

When potassium chromate is added, only barium will be precipitated as barium chromate while the calcium and strontium remain in solution.

$$Ba(CH_3COO)_2 + K_2CrO_4 \longrightarrow BaCrO_4 + 2CH_3COOK$$

2. After separation of barium chromate precipitate, if ammonium sulphate is added to the filtrate, strontium will be precipitated as strontium sulphate while calcium ion will not be precipitated as it is present in the form of double salt $(NH_4)_2SO_4 \cdot CaSO_4 \cdot H_2O$

$$Sr(CH_3COO)_2 + (NH_4)_2SO_4 \rightarrow SrSO_4 + 2CH_3COONH_4$$

16.13.2 Reactions of Barium Ion Ba^{2+}

1. Ammonia Solution: No precipitate of barium hydroxide because of its relatively high solubility

2. Ammonium Carbonate: White precipitate of barium carbonate, soluble in acetic acid and mineral acids

$$BaCl_2 + (NH_4)_2CO_3 \longrightarrow BaCO_3 + 2NH_4Cl$$

3. Ammonium Oxalate Solution: White precipitate of barium oxalate BaC_2O_4, slightly soluble in water but readily dissolved by hot dilute acetic acid (distinction from calcium) and by mineral acids

$$BaCl_2 + (NH_4)_2C_2O_4 \longrightarrow BaC_2O_4 + 2NH_4Cl$$

4. Dilute Sulphuric Acid: White finely divided precipitate of barium sulphate. Precipitate of barium sulphate practically insoluble in water, almost insoluble in mineral acids and ammonium sulphate solution

$$BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HCl$$

5. Potassium Chromate Solution: Yellow precipitate of barium chromate insoluble in water and in dilute acetic acid, but readily soluble in mineral acids

$$BaCl_2 + K_2CrO_4 \longrightarrow BaCrO_4 + 2KCl$$

In the presence of mineral acid yellow colour changes to orange red due to conversion of chromate to dichromate.

$$2CrO_4^{2-} + 2H^+ \rightleftharpoons 2HCrO_4^- \rightleftharpoons Cr_2O_7^{2-} + H_2O$$

The solubility products of $SrCrO_4$ and $CaCrO_4$ are much larger than $BaCrO_4$ and hence they require a larger CrO_4^{2-} ion concentration to precipitate them. The addition of acetic acid to the K_2CrO_4 solution lower the CrO_4^{2-} ion concentration sufficiently to prevent the precipitation of $SrCrO_4$ and $CaCrO_4$ but is maintained high enough to precipitate $BaCrO_4$.

16.13.3 Reactions of Strontium Ion Sr^{2+}

1. Ammonia Solution: No precipitate
2. Ammonium Carbonate: White precipitate of strontium carbonate $SrCO_3$, less soluble in water than calcium carbonate

$$SrCl_2 + Na_2CO_3 \longrightarrow SrCO_3 + 2NaCl$$

3. Dilute Sulphuric Acid: White precipitate of strontium sulphate $SrSO_4$, very sparingly soluble in water, insoluble in ammonium sulphate solution even on boiling (distinction from calcium)

$$SrCl_2 + H_2SO_4 \longrightarrow SrSO_4 + 2HCl$$

It almost completely converted into the corresponding carbonate by boiling with a concentrated solution of sodium carbonate.

$$SrSO_4 + Na_2CO_3 \longrightarrow SrCO_3 + Na_2SO_4$$

4. Ammonium Oxalate Solution: White precipitate of strontium oxalate sparingly soluble in water, and in acetic acid but soluble in mineral acids

$$SrCl_2 + (NH_4)_2C_2O_4 \longrightarrow SrC_2O_4 + 2NH_4Cl$$

16.13.4 Reactions of Calcium Ion Ca^{2+}

1. Ammonia Solution: No precipitate

2. Ammonium Carbonate Solution: White amorphous precipitate of calcium carbonate which becomes crystalline on boiling. The precipitate is soluble in water containing excess carbonic acid (i.e., passing excess carbon dioxide into solution.

$$CaCl_2 + (NH_4)_2CO_3 \longrightarrow CaCO_3 + 2NH_4Cl$$

$$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$$

Similarly $BaCO_3$ and $SrCO_3$ also dissolves in excess of carbonic acid.

3. Dilute Sulphuric Acid: White precipitate of calcium sulphate $CaSO_4 \cdot 2H_2O$ from concentrated solution. The precipitate is appreciably soluble in water, is more soluble in acids than either strontium and barium sulphates and is readily soluble in hot concentrated ammonium sulphate solution owing to the formation of a complex salt (distinction from strontium)

$$CaCl_2 + H_2SO_4 \longrightarrow CaSO_4 + 2HCl$$

$$CaSO_4 + (NH_4)_2SO_4 \longrightarrow (NH_4)_2[Ca(SO_4)_2]$$

4. Ammonium Oxalate Solution: White precipitate of calcium oxalate $CaC_2O_4.H_2O$ is formed. It is insoluble in water and in acetic acid but readily soluble in mineral acids

$$CaCl_2 + (NH_4)_2C_2O_4 \longrightarrow CaC_2O_4 + 2NH_4Cl$$

The calcium oxalate precipitate when heated with acidified potassium permanganate it will be decolourized

$$2KMnO_4 + 5CaC_2O_4 + 8H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 5CaSO_4 + 10CO_2 + 8H_2O$$

5. Potassium Chromate Solution: No precipitate in dilute solutions (solubility 23 grams per litr) nor from concentrated solutions containing acetic acid.

6. Potassium Ferrocyanide Solution: White precipitate of calcium potassium ferrocyanide $CaK_2[Fe(CN)_6]$ is produced by excess of reagent.

$$CaCl_2 + K_4[Fe(CN)_6] \longrightarrow CaK_2[Fe(CN)_6] + 2KCl$$

16.14 CATIONS OF GROUP VI

Reactions of Group VI cations: Group VI contain magnesium, sodium, potassium and ammonium ions.

No group reagents are necessary. Tests for ions are carried directly for Mg^{2+}, Na^+ and K^+ from the filtrate of V group. Reactions of ammonium are studied directly with the given compound.

16.14.1 Reaction of Magnesium Ion

1. Ammonia Solution: Partial precipitation of white gelatinous magnesium hydroxide $Mg(OH)_2$ very less soluble in water but readily soluble in solution of ammonium salts

$$MgSO_4 + 2NH_4OH \longrightarrow Mg(OH)_2 + (NH_4)_2SO_4$$

2. Sodium Hydroxide Solution: White precipitate of magnesium hydroxide insoluble in excess of the reagent but readily soluble in solutions of ammonium salts

3. Ammonium Carbonate Solution: White precipitate of basic magnesium carbonate, often only on boiling or on long standing. No precipitate is obtained in the presence of ammonium salts of strong acids

$$5MgSO_4 + 5(NH_4)_2CO_3 + H_2O \longrightarrow 4MgCO_3 \cdot Mg(OH)_2 + 5(NH_4)_2SO_4 + CO_2$$

4. Sodium Carbonate Solution: White voluminous precipitate of basic carbonate $4MgCO_3\,Mg(OH)_2\,5H_2O$ or $Mg\left[(MgCO_3)_4\right](OH)_2$ insoluble in solutions of bases but soluble in acids and in solutions of ammonium salts

$$5MgSO_4 + 5Na_2CO_3 + H_2O \longrightarrow 4MgCO_3\,Mg(OH)_2 + 5Na_2SO_4 + CO_2$$

5. Sodium Phosphate Solution: White crystalline precipitate of magnesium ammonium phosphate $Mg(NH_4)PO_4 \cdot 6H_2O$ in the presence of ammonium chloride (to prevent the precipitation of magnesium hydroxides and ammonia solutions)

$$MgSO_4 + Na_2HPO_4 + NH_3 \longrightarrow Mg(NH_4)PO_4 + Na_2SO_4$$

A white flocculent precipitate of magnesium hydrogen phosphate $MgHPO_4$ is produced with Na_2HPO_4 in neutral medium. This on heating converts into crystalline magnesium pyrophosphate

$$MgSO_4 + Na_2HPO_4 \longrightarrow MgHPO_4 + Na_2SO_4$$

$$2MgHPO_4 \xrightarrow{\Delta} Mg_2P_2O_7 + H_2O$$

16.14.2 Reactions of Ammonium Ion NH_4^+

1. Sodium Hydroxide Solution: Ammonia gas is evolved on warning. This may be identified

(a) by its odour
(b) by the formation of white fumes of ammonium chloride when a glass rod moistened with concentrated hydrochloric acid is held in the vapour
(c) by its turning moistened red litmus paper blue or turmeric paper brown.
(d) by its ability to turn filter paper moistened with mercurous nitrate solution black (more trust worthy test)
(e) filter paper moistened with a solution of manganous sulphate and hydrogen peroxide gives brown colour due to the oxidation of the manganese by the alkaline solution thus formed.

$$NH_4Cl + NaOH \longrightarrow NaCl + H_2O + NH_3$$

$$Hg_2(NO_3)_2 + 2NH_3 \longrightarrow \underbrace{Hg(NH_2)NO_3 + Hg}_{\text{black}} + NH_4NO_3$$

$$2MnSO_4 + H_2O_2 + 4NH_3 + 4H_2O \longrightarrow 2Mn(OH)_3 + 2(NH_4)_2SO_4$$

2. Nessler's Reagent: Brown precipitate or brown or yellow colouration is produced according to the amount of ammonia or ammonium ions.

$$NH_4^+ + 2[HgI_4]^{2-} + 4OH^- \longrightarrow HgO\cdot Hg(NH_2)I \downarrow 7I^- + 3H_2O$$

Single Answer Questions

1. On heating with concentrated H_2SO_4 a mixture of solid NaCl and Kl produces
(a) Cl_2, I_2, and SO_2 (b) HCl, HI and I_2
(c) HCl, I_2 and SO_2 (d) HCl and HI

2. In a charcoal cavity test in an oxidizing flame (using cobalt nitrate) salts of aluminium, zinc and magnesium metals produce residues with specific colours which of the following gives the correct match of the colour
(a) Thenard's blue $\rightarrow CoAl_2O_4$
(b) Rimmann's green $\rightarrow CoZnO_2$
(c) Pale pink$\rightarrow CoMgO_2$
(d) All of these

3. In the borax-bead test a colourless bead becomes coloured on heating with a transition metal salt. This is due to the formation of coloured
(a) Borate and meta borate of the metal
(b) Boric oxide
(c) Othoborate of the metal
(d) Hexaborate of the metal

4. Which of the following pairs of compounds form a precipitate when their aqueous solutions are mixed together
(a) $FeSO_4 - BaCl_2$ (b) $FeSO_4 + NaCl$
(c) $MgCl_2 - Na_2SO_4$ (d) $Al_2(SO_4)_3 + NaNO_3$

5. An excess of aqueous ammonia solution is added gradually to an aqeous solution $Al_2(SO_4)_3$ which of the following happens
(a) A white precipitate is formed which dissolves in the excess of ammonia solution
(b) A white gelatinous precipitate is formed which does not dissolve in the excess of ammonia solution
(c) A red colouration is developed
(d) No observable change takes place

6. Fe^{3+} can be distiguished from Fe^{2+} by using a reagent (A) which develops a red colour with Fe^{3+} due to the formation of a compound (B). Fe^{2+} in the pure state does not develop any colour with the reagent. The reagent (A) and the product (B) are respectively
(a) $K_4[Fe(CN)_6]$ and $Fe_4[Fe(CN)_6]_3$
(b) NH_4CNS and $[Fe(SCN)]^{2+}$
(c) $K_3[Fe(CN)_6$ and $K_2Fe[Fe(CN)_6]$
(d) Na_2HPO_4 and $FePO_4$

7. What happens when an aqueous solution of $FeSO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ and chrome alum is treated with excess of Na_2O_2 and filtred.
(a) A colourless filtrate and a green residue are obtained
(b) A yellow filtrate and a green residue are obtained
(c) A yellow filtrate and a brown residue are obtained
(d) A green filtrate and a brown residue are obtained

8. During the separation of group V metal (Ba^{2+}, Sr^{2+} and Ca^{2+}) ion as insoluble carbonates by adding a saturated solution of $(NH_4)_2CO_3$, to the salt sample in ammoniacal medium containing an excess of NH_4Cl magnesium is not precipitated either as $Mg(OH)_2$ or $MgCO_3$ because
(a) the concentrations of OH^- and CO_3^{2-} are so low due to common-ion effect that the ionic product values of $Mg(OH)_2$ and $MgCO_3$ cannot exceed their respective solubility product values.
(b) the solubility product values of $Mg(OH)_2$ and $MgCO_3$ decrease due to the common ion effect
(c) the solubility product values of $Mg(OH)_2$ and $MgCO_3$ increase due to common ion effect
(d) the initially formed precipitate of $Mg(OH)_2$ and $MgCO_3$ combine to form a soluble basic carbonate $4MgCO_3 \cdot Mg(OH)_2$

9. A gas X is passed through water to form a saturated solution. The aqueous solution on treatment with a solution of $AgNO_3$, gives a white precipitate. The aqueous solution also dissolves magnesium ribbon with the evolution of a colourless gas Y identify X and Y

(a) $X = CO_2, Y = Cl_2$
(b) $X = Cl_2, Y = HCl$
(c) $X = Cl_2, Y = H_2$
(d) $X = H_2, Y = Cl_2$

10. When dil H_2SO_4 is added to salt 'D', the gas evolved turns lime water milky and also turns acidified dichromate paper green. The salt solution of 'D' gives white ppt with $BaCl_2$ solution, which is soluble in conc. HCl. The salt 'D' is
(a) SO_4^{2-} (b) SO_3^{2-} (c) CO_3^{2-} (d) S^{2-}

11. The ring test is not reliable in presence of
(a) Br^- ions (b) SO_4^{2-} ions
(c) Zn^{2+} ions (d) Al^{3+} ions.

12. Which of the following is insoluble in ammonium hydroxide?
(a) $Fe(OH)_3$ (b) $Cu(OH)_2$
(c) AgOH (d) All of these.

13. Which of the following gives pink colour with red lead?
(a) $Mn(NO_3)_2$ (b) $NaNO_3$
(c) $Cu(NO_3)_2$ (d) $AgNO_3$

14. A metal X on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour. Y is
(a) $Mg(NO_3)_2$ (b) MgO
(c) Mg_3N_2 (d) NH_3

15. The original solution should not be prepared in concentrated H_2SO_4 because.
(a) It is an oxidizing agent which oxidizes H_2S into SO_2, in second group.
(b) It is an oxidizing agent which oxidizes H_2S in to S in second group.
(c) It is a dehydrating reagent.
(d) None of these.

16. When a metal oxalate is heated with conc. H_2SO_4 a mixture of two gases A and B is obtained gas A burns with blue flame. The gases A and B are respectively.
(a) NO_2, CO_2 (b) CO_2, CO
(c) CO, CO_2 (d) CO, NO

17. Zinc sulphide is not precipitated when H_2S is passed through $ZnCl_2$ solution. Because
(a) $ZnCl_2$ is soluble in hydrochloric acid.
(b) Zinc sulphide is soluble in hydrochloric acid.
(c) Zinc sulphide is insoluble in hydrochloric acid.
(d) Zinc chloride does not react with hydrogen sulphide.

18. Which of the following pairs of compounds can be separated by sodium hydroxide?
(a) Al $(OH)_3$, KOH (b) $Zn(OH)_2$, AgOH
(c) $Zn(OH)_2$, $Al(OH)_3$ (d) None of these.

19. A white crystalline solid 'A' on boiling with caustic soda solution gave a gas 'B' which when passed through an alkaline solution of K_2HgI_4 gave a brown ppt. The substance 'A' on heating gave a gas 'C' which rekindled a glowing splinter but does not give brown fumes of nitric oxide. The gas 'C' is.
(a) N_2O (b) NH_3 (c) O_2 (d) O_3

20. Potassium chromate solution is added to an aqueous solution of metal chloride. The yellow precipitate formed is insoluble in water and acid. When flame test is performed with the precipitate the colour produced is.
(a) Lilac (b) apple green
(c) crimson red (d) golden yellow

21. Which of the following is formed when $K_4[Fe(CN)_6]$ is added in excess to Fe^{3+} salt ?
(a) $Fe_4[Fe(CN)_6]_3$ Prussian blue
(b) $KFe[Fe(CN)_6]$ intense blue colour
(c) $Fe_3[Fe(CN)_6]_2$
(d) $Fe(OH)_3$

22. An original salt solution in acidic medium did not give any precipitate on passing H_2S gas. Such a solution was boiled, reboiled after dilution, 3 times. To such a solution, two drops of conc. HNO_3 was added, then heated and water was added. To this resulting solution, NH_4C1 was first added followed by excess of NH_4OH. Finally, a green precipitate was obtained. Hence the cation may be.
(a) $A1^{3+}$ (b) Fe^{2+} (c) Fe^{3+} (d) Cr^{3+}

23. $CaSO_4$ is insoluble in water but not precipitated when excess of $(NH_4)_2SO_4$ is added to calcium chloride because.
(a) Ca deposited from the solution
(b) Forms a soluble calcium hydroxide in solution.
(c) Forms a soluble nitride in solution
(d) Forms a soluble complex in solution.

24. A salt on treatment with dil HCl gives a pungent smelling gas and a yellow ppt. The salt gives green flame when burnt. The salt solution gives a yellow ppt. with potassium chromate. The salt is
(a) BaS_2O_3 (b) PbS_2O_3
(c) $CuSO_4$ (d) $NiSO_4$

25. Mercury (II) forms a complex with
(a) KI (b) H_2S
(c) $SnCl_2$ (d) NaOH

26. $[X] + H_2SO_4 \longrightarrow [Y]$. a colourless gas with irritating smell $[Y] + K_2Cr_2O_7 + H_2SO_4 \longrightarrow$ green solution [X] and [Y] are
(a) SO_3^{2-}, SO_2 (b) Cr^-, HCl
(c) S^-, H_2S (d) CO_3^-, CO_2

27. A + dil. $HNO_3 \rightarrow B + NO + S + H_2O$; $B + NH_4OH \rightarrow$ Deep blue solution; Identify 'B'
(a) $Pb(NO_3)_2$ (b) $Bi(NO_3)_3$
(c) $Cu(NO_3)_2$ (d) $Cd(NO_3)_2$

28. A metal salt solution forms a yellow precipitate with potassium chromate in acetic acid, a white precipitate with dil H_2SO_4, but gives no precipitate with NaCl. The white precipitate obtained when $(NH_4)_2CO_3$ is added to the metal, salt solution will consist of
(a) $PbCO_3$ (b) $BaCO_3$
(c) $MgCO_3$ (d) $CaCO_3$

29. The use of H_2S as a reagent for qualitative analysis of cations is based on the fact that
(a) H_2S form coloured sulphides
(b) H_2S is a weak acid and we can control the degree of ionization of H_2S by adding HCI or NH_4OH
(c) H_2S is a weak acid
(d) The colour of the metal sulphide precipitated depends upon the pH of the solution

30. A metal nitrate on reaction with excess sodium hydroxide solution gives a white precipitate, but it rapidly turns to brown on adding bromine water. Aqueous solution of metal nitrate produces a pink precipitate with disodium hydrogen phosphate solution in the presence ammonia. The cation is
(a) Cu^{2+} (b) Mn^{2+} (c) Mg^{2+} (d) Zn^{2+}

31. (i) $A + Na_2CO_3 \rightarrow B + C$, (ii) $A \xrightarrow{CO_2} (\text{Milky})C$
The chemical formula of A and B are respectively:
(a) NaOH and $Ca(OH)_2$ (b) $Ca(OH)_2$ and NaOH
(c) NaOH and CaO (d) CaO and $Ca(OH)_2$

32. When a KI solution is added to a metal nitrate, a black precipitate is produced which dissolves in an excess of KI to give an orange solution. The metal ion is:
(a) Hg^{2+} (b) Bi^{3+}
(c) Cu^{2+} (d) Pb^{2+}

33. Which is not easily precipitated from aqueous solution?
(a) Cl^- (b) SO_4^{-2}
(c) NO_3^- (d) CO_3^{2-}

34. Soda extract is useful when given mixture has any insoluble salt, it is prepared by:
(a) Fusing soda and mixture and then extracting with water
(b) Dissolving $NaHCO_3$ and mixture in dil. HCl
(c) Boiling Na_2CO_3 and mixture in dil. HCl
(d) boiling Na_2CO_3 and mixture in distilled water

35. An aqueous solution of a substance, on treatment with dilute HCl, gives a white precipitate soluble in hot water. When H_2S is passed through the hot acidic solution, a black precipitate is formed. The substance is:
(a) Hg_2^{2+} salt (b) Cu^{2+} salt
(c) Ag^+ salt (d) Pb^{2+} salt

36. $CrCl_3 \xrightarrow[NH_4OH]{NH_4Cl} (A) \xrightarrow[H_2O]{Na_2O_2} (B) \xrightarrow[\text{acetate}]{\text{Lead}} (C)$
In this reaction sequence, the compound (C) is:
(a) Na_2CrO_4 (b) $Na_2Cr_2O_7$
(c) $Cr(OH)_3$ (d) $PbCrO_4$

37. Identify the correct order of solubility of Na_2S, CuS and ZnS:
(a) CuS > ZnS > Na_2S (b) ZnS > Na_2S > CuS
(c) Na_2S > CuS > ZnS (d) Na_2S > ZnS > CuS

38. $2Cu^{2+} + 5I^- \longrightarrow 2CuI\downarrow + [X]$
$[X] + 2S_2O_3^{2-} \longrightarrow 3[Y] + S_4O_6^{2-}$; X and Y are :
(a) I_3^- and I^- (b) I_2 and I_3^-
(c) I_2 and I^- (d) I_3^- and I_2

39. In Nessler's reagent, the ion present is:
(a) HgI^{2-} (b) HgI_4^{2-}
(c) Hg^+ (d) Hg^{2+}

40. A reddish pink substance on heating gives off a vapour which condenses on the sides of the test tube and the substance turns blue. It on cooling and water is added to the residue it turns to its original colour. The substance is:
(a) Iodine crystals
(b) Copper sulphate crystals
(c) Cobalt chloride crystals
(d) Zinc oxide

41. Oxalate + MnO_2 + Dil. $H_2SO_4 \rightarrow$ Gas
The gas evolved is:
(a) CO_2 (b) CO (c) SO_2 (d) O_2

42. To avoid the precipitation of hydroxides of Ni^{2+}, Co^{2+}, Zn^{2+} and Mn^{2+} along with those of Fe^{3+}, Al^{3+} and Cr^{3+} the third group solution should be:
(a) Heated with a few drops of conc. HNO_3
(b) Treated with excess of NH_4Cl
(c) Concentrated
(d) None of these

43. Which set are yellow?
(a) KO_3, Sb_2S_3, CdS
(b) Sb_2S_3, CdS, $PbCrO_4$
(c) $PbCrO_4$, As_2S_3, SnS_2
(d) SnS_2, As_2S_3, $PbCrO_4$, PbO

44. Which of the following reagents can separate a mixture of AgCl and AgI?
(a) KCN (b) $Na_2S_2O_3$
(c) HNO_3 (d) NH_3

45. Which nitrate on decomposition will give metal?
(a) $Cu(NO_3)$ (b) $NaNO_3$
(c) KNO_3 (d) $AgNO_3$

46. Which of the following compounds does not exist?
(a) CrO_2Br_2 (b) CrO_2Cl_2
(c) $POCl_3$ (d) BiOCl

47. Which one among the following pairs of ions cannot be separated by H_2S in dilute HCl?
(a) Bi^{3+}, Sn^{2+} (b) Al^{3+}, Hg^{2+}
(c) Zn^{2+}, Cu^{2+} (d) Ni^{2+}, Cu^{2+}

48. Salt (A) gives brick red fumes (B) with conc. H_2SO_4 and $K_2Cr_2O_7$ which gives yellow solution (C) with NaOH and it gives yellow ppt. (D) with acetic acid and lead acetate. What is (C)?

(a) Na_2CrO_4 (b) CrO_2Cl_2
(c) $PbCrO_4$ (d) NaCl

49. When a nitrate is warmed with zinc powder and an NaOH solution, a gas is evolved. Which of the following reagents will be turned brown by the gas?

(a) Sodium nitroprusside
(b) Sodium cobalt nitrite
(c) Nessler's reagent
(d) Barium chloride

50. On adding KI solution in excess to a solution of $CuSO_4$ we get a precipitate 'P' and another liquor 'M'. Select the correct pairs:

(a) P is Cu_2I_2 and M is I_2 solution
(b) P is CuI_2 and M is I_2 solution
(c) P is Cu_2I_2 and M is KI_3 solution
(d) P is CuI_2 and M is KI_3 solution

51. Which of the following ions is responsible for the brown colour in the ring test for a nitrate?

(a) $[Fe(H_2O)_5NO]^{2+}$ (b) $[Fe(CN)_5NO]^{2-}$
(c) $[Fe(NO_2)_6]^{4-}$ (d) $[Fe(H_2O)_5NO_2]^{+}$

52. There is mixture of Cu(II) chloride and Fe(II) sulphate. The best way to separate the metal ions from this mixture in qualitative analysis is:

(a) Hydrogen sulphide in acidic medium, where only Cu(II) sulphide will be precipitated
(b) Ammonium hydroxide buffer, where only Fe(II) hydroxide will be precipitated
(c) Hydrogen sulphide in acidic medium, where only Fe(II) sulphide will be precipitated
(d) Ammonium hydroxide buffer, where only Cu(II) hydroxide will be precipitated

53. Which of the following reagents can be used to distinguish between a sulphite and a sulphate in solution?

(a) $FeSO_4$ (b) $Na_2[Fe(CN)_5NO]$
(c) $BaCl_2$ (d) $Na_3[Co(NO_2)_6]$

54. A doctor by mistake administers a $Ba(NO_3)_2$ solution to a patient for radiography investigations. Which of the following should be given as the best to prevent the absorption of soluble barium?

(a) NaCl (b) Na_2SO_4 (c) Na_2CO_3 (d) NH_4C1

55. A colourless water soluble solid 'X' on heating gives equimolar quantities of Y and Z. Y gives dense white fumes HCl and Z does so with NH_3. Y gives brown precipitate with Nessler's reagent and Z gives white precipitate with nitrates of Ag^+, Pb^{2+} and Hg^+. 'X' is:

(a) NH_4Cl (b) NH_4NO_3
(c) NH_4NO_2 (d) $FeSO_4$

56. The colour of the iodine solution is discharged by shaking with:

(a) Sodium sulphate
(b) Sodium chloride
(c) Aqueous sulphur dioxide
(d) Sodium bromide

57. Three separate samples of a solution of a single salt gave these results. One formed a white precipitate with excess ammonia solution, one formed a white precipitate with dil. NaCl solution and one formed a black precipitate with H_2S. The salt could be:

(a) $AgNO_3$ (b) $Pb(NO_3)_2$
(c) $Hg(NO_3)_2$ (d) $MnSO_4$

58. In an alkaline solution sodium nitroprusside gives a violet colour with:

(a) S^{2-} (b) SO_3^{2-} (c) SO_4^{2-} (d) NO_2^-

59. A pale yellow precipitate and a gas with pungent odour are formed on warming dilute hydrochloric acid with an aqueous solution containing

(a) sulphate ion (b) sulphide ion
(c) thiosulphate ion (d) sulphite ion

60. Which of the following compounds does magnesium precipitate when you test for it?

(a) $MgCO_3 \cdot MgO$ (b) $MgCO_3$
(c) $Mg(OH)_2$ (d) $MgNH_4PO_4 \cdot 6H_2O$

61. An inorganic salt, when warmed with NaOH solution, gives off a gas that turns a filter paper soaked with an alkaline solution of $K_2[HgI_4]$ brown. After complete evolution of gas, the reaction mixture when warmed with Al powder and NaOH solution a gas is evolved that gives white fumes with a glass rod wet with HCl solution. The salt responds to the brown ring test when acetic acid is used in place of conc. H_2SO_4. The cation and anion which may be present in the salt are, respectively,

(a) NH_4^+ and NO_3^-
(b) NH_4^+ and NO_2^-
(c) Any cations and NO_3^-
(d) Any cation and NO_2^-

62. Which of the following does not respond to chromyl chloride test?.

(a) NH_4C1 (b) KC1 (c) $CuCl_2$ (d) $SnCl_4$

63. Addition of solution of oxalate to an aqueous solution of mixture of Ba^{2+} Sr^{2+} and Ca^{2+} will precipitate

(a) Ca^{2+} (b) Ca^{2+} and Sr^{2+}
(c) Ba^{2+} and Sr^{2+} (d) All of these

64. Reagent A is $H_2S(g)$ dissolved in a buffer solution of pH = 3 and reagent B is $H_2S(g)$ dissolved in a buffer solution with pH = 8. Pick up the incorrect statement

(a) Reagent A gives a black PPt with aq $CuSO_4$
(b) Reagent A gives an yellow PPt with aq $CdSO_4$
(c) Reagent B gives a black PPt with aq $NiSO_4$
(d) Reagent B gives a black PPt with aq $ZnSO_4$

65. Green coloured solute

$$A \xrightarrow{AgNO_3} \text{white ppt} + B$$

$$A + H_2O_2 + KOH \xrightarrow{\Delta} \text{Yellow solution(c)}$$

$$C \xrightarrow[(CH_3COOH)]{(CH_3COO)_2Pb} \text{Yellow ppt D}$$

Which is correct

	A	B	C	D
a	AgCl	KCl	PbI_2	PbI_2
b	AgCl	KCl	$PbCrO_4$	$Pb_2Cr_2O_7$
c	AgCl	$CrCl_3$	$PbCrO_4$	PbI_2
d	$CrCl_3$	AgCl	K_2CrO_4	$PbCrO_4$

66. Some pale green crystals are strongly heated. The gases given off are passed into a container surrounded by ice and then through a solution of acidified $KMnO_4$. The $KMnO_4$ is decolourized, a waxy white solid is formed in the ice container; this is dissolved in water. The solution will:
(a) Give the precipitate with silver nitrate solution
(b) Give a precipitate with barium chloride solution
(c) Turn red litmus blue
(d) Give blue colour with starch solution.

67. To a solution of a substance, gradual addition of ammonium hydroxide results in a brown precipitate, which does not dissolve in excess of NH_4OH. However, when HCl is added to the original solution a white precipitate is formed. The solution contained:
(a) Lead salt (b) Silver salt
(c) Mercurous salt (d) Copper salt.

68. An aqueous solution contains Hg^{2+}, Hg_2^{2+} Pb^{2+} and Cd^{2+}. The addition of HCl(6N) will precipitate:
(a) Hg_2Cl_2 only (b) $PbCl_2$ only
(c) $PbCl_2$ and Hg_2Cl_2 (d) $PbCl_2$ and $HgCl_2$

69. When H_2S passed through the aqueous acidic solution containing Cu^{+2}, Zn^{+2}, Mn^{+2}, Ni^{+2}, Bi^{+3}, Pb^{+2} which set is ppted
(a) Zn^{+2}, Cu^{+2}, Mn^{+2} (b) Zn^{+2}, Ni^{+2}, Pb^{+2}
(c) Cu^{+2}, Bi^{+3}, Pb^{+2} (d) Cu^{+2}, Ni^{+2}, Bi^{+3}

70. A coloured solution known to contain two metal ions, was treated with excess cold sodium hydroxide solution. When filtered a whitish solid, slowly changing to brown, was retained on the filter paper and a colourless solution collected as filtarate. Dropwise addition of hydrochloric acid to the filtrate produced a white precipitate which dissolved in excess acid Treatment of the residue from the filter paper with a solution of strong oxidizer produced a reddish-violet solution.
Indicate any pairs of ions which on testing the above leads to the observed changes
(a) Zn^{2+} and Mn^{2+} ions (b) Mg^{2+} and Zn^{2+} ions
(c) Mn^{2+} and Mg^{2+} ions (d) Fe^{2+} and Zn^{2+} ions

71. Generation of a blue colour which is not due to metal or metal ion-ammonia interaction
(a) Sodium is dissolved in liquid ammonia
(b) Copper (II) sulphate is reacted with ammonium hydroxide
(c) Cobalt (II) chloride is reacted with ammonium hydroxide
(d) Alumina reacted with cobalt nitrate

72. Which of the following compounds will not respond to the borax-bead test
(a) $FeSO_4$ (b) $A1_2(SO_4)_3$
(c) $Co(NO_3)_2$ (d) $Cr_2(SO_4)_3$

73. A reagent is added to a solution of manganous salt in cold dilute HNO_3 A purple colour appeared due to the formation of $HMnO_4$. The reagent is
(a) $NaBiO_3$ (b) KNO_3
(c) $(NH_4)_2\,C_2O_4$ (d) H_2O_2

74. Which of the following pairs of ions can be separated from each other using a concentrated NaOH solution
(a) Al^{3+} and Sn^{2+} (b) $CuSO_4$ and $ZnSO_4$
(c) Al^{3+} and Zn^{2+} (d) Zn^{2+} and Pb^{2+}

75. A metal cation gives black ppt with H_2S which dissolves in hot conc. HNO_3 to give white ppt. When cation reacts with KI it gives intense brown solution on further treating with thioslphate becomes colourless and white ppt becomes visible metal ion also reacts with KCN to form poisonous gas
(a) Fe^{3+} (b) Co^{2+}
(c) Cd^{2+} (d) Cu^{2+}

76. A mixture contains Cu^{2+}, $A1^{3+}$ and Ni^{2+}. Following steps have been adopted for separation but written in disorder.
I. Filter, boil off H_2S gas and NH_4C1, heat and add NH_4OH
II. Filter, add NH_4OH and pass H_2S gas
III. Pass H_2S gas into acidified solution of mixture
steps will be used in the following order
(a) I, II, III (b) III, I, II
(c) III, II, I (d) I, III, II

77. An orange colour mixture changes to green on acidification. Mixture may contain
(a) HgI_2, CrO_4^{2-} (b) $Cr_2O_7^{2-}$, Fe^{2+}
(c) SO_3^{2-}, MnO_4^- (d) Fe^{2-}, CrO_4^{2-}

78. A white crystalline substance dissolves in water. On passing H_2S gas in this solution, a black ppt is obtained. The black ppt dissolves completely in hot HNO_3. On adding a few drops of conc. H_2SO_4, a white ppt is obtained. This ppt is that of
(a) $BaSO_4$ (b) $SrSO_4$
(c) $Pb(NO_3)_2$ (d) $CdSO_4$

79. A pale green crystalline metal salt of M dissolves freely in water. On standing it gives a brown ppt on addition of aqueous NaOH. The metal salt solution also gives a black ppt on bubbling H_2S in basic medium. An aqueous solution of the metal salt decolourizes the pink colour of the permanganate solution. The metal in the metal salt solution is

(a) Coppe (b) Aluminium
(c) Lead (d) Iron

80. The compound which does not respond to ring test

(a) $Pb(NO_3)_2$ (b) $LiNO_3$
(c) $NaNO_3$ (d) KNO_3

81. Which of the following gives a precipitate with $Pb(NO_3)_2$ but not with $Ba(NO_3)_2$?

(a) Sodium chloride
(b) Sodium acetate
(c) Sodium nitrate
(d) Sodium hydrogen phosphate

82. In the separation of Cu^{2+} and Cd^{2+} in 2nd group qualitative analysis of cations tetrammine copper (II) sulphate and tetrammine cadmium (II) sulphate react with KCN to form the corresponding cyano complexes. Which one of the following pairs of the complexes and their relative stability enables the separation of Cu^{2+} and Cd^{2+}?

(a) $K_3[Cu(CN)_4$ more stable and $K_2[Cd\ (CN)_4]$ less stable
(b) $K_2[Cu(CN)_4$ less stable and $K_2[Cd\ (CN)_4]$ more stable
(c) $K_2[Cu(CN)_4$ more stable and $K_2[Cd\ (CN)_4]$ less stable
(d) $K_3[Cu(CN)_4$ less stable and $K_2[Cd\ (CN)_4]$ more stable

83. Read of the following statements and choose the correct code w.r.t true (T) and false (F).

I. manganese salts give a violet borax bead test in reducing flame
II. from a mixed precipitate of AgCl and AgI, ammonia solution dissolves only AgCl
III. ferric ions give a deep green precipitate, on adding potassium ferrocyanide solution
IV. on boiling the solution having K^+, Ca^{2+} and HCO_3^- we get a precipitate of $K_2Ca(CO_3)_2$

(a) TTFF (b) FTFT
(c) FTFF (d) TTFT

84. Cationic part of chromyl chloride is

(a) Cr^{3+} (b) CrO_2^+ (c) CrO_2^{2+} (d) CrO_4^{2+}

85. Acidified $K_2Cr_2O_7$ will fail to disstinguish between

(a) CO and CO_2 (b) $C_2O_4^{2-}$ and CO_3^{2-}
(c) CO and SO_2 (d) $C_2O_4^{2-}$ and F^-

86. If HCl is not added before passing H_2S in the second group it may result in the

(a) In complete precipitation of second group sulphide
(b) Precipitation of sulphides of cations belonging to subsequent groups
(c) Precipitation of sulphur takes place
(d) Precipitation lead as lead sulphide.

87. Some white crystals are heated. A cracking sound is heard and brown fumes are given of the residue after heating is seen to be yellow-brown in colour. The incorrect statement regarding these observations is

(a) The residue is litharge.
(b) The fumes rekindles the glowing splinter.
(c) The brown colour is due to bromine
(d) The brown fumes is a mixture of the gases

88. In qualitative analysis of 1 group radicals a white precipitate is formed which is insoluble in boiling water but when treated with NH_4OH it turns black; the precipitation may be

(a) $PbCl_2$ (b) AgCl
(c) $HgCl_2$ (d) Hg_2Cl_2

89. Two test tubes containing nitrate and bromide are treated separately with conc. H_2SO_4; brown fumes evolved are passed in water. The water will be coloured by vapours evolved from the test tube containing

(a) Nitrate (b) Bromide
(c) Both a and b (d) None of these

90. A white powder when strongly heated gives off brown fumes. A solution of this powder gives a yellow precipitate with KI. When a solution of barium chloride is added to a solution of powder, a white precipitate results. This white powder may be

(a) A soluble sulphate (b) KBr or NaBr
(c) $Ba(NO_3)_2$ (d) $AgNO_3$

91. Two colourless solutions are mixed, the mixture is densy coloured white. The two solutions are of

(a) Copper sulphate and sodium carbonate
(b) Sodium carbonate and calcium nitrate
(c) Ferrous sulphate and barium nitrate
(d) Sodium chloride and sodium carbonate

92. Which of the following statements is not correct?

(a) Lead (II) chloride is soluble in hot water and reappears on cooling
(b) In dilute HCl the solubility of $PbCl_2$ is diminished in comparison to that in water
(c) ln concentrated HCl the solubility of $PbCl_2$ is very much diminshed in comparison to that in water
(d) Lead (II) chloride forms complex $[(NH_4)_2\ PbCl_4]$

93. Brown precipitate (A) dissolve in HNO, gives (B) which gives white ppt (C) with NH_4OH'(C) on reaction with HCl gives solution (D), which gives white turbidity on addition of water. What is the turbidity?

(a) $BiO(NO_3)$ (b) $Bi(OH)_3$
(c) BiOCl (d) $Bi(NO_3)_3$

94. A mixture of Na_2CO_3 and Na_2SO_3 is treated with dil H_2SO_4 in a set up such that the gaseous mixture emerging can pass through a solution of $BaCl_2$ and then through $K_2Cr_2O_7$ acidified with dil H_2SO_4. Which of the following will you observe.
(a) The $BaCl_2$ solution remain un affected and acidified dichromate solution turns green.
(b) $BaCl_2$ solution gives white precipitate and acidified dichromate solution remains unaffected
(c) The $BaCl_2$ solution gives a white precipitate and acidified dichromate solution turns green.
(d) Both the solution remain unaffected.

95. $CuSO_4$ solution reacts with excess KCN to give
(a) $Cu(CN)_2$ (b) CuCN
(c) $K_2[Cu(CN)_2]$ (d) $K_3[Cu\ (CN)_4]$

96. In the precipitation of the iron group in qualitative analysis, ammonium chloride is added before adding ammonium hydroxide to
(a) decrease concentration of OH^- ions
(b) prevent interference by phosphate ions
(c) increase concentration of Cl^- ions
(d) increase concentration of OH^- ions

97. When KCN is added to blue copper sulphate solution, its colour is decolourized because of the formation of
(a) $[Cu(CN)_4]^{2-}$
(b) Cu^{2+} get reduced to form $[Cu\ (CN)_4]^{4-}$
(c) $Cu(CN)_2$
(d) CuCN

98. A blue colouration is not obtained when
(a) Ammonium hydroxide dissolves in copper sulphate
(b) Copper sulphate solution reacts with $K_4[Fe(CN)_6]$
(c) Ferric chloride reacts with sodium ferrocyanide
(d) Ferrous sulphate reacts with $K_3\ [Fe(CN)_6]$

99. An aqueous solution of colourless metal sulphate of M gives a white ppt with NH_4OH. This was soluble in excess of NH_4OH on passing H_2S through this solution a white ppt is formed the metal M in the salt is
(a) Ca (b) Ba (c) Al (d) Zn

100. A mixture of chlorides of copper, cadmium, chromium, iron and aluminium was dissolved in water acidified with HCl, and H_2S gas was passed for sufficient time. It was filtered, boiled and a few drops of nitric acid were added while boiling. To this solution ammonium chloride and sodium hydroxide were added in excess and filtered. The filtrate shall give test for
(a) Sodium and iron ion
(b) Sodium, chromium, aluminium ions
(c) Aluminium and iron ion
(d) Sodium, iron, cadmium and aluminium ion.

101. A pale yellow crystalline solid insoluble in water but soluble in CS_2 is allowed to react with nitric oxide to give X and Y. X is a colourless gas with pungent odour. X is further to react in aqueous medium with nitric oxide to yield Z and T compounds X, Z and T are
(a) SO_3, H_2SO_3, N_2O
(b) SO_2, H_2SO_4, N_2O
(c) SO_2, H_2SO_4, N_2
(d) SO_3, H_2SO_3, N_2

102. A mixture of two salts is not water soluble but dissolves completely in hot dil HCl to form colourless solution. The mixture could be
(a) $AgNO_3$ and KBr
(b) $BaCO_3$ and ZnS
(c) $FeCl_3$ and $CaCO_3$
(d) $Mn(NO_3)_2$ and $MgSO_4$

103. A cation M^{+x} forms violet colouration $[M(H_2O)_6]^{+x}$ in aqueous medium
M^{+x} + NH_3 solution $\rightarrow$ grey, blue gelationous precipitate soluble in excess of precipitant
$M^{+x} + S_2O_8{}^{-2} \xrightarrow{H^+}$ yellow solution in presence of one drop of dilute $AgNO_3$ which turns blue
When the solution is acidified and 3 ml of ether added followed by the addition of H_2O_2 is done. M^{+x} in the above analysis
(a) Fe^{+3} (b) Cr^{+3} (c) Al^{+3} (d) Cr^{+2}

104. On passing H_2S gas into the aqueous solution of an inorganic salt gives a black precipitate which dissolves in aquaregia. The solution is evaporated to dryness and residue is extracted with dilute HCl. When hot nitrite solution is added along with the excess of acetic acid into this solution, a yellow coloured precipitate is obtained. Hence the inorganic salt may be
(a) $NiCl_2$ (b) $Hg\ (NO_3)_2$
(c) $CoCl_2$ (d) $MnCl_2$

105. The best explanation for the solubility of MnS in dilute HCl is that
(a) Solubility product of $MnCl_2$ is less than that of MnS
(b) Concentration of Mn^{2+} is lowered by the formation of complex ions with chloride ions
(c) Concentration of S^{2-} ions is lowered by oxidation of free sulphur
(d) Concentration of S^{2-} ions is lowered by the formation of H_2S

106. Which of the following statements is wrong?
(a) Cu^{2+} salts give a borax bead test
(b) From a mixed precipitate of Ag*Cl* and AgI, ammonia solution dissolves only AgI
(c) Ferric ions give deep blue colouration on adding potassium ferrocyanide solution
(d) On boiling a solution having K^+, Ca^{2+} and HCO_3^- ions, we can get a precipitate of $K_2Ca\ (CO_3)_2$

107. How do we differentiate between Fe^{3+} and Cr^{3+} in group - III?

(a) By adding excess of NH_4OH solution
(b) By increasing NH^+_4 ion concentration
(c) By decreasing OH^- ion concentration
(d) Both b and c

108. When $KMnO_4$ solution is added to oxalic acid (hot) solution, the decolourization is slow in the begining but becomes very fast after few minutes. This is because
(a) Mn^{2+} acts as autocatalyst
(b) CO_2 is formed
(c) Reaction is exothermic
(d) Still reason is not known

109. NO_3^- ion in presence of nitrite cannot be identified by brown ring test. Nitrate is decomposed and brown ring test performed for identification of nitrate. Which of the following preferred reagent for decomposition of NO_2^-?
(a) Urea (b) NH_4Cl
(c) $(NH_2)_2\,C = S$ (d) NH_2SO_3H

More than Answer Type Questions

1. The reagent NH_4Cl and NH_4OH will not precipitate
(a) Ca^{2+} (b) Al^{3+} (c) Bi^{3+} (d) Mg^{2+}

2. Mark the *incorrect* statement(s)
(a) Group I radicals are precipitated as chloride
(b) Group IV radicals are precipitated as sulphide
(c) Group V radicals are precipitated as hydroxide
(d) Group III radicals are precipitated as chloride

3. Salts 'A' and 'B' on reaction with dil H_2SO_4 liberate gases 'X' and 'Y' respectively. Both turn lime water milky and milkyness disappears when excess of gases are passed 'X' has pungent suffocating smell and turns $K_2Cr_2O_7$ (acidified) paper green whereas 'Y' is colourless, odourless gas. Salts 'A' and 'B' are
(a) Na_2CO_3 (b) Na_2SO_3
(c) Na_2S (d) $(COO)_2(NH_4)_2$

4. Cone. HNO_3 gives brown gas with.
(a) NO_3^- (b) Br^- (c) I^- (4) Cl^-

5. Which of the sulphates are insoluble in water?
(a) $BaSO_4$ (b) $PbSO_4$
(c) $Bi(SO_4)_3$ (d) $CuSO_4$

6. Which of the following cannot be used instead of NH_4Cl in group III analysis?
(a) NH_4NO_3 (b) $(NH_4)_2SO_4$
(c) $(NH_4)_2CO_3$ (d) NaCl

7. Which of the following are soluble in Na_2CO_3 solution?
(a) Al_2O_3 (b) $ZnSO_4$
(c) $FeSO_4$ (d) All of these

8. Which of the following is used to identify sulphide ions in solution?
(a) Sodium nitroprusside test
(b) H_2S test
(c) Lead acetate test
(d) Cadmium chloride test

9. Which of the following gives borax bead test?
(a) Copper salt (b) Nickel salt
(c) Cobalt salt (d) Aluminium salt.

10. Which of the following radicals gives positive test with $K_4[Fe(CN)_6]$?
(a) Zn^{2+} (b) Fe^{3+} (c) Cu^{2+} (d) Cd^{2+}

11. Which of the following pairs of reagent gives colloidal sulphur?
(a) H_2S, SO_2 (b) H_2S, HNO_3
(c) H_2S, As_2O_3 (d) None of these.

12. Which of the following statements(s) is/are correct?
(a) Prussian blue and turnbull's blue are identical in structure.
(b) NH_4Cl is a strong electrolyte which decreases the ionization of NH_4OH by common ion effect so as to precipitate less soluble hydroxides of Al^{3+}, Cr^{3+}, Fe^{3+}.
(c) $Fe(OH)_3$ and $Al(OH)_3$ can be separated by adding NaOH.
(d) A yellow precipitate is obtained on adding KI in aqueous solutions of Pb^{2+}, Ag^+.

13. In the group separation operation before precipitating out 3rd group metal ions as hydroxides it is necessary to boil the solution of the salt mixture with few drops of concentrated HNO_3. This is done to bring about one conversion but not other which are.
(a) Fe^{2+} to Fe^{3+} (b) Cr^{3+} to CrO_4^{2-}
(c) Mn^{2+} to MnO_4^- (d) Al^{3+} to AlO_2^-

14. Which of the following changes do not occur when a solution containing Mn^{2+} and Cr^{3+} is heated with NaOH and H_2O_2?
(a) Hydrated manganese dioxide and Na_2CrO_4 are formed.
(b) Mn $(OH)_2$ and Cr $(OH)_3$ are precipitated.
(c) Na_2MnO_4 and Na_2CrO_4 are formed.
(d) Na_2MnO_4 and Cr $(OH)_3$ are formed.

15. In the microcosmic salt bead test.
(a) First a colourless transparent bead of sodium metaphosphate is formed.
(b) Sodium metaphosphate then combines with metallic oxides to from coloured beads of orthophosphates
(c) Copper gives green coloured bead in oxidizing flame in hot condition while a blue coloured bead is formed in cold condition.
(d) Copper gives colourless bead in reducing flame in hot condition and red coloured bead in cold condition.

16. In the borax bead test:
(a) Sodium metaborate on heating gives a glassy bead of B_2O_3.
(b) Sodium tetraborate on heating gives a glassy bead of sodium metaborate and boric anhydride.

(c) When borax is heated with any transition metal salt then the glassy bead forms a coloured bead of transition metal metaborate.
(d) The metaborate of different metals have different colours in oxidizing and reducing flames.

17. Which of the following is correct statement(s)?
(a) Oxidation number of iron in brown ring complex is +1.
(b) Ring test is not reliable in presence of nitrite ion.
(c) Oxidation number of iron in brown ring complex is +2.
(d) Sulphate does not perform ring test.

18. The only cations present in a slightly acid solution are. Fe^{3+}, Zn^{2+} and Cu^{2+}. The reagent which when added in excess to this solution would identify and separate Fe^{3+} in one step is.
(a) 2MHCl (b) $6MNH_3$
(c) 6MNaOH (d) H_2S gas

19. Which of the following precipitate is obtained in the form of hydroxide when NH_4OH is added in presence of NH_4Cl?
(a) $Fe(OH)_3$ (b) $Al(OH)_3$
(c) $Zn(OH)_3$ (d) $Cr(OH)_3$

20. A gas 'X' is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white ppt. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas 'Y'. Identify 'X' and 'Y'.
(a) $X = CO_2$, (b) $Y = CO_2$
(c) $X = Cl_2$ (d) $Y = H_2$

21. $A(\text{Black}) \xrightarrow{H_2SO_4} B(\text{gas}) \xrightarrow{HNO_3} \underset{(\text{Whiteturbid})}{\text{Colloidal Sulphur}},$

$B + D \xrightarrow{HCl} E + H_2SO_4$; Identify A & E
(a) A = FeS (b) A = ZnS
(c) E = CuS (d) E = HgS

22. Which of the following statement(s) is (are) correct with reference to the ferrous and ferric ions?
(a) Fe^{3+} gives brown colour with potassium ferricyanide.
(b) Fe^{2+} gives blue precipitate with potassium ferricyanide.
(c) Fe^{3+} gives red colour with potassium thiocyanate.
(d) Fe^{2+} gives brown colour with ammonium thiocyanate.

23. Carbon dioxide (CO_2) when passed through lime water, it first turns milky and then becomes clean. The turbidity reappears if
(a) the clear solution is boiled
(b) more CO_2 gas is bubbled through it
(c) fresh lime water **is** added to it
(d) none of the above is correct

24. Which of the following mixtures cannot be separated bypassing H_2S through their solutions containing dilute HCl?
(a) Cu^{2+} and Sb^{3+} (b) Pb^{2+} and Cd^{2+}
(c) Pb^{2+} and Al^{3+} (d) Zn^{2+} and Mn^{2+}

25 When a mixture of NH_4Cl, NH_4OH and NaH_2PO_4 was added to a solution containing Mg^{2+}, a white precipitate (A) was formed. When A was heated strongly the residue B was obtained. A and B are
(a) $Mg(NH_4)PO_4$ (b) $Mg_2P_2O_7$
(c) $Mg_3(PO_4)_2$ (d) MgO

26. A solution of salt/salts in HCl when diluted with water turns milky. It indicates the presence of
(a) Sn salt (b) Bi salt
(c) Cr salt (d) Cu salt

27. Aqueous solution of KI is used to detect
(a) Hg^{2+} (b) Pb^{2+}
(c) Ag^+ (d) Ba^{2+}

28. Potassium chromate (K_2CrO_4) is used to identify
(a) Pb^{2+} (b) Ba^{2+}
(c) Ag^+ (d) Ca^{2+}

29. To a mixture of two salts when sodium hydroxide is added dropwise precipitate is formed but on adding excess of sodium hydroxide the amount of precipitate decreases which of the following pair of compounds present in the mixture.
(a) Magnesium chloride and Aluminium chloride
(b) Manganous sulphate and Zinc sulphate
(c) Copper sulphate and Zinc sulphate
(d) Ferrous sulphate and copper sulphate

30. Which of the following statements are correct when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?
(a) A deep red vapour is evolved
(b) The vapour when passed into NaOH solution gives a yellow solution of Na_2CrO_4
(c) Chlorine gas evolved
(d) Chromyl chloride is formed

31. Which of the following will be precipitates when H_2S gas is passed through their solutions at pH = 10.0?
(a) Ag^+ (b) Bi^{3+}
(c) Zn^{2+} (d) Ca^{2+}

32. Which of the following statements regarding chromyl chloride test are correct.
(a) Chromyl chloride test cannot be performed in the presence of chlorates.
(b) Bromides and iodides also give chromyl chloride test.
(c) Chromyl chloride test cannot be performed to hydrochloric acid that contain chloride ion
(d) Chromyl chloride test should be carried only in dry test tube.

33. Which of the following ions can be separated by using NH_4Cl and $NH_4\ OH$?
(a) Fe^{3+} and Cr^{3+} (b) Cr^{3+} and Co^{2+}
(c) Cr^{3+} and Al^{3+} (d) Al^{3+} and Ba^{2+}

34. Which of the following species will be decomposed on acidification?

(a) $[Ag(NH_3)_2]^+$ (b) $[Cu(NH_3)_4]^{2+}$

(c) $[Zn(OH)_4]^{2-}$ (d) $[Pb(OH)_4]^{2-}$

35. In which of the following cases will a violet colouration be observed?

(a) An alkaline solution of sodium nitroprusside is treated with a solution of Na_2S.

(b) A solution of sodium cobalt nitrate is treated with KCl.

(c) A solution of $Mn(NO_3)_2$ is treated with sodium bismuthate or red lead in the presence of concentrated HNO_3.

(d) A solution of sodium nitroprusside in aqueous NaOH is treated with Na_2SO_3.

36. Which of the following is/are correct?

(a) Lime water is used for testing carbonate radical

(b) Addition of Na_2CO_3 to an aqueous solution of Ba^{2+} and Mg^{2+} will precipitate both

(c) Ammonium thiocyanate forms coloured ppt. with Co^{2+}

(d) Ammonium sulphate can be used in place of ammonium chloride in the 3rd group

37. Which of the following statement is/are correct?

(a) HgS dissolves in aquaregia but does not dissolve in dil. HNO_3

(b) As_2S_3 dissolves in yellow ammonium sulphide but does not dissolve in dil. HCl

(c) Alkali and alkaline bicarbonates are solids while all other exist in solution form

(d) If nitrate ion is present, a light yellow precipitate will be formed in II group

38. In qualitative analysis of chromium (III) ion, it is oxidized to chromate and thus can be detected. Which of the following reagents can be used for oxidation of chromium (III) ion to chromate ion?

(a) Excess NaOH + 10% H_2O_2

(b) Excess of NaOH + $NaBiO_3$

(c) Br_2 water in alkaline solution

(d) $(NH_4)_2\ S_2O_8$ Or $K_2S_2O_8$

Comprehension Type Questions

Passage-I

A colourless salt solution on passing H_2S in basic medium gave a buff coloured precipitate, which dissolves in HCl, further treating with NaOH, gave a precipitate finally, which was brown in colour. The brown precipitate on heating strongly with concentrated nitric acid and Pb_3O_4 gave a pink coloured solution finally.

1. The buff coloured precipitate dissolves in HCl to form a colourless solution. The compound in solution is

(a) $Mn(OH)_2$ (b) $MnOCl_2$

(c) $MnCl_4$ (d) $MnCl_2$

2. Which of the following is *correct* statement?

(a) The buff coloured precipitate of MnS dissolves in HCl because $MnCl_2$ is formed which is soluble

(b) MnS dissolves in HCl because $MnCl_2$, and H_2S are formed and HCl supresses the dissociation of H_2S and thereby prevents $MnCl_2$, from reacting with H_2S.

(c) MnS dissolves in HCl because when $MnCl_2$ and H_2S are formed, H_2S escapes in air as H_2S is a gas while HCl is a liquid

(d) MnS dissolves in HCl forming a complex

3. The colourless solution obtained after dissolving MnS in HCl, on heating with NaOH gives MnO_2, which means Mn^{2+} got oxidized to Mn^{4+}. This shows that

(a) Na^+ is an oxidizing agent

(b) OH^- is an oxidizing agent

(c) $MnCl_2$ reacts with NaOH to form $Mn(OH)_2$ which in the presence of air gets oxidized to MnO_2, on heating.

(d) $Mn(OH)_2$ which is formed is unstable and gets converted to MnO_2 and H_2O.

Passage-2

A bluish green coloured compound 'A' on heating gives two products 'B' and 'C' A metal 'D' is deposited on passing H_2 through heated 'B'. The compound 'A' and 'B' are insoluble in water.'B' is black in colour, dissolves in HCl acid. On treatment with $K_4[Fe(CN)_6]$ gives a chocolate brown ppt of compound 'E'. 'C' is colourless, odourless gas and turns lime water milky.

1. Compound 'A' is

(a) $CuSO_4$ (b) $CuCO_3$

(c) $FeSO_4$ (d) $CrCl_3$

2. The compounds 'B' and 'C' are respectively.

(a) CuS, SO_2 (b) CuO, CO_2

(c) FeO, H_2S (d) Cr_2O_3, CO

3. The compounds 'D' and 'E' are respectively.

(a) Cu,Cu_2 $[Fe(CN)_6]$ (b) Fe,$Cu_2[Fe(CN)_6]$

(c) Cu,$CuCO_3$ (d) Zn, CuO.

Passage-3

Identify the following compounds A, B, C, D:

$$Na_2CO_3 \xrightarrow[H_2O]{SO_2} A \xrightarrow{Na_2CO_3} B$$

$$B \xrightarrow[S]{\Delta} C \xrightarrow{I_2} D$$

1. When Na_2CO_3 is reacted with SO_2, the reaction suggests.
   (a) SO_2 a powerful oxidizing agent.
   (b) SO_2 more acidic than CO_2
   (c) SO_2 a powerful reducing agent than CO_2
   (d) SO_2 is less volatile than CO_2
2. The compound C in the series is
   (a) Na_2SO_3 (b) $Na_2S_2O_3$
   (c) Na_2SO_4 (d) Na_2CO_3
3. The reaction sequence from C to D suggests
   (a) Sulphur oxidized from + 4 to +5 state
   (b) Sulphur oxidized from +2 to +2.5 state
   (c) I_2 is a reducing agent.
   (d) Sulphur is reduced from + 4 to + 3 state.

Passage-4

4.96 g of a hydrated salt (X) on heating gives 3.16 g of the anhydrous salt (Y). Also (X) decolourizes I_2, in KI and gives white turbidity with dil. HCl.(X) also dissolves unreacted AgBr.

1. The molecular weight of (X) is.
   (a) 158 (b) 248 (c) 160 (d) 200
2. The water molecules of crystallization in (X) is.
   (a) 2 (b) 3 (c) 4 (d) 5
3. Compound formed when (X) is reacted with I_2 is.
   (a) $Na_3[Ag(S_2O_3)_2]$ (b) $Na_2S_2O_7$
   (c) $Na_2S_2O_3$ (d) Na_2SO_3

Passage-5

A mixture of two salts with NaOH liberated a gas which gives a reddish brown precipitate with an alkaline solution of K_2HgI_4. The aqueous solution of mixture on treatment with $BaCl_2$ gives a white precipitate which is sparingly soluble in conc. HCl. On heating the mixture with $K_2Cr_2O_7$ and conc. H_2,SO_4, red vapours (A) are produced. The aqueous solution of mixture gives a deep blue colouration (B) with potassium ferricyanide.

1. The salt mixture contains.
   (a) SO_4^{2-} and as Br^- as acid radicals.
   (b) SO_4^{2-} and Cl^- as acid radicals.
   (c) Br^- and Cl^- as acid radicals.
   (d) SO_4^{2-} and SO_3^{2-} as acid radicals.
2. The salt mixture contains.
   (a) Fe^{2+} and NH_4^+ as basic radicals
   (b) Fe^{3+} and NH_4^+ as basic radicals.
   (c) Fe^{2+} and Na^+ as basic radicals.
   (d) Fe^{2+} and Mg^{2+} as basic radicals.
3. This mixture containing one mole of each component will consume
   (a) 0.2 moles of $KMnO_4$ in acidic medium
   (b) 0.6 moles of $KMnO_4$ in acidic medium
   (c) 0.3 moles of $KMnO_4$ in acidic medium
   (d) 0.4 moles of $KMnO_4$in acidic medium.

Passage-6

A white compound (P) reacts with dilute H,SO_4 to produce a colourless gas (Q) and a colourless solution (R). The reaction between (Q) and acidified $K_2Cr_2O_7$ solution produces a green solution and a slightly coloured precipitate (S). The substance (S) burns in air to produce a gas (T) which reacts with (Q) to yield (S) and a colourless liquid. Anhydrous copper sulphate is turned blue on addition of this colourless liquid. Addition of aqueous NH_3 or NaOH to (R) produces first a precipitate, which dissolves in the excess of respective reagent to produce a clear solution in each case.

1. The salts 'P' and 'R' are.
   (a) $ZnCl_2$ and $ZnSO_4$ respectively.
   (b) ZnS and $ZnSO_3$ respectively.
   (c) ZnS and $ZnSO_4$ respectively.
   (d) ZnO and $ZnSO_4$ respectively.
2. The solid 'S' and gas 'T' are.
   (a) $ZnCl_2$, and SO_2, respectively.
   (b) S and SO_2 respectively.
   (c) Zn $(OH)_2$ and SO_2, respectively.
   (d) ZnO and O_2, respectively.
3. The clear solution formed when (R) is treated with excess of NH_4OH is.
   (a) $Zn(OH)_2$ (b) $[Zn(OH)_4H_2O]^{2-}$
   (c) $[Zn(NH_3)_4]^{2+}$ (d) $[Zn(NH_3)_2]^{2+}$

Passage-7

A series of chemical reactions was carried out to study the chemistry of lead. Reaction 1: Initially, 15.0 mL of 0.300M $Pb(NO_3)_{2(aq)}$ was mixed with 15.0mL of 0.300 M $Na_2SO_{4(aq)}$. All the $Pb(NO_3)_2$ reacted to form compound A was removed by filtration. Reaction 2: Next 15.0 mL of 3.00 M $KI_{(aq)}$ was added to Compound A. The mixture was agitated and some of compound A dissolved. In addition, a yellow precipitate of $Pbl_{2(s)}$ was formed

Reaction 3: The $Pbl_{2(s)}$ was separated and mixed with 15.0 mL of 0.300 M $Na_2CO_{3(aq)}$. A white precipitate of $PbCO_{3(S)}$ formed. All of the $Pbl_{2(s)}$ was converted into $PbCO_{3(S)}$.

Reaction 4: The $PbCO_{3(s)}$ was removed by filtration and a small sample gave off a gas when treated with dilute HCl.

1. Which of the following reaction depicts the formation of the gas in Reaction 4?
   (a) $PbCO_{3(S)} + 2HCl_{(aq)} \rightarrow PbCl_{2(s)} + CO_{2(g)} + H_2O_{3(l)}$
   (b) $Na_2CO_{3(aq)} + 2HCl_{(aq)} \rightarrow 2NaCl_{[aq)} + CO_{2(g)}\ H_2O(l)$
   (c) $PbCO_{3(s)} + 2HCl_{(aq)} \rightarrow PbCl_{2\{s)} + Cl_{2(g)} + H_2O(l)$
   (d) $Pb_2l_{2(s)} + HCl_{(aq)} \rightarrow PbCl_{2(s)} + HI_{(s)}$

2. The Compound A is
(a) $Pb(NO_3)_2$ (b) PbI_2
(c) $NaNO_3$ (d) $PbSO_4$

3. $Pb(OH)_{2(s)}$ is slightly soluble in water, how would the amount of $Pb(OH)_{2(S)}$ that normally dissolves in 1 L of water be affected if the pH were 9.0?
(a) Less would dissolve
(b) The same amount would dissolve
(c) More would dissolve
(d) There is no way to predict the effect of the change in pH of the water

Passage-8

An unknown mixture contains one or two of the following: $CaCO_3$, $BaCl_2$, $AgNO_3$, Na_2SO_4, $ZnSO_4$ and NaOH. The mixture is completely soluble in water and solution gives pink colour with phenolphthalein. When dilute hydrochloric acid is gradually added to the solution, a precipitate is formed which dissolved with further addition of the acid

1. Which of the following combination of compounds is soluble in water?
(a) $BaCl_2$ and $AgNO_3$
(b) $AgNO_3$ and NaOH
(c) $BaCl_2$ and Na_2SO_4
(d) $ZnSO_4$ and excess NaOH

2. The aqueous solution of mixture gives white precipitate with dil. HCl which dissolves in excess of dil. HCl. It confirms
(a) $BaCl_2$ + NaOH (b) Na_2SO_4 + NaOH
(c) $ZnSO_4$ + NaOH (d) $AgNO_3$ + NaOH

3. The white precipitate is:
(a) $ZnSO_4$ (b) Na_2ZnO_2
(c) $Zn(OH)_2$ (d) $ZnCl_2$

Passage-9

A halide of a metal is represented as MX. The halide MX on reaction with conc. H_2SO_4. gave a colourless gas (Y) which gives dense white fumes when a rod of NH_3 is taken near to it and also a salt solution (Z)

(Z) on heating with $CaCO_3$ and carbon give compound (A), gas (P) and CaS. When aqueous solution of salt (A) is made to react with CO_2 gas, a compound (B) is formed which on heating gave salt (A). The salt MX gave lilac colour to the flame when flame test was performed.

1. When compound (B) is heated then.
(a) compound (Y) is formed
(b) compound (A) is formed
(c) compound (Z) is formed
(d) compound (MX) is formed

2. When (Z) is heated with $CaCO_3$ and carbon, then apart from compound (A)
(a) CO_2 and $MgCO_3$ are formed
(b) CO_2 and $MgCO_3$ are formed
(c) CO_2 and CaS are formed
(d) CO_2 and K_2S are formed

3. The compound (MX) is
(a) KCl (b) K_2SO_4 (c) KBr (d) K_2O

Passage-10

A coloured solution known to contain two metal ions was treated with excess cold sodium hydroxide solution. When filtered a whitish solid slowly changing to brown was retained on the filter paper and a colourless solution collected as the filtrate. Drop wise addition of hydrochloric acid to the filtrate produced a white precipitate, which dissolved in excess acid. Treatment of the residue from the filter paper with a solution of a strong oxidizer produced a reddish violet solution.

1. Indicate only pairs of ions which on testing as above leads to the observed changes
(a) Zn^{2+} and Mn^{2+} ions
(b) Mg^{2+} and Zn^{2+} ions
(c) Mn^{2+} and Mg^{2+} ions
(d) Fe^{2+} and Zn^{2+} ions

2. Filtrate obtained after separation of white solid contains
(a) ZnO (b) Na_2ZnO_2
(c) MnO (d) Na_2MnO_2

3. White solid changing to brown is due to formation of
(a) $Mn(OH)_3$ (b) MgO
(c) $Zn(OH)_2$ (d) $MnO(OH)_2$

4. Reddish violet solution obtained due to formation of
(a) ZnO_2^{2-} (b) MnO_4^{2-}
(c) MnO_4^- (d) MnO_2

5. Reddish violet solution is decolourized by
(a) $SO_3^{2-}, C_2O_4^{2-}, Fe^{2+}$ (b) $SO_3^{2-}, HCO_3^-, Fe^{3+}$
(c) $NO_2^-, SO_4^{2-}, Fe^{2+}$ (d) $H_2O_2, C_2O_4^{2-}, CO_3^{2-}$

Passage-11

An aqueous solution of a salt (A) gives a white crystalline precipitate (B) with NaCl solution. The filtrate gives a black precipitate (C) when H_2S is passed into it. Compound (B) dissolves in hot water and the solution gives yellow precipitate (D) on treatment with sodium iodide and cooling. The compound (A) does not give any gas with dilute HCl but liberates a reddish brown gas on heating. On the basis of above work-up answer the following questions:

1. Compound B is
   (a) AgCl (b) $PbCI_2$ (c) BiOCl (d) SbOCl
2. Compound A is
   (a) $PbNO_3$ (b) $PbNO_2$ (c) $AgNO_3$ (d) $AgNO_2$
3. Compound C is
   (a) Ag_2S (b) PbS (c) Bi_2S_3 (d) Sb_2S_3

Passage-12

A water soluble salt 'A' on strong heating gives a reddish brown coloured gas and a colourless gas leaving a grey residue. On addition of dilute HCl to an aqueous solution of A, a white precipitate 'B' is formed. 'B' is insoluble in both cold & hot water. The precipitate 'B' when treated with NH_4OH forms a black residue 'C'. 'C' is soluble in aquaregia and forms a compound 'D' with formation of NH_4Cl. When KI is added drop wise to a solution of 'D' a red precipitate 'E' is formed which is soluble in excess KI due to the formation of a soluble compound 'F'

1. The compound 'D' is
   (a) Hg_2Cl_2 (b) $HgCl_2$ (c) AgCl (d) $PbCl_2$
2. When alkaline solution of 'F' is added to a solution of $NH^+{}_4$ ions
   (a) A white precipitate is formed
   (b) A brown precipitate is formed
   (c) White precipitate is formed and dissolves in excess 'F'
   (d) Brown precipitate is formed and dissolves in excess 'F'
3. When 0.01 moles of 'E' is dissolved in a solution containing 0.02 moles of KI in l00g, of H_2O, the freezing point depression of the solution is: w.r.t water (K_f of H_2O = 1.86 K. kg $mole^{-1}$)
   (a) 0.186 K (b) 0.558K
   (c) 0.372 K (d) 0.744

Passage-13

Compound A $\xrightarrow{\Delta}$ Initially swelled $\xrightarrow[\text{strong heating}]{\Delta}$ Amorphous powder $\longrightarrow$ Lilac flame in the flame test

Compound A (aq. solution) $\xrightarrow{\text{excess NaOH}}$ 'B' (No change in colour) $\xrightarrow{H_2O_2}$ C (Yellow solutions)

1. Compound 'A' is having water of crystallization by the number of
   (a) 6 (b) 12 (c) 24 (d) 36
2. The metal ion in compound B is having oxidation state of
   (a) Zero (b) II (c) III (d) IV
3. The hybridization of the central atom in compound C is
   (a) sp^3 (b) sp^3d (c) d^2sp^3 (d) d^3s

Passage-14

A white solid (A) reacts with dilute H_2SO_4 to produce a colourless gas (B) and a colourless solution (C). The reaction between (B) and acidified dichromate yield a green solution and a slightly coloured precipitate (D). The substance (D) when burnt in air gives a gas (E) which reacts with (B) to yield (D) and a colourless liquid. Anhydrous copper sulphate turns blue with this colourless liquid. The addition of aqueous NH_3, or NaOH to (C) produces a precipitate that dissolves in an excess of the reagent to form a clear solution

1. Which of the following gases are (B) and (E) respectively?
   (a) CO_2 and SO_2
   (b) SO_2 and H_2S
   (c) H_2S and SO_2
   (d) CO_2 and H_2S
2. What would appear if the gas (B) is passed through an aqueous solution of Pb $(NO_3)_2$?
   (a) White precipitate soluble in hot dil HNO_3
   (b) A black precipitate soluble in hot dil HNO_3
   (c) A black precipitate insoluble in hot dil HNO_3
   (d) A yellow precipitate soluble in hot conc HNO_3
3. Suppose the solution obtained by the treatment of the solution (C) with an excess of NaOH is acidified with acetic acid and the gas B is passed through it which of the following will be obtained?
   (a) Colourless solution (b) Yellow precipitate
   (c) Black precipitate (d) White precipitate

Passage-15

A white solid (A) reacts with dil H_2SO_4 to produce a colourless gas (B) and a colourless solution (C). The reaction between (B) and acidified dichromate yields a green solution and a slightly coloured precipitate (D). The substance (D) when burnt in air gives a gas (E) which reacts with (B) to yield (D) and a colourless liquid. Anhydrous copper sulphate turns blue with this colourless liquid. The addition of aqueous NH_3 or NaOH solution to (C) produces a precipitate that dissolves in an excess of both the reagents to form a clear solution

1. Excess of precipitate (D) is added to hot NaOH solution then the main product that will be formed in the reaction is
   (a) Na_2S (b) Na_2S_5 (c) $Na_2S_4O_6$ (d) SO_2
2. When NH_4OH is added to the solution (C) first a white precipitate is obtained which dissolves in excess of

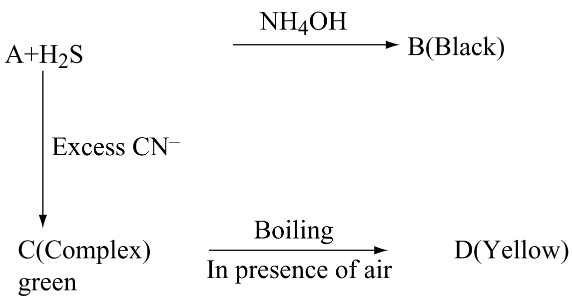
reagent due to the formation of (X) hence, compound (X) is

(a) $NaA1O_2$ (b) $(NH_4)_2[Zn(SO)_4]$
(c) $Zn(OH)_2$ (d) $[Zn(NH_3)_4]SO_4$

3. If 3 moles of gas (E) are mixed with 4 moles of gas (B), then the no. of moles of (D) formed will be: (assume 100% efficiency of the reaction)

(a) 6 (b) 0.75 (c) 4 (d) 7.5

Passage-16



1. Compound B is

(a) NiS (b) FeS
(c) CoS (d) PbS

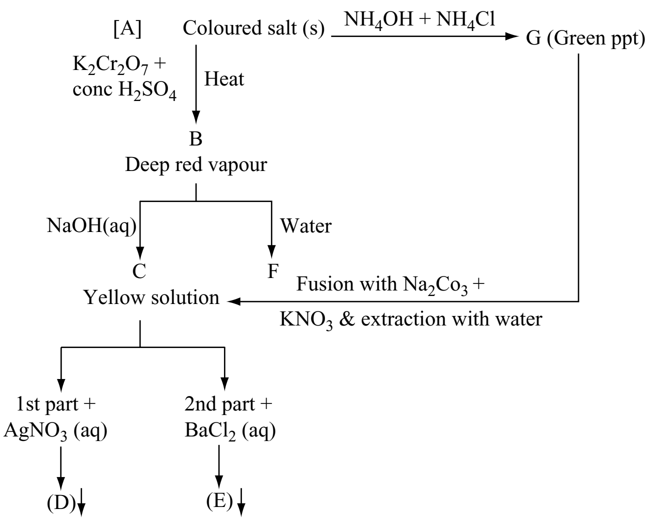
2. The compound C is

(a) $[Co(CN)_5]^{-3}$ (b) $[Co(CN)_6]^{-4}$
(c) $[Fe(CN)_6]^{-4}$ (d) $[Ni\ (CN)_4]^{2-}$

3. The hybridization of central metal atom in D is

(a) dsp^2 (b) sp^2 (c) sp^3d^2 (d) d^2sp^3

Passage-17



1. The colour of the precipitates (D) and (E) are

(a) White and yellow
(b) Yellow
(c) Brick red and yellow
(d) Yellow and brick red

2. The compound (A) is

(a) $FeCl_3$ (b) H_2CrO_4
(c) $CrCl_3$ (d) $AlCl_3$

3. $[A](S) + MnO_2 + H_2SO_4(conc) \longrightarrow [x]$ greenish yellow gas.

Select the correct choice (x)

(a) It gives yellow precipitate with $AgNO_3$
(b) It liberates I_2 form KI solution
(c) It turns starch paper orange red
(d) It turns titan yellow solution red

Passage-18

A colourless solid (A) on hydrolysis produces a heavy white precipitate. Compound (A) gives a clear solution in concentrated HCl, however, when added to large amount of water, it gives precipitate of (B), when H_2S is passed through a suspension of (A) or (B), brown black ppt of (C) is obtained. Compound (A) liberates a gas (D) on treating with H_2SO_4, the gas (D) is water soluble and gives white ppt (E) with solution of mercurous salt but not with mercuric salt.

1. Compound (A) would be:

(a) $SnCl_2$ (b) AgCl (c) $BiCl_3$ (d) $PbCl_2$

2. The gas (D) must be

(a) H_2S (b) HCl (c) Cl_2 (d) NH_3

3. White ppt (E) is

(a) $HgCl_2$ (b) Hg_2Cl_2 (c) Hg (d) $Hg(OH)_2$

Matching Type Questions

1. Column-I	**Column-II**
(a) $AlCl_3$	(p) Gives white ppt. with Na_2HPO_4 in presence of NH_4Cl and NH_4OH
(b) $FeSO_4$	(q) Gives reddish brown ppt. with NH_4OH
(c) $FeCl_3$	(r) Gives white ppt. with NH_4OH which dissolves in excess of NaOH
(d) $MgCl_2$	(s) Gives dirty green ppt. with NH_4OH as well as with NaOH

2. Column-I	**Column-II**
(a) $Na_2C_2O_4 + CaCl_2 \longrightarrow$	(p) Brown ring
(b) $H_2SO_4 + BaCl_2 \longrightarrow$	(q) Bromide.
(c) Cl_2 water gives brown layer to chloroform	(r) White precipitate.
(d) $FeSO_4 + NO \longrightarrow$	(s) Sulphate ion.

3. **Column-I**	**Column-II**
(a) Yellow ppt. of $PbCrO_4$	(p) Blue or green colour to flame.
(b) Sodium nitroprusside.	(q) Brown ring test.
(c) Nitrate ions	(r) Chromyl chloride test.
(d) Cu salt and BO_3^{3-} ion.	(s) Violet colour with Na_2S.

4. **Column-I**	**Column-II**
(a) HCl	(p) common ion effect
(b) NH_4Cl	(q) decrease ionization of H_2S
(c) NH_4OH	(r) increase ionization H_2S
(d) HNO_3	(s) oxidize Fe^{2+} to Fe^{3+}
	(t) Suppress ionization of NH_4OH

5. Match the test (List-I) with the reagents (List-II)

List-I	**List-II**
(a) Chromyl chloride test	(p) Nessler's reagent
(b) Prussian blue test	(q) $K_2Cr_2O_7$
(c) Chocolate brown test	(r) NaOH
(d) NH_4Cl	(s) $K_4[Fe(CN)_6]$
	(t) Copper chloride

6. Match the following Column-I with Column-II

Column-I	**Column-II**
(a) $FeCl_3(aq) + H_2S(g) \longrightarrow$	(p) Colourless solution is obtained
(b) $SnCl_2(aq) + NaOH(aq) \longrightarrow$	(q) White precipitate is formed
(c) $BiCl_3(s) + H_2O(l) \longrightarrow$	(r) Green Coloured solution is obtained
(d) $CaCl_2 + NaHCO_3 \xrightarrow{\Delta}$	(s) None-redox reaction
	(t) Colourless gas is evolved

7. Match the following Column-I with Column-II

Column-I (Inorganic ions)	**Column-II (can be tested using reagent)**
(a) Co^{2+}	(p) $K_4[Fe(CN)_6]$
(b) Cu^{2+}	(q) KSCN
(c) Fe^{3+}	(r) $K_3[Fe(CN)_6]$
(d) Zn^{2+}	(s) $KNO_2 + CH_3CO_2H$
	(t) $K_2[Hg(SCN)_4]$

8. Match the following Column-I with Column-II

Column-I	**Column-II**
(a) SO_2	(p) Turns acidified $K_2Cr_2O_7$ solution green
(b) CO_2	(q) Turns blue litmus red
(c) H_2S	(r) Turns lime water milky
d) H_2O_2	(s) Used for restoring old paintings in the museums

9. Match the pair of salts given in Column-I with the reagents used to distinguish them given in Column-II

Column-I	**Column-II**
(a) $FeSO_4$ & $FeCl_3$	(p) Sodium hydroxide
(b) $MnSO_4$ & $ZnSO_4$	(q) Ammonium hydroxide
(c) $CuSO_4$ & $FeSO_4$	(r) Potassium ferrocyanide
(d) $Al_2(SO_4)_3$ & $ZnSO_4$	(s) Lead dioxide and Conc. HNO_3

10. **List-I**	**List-II**
(a) ZnO	(p) Yellow when cold and yellowish brown when hot
(b) SnO_2, Bi_2O_3	(q) Yellow coloured
(c) Fe_2O_3	(r) White when cold and yellow when hot
(d) PbO	(s) Brown when cold and black/red when hot

11. Match the reagent given in List-II used for the detection of cation in List-I

List-I	**List-II**
(a) Pb^{2+}	(p) $K_2Cr_2O_7$
(b) Ba^{2+}	(q) $(NH_4)_2SO_4$
(c) Cu^{2+}	(r) $K_4[Fe(CN)_6]$
(d) Zn^{2+}	(s) NH_4OH
(e) Fe^{2+}	(t) H_2S

12. **Column-I**	**Column-II**
(a) Precipitate with KCN which is soluble in excess of reagent	(p) Fe^{2+}
(b) Precipitate with NaOH and NH_4OH which is soluble in excess of NaOH	(q) Hg^{2+}
(c) Coloured ppt with KI which is soluble in excess of reagent	(r) Pb^{2+}
(d) Black ppt with H_2S which is soluble in hot and dil HNO_3	(s) Ag^+

13. Column-I	Column-II
(a) Soluble in a concentrated NH_3 solution	(p) Ag_2S
(b) Soluble in excess KCN solution	(q) $Cu(OH)_2$
(c) Soluble in excess hypo solution	(r) AgBr
(d) Soluble in concentrated HCl	(s) AgCl

14. Column-I	Column-II
(a) Amphoteric metal oxide	(p) Pb
(b) Metal acetate $\xrightarrow{\Delta}$ acetone + metal carbonate	(q) Zn
(c) Metal carbonate $\xrightarrow{\Delta}$ metal oxide + CO_2	(r) Na
(d) Metal nitrate $\xrightarrow{\Delta}$ metal oxide + NO_2 + O_2	(s) Li

15. Column-I	Column-II
(a) S^{2-}	(p) white ppt with $AgNO_3$
(b) NO^{-}_2	(q) Evolution of pungent smell gas with Al + NaOH
(c) SO_3^{2-}	(r) Brown fumes with conc H_2SO_4
(d) CH_3COO^-	(s) Decolourizes acidified $KMnO_4$

16. Match the reactions in Column-I with their corresponding products in Column-II

Column-I (Reactions)	Column-II (Products)
(a) $NH_3 + NaOCl \xrightarrow[\text{glue}]{\Delta}$	(p) N_2H_4
(b) $NH_4NO_2 + NaOH \rightarrow$	(q) N_2
(c) $NH_4NO_3 + Ca(OH)_2 \rightarrow$	(r) NH_3
(d) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$	(s) H_2O

17. Matching type questions

Column-I	Column-II
(a) Mn^{2+}	(p) Flame test
(b) Pb^{2+}	(q) With aq. NH_3 gives white precipitate
(c) Ag^+	(r) With neutral solution of disodium hydrogen phosphate gives yellow precipitate
(d) Cu^{2+}	(s) Black precipitate with KCNS which slowly changes to white
	(t) Gives ppt. with dil. HCl

18. Match the following Column-I with Column-II

Column-I	Column-II (Colour of product)
(a) $AgNO_3$ test using NaCl / KI	(P) White
(b) $Hg_2I_2 \xrightarrow{\text{Water}}$	(q) Yellow
(c) $PbCl_2$ using K_2CrO_4/KI	(r) Red
(d) ZnO heated with cobalt nitrate	(s) Black
	(t) Green

19. Match the following column-I Column-II

Column-I	Column-II
(a) $CuSO_4$	(p) Brown precipitate with $K_4[Fe(CN)_6]$
(b) NaBr	(q) Colourless gas evolved with dil. H_2SO_4
(c) KNO_2	(r) Gives yellow turbidity with dil. HCl
(d) CaS_2O_3	(s) Gives brown fumes with conc H_2SO_4

Numerical Type Questions

1. A group II cation of inorganic cation radical analysis can oxidize CN^- as well as I^- in aqueous solution. It combine with excess of hypo to form a complex. Find the number of ions produced by each formula unit of complex.
2. Iodate ions (IO_3^-) can be reduced to iodine by iodide ions. How many moles of iodine are produced for every mole of iodate ions consumed in the reaction.

Single Answer Questions

1. c	**2.** d	**3.** a	**4.** a	**5.** b	**6.** b
7. c	**8.** a	**9.** c	**10.** b	**11.** a	**12.** a
13. a	**14.** c	**15.** b	**16.** c	**17.** b	**18.** b
19. a	**20.** b	**21.** a	**22.** d	**23.** d	**24.** a
25. a	**26.** a	**27.** c	**28.** b	**29.** b	**30.** b
31. b	**32.** b	**33.** c	**34.** d	**35.** d	**36.** d
37. d	**38.** a	**39.** b	**40.** c	**41.** a	**42.** b
43. c	**44.** d	**45.** d	**46.** a	**47.** a	**48.** a
49. c	**50.** c	**51.** a	**52.** a	**53.** c	**54.** b
55. a	**56.** c	**57.** b	**58.** a	**59.** c	**60.** d
61. b	**62.** d	**63.** d	**64.** d	**65.** d	**66.** b
67. c	**68.** c	**69.** c	**70.** a	**71.** d	**72.** b
73. a	**74.** b	**75.** d	**76.** b	**77.** b	**78.** c
79. d	**80.** a	**81.** a	**82.** a	**83.** b	**84.** c
85. a	**86.** b	**87.** c	**88.** d	**89.** b	**90.** d
91. c	**92.** c	**93.** c	**94.** b	**95.** d	**96.** a
97. d	**98.** b	**99.** d	**100.** b	**101.** b	**102.** b
103. b	**104.** c	**105.** d	**106.** b	**107.** c	**108.** a
109. d					

More than One Answer

1. a, d	**2.** c, d	**3.** a, b	**4.** b, c
5. a, b	**6.** b, c, d	**7.** a	**8.** a, b, c, d
9. a, b, c	**10.** a, b, c	**11.** a, b	**12.** a, b, c, d
13. a, b, c, d	**14.** b, c, d	**15.** a, b, c, d	**16.** b, c, d
17. a, b	**18.** b	**19.** a, b, d	**20.** c, d
21. a, b, c	**22.** a, b, c	**23.** a, c	**24.** a, b, d
25. a, b	**26.** a, b	**27.** a, b, c	**28.** a, b, c
29. a, b, c	**30.** a, b, d	**31.** a, b, c	**32.** a, c, d
33. b, d	**34.** a, b, c, d	**35.** a, c	**36.** a, b, c
37. a, b, d	**38.** a, b, c, d		

Comprehension Type Questions

Passage 1	**1.** d	**2.** b	**3.** c		
Passage 2	**1.** b	**2.** b	**3.** a		
Passage 3	**1.** d	**2.** b	**3.** b		
Passage 4	**1.** b	**2.** d	**3.** b		
Passage 5	**1.** b	**2.** a	**3.** a		
Passage 6	**1.** c	**2.** b	**3.** c		
Passage 7	**1.** a	**2.** d	**3.** a		
Passage 8	**1.** d	**2.** c	**3.** c		
Passage 9	**1.** b	**2.** c	**3.** a		
Passage 10	**1.** a	**2.** b	**3.** d	**4.** c	**5.** a
Passage 11	**1.** b	**2.** a	**3.** b		
Passage 12	**1.** b	**2.** b	**3.** b		
Passage 13	**1.** c	**2.** c	**3.** d		
Passage 14	**1.** c	**2.** b	**3.** d		
Passage 15	**1.** b	**2.** d	**3.** b		
Passage 16	**1.** a	**2.** d	**3.** c		
Passage 17	**1.** c	**2.** c	**3.** b		
Passage 18	**1.** c	**2.** b	**3.** b		

Match the Type Questions

1.	a-r	b-s	c-q	d-p	
2.	a-r	b-r, s	c-q	d-p	
3.	a-r	b-s	c-q	d-p	
4.	a-p, q	b-p, t	c-r	d-q, s	
5.	a-q, r, t	b-s	c-s, t	d-p, r	
6.	a-q, r	b-p, s	c-q, s	d-q, s, t	
7.	a-q, s, t	b-p	c-q	d-p	
8.	a-p, q, r	b-q, r	c-p, q	d-p, s	
9.	a-p, q, r	b-p, q, r, s	c-p, q, r	d-q, r	
10.	a-r	b-p	c-s	d-q	
11.	a-p, q, s, t	b-p, q	c-r, s, t	d-r, s, t	e-p, s
12.	a-p, s	b-r	c-q, r	d-r, s	
13.	a-q, r, s	b-p, q, r, s	c-q, r, s	d-q, s	
14.	a-p, q	b-r	c-p, q, s	d-p, q, s	
15.	a-s	b-q, r, s	c-p, s	d-p	
16.	a-p	b-r, s	c-r, s	d-q, s	
17.	a-p, q	b-q, t	c-r, t	d-p, s	
18.	a-p, q	b-r, s	c-q	d-t	
19.	a-p,	b-q, s	c-q, s	d-q, r	

Numerical Type Questions

1. 5 **2.** 5

Analytical Chemistry Hints

Single Answer Questions

1. $2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$

$2KI + H_2SO_4 \longrightarrow K_2SO_4 + 2HI$

$2HI + H_2SO_4 \longrightarrow 2H_2O + SO_2 + I_2$

6. NH_4CNS gives blood red colour with ferric salts but not with ferrous salts

7. $Na_2O_2 + 2H_2O \longrightarrow 2NaOH + H_2O_2$

$$2FeSO_4 + Na_2O_2 + 2NaOH + 2H_2O \longrightarrow \underset{\text{Brown}}{2Fe(OH)_3} + Na_2SO_4$$

$$Al_2(SO_4)_3 + 8NaOH \longrightarrow 2NaAlO_2 + 3Na_2SO_4 + 4H_2O$$

$$Cr_2(SO_4)_3 + 10NaOH + 3H_2O_2 \longrightarrow 2Na_2CrO_4 + 3Na_2SO_4 + 8H_2O$$

$Fe(OH)_3$ is brown residue $NaAlO_2$ is colourless solution and Na_2CrO_4 is yellow solution

9. $H_2O + Cl_2 \to HCl + HOCl$

$Mg + 2HCl \to MgCl_2 + H_2(s)$

11. Conc H_2SO_4 oxidizes Br^- ion to orange coloured Br_2 gas which interfere with brown ring test

13. Red lead (Pb_3O_4) oxidizes Mn^{2+} to purple $HMnO_4$

15. $H_2SO_4 + H_2S \longrightarrow 2H_2O + SO_2 + S$

19. $NH_4NO_3 + NaOH \xrightarrow{\Delta} NaNO_3 + NH_3 + H_2O$

$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

20. $BaCl_2 + K_2CrO_4 \longrightarrow BaCrO_4 + 2KCl$

22. Green precipitate is $Cr(OH)_3 \cdot Fe^{2+}$ will be oxidized to Fe^{3+} with conc HNO_3 forming Brown ppt

23. $CaSO_4$ dissolves more in water in the presence of $(NH_4)_2SO_4$ due to formation of double salt $(NH_4)SO_4$ $CaSO_4 \cdot H_2O$ or complex salt

24. $BaS_2O_3 + 2HCl \longrightarrow BaCl_2 + H_2O + SO_2\uparrow + S\downarrow$

$$BaS_2O_3 + K_2CrO_4 \longrightarrow \underset{\text{Yellow}}{BaCrO_4\downarrow} + K_2S_2O_3$$

25. $Hg^{2+} + 4KI \longrightarrow K_2HgI_4 + 2K^+$

27. $3CuS + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO + 3S$

30. $Mn(NO_3)_2 + 2NaOH \longrightarrow Mn(OH)_2 + 2NaNO_3$

$H_2O + Br_2 \longrightarrow 2HBr + O$

$2Mn(OH)_2 + H_2O + O \longrightarrow 2Mn(OH)_3$

$Mn(OH)_3 + Na_2HPO_4 \longrightarrow \underset{\text{Pink.}}{Mn\,PO_4\downarrow} + 2NaOH + H_2O$

31. $\underset{(A)}{Ca(OH)_2} + Na_2CO_3 \longrightarrow \underset{(C)}{CaCO_3} + \underset{(B)}{2NaOH}$

$Ca(OH)_2 + CO_2 \longrightarrow \underset{(C)}{CaCO_3} + H_2O$

32. $Bi(NO_3)_3 + 3KI \longrightarrow \underset{\text{Black}}{BiI_3\downarrow} + 3KNO_3$

$BiI_3 + KI \longrightarrow \underset{\text{Orange}}{K[BiI_4]}$

33. Nitrates are highly soluble.

38. $2Cu^{2+} + 5I^- \longrightarrow Cu_2I_2 + I_3^-$

$I_3^- + 2S_2O_3^{2-} \longrightarrow 3I^- + S_4O_6^{2-}$

40. Hydrated cobalt chloride is pink. On heating, losing water of crystallization becomes colourless. But on cooling it absorbs moisture again and get pink colour.

41. $H_2C_2O_4 + MnO_2 + H_2SO_4 \longrightarrow MnSO_4 + 2CO_2 + 2H_2O$

45. $2AgNO_3 \longrightarrow 2Ag + 2NO_2 + O_2$

46. Dichromate oxidizes bromide to bromine but do not give CrO_2Br_2 similar CrO_2Cl_2

49. Metal nitrates are reduced to ammonia with zinc in alkaline medium which turns. Nessler's reagent to brown.

53. Both sulphite and sulphates give white precipitates with $BaCl_2$ solution but $BaSO_3$ is soluble by adding dil HCl where as $BaSO_4$ is insoluble.

54. Na_2SO_4 converts the $Ba(NO_3)_2$ to insoluble $BaSO_4$

61. Brown ring test is given by nitrate ion only with conc. H_2SO_4 but nitrite ion can respond even with acetic acid.

62. Insoluble and covalent chlorides do not respond to chromyl chloride test. $SnCl_4$ is covalent.

63. $FeSO_4 \cdot 7H_2O \xrightarrow{\Delta} FeSO_4 + 7H_2O$

$2FeSO_4. \xrightarrow{\Delta} Fe_2O_3 + SO_2 + SO_3$

SO_2 reduces acidified K_2CrO_7 but SO_3 with $BaCl_2$ solution gives $BaSO_4$ precipitate

67. Mercurous salts give blackish brown ppt with ammonia due to formation of Hg + NH_2HgCl. with HCl mercurous salts give insoluble Hg_2Cl_2

69. When H_2S is passed in acid medium the IInd group cations Cu^{2+}, Bi^{3+} and Pb^{2+} will be precipitated as sulphides

70. $MnSO_4 + 2NaOH \longrightarrow Mn(OH)_2 + Na_2SO_4$

$2Mn(OH)_2 + O_2 \longrightarrow 2MnO(OH)_2$

$ZnSO_4 + 2NaOH \longrightarrow Zn(OH)_2 + Na_2SO_4$

$Zn(OH)_2 + 2NaOH \longrightarrow \underset{\text{Soluble}}{Na_2ZnO_2} + 2H_2O$

$MnO(OH)_2$ on oxidation with strong oxidizing agent converts to violet solution of permanganate.

71. Aluminium gives Thenard blue with cobalt nitrate which is not due to ammonia.

73. Sodium bismuthate can oxidize colourless manganous salt to purple permanganate

76. 1. Passing H_2S in acid medium precipitate CuS but not Al^{3+} and Ni^{2+}. 2. Addition of NH_4Cl and NH_4OH now precipitate $Al(OH)_3$ but not Ni^{2+} · Ni^{2+} is precipitated as NiS in alkaline medium.

77. Fe^{2+} ion reduces orange red $Cr_2O_7^{2-}$ ion to green Cr^{3+} ion.

78. Ba^{2+} and Sr^{2+} do not give precipitates with H_2S · Cadmium give yellow CdS.

79. Ferrous salts give first green precipitate with NaOH which slowly changes to brown due to oxidation by air; and give black FeS ppt in alkaline medium. Also reduces permanaganate.

82. $K_3[Cu(CN)_4]$ is stable but $K_2[Cd(CN)_4]$ is less stable and dissociate to give free Cd^{2+} ions which give yellow ppt with H_2S.

84. $CrO_2Cl_2 \longrightarrow CrO_2^{2+} + 2Cl^-$

85. CO and CO_2 have no reaction with $K_2Cr_2O_7$.

87. The compound is $Pb(NO_3)_2$ The brown fumes are due to NO_2 not due to Br_2.

89. NO_2 when dissolved in water forms colourless HNO_2 and HNO_3 but Br_2 when dissolved in water forms reddish brown liquid.

90. $\underset{\text{White}}{2AgNO_3} \longrightarrow 2Ag + \underset{\text{Brown}}{2NO_2} + O_2$

$2AgNO_3 + BaCl_2 \longrightarrow Ba(NO_3)_2 + 2AgCl\downarrow$

91. Barium sulphate precipitates.

92. In conc. HCl, $PbCl_2$ dissolves due to the formation H_2PbCl_4 complex.

93. $\underset{\text{Brown}}{Bi_2S_3} + 8HNO_3 \longrightarrow 2Bi(NO_3)_3 + 4H_2O + 2NO + 3S$

$Bi(NO_3)_3 + 3NH_4OH \longrightarrow Bi(OH)_3 + 3NH_4NO_3$

$Bi(OH)_3 + 3HCl \longrightarrow BiCl_3 + 3H_2O$

$BiCl_3 + H_2O \longrightarrow \underset{\text{White}}{BiOCl\downarrow} + 2HCl$

94. Both CO_2 and SO_2 liberated form Na_2CO_3 and Na_2SO_3 will be absorbed completely by $BaCl_2$. So $K_2Cr_2O_7$ is not effected.

100. With H_2S Cu^{2+} and Cd^{2+} are precipitated in acid medium. Filtrate contain Cr^{3+}, Fe^{3+} (due to oxidation by HNO_3) and Al^{3+}. This when boiled with ammonium chloride and sodium hydroxide, Al^{3+} dissolves due to formation of $NaAlO_2$ and Cr^{3+} dissolve due to formation of $NaCrO_2$. So filtrate gives Test for Na^+, Al^{3+} and Cr^{3+}

101. $S + 2NO \rightarrow SO_2 + N_2$

$SO_2 + 2NO + H_2O \rightarrow H_2SO_4 + N_2O$

102. $BaCO_3 + 2HCl \rightarrow BaCl_2 + H_2O + CO_2$

$ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$

More than One Answer

3. $Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O + CO_2$

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$

$Na_2SO_3 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O + SO_2$

$Ca(OH)_2 + SO_2 \rightarrow CaSO_3 + H_2O$

$CaSO_3 + H_2O + SO_2 \rightarrow Ca(HSO_3)_2$

SO_2 has pungent suffocating smell and reduces acidified orange red dichromate to green chromic sulphate.

4. Conc. HNO_3 oxidizes Br^- and I^- to Br_2 and I_2 respectively and itself reduces to brown NO_2.

6. If $(NH_4)_2SO_4$ or $(NH_4)_2CO_3$ are used in Group III analysis the Vth group cations Ba^{2+}, Sr^{2+} will be precipitated as sulphaters or carbonates.

7. Only Al_2O_3 dissolves in Na_2CO_3 due to formation of $NaAlO_2$ $ZnSO_4$: precipitates as basic zinc carbonate while $FeSO_4$ precipitates as $FeCO_3$.

8. Sulphides give a violet colour with sodium nitroprusside, H_2S with rotten eggs smell, turns lead acetate to black, give yellow precipitate with $CdCl_2$

10. With $K_4[Fe(CN)_6]$, Zn^{2+} gives white, Fe^{3+} gives blue and Cu^{2+} give chocolate brown precipitate

14. $MnCl_2 + 2NaOH + H_2O_2 \rightarrow MnO_2 + 6NaCl + 2H_2O$

$2CrCl_3 + 3H_2O_2 + 10NaOH \rightarrow 2Na_2CrO_4 + 6NaCl + 8H_2O$

15. $Na(NH_4)HPO_4 \rightarrow NaPO_3 + NH_3 + H_2O$

$NaPO_3 + MO \rightarrow NaMPO_4$ $(M = Cu; Co, etc)$

17. Due to transfer of electron form NO to Fe^{2+} oxidation number of Fe^{2+} changes from +2 to +1. When nitrite present it also liberate NO gas with acid.

18. In 6M NH_3 both Cu^{2+} and Zn^{2+} form soluble complexes.

20. $H_2O + Cl_2 \rightarrow HCl + HOCl$

$2HCl + Mg \rightarrow MgCl_2 + H_2$

21. $FeS + H_2SO_4 \rightarrow FeSO_4 + H_2S$

$H_2S + CuSO_4 \rightarrow CuS + H_2SO_4$

$H_2S + HgCl_2 \rightarrow \underset{\text{Black}}{HgS} + 2HCl$

22. $FeCl_3 + K_3[Fe(CN)_6] \rightarrow \underset{\text{Brown Colouration}}{Fe[Fe(CN)_6]} + 3KCl$

$3FeSO_4 + 2K_3[Fe(CN)_6] \rightarrow Fe_3[Fe(CN)_6]_2 + 3K_2SO_4$

$FeCl_3 + 3KSCN \rightarrow \underset{\text{Blood red}}{Fe(SCN)_3} + 3KCl$

$FeSO_4 + KSCN \rightarrow$ No colour

23. $Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + H_2O + CO_2$

$Ca(HCO_3)_2 + Ca(OH)_2 \longrightarrow 2CaCO_3 + 2H_2O$

29. Both $Mn(OH)_2$ and $Zn(OH)_2$ are precipitated by adding NaOH but $Zn(OH)_2$ dissolves in excess of NaOH but not Mn $(OH)_2$

32. Chromyl chloride hydrolyse in water forming CrO_3 So CrO_2Cl_2 vapour are not liberated when carried in wet test tube or with hydrochloric acid.

34. Ammine complexes decomposes in acid medium while hydroxyl complexes will be neutralized by acid.

Comprehensive Type Questions

Passage-I

$MnCl_2 + H_2S \longrightarrow \underset{\text{Buff Colour}}{MnS} + 2HCl$

$MnS + \underset{\text{hot}}{2HCl} \longrightarrow MnCl_2 + H_2S\uparrow$

$MnCl_2 + 2NaOH \longrightarrow Mn(OH)_2 + 2NaCl$

$2Mn(OH)_2 + O_2 \longrightarrow 2MnO(OH)_2$

$MnO(OH)_2 \longrightarrow MnO_2 + H_2O$

Passage-2

$\underset{A}{CuCO_3} \longrightarrow \underset{B}{CuO} + \underset{C}{CO_2}$

$CuO + H_2 \longrightarrow \underset{D}{Cu} + H_2O$

$CuO + H_2SO_4 \longrightarrow CuSO_4 + H_2O$

$2CuSO_4 + K_4[Fe(CN)_6] \longrightarrow Cu_2[Fe(CN)_6] + 2K_2SO_4$

Passage-3

$$Na_2CO_3 + H_2O + 2SO_2 \longrightarrow \underset{(A)}{2NaHSO_3} + CO_2$$

$$2NaHSO_3 + Na_2CO_3 \longrightarrow \underset{(B)}{2Na_2SO_3} + H_2O + CO_2$$

$$Na_2SO_3 + S \longrightarrow \underset{(C)}{Na_2S_2O_3}$$

$$2Na_2S_2O_3 + I_2 \longrightarrow \underset{(D)}{Na_2S_4O_6} + 2NaI$$

Passage-4

The salt is hypo

Passage-6

$$\underset{P}{ZnS} + H_2SO_4 \longrightarrow \underset{R}{ZnSO_4} + \underset{Q}{H_2S}$$

$$K_2Cr_2O_7 + 4H_2SO_4 + 3H_2S \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + \underset{(S)}{3S}$$

$$S + O_2 \longrightarrow \underset{T}{SO_2}$$

$$SO_2 + 2H_2S \longrightarrow 3S + 2H_2O$$

$$\underset{\text{White}}{CuSO_4} + 5H_2O \longrightarrow \underset{\text{Blue}}{CuSO_4 \cdot 5H_2O}$$

$$ZnSO_4 + 4NH_3 \longrightarrow \left[Zn(NH_3)_4\right]SO_4$$

Passage-9

$$2KCl + H_2SO_4 \longrightarrow \underset{Z}{K_2SO_4} + \underset{Y}{2HCl}$$

$$NH_3 + HCl \longrightarrow NH_4Cl$$

$$K_2SO_4 + CaCO_3 + 4C \longrightarrow \underset{A}{K_2CO_3} + \underset{P}{CaS} + 4CO$$

$$\underset{A}{K_2CO_3} + H_2O + CO_2 \longrightarrow \underset{B}{2KHCO_3}$$

$$2KHCO_3 \xrightarrow{\Delta} K_2CO_3 + H_2O + CO_2$$

Passage-10

$$Zn^{2+} + 2NaOH \longrightarrow Zn(OH)_2 + 2Na^+$$

$$Zn(OH)_2 + 2NaOH \longrightarrow Na_2ZnO_2 + 2H_2O$$

$$Mn^{2+} + 2NaOH \longrightarrow Mn(OH)_2 + 2Na^+$$

$$\underset{\text{White}}{2Mn(OH)_2} + O_2 \longrightarrow \underset{\text{Brown}}{2MnO(OH)_2} \downarrow$$

With an oxidizing agent $MnO(OH)_2$ will be oxidized to MnO_4^- having violet colour

Permanganate can be reduced to colourless Mn^{2+} by $SO_3^{2-}, C_2O_4^{2-}$ and Fe^{2+}

Passage-11

$$\underset{A}{Pb(NO_3)_2} + 2NaCl \longrightarrow \underset{B}{PbCl_2} \downarrow + 2NaNO_3$$

$PbCl_2$ is partially soluble in water. So the filtrate gives black precipitate with H_2S

$$PbCl_2 + H_2S \longrightarrow \underset{\substack{\text{Black}\\ C}}{PbS} + 2HCl$$

Passage-12

$$\underset{A}{Hg_2(NO_3)_2} \xrightarrow{\Delta} 2Hg + 2NO_2 + O_2$$

$$Hg_2(NO_3)_2 + 2HCl \longrightarrow Hg_2Cl_2 + 2HNO_3$$

$$Hg_2Cl_2 + 2NH_4OH \longrightarrow Hg + NH_2HgCl + NH_4Cl + 2H_2O$$

$$2HNO_3 \longrightarrow H_2O + 2NO_2 + O$$

$$Hg + 2HCl + O \longrightarrow HgCl_2 + NH_4Cl$$

$$HgCl_2 + 2KI \longrightarrow HgI_2 + 2KCl$$

$$HgI_2 + 2KI \longrightarrow K_2HgI_4$$

$$2K_2HgI_4 + 3KOH + NH_3 \longrightarrow \underset{\text{Brown PPt}}{NH_2\,HgO\,HgI} + 7KI + 2H_2O$$

3. $$\underset{0.01}{HgI_2} + \underset{0.02}{2KI} \longrightarrow \underset{0.01}{K_2HgI_4}$$

$$K_2HgI_4 \longrightarrow 2K^+ + HgI_4^{2-} \quad \text{vant Hoff factor } i = 3$$

$$\Delta T_f = K_f \times i \times m$$

$$= 1.86 \times 3 \times \frac{0.01}{100} \times 1000 = 0.558$$

$\therefore$ Depression = 0.558K

Passage-13

Compound A is $K_2SO_4\, Cr_2(SO_4)_3 \cdot 24\,H_2O$
K gives lilac flame colour
When boiled with H_2O_2 and NaOH Cr^{3+} changes to yellow Na_2CrO_4
In CrO_4^{2-} ion the hybridization of Chromium is d^3s

Passage-14

$$\underset{A}{ZnS} + H_2SO_4 \longrightarrow \underset{B}{ZnSO_4} + H_2S$$

H_2S reduce orange red $K_2Cr_2O_7$ to green Cr^{3+} along with Sulphur on burning in air gives SO_2. Sulphur dioxide and H_2S react giving sulphur and H_2O which turns colourness $CuSO_4$ to blue $CuSO_4 \cdot 5H_2O \cdot ZnSO_4$ gives precipitates of $Zn(OH)_2$ with both NaOH and NH_4OH but dissolves in excess of NaOH and NH_4OH due to formation of Na_2ZnO_2 and $[Zn(NH_3)_4]SO_4$. Zinc will be precipitated with H_2S as ZnS because during neutralization of NaOH the salt formed is sodium acetate which produces alkalinity due to hydrolysis.

Passage-15

1.

Sol: (A) = ZnS (B) = H_2S (C) = $ZnSO_4$ (D) S (E) = SO_4
$4S + 6NaOH \rightarrow 2Na_2S + Na_2S_2O_3 + 3H_2O$
$Na_2S + 3S \rightarrow Na_2S_5$

2.

Sol: $ZnSO_4 + 2NH_4OH \rightarrow Zn(OH)_2 \downarrow + (NH_4)_2SO_4$
$Zn(OH)_2 + (NH_4)_2SO_4 + 2NH_3aq \rightarrow [Zn(NH_3)_4]SO_4 + 2H_2O$

3.

Sol: $SO_2 + 2H_2S \rightarrow \frac{3}{8} S_8 + 2H_2O$

Passage-17

4. $2AgNO + Na_2CrO_4 \rightarrow \underset{\text{Brick red}}{Ag_2CrO_4\downarrow} + 2NaNO_3$

$BaCl, + Na_2CrO_4 \rightarrow \underset{\text{yellow}}{BaCrO_4\downarrow} + 2NaCl$

5. $CrCl_3 + MnO_2 + 4H_{2v}SO_4 \longrightarrow Cr_2(SO_4)_3 + MnSO_4 + 2H_2O + 3Cl_2$
$2KI + Cl_2 \longrightarrow 2KCl + I_2$

Passage-18

6.

Sol: $\underset{(A)}{BiCl_3} + H_2O \longrightarrow \underset{(B)}{BiOCl} + 2HCl$

$2Bi^{3+} + 3H_2S \longrightarrow \underset{(C)}{Bi_2S_3} + 6H^+$

$2BiCl_3 + 3H_2SO_4 \longrightarrow \underset{(D)}{Bi_2(SO_4)_3} + 6HCl(g)$

$2HCl + 2Hg^+ \longrightarrow Hg_2Cl_2 + 2H^+$

Hints to Matching Type Questions

7.

Sol: Mn^{2+} gives green flashes in flame test. Also it gives white ppt. with aq. NH_3
$MnCl_2 + 2NH_4OH \longrightarrow Mn(OH)_4 + 2NH_4Cl$
Pb^{2+} gives white precipitate with aqueous ammonia and dil. HCl

$Pb^{2+} + 2OH^- \longrightarrow \underset{\text{White}}{Pb(OH)_2\downarrow}$

$Pb^{2+} + 2Cl^- \longrightarrow \underset{\text{White}}{PbCl_2\downarrow}$

$Ag^+ + HPO_4^{2-} \longrightarrow \underset{\text{Yellow ppt}}{Ag_3PO_4\downarrow}$

$Cu^{2+} + SCN^- \longrightarrow \underset{\text{Yellow}}{Cu(SCN)_2\downarrow} + H^+$

$2Cu(SCN)_2 \longrightarrow \underset{\text{White}}{2CuSCN\downarrow} + (SCN)_2$

8.

Sol: $\underset{\text{red}}{Hg_2I_2} \xrightarrow[\Delta]{H_2O} \underset{\text{black}}{HgI_2} + Hg$

$Pb^{2+} + K_2CrO_4 \longrightarrow PbCrO_4$ (yellow)
$Pb^{2+} + 2KI \longrightarrow PbI_2$ (yellow)
$ZnO + CoO \longrightarrow CoZnO_2$ (Rinmann's green)

9.

$CuSO_4 + K_4[Fe(CN)_6] \longrightarrow \underset{\text{chocolate brown}}{Cu_2[Fe(CN)_6]\downarrow s}$

(a) $CuSO_4 + BaCl_2 \longrightarrow \underset{\text{white}}{BaSO_4\downarrow} CuCl_2$

(b) $NaBr + H_2SO_4 \longrightarrow \underset{\text{Brown fumes}}{Br_2\uparrow} + NaHSO_4 + H_2O + \underset{\text{colourless gas}}{SO_2}$

(c) $2KNO_2 + H_2SO_4 \longrightarrow 2HNO_2 \longrightarrow NO + NO_2 + H_2O + K_2SO_4$

(d) $CaS_2O_3 + HCl \longrightarrow CaCl_2 + \underset{\text{yellow turbity}}{S} + SO_2$ colourless gas

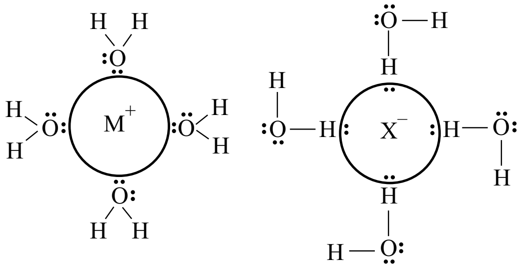
$CaS_2O_3 + BaCl_2 \longrightarrow \underset{\text{white}}{BaS_2O_3\downarrow} + CaCl_2$

APPENDICES

Appendix I

Hydration, Hydrolysis and Solubility

In ionic compounds, when dissolved in polar solvents, the ions (both positive and negative) get solvated (hydrated if the solvent is water). While writing the chemical equations for reactions of ions in solution, we often write ions as if they were simple particles in solution. For example, we may write the sodium ion as Na^+ or perhaps Na^+ (aq). But there are definite reactions between ions and molecules of the polar solvent water. Since the water molecule being polar, its negative charged end (oxygen end) will be attracted by the cation while the positive charge end (hydrogen end) will be attracted by the anion. These ions are called hydrated ions.



The amount of heat liberated during the process is called as *heat of hydration* or *hydration energy.*

As the electronegativities of metals are not too high, Latimer calculated the hydration energies of metal ions by using the formula (neglecting their electronegativities)

$$\Delta H_{hyd} = -\frac{60,900 \times z^2}{(r+50)} \text{ KJ mol}^{-1}$$

where z is the charge on the cation and r is the cationic radius (in picometers).

The hydration enthalpies in Table A1 show that the hydration enthalpies of metal ions with electronegativity >1.5 are much higher than those of the metal ions with electronegativity <1.5 which have comparable radius and charge. This indicates that for the metal ions having electronegativity greater than 1.5, there is not only just an electrostatic attraction between the metal ion and the negative end of the water molecule, but there also may be some degree of covalent bond formation in which an unshared electron pair on water is shared with the metal.

If the cation size is small and its charge is more, the unshared electron pair of the water molecule is pulled to (or

Table A1 Hydration enthalpies of metal cations KJ mol^{-1}

Electronegativity < 1.5			Electronegativity >1.5		
Ion	**Radius**	**ΔH_{hyd}**	**Ion**	**Radius**	**ΔH_{hyd}**
		+1 ions			
Cs	181	–263	Tl	164	–326
Rb	166	–296	Ag	129	–475
K	152	–321	Cu	91	–594
Na	116	–405			
Li	90	–515			
H		–1091			
		+2 ions			
Ra		–1259	Pb	133	–1480
Ba	149	–1304	Sn		–1554
Eu	131	–1458	Hg	116	–1824
Sr	132	–1445	Cd	109	–1806
No	124	–1485	Ag	108	–1931
Yb	116	–1594	Ti	100	–1862
Ca	114	–1592	V	93	–1918
Mg	86	–1922	Cr	94	–1850
			Mn	97	–1845
			Fe	92	–1920
			Co	88	–2054
			Ni	83	–2106
			Cu	91	–2110
			Zn	88	–2044
			Be	59	–2487
		+3 ions			
Pu	114	–3441	T1	102	–4184
La	117	–3283	In	94	–4109
Lu	100	–3758	ba	76	–4685
Y	104	–3620	Ti	81	–4154
Sc	88	–3960	V	78	–4375
			Cr	75	–4402
			Mn	78	–4544
			Fe	78	–4376
			Co	75	–4651
			Al	67	–4660
		+4 ions			
Th	108	–6136			
U	103	–6470			
Ce	101	–6489			
Zr	86	–6593			
Hf	85	–7120			

more even shared with) the metal ion, thus polarizing the water molecule more, releasing the proton and combining with hydroxide ion.

$$M^{n+}(aq) + H_2O \rightleftharpoons M\,(OH)^{(n-1)+}(aq) + H^+(aq)$$

This reaction is almost similar to the equation for the equilibrium process of ionization of a weak acid such as acetic acid.

$$CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$$

Table A2 Hydrolysis constants for metal cation

Electronegativity < 1.5				Electronegativity > 1.5				
Ion	**Radius**	**(a)**	**pKa**	**Ion**	**Radius**	**(a)**	**(b)**	**pKa**
				+1 ions				
K	152	0.007	14.5	Tl	164	0.006	0.016	13.2
Na	116	0.009	14.2	Ag	129	0.008	0.049	12.0
Li	90	0.011	13.6					
				+2 ions				
Ba	149	0.027	13.5	Pb	133	0.030	0.066	7.7
Sr	132	0.030	13.3	Sn				3.4
Ca	114	0.035	12.8	Hg	116	0.034	0.082	3.4
Mg	86	0.047	11.4	Cd	109	0.037	0.055	10.1
				Cr	94	0.043	0.043	10.0
				Mn	97	0.041	0.046	10.6
				Fe	92	0.043	0.075	9.5
				Co	88	0.045	0.082	9.6
				Ni	83	0.048	0.088	9.9
				Zn	88	0.045	0.060	9.0
				Be	59	0.068	0.074	6.2
				+3 ions				
Pu	114	0.079	7.0	Bi	117	0.077	0.127	1.1
La	117	0.077	8.5	Tl	102	0.088	0.140	0.6
Lu	100	0.090	7.6	Au	99	0.091	0.191	–1.5
Y	104	0.086	7.7	In	94	0.096	0.123	4.0
Sc	88	0.102	4.3	Ti	81	0.111	0.115	2.2
				Ga	76	0.118	0.148	2.6
				Fe	78	0.115	0.147	2.2
				Cr	75	0.120	0.135	4.0
				Al	67	0.134	0.145	5.0
				+4 ions				
Th	108	0.148	3.2	Sn	83	0.193	0.222	–0.6
Pa	104	0.154	–0.8	Ti	74	0.216	0.220	–4.0
U	103	0.155	0.6					
Np	101	0.158	1.5					
Pu	100	0.160	0.5					
Ce	101	0.158	–1.1					
Hf	85	0.188	0.2					
Zr	86	0.186	–0.3					

(a) is $\frac{z^2}{r}$ ratio for the cation

(b) is $\frac{z^2}{r + 0.096(\chi - 1.5)}$ for the cation

The equilibrium constant expression for the acid-producing reaction of metal ions is commonly called as hydrolysis. Thus hydration is a simple ion-dipole attraction between ions and water molecules while hydrolysis is the chemical reaction between ion/molecule with water molecule.

The equilibrium constant K_h or since the reaction produces acid as Ka may be measured. The pKa values of different cations are given in Table A2. Lesser the pKa values, higher the degree of hydrolysis.

The hydrolysis may be continued in stepwise until the charge on the metal ion is completely neutralized, a metal hydroxide results. This species is neither an ion nor a hydrate and usually insoluble, hence, a precipitate appears in the solution. Often the insoluble metal hydroxides will subsequently lose molecules of water to give insoluble metal oxides.

$$M(OH)_n(s) \longrightarrow MO_{n/2}(s) + \frac{n}{2}H_2O$$

Since it is difficult for us to tell when this has happened, we will not attempt to distinguish metal hydroxide and metal oxide precipitates.

It can be shown that a metal hydroxide or oxide will precipitate at *a* pH that is fairly close to the pKa value of the metal ion.

$$pH = pKa - \frac{1}{n}\log\,[M^{n+}] - \frac{5.6}{n}$$

However, the exact pH at which the precipitated oxide or hydroxide becomes the predominant form of the metal depend on concentration to some extent. In general, for soluble species the range of pKa at which precipitation occurs increases with dilution while for insoluble species it decreases.

The cations may be grouped into different *categories of acidity.* The ions in a given category of acidity share several important chemical properties such as the degree to which the ions react with water, the solubility or insolubility of salts formed by these ions and the properties of the compounds formed by them in the absence of water.

1. Non-acidic Cations: Cations with pKa values of 14 or greater do not show any hydrolysis reactions, e.g., Rb^+ and Cs^+. These cannot be precipitated from a 1M solution at any pH.

2. Feebly acidic Cations: Cations with pKa values between 11.5 and 14 produce acidities but cannot be directly detectable. They occasionally show some significant consequences, e.g., Li^+, Ba^{2+}, Sr^{2+} and Ca^{2+}. These are precipitated at pH values above their pKa values.

3. Weakly acidic Cations: Cations with pKa values 6 and 11.5 are termed as weakly acidic cations, e.g., Mg^{2+} and the d-block metal ions with +2 charge (M^{2+}). These ions show enough acidity to precipitate insoluble metal hydroxides at pH values over the pKa, i.e., in neutral or just slightly basic solution.

4. Moderately acidic Cations: Cations such as Al^{3+} and the +3 charged d-block ions have pKa values 1–6. These are acidic with a detectable pH value lower than even that of water containing dissolved carbon dioxide (H_2CO_3). The pKa values of these ions are comparable to those of typical organic acids such as acetic acid and liberate CO_2 when sodium carbonate is added to them. Thus they will precipitate as hydroxides instead of metal carbonates on addition of sodium carbonate. These ions exist in solution at quite low pH values. If the solutions of these ions are not kept highly acidic, metal hydroxides will precipitate. Hence, their solutions are prepared in a slightly acidic medium.

5. Strongly acidic Cations: These have pKa values between –4 and 1, e.g., Ti^{4+}. These react with water giving a strongly acidic solution and precipitate as an insoluble metal oxide or hydroxide. In very high concentrated acids, i.e., $[H^+] > 1M$, or pH < O only, these ions exist in solution.

6. Very strongly acidic Cations: These cations presumably have pKa value below –4. These ions react irreversibly with water producing an oxide or hydroxide and a high concentration of hydrogen ion. The concentration of the acid produced by these ions is *levelled,* i.e., characteristic of the acid of water solutions H_3O^+. Thus no acid stronger than H_3O^+ can persist in water and there is no pH at which these "cations" can predominate or even exist in water. The cations of the purely non-metallic elements of the periodic table have pKa ratios in this range, which is to say that they cannot exist in water as cations.

Effect of Charge and Radius of Cation on its acidic Character: With increase in the number of charges on the cation and decrease in its radius, its acidic character increases. It can be expressed as

$$\text{pKa} = (15.14 - 88.16) \times \frac{z^2}{r}$$

If we assume the electronegativity of the element in the cation is <1.50, we can classify the elements into the six categories as explained above in the following manner.

If the $\frac{z^2}{r}$ ratio of the cation is <0.01, it is non-acidic.

If the $\frac{z^2}{r}$ is between 0.01 and 0.04, the cation is feebly acidic.

If the $\frac{z^2}{r}$ is between 0.04 and 0.10, the cation is weakly acidic.

If the $\frac{z^2}{r}$ is between 0.10 and 0.16, the cation is moderately acidic.

If the $\frac{z^2}{r}$ is between 0.16 and 0.22, the cation is strongly acidic.

If the $\frac{z^2}{r}$ is over 0.22, the cation is very strongly acidic.

Effect of Electronegativity: Electronegativity of the elements in cation also play an important role in the acidity of the cation. If the electronegativity of metal ion in Pauling scale (χ_p) is more than 1.5 they will have more acidic character than the metal ions of similar charge and size. In such a case, for the excess acidity, the equation is modified as

$$\text{pKa} = (15.14 - 88.16)\left[\frac{z^2}{r + 0.096(\chi_p - 1.5)}\right]$$

Table A3 Relationship between pKa, acidity, and $\frac{z^2}{r}$ ratio of the metal ions

pKa Range	Category	$\frac{z^2}{r}$ ratio	χ_p	Examples
14–15	Non-acidic	0.00–0.01	<1.8	Most +1 ions of the s-block
11.5–14	Feebly acidic	0.00–0.01	>1.8	Tl^+
11.5–14	Feebly acidic	0.01–0.04	<1.8	Most +2 ions of s-, f-blocks
6–11.5	Weakly acidic	0.01–0.04	>1.8	Most +2 ions of the d-block
6–11.5	Weakly acidic	0.04–0.10	<1.8	All +3 ions of the f-block
1–6	Moderately acidic	0.04–0.10	>1.8	Most +3 ions of the d-block
1–6	Moderately acidic	0.10–0.16	<1.8	Most +4 ions of the f-block
(–4)–1	Strongly acidic	0.10–0.16	>1.8	Most +4 ions of the d-block
(–4)–1	Strongly acidic	0.16–0.22	<1.8	
<(–4)	Very strongly acidic	0.16 and up	>1.8	
<(–4)	Very strongly acidic	0.22 and up	<1.8	

This equation should be used only when the electronegativity in Pauling scale of the metal exceeds 1.5.

If the electronegativity of the metal ion is 1.8 or greater move it up one category in acidity of each category explained earlier.

Compounds that Fail to Undergo Hydrolysis and Oxocations

From the discussion made so far carbon tetrachloride should be one of the most reactive of all the chlorides of the elements since $\frac{z^2}{r}$ for C^{4+} is quite impressive (0.533). However, this expected reaction do not take place. The reasons are

1. The maximum coordination of the second period elements is only 4 and cannot expand their octet due to the absence of d-orbitals.
2. Steric crowding due to the presence of Cl atoms around C atom which do not allow water molecule to approach antibonding orbitals of C–Cl bonds.
3. The negative end of water molecule also cannot attack the vacant 3d-orbital of Cl atom because of polarity of the bond. The negative charge (δ-) developed on Cl atoms repels the incoming negative end of water molecule.

But at high temperatures the water molecule can form a bond with the next higher energy level, i.e., 3s-orbital. So, hydrolysis of $CC1_4$ take place at very high temperatures only. This happens when $CC1_4$ is used as a fire extinguisher along with water on a very hot fire, forming an intermediate deadly poisonous phosgene, $COCl_2$.

Another case of "unexpected" non-reactivity of a halide is that of sulphur hexafluoride, SF_6, which resists reacting with steam at 500°C and resists molten KOH. This is again due to the steric crowding of six fluorine atoms around sulphur atom which do not allow water molecule to approach sulphur atom. Also, the S–F bonds have some double bond character due to back bonding from fluorine atom to the vacant 3d-orbitals of sulphur. The atoms get larger as we go down Group 16 (VI), however, SeF_6 is unreactive with water at 25°C but is hydrolyzed in the respiratory tract. TeF_6 reacts slowly with cold water.

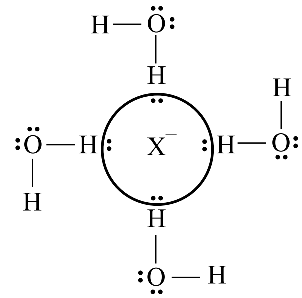
Certain ions near the centre of the fifth and sixth periods (e.g., Ag^+, Sn^{2+}, Hg^{2+} and Au^+) the preferred coordination number is often quite small—as low as 2. There is some anomalous chemistry in this part of the periodic table. Here, the relativistic effects also play some important role. The acidity of hydrate Hg^{2+} and Sn^{2+} ions is much higher than expected. Since less number of water molecules are coordinated to the metal ion, the electrostatic pulling on the electrons of each water molecule is stronger than in more normal hydrated ions. Similarly, the smallest hydrated ion Be^{2r} is also more acidic than one would expect.

Sometimes, when two or four hydroxide groups are attached to a (highly charged) cation, they may lose molecules of water and form *oxocations.* Such species are particularly persistant for the +5 and +6 oxidation states of the f-block elements like U, Np, Pu and Am; the most common form in which uranium is in salts of the yellow uranyl ion is UO_2^{2+}. Such species are very common in the chemistry of molybdenum (V) and (VI) and vanadium (IV) and (V).

Many non-metal halides on partial hydrolysis form species that resembles these compounds, e.g., $POCl_3$. These compounds generally hydrolyze further with more water. The precipitates formed in the hydrolysis of $SbCl_3$ and $BiCl_3$ are SbOCl and BiOCl but not the oxides.

Hydration and Hydrolysis of Monoatomic Non-Metallic Anions

Just like cations are hydrated and hydrolyzed in water, the anions also get hydrated and hydrolyzed. The positive end of the dipolar water molecule, i.e., one of the hydrogen atom is attracted to each of the unshared pairs on the anion forming hydrogen bond to the anion.



The hydration energies of anions can be calculated from the relation

$$\Delta H_{hyd} \approx -57000 \times \frac{z^2}{r} \text{ KJ mol}^{-1}$$

Table A4 Hydration enthalpies of some anions (KJ mol^{-1})

Anion	Radius	Hydration energy
F^-	119	–513
Cl^-	167	–370
Br^-	182	–339
I^-	206	–294
S^{2-}	170	–1372
OH^-	119	–460
NO_3^-	165	–314
CN^-	177	–342
N_3^-	181	–298
SH^-	193	–336
BF_4^-	215	–223
SO_4^{2-}	244	–1059

Anion	Radius	Hydration energy
ClO_4^-	226	–235
CO_3^{2-}	164	–1314
SO_4^{2-}	244	–1059

As we would expect, these energies increase with increasing charge and decreasing size of the anion.

The interaction between an ion (the anion) and its water of hydration frequently is great enough that one or more of water molecules may be pulled apart in the process of hydrolysis. In this case the partially positively charged hydrogen atom of the water bonds to the anion, releasing the reminder of the water molecule—a hydroxide ion—and produce basic solution.

$$X^{n-} + H_2O \rightleftharpoons XH^{(n-1)-} + OH^-$$

$$S^{2-} + H_2O \rightleftharpoons SH^- + OH^-$$

An equilibrium constant can be written for the above reaction, i.e., the hydrolysis constant K_b. Higher the pK_b value, the weaker the base, i.e., the anion is acting as a weak base producing a solution with a pH not much above that of distilled water. Small or negative pK_b values indicates a very reactive, very basic anion, which causes a strong change in the pH of the solution. Most of the values given in Table A5 should be treated as illustrative, rather than quantitatively reliable.

Table A5 Basic character of monoatomic anions and their protonated forms in aqueous solution

Monoatomic anions			
C^{4-} very strong	N^{3-} very strong	O^{2-} very strong $pK_1 = -22$	F^- weak $pK_1 = -10.85$
Si^{4-} very strong	P^{3-} very strong	S^{2-} strong $pK_1 = -4$	Cl^- non-basic $pK_1 = 20.3$
Ge^{4-} very strong	As^{3-} very strong	Se^{2-} strong $pK_1 = -1$	Br^- non-basic $pK_1 = 22.7$
		Te^{2-} moderate $pK_1 = 3.0$	I^- non-basic $pK_1 = 23.3$
Partially protonated anions			
CH_3^- very strong $pK_4 = -30$	NH_2^- very strong $pK_3 = -25$	OH^- strong $pK_2 = -1.74$	
SiH_3^- very strong $pK_4 = -21$	PH_2 very strong $pK_3 = -13$	SH^- moderate $pK_2 = 7.11$	
GeH_3^- very strong $pK_4 = -11$	As H_2 very strong $pK_3 = -9$	SeH^- weak $pK_2 = 10.3$	
		TeH^- feeble $pK_2 = 11.4$	

It is useful to categorize the anions in categories of basicity that match those used for acidity of cations. This categorization can be considered in terms of acid-base reactivity.

1. Non-Basic Anions: The simple anions that have no detectable basicity belong to this category. Their pK_b values are above 14, e.g., Cl^-, Br^-, I^-, etc. These ions cannot take up H^+ ion from water molecule and their basicity is sufficiently less than that of water molecules.

2. Feebly Basic Anions: These have pK_b values between 11.5 and 14. There are no examples for this category.

3. Weakly Basic Anions: These have pK_b values in a range 6–11.5.

4. Moderately Basic Anions: These have pKb values in a range 1–6, e.g., Te^{2-}.

5. Strongly Basic Anions: These have pK_b values in a range (–4) –1, e.g., Se^{-2} and S^{2-}.

6. Very Strongly Basic Anions: These have pK_b values below –4. They react with water irreversibly, e.g., O^{2-}.

$$O^{2-} + H_2O \rightarrow 2OH^-$$

This reaction is an illustration of the levelling properties of very strong bases.

Acid–Base Reactivity Increases with Increasing Charge and Decreasing Radius

Solubility Rules for Salts of Oxo and Fluro Anions

Rule 1 The reaction of acidic cations and basic anions gives insoluble salts: Here acidic cation means weakly, moderately, strongly or very strongly acidic cations. The term basic anions refers to any anion that are weakly, moderately, strongly or very strongly basic.

The above generalized solubility rule approximately sums up two commonly given solubility rules (i) all carbonates are insoluble except those of the Group-1 elements and NH_4^+ and (ii) all hydroxides are insoluble except those of the Group-1 elements Sr^{2+} and Ba^{2+}. The generalized solubility rule give up some exactness in terms of the slightly different behaviour of hydroxides and carbonates, but it is clearly applicable to the salts of a number of other important oxoanions, such as phosphate, arsenate, silicate and borate.

Rule 2 Cross combination give soluble salts: Cross combination of ions (i) Non-acidic cations when combine with all other basic anions including even feebly basic anions and (ii) acidic cations including feebly acidic cations when combine with non-basic anions give soluble salts.

The most common solubility rule related to this rule of thumb is that for nitrate. All nitrates are soluble. Since most cations are acidic, their soluble oxosalts will be of non-basic anions such as nitrate, perchlorate and the fluoroanions provided that the principles of hard acid, hard base and soft acid, soft base do not interfere, this rule also covers the solubility of most chlorides, bromides and iodides.

Rule 3 Solubility Tendency: The generalized solubility tendency for the salts formed by the combination of non-acidic cation with non-basic anions is that large non-acidic cations and large non-basic anions give insoluble salts.

This is a tendency rather than a rule, since the driving force here is not strong enough to give highly insoluble salts. Precipitation often does not immediately begin which allows time for the growth of large crystals.

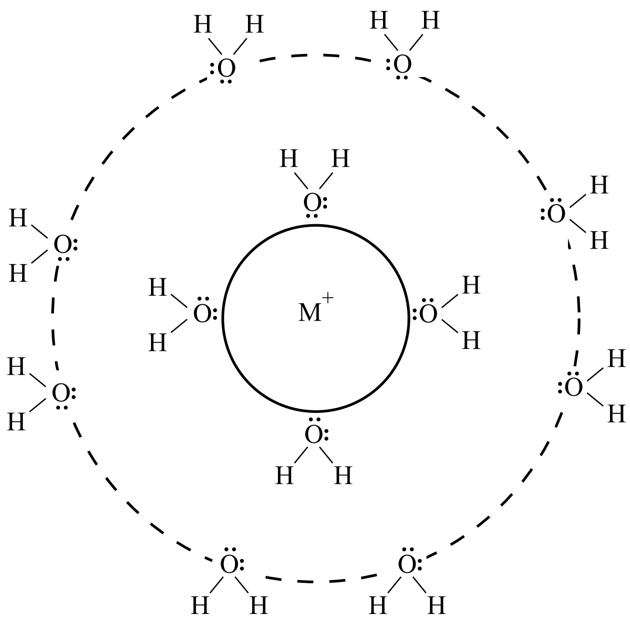
This tendency to crystallize out of solution or precipitate is almost completely absent for the smallest of the non-acidic cations, Na^+ (hence the specific solubility rule that all sodium salts are soluble) and the smallest of the non-basic anions NO_3^- Cl^-, Br^- and I^-. This tendency becomes more pronounced with Cs^+ salts and perchlorates ($CsClO_4$) itself precipitate readily, and it becomes most pronounced with the salts of complex cations and complex anions which are very large and non-acidic and very large and non-basic, respectively.

Rule 4 Solubility Tendency: Salts formed by the combination of feebly acidic cation with moderately basic and very strongly basic anions precipation usually occurs, but oxides and hydroxides of feebly acidic cations are (at least somewhat) soluble. Salts formed by the combination of weakly acidic, moderately acidic, strongly acidic and very strongly acidic cations with feebly basic anions are soluble, but for example sulphates of some large, weakly acidic cations also show low solubility. Large feebly acidic cations (such as Ba^{2+}) give precipitates with large feebly basic anions (such as SO_4^{-2} and SeO_4^{-2}). This tendency applies somewhat to the fluoride ion itself, although with $pK_b = 10.85$ it is actually just barely weakly basic. LiF is insoluble and the most insoluble fluoride is CaF_2.

In addition, it should be noted that *the solution of insoluble salts of basic anions will be enhanced in solutions of strong acids.* This solubility enhancement is not a true case of solubility, perhaps, but a reaction. A basic anion tends to react more completely with strong acid, H_3O^+ than with a less acidic metal ion, thus generating a new salt of that metal ion will be especially enhanced if the weak oxo-acid decomposes. Thus carbonates and sulphites readily "dissolve" in strong acids since the carbonic and sulphurous acids produced decompose to carbon dioxide and sulphur dioxide, gases that escape from the solution and allow the equilibrium to shift further towards dissolution.

Structure of Hydrated Ions

Certainly, all hydrated metal ions and non-metal or oxo-anions have a layer of water molecules surrounding the ion. This layer is referred to as the *primary hydration sphere* or *inner sphere* of the hydrated ion. Larger cation or anion can accommodate more number of water molecules in this primary hydration sphere. But ions that are at all acidic or basic exert such a strong attraction for the water molecules in their primary hydration sphere, pulling electron pairs towards a cation or hydrogens towards an anion, that these water molecules exert an enhanced hydrogen bonding attraction for other water molecules and organize them into a *secondary hydration sphere* or *outer sphere* around the first layer and without the moderating effect of this secondary hydration sphere most metal cations would be strongly or very strongly acidic. Note that this phenomenon has the same cause as does the phenomenon of hydrolysis that we discussed earlier. In this case, however, the attraction does not have to be strong enough to pull the water molecule of the second layer apart—it only needs to attach it by strengthened hydrogen bonds. Thus we can see why this phenomenon should be directly related to the acidity or basicity of the cation or anion.



The secondary hydration sphere may consist of more than one layer of water molecules—the stronger the attraction of the bare ion for water molecules (i.e., the greater its acidity or basicity), the more the water molecules that may be attached in additional layer. Each subsequent layers would be expected to involve weaker attractions, however, so water molecules would stay immobilized by hydrogen bonds for shorter and shorter periods. Many methods have been used to study this phenomenon of multiple hydration layers, but these diverse methods of measuring the number of water molecules attached or the radius of the hydrated ion give inconsistent results, since each method requires a different amount of time for the water

Table A6 Hydration numbers and hydrated radii of some hydrated ions

Ion	$\frac{Z^2}{r}$	Hydration number	Hydrated radius (Pm)
Cs^+	0.0055	6	228
K^+	0.0066	7	232
Na^+	0.0088	13	276
Li^+	0.0111	22	340
Ba^{2+}	0.0268	28	
Sr^{2+}	0.0303	29	
Ca^{2+}	0.0351	29	
Mg^{2+}	0.0465	36	
Cd^{2+}	0.0549	39	
Zn^{2+}	0.0599	44	

molecules to remain attached in order to be counted. But nearly all measurements give consistent trends in hydration numbers among series of ions such as the results for the ions in Table A6 obtained from transference measurements by way of *Stokes radii* or *hydrated radii* of hydrated ions.

Metal ions can migrate across cell membranes which consist largely of lipids in which metal ions are quite insoluble. One model of this process is called the *pre-model.* The cell membrane contains pores through which hydrated ions can pass without having to dissolve in the lipids. The antibiotic gramicidin A in fact contains a hollow centre of about 400-pm diameter that when in biological membrane causes cells to pass ions readily. The relative rate of passage of ions through this pore is $Cs^+ > Rb^+ > K^+ > Na^+ > Li^+$ which indicates that the hydrated cesium ion is indeed the smallest hydrated ion in this series.

Appendix II

Strength of Acids

(i) Binary hydrogen compounds (Hydraacids): In the periodic table, acidity among binary hydrogen compounds increases from left to right.

$NH_3 << H_2O << HF; PH_3 << H_2S << HCl$

Along a group, acidity increases as one passes downward. Thus

$H_2O < H_2S < H_2Se < H_2Te$; (or) $HF < HCl < HBr < HI$

The increase in acidity along a period may be explained from the increasing electronegativity of the elements linked to hydrogen. Here size of atom does not change appreciably from one element to the next; the more electronegativity of an element makes proton release easily due to greater shift of the bonding electron cloud towards it. So, the activity increases.

The consideration of electronegativity alone does not explain the increase of acidity down a group. The consideration of electronegativity leads to opposite expectations. In a group the atomic size increases rapidly from one element to the next and upsets the electronegativity factor. As the size of atoms increases, the bond with hydrogen becomes weaker (due to poorer overlap of atomic orbital of bigger atom with the small orbital of hydrogen). This causes the increase in acid strength which parallels the decrease in bond strength.

For example, nitrogen and chlorine have almost equal electronegativity. Yet, the hydrides NH_3 and HCl are quite opposite in their acid–base behaviour in water. We may understand this difference from a comparison of enthalpy changes (ΔH) for dissolving the gaseous hydrides in water. This can be calculated by using Born–Haber cycle as follows.

$$\begin{tikzcd}
HX(g) + H_2O(l) \arrow[r, "\Delta H"] \arrow[d, "D"] & H_3O^+_{(aq)} & X^-_{(aq)} \\
H(g) + X(g) + H_2O(l) \arrow[r, "I_H - EA_X"] & H^+(g) \arrow[u, "\Delta H_{hyd}"] & X^-_{(g)} + H_2O(l) \arrow[u, "\Delta H_{hyd}"]
\end{tikzcd}$$

$$\Delta H = D + I_H - EA_x + \Delta H_{hyd}(H^+) + \Delta H_{hyd}(X^-)$$

$$I_H = +1312\ \text{kJ mol}^{-1}; \Delta H_{hyd}(H^+) = 1130\ \text{kJ mol}^{-1}$$

For HCl (X - Cl)

$$D = 431; EA(Cl) = 349; \Delta H_{hyd}(Cl^-) = -322\ (\text{all kJ mol}^{-1})$$

$$\therefore \Delta H = 431 + 1312 - 349 - 1130 - 322 = -58\ \text{kJ mol}^{-1}$$

For NH_3($X = NH_2$)

$$D = 389; EA(NH_2) = 71; \Delta H_{hyd}(NH_2^-) = -460\ (\text{all kJ mol}^{-1})$$

$$\therefore \Delta H = 389 + 1312 - 71 - 1130 - 460 = +40\ \text{kJ mol}^{-1}$$

In both cases ΔS is negative due to higher ordering in solution but their contributions (in Joules) are expected to be similar. Hence the ΔG terms for the processes may be compared from the ΔH values calculated above. We find that dissolution of NH_3(g) in water to furnish $H^+_{(aq)}$ is favoured by the bond dissociation energy (D) and enthalpy of hydration of NH_2^- (g) but the very low e.a. value of NH_3(g) makes ΔH positive. Similarly the strengths of halogen hydracids (HX) in water in terms of energy changes may be compared (see Chapter 11, Halogens).

(ii) Oxoacids: For oxoacids of the type X – O – H, acidity decreases as the element X appears lower in a group of the periodic table. For example, the strength of oxoacids of the halogens decreases as HOCl > HOBr > HOI.

The acidic hydrogen is linked to the halogen atom via oxygen and the acid strength is determined by the ease of ionization of O–H bond. The electronegativity of the atom X plays the important role in the matter, by pulling the charge cloud of the X–O bond to itself, the effect is now relayed by the oxygen atom to the O–H bond (inductive effect) facilitating release of the acidic hydrogen. This is why acidity decreases down the group with decreasing electronegativity of the element X.

Acid	Order of K_a	Acid	Order of K_a
HOC*l*	10^{-8}	H_2SO_3	$K_1 = 1.6 \times 10^{-2}$; $K_2 = 1 \times 10^{-7}$
HOBr	10^{-9}	H_2SeO_3	$K_1 \sim 3.5 \times 10^{-3}$; $K_2 \sim 5 \times 10^{-8}$
HOI	10^{-13}	H_2TeO_3	$K_1 \sim 3.0 \times 10^{-3}$; $K_2 \sim 2 \times 10^{-8}$

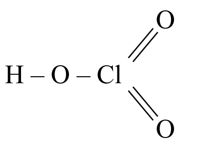
When the element X has additional oxygen atoms attached to it, inductive effect of these oxygen atoms makes the acid stronger—acidity increases with the number of oxygen atoms.

H – O – Cl

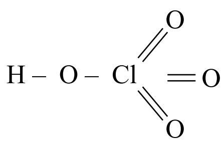
O.S. of Cl = +1
Ka = 10^{-8}

H – O – Cl = O

O.S. of Cl = +3
Ka = 10^{-2}



O.S. of Cl = +5
Ka = large



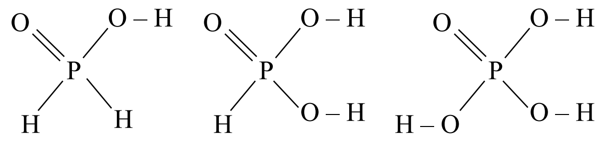
O.S. of Cl = +7
Ka = Very large

Similarly nitric acid (HO–NO_2) is stronger than nitrous acid (HO–NO) and sulphuric acid (HO · SO_2 · OH) is stronger acid than sulphurous acid (HO · SO · OH) since nitrogen and sulphur atoms in nitric acid and sulphuric acid are having more coordinated oxygen atoms.

Similarly, an oxoacid will be stronger proton donor when the rest of the molecule exerts greater electron withdrawing role. This is favoured by

(i) A high formal oxidation state of the central atom.
(ii) Presence of a large number of terminal O-atoms or other electronegative atoms like F or Cl.

If the number of unprotonated (terminal) oxygen atoms are equal in number but with increase in the number of –OH groups the acidic strength decreases. For example, in the oxoacids of phosphorous H_3PO_2, H_3PO_3 and H_3PO_4 the acidic character decreases in the order $H_3PO_2 > H_3PO_3 \geq H_3PO_4$. The structures of these acids are



H_3PO_2
$pK_a = 1.244$

H_3PO_3
pK_{a_1} **= 1.257**
pK_{a_2} **= 6.7**

H_3PO_4
pK_{a_1} **= 2.15**
pK_{a_2} **= 7.20**
pK_{a_3} **= 12.37**

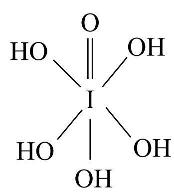
Acidity arises from the –O–H group, one in H_3PO_2, two in H_3PO_3 and three in H_3PO_4. The inductive effect of the single oxygen atom acts on the single –O–H bond in H_3PO_2, but it is distributed over two and three –O–H bonds in H_3PO_3 and in H_3PO_4, respectively. Accordingly, acidity decreases from H_3PO_2 to H_3PO_3.

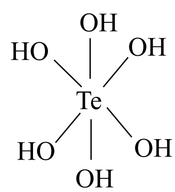
In the case of oxoacids containing the same number of –O–H groups and same number of unprotonated (terminal) oxygen atoms the acidic strength should increase with the decrease in electronegativity of central atom as explained.

So, among $HClO_3$, $HBrO_3$ and HIO_3 the ionic character of O–H group must be more in HIO_3 and least in $HClO_3$. So, the expected order of relative strength is $HOClO_2 < HOBrO_2 < HOIO_2$. But surprisingly it was found that both chloric and bromic acids are strong acids of nearly equal strength with $pK_a \leq O$, whereas iodic acid is slightly weaker with $pK_a = 0.804$. In concentrated solutions of HIO_3, the iodate ions formed by deprotonation react with undissociated acid according to equilibrium.

$$IO_3^- + HIO_3 \rightleftharpoons [H(IO_3)_2]^-; K_a = 41 \text{mol}^{-1}$$

In the case of perhalic acids due to bigger size and more positive charge developed on iodine, it allows more coordination number. So, HIO_4 in aqueous solution exists as hydrated H_5IO_6 causing the decrease in unprotonated oxygen atoms.





So, its acidic strength becomes less than perchloric acid. Similarly among H_2SO_4, H_2SeO_4 and H_2TeO_4, the

H_2TeO_4 becomes a weaker acid as it exists as H_6TeO_6 in water due to hydration.

Several attempts have been made to correlate the pK_a values of oxoacids of the formula type $O_nX(OH)_m$ with the number of –OH groups and unprotonated oxygen atoms (n). Among these, the simplest is due to Pauling. According to Pauling the strengths of mononuclear acids can be predicted basing on the following two rules.

(i) For an oxoacid of type $O_n X (OH)_m$, $pK_a \simeq 9 - 7n$.

Subsequently, it was found that the relation pKa = 8 – 5n fits better with some acids.

(ii) For polyprotic acids (m > 1), the sucessive pK_a values increase by five units.

Estimated pK_a values for different oxoacids (experimental values in parentheses).

n	$pK_a = 9 - 7n$	$pK_a = 8 - 5n$	Examples
0	9	8	HOCl (72); $Si(OH)_4$ (10)
1	2	3	H_2CO_3 (3–6); $HClO_2$(2.0)
	$pK_{a_1} = 2$	$pK_{a_1} = 3$	H_3PO_4 (2.15; 7.20; 12.37)
	$pK_{a_2} = 7$	$pK_{a_2} = 8$	H_3PO_3 (1.26; 6.7)
	$pK_{a_3} = 12$	$pK_a = 13$	H_3PO_2 (1.24)
2	–5	–2	HNO_3 (–1.4); H_2SO_4 (–2.0; 1.9)
			$HClO_3$ (–1.0)
3	–12	–7	$HClO_4$ (–10)

The importance of electronegativity is inherent in all these approaches. The highly electronegative unprotonated oxygen(s) draws electrons always from the central atom and thereby increases the positive charge on the central atom. This enhanced positive charge in turn draws the electrons from the $^-$OH bond and facilitates the release of H^+. Thus the acid strength of a molecule increases with increase in the number of unprotonated oxygen atoms. But in the case of oxyacids containing a large central atom with more positive charge, such as H_2TeO_4 or HIO_4 the central atom allows more coordination number by hydration. This decreases the number of unprotonated oxygen atoms resulting in the decrease of inductive effect of unprotonated oxygen atoms causing the decrease in the strength of acid.

Similar conclusions are also reached if we consider the conjugate base. Unprotonated oxygen atom help delocalization of the negative charge on the conjugate base and stabilize it, making it a weak base.

So far we have seen the acidity or basicity of a Bronsted acid (or base) in solution. A comparison of acid–base strength of Bronsted acids or bases in the gas phase is provided by the gas phase proton affinity. Analogous to electron affinity, the proton affinity (P. A.) is the energy released when a proton is added to an isolated atom or molecule (or ion).

$$B(g) + H^+(g) \longrightarrow BH^+(g)$$

The thermodynamic quantity of proton attachment enthalpy is equal to P. A. and is equal to ΔH for the reverse process.

$$BH^+(g) \longrightarrow B(g) + H^+(g)$$

The proton affinity is thus a measure for both the acidity of BH^+ and basicity of B in the gas phase. A large proton affinity means that removal of the hydrogen ion from $BH^+(g)$ is difficult, that is, B(g) is a stronger base or BH^+ is a weak acid.

Some trends of gas phase acidic (and basic) character are interesting, and therefore some of the examples may be taken for comparing their relative strengths. The proton affinities (kJ mol^{-1}) are given in parenthesis.

(i) C_2H_5OH (797) > CH_3OH (777)

(ii) $CHCl_3 > CHF_3$ or CCl_3^- (1515) < CF_3^- (1572)

(iii) CH_3OH (777) > H_2O (696)

(iv) CH_3SH (793) > H_2S (739)

We may expect that the acidic character increases with substitution by a more electron withdrawing group. For a species HXR_n (R = substituent), the process

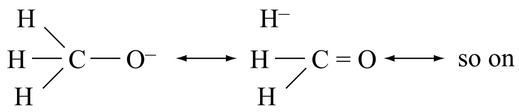
$$HXR_n(g) \longrightarrow H^+(g) + XR_n^-(g)$$

should be favoured by increasing electronegativity of R-groups, i.e., increasing positive charge on X. But this consideration alone cannot explain the acidity trends discussed so far. The energy of the ionization process, as written above, is nearly the coulombic interaction energy between the proton and the anion. The actual energy term involves a second (hypothetical) factor—electronic relaxation of the anion XR_n^- and this involves redistribution of charge within the anion. The more readily the substituents can delocalize the negative charge on the anion, the greater will be its stability. This electronic relaxation, usually exothermic, will make a definite contribution to the total energy of the ionization process. Combination of these two effects now help us to understand the acidity trends given above.

(i) As the size of the alkyl group increases, it becomes more polarizable. This releases higher electronic relaxation energy in $C_2H_5O^-$ and makes it more stable. Hence C_2H_5OH is stronger acid than CH_3OH.

(ii) In the case of CCl_3^- and CF_3^- ions the negative formal charge on the carbon is extensively delocalized on the larger CCl_3^- than in the smaller CF_3^-. This increases the electronic relaxation energy and hence offsets the trend expected from electronegativity alone.

(iii) In CH_3OH, there is greater electronic relaxation energy of CH_3O^- by way of hyperconjugation.



This effect must overcome the greater inductive effect of the methyl group, as evident from the acidity trend. But for CH_3SH, the sizes of carbon and sulphur are greatly dissimilar, and the delocalization is not sufficient to overcome the greater electron donating ability of the methyl group. Therefore the electrostatic energy term dominates over the small electronic relaxation energy.

Gas phase strengths also show trends different from those observed in aqueous solution. Thus, LiOH, NaOH and KOH are equally strong bases in water. But in gas phase, LiOH < NaOH < KOH; therefore, this order follows the electron releasing tendency of the cation to bind an extra proton.

Gas phase proton affinities are of limited applicability because they do not take into account the interaction with solvent, a factor that is highly significant in common reactions. In fact, gas phase proton affinity data suggest that transfer of a hydrogen ion from HCl(g) to $H_2O(g)$ would be highly endothermic and thus thermodynamically unfavourable.

$$HCl(g) + H_2O(g) \longrightarrow H_3O^+(g) + Cl^-(g);$$
$$\Delta H = 1393 - 686 = +707 \text{ kJ mol}^{-1}$$

Appendix III

Atomic Weights

Name	Symbol	Atomic number	Atomic weight
Actinium	Ac	89	(227)
Aluminum	Al	13	26.981539
Americium	Am	95	(243)
Antimony	Sb	51	121.757
Argon	Ar	18	39.948
Arsenic	As	33	74.92159
Astatine	At	85	(210)
Barium	Ba	56	137.327
Berkelium	Bk	97	(247)
Beryllium	Be	4	9.012182
Bismuth	Bi	83	208.98037
Boron	B	5	10.811
Bromine	Br	35	79.904
Cadmium	Cd	48	112.411
Calcium	Ca	20	40.078
Californium	Cf	98	(251)
Carbon	C	6	12.001
Cerium	Ce	58	140.115
Caesium	Cs	55	132.90543
Chlorine	Cl	17	35.4527
Chromium	Cr	24	51.9961
Cobalt	Co	27	58.93320
Copper (Cuprum)	Cu	29	63.546
Curium	Cm	96	(247)
Dysprosium	Dy	66	162.50
Einsteinium	Es	99	(252)
Erbium	Er	68	167.26
Europium	Eu	63	151.965
Fermium	Fm	100	(257)
Fluorine	F	9	18.9984032
Francium	Fr	87	(223)
Gadolinium	Gd	64	157.25
Gallium	Ga	31	69.723
Germanium	Ge	32	72.61
Gold (Aurum)	Au	79	196.96654
Hafnium	Hf	72	178.49
Helium	He	2	4.002602
Holmium	Ho	67	164.93032
Hydrogen	H	1	1.00794
Indium	In	49	114.82
Iodine	I	53	126.90447
Iridium	Ir	77	192.22
Iron (ferrum)	Fe	26	55.847
Krypton	Kr	36	83.80
Lanthanum	La	57	138.9055
Lawrencium	Lr	103	(262)
Lead (plumbum)	Pb	82	207.2
Lithium	Li	3	6.941
Lutetium	Lu	71	174.967
Magnesium	Mg	12	24.3050
Manganese	Mn	25	54.93805
Mendelevium	Md	101	(258)
Mercury	Hg	80	200.59
Molybdenum	Mo	42	95.94
Neodymium	Nd	60	144.24
Neon	Ne	10	20.1797
Neptunium	Np	93	(237)
Nickel	Ni	28	58.6934

Name	Symbol	Atomic number	Atomic weight
Niobium	Nb	41	92.90638
Nitrogen	N	7	14.00674
Nobilium	No	102	(259)
Osmium	Os	76	190.2
Oxygen	O	8	15.9994
Palladium	Pb	46	106.42
Phosphorus	P	15	30.973762
Platinum	Pt	78	195.08
Plutonium	Pu	94	(244)
Polonium	Po	84	(209)
Potassium	K	19	39.0983
Praseodymium	Pr	59	140.90765
Promethium	Pm	61	(145)
Protactinium	Pa	91	231.03588
Radium	Ra	88	(226)
Radon	Rn	86	(222)
Rhenium	Re	75	186.207
Rhodium	Rh	45	102.90550
Rubidium	Rb	37	85.4678
Ruthenium	Ru	44	101.07
Samarium	Sm	62	150.36
Scandium	Sc	21	44.955910
Selenium	Se	34	78.96
Silicon	Si	14	28.0855
Silver (argentum)	Ag	47	107.8682
Sodium	Na	11	22.989768
Strontium	Sr	38	87.62
Sulfur	S	16	32.066
Tantalum	Ta	73	180.9479
Technetium	Tc	43	(98)
Tellurium	Te	52	127.60
Terbium	Tb	65	158.92534
Thallium	T*l*	81	204.3833
Thorium	Th	90	232.0381
Thulium	Tm	69	168.93421
Tin (stannum)	Sn	50	118.710
Titanium	Ti	22	47.88
Tungsten (wolfram)	W	74	183.85
Unnilennium	Une	109	(267)
Unnilhexium	Unh	106	(263)
Unniloctium	Uno	108	(265)
Unnilpentium	Unp	105	(262)
Unnilquadrum	Unq	104	(261)
Unnilseptium	Uns	107	(262)
Uranium	U	92	238.0289
Vanadium	V	23	50.9415
Xenon	Xe	54	131.29
Ytterbium	Yb	70	173.04
Yttrium	Y	39	88.90585
Zinc	Zn	30	65.39
Zirconium	Zr	40	91.224

Appendix IV

Electron Affinities (KJ mol^{-1})

Z	Element	Value
1	H	72.775
2	He	0
3	Li	59.63
4	Be	0
5	B	26.7
6	C	153.89
7	$N \rightarrow N^{1-}$	7
	$N^{1-} \rightarrow N^{2-}$	–673
	$N^{2-} \rightarrow N^{3-}$	–1070
8	$O \rightarrow O^{1-}$	140.986
	$O^{1-} \rightarrow O^{2-}$	–744
9	F	328.0
10	Ne	0
11	Na	52.871
12	Mg	0
13	Al	42.5
14	Si	133.6
15	$P \rightarrow P^{1-}$	72.02
	$P^{1-} \rightarrow P^{2-}$	–468
	$P^{2-} \rightarrow P^{3-}$	–886
16	$S \rightarrow S^{1-}$	200.42
	$S^{1-} \rightarrow S^{2-}$	–456
17	C*l*	349.0
18	Ar	0
19	K	48.387
20	Ca	0
21	Sc	18.1
22	Ti	7.62
23	V	50.6
24	Cr	64.26

Z	Element	Value
25	Mn	0
26	Fe	15.7
27	Co	63.7
28	Ni	111.5
29	Cu	118.4
30	Zn	0
31	Ga	28.9
32	Ge	119.0
33	$As \rightarrow As^{1-}$	78
	$As^{1-} \rightarrow As^{2-}$	–435
	$As^{2-} \rightarrow As^{3-}$	–802
34	$Se \rightarrow Se^{1-}$	194.980
	$Se^{1-} \rightarrow Se^{2-}$	–410
35	Br	324.6
36	Kr	0
37	Rb	46.887
38	Sr	0
39	Y	29.6
40	Zr	41.1
41	Nb	86.1
42	Mo	71.9
43	Tc	53
44	Ru	101.3
45	Rh	109.7
46	Pd	53.7
47	Ag	125.6

Z	Element	Value
48	Cd	0
49	In	28.9
50	Sn	107.3
51	Sb	103.2
52	Te	190.16
53	I	295.18
54	Xe	0
55	Cs	45.509
56	Ba	0
57	La	48
58–71	Ln	50
72	Hf	0
73	Ta	31.06
74	W	78.63
75	Re	14.47
76	Os	106.1
77	Ir	151.0
78	Pt	205.3
79	Au	222.76
80	Hg	0
81	Tl	19.2
82	Pb	35.1
83	Bi	91.2
84	Po	183.3
85	At	270.1
86	Rn	0

Appendix V

Ionization Energies (MJ mol^{-1})

To obtain values in electron volts multiply table values by 10.364

Z	Element	I	II	III	IV	V	VI	VII	VIII	IX	X
1	H	1.3120									
2	He	2.3723	5.2504								
3	Li	0.5203	7.2981	11.8149							
4	Be	0.8995	1.7571	14/8487	21.0065						
5	B	0.8006	2.4270	3.6598	25.0257	32.8266					
6	C	1.0864	2.3526	4.6205	6.2226	37.8304	47.2769				
7	N	1.4023	2.8561	4.5781	7.4751	9.4449	53.2664	64.3598			
8	O	1.3140	3.3882	5.3004	7.4693	10.9895	13.3264	71.3345	84.0777		
9	F	1.6810	3.3742	6.0504	8.4077	11.0227	15.1640	17.8677	92.0378	106.4340	
10	Ne	2.0807	3.9523	6.122	9.370	12.178	15.238	19.999	23.069	115.3791	131.4314
11	Na	0.4958	4.5624	6.912	9.544	13.353	16.610	20.115	25.490	28.934	141.3626
12	Mg	0.7377	1.4507	7.7328	10.540	13.628	17.995	21.704	25.656	31.643	35.462
13	Al	0.5776	1.8167	2.7448	11.578	14.831	18.378	23.295	27.459	31.861	38.457

Z	Element	I	II	III	IV	V	VI	VII	VIII	IX	X
14	Si	0.7865	1.5771	3.2316	4.3555	16.091	19.785	23.786	29.252	33.877	38.733
15	P	1.0118	1.9032	2.912	4.957	6.2739	21.269	25.397	29.854	35.867	40.959
16	S	0.9996	2.251	3.361	4.564	7.013	8.4956	27.106	31.670	36.578	43.138
17	Cl	1.2511	2.297	3.822	5.158	6.54	9.362	11.0182	33.605	38.598	43.962
18	Ar	1.5205	2.6658	3.931	5.771	7.238	8.7810	11.9952	13.8417	40.760	46.187
19	K	0.4189	3.0514	4.411	5.877	7.976	9.649	11.343	14.942	16.964	48.576
20	Ca	0.5898	1.1454	4.9120	6.474	8.144	10.496	12.32	14,207	18.192	20.3849
21	Sc	0.631	1.235	2.389	7.089	8.844	10.72	13.32	15.31	17.370	21.741
22	Ti	0.658	1.310	2.6525	4.1746	9.573	11.517	13.59	16.26	18.64	20.833
23	V	0.650	1.414	2.8280	4.5066	6.299	12.362	14.489	16.760	19.86	22.24
24	Cr	0.6528	1.496	2.987	4.74	6.69	8.738	15.54	17.82	20.19	23.58
25	Mn	0.7174	1.5091	3.2484	4.94	6.99	9.2	11.508	18.956	21.40	23.96
26	Fe	0.7594	1.561	2.9574	5.29	7.24	9.6	12.1	14.575	22.678	25.29
27	Co	0.758	1.646	3.232	4.95	7.67	9.84	12.4	15.1	17.959	26.6
28	Ni	0.7367	1.7530	3.393	5.30	7.28	10.4	12.8	15.6	18.6	21.66
29	Cu	0.7455	1.9579	3.554	5.33	7.71	9.94	13.4	16.0	19.2	22.4
30	Zn	0.9064	1.7333	3.8327	5.73	7.97	10.4	12.9	16.8	19.6	23.0
31	Ga	0.5788	1.979	2.963	6.2						
32	Ge	0.7622	1.5372	3.302	4.410	9.02					
33	As	0.944	1.7978	2.7355	4.837	6.043	12.31				
34	Se	0.9409	2.045	2.9737	4.1435	6.59	7.883	14.99			
35	Br	1.1399	2.10	3.5	4.56	5.76	8.55	9.938	18.60		
36	Kr	1.3507	2.3503	3.565	5.07	6.24	7.57	10.71	12.2	22.28	
37	Rb	0.4030	2.633	3.9	5.08	6.85	8.14	9.57	13.1	14.5	26.74
38	Sr	0.5495	1.0643	4.21	5.5	6.91	8.76	10.2	11.80	15.6	17.1
39	Y	0.616	1.181	1.980	5.96	7.43	8.97	11.2	12.4	14.11	18.4
40	Zr	0.660	1.267	2.218	3.313	7.86					
41	Nb	0.664	1.382	2.416	3.69	4.877	9.900	12.1			
42	Mo	0.6850	1.558	2.621	4.477	5.91	6.6	12.23	14.8		
43	Tc	0.702	1.472	2.850							
44	Ru	0.711	1.617	2.747							
45	Rh	0.720	1.744	2.997							
46	Pd	0.805	1.875	3.177							
47	Ag	0.7310	2.074	3.361							
48	Cd	0.8677	1.6314	3.616							
49	In	0.5583	1.8206	2.705	5.2						
50	Sn	0.7086	1.4118	2.9431	3.9303	6.974					
51	Sb	0.8316	1.595	2.44	4.26	5.4	10.4				
52	Te	0.8693	1.79	2.698	3.610	5.669	6.82	13.2			
53	I	1.0084	1.8459	3.2							
54	Xe	1.1704	2.046	3.10							
55	Cs	0.3757	2.23								
56	Ba	0.5029	0.96526								
57	La	0.5381	1.067	1.8503	4.820						
58	Ce	0.528	1.047	1.949	3.543						
59	Pr	0.523	1.018	2.086	3.761	5.552					
60	Nd	0.530	1.034	2.13	3.900	5.790					
61	Pm	0.536	1.052	2.15	3.97	5.953					

Z	Element	I	II	III	IV	V	VI	VII	VIII	IX	X
62	Sm	0.543	1.068	2.26	4.00	6.046					
63	Eu	0.547	1.085	2,40	4.11	6.101					
64	Gd	0.592	1.17	1.99	4.24	6.249					
65	Tb	0.564	1.112	2.11	3.84	6.413					
66	Dy	0.572	1.126	2.20	4.00	5.990					
67	Ho	0.581	1.139	2.20	4.10	6.169					
68	Er .	0.589	1.151	2.19	4.11	6.282					
69	Tm	0.5967	1.163	2.284	4.12	6.313					
70	Yb	0.6034	1.175	2.415	4.22	6.328					
71	Lu	0.5235	1.34	2.022	4.36	6.445					
72	Hf	0.654	1.44	2.25	3.21	6.596					
73	Ta	0.761									
74	W	0.770									
75	Re	0.760									
76	Os	0.84									
77	Ir	0.88									
78	Pt	0.87	1.7911								
79	Au	0.8901	1.98								
80	Hg	1.0070	1.8097	3.30							
81	Tl	0.5893	1.9710	2.878							
82	Pb	0.7155	1.4504	3.0815	4.083	6.64					
83	Bi	0.7033	1.610	2.466	4.37	5.40	8.62				
84	Po	0.812									
85	At										
86	Rn	1.0370									
87	Fr										
88	Ra	0.5094	0.97906								
89	Ac	0.49	1.17								
90	Th	0.59	1.11	1.93	2.78						
91	Pa	0.57									
92	U	0.59									
93	Np	0.60									
94	Pu	0.585									
95	Am	0.578									
98	Cm	0.581									
97	Bk	0.601									
98	Cf	0.608									
99	Es	0.619									
100	Fm	0.627									
101	Md	0.635									
102	No	0.642									

Appendix VI

UNITS AND CONVERSION FACTORS

The International System of Units (SI)

SI Base Units

Physical quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	k
Amount of substance	mole	mol
Luminous intensity	candela	cd

Common Derived Units

Physical quantity	Unit	Symbol	Definition
Frequency	hertz	Hz	s^{-1}
Energy	joule	J	$kg\ m^2s^{-2}$
Force	newton	N	$J\ m^{-1}$
Pressure	pascal	Pa	Nm^{-2}
Power	watt	W	Js^{-1}
Electric charge	coulomb	C	As
Electric potential difference	volt	V	$J\ A^{-1}\ S^{-1}$
Electric resistance	ohm	Ω	VA^{-1}
Electric capacitance	farad	F	AsV^{-1}
Magnetic flux	weber	Wb	V s
Inductance	henry	H	$Vs\ A^{-1}$
Magnetic flux density	tesla	T	Vsm^{-2}

Prefixes

Prefix	Symbol	Multiply by
atto	a	10^{-18}
fempto	f	10^{-15}
pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}
deci	d	10^{-1}
deka	da	10
hecto	h	10^{2}
kilo	k	10^{3}
mega	M	10^{6}
giga	G	10^{9}
tera	T	10^{12}
peta	P	10^{15}
exa	E	10^{18}

Physical and Chemical Constants

Electronic charge	e	=	1.60217733	×	10^{-19}C
Planck constant	h	=	6.6260755	×	10^{-34} Js,
			6.6260755	×	10^{-27} erg s
Speed of light	c	=	2.997925458	×	10^{8} ms^{-1}
Rydberg constant	R	=	1.0973731534	×	10^{5} cm^{-1}
Boltzmann constant	k	=	1.380658	×	10^{-23} $J\ K^{-1}$
Gas constant	R	=	8.314570JK^{-1} mol^{-1}		
Avogadro number	N_A	=	6.0221367	×	10^{23} mol^{-1}
Faraday constant	*f*	=	9.6485309	×	10^{4}C mol^{-1}
Electronic rest mass	m_e	=	9.1093897	×	10^{-28} g mol^{-1}
Proton mass	m_p	=	1.672623	×	10^{-24} g
Bohr radius	a_0	=	52.91771249 pm		
Bohr magneton	μ_B	=	9.274096	×	$10^{-24}JT^{-1}$;
			9.274096	×	10^{-21} erg $gauss^{-1}$
Permittivity of vacuum	ε_0	=	8.854187817	×	10^{-12} C^2 m^{-1} J^{-1}
Pi	π	=	3.1415926536		
Base natural logarithms	e	=	2.71828		

Conversion Factors

Multiply length	By	To obtain
cm	10^{8}	Å
cm	10^{7}	nm
cm	10^{10}	pm
Å	100	pm
Energy		
kcal mol^{-1}	4.184	kJ mol^{-1}
eV	96.49	kJ mol^{-1}
erg	10^{-7}	J
wave numbers (cm^{-1})	1.1962×10^{-2}	kJ mol^{-1}
kJ mol^{-1}	83.59	cm^{-1}
eV	23.06	kcal mol^{-1}
Dipole moments		
Debye	3.336×10^{-30}	Cm
Cm	0.300×10^{30}	D
Pressure		
atmosphere	1.013×10^{5}	Pa
mm Hg (torr)	133.3	Pa
pascal	9.869×10^{-6}	atm
pascal	7.501×10^{-3}	mm Hg (torr)